

UTIs. In addition, males who require temporary catheterization following surgery can be given antibiotics to lower the risk of UTIs.

Resources

PERIODICALS

Harrar, Sari. "Bladder Infection Protection." *Prevention* November 2003: 174.

Jancin, Bruce. "Presumed Cystitis Well Managed Via Telephone: Large Kaiser Experience." *Family Practice News* November 1, 2003: 41.

Prescott, Lawrence M. "Presumed Quinolone Gets the Nod for Uncomplicated Cystitis." *Urology Times* November 2003: 11.

Rebecca J. Frey, PhD
Teresa G. Odle

Cystometry

Definition

Cystometry is a test of bladder function in which pressure and volume of fluid in the bladder is measured during filling, storage, and voiding.

Purpose

The urinary bladder stores urine produced by the kidneys. The main muscle of the bladder wall, the detrusor, relaxes to allow expansion of the bladder during filling. The urethra, the tube through which urine exits, is held closed by a ring of muscle, known as the urethral sphincter. As volume increases, stretching of the detrusor and pressure on the sphincter sends signals to the brain, indicating the need for urination, or voiding. Voluntary relaxation of the sphincter and automatic contractions of the detrusor allow successful and virtually complete voiding.

A cystometry study is performed to diagnose problems with urination, including incontinence, urinary retention, and recurrent urinary tract infections. Urinary difficulties may occur because of weak or hyperactive sphincter or detrusor, or incoordination of their two activities. Infection of the bladder or urethra may cause incontinence, as can obstruction of the urethra from scar tissue, prostate enlargement, or other benign or cancerous growths. Loss of sensation due to nerve damage can lead to chronic overfilling.

Precautions

The mild irritation of the urinary tract necessary for insertion of the catheter may occasionally cause flushing, sweating, and **nausea**.

Description

The patient begins by emptying the bladder as much as possible. A thin plastic catheter is then slowly inserted into the urethra until it reaches the bladder. Measurements are taken of the residual urine volume and bladder pressure. Pressure measurements may require a rectal probe to account for the contribution of the abdominal muscles to the pressure recording.

The bladder is then gradually filled with either warm water, room temperature water, saline solution, carbon dioxide gas, or a contrast solution for x-ray analysis, depending on the type of study being done. The patient is asked to describe sensations during filling, including temperature sensations and when the first feeling of bladder fullness occurs. Once the bladder is completely full, the patient is asked to begin voiding, and measurements are again made of pressure and volume, as well as flow rate and pressure.

Preparation

There is no special preparation needed for this test. The patient may be asked to stop taking certain medications in advance of the test, including sedatives, cholinergics, and anticholinergics.

Aftercare

Cystometry can be somewhat uncomfortable. The patient may wish to reserve an hour or so afterward to recover. Urinary frequency or urgency, and some reddening of the urine, may last for a day. Increasing fluid intake helps to flush out the bladder, but caffeinated, carbonated, or alcoholic beverages are discouraged, because they may irritate the bladder lining. Signs of infection, such as **fever**, chills, low back pain, or persistent blood in the urine, should be reported to the examining physician.

Risks

There is a slight risk of infection due to tearing of the urethral lining.

Normal results

The normal bladder should not begin contractions during filling and should initially expand without

KEY TERMS

Detrusor—Muscle of the bladder wall.

Sphincter—Ring of muscle between the bladder and the urethra that functions to close off the urethra.

Urethra—Tube that empties urine from the bladder to the exterior of the body.

resistance. A feeling of fullness occurs with a volume of 100–200 ml. The adult bladder capacity is 300–500 ml. The sphincter should relax and open when the patient wills it, accompanied by detrusor contractions. During voiding, detrusor contraction should be smooth and lead to a steady urine stream.

Abnormal results

Inability of the bladder to relax during filling, or low bladder volume, may indicate interstitial **cystitis**, prostate enlargement, or **bladder cancer**. Contraction of the bladder during filling may be due to irritation from infection or cysts, obstruction of the bladder outlet, or neurological disease such as stroke, **multiple sclerosis**, or spinal cord injury. Diminished sensation may occur with nerve lesions, **peripheral neuropathy**, or chronic overfilling.

Resources

OTHER

“Cystometrogram, Simple and Complex.” *HealthGatePage*.
< <http://www.healthgate.com/HealthGate/free/dph/static/dph.0085.shtml> > .

Richard Robinson

Cystoscopy

Definition

Cystoscopy (cystourethroscopy) is a diagnostic procedure that is used to look at the bladder (lower urinary tract), collect urine samples, and examine the prostate gland. Performed with an optic instrument known as a cystoscope (urethroscope), this instrument uses a lighted tip for guidance to aid in diagnosing urinary tract disease and prostate disease. Performed by a urologist, this surgical test also enables biopsies to

be taken or small stones to be removed by way of a hollow channel in the cystoscope.

Purpose

Categorized as an endoscopic procedure, cystoscopy is used by urologists to examine the entire bladder lining and take biopsies of any areas that look questionable. This test is not used on a routine basis, but may benefit the urologist who is needing further information about a patient who displays the following symptoms or diagnosis:

- blood in the urine (also known as hematuria)
- incontinence or the inability to control urination
- a urinary tract infection
- a urinary tract which display signs of congenital abnormalities
- tumors located in the bladder
- the presence of bladder or **kidney stones**
- a stiffness or strained feeling of the urethra or ureters
- symptoms of an **enlarged prostate**

Blood and urine studies, in addition to x rays of the kidneys, ureters and bladder may all occur before a cystoscopy. At the time of surgery, a retrograde pyelogram may also be performed. Additional blood studies may be needed immediately following surgery.

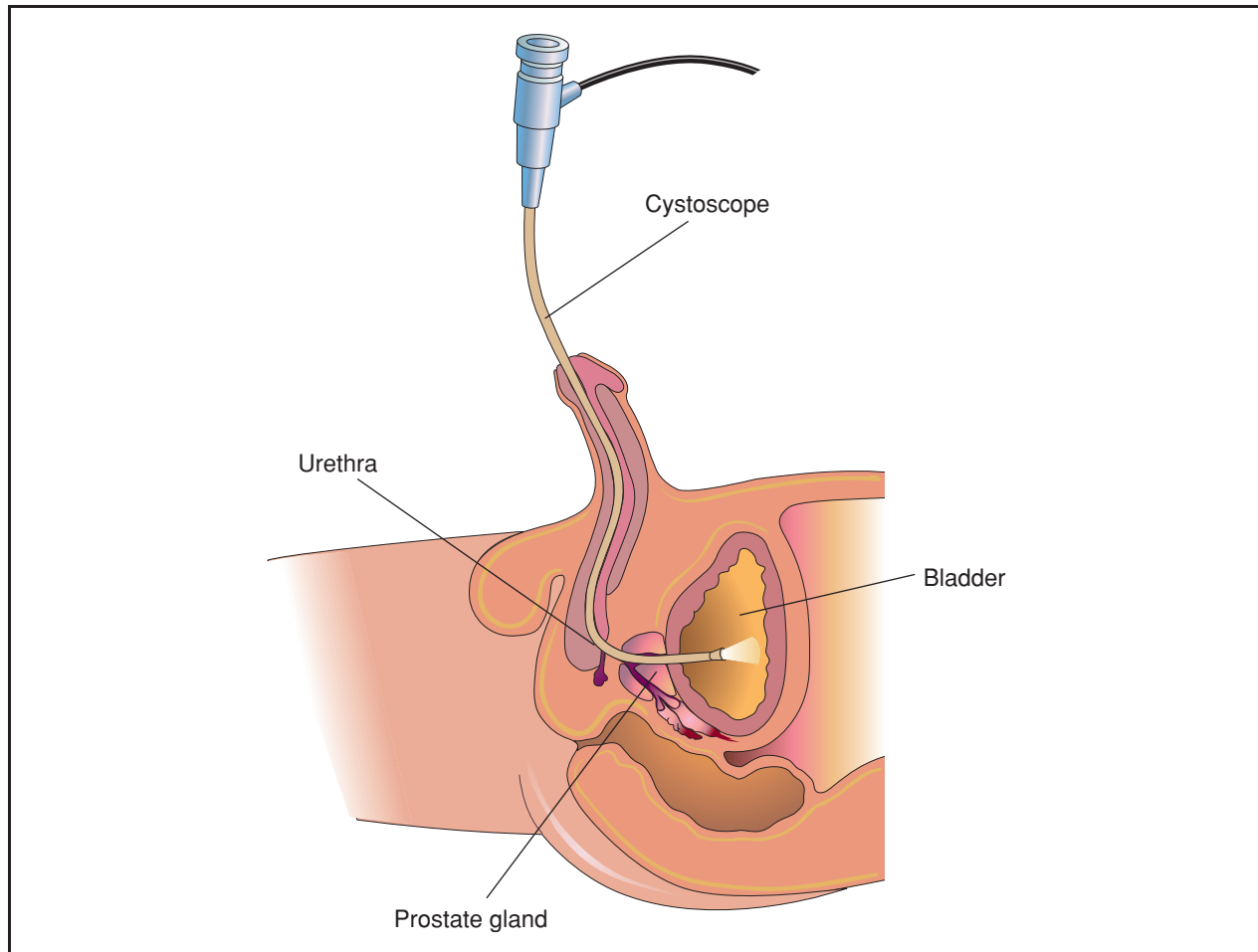
Precautions

While the cystoscopy procedure is commonly relied upon to gather additional diagnostic information, it is an invasive surgical technique that may involve risks for certain patients. Those who are extremely overweight (obese), smoke, are recovering from a recent illness, or are treating a chronic condition may face additional risks from surgery.

Surgical risk also increases in patients who are currently using certain drugs including antihypertensives; **muscle relaxants**; tranquilizers; sleep inducers; insulin; sedatives; **beta blockers**; or cortisone. Those who use mind-altering drugs also put themselves at increased risk of complications during surgery. The following mind-altering drugs should be avoided: **narcotics**; psychedelics; hallucinogens; marijuana; sedatives; hypnotics; or **cocaine**.

Description

Depending on the type of information needed from a cystoscopy, the procedure typically takes



Cystoscopy is a diagnostic procedure which is used to view the bladder, collect urine samples, and examine the prostate gland. This procedure also enables biopsies to be taken. The primary instrument used in cystoscopy is the cystoscope, a tube which is inserted through the penis into the urethra, and ultimately into the bladder. (Illustration by Electronic Illustrators Group.)

10–40 minutes to complete. The patient will be asked to urinate before surgery which allows an accurate measurement of the remaining urine in the bladder. A well-lubricated cystoscope is inserted through the urethra into the bladder where a urine sample is taken. Fluid is then pushed in to inflate the bladder and allow the urologist to examine the entire bladder wall.

During an examination, the urologist may take the following steps: remove either bladder or kidney stones; gather tissue samples; and treat any suspicious lesions. In order to perform x-ray studies (retrograde pyelogram), a harmless dye is injected into the ureters by way of a catheter that is passed through the previously placed cystoscope. After completion of all needed tests, the cystoscope is removed.

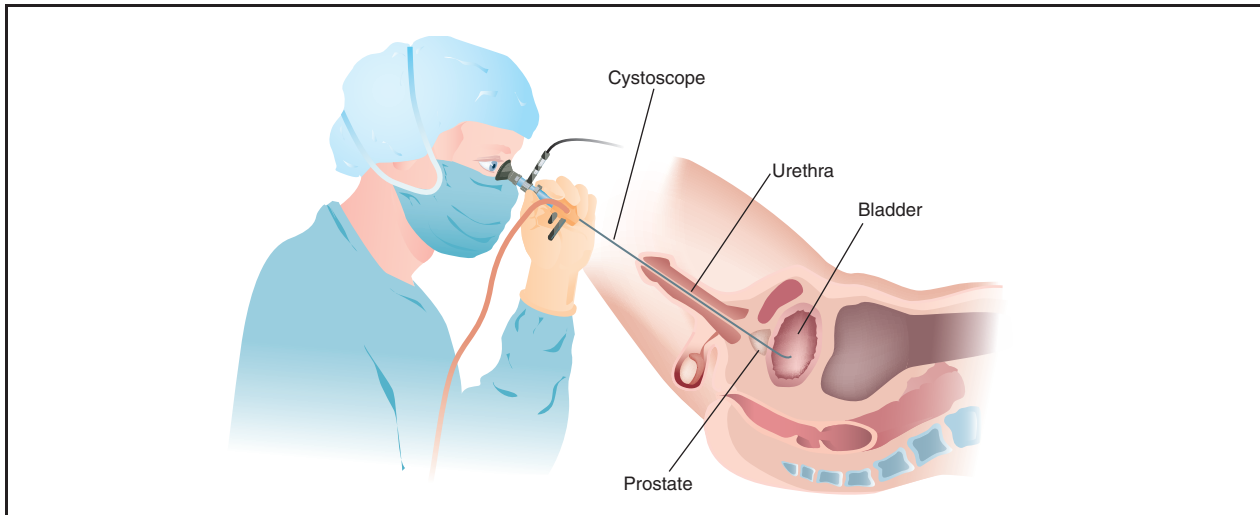
Preparation

As procedure that can be completed in a hospital, doctor's office, or outpatient surgical facility, an injection of spinal or **general anesthesia** may be used prior to a cystoscopy. While this test is typically performed on an outpatient basis, a patient may require up to three days of recovery in the hospital.

Aftercare

Patients who have undergone a cystoscopy will be instructed to follow these steps to ensure a quick recovery:

- due to soreness or discomfort that may occur in the urethra, especially while urinating, several warm baths a day are recommended to relieve any pain



A cystoscope helps the doctor examine the urethra, bladder, and prostate. (Illustration by Argosy Inc.)

- allow four days for recovery
- blood may appear in the urine—this is common, and soon clears up in one to two days following the procedure
- avoid strenuous **exercise** for a minimum of two weeks following surgery
- sexual relations may continue when the urologist determines that healing is complete
- wait at least two days after surgery before driving

Patients may also be prescribed **pain** relievers and **antibiotics** following surgery. Minor pain may also be treated with over-the-counter, non-prescription drugs such as **acetaminophen**.

Risks

As with any surgical procedure, there are some risks involved with a cystoscopy. Complications may include: profuse bleeding; a damaged urethra; a perforated bladder; a urinary tract infection; or an injured penis.

Patients should also contact their physician if they experience any of the following symptoms following surgery: pain, redness, swelling, drainage, or bleeding from the surgical site; signs of infection that may include **headache**, muscle aches, dizziness or an overall ill feeling and **fever**; **nausea** or **vomiting**; strenuous or painful urination; or symptoms that may result as side-effects from the medication.

Normal results

A successful cystoscopy includes a thorough examination of the bladder and collection of urine

KEY TERMS

Endoscopy—Examination of body organs or cavities through the use of an endoscope (a lighted optical instrument used to see inside body cavities), such as a cystoscope used to complete a cystoscopy.

Retrograde pyelogram—A pyelography or x-ray technique where radiopaque dye is injected into the kidneys from below, by way of the ureters, allowing further examination of the kidneys.

Ureter—The tube that carries urine from the kidney to the bladder, with each kidney having one ureter.

Urethra—A passageway from the bladder to the outside for the discharge of urine. In the female this canal lies between the vagina and clitoris; in the male the urethra travels through the penis, opening at the tip.

samples for cultures. If no abnormalities are seen, the results are indicated as normal.

Abnormal results

Cystoscopy allows the urologist to detect inflammation of the bladder lining, prostatic enlargement, or tumors. If these are seen, further evaluation or biopsies may be needed in addition to the removal of some tumors.

Resources

ORGANIZATIONS

American Cancer Society. 1599 Clifton Rd., NE, Atlanta, GA 30329-4251. (800) 227-2345. <<http://www.cancer.org>> .

Beth A. Kapes

Cystourethroscopy see **Cystoscopy**

Cytomegalic inclusion disease see
Cytomegalovirus infection

Cytomegalovirus antibody screening test

Definition

Cytomegalovirus (CMV) is a common human virus. Antibodies to CMV are evidence of a current or past infection.

Purpose

Consequences of a CMV infection can be devastating in a pregnant woman, a transplant patient, or a person with human **immunodeficiency** virus (HIV). Antibody screening helps control the infection risk for these groups.

In a healthy, nonpregnant person, CMV infection is almost never serious. Symptoms, if present, are mild, often resembling **infectious mononucleosis** due to Epstein-Barr virus. Antibody screening distinguishes between these two infections.

Description

When first exposed to CMV, a person's immune system is triggered and quickly makes antibodies to fight the virus. Antibodies are special proteins designed to attack and destroy foreign material, in this case, the cytomegalovirus.

The test combines a person's serum with a substance to which CMV antibodies attach. This antibody-antigen complex is measured and the amount of original antibody determined. If positive for antibodies, the serum is diluted, or titered, and the test repeated until the serum is so dilute it no longer gives a positive result. The last dilution that gives a positive result is the titer reported.

A test positive for CMV antibodies means the person has been infected with the virus, either currently or in the past; it does not mean the person has lifetime immunity. After an infection, this virus, like all members of the herpes virus group, can stay hidden inside a person and cause infection if the person's immune system later weakens and antibody protection decreases. In fact, reactivation of such hidden (or latent) infection is not at all uncommon and usually occurs without symptoms.

Transplant patients and people with weakened immune systems, including those with HIV, are vulnerable to infection from several routes, including from another person, from a donated organ or transfused blood, or from reactivation of a past infection. Before transplant, both the recipient and donor are usually tested for antibodies. A recipient who has never had CMV (negative for antibodies), should not receive an organ from a donor who has had CMV (positive for antibodies). CVM infection can be associated with organ rejection, or can cause illness such as pneumonia, hepatitis, or death. Similarly, blood is usually screened for CMV antibodies before being transfused into a person with a weakened immune system.

CMV infection is the most common congenital infection (existing at birth). The infection, passed from mother to baby, can cause permanent mental or physical damage, or **death**. The antibody screening test tells a woman whether or not she has antibody protection against the virus in case she is exposed during **pregnancy**. Pregnant women 25 years and older who are immune to CMV are much less likely to pass the virus to their babies than younger women who have never been exposed to CMV.

Tests that measure a specific type of antibody help tell the difference between a current and a past infection. Immunoglobulin M (IgM) antibodies appear at the beginning of an infection and last only weeks. Immunoglobulin G (IgG) antibodies appear 10–14 days later and can last a lifetime. A person suspected of having a current infection should be tested at the beginning of the infection and again 10–14 days later.

The CMV antibody screening test is also called the transplant reaction screening test. Results are usually available the following day.

It may be possible to test fetal blood for certain antibodies to CMV virus by drawing a blood sample from the umbilical cord. This may be an important test to add to prenatal care, since newborn babies with CMV often show no symptoms.

KEY TERMS

Antibody—A special protein built by the body as a defense against foreign material entering the body.

Cytomegalovirus (CMV)—A common human virus causing mild or no symptoms in healthy people, but permanent damage or death to an infected fetus, a transplant patient, or a person with HIV.

Titer—A dilution of a substance with an exact known amount of fluid. For example, one part of serum diluted with four parts of saline is a titer of 1:4.

Preparation

The adult CMV antibody screening test requires 5 mL of blood. Collection of the sample takes only a few minutes.

Aftercare

Discomfort or bruising may occur at the puncture site or the person may feel dizzy or faint. Pressure to the puncture site until the bleeding stops reduces bruising. Warm packs to the puncture site relieve discomfort.

Normal results

A person without previous exposure to CMV will test negative.

Abnormal results

The presence of antibodies means the person has been infected with CMV, either now or in the past. An antibody titer at least four times higher at the end of the illness than at the beginning, or the presence of IgM antibodies, indicates a recent or current first time infection.

People with weak immune systems may not generate antibodies against CMV. A current infection in a transplant patient or a person with HIV is confirmed with other tests, such as viral culture.

Resources

PERIODICALS

Fowler, Karen B., Sergio Stagno, and Robert F. Pass.

“Maternal Immunity and Prevention of Congenital Cytomegalovirus Infection.” *JAMA, The Journal of the American Medical Association* February 26, 2003: 1008.

Gerber, Stefan, et al. “Prenatal Diagnosis of Congenital Cytomegalovirus Infection by Detection of Immunoglobulin M Antibodies to the 70-d Heat Shock Protein in Fetal Serum.” *American Journal of Obstetrics and Gynecology* October 2002: 955.

Nancy J. Nordenson
Teresa G. Odle

Cytomegalovirus infection

Definition

Cytomegalovirus (CMV) is a virus related to the group of herpes viruses. Infection with CMV can cause no symptoms, or can be the source of serious illness in people with weak immune systems. CMV infection is also an important cause of birth defects.

Description

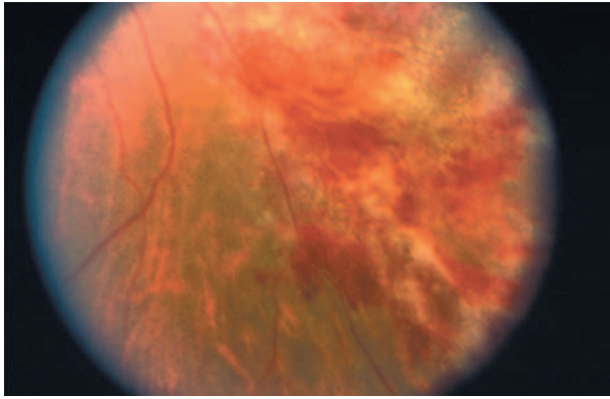
CMV is an extremely common organism worldwide. It is believed that about 85% of the adults in the United States have been infected by CMV at some point in their lives. CMV is found in almost all of the body's organs. It is also found in body fluids, including semen, saliva, urine, feces, breast milk, blood, and secretions of the cervix (the narrow, lower section of the uterus).

CMV is also able to cross the placenta (the organ that provides oxygen and nutrients to the unborn baby in the uterus). Because CMV can cross the placental barrier, initial infection in a pregnant woman can lead to infection of the developing baby.

Causes and symptoms

CMV is passed between people through contact with body fluids. CMV also can be passed through sexual contact. Babies can be born infected with CMV, either becoming infected in the uterus (congenital infection) or during birth (from infected cervical secretions).

Like other herpes viruses, CMV remains inactive (dormant) within the body for life after the initial infection. Some of the more serious types of CMV infections occur in people who have been harboring the dormant virus, only to have it reactivate when their immune system is stressed. Immune systems may be weakened because of **cancer chemotherapy**, medications given after organ transplantation, or diseases



An infected retina of an AIDS patient. Cytomegaloviruses are herpes viruses that can, among other problems, act as opportunistic infectious agents in suppressed immune systems, a common problem with AIDS sufferers. (Custom Medical Stock Photo. Reproduced by permission.)

that significantly lower immune resistance like acquired **immunodeficiency syndrome (AIDS)**.

In a healthy person, initial CMV infection often occurs without symptoms and is rarely noticed. Occasionally, a first-time infection with CMV may cause a mild illness called mononucleosis. Symptoms include swollen glands, liver, and spleen; **fever**; increased white blood cells; headache; fatigue; and **sore throat**. About 8% of all mononucleosis cases are due to CMV infection. A similar infection, though slightly more serious, may occur two to four weeks after receiving a blood **transfusion** containing CMV.

In people with weakened immune systems, CMV infection can cause more serious and potentially life-threatening illnesses. These illnesses include **pneumonia**, and inflammations of the liver (hepatitis), brain (**encephalitis**), esophagus (esophagitis), large intestine (colitis), and retina of the eye (retinitis).

Babies who contract CMV from their mothers during birth rarely develop any illness from these infections. Infants born prematurely who become CMV infected during birth have a greater chance of complications, including pneumonia, hepatitis, decreased blood platelets.

However, an unborn baby is at great risk for serious problems when the mother becomes infected with CMV for the first time while pregnant. About 10% of these babies will be born with obvious problems, including **prematurity**, lung problems, an enlarged liver and spleen, **jaundice**, anemia, low birth weight, small head size, and inflammation of the retina. About 90% of these babies may appear perfectly normal at birth. Unfortunately, about 20% will

KEY TERMS

Cervix—The narrowed, lowest part of the uterus through which a baby must pass in order to enter the birth canal.

Congenital—A condition that exists before birth and at birth.

Placenta—The organ that provides oxygen and nutrition from the mother to the unborn baby during pregnancy. The placenta is attached to the wall of the uterus and leads to the unborn baby via the umbilical cord.

later develop severe hearing impairments and **mental retardation**. A 2003 report found that pregnant women 25 years and older who are immune to CMV are much less likely to pass the virus to their babies than younger women who have never been exposed to CMV.

Diagnosis

Body fluids or tissues can be tested to reveal CMV infection. However, this information is not always particularly helpful because CMV stays dormant in the cells for life. Tests to look for special immune cells (antibodies) directed specifically against CMV are useful in proving that a person has been infected with CMV. However, these tests do not give any information regarding when the CMV infection first occurred.

Treatment

Ganciclovir and foscarnet are both antiviral medications that have been used to treat patients with weak immune systems who develop a serious illness from CMV (including retinitis). As of 1998, research was still being done to try to find useful drugs to treat newborn babies suffering from congenital infection with CMV. **Antiviral drugs** are not used to treat CMV infection in otherwise healthy patients because the drugs have significant side effects that outweigh their benefits. In 2003, researchers in Europe announced a new compound that appeared to be highly effective against CMV infections. The new drug acted earlier in the viral replication of the infection and showed promise, however, clinical trials were continuing.

Prognosis

Prognosis in healthy people with CMV infection is excellent. About 0.1% of all newborn babies will have

serious damage from CMV infection occurring while they were developing in the uterus. About 50% of all transplant patients will develop severe illnesses due to reactivation of dormant CMV infection. These illnesses have a high rate of serious complications and **death**.

Prevention

Prevention of CMV infection in the normal, healthy person involves good handwashing. Blood products can be screened or treated to insure that they do not contain CMV. In 2003, a new high-dose prophylactic (preventive) treatment was being tested to reduce CMV risk in stem cell transplant recipients.

Resources

PERIODICALS

Fowler, Karen B., Sergio Stagno, and Robert F. Pass.
“Maternal Immunity and Prevention of Congenital

Cytomegalovirus Infection.” *JAMA, The Journal of the American Medical Association* February 26, 2003: 1008.

“High-Dose Acyclovir May Reduce Cytomegalovirus Infection Risk.” *Virus Weekly* July 15, 2003: 16.

“Novel Compound Highly Effective Against Cytomegalovirus Infection.” *AIDS Weekly* November 25, 2003: 17.

ORGANIZATIONS

Baylor College of Medicine. 1 Baylor Plaza, Houston, TX 77030. (713) 798-4951. <<http://public.bcm.tmc.edu>> .

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>> .

March of Dimes Birth Defects Foundation. 1275 Mamaroneck Ave., White Plains, NY 10605. (914) 428-7100. resourcecenter@modimes.org. <<http://www.modimes.org>> .

Rosalyn Carson-DeWitt, MD
Teresa G. Odle

D

D & C see **Dilatation and curettage**

Dacryocystitis

Definition

Dacryocystitis is an inflammation of the tear sac (lacrimal sac) at the inner corner of the eye.

Description

Tears drain into little openings (puncta) in the inner corners of the eyelids. From there, the tears travel through little tube-like structures (canaliculi) to the lacrimal sac. The nasolacrimal ducts then take the tears from the lacrimal sac to the nose. That's why people need to blow their nose when they cry a lot.

Dacryocystitis is usually caused by a blockage of the nasolacrimal duct, which allows fluid to drain into the nasal passages. When the lacrimal sac does not drain, bacteria can grow in the trapped fluid. This condition is most common in infants and people over 40 years old.

Causes and symptoms

In newborn infants, the nasolacrimal duct may fail to form an opening—a condition called dacryostenosis. The cause of dacryocystitis in adults is usually associated with inflammation and infection in the nasal region. Dacryocystitis can be acute, having a sudden onset, or it can be chronic, with symptoms occurring over the course of weeks or months. Symptoms of acute dacryocystitis can include **pain**, redness, tearing, and swelling at the inner corner of the eye by the nose. In chronic dacryocystitis, the eye area may be swollen, watery or teary, and, when

pressure is applied to the area, there may be a discharge of pus or mucus through the punctum.

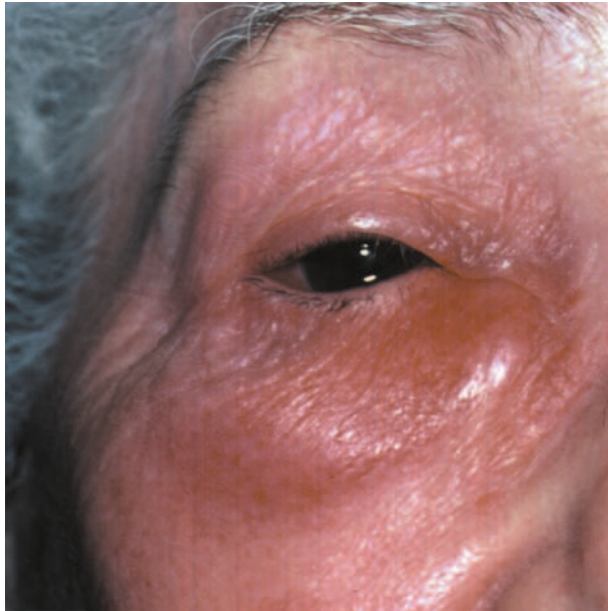
Diagnosis

Dacryocystitis usually occurs in only one eye. As mentioned, the symptoms can range from watery eyes, pain, swelling, and redness to a discharge of pus when pressure is applied to the area between the bridge of the nose and the inner eyelids. A sample of the pus may be collected on a swab or in a tube for laboratory analysis. The type of antibiotic and treatment may depend on which bacteria is present. In the acute form, a blood test may reveal an elevated white blood cell (WBC) count; with a chronic infection, the WBC count is usually normal. To identify the exact location of the blockage, an x ray can be taken after a dye is injected into the duct in a procedure called dacryocystography.

Treatment

A warm compress applied to the area can help relieve pain and promote drainage. Topical and oral **antibiotics** may be prescribed if an infection is present. Intravenous antibiotics may be needed if the infection is severe. In some cases, a tiny tube (cannula) is inserted into the tear duct which is then flushed with a sterile salt water solution (sterile saline). If other treatments fail to clear up the symptoms, surgery (dacryocystorhinostomy) to drain the lacrimal sac into the nasal cavity can be performed. In extreme cases, the lacrimal sac will be removed completely.

In infants, gentle massage of the lacrimal sac four times daily for up to nine months can drain the sac and sometimes clear a blockage. As the infant grows, the duct may open by itself. If the duct does not open, it may need to be dilated with a minor surgical procedure.



Dacryocystitis of the right eye. The inner corner of the lower lid is bulging from an inflamed tear sac. Blockage of the tear duct causes fluid to be trapped in the tear sac, which becomes infected. (Custom Medical Stock Photo. Reproduced by permission.)

KEY TERMS

Canaliculi—Also known as lacrimal ducts, these tube-like structures carry the tears from the eyes to the lacrimal sac.

Cannula—A narrow tube that can be inserted into a duct.

Dacryocystography—An x ray of the tear duct after injection of a dye that is used to help locate a blockage in the duct.

Dacryocystorhinostomy—A surgical procedure to drain the tear sac into the nasal passage.

Dacryostenosis—Obstruction or narrowing of the nasolacrimal duct. May be present at birth.

Nasoacrimonial duct—The tube that carries the tears from the lacrimal sac to the nose.

Punctum—Tiny opening at the inner corners of the upper and lower lids. The area for the beginning of tear drainage.

Prognosis

Treatment of dacryocystitis with antibiotics is usually successful in clearing the infection that is

present. If there is a permanent blockage that prevents drainage, infection may recur and surgery may be required to open the duct. If left untreated, the infected sac can rupture, forming an open, draining sore.

Prevention

There are no specific recommendations for the prevention of dacryocystitis, however, good hygiene may decrease the chances of infection.

Resources

BOOKS

Gorbach, Sherwood L., John G. Bartlett, and Neil R. Blacklow, editors. "Dacryocystitis." In *Infectious Diseases*. 2nd ed. Philadelphia: W. B. Saunders Co., 1998.

Altha Roberts Edgren

Dandruff see **Seborrheic dermatitis**

Death

Definition

Death is defined as the cessation of all vital functions of the body including the heartbeat, brain activity (including the brain stem), and breathing.

Description

Death comes in many forms, whether it be expected after a diagnosis of terminal illness or an unexpected accident or medical condition.

Terminal illness

When a terminal illness is diagnosed, a person, family, friends, and physicians are all able to prepare for the impending death. A terminally ill individual goes through several levels of emotional acceptance while in the process of dying. First, there is denial and isolation. This is followed by anger and resentment. Thirdly, a person tries to escape the inevitable. With the realization that death is eminent, most people suffer from depression. Lastly, the reality of death is realized and accepted.

Causes and symptoms

The two leading causes of death for both men and women in the United States are heart disease

ELISABETH KÜBLER-ROSS (1926–2004)

A contemporary physician who was a world authority on the subject of death and after-death states. Born in Switzerland on July 8, 1926, she worked as a country doctor before moving to the United States. During World War II she spent weekends at the Kantonspital (Cantonai Hospital) in Zürich, where she volunteered to assist escaped refugees. After the war she visited Majdanek concentration camp, where the horrors of the death chambers stimulated in her a desire to help people facing death and to understand the human impulses of love and destruction. She extended her medical background by becoming a practicing psychiatrist. Her formal work with dying patients began in 1965 when she was a faculty member at the University of Chicago. She also conducted research on basic questions concerning life after death at the Manhattan State Hospital, New York. Her studies of death and dying involved accounts by patients who reported out-of-the-body travel. Her research tends to show that while dying can be painful, death itself is a peaceful condition. Her 1969 text, *On Death and Dying*, was hailed by her colleagues and also became a popular best-seller.

In 1978 Kübler-Ross helped to found Shanti Nilaya (Final Home of Peace), a healing and growth center in Escondido, California. This was an extension of her well-known "Life-Death and Transition" workshops conducted in various parts of the United States and Canada, involving physicians, nurses, social workers, laypeople, and terminally ill patients. Much of Kübler-Ross's later research was directed toward proving the existence of life after death. Her publication *To Live Until We Say Good-bye* (1979) was both praised as a "celebration of life" and criticized as "prettifying" the real situation. She also dealt with issues such as AIDS and "near death" experiences. In the mid-1980s, Shanti Nilaya moved from San Diego County, California, to Head Waters, Virginia, where it continues to offer courses and short- and long-term therapeutic sessions.

and **cancer**. Accidental death was a distant third followed by such problems as **stroke**, chronic lung disorders, pneumonia, **suicide**, cirrhosis, diabetes mellitus, and murder. The order of these causes of death varies among persons of different age, ethnicity, and gender.

Diagnosis

In an age of organ transplantation, identifying the moment of death may now involve another life. It thereby takes on supreme legal importance. It is largely due to the need for transplant organs that death has been so precisely defined.

The official signs of death include the following:

- no pupil reaction to light
- no response of the eyes to caloric (warm or cold) stimulation
- no jaw reflex (the jaw will react like the knee if hit with a reflex hammer)
- no gag reflex (touching the back of the throat induces **vomiting**)
- no response to **pain**
- no breathing
- a body temperature above 86 °F (30 °C), which eliminates the possibility of resuscitation following cold-water drowning
- no other cause for the above, such as a head injury
- no drugs present in the body that could cause apparent death
- all of the above for 12 hours
- all of the above for six hours and a flat-line electroencephalogram (brain wave study)
- no blood circulating to the brain, as demonstrated by **angiography**

Current ability to resuscitate people who have "died" has produced some remarkable stories. Drowning in cold water (under 50 °F/10 °C) so effectively slows metabolism that some persons have been revived after a half hour under water.

Treatment

Only recently has there been concerted public effort to address the care of the dying in an effort to improve their comfort and lessen their alienation from those still living. Hospice care represents one of the greatest advances made in this direction. There has also been a liberalization of the use of **narcotics** and other drugs for symptomatic relief and improvement in the quality of life for the dying.

Living will

One of the most difficult issues surrounding death in the era of technology is that there is now a choice, not of the event itself, but of its timing. When to die, and more often, when to let a loved one die, is coming within people's power to determine. This is both a blessing and a dilemma. Insofar as the decision can be made ahead of time, a living will is an attempt to address this dilemma. By outlining the conditions under which one would rather be allowed to die, a person can contribute significantly to that final decision, even if not competent to do

KEY TERMS

Angiography—X rays of blood vessels filled with a contrast agent.

Caloric testing—Flushing warm and cold water into the ear stimulates the labyrinth and causes vertigo and nystagmus if all the nerve pathways are intact.

Electroencephalogram—Recording of electrical activity in the brain.

Hospice—Systematized care of dying persons.

Living will—A legal document detailing a person's wishes during the end of life, to be carried out by designated decision makers.

Stroke—Interruption of blood flow to a part of the brain with consequent brain damage, also known as a cerebrovascular accident (CVA).

so at the time of actual death. The problem is that there are uncertainties surrounding every severely ill person. Each instance presents a greater or lesser chance of survival. The chance is often greater than zero. The best living will follows an intimate discussion with decision makers covering the many possible scenarios surrounding the end of life. This discussion is difficult, for few people like to contemplate their own demise. However, the benefits of a living will are substantial, both to physicians and to loved ones who are faced with making final decisions. Most states have passed living will laws, honoring instructions on artificial **life support** that were made while a person was still mentally competent.

Euthanasia

Another issue that has received much attention is assisted suicide (euthanasia). In 1997, the State of Oregon placed the issue on the ballot, amid much consternation and dispute. Perhaps the main reason euthanasia has become front page news is because Dr. Jack Kevorkian, a pathologist from Michigan, is one of its most vocal advocates. The issue highlights the many new problems generated by increasing ability to intervene effectively in the final moments of life and unnaturally prolong the process of dying. The public appearance of euthanasia has also stimulated discussion about more compassionate care of the dying.

Prevention

Autopsy after death is a way to precisely determine a cause of death. The word autopsy is derived from

Greek meaning to see with one's own eyes. A pathologist extensively examines a body and submits a detailed report to an attending physician. Although an autopsy can do nothing for an individual after death, it can benefit the family and, in some cases, medical science. Hereditary disorders and disease may be found. This knowledge could be used to prevent illness in other family members. Information culled from an autopsy can be used to further medical research. The link between **smoking** and lung cancer was confirmed from data gathered through autopsy. Early information about **AIDS** was also compiled through autopsy reports.

Resources

BOOKS

Finkbeiner, J. *Autopsy: A Manual & Atlas*. Philadelphia: Saunders, 2001.

Iseron, Kenneth B. *Death to Dust: What Happens to Dead Bodies?* Tucson: Galen Press Ltd, 2001.

Mount, Balfour M. "Care of Dying Patients and Their Families." In *Cecil Textbook of Medicine*, edited by Lee Goldman, et al., 21st ed. Philadelphia: W.B. Saunders, 2000.

Sheaff, Michael T., and Deborah J. Hopster. *Post Mortem Technique Handbook*. New York: Springer Verlag, 2001.

PERIODICALS

Roger, V. L., et al. "Time Trends in the Prevalence of Atherosclerosis: A Population-based Autopsy Study." *American Journal of Medicine* 110, no. 4 (2001): 267-273.

Targonski, P., et al. "Referral to Autopsy: Effect of Autemortem Cardiovascular Disease. A Population-based Study in Olmsted County, Minnesota." *Annals of Epidemiology* 11, no. 4 (2001): 264-270.

ORGANIZATIONS

American Academy of Family Physicians. 11400 Tomahawk Creek Parkway, Leawood, KS 66211-2672. (913) 906-6000. <<http://www.aafp.org>>.

American Medical Association. 515 N. State Street, Chicago, IL 60610. (312) 464-5000. <<http://www.ama-assn.org>>.

American Society of Clinical Pathologists. 2100 West Harrison Street, Chicago, IL 60612. (312) 738-1336. <<http://www.ascp.org/index.asp>>.

College of American Pathologists. 325 Waukegan Road, Northfield, IL 60093. (800) 323-4040. <<http://www.cap.org>>.

Hospice Foundation of America. 2001 S St. NW Suite 300, Washington, DC 20009. (800) 854-3402. <<http://www.hospicefoundation.org>>.

OTHER

American Association of Retired Persons. <<http://www.aarp.org>>.

Association for Death Education and Counseling. <<http://www.adec.org>>.
Death and Dying Grief Support. <<http://www.death-dying.com>>.
National Center for Health Statistics. <<http://www.cdc.gov/nchs>>.

L. Fleming Fallon, Jr., MD, DrPH

Debridement

Definition

Debridement is the process of removing non-living tissue from pressure ulcers, **burns**, and other **wounds**.

Purpose

Debridement speeds the healing of pressure ulcers, burns, and other wounds. Wounds that contain non-living (necrotic) tissue take longer to heal. The necrotic tissue may become colonized with bacteria, producing an unpleasant odor. Though the wound is not necessarily infected, the bacteria can cause inflammation and strain the body's ability to fight infection. Necrotic tissue may also hide pockets of pus called abscesses. Abscesses can develop into a general infection that may lead to **amputation** or **death**.

Precautions

Not all wounds need debridement. Sometimes it is better to leave a hardened crust of dead tissue, called an eschar, than to remove it and create an open wound, particularly if the crust is stable and the wound is not inflamed. Before performing debridement, the physician will take a medical history with attention to factors that might complicate healing, such as medications being taken and **smoking**. The physician will also note the cause of the wound and the ways it has been treated. Some ulcers and other wounds occur in places where blood flow is impaired, for example, the foot ulcers that can accompany **diabetes mellitus**. In such cases, the physician or nurse may decide not to debride the wound because blood flow may be insufficient for proper healing.

Description

In debridement, dead tissue is removed so that the remaining living tissue can adequately heal. Dead

tissue exposed to the air will form a hard black crust, called an eschar. Deeper tissue will remain moist and may appear white, or yellow and soft, or flimsy. The four major debridement techniques are surgical, mechanical, chemical, and autolytic.

Surgical debridement

Surgical debridement (also known as sharp debridement) uses a scalpel, scissors, or other instrument to cut dead tissue from a wound. It is the quickest and most efficient method of debridement. It is the preferred method if there is rapidly developing inflammation of the body's connective tissues (**cellulitis**) or a more generalized infection (sepsis) that has entered the bloodstream. The procedure can be performed at a patient's bedside. If the target tissue is deep or close to another organ, however, or if the patient is experiencing extreme **pain**, the procedure may be done in an operating room. Surgical debridement is generally performed by a physician, but in some areas of the country an advance practice nurse or physician assistant may perform the procedure.

The physician will begin by flushing the area with a saline (salt water) solution, and then will apply a topical anesthetic gel to the edges of the wound to minimize pain. Using a forceps to grip the dead tissue, the physician will cut it away bit by bit with a scalpel or scissors. Sometimes it is necessary to leave some dead tissue behind rather than disturb living tissue. The physician may repeat the process again at another session.

Mechanical debridement

In mechanical debridement, a saline-moistened dressing is allowed to dry overnight and adhere to the dead tissue. When the dressing is removed, the dead tissue is pulled away too. This process is one of the oldest methods of debridement. It can be very painful because the dressing can adhere to living as well as nonliving tissue. Because mechanical debridement cannot select between good and bad tissue, it is an unacceptable debridement method for clean wounds where a new layer of healing cells is already developing.

Chemical debridement

Chemical debridement makes use of certain enzymes and other compounds to dissolve necrotic tissue. It is more selective than mechanical debridement. In fact, the body makes its own enzyme, collagenase, to break down collagen, one of the major building blocks of skin. A pharmaceutical version of collagenase is available and is highly effective as a debridement



A burn sufferer undergoes debridement (the removal of dead skin). The patterns on his chest are from skin grafts. (Photograph by Ann Chawatsky, Phototake NYC. Reproduced by permission.)

agent. As with other debridement techniques, the area first is flushed with saline. Any crust of dead tissue is etched in a cross-hatched pattern to allow the enzyme to penetrate. A topical antibiotic is also applied to prevent introducing infection into the bloodstream. A moist dressing is then placed over the wound.

Autolytic debridement

Autolytic debridement takes advantage of the body's own ability to dissolve dead tissue. The key to the technique is keeping the wound moist, which can be accomplished with a variety of dressings. These dressings help to trap wound fluid that contains growth factors, enzymes, and immune cells that promote wound healing. Autolytic debridement is more selective than any other debridement method, but it also takes the longest to work. It is inappropriate for wounds that have become infected.

Preparation

The physician or nurse will begin by assessing the need for debridement. The wound will be examined,

KEY TERMS

Eschar—A hardened black crust of dead tissue that may form over a wound.

Pressure ulcer—Also known as a decubitus ulcer, pressure ulcers are open wounds that form whenever prolonged pressure is applied to skin covering bony outcrops of the body. Patients who are bedridden are at risk of developing pressure ulcers. Pressure ulcers are commonly known as bedsores.

Sepsis—A severe systemic infection in which bacteria have entered the blood stream.

frequently by inserting a gloved finger into the wound to estimate the depth of dead tissue and evaluate whether it lies close to other organs, bone, or important body features. The area may be flushed with a saline solution before debridement begins, and a topical anesthetic gel or injection may be applied if surgical or mechanical debridement is being performed.

Aftercare

After surgical debridement, the wound will be packed with a dry dressing for a day to control bleeding. Afterward, moist dressings are applied to promote wound healing. Moist dressings are also used after mechanical, chemical, and autolytic debridement. Many factors contribute to wound healing, which frequently can take considerable time. Debridement may need to be repeated.

Risks

It is possible that underlying tendons, blood vessels or other structures will be damaged during the examination of the wound and during surgical debridement. Surface bacteria may also be introduced deeper into the body, causing infection.

Normal results

Removal of dead tissue from pressure ulcers and other wounds speeds healing. Although these procedures cause some pain, they are generally well tolerated by patients and can be managed more aggressively. It is not uncommon to debride a wound again in a subsequent session.

Resources

ORGANIZATIONS

American Academy of Wound Management. 1255 23rd St., NW, Washington, DC 20037. (202) 521-0368. <<http://www.aawm.org>>.

Wound Care Institute. 1100 N.E. 163rd Street, Suite #101, North Miami Beach, FL 33162. (305) 919-9192. <<http://woundcare.org>>.

Richard H. Camer

Decompression sickness

Definition

Decompression sickness (DCS) is a dangerous and occasionally lethal condition caused by nitrogen bubbles that form in the blood and other tissues of scuba divers who surface too quickly.

Description

According to the Divers Alert Network (DAN), a worldwide organization devoted to safe-diving research

and promotion, less than 1% of divers fall victim to DCS or the rarer bubble problem called **gas embolism**, **air embolism**, or arterial gas embolism (AGE). A study of the United States military community in Okinawa, where tens of thousands of sport and military dives are made each year, identified 84 DCS and 10 AGE cases in 1989–95, including nine deaths. This translated into estimates of one case in every 7,400 dives and one **death** in every 76,900 dives. DCS symptoms can be quite mild, however, and many cases certainly go unnoticed by divers.

At times the terminology adopted by writers on DCS can be confusing. Some substitute the term decompression illness (DCI) for DCS. Others treat DCI as a label encompassing both DCS and AGE. An older term for DCS is caisson disease, coined in the nineteenth century when it was discovered that bridge construction crews working at the bottom of lakes and rivers in large pressurized enclosures (caissons) were experiencing joint **pain** (a typical DCS symptom) on returning to the surface.

Causes and symptoms

The air we breathe is mostly a mixture of two gases, nitrogen (78%) and oxygen (21%). Unlike oxygen, nitrogen is a biologically inert gas, meaning that it is not metabolized (converted into other substances) by the body. For this reason, most of the nitrogen we inhale is expelled when we exhale, but some is dissolved into the blood and other tissues. During a dive, however, the lungs take in more nitrogen than usual. This happens because the surrounding water pressure is greater than the air pressure at sea level (twice as great at 33 ft [10 m], for instance). As the water pressure increases, so does the pressure of the nitrogen in the compressed air inhaled by the diver. Because increased pressure causes an increase in gas density, the diver takes in more nitrogen with each breath than he or she would at sea level. Instead of being exhaled, however, the extra nitrogen safely dissolves into the tissues, where it remains until the diver begins his or her return to the surface (under some circumstances the extra nitrogen can cause nitrogen narcosis, but that condition is distinct from DCS). On the way up, decompression occurs (in other words, the water pressure drops), and with the change in pressure, the extra nitrogen gradually diffuses out of the tissues and is delivered by the bloodstream to the lungs, which expel it from the body. If the diver surfaces too quickly, however, potentially dangerous nitrogen bubbles can form in the tissues and cause DCS. These bubbles can compress nerves, obstruct arteries, veins, and lymphatic vessels, and trigger harmful chemical reactions in the blood. The precise reasons for bubble formation remain unclear.

How much extra nitrogen enters the tissues varies with the dive's depth and duration. Dive tables prepared by the U.S. Navy and other organizations specify how long most divers can safely remain at a particular depth. If the dive table limits are exceeded, the diver must pause on the way up to allow the nitrogen to diffuse into the bloodstream without forming bubbles; these pauses are called decompression stops, and are carefully calibrated. DCS can occur, however, even when a diver obeys safe diving rules. In such cases, the predisposing factors include **fatigue**, **obesity**, **dehydration**, **hypothermia**, and recent alcohol use. People who fly or travel to high-altitude locations without letting 12–24 hours pass after their last dive are at risk for DCS as well because their bodies undergo further decompression. This is true even when flying in commercial aircraft. Many travelers are unaware that to save money on fuel the cabin pressure in commercial aircraft is set much lower than the pressure at sea level. At 30,000 ft (9,144 m), for instance, cabin pressure is usually equivalent to the pressure at 7,000–8,000 ft (2,133–2,438 m) above sea level, a safe setting for everyone but recent divers. Exactly how long a diver should wait before flying or traveling to a high-altitude location depends on how much diving he or she has done and other considerations. If there is uncertainty about the appropriate waiting period, the sensible course of action is to let the full 24 hours pass.

Because the nitrogen bubbles that cause DCS can affect any of the body's tissues, including the blood, bones, nerves, and muscles, many kinds of symptoms are possible. Symptoms can appear minutes after a diver surfaces, and in about 80% of cases do so within eight hours. Pain is often the only symptom; this is sometimes called the bends, although many people incorrectly use that term as a synonym for DCS itself. The pain, which ranges from mild to severe, is usually limited to the joints, but can be felt anywhere. Severe itching (pruritis), skin **rashes**, and skin mottling (cutis marmorata) are other possible symptoms. All of these are sometimes classified as manifestations of type 1 or "mild" DCS. Type 2 or "serious" DCS can lead, among other things, to **paralysis**, brain damage, heart attacks, and death. Many DCS victims, however, experience both type 1 and type 2 symptoms.

Diagnosis

Diagnosis requires taking a medical history (questioning the patient about his or her health and recent activities) and conducting a physical examination.

Treatment

DCS is treated by giving the patient oxygen and placing him or her in a hyperbaric chamber, an

KEY TERMS

Gas embolism—The presence of a gas bubble in the bloodstream that obstructs circulation.

Hyperbaric chamber—A sealed compartment in which air pressure is gradually increased and then gradually decreased, allowing nitrogen bubbles to shrink and the nitrogen to safely diffuse out of body tissue.

Lymphatic vessels—Vessels that carry a fluid called lymph from the tissues to the bloodstream.

Nitrogen narcosis—Also called "rapture of the deep," the condition is caused by increased nitrogen pressure at depth and is characterized by symptoms similar to alcohol intoxication.

enclosure in which the air pressure is first gradually increased and then gradually decreased. This shrinks the bubbles and allows the nitrogen to safely diffuse out of the tissues. Hyperbaric chamber facilities exist throughout the United States. No matter how mild one's symptoms may appear, immediate transportation to a facility is essential. Treatment is necessary even if the symptoms clear up before the facility is reached, because bubbles may still be in the bloodstream and pose a threat. DAN maintains a list of facilities and a 24-hour hotline that can provide advice on handling DCS and other diving emergencies.

Prognosis

DCS sufferers who undergo chamber treatment within a few hours of symptom onset usually enjoy a full recovery. If treatment is delayed the consequences are less predictable, although many people have been helped even after several days have passed. A 1992 DAN report on diving accidents indicated that full recovery following chamber treatment was immediate for about 50% of divers. Some people, however, suffer **numbness**, **tingling**, or other symptoms that last weeks, months, or even a lifetime. In the Okinawa study, six of the 94 patients experienced "long-lasting" symptoms even after repeated chamber treatments.

Prevention

The obvious way to minimize the risk of falling victim to DCS is to follow the rules on safe diving and air travel after a dive. People who are obese, suffer from lung or heart problems, or are otherwise in poor health should not dive. And because the effect of

nitrogen diffusion on the fetus remains unknown, diving while pregnant is not recommended.

Resources

ORGANIZATIONS

American College of Hyperbaric Medicine. PO Box 25914-130, Houston, Texas 77265. (713) 528-0657. <<http://www.hyperbaricmedicine.org>>.

Divers Alert Network. The Peter B. Bennett Center, 6 West Colony Place, Durham, NC 27705. (800) 446-2671. <<http://www.diversalertnetwork.org>>.

Undersea and Hyperbaric Medical Society. 10531 Metropolitan Ave., Kensington, MD 20895. (301) 942-2980. <<http://www.uhms.org>>.

Howard Baker

Decongestants

Definition

Decongestants are medicines used to relieve nasal congestion (stuffy nose).

Purpose

A congested or stuffy nose is a common symptom of colds and **allergies**. This congestion results when membranes lining the nose become swollen. Decongestants relieve the swelling by narrowing the blood vessels that supply the nose. This reduces the blood supply to the swollen membranes, causing the membranes to shrink.

These medicines do not cure colds or reverse the effects of histamines—chemicals released as part of the allergic reaction. They will not relieve all of the symptoms associated with colds and allergies, only the stuffiness.

When considering whether to use a decongestant for cold symptoms, keep in mind that most colds go away with or without treatment and that taking medicine is not the only way to relieve a stuffy nose. Drinking hot tea or broth or eating chicken soup may help. There are also adhesive strips that can be placed on the nose to help widen the nasal passages, making breathing through the nasal passages a bit easier when congestion is present.

Precautions

Decongestant nasal sprays and nose drops may cause a problem called rebound congestion if used

repeatedly over several days. When this happens, the nose remains stuffy or gets worse with every dose. The only way to stop the cycle is to stop using the drug. The stuffiness should then go away within about a week. Anyone who shows signs of severe rebound congestion should also contact his or her physician.

Do not use decongestant nasal sprays for more than three days. Decongestants taken by mouth should not be used for more than seven days. If the congestion has not gone away in this time, or if the symptoms are accompanied by **fever**, call a physician.

Do not use a decongestant nasal spray after the product's expiration date. If the product has become cloudy or discolored, throw it away and do not use it. Do not share droppers or spray bottles with anyone else, as this could spread infection. Do not let droppers and bottle tips touch countertops or other surfaces.

Some decongestants cause drowsiness. People who take these drugs should not drive, use machines or do anything else that might be dangerous until they have found out how the drugs affect them.

In general, older people may be more sensitive to the effects of decongestants and may need to take lower doses to avoid side effects. People in this age group should not take long-acting (extended release) forms of decongestants unless they have previously taken a short-acting form with no ill effects.

Children may also be more sensitive to the effects of decongestants. Before giving any decongestant to a child, check the package label carefully. Some of these medicines are too strong for use in children. Serious side effects are possible if they are given large amounts of these drugs or if they swallow nose drops, nasal spray or eye drops. If this happens, call a physician or poison center immediately.

Special conditions

People with certain medical conditions or who are taking certain other medicines can have problems if they take decongestants. Before taking these drugs, be sure to let the physician know about any of these conditions:

ALLERGIES. Anyone who has had unusual reactions to decongestants in the past should let his or her physician know before these drugs or any similar drugs are prescribed. The physician should also be told about any allergies to foods, dyes, preservatives, or other substances.

PREGNANCY. In studies of laboratory animals, some decongestants have had unwanted effects on fetuses. However, it is not known whether such effects

also occur in people. Women who are pregnant or who plan to become pregnant should check with their physicians before taking decongestants.

BREASTFEEDING. Some decongestants pass into breast milk and may have unwanted effects on nursing babies whose mothers take the drugs. Women who are breastfeeding should check with their physicians before using decongestants. If they need to take the medicine, it may be necessary to bottle feed the baby with formula while taking it.

OTHER MEDICAL CONDITIONS. Anyone with heart or blood vessel disease, high blood pressure, diabetes, **enlarged prostate**, or overactive thyroid should not take decongestants unless under a physician's supervision. The medicine can increase blood sugar in people with diabetes. It can be especially dangerous in people with high blood pressure, as it may increase blood pressure.

Before using decongestants, people with any of these medical problems should make sure their physicians are aware of their conditions:

- glaucoma
- history of mental illness

Decongestants may have a variety of side effects, and may also interact with other medications the patient is taking.

Side effects

DECONGESTANT NASAL SPRAYS AND NOSE DROPS. The most common side effects from decongestant nasal sprays and nose drops are sneezing and temporary burning, stinging, or dryness. These effects are usually temporary and do not need medical attention. If any of the following side effects occur after using a decongestant nasal spray or nose drops, stop using the medicine immediately and call the physician:

- increased blood pressure
- **headache**
- fast, slow, or fluttery heartbeat
- nervousness
- **dizziness**
- **nausea**
- sleep problems

DECONGESTANTS TAKEN BY MOUTH. The most common side effects of decongestants taken by mouth are nervousness, restlessness, excitability, dizziness, drowsiness, headache, nausea, weakness, and sleep problems. Anyone who has these symptoms

while taking decongestants should stop taking them immediately.

Patients who have these symptoms while taking decongestants should call the physician immediately:

- increased blood pressure
- fast, irregular, or fluttery heartbeat
- severe headache
- tightness or discomfort in the chest
- breathing problems
- fear or anxiety
- **hallucinations**
- trembling or shaking
- convulsions (seizures)
- pale skin
- painful or difficult urination

Other side effects may occur. Anyone who has unusual symptoms after taking a decongestant should get in touch with his or her physician.

Interactions with other medicines

Decongestants may interact with a variety of other medicines. When this happens, the effects of one or both of the drugs may change or the risk of side effects may be greater. Do not take decongestants at the same time as these drugs:

- Monoamine oxidase inhibitors (MAO inhibitors) such as phenzeline (Nardil) or tranylcypromine (Parnate), used to treat conditions including depression and Parkinson's disease. Do not take decongestants at the same time as a MAO inhibitor or within two weeks of stopping treatment with an MAO inhibitor unless a physician approves.
- Other products containing the same or other decongestants
- Caffeine.

In addition, anyone who takes decongestants should let the physician know all other medicines he or she is taking. Among the drugs that may interact with decongestants are:

- tricyclic antidepressants such as imipramine (Tofranil) or desipramine (Norpramin)
- the antidepressant maprotiline (Ludiomil)
- amantadine (Symmetrel)
- amphetamines
- medicine to relieve **asthma** or other breathing problems
- methylphenidate (Ritalin)

KEY TERMS

Fetus—A developing baby inside the womb.

Hallucination—A false or distorted perception of objects, sounds, or events that seems real. Hallucinations usually result from drugs or mental disorders.

- appetite suppressants
- other medicine for colds, sinus problems, hay fever or other allergies
- beta-blockers such as atenolol (Tenormin) and propranolol (Inderal)
- digitalis glycosides, used to treat heart conditions

The list above does not include every drug that may interact with decongestants. Be sure to check with a physician or pharmacist before combining decongestants with any other prescription or nonprescription (over-the-counter) medicine.

Description

Decongestants are sold in many forms, including tablets, capsules, caplets, gelcaps, liqui-caps, liquids, nasal sprays, and nose drops. These drugs are sometimes combined with other medicines in cold and allergy products designed to relieve several symptoms. Some decongestant products require a physician's prescription, but there are also many nonprescription (over-the-counter) products. Ask a physician or pharmacist about choosing an appropriate decongestant.

Commonly used decongestants include oxymetazoline (Afrin and other brands) and pseudoephedrine (Sudafed, Actifed, and other brands). The decongestant oxymetazoline is also used in some eye drops to relieve redness and **itching**.

The recommended dosage depends on the drug. Check with the physician who prescribed the drug or the pharmacist who filled the prescription for the correct dosage, and always take the medicine exactly as directed. If using nonprescription (over-the-counter) types, follow the directions on the package label or ask a pharmacist for assistance. Never take larger or more frequent doses, and do not take the drug for longer than directed.

Risks

Anyone considering taking a decongestant should take a close look at the labels of any already in their

medicine cabinet. In 2000, the Food and Drug Administration prohibited over-the-counter sales of medicines containing the decongestant phenylpropanolamine. The medicine is associated with an increased risk of **stroke** in people ages 18 to 49, especially women. Many cold remedies contained this medicine. Contact a pharmacist if there is any question about the ingredients in a medication. Over-the-counter remedies containing phenylpropanolamine should be discarded.

Normal results

The desired result when taking decongestants is the short-term relief of nasal congestion.

Resources

PERIODICALS

Henderson, Charles W. "Voluntary Withdrawal of Cold and Allergy Products Announced." *Medical Letter on the CDC and FDA*, November 26, 2000.

"An Ingredient Under Fire: Drugmakers are Jittery After an FDA Panel Ruling." *Newsweek*, October 30, 2000: 59.

OTHER

Medline Plus Health Information. U.S. National Library of Medicine. <<http://www.nlm.nih.gov/medlineplus>>.

Deanna M. Swartout-Corbeil, R.N.

Decubitus ulcers see **Bedsore**s

Deep vein thrombosis

Definition

Deep vein thrombosis (DVT) is a blood clot in a major vein, usually in the legs and/or pelvis.

Description

Deep vein thrombosis is a common but difficult to detect illness that can be fatal if not treated effectively. According to the American Heart Association, more than two million Americans develop deep vein thrombosis annually. An estimated 600,000 of these develop pulmonary **embolism**, a potentially fatal complication where the blood clots break off and form pulmonary emboli, plugs that block the lung arteries. Sixty thousand people die of **pulmonary embolism** each year. Deep vein thrombosis is also called venous thromboembolism, **thrombophlebitis** or phlebothrombosis.

Deep vein thrombosis is a major complication in patients who have had **orthopedic surgery** or pelvic, abdominal, or thoracic surgery. Patients with **cancer** and other chronic illnesses (including congestive **heart failure**), as well as those who have suffered a recent myocardial infarction, are also at high risk for developing DVT. Deep vein thrombosis can be chronic, with recurrent episodes.

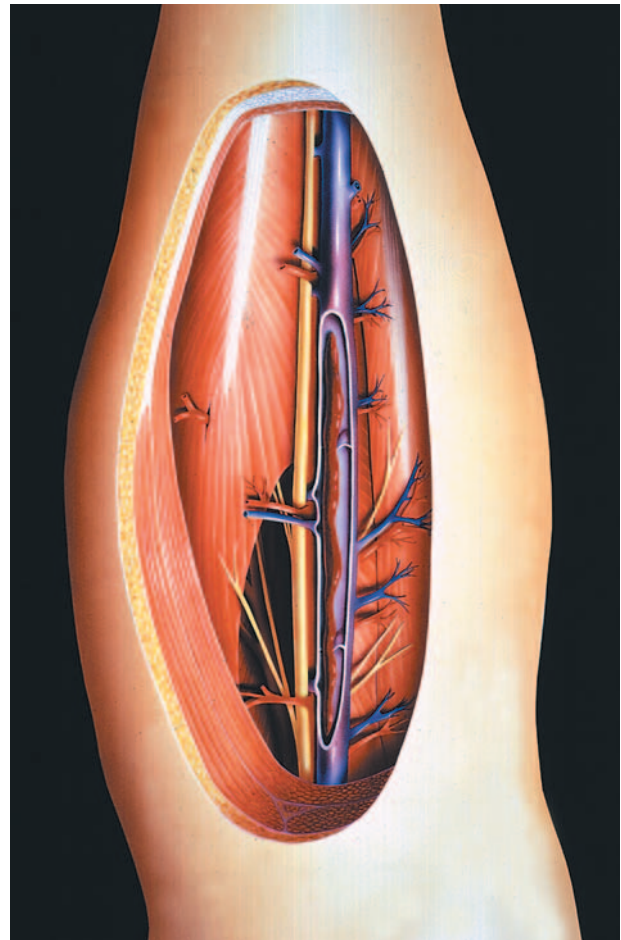
Causes and symptoms

Deep vein thrombosis is caused by **blood clots** in blood vessels that form in veins where blood flow is sluggish or has been disturbed, in pockets in the calf's deep veins, or in veins that have been traumatized. Symptoms include swelling and tenderness of the calf or thigh, and possibly warmth. Only 23–50% of patients experience symptoms, so it's often "silent." Some individuals and families have underlying clotting tendencies that can be tested for.

Diagnosis

Deep vein thrombosis can be detected through venography and radionuclide **venography**, **Doppler ultrasonography**, and impedance plethysmography. Venography is the most accurate test, but it is not used much, because it is often painful, expensive, exposes the patient to radiation, and can cause reactions and complications. Venography identifies the location, extent, and degree of attachment of the blood clots, and enables the condition of the deep leg veins to be assessed. A contrast solution is injected into a foot vein through a catheter. The physician observes the movement of the solution through the vein with a fluoroscope while a series of x rays are taken. Venography takes 30–45 minutes and can be done in a physician's office, a laboratory, or a hospital. Radionuclide venography, in which a radioactive isotope is injected, is occasionally used, especially if a patient has had reactions to contrast solutions.

Doppler ultrasonography is usually the preferred procedure for detecting deep vein thrombosis. This technique uses sound waves to measure blood flow through leg veins and arteries. A blood pressure cuff is wrapped around the patient's ankle and a transducer with gel on it is placed over pulse points of the foot and lower leg. High-frequency sounds bounce off the soft tissue, and the echoes are converted into images on a monitor. It is very accurate in detecting clots above the knee that can become pulmonary embolisms. Usually performed in a physician's office



This illustration features a dissected human lower leg showing clot formation (thrombosis) along the length of a vein. (Custom Medical Stock Photo. Reproduced by permission.)

or hospital outpatient diagnostic center, Doppler ultrasound usually takes 30–45 minutes.

Impedance plethysmography records changes in blood volume and vessel resistance. A blood pressure cuff is wrapped around the leg above the knee, four electrodes are placed near the knee and the ankle, and the cuff is inflated. How efficiently the veins return to normal is measured. Performed in a physician's office, it takes about 15 minutes.

Treatment

Deep vein thrombosis can be treated with drug therapy, bed rest, and gradient elastic stockings. Medications include anticoagulants that "thin" blood to prevent further growth of blood clots, as well as clot-dissolving drugs. Heparin is a common injectable anticoagulant, and is usually followed by

KEY TERMS

Pulmonary embolism—An obstruction of a blood vessel in the lungs, usually caused by a blood clot that blocks a coronary artery. Pulmonary embolism can be very serious and, in some cases, fatal.

Thrombosis—The development of a blood clot inside a blood vessel.

coumadin tablets for at least three months. Bed rest with the patient's legs elevated is necessary until the condition improves. Gradient elastic stockings should then be worn, and standing for long periods of time avoided. In some cases, a filter is placed in the major vein (the inferior vena cava) to trap emboli or clots before they get to the heart and lungs.

Alternative treatment

Deep vein thrombosis can be life-threatening and must be treated with conventional medical therapies. However, there are alternative therapies that can be used in conjunction with emergency treatments to dissolve the clot that help support the body and prevent recurrence. A trained alternative health care practitioner should be consulted due to the severity of this condition.

Prognosis

In many cases, deep vein thrombosis can be successfully treated if diagnosed early.

Prevention

Deep vein thrombosis can be prevented through prophylactic **anticoagulant drugs** and venous stasis prevention with gradient elastic stockings and intermittent pneumatic compression of the legs. High-risk patients often need to remain on anticoagulants like Coumadin indefinitely.

Resources

PERIODICALS

Davidson, Bruce L., and Eric J. Deppert. "Ultrasound for the Diagnosis of Deep Vein Thrombosis: Where to Now?" *British Medical Journal* 316 (January 3, 1998): 2.

Lori De Milto

Deer-fly fever see **Tularemia**

Defibrillation

Definition

Defibrillation is a process in which an electronic device sends an electric shock to the heart to stop an extremely rapid, irregular heartbeat, and restore the normal heart rhythm.

Purpose

Defibrillation is performed to correct life-threatening fibrillations of the heart, which could result in cardiac arrest. It should be performed immediately after identifying that the patient is experiencing a cardiac emergency, has no pulse, and is unresponsive.

Precautions

Defibrillation should not be performed on a patient who has a pulse or is alert, as this could cause a lethal heart rhythm disturbance or cardiac arrest. The paddles used in the procedure should not be placed on a woman's breasts or over a pacemaker.

Description

Fibrillations cause the heart to stop pumping blood, leading to brain damage and/or cardiac arrest. About 10% of the ability to restart the heart is lost with every minute that the heart stays in fibrillation. **Death** can occur in minutes unless the normal heart rhythm is restored through defibrillation. Because immediate defibrillation is crucial to the patient's survival, the American Heart Association has called for the integration of defibrillation into an effective emergency cardiac care system. The system should include early access, early **cardiopulmonary resuscitation**, early defibrillation, and early advanced cardiac care.

Defibrillators deliver a brief electric shock to the heart, which enables the heart's natural pacemaker to regain control and establish a normal heart rhythm. The defibrillator is an electronic device with electrocardiogram leads and paddles. During defibrillation, the paddles are placed on the patient's chest, caregivers stand back, and the electric shock is delivered. The patient's pulse and heart rhythm are continually monitored. Medications to treat possible causes of the abnormal heart rhythm may be administered. Defibrillation continues until the patient's condition stabilizes or the procedure is ordered to be discontinued.



Defibrillation by paddles. (Photograph by Patricia Barber, RBP, Custom Medical Stock Photo. Reproduced by permission.)

KEY TERMS

Cardiac arrest—A condition in which the heart stops functioning. Fibrillation can lead to cardiac arrest if not corrected quickly.

Fibrillation—Very rapid contractions or twitching of small muscle fibers in the heart.

Pacemaker—A surgically implanted electronic device that sends out electrical impulses to regulate a slow or erratic heartbeat.

Early defibrillators, about the size and weight of a car battery, were used primarily in ambulances and hospitals. The American Heart Association now advocates public access defibrillation; this calls for placing automated external defibrillators (AEDs) in police vehicles, airplanes, and at public events, etc. The AEDs are smaller, lighter, less expensive, and easier to use than the early defibrillators. They are computerized to provide simple, verbal instructions to the operator and to make it impossible to deliver a shock to a patient whose heart is not fibrillating. The placement of AEDs is likely to expand to many public locations.

Preparation

After help is called for, **cardiopulmonary resuscitation (CPR)** is begun and continued until the caregivers arrive and set up the defibrillator. Electrocardiogram leads are attached to the patient's chest. Gel or paste is applied to the defibrillator paddles, or two gel pads are placed on the patient's chest. The caregivers verify lack of a pulse, and select a charge.

Aftercare

After defibrillation, the patient's cardiac status, breathing, and vital signs are monitored until he or she is stable. Typically, this monitoring takes place after the patient has been removed to an intensive care or cardiac care unit in a hospital. An electrocardiogram and **chest x ray** are taken. The patient's skin is cleansed to remove gel or paste, and, if necessary, ointment is applied to **burns**. An intravenous line provides additional medication, as needed.

Risks

Skin burns from the defibrillator paddles are the most common complication of defibrillation. Other risks include injury to the heart muscle, abnormal heart rhythms, and **blood clots**.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

Lori De Milto

Definitive cancer therapy see **Cancer therapy, definitive**

Degenerative arthritis see **Osteoarthritis**

Dehydration

Definition

Dehydration is the loss of water and salts essential for normal body function.

Description

Dehydration occurs when the body loses more fluid than it takes in. This condition can result from illness; a hot, dry climate; prolonged exposure to sun or high temperatures; not drinking enough water; and overuse of **diuretics** or other medications that increase urination. Dehydration can upset the delicate fluid-salt balance needed to maintain healthy cells and tissues.

Water accounts for about 60% of a man's body weight. It represents about 50% of a woman's weight. Young and middle-aged adults who drink when

they're thirsty do not generally have to do anything more to maintain their body's fluid balance. Children need more water because they expend more energy, but most children who drink when they are thirsty get as much water as their systems require.

Age and dehydration

Adults over the age of 60 who drink only when they are thirsty probably get only about 90% of the fluid they need. Developing a habit of drinking only in response to the body's thirst signals raises an older person's risk of becoming dehydrated. Seniors who have relocated to areas where the weather is warmer or dryer than the climate they are accustomed to are even likelier to become dehydrated unless they make it a practice to drink even when they are not thirsty.

Dehydration in children usually results from losing large amounts of fluid and not drinking enough water to replace the loss. This condition generally occurs in children who have stomach flu characterized by **vomiting** and **diarrhea**, or who can not or will not take enough fluids to compensate for excessive losses associated with **fever** and sweating of acute illness. An infant can become dehydrated only hours after becoming ill. Dehydration is a major cause of infant illness and **death** throughout the world.

Types of dehydration

Mild dehydration is the loss of no more than 5% of the body's fluid. Loss of 5–10% is considered moderate dehydration. Severe dehydration (loss of 10–15% of body fluids) is a life-threatening condition that requires immediate medical care.

Complications of dehydration

When the body's fluid supply is severely depleted, hypovolemic **shock** is likely to occur. This condition, which is also called physical collapse, is characterized by pale, cool, clammy skin; rapid heartbeat; and shallow breathing.

Blood pressure sometimes drops so low it can not be measured, and skin at the knees and elbows may become blotchy. **Anxiety**, restlessness, and thirst increase. After the patient's temperature reaches 107 °F (41.7 °C) damage to the brain and other vital organs occurs quickly.

Causes and symptoms

Strenuous activity, excessive sweating, high fever, and prolonged vomiting or diarrhea are common causes of dehydration. So are staying in the sun too

long, not drinking enough fluids, and visiting or moving to a warm region where it doesn't often rain. Alcohol, **caffeine**, and diuretics or other medications that increase the amount of fluid excreted can cause dehydration.

Reduced fluid intake can be a result of:

- appetite loss associated with acute illness
- excessive urination (polyuria)
- nausea
- bacterial or viral infection or inflammation of the pharynx (pharyngitis)
- inflammation of the mouth caused by illness, infection, irritation, or vitamin deficiency (stomatitis)

Other conditions that can lead to dehydration include:

- disease of the adrenal glands, which regulate the body's water and salt balance and the function of many organ systems
- diabetes mellitus
- eating disorders
- **kidney disease**
- chronic lung disease.

An infant who does not wet a diaper in an eight-hour period is dehydrated. The soft spot on the baby's head (fontanel) may be depressed. Symptoms of dehydration at any age include cracked lips, dry or sticky mouth, lethargy, and sunken eyes. A person who is dehydrated cries without shedding tears and does not urinate very often. The skin is less elastic than it should be and is slow to return to its normal position after being pinched.

Dehydration can cause confusion, **constipation**, discomfort, drowsiness, fever, and thirst. The skin turns pale and cold, the mucous membranes lining the mouth and nose lose their natural moisture. The pulse sometimes races and breathing becomes rapid. Significant fluid loss can cause serious neurological problems.

Diagnosis

The patient's symptoms and medical history usually suggest dehydration. **Physical examination** may reveal shock, rapid heart rate, and/or low blood pressure. Laboratory tests, including blood tests (to check electrolyte levels) and urine tests (e.g., urine specific gravity and creatinine), are used to evaluate the severity of the problem. Other laboratory tests may be ordered to determine the underlying condition

(such as diabetes or an adrenal gland disorder) causing the dehydration.

Treatment

Increased fluid intake and replacement of lost electrolytes are usually sufficient to restore fluid balances in patients who are mildly or moderately dehydrated. For individuals who are mildly dehydrated, just drinking plain water may be all the treatment that is needed. Adults who need to replace lost electrolytes may drink sports beverages (e.g., Gatorade or Recharge) or consume a little additional salt. Parents should follow label instructions when giving children Pedialyte or other commercial products recommended to relieve dehydration. Children who are dehydrated should receive only clear fluids for the first 24 hours.

A child who is vomiting should sip one or two teaspoons of liquid every 10 minutes. A child who is less than a year old and who is not vomiting should be given one tablespoon of liquid every 20 minutes. A child who is more than one year old and who is not vomiting should take two tablespoons of liquid every 30 minutes. A baby who is being breast-fed should be given clear liquids for two consecutive feedings before breastfeeding is resumed. A bottle-fed baby should be given formula diluted to half its strength for the first 24 hours after developing symptoms of dehydration.

In order to accurately calculate fluid loss, it's important to chart weight changes every day and keep a record of how many times a patient vomits or has diarrhea. Parents should note how many times a baby's diaper must be changed.

Children and adults can gradually return to their normal diet after they have stopped vomiting and no longer have diarrhea. Bland foods should be reintroduced first, with other foods added as the digestive system is able to tolerate them. Milk, ice cream, cheese, and butter should not be eaten until 72 hours after symptoms have disappeared.

Medical care

Severe dehydration can require hospitalization and intravenous fluid replacement. If an individual's blood pressure drops enough to cause or threaten the development of shock, medical treatment is usually required. A doctor should be notified whenever an infant or child exhibits signs of dehydration or a parent is concerned that a stomach virus or other acute illness may lead to dehydration.

a doctor should also be notified if:

- a child less than three months old develops a fever higher than 100 °F (37.8 °C)
- a child more than three months old develops a fever higher than 102 °F (38.9 °C)
- symptoms of dehydration worsen
- an individual urinates very sparingly or does not urinate at all during a six-hour period
- dizziness, listlessness, or excessive thirst occur
- a person who is dieting and using diuretics loses more than 3 lb (1.3 kg) in a day or more than 5 lb (2.3 kg) a week

When treating dehydration, the underlying cause must also be addressed. For example, if dehydration is caused by vomiting or diarrhea, medications may be prescribed to resolve these symptoms. Patients who are dehydrated due to diabetes, kidney disease, or adrenal gland disorders must receive treatment for these conditions as well as for the resulting dehydration.

Alternative treatment

Gelatin water can be substituted for electrolyte-replacement solutions. It is made by diluting a 3-oz package in a quart of water or by adding one-quarter teaspoon of salt and a tablespoon of sugar to a pint of water.

Prognosis

Mild dehydration rarely results in complications. If the cause is eliminated and lost fluid is replaced, mild dehydration can usually be cured in 24–48 hours.

Vomiting and diarrhea that continue for several days without adequate fluid replacement can be fatal. The risk of life-threatening complications is greater for young children and the elderly. However, dehydration that is rapidly recognized and treated has a good outcome.

Prevention

Patients who are vomiting or who have diarrhea can prevent dehydration by drinking enough fluid for their urine to remain the color of pale straw. Ensuring that patients always drink adequate fluids during an illness will help prevent dehydration. Infants and young children with diarrhea and vomiting can be given electrolyte solutions such as Pedialyte to help prevent dehydration. People who are not ill can

KEY TERMS

Electrolytes—Mineral salts, such as sodium and potassium, dissolved in body fluid.

maintain proper fluid balance by drinking several glasses of water before going outside on a hot day. It is also a good idea to avoid coffee and tea, which increase body temperature and water loss.

Patients should know whether any medication they are taking can cause dehydration and should get prompt medical care to correct any underlying condition that increases the risk of dehydration.

Other methods of preventing dehydration and ensuring adequate fluid intake include:

- eating more soup at mealtime
- drinking plenty of water and juice at mealtime and between meals
- keeping a glass of water nearby when working or relaxing

Resources

OTHER

“Hydration—Getting Enough Water.” *Loyola University Health System*. May 13, 1998. <<http://www.luhs.org>>.

Maureen Haggerty

Delavirdine see **Non-nucleoside reverse transcriptase inhibitors**

Delayed hypersensitivity skin test

Definition

A delayed hypersensitivity test (DHT) is an immune function test measuring the presence of activated T cells that recognize a certain substance.

Purpose

The immune system protects against infection by viruses, bacteria, fungi, and parasites. After initial exposure to a foreign substance, or antigen, the immune system creates both antibodies and sensitized

T cells. Both these immune agents respond when the body is reexposed to the antigen. Antibodies, which are circulating proteins, respond within minutes, to give what, is termed an immediate hypersensitivity reaction. T cells responses occur over several days, and are thus called delayed hypersensitivity reactions. The cascade of events initiated by the T cells leads to hardening (induration) and redness (erythema) at the injection site.

A DHT is performed for one of three reasons:

- To test for exposure to specific diseases, such as **tuberculosis** (TB). Tuberculosis testing is done by injecting into the skin a small volume of TB antigen, which contains no organisms (live or dead) but can still provoke an immune response.
- To test for allergic sensitivity to potential skin irritants, such as poison ivy. Skin allergy testing is usually done by placing a series of adhesive patches on the skin containing potential allergens, or allergy-causing substances.
- To assess the vitality of the T cell response as part of the evaluation of immune system health in infection, **cancer**, immune disorders, pre-transplantation screening, **aging**, and malnutrition. DHT can help predict survival in immunocompromised patients, and evaluate the success of restorative therapy. Antigens used for these tests must be ones the patient has been exposed to before, and, therefore, include inactivated antigens from common infectious agents to which the patient might have been exposed, such as **mumps**, *Candida albicans*, **tetanus** toxoid, and trichophyton (a skin fungus).

Precautions

No special precautions are necessary for most patients. Those with known hypersensitivity to certain skin irritants should alert the clinician performing the test. Some commercial preparations of fungal antigens contain mercury, a source of irritation to some patients.

Description

The most accurate TB test is the Mantoux test, in which a small amount of TB antigen is injected into the skin. The area is examined 48–72 hours after the injection.

In the patch test, 20–30 adhesive patches are usually placed on the upper back. The patches are kept in place and the area is kept dry for 48 hours. The patches are then removed, and the skin is

examined 24 hours afterward, and possibly again a day or more following that. Patch testing is usually performed following a patient complaint of skin irritation from an unknown substance. Testing may suggest several candidates; identifying the right one requires careful review of the patient's possible exposure.

The test of overall T cell responsiveness is performed with several injections. Each area injected is circled and marked. Results are read 48 hours after the injection.

Preparation

No special preparation is necessary.

Aftercare

Patches should be kept dry. Injection sites may be washed, but excessive rubbing should be avoided. Patches and injection sites may become reddened or irritated. If a patch causes severe itching or discomfort, the patient should remove it immediately.

Risks

DHT is quite safe for virtually all people. There is no risk of infection from the agents injected, since they are purified antigens, not whole organisms. Life threatening, hypersensitive reactions (**anaphylaxis**) are a very small risk; patients should notify the administering physician immediately if signs of wheezing, swelling, or diffuse redness of the skin develops.

Normal results

Absence of exposure to TB is indicated by absent or very little skin reaction; redness or hardness smaller than 5 mm (about 0.25 in) is considered normal for a person not exposed or infected with TB.

Patch test sites should be normal or only slightly red.

T cell responsiveness tests should be positive; that is, the injected areas should be reddened and hard. Two affected areas of 2 mm or more is considered a positive result.

Abnormal results

TB exposure is indicated by a reaction of 10 mm or more. The degree of redness is not important. A 5–10 mm area could indicate exposure if there is an underlying risk to TB.

KEY TERMS

Allergen—A foreign substance that provokes an immune reaction in some sensitive people but not in most others.

Anaphylaxis—An exaggerated, life-threatening hypersensitivity reaction to a previously encountered antigen.

Antibody—An immune system protein made to fight infection.

Antigen—A foreign substance detected that provokes an immune reaction.

Patch test areas that become reddened and irritated indicate reaction to the substance in the patch.

Absence of any reaction to injected areas indicates lack of T cell responsiveness, a condition called anergy. T cell anergy is seen in immune deficiency diseases including **AIDS**, some cases of infectious diseases, malignancies, immunosuppressive therapy (including corticosteroid treatment), some autoimmune diseases, **malnutrition**, major surgery, and some viral immunizations.

Resources

BOOKS

Lawlor Jr., G. J., et. al. *Manual of Allergy and Immunology*. Little, Brown and Co., 1995.

Richard Robinson

Delirium

Definition

Delirium is a state of mental confusion that develops quickly and usually fluctuates in intensity.

Description

Delirium is a syndrome, or group of symptoms, caused by a disturbance in the normal functioning of the brain. The delirious patient has a reduced awareness of and responsiveness to the environment, which may be manifested as disorientation, incoherence, and memory disturbance. Delirium is often marked by **hallucinations**, **delusions**, and a dream-like state.

Delirium affects at least one in 10 hospitalized patients, and is a common part of many terminal illnesses. Delirium is more common in the elderly than in the general population. While it is not a specific disease itself, patients with delirium usually fare worse than those with the same illness who do not have delirium.

Causes and symptoms

Causes

There are a large number of possible causes of delirium. Metabolic disorders are the single most common cause, accounting for 20–40% of all cases. This type of delirium, termed “metabolic encephalopathy,” may result from organ failure, including liver or kidney failure. Other metabolic causes include **diabetes mellitus**, **hyperthyroidism** and **hypothyroidism**, vitamin deficiencies, and imbalances of fluids and electrolytes in the blood. Severe **dehydration** can also cause delirium.

Drug intoxication (“intoxication confusional state”) is responsible for up to 20% of delirium cases, either from side effects, overdose, or deliberate ingestion of a mind-altering substance. Medicinal drugs with delirium as a possible side effect or result of overdose include:

- anticholinergics, including atropine, scopolamine, chlorpromazine (an antipsychotic), and diphenhydramine (an antihistamine)
- sedatives, including **barbiturates**, **benzodiazepines**, and ethanol (drinking alcohol)
- antidepressant drugs
- anticonvulsant drugs
- nonsteroidal anti-inflammatory drugs (NSAIDs), including ibuprofen and **acetaminophen**
- corticosteroids, including prednisone
- anticancer drugs, including methotrexate and procarbazine
- lithium
- cimetidine
- antibiotics
- L-dopa

Delirium may result from ingestion of legal or illegal psychoactive drugs, including:

- ethanol (drinking alcohol)
- marijuana
- **LSD (lysergic acid diethylamide)** and other hallucinogens

- amphetamines
- cocaine
- opiates, including heroin and morphine
- PCP (phencyclidine)
- inhalants

Drug withdrawal may also cause delirium. Delirium tremens, or “DTs,” may occur during alcohol withdrawal after prolonged or intense consumption. Withdrawal symptoms are also possible from many of the psychoactive prescription drugs.

Poisons may cause delirium (“toxic encephalopathy”), including:

- solvents, such as gasoline, kerosene, turpentine, benzene, and alcohols
- carbon monoxide
- refrigerants (Freon)
- heavy metals, such as lead, mercury, and arsenic
- insecticides, such as Parathion and Sevin
- mushrooms, such as *Amanita* species
- plants such as jimsonweed (*Datura stramonium*) and morning glory (*Ipomoea* spp.)
- animal venoms

Other causes of delirium include:

- infection
- fever
- head trauma
- epilepsy
- brain hemorrhage or infarction
- brain tumor
- low blood oxygen (hypoxemia)
- high blood carbon dioxide (hypercapnia)
- post-surgical complication

Symptoms

The symptoms of delirium come on quickly, in hours or days, in contrast to those of dementia, which develop much more slowly. Delirium symptoms typically fluctuate through the day, with periods of relative calm and lucidity alternating with periods of florid delirium. The hallmark of delirium is a fluctuating level of consciousness. Symptoms may include:

- decreased awareness of the environment
- confusion or disorientation, especially of time
- memory impairment, especially of recent events

KEY TERMS

Dementia—A loss of mental ability severe enough to interfere with functioning. While dementia and delirium have some of the same symptoms, dementia has a much slower onset.

Electroencephalogram (EEG)—A chart of the brain wave patterns picked up by electrodes placed on the scalp. This is useful for diagnosing central nervous system disorders.

Encephalopathy—A brain dysfunction or disorder.

- hallucinations
- illusions and misinterpreted stimuli
- increased or decreased activity level
- mood disturbance, possibly including **anxiety**, euphoria or depression
- language or speech impairment

Diagnosis

Delirium is diagnosed through the medical history and recognition of symptoms during mental status examination. The most important part of diagnosis is determining the cause of the delirium. Tests may include blood and urine analysis for levels of drugs, fluids, electrolytes, and blood gases, and to test for infection; lumbar puncture (“spinal tap”) to test for central nervous system infection; x ray, computed tomography scans (CT), or **magnetic resonance imaging (MRI)** scans to look for tumors, hemorrhage, or other brain abnormality; thyroid tests; **electroencephalography (EEG)**; **electrocardiography (ECG)**; and possibly others as dictated by the likely cause.

Treatment

Treatment of delirium begins with recognizing and treating the underlying cause. Delirium itself is managed by reducing disturbing stimuli, or providing soothing ones; use of simple, clear language in communication; and reassurance, especially from family members. Physical restraints may be needed if the patient is a danger to himself or others, or if he insists on removing necessary medical equipment such as intravenous lines or monitors. Sedatives or **antipsychotic drugs** may be used to reduce anxiety, hallucinations, and delusions.

Prognosis

Persons with delirium usually have a worse prognosis for the underlying disease than the person without delirium. Nonetheless, those without terminal illness usually recover from delirium. They may not, however, regain all their original cognitive abilities, and may be left with some permanent impairments, including **fatigue**, irritability, difficulty concentrating, or mood changes.

Prevention

Prevention of delirium is focused on treating or avoiding its underlying causes. The most preventable forms are those induced by drugs. Strategies for reducing delirium include following prescriptions, consulting the prescribing physician immediately if symptoms occur, and consulting the physician before discontinuing the drug, even if it has been ineffective; avoiding intoxication with legal or illegal drugs, and seeking professional assistance before suddenly discontinuing an addictive drug such as alcohol or heroin; maintaining good **nutrition**, which promotes general health and can minimize the likelihood of delirium from alcohol intoxication and withdrawal; and avoiding exposure to solvents, insecticides, heavy metals, or biological poisons in the home or workplace.

Resources

BOOKS

Guze, Samuel, editor. *Adult Psychiatry*. Mosby Year Book, 1997.

Richard Robinson

Delta virus hepatitis see **Hepatitis D**

Delusions

Definition

A delusion is an unshakable belief in something untrue. These irrational beliefs defy normal reasoning, and remain firm even when overwhelming proof is presented to dispute them. Delusions are often accompanied by **hallucinations** and/or feelings of **paranoia**, which act to strengthen confidence in the delusion. Delusions are distinct from culturally or religiously based beliefs that may be seen as untrue by outsiders.

Description

Delusions are a common symptom of several mood and personality-related mental illnesses, including **schizoaffective disorder**, **schizophrenia**, shared psychotic disorder, major depressive disorder, and **bipolar disorder**. They are also the major feature of delusional disorder. Individuals with delusional disorder suffer from long-term, complex delusions that fall into one of six categories: persecutory, grandiose, jealousy, erotomanic, somatic, or mixed. There are also delusional disorders such as **dementia** that clearly have organic or physical causes.

Persecutory

Individuals with persecutory delusional disorder are plagued by feelings of paranoia and an irrational yet unshakable belief that someone is plotting against them, or out to harm them.

Grandiose

Individuals with grandiose delusional disorder have an inflated sense of self-worth. Their delusions center on their own importance, such as believing that they have done or created something of extreme value or have a “special mission.”

Jealousy

Jealous delusions are unjustified and irrational beliefs that an individual’s spouse or significant other has been unfaithful.

Erotomanic

Individuals with erotomanic delusional disorder believe that another person, often a stranger, is in love with them. The object of their affection is typically of a higher social status, sometimes a celebrity. This type of delusional disorder may lead to stalking or other potentially dangerous behavior.

Somatic

Somatic delusions involve the belief that something is physically wrong with the individual. The delusion may involve a medical condition or illness or a perceived deformity. This condition differs from **hypochondriasis** in that the deformity is perceived as a fixed condition not a temporary illness.

Mixed

Mixed delusions are those characterized by two or more of persecutory, grandiose, jealousy, erotomanic, or somatic themes.

KEY TERMS

Hallucinations—False or distorted sensory experiences that appear to be real perceptions.

Paranoia—An unfounded or exaggerated distrust of others.

Shared psychotic disorder—Also known as folie à deux; shared psychotic disorder is an uncommon disorder in which the same delusion is shared by two or more individuals.

Causes and symptoms

Some studies have indicated that delusions may be generated by abnormalities in the limbic system, the portion of the brain on the inner edge of the cerebral cortex that is believed to regulate emotions. The exact source of delusions has not been conclusively found, but potential causes include genetics, neurological abnormalities, and changes in brain chemistry. Delusions are also a known possible side effect of drug use and **abuse** (e.g., amphetamines, **cocaine**, PCP).

Diagnosis

Patients with delusional symptoms should undergo a thorough **physical examination** and patient history to rule out possible organic causes (such as dementia). If a psychological cause is suspected, a mental health professional will typically conduct an interview with the patient and administer one of several clinical inventories, or tests, to evaluate mental status.

Treatment

Delusions that are symptomatic of delusional disorder should be treated by a psychologist and/or psychiatrist. Though **antipsychotic drugs** are often not effective, antipsychotic medication such as thioridazine (Mellaril), haloperidol (Haldol), chlorpromazine (Thorazine), clozapine (Clozaril), or risperidone (Risperdal) may be prescribed, and cognitive therapy or psychotherapy may be attempted.

If an underlying condition such as schizophrenia, depression, or drug abuse is found to be triggering the delusions, an appropriate course of medication and/or psychosocial therapy is employed to treat the primary disorder. The medication, typically, will include an antipsychotic agent.

Prognosis

Delusional disorder is typically a chronic condition, but with appropriate treatment, a remission of delusional symptoms occurs in up to 50% of patients. However, because of their strong belief in the reality of their delusions and a lack of insight into their condition, individuals with this disorder may never seek treatment, or may be resistant to exploring their condition in psychotherapy.

Resources

ORGANIZATIONS

American Psychiatric Association. 1400 K Street NW, Washington DC 20005. (888) 357-7924. <<http://www.psych.org>>.

American Psychological Association (APA). 750 First St. NE, Washington, DC 20002-4242. (202) 336-5700. <<http://www.apa.org>>.

National Alliance for the Mentally Ill (NAMI). Colonial Place Three, 2107 Wilson Blvd., Ste. 300, Arlington, VA 22201-3042. (800) 950-6264. <<http://www.nami.org>>.

Paula Anne Ford-Martin

Dementia

Definition

Dementia is a loss of mental ability severe enough to interfere with normal activities of daily living, lasting more than six months, not present since birth, and not associated with a loss or alteration of consciousness.

Description

Dementia is a group of symptoms caused by gradual death of brain cells. The loss of cognitive abilities that occurs with dementia leads to impairments in memory, reasoning, planning, and behavior. While the overwhelming number of people with dementia are elderly, it is not an inevitable part of **aging**. Instead, dementia is caused by specific brain diseases. **Alzheimer's disease** (AD) is the most common cause, followed by vascular or multi-infarct dementia.

The prevalence of dementia has been difficult to determine, partly because of differences in definition among different studies, and partly because there is some normal decline in functional ability with age. Dementia affects 5–8% of all people between ages 65 and 74, and up to 20% of those between 75 and 84.

Estimates for dementia in those 85 and over range from 30–47%. Between two and four million Americans have AD; that number is expected to grow to as many as 14 million by the middle of the twenty-first century as the population ages.

The cost of dementia can be considerable. While most people with dementia are retired and do not suffer income losses from their disease, the cost of care often is enormous. Financial burdens include lost wages for family caregivers, medical supplies and drugs, and home modifications to ensure safety. Nursing home care may cost several thousand dollars a month or more. The psychological cost is not as easily quantifiable but can be even more profound. The person with dementia loses control of many of the essential features of his life and personality, and loved ones lose a family member even as they continue to cope with the burdens of increasing dependence and unpredictability.

Causes and symptoms

Causes

Dementia usually is caused by degeneration in the cerebral cortex, the part of the brain responsible for thoughts, memories, actions and personality. Death of brain cells in this region leads to the cognitive impairment that characterizes dementia.

The most common cause of dementia is AD, accounting for one-half to three-fourths of all cases. The brain of a person with AD becomes clogged with two abnormal structures, called neurofibrillary tangles and senile plaques. Neurofibrillary tangles are twisted masses of protein fibers inside nerve cells, or neurons. Senile plaques are composed of parts of neurons surrounding a group of proteins called beta-amyloid deposits. Why these structures develop is unknown. Current research indicates possible roles for inflammation, blood flow restriction, and toxic molecular fragments known as free radicals. Several genes have been associated with higher incidences of AD, although the exact role of these genes still is unknown.

Vascular dementia is estimated to cause from 5–30% of all dementias. It occurs from decrease in blood flow to the brain, most commonly due to a series of small strokes (multi-infarct dementia). Other cerebrovascular causes include: **vasculitis** from **syphilis**, **Lyme disease**, or systemic lupus erythematosus; subdural hematoma; and subarachnoid hemorrhage. Because of the usually sudden nature of its cause, the symptoms of vascular dementia tend to begin more

abruptly than those of Alzheimer's dementia. Symptoms may progress stepwise with the occurrence of new strokes. Unlike AD, the incidence of vascular dementia is lower after age 75.

Other conditions that may cause dementia include:

- AIDS
- Parkinson's disease
- Lewy body disease
- Pick's disease
- Huntington's disease
- Creutzfeldt-Jakob disease
- brain tumor
- **hydrocephalus**
- head trauma
- multiple sclerosis
- prolonged **abuse** of alcohol or other drugs
- vitamin deficiency: thiamin, niacin, or B₁₂
- hypothyroidism
- hypercalcemia

Symptoms

Dementia is marked by a gradual impoverishment of thought and other mental activities. Losses eventually affect virtually every aspect of mental life. The slow progression of dementia is in contrast with **delirium**, which involves some of the same symptoms, but has a very rapid onset and fluctuating course with alteration in the level of consciousness. However, delirium may occur with dementia, especially since the person with dementia is more susceptible to the delirium-inducing effects of many types of drugs.

Symptoms include:

- Memory losses. Memory loss usually is the first symptom noticed. It may begin with misplacing valuables such as a wallet or car keys, then progress to forgetting appointments, where the car was left, and the route home, for instance. More profound losses follow, such as forgetting the names and faces of family members.
- Impaired abstraction and planning. The person with dementia may lose the ability to perform familiar tasks, to plan activities, and to draw simple conclusions from facts.
- Language and comprehension disturbances. The person may be unable to understand instructions, or follow the logic of moderately complex sentences.

Later, he or she may not understand his or her own sentences, and have difficulty forming thoughts into words.

- Poor judgment. The person may not recognize the consequences of his or her actions or be able to evaluate the appropriateness of behavior. Behavior may become rude, overly friendly, or aggressive. Personal hygiene may be ignored.
- Impaired orientation ability. The person may not be able to identify the time of day, even from obvious visual clues; or may not recognize his or her location, even if familiar. This disability may stem partly from losses of memory and partly from impaired abstraction.
- Decreased attention and increased restlessness. This may cause the person with dementia to begin an activity and quickly lose interest, and to wander frequently. Wandering may cause significant safety problems, when combined with disorientation and memory losses. The person may begin to cook something on the stove, then become distracted and wander away while it is cooking.
- Behavioral changes and **psychosis**. The person with dementia may lose interest in once-pleasurable activities, and become more passive, depressed, or anxious. Delusions, suspicion, **paranoia**, and hallucinations may occur later in the disease. Sleep disturbances may occur, including **insomnia** and sleep interruptions.

Diagnosis

Since dementia usually progresses slowly, diagnosing it in its early stages can be difficult. However, prompt intervention and treatment has been shown to help slow the effects of dementia, so early diagnosis is important. Several office visits over several months or more may be needed. Diagnosis begins with a thorough physical exam and complete medical history, usually including comments from family members or caregivers. A family history of either AD or cerebrovascular disease may provide clues to the cause of symptoms. Simple tests of mental function, including word recall, object naming, and number-symbol matching, are used to track changes in the person's cognitive ability.

Depression is common in the elderly and can be mistaken for dementia; therefore, ruling out depression is an important part of the diagnosis. Distinguishing dementia from the mild normal cognitive decline of advanced age also is critical. The medical history includes a complete listing of drugs being

taken, since a number of drugs can cause dementia-like symptoms.

Determining the cause of dementia may require a variety of medical tests, chosen to match the most likely etiology. Cerebrovascular disease, hydrocephalus, and tumors may be diagnosed with x rays, CT or MRI scans, and vascular imaging studies. Blood tests may reveal nutritional deficiencies or hormone imbalances.

Treatment

Treatment of dementia begins with treatment of the underlying disease, where possible. The underlying causes of nutritional, hormonal, tumor-caused and drug-related dementias may be reversible to some extent. Treatment for stroke-related dementia begins by minimizing the risk of further strokes, through **smoking** cessation, **aspirin** therapy, and treatment of hypertension, for instance. No therapies can reverse the progression of AD. Aspirin, estrogen, vitamin E, and selegiline have been evaluated for their ability to slow the rate of progression. However, none of these have been proven effective. In fact, in 2002 and 2003, research revealed that non-steroidal anti-inflammatory agents (NSAIDs) did not help prevent AD and dementia. In the same two years, the Women's Health Initiative, a large clinical trial, was halted because of detrimental effects of combined estrogen and progestin therapy, or **hormone replacement therapy** (HRT). Not only was HRT found to increase risk of **breast cancer**, **stroke**, and other heart disease, but the risk of probable dementia was twice that for women taking HRT than for those taking a placebo. Further, those taking HRT had a substantial and clinically important decline in indicators of cognitive ability. Studies still debate the effects of vitamin E on slowing the progression of moderately severe AD.

Care for a person with dementia can be difficult and complex. The patient must learn to cope with functional and cognitive limitations, while family members or other caregivers assume increasing responsibility for the person's physical needs. In progressive dementias such as AD, the person may ultimately become completely dependent. Education of the patient and family early in the disease progression can help them anticipate and plan for inevitable changes.

Symptoms of dementia may be treated with a combination of psychotherapy, environmental modifications, and medication. Drug therapy can be complicated by forgetfulness, especially if the prescribed drug must be taken several times daily.

Behavioral approaches may be used to reduce the frequency or severity of problem behaviors, such as aggression or socially inappropriate conduct. Problem behavior may be a reaction to frustration or overstimulation; understanding and modifying the situations that trigger it can be effective. Strategies may include breaking down complex tasks, such as dressing or feeding, into simpler steps, or reducing the amount of activity in the environment to avoid confusion and agitation. Pleasurable activities, such as crafts, games, and music, can provide therapeutic stimulation and improve mood.

Modifying the environment can increase safety and comfort while decreasing agitation. Home modifications for safety include removal or lock-up of hazards such as sharp knives, dangerous chemicals, and tools. Child-proof latches or Dutch doors may be used to limit access as well. Lowering the hot water temperature to 120 °F (48.9 °C) or less reduces the risk of scalding. Bed rails and bathroom safety rails can be important safety measures, as well. Confusion may be reduced with simpler decorative schemes and presence of familiar objects. Covering or disguising doors (with a mural, for example) may reduce the tendency to wander. Positioning the bed in view of the bathroom can decrease incontinence.

Two drugs, tacrine (Cognex) and donepezil (Aricept), are commonly prescribed for AD. These drugs inhibit the breakdown of acetylcholine in the brain, prolonging its ability to conduct chemical messages between brain cells. They provide temporary improvement in cognitive functions for some patients with mild to moderate AD and help delay disease progression.

Psychotic symptoms, including paranoia, **delusions**, and **hallucinations**, may be treated with antipsychotic drugs, such as haloperidol, chlorpromazine, risperidone, and clozapine. Side effects of these drugs can be significant. **Antianxiety drugs** such as Valium may improve behavioral symptoms, especially agitation and **anxiety**, although BuSpar has fewer side effects. The anticonvulsant carbamazepine also is sometimes prescribed for agitation. Depression is treated with antidepressants, usually beginning with selective serotonin reuptake inhibitors (SSRIs) such as Prozac or Paxil, followed by monoamine oxidase inhibitors or tricyclic antidepressants. In general, medications should be administered cautiously to demented patients, in the lowest possible effective doses, to minimize side effects. Supervision of taking medications is generally required.

Long-term institutional care may be needed for the person with dementia, as profound cognitive losses

often precede death by a number of years. Early planning for the financial burden of nursing home care is critical. Useful information about financial planning for long-term care is available through the Alzheimer's Association.

Family members or others caring for a person with dementia often are subject to extreme stress, and may develop feelings of anger, resentment, guilt, and hopelessness, in addition to the sorrow they feel for their loved one and for themselves. Depression is an extremely common consequence of being a full-time caregiver for a person with dementia. Support groups can be an important way to deal with the stress of caregiving. The location and contact numbers for caregiver support groups are available from the Alzheimer's Association; they may also be available through a local social service agency or the patient's physician. Medical treatment for depression may be an important adjunct to group support.

Alternative treatment

Several drugs are currently being tested for their ability to slow the progress of AD. These include acetyl-L-carnitine, which acts on the cellular energy structures known as mitochondria; propentofylline, which may aid circulation; milameline, which acts similarly to tacrine and donepezil; and ginkgo extract.

Ginkgo extract, derived from the leaves of the *Ginkgo biloba* tree, interferes with a circulatory protein called platelet activating factor. It also increases circulation and oxygenation to the brain. Ginkgo extract has been used for many years in China and is widely prescribed in Europe for treatment of circulatory problems. A 1997 study of patients with dementia seemed to show that ginkgo extract could improve their symptoms, though the study was criticized for certain flaws in its method.

Prognosis

The prognosis for dementia depends on the underlying disease. On average, people with Alzheimer's disease live eight years past their diagnosis, with a range from one to 20 years. Vascular dementia usually is progressive, with death from stroke, infection, or heart disease.

Prevention

There is no known way to prevent Alzheimer's disease, although several drugs under investigation may reduce its risk or slow its progression. The risk of developing multi-infarct dementia may be reduced by reducing the risk of stroke. Various studies

KEY TERMS

Donepezil—A drug commonly prescribed for Alzheimer's disease that provides temporary improvement in cognitive functions for some patients with mild-to-moderate forms of the disease.

Ginkgo extract—Made from the leaves of the *Ginkgo biloba* tree, this extract, used in other countries to treat circulatory problems, may improve the symptoms of patients with dementia.

Neurofibrillary tangles—Abnormal structures, composed of twisted masses of protein fibers within nerve cells, found in the brains of people with Alzheimer's disease.

Senile plaques—Abnormal structures, composed of parts of nerve cells surrounding protein deposits, found in the brains of people with Alzheimer's disease.

Tacrine—A drug commonly prescribed for Alzheimer's disease that provides temporary improvement in cognitive functions for some patients with mild-to-moderate forms of the disease.

continue to determine ways to lower risk of AD and dementia. For example, a 2003 study in the New England Journal of Medicine reported that people over age 75 who participated in leisure activities such as playing board games, reading, dancing, and playing musical instruments were less likely to have dementia after five years than others their age.

Resources

PERIODICALS

"Antioxidants Don't Prevent Dementia." *JAAPA* — *Journal of the American Academy of Physicians Assistants*, May 2003: 25.

MacReady, Norra. "Prompt Intervention May Help Slow Dementia." *Clinical Psychiatry News*, May 2003: 38.

"Research Breifs: Play Keeps Dementia Away." *GP*, June 23, 2003: 04.

"Risks of Hormone Treatment." *The Lancet*, May 31, 2003: 1877.

ORGANIZATIONS

Alzheimer's Association. 919 North Michigan Ave., Suite 1000, Chicago, IL 60611. (800) 272-3900. <<http://www.alz.org>>.

Richard Robinson
Teresa G. Odle

Demyelinating disease see **Multiple sclerosis**

Dengue fever

Definition

Dengue **fever** is a disease caused by one of a number of viruses that are carried by mosquitoes. These mosquitoes then transmit the virus to humans.

Description

The virus that causes dengue fever is called an arbovirus, which stands for arthropod-borne virus. Mosquitoes are a type of arthropod. In a number of regions, mosquitoes carry this virus and are responsible for passing it along to humans. These regions include the Middle East, the far East, Africa, and the Caribbean Islands. In these locations, the dengue fever arbovirus is endemic, meaning that the virus naturally and consistently lives in that location. The disease only shows up in the United States sporadically.

In order to understand how dengue fever is transmitted, several terms need to be defined. The word “host” means an animal (including a human) that can be infected with a particular disease. The word “vector” means an organism that can carry a particular disease-causing agent (like a virus or bacteria) without actually developing the disease. The vector can then pass the virus or bacteria on to a new host.

Many of the common illnesses in the United States (including the **common cold**, many viral causes of diarrhea, and **influenza** or “flu”) are spread because the viruses that cause these illness can be passed directly from person to person. However, dengue fever cannot be passed directly from one infected person to another. Instead, the virus responsible for dengue fever requires an intermediate vector, a mosquito, that carries the virus from one host to another. The mosquito that carries the arbovirus responsible for dengue fever is the same type of mosquito that can transmit other diseases, including **yellow fever**. This mosquito is called *Aedes aegypti*. The most common victims are children younger than 10 years of age.

Causes and symptoms

Dengue fever can occur when a mosquito carrying the arbovirus bites a human, passing the virus on to the new host. Once in the body, the virus travels to various glands where it multiplies. The virus can then enter the bloodstream. The presence of the virus within the blood vessels, especially those feeding the skin, causes changes to these blood vessels. The vessels swell and leak. The spleen and lymph nodes become

enlarged, and patches of liver tissue die. A process called disseminated intravascular coagulation (DIC) occurs, where chemicals responsible for clotting are used up and lead to a risk of severe bleeding (hemorrhage).

After the virus has been transmitted to the human host, a period of incubation occurs. During this time (lasting about five to eight days) the virus multiplies. Symptoms of the disease appear suddenly and include high fever, chills, **headache**, eye **pain**, red eyes, enlarged lymph nodes, a red flush to the face, lower back pain, extreme weakness, and severe aches in the legs and joints.

This initial period of illness lasts about two or three days. After this time, the fever drops rapidly and the patient sweats heavily. After about a day of feeling relatively well, the patient’s temperature increases again, although not as much as the first time. A rash of small red bumps begins on the arms and legs, spreading to the chest, abdomen, and back. It rarely affects the face. The palms of the hands and the soles of the feet become swollen and turn bright red. The characteristic combination of fever, rash, and headache are called the “dengue triad.” Most people recover fully from dengue fever, although weakness and fatigue may last for several weeks. Once a person has been infected with dengue fever, his or her immune system keeps producing cells that prevent reinfection for about a year.

More severe illness may occur in some people. These people may be experiencing dengue fever for the first time. However, in some cases a person may have already had dengue fever at one time, recovered, and then is reinfected with the virus. In these cases, the first infection teaches the immune system to recognize the presence of the arbovirus. When the immune cells encounter the virus during later infections, the immune system over-reacts. These types of illnesses, called dengue hemorrhagic fever (DHF) or dengue **shock** syndrome (DSS), involve more severe symptoms. Fever and headache are the first symptoms, but the other initial symptoms of dengue fever are absent. The patient develops a **cough**, followed by the appearance of small purplish spots (petechiae) on the skin. These petechiae are areas where blood is leaking out of the vessels. Large bruised areas appear as the bleeding worsens and abdominal pain may be severe. The patient may begin to vomit a substance that looks like coffee grounds. This is actually a sign of bleeding into the stomach. As the blood vessels become more damaged, they leak more and continue to increase in diameter (dilate), causing a decrease in blood flow to all tissues of the body. This state of low blood flow is called shock. Shock can result in damage to the body’s

KEY TERMS

Endemic—Naturally and consistently present in a certain geographical region.

Host—The organism (such as a monkey or human) in which another organism (such as a virus or bacteria) is living.

Vector—A carrier organism (such as a fly or mosquito) that delivers a virus (or other agent of infection) to a host.

organs (especially the heart and kidneys) because low blood flow deprives them of oxygen.

Diagnosis

Diagnosis should be suspected in endemic areas whenever a high fever goes on for two to seven days, especially if accompanied by a bleeding tendency. Symptoms of shock should suggest the progression of the disease to DSS.

The arbovirus causing dengue fever is one of the few types of arbovirus that can be isolated from the serum of the blood. The serum is the fluid in which blood cells are suspended. Serum can be tested because the phase in which the virus travels throughout the bloodstream is longer in dengue fever than in other arboviral infections. A number of tests are used to look for reactions between the patient's serum and laboratory-produced antibodies. Antibodies are special cells that recognize the markers (or antigens) present on invading organisms. During these tests, antibodies are added to a sample of the patient's serum. Healthcare workers then look for reactions that would only occur if viral antigens were present in that serum.

Treatment

There is no treatment available to shorten the course of dengue fever, DHF, or DSS. Medications can be given to lower the fever and to decrease the pain of muscle aches and headaches. Fluids are given through a needle in a vein to prevent **dehydration**. Blood transfusions may be necessary if severe hemorrhaging occurs. Oxygen should be administered to patients in shock.

Prognosis

The prognosis for uncomplicated dengue fever is very good, and almost 100% of patients fully recover.

However, as many as 6–30% of all patients die when DHF occurs. The death rate is especially high among the youngest patients (under one year old). In places where excellent medical care is available, very close monitoring and immediate treatment of complications lowers the **death** rate among DHF and DSS patients to about 1%.

Prevention

Prevention of dengue fever means decreasing the mosquito population. Any sources of standing water (buckets, vases, etc.) where the mosquitoes can breed must be eliminated. Mosquito repellant is recommended for those areas where dengue fever is endemic. To help break the cycle of transmission, sick patients should be placed in bed nets so that mosquitoes cannot bite them and become arboviral vectors.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Rosalyn Carson-DeWitt, MD

Dental caries see **Tooth decay**

Dental cavity see **Tooth decay**

Dental hygiene see **Oral hygiene**

Dental injuries see **Dental trauma**

Dental trauma

Definition

Dental trauma is injury to the mouth, including teeth, lips, gums, tongue, and jawbones. The most common dental trauma is a broken or lost tooth.

Description

Dental trauma may be inflicted in a number of ways: contact sports, motor vehicle accidents, fights, falls, eating hard foods, drinking hot liquids, and other such mishaps. As oral tissues are highly sensitive, injuries to the mouth are typically very painful. Dental trauma should receive prompt treatment from a dentist.

Causes and symptoms

Soft tissue injuries, such as a “fat lip,” a burned tongue, or a cut inside the cheek, are characterized by **pain**, redness, and swelling with or without bleeding. A broken tooth often has a sharp edge that may cut the tongue and cheek. Depending on the position of the fracture, the tooth may or may not cause **toothache** pain. When a tooth is knocked out (evulsed), the socket is swollen, painful, and bloody. A jawbone may be broken if the upper and lower teeth no longer fit together properly (**malocclusion**), or if the jaws have pain with limited ability to open and close (mobility), especially around the temporomandibular joint (TMJ).

Diagnosis

Dental trauma is readily apparent upon examination. Dental x rays may be taken to determine the extent of the damage to broken teeth. More comprehensive x rays are needed to diagnose a broken jaw.

Treatment

Soft tissue injuries may require only cold compresses to reduce swelling. Bleeding may be controlled with direct pressure applied with clean gauze. Deep lacerations and punctures may require stitches. Pain may be managed with **aspirin** or **acetaminophen** (Tylenol, Aspirin Free Excedrin) or ibuprofen (Motrin, Advil).

Treatment of a broken tooth will vary depending on the severity of the fracture. For immediate first aid, the injured tooth and surrounding area should be rinsed gently with warm water to remove dirt, then covered with a cold compress to reduce swelling and ease pain. A dentist should examine the injury as soon as possible. Any pieces from the broken tooth should be saved and brought along.

If a piece of the outer tooth has chipped off, but the inner core (pulp) is undisturbed, the dentist may simply smooth the rough edges or replace the missing section with a small composite filling. In some cases, a fragment of broken tooth may be bonded back into place. If enough tooth is missing to compromise the entire tooth structure, but the pulp is not permanently damaged, the tooth will require a protective coverage with a gold or porcelain crown. If the pulp has been seriously damaged, the tooth will require **root canal treatment** before it receives a crown. A tooth, that is vertically fractured or fractured below the gumline will require root canal treatment and protective restoration. A tooth that no longer has enough remaining structure to retain a crown may have to be extracted (surgically removed).

When a permanent tooth has been knocked out, it may be saved with prompt action. The tooth must be found immediately after it has been lost. It should be picked up by the natural crown (the top part covered by hard enamel). It must not be handled by the root. If the tooth is dirty, it may be gently rinsed under running water. It should never be scrubbed, and it should never be washed with soap, toothpaste, mouthwash, or other chemicals. The tooth should not be dried or wrapped in a tissue or cloth. It must be kept moist at all times.

The tooth may be placed in a clean container of milk, cool water with or without a pinch of salt, or in saliva. If possible, the patient and the tooth should be brought to the dentist within 30 minutes of the tooth loss. Rapid action improves the chances of successful re-implantation; however, it is possible to save a tooth after 30 minutes, if the tooth has been kept moist and handled properly.

The body usually rejects re-implantation of a primary (baby) tooth. In this case, the empty socket is treated as a soft tissue injury and monitored until the permanent tooth erupts.

A broken jaw must be set back into its proper position and stabilized with wires while it heals. Healing may take six weeks or longer, depending on the patient's age and the severity of the fracture.

Alternative treatment

There is no substitute for treatment by a dentist or other medical professional. There are, however, homeopathic remedies and herbs that can be used simultaneously with dental care and throughout the healing process. Homeopathic arnica (*Arnica montana*) should be taken as soon as possible after the injury to help the body deal with the trauma. Repeating a dose several times daily for the duration of healing is also useful. Homeopathic hypericum (*Hypericum perforatum*) can be taken if nerve pain is involved, especially with a **tooth extraction** or root canal. Homeopathic comfrey (officinale) *Symphytum* may be helpful in treating pain due to broken jaw bones, but should only be used after the bones have been reset. Calendula (*Calendula officinalis*) and plantain (*Plantago major*) can be used as a mouth rinse to enhance tissue healing. These herbs should not be used with deep lacerations that need to heal from the inside first.

Prognosis

When dental trauma receives timely attention and proper treatment, the prognosis for healing is good. As

KEY TERMS

Crown—The natural part of the tooth covered by enamel. A restorative crown is a protective shell that fits over a tooth.

Eruption—The process of a tooth breaking through the gum tissue to grow into place in the mouth.

Evulsion—The forceful, and usually accidental, removal of a tooth from its socket in the bone.

Extraction—The surgical removal of a tooth from its socket in the bone.

Malocclusion—A problem in the way the upper and lower teeth fit together in biting or chewing.

Pulp—The soft innermost layer of a tooth containing blood vessels and nerves.

Root canal treatment—The process of removing diseased or damaged pulp from a tooth, then filling and sealing the pulp chamber and root canals.

Temporomandibular joint (TMJ)—The jaw joint formed by the mandible (lower jaw bone) moving against the temporal (temple and side) bone of the skull.

with other types of trauma, infection may be a complication, but a course of antibiotics is generally effective.

Prevention

Most dental trauma is preventable. Car seat belts should always be worn, and young children should be secured in appropriate car seats. Homes should be monitored for potential tripping and slipping hazards. Child-proofing measures should be taken, especially for toddlers. In addition to placing gates across stairs and padding sharp table edges, electrical cords should be tucked away. Young children may receive severe oral **burns** from gnawing on live power cords.

Everyone who participates in contact sports should wear a mouthguard to avoid dental trauma. Athletes in football, ice hockey, wrestling, and boxing commonly wear mouthguards. The mandatory use of mouthguards in football prevents about 200,000 oral injuries annually. Mouthguards should also be worn along with helmets in noncontact sports such as skateboarding, in-line skating, and bicycling. An athlete who does not wear a mouthguard is 60 times more likely to sustain dental trauma than one who does. Any activity involving speed, an increased chance of falling, and potential contact with a hard piece of

equipment has the likelihood of dental trauma that may be prevented or substantially reduced in severity with the use of mouthguards.

Resources

ORGANIZATIONS

American Academy of Pediatric Dentistry. 211 East Chicago Ave., Ste. 700, Chicago, IL 60611-2616. (312) 337-2169. <<http://www.aapd.org>>.

American Association of Endodontists. 211 East Chicago Ave., Ste. 1100, Chicago, IL 60611-2691. (800) 872-3636. <<http://www.aae.org>>.

American Association of Oral and Maxillofacial Surgeons. 9700 West Bryn Mawr Ave., Rosemont, IL 60018-5701. (847) 678-6200. <<http://www.aaoms.org>>.

American Dental Association. 211 E. Chicago Ave., Chicago, IL 60611. (312) 440-2500. <<http://www.ada.org>>.

Bethany Thivierge

Depersonalization disorder see **Dissociative disorders**

Depo-Provera/Norplant

Definition

Norplant is a long-acting hormone that is inserted under the skin and prevents conception for up to five years. Depo-Provera is also a hormone, but is administered by intramuscular injection and provides protection against **pregnancy** for three months. Lunelle is another injectable contraceptive that is administered monthly (every 28 to 30 days); it was approved by the Food and Drug Administration (FDA) in October 2000. The hormone in Norplant and Depo-Provera is progestin, a synthetic hormone similar to one found naturally in a woman's body; Lunelle contains the hormones progestin and estrogen.

Purpose

The purpose of these hormones is to prevent pregnancy; they are about 99% effective in achieving this goal. No hormonal contraceptive methods provide protection from **AIDS** or other sexually transmitted diseases.

Depo-Provera and Lunelle are given as an injection and work in several ways to prevent conception. First, the egg (ovum) is prevented from maturing and being released. The mucus in the cervix (opening into

the uterus or womb) becomes thicker, making it difficult for the sperm to enter. Depo-Provera and Lunelle also cause the lining of the uterus to become thinner, making implantation of a fertilized egg unlikely.

An injection of Depo-Provera or Lunelle must be given within the first five days of a normal period. Depo-Provera provides protection against pregnancy for three months, while Lunelle provides similar protection for one month. Ovulation (release of a mature egg) typically occurs within 60 days of the last injection of Lunelle, about twice as fast after use of Depo-Provera. Also, because Lunelle is a combined hormone contraceptive as opposed to progestin-only Depo-Provera and Norplant, it is less likely to cause irregular or absent menstruation.

Norplant capsules contain a synthetic hormone that is slowly released over a period of up to five years. It functions like Depo-Provera in that it prevents the ovaries from producing ova (eggs) and also results in thicker mucus in the cervix, which prevents the sperm from passing through the cervix. Norplant can be inserted at any time.

Preparation

The woman being considered for Depo-Provera or Lunelle will have a pelvic and breast examination, a **Pap test** (a microscopic examination of cell samples taken from the cervix), blood pressure check, weight check, and a review of her medical history. Women who have **diabetes mellitus**, major depression, blood clotting problems, **liver disease**, or weight problems should use these methods only under strict medical supervision. Depo-Provera or Lunelle should not be used if the woman is pregnant, has unexplained vaginal bleeding, suffers from severe liver disease, has breast cancer, or has a history of **blood clots** or **stroke**.

Individuals who select Norplant will receive the same basic physical examination. If approved for this method, a site of implantation will be selected (usually the inside of the upper arm), and the area prepared for minor surgery. The skin will be washed with soap and water, and an antiseptic, such as iodine solution, will be applied. The physician will use a local anesthetic to numb the area, a small incision will be made, the six Norplant capsules will be inserted, and the incision sewn up (sutured). Protection against pregnancy normally begins within 24 hours. If necessary, the implants can be removed in 15–20 minutes. Norplant should not be used by women who are pregnant, have blood clotting problems, or have unexplained vaginal bleeding. Advantages include light periods with less cramping and decreased anemia. This form of birth



A physician inserts a contraceptive implant under the skin of a woman's arm. (Photo Researchers, Inc. Reproduced by permission.)

KEY TERMS

Hormone—A chemical produced in a gland or organ and transported by the blood to another area of the body where it produces a specific effect.

Pap test—A microscopic examination of cell samples taken from the cervix.

control may also be protective against endometrial cancer.

Because Depo-Provera and Norplant use only the hormone progestin, they may provide an alternative for women who can not use estrogen-containing birth control pills. One benefit of Lunelle, however, is that its effects wear off more quickly than Depo-Provera, an important factor in the event that a woman has serious side effects or wants to become pregnant.

Risks

The most common side effects associated with Depo-Provera and Lunelle are yellowing of the skin, **headache**, nervousness, **dizziness**, abdominal **pain**, hair loss, rash, increase in the number of migraine headaches, increased or decreased interest in sexual intercourse, the development of dark spots on the skin, depression, and weakness. Danger signs that need to be reported immediately include weight gain, heavy vaginal bleeding, frequent urination, blurred vision, **fainting**, severe abdominal pain, and coughing up blood. Because the effects of Depo-Provera may last up to 12 weeks, it may take a longer time for women trying

to conceive to become pregnant after discontinuing the injections.

The main reactions to Norplant include headache, weight gain, irregular periods or no period at all, breast tenderness, **acne**, gain or loss of facial hair, color changes of the skin over the area of insertion, and ovarian cysts. The doctor should be notified immediately of lumps in the breast, heavy vaginal bleeding, yellowing of the skin or eyes, or infection of the incision. Women who use Norplant are discouraged from **smoking**.

Normal results

These hormone contraceptive methods normally result in a success rate of 99%.

Resources

OTHER

- “Depo Provera.” *Planned Parenthood of Western Washington*. <<http://www.ppww.org/depo.htm>> .
- “Is Depo-Provera For You?” *Planned Parenthood Federation of America, Inc.* <<http://www.planned-parenthood.org/Library/birthcontrol/depoforyou.html>> .
- “Lunelle.” *Food and Drug Administration (FDA) Page*. <<http://www.fda.gov/cder/foi/label/2000/20874lbl.pdf>> .

Donald G. Barstow, RN

Depression see **Bipolar disorder; Postpartum depression**

Depressive disorders

Definition

Depression or depressive disorders (unipolar depression) are mental illnesses characterized by a profound and persistent feeling of sadness or despair and/or a loss of interest in things that once were pleasurable. Disturbance in sleep, appetite, and mental processes are a common accompaniment.

Description

Everyone experiences feelings of unhappiness and sadness occasionally. But when these depressed feelings start to dominate everyday life and cause physical and mental deterioration, they become what are known as depressive disorders. Each year in the

United States, depressive disorders affect an estimated 17 million people at an approximate annual direct and indirect cost of \$53 billion. One in four women is likely to experience an episode of severe depression in her lifetime, with a 10–20% lifetime prevalence, compared to 5–10% for men. The average age a first depressive episode occurs is in the mid-20s, although the disorder strikes all age groups indiscriminately, from children to the elderly.

There are two main categories of depressive disorders: major depressive disorder and dysthymic disorder. Major depressive disorder is a moderate to severe episode of depression lasting two or more weeks. Individuals experiencing this major depressive episode may have trouble sleeping, lose interest in activities they once took pleasure in, experience a change in weight, have difficulty concentrating, feel worthless and hopeless, or have a preoccupation with death or **suicide**. In children, the major depression may appear as irritability.

While major depressive episodes may be acute (intense but short-lived), dysthymic disorder is an ongoing, chronic depression that lasts two or more years (one or more years in children) and has an average duration of 16 years. The mild to moderate depression of dysthymic disorder may rise and fall in intensity, and those afflicted with the disorder may experience some periods of normal, non-depressed mood of up to two months in length. Its onset is gradual, and dysthymic patients may not be able to pinpoint exactly when they started feeling depressed. Individuals with dysthymic disorder may experience a change in sleeping and eating patterns, low self-esteem, **fatigue**, trouble concentrating, and feelings of hopelessness.

Depression also can occur in **bipolar disorder**, an affective mental illness that causes radical emotional changes and mood swings, from manic highs to depressive lows. The majority of bipolar individuals experience alternating episodes of mania and depression.

Causes and symptoms

The causes behind depression are complex and not yet fully understood. While an imbalance of certain neurotransmitters—the chemicals in the brain that transmit messages between nerve cells—is believed to be key to depression, external factors such as upbringing (more so in dysthymia than major depression) may be as important. For example, it is speculated that, if an individual is abused and neglected throughout childhood and adolescence, a

Signs of Depression

Lack of interest or pleasure in daily activities
 Significant weight loss (without dieting) or weight gain
 Difficulty sleeping or excessive sleeping
 Loss of energy
 Feelings of worthlessness or guilt
 Difficulty in making decisions
 Restlessness
 Recurrent thoughts of death

pattern of low self-esteem and negative thinking may emerge. From that, a lifelong pattern of depression may follow. A 2003 study reported that two-thirds of patients with major depression say they also suffer from chronic **pain**. A 2004 study linked severe **obesity** with major depression. Another study showed a strong relationship between **smoking** and depression among teens.

Heredity seems to play a role in who develops depressive disorders. Individuals with major depression in their immediate family are up to three times more likely to have the disorder themselves. It would seem that biological and genetic factors may make certain individuals pre-disposed or prone to depressive disorders, but environmental circumstances often may trigger the disorder.

External stressors and significant life changes, such as chronic medical problems, **death** of a loved one, divorce or estrangement, **miscarriage**, or loss of a job, also can result in a form of depression known as adjustment disorder. Although periods of adjustment disorder usually resolve themselves, occasionally they may evolve into a major depressive disorder.

Major depressive episode

Individuals experiencing a major depressive episode have a depressed mood and/or a diminished interest or pleasure in activities. Children experiencing a major depressive episode may appear or feel irritable rather than depressed. In addition, five or more of the following symptoms will occur on an almost daily basis for a period of at least two weeks:

- Significant change in weight.
- **Insomnia** or hypersomnia (excessive sleep).
- Psychomotor agitation or retardation.
- Fatigue or loss of energy.
- Feelings of worthlessness or inappropriate guilt.

- Diminished ability to think or to concentrate, or indecisiveness.
- Recurrent thoughts of death or suicide and/or suicide attempts.

Dysthymic disorder

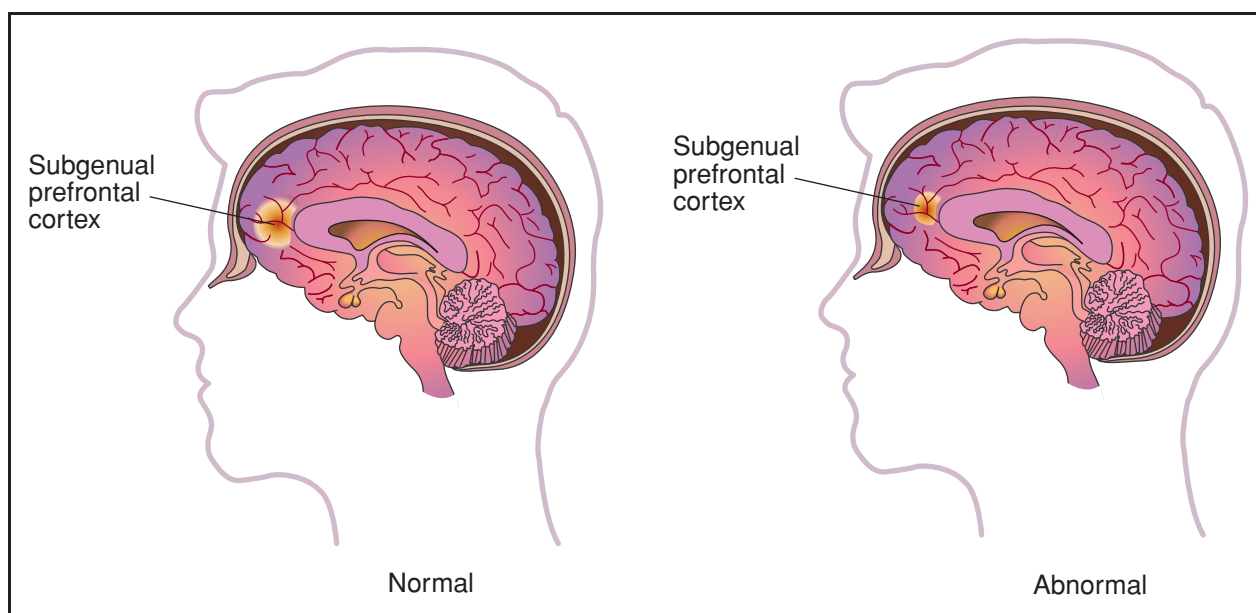
Dysthymia commonly occurs in tandem with other psychiatric and physical conditions. Up to 70% of dysthymic patients have both dysthymic disorder and major depressive disorder, known as double depression. **Substance abuse**, panic disorders, personality disorders, social **phobias**, and other psychiatric conditions also are found in many dysthymic patients. Dysthymia is prevalent in patients with certain medical conditions, including **multiple sclerosis**, **AIDS**, **hypothyroidism**, **chronic fatigue syndrome**, Parkinson's disease, diabetes, and post-cardiac transplantation. The connection between dysthymic disorder and these medical conditions is unclear, but it may be related to the way the medical condition and/or its pharmacological treatment affects neurotransmitters. Dysthymic disorder can lengthen or complicate the recovery of patients also suffering from medical conditions.

Along with an underlying feeling of depression, people with dysthymic disorder experience two or more of the following symptoms on an almost daily basis for a period for two or more years (most suffer for five years), or one year or more for children:

- under or overeating
- insomnia or hypersomnia
- low energy or fatigue
- low self-esteem
- poor concentration or trouble making decisions
- feelings of hopelessness

Diagnosis

In addition to an interview, several clinical inventories or scales may be used to assess a patient's mental status and determine the presence of depressive symptoms. Among these tests are: the Hamilton Depression Scale (HAM-D), Child Depression Inventory (CDI), Geriatric Depression Scale (GDS), Beck Depression Inventory (BDI), and the Zung Self-Rating Scale for Depression. These tests may be administered in an outpatient or hospital setting by a general practitioner, social worker, psychiatrist, or psychologist.



Recent scientific research has indicated that the size of the subgenual prefrontal cortex of the brain (located behind the bridge of the nose) may be a determining factor in hereditary depressive disorders. (Illustration by Electronic Illustrators Group.)

Treatment

Major depressive and dysthymic disorders are typically treated with antidepressants or psychosocial therapy. Psychosocial therapy focuses on the personal and interpersonal issues behind depression, while antidepressant medication is prescribed to provide more immediate relief for the symptoms of the disorder. When used together correctly, therapy and antidepressants are a powerful treatment plan for the depressed patient.

Antidepressants

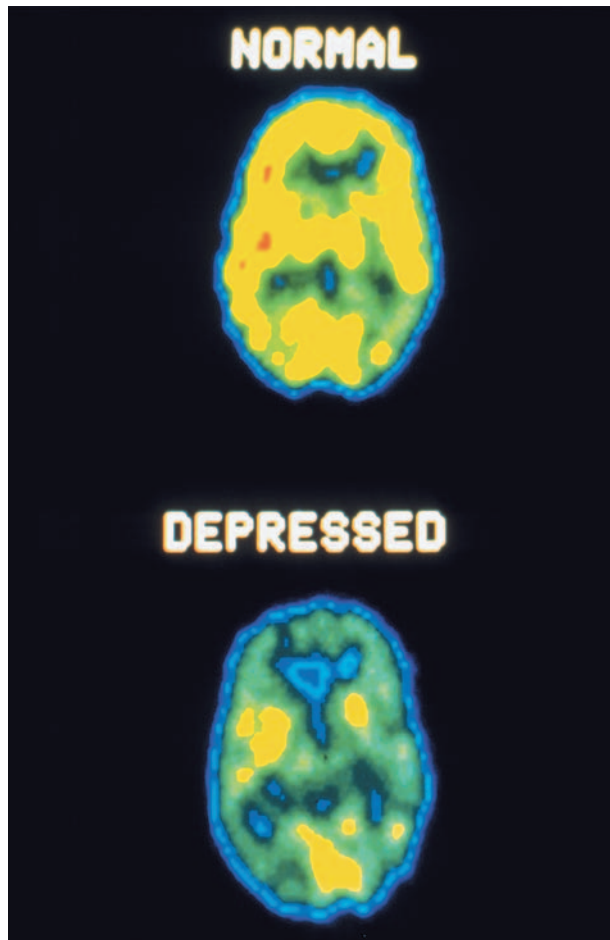
Selective serotonin reuptake inhibitors (SSRIs) such as fluoxetine (Prozac) and sertraline (Zoloft) reduce depression by increasing levels of serotonin, a neurotransmitter. Some clinicians prefer SSRIs for treatment of dysthymic disorder. **Anxiety, diarrhea, drowsiness, headache, sweating, nausea, poor sexual functioning, and insomnia** all are possible side effects of SSRIs. In early 2004, a joint panel of the U.S. Food and Drug Administration (FDA) issued stronger warnings to physicians and parents about increased risk of suicide among children and adolescents taking SSRIs.

Tricyclic antidepressants (TCAs) are less expensive than SSRIs, but have more severe side-effects, which may include persistent **dry mouth, sedation,**

dizziness, and cardiac **arrhythmias**. Because of these side effects, caution is taken when prescribing TCAs to elderly patients. TCAs include amitriptyline (Elavil), imipramine (Tofranil), and nortriptyline (Aventyl, Pamelor). A 10-day supply of TCAs can be lethal if ingested all at once, so these drugs may not be a preferred treatment option for patients at risk for suicide.

Monoamine oxidase inhibitors (MAOIs) such as tranylcypromine (Parnate) and phenelzine (Nardil) block the action of monoamine oxidase (MAO), an enzyme in the central nervous system. Patients taking MAOIs must cut foods high in tyramine (found in aged cheeses and meats) out of their diet to avoid potentially serious hypertensive side effects.

Heterocyclics include bupropion (Wellbutrin) and trazodone (Desyrel). Bupropion should not be prescribed to patients with a **seizure disorder**. Side effects of the drug may include agitation, anxiety, confusion, tremor, dry mouth, fast or irregular heartbeat, headache, low blood pressure, and insomnia. Because trazodone has a sedative effect, it is useful in treating depressed patients with insomnia. Other possible side effects of trazodone include dry mouth, gastrointestinal distress, dizziness, and headache. In 2003, Wellbutrin's manufacturer released a once-daily version of the drug that offered low risk of sexual side effects or weight gain.



Positron emission tomography (PET) scans comparing a normal brain with that of someone with a depressed mental disorder. (Photo Researchers, Inc. Reproduced by permission.)

Psychosocial therapy

Psychotherapy explores an individual's life to bring to light possible contributing causes of the present depression. During treatment, the therapist helps the patient to become self-aware of his or her thinking patterns and how they came to be. There are several different subtypes of psychotherapy, but all have the common goal of helping the patient develop healthy problem solving and coping skills.

Cognitive-behavioral therapy assumes that the patient's faulty thinking is causing the current depression and focuses on changing the depressed patient's thought patterns and perceptions. The therapist helps the patient identify negative or distorted thought patterns and the emotions and behavior that accompany them, and then retrains the depressed individual to recognize the thinking and react differently to it.

Electroconvulsant therapy

ECT, or **electroconvulsive therapy**, usually is employed after all therapy and pharmaceutical treatment options have been explored. However, it is sometimes used early in treatment when severe depression is present and the patient refuses oral medication, or when the patient is becoming dehydrated, extremely suicidal, or psychotic.

The treatment consists of a series of electrical pulses that move into the brain through electrodes on the patient's head. ECT is given under **general anesthesia** and patients are administered a muscle relaxant to prevent convulsions. Although the exact mechanisms behind the success of ECT therapy are not known, it is believed that the electrical current modifies the electrochemical processes of the brain, consequently relieving depression. Headaches, muscle soreness, nausea, and confusion are possible side effects immediately following an ECT procedure. Memory loss, typically transient, also has been reported in ECT patients.

Alternative treatment

St. John's wort (*Hypericum perforatum*) is used throughout Europe to treat depressive symptoms. Unlike traditional prescription antidepressants, this herbal antidepressant has few reported side effects. Despite uncertainty concerning its effectiveness, a 2003 report said acceptance of the treatment continues to increase. A poll showed that about 41% of 15,000 science professionals in 62 countries said they would use St. John's wort for mild to moderate depression. Although St. John's wort appears to be a safe alternative to conventional antidepressants, care should be taken, as the herb can interfere with the actions of some pharmaceuticals. The usual dose is 300 mg three times daily.

Homeopathic treatment also can be therapeutic in treating depression. Good nutrition, proper sleep, exercise, and full engagement in life are very important to a healthy mental state.

In several small studies, S-adenosyl-methionine (SAM, SAME) was shown to be more effective than placebo and equally effective as tricyclic antidepressants in treating depression. The usual dosage is 200 mg to 400 mg twice daily. In 2003, a U.S. Department of Health and Human Services team reviewed 100 clinical trials on SAME and concluded that it worked as well as many prescription medications without the side effects of stomach upset and decreased sexual desire.

KEY TERMS

Hypersomnia—The need to sleep excessively; a symptom of dysthymic and major depressive disorder.

Neurotransmitter—A chemical in the brain that transmits messages between neurons, or nerve cells. Changes in the levels of certain neurotransmitters, such as serotonin, norepinephrine, and dopamine, are thought to be related to depressive disorders.

Psychomotor agitation—Disturbed physical and mental processes (e.g., fidgeting, wringing of hands, racing thoughts); a symptom of major depressive disorder.

Psychomotor retardation—Slowed physical and mental processes (e.g., slowed thinking, walking, and talking); a symptom of major depressive disorder.

In 2003, a report from Great Britain emphasized that more physicians should encourage alternative treatments such as behavioral and self-help programs, supervised **exercise** programs, and watchful waiting before subscribing antidepressant medications for mild depression.

Prognosis

Untreated or improperly treated depression is the number one cause of suicide in the United States. Proper treatment relieves symptoms in 80–90% of depressed patients. After each major depressive episode, the risk of recurrence climbs significantly—50% after one episode, 70% after two episodes, and 90% after three episodes. For this reason, patients need to be aware of the symptoms of recurring depression and may require long-term maintenance treatment of antidepressants and/or therapy.

Research has found that depression may lead to other problems as well. Increased risk of heart disease has been linked to depression, particularly in postmenopausal women. And while chronic pain may cause depression, a 2004 study in Canada revealed that depression also may lead to back pain.

Prevention

Patient education in the form of therapy or self-help groups is crucial for training patients with

depressive disorders to recognize symptoms of depression and to take an active part in their treatment program. Extended maintenance treatment with antidepressants may be required in some patients to prevent relapse. Early intervention for children with depression is effective in arresting development of more severe problems.

Resources

PERIODICALS

- “Depression Can Lead to Back Pain.” *Biotech Week*, March 24, 2004: 576.
- “Depression May Be a Risk Factor for Heart Disease, Death in Older Women.” *Women’s Health Weekly*, March 4, 2004: 90.
- “FDA Approves Once-daily Supplement.” *Biotech Week*, September 24, 2003: 6.
- “FDA Panel Urges Stronger Warnings of Child Suicide.” *SCRIP World Pharmaceutical News*, February 6, 2004: 24.
- Jancin, Bruce. “Chronic Pain Affects 67% of Patients With Depression: ‘Stunning’ Finding in Primary Care Study.” *Internal Medicine News*, September 15, 2003: 4.
- “National Study Indicates Obesity Is Linked to Major Depression.” *Drug Week*, February 13, 2004: 338.
- “A Natural Mood-booster that Really Works: a Group of Noted Researchers Found that the Supplement SAME Works as Well as Antidepressant Drugs.” *Natural Health*, July 2003: 22.
- “Researchers See Link Between Depression, Smoking.” *Mental Health Weekly*, March 1, 2004: 8.
- “St. John’s Wort Healing Reputation Upheld.” *Nutraceuticals International*, September 2003.
- “Try Alternatives Before Using Antidepressants.” *GP*, September 29, 2003: 12.

ORGANIZATIONS

- American Psychiatric Association. 1400 K Street NW, Washington DC 20005. (888) 357-7924. <<http://www.psych.org>>.
- American Psychological Association (APA). 750 First St. NE, Washington, DC 20002-4242. (202) 336-5700. <<http://www.apa.org>>.
- National Alliance for the Mentally Ill (NAMI). Colonial Place Three, 2107 Wilson Blvd., Ste. 300, Arlington, VA 22201-3042. (800) 950-6264. <<http://www.nami.org>>.
- National Depressive and Manic-Depressive Association (NDMDA). 730 N. Franklin St., Suite 501, Chicago, IL 60610. (800) 826-3632. <<http://www.ndmda.org>>.
- National Institute of Mental Health. Mental Health Public Inquiries, 5600 Fishers Lane, Room 15C-05, Rockville, MD 20857. (888) 826-9438. <<http://www.nimh.nih.gov>>.

Paula Anne Ford-Martin
Teresa G. Odle

Dermabrasion see **Skin resurfacing**

Dermatitis

Definition

Dermatitis is a general term used to describe inflammation of the skin.

Description

Most types of dermatitis are characterized by an itchy pink or red rash.

Contact dermatitis is an allergic reaction to something that irritates the skin and is manifested by one or more lines of red, swollen, blistered skin that may itch or seep. It usually appears within 48 hours after touching or brushing against a substance to which the skin is sensitive. The condition is more common in adults than in children.

Contact dermatitis can occur on any part of the body, but it usually affects the hands, feet, and groin. Contact dermatitis usually does not spread from one person to another, nor does it spread beyond the area exposed to the irritant unless affected skin comes into contact with another part of the body. However, in the case of some irritants, such as poison ivy, contact dermatitis can be passed to another person or to another part of the body.

Stasis dermatitis is characterized by scaly, greasy looking skin on the lower legs and around the ankles. Stasis dermatitis is most apt to affect the inner side of the calf.

Nummular dermatitis, which is also called nummular eczematous dermatitis or nummular eczema, generally affects the hands, arms, legs, and buttocks of men and women older than 55 years of age. This stubborn inflamed rash forms circular, sometimes itchy, patches and is characterized by flares and periods of inactivity.

Atopic dermatitis is characterized by **itching**, scaling, swelling, and sometimes blistering. In early childhood it is called infantile eczema and is characterized by redness, oozing, and crusting. It is usually found on the face, inside the elbows, and behind the knees.

Seborrheic dermatitis may be dry or moist and is characterized by greasy scales and yellowish crusts on the scalp, eyelids, face, external surfaces of the ears, underarms, breasts, and groin. In infants it is called "cradle cap."

Causes and symptoms

Allergic reactions are genetically determined, and different substances cause contact dermatitis to

develop in different people. A reaction to resin produced by poison ivy, poison oak, or poison sumac is the most common source of symptoms. It is, in fact, the most common allergy in this country, affecting one of every two people in the United States.

Flowers, herbs, and vegetables can also affect the skin of some people. **Burns** and **sunburn** increase the risk of dermatitis developing, and chemical irritants that can cause the condition include:

- chlorine
- cleansers
- detergents and soaps
- fabric softeners
- glues used on artificial nails
- perfumes
- topical medications

Contact dermatitis can develop when the first contact occurs or after years of use or exposure.

Stasis dermatitis, a consequence of poor circulation, occurs when leg veins can no longer return blood to the heart as efficiently as they once did. When that happens, fluid collects in the lower legs and causes them to swell. Stasis dermatitis can also result in a rash that can break down into sores known as stasis ulcers.

The cause of nummular dermatitis is not known, but it usually occurs in cold weather and is most common in people who have dry skin. Hot weather and stress can aggravate this condition, as can the following:

- **allergies**
- fabric softeners
- soaps and detergents
- wool clothing
- bathing more than once a day

Atopic dermatitis can be caused by allergies, **asthma**, or stress, and there seems to be a genetic predisposition for atopic conditions. It is sometimes caused by an allergy to nickel in jewelry.

Seborrheic dermatitis (for which there may also be a genetic predisposition) is usually caused by overproduction of the oil glands. In adults it can be associated with **diabetes mellitus** or gold allergy. In infants and adults it may be caused by a biotin deficiency.

Diagnosis

The diagnosis of dermatitis is made on the basis of how the rash looks and its location. The doctor may scrape off a small piece of affected skin for microscopic examination or direct the patient to discontinue use of any potential irritant that has recently come into contact with the affected area. Two weeks after the rash disappears, the patient may resume use of the substances, one at a time, until the condition recurs. Eliminating the substance most recently added should eliminate the irritation.

If the origin of the irritation has still not been identified, a dermatologist may perform one or more patch tests. This involves dabbing a small amount of a suspected irritant onto skin on the patient's back. If no irritation develops within a few days, another patch test is performed. The process continues until the patient experiences an allergic reaction at the spot where the irritant was applied.

Treatment

Treating contact dermatitis begins with eliminating or avoiding the source of irritation. Prescription or over-the-counter corticosteroid creams can lessen inflammation and relieve irritation. Creams, lotions, or ointments not specifically formulated for dermatitis can intensify the irritation. Oral **antihistamines** are sometimes recommended to alleviate itching, and **antibiotics** are prescribed if the rash becomes infected. Medications taken by mouth to relieve symptoms of dermatitis can make skin red and scaly and cause hair loss.

Patients who have a history of dermatitis should remove their rings before washing their hands. They should use bath oils or glycerine-based soaps and bathe in lukewarm saltwater.

Patting rather than rubbing the skin after bathing and thoroughly massaging lubricating lotion or non-prescription cortisone creams into still-damp skin can soothe red, irritated nummular dermatitis. Highly concentrated cortisone preparations should not be applied to the face, armpits, groin, or rectal area. Periodic medical monitoring is necessary to detect side effects in patients who use such preparations on **rashes** covering large areas of the body.

Coal-tar salves can help relieve symptoms of nummular dermatitis that have not responded to other treatments, but these ointments have an unpleasant odor and stain clothing.

Patients who have stasis dermatitis should elevate their legs as often as possible and sleep with a pillow between the lower legs.



Dermatitis on hands and fingers. (Custom Medical Stock Photo. Reproduced by permission.)

Tar or zinc paste may also be used to treat stasis dermatitis. Because these compounds must remain in contact with the rash for as long as two weeks, the paste and bandages must be applied by a nurse or a doctor.

Coal-tar shampoos may be used for seborrheic dermatitis that occurs on the scalp. Sun exposure after the use of these shampoos should be avoided because the risk of sunburn of the scalp is increased.

Alternative treatment

Some herbal therapies can be useful for skin conditions. Among the herbs most often recommended are:

- Burdock root (*Arctium lappa*)
- Calendula (*Calendula officinalis*) ointment
- Chamomile (*Matricaria recutita*) ointment
- Cleavers (*Galium ssp.*)
- Evening primrose oil (*Oenothera biennis*)
- Nettles (*Urtica dioica*)

Contact dermatitis can be treated botanically and homeopathically. Grindelia (*Grindelia* spp.) and sassafras (*Sassafras albidum*) can help when applied topically. Determining the source of the problem and eliminating it is essential. Oatmeal baths are very helpful in relieving the itch. Bentonite clay packs or any mud pack draws the fluid out and helps dry up the lesions. Cortisone creams are not recommended.

Stasis dermatitis should be treated by a trained practitioner. This condition responds well to topical herbal therapies, however, the cause must also be addressed. Selenium-based shampoos, topical applications of flax oil and/or olive oil, and biotin

KEY TERMS

Allergic reaction—An inappropriate or exaggerated genetically determined reaction to a chemical that occurs only on the second or subsequent exposures to the offending agent, after the first contact has sensitized the body.

Corticosteriod—A group of synthetic hormones that are used to prevent or reduce inflammation. Toxic effects may result from rapid withdrawal after prolonged use or from continued use of large doses.

Patch test—A skin test that is done to identify allergens. A suspected substance is applied to the skin. After 24–48 hours, if the area is red and swollen, the test is positive for that substance. If no reaction occurs, another substance is applied. This is continued until the patient experiences an allergic reaction where the irritant was applied to the skin.

Rash—A spotted, pink or red skin eruption that may be accompanied by itching and is caused by disease, contact with an allergen, food ingestion, or drug reaction.

Ulcer—An open sore on the skin, resulting from tissue destruction, that is usually accompanied by redness, pain, or infection.

supplementation are among the therapies recommended for seborrheic dermatitis.

Prognosis

Dermatitis is often chronic, but symptoms can generally be controlled.

Prevention

Contact dermatitis can be prevented by avoiding the source of irritation. If the irritant cannot be avoided completely, the patient should wear gloves and other protective clothing whenever exposure is likely to occur.

Immediately washing the exposed area with soap and water can stem allergic reactions to poison ivy, poison oak, or poison sumac, but because soaps can dry the skin, patients susceptible to dermatitis should use them only on the face, feet, genitals, and underarms.

Clothing should be loose fitting and 100% cotton. New clothing should be washed in dye-free, unscented detergent before being worn.

Injury to the lower leg can cause stasis dermatitis to ulcerate (form open sores). If stasis ulcers develop, a doctor should be notified immediately.

Yoga and other relaxation techniques may help prevent atopic dermatitis caused by **stress**.

Avoidance of sweating may aid in preventing seborrheic dermatitis.

A patient who has dermatitis should also notify a doctor if any of the following occurs:

- fever develops
- skin oozes or other signs of infection appear
- symptoms do not begin to subside after seven days' treatment
- he/she comes into contact with someone who has a wart, cold sore, or other viral skin infection

Resources

OTHER

"Allergic Contact Dermatitis." *The Skin Site*. April 10, 1998 (January 11, 2006). <http://www.skincare.com/info_allergic.htm>.

Maureen Haggerty

Dermatomyositis

Definition

Dermatomyositis (DM) is a rare inflammatory muscle disease that leads to destruction of muscle tissue usually accompanied by **pain** and weakness.

Description

Dermatomyositis is one of a group of three related diseases called inflammatory **myopathies**. The other two are **polymyositis** and inclusion-body **myositis**. These diseases are rare; only about 20,000 people in the United States have dermatomyositis. Another estimates suggest that DM occurs in about 5.5 individuals out of every one million. The disease is of unknown origin and can develop in children and adults. Most often individuals either develop DM either between the ages of five and 14 or they do not develop it until they are over age 45. In all age groups, females are twice as likely to develop the disease than males. Although DM causes pain and weakness, it is not necessarily life threatening. However, adults, but not children, who develop DM have an increased risk of

developing **cancer** and should be screened for malignancies regularly.

Causes and symptoms

The exact cause of dermatomyositis is unknown. It is an autoimmune disease. In a healthy body, cells of immune system attack only foreign or defective cells in the body to protect it from disease. In an autoimmune disease, the immune system attacks normal body cells. In the case of DM, immune system cells attack healthy cells of small blood vessels in the muscle and skin. Over time, this causes muscle fiber to shrink and sometimes cuts off blood supply to the muscle. DM tends to develop in muscles closest to the center of the body.

As yet, there is no clear explanation of what causes an individual to develop DM. It is thought that the disease may be triggered by a virus or exposure to certain drugs or vaccines. According to the **Muscular Dystrophy** Association, recent research suggests developing DM may be related to the mixing of blood cells that sometimes occurs between the mother and fetus during **pregnancy**. The disease is not directly inherited, although there may be some genetic sensitivity toward whatever triggers it.

Often the first sign of DM is the development of a patchy, scaly, violet to dark red skin rash on the face, neck, shoulders, upper chest, knees, or back. Often the rash appears before any signs of illness or muscle weakness. About 40% of children and teens develop hard, painful bumps under the skin that are deposits of calcium, a mineral used in bone formation. This condition, called calcinosis, is much less common in adults.

Muscle weakness, especially in the upper arms, hips, thighs, and neck, becomes apparent in activities such as climbing stairs or reaching up over the head. This weakness develops after the rash appears. Some people have difficulty swallowing and chewing when the muscles of the face and esophagus are affected. Individuals may also feel tired, weak, have a low-grade **fever**, weight loss, and joint stiffness. Some individuals have the rash for years before they progress to these symptoms, while in others the onset of symptoms is rapid. In children the development of symptoms is almost always gradual, making diagnosis especially difficult.

Diagnosis

DM can be difficult to diagnose, and often the first doctor an individual sees is a dermatologist for treatment of the rash and then is referred to a

rheumatologist, specialist in internal medicine or neurologist when DM is suspected. Many tests may be done to rule out other diseases before a firm diagnosis is made. A blood test is done to measure the level of creatine kinase. Creatine kinase is an enzyme found in muscle tissue. When muscle is damaged, this enzyme leaks out into the blood. An increased level of creatine kinase in the blood suggests DM as a possible diagnosis. Another blood test may be done to test for specific immune system antibodies. Antibodies are proteins made in response to material the body thinks is foreign.

An electromyogram (EMG) is a test that measures electrical activity in muscles as they contract. Individuals with inflammatory myopathies usually have distinct patterns of electrical activity in the affected muscles. However, up to 15% of people with DM have normal electromyogram readings, so this test is not definitive. The definitive test is a muscle biopsy. The doctor takes a small sample of muscle tissue and examines it under a microscope. From this sample, the doctor can differentiate DM from other inflammatory myopathies and other muscle wasting diseases.

Treatment

The goal of treatment is to improve muscle strength and allow the individual to participate in normal daily activities. Individuals are given steroid drugs (prednisone, **corticosteroids**) that suppress the immune system. Over time, these drugs often produce undesirable side effects, so treatment is usually begun with a large dose, then tapered to the minimum dose needed for maintenance. People who do not respond well to steroid treatment may be treated with other immunosuppressive drugs or intravenous immunoglobulin. Individuals with DM are advised to avoid exposure to the sun, as sunlight worsens the skin rash. Physical therapy is often helpful in keeping joints from stiffening and freezing. Moderate **exercise** is also recommended.

Alternative treatment

A healthy diet high is recommended for all individuals with supplemental protein for those with severe muscle damage.

Prognosis

The course of DM is highly variable. In about 20% of people, the disease spontaneously goes into remission and individuals are able to lead symptom-

KEY TERMS

Immunoglobulin—Material containing specific antibodies to fight disease that can be injected into an individual to fight infection.

Inflammation—An infection or irritation of a tissue.

Myopathy—Relating to muscle tissue.

free lives for long periods. On the other hand, in about 5% of individuals the disease progresses to **death** because of heart and lung involvement. The majority of people continue to have some symptoms and require long-term treatment, but their degree of daily activity varies greatly.

Serious complications from DM include involvement of the muscles of the heart and lungs, difficulty eating and swallowing, and a tendency to develop cancer. This association is seen only in adults and not in children. Individuals over age 60 are more likely to have serious complications than younger individuals.

Prevention

There is no known way to prevent this disease.

Resources

PERIODICALS

Koler, Ric A. and Andrew Montemarano. "Dermatomyositis." *American Family Physician*, 24, no. 9 (1 November 2001) 1565-1574 [cited 16 February 2005]. < <http://www.aafp.org/afp/2001101/1565.html> >.

ORGANIZATIONS

American Autoimmune Related Disease Association. 22100 Gratiot Avenue, Eastpointe, East Detroit, MI 48201-2227. 800-598-4668. < <http://www.aarda.org> >.

Muscular Dystrophy Association. 3300 East Sunrise Drive, Tucson, AZ 85718-3208. 800-572-1717. < <http://www.mdausa.org> >.

Myositis Association. 1233 20th Street, NW, Washington, DC 20036. 800-821-7356. < <http://www.myositis.org> >.

National Organization for Rare Disorders (NORD). P. O. Box 1968, Danbury, CT 06813-1968. 800-999-NORD. < <http://www.rarediseases.org> >.

OTHER

Callen, Jeffrey P. *Dermatomyositis*, 5 December 2002 [cited 16 February 2005]. < <http://www.emedicine.com/derm/topic98.htm> >.

Hashmat, Aamir and Zaineb Daud. *Dermatomyositis/Polymyositis*, 16 January 2004 [cited 16 February 2005]. < <http://www.emedicine.com/neuro/topic85.htm> >.

Tish Davidson, A.M.

Dermatophyte infections see **Ringworm**

DES exposure

Definition

DES (diethylstilbestrol) is a hormone that was prescribed for pregnant women in the 1950s and early 1960s. Many years later, doctors discovered that the daughters of the women who received DES were at high risk for a variety of problems, including **infertility**, **premature labor**, and **cancer** of the vagina and cervix.

Description

In the 1950s and early 1960s, several drug companies claimed that DES (diethylstilbestrol) could prevent miscarriages. DES is a synthetic hormone, related to estrogen. Since up to 20% of all pregnancies end in **miscarriage**, this seemed like an important breakthrough and DES was prescribed for many women who had bleeding in early **pregnancy**. Ultimately, it was found to have no effect on miscarriages and the practice of prescribing DES was stopped in the 1960s. Almost 10 years later, the daughters of women who had taken DES during pregnancy began to develop unusual symptoms.

Doctors discovered that when these young women reached their teens, they were at higher risk for a variety of problems, including:

- clear cell adenocarcinoma of the vagina and cervix
- infertility
- premature labor and other problems in pregnancy

It is estimated that five to 10 million people in the United States were exposed to DES between 1938 and 1971.

Causes and symptoms

DES has affected a very specific group of women. These are women who were exposed to DES in utero before 18 weeks of pregnancy. In other words, their mothers must have taken DES within the first four-

five months of pregnancy. It is now known that the female reproductive organs are formed during that time. DES appears to interfere with proper growth and development of the uterus, cervix, vagina, and fallopian tubes. In 2003, new research showed that DES also was associated with increased risk of **breast cancer**.

In the early 1970s, there was an increase in a rare form of cancer, clear cell adenocarcinoma of the vagina and cervix. Up until that time, doctors had seen these cancers only in elderly women. Suddenly, young women who had the disease appeared.

This was so unusual that researchers studied these women to see if they had anything in common. After a great deal of questioning and examination, it was found that they all had one factor in common. All of the young women had been exposed to DES in utero in the early weeks of pregnancy.

Today, it is difficult to imagine how shocking this discovery was. Doctors had only recently recognized that medications and exposure to chemicals during pregnancy could cause birth defects. This was a birth defect that had gone undetected for almost two decades.

Since then, doctors have studied DES daughters very carefully. Fortunately, the risk of clear cell adenocarcinoma is actually quite low. In fact, it appears that if a DES daughter has not developed this cancer by age 30, she will not develop it. Since all DES daughters are now over age 30, there should be no further cases related to DES exposure. However, there are a number of other symptoms and problems associated with DES exposure.

- **Cervix and vagina.** DES daughters often have distinctive changes of the cervix and vagina that can be seen during a **pelvic exam**. These changes include a cervical hood (a vaginal fold draped over the cervix), cockscomb cervix (an abnormally shaped cervix), and adenosis (glandular cells normally located within the cervix that appear on the outside of the cervix and in the vagina).
- **Fallopian tubes.** Some DES daughters have fallopian tube abnormalities that lead to infertility.
- **Uterus.** Many DES daughters have a uterus that is abnormal in size and shape. The classic sign is the T-shaped uterus. In the normal uterus, the cavity (hollow space inside) is rounded. In a T-shaped uterus, the cavity is reduced to a thin T. The abnormal shape of the inside of the uterus makes it harder for a woman to get pregnant and leads to a higher risk of premature labor and birth.

KEY TERMS

Cervix—The opening at the bottom of the uterus.

Colposcopy—A special examination of the cervix using a magnifying scope. This is a procedure that can be done in the doctor's office.

Fallopian tubes—The tubes that carry the ovum (egg) from the ovary to the uterus.

Pap smear—A screening test for precancerous and cancerous cells on the cervix. This simple test is done during a routine pelvic exam and involves scraping cells from the cervix.

Diagnosis

Women who have been exposed to DES should have a pelvic exam at least once a year. In addition to the usual pelvic exam and Pap smear, DES daughters also should have Pap smears of the vagina and, if possible, **colposcopy**. During colposcopy, the doctor looks at the cervix and vagina through a special magnifying scope. In this way, tiny areas of abnormal cells can be seen. This procedure is easily performed in the doctor's office.

When DES daughters get pregnant, they may be at high risk for premature labor and birth and should be monitored very carefully.

Not all women who were exposed to DES develop problems in pregnancy. However, if problems like infertility or miscarriage occur, the doctor may recommend a special x-ray test to check the woman's fallopian tubes and uterus. This special test is called a hysterosalpingogram.

Treatment

There is no treatment for the abnormalities of the fallopian tubes and uterus caused by DES exposure. Fortunately, there are treatments that can help with infertility and premature labor. Clear cell adenocarcinoma of the vagina or cervix must be treated with surgery and, possibly, **chemotherapy**.

Resources

PERIODICALS

Kruse, Kelly, Diane Lauver, and Karen Hanson. "Clinical Implications of DES." *Nurse Practitioner*, July 2003: 26–29.

Morantz, Carrie, and Brian Torrey. "CDC Web site on Diethylstilbestrol." *American Family Physician*, November 15, 2003: 2088.

OTHER

Centers for Disease Control and Prevention DES Web Site.
DES Update Home <<http://www.cdc.gov/DES>>.

Amy B. Tuteur, MD
 Teresa G. Odle

Detached retina see **Retinal detachment**

Detoxification

Definition

Detoxification is one of the more widely used treatments and concepts in alternative medicine. It is based on the principle that illnesses can be caused by the accumulation of toxic substances (toxins) in the body. Eliminating existing toxins and avoiding new toxins are essential parts of the healing process. Detoxification utilizes a variety of tests and techniques.

Purpose

Detoxification is helpful for those patients suffering from many chronic diseases and conditions, including **allergies**, **anxiety**, arthritis, **asthma**, chronic infections, depression, diabetes, headaches, heart disease, high cholesterol, low blood sugar levels, digestive disorders, mental illness, and **obesity**. It is helpful for those with conditions that are influenced by environmental factors, such as **cancer**, as well as for those who have been exposed to high levels of toxic materials due to accident or occupation. Detoxification therapy is useful for those suffering from allergies or immune system problems that conventional medicine is unable to diagnose or treat, including chronic fatigue syndrome, environmental illness/multiple chemical sensitivity, and **fibromyalgia**. Symptoms for those suffering these conditions may include unexplained **fatigue**, increased allergies, hypersensitivity to common materials, intolerance to certain foods and indigestion, aches and pains, low grade **fever**, headaches, **insomnia**, depression, sore throats, sudden weight loss or gain, lowered resistance to infection, general malaise, and disability. Detoxification can be used as a beneficial preventative measure and as a tool to increase overall health, vitality, and resistance to disease.

Description

Origins

Detoxification methods of healing have been used for thousands of years. **Fasting** is one of the oldest therapeutic practices in medicine. Hippocrates, the ancient Greek known as the “Father of Western medicine,” recommended fasting as a means for improving health. **Ayurvedic medicine**, a traditional healing system that has developed over thousands of years, utilizes detoxification methods to treat many chronic conditions and to prevent illness.

Detoxification treatment has become one of the cornerstones of alternative medicine. Conventional medicine notes that environmental factors can play a significant role in many illnesses. Environmental medicine is a field that studies exactly how those environmental factors influence disease. Conditions such as asthma, cancer, **chronic fatigue syndrome**, **multiple chemical sensitivity**, and many others are strongly influenced by exposure to toxic or allergenic substances in the environment. The United States Centers for Disease Control estimate that over 80% of all illnesses have environmental and lifestyle causes.

Detoxification has also become a prominent treatment as people have become more aware of environmental pollution. It is estimated that one in every four Americans suffers from some level of **heavy metal poisoning**. Heavy metals, such as lead, mercury, cadmium, and arsenic, are by-products of industry. Synthetic agriculture chemicals, many of which are known to cause health problems, are also found in food, air, and water. American agriculture uses nearly 10lb (4.5 kg) of pesticides per person on the food supply each year. These toxins have become almost unavoidable. Pesticides that are used only on crops in the southern United States have been found in the tissue of animals in the far north of Canada. DDT, a cancer-causing insecticide that has been banned for decades, is still regularly found in the fatty tissue of animals, birds, and fish, even in extremely remote regions such as the North Pole.

The problem of toxins in the environment is compounded because humans are at the top of the food chain and are more likely to be exposed to an accumulation of toxic substances in the food supply. For instance, pesticides and herbicides are sprayed on grains that are then fed to farm animals. Toxic substances are stored in the fatty tissue of those animals. In addition, those animals are often injected with synthetic hormones, **antibiotics**, and other chemicals. When people eat meat products, they are exposed to the full range of chemicals and additives used along

Common Herbs Used for Detoxification

Antibiotics	Anticatarrhals (Help Eliminate Mucus)	Blood Cleaners
Clove	Boneset	Burdock root
Echinacea	Echinacea	Dandelion root
Eucalyptus	Garlic	Echinacea
Garlic	Goldenseal root	Oregon grape root
Myrrh	Hyssop	Red clover blossoms
Prickly ash bark	Sage	Yellow dock root
Propolis	Yarrow	
Wormwood		

Diaphoretics/Skin Cleaners	Diuretics	Laxatives
Boneset	Cleavers	Buckthorn
Burdock root	Corn silk	Cascara sagrada
Cayenne pepper	Horsetail	Dandelion root
Elder flowers	Juniper berries	Licorice root
Ginger root	Parsley leaf	Rhubarb root
Goldenseal root	Uva ursi	Senna leaf
Peppermint	Yarrow dock	Yellow dock
Oregon grape root		
Yellow dock		

the entire agricultural chain. Detoxification specialists call this build up of toxins *bioaccumulation*. They assert that the bioaccumulation of toxic substances over time is responsible for many physical and mental disorders, especially ones that are increasing rapidly (like asthma, cancer, and mental illness). As a result, detoxification therapies are increasing in importance and popularity.

Toxins in the body include heavy metals and various chemicals such as pesticides, pollutants, and food additives. Drugs and alcohol have toxic effects in the body. Toxins are produced as normal by-products in the intestines by the bacteria that break down food. The digestion of protein also creates toxic by-products in the body.

The body has natural methods of detoxification. Individual cells get detoxified in the lymph and circulatory system. The liver is the principle organ of detoxification, assisted by the kidneys and intestines. Toxins can be excreted from the body by the kidneys, bowels, skin, and lungs. Detoxification treatments become necessary when the body's natural detoxification systems become overwhelmed. This can be caused by long-term effects of improper diet, **stress**, overeating, sedentary lifestyles, illness, and poor health habits in general. When a build up of toxic substances in the body creates illness, it's called toxemia. Some people's digestive tracts become unable to digest food properly, due to years of overeating and diets that are high in fat

and processed foods and low in fiber (the average American diet). When this happens, food cannot pass through the digestive tract efficiently. Instead of being digested properly or eliminated from the bowel, food can literally rot inside the digestive tract and produce toxic by-products. This state is known as toxic colon syndrome or intestinal toxemia.

Detoxification therapies try to activate and assist the body's own detoxification processes. They also try to eliminate additional exposure to toxins and strengthen the body and immune system so that toxic imbalances won't occur in the future.

Testing for toxic substances

Detoxification specialists use a variety of tests to determine the causes contributing to toxic conditions. These causes include infections, allergies, addictions, toxic chemicals, and digestive and organ dysfunction. Blood, urine, stool, and hair analyses, as well as allergy tests, are used to measure a variety of bodily functions that may indicate problems. Detoxification therapists usually have access to laboratories that specialize in sophisticated diagnostic tests for toxic conditions.

People who have toxemia are often susceptible to infection because their immune systems are weakened. Infections can be caused by parasites, bacteria, viruses, and a common yeast. Therapists will screen patients for underlying infections that may be contributing to illness.

Liver function is studied closely with blood and urine tests because the liver is the principle organ in the body responsible for removing toxic compounds. When the liver detoxifies a substance from the body, it does so in two phases. Tests are performed that indicate where problems may be occurring in these phases, which may point to specific types of toxins. Blood and urine tests can also be completed that screen for toxic chemicals such as PCBs (environmental poisons), formaldehyde (a common preservative), pesticides, and heavy metals. Another useful blood test is a test for zinc deficiency, which may reveal heavy metal **poisoning**. Hair analysis is used to test for heavy metal levels in the body. Blood and urine tests check immune system activity, and hormone levels can also indicate specific toxic compounds. A 24-hour urine analysis, where samples are taken around the clock, allows therapists to determine the efficiency of the digestive tract and kidneys. Together with stool analysis, these tests may indicate toxic bowel syndrome and digestive system disorders. Certain blood and urine tests may point to nutritional

deficiencies and proper recovery **diets** can be designed for patients as well.

Detoxification therapists may also perform extensive allergy and hypersensitivity tests. Intradermal (between layers of the skin) and sublingual (under the tongue) **allergy tests** are used to determine a patient's sensitivity to a variety of common substances, including formaldehyde, auto exhaust, perfume, tobacco, chlorine, jet fuel, and other chemicals.

Food allergies require additional tests because these allergies often cause reactions that are delayed for several days after the food is eaten. The RAST (radioallergosorbent test) is a blood test that determines the level of antibodies (immunoglobulins) in the blood after specific foods are eaten. The cytotoxic test is a blood test that determines if certain substances affect blood cells, including foods and chemicals. The ELISA-ACT (enzyme-linked immunoserological assay activated cell test) is considered to be one of the most accurate tests for allergies and hypersensitivity to foods, chemicals, and other agents. Other tests for food allergies are the elimination and rotation diets, in which foods are systematically evaluated to determine the ones that are causing problems.

Detoxification therapists usually interview and counsel patients closely to determine and correct lifestyle, occupational, psychological, and emotional factors that may also be contributing to illness.

Detoxification therapies

Detoxification therapists use a variety of healing techniques after a diagnosis is made. The first step is to eliminate a patient's exposure to all toxic or allergenic substances. These include heavy metals, chemicals, radiation (from x rays, power lines, cell phones, computer screens, and microwaves), smog, polluted water, foods, drugs, caffeine, alcohol, perfume, excess noise, and stress. If mercury poisoning has been determined, the patient will be advised to have mercury fillings from the teeth removed, preferably by a holistic dentist.

Specific treatments are used to stimulate and assist the body's detoxification process. Dietary change is immediately enacted, eliminating allergic and unhealthy foods, and emphasizing foods that assist detoxification and support healing. Detoxification diets are generally low in fat, high in fiber, and vegetarian with a raw food emphasis. Processed foods, alcohol, and **caffeine** are avoided. **Nutritional supplements** such as **vitamins**, **minerals**,

antioxidants, amino acids, and essential fatty acids are often prescribed. Spirulina is a sea algae that is frequently given to assist in eliminating heavy metals. Lipotropic agents are certain vitamins and nutrients that promote the flow of bile and fat from the liver.

Many herbal supplements are used in detoxification therapies as well. Milk thistle extract, called silymarin, is one of the more potent herbs for detoxifying the liver. Naturopathy, Ayurvedic medicine, and **traditional Chinese medicine** (TCM) recommend numerous herbal formulas for detoxification and immune strengthening. If infections or parasites have been found, these are treated with herbal formulas and, in difficult cases, antibiotics.

For toxic bowel syndrome and digestive tract disorders, herbal **laxatives** and high fiber foods such as psyllium seeds may be given to cleanse the digestive tract and promote elimination. Colonics are used to cleanse the lower intestines. Digestive enzymes are prescribed to improve digestion, and acidophilus and other friendly bacteria are reintroduced into the system with nutritional supplements.

Fasting is another major therapy in detoxification. Fasting is one of the quickest ways to promote the elimination of stored toxins in the body and to prompt the healing process. People with severe toxic conditions are supervised closely during fasting because the number of toxins in the body temporarily increases as they are being released.

Chelation therapy is used by detoxification specialists to rid the body of heavy metals. Chelates are particular substances that bind to heavy metals and speed their elimination. Homeopathic remedies have also been shown to be effective for removing heavy metals.

Sweating therapies can also detoxify the body because the skin is a major organ of elimination. Sweating helps release those toxins that are stored in the subcutaneous (under the skin) fat cells. Saunas, **therapeutic baths**, and **exercise** are some of these treatments. Body therapies may also be prescribed, including massage therapy, **acupressure**, **shiatsu**, manual lymph drainage, and polarity therapy. These body therapies seek to improve circulatory and structural problems, reduce stress, and promote healing responses in the body. Mind/body therapies such as psychotherapy, counseling, and stress management techniques may be used to heal the psychological components of illness and to help patients overcome their negative patterns contributing to illness.

KEY TERMS

Allergen—A foreign substance, such as mites in house dust or animal dander, that when inhaled causes the airways to narrow and produces symptoms of asthma.

Antibody—A protein, also called immunoglobulin, produced by immune system cells to remove antigens (the foreign substances that trigger the immune response).

Fibromyalgia—A condition of debilitating pain, among other symptoms, in the muscles and the myofascia (the thin connective tissue that surrounds muscles, bones, and organs).

Hypersensitivity—The state where even a tiny amount of allergen can cause severe allergic reactions.

Multiple chemical sensitivity—A condition characterized by severe and crippling allergic reactions to commonly used substances, particularly chemicals. Also called environmental illness.

Practitioners and treatment costs

The costs of detoxification therapies can vary widely, depending on the number of tests and treatments required. Detoxification treatments can be lengthy and involved since illnesses associated with toxic conditions usually develop over many years and may not clear up quickly. Detoxification treatments may be lengthy because they often strive for the holistic healing of the body, mind, and emotions.

Practitioners may be conventionally trained medical doctors with specialties in environmental medicine or interests in alternative treatment. The majority of detoxification therapists are alternative practitioners, such as naturopaths, homeopaths, ayurvedic doctors, or traditional Chinese doctors. Insurance coverage varies, depending on the practitioner and the treatment involved. Consumers should review their individual insurance policies regarding treatment coverage.

Preparations

Patients can assist diagnosis and treatment by keeping detailed diaries of their activities, symptoms, and contact with environmental factors that may be affecting their health. Reducing exposure to

environmental toxins and making immediate dietary and lifestyle changes may speed the detoxification process.

Side effects

During the detoxification process, patients may experience side effects of fatigue, malaise, aches and pains, emotional duress, **acne**, headaches, allergies, and symptoms of colds and flu. Detoxification specialists claim that these negative side effects are part of the healing process. These reactions are sometimes called *healing crises*, which are caused by temporarily increased levels of toxins in the body due to elimination and cleansing.

Research and general acceptance

Although environmental medicine is gaining more respect within conventional medicine, detoxification treatment is scarcely mentioned by the medical establishment. The research that exists on detoxification is largely testimonial, consisting of individual personal accounts of healing without statistics or controlled scientific experiments. In the alternative medical community, detoxification is an essential and widely accepted treatment for many illnesses and chronic conditions.

Resources

PERIODICALS

Alternative Therapies Magazine. P.O. Box 17969, Durham, NC 27715. (919) 668-8825. <<http://www.alternative-therapies.com>>.

Journal of Occupational and Environmental Medicine. 1114 N. Arlington Heights Rd., Arlington Heights, IL 60004. (847) 818-1800.

ORGANIZATIONS

American Holistic Medical Association. 4101 Lake Boone Trail, Suite 201, Raleigh, NC 27607. <<http://www.holisticmedicine.org/index.html>>.

Cancer Prevention Coalition. 2121 West Taylor St., Chicago, IL 60612. (312) 996-2297. <<http://www.preventcancer.com>>.

Center for Occupational and Environmental Medicine. 7510 Northforest Dr., North Charleston, SC 29420. (843) 572-1600. <<http://www.coem.com>>.

Northeast Center for Environmental Medicine. P.O. Box 2716, Syracuse, NY 13220. (800) 846-ONUS.

Northwest Center for Environmental Medicine. 177 NE 102nd St., Portland, OR 97220. (503) 561-0966.

OTHER

A Citizens Toxic Waste Manual. Greenpeace USA, 1436 U St. NW, Washington, DC 20009. (202) 462-1177.

Douglas Dupler, MA

Deviated septum

Definition

The nasal septum is a thin structure, separating the two sides of the nose. If it is not in the middle of the nose, then it is deviated.

Description

The nasal septum is composed of two parts. Toward the back of the head the nasal septum is rigid bone, but further forward the bone becomes cartilage. With one finger in each nostril this cartilage can easily be bent back and forth. If the nasal septum is sufficiently displaced to one side, it will impede the flow of air and mucus through the nose. This condition, called a deviated septum, can cause symptoms and disease.

Causes and symptoms

A deviated septum can be a simple variation in normal structure or the result of a broken nose. Any narrowing of the nasal passageway that it causes will threaten the drainage of secretions from the sinuses, which must pass through the nose. It is a general rule of medicine that when flow is obstructed, whether it is mucus from the sinuses or bile from the gall bladder, infection results. People with **allergic rhinitis** (hay fever) are at greater risk of obstruction because their nasal passageways are already narrowed by the swollen membranes lining them. The result is **sinusitis**, which can be acute and severe or chronic and lingering.

Diagnosis

It is easy to see that a septum is deviated. It is more difficult to determine if that deviation needs correction. It is common for a patient to complain that he/she can breathe through only one nostril. Then the diagnosis is easy. A deviated septum may also contribute to **snoring**, **sleep apnea**, and other breathing disorders.

Treatment

The definitive treatment is surgical repositioning of the septum, accomplished by breaking it loose and fixing it in a proper place while it heals. Decongestants like pseudoephedrine or phenylpropanolamine will shrink the membranes and thereby enlarge the passages. **Antihistamines**, nasal cortisone spray, and other allergy treatments may also be temporarily beneficial.



A close-up of person with a deviated septum. (Custom Medical Stock Photo. Reproduced by permission.)

Alternative treatment

As a palliative, saline drops and sprays are very helpful in loosening mucus in the obstructed side and preventing drying in the other side, where all the air blows. Hot peppers, such as jalapenos, can produce enough tears and discharge to flush out a stopped-up nose. An even more effective treatment is called a nasal lavage, often done using a small pot with a spout. Saline solution is poured into one nostril and allowed to flow out the other nostril. Then, the process is repeated in reverse. These therapies are all useful to take care of symptoms, but do not correct the problem. Nasospecific, a procedure where a deflated balloon is inserted in the nostril and inflated to a large enough degree to adjust the septal deviation, can be an alternative to surgery. A trained practitioner in the nasospecific procedure is necessary.

KEY TERMS

Allergen—Any substance that irritates people sensitive (allergic) to it.

Allergic rhinitis—Swelling and inflammation of the nasal membranes caused by sensitivity to airborne matter like pollen or cat hair.

Saline—A salt solution in water. Normal saline has the same salt concentration as the body, 0.9%.

Sinuses—The nasal sinuses, air-filled cavities surrounding the eyes and nose, like the nose itself are lined with mucus-producing membranes. They provide cleansing to the nose, resonance to the voice, and structure to the face.

Sinusitis—Infection of the sinuses.

Sleep apnea—A condition in which breathing is temporarily interrupted during sleep. It leads to high blood pressure, sleepiness, and a variety of other problems.

Prognosis

Surgical repair is curative and carries little risk. Chronic infection can be painful and lead to complications until it is resolved. If there is continued obstruction, the infection will very likely return.

Prevention

Avoidance of virus colds, airborne dusts, air pollution, and known allergens will minimize the irritation and swelling of the membranes lining the nasal passages.

Resources

BOOKS

Ballenger, John Jacob. *Disorders of the Nose, Throat, Ear, Head, and Neck*. Philadelphia: Lea & Febiger, 1991.

J. Ricker Polsdorfer, MD

Dextromethorphan see **Cough suppressants**

Diabetes insipidus

Definition

Diabetes insipidus (DI) is a disorder that causes the patient to produce tremendous quantities of urine.

The massively increased urine output is usually accompanied by intense thirst.

Description

The balance of fluid within the body is maintained through a number of mechanisms. One important chemical involved in fluid balance is called antidiuretic hormone (ADH). ADH is produced by the pituitary, a small gland located at the base of the brain. In a healthy person and under normal conditions, ADH is continuously released. ADH influences the amount of fluid that the kidneys reabsorb into the circulatory system and the amount of fluid that the kidneys pass out of the body in the form of urine.

Production of ADH is regulated by the osmolality of the circulating blood. Osmolality refers to the concentration of dissolved chemicals (such as sodium, potassium, and chloride; together called solute) circulating in the fluid base of the blood (plasma). When there is very little fluid compared to the concentration of solute, the pituitary will increase ADH production. This tells the kidneys to retain more water and to decrease the amount of urine produced. As fluid is retained, the concentration of solute will normalize. At other times, when the fluid content of the blood is high in comparison to the concentration of solute, ADH production will decrease. The kidneys are then free to pass an increased amount of fluid out of the body in the urine. Again, this will allow the plasma osmolality to return to normal.

Diabetes insipidus occurs when either the amount of ADH produced by the pituitary is below normal (central DI), or the kidneys' ability to respond to ADH is defective (nephrogenic DI). In either case, a person with DI will pass extraordinarily large quantities of urine, sometimes reaching 10 or more liters each day. At the same time, the patient's blood will be very highly concentrated, with low fluid volume and high concentrations of solute.

DI occurs on average when a person is about 24 years old, and occurs more frequently in males than in females.

Causes and symptoms

DI may run in families. The cause of this type of DI is unknown. Other times, central DI can be caused by:

- an injury to the head
- brain surgery
- cancers that have spread to the pituitary gland (most commonly occurring with **breast cancer**)

- sarcoidosis (or other related disorders), causing destruction of the pituitary gland
- any condition or illness that causes decreased oxygen delivery to the brain
- the use of certain medications that decrease ADH production (like the antiseizure drug phenytoin)
- the excessive use of alcohol

Central DI may also occur in women who are pregnant or have just given birth, and in patients with **AIDS** who have suffered certain types of brain infections. Nephrogenic DI sometimes occurs in patients who are taking the medication lithium, patients who have high levels of blood calcium, and patients who are pregnant.

DI is easily confused with an entirely unrelated disorder, psychogenic polydipsia. Polydipsia refers to drinking large amounts of water. Psychogenic polydipsia is a psychiatric problem that makes a person drink huge quantities of water uncontrollably.

Symptoms of DI include extreme thirst and the production of tremendous quantities of urine. Patients with DI typically drink huge amounts of water, and usually report a specific craving for cold water. When the amount of water passed in the urine exceeds the patient's ability to drink ample replacement water, the patient may begin to suffer from symptoms of **dehydration**. These symptoms include weakness, **fatigue**, **fever**, low blood pressure, increased heart rate, **dizziness**, and confusion. If left untreated, the patient could lapse into unconsciousness and die.

Diagnosis

Diagnosis should be suspected in any patient with sudden increased thirst and urination. Laboratory examination of urine will reveal very dilute urine, made up mostly of water with no solute. Examination of the blood will reveal very concentrated blood, high in solute and low in fluid volume.

A water deprivation test may be performed. This test requires a patient to stop all fluid intake. The patient is weighed just before the test begins, and urine is collected and examined hourly. The test is stopped when:

- the patient has lost more than 5% of his or her original body weight
- the patient has reached certain limits of low blood pressure and increased heart rate

KEY TERMS

Concentration—Refers to the amount of solute present in a solution, compared to the total amount of solvent.

Dilute—A solution that has comparatively more fluid in it, relative to the quantity of solute.

Osmolality—A measure of the solute-to-solvent concentration of a solution.

Solute—Solid substances that are dissolved in liquid in order to make a solution.

- the urine is no longer changing significantly from one sample to the next in terms of solute concentration.

The next step of the test involves injecting a synthetic form of ADH, with one last urine sample examined 60 minutes later. Comparing plasma and urine osmolality allows the doctor to diagnose either central DI, nephrogenic DI, partial DI, or psychogenic polydipsia.

Treatment

A number of medications can be given to decrease the quantity of fluid passed out into the urine. These include rasopressin (Pitressin) injected and desmopressin acetate (DDAVP) inhaled through the nose. Other medications that may be given include some antidiuretic drugs (chlorpropamide, clofibrate, carbamazepine). Patients with nephrogenic DI, however, will also require special **diets** that restrict the amount of solute taken in. These patients are also treated with a type of medication called a thiazide diuretic.

Prognosis

Uncomplicated diabetes insipidus is controllable with adequate intake of water and most patients can lead normal lives.

Resources

PERIODICALS

Singer, Irwin, et al. "The Management of Diabetes Insipidus in Adults." *Archives of Internal Medicine* 157, no. 12 (June 23, 1997): 1293+.

Rosalyn Carson-DeWitt, MD

Diabetes mellitus

Definition

Diabetes mellitus is a condition in which the pancreas no longer produces enough insulin or cells stop responding to the insulin that is produced, so that glucose in the blood cannot be absorbed into the cells of the body. Symptoms include frequent urination, lethargy, excessive thirst, and hunger. The treatment includes changes in diet, oral medications, and in some cases, daily injections of insulin.

Description

Diabetes mellitus is a chronic disease that causes serious health complications including renal (kidney) failure, heart disease, **stroke**, and blindness. Approximately 17 million Americans have diabetes. Unfortunately, as many as one-half are unaware they have it.

Background

Every cell in the human body needs energy in order to function. The body's primary energy source is glucose, a simple sugar resulting from the digestion of foods containing carbohydrates (sugars and starches). Glucose from the digested food circulates in the blood as a ready energy source for any cells that need it. Insulin is a hormone or chemical produced by cells in the pancreas, an organ located behind the stomach. Insulin bonds to a receptor site on the outside of cell and acts like a key to open a doorway into the cell through which glucose can enter. Some of the glucose can be converted to concentrated energy sources like glycogen or fatty acids and saved for later use. When there is not enough insulin produced or when the doorway no longer recognizes the insulin key, glucose stays in the blood rather entering the cells.

The body will attempt to dilute the high level of glucose in the blood, a condition called hyperglycemia, by drawing water out of the cells and into the bloodstream in an effort to dilute the sugar and excrete it in the urine. It is not unusual for people with undiagnosed diabetes to be constantly thirsty, drink large quantities of water, and urinate frequently as their bodies try to get rid of the extra glucose. This creates high levels of glucose in the urine.

At the same time that the body is trying to get rid of glucose from the blood, the cells are starving for glucose and sending signals to the body to eat more food, thus making patients extremely hungry. To

provide energy for the starving cells, the body also tries to convert fats and proteins to glucose. The breakdown of fats and proteins for energy causes acid compounds called ketones to form in the blood. Ketones also will be excreted in the urine. As ketones build up in the blood, a condition called ketoacidosis can occur. This condition can be life threatening if left untreated, leading to **coma** and **death**.

Types of diabetes mellitus

Type I diabetes, sometimes called juvenile diabetes, begins most commonly in childhood or adolescence. In this form of diabetes, the body produces little or no insulin. It is characterized by a sudden onset and occurs more frequently in populations descended from Northern European countries (Finland, Scotland, Scandinavia) than in those from Southern European countries, the Middle East, or Asia. In the United States, approximately three people in 1,000 develop Type I diabetes. This form also is called insulin-dependent diabetes because people who develop this type need to have daily injections of insulin.

Brittle diabetics are a subgroup of Type I where patients have frequent and rapid swings of blood sugar levels between hyperglycemia (a condition where there is too much glucose or sugar in the blood) and **hypoglycemia** (a condition where there are abnormally low levels of glucose or sugar in the blood). These patients may require several injections of different types of insulin during the day to keep the blood sugar level within a fairly normal range.

The more common form of diabetes, Type II, occurs in approximately 3–5% of Americans under 50 years of age, and increases to 10–15% in those over 50. More than 90% of the diabetics in the United States are Type II diabetics. Sometimes called age-onset or adult-onset diabetes, this form of diabetes occurs most often in people who are overweight and who do not **exercise**. It is also more common in people of Native American, Hispanic, and African-American descent. People who have migrated to Western cultures from East India, Japan, and Australian Aboriginal cultures also are more likely to develop Type II diabetes than those who remain in their original countries.

Type II is considered a milder form of diabetes because of its slow onset (sometimes developing over the course of several years) and because it usually can be controlled with diet and oral medication. The consequences of uncontrolled and untreated Type II diabetes, however, are the just as serious as those for Type I. This form is also called noninsulin-dependent

diabetes, a term that is somewhat misleading. Many people with Type II diabetes can control the condition with diet and oral medications, however, insulin injections are sometimes necessary if treatment with diet and oral medication is not working.

Another form of diabetes called **gestational diabetes** can develop during **pregnancy** and generally resolves after the baby is delivered. This diabetic condition develops during the second or third trimester of pregnancy in about 2% of pregnancies. In 2004, incidence of gestational diabetes were reported to have increased 35% in 10 years. Children of women with gestational diabetes are more likely to be born prematurely, have hypoglycemia, or have severe **jaundice** at birth. The condition usually is treated by diet, however, insulin injections may be required. These women who have diabetes during pregnancy are at higher risk for developing Type II diabetes within 5–10 years.

Diabetes also can develop as a result of pancreatic disease, **alcoholism**, **malnutrition**, or other severe illnesses that **stress** the body.

Causes and symptoms

Causes

The causes of diabetes mellitus are unclear, however, there seem to be both hereditary (genetic factors passed on in families) and environmental factors involved. Research has shown that some people who develop diabetes have common genetic markers. In Type I diabetes, the immune system, the body's defense system against infection, is believed to be triggered by a virus or another microorganism that destroys cells in the pancreas that produce insulin. In Type II diabetes, age, **obesity**, and family history of diabetes play a role.

In Type II diabetes, the pancreas may produce enough insulin, however, cells have become resistant to the insulin produced and it may not work as effectively. Symptoms of Type II diabetes can begin so gradually that a person may not know that he or she has it. Early signs are lethargy, extreme thirst, and frequent urination. Other symptoms may include sudden weight loss, slow wound healing, urinary tract infections, gum disease, or blurred vision. It is not unusual for Type II diabetes to be detected while a patient is seeing a doctor about another health concern that is actually being caused by the yet undiagnosed diabetes.

Individuals who are at high risk of developing Type II diabetes mellitus include people who:

- are obese (more than 20% above their ideal body weight)

- have a relative with diabetes mellitus
- belong to a high-risk ethnic population (African-American, Native American, Hispanic, or Native Hawaiian)
- have been diagnosed with gestational diabetes or have delivered a baby weighing more than 9 lbs (4 kg)
- have high blood pressure (140/90 mmHg or above)
- have a high density lipoprotein cholesterol level less than or equal to 35 mg/dL and/or a triglyceride level greater than or equal to 250 mg/dL
- have had impaired glucose tolerance or impaired fasting glucose on previous testing

Several common medications can impair the body's use of insulin, causing a condition known as secondary diabetes. These medications include treatments for high blood pressure (furosemide, clonidine, and thiazide **diuretics**), drugs with hormonal activity (**oral contraceptives**, thyroid hormone, progestins, and glucocorticoids), and the anti-inflammation drug indomethacin. Several drugs that are used to treat **mood disorders** (such as **anxiety** and depression) also can impair glucose absorption. These drugs include haloperidol, lithium carbonate, phenothiazines, **tricyclic antidepressants**, and adrenergic agonists. Other medications that can cause diabetes symptoms include isoniazid, nicotinic acid, cimetidine, and heparin. A 2004 study found that low levels of the essential mineral chromium in the body may be linked to increased risk for diseases associated with **insulin resistance**.

Symptoms

Symptoms of diabetes can develop suddenly (over days or weeks) in previously healthy children or adolescents, or can develop gradually (over several years) in overweight adults over the age of 40. The classic symptoms include feeling tired and sick, frequent urination, excessive thirst, excessive hunger, and weight loss.

Ketoacidosis, a condition due to **starvation** or uncontrolled diabetes, is common in Type I diabetes. Ketones are acid compounds that form in the blood when the body breaks down fats and proteins. Symptoms include abdominal **pain**, **vomiting**, rapid breathing, extreme lethargy, and drowsiness. Patients with ketoacidosis will also have a sweet breath odor. Left untreated, this condition can lead to coma and death.

With Type II diabetes, the condition may not become evident until the patient presents for medical treatment for some other condition. A patient may have heart disease, chronic infections of the gums

and urinary tract, blurred vision, **numbness** in the feet and legs, or slow-healing wounds. Women may experience genital **itching**.

Diagnosis

Diabetes is suspected based on symptoms. Urine tests and blood tests can be used to confirm a diagnosis of diabetes based on the amount of glucose found. Urine can also detect ketones and protein in the urine that may help diagnose diabetes and assess how well the kidneys are functioning. These tests also can be used to monitor the disease once the patient is on a standardized diet, oral medications, or insulin.

Urine tests

Clinistix and Diastix are paper strips or dipsticks that change color when dipped in urine. The test strip is compared to a chart that shows the amount of glucose in the urine based on the change in color. The level of glucose in the urine lags behind the level of glucose in the blood. Testing the urine with a test stick, paper strip, or tablet that changes color when sugar is present is not as accurate as blood testing, however it can give a fast and simple reading.

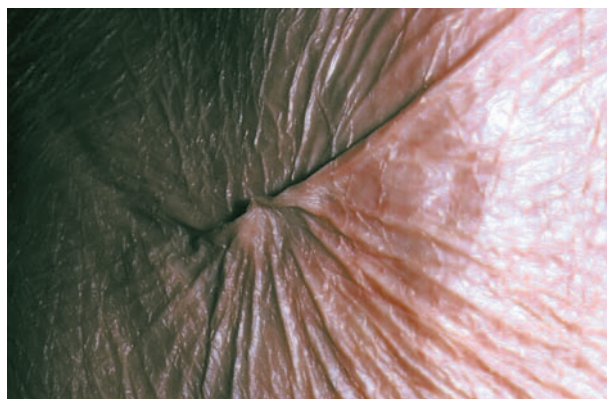
Ketones in the urine can be detected using similar types of dipstick tests (Acetest or Ketostix). Ketoacidosis can be a life-threatening situation in Type I diabetics, so having a quick and simple test to detect ketones can assist in establishing a diagnosis sooner.

Another dipstick test can determine the presence of protein or albumin in the urine. Protein in the urine can indicate problems with kidney function and can be used to track the development of renal failure. A more sensitive test for urine protein uses radioactively tagged chemicals to detect microalbuminuria, small amounts of protein in the urine, that may not show up on dipstick tests.

Blood tests

FASTING GLUCOSE TEST. Blood is drawn from a vein in the patient's arm after a period at least eight hours when the patient has not eaten, usually in the morning before breakfast. The red blood cells are separated from the sample and the amount of glucose is measured in the remaining plasma. A plasma level of 7.8 mmol/L (200 mg/L) or greater can indicate diabetes. The **fasting** glucose test is usually repeated on another day to confirm the results.

POSTPRANDIAL GLUCOSE TEST. Blood is taken right after the patient has eaten a meal.



Wrinkled, dehydrated skin of a person in a diabetic coma. Untreated diabetes mellitus results in elevated blood glucose levels, causing a variety of symptoms that can culminate in a diabetic coma. (Photo Researchers, Inc. Reproduced by permission.)

ORAL GLUCOSE TOLERANCE TEST. Blood samples are taken from a vein before and after a patient drinks a thick, sweet syrup of glucose and other sugars. In a non-diabetic, the level of glucose in the blood goes up immediately after the drink and then decreases gradually as insulin is used by the body to metabolize, or absorb, the sugar. In a diabetic, the glucose in the blood goes up and stays high after drinking the sweetened liquid. A plasma glucose level of 11.1 mmol/L (200 mg/dL) or higher at two hours after drinking the syrup and at one other point during the two-hour test period confirms the diagnosis of diabetes.

A diagnosis of diabetes is confirmed if there are symptoms of diabetes and a plasma glucose level of at least 11.1 mmol/L, a fasting plasma glucose level of at least 7 mmol/L; or a two-hour plasma glucose level of at least 11.1 mmol/L during an oral glucose tolerance test.

Home blood glucose monitoring kits are available so patients with diabetes can monitor their own levels. A small needle or lancet is used to prick the finger and a drop of blood is collected and analyzed by a monitoring device. Some patients may test their blood glucose levels several times during a day and use this information to adjust their doses of insulin.

Treatment

There is currently no cure for diabetes. The condition, however, can be managed so that patients can live a relatively normal life. Treatment of diabetes focuses on two goals: keeping blood glucose within normal range and preventing the development of long-term complications. Careful monitoring of diet,

exercise, and blood glucose levels are as important as the use of insulin or oral medications in preventing complications of diabetes. In 2003, the American Diabetes Association updated its Standards of Care for the management of diabetes. These standards help manage health care providers in the most recent recommendations for diagnosis and treatment of the disease.

Dietary changes

Diet and moderate exercise are the first treatments implemented in diabetes. For many Type II diabetics, weight loss may be an important goal in helping them to control their diabetes. A well-balanced, nutritious diet provides approximately 50–60% of calories from carbohydrates, approximately 10–20% of calories from protein, and less than 30% of calories from fat. The number of calories required by an individual depends on age, weight, and activity level. The calorie intake also needs to be distributed over the course of the entire day so surges of glucose entering the blood system are kept to a minimum.

Keeping track of the number of calories provided by different foods can become complicated, so patients usually are advised to consult a nutritionist or dietitian. An individualized, easy to manage diet plan can be set up for each patient. Both the American Diabetes Association and the American Dietetic Association recommend **diets** based on the use of food exchange lists. Each food exchange contains a known amount of calories in the form of protein, fat, or carbohydrate. A patient's diet plan will consist of a certain number of exchanges from each food category (meat or protein, fruits, breads and starches, vegetables, and fats) to be eaten at meal times and as snacks. Patients have flexibility in choosing which foods they eat as long as they stick with the number of exchanges prescribed.

For many Type II diabetics, weight loss is an important factor in controlling their condition. The food exchange system, along with a plan of moderate exercise, can help them lose excess weight and improve their overall health.

Oral medications

Oral medications are available to lower blood glucose in Type II diabetics. In 1990, 23.4 outpatient prescriptions for oral antidiabetic agents were dispensed. By 2001, the number had increased to 91.8 million prescriptions. Oral antidiabetic agents accounted for more than \$5 billion dollars in worldwide retail sales per year in the early twenty-first century and were the fastest-

growing segment of diabetes drugs. The drugs first prescribed for Type II diabetes are in a class of compounds called sulfonylureas and include tolbutamide, tolazamide, acetohexamide, and chlorpropamide. Newer drugs in the same class are now available and include glyburide, glimeperide, and glipizide. How these drugs work is not well understood, however, they seem to stimulate cells of the pancreas to produce more insulin. New medications that are available to treat diabetes include metformin, acarbose, and troglitazone. The choice of medication depends in part on the individual patient profile. All drugs have side effects that may make them inappropriate for particular patients. Some for example, may stimulate weight gain or cause stomach irritation, so they may not be the best treatment for someone who is already overweight or who has stomach ulcers. Others, like metformin, have been shown to have positive effects such as reduced cardiovascular mortality, but but increased risk in other situations. While these medications are an important aspect of treatment for Type II diabetes, they are not a substitute for a well planned diet and moderate exercise. Oral medications have not been shown effective for Type I diabetes, in which the patient produces little or no insulin.

Constant advances are being made in development of new oral medications for persons with diabetes. In 2003, a drug called Metaglip combining glipizide and metformin was approved in a dingle tablet. Along with diet and exercise, the drug was used as initial therapy for Type 2 diabetes. Another drug approved by the U.S. Food and Drug Administration (FDA) combines metformin and rosiglitazone (Avandia), a medication that increases muscle cells' sensitivity to insulin. It is marketed under the name Avandamet. So many new drugs are under development that it is best to stay in touch with a physician for the latest information; physicians can find the best drug, diet and exercise program to fit an individual patient's need.

Insulin

Patients with Type I diabetes need daily injections of insulin to help their bodies use glucose. The amount and type of insulin required depends on the height, weight, age, food intake, and activity level of the individual diabetic patient. Some patients with Type II diabetes may need to use insulin injections if their diabetes cannot be controlled with diet, exercise, and oral medication. Injections are given subcutaneously, that is, just under the skin, using a small needle and syringe. Injection sites can be anywhere on the body where there is looser skin, including the upper arm, abdomen, or upper thigh.

Purified human insulin is most commonly used, however, insulin from beef and pork sources also are available. Insulin may be given as an injection of a single dose of one type of insulin once a day. Different types of insulin can be mixed and given in one dose or split into two or more doses during a day. Patients who require multiple injections over the course of a day may be able to use an insulin pump that administers small doses of insulin on demand. The small battery-operated pump is worn outside the body and is connected to a needle that is inserted into the abdomen. Pumps can be programmed to inject small doses of insulin at various times during the day, or the patient may be able to adjust the insulin doses to coincide with meals and exercise.

Regular insulin is fast-acting and starts to work within 15–30 minutes, with its peak glucose-lowering effect about two hours after it is injected. Its effects last for about four to six hours. NPH (neutral protamine Hagedorn) and Lente insulin are intermediate-acting, starting to work within one to three hours and lasting up to 18–26 hours. Ultra-lente is a long-acting form of insulin that starts to work within four to eight hours and lasts 28–36 hours.

Hypoglycemia, or low blood sugar, can be caused by too much insulin, too little food (or eating too late to coincide with the action of the insulin), alcohol consumption, or increased exercise. A patient with symptoms of hypoglycemia may be hungry, cranky, confused, and tired. The patient may become sweaty and shaky. Left untreated, the patient can lose consciousness or have a seizure. This condition is sometimes called an insulin reaction and should be treated by giving the patient something sweet to eat or drink like a candy, sugar cubes, juice, or another high sugar snack.

Surgery

Transplantation of a healthy pancreas into a diabetic patient is a successful treatment, however, this transplant is usually done only if a kidney transplant is performed at the same time. Although a pancreas transplant is possible, it is not clear if the potential benefits outweigh the risks of the surgery and drug therapy needed.

Alternative treatment

Since diabetes can be life-threatening if not properly managed, patients should not attempt to treat this condition without medical supervision. A variety of alternative therapies can be helpful in managing the symptoms of diabetes and supporting patients with the disease. **Acupuncture** can help relieve the pain associated with **diabetic neuropathy** by stimulation of

certain points. A qualified practitioner should be consulted. Herbal remedies also may be helpful in managing diabetes. Although there is no herbal substitute for insulin, some herbs may help adjust blood sugar levels or manage other diabetic symptoms. Some options include:

- fenugreek (*Trigonella foenum-graecum*) has been shown in some studies to reduce blood insulin and glucose levels while also lowering cholesterol
- bilberry (*Vaccinium myrtillus*) may lower blood glucose levels, as well as helping to maintain healthy blood vessels
- garlic (*Allium sativum*) may lower blood sugar and cholesterol levels
- onions (*Allium cepa*) may help lower blood glucose levels by freeing insulin to metabolize them
- cayenne pepper (*Capsicum frutescens*) can help relieve pain in the peripheral nerves (a type of diabetic neuropathy)
- ginkgo (*Ginkgo biloba*) may maintain blood flow to the retina, helping to prevent diabetic retinopathy

Any therapy that lowers stress levels also can be useful in treating diabetes by helping to reduce insulin requirements. Among the alternative treatments that aim to lower stress are hypnotherapy, **biofeedback**, and meditation.

Prognosis

Uncontrolled diabetes is a leading cause of blindness, end-stage renal disease, and limb amputations. It also doubles the risks of heart disease and increases the risk of stroke. Eye problems including **cataracts**, **glaucoma**, and diabetic retinopathy also are more common in diabetics.

Diabetic **peripheral neuropathy** is a condition where nerve endings, particularly in the legs and feet, become less sensitive. Diabetic foot ulcers are a particular problem since the patient does not feel the pain of a blister, callous, or other minor injury. Poor blood circulation in the legs and feet contribute to delayed wound healing. The inability to sense pain along with the complications of delayed wound healing can result in minor injuries, blisters, or callouses becoming infected and difficult to treat. In cases of severe infection, the infected tissue begins to break down and rot away. The most serious consequence of this condition is the need for **amputation** of toes, feet, or legs due to severe infection.

Heart disease and **kidney disease** are common complications of diabetes. Long-term complications

KEY TERMS

Cataract—A condition where the lens of the eye becomes cloudy.

Diabetic peripheral neuropathy—A condition where the sensitivity of nerves to pain, temperature, and pressure is dulled, particularly in the legs and feet.

Diabetic retinopathy—A condition where the tiny blood vessels to the retina, the tissues that sense light at the back of the eye, are damaged, leading to blurred vision, sudden blindness, or black spots, lines, or flashing lights in the field of vision.

Glaucoma—A condition where pressure within the eye causes damage to the optic nerve, which sends visual images to the brain.

Hyperglycemia—A condition where there is too much glucose or sugar in the blood.

Hypoglycemia—A condition where there is too little glucose or sugar in the blood.

Insulin—A hormone or chemical produced by the pancreas, insulin is needed by cells of the body in order to use glucose (sugar), the body's main source of energy.

Ketoacidosis—A condition due to starvation or uncontrolled Type I diabetes. Ketones are acid compounds that form in the blood when the body breaks down fats and proteins. Symptoms include abdominal pain, vomiting, rapid breathing, extreme tiredness, and drowsiness.

Kidney dialysis—A process where blood is filtered through a dialysis machine to remove waste products that would normally be removed by the kidneys. The filtered blood is then circulated back into the patient. This process also is called renal dialysis.

Pancreas—A gland located behind the stomach that produces insulin.

may include the need for **kidney dialysis** or a kidney transplant due to kidney failure.

Babies born to diabetic mothers have an increased risk of birth defects and distress at birth.

Prevention

Research continues on diabetes prevention and improved detection of those at risk for developing diabetes. While the onset of Type I diabetes is

unpredictable, the risk of developing Type II diabetes can be reduced by maintaining ideal weight and exercising regularly. The physical and emotional stress of surgery, illness, pregnancy, and alcoholism can increase the risks of diabetes, so maintaining a healthy lifestyle is critical to preventing the onset of Type II diabetes and preventing further complications of the disease.

Resources

PERIODICALS

- Crutchfield, Diane B. "Oral Antidiabetic Agents: Back to the Basics." *Geriatric Times*, May 1, 2003: 20.
- "Gestational Diabetes Increases 35% in 10 Years." *Health & Medicine Week*, March 22, 2004: 220.
- Kordella, Terri. "New Combo Pills." *Diabetes Forecast*, March 2003: 42.
- "New Drugs." *Drug Topics*, November 18, 2002: 73.
- "Research: Lower Chromium Levels Linked to Increased Risk of Disease." *Diabetes Week*, March 29, 2004: 21.
- "Standards of Medical Care for Patients with Diabetes Mellitus: American Diabetes Association." *Clinical Diabetes*, Winter 2003: 27.
- "Wider Metformin Use Recommended." *Chemist & Druggist*, January 11, 2003: 24.

ORGANIZATIONS

- American Diabetes Association. 1701 North Beauregard Street, Alexandria, VA 22311. (800) 342-2383. <<http://www.diabetes.org>>.
- American Dietetic Association. 216 W. Jackson Blvd., Chicago, IL 60606-6995. (312) 899-0040. <<http://www.eatright.org>>.
- Juvenile Diabetes Foundation. 120 Wall St., 19th Floor, New York, NY 10005. (800) 533-2873. <<http://www.jdf.org>>.
- National Diabetes Information Clearinghouse. 1 Information Way, Bethesda, MD 20892-3560. (800) 860-8747. Ndic@info.niddk.nih.gov. <<http://www.niddk.nih.gov/health/diabetes/ndic.htm>>.

OTHER

- Centers for Disease Control*. <<http://www.cdc.gov/nccdphp/ddt/ddthome.htm>>.
- "Insulin-Dependent Diabetes." National Institute of Diabetes and Digestive and Kidney Diseases. National Institutes of Health, NIH Publication No.94-2098.
- "Noninsulin-Dependent Diabetes." National Institute of Diabetes and Digestive and Kidney Diseases. National Institutes of Health, NIH Publication No.92-241.

Altha Roberts Edgren
Teresa G. Odle

Diabetic control index see **Glycosylated hemoglobin test**

Diabetic foot infections

Definition

Diabetic foot infections are infections that can develop in the skin, muscles, or bones of the foot as a result of the nerve damage and poor circulation that is associated with diabetes.

Description

People who have diabetes have a greater-than-average chance of developing foot infections. Because a person who has diabetes may not feel foot **pain** or discomfort, problems can remain undetected until **fever**, weakness, or other signs of systemic infection appear. As a result, even minor irritations occur more often, heal more slowly, and are more likely to result in serious health problems.

With diabetes, foot infections occur more frequently because the disease causes nervous system changes and poor circulation. Because the nerves that control sweating no longer work, the skin of the feet can become very dry and cracked, and calluses tend to occur more frequently and build up faster. If not trimmed regularly, these calluses can turn into open sores or ulcers. Because diabetic nerve damage can cause a loss of sensation (neuropathy), if the feet are not regularly inspected, an ulcer can quickly become infected and, if not treated, may result in the death of tissue (**gangrene**) or **amputation**.

The risk of infection is greatest for people who are over the age of 60 and for those who have one or more of the following:

- poorly controlled diabetes
- foot ulcers
- laser treatment for changes in the retina
- kidney or vascular disease
- loss of sensation (neuropathy)

Causes and symptoms

Bacteria can cause an infection through small cracks (fissures) that can develop in the dry skin around the heel and on other parts of the foot or through corns, calluses, blisters, hangnails, or ulcers. If not treated, the bacterial infection can destroy skin, tissue, and bone or spread throughout the body.

Common sites of diabetic foot infections include the following:



Persons with diabetes often suffer from foot ulcers, as shown above. (Custom Medical Stock Photo. Reproduced by permission.)

- blisters, corns, or callouses that bleed beneath the skin
- bunions, hammertoes, or other abnormalities in the bones of the foot
- scar tissue that has grown over the site of an earlier infection
- foot ulcers caused by pressure, nerve damage, or poor circulation (Ulcers occur most often over the ball of the foot, on the bottom of the big toe, or on the sides of the foot due to poorly fitting shoes.)
- injuries that tear or puncture the skin

Diagnosis

A physician who specializes in the treatment of the foot (podiatrist) or the doctor who normally treats the patient's diabetes will treat the infection. An x ray of the foot will be taken to determine whether the bone has become infected. A sample from the wound will be cultured to identify the organism that is causing the infection so that the appropriate antibiotic can be selected.

Treatment

From the results of the culture, the appropriate antibiotic will be prescribed. Any dead or infected tissue will be surgically removed and, if necessary, a cast and/or special shoes may be used to protect the area. In addition, the patient will be instructed to keep off their feet. If the ulcer does not heal, the physician may perform surgery to increase blood flow to the foot. It is also important for the patient to practice good diabetes control and keep blood glucose levels from getting too high.

KEY TERMS

Fissure—A deep crack.

Neuropathy—An abnormality of the nerves outside the brain and spinal cord.

Ulcer—A sore or lesion.

Alternative treatment

Acupuncture and vitamin C can boost the body's infection-fighting ability. A variety of other **vitamins** and herbs may improve general health and diabetes control. Because diabetes is a potentially deadly disease, it can be dangerous to try alternative approaches without a doctor's approval or without consulting a trained practitioner of alternative medicine.

Prognosis

Without proper treatment, diabetic foot infections can lead to serious illness, gangrene, amputation, and even death if the infection spreads throughout the body. If treated properly and the patient practices good **foot care**, the prognosis is generally optimistic.

Prevention

There are many things that a diabetic individual can do to prevent the occurrence of foot infections, including the following:

- control blood glucose and do not allow it to get too high
- avoid **smoking**
- keep blood pressure and cholesterol under control
- exercise to stimulate blood flow
- keep feet clean, dry, and warm
- check your feet every day for blisters, scratches, and skin that is hard, broken, inflamed, or feels hot or cold when touched
- after bathing, carefully dry feet and apply thin coat of petroleum jelly or hand cream to prevent dry skin from cracking
- use a pumice stone and emery board to trim calluses
- do not neglect an ulcer, should one develop

Resources

ORGANIZATIONS

American Diabetes Association. 1701 North Beauregard Street, Alexandria, VA 22311. (800) 342-2383. <<http://www.diabetes.org>>.

Juvenile Diabetes Foundation. 120 Wall St., 19th Floor, New York, NY 10005. (800) 533-2873. <<http://www.jdf.org>>.

National Diabetes Information Clearinghouse. 1 Information Way, Bethesda, MD 20892-3560. (800) 860-8747. <<http://www.niddk.nih.gov/health/diabetes/ndic.htm>>.

Maureen Haggerty

Diabetic ketoacidosis

Definition

Diabetic ketoacidosis is a dangerous complication of diabetes mellitus in which the chemical balance of the body becomes far too acidic.

Description

Diabetic ketoacidosis (DKA) always results from a severe insulin deficiency. Insulin is the hormone secreted by the body to lower the blood sugar levels when they become too high. Diabetes mellitus is the disease resulting from the inability of the body to produce or respond properly to insulin, required by the body to convert glucose to energy. In childhood diabetes, DKA complications represent the leading cause of **death**, mostly due to the accumulation of abnormally large amounts of fluid in the brain (cerebral **edema**). DKA combines three major features: hyperglycemia, meaning excessively high blood sugar levels; hyperketonemia, meaning an overproduction of ketones by the body; and acidosis, meaning that the blood has become too acidic.

Insulin deficiency is responsible for all three conditions: the body glucose goes largely unused since most cells are unable to transport glucose into the cell without the presence of insulin; this condition makes the body use stored fat as an alternative source instead of the unavailable glucose for energy, a process that produces acidic ketones, which build up because they require insulin to be broken down. The presence of excess ketones in the bloodstream in turn causes the blood to become more acidic than the body tissues, which creates a toxic condition.

Causes and symptoms

DKA is most commonly seen in individuals with type I diabetes, under 19 years of age and is usually

KEY TERMS

Acidosis—A condition that causes the pH of the blood to drop and become more acidic.

Diabetes mellitus—Disease characterized by the inability of the body to produce or respond properly to insulin, required by the body to convert glucose to energy.

Edema—The presence of abnormally large amounts of fluid in the intercellular tissue spaces of the body.

Glucose—The type of sugar found in the blood.

Hyperglycemia—Condition characterized by excessively high levels of glucose in the blood, and occurs when the body does not have enough insulin or cannot use the insulin it does have to turn glucose into energy. Hyperglycemia is often indicative of diabetes that is out of control.

Hyperketonemia—Condition characterized by an overproduction of ketones by the body.

Hypoglycemia—Lower than normal levels of glucose in the blood.

Hypokalemia—A deficiency of potassium in the blood.

Insulin—A hormone secreted by the pancreas in response to high blood sugar levels that induces hypoglycemia. Insulin regulates the body's use of

glucose and the levels of glucose in the blood by acting to open the cells so that they can intake glucose.

Ketones—Poisonous acidic chemicals produced by the body when fat instead of glucose is burned for energy. Breakdown of fat occurs when not enough insulin is present to channel glucose into body cells.

Lactic acidosis—A serious condition caused by the build up of lactic acid in the blood, causing it to become excessively acidic. Lactic acid is a by-product of glucose metabolism.

Metabolism—The sum of all chemical reactions that occur in the body resulting in growth, transformation of foodstuffs into energy, waste elimination, and other bodily functions.

Polyuria—Excessive secretion of urine.

Type I diabetes—Also called juvenile diabetes. Type I diabetes typically begins early in life. Affected individuals have a primary insulin deficiency and must take insulin injections.

Type II diabetes—Type II diabetes is the most common form of diabetes and usually appears in middle aged adults. It is often associated with obesity and may be delayed or controlled with diet and exercise.

caused by the interruption of their insulin treatment or by acute infection or trauma. A small number of people with type II diabetes also experience ketoacidosis, but this is rare given the fact that type II diabetics still produce some insulin naturally. When DKA occurs in type II patients, it is usually caused by a decrease in food intake and an increased insulin deficiency due to hyperglycemia.

Some common DKA symptoms include:

- high blood sugar levels
- frequent urination (polyuria) and thirst
- fatigue and lethargy
- nausea
- **vomiting**
- abdominal pain
- fruity odor to breath
- rapid, deep breathing
- muscle stiffness or aching
- **coma**

Diagnosis

Diagnosis requires the demonstration of hyperglycemia, hyperketonemia, and acidosis. DKA is established if the patient's urine or blood is strongly positive for glucose and ketones. Normal glucose levels in a non-diabetic person on average range from 80–110 mg/dl. A person with diabetes will typically fluctuate outside those parameters. DKA glucose levels exceed 250 mg/dl and can reach 400 to 800 mg/dL. A low serum bicarbonate level (usually below 15 mEq/L) is also present, indicative of acidosis.

A blood test or **urinalysis** can quickly determine the concentration of glucose in the bloodstream. Test strips are available to patients commercially can submerge in urine to detect the presence or concentration of ketones.

Treatment

Ketoacidosis is treated under medical supervision and usually in a hospital setting.

Basic treatment includes:

- administering insulin to correct the hyperglycemia and hyperketonemia
- replacing fluids lost through excessive urination and vomiting intravenously
- balancing electrolytes to re-establish the chemical equilibrium of the blood and prevent potassium deficiency (**hypokalemia**) during treatment
- treatment for any associated bacterial infection

Prognosis

With proper medical attention, DKA is almost always successfully treated. The DKA mortality rate is about 10%. Coma on admission adversely affects the prognosis. The major causes of death are circulatory collapse, hypokalemia, infection, and cerebral edema.

Prevention

Once diabetes has been diagnosed, prevention measures to avoid DKA include regular monitoring of blood glucose, administration of insulin, and lifestyle maintenance. Glucose monitoring is especially important during periods of stress, infection, and trauma when glucose concentrations typically increase as a response to these situations. Ketone tests should also be performed during these periods or when glucose is elevated.

Resources

ORGANIZATIONS

American Diabetes Association. 1701 North Beauregard Street, Alexandria, VA 22311. (800) DIABETES (800-342-2383). <<http://www.diabetes.org/>>.

Juvenile Diabetes Foundation. 120 Wall St., 19th Floor, New York, NY 10005. (800) 533-CURE. <<http://www.jdf.org/>>.

National Institute of Diabetes and Digestive and Kidney Disorders (NIDDK). 31 Center Drive, MSC 2560, Bethesda, MD 20892-2560. <<http://www.niddk.nih.gov>>.

Gary Gilles

affecting several parts of the body, or focal, affecting a specific nerve and part of the body.

Description

The nervous system consists of two major divisions: the central nervous systems (CNS) which includes the brain, the cranial nerves, and the spinal cord, and the peripheral nervous system (PNS) which includes the nerves that link the CNS with the sensory organs, muscles, blood vessels, and glands of the body. These peripheral nerves are either motor, meaning that they are involved in motor activity such as walking, or sensory, meaning that they carry sensory information back to the CNS. The PNS also works with the CNS to regulate involuntary (autonomic) processes such as breathing, heartbeat, blood pressure, etc.

There are two types of diffuse diabetic neuropathy that affect different nervous system functions. Diffuse **peripheral neuropathy** primarily affects the limbs, damaging the nerves of the feet and hands. Autonomic neuropathy is the other form of diffuse neuropathy and it affects the heart and other internal organs.

Focal—or localized—diabetic neuropathy affects specific nerves, most commonly in the torso, leg, or head.

Diabetic neuropathy can lead to muscular weakness, loss of feeling or sensation, and loss of autonomic functions such as digestion, erection, bladder control, and sweating among others.

The longer a person has diabetes, the more likely the development of one or more forms of neuropathy. Approximately 60–70% of patients with diabetes have neuropathy, but only about 5% will experience painful symptoms.

Causes and symptoms

The exact cause of diabetic neuropathy is not known. Researchers believe that the process of nerve damage is related to high glucose concentrations in the blood that could cause chemical changes in nerves, disrupting their ability to effectively send messages. High blood glucose is also known to damage the blood vessels that carry oxygen and other nutrients to the nerves. In addition, some people may have a genetic predisposition to develop neuropathy.

There is a wide range of symptoms associated with diabetic neuropathy, and they depend on which nerves and parts of the body are affected and also on the type of neuropathy present. Some patients have very mild symptoms, while others are severely disabled.

Diabetic neuropathy

Definition

Diabetic neuropathy is a nerve disorder caused by diabetes mellitus. Diabetic neuropathy may be diffuse,

Common symptoms of diffuse peripheral neuropathy include:

- numbness and feelings of **tingling** or burning
- insensitivity to **pain**
- needle-like jabs of pain
- extreme sensitivity to touch
- loss of balance and coordination

Common symptoms of diffuse autonomic neuropathy include:

- impaired urination and sexual function
- bladder infections
- stomach disorders, due to the impaired ability of the stomach to empty (gastric stasis)
- nausea, **vomiting** and bloating
- dizziness, lightheadedness, and **fainting** spells
- loss of appetite

Common symptoms of focal neuropathy include:

- pain in the front of a thigh
- severe pain in the lower back
- pain in the chest or stomach
- ache behind an eye
- double vision
- paralysis on one side of the face

In severe diabetic neuropathy loss of sensation can lead to injuries that are unnoticed, progressing to infections, ulceration and possibly **amputation**.

Diagnosis

The diagnosis of neuropathy is based on the symptoms that present during a physical exam. Pain assessment is usually the first step. Patients may have more than one type of pain, and the history helps the doctor determine whether the pain has a neuropathic cause.

The exam may include:

- a screening test for lost sensation
- nerve conduction studies to check the flow of electric current through a nerve
- electromyography (EMG) to see how well muscles respond to electrical impulses transmitted by nearby nerves.
- ultrasound to show how the bladder and other parts of the urinary tract are functioning
- sometimes a nerve biopsy may be performed.

KEY TERMS

Central nervous system (CNS)—Part of the nervous system consisting of the brain, cranial nerves, and spinal cord. The brain is the center of higher processes, such as thought and emotion, and is responsible for the coordination and control of bodily activities and the interpretation of information from the senses. The cranial nerves and spinal cord link the brain to the peripheral nervous system.

Diabetes mellitus—Disease characterized by the inability of the body to produce or respond properly to insulin, required by the body to convert glucose to energy.

Glucose—The type of sugar found in the blood.

Peripheral nervous system (PNS)—One of the two major divisions of the nervous system. PNS nerves link the central nervous system with sensory organs, muscles, blood vessels, and glands.

Specialists who treat diabetic neuropathy include:

- neurologists: specialists in nervous system disorders
- urologists: specialists in urinary tract disorder
- gastroenterologists: specialists in digestive disorders
- podiatrists: specialists in caring for the feet

Treatment

Treatment of diabetic neuropathy is usually focused on treating the symptoms associated with the neuropathy and addressing the underlying cause by improving the control of blood sugar levels, which may heal the early stages of neuropathy.

There is no cure for the permanent nerve damage caused by neuropathy. To help control pain, the choice of proven drug therapies has broadened during the past decade. Pain medication, such as the topical skin cream capsaicin, is usually no stronger than codeine because of the potential for **addiction** with long-term use of such drugs. Four main classes of drugs are available for **pain management**, alone or in combination: **tricyclic antidepressants** (Imipramine, Nortriptyline), narcotic **analgesics** (Morphine), anticonvulsants (Carbamazepine, Gabapentin), and antiarrhythmics.

Prognosis

Early stage diabetic neuropathy can usually be reversed with good glucose control. Once nerve

damage has occurred it cannot be reversed. The prognosis is largely dependent on the management of the underlying condition, diabetes, which may halt the progression of the neuropathy and improve symptoms. Recovery, if it occurs, is slow.

Prevention

Tight glucose control and the avoidance of alcohol and cigarettes help protect nerves from damage.

Resources

ORGANIZATIONS

American Diabetes Association. 1701 North Beauregard Street, Alexandria, VA 22311. (800) DIABETES (800-342-2383). <<http://www.diabetes.org/>>.

Juvenile Diabetes Foundation. 120 Wall St., 19th Floor, New York, NY 10005. (800) 533-CURE. <<http://www.jdf.org/>>.

Gary Gilles

Dialysis, kidney

Definition

Dialysis treatment replaces the function of the kidneys, which normally serve as the body's natural filtration system. Through the use of a blood filter and a chemical solution known as dialysate, the treatment removes waste products and excess fluids from the bloodstream, while maintaining the proper chemical balance of the blood. There are two types of dialysis treatment: hemodialysis and peritoneal dialysis.

Purpose

Dialysis can be used in the treatment of patients suffering from **poisoning** or overdose, in order to quickly remove drugs from the bloodstream. Its most prevalent application, however, is for patients with temporary or permanent kidney failure. For patients with end-stage renal disease (ESRD), whose kidneys are no longer capable of adequately removing fluids and wastes from their body or of maintaining the proper level of certain kidney-regulated chemicals in the bloodstream, dialysis is the only treatment option available outside of **kidney transplantation**. In 1996 in the United States, over 200,000 people underwent regular dialysis treatments to manage their ESRD.

Precautions

Blood pressure changes associated with hemodialysis may pose a risk for patients with heart problems. Peritoneal dialysis may be the preferred treatment option in these cases.

Peritoneal dialysis is not recommended for patients with abdominal **adhesions** or other abdominal defects, such as a **hernia**, that might compromise the efficiency of the treatment. It is also not recommended for patients who suffer frequent bouts of diverticulitis, an inflammation of small pouches in the intestinal tract.

Description

There are two types of dialysis treatment: hemodialysis and peritoneal dialysis:

Hemodialysis

Hemodialysis is the most frequently prescribed type of dialysis treatment in the United States. The treatment involves circulating the patient's blood outside of the body through an extracorporeal circuit (ECC), or dialysis circuit. Two needles are inserted into the patient's vein, or access site, and are attached to the ECC, which consists of plastic blood tubing, a filter known as a dialyzer (artificial kidney), and a dialysis machine that monitors and maintains blood flow and administers dialysate. Dialysate is a chemical bath that is used to draw waste products out of the blood.

Since the 1980s, the majority of hemodialysis treatments in the United States have been performed with hollow fiber dialyzers. A hollow fiber dialyzer is composed of thousands of tube-like hollow fiber strands encased in a clear plastic cylinder several inches in diameter. There are two compartments within the dialyzer (the blood compartment and the dialysate compartment). The membrane that separates these two compartments is semipermeable. This means that it allows the passage of certain sized molecules across it, but prevents the passage of other, larger molecules. As blood is pushed through the blood compartment in one direction, suction or vacuum pressure pulls the dialysate through the dialysate compartment in a countercurrent, or opposite direction. These opposing pressures work to drain excess fluids out of the bloodstream and into the dialysate, a process called ultrafiltration.

A second process called diffusion moves waste products in the blood across the membrane into the dialysate compartment, where they are carried out of the body. At the same time, electrolytes and other

chemicals in the dialysate solution cross the membrane into the blood compartment. The purified, chemically balanced blood is then returned to the body.

Most hemodialysis patients require treatment three times a week, for an average of three–four hours per dialysis “run.” Specific treatment schedules depend on the type of dialyzer used and the patient’s current physical condition. While the treatment prescription and regimen is usually overseen by a nephrologist (a doctor that specializes in the kidney), dialysis treatments are typically administered by a nurse or patient care technician in outpatient clinics known as dialysis centers, or in hospital-based dialysis units. In-home hemodialysis treatment is also an option for some patients, although access to this type of treatment may be limited by financial and lifestyle factors. An investment in equipment is required and another person in the household should be available for support and assistance with treatments.

Peritoneal dialysis

In peritoneal dialysis, the patient’s peritoneum, or lining of the abdomen, acts as a blood filter. A catheter is surgically inserted into the patient’s abdomen. During treatment, the catheter is used to fill the abdominal cavity with dialysate. Waste products and excess fluids move from the patient’s bloodstream into the dialysate solution. After a waiting period of six to 24 hours, depending on the treatment method used, the waste-filled dialysate is drained from the abdomen, and replaced with clean dialysate.

There are three types of peritoneal dialysis:

- Continuous ambulatory peritoneal dialysis (CAPD). A continuous treatment that is self-administered and requires no machine. The patient inserts fresh dialysate solution into the abdominal cavity, waits four to six hours, and removes the used solution. The solution is immediately replaced with fresh dialysate. A bag attached to the catheter is worn under clothing.
- Continuous cyclic peritoneal dialysis (CCPD). An overnight treatment that uses a machine to drain and refill the abdominal cavity, CCPD takes 10–12 hours per session.
- Intermittent peritoneal dialysis (IPD). This hospital-based treatment is performed several times a week. A machine administers and drains the dialysate solution, and sessions can take up to 24 hours.

Peritoneal dialysis is often the treatment option of choice in infants and children, whose small size can

make vascular (through a vein) access difficult to maintain. Peritoneal dialysis can also be done outside of a clinical setting, which is more conducive to regular school attendance.

Preparation

Patients are weighed immediately before and after each hemodialysis treatment to assess their fluid retention. Blood pressure and temperature are taken and the patient is assessed for physical changes since their last dialysis run. Regular blood tests monitor chemical and waste levels in the blood. Prior to treatment, patients are typically administered a dose of heparin, an anticoagulant that prevents blood clotting, to ensure the free flow of blood through the dialyzer and an uninterrupted dialysis run for the patient.

Aftercare

Both hemodialysis and peritoneal dialysis patients need to be vigilant about keeping their access sites and catheters clean and infection-free during and between dialysis runs.

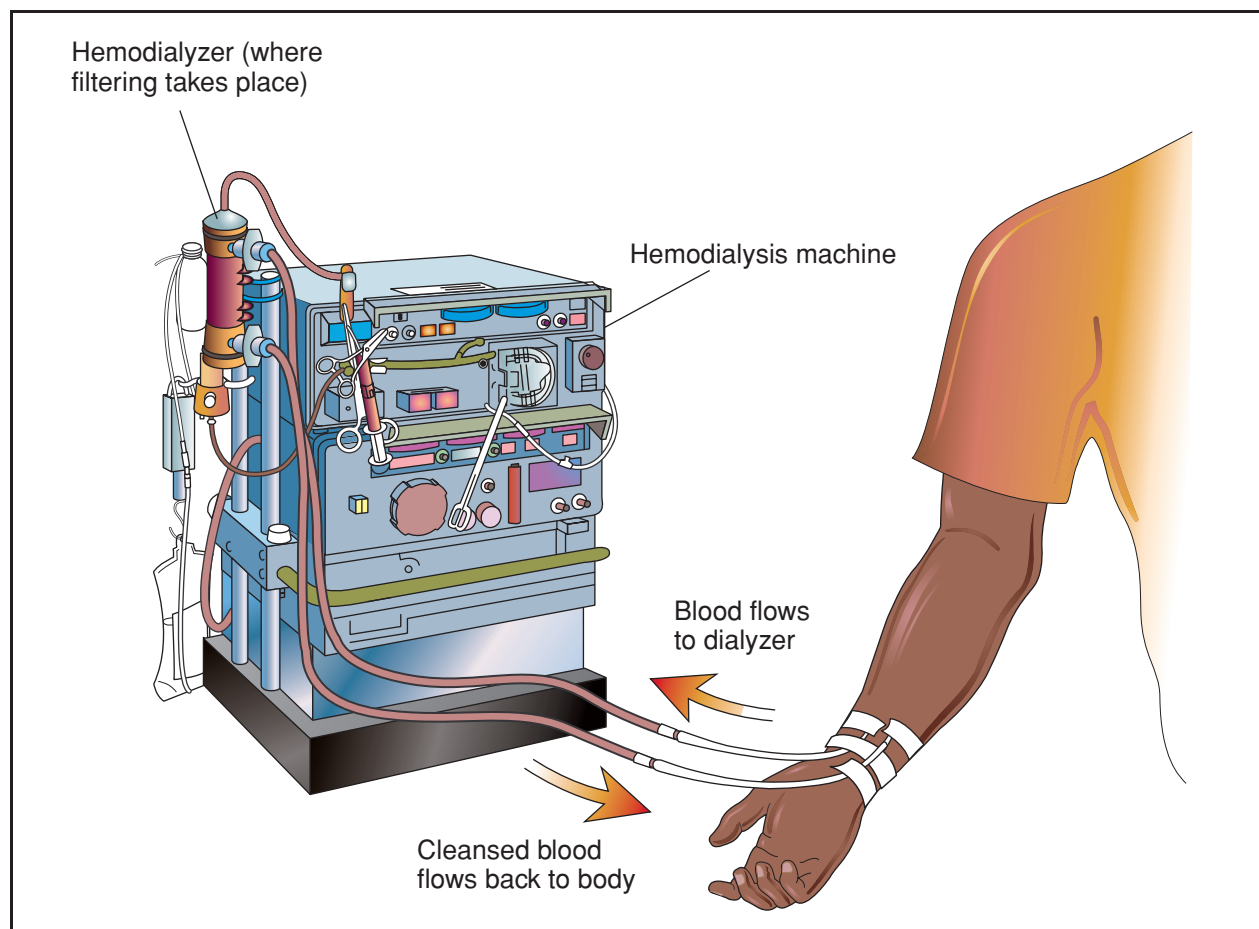
Dialysis is just one facet of a comprehensive treatment approach for ESRD. Although dialysis treatment is very effective in removing toxins and fluids from the body, there are several functions of the kidney it cannot mimic, such as regulating high blood pressure and red blood cell production. Patients with ESRD need to watch their diet and fluid intake carefully and take medications as prescribed to manage their disease.

Risks

Many of the risks and side effects associated with dialysis are a combined result of both the treatment and the poor physical condition of the ESRD patient. Dialysis patients should always report side effects to their healthcare provider.

Anemia

Hematocrit (Hct) levels, a measure of red blood cells, are typically low in ESRD patients. This deficiency is caused by a lack of the hormone erythropoietin, which is normally produced by the kidneys. The problem is elevated in hemodialysis patients, who may incur blood loss during hemodialysis treatments. Epoetin alfa, or EPO (sold under the trade name Epogen), a hormone therapy, and intravenous or



Hemodialysis is the most frequently prescribed type of dialysis treatment in the United States. This treatment involves circulating the patient's blood outside of the body through a dialysis circuit. The blood is filtered and cleansed inside the hemodialyzer and returned to the body. (Illustration by Electronic Illustrators Group.)

oral iron supplements are used to manage anemia in dialysis patients.

Cramps, nausea, vomiting, and headaches

Some hemodialysis patients experience cramps and flu-like symptoms during treatment. These can be caused by a number of factors, including the type of dialysate used, composition of the dialyzer membrane, water quality in the dialysis unit, and the ultrafiltration rate of the treatment. Adjustment of the dialysis prescription often helps alleviate many symptoms.

Hypotension

Because of the **stress** placed on the cardiovascular system with regular hemodialysis treatments, patients are at risk for **hypotension**, a sudden drop in blood pressure. This can often be controlled by medication and adjustment of the patient's dialysis prescription.

Infection

Both hemodialysis and peritoneal dialysis patients are at risk for infection. Hemodialysis patients should keep their access sites clean and watch for signs of redness and warmth that could indicate infection. Peritoneal dialysis patients must follow the same precautions with their catheter. **Peritonitis**, an infection of the peritoneum, causes flu-like symptoms and can disrupt dialysis treatments if not caught early.

Infectious diseases

Because there is a great deal of blood exposure involved in dialysis treatment, a slight risk of contracting **hepatitis B** and hepatitis C exists. The hepatitis B **vaccination** is recommended for most hemodialysis patients. As of 1997, there had only been one documented case of HIV being transmitted in a United States dialysis unit to a staff member, and no documented cases of HIV ever

KEY TERMS

Access site—The vein tapped for vascular access in hemodialysis treatments. For patients with temporary treatment needs, access to the bloodstream is gained by inserting a catheter into the subclavian vein near the patient's collarbone. Patients in long-term dialysis require stronger, more durable access sites, called fistulas or grafts, that are surgically created.

Dialysate—A chemical bath used in dialysis to draw fluids and toxins out of the bloodstream and supply electrolytes and other chemicals to the bloodstream.

Dialysis prescription—The general parameters of dialysis treatment that vary according to each patient's individual needs. Treatment length, type of dialyzer and dialysate used, and rate of ultrafiltration are all part of the dialysis prescription.

Dialyzer—An artificial kidney usually composed of hollow fiber which is used in hemodialysis to eliminate waste products from the blood and remove excess fluids from the bloodstream.

Erythropoietin—A hormone produced by the kidneys that stimulates the production of red blood cells by bone marrow.

ESRD—End-stage renal disease; chronic or permanent kidney failure.

Extracorporeal circuit (ECC)—The path the hemodialysis patient's blood takes outside of the body. It typically consists of plastic tubing, a hemodialysis machine, and a dialyzer.

Hematocrit (Hct) level—A measure of red blood cells.

Peritoneum—The abdominal cavity; the peritoneum acts as a blood filter in peritoneal dialysis.

being transmitted between dialysis patients in the United States. The strict standards of **infection control** practiced in modern hemodialysis units makes the chance of contracting one of these diseases very small.

Normal results

Puffiness in the patient related to **edema**, or fluid retention, may be relieved after dialysis treatment. The patient's overall sense of physical well-being may also be improved. Because dialysis is an ongoing treatment process for many patients, a baseline for normalcy can be difficult to gauge.

Resources

ORGANIZATIONS

American Association of Kidney Patients. 100 S. Ashley Dr., #280, Tampa, FL 33602. (800) 749-2257. <<http://www.aakp.org>>.

American Kidney Fund (AKF). Suite 1010, 6110 Executive Boulevard, Rockville, MD 20852. (800) 638-8299. <<http://216.248.130.102/Default.htm>>.

National Kidney Foundation. 30 East 33rd St., New York, NY 10016. (800) 622-9010. <<http://www.kidney.org>>.

United States Renal Data System (USRDS). The University of Michigan, 315 W. Huron, Suite 240, Ann Arbor, MI 48103. (734) 998-6611. <<http://www.med.umich.edu/usrds>>.

Paula Anne Ford-Martin

Diaper rash

Definition

Dermatitis of the buttocks, genitals, lower abdomen, or thigh folds of an infant or toddler is commonly referred to as diaper rash.

Description

The outside layer of skin normally forms a protective barrier that prevents infection. One of the primary causes of dermatitis in the diaper area is prolonged skin contact with wetness. Under these circumstances, natural oils are stripped away, the outer layer of skin is damaged, and there is increased susceptibility to infection by bacteria or yeast.

Diaper rash is a term that covers a broad variety of skin conditions that occur on the same area of the body. Some babies are more prone to diaper rash than others.

Causes and symptoms

Frequently a flat, red rash is caused by simple chafing of the diaper against tender skin, initiating a friction rash. This type of rash is not seen in the skin folds. It may be more pronounced around the edges of the diaper, at the waist and leg bands. The baby generally doesn't appear to experience much discomfort. Sometimes the chemicals or detergents in the diaper are contributing factors and may result in **contact dermatitis**. These **rashes** should clear up easily with

proper attention. Ignoring the condition may lead to a secondary infection that is more difficult to resolve.

Friction of skin against itself can cause a rash in the baby's skin folds, called intertrigo. This rash appears as reddened areas that may ooze and is often uncomfortable when the diaper is wet. Intertrigo can also be found on other areas of the body where there are deep skin folds that tend to trap moisture.

Seborrheic dermatitis is the diaper area equivalent of cradle cap. It is scaly and greasy in appearance and may be worse in the folds of the skin.

Yeast, or candidal dermatitis, is the most common infectious cause of diaper rash. The affected areas are raised and quite red with distinct borders, and satellite lesions may occur around the edges. Yeast is part of the normal skin flora, and is often an opportunistic invader when simple diaper rash is untreated. It is particularly common after treatment with **antibiotics**, which kill the good bacteria that normally keep the yeast population in check. Usual treatments for diaper rash will not clear it up. Repeated or difficult to resolve episodes of yeast infection may warrant further medical attention, since this is sometimes associated with diabetes or immune problems.

Another infectious cause of diaper rash is **impetigo**. This bacterial infection is characterized by blisters that ooze and crust.

Diagnosis

The presence of **skin lesions** in the diaper area means that the baby has diaper rash. However, there are several types of rash that may require specific treatment in order to heal. It is useful to be able to distinguish them by appearance as described above.

A baby with a rash that does not clear up within two to three days or a rash with blisters or bleeding should be seen by a healthcare professional for further evaluation.

Treatment

Antibiotics are generally prescribed for rashes caused by bacteria, particularly impetigo. This may be a topical or oral formulation, depending on the size of the area involved and the severity of the infection.

Over-the-counter antifungal creams, such as Lotrimin, are often recommended to treat a rash resulting from yeast. If topical treatment is not effective, an oral antifungal may be prescribed.



Baby with severe diaper rash. (Custom Medical Stock Photo. Reproduced by permission.)

Mild steroid creams, such as 0.5–1% hydrocortisone, can be used for seborrheic dermatitis and sometimes intertrigo. Prescription strength creams may be needed for short-term treatment of more stubborn cases.

Alternative treatment

Good diaper hygiene will prevent or clear up many simple cases of diaper rash. Diapers should be checked very frequently and changed as soon as they are wet or soiled. Good air circulation is also important for healthy skin. Babies should have some time without wearing a diaper, and a waterproof pad can be used to protect the bed or other surface. Rubber pants, or other occlusive fabrics, should not be used over the diaper area. Some cloth-like disposable diapers promote better air circulation than plastic-type diapers. It may be necessary for mothers to experiment with diaper types to see if the baby's skin reacts better to cloth or disposable ones. If disposable diapers are used, the baby's skin may react differently to various brands. If the baby is wearing cloth diapers, they should be washed in a mild detergent and double rinsed.

The diaper area should be cleaned with something mild, even plain water. Some wipes contain alcohol or chemicals that can be irritating for some babies. Plain water may be the best cleansing substance when there is a rash. Using warm water in a spray bottle (or giving a quick bath) and then lightly patting the skin dry can produce less skin trauma than using wipes. In the event of suspected yeast, a tablespoon of cider vinegar can be added to a cup of warm water and used as a cleansing solution. This is dilute enough that it should not burn, but acidifies the skin pH enough to hamper the yeast growth.

KEY TERMS

Dermatitis—Inflammation of the skin.

Barrier ointments can be valuable to treat rashes. Those that contain zinc oxide are especially effective. These creams and ointments protect already irritated skin from the additional insult of urine and stool, particularly if the baby has **diarrhea**. Cornstarch powder may be used on rashes that are moist, such as impetigo.

Nutrition

What the baby eats can make a difference in stool frequency and acidity. Typically, breast-fed babies will have fewer problems with rashes. When adding a new food to the diet, the baby should be observed closely to see whether rashes are produced around the baby's mouth or anus. If this occurs, the new food should be discontinued.

Babies who are taking antibiotics are more likely to get rashes due to yeast. To help bring the good bacterial counts back to normal, *Lactobacillus bifidus* can be added to the diet. It is available in powder form from most health food stores.

Herbal treatment

Some herbal preparations can be useful for diaper rash. Calendula reduces inflammation, tightens tissues, and disinfects. It has been recommended for seborrheic dermatitis as well as for general inflammation of the skin. The ointment should be applied at each diaper change. Chickweed ointment can also be soothing for irritated skin and may be applied once or twice daily.

Prognosis

Treated appropriately, diaper rash will resolve fairly quickly if there is no underlying health problem or skin disease.

Prevention

Frequent diaper changes are important to keep the skin dry and healthy. Application of powders and ointments is not necessary when there is no rash. Finding the best combination of cleansing and diapering products for the individual baby will also help to prevent diaper rash.

Resources

OTHER

Greene, Alan. "Diaper Rash." *Dr. Greene's House Calls*. 1996. <<http://drgreene.com/960430.asp>>.

Judith Turner

Diaphragm (birth control)

Definition

Diaphragms are dome-shaped barrier methods of **contraception** that block sperm from entering the uterus. They are made of latex (rubber) and formed like a shallow cup. Since vaginas vary in size, each patient will need to be fitted by a doctor or nurse with a diaphragm that conforms to the shape and contour of the vagina as well as the strength of the muscles in the vaginal walls. Diaphragms must be used with spermicidal cream or jelly. The device should cause no discomfort, and neither the woman nor her partner should feel that it is there.

Purpose

The purpose of a diaphragm is to prevent access to the womb (uterus) by the sperm and thus prevent conception. The level of effectiveness is about 95%.

Precautions

Each client will undergo a **physical examination** and a Pap smear. If these are normal, the physician will fit the patient for the device and give instructions on how to insert, remove, and clean the object. She will also be taught the signs and symptoms of potential complications.

Description

Prior to insertion, the inside of the dome and the rim are covered with a thick layer (perhaps a tablespoon) of a spermicide that is compatible with the diaphragm being used. The domed area covers the opening into the uterus (cervix) and keeps the spermicide in place. As a result, any sperm that might get under the diaphragm will be destroyed.

Diaphragms may be inserted two–three hours prior to intercourse, and must be left in place for six to eight hours following sexual relations. During this time the woman may not swim, bathe, or douche, but she may

KEY TERMS

Spermicide—A substance that kills sperm.

Toxic shock syndrome—An uncommon, but potentially fatal, disease that has been associated with the use of diaphragms and vaginal tampons. The symptoms include high fever, vomiting, and diarrhea.

shower. If she desires to have intercourse again before the six to eight hours have passed, the diaphragm should not be removed. Instead, an applicator full of spermicide should be deposited into the vagina.

A diaphragm will last for a year or more. It should be examined weekly for holes. This can be done by holding it up to the light or filling it with water.

Preparation

Before inserting the diaphragm, the woman should empty her bladder and wash her hands with soap and water. The device should be checked for leaks by filling it with water or holding it up to the light. A spermicidal jelly is then applied to the inside and outside, and especially around the rim. While standing with one foot elevated on a chair or step, lying down, or squatting, the woman folds the diaphragm inward toward the middle and inserts it into the vagina as far as it will go.

Aftercare

When removed, the diaphragm should be washed with a mild soap and water. After being dried, it can be dusted with corn starch before being returned to its container. The diaphragm should always be stored away from sunlight and heat in a cool, dry place. It should not be washed with harsh or perfumed soaps or used with perfumed powders because either of these substances can damage the diaphragm.

Risks

Although rare, wearing the diaphragm longer than the recommended time can result in toxic **shock syndrome**. The signs and symptoms of this serious illness include sudden onset of high **fever**, **vomiting**, **diarrhea**, **dizziness**, faintness, weakness, aching muscles and joints, and rash. The doctor must be notified immediately if any of these conditions appear. An allergic reaction to the spermicide or the material

from which the device is made is also possible. Diaphragm use is also associated with an increased risk of bladder infections.

It should be noted that the diaphragm can become dislodged during intercourse, which could result in an unwanted **pregnancy**. To ensure a secure fit, a woman should be examined for a refitting if she gains or loses more than 10 lbs (4.5 kg), or after she gives birth.

Normal results

Consumers can expect an efficiency rate of about 95% in preventing pregnancy. Using a male **condom** in conjunction with the diaphragm decreases the potential for pregnancy. Diaphragms provide no protection against **AIDS** or other sexually transmitted diseases.

Resources

ORGANIZATIONS

Planned Parenthood Federation of America, Inc. 810 Seventh Ave., New York, NY, 10019. (800) 669-0156. < <http://www.plannedparenthood.org> >.

OTHER

“The Diaphragm.” *Cincinnati Women’s Services*. < <http://gynpages.com/cws/8.html> >.

“Guide to Safer Sex.” *Sexual Health InfoCenter*. < <http://www.sexhealth.org/infocenter/GuideSS/diaphragm.htm> >.

Donald G. Barstow, RN

Diaphragmatic hernia see **Hernia**

Diarrhea

Definition

To most individuals, diarrhea means an increased frequency or decreased consistency of bowel movements; however, the medical definition is more exact than this. In many developed countries, the average number of bowel movements is three per day. However, researchers have found that diarrhea best correlates with an increase in stool weight; stool weights above 10oz (300 gs) per day generally indicates diarrhea. This is mainly due to excess water, which normally makes up 60–85% of fecal matter. In this way, true diarrhea is distinguished from diseases that cause only an increase in the number of bowel movements (hyperdefecation) or incontinence (involuntary loss of bowel contents).

Diarrhea is also classified by physicians into acute, which lasts one or two weeks, and chronic, which continues for longer than 2 or 3 weeks. Viral and bacterial infections are the most common causes of acute diarrhea.

Description

In many cases, acute infectious diarrhea is a mild, limited annoyance. However, worldwide acute infectious diarrhea has a huge impact, causing over five million deaths per year. While most deaths are among children under five years of age in developing nations, the impact, even in developed countries, is considerable. For example, over 250,000 individuals are admitted to hospitals in the United States each year because of one of these episodes. Rapid diagnosis and proper treatment can prevent much of the suffering associated with these devastating illnesses.

Chronic diarrhea also has a considerable effect on health, as well as on social and economic well being. Patients with **celiac disease**, inflammatory bowel disease, and other prolonged diarrheal illnesses develop nutritional deficiencies that diminish growth and immunity. They affect social interaction and result in the loss of many working hours.

Causes and symptoms

Diarrhea occurs because more fluid passes through the large intestine (colon) than that organ can absorb. As a rule, the colon can absorb several times more fluid than is required on a daily basis. However, when this reserve capacity is overwhelmed, diarrhea occurs.

Diarrhea is caused by infections or illnesses that either lead to excess production of fluids or prevent absorption of fluids. Also, certain substances in the colon, such as fats and bile acids, can interfere with water absorption and cause diarrhea. In addition, rapid passage of material through the colon can also do the same.

Symptoms related to any diarrheal illness are often those associated with any injury to the gastrointestinal tract, such as **fever**, **nausea**, **vomiting**, and abdominal **pain**. All or none of these may be present depending on the disease causing the diarrhea. The number of bowel movements can vary—up to 20 or more per day. In some patients, blood or pus is present in the stool. Bowel movements may be difficult to flush (float) or contain undigested food material.

The most common causes of acute diarrhea are infections (the cause of **traveler's diarrhea**), **food**

poisoning, and medications. Medications are a frequent and often over-looked cause, especially **antibiotics** and **antacids**. Less often, various sugar free foods, which sometimes contain poorly absorbable materials, cause diarrhea.

Chronic diarrhea is frequently due to many of the same things that cause the shorter episodes (infections, medications, etc.); symptoms just last longer. Some infections can become chronic. This occurs mainly with parasitic infections (such as *Giardia*) or when patients have altered immunity (**AIDS**).

The following are the more usual causes of chronic diarrhea:

- **AIDS**
- colon **cancer** and other bowel tumors
- endocrine or hormonal abnormalities (thyroid, diabetes mellitus, etc.)
- food allergy
- inflammatory bowel disease (**Crohn's disease** and ulcerative colitis)
- lactose intolerance
- malabsorption syndromes (celiac and Whipple's disease)
- other (alcohol, microscopic colitis, radiation, surgery)

Complications

The major effects of diarrhea are **dehydration**, **malnutrition**, and weight loss. Signs of dehydration can be hard to notice, but increasing thirst, **dry mouth**, weakness or lightheadedness (particularly if worsening on standing), or a darkening/decrease in urination are suggestive. Severe dehydration leads to changes in the body's chemistry and could become life-threatening. Dehydration from diarrhea can result in kidney failure, neurological symptoms, arthritis, and skin problems.

Diagnosis

Most cases of acute diarrhea never need diagnosis or treatment, as many are mild and produce few problems. But patients with fever over 102 °F (38.9 °C), signs of dehydration, bloody bowel movements, severe abdominal pain, known immune disease, or prior use of antibiotics need prompt medical evaluation.

When diagnostic studies are needed, the most useful are stool culture and examination for parasites; however these are often negative and a cause cannot be found in a large number of patients. The earlier

cultures are performed, the greater the chance of obtaining a positive result. For those with a history of antibiotic use in the preceding two months, stool samples need to be examined for the toxins that cause **antibiotic-associated colitis**. Tests are also available to check stool samples for microscopic amounts of blood and for cells that indicate severe inflammation of the colon. Examination with an endoscope is sometimes helpful in determining severity and extent of inflammation. Tests to check changes in blood chemistry (potassium, magnesium, etc.) and a complete **blood count** (CBC) are also often performed.

Chronic diarrhea is quite different, and most patients with this condition will receive some degree of testing. Many exams are the same as for an acute episode, as some infections and parasites cause both types of diarrhea. A careful history to evaluate medication use, dietary changes, family history of illnesses, and other symptoms is necessary. Key points in determining the seriousness of symptoms are weight loss of over 10 lb (4.5 kg), blood in the stool, and nocturnal diarrhea (symptoms that awaken the patient from sleep).

Both prescription and over-the-counter medications can contain additives, such as lactose and sorbitol, that will produce diarrhea in sensitive individuals. Review of **allergies** or skin changes may also point to a cause. Social history may indicate if **stress** is playing a role or identify activities which can be associated with diarrhea (for example, diarrhea that occurs in runners).

A combination of stool, blood, and urine tests may be needed in the evaluation of chronic diarrhea; in addition a number of endoscopic and x-ray studies are frequently required.

Treatment

Treatment is ideally directed toward correcting the cause; however, the first aim should be to prevent or treat dehydration and nutritional deficiencies. The type of fluid and nutrient replacement will depend on whether oral feedings can be taken and the severity of fluid losses. Oral rehydration solution (ORS) or intravenous fluids are the choices; ORS is preferred if possible.

A physician should be notified if the patient is dehydrated, and if oral replacement is suggested then commercial (Pedialyte and others) or homemade preparations can be used. The World Health Organization (WHO) has provided this easy recipe for home preparation, which can be taken in small frequent sips:

- Table salt—3/4 tsp
- Baking powder—1 tsp
- Orange juice—1 c
- Water—1 qt (1l)

When feasible, food intake should be continued even in those with acute diarrhea. A physician should be consulted as to what type and how much food is permitted.

Anti-motility agents (loperamide, diphenoxylate) are useful for those with chronic symptoms; their use is limited or even contraindicated in most individuals with acute diarrhea, especially in those with high fever or bloody bowel movements. They should not be taken without the advice of a physician.

Other treatments are available, depending on the cause of symptoms. For example, the bulk agent psyllium helps some patients by absorbing excess fluid and solidifying stools; cholestyramine, which binds bile acids, is effective in treating bile salt induced diarrhea. Low fat diets or more easily digestible fat is useful in some patients. New antidiarrheal drugs that decrease excessive secretion of fluid by the intestinal tract is another approach for some diseases. Avoidance of medications or other products that are known to cause diarrhea (such as lactose) is curative in some, but should be discussed with a physician.

Alternative treatment

It is especially important to find the cause of diarrhea, since stopping diarrhea when it is the body's way of eliminating something foreign is not helpful and can be harmful in the long run.

One effective alternative approach to preventing and treating diarrhea involves oral supplementation of aspects of the normal flora in the colon with the yeasts *Lactobacillus acidophilus*, *L. bifidus*, or *Saccharomyces boulardii*. In clinical settings, these "biotherapeutic" agents have repeatedly been helpful in the resolution of diarrhea, especially antibiotic-associated diarrhea. Their effectiveness is also supported by the results of a research study published in the *Journal of the American Medical Association* in 1996.

Nutrient replacement also plays a role in preventing and treating episodes of diarrhea. Zinc especially appears to have an effect on the immune system, and deficiency of this mineral can lead to chronic diarrhea. Also, zinc replacement improves growth in young patients. Plenty of fluids, especially water, should be taken by individuals suffering from diarrhea to prevent dehydration. The BRAT diet also can be useful in

KEY TERMS

Anti-motility medications—Medications such as loperamide (Imodium), diphenoxylate (Lomotil), or medications containing codeine or narcotics that decrease the ability of the intestine to contract. These can worsen the condition of a patient with dysentery or colitis.

Colitis—Inflammation of the colon.

Endoscope—An endoscope, as used in the field of gastroenterology, is a thin flexible tube that uses a lens or miniature camera to view various areas of the gastrointestinal tract. Both diagnosis, through biopsies or other means, and therapeutic procedures can be done with this instrument.

Endoscopy—The performance of an exam using an endoscope is known generally as endoscopy.

Lactose intolerance—An inability to properly digest milk and dairy products.

Oral rehydration solution (ORS)—A liquid preparation developed by the World Health Organization that can decrease fluid loss in persons with diarrhea. Originally developed to be prepared with materials available in the home, commercial preparations have recently come into use.

Steatorrhea—Excessive amounts of fat in the feces.

helping to resolve diarrhea. This diet limits food intake to bananas, rice, applesauce, and toast. These foods provide soluble and insoluble fiber without irritation. If the toast is slightly burnt, the charcoal can help sequester toxins and pull them from the body.

Acute homeopathic remedies can be very effective for treating diarrhea especially in infants and young children.

Prognosis

Prognosis is related to the cause of the diarrhea; for most individuals in developed countries, a bout of acute, infectious diarrhea is at best uncomfortable. However, in both industrialized and developing areas, serious complications and **death** can occur.

For those with chronic symptoms, an extensive number of tests are usually necessary to make a proper diagnosis and begin treatment; a specific diagnosis is found in 90% of patients. In some, however, no specific cause is found and only treatment with bulk agents or anti-motility agents is indicated.

Prevention

Proper hygiene and food handling techniques will prevent many cases. Traveler's diarrhea can be avoided by use of Pepto-Bismol and/or antibiotics, if necessary. The most important action is to prevent the complications of dehydration.

Resources

ORGANIZATIONS

World Health Organization, Division of Emerging and Other Communicable Diseases Surveillance and Control.
Avenue Appia 20, 1211 Geneva 27, Switzerland. (+00 41 22) 791 21 11. <<http://www.who.int>>.

OTHER

“Directory of Digestive Diseases Organizations for Patients.” *National Institute of Diabetes and Digestive and Kidney Disease*. <<http://www.niddk.nih.gov>>.
“A Neglected Modality for the Treatment and Prevention of Selected Intestinal and Vaginal Infections.” *JAMA*. <<http://pubs.ama-assn.org>>.
Selected publications and documents on diarrhoeal diseases (including cholera). *World Health Organization (WHO)*. <<http://www.who.ch/chd/pub/cdd/cddpub.htm>>.

David Kaminstein, MD

Diazepam see **Benzodiazepines**

Diclofenac see **Nonsteroidal anti-inflammatory drugs**

Dicyclomine see **Antispasmodic drugs**

Didanosine see **Antiretroviral drugs**

Diets

Definition

Humans may alter their usual eating habits for many reasons, including weight loss, disease prevention or treatment, removing toxins from the body, or to achieve a general improvement in physical and mental health. Others adopt special diets for religious reasons. In the case of some vegetarians and vegans, dietary changes are made out of ethical concerns for the rights of animals.

Purpose

People who are moderately to severely overweight can derive substantial health benefits from a

weight-loss diet. A weight reduction of just 10–20 pounds can result in reduced cholesterol levels and lower blood pressure. Weight-related health problems include heart disease, diabetes, high blood pressure, and high levels of blood sugar and cholesterol.

In individuals who are not overweight, dietary changes also may be useful in the prevention or treatment of a range of ailments including acquired **immuno deficiency syndrome (AIDS)**, **cancer**, **osteoporosis**, inflammatory bowel disease, chronic pulmonary disease, renal disease, Parkinson's disease, seizure disorders, and food allergies and intolerances.

Description

Origins

The practice of altering diet for special reasons has existed since antiquity. For example, Judaism has included numerous dietary restrictions for thousands of years. One ancient Jewish sect, the Essenes, is said to have developed a primitive **detoxification** diet aimed at preparing the bodies, minds, and spirits of its members for the coming of a “messiah” who would deliver them from their Roman captors. Preventive and therapeutic diets became popular during the late twentieth century. Books promoting the latest dietary plan continue to make the bestseller lists, although not all of the information given is considered authoritative.

The idea of a healthful diet is to provide all of the calories and nutrients needed by the body for optimal performance, at the same time ensuring that neither nutritional deficiencies nor excesses occur. Diet plans that claim to accomplish those objectives are so numerous they are virtually uncountable. These diets employ a variety of approaches, including the following:

- **Fixed-menu:** Offers little choice to the dieter. Specifies exactly which foods will be consumed. Easy to follow, but may be considered boring to some dieters.
- **Formula:** Replaces some or all meals with a nutritionally balanced liquid formula or powder.
- **Exchange-type:** Allows the dieter to choose between selected foods from each food group.
- **Flexible:** Doesn't concern itself with the overall diet, simply with one aspect such as fat or energy.

Diets also may be classified according to the types of foods they allow. For example, an omnivorous diet consists of both animal and plant foods, whereas a lacto-ovo-vegetarian diet permits no animal flesh, but includes eggs, milk, and dairy products. A vegan

diet is a stricter form of **vegetarianism** in which eggs, cheese, and other milk products are prohibited.

A third way of classifying diets is according to their purpose: religious, weight-loss, detoxification, lifestyle-related, or aimed at prevention or treatment of a specific disease.

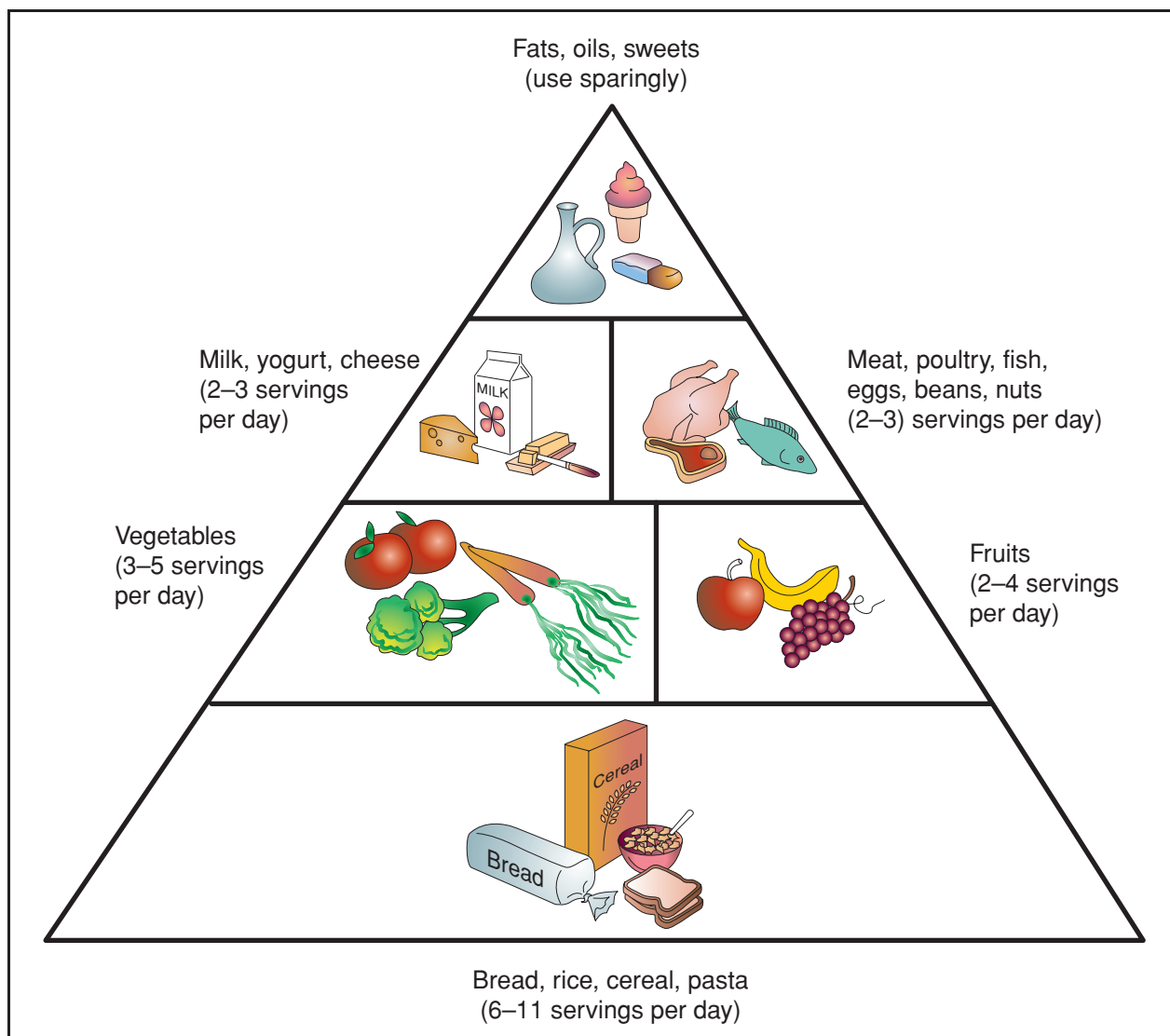
Precautions

Dieters should be cautious about plans that severely restrict the size of food portions, or that eliminate entire food groups from the diet. It is highly probable that they will become discouraged and drop out of such programs. The best diet is one that can be maintained indefinitely without ill effects, that offers sufficient variety and balance to provide everything needed for good health, and that is considerate of personal food preferences. Many controversies have arisen in the past over the benefits and risks of high-protein, low carbohydrate diets such as the **Atkins diet**. Most physician groups and health organizations have spoken out negatively against the program. In 2003, these statements were largely supported. Though clinical trials showed that these types of diets worked in lowering weight without raising cholesterol for the short-term, many of the participants gained a percentage of the weight back after only one year. A physician group also spoke out about high protein diets' dangers for people with decreased kidney function and the risk of bone loss due to decreased calcium intake.

Low-fat diets are not recommended for children under the age of two. Young children need extra fat to maintain their active, growing bodies. Fat intake may be gradually reduced between the ages of two and five, after which it should be limited to a maximum of 30% of total calories through adulthood. Saturated fat should be restricted to no more than 10% of total calories.

Weight-loss dieters should be wary of the “yo-yo” effect that occurs when numerous attempts are made to reduce weight using high-risk, quick-fix diets. This continued “cycling” between weight loss and weight gain can slow the basal metabolic rate and can sometimes lead to eating disorders. The dieter may become discouraged and frustrated by this success/failure cycle. The end result of yo-yo dieting is that it becomes more difficult to maintain a healthy weight.

Caution also should be exercised about weight loss diets that require continued purchases of special prepackaged foods. Not only do these tend to be costly and over-processed, they also may prevent dieters from learning the food-selection and preparation skills



Suggested daily food servings appeared in the food pyramid up until 2005 when it was revised. The current food pyramid does not contain recommended portions. (Illustration by Electronic Illustrators Group.)

essential to maintenance of weight loss. Further, dieters should consider whether they want to carry these special foods to work, restaurants, or homes of friends.

Concern has been expressed about weight-loss diet plans that do not include **exercise**, considered essential to long-term weight management. Some diets and supplements may be inadvisable for patients with special conditions or situations. In fact, use of the weight loss supplement ephedra was found to cause serious conditions such as **heart attack** and **stroke**. In 2003, the U.S. Food and Drug Administration (FDA) was considering controlling or banning the supplement. In short, most physician organizations see fad diets as distracting from learning how to achieve

weight control over the long term through healthy lifestyle changes such as eating smaller, more balanced meals and exercising regularly.

Certain fad diets purporting to be official diets of groups such as the American Heart Association and the Mayo Clinic are in no way endorsed by those institutions. People thinking of starting such a diet should check with the institution to ensure its name has not been misappropriated by an unscrupulous practitioner.

Side effects

A wide range of side effects (some quite serious) can result from special diets, especially those that are

nutritionally unbalanced. Further problems can arise if the dieter is taking high doses of dietary supplements. Food is essential to life, and improper **nutrition** can result in serious illness or **death**.

Research and general acceptance

It is agreed among traditional and complementary practitioners that many patients could substantially benefit from improved eating habits. Specialized diets have proved effective against a wide variety of conditions and diseases. However, dozens of unproved but widely publicized fad diets emerge each year, prompting widespread concerns about their usefulness, cost to the consumer, and their safety.

Resources

PERIODICALS

"American College of Preventive Medicine Weighs in Against Fad Diets." *Obesity and Diabetes Week*, March 17, 2003: 7.

"Atkins Diet Vindicated But Long-term Success Questionable." *Obesity, Fitness and Wellness Week*, June 14, 2003: 25.

"High-protein Diets Risky for Bones and Kidneys." *Health Science*, Spring 2003: 9.

Kirn, Timothy F. "FDA Probes Ephedra, Proposes Warning Label (Risk of Heart Attack, Seizure, Stroke)." *Clinical Psychiatry News*, April 2003: 49.

ORGANIZATIONS

American Dietetic Association. 216 West Jackson Blvd., Chicago, IL 60606-6995. (312) 899-0040. <<http://www.eatright.org>>.

David Helwig
Teresa G. Odle

Diffuse esophageal spasm

Definition

Diffuse esophageal spasm is a term used to define an uncoordinated or spastic esophagus.

Description

The esophagus is a muscular tube that actively transports food from the throat to the stomach by rhythmic contractions known as peristalsis. The actual mechanism and anatomy are quite complex, involving three distinct segments and allowing a person to swallow even when upside-down. Diffuse esophageal

KEY TERMS

Contrast agent—A substance that produces shadows on x rays.

Manometry—Measurement of pressure.

Peristalsis—Slow, rhythmic contractions of the muscles in a tubular organ, such as the intestines, that propel the contents along.

spasm describes a condition where the entire esophagus is spastic—along its entire length, the muscular activity is increased and uncoordinated. The name corkscrew esophagus describes perfectly the appearance of this disorder on x rays.

X rays may reveal a slightly different appearance and result in the designation rosary bead esophagus, but the cause is still diffuse spasm, and the two entities behave in the same way.

Causes and symptoms

The cause appears to be disruption of the complex system of nerves that coordinates the muscular activity. The result is difficulty swallowing (dysphagia) and **pain** that feels like a **heart attack** and can involve the entire chest, jaw, and arms.

Diagnosis

Swallowing problems usually call for esophagograms. In the x-ray department, the patient is given a contrast agent to drink. During swallowing, x rays record the passage of the agent down the esophagus and into the stomach. Instead of a straight tube with well-coordinated waves of contraction, the resulting x rays show a writhing organ resembling a giant corkscrew.

Another test that is used in many disorders of esophageal motility is manometry. Pressures inside the esophagus are measured every inch or so using a balloon device that is passed all the way down to the stomach. The result is a precise record of its activity that yields a specific diagnosis.

Treatment

Soft and liquid foods pass more easily than solid pieces. Medications of several types are helpful—nifedipine, hydralazine, isoproterenol, and nitrates being the most successful. Several other treatments

have uncertain results. For severe cases, relief is obtained two-thirds of the time by cutting the muscles along the entire length of the esophagus. This is a major surgical procedure.

Prognosis

This condition does not go away, nor is treatment entirely satisfactory. Patients need to be careful of what they eat and continue on medication if a beneficial one is found. Fortunately, the condition does not get progressively worse as time passes.

Resources

BOOKS

Goyal, Raj K. "Diseases of the Esophagus." In *Harrison's Principles of Internal Medicine*, edited by Anthony S. Fauci, et al. New York: McGraw-Hill, 1997.

J. Ricker Polsdorfer, MD

DiGeorge syndrome

Definition

DiGeorge syndrome (also called 22q11 deletion syndrome, congenital thymic hypoplasia, or third and fourth pharyngeal pouch syndrome) is a birth defect that is caused by an abnormality in chromosome 22 and affects the baby's immune system. The disorder is marked by absence or underdevelopment of the thymus and parathyroid glands. It is named for Angelo DiGeorge, the pediatrician who first described it in 1965. Some researchers prefer to call it DiGeorge anomaly, or DGA, rather than DiGeorge syndrome, on the grounds that the defects associated with the disorder represent the failure of a part of the human embryo to develop normally rather than a collection of symptoms caused by a single disease.

Description

The prevalence of DiGeorge syndrome is debated; the estimates range from 1:4000 to 1:6395. Because the symptoms caused by the chromosomal abnormality vary somewhat from patient to patient, the syndrome probably occurs much more often than was previously thought. DiGeorge syndrome is sometimes described as one of the "CATCH 22" disorders, so named because of their characteristics—cardiac defects, abnormal facial features, thymus underdevelopment,

cleft palate, and hypocalcemia—caused by a deletion of several genes in chromosome 22. The specific facial features associated with DiGeorge syndrome include low-set ears, wide-set eyes, a small jaw, and a short groove in the upper lip. The male/female ratio is 1:1. The syndrome appears to be equally common in all racial and ethnic groups.

Causes and symptoms

DiGeorge syndrome is caused either by inheritance of a defective chromosome 22 or by a new defect in chromosome 22 in the fetus. The type of defect that is involved is called deletion. A deletion occurs when the genetic material in the chromosomes does not recombine properly during the formation of sperm or egg cells. The deletion means that several genes from chromosome 22 are missing in DiGeorge syndrome patients. Although efforts have been made in the early 2000s to identify individual candidate genes for DGA, it appears that a combination of several genes in the deleted area is responsible for the disorder. Detailed genetic mapping of chromosome 22 has, however, identified a so-called DiGeorge critical region (DGCR), which has been completely sequenced.

According to a 1999 study, 6% of children with DiGeorge syndrome inherited the deletion from a parent, while 94% had a new deletion. Other conditions that are associated with DiGeorge syndrome are diabetes (a condition where the pancreas no longer produces enough insulin) in the mother and fetal alcohol syndrome (a pattern of **birth defects**, and learning and behavioral problems affecting individuals whose mothers consumed alcohol during **pregnancy**). Other chromosomal abnormalities that have been found in patients diagnosed with DGA include deletions on chromosomes 10p13, 17p13, and 18q21.

The loss of the genes in the deleted material means that the baby's third and fourth pharyngeal pouches fail to develop normally during the twelfth week of pregnancy. This developmental failure results in a completely or partially absent thymus gland and parathyroid glands. In addition, 74% of fetuses with DiGeorge syndrome have severe heart defects. The child is born with a defective immune system and an abnormally low level of calcium in the blood. Some children with DGA are also born with malformations of the genitals or urinary tract.

These defects usually become apparent within 48 hours of birth. The infant's heart defects may lead to **heart failure**, or there may be seizures and other evidence of a low level of calcium in the blood (**hypocalcemia**).

DiGeorge syndrome is also associated with an increased risk of **autoimmune disorders**. Cases have been reported of DGA in association with Graves' disease, immune thrombocytopenic purpura, juvenile **rheumatoid arthritis**, and severe eczema.

Diagnosis

Diagnosis of DiGeorge syndrome can be made by ultrasound examination around the eighteenth week of pregnancy, when abnormalities in the development of the heart or the palate can be detected. Another technique that is used to diagnose the syndrome before birth is called fluorescence in situ hybridization, or FISH. This technique uses DNA probes from the DiGeorge region on chromosome 22. FISH can be performed on cell samples obtained by **amniocentesis** as early as the fourteenth week of pregnancy. It confirms about 95% of cases of DiGeorge syndrome.

If the mother has not had prenatal testing, the diagnosis of DiGeorge syndrome is sometimes suggested by the child's facial features at birth. In other cases, the doctor makes the diagnosis during heart surgery when he or she notices the absence or abnormal location of the thymus gland. The diagnosis can be confirmed by blood tests for calcium, phosphorus, and parathyroid hormone levels, and by the sheep cell test for immune function.

Treatment

Hypocalcemia

Hypocalcemia in DiGeorge patients is unusually difficult to treat. Infants are usually given calcium and vitamin D by mouth. Severe cases have been treated by transplantation of fetal thymus tissue or bone marrow.

Heart defects

Infants with life-threatening heart defects are treated surgically.

Defective immune function

Children with DiGeorge syndrome should be kept on low-phosphorus **diets** and kept away from crowds or other sources of infection. They should not be immunized with vaccines made from live viruses or given **corticosteroids**.

Prognosis

The prognosis is variable; many infants with DiGeorge syndrome die from overwhelming infection,

KEY TERMS

Deletion—A genetic abnormality in which a segment of a chromosome is lost. DiGeorge syndrome is caused by a deletion on human chromosome 22.

Fetal alcohol syndrome—A cluster of birth defects that includes abnormal facial features and mental retardation, caused by the mother's consumption of alcoholic beverages during pregnancy.

Fluorescence in situ hybridization (FISH)—A technique for diagnosing DiGeorge syndrome before birth by analyzing cells obtained by amniocentesis with DNA probes. FISH is about 95% accurate.

Hypocalcemia—An abnormally low level of calcium in the blood.

Hypoplasia—A deficiency or underdevelopment of a tissue or body structure.

T cells—A type of white blood cell produced in the thymus gland. T cells are an important part of the immune system. Infants born with an underdeveloped or absent thymus do not have a normal level of T cells in their blood.

seizures, or heart failure within the first year. One study of a series of 558 patients reported 8% mortality within six months of birth, with heart defects accounting for all but one of the deaths. Infections resulting from severe immune deficiency are the second most common cause of **death** in patients with DGA. Advances in heart surgery indicate that the prognosis is most closely linked to the severity of the heart defects and the partial presence of the thymus gland. In most children who survive, the number of T cells, a type of white blood cell, in the blood rises spontaneously as they mature. Survivors are likely to be mentally retarded, however, and to have other developmental difficulties, including seizures or other psychiatric and neurological problems in later life.

Prevention

Genetic counseling is recommended for parents of children with DiGeorge syndrome because the disorder can be detected prior to birth. Although most children with DiGeorge syndrome did not inherit the chromosome deletion from their parents, they have a 50% chance of passing the deletion on to their own children.

Because of the association between DiGeorge syndrome and fetal alcohol syndrome, pregnant women should avoid drinking alcoholic beverages.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD, editors.
“Immunodeficiency Diseases.” Section 12, Chapter 147
In *The Merck Manual of Diagnosis and Therapy*.
Whitehouse Station, NJ: Merck Research
Laboratories, 2004.

McDonald-McGinn, Donna M., et al. *22q11 Deletion Syndrome*. Philadelphia: The Children’s Hospital of Philadelphia, 1999.

PERIODICALS

Guduri, Sridhar, MD, and Iftikhar Hussain, MD.
“DiGeorge Syndrome.” *eMedicine* May 28, 2002.
< <http://www.emedicine.com/med/topic567.htm> >.

Verri, A., P. Maraschio, K. Devriendt, et al. “Chromosome 10p Deletion in a Patient with Hypoparathyroidism, Severe Mental Retardation, Autism and Basal Ganglia Calcifications.” *Annales de génétique* 47 (July-September 2004): 281–287.

Yatsenko, S. A., A. N. Yatsenko, K. Szigeti, et al.
“Interstitial Deletion of 10p and Atrial Septal Defect in DiGeorge 2 Syndrome.” *Clinical Genetics* 66 (August 2004): 128–136.

ORGANIZATIONS

Canadian 22q Group. 320 Cote Street Antoine, West
Montreal, Quebec H3Y 2J4.

Chromosome Deletion Outreach, Inc. P.O. Box 724, Boca
Raton, FL 33429-0724. (888) 236-6680.

International DiGeorge/VCF Support Network, c/o Family
Voices of New York. 46 1/2 Clinton Avenue, Cortland,
NY 13045. (607) 753-1250.

National Organization for Rare Disorders (NORD).
55 Kenosia Avenue, P. O. Box 1968, Danbury, CT
06813-1968. (203) 744-0100. Fax: (203) 798-2291.
< <http://www.rarediseases.org> >.

Rebecca J. Frey, PhD

Digital rectal examination see **Rectal examination**

Digitalis drugs

Definition

Digitalis drugs are medicines made from a type of foxglove plant (*Digitalis purpurea*) that have a stimulating effect on the heart.

Purpose

Digitalis drugs are used to treat heart problems such as congestive heart failure and irregular heartbeat.

These medicines help make the heart stronger and more efficient. This, in turn, improves blood circulation and helps relieve the swelling of the hands and ankles that is common in people with heart problems.

Description

Digitalis drugs, also known as digitalis glycosides, are available only with a physician’s prescription. They are sold in tablet, capsule, liquid, and injectable forms. Commonly used digitalis drugs are digitoxin (Crystodigin) and digoxin (Lanoxin).

Recommended dosage

The recommended dosage is different for each patient. The physician who prescribes the medicine will determine the correct dose. Taking exactly the right amount of medicine and taking it exactly as directed are very important. Never take larger or more frequent doses. During treatment with a digitalis heart medicine, the physician will monitor blood levels of the drug and will decide whether the dose needs to be changed. Patients should never change the dose of this medicine unless told to do so by their physicians.

Precautions

Seeing a physician regularly while taking digitalis drugs is very important. The physician will check to make sure the medicine is working as it should and will make any necessary changes in dosage or in instructions for taking the medicine.

Patients taking digitalis drugs should learn to take their pulse and should check it regularly while under treatment with this medicine. Changes in pulse rate, rhythm, or force could be signs of side effects.

Do not stop taking this medicine suddenly without checking with the physician who prescribed it. This could cause a serious change in heart function.

Digitalis drugs are responsible for many accidental poisonings in children. Keep this medicine out of the reach of children.

Be alert to the signs of overdose. Overdosing is a serious concern with digitalis drugs, because the amount of medicine that most people need to help their heart problems is very close to the amount that can cause problems from overdose. If any of these signs of overdose occur, check with a physician as soon as possible:

- loss of appetite
- nausea

- vomiting
- pain in the lower stomach
- diarrhea
- extreme tiredness or weakness
- extremely slow or irregular heartbeat (or fast heart-beat in children)
- blurred vision or other vision changes
- drowsiness
- confusion or depression
- headache
- fainting

Anyone who is taking digitalis drugs should be sure to tell the health care professional in charge before having any surgical or dental procedures or receiving emergency treatment. Physicians may advise people taking digitalis drugs to wear or carry medical identification indicating that they are taking this medicine.

Patients need to be very careful not to accidentally take this medicine in place of another medicine that looks similar. Patients who are taking other medicines that look like their digitalis medicine should ask their pharmacists for suggestions on how to avoid mix-ups.

Anyone who has had unusual reactions to digitalis drugs in the past should let his or her physician know before taking the drugs again. The physician should also be told about any allergies to foods, dyes, preservatives, or other substances.

Women who are pregnant or breastfeeding or who may become pregnant should check with their physicians before using digitalis drugs.

Older people may be especially sensitive to the effects of digitalis drugs, which may increase the chance of overdose.

Before using digitalis drugs, people with any of the following the medical problems should make sure their physicians are aware of their conditions:

- heart disease
- heart rhythm problems
- severe lung disease
- kidney disease
- liver disease
- thyroid disease



Digitalis purpurea. (Photo Researchers, Inc. Reproduced by permission.)

Side effects

Side effects are rare with this medicine. Check with a physician as soon as possible if a skin rash, **hives**, or any other unusual or troublesome symptoms occur. Watch for signs of overdose.

Interactions

Digitalis drugs may interact with a number of other medicines. When this happens, the effects of one or both of the drugs may change or the risk of side effects may be greater. For example:

- Taking digitalis drugs with other heart medicines, amphetamines, or diet pills could increase the risk of heart rhythm problems.
- Calcium channel blockers, used to treat high blood pressure, may cause higher than usual levels of digitalis drugs in the body that could lead to symptoms of overdose as covered in the above section.

- Diuretics (water pills) or other medicines that lower the amount of potassium in the body may increase the side effects of digitalis drugs.
- Medicines that increase the amount of potassium in the body may raise the risk of serious heart rhythm problems when taken with digitalis drugs.
- Diarrhea medicine or cholesterol-lowering drugs such as cholestyramine (Questran) and colestipol (Colestid) may keep digitalis medicines from being absorbed into the body. To prevent this problem, digitalis drugs should be taken several hours before or after taking these medicines.

The list above does not include every drug that may interact with digitalis drugs. Be sure to check with a physician or pharmacist before taking any other prescription or nonprescription (over-the-counter) medicine.

In addition, a diet high in fiber may interfere with the effects of digitalis drugs by preventing the medicine from being absorbed into the body. To avoid this problem, eat high fiber foods (such as bran products, whole wheat bread, and fresh fruits and vegetables) several hours before or after taking digitalis medicine.

Nancy Ross-Flanigan

Digoxin see **Digitalis drugs; Antiarrhythmic drugs**

Dilatation and curettage

Definition

Dilatation and curettage (D & C) is a gynecological procedure in which the lining of the uterus (endometrium) is scraped away.

Purpose

D & C is commonly used to obtain tissue for microscopic evaluation to rule out cancer. D & C may also be used to diagnose and treat heavy menstrual bleeding, and to diagnose endometrial polyps and uterine fibroids. A D & C can be used as a treatment as well, to remove **pregnancy** tissue after a miscarriage, incomplete abortion, or **childbirth**. Endometrial polyps may be removed, and sometimes benign uterine tumors (fibroids) may be scraped away. D & C can also be used as an early abortion technique up to 16 weeks.

Description

D & C is usually performed under **general anesthesia**, although local or epidural anesthesia can also be used. A local lessens risk and costs, but the woman will feel cramping during the procedure. The type of anesthesia used often depends upon the reason for the D & C.

In the procedure (which takes only minutes to perform), the doctor inserts an instrument to hold open the vaginal walls, and then stretches the opening of the uterus to the vagina (the cervix) by inserting a series of tapering rods, each thicker than the previous one, or by using other specialized instruments. This process of opening the cervix is called dilation.

Once the cervix is dilated, the physician inserts a spoon-shaped surgical device called a curette into the uterus. The curette is used to scrape away the uterine lining. One or more small tissue samples from the lining of the uterus or the cervical canal are sent for analysis by microscope to check for abnormal cells.

Although simpler, less expensive techniques such as a vacuum aspiration are quickly replacing the D & C as a diagnostic method, it is still often used to diagnose and treat a number of conditions.

Preparation

Because opening the cervix can be painful, sedatives may be given before the procedure begins. Deep breathing and other relaxation techniques may help ease cramping during cervical dilation.

Aftercare

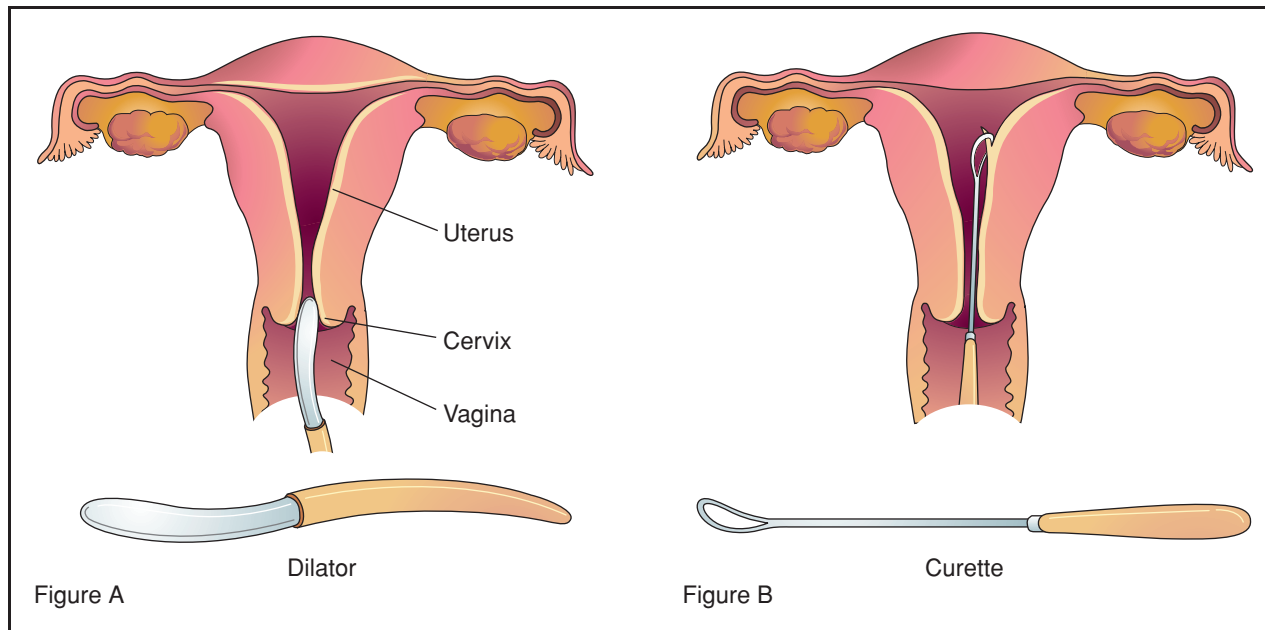
A woman who has had a D & C performed in a hospital can usually go home the same day or the next day. Many women experience backache and mild cramps after the procedure, and may pass small **blood clots** for a day or so. Vaginal staining or bleeding may continue for several weeks.

Most women can resume normal activities almost immediately. Patients should avoid sexual intercourse, douching, and tampon use for at least two weeks to prevent infection while the cervix is closing and to allow the endometrium to heal completely.

Risks

The primary risk after the procedure is infection. Signs of infection include:

- fever
- heavy bleeding



Dilatation and curettage (D & C) is used primarily to diagnose and treat heavy menstrual bleeding and to diagnose endometrial polyps, uterine fibroids, uterine cancer and cervical cancer. When performing a D & C, the physician inserts a speculum to separate and hold the vaginal walls, then stretches open the cervix with a dilator. Once the cervix is dilated, the physician will insert a curette into the uterus and scrape away small portions of the uterine lining for laboratory analysis. (Illustration by Electronic Illustrators Group.)

KEY TERMS

Endometrial polyps—A growth in the lining of the uterus (endometrium) that may cause bleeding and can develop into cancer.

Epidural anesthesia—A type of anesthesia that is injected into the epidural space of the spinal cord to numb the nerves leading to the lower half of the body.

Uterine fibroid—A noncancerous tumor of the uterus that can range from the size of a pea to the size of a grapefruit. Small fibroids require no treatment, but those causing serious symptoms may need to be removed.

complications include puncture of the uterus (which usually heals on its own) or puncture of the bowel or bladder (which require further surgery to repair).

Normal results

Removal of the uterine lining causes no side effects, and may be beneficial if the lining has thickened so much that it causes heavy periods. The uterine lining soon grows again normally, as part of the menstrual cycle.

Resources

BOOKS

Carlson, Karen J., Stephanie A. Eisenstat, and Terra Ziporyn. *The Harvard Guide to Women's Health*. Cambridge, MA: Harvard University Press, 1996.

Carol A. Turkington

- severe cramps
- foul-smelling vaginal discharge

A woman should report any of these symptoms to her doctor, who can treat the infection with **antibiotics** before it becomes serious.

D & C is a surgical operation, which carries certain risks associated with general anesthesia. Rare

Dilated cardiomyopathy see **Congestive cardiomyopathy**

Diltiazem see **Calcium channel blockers**

Dilution test see **Kidney function tests**

Diphenhydramine see **Antihistamines**

Diphtheria

Definition

Diphtheria is a potentially fatal, contagious disease that usually involves the nose, throat, and air passages, but may also infect the skin. Its most striking feature is the formation of a grayish membrane covering the tonsils and upper part of the throat.

Description

Like many other upper respiratory diseases, diphtheria is most likely to break out during the winter months. At one time it was a major childhood killer, but it is now rare in developed countries because of widespread immunization. Since 1988, all confirmed cases in the United States have involved visitors or immigrants. In countries that do not have routine immunization against this infection, the mortality rate varies from 1.5–25%.

Persons who have not been immunized may get diphtheria at any age. The disease is spread most often by droplets from the coughing or sneezing of an infected person or carrier. The incubation period is two to seven days, with an average of three days. It is vital to seek medical help at once when diphtheria is suspected, because treatment requires emergency measures for adults as well as children.

Causes and symptoms

The symptoms of diphtheria are caused by toxins produced by the diphtheria bacillus, *Corynebacterium diphtheriae* (from the Greek for “rubber membrane”). In fact, toxin production is related to infections of the bacillus itself with a particular bacteria virus called a phage (from bacteriophage; a virus that infects bacteria). The intoxication destroys healthy tissue in the upper area of the throat around the tonsils, or in open **wounds** in the skin. Fluid from the dying cells then coagulates to form the telltale gray or grayish green membrane. Inside the membrane, the bacteria produce an exotoxin, which is a poisonous secretion that causes the life-threatening symptoms of diphtheria. The exotoxin is carried throughout the body in the bloodstream, destroying healthy tissue in other parts of the body.

The most serious complications caused by the exotoxin are inflammations of the heart muscle (**myocarditis**) and damage to the nervous system. The risk of serious complications is increased as the time between onset of symptoms and the administration

of antitoxin increases, and as the size of the membrane formed increases. The myocarditis may cause disturbances in the heart rhythm and may culminate in **heart failure**. The symptoms of nervous system involvement can include seeing double (diplopia), painful or difficult swallowing, and slurred speech or loss of voice, which are all indications of the exotoxin’s effect on nerve functions. The exotoxin may also cause severe swelling in the neck (“bull neck”).

The signs and symptoms of diphtheria vary according to the location of the infection:

Nasal

Nasal diphtheria produces few symptoms other than a watery or bloody discharge. On examination, there may be a small visible membrane in the nasal passages. Nasal infection rarely causes complications by itself, but it is a public health problem because it spreads the disease more rapidly than other forms of diphtheria.

Pharyngeal

Pharyngeal diphtheria gets its name from the pharynx, which is the part of the upper throat that connects the mouth and nasal passages with the voice box. This is the most common form of diphtheria, causing the characteristic throat membrane. The membrane often bleeds if it is scraped or cut. It is important not to try to remove the membrane because the trauma may increase the body’s absorption of the exotoxin. Other signs and symptoms of pharyngeal diphtheria include mild **sore throat**, **fever** of 101–102 °F (38.3–38.9 °C), a rapid pulse, and general body weakness.

Laryngeal

Laryngeal diphtheria, which involves the voice box or larynx, is the form most likely to produce serious complications. The fever is usually higher in this form of diphtheria (103–104 °F or 39.4–40 °C) and the patient is very weak. Patients may have a severe **cough**, have difficulty breathing, or lose their voice completely. The development of a “bull neck” indicates a high level of exotoxin in the bloodstream. Obstruction of the airway may result in respiratory compromise and **death**.

Skin

This form of diphtheria, which is sometimes called cutaneous diphtheria, accounts for about 33% of diphtheria cases. It is found chiefly among people

with poor hygiene. Any break in the skin can become infected with diphtheria. The infected tissue develops an ulcerated area and a diphtheria membrane may form over the wound but is not always present. The wound or ulcer is slow to heal and may be numb or insensitive when touched.

Diagnosis

Because diphtheria must be treated as quickly as possible, doctors usually make the diagnosis on the basis of the visible symptoms without waiting for test results.

In making the diagnosis, the doctor examines the patient's eyes, ears, nose, and throat in order to rule out other diseases that may cause fever and sore throat, such as **infectious mononucleosis**, a sinus infection, or **strep throat**. The most important single symptom that suggests diphtheria is the membrane. When a patient develops skin infections during an outbreak of diphtheria, the doctor will consider the possibility of cutaneous diphtheria and take a smear to confirm the diagnosis.

Laboratory tests

The diagnosis of diphtheria can be confirmed by the results of a culture obtained from the infected area. Material from the swab is put on a microscope slide and stained using a procedure called Gram's stain. The diphtheria bacillus is called Gram-positive because it holds the dye after the slide is rinsed with alcohol. Under the microscope, diphtheria bacilli look like beaded rod-shaped cells, grouped in patterns that resemble Chinese characters. Another laboratory test involves growing the diphtheria bacillus on a special material called Loeffler's medium.

Treatment

Diphtheria is a serious disease requiring hospital treatment in an intensive care unit if the patient has developed respiratory symptoms. Treatment includes a combination of medications and supportive care:

Antitoxin

The most important step is prompt administration of diphtheria antitoxin, without waiting for laboratory results. The antitoxin is made from horse serum and works by neutralizing any circulating exotoxin. The doctor must first test the patient for sensitivity to animal serum. Patients who are sensitive (about 10%) must be desensitized with diluted antitoxin, since the antitoxin is the only specific substance

that will counteract diphtheria exotoxin. No human antitoxin is available for the treatment of diphtheria.

The dose ranges from 20,000–100,000 units, depending on the severity and length of time of symptoms occurring before treatment. Diphtheria antitoxin is usually given intravenously.

Antibiotics

Antibiotics are given to wipe out the bacteria, to prevent the spread of the disease, and to protect the patient from developing **pneumonia**. They are not a substitute for treatment with antitoxin. Both adults and children may be given penicillin, ampicillin, or erythromycin. Erythromycin appears to be more effective than penicillin in treating people who are carriers because of better penetration into the infected area.

Cutaneous diphtheria is usually treated by cleansing the wound thoroughly with soap and water, and giving the patient antibiotics for 10 days.

Supportive care

Diphtheria patients need bed rest with intensive nursing care, including extra fluids, oxygenation, and monitoring for possible heart problems, airway blockage, or involvement of the nervous system. Patients with laryngeal diphtheria are kept in a **croup** tent or high-humidity environment; they may also need throat suctioning or emergency surgery if their airway is blocked.

Patients recovering from diphtheria should rest at home for a minimum of two to three weeks, especially if they have heart complications. In addition, patients should be immunized against diphtheria after recovery, because having the disease does not always induce antitoxin formation and protect them from reinfection.

Prevention of complications

Diphtheria patients who develop myocarditis may be treated with oxygen and with medications to prevent irregular heart rhythms. An artificial pacemaker may be needed. Patients with difficulty swallowing can be fed through a tube inserted into the stomach through the nose. Patients who cannot breathe are usually put on mechanical respirators.

Prognosis

The prognosis depends on the size and location of the membrane and on early treatment with antitoxin; the longer the delay, the higher the death rate. The most vulnerable patients are children under age 15 and

KEY TERMS

Antitoxin—An antibody against an exotoxin, usually derived from horse serum.

Bacillus—A rod-shaped bacterium, such as the diphtheria bacterium.

Carrier—A person who may harbor an organism without symptoms and may transmit it to others.

Cutaneous—Located in the skin.

Diphtheria-tetanus-pertussis (DTP)—The standard preparation used to immunize children against diphtheria, tetanus, and whooping cough. A so-called “acellular pertussis” vaccine (aP) is usually used since its release in the mid-1990s.

Exotoxin—A poisonous secretion produced by bacilli which is carried in the bloodstream to other parts of the body.

Gram’s stain—A dye staining technique used in laboratory tests to determine the presence and type of bacteria.

Loeffler’s medium—A special substance used to grow diphtheria bacilli to confirm the diagnosis.

Myocarditis—Inflammation of the heart tissue.

Toxoid—A preparation made from inactivated exotoxin, used in immunization.

those who develop pneumonia or myocarditis. Nasal and cutaneous diphtheria are rarely fatal.

Prevention

Prevention of diphtheria has four aspects:

Immunization

Universal immunization is the most effective means of preventing diphtheria. The standard course of immunization for healthy children is three doses of DPT (diphtheria-tetanus-pertussis) preparation given between two months and six months of age, with booster doses given at 18 months and at entry into school. Adults should be immunized at 10 year intervals with Td (tetanus-diphtheria) toxoid. A toxoid is a bacterial toxin that is treated to make it harmless but still can induce immunity to the disease.

Isolation of patients

Diphtheria patients must be isolated for one to seven days or until two successive cultures show that

they are no longer contagious. Children placed in **isolation** are usually assigned a primary nurse for emotional support.

Identification and treatment of contacts

Because diphtheria is highly contagious and has a short incubation period, family members and other contacts of diphtheria patients must be watched for symptoms and tested to see if they are carriers. They are usually given antibiotics for seven days and a booster shot of diphtheria/tetanus toxoid.

Reporting cases to public health authorities

Reporting is necessary to track potential epidemics, to help doctors identify the specific strain of diphtheria, and to see if resistance to penicillin or erythromycin has developed.

Resources

BOOKS

Chambers, Henry F. “Infectious Diseases: Bacterial & Chlamydial.” In *Current Medical Diagnosis and Treatment*, 1998, edited by Stephen McPhee, et al., 37th ed. Stamford: Appleton & Lange, 1997.

Rebecca J. Frey, PhD

Diplegia see **Paralysis**

Direct Coombs’ test see **Coombs’ tests**

Direct laryngoscopy see **Laryngoscopy**

Discoid lupus erythematosus

Definition

Discoid lupus erythematosus (DLE) is a disease in which coin-shaped (discoid) red bumps appear on the skin.

Description

The disease called discoid lupus erythematosus only affects the skin, although similar discoid **skin lesions** can occur in the serious disease called **systemic lupus erythematosus** (SLE). Only about 10% of all patients with DLE will go on to develop the multi-organ disease SLE.

The tendency to develop DLE seems to run in families. Although men or women of any age can

develop DLE, it occurs in women three times more frequently than in men. The typical DLE patient is a woman in her 30s.

Causes and symptoms

The cause of DLE is unknown. It is thought that DLE (like SLE) may be an autoimmune disorder. **Autoimmune disorders** are those that occur when cells of the immune system are misdirected against the body. Normally, immune cells work to recognize and help destroy foreign invaders like bacteria, viruses, and fungi. In autoimmune disorders, these cells mistakenly recognize various tissues of the body as foreign invaders, and attack and destroy these tissues. In SLE, the misdirected immune cells are antibodies. In DLE, the damaging cells are believed to be a type of white blood cell called a T lymphocyte. The injury to the skin results in inflammation and the characteristic discoid lesions.

In DLE, the characteristic skin lesion is circular and raised. The reddish rash is about 5–10 mm in diameter, with the center often somewhat scaly and lighter in color than the darker outer ring. The surface of these lesions is sometimes described as “warty.” There is rarely any **itching** or **pain** associated with discoid lesions. They tend to appear on the face, ears, neck, scalp, chest, back, and arms. As DLE lesions heal, they leave thickened, scarred areas of skin. When the scalp is severely affected, there may be associated hair loss (**alopecia**).

People with DLE tend to be quite sensitive to the sun. They are more likely to get a **sunburn**, and the sun is likely to worsen their discoid lesions.

Diagnosis

Diagnosis of DLE usually requires a **skin biopsy**. A small sample of a discoid lesion is removed, specially prepared, and examined under a microscope. Usually, the lesion has certain microscopic characteristics that allow it to be identified as a DLE lesion. Blood tests will not reveal the type of antibodies present in SLE, and **physical examination** usually does not reveal anything other than the skin lesions. If antibodies exist in the blood, or if other symptoms or physical signs are found, it is possible that the discoid lesions are a sign of SLE rather than DLE.

Treatment

Treatment of DLE primarily involves the use of a variety of skin creams. **Sunscreens** are used for



Discoloration of the hands is one characteristic of discoid lupus erythematosus. (Custom Medical Stock Photo. Reproduced by permission.)

KEY TERMS

Antibody—Specialized cells of the immune system that can recognize organisms invading the body (like bacteria, viruses, and fungi). The antibodies are then able to start a complex chain of events designed to kill these foreign invaders.

Autoimmune disorder—A disorder in which the body’s antibodies mistake the body’s own tissues for foreign invaders. The immune system then attacks and causes damage to these tissues.

Immune system—The system of specialized organs, lymph nodes, and blood cells throughout the body that work together to defend the body against foreign invaders (bacteria, viruses, fungi, etc.).

protection. Steroid creams can be applied to decrease inflammation. Occasionally, small amounts of a steroid preparation will be injected with a needle into a specific lesion. Because of their long list of side effects, steroid preparations taken by mouth are avoided. Sometimes, short-term treatment with oral steroids will be used for particularly severe DLE outbreaks. Medications used to treat the infectious disease **malaria** are often used to treat DLE.

Alternative treatment

Alternative treatments for DLE include eating a healthy diet, low in red meat and dairy products and high in fish containing **omega-3 fatty acids**. These types of fish include mackerel, sardines, and salmon. Following a healthy diet is thought to decrease

inflammation. Dietary supplements believed to be helpful include **vitamins** B, C, E, and selenium. Vitamin A is also recommended to improve DLE lesions. Constitutional homeopathic treatment can help heal DLE as well as help prevent it developing into SLE.

Prognosis

For the most part, the prognosis for people with DLE is excellent. While the lesions may be cosmetically unsightly, they are not life threatening and usually do not cause a patient to change his or her lifestyle. Only about 10% of patients with DLE will go on to develop SLE.

Prevention

DLE cannot be prevented. Recommendations to prevent flares of DLE in patients with the disease include avoiding exposure to sun and consistently using sunscreen.

Resources

ORGANIZATIONS

The American College of Rheumatology. 1800 Century Place, Suite 250, Atlanta, GA 30345. (404) 633-3777. < <http://www.rheumatology.org> >.

Lupus Foundation of America. 1300 Piccard Dr., Suite 200, Rockville, MD 20850. (800) 558-0121. < <http://www.lupus.org> >.

Rosalyn Carson-DeWitt, MD

Disk removal

Definition

One of the most common types of back surgery is disk removal (discectomy), the removal of an intervertebral disk, the flexible plate that connects any two adjacent vertebrae in the spine. Intervertebral disks act as shock absorbers, protecting the brain and spinal cord from the impact produced by the body's movements.

Purpose

About 150,000 Americans undergo disk removal each year in the United States. Removing the intervertebral disk is performed to treat back **pain** that has lasted at least six weeks as a result of an abnormal disk

and that has not responded to conservative treatment. Surgery is also performed if there is pressure on the lumbosacral nerve roots that causes weakness or bowel or bladder disfunction.

As a person ages, the disks between the vertebrae degenerate and dry out, and the fibers holding them in place tear. Eventually, the disk can form a blister-like bulge, compressing nerves in the spine and causing pain. This is called a “prolapsed” (or herniated) disk. If such a disk causes muscle weakness or interferes with bladder or bowel function because it is pressing on a nerve root, immediate surgery to remove the disk may be needed.

The aim of the surgery is to try to relieve all pressure on nerve roots by removing the pulpy material from the disk, or the disk itself. If it is necessary to remove material from several nearby vertebrae, the spine may become unsteady. In this case, the surgeon will perform a spinal fusion, removing all the disks between two or more vertebrae and roughening the bones so that the vertebrae heal together. Bone strips taken from the patient's leg or hip may be used to help hold the vertebrae together. Spinal fusion decreases pain but it also decreases spinal mobility.

Precautions

The doctor will obtain x rays, neuroimaging studies, including **computed tomography scan** (CT scan) myelogram and **magnetic resonance imaging** (MRI), and clinical exams to determine the precise location of the affected disk.

Description

The surgery is done under general anaesthesia, which puts the patient to sleep and affects the whole body. Operating on the patient's back, the neurosurgeon or orthopedic surgeon makes an opening into the vertebral canal, and then moves the dura and the bundle of nerves called the “cauda equina” (horse's tail) aside, which exposes the disk. If a portion of the disk has moved from between the vertebrae out into the nerve canal, it is simply removed. If the disk itself has become fragmented and partially displaced, or not fragmented but bulging extensively, the surgeon will remove the bulging or displaced part of the disk and the part that lies in the space between the vertebrae.

Preparation

The patient is given an injection an hour before the surgery to dry up internal fluids and encourage drowsiness.

KEY TERMS

Discectomy—The surgical removal of a portion of an intervertebral disk.

Dura—The strongest and outermost of three membranes that protect the brain, spinal cord, and nerves of the cauda equina.

Herniated disk—A blisterlike bulging or protrusion of the contents of the disk out through the fibers that normally hold them in place. It is also called a ruptured disk, slipped disk, or displaced disk.

Intervertebral disk—Cylindrical elastic-like gel pads that separate and join each pair of vertebrae in the spine.

Laminectomy—An operation in which the surgeon cuts through the covering of a vertebra to reach a herniated disk in order to remove it.

Vertebra—The bones that make up the back bone (spine).

Aftercare

After the operation, the patient will awaken lying flat and face down, and must remain this way for several days, changing position only to avoid bedsores. There may be slight pain or stiffness in the back area.

Patients should sleep on a firm mattress and avoid bending at the waist, lifting heavy weights, or sitting in one spot for a long time (such as riding in a car).

After surgery, patients can usually leave the hospital on the fourth or fifth day. They must:

- avoid sitting for more than 15–20 minutes
- use a reclined chair
- avoid bending, twisting, or lifting
- begin gentle walking (indoors or outdoors), gradually increasing
- begin stationary biking or gentle swimming after two weeks
- continue **exercise** for the next four weeks
- slow down if they experience more than minor pain in the back or leg

Risks

All surgery carries some risk due to heart and lung problems or the anesthesia itself, but this risk is

generally extremely small. (The risk of **death** from general anesthesia for all types of surgery, for example, is only about 1 in 1,600.)

The most common risk of the surgery is infection, which occurs in 1–2% of cases. Rarely, the surgery can damage nerves in the lower back or major blood vessels in front of the disk. Occasionally, there may be some residual **paralysis** of a particular leg or bladder muscle after surgery, but this is the result of the disk problem that necessitated the surgery, not the operation itself.

While disk removals can relieve pain in 90% of cases, there are some people who do not get pain relief, depending on how long they had the condition requiring surgery and other factors.

Normal results

After about five days, most patients can leave the hospital. They can resume all normal activities, including work, after four to six weeks of recuperation at home.

In properly evaluated patients, there is a very good chance that disk removal will be successful in easing pain. Even in patients over age 60, disk surgery has a “good to excellent” result for 87% of patients. Disk surgery can relieve both back and leg pain, but the greatest pain relief will occur with the leg pain.

Resources

BOOKS

Younson, Robert M., et al., editors. *The Surgery Book: An Illustrated Guide to 73 of the Most Common Operations*. New York: St. Martin's Press, 1993.

Carol A. Turkington

Discectomy see **Disk removal**

Dislocations and subluxations

Definition

In medicine, the terms dislocation and subluxation refer to the displacement of bones that form a joint. These conditions affecting the joint most often result from trauma that causes adjoining bones to no longer align with each other. A partial or incomplete dislocation is called a subluxation.

Description

In a healthy joint, the bones are normally held together with tough, fibrous bands called ligaments. These ligaments are attached to each bone along with a fibrous sac surrounding the joint called the articular capsule or joint capsule. The ligaments and joint capsule are relatively strong and nonelastic but permit movement within normal limits for each particular joint. In the event of a dislocation, one of the bones making up the joint is forced out of its natural alignment from excessive stretching and tearing of the joint ligaments and capsule. Muscles and tendons surrounding the joint are usually stretched and injured to some degree.

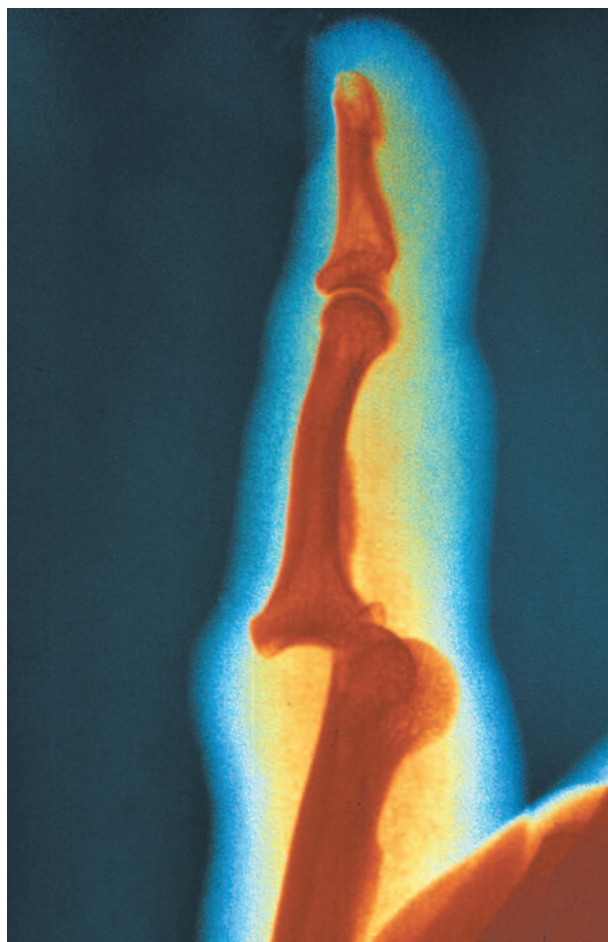
Causes and symptoms

A violent movement at the joint that exceeds normal limits usually causes a joint dislocation. Although dislocations often result from trauma, they sometimes occur as a result of disease affecting the joint structures. In the process of the dislocation, there is tearing of the ligaments and the articular capsule, which are vital structures for connecting the bone. Following a dislocation, the bones affected are often immobile and the affected limb may be locked in an abnormal position; **fractures** are also a concern with severe dislocations.

Important factors in recognizing a dislocation or subluxation include a history of experiencing a fall or receiving a blow in a particular joint followed by the sudden onset of loss of function to the involved limb. Immediately after the dislocation, the joint almost always swells significantly and feels painful when pressure is applied (point tenderness). If trauma to the joint causing the dislocation or subluxation is violent in nature, small chips of bone can be torn away with the supporting structures. Chronic recurrent dislocations may take place without severe pain because of the somewhat slack condition of the surrounding muscles and other supporting tissues. A first-time dislocation is considered and treated as a possible fracture. Risk factors that can increase susceptibility of joint dislocation and subluxation are shallow or abnormally formed joint surfaces present at birth (congenital) and/or other diseases of ligaments and tissue around a joint. Some infants are born with a hip dislocation. Both sexes and all ages are affected.

Diagnosis

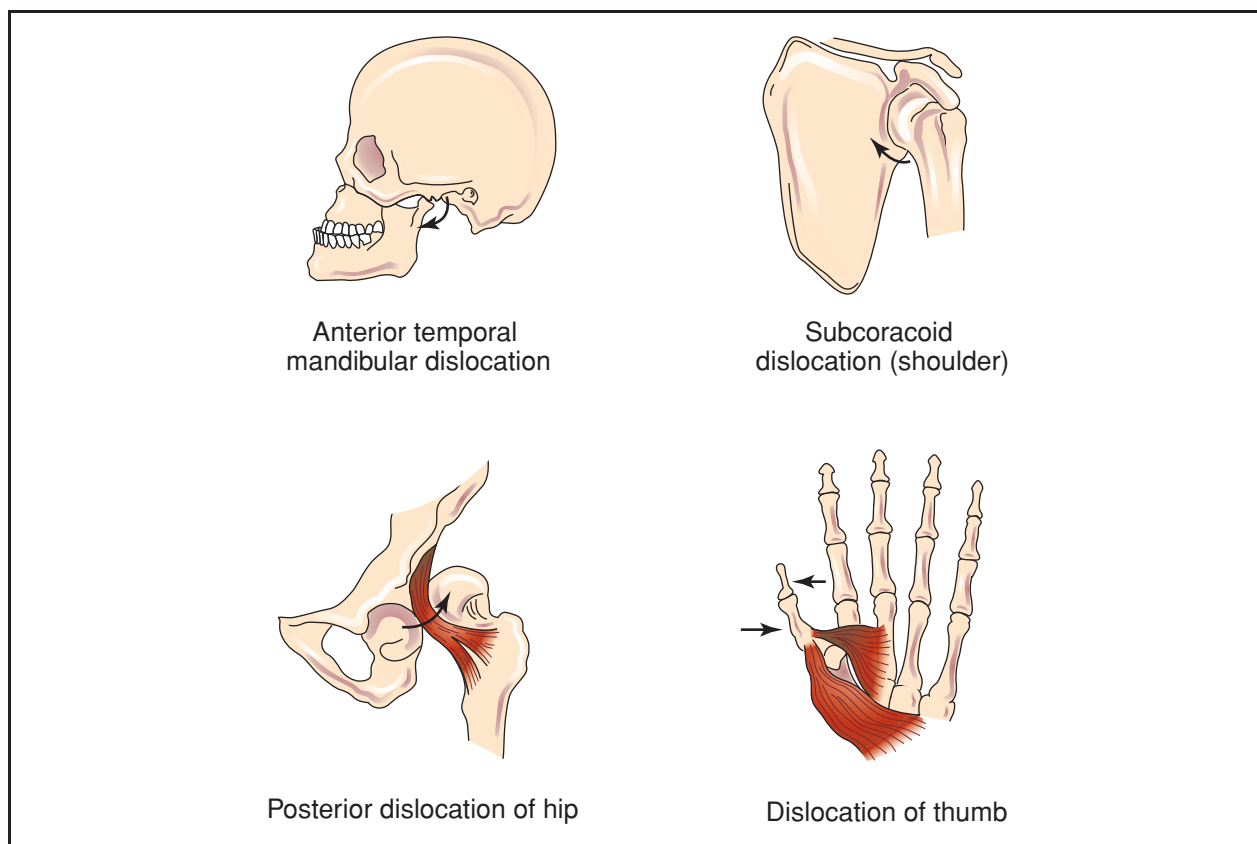
A thorough medical history and physical exam by a physician is the first step in the correct diagnosis of dislocations and subluxations. X rays of the joint and adjacent bones can locate and help determine the extent of dislocated joints.



This x ray shows the dislocation between two bones in a finger. (Photo Researchers, Inc. Reproduced by permission.)

Treatment

Immediately after the dislocation, the application of ice is helpful to control swelling and decrease **pain**. If the patient needs to be transported, it is important to prevent the joint from moving (**immobilization**). At times, a cast or splint may be used to immobilize the joint and ensure proper alignment and healing. The treatment of realigning bones following a dislocation is called reduction. This may include simple maneuvers that manipulate the joint to reposition the bones or surgical procedures to restore the joint to its normal position. A general anesthesia or muscle relaxant may be used to help make joint reduction possible by relaxing surrounding muscles in spasm. **Acetaminophen** or aspirin are sometimes used to control moderate pain, and narcotics may be prescribed by the physician if the pain is severe. Recurring dislocation may require surgical reconstruction or replacement of the joint. It is not recommended to attempt to reset a dislocated joint



Dislocations and subluxations refer to the displacement of bones that form a joint. Such conditions most often result from trauma causing adjoining bones to no longer touch each other. A partial or incomplete dislocation is called a subluxation. The illustrations above indicate dislocation of the jaw bone, shoulder blade, hip bone, and the thumb. (Illustration by Electronic Illustrators Group.)

outside of a medical environment with experienced medical personnel, because a fracture may be present.

Alternative treatment

Chiropractic care has been shown to be effective for joint subluxation and dislocation, especially in the spine. Swelling can be addressed using botanical therapies. Bromelain, a pineapple enzyme, and turmeric (*Curcuma longa*) are the most potent botanical remedies for this purpose. Acute homeopathic care with *arnica* (*Arnica montana*) can reduce the trauma to the body. Ligament and tendon strengthening can be assisted both botanically and homeopathically.

Prognosis

Joint ligaments have poor blood supply and, therefore, heal slowly. This healing process continues long after the symptoms of the dislocation injury have diminished. Once a joint has been either subluxated or

completely dislocated, the connective tissue binding or holding it in correct alignment is stretched to such an extent that the joint becomes extremely vulnerable to repeated dislocations. However, this chance of recurrent dislocation and subluxation will decrease if a proper **rehabilitation** program is implemented to strengthen surrounding muscles of the joint. Most joint dislocations are curable with prompt treatment. After the dislocation has been corrected, the joint may require immobilization with a cast or sling for two to eight weeks.

Prevention

When an individual is involved in strenuous sports or heavy work, involved joints may be protected by elastic bandage wraps, tape wraps, knee and shoulder pads, or special support stockings. Keeping the muscles surrounding the joint strong will also help prevent dislocations. Long-term problems may also be prevented by allowing an adequate

KEY TERMS

Articular capsule—An envelope of tissue that surrounds a free moving joint, composed of an external layer of white fibrous tissue and an external synovial membrane that secretes a lubricant into the joint.

amount of time for an injured joint to rest and heal prior to resuming full activity.

Resources

OTHER

Griffith, H. Winter. "Dislocations or Subluxation."
Thrive Online. 1998. [cited March 25, 1998]. < <http://thriveonline.oxygen.com> >.

Jeffrey P. Larson, RPT

Disopyramide see **Antiarrhythmic drugs**

Disproportionate dwarfism see
Achondroplasia

Dissecting aneurysm see **Aortic dissection**

Dissecting hematoma see **Aortic dissection**

Disseminated lupus erythematosus see
Systemic lupus erythematosus

Dissociative disorders

Definition

The dissociative disorders are a group of mental disorders that affect consciousness defined as causing significant interference with the patient's general functioning, including social relationships and employment.

Description

In order to have a clear picture of these disorders, dissociation should first be understood. Dissociation is a mechanism that allows the mind to separate or compartmentalize certain memories or thoughts from normal consciousness. These split-off mental contents are not erased. They may resurface spontaneously or be triggered by objects or events in the person's environment.

Dissociation is a process that occurs along a spectrum of severity. It does not necessarily mean that a person has a dissociative disorder or other mental illness. A mild degree of dissociation occurs with some physical stressors; people who have gone without sleep for a long period of time, have had "laughing gas" for dental surgery, or have been in a minor accident often have brief dissociative experiences. Another commonplace example of dissociation is a person becoming involved in a book or movie so completely that the surroundings or the passage of time are not noticed. Another example might be driving on the highway and taking several exits without noticing or remembering. Dissociation is related to hypnosis in that hypnotic trance also involves a temporarily altered state of consciousness. Most patients with dissociative disorders are highly hypnotizable.

People in other cultures sometimes have dissociative experiences in the course of religious (in certain trance states) or other group activities. These occurrences should not be judged in terms of what is considered "normal" in the United States.

Moderate or severe forms of dissociation are caused by such traumatic experiences as childhood **abuse**, combat, criminal attacks, brainwashing in hostage situations, or involvement in a natural or transportation disaster. Patients with **acute stress disorder**, post-traumatic stress disorder (PTSD), or conversion disorder and somatization disorder may develop dissociative symptoms. Recent studies of trauma indicate that the human brain stores traumatic memories in a different way than normal memories. Traumatic memories are not processed or integrated into a person's ongoing life in the same fashion as normal memories. Instead they are dissociated, or "split off," and may erupt into consciousness from time to time without warning. The affected person cannot control or "edit" these memories. Over a period of time, these two sets of memories, the normal and the traumatic, may coexist as parallel sets without being combined or blended. In extreme cases, different sets of dissociated memories may alter subpersonalities of patients with dissociative identity disorder (multiple personality disorder).

The dissociative disorders vary in their severity and the suddenness of onset. It is difficult to give statistics for their frequency in the United States because they are a relatively new category and are often misdiagnosed. Criteria for diagnosis require significant impairment in social or vocational functioning.

Dissociative amnesia

Dissociative **amnesia** is a disorder in which the distinctive feature is the patient's inability to

remember important personal information to a degree that cannot be explained by normal forgetfulness. In many cases, it is a reaction to a traumatic accident or witnessing a violent crime. Patients with dissociative amnesia may develop depersonalization or trance states as part of the disorder, but they do not experience a change in identity.

Dissociative fugue

Dissociative fugue is a disorder in which a person temporarily loses his or her sense of personal identity and travels to another location where he or she may assume a new identity. Again, this condition usually follows a major stressor or trauma. Apart from inability to recall their past or personal information, patients with dissociative fugue do not behave strangely or appear disturbed to others. Cases of dissociative fugue are more common in wartime or in communities disrupted by a natural disaster.

Depersonalization disorder

Depersonalization disorder is a disturbance in which the patient's primary symptom is a sense of detachment from the self. Depersonalization as a symptom (not as a disorder) is quite common in college-age populations. It is often associated with sleep deprivation or "recreational" drug use. It may be accompanied by "derealization" (where objects in an environment appear altered). Patients sometimes describe depersonalization as feeling like a robot or watching themselves from the outside. Depersonalization disorder may also involve feelings of **numbness** or loss of emotional "aliveness."

Dissociative identity disorder (DID)

Dissociative identity disorder (DID) is the newer name for multiple personality disorder (MPD). DID is considered the most severe dissociative disorder and involves all of the major dissociative symptoms.

Dissociative disorder not otherwise specified (DDNOS)

DDNOS is a diagnostic category ascribed to patients with dissociative symptoms that do not meet the full criteria for a specific dissociative disorder.

Causes and symptoms

The moderate to severe dissociation that occurs in patients with dissociative disorders is understood to result from a set of causes:

- an innate ability to dissociate easily
- repeated episodes of severe physical or sexual abuse in childhood
- the lack of a supportive or comforting person to counteract abusive relative(s)
- the influence of other relatives with dissociative symptoms or disorders

The relationship of dissociative disorders to childhood abuse has led to intense controversy and lawsuits concerning the accuracy of childhood memories. The brain's storage, retrieval, and interpretation of memories are still not fully understood. Controversy also exists regarding how much individuals presenting dissociative disorders have been influenced by books and movies to describe a certain set of symptoms (scripting).

The major dissociative symptoms are:

Amnesia

Amnesia in a dissociative disorder is marked by gaps in a patient's memory for long periods of time or for traumatic events. Doctors can distinguish this type of amnesia from loss of memory caused by head injuries or drug intoxication, because the amnesia is "spotty" and related to highly charged events and feelings.

Depersonalization

Depersonalization is a dissociative symptom in which the patient feels that his or her body is unreal, is changing, or is dissolving. Some patients experience depersonalization as being outside their bodies or watching a movie of themselves.

Derealization

Derealization is a dissociative symptom in which the external environment is perceived as unreal. The patient may see walls, buildings, or other objects as changing in shape, size, or color. In some cases, the patient may feel that other persons are machines or robots, though the patient is able to acknowledge the unreality of this feeling.

Identity disturbances

Patients with dissociative fugue, DDNOS, or DID often experience confusion about their identities or even assume new identities. Identity disturbances result from the patient having split off entire personality traits or characteristics as well as memories. When a stressful or traumatic experience triggers the

reemergence of these dissociated parts, the patient may act differently, answer to a different name, or appear confused by his or her surroundings.

Diagnosis

When a doctor is evaluating a patient with dissociative symptoms, he or she will first rule out physical conditions that sometimes produce amnesia, depersonalization, or derealization. These physical conditions include epilepsy, head injuries, brain disease, side effects of medications, **substance abuse**, intoxication, **AIDS**, **dementia** complex, or recent periods of extreme physical **stress** and sleeplessness. In some cases, the doctor may give the patient an electroencephalogram (EEG) to exclude epilepsy or other seizure disorders.

If the patient appears to be physically normal, the doctor will rule out psychotic disturbances, including **schizophrenia**. In addition, doctors can use some **psychological tests** to narrow the diagnosis. One is a screener, the Dissociative Experiences Scale (DES). If the patient has a high score on this test, he or she can be evaluated further with the Dissociative Disorders Interview Schedule (DDIS) or the Structured Clinical Interview for *DSM-IV* Dissociative Disorders (SCID-D). It is also possible for doctors to measure a patient's hypnotizability as part of a diagnostic evaluation.

Treatment

Treatment of the dissociative disorders often combines several methods.

Psychotherapy

Patients with dissociative disorders often require treatment by a therapist with some specialized understanding of dissociation. This background is particularly important if the patient's symptoms include identity problems. Many patients with dissociative disorders are helped by group as well as individual treatment.

Medications

Some doctors will prescribe tranquilizers or anti-depressants for the **anxiety** and/or depression that often accompany dissociative disorders. Patients with dissociative disorders are, however, at risk for abusing or becoming dependent on medications. As of 2001, there is no drug that can reliably counteract dissociation itself.

Hypnosis

Hypnosis is frequently recommended as a method of treatment for dissociative disorders, partly because

KEY TERMS

Amnesia—A general medical term for loss of memory that is not due to ordinary forgetfulness. Amnesia can be caused by head injuries, brain disease, or epilepsy, as well as by dissociation.

Depersonalization—A dissociative symptom in which the patient feels that his or her body is unreal, is changing, or is dissolving.

Derealization—A dissociative symptom in which the external environment is perceived as unreal.

Dissociation—A psychological mechanism that allows the mind to split off traumatic memories or disturbing ideas from conscious awareness.

Fugue—A dissociative experience during which a person travels away from home, has amnesia for their past, and may be confused about their identity but otherwise appear normal.

Hypnosis—The means by which a state of extreme relaxation and suggestibility is induced: used to treat amnesia and identity disturbances that occur in dissociative disorders.

Multiple personality disorder (MPD)—An older term for dissociative identity disorder (DID).

Trauma—A disastrous or life-threatening event that can cause severe emotional distress, including dissociative symptoms and disorders.

hypnosis is related to the process of dissociation. Hypnosis may help patients recover repressed ideas and memories. Therapists treating patients with DID sometimes use hypnosis in the process of “fusing” the patient's alternate personalities.

Prognosis

Prognoses for dissociative disorders vary. Recovery from dissociative fugue is usually rapid. Dissociative amnesia may resolve quickly, but can become a chronic disorder in some patients. Depersonalization disorder, DDNOS, and DID are usually chronic conditions. DID usually requires five or more years of treatment for recovery.

Prevention

Since the primary cause of dissociative disorders is thought to involve extended periods of humanly inflicted trauma, prevention depends on the

elimination of child abuse and psychological abuse of adult prisoners or hostages.

Resources

BOOKS

Eisendrath, Stuart J. "Psychiatric Disorders." In *Current Medical Diagnosis and Treatment*, 1998, edited by Stephen McPhee, et al., 37th ed. Stamford: Appleton & Lange, 1997.

Rebecca J. Frey, PhD

Dissociative identity disorder see **Multiple personality disorder**

Diuretics

Definition

Diuretics are medicines that help reduce the amount of water in the body.

Purpose

Diuretics are used to treat the buildup of excess fluid in the body that occurs with some medical conditions such as congestive **heart failure**, liver disease, and **kidney disease**. Some diuretics are also prescribed to treat high blood pressure. These drugs act on the kidneys to increase urine output. This reduces the amount of fluid in the bloodstream, which in turn lowers blood pressure.

Description

There are several types of diuretics, also called water pills:

- Loop diuretics, such as bumetanide (Bumex) and furosemide (Lasix), get their name from the loop-shaped part of the kidneys where they have their effect.
- Thiazide diuretics include such commonly used diuretics as hydrochlorothiazide (HydroDIURIL, Esidrix), chlorothiazide (Diuril), and chlorthalidone (Hygroton).
- Potassium-sparing diuretics prevent the loss of potassium, which is a problem with other types of diuretics. Examples of potassium-sparing diuretics are amiloride (Midamor) and triamterene (Dyrenium).

In addition, some medicines contain combinations of two diuretics. The brands Dyazide and Maxzide, for example, contain the thiazide diuretic hydrochlorothiazide with the potassium-sparing diuretic triamterene.

Some nonprescription (over-the-counter) medicines contain diuretics. However, the medicines described here cannot be bought without a physician's prescription. They are available in tablet, capsule, liquid, and injectable forms.

Recommended dosage

The recommended dosage depends on the type of diuretic and may be different for different patients. Check with the physician who prescribed the drug or the pharmacist who filled the prescription for the correct dosage, and take the medicine exactly as directed.

Precautions

Seeing a physician regularly while taking a diuretic is important. The physician will check to make sure the medicine is working as it should and will watch for unwanted side effects.

Some people feel unusually tired when they first start taking diuretics. This effect usually becomes less noticeable over time, as the body adjusts to the medicine.

Because diuretics increase urine output, people who take this medicine may need to urinate more often, even during the night. Health care professionals can help patients schedule their doses to avoid interfering with their sleep or regular activities.

For patients taking the kinds of diuretics that rob potassium from the body, physicians may recommend adding potassium-rich foods or drinks, such as citrus fruits and juices, to the diet. Or they may suggest taking a potassium supplement or taking another medicine that keeps the body from losing too much potassium. If the physician recommends any of these measures, be sure to closely follow his or her directions. Do not make other diet changes without checking with the physician. People who are taking potassium-sparing diuretics should not add potassium to their diets, as too much potassium may be harmful.

People who take diuretics may lose too much water or potassium when they get sick, especially if they have severe **vomiting** and **diarrhea**. They should check with their physicians if they become ill.

These medicines make some people feel light-headed, dizzy, or faint when they get up after sitting

or lying down. Older people are especially likely to have this problem. Drinking alcohol, exercising, standing for long periods, or being in hot weather may make the problem worse. To lessen the problem, get up gradually and hold onto something for support if possible. Avoid drinking too much alcohol and be careful in hot weather or when exercising or standing for a long time.

Anyone who is taking a diuretic should be sure to tell the health care professional in charge before having surgical or dental procedures, medical tests, or emergency treatment.

Some diuretics make the skin more sensitive to sunlight. Even brief exposure to sun can cause a severe **sunburn**, **itching**, a rash, redness, or other changes in skin color. While being treated with this medicine, avoid being in direct sunlight, especially between 10 a.m. and 3 p.m.; wear a hat and tightly woven clothing that covers the arms and legs; use a sunscreen with a skin protection factor (SPF) of at least 15; protect the lips with a sun block lipstick; and do not use tanning beds, tanning booths, or sunlamps. People with fair skin may need to use a sunscreen with a higher skin protection factor.

Special conditions

People who have certain medical conditions or who are taking certain other medicines may have problems if they take diuretics. Before taking these drugs, be sure to let the physician know about any of these conditions:

ALLERGIES. Anyone who has had unusual reactions to diuretics or **sulfonamides** (sulfa drugs) in the past should let his or her physician know before using a diuretic. The physician should also be told about any **allergies** to foods, dyes, preservatives, or other substances.

PREGNANCY. Diuretics will not help the swelling of hands and feet that some women have during pregnancy. In general, pregnant women should not use diuretics unless a physician recommends their use. Although studies have not been done on pregnant women, studies of laboratory animals show that some diuretics can cause harmful effects when taken during pregnancy.

BREASTFEEDING. Some diuretics pass into breast milk, but no reports exist of problems in nursing babies whose mothers use this medicine. However, thiazide diuretics may decrease the flow of breast milk. Women who are breastfeeding and need to use a diuretic should check with their physicians.

OTHER MEDICAL CONDITIONS. Side effects of some diuretics may be more likely in people who have had a recent heart attack or who have **liver disease** or severe kidney disease. Other diuretics may not work properly in people with liver disease or severe kidney disease. Diuretics may worsen certain medical conditions, such as gout, kidney stones, **pancreatitis**, lupus erythematosus, and hearing problems. In addition, people with diabetes should be aware that diuretics may increase blood sugar levels. People with heart or blood vessel disease should know that some diuretics increase cholesterol or triglyceride levels. The risk of an allergic reaction to certain diuretics is greater in people with bronchial **asthma**. Before using diuretics, people with any of these medical problems should make sure their physicians are aware of their conditions. Also, people who have trouble urinating or who have high potassium levels in their blood may not be able to take diuretics and should check with a physician before using them.

USE OF CERTAIN MEDICINES. Taking diuretics with certain other drugs may affect the way the drugs work or may increase the chance of side effects.

Side effects

Some side effects, such as loss of appetite, **nausea** and vomiting, stomach cramps, diarrhea, and dizziness, usually lessen or go away as the body adjusts to the medicine. These problems do not need medical attention unless they continue or interfere with normal activities.

Patients taking potassium-sparing diuretics should know the signs of too much potassium and should check with a physician as soon as possible if any of these symptoms occur:

- irregular heartbeat
- breathing problems
- numbness or **tingling** in the hands, feet, or lips
- confusion or nervousness
- unusual tiredness or weakness
- weak or heavy feeling in the legs

Patients taking diuretics that cause potassium loss should know the signs of too little potassium and should check with a physician as soon as possible if they have any of these symptoms:

- fast or irregular heartbeat
- weak pulse
- nausea or vomiting
- dry mouth

KEY TERMS

Inflammation—Pain, redness, swelling, and heat that usually develop in response to injury or illness.

Lupus erythematosus—A chronic disease that affects the skin, joints, and certain internal organs.

Pancreas—A gland located beneath the stomach. The pancreas produces juices that help break down food.

Potassium—A mineral found in whole grains, meat, legumes, and some fruits and vegetables. Potassium is important for many body processes, including proper functioning of the nerves and muscles.

Triglyceride—A substance formed in the body from fat in the diet. Triglycerides are the main fatty materials in the blood. Together with protein, they make up high- and low-density lipoproteins (HDLs and LDLs). Triglyceride levels are important in the diagnosis and treatment of many diseases including high blood pressure, diabetes, and heart disease.

- excessive thirst
- muscle cramps or **pain**
- unusual tiredness or weakness
- mental or mood changes

Interactions

Diuretics may interact with other medicines. When this happens, the effects of one or both of the drugs may change or the risk of side effects may be greater. Anyone who takes a diuretic should let the physician know all other medicines he or she is taking and should ask whether the possible interactions can interfere with drug therapy. Among the drugs that may interact with diuretics are:

- Angiotensin-converting enzyme (ACE) inhibitors, such as benazepril (Lotensin), captopril (Capoten), and enalapril (Vasotec), used to treat high blood pressure. Taking these drugs with potassium-sparing diuretics may cause levels of potassium in the blood to be too high, increasing the chance of side effects.
- Cholesterol-lowering drugs such as cholestyramine (Questran) and colestipol (Colestid). Taking these drugs with combination diuretics such as Dyazide and Maxzide may keep the diuretic from working.

Take the diuretic at least one hour before or four hours after the cholesterol-lowering drug.

- Cyclosporine (Sandimmune), a medicine that suppresses the immune system. Taking this medicine with potassium-sparing diuretics may increase the chance of side effects by causing levels of potassium in the blood to be too high.
- Potassium supplements, other medicines containing potassium, or salt substitutes that contain potassium. Taking these with potassium-sparing diuretics may lead to too much potassium in the blood, increasing the chance of side effects.
- Lithium, used to treat **bipolar disorder** (manic-depressive illness). Using this medicine with potassium-sparing diuretics may allow lithium to build up to poisonous levels in the body.
- Digitalis heart drugs, such as digoxin (Lanoxin). Using this medicine with combination diuretics such as triamterene-hydrochlorothiazide (Dyazide, Maxzide) may cause blood levels of the heart medicine to be too high, making side effects such as changes in heartbeat more likely.

The list above does not include every drug that may interact with diuretics. Check with a physician or pharmacist before combining diuretics with any other prescription or nonprescription (over-the-counter) medicine.

Nancy Ross-Flanigan

Diverticulitis see **Diverticulosis and diverticulitis**

Diverticulosis and diverticulitis

Definition

Diverticulosis refers to a condition in which the inner, lining layer of the large intestine (colon) bulges out (herniates) through the outer, muscular layer. These outpouchings are called diverticula. Diverticulitis refers to the development of inflammation and infection in one or more diverticula.

Description

Diverticula tend to occur most frequently in the last segment of the large intestine, the sigmoid colon. They occur with decreasing frequency as one examines

further back toward the beginning of the large intestine. The chance of developing diverticula increases with age, so that by the age of 50, about 20–50% of all people will have some diverticula. By the age of 90, virtually everyone will have developed some diverticula. Most diverticula measure about 3 mm to just over 3 cm in diameter. Larger diverticula, termed giant diverticula, are quite infrequent, but may measure as large as 15 cm in diameter.

Causes and symptoms

Diverticula are believed to be caused by overly forceful contractions of the muscular wall of the large intestine. As areas of this wall spasm, they become weaker and weaker, allowing the inner lining to bulge through. The anatomically weakest areas of the intestinal wall occur next to blood vessels which course through the wall, so diverticula commonly occur in this location.

Diverticula are most common in the developed countries of the West (North America, Great Britain, northern and western Europe). This is thought to be due to the diet of these countries, which tends to be quite low in fiber. A diet low in fiber results in the production of smaller volumes of stool. In order to move this smaller stool along the colon and out of the rectum, the colon must narrow itself significantly, and does so by contracting down forcefully. This causes an increase in pressure, which, over time, weakens the muscular wall of the intestine and allows diverticular pockets to develop.

The origin of giant diverticula development is not completely understood, although one theory involves gas repeatedly entering and becoming trapped in an already-existing diverticulum, causing stretching and expansion of that diverticulum.

The great majority of people with diverticulosis will remain symptom-free. Many diverticula are quite accidentally discovered during examinations for other conditions of the intestinal tract.

Some people with diverticulosis have symptoms such as **constipation**, cramping, and bloating. It is unclear whether these symptoms are actually caused by the diverticula themselves, or whether some other gastrointestinal condition (such as **irritable bowel syndrome**) might be responsible. A complication of diverticulosis occurs because many diverticula develop in areas very near blood vessels. Therefore, one serious risk of diverticulosis involves bleeding. Although an infrequent complication, the bleeding can be quite severe. Seventy-five percent of such bleeding episodes occur due to diverticula located on the right side of the

colon. About 50% of the time, such bleeding will stop on its own.

One of the most common and potentially serious complications of diverticulosis is inflammation and infection of a particular diverticulum, called diverticulitis.

Diverticulitis is three times more likely to occur in the left side of the large intestine. Since most diverticula are located in the sigmoid colon (the final segment of the large intestine which empties into the rectum), most diverticulitis also takes place in the sigmoid. The elderly have the most serious complications from diverticulitis, although very severe infections can also occur in patients under the age of 50. Men are three times as likely as women to be stricken with diverticulitis.

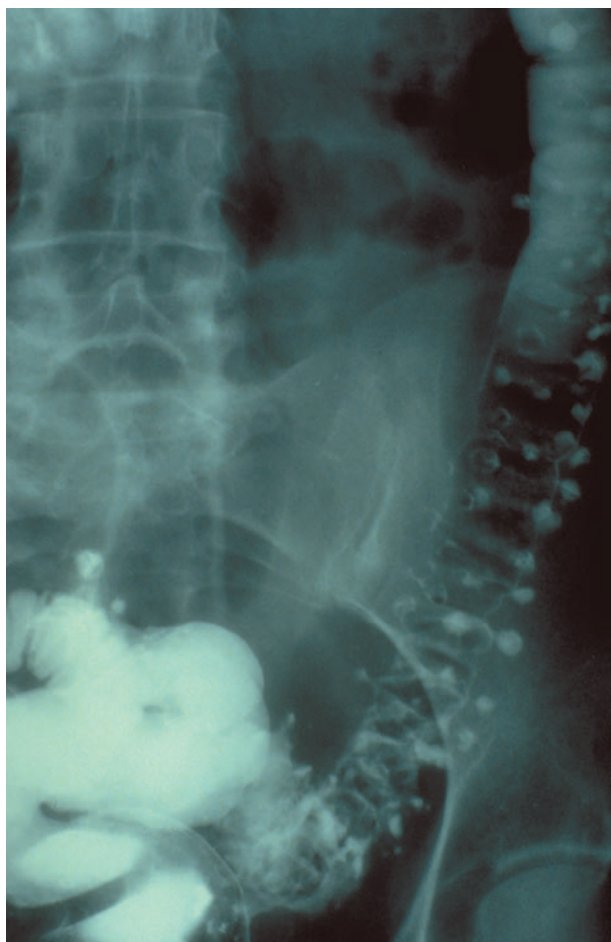
Diverticulitis is believed to occur when a hardened piece of stool, undigested food, and bacteria (called a fecalith) becomes lodged in a diverticulum. This blockage interferes with the blood supply to the area, and infection sets in.

An individual with diverticulitis will experience pain (especially in the lower left side of the abdomen) and **fever**. In response to the infection and the irritation of nearby tissues within the abdomen, the abdominal muscles may begin to spasm. About 25% of all patients with diverticulitis will have some rectal bleeding, although this rarely becomes severe. Walled-off pockets of infection, called abscesses, may appear within the wall of the intestine, or even on the exterior surface of the intestine. When a diverticulum weakens sufficiently, and is filled to bulging with infected pus, a perforation in the intestinal wall may develop. When the infected contents of the intestine spill out into the abdomen, the severe infection called **peritonitis** may occur. Peritonitis is an infection and inflammation of the lining of the abdominal cavity, the peritoneum. Other complications of diverticulitis include the formation of abnormal connections between two organs that normally do not connect (fistulas; for example, the intestine and the bladder), and scarring outside of the intestine which squeezes off a portion of the intestine, obstructing it.

Diagnosis

As mentioned, the majority of diverticula do not cause any symptoms, and are often found by coincidence during an examination being performed for some other medical condition.

When diverticula are suspected because a patient begins to have sudden rectal bleeding, the location of



A barium study x ray showing colonic diverticulosis. (Custom Medical Stock Photo. Reproduced by permission.)

the bleeding can be studied by performing an **angiography**. Angiography involves inserting a tiny tube through an artery in the leg, and moving it up into one of the major arteries of the gastrointestinal system. A particular chemical (contrast medium) which will show up on x-ray films is injected, and the area of bleeding is located by looking for an area where the contrast is leaking into the interior (lumen) of the intestine.

A procedure called endoscopy provides another method for examining the colon and locating the site of bleeding. In endoscopy, a small, flexible scope (endoscope) is inserted through the rectum and into the intestine. The scope usually bears a fiber-optic camera, which allows the view through this endoscope to be projected onto a television screen. The operator can introduce the endoscope further and further through the intestine to find the location of the bleeding.

Diagnosis of diverticulitis is not difficult in patients with previously diagnosed diverticulosis. The presence of abdominal **pain** and fever in such an individual would make the suspicion of diverticulitis quite high. Examination of the abdomen will usually reveal tenderness to touch, with the patient's abdominal muscles contracting strongly to protect the tender area. During a rectal exam (performed by inserting a finger into the rectum), a doctor may be able to feel an abnormal mass. Touching this mass may prove painful to the patient.

When a practitioner is suspicious of diverticulitis as the cause for the patient's symptoms, he or she will most likely avoid the types of tests usually used to diagnose gastrointestinal disorders. These include **barium enema** and endoscopy. The concern is that the increased pressure exerted on the intestine during these exams may increase the likelihood of intestinal perforation. After medical treatment for the diverticulitis, these examinations may be performed in order to learn the extent of the patient's disease.

Treatment

Only about 20% of patients with diverticulosis ever have symptoms which lead them to seek medical help. Most people never know that they have diverticula. For those individuals who have cramping pain and constipation believed to be due to diverticulosis, the usual prescription involves increasing the fiber in the diet. This can be done by adding special diet supplements of bran or psyllium seed, which increase stool volume. Bleeding diverticula can usually be treated by bed rest, with blood **transfusion** needed for more severe bleeding (hemorrhaging). In cases of very heavy hemorrhaging, medications which encourage clotting can be injected during the course of a diagnostic angiography.

While there are almost no situations when uncomplicated diverticulosis requires surgery, giant diverticula always require removal. This is due to the very high chance of infection and perforation of these diverticula. When giant diverticula are diagnosed, the usual treatment involves removing that portion of the intestine.

Treatment for uncomplicated diverticulitis usually requires hospitalization. "Resting the bowel" is a mainstay of treatment, and involves keeping the patient from eating or sometimes even drinking anything by mouth. Therefore, the patient will need to receive fluids through a needle in the vein (intravenous or IV fluids). **Antibiotics** will also be administered through the IV. Some physicians will agree to try

treatment at home for very mildly ill patients. These patients will be put on a liquid diet and receive oral antibiotics.

The various complications of diverticulitis need to be treated aggressively, because the death rate from such things as perforation and peritonitis is quite high. Abscesses can be drained of their infected contents by inserting a needle through the skin of the abdomen and into the **abscess**. When this is unsuccessful, open abdominal surgery will be required to remove the piece of the intestine containing the abscess. Fistulas require surgical repair, including the removal of the length of intestine containing the origin of the **fistula**, followed by immediate reconnection of the two free ends of intestine. Peritonitis requires open surgery. The entire abdominal cavity is cleaned by being irrigated (washed) with a warmed sterile saltwater solution, and the damaged piece of intestine is removed. Obstructions require immediate surgery to prevent perforation. Massive, uncontrollable bleeding, while rare, may require removal of part or all of the large intestine.

During any of these types of operations, the surgeon must make an important decision regarding the quantity of intestine which must be removed. When the amount of intestine removed is great, it may be necessary to perform a **colostomy**. A colostomy involves pulling the end of the remaining intestine through the abdominal wall, to the outside. This bit of intestine is then fashioned so that a bag can be fit over it. The patient's waste (feces) collect in the bag, because the intestine no longer connects with the rectum. This colostomy may be temporary, in which case another operation will be required to reconnect the intestine, after some months of substantial healing has occurred. Other times, the colostomy will need to be permanent, and the patient will have to adjust to living permanently with the colostomy bag. Most people with colostomies are able to go on with a very active life.

Occasionally, a patient will have such severe diverticular disease that a surgeon recommends planning ahead, and schedules removal of a portion of the colon. This is done to avoid the high risk of surgery performed after a complication has set in. Certain developments in a patient will identify those patients who are at very high risk of experiencing dangerous complications. Such elective surgery may be recommended:

- when an older individual has had several attacks of diverticulitis
- when someone under the age of 50 has had even one attack

KEY TERMS

Angiography—An x-ray study of the arteries in a particular part of the body. Angiography is often performed in order to localize internal bleeding.

Bowel obstruction—A blockage in the intestine which prevents the normal flow of waste down the length of the intestine.

Colostomy—A procedure performed when a large quantity of intestine is removed. The end piece of the intestine leading to the rectum is closed.

Diverticula—Outpouchings in the large intestine caused when the inner, lining layer of the large intestine (colon) bulges out (herniates) through the outer, muscular layer.

Endoscopy—Examination of an area of the gastrointestinal tract by putting a lighted scope, usually bearing a fiber-optic camera, into the rectum, and passing it through the intestine.

Fistula—An abnormal connection formed between two organs that usually have no connection whatsoever.

Sigmoid colon—The final portion of the large intestine that empties into the rectum.

- when treatment does not get rid of a painful mass
- when the intestine appears to be narrowing on x-ray examination (this could suggest the presence of **cancer**)
- when certain patients begin to regularly experience painful urination or urinary infections (this suggests that there may be a connection between the intestine and the bladder)
- when there is any question of cancer
- when the diverticular disease appears to be progressing rapidly

Prognosis

The prognosis for people with diverticula is excellent, with only 20% of such patients ever seeking any medical help for their condition.

While diverticulitis can be a difficult and painful disease, it is usually quite treatable. Prognosis is worse for individuals who have other medical problems, particularly those requiring the use of steroid medications, which increase the chances of developing a serious infection. Prognosis is also worse in the elderly.

Prevention

While there is no absolutely certain way to prevent the development of diverticula, it is believed that high-fiber **diets** are of help. Foods that are recommended for their high fiber content include whole grain breads and cereals, and all types of fruits and vegetables. Most experts suggest that individuals take in about 0.71–1.23 oz (20–35) gs of fiber daily. If this is not possible to achieve through a person's diet, there are fiber products which can be mixed into 8 oz (237l) of water or juice, and which provide about 0.13–19 oz (4–6 gs) of fiber.

Resources

ORGANIZATIONS

National Digestive Diseases Information Clearinghouse.
2 Information Way, Bethesda, MD 20892-3570. (800)
891-5389. < <http://www.niddk.nih.gov/health/digest/nddic.htm> >.

Rosalyn Carson-DeWitt, MD

Dizziness

Definition

As a disorder, dizziness is classified into three categories—vertigo, syncope, and nonsyncope nonvertigo. Each category has a characteristic set of symptoms, all related to the sense of balance. In general, syncope is defined by a brief loss of consciousness (**fainting**) or by dimmed vision and feeling uncoordinated, confused, and lightheaded. Many people experience a sensation like syncope when they stand up too fast. Vertigo is the feeling that either the individual or the surroundings are spinning. This sensation is like being on a spinning amusement park ride. Individuals with nonsyncope nonvertigo dizziness feel as though they cannot keep their balance. This feeling may become worse with movement.

Description

The brain coordinates information from the eyes, the inner ear, and the body's senses to maintain balance. If any of these information sources is disrupted, the brain may not be able to compensate. For example, people sometimes experience motion sickness because the information from their body tells the brain that they are sitting still, but information from

the eyes indicates that they are moving. The messages do not correspond and dizziness results.

Vision and the body's senses are the most important systems for maintaining balance, but problems in the inner ear are the most frequent cause of dizziness. The inner ear, also called the vestibular system, contains fluid that helps fine tune the information the brain receives from the eyes and the body. When fluid volume or pressure in one inner ear changes, information about balance is altered. The discrepancy gives conflicting messages to the brain about balance and induces dizziness.

Certain medical conditions can cause dizziness, because they affect the systems that maintain balance. For example, the inner ear is very sensitive to changes in blood flow. Because medical conditions such as high blood pressure or low blood sugar can affect blood flow, these conditions are frequently accompanied by dizziness. Circulation disorders are the most common causes of dizziness. Other causes are **head injury**, ear infection, allergies, and nervous system disorders.

Dizziness often disappears without treatment or with treatment of the underlying problem, but it can be long term or chronic. According to the National Institutes of Health, 42% of Americans will seek medical help for dizziness at some point in their lives. The costs may exceed a billion dollars and account for five million doctor visits annually. Episodes of dizziness increase with age. Among people aged 75 or older, dizziness is the most frequent reason for seeing a doctor.

Causes and symptoms

Careful attention to symptoms can help determine the underlying cause of the dizziness. Underlying problems may be benign and easily treated or they may be dangerous and in need of intensive therapy. Not all cases of dizziness can be linked to a specific cause. More than one type of dizziness can be experienced at the same time and symptoms may be mixed. Episodes of dizziness may last for a few seconds or for days. The length of an episode is related to the underlying cause.

The symptoms of syncope include dimmed vision, loss of coordination, confusion, lightheadedness, and sweating. These symptoms can lead to a brief loss of consciousness or fainting. They are related to a reduced flow of blood to the brain; they often occur when a person is standing up and can be relieved by sitting or lying down. Vertigo is characterized by a sensation of spinning or turning, accompanied by

nausea, vomiting, ringing in the ears, **headache**, or **fatigue**. An individual may have trouble walking, remaining coordinated, or keeping balance. Nonsyncope nonvertigo dizziness is characterized by a feeling of being off balance that becomes worse if the individual tries moving or performing detail-intense tasks.

A person may experience dizziness for many reasons. Syncope is associated with low blood pressure, heart problems, and disorders in the autonomic nervous system, the system of involuntary functions such as breathing. Syncope may also arise from emotional distress, pain, and other reactions to outside stressors. Nonsyncope nonvertigo dizziness may be caused by rapid breathing, low blood sugar, or **migraine headache**, as well as by more serious medical conditions.

Vertigo is often associated with inner ear problems called vestibular disorders. A particularly intense vestibular disorder, Ménière's disease, interferes with the volume of fluid in the inner ear. This disease, which affects approximately one in every 1,000 people, causes intermittent vertigo over the course of weeks, months, or years. Ménière's disease is often accompanied by ringing or buzzing in the ear, **hearing loss**, and a feeling that the ear is blocked. Damage to the nerve that leads from the ear to the brain can also cause vertigo. Such damage can result from head injury or a tumor. An **acoustic neuroma**, for example, is a benign tumor that wraps around the nerve. Vertigo can also be caused by disorders of the central nervous system and the circulatory system, such as hardening of the arteries (arteriosclerosis), **stroke**, or **multiple sclerosis**.

Some medications cause changes in blood pressure or blood flow. These medications can cause dizziness in some people. Prescription medications carry warnings of such side effects, but common drugs, such as **caffeine** or nicotine, can also cause dizziness. Certain **antibiotics** can damage the inner ear and cause hearing loss and dizziness.

Diet may cause dizziness. The role of diet may be direct, as through alcohol intake. It may be also be indirect, as through arteriosclerosis caused by a high-fat diet. Some people experience a slight dip in blood sugar and mild dizziness if they miss a meal, but this condition is rarely dangerous unless the person is diabetic. Food sensitivities or **allergies** can also be a cause of dizziness. Chronic conditions, such as heart disease, and serious acute problems, such as seizures and strokes, can cause dizziness. However, such conditions usually exhibit other characteristic symptoms.

Diagnosis

During the initial medical examination, an individual with dizziness should provide a detailed description of the type of dizziness experienced, when it occurs, and how often each episode lasts. A diary of symptoms may help track this information. Report any symptoms that accompany the dizziness, such as a ringing in the ear or nausea, any recent injury or infection, and any medication taken.

Blood pressure, pulse, respiration, and body temperature are checked, and the ear, nose, and throat are scrutinized. The sense of balance is assessed by moving the individual's head to various positions or by tilt-table testing. In tilt-table testing, the person lies on a table that can be shifted into different positions and reports any dizziness that occurs.

Further tests may be indicated by the initial examination. Hearing tests help assess ear damage. X rays, computed tomography scan (CT scan), and magnetic resonance imaging (MRI) can pinpoint evidence of nerve damage, tumor, or other structural problems. If a vestibular disorder is suspected, a technique called electronystagmography (ENG) may be used. ENG measures the electrical impulses generated by eye movements. Blood tests can determine diabetes, **high cholesterol**, and other diseases. In some cases, a heart evaluation may be useful. Despite thorough testing, an underlying cause cannot always be determined.

Treatment

Treatment is determined by the underlying cause. If an individual has a cold or **influenza**, a few days of bed rest is usually adequate to resolve dizziness. Other causes of dizziness, such as mild vestibular system damage, may resolve without medical treatment.

If dizziness continues, drug therapy may prove helpful. Because circulatory problems often cause dizziness, medication may be prescribed to control blood pressure or to treat arteriosclerosis. Sedatives may be useful to relieve the tension that can trigger or aggravate dizziness. Low blood sugar associated with diabetes sometimes causes dizziness and is treated by controlling blood sugar levels. An individual may be asked to avoid caffeine, nicotine, alcohol, and any substances that cause allergic reactions. A low-salt diet may also help some people.

When other measures have failed, surgery may be suggested to relieve pressure on the inner ear. If the dizziness is not treatable by drugs, surgery, or other means, physical therapy may be used and the patient may be taught coping mechanisms for the problem.

Alternative treatment

Because dizziness may arise from serious conditions, it is advisable to seek medical treatment. Alternative treatments can often be used alongside conventional medicine without conflict. Relaxation techniques, such as **yoga** and **massage therapy** that focus on relieving tension, are popularly recommended methods for reducing **stress**. Aromatherapists recommend a warm bath scented with essential oils of lavender, geranium, and sandalwood.

Homeopathic therapies can work very effectively for dizziness, and are especially applicable when no organic cause can be identified. An osteopath or chiropractor may suggest adjustments of the head, jaw, neck, and lower back to relieve pressure on the inner ear. Acupuncturists also offer some treatment options for acute and chronic cases of dizziness. Nutritionists may be able to offer advice and guidance in choosing dietary supplements, identifying foods to avoid, and balancing nutritional needs.

Prognosis

Outcome depends on the cause of dizziness. Controlling or curing the underlying factors usually relieves dizziness. In some cases, dizziness disappears without treatment. In a few cases, dizziness can become a permanent disabling condition and a person's options are limited.

Prevention

Most people learn through experience that certain activities will make them dizzy and they learn to avoid them. For example, if reading in a car produces **motion sickness**, an individual leaves reading materials for after the trip. Changes to the diet can also cut down on episodes of dizziness in susceptible people. Relaxation techniques can help ward off tension and **anxiety** that can cause dizziness.

These techniques can help minimize or even prevent dizziness for people with chronic diseases. For example, persons with Ménière's disease may avoid episodes of vertigo by leaving salt, alcohol, and caffeine out of their diets. Reducing blood cholesterol can help diminish arteriosclerosis and indirectly treat dizziness.

Some cases of dizziness cannot be prevented. Acoustic neuromas, for example, are not predictable or preventable. When the underlying cause of dizziness cannot be discovered, it may be difficult to recommend preventive measures. Alternative approaches designed to rebalance the body's energy flow, such as **acupuncture** and constitutional homeopathy, may be helpful in cases where the cause of dizziness cannot be pinpointed.

KEY TERMS

Acoustic neuroma—A benign tumor that grows on the nerve leading from the inner ear to the brain. As the tumor grows, it exerts pressure on the inner ear and causes severe vertigo.

Arteriosclerosis—Hardening of the arteries caused by high blood cholesterol and high blood pressure.

Autonomic nervous system—The part of the nervous system that controls involuntary functions such as breathing and heart beat.

Computed tomography (CT)—An imaging technique in which cross-sectional x rays of the body are compiled to create a three-dimensional image of the body's internal structures.

Electronystagmography—A method for measuring the electricity generated by eye movements. Electrodes are placed on the skin around the eye and the individual is subjected to a variety of stimuli so that the quality of eye movements can be assessed.

Magnetic resonance imaging (MRI)—An imaging technique that uses a large circular magnet and radio waves to generate signals from atoms in the body. These signals are used to construct images of internal structures.

Vestibular system—The area of the inner ear that helps maintain balance.

Resources

ORGANIZATIONS

Ménière's Network. 1817 Patterson St., Nashville, TN 37203. (800) 545-4327. <<http://www.earfoundation.org>>.

Vestibular Disorders Association. PO Box 4467, Portland, OR 97208-4467. (503) 229-7705. <<http://www.teleport.com/~veda>>.

Julia Barrett

DKA see **Diabetic ketoacidosis**

DLE see **Discoid lupus erythematosus**

Domestic violence see **Abuse**

Donovanosis see **Granuloma inguinale**

Doppler echocardiography see **Echocardiography**

Doppler ultrasonography

Definition

Doppler ultrasonography is a non-invasive diagnostic procedure that changes sound waves into an image that can be viewed on a monitor.

Purpose

Doppler ultrasonography can detect the direction, velocity, and turbulence of blood flow. It is frequently used to detect problems with heart valves or to measure blood flow through the arteries. Specifically, it is useful in the work up of **stroke** patients, in assessing blood flow in the abdomen or legs, and in viewing the heart to monitor carotid artery diseases.

Precautions

The test is widely used because it is noninvasive, uses no x rays, and gives excellent images. It is harmless, painless, and widely available.

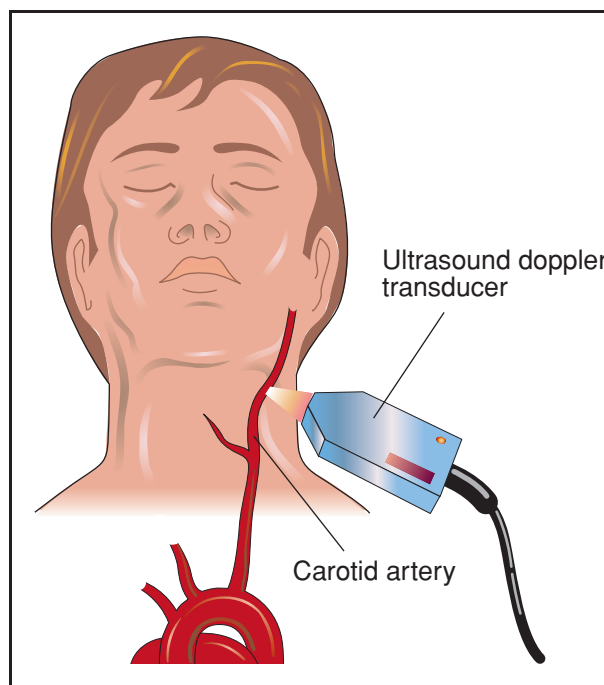
Description

Doppler ultrasonography makes use of two different principles. The ultrasound principle is this: when a high-frequency sound is produced and aimed at a target, it will be reflected by its target and the reflected sound can be detected back at its origin. In addition, it is known that certain crystals (called piezoelectric crystals) produce an electrical pulse when vibrated by a returning sound.

The Doppler principle is simply that sound pitch increases as the source moves toward the listener and decreases as it moves away.

Medical science utilizes these two principles in the following way. A transducer (sometimes called a probe) containing piezoelectric crystals sends a series of short sound pulses into the body and pauses between each pulse to listen for the returning sounds. The machine then determines the direction and depth of each returning sound and converts this into a point of light on a television monitor. Thousands of these pulses are computed and displayed every second to produce an image of the organ being studied. The image allows the doctor to see the organ functioning in real time.

The newest addition to this test is the addition of color. Adding color to the image shows the direction and rate of blood flow more clearly.



Doppler ultrasonography can detect the direction, velocity, and turbulence of blood flow. Because it is non-invasive and uses no x rays, doppler ultrasonography is widely used for numerous diagnostic procedures. (Illustration by Electronic Illustrators Group.)

During a Doppler ultrasonography procedure the technician will apply a gel to the skin, then place the transducer against the skin at various angles. The transducer sends the information it receives to a television monitor that shows a moving image of the organ being studied. The technician can save these images either on video tape, paper, or x-ray film for further study.

Preparation

There is no special preparation needed for this test. The ultrasound technician may apply a clear gel to the skin in order to help the transducer more freely over the body.

Aftercare

No aftercare is necessary.

Normal results

A Doppler ultrasonography test showing no restricted blood flow is a normal finding.

KEY TERMS

Doppler effect—The principle that the sound of an object moving toward you has a higher pitch than the sound when it is moving away from you.

Transducer—The part of a machine that changes signals in one form into another form.

Ultrasound—Sound that is too high for the human ear to hear.

Abnormal results

Disrupted or obstructed blood flow through the neck arteries may indicate the person is at risk of having a stroke. (Narrowed arterial flow in the legs does not necessarily indicate a risk of stroke.)

Resources

BOOKS

Samuels, Martin, and Steven Feske, editors. *Office Practice of Neurology*. New York: Churchill Livingstone, 1996.

Dorothy Elinor Stonely

Down syndrome

Definition

Down syndrome (DS) is the most common cause of **mental retardation** and malformation in a newborn. It occurs because of the presence of an extra chromosome. It was first described in 1866 by Dr. John L. H. Down (1828–1896), an English physician.

Down syndrome occurs about once in every 800 births. It is estimated that about 6,000 children are born with DS each year in the United States.

Description

Chromosomes are the units of genetic information that exist within every cell of the body. Twenty-three distinctive pairs, or 46 total chromosomes, are located within the nucleus (central structure) of each cell. When a baby is conceived by the combining of one sperm cell with one egg cell, the baby receives 23 chromosomes from each parent, for a total of 46 chromosomes. Sometimes, an accident in the production of a sperm or egg cell causes that cell to

contain 24 chromosomes. This event is referred to as nondisjunction. When this defective cell is involved in the conception of a baby, that baby will have a total of 47 chromosomes. The extra chromosome in Down syndrome is labeled number 21. For this reason, the existence of three such chromosomes is sometimes referred to as trisomy 21.

In a very rare number of Down syndrome cases (about 1–2%), the original egg and sperm cells are completely normal. The problem occurs sometime shortly after fertilization; during the phase where cells are dividing rapidly. One cell divides abnormally, creating a line of cells with an extra chromosome 21. This form of genetic disorder is called a mosaic. The individual with this type of Down syndrome has two types of cells: those with 46 chromosomes (the normal number), and those with 47 chromosomes (as occurs in Down syndrome). Some researchers have suggested that individuals with this type of mosaic form of Down syndrome have less severe signs and symptoms of the disorder.

Another relatively rare genetic accident which can cause Down syndrome is called translocation. During cell division, the number 21 chromosome somehow breaks. A piece of the 21 chromosome then becomes attached to another chromosome. Each cell still has 46 chromosomes, but the extra piece of chromosome 21 results in the signs and symptoms of Down syndrome. Translocations occur in about 3–4% of cases of Down syndrome.

Down syndrome occurs in about one in every 800–1,000 births. It affects an equal number of boys and girls. Less than 25% of Down syndrome cases occur due to an extra chromosome in the sperm cell. The majority of cases of Down syndrome occur due to an extra chromosome 21 within the egg cell supplied by the mother (nondisjunction). As a woman's age (maternal age) increases, the risk of having a Down syndrome baby increases significantly. For example, at younger ages, the risk is about one in 4,000. By the time the woman is age 35, the risk increases to one in 400; by age 40 the risk increases to one in 110; and by age 45 the risk becomes one in 35. There is no increased risk of either mosaicism or translocation with increased maternal age.

Down syndrome sometimes occurs together with such other developmental disorders as Rett syndrome. Although such double diagnoses are very rare, it is important for parents to recognize that the presence of one chromosomal abnormality does not exclude the possibility that their child may have a second anomaly.

Causes and symptoms

While Down syndrome is a chromosomal disorder, a baby is usually identified at birth through observation of a set of common physical characteristics. Babies with Down syndrome tend to be overly quiet, less responsive, with weak, floppy muscles. Furthermore, a number of physical signs may be present. These include:

- flat appearing face
- small head
- flat bridge of the nose
- smaller than normal, low-set nose
- small mouth, which causes the tongue to stick out and to appear overly large
- upward slanting eyes
- extra folds of skin located at the inside corner of each eye, near the nose (called epicanthal folds)
- rounded cheeks
- small, misshapen ears
- small, wide hands
- an unusual, deep crease across the center of the palm (called a simian crease)
- a malformed fifth finger
- a wide space between the big and the second toes
- unusual creases on the soles of the feet
- overly-flexible joints (sometimes referred to as being double-jointed)
- shorter than normal height

Other types of defects often accompany Down syndrome. About 30–50% of all children with Down syndrome are found to have heart defects. A number of different heart defects are common in Down syndrome, including abnormal openings (holes) in the walls that separate the heart's chambers (**atrial septal defect**, ventricular septal defect). These result in abnormal patterns of blood flow within the heart. The abnormal blood flow often means that less oxygen is sent into circulation throughout the body. Another heart defect that occurs in Down syndrome is called Tetralogy of Fallot. **Tetralogy of Fallot** consists of a hole in the heart, along with three other major heart defects.

Malformations of the gastrointestinal tract are present in about 5–7% of children with Down syndrome. The most common malformation is a narrowed, obstructed duodenum (the part of the intestine into which the stomach empties). This disorder, called duodenal atresia, interferes with the baby's

milk or formula leaving the stomach and entering the intestine for digestion. The baby often vomits forcibly after feeding, and cannot gain weight appropriately until the defect is repaired.

Other medical conditions that occur in patients with Down syndrome include an increased chance of developing infections, especially ear infections and **pneumonia**; certain kidney disorders; thyroid disease (especially low or hypothyroid); **hearing loss**; vision impairment requiring glasses (corrective lenses); and a 20-times greater chance of developing leukemia (a blood disorder).

Development in a baby and child with Down syndrome occurs at a much slower than normal rate. Because of weak, floppy muscles (hypotonia), babies learn to sit up, crawl, and walk much later than their normal peers. Talking is also quite delayed. The level of mental retardation is considered to be mild-to-moderate in Down syndrome. The actual IQ range of Down syndrome children is quite varied, but the majority of such children are in what is sometimes known as the trainable range. This means that most people with Down syndrome can be trained to do regular self-care tasks, function in a socially appropriate manner in a normal home environment, and even hold simple jobs.

As people with Down syndrome age, they face an increased chance of developing the brain disease called Alzheimer's (sometimes referred to as **dementia** or senility). Most people have a six in 100 risk of developing Alzheimer's, but people with Down syndrome have a 25 in 100 chance of the disease. In addition to an increased risk of developing Alzheimer's, patients with DS show the first signs of the disease much earlier than most people, often in their early 40s. Alzheimer's disease causes the brain to shrink and to break down. The number of brain cells decreases, and abnormal deposits and structural rearrangements occur. This process results in a loss of brain functioning. People with Alzheimer's have strikingly faulty memories. Over time, people with **Alzheimer's disease** will lapse into an increasingly unresponsive state. Some researchers have shown that even Down syndrome patients who do not appear to have Alzheimer's disease have the same changes occurring to the structures and cells of their brains. A new questionnaire was published in 2004 to help doctors evaluate adults with Down syndrome for symptoms of Alzheimer's-related dementia.

As people with Down syndrome age, they also have an increased chance of developing a number of other illnesses, including **cataracts**, thyroid problems, diabetes, and seizure disorders.

Diagnosis

Diagnosis is usually suspected at birth, when the characteristic physical signs of Down syndrome are noted. Once this suspicion has been raised, genetic testing (chromosome analysis) can be undertaken in order to verify the presence of the disorder. This testing is usually done on a blood sample, although chromosome analysis can also be done on other types of tissue, including skin. The cells to be studied are prepared in a laboratory. Chemical stain is added to make the characteristics of the cells and the chromosomes stand out. Chemicals are added to prompt the cells to go through normal development, up to the point where the chromosomes are most visible, prior to cell division. At this point, they are examined under a microscope and photographed. The photograph is used to sort the different sizes and shapes of chromosomes into pairs. In most cases of Down syndrome, one extra chromosome 21 will be revealed. The final result of such testing, with the photographed chromosomes paired and organized by shape and size, is called the individual's karyotype.

Treatment

No treatment is available to cure Down syndrome. Treatment is directed at addressing the individual concerns of a particular patient. For example, heart defects will many times require surgical repair, as will duodenal atresia. Many Down syndrome patients will need to wear glasses to correct vision. Patients with hearing impairment benefit from hearing aids.

A drug known as piracetam received some attention in the treatment of Down syndrome patients in the mid-1990s. Piracetam is a so-called "smart drug" that is marketed in Europe and Japan to normal adults hoping to increase their cognitive abilities. It is also sold to skiers and mountain climbers as a remedy for loss of concentration at high altitudes. Although some European researchers have studied piracetam as a possible treatment for dementia in Alzheimer's disease, none of these trials have shown as of 2004 that the drug is of any benefit to Alzheimer's patients. Piracetam is not approved for use in the United States; several large shipments of it were seized by the Food and Drug Administration in 2004. Piracetam can be obtained via the Internet but is not recommended by mainstream medical practitioners.

In 1998 the European company licensed to produce piracetam, UCB Pharma in Belgium, issued a statement discouraging its use in children with Down syndrome. The company obtained orphan drug status for piracetam from the FDA in the early 2000s and has

conducted a controlled trial of the drug as a possible treatment for **muscle spasms** (myoclonus) in children.

While some decades ago, all Down syndrome children were quickly placed into institutions for lifelong care. Research shows very clearly that the best outlook for children with Down syndrome is a normal family life in their own home. This approach, however, requires careful support and education of the parents and the siblings. It is a life-changing event to learn that a new baby has a permanent condition that will effect essentially all aspects of his or her development. Some community groups exist to help families deal with the emotional effects of this new information, and to help plan for the baby's future. Schools are required to provide services for children with Down syndrome, sometimes in separate special education classrooms, and sometimes in regular classrooms. This educational practice is called mainstreaming or inclusion.

Prognosis

The prognosis in Down syndrome is quite variable, depending on the types of complications (heart defects, susceptibility to infections, development of leukemia) of each individual baby. The severity of the retardation can also vary significantly. Without the presence of heart defects, about 90% of children with Down syndrome live into their teens. People with Down syndrome appear to go through the normal physical changes of **aging** more rapidly, however. The average age of **death** for an individual with Down syndrome is about 50–55 years. The most common cause of death is heart disease.

Still, the prognosis for a baby born with Down syndrome in the early 2000s is better than ever before. Because of modern medical treatments, including **antibiotics** to treat infections and surgery to treat heart defects and duodenal atresia, life expectancy has greatly increased. Community and family support allows people with Down syndrome to have rich, meaningful relationships. Because of educational programs, some people with Down syndrome are able to hold jobs.

Men with Down syndrome appear to be uniformly sterile (meaning that they are unable to have offspring). Women with Down syndrome, however, are fully capable of having babies. About 50% of these babies, however, will also be born with Down syndrome.

Prevention

Efforts at prevention of Down syndrome are aimed at genetic counseling of couples who are

KEY TERMS

Chromosome—The structures that carry genetic information. Chromosomes are located within every cell, and are responsible for directing the development and functioning of all the cells in the body. The normal number is 46 (23 pairs).

Karyotype—The specific chromosomal makeup of a particular cell.

Mental retardation—A condition where an individual has a lower-than-normal IQ, and thus is developmentally delayed.

Mosaic—A term referring to a genetic situation, in which an individual's cells do not have the exact same composition of chromosomes. In Down syndrome, this may mean that some of the individual's cells have a normal 46 chromosomes, while other cells have an abnormal 47 chromosomes.

Nondisjunction—A genetic term referring to an event which takes place during cell division, in which a genetic accident causes an egg or sperm cell to have 24 chromosomes, rather than the normal 23.

Orphan drug—A term for a drug that treats a rare disease, defined by the Food and Drug Administration (FDA) as one that affects fewer than 200,000 Americans. The FDA has an Office of Orphan Products Development (OOPD), which offers grants to researchers to develop these products.

Translocation—A genetic term referring to a situation during cell division in which a piece of one chromosome breaks off and sticks to another chromosome.

Trisomy—The condition of having three identical chromosomes instead of the normal two.

preparing to have babies. A counselor needs to inform a woman that her risk of having a baby with Down syndrome increases with her increasing age. Two types of testing is available during a **pregnancy** to determine if the baby being carried has Down syndrome.

Screening tests are used to estimate the chance that an individual woman will have a baby with Down syndrome. At 14–17 weeks of pregnancy, measurements of a substance called AFP (alpha-fetoprotein) can be performed. AFP is normally found circulating in the blood of a pregnant woman, but may be unusually high or low with certain

disorders. Carrying a baby with Down syndrome often causes AFP to be lower than normal. This information alone, or along with measurements of two other hormones, is considered along with the mother's age to calculate the risk of the baby being born with Down syndrome. These results are only predictions, and are only correct about 60% of the time.

The only way to definitively establish (with about 98–99% accuracy) the presence or absence of Down syndrome in a developing baby, is to test tissue from the pregnancy itself. This is usually done either by **amniocentesis** or **chorionic villus sampling** (CVS). In amniocentesis, a small amount of the fluid in which the baby is floating is withdrawn with a long, thin needle. In chorionic villus sampling, a tiny tube is inserted into the opening of the uterus to retrieve a small sample of the placenta (the organ that attaches the growing baby to the mother via the umbilical cord, and provides oxygen and nutrition). Both amniocentesis and CVS allow the baby's own karyotype to be determined. A couple must then decide whether to use this information in order to begin to prepare for the arrival of a baby with Down syndrome, or to terminate the pregnancy.

Once a couple has had one baby with Down syndrome, they are often concerned about the likelihood of future offspring also being born with the disorder. Most research indicates that this chance remains the same as for any woman at a similar age. However, when the baby with Down syndrome has the type that results from a translocation, it is possible that one of the two parents is a carrier of that defect. A carrier conveys the genetic defect to the next generation but does not actually have the disorder. When one parent is a carrier of a translocation, the chance of future offspring having Down syndrome is greatly increased. The specific risk requires evaluation by a genetic counselor.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD, editors. "Congenital Anomalies." Section 19, Chapter 261 In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

Tierney, Lawrence, et al. *Current Medical Diagnosis and Treatment*. Los Altos, CA: Lange Medical Publications, 2001.

PERIODICALS

Chen, Harold, MD. "Down Syndrome." *eMedicine* December 6, 2004. <<http://www.emedicine.com/med/topic567.htm>>.

- Egan, J. F., P. A. Benn, C. M. Zelop, et al. "Down Syndrome Births in the United States from 1989 to 2001." *American Journal of Obstetrics and Gynecology* 191 (September 2004): 1044–1048.
- Evans, J. G., G. Wilcock, and J. Birks. "Evidence-Based Pharmacotherapy of Alzheimer's Disease." *International Journal of Neuropsychopharmacology* 7 (September 2004): 351–369.
- Leonard, H., L. Weaving, P. Eastaugh, et al. "Trisomy 21 and Rett Syndrome: A Double Burden." *Journal of Paediatrics and Child Health* 40 (July 2004): 406–409.
- Prasher, V., A. Farooq, and R. Holder. "The Adaptive Behaviour Dementia Questionnaire (ABDQ): Screening Questionnaire for Dementia in Alzheimer's Disease in Adults with Down Syndrome." *Research in Developmental Disabilities* 25 (July-August 2004): 385–397.
- Tanner, Lindsey. "Study: Drug May Hurt Syndrome Kids." *Chicago: Associated Press* April 12, 2001.
- Tyler, C., and J. C. Edman. "Down Syndrome, Turner Syndrome, and Klinefelter Syndrome: Primary Care throughout the Life Span." *Primary Care* 31 (September 2004): 627–648.

ORGANIZATIONS

- National Down Syndrome Congress. 1605 Chantilly Drive, Suite 250, Atlanta, GA 30324-3269. (800) 232-6372.
- National Down Syndrome Society. 666 Broadway, 8th Floor, New York, NY 10012-2317. (800) 221-4602. <<http://www.ndss.org>>.
- National Organization for Rare Disorders (NORD). 55 Kenosia Avenue, P. O. Box 1968, Danbury, CT 06813-1968. (203) 744-0100. Fax: (203) 798-2291. <<http://www.rarediseases.org>>.
- United States Food and Drug Administration (FDA). 5600 Fishers Lane, Rockville, MD 20857-0001. (888) INFO-FDA. <<http://www.fda.gov>>.

OTHER

- Food and Drug Administration (FDA). "Grants Awarded by the OOPD Program." <<http://www.fda.gov/orphan/grants/previous.htm>>.
- Food and Drug Administration (FDA). "Refusal Actions by FDA as Recorded in OASIS for China (Mainland). August 2004." <http://www.fda.gov/ora/oasis/1/ora_oasis_c_cn.html>.
- Leshin, Lem, MD. "Piracetam and Down Syndrome." <http://www.ds-health.com/piracet.html>.

Kim A. Sharp, M.Ln.
Rebecca J. Frey, PhD

- Down's syndrome see **Down syndrome**
- Doxazosin see **Alpha₁-adrenergic blockers**
- Doxepin see **Antidepressants, tricyclic**
- Doxycycline see **Tetracyclines**

- Dracontiasis see **Guinea worm infection**
- Dracunculiasis see **Guinea worm infection**
- Drooping eyelid see **Ptoxis**
- Drowning see **Near-drowning**
- Drug abuse see **Substance abuse and dependence**
- Drug addiction see **Substance abuse and dependence**
- Drug dependence see **Substance abuse and dependence**

Drug metabolism/interactions

Definition

Drug metabolism is the process by which the body breaks down and converts medication into active chemical substances.

Precautions

Drugs can interact with other drugs, foods, and beverages. Interactions can lessen or magnify the desired therapeutic effect of a drug, or may cause unwanted or unexpected side effects. There are thousands of possible drug-to-drug and drug-to-food interactions, and many medications and supplements are contraindicated (not recommended) under certain conditions or in patients with specific diseases and disorders. This is why it is imperative that patients always keep their physician fully informed about all drugs and dietary supplements (including herbal remedies) they are taking.

Description

The primary site of drug metabolism is the liver, the organ that plays a major role in metabolism, digestion, **detoxification**, and elimination of substances from the body. Enzymes in the liver are responsible for chemically changing drug components into substances known as metabolites. Metabolites are then bound to other substances for excretion through the lungs, or bodily fluids such as saliva, sweat, breast milk, and urine, or through reabsorption by the intestines. The primary mode of excretion is through the kidneys.

The family of liver isoenzymes known as cytochrome P-450 are crucial to drug metabolism. These enzymes (labeled CYP1A2, CYP2C9, CYP2C19, CYP2D6, and CYP3A4) have a catabolic action on substances, breaking them down into metabolites. Consequently, they also act to lower the concentration of medication in the bloodstream.

Drug interactions can occur when one drug inhibits or induces a P-450 that acts on another drug. An example is nicotine, a drug contained in tobacco, and known to induce P-450s. Individuals with **liver disease** (e.g., **cirrhosis**) may also have insufficient levels of P-450 enzymes. As a result, the concentration of drugs metabolized by these enzymes (e.g., amprenavir and other **protease inhibitors**) remains high and can build up to toxic levels in the bloodstream. In addition, certain medications and foods, such as grapefruit juice, can inactivate or lessen the metabolic activity of P-450s. Changing the drug dosage can alleviate the problem in some cases.

The metabolic rate can vary significantly from person to person, and drug dosages that work quickly and effectively in one individual may not work well for another. Factors such as genetics, environment, **nutrition**, and age also influence drug metabolism; infants and elderly patients may have a reduced capacity to metabolize certain drugs, and may require adjustments in dosage.

Causes and symptoms

Drugs that commonly interact with other medications include:

- **Diuretics.** Diuretics such as hydrochlorothiazide can reduce serum potassium and sodium electrolyte levels when taken with digoxin and lithium, respectively.
- **Monoamine oxidase inhibitors (MAOIs).** MAOI antidepressants can cause convulsions and other serious side effects when used with **tricyclic antidepressants** (e.g., Imipramine, Nortriptyline), **selective serotonin reuptake inhibitors** (SSRIs), or sympathomimetic drugs (e.g., amphetamines).
- **Antibiotics.** Antibiotics may reduce the efficiency of oral contraceptives.
- **Metals.** Medications containing metals, such as **antacids** with aluminum additives and iron supplements, can reduce the absorption of **tetracyclines** and fluoroquinolones.
- **Drugs that inhibit liver enzyme function.** Drugs that slow drug metabolism include ciprofloxacin,

erythromycin, fluoxetine, nefazodone, paroxetine, and ritonavir. The therapeutic effect of other medications taken with these drugs may be amplified. Warfarin, a blood thinner, should be used with great caution in individuals taking these drugs.

Foods and beverages that may interact with drugs include:

- **Grapefruit juice.** Grapefruit juice inhibits the metabolism of many medications, including cyclosporine, felodipine, nifedipine, nitrendipine, nisoldipine, carbamazepine, triazolam, and midazolam.
- **Foods and beverages with tyramines.** Red wine, malted beers, smoked foods (e.g., fish and meats), dried fruits, and aged cheeses may contain tyramines, and can cause a severe and dangerous elevation in blood pressure when taken with MAOI inhibitors (a class of antidepressants).
- **Dairy products.** Milk, cream, and other dairy products containing calcium can prevent the absorption of antibiotics such as tetracycline, doxycycline, and ciprofloxacin when they are taken with the drug. In addition, whole milk with vitamin D can cause milk-alkali syndrome in patients taking aluminum hydroxide antacids.
- **Caffeinated beverages.** The **caffeine** contained in coffee and colas can influence drug metabolism.
- **Alcohol.** Alcohol is a central nervous system depressant, and should not be taken with other CNS depressants (e.g., antipsychotics, **antihistamines**). In addition, certain fermented beverages may contain tyramines.

This list is not all-inclusive and individuals should always let their doctor and pharmacist know when they are taking other medications, herbal remedies, or dietary supplements. Anyone who experiences a serious reaction to a drug that is not consistent with its product labeling should report the event to their doctor and/or the MedWatch adverse event reporting system of the United States Food and Drug Administration (FDA).

Alternative treatment

The growing use of herbal supplements has also increased the opportunity for adverse drug and herbal interactions. In 2000, the FDA issued a warning on the popular herb **St. John's wort** (*Hypericum perforatum*). The supplement was found to inhibit the effect of indinavir, a protease inhibitor used in the treatment of HIV. It may also affect the action of cyclosporine and other protease inhibitors (e.g., amprenavir,

ritonavir). Further clinical studies are still necessary to determine the full metabolic effects of the herb.

Other herbs which may interact with allopathic medications include ginkgo bilboa, ginseng, and garlic, which may all heighten the blood thinning effect of the anticoagulant warfarin. Because herbs are regulated by the FDA as dietary supplements, they do not require the same extensive clinical trials and premarket testing as drugs do before they are cleared for sale in the United States. As such, there is still much to learn about the potential interactions and adverse effects associated with herbal supplements. Individuals who experience serious side effects from dietary supplements should report them to FDA's MedWatch program.

Diagnosis

Drug interactions can be difficult to detect. In some cases, adverse reactions may closely resemble the symptoms of the disease or condition the medication was prescribed to treat. Patients who take a number of medications or self-treat with over-the-counter drugs and/or herbal remedies may not be able to determine which drug actually triggered the interaction. A 2001 study by University of Florida researchers found that less than half of the women participating disclosed their use of herbal therapies to their healthcare providers. In cases where a serious drug or herb interaction occurs, withholding this information can delay diagnosis and put the patient at increased risk.

Treatment

Treatment of a drug interaction is dependant on a number of factors, including the medication(s) or supplements used and the medical history of the patient. A dosage adjustment may reverse the effects of some interactions. Serious or life-threatening interactions will require more aggressive therapies.

Prevention

Patients with chronic health conditions, particularly those with liver disorders, should always inform their healthcare professional before taking any over-the-counter (OTC) medications or dietary supplements. Because of the risk for a drug-to-drug interaction, individuals should also let their doctor know if they are taking drugs prescribed by other physicians. Individuals should closely follow instructions for use and package directions on both prescription and over-the-counter drugs.

KEY TERMS

Catabolism—A process of metabolism that breaks down complex substances into simple ones.

Cirrhosis—Liver disease characterized by the widespread disruption of the normal liver structure and function.

CNS depressant—Anything that depresses, or slows, the sympathetic impulses of the central nervous system (i.e., respiratory rate, heart rate).

Drug interaction—A chemical or physiological reaction that can occur when two different drugs are taken together.

Enzymes—Organic substances (proteins) composed of amino acids that trigger and regulate chemical reactions in the body. There are over 700 identified human enzymes.

Liver—A solid organ located on the right in the upper abdomen. It plays a major role in metabolism, digestion, detoxification, and elimination of substances from the body.

Metabolism—The sum of all the physical and chemical processes occurring in the body to organize and maintain life.

Metabolites—Substances produced by metabolism or by a metabolic process.

Milk-alkali syndrome—Elevated blood calcium levels and alkalosis caused by excessive intake of milk and alkalis. Usually occurs in the treatment of peptic ulcer.

Consulting with a pharmacist and/or physician may be beneficial if package directions are unclear to the patient.

As a rule, grapefruit juice should not be taken with medication unless recommended by a doctor. Patients taking MAOI inhibitors should always check food and beverage labels to ensure tyramines are not included, and should avoid all fermented drinks.

Resources

BOOKS

Beers, Mark H., and Robert Berkow. *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck & Co., Inc., 2004.

Medical Economics Company. *The Physicians Desk Reference (PDR)*. 55th ed. Montvale, NJ: Medical Economics Company, 2001.

PERIODICALS

Hardy, Mary L. "Herb-Drug Interactions: An Evidence-Based Table." *Internal Medicine Alert* 23 (January 29, 2001): 1.

ORGANIZATIONS

United States Food and Drug Administration (FDA).
MedWatch Adverse Events Reporting Program. 5600
Fishers Lane, Rockville, MD 20852-9787. 800-FDA-
1088. <<http://www.fda.gov/medwatch/>>.

Paula Anne Ford-Martin

Drug overdose

Definition

A drug overdose is the accidental or intentional use of a drug or medicine in an amount that is higher than is normally used.

Description

All drugs have the potential to be misused, whether legally prescribed by a doctor, purchased over-the-counter at the local drug store, or bought illegally on the street. Taken in combination with other drugs or with alcohol, even drugs normally considered safe can cause **death** or serious long term consequences. Children are particularly at risk for accidental overdose, accounting for over one million poisonings each year from drugs, alcohol, and other chemicals and toxic substances. People who suffer from depression and who have suicidal thoughts are also at high risk for drug overdose.

Causes and symptoms

Accidental drug overdose may be the result of misuse of prescription medicines or commonly used medications like **pain** relievers and cold remedies. Symptoms differ depending on the drug taken. Some of the drugs commonly involved in overdoses are listed below along with symptoms and outcomes.

Acetaminophen is the generic name for the commonly used pain reliever Tylenol. Overdose of this drug causes liver damage with symptoms that include loss of appetite, tiredness, **nausea and vomiting**, paleness, and sweating. The next stage of symptoms indicates liver failure and includes abdominal pain and tenderness, swelling of the liver, and abnormal blood tests for liver enzymes. In the last stage of this

poisoning, liver failure advances and the patient becomes jaundiced, with yellowing of the skin and whites of the eyes. They may also experience kidney failure, bleeding disorders, and encephalopathy (swelling of the brain).

Anticholinergic drugs (drugs that block the action of acetylcholine, a neurotransmitter) like atropine, scopolamine, belladonna, **antihistamines**, and antipsychotic agents cause the skin and moist tissues (like in the mouth and nose) to become dry and flushed. Dilated pupils, an inability to urinate, and mental disturbances are also symptoms. Severe toxicity can lead to seizures, abnormal heart rhythms, extremely high blood pressure, and **coma**.

Antidepressant drugs like amitriptyline, desipramine, and nortriptyline can cause irregular heart rate, **vomiting**, low blood pressure, confusion, and seizures. An overdose of antidepressants also causes symptoms similar to those seen with anticholinergic drug overdoses.

Cholinergic drugs (drugs that stimulate the parasympathetic nervous system) like carbamate and pilocarpine cause **nausea, diarrhea**, increased secretion of body fluids (sweat, tears, saliva, and urine), **fatigue**, and muscle weakness. Convulsions are possible. Death can occur due to **respiratory failure** and **heart failure**.

Cocaine and crack cocaine overdoses cause seizures, high blood pressure, increased heart rate, **paranoia**, and other changes in behavior. **Heart attack** or **stroke** are serious risks within three days after cocaine overdose.

Depressant drugs (tranquilizers, **antianxiety drugs**, sleeping pills) cause sleepiness, slowed or slurred speech, difficulty walking or standing, blurred vision, impaired ability to think, disorientation, and mood changes. Overdose symptoms can include slowed breathing, very low blood pressure, stupor, coma, **shock**, and death.

Digoxin, a drug used to regulate the heart, can cause irregular heart beats, nausea, confusion, loss of appetite, and blurred vision.

Narcotics or opiates are drugs like heroin, morphine, and codeine. Clonidine and diphenoxylate (Lomotil) are also in this category. Overdose with opiate drugs causes **sedation** (sleepiness), low blood pressure, slowed heart rate, and slowed breathing. Pinpoint pupils, where the black centers of the eyes become smaller than normal, are common in opiate overdose. However, if other drugs are taken at the same time as the opiates, they may counteract this

effect on the pupils. A serious risk is that the patient will stop breathing.

Salicylates are found in **aspirin** and some creams or ointments used for muscle and joint pain (like Ben-Gay), and creams for **psoriasis**, a skin condition. Initial symptoms are gastrointestinal irritation, **fever**, and vomiting, possibly with blood in the vomit. This overdose will cause **metabolic acidosis** and **respiratory alkalosis**, conditions where the body's acid/base balance is malfunctioning. Symptoms include rapid heart beat and fast breathing. Nervous system symptoms include confusion, **hallucinations**, tiredness, and ringing in the ears. An increased tendency to bleed is also common. Serious complications include acute renal failure, coma, and heart failure. Acute salicylate poisoning can lead to death.

Diagnosis

Diagnosis of a drug overdose may be based on the symptoms that develop, however, the drug may do extensive damage to the body before significant symptoms develop. If the patient is conscious, he or she may be able to tell what drugs were taken and in what amounts. The patient's recent medical and social history may also help in a diagnosis. For example, a list of medications that the patient takes, whether or not alcohol was consumed recently, even if the patient has eaten in the last few hours before the overdose, can be valuable in determining what was taken and how fast it will be absorbed into the system.

Different drugs have varying effects on the body's acid/base balance and on certain elements in the blood like potassium and calcium. Blood tests can be used to detect changes in body chemistry that may give clues to what drugs were taken. Blood can also be screened for various drugs in the system. Once the overdose drug is identified, blood tests can be used to monitor how fast the drug is being cleared out of the body. Urine tests can also be used to screen for some drugs and to detect changes in the body's chemistry. Blood and urine tests may show if there is damage to the liver or kidneys as a result of the overdose.

Treatment

Immediate care

If a drug overdose is discovered or suspected, and the person is unconscious, having convulsions, or is not breathing, call for emergency help immediately. If the person who took the drug is not having symptoms, do not wait to see if symptoms develop; call a poison control center immediately. Providing as much

information as possible to the poison control center can help determine what the next course of action should be.

The poison control center, paramedics, and emergency room staff will want to know:

- What drug(s) were taken try to locate the drug's container?
- How much of the drug was taken?
- When was the drug taken?
- Was the drug taken with alcohol or any other drugs or chemicals?
- What is the age of the patient?
- What symptoms are the patient experiencing?
- Is the patient conscious?
- Is the patient breathing?

The poison control center may recommend trying to get the patient to vomit. A liquid called **ippecac** syrup, which is used to induce vomiting, is available from pharmacies without a prescription. Pediatricians may recommend that families keep ippecac syrup on hand in households with children. This medication should be used only on the advice of a medical professional. Vomiting should not be induced if the patient is unconscious.

Emergency care

Emergency medical treatment may include:

- Assessment of the patient's airway and breathing to making sure that the trachea, the passage to the lungs, is not blocked. If needed, a tube may be inserted through the mouth and into the trachea to help the patient breath. This procedure is called intubation.
- Assessment of the patient's heart rate, blood pressure, body temperature, and other physical signs that might indicate the effects of the drug.
- Blood and urine samples may be collected to test for the presence of the suspected overdose drug, and any other drugs or alcohol that might be present.
- Elimination of the drug that has not yet been absorbed is attempted. Vomiting may be induced using ippecac syrup or other drugs that cause vomiting. Ipecac syrup should not be given to patients who overdosed with **tricyclic antidepressants**, theophylline, or any drug that causes a significant change in mental status. If a patient vomits while unconscious, there is a serious risk of **choking**.

- Gastric lavage, or washing out the stomach, may be attempted. For this procedure a tube flexible tube is inserted through the nose, down the throat, and into the stomach. The contents of the stomach are then suctioned out through the tube. A solution of saline (salt water) is injected into the tube to rinse out the stomach. This solution is then suctioned out. This is the process used when someone has his/her stomach pumped.
- Activated charcoal is sometimes given to absorb the drug.
- Medication to stimulate urination or defecation may be given to try to flush the excess drug out of the body faster.
- Intravenous (IV) fluids may be given. An intravenous line, a needle inserted into a vein, may be put into the arm or back of the hand. Fluids, either sterile saline (salt water solution) or dextrose (sugar water solution), can be administered through this line. Increasing fluids can help to flush the drug out of the system and to reestablish balance of fluids and **minerals** in the body. The pH (acid/base balance) of the body may need to be corrected by administering electrolytes like sodium, potassium, and bicarbonate through this IV line. If drugs need to be administered quickly, they can also be injected directly into the IV line.
- Hemodialysis is a procedure where blood is circulated out of the body, pumped through a dialysis machine, then reintroduced back into the body. This process can be used to filter some drugs out of the blood. It may also be used temporarily or long term if the kidneys are damaged due to the overdose.
- Antidotes are available for some drug overdoses. An antidote is another drug that counteracts or blocks the overdose drug. For example, acetaminophen overdose can be treated with an oral medication, N-acetylcysteine (Mucomyst), if the level of acetaminophen found in the blood is extremely high. Naloxone is an anti-narcotic drug that is given to counteract narcotic poisoning. Nalmefen or **methadone** may also be used.
- Psychiatric evaluation may be recommended if the drug overdose was taken deliberately.

Prognosis

While many victims of drug overdose recover without long term effects, there can be serious consequences. Some drug overdoses cause the failure of major organs like the kidneys or liver, or failure of whole systems like the respiratory or circulatory systems. Patients who survive drug overdose may need **kidney dialysis**, kidney or liver transplant, or ongoing

KEY TERMS

Gastric lavage—Also called a stomach pump. For this procedure, a flexible tube is inserted through the nose, down the throat, and into the stomach and the contents of the stomach are suctioned out. The inside of the stomach is rinsed with a saline (salt water) solution.

Intubation—A procedure where a tube is inserted through the mouth and into the trachea keep the airway open and to help the patient breathe.

care as a result of heart failure, stroke, or coma. Death can occur in almost any drug overdose situation, particularly if treatment is not started immediately.

Prevention

To protect children from accidental drug overdose, all medications should be stored in containers with child resistant caps. All drugs should be out of sight and out of reach of children, preferably in a locked cabinet. Prescription medications should be used according to directions and only by the person whose name is on the label. Threats of **suicide** need to be taken seriously and appropriate help sought for people with depression or other mental illness that may lead to suicide.

Resources

OTHER

“Drug Overdose.” *American Institute of Preventative Medicine*. <<http://www.healthy.net>> .

Graber, Mark A., and Rhea Allen. “Emergency Medicine: Overdose and Toxidromes.” *University of Iowa Family Practice Handbook*. <<http://www.vh.org/Providers/ClinRef/FPHandbook/Chapter01/20-1.html>> .

Altha Roberts Edgren

Drug therapy monitoring

Definition

Drug therapy monitoring, also known as Therapeutic Drug Monitoring (TDM), is a means of monitoring drug levels in the blood.

Purpose

TDM is employed to measure blood drug levels so that the most effective dosage can be determined, with toxicity prevented. TDM is also utilized to identify noncompliant patients (those patients who, for whatever reason, either cannot or will not comply with drug dosages as prescribed by the physician).

Precautions

Because so many different factors influence blood drug levels, the following points should be taken into consideration during TDM: the age and weight of the patient; the route of administration of the drug; the drug's absorption rate, excretion rate, delivery rate, and dosage; other medications the patient is taking; other diseases the patient has; the patient's compliance regarding the drug treatment regimen; and the laboratory methods used to test for the drug.

Description

TDM is a practical tool that can help the physician provide effective and safe drug therapy in patients who need medication. Monitoring can be used to confirm a blood drug concentration level that is above or below the therapeutic range, or if the desired therapeutic effect of the drug is not as expected. If this is the case, and dosages beyond normal then have to be prescribed, TDM can minimize the time that elapses.

TDM is important for patients who have other diseases that can affect drug levels, or who take other medicines that may affect drug levels by interacting with the drug being tested. As an example, without drug monitoring, the physician cannot be sure if a patient's lack of response to an antibiotic reflects bacterial resistance, or is the result of failure to reach the proper therapeutic range of antibiotic concentration in the blood. In cases of life-threatening infections, timing of effective antibiotic therapy is critical to success. It is equally crucial to avoid toxicity in a seriously ill patient. Therefore, if toxic symptoms appear with standard dosages, TDM can be used to determine changes in dosing.

Drawn blood, used for TDM, demonstrates a drug action in the body at any specific time, whereas drug levels examined from urine samples reflect the presence of a drug over many days (depending on the rate of excretion). Therefore, blood testing is the procedure of choice when definite data are required.

Therapeutic Drug Monitoring: Therapeutic And Toxic Range

Drug Level*	Use	Therapeutic Level*	Toxic
Acetaminophen mg/ml	Analgesic, antipyretic	Depends on use	>250
Amikacin mg/ml	Antibiotic	12–25 mg/ml**	>25
Aminophylline ng/ml	Bronchodilator	10–20 mg/ml	>20
Amitriptyline ng/ml	Antidepressant	120–150 ng/ml	>500
Carbamazepine mg/ml	Anticonvulsant	5–12 mg/ml	>12
Chloramphenicol mg/ml	Antibiotic	10–20 mg/ml	>25
Digoxin ng/ml	Cardiotonic	0.8–2.0 ng/ml	>2.4
Gentamicin	Antibiotic	4–12 mg/L	>12 mg/L
Lidocaine	Antiarrhythmic	1.5–5.0 mg/ml	>5 mg/ml
Lithium mEq/L	Antimanic	0.7–2.0 mEq/L	>2.0
Nortriptyline ng/ml	Antidepressant	50–150 ng/ml	>500
Phenobarbital mg/ml	Anticonvulsant	10–30 mg/ml	>40
Phenytoin mg/ml	Anticonvulsant	7–20 mg/ml	>30
Procainamide mg/ml	Antiarrhythmic	4–8 mg/ml	>16
Propranolol ng/ml	Antiarrhythmic	50–100 ng/ml	>150
Quinidine mg/ml	Antiarrhythmic	1–4 mg/ml	>10
Theophylline mg/ml	Bronchodilator	10–20 mg/ml	>20
Tobramycin mg/ml	Antibiotic	4–12 mg/ml**	>12
Valproic acid mg/ml	Anticonvulsant	50–100 mg/ml	>100

*Values are laboratory-specific

**Concentration obtained 30 minutes after the end of a 30-minute infusion.

However, for adequate absorption and therapeutic levels to be accurate, it is important to allow for sufficient time to pass between the administration of the medication and the collection of the blood sample.

Blood specimens for drug monitoring can be taken at two different times: during the drug's highest therapeutic concentration ("peak" level), or its lowest ("trough" level). Occasionally called residual levels, trough levels show sufficient therapeutic levels; whereas peak levels show poisoning (toxicity). Peak and trough levels should fall within the therapeutic range.

Preparation

In preparing for this test, the following guidelines should be observed:

- Depending on the drug to be tested, the physician should decide if the patient is to be fasting (nothing to eat or drink for a specified period of hours) before the test.
- For patients suspected of symptoms of drug toxicity, the best time to draw the blood specimen is when the symptoms are occurring.
- If there is a question as to whether an adequate dose of the drug is being achieved, it is best to obtain trough (lowest therapeutic concentration) levels.
- Peak (highest concentration) levels are usually obtained one to two hours after oral intake, approximately one hour after intramuscular (IM) administration (a shot in the muscle), and approximately 30 minutes after intravenous (IV) administration. Residual, or trough, levels are usually obtained within 15 minutes of the next scheduled dose.

Risks

Risks for this test are minimal, but may include slight bleeding from the blood-drawing site, fainting or feeling lightheaded after blood is drawn, or accumulation of blood under the puncture site (hematoma).

Resources

BOOKS

Pagana, Kathleen Deska. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis: Mosby, Inc., 1998.

Janis O. Flores

Drugs used in labor

Definition

These drugs are used to induce (start) or continue labor.

Purpose

The drug described here, oxytocin, makes the uterus (womb) contract. Physicians use it to deliberately start labor. Because there are some risks with using oxytocin, this should be done only when there are good medical reasons. Any woman who is being given oxytocin should make sure she has discussed the benefits and risks with her physician.

Oxytocin also may be used to control bleeding after delivery or to help make the milk flow in women who are breastfeeding their babies.

Description

Oxytocin is a hormone and is available only with a physician's prescription. When used to start or continue labor, it is slowly injected into a vein. A nasal spray form is used to increase milk flow in breastfeeding. Some commonly used brand names are Pitocin and Syntocinon.

Recommended dosage

The dosages given here are average doses. However, doses may be different for different patients. Follow the orders of the physician who prescribed the drug.

For increasing milk production:

One spray into one or both nostrils, two–three minutes before nursing or using a breast pump.

For starting or continuing labor:

The physician in charge will determine the appropriate dose.

Precautions

Oxytocin does not help increase or continue labor in all patients. When it does not help, the physician may deliver the baby by **cesarean section**.

In women who are especially sensitive to oxytocin, the drug may cause contractions to become too strong. This could tear the uterus or deprive the fetus of blood and oxygen during labor.

Oxytocin does not help improve milk flow in all women who are breastfeeding. Check with a physician if the drug does not seem to be working.

Women with heart disease, high blood pressure, or **kidney disease** should let their physicians know about these conditions before taking oxytocin. Also, anyone who has had an unusual reaction to oxytocin in the past should inform their physician.

Side effects

Oxytocin has caused irregular heartbeat and increased bleeding in some women after delivery. It may also cause **jaundice** (yellowing of the eyes and skin) in newborns.

KEY TERMS

Cesarean section—The delivery of a baby through a surgical procedure.

Fetus—A developing baby inside the womb.

Hormone—A substance that is produced in one part of the body, then travels through the bloodstream to another part of the body where it has its effect.

Other side effects are rare, but may include **nausea**, **vomiting**, confusion, **dizziness**, convulsions, breathing problems, **headache**, **hives**, skin rash, **itching**, pelvic or abdominal **pain**, and weakness. The nasal spray form may cause watery eyes or irritation of the nose.

Interactions

Anyone who takes oxytocin should let the physician know all other medicines she is taking.

Nancy Ross-Flanigan

Dry mouth

Definition

Dry mouth, known medically as xerostomia, is the abnormal reduction of saliva due to medication, disease, or medical therapy.

Description

Dry mouth due to the lack of saliva can be a serious medical problem. Decreased salivation can make swallowing difficult, can decrease taste sensation, and can promote **tooth decay**.

Causes and symptoms

Dry mouth, resulting from thickened or reduced saliva flow, can be caused by a number of factors: medications, both prescription and over-the-counter; such systemic diseases as anemia, HIV infection, or diabetes; manifestations of **Sjögren's syndrome** (as rheumatoid arthritis, lupus, chronic hardening and thickening of the skin, or chronic and progressive

inflammation of skeletal muscles); infections of the salivary glands; blockage of the salivary ducts caused by stones or tumors forming in the ducts through which the saliva passes; **dehydration**; such medical therapies as local surgery or radiation; secretion reduction normally involved in the aging process; and emotional **stress**.

Diagnosis

The diagnosis of dry mouth is not difficult. The patient will state that his or her saliva is very thick or nonexistent. Finding the cause of dry mouth may be more difficult and require some laboratory testing. Salivary gland biopsy for stones or tumors should be performed if indicated.

Treatment

The treatment of dry mouth involves the management of the condition causing it. If dry mouth is caused by medication, the medication should be changed. If dry mouth is caused by blockage of the salivary ducts, the cause of the blockage should be investigated. When systemic diseases, such as diabetes and anemia, are brought under control dry mouth problems may decrease.

The use of caffeine-containing beverages, alcoholic beverages, and mouthwashes containing alcohol should be minimized. The drinking of water and fruit juices will decrease dry mouth problems. Chewing gum and lemon drops can be used to stimulate saliva flow. Bitters also can initiate salivary flow as long as the salivary glands and ducts are functional. Commercial saliva substitutes are available without prescription and can be used as frequently as needed. Use of a humidifier in the bedroom reduces nighttime oral dryness.

Dry mouth caused by the **aging** process or **radiation therapy** for **cancer** can be treated by such oral medications as pilocarpine (Salagen). Drugs that are given to increase the flow of saliva are known as sialogogues.

Prognosis

The prognosis for patients with xerostomia due to medication problems is good, if the offending agent can be changed. Dry mouth due to systemic problems may be eliminated or improved once the disease causing the dry mouth is under control. Persistent xerostomia can be managed well with saliva substitutes.

KEY TERMS

Salivary duct—Tube through which saliva is carried from the salivary gland to the mouth.

Salivary gland—Gland in which saliva forms.

Sialogogue—A medication given to increase the flow of saliva.

Xerostomia—The medical term for dry mouth.

Prevention

A patient needs to ask his or her health care provider if any medication to be prescribed will cause dry mouth. Patients with persistent xerostomia need to practice good **oral hygiene** and visit a dentist on a regular basis; the lack of adequate saliva can cause severe dental decay. The salivary glands are very sensitive to radiation, so any patient scheduled for radiation therapy of the head and neck needs to discuss with the radiation therapist ways to minimize exposure of the salivary glands to radiation.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD, editors. "Dentistry in Medicine." Section 9, Chapter 103 In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

PERIODICALS

Bruce, S. D. "Radiation-Induced Xerostomia: How Dry Is Your Patient?" *Clinical Journal of Oncology Nursing* 8 (February 2004): 61–67.

Nagler, R. M. "Salivary Glands and the Aging Process: Mechanistic Aspects, Health-Status and Medicinal - Efficacy Monitoring." *Biogerontology* 5 (March 2004): 223–233.

Pinto, A., and S. S. De Rossi. "Salivary Gland Disease in Pediatric HIV Patients: An Update." *Journal of Dentistry for Children (Chicago)* 71 (January-April 2004): 33–37.

Porter, S. R., C. Scully, and A. M. Hegarty. "An Update of the Etiology and Management of Xerostomia." *Oral Surgery, Oral Medicine, Oral Pathology, Oral Radiology, and Endodontics* 97 (January 2004): 28–46.

ORGANIZATIONS

American Dental Association. 211 E. Chicago Ave., Chicago, IL 60611. (312) 440-2500. <<http://www.ada.org>>.

American Medical Association. 515 N. State St., Chicago, IL 60612. (312) 464-5000. <<http://www.ama-assn.org>>.

Joseph Knight, PA
Rebecca J. Frey, PhD

Dry skin see **Ichthyosis**

Dual energy x-ray absorptiometry (DXA) scan see **Bone density test**

DUB see **Dysfunctional uterine bleeding**

Duchenne muscular dystrophy see **Muscular dystrophy**

Duodenal atresia see **Duodenal obstruction**

Duodenal obstruction

Definition

Duodenal obstruction is a failure of food to pass out of the stomach either from a complete or partial obstruction.

Description

The duodenum is the first part of the intestine, into which the stomach, the gall bladder, and the pancreas empty their contents. The pylorus connects the duodenum with the stomach and contains the valve that regulates stomach emptying. Obstruction usually occurs right at this outlet, so that the gall bladder and pancreas are unable to drain their secretions without hindrance.

Causes and symptoms

Obstruction of the duodenum occurs in adults and infants, each for a different set of reasons. In adults, the usual cause is a peptic ulcer of such antiquity that repeated cycles of injury and scarring have narrowed the passageway. Medical treatment of ulcers has progressed to the point where such obstinate ulcer disease is rarely seen any more. In infants, the conditions are congenital—either the channel is underdeveloped or the pylorus is overdeveloped. The first type is called duodenal hypoplasia and the second is termed hypertrophic **pyloric stenosis**. In rare cases, the channel may be missing altogether, a condition called duodenal atresia. To say that these anomalies are

congenital is not to say their cause is understood. As with most **birth defects**, the specific cause is not known.

Food that cannot exit the stomach in the forward direction will return whence it came. **Vomiting** is the constant symptom of duodenal obstruction. It may be preceded by **indigestion** and **nausea** as the stomach attempts to squeeze its contents through an ever narrowing outlet.

Hypertrophic pyloric stenosis appears soon after birth. The infant will vomit feedings, lose weight, and be restless and irritable.

Diagnosis

X rays taken with contrast material in the stomach readily demonstrate the site of the blockage and often the ulcer that caused it. Gastroscopy is another way to evaluate the problem. In infants, x rays may not be necessary to detect pyloric stenosis. It is often possible to feel the enlarged pylorus, like an olive, deep under the ribs and see the stomach rippling as it labors to force food through.

Treatment

Bowel obstruction requires a surgeon, sometimes immediately. Newer surgical techniques constantly improve the outcome, but obstruction is a mechanical problem that needs a mechanical solution. Most adults who come to surgery for obstruction have suffered for years from peptic ulcer disease. They will usually benefit from **ulcer surgery** at the same time their obstruction is relieved. The surgeon will therefore select a procedure that combines relief of obstruction with remedy for ulcer disease. There are many choices. In fact, even without obstruction, functional considerations require ulcer surgery to include enhancement of stomach emptying.

To treat an infant with hypertrophic pyloric stenosis, some surgeons have had success with forceful balloon dilation of the pylorus done through a gastroscope, but the standard procedure is to cut across the overdeveloped circular muscle that is constricting the stomach outlet. There are reports of infant hypertrophic pyloric stenosis remitting without surgery following a very careful feeding schedule, but mortality is unacceptably high.

Prognosis

A functioning and unrestricted intestine is a prerequisite for living independent of the most

KEY TERMS

Atresia—Failure to develop; complete absence.

Contrast agent—A substance that produces shadows on an x ray so that hollow structures can be more easily seen.

Gastroscopy—Looking into the stomach with a flexible viewing instrument called a gastroscope.

Hypoplasia—Incomplete development.

Peptic ulcer—A wound in the lower stomach and duodenum caused by stomach acid and a newly discovered germ called *Helicobacter pylori*.

advanced and continuous medical care available. Achieving this desirable goal is the rule with surgery for duodenal obstructions of all types. The bowel is so malleable that there is a rearrangement to suit every occasion. The variety of possible configurations is limited only by the surgeon's imagination.

Prevention

Prompt and effective treatment of peptic ulcers will prevent chronic scarring and narrowing. Drugs developed over the past few decades have all but eliminated the need for ulcer surgery.

Resources

BOOKS

Redel, Carol A., and R. Jeff Zeiwner. "Anatomy and Anomalies of the Stomach and Duodenum." In *Sleisenger & Fordtran's Gastrointestinal and Liver Disease*, edited by Mark Feldman, et al. Philadelphia: W. B. Saunders Co., 1997.

J. Ricker Polsdorfer, MD

Duodenal stenosis see **Duodenal obstruction**

Duodenal ulcers see **Ulcers (digestive)**

Duodenum x rays see **Hypotonic duodenography**

Duplicated ureter see **Congenital ureter anomalies**

Dwarfism see **Achondroplasia; Pituitary dwarfism**

Dysentery

Definition

Dysentery is a general term for a group of gastrointestinal disorders characterized by inflammation of the intestines, particularly the colon. Characteristic features include abdominal **pain** and cramps, straining at stool (tenesmus), and frequent passage of watery **diarrhea** or stools containing blood and mucus. The English word dysentery comes from two Greek words meaning “ill” or “bad” and “intestine.”

It should be noted that some doctors use the word “dysentery” to refer only to the first two major types of dysentery discussed below, while others use the term in a broader sense. For example, some doctors speak of **schistosomiasis**, a disease caused by a parasitic worm, as bilharzial dysentery, while others refer to acute diarrhea caused by viruses as viral dysentery.

Description

Dysentery is a common but potentially serious disorder of the digestive tract that occurs throughout the world. It can be caused by a number of infectious agents ranging from viruses and bacteria to protozoa and parasitic worms; it may also result from chemical irritation of the intestines. Dysentery is one of the oldest known gastrointestinal disorders, having been described as early as the Peloponnesian War in the fifth century B.C. Epidemics of dysentery were frequent occurrences aboard sailing vessels as well as in army camps, walled cities, and other places in the ancient world where large groups of human beings lived together in close quarters with poor sanitation. As late as the eighteenth and nineteenth centuries, sailors and soldiers were more likely to die from the “bloody flux” than from injuries received in battle. It was not until 1897 that a bacillus (rod-shaped bacterium) was identified as the cause of one major type of dysentery.

Dysentery in the modern world is most likely to affect people in the less developed countries and travelers who visit these areas. According to the Centers for Disease Control and Prevention (CDC), most cases of dysentery in the United States occur in immigrants from the developing countries and in persons who live in inner-city housing with poor sanitation. Other groups of people at increased risk of dysentery are military personnel stationed in developing countries, frequent travelers, children in day care centers, people in nursing homes, and men who have sex with other men.

Causes & symptoms

Causes

The most common types of dysentery and their causal agents are as follows:

- **Bacillary dysentery.** Bacillary dysentery, which is also known as **shigellosis**, is caused by four species of the genus *Shigella*: *S. dysenteriae*, the most virulent species and the one most likely to cause epidemics; *S. sonnei*, the mildest species and the most common form of *Shigella* found in the United States; *S. boydii*; and *S. flexneri*. *S. flexneri* is the species that causes Reiter's syndrome, a type of arthritis that develops as a late complication of shigellosis. About 15,000 cases of shigellosis are reported to the CDC each year for the United States; however, the CDC maintains that the true number of annual cases may be as high as 450,000, since the disease is vastly underreported. About 85 percent of cases in the United States are caused by *S. sonnei*. The *Shigella* organisms cause the diarrhea and pain associated with dysentery by invading the tissues that line the colon and secreting an enterotoxin, or harmful protein that attacks the intestinal lining.
- **Amebic dysentery.** Amebic dysentery, which is also called intestinal **amebiasis** and amebic colitis, is caused by a protozoon, *Entamoeba histolytica*. *E. histolytica*, whose scientific name means “tissue-dissolving,” is second only to the organism that causes **malaria** as a protozoal cause of **death**. *E. histolytica* usually enters the body during the cyst stage of its life cycle. The cysts may be found in food or water contaminated by human feces. Once in the digestive tract, the cysts break down, releasing an active form of the organism called a trophozoite. The trophozoites invade the tissues lining the intestine, where they are usually excreted in the patient's feces. They sometimes penetrate the lining itself, however, and enter the bloodstream. If that happens, the trophozoites may be carried to the liver, lung, or other organs. Involvement of the liver or other organs is sometimes called metastatic amebiasis.
- **Balantidiasis, giardiasis, and cryptosporidiosis.** These three intestinal infections are all caused by protozoa, *Balantidium coli*, *Giardia lamblia*, and *Cryptosporidium parvum* respectively. Although most people infected with these protozoa do not become severely ill, the disease agents may cause dysentery in children or immunocompromised individuals. There are about 3,500 cases of cryptosporidiosis reported to the CDC each year in the United States, and about 22,000 cases of giardiasis.

- **Viral dysentery.** Viral dysentery, which is sometimes called traveler's diarrhea or viral **gastroenteritis**, is caused by several families of viruses, including rotaviruses, caliciviruses, astroviruses, **noroviruses**, and adenoviruses. There are about 3.5 million cases of viral dysentery in infants in the United States each year, and about 23 million cases each year in adults. The CDC estimates that viruses are responsible for 9.2 million cases of dysentery related to **food poisoning** in the United States each year. Whereas most cases of viral dysentery in infants are caused by rotaviruses, caliciviruses are the most common disease agents in adults. Noroviruses were responsible for about half of the outbreaks of dysentery on cruise ships reported to the CDC in 2002.
- **Dysentery caused by parasitic worms.** Both whipworm (trichuriasis) and flatworm or fluke (schistosomiasis) infestations may produce the violent diarrhea and abdominal cramps associated with dysentery. Schistosomiasis is the second most widespread tropical disease after malaria. Although the disease is rare in the United States, travelers to countries where it is endemic may contract it. The World Health Organization (WHO) estimates that about 200 million people around the world carry the parasite in their bodies, with 20 million having severe disease.

Symptoms

In addition to the characteristic bloody and/or watery diarrhea and abdominal cramps of dysentery, the various types have somewhat different symptom profiles:

- **Bacillary dysentery.** The symptoms of shigellosis may range from the classical bloody diarrhea and tenesmus characteristic of dysentery to the passage of nonbloody diarrhea that resembles the loose stools caused by other intestinal disorders. The high **fever** associated with shigellosis begins within one to three days after exposure to the organism. The patient may also have pain in the rectum as well as abdominal cramping. The acute symptoms last for three to seven days, occasionally for as long as a month. Bacillary dysentery may lead to two potentially fatal complications outside the digestive tract: **bacteremia** (bacteria in the bloodstream), which is most likely to occur in malnourished children; and hemolytic uremic syndrome, a type of kidney failure that has a mortality rate above 50 percent.
- **Amebic dysentery.** Amebic dysentery often has a slow and gradual onset; most patients with amebiasis visit the doctor after several weeks of diarrhea and bloody stools. Fever is unusual with amebiasis unless the patient has developed a liver **abscess** as a complication of the infection. The most serious complication of amebic dysentery, however, is fulminant or necrotizing colitis, which is a severe inflammation of the colon characterized by **dehydration**, severe abdominal pain, and the risk of perforation (rupture) of the colon.
- **Dysentery caused by other protozoa.** Dysentery associated with giardiasis begins about 1–3 weeks after infection with the organism. It is characterized by bloating and foul-smelling flatus, **nausea and vomiting**, headaches, and low-grade fever. These acute symptoms usually last for three or four days. The symptoms of cryptosporidiosis are mild in most patients but are typically severe in patients with **AIDS**. Diarrhea usually starts between seven and 10 days after exposure to the organism and may be copious. The patient may have pain in the upper right abdomen, **nausea**, and **vomiting**, but fever is unusual.
- **Viral dysentery.** Viral dysentery has a relatively rapid onset; symptoms may begin within hours of infection. The patient may be severely dehydrated from the diarrhea but usually has only a low-grade fever. The diarrhea itself may be preceded by one to three days of nausea and vomiting. The patient's abdomen may be slightly tender but is not usually severely painful.
- **Dysentery caused by parasitic worms.** Patients with intestinal schistosomiasis typically have a gradual onset of symptoms. In addition to bloody diarrhea and abdominal pain, these patients usually have **fatigue**. An examination of the patient's colon will usually reveal areas of ulcerated tissue, which is the source of the bloody diarrhea.

Diagnosis

Patient history and physical examination

The **physical examination** in the primary care doctor's office will not usually allow the doctor to determine the specific parasite or other disease agent that is causing the bloody diarrhea and other symptoms of dysentery, although the presence or absence of fever may help to narrow the diagnostic possibilities. The patient's age and history are usually better sources of information. The doctor may ask about such matters as the household water supply and food preparation habits, recent contact with or employment in a nursing home or day care center, recent visits to tropical countries, and similar questions. The doctor

will also need to know when the patient first noticed the symptoms.

The doctor will also evaluate the patient for signs of dehydration resulting from the loss of fluid through the intestines. Fatigue, drowsiness, dryness of the mucous membranes lining the mouth, low blood pressure, loss of normal skin tone, and rapid heartbeat (above 100 beats per minute) may indicate that the patient is dehydrated.

Laboratory tests

The most common laboratory test to determine the cause of dysentery is a stool sample. The patient should be asked to avoid using over-the-counter **antacids** or antidiarrheal medications until the sample has been collected, as these preparations can interfere with the test results. The organisms that cause cryptosporidiosis, bacillary dysentery, amebic dysentery, and giardiasis can be seen under the microscope, as can the eggs produced by parasitic worms. In some cases repeated stool samples, a sample of mucus from the intestinal lining obtained through a proctoscope, or a tissue sample from the patient's colon may be necessary to confirm the diagnosis. Antigen testing of a stool sample can be used to diagnose a rotavirus infection as well as parasitic worm infestations.

The doctor will also usually order a blood test to evaluate the electrolyte levels in the patient's blood in order to assess the need for rehydration.

Imaging studies

Imaging studies (usually CT scans, x rays, or ultrasound) may be performed in patients with amebic dysentery to determine whether the lungs or liver have been affected. They may also be used to diagnose schistosomiasis, as the eggs produced by the worms will show up on ultrasound or MRI studies of the liver, intestinal wall, or bladder.

Treatment

Medications are the primary form of treatment for dysentery:

- **Bacillary dysentery.** Dysentery caused by *Shigella* is usually treated with such **antibiotics** as trimethoprim-sulfamethoxazole (Bactrim, Septra), nalidixic acid (NegGram), or ciprofloxacin (Cipro, Ciloxan). Because the various species of *Shigella* are becoming resistant to these drugs, however, the doctor may prescribe one of the newer drugs described below. Patients with bacillary dysentery should not be given antidiarrheal medications, including loperamide

(Imodium), paregoric, and diphenolate (Lomotil), because they may make the illness worse.

- **Amebic dysentery.** The most common drugs given for amebiasis are diloxanide furoate (Diloxide), iodoquinol (Diquinol, Yodoxin), and metronidazole (Flagyl). Metronidazole should not be given to pregnant women but paromomycin (Humatin) may be used instead. Patients with very severe symptoms may be given emetine dihydrochloride or dehydroemetine, but these drugs should be stopped once the patient's symptoms are controlled.
- **Dysentery caused by other protozoa.** Balantidiasis, giardiasis, and cryptosporidiosis are treated with the same drugs as amebic dysentery; patients with giardiasis resistant to treatment may be given albendazole (Zentel) or furazolidone (Furoxone).
- **Viral dysentery.** The primary concern in treating viral dysentery, particularly in small children, is to prevent dehydration. Antinausea and antidiarrhea medications should not be given to small children. Probiotics, including *Lactobacillus casei* and *Saccharomyces boulardii*, have been shown to reduce the duration and severity of viral diarrhea in small children by 30–70 percent.
- **Dysentery caused by parasitic worms.** Whipworm infestations are usually treated with anthelmintic medications, most commonly mebendazole (Vermox). Schistosomiasis may be treated with praziquantel (Biltricide), metrifonate (Trichlorfon), or oxamniquine, depending on the species causing the infestation.

Newer drugs that have been developed to treat dysentery include tinidazole (Tindamax, Fasigyn), an antiprotozoal drug approved by the Food and Drug Administration (FDA) in 2004 to treat giardiasis and amebiasis in adults and children over the age of three years. This drug should not be given to women in the first three months of **pregnancy**. In addition, adults taking tinidazole should not drink alcoholic beverages while using it, or for three days after the end of treatment. The other new drug is nitazoxanide (Alinia), another antiprotozoal medication that has the advantage of lacking the bitter taste of metronidazole and tinidazole.

Fluid replacement is given if the patient has shown signs of dehydration. The most common treatment is an oral rehydration fluid containing a precise amount of salt and a smaller amount of sugar to replace electrolytes as well as water lost through the intestines. Infalyte and Pedialyte are oral rehydration fluids formulated for the special replacement needs of infants and young children.

Surgery

Surgery is rarely necessary in treating dysentery, but may be required in cases of fulminant colitis, particularly if the patient's colon has perforated. Patients with liver abscesses resulting from amebic dysentery may also require emergency surgery if the abscess ruptures. In some cases exploratory surgery may be needed to determine whether severe abdominal pain is caused by schistosomiasis, amebic dysentery, or **appendicitis**.

Alternative treatments

There are a number of alternative treatments for dysentery, most of which are derived from plants used by healers for centuries. Because dysentery was known to ancient civilizations as well as modern societies, such alternative systems as **traditional Chinese medicine** (TCM) and **Ayurvedic medicine** developed treatments for it.

Ayurvedic medicine

Ayurvedic medicine recommends fruits and herbs, specifically cumin seed, bael fruit (*Aegle marmelos*, also known as Bengal quince), and arjuna (*Terminalia arjuna*) bark for the treatment of dysentery. Ayurvedic practitioners may also give the patient dietary supplements known as Isabbael, Lashunadi Bati, and Bhuwaneshwar Ras. To rehydrate the body, adult patients may be given a combination of slippery elm water and barley to drink, at least a pint per day.

Traditional Chinese medicine

To treat dysentery, traditional Chinese doctors use astringent drugs, which are intended to constrict or tighten mucous membranes and other body tissues to slow down fluid loss. Myrobalan fruit (*Terminalia chebula*), nut galls (swellings produced on the leaves and stems of oak trees by the secretions of certain insects), and opium extracted from the opium poppy (*Papaver somniferum*) are the natural materials most commonly used. Paregoric, a water-based solution of morphine that is still used in the West to treat diarrhea, is derived from the opium poppy.

Other plant-based remedies

Researchers in Mexico reported in early 2005 that the roots of *Geranium mexicanum*, a plant that produces a sap traditionally used to treat coughs or diarrhea, contains compounds that are active against both *Giardia lamblia* and *Entamoeba histolytica*. Plant biologists in Africa are studying the effectiveness of

African mistletoe (*Tapinanthus dodoneifolius*), a traditional remedy for dysentery among the Hausa and Fulani tribes of Nigeria.

Dietary supplements

A study published in the *American Journal of Clinical Nutrition* in early 2005 reported that supplemental zinc (twice the recommended daily dietary allowance) boosts the body's immune response during acute shigellosis.

Homeopathy

There are at least ten different homeopathic remedies used to treat diarrhea. Contemporary homeopaths, however, distinguish between diarrhea that can be safely treated at home with such homeopathic remedies as *Podophyllum*, *Veratrum album*, *Bryonia*, and *Arsenicum*, and diarrhea that indicates dysentery and should be referred to a physician. Signs of dehydration (loss of normal skin texture, **dry mouth**, sunken eyes), severe abdominal pain, blood in the stool, and unrelieved vomiting are all indications that mainstream medical care is required.

Prognosis

Most adults in developed countries recover completely from an episode of dysentery. Children are at greater risk of becoming dehydrated, however; bacillary dysentery in particular can lead to a child's death from dehydration in as little as 12–24 hours.

- **Bacillary dysentery.** Most patients recover completely from shigellosis, although their bowel habits may not become completely normal for several months. About 3 percent of people infected by *S. flexneri* will develop Reiter's syndrome, which may lead to a chronic form of arthritis that is difficult to treat. Elderly patients or those with weakened immune systems sometimes develop secondary bacterial infections after an episode of shigellosis.
- **Amebic dysentery.** Most people in North America who become infected with *E. histolytica* do not become severely ill. Patients who develop a severe case of amebic dysentery, however, are at increased risk for such complications as fulminant colitis or liver abscess. About 0.5 percent of patients with amebic dysentery develop fulminant colitis, but almost half of these patients die. Between 2 and 7 percent of cases of amebic liver abscess result in rupture of the abscess with a high mortality rate. Men are 7–12 times more likely to develop a liver abscess than women. Any patient diagnosed with amebic dysentery should have stool samples examined for relapse

1, 3, and 6 months after treatment with medications whether or not they have developed complications.

- Dysentery caused by other protozoa. Cryptosporidiosis may lead to respiratory infections or **pancreatitis** in patients with AIDS. The risk of these complications, however, is reduced in AIDS patients who are receiving highly active antiretroviral therapy (HAART).
- Viral dysentery. Most people in North America recover completely without complications unless they become severely dehydrated. Viral dysentery in children in developing countries, however, is a major cause of mortality.
- Dysentery caused by parasitic worms. Untreated whipworm infections can lead to loss of appetite, chronic diarrhea, and retarded growth in children. Untreated schistosomiasis can develop into a chronic intestinal disorder in which fibrous tissue, small growths, or strictures (abnormal narrowing) may form inside the intestine. Patients treated for schistosomiasis should have stool samples checked for the presence of worm eggs 3 and 6 months after the end of treatment.

Prevention

The disease agents that cause dysentery do not confer immunity against reinfection at a later date. As of 2005 there are no vaccines for bacillary dysentery or amebic dysentery; however, a vaccine against schistosomiasis is under investigation. An oral vaccine against **rotavirus infections** was developed for small children but was withdrawn in 2004 because it was associated with an increased risk of small-bowel disorders. Newer vaccines against rotaviruses and caliciviruses are being developed as of 2005.

Public health measures

Public health measures to control the spread of dysentery include the following:

- Requiring doctors to report cases of disease caused by *Shigella*, *Entamoeba histolytica*, and other parasites that cause dysentery. Careful reporting allows the CDC and state public health agencies to investigate local outbreaks and plan prevention efforts.
- Posting advisories for travelers about outbreaks of dysentery and other health risks in foreign countries. The Travelers' Health section of the CDC website (<http://www.cdc.gov/travel/>) is a good source of up-to-date information.
- Instructing restaurant workers and other food handlers about proper methods of hand washing, food storage, and food preparation.

KEY TERMS

Anthelmintic (also spelled anthelmintic)—A type of drug or herbal preparation given to destroy parasitic worms or expel them from the body.

Bacillus—A rod-shaped bacterium. One common type of dysentery is known as bacillary dysentery because it is caused by a bacillus.

Enterotoxin—A type of harmful protein released by bacteria and other disease agents that affects the tissues lining the intestines.

Fulminant—Occurring or flaring up suddenly and with great severity. A potentially fatal complication of amebic dysentery is an inflammation of the colon known as fulminant colitis.

Probiotics—Food supplements containing live bacteria or other microbes intended to improve or restore the normal balance of microorganisms in the digestive tract.

Proctoscope—An instrument consisting of a thin tube with a light source, used to examine the inside of the rectum.

Protozoan (plural, protozoa)—A member of the simplest form of animal life, a one-celled organism. Amebic dysentery is caused by a protozoan.

Reiter's syndrome—A group of symptoms that includes arthritis, inflammation of the urethra, and conjunctivitis, and develops as a late complication of infection with *Shigella flexneri*. The syndrome was first described by a German doctor named Hans Reiter in 1918.

Tenesmus—Straining to urinate or defecate without being able to do so. Tenesmus is a characteristic feature of bacillary dysentery.

Trophozoite—The active feeding stage of a protozoal parasite, as distinct from its encysted stage.

- Instructing workers in day care centers and nursing homes about the proper methods for changing and cleaning soiled diapers or bedding.
- Inspecting wells, other sources of drinking water, and swimming pools for evidence of fecal contamination.

Personal precautions

Individuals can lower their risk of contracting dysentery by the following measures:

- Not allowing anyone in the household who has been diagnosed with amebic or bacillary dysentery to

prepare food or pour water for others until their doctor confirms that they are no longer carrying the disease agent.

- Avoiding anal sex or oral-genital contacts.
- Washing the hands carefully with soap and water after using the bathroom, and supervising the hand-washing of children in day care centers or those at home who are not completely toilet-trained.
- When traveling, drinking only boiled or treated water, and eating only cooked hot foods or fruits that can be peeled by the traveler.
- Avoiding swimming in fresh water in areas known to have outbreaks of schistosomiasis.

Resources

BOOKS

- Cummings, Stephen, MD, and Dana Ullman, MPH. *Everybody's Guide to Homeopathic Medicines*, revised and expanded. New York: Jeremy P. Tarcher, 1991.
- "Enterobacteriaceae Infections." Section 13, Chapter 161 in *The Merck Manual of Diagnosis and Therapy*, edited by Mark H. Beers, MD, and Robert Berkow, MD. Whitehouse Station, NJ: Merck Research Laboratories, 2004.
- "Intestinal Protozoa." Section 13, Chapter 161 in *The Merck Manual of Diagnosis and Therapy*, edited by Mark H. Beers, MD, and Robert Berkow, MD. Whitehouse Station, NJ: Merck Research Laboratories, 2004.
- Pelletier, Kenneth R., MD. *The Best Alternative Medicine*. New York: Simon & Schuster, 2002.
- Reid, Daniel P. *Chinese Herbal Medicine*. Boston: Shambhala, 1993.

PERIODICALS

- Calzada, F., J. A. Cervantes-Martinez, and L. Yopez-Mulia. "In vitro Antiprotozoal Activity from the Roots of *Geranium mexicanum* and Its Constituents on *Entamoeba histolytica* and *Giardia lamblia*." *Journal of Ethnopharmacology* 98 (April 8, 2005): 191–193.
- Chijide, Valda M., MD, and Keith F. Woeltje, MD. "Balantidiasis." *eMedicine*, 12 March 2002. <<http://www.emedicine.com/med/topic203.htm>>.
- Deeni, Y. Y., and N. M. Sadiq. "Antimicrobial Properties and Phytochemical Constituents of the Leaves of African Mistletoe (*Tapinanthus dodoneifolius* (DC) Danser) (Loranthaceae): An Ethnomedicinal Plant of Hausaland, Northern Nigeria." *Journal of Ethnopharmacology* 83 (December 2002): 235–240.
- Eisen, Damon, MD. "Cryptosporidiosis." *eMedicine*, 18 November 2004. <<http://www.emedicine.com/med/topic484.htm>>.
- Goodgame, Richard W., MD. "Gastroenteritis, Viral." *eMedicine*, 14 June 2004. <<http://www.emedicine.com/MED/topic856.htm>>.
- Hlavsa, M. C., J. C. Watson, and M. J. Beach. "Cryptosporidiosis Surveillance—United States

- 1999–2002." *Morbidity and Mortality Weekly Report, Surveillance Summaries* 54 (January 28, 2005): 1–8.
- Hlavsa, M. C., J. C. Watson, and M. J. Beach. "Giardiasis Surveillance—United States, 1998–2002." *Morbidity and Mortality Weekly Report, Surveillance Summaries* 54 (January 28, 2005): 9–16.
- Hu, F., R. Lu, B. Huang, and M. Liang. "Free Radical Scavenging Activity of Extracts Prepared from Fresh Leaves of Selected Chinese Medicinal Plants." *Fitoterapia* 75 (January 2004): 14–23.
- Kroser, Joyann A., MD. "Shigellosis." *eMedicine*, 17 May 2002. <<http://www.emedicine.com/med/topic2112.htm>>.
- Nachimuthu, Senthil, MD, and Paul Piccione, MD. "Food Poisoning." *eMedicine*, 10 January 2005. <<http://www.emedicine.com/med/topic807.htm>>.
- Pennardt, Andre, MD. "Giardiasis." *eMedicine*, 25 June 2004. <<http://www.emedicine.com/emerg/topic215.htm>>.
- Rahman, M. J., P. Sarker, S. K. Roy, et al. "Effects of Zinc Supplementation as Adjunct Therapy on the Systemic Immune Responses in Shigellosis." *American Journal of Clinical Nutrition* 81 (February 2005): 495–502.
- Scoggins, Thomas, MD, and Igor Boyarsky, DO. "Reiter Syndrome." *eMedicine*, 7 December 2004. <<http://www.emedicine.com/EMERG/topic498.htm>>.
- Swords, Robert, MD, and J. Robert Cantey, MD. "Amebiasis." *eMedicine*, 22 February 2002. <<http://www.emedicine.com/med/topic116.htm>>.
- White, C. A. Jr. "Nitazoxanide: A New Broad-Spectrum Antiparasitic Agent." *Expert Review of Anti-Infective Therapy* 2 (February 2004): 43–49.
- Wingate, D., S. F. Phillips, S. J. Lewis, et al. "Guidelines for Adults on Self-Medication for the Treatment of Acute Diarrhea." *Alimentary Pharmacology and Therapeutics* 15 (June 2001): 773–782.

ORGANIZATIONS

- Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>
- Infectious Diseases Society of America (IDSA). 66 Canal Center Plaza, Suite 600, Alexandria, VA 22314. (703) 299-0200. Fax: (703) 299-0204. <<http://www.idsociety.org>>.
- World Health Organization (WHO). <<http://www.who.int/en/>>.

OTHER

- Centers for Disease Control and Prevention. Disease Information. "Shigellosis." <http://www.cdc.gov/ncidod/dbmd/diseaseinfo/shigellosis_t.htm>
- Centers for Disease Control and Prevention, Division of Parasitic Diseases. Fact Sheet. "Amebiasis." <http://www.cdc.gov/ncidod/dpd/parasites/amebiasis/factsht_amebiasis.htm>
- Centers for Disease Control and Prevention, National Center for Infectious Diseases, Travelers' Health. "New Medication Approved for Treatment of Giardiasis and

Amebiasis.” < http://www.cdc.gov/travel/other/tinidazole_approval_2004.htm >

World Health Organization. “Shigella.” < <http://www.who.int/topics/shigella/en/> >.

Rebecca Frey, PhD

Dysfunctional uterine bleeding

Definition

Dysfunctional uterine bleeding is irregular, abnormal uterine bleeding that is not caused by a tumor, infection, or **pregnancy**.

Description

Dysfunctional uterine bleeding (DUB) is a disorder that occurs most frequently in women at the beginning and end of their reproductive lives. About half the cases occur in women over 45 years of age, and about one fifth occur in women under age 20.

Dysfunctional uterine bleeding is diagnosed when other causes of uterine bleeding have been eliminated. Failure of the ovary to release an egg during the menstrual cycle occurs in about 70% of women with DUB. This is probably related to a hormonal imbalance.

DUB is common in women who have polycystic ovary syndrome (cysts on the ovaries). Women who are on dialysis may also have heavy or prolonged periods. So do some women who use an intrauterine device (**IUD**) for birth control.

DUB is similar to several other types of uterine bleeding disorders and sometimes overlaps these conditions.

Menorrhagia

Menorrhagia, sometimes called hypermenorrhea, is another term for abnormally long, heavy periods. This type of period can be a symptom of DUB, or many other diseases or disorders. In menorrhagia, menstrual periods occur regularly, but last more than seven days, and blood loss exceeds 3 oz (88.7 ml). Passing **blood clots** is common. Between 15–20% of healthy women experience debilitating menorrhagia that interferes with their normal activities. Menorrhagia may or may not signify a serious underlying problem.

Metrorrhagia

Metrorrhagia is bleeding between menstrual periods. Bleeding is heavy and irregular as opposed to ovulatory spotting which is light bleeding, in mid-cycle, at the time of ovulation.

Polymenorrhea

Polymenorrhea describes the condition of having too frequent periods. Periods occur more often than every 21 days, and ovulation usually does not occur during the cycle.

Causes and symptoms

Dysfunctional uterine bleeding often occurs when the endometrium, or lining of the uterus, is stimulated to grow by the hormone estrogen. When exposure to estrogen is extended, or not balanced by the presence of progesterone, the endometrium continues to grow until it outgrows its blood supply. Then it sloughs off, causing irregular bleeding. If the bleeding is heavy enough and frequent enough, anemia can result.

Menorrhagia is representative of DUB. It is caused by many conditions including some outside the reproductive system. Causes of menorrhagia include:

- adenomyosis (a benign condition characterized by growths in the area of the uterus)
- imbalance between the hormones estrogen and progesterone
- fibroid tumors
- pelvic infection
- endometrial **cancer** (cancer of the inner mucous membrane of the uterus)
- endometrial polyps
- endometriosis (a condition in which endometrial or endometrial-like tissue appears outside of its normal place in the uterus)
- use of an intrauterine device (IUD) for contraception
- hypothyroidism
- blood clotting problems (rare)
- lupus erythematosus
- pelvic inflammatory disease
- steroid therapy
- advanced **liver disease**
- renal (kidney) disease
- chemotherapy (cancer treatment with chemicals)

To diagnose dysfunctional uterine bleeding, many of the potential causes mentioned above must be eliminated. When all potential causes connected with pregnancy, infection, and tumors (benign or malignant) are eliminated, then menorrhagia is presumed to be caused by dysfunctional uterine bleeding.

Diagnosis

Diagnosis of any menstrual irregularity begins with the patient herself. The doctor will ask for a detailed description of the problem, and take a history of how long it has existed, and any patterns the patient has observed. A woman can assist the doctor in diagnosing the cause of abnormal uterine bleeding by keeping a record of the time, frequency, length, and quantity of bleeding. She should also tell the doctor about any illnesses, including long-standing conditions, like **diabetes mellitus**. The doctor will also inquire about sexual activity, use of contraceptives, current medications, and past surgical procedures.

Laboratory tests

After taking the woman's history, the gynecologist or family practitioner does a pelvic examination and Pap smear. To rule out specific causes of abnormal bleeding, the doctor may also do a pregnancy test and blood tests to check the level of thyroid hormone. Based on the initial test results, the doctor may want to do tests to determine the level of other hormones that play a role in reproduction. A test of blood clotting time and an adrenal function test are also commonly done.

Imaging

Imaging tests are important diagnostic tools for evaluating abnormal uterine bleeding. Ultrasound examination of the pelvic and abdominal area is used to help locate **uterine fibroids**, also called uterine leiomyoma, a type of tumor. Visual examination through hysteroscopy—where a camera inside a thin tube is inserted directly into the uterus so that the doctor can see the uterine lining—is also used to assess the condition of the uterus.

Hysterosalpingography can help outline endometrial polyps and fibroids and help detect endometrial cancer. In this procedure an x ray is taken after contrast media has been injected into the cervix. **Magnetic resonance imaging** (MRI) of the pelvic region can also be used to locate fibroids and tumors.

Invasive procedures

Endometrial biopsy (the removal and examination of endometrial tissue) is the most important testing procedure. It allows the doctor to sample small areas of the uterine lining, while cervical biopsy allows the cervix to be sampled. Tissues are then examined for any abnormalities.

Dilation and curettage (D & C), once common is rarely done today for diagnosis of DUB. It is done while the patient is under either general or regional anesthesia. Women over 30 are more likely to need a D & C, as part of the diagnostic procedure, than younger women.

Because DUB is diagnosed by eliminating other possible disorders, diagnosis can take a long time and involve many tests and procedures. Older women are likely to need more extensive tests than adolescents because the likelihood of reproductive cancers is greater in this age group, and therefore must be definitively eliminated before treating bleeding symptoms.

Treatment

Treatment of DUB depends on the cause of the bleeding and the age of the patient. When the underlying cause of the disorder is known, that disorder is treated. Otherwise the goal of treatment is to relieve the symptoms to a degree that uterine bleeding does not interfere with a woman's normal activities or cause anemia.

Generally the first approach to controlling DUB is to use oral contraceptives that provide a balance between the hormones estrogen and progesterone. **Oral contraceptives** are often very effective in adolescents and young women in their twenties. **NSAIDs (nonsteroidal anti-inflammatory drugs)**, like Naprosyn and Motrin, are also used to treat DUB.

When bleeding cannot be controlled by hormone treatment, surgery may be necessary. Dilation and curettage sometimes relieves the symptoms of DUB. If that fails, endometrial ablation removes the uterine lining, but preserves a woman's uterus. This procedure is sometimes be used instead of **hysterectomy**. However, as it affects the uterus, it can only be used when a woman has completed her childbearing years. The prescription of iron is also important to decrease the risk of anemia.

Until the 1980s, hysterectomy often was used to treat heavy uterine bleeding. Today hysterectomy is used less frequently to treat DUB, and then only after other methods of controlling the symptoms have failed. A hysterectomy leaves a woman unable to bear children,

KEY TERMS

Dilation and curettage (D & C)—A procedure performed under anesthesia during which the cervix is dilated, and tissue lining the uterus is scraped out with a metal spoon-shaped instrument or a suction tube. The procedure can be either diagnostic, or to remove polyps.

Endometrial biopsy—The removal of tissue either by suction or scraping of samples of tissue from the uterus. The cervix is not dilated. The procedure has a lower rate of diagnostic accuracy than a D & C, but can be done as an office procedure under local anesthesia.

Endometrial cancer—Cancer of the inner mucous membrane of the uterus.

Fibroids, or fibroid tumors—Fibroid tumors are non-cancerous (benign) growths in the uterus. They occur in 30–40% of women over age 40, and do not need to be removed unless they are causing symptoms that interfere with a woman's normal activities.

Hypothyroidism—A disorder in which the thyroid gland produces too little thyroid hormone causing a decrease in the rate of metabolism with associated effects on the reproductive system.

Lupus erythematosus—A chronic inflammatory disease in which inappropriate immune system reactions cause abnormalities in the blood vessels and connective tissue.

Progesterone—A hormone naturally secreted by the ovary, or manufactured synthetically, that prepares the uterus for implantation of a fertilized egg.

Prostaglandins—A group of chemicals that mediate, or determine the actions of other chemicals in the cell or body.

and, therefore, is limited largely to women who are unable to, or uninterested in, bearing children. Still, hysterectomy is a common treatment for long-standing DUB in women done with childbearing.

Alternative treatment

Alternative practitioners concentrate on good **nutrition** as a way to prevent heavy periods that are not caused by uterine fibroids, endometrial polyps, endometriosis, or cancer. Iron supplementation (100 mg per day) not only helps prevent anemia, but

also appears to reduce menorrhagia in many women. Other recommended dietary supplements include **vitamins** A and C. Vitamin C improves capillary fragility and enhances iron uptake.

Vitamin E and bioflavonoid supplements are also recommended. Vitamin E can help reduce blood flow, and bioflavonoids help strengthen the capillaries. Vitamin K is known to play a role in clotting and is helpful in situations where heavy bleeding may be due to clotting abnormalities.

Botanical medicines used to assist in treating abnormal bleeding include spotted cranesbill (*Geranium maculatum*), birthroot (*Trillium pendulum*), blue cohosh (*Caulophyllum thalictroides*), witch hazel (*Hamamelis virginiana*), shepherd's purse (*Capsella bursa-pastoris*), and yarrow (*Achillea millifolia*). These are all stiptic herbs that act to tighten blood vessels and tissue. Hormonal balance can also be addressed with herbal formulations containing phytoestrogens and progestosterone.

Prognosis

Response to treatment for DUB is highly individual and is not easy to predict. The outcome depends largely on the woman's medical condition and her age. Many women, especially adolescents, are successfully treated with hormones (usually oral contraceptives). As a last resort, hysterectomy removes the source of the problem by removing the uterus, but this operation is not without risk, or the possibility of complications.

Prevention

Dysfunctional uterine bleeding is not a preventable disorder.

Resources

OTHER

"Menorrhagia." *The Wellness Web*. <<http://wellweb.com/INDEX/MENORRHAGIA.htm>>.

Tish Davidson, A.M.

Dyslexia

Definition

Dyslexia is a learning disability characterized by problems in reading, spelling, writing, speaking, or

listening. In many cases, dyslexia appears to be inherited.

Description

The word dyslexia is derived from the Greek word, *dys* (meaning poor or inadequate) and the word *lexis* (meaning words or language).

The National Institutes of Health estimates that about 15% of the United States population is affected by learning disabilities, mostly with problems in language and reading. The condition appears in all ages, races, and income levels. Dyslexia is not a disease, but describes rather a different kind of mind that learns in a different way from other people. Many people with the condition are gifted and very productive; dyslexia is not at all linked to low intelligence. In fact, intelligence has nothing to do with dyslexia.

Dyslexic children seem to have trouble learning early reading skills, problems hearing individual sounds in words, analyzing whole words in parts, and blending sounds into words. Letters such as “d” and “b” may be confused.

When a person is dyslexic, there is often an unexpected difference between achievement and aptitude. However, each person with dyslexia has different strengths and weaknesses, although many have unusual talents in art, athletics, architecture, graphics, drama, music, or engineering. These special talents are often in areas that require the ability to integrate sight, spatial skills, and coordination.

Often, a person with dyslexia has a problem translating language into thought (such as in listening or reading), or translating thought into language (such as in writing or speaking).

Common characteristics include problems with:

- identifying single words
- understanding sounds in words, sound order, or rhymes
- spelling
- transposing letters in words
- handwriting
- reading comprehension
- delayed spoken language
- confusion with directions, or right/left handedness
- confusion with opposites (up/down, early/late, and so on)
- mathematics

Causes and symptoms

The underlying cause of dyslexia is not known, although research suggests the condition is often inherited. In 1999, The Centre for Reading Research in Norway presented the first research to study the largest family with reading problems ever known. By studying the reading and writing abilities of close to 80 family members across four generations the researchers reported, for the first time, that chromosome 2 can be involved in the inheritability of dyslexia. When a fault occurs on this gene it leads to difficulties in processing written language. Previous studies have pointed out linkages of other potential dyslexia genes to chromosome 1, chromosome 15 (DYX1 gene), and to chromosome 6 (DYX2 gene). The researchers who pinpointed the newly localized gene on chromosome 2 (DYX3) hope that this finding will lead to earlier and more precise diagnoses of dyslexia.

New research suggests a possible link with a subtle visual problem that affects the speed with which affected people can read. Other experts believe that dyslexia is related to differences in the structure and function of the brain that manifests differently in different people.

Diagnosis

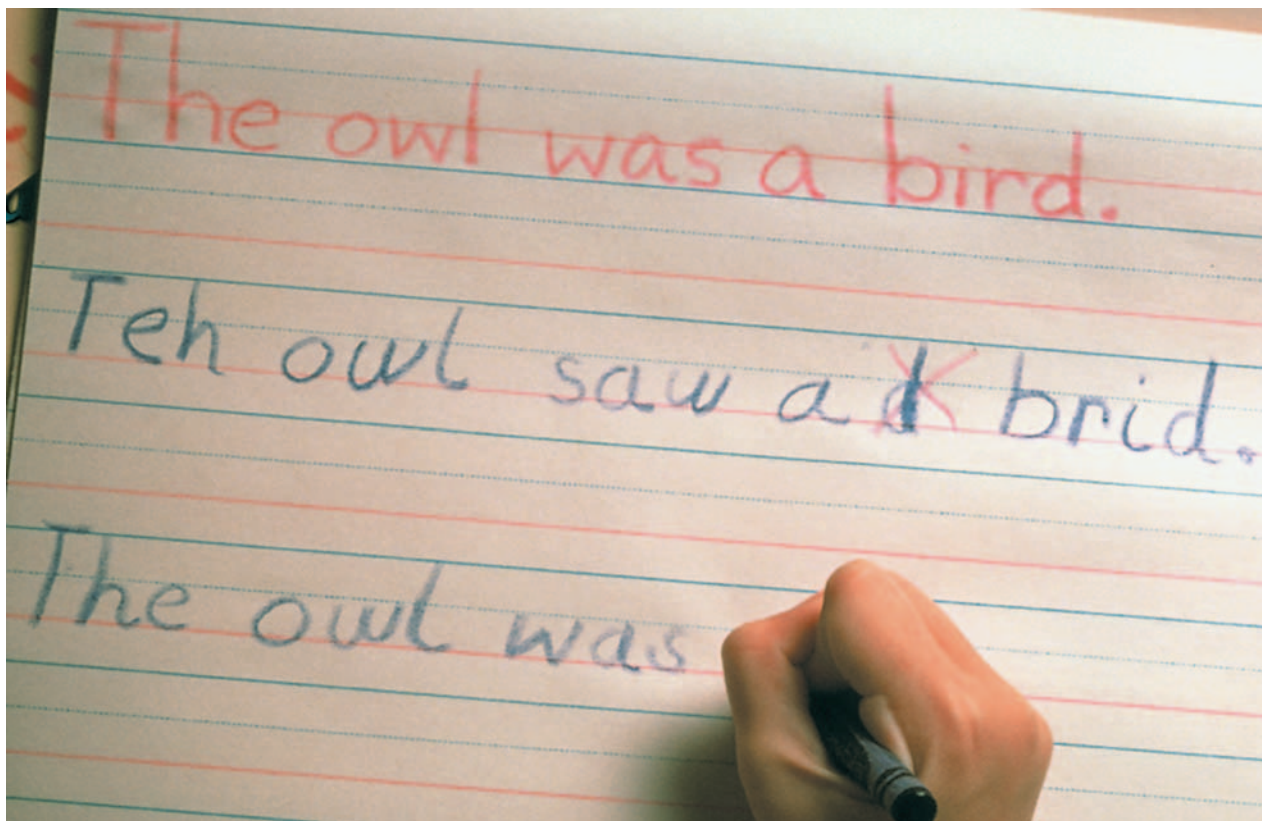
Anyone who is suspected to have dyslexia should have a comprehensive evaluation, including hearing, vision, and intelligence testing. The test should include all areas of learning and learning processes, not just reading.

As further research pinpoints the genes responsible for some cases of dyslexia, there is a possibility that earlier testing will be established to allow for timely interventions to prevent the onset of the condition and to treat it when it does occur. Unfortunately, in many schools, a child is not identified as having dyslexia until after repeated failures.

Treatment

If a child is diagnosed with dyslexia, the parents should find out from the school or the diagnostician exactly what the problem is, and what method of teaching is recommended and why. No single method will work with every child, and experts often disagree as to the best method to use.

The primary focus of treatment is aimed at helping the specific learning problem of each affected person. Most often, this may include modifying teaching methods and the educational environment, since



A student with dyslexia has difficulty copying words. (Photograph by Will & Deni McIntyre, Photo Researchers, Inc. Reproduced by permission.)

KEY TERMS

Spatial skills—The ability to locate objects in three dimensional world using sight or touch.

traditional educational methods will not always work with a dyslexic child.

People with dyslexia need a structured language program, with direct instruction in the letter-sound system. Teachers must give the rules governing written language. Most experts agree that the teacher should emphasize the association between simple phonetic units with letters or letter groups, rather than an approach that stresses memorizing whole words.

It is important to teach these students using all the senses: hearing, touching, writing, and speaking, provided by an instructor who is specifically trained in a program that is effective for dyslexic students.

Prognosis

Many successful and even famous people have dyslexia. How well a person with dyslexia functions in life depends on the way the disability affects that person. There is a great deal of variation among different people with dyslexia, producing different symptoms and different degrees of severity.

Prognosis is usually good if the condition is diagnosed early, and if the person has a strong self image with supportive family, friends, and teachers. It is imperative for a good outcome that the person be involved in a good remedial program.

Resources

PERIODICALS

Fagerheim, Toril, et al. "A New Gene (DYX3) for Dyslexia is Located on Chromosome 2." *Journal of Medical Genetics* 36 (September 1999): 664-669.

Beth A. Kapes

Dyslipidemia see **Hyperlipoproteinemia**

Dysmenorrhea

Definition

Dysmenorrhea is the occurrence of painful cramps during menstruation.

Description

More than half of all girls and women suffer from dysmenorrhea (cramps), a dull or throbbing **pain** that usually centers in the lower mid-abdomen, radiating toward the lower back or thighs. Menstruating women of any age can experience cramps.

While the pain may be only mild for some women, others experience severe discomfort that can significantly interfere with everyday activities for several days each month.

Causes and symptoms

Dysmenorrhea is called “primary” when there is no specific abnormality, and “secondary” when the pain is caused by an underlying gynecological problem. It is believed that primary dysmenorrhea occurs when hormone-like substances called “prostaglandins” produced by uterine tissue trigger strong muscle contractions in the uterus during menstruation. However, the level of prostaglandins does not seem to have anything to do with how strong a woman’s cramps are. Some women have high levels of prostaglandins and no cramps, whereas other women with low levels have severe cramps. This is why experts assume that cramps must also be related to other things (such as genetics, **stress**, and different body types) in addition to prostaglandins. The first year or two of a girl’s periods are not usually very painful. However, once ovulation begins, the blood levels of the prostaglandins rise, leading to stronger contractions.

Secondary dysmenorrhea may be caused by endometriosis, fibroid tumors, or an infection in the pelvis.

The likelihood that a woman will have cramps increases if she:

- has a family history of painful periods
- leads a stressful life
- does not get enough **exercise**
- uses **caffeine**
- has pelvic inflammatory disease

Symptoms include a dull, throbbing cramping in the lower abdomen that may radiate to the lower back and thighs. In addition, some women may experience

nausea and vomiting, **diarrhea**, irritability, sweating, or dizziness. Cramps usually last for two or three days at the beginning of each menstrual period. Many women often notice their painful periods disappear after they have their first child, probably due to the stretching of the opening of the uterus or because the birth improves the uterine blood supply and muscle activity.

Diagnosis

A doctor should perform a thorough **pelvic exam** and take a patient history to rule out an underlying condition that could cause cramps.

Treatment

Secondary dysmenorrhea is controlled by treating the underlying disorder.

Several drugs can lessen or completely eliminate the pain of primary dysmenorrhea. The most popular choice are the nonsteroidal anti-inflammatory drugs (NSAIDs), which prevent or decrease the formation of prostaglandins. These include **aspirin**, ibuprofen (Advil), and naproxen (Aleve). For more severe pain, prescription strength ibuprofen (Motrin) is available. These drugs are usually begun at the first sign of the period and taken for a day or two. There are many different types of NSAIDs, and women may find that one works better for them than the others.

If an NSAID is not available, **acetaminophen** (Tylenol) may also help ease the pain. Heat applied to the painful area may bring relief, and a warm bath twice a day also may help. While birth control pills will ease the pain of dysmenorrhea because they lead to lower hormone levels, they are not usually prescribed just for pain management unless the woman also wants to use them as a birth control method. This is because these pills may carry other more significant side effects and risks.

New studies of a drug patch containing glyceryl trinitrate to treat dysmenorrhea suggest that it also may help ease pain. This drug has been used in the past to ease preterm contractions in pregnant women.

Alternative treatment

Simply changing the position of the body can help ease cramps. The simplest technique is assuming the fetal position, with knees pulled up to the chest while hugging a heating pad or pillow to the abdomen. Likewise, several **yoga** positions are popular ways to ease menstrual pain. In the “cat stretch,” position, the woman rests on her hands and knees, slowly arching the back. The pelvic tilt is another popular yoga

KEY TERMS

Endometriosis—The growth of uterine tissue outside the uterus.

Hormone—A chemical messenger secreted by a gland and released into the blood, which allows it to travel to distant cells where it exerts an effect.

Ovary—One of the two almond-shaped glands in the female body that produces the hormones estrogen and progesterone.

Ovulation—The monthly release of an egg from an ovary.

Progesterone—The hormone produced by the ovary after ovulation that prepares the uterine lining for a fertilized egg.

Uterus—The female reproductive organ that contains and nourishes a fetus from implantation until birth.

position, in which the woman lies with knees bent, and then lifts the pelvis and buttocks.

Dietary recommendations to ease cramps include increasing fiber, calcium, and complex carbohydrates, cutting fat, red meat, dairy products, caffeine, salt, and sugar. **Smoking** also has been found to worsen cramps. Recent research suggests that vitamin B supplements, primarily vitamin B₆ in a complex, magnesium, and fish oil supplements (**omega-3 fatty acids**) also may help relieve cramps.

Other women find relief through visualization, concentrating on the pain as a particular color and gaining control of the sensations. **Aromatherapy** and massage may ease pain for some women. Others find that imagining a white light hovering over the painful area can actually lessen the pain for brief periods.

Exercise may be a way to reduce the pain of menstrual cramps through the brain's production of endorphins, the body's own painkillers. And orgasm can make a woman feel more comfortable by releasing tension in the pelvic muscles.

Acupuncture and Chinese herbs are another popular alternative treatments for cramps.

Prognosis

Medication should lessen or eliminate pain.

Prevention

NSAIDs taken a day before the period begins should eliminate cramps for some women.

Resources

PERIODICALS

McDonald, Claire, and Susan McDonald. "A Woman's Guide to Self-care." *Natural Health* January-February 1998:121-142.

ORGANIZATIONS

National Women's Health Network. 514 10th St. NW, Suite 400, Washington, DC 20004. (202) 628-7814. <<http://www.womenshealthnetwork.org>>.

Carol A. Turkington

Dysmetria see **Movement disorders**

Dyspepsia

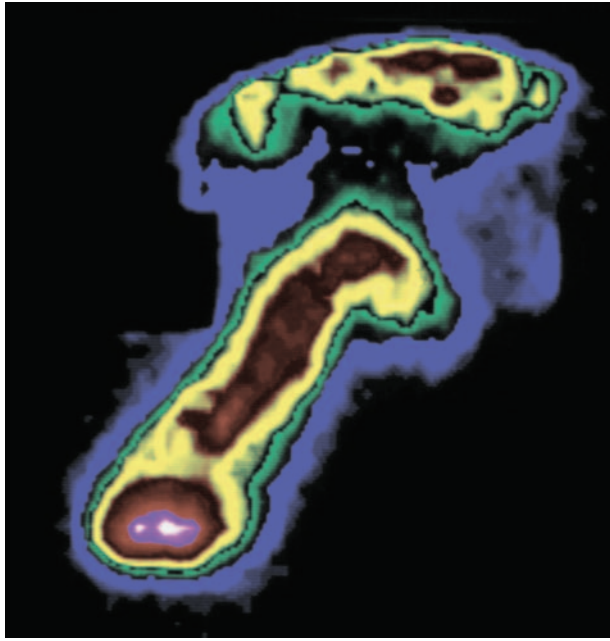
Definition

Dyspepsia can be defined as painful, difficult, or disturbed digestion, which may be accompanied by symptoms such as **nausea and vomiting**, **heartburn**, bloating, and stomach discomfort.

Causes and symptoms

The digestive problems may have an identifiable cause, such as bacterial or viral infection, peptic ulcer, gallbladder, or **liver disease**. The bacteria *Helicobacter pylori* is often found in those individuals suffering from duodenal or gastric ulcers. Investigation of recurrent **indigestion** should rule out these possible causes.

Often, there is no organic cause for the problem, in which case dyspepsia is classified as functional or nonulcer dyspepsia. There is evidence that functional dyspepsia may be related to abnormal motility of the upper gastrointestinal tract (a state known as dysmotility in which the esophagus, stomach, and upper intestine behave abnormally). These patients may respond to a group of drugs called prokinetic agents. A review of eating habits (e.g., chewing with the mouth open, gulping food, or talking while chewing) may reveal a tendency to swallow air. This may contribute to feeling bloated, or to excessive belching. Smoking, caffeine, alcohol, or carbonated beverages may contribute to the discomfort. When there is sensitivity or allergy to certain food substances, eating those foods may cause gastrointestinal distress. Some medications are associated with indigestion. Stomach problems may also be a response to **stress** or emotional unrest.



A false-color gamma scan of a human stomach with dyspepsia, or indigestion, during tests to study its rate of emptying. (Photograph by Jean-Perrin, Custom Medical Stock Photo. Reproduced by permission.)

Diagnosis

A **physical examination** by a health care professional may reveal mid-abdominal **pain**. A rectal examination may be done to rule out bleeding. If blood is found on rectal exam, laboratory studies, including a **blood count** may be ordered. Endoscopy and barium studies may be used to rule out underlying gastrointestinal disease. Upper gastrointestinal x-ray studies using barium may allow for visualization of abnormalities. Endoscopy permits collection of tissue and culture specimens which may be used to further confirm a diagnosis.

Treatment

The treatment of dyspepsia is based on assessment of symptoms and suspected causative factors. Clinical evaluation is aimed at distinguishing those patients who require immediate diagnostic work-ups from those who can safely benefit from more conservative initial treatment. Some of the latter may require only reassurance, dietary modifications, or antacid use. Medications to block production of stomach acids, prokinetic agents, or antibiotic treatment may be considered. Further diagnostic investigation is indicated if there is severe abdominal pain, pain radiating to the back, unexplained weight loss, difficulty swallowing, a

KEY TERMS

Anemia—Diagnosed through laboratory study of the blood, a deficiency in hemoglobin or red blood cells, often associated with paleness or loss of energy.

Endoscopy—A diagnostic procedure using a lighted instrument to examine a body cavity or internal organ. Endoscopy permits collection of tissue and culture specimens.

palpable mass, or anemia. Additional work-up is also indicated if a patient does not respond to prescribed medications.

Prognosis

Statistics show an average of 20% of patients with dyspepsia have duodenal or gastric ulcer disease, 20% have **irritable bowel syndrome**, fewer than 1% of patients had **cancer**, and the range for functional, or non-ulcer dyspepsia (**gastritis** or superficial erosions), was from 5–40%.

Resources

PERIODICALS

Talley, N. J. "Non-ulcer Dyspepsia: Current Approaches to Diagnosis and Management." *American Family Physician* May 1993: 1407-1416.

OTHER

"Clinical Economics: Gastrointestinal Disease in Primary Care." April 23, 1998. <<http://www.avicenna.com>>.

Kathleen D. Wright, RN

Dysphasia

Definition

Dysphasia is a partial or complete impairment of the ability to communicate resulting from brain injury.

Description

Approximately one million Americans currently suffer from one of the various forms of dysphasia, and an additional 80,000 new cases occur annually. The term "dysphasia" is more frequently used by

European health professionals, whereas in North America the term, **aphasia** is more commonly preferred. These two terms, however, can be and are used interchangeably. They both refer to the full or partial loss of verbal communication skills due to damage or degeneration of the brain's language centers. Developmental Dysphasia is considered to be a learning disability, but will not be the focus of this article.

Verbal communication is derived from several regions located in the language-dominant hemisphere of the brain. These include the adjacent inferior parietal lobe, the inferolateral lobe, and the posterosuperior temporal lobe, as well as the subcortical connection between these areas. Disease, direct trauma, lesion, or infarction involving one or more of these regions can disrupt or prevent proper language function. Dysphasia does not necessarily prevent proper cognitive function, so the patient can think and feel with perfect clarity. This can be extremely frustrating for the patient, as they cannot express these thoughts and feelings to others.

Dysphasia can occur in a variety of forms, depending on how the communicative disruption manifests. Classically, dysphasia can affect one or more of the basic language functions: comprehension (understanding spoken language), naming (identifying items with words), repetition (repeating words or phrases), and speech. Although there are several subtypes of dysphasias, they most commonly manifest in one of three syndromes: expressive dysphasia, receptive dysphasia, or global dysphasia.

Expressive Dysphasia

Expressive dysphasia, also known as motor dysphasia, produces a conscious and recognizable disruption of a patient's speech production and language output. This includes the impairment of speech initiation, proper grammatical sequencing, and proper word forming and articulation. Although patients can perfectly understand what is said to them, they have great difficulty communicating their thoughts.

BROCA'S DYSPHASIA. Broca's dysphasia is the most common type of expressive dysphasia. It is caused by damage to the lower area of the premotor cortex, located just in front of the primary motor cortex. This region is most commonly referred to as the Broca's area. Speech for patients suffering from Broca's dysphasia may be completely impossible. Others may be able to form single words or full sentences, but only through great effort. "Telegraphing," the omission of articles and conjunctions, may also be exhibited.

TRANSCORTICAL DYSPHASIA. Also known as isolation syndrome, transcortical dysphasia is caused by damage to the language-dominant brain that separates all or parts of the central region from the rest of the brain. There are three sub-classes of transcortical dysphasia, which define the impairments to a patient's ability to repeat words, sentences, and phrases: transcortical motor dysphasia, transcortical sensory dysphasia, and mixed transcortical dysphasia. Additional impairments may occur depending on the extent and location of the damage.

Receptive Dysphasia

Receptive dysphasia, also known as sensory dysphasia, impairs the patient's comprehension and meaning of language. Unlike expressive dysphasia, the patient can speak fluently and articulately, but will utilize meaningless words, nonsensical grammar, and unnecessary phrases to the point of becoming incomprehensible. However, they will be completely unaware of their mistakes. Additionally, the patient will find it difficult to comprehend spoken language and/or word-object relation.

WERNICKE'S DYSPHASIA. Also known as semantic dysphasia, Wernicke's dysphasia is the most common of the receptive dysphasias. It is caused by damage to the Wernicke's area, located in the posterior superior temporal lobe of the language-dominant hemisphere. Although the patient can speak clearly and at length, many of their words, phrases, and sentences will be nonsensical in nature. Additionally, they will experience difficulty in understanding spoken language, if not suffer a complete lack of comprehension. Semantic distinctions between words may become mixed up and jumbled, furthering confusion.

ANOMIC DYSPHASIA. Anomic dysphasia, also referred to as amnesic dysphasia, is caused by damage to the temporal parietal area and/or the angular gyrus region. Although very similar to Wernicke's dysphasia, anomic dysphasia is distinguished by its disruption of a patient's word-retrieval skills. They will be unable to correctly name people or objects, causing them to pause or substitute generalized words (like "thing"). Otherwise, the patient will exhibit few, if any, language impairments.

CONDUCTION DYSPHASIA. Also known as associative dysphasia, conduction dysphasia is a relatively uncommon disease (representing only 10% of the cases). Damage to the upper temporal lobe, lower parietal, or connection between the Wernicke's and Broca's areas can result in the inability to repeat words, phrases, or sentences. The patient may also

suffer the inability to describe people or objects in the proper terms.

Global Dysphasia

Global dysphasia, the third most common form of dysphasia, results from damage to both the anterior and posterior regions of the language-dominant hemisphere. In global dysphasia, all of the patient's language skills are disrupted; however, some may be disrupted more severely than others.

Causes & symptoms

Currently, over one million people in the United States suffer a permanent type of dysphasia. Although dysphasia may manifest in several ways, the common cause for its onset is damage or trauma to the brain. **Stroke**, in particular, is the most common cause for dysphasia. Of the half million stroke victims reported annually in the United States, approximately 100,000 will suffer some form of dysphasia. Infection, direct trauma, **transient ischemic attack** (TIA), brain tumors, and degeneration can also instigate the onset of dysphasia.

Symptoms of dysphasia will quickly manifest after damage to the brain has occurred, and will present in accordance to the particular type of dysphasia suffered. Due to the proximity to areas of the brain that control motor function, expressive dysphasias can be accompanied by noticeable motor impairment. The majority of symptoms will be language related, including:

- Difficulty remembering words
- Difficulty naming objects and/or people
- Difficulty speaking in complete and/or meaningful sentences
- Difficulty speaking in any fashion
- Difficulty reading or writing
- Difficulty expressing thoughts and feelings
- Difficulty understanding spoken language
- Using incorrect or jumbled words
- Using words in the wrong order

Diagnosis

Dysphasia is frequently diagnosed while the patient is being treated for injury to the brain, be it from trauma or disease. The health professional, typically a neurologist, will conduct standard cognitive tests, including tests to determine whether the

patient's language centers have been affected. If the patient exhibits signs of difficulty communicating, they will often be referred to a speech-language pathologist. In turn, the pathologist will conduct a comprehensive examination of the patient's ability language and comprehension skills. This examination may begin with evaluating the patient's ability to repeat words and phrases, recognize and describe objects, and comprehend what is said to them. More extensive and standardized language-based tests may be required, including the Porch Index of Speech Ability and the Boston Diagnostic Aphasia Examination. Based on the result of the examinations, the health professional will be able to determine the type of dysphasia inflicting the patient. More extensive damage may require the use of computed tomography or **magnetic resonance imaging** for an effective diagnosis.

Treatment

Initially it is necessary to treat and stabilize the injury underlying the development of the patient's dysphasia. In some cases, such as with damage caused by TIA, a full recovery can be expedient and take only a few days. Unfortunately, most dysphasias can take months, if not years, to recover from. Even after prolonged therapy, many patients never achieve a full recovery. Efficacy of treatment greatly depends on the promptness with which it begins. For this reason, many medical facilities have speech-language pathologists on staff to begin the initial treatment process as quickly as possible.

There is no medical or surgical cure for dysphasia. Treatment, instead, relies strongly upon the use of various speech therapies. Much like physical therapy strengthens muscles and bones back to normalcy, speech therapy allow the patient to regain language function, as well as rebuild their communications skills. Treatment is typically conducted with a trained speech therapist. However, group sessions are common and allow the patient to practice their language skills in a non-threatening environment with others sharing their disability. Although much of therapeutic work is conducted by a speech therapist, friends and family also play a vital role in the patient's recovery. They can help the patient continually practice and **exercise** language skills while outside the therapeutic setting. Many times, family members are included on therapy sessions to teach them how to communicate with and understand the patient.

There are several treatments available, which utilize the patient's remaining language abilities to rebuild and compensate for those that were lost.

KEY TERMS

Transient ischemic attack—Also known as a mini-stroke, a transient ischemic attack is caused by a temporary interruption of blood flow in an area of the brain. Unlike in a true stroke, normal brain function will return with 24 hours.

These include out-put focused therapy (stimulation-response), psycholinguistic therapy (cognitive), cognitive neurorehabilitation, and combinations thereof. Although these treatments approach aphasia differently, they all share a common thread by identifying the specific communication deficits and then targeting them with various modalities (computer-aided therapy, picture cards, reading and writing exercises, speech practice, etc.). These techniques stimulate the various parts of the brain associated with language, memory, and understanding, and thus allow it to heal.

Prognosis

Fortunately, about half of patients will suffer from transient dysphasia, in which the symptoms fade completely after only a few days. However, a patient's prognosis will greatly depend on several factors, such as the location and extent of the underlying damage. Additional factors of importance are the patient's age, general health, and mental health and motivation. Handedness may also be an indicator for recovery, as left-handed individuals have language centers located in both hemispheres of the brain (not just the left). As such, left-handed patients have access to language skills from either side of the brain, which can expedite their recovery. Even with therapy, dysphasia may take several years to overcome. Indeed, some patients will never regain their pre-trauma skill level of communication and speech.

Prevention

Dysphasia can be prevented by avoiding the causes of brain injury and stroke, such as high blood

pressure. In particular, eating a healthy diet and not **smoking** to maintain proper blood pressure will help prevent damaging strokes. Although it is impossible to predict head trauma, the use of head protection while participating in dangerous sports or activities can reduce the risk of serious brain damage.

Resources

BOOKS

- Brookshire, R. *Introduction to Neurogenic Communication Disorders (6th edition)* St. Louis, MO: Mosby, 2003.
 Darley, F. *Aphasia*. Philadelphia, PA: WB Saunders, 1982.
 Newman, S., and R. Epstein (eds). *Current Perspectives in Dysphasia*. New York: Churchill Livingstone, 1985.

PERIODICALS

- Albert, M.L.. "Treatment of Aphasia." *Archives of Neurology* 55 (November, 1998): 1417-1419.

ORGANIZATIONS

- National Aphasia Association. 29 John Street, Suite 1103, New York, NY 21108. (800) 922-4622. <<http://www.aphasia.org>>.
 Speakability. 1 Royal Street, London, UK SE1 7LL. 020-7261-9572. <<http://www.speakability.org.uk/>>.

OTHER

- "Aphasia." *The Merck Manual (Section 14. Neurologic Disorders)* <<http://www.merck.com/mrkshared/mmanual/section14/chapter169/169b.jsp>>.
 "Aphasia." *National Institute on Deafness and Other Communication Disorders* <<http://www.nidcd.nih.gov/health/voice/aphasia.asp>>.
 "CMSD 336 Neuropathologies of Language and Cognition." *The Neuroscience on the Web Series* <<http://www.csuchico.edu/~pmccaff/syllabi/SPPA336/336unit5.html>>.

Jason Fryer

Dysphasia see **Aphasia**

Dyspnea see **Shortness of breath**

Dysthymic disorder see **Depressive disorders**

Dystonia see **Movement disorders**

E

E. coli see **Escherichia coli**

E. coli infection see **Enterobacterial infections**

E. coli O157:H7 infection see **Escherichia coli**

Ear canal infection see **Otitis externa**

Ear exam with an otoscope

Definition

An otoscope is a hand-held instrument with a tiny light and a cone-shaped attachment called an ear speculum, which is used to examine the ear canal. An ear examination is a normal part of most physical examinations by a doctor or nurse. It is also done when an ear infection or other type of ear problem is suspected.

Purpose

An otoscope is used to look into the ear canal to see the ear drum. Redness or fluid in the eardrum can indicate an ear infection. Some otoscopes can deliver a small puff of air to the eardrum to see if the eardrum will vibrate (which is normal). This type of ear examination with an otoscope can also detect a build up of wax in the ear canal, or a rupture or puncture of the eardrum.

Precautions

No special precautions are required. However, if an ear infection is present, an ear examination may cause some discomfort or **pain**.

Description

An ear examination with an otoscope is usually done by a doctor or a nurse as part of a complete **physical examination**. The ears may also be examined if an ear infection is suspected due to **fever**, ear pain, or **hearing loss**. The patient will often be asked to tip the head slightly toward the shoulder so the ear to be examined is pointing up. The doctor or nurse may hold the ear lobe as the speculum is inserted into the ear, and may adjust the position of the otoscope to get a better view of the ear canal and eardrum. Both ears are usually examined, even if there seems to be a problem with just one ear.

Preparation

No special preparation is required prior to an ear examination with an otoscope. The ear speculum, which is inserted into the ear, is cleaned and sanitized before it is used. The speculums come in various sizes, and the doctor or nurse will select the size that will be most comfortable for the patient's ear.

Aftercare

If an ear infection is diagnosed, the patient may require treatment with **antibiotics**. If there is a buildup of wax in the ear canal, it might be rinsed or scraped out.

Risks

This type of ear examination is simple and generally harmless. Caution should always be used any time an object is inserted into the ear. This process could irritate an infected external ear canal and could rupture an eardrum if performed improperly or if the patient moves.

Normal results

The ear canal is normally skin-colored and is covered with tiny hairs. It is normal for the ear canal to have some yellowish-brown earwax. The eardrum is

KEY TERMS

Ear speculum—A cone- or funnel-shaped attachment for an otoscope which is inserted into the ear canal to examine the eardrum.

Otoscope—A hand-held instrument with a tiny light and a funnel-shaped attachment called an ear speculum, which is used to examine the ear canal and eardrum.

typically thin, shiny, and pearly-white to light gray in color. The tiny bones in the middle ear can be seen pushing on the eardrum membrane like tent poles. The light from the otoscope will reflect off of the surface of the ear drum.

Abnormal results

An ear infection will cause the eardrum to look red and swollen. In cases where the eardrum has ruptured, there may be fluid draining from the middle ear. A doctor may also see scarring, retraction of the eardrum, or bulging of the eardrum.

Resources

ORGANIZATIONS

American Academy of Otolaryngology-Head and Neck Surgery, Inc. One Prince St., Alexandria VA 22314-3357. (703) 836-4444. <<http://www.entnet.org>>.

Ear Foundation. 1817 Patterson St., Nashville, TN 37203. (800) 545-4327. <<http://www.earfoundation.org>>.

OTHER

“Ear Test.” HealthAnswers.com. <<http://www.healthanswers.com>>.

Altha Roberts Edgren

Ear surgery

Definition

Ear surgery is the treatment of diseases, injuries, or deformations of the ear by operation with instruments.

Purpose

Ear surgery is performed to correct certain types of hearing loss, and to treat diseases of, injuries to, or

deformities of the ear's auditory tube, middle ear, inner ear, and auditory and vestibular systems. Ear surgery is commonly performed to treat conductive **hearing loss**, persistent ear infections, unhealed perforated eardrums, congenital ear defects, and tumors.

Ear surgery is performed on children and adults. In some cases, surgery is the only treatment; in others, it is used only when more conservative medical treatment fails.

Precautions

The precautions vary, depending on the type of ear surgery under consideration. For example, **stapedectomy** (removal of parts of the middle ear and insertion of prosthesis parts) should not be performed on people with external or middle ear infection or inner ear disease. For people with complete hearing loss in the other ear, it should be performed cautiously. Microsurgery for the removal of a cholesteatoma (a cyst-like mass of cells in the middle ear) should not be performed on patients who are extremely ill or have other medical conditions. Tympanoplasty (any surgical procedure on the eardrum or middle ear) should not be performed on patients with chronic sinus or nasal problems or with medical problems such as poorly controlled diabetes and heart disease. Surgery for congenital microtia and atresia (absence of normal bodily openings, such as the outer ear canal) should not be performed if the middle ear space is totally or almost totally absent.

Description

Most ear surgery is microsurgery, performed with an operating microscope to enable the surgeon to view the very small structures of the ear. The use of minimally invasive **laser surgery** for middle ear procedures is growing. Laser surgery reduces the amount of trauma due to vibration, enhances coagulation, and enables surgeons to access hard to reach places in the middle ear. Laser surgery can be performed in an office operating suite. Types of ear surgery include stapedectomy, tympanoplasty, **myringotomy** and ear tube surgery, ear surgery to repair a **perforated eardrum**, cochlear implants, and **tumor removal**.

Stapedectomy

To restore hearing loss, which is usually due to **otosclerosis**, stapedectomy is performed. Stapedectomy is the removal of all or part of the stapes, one of the bones in the middle ear, and replacement with a tiny prosthesis. An incision is made in the middle ear, the small bones are identified, and the stapes is

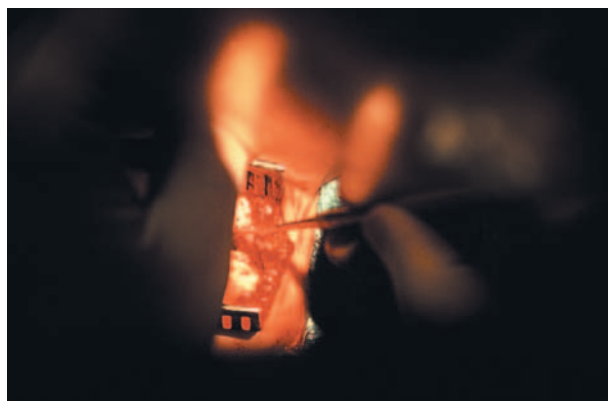
removed. The stainless steel wire and cellulose sponge prosthesis is inserted, blood and fluid are drained, and the wound is closed. Performed in a hospital or outpatient surgical facility under local or general anesthetic, full recovery takes about three weeks but hearing should improve immediately.

Tympanoplasty

Tympanoplasty is performed to reconstruct the eardrum after partial or total conductive hearing loss, usually caused by chronic middle ear infections, or perforations that do not heal. This is usually a same day surgery, performed under either local or **general anesthesia**. After making an incision in the ear to view the perforation, the ear drum is elevated away from the ear canal and lifted forward. If the bones of hearing (ossicular chain) are functioning, tissue is taken from the ear and grafted to the eardrum to close the perforation. A thin sheet of silastic and Gelfoam hold the graft in place. The ear is stitched together, and a sterile patch is placed on the outside of the ear canal. Tympanoplasty is successful in over 90% of all cases. The need for ossicular reconstruction (reconstruction of tiny bones of the middle ear) is sometimes known before surgery and even when identified during surgery, can usually be done while reconstructing the eardrum. If the gap between the anvil bone and the stapes is small, a small piece of bone or cartilage from the patient can be inserted; if it is large, the incus bone is removed, modelled into a prosthesis, and reinserted between the stapes and the malleus. Reconstruction could also be achieved by inserting a strut made from artificial bone. For tympanoplasty with ossicular reconstruction, the patient usually stays in the hospital overnight. The recovery period is about four weeks.

Myringotomy and ear tube surgery

Myringotomy and ear tube surgery is performed to drain ear fluid and prevent ear infections when **antibiotics** don't work or when ear infections are chronic. The process normalizes pressure in the middle ear and decreases fluid accumulation. It is most commonly performed on infants and children, in whom ear infections are most frequent, and may be done on one or both ears. The surgeon makes a small hole in the ear drum, then uses suction to remove fluid. A small ear tube of metal or plastic is inserted into the ear drum to allow continual drainage. The tube prevents infections as long as it stays in place, which varies from six months to three years. When the tube falls out, the hole grows over. As many of 25% of children under the age of two who need **ear tubes** may need them again. Myringotomy and ear tube



Microsurgery being performed in the inner ear. (Photograph by Hans Halberstadt, Photo Researchers, Inc. Reproduced by permission.)

surgery is performed in a hospital, using a general anesthetic for most children and a local anesthetic for older children or adults. No anesthetic may be used for infants. The procedure usually takes about two hours. Most patients can go home the same day; children under three years of age and those with chronic diseases usually stay overnight.

Ear surgery for a perforated eardrum

Ear surgery for a perforated eardrum is only performed in rare cases where it does not heal on its own. In most cases, this is performed in a surgeon's office using a topical anesthetic. The surgeon scratches the undersurface of the eardrum, stimulating the skin to heal and the eardrum to close. A thin patch placed on the eardrum's outer surface allows the skin under the eardrum to heal.

Cochlear implants

Cochlear implants stimulate nerve ends within the inner ear, enabling deaf children to hear. The device has a microphone that remains outside the ear, a processor that selects and codes speech sounds, and a receiver/stimulator to convert the coded sounds to electric signals that stimulate the hearing nerve and are recognized by the brain as sound. During surgery, an incision is made behind and slightly above the ear. A circular hole is drilled in the bone to receive the device's internal coil. The mastoid bone leading to the middle ear is opened to receive the electrodes. The internal coil is inserted and secured, followed by the electrodes. The wound is stitched up and when it heals, an external unit comprised of a stimulator with a microphone is worn behind the ear. Performed in a hospital under general anesthesia, the operation takes about two hours and

usually requires a hospital stay overnight. The patient can resume normal activities in two to three weeks.

Ear surgery for tumors

Some ear tumors can be very serious and should be removed surgically. For a tumor on the skin of the ear canal, the skin is removed surgically, the bone beneath it is drilled away and a skin graft is placed in the ear canal. If the tumor is near the eardrum, the skin of the ear canal and the eardrum are removed along with the bone surrounding the ear canal. A skin graft is placed on the bare bone. For basal cell cancers and low grade glandular malignancies, surgical resection of the ear canal is adequate. Squamous cell carcinoma, a serious form of **cancer**, of the external ear canal requires radical surgery, followed by radiation therapy. Cholesteatoma, a benign tumor caused by an infection in a perforated eardrum that did not heal properly and can destroy the bones of hearing, is removed with microsurgery. **Mastoidectomy** is performed for mastoiditis, an inflammation of the middle ear, if medical therapy does not work. Petrous apicectomy is performed to drain the petrous apicitis, the bone between the middle ear and the clivus.

Ear surgery for congenital ear defects

Congenital atresia, the absence of the external ear canal, and congenital microtia, abnormal growth of the external ear, often occur together, although atresia can occur without microtia. Surgery to reconstruct the ear usually takes place when the child is four or five years old and may require several operations. A facial plastic surgeon and an ear surgeon work together, repairing the microtia first and then the atresia. During surgery, a bony opening is created over the bones of hearing. The surfaces of the bony ear canal are then relined with a skin graft from the thigh or abdomen. Tissue from behind the eardrum is used to create a new eardrum. In many cases, the middle ear will also need to be reconstructed. Surgery is performed in a hospital under general anesthesia.

Other types of ear surgery

Surgery may also be appropriate to remove multiple bony overgrowths of the ear canal or in rare cases of compromised auditory tube function, to narrow the tube.

Preparation

The preparation depends upon the type of ear surgery performed. For many procedures, blood and urine studies and hearing tests are conducted.

KEY TERMS

Auditory—Relating to the sense of the organs of hearing.

Cholesteatoma—A cystic mass of cells in the middle ear, occurring as a congenital defect or as a serious complication of a disease or traumatic condition of the ear.

Otologic—Relating to the study, diagnosis, and treatment of diseases of the ear and related structures.

Aftercare

The type of aftercare depends upon the type of surgery performed. In most cases, the ear(s) should be kept dry and warm. Non-prescription drugs such as **acetaminophen** can be used for **pain**.

Risks

The type of risk depends on the type of surgery performed. Total hearing loss is rare.

Resources

ORGANIZATIONS

- American Academy of Otolaryngology-Head and Neck Surgery, Inc. One Prince St., Alexandria VA 22314-3357. (703) 836-4444. < <http://www.entnet.org> > .
- American Hearing Research Foundation. 55 E. Washington St., Suite 2022, Chicago, IL 60602. (312) 726-9670. < <http://www.american-hearing.org/> > .
- American Speech-Language-Hearing Association. 10801 Rockville Pike, Rockville, MD 20852. (800) 638-8255. < <http://www.asha.org> > .

Lori De Milto

Ear tubes see **Myringotomy and ear tubes**

Ear wax impaction see **Cerumen impaction**

Eardrum perforation see **Perforated eardrum**

Eastern equine encephalitis see **Arbovirus encephalitis**

Eating disorders see **Anorexia nervosa; Bulimia nervosa**

Eaton agent pneumonia see **Mycoplasma infections**

Ebola virus infection see **Hemorrhagic fevers**

Ecchymosis see **Bruises**

ECG see **Electrocardiography**

Echinacea

Definition

Echinacea, or purple coneflower, is a perennial herb of the Composite family, commonly known as the daisy family. Most often referred to as the purple coneflower, this hardy plant also known as Sampson root, Missouri snakeroot, and rudbeckia. The prominent, bristly seed head inspired the generic name of the plant, taken from the Greek word *echinos* meaning hedgehog.

Description

Echinacea is a North American prairie native, abundant in the Mid-west, and cultivated widely in ornamental and medicinal gardens. The purple-pink rays of the blossom droop downward from a brassy hued center cone composed of many small, tubular florets. The conspicuous flowers bloom singly on stout, prickly stems from mid-summer to autumn. Flower heads may grow to 4 in (10.16 cm) across. The dark green leaves are opposite, entire, lanceolate, toothed, and hairy with three prominent veins. The narrow upper leaves are attached to the stem with stalks. The lower leaves are longer, emerging from the stem without a leaf stalk, and growing to 8 in (20.32 cm) in length. The plant develops deep, slender, black roots. Echinacea propagates easily from seed or by root cuttings. However, due to its increasing popularity as an herbal supplement, echinacea is numbered among the 19 medicinal plants considered at risk by the Vermont nonprofit organization, United Plant Savers.

Purpose

Three species of echinacea are useful medicinally: *Echinacea augustifolia*, *Echinacea purpurea*, and *Echinacea pallida*. The entire plant has numerous medicinal properties that act synergistically to good effect. Echinacea is most often used to boost the immune system and fight infection. Research has shown that echinacea increases production of interferon in the body. It is antiseptic and antimicrobial, with properties that act to increase the number of white blood cells available to destroy bacteria and

slow the spread of infection. As a depurative, the herbal extract cleanses and purifies the bloodstream, and has been used effectively to treat **boils**. Echinacea is vulnerary, promoting wound healing through the action of a chemical substance in the root known as caffeic acid glycoside. As an alterative and an immuno-modulator, echinacea acts gradually to promote beneficial change in the entire system. It has also been used to treat urinary infection and *Candida albicans* infections. Echinacea is a febrifuge, useful in reducing fevers. It is also useful in the treatment of hemorrhoids. A tincture, or a strong decoction of echinacea serves as an effective mouthwash for the treatment of pyorrhea and gingivitis.

Native American plains Indians relied on echinacea as an all-purpose antiseptic. The Sioux tribe valued the root as a remedy for snake bite, the Cheyenne tribe chewed the root to quench thirst, and another tribe washed their hands in a decoction of echinacea to increase their tolerance of heat. European settlers learned of the North American herb's many uses, and soon numerous echinacea-based remedies were commercially available from pharmaceutical companies in the United States. Echinacea was a popular remedy in the United States through the 1930s. It was among many medicinal herbs listed in the *U.S. Pharmacopoeia*, the official United States government listing of pharmaceutical raw materials and recipes. The herb fell out of popular use in the United States with the availability of **antibiotics**. In West Germany, over 200 preparations are made from the species *E. purpurea*. Commercially prepared salves, tinctures, teas, and extracts are marketed using standardized extracts. Echinacea is regaining its status in the United States as a household medicine-chest staple in many homes. It is one of the best-selling herbal supplements in United States health food stores.

Clinical studies have found that the entire plant possesses medicinal properties with varying levels of effectiveness. Echinacea is of particular benefit in the treatment of upper respiratory tract infections. Some research has shown that echinacea activates the macrophages that destroy **cancer** cells and pathogens. When taken after cancer treatments, an extract of the root has been found to increase the body's production of white blood cells. Echinacea has been shown to be most effective when taken at the first sign of illness, rather than when used as a daily preventative. Other research has demonstrated the significant effect of *E. purpurea* root on reducing the duration and severity of colds and flu. Some herbal references list only the root as the medicinal part, others include the

aerial parts of the plant, particularly the leaf. Research studies in Europe and the United States have concluded that the entire plant is medicinally effective. Most research has been done on the species *E. pallida* and *E. purpurea*. All three species of echinacea are rich in **vitamins** and **minerals**. Echinacea is an herbal source of niacin, chromium, iron, manganese, selenium, silicon, and zinc.

Preparations

The quality of any herbal supplement depends greatly on the conditions of weather and soil where the herb was grown, the timing and care in harvesting, and the manner of preparation and storage.

Decoction is the best method to extract the mineral salts and other healing components from the coarser herb materials, such as the root, bark, and stems. It is prepared by adding 1 oz (28.4 g) of the dried plant materials, or 2 oz (56.7 g) of fresh plant parts, to 1 pt (0.47 l) of pure, unchlorinated, boiled water in a non-metallic pot. Simmer for about one half hour. Strain and cover. A decoction may be refrigerated for up to two days and retain its healing qualities.

An infusion is the method used to derive benefits from the leaves, flowers, and stems in the form of an herbal tea. Use twice as much fresh, chopped herb as dried herb. Steep in 1 pt (0.47 l) of boiled, unchlorinated water for 10–15 minutes. Strain and cover. Drink warm, sweetened with honey if desired. A standard dose is three cups per day. An infusion will keep for up to two days in the refrigerator and retain its healing qualities.

A tincture is the usual method to prepare a concentrated form of the herbal remedy. Tinctures, properly prepared and stored, will retain medicinal potency for two years or more. Combine 4 oz (114 g) of finely cut fresh or powdered dry herb with 1 pt (0.47 l) of brandy, gin, or vodka in a glass container. The alcohol should be enough to cover the plant parts and have a 50/50 ratio of alcohol to water. Place the mixture away from light for about two weeks, shaking several times each day. Strain and store in a tightly capped, dark glass bottle. A standard dose is 0.14 oz (4 ml) of the tincture three times a day.

Precautions

Echinacea is considered safe in recommended doses. Pregnant or lactating women, however, are advised not to take echinacea in injection form.

KEY TERMS

Alterative—A medicinal substance that acts gradually to nourish and improve the system.

Antimicrobial—A plant substance that acts to inhibit the growth of harmful microorganisms, or acts to destroy them.

Febrifuge—A plant substance that acts to prevent or reduce fever.

Glycoside—An herbal carbohydrate that exerts powerful effect on hormone-producing tissues. The glycoside breaks down into a sugar and a non-sugar component.

Lanceolate—Narrow, leaf shape that is longer than it is wide, and pointed at the end.

Macrophage—Specialized cells present throughout the lymphoid tissues of the body that circulate in the bloodstream. Macrophages have a surface marker that stimulates other cells to react to an antigen.

Because the plant has proven immuno-modulating properties, individuals with systemic lupus erythematosus, **rheumatoid arthritis**, **tuberculosis**, leukemia, multiple sclerosis, or **AIDS** should consult their physician before using echinacea. Echinacea should not be given to children under two years of age, and it should only be given to children over two in consultation with a physician. Research indicates that echinacea is most effective when taken at first onset of symptoms of cold or flu, and when usage is continued no longer than eight weeks. There is some indication that the herb loses its effectiveness when used over a long period of time. It is necessary to interrupt use for a minimum of several weeks in order to give the body's immune system the opportunity to rest and adjust.

Side effects

No side effects are reported with oral administration of echinacea, either in tincture, capsule, or as a tea, when taken according to recommended doses. Chills, **fever**, and allergic reactions have been reported in some research studies using an injection of the plant extract.

Interactions

None reported. When used in combination with other herbs, dosage should be lowered.

Resources

PERIODICALS

Deneen, Sally, and Tracey C. Rembert. "Stalking Medicinal Plants, An International Trade Imperils Wild Herbs." *E Magazine* July-August 1999.

OTHER

Herb World News Online, Research Reviews. Herb Research Foundation. 1999. <<http://www.herbs.org>>.

Clare Hanrahan

Echinococcosis

Definition

Echinococcosis (Hydatid disease) refers to human infection by the immature (larval) form of tapeworm, *Echinococcus*. One of three forms of the *Echinococcus* spp., *E. granulosus*, lives on dogs and livestock, and infects humans through contact with these animals. Allergic reactions and damage to various organs from cyst formation are the most common forms of disease in humans.

Description

E. granulosus is found in many areas of Africa, China, South America, Australia, New Zealand, and Mediterranean and eastern Europe, as well as in parts of the western United States. The parasite lives in regions where dogs and livestock cohabitate. Direct exposure to infectious dogs, as well as parasitic eggs released into the environment during shedding, are both sources of human infection.

In humans, cysts containing the larvae develop after ingestion of eggs. Cysts form primarily in the lungs and liver. Cysts developing in the liver are responsible for about two-thirds of echinococcosis cases. Echinococcosis is a significant public health problem in many areas of the world, but control programs have decreased the rate of infection in some regions. In Kenya alone, the numbers of persons infected each year is as high as 220 per 100,000 population.

Causes and symptoms

After ingestion, the eggs develop into embryos within the intestines and then travel to the liver and lungs through major blood vessels. The embryos then begin to form cysts within the liver and lungs, causing

damage as they enlarge over a period of five to 20 years. Cysts may become over 8 in (20.3 cm) or more in size and contain a huge amount of highly allergenic fluid. Studies show that while the liver is most often targeted, lungs, brain, heart, and bone can also be affected.

The major symptoms are due to compression damage, blockage of vessels and ducts (such as the bile ducts), and leakage of fluid from cysts. The following symptoms are frequent.

- Liver involvement causes **pain** and eventually **jaundice** or **cholangitis** due to blockage of bile ducts. Infection of cysts leads to abscesses in up to 20%.
- Lung cysts cause **cough** and chest pain.
- Bone cysts cause **fractures** and damage to bone tissue.
- Heart involvement leads to irregularities of heart beat and inflammation of the covering of the heart (pericardium).
- Allergic reactions occur from leakage of cyst fluid that contains antigens. **Itching**, **fever**, and **rashes** are frequent, and fatal allergic reactions (**anaphylaxis**) have been reported. Eosinophils, which are blood cells involved in allergic reactions, are increased in many patients.

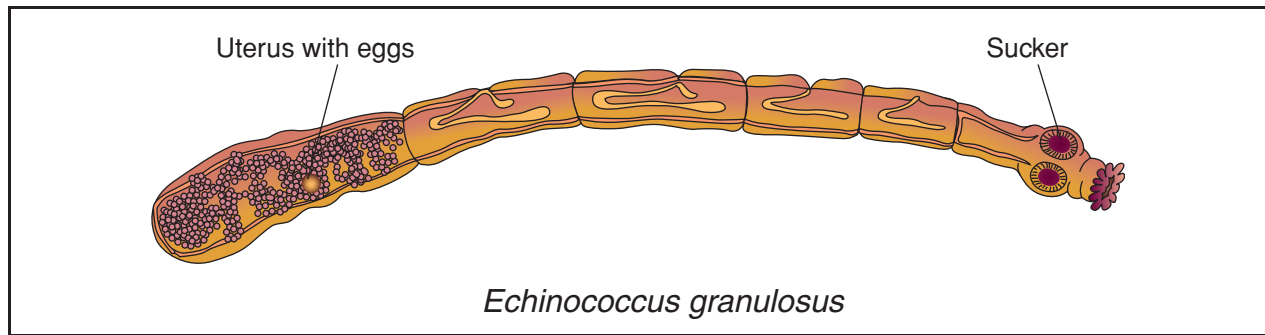
Diagnosis

X rays, **computed tomography scans** (CT scans), and ultrasound are very helpful in detecting cysts. Some cysts will develop characteristic hardening of organ tissues from calcium deposits (calcifications). Blood tests to detect antibodies are useful when positive, but up to 50% of patients have negative results. Examination of aspirated cyst fluid for parasites can be diagnostic, but carries the danger of a fatal allergic reaction. Treatment with anti-parasitic medications before aspiration is reported to decrease allergic complications and decrease the risk of spread during the procedure.

Treatment

Treatment depends on the size and location of cysts, as well as the symptoms they are producing. Surgical removal of cysts and/or surrounding tissue is the accepted method of treatment, but carries a risk of cyst rupture with spread or allergic reactions. Recent studies using medication alongside aspiration and drainage of cysts instead of surgery are very encouraging.

The medication albendazole can be taken before or after surgery or alone without surgery. However, its



Infection with the larva of *Echinococcus granulosus* (shown above) is responsible for the disease echinococcosis. (Illustration by Electronic Illustrators Group.)

KEY TERMS

Allergenic—A substance capable of causing an allergic reaction.

Cholangitis—Infection or inflammation of the bile ducts; often causes abdominal pain, fever, and jaundice.

Computed tomography (CT) scan—A specialized x-ray procedure in which cross-sections of the area in question can be examined in detail.

Cyst—A protective sac that includes either fluid or the cell of an organism. The cyst enables many organisms to survive in the environment for long periods of time without need for food or water.

Embryo—The very beginning stages of development of an organism.

Jaundice—The yellow-greenish coloring of the skin and eyes due to the presence of bile pigments. The presence of jaundice is usually, but not always, a sign of liver disease.

Tapeworm—An intestinal parasite that attaches to the intestine or travels to other organs such as the liver and lungs.

Ultrasound—A noninvasive procedure based on changes in sound waves of a frequency that cannot be heard, but respond to changes in tissue composition.

effectiveness as a single treatment is still not known. Multiple courses of medication are often necessary, with cure rates of only about 30%. Response to treatment is best monitored by serial CT scans or similar x-ray studies.

Prevention

Good hand washing, treating infected dogs, and preventing dogs' access to slaughter houses discourage spread of the disease. Limiting the population of stray dogs has also been helpful.

Resources

OTHER

“Percutaneous Drainage Compared with Surgery for HepaticHydatid Cysts.” *New England Journal of Medicine Online*. <<http://content.nejm.org>> .

David Kaminstein, MD

Echinococcus granulosus infection see
Echinococcosis

Echocardiography

Definition

Echocardiography is a diagnostic test that uses ultrasound waves to create an image of the heart muscle. Ultrasound waves that rebound or echo off the heart can show the size, shape, and movement of the heart's valves and chambers as well as the flow of blood through the heart. Echocardiography may show such abnormalities as poorly functioning heart valves or damage to the heart tissue from a past **heart attack**.

Purpose

Echocardiography is used to diagnose certain cardiovascular diseases. In fact, it is one of the most

widely used diagnostic tests for heart disease. It can provide a wealth of helpful information, including the size and shape of the heart, its pumping strength, and the location and extent of any damage to its tissues. It is especially useful for assessing diseases of the heart valves. It not only allows doctors to evaluate the heart valves, but it can detect abnormalities in the pattern of blood flow, such as the backward flow of blood through partly closed heart valves, known as regurgitation. By assessing the motion of the heart wall, echocardiography can help detect the presence and assess the severity of **coronary artery disease**, as well as help determine whether any chest **pain** is related to heart disease. Echocardiography can also help detect hypertrophic cardiomyopathy, in which the walls of the heart thicken in an attempt to compensate for heart muscle weakness. The biggest advantage to echocardiography is that it is noninvasive (does not involve breaking the skin or entering body cavities) and has no known risks or side effects.

Precautions

Echocardiography is an extremely safe procedure and no special precautions are required.

Description

Echocardiography creates an image of the heart using ultra-high-frequency sound waves—sound waves that are too high in frequency to be heard by the human ear. The technique is very similar to ultrasound scanning commonly used to visualize the fetus during **pregnancy**.

An echocardiography examination generally lasts between 15–30 minutes. The patient lies bare-chested on an examination table. A special gel is spread over the chest to help the transducer make good contact and slide smoothly over the skin. The transducer, a small hand-held device at the end of a flexible cable, is placed against the chest. Essentially a modified microphone, the transducer directs ultrasound waves into the chest. Some of the waves get echoed (or reflected) back to the transducer. Since different tissues and blood all reflect ultrasound waves differently, these sound waves can be translated into a meaningful image of the heart, which can be displayed on a monitor or recorded on paper or tape. The patient does not feel the sound waves, and the entire procedure is painless. In fact, there are no known side effects.

Occasionally, variations of the echocardiography test are used. For example, Doppler echocardiography



A patient getting an EKG. (Photo Researchers. Reproduced by permission.)

employs a special microphone that allows technicians to measure and analyze the direction and speed of blood flow through blood vessels and heart valves. This makes it especially useful for detecting and evaluating regurgitation through the heart valves. By assessing the speed of blood flow at different locations around an obstruction, it can also help to precisely locate the obstruction.

An **exercise** echocardiogram is an echocardiogram performed during exercise, when the heart muscle must work harder to supply blood to the body. This allows doctors to detect heart problems that might not be evident when the body is at rest and needs less blood. For patients who are unable to exercise, certain drugs can be used to mimic the effects of exercise by dilating the blood vessels and making the heart beat faster.

Preparation

The patient removes any clothing and jewelry above the chest.

Aftercare

No special measures need to be taken following echocardiography.

Risks

There are no known risks associated with the use of echocardiography.

Normal results

A normal echocardiogram shows a normal heart structure and the normal flow of blood through the

KEY TERMS

Noninvasive—Pertaining to a diagnostic procedure or treatment that does not require the skin to be broken or a body cavity to be entered.

Regurgitation—Backward flow of blood through a partly closed heart valve.

Transducer—A device that converts electrical signals into ultrasound waves and ultrasound waves back into electrical impulses.

Ultrasound—Sound waves at a frequency of over 20,000 kHz, often used for diagnostic imaging.

heart chambers and heart valves. However, a normal echocardiogram does not rule out the possibility of heart disease.

Abnormal results

An echocardiogram may show a number of abnormalities in the structure and function of the heart, such as:

- thickening of the wall of the heart muscle (especially the left ventricle)
- abnormal motion of the heart muscle
- blood leaking backward through the heart valves (regurgitation)
- decreased blood flow through a heart valve (stenosis)

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

National Heart, Lung and Blood Institute. PO Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

Robert Scott Dinsmoor

Echovirus infections see **Enterovirus infections**

Eclampsia see **Preeclampsia and eclampsia**

ECT see **Electroconvulsive therapy**

Ectopic orifice of the ureter see **Congenital ureter anomalies**

Ectopic pregnancy

Definition

In an ectopic **pregnancy**, the fertilized egg implants in a location outside the uterus and tries to develop there. The word ectopic means “in an abnormal place or position.” The most common site is the fallopian tube, the tube that normally carries eggs from the ovary to the uterus. However, ectopic pregnancy can also occur in the ovary, the abdomen, and the cervical canal (the opening from the uterus to the vaginal canal). The phrases tubal pregnancy, ovarian pregnancy, cervical pregnancy, and abdominal pregnancy refer to the specific area of an ectopic pregnancy.

Description

Once a month, an egg is produced in a woman’s ovary and travels down the fallopian tube where it meets the male’s sperm and is fertilized. In a normal pregnancy the fertilized egg, or zygote, continues on its passage down the fallopian tube and enters the uterus in three to five days. The zygote continues to grow, implanting itself securely in the wall of the uterus. The zygote’s cells develop into the embryo (the organism in its first two months of development) and placenta (a spongy structure that lines the uterus and nourishes the developing organism).

In a tubal ectopic pregnancy, the fertilized egg cannot make it all the way down the tube because of scarring or obstruction. The fallopian tube is too narrow for the growing zygote. Eventually the thin walls of the tube stretch and may burst (rupture), resulting in severe bleeding and possibly the **death** of the mother. More than 95% percent of all ectopic pregnancies occur in the fallopian tube. Only 1.5% develop in the abdomen; less than 1% develop in the ovary or the cervix.

Causes and symptoms

As many as 50% of women with ectopic pregnancies have a history of **pelvic inflammatory disease** (PID). This is an infection of the fallopian tubes (salpingitis) that can spread to the uterus or ovaries. It is most commonly caused by the organisms *Gonorrhea* and *Chlamydia* and is usually transmitted by sexual intercourse.

Other conditions also increase the risk of ectopic pregnancy. They include:

- **Endometriosis**. A condition in which the tissue that normally lines the uterus is found outside the uterus, and can block a fallopian tube.

- Exposure to diethylstilbestrol (DES) as a fetus. If a woman's mother took DES (a synthetic version of the hormone estrogen) during pregnancy, the woman may have abnormalities in her fallopian tubes that can make ectopic pregnancy more likely.
- Taking hormones. Estrogen and progesterone are hormones that regulate the menstrual cycle and may be in medications prescribed by a doctor for birth control or other reasons. Taking these hormones can affect the interior lining of the fallopian tubes and slow the movement of the fertilized egg down the tube. Women who become pregnant in spite of taking some progesterone-only contraceptives have a greater chance of an ectopic pregnancy. Ectopic pregnancy is also more likely when the ovaries are artificially stimulated with hormones to produce eggs for **in vitro fertilization** (a procedure in which eggs are taken from a woman's body, fertilized, and then placed in the uterus in an attempt to conceive a child).
- Use of an intrauterine device (**IUD**). These contraceptive devices are designed to prevent fertilized eggs from becoming implanted in the uterus, but they have only a minimal effect on preventing ectopic pregnancies. Therefore, if a woman becomes pregnant while using an IUD for **contraception**, the fertilized egg is more likely to be implanted someplace other than the uterus. For example, among women who become pregnant while using a progesterone-bearing IUD, about 15% have ectopic pregnancies.
- Surgery on a fallopian tube. The risk of ectopic pregnancy can be as high as 60% after undergoing elective tubal sterilization, a procedure in which the fallopian tubes are severed to prevent pregnancy. Women who have successful surgery to reverse the procedure are also more likely to have an ectopic pregnancy.

Early symptoms

In an ectopic pregnancy all the hormonal changes associated with a normal pregnancy may occur. The early symptoms include: **fatigue**; **nausea**; a missed period; breast tenderness; **low back pain**; mild cramping on one side of the pelvis; and abnormal vaginal bleeding, usually spotting.

Later symptoms

As the embryo grows too large for the confined space in the tube, the first sign that something is wrong may be a stabbing **pain** in the pelvis or abdomen. If the tube has ruptured, blood may irritate the diaphragm and cause shoulder pain. Other warning signs are lightheadedness and **fainting**.

Diagnosis

To confirm an early diagnosis of ectopic pregnancy, the doctor must determine first that the patient is pregnant and that the location of the embryo is outside the uterus. If an ectopic pregnancy is suspected, the doctor will perform a pelvic examination to locate the source of pain and to detect a mass in the abdomen.

Several laboratory tests of the patient's blood provide information for diagnosis. Measurement of the human chorionic gonadotropin (hCG) level in the patient's blood serum is the most useful laboratory test in the early stages. In a normal pregnancy, the level of this hormone doubles about every two days during the first 10 weeks. In an ectopic pregnancy, the rate of the increase is much slower and the low hCG for the stage of the pregnancy is a strong indication that the pregnancy is abnormal. (It could also represent a **miscarriage** in progress.) The level is usually tested several times over a period of days to determine whether or not it is increasing at a normal rate.

Progesterone levels in the blood are also measured. Lower than expected levels can indicate that the pregnancy is not normal.

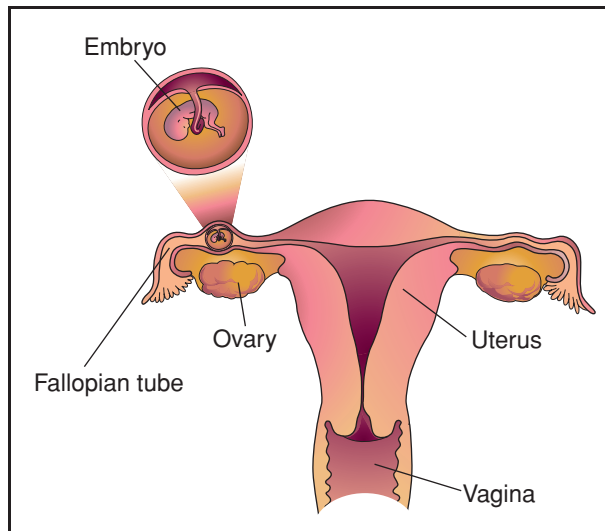
An ultrasound examination may provide information about whether or not the pregnancy is ectopic. A device called a transducer, which emits high frequency sound waves, is moved over the surface of the patient's abdomen or inserted into the vagina. The sound waves bounce off of the internal organs and create an image on a screen. The doctor should be able to see whether or not there is a fetus developing in the uterus after at least five weeks of gestation. Before that point, a normal pregnancy is too small to see.

A culdocentesis may also help confirm a diagnosis. In this procedure a needle is inserted into the space at the top of the vagina, behind the uterus and in front of the rectum. Blood in this area may indicate bleeding from a ruptured fallopian tube.

A **laparoscopy** will enable the doctor to see the patient's reproductive organs and examine an ectopic pregnancy. In this technique, a hollow tube with a light on one end is inserted through a small incision in the abdomen. Through this instrument the internal organs can be observed.

Treatment

Ectopic pregnancy requires immediate treatment. The earlier the condition is treated, the better the chance to preserve the fallopian tube intact for future normal pregnancies.



In an ectopic pregnancy, the fertilized egg implants in a location outside the uterus and attempts to develop at that site. The most common site of an ectopic pregnancy is the fallopian tube, but it can occur in the ovary, the abdomen, and the cervical wall. More than 95% of all ectopic pregnancies occur in the fallopian tube. (Illustration by Electronic Illustrators Group.)

Medical

If the ectopic pregnancy is discovered in a very early stage of development, the drug methotrexate may be given. The best results are obtained when the pregnancy is less than six weeks old and the tubal mass is no more than 1.4 in (3.5 cm) in diameter. Methotrexate, which has been used successfully since 1987, works by inhibiting the growth of rapidly growing cells. (It is also used to treat some cancers.) Most side effects are mild and temporary, but the patient must be monitored after treatment. Usually the medication is injected into the muscle in a single dose, but may also be given intravenously or injected directly into the fallopian tube to dissolve the embryonic tissue. Methotrexate has also been used to treat ovarian, abdominal, and cervical pregnancies that are discovered in the early stages.

Surgical

When a laparoscopy is done to visualize the ectopic pregnancy, the scope can be fitted with surgical tools and used to remove the ectopic mass immediately after it is identified. The affected fallopian tube can be repaired or removed as necessary. This procedure can be done without requiring the patient to stay in the hospital overnight.

When the pregnancy has ruptured, a surgical incision into the abdomen, or laparotomy, is

KEY TERMS

Embryo—In humans, the developing organism from conception until approximately the end of the second month.

Fallopian tube—The tube that carries the egg from the ovary to the uterus.

Human chorionic gonadotropin (hCG)—A hormone excreted during the development of an embryo or fetus.

Laparoscopy—Examination of the contents of the abdominal cavity with a fiberoptic tube inserted through a small incision.

Laparotomy—Surgical incision into the abdomen to locate, repair, and/or remove injured or diseased tissues.

Pelvic inflammatory disease (PID)—Acute or chronic inflammation in the pelvic cavity, particularly inflammation of the fallopian tubes (salpingitis) and its complications.

Rupture—A breaking apart of an organ or tissue.

Salpingitis—Inflammation of the fallopian tube.

Tubal pregnancy—Pregnancy in one of the fallopian tubes.

Zygote—The fertilized egg.

performed to stop the immediate loss of blood and to remove the embryo. This usually requires **general anesthesia** and a hospital stay. Every effort is made to preserve and repair the injured fallopian tube. However, if the fallopian tube has already ruptured, repair is extremely difficult and the tube is usually removed.

Alternative treatment

Ectopic pregnancy was first described in the eleventh century and was a potentially fatal condition until the advent of surgery and blood transfusions in the early twentieth century. The sophisticated diagnostic tools and surgical procedures developed since the 1970s have equipped modern medicine with the tools to not only save a woman's life, but also to preserve her future fertility.

Although there are herbal remedies for the temporary relief of the common symptoms of **anxiety** and abdominal discomfort, prompt medical treatment is the only sure remedy for ectopic pregnancy.

Prognosis

Ectopic pregnancies are the leading cause of pregnancy-related deaths in the first trimester and account for 9% of all pregnancy-related deaths in the United States. More than 1% of pregnancies are ectopic, and they are becoming more common. The reason for this increase is not clearly understood, though it is thought that the dramatic increase in **sexually transmitted diseases (STD)** is at least partly responsible.

The earlier an ectopic pregnancy is diagnosed and treated, the better the outcome. The chances of having a successful pregnancy are lower after an ectopic pregnancy, but depend on the extent of permanent fallopian tube damage. If the tube has been spared, chances are as high as 60%. The chances of a successful pregnancy after the removal of one tube are 40%.

Prevention

Many forms of ectopic pregnancy cannot be prevented. However, tubal pregnancies, which make up the majority of ectopic pregnancies, may be prevented by avoiding conditions that cause damage to the fallopian tubes. Since half of all women who experience ectopic pregnancy have a history of PID, avoiding this infection or getting early diagnosis and treatment for sexually transmitted diseases will decrease the risk of a future problem.

Resources

ORGANIZATIONS

Resolve. 1310 Broadway, Somerville, MA 02144-1731. (617) 623-0744. <<http://www.resolve.org>>.

Karen Ericson, RN

Eczema see **Dermatitis**

ED see **Impotence**

Description

Normally the body maintains a balance of fluid in tissues by ensuring that the same amount of water entering the body also leaves it. The circulatory system transports fluid within the body via its network of blood vessels. The fluid, which contains oxygen and nutrients needed by the cells, moves from the walls of the blood vessels into the body's tissues. After its nutrients are used up, fluid moves back into the blood vessels and returns to the heart. The lymphatic system (a network of channels in the body that carry lymph, a colorless fluid containing white blood cells to fight infection) also absorbs and transports this fluid. In edema, either too much fluid moves from the blood vessels into the tissues, or not enough fluid moves from the tissues back into the blood vessels. This fluid imbalance can cause mild to severe swelling in one or more parts of the body.

Causes and symptoms

Many ordinary factors can upset the balance of fluid in the body to cause edema, including:

- **Immobility.** The leg muscles normally contract and compress blood vessels to promote blood flow with walking or running. When these muscles are not used, blood can collect in the veins, making it difficult for fluid to move from tissues back into the vessels.
- **Heat.** Warm temperatures cause the blood vessels to expand, making it easier for fluid to cross into surrounding tissues. High humidity also aggravates this situation.
- **Medications.** Certain drugs, such as steroids, hormone replacements, **nonsteroidal anti-inflammatory drugs (NSAIDs)**, and some blood pressure medications may affect how fast fluid leaves blood vessels.
- **Intake of salty foods.** The body needs a constant concentration of salt in its tissues. When excess salt is taken in, the body dilutes it by retaining fluid.
- **Menstruation and pregnancy.** The changing levels of hormones affect the rate at which fluid enters and leaves the tissues.

Some medical conditions may also cause edema, including:

- **Heart failure.** When the heart is unable to maintain adequate blood flow throughout the circulatory system, the excess fluid pressure within the blood vessels can cause shifts into the interstitial spaces. Left-sided heart failure can cause **pulmonary edema**, as fluid shifts into the lungs. The patient may develop rapid, shallow respirations, **shortness of breath**, and a **cough**. Right-sided heart failure can cause pitting

Edema

Definition

Edema is a condition of abnormally large fluid volume in the circulatory system or in tissues between the body's cells (interstitial spaces).

edema, a swelling in the tissue under the skin of the lower legs and feet. Pressing this tissue with a finger tip leads to a noticeable momentary indentation.

- **Kidney disease.** The decrease in sodium and water excretion can result in fluid retention and overload.
- **Thyroid or liver disease.** These conditions can change the concentration of protein in the blood, affecting fluid movement in and out of the tissues. In advanced liver disease, the liver is enlarged and fluid may build-up in the abdomen.
- **Malnutrition.** Protein levels are decreased in the blood, and in an effort to maintain a balance of concentrations, fluid shifts out of the vessels and causes edema in tissue spaces.

Some conditions that may cause swelling in just one leg include:

- **Blood clots.** Clots can cause pooling of fluid and may be accompanied by discoloration and **pain**. In some instances, clots may cause no pain.
- **Weakened veins.** **Varicose veins**, or veins whose walls or valves are weak, can allow blood to pool in the legs. This is a common condition.
- **Infection and inflammation.** Infection in leg tissues can cause inflammation and increasing blood flow to the area. Inflammatory diseases, such as **gout** or arthritis, can also result in swelling.
- **Lymphedema.** Blocked lymph channels may be caused by infection, scar tissue, or hereditary conditions. Lymph that can't drain properly results in edema. **Lymphedema** may also occur after **cancer** treatments, when the lymph system is impaired by surgery, radiation, or **chemotherapy**.
- **Tumor.** Abnormal masses can compress leg vessels and lymph channels, affecting the rate of fluid movement.

Symptoms vary depending on the cause of edema. In general, weight gain, puffy eyelids, and swelling of the legs may occur as a result of excess fluid volume. Pulse rate and blood pressure may be elevated. Hand and neck veins may be observed as fuller.

Diagnosis

Edema is a sign of an underlying problem, rather than a disease unto itself. A diagnostic explanation should be sought. Patient history and presenting symptoms, along with laboratory blood studies, if indicated, assist the health professional in determining the cause of the edema.

Treatment

Treatment of edema is based on the cause. Simple steps to lessen fluid build-up may include:

- **Reducing sodium intake.** A high sodium level causes or aggravates fluid retention.
- **Maintaining proper weight.** Being overweight slows body fluid circulation and puts extra pressure on the veins.
- **Exercise.** Regular exercise stimulates circulation.
- **Elevation of the legs.** Placing the legs at least 12 in (30.5 cm) above the level of the heart for 10–15 minutes, three to four times a day, stimulates excess fluid re-entry into the circulatory system.
- **Use of support stocking.** Elastic stockings, available at most medical supply or drug stores, will compress the leg vessels, promoting circulation and decreasing pooling of fluid due to gravity.
- **Massage.** Massaging the body part can help to stimulate the release of excess fluids, but should be avoided if the patient has blood clots in the veins.
- **Travel breaks.** Sitting for long periods will increase swelling in the feet and ankles. Standing and/or walking at least every hour or two will help stimulate blood flow.

The three “Ds”—diuretics, digitalis, and diet—are frequently prescribed for medical conditions that result in excess fluid volume. **Diuretics** are medications that promote urination of sodium and water. Digoxin is a digitalis preparation that is sometimes needed to decrease heart rate and increase the strength of the heart's contractions. Dietary recommendations include less sodium in order to decrease fluid retention. Consideration of adequate protein intake is also made.

For patients with lymphedema, a combination of therapies may prove effective. Combined decongestive therapy includes the use of manual lymph drainage (MLD), compression bandaging, garments and pumps, and physical therapy. MLD involves the use of light massage of the subcutaneous tissue where the lymph vessels predominate. Massage begins in an area of the body trunk where there is normal lymph function and proceeds to areas of lymphatic insufficiency, in an effort to stimulate new drainage tract development. (MLD should not be used for patients with active cancer, deep vein clots, congestive heart failure, or cellulitis.) MLD sessions are followed by application of compression garments or pumps. Physical therapy is aimed at strengthening the affected limb and increasing joint mobility.



Gross lymphedema in the arm of an elderly woman following radiotherapy treatment for breast cancer. (Photograph by Dr. P. Marazzi. Photo Researchers, Inc. Reproduced by permission.)

KEY TERMS

Digitalis—A naturally occurring compound used in the preparation of the medication, digoxin, prescribed to increase the heart rate and strengthen the force of the heart's contractions.

Diuretics—Medications used in the treatment of fluid overload, to promote excretion of sodium and water.

Interstitial spaces—Areas of the body occurring outside the vessels or organs, between the cells.

Pitting edema—A swelling in the tissue under the skin, resulting from fluid accumulation, that is measured by the depth of indentation made by finger pressure over a bony prominence.

edema. Foods that worsen edema, such as alcohol, **caffeine**, sugar, dairy products, soy sauce, animal protein, chocolate, olives, and pickles, should be avoided. Diuretic herbs can also help relieve edema. One of the best herbs for this purpose is dandelion (*Taraxacum mongolicum*), since, in addition to its diuretic action, it is a rich source of potassium. (Diuretics flush potassium from the body and it must be replaced to avoid potassium deficiency.) **Hydrotherapy** using daily contrast applications of hot and cold (either compresses or immersion) may also be helpful.

Resources

ORGANIZATIONS

Lymphedema and Wound Care Clinic of Austin. 5750 Balcones Dr., Ste. 110, Austin, TX 78731. (512) 453-1930.

Alternative treatment

Dietary changes, in addition to cutting back the amount of sodium eaten, may also help reduce

Kathleen D. Wright, RN

Edrophonium test see **Tensilon test**

Edwards' syndrome

Definition

Edwards' syndrome is caused by an extra copy of chromosome 18. For this reason, it is also called trisomy 18 syndrome. The extra chromosome is lethal for most babies born with this condition. It causes major physical abnormalities and severe mental retardation, and very few children afflicted with this disease survive beyond a year.

Description

Humans normally have 23 pairs of chromosomes. Chromosomes are numbered 1–22, and the 23rd pair is composed of the sex chromosomes, X and Y. A person inherits one set of 23 chromosomes from each parent. Occasionally, a genetic error occurs during egg or sperm cell formation. A child conceived with such an egg or sperm cell may inherit an incorrect number of chromosomes.

In the case of Edwards' syndrome, the child inherits three, rather than two, copies of chromosome 18. Trisomy 18 occurs in approximately one in every 3,000 newborns and affects girls more often than boys. Women older than their early thirties have a greater risk of conceiving a child with trisomy 18, but it can occur in younger women.

Causes and symptoms

A third copy of chromosome 18 causes numerous abnormalities. Most children born with Edwards' syndrome appear weak and fragile, and they are often underweight. The head is unusually small and the back of the head is prominent. The ears are malformed and low-set, and the mouth and jaw are small. The baby may also have a **cleft lip** or **cleft palate**. Frequently, the hands are clenched into fists, and the index finger overlaps the other fingers. The child may have clubfeet and toes may be webbed or fused.

Numerous problems involving the internal organs may be present. Abnormalities often occur in the lungs and diaphragm (the muscle that controls breathing), and heart defects and blood vessel malformations are common. The child may also have malformed kidneys and abnormalities of the urogenital system.

Diagnosis

Physical abnormalities point to Edwards' syndrome, but definitive diagnosis relies on karyotyping.

KEY TERMS

Aminocentesis—A procedure in which a needle is inserted through a pregnant woman's abdomen and into her uterus to withdraw a small sample of amniotic fluid. The amniotic fluid can be examined for signs of disease or other problems afflicting the fetus.

Chorionic villus sampling—A medical test that is best done during weeks 10–12 of a pregnancy. The procedure involves inserting a needle into the placenta and withdrawing a small amount of the chorionic membrane for analysis.

Chromosome—A structure composed of deoxyribonucleic acid (DNA) contained within a cell's nucleus (center) where genetic information is stored. Humans have 23 pairs of chromosomes, each of which has recognizable characteristics (such as length and staining patterns) that allow individual chromosomes to be identified. Identification is assigned by number (1–22) or letter (X or Y).

Karyotyping—A laboratory test used to study an individual's chromosome make-up. Chromosomes are separated from cells, stained, and arranged in order from largest to smallest so that their number and structure can be studied under a microscope.

Maternal serum analyte screening—A medical procedure in which a pregnant woman's blood is drawn and analyzed for the levels of certain hormones and proteins. These levels can indicate whether there may be an abnormality in the unborn child. This test is not a definitive indicator of a problem and is followed by more specific testing such as amniocentesis or chorionic villus sampling.

Trisomy—A condition in which a third copy of a chromosome is inherited. Normally only two copies should be inherited.

Ultrasound—A medical test that is also called ultrasonography. Sound waves are directed against internal structures in the body. As sound waves bounce off the internal structure, they create an image on a video screen. An ultrasound of a fetus at weeks 16–20 of a pregnancy can be used to determine structural abnormalities.

Karyotyping involves drawing the baby's blood or bone marrow for a microscopic examination of the chromosomes. Using special stains and microscopy,

individual chromosomes are identified, and the presence of an extra chromosome 18 is revealed.

Trisomy 18 can be detected before birth. If a pregnant woman is older than 35, has a family history of genetic abnormalities, has previously conceived a child with a genetic abnormality, or has suffered earlier miscarriages, she may undergo tests to determine whether her child carries genetic abnormalities. Potential tests include maternal serum analysis or screening, ultrasonography, **amniocentesis**, and chorionic villus sampling.

Treatment

There is no cure for Edwards' syndrome. Since trisomy 18 babies frequently have major physical abnormalities, doctors and parents face difficult choices regarding treatment. Abnormalities can be treated to a certain degree with surgery, but extreme invasive procedures may not be in the best interests of an infant whose lifespan is measured in days or weeks. Medical therapy often consists of supportive care with the goal of making the infant comfortable, rather than prolonging life.

Prognosis

Most children born with trisomy 18 die within their first year of life. The average lifespan is less than two months for 50% of the children, and 90–95% die before their first birthday. The 5–10% of children who survive their first year are severely mentally retarded. They need support to walk, and learning is limited. Verbal communication is also limited, but they can learn to recognize and interact with others.

Prevention

Edwards' syndrome cannot be prevented.

Resources

ORGANIZATIONS

Chromosome 18 Registry & Research Society. 6302 Fox Head, San Antonio, TX 78247. (210) 657-4968. <<http://www.chromosome18.org>>.

Support Organization for Trisomy 18, 13, and Related Disorders (SOFT). 2982 South Union St., Rochester, NY 14624. (800) 716-7638. <<http://www.trisomy.org>>.

Julia Barrett

EEG see **Electroencephalography**

Egyptian conjunctivitis see **Trachoma**

Ehlers-Danlos syndrome

Definition

The Ehlers-Danlos syndromes (EDS) refer to a group of inherited disorders that affect collagen structure and function. Genetic abnormalities in the manufacturing of collagen within the body affect connective tissues, causing them to be abnormally weak.

Description

Collagen is a strong, fibrous protein that lends strength and elasticity to connective tissues such as the skin, tendons, organ walls, cartilage, and blood vessels. Each of these connective tissues requires collagen tailored to meet its specific purposes. The many roles of collagen are reflected in the number of genes dedicated to its production. There are at least 28 genes in humans that encode at least 19 different types of collagen. Mutations in these genes can affect basic construction as well as the fine-tuned processing of the collagen.

EDS was originally described by Dr. Van Meekeren in 1682. Dr. Ehlers and Dr. Danlos further characterized the disease in 1901 and 1908, respectively. Today, according to the Ehlers-Danlos National Foundation, one in 5,000 to one in 10,000 people are affected by some form of EDS.

EDS is a group of genetic disorders that usually affects the skin, ligaments, joints, and blood vessels. Classification of EDS types was revised in 1997. The new classification involves categorizing the different forms of EDS into six major sub-types, including classical, hypermobility, vascular, kyphoscoliosis, arthrochalasia, and dermatosparaxis, and a collection of rare or poorly defined varieties. This new classification is simpler and based more on descriptions of the actual symptoms.

Classical type

Under the old classification system, EDS classical type was divided into two separate types: type I and type II. The major symptoms involved in EDS classical type are the skin and joints. The skin has a smooth, velvety texture and **bruises** easily. Affected individuals typically have extensive scarring, particularly at the knees, elbows, forehead, and chin. The joints are hyper-extensible, giving a tendency towards dislocation of the hip, shoulder, elbow, knee, or clavicle. Due to decreased muscle tone, affected infants may experience

a delay in reaching motor milestones. Children may have a tendency to develop hernias or other organ shifts within the abdomen. **Sprains** and partial or complete joint dislocations are also common. Symptoms can range from mild to severe. EDS classical type is inherited in an autosomal dominant manner.

There are three major clinical diagnostic criteria for EDS classical type. These include skin hyperextensibility, unusually wide **scars**, and joint hypermobility. At this time there is no definitive test for the diagnosis of classical EDS. Both DNA and biochemical studies have been used to help identify affected individuals. In some cases, a skin biopsy has been found to be useful in confirming a diagnosis. Unfortunately, these tests are not sensitive enough to identify all individuals with classical EDS. If there are multiple affected individuals in a family, it may be possible to perform prenatal diagnosis using a DNA information technique known as a linkage study.

Hypermobility type

Excessively loose joints are the hallmark of this EDS type, formerly known as EDS type III. Both large joints, such as the elbows and knees, and small joints, such as toes and fingers, are affected. Partial and total joint **dislocations** are common, and particularly involve the jaw, knee, and shoulder. Many individuals experience chronic limb and joint **pain**, although x rays of these joints appear normal. The skin may also bruise easily. **Osteoarthritis** is a common occurrence in adults. EDS hypermobility type is inherited in an autosomal dominant manner.

There are two major clinical diagnostic criteria for EDS hypermobility type. These include skin involvement (either hyperextensible skin or smooth and velvety skin) and generalized joint hypermobility. At this time there is no test for this form of EDS.

Vascular type

Formerly called EDS type IV, EDS vascular type is the most severe form. The connective tissue in the intestines, arteries, uterus, and other hollow organs may be unusually weak, leading to organ or blood vessel rupture. Such ruptures are most likely between ages 20 and 40, although they can occur any time, and may be life-threatening.

There is a classic facial appearance associated with EDS vascular type. Affected individuals tend to have large eyes, a thin pinched nose, thin lips, and a slim body. The skin is thin and translucent, with veins dramatically visible, particularly across the chest.

The large joints have normal stability, but small joints in the hands and feet are loose, showing hyperextensibility. The skin bruises easily. Other complications may include collapsed lungs, premature **aging** of the skin on the hands and feet, and ruptured arteries and veins. After surgery there tends to be poor wound healing, a complication that tends to be frequent and severe. **Pregnancy** also carries the risk complications. During and after pregnancy there is an increased risk of the uterus rupturing and of arterial bleeding. Due to the severe complications associated with EDS type IV, **death** usually occurs before the fifth decade. A study of 419 individuals with EDS vascular type, completed in 2000, found that the median survival rate was 48 years, with a range of six to 73 years. EDS vascular type is inherited in an autosomal dominant manner.

There are four major clinical diagnostic criteria for EDS vascular type. These include thin translucent skin, arterial/intestinal/uterine fragility or rupture, extensive bruising, and characteristic facial appearance. EDS vascular type is caused by a change in the gene COL3A1, which codes for one of the collagen chains used to build Collagen type III. Laboratory testing is available for this form of EDS. A **skin biopsy** may be used to demonstrate the structurally abnormal collagen. This type of biochemical test identifies more than 95% of individuals with EDS vascular type. Laboratory testing is recommended for individuals with two or more of the major criteria.

DNA analysis may also be used to identify the change within the COL3A1 gene. This information may be helpful for **genetic counseling** purposes. Prenatal testing is available for pregnancies in which an affected parent has been identified and their DNA mutation is known or their biochemical defect has been demonstrated.

Kyphoscoliosis type

The major symptoms of kyphoscoliosis type, formerly called EDS type VI, are general joint looseness. At birth, the muscle tone is poor, and motor skill development is subsequently delayed. Also, infants with this type of EDS have an abnormal curvature of the spine (**scoliosis**). The scoliosis becomes progressively worse with age, with affected individuals usually unable to walk by age 20. The eyes and skin are fragile and easily damaged, and blood vessel involvement is a possibility. The bones may also be affected as demonstrated by a decrease in bone mass. Kyphoscoliosis type is inherited in an autosomal recessive manner.

There are four major clinical diagnostic criteria for EDS kyphoscoliosis type. These include generally loose joints, low muscle tone at birth, scoliosis at birth (which worsens with age), and a fragility of the eyes, which may give the white area of the eye a blue tint or cause the eye to rupture. This form of EDS is caused by a change in the PLOD gene on chromosome 1, which encodes the enzyme lysyl hydroxylase. A laboratory test is available in which urinary hydroxylysyl pyridinoline is measured. This test, performed on urine is extremely sensitive and specific for EDS kyphoscoliosis type. Laboratory testing is recommended for infants with three or more of the major diagnostic criteria.

Prenatal testing is available if a pregnancy is known to be at risk and an identified affected family member has had positive laboratory testing. An **amniocentesis** may be performed in which fetal cells are removed from the amniotic fluid and enzyme activity is measured.

Arthrochalasia type

Dislocation of the hip joint typically accompanies arthrochalasia type EDS, formerly called EDS type VIIB. Other joints are also unusually loose, leading to recurrent partial and total dislocations. The skin has a high degree of stretchability and bruises easily. Individuals with this type of EDS may also experience mildly diminished bone mass, scoliosis, and poor muscle tone. Arthrochalasia type is inherited in an autosomal dominant manner.

There are two major clinical diagnostic criteria for EDS arthrochalasia type. These include severe generalized joint hypermobility and bilateral hip dislocation present at birth. This form of EDS is caused by a change in either of two components of Collagen type I, called proa1(I) type A and proa2(I) type B. A skin biopsy may be performed to demonstrate an abnormality in either components. Direct DNA testing is also available.

Dermatosparaxis type

Individuals with this type of EDS, once called type VIIC, have extremely fragile skin that bruises easily but does not scar excessively. The skin is soft and may sag, leading to an aged appearance even in young adults. Individuals may also experience hernias. Dermatosparaxis type is inherited in an autosomal recessive manner.

There are two major clinical diagnostic criteria for EDS dermatosparaxis type. These include severe skin fragility and sagging or aged appearing skin. This



Elasticity of the skin is one characteristic of this rare disorder. (Photograph by Biophoto Associates, Photo Researchers, Inc. Reproduced by permission.)

form of EDS is caused by a change in the enzyme called procollagen I N-terminal peptidase. A skin biopsy may be performed for a definitive diagnosis of Dermatosparaxis type.

Other types

There are several other forms of EDS that have not been as clearly defined as the aforementioned types. Forms of EDS within this category may present with soft, mildly stretchable skin, shortened bones, chronic **diarrhea**, joint hypermobility and dislocation, bladder rupture, or poor wound healing. Inheritance patterns within this group include X-linked recessive, autosomal dominant, and autosomal recessive.

Causes and symptoms

There are numerous types of EDS, all caused by changes in one of several genes. The manner in which EDS is inherited depends on the specific gene involved. There are three patterns of inheritance for EDS: autosomal dominant, autosomal recessive, and X-linked (extremely rare).

Chromosomes are made up of hundreds of small units known as genes, which contain the genetic material necessary for an individual to develop and function. Humans have 46 chromosomes, which are matched into 23 pairs. Because chromosomes are inherited in pairs, each individual receives two copies of each chromosome and likewise two copies of each gene.

Changes or mutations in genes can cause genetic diseases in several different ways, many of which are represented within the spectrum of EDS.

In autosomal dominant EDS, only one copy of a specific gene must be changed for a person to have EDS. In autosomal recessive EDS, both copies of a specific gene must be changed for a person to have EDS. If only one copy of an autosomal recessive EDS gene is changed the person is referred to as a carrier, meaning they do not have any of the signs or symptoms of the disease itself, but carry the possibility of passing on the disorder to a future child. In X-linked EDS a specific gene on the X chromosome must be changed. However, this affects males and females differently because males and females have a different number of X chromosomes.

The few X-linked forms of EDS fall under the category of X-linked recessive. As with autosomal recessive, this implies that both copies of a specific gene must be changed for a person to be affected. However, because males only have one X-chromosome, they are affected if an X-linked recessive EDS gene is changed on their single X-chromosome. That is, they are affected even though they have only one changed copy. On the other hand, that same gene must be changed on both of the X-chromosomes in a female for her to be affected.

Although there is much information regarding the changes in genes that cause EDS and their various inheritance patterns, the exact gene mutation for all types of EDS is not known.

Diagnosis

Clinical symptoms such as extreme joint looseness and unusual skin qualities, along with family history, can lead to a diagnosis of EDS. Specific tests, such as skin biopsies are available for diagnosis of certain types of EDS, including vascular, arthrochalasia, and dermatosparaxis types. A skin biopsy involves removing a small sample of skin and examining its microscopic structure. A urine test is available for the Kyphoscoliosis type.

Management of all types of EDS may include genetic counseling to help the affected individual and their family understand the disorder and its impact on other family members and future children.

If a couple has had a child diagnosed with EDS the chance that they will have another child with the same disorder depends on with what form of EDS the child has been diagnosed and if either parent is affected by the same disease or not.

Individuals diagnosed with an autosomal dominant form of EDS have a 50% chance of passing the

same disorder on to a child in each pregnancy. Individuals diagnosed with an autosomal recessive form of EDS have an extremely low risk of having a child with the same disorder.

X-linked recessive EDS is accompanied by a slightly more complicated pattern of inheritance. If a father with an X-linked recessive form of EDS passes a copy of his X chromosome to his children, the sons will be unaffected and the daughters will be carriers. If a mother is a carrier for an X-linked recessive form of EDS, she may have affected or unaffected sons, or carrier or unaffected daughters, depending on the second sex chromosome inherited from the father.

Prenatal diagnosis is available for specific forms of EDS, including kyphoscoliosis type and vascular type. However, prenatal testing is only a possibility in these types if the underlying defect has been found in another family member.

Treatment

Medical therapy relies on managing symptoms and trying to prevent further complications. There is no cure for EDS.

Braces may be prescribed to stabilize joints, although surgery is sometimes necessary to repair joint damage caused by repeated dislocations. Physical therapy teaches individuals how to strengthen muscles around joints and may help to prevent or limit damage. Elective surgery is discouraged due to the high possibility of complications.

Alternative treatment

There are anecdotal reports that large daily doses 0.04–0.14 oz (1–4 g) of vitamin C may help decrease bruising and aid in wound healing. Constitutional homeopathic treatment may be helpful in maintaining optimal health in persons with a diagnosis of EDS. An individual with EDS should discuss these types of therapies with their doctor before beginning them on their own. Therapy that does not require medical consultation involves protecting the skin with sunscreen and avoiding activities that place **stress** on the joints.

Prognosis

The outlook for individuals with EDS depends on the type of EDS with which they have been diagnosed. Symptoms vary in severity, even within one sub-type, and the frequency of complications

KEY TERMS

Arthrochalasia—Excessive looseness of the joints.

Blood vessels—General term for arteries, veins, and capillaries that transport blood throughout the body.

Cartilage—Supportive connective tissue that cushions bone at the joints or which connects muscle to bone.

Collagen—The main supportive protein of cartilage, connective tissue, tendon, skin, and bone.

Connective tissue—A group of tissues responsible for support throughout the body; includes cartilage, bone, fat, tissue underlying skin, and tissues that support organs, blood vessels, and nerves throughout the body.

Dermatosparaxis—Skin fragility caused by abnormal collagen.

Hernia—A rupture in the wall of a body cavity, through which an organ may protrude.

Homeopathic—A holistic and natural approach to healthcare.

Hyperextensibility—The ability to extend a joint beyond the normal range.

Hypermobility—Unusual flexibility of the joints, allowing them to be bent or moved beyond their normal range of motion.

Joint dislocation—The displacement of a bone.

Kyphoscoliosis—Abnormal front-to-back and side-to-side curvature of the spine.

Ligament—A type of connective tissue that connects bones or cartilage and provides support and strength to joints.

Osteoarthritis—A degenerative joint disease that causes pain and stiffness.

Scoliosis—An abnormal, side-to-side curvature of the spine.

Tendon—A strong connective tissue that connects muscle to bone.

Uterus—A muscular, hollow organ of the female reproductive tract. The uterus contains and nourishes the embryo and fetus from the time the fertilized egg is implanted until birth.

Vascular—Having to do with blood vessels.

changes on an individual basis. Some individuals have negligible symptoms while others are severely restricted in their daily life. Extreme joint instability and scoliosis may limit a person's mobility. Most

individuals will have a normal lifespan. However, those with blood vessel involvement, particularly those with EDS vascular type, have an increased risk of fatal complications.

EDS is a lifelong condition. Affected individuals may face social obstacles related to their disease on a daily basis. Some people with EDS have reported living with fears of significant and painful skin ruptures, becoming pregnant (especially those with EDS vascular type), their condition worsening, becoming unemployed due to physical and emotional burdens, and social stigmatization in general.

Constant bruises, skin **wounds**, and trips to the hospital take their toll on both affected children and their parents. Prior to diagnosis parents of children with EDS have found themselves under suspicion of **child abuse**.

Some people with EDS are not diagnosed until well into adulthood and, in the case of EDS vascular type, occasionally not until after death due to complications of the disorder. Not only may the diagnosis itself be devastating to the family, but in many cases other family members find out for the first time they are at risk for being affected.

Although individuals with EDS face significant challenges, it is important to remember that each person is unique with their own distinguished qualities and potential. Persons with EDS go on to have families, to have careers, and to be accomplished citizens, surmounting the challenges of their disease.

Resources

PERIODICALS

"Clinical and Genetic Features of Ehlers-Danlos Syndrome Type IV, the Vascular Type." *The New England Journal of Medicine* 342, no. 10 (2000).

"Living a Restricted Life with Ehlers-Danlos Syndrome." *International Journal of Nursing Studies* 37 (2000): 111–118.

ORGANIZATIONS

Ehlers-Danlos Support Group - UK. PO Box 335, Farnham, Surrey, GU10 1XJ. UK 01252 690 940. <<http://www.atv.ndirect.co.uk>>.

Ehlers-Danlos National Foundation. 6399 Wilshire Blvd., Ste 203, Los Angeles, CA 90048 (323) 651-3038. Fax: (323) 651-1366. <<http://www.ednf.org>>.

OTHER

GeneClinics. <<http://www.geneclinics.org>>.

Java O. Solis, MS

Ehrlichiosis

Definition

Ehrlichiosis is a bacterial infection that is spread by ticks. Symptoms include **fever**, chills, **headache**, muscle aches, and tiredness.

Description

Ehrlichiosis is a tick-borne disease caused by infection with *Ehrlichia* bacteria. Ticks are small, blood-sucking arachnids. Although some ticks carry disease-causing organisms, most do not. When an animal or person is bitten by a tick that carries bacteria, the bacteria are passed to that person or animal during the tick's feeding process. It is believed that the tick must remain attached to the person or animal for at least 24 hours to spread the infection.

There are two forms of ehrlichiosis in the United States; human monocytic ehrlichiosis and human granulocytic ehrlichiosis. Monocytic ehrlichiosis is caused by *Ehrlichia chaffeensis*, which is spread by the Lone Star tick, *Amblyomma americanum*. As of early 1998, about 400 cases of monocytic ehrlichiosis had been reported in 30 states, primarily in the southeastern and south central United States. The bacteria that causes granulocytic ehrlichiosis is not known, but suspected to be either *Ehrlichia equi* or *Ehrlichia phagocytophila*. Granulocytic ehrlichiosis is probably spread by the blacklegged tick *Ixodes scapularis* (which also spreads **Lyme disease**). About 100 cases of granulocytic ehrlichiosis have been reported in Connecticut, Massachusetts, Rhode Island, Minnesota, New York, and Wisconsin.

Causes and symptoms

Both forms of ehrlichiosis have similar symptoms, and the illnesses can range from mild to severe and life-threatening. Risk factors include old age and exposure to ticks through work or recreation. Symptoms occur seven to 21 days following a tick bite although patients may not recall being bitten. Fever, tiredness, headache, muscle aches, chills, loss of appetite, confusion, nausea, and **vomiting** are common to both diseases. A rash may occur.

Diagnosis

Ehrlichiosis may be diagnosed and treated by doctors who specialize in blood diseases (hematologists) or an infectious disease specialist. Because ehrlichiosis is not very common and the symptoms are

KEY TERMS

Tick-borne disease—A disease that is spread to animals by the bite of an infected tick.

not unique, it may be misdiagnosed. A recent history of a tick bite is helpful in the diagnosis. Blood tests will be done to look for antibodies to *Ehrlichia*. Staining and microscopic examination of the blood sample may show *Ehrlichia* bacteria inside white blood cells. Another test, called polymerase chain reaction (PCR), is a very sensitive assay to detect bacteria in the blood sample, but it is not always available.

Treatment

Antibiotic treatment should begin immediately if ehrlichiosis is suspected, even if laboratory results are not available. Treatment with either tetracycline (Sumycin, Achromycin V) or doxycycline (Monodox, Vibramycin) is recommended. Many patients with ehrlichiosis are admitted to the hospital for treatment.

Prognosis

For otherwise healthy people, a full recovery is expected following treatment for ehrlichiosis. Elderly patients are at a higher risk for severe disease, which may be fatal. Serious complications include lung or gastrointestinal bleeding. Two to 10 patients out of 100 die from the disease.

Prevention

The only prevention for ehrlichiosis is to minimize exposure to ticks by staying on the trail when walking through the woods, avoiding tall grasses, wearing long sleeves and tucking pant legs into socks, wearing insect repellent, and checking for ticks after an outing. Remove a tick as soon as possible by grasping the tick with tweezers and gently pulling.

Resources

BOOKS

McDade, Joseph E., and James G. Olsen. "Ehrlichiosis, Q Fever, Typhus, Rickettsialpox, and Other Rickettsioses." In *Infectious Diseases*. 2nd ed. Philadelphia: W. B. Saunders Co., 1998.

Belinda Rowland, PhD

EKG see **Electrocardiography**

Elder Abuse

Definition

Elder **abuse** is a general term used to describe harmful acts toward an elderly adult, such as physical abuse, sexual abuse, emotional or psychological abuse, financial exploitation, and neglect, including self-neglect.

Description

Results from the National Elder Abuse Incidence Study, funded in part by the Administration on Aging, suggest that over 500,000 people 60 years of age and older are abused or neglected each year in the United States. It was also found that four times as many incidents of abuse, neglect, or self-neglect are never reported, causing researchers to estimate that as many as two million elderly persons in the United States are abused each year. In 90% of the cases, the abusers were found to be family members and most often were the adult children or spouses of those abused. In addition, equal numbers of men and women have been identified as the abusers. However, women, especially those over 80 years of age, tend to be victimized more than men.

Elder abuse can take place anywhere, but the two main settings addressed by law are domestic settings, such as the elder's home or the caregiver's home, and institutional settings, such as a nursing home or group home. In general, there are five basic types of elderly abuse: physical, sexual, emotional or psychological, financial, and neglect. Data from National Center on Elder Abuse indicates that more than half of the cases reported involve some kind of neglect, whereas 1 in 7 cases involve physical abuse. It is considered neglect when a caretaker deprives an elderly person of the necessary care needed in order to avoid physical or mental harm. Sometimes the behavior of an elderly person threatens his or her own health; in those cases, the abuse is called self-neglect. Physical abuse refers to physical force that causes bodily harm to an elderly person, such as slapping, pushing, kicking, pinching, or burning.

About 1 in 8 cases of elderly abuse involve some form of financial exploitation, which is defined as the use of an elderly person's resources without his or her consent. The National Center on Elder Abuse defines emotional and psychological abuse of a senior as causing anguish, **pain**, or distress through verbal or nonverbal acts, such as verbal assaults, insults, intimidation, and humiliation, for example. Isolating elderly

persons from their friends and family as well as giving them the silent treatment are two other forms of emotional and psychological abuse. Any kind of non-consensual sexual contact with an elderly person that takes place without his or her consent is considered sexual abuse.

Causes and symptoms

Elder abuse is a complex problem that can be caused by many factors. According to the National Center on Elder Abuse, social isolation and mental impairment are two factors of elder abuse. Studies show that people advanced in years, such as in their eighties, with a high level of frailty and dependency are more likely to be victims of elder abuse than people who are younger and better equipped to stand up for themselves. Because spouses make up a large percentage of elder abusers, at least 40% statistically, some research has been done in the area, which shows that a pattern of domestic violence is associated with many of the cases. The risk of elder abuse appears to be especially high when adult children live with their elderly parents for financial reasons or because they have personal problems, such as drug dependency or mental illness. Some experts have speculated that elderly people living in rural areas with their caretakers may have a higher risk of being abused than city dwellers. The idea behind this theory is that the opportunity exists for the abuse to occur, but there is less likelihood that the abuser will be caught. More research in this very important area is needed in order to illuminate the relationship between these factors.

The National Center on Elder Abuse identifies the following as signs of elder abuse:

- Bruises, pressure marks, broken bones, abrasions, and **burns** may indicate physical abuse or neglect.
- Unexplained withdrawal from normal activities and unusual depression may be indicators of emotional abuse.
- Bruises around the breasts or genital area, as well as unexplained bleeding around the genital area, may be signs of sexual abuse.
- Large withdrawals of money from an elder's bank account, sudden changes in a will, and the sudden disappearance of valuable items may be indications of financial exploitation.
- Bedsores, poor hygiene, unsanitary living conditions, and unattended medical needs may be signs of neglect.

- Failure to take necessary medicines, leaving a burning stove unattended, poor hygiene, confusion, unexplained weight loss, and **dehydration** may all be signs of self-neglect.

Diagnosis and Treatment

The National Committee for the Prevention of Elder Abuse notes that Adult Protective Services (APS) caseworkers are often on the front lines when it comes to elderly abuse. People being abused or those who believe abuse is taking place can turn to their local APS office for help. The APS routinely screens calls, keeps all information confidential, and, if necessary, sends a caseworker out to conduct an investigation. In the event that a crisis intervention is needed, the APS caseworker can arrange for any necessary emergency treatment. If it is unclear whether elder abuse has taken place, the APS caseworker can serve as a liaison between the elderly person and other community agencies.

According to the National Committee for the Prevention of Elder Abuse, “professionals in the field of aging are often the first to discover signs of elder abuse.” Providing encouragement and advice, they play a critical role in educating others with regard to the needs of the elderly. They not only provide valuable support to the victims of abuse, but they also monitor high-risk situations and gather important information that can help validate that abuse has taken place.

Some people might think that a person who has cognitive impairment might be unable to describe mistreatment; however, that is not the case. In fact, guidelines set by the American Medical Association call for “routine questions about abuse and neglect even among patients with cognitive impairment in order to improve the identification of cases and implement appropriate treatment and referral.” Rather than an inability to describe mistreatment, what might stop an elderly person from reporting abuse is a sense of embarrassment or fear of retaliation. To complicate matters, differences exist among cultural groups regarding what defines abuse.

Therefore, most states have established laws that define elder abuse and require health care providers to report any cases they encounter with penalties attached for failing to do so. Indeed, statistics show that health care providers, for example, report almost 25% of the known cases of elder abuse. Therefore, physicians play a very important role in identifying and treating elders who have been abused. And yet, in an article published by the *Journal of the American*

Geriatrics Society, Dr. Conlin pointed out that only 1 of every 13 cases of elder abuse are reported by physicians. There may be several reasons for this. In some cases, the problem may simply go unnoticed, especially if the physician has no obvious reason to suspect any wrongdoing. In other cases, the patient may hide or deny the problem.

In recent years, much media attention has been focused on elderly abuse that takes place in institutional settings. Anyone who believes that a loved one is being abused while in a nursing home or other institutional setting should contact the authorities for assistance immediately.

Prognosis

The mortality rate of an elderly person who has been mistreated is higher than the mortality rate of an elderly person who has not experienced abuse. Nonetheless, numerous success stories exist regarding successful interventions. Social workers and health care professionals, as well as concerned citizens from a variety of backgrounds, have played a key role in identifying and obtaining treatment for abused elders.

Prevention

Planning for the future is one of the best ways to avoid elder abuse. Consider a variety of retirement options, ones that will encourage safety as well as independence. It is important to stay active in the community. Avoiding isolation minimizes the likelihood that abuse will occur. Seek professional counsel when necessary; it is important for everyone to know their rights and to be advocates on their own behalf.

Resources

PERIODICALS

- Clarke, M. E., Pierson, W. “Management of elder abuse in the emergency department.” *Emergency Medical Clinics of North America* 17 (1999): 631–644.
- Conlin, M. “Silent suffering: a case study of elder abuse and neglect.” *Journal of the American Geriatrics Society* 43 (1995): 1303–1308.
- Lachs, M. S., Willimas, C. S., O’Brien, S., Pillemer, K. A., Charlson, M. E. “The mortality of elderly mistreatment.” *Journal of the American Medical Association* 280 (1998): 429–432.

OTHER

- Administration on Aging “Elder Rights & Resources. Elder Abuse.” *Administration on Aging* 10 December 2004 Administration on Aging, Department of

Health and Human Services. 1 April 2005 <http://www.aoa.gov/eldfam/Elder_Rights/Elder_Abuse/Elder_Abuse_pf.asp> .

American Medical Association "Featured CSA Report: AMA Data on Violence Between Intimates (I-00): Elder Abuse." *American Medical Association* January 2005 American Medical Association. 1 April 2005 <<http://www.ama-assn.org/ama/pub/category/13577.html>> .

National Center on Elder Abuse "The Basics: Major Types of Elder Abuse." *National Center on Elder Abuse* 15 May 2003 National Center on Elder Abuse. 1 April 2005 <<http://www.elderabusecenter.org/>> .

National Center on Elder Abuse "Elder Abuse: Frequently Asked questions." *National Center on Elder Abuse* 23 March 2005 National Center on Elder Abuse. 1 April 2005 <<http://www.elderabusecenter.org/>> .

National Committee for the Prevention of Elder Abuse "The Role of Professionals and Concerned Citizens." *National Committee for the Prevention of Elder Abuse* March 2003 National Committee for the Prevention of Elder Abuse. 1 April 2005 <<http://www.elderabusecenter.org/>> .

Lee Ann Paradise

Electric shock injuries

Definition

Electric shock injuries are caused by lightning or electric current from a mechanical source passing through the body.

Description

Electric shocks are responsible for about 1,000 deaths in the United States each year, or about 1% of all accidental deaths.

Causes and symptoms

The severity of injury depends on the current's pressure (voltage), the amount of current (amperage), the type of current (direct vs. alternating), the body's resistance to the current, the current's path through the body, and how long the body remains in contact with the current. The interplay of these factors can produce effects ranging from barely noticeable **tingling** to instant **death**; every part of the body is vulnerable. Although the severity of injury is determined primarily by the voltage, low voltage

can be just as dangerous as high voltage under the right circumstances. People have been killed by shocks of just 50 volts.

How electric shocks affect the skin is determined by the skin's resistance, which in turn is dependent upon the wetness, thickness, and cleanliness of the skin. Thin or wet skin is much less resistant than thick or dry skin. When skin resistance is low, the current may cause little or no skin damage but severely burn internal organs and tissues. Conversely, high skin resistance can produce severe skin **burns** but prevent the current from entering the body.

The nervous system (the brain, spinal cord, and nerves) is particularly vulnerable to injury. In fact, neurological problems are the most common kind of nonlethal harm suffered by electric shock victims. Some neurological damage is minor and clears up on its own or with medical treatment, but some is severe and permanent. Neurological problems may be apparent immediately after the accident, or gradually develop over a period of up to three years.

Damage to the respiratory and cardiovascular systems is most acute at the moment of injury. Electric shocks can paralyze the respiratory system or disrupt heart action, causing instant death. Also at risk are the smaller veins and arteries, which dissipate heat less easily than the larger blood vessels and can develop **blood clots**. Damage to the smaller vessels is probably one reason why **amputation** is often required following high-voltage injuries.

Many other sorts of injuries are possible after an electric shock, including **cataracts**, kidney failure, and substantial destruction of muscle tissue. The victim may suffer a fall or be hit by debris from exploding equipment. An electric arc may set clothing or nearby flammable substances on fire. Strong shocks are often accompanied by violent **muscle spasms** that can break and dislocate bones. These spasms can also freeze the victim in place and prevent him or her from breaking away from the source of the current.

Diagnosis

Diagnosis relies on gathering information about the circumstances of the accident, a thorough **physical examination**, and monitoring of cardiovascular and kidney activity. The victim's neurological condition can fluctuate rapidly and requires close observation. A computed tomography scan (CT scan) or **magnetic resonance imaging** (MRI) may be necessary to check for brain injury.

Treatment

When an electric shock accident happens at home or in the workplace, the main power should immediately be shut off. If that cannot be done, and current is still flowing through the victim, the alternative is to stand on a dry, nonconducting surface such as a folded newspaper, flattened cardboard carton, or plastic or rubber mat and use a nonconducting object such as a wooden broomstick (never a damp or metallic object) to push the victim away from the source of the current. The victim and the source of the current must not be touched while the current is still flowing, for this can electrocute the rescuer. Emergency medical help should be summoned as quickly as possible. People who are trained to perform **cardiopulmonary resuscitation (CPR)** should, if appropriate, begin first aid while waiting for emergency medical help to arrive.

Burn victims usually require treatment at a burn center. Fluid replacement therapy is necessary to restore lost fluids and electrolytes. Severely injured tissue is repaired surgically, which can involve **skin grafting** or amputation. **Antibiotics** and antibacterial creams are used to prevent infection. Victims may also require treatment for kidney failure. Following surgery, physical therapy to facilitate recovery, and psychological counseling to cope with disfigurement, may be necessary.

Prognosis

Electric shocks cause death in 3–15% of cases. Many survivors require amputation or are disfigured by their burns. Injuries from household appliances and other low-voltage sources are less likely to produce extreme damage.

Prevention

Parents and other adults need to be alert to possible electric dangers in the home. Damaged electric appliances, wiring, cords, and plugs should be repaired or replaced. Electrical repairs should be attempted only by people with the proper training. Hair dryers, radios, and other electric appliances should never be used in the bathroom or anywhere else they might accidentally come in contact with water. Young children need to be kept away from electric appliances and should be taught about the dangers of electricity as soon as they are old enough. Electric outlets require safety covers in homes with young children.

KEY TERMS

Antibiotics—Substances used against microorganisms that cause infection.

Cataract—Clouding of the lens of the eye or its capsule (surrounding membrane).

Computed tomography scan (CT scan)—A process that uses x rays to create three-dimensional images of structures inside the body.

Electrolytes—Substances that conduct electric current within the body and are essential for sustaining life.

Magnetic resonance imaging (MRI)—The use of electromagnetic energy to create images of structures inside the body.

Skin grafting—A technique in which a piece of healthy skin from the patient's body (or a donor's) is used to cover another part of the patient's body that has lost its skin.

During thunderstorms, people should go indoors immediately, even if no rain is falling, and boaters should return to shore as rapidly as possible. People who cannot reach indoor shelter should move away from metallic objects such as golf clubs and fishing rods and lie down in low-ground areas. Standing or lying under or next to tall or metallic structures is unsafe. An automobile is appropriate cover, as long as the radio is off. Telephones, computers, hair dryers, and other appliances that can act as conduits for lightning should not be used during thunderstorms.

Resources

BOOKS

Dimick, Alan R. "Electrical Injuries." In *Harrison's Principles of Internal Medicine*, edited by Anthony S. Fauci, et al. New York: McGraw-Hill, 1997.

Howard Baker

Electrical nerve stimulation

Definition

Electrical nerve stimulation, also called transcutaneous electrical nerve stimulation (TENS), is a

noninvasive, drug-free **pain management** technique. By sending electrical signals to underlying nerves, the battery-powered TENS device can relieve a wide range of chronic and acute **pain**.

Purpose

TENS is used to relieve pain caused by a variety of chronic conditions, including:

- neck and lower back pain
- headache/migraine
- arthritis
- post-herpetic **neuralgia** (lingering chronic pain after an attack of **shingles**)
- sciatica (pain radiating from lower back, through the legs, to the foot)
- temporomandibular joint pain
- osteoarthritis
- amputation (phantom limb)
- fibromyalgia (a condition causing aching and stiffness throughout the body)

The device is also effective against short-term pain, such as:

- shingles (painful skin eruptions along the nerves)
- bursitis (inflammation of tissue surrounding a joint)
- childbirth
- post-surgical pain
- fractures
- muscle and joint pain
- sports injuries
- menstrual cramps

Precautions

Because TENS may interfere with pacemaker function, patients with **pacemakers** should consult a cardiologist before using a TENS unit. Patients should also avoid electrical stimulation in the front of the neck, which can be hazardous. The safety of the device during **pregnancy** has not been established.

TENS doesn't cure any condition; it simply eases pain. Patients who are not sure what is causing their pain should consult a physician before using TENS.

KEY TERMS

Fibromyalgia—A condition characterized by aching and stiffness, fatigue and poor sleep, as well as tenderness at various sites on the body.

Osteoarthritis—A painful joint disease aggravated by mechanical stress.

Phantom limb—The perception that a limb is present (and throbbing with pain) after it has been amputated.

Post-herpetic neuralgia—Lingering pain that can last for years after an attack of shingles.

Sciatica—Pain that radiates along the sciatic nerve, extending from the buttock down the leg to the foot.

Temporomandibular joint pain (TMJ)—Pain and other symptoms affecting the head, jaw, and face that are caused when the jaw joints and muscles controlling them don't work together correctly.

Description

The TENS device is a small battery-powered stimulator that produces low-intensity electrical signals through electrodes on or near a painful area, producing a **tingling** sensation that reduces pain. There is no dosage limitation, and the patient controls the amount of pain relief.

Some experts believe TENS works by blocking pain signals in the spinal cord, or by delivering electrical impulses to underlying nerve fibers that lessen the experience of pain. Others suspect that the electrical stimulation triggers the release of natural painkillers in the body.

Patients can rent a TENS unit before buying one, to see if it is effective against their pain.

Preparation

After TENS has been prescribed, a doctor will refer the patient to a TENS specialist, who will explain how to use the machine. The specialist works with the patient to determine the settings and electrode placements for the best pain relief.

Risks

TENS is nonaddictive and completely safe. The only side effect may be a slight skin irritation or redness in some people, which can be prevented by using different gels or electrodes.

Normal results

The amount of relief a person gets using TENS depends on the underlying cause of the pain, a person's mental state, and whether or not medication is also used. At least one study found that both a real TENS machine and a placebo were equally effective in reducing pain. This suggests that at least part of its effectiveness may be due to the patient's belief in its ability to ease pain.

Carol A. Turkington

Electrical stimulation of the brain

Definition

Electrical stimulation of the brain (ESB) is a relatively new technique used to treat chronic pain and **tremors** associated with Parkinson disease. ESB is administered by passing an electrical current through an electrode implanted in the brain.

Purpose

While the implantation of electrodes in the brain is used to treat or diagnose several disorders, the term ESB is limited here to the treatment of tremors, and as a pain management tool for patients suffering from back problems and other chronic injuries and illnesses.

Precautions

An ESB tremor control device, used in treating Parkinson patients, may interfere with or be affected by cardiac **pacemakers** and other medical equipment. As a result, patients with other implanted medical equipment may not be good candidates for the therapy.

Description

Electrical stimulation of the brain, or deep brain stimulation, is effective in treating tremor in up to 88% of **Parkinson disease** patients. An electrode is implanted into the thalamus (part of the brain) of the patient, and attached to an electric pulse generator via an extension wire. The pulse generator is implanted into the patient's pectoral, or chest area, and the extension wire is tunneled under the skin. The pulse generator sends out intermittent electrical stimulation

to the electrode in the thalamus, which inhibits or partially relieves the tremor. The generator can be turned on and off with a magnet, and needs to be replaced every three to five years.

Similar methods have been used to treat chronic **pain** that responded unfavorably to conventional therapies. A remote transmitter allows these patients to trigger electric stimulation to relieve their symptoms on an as-needed basis. Patients with failed back syndrome, trigeminal neuropathy (pertaining to the fifth cranial nerve), and peripheral neuropathy fared well for pain control with this treatment, while patients with spinal cord injury and postherpetic neuralgia (pain along the nerves following herpes) did poorly.

Preparation

The patient should be free of any type of infection before undergoing an ESB procedure. He or she may be advised to discontinue any medication for a prescribed period of time before surgery.

Aftercare

After neurosurgery, patients should undergo regular head dressing changes, minimize exposure to others, and practice good personal hygiene in order to prevent a brain infection. The head may also be kept elevated for a prescribed period of time in order to decrease swelling of the brain.

Risks

The implantation of electrodes into the brain carries risks of hemorrhage, infarction, infection, and cerebral **edema**. These complications could cause irreversible neurological damage.

Patients with an implanted ESB tremor control device may experience headaches, disequilibrium (a disturbance of the sense of balance), burning or **tingling** of the skin, or partial paralysis.

Normal results

ESB is effective in pain control for specific conditions. It can provide long-term pain relief with few side effects or complications.

For the control of tremors a deep brain stimulator does provide some relief. It is recommended for patients with tremors severe enough to affect their quality of life.

KEY TERMS

Infarction—A sudden insufficiency of local blood supply.

Neuralgia—Pain extending along one or more nerves.

Neuropathy—A functional disturbance or change in the nervous system.

Parkinson disease—A chronic neurological illness that causes tremors, stiffness, and difficulty in moving and walking.

Resources

OTHER

The Parkinson's Web. < <http://pdweb.mgh.harvard.edu> > .

University of Southern California. *The*

ANGELN eurosurgical Information Resource. < <http://www.usc.edu/hsc/neurosurgery/angel.html> > .

Paula Anne Ford-Martin

Electrocardiography

Definition

Electrocardiography is a commonly used, non-invasive procedure for recording electrical changes in the heart. The record, which is called an electrocardiogram (ECG or EKG), shows the series of waves that relate to the electrical impulses which occur during each beat of the heart. The results are printed on paper or displayed on a monitor. The waves in a normal record are named P, Q, R, S, and T and follow in alphabetical order. The number of waves may vary, and other waves may be present.

Purpose

Electrocardiography is a starting point for detecting many cardiac problems. It is used routinely in physical examinations and for monitoring the patient's condition during and after surgery, as well as during intensive care. It is the basic measurement used for tests such as exercise tolerance. It is used to



An EKG strip indicating atrial flutter. (Custom Medical Stock Photo. Reproduced by permission.)

evaluate causes of symptoms such as chest **pain**, shortness of breath, and **palpitations**.

Precautions

No special precautions are required.

Description

The patient disrobes from the waist up, and electrodes (tiny wires in adhesive pads) are applied to specific sites on the arms, legs, and chest. When attached, the electrodes are called leads; three to 12 leads may be employed.

Muscle movement may interfere with the recording, which lasts for several beats of the heart. In cases where rhythm disturbances are suspected to be infrequent, the patient may wear a small Holter monitor in order to record continuously over a 24-hour period; this is known as ambulatory monitoring.

Preparation

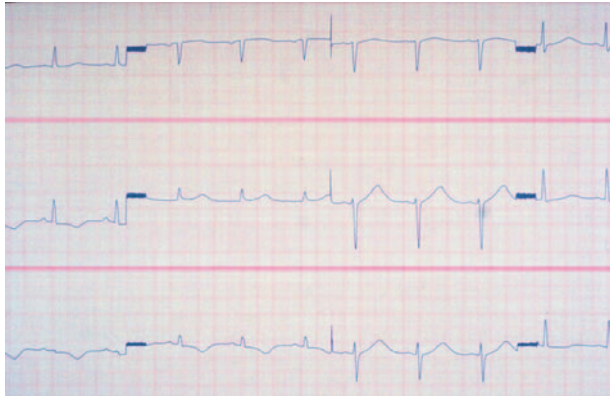
The skin is cleaned to obtain good electrical contact at the electrode positions.

Aftercare

To avoid skin irritation from the salty gel used to obtain good electrical contact, the skin should be thoroughly cleaned after removal of the electrodes.

Risks

No complications from this procedure have been observed.



This EKG strip shows evidence of Wolff-Parkinson-White syndrome. (Custom Medical Stock Photo. Reproduced by permission.)



A patient undergoing electrocardiography. (Russell Curtis, Photo Researchers. Reproduced by permission.)

Normal results

When the heart is operating normally, each part contracts in a specific order. Contraction of the muscle is triggered by an electrical impulse. These electrical impulses travel through specialized cells that form a conduction system. Following this pathway ensures that contractions will occur in a coordinated manner.

When the presence of all waves is observed in the electrocardiogram and these waves follow the order defined alphabetically, the heart is said to show a normal sinus rhythm, and impulses may be assumed to be following the regular conduction pathway.

The heart is described as showing arrhythmia or dysrhythmia when time intervals between waves, the order, or the number of waves do not fit this pattern.

Other features that may be altered include the direction of wave deflection and wave widths.

In the normal heart, electrical impulses—at a rate of 60–100 times per minute—originate in the sinus node. The sinus node is located in the first chamber, known as the right atrium, where blood re-enters the heart. After traveling down to the junction between the upper and lower chambers, the signal stimulates the atrioventricular node. From here, after a delay, it passes by specialized routes through the lower chambers or ventricles. In many disease states, the passage of the electrical impulse can be interrupted in a variety of ways, causing the heart to perform less efficiently.

Abnormal results

Special training is required for interpretation of the electrocardiogram. To summarize the features used in interpretations in the simplest manner, the P wave of the electrocardiogram is associated with the contraction of the atria. The QRS series of waves, or QRS complex, is associated with ventricular contraction, with the T wave coming after the contraction. Finally, the P-Q or P-R interval gives a value for the time taken for the electrical impulse to travel from the atria to the ventricle (normally less than 0.2 sec).

The cause of dysrhythmia is ectopic beats. Ectopic beats are premature heart beats that arise from a site other than the sinus node—commonly from the atria, atrioventricular node, or the ventricle. When these dysrhythmias are only occasional, they may produce no symptoms, or a feeling of the heart turning over or “flip-flopping” may be experienced. These occasional dysrhythmias are common in healthy people, but they also can be an indication of heart disease.

The varied sources of dysrhythmias provide a wide range of alterations in the form of the electrocardiogram. Ectopic beats that start in the ventricle display an abnormal QRS complex. This can indicate disease associated with insufficient blood supply to the muscle (myocardial ischemia). Multiple ectopic sites lead to rapid and uncoordinated contractions of the atria or ventricles. This condition is known as fibrillation. In atrial fibrillation, P waves are absent, and the QRS complex appears at erratic intervals, or “irregularly irregular.”

When the atrial impulse fails to reach the ventricle, a condition known as **heart block** results. If this is partial, the P-R interval (the time for the impulse to

KEY TERMS

Ambulatory monitoring—ECG recording over a prolonged period during which the patient can move around.

Arrhythmia or dysrhythmia—Abnormal rhythm in hearts that contract in an irregular way.

ECG or EKG—A record of the waves that relates to the electrical impulses produced at each beat of the heart.

Electrodes—Tiny wires in adhesive pads that are applied to the body for ECG measurement.

Fibrillation—Rapid, uncoordinated contractions of the upper or the lower chambers of the heart.

Lead—Name given the electrode when it is attached to the skin.

reach the ventricle) is prolonged. If complete, the ventricles beat independently of the atria at about 40 beats per minute, and the QRS complex is mostly dissociated from the P wave.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

Alison M. Grant

Electroconvulsive therapy

Definition

Electroconvulsive therapy (ECT) is a medical treatment for severe mental illness in which a small, carefully controlled amount of electricity is introduced into the brain. This electrical stimulation, used in conjunction with anesthesia and muscle relaxant medications, produces a mild generalized seizure or convulsion. While used to treat a variety of psychiatric disorders, it is most effective in the treatment of severe depression, and provides the most rapid relief currently available for this illness.

Purpose

The purpose of electroconvulsive therapy is to provide relief from the signs and symptoms of mental illnesses such as severe depression, **mania**, and schizophrenia. ECT is indicated when patients need rapid improvement because they are suicidal, self-injurious, refuse to eat or drink, cannot or will not take medication as prescribed, or present some other danger to themselves. Antidepressant medications, while effective in many cases, may take two–six weeks to produce a therapeutic effect. Antipsychotic medications used to treat mania and **schizophrenia** have many uncomfortable and sometimes dangerous side effects, limiting their use. In addition, some patients develop **allergies** and therefore are unable to take their medicine.

Precautions

The most common risks associated with ECT are disturbances in heart rhythm. Broken or dislocated bones occur very rarely.

Description

The treatment of severe mental illness, such as schizophrenia, using electroconvulsive therapy was introduced in 1938 by two Italian doctors named Cerletti and Bini. In those days many doctors believed that convulsions were incompatible with schizophrenia since, according to their observations, this disease rarely occurred in individuals suffering from epilepsy. They concluded, therefore, that if convulsions could be artificially produced in patients with schizophrenia, the illness could be cured. Some doctors were already using a variety of chemicals to produce seizures, but many of their patients died or suffered severe injuries because the strength of the convulsions could not be well controlled.

Electroconvulsive therapy is among the most controversial of all procedures used to treat mental illness. When it was first introduced, many people were frightened simply because it was called “shock treatment.” Many assumed the procedure would be painful, others thought it was a form of electrocution, and still others believed it would cause brain damage. Unfortunately, unfavorable publicity in newspapers, magazines, and movies added to these fears.

Indeed, in those early years, patients and families were rarely educated by doctors and nurses regarding this or other forms of psychiatric treatment. In addition, no anesthesia or muscle relaxants were used. As a result,

patients had violent seizures, and even though they did not remember them, the procedure itself was frightening.

The way these treatments are given today is very different from the procedures used in the past. Currently, ECT is offered on both an inpatient and outpatient basis. Hospitals have specially equipped rooms with oxygen, suction, and cardiopulmonary resuscitation (**CPR**) in order to deal with the rare emergency.

The treatment is carried out as follows: approximately 30 minutes before the scheduled treatment time, the patient may receive an injection of a medication (such as atropine) that keeps the pulse rate from decreasing too much during the convulsion. Next, the patient is placed on a cot and hooked up to a machine that automatically takes and displays vital signs (temperature, pulse, respiration, and blood pressure) on a television-like monitor. A mild anesthetic is then injected into a vein, followed by a medication (such as Anectine) that relaxes all of the muscles in the body so that the seizure is mild, and the risk of broken bones is virtually eliminated.

When the patient is both relaxed and asleep, an airway is placed in the mouth to aid with breathing. Electrodes are placed on the sides of the head in the temple areas. An electric current is passed through the brain by means of a machine specifically designed for this purpose. The usual dose of electricity is 70–150 volts for 0.1–0.5 seconds. In the first stage of the seizure (tonic phase), the muscles in the body that have not been paralyzed by medication contract for a period of five to 15 seconds. This is followed by the second stage (clonic phase) that is characterized by twitching movements, usually visible only in the toes or in a non-paralyzed arm or leg. These are caused by alternating contraction and relaxation of these same muscles. This stage lasts approximately 10–60 seconds. The entire procedure, from beginning to end, lasts about 30 minutes.

The total number of treatments a patient will receive depends upon many factors such as age, diagnosis, the history of illness, family support, and response to therapy. Patients with depression, for example, usually require six to 12 treatments. Treatments are usually administered every other day, three times a week.

The electrodes may be placed on both sides of the head (bilateral) or one side (unilateral). While bilateral ECT appears to be somewhat more effective, unilateral ECT is preferred for individuals who experience prolonged confusion or forgetfulness following treatment. Many doctors begin treatment with unilateral ECT, then change to bilateral if the patient is not improving.

Post-treatment confusion and forgetfulness are common, though disturbing symptoms associated with ECT. Doctors and nurses must be patient and supportive by providing patients with factual information about recovery. Elderly patients, for example, may become increasingly confused and forgetful as the treatments continue. These symptoms usually subside with time, but a small minority of patients state that they have never fully recovered from these effects.

With the introduction of antipsychotics in the 1950s, the use of ECT became less frequent. These new medications provided relief for untold thousands of patients who suffered greatly from their illness. However, there are a number of side effects associated with these drugs, some of which are irreversible. Another drawback is that some medications do not produce a therapeutic effect for two–six weeks. During this time the patient may present a danger to himself or others. In addition, there are patients who do not respond to medicine or who have severe allergic reactions. For these individuals, ECT may be the only treatment that will help.

Preparation

Patients and relatives are prepared for ECT by being shown video tapes that explain both the procedure and the risks involved. The physician then answers any questions these individuals may have, and the patient is asked to sign an “Informed Consent Form.” This gives the doctor and the hospital permission to administer the treatment.

Once the form is signed, the doctor performs a complete physical examination, and orders a number of tests that can help identify any potential problem. These tests may include a **chest x ray**, an electrocardiogram (ECG), urinalysis, spinal x ray, brain wave (EEG), and complete blood count (CBC).

Some medications, such as lithium and a type of antidepressant known as **monoamine oxidase inhibitors**, should be discontinued for some time before treatment. Patients are instructed not to eat or drink for at least eight hours prior to the procedure in order to reduce the possibility of **vomiting** and **choking**.

Aftercare

After the treatment, patients are moved to a recovery area. Vital signs are recorded every five minutes until the patient is fully awake, which may take 15–30 minutes. Some initial confusion may be present but usually disappears in a matter of minutes. There

KEY TERMS

Mania—A mood disorder in which a person experiences prolonged elation or irritability characterized by overactivity that can lead to exhaustion and medical emergencies.

Relapse—A return of the signs and symptoms of an illness.

Schizophrenia—A severe mental illness in which a person has difficulty distinguishing what is real from what is not real. It is often characterized by hallucinations, delusions, and withdrawal from people and social activities.

may be complaints of **headache**, muscle **pain**, or back pain. Such discomfort is quickly relieved by mild medications such as **aspirin**.

Risks

Advanced medical technology has substantially reduced the complications associated with ECT. These include slow heart beat (bradycardia), rapid heart beat (tachycardia), memory loss, and confusion. Persons at high risk for ECT include those with recent heart attack, uncontrolled blood pressure, brain tumors, and previous spinal injuries.

Normal results

ECT often produces dramatic improvement in the signs and symptoms of major depression, especially in elderly individuals, sometimes during the first week of treatment. While it is estimated that 50% of these patients will experience a future return of symptoms, the prognosis for each episode of illness is good. Mania also often responds well to treatment. The picture is not as bright for schizophrenia, which is more difficult to treat and is characterized by frequent relapses.

A few patients are placed on maintenance ECT. This means they return to the hospital every one–two months, as needed, for an additional treatment. These individuals are thus able to keep their illness under control and lead a normal and productive life.

Resources

BOOKS

Stuart, Gail W., and Michele T. Laraia. *Principles and Practice of Psychiatric Nursing*. St. Louis: Mosby-Year Book, Inc., 1998.

ORGANIZATIONS

National Institutes of Health. 5600 Fishers Lane. Room 7C02, Rockville, MD 20857. (301) 496-4000. <<http://www.nih.gov>>.

Donald G. Barstow, RN

Electrocution see **Electric shock injuries**

Electroencephalography

Definition

Electroencephalography, or EEG, is a neurological test that uses an electronic monitoring device to measure and record electrical activity in the brain.

Purpose

The EEG is a key tool in the diagnosis and management of epilepsy and other seizure disorders. It is also used to assist in the diagnosis of brain damage and disease (e.g., **stroke**, tumors, **encephalitis**), **mental retardation**, sleep disorders, degenerative diseases such as **Alzheimer's disease** and Parkinson's disease, and certain mental disorders (e.g., **alcoholism**, **schizophrenia**, **autism**).

An EEG may also be used to monitor brain activity during surgery and to determine brain death.

Precautions

Electroencephalography should be administered and interpreted by a trained medical professional only. Data from an EEG is only one element of a complete medical and/or psychological patient assessment, and should never be used alone as the sole basis for a diagnosis.

Description

Before the EEG begins, a nurse or technician attaches approximately 16–20 electrodes to the patient's scalp with a conductive, washable paste. Depending on the purpose for the EEG, implantable or invasive electrodes are occasionally used. Implantable electrodes include sphenoidal electrodes, which are fine wires inserted under the zygomatic arch, or cheekbone; and depth electrodes, which are surgically-implanted into the brain. The EEG electrodes are painless, and are used to measure the electrical activity in various regions of the brain.



This woman is undergoing an electroencephalogram (EEG) to diagnose Alzheimer's disease. On the computer screen at the right are the colored scans of the electrical activity in her brain. Alzheimer's patients show a specific abnormality in their EEGs. (Photograph by Catherine Pouedras, Photo Researchers, Inc. Reproduced by permission.)

For the test, the patient lies on a bed, padded table, or comfortable chair and is asked to relax and remain still during the EEG testing period. An EEG usually takes no more than one hour. During the test procedure, the patient may be asked to breathe slowly or quickly; visual stimuli such as flashing lights or a patterned board may be used to stimulate certain types of brain activity. Throughout the procedure, the electroencephalograph machine makes a continuous graphic record of the patient's brain activity, or brainwaves, on a long strip of recording paper or on a computer screen. This graphic record is called an electroencephalogram.

The sleep EEG uses the same equipment and procedures as a regular EEG. Patients undergoing a sleep EEG are encouraged to fall asleep completely rather than just relax. They are typically provided a bed and a quiet room conducive to sleep. A sleep EEG lasts up to three hours.

In an ambulatory EEG, patients are hooked up to a portable cassette recorder. They then go about their normal activities, and take their normal rest and sleep for a period of up to 24 hours. During this period, the

patient and patient's family record any symptoms or abnormal behaviors, which can later be correlated with the EEG to see if they represent seizures.

Many insurance plans provide reimbursement for EEG testing. Costs for an EEG range from \$100 to more than \$500, depending on the purpose and type of test (i.e., asleep or awake, and invasive or non-invasive electrodes). Because coverage may be dependent on the disorder or illness the EEG is evaluating, patients should check with their individual insurance plan.

Preparation

Full instructions should be given to EEG patients when they schedule their test. Typically, individuals on medications that affect the central nervous system, such as anticonvulsants, stimulants, or antidepressants, are told to discontinue their prescription for a short time prior to the test (usually one to two days). Patients may be asked to avoid food and beverages that contain caffeine, a central nervous system stimulant. However, any such request should be

KEY TERMS

Epilepsy—A neurological disorder characterized by recurrent seizures with or without a loss of consciousness.

Ictal EEG—Used to measure brain activity during a seizure. May be useful in learning more about patients who aren't responding to conventional treatments.

cleared by the treating physician. Patients may also be asked to arrive for the test with clean hair free of spray or other styling products.

Patients undergoing a sleep EEG may be asked to remain awake the night before their test. They may be given a sedative prior to the test to induce sleep.

Aftercare

If the patient has suspended regular medication for the test, the EEG nurse or technician should advise him when he can begin taking it again.

Risks

Being off medication for one–two days may trigger seizures. Certain procedures used during EEG may trigger seizures in patients with epilepsy. Those procedures include flashing lights and deep breathing. If the EEG is being used as a diagnostic for epilepsy (i.e., to determine the type of seizures an individual is suffering from), this may be a desired effect, although the patient needs to be monitored closely so that the seizure can be aborted if necessary. This type of test is known as an ictal EEG.

Normal results

In reading and interpreting brainwave patterns, a neurologist or other physician will evaluate the type of brainwaves and the symmetry, location, and consistency of brainwave patterns. He will also look at the brainwave response to certain stimuli presented during the EEG test (such as flashing lights or noise). There are four basic types of brainwaves: alpha, beta, theta, and delta. “Normal” brainwave patterns vary widely, depending on factors of age and activity. For example, awake and relaxed individuals typically register an alpha wave pattern of eight to 13 cycles per second. Young children and sleeping adults may

have a delta wave pattern of under four cycles per second.

Abnormal results

The EEG readings of patients with epilepsy or other seizure disorders display bursts or spikes of electrical activity. In focal epilepsy, spikes are restricted to one hemisphere of the brain. If spikes are generalized to both hemispheres of the brain, multifocal epilepsy may be present.

The diagnostic brainwave patterns of other disorders varies widely. The appearance of excess theta waves (four to eight cycles per second) may indicate brain injury. Brain wave patterns in patients with brain disease, mental retardation, and brain injury show overall slowing. A trained medical specialist should interpret EEG results in the context of the patient's medical history, and other pertinent medical test results.

Resources

BOOKS

Restak, Richard M. *Brainscapes: An Introduction to What Neuroscience Has Learned About the Structure, Function, and Abilities of the Brain*. New York: Hyperion, 1995.

Paula Anne Ford-Martin

Electrolyte disorders

Definition

An electrolyte disorder is an imbalance of certain ionized salts (i.e., bicarbonate, calcium, chloride, magnesium, phosphate, potassium, and sodium) in the blood.

Description

Electrolytes are ionized molecules found throughout the blood, tissues, and cells of the body. These molecules, which are either positive (cations) or negative (anions), conduct an electric current and help to balance pH and acid-base levels in the body. Electrolytes also facilitate the passage of fluid between and within cells through a process known as *osmosis* and play a part in regulating the function of the neuromuscular, endocrine, and excretory systems.

The serum electrolytes include:

- Sodium (Na). A positively charged electrolyte that helps to balance fluid levels in the body and facilitates neuromuscular functioning.
- Potassium (K). A main component of cellular fluid, this positive electrolyte helps to regulate neuromuscular function and osmotic pressure.
- Calcium (Ca). A cation, or positive electrolyte, that affects neuromuscular performance and contributes to skeletal growth and blood coagulation.
- Magnesium (Mg). Influences muscle contractions and intracellular activity. A cation.
- Chloride (Cl). An anion, or negative electrolyte, that regulates blood pressure.
- Phosphate (HPO₄). Negative electrolyte that impacts metabolism and regulates acid-base balance and calcium levels.
- Bicarbonate (HCO₃). A negatively charged electrolyte that assists in the regulation of blood pH levels. Bicarbonate insufficiencies and elevations cause acid-base disorders (i.e., acidosis, alkalosis).

Medications, chronic diseases, and trauma (for example, **burns**, or fractures etc.) may cause the concentration of certain electrolytes in the body to become too high (hyper-) or too low (hypo-). When this happens, an electrolyte imbalance, or disorder, results.

Causes and symptoms

Sodium

HYPERNATREMIA. Sodium helps the kidneys to regulate the amount of water the body retains or excretes. Consequently, individuals with elevated serum sodium levels also suffer from a loss of fluids, or dehydration. **Hyponatremia** can be caused by inadequate water intake, excessive fluid loss (i.e., diabetes insipidus, **kidney disease**, severe burns, and prolonged **vomiting** or **diarrhea**), or sodium retention (caused by excessive sodium intake or aldosteronism). In addition, certain drugs, including loop diuretics, corticosteroids, and antihypertensive medications may cause elevated sodium levels.

Symptoms of hypernatremia include:

- thirst
- orthostatic hypotension
- dry mouth and mucous membranes
- dark, concentrated urine
- loss of elasticity in the skin

- irregular heartbeat (tachycardia)
- irritability
- fatigue
- lethargy
- heavy, labored breathing
- muscle twitching and/or seizures

HYPONATREMIA. Up to 1% of all hospitalized patients and as many as 18% of nursing home patients develop hyponatremia, making it one of the most common electrolyte disorders. A 2004 study questioned the routine make-up of fluids prescribed for children and delivered intravenously (through a needle into a vein) in hospitals today. The authors recommended only using IV fluids when necessary and then using isotonic saline. **Diuretics**, certain psychoactive drugs (i.e., fluoxetine, sertraline, haloperidol), specific antipsychotics (lithium), vasopressin, chlorpropamide, the illicit drug “ecstasy,” and other pharmaceuticals can cause decreased sodium levels, or **hyponatremia**. Low sodium levels may also be triggered by inadequate dietary intake of sodium, excessive perspiration, water intoxication, and impairment of adrenal gland or kidney function.

Symptoms of hyponatremia include:

- nausea, abdominal cramping, and/or vomiting
- headache
- edema (swelling)
- muscle weakness and/or tremor
- paralysis
- disorientation
- slowed breathing
- seizures
- **coma**

Potassium

HYPERKALEMIA. **Hyperkalemia** may be caused by ketoacidosis (diabetic coma), myocardial infarction (**heart attack**), severe burns, kidney failure, **fasting**, bulimia nervosa, gastrointestinal bleeding, adrenal insufficiency, or Addison’s disease. Diuretic drugs, cyclosporin, lithium, heparin, ACE inhibitors, **beta blockers**, and trimethoprim can increase serum potassium levels, as can heavy **exercise**. The condition may also be secondary to hypernatremia (low serum concentrations of sodium). Symptoms may include:

- weakness
- nausea and/or abdominal pain

- irregular heartbeat (arrhythmia)
- diarrhea
- muscle pain

HYPOKALEMIA. Severe **dehydration**, aldosteronism, Cushing's syndrome, kidney disease, long-term diuretic therapy, certain penicillins, laxative **abuse**, congestive **heart failure**, and adrenal gland impairments can all cause depletion of potassium levels in the bloodstream. A substance known as glycyrrhetic acid, which is found in licorice and chewing tobacco, can also deplete potassium serum levels. Symptoms of **hypokalemia** include:

- weakness
- paralysis
- increased urination
- irregular heartbeat (arrhythmia)
- orthostatic hypotension
- muscle pain
- tetany

Calcium

HYPERCALCEMIA. Blood calcium levels may be elevated in cases of thyroid disorder, **multiple myeloma**, metastatic **cancer**, multiple bone **fractures**, milk-alkali syndrome, and Paget's disease. Excessive use of calcium-containing supplements and certain over-the-counter medications (i.e., **antacids**) may also cause **hypercalcemia**. In infants, lesser known causes may include blue diaper syndrome, Williams syndrome, secondary **hyperparathyroidism** from maternal **hypocalcemia**, and dietary phosphate deficiency. Symptoms include:

- fatigue
- constipation
- depression
- confusion
- muscle pain
- nausea and vomiting
- dehydration
- increased urination
- irregular heartbeat (arrhythmia)

HYPOCALCEMIA. Thyroid disorders, kidney failure, severe burns, sepsis, vitamin D deficiency, and medications such as heparin and glucocorticoids can deplete blood calcium levels. Lowered levels cause:

- muscle cramps and spasms
- tetany and/or convulsions

- mood changes (depression, irritability)
- dry skin
- brittle nails
- facial twitching

Magnesium

HYPERMAGNESEMIA. Excessive magnesium levels may occur with end-stage renal disease, **Addison's disease**, or an overdose of magnesium salts. Hypermagnesemia is characterized by:

- lethargy
- hypotension
- decreased heart and respiratory rate
- muscle weakness
- diminished tendon reflexes

HYPOMAGNESEMIA. Inadequate dietary intake of magnesium, often caused by chronic **alcoholism** or **malnutrition**, is a common cause of hypomagnesemia. Other causes include malabsorption syndromes, **pancreatitis**, aldosteronism, burns, hyperparathyroidism, digestive system disorders, and diuretic use. Symptoms of low serum magnesium levels include:

- leg and foot cramps
- weight loss
- vomiting
- muscle spasms, twitching, and tremors
- seizures
- muscle weakness
- arrhythmia

Chloride

HYPERCHLOREMIA. Severe dehydration, kidney failure, hemodialysis, traumatic brain injury, and aldosteronism can also cause hyperchloremia. Drugs such as boric acid and ammonium chloride and the intravenous (IV) infusion of sodium chloride can also boost chloride levels, resulting in hyperchloremic **metabolic acidosis**. Symptoms include:

- weakness
- headache
- nausea
- cardiac arrest

HYPCHLOREMIA. Hypochloremia usually occurs as a result of sodium and potassium depletion (i.e., hyponatremia, hypokalemia). Severe depletion of

serum chloride levels causes *metabolic alkalosis*. This alkalization of the bloodstream is characterized by:

- mental confusion
- slowed breathing
- paralysis
- muscle tension or spasm

Phosphate

HYPERPHOSPHATEMIA. Skeletal fractures or disease, kidney failure, **hypoparathyroidism**, hemodialysis, diabetic ketoacidosis, acromegaly, systemic infection, and intestinal obstruction can all cause phosphate retention and build-up in the blood. The disorder occurs concurrently with hypocalcemia. Individuals with mild hyperphosphatemia are typically asymptomatic, but signs of severe hyperphosphatemia include:

- tingling in hands and fingers
- muscle spasms and cramps
- convulsions
- cardiac arrest

HYPOPHOSPHATEMIA. Serum phosphate levels of 2 mg/dL or below may be caused by hypomagnesemia and hypokalemia. Severe burns, alcoholism, **diabetic ketoacidosis**, kidney disease, hyperparathyroidism, **hypothyroidism**, Cushing's syndrome, malnutrition, hemodialysis, **vitamin D deficiency**, and prolonged diuretic therapy can also diminish blood phosphate levels. There are typically few physical signs of mild phosphate depletion. Symptoms of severe hypophosphatemia include:

- muscle weakness
- weight loss
- bone deformities (osteomalacia)

Diagnosis

Diagnosis is performed by a physician or other qualified healthcare provider who will take a medical history, discuss symptoms, perform a complete physical examination, and prescribe appropriate laboratory tests. Because electrolyte disorders commonly affect the neuromuscular system, the provider will test reflexes. If a calcium imbalance is suspected, the physician will also check for Chvostek's sign, a reflex test that triggers an involuntary facial twitch, and Trousseau's sign, a muscle spasm that occurs in response to pressure on the upper arm.

Serum electrolyte imbalances can be detected through blood tests. Blood is drawn from a vein on the back of the hand or inside of the elbow by a medical technician, or phlebotomist, and analyzed at a lab.

Normal levels of electrolytes are:

- Sodium. 135–145 mEq/L (serum)
- Potassium. 3.5–5.5 mEq/L (serum)
- Calcium. 8.8–10.4 mg/dL (total Ca; serum); 4.7–5.2 mg/dL (unbound Ca; serum)
- Magnesium. 1.4–2.1 mEq/L (plasma)
- Chloride. 100–108 mEq/L (serum)
- Phosphate. 2.5–4.5 mg/dL (plasma; adults)

Standard ranges for test results may vary due to differing laboratory standards and physiological variances (gender, age, and other factors). Other blood tests that determine pH levels and acid-base balance may also be performed.

Treatment

Treatment of electrolyte disorders depends on the underlying cause of the problem and the type of electrolyte involved. If the disorder is caused by poor diet or improper fluid intake, nutritional changes may be prescribed. If medications such as diuretics triggered the imbalance, discontinuing or adjusting the drug therapy may effectively treat the condition. Fluid and electrolyte replacement therapy, either intravenously or by mouth, can reverse electrolyte depletion.

Hemodialysis treatment may be required to reduce serum potassium levels in hyperkalemic patients with impaired kidney function. It may also be recommended for renal patients suffering from severe hypermagnesemia.

Prognosis

A patient's long-term prognosis depends upon the root cause of the electrolyte disorder. However, when treated quickly and appropriately, electrolyte imbalances in and of themselves are usually effectively reversed.

When they are mild, some electrolyte imbalances have few to no symptoms and may pass unnoticed. For example, transient hyperphosphatemia is usually fairly benign. However, long-term elevations of blood phosphate levels can lead to potentially fatal soft tissue and vascular calcifications and bone disease, and severe serum phosphate deficiencies (hypophosphatemia) can cause encephalopathy, coma, and **death**.

Severe hyponatremia has a mortality rate of 40–60%. Death is commonly due to cerebrovascular damage and hemorrhage resulting from dehydration and shrinkage of the brain cells.

KEY TERMS

Acid-base balance—A balance of acidity and alkalinity of fluids in the body that keeps the pH level of blood around 7.35–7.45.

Aldosteronism—A condition defined by high serum levels of aldosterone, a hormone secreted by the adrenal gland that is responsible for increasing sodium reabsorption in the kidneys.

Addison's disease—A disease characterized by a deficiency in adrenocortical hormones due to destruction of the adrenal gland.

Bulimia nervosa—An eating disorder characterized by bingeing and purging (self-induced vomiting) behaviors.

Milk-alkali syndrome—Elevated blood calcium levels and alkalosis caused by excessive intake of milk and alkalis. Usually occurs in the treatment of peptic ulcer.

Orthostatic hypotension—A drop in blood pressure that causes faintness or dizziness and occurs when one rises to a standing position. Also known as postural hypotension.

Osmotic pressure—Pressure that occurs when two solutions of differing concentrations are separated by a semipermeable membrane, such as a cellular wall, and the lower concentration solute is drawn across the membrane into the higher concentration solute (osmosis).

Tetany—A disorder of the nervous system characterized by muscle cramps, spasms of the arms and legs, and numbness of the extremities.

Prevention

Physicians should use caution when prescribing drugs known to affect electrolyte levels and acid-base balance. Individuals with kidney disease, thyroid problems, and other conditions that may place them at risk for developing an electrolyte disorder should be educated on the signs and symptoms.

Resources

BOOKS

Post, Theodore, and Burton Rose. *Clinical Physiology of Acid-Base and Electrolyte Disorders*. 5th ed. New York: McGraw-Hill Professional, 2001.

PERIODICALS

Cohn, Jay N., et al. "New Guidelines for Potassium Replacement in Clinical Practice: A Contemporary

Review by the National Council on Potassium in Clinical Practice." *Archives of Internal Medicine* 160, no.16 (September 11, 2000): 2429-36.

Goh, Kian Ping. "Management of Hyponatremia." *American Family Physician* May 15, 2004: 2387.

Moritz, Michael L., Juan Carlos Ayus. "Hospital-acquired Hyponatremia: Why are There Still Deaths?" *Pediatrics* May 2004: 1395-1397.

Springate, James E., Mary F. Carroll. "HAdditional Causes of Hypercalcemia in Infants." *American Family Physician* June 15, 2004: 2766.

Paula Anne Ford-Martin
Teresa G. Odle

Electrolyte supplements

Definition

Electrolyte supplements are a varied group of prescription and nonprescription preparations used to correct imbalances in the body's electrolyte levels. Electrolytes themselves are substances that dissociate into ions (electrically charged atoms or atom groups) when they melt or are dissolved, thus serving to conduct electricity. In the human body, electrolytes are thus critical to the proper distribution of water, muscle contraction and expansion, transmission of nerve impulses, delivery of oxygen to body tissues, heart rate and rhythm, acid-base balance, and other important functions or conditions.

The ions that are formed when electrolytes are dissolved in body fluids are either positively or negatively charged. Positively charged ions are called cations, and are formed when an atom or atom group loses electrons. The most important cations in the human body are sodium, potassium, magnesium, and calcium ions. Negatively charged ions are called anions, and are formed when an atom or atom group gains electrons. The principal anions in the body include bicarbonate, chloride, phosphate, and sulfate ions, as well as ions formed by certain protein compounds or organic acids.

About 60 percent of an adult human male's total body weight is water. In adult women, the figure is about 55 percent, and is even lower in the elderly and in obese people. Two-thirds of total body water (TBW) lies inside cells and is known as intracellular fluid or ICF. The remaining third of TBW lies outside the cells and is called extracellular fluid or ECF. About 75 percent of ECF lies in connective tissue or the spaces between tissues outside the blood vessels (interstitial

spaces), while the remaining 25 percent is within the blood vessels. In addition to representing different proportions of TBW, ICF and ECF differ significantly in their electrolyte content. Whereas the major cation in ICF is potassium, the most important cation in ECF is sodium. These differences in electrolyte levels help to regulate the movement of water between ICF and ECF.

Children are more vulnerable than adults to fluid and electrolyte imbalances, in part because they have different ratios of TBW to total body weight, and of ICF to ECF. A newborn baby carried to full term has a TBW ratio between 75 and 80 percent. The baby's total body water ratio decreases by 4–5 percent during the first week after birth and reaches the adult level of 60 percent by twelve months of age. Similarly, a newborn has an ICF: ECF ratio of 55: 45, which falls to the adult ratio of 70: 30 during the first year of life. In addition to these different fluid ratios, children's kidneys are less efficient than adults in regulating water balance; children have smaller organ systems that dissipate body heat less efficiently; and their core body temperature rises faster than that of an adult when they become dehydrated. All these factors help to explain why some electrolyte supplements are formulated specifically for children.

Purpose

The purpose of electrolyte supplements is to restore the proper ratio of total body water to total body weight and the correct proportions of the various electrolytes in body fluids. Electrolyte imbalances may result from excessive intake or inadequate elimination of electrolytes on the one hand or by insufficient intake or excessive elimination on the other hand.

Body regulation of water and electrolytes

Under normal conditions, the water and electrolyte content of the body is regulated by the kidneys, the secretion of antidiuretic hormone, and the sensation of thirst. The average adult needs to take in about 700–800 mL (about 1.5–1.7 pints) of water per day in order to match the water lost through perspiration, breathing, and excretion of waste products. The water that is taken in by mouth is added to the 200–300 mL (0.42–0.63 pints) of water that are formed in the body each day through tissue breakdown.

The amount of water needed to match fluid losses, however, may be considerably greater than the average during **exercise** or in patients with **fever**, severe **vomiting**, or **diarrhea**. Adults with fever typically lose an additional .75–1.0 ounces of fluid per day for each degree that their temperature rises above normal. With regard to

diarrhea, adults with **cholera** have been reported to lose as much as a quart of fluid per hour in their stools. The fluid lost in this way also contains sodium, potassium, and chloride, resulting in electrolyte imbalances in cholera patients as well as **dehydration**.

Exercise raises the total metabolism of the body to 5–15 times the resting rate. Most of this energy (70–90 percent) is released as heat, which is partially dissipated by the evaporation of sweat. Depending on weather conditions, the type and weight of clothing being worn, and the intensity of exercise or physical work performed, adults may lose anywhere from 1 to 2.5 quarts of fluid per hour through perspiration. Sweat, however, contains sodium chloride as well as smaller amounts of potassium, calcium, and magnesium. In order to maintain the proper balance of electrolytes in the body as well as fluid, athletes or people employed in outdoor work during warm weather may need to replace the electrolytes lost in sweat by taking capsules or drinking beverages containing supplemental electrolytes.

With regard to the sense of thirst, it is not always an accurate indication of the body's need for water. Researchers have found that many people do not feel thirsty until they have already lost about 2 percent of their total body weight through fluid losses. As a result, most people will not replace enough fluid during exercise or hot weather simply by drinking water until they no longer feel thirsty. In addition, the **aging** process, certain mental disorders, or drugs may affect a person's sense of thirst.

At the other extreme of water intake, a person may drink excessive amounts of water due to misunderstandings about their need for extra fluid during exercise. This condition is known as water intoxication or hyperhydration. It leads to abnormally low levels of sodium in the blood, a condition known as **hyponatremia**. Water intoxication may lead to swelling of the brain, confusion, disorientation, and eventually **coma** or **death**. Several marathon runners have died from water intoxication since 2002, as have teenagers who consumed large amounts of water after taking doses of Ecstasy (MDMA), a so-called “club drug.” Other persons at risk for water intoxication include people with eating disorders and children with **mental retardation**. An important article published in the *New England Journal of Medicine* in April 2005 reported that as many as 13 percent of marathon runners developed hyponatremia during the course of a race as a result of drinking too much water, usually 3 quarts or more. Female athletes appear to be at greater risk of water intoxication and hyponatremia than male athletes.

Conditions associated with fluid and electrolyte imbalance

The most common conditions leading to fluid and electrolyte imbalance are as follows:

- Exposure to extended periods of extremely hot weather.
- High levels of athletic activity, military training, or outdoor work in such fields as construction, agriculture, forestry, fishing, and certain types of manufacturing.
- Extreme changes in diet.
- Reduced fluid intake.
- Medication side effects. Certain drugs, particularly **diuretics**, beta-blockers, and **vasodilators**, may increase the loss of electrolytes in urine and/or interfere with the body's ability to regulate its temperature during exercise or in hot weather.
- Severe illnesses characterized by high fever, recurrent diarrhea, and/or frequent vomiting. Such illnesses include cholera, viral **gastroenteritis** ("stomach flu"), **shigellosis**, and amebic dysentery.
- Severe **burns** covering more than 10 percent of the body.
- Surgical creation of a stoma or urinary diversion. These operations sometimes lead to an increased loss of body fluids while the patient's body is adjusting to the changes in urination and excretion resulting from the surgery. In addition, some forms of weight loss surgery intended to bypass parts of the small intestine in which food absorption occurs have a 70-percent rate of electrolyte imbalances as a complication of the operation.
- Diseases affecting the kidneys. These include **diabetes mellitus**, **diabetes insipidus**, and syndrome of inappropriate antidiuretic hormone secretion (SIADH) as well as **cancer** or infections of the kidneys.
- In infants, premature birth.

Description

The various electrolyte supplements used in the United States and Canada as of 2005 are intended to prevent or treat electrolyte imbalances in very different situations or groups of patients. They range from sports drinks and other supplements used by amateur or professional athletes to prevent **muscle cramps** and improve athletic performance, to liquids used at home to prevent dehydration in children with diarrhea, to injections administered as part of enteral (feeding

through a tube or stoma directly into the small intestine) or parenteral **nutrition** (intravenous feeding that bypasses the digestive tract).

The major categories of electrolyte supplements are as follows:

- **Sports drinks.** Sports drinks are beverages specially formulated to contain appropriate amounts of electrolytes and carbohydrates as well as water to replace the fluid and sodium lost through sweat during athletic activities. These beverages are popular with athletes at the college level, being used by about $\frac{3}{4}$ of the students on varsity teams at major universities. According to the American College of Sports Medicine as well as American and Canadian dietitians' associations, sports drinks are effective in supplying food energy for the muscles, maintaining proper levels of blood sugar, maintaining the proper functioning of the thirst mechanism, and lowering the risk of dehydration or hyponatremia. Other researchers have noted that the flavoring added to sports drinks encourages athletes to drink more during periods of exercise and thus maintain proper levels of hydration. Sports drinks can be purchased in supermarkets as well as health food stores; they include such well-known beverages as Gatorade, Lucozade, Red Devil, Powerade, and Red Alert. Some of these popular supplements come in a semi-solid form known as energy gels, some of which contain **caffeine** or various herbal compounds as well as carbohydrates and electrolytes.
- **Over-the-counter powders and tablets.** Some athletes—particularly those who participate in such sports as long-distance running or endurance cycling—prefer capsules or concentrated powders to maintain their electrolyte balance during exercise. The powders are mixed with 12 or 16 ounces of cold water prior to drinking, while the capsules can be taken before, during, and after exercise. Most contain flavorings to mask the naturally salty or bitter taste of the electrolytes themselves. Common brand names include eForce, NutriBiotic, and Endurolytes; prices range from \$11 for 100 capsules to \$35 for a 30-serving jar of powder. These products are regarded by the Food and Drug Administration (FDA) as dietary supplements.
- **Over-the-counter electrolyte replenishers for children.** As has been mentioned earlier, infants and young children are more vulnerable to dehydration than adults, particularly from severe gastroenteritis or diarrhea. A child may become dehydrated in less than a day from recurrent vomiting or episodes of diarrhea. Some doctors recommend that parents

keep oral rehydration fluids containing mixtures of carbohydrates and electrolytes specially formulated for children in the medicine chest at home in case the child becomes dehydrated from a stomach virus or similar illness. Common brand names for these products, which are regulated by the FDA as “medical foods,” include Pedialyte, Infalyte, Naturalyte, and Rehydralyte. Most come in a powdered form to be mixed with water as well as liquid forms; Pedialyte is also available as fruit-flavored freezer pops. Typical prices range from \$6 for a box of 16 freezer pops to \$34 for a quart of oral electrolyte solution.

- Oral rehydration formulae for children and adults. Oral rehydration salts, which are also known as ORS, have been a staple of treatment for cholera and other diseases accompanied by severe diarrhea in developing countries for almost half a century. First researched in the 1940s, oral rehydration salts were adopted by the World Health Organization (WHO) in 1978 in order to reduce the risk of death from dehydration caused by cholera-related diarrhea. Since the introduction of ORS, the number of children around the world who die from acute diarrhea has been reduced from 5 million per year to 1.3 million. Reformulated by WHO in 2002, the ORS salts come in packets to be kept in the home and mixed with water as soon as a child (or adult) falls ill. The new formula is a low-glucose and low-sodium mixture. If the WHO packets are unavailable, a comparable form of oral rehydration solution can be made by adding 8 tsp of table sugar, $\frac{1}{2}$ tsp of salt, $\frac{1}{2}$ tsp of baking soda (bicarbonate of soda), and $\frac{1}{3}$ tsp of potassium chloride to a liter (1.05 quarts) of water. In an emergency, a solution prepared from 1 tbsp of sugar and $\frac{1}{2}$ tsp of salt added to 1 liter of water can be used to treat diarrhea.
- Multiple electrolyte injections. Various mixtures of electrolytes are available by prescription in injectable form to be added to enteral or parenteral nutrition formulae. These forms of feeding are used in patients who require supplementation or complete replacement of feeding by mouth, including patients with various intestinal disorders, AIDS, or severe burns. Basic solutions for **total parenteral nutrition**, or TPN, contain the electrolytes sodium, potassium, chloride, phosphate, and magnesium, although the exact proportion of electrolytes can be tailored to an individual patient's needs. Some injectable formulae contain dextrose, a sugar, and acetate or lactate as well as the five major electrolytes. Common brand names include Hyperlite, TPN Electrolytes, Lypholyte, Nutrilite, Plasma-Lyte 148, and others. Some patients are taught to use these injectable formulae at home.

Recommended dosage

Recommended dosages for electrolyte supplements are as follows:

- Sports drinks. Since sports drinks and energy gels are not medications in the strict sense, the amount consumed will vary not only from person to person but also in a given individual from day to day depending on weather conditions, level of athletic conditioning, length of activity, and other factors. To lower the risk of dehydration in adults in hot weather, the American College of Sports Medicine recommends taking 20 ounces of a sports drink 2–3 hours before exercising; another 10 ounces 20 minutes before exercising; 10 ounces every 10–20 minutes during exercise; and 20 ounces per pound of weight lost during exercise after the activity is over. A group of researchers at Texas A & M University suggests that athletes competing in endurance events should drink about 500 mL (1 pint) of a sports drink containing 4–8 percent of carbohydrates and electrolytes 1–2 hours before an event. For events lasting longer than an hour, they should drink between 600 and 1200 mL (1.25 to 2.5 pints) per hour of a sports drink containing carbohydrate plus 0.5–0.7 grams of sodium per quart.
- Over-the-counter powders and tablets. The usual recommended dose of powdered electrolytes is one scoopful (or prepackaged envelope) of powder dissolved in 12–16 ounces of water before exercising. Capsules may be taken as follows: 1–3 capsules 30–60 minutes before exercising; 1–6 capsules per hour during the workout; and 1–3 capsules after exercising.
- Over-the-counter electrolyte replenishers for children. Dosages for Pedialyte and similar oral rehydration solutions for children are usually based on the child's age and weight. The child's doctor should determine the quantity to be given if the child is younger than 12 months of age. Children between the ages of 1 and 2 years are usually given 34 mL of electrolyte solution per pound of body weight during the first eight hours of treatment and 75 mL per pound of body weight during the next 16 hours, although the doctor may adjust the dose if the child is very thirsty. Children between the ages of 2 and 10 are given 23 mL of electrolyte solution per pound of body weight for the first four to six hours of treatment, followed by 45 mL per pound taken over the next 18–24 hours. Freezer pops may be given to children older than 1 year as often as the child desires.
- Oral rehydration formulae. The WHO form of oral rehydration liquid is made by adding the full

contents of one packet of powdered oral rehydration salts to a quart of drinking water. The solution should not be boiled. A fresh quart of solution should be mixed each day. Infants and young children should be given the solution in small amounts by spoon as often as possible. Adults and teenagers should take the WHO formula according to the doctor's directions.

- **Multiple electrolyte injections.** Basic TPN solutions are usually made up in liter batches and adjusted to each individual patient's needs. The standard adult dosage is 2 liters per day, usually administered by drip through a needle or catheter placed in the patient's vein for a 10–12-hour period once a day or 5 days per week. The patient may be given several units of premixed TPN fluid to store at home in the refrigerator or freezer. Each dose should be taken from the refrigerator 4–6 hours prior to use to allow it to warm to room temperature. TPN solution stored in a freezer should be moved to a refrigerator 24 hours before use.

Precautions

- **Sports drinks.** Sports drinks should not be given to rehydrate children with vomiting or diarrhea, as they do not contain the proper balance of carbohydrates and electrolytes needed by children's bodies.
- **Over-the-counter powders and tablets.** These products should always be taken with adequate amounts of water and kept out of the reach of children.
- **Over-the-counter electrolyte replenishers for children.** These preparations should be stored out of the reach of children and away from heat and direct sunlight. In addition, they should not be given to patients with intestinal blockage.
- **Oral rehydration formulae.** WHO oral rehydration salts and packets of similar formulae should not be stored in damp places, as moisture can cause the contents to lose their effectiveness. These products should also be kept away from heat or direct sunlight. Unused oral rehydration solution should be discarded at the end of each day. As with electrolyte replenishers for children, oral rehydration formulae should not be given to patients with intestinal blockage.
- **Multiple electrolyte injections.** Patients using multiple electrolyte injections as part of total parenteral nutrition should have their blood and urine checked at regular intervals while they are receiving these medicines. They should also be taught to recognize the signs of infection at the injection site (**pain**, swelling, redness, or a cold sensation). In addition, these

KEY TERMS

Anion—An ion carrying a negative charge owing to a surplus of electrons. Anions in the body include bicarbonate, chloride, phosphate, sulfate, certain organic acids, and certain protein compounds.

Cation—An ion carrying a positive charge due to a loss of electrons. Cations in the body include sodium, potassium, magnesium, and calcium ions.

Cholera—A severe bacterial infection of the small intestine characterized by profuse diarrhea and eventual dehydration. Cholera is still a frequent cause of death among children in developing countries.

Electron—An elementary particle carrying a negative charge. Electrons may exist either independently or as components of an atom outside its nucleus.

Enteral nutrition—Nourishment given through a tube or stoma directly into the small intestine, thus bypassing the upper digestive tract.

Hyponatremia—Insufficient sodium in the blood.

Interstitial spaces—Spaces within body tissues that are outside the blood vessels. Interstitial spaces are also known as interstitial compartments.

Ion—An atom or group of atoms that acquires an electrical charge by the gain or loss of electrons.

Metabolism—The sum of an organism's physical and chemical processes that produce and maintain living tissue, and make energy available to the organism. Insulin resistance is a disorder of metabolism.

Parenteral nutrition—Nutrition supplied intravenously, thus bypassing the patient's digestive tract entirely.

Stoma—A surgical opening made in the abdominal wall to allow waste products to pass directly to the outside.

Water intoxication—A potentially life-threatening condition caused by drinking too much water, which leads to hyponatremia and may result in seizures, coma, and death.

patients should not use sports drinks, other electrolyte supplements, or over-the-counter medications (including herbal preparations) without consulting their doctor. The injections should not be used if the fluid looks cloudy, has solid particles floating in it, or has separated. The injections should be stored away from sunlight and moisture. In addition, patients receiving multiple electrolyte injections

should not stop them suddenly without telling their doctor, as the dosage may need to be reduced slowly before the TPN is discontinued.

Side effects

- Sports drinks. Some persons do not like the salty taste of many sports drinks. They may wish to consider products containing glycine, which is an amino acid that neutralizes the salty taste of the electrolytes themselves. A more serious side effect of sports drinks, however, is **tooth decay**. An article published by researchers at the University of Maryland Dental School in 2005 showed that sports drinks erode tooth enamel at a rate three to 11 times faster than cola-based soft drinks.
- Over-the-counter powders and tablets. No side effects have been reported for these products.
- Over-the-counter electrolyte replenishers for children. Side effects may include allergic reactions, including **hives**, swelling of the face or hands, trouble breathing, **tingling** in the mouth or throat. Other side effects may include signs of too much sodium in the body, such as **dizziness**, seizures, muscle twitching, or restlessness. The doctor should be notified *at once* if any of these side effects occur. A less serious side effect that occurs in some children is mild vomiting.
- Oral rehydration formulae. May produce the same side effects as electrolyte replenishers for children.
- Multiple electrolyte injections. Minor side effects may include increased frequency of urination, **dry mouth**, increased thirst, or drowsiness. Serious side effects include rapid weight gain, yellowing of the skin or eyes, fruity odor on the breath, **numbness** or tingling in the hands or feet, uneven heartbeat, **shortness of breath**, confusion, or weakness with muscle twitching. Patients should notify their doctor *at once* if they notice any of these side effects.
- Oral rehydration formulae for adults. No interactions with other medications have been reported; however, the doctor should be informed of all other medications that the patient is taking in case a dosage adjustment may be necessary.
- Multiple electrolyte injections. Sports drinks and other electrolyte supplements may raise total blood electrolyte levels in patients receiving these injections.

Resources

BOOKS

- “Childhood Infections: Acute Infectious Gastroenteritis.” Section 19, Chapter 265 in *The Merck Manual of Diagnosis and Therapy*, edited by Mark H. Beers, MD, and Robert Berkow, MD. Whitehouse Station, NJ: Merck Research Laboratories, 2004.
- “Fluid and Electrolyte Disorders in Infants and Children.” Section 19, Chapter 259 in *The Merck Manual of Diagnosis and Therapy*, edited by Mark H. Beers, MD, and Robert Berkow, MD. Whitehouse Station, NJ: Merck Research Laboratories, 2004.
- “Nutritional Support.” Section 1, Chapter 1 in *The Merck Manual of Diagnosis and Therapy*, edited by Mark H. Beers, MD, and Robert Berkow, MD. Whitehouse Station, NJ: Merck Research Laboratories, 2004.
- “Water, Electrolyte, Mineral, and Acid-Base Metabolism.” Section 2, Chapter 12 in *The Merck Manual of Diagnosis and Therapy*, edited by Mark H. Beers, MD, and Robert Berkow, MD. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

PERIODICALS

- Almond, Christopher S. D., MD, MPH, Andrew Y. Shin, MD, Elizabeth B. Fortescue, MD, et al. “Hyponatremia among Runners in the Boston Marathon.” *New England Journal of Medicine* 352 (April 14, 2005): 1550–1556.
- Bovill, M. E., W. J. Tharion, and H. R. Lieberman. “Nutrition Knowledge and Supplement Use among Elite U. S. Army Soldiers.” *Military Medicine* 168 (December 2003): 997–1000.
- Comeau, Matthew J., PhD. “A Hot Issue for Summer Exercisers.” *ACSM Fit Society Page* (Summer 2001): 4.
- Froiland, K., W. Koszewski, J. Hingst, and L. Kopecky. “Nutritional Supplement Use among College Athletes and Their Sources of Information.” *International Journal of Sport Nutrition and Exercise Metabolism* 14 (February 2004): 104–120.
- “Position of Dietitians of Canada, the American Dietetic Association, and the American College of Sports Medicine: Nutrition and Athletic Performance.” *Canadian Journal of Dietetic Practice and Research* 61 (Winter 2000): 176–192.
- Rao, M. C. “Oral Rehydration Therapy: New Explanations for an Old Remedy.” *Annual Review of Physiology* 66 (2004): 385–417.

Interactions

- Sports drinks. Sports drinks may raise blood electrolyte levels in patients receiving total parenteral nutrition.
- Over-the-counter powders and tablets. No interactions have been reported.
- Over-the-counter electrolyte replenishers for children. Children receiving premixed forms of these preparations should not eat food with added salt or drink fruit juices until the diarrhea has stopped.

- Rice, Henry, MD. "Fluid Therapy for the Pediatric Surgical Patient." *eMedicine*, 17 September 2004. < <http://www.emedicine.com/ped/topic2954> > .
- Sawka, Michael N., and Scott J. Montain. "Fluid and Electrolyte Supplementation for Exercise Heat Stress." *American Journal of Clinical Nutrition* 72 (August 2000): 564S–572S.
- von Duvillard, S. P., W. A. Braun, M. Markofski, et al. "Fluids and Hydration in Prolonged Endurance Performance." *Nutrition* 20 (July-August 2004): 651–656.
- von Fraunhofer, J. A., and M. M. Rogers. "Effects of Sports Drinks and Other Beverages on Dental Enamel." *General Dentistry* 53 (January-February 2005): 28–31.
- Wexler, Randell K., MD. "Evaluation and Treatment of Heat-Related Illnesses." *American Family Physician* 65 (June 1, 2002): 2307–2320.

ORGANIZATIONS

- American College of Sports Medicine (ACSM). 401 West Michigan Street, Indianapolis, IN 46202-3233. (317) 637-9200. Fax: (317) 634-7817. < <http://www.acsm.org> > .
- American Society of Health-System Pharmacists (ASHP). 7272 Wisconsin Avenue, Bethesda, MD 20814. (301) 657-3000. < www.ashp.org > .
- Rehydration Project. P. O. Box 1, Samara, 5235, Costa Rica. + 506 656-0504. Fax: + 1 603 849-5656. < <http://rehydrate.org> > .
- United States Food and Drug Administration (FDA). 5600 Fishers Lane, Rockville, MD 20857-0001. (888) INFO-FDA. < www.fda.gov > .

OTHER

- Goodall, Roger M. "Oral Rehydration Therapy: How It Works." < http://rehydrate.org/ors/ort_how_it_works.htm > .
- World Health Organization (WHO) Media Centre press release, 8 May 2002. "New Formula for Oral Rehydration Salts Will Save Millions of Lives." < <http://www.who.int/mediacentre/news/releases/release35/en> > .

Rebecca J. Frey, PhD

Electrolyte tests

Definition

Electrolytes are positively and negatively charged molecules, called ions, that are found within cells, between cells, in the bloodstream, and in other fluids

throughout the body. Electrolytes with a positive charge include sodium, potassium, calcium, and magnesium; the negative ions are chloride, bicarbonate, and phosphate. The concentrations of these ions in the bloodstream remain fairly constant throughout the day in a healthy person. Changes in the concentration of one or more of these ions can occur during various acute and chronic disease states and can lead to serious consequences.

Purpose

Tests that measure the concentration of electrolytes are useful in the emergency room and to obtain clues for the diagnosis of specific diseases. Electrolyte tests are used for diagnosing dietary deficiencies, excess loss of nutrients due to urination, **vomiting**, and **diarrhea**, or abnormal shifts in the location of an electrolyte within the body. When an abnormal electrolyte value is detected, the physician may either act to immediately correct the imbalance directly (in the case of an emergency) or run further tests to determine the underlying cause of the abnormal electrolyte value. Electrolyte disturbances can occur with malfunctioning of the kidney (renal failure), infections that produce severe and continual diarrhea or vomiting, drugs that cause loss of electrolytes in the urine (**diuretics**), **poisoning** due to accidental consumption of electrolytes, or diseases involving hormones that regulate electrolyte concentrations.

Precautions

Electrolyte tests are performed from routine blood tests. The techniques are simple, automated, and fairly uniform throughout the United States. During the preparation of blood plasma or serum, health workers must take care not to break the red blood cells, especially when testing for serum potassium. Because the concentration of potassium within red blood cells is much higher than in the surrounding plasma or serum, broken cells would cause falsely elevated potassium levels.

Description

Electrolyte tests are typically conducted on blood plasma or serum, urine, and diarrheal fluids. Electrolytes can be classified in at least five different ways. One way is that some electrolytes tend to exist mostly inside cells, or are intracellular, while others tend to be outside cells, or are extracellular. Potassium, phosphate, and magnesium occur at much greater levels inside the cell than outside, while sodium and chloride occur at much greater levels

extracellularly. A second classification distinguishes those electrolytes that participate directly in the transmission of nerve impulses and those that do not. Sodium, potassium, and calcium are the important electrolytes involved in nerve impulses, and disorders affecting them are most closely associated with neurological disorders. A third classification focuses on electrolytes that are able to form a tight union, or complex, with one another. Calcium and phosphate have the greatest tendency to form complexes with each other. Disorders that cause an increase in either plasma calcium or phosphate can result in the deposit of calcium-phosphate crystals in the soft tissues of the body. A fourth classification concerns those electrolytes that influence the acidity or alkalinity of the bloodstream, also known as the pH. The pH of the bloodstream is normally in the range of 7.35–7.45. A decrease below this range is called acidosis, while a pH above this range is called alkalosis. The electrolytes most closely associated with the pH of the bloodstream are bicarbonate, chloride, and phosphate.

Preparation

All electrolyte tests can be performed on plasma or serum. Plasma is prepared by withdrawing a blood sample and placing it in a test tube containing a chemical that prevents blood from clotting (an anticoagulant). Serum is prepared by withdrawing a blood sample, placing it in a test tube, and allowing it to clot. The blood spontaneously clots within a minute of withdrawing the blood from a vein. The serum or plasma is then rapidly spun with a centrifuge in order to remove the blood cells or clot.

Normal results

Electrolyte concentrations are similar whether measured in serum or plasma. Values can be expressed in terms of weight per unit volume (mg/deciliter; mg/dL) or in the number of molecules in a volume, or molarity (**moles** or millimoles/liter; M or mM). The range of normal values sometimes varies slightly between different age groups, for males and females, and between different analytical laboratories.

The normal level of serum sodium is in the range of 136–145 mM. The normal levels of serum potassium are 3.5–5.0 mM. Note that sodium occurs at a much higher concentration than potassium. The normal concentration of total serum calcium (bound calcium plus free calcium) is in the range of 8.8–10.4 mg/dL. About 40% of the total calcium in the plasma is loosely bound to proteins; this calcium is

referred to as bound calcium. The normal range of free calcium is 4.8–5.2 mg/dL. The normal concentration of serum magnesium is in the range of 2.0–3.0 mg/dL.

The normal concentration range of chloride is 350–375 mg/dL or 98–106 mM. The normal level of phosphate, as expressed as the concentration of phosphorus, is 2.0–4.3 mg/dL. Bicarbonate is an electrolyte that is freely and spontaneously interconvertable with carbonic acid and carbon dioxide. The normal concentration of carbonic acid (H_2CO_3) is about 1.35 mM. The normal concentration of bicarbonate (HCO_3^-) is about 27 mM. The concentration of total carbon dioxide is the sum of carbonic acid and bicarbonate; this sum is normally in the range of 26–28 mM. The ratio of bicarbonate/carbonic acid is more significant than the actual concentrations of these two forms of carbon dioxide. Its normal value is 27/1.35 (equivalent to 20/1).

Abnormal results

Positively charged electrolytes

High serum sodium levels (**hypernatremia**) occur at sodium concentrations over 145 mM, with severe hypernatremia over 152 mM. Hypernatremia is usually caused by diseases that cause excessive urination. In these cases, water is lost, but sodium is still retained in the body. The symptoms include confusion and can lead to convulsions and **coma**. Low serum sodium levels (**hyponatremia**) are below 130 mM, with severe hyponatremia at or below 125 mM. Hyponatremia often occurs with severe diarrhea, with losses of both water and sodium, but with sodium loss exceeding water loss. Hyponatremia provokes clinical problems only if serum sodium falls below 125 mM, especially if this has occurred rapidly. The symptoms can be as mild as tiredness but may lead to convulsions and coma.

High serum potassium (**hyperkalemia**) occurs at potassium levels above 5.0 mM; it is considered severe over 8.0 mM. Hyperkalemia is relatively uncommon, but sometimes occurs in patients with kidney failure who take potassium supplements. Hyperkalemia can result in abnormal beating of the heart (cardiac **arrhythmias**). Low serum potassium (**hypokalemia**) occurs when serum potassium falls below 3.0 mM. It can result from low dietary potassium, as during **starvation** or in patients with anorexia nervosa; from excessive losses via the kidneys, as caused by diuretic drugs; or by diseases of the adrenal or pituitary glands. Mild hypokalemia causes muscle weakness, while severe hypokalemia can cause paralysis, the inability to breathe, and cardiac arrhythmias.

High levels of calcium ions (**hypercalcemia**) occur at free calcium ion concentrations over 5.2 mg/dL or total serum calcium above 10.4 mg/dL. Hypercalcemia usually occurs when the body dissolves bone at an abnormally fast rate, increasing both serum calcium and serum phosphate. Sudden hypercalcemia can cause vomiting and coma, while prolonged and moderate hypercalcemia results in the deposit of calcium phosphate crystals in the kidneys and eye. **Hypocalcemia** occurs when serum free calcium ions fall below 4.4 mg/dL, or when total serum calcium falls below 8.8 mg/dL. Hypocalcemia can result from **hypoparathyroidism** (low parathyroid hormone), from failure to produce 1,25-dihydroxyvitamin D, from low levels of plasma magnesium, and from phosphate poisoning (the phosphate enters the bloodstream and forms a complex with the free serum calcium). Hypocalcemia can cause depression and muscle spasms.

Hypermagnesemia occurs at serum magnesium levels over 25 mM (60 mg/dL). Hypermagnesemia is rare but can occur with the excessive consumption of magnesium salts. Hypomagnesemia occurs when serum magnesium levels fall below 0.8 mM, and can result from poor **nutrition**. Chronic alcoholism is the most common cause of hypomagnesemia, in part because of poor diet. Magnesium levels below 0.5 mM (1.2 mg/dL) cause serum calcium levels to decline. Some of the symptoms of hypomagnesemia, including twitching and convulsions, actually result from the concurrent hypocalcemia. Hypomagnesemia can also result in hypokalemia and thereby cause cardiac arrhythmias.

Negatively charged electrolytes

Serum chloride levels sometimes increase to abnormal levels as an undesirable side effect of medical treatment with sodium chloride or ammonium chloride. The toxicity of chloride results not from the chloride itself, but from the fact that the chloride occurs as the acid, hydrogen chloride (more commonly known as hydrochloric acid, or HCl). An overdose of chloride may cause the accumulation of hydrochloric acid in the bloodstream, with consequent acidosis. **Renal tubular acidosis**, one of many kidney diseases, involves the failure to release acid into the urine. The acidosis produces weakness, **headache**, **nausea**, and cardiac arrest. Low plasma chloride leads to the opposite situation: a decline in the acid content of the bloodstream. This is known as alkalization of the bloodstream, or alkalosis. Hydrochloric acid, originally from extracellular fluids, can be lost by vomiting. At its most severe, alkalosis results in **paralysis** (tetany).

Hyperphosphatemia occurs at serum phosphate levels above 5 mg/dL. It can result from the failure of the kidneys to excrete phosphate into the urine, causing phosphate to accumulate in the bloodstream. Hyperphosphatemia can also be caused by the impaired action of parathyroid hormone and by phosphate poisoning. Severe hyperphosphatemia can cause paralysis, convulsions, and cardiac arrest. These symptoms result because the phosphate, occurring in elevated levels, complexes with free serum calcium, resulting in hypocalcemia. Tests for heart function (an electrocardiogram) and parathyroid hormone levels are used in the diagnosis of hyperphosphatemia. Hypophosphatemia occurs if serum phosphorus falls to 2.0 mg/dL or lower. It often results from a shift of inorganic phosphate from the bloodstream to various organs and tissues. This shift can be caused by a rise in pH (alkalization) of the bloodstream, which can occur during hyperventilation, a reaction in various disease states. A shift in phosphate to intracellular tissues may draw calcium away from the bloodstream via the formation of insoluble calcium phosphate crystals within cells, with consequent hypocalcemia. Thus, tests for abnormalities in phosphate metabolism also involve tests for serum calcium.

Bicarbonate metabolism involves several compounds. When dietary starches, sugars, and fats are broken down for energy production, carbon dioxide is created. Much of this carbon dioxide (CO₂) spontaneously converts to carbonic acid (H₂CO₃), and some of the carbonic acid spontaneously converts to bicarbonate (HCO₃⁻) plus a hydrogen ion (H⁺). Eventually, almost every molecule of carbon dioxide produced in the body, whether in the form of carbon dioxide, carbonic acid, or bicarbonate, must convert back to carbon dioxide in order to leave via the lungs during normal breathing.

If one holds one's breath, carbon dioxide cannot escape from the lungs, but continues to be generated within the body. This results in an increase in production of carbonic acid. A portion of the carbonic acid breaks apart (dissociates), causing an increase in hydrogen ions in the plasma, with a resulting acidosis. Tests for serum bicarbonate levels are accompanied by tests for acidosis (pH test). Conversely, when one breathes too rapidly (hyperventilation), the carbon dioxide is drawn off from the bloodstream and expelled in the breath at an increased rate. This results in an increase in the rate of combination of bicarbonate with hydrogen ions, resulting in alkalosis. Acidosis and alkalosis can be produced by means other than by altering the rate of breathing. The carbonic acid and bicarbonate in the bloodstream

minimize (or buffer) any trend to acidosis or alkalosis. Tests for bicarbonate are generally accompanied by tests for blood pH and possibly tests for kidney malfunction, abnormal hormone function, or gastrointestinal disorders.

Resources

PERIODICALS

Fried, L. F., and P. M. Palevsky. "Hyponatremia and hypernatremia." *Medical Clinics of North America* 81 (1997): 585-609.

Tom Brody, PhD

Electromyography

Definition

Electromyography (EMG) is an electrical recording of muscle activity that aids in the diagnosis of neuromuscular disease.

Purpose

Muscles are stimulated by signals from nerve cells called motor neurons. This stimulation causes electrical activity in the muscle, which in turn causes contraction. This electrical activity is detected by a needle electrode inserted into the muscle and connected to a recording device. Together, the electrode and recorder are called an electromyography machine. EMG can determine whether a particular muscle is responding appropriately to stimulation, and whether a muscle remains inactive when not stimulated.

EMG is performed most often to help diagnose different diseases causing weakness. Although EMG is a test of the motor system, it may help identify abnormalities of nerves or spinal nerve roots that may be associated with **pain** or **numbness**. Other symptoms for which EMG may be useful include numbness, atrophy, stiffness, fasciculation, cramp, deformity, and spasticity. EMG results can help determine whether symptoms are due to a muscle disease or a neurological disorder, and, when combined with clinical findings, usually allow a confident diagnosis.

EMG can help diagnose many muscle and nerve disorders, including:

- muscular dystrophy
- congenital **myopathies**

- mitochondrial myopathies
- metabolic myopathies
- myotonias
- peripheral neuropathies
- radiculopathies
- nerve lesions
- amyotrophic lateral sclerosis
- polio
- spinal muscular atrophy
- Guillain-Barré syndrome
- ataxias
- myasthenias

Precautions

No special precautions are needed for this test. Patients with a history of bleeding disorder should consult with their treating physician before the test. If a muscle biopsy is planned as part of the diagnostic work-up, EMG should not be performed at the same site, as it may effect the microscopic appearance of the muscle.

Description

During an EMG test, a fine needle is inserted into the muscle to be tested. This may cause some discomfort, similar to that of an injection. Recordings are made while the muscle is at rest, and then during the contraction. The person performing the test may move the limb being tested, and direct the patient to move it with various levels of force. The needle may be repositioned in the same muscle for further recording. Other muscles may be tested as well. A typical session lasts from 30–60 minutes.

A slightly different test, the *nerve conduction velocity test*, is often performed at the same time with the same equipment. In this test, stimulating and recording electrodes are used, and small electrical shocks are applied to measure the ability of the nerve to conduct electrical signals. This test may cause mild **tingling** and discomfort similar to a mild shock from static electricity. Evoked potentials may also be performed for additional diagnostic information. Nerve conduction velocity and evoked potential testing are especially helpful when pain or sensory complaints are more prominent than weakness.

KEY TERMS

Motor neurons—Nerve cells that transmit signals from the brain or spinal cord to the muscles.

Motor unit action potentials—Spikes of electrical activity recorded during an EMG that reflect the number of motor units (motor neurons and the muscle fibers they transmit signals to) activated when the patient voluntarily contracts a muscle.

Preparation

No special preparation is needed. The doctor supervising and interpreting the test should be given information about the symptoms, medical conditions, suspected diagnosis, neuroimaging studies, and other test results.

Aftercare

Minor pain and bleeding may continue for several hours after the test. The muscle may be tender for a day or two.

Risks

There are no significant risks to this test, other than those associated with any needle insertion (pain, bleeding, bruising, or infection).

Normal results

There should be some brief EMG activity during needle insertion. This activity may be increased in diseases of the nerve and decreased in long-standing muscle disorders where muscle tissue is replaced by fibrous tissue or fat. Muscle tissue normally shows no EMG activity when at rest or when moved passively by the examiner. When the patient actively contracts the muscle, spikes (motor unit action potentials) should appear on the recording screen, reflecting the electrical activity within. As the muscle is contracted more forcefully, more groups of muscle fibers are recruited or activated, causing more EMG activity.

Abnormal results

The interpretation of EMG results is not a simple matter, requiring analysis of the onset, duration, amplitude, and other characteristics of the spike patterns.

Electrical activity at rest is abnormal; the particular pattern of firing may indicate denervation (for example, a nerve lesion, radiculopathy, or lower motor neuron degeneration), myotonia, or inflammatory myopathy.

Decreases in the amplitude and duration of spikes are associated with muscle diseases, which also show faster recruitment of other muscle fibers to compensate for weakness. Recruitment is reduced in nerve disorders.

Resources

OTHER

Falck, B., E. Stalberg, and L. Korpinen. *The Expert Electromyographer*. <<http://www.tut.fi/~korpinen/EMG.htm>>.

Richard Robinson

Electronic fetal monitoring

Definition

Electronic fetal monitoring (EFM) is a method for examining the condition of a baby in the uterus by noting any unusual changes in its heart rate. Electronic fetal monitoring is performed late in **pregnancy** or continuously during labor to ensure normal delivery of a healthy baby. EFM can be utilized either externally or internally in the womb.

Purpose

The heart rate of a fetus undergoes constant adjustment as it responds to its environment and other stimuli. The fetal monitor records an unborn baby's heart rate and graphs it on a piece of paper. Electronic fetal monitoring is usually advised for high-risk pregnancies, when the baby is in danger of distress. Specific reasons for EFM include: babies in a breech position, **premature labor**, and induced labor, among others.

When electronic fetal monitoring was originally introduced in the 1960s and 1970s, the hope was that it would help physicians diagnose fetal hypoxia, or lack of oxygen, in time to prevent damage to the baby. This lack of oxygen, also known as perinatal asphyxia or birth asphyxia, is an important cause of **stillbirth** and newborn deaths. It occurs when there are less than normal amounts of oxygen delivered to the body or

an organ and there is build-up of carbon dioxide in the body or tissue. A lack of blood flow to an organ can cause asphyxia. Perinatal asphyxia can occur a long time before birth, shortly before birth, during delivery, or after birth. If the interruption to the supply of oxygen is short, the baby may recover without any damage. If the time is longer, there may be some injury that is reversible. If the time period without oxygen is very long, there may be permanent injury to one or more organs of the body. It is important to detect any signs of asphyxia as soon as possible. One of the signs is an abnormal heart rate and rhythm in the unborn baby, which can be detected by electronic fetal monitoring.

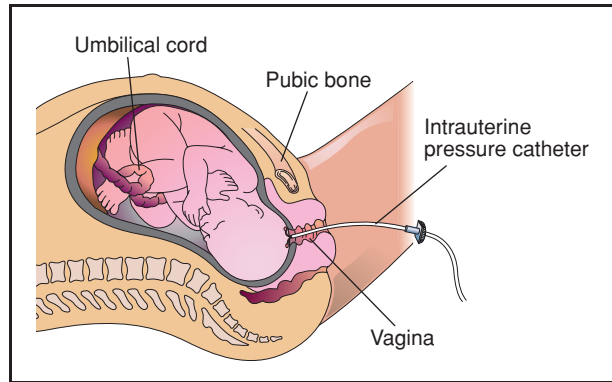
The fetal monitor is a more intricate version of the machine that a health care provider uses to listen to a baby's heartbeat. The monitor that is used during prenatal visits just picks up the sound of the baby's heart beating. The fetal monitor also keeps a continuous paper record of the heart rate. In addition, the fetal monitor can record uterine contractions on the lower part of the paper strip. This helps the doctor or midwife determine how a baby is handling the **stress** of contractions. The normal pattern is for the baby's heartbeat to drop slightly during a contraction and then go back to normal after the contraction is over. EFM looks for any changes from this normal pattern, particularly if there is a drastic drop in the baby's heart beat or if the heart rate does not recover immediately after a contraction.

Because it is an indirect test, it is not perfect. When an adult complains to a provider about not feeling well, checking the heart rate is only one of many things that the doctor will do. With an unborn baby, however, checking the heart rate is basically the only thing that a doctor or midwife can do.

Fetal monitoring can be helpful in a variety of different situations. During pregnancy, fetal monitoring can be used as a part of **antepartum testing**. If the practitioner feels that a baby may be at increased risk of problems toward the end of pregnancy, a baby can be checked every week or every other week with a non-stress test. In this test, changes in the baby's heart rate are measured along with the fetus' own movements. The heart rate of a healthy baby should go up whenever she or he moves.

Fetal monitoring is used on and off during early labor. As labor progresses, more monitoring is often needed. Usually, as the time for delivery nears, the monitor is left on continuously since the end of labor tends to be the most stressful time for the baby.

A baby who is having trouble in labor will show characteristic changes in heart rate after a contraction



Electronic fetal monitoring (EFM) is performed late in pregnancy or continuously during labor to ensure normal delivery of a healthy baby. EFM can be utilized either externally or internally in the womb. The illustration above shows the internal procedure, in which an electrode is attached directly to the baby's scalp to monitor the heart rate. Uterine contractions are recorded using an intrauterine pressure catheter which is inserted through the cervix into the uterus. (Illustration by the Electronic Illustrators Group.)

(late decelerations). If a baby is not receiving enough oxygen to withstand the stress of labor and delivery is many hours away, a cesarean section (C-section) may be necessary.

Description

Using the external fetal monitor is simple and painless. Two elastic belts are placed around the mother's abdomen. One belt holds a listening device in place while the other belt holds the contraction monitor. The nurse or midwife adjusts the belts to get the best readings from each device.

Sometimes, it is difficult to hear the baby's heart-beat with the external monitoring device. Other times, the monitor may show subtle signs of a developing problem. In either case, the doctor or midwife may recommend that the external belt be replaced with an internal monitor.

The internal monitor is an electronic wire that rests directly on the baby's head. The provider can place it on the baby's head during an internal exam. The internal monitor can only be used when the cervix is already open. This device provides a more accurate record of the baby's heart rate.

Preparation

There are no special preparations needed for fetal monitoring.

Risks

External EFM poses no direct risks to the baby. However, because of being connected to the machine, the mother cannot walk around. This inactivity may prolong labor and reduce oxygen levels in the mother's blood, both of which can be detrimental to the unborn baby. Another problem is that electronic fetal monitoring seems to be associated with an increase in caesarian deliveries. There is a concern that EFM can give false alarms of distress in the baby, and that this can lead to unneeded caesarians. With internal monitoring, there is a higher risk for infection. For these and other reasons, the United States Preventive Services Task Force states that there is some evidence that using electronic fetal monitoring on low-risk women in labor might not be indicated. Many physicians, however, continue to use EFM routinely, and believe it to be of value in both low-risk and high-risk labors.

Normal results

An unborn baby's heart rate normally ranges from 120–160 beats per minute (bpm). A baby who is receiving enough oxygen through the placenta will move around. The monitor strip will show the baby's heart rate rising briefly as he/she moves (just as an adult's heart rate rises when he/she moves).

The baby's monitor strip is considered to be reactive when the baby's heart rate rises at least 20 bpm above the baseline heart rate for at least 20 seconds. This must occur at least twice in a 20-minute period. A reactive heart rate tracing (also known as a reactive non-stress test) is considered a sign of the baby's well being.

Abnormal results

If the baby's heart rate drops very low or rises very high, this signals a serious problem. In either of these cases it is obvious that the baby is in distress and must be delivered soon. However, many babies who are having problems do not give such clear signs.

During a contraction, the flow of oxygen (from the mother) through the placenta (to the baby) is temporarily stopped. It is as if the baby has to hold its breath during each contraction. Both the placenta and the baby are designed to withstand this condition. Between contractions, the baby should be receiving more than enough oxygen to do well during the contraction.

The first sign that a baby is not getting enough oxygen between contractions is often a drop in the

KEY TERMS

Breech presentation—Fetal position in which the buttocks come first.

Cesarean section c-section—Delivery of a baby through an incision in the mother's abdomen instead of through the vagina.

Hypoxia—An oxygen deficiency.

baby's heart rate after the contraction (late deceleration). The baby's heart rate recovers to a normal level between contractions, only to drop again after the next contraction. This is also a more subtle sign of distress.

These babies will do fine if they are delivered in a short period of time. Sometimes, these signs develop long before delivery is expected. In that case, a C-section may be necessary.

Resources

PERIODICALS

Kripke, Clarissa C. "Why Are We Using Electronic Fetal Monitoring?" *American Family Physician* May 1, 1999.

Sweha, Amir, et al. "Interpretation of the Electronic Fetal Heart Rate During Labor" *American Family Physician* May 1, 1999.

Deanna M. Swartout-Corbeil, R.N.

Electrophysiology study of the heart

Definition

An electrophysiology (EP) study of the heart is a nonsurgical analysis of the electrical conduction system (normal or abnormal) of the heart. The test employs cardiac catheters and sophisticated computers to generate electrocardiogram (EKG) tracings and electrical measurements with exquisite precision from within the heart chambers.

The EP study can be performed solely for diagnostic purposes. It also is performed to pinpoint the exact location of electrical signals (cardiac mapping) in conjunction with a therapeutic procedure called **catheter ablation**.

The test is simple, not painful, and performed in a special laboratory under controlled clinical circumstances by cardiologists and nurses who subspecialize in electrophysiology.

Purpose

A cardiologist may recommend an EP study when the standard EKG, Holter monitor, event recorder, **stress test**, echocardiogram, or angiogram cannot provide enough information to evaluate an abnormal heart rhythm, called an arrhythmia.

An EP study also may be beneficial in diagnosing a suspected arrhythmia in a patient who shows symptoms of an arrhythmia but in whom it could not be detected from other tests.

The purpose and great value of an EP study is that it offers more detailed information to the doctor about the electrical activity in the heart than the aforementioned noninvasive tests because electrodes are placed directly *on* heart tissue. This allows the electrophysiologist to determine the specific location of an arrhythmia and, oftentimes, correct it during the same procedure. This corrective treatment is permanent and considered a cure, and, in many cases, the patient may not need to take heart medications.

EP studies may be helpful in assessing:

- certain tachycardias or bradycardias of unknown cause
- patients who have been resuscitated after experiencing sudden cardiac death
- various symptoms of unknown cause, such as chest pain, shortness of breath, **fatigue**, or syncope (dizziness/fainting)
- response to anti-arrhythmic therapy

Precautions

Pregnant patients should not undergo an EP study because of exposure to radiation during the study, which may be harmful to the growing baby.

Patients who have **coronary artery disease** may need to have that treated before having an EP study.

Description

The rhythmic pumping action of the heart, which is essentially a muscle, is the result of electrical impulses traveling throughout the walls of the four heart chambers. These impulses originate in the

sinoatrial (SA) node, which are specialized cells situated in the top right chamber of the heart: the right atrium. Normally, the SA node, acting like a spark plug, spontaneously generates the impulses, which travel through specific pathways throughout the atria to the atrioventricular (AV) node. The AV node is a relay station, sending the impulses to more specialized muscle fibers throughout the bottom chambers of the heart: the ventricles. If these pathways become damaged or blocked or if extra (abnormal) pathways exist, the heart's rhythm may be altered (perhaps too slow, too fast, or irregular), which can seriously affect the heart's pumping ability.

The patient is transported to the x-ray table in the EP lab and connected to various monitors. Sterile sheets are placed over him or her. A minimum of two catheters are inserted into the right femoral (thigh) vein in the groin area. Depending on the type of arrhythmia, the number of catheters used in an EP test and their route to the heart may vary. For certain tachycardias, two more catheters may be inserted in the left groin and one in the internal jugular (neck) vein or in the subclavian (below the clavicle) vein. The catheters are about 0.08 in (2 mm) in diameter, about the size of a spaghetti noodle. The catheters used in catheter ablation are slightly larger.

With the help of fluoroscopy (x rays on a television screen), all the catheters are guided to several specific locations in the heart. Typically, four to 10 electrodes are located on the end of the catheters, which have the ability to send electrical signals to stimulate the heart (called pacing) and to receive electrical signals from the heart—but not at the same time (just as a walkie-talkie cannot send and receive messages at the same time).

First, the electrodes are positioned to receive signals from inside the heart chambers. This allows the doctor to measure how fast the electrical impulses travel currently in the patient's heart. These measurements are called the patient's baseline measurements. Next, the electrodes are positioned to pace: The EP team actually tries to induce (sometimes in combination with various heart drugs) the arrhythmia that the patient has previously experienced so the team can observe it in a controlled environment, compare it to the patient's clinical or spontaneous arrhythmia, and decide how to treat it.

Once the arrhythmia is induced and the team determines it can be treated with catheter ablation, cardiac mapping is performed to locate precisely the origin and route of the abnormal pathway. When this



An electrophysiologist nurse monitors a patient's heart rhythm during an electrophysiology study for tachycardia. (Photograph by Collette Placek. Reproduced by permission.)

is accomplished, the ablating electrode catheter is positioned directly against the abnormal pathway, and high radio-frequency energy is delivered through the electrode to destroy (burn) the tissue in this area.

Preparation

The following preparations are made for an EP study:

- the patient may be advised to stop taking certain medications, especially heart drugs, that may interfere with the test results.
- blood tests usually are ordered the week before the test.
- the patient undergoes conscious **sedation** (awake but relaxed) during the test. This is accomplished quite often with the anesthetic drugs VersedR (Roche laboratories) and fentanyl.
- a local anesthetic is injected at the site of catheter insertion.

Aftercare

The patient needs to rest flat in bed for several hours after the procedure to allow healing at the catheter insertion sites.

The patient often returns home either the same day of the test or the next day. Someone should drive him or her home.

The doctor may prescribe drugs and/or insert an AFCD to treat the arrhythmia and may do a possible follow-up EP study.

Risks

The EP diagnostic study and catheter ablation are low-risk procedures. There is a small risk of bleeding and/or infection at the site of catheter insertion, but this occurs less than 1% of the time. Blood clot formation occurs only two in 1,000 instances and is minimized with blood thinner medications

KEY TERMS

Ablation—Remove or destroy, such as by burning or cutting.

Angiogram—X ray of a blood vessel after special x-ray dye has been injected into it.

Bradycardia—Slow heartbeat.

Cardiac catheter—Long, thin, flexible tube that is threaded into the heart through a blood vessel.

Cardiologist—Doctor who specializes in diagnosing and treating heart diseases.

Echocardiogram—Ultrasound image of the heart.

Electrocardiogram—Tracing of the electrical activity of the heart.

Electrode—Medium for conducting an electrical current—in this case, platinum wires.

Electrophysiology—Study of how electrical signals in the body relate to physiologic function.

Event recorder—A small machine, worn by a patient usually for several days or weeks, that is activated by the patient to record his or her EKG when a symptom is detected.

Fibrillation—Rapid, random contraction (quivering).

Holter monitor—A small machine, worn by a patient usually for 24 hours, that continuously records the patient's EKG during usual daily activity.

Stress test—Recording a patient's EKG during exercise.

Supraventricular tachycardia—A fast heart beat that originates above the ventricles.

Tachycardia—Fast heartbeat.

Vascular—Pertaining to blood vessels.

administered during the procedure. Vascular injuries causing hemorrhage or thrombophlebitis are possible but occur less than 0.7% of the time. Cardiac perforations occur only in one or two per 1,000 instances. If the right internal jugular vein is accessed, the small possibility of puncturing the lung with the catheter exists, which, at worst, could cause a collapsed lung.

Because **ventricular tachycardia** or fibrillation (lethal arrhythmias) may be induced in the patient, the EP lab personnel must be prepared to defibrillate the patient as necessary.

Normal results

The heart initiates and conducts electrical impulses normally.

Abnormal results

Confirmation of **arrhythmias**, such as:

- supraventricular tachycardias
- ventricular arrhythmias
- accessory (extra) pathways
- bradycardias

Resources

ORGANIZATIONS

Cardiac Arrhythmia Research and Education Foundation (C.A.R.E.). 2082 Michelson Dr. #301 Irvine, CA 92612 1-800-404-9500. < <http://www.longqt.com> > .

Medtronics Manufacturer of Therapeutic Devices. 710 Medtronic Parkway NE, Minneapolis, MN 55432-5604. (800) 328-2518. < <http://www.medtronic.com> > .

Midwest Heart Specialists. Physician Office Building, 3825 Highland Ave., Tower 2, Ste. 400, Downers Grove, Illinois 60515. (630) 719-4799. < <http://www.midwestheart.com> > .

United States Catheter Instruments (USCI). 129 Concord Road Billerica, MA 01821. (800) 826-2273.

Collette L. Placek

Electroshock therapy see **Electroconvulsive therapy**

Elephantiasis

Definition

The word elephantiasis is a vivid and accurate term for the syndrome it describes: the gross (visible) enlargement of the arms, legs, or genitals to elephantoid size.

Description

True elephantiasis is the result of a parasitic infection caused by three specific kinds of round worms. The long, threadlike worms block the body's lymphatic system—a network of channels, lymph nodes, and organs that helps maintain proper fluid levels in the body by draining lymph from tissues into the bloodstream. This blockage causes fluids to collect in the tissues, which can lead to great swelling, called

“lymphedema.” Limbs can swell so enormously that they resemble an elephant’s foreleg in size, texture, and color. This is the severely disfiguring and disabling condition of elephantiasis.

There are a few different causes of elephantiasis, but the agents responsible for most of the elephantiasis in the world are filarial worms: white, slender round worms found in most tropical and subtropical places. They are transmitted by particular kinds (species) of mosquitoes, that is, bloodsucking insects. Infection with these worms is called “lymphatic filariasis” and over a long period of time can cause elephantiasis.

Lymphatic **filariasis** is a disease of underdeveloped regions found in South America, Central Africa, Asia, the Pacific Islands, and the Caribbean. It is a disease that has been present for centuries, as ancient Persian and Indian writings clearly described elephant-like swellings of the arms, legs, and genitals. It is estimated that 120 million people in the world have lymphatic filariasis. The disease appears to be spreading, in spite of decades of research in this area.

Other terms for elephantiasis are Barbados leg, elephant leg, morbus herculeus, mal de Cayenne, and myelolymphangioma.

Other situations that can lead to elephantiasis are:

- a protozoan disease called **leishmaniasis**
- a repeated streptococcal infection
- the surgical removal of lymph nodes (usually to prevent the spread of **cancer**)
- a hereditary birth defect

Causes and symptoms

Three kinds of round worms cause elephantiasis filariasis: *Wuchereria bancrofti*, *Brugia malayi*, and *Brugia timori*. Of these three, *W. bancrofti* makes up about 90% of the cases. Man is the only known host of *W. bancrofti*.

Culex, *Aedes*, and *Anopheles* mosquitoes are the carriers of *W. bancrofti*. *Anopheles* and *Mansonia* mosquitoes are the carriers of *B. malayi*. In addition, *Anopheles* mosquitoes are the carriers of *B. timori*.

Infected female mosquitoes take a blood meal from a human, and in doing so, introduce larval forms of the particular parasite they carry to the person. These larvae migrate toward a lymphatic channel, then travel to various places within the lymphatic system, usually positioning themselves in or near

lymph nodes throughout the body. During this time, they mature into more developed larvae and eventually into adult worms. Depending upon the species of round worm, this development can take a few months or more than a year. The adult worms grow to about 1 in (2.5 cm) to 4 in (10 cm) long.

The adult worms can live from about three to eight years. Some have been known to live to 20 years, and in one case 40 years. The adult worms begin reproducing numerous live embryos, called microfilariae. The microfilariae travel to the bloodstream, where they can be ingested by a mosquito when it takes a blood meal from the infected person. If they are not ingested by a mosquito, the microfilariae die within about 12 months. If they are ingested by a mosquito, they continue to mature. They are totally dependent on their specific species of mosquito to develop further. The cycle continues when the mosquito takes another blood meal.

Most of the symptoms an infected person experiences are due to the blockage of the lymphatic system by the adult worms and due to the substances (excretions and secretions) produced by the worms.

The body’s allergic reactions may include repeated episodes of **fever**, shaking chills, sweating, headaches, **vomiting**, and **pain**. Enlarged lymph nodes, swelling of the affected area, skin ulcers, bone and joint pain, tiredness, and red streaks along the arm or leg also may occur. Abscesses can form in lymph nodes or in the lymphatic vessels. They may appear at the surface of the skin as well.

Long-term infection with lymphatic filariasis can lead to **lymphedema**, hydrocele (a buildup of fluid in any saclike cavity or duct) in the scrotum, and elephantiasis of the legs, scrotum, arms, penis, breasts, and vulvae. The most common site of elephantiasis is the leg. It typically begins in the ankle and progresses to the foot and leg. At first the swollen leg may feel soft to the touch but eventually becomes hard and thick. The skin may appear darkened or warty and may even crack, allowing bacteria to infect the leg and complicate the disease. The microfilariae usually don’t cause injury. In some instances, they cause “eosinophilia,” an increased number of eosinophils (a type of white blood cells) in the blood.

This disease is more intense in people who never have been exposed to lymphatic filariasis than it is in the native people of tropical areas where the disease occurs. This is because many of the native people often are immunologically tolerant.

Diagnosis

The only sure way to diagnose lymphatic filariasis is by detecting the parasite itself, either the adult worms or the microfilariae.

Microscopic examination of the person's blood may reveal microfilariae. But many times, people who have been infected for a long time do not have microfilariae in their bloodstream. The absence of them, therefore, does not mean necessarily that the person is not infected. In these cases, examining the urine or hydrocele fluid or performing other clinical tests is necessary.

Collecting blood from the individual for microscopic examination should be done during the night when the microfilariae are more numerous in the bloodstream. (Interestingly, this is when mosquitoes bite most frequently.) During the day microfilariae migrate to deeper blood vessels in the body, especially in the lung. If it is decided to perform the blood test during the day, the infected individual may be given a "provocative" dose of medication to provoke the microfilariae to enter the bloodstream. Blood then can be collected an hour later for examination.

Detecting the adult worms can be difficult because they are deep within the lymphatic system and difficult to get to. Biopsies usually are not performed because they usually don't reveal much information.

Treatment

The drug of choice in treating lymphatic filariasis is diethylcarbamazine (DEC). The trade name in the United States is Hetrazan.

The treatment schedule is typically 2 mg/kg per day, three times a day, for three weeks. The drug is taken in tablet form.

DEC kills the microfilariae quickly and injures or kills the adult worms slowly, if at all. If all the adult worms are not killed, remaining paired males and females may continue to produce more larvae. Therefore, several courses of DEC treatment over a long time period may be necessary to rid the individual of the parasites.

DEC has been shown to reduce the size of enlarged lymph nodes and, when taken long-term, to reduce elephantiasis. In India, DEC has been given in the form of a medicated salt, which helps prevent spread of the disease.

The side effects of DEC almost all are due to the body's natural allergic reactions to the dying parasites rather than to the DEC itself. For this reason, DEC must be given carefully to reduce the danger to the



Man suffering from elephantiasis. (Photograph by C. James Webb, Phototake NYC. Reproduced by permission.)

individual. Side effects may include fever, chills, **head-ache**, **dizziness**, nausea and vomiting, **itching**, and joint pain. These side effects usually occur within the first few days of treatment. These side effects usually subside as the individual continues taking the drug.

There is an alternate treatment plan for the use of DEC. This plan is designed to kill the parasites slowly (to reduce allergic reactions to the dead microfilariae and dying adult worms within the body). Lower doses of DEC are taken for the first few days, followed by the higher dose of 2 mg/kg per day for the remaining three weeks. In addition, steroids may be prescribed to prevent the individual's body from reacting severely to the dead worms.

Another drug used is Ivermectin. Early research studies of Ivermectin show that it is excellent in killing microfilariae, but the effects of this drug on the adult worms are still being investigated. It is probable that patients will need to continue using DEC to kill the

KEY TERMS

Antigen—Any substance (usually a protein) that causes an immune response by the body to produce antibodies.

Filarial—Threadlike. The word “filament” is formed from the same root word.

Host—A person or animal in which a parasite lives, is nourished, grows, and reproduces.

Lymph—A watery substance that collects in the tissues and organs of the body and eventually drains into the bloodstream.

Lymphatic system—A network composed of vessels, lymph nodes, the tonsils, the thymus gland, and the spleen. It is responsible for transporting fluid and nutrients to the bloodstream and for maturing certain blood cells that are part of the body's immune system.

Lymphedema—The unnatural accumulation of lymph in the tissues of the body, which results in swelling in that area.

Protozoa—(Plural form of protozoan) Single-celled organisms (not bacteria) of which about 30 kinds cause disease in humans.

Streptococcal—Pertaining to any of the *Streptococcus* bacteria. These organisms can cause pneumonia, skin infections, and many other diseases.

adult worms. Mild side effects of Ivermectin include headache, fever, and myalgia.

Other means of managing lymphatic filariasis are pressure bandages to wrap the swollen limb and elastic stockings to help reduce the pressure. Exercising and elevating a bandaged limb also can help reduce its size.

Surgery can be performed to reduce elephantiasis by removing excess fatty and fibrous tissue, draining the swelled area, and removing the dead worms.

Prognosis

With DEC treatment, the prognosis is good for early and mild cases of lymphatic filariasis. The prognosis is poor, however, for heavy parasitic infestations.

Prevention

The two main ways to control this disease are to take DEC preventively, which has shown to be

effective, and to reduce the number of carrier insects in a particular area.

Avoiding mosquito bites with insecticides and insect repellents is helpful, as is wearing protective clothing and using bed netting.

Much effort has been made in cleaning the breeding sites (stagnant water) of mosquitoes near people's homes in areas where filariasis is found.

Before visiting countries where lymphatic filariasis is found, it would be wise to consult a travel physician to learn about current preventative measures.

Resources

ORGANIZATIONS

National Lymphedema Network. 2211 Post St., Suite 404, San Francisco, CA 94115. (800) 541-3259. < <http://www.hooked.net> > .

National Organization for Rare Disorders. PO Box 8923, New Fairfield, CT 06812-8923. (800) 999-6673. < <http://www.rarediseases.org> > .

Collette L. Placek

ELISA (Enzyme-linked Immunosorbant see **AIDS tests**)

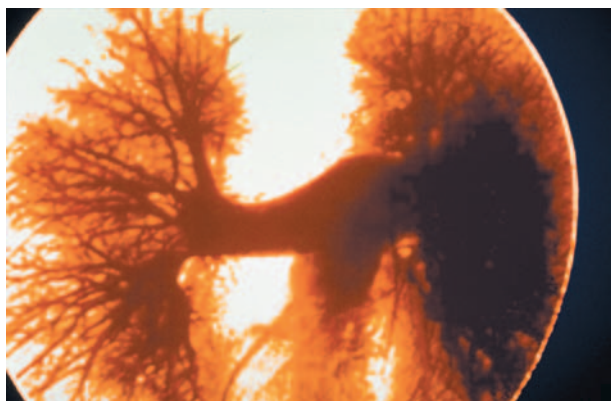
Embolism

Definition

An embolism is an obstruction in a blood vessel due to a blood clot or other foreign matter that gets stuck while traveling through the bloodstream. The plural of embolism is emboli.

Description

Emboli have moved from the place where they were formed through the bloodstream to another part of the body, where they obstruct an artery and block the flow of blood. The emboli are usually formed from **blood clots** but are occasionally comprised of air, fat, or tumor tissue. Embolic events can be multiple and small, or single and massive. They can be life-threatening and require immediate emergency medical care. There are three general categories of emboli: arterial, gas, and pulmonary. Pulmonary emboli are the most common.



A close up view of a pulmonary embolism. (*Custom Medical Stock Photo. Reproduced by permission.*)

Arterial embolism

In arterial emboli, blood flow is blocked at the junction of major arteries, most often at the groin, knee, or thigh. Arterial emboli are generally a complication of heart disease. An **arterial embolism** in the brain (cerebral embolism) causes **stroke**, which can be fatal. An estimated 5–14% of all strokes are caused by cerebral emboli. Arterial emboli to the extremities can lead to tissue death and **amputation** of the affected limb if not treated effectively within hours. Intestines and kidneys can also suffer damage from emboli.

Gas embolism

Gas emboli result from the compression of respiratory gases into the blood and other tissues due to rapid changes in environmental pressure, for example, while flying or scuba diving. As external pressure decreases, gases (like nitrogen) that are dissolved in the blood and other tissues become small bubbles that can block blood flow and cause organ damage.

Pulmonary embolism

In a **pulmonary embolism**, a common illness, blood flow is blocked at a pulmonary artery. When emboli block the main pulmonary artery, and in cases where there are no initial symptoms, a pulmonary embolism can quickly become fatal. According to the American Heart Association, an estimated 600,000 Americans develop pulmonary emboli annually and 60,000 die from it.

A pulmonary embolism is difficult to diagnose. Less than 10% of patients who die from a pulmonary embolism were diagnosed with the condition. More than 90% of cases of pulmonary emboli are

complications of **deep vein thrombosis**, blood clots in the deep vein of the leg or pelvis.

Causes and symptoms

Arterial emboli are usually a complication of heart disease where blood clots form in the heart's chambers. Gas emboli are caused by rapid changes in environmental pressure that could happen when flying or scuba diving. A pulmonary embolism is caused by blood clots that travel through the blood stream to the lungs and block a pulmonary artery. More than 90% of the cases of pulmonary embolism are a complication of deep vein thrombosis, which typically occurs in patients who have had **orthopedic surgery** and patients with **cancer** or other chronic illnesses like congestive **heart failure**.

Risk factors for arterial and pulmonary emboli include: prolonged bed rest, surgery, **childbirth**, **heart attack**, stroke, congestive heart failure, cancer, **obesity**, a broken hip or leg, **oral contraceptives**, sickle cell anemia, chest trauma, certain congenital heart defects, and old age. Risk factors for gas emboli include: scuba diving, amateur plane flight, **exercise**, injury, obesity, **dehydration**, excessive alcohol, colds, and medications such as **narcotics** and **antihistamines**.

Symptoms of an arterial embolism include:

- severe **pain** in the area of the embolism
- pale, bluish cool skin
- numbness
- tingling
- muscular weakness or **paralysis**

Common symptoms of a pulmonary embolism include:

- labored breathing, sometimes accompanied by chest pain
- a rapid pulse
- a **cough** that may produce sputum
- a low-grade **fever**
- fluid build-up in the lungs

Less common symptoms include:

- coughing up blood
- pain caused by movement or breathing
- leg swelling
- bluish skin
- fainting
- swollen neck veins

Diagnosis

An embolism can be diagnosed through the patient's history, a physical exam, and diagnostic tests. The use of various tests may change, as physicians and clinical guidelines evaluate the most effective test in terms of accuracy and cost. For arterial emboli, cardiac ultrasound and/or arteriography are ordered. For a pulmonary embolism, a **chest x ray**, lung scan, pulmonary **angiography**, **electrocardiography**, arterial blood gas measurements, and **venography** or venous ultrasound could be ordered.

Diagnosing an arterial embolism

Ultrasound uses sound waves to create an image of the heart, organs, or arteries. The technologist applies gel to a hand-held transducer, then presses it against the patient's body. The sound waves are converted into an image that can be displayed on a monitor. Performed in an outpatient diagnostic laboratory, the test takes 30–60 minutes.

An arteriogram is an x ray in which a contrast medium is injected to make the arteries visible. It can be performed in a radiology unit, outpatient clinic, or diagnostic center of a hospital.

Diagnosing a pulmonary embolism

A chest x ray can show fluid build-up and detect other respiratory diseases. The perfusion lung scan shows poor flow of blood in areas beyond blocked arteries. The patient inhales a small amount of radiopharmaceutical and pictures of airflow into the lungs are taken with a gamma camera. Then a different radiopharmaceutical is injected into an arm vein and lung blood flow is scanned. A normal result essentially rules out a pulmonary embolism. A lung scan can be performed in a hospital or an outpatient facility and takes about 45 minutes.

Pulmonary angiography is one of the most reliable tests for diagnosing a pulmonary embolism. Pulmonary angiography is a radiographic test that involves injection of a radio contrast agent to show the pulmonary arteries. A cinematic camera records the blood flow through the patient, who lies on a table. Pulmonary angiography is usually performed in a hospital's radiology department and takes 30–60 minutes.

An electrocardiograph shows the heart's electrical activity and helps distinguish a pulmonary embolism from a heart attack. Electrodes covered with conducting jelly are placed on the patient's chest, arms, and

legs. Impulses of the heart's activity are traced on paper. The test takes about 10 minutes.

Arterial blood gas measurements are sometimes helpful but, alone, they are not diagnostic for pulmonary embolism. Blood is taken from an artery instead of a vein, usually in the wrist.

Venography is used to look for the most likely source of a pulmonary embolism, deep vein thrombosis. It is very accurate, but it is not used often, because it is painful, expensive, exposes the patient to a fairly high dose of radiation, and can cause complications. Venography identifies the location, extent, and degree of attachment of the blood clots and enables the condition of the deep leg veins to be assessed. A contrast solution is injected into a foot vein through a catheter. The physician observes the movement of the solution through the vein with a fluoroscope while a series of x rays are taken. Venography takes between 30–45 minutes and can be done in a physician's office, a laboratory, or a hospital. Radionuclide venography, in which a radioactive isotope is injected, is occasionally used, especially if a patient has had reactions to contrast solutions. Venous ultrasound is the preferred evaluation of leg veins.

As noninvasive methods such as high-speed computed tomography (CT) scanning improve, they may be used to diagnose emboli. For instance, spiral (also called helical) CT scans may be the preferred tool for diagnosing pulmonary embolism in pregnant women.

Treatment

Patients with emboli require immediate hospitalization. They are generally treated with clot-dissolving and/or clot-preventing drugs. **Thrombolytic therapy** to dissolve blood clots is the definitive treatment for a severe pulmonary embolism. Streptokinase, urokinase, and recombinant tissue plasminogen activator (TPA) are used. Heparin has been the anticoagulant drug of choice for preventing formation of blood clots. A new drug has been approved for treatment of acute pulmonary emboli. Called fondaparinux (Arixtra), it usually is administered with Warfarin, an oral anticoagulant. Warfarin is sometimes used with other drugs to treat acute embolism events and is usually continued after the hospitalization to help prevent future emboli. Arixtra also has been used on an ongoing basis to prevent pulmonary emboli.

In the case of an arterial embolism, the affected limb is placed in a dependent position and kept warm. Embolectomy is the treatment of choice in the majority of early cases of arterial emboli in the extremities.

In this procedure, a balloon-tipped catheter is inserted into the artery to remove thromboembolic matter.

With a pulmonary embolism, **oxygen therapy** is often used to maintain normal oxygen concentrations. For people who can't take anticoagulants and in some other cases, surgery may be needed to insert a device that filters blood returning to the heart and lungs.

Prognosis

Of patients hospitalized with an arterial embolism, 25–30% die, and 5–25% require amputation of a limb. About 10% of patients with a pulmonary embolism die suddenly within the first hour of onset of the condition. The outcome for all other patients is generally good; only 3% of patients die who are properly diagnosed early and treated. In cases of an undiagnosed pulmonary embolism, about 30% of patients die.

Prevention

Embolism can be prevented in high risk patients through antithrombotic drugs such as heparin, venous interruption, gradient elastic stockings, and intermittent pneumatic compression of the legs. The combination of graduated compression stockings and low-dose heparin is significantly more effective than low-dose heparin alone.

Gradient elastic stockings, also called anti-embolism stockings, decrease the risk of blood clots by compressing superficial leg veins and forcing blood into the deep veins. They can be knee-, thigh-, or waist-length. Many physicians order the use of stockings before surgery and until there is no longer an elevated risk of developing blood clots. The risk of deep vein thrombosis after surgery is reduced 50% with the use of these stockings. The American Heart Association recommends that the use of graduated compression stockings be considered for all high-risk surgical patients.

Intermittent pneumatic compression involves wrapping knee- or thigh-high cuffs around the legs to prevent blood clots. The cuffs are connected to a pump that inflates and deflates, mimicking the heart's normal pumping action and reducing the pooling of blood. Intermittent pneumatic compression can be used during surgery and recovery and continues until there is no longer an elevated risk of developing blood clots. The American Heart Association recommends the use of intermittent pneumatic compression for patients who cannot take anticoagulants, for example, spinal cord and brain trauma patients.

KEY TERMS

Anticoagulants—Drugs that suppress, delay, or prevent blood clots. Anticoagulants are used to treat embolisms.

Artery—A blood vessel that carries blood from the heart to other body tissues. Embolisms obstruct arteries.

Deep vein thrombosis—A blood clot in the calf's deep vein. This frequently leads to pulmonary embolism if untreated.

Emboli—Clots or other substances that travel through the blood stream and get stuck in an artery, blocking circulation.

Thrombolytics—Drugs that dissolve blood clots. Thrombolytics are used to treat embolisms.

Resources

PERIODICALS

Doyle, Nora M., et al. "Diagnosis of Pulmonary Embolism: A Cost-effective Analysis." *American Journal of Obstetrics and Gynecology* September 2004: 1019–1024.

Truelove, Christiane. "First for Pulmonary Embolism." *Med Ad News* August 2004: 82.

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

Lori De Milto
Teresa G. Odle

Emergency contraception

Definition

Emergency **contraception** or emergency birth control uses either emergency contraceptive pills (ECPs) or a Copper-T intrauterine device (**IUD**) to help prevent **pregnancy** following unprotected vaginal intercourse.

Purpose

Emergency contraception may be used to prevent pregnancy after vaginal intercourse when:

- A birth control method was not used. Young people, in particular, may not be prepared for their first experience of sexual intercourse.
- A **condom** broke or slipped and ejaculation occurred within the woman's vagina.
- The male failed to withdraw from the vagina before ejaculation.
- A woman failed to take her birth control pills.
- A **diaphragm**, cap, or shield slipped out of place, followed by ejaculation within the vagina.
- A woman's "safe days" were miscalculated.
- A woman was raped or otherwise forced to have unprotected intercourse.

Women who missed taking their **oral contraceptives** may consider emergency contraception if:

- A new packet of pills was started at least two days late.
- Two to four of the first seven active (hormone-containing) pills (days 1–7) were missed.
- Five or more active pills were missed consecutively.

On average eight out of every 100 fertile women will become pregnant after having one episode of unprotected vaginal intercourse during the second or third week of their menstrual cycle. Following treatment with combined ECPs, only two of those 100 women will become pregnant—a 75% reduction. Following treatment with progestin-only ECPs, only one woman out of the 100 will become pregnant—an 89% reduction. Following emergency insertion of an IUD there is a 99.9% reduction in the risk of pregnancy.

Precautions

Emergency contraception does not work after the onset of pregnancy; nor should it be used as a regular method of birth control. ECPs do not prevent pregnancy from intercourse that occurs following the treatment; another birth control method must be used to prevent pregnancy. Although ECPs will not affect an existing pregnancy and will not harm the fetus, emergency contraception should not be used if a woman is already pregnant.

Frequent use of ECPs can result in irregular or unpredictable menstrual periods. Additional doses of ECPs usually do not reduce the risk of pregnancy and they increase the risk of side effects including **nausea and vomiting**.

Almost all women can use emergency contraception safely, even those who cannot use oral contraceptives as a regular method of birth control because of heart disease, **blood clots**, **stroke**, or other cardiovascular problems. The anti-convulsive medication Dilantin may reduce the effectiveness of ECPs. Some physicians recommend doubling the first of the two ECP doses if taken with Dilantin.

Progestin-only ECPs (POPs) are not recommended for women who:

- may be pregnant already
- have a hypersensitivity to any component of the medication
- have abnormal, undiagnosed genital bleeding.

Copper-T IUDs should not be used for emergency contraception if a woman:

- is pregnant
- has a history of **pelvic inflammatory disease** (PID) that has impaired her fertility
- has one of numerous other conditions affecting her reproductive system
- has—or is currently at risk for contracting—a sexually transmitted disease (STD) such as HIV/AIDS, chlamydia, or **gonorrhea**, since IUD insertion can introduce infectious agents into the sterile uterine cavity.

Those at risk for contracting an STD include women who:

- have been raped
- have had unprotected sex with a new partner
- are in a non-monogamous relationship
- use intravenous drugs
- have partners who use intravenous drugs

Description

Although emergency contraception—sometimes called post-coital or morning-after contraception—has been available for over a quarter of a century, almost one-half of the 6.3 million pregnancies in the United States each year are unintended. Among teen pregnancies 80% are unintentional. About one-half of unintended pregnancies are caused by contraceptive failure, either a failure of the method or a mistake by the user. The remainder of unintended pregnancies occurs because birth control was not employed. Emergency contraception could help prevent some of the 1.4 million abortions that take place in the United States every year.

Emergency contraception prevents pregnancy by one of the following methods:

- delaying or inhibiting ovulation—the release of eggs from the ovary
- altering the transport of the sperm or egg, thereby preventing fertilization of the egg by a sperm
- altering the endometrium or uterine lining, thereby preventing implantation—the attachment of the fertilized egg to the wall of the uterus

The mechanism by which ECPs prevent pregnancy depends on the stage of the woman's menstrual cycle. In most cases ECPs delay or inhibit ovulation and have no effect on implantation. IUDs used as emergency contraception appear to interfere with implantation of the fertilized egg; although they also may prevent fertilization, as they are thought to do when they are used as a regular method of birth control.

Emergency contraceptive pills (ECPs)

ECPs contain synthetic hormones that mimic the hormones produced by a woman's body. Many common brands of birth control pills can be used for emergency contraception even though they are not labeled for that use. Any of the first 21 pills in a regular 28-pill package of oral contraceptives can be used for emergency contraception. The last seven pills in 28-pill packs do not contain hormones. The number of pills that constitute an emergency contraceptive dose depends on the brand of pill. The same brand should be used for both doses of ECPs. Many ECPs are available outside of the United States, where they are packaged, labeled, and sold for emergency contraceptive purposes.

COMBINED ECPS. Combined ECPs available in the United States contain 100 micrograms of the synthetic estrogen, ethinyl estradiol, and 0.5–0.6 mg of the synthetic progestin levonorgestrel per dose. Combined ECPs are taken according to the Yuzpe Regimen, named after A. Albert Yuzpe, the Canadian researcher who first demonstrated their safety and effectiveness in 1974. With the Yuzpe Regimen, the first dose of combined ECPs is taken as soon as possible after unprotected intercourse and the second dose is taken 12 hours later. However the timing of the second dose can vary by a few hours without diminishing its effectiveness. The Preven Emergency Contraceptive Kit—the first product to be specifically labeled and marketed for emergency contraception—is no longer available.

Combined ECPs available in the United States include:

- Alesse, manufactured by Wyeth-Ayerst; five pink pills per dose
- Aviane, manufactured by Duramed; five orange pills per dose
- Cryelle, manufactured by Barr; four white pills per dose
- Enpresse from Barr; four orange pills per dose
- Lessina from Barr; five pink pills per dose
- Levlen from Berlex; four light orange pills per dose
- Levlite from Berlex; five pink pills per dose
- Levora from Watson; four white pills per dose
- Lo/Ovral from Wyeth-Ayerst; four white pills per dose
- Low-Ogestrel from Watson; four white pills per dose
- Luteru from Watson; five white pills per dose
- Nordette from Wyeth-Ayerst; four light orange pills per dose
- Ogestrel from Watson; two white pills per dose
- Ovral from Wyeth-Ayerst; two white pills per dose
- Portia from Barr; four pink pills per dose
- Seasonale from Barr; four pink pills per dose
- Tri-Levlen from Berlex; four yellow pills per dose
- Triphasil from Wyeth-Ayerst; four yellow pills per dose
- Trivora from Watson; four pink pills per dose

PROGESTIN-ONLY ECPS. Progestin-only ECPs (POPs) are prescribed frequently, particularly for women who cannot take estrogen or who are breastfeeding. POPs contain 0.75 mg of levonorgestrel per dose. They are equally effective regardless of whether the two doses are taken simultaneously or 12–24 hours apart. POPs are most effective if taken within 72 hours of unprotected intercourse; however they reduce the risk of pregnancy if taken within 120 hours.

Progestin-only pills include:

- Plan B from Barr is the only drug available in the United States that is specifically designed and designated as an ECP—one white pill per dose.
- Ovrette from Wyeth-Ayerst requires swallowing 20 yellow pills for each dose.

The Copper-T IUD

The Copper-T 380A IUD (ParaGard) is a T-shaped device that provides emergency contraception if inserted into the uterus by a healthcare provider within seven days after unprotected intercourse. It can be removed by the healthcare provider after the woman's next menstrual period begins or it can remain

in place for up to 10–12 years as an effective method of birth control.

Availability

In most of the United States, emergency contraception requires a special prescription or a prescription for a monthly supply of an appropriate oral contraceptive. Most physicians do not routinely discuss the use of emergency contraception with their patients and some pharmacies refuse to carry ECPs.

Emergency contraception is available from:

- public and college health clinics
- women's health centers
- Planned Parenthood clinics
- private doctors
- hospital emergency rooms, except those affiliated with a religion that opposes the use of birth control
- pharmacists directly, in a small number of states.

Some healthcare providers may prescribe ECPs over the telephone. **Sexual assault** victims may be offered ECPs in the hospital emergency room.

In many countries ECPs are available without a prescription. However in the United States emergency contraception remains controversial. In September of 2004, the U. S. Department of Justice released guidelines for the treatment of sexual assault victims without mentioning the option of emergency contraception. As of early 2005, the U.S. Food and Drug Administration (FDA) had delayed approval of over-the-counter (OTC) status for Plan B. However many professional healthcare organizations and advocacy groups for women's reproductive rights were working to make ECPs available without a prescription in the United States.

Costs

The cost of emergency contraception varies greatly according to region and location and any additional required services. Family-planning clinics and public healthcare centers may provide lower-cost emergency contraception or charge according to an income-based sliding scale.

As of 2005, estimated costs for emergency contraception were:

- \$8–\$35 for Plan B
- \$20–\$50 for combined ECPs
- \$50–\$70 for other progestin-only ECPs

- \$35–\$150 for a visit to a healthcare provider
- \$10–\$20 for a pregnancy test
- about \$400 for an exam, IUD, and insertion; however the IUD can remain in place for up to 12 years.

Preparation

For emergency contraception to be effective, it must be used as soon as possible following unprotected intercourse. Some healthcare providers and women's health centers prescribe or supply packets of ECPs—called EC-to-Go—so that they are available immediately if required. Supplies of ECPs are particularly important for women who are at high risk for having unprotected intercourse. EC-to-Go also avoids the cost of an extra visit to a healthcare provider.

Studies have found that neither the use of ECPs, nor having a supply of ECPs on hand, reduce the likelihood that women, including teenagers, will use conventional contraceptive methods. In fact it has been shown that the use of ECPs often increases the likelihood that a regular birth control method will be employed.

If an office visit is required, a healthcare provider may take a medical history, perform a pregnancy test on a urine sample, and—provided that pregnancy has not occurred—discuss the appropriate type of emergency contraception.

Aftercare

For about 10–15% of women who take ECPs, the timing, duration, and/or amount of bleeding for their next menstrual period may be different than usual. About 50% of women have their first post-ECP menstrual period one to three days earlier or later than expected. Most often it is earlier than expected. Bleeding may be normal or heavier, lighter, or more spotty than usual.

Following IUD insertion, a woman may need to be escorted or driven home and she may require rest.

Risks

Emergency contraception is considered to be both safe and effective for teenagers as well as adult women. However emergency contraception may not prevent an ectopic pregnancy—a pregnancy outside of the uterus, in the fallopian tubes or abdomen. Ectopic pregnancies are medical emergencies and can be fatal.

Side effects of ECPs

About 50% of women feel sick to their stomachs for approximately 24 hours after taking combined ECPs. **Nausea** occurs in 30–50% of women and 15–25% of women experience **vomiting**. Only 23% of women who take progestin-only ECPs experience nausea and only 6% have vomiting.

If vomiting occurs within one hour of taking ECPs, the dose may have to be repeated. OTC medications such as Dramamine II, Bonine, or their generic equivalents, taken one hour before the ECPs, reduce the risk of nausea and vomiting, although they may cause drowsiness. Two 25-mg tablets of Meclizine, taken one hour before the ECPs, reduce the risk of nausea by 27% and the risk of vomiting by 64%; however there is about a 30% risk of drowsiness. If vomiting occurs after the first dose of an ECP, anti-nausea medication should be taken one hour before the second dose. The second dose also may be taken as a vaginal suppository by placing the pills as far as possible into the vagina for absorption through the vaginal tissue.

Other side effects of ECPs can include:

- breast tenderness
- abdominal pain
- irregular bleeding
- dizziness
- headaches
- fatigue.

Side effects usually last only one to two days and are far less frequent with progestin-only ECPs as compared with combined ECPs.

Side effects of IUD insertion

Side effects of IUD insertion may include:

- abdominal discomfort
- vaginal bleeding or spotting
- infection.

However the risk of pelvic infection is very small among women who are not at risk for STDs.

Other possible side effects of IUD insertion include:

- heavy menstrual flow
- cramping
- infertility
- uterine puncture.

KEY TERMS

Emergency contraceptive pills; ECPs—Medication containing synthetic hormones for preventing pregnancy after unprotected vaginal intercourse.

Endometrium—The lining of the uterus.

Ethinyl estradiol—A semi-synthetic derivative of estradiol—an estrogen or female sex hormone—used in birth control pills and combined ECPs.

Implantation—The embedding of a fertilized egg in the inner wall of the uterus.

Intrauterine device; IUD—A device inserted into the uterus to prevent pregnancy.

Levonorgestel—A synthetic progestin used in ECPs.

Ovulation—The discharge of an ovum (egg) from the mature follicle of the ovary.

Progestin—A synthetic or natural drug that acts on the uterine lining.

Yuzpe Regimen—A two-dose treatment with combined ECPs to prevent pregnancy after unprotected intercourse; the first dose is taken as soon as possible and the second dose is taken 12 hours after the first.

Normal results

The effectiveness of emergency contraception depends both on the stage of the woman's menstrual cycle and on how soon the emergency contraception is used following unprotected vaginal intercourse. The closer a woman is to ovulation—her fertile period during which eggs are released from the ovary—the less effective emergency contraception will be.

ECPs are less effective than the most popular birth control methods:

- If taken within 72 hours of unprotected intercourse, combined ECPs are about 75% effective for preventing pregnancy.
- Progestin-only ECPs are 95% effective if taken within 24 hours of unprotected intercourse and about 89% effective if taken within 72 hours.
- A Copper-T IUD is 99.9% effective if inserted within seven days of unprotected intercourse.

If a normal menstrual period does not begin within three weeks after taking ECPs, or if signs of pregnancy develop, a healthcare provider should be consulted immediately.

Signs of pregnancy include:

- a missed menstrual period
- nausea
- unexplained fatigue
- enlarged or sore breasts
- headaches
- frequent urination

The majority of women express satisfaction with emergency contraception. One study of 235 women who had used ECPs found that 91% were satisfied with the method and 97% would recommend it to others.

Resources

BOOKS

Emergency Contraception: A Medical Dictionary, Bibliography, and Annotated Research Guide to Internet References. Icon Health Publications, 2004.

The Essential Guide for Emergency Contraception. American Healthcare Consultants, 2004.

PERIODICALS

Brody, Jane E. "The Politics of Emergency Contraception." *New York Times* August 24, 2004: F.7.

Cantor, Julie, and Ken Baum. "The Limits of Conscientious Objection—May Pharmacists Refuse to Fill Prescriptions for Emergency Contraception?" *New England Journal of Medicine* 351, no. 19 (November 4, 2004): 2008–12.

Raine, Tina R., et al. "Direct Access to Emergency Contraception Through Pharmacies and Effect on Unintended Pregnancy and STIs." *Journal of the American Medical Association* 293, no. 1 (January 5, 2005): 54–62.

Weismiller, David G. "Emergency Contraception." *American Family Physician* 70, no. 4 (August 15, 2004): 707–14.

ORGANIZATIONS

Association of Reproductive Health Professionals. 2401 Pennsylvania Avenue NW, Suite 350, Washington, DC 20037. 202-466-3825. < <http://arhp.org> > .

The Emergency Contraception for Diverse Communities Project, Program for Appropriate Technology in Health (PATH). 1455 NW Leary Way, Seattle, WA 98107-5136. 206-285-3500. < <http://www.path.org> > .

National Women's Health Information Center, Office on Women's Health, U.S. Department of Health and Human Services. 8550 Arlington Blvd., Suite 300, Fairfax, VA 22031. 800-994-WOMAN (9662). < <http://www.4woman.gov> > .

Office of Population Research, Princeton University. Wallace Hall, Princeton, NJ 08544. 609-258-4870. < <http://opr.princeton.edu> > .

Planned Parenthood Federation of America, Inc. 434 West 33rd Street, New York, NY 10001. 800-230-7562 (PLAN). < <http://www.plannedparenthood.org> > .
U.S. Food and Drug Administration. 5600 Fishers Lane, Rockville MD 20857-0001. 888-INFO-FDA (888-463-6332). < <http://www.fda.gov> > .

OTHER

Emergency Contraception. Planned Parenthood. February 2005 [cited March 6, 2005]. < <http://www.plannedparenthood.org/pp2/portal/medicalinfo/ec/pub-emergency-contraception.xml> > .

Emergency Contraception. Planned Parenthood. June 2004 [cited March 6, 2005]. < <http://www.plannedparenthood.org/pp2/portal/medicalinfo/ec/fact-emergency-contraception.xml> > .

Frequently Asked Questions About Emergency Contraception. The National Women's Health Information Center. November 2002 [cited March 6, 2005]. < <http://www.4woman.gov/faq/econtracep.htm> > .

Not-2-Late.com. The Emergency Contraception Website. Office of Population Research, Princeton University and the Association of Reproductive Health Professionals. [Cited March 6, 2005]. < <http://ec.princeton.edu> > .

Margaret Alic, Ph.D.

EMG see **Electromyography**

Emollient bath see **Therapeutic baths**

Emphysema

Definition

Emphysema is a chronic respiratory disease where there is over-inflation of the air sacs (alveoli) in the lungs, causing a decrease in lung function, and often, breathlessness.

Description

Emphysema is the most common cause of **death** from respiratory disease in the United States, and is the fourth most common cause of death overall. There are 1.8 million Americans with the disease, which ranks fifteenth among chronic conditions that cause limitations of activity. The disease is usually caused by **smoking**, but a small number of cases are caused by an inherited defect.

Normally functioning lungs are elastic, efficiently expanding and recoiling as air passes freely through

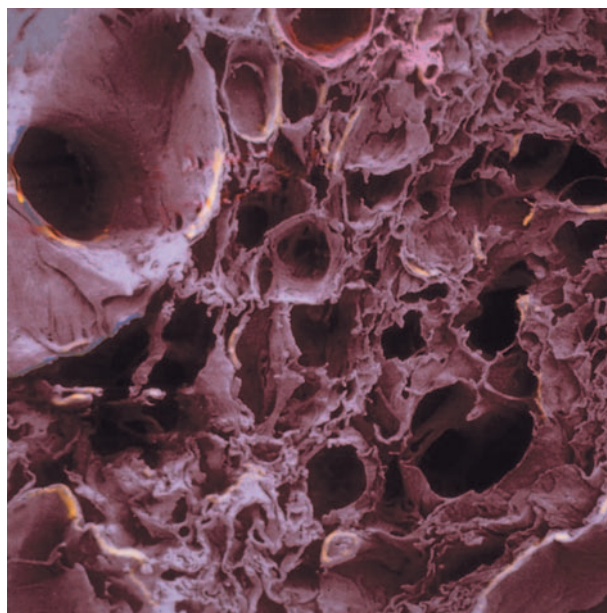
the bronchus to the alveoli, where oxygen is moved into the blood and carbon dioxide is filtered out. When a person inhales cigarette smoke or certain other irritants, his or her immune system responds by releasing substances that are meant to defend the lungs against the smoke. These substances can also attack the cells of the lungs, but the body normally inhibits such action with the release of other substances. In smokers and those with the inherited defect, however, no such prevention occurs and the lung tissue is damaged in such a way that it loses its elasticity. The small passageways (bronchioles) leading to the alveoli collapse, trapping air within the alveoli. The alveoli, unable to recoil efficiently and move the air out, over expand and rupture. As the disease progresses, coughing and shortness of breath occur. In the later stages, the lungs cannot supply enough oxygen to the blood. Emphysema often occurs with other respiratory diseases, particularly chronic **bronchitis**. These two diseases are often referred to as one disorder called chronic obstructive pulmonary disease (COPD).

Emphysema is most common among people aged 50 and older. Those with inherited emphysema may experience the onset as early as their thirties or forties. Men are more likely than women to develop emphysema, but female cases are increasing as the number of female smokers rises.

Causes and symptoms

Heavy cigarette smoking causes about 80–90% of all emphysema cases. However a few cases are the result of an inherited deficiency of a substance called alpha-1-antitrypsin (AAT). The number of Americans with this deficiency is relatively small, probably no greater than 70,000. Pipe, cigar, and **marijuana** smoking can also damage the lungs. While a person may be less likely to inhale cigar and pipe smoke, these types of smoke can also impair lung function. Marijuana smoke may be even more damaging because it is inhaled deeply and held in by the smoker.

The symptoms of emphysema develop gradually over many years. It is a common occurrence for many emphysema patients to have lost over half of their functioning lung tissue before they become aware that something is wrong. **Shortness of breath**, a chronic mild **cough** (which may be productive of large amounts of dark, thick sputum, and often dismissed as “smoker’s cough”), and sometimes weight loss are associated with emphysema. Initially, a patient may only notice shortness of breath when he or she is exercising. However, as the disease progresses, it will occur with less exertion or no exertion at all. Emphysema patients



A scanning electron microscopy (SEM) of lung tissue indicating emphysema. (Photograph by Hossler, Ph.D., Custom Medical Stock Photo. Reproduced by permission.)

may also develop an enlarged, or “barrel,” chest. Other symptoms may be skipped breaths, difficulty sleeping, morning headaches, increased difficulty breathing while lying down, chronic fatigue, and swelling of the feet, ankles, or legs. Those with emphysema are at risk for a variety of other complications resulting from weakened lung function, including **pneumonia**.

Diagnosis

A variety of pulmonary function tests may be ordered. In the early stages of emphysema, the only result may be dysfunction of the small airways. Patients with emphysema may show an increase in the total amount of air that is in the lungs (total lung capacity), but a decrease in the amount of air that can be breathed out after taking a deep breath (vital capacity). With severe emphysema, vital capacity is substantially below normal. Spirometry, a procedure that measures air flow and lung volume, helps in the diagnosis of emphysema.

A **chest x ray** is often ordered to aid in the diagnosis of emphysema, though patients in the early stages of the disease may have normal findings. Abnormal findings on the chest x ray include over-inflation of the lungs and an abnormally increased chest diameter. The diaphragm may appear depressed or flattened. In addition, patients with advanced



X ray showing emphysema in the lungs. (Photo Researchers. Reproduced by permission.)

emphysema may show a smaller or vertical heart. The physician may observe blisters in the lungs and bulging of the accessory muscles of the respiratory system. Late in the disease, an EKG will show signs of right ventricular failure in the heart and increased hemoglobin due to lower levels of oxygen in the patient's blood.

Treatment

Treatment methods for emphysema do not cure or reverse the damage to the lungs. However, they may slow the progression of the disease, relieve symptoms, and help control possibly fatal complications. The first step in treatment for smokers is to quit, so as to prevent any further deterioration of breathing ability. Smoking cessation programs may be effective. Consistent encouragement along with the help of health care professionals as well as family and friends can help increase the success rate of someone attempting to quit.

If the patient and the health care team develop and maintain a complete program of respiratory care, disability can be decreased, acute episodes of illness may be prevented, and the number of hospitalizations reduced. However, only quitting smoking has been shown to slow down the progression of the disease, and among all other treatments, only **oxygen therapy** has shown an increase in the survival rate.

Home oxygen therapy may improve the survival times in those patients with advanced emphysema who also have low blood oxygen levels. It may improve the patient's tolerance of exercise, as well as improve their performance in certain aspects of brain function and muscle coordination. The functioning of the heart may also improve with an increased concentration of oxygen in the blood. Oxygen may also decrease **insomnia** and headaches. Some patients may only receive oxygen at night, but studies have illustrated that it is most effective when administered at least 18, but preferably 24 hours per day. Portable oxygen tanks prescribed to patients carry a limited supply and must be refilled on a regular basis by a home health provider. Medicare and most insurance companies cover a large proportion of the cost of home oxygen therapy. Patients should be instructed regarding special safety issues involving the transport and presence of oxygen in the home.

A variety of medications may be used in the treatment of emphysema. Usually the patient responds best to a combination of medicines, rather than one single drug.

Bronchodilators are sometimes used to help alleviate the patient's symptoms by relaxing and opening the airways. They can be inhaled, taken by mouth, or injected. Another category of medication often used is **corticosteroids** or steroids. These help to decrease the inflammation of the airway walls. They are occasionally used if bronchodilators are ineffective in preventing airway obstruction. Some patients' lung function improves with corticosteroids, and inhaled steroids may be beneficial to patients with few side effects. A variety of **antibiotics** are frequently given at the first sign of a respiratory infection, such as increased amounts of sputum, or if there has been a change in the color of the sputum. **Expectorants** can help to loosen respiratory secretions, enabling the patient to more easily expel them from the airways.

Many of the medications prescribed involve the use of a metered dose inhaler (MDI) that may require special instruction to be used correctly. MDIs are a convenient and safe method of delivering medication to the lungs. However, if they are used incorrectly the medication will not get to the right place. Proper technique is essential for the medication to be effective.

For some patients, surgical treatment may be the best option. Lung volume reduction surgery is a surgical procedure in which the most diseased parts of the lung are removed to enable the remaining lung and breathing muscles to work more efficiently. Preliminary studies suggest improved survival rates and better functioning with the surgery. Another surgical procedure used for emphysema patients is **lung**

KEY TERMS

Alveoli—Small cells or cavities. In the lungs, these are air sacs where oxygen enters the blood and carbon dioxide is filtered out.

Pulmonary—Related to or associated with the lungs.

transplantation. Transplantation may involve one or both lungs. However, it is a risky and expensive procedure, and donor organs may not be available.

For those patients with advanced emphysema, keeping the air passages reasonably clear of secretions can prove difficult. Some common methods for mobilizing and removing secretions include:

- **Postural drainage.** This helps to remove secretions from the airways. The patient lies in a position that allows gravity to aid in draining different parts of the lung. This is often done after the patient inhales an aerosol medication. The basic position involves the patient lying on the bed with his chest and head over the side and the forearms resting on the floor.
- **Chest percussion.** This technique involves lightly clapping the back and chest, and may help to loosen thick secretions.
- **Coughing and deep breathing.** These techniques may aid the patient in bringing up secretions.
- **Aerosol treatments.** These treatments may involve solutions of saline, often mixed with a bronchodilator, which are then inhaled as an aerosol. The aerosols thin and loosen secretions. A treatment normally takes 10 to 15 minutes, and is given three or four times a day.

Patients with COPD can learn to perform a variety of self-help measures that may help improve their symptoms and their ability to participate in everyday activities. These measures include:

- Avoiding any exposure to dusts and fumes.
- Avoiding air pollution, including the cigarette smoke of others.
- Avoiding other people who have infections like the cold or flu. Get a pneumonia **vaccination** and a yearly flu shot.
- Drinking plenty of fluids. This helps to loosen respiratory secretions so they can be brought up more easily through coughing.

- Avoiding extreme temperatures of heat or cold. Also avoiding high altitudes. (Special precautions can be taken that may enable the emphysema patient to fly on a plane.)
- Maintaining adequate nutritional intake. Normally a high protein diet taken in many small feedings is recommended.

Alternative treatment

Many patients are interested in whether any alternative treatments for emphysema are available. Some practitioners recommend supplements of antioxidant nutrients. There have also been some studies indicating a correlation between a low Vitamin A levels and COPD, with suggestions that supplements of vitamin A might be beneficial. Aromatherapists have used essential oils like eucalyptus, lavender, pine, and rosemary to help relieve nasal congestion and make breathing easier. The herb elecampane may act as an expectorant to help patients clear mucus from the lungs. The patient should discuss these remedies with their health care practitioner prior to trying them, as some may interact with the more traditional treatments that are already being used.

Prognosis

Emphysema is a serious and chronic disease that cannot be reversed. If detected early, the effects and progression can be slowed, particularly if the patient stops smoking immediately. Complications of emphysema include higher risks for pneumonia and acute bronchitis. Overall, the prognosis for patients with emphysema is poor, with a survival rate for all those with COPD of four years, and even less for emphysema. However, individual cases vary and many patients can live much longer with supplemental oxygen and other treatment measures.

Prevention

The best way to prevent emphysema is to avoid smoking. Even patients with inherited emphysema should avoid smoking, as it especially worsens the onset and severity. If patients quit smoking as soon as evidence of small airway obstruction begins, they can significantly improve their prognosis.

Resources

BOOKS

Beers, Mark H., and Robert Berkow, editors. *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck and Company, Inc., 2004.

PERIODICALS

"Data Mounting on Merits of Lung Volume Reduction Surgery." *Family Practice News* February 15, 2001: 5.

Lewis, Laurie. "Optimal Treatment for COPD." *PatientCare* May 30, 2000: 60.

ORGANIZATIONS

American Lung Association. 1740 Broadway New York, NY 10019. (212) 315-8700. <<http://www.lungusa.org>>.

National Emphysema Foundation. 15 Stevens St. Norwalk, CT 06856. <<http://www.emphysemafoundation.org>>.

National Heart, Lung and Blood Institute. <<http://www.nhlbi.nih.gov>>.

Deanna M. Swartout-Corbeil, R.N.

Empyema

Definition

Empyema is a condition in which pus and fluid from infected tissue collects in a body cavity. The name comes from the Greek word *empyein* meaning pus-producing (suppurate). Empyema is most often used to refer to collections of pus in the space around the lungs (pleural cavity), but sometimes refers to similar collections in the gall bladder or the pelvic cavity. Empyema in the pleural cavity is sometimes called empyema thoracis, or empyema of the chest, to distinguish it from empyema elsewhere in the body.

Description

Empyema may have a number of causes but is most frequently a complication of **pneumonia**. Its development can be divided into three phases: an acute phase in which the body cavity fills with a thin fluid containing some pus; a second stage in which the fluid thickens and a fibrous, coagulation protein (fibrin) begins to accumulate within the cavity; and a third or chronic stage in which the lung or other organ is encased within a thick covering of fibrous material.

Causes and symptoms

Empyema thoracis can be caused by a number of different organisms, including bacteria, fungi, and amebas, in connection with pneumonia, chest wounds, chest surgery, lung abscesses, or a ruptured esophagus. The infective organism can get into the pleural cavity either

through the bloodstream or other circulatory system, in secretions from lung tissue, or on the surfaces of surgical instruments or objects that cause open chest **wounds**. The most common organisms that cause empyema are the following bacteria: *Streptococcus pneumoniae*, *Haemophilus influenzae*, and *Staphylococcus aureus*. *S. aureus* is the most common cause in all age groups, accounting for 90% of cases of empyema in infants and children. Pelvic empyema in women is most often caused by *Bacteroides* strains or *Pseudomonas aeruginosa*. In elderly, chronically ill, or alcoholic patients, empyema is often caused by *Klebsiella pneumoniae* species of bacteria.

When the disease organisms arrive in the cavity surrounding the lungs, they infect the tissues that cover the lungs and line the chest wall. As the body attempts to fight off the infection, the cavity fills up with tissue fluid, pus, and dead tissue cells. Empyema of the gall bladder or pelvis results from similar reactions to infection in those parts of the body.

The signs and symptoms of empyema vary somewhat according to the location of the infection and its severity. In empyema thoracis, patients usually exhibit symptoms of pneumonia, including **fever**, **cough**, **fatigue**, **shortness of breath**, and chest pain. They may prefer to lie on the side of the body affected by the empyema. Family members may notice **bad breath**. In severe cases, the patient may become dehydrated, cough up blood or greenish-brown sputum, run a fever as high as 105°F (40.6°C), or fall into a **coma**.

Patients with thoracic empyema may develop potentially life-threatening complications if the condition is not treated. The infected tissues may develop large collections of pus (abscesses) that can rupture into the patient's airway, or the infection may spread to the tissues surrounding the heart. In extreme cases the empyema may spread to the brain by means of bacteria carried in the bloodstream.

In pelvic empyema, the infection produces large amounts of thick, foul-smelling pus that is rapidly replaced even after drainage. Empyema of the gallbladder is marked by intense **pain** on the upper right side of the abdomen, high fever, and rigidity of the muscles over the infected area.

Diagnosis

A physician may consider the possibility of empyema thoracis in patients with pneumonia or other symptoms of lung infection. When listening to sounds within the patient's chest with a stethoscope, the sounds of breathing will be partly muffled and harder to hear in the patients with empyema. The

area of the chest over the infection will sound dull when tapped or thumped (percussed). On an x ray, empyema thoracis will appear as a cloudy or opaque area. The amount of fluid present in the pleural cavity can be estimated using an ultrasound imaging procedure. The diagnosis of empyema, however, has to be confirmed with laboratory tests because its symptoms can be caused by other disease conditions.

The diagnosis of empyema is usually confirmed by analyzing a sample of fluid taken from the pleural cavity. The sample is obtained by a procedure called **thoracentesis**. In this procedure, the patient is given a local anesthetic, a needle is inserted into the pleural cavity through the back between the ribs on the infected side, and a sample of fluid is withdrawn. If the patient has empyema, there will be a very high level of one particular kind of immune cell (white blood cells), a high level of protein, and a very low level of blood sugar. The fluid can also be tested for the specific disease organism by staining or tissue cultures. In some cases, the color, smell, or consistency of the tissue fluid also helps to confirm the diagnosis.

Treatment

Empyema is treated using a combination of medications and surgical techniques. Treatment with medication involves intravenously administering a two-week course of **antibiotics**. It is important to give antibiotics as soon as possible to prevent first-stage empyema from progressing to its later stages. The antibiotics most commonly used are penicillin and vancomycin. Patients experiencing difficulty breathing are also given oxygen therapy.

Surgical treatment of empyema has two goals: drainage of the infected fluid and closing up of the space left in the pleural cavity. If the infection is still in its early stages, the fluid can be drained by thoracentesis. In second-stage empyema, the surgeon will insert a chest tube in the patient's rib cage or remove part of a rib (rib resection) in order to drain the fluid. In third-stage empyema, the surgeon may cut or peel away the thick fibrous layer coating the lung. This procedure is called decortication. When the fibrous covering is removed, the lung will expand to fill the space in the chest cavity. The doctor can use video-assisted thoracic surgery (VATS) techniques to position the chest tube or to perform a limited decortication. The VATS technique allows a physician to see within the body during certain surgical procedures. Empyema of the gallbladder is a serious condition that is treated with intravenous antibiotics and surgical removal of the gallbladder.

KEY TERMS

Abscess—An area of inflamed and injured body tissue that fills with pus.

Decortication—Surgical removal of the fibrous peel that covers the lungs in third-stage empyema.

Empyema—The collection of pus in a body cavity, particularly the lung or pleural cavity.

Fibrin—A fibrous blood protein vital to coagulation and blood clot formation.

Percussion—A diagnostic technique in which the back, chest, or abdomen is tapped to determine whether body cavities contain abnormal fluid.

Pleural cavity—The space surrounding the lungs, including the membranes covering the lungs and lining the inside of the chest wall.

Pneumonia—Inflammation of the lungs usually caused by a virus, bacteria, or other organism.

Resection—The surgical removal of part of an organ or body structure, as in rib resection.

Suppurate—To produce or discharge pus.

Thoracentesis—A procedure in which fluid is withdrawn from the pleural cavity through a needle inserted between the ribs. The fluid may be withdrawn either for diagnostic tests or to drain the cavity.

Video-assisted thoracic surgery (VATS)—A technique used to aid in the placement of chest tubes or when performing decortications when treating advanced empyema.

Prognosis

The prognosis for recovery is generally good, except in those cases with complications, such as a **brain abscess** or blood **poisoning**, or cases caused by certain types of streptococci.

Resources

BOOKS

Stauffer, John L. "Lung." In *Current Medical Diagnosis and Treatment*, 1998, edited by Stephen McPhee, et al., 37th ed. Stamford: Appleton & Lange, 1997.

Rebecca J. Frey, PhD

Enalapril see **Angiotensin-converting enzyme inhibitors**

Encephalitis

Definition

Encephalitis is an inflammation of the brain, usually caused by a direct viral infection or a hypersensitivity reaction to a virus or foreign protein. Brain inflammation caused by a bacterial infection is sometimes called cerebritis. When both the brain and spinal cord are involved, the disorder is called encephalomyelitis. An inflammation of the brain's covering, or meninges, is called **meningitis**.

Description

Encephalitis is an inflammation of the brain. The inflammation is a reaction of the body's immune system to infection or invasion. During the inflammation, the brain's tissues become swollen. The combination of the infection and the immune reaction to it can cause **headache** and a **fever**, as well as more severe symptoms in some cases.

Approximately 2,000 cases of encephalitis are reported to the Centers for Disease Control in Atlanta, GA each year. The viruses causing primary encephalitis can be epidemic or sporadic. The **polio** virus is an epidemic cause. Arthropod-borne viral encephalitis is responsible for most epidemic viral encephalitis. The viruses live in animal hosts and mosquitos that transmit the disease. The most common form of non-epidemic or sporadic encephalitis is caused by the herpes simplex virus, type 1 (HSV-1) and has a high rate of death. **Mumps** is another example of a sporadic cause.

Causes and symptoms

Causes

There are more than a dozen viruses that can cause encephalitis, spread by either human-to human contact or by animal bites. Encephalitis may occur with several common viral infections of childhood. Viruses and viral diseases that may cause encephalitis include:

- **chickenpox**
- **measles**
- **mumps**
- **Epstein-Barr virus (EBV)**
- **cytomegalovirus infection**
- **HIV**
- **herpes simplex**
- **herpes zoster (shingles)**

- **herpes B**
- **polio**
- **rabies**
- **mosquito-borne viruses (arboviruses)**

Primary encephalitis is caused by direct infection by the virus, while secondary encephalitis is due to a post-infectious immune reaction to viral infection elsewhere in the body. Secondary encephalitis may occur with measles, chickenpox, mumps, **rubella**, and EBV. In secondary encephalitis, symptoms usually begin five to 10 days after the onset of the disease itself and are related to the breakdown of the myelin sheath that covers nerve fibers.

In rare cases, encephalitis may follow **vaccination** against some of the viral diseases listed above. Creutzfeldt-Jakob disease, a very rare brain disorder caused by an infectious particle called a prion, may also cause encephalitis.

Mosquitoes spread viruses responsible for equine encephalitis (eastern and western types), St. Louis encephalitis, California encephalitis, and Japanese encephalitis. **Lyme disease**, spread by ticks, can cause encephalitis, as can Colorado tick fever. Rabies is most often spread by animal bites from dogs, cats, mice, raccoons, squirrels, and bats and may cause encephalitis.

Equine encephalitis is carried by mosquitoes that do not normally bite humans but do bite horses and birds. It is occasionally picked up from these animals by mosquitoes that do bite humans. **Japanese encephalitis** and St. Louis encephalitis are also carried by mosquitoes. The risk of contracting a mosquito-borne virus is greatest in mid- to late summer, when mosquitoes are most active, in those rural areas where these viruses are known to exist. Eastern equine encephalitis occurs in eastern and southeastern United States; western equine and California encephalitis occur throughout the West; and St. Louis encephalitis occurs throughout the country. Japanese encephalitis does not occur in the United States, but is found throughout much of Asia. The viruses responsible for these diseases are classified as arbovirus and these diseases are collectively called **arbovirus encephalitis**.

Herpes simplex encephalitis, the most common form of sporadic encephalitis in western countries, is a disease with significantly high mortality. It occurs in children and adults and both sides of the brain are affected. It is theorized that brain infection is caused by the virus moving from a peripheral location to the brain via two nerves, the olfactory and the trigeminal (largest nerves in the skull).

Herpes simplex encephalitis is responsible for 10% of all encephalitis cases and is the main cause of sporadic, fatal encephalitis. In untreated patients, the rate of **death** is 70% while the mortality is 15–20% in patients who have been treated with acyclovir. The symptoms of herpes simplex encephalitis are fever, rapidly disintegrating mental state, headache, and behavioral changes.

Symptoms

The symptoms of encephalitis range from very mild to very severe and may include:

- headache
- fever
- lethargy (sleepiness, decreased alertness, and fatigue)
- malaise
- nausea and **vomiting**
- visual disturbances
- tremor
- decreased consciousness (drowsiness, confusion, delirium, and unconsciousness)
- stiff neck
- seizures

Symptoms may progress rapidly, changing from mild to severe within several days or even several hours.

Diagnosis

Diagnosis of encephalitis includes careful questioning to determine possible exposure to viral sources. Tests that can help confirm the diagnosis and rule out other disorders include:

- Blood tests. These are to detect antibodies to viral antigens, and foreign proteins.
- Cerebrospinal fluid analysis (spinal tap). This detects viral antigens, and provides culture specimens for the virus or bacteria that may be present in the cerebrospinal fluid.
- Electroencephalogram (EEG).
- CT and MRI scans.

A **brain biopsy** (surgical gathering of a small tissue sample) may be recommended in some cases where treatment to date has been ineffective and the cause of the encephalitis is unclear. Definite diagnosis by biopsy may allow specific treatment that would otherwise be too risky.

Treatment

Choice of treatment for encephalitis will depend on the cause. Bacterial encephalitis is treated with **antibiotics**. Viral encephalitis is usually treated with antiviral drugs including acyclovir, ganciclovir, foscarnet, ribovarin, and AZT. Viruses that respond to acyclovir include herpes simplex, the most common cause of sporadic (non-epidemic) encephalitis in the United States.

The symptoms of encephalitis may be treated with a number of different drugs. **Corticosteroids**, including prednisone and dexamethasone, are sometimes prescribed to reduce inflammation and brain swelling. Anticonvulsant drugs, including dilantin and phenytoin, are used to control seizures. Fever may be reduced with **acetaminophen** or other fever-reducing drugs.

A person with encephalitis must be monitored carefully, since symptoms may change rapidly. Blood tests may be required regularly to track levels of fluids and salts in the blood.

Prognosis

Encephalitis symptoms may last several weeks. Most cases of encephalitis are mild, and recovery is usually quick. Mild encephalitis usually leaves no residual neurological problems. Overall, approximately 10% of those with encephalitis die from their infections or complications such as secondary infection. Some forms of encephalitis have more severe courses, including herpes encephalitis, in which mortality is 15–20% with treatment, and 70–80% without. Antiviral treatment is ineffective for eastern equine encephalitis, and mortality is approximately 30%.

Permanent neurological consequences may follow recovery in some cases. Consequences may include personality changes, memory loss, language difficulties, seizures, and partial paralysis.

Prevention

Because encephalitis is due to infection, it may be prevented by avoiding the infection. Minimizing contact with others who have any of the viral illness listed above may reduce the chances of becoming infected. Most infections are spread by hand-to-hand or hand-to-mouth contact; frequent hand washing may reduce the likelihood of infection if contact cannot be avoided.

Mosquito-borne viruses may be avoided by preventing mosquito bites. Mosquitoes are most active at dawn and dusk, and are most common in moist areas

KEY TERMS

Cerebrospinal fluid analysis—A analysis that is important in diagnosing diseases of the central nervous system. The fluid within the spine will indicate the presence of viruses, bacteria, and blood. Infections such as encephalitis will be indicated by an increase of cell count and total protein in the fluid.

Computerized tomography (CT) Scan—A test to examine organs within the body and detect evidence of tumors, blood clots, and accumulation of fluids.

Electroencephalogram (EEG)—A chart of the brain waves picked up by the electrodes placed on the scalp. Changes in brain wave activity can be an indication of nervous system disorders.

Inflammation—A response from the immune system to an injury. The signs are redness, heat, swelling, and pain.

Magnetic Resonance Imaging (MRI)—MRI is diagnostic radiography using electromagnetic energy to create an image of the central nervous system (CNS), blood system, and musculoskeletal system.

Vaccine—A preparation containing killed or weakened microorganisms used to build immunity against infection from that microorganism.

Virus—A very small organism that can only live within a cell. They are unable to reproduce outside that cell.

with standing water. Minimizing exposed skin and use of mosquito repellents on other areas can reduce the chances of being bitten.

Vaccines are available against some viruses, including polio, herpes B, Japanese encephalitis, and equine encephalitis. Rabies vaccine is available for animals; it is also given to people after exposure. Japanese encephalitis vaccine is recommended for those traveling to Asia and staying in affected rural areas during transmission season.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Richard Robinson

Encephalocele see **Congenital brain defects**

Encopresis

Definition

Encopresis is repeatedly having bowel movements in places other than the toilet after the age when bowel control can normally be expected.

Description

Most children have established bowel control by the time they are four years old. After that age, when they repeatedly have bowel movements in inappropriate places, they may have encopresis. In the United States, encopresis affects 1–2% of children under age 10. About 80% of these are boys.

Encopresis can be either involuntary or voluntary. Involuntary encopresis is related to **constipation**, passing hard painful feces, and difficult bowel movements. Often children with involuntary encopresis stain their underpants with liquid feces. They are usually unaware that this has happened. Voluntary encopresis is much less common and is associated with behavioral or psychological problems. Both types of encopresis occur most often when the child is awake, rather than at night.

Causes and symptoms

Although a few children experience encopresis because of malformations of the lower bowel and anus or irritable bowel disease, most have no physical problems to explain this disorder. Constipation is present in about 80% of children who experience involuntary encopresis. As feces moves through the large intestine, water is removed. The longer the feces stays in the large intestine, the more water is removed, and the harder the feces becomes. The result can be hard or painful bowel movements. In response, children may start to hold back when they feel the urge to eliminate in order to avoid **pain**. This starts a cycle of constipation that results in retentive encopresis.

Once elimination is avoided, the bowel becomes full of hard feces. This stretches the large intestine. Eventually the intestine becomes so stretched that liquid feces backed up behind the blockage is able to leak around the hard feces. Children with this type of encopresis do not feel the urge to have a bowel movement and are often surprised when their pants are stained with foul smelling liquid feces. This leakage of feces is called overflow incontinence. Parents sometimes mistake this soiling for **diarrhea**, because the

feces expelled is liquid. Every so often, children with involuntary encopresis may pass large stools, sometimes with volumes big enough to clog the toilet, but the relief this brings is temporary.

Although about 95% of encopresis is involuntary, some children intentionally withhold bowel movements. The American Psychiatric Association (APA) recognizes voluntary encopresis without constipation as a psychological disorder. This disorder is said to occur when a child who has control over his bowel movements chooses to have them in an inappropriate place. The feces is a normal consistency, not hard. Sometimes it is smeared in an obvious place, but it may also be hidden from adults.

Voluntary encopresis may result from a power struggle between caregivers and the child during toilet training, or the child may have developed an unusual fear of the toilet. It is also associated with **oppositional defiant disorder (ODD)**, **conduct disorder**, sexual abuse, and high levels of psychological stress. For example, children who were separated from their parents during World War II were reported to have a high rate of encopresis. However, parents and caregivers should be aware that very few children soil intentionally and most do not have a behavioral or psychological problem and should not be punished for their soiling accidents.

Diagnosis

Diagnosis is based primarily on the child's history of inappropriate bowel movements. Physical examinations are almost always normal, except for a mass of hard feces blocking the lower intestine. Other physical causes of soiling, such as illness, reaction to medication, **food allergies**, and physical disabilities, may also be ruled out through history and a **physical examination**. In addition, to be diagnosed with encopresis the child must be old enough to establish regular bowel control—usually chronologically and developmentally at least four years of age.

Treatment

The goal of treatment is to establish regular, soft, pain free bowel movements in the toilet. First the physician tries to determine the cause of encopresis, whether physical or psychological. Regardless of the cause, the bowel must be emptied of hard, impacted feces. This can be done using an enema, **laxatives**, and/or stool softeners such as mineral oil. **Enemas** and laxatives should be used only at a doctor's recommendation.

KEY TERMS

Feces—Waste products eliminated from the large intestine; excrement.

Incontinence—The inability to control the release of urine or feces.

Laxative—Material that encourages a bowel movement.

Stools—feces, bowel movements.

Next, the child is given stool softeners to keep feces soft and to give the stretched intestine time to shrink back to its normal size. This shrinking process may take several months, during which time stool softeners may need to be used regularly. Children also need two or three regularly scheduled toilet sits daily in an effort to establish consistent bowel habits. These toilet sits are often more effective if done after meals. Maintaining soft, easy-to-pass stools is also important if the child is afraid of the toilet because of past painful bowel movements. A child psychologist or psychiatrist can suggest treatment for the rare child with serious behavioral problems such as smearing or hiding feces.

Alternative treatment

Many herbal stool softeners and laxatives are available as both tablets and liquids. Psyllium, the seed of several plants of the genus *Plantago* is one of the most effective. Other natural remedies for constipation include castor seed oil (*Ricinus communis*), senna (*Cassia senna* or *Senna alexandrina*), and dong quai *Angelica polymorpha* or *Angelica sinensis*.

Prognosis

For almost all children, once constipation is controlled, the problem of soiling disappears. This may take several months, and relapses may occur, but with effective prevention strategies, encopresis can be eliminated. Children who are in a power struggle over toileting usually outgrow their desire to have bowel movements in inappropriate places. The prognosis for children with serious behavioral and psychological problems that result in smearing or hiding feces depends largely on resolving the underlying problems.

Prevention

The best way to prevent encopresis is to prevent constipation. Methods of preventing constipation include:

- increasing the amount of liquids, especially water, the child drinks
- adding high fiber foods to the diet (e.g. dried beans, fresh fruits and vegetables, whole wheat bread and pasta, popcorn)
- establishing regular bowel habits
- limiting the child's intake of dairy products (e.g. milk, cheese, yogurt, ice cream) that promote constipation.
- treating constipation promptly with stool softeners, so that it does not become worse.

Resources

BOOKS

American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders*, 4th ed. text revision. Washington D.C.: American Psychiatric Association, 2000.

PERIODICALS

Kuhn, Brett R., Bethany A. Marcus, and Sheryl L. Pitner. "Treatment Guidelines for Primary Nonretentive Encopresis and Stool Toileting Refusal." *American Family Physician*, 59, no. 8 (15 April 1999) 2171-2183. [cited 16 February 2005]. <<http://www.aafp.org/afp/2001101/1565.html>> .

ORGANIZATIONS

American Academy of Child and Adolescent Psychiatry, P. O. Box 96106, Washington, D.C. 20090. 800-333-7636. <www.aacap.org> .

OTHER

Borowitz, Stephen. *Encopresis*, 14 June 2004 [cited 20 February 2005]. <<http://www.emedicine.com/ped/topics670.html>> .

Tish Davidson, A.M.

Endarterectomy

Definition

Endarterectomy is an operation to remove or bypass the fatty deposits, or blockage, in an artery narrowed by the buildup of fatty tissue (**atherosclerosis**).

Purpose

Removing the fatty deposits restores normal blood flow to the part of the body supplied by the

artery. An endarterectomy is performed to treat cerebrovascular disease in which there is a serious reduction of blood supply to the brain (carotid endarterectomy), or to treat **peripheral vascular disease** (impaired blood supply to the legs).

Endarterectomy is most often performed on one of the two main arteries in the neck (the carotids) opening the narrowed arteries leading to the brain. When performed by an experienced surgeon, the practice is extremely effective, reducing the risk of **stroke** by up to 70%. Recent studies indicate it is effective in preventing stroke, even among those patients who had no warning signs except narrowed arteries detected by their doctors on a routine exam.

Precautions

Before the surgery, a full medical exam is usually done to assess any specific health problems, such as diabetes, high blood pressure, heart disease, or stroke. If possible, reversible health problems, such as cigarette **smoking** or being overweight, should be corrected.

Description

Carotid artery disease

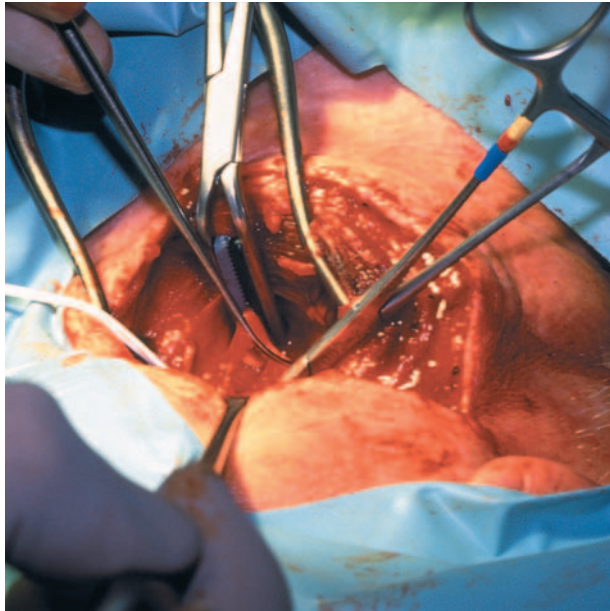
Every person has four carotid arteries (the internal and external carotids on each side of the neck) through which blood from the heart moves into the brain. If one of these arteries becomes blocked by fat and cholesterol, the patient may have a range of symptoms, including:

- weakness in one arm, leg, half of the face, or one entire side of the body
- numbness or **tingling**
- **paralysis** of an arm, leg, or face
- slurred speech
- dizziness
- confusion, **fainting**, or coma
- stroke

Removing this fatty buildup, or bypassing a blocked segment, may restore blood flow to the brain, eliminate or decrease the symptoms, and lessen the risk of a stroke.

Peripheral vascular disease

When the blood vessels in the legs (and sometimes the arms) become narrowed, this can restrict blood flow and cause **pain** in the affected area. In severe cases, the tissue may die, requiring **amputation**.



In this procedure, surgeons are removing plaque from the carotid artery. (Custom Medical Stock Photo. Reproduced by permission.)

The narrowing is usually caused by buildup of fatty plaques in the vessels, often as the result of smoking, high blood pressure, or poorly-controlled diabetes mellitus. The vessels usually narrow slowly, but it's possible for a blood clot to form quickly, causing sudden severe pain in the affected leg or arm.

Procedure

Endarterectomy is a delicate operation that may require several hours. The surgeon begins by making an incision over the blocked artery and inserting a tube above and below the blockage to redirect the blood flow while the artery is opened.

Next, the surgeon removes the fat and cholesterol buildup, along with any **blood clots** that have formed, with a blunt dissecting instrument. Then the surgeon bathes the clean wall in salt solution combined with heparin, an anticoagulant. Then the surgeon stitches the artery just enough so that the bypass shunt tube can be removed, and then he/she stitches the artery completely closed. After checking to make sure no blood is leaking, the surgeon next closes the skin incision with stitches.

The operation should improve symptoms, although its long-term effects may be more limited, since arterial narrowing is rarely confined to one area of one artery. If narrowing is a problem throughout

the body, arterial **reconstructive surgery** may be required.

The total cost of an endarterectomy, including diagnostic tests, surgery, hospitalization, and follow-up care, will vary according to hospital, doctor, and area of the country where the operation is performed, but a patient can expect to pay in the range of \$15,000. Patients who are very young, very old, or very ill, or who need more extensive surgery, may require more expensive treatment.

Preparation

Before surgery, the doctor pinpoints the location of the narrowed artery with an x-ray procedure called **angiography**. For surgery to be effective, the degree of narrowing should be at least 70%, but it should not be total. Patients undergoing angiography are given a local anesthetic, but the endarterectomy itself requires the use of a general anesthesia.

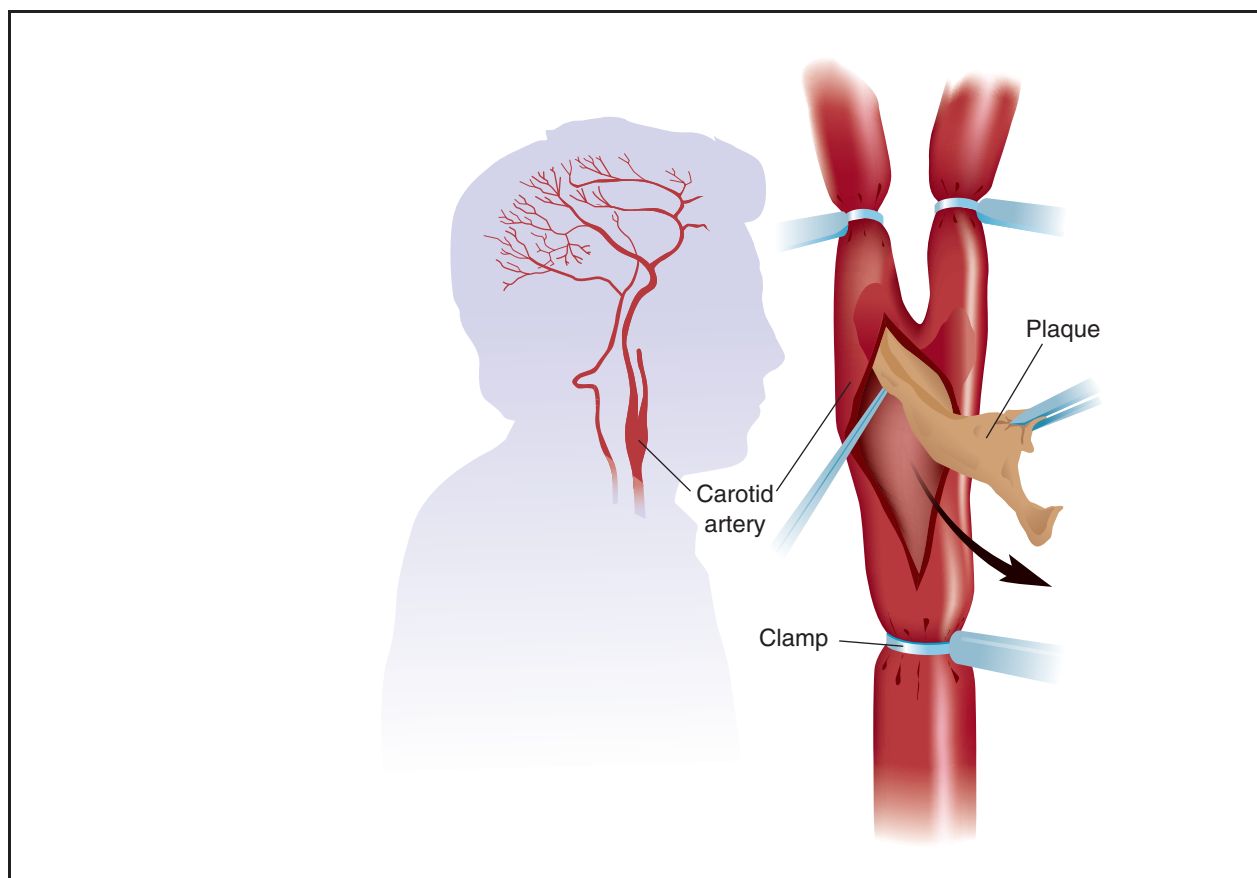
Aftercare

After the surgery, the patient spends the first two days lying flat in bed. Patients who have had carotid endarterectomy should not bend the neck sharply during this time. Because the blood flow to the brain is now greatly increased, patients may experience a brief but severe **headache**, or lightheadedness. There may be a slight loss of sensation in the skin, or maybe a droop in the mouth, if any of the nerves in the neck were lightly bruised during surgery. In time, this should correct itself.

Risks

The amount of risk depends on the hospital, the skill of the surgeon, and the severity of underlying disease. Patients who have just had an acute stroke are at greatest risk. During carotid artery surgery, blood flow is interrupted through the artery, so that paralysis and other stroke symptoms may occur. These may resolve after surgery, or may result in permanent stroke. Paralysis is usually one-sided; other stroke symptoms may include loss of half the field of vision, loss of sensation, double vision, speech problems, and personality changes. Risks of endarterectomy to treat either carotid artery or peripheral vascular disease include:

- reactions to anesthesia
- bleeding
- infection
- blood clots



Plaque is removed from the carotid artery by clamping the artery, cutting the plaque out, and closing the opening back up.
(Illustration by Argosy Inc.)

KEY TERMS

Carotid arteries—The four principal arteries of the neck and head. There are two common carotid arteries, each of which divides into the two main branches (internal and external).

Diabetes mellitus—A disorder in which the pancreas doesn't produce enough (or any) insulin. As a result, the blood levels of sugar become very high. Among other things, diabetes can lead to the breakdown of small blood vessels and a high risk of atherosclerosis and high blood pressure.

Stroke—Damage to the part of the brain caused by an interruption of the blood supply. In some cases, small pieces of plaque in the carotid artery may break loose and block an artery in the brain. A narrowed carotid artery also can be the source of blood clots travelling to the brain, or the artery can become completely clogged, blocking all blood flow to the brain.

Normal results

The results after successful surgery are usually striking. The newly opened artery should help to restore normal blood flow. In carotid endarterectomy, surgery should prevent the risk of brain damage and stroke. However, the buildup of fat and cholesterol usually affects all arteries, not just the one that was operated on. Affected arteries in other parts of the body may be equally clogged and potentially dangerous. Even arteries that were operated electively will likely begin to clog up again after the surgery.

For this reason, lifestyle changes (no smoking, low fat, low cholesterol diet) are important, especially if diet and lifestyle contributed to the development of the problem in the first place.

Resources

PERIODICALS

"Better Blood Flow: Surgery May Strike Down Stroke Risk." *Prevention* 47 (February 1, 1995): 50-52.

ORGANIZATIONS

National Institute of Neurological Disorders and Stroke. PO Box 5801, Bethesda, MD 20824. (800) 352-9424. <<http://www.ninds.nih.gov/index.htm>>.

National Institute of Neurological Disorders at the Neurology Institute. PO Box 5801, Bethesda, MD 20824.

Carol A. Turkington

Endemic syphilis see **Bejel**

Endocardial resection see **Myocardial resection**

Endocarditis

Definition

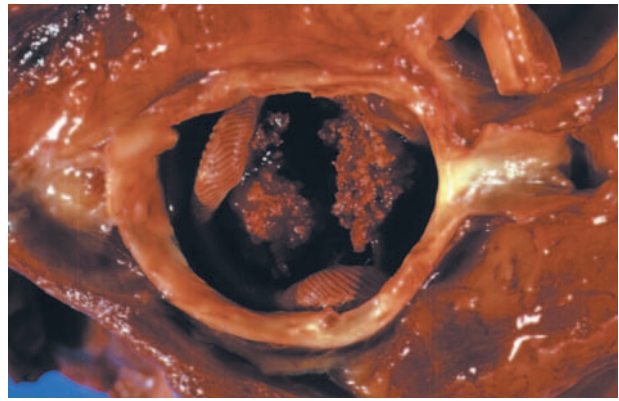
The endocardium is the inner lining of the heart muscle, which also covers the heart valves. When the endocardium becomes damaged, bacteria from the blood stream can become lodged on the heart valves or heart lining. The resulting infection is known as endocarditis.

Description

The endocardium lines all four chambers of the heart—two at the top (the right and left atria) and two at the bottom (the right and left ventricles)—through which blood passes as the heart beats. It also covers the four valves (the tricuspid valve, the pulmonary valve, the mitral valve, and the aortic valve), which normally open and close to allow the blood to flow in only one direction through the heart during each contraction.

For the heart to pump blood efficiently, the four chambers must contract and relax, and the four valves must open and close, in a well coordinated fashion. By damaging the valves or the walls of the heart chambers, endocarditis can interfere with the ability of the heart to do its job.

Endocarditis rarely occurs in people with healthy, normal hearts. Rather, it most commonly occurs when there is damage to the endocardium. The endocardium may be affected by a congenital heart defect, such as **mitral valve prolapse**, in which blood leaks through a poorly functioning mitral valve back into the heart. It may also be damaged by a prior scarring of the heart muscle, such as **rheumatic fever**, or replacement of a heart valve. Any of these conditions can damage the

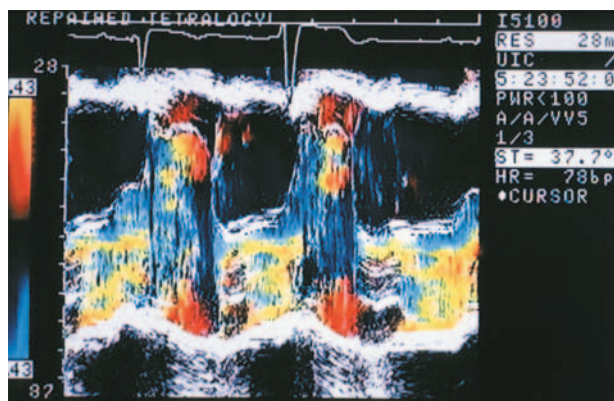


A close-up view of an infected artificial heart valve showing bacterial endocarditis (the granulated tissue at center of image). When infection occurs early after surgery, it is likely that organisms have gained entry during the operative period. This type of infection is usually caused by *Staphylococcus epidermidis* and *S. aureus* and is treated with antibiotic drugs. (Photograph by Dr. E. Walker, Photo Researchers, Inc. Reproduced by permission.)

endocardium and make it more susceptible to infection.

Bacteria can get into the blood stream (a condition known as **bacteremia**) in a number of different ways: It may spread from a localized infection such as a urinary tract infection, **pneumonia**, or skin infection or get into the blood stream as a result of certain medical conditions, such as severe periodontal disease, colon cancer, or inflammatory bowel disease. It can enter the blood stream during minor procedures, such as periodontal surgery, tooth extractions, teeth cleaning, tonsil removal, prostate removal, or endoscopic examination. It can also be introduced through in-dwelling catheters, which are used for intravenous medications, intravenous feeding, or dialysis. In people who use intravenous drugs, the bacteria can enter the blood stream through unsterilized, contaminated needles and syringes. (People who are prone to endocarditis generally need to take prescribed **antibiotics** before certain surgical or dental procedures to help prevent this infection.)

If not discovered and treated, infective endocarditis can permanently damage the heart muscle, especially the valves. For the heart to work properly, all four valves must be functioning well, opening at the right time to let blood flow in the right direction and closing at the right time to keep the blood from flowing in the wrong direction. If the valve is damaged, this may allow blood to flow backward—a condition known as regurgitation. As a result of a poorly functioning valve, the heart muscle has to work harder to pump blood and may become weakened, leading to



This echocardiogram shows an aortic regurgitation due to endocarditis, an infection of the lining membrane of the cardiac chambers. (Custom Medical Stock Photo. Reproduced by permission.)

heart failure. **Heart failure** is a chronic condition in which the heart is unable to pump blood well enough to supply blood adequately to the body.

Another danger associated with endocarditis is that the vegetation formed by bacteria colonizing on heart valves may break off, forming emboli. These emboli may travel through the circulation and become lodged in blood vessels. By blocking the flow of blood, emboli can starve various tissues of nutrients and oxygen, damaging them. For instance, an embolus lodged in the blood vessels of the lungs may cause pneumonia-like symptoms. An embolus may also affect the brain, damaging nerve tissue, or the kidneys, causing **kidney disease**. Emboli may also weaken the tiny blood vessels called capillaries, causing hemorrhages (leaking blood vessels) throughout the body.

Causes and symptoms

Most cases of infective endocarditis occur in people between the ages of 15 and 60, with a median age at onset of about 50 years. Men are affected about twice as often as women are. Other factors that put people at increased risk for endocarditis are congenital heart problems, heart surgery, previous episodes of endocarditis, and intravenous drug use.

While there is no single specific symptom of endocarditis, a number of symptoms may be present. The most common symptom is a mild **fever**, which rarely goes above 102°F (38.9°C). Other symptoms include chills, weakness, **cough**, trouble breathing, headaches, aching joints, and loss of appetite.

Emboli may also cause a variety of symptoms, depending on their location. Emboli throughout the body may cause Osler's nodes, small, reddish, painful bumps most commonly found on the inside of fingers and toes. Emboli may also cause petechiae, tiny purple or red spots on the skin, resulting from hemorrhages under the skin's surface. Tiny hemorrhages resembling splinters may also appear under the fingernails or toenails. If emboli become lodged in the blood vessels of the lungs, they may cause coughing or **shortness of breath**. Emboli lodged in the brain may cause symptoms of a mini-stroke, such as **numbness**, weakness, or paralysis on one side of the body or sudden vision loss or double vision. Emboli may also damage the kidneys, causing blood to appear in the urine. Sometimes the capillaries on the surface of the spleen rupture, causing the spleen to become enlarged and tender to the touch. Anyone experiencing any of these symptoms should seek medical help immediately.

Diagnosis

Doctors begin the diagnosis by taking a history, asking the patient about the symptoms mentioned above. During a **physical examination**, the doctor may also uncover signs such as fever, an enlarged spleen, signs of kidney disease, or hemorrhaging. Listening to the patient's chest with a stethoscope, the doctor may also hear a heart murmur. A heart murmur may indicate abnormal flow of blood through one of the heart chambers or valves.

Doctors take a sample of the patient's blood to test it for bacteria and other microorganisms that may be causing the infection. They usually also use a test called **echocardiography**, which uses ultrasound waves to make images of the heart, to check for abnormalities in the structure of the heart wall or valves. One of the tell-tale signs they look for in echocardiography is vegetation, the abnormal growth of tissue around a valve composed of blood platelets, bacteria, and a clotting protein called fibrin. Another tell-tale sign is regurgitation, or the backward flow of blood, through one of the heart valves. A normal echocardiogram does not exclude the possibility of endocarditis, but an abnormal echocardiogram can confirm its presence. If an echocardiogram cannot be done or its results are inconclusive, a modified technique called **transesophageal echocardiography** is sometimes performed. Transesophageal echocardiography involves passing an ultrasound device into the esophagus to get a clearer image of the heart.

KEY TERMS

Aortic valve—The valve between the left ventricle of the heart and the aorta.

Bacteremia—An infection caused by bacteria in the blood.

Congestive heart failure—A condition in which the heart muscle cannot pump blood as efficiently as it should.

Echocardiography—A diagnostic test using reflected sound waves to study the structure and motion of the heart muscle.

Embolus—A bit of foreign material, such as gas, a piece of tissue, or tiny clot, that travels in the circulation until it becomes lodged in a blood vessel.

Endocardium—The inner wall of the heart muscle, which also covers the heart valves.

Mitral valve—The valve between the left atrium and the left ventricle of the heart.

Osler's nodes—Small, raised, reddish, tender areas associated with endocarditis, commonly found inside the fingers or toes.

Petechiae—Tiny purple or red spots on the skin associated with endocarditis, resulting from hemorrhages under the skin's surface.

Pulmonary valve—The valve between the right ventricle of the heart and the pulmonary artery.

Transducer—A device that converts electrical signals into ultrasound waves and ultrasound waves back into electrical impulses.

Transesophageal echocardiography—A diagnostic test using an ultrasound device, passed into the esophagus of the patient, to create a clear image of the heart muscle.

Tricuspid valve—The valve between the right atrium and the right ventricle of the heart.

Vegetation—An abnormal growth of tissue around a valve, composed of blood platelets, bacteria, and a protein involved in clotting.

Treatment

When doctors suspect infective endocarditis, they will admit the patient to a hospital and begin treating the infection before they even have the results of the blood culture. Their choice of antibiotics depends on what the most likely infecting microorganism is. Once

the results of the **blood culture** become available, the doctor can adjust the medications, using specific antibiotics known to be effective against the specific microorganism involved.

Unfortunately, in recent years, the treatment of endocarditis has become more complicated as a result of antibiotic resistance. Over the past few years, especially as antibiotics have been overprescribed, more and more strains of bacteria have become increasingly resistant to a wider range of antibiotics. For this reason, doctors may need to try a few different types of antibiotics—or even a combination of antibiotics—to successfully treat the infection. Antibiotics are usually given for about one month, but may need to be given for an even longer period of time if the infection is resistant to treatment.

Once the fever and the worst of the symptoms have gone away, the patient may be able to continue antibiotic therapy at home. During this time, the patient should make regular visits to the health care team for further testing and physical examination to make sure that the antibiotic therapy is working, that it is not causing adverse side effects, and that there are no complications such as emboli or heart failure. The patient should alert the health-care team to any symptoms that could indicate serious complications: For instance, trouble breathing or swelling in the legs could indicate congestive heart failure. **Headache**, joint **pain**, blood in the urine, or **stroke** symptoms could indicate an embolus, and fever and chills could indicate that the treatment is not working and the infection is worsening. Finally, **diarrhea**, rash, **itching**, or joint pain may suggest a bad reaction to the antibiotics. Anyone experiencing any of these symptoms should alert the health care team immediately.

In some cases, surgery may be needed. These include cases of congestive heart failure, recurring emboli, infection that doesn't respond to treatment, poorly functioning heart valves, and endocarditis involving prosthetic (artificial) valves. The most common surgical treatment involves cutting away (debriding) damaged tissue and replacing the damaged valve.

Prognosis

If left untreated, infective endocarditis continues to progress and is always fatal. However, if it is diagnosed and properly treated within the first six weeks of infection, the infection can be completely cured in about 90% of the cases. The prognosis depends on a number of factors, such as the patient's age and overall physical condition, the severity of the diseases involved, the exact site of the infection, how

vulnerable the microorganisms are to antibiotics, and what kind of complications the endocarditis may be causing.

Prevention

Some people are especially prone to endocarditis. These include people with past episodes of endocarditis, those with congenital heart problems or heart damage from rheumatic fever, and those with artificial heart valves. Intravenous drug users are also at increased risk. Anyone who falls into a high-risk category should alert his or her health-care professionals before undergoing any surgical or dental procedures. High-risk patients must be treated in advance with antibiotics before these procedures to minimize the risk of infection.

Resources

BOOKS

Zaret, Barry L., et al., editors. *The Patient's Guide to Medical Tests*. Boston: Houghton Mifflin, 1997.

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

National Heart, Lung and Blood Institute. PO Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

Robert Scott Dinsmoor

Endocrine pancreatic cancer see **Pancreatic cancer, endocrine**

Endometrial biopsy

Definition

Endometrial biopsy is a procedure in which a sample of the endometrium (tissue lining the inside of the uterus) is removed for microscopic examination.

Purpose

The test is most often performed to find out the cause of abnormal uterine bleeding. Abnormal bleeding includes bleeding between menstrual periods, excessive bleeding during a menstrual period, or bleeding after **menopause**. Since abnormal uterine bleeding can indicate **cancer**, an endometrial biopsy is done to

rule out **endometrial cancer** or hyperplasia (a potentially precancerous condition).

Endometrial biopsies are also done as a screening test for endometrial cancer in postmenopausal women on **hormone replacement therapy**. Hormone replacement therapy usually requires a woman to take estrogen and progesterone. An endometrial biopsy is particularly useful in cases where postmenopausal women take estrogen, but cannot take progesterone. Estrogen in the system without the balancing effect of progesterone has been linked to an increased risk of endometrial cancer.

An endometrial biopsy can also be used as part of an **infertility** exam to rule out problems with the development of the endometrium. This condition is called luteal phase defect and can cause the endometrium to not support a pregnancy. An endometrial biopsy can also be used to evaluate the problem of repeated early miscarriages.

Precautions

If the endometrial biopsy is being done to investigate why a woman is unable to get pregnant, the test must be performed at a specific time during the menstrual cycle. Since the test evaluates whether the endometrium is developed adequately to support implantation and growth of a fertilized egg, it is critical to perform the test approximately three days before the expected menstrual period.

Description

The test is performed by a doctor who specializes in women's reproductive health (an obstetrician/gynecologist). The test is performed either in the doctor's office or in a local hospital. The patient may be asked to take **pain** medication (like Motrin or Aleve) an hour or so before the procedure. A local anesthetic may be injected into the cervix in order to decrease pain and discomfort during the procedure.

The woman will be asked to lie on her back with knees apart and feet in stirrups. The doctor will first conduct a thorough exam of the pelvic region, including the vulva (the external genitals), vagina, and uterus. A speculum (an instrument that is used to hold the walls of the vagina open) will be inserted into the vagina. A small, hollow plastic tube is then passed into the uterine cavity. A small piece of the uterine lining is sucked out with a plunger that is attached to the tube. Once the sample is obtained, the instruments are removed. The sample is sent to the laboratory for microscopic examination.



A micrograph image of the internal human uterine wall depicting the mucosa, or endometrium. (Photograph by Professors P.M. Motta and S. Makabe, Custom Medical Stock Photo. Reproduced by permission.)

The patient may experience some pain when the cervix is grasped. The patient may also feel some cramping, pressure, and discomfort when the instruments are inserted into the uterus and the tissue sample is collected.

Preparation

For the small number of endometrial biopsies that are done as part of infertility testing, a pregnancy test is also often performed before the procedure. Since the biopsy is performed late in the menstrual cycle, it is possible that the woman may be pregnant.

Aftercare

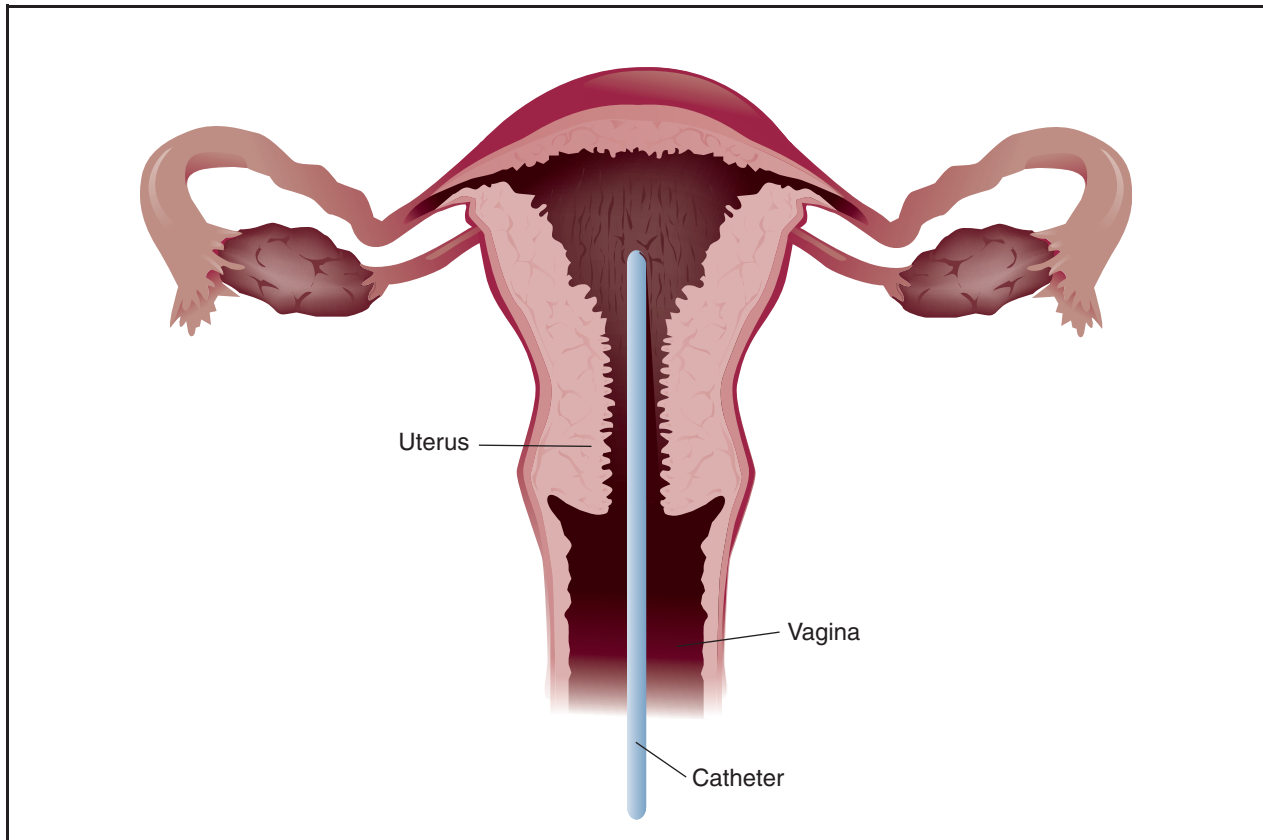
The biopsy may cause a small amount of bleeding (spotting). The woman can resume normal activities

right away. If cramping becomes severe, heavy bleeding occurs, or the woman develops a high temperature, the doctor should be notified immediately.

If the test is being done to determine the cause of infertility, the onset of the menstrual period following the biopsy should be reported to the doctor. This will allow the doctor to correctly predict if the endometrium has been developing at the expected rate.

Risks

The risks of an endometrial biopsy are very small. There is a possibility that prolonged bleeding may occur after the procedure. There is also a slight chance of an infection. Very rarely, there are instances when the uterus is pierced (perforated) or the cervix is torn because of the biopsy.



A catheter is inserted into the uterus to remove uterine cells for further examination. (Illustration by Argosy Inc.)

KEY TERMS

Biopsy—The surgical removal and microscopic examination of living tissue for diagnostic purposes.

Cervix—The opening of the uterus extending into the vagina.

Endometrium—The layer lining the inner cavity of the uterus; this layer changes daily throughout the menstrual cycle.

Uterus—The hollow, muscular female organ that supports the development and nourishment of the unborn baby during pregnancy.

Normal results

Most biopsies are done to rule out endometrial cancer or endometrial hyperplasia. A normal result shows no cancerous or precancerous cells. Normal results also show that the uterine lining is changing at the proper rate. If it is, then the results of the biopsy

are said to be “in-phase” because the tissue looks appropriate and has developed normally for the late phase of the menstrual cycle.

Abnormal results

If the endometrium is not developing at the appropriate rate, the results are said to be “out-of-phase” or abnormal. The endometrium has not developed appropriately and cannot support a **pregnancy**. This condition is called luteal phase defect and may need to be treated with progesterone.

Abnormal appearance of the cells forming the uterine tissue could also indicate uterine cancer, or the presence of fibroids or polyps in the uterus.

Resources

ORGANIZATIONS

American Cancer Society. 1599 Clifton Rd., NE, Atlanta, GA 30329-4251. (800) 227-2345. <<http://www.cancer.org>> .

Cancer Research Institute. 681 Fifth Ave., New York, N.Y. 10022. (800) 992-2623. <<http://www.cancerresearch.org>> .

Gynecologic Cancer Foundation. 401 North Michigan Ave., Chicago, IL 60611. (800) 444-4441.

National Cancer Institute. Building 31, Room 10A31, 31 Center Drive, MSC 2580, Bethesda, MD 20892-2580. (800) 422-6237. <<http://www.nci.nih.gov>>.

Lata Cherath, PhD

Endometrial cancer

Definition

Endometrial **cancer** develops when the cells that make up the inner lining of the uterus (the endometrium) become abnormal and grow uncontrollably.

Description

Endometrial cancer (also called uterine cancer) is the fourth most common type of cancer among women and the most common gynecologic cancer. Approximately 34,000 women are diagnosed with endometrial cancer each year. In 1998, approximately 6,300 women died from this cancer. Although endometrial cancer generally occurs in women who have gone through menopause and are 45 years of age or older, 30% of the women with endometrial cancer are younger than 40 years of age. The average age at diagnosis is 60 years old.

The uterus, or womb, is the hollow female organ that supports the development of the unborn baby during **pregnancy**. The uterus has a thick muscular wall and an inner lining called the endometrium. The endometrium is very sensitive to hormones and it changes daily during the menstrual cycle. The endometrium is designed to provide an ideal environment for the fertilized egg to implant and begin to grow. If pregnancy does not occur, the endometrium is shed causing the menstrual period.

More than 95% of uterine cancers arise in the endometrium. The most common type of uterine cancer is adenocarcinoma. It arises from an abnormal multiplication of endometrial cells (atypical adenomatous hyperplasia) and is made up of mature, specialized cells (well-differentiated). Less commonly, endometrial cancer arises without a preceding hyperplasia and is made up of poorly differentiated cells. The more common of these types are the papillary serous and clear cell carcinomas. Poorly differentiated endometrial cancers are often associated with a less promising prognosis.

The highest incidence of endometrial cancer in the United States is in Caucasians, Hawaiians, Japanese, and African Americans. American Indians, Koreans, and Vietnamese have the lowest incidence. African American and Hawaiian women are more likely to be diagnosed with advanced cancer and, therefore, have a higher risk of dying from the disease.

Causes and symptoms

Although the exact cause of endometrial cancer is unknown, it is clear that high levels of estrogen, when not balanced by progesterone, can lead to abnormal growth of the endometrium. Factors that increase a woman's risk of developing endometrial cancer are:

- **Age.** The risk is considerably higher in women who are over the age of 50 and have gone through **menopause**.
- **Obesity.** Being overweight is a very strong risk factor for this cancer. Fatty tissue can change other normal body chemicals into estrogen, which can promote endometrial cancer.
- **Estrogen replacement therapy.** Women receiving estrogen supplements after menopause have a 12 times higher risk of getting endometrial cancer if progesterone is not taken simultaneously.
- **Diabetes.** Diabetics have twice the risk of getting this cancer as nondiabetic women. It is not clear if this risk is due to the fact that many diabetics are also obese and hypertensive. One 1998 study found that women who were obese and diabetic were three times more likely to develop endometrial cancer than women who were obese but nondiabetic. This study also found that nonobese diabetics were not at risk of developing endometrial cancer.
- **Hypertension.** High blood pressure (or hypertension) is also considered a risk factor for uterine cancer.
- **Irregular menstrual periods.** During the menstrual cycle, there is interaction between the hormones estrogen and progesterone. Women who do not ovulate regularly are exposed to high estrogen levels for longer periods of time. If a woman does not ovulate regularly, this delicate balance is upset and may increase her chances of getting uterine cancer.
- **Early first menstruation or late menopause.** Having the first period at a young age (the mean age of menses is 12.16 years in African American girls and 12.88 years in caucasian girls) or going through menopause at a late age (over age 51) seem to put women at a slightly higher risk for developing endometrial cancer.

- **Tamoxifen.** This drug, which is used to treat or prevent breast cancer, increases a woman's chance of developing endometrial cancer. Tamoxifen users tend to have more advanced endometrial cancer with an associated poorer survival rate than those who do not take the drug. In many cases, however, the value of tamoxifen for treating breast cancer and for preventing the cancer from spreading far outweighs the small risk of getting endometrial cancer.
- **Family history.** Some studies suggest that endometrial cancer runs in certain families. Women with inherited mutations in the BRCA1 and BRCA2 genes are at a higher risk of developing breast, ovarian, and other gynecologic cancers. Those with the hereditary nonpolyposis colorectal cancer gene have a higher risk of developing endometrial cancer.
- **Breast, ovarian, or colon cancer.** Women who have a history of these other types of cancer are at an increased risk of developing endometrial cancer.
- **Low parity or nulliparity.** Endometrial cancer is more common in women who have born few (low parity) or no (nulliparity) children. The high levels of progesterone produced during pregnancy has a protective effect against endometrial cancer. The results of one study suggest that nulliparity is associated with a lower survival rate.
- **Infertility.** Risk is increased due to nulliparity or the use of fertility drugs.
- **Polycystic ovary syndrome.** The increased level of estrogen associated with this abnormality raises the risk of cancers of the breast and endometrium.

The most common symptom of endometrial cancer is unusual vaginal spotting, bleeding, or discharge. In women who are near menopause (perimenopausal), symptoms of endometrial cancer could include bleeding between periods (intermenstrual bleeding), heavy bleeding that lasts for more than seven days, or short menstrual cycles (fewer than 21 days). For women who have gone through menopause, any vaginal bleeding or abnormal discharge is suspect. **Pain** in the pelvic region and the presence of a lump (mass) are symptoms that occur late in the disease.

Diagnosis

If endometrial cancer is suspected, a series of tests will be conducted to confirm the diagnosis. The first step will involve taking a complete personal and family medical history. A physical examination, which will include a thorough pelvic examination, will also be done.

The doctor may order an **endometrial biopsy**. This is generally performed in the doctor's office and does not require anesthesia. A thin, flexible tube is inserted through the cervix and into the uterus. A small piece of endometrial tissue is removed. The patient may experience some discomfort, which can be minimized by taking an anti-inflammatory medication (like Advil or Motrin) an hour before the procedure.

If an adequate amount of tissue was not obtained by the endometrial biopsy, or if the biopsy tissue looks abnormal but confirmation is needed, the doctor may perform a **dilatation and curettage (D & C)**. This procedure is done in the outpatient surgery department of a hospital and takes about an hour. The patient may be given general anesthesia. The doctor dilates the cervix and uses a special instrument to scrape tissue from inside the uterus.

The tissue that is obtained from the biopsy or the D & C is sent to a laboratory for examination. If cancer is found, then the type of cancer will be determined. The treatment and prognosis depends on the type and stage of the cancer.

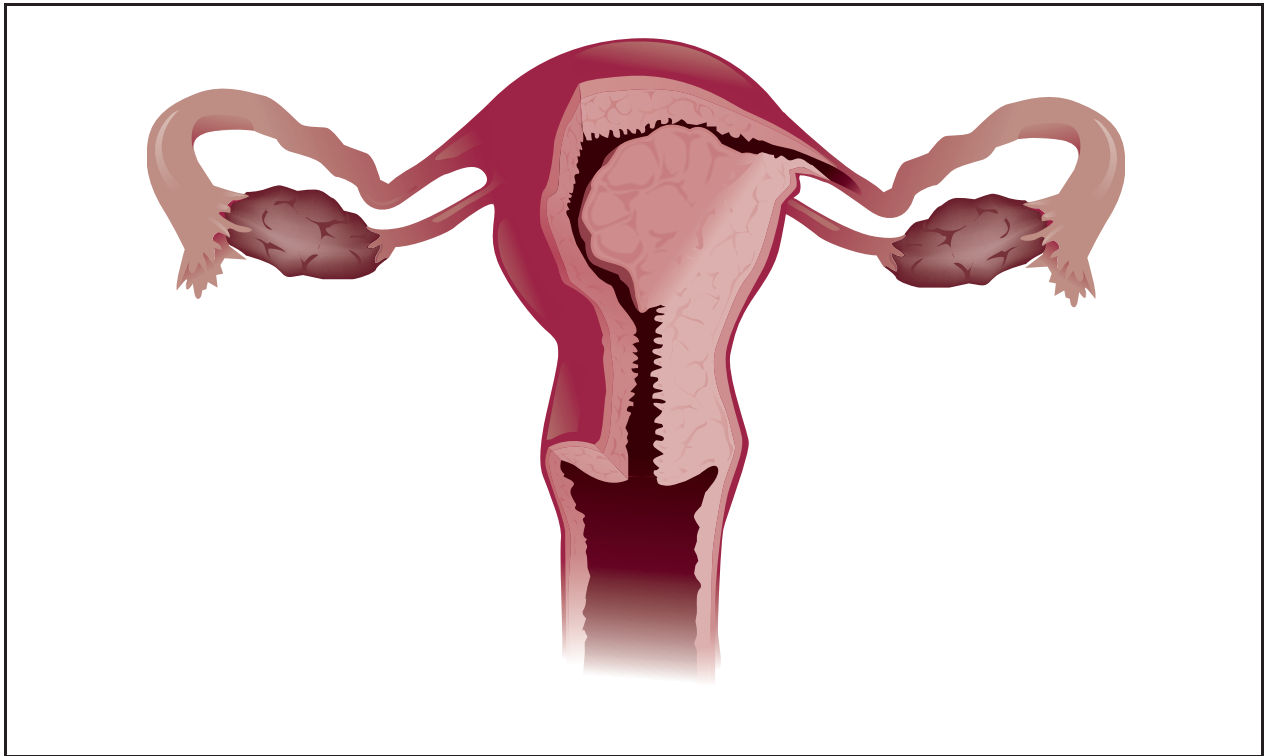
Trans-vaginal ultrasound may be used to measure the thickness of the endometrium. For this painless procedure, a wand-like ultrasound transducer is inserted into the vagina to enable visualization and measurement of the uterus, the thickness of the uterine lining, and other pelvic organs.

Other possible diagnostic procedures include sonohysterography and **hysteroscopy**. For sonohysterography, a small tube is passed through the cervix and into the uterus. A small amount of a salt water (saline) solution is injected through the tube to open the space within the uterus and allow ultrasound visualization of the endometrium. For hysteroscopy, a wand-like camera is passed through the cervix to allow direct visualization of the endometrium. Both of these procedures cause discomfort, which may be reduced by taking an anti-inflammatory medication prior to the procedure.

Treatment

Clinical staging

The International Federation of Gynecology and Obstetrics (FIGO) has adopted a staging system for endometrial cancer. The stage of cancer is determined after surgery. Endometrial cancer is categorized into four stages (I, II, III, and IV) that are subdivided (A, B, and possibly C) based on the depth or spread of cancerous tissue. Seventy percent of all uterine cancers are stage I, 10–15% are stage II, and the



Cancer located in the uterus. (Illustration by Argosy Inc.)

remainder are stages III and IV. The cancer is also graded (G1, G2, and G3) based upon microscopic analysis of the aggressiveness of the cancer cells.

The FIGO stages for endometrial cancer are:

- Stage I. Cancer is limited to the uterus.
- Stage II. Cancer involves the uterus and cervix.
- Stage III. Cancer has spread out of the uterus but is restricted to the pelvic region.
- Stage IV. Cancer has spread to the bladder, bowel, or other distant locations.

The mainstay of treatment for most stages of endometrial cancer is surgery. **Radiation therapy**, hormonal therapy, and **chemotherapy** are additional treatments (called adjuvant therapy). The necessity of adjuvant therapy is a controversial topic which should be discussed with the patient's treatment team.

Surgery

Most women with endometrial cancer, except those with stage IV disease, are treated with a **hysterectomy**. A simple hysterectomy involves the removal of the uterus. In a bilateral **salpingo-oophorectomy** with total hysterectomy, the ovaries, fallopian tubes, and uterus are removed. This may be necessary

because endometrial cancer often spreads to the ovaries first. The lymph nodes in the pelvic region may also be biopsied or removed to check for metastasis. Hysterectomy is traditionally performed through an incision in the abdomen (laparotomy), however, endoscopic surgery (**laparoscopy**) with vaginal hysterectomy is also being used. Women with stage I disease may require no further treatment. However, those with higher grade disease will receive adjuvant therapy.

Radiation therapy

The decision to use radiation therapy depends on the stage of the disease. Radiation therapy may be used before surgery (preoperatively) and/or after surgery (postoperatively). Radiation given from a machine that is outside the body is called external radiation therapy. Sometimes applicators containing radioactive compounds are placed inside the vagina or uterus. This is called internal radiation therapy or brachytherapy and requires hospitalization.

Side effects are common with radiation therapy. The skin in the treated area may become red and dry. **Fatigue**, upset stomach, **diarrhea**, and **nausea** are also common complaints. Radiation therapy in the pelvic area may cause the vagina to become narrow (vaginal

stenosis), making intercourse painful. Premature menopause and some problems with urination may also occur.

Chemotherapy

Chemotherapy is usually reserved for women with stage IV or recurrent disease because this therapy is not a very effective treatment for endometrial cancer. The **anticancer drugs** are given by mouth or intravenously. Side effects include stomach upset, **vomiting**, appetite loss, hair loss, mouth or vaginal sores, fatigue, menstrual cycle changes, and **premature menopause**. There is also an increased chance of infections.

Hormonal therapy

Hormonal therapy uses drugs like progesterone to slow the growth of endometrial cells. These drugs are usually available as pills. This therapy is usually reserved for women with advanced or recurrent disease. Side effects include fatigue, fluid retention, and appetite and weight changes.

Alternative treatment

Although alternative and complementary therapies are used by many cancer patients, very few controlled studies on the effectiveness of such therapies exist. Mind-body techniques, such as prayer, **biofeedback**, visualization, **meditation**, and **yoga**, have not shown any effect in reducing cancer, but they can reduce **stress** and lessen some of the side effects of cancer treatments. Clinical studies of hydrazine sulfate found that it had no effect on cancer and even worsened the health and well-being of the study subjects. One clinical study of the drug amygdalin (Laetrile) found that it had no effect on cancer. Laetrile can be toxic and has caused deaths. Shark cartilage, although highly touted as an effective cancer treatment, is an improbable therapy that has not been the subject of clinical study.

The American Cancer Society has found that the “metabolic diets” pose serious risk to the patient. The effectiveness of the macrobiotic, Gerson, and Kelley **diets** and the Manner metabolic therapy has not been scientifically proven. The FDA was unable to substantiate the anticancer claims made about the popular Cancell treatment.

There is no evidence for the effectiveness of most over-the-counter herbal cancer remedies. Some herbs have shown an anticancer effect. As shown in clinical studies, Polysaccharide krestin, from the mushroom *Coriolus versicolor*, has significant effectiveness against cancer. In a small study, the green

KEY TERMS

Adjuvant therapy—A treatment done when there is no evidence of residual cancer in order to aid the primary treatment. Adjuvant treatments for endometrial cancer are radiation therapy, chemotherapy, and hormone therapy.

Atypical adenomatous hyperplasia—The overgrowth of the endometrium. This precancerous condition is estimated to progress to cancer in one third of the cases.

Dilation and curettage (D & C)—A procedure in which the doctor opens the cervix and uses a special instrument to scrape tissue from the inside of the uterus.

Endometrial biopsy—A procedure in which a sample of the endometrium is removed and examined under a microscope.

Endometrium—The mucosal layer lining the inner cavity of the uterus. The endometrium’s structure changes with age and with the menstrual cycle.

Estrogen—A female hormone responsible for stimulating the development and maintenance of female secondary sexual characteristics.

Estrogen replacement therapy (ERT)—A treatment in which estrogen is used therapeutically during menopause to alleviate certain symptoms such as hot flashes. ERT has also been shown to reduce the risk of osteoporosis and heart disease in women.

Progesterone—A female hormone that acts on the inner lining of the uterus and prepares it for implantation of the fertilized egg.

Progestins—A female hormone, like progesterone, that acts on the inner lining of the uterus.

alga *Chlorella pyrenoidosa* has been shown to have anticancer activity. In a few small studies, evening primrose oil has shown some benefit in the treatment of cancer.

Prognosis

Because it is possible to detect endometrial cancer early, the chances of curing it are excellent. The five year survival rates for endometrial cancer by stage are: 90%, stage I; 60%, stage II; 40%, stage III; and 5%, stage IV. Endometrial cancer most often spreads to the lungs, liver, bones, brain, vagina, and certain lymph nodes.

Prevention

Women (especially postmenopausal women) should report any abnormal vaginal bleeding or discharge to the doctor. Controlling obesity, blood pressure, and diabetes can help to reduce the risk of this disease. Women on estrogen replacement therapy have a substantially reduced risk of endometrial cancer if progestins are taken simultaneously. Long term use of birth control pills has been shown to reduce the risk of this cancer. Women who have irregular periods may be prescribed birth control pills to help prevent endometrial cancer. Women who are taking tamoxifen and those who carry the hereditary nonpolyposis colorectal cancer gene should be screened regularly, receiving annual pelvic examinations.

Resources

BOOKS

- Bruss, Katherine, Christina Salter, and Esmeralda Galan, editors. *American Cancer Society's Guide to Complementary and Alternative Cancer Methods*. Atlanta: American Cancer Society, 2000.
- Burke, Thomas, Patricia Eifel, and Muggia Franco. "Cancers of the Uterine Body." In *Cancer: Principles & Practice of Oncology*, edited by Vincent DeVita, Samuel Hellman, and Steven Rosenberg. Philadelphia: Lippincott Williams & Wilkins, 2001, pp. 1573–86.
- Long, Harry. "Carcinoma of the Endometrium." In *Current Therapy in Cancer*, edited by John Foley, Julie Vose, and James Armitage. Philadelphia: W. B. Saunders Company, 1999, pp. 162–66.
- Primack, Aron. "Complementary/Alternative Therapies in the Prevention and Treatment of Cancer." In *Complementary/Alternative Medicine: An Evidence-Based Approach*, edited by John Spencer and Joseph Jacobs. St. Louis: Mosby, 1999, pp. 123–69.

PERIODICALS

- Bristow, Robert. "Endometrial Cancer." *Current Opinion in Oncology* 11 (September 1999): 388–393.
- Canavan, Timothy, and Nipa Doshi. "Endometrial Cancer." *American Family Physician* 59 (June 1999): 3069–3077.
- Elit, Laurie. "Endometrial Cancer: Prevention, Detection, Management, and Follow up." *Canadian Family Physician* 46 (April 2000): 887–892.
- Hogberg, Thomas, Margareta Fredstorp, and Anuja Jhingran. "Indications for Adjuvant Radiotherapy in Endometrial Carcinoma." *Hematology/Oncology Clinics of North America: Current Therapeutic Issues in Gynecologic Cancer* 13 (February 1999): 189–209.

ORGANIZATIONS

- American Cancer Society. 1599 Clifton Rd. NE, Atlanta, GA 30329. (800) 227-2345. <<http://www.cancer.org/>>.
- Cancer Research Institute, National Headquarters. 681 Fifth Ave., New York, NY 10022. (800) 992-2623. <<http://www.cancerresearch.org>>.

Gynecologic Cancer Foundation. 401 North Michigan Ave., Chicago, IL 60611. (800) 444-4441. <<http://www.wcn.org>>.

National Cancer Institute, National Institutes of Health. 9000 Rockville Pike, Bethesda, MD 20892. (800) 422-6237. <<http://cancernet.nci.nih.gov/>>.

OTHER

"Cancer of the Uterus." *Cancernet*. December 2000. [cited March 13, 2001]. <http://cancernet.nci.nih.gov/wyntk_pubs/uterus.htm>.

Lata Cherath, PhD
Belinda Rowland, PhD

Endometriosis

Definition

Endometriosis is a condition in which bits of the tissue similar to the lining of the uterus (endometrium) grow in other parts of the body. Like the uterine lining, this tissue builds up and sheds in response to monthly hormonal cycles. However, there is no natural outlet for the blood discarded from these implants. Instead, it falls onto surrounding organs, causing swelling and inflammation. This repeated irritation leads to the development of scar tissue and **adhesions** in the area of the endometrial implants.

Description

Endometriosis is estimated to affect 7% of women of childbearing age in the United States. It most commonly strikes between the ages of 25 and 40. Endometriosis can also appear in the teen years, but never before the start of menstruation. It is seldom seen in postmenopausal women.

Endometriosis was once called the "career woman's disease" because it was thought to be a product of delayed childbearing. The statistics defy such a narrow generalization; however, pregnancy may slow the progress of the condition. A more important predictor of a woman's risk is if her female relatives have endometriosis. Another influencing factor is the length of a woman's menstrual cycle. Women whose periods last longer than a week with an interval of less than 27 days between them seem to be more prone to the condition.

Endometrial implants are most often found on the pelvic organs—the ovaries, uterus, fallopian tubes, and in the cavity behind the uterus. Occasionally,

this tissue grows in such distant parts of the body as the lungs, arms, and kidneys. Newly formed implants appear as small bumps on the surfaces of the organs and supporting ligaments and are sometimes said to look like “powder burns.” **Ovarian cysts** may form around endometrial tissue (endometriomas) and may range from pea to grapefruit size. Endometriosis is a progressive condition that usually advances slowly, over the course of many years. Doctors rank cases from minimal to severe based on factors such as the number and size of the endometrial implants, their appearance and location, and the extent of the scar tissue and adhesions in the vicinity of the growths.

Causes and symptoms

Although the exact cause of endometriosis is unknown, a number of theories have been put forward. Some of the more popular ones are:

- **Implantation theory.** Originally proposed in the 1920s, this theory states that a reversal in the direction of menstrual flow sends discarded endometrial cells into the body cavity where they attach to internal organs and seed endometrial implants. There is considerable evidence to support this explanation. Reversed menstrual flow occurs in 70–90% of women and is thought to be more common in women with endometriosis. However, many women with reversed menstrual flow do not develop endometriosis.
- **Vascular-lymphatic theory.** This theory suggests that the lymph system or blood vessels (vascular system) is the vehicle for the distribution of endometrial cells out of the uterus.
- **Coelomic metaplasia theory.** According to this hypothesis, remnants of tissue left over from prenatal development of the woman’s reproductive tract transforms into endometrial cells throughout the body.
- **Induction theory.** This explanation postulates that an unidentified substance found in the body forces cells from the lining of the body cavity to change into endometrial cells.

In addition to these theories, the following factors are thought to influence the development of endometriosis:

- **Heredity.** A woman’s chance of developing endometriosis is seven times greater if her mother or sisters have the disease.
- **Immune system function.** Women with endometriosis may have lower functioning immune systems that have trouble eliminating stray endometrial cells. This would explain why a high percentage of women

experience reversed menstrual flow while relatively few develop endometriosis.

- **Dioxin exposure.** Some research suggests a link between the exposure to dioxin (TCDD), a toxic chemical found in weed killers, and the development of endometriosis.

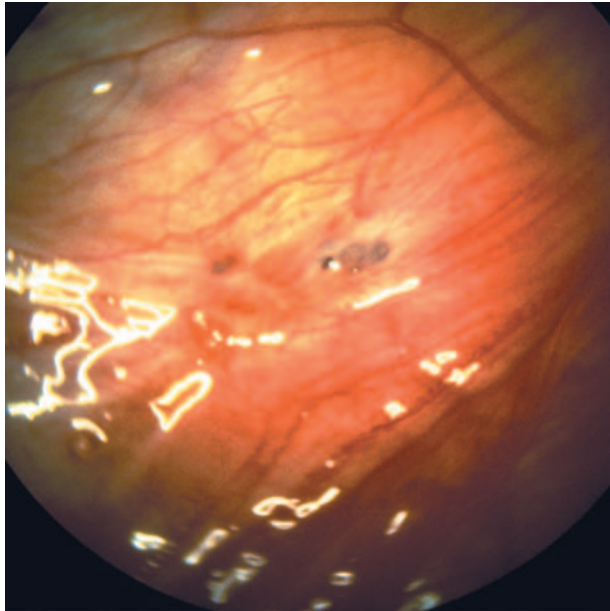
While many women with endometriosis suffer debilitating symptoms, others have the disease without knowing it. Paradoxically, there does not seem to be any relation between the severity of the symptoms and the extent of the disease. The most common symptoms are:

- **Menstrual pain.** Pain in the lower abdomen that begins a day or two before the menstrual period starts and continues through to the end is typical of endometriosis. Some women also report lower back aches and pain during urination and bowel movement, especially during their periods.
- **Painful sexual intercourse.** Pressure on the vagina and cervix causes severe pain for some women.
- **Abnormal bleeding.** Heavy menstrual periods, irregular bleeding, and spotting are common features of endometriosis.
- **Infertility.** There is a strong association between endometriosis and infertility, although the reasons for this have not been fully explained. It is thought that the build up of scar tissue and adhesions blocks the fallopian tubes and prevents the ovaries from releasing eggs. Endometriosis may also affect fertility by causing hormonal irregularities and a higher rate of early **miscarriage**.

Diagnosis

If a doctor suspects endometriosis, the first step will be to perform a **pelvic exam** to try to feel if implants are present. Very often there is no strong evidence of endometriosis from a physical exam. The only way to make a definitive diagnosis is through minor surgery called a **laparoscopy**. A laparoscope, a slender scope with a light on the end, is inserted into the woman’s abdomen through a small incision near her belly button. This allows the doctor to examine the internal organs for endometriotic growths. Often, a sample of tissue is taken for later examination in the laboratory. Endometriosis is sometimes discovered when a woman has abdominal surgery for another reason such as **tubal ligation** or **hysterectomy**.

Various imaging techniques such as ultrasound, computed tomography scan (CT scan), or magnetic resonance imaging (MRI) can offer additional



An endoscopic view of endometriosis on pelvic wall. (Custom Medical Stock Photo. Reproduced by permission.)

information but aren't useful in making the initial diagnosis. A blood test may also be ordered because women with endometriosis have higher levels of the blood protein CA125. Testing for this substance before and after treatment can predict a recurrence of the disease, but the test is not reliable as a diagnostic tool.

Treatment

How endometriosis is treated depends on the woman's symptoms, her age, the extent of the disease, and her personal preferences. The condition cannot be fully eradicated without surgery. Conservative treatment focuses on managing the pain, preserving fertility, and delaying the progress of the condition.

Pain relief

Over-the-counter pain relievers such as **aspirin** and acetaminophen (Tylenol) are useful for mild cramping and menstrual pain. Prescription-strength and over-the-counter nonsteroidal anti-inflammatory drugs (NSAIDs), such as ibuprofen (Motrin, Advil) and naproxen (Naprosyn), are also effective. If pain is severe, a doctor may prescribe narcotic medications, although these can be addictive and are rarely used.

Hormonal treatments

Hormonal therapies effectively tame endometriosis but also act as contraceptives. A woman who is

hoping to become pregnant would take these medications for a period of time, then try to conceive within several months of discontinuing treatment.

- **Oral contraceptives.** Continuously taking estrogen-progestin pills tricks the body into thinking it is pregnant. This state of pseudopregnancy means reduced pelvic pain and a temporary withering of endometrial implants.
- **Danazol (Danocrine) and gestrinone** are synthetic male hormones that lower estrogen levels, prevent menstruation, and shrink endometrial tissues. On the downside, they lead to weight gain and menopause-like symptoms, and cause some women to develop masculine characteristics.
- **Progestins.** Medroxyprogesterone (**Depo-Provera**) and related drugs may also be used in treating endometriosis. They have been proven effective in minimizing pain and halting the progress of the condition, but are rarely used because of the high rate of side effects.
- **Gonadotropin-releasing hormone (GnHR) agonists.** These estrogen-inhibiting drugs successfully limit pain and prevent the growth of endometrial implants. They can cause **menopause** symptoms, however, and doses have to be regulated to prevent bone loss associated with low estrogen levels.

Surgery

Removing the uterus, ovaries, and fallopian tubes is the only permanent method of eliminating endometriosis. This is an extreme measure that deprives a woman of her ability to bear children and forces her body into menopause. Endometrial implants and ovarian cysts can be removed with **laser surgery** performed through a laparoscope. For women with minimal endometriosis, this technique is usually successful in reducing pain and slowing the condition's progress. It may also help infertile women increase their chances of becoming pregnant.

Alternative treatment

Although severe endometriosis should not be self-treated, many women find they can help their condition through alternative therapies. Taking vitamin B complex combined with **vitamins C, E**, and the **minerals** calcium, magnesium, and selenium can help the depression and lack of energy that may accompany endometriosis. B vitamins also counteract the side effects of hormonal drugs. Other women have found relief when they turned to a macrobiotic diet. Less extreme **diets** that cut out sugar, salt, and processed

KEY TERMS

Adhesions—Web-like scar tissue that may develop as a result of endometriosis and bind organs to one another.

Dioxin—A toxic chemical found in weed killers that has been linked to the development of endometriosis.

Endometrial implants—Growths of endometrial tissue that attach to organs, primarily in the pelvic cavity.

Endometrium—The tissue lining the uterus that grows and sheds each month during a woman's menstrual cycle.

Estrogen—A female hormone that promotes the growth of endometrial tissue.

Hormonal therapy—Use of hormone medications to inhibit menstruation and relieve the symptoms of endometriosis.

Laparoscopy—A diagnostic procedure for endometriosis performed by inserting a slender, wand-like instrument through a small incision in the woman's abdomen.

Menopause—The end of a woman's menstrual periods when the body stops making estrogen.

Retrograde menstruation—Menstrual flow that travels into the body cavity rather than being expelled through the uterus.

foods are sometimes helpful, as well. Mind-body therapies such as relaxation and visualization help women cope with pain. Other avenues to combat pain include **acupuncture** and **biofeedback** techniques. Still other women report positive results after being treated by chiropractors or homeopathic doctors.

Prognosis

Most women who have endometriosis have minimal symptoms and do well. Overall, endometriosis symptoms come back in an average of 40% of women over the five years following treatment. With hormonal therapy, pain returned after five years in 37% of patients with minimal symptoms and 74% of those with severe cases. The highest success rate from conservative treatment followed complete removal of implants using laser surgery. Eighty percent of these women were still pain-free five years later. In cases that don't respond to these treatments, a woman and

her doctor may consider surgery to remove her reproductive organs.

Prevention

There is no proven way to prevent endometriosis. One study, however, indicated that girls who begin participating in aerobic **exercise** at a young age are less likely to develop the condition.

Resources

ORGANIZATIONS

Endometriosis Association International Headquarters.
8585 North 76th Place, Milwaukee, WI 53223. (800)
992-3636. <<http://EndometriosisAssn.org>> .

Stephanie Slon

Endometritis see **Pelvic inflammatory disease**

Endorectal ultrasound

Definition

Endorectal ultrasound (ERUS) is a procedure where a probe is inserted into the rectum and high frequency sound waves (ultrasound waves) are generated. The pattern of echoes as they bounce off tissues is converted into a picture (sonogram) on a television screen.

Purpose

ERUS is used as a diagnostic procedure in **rectal cancer** to determine stage of the tumor and as a post-radiation, presurgical examination to assess extent of tumor shrinkage. ERUS can also be used in cases of anal **fistula** (an abnormal passage) and problems with the anal sphincter muscles (muscles that control the opening and closing of the anus).

Precautions

Normal precautions should be taken with any diagnostic procedure. Since the population in which this procedure is normally done is elderly, the imaging staff should be extra cautious about stressing the patient. The procedure is invasive and may be embarrassing to some. Other patients may be anxious about their medical condition since endorectal ultrasounds are not routine. This places an added burden on

KEY TERMS

Anal sphincter muscles—Muscles that control the opening and closing of the anus.

Fistula—An abnormal passage.

Sonogram—The picture formed by the pattern of echoes from an ultra sound.

Ultrasound waves—High frequency sound waves.

already stressed hearts and nervous systems. Physicians, nurses, and technicians may need to be prepared for **stress** reactions that could include the heart, **asthma**, or anxious behaviors.

Description

ERUS has been used as a means to determine the depth of rectal cancers and to assess whether the tumor has affected surrounding tissues. This pre-treatment procedure has proven to be an accurate tool for tailoring surgery for patients.

Problems with interpretation of the sonograms after radiation and before surgery have resulted in tumors being identified that were merely the formation of fibrous tissues that remained after the tumors had been eliminated by the radiation. Yet, some of the fibrous areas actually hid residual tumors. Rectal anatomy itself can affect the accuracy of ultrasound reading. This makes ERUS problematic in determining the amount of tumor reduction a patient has after **radiation therapy**.

Preparation

The patient must evacuate the bowels completely before the procedure is done. This usually is assisted though the use of several **enemas**. The patient may be told to adhere to a liquid diet the day prior to doing this procedure. The probe is inserted, usually with little discomfort for the patient since it will only be examining the first few inches of the colon.

Aftercare

Since ERUS is a minor invasive procedure, there is no aftercare.

Risks

There are no risks to having an ultrasound.

Normal results

Normal results after an endorectal ultrasound are normal, healthy tissues.

Abnormal results

Abnormal results range from any number of congenital deformities in the lining of the rectum to serious rectal cancers.

Resources

BOOKS

Johnston, Lorraine. *Colon and Rectal Cancer: A Comprehensive Guide for Patients and Families*. Sebastopol, CA: O'Reilly, 2000.

Levin, Bernard. *Colorectal Cancer: A Thorough and Compassionate Resource for Patients and Their Families*. New York: Villard, 1999.

PERIODICALS

Gavioli, M. A. Bagni, I. Piccagli, S. Fundaro, and G. Natalini. "Usefulness of Endorectal Ultrasound after Preoperative Radiotherapy in Rectal Cancer: Comparison between Sonographic and Histopathologic Changes." *Dis ColonRectum* (August 2000): 1075-83.

OTHER

American Society of Colon and Rectal Surgeons. "Practice Parameters for the Treatment of Rectal Carcinoma." May 9, 2001. < http://www.asco.org/prof/me/html/abstracts/gasc/m_969.htm > .

National Cancer Institute. "NCI/PDQ Patient Statement:Rectal Cancer Updated 11/2000." *OncoLink*. May 9, 2001. < http://www.oncolink.upenn.edu/pdq_html/2/engl/200076.html > .

Janie F. Franz

Endoscopic retrograde cholangiopancreatography

Definition

Endoscopic retrograde cholangiopancreatography (ERCP) is a technique in which a hollow tube called an endoscope is passed through the mouth and stomach to the duodenum (the first part of the small intestine). This procedure was developed to examine abnormalities of the bile ducts, pancreas, and gallbladder. It was developed during the late 1960s and is used today to diagnose and treat blockages of the bile and pancreatic ducts.

The term has three parts to its definition:

- endoscopic refers to the use of an endoscope
- retrograde refers to the injection of dye up into the bile ducts in a direction opposing, or against, the normal flow of bile down the ducts
- cholangiopancreatography means visualization of the bile ducts (cholangio) and pancreas (pancreato)

Purpose

Until the 1970s, methods to visualize the bile ducts produced images that were of relatively poor quality and often misleading; in addition, the pancreatic duct could not be examined at all. Patients with symptoms related to the bile ducts or pancreatic ducts frequently needed surgery to diagnose and treat their conditions.

Using ERCP, physicians can obtain high-quality x rays of these structures and identify areas of narrowing (strictures), cancers, and **gallstones**. This procedure can help determine whether bile or pancreatic ducts are blocked; it also identifies where they are blocked along with the cause of the blockage. ERCP may then be used to relieve the blockage. For patients requiring surgery or additional procedures for treatment, ERCP outlines the anatomical changes for the surgeon.

Precautions

The most important precaution is that the examination should be performed by an experienced physician. The procedure is much more technically difficult than many other gastrointestinal endoscopic studies. Patients should seek physicians with experience performing ERCP. Patients should inform the physician about any allergies (including **allergies** to contrast dyes, iodine, or shellfish), medication use, and medical problems. Occasionally, patients may need to be admitted to the hospital after the procedure.

Description

After **sedation**, a specially adapted endoscope is passed through the mouth, through the stomach, then into the duodenum. The opening to ducts that empty from the liver and pancreas is identified, and a plastic tube or catheter is placed into the orifice (opening). Contrast dye is then injected into the ducts, and with the assistance of a radiologist, pictures are taken.

Preparation

The upper intestinal tract must be empty for the procedure, so patients should not eat or drink for at

KEY TERMS

Endoscope, endoscopy—An endoscope used in the field of gastroenterology is a hollow, thin, flexible tube that uses a lens or miniature camera to view various areas of the gastrointestinal tract. When the procedure is performed to examine the bile ducts or pancreas, the organs are not viewed directly, but rather indirectly through the injection of contrast. The performance of an exam using an endoscope is referred to as endoscopy. Diagnosis through biopsies or other means and therapeutic procedures can also be done using these instruments.

Visualization—The process of making an internal organ visible. A radiopaque substance is introduced into the body, then an x-ray picture of the desired area is taken.

least six to 12 hours before the exam. Patients should ask the physician about taking their medications before the procedure.

Aftercare

Someone should be available to take the person home after the procedure and stay with them for a while; patients will not be able to drive themselves because they undergo sedation during this test. **Pain** or any other unusual symptoms should be reported to the physician.

Risks

ERCP-related complications can be broken down into those related to medications used during the procedure, the diagnostic part of the procedure, and those related to endoscopic therapy. The overall complication rate is 5–10%; most of those occur when diagnostic ERCP is combined with a therapeutic procedure. During the exam, the endoscopist can cut or stretch structures (such as the muscle leading to the bile duct) to treat the cause of the patient's symptoms. Although the use of sedatives carries a risk of decreasing cardiac and respiratory function, it is very difficult to perform these procedures without these drugs.

The major complications related to diagnostic ERCP are **pancreatitis** (inflammation of the pancreas) and **cholangitis** (inflammation of the bile ducts). **Bacteremia** (the passage of bacteria into the blood stream) and perforation (hole in the intestinal tract) are additional risks.

Normal results

Because certain standards have been set for the normal diameter or width of the pancreatic duct and bile ducts, measurements using x rays are taken to determine if the ducts are too large (dilated) or too narrow (strictured). The ducts and gallbladder should be free of stones or tumors.

Abnormal results

When areas in the pancreatic or bile ducts (including those in the liver) are too wide or too narrow compared with the standard, the test is considered abnormal. Once these findings are demonstrated using ERCP, symptoms are usually present; they generally do not change without treatment. Stones, identified as opaque or solid structures within the ducts, are also considered abnormal. Masses or tumors may also be seen, but sometimes the diagnosis is made not by direct visualization of the tumor, but by indirect signs, such as a single narrowing of one of the ducts. Overall, ERCP has an excellent record in diagnosing these abnormalities.

Resources

OTHER

Endoscopic Retrograde Cholangiopancreatography. [cited June 21, 2004]. <<http://www.asge.org>>.

Measuring Procedural Skills. [cited June 21, 2004]. <<http://www.acponline.org/journals/annals/15dec96/procskil.htm>>.

Treatment of Acute Biliary Pancreatitis. [cited June 21, 2004]. <<http://content.nejm.org>>.

David Kaminstein, MD

Endoscopic sclerotherapy see **Sclerotherapy for esophageal varices**

Endoscopic sphincterotomy

Definition

Endoscopic sphincterotomy or endoscopic retrograde sphincterotomy (ERS) is a relatively new endoscopic technique developed to examine and treat abnormalities of the bile ducts, pancreas and gallbladder. The procedure was developed as an extension to the diagnostic examination, ERCP (endoscopic retrograde cholangiopancreatography); with the addition of “sphincterotomy,” abnormalities found during the

study could be treated at the same time without the need for invasive surgery.

The term ERS has three parts to its definition;

- endoscopic refers to the use of an endoscope
- retrograde refers to the insertion of the endoscope *up* into the ducts in a direction opposite to or against the normal flow of bile *down* the ducts
- sphincterotomy, which means cutting of the sphincter or muscle that lies at the juncture of the intestine with both the bile and pancreatic ducts.

Purpose

Until the 1970s, patients with symptoms related to disease of the bile ducts or pancreas frequently needed surgery to diagnose the cause and treat any abnormalities. ERCP allowed physicians for the first time to obtain high quality x rays of the common bile and pancreatic ducts, and detect areas of narrowing (strictures), stones, and tumors. ERCP was not initially designed for treatment. ERS was developed shortly after and enabled physicians to treat the abnormalities identified by the injection of dye and x rays.

The revolutionary technique made possible the endoscopic removal of stones and stretching of areas of narrowing (strictures). It has since been expanded to include drainage of bile from blocked ducts and treatment of various abnormalities of the pancreas.

Precautions

The most important precaution related to both ERCP and ERS is to have the procedure performed by an experienced physician. ERS is technically more difficult than many other gastrointestinal endoscopic studies, including ERCP. Patients should inquire as to the physician’s experience with the procedure. The physician should also be informed of any **allergies**, medication use, and medical problems.

Description

ERS is generally performed only after ERCP has been successfully accomplished and detail of the anatomy and abnormalities is known. During ERS, a number of various instruments are inserted through the endoscope in order to “cut” or stretch the sphincter. Once this is done, additional instruments are passed that enable the removal of stones and the stretching of narrowed regions of the ducts. Drains (stents) can also be used to prevent a narrowed area from rapidly returning to its previously narrowed state.

KEY TERMS

Endoscope, Endoscopy—An endoscope as used in the field of gastroenterology is a thin flexible tube which uses a lens or miniature camera to view various areas of the gastrointestinal tract. When the procedure is performed to examine certain organs such as the bile ducts or pancreas, the organs are not viewed directly, but rather indirectly through the injection of x-ray dye. The performance of an exam using an endoscope is referred by the general term endoscopy. Diagnosis through biopsies or other means and therapeutic procedures can be done with these instruments.

NSAIDS—This abbreviation stands for non-steroidal anti-inflammatory drugs, which are medications such as Ibuprofen that are used to control pain and inflammation. Most may be purchased over the counter. One of their major side effects is that they decrease the effect of the normal blood clotting factors in blood. In patients undergoing surgical or endoscopic procedures, this can lead to an increased risk of bleeding.

Preparation

The upper intestinal tract must be empty for the procedure, so patients must not eat or drink for at least six to 12 hours before the exam. Patients need to inquire about taking their medications before the procedure. Some patients may require **antibiotics** before and/or after the procedure. When possible, **aspirin** or NSAIDS should not be taken within several days before the procedure, because they interfere with blood clotting.

Aftercare

When ERS is performed, physicians will usually want to observe the patient closely for several hours to ensure that there are no signs of complications. **Pain** or any other unusual symptoms should be reported. Admission to the hospital may be advised.

Risks

ERS complications are related either to the drugs used during the procedure, or the results of dye injection or cutting of tissue. The overall complication rate is 5–10%. During the exam, the endoscopist can cut or stretch structures (such as the muscle leading to the

bile duct) to treat the cause of the patient's symptoms. Cutting or stretching of these structures can sometimes cause a hole or perforation. The use of sedatives also carries a risk of decreasing cardiac and respiratory function, however, it is very difficult to perform these procedures without these drugs.

Other major complications related to ERCP or ERS are **pancreatitis** (inflammation of the pancreas) and **cholangitis** (inflammation of the bile ducts). **Bacteremia** (the passage of bacteria into the blood stream) and bleeding are also risks.

Normal results

Certain standards have been set for the diameter or width of the pancreatic and bile ducts. Measurements by x ray are used to determine if the ducts are too large (dilated) or too narrow (strictured). Lastly, the ducts and gallbladder should be free of any solid particles, such as stones, and free of areas of narrowing.

Resources

OTHER

“Endoscopic Retrograde

Cholangiopancreatography.” *American Society for Gastrointestinal Endoscopy*. <<http://www.asge.org>> .

“Treatment of Acute Biliary Pancreatitis.” *New England Journal of Medicine Online*. <<http://content.nejm.org>> .

David Kaminstein, MD

Enemas

Definition

An enema is the insertion of a solution into the rectum and lower intestine.

Purpose

Enemas may be given for the following purposes:

- to remove feces when an individual is constipated or impacted,
- to remove feces and cleanse the rectum in preparation for an examination,
- to remove feces prior to a surgical procedure to prevent contamination of the surgical area,
- to administer drugs or anesthetic agents.

Precautions

The rectal tube used for infusion of the enema solution should be smooth and flexible to decrease the possibility of damage to the mucous membrane that lines the rectum. Tap water is commonly used for adults but should not be used for infants because of the danger of electrolyte (substance that conducts electric current within the body and is essential for sustaining life) imbalance. The colon absorbs water, and repeated tap water enemas can cause cardiovascular overload and electrolyte imbalance. Similarly, repeated saline enemas can cause increased absorption of fluid and electrolytes into the bloodstream, resulting in overload. Individuals receiving frequent enemas should be observed for overload symptoms that include **dizziness**, sweating, or **vomiting**.

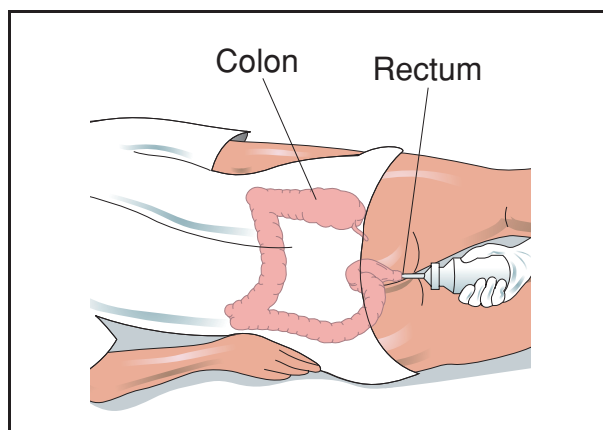
Soap suds and saline used for cleansing enemas can cause irritation of the lining of the bowel, with repeated use or a solution that is too strong. Only white soap should be used; the bar should not have been previously used, to prevent infusing undesirable organisms into the individual receiving the enema. Common household detergents are considered too strong for the rectum and bowel. The commercially prepared castile soap is preferred, and should be used in concentration no greater than 5 cc soap to 1,000 cc of water.

Description

Cleansing enemas act by stimulation of bowel activity through irritation of the lower bowel, and by distention with the volume of fluid instilled. When the enema is administered, the individual is usually lying on the left side, which places the sigmoid colon (lower portion of bowel) below the rectum and facilitates infusion of fluid. The length of time it takes to administer an enema depends on the amount of fluid to be infused. The amount of fluid administered will vary depending on the age and size of the person receiving the enema, however general guidelines would be:

- Infant: 250 cc or less
- Toddler and preschooler: 500 cc or less
- School-aged child: 500–1,000 cc
- Adult: 750–1,000 cc

Some may differentiate between high and low enemas. A high enema, given to cleanse as much of the large bowel as possible, is usually administered at higher pressure and with larger volume (1,000 cc), and the individual changes position several times in order for the fluid to flow up into the bowel. A low



Enemas may be given for the following purposes: to remove feces when an individual is constipated, or to remove feces and cleanse the rectum in preparation for an examination, or prior to surgery to prevent contamination. There are two types of enemas: the high enema, given to cleanse the large bowel, and the low enema, to cleanse only the lower bowel. (Illustration by Electronic Illustrators Group.)

enema, intended to cleanse only the lower bowel, is administered at lower pressure, using about 500 cc of fluid.

Oil retention enemas serve to lubricate the rectum and lower bowel, and soften the stool. For adults, about 150–200 cc of oil is instilled, while in small children, 75–150 cc of oil is considered adequate. Salad oil or liquid petrolatum are commonly used at a temperature of 91°F (32.8°C). There are also commercially prepared oil retention enemas. The oil is usually retained for one to three hours before it is expelled.

The rectal tube used for infusion of the solution, usually made of rubber or plastic, has two or more openings at the end through which the solution can flow into the bowel. The distance to which the tube must be inserted is dependent upon the age and size of the patient. For adult, insertion is usually 3–4 in (7.5–10 cm); for children, approximately 2–3 in (5–7.5 cm); and for infants, only 1–1.5 in (2.5–3.75 cm). The rectal tube is lubricated before insertion with a water soluble lubricant to ease insertion and decrease irritation to the rectal tissues.

The higher the container of solution is placed, the greater the force in which the fluid flows into the patient. Routinely, the container should be no higher than 12 in (30 cm) above the level of the bed; for a high cleansing enema, the container may be 12–18 in (30–45 cm) above the bed level, because the fluid is to be instilled higher into the bowel.

KEY TERMS

Electrolyte—A substance that conducts electric current within the body and is essential for sustaining life.

Intestine—Also called the bowels and divided into large and small intestine, they extend from the stomach to the anus, where waste products exit the body. The small intestine is about 20 ft (6.1m) long and the large intestine, about 5 ft (1.5m) long.

Rectum—The portion of bowel just before the anus. The prefix *recto* is used with a variety of words in relation to conditions that affect the rectum.

Preparation

The solution used in the procedure is measured, mixed, and warmed before administration of the enema.

Aftercare

If necessary, a specimen will be collected for diagnostic evaluation. If the enema was given to alleviate **constipation**, the better approach to combatting constipation in the future is with a high fiber diet (five to six servings of whole grain foods) and adequate fluid intake (seven to eight glasses of water per day). Regular **exercise** and going to the bathroom when necessary will also help. If constipation is a chronic problem, medical help should be consulted to determine if there is underlying disorder.

Risks

Habitual use of enemas as a means to combat constipation can make the problem even more severe when their use is discontinued. Enemas should be used only as a last resort for treatment of constipation and with a doctor's recommendation. Enemas should not be administered to individuals who have recently had colon or rectal surgery, a heart attack, or who suffer from an unknown abdominal condition or an irregular heartbeat.

Resources

OTHER

Eller, D. "Spring Cleaning from the Inside Out." *Remedy* May/June 1997. [cited May 28, 1998]. <<http://thriveonline.oxygen.com>>.

Kathleen D. Wright, RN

Enlarged prostate

Definition

A non-cancerous condition that affects many men past 50 years of age, enlarged prostate makes urinating more difficult by narrowing the urethra, a tube running from the bladder through the prostate gland. It can be effectively treated by surgery and, today, by certain drugs.

Description

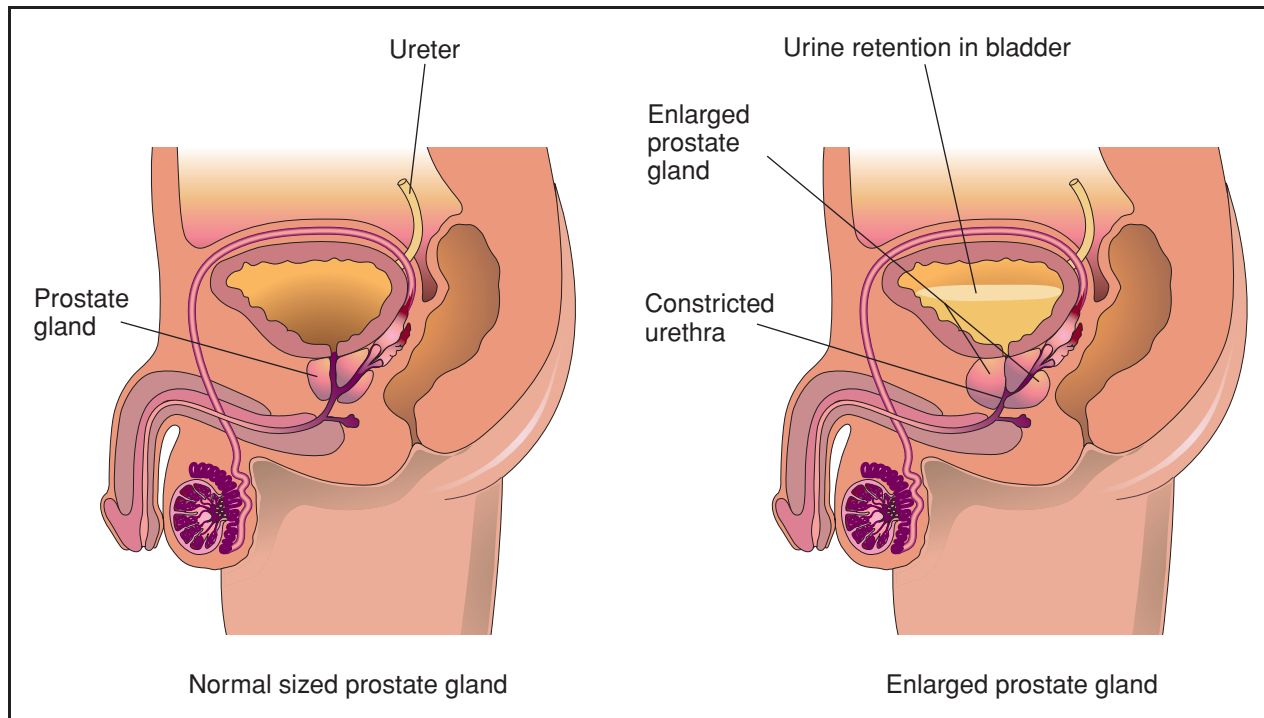
The common term for enlarged prostate is BPH, which stands for benign (non-cancerous) prostatic hyperplasia or hypertrophy. Hyperplasia means that the prostate cells are dividing too rapidly, increasing the total number of cells, and, therefore, the size of the organ itself. Hypertrophy simply means "enlargement." BPH is part of the aging process. The actual changes in the prostate may start as early as the 30s but take place very gradually, so that significant enlargement and symptoms usually do not appear until after age 50. Past this age the chances of the prostate enlarging and causing urinary symptoms become progressively greater. More than 40% of men in their 70s have an enlarged prostate. Symptoms generally appear between ages 55–75. About 10% of all men eventually will require treatment for BPH.

BPH has been viewed as a rare condition in African, Chinese, and other Asian peoples for reasons that are not clear.

Causes and symptoms

The cause of BPH is a mystery, but age-related changes in the levels of hormones circulating in the blood may be a factor. Whatever the cause, an enlarging prostate gradually narrows the urethra and obstructs the flow of urine. Even though the muscle in the bladder wall becomes stronger in an attempt to push urine through the smaller urethra, in time, the bladder fails to empty completely at each urination. The urine that collects in the bladder can become infected and lead to stone formation. The kidneys themselves may be damaged by infection or by urine constantly "backing up."

When the enlarging prostate gland narrows the urethra, a man will have increasing trouble starting the urine stream. Because some urine remains behind in the bladder, he will have to urinate more often, perhaps two or three times at night (nocturia). The need to urinate can become very urgent and, in time, urine may dribble out to stain a man's clothing. Other symptoms of BPH are a weak and sometimes a split



An enlarged prostate is a non-cancerous condition in which the narrowing of the urethra makes the elimination of urine more difficult. It most often occurs in men over age 50. (Illustration by Electronic Illustrators Group.)

stream and general aching or pain in the perineum (the area between the scrotum and anus). Some men may have considerable enlargement of the prostate before even mild symptoms develop.

If a man must strain hard to force out the urine, small veins in the bladder wall and urethra may rupture, causing blood to appear in the urine. If the urinary stream becomes totally blocked, the urine collecting in the bladder may cause severe discomfort, a condition called acute urinary retention. Urine that stagnates in the bladder can easily become infected. A burning feeling during urination and **fever** are clues that infection may have developed. Finally, if urine backs up long enough it may increase pressure in the kidneys, though this rarely causes permanent kidney damage.

Diagnosis

When a man's symptoms point to BPH, the first thing the physician will want to do is a digital **rectal examination**, inserting a finger into the anus to feel whether—and how much—the prostate is enlarged. A smooth prostate surface suggests BPH, whereas a distinct lump in the gland might mean prostate cancer. The next step is a blood test for a substance called prostate-specific antigen or PSA. Between 30–50% of men with

BPH have an elevated PSA level. This does not mean **cancer** by any means, but other measures are needed to make sure that the prostate enlargement is in fact benign. An ultrasound exam of the prostate, which is entirely safe and delivers no radiation, can show whether it is enlarged and may show that cancer is present.

If digital or ultrasound examination of the prostate raises the suspicion of cancer, most urologists will recommend that a prostatic tissue biopsy be performed. This is usually done using a lance-like instrument that is inserted into the rectum. It pierces the rectal wall and, guided by the physician's finger, obtains six to eight pieces of prostatic tissue that are sent to the laboratory for microscopic examination. If cancer is present, the prognosis and treatment are changed accordingly.

A catheter placed through the urethra and into the bladder can show how much urine remains in the bladder after the patient urinates—a measure of how severe the obstruction is. Another and very simple test for obstruction is to have the man urinate into a uroflowmeter, which measures the rate of urine flow. A very certain—though invasive—way of confirming obstruction from an enlarged prostate is to pass a special viewing instrument called a cystoscope into the bladder, but this is not often necessary.

It is routine to check a urine sample for an increased number of white blood cells, which may mean there is infection of the bladder or kidneys. The same sample may be cultured to show what type of bacterium is causing the infection, and which **antibiotics** will work best. The state of the kidneys may be checked in two ways: imaging by either ultrasound or injecting a dye (the intravenous urogram, or pyelogram); or a blood test for creatinine, which collects in the blood when the kidneys cannot eliminate it.

Treatment

Drugs

A class of drugs called alpha-adrenergic blockers, which includes phenoxybenzamine and doxazosin, relax the muscle tissue surrounding the bladder outlet and lining the wall of the urethra to permit urine to flow more freely. These drugs improve obstructive symptoms, but do not keep the prostate from enlarging. Other drugs (**finasteride** is a good example) do shrink the prostate and may delay the need for surgery. Symptoms may not, however, improve until the drug has been used for three months or longer. Antibiotic drugs are given promptly whenever infection is diagnosed. Some medications, including **antihistamines** and some **decongestants**, can make the symptoms of BPH suddenly worse and even cause acute urinary retention, and therefore should be avoided.

Intermediate treatments

When drugs have failed to control symptoms of BPH but the physician does not believe that conventional surgery is yet needed, a procedure called transurethral needle ablation may be tried. In the office and using **local anesthesia**, a needle is inserted into the prostate and radiofrequency energy is applied to destroy the tissue that is obstructing urine flow. Another new approach is microwave hyperthermia, using a device called the Prostatron to deliver microwave energy to the prostate through a catheter. This procedure is done at an outpatient surgery center.

Surgery

For many years the standard operation for BPH has been transurethral resection (TUR) of the prostate. Under general or spinal anesthesia, a cystoscope is passed through the urethra and prostate tissue surrounding the urethra is removed using either a cutting

instrument or a heated wire loop. The small pieces of prostate tissue are washed out through the scope. No incision is needed for TUR. There normally is some blood in the urine for a few days following the procedure. In a few men—less than 5% of all those having TUR—urine will continue to escape unintentionally. Other uncommon complications include a temporary rise in blood pressure with mental confusion, which is treated by giving salt solution. Impotence—the inability to achieve lasting penile erections—does occur, but probably in fewer than 10% of patients. A narrowing or stricture rarely develops in the urethra, but this can be treated fairly easily.

Alternatives to TUR, some only recently introduced, include:

- Laser ablation of the prostate. Laser energy is applied to the prostate through a special fiber passed through a cystoscope. The procedure is done in an operating room, and several patients have retained urine postoperatively.
- Transurethral incision of the prostate. Less invasive than standard TUR, an incision is made through the prostate to open up the part of the urethra passing through it. This may work well in men whose prostate is not grossly enlarged.
- Transurethral vaporization. A small roller ball is used to break up and vaporize the obstructing prostatic tissue, rather than cutting it away as in standard TUR. This is equally successful but patients usually can leave the hospital within 24 hours, and there is less blood loss.
- If the prostate is greatly enlarged—as is the case in about 5–10% of those diagnosed – an incision is made to perform an open **prostatectomy**, removing the entire gland under direct vision.

Alternative treatment

An extract of the **saw palmetto** (*Serenoa repens* or *S. serrulata*) has been shown to stop or decrease the hyperplasia of the prostate. Symptoms of BPH will improve after taking the herb for one to two months, but continued use is recommended.

Prognosis

In a man without symptoms whose prostate is enlarged, it is hard to predict when urinary symptoms will develop and how rapidly they will progress. For this reason some specialists (urologists) advise a period of “watchful waiting.” When BPH is treated by conventional TUR, there is a small risk of

KEY TERMS

Catheter—A rubber or plastic tube placed through the urethra into the bladder to remove excess urine when the flow of urine is cut off, or to prevent urinary infection.

Creatinine—One of the “waste” substances normally excreted by the kidneys into the urine. When urine flow is slowed, creatinine may collect in the blood and cause toxic effects.

Hyperplasia—A condition where cells, such as those making up the prostate gland, rapidly divide abnormally and cause the organ to become enlarged.

Hypertrophy—A technical term for enlargement, as in BPH (benign prostatic hypertrophy).

Urethra—In males, the tube that conducts urine from the bladder through the penis to the outside of the body. When narrowed by an enlarging prostate, symptoms of BPH develop.

Urinary retention—The result of progressive obstruction of the urethra by an enlarging prostate, causing urine to remain in the bladder even after urination.

complications but, in the great majority of men, urinary symptoms will be relieved and their quality of life will be much enhanced. In the future, it is possible that the less invasive forms of surgical treatment will be increasingly used to achieve results as good as those of the standard operation. It also is possible that new medications will be developed that shrink the prostate and eliminate obstructive symptoms so that surgery can be avoided altogether.

Prevention

Whether or not BPH is caused by hormonal changes in **aging** men, there is no known way of preventing it. Once it does develop and symptoms are present that interfere seriously with the patient's life, timely medical or surgical treatment will reliably prevent symptoms from getting worse. Also, if the condition is treated before the prostate has become grossly enlarged, the risk of complications is minimal. One of the potentially most serious complications of BPH, urinary infection (and possible infection of the kidneys), can be prevented by using a catheter to drain excess urine out of the bladder so that it does not collect, stagnate, and become infected.

Resources

ORGANIZATIONS

Prostate Health Council. American Foundation for Urologic Disease. 1128 N. Charles St., Baltimore, MD 21201. (800) 242-AFUD.

David A. Cramer, MD

Entamoeba histolytica infection see
Amebiasis

Enteric fever see **Typhoid fever**

Enterically transmitted non-A non-B see
Hepatitis E

Enterobacterial infections

Definition

Enterobacterial infections are disorders of the digestive tract and other organ systems produced by a group of gram-negative, rod-shaped bacteria called Enterobacteriaceae. Gram-negative means that the organisms do not retain the violet color of the dye used to make Gram stains. The most troublesome organism in this group is *Escherichia coli*. Other enterobacteria are species of *Salmonella*, *Shigella*, *Klebsiella*, *Enterobacter*, *Serratia*, *Proteus*, and *Yersinia*.

Description

Enterobacterial infections can be produced by bacteria that normally live in the human digestive tract without causing serious disease, or by bacteria that enter from the outside. In many cases these infections are nosocomial, which means that they can be acquired in the hospital. *Klebsiella* and *Proteus* sometimes cause **pneumonia**, ear and sinus infections, and urinary tract infections. *Enterobacter* and *Serratia* often cause bacterial infection of the blood (**bacteremia**), particularly in patients with weakened immune systems.

Diarrhea caused by enterobacteria is a common problem in the United States. It is estimated that each person in the general population has an average of 1.5 episodes of diarrhea each year, with higher rates in children, institutionalized people, and Native Americans. This type of enterobacterial infection can range from a minor nuisance to a life-threatening disorder, especially in infants, elderly persons, **AIDS**

patients, and malnourished people. Enterobacterial infections are one of the two leading killers of children in developing countries.

Causes and symptoms

Causes

Enterobacterial infections in the digestive tract typically start when the organisms invade the mucous tissues that line the digestive tract. They may be bacteria that are already present in the stomach and intestines, or they may be transmitted by contaminated food and water. It is also possible for enterobacterial infections to spread by person-to-person contact. The usual incubation period is 12–72 hours.

ESCHERICHIA COLI INFECTIONS. *E. coli* infections cause most of the enterobacterial infections in the United States. The organisms are categorized according to whether they are invasive or noninvasive. Noninvasive types of *E. coli* include what are called enteropathogenic *E. coli*, or EPEC, and enterotoxigenic *E. coli*, or ETEC. EPEC and ETEC types produce a bacterial poison (toxin) in the stomach that interacts with the digestive juices and causes the patient to lose large amounts of water through the intestines.

The invasive types of *E. coli* are called enterohemorrhagic *E. coli*, or EHEC, and enteroinvasive *E. coli*, or EIEC. These subtypes invade the stomach tissues directly, causing tissue destruction and bloody stools. EHEC can produce complications leading to **hemolytic-uremic syndrome (HUS)**, a potentially fatal disorder marked by the destruction of red blood cells and kidney failure. EHEC has become a growing problem in the United States because of outbreaks caused by contaminated food. A particular type of EHEC known as O157:H7 has been identified since 1982 in undercooked hamburgers, unpasteurized milk, and apple juice. Between 2–7% of infections caused by O157:H7 develop into HUS.

Symptoms

The symptoms of enterobacterial infections are sometimes classified according to the type of diarrhea they produce.

WATERY DIARRHEA. Patients infected with ETEC, EPEC, some types of *Salmonella*, and some types of *Shigella* develop a watery diarrhea. These infections are located in the small intestine, result from bacterial toxins interacting with digestive juices, do not produce inflammation; and do not usually need treatment with **antibiotics**.

BLOODY DIARRHEA (DYSENTERY). Bloody diarrhea is sometimes called **dysentery**. It is produced by EHEC, EIEC, some types of *Salmonella*, some types of *Shigella*, and *Yersinia*. In dysentery, the infection is located in the colon, cells and tissues are destroyed, inflammation is present, and antibiotic therapy is usually required.

NECROTIZING ENTEROCOLITIS (NEC). **Necrotizing enterocolitis (NEC)** is a disorder that begins in newborn infants shortly after birth. Although NEC is not yet fully understood, it is thought that it results from a bacterial or viral invasion of damaged intestinal tissues. The disease organisms then cause the death (necrosis) of bowel tissue or **gangrene** of the bowel. NEC is primarily a disease of **prematurity**; 60–80% of cases occur in high-risk preterm infants. NEC is responsible for 2–5% of cases in newborn intensive care units (NICU). Enterobacteriaceae that have been identified in infants with NEC include *Salmonella*, *E. coli*, *Klebsiella*, and *Enterobacter*.

Diagnosis

Patient history

The diagnosis of enterobacterial infections is complicated by the fact that viruses, protozoa, and other types of bacteria can also cause diarrhea. In most cases of mild diarrhea, it is not critical to identify the organism because the disorder is self-limiting. Some groups of patients, however, should have stool tests. They include:

- patients with bloody diarrhea,
- patients with watery diarrhea who have become dehydrated,
- patients with watery diarrhea that has lasted longer than three days without decreasing in amount,
- patients with disorders of the immune system.

The patient history is useful for public health reasons as well as helping the doctor determine what type of enterobacterium may be causing the infection. The doctor will ask about the frequency and appearance of the diarrhea as well as other digestive symptoms. If the patient is nauseated and **vomiting**, the infection is more likely to be located in the small intestine. If the patient is running a **fever**, a diagnosis of dysentery is more likely. The doctor will also ask if anyone else in the patient's family or workplace is sick. Some types of enterobacteriaceae are more likely to cause group outbreaks than others. Other questions include the patient's food intake over the last few days and whether he or she has recently traveled to countries with **typhoid fever** or **cholera** outbreaks.

Physical examination

The most important parts of the **physical examination** are checking for signs of severe fluid loss and examining the abdomen to rule out typhoid fever. The doctor will look at the inside of the patient's mouth and evaluate the skin for signs of **dehydration**. The presence of a skin rash and an enlarged spleen suggests typhoid rather than a bacterial infection. If the patient's abdomen hurts when the doctor examines it, a diagnosis of dysentery is more likely.

Laboratory tests

The most common test that is used to identify the cause of diarrhea is the stool test. Examining a stool sample under a microscope can help to rule out parasitic and protozoal infections. Routine stool cultures, however, cannot be used to identify any of the four types of *E. coli* that cause intestinal infections. ETEC, EPEC, and EIEC are unusual in the United States and can usually be identified only by specialists in research laboratories. Because of concern about EHEC outbreaks, however, most laboratories in the United States can now screen for O157:H7 with a test that identifies its characteristic toxin. All patients with bloody diarrhea should have a stool sample tested for *E. coli* O157:H7.

Treatment

The initial treatment of enterobacterial diarrhea is usually empiric. Empiric means that the doctor treats the patient on the basis of the visible symptoms and professional experience in treating infections, without waiting for laboratory test results. Since the results of stool cultures can take as long as two days, it is important to prevent dehydration. The patient will be given fluids to restore the electrolyte balance and paregoric to relieve abdominal cramping.

Newborn infants and patients with immune system disorders will be given antibiotics intravenously once the organism has been identified. Gentamicin, tobramycin, and amikacin are being used more frequently to treat enterobacterial infections because many of the organisms are becoming resistant to ampicillin and cephalosporin antibiotics.

Alternative treatment

Alternative treatments for diarrhea are intended to relieve the discomfort of abdominal cramping. Most alternative practitioners advise consulting a medical doctor if the patient has sunken eyes, dry eyes or mouth, or other signs of dehydration.

KEY TERMS

Dysentery—A type of diarrhea caused by infection and characterized by mucus and blood in the stools.

Empirical treatment—Medical treatment that is given on the basis of the doctor's observations and experience.

Escherichia coli—A type of enterobacterium that is responsible for most cases of severe bacterial diarrhea in the United States.

Hemolytic-uremic syndrome (HUS)—A potentially fatal complication of *E. coli* infections characterized by kidney failure and destruction of red blood cells.

Necrotizing enterocolitis (NEC)—A disorder in newborns caused by bacterial or viral invasion of vulnerable intestinal tissues.

Nosocomial infections—Infections acquired in hospitals.

Toxin—A poison produced by certain types of bacteria.

Herbal medicine

Herbalists may recommend cloves taken as an infusion or ginger given in drop doses to control intestinal cramps, eliminate gas, and prevent vomiting. Peppermint (*Mentha piperita*) or chamomile (*Matricaria recutita*) tea may also ease cramps and intestinal spasms.

Homeopathy

Homeopathic practitioners frequently recommend *Arsenicum album* for diarrhea caused by contaminated food, and *Belladonna* for diarrhea that comes on suddenly with mucus in the stools. *Veratrum album* would be given for watery diarrhea, and *Podophyllum* for diarrhea with few other symptoms.

Prognosis

The prognosis for most enterobacterial infections is good; most patients recover in about a week or 10 days without needing antibiotics. HUS, on the other hand, has a mortality rate of 3–5% even with intensive care. About a third of the survivors have long-term problems with kidney function, and another 8%

develop high blood pressure, seizure disorders, and blindness.

Prevention

The World Health Organization (WHO) offers the following suggestions for preventing enterobacterial infections, including *E. coli* O157:H7 dysentery:

- Cook ground beef or hamburgers until the meat is thoroughly done. Juices from the meat should be completely clear, not pink or red. All parts of the meat should reach a temperature of 70°C (158°F) or higher.
- Do not drink unpasteurized milk or use products made from raw milk.
- Wash hands thoroughly and frequently, especially after using the toilet.
- Wash fruits and vegetables carefully, or peel them. Keep all kitchen surfaces and serving utensils clean.
- If drinking water is not known to be safe, boil it or drink bottled water.
- Keep cooked foods separate from raw foods, and avoid touching cooked foods with knives or other utensils that have been used with raw meat.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>> .

Rebecca J. Frey, PhD

Enterobiasis

Definition

Enterobiasis, or pinworm infection as it is commonly called, is an intestinal infection caused by the parasitic roundworm called *Enterobius vermicularis*. The most common symptom of this irritating, but not particularly dangerous, disease is **itching** around the anal area.

Description

Enterobiasis is also called seatworm infection or oxyuriasis. In the United States, enterobiasis is the most common worm infection, and some estimate that approximately 10% of the United States

population is infected. Worldwide, approximately 200 million people are infected. Enterobiasis can affect people of any age, but is most common among children ages 5–14 and particularly affects those in the daycare setting.

Causes and symptoms

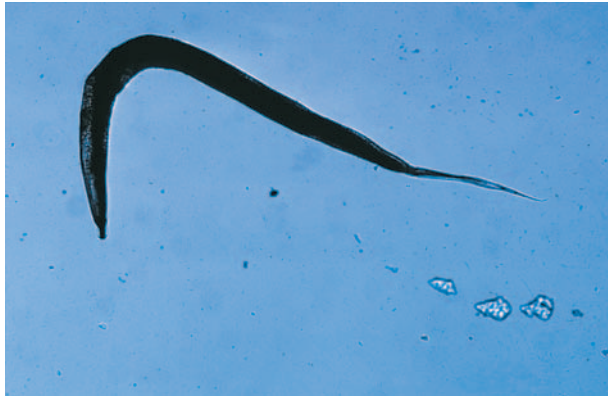
The disease is highly contagious and is caused by a parasitic worm called *Enterobius vermicularis*. The adult female worm is about the size of a staple (approximately 0.4 in [1 cm long] and 0.02 in [0.5 mm] wide) and has a pointed tip. The disease is transmitted by ingesting the eggs of the pinworm. These eggs travel to the small intestine where, after approximately one month, they hatch and mature into adult worms. During the night, the female adult worms travel to the area around the anus and deposit eggs in the folds of the anal area. A single female pinworm can lay 10,000 eggs and, after laying eggs, dies. The eggs are capable of causing infection after six hours at body temperature.

Significant itching in the anal region is caused by the movement of the adult worm as the eggs are deposited. When an individual scratches the anal region, the tiny eggs get under the finger nails and in the underwear and night clothes. Anything the individual touches with the contaminated fingers, for example, toys, bedding, blankets, bathroom door knobs, or sinks, becomes contaminated. The eggs are very hardy and can live on surfaces for two to three weeks. Anyone touching these contaminated surfaces can ingest the eggs and become infected. An individual can also become infected by inhaling and swallowing the eggs, for example, when the bedcovers are shaken.

Many individuals with enterobiasis exhibit no symptoms. When present, however, symptoms of the infection begin approximately two weeks after ingesting the pinworm eggs. The main symptom is itching around the anus. Because the itching intensifies at night, when the female worms comes to the anus to lay eggs, it often leads to disrupted sleep and irritability. Poor sleeping at night in small children can be related to pinworms. Occasionally, the itching causes some bleeding and bruising in the region, and secondary bacterial infections can occur. In females, the itching may spread to the vagina and sometimes causes an infection of the vaginal region (vaginitis). Enterobiasis usually lasts one to two months.

Diagnosis

First, a physician will rule out other potential causes of the itching, such as **hemorrhoids**, lice, or



The pinworm of the genus *Enterobius* pictured above is the source of this infestation occurring in children. (Photo Researchers, Inc. Reproduced by permission.)

fungal or bacterial infection. Once these have been ruled out, an accurate diagnosis of enterobiasis will require that either the eggs or the adult worms are detected. Rarely, the adult worms are seen as thin, yellowish-white threads, about 0.4 in (1 cm) long, in the stools of the infected person. Usually, an hour or so after the individual goes to sleep, the adult female worms may be seen moving around laying eggs if a flashlight is shone at the rectal area.

An easier method is to observe the eggs under the microscope. In order to collect a specimen for laboratory diagnosis, the physician may provide a paddle with a sticky adhesive on one side, or an individual may be instructed to place a piece of shiny cellophane tape sticky side down against the anal opening. The best time to perform this test is at night or as soon as the individual wakes up in the morning, before having a bowel movement or taking a bath or shower. The pinworm eggs will stick to the tape, which can then be placed on a specimen slide. When under a microscope in the laboratory, the eggs will be clearly visible.

Treatment

In order to treat the disease, either mebendazole (Vermox) or pyrantel pamoate (Pin-X) will be given in two oral doses spaced two weeks apart. These medications eradicate the infection in approximately 90% of cases. Re-infection is common and several treatments may be required. Because the infection is easily spread through contact with contaminated clothing or surfaces, it is recommended that all family members receive the therapeutic dose. Sometimes a series of six

KEY TERMS

Anus—The opening through which feces are eliminated.

Hemorrhoid—An area around the anus where veins become dilated and the tissue swells, causing itching and pain.

Rectum—The end of the large intestine in which feces collects for elimination through the anus.

Vaginitis—Inflammation of the vagina.

treatments are given, each spaced two weeks apart. If family members continue to be infected, a source outside the house may be responsible.

To relieve the rectal itching, a shallow warm bath with either half a cup of table salt, or Epsom salts is recommended. Also, application of an ointment containing zinc oxide or regular petroleum jelly can be used to relieve rectal itching.

Prognosis

Pinworms cause little damage and can be easily eradicated with proper treatment. Full recovery is expected.

Prevention

The disease can be prevented by treating all the infected cases and thus eliminating the source of infection. Some ways to keep from catching or spreading the disease include the following recommendations:

- wash hands thoroughly before handling food and eating
- keep finger nails short and clean
- avoiding scratching the anal area
- take early morning showers to wash away eggs deposited overnight
- once the infection has been identified, and treatment is started, change the bed linen, night clothes, and underwear daily
- machine wash linens in hot water and dry with heat to kill any eggs
- open the blinds or curtains since eggs are sensitive to sunlight

Resources

BOOKS

Fauci, Anthony S., et al., editors. *Harrison's Principles of Internal Medicine*. New York: McGraw-Hill, 1997.

Lata Cherath, PhD

Enterohemorrhagic *E. coli* see **Escherichia coli**

Enterostomy

Definition

An enterostomy is an operation in which the surgeon makes a passage into the patient's small intestine through the abdomen with an opening to allow for drainage or to insert a tube for feeding. The opening is called a stoma, from the Greek word for mouth. Enterostomies may be either temporary or permanent. They are classified according to the part of the intestine that is used to create the stoma. If the ileum, which is the lowest of the three sections of the small intestine, is used to make the stoma, the operation is called an ileostomy. If the jejunum, which is the middle section of the small intestine, is used, the operation is called a jejunostomy. Some people use the word *ostomy* as a word that covers all types of enterostomies.

Purpose

Enterostomies are performed in order to create a new opening for the passage of fecal matter when normal intestinal functioning is interrupted or when diseases of the intestines cannot be treated by medications or less radical surgery. Some situations that may require enterostomies include:

- Healing of inflamed bowel segments. Enterostomies performed for this reason are usually temporary.
- Emergency treatment of gunshot or other penetrating wounds of the abdomen. An enterostomy is needed to prevent the contents of the intestine from causing a serious inflammation of the inside of the abdominal cavity (**peritonitis**). These enterostomies are also often temporary.
- Placement of a tube for enteral feeding. Enteral feeding is a method for conveying nutritional solutions directly into the stomach or jejunum through a tube. Tube enterostomies may be long-term but are not permanent.

- Removal of diseased sections of the intestines. Ileostomies performed for this reason are permanent. The most common disorders requiring permanent ileostomy are **Crohn's disease**, **familial polyposis**, and ulcerative colitis. Familial polyposis and **ulcerative colitis** are serious health risks because they can develop into **cancer**.
- Treatment of advanced cancer or other causes of intestinal obstruction.

Precautions

Enterostomies are usually performed only as emergency treatments for traumatic injuries in the abdomen or as final measures for serious disorders of the intestines. Most patients do not refuse to have the operation performed when the need for it is explained to them. A small minority, however, refuse enterostomies because of strong psychological reactions to personal disfigurement and the need to relearn bowel habits.

Description

Ileostomy

Ileostomies represent about 25% of enterostomies. They are performed after the surgeon removes a diseased colon and sometimes the rectum as well. The most common ileostomy is called a Brooke ileostomy after the English surgeon who developed it. In a Brooke ileostomy, the surgeon makes the stoma in the lower right section of the abdomen. The ileum is pulled through an opening (incision) in the muscle layer. The surgeon then turns the cut end of the intestine inside out and sews it to the edges of the hole. He or she then positions an appliance for collecting the fecal material. The appliance consists of a plastic bag that fits over the stoma and lies flat against the abdomen. The patient is taught to drain the bag from time to time during the day. Ileostomies need to be emptied frequently because the digested food contains large amounts of water. Shortly after the operation, the ileostomy produces 1–2 qt. (0.9–1.9 l) of fluid per day; after a month or two of adjustment, the volume decreases to 1–2 pt (0.5–0.9) per day.

KOCK POUCH (CONTINENT ILEOSTOMY). The Kock pouch is a variation of the basic ileostomy and is named for its Swedish inventor. In the Kock technique, the surgeon forms a pouch inside the abdominal cavity behind the stoma that collects the fecal material. The stoma is shaped into a valve to prevent fluid from leaking onto the patient's abdomen. The patient then empties the pouch several times daily by inserting a tube (catheter) through the valve. The Kock technique

is sometimes called a continent ileostomy because the fluid is contained inside the abdomen. It is successful in 70–90% of patients who have it done.

Jejunostomy

A jejunostomy is similar to an ileostomy except that the stoma is placed in the second section of the small intestine rather than the third. Jejunostomies are performed less frequently than ileostomies. They are almost always temporary procedures.

Tube enterostomies

Tube enterostomies are operations in which the surgeon makes a stoma into the stomach itself or the jejunum in order to insert a tube for liquid nutrients. Tube enterostomies are performed in patients who need tube feeding for longer than six weeks, or who have had recent mouth or nose surgery. As long as the patient's intestinal tract can function, tube feedings are considered preferable to intravenous feeding. Enteral **nutrition** is safer than intravenous fluids and helps to keep the patient's digestive tract functioning.

Preparation

Preoperative preparation includes both patient education and physical preparation.

Patient education

If the patient is going to have a permanent ileostomy, the doctor will explain what will happen during the operation and why it is necessary. Most patients are willing to accept an ostomy as an alternative to the chronic **pain** and **diarrhea** of ulcerative colitis or the risk of cancer from other intestinal disorders. The patient can also meet with an enterostomal therapist (ET) or a member of the United **Ostomy** Association, which is a support group for people with ostomies.

Medical preparation

The patient is prepared for surgery with an evaluation of his or her nutritional status, possible need for blood transfusions, and **antibiotics** if necessary. If the patient does not have an intestinal obstruction or severe inflammation, he or she may be given a large quantity of a polyethylene glycol (PEG) solution to cleanse the intestines before surgery.

Aftercare

Aftercare of an enterostomy is both psychological and medical.

Medical aftercare

If the enterostomy is temporary, aftercare consists of the usual monitoring of surgical wounds for infection or bleeding. If the patient has had a permanent ileostomy, aftercare includes learning to use the appliance or empty the Kock pouch; learning to keep the stoma clean; and readjusting bathroom habits. Recovery takes a long time because major surgery is a shock to the system and the intestines take several days to resume normal functioning. The patient's fluid intake and output will be checked frequently to minimize the risk of dehydration.

Patient education

Ileostomy patients must learn to watch their fluid and salt intake. They are at greater risk of becoming dehydrated in hot weather, from **exercise**, or from diarrhea. In some cases they may need extra bananas or orange juice in the diet to keep up the level of potassium in the blood.

Patient education includes social concerns as well as physical self-care. Many ileostomy patients are worried about the effects of the operation on their close relationships and employment. If the patient has not seen an ET before the operation, the aftercare period is a good time to find out about self-help and support groups. The ET can also evaluate the patient's emotional reactions to the ostomy.

Risks

Enterostomies are not considered high-risk operations by themselves. About 40% of ileostomy patients have complications afterward, however; about 15% require minor surgical corrections. Possible complications include:

- skin irritation caused by leakage of digestive fluids onto the skin around the stoma; Irritation is the most common complication of ileostomies
- diarrhea
- the development of abscesses
- gallstones or stones in the urinary tract
- inflammation of the ileum
- odors can often be prevented by a change in diet
- intestinal obstruction
- a section of the bowel pushing out of the body (prolapse)

KEY TERMS

Crohn's disease—A disease of the intestines that causes inflammation leading to scarring, thickening of the walls of the intestine, and eventual obstruction.

Duodenum—The first of the three segments of the small intestine. The duodenum connects the stomach and the jejunum.

Enteral nutrition—A technique for feeding patients with liquid formulas conveyed directly into the stomach or jejunum through tubes.

Enterostomal therapist (ET)—A specialized counselor, usually a registered nurse, who provides ostomy patients with education and counseling before the operation. After surgery, the ET helps the patient learn to take care of the stoma and appliance, and offers long-term emotional support.

Familial polyposis—A disease that runs in families in which lumps of tissue (polyps) form inside the colon. Familial polyposis may develop into cancer.

Ileum—The third segment of the small intestine, connecting the jejunum and the large intestine.

Jejunum—The second of the three segments of the small intestine, connecting the duodenum and the ileum.

Kock pouch—A type of ileostomy in which the surgeon forms an artificial rectum from a section of the ileum. A Kock pouch is sometimes called a continent ileostomy because it is drained with a tube.

Ostomy—A common term for all types of enterostomies.

Stoma—The surgically constructed mouth or passage between the intestine and the outside of the patient's body.

Tube enterostomy—An enterostomy performed to allow the insertion of a feeding tube into the jejunum or stomach.

Ulcerative colitis—A disease of the colon characterized by inflammation of the mucous lining, ulcerated areas of tissue, and bloody diarrhea.

Normal results

Normal results include recovery from the surgery with few or no complications. About 95% of people with ostomies recover completely, are able to return to work, and consider themselves to be in good health.

Many ileostomy patients enjoy being able to eat a full range of foods rather than living on a restricted diet. Some patients, however, need to be referred to psychotherapists to deal with depression or other emotional problems after the operation.

Resources

ORGANIZATIONS

United Ostomy Association, Inc. (UOA). 19772 MacArthur Blvd., Suite 200, Irvine, CA 92612-2405. (800) 826-0826. <<http://www.uoa.org>> .

Rebecca J. Frey, PhD

Enterovirus infections

Definition

Enteroviruses are so named because they reproduce initially in the gastrointestinal tract after infection occurs. Despite this, they usually do not lead to intestinal symptoms; rather it is their spread to organs, such as the nervous system, heart, skin, and others that causes disease. Enteroviruses are part of a larger group of viruses known as Picornaviruses. The word comes from the combination of “pico” (Spanish, meaning “a little bit”), and RNA (ribonucleic acid, an important component of genetic material).

Description

There are four groups of enteroviruses: Coxsackievirus, Echovirus, ungrouped Enterovirus, and Y Poliovirus.

Viruses are generally divided into those that use DNA (deoxyribonucleic acid) or RNA as their genetic material; all enteroviruses are RNA viruses. They are found worldwide, but infection is more common in areas of poor hygiene and overcrowding.

Although most cases of enterovirus do not produce symptoms, some five to 10 million individuals in the United States each year suffer from one of the enteroviral diseases. Illness is more common in the very young. While there are close to 70 different strains of enteroviruses, over 70% of infections are caused by only 10 types.

The virus is most commonly transmitted by the fecal-oral route (contamination of fingers or objects by human waste material); in some instances transmission is through contaminated food or water.

Passage of some strains of virus by way of air droplets can lead to respiratory illness. Infection of fetuses by way of the placenta also has been documented. Breast milk contains antibodies which can protect newborns.

The incubation period for most enteroviruses ranges from two to 14 days. In areas of temperate climate, infections occur mainly in the summer and fall.

Causes and symptoms

Enteroviruses are believed to be the cause of at least 10 distinct illnesses. Once they enter the body, they multiply in the cells that line the gastrointestinal tract, and eventually reach sites of lymphatic tissue (such as the tonsils). While most of these diseases are of short duration and do not cause significant injury, some can produce severe illness. Each presents its own unique symptoms. And a 2003 report to the Infectious Diseases Society of America reminded physicians that infants with enteroviral infections often present early in their illnesses with no signs of **fever**, complicating diagnosis.

The main syndromes caused by the various enteroviruses are the following:

- Summer gripe (nonspecific febrile illness). This is the most common syndrome, and is characterized by flu-like symptoms of fever, **headache**, and weakness, that typically last three to four days. Many patients also develop upper respiratory symptoms and some **nausea and vomiting**. One of the major ways to distinguish this disease from **influenza**, is the fact that gripe most often occurs in the summer.
- Generalized disease of the newborn is a potentially serious infection in which infants from one week to three months of age develop a syndrome that can be difficult to distinguish from a severe bacterial infection. Fever, irritability, and decreased responsiveness or excessive sleepiness are the major symptoms. Inflammation of heart muscle (**myocarditis**), low blood pressure, hepatitis, and **meningitis** sometimes complicate the illness.
- Aseptic meningitis **encephalitis** is a well known syndrome caused by this group of viruses. In fact, enteroviruses are responsible for over 90% of cases of aseptic meningitis, and most often hit children and young adults. Headache, fever, avoidance of light, and eye **pain** are characteristic. Drowsiness may be prominent, and other symptoms include **sore throat**, **cough**, muscle pain, and rash. Occasionally, not only the meninges;mdash;the covering around the brain and spinal cord;mdash;is infected, but also brain tissue itself, producing encephalitis. The illness resolves after about a week or so, and permanent damage is unusual. Enteroviruses can also produce the Guillian-Barré syndrome, which involves weakness and **paralysis** of the extremities and even the muscles of respiration.
- Pleurodynia (Bornholm's disease) is due to viral infection and inflammation of the chest and abdominal muscles used for breathing. Pain occurs as acute episodes, lasting 30 minutes or so. Coxsackie B virus is the usual cause of the illness.
- Myocarditis and/or **pericarditis** involves infection of the heart muscle (myocardium) and the covering around the heart (pericardium). Infants and young adults are the most susceptible, and for some reason, more than two-thirds of cases occur in males. The disease usually begins as an upper respiratory tract infection with cough, **shortness of breath**, and fever. Chest pain, increasing shortness of breath, irregularities of cardiac rhythm, and **heart failure** sometimes develop. Some patients wind up with long-term heart failure if the heart muscle is significantly affected.
- Exanthems is the medical term for **rashes**, and enterovirus is the number one cause of summer and fall rashes in children. They occur anywhere on the body, and often resemble diseases such as **measles**.
- Hand-foot-and-mouth disease occurs initially as a sore throat (often involving the tongue as well), and is followed by a rash on the hands, and sometimes the feet. The rash often forms small blisters, which lead to ulcers. Symptoms generally resolve within a week. A specific Coxsackievirus (A16) is the most frequent cause of this highly infectious disease.
- Herpangina is most often caused by one of the Coxsackie A viruses, and appears as the acute onset of fever and sore throat. This last symptom is particularly severe, as the virus produces multiple ulcers in the throat. Swallowing becomes very painful; symptoms can persist for several weeks.
- Acute hemorrhagic **conjunctivitis** involves viral infection of the conjunctiva, which is a covering around the eye. Pain, blurred vision, aversion to light, and a discharge from the eye are the main symptoms. Headache and fever occur in about one in five patients. The disease runs its course in about 10 days.

A number of other illnesses have been attributed to enteroviruses, including **pneumonia** and other respiratory infections, **myositis** or muscle inflammation, arthritis, and acute inflammation of the kidneys.

KEY TERMS

Antibodies—Proteins that are formed by the body and play a role in defense against infection.

Antibiotic—A medication that is designed to kill or weaken bacteria.

Meninges—Outer covering of the spinal cord and brain. Infection is called meningitis, which can lead to damage to the brain or spinal cord and lead to death.

It is clear then that these viruses produce a number of various illnesses, most often in younger age groups.

Diagnosis

In the majority of cases, diagnosis is based on the characteristic symptoms that the virus produces (such as the chest pain in pleurodynia). Rarely is it necessary to identify a specific strain of virus causing the illness. It is more important to be certain that the infection is due to a virus that does not require treatment with **antibiotics**.

Culture, or growing the organism outside of the body, is helpful only when obtained from areas that tend to indicate recent infection, such as from swollen joints, cerebrospinal fluid, or blood. Cultures from other areas, such as the throat, can be misleading. This is because the virus may remain for long periods of time in places with a large amount of lymphatic tissue. As a rule, cultures done early in the illness are more likely to identify the virus.

New techniques that involve identification of viral genetic material (PCR) are useful in certain cases, but are not indicated for routine testing.

Treatment

As noted above, enterovirus is capable of attacking many different organs and producing a variety of symptoms. Most infections are mild and improve without complications, requiring no specific therapy. When the virus attacks critical organs however, such as the heart, respiratory muscles, nervous system, etc., specialized care is often needed.

As of 2001, no effective antiviral medication for enterovirus has undergone investigation in patients, though some drugs appeared promising for the future. In some patients who are unable to produce antibodies

(hypogammaglobulinemia), administering antibodies themselves is helpful.

Prognosis

The overall outlook for enterovirus infection depends on the organs involved, and the immune condition of the individual patient. Unless vital organs are involved or immunity is abnormal, infection causes few problems. On the other hand, patients who have diseases that affect antibody production can develop chronic infection of the brain or meninges. A 2003 study found that enterovirus infections can increase the risk of type 1 diabetes in children who are genetically predisposed to diabetes.

Prevention

In the hospital setting, the best means of avoiding transmission of infection is the use of good hand-washing practices and other appropriate precautions (gowns and gloves for hospital staff). The virus is found in feces for up to one week after infection; therefore precautions that isolate waste material (enteric precautions) will help decrease the chance of spreading the illness.

Resources

PERIODICALS

“Enterovirus Infections Increase Risk of Type 1 Diabetes in High-Risk Children.” *Diabetes Week* June 16, 2003: 22.

Tucker, Miriam E. “Fever Often Absent in Early Enteroviral Illness (Severe Cases).” *Pediatric News* January 2003: 20–22.

OTHER

“Hand, Foot and Mouth Disease.” *The Picornavirus Home Page*. <<http://www.iah.bbsrc.ac.uk/virus/Picornaviridae/picornavirus.htm>> .

“Weekly Clinicopathological Exercises: Case 47- 1993: A 28-Year-Old Man with Recurrent Ventricular Tachycardia and Dysfunction of Multiple Organs.” *New England Journal of Medicine Online*. <<http://content.nejm.org>> .

David Kaminstein, MD
Teresa G. Odle

Entropy see **Eyelid disorders**

Enuresis see **Bed-wetting**

Environmental medicine see **Wilderness medicine**

Enzyme therapy

Definition

Enzyme therapy is a plan of dietary supplements of plant and animal enzymes used to facilitate the digestive process and improve the body's ability to maintain balanced metabolism.

Purpose

In traditional medicine, enzyme supplements are often prescribed for patients suffering from disorders that affect the digestive process, such as cystic fibrosis, Gaucher's disease, and **celiac disease**. A program of enzyme supplementation is rarely recommended for healthy patients. However, proponents of enzyme therapy believe that such a program is beneficial for everyone. They point to enzymes' ability to purify the blood, strengthen the immune system, enhance mental capacity, cleanse the colon, and maintain proper pH balance in urine. They feel that by improving the digestive process, the body is better able to combat infection and disease.

Some evidence exists that pancreatic enzymes derived from animal sources are helpful in cancer treatment. The enzymes may be able to dissolve the coating on **cancer** cells and may make it easier for the immune system to attack the cancer.

A partial list of the wide variety of complaints and illnesses that can be treated by enzyme therapy includes:

- AIDS
- anemia
- alcohol consumption
- anxiety
- acute inflammation
- back pain
- cancer
- colds
- chronic **fatigue** syndrome
- colitis
- constipation
- diarrhea
- food **allergies**
- gastritis
- gastric duodenal ulcer
- gout
- headaches

- hepatitis
- hypoglycemia
- infections
- mucous congestion
- multiple sclerosis
- nervous disorders
- nutritional disorders
- obesity
- premenstrual syndrome (PMS)
- stress

Description

Origins

Enzymes are protein molecules used by the body to perform all of its chemical actions and reactions. The body manufactures several thousands of enzymes. Among them are the digestive enzymes produced by the stomach, pancreas, small intestine, and the salivary glands of the mouth. Their energy-producing properties are responsible for not only the digestion of nutrients, but their absorption, transportation, metabolization, and elimination as well.

Enzyme therapy is based on the work of Dr. Edward Howell in the 1920s and 1930s. Howell proposed that enzymes from foods work in the stomach to pre-digest food. He advocated the consumption of large amounts of plant enzymes, theorizing that if the body had to use less of its own enzymes for digestion, it could store them for maintaining metabolic harmony. Four categories of plant enzymes are helpful in pre-digestion: protease, amylase, lipase, and cellulase. Cellulase is particularly helpful because the body is unable to produce it.

Animal enzymes, such as pepsin extracted from the stomach of pigs, work more effectively in the duodenum. They are typically used for the treatment of nondigestive ailments.

The seven categories of food enzymes and their activities

- amylase breaks down starches
- cellulase breaks down fibers
- lactase breaks down dairy products
- lipase breaks down fats
- maltase breaks down grains
- protease breaks down proteins
- sucrase breaks down sugars

Enzyme theory generated further interest as the human diet became more dependent on processed and cooked foods. Enzymes are extremely sensitive to heat, and temperatures above 118°F (48°C) destroy them. Modern processes of pasteurization, canning, and microwaving are particularly harmful to the enzymes in food.

Enzyme supplements are extracted from plants like pineapple and papaya and from the organs of cows and pigs. The supplements are typically given in tablet or capsule form. Pancreatic enzymes may also be given by injection. The dosage varies with the condition being treated. For nondigestive ailments, the supplements are taken in the hour before meals so that they can be quickly absorbed into the blood. For digestive ailments, the supplements are taken immediately before meals accompanied by a large glass of fluids. Pancreatic enzymes may be accompanied by doses of vitamin A.

Preparations

No special preparations are necessary before beginning enzyme therapy. However, it is always advisable to talk to a doctor or pharmacist before purchasing enzymes and beginning therapy.

Precautions

People with allergies to beef, pork, pineapples, and papaya may suffer allergic reactions to enzyme supplements. Tablets are often coated to prevent them from breaking down in the stomach, and usually shouldn't be chewed or crushed. People who have difficulty swallowing pills can request enzyme supplements in capsule form. The capsules can then be opened and the contents sprinkled onto soft foods like applesauce.

Side effects

Side effects associated with enzyme therapy include **heartburn**, **nausea and vomiting**, diarrhea, bloating, gas, and **acne**. According to the principles of therapy, these are temporary cleansing symptoms. Drinking eight to 10 glasses of water daily and getting regular **exercise** can reduce the discomfort of these side effects. Individuals may also experience an increase in bowel movements, perhaps one or two per day. This is also considered a positive effect.

Plant enzymes are safe for pregnant women, although they should always check with a doctor before using enzymes. Pregnant women should avoid animal enzymes. In rare cases, extremely high doses of

KEY TERMS

Celiac disease—A chronic disease characterized by defective digestion and use of fats.

Cystic fibrosis—A genetic disease that causes multiple digestive, excretion, and respiratory complications. Among the effects, the pancreas fails to provide secretions needed for the digestion of food.

Duodenum—The first part of the small intestine.

Gaucher's disease—A rare genetic disease caused by a deficiency of enzymes needed for the processing of fatty acids.

Metabolism—The system of chemical processes necessary for living cells to remain healthy.

enzymes can result in a build up of uric acid in the blood or urine and can cause a break down of proteins.

Research and general acceptance

In the United States, the Food and Drug Administration (FDA) has classified enzymes as a food. Therefore, they can be purchased without a prescription. However, insurance coverage is usually dependent upon the therapy resulting from a doctor's orders.

Resources

OTHER

Enzyme Therapy for Your Health. <<http://members.tripod.com/~colloid/enzyme.htm>> .

Questions and Answers about Food Enzymes and Nutrition. <<http://www.enzymes.com>> .

Therapies: Enzyme Therapy. <<http://library.thinkquest.org/24206/enzyme-therapy.html>> .

Mary McNulty

Eosinophilic granuloma see **Histiocytosis X**

Eosinophilic pneumonia

Definition

Eosinophilic **pneumonia** is a group of diseases in which there is an above normal number of eosinophils in the lungs and blood.

Description

Eosinophilia is an increase in the number of eosinophils. Eosinophilic pneumonia is characterized by a large number of eosinophils in the lungs, usually in the absence of an infectious disease. Eosinophils are one of the white blood cells and are classified as a granulocyte. They are part of the non-specific immune system and participate in inflammatory reactions. Eosinophils contain cationic molecules that are useful for destroying infectious agents, especially helminthic parasites (worms). There are several types of eosinophilic pneumonia. Löffler's pneumonia is a temporary infiltration of eosinophils into the lungs. The patient will feel tired, have a **cough**, spasms of the bronchial airway, and difficulty breathing. Löffler's pneumonia will clear spontaneously, but slowly over the course of about a month. Another form of eosinophilic pneumonia, pulmonary infiltrates with eosinophilia (PIE), is a more serious and potentially fatal disease. In PIE, the patient experiences asthma, pulmonary infiltrates, disorders of the peripheral nervous system, central nervous systems symptoms, and periarteritis nodosa.

Causes and symptoms

Pneumonia with eosinophils occurs as part of a hypersensitivity reaction. A hypersensitivity reaction is an over-reaction of the immune system to a particular stimulus. As part of the hypersensitive reaction, cells of the immune system are produced in increased numbers and migrate into areas targeted by the hypersensitivity reaction. In the case of eosinophilic pneumonia, the lungs are the target. Generally, eosinophilia pneumonia is not a reaction to an infection. There is a correlation between **asthma** and eosinophilic pneumonia. Eosinophilic pneumonia can also be caused by drugs and, in some people, by polluted air. The symptoms range from mild (coughing, wheezing, and shortness of breath) to severe and life threatening (severe **shortness of breath** and difficulty getting enough oxygen). The symptoms may resolve spontaneously or can persist for long periods of time. In a few cases, the disease may rapidly produce life-threatening pneumonia.

Diagnosis

Since eosinophilia is common to a number of conditions, the physician must rule out asthma and infection by helminths when diagnosing eosinophilic pneumonia. A whole **blood count** will reveal an increased number of eosinophils in the blood. An x ray of the lungs may show the presence of infiltrates

KEY TERMS

Infiltrates—Cells or body fluids that have passed into a tissue or body cavity.

Sputum—Material coughed up from the throat or lungs.

(the eosinophils and fluid). If sputum is produced in coughing, eosinophils will be seen instead of the more normal profile of granulocytes seen when an infectious agent is present.

Treatment

Eosinophilic pneumonia may not respond to drugs used to treat asthma. Eosinophilic pneumonia is usually treated with steroids, particularly glucocorticosteroids. Steroids are not effective against infectious agents, but the main disease process in eosinophilic pneumonia is an inflammatory reaction, not a response to infection. When eosinophilia is produced as a consequence of asthma or an infection by helminths, treatment of the asthma or helminths will reduce the eosinophilia.

Resources

BOOKS

Berkow, Robert, editor. *Merck Manual of Medical Information*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

John T. Lohr, PhD

Ephedrine see **Bronchodilators**

Epicondylitis see **Tennis elbow**

Epidemic icterus see **Hepatitis A**

Epidemic typhus see **Typhus**

Epidemic viral gastroenteritis see **Rotavirus infections**

Epidermolysis bullosa

Definition

Epidermolysis bullosa (EB) is a group of rare inherited skin diseases that are characterized by the development of blisters following minimal

pressure to the skin. Blistering often appears in infancy in response to simply being held or handled. In rarer forms of the disorder, EB can be life-threatening. There is no cure for the disorder. Treatment focuses on preventing and treating wounds and infection.

Description

Epidermolysis bullosa has three major forms and at least 16 subtypes. The three major forms are EB simplex, junctional EB, and dystrophic EB. These can range in severity from mild blistering to more disfiguring and life-threatening disease. Physicians diagnose the form of the disease based on where the blister forms in relation to the epidermis (the skin's outermost layer) and the deeper dermis layer.

The prevalence of epidermolysis varies among different populations. A study in Scotland estimated the prevalence to be one in 20,400. Researchers in other parts of the world estimate the prevalence to be one in 100,000. This variance is due to the variability of expression. Many cases of epidermolysis bullosa are often not accurately diagnosed and thus, are not reported.

Causes and symptoms

EB can be inherited as the result of a dominant genetic abnormality (only one parent carries the abnormal gene) or a recessive genetic abnormality (both parents carry the abnormal gene).

EB simplex results from mutations in genes responsible for keratin 5 and 14, which are proteins that give cells of the epidermis its structure. EB simplex is transmitted in an autosomal dominant fashion.

Dystrophic EB is caused by mutations in genes for type VII collagen, the protein contained in the fibers anchoring the epidermis to the deeper layers of the skin. The genetic mutations for junctional EB are found in the genes responsible for producing the protein Laminin-5. Dystrophic EB is an autosomal disorder and will only result if both parents transmit an abnormal gene during conception.

EB simplex, the most common form of EB, is the least serious form of the disease. In most affected individuals, the blisters are mild and do not scar after they heal. Some forms of EB simplex affect just the hands and feet. Other forms of EB simplex can lead to more widespread blistering, as well as hair loss and missing teeth. Recurrent blistering is annoying but not life threatening.

KEY TERMS

Collagen—The main supportive protein of cartilage, connective tissue, tendon, skin, and bone.

Dermis—The layer of skin beneath the epidermis.

Epidermis—The outermost layer of the skin.

Keratin—A tough, nonwater-soluble protein found in the nails, hair, and the outermost layer of skin. Human hair is made up largely of keratin.

The second, or junctional, form of EB does not lead to scarring. However, skin on the areas prone to blistering, such as elbows and knees, often shrinks. In one variation of junctional EB, called gravis junctional EB of Herlitz, the blistering can be so severe that affected infants may not survive due to massive infection and **dehydration**.

The third form of EB, dystrophic EB, varies greatly in terms of severity, but more typically affects the arms and legs. In one variation, called Hallopeau-Siemens EB, repeated blistering and scarring of the hands and feet causes the fingers and toes to fuse, leaving them dysfunctional and with a mitten-like appearance.

Diagnosis

Physicians and researchers distinguish between the three major subtypes of EB based on which layer of the epidermis separates from the deeper dermis layer of the skin below. Patients suspected of having EB should have a fresh blister biopsied for review. This sample of tissue is examined under an electron microscope or under a conventional microscope using a technique called immunofluorescence, which helps to map the underlying structure.

Knowing that a family member has EB can help establish the diagnosis, but it is possible that parents or siblings will show no sign of the disease, either because it is caused by a new genetic mutation, or because the parents are carriers of the recessive trait and do not display the disease.

Treatment

The most important treatment for EB is daily wound care. Because the skin is very fragile, care must be taken to be certain that dressing changes do not cause further damage. Tape should not be applied directly to skin and bandages should be soaked off.

Infection is a major concern, so a topical antibiotic, such as bacitracin, mupirocin, or sulfadiazine, should be routinely applied. Among persons with recessive dystrophic EB, the anticonvulsant phenytoin is sometimes effective because it decreases production of an enzyme that breaks down collagen.

Prognosis

The prognosis of EB varies depending on the subtype of the disease. Individuals with EB simplex can live long, fulfilling lives. The severity of the junctional and dystrophic forms of EB can vary greatly. Infants affected with some forms of the disease often do not survive infancy; other forms can lead to severe scarring and disfigurement.

Resources

BOOKS

- Fine, Jo-David, et al. *Epidermolysis Bullosa: Clinical, Epidemiologic, and Laboratory Advances, and the Findings of the National Epidermolysis Bullosa Registry*. Baltimore: Johns Hopkins Univ Press, 1999.
- Fitzpatrick, Thomas B., Richard A. Johnson, Wolff Klaus, and Dick Suurmond. *Color Atlas and Synopsis of Clinical Dermatology*. 4th ed. New York: McGraw-Hill, 2000.
- Mallory, S.B. *Atlas of Pediatric Dermatology*. Pearl River, NY: Parthenon, 2001.

PERIODICALS

- Cotell, S., N. D. Robinson, and L. S. Chan. "Autoimmune Blistering Skin Diseases." *American Journal of Emerging Medicine* 18, no. 3 (2000): 288–99.

ORGANIZATIONS

- American Academy of Dermatology. PO Box 4014, 930 N. Meacham Rd., Schaumburg, IL 60168-4014. (847) 330-0230. Fax: (847) 330-0050. <<http://www.aad.org>>.

OTHER

- Dermatology Online Atlas. <<http://www.dermis.net/doia/diagnose.asp?zugr=d&lang=e&diagnr=757320&topic=t>>.
- Dystrophic Epidermolysis Bullosa Research Association International. <<http://debra-international.org/index1.htm>>.
- Epidermolysis Bullosa Medical Research Foundation. <<http://www.med.stanford.edu/school/dermatology/ebmrf/>>.
- Oregon Health Sciences University. <<http://www.ohsu.edu/clinweb/C17/C17.800.865.410.html>>.
- University of Iowa College of Medicine. <<http://tray.dermatology.uiowa.edu/EBA-001.htm>>.

L. Fleming Fallon, Jr., MD, PhD, DrPH

Epididymitis

Definition

Epididymitis is inflammation or infection of the epididymis. In this long coiled tube attached to the upper part of each testicle, sperm mature and are stored before ejaculation.

Description

Epididymitis is the most common cause of **pain** in the scrotum. The acute form is usually associated with the most severe pain and swelling. If symptoms last for more than six weeks after treatment begins, the condition is considered chronic.

Epididymitis can occur any time after the onset of puberty but is most common between the ages of 18 and 40. It is especially common among members of the military who **exercise** for extended periods without emptying their bladders.

Factors that increase the risk of developing epididymitis include:

- infection of the bladder, kidney, prostate, or urinary tract
- other recent illness
- narrowing of the urethra (the tube that drains urine from the bladder)
- use of a urethral catheter

Causes and symptoms

Although epididymitis can be caused by the same organisms that cause some **sexually transmitted diseases** (STDs) or occur after prostate surgery, the condition is generally due to pus-generating bacteria associated with infections in other parts of the body.

Epididymitis can also be caused by injury or infection of the scrotum or by irritation from urine that has accumulated in the vas deferens (the duct through which sperm travels after leaving the epididymis).

Epididymitis is characterized by sudden redness and swelling of the scrotum. The affected testicle is hard and sore, and the other testicle may feel tender. The patient has chills and fever and usually has acute **urethritis** (inflammation of the urethra).

Enlarged lymph nodes in the groin cause scrotal pain that intensifies throughout the day and may become so severe that walking normally becomes impossible.

Diagnosis

Laboratory tests used to diagnose epididymitis include:

- urinalysis and urine culture
- examination of discharges from the urethra and prostate gland
- blood tests to measure white-cell counts

Treatment

Because epididymitis that affects both testicles can make a man sterile, antibiotic therapy must be initiated as soon as symptoms appear. To prevent reinfection, medication must be taken exactly as prescribed, even if the patient's symptoms disappear or he begins to feel better. Over-the-counter anti-inflammatories can relieve pain but should not be used without the approval of a family physician or urologist.

Bed rest is recommended until symptoms subside, and patients are advised to wear athletic supporters when they resume normal activities. If pain is severe, a local anesthetic like lidocaine (Xylocaine) may be injected directly into the spermatic cord.

Self-care

A patient who has epididymitis should not drink beverages that contain **caffeine**. To prevent **constipation**, he should use stool softeners or eat plenty of fruit, nuts, whole grain cereals, and other foods with laxative properties.

An ice bag wrapped in a towel can reduce pain and swelling but should be removed from the inflamed area for a few minutes every hour to prevent **burns**.

Strenuous activity should be avoided until symptoms disappear. Sexual activity should not be resumed until a month after symptoms disappear.

If a second course of treatment doesn't eradicate stubborn symptoms, longterm anti-inflammatory therapy may be recommended. In rare instances, chronic symptoms require surgery.

Surgery

Each of the surgical procedures used to treat epididymitis is performed under local anesthesia on an outpatient basis. Both of them cause sterility.

Epididymectomy involves removing the inflamed section of the epididymitis through a small incision in the scrotum.

Bilateral **vasectomy** prevents fluid and sperm from passing through the epididymis. This procedure is usually performed on men who have chronic epididymitis or on elderly patients undergoing prostate surgery.

Prognosis

Pain generally subsides 24–72 hours after treatment begins. Complete healing may take weeks or months.

Prevention

Using condoms and not having sex with anyone who has an STD can prevent some cases of epididymitis.

Resources

OTHER

"Epididymitis." *Digital Urology Journal*. June 7, 1998.
< <http://www.duj.com/epididymitis.html> > .

Maureen Haggerty

Epidural abscess see **Central nervous system infections**

Epidural anesthetic see **Anesthesia, local**

Epididymitis

Definition

Epididymitis is an infection of the epididymis, which can lead to severe airway obstruction.

Description

When air is inhaled (inspired), it passes through the nose and the nasopharynx or through the mouth and the oropharynx. These are both connected to the larynx, a tube made of cartilage. The air continues down the larynx to the trachea. The trachea then splits into two branches, the left and right bronchi (bronchial tubes). These bronchi branch into smaller air tubes that run within the lungs, leading to the small air sacs of the lungs (alveoli).

Either food, liquid, or air may be taken in through the mouth. While air goes into the larynx and the respiratory system, food and liquid are directed into the tube leading to the stomach, the esophagus.

Because food or liquid in the bronchial tubes or lungs could cause a blockage or lead to an infection, the airway is protected. The epiglottis is a leaf-like piece of cartilage extending upwards from the larynx. The epiglottis can close down over the larynx when someone is eating or drinking, preventing these food and liquids from entering the airway.

Epiglottitis is an infection and inflammation of the epiglottis. Because the epiglottis may swell considerably, there is a danger that the airway will be blocked off by the very structure designed to protect it. Air is then unable to reach the lungs. Without intervention, epiglottitis has the potential to be fatal.

Epiglottitis is primarily a disease of two to seven-year-old children, although older children and adults can also contract it. Boys are twice as likely as girls to develop this infection. Because epiglottitis involves swelling and infection of tissues, which are all located at or above the level of the epiglottis, it is sometimes referred to as supraglottitis (*supra*, meaning above). About 25% of all children with this infection also have pneumonia.

Causes and symptoms

The most common cause of epiglottitis is infection with the bacteria called *Haemophilus influenzae type b*. Other types of bacteria are also occasionally responsible for this infection, including some types of *Streptococcus* bacteria and the bacteria responsible for causing diphtheria.

A patient with epiglottitis typically experiences a sudden **fever**, and begins having severe throat and neck **pain**. Because the swollen epiglottis interferes significantly with air movement, every breath creates a loud, harsh, high-pitched sound referred to as **stridor**. Because the vocal cords are located in the larynx just below the area of the epiglottis, the swollen epiglottis makes the patient's voice sound muffled and strained. Swallowing becomes difficult, and the patient may drool. The patient often leans forward and juts out his or her jaw, while struggling for breath.

Epiglottitis strikes suddenly and progresses quickly. A child may begin complaining of a sore throat, and within a few hours be suffering from extremely severe airway obstruction.

Diagnosis

Diagnosis begins with a high level of suspicion that a quickly progressing illness with fever, **sore throat**, and airway obstruction is very likely to be

epiglottitis. If epiglottitis is suspected, no efforts should be made to look at the throat, or to swab the throat in order to obtain a culture for identification of the causative organism. These maneuvers may cause the larynx to go into spasm (laryngospasm), completely closing the airway. These procedures should only be performed in a fully-equipped operating room, so that if laryngospasm occurs, a breathing tube can be immediately placed in order to keep the airway open.

An instrument called a laryngoscope is often used in the operating room to view the epiglottis, which will appear cherry-red and quite swollen. An x-ray picture taken from the side of the neck should also be obtained. The swollen epiglottis has a characteristic appearance, called the "thumb sign."

Treatment

Treatment almost always involves the immediate establishment of an artificial airway: inserting a breathing tube into the throat (intubation); or making a tiny opening toward the base of the neck and putting a breathing tube into the trachea (tracheostomy). Because the patient's apparent level of distress may not match the actual severity of the situation, and because the disease's progression can be quite surprisingly rapid, it is preferable to go ahead and place the artificial airway, rather than adopting a wait-and-see approach.

Because epiglottitis is caused by a bacteria, **antibiotics** such as cefotaxime, ceftriaxone, or ampicillin with sulbactam should be given through a needle placed in a vein (intravenously). This prevents the bacteria that are circulating throughout the bloodstream from causing infection elsewhere in the body.

Prognosis

With treatment (including the establishment of an artificial airway), only about 1% of children with epiglottitis die. Without the artificial airway, this figure jumps to 6%. Most patients recover from the infection, and can have the breathing tube removed (extubation) within a few days.

Prevention

Prevention involves the use of a vaccine against *H. influenzae type b* (called the Hib vaccine). It is given to babies at two, four, six, and 15 months. Use of this vaccine has made epiglottitis a very rare occurrence.

KEY TERMS

Epiglottitis—A leaf-like piece of cartilage extending upwards from the larynx, which can close like a lid over the trachea to prevent the airway from receiving any food or liquid being swallowed.

Extubation—Removal of a breathing tube.

Intubation—Putting a breathing tube into the airway.

Laryngospasm—Spasm of the larynx.

Larynx—The part of the airway lying between the pharynx and the trachea.

Nasopharynx—The part of the airway into which the nose leads.

Oropharynx—The part of the airway into which the mouth leads.

Supraglottitis—Another term for epiglottitis.

Trachea—The part of the airway that leads into the bronchial tubes.

Tracheostomy—A procedure in which a small opening is made in the neck and into the trachea. A breathing tube is then placed through this opening.

Resources

ORGANIZATIONS

American Academy of Otolaryngology-Head and Neck Surgery, Inc. One Prince St., Alexandria VA 22314-3357. (703) 836-4444. <<http://www.entnet.org>>.

Rosalyn Carson-DeWitt, MD

Epilepsy see **Seizure disorder**

Epinephrine see **Bronchodilators**

Episiotomy

Definition

An episiotomy is a surgical incision made in the area between the vagina and anus (perineum). This is done during the last stages of labor and delivery to expand the opening of the vagina to prevent tearing during the delivery of the baby.

Purpose

This procedure is usually done during the delivery or birthing process when the vaginal opening does not stretch enough to allow the baby to be delivered without tearing the surrounding tissue.

Precautions

Prior to the onset of labor, pregnant women may want to discuss the use of episiotomy with their care providers. It is possible that, with adequate preparation and if the stages of labor and delivery are managed with adequate coaching and support, the need for an episiotomy may be reduced.

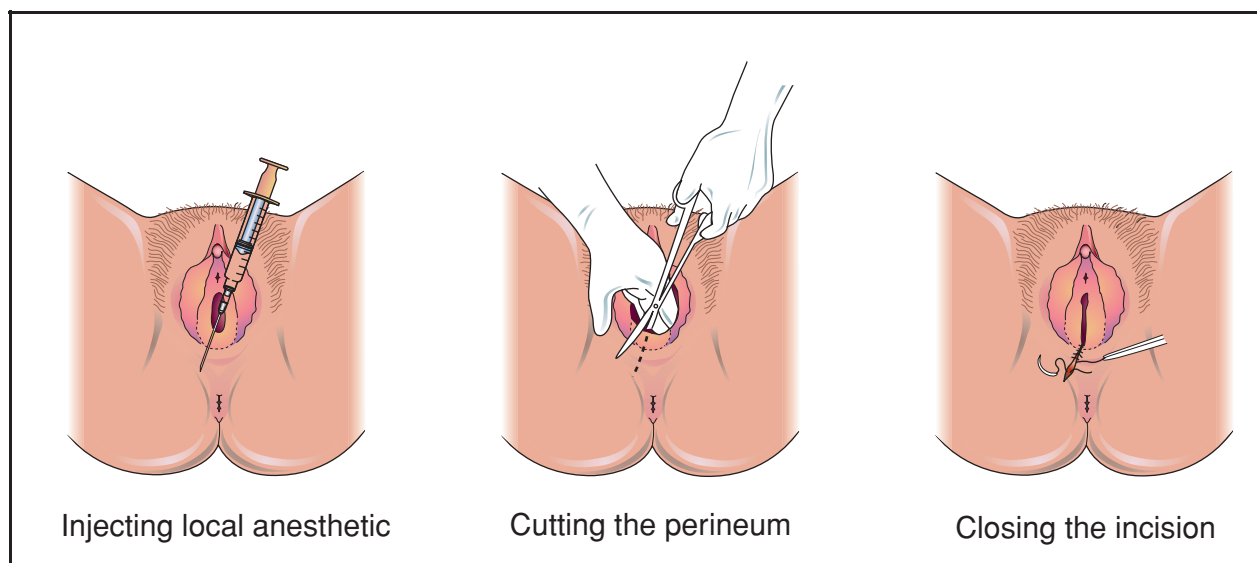
Description

An episiotomy is a surgical incision, usually made with sterile scissors, in the perineum as the baby's head is being delivered. This procedure may be used if the tissue around the vaginal opening begins tearing or does not seem to be stretching enough to allow the baby to be delivered.

In most cases, the physician makes a midline incision along a straight line from the lowest edge of the vaginal opening to toward the anus. In other cases, the episiotomy is performed by making a diagonal incision across the midline between the vagina and anus. This method is used much less often, may be more painful, and may require more healing time than the midline incision. After the baby is delivered through the extended vaginal opening, the incision is closed with stitches. A local anesthetic agent may be applied or injected to numb the area before it is sewn up (sutured).

Several reasons are cited for performing episiotomies. Some experts believe that an episiotomy speeds up the birthing process, making it easier for the baby to be delivered. This can be important if there is any sign of distress that may harm the mother or baby. Because tissues in this area may tear during the delivery, another reason for performing an episiotomy is that a clean incision is easier to repair than a jagged tear and may heal faster. Although the use of episiotomy is sometimes described as protecting the pelvic muscles and possibly preventing future problems with **urinary incontinence**, it is not clear that the procedure actually helps.

The use of episiotomy during the birthing process is fairly widespread in the United States. Estimates of episiotomy use in hospitals range from 65–95% of deliveries, depending on how many times the mother



An episiotomy is a surgical incision made in the perineum, the area of tissue between the vaginal opening and the anus, during the birthing process. This procedure may be used if the tissue around the vaginal opening begins to tear or is not stretching enough to allow the baby to be delivered vaginally. In the United States, the rate of episiotomies being performed is estimated at 65–95%. (Illustration by Electronic Illustrators Group.)

has given birth previously. This routine use of episiotomy is being reexamined in many hospitals and health care settings. However, an episiotomy is always necessary during a forceps delivery because of the size of the forceps.

Preparation

It may be possible to avoid the need for an episiotomy. Pregnant women may want to talk with their care providers about the use of episiotomy during the delivery. Kegel exercises are often recommended during the **pregnancy** to help strengthen the pelvic floor muscles. Prenatal perineal massage may help to stretch and relax the tissue around the vaginal opening. During the delivery process, warm compresses can be applied to the area along with the use of perineal massage. Coaching and support are also important during the delivery process. A slowed, controlled pushing during the second stage of labor (when the mother gets the urge to push) may allow the tissues to stretch rather than tear. Also, an upright birthing position (rather than one where the mother is lying down) may decrease the need for an episiotomy.

Aftercare

The area of the episiotomy may be uncomfortable or even painful for several days. Several practices can relieve some of the **pain**. Cold packs can be applied to

the perineal area to reduce swelling and discomfort. Use of the Sitz bath available at the hospital or birth center can ease the discomfort, too. This unit circulates warm water over the area. A squirt bottle with water can be used to clean the area after urination or defecation rather than wiping with tissue. Also, the area should be patted dry rather than wiped. Cleansing pads soaked in witch hazel (such as Tucks) are very effective for cleaning the area and also feel soothing.

Risks

Several side effects of episiotomy have been reported, including infection, increased pain, prolonged healing time, and increased discomfort once sexual intercourse is resumed. There is also the risk that the episiotomy incision will be deeper or longer than is necessary to permit the birth of the infant. There is a risk of increased bleeding.

Normal results

In a normal and well managed delivery, an episiotomy may be avoided altogether. If an episiotomy is deemed to be necessary, a simple midline incision will be made to extend the vaginal opening without additional tearing or extensive trauma to the perineal area. Although there may be some pain associated with the healing of the episiotomy incision, relief can usually

KEY TERMS

Kegel exercises—A series of contractions and relaxations of the muscles in the perineal area. These exercises are thought to strengthen the pelvic floor and may help prevent urinary incontinence in women.

Perineum—The area between the opening of the vagina and the anus in a woman, or the area between the scrotum and the anus in a man.

Sitz bath—A shallow tub or bowl, sometimes mounted above a toilet, that allows the perineum and buttocks to be immersed in circulating water.

Urinary incontinence—The inability to prevent the leakage or discharge of urine. This situation becomes more common as people age, and is more common in women who have given birth to more than one child.

be provided with mild pain relievers and supportive measures, such as the application of cold packs.

Abnormal results

An episiotomy incision that is too long or deep may extend into the rectum, causing more bleeding and an increased risk of infection. Additional tearing or tissue damage may occur beyond the episiotomy incision, leaving a cut and a tear to be repaired.

Resources

OTHER

Childbirth.org. <<http://www.childbirth.org>>.

Altha Roberts Edgren

Epispadias see **Hypospadias and epispadias**

Epistaxis see **Nosebleed**

EPS see **Electrophysiology study of the heart**

Epstein-Barr virus test

Definition

The Epstein-Barr virus test is a blood test, or group of tests, to determine the presence or absence of antibodies in the blood stream directed against

proteins of the Epstein-Barr virus, the cause of **infectious mononucleosis**.

Purpose

The test is primarily used to detect whether first time infection (called primary infection) with the Epstein-Barr virus is currently occurring, or has occurred within a short period of time. The pattern of the antibodies detected can, however, tell if the person has never been infected with the Epstein-Barr virus, or if the infection occurred in the more distant past. These tests are mostly utilized in the diagnosis of Epstein-Barr virus-associated infectious mononucleosis when the more common diagnostic test, the heterophile antibody, is negative, or in situations where the infection is manifesting unusual symptoms. Therefore, the tests are often not needed in a situation where a doctor believes that a person has mononucleosis and the heterophile test (also called the monospot test) is positive.

In addition, Epstein-Barr virus testing is usually not needed in the evaluation of a patient who has long-lasting **fatigue**, and may have the **chronic fatigue syndrome**. Initially, it was thought that discovering a particular pattern of antibodies to this virus was helpful in the diagnosis of chronic fatigue syndrome, but this no longer appears to be the case.

Precautions

As in any blood test, standard precautions should be performed to prevent infection at the site where the blood is obtained, and to prevent excess bleeding. Normally, the site is cleaned with an antiseptic liquid prior to the blood being obtained; a sterile non-reusable needle and syringe are used; and, once the needle is removed, pressure is placed at the site until bleeding has stopped.

Description

These tests are more often performed in a consulting laboratory than at a physician's office or in a hospital laboratory. Like most antibody tests, they are performed on serum, the liquid part of the blood obtained after the whole blood is allowed to clot in a tube. Antibodies can be detected against several components of the Epstein-Barr virus (EBV). These components are the EBV early antigen (EA), the viral capsid antigen (VCA), and the nuclear antigen (EBNA). These several antigens are different proteins that are produced in the process (stages) of the virus' growth.

At the time of infection with Epstein-Barr virus, antibodies to EA are found and usually last for four to six months only. This antibody, however, persists substantially longer in about 10% of persons who have had EBV infection in the more remote past. The absence of antibody to EA when other EBV antibodies are present strongly suggests that first time infection with EBV occurred in the past.

Antibody to VCA is found both early and late in EBV infection. At the time of infection, antibody of both the IgM and IgG types are detectable. After four to six months, usually, only the IgG antibody against VCA can be found.

Unlike antibodies to EA and VCA, antibody to EBNA does not usually develop until recovery from first time infection of this virus. Therefore, finding detectable amounts of antibody to EBNA during an illness which might be caused by EBV makes the causal relationship very unlikely.

Preparation

The skin area from which the blood sample will be obtained is wiped with an antiseptic such as alcohol or iodine.

Aftercare

The aftercare is similar to that for any blood test. Usually, pressure is applied to the area for several moments until bleeding stops. If the results are difficult to interpret, it may be necessary to re-test later, after waiting one to three weeks. The change in the amounts of antibody detected between the two tests can be particularly useful, at times, in helping to make a diagnosis.

Risks

There are no risks over and above those of having blood drawn for any other purpose. These tests are more expensive than many other blood tests but are usually covered by medical insurance.

Normal results

The pattern of the three antibodies can be used to determine whether the person has not had infection with EBV to this point (is susceptible to infection); is currently, or recently, infected with EBV for the first time; or has had first time infection with EBV sometime in the past (more than six months ago).

If one defines "normal" results as either not having EBV in the past, and call that category one;

or having had it in the past, and call that category two. Most young children below the age of five will fall into category one, while most adults over the age of 20 years will fall into category two.

The results for susceptibility are:

- antibody to EA = negative
- antibody to VCA (either IgM or IgG) = negative
- antibody to EBNA = negative

The results for past infection are:

- antibody to EA = negative (90% of time)
- antibody to VCA IgM = negative
- Antibody to VCA IgG = positive
- Antibody to EBNA = Positive.

It is important to realize that the Epstein-Barr virus, like all the human herpes viruses, does not totally leave the body after the patient recovers from illness. With EBV, the virus will intermittently recur in the saliva of people without any symptoms. Such people will have a test pattern of previous infection. It is this group of people who can transmit EBV to others without themselves being ill.

Abnormal results

The results for current or recent infection are:

- antibody to EA = positive
- antibody to VCA IgM = positive
- antibody to VCA IgG = positive
- antibody to EBNA = negative.

Without the pattern of the three antibodies, it can be difficult to be accurate in interpretation. The presence of antibody to VCA IgM is the best single test for current or recent first time infection.

Resources

PERIODICALS

Henle, G., W. Henle, and C. A. Horowitz. "Epstein-Barr Virus Specific Diagnosis: Tests in Infectious Mononucleosis." *Human Pathology* 5 (1997): 551-558.

Larry I. Lutwick, MD, FACP

ERCP see **Endoscopic retrograde cholangiopancreatography**

Erectile disorder see **Impotence**

Erectile dysfunction

Definition

Erectile dysfunction (ED), formerly known as **impotence**, is the inability to achieve or maintain an erection long enough to engage in sexual intercourse.

Description

Under normal circumstances, when a man is sexually stimulated, his brain sends a message down the spinal cord and into the nerves of the penis. The nerve endings in the penis release chemical messengers, called neurotransmitters, that signal the arteries that supply blood to the corpora cavernosa (the two spongy rods of tissue that span the length of the penis) to relax and fill with blood. As they expand, the corpora cavernosa close off other veins that would normally drain blood from the penis. As the penis becomes engorged with blood, it enlarges and stiffens, causing an erection. Problems with blood vessels, nerves, or tissues of the penis can interfere with an erection.

Causes and symptoms

It is estimated that up to 30 million American men frequently suffer from ED and that it strikes up to half of all men between the ages of 40 and 70. Doctors used to think that most cases of ED were psychological in origin, but they now recognize that, at least in older men, physical causes may play a primary role in 60% or more of all cases. In men over the age of 60, the leading cause is **atherosclerosis**, or narrowing of the arteries, which can restrict the flow of blood to the penis. Injury or disease of the connective tissue, such as **Peyronie's disease**, may prevent the corpora cavernosa from completely expanding. Damage to the nerves of the penis from certain types of surgery or neurological conditions, such as Parkinson's disease or **multiple sclerosis**, may also cause ED. Men with diabetes are especially at risk for erectile dysfunction because of their high risk of both atherosclerosis and a nerve disease called **diabetic neuropathy**.

Some drugs, including certain types of blood pressure medications, **antihistamines**, tranquilizers (especially before intercourse), and antidepressants known as **selective serotonin reuptake inhibitors** (SSRIs, including Prozac and Paxil) can interfere with erections. **Smoking**, excessive alcohol consumption, and illicit drug use may also contribute. In some cases, low levels of the male hormone testosterone may

contribute to erectile failure. Finally, psychological factors, such as **stress**, guilt, or **anxiety**, may also play a role, even when the ED is primarily due to organic causes.

Diagnosis

When diagnosing the underlying cause of erectile dysfunction, the doctor begins by asking the man a number of questions about when the problem began, whether it only happens with specific sex partners, and whether he ever wakes up with an erection. (Men whose dysfunction occurs only with certain partners or who wake up with erections are more likely to have a psychological cause for their ED.) Sometimes, the man's sex partner is also interviewed. In some cases, domestic discord may be a factor.

The doctor also obtains a thorough medical history to find out about past pelvic surgery, diabetes, cardiovascular disease, **kidney disease**, and any medications the man may be taking. The **physical examination** should include a genital examination, hormone tests, and a glucose test for diabetes. Sometimes a measurement of blood flow through the penis may be taken.

Treatment

Years ago, the standard treatment for erectile dysfunction was a penile implant or long-term psychotherapy. Although physical causes are now more readily diagnosed and treated, individual or marital counseling is still an effective treatment for ED when emotional factors play a role.

There are three prescription medications available in the United States to treat the physical causes of ED: **sildenafil citrate** (Viagra), vardenafil (Levitra), and tadalafil (Cialis). They help about three-fourths of all men who try them in the general population. Several studies have indicated that their success rate in diabetic men may be slightly lower, averaging around 60–65%. All three pills work by enhancing the effects of nitric oxide. This chemical relaxes muscles in the penis to allow more blood to flow in. These pills do not cause an erection by themselves. Sexual stimulation is also required. Sildenafil or vardenafil should be taken about an hour before sex. Each is effective for roughly four hours. Tadalafil lasts for up to 36 hours. Men should not have sex more than once every 24 hours after using these drugs. Before Viagra's approval in 1998, drug treatment of erectile dysfunction was limited to alprostadil (prostaglandin [E.sub.1]), either injected

into the penis or inserted as a pellet into the urethra. Sales of the three newer ED drugs reached \$3.4 billion in 2004, according to the research firm IMS Health. In 2004, Viagra had about 66% of the market share for ED drugs compared to 19% for Levitra and 14% for Cialis.

Viagra

Sildenafil citrate was originally developed in 1991 as a treatment for **angina**, or chest **pain**. The drug, marketed under the name Viagra, received FDA approval as a treatment for erectile dysfunction in March 1998, and since that time it has been prescribed for more than 20 million men worldwide. It was the first oral medication approved for ED treatment. Viagra is a vasodilator, a drug that has the effect of dilating the blood vessels. It works by improving blood circulation to the penis, and by enhancing the effects of nitric oxide, the agent that relaxes the smooth muscle of the penis and regulates blood vessels during sexual stimulation, allowing the penis to become engorged and achieve an erection.

The average recommended dose of Viagra is 50 mg. It comes in doses of 25 mg., 50 mg., and 100 mg. The medication is taken approximately one hour before sexual activity is planned, and may remain effective for up to four hours. One drawback is that to be effective, it should be taken on an empty stomach. Also, high-fat foods can interfere with the absorption of Viagra. Viagra does not increase sexual desire. Sexual stimulation and arousal are required for it to be effective.

Many insurance plans cover the cost of Viagra, provided it is prescribed to treat erectile dysfunction. The pills cost approximately \$10 each, and insurers may limit coverage to a specific number of pills each month.

The primary drawback to Viagra, which works about an hour after it is taken, is that the FDA cautions men with heart disease or low blood pressure to be thoroughly examined by a physician before obtaining a prescription. At least 130 men have died while taking Viagra. However, the FDA said the men—most over the age of 64—died of **heart attack** or **stroke** due to health problems exaggerated by sexual activity, not the drug itself.

Levitra

In early 2003, a second prescription drug to treat erectile dysfunction, Levitra, was approved by the FDA. Like Viagra, Levitra helps increase blood flow

to the penis and may help men with ED get and keep an erection. Once a man has completed sexual activity, blood flow to the penis should decrease and the erection should go away. Levitra should be taken approximately 60 minutes prior to sexual activity. In clinical trials, most patients were able to begin sexual activity before that time. A 2004 study showed that about 50% of men taking Levitra experienced a firm erection within 25 minutes and a small percentage in as quickly as 10 minutes. Studies also showed that Levitra improved erectile function in men who had other health factors, such as diabetes or prostate surgery.

Men taking nitrate drugs, often used to control chest pain (also known as angina), should not take Levitra. Men who use alpha-blockers, sometimes prescribed for high blood pressure or prostate problems, also should not take Levitra. Such combinations could cause blood pressure to drop to an unsafe level. Levitra is available in 2.5 mg, 5 mg, 10 mg, and 20 mg tablets and should be taken no more than once a day. The average cost per pill is about \$10 and is covered by many insurance plans.

Cialis

Cialis is the third oral drug prescribed to treat erectile dysfunction, approved by the FDA in November 2003. Its most notable difference from Viagra and Levitra, which work for about four hours, is that Cialis works for up to 36 hours. Cialis helps increase blood flow in the penis when a man is sexually stimulated. It can help men with ED get and keep an erection satisfactory for sexual activity. Once a man has completed sexual activity, blood flow to his penis decreases, and his erection goes away. Cialis is clinically proven to improve erectile function in most men with ED, including those with mild, moderate or severe ED.

The most common side effects with Cialis are **headache**, upset stomach, back pain, and muscle aches. These side effects usually go away after a few hours. Patients who get back pain and muscle aches usually get it 12 to 24 hours after taking Cialis. Back pain and muscle aches usually go away by themselves within 48 hours.

Cialis comes in 5 mg., 10 mg. (the recommended starting dose), and 20 mg. tablets. The average cost per tablet is about \$10 and it is covered by many insurance plans. Since the absorption of Cialis is not affected by food or high-fat foods, it does not need to be taken on an empty stomach. Studies

show that in most men, Cialis begins working in about 30 minutes and may be taken up to once per day by most patients.

Priapism, a prolonged erection, is a very rare potential side effect of all prescription ED medications. Persons taking Viagra, Levitra, or Cialis who have a prolonged erection, lasting more than four hours, should seek immediate medical attention. Priapism can cause damage to the penis potentially leading to the permanent inability to have an erection.

Because sexual activity can stress the heart, men who have heart problems should check with their physician to see if sexual activity is recommended. Erectile dysfunction drugs may trigger temporary **hypotension** (low blood pressure) and is known to increase cardiovascular nerve activity, so physicians should prescribe them with caution in men with a history of heart attack, atherosclerosis (hardening of the arteries), angina, arrhythmia, and chronic low blood pressure problems. ED drugs are not labeled or approved for use by women or children, or by men without erectile dysfunction. Anyone experiencing cardiovascular symptoms such as **dizziness**, chest or arm pain, and **nausea** when participating in sexual activity after taking an ED medication should stop the encounter. They should also not take any ED drug again until they have discussed the episode with their healthcare provider. It is recommended that men with kidney or liver impairments, and men over age 65, start at the lowest possible dosage of ED medications.

Other treatment options

Although the commercial availability of Viagra, Levitra, and Cialis has been useful in many men, **prostate cancer** patients and ED caused by psychological problems often require alternative treatment. A commonly used alternative consists of a three-drug injection containing alprostadil, papaverine hydrochloride, and phentolamine mesylate. Though it is commonly referred to as the “Knoxville formula,” apparently for the city of its original introduction, a number of slightly varying formulas have been in use around the country. The three-drug preparation is administered by injection into the corpora cavernosa to induce erection.

Other traditional therapies for ED include vacuum pump therapy, injection therapy involving injecting a substance into the penis to enhance blood flow, and a penile implantation device. In rare cases, if narrowed or diseased veins are responsible for ED,

surgeons may reroute the blood flow into the corpora cavernosa or remove leaking vessels.

In vacuum pump therapy, a man inserts his penis into a clear plastic cylinder and uses a pump to force air out of the cylinder. This forms a partial vacuum around the penis, which helps to draw blood into the corpora cavernosa. The man then places a special ring over the base of the penis to trap the blood inside it. The only side effect with this type of treatment is occasional bruising if the vacuum is left on too long.

Injection therapy involves injecting a substance into the penis to enhance blood flow and cause an erection. The FDA approved a drug called alprostadil (Caverject) for this purpose in 1995. Alprostadil relaxes smooth muscle tissue to enhance blood flow into the penis. It must be injected shortly before intercourse. Another, similar drug that is sometimes used is papaverine. Either drug may sometimes cause painful erections or priapism that must be treated with a shot of epinephrine. Alprostadil may also be administered into the urethral opening of the penis. In MUSE (medical urethral system for erection), the man inserts a thin tube the width of a spaghetti noodle into his urethral opening and presses down on a plunger to deliver a tiny pellet containing alprostadil into his penis. The drug takes about 10 minutes to work and the erection lasts about an hour. The main side effect is a sensation of pain and burning in the urethra, which can last about five to 15 minutes. The injection process itself is often painful.

Implantable **penile prostheses** are usually considered a last resort for treating erectile dysfunction. They are implanted in the corpora cavernosa to make the penis rigid without the need for blood flow. The semi-rigid type of prosthesis consists of a pair of flexible silicone rods that can be bent up or down. This type of device has a low failure rate but, unfortunately, it causes the penis to always be erect, which can be difficult to conceal under clothing.

The inflatable type of device consists of cylinders that are implanted in the corpora cavernosa, a fluid reservoir implanted in the abdomen, and a pump placed in the scrotum. The man squeezes the pump to move fluid into the cylinders and cause them to become rigid. (He reverses the process by squeezing the pump again.) While these devices allow for intermittent erections, they have a slightly higher malfunction rate than the silicon rods. Men can return to sexual activity six to eight weeks after implantation surgery. Since implants affect the corpora cavernosa,

they permanently take away a man's ability to have a natural erection.

Alternative treatment

A number of herbs have been promoted for treating erectile dysfunction. The most widely touted is yohimbe (*Corynanthe yohimbe*), derived from the bark of the yohimbe tree native to West Africa. It has been used in Europe for about 75 years to treat ED. The FDA approved yohimbe as a treatment for ED in the late 1980s. It is sold as an over-the-counter dietary supplement and as a prescription drug under brand names such as Yocon, Aphrodyne, Erex, Yohimex, Testomar, Yohimbe, and Yovital.

There is no clear medical research that indicates exactly how or why yohimbe works in treating ED. It is generally believed that yohimbe dilates blood vessels and stimulates blood flow to the penis, causing an erection. It also prevents blood from flowing out of the penis during an erection. It may also act on the central nervous system, specifically the lower spinal cord area where sexual signals are transmitted. Studies show it is effective to some degree in 30–40% of men with ED. It is primarily effective in men with ED caused by vascular, psychogenic (originating in the mind), or diabetic problems. It usually does not work in men whose dysfunction is caused by organic nerve damage. In healthy men without ED, yohimbe in some cases appears to increase sexual stamina and prolong erections. The usual dosage of yohimbine (yohimbe extract) to treat ED is 5.4 mg three times a day. It may take three to six weeks for it to take effect.

Ginkgo *Ginkgo biloba*, is also used to treat erectile dysfunction, although it has not been shown to help the condition in controlled studies, and probably has more of a psychological effect. In addition, ginkgo carries some risk of abnormal blood clotting and should be avoided by men taking blood thinners, such as coumadin. Other herbs promoted for treating ED include true unicorn root *Aletris farinosa*, **saw palmetto** *Serenoa repens*, ginseng *Panax ginseng*, and Siberian ginseng *Eleutherooccus senticosus*. Nux vomica *Strychnos nux-vomica* has been recommended, especially when ED is caused by excessive alcohol, cigarettes, or dietary indiscretions. Nux vomica can be very toxic if taken improperly, so it should be used only under the strict supervision of a physician trained in its use.

There are quite a few Chinese herbal remedies for erectile dysfunction usually combinations of

KEY TERMS

Angina—A condition in which lack of blood to the heart causes severe chest pain.

Antihistamines—A drug that blocks cell receptors for histamine, usually to prevent allergic effects such as sneezing and itching.

Arrhythmia—An irregularity in the normal rhythm or force of the heartbeat.

Atherosclerosis—An arterial disease in which raised areas of degeneration and cholesterol deposits (plaques) form on the inner surfaces of the arteries. Also called hardening of the arteries.

Corpora cavernosa—Either of a pair of columns of erectile tissue at either side of the penis that, together with the corpus spongiosum, produce an erection when filled with blood.

Diabetic neuropathy—A disease or disorder, especially a degenerative one, caused by diabetes that affects the nervous system.

Neurotransmitter—A chemical that carries messages between different nerve cells or between nerve cells and muscles.

Peyronie's disease—Local fibrous scarring causing the erect penis to be bent to such a degree that it interferes with sexual intercourse.

Priapism —A prolonged erection lasting more than four hours.

SSRIs—Selective serotonin reuptake inhibitors, a class of medication used to treat depression.

Vasodilator—An agent, such as a nerve or hormone, that widens the blood vessels, which in turn decreases resistance to blood flow and lowers blood pressure.

herbs and sometimes animal parts such as deer antler and sea horse. **Acupuncture** is also used to treat ED, although Western doctors question its effectiveness.

Prognosis

With proper diagnosis, erectile dysfunction can nearly always be treated or coped with successfully. Unfortunately, fewer than 10% of men with ED seek treatment. However, with the heavy advertising and marketing associated with Viagra,

Levitra, and Cialis, this number is expected to rise dramatically.

Prevention

There is no specific treatment to prevent erectile dysfunction. Perhaps the most important measure is to maintain general good health and avoid atherosclerosis by exercising regularly, controlling weight, controlling **hypertension** and **high cholesterol** levels, and not smoking. Avoiding excessive alcohol intake may also help.

Resources

BOOKS

Icon Health Publications. *The Official Patient's Sourcebook on Erectile Dysfunction: A Revised and Updated Directory for the Internet Age*. San Diego: Icon Health Publications, 2005.

Kloner, Robert A. *Heart Disease and Erectile Dysfunction*. Totowa, NJ: Humana Press, 2004.

Lue, Tom F. *An Atlas of Erectile Dysfunction, Second Edition*. London: Taylor & Francis Group, 2003.

Metz, Michael E. and Barry W. McCarthy. *Coping With Erectile Dysfunction: How to Regain Confidence and Enjoy Great Sex*. Oakland, CA: New Harbinger Publications, 2004.

PERIODICALS

Roberts, Shauna S. "Options Increase for Men With Erectile Dysfunction: Doctors Now Have an Array of Treatments for Men's Sexual Problems." *Diabetes Forecast* (September 2004): 55-58.

Sadovsky, Richard. "Sildenafil is Safe and Effective in Men With Stable CAD." *American Family Physician* (September 1, 2004): 955.

Tomlinson, John. "The Patient With Erectile Dysfunction." *The Practitioner* (February 9, 2005): 104.

Trissel, Lawrence A. "Long-Term Stability of Trimix: A Three-Drug Injection Used to Treat Erectile Dysfunction." *International Journal of Pharmaceutical Compounding* (May-June 2004): 231-235.

Wooten, James M. "Erectile Dysfunction: There are Now Three Effective Oral Drugs for Treating ED, Paving the Way for Nurses to Put Patients With This Condition on the Road to Good Sexual Health." *RN* (October 2004): 40.

Zoler, Mitchel L. "Drug Update: Erectile Dysfunction." *OB/GYN News* (April 15, 2004): 92.

ORGANIZATIONS

Erectile Dysfunction Information Center. 10949 Bren Road East, Minnetonka, MN 55343-9613. (866) 294-7508. publisher@erectile-dysfunction-impotence.org. <http://www.cure-ed.org>. 952-852-5560

OTHER

December 2003. National Kidney and Urologic Diseases Information Clearinghouse *Erectile Dysfunction* <http://kidney.niddk.nih.gov/kudiseases/pubs/impotence/> (Accessed March 24, 2005).

Ken R. Wells

Erectile dysfunction treatment

Definition

Drugs and devices that treat **erectile dysfunction** (ED), the inability to achieve or maintain an erection long enough to engage in sexual intercourse.

Purpose

The purpose of ED treatment is to allow men to achieve and maintain an erection of sufficient strength and duration to engage in sexual intercourse.

Precautions

Because sexual activity can **stress** the heart, men who have heart problems should check with their physician to see if sexual activity is recommended. Erectile dysfunction drugs may trigger temporary **hypotension** (low blood pressure) and is known to increase cardiovascular nerve activity, so physicians should prescribe them with caution in men with a history of **heart attack**, **atherosclerosis** (hardening of the arteries), **angina**, arrhythmia, and chronic low blood pressure problems. ED drugs are not labeled or approved for use by women or children, or by men without erectile dysfunction. Anyone experiencing cardiovascular symptoms such as **dizziness**, chest or arm **pain**, and **nausea** when participating in sexual activity after taking an ED medication should stop the encounter. They should also not take any ED drug again until they have discussed the episode with their healthcare provider. It is recommended that men with kidney or liver impairments, and men over age 65, start at the lowest possible dosage of ED medications.

Description

There are three prescription medications available in the United States to treat the physical causes of ED: **sildenafil citrate** (Viagra), **vardeafil** (Levitra), and **tadalafil** (Cialis). They help about three-fourths of all

men who try them in the general population. Several studies have indicated that their success rate in diabetic men may be slightly lower, averaging around 60-65%. All three pills work by enhancing the effects of nitric oxide. This chemical relaxes muscles in the penis to allow more blood to flow in. These pills do not cause an erection by themselves. Sexual stimulation is also required. Sildenafil or vardenafil should be taken about an hour before sex. Each is effective for roughly four hours. Tadalafil lasts for up to 36 hours. Men should not have sex more than once every 24 hours after using these drugs. Before Viagra's approval in 1998, drug treatment of erectile dysfunction was limited to alprostadil (prostaglandin), either injected into the penis or inserted as a pellet into the urethra. Sales of the three newer ED drugs reached \$3.4 billion in 2004, according to the research firm IMS Health. In 2004, Viagra had about 66% of the market share for ED drugs compared to 19% for Levitra and 14% for Cialis.

Viagra

Sildenafil was originally developed in 1991 as a treatment for angina, or chest pain. The drug, marketed under the name Viagra, received FDA approval as a treatment for erectile dysfunction in March 1998, and since that time it has been prescribed for more than 20 million men worldwide. It was the first oral medication approved for ED treatment. Viagra is a vasodilator, a drug that has the effect of dilating the blood vessels. It works by improving blood circulation to the penis, and by enhancing the effects of nitric oxide, the agent that relaxes the smooth muscle of the penis and regulates blood vessels during sexual stimulation, allowing the penis to become engorged and achieve an erection.

The average recommended dose of Viagra is 50 mg. It comes in doses of 25 mg., 50 mg., and 100 mg to. The medication is taken approximately one hour before sexual activity is planned, and may remain effective for up to four hours. One drawback is that to be effective, it should be taken on an empty stomach. Also, high-fat foods can interfere with the absorption of Viagra. Viagra does not increase sexual desire. Sexual stimulation and arousal are required for it to be effective.

Levitra

In early 2003, a second prescription drug to treat erectile dysfunction, Levitra, was approved by the FDA. Like Viagra, Levitra helps increase blood flow to the penis and may help men with ED get and keep an erection. Once a man has completed sexual activity,

blood flow to the penis should decrease and the erection should go away. Levitra should be taken approximately 60 minutes prior to sexual activity. In clinical trials, most patients were able to begin sexual activity before that time. A 2004 study showed that about 50% of men taking Levitra experienced a firm erection within 25 minutes and a small percentage in as quickly as 10 minutes. Studies also showed that Levitra improved erectile function in men who had other health factors, such as diabetes or prostate surgery.

Men taking nitrate drugs, often used to control chest pain (also known as angina), should not take Levitra. Men who use alpha-blockers, sometimes prescribed for high blood pressure or prostate problems, also should not take Levitra. Such combinations could cause blood pressure to drop to an unsafe level. Levitra is available in 2.5 mg, 5 mg, 10 mg, and 20 mg tablets.

Cialis

Cialis is the third oral drug prescribed to treat erectile dysfunction, approved by the FDA in November 2003. Its most notable difference from Viagra and Levitra, which work for about four hours, is that Cialis works for up to 36 hours. Cialis helps increase blood flow in the penis when a man is sexually stimulated. It can help men with ED get and keep an erection satisfactory for sexual activity. Once a man has completed sexual activity, blood flow to his penis decreases, and his erection goes away. Cialis is clinically proven to improve erectile function in most men with ED, including those with mild, moderate or severe ED.

The most common side effects with Cialis are **headache**, upset stomach, back pain, and muscle aches. These side effects usually go away after a few hours. Patients who get back pain and muscle aches usually get it 12 to 24 hours after taking Cialis. Back pain and muscle aches usually go away by themselves within 48 hours.

Cialis comes in 5 mg., 10 mg. (the recommended starting dose), and 20 mg. tablets. Since the absorption of Cialis is not affected by food or high-fat foods, it does not need to be taken on an empty stomach. Studies show that in most men, Cialis begins working in about 30 minutes and may be taken up to once per day by most patients.

The cost of Viagra, Levitra, and Cialis is about \$10 per dose and is covered by most insurance plans.

Although the commercial availability of Viagra, Levitra, and Cialis has been useful in many men, **prostate cancer** patients and ED caused by psychological problems often require alternative treatment. A

commonly used alternative consists of a three-drug injection containing alprostadil, papaverine hydrochloride, and phentolamine mesylate. Though it is commonly referred to as the “Knoxville formula,” apparently for the city of its original introduction, a number of slightly varying formulas have been in use around the country. The three-drug preparation is administered by injection into the corpora cavernosa to induce erection.

Other traditional therapies for ED include vacuum pump therapy, injection therapy involving injecting a substance into the penis to enhance blood flow, and a penile implantation device. In rare cases, if narrowed or diseased veins are responsible for ED, surgeons may reroute the blood flow into the corpora cavernosa or remove leaking vessels.

In vacuum pump therapy, a man inserts his penis into a clear plastic cylinder and uses a pump to force air out of the cylinder. This forms a partial vacuum around the penis, which helps to draw blood into the corpora cavernosa. The man then places a special ring over the base of the penis to trap the blood inside it. The only side effect with this type of treatment is occasional bruising if the vacuum is left on too long.

Injection therapy involves injecting a substance into the penis to enhance blood flow and cause an erection. The FDA approved a drug called alprostadil (Caverject) for this purpose in 1995. Alprostadil relaxes smooth muscle tissue to enhance blood flow into the penis. It must be injected shortly before intercourse. Another, similar drug that is sometimes used is papaverine. Either drug may sometimes cause painful erections or **priapism** that must be treated with a shot of epinephrine. Alprostadil may also be administered into the urethral opening of the penis. In MUSE (medical urethral system for erection), the man inserts a thin tube the width of a spaghetti noodle into his urethral opening and presses down on a plunger to deliver a tiny pellet containing alprostadil into his penis. The drug takes about 10 minutes to work and the erection lasts about an hour. The main side effect is a sensation of pain and burning in the urethra, which can last about five to 15 minutes. The injection process itself is often painful.

Implantable **penile prostheses** are usually considered a last resort for treating erectile dysfunction. They are implanted in the corpora cavernosa to make the penis rigid without the need for blood flow. The semi-rigid type of prosthesis consists of a pair of flexible silicone rods that can be bent up or down. This type of device has a low failure rate

KEY TERMS

Angina—A condition in which lack of blood to the heart causes severe chest pain.

Arrhythmia—An irregularity in the normal rhythm or force of the heartbeat.

Atherosclerosis—An arterial disease in which raised areas of degeneration and cholesterol deposits (plaques) form on the inner surfaces of the arteries. Also called hardening of the arteries.

Corpora cavernosa—Either of a pair of columns of erectile tissue at either side of the penis that, together with the corpus spongiosum, produce an erection when filled with blood.

Priapism —A prolonged erection lasting more than four hours.

Vasodilator—An agent, such as a nerve or hormone, that widens the blood vessels, which in turn decreases resistance to blood flow and lowers blood pressure.

but, unfortunately, it causes the penis to always be erect, which can be difficult to conceal under clothing.

The inflatable type of device consists of cylinders that are implanted in the corpora cavernosa, a fluid reservoir implanted in the abdomen, and a pump placed in the scrotum. The man squeezes the pump to move fluid into the cylinders and cause them to become rigid. (He reverses the process by squeezing the pump again.) While these devices allow for intermittent erections, they have a slightly higher malfunction rate than the silicon rods. Men can return to sexual activity six to eight weeks after implantation surgery. Since implants affect the corpora cavernosa, they permanently take away a man’s ability to have a natural erection.

Preparation

No preparation is needed for Viagra, Levitra, or Cialis.

Aftercare

The primary aftercare is to monitor blood pressure following use of ED medications.

Risks

Priapism, a prolonged erection, is a very rare potential side effect of all prescription ED

medications. Persons taking any erectile dysfunction medication who have a prolonged erection, lasting more than four hours, should seek immediate medical attention. Priapism can cause damage to the penis potentially leading to the permanent inability to have an erection.

Normal results

The goal of all ED treatments is to achieve and maintain an erection strong enough to engage in sexual intercourse, which is usually 30-60 minutes.

Abnormal results

The most common abnormal response is the failure to achieve and maintain an erection strong enough and long enough to engage in sexual intercourse.

Resources

BOOKS

Icon Health Publications. *The Official Patient's Sourcebook on Erectile Dysfunction: A Revised and Updated Directory for the Internet Age*. San Diego: Icon Health Publications, 2005.

Kloner, Robert A. *Heart Disease and Erectile Dysfunction*. Totowa, NJ: Humana Press, 2004.

Lue, Tom F. *An Atlas of Erectile Dysfunction, Second Edition*. London: Taylor & Francis Group, 2003.

Metz, Michael E. and Barry W. McCarthy. *Coping With Erectile Dysfunction: How to Regain Confidence and Enjoy Great Sex*. Oakland, CA: New Harbinger Publications, 2004.

PERIODICALS

Roberts, Shauna S. "Options Increase for Men With Erectile Dysfunction: Doctors Now Have an Array of Treatments for Men's Sexual Problems." *Diabetes Forecast* (September 2004): 55-58.

Sadovsky, Richard. "Sildenafil is Safe and Effective in Men With Stable CAD." *American Family Physician* (September 1, 2004): 955.

Tomlinson, John. "The Patient With Erectile Dysfunction." *The Practitioner* (February 9, 2005): 104.

Trissel, Lawrence A. "Long-Term Stability of Trimix: A Three-Drug Injection Used to Treat Erectile Dysfunction." *International Journal of Pharmaceutical Compounding* (May-June 2004): 231-235.

Wooten, James M. "Erectile Dysfunction: There are Now Three Effective Oral Drugs for Treating ED, Paving the Way for Nurses to Put Patients With This Condition on the Road to Good Sexual Health." *RN* (October 2004): 40.

Zoler, Mitchel L. "Drug Update: Erectile Dysfunction." *OB/GYN News* (April 15, 2004): 92.

ORGANIZATIONS

Erectile Dysfunction Information Center. 10949 Bren Road East, Minnetonka, MN 55343-9613. (866) 294-7508. publisher@erectile-dysfunction-impotence.org. <http://www.cure-ed.org>. 952-852-5560

OTHER

National Kidney and Urologic Diseases Information Clearinghouse *Erectile Dysfunction*. December 2003. <http://kidney.niddk.nih.gov/kudiseases/pubs/impotence/> (cited March 31, 2005).

Ken R. Wells

Ergotamine see **Antimigraine drugs**

Erosive gastritis see **Gastritis**

Erysipelas

Definition

Erysipelas is a skin infection that often follows strep throat.

Description

Erysipelas, also called St. Anthony's fire, is caused by infection by Group A *Streptococci*. This same type of bacteria is responsible for such infections as strep throat, and infections of both surgical and other kinds of **wounds** in the skin. The infection occurs most often in young infants and the elderly.

Causes and symptoms

Erysipelas usually occurs rather abruptly. When the preceding infection was strep throat, the rash begins on the face. Occasionally, when the preceding infection was of a wound from an injury or operation, the rash will appear on an arm or leg.

Classically, the usual presentation is a bright-red, butterfly-shaped rash appearing across the bridge of the nose and the cheeks. It is hot to the touch, painful, shiny, and swollen, with clearly defined margins. The edges of the rash are a raised ridge, hard to the touch. There may be fluid-filled bumps scattered along the area. The rash spreads rapidly. Some patients have swelling of the eyelids, sometimes so severe that their eyes swell shut. The patient may have **fever**, chills, loss of energy, **nausea** and vomiting, and swollen, tender lymph nodes. In severe

KEY TERMS

Bacteremia—The presence of bacteria in the blood.

Streptococcus—A bacteria that causes erysipelas and strep throat, as well as other infections.

cases, walled-off areas of pus (abscesses) may develop beneath the skin. If left untreated, the streptococcal bacteria may begin circulating in the bloodstream (a condition called **bacteremia**). A patient may then develop an overwhelming, systemic infection called **sepsis**, with a high risk of **death**.

Diagnosis

The rash of erysipelas is very characteristic, raising the practitioner's suspicion towards that diagnosis, especially when coupled with a history of recent strep infection. Attempts to culture (grow) the bacteria from a sample of the rash usually fail. When the bacteria are present in the blood, they may be grown in a laboratory, and identified under a microscope. Other laboratory tests involve reacting fluorescently-tagged antibodies with a sample of the patient's infected tissue. This type of test may be successful in positively identifying the streptococcal bacteria.

Treatment

Penicillin is the drug of choice for treating erysipelas. It can usually be given by mouth, although in severe cases (or in cases of diagnosed bacteremia) it may be given through a needle placed in a vein (intravenously).

Even with antibiotic treatment, swelling may continue to spread. Other symptoms, such as fever, **pain**, and redness, usually decrease rapidly after penicillin is started. Cold packs and pain relievers may help decrease discomfort. Within about five to 10 days, the affected skin may begin drying up and flaking off.

Prognosis

With prompt treatment, the prognosis from erysipelas is excellent. Delay of treatment, however, increases the chance for bacteremia and the potential for death from overwhelming sepsis. This is particularly true of people with weakened immune systems (babies, the elderly, and people ill with other diseases, especially Acquired **Immunodeficiency**

Syndrome, or **AIDS**). Frequently, an individual who has had erysipelas will have it occur again in the same location.

Prevention

Prevention involves appropriate and complete treatment of streptococcal infections, including **strep throat** and wound infections.

Resources

PERIODICALS

Huerter, Christopher, et al. "Helpful Clues to Common Rashes." *Patient Care* 31, no. 8 (April 30, 1997): 9+.

Rosalyn Carson-DeWitt, MD

Erythema infectiosum *see* **Fifth disease**

Erythema multiforme

Definition

Erythema multiforme is a skin disease that causes lesions and redness around the lesions.

Description

Erythema multiforme appears on the skin and the mucous membranes (the lining of the mouth, digestive tract, vagina, and other organs). Large, symmetrical red blotches appear all over the skin in a circular pattern. On mucous membranes, it begins as blisters and progresses to ulcers. A more advanced form, called Stevens-Johnson syndrome, can be severe and even fatal.

Causes and symptoms

Erythema multiforme has many causes, most commonly are drugs. Penicillin, **sulfonamides**, certain epilepsy drugs, aspirin, and **acetaminophen** are the most likely medication-induced causes. Erythema multiforme can also be caused by certain diseases. Herpes virus and mycoplasma pneumonia are likely infectious causes.

Diagnosis

The appearance of the rash is sufficiently unique to identify it on sight. Having identified it, the physician will determine the underlying cause.

KEY TERMS

Herpes virus—Viruses that can infect the skin, mucous membranes, and brain, and they are responsible for such diseases as herpes simplex, chicken pox, and shingles.

Mycoplasma pneumoniae—An incomplete bacterium that infects the lung.

Treatment

Erythema multiforme is inadvertently treated when the causative agent, whether it be a drug or a disease, is treated. In severe cases, cortisone-like medication is often used along with general supportive measures and prevention of infection.

Prognosis

As a rule, the rash abates by itself without damaging the skin. Only in the case of infection, severe blistering, or continued use of an offending drug does complications occur.

Resources

BOOKS

Fauci, Anthony S., et al., editors. *Harrison's Principles of Internal Medicine*. New York: McGraw-Hill, 1997.

J. Ricker Polsdorfer, MD

Erythema nodosum

Definition

Erythema nodosum is a skin disorder characterized by painful red nodules appearing mostly on the shins.

Description

Erythema nodosum is an eruption of tender red lumps on both shins and occasionally the arms and face. Bruising often accompanies the nodule formation. Erythema nodosum is most prevalent in young adults.

Causes and symptoms

Erythema nodosum can be caused by many important and treatable diseases. Among them are

KEY TERMS

Biopsy—Surgical removal of tissue for diagnostic purposes.

Panniculitis—Inflammation of fatty tissue.

tuberculosis, several fungal lung infections, **leprosy**, inflammatory bowel disease, and some potentially dangerous bacterial infections. Drugs can also induce erythema nodosum. The most common are penicillin, **sulfonamides**, and birth control pills.

Diagnosis

There are a few other skin eruptions that mimic erythema nodosum, so the physician may have to perform a biopsy to sort them out. There are a few types of *panniculitis*, fat inflammation, that may signal a **cancer** somewhere in the body, and there are other kinds of inflammation that may confuse the diagnosis.

Once the skin problem has been diagnosed, its underlying cause must then be identified. A lengthy evaluation may ensue, and often times the cause remains unknown.

Treatment

Painful nodules can be treated with mild **pain** killers and local application of ice packs. Medical attention will be directed toward the underlying disease.

The nodules will eventually disappear, leaving no trace behind.

Resources

BOOKS

Bennett, J. Claude, and Fred Plum, editors. *Cecil Textbook of Medicine*. Philadelphia: W. B. Saunders Co., 1996.

J. Ricker Polsdorfer, MD

Erythremia see **Polycythemia vera**

Erythroblastosis fetalis

Definition

Erythroblastosis fetalis refers to two potentially disabling or fatal blood disorders in infants: Rh

incompatibility disease and ABO incompatibility disease. Either disease may be apparent before birth and can cause fetal **death** in some cases. The disorder is caused by incompatibility between a mother's blood and her unborn baby's blood. Because of the incompatibility, the mother's immune system may launch an immune response against the baby's red blood cells. As a result, the baby's blood cells are destroyed, and the baby may suffer severe anemia (deficiency in red blood cells), brain damage, or death.

Description

Red blood cells carry several types of proteins, called antigens, on their surfaces. The A, B, and O antigens are used to classify a person's blood as type A, B, AB, or O. Each parent passes one A, B, or O antigen gene to their child. How the genes are paired determines the person's blood type.

A person who inherits an A antigen gene from each parent has type A blood; receiving two B antigen genes corresponds with type B blood; and inheriting A and B antigen genes means a person has type AB blood. If the O antigen gene is inherited from both parents, the child has type O blood; however, the pairing of A and O antigen genes corresponds with type A blood; and if the B antigen gene is matched with the O antigen gene, the person has type B blood.

Another red blood cell antigen, called the Rh factor, also plays a role in describing a person's blood type. A person with at least one copy of the gene for the Rh factor has Rh-positive blood; if no copies are inherited, the person's blood type is Rh-negative. In **blood typing**, the presence of A, B, and O antigens, plus the presence or absence of the Rh-factor, determine a person's specific blood type, such as A-positive, B-negative, and so on.

A person's blood type has no effect on health. However, an individual's immune system considers only that person's specific blood type, or a close match, acceptable. If a radically different blood type is introduced into the bloodstream, the immune system produces antibodies, proteins that specifically attack and destroy any cell carrying the foreign antigen.

Determining a person's blood type is very important if she becomes pregnant. Blood cells from the unborn baby (fetal red blood cells) can cross over into the mother's bloodstream, especially at delivery. If the mother and her baby have compatible blood types, the crossover does not present any danger. However, if the blood types are incompatible, the

mother's immune system manufactures antibodies against the baby's blood.

Usually, this incompatibility is not a factor in a first **pregnancy**, because few fetal blood cells reach the mother's bloodstream until delivery. The antibodies that form after delivery cannot affect the first child. In later pregnancies, fetuses and babies may be in grave danger. The danger arises from the possibility that the mother's antibodies will attack the fetal red blood cells. If this happens, the fetus or baby can suffer severe health effects and may die.

There are two types of incompatibility diseases: Rh incompatibility disease and ABO incompatibility disease. Both diseases have similar symptoms, but Rh disease is much more severe, because anti-Rh antibodies cross over the placenta more readily than anti-A or anti-B antibodies. (The immune system does not form antibodies against the O antigen.) Therefore, a greater percentage of the baby's blood cells are destroyed by Rh disease.

Both incompatibility diseases are uncommon in the United States due to medical advances over the last 50 years. For example, prior to 1946 (when newborn blood transfusions were introduced) 20,000 babies were affected by Rh disease yearly. Further advances, such as suppressing the mother's antibody response, have reduced the incidence of Rh disease to approximately 4,000 cases per year.

Rh disease only occurs if a mother is Rh-negative and her baby is Rh-positive. For this situation to occur, the baby must inherit the Rh factor gene from the father. Most people are Rh-positive. Only 15% of the Caucasian population is Rh-negative, compared to 5–7% of the African-American population and virtually none of Asian populations.

ABO incompatibility disease is almost always limited to babies with A or B antigens whose mothers have type O blood. Approximately one third of these babies show evidence of the mother's antibodies in their bloodstream, but only a small percentage develop symptoms of ABO incompatibility disease.

Cause and symptoms

Rh disease and ABO incompatibility disease are caused when a mother's immune system produces antibodies against the red blood cells of her unborn child. The antibodies cause the baby's red blood cells to be destroyed and the baby develops anemia. The baby's body tries to compensate for the anemia by releasing immature red blood cells, called erythroblasts, from the bone marrow.

The overproduction of erythroblasts can cause the liver and spleen to become enlarged, potentially causing liver damage or a ruptured spleen. The emphasis on erythroblast production is at the cost of producing other types of blood cells, such as platelets and other factors important for blood clotting. Since the blood lacks clotting factors, excessive bleeding can be a complication.

The destroyed red blood cells release the blood's red pigment (hemoglobin) which degrades into a yellow substance called bilirubin. Bilirubin is normally produced as red blood cells die, but the body is only equipped to handle a certain low level of bilirubin in the bloodstream at one time. Erythroblastosis fetalis overwhelms the removal system, and high levels of bilirubin accumulate, causing hyperbilirubinemia, a condition in which the baby becomes jaundiced. The **jaundice** is apparent from the yellowish tone of the baby's eyes and skin. If hyperbilirubinemia cannot be controlled, the baby develops kernicterus. The term kernicterus means that bilirubin is being deposited in the brain, possibly causing permanent damage.

Other symptoms that may be present include high levels of insulin and low blood sugar, as well as a condition called hydrops fetalis. Hydrops fetalis is characterized by an accumulation of fluids within the baby's body, giving it a swollen appearance. This fluid accumulation inhibits normal breathing, because the lungs cannot expand fully and may contain fluid. If this condition continues for an extended period, it can interfere with lung growth. Hydrops fetalis and anemia can also contribute to heart problems.

Diagnosis

Erythroblastosis fetalis can be predicted before birth by determining the mother's blood type. If she is Rh-negative, the father's blood is tested to determine whether he is Rh-positive. If the father is Rh-positive, the mother's blood will be checked for antibodies against the Rh factor. A test that demonstrates no antibodies is repeated at week 26 or 27 of the pregnancy. If antibodies are present, treatment is begun.

In cases in which incompatibility is not identified before birth, the baby suffers recognizable characteristic symptoms such as anemia, hyperbilirubinemia, and hydrops fetalis. The blood incompatibility is uncovered through blood tests such as the Coombs test, which measures the level of maternal antibodies attached to the baby's red blood cells. Other blood tests reveal anemia, abnormal blood counts, and high levels of bilirubin.

Treatment

When a mother has antibodies against her unborn infant's blood, the pregnancy is watched very carefully. The antibodies are monitored and if levels increase, **amniocentesis**, fetal umbilical cord blood sampling, and ultrasound are used to assess any effects on the baby. Trouble is indicated by high levels of bilirubin in the amniotic fluid or baby's blood, or if the ultrasound reveals hydrops fetalis. If the baby is in danger, and the pregnancy is at least 32–34 weeks along, labor is induced. Under 32 weeks, the baby is given blood transfusions while still in the mother's uterus.

There are two techniques that are used to deliver a blood **transfusion** to a baby before birth. In the first, a needle is inserted through the mother's abdomen and uterus, and into the baby's abdomen. Red blood cells injected into the baby's abdominal cavity are absorbed into its bloodstream. In early pregnancy or if the baby's bilirubin levels are gravely high, cordocentesis is performed. This procedure involves sliding a very fine needle through the mother's abdomen and, guided by ultrasound, into a vein in the umbilical cord to inject red blood cells directly into the baby's bloodstream.

After birth, the severity of the baby's symptoms are assessed. One or more transfusions may be necessary to treat anemia, hyperbilirubinemia, and bleeding. Hyperbilirubinemia is also treated with **phototherapy**, a treatment in which the baby is placed under a special light. This light causes changes in how the bilirubin molecule is shaped, which makes it easier to excrete. The baby may also receive oxygen and intravenous fluids containing electrolytes or drugs to treat other symptoms.

Prognosis

In many cases of blood type incompatibility, the symptoms of erythroblastosis fetalis are prevented with careful monitoring and blood type screening. Treatment of minor symptoms is typically successful and the baby will not suffer long-term problems.

Nevertheless, erythroblastosis is a very serious condition for approximately 4,000 babies annually. In about 15% of cases, the baby is severely affected and dies before birth. Babies who survive pregnancy may develop kernicterus, which can lead to deafness, speech problems, cerebral palsy, or mental retardation. Extended hydrops fetalis can inhibit lung growth and contribute to heart failure. These serious complications are life threatening, but with good medical

KEY TERMS

Amniocentesis—A procedure in which a needle is inserted through a pregnant woman's abdomen and into her uterus to withdraw a small sample of amniotic fluid. The amniotic fluid can be examined for sign of disease or other problems afflicting the fetus.

Amniotic fluid—The fluid that surrounds a fetus in the uterus.

Anemia—A condition in which there is an abnormally low number of red blood cells in the bloodstream. Major symptoms are paleness, shortness of breath, unusually fast or strong heart beats, and tiredness.

Antibody—A protein molecule produced by the immune system in response to a protein that is not recognized as belonging in the body.

Antigen—A protein that can elicit an immune response in the form of antibody formation. With regard to red blood cells, the major antigens are A, B, O, and the Rh factor.

Bilirubin—A yellow-colored end-product of hemoglobin degradation. It is normally present at very low levels in the bloodstream; at high levels, it produces jaundice.

Cordocentesis—A procedure for delivering a blood transfusion to a fetus. It involves a fine needle being threaded through a pregnant woman's abdomen and into the umbilical cord with the aid of ultrasound imaging.

Hemoglobin—A molecule in red blood cells that transports oxygen and gives the cells their characteristic color.

Hydrops fetalis—A condition in which a fetus or newborn baby accumulates fluids, causing swollen arms and legs and impaired breathing.

Hyperbilirubinemia—A condition in which bilirubin accumulates to abnormally high levels in the bloodstream.

Placenta—A protective membrane that surrounds and protects the fetus during pregnancy.

Platelet—A blood factor that is important in forming blood clots.

Rh factor—An antigen that is found on the red blood cells of most people. If it is present, the blood type is referred to as Rh-positive; if absent, the blood type is Rh-negative.

treatment, the fatality rate is very low. According to the U.S. Centers for Disease Control and Prevention, there were 21 infant deaths in the United States during 1996 that were attributable to hemolytic disease (erythroblastosis fetalis) and jaundice.

Prevention

With any pregnancy, whether it results in a live birth, **miscarriage**, **stillbirth**, or abortion, blood typing is a universal precaution against blood compatibility disease. Blood types cannot be changed, but adequate forewarning allows precautions and treatments that limit the danger to unborn babies.

If an Rh-negative woman gives birth to an Rh-positive baby, she is given an injection of immunoglobulin G, a type of antibody protein, within 72 hours of the birth. The immunoglobulin destroys any fetal blood cells in her bloodstream before her immune system can react to them. In cases where this precaution is not taken, antibodies are created and future pregnancies may be complicated.

Resources

PERIODICALS

Bowman, John. "The Management of Hemolytic Disease in the Fetus and Newborn." *Seminars in Perinatology* 21, no. 1 (February 1997): 39.

Julia Barrett

Erythrocyte sedimentation rate

Definition

The erythrocyte sedimentation rate (ESR), or sedimentation rate (sed rate), is a measure of the settling of red blood cells in a tube of blood during one hour. The rate is an indication of inflammation and increases in many diseases.

Purpose

ESR is increased in rheumatoid diseases, most infections, and in **cancer**. An advanced rate doesn't diagnose a specific disease, but it does indicate that an underlying disease may be present.

A physician can use ESR to monitor a person with an associated disease. When the disease worsens, the

ESR increases; when the disease improves, the ESR decreases. The ESR doesn't always follow the course of cancer.

ESR is called an acute-phase reactant test, meaning that it reacts to acute conditions in the body, such as infection or trauma. The rate increase follows a rise in temperature and white blood cells count, peaks after several days, and usually lasts longer than the elevated temperature or white blood cells count.

Precautions

The ESR should not be used to screen healthy persons for disease.

Description

The ESR test is a simple test dating back to the ancient Greeks. A specific amount of diluted, unclotted blood is placed in a special narrow tube and left undisturbed for exactly one hour. The red cells settle towards the bottom of the tube, and the pale yellow liquid (plasma) rises to the top. After 60 minutes, measurements are taken of the distance the red cells traveled to settle at the bottom of the tube. Two methods, the Westergren and the Wintrobe, are used by laboratories; each method produces slightly different results. Most laboratories use the Westergren method.

Normally red cells don't settle far toward the bottom of the tube. Many diseases make extra or abnormal proteins that cause the red cells to move close together, stack up, and form a column (rouleaux). In a group, red cells are heavier and fall faster. The faster they fall, the further they settle, and the higher the ESR.

The ESR test is covered by insurance when medically necessary. Results are usually available the same or following day.

Preparation

This test requires 7mL–10 mL of blood. A health-care worker ties a tourniquet on the patient's upper arm, locates a vein in the inner elbow region, and inserts a needle into that vein. Vacuum action draws the blood through the needle into an attached tube. Collection of the sample takes only a few minutes.

Aftercare

Discomfort or bruising may occur at the puncture site. Pressure applied to the puncture site until the

KEY TERMS

Acute phase reactant—A substance in the blood that increases as a response to an acute conditions such as infection, injury, tissue destruction, some cancers, burns, surgery, or trauma.

Erythrocyte sedimentation rate (ESR)—The distance that red blood cells settle in a tube of blood in one hour. It is an indication of inflammation.

Rouleaux—The stacking up of red blood cells, caused by extra or abnormal proteins in the blood that decrease the normal distance red cells maintain between each other.

bleeding stops reduces bruising. Warm packs to the puncture site relieve discomfort. The patient may feel dizzy or faint.

Normal results

A normal value does not rule out disease. Normal values for the Westergren method are: Men 0 mm/hour–15 mm/hour; women 0 mm/hour–20 mm/hour; and children 0 mm/hour–10 mm/hour.

Abnormal results

The highest ESR levels are usually seen in a cancer of a certain type of white blood cell (**multiple myeloma**) and rheumatoid disease, such as **rheumatoid arthritis**. Many other diseases also increase the ESR: infection, **kidney disease**, anemia, diseases involving white blood cells, cancer, and autoimmune and inflammatory diseases.

Any disease that changes the shape and size of red blood cells decreases the ESR. Distorted cells, such as with **sickle cell disease**, do not stack, and consequently do not settle far, even in the presence of an ESR-associated disease. Diseases that cause the body to make less protein or extra red blood cells also decrease the ESR.

Resources

PERIODICALS

Saadeh, Constantine. "The Erythrocyte Sedimentation Rate: Old and New Clinical Applications." *Southern Medical Journal* March 1998: 220-255.

Nancy J. Nordenson

Erythromycins

Definition

Erythromycins are medicines that kill bacteria or prevent their growth.

Purpose

Erythromycins are **antibiotics**, medicines used to treat infections caused by microorganisms. Physicians prescribe these drugs for many types of infections caused by bacteria, including **strep throat**, sinus infections, **pneumonia**, ear infections, **tonsillitis**, **bronchitis**, **gonorrhea**, **pelvic inflammatory disease** (PID), and urinary tract infections. Some medicines in this group are also used to treat **Legionnaires' disease** and ulcers caused by bacteria. These drugs will *not* work for colds, flu, and other infections caused by viruses.

Description

The drugs described here include erythromycins (Erythrocin, Ery-C, E-Mycin, and other brands) and medicines that are chemically related to erythromycins, such as azithromycin (Zithromax) and clarithromycin (Biaxin). They are available only with a physician's prescription and are sold in capsule, tablet (regular and chewable), liquid, and injectable forms.

Recommended dosage

The recommended dosage depends on the type of erythromycin, the strength of the medicine, and the medical problem for which it is being taken. Check with the physician who prescribed the drug or the pharmacist who filled the prescription for the correct dosage.

Always take erythromycins exactly as directed. Never take larger, smaller, more frequent, or less frequent doses. To make sure the infection clears up completely, it is very important to take the medicine for as long as it has been prescribed. Do not stop taking the drug just because symptoms begin to improve. This is important with all types of infections, but it is especially important in "strep" infections, which can lead to serious heart problems if they are not cleared up completely.

Erythromycins work best when they are at constant levels in the blood. To help keep levels constant, take the medicine in doses spaced evenly through the day and night. Do not miss any doses. Some of these medicines are most effective when taken with a full glass of water on an empty stomach, but they may be

taken with food if stomach upset is a problem. Others work equally well when taken with or without food. Check package directions or ask the physician or pharmacist for instructions on how to take the medicine.

Precautions

Symptoms should begin to improve within a few days of beginning to take this medicine. If they do not, or if they get worse, check with the physician who prescribed the medicine.

Erythromycins may cause mild **diarrhea**, that usually goes away during treatment. However, severe diarrhea could be a sign of a very serious side effect. Anyone who develops severe diarrhea while taking erythromycin or related drugs should stop taking the medicine and call a physician immediately.

Special conditions

Taking erythromycins may cause problems for people with certain medical conditions or people who are taking certain other medicines. Before taking these drugs, be sure to let the physician know about any of these conditions:

ALLERGIES. Anyone who has had unusual reactions to erythromycins, azithromycin, or clarithromycin in the past should let his or her physician know before taking the drugs again. The physician should also be told about any **allergies** to foods, dyes, preservatives, or other substances.

PREGNANCY. Some medicines in this group may cause problems in pregnant women and have the potential to cause **birth defects**. Women who are pregnant or who may become pregnant should check with their physicians before taking these drugs.

BREASTFEEDING. Erythromycins pass into breast milk. Mothers who are breastfeeding and who need to take this medicine should check with their physicians.

OTHER MEDICAL CONDITIONS. Before using erythromycins, people with any of these medical problems should make sure their physicians are aware of their conditions:

- heart disease
- liver disease
- hearing loss

USE OF CERTAIN MEDICINES. Taking erythromycins with certain other drugs may affect the way the drugs work or may increase the chance of side effects.

KEY TERMS

Bronchitis—Inflammation of the air passages of the lungs.

Gonorrhea—A sexually transmitted disease (STD) that causes infection in the genital organs and may cause disease in other parts of the body.

Inflammation—Pain, redness, swelling, and heat that usually develop in response to injury or illness.

Legionnaire's disease—A lung disease caused by a bacterium.

Microorganism—An organism that is too small to be seen with the naked eye.

Pelvic inflammatory disease (PID)—Inflammation of the female reproductive tract, caused by any of several microorganisms. Symptoms include severe abdominal pain, high fever, and vaginal discharge. Severe cases can result in sterility.

Pneumonia—A disease in which the lungs become inflamed. Pneumonia may be caused by bacteria, viruses, or other organisms, or by physical or chemical irritants.

Sinus—Any of several air-filled cavities in the bones of the skull.

Strep throat—A sore throat caused by infection with *Streptococcus* bacteria. Symptoms include sore throat, chills, fever, and swollen lymph nodes in the neck.

Tonsillitis—Inflammation of a tonsil, a small mass of tissue in the throat.

Urinary tract—The passage through which urine flows from the kidneys out of the body.

Side effects

The most common side effects are mild diarrhea, **nausea**, **vomiting**, and stomach or abdominal cramps. These problems usually go away as the body adjusts to the drug and do not require medical treatment. Less common side effects, such as sore mouth or tongue and vaginal itching and discharge also may occur and do not need medical attention unless they persist or are bothersome.

More serious side effects are not common, but may occur. If any of the following side effects occur, check with a physician immediately:

- severe stomach **pain**, nausea, vomiting, or diarrhea
- fever

- skin rash, redness, or **itching**
- unusual tiredness or weakness

Although rare, very serious reactions to azithromycin (Zithromax) are possible, including extreme swelling of the lips, face, and neck, and **anaphylaxis** (a violent allergic reaction). Anyone who develops these symptoms after taking azithromycin should stop taking the medicine and get immediate medical help.

Other rare side effects may occur with erythromycins and related drugs. Anyone who has unusual symptoms after taking these medicines should get in touch with his or her physician.

Interactions

Erythromycins may interact with many other medicines. When this happens, the effects of one or both of the drugs may change or the risk of side effects may be greater. Anyone who takes erythromycins should let the physician know all other medicines he or she is taking. Among the drugs that may interact with erythromycins are:

- acetaminophen (Tylenol)
- medicine for overactive thyroid
- male hormones (androgens)
- female hormones (estrogens)
- other antibiotics
- blood thinners
- disulfiram (Antabuse), used to treat alcohol **abuse**
- antiseizure medicines such as valproic acid (Depakote, Depakene)
- caffeine
- the **antihistamines** astemizole (Hismanal)
- antiviral drugs such as (zidovudine) Retrovir

The list above does not include every drug that may interact with erythromycins. Be sure to check with a physician or pharmacist before combining erythromycins with any other prescription or nonprescription (over-the-counter) medicine.

Nancy Ross-Flanigan

Erythropoietin see **Cancer therapy, supportive; Immunologic therapies**

Erythropoietin test

Definition

Erythropoietin, also called EPO, is a type of protein called a glycoprotein that is formed mainly in the kidneys to stimulate the production of red blood cells.

Purpose

The erythropoietin (EPO) test is used to determine if hormonal secretion is causing changes in the red blood cells. The test has great value in evaluating low hemoglobin (anemia), and another disorder called polycythemia, in which unusually large numbers of red blood cells are found in the blood. The EPO test is also used to identify kidney tumors and identify or assess **kidney disease**. It also may be used to evaluate **abuse** by athletes who believe commercially prepared erythropoietin enhances performance.

Precautions

Not every laboratory is equipped to evaluate EPO, so the reference laboratory (a large commercial lab that does tests for hospitals not equipped to do them) performing the test may require as many as four days to complete the analysis. It should also be noted that EPO values increase in **pregnancy**, in which significantly higher levels are found before the twenty-fourth week.

Description

Erythropoietin is produced primarily in the kidneys but interacts with other factors in the bone marrow to increase red cell production. EPO is unique among the blood cell growth factors, because it is the only one that behaves like a hormone.

Erythropoietin acts as the principal regulator in the production of red blood cells (erythrocytes) by controlling the number, the kinds, and the survival of the cells. Because of this ability, it is being investigated for use in **cancer** patients to prevent anemia (hemoglobin concentration in the blood is lower than normal), or to treat anemia that has been induced by **chemotherapy** and bone marrow transplantation (BMT).

The correction of anemia can result in reduced transfusion requirements, so the erythropoietin test is used to diagnose anemia, including the anemia of end-

stage renal disease. Erythropoietin determination is also valuable in diagnosing a condition known as polycythemia, when increased numbers of red blood cells occur. Levels of erythropoietin are extremely low in **polycythemia vera** but are normal or high in **secondary polycythemia**. It happens rarely, but cysts in the liver or kidneys, as well as tumors in the kidneys or brain, can also produce erythropoietin. Patients with these conditions can have high levels of erythropoietin and may develop secondary polycythemia.

Kidney disease can cause anemia and many patients on **kidney dialysis** will require monthly EPO tests to check their hemoglobin levels.

Some athletes use EPO to enhance performance, as the increased red cell volume adds more oxygen-carrying capacity to the blood. Adverse reactions to this practice can include clotting abnormalities, **headache**, seizures, high blood pressure, nausea, vomiting, **diarrhea**, and rash.

Preparation

The EPO test requires a blood sample. The patient is to fast with nothing to eat or drink for at least eight hours before the test. It is also suggested that the patient lie down for 30 minutes before the test.

Risks

Risks for this test are minimal, but may include slight bleeding from the blood-drawing site, fainting or feeling lightheaded after venipuncture, and hematoma (blood accumulating under the puncture site).

Normal results

Reference values vary from laboratory to laboratory, but a general normal range is 11–48 mU/ml (milliunits per milliliter).

Abnormal results

Low levels of EPO are found in anemic patients with inadequate or absent production of erythropoietin. Severe kidney disease may decrease production of EPO, and congenital absence of EPO can occur.

Elevated levels of EPO can be found in some **anemias** when the body tries to overcompensate for reduced blood volume. Elevated levels are also seen in polycythemia, and erythropoietin-secreting tumors.

KEY TERMS

Anemia—A condition in which the hemoglobin concentration in the blood is below normal.

Polycythemia vera—A condition characterized by an unusually large number of red blood cells in the blood due to increased production by the bone marrow. Symptoms include headaches, blurred vision, high blood pressure, dizziness, and night sweats.

Secondary polycythemia—Secondary polycythemia occurs when the excess of red blood cells is caused by a condition other than polycythemia vera. For example, when low levels of oxygen in the blood stimulate the bone marrow to produce more red blood cells, as in chronic lung disease.

Resources

PERIODICALS

“GP Clinical: Anemia in Kidney Disease.” *GP* November 5, 2004: 66.

Janis O. Flores
Teresa G. Odle

ESB see **Electrical stimulation of the brain**

Escherichia coli

Definition

E. coli (*Escherichia coli*) is one of several types of bacteria that normally inhabit the intestine of humans and animals (commensal organism). Some strains of *E. coli* are capable of causing disease under certain conditions when the immune system is compromised or disease may result from an environmental exposure.

Description

E. coli bacteria may give rise to infections in **wounds**, the urinary tract, biliary tract, and abdominal cavity (**peritonitis**). This organism may cause septicemia, neonatal **meningitis**, infantile **gastroenteritis**, tourist diarrhea, and hemorrhagic **diarrhea**. An *E. coli* infection may also arise due to environmental exposure. Infections with this type of bacteria pose a serious threat to public health with outbreaks arising

from food and water that has been contaminated with human or animal feces or sewage. This type of bacteria has been used as a biological indicator for safety of drinking water since the 1890s. Exposure may also occur during hospitalization, resulting in **pneumonia** in immunocompromised patients or those on a ventilator

Causes and symptoms

The symptoms of infection and resulting complications are dependent upon the strain of *E. coli* and the site of infection. These bacteria produce toxins that have a wide range of effects. Symptoms caused by some *E. coli* infections range from mild to severe, bloody diarrhea, acute abdominal **pain**, **vomiting**, and **fever**. Gastrointestinal complications that can cause *E. coli* infections include **irritable bowel syndrome** (IBS) ischemic colitis, **appendicitis**, perforation of the large bowel, and in some instances **gangrene** in the colon. Other known *E. coli*-causing infections may include chronic renal failure, **pancreatitis**, and **diabetes mellitus**. Some neurological symptoms such as drowsiness, seizure and **coma** may occur. In infants, *E. coli* infections are present in cases of infantile gastroenteritis and neonatal meningitis.

Strains of *E. coli* that produce diarrhea were initially distinguished by their O (somatic) antigens found on the bacterial surface. Although there is an overlap in characteristics between strains, they may be classified into four main groups; enterohemorrhagic (O157), enteropathogenic (O55,O111), enterotoxigenic (O6,O78), and enteroinvasive (O124,O164).

E. coli O157 (VTEC)

The O157:H7 strain is the member of the group most often associated with a particularly severe form of diarrhea. (The O indicates the somatic antigen, while the H denotes the flagellar antigen, both of which are found on the cell surface of the bacteria.) The bacterium was discovered in 1977, and first reports of infections followed in 1982. *E. coli* O157:H7, as it is frequently referred to by researchers, causes bloody diarrhea in many infected patients. It accounts for about 2% of all cases of diarrhea in the western world, and at least one-third of cases of hemorrhagic colitis, or about 20,000 cases per year

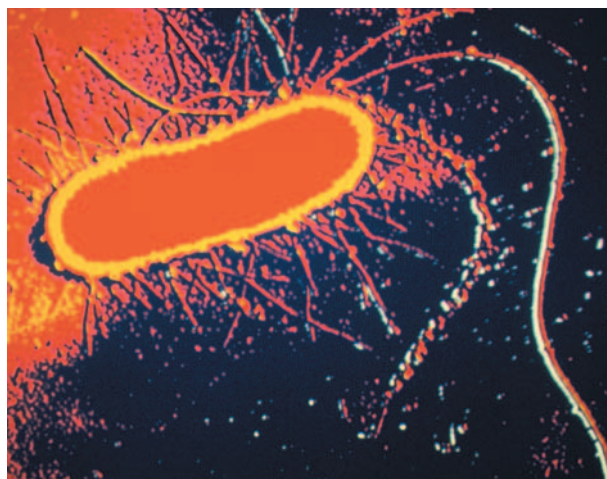
E. coli O157:H7 is also the most common cause of unique syndromes, known as the **Hemolytic-Uremic Syndrome** (HUS) and thrombocytopenic purpura (TTP), which causes kidney failure, hemolytic anemia, and **thrombocytopenia**. Usually, infection with this strain of bacteria will subside without further

complications. However, about 5% of people who are infected will develop HUS/TTP. This infection also accounts for the majority of episodes of HUS, especially in children.

This strain of bacteria produces a potent toxin called verotoxin, named for toxin's ability to kill green monkey kidney or "vero" cells. Bacteria that produce verotoxin are referred to as Verotoxin-producing *E. coli* (VTEC). The numbers of bacteria that are necessary to reproduce infectious levels of bacteria are quite small, estimated at 10-100 viable bacteria. These toxins are lethal for intestinal cells and those that line vessels (endothelial cells), inhibiting protein synthesis causing cell death. It is believed that the damage to blood vessels results in the formation of clots, which eventually leads to the Hemolytic-Uremic Syndrome. HUS/TTP is a serious, often fatal, syndrome that has other causes in addition to *E. coli* O157:H7; it is characterized by the breaking up of red blood cells (hemolysis) and kidney failure (uremia). The syndrome occurs most often in the very young and very old.

E. coli O157:H7 is commonly found in cattle and poultry, and outbreaks have of disease have been associated with cattle and bovine products. There are reports of contamination from unpasteurized apple juice, hamburger meat, radish sprouts, lettuce, and potatoes, as well as other food sources. Environmental contamination may occur in water drained from cattle pastures or water containing human sewage used for drinking or swimming. Human to human transmission, through contact with fecal matter, has also been identified in daycare centers.

After an incubation period of three to four days on average, watery diarrhea begins, which rapidly progresses to bloody diarrhea in many victims, in which case the bowel movement may be mostly blood. **Nausea**, vomiting, and low-grade fever are also frequently present. Gastrointestinal symptoms last for about one week, and recovery is often spontaneous. Symptomatic infection may occur in about 10% of infected individuals. About 5-10% of individuals, usually at the extremes of age or elevated leukocyte count, develop HUS/TTP, and ultimately, kidney failure. Patients taking antibiotics or medications for gastric acidity may also be at risk. Neurological symptoms can also occur as part of HUS/TTP and consist of seizures, **paralysis**, and coma. **Rectal prolapse** may also be a complication, and in some cases colitis, appendicitis, perforation of the large bowel, and gangrene in the bowel. Systemically, the most prevalent complications of *E. coli* 157 infections are HUS and TTP.



A magnified image of the *E. coli* bacterium. (The Stock Market. Reproduced by permission.)

E. coli non-O157 (VTEC)

These strains of *E. coli* produce verotoxin, but are strains other than O157. There have been as many as one hundred different types implicated in the development of disease. Strain OH111 was found to be involved in outbreaks in Australia, Japan, and Italy. The O128, O103, and O55 groups have also been implicated in diarrhea outbreaks. In Britain, cases of infantile gastroenteritis in maternity hospitals and neonatal units have been attributed to the *E. coli* non-O157 group. Many of these organisms have been identified in cattle.

Enterotoxigenic *E. coli*

Two toxins may be produced by this group, the heat-labile enterotoxin (LT) that can produce enteritis in infants, and a heat stable enterotoxin (ST), the action of which has yet to be determined.

Enteroinvasive *E. coli*

Some strains of the enteroinvasive *E. coli* have been involved in the development of gastroenteritis in infants. These organisms do not produce an enterotoxin. The cells of the intestine are affected, with the development of symptoms that are typical of a shigellosis infection.

Diagnosis

Diagnosis of a specific type of infection is dependent upon the characteristics of the particular strain of the organism.

***E. coli* O157:H7 (HUS)**

This particular strain of *E. coli* is suspected when bloody diarrhea, bloody stools, lack of fever, elevated leukocyte count, and abdominal tenderness are present. Stool cultures are used to tentatively identify the bacteria. Unfortunately, cultures are often negative or inconclusive if done after 48 hours of symptoms. Further tests are usually needed, however, for confirmation of infection. This may include a full blood count, blood film, and tests to determine urea, electrolyte, and LDH (lactate dehydrogenase) levels. Damaged red blood cells, and elevated levels of creatinine, urea, and LDH with a drop in **platelet count** may indicate that HUS will develop. Immunomagnetic separation is now being used for diagnosis as well.

***E. coli* non-O157 (VTEC)**

Diagnosis is often difficult for these types of bacteria, but production of enterohemolysin (Ehly) is used as an indicator. Other diagnostic tests are used to detect verotoxins, including ELISA (enzyme-linked immunosorbent assays), colony immunoblotting, and DNA-based tests.

***E. coli* 0157 STEC**

Methods for detection of this type of bacteria are under development, including culture growth media selective for this organism. Immunomagnetic separation and specific ELISA, latex agglutination tests, colony immunoblot assays, and other immunological-based detection methods are being explored.

Treatment

Uncomplicated cases of the *E. coli* O157:H7 the infection clear up within ten days. It is not certain that **antibiotics** are helpful in treating *E. coli* O157:H7 and there is some evidence that they may be harmful. **Dehydration** resulting from diarrhea must be treated with either Oral Rehydration Solution (ORS) or intravenous fluids. Anti-motility agents that decrease the intestines' ability to contract, should not be used in any patient with bloody diarrhea. Treatment of HUS, if it develops, involves correction of clotting factors, plasma exchange, and kidney dialysis. Blood transfusions may be required. Treatment methods for other *E. coli* infections are similar. Antibiotics are often used in the treatment of *E. coli* infections, but their role is controversial. Some antibiotics may enhance the development of HUS/TTP depending upon their action, as well as the use of anti diarrhea medications that should be avoided. Phosphoenolpyruvate

analogues may be helpful. Gentamicin, ampicillin, ceftazidime, or beta-lactamase-stable cephalosporin may be administered for neonatal meningitis. Antibiotic therapy is further complicated by the presence of antibiotic resistant organisms.

Alternative treatment

Studies have been conducted to determine if diarrhea symptoms can be reduced by alternative therapies such as the consumption of herbal teas, psyllium, and **acupuncture**. Patients should consult their doctors before using any alternative treatments, as *E. coli* can be life threatening and should be closely monitored.

Prognosis

In most cases of O157:H7, symptoms last for about a week and recovery is often spontaneous. Ten percent of individuals with *E. coli* O157:H7 infection develop HUS; 5% of those will die of the disease. Some who recover from HUS will be left with some degree of kidney damage and possibly irritable bowel syndrome. Additionally, there is a possibility of chronic *E. coli* infection.

Infants that develop *E. coli* infections may be permanently affected. Gastroenteritis may leave the child with lactose intolerance. Neonates developing meningitis from *E. coli* strains have a high morbidity and mortality rate

Prevention

Thorough cooking of all meat and poultry products and adhering to proper food preparation is the most effective way to avoid infection. More studies are needed to determine the appropriate safety margins for killing these bacteria. Food irradiation methods are also being developed to sanitize food. Vaccinations to *E. coli* 0157 are under development, as are medications aimed at limiting the effects of the verotoxin. The enforcement of regulations for meat production and water are critical. Steam pasteurization is used in the United States and is being explored in other countries.

Prevention of *E. coli* gastroenteritis in infants is best achieved by breast-feeding. The breast milk contains antibodies that combat the infection. For bottle-fed infants, care should be taken in the preparation of the milk and bottles. Good hygiene of the umbilical cord area is important. Keeping this area clean and dry may reduce infection.

KEY TERMS

Antigen—A substance, usually a protein, that causes the formation of an antibody and reacts specifically with that antibody.

Anti-motility medications—Medications such as loperamide (Imodium), dephenoxylate (Lomotil), or medications containing Codeine or narcotics which decrease the ability of the intestine to contract. This can worsen the condition of a patient with dysentery or colitis.

Colitis—Inflammation of the colon or large intestine, usually causing diarrhea that may be bloody.

Food irradiation methods—A process using radiant energy to kill microorganisms in food, to extend the amount of time in which food can be sold and eaten safely.

Oral Rehydration Solution (ORS)—A liquid preparation developed by the World Health Organization that can decrease fluid loss in persons with diarrhea. Originally developed to be prepared with materials available in the home, commercial preparations have recently come into use.

Urea—Chemical formed during the body's metabolism of nitrogen and normally excreted by the kidney. Urea levels rise in the blood when kidney failure occurs.

Resources

BOOKS

Shanson, D. C. *Microbiology in Clinical Practice*. Woburn: Butterworth-Heinemann, 1999.

PERIODICALS

Chart, H., M. Sussman, and D. E. S. Stewart-Tull, eds.

“*E. coli*-Friend or Foe?” *Journal of Applied Microbiology* The Society for Applied Microbiology Symposium Series No. 29.

Long K., E. Vasquez-Garibay, J. Mathewson, J. de la Cabada, and H. DuPont. “The Impact of Infant Feeding Patterns on Infection and Diarrheal Disease Due to Enterotoxigenic *Escherichia coli*.” *Salud Publica Mex* July–August 1999: 263-70.

OTHER

Centers for Disease Control and Prevention. “Preventing Foodborne Illness: *Escherichia coli* 0157:H7.” August 9, 1996. [cited May 30, 2004]. < http://www.cdc.gov/ncidod/dbmd/diseaseinfo/escherichiacoli_g.htm > .

Jill Granger, MS
David Kaminstein, MD

Esophageal acidity test see **Esophageal function tests**

Esophageal aperistalsis see **Achalasia**

Esophageal atresia

Definition

Esophageal atresia is a serious birth defect in which the esophagus, the long tube that connects the mouth to the stomach, is segmented and closed off at any point. This condition usually occurs with **tracheoesophageal fistula**, a condition in which the esophagus is improperly attached to the trachea, the nearby tube that connects the nasal area to the lungs. Esophageal atresia occurs in approximately 1 in 4,000 live births.

Description

Failure of an unborn child (fetus) to develop properly results in **birth defects**. Many of these defects involve organs that do not function, or function only incidentally, before birth, and, as a result, go undetected until the baby is born. In this case, the digestive tract is unnecessary for fetal growth, since all **nutrition** comes from the mother through the placenta and umbilical cord.

During fetal development, the esophagus and the trachea arise from the same original tissue. Normally, the two tubes would form separately (differentiate); however, in cases of esophageal atresia and tracheoesophageal fistulas, they do not, resulting in various malformed configurations. The most common configuration is the “C” type, in which the upper part of the esophagus abruptly ends in a blind pouch, while the lower part attaches itself to the trachea. This configuration occurs in 85–90% of cases. Esophageal atresia without involvement of the trachea occurs in only 8% of cases.

Causes and symptoms

The cause of esophageal atresia, like that of most birth defects, is unknown.

An infant born with this defect will at first appear all right, swallowing normally. However, the blind pouch will begin to fill with mucus and saliva that would normally pass through the esophagus to the stomach. These secretions back up into the mouth

KEY TERMS

Fetal—Refers to the fetus, also known in the first two months after conception as an embryo.

Fistula—Unnatural connection between two hollow organs or one organ and the outside.

and nasal area, causing the baby to drool excessively. When fed, the baby will also immediately regurgitate what he or she has eaten. **Choking** and coughing may also occur as the baby breaths in the fluid backing up from the esophagus. Aspiration **pneumonia**, an infection of the respiratory system caused by inhalation of the contents of the digestive tract, may also develop.

Diagnosis

Physicians who suspect esophageal atresia after being presented with the above symptoms diagnose the condition using x-ray imaging or by passing a catheter through the nose and into the esophagus. Esophageal atresia is indicated if the catheter hits an obstruction 4–5 in (10–13 cm) from the nostrils.

Treatment

Infants with esophageal atresia are unlikely to survive without surgery to reconnect the esophagus. The procedure is done as soon as possible; however, **prematurity**, the presence of other birth defects, or complications of aspiration pneumonia may delay surgery. Once diagnosed, the baby will be fed intravenously until he or she has recovered sufficiently from the operation. Mucus and saliva will also be continuously removed via a catheter until recovery has occurred. When surgery is performed, the esophagus is reconnected and, if necessary, separated from the trachea. If the two ends of the esophagus are too far apart to be reattached, tissue from the large intestine is used to join them.

Prognosis

Surgery to correct esophageal atresia is usually successful. Post-operative complications may include difficulty swallowing, since the esophagus may not contract efficiently, and gastrointestinal reflux, in which the acidic contents of stomach back up into the lower part of the esophagus, possibly causing ulcers.

Resources

BOOKS

Long, John D., and Roy Orlando. "Anatomy and Development and Acquired Anomalies of the Esophagus." In *Sleisenger & Fordtran's Gastrointestinal and Liver Disease*, edited by Mark Feldman, et al. Philadelphia: W. B. Saunders Co., 1998.

J. Ricker Polsdorfer, MD

Esophageal cancer

Definition

Esophageal **cancer** is a malignancy that develops in tissues of the hollow, muscular canal (esophagus) along which food and liquid travel from the throat to the stomach.

Description

Esophageal cancer usually originates in the inner layers of the lining of the esophagus and grows outward. In time, the tumor can obstruct the passage of food and liquid, making swallowing painful and difficult. Since most patients are not diagnosed until the late stages of the disease, esophageal cancer is associated with poor quality of life and low survival rates.

Squamous cell carcinoma is the most common type of esophageal cancer, accounting for 95% of all esophageal cancers worldwide. The esophagus is normally lined with thin, flat squamous cells that resemble tiny roof **shingles**. Squamous cell carcinoma can develop at any point along the esophagus but is most common in the middle portion.

Adenocarcinoma has surpassed squamous cell carcinoma as the most common type of esophageal cancer in the United States. Adenocarcinoma originates in glandular tissue not normally present in the lining of the esophagus. Before adenocarcinoma can develop, glandular cells must replace a section of squamous cells. This occurs in Barrett's esophagus, a precancerous condition in which chronic acid reflux from the stomach stimulates a transformation in cell type in the lower portion of the esophagus.

A very small fraction of esophageal cancers are melanomas, **sarcomas**, or lymphomas.

There is great variability in the incidence of esophageal cancer with regard to geography, ethnicity, and gender. The overall incidence is increasing.

About 13,000 new cases of esophageal cancer are diagnosed in the United States each year. During the same 12-month period, 12,000 people die of this disease. It strikes between five and ten North Americans per 100,000. In some areas of China the cancer is endemic.

Squamous cell carcinoma usually occurs in the sixth or seventh decade of life, with a greater incidence in African-Americans than in others. Adenocarcinoma develops earlier and is much more common in white patients. In general, esophageal cancer occurs more frequently in men than in women.

Causes and symptoms

The exact cause of esophageal cancer is unknown, although many investigators believe that chronic irritation of the esophagus is a major culprit. Most of the identified risk factors represent a form of chronic irritation. However, the wide variance in the distribution of esophageal cancer among different demographic groups raises the possibility that genetic factors also play a role.

Several risk factors are associated with esophageal cancer.

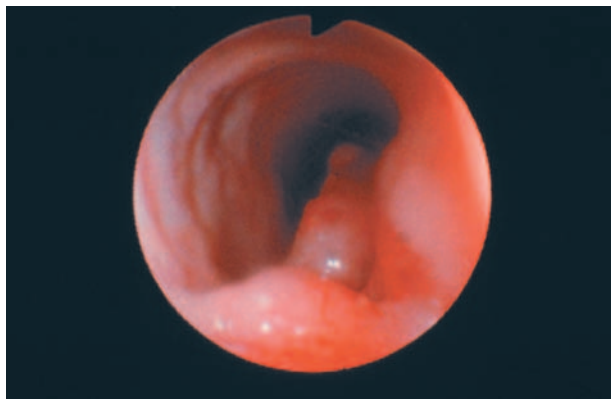
- Tobacco and alcohol consumption are the major risk factors, especially for squamous cell carcinoma. **Smoking** and alcohol **abuse** each increase the risk of squamous cell carcinoma by five-fold. The effects of the two are synergistic, in that the combination of smoking and alcohol increases the risk by 25- to 100- fold. It is estimated that drinking about 13 ounces of alcohol every day for an extended period of time raises the risk of developing esophageal cancer by 18%. That likelihood increases to 44% in individuals who also smoke one or two packs of cigarettes a day. Smokeless tobacco also increases the risk for esophageal cancer.
- Gastroesophageal reflux is a condition in which acid from the stomach refluxes backwards into the lower portion of the esophagus, sometimes causing symptoms of **heartburn**. In some cases of gastroesophageal reflux, the chronic exposure to acid causes the inner lining of the lower esophagus to change from squamous cells to glandular cells. This is called Barrett's esophagus. Patients with Barrett's esophagus are roughly 30 to 40 times more likely than the general population to develop adenocarcinoma of the esophagus.
- A diet low in fruits, vegetables, zinc, riboflavin, and other **vitamins** can increase risk of developing to esophageal cancer.
- Caustic injury to the esophagus inflicted by swallowing lye or other substances that damage esophageal cells can lead to the development of squamous cell esophageal cancer in later life.
- Achalasia is a condition in which the lower esophageal sphincter (muscle) cannot relax enough to let food pass into the stomach. Squamous cell esophageal cancer develops in about 6% of patients with achalasia.
- Tylosis is a rare inherited disease characterized by excess skin on the palms and soles. Affected patients have a much higher probability of developing esophageal cancer than the general population. They should have regular screenings to detect the disease in its early, most curable stages.
- Esophageal webs, which are protrusions of tissue into the esophagus, and diverticula, which are out-pouchings of the wall of the esophagus, are associated with a higher incidence of esophageal cancer.

Symptoms

Unfortunately, symptoms generally don't appear until the tumor has grown so large that the patient cannot be cured. Dysphagia (trouble swallowing or a sensation of having food stuck in the throat or chest) is the most common symptom. Swallowing problems may occur occasionally at first, and patients often react by eating more slowly and chewing their food more carefully and, as the tumor grows, switching to soft foods or a liquid diet. Without treatment, the tumor will eventually prevent even liquid from passing into the stomach. A sensation of burning or slight mid-chest pressure is a rare, often-disregarded symptom of esophageal cancer. Painful swallowing is usually a symptom of a large tumor obstructing the opening of the esophagus. It can lead to regurgitation of food, weight loss, physical wasting, and **malnutrition**. Anyone who has trouble swallowing, loses a significant amount of weight without dieting, or cannot eat solid food because it is too painful to swallow should see a doctor.

Diagnosis

A barium swallow is usually the first test performed on a patient whose symptoms suggest esophageal cancer. After the patient swallows a small amount of barium, a series of x rays can highlight any bumps or flat raised areas on the normally smooth surface of the esophageal wall. It can also detect large, irregular areas that narrow the esophagus in patients with advanced cancer, but it cannot provide information



A close-up view of a cancerous esophageal tumor. (Custom Medical Stock Photo. Reproduced by permission.)

about disease that has spread beyond the esophagus. A double contrast study is a barium swallow with air blown into the esophagus to improve the way the barium coats the esophageal lining. Endoscopy is a diagnostic procedure in which a thin lighted tube (endoscope) is passed through the mouth, down the throat, and into the esophagus. Cells that appear abnormal are removed for biopsy. Once a diagnosis of esophageal cancer has been confirmed through biopsy, staging tests are performed to determine whether the disease has spread (metastasized) to tissues or organs near the original tumor or in other parts of the body. These tests may include computed tomography, endoscopic ultrasound, **thoracoscopy**, **laparoscopy**, and **positron emission tomography**.

Treatment

Treatment for esophageal cancer is determined by the stage of the disease and the patient's general health. The most important distinction to make is whether the cancer is curable. If the cancer is in the early stages, cure may be possible. If the cancer is advanced or if the patient will not tolerate major surgery, treatment is usually directed at palliation (relief of symptoms only) instead of cure.

Staging

Stage 0 is the earliest stage of the disease. Cancer cells are confined to the innermost lining of the esophagus. Stage I esophageal cancer has spread slightly deeper, but still has not extended to nearby tissues, lymph nodes, or other organs. In Stage IIA, cancer has invaded the thick, muscular layer of the esophagus that propels food into the stomach and may involve connective tissue covering the outside of the

esophagus. In Stage IIB, cancer has spread to lymph nodes near the esophagus and may have invaded deeper layers of esophageal tissue. Stage III esophageal cancer has spread to tissues or lymph nodes near the esophagus or to the trachea (windpipe) or other organs near the esophagus. Stage IV cancer has spread to distant organs like the liver, bones, and brain. Recurrent esophageal cancer is disease that develops in the esophagus or another part of the body after initial treatment.

Surgery

The most common operations for the treatment of esophageal cancer are esophagectomy and esophagogastrectomy. Esophagectomy is the removal of the cancerous part of the esophagus and nearby lymph nodes. This procedure is performed only on patients with very early cancer that has not spread to the stomach. Esophagogastrectomy is the removal of the cancerous part of the esophagus, nearby lymph nodes, and the upper part of the stomach. The resected esophagus is replaced with the stomach or parts of intestine so the patient can swallow. These procedures can significantly relieve symptoms and improve the nutritional status of more than 80% of patients with dysphagia. Although surgery can cure some patients whose disease has not spread beyond the esophagus, but more than 75% of esophageal cancers have spread to other organs before being diagnosed. Less extensive surgical procedures can be used for palliation.

Chemotherapy

Oral or intravenous **chemotherapy** alone will not cure esophageal cancer, but pre-operative treatments can shrink tumors and increase the probability that cancer can be surgically eradicated. Palliative chemotherapy can relieve symptoms of advanced cancer but will not alter the outcome of the disease.

Radiation

External beam or internal radiation, delivered by machine or implanted near cancer cells inside the body, is only rarely used as the primary form of treatment. Post-operative radiation is sometimes used to kill cancer cells that couldn't be surgically removed. Palliative radiation is effective in relieving dysphagia in patients who cannot be cured. However, radiation is most useful when combined with chemotherapy as either the definitive treatment or preoperative treatment.

KEY TERMS

Computed tomography—A radiology test by which images of cross-sectional planes of the body are obtained.

Endoscopic ultrasound—A radiology test utilizing high frequency sound waves, conducted via an endoscope.

Laparoscopy—Examination of the contents of the abdomen through a thin, lighted tube passed through a small incision.

Positron emission tomography—A radiology test by which images of cross-sectional planes of the body are obtained, utilizing the properties of the positron. The positron is a subatomic particle of equal mass to the electron, but of opposite charge.

Synergistic—The combined action of two or more processes is greater than the sum of each acting separately.

Thoracoscopy—Examination of the contents of the chest through a thin, lighted tube passed through a small incision.

Palliation

In addition to surgery, chemotherapy, and radiation, other palliative measures can provide symptomatic relief. Dilatation of the narrowed portion of the esophagus with soft tubes can provide short-term relief of dysphagia. Placement of a flexible, self-expanding stent within the narrowed portion is also useful in allowing more food intake.

Follow-up treatments

Regular barium swallows and other imaging studies are necessary to detect recurrence or spread of disease or new tumor development.

Alternative treatment

Photodynamic therapy (PDT) involves intravenously injecting a drug that is absorbed by cancer cells and kills them after they are exposed to specific laser beams. PDT can be used for palliation, but it also cured some early esophageal cancers during preliminary studies. Researchers are comparing its benefits with those of more established therapies.

Endoscopic laser therapy involves delivering short, powerful laser treatments to the tumor through

an endoscope. It can improve dysphagia, but multiple treatments are required, and the benefit is seldom long-lasting.

Prognosis

Since most patients are diagnosed when the cancer has spread to lymph nodes or other structures, the prognosis for esophageal cancer is poor. Generally, no more than half of all patients are candidates for curative treatment. Even if cure is attempted, the cancer can recur.

Prevention

There is no known way to prevent esophageal cancer.

Resources

BOOKS

Heitmiller, Richard F., Arlene A. Forastiere, and Lawrence R. Kleinberg. "Esophagus." In *Clinical Oncology*, edited by Martin D. Abeloff, 2nd ed. New York: Churchill Livingstone, 2000.

Zwischenberger, Joseph B., Scott K. Alpard, and Mark B. Orringer. "Esophageal Cancer." In *Sabiston Textbook of Surgery*, edited by Courtney Townsend, Jr., 16th ed. Philadelphia: W. B. Saunders Company, 2001.

ORGANIZATIONS

American Cancer Society. 1599 Clifton Road NE, Atlanta, GA 30329. (800)ACS-2345. <<http://www.cancer.org>>.

National Coalition for Cancer Survivorship. 1010 Wayne Avenue, 5th Floor, Suite 300, Silver Spring, MD 20910. Telephone: 1-888-650-9127.

Maureen Haggerty
Kevin O. Hwang, M.D.

Esophageal disorders

Definition

The esophagus is a tube that connects the back of the mouth to the stomach. Abnormalities of the esophagus generally fall into one of four categories: structural abnormalities, motility disorders, inflammatory disorders, and malignancies.

Description

The main function of the esophagus is to move food from the back of the mouth to the stomach. The adult esophagus is about 10 in (25 cm) long. It consists of a layer of cells that secretes mucus and two layers of muscle, one circular and one longitudinal. This combination of muscles allows the esophagus to contract and propel food from the mouth the stomach. This rhythmic contraction is called peristalsis. At the end of the esophagus nearest the mouth is a ring of muscle called the upper esophageal sphincter (UES). A similar muscular ring called the lower esophageal sphincter (LES) is found 1–1.5 in (2–4 cm) above the point where the esophagus enters the stomach. The LES contracts to prevent the contents of the stomach from backflowing into the lower end of the esophagus.

Structural abnormalities

Structural abnormalities of the esophagus can be either congenital or acquired. Congenital abnormalities occur in about 1 of every 3,000–5,000 births. The two most common congenital esophageal abnormalities are **esophageal atresia** (EA) and **tracheoesophageal fistula** (TEF).

EA is a condition in which the esophagus is interrupted and the portion of the tube near the mouth is not connected to the portion that goes into the stomach. Usually the upper part of the tube ends in a blind pouch. This creates a life-threatening condition for the newborn who is unable to eat.

TEF is a condition in which the esophagus is connected to the trachea (windpipe). The trachea and the esophagus lie parallel to each other in the neck. Sometimes during fetal development, a connection called a **fistula** develops between these two tubes. This allows food to enter the trachea and be inhaled into the lungs causing a life-threatening condition called aspiration **pneumonia**. Often TEF and EA are present in the same infant. Both these conditions must be surgically corrected for the infant to survive.

Other, less common congenital structural abnormalities include webs, stenosis, cysts, and diverticula. Webs are thin membranes that lie across the esophagus and cause a partial obstruction. Stenosis is the abnormal reduction in the diameter of the esophagus due to thickening of the esophageal wall. Diverticula are pouches of tissue that extend off the esophagus. Both diverticula and stenosis can be either congenital or acquired later in life.

Acquired structural abnormalities of the esophagus include Schatzki ring and hiatal **hernia**. Schatzki

ring, sometimes called a **lower esophageal ring**, is a circular band of tissue located where the esophagus empties into the stomach. This ring is found in 6–14% of individuals, and for most people the presence of this ring does not create symptoms. Schatzki ring is found equally in all races and in both men and women. Schatzki rings that cause symptoms usually occur in middle age individuals. The ring can cause intermittent problems swallowing food or food impaction where the esophagus enters the stomach.

The esophagus passes a gap in the diaphragm called the diaphragmatic hiatus in order to reach the stomach. Hiatal hernia (also called hiatus hernia) is a condition that occurs when a portion of the stomach pushes up through this gap next to the esophagus. Although a hiatus hernia is not a direct structural abnormality of the esophagus, it is associated with gastroesophageal reflux disease (GERD) or **heartburn** in which the acidic stomach contents backflow into the lower part of the esophagus and erode the cell lining. Hiatal hernia is very common and often causes no symptoms. It is treated as a separate entry.

Lacerations, tears and ruptures of the esophagus, known as **Mallory-Weiss syndrome** and Boerhaave syndrome, are life-threatening disorders. Mallory-Weiss syndrome usually occurs in alcoholics. In both conditions, tears result from **vomiting** and retching. The resulting bleeding creates a medical emergency that can be fatal.

Motility abnormalities

Motility abnormalities create difficulty in swallowing, called dysphagia. Dysphagia is a symptom of several esophageal motility disorders as well as several obstructive disorders such as esophageal webs or Schatzki ring.

Achalasia is an esophageal motility disorder caused by uncoordinated contractions of the two muscular layers that make up the esophagus. Because muscular contractions are disorganized, peristalsis and the orderly movement of food down the esophagus does not occur. In addition, with achalasia the lower esophageal sphincter remains contracted when food is present in the esophagus which prevents the food from entering the stomach. This causes the esophagus to bulge above the LES, a condition called megaesophagus.

Achalasia is caused by destruction of some of the nerve cells that control muscular contraction of the esophagus. This disorder generally begins in young adults and becomes progressively worse as the individual ages. Individuals with achalasia also have a

higher risk of developing **esophageal cancer** at an earlier than usual age.

Individuals can also develop esophageal motility disorders secondary to other muscle diseases. **Scleroderma** is a disorder in which smooth muscle begins to atrophy. The smooth muscle in the esophagus can be affected just like other smooth muscle in the body, making swallowing difficult. Scleroderma esophagus is also associated with GERD and increased risk of **cancer** of the esophagus. Other conditions such as **diabetes mellitus**, **alcoholism**, and some psychiatric disorders can also produce secondary esophageal motility disorders.

Inflammatory disorders

Inflammatory esophageal disorders fall under the general name of esophagitis. Esophagitis causes the esophagus to become swollen and the lining of the esophagus becomes eroded and sore. It is present in about 5% of the population in the United States. There are four main types of esophagitis: reflux, infection, corrosive, and radiation. Reflux esophagitis is caused by GERD when the lower esophageal sphincter does not close tightly and the acidic contents of the stomach enter esophagus. GERD is common and is treated in depth in a separate entry. Infectious esophagitis can be caused fungal, viral, or bacterial infections. Infectious esophagitis occurs frequently in individuals with compromised immune systems, such as those with **AIDS** or leukemia. Corrosive esophagitis occurs when an individual either intentionally or accidentally swallows harsh chemicals such as lye. Radiation esophagitis is a complication of radiation treatments for cancer of the esophagus or lung.

Malignancies

Barrett's esophagus is a pre-cancerous condition which has a high risk of developing into esophageal cancer. It is found most commonly in white males in their 50s and 60s and is usually associated with years of chronic GERD.

Cancers of the esophagus tend to be aggressive and have poor outcomes. Adenocarcinoma is the primary cancer of the esophagus. Esophageal cancers are treated in detail in a separate entry.

Causes and symptoms

The causes of esophageal disorders depend on the type of disorder. Congenital defects are caused by errors in development. It is not clear why some structural disorders, such as Schatzki ring and hiatal

hernia, occur. Many more people have these defects than develop symptoms or seek medical care, so that the presence of these asymptomatic structural defects is found only during autopsies. Other individuals develop symptoms that require medical attention. **Obesity** and advancing age are thought to be contributing factors in developing symptoms.

Achalasia is caused by death of nerve cells that control the muscles that make peristalsis possible. These nerve cells are destroyed by T cells that are part of the body's immune system. It is not clear what triggers these T cells to attack inappropriately. Difficulty swallowing develops slowly, usually beginning in young adults, although the disorder can occur in children. As nerve control is lost, the LES fails to relax, preventing food from entering the stomach. As a result, the lower part of the esophagus becomes stretched creating a condition called megaesophagus. At night, food is often regurgitated and can be inhaled into the lungs, creating the risk of aspiration pneumonia. Achalasia can also be caused by **Chagas' disease**, a disease rare in North America, but common in Central and South America. Individuals with achalasia are also at higher risk to develop esophageal cancer, esophageal infections, and esophageal rupture.

Inflammatory esophagitis is most often caused by GERD. Infectious esophagitis can be caused by fungi, usually *Candida albicans*, bacteria, or viruses. Fungal infections usually occur in individuals who have diabetes, a weakened immune system, or who are taking **antibiotics**. Antibiotics change the balance of naturally occurring bacteria in the esophagus and allow fungi, which are normally present in the digestive tract, to grow unchecked. The most common causes of viral esophagitis are cytomegalovirus (CMV) and *Herpes simplex*. These are usually opportunistic infections in individuals with HIV/AIDS.

Corrosive esophagitis is usually caused by swallowing harsh chemicals, but it can also be caused by certain medications. Radiation esophagus is a side effect of **radiation therapy** for cancer.

Diagnosis

EA and TEF can sometimes be diagnosed in fetal ultrasounds before birth. If not, these defects become obvious soon after birth, because the infant is unable to eat. The inability to pass a tube from the mouth to the stomach is a definite diagnosis for EA. TEF can be detected through x-rays.

A barium swallow x ray with video is the basic method of diagnosing most esophageal disorders. For

a barium swallow x ray, the individual drinks a barium, a material that coats the esophagus and shows up on x-ray film. A video camera records the passage of the barium down the esophagus in order to detect **swallowing disorders** or pockets and pouches (diverticula) bulging from the esophagus. A barium swallow is also used to detect Schatzki rings.

Upper gastrointestinal endoscopy is often used in conjunction with a barium swallow to diagnose esophageal disorders. In an endoscopy, a thin, fiberoptic tube with a tiny camera is inserted into the esophagus. This allows the physician to see the lining of the esophagus. Endoscopes are equipped to take samples (biopsies) of any areas that may appear pre-cancerous or cancerous or to collect samples to test for the organism causing infectious esophagitis.

GERD can often be diagnosed from symptoms such as heartburn and regurgitation. Mallory-Weiss tears and Boerhaave syndrome are difficult to diagnose. Individuals with these disorders are often severely ill and have intense chest **pain** and vomiting, however chest x rays are normal 10–15% of the time. CT scans may be used in conjunction with chest x rays.

Treatment

Surgery is the only treatment for EA and TEF. It is done as soon as possible, based on the condition of the infant and any other **birth defects** that may be present that could affect the surgery.

Schatzki rings and hiatal hernias often cause no or mild symptoms and need no treatment. In severe cases, Schatzki rings are treated with bougienage. In this treatment, a series of tubes of ever-increasing diameter are inserted through the esophagus to stretch the ring. Stretching can also be done with balloon dilation. Surgery is done when no other treatment succeeds in relieving symptoms. Large hiatal hernias can be repaired surgically, but often there is not need for treatment. GERD accompanies many hiatal hernias. GERD can be treated with drugs to block acid production in the stomach (H2 blockers or **proton pump inhibitors**) and changes in diet. In severe cases of GERD stomach surgery may be necessary.

Mallory-Weiss syndrome and Boerhaave syndrome are medical emergencies. The individual is stabilized and the tear or rupture is repaired surgically. The chance of infection (**sepsis**) is high, so individuals are admitted to intensive care and do not take any food or liquid by mouth for 7–10 days. Hospital stays can last months, and repeated tears are possible.

Achalasia is treated with drugs that relax the smooth muscle and allow the LES to relax and open. When this fails, surgery to may be needed. Individuals that are not good candidates for surgery (the elderly or frail) may be treated by injecting botulinum toxin (botox) into the LES to prevent it from closing. The disadvantage of this treatment is its expense and the fact that more than one injection is needed.

Infectious esophagitis is treated by treating the underlying cause of the disease with antifungal, antiviral or antibiotic medications. These can be given either by mouth or intravenously (IV) depending on the severity of the disease.

Malignancies are treated with **chemotherapy** and radiation. See the entry on esophageal cancer for specific details.

Prognosis

The outcome of treatment depends on the type of disorder, severity, age, and general health of the individual. EA and TEF surgeries are often successful, but infants born with these conditions frequently have other congenital abnormalities that compromise their health.

When needed, treatment for Schatzki rings produces relief of symptoms, but almost always has to be repeated periodically.

Mallory-Weiss and Boerhaave syndromes are often fatal. Thirty to fifty percent of individuals die from these disorders even if they are diagnosed promptly. If diagnosis is delayed, the death rate can be as high as 90%.

Achalasia and scleroderma esophagus are progressive diseases that need continued therapy. They frequently lead to serious weight loss and **malnutrition**.

The outcome for treatment of inflammatory esophagitis depends almost entirely on the success of treating the underlying cause. Where individuals have a weakened immune system, infectious esophagitis can be an ongoing problem. When inflammatory esophagitis is caused by GERD, treatment along with lifestyle modification is usually successful in providing relief.

Esophageal cancers are aggressive and have generally poor outcomes.

Prevention

Many symptoms of esophageal disorders can be prevented or alleviated by lifestyle changes that include:

KEY TERMS

Atrophy—To wither and become unresponsive.

Congenital—Present at birth.

Diaphragm—A muscle that separates the cavity containing the lungs from the abdomen.

Diverticula—Abnormal pouches of tissue that bulge off the main part of the digestive system.

Peristalsis—A wave of contractions passing through a hollow muscular tube such as the esophagus or intestine.

- weight loss to control obesity
- eating slowly and chewing food well
- eating smaller and more frequent meals
- not eating several hours before going to bed
- limiting the use of alcohol and caffeine

Resources

OTHER

Ansari, Sajid and Sandeep Mukherjee. *Esophagitis*, 22 November 2004 [cited 1 March 2005]. < <http://www.emedicine.com/med/topic735.htm> > .

Carey, Martin J. *Esophageal Perforation, Rupture and Tears*, 26 July 2002 [cited 1 March 2005]. < <http://www.emedicine.com/emerg/topic176.htm> > .

“Esophageal Disorders.” *The Merck Manual*. Eds. Mark H. Beers and Robert Berkow. 1995-2005 [cited 1 March 2005]. < <http://www.merck.com/mrksshared/mmanual/section3/chapter20/20a.jsp> > .

Fayyad, Abdullah and Eric Gaumnitz. *Esophageal Motility Disorders*, 3 September 2004 [cited 1 March 2005]. < <http://www.emedicine.com/med/topic740.htm> > .

Minkes, Robert K. and Alison Snyder. *Congenital Anomalies of the Esophagus*, 14 June 2004 [cited 1 March 2005]. < <http://www.emedicine.com/ped/topic2934.htm> > .

Paik, Nam-Jong. *Dysphagia*, 19 August 2004 [cited 1 March 2005]. < <http://www.emedicine.com/pmr/topic194.htm> > .

Patti, Marco. *Gastroesophageal Reflux Disease*, 29 December 2004 [cited 1 March 2005]. < <http://www.emedicine.com/med/topic857.htm> > .

Qureshi, Wagar A. *Hiatal Hernia*, 29 December 2004 [cited 1 March 2005]. < <http://www.emedicine.com/pmr/topic1012.htm> > .

Vossough, Arastoo and Stephen E. Rubesin. *Schatzki Ring*, 14 April 2003 [cited 1 March 2005]. < <http://www.emedicine.com/radio/topic620.htm> > .

Tish Davidson, A.M.

Esophageal diverticula see **Esophageal pouches**

Esophageal function tests

Definition

The esophagus is the swallowing tube through which food passes on its way from the mouth to the stomach. The main function of this organ is to propel food down into the stomach. There is also a mechanism to prevent food from coming back up or “refluxing” from the stomach into the esophagus. Esophageal function tests are used to determine if these processes are normal or abnormal.

Purpose

The esophagus is a long, muscular tube that also has two muscles (or sphincters) at the top and bottom. All of these muscular areas must contract in an exact sequence for swallowing to proceed normally. There are three main symptoms that occur when esophageal function is abnormal: difficulty with swallowing (dysphagia), **heartburn**, and chest **pain**.

Doctors perform a variety of tests to evaluate these symptoms. Endoscopy, which is not a test of esophageal function, is often used to determine if the lining of the esophagus has any ulcers, tumors, or areas of narrowing (strictures). Many times, however, endoscopy only shows the doctor if there is injury to the esophageal lining, and the procedure gives no information about the cause of the problem.

Therefore, in addition to endoscopy, several studies are available that measure esophageal function. There are three basic types of tests used to assess esophageal function:

- Manometry is used to study the way the muscles of the esophagus contract, and is most useful for the investigation of difficulty with swallowing.
- Esophageal pH monitoring measures changes in esophageal acidity, and is valuable for evaluating patients with heartburn or gastroesophageal reflux disease (GERD).

- X-ray studies investigate swallowing difficulties. They either follow the progress of barium during swallowing using a fluoroscope, or they use radioactive scanning techniques.

Precautions

Pregnant patients undergoing x-ray exams should carefully review the risks and benefits with their doctors. Most x-ray exams of the gastrointestinal tract do not involve radiation levels that are harmful to the unborn baby.

Description

Manometry

This study is designed to measure the pressure changes produced by contraction of the muscular portions of the esophagus. An abnormality in the function of any one of the segments of the swallowing tube causes difficulty in swallowing. Doctors call this symptom dysphagia. This exam is most useful in evaluating those patients whose endoscopy is negative.

During manometry, the patient swallows a thin tube carrying a device that senses changes in pressures in the esophagus. Readings are taken at rest and during swallowing. Medications are sometimes given during the study to help in the diagnosis. The results are then transmitted to recording equipment. Manometry can best identify diseases that produce disturbances of motility or contractions of the esophagus.

ESOPHAGEAL PH MONITORING. This procedure involves measuring the esophagus' exposure to acid that has "refluxed" from the stomach. The test is ideal for evaluating recurring heartburn or GERD. Too much acid produces not only heartburn, but also ulcers that can bleed or produce areas of narrowing (strictures) when they heal.

Normally, acid refluxes into the esophagus in only small amounts for short periods of time. A muscle called the lower esophageal sphincter prevents excessive reflux. Spontaneous contractions that increase esophageal emptying and production of saliva are other important protective mechanisms.

"pH" is the scientific term that tells just how acidic or alkaline a substance is. Researchers have shown that in the esophagus, the presence of acid is damaging only if it persists for prolonged periods. Therefore, the test has been designed to monitor the level of acidity over 24 hours, usually in the home. In this way, patients maintain their daily routine, documenting their symptoms, and at what point in their activities they

occurred. During this period, a thin tube with a pH monitor remains in the esophagus to record changes. After the study, a computer is used to compare changes in acidity with symptoms reported by the patient.

Surgery is an effective and long-lasting treatment for symptoms of recurrent reflux and is the choice of many patients and doctors. pH monitoring is usually performed before surgery to confirm the diagnosis and to judge the effects of drug therapy.

X-RAY TESTS. These fall into two categories: (1) those done with the use of barium and a fluoroscope; and (2) those performed with radioactive materials.

Studies performed with fluoroscopy are of greatest value in identifying a structural abnormality of the esophagus. Although this is not truly an esophageal function test, it does allow doctors to consider other diagnostic possibilities. Often a sandwich or marshmallow coated with barium is used to identify the site of an obstruction.

During fluoroscopy, the radiologist can observe the passage of material through the esophagus in real time, and video recordings can also be done. This is particularly useful when the swallowing symptoms appear to involve mainly the upper region of the esophagus. The most common cause of swallowing difficulties is a previous stroke, although other diseases of the neuromuscular system (like myasthenia gravis) can produce the same symptoms.

Scans using low-dose radioactive materials are useful because they are able not only to demonstrate that food passes through the esophagus more slowly than normal, but also how slow. These studies involve swallowing food coated with material that is followed by a nuclear medicine scanner. Scans are best used when other methods have failed to make a diagnosis, or if it is necessary to determine the degree of the abnormality. As of 1997, scans mainly served as research tools.

Preparation

Patients should not eat or drink for several hours before the exam. Many medications affect the esophagus; doses sometimes need to be adjusted or even stopped for a while. Patients must inform doctors of all medications taken, including over-the-counter medications (purchased without a doctor's prescription), and any known **allergies**.

Aftercare

For most of these studies, no special care is needed after the procedure. Patients can often go about

normal daily activities following any of these tests. One exception is for those who undergo an x-ray exam with the use of barium. This can have a constipating effect and patients should ask about using a mild laxative later on.

Risks

Exposure of a fetus to x rays, especially in the first three months, is a potential risk.

Other studies of esophageal function are essentially free of any significant risk. The tubes passed during these procedures are small, and most patients adjust to them quite well. However, since medications cannot be used to relax patients, some may not tolerate the exam.

Abnormal results

Manometry is used to diagnose abnormalities related to contraction or relaxation of the various muscular regions of the esophagus. These studies cannot distinguish whether injury to either the muscle or nerves of the esophagus is producing the abnormal results. Only the final effect on esophageal muscle is identified. Results should be interpreted in light of the patient's entire medical history.

For example, there are many diseases that cause poor relaxation of the lower esophageal sphincter. When no cause is found, the disease is called **achalasia**.

Abnormal results of pH tests can confirm symptoms of heartburn or indicate a cause of chest pain (or rarely, swallowing difficulties). Doctors may want to start or change medications based on these results, or even repeat the test using different doses of medication. As noted above, these studies are indicated before surgical treatment of GERD.

X-ray tests can only serve to document an abnormality, and they are far from perfect. If they are negative, then other studies are often needed.

Resources

PERIODICALS

Mittal, Ravinder K., and David H. Balaban. "The Esophagogastric Junction." *New England Journal of Medicine* 336 (27 Mar. 1997): 924-932.

David Kaminstein, MD

Esophageal laceration see **Mallory-Weiss syndrome**

Esophageal manometry see **Esophageal function tests**

Esophageal pouches

Definition

Esophageal pouches, also known as esophageal diverticula, are pocket-like structures formed when the interior space of the esophagus, the tube that connects the mouth to the stomach, protrudes into the walls that surround it.

Description

The esophagus is a muscular tube that propels food into the stomach. A defect in the wall of the esophagus may allow the lining to herniate, creating a space where food can be caught. Pouches can appear anywhere between the throat and the stomach. They occur primarily in men and usually later in life.

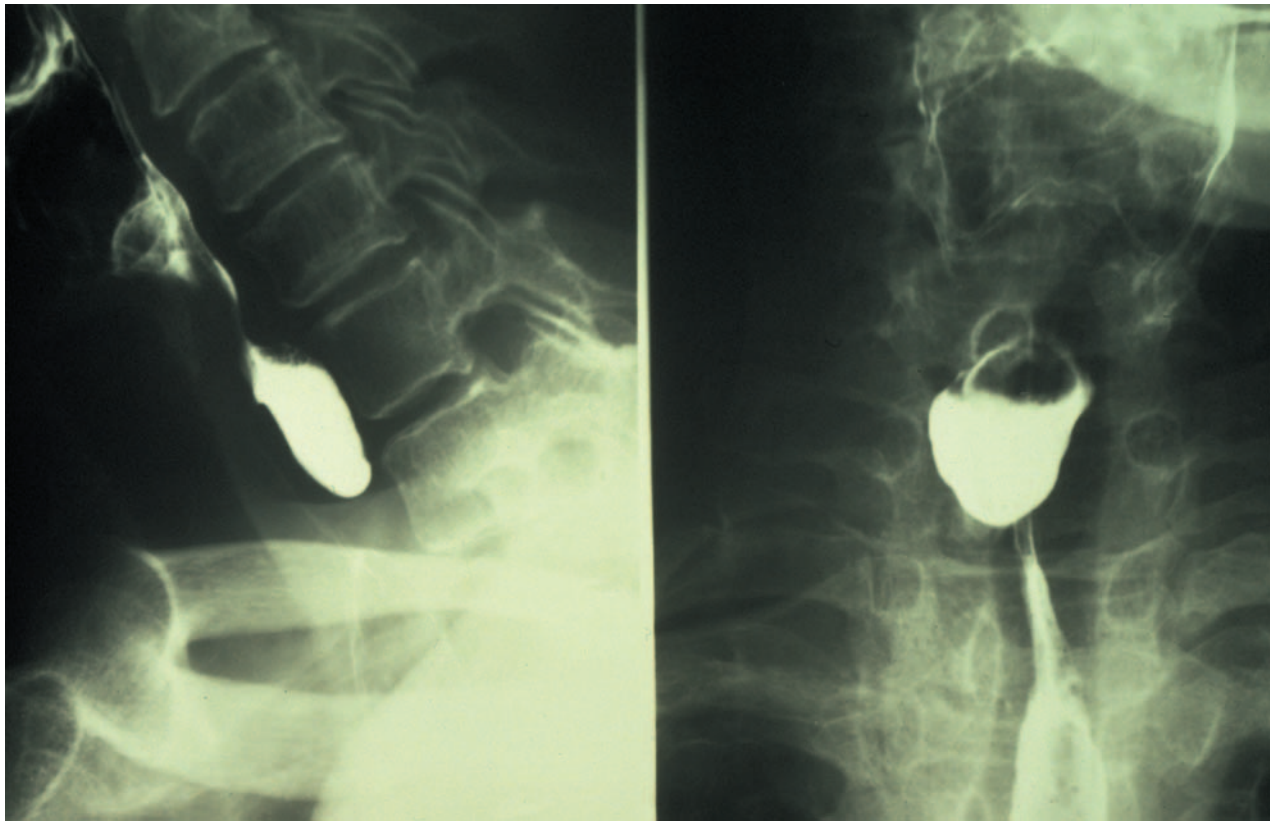
Different names for the condition apply to different locations along the esophagus:

- Zenker's diverticula are pharyngeal pouches, or ones that occur in the upper neck area at the top of the esophagus.
- Traction diverticula are a type of mid-esophageal pouch.
- Epiphrenic diverticula occur at the bottom of the esophagus near where it enters the stomach.

Causes and symptoms

To propel food into the stomach (or out of it during **vomiting**) the esophagus generates internal pressure just like the bowel. Under certain circumstances, that pressure can herniate the esophageal lining through a weakness in the wall, creating a pouch (a balloon squeezed in the hand will herniate through the fingers in the same way). Pouches are more common in people who have motility disorders of the esophagus, swallowing that is not well coordinated and may be spastic. A traction diverticulum can develop from a scar that pulls the esophagus out of shape. Food and saliva can collect in all of these pouches.

Pouches in the neck usually cause **bad breath** (halitosis) and the regurgitation of swallowed food and saliva. Some patients with Zenker's diverticula can push on their neck and make old food appear in their mouths. Pouches near the stomach may cause swallowing problems, conditions known as *achalasia* or *dysphagia*. Mid-esophageal pouches usually cause no symptoms.



A split x-ray image of the upper chest, neck, and esophagus (left), and chest and esophagus (right). (Custom Medical Stock Photo. Reproduced by permission.)

KEY TERMS

Achalasia—Failure of the lower end of the esophagus (or another tubular valve) to open, resulting in obstruction, either partial or complete.

Contrast agent—A substance that produces shadows on an x ray so that hollow structures can be more easily seen.

Dysphagia—Difficult swallowing.

Esophagoscopy—Looking down the esophagus with a flexible viewing instrument.

Herniate—To protrude beyond usual limits.

Manometry—Pressure measurement.

In the most serious cases, a person may be unable to swallow because the esophagus is obstructed, or the esophagus may rupture, spilling its contents into the chest or neck.

Diagnosis

Difficulty swallowing, bad breath, or food reappearing in the back of the mouth are among the signs physicians look for when diagnosing this condition. Sometimes the patient may also experience **pain** in the chest resembling a heart attack. A series of x rays taken while swallowing a contrast agent usually demonstrates the diverticulum clearly. An esophagoscopy may also be needed to gather more detail. Manometry, measuring pressures inside the esophagus using a balloon that is passed down it, may help determine the cause of the diverticula.

Treatment

Treatment for this condition is primarily aimed at alleviating symptoms. Physicians direct the patient to eat a bland diet, to chew his or her food thoroughly, and to drink water after eating to clean out the pouches. If the condition is severe, several types of surgery are available to remove the pouches and repair the defects. If a pouch is due to a stenosis (narrowing) in the

esophagus it may be possible to relieve it by passing a dilator through it, a process called bougienage.

Prognosis

The two complications that can render these nuisances dangerous, obstruction and rupture, are emergencies. Both require immediate medical attention. Other than that, diverticula will usually grow slowly over the years, gradually increasing the symptoms they cause.

Resources

BOOKS

Goyal, Raj K. "Diseases of the Esophagus." In *Harrison's Principles of Internal Medicine*, edited by Anthony S. Fauci, et al. New York: McGraw-Hill, 1997.

J. Ricker Polsdorfer, MD

Esophageal ulcers see **Ulcers (digestive)**

Esophagogastroduodenoscopy

Definition

An endoscope as used in the field of gastroenterology (the medical study of the stomach and intestines) is a thin, flexible tube that uses a lens or miniature camera to view various areas of the gastrointestinal tract. When the procedure is limited to the examination of the inside of the gastrointestinal tract's upper portion, it is called upper endoscopy or esophagogastroduodenoscopy (EGD). With the endoscope, the esophagus (swallowing tube), stomach, and duodenum (first portion of the small intestine) can be easily examined, and abnormalities frequently treated. Patients are usually sedated during the exam.

Purpose

EGD is performed to evaluate or treat symptoms relating to the upper gastrointestinal tract, such as:

- upper abdominal or chest **pain**
- nausea or **vomiting**
- difficulty swallowing (dysphagia)
- bleeding from the upper intestinal tract
- anemia (low **blood count**). EGD can be used to treat certain conditions, such as an area of narrowing or bleeding in the upper gastrointestinal tract

Upper endoscopy is more accurate than x rays for detecting inflammation, ulcers, or tumors. It is used to diagnose early **cancer** and can frequently determine whether a growth is benign (not cancerous) or malignant (cancerous).

Biopsies (small tissue samples) of inflamed or "suspicious" areas can be obtained and examined by a pathologist. Cell scrapings can also be taken by the introduction of a small brush; this helps in the diagnosis of cancer or infections.

When treating conditions in the upper gastrointestinal tract, small instruments are passed through the endoscope that can stretch narrowed areas (strictures), or remove swallowed objects (such as coins or pins). In addition, bleeding from ulcers or vessels can be treated by a number of endoscopic techniques.

Recent studies have shown the usefulness of endoscopic removal of early tumors of the esophagus or stomach. This is done either with injection of certain materials (like alcohol), or with the use of instruments (like lasers) that burn the tumor. Other techniques combining medications and lasers also show promise.

Precautions

Patients should inquire as to the doctor's expertise with these procedures, especially when therapy is the main goal. The doctor should be informed of any **allergies**, medication use, and medical problems.

Description

First, a "topical" (local) medication to numb the gag reflex is given either by spray or is gargled. Patients are usually sedated for the procedure (though not always) by injection of medications into a vein. The endoscopist then has the patient swallow the scope, which is passed through the upper gastrointestinal tract. The lens or camera at the end of the instrument allows the endoscopist to examine each portion of the upper gastrointestinal tract; photos can be taken for reference. Air is pumped in through the instrument to allow proper observation. Biopsies and other procedures can be performed without any significant discomfort.

Preparation

The upper intestinal tract must be empty for the procedure, so it is necessary NOT to eat or drink for at least 6–12 hours before the exam. Patients need to inquire about taking their medications before the procedure.



Esophagogastroduodenoscopy (EGD) is performed to evaluate or treat symptoms relating to the upper gastrointestinal tract. By inserting an endoscope into the mouth and guiding it through the gastrointestinal tract, the esophagus, stomach, and duodenum can be examined and abnormalities treated. (Illustration by Electronic Illustrators Group.)

KEY TERMS

Pathologist—A doctor who specializes in the anatomic (structural) and chemical changes that occur with diseases. These doctors function in the laboratory, examining biopsy specimens, and regulating studies performed by the hospital laboratories (blood tests, urine tests, etc). Pathologists also perform autopsies.

Aftercare

Someone should be available to take the person home after the procedure and stay with them for a while; patients will not be able to drive themselves due to **sedation**. Pain or any other unusual symptoms should be reported immediately.

It is important to recognize early signs of any possible complication. The doctor should be notified if the patient has **fever**, trouble swallowing, or increasing throat, chest, or abdominal pain.

Risks

EGD is safe and well tolerated; however, complications can occur as with any procedure. These are most often due to medications used during the procedure, or are related to endoscopic therapy. The overall complication rate of EGD is less than 2%, and many of these complications are minor (such as inflammation of the vein through which medication is given). However, serious ones can and do occur, and almost half of them are related to the heart or lungs. Bleeding or perforations (holes in the gastrointestinal tract) are also reported, especially when tumors or narrowed areas are treated or biopsied. Infections have also been rarely transmitted; improved cleaning techniques should be able to prevent them.

Resources

OTHER

“Understanding Upper Endoscopy.” *American Society for Gastrointestinal Endoscopy*. <<http://www.asge.org>>.

David Kaminstein, MD

Essential tremor see **Tremors**

Estradiol see **Hormone replacement therapy**

Estrogen see **Hormone replacement therapy**

Estrogen fractions test see **Sex hormones tests**

Estrogen replacement therapy see **Hormone replacement therapy**

Ethambutol see **Antituberculosis drugs**

Etodolac see **Nonsteroidal anti-inflammatory drugs**

Evoked potential studies

Definition

Evoked potential studies are a group of tests of the nervous system that measure electrical signals along the nerve pathways.

Purpose

Nerves convey information to the body by sending electrical signals down the length of the nerve. These signals can be recorded by wires placed over the nerves on the surface of the skin, in a procedure called an evoked potential (EP) study. The person conducting the test evokes the patient's neural activity by visual or auditory stimulation or using a mild electrical shock. This causes changes in the electrical potential in the nerves. Analysis of the signals can provide information about the condition of nerve pathways, especially those in the brain and spinal cord. They can indicate the presence of disease or degeneration, and can help determine the location of nerve lesions.

There are three major types of EP studies used regularly:

- Visual evoked potentials are used to diagnose visual losses due to optic nerve damage, especially from **multiple sclerosis**. They are also useful to diagnose

“hysterical blindness,” in which loss of vision is not due to any nerve damage.

- Auditory evoked potentials are used to diagnose hearing losses. They can distinguish damage to the acoustic nerve (which carries signals from the ear to the brain stem) from damage to the auditory pathways within the brainstem. Most auditory EPs record activity from the brainstem, and are therefore called “brainstem auditory evoked potentials.” Disorders diagnosed with auditory EPs include **acoustic neuroma** (tumors of the inner ear) and multiple sclerosis (chronic disease in which nerves lose patches of their outer covering). They may also be used to assess high frequency hearing ability, to determine brain **death**, and to monitor brainstem function during surgery
- Somatosensory evoked potentials record transmission of nerve impulses from the limbs to the brain, and can be used to diagnose nerve damage or degeneration within the spinal cord or nerve roots from multiple sclerosis, trauma, or other degenerative disease. Somatosensory EPs can be used to distinguish central versus peripheral nerve disease, when combined with results from a nerve conduction velocity test, which measures nerve function in the extremities.

Precautions

Evoked potential studies are painless, noninvasive, and without any significant risk. Somatosensory EP tests involve very mild electric shocks, usually felt as a **tingling**.

Description

The person performing the test locates and marks specific spots on the patient's head for placement of electrodes. These spots are cleaned, and an adhesive conducting paste is applied. Cup electrodes are attached. For somatosensory EP, spots on the arm or leg are also marked and cleaned; electrodes may be taped in place. The patient sits or reclines in a chair throughout the tests.

For a visual EP, the patient focuses on a TV screen which displays a checkerboard pattern. The eye not being tested is covered with a patch. For children or others whose attention may wander, goggles are used which show the pattern to one eye at a time. Each eye is usually tested twice, and the entire procedure takes approximately 30–45 minutes.

For auditory EP, headphones are used to deliver a series of clicks to one ear at a time. A masking or static

KEY TERMS

Nerve conduction velocity test—A test of the speed of conduction of nerves, performed on the nerves in the arm and leg.

sound is played into the other ear. Each ear is usually tested twice, and the entire procedure takes approximately 30–45 minutes.

For somatosensory EP, mild electrical shocks are delivered to the arm or leg. This may cause some twitching and tingling. The stimulus lasts for about two minutes at a time, and the entire procedure takes approximately 30 minutes.

After the tests, the electrodes are removed with acetone and the scalp is cleaned.

Preparation

Hair must be clean, dry, and free of any braids, pins, or jewelry. The patient should shampoo before the test, and must not use any hair spray, gel, or other hair care products after shampooing. Clothing should be loose and comfortable. The patient may eat and take some medications as usual before the test, although sedative medications should be avoided on the day of the test, if possible. It is best to check with the physician supervising the test for specific instructions.

Aftercare

This test is painless and has no residual effects. The patient may return to work or other activities immediately afterward.

Normal results

EP test results are displayed as jagged electrical tracings (wave forms), which have characteristic shapes, heights, and lengths, indicating the speed and intensity of signal transmission. Results are read by someone trained in evoked potential studies.

Abnormal results

Changes in the electrical tracings may indicate damage to or degeneration of nerve pathways to the brain from the eyes, ears, or limbs. Absence of any activity may mean complete loss of nerve function in

that pathway. Other changes may provide evidence of the type and location of nerve damage.

Resources

BOOKS

Samuels, Martin, and Seven Feske, editors. *Office Practice of Neurology*. New York: Churchill Livingstone, 1996.

Richard Robinson

Evoked responses see **Evoked potential studies**

Exanthema subitum see **Roseola**

Exercise

Definition

Exercise is physical activity that is planned, structured, and repetitive for the purpose of conditioning any part of the body. Exercise is utilized to improve health, maintain fitness and is important as a means of physical **rehabilitation**.

Purpose

Exercise is useful in preventing or treating coronary heart disease, **osteoporosis**, weakness, diabetes, **obesity**, and depression. Range of motion is one aspect of exercise important for increasing or maintaining joint function. Strengthening exercises provide appropriate resistance to the muscles to increase endurance and strength. **Cardiac rehabilitation** exercises are developed and individualized to improve the cardiovascular system for prevention and rehabilitation of cardiac disorders and diseases. A well-balanced exercise program can improve general health, build endurance, and delay many of the effects of **aging**. The benefits of exercise not only improve physical health, but also enhance emotional well-being.

A study released in 2003 reported that exercise combined with behavioral therapy may even help manage the symptoms experienced by Gulf War veterans. Specifically, exercise helped improve symptoms related to **fatigue**, distress, cognitive problems and mental health functioning. In the same year, the American Heart Association released a statement saying that exercise was beneficial even for patients awaiting heart transplants. Another study showed that women who participated in strenuous physical activity over a number of years could reduce their risk for

breast cancer. Finally, research showed that men and women age 40 to 50 who exercised moderately for 60 to 90 minutes a day were less likely to catch a cold than those who sat around.

Precautions

Before beginning any exercise program, an evaluation by a physician is recommended to rule out any potential health risks. Once health and fitness are determined, and any or all physical restrictions identified, an individual's exercise program should be under the supervision of a health care professional. This is particularly true when exercise is used as a form of rehabilitation. If symptoms of **dizziness**, **nausea**, excessive **shortness of breath**, or chest **pain** are present during any exercise program, an individual should stop the activity and inform a physician about these symptoms before resuming activity. Exercise equipment must be checked to determine if it can bear the weight of people of all sizes and shapes.

Description

Range of motion exercise

Range of motion exercise refers to activity aimed at improving movement of a specific joint. This motion is influenced by several structures: configuration of bone surfaces within the joint, joint capsule, ligaments, and muscles and tendons acting on the joint. There are three types of range of motion exercises: passive, active, and active assists. Passive range of motion is movement applied to a joint solely by another person or persons or a passive motion machine. When passive range of motion is applied, the joint of an individual receiving exercise is completely relaxed while the outside force moves the body part, such as a leg or arm, throughout the available range. Injury, surgery, or **immobilization** of a joint may affect the normal joint range of motion. Active range of motion is movement of a joint provided entirely by the individual performing the exercise. In this case, there is no outside force aiding in the movement. Active assist range of motion is described as a joint receiving partial assistance from an outside force. This range of motion may result from the majority of motion applied by an exerciser or by the person or persons assisting the individual. It also may be a half-and-half effort on the joint from each source.

Strengthening exercise

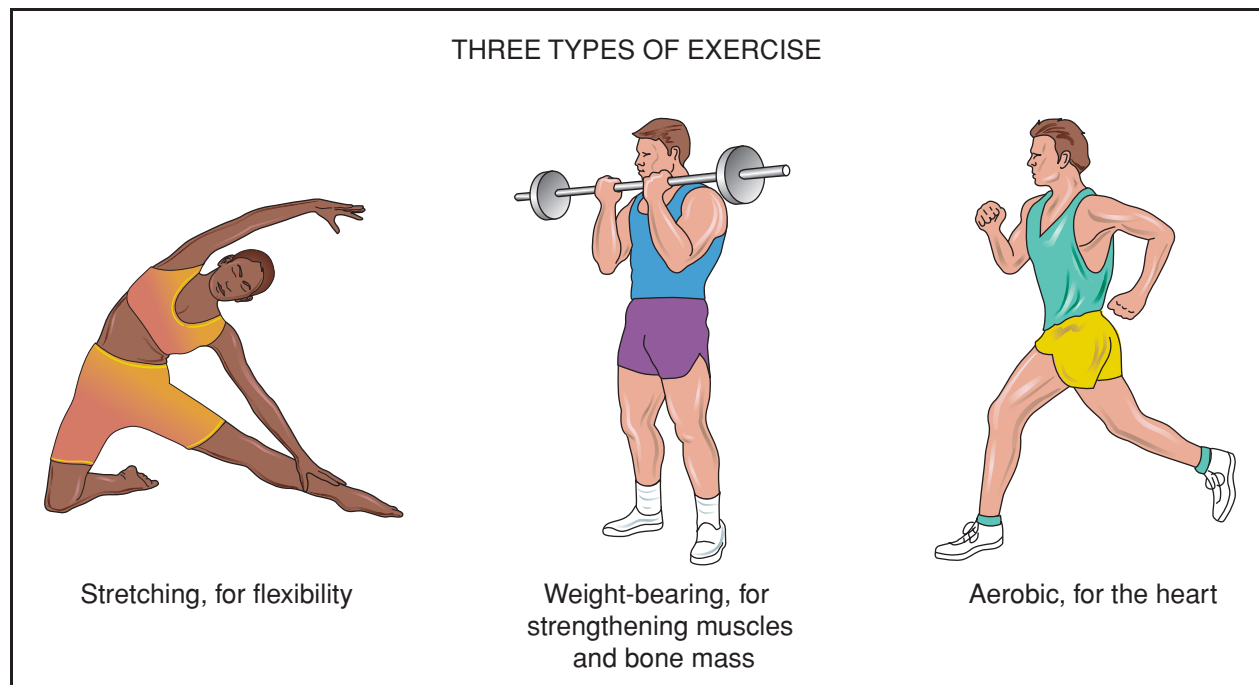
Strengthening exercise increases muscle strength and mass, bone strength, and the body's metabolism.

It can help attain and maintain proper weight and improve body image and self-esteem. A certain level of muscle strength is needed to do daily activities, such as walking, running and climbing stairs. Strengthening exercises increase this muscle strength by putting more strain on a muscle than it is normally accustomed to receiving. This increased load stimulates the growth of proteins inside each muscle cell that allow the muscle as a whole to contract. There is evidence indicating that strength training may be better than aerobic exercise alone for improving self-esteem and body image. Weight training allows one immediate feedback, through observation of progress in muscle growth and improved muscle tone. Strengthening exercise can take the form of isometric, isotonic and isokinetic strengthening.

ISOMETRIC EXERCISE. During isometric exercises, muscles contract. However, there is no motion in the affected joints. The muscle fibers maintain a constant length throughout the entire contraction. The exercises are usually performed against an immovable surface or object such as pressing one's hand against a wall. The muscles of the arm are contracting but the wall is not reacting or moving as a result of the physical effort. Isometric training is effective for developing total strength of a particular muscle or group of muscles. It often is used for rehabilitation since the exact area of muscle weakness can be isolated and strengthening can be administered at the proper joint angle. This kind of training can provide a relatively quick and convenient method for overloading and strengthening muscles without any special equipment and with little chance of injury.

ISOTONIC EXERCISE. Isotonic exercise differs from isometric exercise in that there is movement of a joint during the muscle contraction. A classic example of an isotonic exercise is weight training with dumbbells and barbells. As the weight is lifted throughout the range of motion, the muscle shortens and lengthens. Calisthenics are also an example of isotonic exercise. These would include chin-ups, push-ups, and sit-ups, all of which use body weight as the resistance force.

ISOKINETIC EXERCISE. Isokinetic exercise utilizes machines that control the speed of contraction within the range of motion. Isokinetic exercise attempts to combine the best features of both isometrics and weight training. It provides muscular overload at a constant preset speed while a muscle mobilizes its force through the full range of motion. For example, an isokinetic stationary bicycle set at 90 revolutions per minute means that despite how hard and fast the exerciser works, the isokinetic properties of the bicycle will allow the exerciser to pedal only as fast as 90



Exercise is utilized to improve health, maintain fitness, and is important as a means of physical rehabilitation. (Illustration by Electronic Illustrators Group.)

revolutions per minute. Machines known as Cybex and Biodex provide isokinetic results; they generally are used by physical therapists.

Cardiac rehabilitation

Exercise can be very helpful in prevention and rehabilitation of cardiac disorders and disease. With an individually designed exercise program set at a level considered safe for the individual, people with symptoms of **heart failure** can substantially improve their fitness levels. The greatest benefit occurs as muscles improve the efficiency of their oxygen use, which reduces the need for the heart to pump as much blood. While such exercise doesn't appear to improve the condition of the heart itself, the increased fitness level reduces the total workload of the heart. The related increase in endurance also should translate into a generally more active lifestyle. Endurance or aerobic routines, such as running, brisk walking, cycling, or swimming, increase the strength and efficiency of the muscles of the heart.

Preparation

A **physical examination** by a physician is important to determine if strenuous exercise is appropriate or detrimental for an individual. Prior to the exercise

program, proper stretching is important to prevent the possibility of soft tissue injury resulting from tight muscles, tendons, ligaments, and other joint-related structures.

Aftercare

Proper cool down after exercise is important in reducing the occurrence of painful **muscle spasms**. It has been documented that proper cool down also may decrease frequency and intensity of muscle stiffness the day following any exercise program.

Risks

Improper warm up can lead to muscle **strains**. Overexertion without enough time between exercise sessions to recuperate also can lead to muscle strains, resulting in inactivity due to pain. Stress **fractures** also are a possibility if activities are strenuous over long periods of time without proper rest. Although exercise is safe for the majority of children and adults, there is still a need for further studies to identify potential risks.

Normal results

Significant health benefits are obtained by including a moderate amount of physical exercise in the form

KEY TERMS

Aerobic—Exercise training that is geared to provide a sufficient cardiovascular overload to stimulate increases in cardiac output.

Calisthenics—Exercise involving free movement without the aid of equipment.

Endurance—The time limit of a person's ability to maintain either a specific force or power involving muscular contractions.

Osteoporosis—A disorder characterized by loss of calcium in the bone, leading to thinning of the bones. It occurs frequently in postmenopausal women.

of an exercise prescription. This is much like a drug prescription in that it also helps enhance the health of those who take it in the proper dosage. Physical activity plays a positive role in preventing disease and improving overall health status. People of all ages, both male and female, benefit from regular physical activity. Regular exercise also provides significant psychological benefits and improves quality of life. Studies released in 2003 showed the actual activity in the brain promoted by regular aerobic exercise. It appears that exercise also improves problem solving and other brain-related abilities.

Abnormal results

There is a possibility of exercise burnout if an exercise program is not varied and adequate rest periods are not taken between exercise sessions. Muscle, joint, and cardiac disorders have been noted among people who exercise. However, they often have had preexisting or underlying illnesses.

Resources

BOOKS

Bookhout, Mark R., and Grenman, Philip. *Principles of Exercise Prescription*. Woburn, MA: Butterworth-Heinemann, 2001.

Harr, Eric. *The Portable Personal Trainer*. New York: Broadway Books, 2001.

McArdle, William D., Frank I. Katch, and Victor L. Katch. *Exercise Physiology: Energy, Nutrition, and Human Performance*. 5th ed. Philadelphia: Lippincott, 2001.

Redding, Morgan. *Physical Fitness : Concepts and Applications*. Dubuque, IA: Kendall/Hunt Publishing, 2001.

Roberts, Matt. *90-Day Fitness Plan*. Littleton, CO: DK Publishers, 2001.

PERIODICALS

Brun, J. F., M. Dumortier, C. Fedou, and J. Mercier. "Exercise Hypoglycemia in Nondiabetic Subjects." *Diabetes and Metabolism* 27 (2001): 92-106.

"Cognitive Behavioral Therapy Plus Exercise May Alleviate Symptoms." *Mental Health Weekly Digest* (March 31, 2003): 3.

Evans, E. M., R. E. Van Pelt, E. F. Binder, D. B. Williams, A. A. Ehsani, and W. M. Kohrt. "Effects of HRT and Exercise Training on Insulin Action, Glucose Tolerance, and Body Composition in Older women." *Journal of Applied Physiology* 90 (2001): 2033-2040.

"Exercise May Help Patients." *Heart Disease Weekly* (March 30, 2003): 44.

Killian, K. J. "Is Exercise Tolerance Limited by the Heart or the Lungs?" *Clinical Investigations in Medicine* 24 (2001): 110-117.

Resnick, B. "Testing a model of exercise behavior in older adults." *Research in Nursing and Health* 24, no.2 (2001): 83-92.

"Stay Active to Stay Cold-Free: A Recent Study Found that You can Ward Off the Sniffle with a Little Exercise." *Natural Health* (March 2003): 30.

"Strenuous Physical Activity Throughout Life can Decrease Risk ." *Cancer Weekly* (March 18, 2003): 32.

"Study is First to Confirm Link Between Exercise and Changes in Brain." *Obesity, Fitness and Wellness Week* (February 22, 2003): 13.

ORGANIZATIONS

American College of Sports Medicine. 401 W. Michigan Street, Indianapolis, IN 46202-3233. (317) 637-9200. Fax: (317) 634-7817. <<http://www.acsm.org/>> . mkeckhaver@acsm.org.

American Medical Association. 515 N. State Street, Chicago, IL 60610. (312) 464-5000. <<http://www.ama-assn.org/>> .

American Physical Therapy Association. 1111 North Fairfax Street Alexandria, VA 22314. (703) 684-2782. <<http://www.apta.org>> .

National Athletic Trainers' Association. 2952 Stemmons Freeway, Dallas, TX 75247-6916. (800) 879-6282 or (214) 637-6282. Fax: (214) 637-2206. <<http://www.nata.org/>> .

OTHER

American Diabetes Association. <<http://www.diabetes.org/exercise>> .

American Heart Association. <<http://www.americanheart.org>> .

American Orthopaedic Society for Sports Medicine. <<http://www.sportsmed.org>> .

American Society of Exercise Physiologists. <<http://www.css.edu/asep>> .

L. Fleming Fallon, Jr., MD, DrPH
Teresa G. Odle

Exercise electrocardiogram see

Stress test

Exercise stress test see **Stress test**

Exhibitionism see **Sexual perversions**

Exocrine pancreatic cancer see **Pancreatic cancer, exocrine**

Exophthalmos

Definition

When there is an increase in the volume of the tissue behind the eyes, the eyes will appear to bulge out of the face. The terms exophthalmos and proptosis apply. Proptosis can refer to any organ that is displaced forward, while exophthalmos refers just to the eyes.

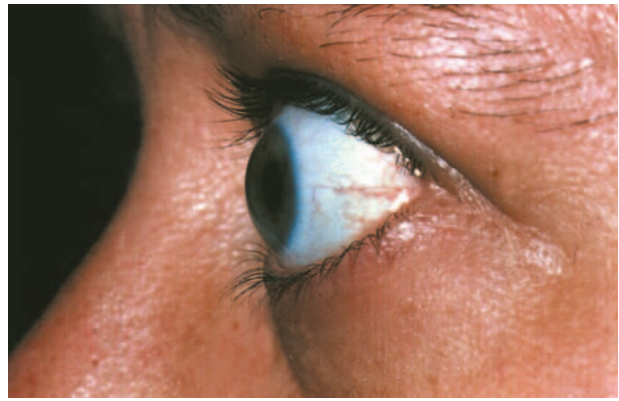
Description

The eye socket (orbit) is made of bone and therefore will not yield to increased pressure within it. Only forward displacement of the eyeball (globe) will allow more room if tissue behind the eye is increasing.

Causes and symptoms

The most common cause of exophthalmos is Graves' disease, overactivity of the thyroid gland. The contents of the orbits swell due to inflammation, forcing the eyes forward. The inflammation affects primarily the muscles. This combination of muscle impairment and forward displacement reduces eye movement, causing double vision and crossed eyes (**strabismus**). The optic nerves can also be affected, reducing vision, and the clear membrane (conjunctiva) covering the white part of the eyes and lining the inside of the eyelids can swell. Finally, the eyes may protrude so far that the eyelids cannot close over them, leading to corneal damage.

Exophthalmos from Graves' disease is bilateral (occurring on both sides), but not necessarily symmetrical. In contrast, exophthalmos from orbital tumors or a blood clot in the brain happens on only one side.



A side view of the bulging eye (exophthalmos) of a person suffering from thyrotoxicosis. Exophthalmos is caused by swelling of the soft tissue in the eye socket, which forces the eyeball to be pushed forward and the eyelids stretched apart. (Photograph by Dr. P. Marazzi, Photo Researchers, Inc. Reproduced by permission.)

Diagnosis

Exophthalmos is obvious when it is advanced enough to cause complications. When there is doubt in the early stages, a mechanical device called an exophthalmometer can measure the protrusion. **Computed tomography scans** (CT scans) are of great value in examining the bony components of the orbit. **Magnetic resonance imaging** (MRI) scanning is equally valuable for displaying the contents of the orbit, because it "sees through" the bone.

Treatment

If a tumor is growing behind the eye, it needs to be removed. If Graves' disease is the cause, it may subside with treatment of the overactive thyroid, but this is not guaranteed. Local care to the front of the eye to keep it moist is necessary if the eyelid cannot close.

Prognosis

Exophthalmos can be progressive. Its progress must be carefully followed, treating complications as they occur.

Prevention

Vision can usually be preserved with attentive treatment. There is currently no way to prevent any of the underlying conditions that lead to exophthalmos.

KEY TERMS

Conjunctivae—The clear membranes that line the inside of the eyelids and cover the white part (sclera) of the eyeballs.

Cornea—The clear, dome-shaped part of the front of the eye, through which light first enters the eye. It is located in front of the colored part of the eye (iris).

Inflammation—The body's reaction to invasion by foreign matter, particularly infection. The result is swelling and redness from an increase in water and blood, and pain from the chemical activity of the reaction.

Strabismus—Any deviation of the eyes from a common direction. Commonly called a turned eye.

Thyroid—A gland in the neck overlying the windpipe that regulates the speed of metabolic processes by producing a hormone, thyroxin.

Resources

BOOKS

Fauci, Anthony S., et al., editors. *Harrison's Principles of Internal Medicine*. New York: McGraw-Hill, 1997.

J. Ricker Polsdorfer, MD

Expectorants

Definition

Expectorants are drugs that loosen and clear mucus and phlegm from the respiratory tract.

Purpose

The drug described here, guaifenesin, is a common ingredient in **cough** medicines. It is classified as an expectorant, a medicine that helps clear mucus and other secretions from the respiratory tract. However, some debate exists about how effectively guaifenesin does this. In addition, some cough medicines contain other ingredients that may cancel out guaifenesin's effects. **Cough suppressants** such as codeine, for example, work against guaifenesin because they discourage coughing up the secretions that the expectorant loosens.

There are other ways to loosen and clear the respiratory secretions associated with colds. These include using a humidifier and drinking six to eight glasses of water a day.

Description

Guaifenesin is an ingredient in many cough medicines, such as the brand names Anti-Tuss, Dristan Cold & Cough, Guaifed, GuaiCough, and some Robitussin products. Some products that contain guaifenesin are available only with a physician's prescription; others can be bought without a prescription. They come in several forms, including capsules, tablets, and liquids.

Recommended dosage

Adults and children 12 and over

200–400 mg every four hours. No more than 2,400 mg in 24 hours.

Children 6–11

100–200 mg every four hours. No more than 1,200 mg in 24 hours.

Children 2–5

50–100 mg every four hours. No more than 600 mg in 24 hours.

Children under two

Not recommended.

Precautions

Do not take more than the recommended daily dosage of guaifenesin.

Guaifenesin is not meant to be used for coughs associated with **asthma**, **emphysema**, chronic **bronchitis**, or **smoking**. It also should not be used for coughs that are producing a large amount of mucus.

A lingering cough could be a sign of a serious medical condition. Coughs that last more than seven days or are associated with **fever**, rash, **sore throat**, or lasting **headache** should have medical attention. Call a physician as soon as possible.

Some studies suggest that guaifenesin causes birth defects. Women who are pregnant or plan to become pregnant should check with their physicians before using any products that contain guaifenesin. Whether guaifenesin passes into breast milk is not

KEY TERMS

Asthma—A disease in which the air passages of the lungs become inflamed and narrowed.

Bronchitis—Inflammation of the air passages of the lungs.

Chronic—A word used to describe a long-lasting condition. Chronic conditions often develop gradually and involve slow changes.

Cough suppressant—Medicine that stops or prevents coughing.

Emphysema—An irreversible lung disease in which breathing becomes increasingly difficult.

Mucus—Thick fluid produced by the moist membranes that line many body cavities and structures.

Phlegm—Thick mucus produced in the air passages.

Respiratory tract—The air passages from the nose into the lungs.

Secretion—A substance, such as saliva or mucus, that is produced and given off by a cell or a gland.

known, but no ill effects have been reported in nursing babies whose mothers used guaifenesin.

Side effects

Side effects are rare, but may include **vomiting**, diarrhea, stomach upset, headache, skin rash, and **hives**.

Interactions

Guaifenesin is not known to interact with any foods or other drugs. However, cough medicines that contain guaifenesin may contain other ingredients that do interact with foods or drugs. Check with a physician or pharmacist for details about specific products.

Nancy Ross-Flanigan

Exstrophy of the urinary bladder *see*

Congenital bladder anomalies

External fetal monitoring *see* **Electronic fetal monitoring**

External otitis *see* **Otitis externa**

External sphincter electromyography

Definition

External sphincter **electromyography** helps physicians determine how well the external urinary sphincter muscle is working by measuring the electrical activity in it during contraction and relaxation.

Purpose

The external sphincter muscle is the ring-like muscle that controls urine release from the bladder. When a patient cannot voluntarily control urination (incontinence), a physician may order this test to determine if the problem is caused by the failure of this muscle. The voluntary contraction or release of a muscle such as the external sphincter involves a complex process in which the nerves controlling the muscle signal it to move through the release and uptake of chemicals called neurotransmitters and the generation of electrical impulses. This test records the electrical impulses given off when the muscle contracts or relaxes and allows the physician to determine if the muscle is working properly, if it has been damaged by disease, or some other condition.

Precautions

Patients who are taking **muscle relaxants** or drugs that act like or have an effect on the neurotransmitter acetylcholine (cholinergic or anti-cholinergic drugs) should tell the doctor since they will change the test results. The results will also be altered if the patient moves during the test or if the electrodes are improperly placed.

Description

The patient puts on a surgical gown and lies down on the examining table. The procedure, which takes between 30–60 minutes, may be conducted one of three ways:

- **Skin electrodes.** This is the most commonly used method of recording information. The skin where the electrodes will be placed is cleaned and shaved and an electrically conductive paste is applied. The electrodes are then taped in place. For female patients, the electrodes are taped around the urethra, while for male patients they are placed between the scrotum and the anus.
- **Needle electrodes.** This is considered the most accurate method, since the electrodes are inserted directly

into the muscle, using needles to guide placement. For male patients, a gloved finger is inserted in the rectum, then needles with wires attached are inserted through the skin between the anus and the scrotum. For female patients, the needles are inserted around the urethra. The discomfort of placing the needles is about the same as that of an injection. The needles are withdrawn, and the wires are taped to the thigh.

- **Anal plug electrodes.** The tip of an anal plug is lubricated and inserted into the rectum as the patient relaxes the anal sphincter. Electrodes are attached to the anal plug.

Once the electrodes are in place and attached to the recording device, the patient is asked to alternately contract and relax the external sphincter muscle. The electrical activity generated during these contractions and relaxations is recorded on a graph called an electromyogram.

Preparation

Before the test, the patient should discuss with the doctor whether it is necessary to temporarily discontinue any medications, and follow the doctor's orders. No changes in diet or activity are necessary.

Aftercare

Women may see some blood in their urine the first time they urinate after the test. Blood in the urine of men or blood in the urine of women after the first urination should be reported the doctor. The patient should take a warm bath and drink plenty of fluids to ease any discomfort after the test.

Risks

Complications of external sphincter electromyography are rare. Occasionally patients report blood in their urine after being tested with needle electrodes. Also, the urethra may become mildly irritated causing a change in the normal frequency of urination.

Normal results

In a normally functioning external sphincter muscle, the electromyogram will show increased electrical activity when the patient tightens the muscle and a little or no electrical activity when it is relaxed.

Abnormal results

A diseased external sphincter muscle will produce an abnormal pattern of electrical activity. Conditions

KEY TERMS

Anti-cholinergic drug—A medication that blocks or subdues the action of the neurotransmitter acetylcholine.

Cholinergic drug—A medication that mimics or enhances the action of the neurotransmitter acetylcholine.

Sphincter—A circular muscle that aids in the opening or closing of an opening in the body.

that affect the external sphincter may include **multiple sclerosis**, **neurogenic bladder**, Parkinson's disease, **spinal cord injury**, and **stress incontinence**. However, additional tests must be done in order to confirm any of these diagnoses.

Resources

BOOKS

Lewis, J. A., editor. "External Sphincter Electromyography." In *Illustrated Guide to Diagnostic Tests*. Springhouse, PA: Springhouse Corp., 1994.

Tish Davidson, A.M.

Extracorporeal membrane oxygenation

Definition

Extracorporeal membrane oxygenation (ECMO) is a special procedure that uses an artificial heart-lung machine to take over the work of the lungs (and sometimes also the heart). ECMO is used most often in newborns and young children, but it also can be used as a last resort for adults whose heart or lungs are failing.

Purpose

In newborns, ECMO is used to support or replace an infant's undeveloped or failing lungs by providing oxygen and removing carbon dioxide waste products so the lungs can rest. Infants who need ECMO may include those with:

- meconium aspiration syndrome, (breathing in of a newborn's first stool by a fetus or newborn, which

can block air passages and interfere with lung expansion)

- persistent **pulmonary hypertension**, (a disorder in which the blood pressure in the arteries supplying the lungs is abnormally high)
- respiratory distress syndrome (a lung disorder usually of premature infants that causes increasing difficulty in breathing, leading to a life-threatening deficiency of oxygen in the blood)
- congenital diaphragmatic **hernia**, (the profusion of part of the stomach through an opening in the diaphragm)
- pneumonia
- blood **poisoning**

ECMO is also used to support a child or adult patient's damaged, infected, or failing lungs for a few hours to allow treatment or healing. It is effective for those patients with severe, but reversible, heart or lung problems who haven't responded to treatment with a ventilator, drugs, or extra oxygen. Adults and children who need ECMO usually have one of these problems:

- heart failure
- pneumonia
- **respiratory failure** caused by trauma or severe infection

The ECMO procedure can help a patient's lungs and heart rest and recover, but it will not cure the underlying disease. Any patient who requires ECMO is seriously ill and will likely die without the treatment. Because there is some risk involved, this method is used only when other means of support have failed.

Precautions

Typically, ECMO patients have daily chest x rays and blood work, and constant vital sign monitoring. They are usually placed on a special rotating bed that is designed to decrease pressure on the skin and help move secretions from the lungs.

After the patient is stable on ECMO, the breathing machine settings will be lowered to "rest" settings, which allows the lungs to rest without the risk of too much oxygen or pressure from the ventilator.

Description

There are two types of ECMO: Venoarterial (V-A) ECMO supports the heart and lungs, and is used for patients with blood pressure or heart functioning problems in addition to respiratory

problems. Venovenous (V-V) ECMO supports the lungs only.

V-A ECMO requires the insertion of two tubes, one in the jugular and one in the carotid artery. In the V-V ECMO procedure, the surgeon places a plastic tube into the jugular vein through a small incision in the neck.

Once in place, the tubes are connected to the ECMO circuit, and then the machine is turned on. The patient's blood flows out through the tube and may look very dark because it contains very little oxygen. A pump pushes the blood through an artificial membrane lung, where oxygen is added and carbon dioxide is removed. The size of the artificial lung depends on the size of the patient; sometimes adults need two lungs. The blood is then warmed and returned to the patient. A steady amount of blood (called the flow rate) is pushed through the ECMO machine every minute. As the patient improves, the flow rate is lowered.

Many patients require heavy **sedation** while they are on ECMO to lessen the amount of oxygen needed by the muscles.

As the patient improves, the amount of ECMO support will be decreased gradually, until the machine is turned off for a brief trial period. If the patient does well without ECMO, the treatment is stopped.

Typically, newborns remain on ECMO for three to seven days, although some babies need more time (especially if they have a diaphragmatic hernia). Once the baby is off ECMO, he or she will still need a ventilator (breathing machine) for a few days or weeks. Adults may remain on ECMO for days to weeks, depending on the condition of the patient, but treatment may be continued for a longer time depending on the type of heart or lung disease, the amount of damage to the lungs before ECMO was begun, and the presence of any other illnesses or health problems.

Preparation

Before ECMO is begun, the patient receives medication to ease **pain** and restrict movement.

Aftercare

Because infants on ECMO may have been struggling with low oxygen levels before treatment, they may be at higher risk for developmental problems. They will need to be monitored as they grow.

KEY TERMS

Carotid artery—Two main arteries (passageway carrying blood from the heart to other parts of the body) that carry blood to the brain.

Congenital diaphragmatic hernia—The profusion of part of the stomach through an opening in the diaphragm.

Meconium aspiration syndrome—Breathing in of meconium (a newborn's first stool) by a fetus or newborn, which can block air passages and interfere with lung expansion.

Membrane oxygenator—The artificial lung that adds oxygen and removes carbon dioxide.

Pulmonary hypertension—A disorder in which the blood pressure in the arteries supplying the lungs is abnormally high.

Respiratory distress syndrome—A lung disorder usually of premature infants that causes increasing difficulty in breathing, leading to a life-threatening deficiency of oxygen in the blood.

Venoarterial (V-A) bypass—The type of ECMO that provides both heart and lung support, using two tubes (one in the jugular vein and one in the carotid artery).

Venovenous (V-V) bypass—The type of ECMO that provides lung support only, using a tube inserted into the jugular vein.

Risks

Bleeding is the biggest risk for ECMO patients, since blood thinners are given to guard against **blood clots**. Bleeding can occur anywhere in the body, but is most serious when it occurs in the brain. This is why doctors periodically perform ultrasound brain scans of anyone on ECMO. Stroke, which may be caused by bleeding or blood clots in the brain, has occurred in some patients undergoing ECMO.

If bleeding becomes a problem, the patient may require frequent blood transfusions or operations to control the bleeding. If the bleeding can't be stopped, ECMO will be withdrawn.

Other risks include infection or vocal cord injury. Some patients develop severe blood infections that cause irreversible damage to vital organs.

There is a small chance that some part of the complex equipment may fail, which could introduce air into the system or affect the patient's blood levels,

causing damage or death of vital organs (including the brain). For this reason, the ECMO circuit is constantly monitored by a trained technologist.

Normal results

Lungs and/or heart return to healthy functioning.

Abnormal results

Lungs and/or heart do not improve while on ECMO.

Resources

ORGANIZATIONS

American Society of Extra-Corporeal Technology. 11480 Sunset Hills Rd., Ste. 210E, Reston, VA 20190. (703) 435-8556. < <http://www.amsect.org> > .

ECMO Moms and Dads. PO Box 53848, Lubbock, TX 79543. (806) 794-0259.

Extracorporeal Life Support Organization. 1327 Jones Dr., Ste. 101, Ann Arbor, MI 48105. (734) 998-6600. < <http://www.else.med.umich.edu> > .

Carol A. Turkington

Extracorporeal shock-wave see **Lithotripsy**

Extrinsic allergic alveolitis see
Hypersensitivity pneumonitis

Eye and orbit sonograms see **Eye and orbit ultrasounds**

Eye and orbit ultrasounds

Definition

Ultrasound imaging equipment allows eye specialists (ophthalmologists) to “see” the eye in great detail without the **pain** and risk of exploratory surgery, or the limitations and uncertainty inherent to traditional visual examination. Ultrasound is used to detect and diagnose many eye diseases and injuries, to measure the eye prior to corrective surgery, and directly as a treatment tool.

Purpose

An ophthalmologist uses ultrasonic imaging to help diagnose the underlying cause(s) of a patient's symptoms, to assess the general condition of an injured eye, and to measure the eye prior to corrective

surgery. Situations that may call for ultrasonic imaging include:

- Excessive tearing or visible infection. These external symptoms could indicate a serious underlying problem such as a tumor, an internal infection, the presence of a deeply lodged irritant (foreign body), or the effects of a previously unrecognized injury. When presented with general symptoms, ultrasound can speed diagnosis if a serious condition is suspected.
- Impaired vision. Fuzzy vision, poor night vision, restricted (tunnel) vision, blind spots, extreme light sensitivity, and even blindness can all stem from inner eye conditions ranging from glaucoma and **cataracts**, to retinitis, detached retina, tumors, or impaired blood circulation. Again, high resolution ultrasound can quickly identify causes and pinpoint their location. A special type of ultrasound, known as Doppler, can even perceive and measure circulation in the tiny blood vessels of the eye.
- Eye trauma. The eye can be damaged by a direct impact or a puncture wound, as a result of a general head trauma, or by intense light exposure. Even when the cause of injury is obvious, ultrasound can reveal the exact type, extent, and location of damage, from deformations and ruptures to internal bleeding, and help to guide emergency care efforts.
- Lens replacement surgery. Exact measurement of the eye's optical dimensions with ultrasound greatly improves the visual outcome for cataract patients receiving permanent synthetic lenses; and for severely myopic patients receiving implanted corrective lenses.

Ophthalmic ultrasound imaging is also used routinely to guide the precise placement of instruments during surgery, and can be used directly for the treatment of **glaucoma** and tumors of the eye.

Precautions

Ultrasound of the eye, properly performed by qualified personnel using appropriate equipment, has no risks. There is no evidence to suggest that the procedure itself poses any threat to a healthy eye, or worsens the condition of a diseased or injured eye.

Description

Ophthalmic ultrasound equipment sends high frequency pulses of sound into the eye, where they bounce off the boundaries between different structures in the eye and produce a distinctive pattern of echoes. This echo pattern is received and interpreted by a

computer to produce an image on a television screen. The time it takes an echo to return to the receiver corresponds to the depth it traveled into the eye.

Single transducer (the sound transmitter/receiver) ultrasound is used to measure distances within the eye. This is A-mode ultrasound. A linear array of transducers in a single small probe, B-mode, provides a picture of a cross section through the eye. Doppler mode ultrasound combines B-mode with the ability to detect and measure the flow of blood in the tiny vessels of the eye.

As a direct treatment tool, the vibrations of high intensity A-mode ultrasound can be used to heat and erode tumors. The same technique can be used to control glaucoma by selectively destroying the cells which produce the fluid that causes the internal pressure of the eye to rise.

The procedure followed in a regular ultrasonic eye examination is relatively simple. The patient relaxes in a comfortable chair in a darkened room. Mild anesthetic eye drops are administered and the head is held secure. The ultrasonic probe, coated with a sterile gel to ensure good contact, is lightly pressed against the eye as the images are made. The probe may be applied to the eyelid or directly to the eye, as necessary. The patient feels nothing else, and the whole office procedure takes about 15 minutes.

Preparation

Preparation by the patient is generally unnecessary, although under special circumstances an ophthalmologist may perform pretest procedures. The ophthalmologist and/or ultrasound technician will conduct all preparations at the time of the test.

Aftercare

Patients may experience partial and temporary blurred vision, as well as "eye strain" headaches. These symptoms usually fade within an hour of the procedure, during which time patients should rest their eyes and avoid all activities that require good eyesight, like driving.

Risks

Improperly focused, high-intensity ultrasound could burn and physically disrupt delicate eye tissue and cause injury. This risk is, however, slight and would arise only from improper use, or as a potential side effect of tumor or glaucoma treatment.

KEY TERMS

Cataracts—A clouding of the lens of the eye or the material immediately surrounding it, causing blurred vision. For many people it occurs naturally with aging, but may also result from injury.

Glaucoma—A common eye disease characterized by increased fluid pressure in the eye that damages the optic nerve, which carries sensations to the brain. Glaucoma can be caused by another eye disorder, such as a tumor or congenital malformation, or appear without obvious cause, but if untreated it generally leads to blindness.

Intraocular—Literally, within the eye.

Ophthalmologist—A medical doctor specializing in eye care who is generally, but not necessarily, an eye surgeon.

Retina—The third and innermost membrane of the eye, which contains the light-sensitive nerve tissue that leads into the optic nerve and is the primary instrument of vision. Inflammation of the retina (retinitis) has many causes, including overexposure to intense light, diabetes, and syphilis.

Normal results

A normal ultrasound scan would indicate a fully healthy eye. For therapeutic ultrasound, a normal result would be an improvement in the targeted condition, such as shrinking of a tumor or lessening of pressure inside the eye of a glaucoma patient.

Abnormal results

Because diagnostic ultrasound is generally used to investigate symptoms, the results of a scan will often be abnormal and they will detect evidence of an underlying condition.

Resources

ORGANIZATIONS

American Academy of Ophthalmology. 655 Beach Street, PO Box 7424, San Francisco, CA 94120-7424. <<http://www.eyenet.org>>.

American Institute of Ultrasound in Medicine. 14750 Sweitzer Lane, Suite 100, Laurel, MD 20707-5906. (800) 638-5352. <<http://www.aium.org>>.

National Eye Institute. 2020 Vision Place, Bethesda, MD 20892-3655. (301) 496-5248. <<http://www.nei.nih.gov>>.

Kurt Richard Sternlof

Eye cancer

Definition

Eye **cancer** refers to a cancerous growth in any part of the eye. Some eye cancers are primary, while others represent metastases from primary cancers elsewhere in the body.

All types of eye cancer are rare in comparison to other cancerous tumors. According to the American Cancer Society, 2,090 people in the United States will be diagnosed with cancer of the eye or orbit in 2004, and 180 persons will die from the disease.

Description

Eye cancers can be grouped into three basic categories according to their location in the eye: tumors of the eyelid and conjunctiva; intraocular tumors; and orbital tumors. This article will focus on **retinoblastoma**, the most common eye cancer in children, and intraocular melanoma, the most common eye cancer in adults.

Retinoblastoma can occur at any age but is most often seen in children younger than five. About 200 children a year are diagnosed with it in the United States. Retinoblastoma starts with a small tumor in the retina, the tissue that lies at the very back of the eye. In growing children, the retina originates from cells called retinoblasts that grow and divide very quickly. These cells eventually become the mature cells of the retina when they stop growing. In the case of retinoblastoma the retinoblasts don't stop growing and form a tumor that can continue to grow and cause further complications if not treated quickly.

Retinoblastoma typically has three classifications: intraocular, extraocular and recurrent retinoblastoma. In the intraocular form the cancer can be found in one or both eyes but not in tissue external of the eye. In the extraocular form the cancer has spread outside the eye. It can spread to the tissue surrounding the eye or it can invade other areas of the body. In the recurrent form the cancer returns after already being treated. It may recur in the eye, its surrounding tissues, or elsewhere in the body.

Intraocular melanoma is a rare cancer overall, yet it is the most common eye cancer seen in adults. Intraocular melanoma occurs when cancer cells are found in the uvea of the eye. The uvea includes the iris (the colored portion of eye), the ciliary body (an eye muscle that focuses the lens) and the choroid (found in the back of the eye next to the retina).

Intraocular cancer of the iris usually grows slowly and usually doesn't spread. The tumor is seen on the iris as a darker spot than the surrounding area. Intraocular cancer of the choroid or ciliary body occurs in the back of the eye. They are classified by size with a small tumor being 2-3 mm or smaller and a medium or large tumor being bigger than 3 mm.

Intraocular cancer can spread and become extraocular as well. If not found and treated early enough it can spread to the surrounding tissues, the optic nerve or into the orbit (eye socket).

Causes and symptoms

Genetics is thought to play a role in eye cancer. In regards to retinoblastoma, it is believed that if a tumor develops only in one eye then it isn't hereditary. However, if a tumor occurs in both eyes then it is hereditary. Those who have hereditary retinoblastoma have a rare risk of developing a tumor in the brain and should be monitored on a regular basis.

The causes of intraocular melanoma are not fully understood as of the early 2000s. Age is a factor as well as genetic inheritance. In 2004, a group of ophthalmologists in Scotland identified mutations in the **BRAF** gene in samples of tissue taken from conjunctival melanomas. Interestingly enough, this type of cancer is seen most often in white people from a northern European descent.

The symptoms of this type of cancer usually begin with blurred vision and tenderness of the eye. Advanced symptoms may include loss of vision. If these symptoms persist a person should make an appointment with an eye specialist.

Diagnosis

The diagnosis of eye cancer is usually made by an ophthalmologist, who is a doctor who specializes in treating eye disorders. In the case of cancerous growths, the doctor is usually able to see the tumor through the pupil or directly on the iris if the cancer is intraocular melanoma of the iris. Because the doctor can usually readily see the tumor a biopsy is rarely needed.

An ultrasound or a fluorescein **angiography** are two tests doctors use to further diagnose eye cancers. In an ultrasound sound waves are pointed at the tumor and depending on how they reflect off the tumor the doctor can better diagnose it. In a fluorescein angiography a fluorescent dye is injected into the patient's arm. When this dye circulates through the body and reaches the eye a series of rapid pictures are taken

through the pupil. The tumor will show up in these photos.

Once a diagnosis has been made, the treatment can begin.

Treatment

The treatment depends on how far advanced the tumor is. If the tumor is in the advanced stages and there is little hope of regaining vision the most effective treatment is an enucleation, the removal of the eye. Enucleation obviously is a drastic treatment and is avoided if possible. Other eye surgeries include the following:

- choroidectomy: removal of part of the choroid
- iridectomy: removal of part of the iris
- iridocyclectomy: removal of parts of the ciliary body and parts of iris
- iridotrabeculectomy: removal of parts of the supporting tissues around the cornea and iris

In eye cancer where the tumor is small and there is a good chance that the vision will be restored less drastic measures than the above surgeries are taken. Radiation and **chemotherapy** are two courses of treatment that help in killing off the existing tumor and preventing its spread into other areas of the body. Specific **anticancer drugs** that are used in treating melanomas of the eye include gemcitabine and treosulfan.

Besides radiation and chemotherapy there are other methods of treating eye cancer. Cryotherapy uses extreme cold to destroy the cancer cells. Thermotherapy uses heat from a laser to destroy the cancer cells. Photocoagulation uses a laser to destroy blood vessels that supply the tumor with nutrients. If the tumor isn't advanced these are good options to treat it in order to avoid losing an eye. The chief drawback of thermotherapy is that some visual acuity may be lost in the treated eye.

A radiation/surgical treatment for eye cancer is brachytherapy. A small plaque with radioactive iodine on one side and gold on the other is stitched to the eye behind the tumor with the radioactive iodine facing the tumor. The gold is used to shield the other tissues from the radiation. It is left there for a period of time depending on the dosage of radiation needed and then it is removed. Although iodine brachytherapy is effective in treating melanoma of the eye, it increases the patient's risk of developing a cataract or **glaucoma**.

KEY TERMS

Brachytherapy—A type of radiation treatment for cancer in which the source of the radiation is applied directly to the surface of the body.

Carcinogen—A substance that is known to cause cancer.

Conjunctiva—The thin membrane that lines the eyelids.

Cornea—The transparent front portion of the exterior cover of the eye.

Enucleation—Surgical removal of the eyeball.

Iris (plural, irides)—The circular pigmented membrane behind the cornea of the eye that gives the eye its color. The iris surrounds a central opening called the pupil.

Ocular melanoma—A malignant tumor that arises within the structures of the eye. It is the most common eye tumor in adults.

Ophthalmology—The branch of medicine that deals with the diagnosis and treatment of eye disorders.

Orbit—The bony cavity that contains the eyeball.

Pupil—The opening in the center of the iris of the eye that allows light to enter the eye.

Uvea—The middle of the three coats of tissue surrounding the eye, comprising the choroid, iris, and ciliary body. The uvea is pigmented and well supplied with blood vessels.

Prognosis

All forms of retinoblastoma and intraocular melanoma are treatable. Enucleation can usually be avoided if the tumor is found early enough. In addition, primary cancers of the eye have a relatively low mortality rate if treated promptly.

Clinical trials

As of 2004, the National Cancer Institute is sponsoring 10 clinical trials of treatments for ocular melanoma in adults and 5 trials for treatments of retinoblastoma in children. These trials allow researchers to investigate new types of **radiation therapy** and chemotherapy, new drugs and drug combinations, biological therapies, ways of combining various types of treatment for eye cancer, side effect reduction, and quality of life. Information on specific clinical

trials can be acquired from the National Cancer Institute at <<http://www.nci.nih.gov>> or (800) 4-CANCER.

Prevention

Retinoblastoma is not preventable. In addition, most other types of eye tumors are thought to be partly genetic as of the early 2000s.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD., editors. "Principles of Cancer Therapy." Section 11, Chapter 144 In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 1999.

PERIODICALS

- Garcia-Valenzuela, Enrique, MD, PhD, and Mauricio E. Pons, MD. "Melanoma, Choroidal." *eMedicine* [cited August 1, 2004]. <<http://www.emedicine.com/oph/topic403.htm>> .
- Gear, H., H. Williams, E. G. Kemp, and F. Roberts. "BRAF Mutations in Conjunctival Melanoma." *Investigative Ophthalmology and Visual Science* 45 (August 2004): 2484–2488.
- Keilholz, U., R. Schuster, A. Schmitt, et al. "A Clinical Phase I Trial of Gemcitabine and Treosulfan in Uveal Melanoma and Other Solid Tumours." *European Journal of Cancer* 40 (September 2004): 2047–2052.
- Kiratli, H., and S. Bilgic. "Transpupillary Thermotherapy in the Management of Choroidal Metastases." *European Journal of Ophthalmology* 14 (September-October 2004): 423–429.
- Mercandetti, Michael, MD, and Adam J. Cohen, MD. "Tumors, Orbital." *eMedicine* [cited December 14, 2004]. <<http://www.emedicine.com/oph/topic758.htm>> .
- Puusaari, I., J. Heikkonen, and T. Kivela. "Ocular Complications after Iodine Brachytherapy for Large Uveal Melanomas." *Ophthalmology* 111 (September 2004): 1768–1777.
- Shields, C. L., H. Demirci, E. Karatza, and J. A. Shields. "Clinical Survey of 1643 Melanocytic and Nonmelanocytic Conjunctival Tumors." *Ophthalmology* 111 (September 2004): 1747–1754.

ORGANIZATIONS

- American Academy of Ophthalmology. P. O. Box 7424, San Francisco, CA 94120-7424. (415) 561-8500. <<http://www.aao.org>> .
- American Cancer Society. 1599 Clifton Rd. NE, Atlanta, GA 30329. (800) 227-2345. <<http://www.cancer.org>> .
- Canadian Ophthalmological Society (COS). 610-1525 Carling Avenue, Ottawa ON K1Z 8R9. <<http://www.eyesite.ca>> .

National Eye Institute. 2020 Vision Place, Bethesda, MD 20892-3655. (301) 496-5248. <<http://www.nei.nih.gov>>.

Ocular Oncology Service, Wills Eye Hospital. 840 Walnut Street, Philadelphia, PA 19107. (215) 928-3000. <<http://www.willseye.org>>.

OTHER

American Cancer Society (ACS). *Cancer Facts & Figures 2004*. <http://www.cancer.org/downloads/STT/CAFF_finalPWSecured.pdf>.

Cancer Net. 2001. <<http://www.cancernet.nci.nih.gov>>.

Eye Cancer Info. Separate web site of the Ocular Oncology Service at Wills Eye Hospital. <<http://eyecancerinfo.com/>>.

JHMI. 2001 <<http://www.med.jhu.edu>>.

National Cancer Institute (NCI) Physician Data Query (PDQ). *Intraocular (Eye) Melanoma: Treatment*. January 2, 2003. <<http://www.nci.nih.gov/cancerinfo/pdq/treatment/intraocularmelanoma/healthprofessional>>.

Thomas Scott Eagan
Ronald Watson, PhD
Rebecca J. Frey, PhD

Eye examination

Definition

An eye examination is a series of tests that measure a person's ocular health and visual status, to detect abnormalities in the components of the visual system, and to determine how well the person can see.

Purpose

An eye examination is performed by an ophthalmologist, (M.D. or D.O. -doctor of osteopathy), or an optometrist (O.D.) to determine if there are any pre-existing or potential vision problems. Eye exams may also reveal the presence of many non-eye diseases. Many systemic diseases can affect the eyes, and since the blood vessels on the retina are observed during the exam, certain problems may be uncovered (e.g., high blood pressure or diabetes).

Infants should be examined by a physician to detect any physical abnormalities. Frequency of eye exams then generally differs with age and the health of the person. Eye exams can be performed in infants, and if a problem is noted the infant can be seen, generally by a pediatric ophthalmologist. A child with no symptoms should have an eye exam at age three. Early

exams are important because permanent decreases in vision (e.g., amblyopia, also called lazy eye) can occur if not treated early (usually by ages 6–9). Again, with no other symptoms, the second exam should take place before first grade. After first grade, the American Optometric Association recommends an eye exam every two years; ages 19–40, every two to three years; ages 41–60, every two years; and annually after that. However, these are recommendations for healthy people with no risk factors. Patients should ask their doctors how often they should come for exams. Some patients have risk factors for eye disease (e.g., people with diabetes or a family history of eye disease; African Americans, who are at higher risk for **glaucoma**) and may need more frequent checkups. Also, if children seem to be having trouble in school, problems with reading, rubbing their eyes when reading, etc., an eye exam may be necessary sooner.

Precautions

The examiner needs to know if the patient is taking any medications or has any existing health conditions. Some medications, even over-the-counter (OTC) medications can affect vision or even interfere with the eyedrops the doctor may use during the exam. Certain eyedrops would not be used if the patient has **asthma**, heart problems, or other conditions.

The patient may need someone to drive them home in case the eyes were dilated. Bringing sunglasses to the exam may also help decrease the glare from light until the dilating drops wear off.

Description

An eye examination, given by an ophthalmologist or optometrist, costs about \$100. It may or may not be covered by insurance. It begins with information from the patient (case history) and continues with a set of primary tests, plus additional specialized tests given as needed, dictated by the outcomes of initial testing and the patient's age. The primary tests can be divided into two groups, those that evaluate the physical state of the eyes and surrounding areas, and those that measure the ability to see.

The order of the tests for the exam may differ from doctor to doctor, however, most exams will include the following procedures:

Information gathering and initial observations

The examiner will take eye and medical histories that include the patient's chief complaint, any past eye disorders, all medications being taken (e.g., OTC

medications, **antibiotics**, and birth control pills), any blood relatives with eye disorders, and any systemic disorders the patient may have. The patient should also tell the doctor about hobbies and work conditions. This information helps in modifying prescriptions and lets the doctor know how the patient uses his or her eyes. For example, using a computer screen vs. construction work, the working distance of a computer screen may affect the prescription; the construction worker needs protective eyewear.

The patient should bring their current pair of glasses to the exam. The doctor can get the prescription from the glasses by using an instrument called a lensometer.

Visual acuity examination

Visual acuity measures how clearly the patient can see. It is measured for each eye separately, with and without the current prescription. It is usually measured with a Snellen eye chart, a poster with lines of different-sized letters, each line with a number at the side denoting the distance from which a person with normal vision can read that line. Other kinds of eye charts with identifiable figures are available for children or anyone unfamiliar with the Roman alphabet. These charts are made to be placed at a certain distance (usually 20 ft) from the person being tested. At this distance, people with normal vision can read a certain line (usually the lowest), marked the 20/20 line; these people are said to have 20/20 vision. For people who can't read the smallest line, the examiner assigns a ratio based on the smallest line they can read. The first number (numerator) of the ratio is the distance between the chart and the patient, and the second number (denominator) is the distance where a person with normal vision would be able to read that line. The ratio 20/40 means the patient can see at 20 ft. what people with normal vision can see at 40 ft. away.

When a patient is unable to read any lines on the chart, they are moved closer until they can read the line with the largest letters. The acuity is still measured the same way. A ratio of 5/200 means the person being tested can see at 5 ft what a normal person can see 200 ft.

When a patient cannot read the chart at all, the examiner may hold up some fingers and ask the patient to count them at various distances, and records the result as "counting fingers" at the distance of recognition. If the patient cannot count the examiner's fingers at any distance, the examiner determines if the patient can see hand movements. If so, the result is recorded as "hand movements." If not, the examiner

determines if the patient can detect light from a penlight. If the patient can detect the light but not its direction, the result is recorded as "light perception." If the patient can recognize its direction, the result is recorded as "light projection." If the patient cannot detect the light at all, the result is recorded as "no light perception."

Eye movement examination and cover tests

The examiner asks the patient to look up and down, and to the right and left to see if the patient can move the eyes to their full extent. The examiner asks the patient to stare at an object, then quickly covers one eye and notes any movement in the eye that remains uncovered. This procedure is repeated with the other eye. This, and another similar cover test, helps to determine if there is an undetected eye turn or problem with fixation. The doctor may also have the patient look at a pen and follow it as it is moved close to the eyes. This checks convergence.

Iris and pupil examination

The doctor checks the pupil's response to light (if it dilates and constricts appropriately). The iris is viewed for symmetry and physical appearance. The iris is checked more thoroughly later using a slit lamp.

Refractive error determination-Refraction

The examiner will determine the refractive error and obtain a prescription for corrective lenses for people whose visual acuity is less than 20/20. An instrument called a phoropter, which the patient sits behind, is generally used (sometimes the refraction can be done with a trial frame that the patient wears). The phoropter is equipped with many lenses that allow the examiner to test many combinations of corrections to learn which correction allows the patient to see the eye chart most clearly. This is the part of the exam when the doctor usually says, "Which is better, one or two?" The phoropter also contains prisms, and sometimes the doctor will intentionally make the patient see double. This may help in determining a slight eye turn. The exam will check vision at distance and near (reading).

A prescription for corrective lenses can also be supplied by automated refracting devices, which measure the necessary refraction by shining a light into the eye and observing the reflected light. Another objective way to obtain a prescription is using a hand-held retinoscope. As in the automated method just mentioned, the doctor shines a light in the patient's eyes



A woman looking through a refractor. (Photograph by John Greim, Photo Researchers. Reproduced by permission.)

and can determine an objective prescription. This is helpful in young children or infants.

Sometimes drops will be instilled in the patient's eyes before this part of the exam. The drops may relax accommodation so that the refraction will be more accurate. This is helpful in children and people who are farsighted.

After the refraction and other visual status tests, for example color tests or binocularity tests (can the patient see 3-D, or have depth perception), the doctor will check the health of the eyes and surrounding areas. The main instruments used are the ophthalmoscope and the slit lamp.

Ophthalmoscopic examination

These observations are best accomplished after dilating the pupils and require an ophthalmoscope. The ophthalmoscope most frequently used is a called a *direct ophthalmoscope*. It is a hand-held illuminated 15X multi-lens magnifier that lets the examiner view the inside back area of the eye (fundus). The retina, blood vessels, optic nerve, and other structures are examined.

Slit lamp examination

The slit lamp is a microscope with a light source that can be adjusted. This magnifies the external and some internal structures of the eyes. The lid and lid margin, cornea, iris, pupil, conjunctiva, sclera, and lens are examined. The slit lamp is also used in contact lens evaluations. A little probe called a tonometer may be used at this time to

check the pressure of the eyes. A colored eyedrop may be instilled immediately prior to this test. The drop has a local anesthetic so the patient won't feel the probe touch the eye. It is a quick procedure.

Visual field measurement

A perimeter, the instrument for measuring visual fields, is a hollow hemisphere, equipped with a light source that projects dots of light over the inside surface. The patient's head is positioned so that the eye being tested is at the center of the sphere and (about 13 in. 33 cm) from all points on the inside surface of the hemisphere. The patient stares straight ahead at an image on the center of the surface and signals whenever he or she detects a flash of light. The perimeter records which flashes are seen and which are missed and maps the patient's field of vision and blindspots.

Intraocular pressure (IOP) measurement

Tonometers are used to measure IOP. Some tonometers measure pressure by expelling a puff of air (noncontact tonometer) towards the eyeball from a very short distance. Other tonometers are placed directly on the cornea. The noncontact tonometers are not as accurate as the contact tonometers and are sometimes used for screenings.

Completing the evaluation with additional tests

Depending upon the results other tests may be necessary. These can include, but are not limited to binocular indirect ophthalmoscopy, gonioscopy, color tests, contrast sensitivity testing, ultrasonography, and others. The patient may have to return for additional visits.

Results

External observations

INITIAL OBSERVATIONS AND SLIT LAMP EXAM. Some general observations the doctor may be looking for include: head tilt; drooping eyelids (**ptosis**); eye turns; red eyes (injection); eye movement; size, shape, and color of the iris; clarity of the cornea, anterior chamber, and lens. The anterior chamber lies behind the cornea and in front of the iris. If it appears cloudy or if cells can be seen in it during the slit lamp exam an inflammation may be present. A narrow anterior chamber may put the patient at risk for glaucoma. A clouding of the normally clear lens is called a cataract.

KEY TERMS

Amblyopia—Decreased visual acuity, usually in one eye, in the absence of any structural abnormality in the eye.

Conjunctiva—The mucous membrane that covers the white part of the eyes (sclera) and lines the eyelids.

Cornea—Clear outer covering of the front of the eye.

Floaters—Translucent specks that float across the visual field, due to small objects floating in the vitreous humor.

Fundus—The inside of an organ. In the eye, refers to the back area that can be seen with the ophthalmoscope.

Glaucoma—There are many types of glaucoma. Glaucoma results in optic nerve damage and a decreased visual field and blindness if not treated. It is usually associated with increased IOP, but that is not always the case. The three factors associated with glaucoma are increased IOP, a change in the optic nerve head, and changes in the visual field.

Gonioscope—An instrument used to inspect the eye (e.g., the anterior chamber). It consists of a magnifier and a lens equipped with mirrors; it's placed on the patient's cornea.

Iris—The colored ring just behind the cornea and in front of the lens that controls the amount of light sent to the retina.

Macula—The central part of the retina where the rods and cones are densest.

Ophthalmoscope—An instrument designed to view structures in the back of the eye.

Optic nerve—The nerve that carries visual messages from the retina to the brain.

Pupil—The circular opening that looks like a black hole in the middle of the iris.

Retina—The inner, light-sensitive layer of the eye containing rods and cones; transforms the image it receives into electrical messages which are then sent to the brain via the optic nerve.

Sclera—The tough, fibrous, white outer protective covering that surrounds the eye.

Slit lamp—A microscope that projects a linear slit beam of light onto the eye; allows viewing of the conjunctiva, cornea, iris, aqueous humor, lens, and eyelid.

Tonometer—An instrument that measures intraocular pressure (IOP).

Ultrasonography—A method of obtaining structural information about internal tissues and organs where an image is produced because different tissues bounce back ultrasonic waves differently.

Internal observations

OPHTHALMOSCOPIC EXAM. The observations include, but are not limited to the retina, blood vessels, and optic nerve. The optic nerve enters the back of the eye and can be checked for swelling or other problems. The blood vessels can be viewed as can the retina. The macula is a 3–5 mm area in the back of the eye and is responsible for central vision. The fovea is a small area located within the macula and is responsible for sharp vision. When a person looks at something, they are pointing the fovea at the object. Changes in the macular area can be observed with the ophthalmoscope. Retinal tears or detachments can also be seen.

Visual ability

VISUAL ACUITY. The refraction will determine the refractive status for each eye for distance and for near. A prescription for glasses is made after taking many things into consideration. The eye doctor may alter a

prescription based upon many factors. Different materials for glasses may be suggested. For example, polycarbonate may be suggested for children or people active in sports because it is very impact resistant. Bifocals, trifocals, single-vision spectacles, and contact lenses are also options.

VISUAL FIELDS. A normal visual field extends about 60° upward, about 75° downward, about 65° toward the nose, and about 100° toward the ear and has one blind spot close to the center. Defects in the visual field signify damage to the retina, optic nerve, or the neurological visual pathway.

Seeing clearly does not necessarily mean the eyes are healthy or that the eyes are working together as a team. Regular checkups can detect abnormalities, hopefully before a problem arises. The eye doctor can suggest ways to help protect the eyes and vision (e.g., safety goggles, ultraviolet (UV) coatings on lenses). A person should also have an eye exam if they notice a change in vision, eyestrain, blur, flashes of light, a sudden onset of floaters (little dots),

distortion of objects, double vision, redness, **pain** or discharge.

Resources

ORGANIZATIONS

American Academy of Ophthalmology. 655 Beach Street, PO Box 7424, San Francisco, CA 94120-7424. <<http://www.eyenet.org>> .

American Optometric Association. 243 North Lindbergh Blvd., St. Louis, MO 63141. (314) 991-4100. <<http://www.aoanet.org>> .

Lorraine Lica, PhD

Eye exercises see **Vision training**

Eye glasses and contact lenses

Definition

Eyeglasses and contact lenses are devices that correct refractive errors in vision. Eyeglass lenses are mounted in frames worn on the face, sitting mostly on the ears and nose, so that the lenses are positioned in front of the eyes. Contact lenses appear to be worn in direct contact with the cornea, but they actually float on a layer of tears that separates them from the cornea.

Purpose

The purpose of eyeglasses and contact lenses is to correct or improve the vision of people with nearsightedness (**myopia**), farsightedness (**hyperopia**), **presbyopia**, and **astigmatism**.

Precautions

People allergic to certain plastics should not wear contact lenses or eyeglass frames or lenses manufactured from that type of plastic. People allergic to nickel should not wear Flexon frames. People at risk of being in accidents that might shatter glass lenses should wear plastic lenses, preferably polycarbonate. (Lenses made from polycarbonate, the same type of plastic used for the space shuttle windshield, are about 50 times stronger than other lens materials.) Also, people at risk of receiving electric shock should avoid metal frames.

People employed in certain occupations may be prohibited from wearing contact lenses, or may be

required to wear safety eyewear over the contact lenses. Some occupations, such as construction or auto repair, may require safety lenses and safety frames. Physicians and employers should be consulted for recommendations.

Description

Eyes are examined by optometrists (O.D.) or by ophthalmologists (M.D. or D.O.—doctor of **osteopathy**). Prescriptions, if necessary, are then given to patients for glasses. The glasses are generally made by an optician. A separate contact lens-fitting exam is necessary if the patient wants contact lenses, because an eyeglass prescription is not the same as a contact lens prescription.

Eyeglasses

More than 140 million people in the United States wear eyeglasses. People whose eyes have refractive errors do not see clearly without glasses, because the light emitted from the objects they are observing does not come into focus on their retinas. For people who are farsighted, images come into focus behind the retina; for people who are nearsighted, images come into focus in front of the retina.

LENSES. Lenses work by changing the direction of light so that images come into focus on the retina. The greater the index of refraction of the lens material and the greater the difference in the curvature between the two surfaces of the lens, the greater the change in direction of light that passes through it, and the greater the correction.

Lenses can be unifocal, with one correction for all distances, or they can be correct for more than one distance (multifocal). One type of multifocal, the bifocal, has an area of the lens (usually at the bottom) that corrects for nearby objects (about 14 in from the eyes); the remainder of the lens corrects for distant objects (about 20 ft from the eyes). Another type of multifocal, a trifocal, has an area in-between that corrects for intermediate distances (usually about 28 in). Conventional bifocals and trifocals have visible lines between the areas of different correction; however, lenses where the correction gradually changes from one area to the other, without visible lines, have been available since the 1970s. Such lenses are sometimes called progressives or no-line bifocals.

To be suitable for eyeglass lenses, a material must be transparent, without bubbles, and have a high index of refraction. The greater the index of refraction, the thinner the lens can be. Lenses are made from either glass or

plastic (hard resin). The advantage of plastic is that it is lightweight and more impact resistant than glass. The advantage of glass is that it is scratch resistant and provides the clearest possible vision.

Glass was the first material to be used for eyeglass lenses, and was used for several hundred years before plastic was introduced.

Optical-quality acrylic was introduced for eyeglass use in the early 1940s, but because it was easily scratched, brittle, and discolored rapidly, it did not supplant glass as the material of choice. Furthermore, it wasn't suitable for people with large refractive errors. A plastic called CR-39, introduced in the 1960s, was more suitable. Today, eyeglass wearers can also choose between polycarbonate, which is the most impact-resistant material available for eyewear, and polyurethane, which has exceptional optical qualities and higher refraction than the conventional plastics even glass. Patients with high prescriptions should ask about high index material options for their lenses. Aspheric lenses are also useful for high prescriptions. They are flatter and lighter than conventional lenses.

There are many lenses and lens-coating options for individual needs, including coatings that block the ultraviolet (UV) light or UV and blue light, which have been found to be harmful to the eyes. Such coatings are not needed on polycarbonate lenses, which already have UV protection. UV coatings are particularly important on sunglasses and ski goggles. Sunglasses, when nonprescription, should be labeled with an indication that they block out 99–100% of both UV-A and UV-B rays.

There are anti-scratch coatings that increase the surface hardness of lenses (an important feature when using plastic lenses) and anti-reflective (AR) coatings that eliminate almost all glare and allow other people to see the eyes of the wearer. AR coatings may be particularly helpful to people who use computers or who drive at night. Mirror coatings that prevent other people from seeing the wearer's eyes are also available. There is a whole spectrum of tints, from light tints to darker tints, used in sunglasses. Tint, however, does not block out UV rays, so a UV coating is needed. Polaroid lenses that block out much of the reflected light also allow better vision in sunny weather and are helpful for people who enjoy boating. Photosensitive (photochromatic) lenses that darken in the presence of bright light are handy for people who don't want to carry an extra set of glasses. Photochromatic lenses are available in glass and plastic.

FRAMES. Frames can be made from metal or plastic, and they can be rimless. There is an almost

unlimited variety of shapes, colors, and sizes. The type and degree of refractive correction in the lens determine to some extent the type of frame most suitable. Some lenses are too thick to fit in metal rims, and some large-correction prescriptions are best suited to frames with small-area lenses.

Rimless frames are the least noticeable type, and they are lightweight because the nosepiece and temples are attached directly to the lenses, eliminating the weight of the rims. They tend to not be as sturdy as frames with rims, so they are not a good choice for people who frequently remove their glasses and put them on again. They are also not very suitable for lenses that correct a high degree of farsightedness, because such lenses are thin at the edges.

Metal frames are less noticeable than plastic, and they are lightweight. They are available in solid gold, gold-filled, anodized aluminum, nickel, silver, stainless steel, and now titanium and titanium alloy. Until the late 1980s, when titanium-nickel alloy and titanium frames were introduced, metal frames were, in general, more fragile than plastic frames. The titanium frames, however, are very strong and lightweight. An alloy of titanium and nickel, called Flexon, is not only strong and lightweight, but returns to its original shape after being twisted or dented. It is not perfect for everyone, though, because some people are sensitive to its nickel. Flexon frames are also relatively expensive.

Plastic frames are durable, can accommodate just about any lens prescription, and are available in a wide range of prices. They are also offered in a variety of plastics (including acrylic, epoxy, cellulose acetate, cellulose propionate, polyamide, and nylon) and in different colors, shapes, and levels of resistance to breakage. Epoxy frames are resilient and return to their original shape after being deformed, so they do not need to be adjusted as frequently as other types. Nylon frames are almost unbreakable. They revert to their original shape after extreme trauma and distortion; because of this property, though, they cannot be readjusted after they are manufactured.

FIT. The patient should have the distance between the eyes (PD) measured, so that the optical centers of the lenses will be in front of the patient's pupils. Bifocal heights also have to be measured with the chosen frame in place and adjusted on the patient. Again, this is so the lenses will be positioned correctly. If not positioned correctly, the patient may experience eyestrain or other problems. This can occur with over-the-counter reading glasses. The distance between the lenses is for a "standard" person. Generally, this will

not be a problem, but if a patient is sensitive or has more closely set eyes, for example, it may pose a problem. Persons buying ready-made sunglasses or reading glasses should hold them up to see if they appear clear. They should also hold the lenses to see an object with straight lines reflected off of the lenses. If the lines don't appear straight, the lenses may be warped or inferior.

Patients may sometimes need a few days to adjust to a new prescription; however, problems should be reported, because the glasses may need to be rechecked.

Contact lenses

More than 32 million people in the United States wear these small lenses that fit on top of the cornea. They provide a field of view unobstructed by eyeglass frames; they do not fog up or get splattered, so it is possible to see well while walking in the rain; and they are less noticeable than any eyeglass style. On the other hand, they take time to get accustomed to; require more measurements for fitting; require many follow-up visits to the eye doctor; can lead to complications such as infections and corneal damage; and may not correct astigmatism as well as eyeglasses, especially if the astigmatism is severe.

Originally, hard contact lenses were made of a material called PMMA. Although still available, the more common types of contact lenses are listed below:

- Rigid gas-permeable (RGP) daily-wear lenses are made of plastic that does not absorb water but allows oxygen to get from the atmosphere to the cornea. (This is important because the cornea has no blood supply and needs to get its oxygen from the atmosphere through the film of tears that moves beneath the lens.) They must be removed and cleaned each night.
- Rigid gas-permeable (RGP) extended-wear lenses are made from plastic that also does not absorb water but is more permeable to oxygen than the plastic used for daily-wear lenses. They can be worn up to a week.
- Daily wear soft lenses are made of plastic that is permeable to oxygen and absorbs water; therefore, they are soft and flexible. These lenses must be removed and cleaned each night, and they do not correct all vision problems. Soft lenses are easier to get used to than rigid lenses, but are more prone to tears and do not last as long.
- Extended-wear soft lenses are highly permeable to oxygen, are flexible by virtue of their ability to absorb water, and can usually be worn for up to one week. They do not correct all vision problems. There is more of a risk of infection with extended-wear lenses than with daily-wear lenses.
- Extended-wear disposable lenses are soft lenses worn continually for up to six days and then discarded, with no need for cleaning.
- Planned-replacement soft lenses are daily wear lenses that are replaced on a regular schedule, which is usually every two weeks, monthly, or quarterly. They must also be cleaned.

Soft contact lenses come in a variety of materials. There are also different kinds of RGP and soft multifocal contact lenses available. Monovision, where one contact lens corrects for distance vision while the other corrects for near vision, may be an option for presbyopic patients. Monovision, however, may affect depth perception and may not be appropriate for everyone. Contact lenses also come in a variety of tints. Soft contacts are available that can make eyes appear a different color. Even though such lenses have no prescription, they must still be fitted and checked to make sure that an eye infection does not occur. People should never wear someone else's contact lenses. This can lead to infection or damage to the eye.

Tiny, surgically implanted contact lenses may one day replace eyeglasses, contact lenses and **laser surgery** for some patients with extreme nearsightedness. Called intraocular lenses, they were still investigational in the spring of 2004, and although they are surgically installed, they can be removed. Researchers expected FDA approval in 2004.

Aftercare

Contact lens wearers must be examined periodically by their eye doctors to make sure that the lenses fit properly and that there is no infection. Infection and lenses that do not fit properly can damage the cornea. Patients can be allergic to certain solutions that are used to clean or lubricate the lenses. For that reason, patients should not randomly switch products without speaking with their doctor. Contact lens wearers should seek immediate attention if they experience eye **pain**, a burning sensation, red eyes, intolerable sensitivity to light, cloudy vision, or an inability to keep the eyes open.

To avoid infection, it is important for contact lens wearers to exactly follow their instructions for lens insertion and removal, as well as cleaning. Soft contact lens wearers should never use tap water to rinse their lenses or to make up solutions. All contact lens

KEY TERMS

Astigmatism—Assymetric vision defects due to irregularities in the cornea.

Cornea—The clear outer covering of the front of the eye.

Index of refraction—A constant number for any material for any given color of light that is an indicator of the degree of the bending of the light caused by that material.

Lens—A device that bends light waves.

Permeable—Capable of allowing substances to pass through.

Polycarbonate—A very strong type of plastic often used in safety glasses, sport glasses, and children's eyeglasses. Polycarbonate lenses have approximately 50 times the impact resistance of glass lenses.

Polymer—A substance formed by joining smaller molecules. For example, plastic, acrylic, cellulose acetate, cellulose propionate, nylon, etc.

Presbyopia—A condition affecting people over the age of 40 where the system of accommodation that allows focusing of near objects fails to work because of age-related hardening of the lens of the eye.

Retina—The inner, light-sensitive layer of the eye containing rods and cones; transforms the image it receives into electrical messages sent to the brain via the optic nerve.

Ultraviolet (UV) light—Part of the electromagnetic spectrum with a wavelength just below that of visible light. It is damaging to living material, especially eyes and DNA.

wearers should also always have a pair of glasses and a carrying case for their contacts with them, in case the contacts have to be removed due to eye irritation.

Risks

Wearing contact lenses increases the risk of corneal damage and eye infections.

Normal results

The normal expectation is that people will achieve 20/20 vision while wearing corrective lenses. A new technology for customized eyeglasses patented in

2004 claims to achieve exceptional vision assessment and 20/10 acuity by using wavefront measurements and precise parameters to produce measurements such as pupil size and distance, along with other customized lens and frame features.

Resources

PERIODICALS

Asp, Karen. "Implanted Contact Lenses." *Prevention* (June 2004): 68.

"Patent Issued for Z-lens Wavefront Guided, Customized Eyeglasses." *Medical Devices & Surgical Technology Week* (April 18, 2004): 150.

ORGANIZATIONS

American Academy of Ophthalmology. 655 Beach Street, PO Box 7424, San Francisco, CA 94120-7424. <<http://www.eyenet.org>>.

American Optometric Association. 243 North Lindbergh Blvd., St. Louis, MO 63141. (314) 991-4100. <<http://www.aoanet.org>>.

Optician Association of America. 7023 Little River Turnpike, Suite 207, Annandale, VA 22003. (703) 916-8856. <<http://www.opticians.org>>.

OTHER

Contact Lens Council. <<http://www.contactlenscouncil.org>>.

Lorraine Lica, PhD
Teresa G. Odle

Eye muscle surgery

Definition

Eye muscle surgery is surgery to weaken, strengthen, or reposition any of the muscles that move the eyeball (the extraocular muscles).

Purpose

The purpose of eye muscle surgery is generally to align the pair of eyes so that they gaze in the same direction and move together as a team, either to improve appearance or to aid in the development of binocular vision in a young child. To achieve binocular vision, the goal is to align the eyes so that the location of the image on the retina of one eye corresponds to the location of the image on the retina of the other eye.

In addition, sometimes eye muscle surgery can help people with other eye disorders (nystagmus and Duane syndrome, for example).

Precautions

Depth perception (stereopsis) develops around the age of three months old. For successful development of binocular vision and the ability to perceive three-dimensionally, the surgery should not be postponed past the age of four. The earlier the surgery the better the outcome, so an early diagnosis is important. Surgery may even be performed before two years old. After surgery, if binocular vision is to develop, corrective lenses and eye exercises (vision therapy) will probably be necessary.

Description

The extraocular muscles attach via tendons to the sclera (the white, opaque, outer protective covering of the eyeball) at different places just behind an imaginary equator circling the top, bottom, left, and right of the eye. The other end of each of these muscles attaches to a part of the orbit (the eye socket in the skull). These muscles enable the eyes to move up, down, to one side or the other, or any angle in between.

Normally both eyes move together, receive the same image on corresponding locations on both retinas, and the brain fuses these images into one three-dimensional image. The exception is in **strabismus** which is a disorder where one or both eyes deviate out of alignment, most often outwardly (exotropia) or toward the nose (esotropia). The brain now receives two different images, and either suppresses one or the person sees double (diplopia). This deviation can be adjusted by weakening or strengthening the appropriate muscles to move the eyes toward the center. For example, if an eye turns upward, the muscle at the bottom of the eye could be strengthened.

Rarely, eye muscle surgery is performed on people with **nystagmus** or Duane syndrome. Nystagmus is a condition where one or both eyes move rapidly or oscillate; it can sometimes be helped by moving the eyes to the position of least oscillation. Duane syndrome is a disorder where there is limited horizontal eye movement; it can sometimes be relieved by surgery to weaken an eye muscle.

There are two methods to alter extraocular muscles. Traditional surgery can be used to strengthen, weaken, or reposition an extraocular muscle. The surgeon first makes an incision in the conjunctiva (the

clear membrane covering the sclera), then puts a suture into the muscle to prevent it from getting lost and loosens the muscle from the eyeball with a surgical hook. During a resection, the muscle is detached from the sclera, a piece of muscle is removed so the muscle is now shorter, and the muscle is reattached to the same place. This strengthens the muscle. In a recession, the muscle is made weaker by repositioning it. More than one extraocular eye muscle might be operated on at the same time.

Another way of weakening eye muscles, using botulinum toxin injected into the muscle, was introduced in the early 1980s. Although the botulinum toxin wears off, the realignment may be permanent, depending upon whether neurological connections for binocular vision were established during the time the toxin was active. This technique can also be used to adjust a muscle after traditional surgery.

The cost of eye muscle surgery is about \$2,000–\$4,000, and about 700,000 surgeries are performed annually in the United States.

Preparation

Patients should make sure their doctors are aware of any medications that they are taking, even over-the-counter medications. Patients should not take aspirin, or any other blood-thinning medications for ten days prior to surgery, and should not eat or drink after midnight the night before.

Aftercare

Patients will need someone to drive them home after their surgery. They should continue to avoid **aspirin** and other non-steroidal anti-inflammatory agents for an additional three days, but they can take **acetaminophen** (e.g., Tylenol). Patients should discuss this with the surgeon to be clear what medications they can or cannot take. **Pain** will subside after two to three days, and patients can resume most normal activities within a few days. Again, this may vary with the patient and the patient should discuss returning to normal activity with the surgeon. They should not get their eyes wet for three to four days and should refrain from swimming for 10 days. Operated eyes will be red for about two weeks.

Risks

As with any surgery, there are risks involved. Eye muscle surgery is relatively safe, but very rarely a cut muscle gets lost and can not be retrieved. This, and

KEY TERMS

Botulinum toxin (botulin)—A neurotoxin made by *Clostridium botulinum*; causes paralysis in high doses, but is used medically in small, localized doses to treat disorders associated with involuntary muscle contraction and spasms, in addition to strabismus.

Conjunctiva—The mucous membrane that covers the eyes and lines the eyelids.

Extraocular muscles—The muscles (lateral rectus, medial rectus, inferior rectus, superior rectus, superior oblique, and inferior oblique) that move the eyeball.

Orbit—The cavity in the skull containing the eyeball; formed from seven bones: frontal, maxillary, sphenoid, lacrimal, zygomatic, ethmoid, and palatine.

Retina—The inner, light-sensitive layer of the eye containing rods and cones; transforms the image it receives into electrical messages sent to the brain via the optic nerve.

Sclera—The tough, fibrous, white outer protective covering of the eyeball.

Strabismus—A disorder where the two eyes do not point in the same direction.

other serious reactions, including those caused by anesthetics, can result in vision loss in the affected eye. Occasionally, retinal or nerve damage occurs. Double vision is not uncommon after eye muscle surgery. As mentioned earlier, glasses or vision therapy may be necessary.

Normal results

Cosmetic improvement is likely with success rate estimates varying from about 65–85%. According to the best statistics as of 1998, binocular vision is improved in young children about 35% of the time. There is no improvement, or the condition worsens 15–35% of the time. A second operation may rectify less-than-perfect outcomes.

Resources

ORGANIZATIONS

American Academy of Ophthalmology. 655 Beach Street, PO Box 7424, San Francisco, CA 94120-7424. <<http://www.eyenet.org>> .

American Academy of Pediatric Ophthalmology and Strabismus (AAPOS). <<http://med-aapos.bu.edu>> .

OTHER

Olitsky, Scott E., and Leonard B. Nelson. *Strabismus WebBook*. May 4, 1998. <<http://www.smbs.buffalo.edu/oph/ped/webbook.htm>> .

Lorraine Lica, PhD

Eye training see **Vision training**

Eyelid disorders

Definition

An eyelid disorder is any abnormal condition that affects the eyelids.

Description

Eyelids consist of thin folds of skin, muscle, and connective tissue. The eyelids protect the eyes and spread tears over the front of the eyes. The inside of the eyelids are lined with the conjunctiva of the eyelid (the palpebral conjunctiva), and the outside of the lids are covered with the body's thinnest skin. Some common lid problems include the following: stye, blepharitis, chalazion, entropion, ectropion, eyelid **edema**, and eyelid tumors.

Stye

A stye is an infection of one of the three types of eyelid glands near the lid margins, at the base of the lashes.

Chalazion

A chalazion is an enlargement of a meibomian gland (an oil-producing gland in the eyelid), usually not associated with an infectious agent. More likely, the gland opening is clogged. Initially, a chalazion may resemble a stye, but it usually grows larger. A chalazion may also be located in the middle of the lid and be internal.

Blepharitis

Blepharitis is the inflammation of the eyelid margins, often with scales and crust. It can lead to eyelash loss, chalazia, styes, ectropion, corneal damage, excessive tearing, and chronic **conjunctivitis**.

Entropion

Entropion is a condition where the eyelid margin (usually the lower one) is turned inward; the eyelashes touch the eye and irritate the cornea.

Ectropion

Ectropion is a condition where one or both eyelid margins turn outward, exposing both the conjunctiva that covers the eye and the conjunctiva that lines the eyelid.

Eyelid edema

Eyelid edema is a condition where the eyelids contain excessive fluid.

Eyelid tumors

Eyelids are susceptible to the same skin tumors as the skin over the rest of the body, including noncancerous tumors and cancerous tumors (basal cell carcinoma, squamous cell carcinoma, **malignant melanoma**, and sebaceous gland carcinoma). Eyelid muscles are susceptible to sarcoma.

Causes and symptoms

Stye

Styes are usually caused by bacterial **staphylococcal infections**. The symptoms are **pain** and inflammation in one or more localized regions near the eyelid margin.

Chalazion

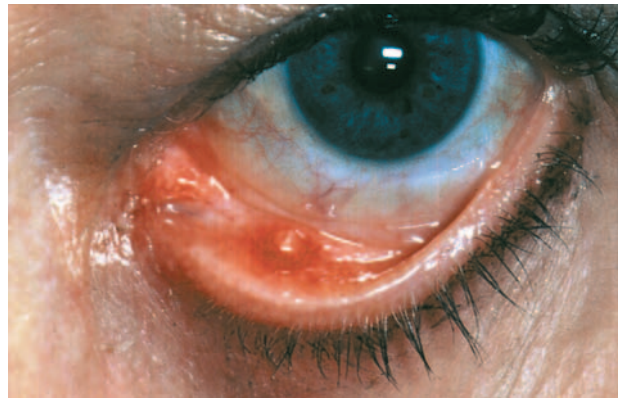
A chalazion is caused by a blockage in the outflow duct of a meibomian gland. Symptoms are inflammation and swelling in the form of a round lump in the lid that may be painful.

Blepharitis

Some cases of blepharitis are caused by bacterial infection and some by head lice, but in some cases, the cause is unclear. It may also be caused by an overproduction of oil by the meibomian glands. Blepharitis can be a chronic condition that begins in early childhood and can last throughout life. Symptoms can include **itching**, burning, a feeling that something is in the eye, inflammation, and scales or matted, hard crusts surrounding the eyelashes.

Entropion

Entropion usually results from **aging**, but sometimes can be due to a congenital defect, a spastic eyelid



A chalazion on the eyelid. This condition is caused by an obstruction of one of the meibomian glands which lubricate the edge of the eyelid. (Photo Researchers, Inc. Reproduced by permission.)

muscle, or a scar on the inside of the lid from surgery, injury, or disease. It is accompanied by excessive tearing, redness, and discomfort.

Ectropion

Similar to entropion, the usual cause of ectropion is aging. It also can be due to a spastic eyelid muscle or a scar, as in entropion. It also can be the result of **allergies**. Symptoms are excessive tearing and hardening of the eyelid conjunctiva.

Eyelid edema

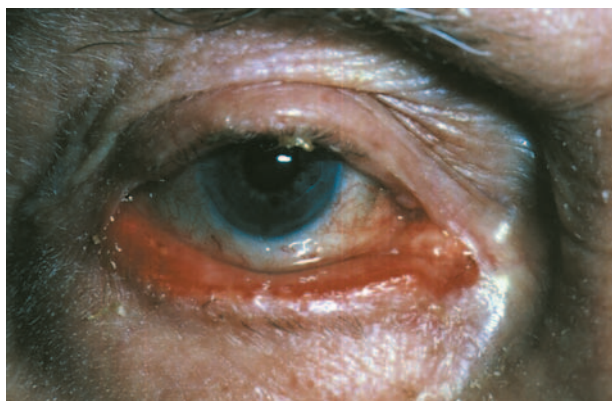
Eyelid edema is most often caused by allergic reactions, for example, allergies to eye makeup, eye-drops or other drugs, or plant allergens such as pollen. **Trichinosis**, a disease caused by eating undercooked meat, also causes eyelid edema. However, swelling can also be caused by more serious causes, such as infection, and can lead to **orbital cellulitis** which can threaten vision. Symptoms can include swelling, itching, redness, or pain.

Eyelid tumors

Tumors found on the eyelids are caused by the same conditions that cause these tumors elsewhere on the body. They are usually painless and may or may not be pigmented. Some possible causes include **AIDS (Kaposi's sarcoma)** or increased exposure to ultraviolet (UV) rays which may lead to skin **cancer**.

Diagnosis

An instrument called a slit lamp is generally used to magnify the structures of the eyes. The doctor may



A close-up of the eye of an elderly patient showing ectropion of the lower eyelid. Ectropion is a condition in which the eyelid turns away from the eye. The most common type is senile ectropion (seen here), in which the droop of the eyelid is due to loss of tissue elasticity in old age and weakness in the muscles surrounding the eye. (Photograph by Dr. P. Marazzi, Photo Researchers, Inc. Reproduced by permission.)

press on the lid margin to see if oil can be expressed from the meibomian glands. The doctor may invert the lid to see the inside of the lid. Biopsy is used to diagnose cancerous tumors.

Treatment

Stye

Styes are treated with warm compresses for 10–15 minutes, three to four times a day. Chloramphenicol ointment may be used as well. Sometimes **topical antibiotics** may be prescribed if the infection is spreading.

Chalazion

About 25% of chalazia will disappear spontaneously, but warm compresses may speed the process. Chloramphenicol ointment may be used as well. Because chalazia are inside the lid, topical medications are generally of no benefit. Medication may need to be injected by the doctor into the chalazion or if that doesn't help the chalazion may need to be excised. If what appears to be a chalazion recurs on the same site as any previous one, the possibility of sebaceous gland carcinoma should be investigated by biopsy.

Blepharitis

Blepharitis is treated with hot compresses, with antibiotic ointment, and by cleaning the eyelids with a moist washcloth and then with baby shampoo. Good hygiene is essential. Patients can try to keep rooms dry, such as by placing a bowl of water on top of a

radiator. Tear film supplements such as hypromellose can help moisten the eyes when dry. If itching, soreness, or redness occurs from the tear film drops, they should be stopped. Topical or systemic **antibiotics** also may be prescribed. If the blepharitis doesn't clear up with treatment or if it seems to be a chronic problem, the patient may have **acne rosacea**. These patients may need to see a dermatologist as well.

Entropion and ectropion

Both entropion and ectropion can be surgically corrected. Prior to surgery, the lower lid of entropion can be taped down to keep the lashes off the eye, and both can be treated with lubricating drops to keep the cornea moist.

Eyelid edema

Patients with swollen eyelids should contact their eye doctor. A severely swollen lid can press on the eye and possibly increase the intraocular pressure. An infection needs to be ruled out. Or, something as simple as an allergy to nail polish and then touching the eyes can cause swelling. The best treatment for allergic eyelid edema is to find and remove the substance causing the allergy. When that is not possible, as in the case of plant allergens, cold compresses and immunosuppressive drugs such as corticosteroid creams are helpful. However, steroids can cause **cataracts** and increase intraocular pressure and patients must be very careful not to get the cream in their eyes. This should not be done unless under a doctor's care. For edema caused by trichinosis, the trichinosis must be treated.

Eyelid tumors

Cancerous tumors should be removed upon discovery, and noncancerous tumors should be removed before they become big enough to interfere with vision or eyelid function. Eyelid tumors require special consideration because of their sensitive location. It is important that treatment not compromise vision, eye movement, or eyelid movement. Accordingly, eyelid reconstruction will sometimes accompany tumor excision.

Prognosis

The prognosis for styes and chalazia is good to excellent. With treatment, blepharitis, ectropion, and entropion usually have good outcomes. The prognosis for nonmalignant tumors, basal cell carcinoma, and squamous cell carcinoma is good once they are

KEY TERMS

Allergen—A substance capable of inducing an allergic response.

Allergic reaction—An immune system reaction to a substance in the environment; symptoms include rash, inflammation, sneezing, itchy watery eyes, and runny nose.

Conjunctiva—The mucous membrane that covers the white part of the eyes and lines the eyelids.

Edema—A condition where tissues contain excessive fluid.

Meibomian gland—Oil-producing glands in the eyelids that open near the eyelid margins.

properly removed. Survival rate for malignant melanoma depends upon how early it was discovered and if it was completely removed. Sebaceous carcinomas are difficult to detect, so poor outcomes are more frequent.

All of these eyelid disorders, if not treated, can lead to other, possibly serious vision problems—dry eye, **astigmatism**, or even vision loss, for example. An ophthalmologist or optometrist should be consulted.

Prevention

Good lid hygiene is very important. Regular eyelid washing with baby shampoo helps prevent styes, chalazia, blepharitis, and eyelid edema. To avoid these problems, it's also important to refrain from touching and rubbing the eyes and eyelids, especially with hands that have not just been washed.

Blepharitis is associated with dandruff, which is caused by a kind of bacteria that is one of the causes of blepharitis. Controlling dandruff by washing the hair, scalp, and eyebrows with shampoo containing selenium

sulfide to kill the bacteria helps control the blepharitis. When using anything near the eyes, it is important to read the label or consult with a doctor first.

Avoiding allergens helps prevent allergic eyelid edema. Staying inside as much as possible when pollen counts are high and eliminating the use of, or at least removing eye makeup thoroughly, or using hypo-allergenic makeup may help if the person is sensitive to those substances.

Sunscreen, UV-blocking sunglasses, and wide brimmed hats can help prevent eyelid tumors.

Entropion and ectropion seem to be unpreventable.

Resources

PERIODICALS

- “At a Glance: Chalazion Versus Stye.” *GP* May 3, 2004: 52.
 “Practical Ophthalmology for GPs: The Treatment of Blepharitis.” *Pulse* (May 10, 2004): 60.

ORGANIZATIONS

- American Academy of Ophthalmology. 655 Beach Street, PO Box 7424, San Francisco, CA 94120-7424. <<http://www.eyenet.org>> .
 American Optometric Association. 243 North Lindbergh Blvd., St. Louis, MO 63141. (314) 991-4100. <<http://www.aoanet.org>> .
 American Society of Ophthalmic Plastic and Reconstructive Surgery. 1133 West Morse Blvd, #201, Winter Park, FL 32789. (407) 647-8839. <<http://www.asoprs.org>> .

OTHER

- “Eyelid Abnormalities.” *Eye Clinic of Fairbanks*. <<http://www.eyeclinikfbks.com/ECFLID.htm>> .
RxMed. <<http://www.rxmed.com>> .

Lorraine Lica, PhD
 Teresa G. Odle

Eyelid edema see **Eyelid disorders**

Eyelid plastic surgery see **Blepharoplasty**

F

Fabry's disease see **Lipidoses**

Face lift

Definition

Face lift surgery is a cosmetic procedure that involves redirecting some of the skin and muscle tissue of the face and neck to counter signs of aging produced by gravity.

Purpose

The purpose of face lift surgery, also known as facialplasty, rhytidoplasty, or cervicofacial rhytidectomy, is to improve the appearance of the face by repositioning the skin and tightening some of the underlying muscle and tissue. The procedure is designed to counter sagging and looseness in skin and muscle tissue caused by gravity as the patient ages. Face lift surgery will not erase all facial wrinkles, as the term rhytidectomy (which literally means “surgical removal of wrinkles”) might imply. Wrinkles around the mouth and eyes, for example, may benefit little from face lift surgery. Other procedures, such as blepharoplasty, chemical peel, or dermabrasion, also may be necessary.

Precautions

Patients with other medical conditions should consult with their primary physician before undergoing face lift surgery. Lung problems, heart disease, and certain other conditions can lead to a higher risk of complications. Patients who take medications that can alter the way their blood clots (including female hormones, **aspirin**, and some non-aspirin pain relievers) should stop these medications prior to surgery to lower the risk that a hematoma will form. A hematoma, a pocket of

blood below the skin, is the most frequent complication of face lift surgery.

Description

Face lift surgery can be performed on an outpatient basis with local anesthetics. Patients typically also receive “twilight anesthesia,” an intravenous sedative that helps to lower their awareness of the procedure being performed.

There are a number of variations of face lift surgery. Which one is used will depend on the patient's facial structure, how much correction is needed, and the preferences of the surgeon performing the procedure. In a typical face lift surgery, the surgeon begins by making an incision within the hairline just above the ear. The incision continues down along the front edge of the ear, around the earlobe, and then up and behind the ear extending back into the hairline. The location of this incision is designed to hide any sign of the procedure later. The same procedure is repeated on the other side of the face. Using various instruments, the surgeon will then work to separate the skin of the face from its underlying tissue, moving down to the cheek and into the neck area and below the chin. Fat deposits over the cheeks and in the neck may be removed surgically or with liposuction at this time. The surgeon will then work to free up and tighten certain bands of muscle and tissue that extend up from the shoulder, below the chin, and up and behind the neck. If these muscles and tissue are not tightened, the looseness and sagging appearance of the skin will return. The surgeon then trims excess skin from the edges of the original incision, pulls the skin back, and staples or sutures it into place.

Preparation

Prior to the procedure, patients meet with their surgeon to discuss the surgery, clarify the results that

can be achieved, and discuss the potential problems that can occur. Having realistic expectations is important in any cosmetic procedure. Patients will learn, for example, that although face lift surgery can improve the contour of the face and neck, other procedures will be necessary to reduce the appearance of many wrinkles. As mentioned earlier, patients will stop taking aspirin, birth control or female hormones, and other medications affecting blood clotting about two weeks before the procedure. Some physicians prescribe vitamin C and K in the belief that this promotes healing. Patients will also be advised to stop **smoking** and to avoid exposure to passive smoke before the procedure and afterward. Some surgeons also recommend antibiotics be taken beforehand to limit the risk of infection. Some surgeons also use a steroid injection before or after the procedure, to reduce swelling.

Aftercare

After the surgery, a pressure bandage will be applied to the face to reduce the risk of hematoma. The patient may spend a few hours resting in a recovery room to ensure no bleeding has occurred. The patient then returns home. Some surgeons recommend that the patient remain reclining for the next 24 hours, consuming a liquid diet, and avoiding any movements that lead the neck to flex. Ice packs for the first few days can help to reduce swelling and lower the risk of hematoma. Patients continue taking an antibiotic until the first stitches come out about five days after the procedure. The bandages are removed seven to ten days later. Many patients return to work and limited activities within two weeks of the procedure.

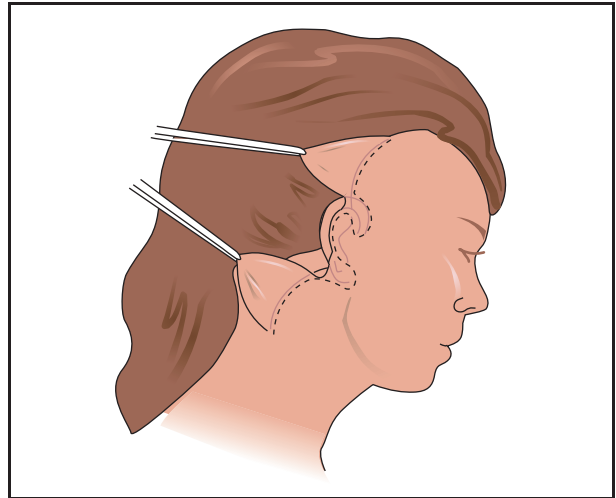
Risks

The major complication seen following face lift surgery is a hematoma. If a hematoma forms, the patient may have to return to have the stitches reopened to find the source of the bleeding. Most hematomas form within 48 hours of surgery. The typical sign is **pain** or swelling affecting one side of the face but not the other.

Another risk of face lift surgery is nerve damage. Sometimes it can affect the patient's ability to raise an eyebrow, or distort his smile, or leave him with limited feeling in his earlobe. Most of these nerve injuries, however, repair themselves within 2–6 months.

Normal results

Some swelling and bruising is normal following face lift surgery. After these disappear, the patient should see



In a typical face lift surgery, the surgeon begins by making an incision within the hairline just above the ear. The incision continues down along the front of the ear, around the earlobe, and then up and behind the ear extending back into the hairline, as shown above. The same procedure is repeated on the other side of the face. The surgeon will then separate the skin from the tissue, remove fat deposits over the cheeks and neck, tighten up muscles and tissues below the chin and upwards behind the neck. The surgeon then trims excess skin from the original incision, pulls the skin back, and sutures it into place. (Illustration by Electronic Illustrators Group.)

KEY TERMS

Hematoma—A complication of surgery in which a collection of blood forms below the skin.

Rhytidectomy—It literally means “wrinkle excision.” It is another, misleading, term for face lift surgery.

Twilight anesthesia—An intravenous mixture of sedatives and other medications that decreases patients' awareness of the procedure being performed.

a noticeable improvement in the contour of his face and neck. Some fine wrinkling of the skin may be improved, but deep wrinkles are likely to require another cosmetic procedure to improve their appearance.

Abnormal results

In addition to the risks outlined above, other complications of face lift surgery include infection, scarring, and hair loss near incision lines.

Resources

ORGANIZATIONS

American Society for Dermatologic Surgery. 930 N. Meacham Road, PO Box 4014, Schaumburg, IL 60168-4014. (847) 330-9830. <<http://www.asds-net.org>>.

American Society of Plastic and Reconstructive Surgeons. 44 E. Algonquin Rd., Arlington Heights, IL 60005. (847) 228-9900. <<http://www.plasticsurgery.org>>.

Richard H. Camer

Factitious disorders

Definition

Factitious disorders are a group of mental disturbances in which patients intentionally act physically or mentally ill without obvious benefits. The name factitious comes from a Latin word that means artificial. These disorders are not **malinger**, which is defined as pretending illness when the “patient” has a clear motive, such as financial gain.

Description

Patients with factitious disorders produce or exaggerate the symptoms of a physical or mental illness by a variety of methods, including contaminating urine samples with blood, taking hallucinogens, injecting themselves with bacteria to produce infections, and other similar behaviors.

There are no reliable statistics on the frequency of factitious disorders, but they are more common in men than in women. The following conditions are sometimes classified as factitious disorders:

Munchausen syndrome

Munchausen syndrome refers to patients whose factitious symptoms are dramatized and exaggerated. Many persons with Munchausen go so far as to undergo major surgery repeatedly, and, to avoid detection, at several locations. Many have been employed in hospitals or in health care professions. The syndrome’s onset is in early adulthood.

Munchausen by proxy

Munchausen by proxy is the name given to factitious disorders in children produced by parents or

other caregivers. The parent may falsify the child’s medical history or tamper with laboratory tests in order to make the child appear sick. Occasionally, they may actually injure the child to assure that the child will be treated.

Ganser’s syndrome

Ganser’s syndrome is an unusual dissociative reaction to extreme **stress** in which the patient gives absurd or silly answers to simple questions. It has sometimes been labeled as psychiatric malingering, but is more often classified as a factitious disorder.

Causes and symptoms

No single explanation of factitious disorders covers all cases. These disorders are variously attributed to underlying **personality disorders**; **child abuse**; the wish to repeat a satisfying childhood relationship with a doctor; and the desire to deceive or test authority figures. Also, the wish to assume the role of patient and be cared for is involved. In many cases, the suffering of a major personal loss has been implicated.

The following are regarded as indications of a factitious disorder:

- dramatic but inconsistent medical history
- extensive knowledge of medicine and/or hospitals
- negative test results followed by further symptom development
- symptoms that occur only when the patient is not being observed
- few visitors
- arguments with hospital staff or similar acting-out behaviors
- eagerness to undergo operations and other procedures

When patients with factitious disorders are confronted, they usually deny that their symptoms are intentional. They may become angry and leave the hospital. In many cases they enter another hospital, which has led to the nickname “hospital hoboes.”

Diagnosis

Diagnosis of factitious disorders is usually based on the exclusion of bona fide medical or psychiatric conditions, together with a combination of the signs listed earlier. In some cases, the

KEY TERMS

Ganser's syndrome—An unusual factitious disorder characterized by dissociative symptoms and absurd answers to direct questions.

Malingering—Pretending to be sick in order to be relieved of an unwanted duty or obtain some other obvious benefit.

Munchausen by proxy—A factitious disorder in children produced by a parent or other caregiver.

Munchausen syndrome—A factitious disorder in which the patient's symptoms are dramatized and exaggerated.

diagnosis is made on the basis of records from other hospitals.

Treatment

Treatment of factitious disorders is usually limited to prompt recognition of the condition and the refusal to give unnecessary medications or to perform unneeded procedures. Factitious disorder patients do not usually remain in the hospital long enough for effective psychiatric treatment. Some clinicians have tried psychotherapeutic treatment for factitious disorder patients, and there are anecdotal reports that antidepressant or antipsychotic medications are helpful in certain cases.

Prognosis

Some patients have only one or two episodes of factitious disorders; others develop a chronic form that may be lifelong. Successful treatment of the chronic form appears to be rare.

Resources

BOOKS

Eisendrath, Stuart J. "Psychiatric Disorders." In *Current Medical Diagnosis & Treatment*, 1998, edited by Stephen J. McPhee, et al., 37th ed. Stamford: Appleton & Lange, 1997.

Rebecca J. Frey, PhD

Factor VIII deficiency see **Hemophilia**

Factor IX deficiency see **Hemophilia**

Failure to thrive

Definition

Failure to thrive (FTT) is used to describe a delay in a child's growth or development. It is usually applied to infants and children up to two years of age who do not gain or maintain weight as they should. Failure to thrive is not a specific disease, but rather a cluster of symptoms which may come from a variety of sources.

Description

Shortly after birth most infants lose some weight. After that expected loss, babies should gain weight at a steady and predictable rate. When a baby does not gain weight as expected, or continues to lose weight, it is not thriving. Failure to thrive may be due to one or more conditions.

Organic failure to thrive (OFTT) implies that the organs involved with digestion and absorption of food are malformed or incomplete so the baby cannot digest its food. Non-organic failure to thrive (NOFTT) is the most common cause of FTT and implies the baby is not receiving enough food due to economic factors or parental neglect, or due to psychosocial problems.

Causes and symptoms

Occasionally, there may be an underlying physical condition that inhibits the baby's ability to take in, digest, or process food. These defects can occur in the esophagus, stomach, small or large intestine, rectum or anus. Usually the defect is an incomplete development of the organ, and it must be surgically corrected. Most physical defects can be detected shortly after birth.

Failure to thrive may also result from lack of available food or the quality of the food offered. This can be due to economic factors in the family, parental beliefs and concepts of **nutrition**, or neglect of the child. In addition, if the baby is being breast fed, the quality or quantity of the mother's milk may be the source of the problem.

Psychosocial problems, often stemming from a lack of nurturing parent-child relations can lead to a failure to thrive. The child may exhibit poor appetite due to depression from insufficient attention from parents.

Infants and toddlers, whose growth is substantially less than expected, are considered to be suffering from FTT.

Diagnosis

Most babies are weighed at birth and that weight is used as a base line for future well-baby check-ups. If the baby is not gaining weight at a predictable rate, the doctor will do a more extensive examination. If there are no apparent physical deformities in the digestive tract, the doctor will examine the child's environment. As part of that examination, the doctor will look at the family history of height and weight. In addition, the parents will be asked about feedings, illnesses, and family routines. If the mother is breastfeeding the doctor will also evaluate her diet, general health, and well being as it affects the quantity and quality of her milk.

Diagnosis of FTT is confirmed by a positive growth and behavioral response to increased nutrition.

Treatment

If there is an underlying physical reason for failure to thrive, such as a disorder of swallowing mechanism or intestinal problems, correcting that problem should reverse the condition. If the condition is caused by environmental factors, the physician will suggest several ways parents may provide adequate food for the child. Maternal education and parental counseling may also be recommended. In extreme cases, hospitalization or a more nurturing home may be necessary.

Prognosis

The first year of life is important as a foundation for growth and physical and intellectual development in the future. Children with extreme failure to thrive in the first year may never catch up to their peers even if their physical growth improves. In about one third of these extreme cases, mental development remains below normal and roughly half will continue to have psychosocial and eating problems throughout life.

When failure to thrive is identified and corrected early, most children catch up to their peers and remain healthy and well developed.

Prevention

Initial failure to thrive caused by physical defects cannot be prevented but can often be corrected before they become a danger to the child. Maternal education and emotional and economic support systems all help to prevent failure to thrive in those cases where there is no physical deformity.

KEY TERMS

Esophagus—The muscular tube which connects the mouth and stomach.

Psychosocial—A term referring to the mind's ability to, consciously or unconsciously, adjust and relate the body to its social environment.

Resources

ORGANIZATIONS

American Humane Association, Children's Division. 63 Inverness Drive East, Englewood, CO 80112-5117. (800) 227-4645. < www.americanhumane.org > .

Federation for Children With Special Needs. 1135 Tremont Street, Suite 420, Boston, MA 02120. (617) 236-7210. < <http://www.fcsn.org> > .

National Digestive Diseases Information Clearinghouse. 2 Information Way, Bethesda, MD 20892-3570. (800) 891-5389. < <http://www.niddk.nih.gov/health/digest/niddic.htm> > .

Dorothy Elinor Stonely

Fainting

Definition

Fainting is loss of consciousness caused by a temporary lack of oxygen to the brain. Known by the medical term "syncope," fainting may be preceded by **dizziness**, **nausea**, or a feeling of extreme weakness.

Description

When a person faints, the loss of consciousness is brief. The person will wake up as soon as normal blood flow is restored to the brain. Blood flow is usually restored by lying flat for a short time. This position puts the head on the same level as the heart so that blood flows more easily to the brain.

A fainting episode may be completely harmless and of no significance, but it can be a symptom of a serious underlying disorder. No matter how trivial it seems, a fainting episode should be treated as a medical emergency until the cause is determined.

Causes and symptoms

Extreme **pain**, fear, or stress may bring on fainting. This type of fainting is caused by overstimulation of the vagus nerve, a nerve connected to the brain that helps control breathing and circulation. In addition, a person who stands still or erect for too long may faint. This type of fainting occurs because blood pools in the leg veins, reducing the amount that is available for the heart to pump to the brain. This type of fainting is quite common in older people or those taking drugs to treat high blood pressure.

When an older person feels faint upon turning the head or looking upward suddenly, the cause could be **osteoarthritis** of the neck bones. Osteoarthritis damages the cartilage between the neck bones and causes pressure on blood vessels leading to the brain.

Fainting can be a symptom of a disease such as Stokes-Adams syndrome, a condition in which blood flow to the brain is temporarily reduced because of an irregular heartbeat. Some people may experience fainting associated with weakness in the limbs or a temporary problem in speaking caused by obstructed blood flow in vessels passing through the neck to the brain. Pregnant women frequently feel faint. Fainting may also occur as a result of low blood sugar. Low blood sugar can occur if a person skips a meal or has diabetes.

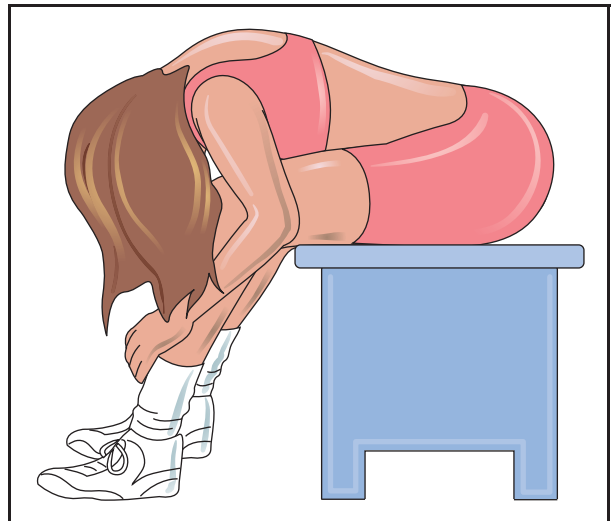
Fainting can also be caused by:

- prolonged coughing
- straining to defecate or urinate
- blowing a wind instrument too hard
- remaining in a stuffy environment with too little oxygen

Sometimes fainting may be caused by a temporary drop in the blood supply to the brain caused by a **transient ischemic attack** (TIA). A TIA, sometimes called a mini-stroke, is a disruption in the blood supply to the brain caused by a blocked or burst blood vessel. Seek help immediately if a fainting spell is followed by one or more of the symptoms listed below:

- numbness or **tingling** in any body part
- blurred vision
- confusion
- difficulty speaking
- loss of movement in arms or legs

A few seconds before fainting, a person may sweat or become pale, feel nauseated or dizzy, and have blurred vision or racing heartbeat. Once the person loses consciousness, the pupils may dilate as the heart



If a person is feeling faint, unconsciousness may be prevented by sitting with the head between the knees, as shown in the illustration above, or by lying flat with the legs raised. (Illustration by Electronic Illustrators Group.)

rate slows down. There may be abnormal movements. Muscles may tighten or the back may arch. These movements do not last long and they are not violent.

In most cases, the patient regains consciousness within a few minutes, but the fainting spell may be followed by nervousness, **headache**, nausea, dizziness, pallor or sweating. The person may faint again, especially if he or she stands up within 30 minutes.

Diagnosis

Most episodes of fainting are a one-time occurrence. When a person experiences repeated fainting spells, a physician should be consulted.

Treatment

Most of the time, a person who faints ends up lying on the floor. If this happens, the patient should be rolled onto his or her back. Because someone who faints often vomits, bystanders should keep the airway open. A person who is fainting should not be held upright or in a sitting position. These positions prevent blood flow to the brain and may bring on a seizure.

Bystanders should check the patient's breathing and pulse rate. The pulse may be weak and slow. If there are no signs of breathing or heart rate, the problem is more serious than fainting, and cardiopulmonary resuscitation (**CPR**) must begin.

If breathing and pulse rates seem normal, the person's legs should be raised above the level of the

KEY TERMS

Osteoarthritis—A disease characterized by damage to the cartilage in the joints. The joints become inflamed, deformed, and enlarged, and movement becomes painful.

Stokes-Adams syndrome—Recurrent episodes of temporary loss of consciousness (fainting) caused by an insufficient flow of blood from the heart to the brain. This syndrome is caused by a very rapid or a very slow heartbeat.

Transient ischemic attack (TIA)—A brief interruption of the blood supply to part of the brain that causes a temporary impairment of vision, speech, or movement. Usually, the episode lasts for just a few moments, but it may be a warning sign for a full-scale stroke.

Vagus nerve—A cranial nerve, that is, a nerve connected to the brain. The vagus nerve has branches to most of the major organs in the body, including the larynx, throat, windpipe, lungs, heart, and most of the digestive system.

head so that gravity can help the blood flow to the brain. Belts, collars or any other constrictive clothing should be loosened.

If the person does not regain consciousness within a minute or two after fainting, medical help should be summoned.

Prognosis

After a fainting spell, the person should regain normal color but may continue to feel weak for a short time. Lying down quietly for a few moments may help.

In most cases, an attack of fainting is not serious. As soon as the underlying pain or stress passes, the danger of repeated episodes also is eliminated.

Prevention

If a person is feeling faint, unconsciousness may be prevented by sitting with the head between the knees or lying flat with the legs raised.

A person who has fainted should lie flat for 10–15 minutes after regaining consciousness to give the system a chance to regain its balance. Standing up too soon may bring on another fainting spell.

Resources

BOOKS

Greenberg, David A., et al. *Clinical Neurology*. 2nd ed. Norwalk, CT: Appleton & Lange, 1993.

Carol A. Turkington

Falciparum malaria see **Malaria**

Fallopian tube ligation see **Tubal ligation**

Fallopian tube removal see **Salpingectomy**

Fallopian tube x rays see

Hysterosalpingography

Famciclovir see **Antiviral drugs**

Familial Mediterranean fever

Definition

Familial Mediterranean **fever** (FMF) is an inherited disorder of the inflammatory response characterized by recurring attacks of fever, accompanied by intense **pain** in the abdomen, chest, or joints. Attacks usually last 12–72 hours, and can occasionally involve a skin rash. **Kidney disease** is a serious concern if the disorder is not treated. FMF is most prevalent in people of Armenian, Sephardic-Jewish, Arabic, and Turkish ancestry.

Description

FMF could be described as a disorder of “inappropriate” inflammation. That is, an event that in a normal situation causes a mild or unnoticeable inflammation might cause a severe inflammatory response in someone with FMF. Certain areas of the body are at risk for FMF-related symptoms. A serosa is a serous (fluid-producing) membrane that can be found inside the abdominal cavity (peritoneum), around the lungs (pleura), around the heart (pericardium), and inside the joints (synovium). The symptoms of FMF are due to inflammation of one or more of the serosal membranes (serositis). Thus, FMF is also sometimes called recurrent polyserositis.

During an attack, large numbers of neutrophils, a type of white blood cell, move into the affected areas causing painful inflammation and fever. These episodes may be accompanied by a skin rash or joint pain. In a few cases, chronic arthritis is a problem. **Amyloidosis** is a potentially serious condition in which proteins called amyloids are mistakenly produced and

deposited in organs and tissues throughout the body. Left untreated, amyloidosis often leads to kidney failure, which is the major long-term health risk in FMF.

In most cases, the attacks of fever and pain are first noticed in childhood or adolescence. The interval between these episodes may be days or months, and is not predictable. However, during these intervals people with FMF typically lead normal lives. It is not entirely clear what brings on an attack, but people with FMF often report mild physical trauma, physical exertion, or emotional **stress** just prior to the onset of symptoms. Treatment for FMF involves an oral medication called colchicine, which is highly effective for the episodes of fever and pain, as well as for amyloidosis and the kidney disease that can result from it.

FMF is most common in certain ethnic groups from the eastern Mediterranean region, but cases in other ethnic groups in other parts of the world are increasingly being reported. FMF is also known by many other names. They include: recurrent hereditary polyserositis, benign paroxysmal **peritonitis**, familial paroxysmal polyserositis, paroxysmal polyserositis, familial recurrent polyserositis, periodic fever, periodic amyloid syndrome, periodic peritonitis syndrome, Reimann periodic disease, Reimann syndrome, Siegel-Cattan-Mamou syndrome, and Armenian syndrome.

Estimates of the incidence of FMF in specific eastern Mediterranean populations range from 1 in 2000 to 1 in 100, depending on the population studied. Specific mutations in the MEFV gene are more common in certain ethnic groups, and may cause a somewhat different course of the disease. A few mutations in the MEFV gene likely became common in a small population in the eastern Mediterranean several thousand years ago. It is postulated that carrying a single copy of a mutated gene produced a modified (but not abnormal) inflammatory response that may have been protective against some infectious agent at that time. Those who carried a single “beneficial” mutation in the MEFV gene were more likely to survive and reproduce, which may explain the high carrier frequency (up to one in five) in some populations. People of Armenian, Sephardic-Jewish, Arabic, and Turkish ancestry are at greatest risk for FMF. However, a better understanding and recognition of the symptoms of FMF in recent years has resulted in more reports of the condition in other ethnic groups, such as Italians and Armenian-Americans.

Causes and symptoms

FMF is a genetic condition inherited in an autosomal recessive fashion. Mutations in the MEFV gene

(short for Mediterranean Fever) on chromosome number 16 are the underlying cause of FMF. Autosomal recessive inheritance implies that a person with FMF has mutations in both copies of the MEFV gene. All genes come in pairs, and one copy of each pair is inherited from each parent. If neither parent of a child with FMF has the condition, it means they carry one mutated copy of the MEFV gene, but also one normal copy, which is enough to protect them from disease. If both parents carry the same autosomal recessive gene, there is a one in four chance in each **pregnancy** that the child will inherit both recessive genes, and thus have the condition.

The MEFV gene carries the instructions for production of a protein called pyrin, named for pyrexia, a medical term for fever. The research group in France that co-discovered the protein named it marenostriin, after ancient Latin words that referred to the Mediterranean Sea. The movement of neutrophils into an area of the body where trauma or infection has occurred is the major cause of inflammation, which is a normal process. Research has shown that pyrin has some function in controlling neutrophils. In a situation where minor trauma or stress occurs, some initial inflammation may follow, but a functional pyrin protein is responsible for shutting-down the response of neutrophils once they are no longer needed. An abnormal pyrin protein associated with FMF may be partly functional, but unstable. In some instances, the abnormal pyrin itself seems to be “stressed”, and loses its ability to regulate neutrophils and inflammation. Left unregulated, a normal, mild inflammation spirals out of control. Exactly what causes pyrin in FMF to lose its ability to control neutrophils in some situations is not known.

The recurrent acute attacks of FMF typically begin in childhood or adolescence. Episodes of fever and painful inflammation usually last 12–72 hours. About 90% of people with FMF have their first attack by age 20. The group of symptoms that characterizes FMF includes the following:

Fever

An FMF attack is nearly always accompanied by a fever, but it may not be noticed in every case. Fevers are typically 100–104 °F (38–40 °C). Some people experience chills prior to the onset of fever.

Abdominal pain

Nearly all people with FMF experience abdominal pain at one point or another, and for most it is the most common complaint. The pain can range from

mild to severe, and can be diffuse or localized. It can mimic **appendicitis**, and many people with undiagnosed FMF have had appendectomies or exploratory surgery of the abdomen done, only to have the fever and abdominal pain return.

Chest pain

Pleuritis, also called **pleurisy**, occurs in up to half of the affected individuals in certain ethnic groups. The pain is usually on one side of the chest. Pericarditis would also be felt as chest pain.

Joint pain

About 50% of people with FMF experience joint pain during attacks. The pain is usually confined to one joint at a time, and often involves the hip, knee, or ankle. For some people, however, the recurrent joint pain becomes chronic arthritis.

Myalgia

Up to 20% of individuals report muscle pain. These episodes typically last less than two days, and tend to occur in the evening or after physical exertion. Rare cases of muscle pain and fever lasting up to one month have been reported.

Skin rash

A rash, described as erysipelas-like erythema, accompanies attacks in a minority of people, and most often occurs on the front of the lower leg or top of the foot. The rash appears as a red, warm, swollen area about 4–6 in (10–15 cm) in diameter.

Amyloidosis

FMF is associated with high levels in the blood of a protein called serum amyloid A (SAA). Over time, excess SAA tends to be deposited in tissues and organs throughout the body. The presence and deposition of excess SAA is known as amyloidosis. Amyloidosis may affect the gastrointestinal tract, liver, spleen, heart, and testes, but effects on the kidneys are of greatest concern. The frequency of amyloidosis varies among the different ethnic groups, and its overall incidence is difficult to determine because of the use of colchicine to avert the problem. Left untreated, however, those individuals who do develop amyloidosis of the kidneys may require a renal transplant, or may even die of renal failure. The frequency and severity of a person's attacks of fever and serositis seem to have no relation to whether they will develop amyloidosis. In fact, a few people with FMF have been

described who have had amyloidosis but apparently no other FMF-related symptoms.

Other symptoms

A small percentage of boys with FMF develop painful inflammation around the testes, headaches are a common occurrence during attacks, and certain types of **vasculitis** (inflammation of the blood vessels) seem to be more common in FMF.

Diagnosis

Individually, the symptoms that define FMF are common. Fevers occur for many reasons, and nonspecific pains in the abdomen, chest, and joints are also frequent ailments. Several infections can result in symptoms similar to FMF (Mallaret **meningitis**, for instance), and many people with FMF undergo exploratory abdominal surgery and ineffective treatments before they are finally diagnosed. Membership in a less commonly affected ethnic group may delay or hinder the correct diagnosis.

In general, symptoms involving one or more of the following broad groups should lead to suspicion of FMF: Unexplained recurrent fevers, polyserositis, skin rash, and/or joint pain; abnormal blood studies (see below); and renal or other disease associated with amyloidosis. A family history of FMF or its symptoms would obviously be an important clue, but the recessive nature of FMF means there usually is no family history. The diagnosis may be confirmed when a person with unexplained fever and pain responds to treatment with colchicine since colchicine is not known to have a beneficial effect on any other condition similar to FMF. Abnormal results on a blood test typically include **leukocytosis** (elevated number of neutrophils in the blood), an increased erythrocyte sedimentation rate (rate at which red blood cells form a sediment in a blood sample), and increased levels of proteins associated with inflammation (called acute phase reactants) such as SAA.

Direct analysis of the MEFV gene for FMF mutations is the only method to be certain of the diagnosis. However, it is not yet possible to detect all MEFV gene mutations that might cause FMF. Thus, if DNA analysis is negative, clinical methods must be relied upon. If both members of a couple were proven to be FMF carriers through genetic testing, highly accurate prenatal diagnosis would be available in any subsequent pregnancy.

Similar syndromes of periodic fever and inflammation include familial Hibernian fever and

hyperimmunoglobulinemia D syndrome, but both are more rare than FMF.

Treatment

Colchicine is a chemical compound that can be used as a medication, and is frequently prescribed for **gout**. Some years ago, colchicine was discovered to also be effective in reducing the frequency and severity of attacks in FMF. Treatment for FMF at this point consists of taking colchicine daily. Studies have shown that about 75% of FMF patients achieve complete remission of their symptoms, and about 95% show marked improvement when taking colchicine. Lower effectiveness has been reported, but there is some question about the number of FMF patients who choose not to take their colchicine between attacks when they are feeling well, and thus lose some of the ability to prevent attacks. Compliance with taking colchicine every day may be hampered by its side effects, which include **diarrhea**, **nausea**, abdominal bloating, and gas. There is a theoretical risk that colchicine use could damage chromosomes in sperms and eggs, or in an embryo during pregnancy, or that it might reduce fertility. However, studies looking at reproduction in men and women who have used colchicine have so far not shown any increased risks. Colchicine is also effective in preventing, delaying, or reversing renal disease associated with amyloidosis.

Other medications may be used as needed to deal with the pain and fever associated with FMF attacks. Dialysis and/or renal transplant might become necessary in someone with advanced kidney disease. Given its genetic nature, there is no cure for FMF, nor is there likely to be in the near future. Any couple that has a child diagnosed with FMF, or anyone with a family history of the condition (especially those in high-risk ethnic groups), should be offered genetic counseling to obtain the most up-to-date information on FMF and testing options.

Prognosis

For those individuals who are diagnosed early enough and take colchicine consistently, the prognosis is excellent. Most will have very few, if any, attacks of fever and polyserositis, and will likely not develop serious complications of amyloidosis. The problem of misdiagnosing FMF continues, but education attempts directed at both the public and medical care providers should improve the situation. Future research should provide a better understanding of the inflammation process, focusing on how neutrophils are genetically regulated. That information

KEY TERMS

Acute phase reactants—Blood proteins whose concentrations increase or decrease in reaction to the inflammation process.

Amyloid—A waxy translucent substance composed mostly of protein, that forms plaques (abnormal deposits) in the brain.

Amyloidosis—Accumulation of amyloid deposits in various organs and tissues in the body such that normal functioning of an organ is compromised.

Colchicine—A compound that blocks the assembly of microtubules—protein fibers necessary for cell division and some kinds of cell movements, including neutrophil migration. Side effects may include diarrhea, abdominal bloating, and gas.

Leukocyte—A white blood cell. The neutrophils are a type of leukocyte.

Leukocytosis—An increase in the number of leukocytes in the blood.

Neutrophil—The primary type of white blood cell involved in inflammation. Neutrophils are a type of granulocyte, also known as a polymorphonuclear leukocyte.

Pericarditis—Inflammation of the pericardium, the membrane surrounding the heart.

Peritonitis—Inflammation of the peritoneum, the membrane surrounding the abdominal contents.

Pleuritis—Inflammation of the pleura, the membrane surrounding the lungs.

Pyrexia—A medical term denoting fevers.

Serositis—Inflammation of a serosal membrane. Polyserositis refers to the inflammation of two or more serosal membranes.

Synovitis—Inflammation of the synovium, a membrane found inside joints.

could then be used to develop treatments for FMF with fewer side effects, and might also assist in developing therapies for other diseases in which abnormal inflammation and immune response are a problem.

Resources

ORGANIZATIONS

National Institute of Arthritis and Musculoskeletal and Skin Diseases. National Institutes of Health, One AMS Circle, Bethesda, MD 20892. <<http://www.nih.gov/niams>> .

National Organization for Rare Disorders (NORD). PO Box 8923, New Fairfield, CT 06812-8923. (203) 746-6518 or (800) 999-6673. Fax: (203) 746-6481. < <http://www.rarediseases.org> > .

National Society of Genetic Counselors. 233 Canterbury Dr., Wallingford, PA 19086-6617. (610) 872-1192. < <http://www.nsgc.org/GeneticCounselingYou.asp> > .

Scott J. Polzin, M.S.

Familial polypsis

Definition

Familial polyposis is an inherited condition which primarily affects the large intestine (colon and rectum). Large numbers of projecting masses of swollen and thickened or tumorous membrane (polyps) develop on the inner lining of this part of the bowel. The polyps eventually become malignant.

Description

Familial polyposis (FP) is known by many synonyms, most include some combination of words which reflect what is known about the disease. As the disease is inherited, the word, family, is often included. Because these mushroom-like growths are the most obvious manifestation of the disorder, the word, polyp, is usually in the term as well. Adenoma refers to the particular kind of polyp that is typically discovered. Some of the names found in medical texts and journals include polyposis coli, familial colonic polyposis, multiple familial polyposis, familial adenomatous colon polyposis, adenomatosis of the colon and rectum (ACR), and familial adenomatous polyposis (FAP). The last term and its abbreviation have been commonly used since the early 1990s. It will be used in this discussion.

Familial polyposis or familial adenomatous polyposis (FAP) is a premalignant disease. This means that a person with FAP, if left untreated, will invariably develop **cancer**. Individuals with this disorder grow hundreds of polyps throughout their large intestines. The polyps, which may also be called adenomas, commonly develop just after **puberty**. Approximately half of all FAP patients will have polyps by age 14. Ninety percent will have detectable polyps by age 25. Usually by age 35–40, one or more of these polyps will become cancerous.

FAP is a rare disease. One in 8,000 people in the United States have FAP. However, it may be very common in affected families. FAP is inherited in an autosomal dominant pattern. This means that a person with FAP has a 50% chance of passing the condition down to each of their children. FAP can also develop in someone with no family history of the disorder, due to a new genetic mutation in that individual. It is thought that approximately one percent of all colorectal cancers in the United States can be attributed to FAP.

Causes and symptoms

FAP is caused by a portion of a gene that mutates or changes. The original cause of the mutation is unknown. Its exact role in FAP is not completely clear. Researchers theorize that the normal gene directs the manufacture of a protein which helps control cell growth. The mutated gene section in FAP generates an abnormal protein which does not perform its normal function. Cells grow out of control, causing the development of multiple, sometimes hundreds, of polyps. One or more of these eventually becomes cancerous.

Many individuals develop polyps without displaying any symptoms. Others experience such gastrointestinal problems as **diarrhea**, **constipation**, abdominal cramps, blood in the stool, or weight loss. FAP patients may also develop nonmalignant tumors (desmoid tumors), and/or some bone and dental abnormalities. In addition, they may exhibit a “spot” on the retina of the eye (congenital hypertrophy of the retinal pigment epithelium, or CHRPE).

Relatives of individuals with diagnosed FAP are at high risk of having the disease themselves. There are no other known risk factors for this condition.

Diagnosis

The abnormal portion of the gene that causes FAP in most patients can be detected. A blood test can then be performed which identifies family members who have the same mutation. They will eventually develop the condition. Children who have a parent with FAP, and siblings of affected patients whose parental history is incomplete, should be evaluated. The polyps characteristic of FAP have been found in children as young as age five. Testing of appropriate individuals should take place as soon as the diagnosis of FAP is established in one member of a family.

Relatives of people with diagnosed FAP should exercise caution regarding where they seek advice and

testing. One study of a commercially available blood test found that less than 20% of patients received any genetic counseling, and almost one third of their physicians misinterpreted the test results.

Registries for FAP patients can be found at many sites in the United States. Such a registry specializes in identification, assistance, and education of people with a particular disease, and is usually a separate department in a research hospital. A team of health professionals who have expertise in the disorder staff the registry.

Testing within a research setting and/or at a facility with a registry of patients with FAP is more likely to safeguard against problems, such as the misunderstanding of test results. As part of a research project, sometimes counseling as well as blood tests are available at no charge to the patient. Insurance coverage varies. Concerns about confidentiality, and future insurance and employment discrimination, may prompt individuals to pay for the examination out of pocket. Commercial blood tests cost approximately \$250 per sample.

If the abnormal gene is found in a family member, annual screening for colon polyps is recommended, beginning at age 11. Flexible **sigmoidoscopy** is used for this examination. It is usually done in a physician's office, or in a hospital department, most often by a gastroenterologist or a surgeon. Food intake may be restricted for 24 hours prior to the procedure. Before the study, the intestine is cleared of stool by one or more small **enemas**. Some physicians prefer to sedate the patient, to help them relax. Then a flexible, lighted, hollow tube (sigmoidoscope) is inserted into the anus and maneuvered into the large intestine. The physician examines the wall of the colon to look for polyps. If polyps are found, one or more may be removed for biopsy.

Most patients report little discomfort during the examination. The procedure itself takes five to fifteen minutes. The patient may be at the facility an hour, or more, if recovery from sedation is needed. If no medication was administered, driving and resumption of normal activities are permitted immediately. The cost of the procedure varies widely, but, as of 1997, it was covered by Medicare, indicating the likelihood of other types of insurance coverage.

In some cases the portion of the gene responsible for FAP cannot be identified. Family members of these patients cannot have a predictive blood test. The current recommendation is for these patients to have the same annual examination with flexible sigmoidoscopy as patients with a diagnosed FAP gene. A noninvasive

screening **eye examination** to detect CHRPE, associated with FAP, may also be performed.

Treatment

The only definitive treatment for FAP is surgical removal of the lower intestine. Since the goal is to prevent cancer, the operation is done as soon as adenomatous polyps are found on sigmoidoscopy. Waiting until a polyp becomes malignant is unsafe, as the cancer may invade surrounding tissues.

There are several choices about the type of surgery to treat this condition. Some authorities advocate removal of the colon, leaving the rectum or lowest portion of the intestine in place. The small intestine can be attached to the rectum, allowing normal bowel function. This is often called ileorectal anastomosis. Others argue that this section is also liable to develop polyps, needs to be monitored regularly, and may require eventual removal.

Excision of the entire lower intestine with preservation of normal bowel function is possible. This entails a more complex surgical procedure. The patient may experience more complications and a longer recovery period. However, the risk of polyp development in this area is very low. Periodic examination of the intestine may not be needed once healing is complete.

The more intricate surgery may be referred to as a J-pouch procedure, an ileal pouch-anal anastomosis, a restorative proctocolectomy, or an ileoanal reservoir procedure. It involves creating a "pouch" of tissue from the small intestine, which is attached to the anus. This serves as a reservoir or holding area for stool, much as the rectum does normally. The surgery is often done in several stages. A temporary ileostomy, which creates an opening of the small intestines onto the abdomen, is required. When all procedures are completed, and after a recuperation period, the patient regains normal bowel function through the anus.

Some researchers suggest that as **genetic testing** becomes more developed, the specific portion of the gene involved may dictate the type of surgery chosen. Those at high risk of developing **rectal polyps** may be advised to have the more complex operation. FAP patients felt to be at lower risk for rectal polyps might be counseled to consider the less radical surgery.

Medical therapy to treat the adenomatous polyps has been attempted. Some **nonsteroidal anti-inflammatory drugs** have been effective in reducing the number and size of the polyps. It is possible that these agents will be used as an additional treatment for FAP, but they are unlikely to replace surgery.

KEY TERMS

Gene—The basic unit of heredity, made of DNA. Each gene occupies certain location on a chromosome.

Mutation—An alteration in a gene, especially one capable of producing a new trait, or a change in function.

Individuals with FAP are at increased risk for cancers of the upper digestive tract including the upper portion of the small bowel (duodenum) and the channels where bile flows (biliary tract). Cancers of the thyroid, pancreas, and adrenal gland are also more commonly found among FAP patients. Periodic examination for the development of malignancy in these areas is considered part of the treatment of FAP. In some cases, such as cancer involving the duodenum, the tests themselves carry a chance of complications. The risk of the study must be weighed against the potential benefits of knowing the results. Nonmalignant growths, called desmoid tumors, also occur more frequently in patients with FAP. Although they are not malignant, they grow quickly into surrounding tissues, causing many difficulties, even **death** in some cases.

Prognosis

The major cause of death in many patients with FAP remains colorectal cancer. One study suggested that even with improved disease recognition, social and emotional factors, such as fear of surgery, may significantly delay a patient's treatment. In recent years, the trend is towards mortality from other causes, such as desmoid tumors or cancers other than colorectal. It has been estimated that a patient with known FAP has a relative risk of dying over three times greater than that of the average population, at a given age.

Prevention

FAP cannot be prevented. Aggressive diagnosis, treatment, and follow-up monitoring are keys to successful management of the disease.

Resources

ORGANIZATIONS

Familial Polyposis Registry. Department of Colorectal Surgery. Cleveland Clinic Foundation. 9500 Euclid Ave., Cleveland OH 44195-5001. (216) 444-6470.

National Organization for Rare Disorders. PO Box 8923, New Fairfield, CT 06812-8923. (800) 999-6673. < <http://www.rarediseases.org> > .

Ellen S. Weber, MSN

Family therapy

Definition

Family therapy is a form of psychotherapy that involves all the members of a nuclear or extended family. It may be conducted by a pair or team of therapists. In many cases the team consists of a man and a woman in order to treat gender-related issues or serve as role models for family members. Although some forms of family therapy are based on behavioral or psychodynamic principles, the most widespread form is based on family systems theory. This approach regards the family, as a whole, as the unit of treatment, and emphasizes such factors as relationships and communication patterns rather than traits or symptoms in individual members.

Family therapy is a relatively recent development in psychotherapy. It began shortly after World War II, when doctors, who were treating schizophrenic patients, noticed that the patients' families communicated in disturbed ways. The doctors also found that the patients' symptoms rose or fell according to the level of tension between their parents. These observations led to considering a family as an organism or system with its own internal rules, patterns of functioning, and tendency to resist change. The therapists started to treat the families of schizophrenic patients as whole units rather than focusing on the hospitalized member. They found that in many cases the family member with **schizophrenia** improved when the "patient" was the family system. (This should not be misunderstood to mean that schizophrenia is caused by family problems, although family problems may worsen the condition.) This approach of involving the entire family in the treatment plan and therapy was then applied to families with problems other than the presence of schizophrenia.

Family therapy is becoming an increasingly common form of treatment as changes in American society are reflected in family structures. It has led to two further developments: couples therapy, which treats relationship problems between marriage partners or gay couples; and the extension of family therapy to religious communities or other groups that resemble families.

Purpose

Family therapy is often recommended in the following situations:

- Treatment of a family member with schizophrenia or multiple personality disorder (MPD). Family therapy helps other family members understand their relative's disorder and adjust to the psychological changes that may be occurring in the relative.
- Families with problems across generational boundaries. These would include problems caused by parents sharing housing with grandparents, or children being reared by grandparents.
- Families that deviate from social norms (common-law relationships, gay couples rearing children, etc.). These families may not have internal problems but may be troubled by outsiders' judgmental attitudes.
- Families with members from a mixture of racial, cultural, or religious backgrounds.
- Families who are scapegoating a member or undermining the treatment of a member in individual therapy.
- Families where the identified patient's problems seem inextricably tied to problems with other family members.
- Blended families with adjustment difficulties.

Most family therapists presuppose an average level of intelligence and education on the part of adult members of the family.

Precautions

Some families are not considered suitable candidates for family therapy. They include:

- families in which one, or both, of the parents is psychotic or has been diagnosed with antisocial or paranoid personality disorder,
- families whose cultural or religious values are opposed to, or suspicious of, psychotherapy,
- families with members who cannot participate in treatment sessions because of physical illness or similar limitations,
- families with members with very rigid personality structures. (Here, members might be at risk for an emotional or psychological crisis),
- families whose members cannot or will not be able to meet regularly for treatment,
- families that are unstable or on the verge of breakup.

Description

Family therapy tends to be short-term treatment, usually several months in length, with a focus on resolving specific problems such as eating disorders, difficulties with school, or adjustments to **bereavement** or geographical relocation. It is not normally used for long-term or intensive restructuring of severely dysfunctional families.

In family therapy sessions, all members of the family and both therapists (if there is more than one) are present at most sessions. The therapists seek to analyze the process of family interaction and communication as a whole; they do not take sides with specific members. They may make occasional comments or remarks intended to help family members become more conscious of patterns or structures that had been previously taken for granted. Family therapists, who work as a team, also model new behaviors for the family through their interactions with each other during sessions.

Family therapy is based on family systems theory, which understands the family to be a living organism that is more than the sum of its individual members. Family therapy uses "systems" theory to evaluate family members in terms of their position or role within the system as a whole. Problems are treated by changing the way the system works rather than trying to "fix" a specific member. Family systems theory is based on several major concepts:

The identified patient

The identified patient (IP) is the family member with the symptom that has brought the family into treatment. The concept of the IP is used by family therapists to keep the family from scapegoating the IP or using him or her as a way of avoiding problems in the rest of the system.

Homeostasis (balance)

The concept of homeostasis means that the family system seeks to maintain its customary organization and functioning over time. It tends to resist change. The family therapist can use the concept of homeostasis to explain why a certain family symptom has surfaced at a given time, why a specific member has become the IP, and what is likely to happen when the family begins to change.

The extended family field

The extended family field refers to the nuclear family, plus the network of grandparents and other

members of the extended family. This concept is used to explain the intergenerational transmission of attitudes, problems, behaviors, and other issues.

Differentiation

Differentiation refers to the ability of each family member to maintain his or her own sense of self, while remaining emotionally connected to the family. One mark of a healthy family is its capacity to allow members to differentiate, while family members still feel that they are “members in good standing” of the family.

Triangular relationships

Family systems theory maintains that emotional relationships in families are usually triangular. Whenever any two persons in the family system have problems with each other, they will “triangle in” a third member as a way of stabilizing their own relationship. The triangles in a family system usually interlock in a way that maintains family homeostasis. Common family triangles include a child and its parents; two children and one parent; a parent, a child, and a grandparent; three siblings; or, husband, wife, and an in-law.

Preparation

In some instances the family may have been referred to a specialist in family therapy by their pediatrician or other primary care provider. It is estimated that as many as 50% of office visits to pediatricians have to do with developmental problems in children that are affecting their families. Some family doctors use symptom checklists or psychological screeners to assess a family’s need for therapy.

Family therapists may be either psychiatrists, clinical psychologists, or other professionals certified by a specialty board in marriage and family therapy. They will usually evaluate a family for treatment by scheduling a series of interviews with the members of the immediate family, including young children, and significant or symptomatic members of the extended family. This process allows the therapist(s) to find out how each member of the family sees the problem, as well as to form first impressions of the family’s functioning. Family therapists typically look for the level and types of emotions expressed, patterns of dominance and submission, the roles played by family members, communication styles, and the locations of emotional triangles. They will

KEY TERMS

Blended family—A family formed by the remarriage of a divorced or widowed parent. It includes the new husband and wife, plus some or all of their children from previous marriages.

Differentiation—The ability to retain one’s identity within a family system while maintaining emotional connections with the other members.

Extended family field—A person’s family of origin plus grandparents, in-laws, and other relatives.

Family systems theory—An approach to treatment that emphasizes the interdependency of family members rather than focusing on individuals in isolation from the family. This theory underlies the most influential forms of contemporary family therapy.

Genogram—A family tree diagram that represents the names, birth order, sex, and relationships of the members of a family. Therapists use genograms to detect recurrent patterns in the family history and to help the members understand their problem(s).

Homeostasis—The tendency of a family system to maintain internal stability and resist change.

Identified patient (IP)—The family member in whom the family’s symptom has emerged or is most obvious.

Nuclear family—The basic family unit, consisting of father, mother, and their biological children.

Triangling—A process in which two family members lower the tension level between them by drawing in a third member.

also note whether these patterns are rigid or relatively flexible.

Preparation also usually includes drawing a genogram, which is a diagram that depicts significant persons and events in the family’s history. Genograms also include annotations about the medical history and major personality traits of each member. Genograms help in uncovering intergenerational patterns of behavior, marriage choices, family alliances and conflicts, the existence of family secrets, and other information that sheds light on the family’s present situation.

Risks

The chief risk in family therapy is the possible unsettling of rigid personality defenses in individuals, or couple relationships that had been fragile before the

beginning of therapy. Intensive family therapy may also be difficult for psychotic family members.

Normal results

Normal results vary, but in good circumstances, they include greater insight, increased differentiation of individual family members, improved communication within the family, loosening of previously automatic behavior patterns, and resolution of the problem that led the family to seek treatment.

Resources

BOOKS

Clark, R. Barkley. "Psychosocial Aspects of Pediatrics & Psychiatric Disorders: Psychosocial Assessment of Children & Families." In *Current Pediatric Diagnosis & Treatment*, edited by William W. Hay Jr., et al. Stamford: Appleton & Lange, 1997.

Rebecca J. Frey, PhD

Famine fever see **Relapsing fever**

Fanconi's syndrome

Definition

Fanconi's syndrome is a set of kidney malfunctions brought about by a variety of seemingly unrelated disorders. Kidney malfunction leads to excessive urine production and excessive thirst, resulting in deficits of water, calcium, potassium, magnesium, and other substances in the body. It often leads to bone disease and stunted growth.

Description

Normally, kidneys cleanse the blood and keep its salt, water, and acidity in balance, leaving what the body needs in the blood and putting what the body doesn't need into the urine, which leaves the body. This task is performed in two steps. First, the blood is filtered through a kidney structure with small holes that keep the cells and large molecules in the blood. Second, some of the small molecules in the filtrate, needed by the body, are reabsorbed and returned to the bloodstream.

This reabsorption step is defective in Fanconi's syndrome. As a consequence, substances that are normally reabsorbed, like glucose, amino acids, small proteins, water, calcium, potassium, magnesium,

bicarbonate, and phosphate, are lost and the body becomes overly acidic.

Fanconi's syndrome is also known as Fanconi syndrome, renal Fanconi syndrome, Fanconi renal tubular syndrome, and Lignac-de Toni-Debré-Fanconi syndrome. Fanconi's anemia is, however, a totally different disease.

Causes and symptoms

Causes

Fanconi's syndrome can be caused by a variety of genetic defects and by certain environmental assaults.

The genetic diseases known to give rise to Fanconi's syndrome are cystinosis (the most common cause in children), **galactosemia**, glycogen storage disease, **hereditary fructose intolerance**, Lowe syndrome, Wilson disease, tyrosinemia, medullary cystic disease, vitamin D dependency, and familial idiopathic Fanconi's syndrome.

Environmental assaults that cause Fanconi's syndrome include exposure to heavy metals (like cadmium, lead, mercury, platinum, uranium), certain drugs (like outdated tetracycline and gentamicin), other substances (like Lysol, paraquat, toluene, the amino acid lysine taken as a nutritional supplement), and **kidney transplantation**.

Symptoms

Fanconi's syndrome symptoms related directly to impaired absorption include excessive urine production and urination; excessive thirst; dehydration; constipation; anorexia nervosa; vomiting; elevated levels of glucose, phosphate, calcium, uric acid, amino acids, and protein (especially beta₂-microglobulin and lysozyme) in the urine; elevated levels of chloride and decreased levels of phosphate and calcium in the blood; and excessively acidic blood.

The most noticeable indirect consequences of impaired reabsorption are the bone diseases, rickets and osteomalacia. **Rickets** affects children and is associated with bone deformities, failure to grow, and difficulty walking. If a person acquires Fanconi's syndrome as an adult, the bone disease is termed osteomalacia and is accompanied by severe bone **pain** and spontaneous **fractures**. Unlike rickets due to **malnutrition**, these diseases cannot be reversed with vitamin D. Muscle weakness and occasional **paralysis** are other indirect consequences of the ineffective reabsorption.

Diagnosis

Diagnosis of Fanconi's syndrome can be made by urine and blood tests. It is also important to find the underlying cause to decide on the best treatment. Other symptoms specific to a particular patient will point to other useful diagnostic tests. For example, high levels of blood galactose in conjunction with symptoms of Fanconi's syndrome indicate the patient is suffering from galactosemia, while high blood levels of cadmium indicate the patient is suffering from cadmium **poisoning**.

Treatment

Fanconi's syndrome is best treated by attacking the underlying cause whenever possible. For example, when cystinosis is treated with the drug cysteamine to lower cystine levels in the body or **Wilson disease** is treated with penicillamine to lower the levels of copper, accompanying symptoms of Fanconi's syndrome will subside. If the patient has acquired the disease from a heavy metal or another toxic agent, all contact with the toxic agent should stop; the condition will then likely disappear.

Nevertheless, additional treatment will be necessary either when it's not possible to treat the underlying cause or while waiting for the kidneys to resume normal function. This is done by restricting sodium chloride (table salt), giving **antacids** to counteract the excessive acidity of the blood, and supplying potassium supplements.

Kidney transplant is the treatment of last resort, used for patients whose kidneys have failed.

Prognosis

Fanconi's syndrome can be reversible. Fanconi's syndrome caused by kidney transplantation usually reverses itself within the first year after transplant surgery. When caused by a toxin in the environment, Fanconi's syndrome generally can be reversed by removing the causative agent from the patient's environment. If it is caused by a genetic disease, it can usually be reversed by treating the disease. However, if Fanconi's syndrome is not treated or if treatment is unsuccessful, the kidneys can fail.

Prevention

Fanconi's syndrome caused secondarily by the genetic diseases galactosemia, glycogen storage disease, hereditary fructose intolerance, and tyrosinemia is prevented by appropriate dietary restrictions to treat the genetic disease, starting in infancy.

KEY TERMS

Acidosis—Condition where the body is more acidic than normal; associated with headache, nausea, vomiting, and visual disturbances.

Fanconi's anemia—An inherited form of aplastic anemia.

Filtrate—The part of filtered material that flows through the filter.

Idiopathic—Refers to a disease of unknown cause.

Polydipsia—Excessive thirst.

Polyuria—Excessive production of urine.

Fanconi's syndrome caused by heavy metals and other toxins can be prevented by avoiding these substances.

Resources

ORGANIZATIONS

The American Society of Nephrology. 2025 M Street NW #800, Washington, DC 20036. (202) 367-1190. <<http://www.asn-online.com>> .

National Kidney Foundation. 30 East 33rd St., New York, NY 10016. (800) 622-9010. <<http://www.kidney.org>> .

OTHER

"Online Mendelian Inheritance in Man." *OMIM Homepage*, <<http://www.ncbi.nlm.nih.gov/Omim>> .

Lorraine Lica, PhD

Farsightedness see **Hyperopia**

FAS see **Fetal alcohol syndrome**

Fasciotomy

Definition

Fasciotomy is a surgical procedure that cuts away the fascia to relieve tension or pressure.

Purpose

The fascia is thin connective tissue covering, or separating, the muscles and internal organs of the body. It varies in thickness, density, elasticity, and composition, and is different from ligaments and tendons.

The fascia can be injured either through constant strain or through trauma. Fasciitis is an inflammation of the fascia. The most common condition for which fasciotomy is performed is plantar fasciitis, an inflammation of the fascia on the bottom of the foot that is sometimes called a heel spur or stone bruise.

Plantar fasciitis is caused by long periods on the feet, being overweight, and wearing shoes that do not support the foot well. Teachers, mail carriers, runners, and others who make heavy use of their feet are especially likely to suffer from plantar fasciitis.

Plantar fasciitis results in moderate to disabling heel **pain**. If nine to twelve months of conservative treatment (reducing time on feet, non-steroid anti-inflammatory drugs, arch supports) under the supervision of a doctor does not result in pain relief, a fasciotomy may be done. Fasciotomy removes a small portion of the fascia to relieve tension and pain. Connective tissue grows back into the cut space left by the cut, effectively lengthening the fascia.

When a fasciotomy is performed on other parts of the body, it is usually done to relieve pressure from a compression injury to a limb. This type of injury often occurs during contact sports. The blood vessels of the limb are damaged. They swell and leak, causing inflammation. Fluid builds up in the area contained by the fascia. A fasciotomy is done to relieve this pressure and prevent tissue death. Similar injury occurs in high voltage electrical **burns** where deep tissue damage occurs.

Precautions

In the case of injury, fasciotomy is done on an emergency basis, and the outcome of the surgery depends largely on the general health of the patient. Plantar fasciotomies are appropriate for most people whose foot problems cannot be resolved in any other way.

Description

Fasciotomy in the limbs is usually done by a surgeon under general or regional anesthesia. An incision is made in the skin, and a small area of fascia is removed where it will best relieve pressure. Then the incision is closed.

Plantar fasciotomy is an endoscopic (performed with the use of an endoscope) procedure. It is done by a foot specialist in a doctor's office or outpatient surgical clinic under local anesthesia and takes

KEY TERMS

Endoscope—A tube that contains a tiny camera and light, that is inserted in the body to allow a doctor to see inside without making a large incision.

20 minutes to one hour. The doctor makes two small incisions on either side of the heel. An endoscope is inserted in one to guide the doctor in where to cut. A tiny knife is inserted in the other. A portion of the fascia is cut from near the heel; then the incisions are closed.

Preparation

Little preparation is done before a fasciotomy. When the fasciotomy is related to burn injuries, the fluid and electrolyte status of the patient are constantly monitored.

Aftercare

Aftercare depends on the reason for the fasciotomy. People who have endoscopic plantar fasciotomy can walk without pain almost immediately, return to wearing their regular shoes within three to five days, and return to normal activities within three weeks. Most will need to wear arch supports in their shoes.

Risks

In endoscopic plantar fasciotomy, the greatest risk is that the arch will drop slightly as a result of this surgery, causing other foot problems. Risks involved with other types of fasciotomy are those associated with the administration of anesthesia and the development of blood clots.

Normal results

Fasciotomy in the limbs reduces pressure, thus reducing tissue death. Endoscopic plantar fasciotomy has a success rate of 90–95%.

Resources

OTHER

“New Treatments for Heel Spur Syndrome.” <<http://www.footspecialist.com/heelspur.html>>.

Tish Davidson, A.M.

Fasting

Definition

Fasting is voluntarily not eating food for varying lengths of time. Fasting is used as a medical therapy for many conditions. It is also a spiritual practice in many religions.

Purpose

Fasting can be used for nearly every chronic condition, including **allergies**, **anxiety**, arthritis, **asthma**, depression, diabetes, headaches, heart disease, high cholesterol, low blood sugar, digestive disorders, mental illness, and **obesity**. Fasting is an effective and safe weight loss method. It is frequently prescribed as a **detoxification** treatment for those with conditions that may be influenced by environmental factors, such as **cancer** and multiple chemical sensitivity. Fasting has been used successfully to help treat people who have been exposed to high levels of toxic materials due to accident or occupation. Fasting is thought to be beneficial as a preventative measure to increase overall health, vitality, and resistance to disease. Fasting is also used as a method of mental and spiritual rejuvenation.

Description

Origins

Used for thousands of years, fasting is one of the oldest therapies in medicine. Many of the great doctors of ancient times and many of the oldest healing systems have recommended it as an integral method of healing and prevention. Hippocrates, the father of Western medicine, believed fasting enabled the body to heal itself. Paracelsus, another great healer in the Western tradition, wrote 500 years ago that “fasting is the greatest remedy, the physician within.” Ayurvedic medicine, the world’s oldest healing system, has long advocated fasting as a major treatment.

Fasting has also been used in nearly every religion in the world, including Christianity, Judaism, Buddhism, and Islam. Many of history’s great spiritual leaders fasted for mental and spiritual clarity, including Jesus, Buddha, and Mohammed. In one of the famous political acts of the last century, the Indian leader Mahatma Gandhi fasted for 21 days to promote peace.

Fasting has been used in Europe as a medical treatment for years. Many spas and treatment centers, particularly those in Germany, Sweden, and Russia,

use medically supervised fasting. Fasting has gained popularity in American alternative medicine over the past several decades, and many doctors feel it is beneficial. Fasting is a central therapy in detoxification, a healing method founded on the principle that the build up of toxic substances in the body is responsible for many illnesses and conditions.

The principle of fasting is simple. When the intake of food is temporarily stopped, many systems of the body are given a break from the hard work of digestion. The extra energy gives the body the chance to heal and restore itself, and burning stored calories gets rid of toxic substances stored in the body.

The digestive tract is the part of the body most exposed to environmental threats, including bacteria, viruses, parasites, and toxins. It requires the most immune system support. When food is broken down in the intestines, it travels through the blood to the liver, the largest organ of the body’s natural detoxification system. The liver breaks down and removes the toxic by-products produced by digestion, including natural ones and the chemicals now present in the food supply. During fasting, the liver and immune system are essentially freed to detoxify and heal other parts of the body.

Many healers claim that fasting is a particularly useful therapy for Americans and for the modern lifestyle, subjected to heavy **diets**, overeating, and constant exposure to food additives and chemicals. Some alternative practitioners have gone so far as to estimate that the average American is carrying 5-10 pounds of toxic substances in their bodies, for which fasting is the quickest and most effective means of removal.

Physiology of fasting

Through evolution, the body became very efficient at storing energy and handling situations when no food was available. For many centuries, fasting was probably a normal occurrence for most people, and the body adapted to it. It is estimated that even very thin people can survive for 40 days or more without food. The body has a special mechanism that is initiated when no food is eaten. Fasting is not **starvation**, but rather the body’s burning of stored energy. Starvation occurs when the body no longer has any stored energy and begins using essential tissues such as organs for an energy source. Therapeutic fasts are stopped long before this happens.

Many physiological changes occur in the body during fasting. During the first day or so, the body uses its glycogen reserves, the sugars that are the basic

energy supply. After these are depleted, the body begins using fat. However, the brain, which has high fuel requirements, still needs glucose (sugars converted from glycogen). To obtain glucose for the brain, the body begins to break down muscle tissue during the second day of the fast. Thus, during fasting some muscle loss will occur. To fuel the brain, the body would need to burn over a pound of muscle a day, but the body has developed another way to create energy that saves important muscle mass. This protein-sparing process is called ketosis, which occurs during the third day of a fast for men and the second day for women. In this highly efficient state, the liver begins converting stored fat and other nonessential tissues into ketones, which can be used by the brain, muscles, and heart as energy. It is at this point in the fast that sensations of hunger generally go away, and many people experience normal or even increased energy levels. Hormone levels and certain functions become more stable in this state as well. The goal of most fasts is to allow the body to reach the ketosis state in order to burn excess fat and unneeded or damaged tissue. Thus, fasts longer than three days are generally recommended as therapy.

Weight loss occurs most rapidly during the first few days of a fast, up to 2 pounds per day. In following days, the figure drops to around 0.5 pound per day. An average weight loss of a pound a day for an entire fast can be expected.

Performing a fast

Fasts can be performed for varying lengths of time, depending on the person and his or her health requirements. For chronic conditions, therapists recommend from two to four weeks to get the most benefits. Seven-day fasts are also commonly performed. A popular fasting program for prevention and general health is a three-day fast taken four times per year, at the change of each season. These can be easily performed over long weekends. Preventative fasts of one day per week are used by many people as well.

Juice fasts are also used by many people, although these are not technically fasts. Juice fasts are less intensive than water fasts because the body doesn't reach the ketosis stage. The advantage of juice fasts is that fruit and vegetable drinks can supply extra energy and nutrients. People can fit a few days of juice fasting into their normal schedules without significant drops in energy. Juice fasts are also said to have cleansing and detoxifying effects. The disadvantage of juice fasts is that the body never gets to the ketosis stage, so these fasters are thought

to lack the deep detoxification and healing effects of the water fast.

Medical supervision is recommended for any fast over three days. Most alternative medicine practitioners, such as homeopaths, naturopathic doctors, and ayurvedic doctors, can supervise and monitor patients during fasts. Those performing extended fasts and those with health conditions may require blood, urine, and other tests during fasting. There are many alternative health clinics that perform medically supervised fasts as well. Some conventional medical doctors may also supervise patients during fasts. Costs and insurance coverage vary, depending on the doctor, clinic, and requirements of the patient.

Preparations

Fasts must be entered and exited with care. To enter a fast, the diet should be gradually lightened over a few days. First, heavy foods such as meats and dairy products should be eliminated for a day or two. Grains, nuts, and beans should then be reduced for several days. The day before a fast, only easily digested foods like fruits, light salads, and soups should be eaten. During the fast, only pure water and occasional herbal teas should be drunk.

Fasts should be ended as gradually as they are entered, going from lighter to heavier foods progressively. The diet after a fast should emphasize fresh, wholesome foods. Fasters should particularly take care not to overeat when they complete a fast.

Precautions

Fasting isn't appropriate for everyone and, in some cases, could be harmful. Any person undertaking a first fast longer than three days should seek medical supervision. Those with health conditions should always have medical support during fasting. Plenty of water should be taken by fasters since **dehydration** can occur. Saunas and sweating therapies are sometimes recommended to assist detoxification, but should be used sparingly. Those fasting should significantly slow down their lifestyles. Taking time off of work is helpful, or at least reducing the work load. Fasters should also get plenty of rest. **Exercise** should be kept light, such as walking and gentle stretching.

Side effects

Those fasting may experience side effects of fatigue, malaise, aches and pains, emotional duress,

KEY TERMS

Ayurvedic medicine—A traditional healing system developed in India.

Toxin—A substance that has poisonous effects on the body.

acne, headaches, allergies, swelling, **vomiting**, **bad breath**, and symptoms of colds and flu. These reactions are sometimes called *healing crises*, which are caused by temporarily increased levels of toxins in the body due to elimination and cleansing. Lower energy levels should be expected during a fast.

Research and general acceptance

The physiology of fasting has been widely studied and documented by medical science. Beneficial effects such as lowered cholesterol and improved general functioning have been shown. Fasting as a treatment for illness and disease has been studied less, although some studies around the world have shown beneficial results. A 1984 study showed that workers in Taiwan who had severe chemical **poisoning** had dramatic improvement after a ten-day fast. In Russia and Japan, studies have demonstrated fasting to be an effective treatment for mental illness. Fasting has been featured on the cover of medical journals, although mainstream medicine has generally ignored fasting and detoxification treatments as valid medical procedures.

The majority of research that exists on fasting is testimonial, consisting of individual personal accounts of healing without statistics or controlled scientific experiments. In the alternative medical community, fasting is an essential and widely accepted treatment for many illnesses and chronic conditions.

Resources

ORGANIZATIONS

Fasting Center International. 32 West Anapurna St., #360, Santa Barbara, CA 93101. <<http://www.fasting.com>>.

Douglas Dupler, MA

Fasting blood sugar test see **Blood sugar tests**

Fasting plasma glucose test see **Blood sugar tests**

Fatigue

Definition

Fatigue is physical and/or mental exhaustion that can be triggered by **stress**, medication, overwork, or mental and physical illness or disease.

Description

Everyone experiences fatigue occasionally. It is the body's way of signaling its need for rest and sleep. But when fatigue becomes a persistent feeling of tiredness or exhaustion that goes beyond normal sleepiness, it is usually a sign that something more serious is amiss.

Physically, fatigue is characterized by a profound lack of energy, feelings of muscle weakness, and slowed movements or central nervous system reactions. Fatigue can also trigger serious mental exhaustion. Persistent fatigue can cause a lack of mental clarity (or feeling of mental "fuzziness"), difficulty concentrating, and in some cases, memory loss.

Causes and symptoms

Fatigue may be the result of one or more environmental causes such as inadequate rest, improper diet, work and home stressors, or poor physical conditioning, or one symptom of a chronic medical condition or disease process in the body. Heart disease, low blood pressure, diabetes, end-stage renal disease, iron-deficiency anemia, **narcolepsy**, and **cancer** can cause long-term, ongoing fatigue symptoms. Acute illnesses such as viral and bacterial infections can also trigger temporary feelings of exhaustion. In addition, mental disorders such as depression can also cause fatigue.

A number of medications, including antihistamines, antibiotics, and blood pressure medications, may cause drowsiness as a side-effect. Individuals already suffering from fatigue who are prescribed one of these medications may wish to check with their healthcare provider about alternative treatments.

Extreme fatigue which persists, unabated, for at least six months, is not the result of a diagnosed disease or illness, and is characterized by flu-like symptoms such as swollen lymph nodes, **sore throat**, and muscle weakness and/or **pain** may indicate a diagnosis of chronic fatigue syndrome. **Chronic fatigue syndrome** (sometimes called chronic fatigue immune deficiency syndrome), is a debilitating illness that

causes overwhelming exhaustion and a constellation of neurological and immunological symptoms. Between 1.5 and 2 million Americans are estimated to suffer from the disorder.

Diagnosis

Because fatigue is a symptom of a number of different disorders, diseases, and lifestyle choices, diagnosis may be difficult. A thorough examination and patient history by a qualified healthcare provider is the first step in determining the cause of the fatigue. A physician can rule out physical conditions and diseases that feature fatigue as a symptom, and can also determine if prescription drugs, poor dietary habits, work environment, or other external stressors could be triggering the exhaustion. Several diagnostic tests may also be required to rule out common physical causes of exhaustion, such as blood tests to check for iron-deficiency anemia.

Diagnosis of chronic fatigue syndrome is significantly more difficult. Because there is no specific biological marker or conclusive blood test to check for the disorder, healthcare providers must rely on the patient's presentation and severity of symptoms to make a diagnosis. In many cases, individuals with chronic fatigue syndrome go through a battery of invasive diagnostic tests and several years of consultation with medical professionals before receiving a correct diagnosis.

Treatment

Conventional medicine recommends the dietary and lifestyle changes outlined above as a first line of defense against fatigue. Individuals who experience occasional fatigue symptoms may benefit from short term use of caffeine-containing central nervous stimulants, which make people more alert, less drowsy, and improve coordination. However, these should be prescribed with extreme caution, as overuse of the drug can lead to serious sleep disorders, like **insomnia**.

Another reason to avoid extended use of **caffeine** is its associated withdrawal symptoms. People who use large amounts of caffeine over long periods build up a tolerance to it. When that happens, they have to use more and more caffeine to get the same effects. Heavy caffeine use can also lead to dependence. If an individual stops using caffeine abruptly, withdrawal symptoms may occur, including headache, fatigue, drowsiness, yawning, irritability, restlessness, **vomiting**, or runny nose. These symptoms can go on for as long as a week.

Alternative treatment

The treatment of fatigue depends on its direct cause, but there are several commonly prescribed treatments for non-specific fatigue, including dietary and lifestyle changes, the use of essential oils and herbal therapies, deep breathing exercises, traditional Chinese medicine, and color therapy.

Dietary changes

Inadequate or inappropriate nutritional intake can cause fatigue symptoms. To maintain an adequate energy supply and promote overall physical well-being, individuals should eat a balanced diet and observe the following nutritional guidelines:

- Drinking plenty of water. Individuals should try to drink 9 to 12 glasses of water a day. Dehydration can reduce blood volume, which leads to feelings of fatigue.
- Eating iron-rich foods (i.e., liver, raisins, spinach, apricots). Iron enables the blood to transport oxygen throughout the tissues, organs, and muscles, and diminished oxygenation of the blood can result in fatigue.
- Avoiding high-fat meals and snacks. High fat foods take longer to digest, reducing blood flow to the brain, heart, and rest of the body while blood flow is increased to the stomach.
- Eating unrefined carbohydrates and proteins together for sustained energy.
- Balancing proteins. Limiting protein to 15-20 grams per meal and two snacks of 15 grams is recommended, but not getting enough protein adds to fatigue. Pregnant or breastfeeding women should get more protein.
- Getting the recommended daily allowance of **B complex vitamins** (specifically, pantothenic acid, folic acid, thiamine, and vitamin B₁₂). Deficiencies in these vitamins can trigger fatigue.
- Getting the recommended daily allowance of selenium, riboflavin, and niacin. These are all essential nutritional elements in metabolizing food energy.
- Controlling portions. Individuals should only eat when they're hungry, and stop when they're full. An overstuffed stomach can cause short-term fatigue, and individuals who are overweight are much more likely to regularly experience fatigue symptoms.

Lifestyle changes

Lifestyle factors such as a high-stress job, erratic work hours, lack of social or family support, or erratic sleep patterns can all cause prolonged fatigue. If stress is an issue, a number of relaxation therapies and techniques are available to help alleviate tension, including massage, yoga, **aromatherapy**, **hydrotherapy**, progressive relaxation exercises, **meditation**, and **guided imagery**. Some individuals may also benefit from individual or family counseling or psychotherapy sessions to work through stress-related fatigue that is a result of family or social issues.

Maintaining healthy sleep patterns is critical to proper rest. Having a set “bedtime” helps to keep sleep on schedule. A calm and restful sleeping environment is also important to healthy sleep. Above all, the bedroom should be quiet and comfortable, away from loud noises and with adequate window treatments to keep sunlight and streetlights out. Removing distractions from the bedroom such as televisions and telephones can also be helpful.

Essential oils

Aromatherapists, hydrotherapists, and other holistic healthcare providers may recommend the use of essential oils of rosemary (*Rosmarinus officinalis*), eucalyptus blue gum (*Eucalyptus globulus*), peppermint, (*Mentha x piperata*), or scots pine oil (*Pinus sylvestris*) to stimulate the nervous system and reduce fatigue. These oils can be added to bathwater or massage oil as a topical application. Citrus oils such as lemon, orange, grapefruit, and lime have a similar effect, and can be added to a steam bath or vaporizer for inhalation.

Herbal remedies

Herbal remedies that act as circulatory stimulants can offset the symptoms of fatigue in some individuals. An herbalist may recommend an infusion of ginger (*Zingiber officinale*) root or treatment with cayenne (*Capsicum annuum*), balmony (*Chelone glabra*), damiana (*Turnera diffusa*), ginseng (*Panax ginseng*), or rosemary (*Rosmarinus officinalis*) to treat ongoing fatigue.

An infusion is prepared by mixing the herb with boiling water, steeping it for several minutes, and then removing the herb from the infusion before drinking. A strainer, tea ball, or infuser can be used to immerse loose herb in the boiling water before steeping and separating it. A second method of infusion is to mix the loose herbal preparation with cold water first, bringing the mixture to a boil in a pan or teapot, and

then separating the tea from the infusion with a strainer before drinking.

Caffeine-containing **central nervous system stimulants** such as tea (*Camellia senensis*) and cola (*Cola nitida*) can provide temporary, short-term relief of fatigue symptoms. However, long-term use of caffeine can cause restlessness, irritability, and other unwanted side effects, and in some cases may actually work to increase fatigue after the stimulating effects of the caffeine wear off. To avoid these problems, caffeine intake should be limited to 300 mg or less a day (the equivalent of 4-8 cups of brewed, hot tea).

Traditional Chinese medicine

Chinese medicine regards fatigue as a blockage or misalignment of *qi*, or energy flow, inside the human body. The practitioner of Chinese medicine chooses **acupuncture** and/or herbal therapy to rebalance the entire system. The Chinese formula Minot Bupleurum soup (or Xiao Chia Hu Tang) has been used for nearly 2,000 years for the type of chronic fatigue that comes after the flu. In this condition, the person has low-grade **fever**, **nausea**, and fatigue. There are other formulas that are helpful in other cases. Acupuncture involves the placement of a series of thin needles into the skin at targeted locations on the body known as acupoints in order to harmonize the energy flow within the human body.

Deep breathing exercises

Individuals under stress often experience fast, shallow breathing. This type of breathing, known as chest breathing, can lead to **shortness of breath**, increased muscle tension, inadequate oxygenation of blood, and fatigue. Breathing exercises can both improve respiratory function and relieve stress and fatigue.

Deep breathing exercises are best performed while laying flat on the back on a hard surface, usually the floor. The knees are bent, and the body (particularly the mouth, nose, and face) is relaxed. One hand should be placed on the chest and one on the abdomen to monitor breathing technique. With proper breathing techniques, the abdomen will rise further than the chest. The individual takes a series of long, deep breaths through the nose, attempting to raise the abdomen instead of the chest. Air is exhaled through the relaxed mouth. Deep breathing can be continued for up to 20 minutes. After the **exercise** is complete, the individual checks again for body tension and relaxation. Once deep breathing techniques have been mastered, an individual can use deep breathing at any time

or place as a quick method of relieving tension and preventing fatigue.

Color therapy

Color therapy, also known as chromatherapy, is based on the premise that certain colors are infused with healing energies. The therapy uses the seven colors of the rainbow to promote balance and healing in the mind and body. Red promotes energy, empowerment, and stimulation. Physically, it is thought to improve circulation and stimulate red blood cell production. Red is associated with the seventh chakra, located at the root; or base of spine. In **yoga**, the chakras are specific spiritual energy centers of the body.

Therapeutic color can be administered in a number of ways. Practitioners of Ayurvedic, or traditional Indian medicine, wrap their patients in colored cloth chosen for its therapeutic hue. Individuals suffering from fatigue would be wrapped in reds and oranges chosen for their uplifting and energizing properties. Patients may also be bathed in light from a color filtered light source to enhance the healing effects of the treatment.

Individuals may also be treated with color-infused water. This is achieved by placing translucent red colored paper or colored plastic wrap over and around a glass of water and placing the glass in direct sunlight so the water can soak up the healing properties and vibrations of the color. Environmental color sources may also be used to promote feelings of stimulation and energy. Red wall and window treatments, furniture, clothing, and even food may be recommended for their energizing healing properties.

Color therapy can be used in conjunction with both hydrotherapy and aromatherapy to heighten the therapeutic effect. Spas and holistic healthcare providers may recommend red color baths or soaks, which combine the benefits of a warm or hot water soak with energizing essential oils and the fatigue-fighting effects of bright red hues used in color therapy.

Prognosis

Fatigue related to a chronic disease or condition may last indefinitely, but can be alleviated to a degree through some of the treatment options outlined here. Exhaustion that can be linked to environmental stressors is usually easily alleviated when those stressors are dealt with properly.

There is no known cure for chronic fatigue syndrome, but steps can be taken to lessen symptoms

KEY TERMS

Aromatherapy—The therapeutic use of plant-derived, aromatic essential oils to promote physical and psychological well-being.

Guided imagery—The use of relaxation and mental visualization to improve mood and/or physical well-being.

Hydrotherapy—Hydrotherapy, or water therapy, is use of water (hot, cold, steam, or ice) to relieve discomfort and promote physical well-being.

and improve quality of life for these individuals while researchers continue to seek a cure.

Prevention

Many of the treatments outlined above are also recommended to prevent the onset of fatigue. Getting adequate rest and maintaining a consistent bedtime schedule are the most effective ways to combat fatigue. A balanced diet and moderate exercise program are also important to maintaining a consistent energy level.

Resources

BOOKS

Davis, Martha, et al. *The Relaxation & Stress Reduction Workbook*. 4th ed. Oakland, CA: New Harbinger Publications, Inc., 1995.

Hoffman, David. *The Complete Illustrated Herbal*. New York: Barnes & Noble Books, 1999.

Paula Anne Ford-Martin

Fatty liver

Definition

Fatty liver is the collection of excessive amounts of triglycerides and other fats inside liver cells.

Description

Also called steatosis, fatty liver can be a temporary or long-term condition, which is not harmful itself, but may indicate some other type of problem. Left untreated, it can contribute to other illnesses. It is

usually reversible once the cause of the problem is diagnosed and corrected. The liver is the organ responsible for changing fats eaten in the diet to types of fat that can be stored and used by the body. Triglycerides are one of the forms of fat stored by the body and used for energy and new cell formation. The break down of fats in the liver can be disrupted by alcoholism, **malnutrition**, **pregnancy**, or **poisoning**. In fatty liver, large droplets of fat, containing mostly triglycerides, collect within cells of the liver. The condition is generally not painful and may go unnoticed for a long period of time. In severe cases, the liver can increase to over three times its normal size and may be painful and tender.

Causes and symptoms

The most common cause of fatty liver in the United States is **alcoholism**. In alcoholic fatty liver, over consumption of alcohol changes the way that the liver breaks down and stores fats. Often, people with chronic alcoholism also suffer from malnutrition by eating irregularly and not consuming a balanced diet. Conditions that can also cause fatty liver are other forms of malnutrition (especially when there is not enough protein in the diet), **obesity**, **diabetes mellitus**, and **Reye's syndrome** in children. Pregnancy can cause a rare, but serious form of fatty liver that starts late in pregnancy and may be associated with **jaundice** and liver failure. Some drug overdoses or toxic chemical poisonings, such as carbon tetrachloride, can also cause fatty liver.

Often, there are no symptoms associated with fatty liver. If there are symptoms, they can include **pain** under the rib cage on the right side of the body, swelling of the abdomen, jaundice, and **fever**. Symptoms that occur less often in alcoholic fatty liver, but more often in pregnancy related fatty liver, are nausea, vomiting, loss of appetite, and abdominal pain.

Diagnosis

During a **physical examination**, a doctor might notice that the liver is enlarged and tender when the abdomen is palpated (examined with the tips of the fingers while the patient lies flat). Blood tests may be used to determine if the liver is functioning properly. A **liver biopsy**, where a small sample of liver tissue is removed with a long needle or through a very small incision, can be used to confirm fatty liver. In pregnant women, the fatty liver condition is usually associated with another serious complication, pre-eclampsia or **eclampsia**. In this condition, the mother has seriously high blood pressure, swelling, and possibly, seizures.

KEY TERMS

Jaundice—A condition where the skin and whites of the eyes take on a yellowish color due to an increase of bilirubin (a compound produced by the liver) in the blood.

Reye's syndrome—A serious, life-threatening illness in children, usually developing after a bout of flu or chickenpox, and often associated with the use of aspirin. In fatal cases, there is evidence of accumulation of fat in the liver.

Triglycerides—A type of fat consumed in the diet and produced by and stored in the body as an energy source.

Laboratory abnormalities include elevations of the SGOT (serum glutamic-oxaloacetic transaminase) and SGPT (serum glutamic pyruvic transaminase). In many cases the alkaline phosphatase will be significantly elevated due to **cholestasis** produced by the fatty infiltration.

Treatment

Treatment involves correcting the condition that caused fatty liver and providing supportive care. In fatty liver caused by alcoholism, the treatment is to give up drinking alcohol and to eat a healthy, well balanced diet. In fatty liver associated with pregnancy, the recommended treatment is to deliver the baby, if the pregnancy is far enough along. Vitamin and mineral supplements along with nutritional support may be useful.

Prognosis

Fatty liver is usually reversible if recognized and treated. There may be some long-term tendency toward other types of liver problems depending on how long and how severe the fatty liver condition was. In pregnant women with the condition, the situation can be life threatening for both the mother and the infant. Left untreated, there is a high risk of **death** for both the mother and baby. Severe liver damage that may require a liver transplant can occur in the mother if the condition is not recognized early.

Prevention

Prevention consists of maintaining a well balanced diet and healthy lifestyle with moderate or

no alcohol consumption. Pregnant women require good prenatal care so that symptoms can be recognized and treated as early as possible. To prevent Reye's syndrome, children should not be given **aspirin** to treat symptoms of the flu or other viruses.

Resources

PERIODICALS

Everson, Gregory T. "Liver Problems in Pregnancy: Part 2, Managing Pre-Existing and Pregnancy-Induced Liver Disease." *Medscape Women's Health* 3, no. 2 (1998).

Altha Roberts Edgren

Febrile agglutination tests see **Fever evaluation tests**

Fecal fat test see **Stool fat test**

Fecal incontinence

Definition

Fecal incontinence is the inability to control the passage of gas or stools (feces) through the anus. For some people fecal incontinence is a relatively minor problem, as when it is limited to a slight occasional soiling of underwear, but for other people it involves a considerable loss of bowel control and has a devastating effect on quality of life and psychological well-being. Fortunately, professional medical treatment is usually able to restore bowel control or at least substantially reduce the severity of the condition.

Description

Fecal incontinence, also called bowel incontinence, can occur at any age, but is most common among people over the age of 65, who sometimes have to cope with **urinary incontinence** as well. It was reported in 1998 that about 2% of adults experience fecal incontinence at least once a week whereas for healthy independent adults over the age of 65 the figure is about 7%. An extensive American survey, published in 1993, found fecal soiling in 7.1% of the surveyed population, with gross incontinence in 0.7%. For men and women the incidence of soiling was the same, but women were almost twice as likely to suffer from gross incontinence.

The wider public health impact of fecal incontinence is considerable. In the United States, more than

\$400 million is spent each year on disposable underwear and other incontinence aids. Fecal incontinence is the second most common reason for seeking a nursing home placement. One-third of the institutionalized elderly suffer from this condition. Incontinence sufferers, however, often hesitate to ask their doctors for help because they are embarrassed or ashamed. The 1993 American survey discovered that only one-sixth of those experiencing soiling had sought medical advice, and only one-half of those afflicted with gross incontinence.

Causes and symptoms

Fecal incontinence can result from a wide variety of medical conditions, including childbirth-related anal injuries, other causes of damage to the anus or rectum, and nervous system problems.

Vaginal-delivery **childbirth** is a major cause of fecal incontinence. In many cases, childbirth results in damage to the anal sphincter, which is the ring of muscle that closes the anus and keeps stools within the rectum until a person can find an appropriate opportunity to defecate. Nerve injuries during childbirth may also be a factor in some cases. An ultrasound study of first-time mothers found sphincter injuries in 35%. About one-third of the injured women developed fecal incontinence or an uncontrollable and powerful urge to defecate (urgency) within six weeks of giving birth. Childbirth-related incontinence is usually restricted to gas, but for some women involves the passing of liquid or solid stools.

The removal of **hemorrhoids** by surgery or other techniques (hemorrhoidectomies) can also cause anal damage and fecal incontinence, as can more complex operations affecting the anus and surrounding areas. Anal and rectal infections as well as Crohn's disease can lead to incontinence by damaging the muscles that control defecation. For some people, incontinence becomes a problem when the anal muscles begin to weaken in midlife or old age.

Dementia, mental retardation, strokes, brain tumors, multiple sclerosis, and other conditions that affect the nervous system can cause fecal incontinence by interfering with muscle function or the normal rectal sensations that trigger sphincter contraction and are necessary for bowel control. One study of **multiple sclerosis** patients discovered that about half were incontinent. Nerve damage caused by long-lasting diabetes mellitus (diabetic neuropathy) is another condition that can give rise to incontinence.

Diagnosis

Medical assessments in cases of fecal incontinence typically involve three steps: asking questions about the patient's past and current health (the medical history); a **physical examination** of the anal region; and testing for objective information regarding anal and rectal function.

Patient history

The medical history relies on questions that allow the doctor to evaluate the nature and severity of the problem and its effect on the patient's life. The doctor asks, for instance, how long the patient has been suffering from incontinence; how often and under what circumstances incontinence occurs; whether the patient has any control over defecation; and whether the patient has obstacles to defecation in his or her everyday surroundings, such as a toilet that can be reached only by climbing a long flight of stairs. For women who have given birth, a detailed obstetric history is also necessary.

Physical examination

The physical examination begins with a visual inspection of the anus and the area lying between the anus and the genitals (the perineum) for hemorrhoids, infections, and other conditions that might explain the patient's difficulties. During this phase of the examination the doctor asks the patient to bear down. Bearing down enables the doctor to check whether rectal prolapse or certain other problems exist. **Rectal prolapse** means that the patient's rectum has been weakened and drops down through the anus. Next, the doctor uses a pin or probe to **stroke** the perianal skin. Normally this touching causes the anal sphincter to contract and the anus to pucker; if it does not, nerve damage may be present. The final phase of the examination requires the doctor to examine internal structures by carefully inserting a gloved and lubricated finger into the anal canal. This allows the doctor to judge the strength of the anal sphincter and a key muscle (the puborectalis muscle) in maintaining continence; to look for abnormalities such as **scars** and rectal masses; and to learn many other things about the patient's medical situation. At this point the doctor performs the anal wink test again and asks the patient to squeeze and bear down.

Laboratory tests

Information from the medical history and physical examination usually needs to be supplemented by tests that provide objective measurements of anal and

rectal function. Anorectal manometry, a common procedure, involves inserting a small tube (catheter) or balloon device into the anal canal or rectum. Manometry measures, among other things, pressure levels in the anal canal, rectal sensation, and anal and rectal reflexes. Tests are also available for assessing nerve damage. An anal ultrasound probe can supply accurate images of the anal sphincter and reveal whether injury has occurred. **Magnetic resonance imaging**, which requires the insertion of a coil into the anal canal, is useful at times.

Treatment

Fecal incontinence arising from an underlying condition such as **diabetic neuropathy** can sometimes be helped by treating the underlying condition. When that does not work, or no underlying condition can be discovered, one approach is to have the patient use a suppository or enema to stimulate defecation at the same time every day or every other day. The goal is to restore regular bowel habits and keep the bowels free of stools. Medications such as loperamide (Imodium) and codeine phosphate are often effective in halting incontinence, but only in less severe cases involving liquid stools or urgency. Dietary changes and exercises done at home to strengthen the anal muscles may also help.

Good results have been reported for **biofeedback** training, although the subject has not been properly researched. In successful cases, patients regain complete control over defecation, or at least improve their control, by learning to contract the external part of the anal sphincter whenever stools enter the rectum. All healthy people have this ability. Biofeedback training begins with the insertion into the rectum of a balloon manometry device hooked up to a pressure monitor. The presence of stools in the rectum is simulated by inflating the balloon, which causes pressure changes that are recorded on the monitor. The monitor also records sphincter contraction. By watching the monitor and following instructions from the equipment operator, the patient gradually learns to contract the sphincter automatically in response to fullness in the rectum. Sometimes one training session is enough, but often several are needed. Biofeedback is not an appropriate treatment in all cases, however. It is used only with patients who are highly motivated; who are able, to some extent, to sense the presence of stools in the rectum; and who have not lost all ability to contract the external anal sphincter. One specialist suggests that possibly two-thirds of incontinence sufferers are candidates for biofeedback.

KEY TERMS

Anus—The opening at the lower end of the rectum.

Colostomy—A surgical procedure in which an opening is made in the wall of the abdomen to allow a part of the large intestine (the colon) to empty outside the body.

Crohn's disease—A disease marked by inflammation of the intestines.

Defecation—Passage of stools through the anus.

Hemorrhoids—Enlarged veins in the anus or rectum. They are sometimes associated with fecal incontinence.

Rectum—The lower section of the large intestine that holds stools before defecation.

Sphincter—A circular band of muscle that surrounds and encloses an opening to the body or to one of its hollow organs. Damage to the sphincter surrounding the anus can cause fecal incontinence.

Stools—Undigested food and other waste that is eliminated through the anus.

Suppository—A solid medication that slowly dissolves after being inserted into the rectum or other body cavity.

Some people may require surgery. Sphincter damage caused by childbirth is often effectively treated with surgery, however, as are certain other kinds of incontinence-related sphincter injuries. Sometimes surgical treatment requires building an artificial sphincter using a thigh muscle (the gracilis muscle). At one time a **colostomy** was necessary for severe cases of incontinence, but is now rarely performed.

Prognosis

Fecal incontinence is a problem that usually responds well to professional medical treatment, even among elderly and institutionalized patients. If complete bowel control cannot be restored, the impact of incontinence on everyday life can still be lessened considerably in most cases. When incontinence remains a problem despite medical treatment, disposable underwear and other commercial incontinence products are available to make life easier. Doctors and nurses can offer advice on coping with incontinence, and people should never be embarrassed about seeking their assistance. Counseling and information are also available from support groups.

Resources

ORGANIZATIONS

International Foundation for Functional Gastrointestinal Disorders. PO Box 17864, Milwaukee, WI 53217. (888) 964-2001. <<http://www.iffgd.org>> .

National Association for Continence. PO Box 8310, Spartanburg, SC 29305-8310. (800) 252-3337. <<http://www.nafc.org>> .

National Digestive Diseases Information Clearinghouse. 2 Information Way, Bethesda, MD 20892-3570. (800) 891-5389. <<http://www.niddk.nih.gov/health/digest/ddic.htm>> .

Howard Baker

Fecal lipids test see **Stool fat test**

Fecal occult blood test

Definition

The fecal occult blood test (FOBT) is performed as part of the routine **physical examination** during the examination of the rectum. It is used to detect microscopic blood in the stool and is a screening tool for colorectal **cancer**.

Purpose

FOBT uses chemical indicators on stool samples to detect the presence of blood not otherwise visible. (The word “occult” in the test’s name means that the blood is hidden from view.) Blood originating from or passing through the gastrointestinal tract can signal many conditions requiring further diagnostic procedures and, possibly, medical treatment. These conditions may be benign or malignant and some of them include:

- **colon cancer**, **rectal cancer**, and gastric cancers
- ulcers
- hemorrhoids
- polyps
- inflammatory bowel disease
- irritations or lesions of the gastrointestinal tract caused by medications (such as **nonsteroidal anti-inflammatory drugs**, also called NSAIDs)
- irritations or lesions of the gastrointestinal tract caused by stomach acid disorders, such as reflux esophagitis

The FOBT is used routinely (in conjunction with a rectal examination performed by a physician) to screen for colorectal cancer, particularly after age 50. The ordering of this test should not be taken as an indication that cancer is suspected. The FOBT must be combined with regular screening endoscopy (such as a **sigmoidoscopy**) to detect cancers at an early stage.

Precautions

Certain foods and medicines can influence the test results. Some fruits contain chemicals that prevent the guaiac, the chemical in which the test paper is soaked, from reacting with the blood. **Aspirin** and some NSAIDs irritate the stomach, resulting in bleeding, and should be avoided prior to the examination. Red meat and many vegetables and fruits containing vitamin C also should be avoided for a specified period of time before the test. All of these factors could result in a false-positive result.

Description

Feces for the stool samples is obtained either by the physician at the **rectal examination** or by the patient at home, using a small spatula or a collection device. In most cases, the collection of stool samples can easily be done at home, using a kit supplied by the physician. The standard kit contains a specially prepared card on which a small sample of stool will be spread, using a stick provided in the kit. The sample is placed in a special envelope and either mailed or brought in for analysis. When the physician applies hydrogen peroxide to the back of the sample, the paper will turn blue if an abnormal amount of blood is present.

Types of fecal occult blood tests

Hemoccult is the most commonly used fecal occult blood test. The Hemoccult test takes less than five minutes to perform and may be performed in the physician's office or in the laboratory. The Hemoccult blood test can detect bleeding from the colon as low as 0.5 mg per day.

Tests that use anti-hemoglobin antibodies (or immunochemical tests) to detect blood in the stool are also used. Immunochemical tests can detect up to 0.7 mg of hemoglobin in the stool and do not require dietary restrictions. Immunochemical tests

- are not accurate for screening for stomach cancer
- are more sensitive than Hemoccult tests in detecting colorectal cancer
- are more expensive than Hemoccult tests.

Hemoquant, another fecal occult blood test, is used to detect as much as 500 mg/g of blood in the stool. Like the Hemoccult, the Hemoquant test is affected by red meat. It is not affected by chemicals in vegetables.

Fecal blood may also be measured by the amount of chromium in the red blood cells in the feces. The stool is collected for three to ten days. The test is used in cases where the exact amount of blood loss required. It is the only test that can exclude blood loss from the gastrointestinal area with accuracy.

Medicare coverage began on January 1, 2004, for a newer fecal occult blood test based on immunoassay. This technique does not rely on guaiac, so it is not influenced by diet or medications used prior to the test. The immunoassay test also requires fewer specimen collections. At a conference of gastroenterologists (physicians who specialize in diseases of the stomach and related digestive systems), a company announced a new fecal occult blood test that was based on DNA and appeared more sensitive than traditional tests. Widespread use of these new tests remains to be seen; the traditional guaiac test has been in place for about 30 years.

Preparation

For 72 hours prior to collecting samples, patients should avoid red meats, NSAIDs (including aspirin), **antacids**, steroids, iron supplements, and vitamin C, including citrus fruits and other foods containing large amounts of vitamin C. Foods like uncooked broccoli, uncooked turnips, cauliflower, uncooked cantaloupe, uncooked radish and horseradish and parsnips should be avoided and not eaten during the 72 hours prior to the examination. Fish, chicken, pork, fruits (other than melons) and many cooked vegetables are permitted in the diet.

Results

Many factors can result in false-positive and false-negative findings.

Positive results

It is important to note that a true-positive finding only signifies the presence of blood—it is not an indication of cancer. The National Cancer Institute states that, in its experience, less than 10% of all positive results were caused by cancer. The FOBT is positive in 1–5% of the unscreened population and 2–10% of those are found to have cancer. The physician will want to follow up on a positive result with further

KEY TERMS

Occult—Not visible or easily detected.

tests, as indicated by other factors in the patient's history or condition.

Negative results

Alternatively, a negative result (meaning no blood was detected) does not guarantee the absence of colon cancer, which may bleed only occasionally or not at all. (Only 50% of colon cancers are FOBT-positive.)

Conclusions

Screening using the FOBT has been demonstrated to reduce colorectal cancer. However, because only half of colorectal cancers are FOBT-positive, FOBT must be combined with regular screening endoscopy to increase the detection of pre-malignant colorectal polyps and cancers. Since, through FOBT, cancer may be detected early, the benefits of possible early detection must be considered along with the likelihood of complications and costs for additional studies.

Resources

BOOKS

- DeVita, Vincent, Samuel Hellman, and Steven Rosenberg. *Cancer: Principles and Practices of Oncology*. Philadelphia: Lippincott Williams & Wilkins, 2001.
- Yamada, Tadetaka, editor. *Textbook of Gastroenterology Volumes One and Two*. Philadelphia: Lippincott Williams & Wilkins, 2001.

PERIODICALS

- "DNA-based Stool Test More Sensitive than Fecal Occult Blood Test in Study." *Health & Medicine Week* November 10, 2003: 194.
- From the Centers for Disease Control and Prevention. "Trends in Screening for Colorectal Cancer—United States, 1997 and 1999." *Journal of the American Medical Association* 28 (March 2001): 12.
- Silverman, Jennifer. "Colorectal Screening Option." *Family Practice News* 33 (December 15, 2003): 49–51.

ORGANIZATIONS

- American Cancer Society. 1599 Clifton Road NE, Atlanta, GA 30329. (800)ACS-2345. <<http://www.cancer.org>>.
- National Cancer Institute (National Institutes of Health). 9000 Rockville Pike, Bethesda, MD 20892. (800) 422-6237. <<http://www.nci.nih.gov>>.

OTHER

- "Colon and Rectum Cancer." *American Cancer Society*. 2000. July 10, 2001. <<http://www.cancer.org>>.
- "Colorectal Cancer Screening." *WebMD*. <<http://my.webmd.com/content/article/2955.291>>.
- "Fecal Occult Blood Test." *Virtual Health Fair*. July 10, 2001. <<http://vfair.com/resources/lab/fecal.htm>>.

Jill S. Lasker
Cheryl Branche, M.D.
Teresa G. Odle

Feldenkrais method

Definition

The Feldenkrais method is an educational system that allows the body to move and function more efficiently and comfortably. Its goal is to re-educate the nervous system and improve motor ability. The system can accomplish much more, relieving pressure on joints and weak points, and allowing the body to heal repetitive strain injuries. Continued use of the method can relieve pain and lead to higher standards of achievement in sports, the martial arts, dancing and other physical disciplines.

Pupils are taught to become aware of their movements and to become aware of how they use their bodies, thus discovering possible areas of **stress** and strain. The goal of Feldenkrais is to take the individual from merely functioning, to functioning well, free of **pain** and restriction of movement. Feldenkrais himself stated that his goal was, "To make the impossible possible, the possible easy, and the easy, elegant."

Purpose

This method of re-educating the nervous system can be beneficial to a wide range of people, including athletes, children, the elderly, martial artists, those who are handicapped, people with special needs, and those suffering from degenerative diseases. It has also proved popular with artists, particularly musicians, a number of whom have used Feldenkrais to improve their performance.

The Feldenkrais Guild of North America (FGNA) states that over half of the those who turn to Feldenkrais practitioners are seeking relief from pain. Many people who have pain from an injury compensate by changing their movements to limit pain. Often these changed movements remain after

MOSHE FELDENKRAIS (1904–1984)

Moshe Feldenkrais was born on the border between Russia and Poland. When he was only a boy of 13, he traveled to Palestine on foot. The journey took a year, and once there, young Feldenkrais worked as a laborer and cartographer, also tutoring others in mathematics. Moving to France in 1933, he graduated in mechanical and electrical engineering from the Ecole des Travaux Publiques de Paris.

Feldenkrais became the first person to open a Judo center in Paris after meeting with Jigaro Kano. He was also one of the first Europeans to become a black belt in Judo, in 1936.

Obtaining his Ph.D. at the Sorbonne, he went on to assist Nobel Prize laureate, Frédéric Joliot-Curie at the Curie Institute. During World War II in England, he worked on the new sonar anti-submarine research.

Prompted by a recurring leg injury, he applied his knowledge of the martial arts and his training as an engineer to devise a method of re-integrating the body. The concept was that more efficient movement would allow for the treatment of pain or disability, and the better-functioning of the body as a whole. Later on, he would begin to teach what he had learned to others in Tel Aviv.

In addition to many books about judo, including *Higher Judo*, he wrote six books on his method.

the pain from the original injury is gone, and new pain may occur. Feldenkrais helps students become aware of the changed movements and allows them to learn new movements that relieve their pain. Apart from the obvious physical benefits of more efficient movement and freedom from pain and restriction, Feldenkrais practitioners assert that there are other positive benefits for overall physical and mental health. Feldenkrais can result in increased awareness, flexibility, and coordination, and better relaxation. Feldenkrais practitioners have also noted other benefits in their students, including improvements in awareness, flexibility, coordination, breathing, digestion, sleep, mood, mental alertness, energy, and range of motion, as well as reduced stress and **hypertension**, and fewer headaches and backaches.

Musicians and athletes can improve their performance in many ways when they learn to use their bodies more efficiently. Feldenkrais can also help injured athletes regain lost potential and free them from pain and restriction of movement.

There are numerous accounts of the remarkable results obtained when Feldenkrais is taught to handicapped children so that they can learn to function despite their limitations. Handicapped people can learn to make full use of whatever potential they have, and to have more confidence in their abilities. Practitioners who specialize in teaching Feldenkrais to those who have handicaps have in many cases allowed the patient to discover ways of performing tasks which were previously thought to be impossible for them.

The elderly, whose movements are often restricted by pain and stiffness, can learn to overcome these obstacles with Feldenkrais instruction. In some instances even severe cases of arthritis have been conquered. Theoretically, Feldenkrais can make possible renewed levels of energy and freedom from restriction.

Description

Origins

Moshe Feldenkrais (1904–1984) was a Russian-born Israeli physicist and engineer who was also an active soccer player and judo master. He devised his system in response to his own recurring knee injury, which had restricted his movement and caused him great pain over a long period of time. Feldenkrais believed that repeated muscle patterns cause the parts of the brain controlling those muscles to stay in a fixed pattern as well. He thought that the more the muscles are used, the more parts of the brain can be activated. He devised a method of re-educating the neuromuscular system and re-evaluating movement to increase efficiency and reduce stress, using his knowledge of mechanics and engineering, and applying some of his martial arts training.

Feldenkrais is described as being a dual system, with two components: “Awareness Through Movement” and “Functional Integration.” The system aims to re-educate the body so that habitual movements that cause strain or pain can be relearned to improve efficiency and eliminate dangerous or painful action.

Feldenkrais helps to translate intention into action. In practice, an individual can learn to achieve his or her highest potential, while at the same time learning to avoid and eliminate stresses, **strains**, and the possibility of injury.

Functional integration

During this session, the patient wears comfortable clothing, and may sit, stand, walk, or lie on a low padded table. The practitioner helps the pupil by guiding him or her through a number of movements. The

practitioner may use touch to communicate with the student, but touch is not used to correct any movements. The purpose of this session is to increase a student's awareness of his or her own movement and become open to different possibilities for movement. The instruction can be focused on a particular activity that the student does every day, or that causes him or her pain. The student can learn to alter habitual movements and re-educate the neuromuscular system. This type of session is particularly useful for those who suffer from limitations originating from misuse, stress, illness, or accident. It can also help athletes and musicians perform to the best of their ability by increasing their possibilities for movement. It offers students the potential for improving their physical and mental performance in addition to heightening the sense of well-being.

Awareness through movement

Feldenkrais's martial arts background can be clearly identified in many of the aspects of Awareness Through Movement (ATM). During group sessions, pupils are taught to become acutely aware of all their movements and to imagine them, so that they can improve the efficiency of their actions in their minds, and put them into practice. Pupils are encouraged to be disciplined about practicing their exercises, to achieve maximum benefit.

Awareness through movement is described as an exploratory, nonjudgmental process through which pupils are encouraged to observe and learn about themselves and their movements. The range of this therapy is wide, and there are thousands of different lessons designed to help specific areas.

Preparations

No preparation is necessary for the practice of Feldenkrais, and all are encouraged to seek help from this system. No condition is considered a preclusion to the benefits of Feldenkrais.

Precautions

As with any therapy or treatment, care should be taken to choose a qualified practitioner. Feldenkrais practitioners stress that the body must not be forced to do anything, and if any movement is painful, or even uncomfortable, it should be discontinued immediately and the patient should seek professional help.

Side effects

No known side effects are associated with the practice of Feldenkrais.

KEY TERMS

Neuromuscular—The body system of nerves and muscles as they function together.

Repetitive strain injury—Injury resulting from a repeated movement such as typing or throwing a ball.

Research and general acceptance

Since Moshe Feldenkrais began to teach his method, it has gradually gained acceptance as an education system. Published research using the method can be found in United States and foreign publications.

Resources

BOOKS

Bratman, Steven. *The Alternative Medicine Sourcebook*. 2nd ed. Chicago: Lowell House, 1999.

Somerville, Robert. *Alternative Medicine: The Definitive Guide*. Tiburon, CA: Future Medicine Publishing, Inc., 1999.

ORGANIZATIONS

Feldenkrais Guild of North America. 3611 SW Hood Ave., Suite 100, Portland, OR 97201. (800) 775-2118. (503) 221-6612. Fax: (503) 221-6616.
< <http://www.feldenkrais.com> > .

Patricia Skinner

Female circumcision see **Female genital mutilation**

Female condom see **Condom**

Female genital mutilation

Definition

Female genital mutilation (FGM) is the cutting, or partial or total removal, of the external female genitalia for cultural, religious, or other non-medical reasons. It is usually performed on girls between the ages of four and 10. It is also called **female circumcision**.

Purpose

FGM results in the cutting or removal of the tissues around the vagina that give women pleasurable

sexual feelings. This procedure is used for social and cultural control of women's sexuality. In its most extreme form, infibulation, where the girl's vagina is sewn shut, the procedure ensures virginity. In some cultures where female circumcision has been a tradition for hundreds of years, this procedure is considered a rite of passage for young girls. Families fear that if their daughters are left uncircumcised, they may not be marriageable. As in most cultures, there is also the fear that the girl might bring shame to the family by being sexually active and becoming pregnant before marriage.

Precautions

It is illegal to perform FGM in many countries, including the United States, Canada, France, Great Britain, Sweden, Switzerland, Egypt, Kenya, and Senegal. This procedure is usually done in the home or somewhere other than a medical setting. Often, it is performed by a family member or by a local "circumciser," using knives, razor blades, or other tools that may not be sterilized before use.

Description

Female circumcision includes a wide range of procedures. The simplest form involves a small cut to the clitoris or labial tissue. A Sunna circumcision removes the prepuce (a fold of skin that covers the clitoris) and/or the tip of the clitoris. A clitoridectomy removes the entire clitoris and some or all of the surrounding tissue; this procedure occurs in approximately 80% of cases. The most extreme form of genital mutilation is excision and infibulation, in which the clitoris and all of the surround tissue are cut away and the remaining skin is sewn together. Only a small opening is left for the passage of urine and menstrual blood. Infibulation accounts for approximately 15% of FGM procedures.

The World Health Organization (WHO) estimates that between 100 million and 140 million girls and women have undergone some form of FGM. As a very deeply rooted cultural and religious tradition still practiced in over 28 African and Asian countries, up to two million girls per year are at risk. The following countries have the highest number of occurrences of FGM: Djibouti (98%), Egypt (97%), Eritrea (95%), Guinea (99%), Mali (94%), Sierra Leone (90%), and Somalia (98-100%). As more people move to Western countries from countries where female circumcision is performed, the practice has come to the attention of health professionals in the United States, Canada, Europe, and Australia.

KEY TERMS

Circumcision—A procedure, usually with religious or cultural significance, where the prepuce or skin covering the tip of the penis on a boy, or the clitoris on a girl, is cut away.

Clitoridectomy—A procedure where the clitoris and possibly some of the surrounding labial tissue at the opening of the vagina is cut away.

Infibulation—A procedure where the tissue around the vagina is sewn shut, leaving only a small opening for the passage of urine and menstrual blood.

In an effort to integrate old customs with modern medical care, some immigrant families have requested that physicians perform the procedure. While trying to be sensitive to cultural traditions, health care providers are sometimes put in the difficult position of choosing to perform this procedure in a medical facility under sanitary conditions, or refusing the request, knowing that it may be done anyway with no medical supervision. Some families who are intent on having this procedure done will take their daughters back to the country they immigrated from in order to have the girls circumcised.

Many national and international medical organizations including the American Medical Association (AMA), Canadian medical organizations, and WHO oppose the practice of female genital mutilation. The United Nations (UN) considers female genital mutilation a violation of human rights. WHO has undertaken a number of projects aimed at decreasing the incidence of FGM. These include the following activities:

- publishing a statement addressing the regional status of FGM and encouraging the development of national policy against its practice,
- organizing training for regional community workers,
- developing educational materials for local health care workers,
- providing alternative occupations for individuals who perform FGM procedures.

Aftercare

A girl or young woman who has recently had the procedure performed may require supportive care to control bleeding and **antibiotics** to prevent infection. Women who were circumcised as children may require

medical care to treat complications. Pregnant women who have been infibulated may have to have the labial tissue cut open to allow the baby to be delivered. Aftercare should be provided with a supportive and nonjudgmental approach towards the girls and women who have undergone this procedure.

Risks

The immediate risks after the procedure are hemorrhage (excessive bleeding), severe pain, and infection (including abscesses, **tetanus**, and **gangrene**). The most severe consequence is **death** due to excessive blood loss. Long term complications include scarring, interference with the drainage of urine and menstrual blood, chronic urinary tract infections, pelvic and back **pain**, and **infertility**. Sexual intercourse can be painful. Complications of **childbirth** are also a risk. It is unclear whether it is related to the procedure itself, or related to the general condition of medical practice, but infant and maternal death rates are generally higher in those communities where female circumcision is practiced.

Resources

OTHER

The Female Genital Mutilation Research Homepage.

< <http://www.hollyfeld.org/fgm> > .

"Female Genital Mutilation." *The World Health Organization.*

< <http://www.who.int/frh-whd/FGM/index.htm> > .

Altha Roberts Edgren

Female infertility *see* **Infertility**

Female sexual arousal disorder

Definition

Female sexual arousal disorder (FSAD) occurs when a woman is continually unable to attain or maintain arousal and lubrication during intercourse, is unable to reach orgasm, or has no desire for sexual intercourse.

Description

The disorder typically affects up to 25 percent of all American women, or an estimated 47 million women. Three-fourths of women with FSAD are postmenopausal. Women describe it as being "unable to

get turned on," or being continually disinterested in sex. It is also called "frigidity." Other terms for the disorder include dyspareunia and vaginismus, both of which involve **pain** during intercourse.

Causes and symptoms

There are numerous causes of this disorder. They include:

- physical problems, such as **endometriosis**, **cystitis**, or vaginitis
- systemic problems, such as diabetes, high blood pressure, or **hypothyroidism**. Even **pregnancy** or the postpartum period (time after delivery of a child) may affect desire. **Menopause** is also known to reduce sexual desire.
- medications, including oral contraceptives, antidepressants, antihypertensives, and tranquilizers
- surgery, such as **mastectomy** or **hysterectomy** which may affect how a woman feels about her sexual self.
- stress
- depression
- use of alcohol, drugs, or cigarette **smoking**

Symptoms vary. A woman may have no desire for sex, or may not be able to maintain arousal, or may be unable to reach orgasm. She may also have pain during sex or orgasm, which interferes with her desire for intercourse.

Diagnosis

To make a diagnosis, a woman's physician - either family doctor, gynecologist, or even urologist - takes a complete medical history to determine when the problem started, how it presents, how severe it is, and what the patient thinks may be causing it. The doctor will also conduct a complete **physical examination**, looking for any abnormalities in the genital region

Treatment

The physician should start by providing education about the disorder and recommending various non-medical treatment strategies. These include:

- use of erotic materials, such as vibrators, books, magazines and videos
- sensual massage, avoiding the genitals
- position changes to reduce pain

KEY TERMS

Dyspareunia— pain in the pelvic area during or after sexual intercourse.

Vaginismus—An involuntary spasm of the muscles surrounding the vagina, making penetration painful or impossible.

- use of lubricants to moisten the vagina and genital area
- kegel exercises to strengthen the vagina and clitoris
- therapy to overcome any relationship or sexual abuse issues

Medical treatments include:

- estrogen replacement therapy, which may help with vaginal dryness, pain and arousal
- testosterone therapy in women who have low levels of this male hormone (Side effects, however, may include deepening voice, hair growth, and acne)
- the EROS clitoral therapy device (EROS-CTD), recently approved by the Food and Drug Administration; a small vacuum pump, placed over the clitoris and gently activated to provide a gentle suction designed to increase blood flow to the region, which, in turn, helps with arousal
- using the herb yohimbine combined with nitric oxide has been found to increase vaginal blood flow in postmenopausal women and thus help with some forms of FSAD

Alternative treatment

Natural estrogens, such as those found in soy products and flax, may be effective. Herbal remedies include belladonna, ginkgo, and motherwort. However, there is no scientific evidence to prove these herbs actually help. Some women squirt vitamin E in their vagina to increase lubrication.

Women may also want to see a sexual therapist for additional help.

Prognosis

Generally, once women seek the appropriate help they are quite likely to find a way to resolve their problems. Often, a holistic approach, using physical as well as emotional therapies, is required for success.

Prevention

Maintaining a close and open relationship with a partner is one way to avoid the emotional pain and isolation that can lead to **sexual dysfunction**. Additionally, women should learn if any medications they take affect sexual function, and should refrain from alcohol and drugs and quit smoking. Women who have anxieties and fears about sexual intercourse, whether because of earlier **abuse**, **rape**, or a prudish upbringing, should deal with those issues through therapy.

Resources

BOOKS

Berman M. D., Jennifer, Berman PhD, Laura, and Elisabeth Bumiller. *For Women Only: A Revolutionary Guide to Overcoming Sexual Dysfunction and Reclaiming Your Sex Life*. Henry Holt & Company, Inc., 2001.

Rako, M. D., Susan *The Hormone of Desire: The Truth About Testosterone, Sexuality, and Menopause*. Three Rivers Press, 1999.

Reichman, Judith. *I'm Not in the Mood: What Every Woman Should Know About Improving Her Libido*. Quill Publishing, 1999.

PERIODICALS

"Consumer Update: Female Sexual Problems." *American Association for Marriage and Family Therapy* June 14, 1999.

"Restoring Sexual Health." *Consumer Reports On Health*. March 2001: 8-10.

ORGANIZATIONS

Female Sexual Medicine Center UCLA Medical Center. 924 Westwood Blvd., Suite 520 Los Angeles, CA 90024. (310) 825-0025 <www.newshe.com>.

National Women's Health Resource Center. 120 Albany Street Suite 820 New Brunswick, NJ 08901. (877) 986-9472. <www.healthywomen.org>.

Debra Gordon

Femoral hernia see **Hernia**

Ferritin test see **Iron tests**

Fetal alcohol syndrome

Definition

Fetal alcohol syndrome (FAS) is a pattern of birth defects, learning, and behavioral problems affecting individuals whose mothers consumed alcohol during pregnancy.

Description

FAS is the most common preventable cause of mental retardation. This condition was first recognized and reported in the medical literature in 1968 in France and in 1973 in the United States. Alcohol is a teratogen, the term used for any drug, chemical, maternal disease or other environmental exposure that can cause **birth defects** or functional impairment in a developing fetus. Some features may be present at birth including low birth weight, **prematurity**, and microcephaly. Characteristic facial features may be present at birth, or may become more obvious over time. Signs of brain damage include delays in development, behavioral abnormalities, and **mental retardation**, but affected individuals exhibit a wide range of abilities and disabilities. It has only been since 1991 that the long-term outcome of FAS has been known. Learning, behavioral, and emotional problems are common in adolescents and adults with FAS. Fetal Alcohol Effect (FAE), a term no longer favored, is sometimes used to describe individuals with some, but not all, of the features of FAS. In 1996, the Institute of Medicine suggested a five-level system to describe the birth defects, learning and behavioral difficulties in offspring of women who drank alcohol during pregnancy. This system contains criteria including confirmation of maternal alcohol exposure, characteristic facial features, growth problems, learning and behavioral problems, and birth defects known to be associated with prenatal alcohol exposure.

The incidence of FAS varies among different populations studied, and ranges from approximately one in 200 to one in 2000 at birth. However, a study reported in 1997, utilizing the Institute of Medicine criteria, estimated the prevalence in Seattle, Washington from 1975–1981 at nearly one in 100 live births. Avoiding alcohol during **pregnancy**, including the earliest weeks of the pregnancy, can prevent FAS. There is no amount of alcohol use during pregnancy that has been proven to be completely safe.

There is no racial or ethnic relation to FAS. Individuals from different genetic backgrounds exposed to similar amounts of alcohol during pregnancy may exhibit different signs or symptoms of FAS. Estimates state that 30–45% of women who consume six or more drinks a day throughout pregnancy will give birth to a child with FAS. The risk of FAS appears to increase as a chronic alcoholic woman progresses in her childbearing years and continues to drink. That is, a child with FAS will often be one of the last born to a chronic alcoholic woman, although older siblings may exhibit milder features

of FAS. Binge drinking, defined as sporadic use of five or more standard alcoholic drinks per occasion, and “moderate” daily drinking (two to four 12 oz bottles of beer, eight to 16 ounces of wine, two to four ounces of liquor) can also result in offspring with features of FAS. Experts say a few binges early in pregnancy—before a woman may even know she is pregnant—may be enough to be dangerous, even if she stops drinking later.

Causes and symptoms

FAS is not a genetic or inherited disorder. It is a pattern of birth defects, learning, and behavioral problems that are the result of maternal alcohol use during the pregnancy. The alcohol freely crosses the placenta and causes damage to the developing embryo or fetus. Alcohol use by the father cannot cause FAS. If a woman who has FAS drinks alcohol during pregnancy, then she may also have a child with FAS. Not all individuals from alcohol exposed pregnancies have obvious signs or symptoms of FAS; individuals of different genetic backgrounds may be more or less susceptible to the damage that alcohol can cause. The dose of alcohol, the time during pregnancy that alcohol is used, and the pattern of alcohol use all contribute to the different signs and symptoms that are found.

Classic features of FAS include short stature, low birthweight and poor weight gain, microcephaly, and a characteristic pattern of facial features. These facial features in infants and children may include small eye openings (measured from inner corner to outer corner), epicanthal folds (folds of tissue at the inner corner of the eye), small or short nose, low or flat nasal bridge, smooth or poorly developed philtrum (the area of the upper lip above the colored part of the lip and below the nose), thin upper lip, and small chin. Some of these features are nonspecific, meaning they can occur in other conditions, or be appropriate for age, racial, or family background. Other major and minor birth defects that have been reported include cleft palate, congenital heart defects, **strabismus**, **hearing loss**, defects of the spine and joints, alteration of the hand creases, small fingernails, and toenails. Since FAS was first described in infants and children, the diagnosis is sometimes more difficult to recognize in older adolescents and adults. Short stature and microcephaly remain common features, but weight may normalize, and the individual may actually become overweight for his/her height. The chin and nose grow proportionately more than the middle part of the face and dental crowding may become a problem. The small eye openings and the appearance of the

upper lip and philtrum may continue to be characteristic. Pubertal changes typically occur at the normal time.

Newborns with FAS may have difficulties with feeding due to a poor suck, have irregular sleep-wake cycles, decreased or increased muscle tone, seizures or **tremors**. Delays in achieving developmental milestones such as rolling over, crawling, walking and talking may become apparent in infancy. Behavior and learning difficulties typical in the preschool or early school years include poor attention span, hyperactivity, poor motor skills, and slow language development. Attention deficit-hyperactivity disorder is a common associated diagnosis. Learning disabilities or mental retardation may be diagnosed during this time. Arithmetic is often the most difficult subject for a child with FAS. During middle school and high school years the behavioral difficulties and learning difficulties can be significant. Memory problems, poor judgment, difficulties with daily living skills, difficulties with abstract reasoning skills, and poor social skills are often apparent by this time. It is important to note that animal and human studies have shown that neurologic and behavioral abnormalities can be present without characteristic facial features. These individuals may not be identified as having FAS, but may fulfill criteria for alcohol-related diagnoses, as set forth by the Institute of Medicine.

In 1991, Streissguth and others reported some of the first long-term follow-up studies of adolescents and adults with FAS. In the approximate 60 individuals they studied, the average IQ was 68, with 70 being the lower limit of the normal range. However, the range of IQ was quite large, as low as 20 (severely retarded) to as high as 105 (normal). The average achievement levels for reading, spelling, and arithmetic were fourth grade, third grade and second grade, respectively. The Vineland Adaptive Behavior Scale was used to measure adaptive functioning in these individuals. The composite score for this group showed functioning at the level of a seven-year-old. Daily living skills were at a level of nine years, and social skills were at the level of a six-year-old.

In 1996, Streissguth and others published further data regarding the disabilities in children, adolescents and adults with FAS. Secondary disabilities, that is, those disabilities not present at birth and that might be preventable with proper diagnosis, treatment, and intervention, were described. These secondary disabilities include: mental health problems; disrupted school experiences; trouble with the law; incarceration for mental health problems, drug **abuse**, or a crime; inappropriate sexual behavior; alcohol and drug

abuse; problems with employment; dependent living; and difficulties parenting their own children. In that study, only seven out of 90 adults were living and working independently and successfully. In addition to the studies by Streissguth, several other authors in different countries have now reported on long-term outcome of individuals diagnosed with FAS. In general, the neurologic, behavioral and emotional disorders become the most problematic for the individuals. The physical features change over time, sometimes making the correct diagnosis more difficult in older individuals, without old photographs and other historical data to review. Mental health problems including attention deficit, depression, panic attacks, **psychosis** and **suicide** threats and attempts, and overall were present in more than 90% of the individuals studied by Streissguth. A 1996 study in Germany reported more than 70% of the adolescents they studied had persistent and severe developmental disabilities and many had psychiatric disorders, the most common of which were emotional disorders, repetitive habits, **speech disorders**, and hyperactivity disorders.

Diagnosis

FAS is a clinical diagnosis, which means that there is no blood, x ray or psychological test that can be performed to confirm the suspected diagnosis. The diagnosis is made based on the history of maternal alcohol use, and detailed physical examination for the characteristic major and minor birth defects and characteristic facial features. It is often helpful to examine siblings and parents of an individual suspected of having FAS, either in person or by photographs, to determine whether findings on the examination might be familial, or if other siblings may also be affected. Sometimes, genetic tests are performed to rule out other conditions that may present with developmental delay or birth defects. Individuals with developmental delay, birth defects or other unusual features are often referred to a clinical geneticist, developmental pediatrician, or neurologist for evaluation and diagnosis of FAS. Psychoeducational testing to determine IQ and/or the presence of learning disabilities may also be part of the evaluation process.

Treatment

There is no treatment for FAS that will reverse or change the physical features or brain damage associated with maternal alcohol use during the pregnancy. Most of the birth defects associated with prenatal alcohol exposure are correctable with

KEY TERMS

Cleft palate—A congenital malformation in which there is an abnormal opening in the roof of the mouth that allows the nasal passages and the mouth to be improperly connected.

Congenital—Refers to a disorder that is present at birth.

IQ—Abbreviation for Intelligence Quotient. Compares an individual's mental age to his/her true or chronological age and multiplies that ratio by 100.

Microcephaly—An abnormally small head.

Miscarriage—Spontaneous pregnancy loss.

Placenta—The organ responsible for oxygen and nutrition exchange between a pregnant mother and her developing baby.

Strabismus—An improper muscle balance of the ocular muscles resulting in crossed or divergent eyes.

Teratogen—Any drug, chemical, maternal disease, or exposure that can cause physical or functional defects in an exposed embryo or fetus.

surgery. Children should have psychoeducational evaluation to help plan appropriate educational interventions. Common associated diagnoses such as attention deficit-hyperactivity disorder, depression, or **anxiety** should be recognized and treated appropriately. The disabilities that present during childhood persist into adult life. However, some of the secondary disabilities mentioned above may be avoided or lessened by early and correct diagnosis, better understanding of the life-long complications of FAS, and intervention. Streissguth has describe a model in which an individual affected by FAS has one or more advocates to help provide guidance, structure and support as the individual seeks to become independent, successful in school or employment, and develop satisfying social relationships.

Prognosis

The prognosis for FAS depends on the severity of birth defects and the brain damage present at birth. **Miscarriage**, **stillbirth** or **death** in the first few weeks of life may be outcomes in very severe cases. Major birth defects associated with FAS are usually treatable with surgery. Some of the factors

that have been found to reduce the risk of secondary disabilities in FAS individuals include diagnosis before the age of six years, stable and nurturing home environments, never having experienced personal violence, and referral and eligibility for disability services. The long-term data helps in understanding the difficulties that individuals with FAS encounter throughout their lifetime and can help families, caregivers and professionals provide the care, supervision, education and treatment geared toward their special needs.

Prevention of FAS is the key. Prevention efforts must include public education efforts aimed at the entire population, not just women of child bearing age, appropriate treatment for women with high-risk drinking habits, and increased recognition and knowledge about FAS by professionals, parents, and caregivers.

Resources

PERIODICALS

Committee of Substance Abuse and Committee on Children with Disabilities. "Fetal Alcohol Syndrome and Alcohol-Related Neurodevelopmental Disorders." *Pediatrics* 106 (August 2000): 358-361.

Cramer, C., and F. Davidhizar. "FAS/FAE: Impact on Children." *Journal of Child Health Care* 3 (Autumn 1999): 31-34.

"Fetal Alcohol Syndrome Is Still a Threat, Says Publication." *Science Letter* September 28, 2004: 448.

Hannigan, J. H., and D. R. Armant. "Alcohol in Pregnancy and Neonatal Outcome." *Seminars in Neonatology* 5 (August 2000): 243-54.

"Prenatal Exposure to Alcohol." *Alcohol Research and Health* 24 (2000): 32-41.

ORGANIZATIONS

Fetal Alcohol Syndrome Family Resource Institute. PO Box 2525, Lynnwood, WA 98036. (253) 531-2878 or (800) 999-3429. <<http://www.fetalalcoholsyndrome.org>> .

Institute of Medicine. National Academy Press, Washington, DC. <<http://www.come-over.to/FAS/IOMsummary.htm>> .

March of Dimes Birth Defects Foundation. 1275 Mamaroneck Ave., White Plains, NY 10605. (888) 663-4637. resourcecenter@modimes.org. <<http://www.modimes.org>> .

Nofas. 216 G St. NE, Washington, DC 20002. (202) 785-4585. <<http://www.nofas.org>> .

Laurie Heron Seaver
Teresa G. Odle

Fetal death see **Stillbirth**

Fetal hemoglobin test

Definition

Fetal hemoglobin (Hemoglobin F), Alkali-resistant hemoglobin, HbF (or Hb F), is the major hemoglobin component in the bloodstream of the fetus. After birth, it decreases rapidly until only traces are found in normal children and adults.

Purpose

The determination of fetal hemoglobin is an aid in evaluating low concentrations of hemoglobin in the blood (anemia), as well as the hereditary persistence of fetal hemoglobin, and a group of inherited disorders affecting hemoglobin, among which are the thalassemias and sickle cell anemia.

Description

At birth, the newborn's blood is comprised of 60%–90% of fetal hemoglobin. The fetal hemoglobin then rapidly decreases to 2% or less after the second to fourth years. By the time of adulthood, only traces (0.5% or less) are found in the bloodstream.

In some diseases associated with abnormal hemoglobin production (see Hemoglobinopathy, below), fetal hemoglobin may persist in larger amounts. When this occurs, the elevation raises the question of possible underlying disease.

For example, HbF can be found in higher levels in hereditary hemolytic **anemias**, in all types of leukemias, in **pregnancy**, diabetes, thyroid disease, and during anticonvulsant drug therapy. It may also reappear in adults when the bone marrow is overactive, as in the disorders of **pernicious anemia**, **multiple myeloma**, and metastatic **cancer** in the marrow. When HbF is increased after age four, it should be investigated for cause.

Hemoglobinopathy

Hemoglobin is the oxygen-carrying pigment found in red blood cells. It is a large molecule made in the bone marrow from two components, heme and globin.

Defects in hemoglobin production may be either genetic or acquired. The genetic defects are further subdivided into errors of heme production (porphyria), and those of globin production (known collectively as the **hemoglobinopathies**).

There are two categories of hemoglobinopathy. In the first category, abnormal globin chains give rise

KEY TERMS

Anemia—A disorder characterized by a reduced blood level of hemoglobin, the oxygen-carrying pigment of blood.

Hemolytic anemia—A form of anemia caused by premature destruction of red cells in the blood stream (a process called hemolysis). Hemolytic anemias are classified according to whether the cause of the problem is inside the red blood cell (in which case it is usually an inherited condition), or outside the cell (usually acquired later in life).

to abnormal hemoglobin molecules. In the second category, normal hemoglobin chains are produced but in abnormal amounts. An example of the first category is the disorder of sickle cell anemia, the inherited condition characterized by curved (sickle-shaped) red blood cells and chronic **hemolytic anemia**. Disorders in the second category are called the thalassemias, which are further divided into types according to which amino acid chain is affected (alpha or beta), and whether there is one defective gene (**thalassemia minor**) or two defective genes (thalassemia major).

Preparation

This test requires a blood sample. The patient is not required to be in a **fasting** state (nothing to eat or drink for a period of hours before the test).

Risks

Risks for this test are minimal, but may include slight bleeding from the blood-drawing site, fainting or feeling lightheaded after venipuncture, or hematoma (blood accumulating under the puncture site).

Normal results

Reference values vary from laboratory to laboratory but are generally found within the following ranges:

- six months to adult: up to 2% of the total hemoglobin
- newborn to six months: up to 75% of the total hemoglobin

Abnormal results

Greater than 2% of total hemoglobin is abnormal.

Resources

BOOKS

Pagana, Kathleen Deska. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis: Mosby, Inc., 1998.

Janis O. Flores

Fetishes see **Sexual perversions**

Fever

Definition

A fever is any body temperature elevation over 100 °F (37.8 °C).

Description

A healthy person's body temperature fluctuates between 97 °F (36.1 °C) and 100 °F (37.8 °C), with the average being 98.6 °F (37 °C). The body maintains stability within this range by balancing the heat produced by the metabolism with the heat lost to the environment. The "thermostat" that controls this process is located in the hypothalamus, a small structure located deep within the brain. The nervous system constantly relays information about the body's temperature to the thermostat, which in turn activates different physical responses designed to cool or warm the body, depending on the circumstances. These responses include: decreasing or increasing the flow of blood from the body's core, where it is warmed, to the surface, where it is cooled; slowing down or speeding up the rate at which the body turns food into energy (metabolic rate); inducing shivering, which generates heat through muscle contraction; and inducing sweating, which cools the body through evaporation.

A fever occurs when the thermostat resets at a higher temperature, primarily in response to an infection. To reach the higher temperature, the body moves blood to the warmer interior, increases the metabolic rate, and induces shivering. The "chills" that often accompany a fever are caused by the movement of blood to the body's core, leaving the surface and extremities cold. Once the higher temperature is achieved, the shivering and chills stop. When the infection has been overcome or drugs such as **aspirin** or **acetaminophen** (Tylenol) have been taken, the thermostat resets to normal and the body's cooling

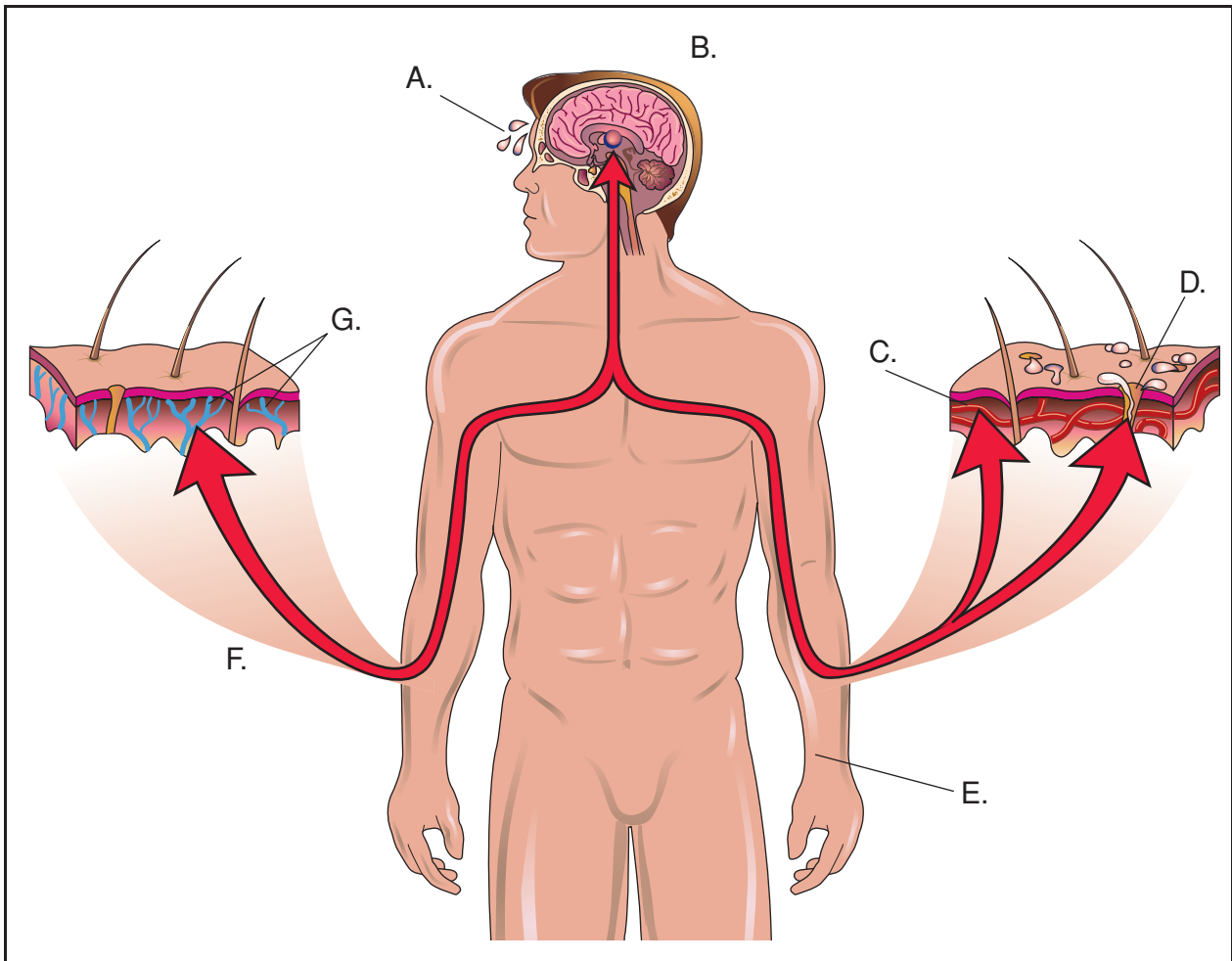
mechanisms switch on: the blood moves to the surface and sweating occurs.

Fever is an important component of the immune response, though its role is not completely understood. Physicians believe that an elevated body temperature has several effects. The immune system chemicals that react with the fever-inducing agent and trigger the resetting of the thermostat also increase the production of cells that fight off the invading bacteria or viruses. Higher temperatures also inhibit the growth of some bacteria, while at the same time speeding up the chemical reactions that help the body's cells repair themselves. In addition, the increased heart rate that may accompany the changes in blood circulation also speeds the arrival of white blood cells to the sites of infection.

Causes and symptoms

Fevers are primarily caused by viral or bacterial infections, such as **pneumonia** or **influenza**. However, other conditions can induce a fever, including allergic reactions; autoimmune diseases; trauma, such as breaking a bone; **cancer**; excessive exposure to the sun; intense exercise; hormonal imbalances; certain drugs; and damage to the hypothalamus. When an infection occurs, fever-inducing agents called pyrogens are released, either by the body's immune system or by the invading cells themselves, that trigger the resetting of the thermostat. In other circumstances, the immune system may overreact (allergic reactions) or become damaged (autoimmune diseases), causing the uncontrolled release of pyrogens. A stroke or tumor can damage the hypothalamus, causing the body's thermostat to malfunction. Excessive exposure to the sun or intensely exercising in hot weather can result in heat **stroke**, a condition in which the body's cooling mechanisms fail. Malignant hyperthermia is a rare, inherited condition in which a person develops a very high fever when given certain anesthetics or **muscle relaxants** in preparation for surgery.

How long a fever lasts and how high it may go depends on several factors, including its cause, the age of the patient, and his or her overall health. Most fevers caused by infections are acute, appearing suddenly and then dissipating as the immune system defeats the infectious agent. An infectious fever may also rise and fall throughout the day, reaching its peak in the late afternoon or early evening. A low-grade fever that lasts for several weeks is associated with autoimmune diseases such as lupus or with some cancers, particularly leukemia and lymphoma.



A dramatic rise in body temperature often includes the following symptoms: **A.** Loss of fluid results in dehydration. **B.** The hypothalamic set-point is increased, raising metabolism. **C.** Blood vessels in skin dilate. **D.** Sweat glands produce excess perspiration. **E.** Increased pulse rate. **F.** Increased hypothalamic set-point may introduce chills and shivering to promote heat production from muscles. **G.** Skin becomes more heat-sensitive. (Illustration by Electronic Illustrators Group.)

Diagnosis

A fever is usually diagnosed using a thermometer. A variety of different thermometers are available, including traditional glass and mercury ones used for oral or rectal temperature readings and more sophisticated electronic ones that can be inserted in the ear to quickly register the body's temperature. For adults and older children, temperature readings are usually taken orally. Younger children who cannot or will not hold a thermometer in their mouths can have their temperature taken by placing an oral thermometer under their armpit. Infants generally have their temperature taken rectally using a rectal thermometer.

As important as registering a patient's temperature is determining the underlying cause of the fever. The

presence or absence of accompanying symptoms, a patient's medical history, and information about what he or she may have ingested, any recent trips taken, or possible exposures to illness help the physician make a diagnosis. Blood tests can aid in identifying an infectious agent by detecting the presence of antibodies against it or providing samples for growth of the organism in a culture. Blood tests can also provide the doctor with white blood cell counts. Ultrasound tests, magnetic resonance imaging (MRI) tests, or computed tomography (CT) scans may be ordered if the doctor cannot readily determine the cause of a fever.

Treatment

Physicians agree that the most effective treatment for a fever is to address its underlying cause, such as

KEY TERMS

Antipyretic—A drug that lowers fever, like aspirin or acetaminophen.

Autoimmune disease—Condition in which a person's immune system attacks the body's own cells, causing tissue destruction.

Febrile seizure—Convulsions brought on by fever.

Malignant hyperthermia—A rare, inherited condition in which a person develops a very high fever when given certain anesthetics or muscle relaxants in preparation for surgery.

Meningitis—A potentially fatal inflammation of the thin membrane covering the brain and spinal cord.

Metabolism—The chemical process by which the body turns food into energy, which can be given off as heat.

Pyrogen—A chemical circulating in the blood that causes a rise in body temperature.

Reye's syndrome—A disorder principally affecting the liver and brain, marked by the rapid development of life-threatening neurological symptoms.

through the administration of **antibiotics**. Also, because a fever helps the immune system fight infection, it usually should be allowed to run its course. Drugs to lower fever (antipyretics) can be given if a patient (particularly a child) is uncomfortable. These include aspirin, acetaminophen (Tylenol), and ibuprofen (Advil). Aspirin, however, should not be given to a child or adolescent with a fever since this drug has been linked to an increased risk of **Reye's syndrome**. Bathing a patient in cool water can also help alleviate a high fever.

A fever requires emergency treatment under the following circumstances:

- newborn (three months or younger) with a fever over 100.5 °F (38 °C)
- infant or child with a fever over 103 °F (39.4 °C)
- fever accompanied by severe **headache**, neck stiffness, mental confusion, or severe swelling of the throat

A very high fever in a small child can trigger seizures (febrile seizures) and therefore should be treated immediately. A fever accompanied by the above symptoms can indicate the presence of a serious infection, such as **meningitis**, and should be brought to the immediate attention of a physician.

Prognosis

Most fevers caused by infection end as soon as the immune system rids the body of the pathogen and do not produce any lasting effects. The prognosis for fevers associated with more chronic conditions, such as autoimmune disease, depends upon the overall outcome of the disorder.

Resources

BOOKS

Gelfand, Jeffrey. "Fever, Including Fever of Unknown Origin." In *Harrison's Principles of Internal Medicine*, edited by Anthony S. Fauci, et al. New York: McGraw-Hill, 1997.

Bridget Travers

Fever blister see **Cold sore**

Fever evaluation tests

Definition

Fever evaluation tests, better known as febrile agglutinins tests, are performed to detect the presence of antibodies in the blood that are sensitive to temperature changes. Antibodies are proteins produced by the immune system in response to specific infectious agents, such as viruses or bacteria. Febrile agglutinins are antibodies that cause red blood cells to clump, but only when the blood is warmed to temperatures higher than the average body temperature of 98.6 °F (37 °C).

Purpose

The febrile agglutinins test is used to confirm the diagnosis of certain infectious diseases that stimulate the body to produce febrile agglutinins. The disease most commonly diagnosed by this test is **brucellosis**, a infection caused by bacteria belonging to the genus *Brucella* and characterized by intermittent fever, sweating, chills, aches, and mental depression. The test is also used to diagnose certain other infectious diseases: salmonellosis, caused by *Salmonella* bacteria and marked by **nausea** and severe **diarrhea**; rickettsial infections, a group of diseases caused by the bacteria *Rickettsia*; and **tularemia**, also called rabbit fever, a bacterial infection characterized by a high fever and swollen lymph nodes. The febrile agglutinins test can also be used to confirm the presence of two types of

cancer, leukemia and lymphoma; however, doctors rarely use the test for this purpose, since other diagnostic tests are more reliable.

Description

A febrile agglutinins test can be performed at a doctor's office or a hospital. A nurse or technician will collect a few drops of blood (about 7ml) in a small tube that has been cooled slightly. The specimen is then taken to a laboratory where it heated and examined for clumping. If the cells clump after warming and unclump as they cool, a febrile agglutinin titer (concentration) of greater than 1:80 is present.

Normal results

The results of febrile agglutinins tests require a doctor's interpretation. In general, however, a normal value is lower than 1:32.

Abnormal results

An value higher than 1:80 suggests a diagnosis for brucellosis or one of the other conditions indicated by this test.

Jill S. Lasker

Fever of unknown origin

Definition

Fever of unknown origin (FUO) refers to the presence of a documented fever for a specified time, for which a cause has not been found after a basic medical evaluation. The classic criteria developed in 1961 included: temperature greater than 101 °F (38.3 °C), for at least three weeks, and inability to find a cause after one week of study. Within the past decade, a revision has been proposed that categorizes FUO into classic, hospital acquired FUO, FUO associated with low white blood counts, and HIV associated FUO (**AIDS** related).

Description

Fever is a natural response of the body that helps in fighting off foreign substances, such as microorganisms, toxins, etc. Body temperature is set by the thermoregulatory center, located in an area in the brain called hypothalamus. Body temperature is not constant all day, but actually is lowest at 6 A.M. and

highest around 4–6 P.M. In addition, temperature varies in different regions of the body; for example, rectal and urine temperatures are about one degree Fahrenheit higher than oral temperature and rectal temperature is higher than urine. It is also important to realize that certain normal conditions can effect body temperature, such as **pregnancy**, food ingestion, age, and certain hormonal changes.

Substances that cause fever are known as “pyrogens.” There are two types of pyrogens; exogenous and endogenous. Those that originate outside the body, such as bacterial toxins, are called “exogenous” pyrogens. Pyrogens formed by the body's own cells in response to an outside stimulus (such as a bacterial toxin) are called “endogenous” pyrogens.

Researchers have discovered that there are several “endogenous” pyrogens. These are made up of small groups of amino acids, the building blocks of proteins. These natural pyrogens have other functions in addition to inducing fever; they have been named “cytokines”. When cytokines are injected into humans, fever and chills develop within an hour. Interferon, tumor necrosis factor, and various interleukins are the major fever producing cytokines.

The production of fever is a very complex process; somehow, these cytokines cause the thermoregulatory center in the hypothalamus to reset the normal temperature level. The body's initial response is to conserve heat by vasoconstriction, a process in which blood vessels narrow and prevent heat loss from the skin and elsewhere. This alone will raise temperature by two to three degrees. Certain behavioral activities also occur, such as adding more clothes, seeking a warmer environment, etc. If the hypothalamus requires more heat, then shivering occurs.

Fever is a body defense mechanism. It has been shown that one of the effects of temperature increase is to slow bacterial growth. However, fever also has some downsides; the body's metabolic rate is increased and with it, oxygen consumption. This can have a devastating effect on those with poor circulation. In addition, fever can lead to seizures in the very young.

When temperature elevation occurs for an extended period of time and no cause is found, the term FUO is then used. The far majority of these patients are eventually found to have one of several diseases.

Causes and symptoms

The most frequent cause of FUO is still infection, though the percentage has decreased in recent years. **Tuberculosis** remains an important cause, especially

when it occurs outside the lungs. The decrease in infections as a cause of FUO is due in part to improved culture techniques. In addition, technological advances have made it easier to diagnose non-infectious causes. For example, tumors and autoimmune diseases in particular are now easier to diagnose. (An autoimmune disease is one that arises when the body tolerance for its own cell antigenic cell markers disappears.)

Allergies to medications can also cause prolonged fever; sometimes patients will have other symptoms suggesting an allergic reaction, such as a rash.

There are many possible causes of FUO; generally though, a diagnosis can be found. About 10% of patients will wind up without a definite cause, and about the same percentage have “factitious fevers” (either self induced or no fever at all).

Some general symptoms tend to occur along with fever; these are called constitutional symptoms and consist of myalgias (muscle aches), chills, and **headache**.

Diagnosis

Few symptoms in medicine present such a diagnostic challenge as fever. Nonetheless, if a careful, logical, and thorough evaluation is performed, a diagnosis will be found in most cases. The patient’s past medical history as well as travel, social, and family history should be carefully searched for important clues.

Usually the first step is to search for an infectious cause. Skin and other screening tests for diseases such as tuberculosis, and examination of blood, urine, and stool, are generally indicated. Antibody levels to a number of infectious agents can be measured; if these are rising, they may point to an active infection.

Various x-ray studies are also of value. In addition to standard examinations, recently developed radiological techniques using ultrasound, computed tomography scan (CT scan) and **magnetic resonance imaging** (MRI) scans are now available. These enable physicians to examine areas that were once accessible only through surgery. Furthermore, new studies using radioactive materials (nuclear medicine), can detect areas of infection and inflammation previously almost impossible to find, even with surgery.

Biopsies of any suspicious areas found on an x-ray exam can be performed by either traditional or newer surgical techniques. Material obtained by biopsy is then examined by a pathologist to look for clues as to the cause of the fever. Evidence of infection, tumor or other diseases can be found in this way. Portions of the biopsy are also sent to the laboratory for culture in an attempt to grow and identify an infectious organism.

KEY TERMS

AIDS—Acquired immune deficiency syndrome is often represented by these initials. The disease is associated with infection by the human immunodeficiency virus (HIV), and has the main feature of repeated infections, due to failure of certain parts of the immune system. Infection by HIV damages part of the body’s natural immunity, and leads to recurrent illnesses.

Antibiotic—A medication that is designed to kill or weaken bacteria.

Computed tomography scan (CT Scan)—A specialized x-ray procedure in which cross-sections of the area in question can be examined in detail. This allows physicians to examine organs such as the pancreas, bile ducts, and others which are often the site of hidden infections.

Magnetic Resonance Imaging (MRI)—This is a new technique similar to CT Scan, but based on the magnetic properties of various areas of the body to compose images.

NSAID—Nonsteroidal anti-inflammatory drugs are medications such as aspirin and ibuprofen that decrease pain and inflammation. Many can now be obtained without a doctor’s prescription.

Ultrasound—A non-invasive procedure based on changes in sound waves of a frequency that cannot be heard, but respond to changes in tissue composition. It is very useful for diagnosing diseases of the gallbladder, liver, and hidden infections, such as abscesses.

Patients with HIV are an especially difficult problem, as they often suffer from many unusual infections. HIV itself is a potential cause of fever.

Treatment

Most patients who undergo evaluation for FUO do not receive treatment until a clear-cut cause is found. **Antibiotics** or medications designed to suppress a fever (such as NSAIDs) will only hide the true cause. Once physicians are satisfied that there is no infectious cause, they may use medications such as NSAIDs, or **corticosteroids** to decrease inflammation and diminish constitutional symptoms.

The development of FUO in certain settings, such as that acquired by patients in the hospital or in those with a low white **blood count**, often needs rapid

treatment to avoid serious complications. Therefore, in these instances patients may be placed on antibiotics after a minimal number of diagnostic studies. Once test results are known, treatment can be adjusted as needed.

Prognosis

The outlook for patients with FUO depends on the cause of the fever. If the basic illness is easily treatable and can be found rather quickly, the potential for a cure is quite good. Some patients continue with temperature elevations for 6 months or more; if no serious disease is found, medications such as NSAIDs are used to decrease the effects of the fever. Careful follow-up and reevaluation is recommended in these cases.

Resources

BOOKS

Gelfand, Jeffrey A., and Charles A. Dinarello. "Fever of Unknown Origin." In *Harrison's Principles of Internal Medicine*, edited by Anthony S. Fauci, et al. New York: McGraw-Hill, 1997.

David Kaminstein, MD

Fiber-modified diet see **Diets**

Fibrin degradation products see **Fibrin split products**

Fibrin split products

Definition

Fibrin split products (FSP) are fragments of protein released from a dissolving clot. The fibrin split products test is one of several tests done to evaluate a person with blood clotting problems (coagulation), particularly disseminated intravascular coagulation (DIC).

Purpose

High levels of FSP in a person's blood are associated with DIC, a serious medical condition that develops when the normal balance between bleeding and clotting is disturbed. Excessive bleeding and clotting injures body organs, and causes anemia or death.

Description

Coagulation begins typically with an injury to some part of the body. The injury sets in motion a cascade of biochemical activities (the coagulation cascade) to stop the bleeding, by forming a clot from a mixture of the blood protein fibrin and platelets.

Once bleeding is stopped, another blood protein dissolves the clot by breaking down the fibrin into fragments. Measurement of these fragments gives information about the clot dissolving portion of coagulation, called fibrinolysis.

In DIC, the coagulation cascade is triggered in an abnormal way. A blood infection, a transfusion reaction, a large amount of tissue damage, such as a burn, a dead fetus, and some cancers can begin the chain of biochemical events leading to blood clots. The coagulation cascade becomes overwhelmed with excessive clotting followed by excessive bleeding. As the large number of clots dissolve, fibrin split products accumulate in the blood and encourage even more bleeding.

Laboratory tests for FSP are done on the yellow liquid portion left over after blood clots (serum). A person's serum is mixed with a substance that binds to FSP. This bound complex is measured, and the original amount of FSP is determined. Some test methods give an actual measurement of FSP; some give a titer, or dilution. Methods that provide a titer look for the presence or absence of FSP. If the serum is positive for FSP, the serum is diluted, or titered, and the test is done again. These steps are repeated until the serum is so dilute that it no longer gives a positive result. The last dilution that gives a positive result is the titer reported.

The FSP test is covered by insurance when medically necessary. Results are usually available within one to two hours. Other names for this test are fibrin degradation products, fibrin breakdown products, or FDP.

Preparation

This test requires 0.17 oz (5/14m) of blood. A healthcare worker ties a tourniquet on the patient's upper arm, locates a vein in the inner elbow region, and inserts a needle into that vein. Vacuum action draws the blood through the needle into an attached tube. Collection of the sample takes only a few minutes.

Aftercare

Discomfort or bruising may occur at the puncture site. Pressure applied to the puncture site until the

KEY TERMS

Coagulation—The entire process of blood clotting.

Coagulation cascade—A sequence of biochemical activities to stop bleeding by forming a clot.

Disseminated intravascular coagulation (DIC)—A serious medical condition that develops when the normal balance between bleeding and clotting is disturbed. Excessive bleeding and clotting injures body organs, and causes anemia or death.

Fibrin split products (FSP)—Pieces of the protein fibrin released from a dissolving clot.

Fibrinolysis—The clot dissolving portion of the coagulation process.

Titer—A dilution of a substance with an exact known amount of fluid. For example, one part of serum diluted with four parts of saline is a titer of 1:4.

bleeding stops reduces bruising. Warm packs to the puncture site relieve discomfort. The patient may feel dizzy or faint.

Risks

People with coagulation problems may bleed longer than normal. The healthcare provider must make sure bleeding has stopped before leaving the patient unattended.

Normal results

Negative at a less than or equal to 1:4 dilution or less than 10 g/mL.

Abnormal results

High levels of FSP indicate DIC. Results of the test must be interpreted by the physician according to the person's clinical symptoms and medical history. Other conditions that increase blood clotting activity also increase FSP: venous thrombosis, surgery and transplants, blood clots in the lung, certain cancers, and **heart attack** (myocardial infarction).

Resources

BOOKS

Miller, Jonathan L. "Blood Coagulation and Fibrinolysis." In *Clinical Diagnosis and Management by Laboratory Methods*, edited by John B. Henry., 19th ed. Philadelphia: W. B. Saunders Co., 1996.

PERIODICALS

Hardaway, Robert M., and Charles H. Williams.

"Disseminated Intravascular Coagulation: An Update." *Comprehensive Therapy* November 1996: 737-743.

Nancy J. Nordenson

Fibrinogen test

Definition

Fibrinogen (Factor I) is a protein that originates in the liver. It is converted to fibrin during the blood-clotting process (coagulation).

Purpose

The fibrinogen test aids in the diagnosis of suspected clotting or bleeding disorders caused by fibrinogen abnormalities.

Precautions

This test is not recommended for patients with active bleeding, acute infection or illness, or in those patients who have received blood transfusions within four weeks.

Drugs that may increase fibrinogen levels include estrogens and oral contraceptives. Drugs that may cause decreased levels include anabolic steroids, androgens, phenobarbital, urokinase, streptokinase, and valproic acid.

Description

Fibrinogen plays two essential roles in the body: it is a protein called an acute-phase reactant that becomes elevated with tissue inflammation or tissue destruction, and it is also a vital part of the "common pathway" of the coagulation process.

In order for blood to clot, fibrinogen must be converted to fibrin by the action of an enzyme called thrombin. Fibrin molecules clump together to form long filaments, which trap blood cells to form a solid clot.

The conversion of fibrinogen to fibrin is the last step of the "coagulation cascade," a series of reactions in the blood triggered by tissue injury and platelet activation. With each step in the cascade, a

coagulation factor in the blood is converted from an inactive to an active form. The active form of the factor then activates several molecules of the next factor in the series, and so on, until the final step, when fibrinogen is converted into fibrin.

The factors involved in the coagulation cascade are numbered I, II, and V through XIII. Factor I is fibrinogen, while factor II (fibrinogen's immediate precursor) is called prothrombin. Most of the coagulation factors are made in the liver, which needs an adequate supply of vitamin K to manufacture the different clotting factors.

When fibrinogen acts as an “acute-phase reactant,” it rises sharply during tissue inflammation or injury. When this occurs, high fibrinogen levels may be a predictor for an increased risk of heart or circulatory disease. Other conditions in which fibrinogen is elevated are cancers of the stomach, breast, or kidney, and inflammatory disorders like **rheumatoid arthritis**.

Reduced fibrinogen levels can be found in **liver disease**, prostate cancer, lung disease, bone marrow lesions, malnourishment, and certain bleeding disorders. The low levels can be used to evaluate disseminated intravascular coagulation (DIS), a serious medical condition that develops when there is a disturbed balance between bleeding and clotting. Other conditions related to decreased fibrinogen levels are those in which fibrinogen is completely absent (congenital afibrinogenemia), conditions in which levels are low (hypofibrinogenemia), and conditions of abnormal fibrinogen (dysfibrinogenemia). Obstetric complications or trauma may also cause low levels. Large-volume blood transfusions cause low levels because banked blood does not contain fibrinogen.

Preparation

This test is performed with a blood sample, which can be drawn at any time of day. The patient does not have to be **fasting** (nothing to eat or drink).

Aftercare

Because a fibrinogen test is often ordered when a bleeding disorder is suspected, the patient should apply pressure or a pressure dressing to the blood-drawn site for a period of time after blood is drawn, and then reexamine the site for bleeding.

Risks

Risks for this test are minimal, but may include slight bleeding from the blood-drawing site, fainting or feeling lightheaded after procedure, or the seeing

KEY TERMS

Fibrin—The last step in the coagulation process. Fibrin forms strands that add bulk to a forming blood clot to hold it in place and help “plug” an injured blood vessel wall.

Platelet—An irregularly shaped cell-like particle in the blood that is an important part of blood clotting. Platelets are activated when an injury causes a blood vessel to break. They change shape from round to spiny, “sticking” to the broken vessel wall and to each other to begin the clotting process.

Prothrombin—A type of protein called a glycoprotein that is converted to thrombin during the clotting process.

Thrombin—An enzyme that converts fibrinogen into strands of fibrin.

the accumulation of blood under the puncture site (hematoma).

Normal results

Normal reference ranges are laboratory-specific, but are usually within the following:

- adult: 200 mg/dL–400 mg/dL
- newborn: 125 mg/dL–300 mg/dL

Abnormal results

Spontaneous bleeding can occur with values less than 100 mg/dL.

Resources

BOOKS

Pagana, Kathleen Deska. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis: Mosby, Inc., 1998.

Janis O. Flores

Fibroadenoma

Definition

Fibroadenomas are benign breast tumors commonly found in young women. Fibroadenoma means “a tumor composed of glandular (related to gland) and fibrous (containing fibers) tissues.”

Description

Breast fibroadenomas, abnormal growths of glandular and fibrous tissues, are most common between the ages of 15 and 30, and are found in 10% of all women (20% of African-American women). They are found rarely in postmenopausal women.

Described as feeling like marbles, these firm, round, movable, and “rubbery” lumps range from 1–5 cm in size. Giant fibroadenomas are larger, lemon-sized lumps. Usually single, from 10–15% of women have more than one.

While some types of breast lumps come and go during the menstrual cycle, fibroadenomas typically do not disappear after a woman’s period, and should be checked by a doctor.

Causes and symptoms

The cause of breast fibroadenomas is unknown. They may be dependent upon estrogen, because they are common in premenopausal women, can be found in postmenopausal women taking estrogen, and because they grow larger in pregnant women.

Fibroadenomas usually cause no symptoms and may be discovered during **breast self-examination**, or during a routine check-up.

Diagnosis

When the doctor takes a complete medical history, they will ask when the lump was first noticed, if there were any symptoms or changes in lump size, and if there is any personal or family history of breast disease.

The doctor thoroughly feels the breasts (palpates). Tests are done, usually including mammography or ultrasound scans, or surgical removal of cells or tissue for examination under a the microscope (biopsy).

Diagnostic tests include:

- Mammogram. An x-ray examination of the breast.
- Ultrasound scan. A technique that uses sound waves to display a two-dimensional image of the breast, showing whether a lump is solid or fluid-filled (cystic).
- Fine-needle aspiration biopsy. A minor procedure wherein fluid or cells are drawn out of the lump through a small needle (aspirated).
- Core biopsy. A procedure wherein a larger piece of tissue is withdrawn from the lump through a larger needle.

KEY TERMS

Aspiration —To withdraw material with a needle and syringe.

Biopsy —To remove cells or tissue for microscopic examination.

Estrogen —Female sex hormone produced by the ovaries.

- Incisional biopsy. A surgical procedure wherein a piece of the lump is removed through an cut (incision).
- Excisional biopsy. A surgical procedure wherein the entire lump is removed through an cut (incision).

Most insurance plans cover the costs of diagnosing and treating fibroadenomas.

Treatment

Performed usually in outpatient settings, breast fibroadenomas are removed by **lumpectomy**, or surgical excision under local or **general anesthesia**. Sometimes lumps in younger women are not removed but are monitored by self-examination, yearly doctor check-ups, and mammograms. Surgery is generally recommended for women over 30, and for lumps that are painful or enlarging.

Alternative treatment

Alternative treatments for breast fibroadenomas include a low-fat, high-fiber, vegetarian-type diet; a reduction in **caffeine** intake; supplementation with evening primrose oil (*Oenothera biennis*), flax oil, or fish oil and **vitamins** E and C; and the application of hot compresses to the breast. In addition, a focus on liver cleansing is important to assist the body in conjugation and elimination of excess estrogens. Botanical remedies can be useful in hormone balancing, as can **acupuncture** and homeopathy. Massaging the breasts with castor oil, straight or infused with herbs or essential oils, can help fibroadenomas reduce and dissipate, as well as keep women in touch with changes in their breast tissue.

Prognosis

Breast fibroadenomas are not cancerous. The lumps recur in up to 20% of women. A small number of lumps disappear on their own.

Prevention

Breast fibroadenomas cannot be prevented. They can be discovered early by regular breast self-examination.

Resources

ORGANIZATIONS

American College of Obstetricians and Gynecologists. 409 12th Street, S.W., PO Box 96920.

Mercedes McLaughlin

Fibrocystic breast disease see **Fibrocystic condition of the breast**

Fibrocystic condition of the breast

Definition

Fibrocystic condition of the breast is a term that may refer to a variety of symptoms: breast lumpiness or tenderness, microscopic breast tissue, and/or the x ray or ultrasound picture of the breast. It has been called a “wastebasket” diagnosis because a wide range of vaguely defined benign breast conditions may be labeled as fibrocystic condition. It is not a **cancer**, and the majority of types of fibrocystic conditions do not increase the risk of **breast cancer**.

Description

There is no such thing as a normal or typical female breast. Breasts come in all shapes and sizes, with varying textures from smooth to extremely lumpy. The tissues of the female breast change in response to hormone levels, normal **aging**, nursing (lactation), weight fluctuations, and injury. To further complicate matters, the breast has several types of tissue; each of these tissue types may respond differently to changes in body chemistry.

Fibrocystic breast condition may be called fibrocystic disease, although it is clearly not a single, specific disease process. Variations or changes in the way the breast feels or looks on x ray may cause the condition to be called “fibrocystic change.” Other names have been used to refer to this imprecise and ill-defined term: mammary dysplasia, mastopathy, chronic cystic **mastitis**, indurative mastopathy, mastalgia, lumpy breasts, or physiologic nodularity.

Estimates vary, but 40–90% of all women have some evidence of “fibrocystic” condition, change, or disease. It is most common among women between the ages 30 and 50, but may be seen at other ages.

Causes and symptoms

Fibrocystic condition of the breast refers to technical findings on diagnostic testing (signs); however, this discussion focuses on symptoms that may fall under the general category of the fibrocystic condition. First, a brief review of the structure and function of the breast may be useful.

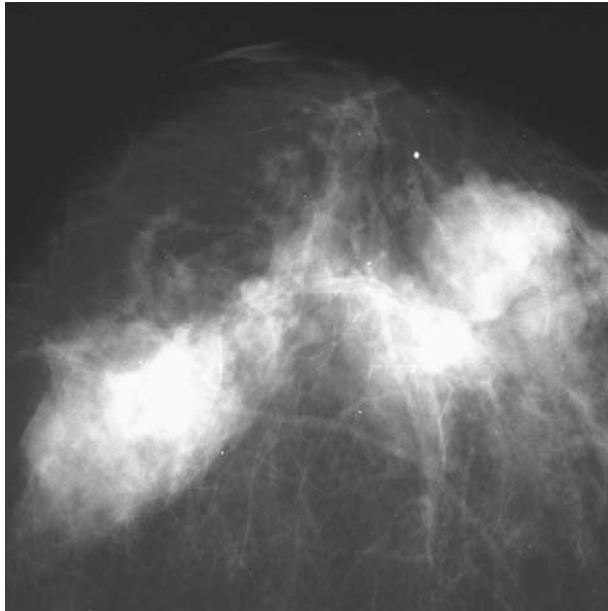
The breast is not supposed to be a soft, smooth organ. It is actually a type of sweat gland. Milk, the breasts’ version of sweat, is secreted when the breast receives appropriate hormonal and environmental stimulation.

The normal breast contains milk glands, with their accompanying ducts, or pipelines, for transporting the milk. These complex structures may not only alter in size, but can increase or decrease in number as needed. Fibrous connective tissue, fatty tissue, nerves, blood and lymph vessels, and lymph nodes, with their different shapes and textures, lie among the ever-changing milk glands. It is no wonder that a woman’s breasts may not feel uniform in texture and that the “lumpiness” may wax and wane.

The fibrocystic condition refers to the tenderness, enlargement, and/or changing “lumpiness” that many women encounter just before or during their menstrual periods. At this time, female hormones are preparing the breasts for **pregnancy**, by stimulating the milk-producing cells, and storing fluid. Each breast may contain as much as three to six teaspoons of excess fluid. Swelling, with increased sensitivity or **pain**, may result. If pregnancy does not occur, the body reabsorbs the fluid, and the engorgement and discomfort are relieved.

Symptoms of fibrocystic breast condition range from mildly annoying in some women to extremely painful in others. The severity of discomfort may vary from month to month in the same woman. Although sometimes distressing, this experience is the body’s normal response to routine hormonal changes.

This cycle of breast sensitivity, pain and/or enlargement, can also result from medications. Some hormone replacement therapies (estrogen and progesterone) used for postmenopausal women can produce these effects. Other medications, primarily, but not exclusively those with hormones may also provoke these symptoms.



A mammogram of a female breast indicating multiple cysts.
(Custom Medical Stock Photo. Reproduced by permission.)

Breast pain unrelated to hormone shifts is called “noncyclic” pain. “Trigger-zone breast pain” is a term that may also be used to describe this area-specific pain. This type of pain may be continuous, or it may be felt intermittently. Trauma, such as a blow to the chest area, a prior breast biopsy, or sensitivity to certain medications may also underlie this type of pain. Fibrocystic condition of the breast may be cited as the cause of otherwise unexplained breast pain.

Lumps, apart from those clearly associated with hormone cycles, may also be placed under the heading of fibrocystic condition. These lumps stand out from enlarged general breast tissue. Although noncancerous lumps may occur, the obvious concern with such lumps is cancer.

Noncancerous breast lumps include:

- **Adenosis.** This condition refers to the enlargement of breast lobules, which contain a greater number of glands than usual. If a group of lobules are found near each other, the affected area may be large enough to be felt.
- **Cysts.** These are fluid-filled sacs in the breast and probably develop as ducts that become clogged with old cells in the process of normal emptying and filling. Cysts usually feel soft and round or oval. However a cyst deep within the breast may feel hard, as it pushes up against firmer breast tissue. A woman with a cyst may experience pain, especially if it increases in size before her menstrual cycle, as is

often the case. Women between the age of 30 and 50 are most likely to develop cysts.

- **Epithelial hyperplasia.** Also called proliferative breast disease, this condition refers to an overgrowth of cells lining either the ducts or the lobules.
- **Fibroadenomas.** These are tumors that form in the tissues outside the milk ducts. The cause of fibroadenomas is unknown. They generally feel smooth and firm, with a somewhat rubber-like texture. Typically a **fibroadenoma** is not attached to surrounding tissue and moves slightly when touched. They are most commonly found in adolescents and women in their early twenties but can occur at any age.
- **Fibrosis.** Sometimes one area of breast tissue persistently feels thicker or more prominent than the rest of the breast. This feeling may be caused by old hardened scar tissue and/or dead fat tissue as a result of surgery or trauma. Often the cause of this type of breast tissue is unknown.
- **Miscellaneous disorders.** A number of other benign (noncancerous) breast problems may be placed under the heading of “fibrocystic condition.” These problems include disorders that may lead to breast inflammation (mastitis), infection, and/or nipple discharge.

Atypical ductal hyperplasia

The condition known as atypical ductal hyperplasia (ADH) is a condition in which the cells lining the milk ducts of the breast are growing abnormally. This condition may appear as spots of calcium salts, or calcifications, on the mammogram. A biopsy removed from the breast would confirm the diagnosis. Atypical ductal hyperplasia is not a cancer. In most women, this condition will cause no problems. However, for some women, especially women with family histories of breast cancer, the risk of developing breast cancer is increased. (One study with over 3,000 female participants indicated that about 20% of the participants with atypical hyperplasia and a family history of breast cancer developed breast cancer, as compared to the 8% of participants who developed the disease with atypical hyperplasia and no family history of breast cancer.) For women with ADH and a family history of breast cancer, more frequent mammograms and closer monitoring may be required.

Diagnosis

Breast cancer is the most common concern of women who feel a breast lump or experience an abnormal breast symptom. For peace of mind, and to rule

out any possibility of cancer, any newly discovered breast lumps should be brought to the attention of a family physician or an obstetrician-gynecologist. He or she will obtain a history and conduct thorough **physical examination** of the area. Depending on the findings on physical examination, the patient is usually referred for tests. The most common of these tests include:

- **Mammography.** A mammogram is an x-ray examination of the breasts. The two major types of abnormalities doctors look for are masses and calcifications; either abnormality may be benign or malignant. The size, shape, and edges of these masses help doctors determine whether or not cancer is present. Sometimes, however, this test may be difficult to interpret, however, due to dense breast tissue.
- **Ultrasonography.** If a suspicious lump is detected during mammography, an ultrasound (the use of high-frequency sound waves to outline the shape of various organs and tissues in the body) is useful (although not definitive) in distinguishing benign from cancerous growths.
- **Ductography.** A ductogram (also called a galactogram) is a test that is sometimes useful in evaluating nipple discharge. A very fine tube is threaded into the opening of the duct onto the nipple. A small amount of dye is injected, outlining the shape of the duct on an x ray, and indicates whether or not there is a mass in the duct.
- **Biopsy.** If a lump cannot be proven benign by mammography and ultrasound, a breast biopsy may be considered. Usually a tissue sample is removed through a needle (fine-needle aspiration biopsy, or FNAB) to obtain a sample of the lump. The sample is examined under the microscope by a pathologist, and a detailed diagnosis regarding the type of benign lesion or cancer is established. In some cases, however, FNAB may not provide a clear diagnosis, and another type of biopsy (such as a surgical biopsy, core-needle biopsy, or other stereotactic biopsy methods—such as the Mammotome or Advanced Breast Biopsy Instrument) may be required.

Other breast conditions such as inflammation or infection are usually recognized on the basis of suspicious history, breastfeeding, or characteristic symptoms such as pain, redness, and swelling. A positive response to appropriate therapies often confirms the diagnosis.

Treatment

Once a specific disorder within the broad category of fibrocystic condition is identified, treatment can be

KEY TERMS

Advanced Breast Biopsy Instrument (ABBI)—Uses a rotating circular knife and thin heated electrical wire to remove a large cylinder of abnormal breast tissue.

Lobules—A small lobe or subdivision of a lobe (often on a gland) that may be seen on the surface of the gland by bumps or bulges.

Lymph nodes—Rounded, encapsulated bodies consisting of an accumulation of lymphatic tissue.

Mammotome—A method for removing breast biopsies using suction to draw tissue into an opening in the side of a cylinder inserted into the breast tissue. A rotating knife then cuts tissue samples from the rest of the breast; also known as a vacuum-assisted biopsy

Stereotactic biopsy—A biopsy taken by precisely locating areas of abnormal growth through the use of delicate instruments.

prescribed. There are a number of treatment options for women with a lump that has been diagnosed as benign. If it is not causing a great deal of pain, the growth may be left in the breast. However, some women may choose to have a lump such as a fibroadenoma surgically removed, especially if it is large. Another option to relieve the discomfort of a painful benign lump is to have the cyst suctioned, or drained. If there is any uncertainty regarding diagnosis, the fluid may be sent to the lab for analysis.

Symptoms of cycle breast sensitivity and engorgement may also be treated with diet, medication, and/or physical modifications. For example,

- Although there is no scientific data to support this claim, many women have reported relief of symptoms when **caffeine** was reduced or eliminated from their diets. Decreasing salt before and during the period when breasts are most sensitive may also ease swelling and discomfort. Low-fat **diets** and elimination of dairy products also appear to decrease soreness for some women. However, it may take several months to realize the effects of these various treatments.
- Over-the-counter **analgesics** such as **acetaminophen** (Tylenol) or ibuprofen (Advil) may be recommended. In some cases, treatment with prescription drugs such as hormones or hormone blockers may prove successful. **Oral contraceptives** may also be prescribed.

- Warm soaks or ice packs may provide comfort. A well-fitted support bra can minimize physical movement and do much to relieve breast discomfort. Breast massage may promote removal of excess fluid from tissues and alleviate symptoms. Massaging the breast with castor oil, straight or infused with herbs or essential oils, can help reduce and dissipate fibroadenomas as well as keep women in touch with changes in their breast tissue.
- Infections are often treated with warm compresses and **antibiotics**. Lactating women are encouraged to continue breastfeeding because it promotes drainage and healing. However, a serious infection may progress to form an **abscess** that may need surgical drainage.
- Some studies of alternative or complementary treatments, although controversial, have indicated that **vitamins** A, B complex and E, and mineral supplements may reduce the risk of developing fibrocystic condition of the breast. Evening primrose oil (*Oenothera biennis*), flaxseed oil, and fish oils have been reported to be effective in relieving cyclic breast pain for some women.

Prognosis

Most benign breast conditions carry no increased risk for the development of breast cancer. However, a small percentage of biopsies uncover overgrowth of tissue in a particular pattern in some women; this pattern indicates a 15–20% increased risk of breast cancer over the next 20 years. Strict attention to early detection measures, such as annual mammograms, is especially important for these women.

Prevention

There is no proven method of preventing the various manifestations of fibrocystic condition from occurring. Some alternative health care practitioners believe that eliminating foods high in methyl xanthines (primarily coffee and chocolate) can decrease or reverse fibrocystic breast changes.

Resources

BOOKS

- Goldmann, David R., and David A. Horowitz, editors. *The American College of Physicians Home Medical Guide: Breast Problems*. New York: Dorling Kindersley, 2000.
- Love, Susan M., with Karen Lindsey. *Dr. Susan Love's Breast Book*. 3rd ed., revised. Reading, MA.: Addison-Wesley, 2000.

Singer, Sydney Ross. *Get It Off! Understanding the Cause of Breast Pain, Cysts, and Cancer*. Paho, HI: ISCD Press, 2000.

PERIODICALS

- Horner, N.K., and J.W. Lampe. "Potential Mechanisms of Diet Therapy for Fibrocystic Breast Conditions Show Inadequate Evidence of Effectiveness." *Journal of the American Dietetic Association* 100 (November 2000): 1368-1380.
- Mannello, F., M. Malatesta, and G. Gazzanelli. "Breast Cancer in Women With Palpable Breast Cysts." *Lancet* 354 (August 1999): 677- 678.
- Morrow, Monica. "The Evaluation of Common Breast Problems." *American Family Physician* 61 (April 15, 2000): 2371-2378, 2385.

ORGANIZATIONS

- American Cancer Society. 1599 Clifton Rd. NE, Atlanta, GA 30329. 1-800-ACS-2345. < <http://www.cancer.org> > .
- American College of Obstetricians and Gynecologists. 409 12th St., S.W., P.O. Box 96920, Washington, DC 20090-6920. < <http://www.acog.org> > .
- Cancer Information Service (CIS). 9000 Rockville Pike, Building 31, Suite 10A18, Bethesda, MD 20892. 1-800-4-CANCER. < <http://www.wicic.nci.nih.gov> > .

OTHER

- National Cancer Institute. *Understanding Breast Changes: A Health Guide for All Women*. July 10, 2001. < http://rex.nci.nih.gov/MAMMOG_WEB/PUBS_POSTERS/UNDRSTNDNG/UNDER_STANDING_CHANGE.html > .

Ellen S. Weber, MSN
Genevieve Slomski, PhD

Fibroids see **Uterine fibroids**

Fibromyalgia

Definition

Fibromyalgia is described as inflammation of the fibrous or connective tissue of the body. Widespread muscle **pain**, fatigue, and multiple tender points characterize these conditions. Fibrositis, fibromyalgia, and fibromyositis are names given to a set of symptoms believed to be caused by the same general problem.

Description

Fibromyalgia is more common than previously thought, with as many as 3–6% of the population

affected by the disorder. Fibromyalgia is more prevalent in adults than children, with more women affected than men, particularly women of childbearing age.

Causes and symptoms

The exact cause of fibromyalgia is not known. Sometimes it occurs in several members of a family, suggesting that it may be an inherited disorder. People with fibromyalgia are most likely to complain of three primary symptoms: muscle and joint pain, stiffness, and **fatigue**.

Pain is the major symptom with aches, tenderness, and stiffness of multiple muscles, joints, and soft tissues. The pain also tends to move from one part of the body to another. It is most common in the neck, shoulders, chest, arms, legs, hips, and back. Although the pain is present most of the time and may last for years, the severity of the pain changes and is dependent on individual patient perception.

Symptoms of fatigue may result from the individual's chronic pain coupled with **anxiety** about the problem and how to find relief. The inflammatory process also produces chemicals that are known to cause fatigue. Other common symptoms are tension headaches, difficulty swallowing, recurrent abdominal pain, **diarrhea**, and **numbness** or **tingling** of the extremities. **Stress**, anxiety, depression, or lack of sleep can increase symptoms. Intensity of symptoms is variable ranging from gradual improvement to episodes of recurrent symptoms.

Diagnosis

Diagnosis is difficult and frequently missed because symptoms of fibromyalgia are vague and generalized. Coexisting nerve and muscle disorders such as rheumatoid arthritis, spinal arthritis, or **Lyme disease** may further complicate the diagnostic process. Presently, there are no tests available to specifically diagnose fibromyalgia. The diagnosis is usually made after ruling out other medical conditions with similar symptoms and using criteria physicians and researchers have defined.

Because of the emotional distress experienced by people with this condition and the influence of stress on the symptoms themselves, fibromyalgia has often been labeled a psychological problem. Recognition of the underlying inflammatory process involved in fibromyalgia has helped promote the validity of this disease.

In 1990, the America College of Rheumatology developed standards for fibromyalgia that health care practitioners can use to diagnose this condition.

According to these standards, a person is thought to have fibromyalgia if he or she has widespread pain in combination with tenderness in at least 11 of the 18 sites known as trigger points. Trigger point sites include the base of the neck, along the backbone, in front of the hip and elbow, and at the rear of the knee and shoulder.

Treatment

There is no known cure for fibromyalgia. Therefore, the goal of treatment is successful symptom management. Treatment usually requires a combination of therapies, **exercise**, proper rest, and diet. A patient's clear understanding of his or her role in the recovery process is imperative for successful management of this condition. In 2004, a study demonstrated that a drug called paroxetine HCl (Paxil CR) in controlled release tablet form significantly reduced symptoms in fibromyalgia patients. As of spring 2004, there were no FDA-approved treatments for fibromyalgia.

Treatments found to be helpful include heat and occasionally cold applications. A regular stretching program is often useful. Aerobic activities focusing on increasing the heart rate are the preferred forms of exercise over most other forms of exertion. Exercise programs need to include good warm-up and cool-down sessions, with special attention given to avoiding exercises causing joint pain. The diet should include a large variety of fruits and vegetables which provide the body with trace elements and **minerals** that are necessary for healthy muscles.

Adequate rest is essential in the treatment of fibromyalgia. Avoidance of stimulating foods or drinks (such as coffee) and medications like **decongestants** prior to bedtime is advised. If diet, exercise, and adequate rest do not relieve the symptoms of fibromyalgia, medications may be prescribed. Medications prescribed and found to have some benefit include **antidepressant drugs**, muscle relaxants, and anti-inflammatory drugs.

People with fibromyalgia often need a rheumatology consultation (a meeting with a doctor who specializes in disorders of the joints, muscles, and soft tissue) to decide the cause of various rheumatic symptoms, to be educated about fibromyalgia and its treatment, and to exclude other rheumatic diseases. A treatment program must be individualized to meet the patient's needs. The rheumatologist, as the team leader, enlists and coordinates the expertise of other health professionals in the care of the patient.

KEY TERMS

Connective tissue—Tissue that supports and binds other body tissue and parts.

Lyme disease—An acute recurrent inflammatory disease involving one or a few joints, believed to be transmitted by a tickborne virus. The condition was originally described in the community of Lyme, Connecticut, but has also been reported in other parts of the United States and other countries. Knees, other large joints are most commonly involved with local inflammation and swelling.

Rheumatology—The study of disorders characterized by inflammation, degeneration of connective tissue, and related structures of the body. These disorders are sometimes collectively referred to as rheumatism.

Alternative treatment

Massage therapy can be helpful, especially when a family member is instructed on specific massage techniques to manage episodes of increased symptoms. Specific attention to mental health, including psychological consultation, may also be important, since depression may precede or accompany fibromyalgia. Other alternative therapies, including hellerwork, **rolfing**, homeopathic medicine, Chinese traditional medicine (both **acupuncture** and herbs), **polarity therapy**, and Western botanical medicine, can assist the person with fibromyalgia to function day to day and can contribute to healing.

Prognosis

Fibromyalgia is a chronic problem. The symptoms sometimes improve and at other times worsen, but they often continue for months to years.

Prevention

There is no known or specific way to prevent fibromyalgia. However, similar to many other medical conditions, remaining as healthy as possible with a good diet, safe exercise, and adequate rest is the best prevention.

Resources

PERIODICALS

“Study: Paroxetine Seems to Reduce Fibromyalgia Symptoms.” *Obesity, Fitness & Wellness Week* June 5, 2004: 803.

ORGANIZATIONS

The American College of Rheumatology. 1800 Century Place, Suite 250, Atlanta, GA 30345. (404) 633-3777. <<http://www.rheumatology.org>> .
Arthritis Foundation. 1300 W. Peachtree St., Atlanta, GA 30309. (800) 283-7800. <<http://www.arthritis.org>> .

Jeffrey P. Larson, RPT
Teresa G. Odle

Fibromyomas see **Uterine fibroids**

Fibrous breast lumps see **Fibroadenoma**

Fifth disease

Definition

Fifth disease is a mild childhood illness caused by the human parvovirus B19 that causes flu-like symptoms and a rash. It is called fifth disease because it was fifth on a list of common childhood illnesses that are accompanied by a rash, including **measles**, **rubella** or German measles, scarlet **fever** (or scarlatina), and scarlatinella, a variant of scarlet fever.

Description

The Latin name for the disease is *erythema infectiosum*, meaning infectious redness. It is also called the “slapped cheek disease” because, when the bright red rash first appears on the cheeks, it looks as if the face has been slapped. Anyone can get the disease, but it occurs more frequently in school-aged children. The disease is usually mild, and both children and adults usually recover quickly without complications. In fact, some individuals exhibit no symptoms and never even feel ill. Outbreaks most often occur in the winter and spring.

Causes and symptoms

Fifth disease is caused by the human parvovirus B19, a member of the Parvoviridae family of viruses, that lives in the nose and throat of the infected person. The virus is spread through the air by coughing and sneezing. Because the virus needs a rapidly dividing cell in order to multiply, it attacks the red blood cells of the body. Once infected, a person is believed to be immune to reinfection.

Symptoms may appear four to 21 days after being exposed to the virus. Initial symptoms are flu-like and

include **headache**, body ache, sore throat, a mild fever of 101 °F (38.3 °C), and chills. It is at this time, prior to the development of the rash, that individuals are contagious. These symptoms last for two to three days. In children, a bright red rash that looks like a slap mark develops suddenly on the cheeks. The rash may be flat or raised and may or may not be itchy. Sometimes, the rash spreads to the arms, legs, and trunk, where it has a lace-like or net-like appearance. The rash can also involve the palms of the hands and soles of the feet. By the time the rash appears, individuals are no longer infectious. On average, the rash lasts for 10–11 days, but may last for as long as five to six weeks. The rash may fade away and then reappear upon exposure to sunlight, hot baths, emotional distress, or vigorous **exercise**.

Adults generally do not develop a rash, but instead may have swollen and painful joints, especially in the hands and feet. In adults, symptoms such as **sore throat**, headache, muscle and joint **pain**, abdominal pain, **diarrhea**, and **vomiting** occur more frequently than in children and are usually more severe. The joint pain can be arthritis-like and last for several months, especially in women, but the disease does not appear to progress to rheumatoid arthritis.

The virus causes the destruction of red blood cells and, therefore, a deficiency in the oxygen-carrying capacity of the blood (anemia) can result. In healthy people, the anemia is mild and only lasts a short while. In people with weakened immune systems, however, either because they have a chronic disease like **AIDS** or **cancer** (immunocompromised), or are receiving medication to suppress the immune system (immunosuppressed), such as organ transplant recipients, this anemia can be severe and last long after the infection has subsided. Symptoms of anemia include **fatigue**, lack of color, lack of energy, and shortness of breath. Some individuals with sickle cell anemia, iron deficiency, a number of different hereditary blood disorders, and those who have received bone marrow transplantations may be susceptible to developing a potentially life-threatening complication called a transient aplastic crisis where the body is temporarily unable to form new red blood cells.

In very rare instances, the virus can cause inflammation of different areas of the body, including the brain (**encephalitis**), the covering of the brain and spinal cord (**meningitis**), the lungs (pneumonitis), the liver (hepatitis), and the heart muscle (**myocarditis**). The virus can also aggravate symptoms for people with an autoimmune disease called systemic lupus erythematosus.



This infant has a rash caused by Fifth disease, or erythema infectiosum. (Custom Medical Stock Photo. Reproduced by permission.)

There is some concern about fifth disease in pregnant women. Although no association with an increased number of **birth defects** has been demonstrated, there is concern that infection during the first three months of pregnancy may lead to a slight increase in the number of miscarriages. There is also some concern that infection later in **pregnancy** may involve a very small risk of premature delivery or stillbirths. As a result, women who get fifth disease while they are pregnant should be monitored closely by a physician.

Diagnosis

Fifth disease is usually suspected based on a patient's symptoms, including the typical appearance of the bright red rash on the cheeks, patient history, age, and the time of year. The physician will also exclude other potential causes for the symptoms and rash, including rubella, infectious mononucleosis, bacterial infections like **Lyme disease**, allergic reactions, and lupus.

In addition, there is a blood test for fifth disease, but it is generally used only for pregnant women and for people who have weakened immune systems or who suffer from blood disorders, such as sickle cell anemia. The test involves measuring for a particular antibody or protein that the body produces in response to infection with the human parvovirus B19. The test is 92–97% specific for this disease.

KEY TERMS

Anemia—A congenital or acquired deficiency in the iron-carrying capacity of the blood.

Antibody—A specific protein produced by the immune system in response to a specific foreign protein or particle called an antigen.

Immunocompromised—A state in which the immune system is weakened or is not functioning properly due to chronic disease.

Immunosuppressed—A state in which the immune system is suppressed by medications during the treatment of other disorders, like cancer, or following an organ transplantation.

Reye's syndrome—A very serious, rare disease, most common in children, that involves an upper respiratory tract infection followed by brain and liver damage.

Sickle cell anemia—A hereditary blood disorder in which the red blood cells are misshapen into crescent or sickle shapes resulting in the reduced oxygen-carrying capacity of the lungs.

Because fifth disease can pose problems for an unborn fetus exposed to the disease through the mother, testing may also be conducted while a fetus is still in the uterus. This test uses fluid collected from the sac around the fetus (amniotic fluid) instead of blood to detect the viral DNA.

Treatment

In general, no specific treatment for fifth disease is required. The symptoms can be treated using over-the-counter medications, such as acetaminophen (Tylenol) or ibuprofen (Motrin, Advil). If the rash itches, calamine lotion can be applied. Aspirin is usually not given to children under the age of 18 to prevent the development of a serious illness called Reye's syndrome.

Patients who are receiving medications to suppress the immune system in the treatment of some other condition may be allowed to temporarily decrease the medications in order to allow the immune system to combat the infection and recover from the anemia. Those with weakened (not suppressed) immune systems, such as AIDS patients, may be given immunoglobulin intravenously to help the immune system fight the infection. People with severe anemia or who experience an aplastic crisis may require hospitalization and blood transfusions.

Prognosis

Generally, fifth disease is mild, and patients tend to improve without any complications. In cases where the patient is either immunocompromised or immunosuppressed, a life-threatening aplastic crisis can occur. With prompt treatment, however, the prognosis is good. Mothers who develop the infection while pregnant can pass the infection on to their fetus, and as such, stand an increased risk of **miscarriage** and **stillbirth**. There are tests and treatments, however, that can be performed on the fetus while still in the uterus that can reduce the risk of anemia or other complications.

Prevention

Currently, there is no vaccine against fifth disease. Avoiding contact with persons who exhibit symptoms of a cold and maintaining good personal hygiene by regularly washing hands may minimize the chances of an infection. Pregnant women should avoid exposure to persons infected with the disease and notify their obstetrician immediately if they are exposed so that they can be tested and monitored closely.

Resources

BOOKS

Berkow, Robert, et al., editors. *Merck Manual of Diagnosis and Therapy*. Rahway, NJ: Merck Research Laboratories, 2004.

Lata Cherath, PhD

Filariasis

Definition

Filiariasis is the name for a group of tropical diseases caused by various thread-like parasitic round worms (nematodes) and their larvae. The larvae transmit the disease to humans through a mosquito bite. Filariasis is characterized by **fever**, chills, headache, and **skin lesions** in the early stages and, if untreated, can progress to include gross enlargement of the limbs and genitalia in a condition called **elephantiasis**.

Description

Approximately 170 million people in the tropical and subtropical areas of southeast Asia, South

America, Africa, and the islands of the Pacific are affected by this debilitating parasitic disease. While filariasis is rarely fatal, it is the second leading cause of permanent and long-term disability in the world. The World Health Organization (WHO) has named filariasis one of only six “potentially eradicable” infectious diseases and has embarked upon a 20-year campaign to eradicate the disease.

In all cases, a mosquito first bites an infected individual then bites another uninfected individual, transferring some of the worm larvae to the new host. Once within the body, the larvae migrate to a particular part of the body and mature to adult worms. Filariasis is classified into three distinct types according to the part of the body that becomes infected: lymphatic filariasis affects the circulatory system that moves tissue fluid and immune cells (lymphatic system); subcutaneous filariasis infects the areas beneath the skin and whites of the eye; and serous cavity filariasis infects body cavities but does not cause disease. Several different types of worms can be responsible for each type of filariasis, but the most common species include the following: *Wucheria bancrofti*, *Brugia malayi* (lymphatic filariasis), *Onchocerca volvulus*, *Loa loa*, *Mansonella streptocerca*, *Dracunculus medinensis* (subcutaneous filariasis), *Mansonella pustans*, and *Mansonella ozzardi* (serous cavity filariasis).

The two most common types of the disease are Bancroftian and Malayan filariasis, both forms of lymphatic filariasis. The Bancroftian variety is found throughout Africa, southern and southeastern Asia, the Pacific islands, and the tropical and subtropical regions of South America and the Caribbean. Malayan filariasis occurs only in southern and southeastern Asia. Filariasis is occasionally found in the United States, especially among immigrants from the Caribbean and Pacific islands.

A larvae matures into an adult worm within six months to one year and can live between four and six years. Each female worm can produce millions of larvae, and these larvae only appear in the bloodstream at night, when they may be transmitted, via an insect bite, to another host. A single bite is usually not enough to acquire an infection, therefore, short-term travelers are usually safe. A series of multiple bites over a period of time is required to establish an infection. As a result, those individuals who are regularly active outdoors at night and those who spend more time in remote jungle areas are at an increased risk of contracting the filariasis infection.

Causes and symptoms

In cases of lymphatic filariasis, the most common form of the disease, the disease is caused by the adult worms actually living in the lymphatic vessels near the lymph nodes where they distort the vessels and cause local inflammation. In advanced stages, the worms can actually obstruct the vessels, causing the surrounding tissue to become enlarged. In Bancroftian filariasis, the legs and genitals are most often involved, while the Malayan variety affects the legs below the knees. Repeated episodes of inflammation lead to blockages of the lymphatic system, especially in the genitals and legs. This causes the affected area to become grossly enlarged, with thickened, coarse skin, leading to a condition called elephantiasis.

In conjunctiva filariasis, the worms' larvae migrate to the eye and can sometimes be seen moving beneath the skin or beneath the white part of the eye (conjunctiva). If untreated, this disease can cause a type of blindness known as onchocerciasis.

Symptoms vary, depending on what type of parasitic worm has caused the infection, but all infections usually begin with chills, **headache**, and fever between three months and one year after the insect bite. There may also be swelling, redness, and **pain** in the arms, legs, or scrotum. Areas of pus (abscesses) may appear as a result of dying worms or a secondary bacterial infection.

Diagnosis

The disease is diagnosed by taking a patient history, performing a **physical examination**, and by screening blood specimens for specific proteins produced by the immune system in response to this infection (antibodies). Early diagnosis may be difficult because, in the first stages, the disease mimics other bacterial skin infections. To make an accurate diagnosis, the physician looks for a pattern of inflammation and signs of lymphatic obstruction, together with the patient's possible exposure to filariasis in an area where filariasis is common. The larvae (microfilariae) can also be found in the blood, but because mosquitos, which spread the disease, are active at night, the larvae are usually only found in the blood between about 10 pm and 2 am.

Treatment

Either ivermectin, albendazole, or diethylcarbamazine is used to treat a filariasis infection by eliminating the larvae, impairing the adult worms' ability to reproduce, and by actually killing adult worms.

Unfortunately, much of the tissue damage may not be reversible. The medication is started at low doses to prevent reactions caused by large numbers of dying parasites.

While effective, the medications can cause severe side effects in up to 70% of patients as a result either of the drug itself or the massive death of parasites in the blood. Diethylcarbamazine, for example, can cause severe allergic reactions and the formation of pus-filled sores (abscesses). These side effects can be controlled using **antihistamines** and anti-inflammatory drugs (**corticosteroids**). Rarely, treatment with diethylcarbamazine in someone with very high levels of parasite infection may lead to a fatal inflammation of the brain (**encephalitis**). In this case, the fever is followed by headache and confusion, then stupor and **coma** caused when massive numbers of larvae and parasites die. Other common drug reactions include dizziness, weakness, and **nausea**.

Symptoms caused by the death of the parasites include fever, headache, muscle pain, abdominal pain, **nausea and vomiting**, weakness, dizziness, lethargy, and **asthma**. Reactions usually begin within two days of starting treatment and may last between two and four days.

No treatment can reverse elephantiasis. Surgery may be used to remove surplus tissue and provide a way to drain the fluid around the damaged lymphatic vessels. Surgery may also be used to ease massive enlargement of the scrotum. Elephantiasis of the legs can also be helped by elevating the legs and providing support with elastic bandages.

Prognosis

The outlook is good in early or mild cases, especially if the patient can avoid being infected again. The disease is rarely fatal, and with continued WHO medical intervention, even gross elephantiasis is now becoming rare.

Prevention

The best method of preventing filariasis is to prevent being repeatedly bitten by the mosquitoes that carry the disease. Some methods of preventing insect bites include the following:

- limit outdoor activities at night, particularly in rural or jungle areas
- wear long sleeves and pants and avoid dark-colored clothing that attracts mosquitoes
- avoid perfumes and colognes

KEY TERMS

Abscess—An area of inflamed and injured body tissue that fills with pus.

Antibody—A specific protein produced by the immune system in response to a specific foreign protein or particle called an antigen.

Conjunctiva—The mucous membrane that lines the inside of the eyelid and the exposed surface of the eyeball.

Elephantiasis—A condition characterized by the gross enlargement of limbs and/or the genitalia that is also accompanied by a hardening and stretching of the overlying skin. Often a result of an obstruction in the lymphatic system caused by infection with a filarial worm.

Encephalitis—Inflammation of the brain.

Lymphatic system—The circulatory system that drains and circulates fluid containing nutrients, waste products, and immune cells, from between cells, organs, and other tissue spaces.

Microfilariae—The larvae and infective form of filarial worms.

Nematode—Round worms.

Subcutaneous—The area directly beneath the skin.

- treat one or two sets of clothing ahead of time with permethrin (Duramon, Permanone).
- wear DEET insect repellent or, especially for children, try citronella or lemon eucalyptus, to repel insects
- if sleeping in an open area or in a room with poor screens, use a bed net to avoid being bitten while asleep
- use air conditioning, the cooler air makes insects less active.

In addition, filariasis can be controlled in highly infested areas by taking ivermectin preventatively before being bitten. Currently, there is no vaccine available, but scientists are working on a preventative vaccine at this time.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>> .

OTHER

“Bacterial Diseases.” “Health touch Online Page.” <<http://www.healthtouch.com>>.

Centers for Disease Control. <<http://www.cdc.gov/nccddphp/ddt/ddthome.htm>>.

International Society of Travel Medicine.
<<http://www.istm.org>>.

King, J. W. *Bug Bytes.* Louisiana State University Medical Center. <<http://www.ccm.lsumc.edu/bugbytes>>.

“Lymphatic Filariasis.” *Centers for Disease Control.*
<<http://www.cdc.gov/travel/yellowbk/page117.htm>>.

Carol A. Turkington

Filgras see **Cancer therapy, supportive; Immunologic therapies**

Finasteride

Definition

Finasteride is a drug that belongs to the class of androgen inhibitors, which means that it blocks the production of male sex hormones. It is sold in the United States and Canada under the brand names Proscar and Propecia.

Purpose

Finasteride has two main purposes: the treatment of urinary problems in men caused by benign prostatic hypertrophy (BPH) or enlargement of the prostate gland; and the stimulation of new hair growth in men with male pattern baldness. Finasteride was first approved by the Food and Drug Administration (FDA) in 1992 under the trade name Proscar as a treatment for BPH. It received a second FDA approval in December 1997 under the trade name Propecia for the treatment of hair loss in men. Finasteride has also been used by some European doctors to treat hair loss in postmenopausal women, although its use in women is considered controversial in the United States. It is considered the most effective nonsurgical treatment for male pattern baldness as of 2005.

Finasteride works to relieve such symptoms of prostate enlargement as urinary urgency, the need to urinate frequently at night (nocturia), inability to completely empty the bladder, incontinence, or painful urination (dysuria) by blocking the production of DHT. DHT causes the prostate gland to grow and increase pressure on the bladder. As the swollen

prostate gradually shrinks, the patient finds it easier to pass urine without discomfort and to empty the bladder completely before going to sleep. Some doctors also prescribe finasteride as pretreatment for prostate surgery, as it lowers the risk of severe bleeding during the operation.

As of early 2005, the National Cancer Institute (NCI) was evaluating finasteride as a possible chemo preventative for **prostate cancer** in selected patients. Researchers were not yet certain, however, which men might benefit most from taking the drug.

Description

Finasteride inhibits the body's production of an enzyme called 5-alpha-reductase, which is needed to convert testosterone to another androgen called 5-alpha-dihydrotestosterone (DHT). Finasteride is a white powder that can be dissolved in alcohol or chloroform but is very difficult to dissolve in water. Both Proscar and Propecia are manufactured as coated tablets to be taken by mouth.

Recommended dosage

- Proscar: Finasteride for treatment of an **enlarged prostate** is taken once a day as a 5-mg tablet. The pill may be crushed or broken if the patient finds it hard to swallow.
- Propecia: Finasteride for hair regrowth is taken once a day as a 1-mg tablet. The drug may be taken with or without meals, as the patient prefers.

Precautions

Finasteride should be stored in dry places and should be kept at a temperature between 59°F and 86°F (15–30°C). Heat and moisture may cause the drug to lose its potency.

The drug can be safely handled by pregnant women as long as the tablets are intact; however, crushed or broken tablets should not be touched by a pregnant woman as the drug can be absorbed through the skin. If the woman is carrying a male fetus, the drug can cause abnormalities in the baby's sex organs. The FDA issued a warning in 2003 that men taking finasteride should not donate blood until one month after the final dose of the drug, on the grounds that their blood could contain high enough levels of the medication to cause **birth defects** in a male baby if given to a pregnant woman.

Patients should be advised that finasteride takes several months to reach its full effect—as long as six

KEY TERMS

Alopecia—The medical term for baldness.

Androgens—A group of hormones that produces masculine characteristics.

Chemo preventative—Description of a drug given in order to prevent the development of a specific disease. Finasteride was in 2005 being evaluated as a possible cancer chemo preventative.

Dysuria—Painful or difficult urination.

Incontinence—Inability to control urination or defecation.

Nocturia—Excessive need to urinate at night.

Prostate—A gland in males below the bladder that surrounds the urethra. Enlargement of the prostate may cause problems in urination.

Urgency—A sudden and powerful need to urinate.

months for BPH and three months for hair regrowth. In addition, the drug's effects on the body are not permanent; the prostate will start to enlarge again or the hair growth will be lost if the patient stops taking the drug.

Proscar can affect the results of a prostate-specific antigen (PSA) test for cancer of the prostate. Between 30 and 50 percent of men taking the drug will have elevated levels of PSA in their blood serum.

Finasteride should be used cautiously by men with liver disorders.

Side effects

As of 2005 reported side effects from using finasteride include:

- impotence or loss of interest in sex
- lumps or **pain** in the breast or a discharge from the nipple
- skin rash, **itching**, or hives
- swelling of the lips or face
- a smaller quantity of ejaculate during intercourse (which does not affect fertility)
- headaches, **dizziness**, or diarrhea
- pain in the testicles

These side effects are more common with the 5-mg dose, but usually go away as soon as the drug is discontinued.

Interactions

As of 2005 finasteride has not been reported to cause significant interactions with other medications.

Resources

BOOKS

"Alopecia (Baldness)." Section 10, Chapter 116 in *The Merck Manual of Diagnosis and Therapy*, edited by Mark H. Beers, and Robert Berkow. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

"Benign Prostatic Hyperplasia (Benign Prostatic Hypertrophy)." Section 17, Chapter 218 in *The Merck Manual of Diagnosis and Therapy*, edited by Mark H. Beers and Robert Berkow. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

Wilson, Billie A., et al. *Nurses Drug Guide 2000*, Stamford, CT: Appleton & Lange, 2000.

PERIODICALS

Arca, E., G. Acikgoz, H. B. Tastan, et al. "An Open, Randomized, Comparative Study of Oral Finasteride and 5% Topical Minoxidil in Male Androgenetic Alopecia." *Dermatology* 209 (2004): 117–125.

Crea, G., G. Sanfilippo, G. Anastasi, et al. "Pre-Surgical Finasteride Therapy in Patients Treated Endoscopically for Benign Prostatic Hyperplasia." *Urologia Internationalis* 74 (January 2005): 51–53.

Haber, R. S. "Pharmacologic Management of Pattern Hair Loss." *Facial Plastic Surgery Clinics of North America* 12 (May 2004): 181–89.

Parnes, H. L., I. M. Thompson, and L. G. Ford. "Prevention of Hormone-Related Cancers: Prostate Cancer." *Journal of Clinical Oncology* 23 (January 10, 2005): 368–77.

Trueb, R. M., and the Swiss Trichology Group. "Finasteride Treatment of Patterned Hair Loss in Normoandrogenic Postmenopausal Women." *Dermatology* 209 (2004): 202–07.

ORGANIZATIONS

American Society of Health-System Pharmacists (ASHP). 7272 Wisconsin Avenue, Bethesda, MD 20814. (301)657-3000. < www.ashp.org > .

United States Food and Drug Administration (FDA). 5600 Fishers Lane, Rockville, MD 20857-0001. (888) INFO-FDA. < www.fda.gov > .

OTHER

Food and Drug Administration (FDA) Medication Deferral List, December 9, 2003. < http://www.fda.gov/ohrms/dockets/ac/03/briefing/4014b1_18_Medication-list%20.doc > .

Rebecca J. Frey, PhD

Fingernail removal see **Nail removal**

Fingertip injuries

Definition

Fingertip trauma covers cuts, accumulation of blood (hematoma), bone breakage, or amputation in the fingertip.

Description

The fingertips are specialized areas of the hand with highly developed sensory and manipulative functions. Large sensory and motor areas located in the brain regulate the precise and delicate functions of fingertips. The fingertip is the site where extensor and flexor tendons insert. Fingertip injuries are extremely common since the hands hold a wide array of objects. In 2001, the approximately 10% of all accidents in the United States referred for Emergency Room consults involve the hand. Hand injuries are frequently the result of job injuries and account for 11–14% of on-the-job injuries and 6% of compensation paid injuries. Injury to the nail bed occurs in approximately 15–24% of fingertip injuries.

Fingertip injuries can result in **amputation** or tissue loss. The injury is assessed whether the bone and underlying tissue are intact and the size of the wound area. The pulp is the area of skin opposite the fingernail and is usually very vulnerable to injury. Pulp injuries commonly occur in persons who use or are in close contact with fast moving mechanical devices. These injuries can crush, cut, and puncture. The fingertips can also be injured by common crushing accidents. This could cause the development of a subungal hematoma (an accumulation of blood under the nail). At the base of the distal phalanx (the first circular skin fold from the tip) injuries can occur that can fracture the underlying bone in the area. Quite commonly a hammer, closing a door, or sport accidents usually cause these injuries. These **fractures** can be simple, requiring little treatment or more complicated involving the joint. The accident may involve the point of insertion of a tendon. Usually this occurs when the terminal joint is being forced to flex while held straight. This motion typically occurs when tucking in sheets during bed making, a common cause of tendon injury. This injury causes a loss of extension (straightening the finger) ability.

Causes and symptoms

Accidental amputations will usually result in profuse bleeding and tissue loss. Injuries to the pulp

can occur as from fast moving mechanical instruments, such as drills. These injuries may puncture the pulp. Injuries such as a subungal hematoma are caused by a crushing type injury. Fractures typically occur as the result of crushing injuries or tendon avulsion. These crushing injuries are frequently caused during sport injury and can be treated by simple interventions such as **immobilization** or more complex procedures if tendons are affected (the trauma is then treated as a tendon injury). Fractures can cause **pain** and, depending on the extent of swelling, there may be some restriction of movement. Tendon injuries can be caused when the terminal joint is exposed to force flexing motion (moving the finger toward the palm) while held straight.

Diagnosis

The attending clinician should evaluate the injury in a careful and systematic manner. The appearance of the hand can provide valuable information concerning presence of fractures, vascular status, and tendon involvement. Bones and joints should be evaluated for motion and tenderness. Nerves should be examined for sensory (feeling sensations) and motor (movement) functioning. Amputations usually profusely bleed and there is tissue loss. The wound is treated based on loss of tissue, bone, and wound area. Injuries to the pulp can be obvious during inspection. Subungal hematoma usually present a purplish-black discoloration under the nail. This is due to a hematoma underneath the nail. Radiographs may be required to assess the alignment of fractures or detect foreign bodies. Patients usually suffer from pain since injuries to the fingertip bone are usually painful and movement may be partially restricted due to swelling of the affected area. Tendon injuries usually result in the loss of ability to straighten or bend the finger.

Treatment

Amputation with bone and underlying tissue intact and a wound area 1 cm or less should be cleaned and treated with a dressing. With these types of wounds healthy tissue will usually grow and replace the injured area. Larger **wounds** may require surgical intervention. Puncture wounds should be cleaned and left open to heal. Patients typically receive antibiotics to prevent infection. A procedure called trephining treats subungal hematomas. This procedure is usually done with a straight cutting needle positioned over the nail. The clinician spins the needle with forefinger and thumb until a hole is made through the nail.

KEY TERMS

Distal—Movement away from the origin.

Flex—To bend.

Laceration—A cut in the skin

Phalanx—A bone of the fingers or toes.

Tendon—A structure that connects a skeletal muscle to bone.

Patients who have extensive crush injuries or subungual hematomas involving laceration to skin folds or nail damage should have the nail removed to examine the underlying tissue (called the matrix). Patients who have a closed subungual hematoma with an intact nail and no other damage (no nail disruption or laceration) are treated conservatively. If the fracture is located two-thirds below the fingertip immobilization using a splint may be needed. Conservative treatment is recommended for crush injuries that fracture the terminal phalanx if a subungual hematoma is not present. Severe fractures near the fist circular skin crease may require surgical correction to prevent irregularity of the joint surface, which can cause difficulty with movement. Injury to a flexor tendon usually requires surgical repair. If this is not possible, the finger and wrist should be placed in a splint with specific positioning to prevent further damage.

Prognosis

Prognosis depends on the extent of traumatic damage to the affected area. Nail lacerations that are not treated may cause nail deformities. When amputation is accompanied with loss of two-thirds of the nail, half of the fingers develop beaking, or a curved nail. Aftercare and follow up are important components of treatment. The patient is advised to keep the hand elevated, check with a clinician two days after treatment, and to splint fractures for two weeks in the extended position. Usually a nail takes about 100 days to fully grow. Healing for an amputation takes about 21 to 27 days. This markedly decreases in elderly patients, primarily due to a compromised circulation normally part of advancing age.

Resources

BOOKS

Townshend, Courtney M., et al. *Sabiston Textbook of Surgery*. 16th ed. W. B. Saunders Company, 2001.

Laith Farid Gulli, M.D.

Fish and shellfish poisoning

Definition

Fish and shellfish **poisoning** is a common but often unrecognized group of illnesses related to food. Three of these illnesses include ciguatera, scombroid, and paralytic shellfish poisoning.

Ciguatera

Definition

Ciguatera (from the Spanish word for a poisonous snail) is a food-related illness that causes abdominal and neurological symptoms.

Causes and symptoms

Ciguatera is caused by eating fish that have a toxin called ciguatoxin. Scientists believe this toxin is acquired by the fish through the food chain, and is originally produced by small algae microorganisms (dinoflagellates). The fish most likely contaminated with ciguatoxin are those that feed close to tropical reefs, including red snapper, grouper, and barracuda. Larger fish are more likely to contain the toxin. Although not as common in the United States, ciguatera is commonly diagnosed on many of the islands in the Pacific Ocean.

Illness from ciguatera can occur in just a few minutes to about 30 hours after eating. Most cases occur one to six hours after eating the contaminated fish. Initial symptoms are abdominal cramps, **nausea**, **vomiting**, or watery **diarrhea**. The most characteristic symptoms of the illness are those involving the nervous system. These include numbness and **tingling** around the lips, tongue, and mouth; itching; **dry mouth**; metallic taste in the mouth; and blurry vision. In more prominent cases, patients may complain of temporary blindness, a slow pulse, and a feeling that their teeth are loose. Patients may also have the strange symptom of reversal of hot and cold sensations on the skin, where cold things feel very hot or painful to the touch. In very severe cases, there may be difficulties in breathing or low blood pressure.

Diagnosis

Ciguatera diagnosis is based on the typical combination of symptoms after eating fish. There are no readily available blood or urine tests to detect the poisoning, but some researchers have developed a test for the toxin left on any remaining fish. A person does not have to be in a tropical area to get ciguatera.

Fish can be caught from one of these distant areas, and can then be shipped and eaten locally. It is important to report suspected cases to local public health officials because more cases may occur from other contaminated fish.

Treatment

The treatment for this illness is general. Patients are given fluids (by mouth or through a vein) and medications to decrease the **itching** or to treat vomiting and/or diarrhea. The neurological symptoms can cause discomfort and treatment with amitriptyline (a medicine that has been used for depression) may be useful. Other medications may also be given.

Prognosis

Although **death** can occur, almost all patients diagnosed with ciguatera will recover. Recovery, however, can be slow and some symptoms can last for weeks or even months. Symptoms can also be aggravated by other illnesses or alcohol.

Prevention

Knowing the kinds of fish linked to ciguatera can help a person avoid eating high-risk fish. However, over 400 different kinds of fish have been linked to the disease, even salmon. A particular fish in a given area may be more likely to cause ciguatera than other fish. For example, red snapper is most often the source of ciguatera in the Pacific, while barracuda is more likely to contain the toxin in Florida. This is why it is illegal to sell barracuda in Florida for human consumption. Cooking the fish does not prevent ciguatera.

Scombroid

Definition

Scombroid is a fish-associated illness caused by eating improperly handled fish. Fish linked to this disease are usually in the Scombridae family, which includes yellowfin tuna, skipjack, bonito, and mackerel.

Causes and symptoms

Scombroid occurs after eating fish that has not been properly refrigerated after capture. Unlike ciguatera, the toxins linked with scombroid are not contracted by the fish from its surroundings. Bacteria that are normally found in fish act directly on a chemical (called histidine) in the flesh of fish that are not properly cooled when stored. This interaction

produces histamine and other chemicals that cause the illness when the fish is eaten.

Symptoms of scombroid occur quickly after eating the fish, as soon as 10 minutes. Since histamine is released by certain cells in the body during an allergic reaction, scombroid can be confused with a fish allergy. Scombroid causes flushing of the face, sweating, a burning feeling in the mouth or throat, vomiting, diarrhea, and headaches. A rash that looks like a **sunburn** may occur, and a small number of patients have **hives**. Some patients have a metallic or peppery taste in their mouths. In more severe cases, rapid pulse, blurred vision, and difficulty breathing can occur. Symptoms usually last about four hours.

Diagnosis

Like ciguatera, scombroid poisoning is diagnosed based on typical symptoms occurring after eating fish. There are usually no available tests for the patient. Experimentally, however, elevated levels of histamine-related products have been found in the urine. It may be possible for public health officials to test any remaining fish flesh for histamine levels. Improperly refrigerated fish caught in both temperate and tropical waters have been linked to the illness. An outbreak of similar cases may be helpful in correctly diagnosing the problem.

Treatment

The treatment for scombroid is usually general. **Antihistamines** like diphenhydramine (Benadryl) may shorten the duration of the illness, but the illness will go away on its own. Some doctors have found that cimetidine (Tagamet) given through a vein may be helpful as well. In rare, more severe cases, epinephrine (adrenaline) may be used.

Prognosis

Although sometimes dramatic and alarming symptoms can occur, scombroid is usually not serious. The patient should be reassured that scombroid is not a fish allergy.

Prevention

Adequate storage of the target fish will always prevent scombroid. Since the fish does not appear spoiled or smell bad, the consumer cannot detect the risk of the illness before eating the fish. Cooking the fish does not prevent scombroid. Suspected cases should be reported to public health officials.

Paralytic shellfish poisoning

Definition

Paralytic shellfish poisoning (PSP) is a nervous system disease caused by eating cooked or raw shellfish that contain environmental toxins. These toxins are produced by a group of algae (dinoflagellates). It is unclear whether these toxins are related to the “blooming” of the algae, also called red tide because the algae can turn the water reddish brown. PSP occurs mostly in May through November.

Causes and symptoms

PSP develops usually within minutes after eating a contaminated shellfish, most commonly a mussel, clam, or oyster. Symptoms include **headache**, a floating feeling, **dizziness**, lack of coordination, and tingling of the mouth, arms, or legs. Muscle weakness causing difficulty swallowing or speaking may occur. Abdominal symptoms such as nausea, vomiting, and diarrhea can also occur. Unlike ciguatera and scombroid, PSP may have a much more serious outcome. PSP may cause difficulty breathing related to weakness or **paralysis** of the breathing muscle. The symptoms may last for six to 12 hours, but a patient may continue to feel weak for a week or more.

Diagnosis

PSP diagnosis is based on symptoms after eating shellfish, even if the shellfish are adequately cooked. No blood or urine test is available to diagnose the illness, but tests in mice to detect the toxin from the eaten fish can be done by public health officials.

Treatment

The treatment of PSP is mostly supportive. If early symptoms are recognized, the doctor will try to flush the toxin from the gastrointestinal tract with medications that create diarrhea. Vomiting may be induced if the patient has no signs of weakness. In cases where the muscles of breathing are weakened, the patient may be placed on a respirator until the weakness goes away. However, this measure is not usually needed. Likewise, the use of a machine to clean the blood (dialysis) has been used in severe cases.

Prognosis

The prognosis for PSP is quite good, especially if the patient has passed the initial 12 hours of illness without needing breathing support. Most deaths occur during this period if breathing help is not available.

KEY TERMS

Algae—Plants that have one cell.

Histamine—A chemical found naturally in the body that produces inflammation and increases blood flow; the uncomfortable symptoms of an allergy attack or an allergic reaction are generally caused by the release of histamine.

Toxin—A poisonous substance usually produced by a living thing.

Prevention

Measures to control PSP require detecting rising numbers of algae in coastal waters by periodic microscopic examination. By law, shellfish beds are closed when levels of the toxin-producing organisms are above acceptable standards. Cooking the shellfish does not prevent this disease. Suspected cases should be reported to public health officials.

Resources

PERIODICALS

Barton, Erik D., Paula Tanner, Steven G. Turchen, et al. “Ciguatera Fish Poisoning: A Southern California Epidemic.” *Western Journal of Medicine* 163, no. 1 (July 1995): 31–35.

Larry I. Lutwick, MD, FACP

Fistula

Definition

A Fistula is a permanent abnormal passageway between two organs in the body or between an organ and the exterior of the body.

Description

Fistulas can arise in any part of the body, but they are most common in the digestive tract. They can also develop between blood vessels and in the urinary, reproductive, and lymphatic systems. Fistulas can occur at any age or can be present at birth (congenital). Some are life-threatening, others cause discomfort, while still others are benign and go undetected or cause few symptoms. Diabetics, individuals with

compromised immune systems (**AIDS**, **cancer**) and individuals with certain gastrointestinal diseases (**Crohn's disease**, inflammatory bowel disease) are at increased risk of developing fistulas.

Fistulas are categorized by the number of openings they have and whether they connect two internal organs or open through the skin. There are four common types:

- Blind fistulas are open on one end only.
- Complete fistulas have one internal opening and one opening on the skin.
- Horseshoe fistulas are complex fistulas with more than one opening on the exterior of the body.
- Incomplete fistulas are tubes of skin that are open on the outside but closed on the inside and do not connect to any internal structure.

Fistulas of the digestive tract

Anal and rectal fistulas develop in the wall of the anus or rectum. They connect the interior of the body to one or several openings in the skin. Anal and rectal fistulas almost always begin as an inflammation in an anal gland. The inflammation then moves into muscle tissue and develops into an **abscess**. In about half of all cases, the abscess develops into a fistula, degrading the muscle until an opening in the skin is created. About 9 people of every 100,000 develop anal fistulas, with men almost twice more likely to develop the condition than women. Although they may develop at any age, the average age for the development of anal fistulas is 38.

Intestinal fistulas can develop in both the large and small intestine. They are commonly associated with diseases such as inflammatory bowel disease (IBD) and Crohn's disease.

Tracheoesophageal fistulas (TEF) are usually **birth defects**. The windpipe, or trachea, is abnormally connected to the esophagus. This allows air to enter the digestive system and makes it possible to breathe food into the lungs (aspiration). In many cases, the esophagus is also incomplete, causing immediate feeding problems. There are several types of TEFs categorized by where the fistula is located and how the esophagus and trachea are connected, but all are life-threatening and require prompt surgery to repair. TEFs occur in about one of every 1,500–3,000 births.

Fistulas of the urinary and reproductive tract

The most common type of fistula involving these systems is a vesicovaginal fistula, in which the

woman's vagina is connected to the urinary bladder. This causes leakage of urine from the vagina and results in frequent vaginal and bladder infections. Fistulas may also develop between the vagina and the large intestine (a enterovaginal fistula) so that feces leaks from the vagina. Although both these types of fistulas are uncommon in the developed world, they are common in poor developing countries and result from long, difficult labor and **childbirth**, especially in very young girls. As a result, they are sometimes referred to as obstetric fistulas.

Some experts suggest that in parts of Africa, as many as 3–4 women develop these fistulas out of every 1,000 births. Others estimate that as many as 2 million women worldwide are living with unrepaired obstetric fistulas. If left unrepaired, obstetric fistulas cause women to constantly leak urine and feces. As a result, they become social outcasts, causing them extreme hardship and psychological trauma.

Fistulas of the circulatory system

Arteriovenous fistulas (AVF) can develop between an artery and a vein in any part of the body. These fistulas vary in size, length, and frequency. Arteries contain blood carrying oxygen to all parts of the body, while veins carry blood that has given up its oxygen back to the lungs. Connections between arteries and veins cause changes in blood pressure that result in abnormal development of the walls of the arteries and abnormal blood flow. Arteriovenous fistulas that are present at birth are sometimes referred to as **arteriovenous malformations** (AVMs). Many arteriovenous fistulas are present, but not evident at birth, and become obvious only after trauma. AVFs can also be acquired from penetrating trauma.

Causes and symptoms

The causes and symptoms of fistulas vary depending on their location. Anal and rectal fistulas are usually caused by an abscess. Symptoms include constant throbbing **pain** and swelling in the rectal area. Pus is sometimes visible draining from the fistula opening on the skin. Many individuals have a **fever** resulting from the infection causing the abscess.

Vaginal fistulas are caused by infection and trauma to the tissue during childbirth. They are easily detected, because the woman smells unpleasant and leaks urine or feces through her vagina. Rarely these fistulas may develop as a complication of **hysterectomy**.

Tracheoesophageal fistulas are the result of errors in the development of the fetus. They are evident at birth, because the infant is unable to swallow or eat normally and are considered a medical emergency that requires surgery if the infant is to survive.

Arteriovenous fistulas are most often congenital defects. Symptoms vary depending on the size and location of the fistula. Often the skin is bright pink or dark red in the area of the fistula. Individuals may complain of pain. The pain is a result of some tissues not receiving enough oxygen because of abnormal blood flow.

Diagnosis

Tests use to determine the presence of a fistula vary with the location of the fistula. When there is an opening to the outside, the physician may be able to see the fistula and probe it. Various imaging studies such as x rays, CT scans, barium **enemas**, endoscopy, and ultrasonography are used to locate less visible fistulas.

Treatment

Anal and rectal fistulas are treated by draining the pus the infected area. The individual also is usually given **antibiotics** to help prevent recurrence of the abscess. If this fails to heal the fistula, surgery may be necessary.

Intestinal fistulas are treated first by reducing the inflammation in the intestine and then, if necessary with surgery. Treatment varies considerably depending on the degree of severity of symptoms the fistula causes. TEFs are always treated with surgery. Obstetric fistulas must also be repaired with surgery. The treatment of arteriovenous fistulas depends on the size and location of the fistula and usually includes surgery.

Alternative treatment

No effective alternative treatments for fistulas are known.

Prognosis

The outcome of fistulas depends on the type and cause of the condition. Surgical repair of obstetric fistulas is almost always successful. Unfortunately, many women in developing countries do not have access to this type of surgery. Treatment of anal and rectal fistulas is almost always successful, although fistulas may recur in up to 18% of individuals. The

KEY TERMS

Abscess—A collection of pus surrounded by inflamed, infected tissue.

Lymphatic system—The part of the circulatory system that carries lymph, a clear fluid that is involved in immune system response.

outcome of surgery on TEFs is highly variable, especially since infants born with this condition often have other developmental abnormalities that may affect the outcome of fistula repair. The degree of successful repair of arteriovenous fistulas depends on their size and location. Uncontrolled bleeding is the most common complication of surgery to repair AVFs.

Prevention

Obstetric fistulas are the only preventable fistulas. These can be prevented with good prenatal and child-birth care and by avoiding **pregnancy** in very young girls. Although anal and rectal fistulas are not preventable, their damage can be minimized by prompt drainage and treatment.

Resources

ORGANIZATIONS

American Society of Colon and Rectal Surgeons. 85 W. Algonquin Road, Suite 550, Arlington Heights, IL 60005. 847-290-9184. < <http://www.facrs.org> > .

OTHER

Legall, Ingrid. *Anal Fistulas and Fissures*, 11 June 2004 [cited 16 February 2005]. < <http://www.emedicine.com/emerg/topic495.htm> > .

“Fistula.” *Medline Plus Medical Encyclopedia* 29 October 2003 [cited 16 February 2005]. < <http://www.nlm.nih.gov/medlineplus/ency/article/002365.htm> > .

Morasch, Mark D. and Dipen Maun. *Arteriovenous Fistulas*, 24 October 2003 [cited 16 February 2005]. < <http://www.emedicine.com/med/topic169.htm> > .

Zagrodnik, Dennis II. *Fistula-in-Ano*, ii June 2004 [cited 3 March 2005]. < <http://www.emedicine.com/med/topic2710.htm> > .

Tish Davidson, A.M.

5p-syndrome see **Cri du chat syndrome**

Flesh-eating disease

Definition

Flesh-eating disease is more properly called necrotizing fasciitis, a rare condition in which bacteria destroy tissues underlying the skin. This tissue death, called necrosis or **gangrene**, spreads rapidly. This disease can be fatal in as little as 12 to 24 hours.

Description

Although the term is technically incorrect, flesh-eating disease is an apt descriptor: the infection appears to devour body tissue. Media reports increased in the middle and late 1990s, but the disease is not new. Hippocrates described it more than three millennia ago and thousands of reports exist from the Civil War. Approximately 500 to 1,500 cases of necrotizing fasciitis occur in the United States each year.

Flesh-eating disease is divided into two types. Type I is caused by anaerobic bacteria, with or without the presence of aerobic bacteria. Type II, also called hemolytic streptococcal gangrene, is caused by group A streptococci; other bacteria may or may not be present. The disease may also be called synergistic gangrene.

Type I fasciitis typically affects the trunk, abdomen, and genital area. For example, Fournier's gangrene is a "flesh-eating" disease in which the infection encompasses the external genitalia. The arms and legs are most often affected in type II fasciitis, but the infection may appear anywhere.

Causes and symptoms

The two most important factors in determining whether or not a person will develop flesh-eating disease are: the virulence (ability to cause disease) of the bacteria and the susceptibility (ability of a person's immune system to respond to infection) of the person who becomes infected with this bacteria.

In nearly every case of flesh-eating disease, a skin injury precedes the disease. As bacteria grow beneath the skin's surface, they produce toxins. These toxins destroy superficial fascia, subcutaneous fat, and deep fascia. In some cases, the overlying dermis and the underlying muscle are also affected.

Initially, the infected area appears red and swollen and feels hot. The area is extremely painful, which is a prominent feature of the disease. Over the course of hours or days, the skin may become blue-gray, and

fluid-filled blisters may form. As nerves are destroyed the area becomes numb. An individual may go into **shock** and develop dangerously low blood pressure. Multiple organ failure may occur, quickly followed by **death**.

Diagnosis

The appearance of the skin, paired with **pain** and **fever** raises the possibility of flesh-eating disease. An x ray, **magnetic resonance imaging** (MRI), or computed tomography scans (CT scans) of the area reveals a feathery pattern in the tissue, caused by accumulating gas in the dying tissue. Necrosis is evident during exploratory surgery, during which samples are collected for bacterial identification.

Treatment

Rapid, aggressive medical treatment, specifically, antibiotic therapy and surgical debridement, is imperative. **Antibiotics** may include penicillin, an aminoglycoside or third-generation cephalosporin, and clindamycin or metronidazole. **Analgesics** are employed for pain control. During surgical **debridement**, dead tissue is stripped away. After surgery, patients are rigorously monitored for continued infection, shock, or other complications. If available, hyperbaric **oxygen therapy** has also be used.

Prognosis

Flesh-eating disease has a fatality rate of about 30%. Diabetes, arteriosclerosis, immunosuppression, **kidney disease**, **malnutrition**, and **obesity** are connected with a poor prognosis. Older individuals and intravenous drug users may also be at higher risk. The infection site also has a role. Survivors may require **plastic surgery** and may have to contend with permanent physical disability and psychological adjustment.

Prevention

Flesh-eating disease, which occurs very rarely, cannot be definitively prevented. The best ways to lower the risk of contracting flesh-eating disease are:

- take care to avoid any injury to the skin that may give the bacteria a place of entry
- when skin injuries do occur, they should be promptly washed and treated with an antibiotic ointment or spray
- people who have any skin injury should rigorously attempt to avoid people who are infected with

KEY TERMS

Aerobic bacteria—Bacteria that require oxygen to live and grow.

Anaerobic bacteria—Bacteria that require the absence of oxygen to live and grow.

CT scan (computed tomography scan)—Cross-sectional x rays of the body are compiled to create a three-dimensional image of the body's internal structures.

Debridement—Surgical procedure in which dead or dying tissue is removed.

Dermis—The deepest layer of skin.

Fascia, deep—A fibrous layer of tissue that envelops muscles.

Fascia, superficial—A fibrous layer of tissue that lies between the deepest layer of skin and the subcutaneous fat.

Gangrene—An extensive area of dead tissue.

Hyperbaric oxygen therapy—A treatment in which the patient is placed in a chamber and breathes oxygen at higher-than-atmospheric pressure. This high-pressure oxygen stops bacteria from growing and, at high enough pressure, kills them.

Magnetic resonance imaging (MRI)—An imaging technique that uses a large circular magnet and radio waves to generate signals from atoms in the body. These signals are used to construct images of internal structures.

Necrosis—Abnormal death of cells, potentially caused by disease or infection.

Subcutaneous—Referring to the area beneath the skin.

streptococci bacteria, a bacteria that causes a simple strep throat in one person may cause flesh-eating disease in another

- have any areas of unexplained redness, pain, or swelling examined by a doctor, particularly if the affected area seems to be expanding

Resources

BOOKS

- Roemmele, Jacqueline A., Donna Batdorff, and Alan L. Bisno. *Surviving the 'Flesh-Eating Bacteria': Understanding, Preventing, Treating, and Living With the Effects of Necrotizing Fascitis*. New York: Avery Penguin Putnam, 2000.

ORGANIZATIONS

National Necrotizing Fascitis Foundation. PO Box 145, Niantic, CT 06357. (616) 261-2538. <<http://www.nnff.org/>>.

Paul A. Johnson, Ed.M.

Flight medicine see **Aviation medicine**

Floppy mitral valve see **Mitral valve prolapse**

Flower remedies

Definition

Flower remedies are specially prepared flower essences, containing the healing energy of plants. They are prescribed according to a patient's emotional disposition, as ascertained by the therapist, doctor, or patients themselves.

Purpose

Flower remedies are more homeopathic than herbal in the way they work, effecting energy levels rather than chemical balances. They have been described as "liquid energy." The theory is that they encapsulate the flowers' healing energy, and are said to deal with and overcome negative emotions, and so relieve blockages in the flow of human energy that can cause illness.

Description

Origins

Perhaps the most famous and widely used system is the Bach flower remedies. This system originated in the 1920s when British physician and bacteriologist, Dr. Edward Bach (1886–1936), noticed that patients with physical complaints often seemed to be suffering from **anxiety** or some kind of negative emotion. He concluded that assessing a patient's emotional disposition and prescribing an appropriate flower essence could treat the physical illness. Bach was a qualified medical doctor, but he also practiced homeopathy.

As a result of his own serious illness in 1917, Bach began a search for a new and simple system of medicine that would treat the whole person. In 1930, he gave up his flourishing practice on Harley Street at the Royal London Homeopathic Hospital and moved to the countryside to devote his life to this research. It

EDWARD BACH (1886–1936)

Edward Bach was a graduate of University College Hospital (M.B., B.S., M.R.C.S.) in England. He left his flourishing Harley Street practice in favor of homeopathy, seeking a more natural system of healing than allopathic medicine. He concluded that healing should be as simple and natural as the development of plants, which were nourished and given healing properties by earth, air, water, and sun.

Bach believed that he could sense the individual healing properties of flowers by placing his hands over the petals. His remedies were prepared by floating summer flowers in a bowl of clear stream water exposed to sunlight for three hours.

He developed 38 remedies, one for each of the negative states of mind suffered by human beings, which he classified under seven group headings: fear, uncertainty, insufficient interest in present circumstances, loneliness, over-sensitivity to influences and ideas, despondency or despair, and overcare for the welfare of others. The Bach remedies can be prescribed for plants, animals, and other living creatures as well as human beings.

is known that at this point, he ceased to dispense the mixture of homeopathy and allopathic medicine that he had been using. Instead, he began investigating the healing properties of plant essences and discovered that he possessed an “intuition” for judging the properties of each flower. Accordingly, he developed the system of treatment that bears his name, and is also the foundation for all other flower-remedy systems.

The Bach Flower Remedies were ostensibly the only system of significance from the 1920s until in the 1970s, when there was a renewed interest in the subject by doctors working in the field of natural medicine. Perhaps the most notable was Dr. Richard Katz, who was seeking new methods of dealing with modern **stress** and the resulting ailments. He focused on the concept of a psychic, psychological effect and chose to pursue this line of research.

In 1979, Katz founded the Flower Essence Society in California, (FES). This society pledged to further the research and development of Bach’s principles. As of 2000, FES hosts a database of over 100 flower essences from more than 50 countries. FES is now an international organization of health practitioners, researchers, students, and others concerned with flower essence therapy.

Bach Flower Remedies

Name	Remedy
Agrimony	Upset by arguments, nonconfrontational, conceals worry and pain
Aspen	Fear of the unknown, anxiety, prone to nightmares, and apprehension
Beech	Critical, intolerant, and negative
Centaury	Submissive and weak-willed
Cerato	Self doubting and overly dependent
Cherry Plum	Emotional thoughts and desparation
Chestnut	Repeats mistakes and has no hindsight
Chicory	Selfish, controlling, attention-seeking, and possessive
Clematis	Absorbed, impractical, and indifferent
Crab Apple	Shame and self-loathing
Elm	Overwelmed and feelings of inadequacy
Gentian	Negative, doubt, and depression
Gorse	Pessimism, hopelessness, and despair
Heather	Self-centered and self-absorbed
Holly	Jealousy, hatred, suspicion, and envy
Honeysuckle	Homesick, living in the past, and nostalgic
Hornbeam	Procrastination, fatigue, and mental exhaustion
Impatiens	Impatience, irritability, and impulsive
Larch	No confidence, inferiority complex, and despondency
Mimulus	Timid, shy, and fear of the unknown
Mustard	Sadness and depression of unknown origin
Oak	Obstinate, inflexible, and overachieving
Olive	Exhaustion
Pine	Guilt and self blame
Red Chesnut	Fear and anxiety for loved ones
Rock Rose	Nightmares, hysteria, terror, and panic
Rock Water	Obsessive, repression, perfectionism, and self denial
Scleranthus	Indecision, low mental clarity, and confusion
Star-of-Bethlehem	Grief and distress
Sweet Chesnut	Despair and hopelessness
Vervain	Overbearing and fanatical
Vine	Arrogant, ruthless, and inflexible
Walnut	Difficulty accepting change
Water Violet	Pride and aloofness
White Chestnut	Worry, preoccupation, and unwanted thoughts
Wild Oat	Dissatisfaction
Wild Rose	Apathy and resignation
Willow	Self pity and bitterness

The Society has connections with an estimated 50,000 active practitioners from around the world, who use flower essence therapy as part of their treatment. FES encourages the study of the plants themselves to determine the characteristics of flower essences. They are compiling an extensive database of case studies and practitioner reports of the use of essences therapeutically, allowing verification and development of the original definitions. They are also engaged in the scientific study of flower essence therapy.

FES says they have developed the theories of Paracelsus and Goethe who researched the “signatures” and “gestures” of botanical specimens, on the premise that the human body and soul are a reflection

of the system of nature. FES plant research interprets the therapeutic properties of flower essences according to these insights.

In this regard, they have devised 12 “windows of perception” for monitoring the attributes of plants. Each of these windows reveals an aspect of the plant’s qualities, although they maintain that what they are seeking is a “whole which is greater than the sum of its parts.” The 12 windows are not considered independent classifications, but more of a blended tapestry of views of the qualities that each plant possesses.

The first window is concerned with the “form” of a plant—its shape classification. The second focuses on its “gesture” or spatial relationship. The third window is a plant’s botanical classification; the Flower Essence Society maintains that considering a plant’s botanical family is essential to obtaining an overview of its properties as a flower essence. The fourth window concerns the time orientation of a particular specimen regarding the daily and seasonal cycles. Why do some flowers bloom at different times of the day, while others, such as the evening primrose, respond to the moon? The fifth window observes a plant’s relationship to its environment. Where a plant chooses to grow, and where it cannot survive, reveals much about its qualities. The sixth window observes a plant’s relationship to the Four Elements and the Four Ethers, as FES maintains that plants exist in one of the elemental or etheric forces in addition to their physical life. “Elements” refers to those developed by the Greeks, as opposed to the modern concept of “molecular building blocks.” It seems that commonly, two elements predominate in a plant, indicating a polarity of qualities, while two can be said to be recessive. The seventh window relates to a plant’s relationship with the other kingdoms of nature: mineral, animal and human, while the eighth relates to the color and color variations of a plant. Katz explains how the language of color tells us so much about the “soul qualities” of a plant. The ninth window concerns all other sensory perceptions of a plant, such as fragrance, texture, and taste. The tenth window involves assessing the chemical substances and properties; the eleventh studies medicinal and herbal uses, as by studying the physical healing properties of plants, we can also understand something of their more subtle effects on the soul. Finally, the twelfth window involves the study of the lore, mythology, folk wisdom, and spiritual and ritual qualities associated with a particular plant. Katz relates how in the past, human beings were more in touch with the natural world, and the remnants of this unconscious plant

wisdom live on in the form of folklore, mythology, and so on.

Because flower remedies operate on approximately the same principles as homeopathy, practitioners quite often prescribe the two therapies in conjunction with each other. They can also be used concurrently with allopathic medicine.

The system consists of 38 remedies, each for a different disposition. The basic theory is that if the remedy for the correct disposition is chosen, the physical illness resulting from the present emotional state can then be cured. There is a rescue remedy made up of five of the essences—cherry plum, clematis, impatiens, rock star, and star of Bethlehem—that is recommended for the treatment of any kind of physical or emotional shock. Therapists recommended that rescue remedy be kept on hand to help with all emergencies.

The 38 Bach Remedies

- agrimony: puts on a cheerful front, hides true feelings, and worries or problems
- aspen: feelings of apprehension, dark foreboding, and premonitions
- beech: critical, intolerant, picky
- centaury: easily comes under the influence of others, weak willed
- cerato: unsure, no confidence in own judgement, intuition, and seeks approval from others
- cherry plum: phobic, fear of being out of control, and tension
- chestnut bud: repeats mistakes, does not learn from experience
- chicory: self-centered, possessive, clingy, demanding, self pity
- clematis: absent minded, dreamy, apathetic, and lack of connection with reality
- crab apple: a “cleanser” for prudishness, self-disgust, feeling unclean
- elm: a sense of being temporarily overwhelmed in people who are usually capable and in control
- gentian: discouraged, doubting, despondent
- gorse: feelings of pessimism, accepting defeat
- heather: need for company, talks about self, and concentrates on own problems
- holly: jealousy, envy, suspicion, anger, and hatred
- honeysuckle: reluctance to enter the present and let the past go

- hornbeam: reluctant to face a new day, weary, can't cope (mental **fatigue**)
- impatiens: impatience, always in a hurry, and resentful of constraints
- larch: feelings of inadequacy and apprehension, lack of confidence and will to succeed
- mimulus: fearful of specific things, shy, and timid
- mustard: beset by "dark cloud" and gloom for no apparent reason
- oak: courageous, persevering, naturally strong but temporarily overcome by difficulties
- olive: for physical and mental renewal, to overcome exhaustion from problems of long-standing
- pine: for self-reproach, always apologizing, assuming guilt
- red chestnut: constant worry and concern for others
- rock rose: panic, intense alarm, dread, horror
- rock water: rigid-minded, self-denial, restriction
- scleranthus: indecision, uncertainty, fluctuating moods
- star of Bethlehem: consoling, following shock or grief or serious news
- sweet chestnut: desolation, despair, bleak outlook
- vervain: insistent, fanatical, over-enthusiastic
- vine: dominating, overbearing, autocratic, tyrannical
- walnut: protects during a period of adjustment or vulnerability
- water violet: proud, aloof, reserved, enjoys being alone
- white chestnut: preoccupation with worry, unwanted thoughts
- wild oat: drifting, lack of direction in life
- wild rose: apathy, resignation, no point in life
- willow bitter: resentful, dissatisfied, feeling life is unfair

Originally, Bach collected the dew from chosen flowers by hand to provide his patients with the required remedy. This became impractical when his treatment became so popular that production could not keep up with demand. He then set about finding a way to manufacture the remedies, and found that floating the freshly picked petals on the surface of spring water in a glass bowl and leaving them in strong sunlight for three hours produced the desired effect. Therapists explain that the water is "potentized" by the essence of the flowers. The potentized water can

then be bottled and sold. For more woody specimens, the procedure is to boil them in a sterilized pan of water for 30 minutes. These two methods produce "mother tinctures" and the same two methods devised by Bach are still used today. Flower essences do not contain any artificial chemical substances, except for alcohol preservative.

Bach remedies cost around \$10 each, and there is no set time limit for treatment. It may take days, weeks, or in some cases months. Flower essences cost around \$6 each, and there is also no set time for the length of treatment, or the amount of essences that may be taken. These treatments are not generally covered by medical insurance.

Precautions

Bach remedies and flower essences are not difficult to understand, and are considered suitable for self administration. The only difficulty may be in finding the correct remedy, as it can sometimes be tricky to pinpoint an individual's emotional disposition. They are even safe for babies, children, and animals. An important aspect of treatment with flower remedies, is that if you feel instinctively that you need a particular remedy, you are encouraged to act on that instinct. However, it is advisable not to continue a particular remedy once you feel you no longer need it, and to try a different one if you feel that progress is not being made.

The remedies are administered from a stoppered bottle and need to be diluted. Individuals sensitive to alcohol can apply the concentrate directly to temples, wrists, behind the ears, or underarms. They should be kept in a cool dark place; like this they should last indefinitely. However, a diluted remedy should not be kept longer than three weeks. Two drops of each diluted remedy should be taken four times a day, including first thing in the morning and last thing at night. If the rescue remedy is being used, four drops should be used instead. Most therapists recommend that they be taken in spring water, but the remedy can be taken directly from the bottle, if care is taken that the dropper does not touch the tongue, as this would introduce bacteria that would spoil the remedy.

It is not recommended that more than six or seven Bach remedies be used at any one time. Instead, it is preferable to divide a larger amount up into two lots to ensure the optimum effectiveness of the remedies. No combination, or amount of combinations of the remedies can cause any harm, rather they become less effective.

KEY TERMS

Aura—Emanation of light from living things (plants and animals) that can be recorded by Kirlian photography.

Essence—The constituent of a plant that determines its characteristics.

Potentize—The process of transferring the healing energy of a plant into spring water.

Window—A perspective adopted to assess the property of a given plant.

Unlike FES, the Bach Centre does not encourage research to “prove” that the remedies work, preferring that people find out for themselves. They strive to keep the use of the Bach remedies as simple as possible, and to this end they do not keep case records. Bach warned before he died that others would try to change his work and make it more complicated. He was determined to keep it simple so that anyone could use it, and that is why he limited the system to only 38 remedies. The Centre points out that many who have used Bach’s research as a starting point have added other remedies to the list, even some that Bach himself rejected.

Side effects

Flower remedies or essences are generally regarded as being totally safe, and there are no known side effects apart from the rare appearance of a slight rash, which is not a reason to discontinue treatment, says the Bach Centre.

Research and general acceptance

Bach flower remedies and flower essences have not yet officially won the support of allopathic medicine, despite the fact that more and more medical doctors are referring patients for such treatments on the strength of personal conviction. However, it is difficult to discount the scores of testimonials. Some practitioners refer skeptics to the research that has been done regarding the “auras” of living things. Theoretically, the stronger the aura, the more alive an organism is. Flower essences have very strong auras.

Resources

BOOKS

Somerville, R. *Flower Remedies*. New York: Time-Life Books.

ORGANIZATIONS

Dr. Edward Bach Centre. Mount Vernon, Bakers Lane, Sotwell, Oxon, OX10 OPX, UK.
centre@bachcentre.com.
< <http://www.bachcentre.com> > .

Flower Essence Society. P.O. Box 459, Nevada City, CA 95959. (800) 736-9222 (US & Canada). (530) 265-9163.
Fax: (530) 265-0584. mail@flowersociety.org.
< <http://www.flowersociety.org> > .

Patricia Skinner

Flu see **Influenza**

Flucona see **Antifungal drugs, systemic**

Fluke infections

Definition

Fluke infections are diseases of the digestive tract and other organ systems caused by several different species of parasitic flatworms (Trematodes) that have complex life cycles involving hosts other than human beings. Trematode comes from a Greek word that means having holes and refers to the external suckers that adult flukes use to draw nourishment from their hosts. Fluke infections are contracted by eating uncooked fish, plants, or animals from fluke-infected waters. Symptoms vary according to the type of fluke infection.

Description

In humans, fluke infections can be classified according to those diseases caused by liver flukes and those caused by lung flukes. Diseases caused by liver flukes include fascioliasis, opisthorchiasis, and clonorchiasis. Cases of liver fluke infection have been reported in Europe and the United States, as well as the Middle East, China, Japan, and Africa. Diseases caused by lung flukes include paragonimiasis. Paragonimiasis is a common infection in the Far East, Southeast Asia, Africa, Central and South America, Indonesia, and the Pacific Islands. It is estimated that between 40 million and 100 million people worldwide suffer from either liver or lung fluke infections.

In their adult stage, liver and lung flukes are symmetrical in shape, ranging between 1/4–1 in in length, and look somewhat like long, plump leaves or blades of grass. They enter through the mouth and can infect any person at any age.

Causes and symptoms

The symptoms of fluke infection differ somewhat according to the type of fluke involved. All forms of liver and lung fluke infection, however, have the following characteristics:

- most persons who get infected do not develop symptoms (asymptomatic)
- the early symptoms of an acute fluke infection are not unique to these diseases alone (nonspecific symptoms)
- infection does not confer immunity against re-infection by the same species or infection by other species of flukes
- infection is usually associated with eating uncooked fish, plants, or animals that live in fresh water

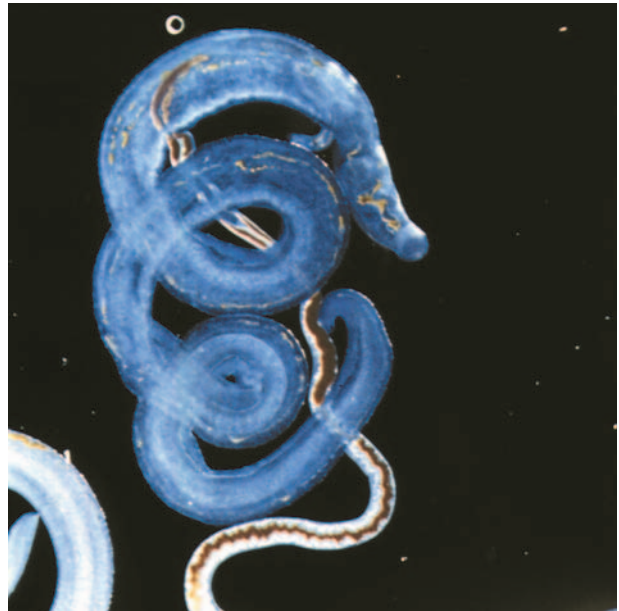
Fascioliasis

Fascioliasis is caused by *Fasciola hepatica*, the sheep liver fluke. The fluke has a three-part life cycle that begins when eggs from a host's feces are deposited in water. The eggs release free-swimming larvae (miracidia) that infect snails. The snails then release free-swimming larvae with tails (cercariae) that form cysts containing larvae in the infective stage (metacercariae) on vegetation growing in fresh water. Humans become infected when they eat watercress, water chestnuts, or other plants covered with the encysted metacercariae.

When a person eats contaminated plants, the cysts are broken open in the digestive system, and the metacercariae leave their cysts, pass through the wall of the intestine, and enter the liver, where they cause inflammation and destroy tissue. After a period of 10–15 weeks in the liver, the adult flukes move to the bile ducts and produce eggs. Acute fascioliasis is marked by abdominal **pain** with **headache**, loss of appetite, anemia, and **vomiting**. Some patients develop hives, muscle pains, or a yellow-color to the skin and whites of the eyes (**jaundice**). Chronic forms of the disease may produce complications, including blockage of the bile ducts or the migration of adult flukes to other parts of the body.

Opisthorchiasis and clonorchiasis

These infections are caused by *Clonorchis sinensis*, the Chinese liver fluke, and *Opisthorchis viverrini* or *O. felinus*. The diseases are widespread, affecting more than 20 million people in Japan, China, Southeast Asia, and India. The life cycle of these liver flukes is similar to that of *F. hepatica* except



A micrograph of adult intestinal blood flukes, *Schistosoma mansoni*. Humans can become infected while bathing or working in contaminated water. (Photo Researchers, Inc. Reproduced by permission.)

that the metacercariae are encysted in freshwater fish rather than on plants. Dogs, cats, and other mammals that eat raw fish can be infected with opisthorchiasis and clonorchiasis.

The symptoms of opisthorchiasis and clonorchiasis are similar to those of fascioliasis and include both acute and chronic forms. In acute infection, the patient may be tired, have a low-grade fever, pains in the joints, a swollen liver, abdominal pain, and a skin rash. The acute syndrome may be difficult to diagnose because the fluke eggs do not appear in the patient's stool for three to four weeks after infection. Patients with the chronic form of the disease experience a loss of appetite, **fatigue**, low-grade fever, diarrhea, and an enlarged liver that feels sore when the abdomen is pressed.

Paragonimiasis

Paragonimiasis is caused by a lung fluke, either *Paragonimus westermani* or *P. skrjabini*. These flukes are larger than liver flukes and infect meat- or fish-eating animals as well as humans. Their life cycle is similar to that of liver flukes except that their encysted larvae infect crabs and crayfish rather than plants or fish. Humans can ingest the encysted metacercariae from drinking contaminated water or eating raw or undercooked crabs and crayfish.

In humans, the metacercariae are released from their cysts in the small intestine and migrate to the lungs or the brain in 1% of cases. In the lungs, the flukes lay their eggs and form areas of inflammation covered with a thin layer of fibrous tissue. These areas of infection may eventually rupture, causing the patient to **cough** up fluke eggs, blood, and inflamed tissue. The period between the beginning of the infection and the appearance of the eggs during coughing is about six weeks. Patients with lung infections may have chest pain and **fever** as well as rust-colored or bloody sputum. Lung infections can lead to lung abscess, **pneumonia**, or **bronchitis**. Patients with fluke infections of the brain may experience seizures or a fatal inflammation of brain tissue called **encephalitis**. Some patients also develop **diarrhea** and abdominal pain or lumps under the skin that contain adult flukes.

Diagnosis

Diagnosis of fluke infections is based on a combination of the patient's history, particularly travel or residence in areas known to have flukes, and identification of the fluke's eggs or adult forms. In some patients, the eggs are found in fluid from the lungs, bile duct, or small intestine. Samples of these fluids can be obtained with a suction instrument (aspirator). Because most types of fluke infections are rare in the United States, stool specimens or body fluid samples may need to be sent to a laboratory with experts in unusual diseases or conditions to identify the specific parasite. In some cases, adult flukes may be found in the patient's stools, vomit, sputum, or skin lumps (for lung flukes). In the case of lung flukes, it is important for the doctor to rule out tuberculosis as a possible diagnosis. A **tuberculosis** skin test and chest x ray will usually be sufficient to do this.

Blood tests may be useful in diagnosing fluke infections, but their usefulness is limited because of cross-reactions. A cross-reaction occurs in blood testing when a particular disease agent reacts with antibodies specific to another disease agent. This result means that the doctor may know that the person is infected by flukes but cannot tell from the blood test alone which specific type of fluke is causing the disease. In addition, blood tests for fluke infections cannot distinguish between past and current infections. In some cases, sophisticated imaging techniques, such as **computed tomography scans** (CT scans) or ultrasound scans of the patient's chest or brain (for lung flukes) or abdomen (for liver flukes), are useful in confirming a diagnosis of fluke infection.

KEY TERMS

Aspirator—A medical instrument that uses suction to withdraw fluids from the lungs, digestive tract, or other parts of the body for laboratory testing.

Asymptomatic—Persons who carry a disease and are usually capable of transmitting the disease but, who do not exhibit symptoms of the disease are said to be asymptomatic.

Cercaria (plural, cercariae)—An intermediate-stage of the fluke larva, released into water by infected snails.

Cross-reaction—A reaction that occurs in blood testing when a disease agent reacts to the specific antibody for another disease agent. Cross-reactions are common in blood tests for fluke infections because the different species are closely related.

Encysted—Enclosed in a cyst or capsule. Flukes spend part of their life cycle as encysted larvae.

Fluke—A parasitic flatworm that has external suckers. Flukes are sometimes called trematodes.

Host—The living animal that supplies nutrition to a parasite.

Jaundice—Yellowing of the skin and the whites of the eyes as a result of excess bile in the blood due to an improperly functioning liver.

Metacercaria (plural, metacercariae)—The encysted stage of a fluke larva that produces infection in human beings.

Miracidium (plural, miracidia)—The free-swimming larval form in the life cycle of the liver fluke.

Parasite—An organism that lives on or inside an animal of a different species and feeds on it or draws nutrients from it.

Trematode—Parasitic flatworms or another name for fluke, taken from a Greek word that means having holes.

Treatment

Liver and lung fluke infections are treated with medications. These include triclabendazole, praziquantel, bithionol, albendazole, and mebendazole. Praziquantel works by paralyzing the flukes' suckers, forcing them to drop away from the walls of the host's blood vessels. In the United States, bithionol is available only from the Centers for Disease Control (CDC). Depending on the species of fluke and the

severity of infection, the course of treatment can vary from several days to several weeks. Cure rates vary from 50–95%. Most patients experience mild temporary side effects from these drugs, including diarrhea, **dizziness**, or headache.

Prognosis

The prognosis for recovery from liver fluke infections is good, although patients with serious infections may be more vulnerable to other diseases, particularly if significant liver damage has occurred. Most patients with lung fluke infections also recover, however, severe infections of the brain can cause **death** from the destruction of central nervous system or brain tissue.

Prevention

No vaccines have been developed that are effective against lung or liver fluke infections. Prevention of these infections includes the following measures:

- boiling or purifying drinking water
- avoiding raw or undercooked fish or salads made from fresh aquatic plants; all food eaten in areas with fluke infestations should be cooked thoroughly; pickling or **smoking** will not kill fluke cysts in fish or shellfish
- control or eradication of the snails that serve as the flukes' intermediate hosts

Resources

BOOKS

Goldsmith, Robert S. "Infectious Diseases: Protozoal & Helminthic." *Current Medical Diagnosis and Treatment*, 1998, edited by Stephen McPhee, et al., 37th ed. Stamford: Appleton & Lange, 1997.

Rebecca J. Frey, PhD

Fluoroquinolones

Definition

Fluoroquinolones are medicines that kill bacteria or prevent their growth.

Purpose

Fluoroquinolones are antimicrobials, medicines used to treat infections caused by microorganisms.

Physicians prescribe these drugs for bacterial infections in many parts of the body. For example, they are used to treat bone and joint infections, skin infections, urinary tract infections, inflammation of the prostate, serious ear infections, **bronchitis**, **pneumonia**, **tuberculosis**, some **sexually transmitted diseases** (STDs), and some infections that affect people with **AIDS**.

Description

Fluoroquinolones are available only with a physician's prescription and are sold in tablet and injectable forms. Examples of these medicines are moxifloxacin (Avelox), ciprofloxacin (Cipro), ofloxacin (Floxin), levofloxacin (Levaquin), lomefloxacin (Maxaquin), norfloxacin (Noroxin), enoxacin (Penetrex), gatifloxacin (Tequin), and sparfloxacin (Zagam).

In the wake of the **anthrax** terrorist attacks in the United States in 2001, ciprofloxacin received extensive media attention because it was the only drug labeled as approved by the Food and Drug Administration (FDA) for both **prophylaxis** and treatment of inhalation anthrax (the most serious form of the disease). However, in late October 2001, the FDA issued a notice clarifying that the antibiotic doxycycline is also approved for anthrax prophylaxis and that doxycycline and amoxicillin are also approved for treatment for all forms of anthrax. The FDA encouraged companies to update labeling of these products with this previously unspecified information.

Recommended dosage

The recommended dosage depends on the type and strength of fluoroquinolone, and the kind of infection for which it is being taken. Check with the physician who prescribed the drug or the pharmacist who filled the prescription for the correct dosage.

To make sure the infection clears up completely, take the medicine for as long as it has been prescribed. Do not stop taking the drug just because symptoms begin to improve. Symptoms may return if the drug is stopped too soon.

Fluoroquinolones work best when they are at constant levels in the blood. To help keep levels constant, take the medicine in doses spaced evenly through the day and night. Do not miss any doses. For best results, take this medicine with a full glass of water and drink several more glasses throughout the day, every day during treatment with the drug. The extra water will help prevent some side effects. Some

fluoroquinolones should be taken on an empty stomach; others may be taken with meals. Check package directions or ask the physician or pharmacist for instructions on how to take the medicine.

Precautions

An important precaution for any antibiotic is that unnecessary use or **abuse of antibiotics** can encourage drug-resistant strains of bacteria to develop and proliferate. These drug-resistant strains then become difficult, or even impossible, to treat. Bacteria found in hospitals appear to have become especially resilient, and are causing increasing difficulty for patients and the doctors treating them. Following the U.S. 2001 anthrax attacks, for example, the American Medical Association urged its members not to prescribe ciprofloxacin unnecessarily. One fear is that the overuse of the drug could reduce its effectiveness against infections such as **typhoid fever**, hospital-acquired pneumonia, and others.

Research suggests that fluoroquinolones may cause bone development problems in children and teenagers. Infants, children, teenagers, pregnant women, and women who are breastfeeding should not take this medicine unless directed to do so by a physician.

Although such side effects are rare, some people have had severe and life-threatening reactions to fluoroquinolones. Call a physician immediately if any of these signs of a dangerous reaction occur:

- swelling of the face and throat
- swallowing problems
- shortness of breath
- rapid heartbeat
- tingling of fingers or toes
- **itching or hives**
- loss of consciousness

Some fluoroquinolones may weaken the tendons in the shoulder, hand, or heel, making the tendons more likely to tear. Anyone who notices **pain** or inflammation in these or other tendon areas should stop taking the medicine immediately and call a physician. Rest and avoid **exercise** until the physician determines whether the tendons are damaged. If the tendons are torn, surgery may be necessary to repair them.

These medicines make some people feel drowsy, dizzy, lightheaded, or less alert. Anyone who takes these drugs should not drive, use machines or do

anything else that might be dangerous until they have found out how the drugs affect them.

This medicine may increase sensitivity to sunlight. Even brief exposure to sun can cause a severe **sunburn** or a rash. While being treated with fluoroquinolones, avoid being in direct sunlight, especially between 10 a.m. and 3 p.m.; wear a hat and tightly woven clothing that covers the arms and legs; use a sunscreen with a skin protection factor (SPF) of at least 15; protect the lips with a sun block lipstick; and do not use tanning beds, tanning booths, or sunlamps.

Do not take **antacids** that contain aluminum, calcium, or magnesium at the same time as fluoroquinolones. The antacids may keep the fluoroquinolones from working as they should. If antacids are needed, take them at least two hours before or two hours after taking norfloxacin or ofloxacin, at least four hours before or two hours after taking ciprofloxacin. Follow the same instructions for taking sucralfate (Carafate), a medicine used to treat stomach ulcers and other irritation in the digestive tract and mouth.

Anyone who has had unusual reactions to fluoroquinolones or related medicines such as cinoxacin (Cinobac) or nalidixic acid (NegGram) in the past should let his or her physician know before taking the drugs again. The physician should also be told about any **allergies** to foods, dyes, preservatives, or other substances.

Before using fluoroquinolones, people with any of these medical problems should make sure their physicians are aware of their conditions:

- kidney disease
- liver disease with kidney disease
- diseases of the brain or spinal cord, including hardening of the arteries in the brain, epilepsy, and other seizure disorders

Taking fluoroquinolones with certain other drugs may affect the way the drugs work or may increase the chance of side effects.

Side effects

The most common side effects are mild **diarrhea**, **nausea**, **vomiting**, stomach or abdominal pain, **dizziness**, drowsiness, lightheadedness, nervousness, sleep problems, and **headache**. These problems usually go away as the body adjusts to the drug and do not require medical treatment unless they are bothersome.

More serious side effects are not common, but may occur. If any of the following side effects occur, check with a physician immediately:

- skin rash or other skin problems such as itching, peeling, hives, or redness
- **fever**
- agitation or confusion
- hallucinations
- shakiness or **tremors**
- seizures or convulsions
- tingling of fingers or toes
- pain where the medicine was injected (lasting after the injection)
- pain in the calves, spreading to the heels
- swelling of the calves or lower legs
- swelling of the face or neck
- swallowing problems
- rapid heartbeat
- shortness of breath
- loss of consciousness

Other rare side effects may occur. Anyone who has unusual symptoms after taking fluoroquinolones should get in touch with his or her physician.

Interactions

Fluoroquinolones may interact with other medicines. When this happens, the effects of one or both of the drugs may change or the risk of side effects may be greater. Anyone who takes fluoroquinolones should let the physician know all other medicines he or she is taking. Among the drugs that may interact with fluoroquinolones are:

- antacids that contain aluminum, calcium, or magnesium
- medicines that contain iron or zinc, including multivitamin and mineral supplements
- sucralfate (Carafate)
- caffeine
- blood thinning drugs such as warfarin (Coumadin)
- airway opening drugs (**bronchodilators**) such as aminophylline, theophylline (Theo-Dur and other brands), and oxtriphylline (choledyl and other brands)
- didanosine (Videx), used to treat HIV infection.

The list above does not include every drug that may interact with fluoroquinolones. Be sure to check with a physician or pharmacist before combining fluoroquinolones with any other prescription or nonprescription (over-the-counter) medicine.

KEY TERMS

Bacteria—Tiny, one-celled forms of life that cause many diseases and infections.

Bronchitis—Inflammation of the air passages of the lungs.

Digestive tract—The stomach, intestines, and other parts of the body through which food passes.

Inflammation—Pain, redness, swelling, and heat that usually develop in response to injury or illness.

Microorganism—An organism that is too small to be seen with the naked eye.

Pneumonia—A disease in which the lungs become inflamed. Pneumonia may be caused by bacteria, viruses, or other organisms, or by physical or chemical irritants.

Prostate—A donut-shaped gland in males below the bladder that contributes to the production of semen.

Sexually transmitted disease (STD)—A disease that is passed from one person to another through sexual intercourse or other intimate sexual contact.

Tendon—A tough band of tissue that connects muscle to bone.

Tuberculosis—An infectious disease that usually affects the lungs, but may also affect other parts of the body. Symptoms include fever, weight loss, and coughing up blood.

Urinary tract—The passage through which urine flows from the kidneys out of the body.

Resources

OTHER

“Fluoroquinolones (Systemic).” National Library of medicine. < <http://www.nlm.nih.gov/medlineplus/druginfo/fluoroquinolonessystemic202656.html> > .

Rosalyn Carson-DeWitt, MD

Fluoxetine see **Selective serotonin reuptake inhibitors**

Flurbiprofen see **Nonsteroidal anti-inflammatory drugs**

Focal glomerulosclerosis see **Nephrotic syndrome**

Folic acid

Definition

Folic acid is a water-soluble vitamin belonging to the B-complex group of **vitamins**. These vitamins help the body break down complex carbohydrates into simple sugars to be used for energy. Excess B vitamins are excreted from the body rather than stored for later use. This is why sufficient daily intake of folic acid is necessary.

Description

Folic acid is also known as folate, or folacin. It is one of the nutrients most often found to be deficient in the Western diet, and there is evidence that deficiency is a problem on a worldwide scale. Folic acid is found in leafy green vegetables, beans, peas and lentils, liver, beets, brussel sprouts, poultry, nutritional yeast, tuna, wheat germ, mushrooms, oranges, asparagus, broccoli, spinach, bananas, strawberries, and cantaloupes. In 1998, the U.S. Food and Drug Administration (FDA) required food manufacturers to add folic acid to enriched bread and grain products to boost intake and to help prevent neural tube defects (NTD).

Purpose

Folic acid works together with vitamin B₁₂ and vitamin C to metabolize protein in the body. It is important for the formation of red and white blood cells. It is necessary for the proper differentiation and growth of cells and for the development of the fetus. It is also used to form the nucleic acid of DNA and RNA. It increases the appetite and stimulates the production of stomach acid for digestion and it aids in maintaining a healthy liver. A deficiency of folic acid may lead to anemia, in which there is decreased production of red blood cells. This reduces the amounts of oxygen and nutrients that are able to get to the tissues. Symptoms may include **fatigue**, reduced secretion of digestive acids, confusion, and forgetfulness. During **pregnancy**, a folic acid deficiency may lead to **preeclampsia**, premature birth, and increased bleeding after birth.

People who are at high risk of strokes and heart disease may greatly benefit by taking folic acid supplements. An elevated blood level of the amino acid **homocysteine** has been identified as a risk factor for some of these diseases. High levels of homocysteine have also been found to contribute to problems with **osteoporosis**. Folic acid, together with vitamins B₆ and

B₁₂, helps break down homocysteine, and may help reverse the problems associated with elevated levels.

Pregnant women have an increased need for folic acid, both for themselves and their child. Folic acid is necessary for the proper growth and development of the fetus. Adequate intake of folic acid is vital for the prevention of several types of **birth defects**, particularly NTDs. The neural tube of the embryo develops into the brain, spinal cord, spinal column, and the skull. If this tube forms incompletely during the first few months of pregnancy a serious, and often fatal, defect results in **spina bifida** or anencephaly. Folic acid, taken from one year to one month before conception through the first four months of pregnancy, can reduce the risk of NTDs by 50–70%. It also helps prevent a **cleft lip and palate**.

Research shows that folic acid can be used to successfully treat cervical dysplasia, a condition diagnosed by a Pap smear, of having abnormal cells in the cervix. This condition is considered to be a possible precursor to **cervical cancer**, and is diagnosed as an abnormal Pap smear. Daily consumption of 1,000 mcg of folic acid for three or more months has resulted in improved cervical cells upon repeat Pap smears.

Studies suggest that long-term use of folic acid supplements may also help prevent lung and **colon cancer**. Researchers have also found that alcoholics who have low folic acid levels face a greatly increased possibility of developing colon **cancer**.

Preparations

To correct a folic acid deficiency, supplements are taken in addition to food. Since the functioning of the B vitamins is interrelated, it is generally recommended that the appropriate dose of B-complex vitamins be taken in place of single B vitamin supplements. The Recommended Dietary Allowances (RDA) for folate is 400 mcg per day for adults, 600 mcg per day for pregnant women, and 500 mcg for nursing women. Medicinal dosages of up to 1,000–2,000 mcg per day may be prescribed.

Precautions

Folic acid is not stable. It is easily destroyed by exposure to light, air, water, and cooking. Therefore, the supplement should be stored in a dark container in a cold, dry place, such as a refrigerator. Many medications interfere with the body's absorption and use of folic acid. This includes sulfa drugs, sleeping pills, estrogen, anti-convulsants, birth control pills, **antacids**, quinine, and some **antibiotics**. Using large

KEY TERMS

Homocysteine—An amino acid involved in the breakdown and absorption of protein in the body.

Preeclampsia—A serious disorder of late pregnancy in which the blood pressure rises, there is a large amount of retained fluids, and the kidneys become less effective and excrete proteins directly into the urine.

Raynaud's disease—A symptom of various underlying conditions affecting blood circulation in the fingers and toes and causing them to be sensitive to cold.

Recommended Dietary Allowance (RDA)—Guidelines for the amounts of vitamins and minerals necessary for proper health and nutrition established by the National Academy of Sciences in 1989.

Water-soluble vitamins—Vitamins that are not stored in the body and are easily excreted. They must, therefore, be consumed regularly as foods or supplements to maintain health.

amounts of folic acid (e.g., over 5,000 mcg per day) can mask a vitamin B₁₂ deficiency and thereby risk of irreversible nerve damage.

Side effects

At levels of 5,000 mcg or less, folic acid is generally safe for use. Side effects are uncommon. However, large doses may cause **nausea**, decreased appetite, bloating, gas, decreased ability to concentrate, and **insomnia**. Large doses may also decrease the effects of phenytoin (Dilantin), a seizure medication.

Interactions

As with all B-complex vitamins, it is best to take folic acid with the other B vitamins. Vitamin C is important to the absorption and functioning of folic acid in the body.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>> .

OTHER

Adams, Suzanne L. *The Art of Cytology: Folic Acid/ B-12 Deficiency*. suzann@concentric.net. <<http://www.concentric.net/~Suza2/page22.htm>> .

“Folic Acid: Coming to A Grocery Store Near You.” <<http://www.mayohealth.org/mayo/9710/htm/folic.htm>> .

“Folic Acid.” <<http://www.cybervitamins.com/folicacid.htm>> .

“Folic acid (oral/injectable).” Dr. Koop.com.Inc. 700 N. Mopac, Suite 400, Austin, TX 48731. <<http://www.drkoop.com/hcr/drugstore/pharmacy/leaflets/english/d00241a1.asp>> .

Pregnancy and Nutrition Update. <<http://www.mayo-health.org/mayo/9601/htm/pregvit.htm>> .

Patience Paradox

Folic acid deficiency anemia

Definition

Folic acid deficiency, an abnormally low level of one of the **B vitamins**, results in anemia characterized by red blood cells that are large in size but few in number.

Description

Folic acid is necessary for growth and cellular repair, since it is a critical component of DNA and RNA as well as essential for the formation and maturation of red blood cells. Folic acid deficiency is one of the most common of all vitamin deficiencies. Although it occurs in both males and females, folic acid deficiency anemia most often affects women over age 30. It becomes increasingly common as age impedes the body's ability to absorb folic acid, a water-soluble vitamin that is manufactured by intestinal bacteria and stored for a short time in the liver. Folic acid deficiency has also been implicated as a cause of neural tube defects in the developing fetus. Recent research has shown that adequate amounts of folic acid can prevent up to one-half of these birth defects, if women start taking folic acid supplements shortly before conception. Research from China in 2004 showed that women who were low in B vitamins and folate before conception, though not technically anemic, still had increased risk of lower birth weight babies and adverse **pregnancy** outcome.

A healthy adult needs at least 400 mcg of folic acid every day. Requirements at least double during pregnancy, and increase by 50% when a woman is breastfeeding. The average American diet, high in fats, sugar, and white flour, provides about 200 mcg of

folic acid, approximately the amount needed to maintain tissue stores of the substance for six to nine months before a deficiency develops. Most of the folic acid in foods (with the exception of the folic acid added to enriched flour and breakfast cereals) occurs as folate. Folate is only about one-half as available for the body to use as is the folic acid in pills and supplements. Folate also is easily destroyed by sunlight, overcooking, or the storing of foods at room temperature for an extended period of time.

Good dietary sources of folate include:

- leafy green vegetables
- liver
- mushrooms
- oatmeal
- peanut butter
- red beans
- soy
- wheat germ

Causes and symptoms

This condition usually results from a diet lacking in foods with high folic acid content, or from the body's inability to digest foods or absorb foods having high folic acid content. Other factors that increase the risk of developing folic acid deficiency anemia are:

- age
- alcoholism
- birth control pills, anticonvulsant therapy, sulfa antibiotics, and certain other medications
- illness
- smoking
- stress

Fatigue is often the first sign of folic acid deficiency anemia. Other symptoms include:

- anorexia nervosa
- pale skin
- paranoia
- rapid heart beat
- sore, inflamed tongue
- weakness
- weight loss

Diagnosis

Diagnostic procedures include blood tests to measure hemoglobin, an iron-containing compound that carries oxygen to cells throughout the body. Symptoms may be reevaluated after the patient has taken prescription folic acid supplements.

Treatment

Folic acid supplements are usually prescribed, and self-care includes avoiding:

- alcohol
- non-herbal tea, **antacids**, and phosphates (contained in beer, ice cream, and soft drinks), which restrict iron absorption
- tobacco

A person with folic acid deficiency anemia should rest as often as necessary until restored energy levels make it possible to resume regular activities. A doctor should be seen if **fever**, chills, muscle aches, or new symptoms develop during treatment, or if symptoms do not improve after two weeks of treatment.

Alternative treatment

Alternative therapies for folic acid deficiency anemia may include **reflexology** concentrated on areas that influence the liver and spleen. Increasing consumption of foods high in folate is helpful. Eating a mixture of yogurt (8 oz) and turmeric (1 tsp) also may help resolve symptoms. A physician should be contacted if the tongue becomes slick or smooth or the patient:

- bruises or tires easily
- feels ill for more than five days
- feels weak or out of breath
- looks pale or jaundiced

Prognosis

Although adequate folic acid intake usually cures this condition in about three weeks, folic acid deficiency anemia can make patients infertile or more susceptible to infection. Severe deficiencies can result in congestive **heart failure**.

Prevention

Eating raw or lightly cooked vegetables every day will help maintain normal folic acid levels, as will taking a folic acid supplement containing at least

400 mcg of this vitamin. Because folic acid deficiency can cause **birth defects**, all women of childbearing age who can become pregnant should consume at least 400 mcg of folic acid daily; a woman who is pregnant should have regular medical checkups, and take a good prenatal vitamin.

Resources

PERIODICALS

Ronnenberg, Alayne G., et al. "IPreconception Hemoglobin and Ferritin Concentrations Are Associated With Pregnancy Outcome in a Prospective Cohort of Chinese Women." *The Journal of Nutrition* October 2004: 2586–2592.

Maureen Haggerty
Teresa G. Odle

Follicle-stimulating hormone test

Definition

The follicle-stimulating hormone (FSH) test measures the amount of FSH in the blood. FSH is a hormone that regulates the growth and development of eggs and sperm, and this test is used to diagnose or evaluate disorders involving the pituitary gland and reproductive system.

Purpose

FSH testing is performed if a physician suspects the patient may have a disorder involving the reproductive system or pituitary gland. The pituitary gland produces FSH, which stimulates the growth of the sacks (follicles) that surround the eggs in a woman's ovaries. This is important for the process of ovulation, in which the egg is released. In men, FSH stimulates production of sperm. If there are abnormal levels of FSH in the blood it may mean that one of several disorders are present. Normal fluctuations occur as a result of **puberty**, the menstrual cycle, **pregnancy**, and **menopause**.

The FSH test is performed more often on women than on men. In women, it is used to determine if menopause has begun, to diagnose **infertility** and menstrual disorders (such as anovulatory bleeding), to measure hormone levels in children who enter puberty at an early age, and to diagnose other disorders. In men, it can be used to determine early puberty,

abnormal tissue growth on one or more of the hormone-secreting (endocrine) glands (called multiple endocrine neoplasia), or to diagnose other disorders.

Description

The FSH test is a blood test. Blood will be drawn from the patient and analyzed in a laboratory.

Preparation

In preparation for the test, there are no food or fluid intake restrictions. Patients may be advised to discontinue certain medications for 48 hours before the test. A menstruating woman having hot flashes or irregular periods should be tested on the second or third day of her menstrual cycle. A woman who has missed a period and is having other menopausal symptoms can be tested at any time.

Aftercare

No aftercare is necessary.

Risks

There are no risks associated with this test.

Normal results

Normal FSH test results vary according to age and sexual maturity. The phase of a woman's menstrual cycle or use of birth-control pills also affects test results.

For an adult male, normal results range from about 4–25 units of FSH in every liter of blood (U/L) or about 5–20 micro-international units in every milliliter.

For a premenopausal woman, normal values range from 4–30 U/L or 5–20 micro-international units per milliliter. In a pregnant woman, FSH levels are too low to measure. After menopause, normal values range from 40–250 U/L or 50–100 micro-international units per milliliter.

FSH levels fluctuate during premenopause. If no other symptoms are present, an elevated FSH level should not be interpreted as proof that menopause has begun.

Abnormal results

Anorexia nervosa and disorders of the hypothalamus or pituitary gland can result in abnormally low FSH levels.

KEY TERMS

Anovulatory bleeding—Bleeding without release of an egg from an ovary.

Hypopituitarism—Underactivity of the pituitary gland.

Hypothalamus—The part of the brain that controls the endocrine system.

Klinefelter's syndrome—Chromosomal abnormality characterized by small testes and male infertility.

Multiple endocrine neoplasia—Abnormal tissue growth on one or more of the endocrine (hormone-secreting) glands.

Polycystic ovary disease—A condition in which a woman has little or no menstruation, is infertile, has excessive body hair, and is obese. The ovaries may contain several cysts.

Turner syndrome—Chromosomal abnormality characterized by immature reproductive organs in women.

Abnormal levels can also indicate:

- infertility
- hypopituitarism
- klinefelter syndrome (in men)
- turner syndrome
- ovarian failure
- polycystic ovary syndrome

Resources

OTHER

"Follicle-Stimulating Hormone Test." *Health Answers.com* February 25, 1998. <<http://www.healthanswers.com>>.

Maureen Haggerty

Follicular cysts see **Ovarian cysts**

Folliculitis

Definition

Folliculitis is inflammation or infection of one or more hair follicles (openings in the skin that enclose hair).

Description

Folliculitis can affect both women and men at any age. It can develop on any part of the body, but is most likely to occur on the scalp, face, or parts of the arms, armpits, or legs not usually covered by clothing.

Small, yellowish-white blister-like lumps (pustules) surrounded by narrow red rings are usually present with both bacterial folliculitis and fungal folliculitis. Hair can grow through or alongside of the pustules, which sometimes ooze blood-stained pus.

Folliculitis can cause **boils** and, in rare instances, serious skin infections. Bacteria from folliculitis can enter the blood stream and travel to other parts of the body.

Causes and symptoms

Folliculitis develops when bacteria, such as *Staphylococcus*, or a fungus enters the body through a cut, scrape, surgical incision, or other break in the skin near a hair follicle. Scratching the affected area can trap fungus or bacteria under the fingernails and spread the infection to hair follicles on other parts of the body.

The bacteria that cause folliculitis are contagious. A person who has folliculitis can infect others who live in the same household.

Factors that increase the risk of developing folliculitis include:

- dermatitis
- diabetes
- dirty, crowded living conditions
- eczema
- exposure to hot, humid temperatures
- infection in the nose or other recent illness
- tight clothing

Diagnosis

Diagnosis is based on the patient's medical history and observations. Laboratory analysis of the substance drained from a pustule can be used to distinguish bacterial folliculitis from fungal folliculitis.

Treatment

Bacterial folliculitis may disappear without treatment, but is likely to recur. Non-prescription **topical antibiotics** like Bacitracin, Mycitracin, or Neomycin, gently rubbed on to affected areas three or four times a



Acne folliculitis. (Custom Medical Stock Photo. Reproduced by permission.)

day, can clear up a small number of bacterial folliculitis pustules. Oral **antibiotics** such as erythromycin (Erythocin) may be prescribed if the infection is widespread. The drug griseofulvin (Fulvicin) and topical antifungal medications are used to treat fungal folliculitis.

A doctor should be notified if:

- pustules spread after treatment has begun or reappear after treatment is completed
- the patient's **fever** climbs above 100 °F (37.8 °C)
- the patient develops boils or swollen ankles
- redness, swelling, warmth, or **pain** indicate that the infection has spread
- unexplained new symptoms appear

Alternative treatment

Eating a balanced diet, including protein, complex carbohydrates, healthy fats, fresh fruits and vegetables, and drinking eight to 10 glasses of water a day may stimulate the body's immune system and shorten the course of the infection. Garlic (*Allium sativum*) and goldenseal (*Hydrastis canadensis*), both antiseptic agents against staph infections, may be taken. The daily dosage would vary from person to person and is based on the severity of the infection. Echinacea (*Echinacea* spp.) is helpful in modulating immune function. Again, the dosage would vary.

Daily doses of 30–50 mg zinc and 1,000–5,000 mg Vitamin C (taken in equal amounts at several times during the day), and 300–2,000 mg bioflavonoids can also strengthen the body's infection-fighting ability. High doses of **vitamins** and **minerals** should not be used without a doctor's approval.

Prognosis

If properly treated, the symptoms of bacterial folliculitis generally disappear in about two weeks. Fungal folliculitis should clear up within six weeks. But it can worsen if the condition is misdiagnosed and inappropriately treated with steroid creams.

Prevention

Anyone who has a tendency to develop folliculitis should cleanse the skin with antibacterial soap twice a day and before shaving and should not use oily skin lotions. Men should not shave while the beard area is infected. When they begin shaving again, they should use a new blade each time. Women who have had fungal folliculitis should use depilatory creams instead of razors. Daily shampooing can help prevent folliculitis in the scalp. The spread of infection can be prevented by not sharing towels or washcloths.

Resources

OTHER

"Folliculitis." *Thrive Online*. April 5, 1998. <<http://thriveonline.oxygen.com>>.

Maureen Haggerty

Food allergies

Definition

Food **allergies** are the body's abnormal responses to harmless foods; the reactions are caused by the immune system's reaction to some food proteins.

Description

Food allergies are often confused with food intolerance. However, the two conditions have different causes and produce different symptoms. A food allergy is also known as food hypersensitivity. The allergy is caused when a person eats something that the immune system incorrectly identifies as harmful.

Food allergies

About 4% of adults have food allergies according to the National Institute of Allergy and Infectious Diseases (NIAID). The condition affects approximately 6 to 8% of children age 4 and younger.

The immune system works to protect the body and creates food-specific antibodies. The antibodies are proteins that battle antigens, substances that are foreign or initially outside the body. The introduction of an antigen produces the immune response. Antibodies are created to destroy the antigen or counteract its effectiveness.

The food that triggered that reaction is called an allergen. The antibodies are like an alarm system coded to detect the food regarded as harmful. The next time the person eats that food, the immune system discharges a large amount of histamine and chemicals. This process meant to protect the body against the allergen causes an allergic reaction that can affect the respiratory tract, digestive tract, skin, and cardiovascular system.

Allergic reactions can occur in minutes or in up to two hours after the person ate the food. Symptoms include swelling of the tongue, **diarrhea**, and **hives**. In severe cases, the allergic reaction can be fatal. The most severe reaction is **anaphylaxis**, which could be life-threatening.

Food intolerance

While food allergies involve the immune system, food intolerance is not related to the immune system. For example, a person who is lactose intolerant has a shortage of lactase, the digestive enzyme that breaks down the sugar in milk and dairy products. That person could experience stomach **pain** or bloating several hours after drinking milk.

People who are food-intolerant can sometimes consume that food and not experience intolerance symptoms. Those diagnosed with food allergies must avoid the foods that produce the allergic reactions.

Allergy-producing foods

Although approximately 160 foods produce allergic reactions, approximately 90% of reactions are caused by some or all items within eight food families. These are milk, eggs, peanuts, tree nuts, fish, shellfish, wheat, and soy. These foods can cause severe reactions. The most adverse reactions are caused by peanuts and tree nuts. According to NIAID, about 0.6% of Americans are impacted by peanut allergies.

Approximately 0.4% of Americans have allergic reactions to tree nuts.

Food allergy demographics

Most children have allergies to eggs, milk, peanuts or tree nuts, and soy, according to the American Dietetic Association (ADA). The young generally outgrow their allergies. They are more likely to outgrow milk and soy allergies, according to NIAID. However, children and adults usually allergic to peanuts and tree nuts for life. The most frequent causes of food allergies in adulthood are peanuts, tree nuts, fish, and shellfish.

Allergies are hereditary. There is a tendency for the immune system to create IgE antibodies in people with family histories of allergies and allergic conditions like hay **fever** and **asthma**, according to NIAID. The likelihood of a child having food allergies increases when both parents are allergic.

Furthermore, people are allergic to the foods that are eaten frequently in their countries. A rice allergy is more common in Japan, and codfish allergies occur more in Scandinavian countries, according to NIAID.

Causes and symptoms

Food allergies are caused by the immune system's reaction to a food item that it believes is harmful. When the food is digested, the immune system responds by creating immunoglobulin E (IgE) antibodies as a defense. The antibodies are proteins found in the bloodstream. Formed to protect the body against harmful substances, the antibodies are created after the person's first exposure to the allergen.

The majority of food allergies are caused by foods in eight families. In some families, every food causes an allergic reaction. In other families like shellfish, a person may be allergic to one species, but able to eat others. The allergy-inducing foods include:

- **Milk.** The dairy family includes milk, ice cream, yogurt, butter, and some margarines. Nondairy foods that contain casein must be avoided. Prepared foods that contain milk range from breads and doughnuts to sausage and soup, according to the ADA.
- **Eggs.** Although a person may be allergic to either the egg white or yolk, the entire egg must be avoided because there is a risk of cross-contamination. Eggs are an ingredient in mayonnaise. Moreover, products such as baked goods, breads, pasta, yogurt, and batter on fried foods may contain eggs. In addition, some egg-substitute products contain egg whites.

- Peanuts grow in the ground and are legumes like lentils and chickpeas. A person with a peanut allergy may not be allergic to other legumes or tree nuts. Products to be avoided include peanuts, peanut butter, peanut oil, and some desserts and candy. In addition, some Asian dishes are prepared with a peanut sauce. Tree nuts include almonds, cashews, pecans, walnuts, Brazil nuts, chestnuts, hazelnuts, macadamia nuts, pine nuts, pistachios, and hickory nuts. Products containing tree nuts include nut oil, nut oil, desserts, candy, crackers, and barbecue sauce. A person may be allergic to one type of nut but able to eat other nuts. That should be determined after consulting with a doctor.
- Fish allergy is generally diagnosed as an allergy to all fish species because the allergen is similar among the different species.
- Shellfish species include lobster, crab, shrimp, clams, oysters, scallops, mollusks, and crawfish. An allergy to one type of shellfish may indicate an allergy to others.
- Wheat is a grain found in numerous foods including breads, cereals, pastas, lunch meats, desserts, and bulgar. It is also found in products such as enriched flour and farina.
- Soy. The soybean is a legume, and people who have this allergy are rarely allergic to peanuts or other legumes. Soy is an ingredient in many processed foods including crackers and baked goods, sauces, and soups. There is also soy in canned tuna, according to the ADA.

The chemical reaction

During the initial exposure, many IgE antibodies are created. These attach to mast cells. These cells are located in tissue throughout the body, especially in areas such as the nose, throat, lungs, skin, and gastrointestinal tract. These are also the areas where allergic reactions occur.

The antibodies are in place, and a reaction is triggered the next time the person eats the food regarded as harmful. As the allergen reacts with the IgE, the body releases histamine and other chemicals. Histamine is a chemical located in the body's cells. When released during an allergic reaction, histamine and other chemicals cause symptoms like inflammation.

The type of allergic reaction depends on where the antibodies are released, according to NIAID. Chemicals released in the ears, nose, and throat could cause the mouth to itch. The person may also

have difficulty breathing or swallowing. If the allergen triggers a reaction in the gastrointestinal tract, the person could experience stomach pain or diarrhea. An allergic reaction that affects skin cells could produce hives. This condition also known as urticaria is an allergic reaction characterized by **itching**, swelling, and the presence of patchy red areas called wheals.

Severe allergic reaction

Anaphylaxis is a severe allergic reaction that is potentially life-threatening. Also known as an anaphylactic reaction, this condition requires immediate medical attention. The reaction occurs within seconds or up to several hours after the person ate the allergy-inducing food.

Symptoms can include difficulty breathing, a **tingling** feeling in the mouth, and a swelling in the tongue and throat. The person may experience hives, **vomiting**, abdominal cramps, and diarrhea. There is also a sudden drop in blood pressure. Anaphylaxis could be fatal if not treated promptly.

Each year, some 150 Americans die from food-induced anaphylaxis, according to NIAID. The casualties are generally adolescents and young adults. The risk increases for people who have allergies and asthma. Also at increased risk are people who experienced previous episodes of a naphylaxis.

The peanut is one of the primary foods that trigger an anaphylactic reaction. Tree nuts also cause the reaction. The nuts generally linked to anaphylaxis are almonds, Brazil nuts, cashews, chestnuts, hazelnuts, macadamia nuts, pecans, pine nuts, pistachios and walnuts. Fish, shellfish, and eggs can also set off the reaction, according to the ADA.

Cross-reactivity

Cross-reactivity is the tendency of a person with one allergy to reaction to another allergen. A person allergic to crab might also be allergic to shrimp. In addition, someone with ragweed sensitivity could experience sensations when trying to eat melons during ragweed pollinating season, according to NIAID. The person's mouth would start itching, and the person wouldn't be able to eat the melon. The cross-reaction happens frequently with cantaloupes. The condition is known as oral allergy syndrome.

Diagnosis

Food allergies are diagnosed by first determining whether a person has an allergy or if symptoms are related to a condition like food intolerance.

The medical professional may be a board-certified allergist, a doctor with education and experience in treating allergies. However, some health plans may require that the patient first see a family practice doctor.

If food allergies are suspected, the doctor will take a detailed case history. The doctor asks the patient if there is a family history of allergies. Other questions are related to the patient's adverse reactions.

The doctor's questions include how the food was prepared, the amount eaten and what time the reaction happened. The patient describes the symptoms and actions taken to relieve them. The doctor also asks if the patient had other similar experiences when eating that food.

The patient receives a physical exam. In addition, the doctor may ask the patient to keep a food diary, a log of what the person eats for one to two weeks. The medical history and the food diary are used in conjunction with testing to diagnose the patient.

Allergy tests

Doctors generally start the testing process with a skin test or a blood test. The prick skin test, which is also known as the scratch test, examines the patient's reaction to a solution containing a protein that triggers allergies.

The doctor places a drop of the substance on the patient's arm or back. The doctor then uses a needle to prick or scratch the skin. This allows the potential allergen to enter the patient's skin. If more than one food allergy is suspected, the test is repeated with other proteins applied to the skin. After about 15 minutes, the doctor can read the reactions on the patient's skin.

If there is no reaction, the patient is probably not allergic to that food. The possibility of an allergy is indicated by the presence of a wheal, a bump that resembles a mosquito bite. The wheal signifies a positive reaction to the test. However, the test may show a false positive, which is a reaction to a food that does not cause allergies.

The skin test is not appropriate for people who are severely allergic or have skin conditions like eczema. Those people are given the RAST (radioallergosorbent test). This test measures the presence of food-specific IgE in the blood. After a sample of the patient's blood is taken, it is sent to a laboratory. The sample is tested with different foods. Levels of antibodies are measured, and the reactions to different proteins are ranked. While measurement systems may vary, a high ranking indicates a high number of

antibodies. Lab results are generally completed within a week.

Results to this test may not be conclusive. A negative test may not have identified antibodies in the patient's blood. Positive results make it probable but not definite that the patient has allergies.

Costs for blood and skin tests will vary, with fees ranging from \$10 to more than \$300. Insurance may cover some of the cost. While both tests are reliable, they aren't 100% accurate. If questions remain, the diagnosis takes into account the patient's medical history and the food diary. If necessary, the patient is put on a special diet.

Elimination diet

If the skin or blood test shows strong positive results, the doctor may put the patient on an elimination diet. This is done when needed to narrow the list of suspected allergens. The person stops eating the foods suspected of causing the allergic reaction. That food is eliminated from the diet for from two to four weeks. If allergy symptoms improve, the food is probably an allergen.

If more confirmation is needed, the doctor may ask the patient to start eating the food again. The elimination diet procedure is generally not utilized if the patient initially had a severe reaction.

Food challenges

Other tests called food challenges may be performed. The challenges are done in a medical setting, with a doctor present. The patient is given capsules that each contain a different food. Some capsules contain allergy-producing foods. Other capsules may be placebos that won't produce a reaction.

The patient swallows the capsule, and the doctor watches for an allergic reaction. In an open food challenge, doctor and patient are aware of the capsule contents. In a single-blind food challenge, only the doctor knows. In a double-blind challenge, neither doctor nor patient knows the contents.

Challenges are rarely authorized by health care providers. Testing is time-consuming and many allergens are difficult to evaluate with the challenges, according to NIAID.

Treatment

The treatment for food allergies is to avoid eating the food that causes the allergy. This preventive treatment includes reading food labels. Manufacturers are

required by the U.S. Food and Drug Administration to list a product's ingredients on the label. However, if there is a question about an ingredient, the person should contact the manufacturer before eating the food. When dining out, people should ask if food contains the allergen or ingredients contain the allergy-inducing foods.

When reading food labels, people with food allergies should know that:

- Words indicating the presence of milk include lactose, ghee, and whey.
- Words signifying eggs in a product include albumin, globulin, and ovomucin.
- While it is apparent that peanuts are an ingredient in a product like peanut butter, there could be peanuts in hydrolyzed plant protein and hydrolyzed vegetable protein.
- People with tree nut allergies should carefully read the labels of products such as cereals and barbecue sauce.
- The American Dietetic Association cautions that surimi, an ingredient in imitation seafood, is made from fish muscle. Furthermore, fish in the form of anchovies is sometimes an ingredient in Worcestershire sauce.
- Words on labels that signal the presence of wheat include gluten, sietan, and vital gluten.

Allergies and children

Parents of children with food allergies need to monitor their children's food choices. They also must know how to care for the child if there is an allergic reaction. Parents need to notify the child's school about the condition. Caregivers should be informed, too. Both the school and caregivers should know how to handle an allergic reaction. Care must be taken because a highly allergic person could react to a piece of food as small as 1/44,000 of a peanut kernel, according to NIAID.

Living with severe allergies

Despite precautions, people may accidentally eat something that causes an allergic reaction. People with severe allergies must be prepared to treat the condition and prevent an anaphylactic reaction. A medical alert bracelet should be worn. This informs people that the person has a food allergy and could have severe reactions.

To reduce the risks from an anaphylactic reaction, the person carries a syringe filled with epinephrine,

which is adrenaline. This is a prescription medication sold commercially as the EpiPen auto injector. While prices vary, one syringe costs about \$50.

The person with allergies must know how to inject the epinephrine. It is helpful for other family members to know how to do this, and parents of an allergic child must be trained in the procedure.

The person is injected at the first sign of a severe reaction. Medical attention is required, and the person should be taken to an emergency room. The person will be treated and monitored because there could be a second severe reaction about four hours after the initial one.

Allergy treatment research

There was no cure for food allergies as of the spring of 2005. That could change, with some relief available for people diagnosed with peanut allergies. According to a study reported on in 2003 in the *New England Journal of Medicine*, 84 people who took the drug TNX-901 had a decrease in their IgE antibody levels.

Organizations including the Food Allergy & Anaphylaxis Network (FAAN) lauded the results of the study that was conducted from July of 1999 through March of 2002. Work on that study was stopped in 2004 when biotechnology companies Genentech, Novartis, and Tanox concentrated efforts instead on use of an asthma medication for treating peanut allergies. Research started in June of 2004 on omalizumab, a medication sold commercially as Xolair. The study of Xolair's effectiveness was expected to take from two to three years.

Alternative treatment

The only treatment for food allergies is for a person to stop eating the food that causes the allergies. Some alternative treatments may be helpful in easing the symptoms caused by allergies. However, people should check with their health care providers before embarking on an alternative treatment.

Prognosis

Food allergies cannot be cured, but they can be managed. The allergen-inducing foods should be avoided. These foods should be replaced with others that provide the **vitamins** and nutrients needed for a healthy diet. Organizations including the American Dietetic Association recommend the following dietetic changes:

- Milk is a source of calcium and vitamins A and D. For people with milk allergies, alternate choices of

calcium include calcium-fortified orange juice and cereal.

- Since eggs are an ingredient in products like bread, egg-free sources of grains are an alternate source of vitamin B.
- Peanuts are a source of vitamin E, niacin, and magnesium. Other sources of these nutrients include other legumes, meat, and grains.
- Fish is a source of protein and nutrients like B vitamins and niacin. Alternate sources of these nutrients should be sought.
- Wheat is a source of many nutrients including niacin and riboflavin. The person allergic to wheat should substitute products made from grains such as oat, corn, rice and barley.
- Although soybeans are rich in nutrients, very little soy is used in commercial products. As a result, a person with this food allergy would not need to find a safe substitute in order to get needed nutrients.

Prevention

People prevent the return of food allergies by following treatment guidelines. These include avoiding the foods that cause allergic reactions, reading food labels, and taking measures to prevent an anaphylactic reaction.

Anaphylaxis is a major concern after a diagnosis of severe food allergies. To reduce the risks associated this reaction, people with food allergies should wear medical alert bracelets and never go anywhere without epinephrine. If possible, family members or friends of adults with allergies should learn how to administer this medication.

The American Dietetic Association advises people to develop an emergency plan. ADA recommendations include preparing a list of the foods the person is allergic to, three emergency contacts, the doctor's name, and a description of how to treat the reaction. This list is kept with the epinephrine syringe.

Resources

BOOKS

Freund, Lee and Rejaunier, Jeanne. *The Complete Idiot's Guide to Food Allergies*. Penguin Group, USA, 2003.
The American Dietetic Association. *Food Allergies: How to Eat Safely and Enjoyably*. John Wiley & Sons., 1998.

PERIODICALS

ORGANIZATIONS

American Dietetic Association. 120 South Riverside Plaza, Suite 2000. Chicago, IL 60606-6995. 800-877-1600.
< <http://www.eatright.org> > .

American Academy of Allergy, Asthma & Immunology. 555 East Wells Street Suite 1100, Milwaukee, WI 53202-3823. 414-272-6071. < <http://www.aaaai.org> > .

The Food Allergy & Anaphylaxis Network. 11781 Lee Jackson Highway, Suite 160, Fairfax, VA 22033. 800-929-4040. < <http://www.foodallergy.org> > .

National Institute of Allergy and Infectious Diseases. 6610 Rockledge Drive, MSC 6612, Bethesda, MD 20892-6612. 301-496-5717. < <http://www.niaid.nih.gov> > .

OTHER

Food Allergy An Overview. National Institute of Allergy and Infectious Diseases. July 2004. [cited March 30, 2005]. < <http://www.niaid.nih.gov/publications/pdf/foodallergy.pdf> > .

Peanut Anti-IgE Study Update. The Food Allergy & Anaphylaxis Network. September 2, 2004 [cited April 5]. < <http://www.foodallergy.org/Research/antiigetherapy.html> > .

Liz Swain

Food poisoning

Definition

Food **poisoning** is a general term for health problems arising from eating contaminated food. Food may be contaminated by bacteria, viruses, environmental toxins, or toxins present within the food itself, such as the poisons in some mushrooms or certain seafood. Symptoms of food poisoning usually involve **nausea**, vomiting and/or **diarrhea**. Some food-borne toxins can affect the nervous system.

Description

Every year millions of people suffer from bouts of **vomiting** and diarrhea each year that they blame on "something I ate." These people are generally correct. Each year in the United States, one to two bouts of diarrheal illness occur in every adult. The Centers for Disease Control and Prevention (CDC) estimates that there are from six to 33 million cases of food poisoning in the United States annually. Many cases are mild and pass so rapidly that they are never diagnosed. Occasionally a severe outbreak creates a newsworthy public health hazard.

Classical food poisoning, sometimes incorrectly called ptomaine poisoning, is caused by a variety of different bacteria. The most common are *Salmonella*, *Staphylococcus aureus*, *Escherichia coli* O157:H7 or

Alice Catherine Evans (1881–1975)



(Corbis. Reproduced by permission.)

Alice Catherine Evans was born on January 29, 1881, in Neath, Pennsylvania. Evans was the second of two children born to Anne Evans and William Howell. Evans taught grade

school for four years because she could not afford to pay college tuition. Following her time as a teacher, Evans enrolled at the Cornell University College of Agriculture, earning her B.S. degree. Evans' professor recommended her for a scholarship, which she received, and she began her master's degree program at the University of Wisconsin where she earned her degree in 1910.

In 1911, Evans took a position with the University of Wisconsin's Dairy Division as a researcher studying cheese-making instead of continuing her education. In 1913, she moved to Washington, D.C., with the division and worked with a team on identifying the cause of contamination in raw cow's milk. By 1917, Evans' research had shown that the bacteria responsible for undulant (Malta) fever was very similar to one found when a cow experienced a spontaneous abortion. When administered to guinea pigs, the two bacteria produced similar results. Her findings were met with much skepticism but, as time went on, Evans' research began to gain support. She continued to document cases of the disease and to argue for the pasteurization process. Finally, after 1930, officials responsible for public health and safety realized the need for this process, which ultimately became a standard procedure. Evans retired from her position with the National Institute of Health in 1945 and died on September 5, 1975.

other *E. coli* strains, *Shigella*, and *Clostridium botulinum*. Each has a slightly different incubation period and duration, but all except *C. botulinum* cause inflammation of the intestines and diarrhea. Sometimes food poisoning is called bacterial **gastroenteritis** or infectious diarrhea. Food and water can also be contaminated by viruses (such as the Norwalk agent that causes diarrhea and the viruses of **hepatitis A** and **E**), environmental toxins (heavy metals), and poisons produced within the food itself (mushroom poisoning or **fish and shellfish poisoning**).

Careless food handling during the trip from farm to table creates conditions for the growth of bacteria that make people sick. Vegetables that are eaten raw, such as lettuce, may be contaminated by bacteria in soil, water, and dust during washing and packing. Home canned and commercially canned food may be improperly processed at too low a temperature or for too short a time to kill the bacteria.

Raw meats carry many food-borne bacterial diseases. The United States Food and Drug Administration (FDA) estimates that 60% or more

of raw poultry sold at retail carry some disease-causing bacteria. Other raw meat products and eggs are contaminated to a lesser degree. Thorough cooking kills the bacteria and makes the food harmless. However, properly cooked food can become recontaminated if it comes in contact with plates, cutting boards, countertops, or utensils that were used with raw meat and not cleaned and sanitized.

Cooked foods can also be contaminated after cooking by bacteria carried by food handlers or from bacteria in the environment. It is estimated that 50% of healthy people have the bacteria *Staphylococcus aureus* in their nasal passages and throat, and on their skin and hair. Rubbing a runny nose, then touching food can introduce the bacteria into cooked food. Bacteria flourish at room temperature, and will rapidly grow into quantities capable of making people sick. To prevent this growth, food must be kept hot or cold, but never just warm.

Although the food supply in the United States is probably the safest in the world, anyone can get food poisoning. Serious outbreaks are rare. When they

Common Pathogens Causing Food Poisoning

Pathogen	Common Host(s)
Campylobacter	Poultry
E.coli O157:H7	Undercooked, contaminated ground beef
Listeria	Found in a variety of raw foods, such as uncooked meats and vegetables, and in processed foods that become contaminated after processing
Salmonella	Poultry, eggs, meat, and milk
Shigella	This bacteria is transmitted through direct contact with an infected person or from food or water that become contaminated by an infected person
Vibrio	Contaminated seafood

occur, the very young, the very old, and those with immune system weaknesses have the most severe and life-threatening cases. For example, this group is 20 times more likely to become infected with the *Salmonella* bacteria than the general population.

Travel outside the United States to countries where less attention is paid to sanitation, water purification, and good food handling practices increases the chances that a person will get food poisoning. People living in institutions such as nursing homes are also more likely to get food poisoning.

Causes and symptoms

The symptoms of food poisoning occur because food-borne bacteria release toxins or poisons as a byproduct of their growth in the body. These toxins (except those from *C. botulinum*) cause inflammation and swelling of the stomach, small intestine and/or large intestine. The result is abdominal muscle cramping, vomiting, diarrhea, **fever**, and the chance of **dehydration**. The severity of symptoms depends on the type of bacteria, the amount consumed, and the individual's general health and sensitivity to the bacterial toxin.

Salmonella

According to a 2001 report from the CDC, *Salmonella* caused almost 50,000 culture-confirmed cases of food poisoning in the United States annually. However, between two and four million probably occur each year. *Salmonella* is found in egg yolks from infected chickens, in raw and undercooked poultry and in other meats, dairy products, fish, shrimp, and many more foods. The CDC estimates that one out of every 50 consumers is exposed to a contaminated egg yolk each year. However, thorough cooking

kills the bacteria and makes the food harmless. *Salmonella* is also found in the feces of pet reptiles such as turtles, lizards, and snakes.

About one out of every 1,000 people get food poisoning from *Salmonella*. Of these, two-thirds are under age 20, with the majority under age nine. Most cases occur in the warm months between July and October.

Symptoms of food poisoning begin eight to 72 hours after eating food contaminated with *Salmonella*. These include traditional food poisoning symptoms of abdominal **pain**, diarrhea, vomiting, and fever. The symptoms generally last one to five days. Dehydration can be a complication in severe cases. People generally recover without antibiotic treatment, although they may feel tired for a week after the active symptoms subside.

Staphylococcus aureus

Staphylococcus aureus is found on humans and in the environment in dust, air, and sewage. The bacteria is spread primarily by food handlers using poor sanitary practices. Almost any food can be contaminated, but salad dressings, milk products, cream pastries, and any food kept at room temperature, rather than hot or cold are likely candidates.

It is difficult to estimate the number of cases of food poisoning from *Staphylococcus aureus* that occur each year, because its symptoms are so similar to those caused by other foodborne bacteria. Many cases are mild and the victim never sees a doctor.

Symptoms appear rapidly, usually one to six hours after the contaminated food is eaten. The acute symptoms of vomiting and severe abdominal cramps without fever usually last only three to six hours and rarely more than 24 hours. Most people recover without medical assistance. Deaths are rare.

Escherichia coli (*E. coli*)

There are many strains of *E. coli*, and not all of them are harmful. The strain that causes most severe food poisoning is *E. coli* O157:H7. Food poisoning by *E. coli* occurs in three out of every 10,000 people. Foodborne *E. coli* is found and transmitted mainly in food derived from cows such as raw milk, raw or rare ground beef and fruit or vegetables that are contaminated.

Symptoms of food poisoning from *E. coli* are slower to appear than those caused by some of the other foodborne bacteria. *E. coli* produces toxins in the large intestine rather than higher up in the

digestive system. This accounts for the delay in symptoms and the fact that vomiting rarely occurs in *E. coli* food poisoning.

One to three days after eating contaminated food, the victim with *E. coli* O157:H7 begins to have severe abdominal cramps and watery diarrhea that usually becomes bloody within 24 hours. There is little or no fever, and rarely does the victim vomit. The bloody, watery diarrhea lasts from one to eight days in uncomplicated cases.

Campylobacter jejuni (*C. jejuni*)

According to the FDA, *C. jejuni* is the leading cause of bacterial diarrhea in the United States. It is responsible for more cases of bacterial diarrhea than *Shigella* and *Salmonella* combined. Anyone can get food poisoning from *C. jejuni*, but children under five and young adults between the ages of 15 and 29 are more frequently infected.

C. jejuni is carried by healthy cattle, chickens, birds, and flies. It is not carried by healthy people in the United States or Europe. The bacteria is also found ponds and stream water. The ingestion of only a few hundred *C. jejuni* bacteria can make a person sick.

Symptoms of food poisoning begin two to five days after eating food contaminated with *C. jejuni*. These symptoms include fever, abdominal pain, nausea, headache, muscle pain, and diarrhea. The diarrhea can be watery or sticky and may contain blood. Symptoms last from seven to 10 days, and relapses occur in about one quarter of people who are infected. Dehydration is a common complication. Other complications such as arthritis-like joint pain and hemolytic-uremic syndrome (HUS) are rare.

Shigella

Shigella is a common cause of diarrhea in travelers to developing countries. It is associated with contaminated food and water, crowded living conditions, and poor sanitation. The bacterial toxins affect the small intestine.

Symptoms of food poisoning by *Shigella* appear 36–72 hours after eating contaminated food. These symptoms are slightly different from those associated with most foodborne bacteria. In addition to the familiar watery diarrhea, nausea, vomiting, abdominal cramps, chills and fever occur. The diarrhea may be quite severe with cramps progressing to classical **dysentery**. Up to 40% of children with severe infections show neurological symptoms. These include

seizures caused by fever, confusion, **headache**, lethargy, and a stiff neck that resembles **meningitis**.

The disease runs its course usually in two to three days but may last longer. Dehydration is a common complication. Most people recover on their own, although they may feel exhausted, but children who are malnourished or have weakened immune systems may die.

Clostridium botulinum (*C. botulinum*)

C. botulinum, which causes both adult **botulism** and infant botulism, is unlike any of the other foodborne bacteria. First, *C. botulinum* is an anaerobic bacterium in that it can only live in the absence of oxygen. Second, the toxins from *C. botulinum* are neurotoxins. They poison the nervous system, causing paralysis without the vomiting and diarrhea associated with other foodborne illnesses. Third, toxins that cause adult botulism are released when the bacteria grows in an airless environment outside the body. They can be broken down and made harmless by heat. Finally, botulism is much more likely to be fatal even in tiny quantities.

Adult botulism outbreaks are usually associated with home canned food, although occasionally commercially canned or vacuum packed foods are responsible for the disease. *C. botulinum* grows well in non-acidic, oxygen-free environments. If food is canned at too low heat or for too brief a time, the bacteria is not killed. It reproduces inside the can or jar, releasing its deadly neurotoxin. The toxin can be made harmless by heating the contaminated food to boiling for ten minutes. However, even a very small amount of the *C. botulinum* toxin can cause serious illness or **death**.

Symptoms of adult botulism appear about 18–36 hours after the contaminated food is eaten, although there are documented times of onset ranging from four hours to eight days. Initially a person suffering from botulism feels weakness and dizziness followed by double vision. Symptoms progress to difficulty speaking and swallowing. Paralysis moves down the body, and when the respiratory muscles are paralyzed, death results from asphyxiation. People who show any signs of botulism poisoning must receive immediate emergency medical care to increase their chance of survival.

Infant botulism is a form of botulism first recognized in 1976. It differs from food-borne botulism in its causes and symptoms. Infant botulism occurs when a child under the age of one year ingests the spores of *C. botulinum*. These spores are found in soil, but a more common source of spores is honey.

The *C. botulinum* spores lodge in the baby's intestinal tract and begin to grow, producing their neurotoxin. Onset of symptoms is gradual. Initially the baby is constipated. This is followed by poor feeding, lethargy, weakness, drooling, and a distinctive wailing cry. Eventually, the baby loses the ability to control its head muscles. From there the paralysis progresses to the rest of the body.

Diagnosis

One important aspect of diagnosing food poisoning is for doctors to determine if a number of people have eaten the same food and show the same symptoms of illness. When this happens, food poisoning is strongly suspected. The diagnosis is confirmed when the suspected bacteria is found in a **stool culture** or a fecal smear from the person. Other laboratory tests are used to isolate bacteria from a sample of the contaminated food. Botulism is usually diagnosed from its distinctive neurological symptoms, since rapid treatment is essential. Many cases of food poisoning go undiagnosed, since a definite diagnosis is not necessary to effectively treat the symptoms. Because it takes time for symptoms to develop, it is not necessarily the most recent food one has eaten that is the cause of the symptoms.

Treatment

Treatment of food poisoning, except that caused by *C. botulinum*, focuses on preventing dehydration by replacing fluids and electrolytes lost through vomiting and diarrhea. Electrolytes are salts and minerals that form electrically charged particles (ions) in body fluids. Electrolytes are important because they control body fluid balance and are important for all major body reactions. Pharmacists can recommend effective, pleasant-tasting, electrolytically balanced replacement fluids that are available without a prescription. When more fluids are being lost than can be consumed, dehydration may occur. Dehydration more likely to happen in the very young, the elderly, and people who are taking **diuretics**. To prevent dehydration, a doctor may give fluids intravenously.

In very serious cases of food poisoning, medications may be given to stop abdominal cramping and vomiting. Anti-diarrheal medications are not usually given. Stopping the diarrhea keeps the toxins in the body longer and may prolong the infection.

People with food poisoning should modify their diet. During period of active vomiting and diarrhea they should not try to eat and should drink only clear

liquids frequently but in small quantities. Once active symptoms stop, they should eat bland, soft, easy to digest foods for two to three days. One example is the BRAT diet of bananas, rice, applesauce, and toast, all of which are easy to digest. Milk products, spicy food, alcohol and fresh fruit should be avoided for a few days, although babies should continue to breastfeed. These modifications are often all the treatment that is necessary.

Severe bacterial food poisonings are sometimes treated with **antibiotics**. Trimethoprim and sulfamethoxazole (Septra, Bactrim), ampicillin (Amcill, Polycill) or ciprofloxacin (Ciloxan, Cipro) are most frequently used.

Botulism is treated in a different way from other bacterial food poisonings. Botulism antitoxin is given to adults, but not infants, if it can be administered within 72 hours after symptoms are first observed. If given later, it provides no benefit.

Both infants and adults require hospitalization, often in the intensive care unit. If the ability to breathe is impaired, patients are put on a mechanical ventilator to assist their breathing and are fed intravenously until the **paralysis** passes.

Alternative treatment

Alternative practitioners offer the same advice as traditional practitioners concerning diet modification. In addition they recommend taking charcoal tablets, *Lactobacillus acidophilus*, *Lactobacillus bulgaricus*, and citrus seed extract. An electrolyte replacement fluid can be made at home by adding one teaspoon of salt and four teaspoons of sugar to one quart of water. For food poisoning other than botulism, two homeopathic remedies, either *Arsenicum album* or *Nux vomica*, are strongly recommended.

Prognosis

Most cases of food poisoning (except botulism) clear up on their own within one week without medical assistance. The ill person may continue feel tired for a few days after active symptoms stop. So long as the ill person does not become dehydrated, there are few complications. Deaths are rare and usually occur in the very young, the very old and people whose immune systems are already weakened.

Complications of *Salmonella* food poisoning include arthritis-like symptoms that occur three to four weeks after infection. Although deaths from *Salmonella* are rare, they do occur. Most deaths

KEY TERMS

Diuretic—Medication that increases the urine output of the body.

Electrolytes—Salts and minerals that produce electrically charged particles (ions) in body fluids. Common human electrolytes are sodium chloride, potassium, calcium, and sodium bicarbonate. Electrolytes control the fluid balance of the body and are important in muscle contraction, energy generation, and almost all major biochemical reactions in the body.

Lactobacillus acidophilus—This bacteria is found in yogurt and changes the balance of the bacteria in the intestine in a beneficial way.

Platelets—Blood cells that help the blood to clot.

caused by *Salmonella* food poisoning have occurred in elderly people in nursing homes.

Adults usually recover without medical intervention, but many children need to be hospitalized as the result of *E. coli* food poisoning. *E. coli* toxins may be absorbed into the blood stream where they destroy red blood cells and platelets. Platelets are important in blood clotting. About 5% of victims develop hemolytic-uremic syndrome which results in sudden kidney failure and makes dialysis necessary. (Dialysis is a medical procedure used to filter the body's waste product when the kidneys have failed).

Botulism is the deadliest of the bacterial food-borne illnesses. With prompt medical care, the death rate is less than 10%.

Prevention

Food poisoning is almost entirely preventable by practicing good sanitation and good food handling techniques. These include:

- keep hot foods hot and cold foods cold
- cook meat to the recommended internal temperature, use a meat thermometer to check and cook eggs until they are no longer runny
- refrigerate leftovers promptly, do not let food stand at room temperature
- avoid contaminating surfaces and other foods with the juices of uncooked meats
- wash fruits and vegetables before using
- purchase pasteurized dairy products and fruit juices

- throw away bulging or leaking cans or any food that smells spoiled
- wash hands well before and during food preparation and after using the bathroom
- sanitize food preparation surfaces regularly

Resources

OTHER

U. S. Food and Drug Administration. Center for Food Safety and Applied Nutrition. *Bad Bug Book*. < <http://vm.cfsan.fda.gov> > .

Suzanne M. Lutwick, MPH

Foot acupressure see **Reflexology**

Foot care

Definition

Foot care involves all aspects of preventative and corrective care of the foot and ankle. Doctors specializing in foot care are called podiatrists.

Purpose

During an average lifetime, each person walks about 115,000 miles and three-quarters of people have foot problems at some point in their lives.

Foot problems can arise from wearing ill-fitting shoes, from general wear and tear, as a result of injury, or as a complication of disease. People with **diabetes mellitus** or circulatory diseases are 20 times more likely to have foot problems than the general public.

Podiatrists are doctors who specialize in treating the foot and ankle. Other doctors who have experience with foot problems are family physicians, orthopedists, sports medicine specialists, and those who care for diabetics. Problems with the feet include foot **pain**, joint inflammation, plantar **warts**, fungal infections (like **athlete's foot**), nerve disorders, torn ligaments, broken bones, bacterial infections, and tissue injuries (like **frostbite**).

Precautions

People with diabetes or circulatory disorders should be alert to even small foot problems. In these people, a break in the skin can lead to infection, **gangrene**, and **amputation**.

Description

Daily foot care for people likely to develop foot problems includes washing the feet in tepid water with mild soap and oiling the feet with vegetable oil or a lanolin-based lotion. Toenails should be cut straight across above the level of the skin after soaking the feet in tepid water. **Corns and calluses** should not be cut. If they need removal, it should be done under the care of a doctor. Athletes foot and plantar warts should also be treated by a doctor if they develop in high risk patients.

Many people with diabetes or circulatory disorders have problems with cold feet. These problems can be reduced by avoiding **smoking** tobacco (smoking constricts the blood vessels), wearing warm socks, not crossing the legs while sitting or not sitting in one position too long, or avoiding constricting stockings.

People with circulatory problems should not use heating pads or hot water bottles on their feet, as even moderate heat can damage the skin if circulation is impaired.

Preparation

No special preparation other than an understanding of the nature of foot problems is necessary to begin routine foot care.

Aftercare

Foot care is preventative and should be ongoing throughout a person's life.

Risks

There are no risks associated with foot care. The risks are in ignoring the feet and allowing problems to develop.

Normal results

With regular care, foot disorders such as infections, skin ulcers, and gangrene can be prevented.

Resources

ORGANIZATIONS

American Diabetes Association. 1701 North Beauregard Street, Alexandria, VA 22311. (800) 342-2383. <<http://www.diabetes.org>> .

American Podiatry Association. 20 Chevy Chase Circle, NW, Washington, D.C. 20015.

Tish Davidson, A.M.

Foreign bodies see **Foreign objects**

Foreign objects

Definition

"Foreign" means "originating elsewhere" or simply "outside the body." Foreign bodies typically become lodged in the eyes, ears, nose, airways, and rectum of human beings.

Description

Both children and adults experience problems caused by foreign objects getting stuck in their bodies. Young children, in particular, are naturally curious and may intentionally put shiny objects, such as coins or button batteries, into their mouths. They also like to stick things in their ears and up their noses. Adults may accidentally swallow a non-food object or inhale a foreign body that gets stuck in the throat. Even if an object like a toothpick successfully passes through the esophagus and into the stomach, it can get stuck inside the rectum. Airborne particles can lodge in the eyes of people at any age.

Foreign bodies can be in hollow organs (like swallowed batteries) or in tissues (like bullets). They can be inert or irritating. If they irritate they will cause inflammation and scarring. They can bring infection with them or acquire it and protect it from the body's immune defenses. They can obstruct passageways either by their size or by the scarring they cause. Some can be toxic.

Causes and symptoms

Eyes

Dust, dirt, sand, or other airborne material can lodge in the eyes, causing minor irritation and redness. More serious damage can be caused by hard or sharp objects that penetrate the surface and become embedded in the cornea or conjunctiva (the mucous membranes around the inner surface of the eyelids). Swelling, redness, bleeding from the surface blood vessels, sensitivity to light, and sudden vision problems are all symptoms of foreign matter in the eyes.

Ears and nose

Children will sometimes put things into their noses, ears, and other openings. Beans, popcorn kernels, raisins, and beads are just a few of the many items that have been found in these bodily cavities. On occasion, insects may also fly into the ears and nose. **Pain, hearing loss,** and a sense of something stuck in



X ray of swallowed spoon and blade in the intestine. (Photo Researchers. Reproduced by permission.)

the ear are symptoms of foreign bodies in the ears. A smelly, bloody discharge from one nostril is a symptom of foreign bodies in the nose.

Airways and stomach

At a certain age children will eat anything. A very partial list of items recovered from young stomachs includes the following: Coins, chicken bones, fish bones, beads, rocks, plastic toys, pins, keys, round stones, marbles, nails, rings, batteries, ball bearings, screws, staples, washers, a heart pendant, a clothespin spring, and a toy soldier. Some of these items will pass right on through and come out the other end. The progress of metal objects has been successfully followed with a metal detector. Others, like sharp bones, can get stuck and cause trouble. Batteries are corrosive and must be removed immediately.

Children eat things and stick things into their bodily openings of their own volition. But they inhale them unwittingly. The most commonly inhaled item is probably a peanut. A crayon and a cockroach have been found in a child's windpipes. These items always cause symptoms (difficulty swallowing and spitting up saliva, for instance) and may elude detection for some time while the child is being treated for **asthma** or recurring **pneumonia**.

Adults are not exempt from unorthodox inedibles. Dental devices are commonly swallowed. Adults with mental illness or subversive motives may swallow inappropriate objects, such as toothbrushes.

Rectum

Sometimes a foreign object will successfully pass through the throat and stomach only to get stuck at the juncture between the rectum and the anal canal.

Items may also be self-introduced to enhance sexual stimulation and then get stuck. Sudden sharp pain during elimination may signify that an object is lodged in the rectum. Other symptoms vary depending upon the size of the object, its location, how long it has been in place, and whether or not infection has set in.

Diagnosis

The symptoms are as diverse as the objects and their locations. The most common manifestation of a foreign object anywhere in the body is infection. Even if the object started out sterile, germs still seem to find it and are able to hide from the body's defenses there. Blockage of passageways—breathing, digestive or excretory—is another result. Pain is common.

Treatment

Eyes

Small particles like sand may be removable without medical help, but if the object is not visible or cannot be retrieved, prompt emergency treatment is necessary. Trauma to the eyes can lead to loss of vision and should never be ignored. Before attempting any treatment, the person should move to a well-lighted area where the object can be more easily spotted. Hands should be washed and only clean, preferably sterile, materials should make contact with the eyes. If the particle is small, it can be dislodged by blinking or pulling the upper lid over the lower lid and flushing out the speck. A clean cloth can also be used to pick out the offending particle. Afterwards, the eye should be rinsed with clean, lukewarm water or an ophthalmic wash.

If the foreign object cannot be removed at home, the eye should be lightly covered with sterile gauze to discourage rubbing. A physician will use a strong light and possibly special eyedrops to locate the object. Surgical tweezers can effectively remove many objects. An antibiotic sterile ointment and a patch may be prescribed. If the foreign body has penetrated the deeper layers of the eye, an ophthalmic surgeon will be consulted for emergency treatment.

Ears and nose

A number of ingenious extraction methods have been devised for removing foreign objects from the nose and ears. A bead in a nostril, for example, can be popped out by blowing into the mouth while holding the other nostril closed. Skilled practitioners have removed peas from the ears by tiny improvised corkscrews; marbles by q-tips with super glue. Tweezers

KEY TERMS

Bronchoscope—An illuminated instrument that is inserted into the airway to inspect and retrieve objects from the bronchial tubes.

Conjunctiva—Mucous membranes around the inner surface of the eyelid.

Cornea—The rounded, transparent portion of the eye that covers the pupil and iris and lets light into the interior

Endoscopy—The surgical use of long, thin instruments that have both viewing and operating capabilities.

Heimlich maneuver—An emergency procedure for removing a foreign object lodged in the airway that is preventing the person from breathing. To perform the Heimlich maneuver on a conscious adult, the rescuer stands behind the victim and encircles his waist. The rescuer makes a fist with one hand and places the other hand on top, positioned below the rib cage and above the waist. The rescuer then applies pressure by a series of upward and inward thrusts to force the foreign object back up the victim's trachea.

often work well, too. Insects can be floated out of the ear by pouring warm (not hot) mineral oil, olive oil, or baby oil into the ear canal. Items that are lodged deep in the ear canal are more difficult to remove because of the possibility of damaging the ear drum. These require emergency treatment from a qualified physician.

Airways and stomach

Mechanical obstruction of the airways, which commonly occurs when food gets lodged in the throat, can be treated by applying the **Heimlich maneuver**. If the object is lodged lower in the airway, a bronchoscope (a special instrument to view the airway and remove obstructions) can be inserted. On other occasions, as when the object is blocking the entrance to the stomach, a fiberoptic endoscope (an illuminated instrument that views the interior of a body cavity) may be used. The physician typically administers a sedative and anesthetizes the throat. The foreign object will then either be pulled out or pushed into the stomach, depending on whether or not the physician thinks it will pass through the digestive tract on its own. Objects in the digestive tract that are neither

irritating, sharp nor large may be followed as they continue on through. Sterile objects that are causing no symptoms may be left in place. Surgical removal of the offending object is necessary if it is causing symptoms.

Rectum

A rectal retractor can remove objects that a physician can feel during **physical examination**. Surgery may be required for objects deeply lodged within the rectum.

Prevention

Using common sense and following safety precautions are the best ways to prevent foreign objects from entering the body. For instance, parents and grandparents should toddler-proof their homes, storing batteries in a locked cabinet and properly disposing of used batteries, so they are not in a location where curious preschoolers can fish them out of a wastebasket. To minimize the chance of youngsters inhaling food, parents should not allow children to eat while walking or playing. Adults should chew food thoroughly and not talk while chewing. Many eye injuries can be prevented by wearing safety glasses while using tools

Resources

PERIODICALS

Gilchrist, B. F., et al. "Pearls and Perils in the Management of Prolonged, Peculiar, Penetrating Esophageal Foreign Bodies in Children." *Journal of Pediatric Surgery* 32, no. 10 (October 1997): 1429-31.

J. Ricker Polsdorfer, MD

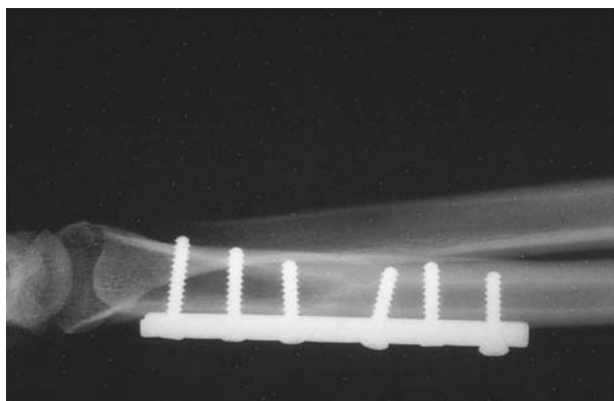
Fourn see **Flesh-eating disease**

47, XXY syndrome see **Klinefelter syndrome**

Fracture repair

Definition

Fracture repair is the process of rejoining and realigning the ends of broken bones. This procedure is usually performed by an orthopedist, general surgeon, or family doctor. In cases of an emergency, first aid measures should be evoked for temporary realignment and **immobilization** until proper medical help is available.



An x-ray image of a healing fracture. (Photograph by Bates, M.D., Custom Medical Stock Photo. Reproduced by permission.)

Purpose

Fracture repair is required when there is a need for restoration of the normal position and function of the broken bone. Throughout the stages of fracture healing, the bones must be held firmly in the correct position. In the event the fracture is not properly repaired, malalignment of the bone may occur, resulting in possible physical dysfunction of the bone or joint of that region of the body.

Precautions

Precautions for fracture repair are anything found to be significant with patients' medical diagnosis and history. This would include an individual's tolerance to anesthesia and the presence of bleeding disorders that may be present to complicate surgery.

Description

Fracture repair is applied by means of **traction**, surgery, and/or by immobilization of the bones. The bone fragments are aligned as close as possible to the normal position without injuring the skin. Metal wires or screws may be needed to align smaller bone fragments. Once the broken ends of the bone are set, the affected area is immobilized for several weeks and kept rigid with a sling, plaster cast, brace or splint. With the use of traction, muscle pull on the fracture site is overcome by weights attached to a series of ropes running over pulleys. Strategically implanted electrical stimulation devices have proven beneficial in healing a fracture site, especially when the fracture is healing poorly and repair by other means is difficult.

Preparation

Emergency splinting may be required to immobilize the body part or parts involved. When fracture repair is necessary, the procedure is often performed in a hospital but can also be successfully done in an outpatient surgical facility, doctor's office or emergency room. Before any surgery for fracture repair, blood and urine studies may be taken from the patient. X rays may follow this if not previously acquired. It has been noted however, that not all **fractures** are immediately apparent on an initial x-ray examination. In this case, where a fracture is definitely suspected the extent of the fracture can be properly diagnosed by repeating the x rays 10–14 days later. Depending upon the situation, local or general anesthesia may be used for fracture repair.

Aftercare

After surgery, x rays may be again taken through the cast or splint to evaluate if rejoined pieces remain in good position for healing. This is usually performed either before the application of the splint or at least before the patient is awakened from the **general anesthesia**. The patient needs to be cautious not to place excess pressure on any part of the cast until it is completely dry. The patient also should avoid excess pressure on the operative site until complete healing has taken place and the injury has been re-examined by the physician. If the cast becomes exposed to moisture it may soften and require repair. The patient should also be instructed to keep the injured region propped up whenever possible to reduce the possibility of swelling.

Risks

Surgical risks of fracture repair are greater in patients over 60 years of age because the bones often taking longer to heal properly. **Obesity** may place extra stress on the healing site, affecting healing and possibly risking reinjury. **Smoking** may slow the healing process after fracture repair, as well as poor **nutrition**, **alcoholism**, and chronic illness. Some medications may affect the fracture site, causing poor union. Such medications include anti-hypertensives and cortisone.

Possible complications following fracture repair include excessive bleeding, improper fit of joined bone ends, pressure on nearby nerves, delayed healing, and a permanent incomplete healing of the fracture. If there is a poor blood supply to the fractured site with one of the portions of broken bone not properly supplied by the blood, the bony portion will die and

KEY TERMS

Compound fracture—A fracture in which the broken end or ends of the bone have torn through the skin. Compound fractures are also known as open fractures

Staphylococcal infection—An infection caused by any of several pathogenic species of staphylococcus, commonly characterized by the formation of abscesses of the skin or other organs.

Streptococcal infection—An infection caused by a pathogenic bacteria of one of several species of the genus streptococcus or their toxins. Almost any organ in the body may be involved.

healing of the fracture will not take place. This is called aseptic necrosis. Poor immobilization of the fracture from improper casting which permits motion between the bone parts may prevent healing and repair of the bone with possible deformity. Infection can interfere with bone repair. This risk is greater in the case of a compound fracture (a bone fracture causing an open wound) where ideal conditions are present for severe streptococcal and staphylococcal infections. Occasionally, fractured bones in the elderly may possibly never heal properly. The risk is increased when nutrition is poor.

Normal results

Once the procedure for fracture repair is completed, the body begins to produce new tissue to bridge the broken pieces. At first, this tissue (called a callus) is soft and easily injured. Later, the body deposits bone **minerals** until the callus becomes a solid piece of bone. The fracture site is thus strengthened further with extra bone. It usually takes about six weeks for a broken bone to heal together. The exact time required for healing depends on the type of fracture and the extent of damage. Before the use of x rays, fracture repair was not always accurate, resulting in crippling deformities. With modern x-ray technology, the physician can view the extent of the fracture, check the setting following the repair, and be certain after the procedure that the bones have not moved from their intended alignment. Children's bones usually heal relatively rapidly.

Abnormal results

Abnormal results of fracture repair include damage to nearby nerves or primary blood vessels.

Improper alignment causing deformity is also an abnormal outcome, however, with today's medical technology it is relatively rare.

Resources

OTHER

Griffith, H. Winter. "Fracture Repair." *ThriveOnline*. 1998. [cited March 3, 1998]. <<http://thriveonline.oxygen.com>> .

Jeffrey P. Larson, RPT

Fractures

Definition

A fracture is a complete or incomplete break in a bone resulting from the application of excessive force.

Description

A fracture usually results from traumatic injury to bones causing the continuity of bone tissues or bony cartilage to be disrupted or broken. Fracture classifications include simple, compound, incomplete and complete. Simple fractures (more recently called "closed") are not obvious as the skin has not been ruptured and remains intact. Compound fractures (now commonly called "open") break the skin, exposing bone and causing additional soft tissue injury and possible infection. A single fracture means that one fracture only has occurred and multiple fractures refer to more than one fracture occurring in the same bone. Fractures are termed complete if the break is completely through the bone and described as incomplete or "greenstick" if the fracture occurs partly across a bone shaft. This latter type of fracture is often the result of bending or crushing forces applied to a bone.

Fractures are also named according to the specific part of the bone involved and the nature of the break. Identification of a fracture line can further classify fractures. Types include linear, oblique, transverse, longitudinal, and spiral fractures. Fractures can be further subdivided by the positions of bony fragments and are described as comminuted, non-displaced, impacted, overriding, angulated, displaced, avulsed, and segmental. Additionally, an injury may be classified as a fracture-dislocation when a fracture involves the bony structures of any joint with associated dislocation of the same joint.

Fractures line identification

Linear fractures have a break that runs parallel to the bone's main axis or in the direction of the bone's shaft. For example, a linear fracture of the arm bone could extend the entire length of the bone. Oblique and transverse fractures differ in that an oblique fracture crosses a bone at approximately a 45 ° angle to the bone's axis. In contrast, a transverse fracture crosses a bone's axis at a 90 ° angle. A longitudinal fracture is similar to a linear fracture. Its fracture line extends along the shaft but is more irregular in shape and does not run parallel to the bone's axis. Spiral fractures are described as crossing a bone at an oblique angle, creating a spiral pattern. This break usually occurs in the long bones of the body such as the upper arm bone (humerus) or the thigh bone (femur).

Bony fragment position identification

Comminuted fractures have two or more fragments broken into small pieces, in addition to the upper and lower halves of a fractured bone. Fragments of bone that maintain their normal alignment following a fracture are described as being non-displaced. An impacted fracture is characterized as a bone fragment forced into or onto another fragment resulting from a compressive force. Overriding is a term used to describe bony fragments that overlap and shorten the total length of a bone. Angulated fragments result in pieces of bone being at angles to each other. A displaced bony fragment occurs from disruption of normal bone alignment with deformity of these segments separate from one another. An avulsed fragment occurs when bone fragments are pulled from their normal position by forceful muscle contractions or resistance from ligaments. Segmental fragmented positioning occurs if fractures in two adjacent areas occur, leaving an isolated central segment. An example of segmental alignment is when the arm bone fractures in two separate places, with displacement of the middle section of bone.

Causes and symptoms

Individuals with high activity levels appear to be at greater risk for fractures. This group includes children and athletes participating in contact sports. Because of an increase in bone brittleness with **aging**, elderly persons are also included in this high-risk population. Up to the age of 50, more men suffer from fractures than women due to occupational hazards. However, after the age of 50, women are

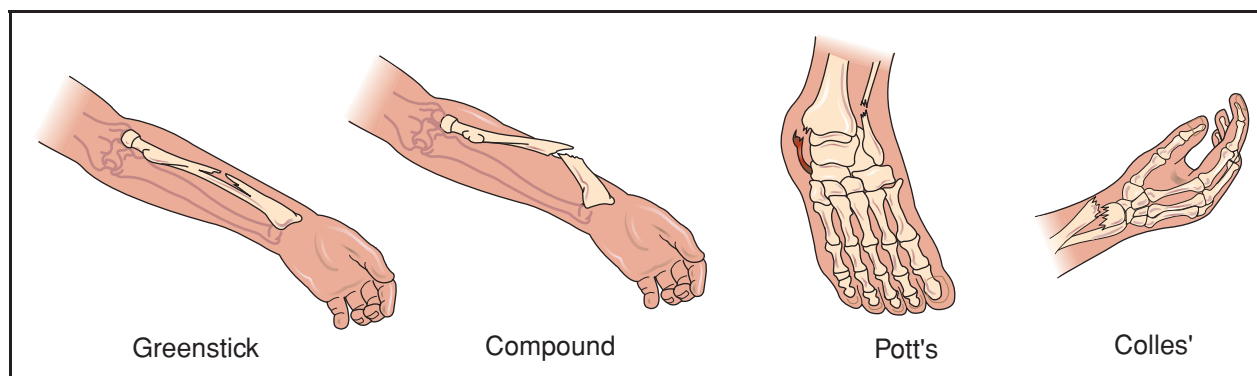
more prone to fractures than men. Specific diseases causing an increased risk for fractures include Paget's disease, **rickets**, **osteogenesis imperfecta**, **osteoporosis**, bone **cancer** and tumors, and prolonged disuse of a nonfunctional body part such as after a **stroke**.

Symptoms of fractures usually begin with **pain** that increases with attempted movement or use of the area and swelling at the involved site. The skin in the area may be pale and an obvious deformity may be present. In more severe cases, there may be a loss of pulse below the fracture site, such as in the extremities, accompanied by **numbness**, **tingling**, or **paralysis** below the fracture. An open or compound fracture is often accompanied by bleeding or bruising. If the lower limbs or pelvis are fractured, pain and resistance to movement usually accompany the injury causing difficulty with weight bearing.

Diagnosis

Diagnosis begins immediately with an individual's own observation of symptoms. A thorough medical history and physical exam by a physician often reveals the presence of a fracture. An x ray of the injured area is the most common test used to determine the presence of a bone fracture. Any x ray series performed involves at least two views of the area to confirm the presence of the fracture because not all fractures are apparent on a single x ray. Some fractures are often difficult to see and may require several views at different angles to see clear fracture lines. In some cases, CT, MRI or other imaging tests are required to demonstrate fracture. Sometimes, especially with children, the initial x ray may not show any fractures but repeat seven to 14 days later may show changes in the bone(s) of the affected area. If a fracture is open and occurs in conjunction with soft tissue injury, further laboratory studies are often conducted to determine if blood loss has occurred.

In the event of exercise-related **stress** fractures (micro-fractures due to excessive stress), a tuning fork can provide a simple, inexpensive test. The tuning fork is a metal instrument with a stem and two prongs that vibrate when struck. If an individual has increased pain when the tuning fork is placed on a bone, such as the tibia or shinbone, the likelihood of a stress fracture is high. Bone scans also are helpful in detecting stress fractures. In this diagnostic procedure, a radioactive tracer is injected into the bloodstream and images are taken of specific areas or the entire skeleton by CT or MRI.



Fractures usually result from a traumatic injury to a bone where the continuity of bone tissues or bony cartilage is disrupted or broken. The illustrations above feature common sites where fractures occur. (Illustration by Electronic Illustrators Group.)

Treatment

Treatment depends on the type of fracture, its severity, the individual's age and general health. The first priority in treating any fracture is to address the entire medical status of the patient. Medical personnel are trained not allow a painful, deformed limb to distract them from potentially life-threatening injury elsewhere or **shock**. If an open fracture is accompanied by serious soft tissue injury, it may be necessary to control bleeding and the shock that can accompany loss of blood.

First aid is the appropriate initial treatment in emergency situations. It includes proper splinting, control of blood loss, and monitoring vital signs such as breathing and circulation.

Immobilization

Immobilization of a fracture site can be done internally or externally. The primary goal of immobilization is to maintain the realignment of a bone long enough for healing to start and progress. Immobilization by external fixation uses splints, casts, or braces. This may be the primary and only procedure for fracture treatment. Splinting to immobilize a fracture can be done with or without **traction**. In emergency situations if the injured individual must be moved by someone other than a trained medical person, splinting is a useful form of fracture management. It should be done without causing additional pain and without moving the bone segments. In a clinical environment, plaster of Paris casts are used for immobilization. Braces are useful as they often allow movement above and below a fracture site. Treatments for stress fractures include rest and decreasing or stopping any activity that causes or increases pain.

Fracture reduction

Fracture reduction is the procedure by which a fractured bone is realigned in normal position. It can be either closed or open. Closed reduction refers to realigning bones without breaking the skin. It is performed with manual manipulation and/or traction and is commonly done with some kind of anesthetic. Open reduction primarily refers to surgery that is performed to realign bones or fragments. Fractures with little or no displacement may not require any form of reduction.

Traction is used to help reposition a broken bone. It works by applying pressure to restore proper alignment. The traction device immobilizes the area and maintains realignment as the bone heals. A fractured bone is immobilized by applying opposing force at both ends of the injured area, using an equal amount of traction and counter-traction. Weights provide the traction pull needed or the pull is achieved by positioning the individual's body weight appropriately. Traction is a form of closed reduction and is sometimes used as an alternative to surgery. Since it restricts movement of the affected limb or body part, it may confine a person to bed rest for an extended period of time.

A person may need open reduction if there is an open, severe, or comminuted fracture. This procedure allows a physician to examine and surgically correct associated soft tissue damage while reducing the fracture and, if necessary, applying internal or external devices. Internal fixation involves the use of metallic devices inserted into or through bone to hold the fracture in a set position and alignment while it heals. Devices include plates,

KEY TERMS

Avulsion fracture—A fracture caused by the tearing away of a fragment of bone where a strong ligament or tendon attachment forcibly pulls the fragment away from the bone tissue.

Axis—A line that passes through the center of the body or body part.

Comminuted fracture—A fracture where there are several breaks in a bone creating numerous fragments.

Compartment syndrome—Compartment syndrome is a condition in which a muscle swells but is constricted by the connective tissue around it, which cuts off blood supply to the muscle.

Contrast hydrotherapy—A series of hot and cold water applications. A hot compress (as hot as an individual can tolerate) is applied for three minutes followed by an ice cold compress for 30 seconds. These applications are repeated three times each and ending with the cold compress.

Osteogenesis imperfecta—A genetic disorder involving defective development of connective tissues, characterized by brittle and fragile bones that are easily fractured by the slightest trauma.

Osteoporosis—Literally meaning “porous bones,” this condition occurs when bones lose an excessive amount of their protein and mineral content, particularly

calcium. Over time, bone mass and strength are reduced leading to increased risk of fractures.

Paget’s disease—Chronic disorder of unknown cause, usually affecting middle aged and elderly people, characterized by enlarged and deformed bones. Excessive breakdown and formation of bone tissue occurs with Paget’s disease and can cause bone to weaken, resulting in bone pain, arthritis, deformities, and fractures.

Reduction—The restoration of a body part to its original position after displacement, such as the reduction of a fractured bone by bringing ends or fragments back into original alignment. The use of local or general anesthesia usually accompanies a fracture reduction. If performed by outside manipulation only, the reduction is described as closed; if surgery is necessary, it is described as open.

Rickets—A condition caused by the dietary deficiency of vitamin D, calcium, and usually phosphorus, seen primarily in infancy and childhood, and characterized by abnormal bone formation.

Traction—The process of placing a bone, limb, or group of muscles under tension by applying weights and pulleys. The goal is to realign or immobilize the part or to relieve pressure on that particular area to promote healing and restore function.

nails, screws, and rods. When healing is complete, the surgeon may or may not remove these devices. Virtually any hip fracture requires open reduction and internal fixation so that the bone will be able to support the patient’s weight.

Alternative treatment

In addition to the importance of calcium for strong bones, many alternative treatment approaches recommend use of mineral supplements to help build and maintain a healthy, resilient skeleton. Some physical therapists use electro-stimulation over a fractured site to promote and expedite healing. Chinese traditional medicine may be helpful by working to reconnect chi through the meridian lines along the line of a fracture. Homeopathy can enhance the body’s healing process. Two particularly useful homeopathic remedies are *Arnica* (*Arnica montana*) and *Symphytum* (*Symphytum officinalis*). If possible, applying contrast **hydrotherapy** to an extremity (e.g., a hand or foot) of a fractured area can assist healing by enhancing circulation.

Prognosis

Fractures involving joint surfaces almost always lead to some degree of arthritis of the joint. Fractures can normally be cured with proper first aid and appropriate aftercare. If determined necessary by a physician, the fractured site should be manipulated, realigned, and immobilized as soon as possible. Realignment has been shown to be much more difficult after six hours. Healing time varies from person to person with the elderly generally needing more time to heal completely. A non-union fracture may result when a fracture does not heal, such as in the case of an elderly person or an individual with medical complications. Recovery is complete when there is no bone motion at the fracture site, and x rays indicate complete healing. Open fractures may lead to bone infections, which delay the healing process. Another possible complication is compartment syndrome, a painful condition resulting from the expansion of enclosed tissue and that may occur when a body part is immobilized in a cast.

Prevention

Adequate calcium intake is necessary for strong bones and can help decrease the risk of fractures. People who do not get enough calcium in their **diets** can take a calcium supplement. **Exercise** can help strengthen bones by increasing bone density, thereby decreasing the risk of fractures from falls. A University of Southern California study reported that older people who exercised one or more hours per day had approximately half the incidence of hip fractures as those who exercised fewer than 30 minutes per day or not at all.

Fractures can be prevented if safety measures are taken seriously. These measures include using seat belts in cars and encouraging children to wear protective sports gear. Estrogen replacement for women past the age of 50 has been shown to help prevent osteoporosis and the fractures that may result from this condition. In one study, elderly women on estrogen replacement therapy demonstrated the lowest occurrence of hip fractures when compared to similar women not on estrogen replacement therapy.

Resources

BOOKS

- Burr, David B. *Musculoskeletal Fatigue and Stress Fracture*. Boca Raton, FL: CRC Press, 2001.
- Jupiter, J. *Fractures and Dislocations of the Hand*. St. Louis: Mosby, 2001.
- Moehring, H. David, and Adam Greenspan. *Fractures: Diagnosis and Treatment*. New York: McGraw Hill, 2000.
- Ogden, John A. *Skeletal Injury in the Child*. New York: Springer Verlag, 2000.
- Schenck, Robert C., and Ronnie P. Barnes. *Athletic Training and Sports Medicine*. 3rd ed. Chicago: American Academy of Orthopaedic Surgery, 1999.

ORGANIZATIONS

- American Academy of Orthopaedic Surgeons. 6300 North River Road, Rosemont, IL 60018-4262. (847) 823-7186 or (800) 346-2267. Fax: (847) 823-8125. <<http://orthoinfo.aaos.org/>> .
- American College of Sports Medicine. 401 W. Michigan St., Indianapolis, IN 46202. (317) 637-9200, Fax: (317) 634-7817.
- Children's Orthopedics of Atlanta. <<http://www.childrensortho.com/fractures.htm>> .
- Nemours Foundation. <http://kidshealth.org/kid/ill_injure/aches/broken_bones.html> .

OTHER

- "About the Human." <<http://orthopedics.about.com/health/orthopedics/blhipfracture.htm>> .
- Family Practice Notebook.com. <<http://www.fpnotebook.com/FRA.htm>> .

National Library of Medicine. <<http://medlineplus.adam.com/ency/article/000001.htm>> .

University of Iowa. <<http://www.vh.org/Providers/ClinRef/FPHandbook/Chapter06/18-6.html>> .

L. Fleming Fallon, Jr., MD, DrPH

Fragile X syndrome

Definition

Fragile X syndrome is the most common form of inherited mental retardation. Individuals with this condition have developmental delay, variable levels of **mental retardation**, and behavioral and emotional difficulties. They may also have characteristic physical traits. Generally, males are affected with moderate mental retardation and females with mild mental retardation.

Description

Fragile X syndrome is also known as Martin-Bell syndrome, Marker X syndrome, and FRAXA syndrome. It is the most common form of inherited mental retardation. Fragile X syndrome is caused by a mutation in the FMR-1 gene, located on the X chromosome. The role of the gene is unclear, but it is probably important in early development.

In order to understand fragile X syndrome it is important to understand how human genes and chromosomes influence this condition. Normally, each cell in the body contains 46 (23 pairs of) chromosomes. These chromosomes consist of genetic material (DNA) needed for the production of proteins, which lead to growth, development, and physical/intellectual characteristics. The first 22 pairs of chromosomes are the same in males and females. The remaining two chromosomes are called the sex chromosomes (X and Y). The sex chromosomes determine whether a person is male or female. Males have only one X chromosome, which is inherited from the mother at conception, and they receive a Y chromosome from the father. Females inherit two X chromosomes, one from each parent. Fragile X syndrome is caused by a mutation in a gene called FMR-1. This gene is located on the X chromosome. The FMR-1 gene is thought to play an important role in the development of the brain, but the exact way that the gene acts in the body is not fully understood.

Fragile X syndrome affects males and females of all ethnic groups. It is estimated that there are about one in 4,000 to one in 6,250 males affected with fragile X syndrome. There are approximately one-half as many females with fragile X syndrome as there are males. The carrier frequency in unaffected females is one in 100 to one in 600, with one study finding a carrier frequency of one in 250.

Causes and symptoms

For reasons not fully understood, the CGG sequence in the FMR-1 gene can expand to contain between 54 and 230 repeats. This stage of expansion is called a premutation. People who carry a premutation do not usually have symptoms of fragile X syndrome, although there have been reports of individuals with a premutation and subtle intellectual or behavioral symptoms. Individuals who carry a fragile X premutation are at risk to have children or grandchildren with the condition. Female premutation carriers may also be at increased risk for earlier onset of menopause; however, premutation carriers may exist through several generations of a family and no symptoms of fragile X syndrome will appear.

The size of the premutation can expand over succeeding generations. Once the size of the premutation exceeds 230 repeats, it becomes a full mutation and the FMR-1 gene is disabled. Individuals who carry the full mutation may have fragile X syndrome. Since the FMR-1 gene is located on the X chromosome, males are more likely to develop symptoms than females. This is because males have only one copy of the X chromosome. Males who inherit the full mutation are expected to have mental impairment. A female's normal X chromosome may compensate for her chromosome with the fragile X gene mutation. Females who inherit the full mutation have an approximately 50% risk of mental impairment. The phenomenon of an expanding trinucleotide repeat in successive generations is called anticipation. Another unique aspect of fragile X syndrome is that mosaicism is present in 15–20% those affected by the condition. Mosaicism is when there is the presence of cells of two different genetic materials in the same individual.

The mutation involves a short sequence of DNA in the FMR-1 gene. This sequence is designated CGG. Normally, the CGG sequence is repeated between six to 54 times. People who have repeats in this range do not have fragile X syndrome and are not at increased risk to have children with fragile X syndrome. Those affected by fragile X syndrome have expanded CGG repeats (over 200) in the first exon of the FMR1 gene (the full mutation)

Fragile X syndrome is inherited in an X-linked dominant manner (characters are transmitted by genes on the X chromosome). When a man carries a premutation on his X chromosome, it tends to be stable and usually will not expand if he passes it on to his daughters (he passes his Y chromosome to his sons). Thus, all of his daughters will be premutation carriers like he is. When a woman carries a premutation, it is unstable and can expand as she passes it on to her children, therefore a man's grandchildren are at greater risk of developing the syndrome. There is a 50% risk for a premutation carrier female to transmit an abnormal mutation with each **pregnancy**. The likelihood for the premutation to expand is related to the number of repeats present; the higher the number of repeats, the greater the chance that the premutation will expand to a full mutation in the next generation. All mothers of a child with a full mutation are carriers of an FMR-1 gene expansion. Ninety-nine percent of patients with fragile X syndrome have a CGG expansion, and less than one percent have a point mutation or deletion on the FMR1 gene.

Individuals with fragile X syndrome appear normal at birth but their development is delayed. Most boys with fragile X syndrome have mental impairment. The severity of mental impairment ranges from learning disabilities to severe mental retardation. Behavioral problems include attention deficit and hyperactivity at a young age. Some may show aggressive behavior in adulthood. Short attention span, poor eye contact, delayed and disordered speech and language, emotional instability, and unusual hand mannerisms (hand flapping or hand biting) are also seen frequently. Characteristic physical traits appear later in childhood. These traits include a long and narrow face, prominent jaw, large ears, and enlarged testes. In females who carry a full mutation, the physical and behavioral features and mental retardation tend to be less severe. About 50% of females who have a full mutation are mentally retarded. Other behavioral characteristics include whirling, spinning, and occasionally **autism**.

Children with fragile X syndrome often have frequent ear and sinus infections. Nearsightedness and lazy eye are also common. Many babies with fragile X syndrome may have trouble with sucking and some experience digestive disorders that cause frequent gagging and vomiting. A small percentage of children with fragile X syndrome may experience seizures. Children with fragile X syndrome also tend to have loose joints which may result in joint dislocations. Some children develop a curvature in the spine, flat feet, and a heart condition known as **mitral valve prolapse**.

Diagnosis

Any child with signs of developmental delay of speech, language, or motor development with no known cause should be considered for fragile X testing, especially if there is a family history of the condition. Behavioral and developmental problems may indicate fragile X syndrome, particularly if there is a family history of mental retardation. Definitive identification of the fragile X syndrome is made by means of a genetic test to assess the number of CGG sequence repeats in the FMR-1 gene. Individuals with the premutation or full mutation may be identified through genetic testing. **Genetic testing** for the fragile X mutation can be done on the developing baby before birth through **amniocentesis** or chorionic villus sampling (CVS), and is 99% effective in detecting the condition due to trinucleotide repeat expansion. Prenatal testing should only be undertaken after the fragile X carrier status of the parents has been confirmed and the couple has been counseled regarding the risks of recurrence. While prenatal testing is possible to do with CVS, the results can be difficult to interpret and additional testing may be required.

Treatment

Presently there is no cure for fragile X syndrome. Management includes such approaches as speech therapy, occupational therapy, and physical therapy. The expertise of psychologists, special education teachers, and genetic counselors may also be beneficial. Drugs may be used to treat hyperactivity, seizures, and other problems. Establishing a regular routine, avoiding over-stimulation, and using calming techniques may also help in the management of behavioral problems. Children with a troubled heart valve may need to see a heart specialist and take medications before surgery or dental procedures. Children with frequent ear and sinus infections may need to take medications or have special tubes placed in their ears to drain excess fluid. Mainstreaming of children with fragile X syndrome into regular classrooms is encouraged because they do well imitating behavior. Peer tutoring and positive reinforcement are also encouraged.

Prognosis

Early diagnosis and intensive intervention offer the best prognosis for individuals with fragile X syndrome. Adults with fragile X syndrome may benefit from vocational training and may need to live in a supervised setting. Life span is typically normal.

KEY TERMS

Amniocentesis—A procedure performed at 16–18 weeks of pregnancy in which a needle is inserted through a woman's abdomen into her uterus to draw out a small sample of the amniotic fluid from around the baby. Either the fluid itself or cells from the fluid can be used for a variety of tests to obtain information about genetic disorders and other medical conditions in the fetus.

CGG or CGG sequence—Shorthand for the DNA sequence: cytosine-guanine-guanine. Cytosine and guanine are two of the four molecules, otherwise called nucleic acids, that make up DNA.

Chorionic villus sampling (CVS)—A procedure used for prenatal diagnosis at 10–12 weeks gestation. Under ultrasound guidance a needle is inserted either through the mother's vagina or abdominal wall and a sample of cells is collected from around the early embryo. These cells are then tested for chromosome abnormalities or other genetic diseases.

Chromosome—A microscopic thread-like structure found within each cell of the body that consists of a complex of proteins and DNA. Humans have 46 chromosomes arranged into 23 pairs. Changes in either the total number of chromosomes or their shape and size (structure) may lead to physical or mental abnormalities.

FMR-1 gene—A gene found on the X chromosome. Its exact purpose is unknown, but it is suspected that the gene plays a role in brain development.

Mitral valve prolapse—A heart defect in which one of the valves of the heart (which normally controls blood flow) becomes floppy. Mitral valve prolapse may be detected as a heart murmur but there are usually no symptoms.

Premutation—A change in a gene that precedes a mutation; this change does not alter the function of the gene.

X chromosome—One of the two sex chromosomes (the other is Y) containing genetic material that, among other things, determine a person's gender.

A 2004 study found that men who are carriers of the fragile X gene but have not have the mutation severe enough to have fragile X syndrome may begin to show signs of tremor disorder, gait instability and memory impairment as they age. The higher prevalence of these symptoms among grandfathers of children with fragile x syndrome was noted so a study was

done to investigate their symptoms compared to men of the same age without the mutation. About 17% of the grandfathers in their 50s had the condition, 37% of those in their 60s, 47% of men in their 70s and 75% of men in their 80s. Often, these men have been diagnosed with other diseases such as Parkinson's or Alzheimer's rather than with fragile X-associated tremor/ataxia syndrome, the name which has been given to these late symptoms from the fragile x mutation.

Resources

PERIODICALS

Kaufmann, Walter E., and Allan L. Reiss. "Molecular and Cellular Genetics of Fragile X Syndrome." *American Journal of Medical Genetics* 88 (1999): 11–24.

Kirn, Timothy F. "New Fragile X Often Misdiagnosed as Parkinson's." *Clinical Psychiatry News* March 2004: 84.

ORGANIZATIONS

Arc of the United States (formerly Association for Retarded Citizens of the US). 500 East Border St., Suite 300, Arlington, TX 76010. (817) 261-6003. <<http://thearc.org>>.

National Fragile X Foundation. PO Box 190488, San Francisco, CA 94119-0988. (800) 688-8765 or (510) 763-6030. Fax: (510) 763-6223. natlfx@sprintmail.com. <<http://nfxf.org>>.

National Fragile X Syndrome Support Group. 206 Sherman Rd., Glenview, IL 60025. (708) 724-8626.

OTHER

"Fragile X Site Mental Retardation 1; FMR1." *Online Mendelian Inheritance in Man*. March 6, 2001. <<http://www3.ncbi.nlm.nih.gov/Omim/>>.

Tarleton, Jack, and Robert A. Saul. "Fragile X Syndrome." *GeneClinics* March 6, 2001. <<http://www.geneclinics.org>>.

Nada Quercia, MS, CCGC
Teresa G. Odle

Frambesia see **Yaws**

Francisella tularensis infection see **Tularemia**

Fresh cell therapy see **Cell therapy**

Friedreich's ataxia

Definition

Friedreich's ataxia (FA) is an inherited, progressive nervous system disorder causing loss of balance and coordination.

Description

Ataxia is a condition marked by impaired coordination. Friedreich's ataxia is the most common inherited ataxia, affecting between 3,000–5,000 people in the United States. FA is an autosomal recessive disease, which means that two defective gene copies must be inherited to develop symptoms, one from each parent. A person with only one defective gene copy will not show signs of FA, but may pass along the gene to offspring. Couples with one child affected by FA have a 25% chance in each **pregnancy** of conceiving another affected child.

Causes and symptoms

Causes

The gene for FA codes for a protein called frataxin. Normal frataxin is found in the cellular energy structures known as mitochondria, where it is thought to be involved in regulating the transport of iron. In FA, the frataxin gene on chromosome 9 is expanded with nonsense information known as a "triple repeat." This extra DNA interferes with normal production of frataxin, thereby impairing iron transport. Normally, there are 10–21 repeats of the frataxin gene. In FA, this sequence may be repeated between 200–900 times. The types of symptoms and severity of FA seems to be associated with the number of repetitions. Patients with more copies have more severe symptomatology. Researchers are still wrestling with how frataxin and the repeats on chromosome 9 are involved in causing FA. One theory suggests that FA develops in part because defects in iron transport prevent efficient use of cellular energy supplies.

The nerve cells most affected by FA are those in the spinal cord involved in relaying information between muscles and the brain. Tight control of movement requires complex feedback between the muscles promoting a movement, those restraining it, and the brain. Without this control, movements become uncoordinated, jerky, and inappropriate to the desired action.

Symptoms

Symptoms of FA usually first appear between the ages of 8 and 15, although onset as early as 18 months or as late as age 25 is possible. The first symptom is usually gait incoordination. A child with FA may graze doorways when passing through, for instance, or trip over low obstacles. Unsteadiness when standing still and deterioration of position sense is common. Foot deformities and walking up off the heels often

results from uneven muscle weakness in the legs. **Muscle spasms and cramps** may occur, especially at night.

Ataxia in the arms follows, usually within several years, leading to decreased hand-eye coordination. Arm weakness does not usually occur until much later. Speech and swallowing difficulties are common. **Diabetes mellitus** may also occur. Nystagmus, or eye tremor, is common, along with some loss of visual acuity. **Hearing loss** may also occur. A side-to-side curvature of the spine (**scoliosis**) occurs in many cases, and may become severe.

Heartbeat abnormalities occur in about two thirds of FA patients, leading to **shortness of breath** after exertion, swelling in the lower limbs, and frequent complaints of cold feet.

Diagnosis

Diagnosis of FA involves a careful medical history and thorough neurological exam. Lab tests include **electromyography**, an electrical test of muscle, and a nerve conduction velocity test. An electrocardiogram may be performed to diagnose heart arrhythmia.

Direct DNA testing is available, allowing FA to be more easily distinguished from other types of ataxia. The same test may be used to determine the presence of the genetic defect in unaffected individuals, such as siblings.

Treatment

There is no cure for FA, nor any treatment that can slow its progress. Amantadine may provide some limited improvement in ataxic symptoms, but is not recommended in patients with cardiac abnormalities. Physical and occupational therapy are used to maintain range of motion in weakened muscles, and to design adaptive techniques and devices to compensate for loss of coordination and strength. Some patients find that using weights on the arms can help dampen the worst of the uncoordinated arm movements.

Heart **arrhythmias** and diabetes are treated with drugs specific to those conditions.

Prognosis

The rate of progression of FA is highly variable. Most patients lose the ability to walk within 15 years of symptom onset, and 95% require a wheelchair for mobility by age 45. Reduction in lifespan from FA complications is also quite variable. Average age at

KEY TERMS

Ataxia—A condition marked by impaired coordination.

Scoliosis—An abnormal, side-to-side curvature of the spine.

death is in the mid-thirties, but may be as late as the mid-sixties. As of mid-1998, the particular length of the triple repeat has not been correlated strongly enough with disease progression to allow prediction of the course of the disease on this basis.

Prevention

There is no way to prevent development of FA in a person carrying two defective gene copies.

Resources

BOOKS

Feldman, Eva L. "Hereditary Cerebellar Ataxias and Related Disorders." In *Cecil Textbook of Medicine*, edited by Russel L. Cecil, et al. Philadelphia: W.B. Saunders Company, 2000.

Isselbacher, Kurt J., et al. "Spinocerebellar Degeneration (Friedreich's Ataxia)." In *Harrison's Principles of Internal Medicine*. New York: McGraw-Hill, 2001.

ORGANIZATIONS

Muscular Dystrophy Association. 3300 East Sunrise Drive, Tucson, AZ 85718. (520) 529-2000 or (800) 572-1717. <<http://www.mdausa.org>> .

Rosalyn Carson-DeWitt, MD

Frostbite and frostnip

Definition

Frostbite is the term for damage to the skin and other tissues caused by freezing. Frostnip is a mild form of cold injury.

Description

In North America, frostbite is largely confined to Alaska, Canada, and the northern states. Recent years have witnessed a substantial decline in the number of cases, probably for several reasons, including better

winter clothing and footwear and greater public understanding of how to avoid cold-weather dangers. At the same time, the nature of the at-risk population has changed as rising numbers of homeless people have made frostbite an urban as well as a rural public health concern. The growing popularity of outdoor winter activities has also expanded the at-risk population.

Causes and symptoms

Frostbite

Skin exposed to temperatures a little below the freezing mark can take hours to freeze, but very cold skin can freeze in minutes or seconds. Air temperature, wind speed, and moisture all affect how cold the skin becomes. A strong wind can lower skin temperature considerably by dispersing the thin protective layer of warm air that surrounds our bodies. Wet clothing readily draws heat away from the skin because water is a potent conductor of heat. The evaporation of moisture on the skin also produces cooling. For these reasons, wet skin or clothing on a windy day can lead to frostbite even if the air temperature is above the freezing mark.

The extent of permanent injury, however, is determined not by how cold the skin and the underlying tissues become but by how long they remain frozen. Consequently, homeless people and others whose self-preservation instincts may be clouded by alcohol or psychiatric illness face a greater risk of frostbite-related **amputation** because they are more likely to stay out in the cold when prudence dictates seeking shelter or medical attention. Alcohol also affects blood circulation in the extremities in a way that can increase the severity of injury (as does **smoking**). A review of 125 Saskatchewan frostbite cases found a tie to alcohol in 46% and to psychiatric illness in 17%. Other risk factors identified by researchers include inadequate clothing, previous cold injury, **fatigue**, wound infection, **atherosclerosis** (an arterial disease), and diabetes. Driving in poor weather can also be dangerous: vehicular failure was a predisposing factor in 15% of the Saskatchewan cases.

Three nearly simultaneous physiological processes underlie frostbite injury: tissue freezing, tissue hypoxia, and the release of inflammatory mediators. Tissue freezing causes ice crystal formation and other changes that damage and eventually kill cells. Much of this harm occurs because the ice produces pressure changes that cause water (crucial for cell survival) to flow out of the cells. Tissue

hypoxia (oxygen deficiency) occurs when the blood vessels in the hands, feet, and other extremities narrow in response to cold. Among its many tasks, blood transfers body heat to the skin, which then dissipates the heat into the environment. Blood vessel narrowing is the body's way of protecting vital internal organs at the expense of the extremities by reducing heat flow away from the core. However, blood also carries life-sustaining oxygen to the skin and other tissues, and narrowed vessels result in oxygen starvation. Narrowing also causes acidosis (an increase in tissue acidity) and increases blood viscosity (thickness). Ultimately, blood stops flowing through the capillaries (the tiny blood vessels that connect the arteries and veins) and **blood clots** form in the arterioles and venules (the smallest arteries and veins). Damage also occurs to the endothelial cells that line the blood vessels. Hypoxia, blood clots, and endothelial damage lead, in turn, to the release of inflammatory mediators (substances that act as links in the inflammatory process), which promote further endothelial damage, hypoxia, and cell destruction.

Frostbite is classified by degree of injury (first, second, third, or fourth), or simply divided into two types, superficial (corresponding to first- or second-degree injury) and deep (corresponding to third- or fourth-degree injury). Most frostbite injuries affect the feet or hands. The remaining 10% of cases typically involve the ears, nose, cheeks, or penis. Once frostbite sets in, the affected part begins to feel cold and, usually, numb; this is followed by a feeling of clumsiness. The skin turns white or yellowish. Many patients experience severe **pain** in the affected part during rewarming treatment and an intense throbbing pain that arises two or three days later and can last days or weeks. As the skin begins to thaw during treatment, **edema** (excess tissue fluid) often accumulates, causing swelling. In second- and higher-degree frostbite, blisters appear. Third-degree cases produce deep, blood-filled blisters and, during the second week, a hard black eschar (scab). Fourth-degree frostbite penetrates below the skin to the muscles, tendons, nerves, and bones. In severe cases of frostbite the dead tissue can mummify and drop off. Infection is also a possibility.

Frostnip

Like frostbite, frostnip is associated with ice crystal formation in the tissues, but no tissue destruction occurs and the crystals dissolve as soon as the skin is warmed. Frostnip affects areas such as the earlobes,

cheeks, nose, fingers, and toes. The skin turns pale and one experiences numbness or **tingling** in the affected part until warming begins.

Diagnosis

Frostbite diagnosis relies on a **physical examination** and may also include conventional radiography (x rays), angiography (x-ray examination of the blood vessels using an injected dye to provide contrast), thermography (use of a heat-sensitive device for measuring blood flow), and other techniques for predicting the course of injury and identifying tissue that requires surgical removal. During the initial treatment period, however, a physician cannot judge how a case will progress. Diagnostic tests only become useful three to five days after rewarming, once the blood vessels have stabilized.

Treatment

Frostbite

Emergency medical help should always be summoned whenever frostbite is suspected. While waiting for help to arrive, one should, if possible, remove wet or tight clothing and put on dry, loose clothing or wraps. A splint and padding are used to protect the injured area. Rubbing the area with snow or anything else is dangerous. The key to prehospital treatment is to avoid partial thawing and refreezing, which releases more inflammatory mediators and makes the injury substantially worse. For this reason, the affected part must be kept away from heat sources such as campfires and car heaters. Experts advise rewarming in the field only when emergency help will take more than two hours to arrive and refreezing can be prevented.

Because the outcome of a frostbite injury cannot be predicted at first, all hospital treatment follows the same route. Treatment begins by rewarming the affected part for 15–30 minutes in water at a temperature of 104–108 °F (40–42.2 °C). This rapid rewarming halts ice crystal formation and dilates narrowed blood vessels. Aloe vera (which acts against inflammatory mediators) is applied to the affected part, which is then splinted, elevated, and wrapped in a dressing. Depending on the extent of injury, blisters may be debrided (cleaned by removing foreign material) or simply covered with aloe vera. A **tetanus** shot and, possibly, penicillin, are used to prevent infection, and the patient is given ibuprofen to combat inflammation. **Narcotics** are needed in most cases to reduce the excruciating



A human hand with frostbite. (Photo Researchers, Inc. Reproduced by permission.)

pain that occurs as sensation returns during rewarming. Except when injury is minimal, treatment generally requires a hospital stay of several days, during which **hydrotherapy** and physical therapy are used to restore the affected part to health. Experts recommend a cautious approach to tissue removal, and advise that 22–45 days must pass before a decision on amputation can safely be made.

Frostnip

Frostnipped fingers are helped by blowing warm air on them or holding them under one's armpits. Other frostnipped areas can be covered with warm hands. The injured areas should never be rubbed.

Alternative treatment

Alternative practitioners suggest several kinds of treatment to speed recovery from frostbite after leaving the hospital. Bathing the affected part in warm water or using contrast hydrotherapy can help enhance circulation. Contrast hydrotherapy involves a series of hot and cold water applications. A hot compress (as hot as the patient can stand) is applied to the affected area for three minutes followed by an ice cold compress for 30 seconds. These applications are repeated three times each, ending with the cold compress. Nutritional therapy to promote tissue growth in damaged areas may also be helpful. Homeopathic and botanical therapies may also assist recovery from frostbite. Homeopathic *Hypericum* (*Hypericum perforatum*) is recommended when nerve ending are affected (especially in the fingers and toes) and *Arnica*

(*Arnica montana*) is prescribed for shock. Cayenne pepper (*Capsicum frutescens*) can enhance circulation and relieve pain. Drinking hot ginger (*Zingiber officinale*) tea also aids circulation. Other possible approaches include **acupuncture** to avoid permanent nerve damage and **oxygen therapy**.

Prognosis

The rapid rewarming approach to frostbite treatment, pioneered in the 1980s, has proved to be much more effective than older methods in preventing tissue loss and amputation. A study of 56 first-, second-, and third-degree frostbite patients treated with rapid rewarming in 1982–85 found that 68% recovered without tissue loss, 25% experienced some tissue loss, and 7% needed amputation. In a comparison group of 98 patients, treatment using older methods resulted in a tissue loss rate of nearly 35% and an amputation rate of nearly 33%. Although the comparison group included a higher proportion of second- and third-degree cases, the difference in treatment results was determined to be statistically significant.

The extreme throbbing pain that many frostbite sufferers endure for days or weeks after rewarming is not the only prolonged symptom of frostbite. During the first weeks or months, people often experience tingling, a burning sensation, or a sensation resembling shocks from an electric current. Other possible consequences of frostbite include skin—color changes, nail deformation or loss, joint stiffness and pain, hyperhidrosis (excessive sweating), and heightened sensitivity to cold. For everyone, a degree of sensory loss lasting at least four years—and sometimes a lifetime—is inevitable.

Prevention

With the appropriate knowledge and precautions, frostbite can be prevented even in the coldest and most challenging environments. Appropriate clothing and footwear are essential. To prevent heat loss and keep the blood circulating properly, clothing should be worn loosely and in layers. Covering the hands, feet, and head is also crucial for preventing heat loss. Outer garments need to be wind and water resistant, and wet clothing and footwear must be replaced as quickly as possible. Alcohol and drugs should be avoided because of their harmful effects on judgment and reasoning. Experts also warn against alcohol use and smoking in the cold because of the circulatory changes they produce. Paying close attention to the weather

report before venturing outdoors and avoiding unnecessary risks such as driving in isolated areas during a blizzard are also important.

Resources

PERIODICALS

Reamy, Brian V. "Frostbite: Review and Current Concepts." *Journal of the American Board of Family Practice* January-February 1998: 34-40.

Howard Baker

Frostnip see **Frostbite and frostnip**

FSH test see **Follicle-stimulating hormone test**

Fugu poisoning

Definition

Fugu **poisoning** occurs when a person eats the flesh of a fugu, also known as a puffer fish, which contains lethal toxins.

Description

Fugu, also known as puffer fish, blowfish, or globe fish, has long been a food delicacy in Japan, but has only been introduced in the United States in the last 30-40 years. The fugu and related species may contain a tetrodotoxin, an extremely potent neurotoxin and one of the most toxic substances known, which produces critical illness and often **death**. Between January 1 and April 1, 2002, at least 10 cases of fugu poisoning were reported in the United States, according to the Centers for Disease Control and Prevention (CDC) in Atlanta. All persons recovered from the poisonings. All of the fish came from the Atlantic Ocean off the coast of Titusville, Florida. Fugu caught in southern U.S. waters, such as the Gulf of Mexico, may also be toxic. Tetrodotoxin has been detected in pufferfish throughout the Pacific Ocean and the Baja California coastal region. Cases of fugu poisoning are sporadically diagnosed, but many more are not recognized or reported. The earliest cases reported to the CDC involved poisonings in Florida during the mid-1970s. Since 1950, only three known fatalities have occurred in the United States, all in Florida.

The dangers of puffer fish consumption have long been recognized. Artifacts recovered from an Egyptian tomb indicate that puffer **fish poisoning** has been known since approximately 2400-2700 B.C. In journals covering expeditions from 1772-1775, Pacific explorer Captain James Cook provided a vivid description of what some believe to be puffer fish poisoning. Fugu are found in waters throughout the world. Scientists have found that toxic fugu have unique exocrine glands for the secretion of tetrodotoxin. The fish appear to actively produce the toxin, rather than passively acquire it from the environment. For these fish, tetrodotoxin may serve as a natural defense mechanism to repel predators. The flesh of the fugu is generally eaten raw in paper-thin slices, known as sashimi. Part of the reported delight in eating fugu is the **tingling** oral sensation induced by minute amounts of tetrodotoxin in the flesh. For this reason, eating fugu is considered an “experience,” rather than just a meal in Japan. The experience is expensive, however, since a plate of this delicacy can cost as much as \$500.

Causes & symptoms

The most common symptoms of fugu poisoning are tingling and burning of the mouth and tongue, **numbness**, drowsiness, and incoherent speech. These symptoms usually occur 30 minutes to two hours after ingestion of the fish, depending on the amount of toxin ingested. In severe cases, ataxia (the inability to coordinate the movements of muscles), muscle weakness, **hypotension** (low blood pressure) and cardiac **arrhythmias** (irregular heartbeat) may develop, followed by muscle twitching and respiratory **paralysis**, and death can occur. In several cases, people died within 17 minutes after eating pufferfish.

Diagnosis

The initial diagnosis is usually made by observation of early symptoms, including an abnormal or unexplained tingling, pricking, or burning sensation on the skin around the mouth and throat. Definitive diagnosis can only be made in a medical laboratory by examination of the ingested fish and identification of the specific toxins. Ill persons should be advised to proceed to a hospital emergency department and contact their local poison control center.

Treatment

There is no antidote for fugu poisoning, therefore treatment is limited to supportive measures

KEY TERMS

Ataxia—A lack of muscle control.

Arrhythmia—An irregularity in the normal rhythm or force of the heartbeat.

Atropine—A poisonous alkaloid obtained from belladonna or related plants, used medically to dilate the pupils of the eyes and to stop spasms.

Endoscopy—The use of a medical instrument consisting of a long tube inserted into the body, usually through a small incision, for diagnostic examination and surgical procedures.

Exocrine—Relating to external secretion glands, such as sweat glands or salivary glands that release a secretion through a duct to the surface of an organ.

Hypotension—Low blood pressure.

Lavage—The washing out of a hollow body organ, for example, the stomach, using a flow of water.

Neurotoxin—A substance that damages, destroys, or impairs the functioning of nerve tissue.

and the removal of the unabsorbed toxin. If spontaneous **vomiting** does not occur, it should be induced. Gastric lavage (stomach washing) with an alkaline solution has been suggested, as well as endoscopy to remove the poison from the proximal small bowel. Following lavage, **activated charcoal** is reported to effectively bind the toxin. Other steps include administration of oxygen, assisted breathing, intravenous atropine for bradycardia (slow heartbeat) and intravenous fluids, along with dopamine, to manage hypotension. Since tetrodotoxins and opiates are similar, use of an opiate antagonist may be useful, according to the American Academy of Family Physicians.

Alternative treatment

There is no alternative medicine treatment for fugu poisoning.

Prognosis

The mortality rate may be as high as 60%. Epidemiologic evidence suggests that recovery can be expected if an affected person survives beyond 24 hours. After 24 hours, a person with fugu poisoning usually makes a full recovery.

Prevention

The only prevention is not to eat any of the species of fugu that contain toxins.

Resources

BOOKS

Olson, Kent R. *Poisoning & Drug Overdose* New York City: McGraw-Hill, 2003.

PERIODICALS

Currie, Bart J. "Marine Antivenoms." *Journal of Toxicology: Clinical Toxicology* (April 2003): 301-308.

"Fugu Fish Sequenced." *Applied Genetics News* (August 2002): 0.

Scully, Mary-Louise. "Tingling Away in Titusville, Florida." *Infectious Disease Alert* (August 1, 2002): 165-167.

ORGANIZATIONS

American Association of Poison Control Centers. 3201 New Mexico Ave., Suite 330, Washington, DC 20016. (202) 362-7217. info@aapcc.org. <http://www.aapcc.org>.

Ken R. Wells

Fugue see **Dissociative disorders**

FUO see **Fever of unknown origin**

Furosemide see **Diuretics**

Furunculosis see **Boils**

Fusobacterium infection see **Anaerobic infections**

G

Galactorrhea

Definition

Galactorrhea is the secretion of breast milk in men, or in women who are not breastfeeding an infant.

Description

Lactation, or the production of breast milk, is a normal condition occurring in women after delivery of a baby. Many women who have had children may even be able to express a small amount of breast milk from the nipple up to two years after **childbirth**. Galactorrhea, or hyperlactation, however, is a rare condition that can occur in both men and women, where a white or grayish fluid is secreted by the nipples of both breasts. While this condition is not serious in itself, galactorrhea can indicate more serious conditions, including hormone imbalances or the presence of tumors.

Causes and symptoms

Causes

Galactorrhea is associated with a number of conditions. The normal production of breast milk is controlled by a hormone called prolactin, which is secreted by the pituitary gland in the brain. Any condition that upsets the balance of hormones in the blood or the production of hormones by the pituitary gland or sexual organs can stimulate the production of prolactin.

Often, a patient with galactorrhea will have a high level of prolactin in the blood. A tumor in the pituitary gland can cause this overproduction of prolactin. At least 30% of women with galactorrhea, menstrual abnormalities, and high prolactin levels have a pituitary gland tumor. Other types of brain tumors, head injuries, or **encephalitis** (an infection of the brain) can also cause galactorrhea.

Tumors or growths in the ovaries or other reproductive organs in women, or in the testicles or related sexual organs of men, can also stimulate the production of prolactin. Any discharge of fluid from the breast after a woman has passed **menopause** may indicate **breast cancer**. However, most often the discharge associated with breast **cancer** will be from one breast only. In galactorrhea both breasts are usually involved. The presence of blood in the fluid discharged from the breast could indicate a benign growth in the breast tissue itself. In approximately 10–15% of patients with blood in the fluid, carcinoma of the breast tissue is present.

A number of medications and drugs can also cause galactorrhea as a side-effect. Hormonal therapies (like **oral contraceptives**), drugs for treatment of depression or other psychiatric conditions, tranquilizers, morphine, heroin, and some medications for high blood pressure can cause galactorrhea.

Several normal physiologic situations can cause production of breast milk. Nipple stimulation in men or women during sexual intercourse may induce lactation, for women particularly during or just after **pregnancy**.

Even after extensive testing, no specific cause can be determined for some patients with galactorrhea.

Symptoms

The primary symptom of galactorrhea is the discharge of milky fluid from both breasts. In women, galactorrhea may be associated with **infertility**, menstrual cycle irregularities, hot flashes, or amenorrhea—a condition where menstruation stops completely. Men may experience loss of sexual interest and **impotence**. Headaches and visual disturbances have also been associated with some cases of galactorrhea.

Diagnosis

Galactorrhea is generally considered a symptom that may indicate a more serious problem. Collection

of a thorough medical history, including pregnancies, surgeries, and consumption of drugs and medications is a first step in diagnosing the cause of galactorrhea. A **physical examination**, along with a breast examination, will usually be conducted. Blood and urine samples may be taken to determine levels of various hormones in the body, including prolactin and compounds related to thyroid function.

A mammogram (an x ray of the breast) or an ultrasound scan (using high frequency sound waves) might be used to determine if there are any tumors or cysts present in the breasts themselves. If a tumor of the pituitary gland is suspected, a series of computer assisted x rays called a computed tomography scan (CT scan) may be done. Another procedure that may be useful is a **magnetic resonance imaging** (MRI) scan to locate tumors or abnormalities in tissues.

Treatment

Treatment for galactorrhea will depend on the cause of the condition and the symptoms. The drug bromocriptine is often prescribed first to reduce the secretion of prolactin and to decrease the size of **pituitary tumors**. This drug will control galactorrhea symptoms and in many cases may be the only therapy necessary. Oral estrogen and progestins (hormone pills, like birth control pills) may control symptoms of galactorrhea for some women. Surgery to remove a tumor may be required for patients who have more serious symptoms of **headache** and vision loss, or if the tumor shows signs of enlargement despite drug treatment. **Radiation therapy** has also been used to reduce tumor size when surgery is not possible or not totally successful. A combination of drug, surgery, and radiation treatment can also be used.

Galactorrhea is more of a nuisance than a real threat to health. While it is important to find the cause of the condition, even if a tumor is discovered in the pituitary gland, it may not require treatment. With very small, slow-growing tumors, some physicians may suggest a “wait and see” approach.

Prognosis

Treatment with bromocriptine is usually effective in stopping milk secretion, however, symptoms may recur if drug therapy is discontinued. Surgical removal or radiation treatment may correct the problem permanently if it is related to a tumor. Frequent monitoring of hormone status and tumor size may be recommended.

KEY TERMS

Amenorrhea—Abnormal cessation of menstruation.

Bromocriptine—Also known as Parlodel, the main drug used to treat galactorrhea by reducing levels of the hormone prolactin.

Hyperlactation—Another term for galactorrhea.

Lactation—The production of breast milk.

Prevention

There is no way to prevent galactorrhea. If the condition is caused by the use of a particular drug, a patient may be able to switch to a different drug that does not have the side-effect of galactorrhea.

Resources

BOOKS

“Galactorrhea.” In *Current Medical Diagnosis & Treatment*, 1998. 37th ed. Stamford: Appleton & Lange, 1997.

Altha Roberts Edgren

Galactosemia

Definition

Galactosemia is an inherited disease in which the transformation of galactose to glucose is blocked, allowing galactose to increase to toxic levels in the body. If galactosemia is untreated, high levels of galactose cause **vomiting**, **diarrhea**, lethargy, low blood sugar, brain damage, **jaundice**, liver enlargement, **cataracts**, susceptibility to infection, and **death**.

Description

Galactosemia is a rare but potentially life-threatening disease that results from the inability to metabolize galactose. Serious consequences from galactosemia can be prevented by screening newborns at birth with a simple blood test.

Galactosemia is an inborn error of metabolism. “Metabolism” refers to all chemical reactions that take place in living organisms. A metabolic pathway is a series of reactions where the product of each step in the series is the starting material for the next step. Enzymes are the chemicals that help

the reactions occur. Their ability to function depends on their structure, and their structure is determined by the deoxyribonucleic acid (DNA) sequence of the genes that encode them. Inborn errors of metabolism are caused by mutations in these genes which do not allow the enzymes to function properly.

Sugars are sometimes called “the energy molecules,” and galactose and glucose are both sugars. For galactose to be utilized for energy, it must be transformed into something that can enter the metabolic pathway that converts glucose into energy (plus water and carbon dioxide). This is important for infants because they typically get most of their nutrient energy from milk, which contains a high level of galactose. Each molecule of lactose, the major sugar constituent of milk, is made up of a molecule of galactose and a molecule of glucose, and so galactose makes up 20% of the energy source of a typical infant’s diet.

Three enzymes are required to convert galactose into glucose-1-phosphate (a phosphorylated glucose that can enter the metabolic pathway that turns glucose into energy). Each of these three enzymes is encoded by a separate gene. If any of these enzymes fail to function, galactose build-up and galactosemia result. Thus, there are three types of galactosemia with a different gene responsible for each.

Every cell in a person’s body has two copies of each gene. Each of the forms of galactosemia is inherited as a recessive trait, which means that galactosemia is only present in individuals with two mutated copies of one of the three genes. This also means that carriers, with only one copy of a gene mutation, will not be aware that they are carrying a mutation (unless they have had a genetic test), as it is masked by the normal gene they also carry and they have no symptoms of the disease. For each step in the conversion of galactose to glucose, if only one of the two copies of the gene controlling that step is normal (i.e. for carriers), enough functional enzyme is made so that the pathway is not blocked at that step. If a person has galactosemia, both copies of the gene coding for one of the enzymes required to convert glucose to galactose are defective and the pathway becomes blocked. If two carriers of the same defective gene have children, the chance of any of their children getting galactosemia (the chance of a child getting two copies of the defective gene) is 25% (one in four) for each **pregnancy**.

Classic galactosemia occurs in the United States about one in every 50,000–70,000 live births.

Causes and symptoms

Galactosemia I

Galactosemia I (also called classic galactosemia), the first form to be discovered, is caused by defects in both copies of the gene that codes for an enzyme called galactose-1-phosphate uridyl transferase (GALT). There are 30 known different mutations in this gene that cause GALT to malfunction.

Newborns with galactosemia I appear normal at birth, but begin to develop symptoms after they are given milk for the first time. Symptoms include vomiting, diarrhea, lethargy (sluggishness or **fatigue**), low blood glucose, jaundice (a yellowing of the skin and eyes), enlarged liver, protein and amino acids in the urine, and susceptibility to infection, especially from gram negative bacteria. Cataracts (a grayish white film on the eye lens) can appear within a few days after birth. People with galactosemia frequently have symptoms as they grow older even though they have been given a galactose-free diet. These symptoms include **speech disorders**, cataracts, ovarian atrophy, and **infertility** in females, learning disabilities, and behavioral problems.

Galactosemia II

Galactosemia II is caused by defects in both copies of the gene that codes for an enzyme called galactokinase (GALK). The frequency of occurrence of galactosemia II is about one in 100,000–155,000 births.

Galactosemia II is less harmful than galactosemia I. Babies born with galactosemia II will develop cataracts at an early age unless they are given a galactose-free diet. They do not generally suffer from liver damage or neurologic disturbances.

Galactosemia III

Galactosemia III is caused by defects in the gene that codes for an enzyme called uridyl diphosphogalactose-4-epimerase (GALE). This form of galactosemia is very rare.

There are two forms of galactosemia III, a severe form, which is exceedingly rare, and a benign form. The benign form has no symptoms and requires no special diet. However, newborns with galactosemia III, including the benign form, have high levels of galactose-1-phosphate that show up on the initial screenings for elevated galactose and galactose-1-phosphate. This situation illustrates one aspect of the importance of follow-up enzyme function tests. Tests showing normal levels of GALT and GALK allow people affected by the benign form of galactosemia III to enjoy a normal diet.

The severe form has symptoms similar to those of galactosemia I, but with more severe neurological problems, including seizures. Only two cases of this rare form had been reported as of 1997.

Diagnosis

The newborn screening test for classic galactosemia is quick and straightforward; all but three states require testing on all newborns. Blood from a baby who is two to three days old is usually first screened for high levels of galactose and galactose-1-phosphate. If either of these compounds is elevated, further tests are performed to find out which enzymes (GALT, GALK, or GALE) are present or missing. DNA testing may also be performed to confirm the diagnosis.

If there is a strong suspicion that a baby has galactosemia, galactose is removed from the diet right away. In this case, an initial screen for galactose or galactose-1-phosphate will be meaningless. In the absence of galactose in the diet, this test will be negative whether the baby has galactosemia or not. In this case, tests to measure enzyme levels must be given to find out if the suspected baby is indeed galactosemic.

In addition, galactosemic babies who are refusing milk or vomiting will not have elevated levels of galactose or galactose phosphate, and their condition will not be detected by the initial screen. Any baby with symptoms of galactosemia (for example, vomiting) should be given enzyme tests.

Treatment

Galactosemia I and II are treated by removing galactose from the diet. Since galactose is a breakdown product of lactose, the primary sugar constituent of milk, this means all milk and foods containing milk products must be totally eliminated. Other foods like legumes, organ meats, and processed meats also contain considerable galactose and must be avoided. Pills that use lactose as a filler must also be avoided. Soy-based and casein hydrolysate-based formulas are recommended for infants with galactosemia.

Treatment of the severe form of galactosemia III with a galactose-restricted diet has been tried, but this disorder is so rare that the long-term effects of this treatment are unknown.

Prognosis

Early detection in the newborn period is the key to controlling symptoms. Long-term effects in untreated babies include severe **mental retardation**, **cirrhosis** of

KEY TERMS

Casein hydrolysate—A preparation made from the milk protein casein, which is hydrolyzed to break it down into its constituent amino acids. Amino acids are the building blocks of proteins.

Catalyst—A substance that changes the rate of a chemical reaction, but is not physically changed by the process.

Enzyme—A protein that catalyzes a biochemical reaction or change without changing its own structure or function.

Galactose—One of the two simple sugars, together with glucose, that makes up the protein, lactose, found in milk. Galactose can be toxic in high levels.

Glucose—One of the two simple sugars, together with galactose, that makes up the protein, lactose, found in milk. Glucose is the form of sugar that is usable by the body to generate energy.

Lactose—A sugar made up of glucose and galactose. It is the primary sugar in milk.

Metabolic pathway—A sequence of chemical reactions that lead from some precursor to a product, where the product of each step in the series is the starting material for the next step.

Metabolism—The total combination of all of the chemical processes that occur within cells and tissues of a living body.

Recessive trait—An inherited trait or characteristic that is outwardly obvious only when two copies of the gene for that trait are present.

the liver, and death. About 75% of the untreated babies die within the first two weeks of life. On the other hand, with treatment, a significant proportion of people with galactosemia I can lead nearly normal lives, although speech defects, learning disabilities, and behavioral problems are common. A 2004 study revealed that children and adolescents with classic galactosemia often have lower quality of life than peers without the disease, exhibiting problems with cognition (thinking and intellectual skills) and social function. In addition, cataracts due to galactosemia II can be completely prevented by a galactose-free diet.

Prevention

Since galactosemia is a recessive genetic disease, the disease is usually detected on a newborn screening

test, since most people are unaware that they are carriers of a gene mutation causing the disease. For couples with a previous child with galactosemia, prenatal diagnosis is available to determine whether a pregnancy is similarly affected. Families in which a child has been diagnosed with galactosemia can have DNA testing which can enable other more distant relatives to determine their carrier status. Prospective parents can then use that information to conduct family planning or to prepare for a child with special circumstances. Children born with galactosemia should be put on a special diet right away, to reduce the symptoms and complications of the disease.

Resources

PERIODICALS

Bosch, Annet M., et al. "Living With Classical Galactosemia: Health-related Quality of Life Consequences." *Pediatrics* May 2004: 1385–1387.

ORGANIZATIONS

Association for Neuro-Metabolic Disorders. 5223 Brookfield Lane, Sylvania, OH 43560. (419) 885-1497.
Metabolic Information Network. PO Box 670847, Dallas, TX 75367-0847. (214) 696-2188 or (800) 945-2188.
Parents of Galactosemic Children, Inc. 2148 Bryton Dr., Powell OH 43065. <<http://www.galactosemia.org/index.htm>> .

OTHER

"GeneCards: Human Genes, Proteins and Diseases." <<http://bioinfo.weizmann.ac.il/cards/>> .
"Vermont Newborn Screening Program: Galactosemia." <http://www.vtmednet.org/~m145037/vhgi_mem/nbsman/galacto.htm> .

Amy Vance, MS, CGC
Teresa G. Odle

Gallbladder cancer

Definition

Cancer of the gallbladder is cancer of the pear-shaped organ that lies on the undersurface of the liver.

Description

Bile from the liver is funneled into the gallbladder by way of the cystic duct. Between meals, the gallbladder stores a large amount of bile. To do this, it must absorb much of the water and electrolytes from the bile. In fact, the inner surface of the gallbladder is the

most absorptive surface in the body. After a meal, the gallbladder's muscular walls contract to deliver the bile back through the cystic duct and eventually into the small intestine, where the bile can help digest food.

Demographics

About 5,000 people are diagnosed with gallbladder cancer each year in the United States, making it the fifth most common gastrointestinal cancer. It is more common in females than males and most patients are elderly. Southwest American Indians have a particularly high incidence— six times that of the general population.

Causes and symptoms

Gallstones are the most significant risk factor for the development of gallbladder cancer. Roughly 75 to 90 percent of patients with gallbladder cancer also have gallstones. Larger gallstones are associated with a higher chance of developing gallbladder cancer. Chronic inflammation of the gallbladder from infection also increases the risk for gallbladder cancer.

Unfortunately, sometimes cancer of the gallbladder does not produce symptoms until late in the disease. When symptoms are evident, the most common is **pain** in the upper right portion of the abdomen, underneath the right ribcage. Patients with gallbladder cancer may also report symptoms such as **nausea**, **vomiting**, weakness, **jaundice**, skin **itching**, **fever**, chills, poor appetite, and weight loss.

Diagnosis

Gallbladder cancer is often misdiagnosed because it mimics other more common conditions, such as gallstones, **cholecystitis**, and **pancreatitis**. But the imaging tests that are utilized to evaluate these other conditions can also detect gallbladder cancer. For example, ultrasound is a quick, noninvasive imaging test that reliably diagnoses gallstones and cholecystitis. It can also detect the presence of gallbladder cancer as well as show how far the cancer has spread. If cancer is suspected, a computed tomography scan is useful in confirming the presence of an abnormal mass and further demonstrating the size and extent of the tumor. Cholangiography, usually performed to evaluate a patient with jaundice, can also detect gallbladder cancer.

There are no specific laboratory tests for gallbladder cancer. Tumors can obstruct the normal flow of

bile from the liver to the small intestine. Bilirubin, a component of bile, builds up within the liver and is absorbed into the bloodstream in excess amounts. This can be detected in a blood test, but it can also manifest clinically as jaundice. Elevated bilirubin levels and clinical jaundice can also occur with other conditions, such as gallstones.

On occasion, gallbladder cancer is diagnosed incidentally. About one percent of all patients who have their gallbladder removed for symptomatic gallstones are found to have gallbladder cancer. The cancer is found either by the surgeon or by the pathologist who inspects the gallbladder with a microscope.

Treatment

Staging of gallbladder cancer is determined by the how far the cancer has spread. The effectiveness of treatment declines as the stage progresses. Stage I cancer is confined to the wall of the gallbladder. Approximately 25% of cancers are at this stage at the time of diagnosis. Stage II cancer has penetrated the full thickness of the wall, but has not spread to nearby lymph nodes or invaded adjacent organs. Stage III cancer has spread to nearby lymph nodes or has invaded the liver, stomach, colon, small intestine, or large intestine. Stage IV disease has invaded very deeply into two or more adjacent organs or has spread to distant lymph nodes or organs by way of metastasis.

Early Stage I cancers involving only the innermost layer of the gallbladder wall can be cured by simple removal of the gallbladder. Cancers at this stage are sometimes found incidentally when the gallbladder is removed in the treatment of gallstones or cholecystitis. The majority of patients have good survival rates. Late Stage I cancers, which involve the outer muscular layers of the gallbladder wall, are generally treated in the same way as Stage II or III cancers. Removal of the gallbladder is not sufficient for these stages. The surgeon also removes nearby lymph nodes as well as a portion of the adjacent liver (radical surgery). Survival rates for these patients are considerably worse than for those with early Stage I disease. Patients with early Stage IV disease may benefit from radical surgery, but the issue is controversial. Late Stage IV cancer has spread too extensively to allow complete excision. Surgery is not an option for these patients.

Other therapies

When long-term survival is not likely, the focus of therapy shifts to improving quality of life. Jaundice and blockage of the stomach are two problems faced by patients with advanced cancer of the gallbladder.

KEY TERMS

Cholangiography—Radiographic examination of the bile ducts after injection with a special dye

Cholecystitis—Inflammation of the gallbladder, usually due to infection

Computed tomography—A radiology test by which images of cross-sectional planes of the body are obtained

Jaundice—Yellowish staining of the skin and eyes due to excess bilirubin in the bloodstream

Metastasis—The spread of tumor cells from one part of the body to another through blood vessels or lymphatic vessels

Pancreatitis—Inflammation of the pancreas

Stent—Slender hollow catheter or rod placed within a vessel or duct to provide support or maintain patency

Ultrasound—A radiology test utilizing high frequency sound waves

These can be treated with surgery, or alternatively, by special interventional techniques employed by the gastroenterologist or radiologist. A stent can be placed across the bile ducts in order to re-establish the flow of bile and relieve jaundice. A small feeding tube can be placed in the small intestine to allow feeding when the stomach is blocked. Pain may be treated with conventional pain medicines or a celiac ganglion nerve block.

Current **chemotherapy** or **radiation therapy** cannot cure gallbladder cancer, but they may offer some benefit in certain patients. For cancer that is too advanced for surgical cure, treatment with chemotherapeutic agents such as 5-fluorouracil may lengthen survival for a few months. The limited benefit of chemotherapy must be weighed carefully against its side effects. Radiation therapy is sometimes used after attempted surgical resection of the cancer to extend survival for a few months or relieve jaundice.

Resources

BOOKS

Abeloff, Martin D. "Gallbladder Carcinoma." In *Clinical Oncology*. 2nd ed. New York: Churchill Livingstone, 2000, pp.1730-1737.

Ahrendt, Steven A., and Henry A. Pitt. "Biliary Tract." In *Sabiston Textbook of Surgery*, edited by Courtney Townsend, Jr., 16th ed. Philadelphia: W.B. Saunders Company, 2001, pp. 1076-1111.

OTHER

National Cancer Institute Cancer Trials web site. < <http://cancertrials.nci.nih.gov/system> > . < <http://www.cancertrials.com> > .

Kevin O. Hwang, MD

Gallbladder disease see **Cholecystitis**

Gallbladder nuclear medicine scan

Definition

A nuclear medicine scan of the gallbladder is used to produce a set of images that look like x rays. The procedure uses a small amount of radioactive dye which is injected into the body. The dye accumulates in the organ, in this case, the gallbladder. A special camera called a scintillation or gamma camera produces images based on how the dye travels through the system and how the radiation is absorbed by the tissues. The procedure is also called cholescintigraphy or a hepatobiliary scan.

Purpose

A nuclear medicine scan can be used to diagnose disease and to find abnormalities in a body organ. A gallbladder scan can detect **gallstones**, tumors, or defects of the gallbladder. It can also be used to diagnose blockages of the bile duct that leads from the gallbladder to the small intestine. Unlike ultrasound, a gallbladder nuclear medicine scan can assess gallbladder function.

Precautions

Women who are pregnant or breastfeeding should tell their doctors before a scan is performed. Some medications or even eating a high fat meal before the procedure can interfere with the results of the scan.

Description

The gallbladder is a small pear-shaped sac located under the liver. The liver produces bile, a yellowish-green mixture of salts, acids, and other chemicals, that are stored in the gallbladder. Bile is secreted into the small intestine to help the body digest fats from foods.

Gallbladder disease, gallstones, **cancer**, or other abnormalities can cause **pain** and other symptoms. A gallbladder condition might be suspected if a patient has chronic or occasional pain in the upper right side of the abdomen. The pain may be stabbing and intense with sudden onset or it may be more of a dull, occasional ache. Loss of appetite, **nausea and vomiting** can also occur. **Fever** may indicate the presence of infection. **Jaundice**, a yellowing of the skin and whites of the eyes, may also indicate that the gallbladder is involved.

A gallbladder nuclear medicine scan may be used to diagnose gallstones, blockage of the bile duct or other abnormalities, and to assess gallbladder functioning and inflammation (**cholecystitis**). The scan is usually performed in a hospital or clinical radiology department. The patient lies on an examination table while a small amount of radioactive dye is injected into a vein in the arm. This dye circulates through the blood and collects in the gallbladder. As the dye moves through the gallbladder, a series of pictures is taken using a special camera called a *scintillation* or *gamma camera*. This procedure produces images that look like x rays. The test usually takes one to two hours to complete, but can last up to four hours.

The results of the scan are read by a radiologist, a doctor specializing in x rays and other types of scanning techniques. A report is sent, usually within 24 hours, to the doctor who will discuss the results with the patient.

Preparation

The patient may be required to withhold food and liquids for up to eight hours before the scan.

Aftercare

No special care is required after the procedure. Once the scan is complete, the patient can return to normal activities.

Risks

Nuclear medicine scans use a very small amount of radioactive material, and the risk of radiation is minimal. Very rarely, a patient may have a reaction to the dye material used.

Normal results

A normal scan shows a gallbladder without gallstones. There will be no evidence of growths or tumors, and no signs of infection or swelling. The normal gallbladder fills with bile and secretes it through the bile duct without blockages.

KEY TERMS

Cholecystitis—Inflammation of the gallbladder.

Cholescintigraphy—Another term for a gallbladder nuclear medicine scan.

Hepatobiliary scan—Another term for a gallbladder nuclear medicine scan.

Scintillation or gamma camera—A camera, somewhat like an x-ray machine, used to photograph internal organs after the patient has been injected with a radioactive material.

Abnormal results

An abnormal scan may show abnormal gallbladder emptying (suggesting gallbladder dysfunction or inflammation), or gallstones in the gallbladder or in the bile duct. The presence of tumors, growths or other types of blockages of the duct or the gallbladder itself could also appear on an abnormal scan.

Resources

OTHER

“Gallbladder Scan, Radioisotope.” Infonet. <<http://infonet.med.cornell.edu>> .

“Nuclear Medicine.” Washington Radiology Associates Page. <<http://www.wrapc.com>> .

Altha Roberts Edgren

Gallbladder surgery see **Cholecystectomy**

Gallbladder x rays

Definition

This is an x-ray exam of the gallbladder (GB), a sac-like organ that stores bile that is located under the liver. The study involves taking tablets containing dye (contrast) which outline any abnormalities when x rays are taken the following day. The test was once the standard for diagnosing diseases of the GB such as **gallstones**, but is used less frequently now. This is due to advances in diagnostic ultrasound, which is quick, accurate and doesn't involve exposure to ionizing radiation. When functional parameters of the gallbladder need to be demonstrated, scintigraphy is now the study of choice. OCG, however, can be useful when a gallbladder is contracted down due to

the presence of many, many gallstones. It can also help determine whether the cystic duct is clear, prior to surgical procedures such as **lithotripsy**. OCG may also be used to evaluate gallbladder disease that doesn't involve gallstones, such as adenomyomatosis of the gallbladder or cholesterosis of the gallbladder.

Purpose

This test, also known as an oral cholecystogram or OCG, is usually ordered to help physicians diagnose disorders of the gallbladder, such as gallstones and tumors, which show up as solid dark structures. It is performed to help in the investigation of patients with upper abdominal **pain**. The test also measures gallbladder function, as the failure of the organ to visualize can signify a non-functioning or diseased gallbladder. The gallbladder may also not visualize if the bilirubin level is over 4 and the study should not be performed under these circumstances.

Precautions

Your physician must be notified if you are pregnant or allergic to iodine. Patients with a history of severe kidney damage, have an increased risk of injury or side effects from the procedure. In those cases, ultrasound is commonly used instead of the x-ray examination. Some people experience side effects from the contrast material (dye tablets), especially **diarrhea**. During preparation for the test, patients should not use any **laxatives**. Diabetics should discuss the need for any adjustment in medication with their physician.

Description

The exam is performed in the radiology department. The night before the test, patients swallow six tablets (one at a time) that contain the contrast (x-ray dye). The following day at the hospital, the radiologist examines the gallbladder with a fluoroscope (a special x ray that projects the image onto a video monitor). Sometimes, patients are then asked to drink a highfat formula that will cause the gallbladder to contract and release bile. X rays will then be taken at various intervals. There is no discomfort from the test. If the gallbladder is not seen, the patient may be asked to return the following day for x rays.

Preparation

The day before the test patients are instructed to eat a high fat lunch (eggs, butter, milk, salad oils, or fatty meats), and a fat-free meal (fruits, vegetables,

KEY TERMS

Bile—A yellow-green liquid produced by the liver, which is released through the bile ducts into the small intestines to help digest fat.

Bilirubin—A reddish-yellow pigment formed from the destruction of red blood cells, and metabolized by the liver. Levels of bilirubin in the blood increase in patients with liver disease or blockage of the bile ducts.

Ultrasound—A non-invasive procedure based on changes in sound waves of a frequency that cannot be heard, but respond to changes in tissue composition. It requires no preparation and no radiation occurs; it has become the “gold standard” for diagnosis of stones in the gallbladder, but is less accurate in diagnosing stones in the bile ducts. Gallstones as small as 2 mm can be identified.

bread, tea or coffee, and only lean meat) in the evening. Two hours after the evening meal, six tablets containing the contrast medium, are taken, one a time. After that, no food or fluid is permitted until after the test.

Aftercare

No special care is required after the study.

Risks

There is a small chance of an allergic reaction to the contrast material. In addition, there is low radiation exposure. X rays are monitored and regulated to provide the minimum amount of radiation exposure needed to produce the image. Most experts feel that the risk is low compared with the benefits. Pregnant women and children are more sensitive to the risks of x rays, and the risk versus benefits should be discussed with the treating physician.

Normal results

The x ray will show normal structures for the age of the patient. The gallbladder should visualize, and be free of any solid structures, such as stones, polyps, etc.

Abnormal results

Abnormal results may show gallstones, tumors, or cholesterol polyps (a tumor growing from the lining that is usually noncancerous). Typically stones will “float” or move around as the patient changes position, whereas tumors will stay in the same place.

Resources

OTHER

“Gall Bladder Exam.” Harvard Medical School. <<http://www.bih.harvard.edu/radiology/Modalities/Xray/xraysSubdivsf/gallbl.html>> .

“Gallstones.” National Institutes of Health. <<http://www.niddk.nih.gov/health/digest/pubs/gallstns/gallstns.htm>> .

“Oral cholecystogram.” Healthanswers.com. <<http://www.healthanswers.com/database/ami/converted/003821.html>> .

Rosalyn Carson-DeWitt, MD

Gallium scan of the body

Definition

A gallium scan of the body is a nuclear medicine test that is conducted using a camera that detects gallium, a form of radionuclide, or radioactive chemical substance.

Purpose

Most gallium scans are ordered to detect cancerous tumors, infections, or areas of inflammation in the body. Gallium is known to accumulate in inflamed, infected, or cancerous tissues. The scans are used to determine whether a patient with an unexplained **fever** has an infection and the site of the infection, if present. Gallium scans also may be used to evaluate **cancer** following **chemotherapy** or **radiation therapy**.

Precautions

Children and women who are pregnant or breast-feeding are only given gallium scans if the potential diagnostic benefits will outweigh the risks.

Description

The patient will usually be asked to come to the testing facility 24–48 hours before the procedure to receive the injection of gallium. Sometimes, the injection will be given only four to six hours before the study or as long as 72 hours before the procedure. The timeframe is based on the area or organs of the body being studied.

For the study itself the patient lies very still for approximately 30–60 minutes. A camera is moved

across the patient's body to detect and capture images of concentrations of the gallium. The camera picks up signals from any accumulated areas of the radionuclide. In most cases, the patient is lying down throughout the procedure. Back (posterior) and front (anterior) views will usually be taken, and sometimes a side (lateral) view is used. The camera may occasionally touch the patient's skin, but will not cause any discomfort. A clicking noise may be heard throughout the procedure; this is only the sound of the scanner registering radiation.

Preparation

The intravenous injection of gallium is done in a separate appointment prior to the procedure. Generally, no special dietary requirements are necessary. Sometimes the physician will ask that the patient have light or clear meals within a day or less of the procedure. Many patients will be given **laxatives** or an enema prior to the scan to eliminate any residual gallium from the bowels.

Aftercare

There is generally no aftercare required following a gallium scan. However, women who are breastfeeding who have a scan will be cautioned against breastfeeding for four weeks following the exam.

Risks

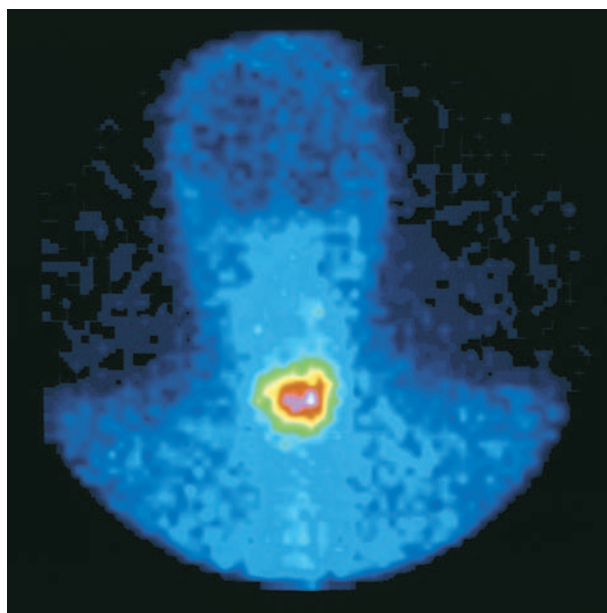
There is a minimal risk of exposure to radiation from the gallium injection, but the exposure from one gallium scan is generally less than exposure from x rays.

Normal results

A radiologist trained in nuclear medicine or a nuclear medicine specialist will interpret the exam results and compare them to other diagnostic tests. It is normal for gallium to accumulate in the liver, spleen, bones, breast tissue, and large bowel.

Abnormal results

An abnormal concentration of gallium in areas other than those where it normally concentrates may indicate the presence of disease. Concentrations may be due to inflammation, infection, or the presence of tumor tissue. Often, additional tests are required to determine if the tumors are malignant (cancerous) or benign.



Gallium scan highlighting the thyroid gland. (Photo Researchers. Reproduced by permission.)

KEY TERMS

Benign—Not cancerous. Benign tumors are not considered immediate threats, but may still require some form of treatment.

Gallium—A form of radionuclide that is used to help locate tumors and inflammation (specifically referred to as GA67 citrate).

Malignant—This term, usually used to describe a tumor, means cancerous, becoming worse and possibly growing.

Nuclear medicine—A subspecialty of radiology used to show the function and anatomy of body organs. Very small amounts of radioactive substances, or tracers, are detected with a special camera as they accumulate in certain organs and tissues.

Radionuclide—A chemical substance, called an isotope, that exhibits radioactivity. A gamma camera, used in nuclear medicine procedures, will pick up the radioactive signals as the substance gathers in an organ or tissue. They are sometimes referred to as tracers.

Even though gallium normally concentrates in organs such as the liver or spleen, abnormally high concentrations will suggest certain diseases and

conditions. For example, Hodgkin's or non-Hodgkin's lymphoma may be diagnosed or staged if there is abnormal gallium activity in the lymph nodes. After a patient receives cancer treatment, such as radiation therapy or chemotherapy, a gallium scan may help to find new or recurring tumors or to record regression of a treated tumor. Physicians can narrow causes of liver problems by noting abnormal gallium activity in the liver. Gallium scans also may be used to diagnose lung diseases or a disease called **sarcoidosis**, in the chest.

Resources

ORGANIZATIONS

American Cancer Society. 1599 Clifton Road NE, Atlanta, GA 30329. (404) 320-3333. <<http://www.cancer.org>>.

American College of Nuclear Medicine. PO Box 175, Landisville, PA 31906. (717) 898-6006.

American Liver Foundation. 1425 Pompton Avenue, Cedar Grove NJ 07009. (800) GO LIVER (465-4837). <<http://www.liverfoundation.org>>.

Society of Nuclear Medicine. 1850 Samuel Morse Drive, Reston, VA 10016. (703) 708-9000. <<http://www.snm.org>>.

OTHER

"A Patient's Guide to Nuclear Medicine." *University of Iowa Virtual Hospital*. July 2, 2001. <<http://www.vh.org/Patients/IHB/Rad/NucMed/PatGuideNucMed/PatGuideNucMed.html>>.

Teresa Odle

Gallstone removal

Definition

Also known as cholelithotomy, gallstone removal is the medical procedure that rids the gallbladder of calculus buildup.

Purpose

The gallbladder is not a vital organ. Its function is to store bile, concentrate it, and release it during digestion. Bile is supposed to retain all of its chemicals in solution, but commonly one of them crystallizes and forms sand, gravel, and finally stones.

The chemistry of **gallstones** is complex and interesting. Like too much sugar in solution, chemicals in bile will form crystals as the gallbladder draws water out of the bile. The solubility of these chemicals is

based on the concentration of three chemicals, not just one—bile acids, phospholipids, and cholesterol. If the chemicals are out of balance, one or the other will not remain in solution. Certain people, in particular the Pima tribe of Native Americans in Arizona, have a genetic predisposition to forming gallstones. Scandinavians also have a higher than average incidence of this disease. Dietary fat and cholesterol are also implicated in their formation. Overweight women in their middle years constitute the vast majority of patients with gallstones in every group.

As the bile crystals aggregate to form stones, they move about, eventually occluding the outlet and preventing the gallbladder from emptying. This creates symptoms. It also results in irritation, inflammation, and sometimes infection of the gallbladder. The pattern is usually one of intermittent obstruction due to stones moving in and out of the way. All the while the gallbladder is becoming more scarred. Sometimes infection fills it with pus—a serious complication.

On occasion a stone will travel down the cystic duct into the common bile duct and get stuck there. This will back bile up into the liver as well as the gallbladder. If the stone sticks at the Ampulla of Vater, the pancreas will also be plugged and will develop **pancreatitis**. These stones can cause a lot of trouble.

Bile is composed of several waste products of metabolism, all of which are supposed to remain in liquid form. The complex chemistry of the liver depends on many chemical processes, which depend in turn upon the chemicals in the diet and the genes that direct those processes. There are greater variations in the output of chemical waste products than there is allowance for their cohabitation in the bile. Incompatible mixes result in the formation of solids.

Gallstones will cause the sudden onset of **pain** in the upper abdomen. Pain will last for 30 minutes to several hours. Pain may move to the right shoulder blade. **Nausea** with or without **vomiting** may accompany the pain.

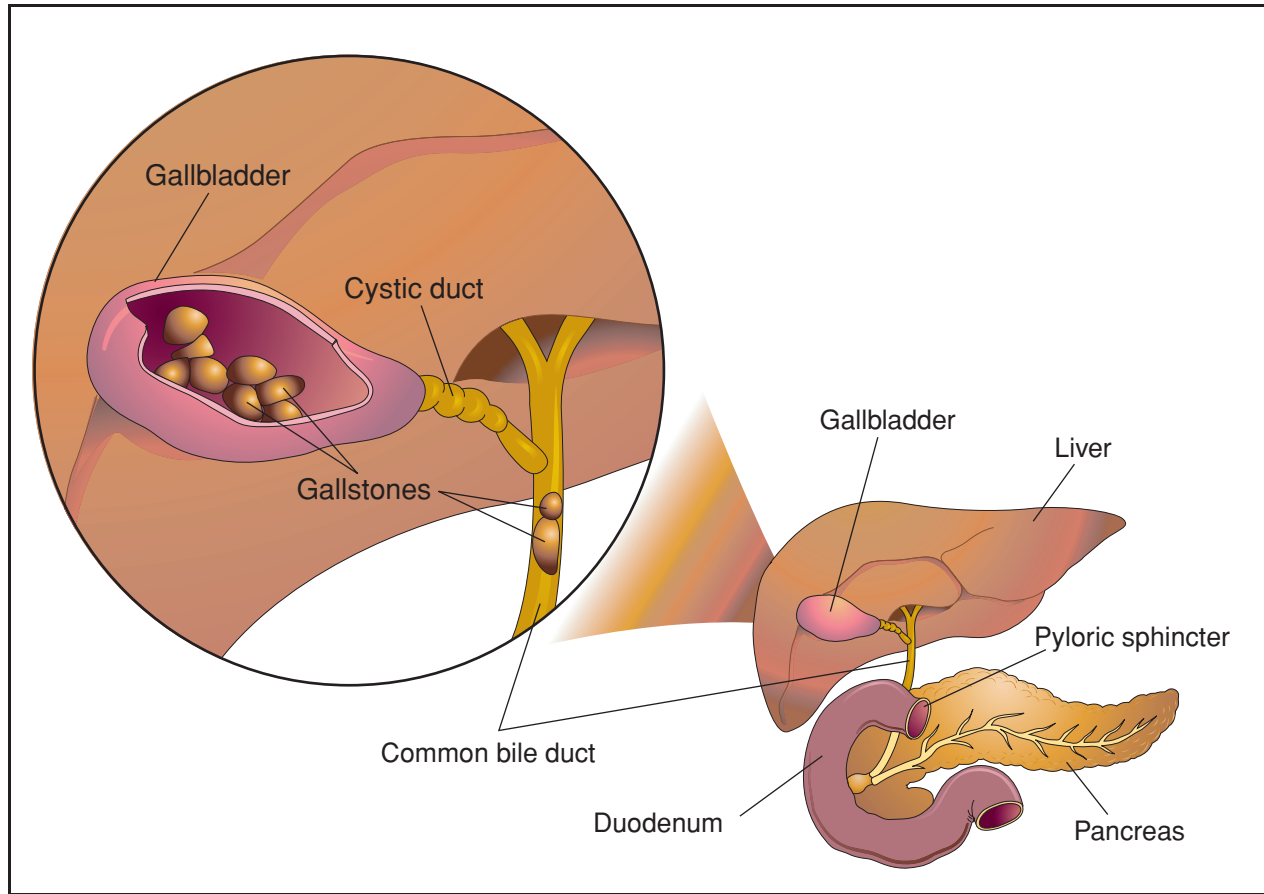
Precautions

Individuals suffering from sickle cell anemia, children, and patients with large stones may seek other treatments.

Description

Laparoscopic cholecystectomy

Surgery to remove the entire gallbladder with all its stones is usually the best treatment, provided the



Gallstone removal, also known as cholelithotomy, usually involves the surgical removal of the entire gallbladder, but in recent years the procedure done by laparoscopy has resulted in smaller surgical incisions and faster recovery time. (Illustration by Electronic Illustrators Group.)

patient is able to tolerate the procedure. Over the past decade, a new technique of removing the gallbladder using a laparoscope has resulted in quicker recovery and much smaller surgical incisions than the six-inch gash under the right ribs that used to be standard. Not everyone is a candidate for this approach.

If a stone is lodged in the bile ducts, additional surgery must be done to remove it. After surgery, the surgeon will ordinarily leave in a drain to collect bile until the system is healed. The drain can also be used to inject contrast material and take x rays during or after surgery.

Endoscopic retrograde cholangiopancreatography (ERCP)

A procedure called endoscopic retrograde cholangiopancreatography (ERCP) allows the removal of some bile duct stones through the mouth, throat, esophagus, stomach, duodenum, and biliary system

without the need for surgical incisions. ERCP can also be used to inject contrast agents into the biliary system, providing superbly detailed pictures.

Cholelithotomy

Rare circumstances require different techniques. Patients too ill for a complete **cholecystectomy** (removal of the gallbladder), sometimes only the stones are removed, a procedure called cholelithotomy. But that does not cure the problem. The liver will go on making faulty bile, and stones will reform, unless the composition of the bile is altered.

Ursodeoxycholic acid

For patients who cannot receive the laparoscopic procedure, there is also a nonsurgical treatment in which ursodeoxycholic acid is used to dissolve the gallstones. Extracorporeal shock-wave **lithotripsy** has also been successfully used to break up gallstones.

KEY TERMS

Cholecystectomy—Surgical removal of the gallbladder.

Cholelithotomy—Surgical incision into the gallbladder to remove stones.

Contrast agent—A substance that causes shadows on x rays (or other images of the body).

Endoscope—One of several instruments designed to enter body cavities. They combine viewing and operating capabilities.

Jaundice—A yellow color of the skin and eyes due to excess bile that is not removed by the liver.

Laparoscopy—Surgery through pencil-sized viewing instruments and tools so that incisions need be less than half an inch long.

During the procedure, high-amplitude sound waves target the stones, slowly breaking them up.

Preparation

There are a number of imaging studies that identify gallbladder disease, but most gallstones will not show up on conventional x rays. That requires contrast agents given by mouth that are excreted into the bile. Ultrasound is very useful and can be enhanced by doing it through an endoscope in the stomach. CT (**computed tomography scans**) and MRI (**magnetic resonance imaging**) scanning are not used routinely but are helpful in detecting common duct stones and complications.

Aftercare

Without a gallbladder, stones rarely reform. Patients who have continued symptoms after their gallbladder is removed may need an ERCP to detect residual stones or damage to the bile ducts caused by the stones before they were removed. Once in a while the Ampulla of Vater is too tight for bile to flow through and causes symptoms until it is opened up.

Resources

BOOKS

Bilhartz, Lyman E., and Jay D. Horton. "Gallstone Disease and Its Complications." In *Sleisenger & Fordtran's Gastrointestinal and Liver Disease*, edited by Mark Feldman, et al. Philadelphia: W. B. Saunders Co., 1998.

J. Ricker Polsdorfer, MD

Gallstones

Definition

A gallstone is a solid crystal deposit that forms in the gallbladder, which is a pear-shaped organ that stores bile salts until they are needed to help digest fatty foods. Gallstones can migrate to other parts of the digestive tract and cause severe **pain** with life-threatening complications.

Description

Gallstones vary in size and chemical structure. A gallstone may be as tiny as a grain of sand or as large as a golf ball. Eighty percent of gallstones are composed of cholesterol. They are formed when the liver produces more cholesterol than digestive juices can liquefy. The remaining 20% of gallstones are composed of calcium and an orange-yellow waste product called bilirubin. Bilirubin gives urine its characteristic color and sometimes causes **jaundice**.

Gallstones are the most common of all gallbladder problems. They are responsible for 90% of gallbladder and bile duct disease, and are the fifth most common reason for hospitalization of adults in the United States. Gallstones usually develop in adults between the ages of 20 and 50; about 20% of patients with gallstones are over 40. The risk of developing gallstones increases with age—at least 20% of people over 60 have a single large stone or as many as several thousand smaller ones. The gender ratio of gallstone patients changes with age. Young women are between two and six times more likely to develop gallstones than men in the same age group. In patients over 50, the condition affects men and women with equal frequency. Native Americans develop gallstones more often than any other segment of the population; Mexican-Americans have the second-highest incidence of this disease.

Definitions

Gallstones can cause several different disorders. Cholelithiasis is defined as the presence of gallstones within the gallbladder itself. Choledocholithiasis is the presence of gallstones within the common bile duct that leads into the first portion of the small intestine (the duodenum). The stones in the duct may have been formed inside it or carried there from the gallbladder. These gallstones prevent bile from flowing into the duodenum. Ten percent of patients with gallstones

have choledocholithiasis, which is sometimes called common-duct stones. Patients who don't develop infection usually recover completely from this disorder.

Cholecystitis is a disorder marked by inflammation of the gallbladder. It is usually caused by the passage of a stone from the gallbladder into the cystic duct, which is a tube that connects the gallbladder to the common bile duct. In 5–10% of cases, however, cholecystitis develops in the absence of gallstones. This form of the disorder is called acalculous cholecystitis. Cholecystitis causes painful enlargement of the gallbladder and is responsible for 10–25% of all gallbladder surgery. Chronic cholecystitis is most common in the elderly. The acute form is most likely to occur in middle-aged adults.

Cholesterolosis or cholesterol polyps is characterized by deposits of cholesterol crystals in the lining of the gallbladder. This condition may be caused by high levels of cholesterol or inadequate quantities of bile salts, and is usually treated by surgery.

Gallstone **ileus**, which results from a gallstone's blocking the entrance to the large intestine, is most common in elderly people. Surgery usually cures this condition.

Narrowing (stricture) of the common bile duct develops in as many as 5% of patients whose gallbladders have been surgically removed. This condition is characterized by inability to digest fatty foods and by abdominal pain, which sometimes occurs in spasms. Patients with stricture of the common bile duct are likely to recover after appropriate surgical treatment.

Causes and symptoms

Gallstones are caused by an alteration in the chemical composition of bile. Bile is a digestive fluid that helps the body absorb fat. Gallstones tend to run in families. In addition, high levels of estrogen, insulin, or cholesterol can increase a person's risk of developing them.

Pregnancy or the use of birth control pills can slow down gallbladder activity and increase the risk of gallstones. So can diabetes, **pancreatitis**, and **celiac disease**. Other factors influencing gallstone formation are:

- infection
- obesity
- intestinal disorders

- coronary artery disease or other recent illness
- multiple pregnancies
- a high-fat, low-fiber diet
- smoking
- heavy drinking
- rapid weight loss

Gallbladder attacks usually follow a meal of rich, high-fat foods. The attacks often occur in the middle of the night, sometimes waking the patient with intense pain that ends in a visit to the emergency room. The pain of a gallbladder attack begins in the abdomen and may radiate to the chest, back, or the area between the shoulders. Other symptoms of gallstones include:

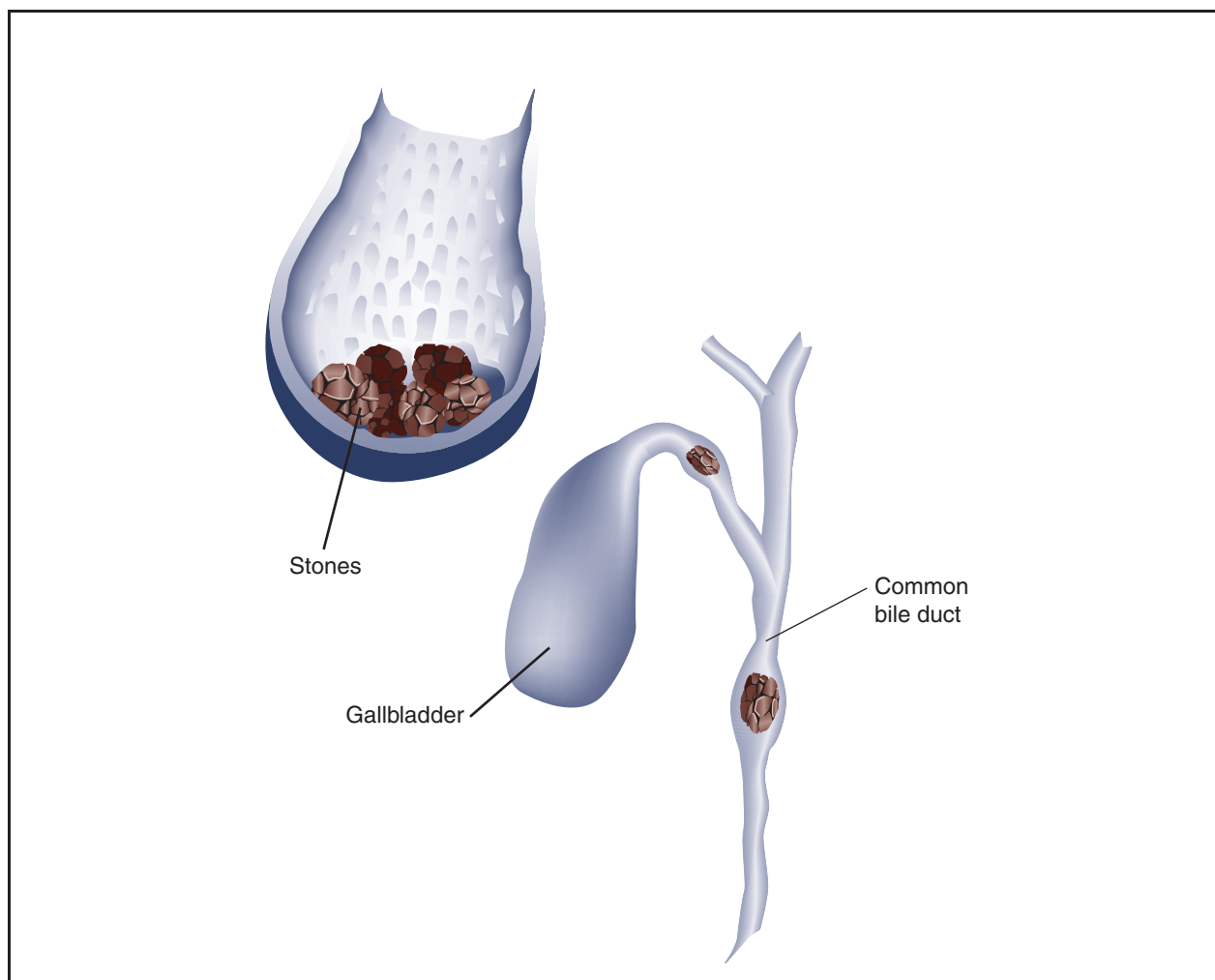
- inability to digest fatty foods
- low-grade **fever**
- chills and sweating
- nausea and **vomiting**
- indigestion
- gas
- belching.
- clay-colored bowel movements

Diagnosis

Gallstones may be diagnosed by a family doctor, a specialist in digestive problems (a gastroenterologist), or a specialist in internal medicine. The doctor will first examine the patient's skin for signs of jaundice and feel (palpate) the abdomen for soreness or swelling. After the basic **physical examination**, the doctor will order blood counts or blood chemistry tests to detect evidence of bile duct obstruction and to rule out other illnesses that cause fever and pain, including stomach ulcers, **appendicitis**, and heart attacks.

More sophisticated procedures used to diagnose gallstones include:

- **Ultrasound imaging.** Ultrasound has an accuracy rate of 96%.
- **Cholecystography** (cholecystogram, gallbladder series, gallbladder x ray). This type of study shows how the gallbladder contracts after the patient has eaten a high-fat meal.
- **Fluoroscopy.** This imaging technique allows the doctor to distinguish between jaundice caused by pancreatic **cancer** and jaundice caused by gallbladder or bile duct disorders.



Gallstones form in the gallbladder but can migrate to other parts of the body via the bile duct. (Illustration by Argosy Inc.)

- Endoscopy (ERCP). ERCP uses a special dye to outline the pancreatic and common bile ducts and locate the position of the gallstones.
- Radioisotopic scan. This technique reveals blockage of the cystic duct.

Treatment

Watchful waiting

One-third of all patients with gallstones never experience a second attack. For this reason many doctors advise watchful waiting after the first episode. Reducing the amount of fat in the diet or following a sensible plan of gradual weight loss may be the only treatments required for occasional mild attacks. A patient diagnosed with gallstones may be able to manage more troublesome episodes by:

- applying heat to the affected area
- resting and taking occasional sips of water
- using non-prescription forms of **acetaminophen** (Tylenol or Anacin-3)

A doctor should be notified if pain intensifies or lasts for more than three hours; if the patient's fever rises above 101 °F (38.3 °C); or if the skin or whites of the eyes turn yellow.

Surgery

Surgical removal of the gallbladder (**cholecystectomy**) is the most common conventional treatment for recurrent attacks. Laparoscopic surgery, the technique most widely used, is a safe, effective procedure that involves less pain and a shorter recovery period than traditional open surgery. In this technique, the

doctor makes a small cut (incision) in the patient's abdomen and removes the gallbladder through a long tube called a laparoscope.

Nonsurgical approaches

LITHOTRIPSY. Shock wave therapy (**lithotripsy**) uses high-frequency sound waves to break up the gallstones. The patient can then take bile salts to dissolve the fragments. Bile salt tablets are sometimes prescribed without lithotripsy to dissolve stones composed of cholesterol by raising the level of bile acids in the gallbladder. This approach requires long-term treatment, since it may take months or years for this method to dissolve a sizeable stone.

CONTACT DISSOLUTION. Contact dissolution can destroy gallstones in a matter of hours. This minimally invasive procedure involves using a tube (catheter) inserted into the abdomen to inject medication directly into the gallbladder.

Alternative treatment

Alternative therapies, like non-surgical treatments, may provide temporary relief of gallstone symptoms. Alternative approaches to the symptoms of gallbladder disorders include homeopathy, Chinese traditional herbal medicine, and **acupuncture**. Dietary changes may also help relieve the symptoms of gallstones. Since gallstones seem to develop more often in people who are obese, eating a balanced diet, exercising, and losing weight may help keep gallstones from forming.

Prognosis

Forty percent of all patients with gallstones have "silent gallstones" that produce no symptoms. Silent stones, discovered only when their presence is indicated by tests performed to diagnose other symptoms, do not require treatment.

Gallstone problems that require treatment can be surgically corrected. Although most patients recover, some develop infections that must be treated with **antibiotics**.

In rare instances, severe inflammation can cause the gallbladder to burst. The resulting infection can be fatal.

Prevention

The best way to prevent gallstones is to minimize risk factors. In addition, a 1998 study suggests that vigorous **exercise** may lower a man's risk of developing gallstones by as much as 28%. The researchers

KEY TERMS

Acalculous cholecystitis—Inflammation of the gallbladder that occurs without the presence of gallstones.

Bilirubin—A reddish-yellow waste product produced by the liver that colors urine and is involved in the formation of some gallstones.

Celiac disease—Inability to digest wheat protein (gluten), which causes weight loss, lack of energy, and pale, foul-smelling stools.

Cholecystectomy—Surgical removal of the gallbladder.

Cholecystitis—Inflammation of the gallbladder.

Choledocholithiasis—The presence of gallstones within the common bile duct.

Cholelithiasis—The presence of gallstones within the gallbladder.

Cholesterolosis—Cholesterol crystals or deposits in the lining of the gallbladder.

Common bile duct—The passage through which bile travels from the cystic duct to the small intestine.

Gallstone ileus—Obstruction of the large intestine caused by a gallstone that has blocked the intestinal opening.

Lithotripsy—A nonsurgical technique for removing gallstones by breaking them apart with high-frequency sound waves.

have not yet determined whether physical activity benefits women to the same extent.

Resources

ORGANIZATIONS

National Digestive Diseases Clearinghouse (NDDIC).
2 Information Way.

National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK). Building 31, Room 9A04, 31 Center Drive, MSC 2560, Bethesda, MD 20879-2560. (301) 496-3583. <<http://www.niddk.nih.gov>> .

Maureen Haggerty

Gamete intrafallopian transfer see **Infertility therapies**

Gamma-glutamyl transferase test see **Liver function tests**

Gammaglobulin

Definition

Gammaglobulin is a type of protein found in the blood. When gammaglobulins are extracted from the blood of many people and combined, they can be used to prevent or treat infections.

Purpose

This medicine is used to treat or prevent diseases that occur when the body's own immune system is not effective against the disease. When disease-causing agents enter the body, they normally trigger the production of antibodies, proteins that circulate in the blood and help fight the disease. Gammaglobulin contains some of these antibodies. When gammaglobulins are taken from the blood of people who have recovered from diseases such as **chickenpox** or hepatitis, they can be given to other people to make them temporarily immune to those diseases. With hepatitis, for example, this is done when someone who has not been vaccinated against hepatitis is exposed to the disease.

Description

Gammaglobulin, also known as immunoglobulin, immune serum globulin or serum therapy, is injected either into a vein or into a muscle. When injected into a vein, it produces results more quickly than when injected into a muscle.

Recommended dosage

Doses are different for different people and depend on the person's body weight and the condition for which he or she is being treated.

Precautions

Anyone who has had unusual reactions to gammaglobulin in the past should let his or her physician know before taking the drugs again. The physician should also be told about any **allergies** to foods, dyes, preservatives, or other substances.

People who have certain medical conditions may have problems if they take gammaglobulins. For example:

- Gammaglobulins may worsen heart problems or deficiencies of immunoglobulin A (IgA, a type of antibody)

KEY TERMS

Hepatitis—Inflammation of the liver caused by a virus, chemical or drugs. There are several different types of hepatitis, including the most common forms: hepatitis A, hepatitis B, and hepatitis C.

Immune system—The body's natural defenses against disease and infection.

Inflammation—Pain, redness, swelling, and heat that usually develop in response to injury or illness.

- Certain patients with low levels of gammaglobulins in the blood (conditions called agammaglobulinemia and hypogammaglobulinemia) may be more likely to have side effects when they take gammaglobulin.

Side effects

Minor side effects such as **headache**, backache, joint or muscle **pain**, and a general feeling of illness usually go away as the body adjusts to this medicine. These problems do not need medical attention unless they continue.

Other side effects, such as breathing problems or a fast or pounding heartbeat, should be brought to a physician's attention as soon as possible.

Anyone who shows the following signs of overdose should check with a physician immediately:

- unusual tiredness or weakness
- dizziness
- nausea
- vomiting
- fever
- chills
- tightness in the chest
- red face
- sweating

Interactions

Anyone who takes gammaglobulin should let the physician know all other medicines he or she is taking and should ask whether interactions with gammaglobulin could interfere with treatment.

Nancy Ross-Flanigan

Ganglion

Definition

A ganglion is a small, usually hard bump above a tendon or in the capsule that encloses a joint. A ganglion is also called a synovial **hernia** or synovial cyst.

Description

A ganglion is a non-cancerous cyst filled with a thick, jelly-like fluid. Ganglions can develop on or beneath the surface of the skin and usually occur between the ages of 20 and 40.

Most ganglions develop on the hand or wrist. This condition is common in people who bowl or who play handball, raquetball, squash, or tennis. Runners and athletes who jump, ski, or play contact sports often develop foot ganglions.

Causes and symptoms

Mild **sprains** or other repeated injuries can irritate and tear the thin membrane covering a tendon, causing fluid to leak into a sac that swells and forms a ganglion.

Ganglions are usually painless, but range of motion may be impaired. Flexing or bending the affected area can cause discomfort, as can continuing to perform the activity that caused the condition.

Cysts on the surface of the skin usually develop slowly but may result from injury or severe strain. An internal ganglion can cause soreness or a dull, aching sensation, but the mass cannot always be felt. Symptoms sometimes become evident only when the cyst causes pressure on a nerve or outgrows the membrane surrounding it.

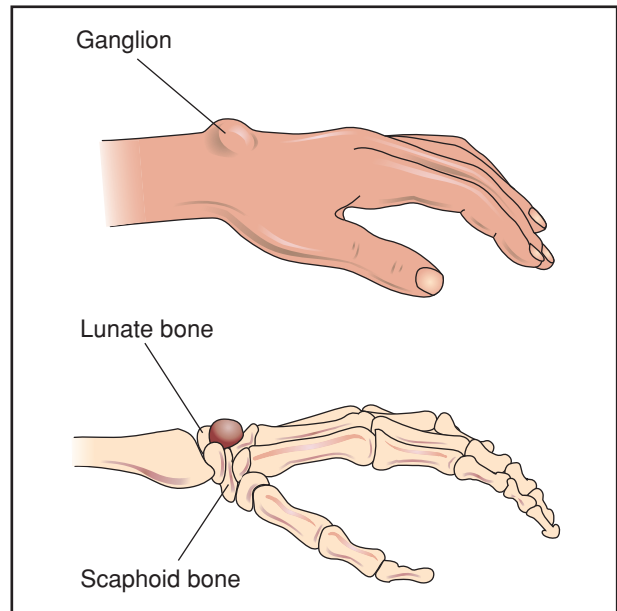
Diagnosis

Diagnosis is usually made through **physical examination** as well as such imaging studies as x ray, ultrasound, and **magnetic resonance imaging** (MRI). Fluid may be withdrawn from the cyst and evaluated.

Treatment

Some ganglions disappear without treatment, and some reappear despite treatment.

Acetaminophen (Tylenol) or other over-the-counter **analgesics** can be used to control mild **pain**. Steroids or local anesthetics may be injected into cysts that cause severe pain or other troublesome



A ganglion is a non-cancerous cyst filled with a thick, jelly-like fluid. Ganglions can develop on or beneath the surface of the skin, most likely on the hand or wrist, although runners and skiers often develop them on the foot. (Illustration by Electronic Illustrators Group.)

symptoms. Surgery performed in a hospital operating room or an outpatient facility, is the only treatment guaranteed to remove a ganglion. The condition can recur if the entire cyst is not removed.

A doctor should be notified if the surgical site drains, bleeds, or becomes

- inflamed
- painful
- swollen or if the patient feels ill or develops:
- head or muscle aches
- dizziness
- fever following surgery

The patient may bathe or shower as usual, but should keep the surgical site dry and covered with a bandage for two or three days after the operation. Patients may resume normal activities as soon as they feel comfortable doing so.

Prognosis

Possible complications include excessive post-operative bleeding and infection of the surgical site. Calcification, or hardening, of the ganglion is rare.

Prevention

Exercises that increase muscle strength and flexibility can prevent ganglions. Warming and cooling down before and after workouts may also decrease the rate of developing ganglions.

Resources

OTHER

“Foot Ganglion.” ThriveOnline. May 25, 1998. <<http://thriveonline.oxygen.com>>.

“Hand or Wrist Ganglion.” ThriveOnline. May 25, 1998. <<http://thriveonline.oxygen.com>>.

Maureen Haggerty

Gangrene

Definition

Gangrene is the term used to describe the decay or death of an organ or tissue caused by a lack of blood supply. It is a complication resulting from infectious or inflammatory processes, injury, or degenerative changes associated with chronic diseases, such as **diabetes mellitus**.

Description

Gangrene may be caused by a variety of chronic diseases and post-traumatic, post-surgical, and spontaneous causes. There are three major types of gangrene: dry, moist, and gas (a type of moist gangrene).

Dry gangrene is a condition that results when one or more arteries become obstructed. In this type of gangrene, the tissue slowly dies, due to receiving little or no blood supply, but does not become infected. The affected area becomes cold and black, begins to dry out and wither, and eventually drops off over a period of weeks or months. Dry gangrene is most common in persons with advanced blockages of the arteries (arteriosclerosis) resulting from diabetes.

Moist gangrene may occur in the toes, feet, or legs after a crushing injury or as a result of some other factor that causes blood flow to the area to suddenly stop. When blood flow ceases, bacteria begin to invade the muscle and thrive, multiplying quickly without interference from the body's immune system.

Gas gangrene, also called myonecrosis, is a type of moist gangrene that is commonly caused by bacterial

infection with *Clostridium welchii*, *Cl. perfringes*, *Cl. septicum*, *Cl. novyi*, *Cl. histolyticum*, *Cl. sporogenes*, or other species that are capable of thriving under conditions where there is little oxygen (anaerobic). Once present in tissue, these bacteria produce gasses and poisonous toxins as they grow. Normally inhabiting the gastrointestinal, respiratory, and female genital tract, they often infect thigh **amputation wounds**, especially in those individuals who have lost control of their bowel functions (incontinence). Gangrene, incontinence, and debility often are combined in patients with diabetes, and it is in the amputation stump of diabetic patients that gas gangrene is often found to occur.

Other causative organisms for moist gangrene include various bacterial strains, including *Streptococcus* and *Staphylococcus*. A serious, but rare form of infection with Group A *Streptococcus* can impede blood flow and, if untreated, can progress to synergistic gangrene, more commonly called necrotizing fasciitis, or infection of the skin and tissues directly beneath the skin.

Chronic diseases, such as diabetes mellitus, arteriosclerosis, or diseases affecting the blood vessels, such as **Buerger's disease** or **Raynaud's disease**, can cause gangrene. Post-traumatic causes of gangrene include compound **fractures**, **burns**, and injections given under the skin or in a muscle. Gangrene may occur following surgery, particularly in individuals with diabetes mellitus or other long-term (chronic) disease. In addition, gas gangrene can be also be a complication of dry gangrene or occur spontaneously in association with an underlying **cancer**.

In the United States, approximately 50% of moist gangrene cases are the result of a severe traumatic injury, and 40% occur following surgery. Car and industrial accidents, crush injuries, and gunshot wounds are the most common traumatic causes. Because of prompt surgical management of wounds with the removal of dead tissue, the incidence of gangrene from trauma has significantly diminished. Surgeries involving the bile ducts or intestine are the most frequent procedures causing gangrene. Approximately two-thirds of cases affect the extremities, and the remaining one-third involve the abdominal wall.

Symptoms

Areas of either dry or moist gangrene are initially characterized by a red line on the skin that marks the border of the affected tissues. As tissues begin to die, dry gangrene may cause some **pain** in the early stages

or may go unnoticed, especially in the elderly or in those individuals with diminished sensation to the affected area. Initially, the area becomes cold, numb, and pale before later changing in color to brown, then black. This dead tissue will gradually separate from the healthy tissue and fall off.

Moist gangrene and gas gangrene are distinctly different. Gas gangrene does not involve the skin as much, but usually only the muscle. In moist or gas gangrene, there is a sensation of heaviness in the affected region that is followed by severe pain. The pain is caused by swelling resulting from fluid or gas accumulation in the tissues. This pain peaks, on average, between one to four days following the injury, with a range of eight hours to several weeks. The swollen skin may initially be blistered, red, and warm to the touch before progressing to a bronze, brown, or black color. In approximately 80% of cases, the affected and surrounding tissues may produce crackling sounds (crepitus), as a result of gas bubbles accumulating under the skin. The gas may be felt beneath the skin (palpable). In wet gangrene, the pus is foul-smelling, while in gas gangrene, there is no true pus, just an almost “sweet” smelling watery discharge.

Fever, rapid heart rate, rapid breathing, altered mental state, loss of appetite, **diarrhea**, **vomiting**, and vascular collapse may also occur if the bacterial toxins are allowed to spread in the bloodstream. Gas gangrene can be a life-threatening condition and should receive prompt medical attention

Diagnosis

A diagnosis of gangrene will be based on a combination of the patient history, a **physical examination**, and the results of blood and other laboratory tests. A physician will look for a history of recent trauma, surgery, cancer, or chronic disease. Blood tests will be used to determine whether infection is present and determine the extent to which an infection has spread.

A sample of drainage from a wound, or obtained through surgical exploration, may be cultured with oxygen (aerobic) and without oxygen (anaerobic) to identify the microorganism causing the infection and to aid in determining which antibiotic will be most effective. The sample obtained from a person with gangrene will contain few, if any, white blood cells and, when stained (with Gram stain) and examined under the microscope, will show the presence of purple (Gram positive), rod-shaped bacteria.

X-ray studies and more sophisticated imaging techniques, such as **computed tomography scans** (CT)



A close-up of gangrene in the toes of a diabetic patient. (Photo Researchers, Inc. Reproduced by permission.)

or **magnetic resonance imaging** (MRI), may be helpful in making a diagnosis since gas accumulation and muscle death (myonecrosis) may be visible. These techniques, however, are not sufficient alone to provide an accurate diagnosis of gangrene.

Precise diagnosis of gas gangrene often requires surgical exploration of the wound. During such a procedure, the exposed muscle may appear pale, beefy-red, or in the most advanced stages, black. If infected, the muscle will fail to contract with stimulation, and the cut surface will not bleed.

Treatment

Gas gangrene is a medical emergency because of the threat of the infection rapidly spreading via the bloodstream and infecting vital organs. It requires immediate surgery and administration of **antibiotics**.

Areas of dry gangrene that remain free from infection (aseptic) in the extremities are most often left to wither and fall off. Treatments applied to the wound externally (topically) are generally not effective without adequate blood supply to support wound healing. Assessment by a vascular surgeon, along with x rays to determine blood supply and circulation to the affected area, can help determine whether surgical intervention would be beneficial.

Once the causative organism has been identified, moist gangrene requires the prompt initiation of intravenous, intramuscular, and/or topical broad-spectrum antibiotic therapy. In addition, the infected tissue must be removed surgically (**debridement**), and amputation of the affected extremity may be necessary. Pain medications (**analgesics**) are prescribed to control discomfort. Intravenous fluids and, occasionally, blood transfusions are indicated to counteract

shock and replenish red blood cells and electrolytes. Adequate hydration and **nutrition** are vital to wound healing.

Although still controversial, some cases of gangrene are treated by administering oxygen under pressure greater than that of the atmosphere (hyperbaric) to the patient in a specially designed chamber. The theory behind using hyperbaric oxygen is that more oxygen will become dissolved in the patient's bloodstream, and therefore, more oxygen will be delivered to the gangrenous areas. By providing optimal oxygenation, the body's ability to fight off the bacterial infection are believed to be improved, and there is a direct toxic effect on the bacteria that thrive in an oxygen-free environment. Some studies have shown that the use of hyperbaric oxygen produces marked pain relief, reduces the number of amputations required, and reduces the extent of surgical debridement required. Patients receiving hyperbaric oxygen treatments must be monitored closely for evidence of oxygen toxicity. Symptoms of this toxicity include slow heart rate, profuse sweating, ringing in the ears, **shortness of breath, nausea and vomiting**, twitching of the lips/cheeks/eyelids/nose, and convulsions.

The emotional needs of the patient must also be met. The individual with gangrene should be offered moral support, along with an opportunity to share questions and concerns about changes in body image. In addition, particularly in cases where amputation was required, physical, vocational, and **rehabilitation** therapy will also be required.

Prognosis

Except in cases where the infection has been allowed to spread through the blood stream, prognosis is generally favorable. Anaerobic wound infection can progress quickly from initial injury to gas gangrene within one to two days, and the spread of the infection in the blood stream is associated with a 20–25% mortality rate. If recognized and treated early, however, approximately 80% of those with gas gangrene survive, and only 15–20% require any form of amputation. Unfortunately, the individual with dry gangrene most often has multiple other health problems that complicate recovery, and it is usually those other system failures that can prove fatal.

Prevention

Patients with diabetes or severe arteriosclerosis should take particular care of their hands and feet because of the risk of infection associated with even a

KEY TERMS

Aerobic—Organism that grows and thrives only in environments containing oxygen.

Anaerobic—Organism that grows and thrives in an oxygen-free environment.

Arteriosclerosis—Build-up of fatty plaques within the arteries that can lead to the obstruction of blood flow.

Aseptic—Without contamination with bacteria or other microorganisms.

Crepitus—A crackling sound.

Gram stain—A staining procedure used to visualize and classify bacteria. The Gram stain procedure allows the identification of purple (Gram positive) organisms and red (Gram negative) organisms.

Hyperbaric oxygen—Medical treatment in which oxygen is administered in specially designed chambers, under pressures greater than that of the atmosphere, in order to treat specific medical conditions.

Incontinence—A condition characterized by the inability to control urination or bowel functions.

Myonecrosis—The destruction or death of muscle tissue.

Sepsis—The spreading of an infection in the bloodstream.

Thrombosis—The formation of a blood clot in a vein or artery that may obstruct local blood flow or may dislodge, travel downstream, and obstruct blood flow at a remote location.

minor injury. Education about proper **foot care** is vital. Diminished blood flow as a result of narrowed vessels will not lessen the body's defenses against invading bacteria. Measures taken towards the reestablishment of circulation are recommended whenever possible. Any abrasion, break in the skin, or infection tissue should be cared for immediately. Any dying or infected skin must be removed promptly to prevent the spread of bacteria.

Penetrating abdominal wounds should be surgically explored and drained, any tears in the intestinal walls closed, and antibiotic treatment begun early. Patients undergoing elective intestinal surgery should receive preventive antibiotic therapy. Use of antibiotics prior to and directly following surgery has been shown to significantly reduce the rate of infection from 20–30% to 4–8%.

Resources

BOOKS

Berktow, Robert, editor. *The Merck Manual of Diagnosis and Therapy*. 17th ed. Rahway, NJ: Merck Research Laboratories, 1997.

PERIODICALS

Basoglu, M., et al. "Fournier's Gangrene: Review of Fifteen Cases." *American Surgeon* November 1997: 1019-1021.

Kathleen D. Wright, RN

Gas embolism

Definition

Gas **embolism**, also called air embolism, is the presence of gas bubbles in the bloodstream that obstruct circulation.

Description

Gas embolism may occur with decompression from increased pressure; it typically occurs in ascending divers who have been breathing compressed air. If a diver does not fully exhale upon ascent, the air in the lungs expands as the pressure decreases, overinflating the lungs and forcing bubbles of gas (emboli) into the bloodstream. When gas emboli reach the arteries to the brain, the blood blockage causes unconsciousness. Gas embolism is second only to drowning as a cause of **death** among divers.

Gas embolism may also result from trauma or medical procedures such as catheterization and open heart surgery that allow air into the circulatory system.

Causes and symptoms

Gas embolism occurs independent of diving depth; it may occur in as little as 6 ft of water. It is frequently caused by a diver holding his breath during ascent. It may also result from an airway obstruction or other condition that prevents a diver from fully exhaling.

The primary sign of gas embolism is immediate loss of consciousness; it may or may not be accompanied by convulsions.

Diagnosis

Any unconscious diver should be assumed to be the victim of gas embolism, regardless of whether

KEY TERMS

Compressed air—Air that is held under pressure in a tank to be breathed underwater by divers. A tank of compressed air is part of a diver's scuba (self-contained underwater breathing apparatus) gear.

Compression—An increase in pressure from the surrounding water that occurs with increasing diving depth.

Decompression—A decrease in pressure from the surrounding water that occurs with decreasing diving depth.

Emboli—Plural of embolus. An embolus is something that blocks the blood flow in a blood vessel. It may be a gas bubble, a blood clot, a fat globule, a mass of bacteria, or other foreign body. It usually forms somewhere else and travels through the circulatory system until it gets stuck.

Hyperbaric chamber—A sealed compartment in which patients are exposed to controlled pressures up to three times normal atmospheric pressure. Hyperbaric treatment may be used to regulate blood gases, reduce gas emboli, and provide higher levels of oxygen more quickly in cases of severe gas poisoning.

Recompression—Restoring the elevated pressure of the diving environment to treat gas embolism by decreasing bubble size.

consciousness was lost during or promptly after ascent. A doctor may also find pockets of air in the chest around the lungs and sometimes a collapsed lung from overinflation and rupture. Coughing up blood or a bloody froth around the mouth are visible signs of lung injury.

Treatment

Prompt **recompression treatment** in a hyperbaric (high-pressure) chamber is necessary to deflate the gas bubbles in the bloodstream, dissolve the gases into the blood, and restore adequate oxygenated blood flow to the brain and other organs. Recompression by returning the diver to deeper water will not work, and should not be attempted. The patient should be kept lying down and given oxygen while being transported for recompression treatment.

Before the diver receives recompression treatment, other lifesaving efforts may be necessary. If the diver isn't breathing, artificial respiration (also called mouth-to-mouth resuscitation or rescue

breathing) should be administered. In the absence of a pulse, **cardiopulmonary resuscitation (CPR)** must be performed.

Prognosis

The prognosis is dependent upon the promptness of recompression treatment and the extent of the damage caused by oxygen deprivation.

Prevention

All divers should receive adequate training in the use of compressed air and a complete evaluation of fitness for diving. People with a medical history of lung cysts or spontaneous collapsed lung (**pneumothorax**), and those with active **asthma** or other lung disease must not dive, for they would be at extreme risk for gas embolism. Patients with conditions such as **alcoholism** and drug **abuse** are also discouraged from diving. Individuals with certain other medical conditions such as diabetes may be able to dive safely with careful training and supervision.

Resources

ORGANIZATIONS

American College of Hyperbaric Medicine. PO Box 25914-130, Houston, Texas 77265. (713) 528-0657. <<http://www.hyperbaricmedicine.org>> .
 Divers Alert Network. The Peter B. Bennett Center, 6 West Colony Place, Durham, NC 27705. (800) 446-2671. <<http://www.diversalertnetwork.org>> .
 Undersea and Hyperbaric Medical Society. 10531 Metropolitan Ave., Kensington, MD 20895. (301) 942-2980. <<http://www.uhms.org>> .

Bethany Thivierge

Gas gangrene see **Gangrene**

Gastrectomy

Definition

Gastrectomy is the surgical removal of all or part of the stomach.

Purpose

Gastrectomy is performed for several reasons, most commonly to remove a malignant tumor or to cure a perforated or bleeding stomach ulcer.

Description

Gastrectomy for cancer

Removal of the tumor, often with removal of surrounding lymph nodes, is the only curative treatment for various forms of gastric (stomach) **cancer**. For many patients, this entails removing not just the tumor but part of the stomach as well. The extent to which lymph nodes should also be removed is a subject of some debate, but some studies show additional survival benefit associated with removal of a greater number of lymph nodes.

Gastrectomy, either total or subtotal (also called partial), is the treatment of choice for gastric adenocarcinomas, primary gastric lymphomas (originating in the stomach), and the rare leiomyosarcomas (also called gastric **sarcomas**). Adenocarcinomas are by far the most common form of **stomach cancer** and are less curable than the relatively uncommon lymphomas, for which gastrectomy offers good odds for survival.

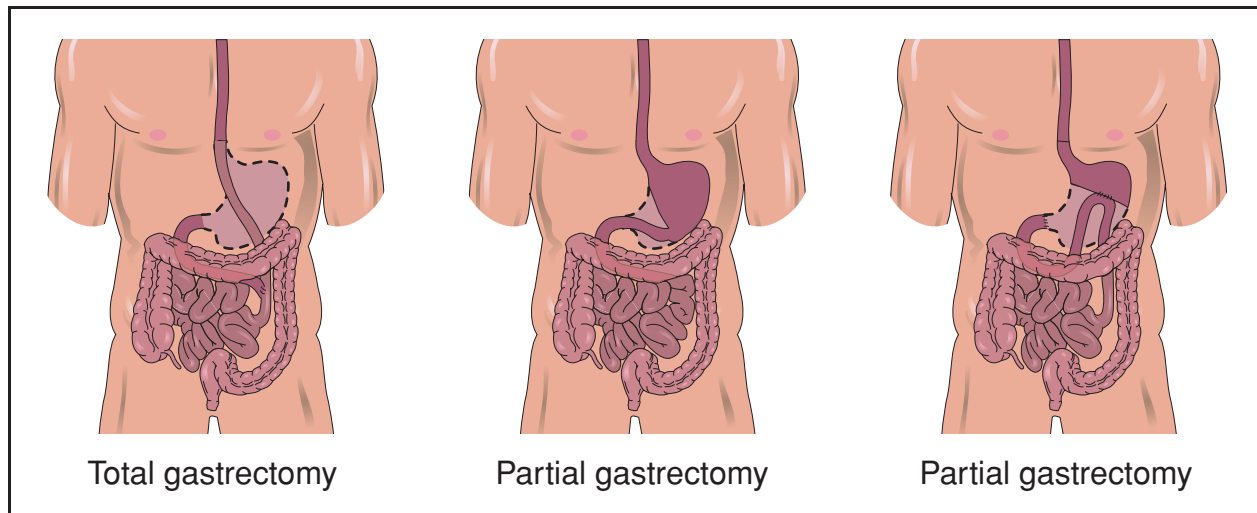
After gastrectomy, the surgeon may “reconstruct” the altered portions of the digestive tract so that it continues to function. Several different surgical techniques are used, but, generally speaking, the surgeon attaches any remaining portion of the stomach to the small intestine.

Gastrectomy for gastric cancer is almost always done by the traditional “open” surgery technique, which requires a wide incision to open the abdomen. However, some surgeons use a laparoscopic technique that requires only a small incision. The laparoscope is connected to a tiny video camera that projects a picture of the abdominal contents onto a monitor for the surgeon’s viewing. The stomach is operated on through this incision.

The potential benefits of laparoscopic surgery include less postoperative **pain**, decreased hospitalization, and earlier return to normal activities. The use of laparoscopic gastrectomy is limited, however. Only patients with early stage gastric cancers or those whose surgery is only intended for palliation—pain and symptomatic relief rather than cure—should be considered for this minimally invasive technique. It can only be performed by surgeons experienced in this type of surgery.

Gastrectomy for ulcers

Gastrectomy is also occasionally used in the treatment of severe peptic ulcer disease or its complications. While the vast majority of peptic ulcers (gastric ulcers in the stomach or duodenal ulcers in the duodenum) are managed with medication, partial gastrectomy is



Gastrectomy, the surgical removal of all or part of the stomach, is performed primarily to remove a malignant tumor or to cure a bleeding stomach ulcer. Following the gastrectomy, the surgeon may reconstruct the altered portions of the digestive tract so that it continues to function. (Illustration by Electronic Illustrators Group.)

sometimes required for peptic ulcer patients who have complications. These include patients who do not respond satisfactorily to medical therapy, those who develop a bleeding or perforated ulcer, and those who develop pyloric obstruction, a blockage to the exit from the stomach.

The surgical procedure for severe ulcer disease is also called an antrectomy, a limited form of gastrectomy in which the antrum, a portion of the stomach, is removed. For duodenal ulcers, antrectomy may be combined with other surgical procedures that are aimed at reducing the secretion of gastric acid, which is associated with ulcer formation. This additional surgery is commonly a **vagotomy**, surgery on the vagus nerve that disables the acid-producing portion of the stomach.

Preparation

Before undergoing gastrectomy, patients may need a variety of tests, such as x rays, **computed tomography scans** (CT scans), ultrasonography, or endoscopic biopsies (microscopic examination of tissue), to assure the diagnosis and localize the tumor or ulcer. **Laparoscopy** may be done to diagnose a malignancy or to determine the extent of a tumor that is already diagnosed. When a tumor is strongly suspected, laparoscopy is often performed immediately before the surgery to remove the tumor; this avoids the need to anesthetize the patient twice and sometimes avoids the need for surgery altogether if the tumor found on laparoscopy is deemed inoperable.

Aftercare

It is important to follow any instructions that have been given for postoperative care. Major surgery usually requires a recuperation time of several weeks.

Risks

Surgery for peptic ulcer is effective, but it may result in a variety of postoperative complications. After gastrectomy, as many as 30% of patients have significant symptoms. An operation called highly selective vagotomy is now preferred for ulcer management, and is safer than gastrectomy.

After a gastrectomy, several abnormalities may develop that produce symptoms related to food intake. This happens largely because the stomach, which serves as a food reservoir, has been reduced in its capacity by the surgery. Other surgical procedures that often accompany gastrectomy for ulcer disease can also contribute to later symptoms: vagotomy, which lessens acid production and slows stomach emptying, and **pyloroplasty**, which enlarges the opening between the stomach and small intestine to facilitate emptying of the stomach.

Some patients experience light-headedness, heart **palpitations** or racing heart, sweating, and **nausea and vomiting** after a meal. These may be symptoms of “dumping syndrome,” as food is rapidly “dumped” into the small intestine from the stomach. This is treated by adjusting the diet and pattern of eating, for example, eating smaller, more frequent meals, and limiting liquids.

KEY TERMS

Antrectomy—A surgical procedure for ulcer disease in which the antrum, a portion of the stomach, is removed.

Laparoscopy—The examination of the inside of the abdomen through a lighted tube, sometimes accompanied by surgery.

Patients who have abdominal bloating and pain after eating, frequently followed by **nausea** and **vomiting**, may have what is called the afferent loop syndrome. This is treated by surgical correction. Patients who have early satiety (feeling of fullness after eating), abdominal discomfort, and vomiting may have bile reflux **gastritis** (also called bilious vomiting), which is also surgically correctable. Many patients also experience weight loss.

Reactive **hypoglycemia** is a condition that results when blood sugar becomes too high after a meal, stimulating the release of insulin, about two hours after eating. A high-protein diet and smaller meals are advised.

Ulcers recur in a small percentage of patients after surgery for peptic ulcer, usually in the first few years. Further surgery is usually necessary.

Vitamin and mineral supplementation is necessary after gastrectomy to correct certain deficiencies, especially vitamin B₁₂, iron, and folate. Vitamin D and calcium are also needed to prevent and treat the bone problems that often occur. These include softening and bending of the bones, which can produce pain, and **osteoporosis**, a loss of bone mass. According to one study, the risk for spinal **fractures** may be as high as 50% after gastrectomy.

Depending on the extent of surgery, the risk for post-operative **death** after gastrectomy for gastric cancer has been reported as 1–3% and the risk of non-fatal complications as 9–18%.

Normal results

Overall survival after gastrectomy for gastric cancer varies greatly by the stage of disease at the time of surgery. For early gastric cancer, the five-year survival rate is up to 80–90%; for late-stage disease, the prognosis is bad. For gastric adenocarcinomas that are amenable to gastrectomy, the five-year survival rate is 10–30%, depending on the location of the tumor. The prognosis for patients with gastric lymphoma is better, with five-year survival rates reported at 40–60%.

Most studies have shown that patients can have an acceptable quality of life after gastrectomy for a potentially curable gastric cancer. Many patients will maintain a healthy appetite and eat a normal diet. Others may lose weight and not enjoy meals as much. Some studies show that patients who have total gastrectomies have more disease-related or treatment-related symptoms after surgery and poorer physical function than patients who have subtotal gastrectomies. There does not appear to be much difference, however, in emotional status or social activity level between patients who have undergone total versus subtotal gastrectomies.

Resources

BOOKS

Feldman, Mark., et al., editors. "Stomach and Duodenum: Complications of Surgery for Peptic Ulcer Disease." In *Steisenger & Fordtran's Gastrointestinal and Liver Disease*. Philadelphia: W. B. Saunders Co., 1998.

Caroline A. Helwick

Gastric acid determination

Definition

Gastric acid determination, also known as stomach acid determination, gastric analysis, or basal gastric secretion, is a procedure to evaluate gastric (stomach) function. The test specifically determines the presence of gastric acid, as well as the amount of gastric acid secreted. It is often done in conjunction with the gastric acid stimulation test, a procedure that measures gastric acid output after injection of a drug to stimulate gastric acid secretion.

Purpose

The purpose of the gastric acid determination is to evaluate gastric function by measuring the amount of acid as suctioned directly from the stomach. The complete gastric acid determination includes the basal gastric secretion test, which measures acid secretion while the patient is in a **fasting** state (nothing to eat or drink), followed by the gastric acid stimulation test, which measures the secretion of gastric acid for one hour after injection of pentagastrin or a similar drug that stimulates gastric acid output. The Gastric acid stimulation test is done when the basal secretion test suggests abnormalities in gastric secretion. It is normally performed immediately afterward.

The basal gastric secretion test is indicated for patients with obscure gastric **pain**, loss of appetite, and weight loss. It is also utilized for suspected peptic (related to the stomach) ulcer, severe stomach inflammation (**gastritis**), and Zollinger-Ellison (Z-E) syndrome (a condition in which a pancreatic tumor, called a **gastrinoma**, stimulates the stomach to secrete excessive amounts of acid, resulting in peptic ulcers). Because external factors like the sight or odor of food, as well as psychological **stress**, can stimulate gastric secretion, accurate testing requires that the patient be relaxed and isolated from all sources of sensory stimulation. Abnormal basal secretion can suggest various gastric and duodenal disorders, so further evaluation requires the gastric acid stimulation test.

The gastric acid stimulation test is indicated when abnormalities are found during the basal secretion test. These abnormalities can be caused by a number of disorders, including duodenal ulcer, **pernicious anemia**, and gastric **cancer**. The test will detect abnormalities, but x rays and other studies are necessary for a definitive diagnosis.

Precautions

Because both the basal gastric secretion test and the gastric acid stimulation test require insertion of a gastric tube (intubation) through the mouth or nasal passage, neither test is recommended for patients with esophageal problems, **aortic aneurysm**, severe gastric hemorrhage, or congestive **heart failure**. The gastric acid stimulation test is also not recommended in patients who are sensitive to pentagastrin (the drug used to stimulate gastric acid output).

Description

This test, whether performed for basal gastric acid secretion, gastric acid stimulation, or both, requires the passage of a lubricated rubber tube, either by mouth or through the nasal passage, while the patient is in a sitting or reclining position on the left side. The tube is situated in the stomach, with proper positioning confirmed by fluoroscopy or x ray.

Basal gastric acid secretion

After a wait of approximately 10–15 minutes for the patient to adjust to the presence of the tube, and with the patient in a sitting position, specimens are obtained every 15 minutes for a period of 90 minutes. The first two specimens are discarded to eliminate gastric contents that might be affected by the stress of the intubation process. The patient is allowed no

liquids during the test, and saliva must be ejected to avoid diluting the stomach contents.

The four specimens collected during the test constitute the *basal acid output*. If analysis suggests abnormally low gastric secretion, the gastric acid stimulation test is performed immediately afterward.

Gastric acid stimulation test

After the basal samples have been collected, the tube remains in place for the gastric acid stimulation test. Pentagastrin, or a similar drug that stimulates gastric acid output, is injected under the skin (subcutaneously). After 15 minutes, a specimen is collected every 15 minutes for one hour. These specimens are called the *poststimulation specimens*. As is the case with the basal gastric secretion test, the patient can have no liquids during this test, and must eject saliva to avoid diluting the stomach contents.

Preparation

The patient should be fasting (nothing to eat or drink after the evening meal) on the day prior to the test, but may have water up to one hour before the test. **Antacids**, anticholinergics, cholinergics, alcohol, H₂-receptor antagonists (Tagamet, Pepcid, Axid, Zantac), reserpine, adrenergic blockers, and adrenocorticosteroids should be withheld for one to three days before the test, as the physician requests. If pentagastrin is to be administered for the gastric acid secretion test, medical supervision should be maintained, as possible side effects may occur.

Aftercare

Complications such as **nausea**, **vomiting**, and abdominal distention or pain are possible following removal of the gastric tube. If the patient has a **sore throat**, soothing lozenges may be given. The patient may also resume the usual diet and any medications that were withheld for the test(s).

Risks

There is a slight risk that the gastric tube may be inserted improperly, entering the windpipe (trachea) and not the esophagus. If this happens, the patient may have a difficult time breathing or may experience a coughing spell until the tube is removed and reinserted properly. Also, because the tube can be difficult to swallow, if a patient has an overactive gag reflex, there may be a transient rise in blood pressure due to **anxiety**.

KEY TERMS

Achlorhydria—An abnormal condition in which hydrochloric acid is absent from the secretions of the gastric glands in the stomach.

Pernicious anemia—One of the main types of anemia, caused by inadequate absorption of vitamin B₁₂. Symptoms include tingling in the hands, legs, and feet, spastic movements, weight loss, confusion, depression, and decreased intellectual function.

Zollinger-Ellison syndrome—A rare condition characterized by severe and recurrent peptic ulcers in the stomach, duodenum, and upper small intestine, caused by a tumor, or tumors, usually found in the pancreas. The tumor secretes the hormone gastrin, which stimulates the stomach and duodenum to produce large quantities of acid, leading to ulceration. Most often cancerous, the tumor must be removed surgically; otherwise total surgical removal of the stomach is necessary.

Normal results

Reference values for the *basal gastric secretion test* vary by laboratory, but are usually within the following ranges:

- men: 1–5 mEq/h
- women: 0.2–3.8 mEq/h

Reference values for the *gastric acid stimulation test* vary by laboratory, but are usually within the following ranges:

- men: 18–28 mEq/h
- women: 11–21 mEq/h

Abnormal results

Abnormal findings in the *basal gastric secretion test* are considered nonspecific and must be evaluated in conjunction with the results of a gastric acid stimulation test. Elevated secretion may suggest different types of ulcers; when markedly elevated, Zollinger-Ellison syndrome is suspected. Depressed secretion can indicate gastric cancer, while complete absence of secretion (achlorhydria) may suggest pernicious anemia.

Elevated gastric secretion levels in the *gastric acid stimulation test* may be indicative of duodenal ulcer; high levels of secretion again suggest Zollinger-Ellison syndrome.

Resources

BOOKS

Pagana, Kathleen Deska. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis: Mosby, Inc., 1998.

Janis O. Flores

Gastric carcinoma see **Stomach cancer**

Gastric emptying scan

Definition

A gastric emptying scan (GES) is an x-ray exam using special radioactive material that allows physicians to identify abnormalities related to emptying of the stomach. Diseases that involve changes in the way the stomach contracts (motility disorders) are best diagnosed by this test.

Purpose

The study is used most frequently to evaluate patients who have symptoms suggestive of decreased, delayed, or rapid gastric emptying, and no visible abnormality to explain their symptoms.

Symptoms pointing to a delay in gastric emptying are non-specific, and may be due to a number of causes, such as ulcers, diabetes, tumors, and others. These symptoms include **nausea**, upper abdominal bloating, and at times **vomiting**. Another significant symptom is called “early satiety,” which means feeling full after eating only a small amount of food. In some patients, weight loss is also present. In addition to symptoms, the finding of a large amount of material in the stomach after an overnight fast suggests abnormal emptying, but does not distinguish between an actual blockage or an irregularity in gastric contractions. It is therefore essential to find out what is causing material to remain in the stomach.

Since many diseases can produce the above symptoms, structural lesions (such as tumors or regions of narrowing or scar tissue) need to be ruled out first. This is usually done by upper gastrointestinal series test or by endoscopy (examination of the inside of an organ, in this instance the stomach, with an instrument that has a light at the end of it and an optical system for examination of the organ). Once it is clear that a mechanical or physical lesion is not the cause of symptoms, attempts to document an abnormality in the nervous or muscular function

KEY TERMS

Endoscopy—The examination of the inside of an organ with an instrument that has a light at the end of it and an optical system for examination of the organ.

Motility—Motility is spontaneous movement. One example is the automatic stomach contractions that move the food content along from the stomach into the intestines. A motility disease is one that involves changes in the way the stomach contracts.

of the stomach is then begun. GES is usually the first step in that evaluation.

Precautions

The exam should not be performed on pregnant women, but is otherwise quite safe. Since eggs are usually used to hold the radioactive material, patients should notify their physician if they are allergic to eggs. However, other materials can be used in place of an egg.

Description

Gastric emptying scans have undergone several changes since the initial studies in the late 1970s. During the study, patients are asked to ingest an egg sandwich containing a radioactive substance (for example, technetium) that can be followed by a special camera. The emptying of the material from the stomach is then followed and displayed both in the form of an image, as well as the percentage emptied over several hours (generally two and four hours). Studies are in progress using substances that are not radioactive, but this procedure is not available to the patient as of yet.

Preparation

The only preparation involved is for the patient to fast overnight before the test.

Risks

The radiation exposure during the study is quite small and safe, unless the patient is pregnant.

Normal results

There are several different measurements considered normal, depending on the radioactive material

and solid meal used. The value is expressed as a percentage of emptying over a period of time. For a technetium-filled egg sandwich, normal emptying is 78 minutes for half the material to leave the stomach, with a variation of 11 minutes either way.

Abnormal results

GES scan studies that show emptying of the stomach in a longer than accepted period is abnormal. Severity of test results and symptoms do not always match; therefore, the physician must carefully interpret these findings. Diabetic injury to the nerves that supply the stomach (called diabetic gastroparesis) is one of the most common causes of abnormal gastric motility. However, up to 30% of patients have no obvious cause to explain the abnormal results and symptoms. These cases are called idiopathic (of unknown cause). GES is often used to follow the effect of medications used for treatment of motility disorders.

Resources

ORGANIZATIONS

American Pseudo-Obstruction and Hirschprung's Disease Society. 158 Pleasant St., North Andover, MA 01845-2797. < <http://www.tiac.net/users/aphs> > .

David Kaminstein, MD

Gastric lavage see **Stomach flushing**

Gastric stapling see **Obesity surgery**

Gastric ulcers see **Ulcers (digestive)**

Gastrinoma

Definition

Gastrinomas are tumors associated with a rare gastroenterological disorder known as Zollinger-Ellison syndrome (ZES). They occur primarily in the pancreas and duodenum (beginning of the small intestine) and secrete large quantities of the hormone gastrin, triggering gastric acid production that produces ulcers. They may be malignant (cancerous) or benign.

Description

Gastrinomas are an integral part of the Zollinger-Ellison syndrome (ZES). In fact, ZES is also known as gastrinoma. This syndrome consists of ulcer disease in the upper gastrointestinal tract, marked increases

in the secretion of gastric acid in the stomach, and tumors of the islet cells in the pancreas. The tumors produce large amounts of gastrin that are responsible for the characteristics of Zollinger-Ellison syndrome, namely severe ulcer disease. Although usually located within the pancreas, they may occur in other organs.

Gastrinomas may occur randomly and sporadically, or they may be inherited as part of a genetic condition called multiple endocrine neoplasia type 1 (MEN-1) syndrome. About half of persons with MEN-1 have gastrinomas, which tend to be more numerous and smaller than tumors in sporadic cases.

About half of ZES patients have multiple gastrinomas, which can vary in size from 1–20 mm. Gastrinomas found in the pancreas are usually much larger than duodenal gastrinomas. About two thirds of gastrinomas are malignant (cancerous). These usually grow slowly, but some may invade surrounding sites rapidly and metastasize (spread) widely. Sometimes, gastrinomas are found only in the lymph nodes, and it is uncertain whether these malignancies have originated in the lymph nodes or have metastasized from a tumor not visible in the pancreas or duodenum.

There is some evidence that the more malignant form of gastrinomas is more frequent in larger pancreatic tumors, especially in females and in persons with a shorter disease symptom duration and higher serum gastrin levels.

Causes and symptoms

Most persons with gastrinomas secrete profound amounts of gastric acid, and almost all develop ulcers, mostly in the duodenum or stomach. Early in the course of the disease, symptoms are typical of peptic ulcers, however once the disease is established, the ulcers become more persistent and symptomatic, and may respond poorly to standard anti-ulcer therapy. Abdominal **pain** is the predominant symptom of ulcer disease. About 40% of patients have **diarrhea** as well. In some patients, diarrhea is the primary symptom of gastrinoma.

Diagnosis

Persons with gastrinomas have many of the same symptoms as persons with ulcers. Their levels of gastric acid, however, are usually far greater than those in common ulcer disease. Gastrinomas are usually diagnosed by a blood test that measures the level of gastrin in the blood. Patients with gastrinomas often have gastrin levels more than 200 pg/mL, which is 4–10

times higher than normal. Serum gastrin levels as high as 450,000 pg/mL have occurred.

When the serum gastrin test does not show these extremely high levels of gastrin, patients may be given certain foods or injections in an attempt to provoke a response that will help diagnose the condition. The most useful of these provocative tests is the secretin injection test (or secretin stimulation or provocative test), which will almost always produce a positive response in persons with gastrinomas but seldom in persons without them.

Surgically, gastrinomas are often difficult to locate, even with careful inspection. They may be missed in at least 10–20% of patients with ZES. Gastrinomas are sometimes found only because they have metastasized and produced symptoms related to the spread of malignancy. Such metastasis may be the most reliable indication of whether the gastrinoma is malignant or benign.

Diagnostic imaging techniques help locate the gastrinomas. The most sophisticated is an x-ray test called radionuclide octreotide scanning (also known as somatostatin receptor scintigraphy or ¹¹¹In pentetreotide SPECT). A study by the National Institutes of Health (NIH) found this test to be superior to other imaging methods, such as computed tomography scan (CT) or **magnetic resonance imaging** (MRI), in pinpointing the location of tumors and guiding physicians in treatment.

Approximately half of all gastrinomas do not show up on imaging studies. Therefore, exploratory surgery is often recommended to try to locate and remove the tumors.

Treatment

Therapy for gastrinomas should be individualized, since patients tend to have varying degrees of disease and symptoms. Treatment is aimed at eliminating the overproduction of gastric acid and removing the gastrin-producing tumors.

Drugs

Gastrinomas may not be easily treated by the standard anti-ulcer approaches. The medical treatment of choice is with drugs called **proton pump inhibitors**, such as omeprazole or lansoprazole, daily. These drugs are potent inhibitors of gastric acid. High doses of H-2 receptor antagonists may also reduce gastric acid secretion, improve symptoms, and induce ulcer healing. These drugs must be continued indefinitely, since even a brief discontinuation will

cause ulcer recurrence. **Antacids** may provide some relief, but it is usually not longlasting or healing.

Surgery

Because of the likelihood that gastrinomas may be malignant, in both sporadic tumors and those associated with the inherited MEN-1 syndrome, surgery to locate and remove gastrinomas is frequently advised. It is now known that complete surgical removal of gastrinomas can cure the overproduction of gastrin, even in patients who have metastases to the lymph nodes. Surgery in patients with MEN-1 and ZES, however, remains controversial since the benefit is less clear.

Freedom from disease after surgery is judged by improved symptoms, reduced gastric acid production, reduced need for drug therapy, normalization of serum gastrin levels, and normalization of results from the secretin stimulation test and imaging studies.

Prognosis

Medical therapy often controls symptoms, and surgery may or may not cure gastrinoma. About 50% of ZES patients in whom gastrinomas are not removed will die from malignant spread of the tumor. In patients with gastrinomas as part of MEN-1 syndrome, the cure rate is extremely low.

A NIH study of patients who had surgical removal of gastrinomas found that 42% were disease-free one year after surgery and 35% were disease-free at five years. Disease recurrences can often be detected with a serum gastrin test or secretin stimulation test.

When gastrinomas are malignant, they often grow slowly. The principal sites of metastasis are the regional lymph nodes and liver, but they may also spread to other structures. About one quarter of patients with gastrinomas have liver metastases at the time of diagnosis. This appears to be more frequent with pancreatic gastrinomas than duodenal gastrinomas.

Metastases of malignant gastrinomas to the liver is very serious. Survival five years after diagnosis is 20–30%, however patients with gastrinomas found only in the lymph nodes have been known to live as long as 25 years after diagnosis, without evidence of further tumor spread. In fact, the life expectancy of patients with gastrinomas that have spread to the lymph nodes is no different from that of patients with gastrinomas that cannot even be found at surgery for about 90%, five years after diagnosis.

KEY TERMS

Gastrin—A hormone secreted in the stomach that is involved in the production of gastric acid. Overproduction of gastric acid contributes to peptic ulcer formation.

Multiple endocrine neoplasia type 1 (MEN-1)—An inherited condition marked by multiple malignancies of the pituitary gland, parathyroid gland, and islet cells of the pancreas. About half of MEN-1 patients with pancreatic islet cell tumors will have gastrinomas, gastrin-producing tumors that lead to ulcer disease.

Peptic ulcer—An eroded area in the stomach lining or in the first part of the duodenum (beginning of the small intestine).

Serum gastrin test—A laboratory test that is performed on a blood sample to determine that level of the hormone gastrin. High levels of gastrin indicate the presence a duodenal ulcer or a gastrinoma.

Sporadic—Occurring at random or by chance, and not as a result of a genetically determined, or inherited, trait.

Resources

ORGANIZATIONS

National Digestive Diseases Information Clearinghouse.
2 Information Way, Bethesda, MD 20892-3570. (800) 891-5389. <<http://www.niddk.nih.gov/health/digest/niddic.htm>>.

Caroline A. Helwick

Gastritis

Definition

Gastritis commonly refers to inflammation of the lining of the stomach, but the term is often used to cover a variety of symptoms resulting from stomach lining inflammation and symptoms of burning or discomfort. True gastritis comes in several forms and is diagnosed using a combination of tests. In the 1990s, scientists discovered that the main cause of true gastritis is infection from a bacterium called *Helicobacter pylori* (*H. pylori*).

Description

Gastritis should not be confused with common symptoms of upper abdominal discomfort. It has been associated with resulting ulcers, particularly peptic ulcers. And in some cases, chronic gastritis can lead to more serious complications.

Nonerosive H. pylori gastritis

The main cause of true gastritis is *H. pylori* infection. *H. pylori* is indicated in an average of 90% of patients with chronic gastritis. This form of nonerosive gastritis is the result of infection with *Helicobacter pylori* bacterium, a microorganism whose outer layer is resistant to the normal effects of stomach acid in breaking down bacteria.

The resistance of *H. pylori* means that the bacterium may rest in the stomach for long periods of times, even years, and eventually cause symptoms of gastritis or ulcers when other factors are introduced, such as the presence of specific genes or ingestion of **non-steroidal anti-inflammatory drugs** (NSAIDs). Study of the role of *H. pylori* in development of gastritis and peptic ulcers has disproved the former belief that **stress** lead to most stomach and duodenal ulcers and has resulted in improved treatment and reduction of stomach ulcers. *H. pylori* is most likely transmitted between humans, although the specific routes of transmission were still under study in early 1998. Studies were also underway to determine the role of *H. pylori* and resulting chronic gastritis in development of gastric **cancer**.

Erosive and hemorrhagic gastritis

After *H. pylori*, the second most common cause of chronic gastritis is use of nonsteroidal anti-inflammatory drugs. These commonly used **pain** killers, including **aspirin**, fenoprofen, ibuprofen and naproxen, among others, can lead to gastritis and peptic ulcers. Other forms of erosive gastritis are those due to alcohol and corrosive agents or due to trauma such as ingestion of foreign bodies.

Other forms of gastritis

Clinicians differ on the classification of the less common and specific forms of gastritis, particularly since there is so much overlap with *H. pylori* in development of chronic gastritis and complications of gastritis. Other types of gastritis that may be diagnosed include:

- Acute stress gastritis—the most serious form of gastritis which usually occurs in critically ill patients,

such as those in intensive care. Stress erosions may develop suddenly as a result of severe trauma or stress to the stomach lining.

- Atrophic gastritis is the result of chronic gastritis which is leading to atrophy, or decrease in size and wasting away, of the gastric lining. Gastric atrophy is the final stage of chronic gastritis and may be a precursor to gastric cancer.
- Superficial gastritis is a term often used to describe the initial stages of chronic gastritis.
- Uncommon specific forms of gastritis include granulomatous, eosinophilic and lymphocytic gastritis.

Causes and symptoms

Nonerosive H. pylori gastritis

H. pylori gastritis is caused by infection from the *H. pylori* bacterium. It is believed that most infection occurs in childhood. The route of its transmission was still under study in 1998 and clinicians guessed that there may be more than one route for the bacterium. Its prevalence and distribution differs in nations around the world. The presence of *H. pylori* has been detected in 86–99% of patients with chronic superficial gastritis. However, physicians are still learning about the link of *H. pylori* to chronic gastritis and peptic ulcers, since many patients with *H. pylori* infection do not develop symptoms or peptic ulcers. *H. pylori* is also seen in 90–100% of patients with duodenal ulcers.

Symptoms of *H. pylori* gastritis include abdominal pain and reduced acid secretion in the stomach. However, the majority of patients with *H. pylori* infection suffer no symptoms, even though the infection may lead to ulcers and resulting symptoms. Ulcer symptoms include dull, gnawing pain, often two to three hours after meals and pain in the middle of the night when the stomach is empty.

Erosive and hemorrhagic gastritis

The most common cause of this form of gastritis is use of NSAIDs. Other causes may be **alcoholism** or stress from surgery or critical illness. The role of NSAIDs in development of gastritis and peptic ulcers depends on the dose level. Although even low doses of aspirin or other nonsteroidal anti-inflammatory drugs may cause some gastric upset, low doses generally will not lead to gastritis. However, as many as 10–30% of patients on higher and more frequent doses of NSAIDs, such as those with chronic arthritis, may develop gastric ulcers. In 1998, studies were underway

to understand the role of *H. pylori* in gastritis and ulcers among patients using NSAIDS.

Patients with erosive gastritis may also show no symptoms. When symptoms do occur, they may include **anorexia nervosa**, gastric pain, **nausea and vomiting**.

Other Forms of Gastritis

Less common forms of gastritis may result from a number of generalized diseases or from complications of chronic gastritis. Any number of mechanisms may cause various less common forms of gastritis and they may differ slightly in their symptoms and clinical signs. However, they all have in common inflammation of the gastric mucosa.

Diagnosis

Nonerosive H. pylori gastritis

H. pylori gastritis is easily diagnosed through the use of the urea breath test. This test detects active presence of *H. pylori* infection. Other serological tests, which may be readily available in a physician's office, may be used to detect *H. pylori* infection. Newly developed versions offer rapid diagnosis. The choice of test will depend on cost, availability and the physician's experience, since nearly all of the available tests have an accuracy rate of 90% or better. Endoscopy, or the examination of the stomach area using a hollow tube inserted through the mouth, may be ordered to confirm diagnosis. A biopsy of the gastric lining may also be ordered.

Erosive or hemorrhagic gastritis

Clinical history of the patient may be particularly important in the diagnosis of this type of gastritis, since its cause is most often the result of chronic use of NSAIDS, alcoholism, or other substances.

Other forms of gastritis

Gastritis that has developed to the stage of duodenal or gastric ulcers usually requires endoscopy for diagnosis. It allows the physician to perform a biopsy for possible malignancy and for *H. pylori*. Sometimes, an upper gastrointestinal x-ray study with barium is ordered. Some diseases such as Zollinger-Ellison syndrome, an ulcer disease of the upper gastrointestinal tract, may show large mucosal folds in the stomach and duodenum on radiographs or in endoscopy. Other tests check for changes in gastric function.

Treatment

H. pylori gastritis

The discovery of *H. pylori*'s role in development of gastritis and ulcers has led to improved treatment of chronic gastritis. In particular, relapse rates for duodenal and gastric ulcers has been reduced with successful treatment of *H. pylori* infection. Since the infection can be treated with **antibiotics**, the bacterium can be completely eliminated up to 90% of the time.

Although *H. pylori* can be successfully treated, the treatment may be uncomfortable for patients and relies heavily on patient compliance. In 1998, studies were underway to identify the best treatment method based on simplicity, patient cooperation and results. No single antibiotic had been found which would eliminate *H. pylori* on its own, so a combination of antibiotics has been prescribed to treat the infection.

DUAL THERAPY. Dual therapy involves the use of an antibiotic and a proton pump inhibitor. **Proton pump inhibitors** help reduce stomach acid by halting the mechanism that pumps acid into the stomach. This also helps promote healing of ulcers or inflammation. Dual therapy has not been proven to be as effective as triple therapy, but may be ordered for some patients who can more comfortably handle the use of less drugs and will therefore more likely follow the two-week course of therapy.

TRIPLE THERAPY. As of early 1998, triple therapy was the preferred treatment for patients with *H. pylori* gastritis. It is estimated that triple therapy successfully eliminates 80–95% of *H. pylori* cases. This treatment regimen usually involves a two-week course of three drugs. An antibiotic such as amoxicillin or tetracycline, and another antibiotic such as clarithromycin or metronidazole are used in combination with bismuth subsalicylate, a substance found in the over-the-counter medication, Pepto-Bismol, which helps protect the lining of the stomach from acid. Physicians were experimenting with various combinations of drugs and time of treatment to balance side effects with effectiveness. Side effects of triple therapy are not serious, but may cause enough discomfort that patients are not inclined to follow the treatment.

OTHER TREATMENT THERAPIES. Scientists have experimented with quadruple therapy, which adds an antisecretory drug, or one which suppresses gastric secretion, to the standard triple therapy. One study showed this therapy to be effective with only a week's course of treatment in more than 90% of patients. Short course therapy was attempted with triple therapy involving antibiotics and a proton pump inhibitor

and seemed effective in eliminating *H. pylori* in one week for more than 90% of patients. The goal is to develop the most effective therapy combination that can work in one week of treatment or less.

MEASURING H. PYLORI TREATMENT EFFECTIVENESS. In order to ensure that *H. pylori* has been eradicated, physicians will test patients following treatment. The breath test is the preferred method to check for remaining signs of *H. pylori*.

Treatment of erosive gastritis

Since few patients with this form of gastritis show symptoms, treatment may depend on severity of symptoms. When symptoms do occur, patients may be treated with therapy similar to that for *H. pylori*, especially since some studies have demonstrated a link between *H. pylori* and NSAIDS in causing ulcers. Avoidance of NSAIDS will most likely be prescribed.

Other forms of gastritis

Specific treatment will depend on the cause and type of gastritis. These may include prednisone or antibiotics. Critically ill patients at high risk for bleeding may be treated with preventive drugs to reduce risk of acute stress gastritis. If stress gastritis does occur, the patient is treated with constant infusion of a drug to stop bleeding. Sometimes surgery is recommended, but is weighed with the possibility of surgical complications or **death**. Once torrential bleeding occurs in acute stress gastritis, mortality is as high as greater than 60%.

Alternative treatment

Alternative forms of treatment for gastritis and ulcers should be used cautiously and in conjunction with conventional medical care, particularly now that scientists have confirmed the role of *H. pylori* in gastritis and ulcers. Alternative treatments can help address gastritis symptoms with diet and **nutritional supplements**, herbal medicine and **ayurvedic medicine**. It is believed that zinc, vitamin A and beta-carotene aid in the stomach lining's ability to repair and regenerate itself. Herbs thought to stimulate the immune system and reduce inflammation include **echinacea** (*Echinacea* spp.) and goldenseal (*Hydrastis canadensis*). Ayurvedic medicine involves **meditation**. There are also certain herbs and nutritional supplements aimed at helping to treat ulcers.

Prognosis

The discovery of *H. pylori* has improved the prognosis for patients with gastritis and ulcers. Since

KEY TERMS

Duodenal—Refers to the duodenum, or the first part of the small intestine.

Gastric—Relating to the stomach.

Mucosa—The mucous membrane, or the thin layer which lines body cavities and passages.

Ulcer—A break in the skin or mucous membrane. It can fester and pus like a sore.

treatment exists to eradicate the infection, recurrence is much less common. As of 1998, the only patients requiring treatment for *H. pylori* were those at high risk because of factors such as NSAIDS use or for those with ulcers and other complicating factors or symptoms. Research will continue into the most effective treatment of *H. pylori*, especially in light of the bacterium's resistance to certain antibiotics. Regular treatment of patients with gastric and duodenal ulcers has been recommended, since *H. pylori* plays such a consistently high role in development of ulcers. It is believed that *H. pylori* also plays a role in the eventual development of serious gastritis complications and cancer. Detection and treatment of *H. pylori* infection may help reduce occurrence of these diseases. The prognosis for patients with acute stress gastritis is much poorer, with a 60 percent or higher mortality rate among those bleeding heavily.

Prevention

The widespread detection and treatment of *H. pylori* as a preventive measure in gastritis has been discussed but not resolved. Until more is known about the routes through which *H. pylori* is spread, specific prevention recommendations are not available. Erosive gastritis from NSAIDS can be prevented with cessation of use of these drugs. An education campaign was launched in 1998 to educate patients, particularly an **aging** population of arthritis sufferers, about risk for ulcers from NSAIDS and alternative drugs.

Resources

PERIODICALS

Podolski, J. L. "Recent Advances in Peptic Ulcer Disease: *H. pylori* Infection and Its Treatment." *Gastroenterology Nursing* 19, no. 4: 128-136.

ORGANIZATIONS

National Digestive Diseases Information Clearinghouse.
2 Information Way, Bethesda, MD 20892-3570.

(800) 891-5389. < <http://www.niddk.nih.gov/health/digest/nddic.htm> > .

OTHER

American College of Gastroenterology Page. < <http://www.acg.org> > .

HealthAnswers.com. < <http://www.healthanswers.com> > .

Teresa Odle

Gastroduodenostomy (Billroth I) see **Ulcer surgery**

Gastroenteritis

Definition

Gastroenteritis is a catchall term for infection or irritation of the digestive tract, particularly the stomach and intestine. It is frequently referred to as the stomach or intestinal flu, although the **influenza** virus is not associated with this illness. Major symptoms include **nausea and vomiting**, **diarrhea**, and abdominal cramps. These symptoms are sometimes also accompanied by **fever** and overall weakness. Gastroenteritis typically lasts about three days. Adults usually recover without problem, but children, the elderly, and anyone with an underlying disease are more vulnerable to complications such as **dehydration**.

Description

Gastroenteritis is an uncomfortable and inconvenient ailment, but it is rarely life-threatening in the United States and other developed nations. However, an estimated 220,000 children younger than age five are hospitalized with gastroenteritis symptoms in the United States annually. Of these children, 300 die as a result of severe diarrhea and dehydration. In developing nations, diarrheal illnesses are a major source of mortality. In 1990, approximately three million deaths occurred worldwide as a result of diarrheal illness.

The most common cause of gastroenteritis is viral infection. Viruses such as rotavirus, adenovirus, astrovirus, and calicivirus and small round-structured viruses (SRSVs) are found all over the world. Exposure typically occurs through the fecal-oral route, such as by consuming foods contaminated by fecal material related to poor sanitation. However, the infective dose can be very low (approximately 100 virus particles), so other routes of transmission are quite probable.

Typically, children are more vulnerable to rotaviruses, the most significant cause of acute watery diarrhea. Annually, worldwide, rotaviruses are estimated to cause 800,000 deaths in children below age five. For this reason, much research has gone into developing a vaccine to protect children from this virus. Adults can be infected with rotaviruses, but these infections typically have minimal or no symptoms.

Children are also susceptible to adenoviruses and astroviruses, which are minor causes of childhood gastroenteritis. Adults experience illness from astroviruses as well, but the major causes of adult viral gastroenteritis are the caliciviruses and SRSVs. These viruses also cause illness in children. The SRSVs are a type of calicivirus and include the Norwalk, Southampton, and Lonsdale viruses. These viruses are the most likely to produce **vomiting** as a major symptom.

Bacterial gastroenteritis is frequently a result of poor sanitation, the lack of safe drinking water, or contaminated food—conditions common in developing nations. Natural or man-made disasters can make underlying problems in sanitation and food safety worse. In developed nations, the modern food production system potentially exposes millions of people to disease-causing bacteria through its intensive production and distribution methods. Common types of bacterial gastroenteritis can be linked to *Salmonella* and *Campylobacter* bacteria; however, *Escherichia coli* 0157 and *Listeria monocytogenes* are creating increased concern in developed nations. **Cholera** and *Shigella* remain two diseases of great concern in developing countries, and research to develop long-term vaccines against them is underway.

Causes and symptoms

Gastroenteritis arises from ingestion of viruses, certain bacteria, or parasites. Food that has spoiled may also cause illness. Certain medications and excessive alcohol can irritate the digestive tract to the point of inducing gastroenteritis. Regardless of the cause, the symptoms of gastroenteritis include diarrhea, **nausea** and vomiting, and abdominal **pain** and cramps. Sufferers may also experience bloating, low fever, and overall tiredness. Typically, the symptoms last only two to three days, but some viruses may last up to a week.

A usual bout of gastroenteritis shouldn't require a visit to the doctor. However, medical treatment is essential if symptoms worsen or if there are complications. Infants, young children, the elderly, and persons with underlying disease require special attention in this regard.

The greatest danger presented by gastroenteritis is dehydration. The loss of fluids through diarrhea and vomiting can upset the body's electrolyte balance, leading to potentially life-threatening problems such as heart beat abnormalities (arrhythmia). The risk of dehydration increases as symptoms are prolonged. Dehydration should be suspected if a **dry mouth**, increased or excessive thirst, or scanty urination is experienced.

If symptoms do not resolve within a week, an infection or disorder more serious than gastroenteritis may be involved. Symptoms of great concern include a high fever (102° F [38.9° C] or above), blood or mucus in the diarrhea, blood in the vomit, and severe abdominal pain or swelling. These symptoms require prompt medical attention.

Diagnosis

The symptoms of gastroenteritis are usually enough to identify the illness. Unless there is an outbreak affecting several people or complications are encountered in a particular case, identifying the specific cause of the illness is not a priority. However, if identification of the infectious agent is required, a stool sample will be collected and analyzed for the presence of viruses, disease-causing (pathogenic) bacteria, or parasites.

Treatment

Gastroenteritis is a self-limiting illness which will resolve by itself. However, for comfort and convenience, a person may use over-the-counter medications such as Pepto Bismol to relieve the symptoms. These medications work by altering the ability of the intestine to move or secrete spontaneously, absorbing toxins and water, or altering intestinal microflora. Some over-the-counter medicines use more than one element to treat symptoms.

If over-the-counter medications are ineffective and medical treatment is sought, a doctor may prescribe a more powerful anti-diarrheal drug, such as motofen or lomotil. Should pathogenic bacteria or parasites be identified in the patient's stool sample, medications such as **antibiotics** will be prescribed.

It is important to stay hydrated and nourished during a bout of gastroenteritis. If dehydration is absent, the drinking of generous amounts of nonalcoholic fluids, such as water or juice, is adequate. **Caffeine**, since it increases urine output, should be avoided. The traditional BRAT diet—bananas, rice, applesauce, and toast—is tolerated by the tender gastrointestinal system, but it is not particularly nutritious. Many, but not all, medical researchers

recommend a diet that includes complex carbohydrates (e.g., rice, wheat, potatoes, bread, and cereal), lean meats, yogurt, fruit, and vegetables. Milk and other dairy products shouldn't create problems if they are part of the normal diet. Fatty foods or foods with a lot of sugar should be avoided. These recommendations are based on clinical experience and controlled trials, but are not universally accepted.

Minimal to moderate dehydration is treated with oral rehydrating solutions that contain glucose and electrolytes. These solutions are commercially available under names such as Naturalyte, Pedialyte, Infalyte, and Rehydralyte. Oral rehydrating solutions are formulated based on physiological properties. Fluids that are not based on these properties—such as cola, apple juice, broth, and sports beverages—are not recommended to treat dehydration. If vomiting interferes with oral rehydration, small frequent fluid intake may be better tolerated. Should oral rehydration fail or severe dehydration occur, medical treatment in the form of intravenous (IV) therapy is required. IV therapy can be followed with oral rehydration as the patient's condition improves. Once normal hydration is achieved, the patient can return to a regular diet.

Alternative treatment

Symptoms of uncomplicated gastroenteritis can be relieved with adjustments in diet, herbal remedies, and homeopathy. An infusion of meadowsweet (*Filipendula ulmaria*) may be effective in reducing nausea and stomach acidity. Once the worst symptoms are relieved, slippery elm (*Ulmus fulva*) can help calm the digestive tract. Of the homeopathic remedies available, *Arsenicum album*, **ippecac**, or *Nux vomica* are three said to relieve the symptoms of gastroenteritis.

Probiotics, bacteria that are beneficial to a person's health, are recommended during the recovery phase of gastroenteritis. Specifically, live cultures of *Lactobacillus acidophilus* are said to be effective in soothing the digestive tract and returning the intestinal flora to normal. *L. acidophilus* is found in live-culture yogurt, as well as in capsule or powder form at health food stores. The use of probiotics is found in folk remedies and has some support in the medical literature. Castor oil packs to the abdomen can reduce inflammation and also reduce spasms or discomfort.

Prognosis

Gastroenteritis is usually resolved within two to three days and there are no long-term effects. If dehydration occurs, recovery is extended by a few days.

KEY TERMS

Dehydration—A condition in which the body lacks the normal level of fluids, potentially impairing normal body functions.

Electrolyte—An ion, or weakly charged element, that conducts reactions and signals in the body. Examples of electrolytes are sodium and potassium ions.

Glucose—A sugar that serves as the body's primary source of fuel.

Influenza—A virus that affects the respiratory system, causing fever, congestion, muscle aches, and headaches.

Intravenous (IV) therapy—Administration of intravenous fluids.

Microflora—The bacterial population in the intestine.

Pathogenic bacteria—Bacteria that produce illness.

Probiotics—Bacteria that are beneficial to a person's health, either through protecting the body against pathogenic bacteria or assisting in recovery from an illness.

Prevention

There are few steps that can be taken to avoid gastroenteritis. Ensuring that food is well-cooked and unspoiled can prevent bacterial gastroenteritis, but may not be effective against viral gastroenteritis.

Resources

PERIODICALS

Hart, C. Anthony, and Nigel A. Cunliffe. "Viral Gastroenteritis." *Current Opinion in Infectious Diseases* 10 (1997): 408.

Moss, Peter J., and Michael W. McKendrick. "Bacterial Gastroenteritis." *Current Opinion in Infectious Diseases* 10 (1997): 402.

Julia Barrett

Gastroesophageal reflux see **Heartburn**

Gastrointestinal bleeding studies see **GI bleeding studies**

Gastrointestinal study see **Liver nuclear medicine scan**

Gastrojejunostomy see **Ulcer surgery**

Gastroschisis see **Abdominal wall defects**

Gastrostomy

Definition

Gastrostomy is a surgical procedure for inserting a tube through the abdomen wall and into the stomach. The tube is used for feeding or drainage.

Purpose

Gastrostomy is performed because a patient temporarily or permanently needs to be fed directly through a tube in the stomach. Reasons for feeding by gastrostomy include **birth defects** of the mouth, esophagus, or stomach, and problems sucking or swallowing.

Gastrostomy is also performed to provide drainage for the stomach when it is necessary to bypass a long-standing obstruction of the stomach outlet into the small intestine. Obstructions may be caused by peptic ulcer scarring or a tumor.

Precautions

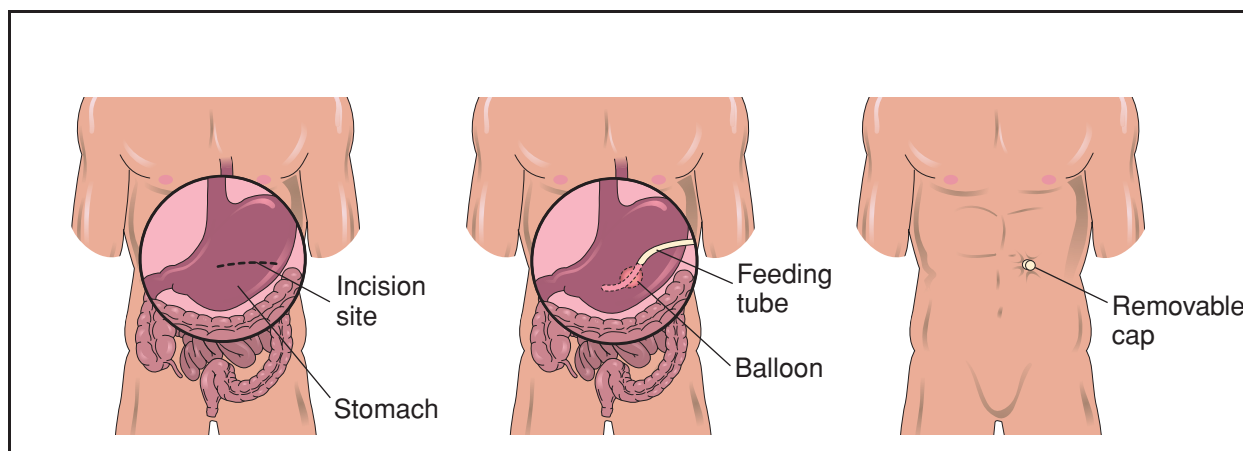
Gastrostomy is a relatively simple procedure. As with any surgery, patients are more likely to experience complications if they are smokers, obese, use alcohol heavily, or use illicit drugs. In addition, some prescription medications may increase risks associated with anesthesia.

Description

Gastrostomy, also called gastrostomy tube insertion, is surgery performed by a general surgeon to give an external opening into the stomach. Surgery is performed either when the patient is under general anesthesia—where the patient feels as if he is in a deep sleep and has no awareness of what is happening—or under **local anesthesia**. With local anesthesia, the patient is awake, but the part of the body cut during the operation is numbed.

A small incision is made on the left side of the abdomen; then, an incision is made through the stomach. A small, flexible, hollow tube, usually made of polyvinylchloride or rubber, is inserted into the stomach. The stomach is stitched closed around the tube, and the incision is closed. The procedure is performed at a hospital or free-standing surgery center.

The length of time the patient needs to remain in the hospital depends on the age of the patient and the patient's general health. In some cases, the hospital stay can be as short as one day, but often is longer.



Gastrostomy is a procedure in which the surgeon makes an opening into the stomach and inserts a feeding tube for feeding or for drainage. (Illustration by Electronic Illustrators Group.)

Normally, the stomach and abdomen heal in five to seven days.

The cost of the surgery varies, depending on the age and health of the patient. Younger, sicker patients require more intensive, thus more expensive, care.

Preparation

Prior to the operation, the doctor will perform endoscopy and take x rays of the gastrointestinal tract. Blood and urine tests will also be performed, and the patient may meet with the anesthesiologist to evaluate any special conditions that might affect the administration of anesthesia.

Aftercare

Immediately after the operation, the patient is fed intravenously for at least 24 hours. Once bowel sounds are heard, indicating that the gastrointestinal system is working, the patient can begin clear liquid feedings through the tube. Gradually feedings are increased.

Patient education concerning use and care of the gastrostomy tube is very important. Patients and their families are taught how to recognize and prevent infection around the tube, how to feed through the tube, how to handle tube blockage, what to do if the tube pulls out, and what normal activities can be continued.

Risks

There are few risks associated with this surgery. The main complications are infection, bleeding, dislodgment of the tube, stomach bloating, **nausea**, and **diarrhea**.

KEY TERMS

Endoscopy—A procedure in which an instrument containing a camera is inserted into the gastrointestinal tract so that the doctor can visually inspect the gastrointestinal system.

Normal results

The patient is able to eat through the gastrostomy tube, or the stomach can be drained through the tube.

Resources

OTHER

“Stomach Tube Insertion.” HealthAnswers.com. < <http://www.healthanswers.com> > .

Tish Davidson, A.M.

Gaucher disease

Definition

Gaucher disease is a rare genetic disorder that results in accumulation of fatty molecules called cerebrosides. It can have serious effects on numerous body organs including the liver, spleen, bones and central

nervous system. Treatments based on molecular biology are becoming available, but are very expensive.

Description

Gaucher disease was first described by the French physician Philippe Gaucher in 1882. It is the most common of a class of diseases called lysosomal storage diseases, each of which is characterized by the accumulation of a specific chemical substance (a different substance depending on the exact disease). Gaucher disease is characterized by a wide array of different symptoms and the severity of the disease ranges from undetectable to lethal.

Three forms of the disease are recognized: Types I, II and III. Type I is by far the most common and shows the mildest symptoms. It is non-neuronopathic, meaning that the nervous system is not attacked. The onset of Type I can occur at any age in childhood or adult life with the average age of onset at about 21 years. Some affected individuals have no symptoms throughout adult life. Type II, the infantile form, accounts for less than 1% of patients with Gaucher disease. It is neuronopathic (attacks the nervous system); nervous system effects are severe, and victims often die within the first year of life. Type III most often has its onset during childhood and has some of the features of both the adult and infantile forms. This affects less than 5% of persons with Gaucher disease.

Gaucher disease is caused by the absence, or near absence, of activity of an enzyme called glucocerebrosidase (GC). The normal action of GC is to break down a common molecule called glucocerebroside. If not broken down, glucocerebroside accumulates in certain cells to levels that can cause damage, especially in the spleen, liver, and bone. The common link among these organs is that they house a cell type called a macrophage. A macrophage is a large cell that surrounds and consumes a foreign substance (such as bacteria) in the body. The cellular structures in which glucocerebroside accumulates are called lysosomes.

The three forms of Gaucher disease also differ in their population genetics. Type I is most common in persons of eastern European (Ashkenazi) Jewish descent. Among this population, the disease occurs at a rate of one in 450 live births and about one in 10 to 15 persons are carriers, making it the most common genetic disease affecting Jewish people. The other two types are equally frequent in all ethnic groups. Type II occurs at a rate of one in 100,000 live births,

while Type III is estimated to occur in one in 50,000 live births.

Causes and symptoms

Lack of the GC enzyme is caused by a mutation in the glucocerebrosidase gene. The gene is located on chromosome 1. As of 2000, there have been over 100 mutations described in this gene that causes Gaucher disease. Gaucher disease is inherited in an autosomal recessive pattern. This means that two defective gene copies must be inherited, one from each parent, for the disease to manifest itself. Persons with only one gene mutation are carriers for the disorder. A person who is a carrier for Gaucher disease does not have any symptoms and does not know he or she is a carrier unless he or she has had specific testing. When both parents are carriers for Gaucher disease, there is a one in four chance (25%) in each **pregnancy** for a child to have Gaucher disease. There is a two in three chance that a healthy sibling of an affected child is a carrier.

The results of Gaucher disease are widespread in the body and include excessive growth of the liver and spleen (hepatosplenomegaly), weakening of bones, and, in acute cases, severe nervous system damage. Many patients experience “bone crises,” which are episodes of extreme **pain** in their bones.

There is a wide array of other problems that occur with Gaucher disease, such as anemia (fewer than normal red blood cells). Just how these other symptoms are caused is not known. Nor is it known why some patients have very mild disease and others have much more significant problems. Even identical twins with the disease can have differing symptoms.

Diagnosis

Diagnosis of Gaucher disease, based initially on the symptoms described above, can be confirmed by microscopic, enzymatic, and molecular tests. Biopsy (surgical removal of tissue from a problem area) of tissue is helpful for microscopic diagnosis. When biopsy tissue is examined under the microscope, cells will appear swollen and will show characteristic features of the cytoplasm (part of the cell body along with the nucleus) and nucleus. Enzyme tests will show deficiency (<30% of normal levels) of the enzyme GC. Molecular analysis of DNA samples looking at four of the more common mutations will show defects in the gene for GC in 95% of Ashkenazi Jewish individuals and in 75% of non-Jewish people. Diagnosis can be performed prenatally (before birth) if the parents’

mutations are known using **amniocentesis** or **chorionic villus sampling**.

Diagnosis as to which of the three types of Gaucher disease an individual has is based on the symptoms, rather than on test results.

Treatment

Until the 1990s, only supportive therapy could be offered. **Analgesics** are used to control pain. Orthopedic treatment is used for bone **fractures**. In some cases, surgical removal of the spleen may be necessary. Several treatments for anemia have been used, including vitamin and iron supplements, blood transfusions, and bone marrow transplants.

The newest form of treatment for Gaucher disease is enzyme replacement therapy, in which GC can be administered intravenously. The enzyme can be prepared either by purification from placentas (alglucerase) or by recombinant DNA manufacturing techniques (imiglucerase). Either way, the cost of treatment ranges from \$100,000 to \$400,000 per year, which can prevent many from obtaining treatment.

Enzyme replacement is effective at reducing most Gaucher symptoms. The notable exception is neurologic damage in Type II disease, which remains unimproved by this treatment. This treatment is not recommended for individuals who are asymptomatic. As of 2000, the efficacy for the treatment of Type III Gaucher disease is not known. Many questions remain about enzyme replacement therapy in regard to dosage, and method and frequency of administration. The treatment program should be individualized for each patient.

Prognosis

A patient's expected lifespan varies greatly with the type of Gaucher disease. Infants with Type II disease have a life span of one to four years. Patients with Types I and III of the disease have highly variable outcomes with some patients dying in childhood and others living full lives. Little is known about the reasons for this variability.

Prevention

Genetic counseling is advised for individuals with Gaucher disease and for their relatives to accurately assess risk and discuss testing options. For couples who previously had a child with Gaucher or in situations where both parents are carriers for known Gaucher mutations, prenatal diagnosis is available to

KEY TERMS

Cerebrosides—Fatty carbohydrates that occur in the brain and nervous system.

Enzymatic replacement therapy—A treatment method used to replace missing enzymes. It is possible to synthesize enzymes and then inject them intravenously into patients.

Glucocerebroside—A cerebroside that contains glucose in the molecule.

determine whether a pregnancy is affected. Families in which a person has been diagnosed with Gaucher disease can have DNA testing, which enables other relatives to determine their carrier status. Prospective parents can then use that information to conduct family planning or to prepare for a child who may have special circumstances.

Families in which both parents are known to be a carrier of a mutation for Gaucher disease could consider preimplantation genetic diagnosis. This relatively new procedure can select an embryo without both Gaucher disease mutations prior to implantation of the embryo into the uterus. This technique is only available at selected genetics centers.

As of the early 2000s, population screening for Gaucher disease is not standard of care.

Resources

PERIODICALS

Beutler, E. "Gaucher Disease." *Archives of Internal Medicine* 159 (1999): 881-2.

ORGANIZATIONS

Alliance of Genetic Support Groups. 4301 Connecticut Ave. NW, Suite 404, Washington, DC 20008. (202) 966-5557. Fax: (202) 966-8553. <<http://www.geneticalliance.org>>.

Children's Gaucher Research Fund. PO Box 2123, Granite Bay, CA 95746-2123. (916) 797-3700. Fax: (916) 797-3707. <<http://www.childrensgaucher.org>>.

National Gaucher Foundation. 11140 Rockville Pike, Suite 350, Rockville, MD 20852-3106. (800) 925-8885. <<http://www.gaucherdisease.org>>.

National Organization for Rare Disorders (NORD). PO Box 8923, New Fairfield, CT 06812-8923. (203) 746-6518 or (800) 999-6673. Fax: (203) 746-6481. <<http://www.rarediseases.org>>.

OTHER

"Cerezyme." Genzyme Therapeutics. <<http://www.cerezyme.com>>.

“Gaucher Disease: Current Issues in Diagnosis and Treatment.” <<http://text.nlm.nih.gov/nih/ta/www/16.html>> .

“Living with Gaucher Disease: A Guide for Patients, Parents, Relatives, and Friends.” <<http://neuro-www3.mgh.harvard.edu/gaucher/living.html>> .

National Foundation for Jewish Genetic Diseases (NFJGD). <<http://www.nfjgd.org/>> .

Amy Vance, MS, CGC

Gay and lesbian health

Definition

Lesbian, gay, bisexual, and transgender (LGBT) individuals are as diverse as the general population in terms of race, ethnicity, age, religion, education, income, and family history. A number of health concerns are unique to or shared by the LGBT community, however, including an increased risk of certain cancers, infectious and **sexually transmitted diseases** (STDs), and mental health disorders; issues relating to **nutrition** and weight, tobacco use, and **substance abuse**; and discrimination by health care and insurance providers.

Description

The definitions of different sexual identities have shifted over the years, as have the perceptions and stereotypes of the general population. Because of the wide range of behaviors and identities that exist in the LGBT community, it is difficult to develop an inclusive definition. It is generally accepted, however, that gay men and lesbians are sexually attracted to or participate in sexual behaviors with individuals of the same gender, while bisexual men and women are sexually attracted to or participate in sexual behaviors with individuals of both genders. Transgender individuals live part- or full-time in a gender role opposite to their genetic sex.

It is estimated that approximately 2.8% of men and 1.4% of women identify as being gay, lesbian, or bisexual while 9.1% of men and 4.3% of women have participated in sexual behavior with someone of the same gender at least once. The true extent of the transgender community has not been well researched in the United States; one study from the Netherlands in 1993 found that one in 11,900 males and one in 30,400 females are transgender.

There are a number of issues that arise when trying to define sexual orientation. Many gay men and lesbians have participated in or continue to participate in sexual activities with members of the opposite sex but choose not to identify as heterosexuals or bisexuals. Others have never participated in sexual activities at all yet still identify as gay, lesbian, or bisexual. Some men and women identifying as bisexuals are in long-term, monogamous relationships with individuals of the same or opposite sex. Male-to-female (MTF) or female-to-male (FTM) transgender individuals may or may not identify themselves as gay or lesbian.

The implications of these identity issues are far-reaching. Misdiagnoses or improper medical recommendations might come from health care providers who have mistakenly assumed sexual behaviors or risks from the patient's stated identity. For example, a provider might incorrectly assume that a lesbian patient has never had sexual intercourse with a male and therefore would not have contracted STDs not normally transmitted by sexual activities between women. It has been difficult to closely estimate the numbers of LGBT individuals in the United States because of varying definitions. Likewise, the statistics in medical or social studies and surveys on LGBT issues might vary widely depending on what definitions were provided for the respondents. Because of this, many researchers have opted for the more inclusive terms of “men who have sex with men” (MSM) and “women who have sex with women” (WSW) to categorize gay, lesbian, and bisexual respondents.

Important health care issues

Many LGBT individuals have difficulty revealing their sexual identity (“coming out”) to their health care providers. They may fear discrimination from providers or believe that their confidentiality might be breached. In some cases health care workers have been poorly trained to address the needs of LGBT individuals or have difficulty communicating with their LGBT patient (one study indicated that 40% of physicians are uncomfortable providing care for gay or lesbian patients). In addition, many questions posed in questionnaires or examinations are heterosexually biased (e.g. asking a lesbian which birth control methods she uses or a gay man if he is married, single, or divorced).

Other reasons why LGBT individuals are often hesitant to share their sexual identity are more logistical. Many insurance companies deny benefits to long-term partners on the basis that they are not

married. LGBT patients may have inadequate access to health care, either because they live in a remote rural area or in the crowded inner city. Some same-sex partners encounter discrimination in hospitals and clinics when they are denied the rights usually given to spouses of a patient such as visiting, making medical decisions, and participating in consultations with physicians.

Some of the health concerns and risk factors that are relevant to LGBT individuals may be shared by the general population, while others are more specific to the LGBT community, and still others are specific to different subgroups of LGBT individuals. These health concerns may be grouped into the following areas of concern:

- Sexual behavior issues: STDs such as human **immunodeficiency virus** (HIV) and acquired immune deficiency syndrome (**AIDS**), **hepatitis A virus** (HAV), **hepatitis B virus** (HBV), **bacterial vaginosis**, **gonorrhea**, chlamydia, and **genital warts** (human papillomavirus or HPV); anal, ovarian, and cervical **cancer**.
- Cultural issues: body image, nutrition, weight, and eating disorders; drug and alcohol **abuse**; tobacco use; parenting and family planning.
- Discrimination issues: inadequate medical care; harassment at work, school, or home; difficulty in obtaining housing, insurance coverage, or child custody; violence.
- Sexual identity issues: conflicts with family, friends, and work mates; psychological issues such as **anxiety**, depression, and **suicide**; economic hardship.

CANCER. Cancer is the second leading cause of **death** in the United States. In 2000, it was estimated that 1,220,100 individuals were diagnosed with cancer and 55,200 lost their lives as a result. LGBT individuals are at an increased risk for certain types of cancers. Some researchers believe that those who do not disclose their sexual identity live with an added **stress** that suppresses the immune system, thus leaving them with an increased risk of tumor growth.

Several studies have indicated that lesbians have higher risk for developing **breast cancer**. This is partially related to higher rates of risk factors such as **obesity**, alcohol use, tobacco use, and nulliparity (not bearing children). It has also been shown that lesbians are less likely to be screened for breast cancer than heterosexual women. Lesbians also have additional risk of developing **ovarian cancer**, due to inadequate access to health care, nulliparity, and not using **oral**

contraceptives (use of oral contraceptives has been shown to decrease the risk of getting ovarian cancer).

Gay and bisexual men (or more generally, men who have sex with men [MSM]) are at higher risk of developing non-Hodgkin's lymphoma, **Hodgkin's disease**, and **anal cancer**. **Kaposi's sarcoma**, an AIDS-associated cancer, used to be found in the gay community at rates thousands of times more than the general population before more effective **antiretroviral drugs** became available for people infected with HIV. Anal cancer is associated with transmission of human papillomavirus (HPV); a 1998 study indicated 73% of HIV-positive and 23% of HIV-negative MSM were infected with more than one type of HPV. The risk factors associated with MSM are also associated with increased rates of anal cancer (i.e. **smoking**, having many sexual partners, and receiving anal intercourse).

AIDS. As of 2001, more than 760,000 individuals had been diagnosed with AIDS in the United States. More than 80% are men; about 16% are women, and 1% are children 12 years old or younger. In 2000, more than 30,000 MSM were diagnosed with AIDS. While this number was down from 1999, it remained the highest risk factor for the disease. Other major risk groups associated with AIDS transmission include intravenous drug users (IDUs) who share needles, heterosexuals who engage in high-risk sexual behaviors, inmates at correctional facilities, and neonates (newborns) whose mothers are infected with HIV.

Gender identity disorder is defined as "a strong and persistent cross-gender identification . . . manifested by symptoms such as a stated desire to be the other sex, frequently passing as the other sex, desire to live or be treated as the other sex, or the conviction that he or she as the typical feelings and reactions of the other sex" (DSM-IV, 302.85). Transvestic fetishism is defined as involving "recurrent, intense sexually arousing fantasies, sexual urges, or behaviors involving cross-dressing" (DSM-IV, 302.3). Both disorders lead to a "disturbance that causes clinically significant distress or impairment in social, occupational, or other important areas of functioning." This last point iterates that transgender individuals are not automatically considered under DSM-IV to have a mental disorder.

NUTRITION AND WEIGHT. Diet and nutritional factors are associated with a number of diseases including cancer, **stroke**, diabetes, heart disease, and **osteoporosis**. It has been shown that lesbians are more likely than heterosexual women to be obese, have a higher body mass index (BMI), and have higher rates of smoking, but are also more likely to have a healthier body image

KEY TERMS

Nulliparity—The condition of being nulliparous, or not bearing offspring.

(42% compared to 21% of heterosexual women). Gay men and adolescents, on the other hand, have been shown to have increased rates of eating disorder behaviors than heterosexual men; examples are binge eating (25% compared to 11%), purging behaviors (12% to 4%), and poor body image (28% to 12%).

SUBSTANCE AND TOBACCO USE. **Marijuana** and **cocaine** use has been shown to be higher among lesbians than heterosexual women. The incidence of the use of some drugs is higher in gay men than heterosexual men; these include marijuana, psychedelic drugs, ecstasy, barbituates, and stimulants such as amyl or butyl nitrate (“poppers”). There is some indication that the use of some illicit drugs speeds up the replication of HIV, although more research needs to be done in this area.

Cigarette smoking is responsible for 430,000 deaths a year in the United States, with an estimated 3,000 nonsmokers dying as a result of exposure to secondhand smoke. In 2004 the rate of smoking among all adults was 28%. In contrast, 50% of gay men, lesbians and bisexuals were noted to be smokers. Lesbians are more than two times as likely to become heavy smokers than heterosexual women. Research shows that lesbians have the toughest time giving up smoking. For many years, efforts to improve health among the LGBT population has focused on AIDS and STDs and smoking cessation has taken a back seat.

Prevention

There are numerous ways that health care providers can improve the access to and experience of health care services for LGBT individuals. These include:

- rewording questionnaires and examinations to be inclusive of LGBT patients
- providing referrals to social service agencies and counseling services that are LGBT-friendly
- taking educational courses that are sensitive to the needs of LGBT patients
- treating the families of LGBT patients as one would the families of heterosexual patients

- maintaining the strictest code of confidentiality
- developing and maintaining health care centers or clinics that address LGBT-specific needs
- asking non-threatening questions to determine if a person is at risk of an STD
- educating patients of risk factors associated with STDs, possible vaccines, and treatments available
- providing services to individuals in the process of disclosing their sexual identity and, if applicable, their families

Resources

PERIODICALS

DuLong, Jessica. “Snuffing Out the Butts: Campaigns to Convince Gay Men and Lesbians to Ditch Cigarettes Tey to Cut Through a Long-held Addiction.” *The Advocate* March 16, 2004: 30-32.

ORGANIZATIONS

Gay and Lesbian Medical Association. 459 Fulton Street, Suite 107, San Francisco, CA 94102. (415) 225-4547. < <http://www.glma.org> > .

Parents, Families, and Friends of Lesbians and Gays. 1726 M Street NW, Suite 400, Washington, DC 20036. (202) 467-8180. < <http://www.pflag.org> > .

OTHER

Albert, Sarah. “Lesbians and Gays More Likely to Smoke, Survey Says.” GayHealth Website. May 16, 2001. < <http://www.gayhealth.com/templates/99022445042572021484300001/news/index.html?record=554> > .

Dean, Laura, Ilan H. Meyer, Kevin Robinson, Randall L. Sell, Robert Sember, Vincent M. B. Silenzio, Deborah J. Bowen, Judith Bradford, Esther Rothblum, Jocelyn White, Patricia Dunn, Ann Lawrence, Daniel Wolfe, and Jessica Xavier. “Lesbian, Gay, Bisexual, and Transgender Health: Findings and Concerns.” *Journal of the Gay and Lesbian Medical Association*. October 2000. < <http://www.glma.org/pub/jglma/vol4/3/index.html> > .

Garbo, Jon. “FDA Approves Combo Vaccine for Hepatitis A and B.” GayHealth Website. May 14, 2001. < <http://www.gayhealth.com/templates/99022462467343139648300002/news?record=553> > .

“Healthy People 2010 Companion Document for Lesbian, Gay, Bisexual, and Transgender (LGBT) Health.” Gay and Lesbian Medical Association. April 2001. < <http://www.glma.org/policy/hp2010/index.html> > .

Stéphanie Dionne
Teresa G. Odle

Gender identity disorder

Definition

The psychological diagnosis gender identity disorder (GID) is used to describe a male or female that feels a strong identification with the opposite sex and experiences considerable distress because of their actual sex.

Description

Gender identity disorder can affect children, adolescents, and adults. Individuals with gender identity disorder have strong cross-gender identification. They believe that they are, or should be, the opposite sex. They are uncomfortable with their sexual role and organs and may express a desire to alter their bodies. While not all persons with GID are labeled as transsexuals, there are those who are determined to undergo sex change procedures or have done so, and, therefore, are classified as transsexual. They often attempt to pass socially as the opposite sex. Transsexuals alter their physical appearance cosmetically and hormonally, and may eventually undergo a sex-change operation.

Children with gender identity disorder refuse to dress and act in sex-stereotypical ways. It is important to remember that many emotionally healthy children experience fantasies about being a member of the opposite sex. The distinction between these children and gender identity disordered children is that the latter experience significant interference in functioning because of their cross-gender identification. They may become severely depressed, anxious, or socially withdrawn.

Causes and symptoms

The cause of gender identity disorder is not known. It has been theorized that a prenatal hormonal imbalance may predispose individuals to the disorder. Problems in the individual's family interactions or family dynamics have also been postulated as having some causal impact.

The *Diagnostic and Statistical Manual of Mental Disorders*, Fourth Edition (*DSM-IV*), the diagnostic reference standard for United States mental health professionals, describes the criteria for gender identity disorder as an individual's strong and lasting cross-gender identification and their persistent discomfort with their biological gender role. This discomfort must

cause a significant amount of distress or impairment in the functioning of the individual.

DSM-IV specifies that children must display at least four of the following symptoms of cross-gender identification for a diagnosis of gender identity disorder:

- a repeatedly stated desire to be, or insistence that he or she is, the opposite sex
- a preference for cross-dressing
- a strong and lasting preference to play make-believe and role-playing games as a member of the opposite sex or persistent fantasies that he or she is the opposite sex
- a strong desire to participate in the stereotypical games of the opposite sex
- a strong preference for friends and playmates of the opposite sex

Diagnosis

Gender identity disorder is typically diagnosed by a psychiatrist or psychologist, who conducts an interview with the patient and takes a detailed social history. Family members may also be interviewed during the assessment process. This evaluation usually takes place in an outpatient setting.

Treatment

Treatment for children with gender identity disorder focuses on treating secondary problems such as depression and **anxiety**, and improving self-esteem. Treatment may also work on instilling positive identifications with the child's biological gender. Children typically undergo psychosocial therapy sessions; their parents may also be referred for family or individual therapy.

Transsexual adults often request hormone and surgical treatments to suppress their biological sex characteristics and acquire those of the opposite sex. A team of health professionals, including the treating psychologist or psychiatrist, medical doctors, and several surgical specialists, oversee this transitioning process. Because of the irreversible nature of the surgery, candidates for sex-change surgery are evaluated extensively and are often required to spend a period of time integrating themselves into the cross-gender role before the procedure begins. Counseling and peer support are also invaluable to transsexual individuals.

KEY TERMS

Cross-dressing—Dressing in clothing that is stereotypical of the opposite sex.

Gender identity disorder (GID)—A strong and lasting cross-gender identification and persistent discomfort with one's biological gender (sex) role. This discomfort must cause a significant amount of distress or impairment in the functioning of the individual.

Transsexual—A person with gender identity disorder who has an overwhelming desire to change anatomic sex; one who seeks hormonal or surgical treatment to change sex.

Prognosis

Long-term follow up studies have shown positive results for many transsexuals who have undergone sex-change surgery. However, significant social, personal, and occupational issues may result from surgical sex changes, and the patient may require psychotherapy or counseling.

Resources

ORGANIZATIONS

American Academy of Child and Adolescent Psychiatry (AACAP). 3615 Wisconsin Ave. NW, Washington, DC 20016. (202) 966-7300. <<http://www.aacap.org>>.

OTHER

The National Transgender Guide. <<http://www.tgguide.com>>.

Paula Anne Ford-Martin

Gender reassignment surgery

Definition

Also known as sex change surgery or sex reassignment surgery, gender reassignment surgery is a procedure that changes a person's external genital organs from those of one gender to those of the other.

Purpose

There are two reasons commonly given for altering the genital organs:

- Newborns with intersex deformities must early on be assigned to one sex or the other. These deformities represent intermediate stages between the primordial female genitals and the change into male caused by male hormone stimulation.
- Both men and women occasionally believe they are physically a different sex than they are mentally and emotionally. This dissonance is so profound that they are willing to be surgically altered. This condition is known as **gender identity disorder**; it is thought to be more common in men than in women.

Although technical considerations in the case of babies with ambiguous genitalia would seem to favor surgical conversion to female structures rather than to male, this conclusion has been challenged in the early 2000s. William G. Reiner, a child and adolescent psychiatrist, undertook a study of 29 genetic males who had been born with defective external genitalia and reared as females following surgery. He found that over half of them identified as males by adolescence in spite of the absence of male genitalia. Dr. Reiner has concluded that psychosexual development in genetic and hormonal males is heavily influenced by exposure to androgens prior to birth, and has called for a reevaluation of the practice of routinely assigning male babies with ambiguous genitalia to the female sex.

Precautions

Sexual identity is probably the most profound characteristic humans have. Assigning it must take place immediately after birth, both for the child's and the parents' comfort. Changing sexual identity may be the most significant change one can experience. It therefore should be done with every care and caution. By the time most adults come to surgery, they have lived for many years with dissonant identity. The average in one study was 29 years. Nevertheless, even then they may not be fully aware of the implications of becoming a member of the other sex.

Gender identity disorder has been officially defined by the American Psychiatric Association in the fourth edition of the *Diagnostic and Statistical Manual of Mental Disorders*, or DSM-IV, as a condition in which the affected individual has "a strong and persistent cross-gender identification" combined with "persistent discomfort about one's assigned sex." Because of widespread social disapproval of surgical gender reassignment, researchers do not know the true prevalence of gender identity disorders in the general population. Early estimates were 1:37,000 males and 1:107,000

females. A recent study in the Netherlands, however, maintains that a more accurate estimation is 1:11,900 males and 1:30,400 females. In any case, the number of surgical procedures is lower than the number of patients diagnosed with gender identity disorders.

Description

Converting male to female anatomy requires removal of the penis, reshaping genital tissue to appear more female, and constructing a vagina. A vagina can be successfully formed from a skin graft or an isolated loop of intestine.

Female to male surgery has achieved lesser success, due to the difficulty of building a functioning penis from the much smaller amount of erectile tissue available in the female genitals. Penis construction is not attempted less than a year after the preliminary surgery to remove the female organs. One study in Singapore found that a third of the patients would not undergo the surgery again. Nevertheless, they were all pleased with the change of sex. Besides the genital organs, the breasts need to be surgically altered for a more male appearance. This can be done quite successfully.

Orgasm, or at least a reasonable degree of erogenous sensitivity, can be experienced by patients after surgery.

Preparation

As of the early 2000s, patients requesting gender reassignment surgery must undergo a lengthy process of physical and psychological evaluation before receiving approval for surgery. The Harry Benjamin International Gender Dysphoria Association (HBIGDA), which is presently the largest worldwide professional association dealing with the treatment of gender identity disorders, has published standards of care that are followed by most surgeons who perform genital surgery for gender reassignment. HBIGDA stipulates that a patient must meet the diagnostic criteria for gender identity disorders as defined by either the Diagnostic and Statistical Manual of Mental Disorders, fourth edition (DSM-IV) or the International Classification of Diseases–10 (ICD-10).

Preparation for gender reassignment surgery includes a period of several years of hormonal treatment to change the body contours, vocal pitch, patterns of facial and body hair, etc. to those of the other sex. As the patient's body and voice begin to change, he or she is also asked to practice dressing and acting as a member of that sex.

KEY TERMS

Chromosomes—The carriers of genes, which determine sex and characteristics.

Gender identity disorder (GID)—A mental disorder in which a person strongly identifies with the other sex and feels uncomfortable with his or her biological sex. It occurs more often in males than in females.

gender reassignment surgery—The surgical alteration and reconstruction of a person's sex organs to resemble those of the other sex as closely as possible. It is sometimes called gender reassignment surgery.

There is some controversy as of the early 2000s regarding the appropriateness of performing gender reassignment surgery on adolescents, as opposed to either infants or older adults, on the grounds that teenagers are often uncertain about many aspects of their sexuality. The recent case of a thirteen-year-old girl who requested gender reassignment surgery in Australia reflects the intensity of this controversy among surgeons as well as members of the general public.

Aftercare

Social support, particularly from the family, is important for readjustment as a member of the opposite sex. If patients were socially or emotionally unstable before the operation, over 30, or had an unsuitable body build for the new sex, they tend to do poorly. Paul McHugh, the former head of the department of psychiatry at Johns Hopkins University, has reported that people who think that gender reassignment surgery will solve all their emotional and social problems are likely to be disappointed. “[These patients] had much the same problems with relationships, work, and emotions as before. The hope that they would emerge now from their emotional difficulties to flourish psychologically had not been fulfilled.”

Risks

All surgery runs the risk of infection, bleeding, and a need to return for repairs. This surgery is irreversible, so the patient must have no doubts about accepting the results.

The most common physical complication of the male to female surgery is narrowing of the new vagina.

There is some risk of patients developing other mental disorders associated with their new sex. Several cases of **anorexia nervosa**, an eating disorder that is far more common in women than in men, have been reported in males who underwent gender reassignment surgery.

Resources

BOOKS

American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders*. 4th ed., revised. Washington, DC: American Psychiatric Association, 2000.

Beers, Mark H., MD, and Robert Berkow, MD., editors. "Psychosexual Disorders." Section 15, Chapter 192 In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2002.

PERIODICALS

McHugh, Paul, MD. "Surgical Sex." *First Things* 147 (November 2004): 34–38.

Reiner, W. G., and J. P. Gearhart. "Discordant Sexual Identity in Some Genetic Males with Cloacal Exstrophy Assigned to Female Sex at Birth." *New England Journal of Medicine* 350 (January 22, 2004): 333–341.

Reiner, W. G. "Psychosexual Development in Genetic Males Assigned Female: The Cloacal Exstrophy Experience." *Child and Adolescent Psychiatry Clinics of North America* 13 (July 2004): 657–674.

Spriggs, M. P. "Ethics and the Proposed Treatment for a 13-Year-Old with Atypical Gender Identity." *Medical Journal of Australia* 181 (September 20, 2004): 319–321.

Winston, A. P., S. Acharya, S. Chaudhuri, and L. Fellowes. "Anorexia Nervosa and Gender Identity Disorder in Biologic Males: A Report of Two Cases." *International Journal of Eating Disorders* 36 (July 2004): 109–113.

ORGANIZATIONS

American Board of Urology (ABU). 2216 Ivy Road, Suite 210, Charlottesville, VA 22903. (434) 979-0059. <<http://www.abu.org>>.

American Psychiatric Association (APA). 1400 K Street, NW, Washington, DC 20005. (888) 357-7924. <<http://www.psych.org>>.

Harry Benjamin International Gender Dysphoria Association, Inc. (HBI-GDA). 1300 South Second Street, Suite 180, Minneapolis, MN 55454. (612) 625-1500. <<http://www.hbigda.org>>.

OTHER

Harry Benjamin International Gender Dysphoria Association (HBI-GDA). *Standards of Care for Gender Identity Disorders*, 6th version. February 2001. <<http://www.hbigda.org/socv6.html>>.

J. Ricker Polsdorfer, MD
Rebecca J. Frey, PhD

Gene therapy

Gene therapy is a rapidly growing field of medicine in which genes are introduced into the body to treat diseases. Genes control heredity and provide the basic biological code for determining a cell's specific functions. Gene therapy seeks to provide genes that correct or supplant the disease-controlling functions of cells that are not, in essence, doing their job. Somatic gene therapy introduces therapeutic genes at the tissue or cellular level to treat a specific individual. Germ-line gene therapy inserts genes into reproductive cells or possibly into embryos to correct genetic defects that could be passed on to future generations. Initially conceived as an approach for treating inherited diseases, like **cystic fibrosis** and Huntington's disease, the scope of potential gene therapies has grown to include treatments for cancers, arthritis, and infectious diseases. Although gene therapy testing in humans has advanced rapidly, many questions surround its use. For example, some scientists are concerned that the therapeutic genes themselves may cause disease. Others fear that germ-line gene therapy may be used to control human development in ways not connected with disease, like intelligence or appearance.

The biological basis of gene therapy

Gene therapy has grown out of the science of genetics or how heredity works. Scientists know that life begins in a cell, the basic building block of all multicellular organisms. Humans, for instance, are made up of trillions of cells, each performing a specific function. Within the cell's nucleus (the center part of a cell that regulates its chemical functions) are pairs of chromosomes. These threadlike structures are made up of a single molecule of DNA (deoxyribonucleic acid), which carries the blueprint of life in the form of codes, or genes, that determine inherited characteristics.

A DNA molecule looks like two ladders with one of the sides taken off both and then twisted around each other. The rungs of these ladders meet (resulting in a spiral staircase-like structure) and are called base pairs. Base pairs are made up of nitrogen molecules and arranged in specific sequences. Millions of these base pairs, or sequences, can make up a single gene, specifically defined as a segment of the chromosome and DNA that contains certain hereditary information. The gene, or combination of genes formed by these base pairs ultimately direct an organism's growth and characteristics through the production of

certain chemicals, primarily proteins, which carry out most of the body's chemical functions and biological reactions.

Scientists have long known that alterations in genes present within cells can cause inherited diseases like cystic fibrosis, sickle-cell anemia, and **hemophilia**. Similarly, errors in the total number of chromosomes can cause conditions such as **Down syndrome** or Turner's syndrome. As the study of genetics advanced, however, scientists learned that an altered genetic sequence also can make people more susceptible to diseases, like **atherosclerosis**, **cancer**, and even **schizophrenia**. These diseases have a genetic component, but also are influenced by environmental factors (like diet and lifestyle). The objective of gene therapy is to treat diseases by introducing functional genes into the body to alter the cells involved in the disease process by either replacing missing genes or providing copies of functioning genes to replace nonfunctioning ones. The inserted genes can be naturally-occurring genes that produce the desired effect or may be genetically engineered (or altered) genes.

Scientists have known how to manipulate a gene's structure in the laboratory since the early 1970s through a process called gene splicing. The process involves removing a fragment of DNA containing the specific genetic sequence desired, then inserting it into the DNA of another gene. The resultant product is called recombinant DNA and the process is genetic engineering.

There are basically two types of gene therapy. Germ-line gene therapy introduces genes into reproductive cells (sperm and eggs) or someday possibly into embryos in hopes of correcting genetic abnormalities that could be passed on to future generations. Most of the current work in applying gene therapy, however, has been in the realm of somatic gene therapy. In this type of gene therapy, therapeutic genes are inserted into tissue or cells to produce a naturally occurring protein or substance that is lacking or not functioning correctly in an individual patient.

Viral vectors

In both types of therapy, scientists need something to transport either the entire gene or a recombinant DNA to the cell's nucleus, where the chromosomes and DNA reside. In essence, vectors are molecular delivery trucks. One of the first and most popular vectors developed were viruses because they invade cells as part of the natural infection process. Viruses have the potential to be excellent vectors because they have a specific relationship with the host in that they colonize certain cell types and tissues in specific organs. As a

result, vectors are chosen according to their attraction to certain cells and areas of the body.

One of the first vectors used was retroviruses. Because these viruses are easily cloned (artificially reproduced) in the laboratory, scientists have studied them extensively and learned a great deal about their biological action. They also have learned how to remove the genetic information that governs viral replication, thus reducing the chances of infection.

Retroviruses work best in actively dividing cells, but cells in the body are relatively stable and do not divide often. As a result, these cells are used primarily for *ex vivo* (outside the body) manipulation. First, the cells are removed from the patient's body, and the virus, or vector, carrying the gene is inserted into them. Next, the cells are placed into a nutrient culture where they grow and replicate. Once enough cells are gathered, they are returned to the body, usually by injection into the blood stream. Theoretically, as long as these cells survive, they will provide the desired therapy.

Another class of viruses, called the adenoviruses, also may prove to be good gene vectors. These viruses can effectively infect nondividing cells in the body, where the desired gene product then is expressed naturally. In addition to being a more efficient approach to gene transportation, these viruses, which cause respiratory infections, are more easily purified and made stable than retroviruses, resulting in less chance of an unwanted viral infection. However, these viruses live for several days in the body, and some concern surrounds the possibility of infecting others with the viruses through sneezing or coughing. Other viral vectors include **influenza** viruses, Sindbis virus, and a herpes virus that infects nerve cells.

Scientists also have delved into nonviral vectors. These vectors rely on the natural biological process in which cells uptake (or gather) macromolecules. One approach is to use liposomes, globules of fat produced by the body and taken up by cells. Scientists also are investigating the introduction of raw recombinant DNA by injecting it into the bloodstream or placing it on microscopic beads of gold shot into the skin with a "gene-gun." Another possible vector under development is based on dendrimer molecules. A class of polymers (naturally occurring or artificial substances that have a high molecular weight and formed by smaller molecules of the same or similar substances), is "constructed" in the laboratory by combining these smaller molecules. They have been used in manufacturing Styrofoam, polyethylene cartons, and Plexiglass. In the laboratory, dendrimers have shown the ability to

transport genetic material into human cells. They also can be designed to form an affinity for particular cell membranes by attaching to certain sugars and protein groups.

The history of gene therapy

In the early 1970s, scientists proposed “gene surgery” for treating inherited diseases caused by faulty genes. The idea was to take out the disease-causing gene and surgically implant a gene that functioned properly. Although sound in theory, scientists, then and now, lack the biological knowledge or technical expertise needed to perform such a precise surgery in the human body.

However, in 1983, a group of scientists from Baylor College of Medicine in Houston, Texas, proposed that gene therapy could one day be a viable approach for treating Lesch-Nyhan disease, a rare neurological disorder. The scientists conducted experiments in which an enzyme-producing gene (a specific type of protein) for correcting the disease was injected into a group of cells for replication. The scientists theorized the cells could then be injected into people with Lesch-Nyhan disease, thus correcting the genetic defect that caused the disease.

As the science of genetics advanced throughout the 1980s, gene therapy gained an established foothold in the minds of medical scientists as a promising approach to treatments for specific diseases. One of the major reasons for the growth of gene therapy was scientists’ increasing ability to identify the specific genetic malfunctions that caused inherited diseases. Interest grew as further studies of DNA and chromosomes (where genes reside) showed that specific genetic abnormalities in one or more genes occurred in successive generations of certain family members who suffered from diseases like intestinal cancer, **bipolar disorder**, **Alzheimer’s disease**, heart disease, diabetes, and many more. Although the genes may not be the only cause of the disease in all cases, they may make certain individuals more susceptible to developing the disease because of environmental influences, like **smoking**, pollution, and **stress**. In fact, some scientists theorize that all diseases may have a genetic component.

On September 14, 1990, a four-year old girl suffering from a genetic disorder that prevented her body from producing a crucial enzyme became the first person to undergo gene therapy in the United States. Because her body could not produce adenosine deaminase (ADA), she had a weakened immune system, making her extremely susceptible to severe, life-threatening

infections. W. French Anderson and colleagues at the National Institutes of Health’s Clinical Center in Bethesda, Maryland, took white blood cells (which are crucial to proper immune system functioning) from the girl, inserted ADA producing genes into them, and then transfused the cells back into the patient. Although the young girl continued to show an increased ability to produce ADA, debate arose as to whether the improvement resulted from the gene therapy or from an additional drug treatment she received.

Nevertheless, a new era of gene therapy began as more and more scientists sought to conduct clinical trial (testing in humans) research in this area. In that same year, gene therapy was tested on patients suffering from melanoma (skin cancer). The goal was to help them produce antibodies (disease fighting substances in the immune system) to battle the cancer.

These experiments have spawned an ever growing number of attempts at gene therapies designed to perform a variety of functions in the body. For example, a gene therapy for cystic fibrosis aims to supply a gene that alters cells, enabling them to produce a specific protein to battle the disease. Another approach was used for brain cancer patients, in which the inserted gene was designed to make the cancer cells more likely to respond to drug treatment. Another gene therapy approach for patients suffering from artery blockage, which can lead to strokes, induces the growth of new blood vessels near clogged arteries, thus ensuring normal blood circulation.

Currently, there are a host of new gene therapy agents in clinical trials. In the United States, both nucleic acid based (*in vivo*) treatments and cell-based (*ex vivo*) treatments are being investigated. Nucleic acid based gene therapy uses vectors (like viruses) to deliver modified genes to target cells. Cell-based gene therapy techniques remove cells from the patient in order to genetically alter them then reintroduce them to the patient’s body. Presently, gene therapies for the following diseases are being developed: cystic fibrosis (using adenoviral vector), HIV infection (cell-based), **malignant melanoma** (cell-based), Duchenne **muscular dystrophy** (cell-based), hemophilia B (cell-based), **kidney cancer** (cell-based), Gaucher’s Disease (retroviral vector), **breast cancer** (retroviral vector), and lung cancer (retroviral vector). When a cell or individual is treated using gene therapy and successful incorporation of engineered genes has occurred, the cell or individual is said to be *transgenic*.

The medical establishment’s contribution to transgenic research has been supported by increased government funding. In 1991, the U.S. government



Early detection of cancer. The researcher's pen marks a band on a DNA sequencing autoradiogram confirming a bladder cancer.
(Custom Medical Stock Photo. Reproduced by permission.)

provided \$58 million for gene therapy research, with increases in funding of \$15-40 million dollars a year over the following four years. With fierce competition over the promise of societal benefit in addition to huge profits, large pharmaceutical corporations have moved to the forefront of transgenic research. In an effort to be first in developing new therapies, and armed with billions of dollars of research funds, such corporations are making impressive strides toward making gene therapy a viable reality in the treatment of once elusive diseases.

Diseases targeted for treatment by gene therapy

The potential scope of gene therapy is enormous. More than 4,200 diseases have been identified as resulting directly from abnormal genes, and countless others that may be partially influenced by a person's genetic makeup. Initial research has concentrated on developing gene therapies for diseases whose genetic origins have been established and for other diseases that can be cured or improved by substances genes produce.

The following are examples of potential gene therapies. People suffering from cystic fibrosis lack a gene needed to produce a salt-regulating protein. This protein regulates the flow of chloride into epithelial cells, (the cells that line the inner and outer skin layers) that cover the air passages of the nose and lungs. Without this regulation, patients with cystic fibrosis build up a thick mucus that makes them prone to lung infections. A gene therapy technique to correct this abnormality might employ an adenovirus to transfer a normal copy of what scientists call the cystic fibrosis transmembrane conductance regulator, or CTRF, gene. The gene is introduced into the patient by spraying it into the nose or lungs. Researchers announced in 2004 that they had, for the first time, treated a dominant neurodegenerative disease called Spinocerebellar ataxia type 1, with gene therapy. This could lead to treating similar diseases such as Huntington's disease. They also announced a single intravenous injection could deliver therapy to all muscles, perhaps providing hope to people with muscular dystrophy.

Familial **hypercholesterolemia** (FH) also is an inherited disease, resulting in the inability to process

cholesterol properly, which leads to high levels of artery-clogging fat in the blood stream. Patients with FH often suffer heart attacks and strokes because of blocked arteries. A gene therapy approach used to battle FH is much more intricate than most gene therapies because it involves partial surgical removal of patients' livers (*ex vivo* transgene therapy). Corrected copies of a gene that serve to reduce cholesterol build-up are inserted into the liver sections, which then are transplanted back into the patients.

Gene therapy also has been tested on patients with **AIDS**. AIDS is caused by the human **immunodeficiency** virus (HIV), which weakens the body's immune system to the point that sufferers are unable to fight off diseases like pneumonias and cancer. In one approach, genes that produce specific HIV proteins have been altered to stimulate immune system functioning without causing the negative effects that a complete HIV molecule has on the immune system. These genes are then injected in the patient's blood stream. Another approach to treating AIDS is to insert, via white blood cells, genes that have been genetically engineered to produce a receptor that would attract HIV and reduce its chances of replicating. In 2004, researchers reported that had developed a new vaccine concept for HIV, but the details were still in development.

Several cancers also have the potential to be treated with gene therapy. A therapy tested for melanoma, or skin cancer, involves introducing a gene with an anticancer protein called tumor necrosis factor (TNF) into test tube samples of the patient's own cancer cells, which are then reintroduced into the patient. In brain cancer, the approach is to insert a specific gene that increases the cancer cells' susceptibility to a common drug used in fighting the disease. In 2003, researchers reported that they had harnessed the cell killing properties of adenoviruses to treat **prostate cancer**. A 2004 report said that researchers had developed a new DNA vaccine that targeted the proteins expressed in **cervical cancer** cells.

Gaucher disease is an inherited disease caused by a mutant gene that inhibits the production of an enzyme called glucocerebrosidase. Patients with Gaucher disease have enlarged livers and spleens and eventually their bones deteriorate. Clinical gene therapy trials focus on inserting the gene for producing this enzyme.

Gene therapy also is being considered as an approach to solving a problem associated with a surgical procedure known as balloon **angioplasty**. In this procedure, a stent (in this case, a type of tubular scaffolding) is used to open the clogged artery. However, in response to the trauma of the stent insertion, the body

initiates a natural healing process that produces too many cells in the artery and results in restenosis, or reclosing of the artery. The gene therapy approach to preventing this unwanted side effect is to cover the outside of the stents with a soluble gel. This gel contains vectors for genes that reduce this overactive healing response.

Regularly throughout the past decade, and no doubt over future years, scientists have and will come up with new possible ways for gene therapy to help treat human disease. Recent advancements include the possibility of reversing **hearing loss** in humans with experimental growing of new sensory cells in adult guinea pigs, and avoiding **amputation** in patients with severe circulatory problems in their legs with angiogenic growth factors.

The Human Genome Project

Although great strides have been made in gene therapy in a relatively short time, its potential usefulness has been limited by lack of scientific data concerning the multitude of functions that genes control in the human body. For instance, it is now known that the vast majority of genetic material does not store information for the creation of proteins, but rather is involved in the control and regulation of gene expression, and is, thus, much more difficult to interpret. Even so, each individual cell in the body carries thousands of genes coding for proteins, with some estimates as high as 150,000 genes. For gene therapy to advance to its full potential, scientists must discover the biological role of each of these individual genes and where the base pairs that make them up are located on DNA.

To address this issue, the National Institutes of Health initiated the Human Genome Project in 1990. Led by James D. Watson (one of the co-discoverers of the chemical makeup of DNA) the project's 15-year goal is to map the entire human genome (a combination of the words gene and chromosomes). A genome map would clearly identify the location of all genes as well as the more than three billion base pairs that make them up. With a precise knowledge of gene locations and functions, scientists may one day be able to conquer or control diseases that have plagued humanity for centuries.

Scientists participating in the Human Genome Project identified an average of one new gene a day, but many expected this rate of discovery to increase. By the year 2005, their goal was to determine the exact location of all the genes on human DNA and the exact sequence of the base pairs that make them up. Some of

the genes identified through this project include a gene that predisposes people to **obesity**, one associated with programmed cell death (apoptosis), a gene that guides HIV viral reproduction, and the genes of inherited disorders like Huntington's disease, Lou Gehrig's disease, and some colon and breast cancers. In April 2003, the finished sequence was announced, with 99% of the human genome's gene-containing regions mapped to an accuracy of 99.9%.

The future of gene therapy

Gene therapy seems elegantly simple in its concept: supply the human body with a gene that can correct a biological malfunction that causes a disease. However, there are many obstacles and some distinct questions concerning the viability of gene therapy. For example, viral vectors must be carefully controlled lest they infect the patient with a viral disease. Some vectors, like retroviruses, also can enter cells functioning properly and interfere with the natural biological processes, possibly leading to other diseases. Other viral vectors, like the adenoviruses, often are recognized and destroyed by the immune system so their therapeutic effects are short-lived. Maintaining gene expression so it performs its role properly after vector delivery is difficult. As a result, some therapies need to be repeated often to provide long-lasting benefits.

One of the most pressing issues, however, is gene regulation. Genes work in concert to regulate their functioning. In other words, several genes may play a part in turning other genes on and off. For example, certain genes work together to stimulate cell division and growth, but if these are not regulated, the inserted genes could cause tumor formation and cancer. Another difficulty is learning how to make the gene go into action only when needed. For the best and safest therapeutic effort, a specific gene should turn on, for example, when certain levels of a protein or enzyme are low and must be replaced. But the gene also should remain dormant when not needed to ensure it doesn't oversupply a substance and disturb the body's delicate chemical makeup.

One approach to gene regulation is to attach other genes that detect certain biological activities and then react as a type of automatic off-and-on switch that regulates the activity of the other genes according to biological cues. Although still in the rudimentary stages, researchers are making headway in inhibiting some gene functioning by using a synthetic DNA to block gene transcriptions (the copying of genetic information). This approach may have implications for gene therapy.

The ethics of gene therapy

While gene therapy holds promise as a revolutionary approach to treating disease, ethical concerns over its use and ramifications have been expressed by scientists and lay people alike. For example, since much needs to be learned about how these genes actually work and their long-term effect, is it ethical to test these therapies on humans, where they could have a disastrous result? As with most clinical trials concerning new therapies, including many drugs, the patients participating in these studies usually have not responded to more established therapies and often are so ill the novel therapy is their only hope for long-term survival.

Another questionable outgrowth of gene therapy is that scientists could possibly manipulate genes to genetically control traits in human offspring that are not health related. For example, perhaps a gene could be inserted to ensure that a child would not be bald, a seemingly harmless goal. However, what if genetic manipulation was used to alter skin color, prevent homosexuality, or ensure good looks? If a gene is found that can enhance intelligence of children who are not yet born, will everyone in society, the rich and the poor, have access to the technology or will it be so expensive only the elite can afford it?

The Human Genome Project, which plays such an integral role for the future of gene therapy, also has social repercussions. If individual genetic codes can be determined, will such information be used against people? For example, will someone more susceptible to a disease have to pay higher insurance premiums or be denied health insurance altogether? Will employers discriminate between two potential employees, one with a "healthy" genome and the other with genetic abnormalities?

Some of these concerns can be traced back to the eugenics movement popular in the first half of the twentieth century. This genetic "philosophy" was a societal movement that encouraged people with "positive" traits to reproduce while those with less desirable traits were sanctioned from having children. Eugenics was used to pass strict immigration laws in the United States, barring less suitable people from entering the country lest they reduce the quality of the country's collective gene pool. Probably the most notorious example of eugenics in action was the rise of Nazism in Germany, which resulted in the Eugenic Sterilization Law of 1933. The law required sterilization for those suffering from certain disabilities and even for some who were simply deemed "ugly." To ensure that this novel science is not abused, many governments have

KEY TERMS

Cell—The smallest living unit of the body that groups together to form tissues and help the body perform specific functions.

Chromosome—A microscopic thread-like structure found within each cell of the body, consisting of a complex of proteins and DNA. Humans have 46 chromosomes arranged into 23 pairs. Changes in either the total number of chromosomes or their shape and size (structure) may lead to physical or mental abnormalities.

Clinical trial—The testing of a drug or some other type of therapy in a specific population of patients.

Clone—A cell or organism derived through asexual (without sex) reproduction containing the identical genetic information of the parent cell or organism.

Deoxyribonucleic acid (DNA)—The genetic material in cells that holds the inherited instructions for growth, development, and cellular functioning.

Embryo—The earliest stage of development of a human infant, usually used to refer to the first eight weeks of pregnancy. The term *fetus* is used from roughly the third month of pregnancy until delivery.

Enzyme—A protein that causes a biochemical reaction or change without changing its own structure or function.

Eugenics—A social movement in which the population of a society, country, or the world is to be improved by controlling the passing on of hereditary information through mating.

Gene—A building block of inheritance, which contains the instructions for the production of a particular

protein, and is made up of a molecular sequence found on a section of DNA. Each gene is found on a precise location on a chromosome.

Gene transcription—The process by which genetic information is copied from DNA to RNA, resulting in a specific protein formation.

Genetic engineering—The manipulation of genetic material to produce specific results in an organism.

Genetics—The study of hereditary traits passed on through the genes.

Germ-line gene therapy—The introduction of genes into reproductive cells or embryos to correct inherited genetic defects that can cause disease.

Liposome—Fat molecule made up of layers of lipids.

Macromolecules—A large molecule composed of thousands of atoms.

Nitrogen—A gaseous element that makes up the base pairs in DNA.

Nucleus—The central part of a cell that contains most of its genetic material, including chromosomes and DNA.

Protein—Important building blocks of the body, composed of amino acids, involved in the formation of body structures and controlling the basic functions of the human body.

Somatic gene therapy—The introduction of genes into tissue or cells to treat a genetic related disease in an individual.

Vectors—Something used to transport genetic information to a cell.

established organizations specifically for overseeing the development of gene therapy. In the United States, the Food and Drug Administration (FDA) and the National Institutes of Health require scientists to take a precise series of steps and meet stringent requirements before proceeding with clinical trials. As of mid-2004, more than 300 companies were carrying out gene medicine developments and 500 clinical trials were underway. How to deliver the therapy is the key to unlocking many of the researchers' discoveries.

In fact, gene therapy has been immersed in more controversy and surrounded by more scrutiny in both the health and ethical arena than most other technologies (except, perhaps, for cloning) that promise to

substantially change society. Despite the health and ethical questions surrounding gene therapy, the field will continue to grow and is likely to change medicine faster than any previous medical advancement.

Resources

PERIODICALS

Abella, Harold. "Gene Therapy May Save Limbs."

Diagnostic Imaging (May 1, 2003): 16.

Christensen R. "Cutaneous Gene Therapy—An Update."

Histochemical Cell Biology (January 2001): 73-82.

"Gene Therapy Important Part of Cancer Research."

Cancer Gene Therapy Week (June 30, 2003): 12.

"Initial Sequencing and Analysis of the Human Genome."

Nature (February 15, 2001): 860-921.

- Kingsman, Alan. "Gene Therapy Moves On." *SCRIP World Pharmaceutical News* (July 7, 2004): 19;ndash;21.
- Nevin, Norman. "What Has Happened to Gene Therapy?" *European Journal of Pediatrics* (2000): S240-S242.
- "New DNA Vaccine Targets Proteins Expressed in Cervical Cancer Cells." *Gene Therapy Weekly* (September 9, 2004): 14.
- "New Research on the Progress of Gene Therapy Presented at Meeting." *Obesity, Fitness & Wellness Week* (July 3, 2004): 405.
- Pekkanen, John. "Genetics: Medicine's Amazing Leap." *Readers Digest* (September 1991): 23-32.
- Silverman, Jennifer, and Steve Perlstein. "Genome Project Completed." *Family Practice News* (May 15, 2003): 50-51.
- "Study Highlights Potential Danger of Gene Therapy." *Drug Week* (June 20, 2003): 495.
- "Study May Help Scientists Develop Safer Mthods for Gene Therapy." *AIDS Weekly* (June 30, 2003): 32.
- Trabis, J. "With Gene Therapy, Ears Grow New Sensory Cells." *Science News* (June 7, 2003): 355.

ORGANIZATIONS

National Human Genome Research Institute. The National Institutes of Health. 9000 Rockville Pike, Bethesda, MD 20892. (301) 496-2433. <<http://www.nhgri.nih.gov>> .

OTHER

Online Mendelian Inheritance in Man. Online genetic testing information sponsored by National Center for Biotechnology Information. <<http://www.ncbi.nlm.nih.gov/Omim/>> .

Katherine S. Hunt, MS
Teresa G. Odle

General adaptation syndrome

Definition

General adaptation syndrome, or GAS, is a term used to describe the body's short-term and long-term reactions to **stress**.

Stressors in humans include such physical stressors as **starvation**, being hit by a car, or suffering through severe weather. Additionally, humans can suffer such emotional or mental stressors as the loss of a loved one, the inability to solve a problem, or even having a difficult day at work.

Description

Originally described by Hans Selye (1907–1982), an Austrian-born physician who emigrated to Canada in 1939, the general adaptation syndrome represents a three-stage reaction to stress. Selye explained his choice of terminology as follows: "I call this syndrome *general* because it is produced only by agents which have a general effect upon large portions of the body. I call it *adaptive* because it stimulates defense. . . . I call it a *syndrome* because its individual manifestations are coordinated and even partly dependent upon each other."

Selye thought that the general adaptation syndrome involved two major systems of the body, the nervous system and the endocrine (or hormonal) system. He then went on to outline what he considered as three distinctive stages in the syndrome's evolution. He called these stages the alarm reaction (AR), the stage of resistance (SR), and the stage of exhaustion (SE).

Stage 1: Alarm reaction (AR)

The first stage of the general adaptation stage, the alarm reaction, is the immediate reaction to a stressor. In the initial phase of stress, humans exhibit a "fight or flight" response, which prepares the body for physical activity. However, this initial response can also decrease the effectiveness of the immune system, making persons more susceptible to illness during this phase.

Stage 2: Stage of resistance (SR)

Stage 2 might also be named the stage of adaptation, instead of the stage of resistance. During this phase, if the stress continues, the body adapts to the stressors it is exposed to. Changes at many levels take place in order to reduce the effect of the stressor. For example, if the stressor is starvation (possibly due to anorexia), the person might experienced a reduced desire for physical activity to conserve energy, and the absorption of nutrients from food might be maximized.

Stage 3: Stage of exhaustion (SE)

At this stage, the stress has continued for some time. The body's resistance to the stress may gradually be reduced, or may collapse quickly. Generally, this means the immune system, and the body's ability to resist disease, may be almost totally eliminated. Patients who experience long-term stress may succumb to heart attacks or severe infection due to their reduced immunity. For example, a person with a stressful job may experience long-term stress that might lead to high blood pressure and an eventual **heart attack**.

Stress, a useful reaction?

The reader should note that Dr. Selye did not regard stress as a purely negative phenomenon; in fact, he frequently pointed out that stress is not only an inevitable part of life but results from intense joy or pleasure as well as fear or **anxiety**. “Stress is not even necessarily bad for you; it is also the spice of life, for any emotion, any activity, causes stress.” Some later researchers have coined the term “eustress” or pleasant stress, to reflect the fact that such positive experiences as a job promotion, completing a degree or training program, marriage, travel, and many others are also stressful.

Selye also pointed out that human perception of and response to stress is highly individualized; a job or sport that one person finds anxiety-provoking or exhausting might be quite appealing and enjoyable to someone else. Looking at one’s responses to specific stressors can contribute to better understanding of one’s particular physical, emotional, and mental resources and limits.

Causes and symptoms

Stress is one cause of general adaptation syndrome. The results of unrelieved stress can manifest as **fatigue**, irritability, difficulty concentrating, and difficulty sleeping. Persons may also experience other symptoms that are signs of stress. Persons experiencing unusual symptoms, such as hair loss, without another medical explanation might consider stress as the cause.

The general adaptation syndrome is also influenced by such universal human variables as overall health and nutritional status, sex, age, ethnic or racial background, level of education, socioeconomic status (SES), genetic makeup, etc. Some of these variables are biologically based and difficult or impossible to change. For example, recent research indicates that men and women respond somewhat differently to stress, with women being more likely to use what is called the “tend and befriend” response rather than the classical “fight or flight” pattern. These researchers note that most of the early studies of the effects of stress on the body were conducted with only male subjects.

Selye’s observation that people vary in their perceptions of stressors was reflected in his belief that the stressors themselves are less dangerous to health than people’s maladaptive responses to them. He categorized certain diseases, ranging from cardiovascular disorders to inflammatory diseases and mental disorders as “diseases of adaptation,” regarding them as “largely due to errors in our adaptive response to stress” rather than the direct result of such outside factors as germs, toxic substances, etc.

Diagnosis

GAS by itself is not an official diagnostic category but rather a descriptive term. A person who consults a doctor for a stress-related physical illness may be scheduled for blood or urine tests to measure the level of cortisol or other stress-related hormones in their body, or imaging studies to evaluate possible abnormalities in their endocrine glands if the doctor thinks that these tests may help to establish or confirm a diagnosis.

The American Psychiatric Association (APA) recognizes stress as a factor in **anxiety disorders**, particularly **post-traumatic stress disorder** (PTSD) and **acute stress disorder** (ASD). These two disorders are defined as symptomatic reactions to extreme traumatic stressors (war, natural or transportation disasters, criminal assault, **abuse**, hostage situations, etc.) and differ chiefly in the time frame in which the symptoms develop. The APA also has a diagnostic category of **adjustment disorders**, which are characterized either by excessive reactions to stressors within the normal range of experience (e.g. academic examinations, relationship breakups, being fired from a job) or by significant impairment in the person’s occupational or social functioning.

Treatment

Treatment of stress-related illnesses typically involves one or more **stress reduction** strategies. Stress reduction strategies generally fall into one of three categories: avoiding stressors; changing one’s reaction to the stressor(s); or relieving stress after the reaction to the stressor(s). Many mainstream as well as complementary or alternative (CAM) strategies for stress reduction, such as exercising, listening to music, **aromatherapy**, and massage relieve stress after it occurs.

Many psychotherapeutic approaches attempt to modify the patient’s reactions to stressors. These approaches often include an analysis of the patient’s individual patterns of response to stress; for example, one commonly used set of categories describes people as “speed freaks,” “worry warts,” “cliff walkers,” “loners,” “basket cases,” and “drifters.” Each pattern has a recommended set of skills that the patient is encouraged to work on; for example, worry **warts** are advised to reframe their anxieties and then identify their core values and goals in order to take concrete action about their worries. In general, persons wishing to improve their management of stress should begin by consulting a medical professional with whom they feel comfortable to discuss which option, or combination of options, they can use.

KEY TERMS

Adjustment disorder—A disorder defined by the development of significant emotional or behavioral symptoms in response to a stressful event or series of events within the normal range of human experience.

Cortisol—A steroid hormone released by the cortex (outer portion) of the adrenal gland when a person is under stress.

Eustress—A term that is sometimes used to refer to positive stress.

Stress management—A set of techniques and programs intended to help people deal more effectively with stress in their lives by analyzing the specific stressors and taking positive actions to minimize their effects. Most stress management programs deal with job stress and workplace issues.

Stressor—A stimulus or event that provokes a stress response in an organism. Stressors can be categorized as acute or chronic, and as external or internal to the organism.

Selye himself recommended an approach to stress that he described as “living wisely in accordance with natural laws.” In his now-classic book *The Stress of Life* (1956), he discussed the following as important dimensions of living wisely:

- Adopting an attitude of gratitude toward life rather than seeking revenge for injuries or slights.
- Acting toward others from altruistic rather than self-centered motives.
- Retaining a capacity for wonder and delight in the genuinely good and beautiful things in life.
- Finding a purpose for one’s life and expressing one’s individuality in fulfilling that purpose.
- Keeping a healthy sense of modesty about one’s goals or achievements.

Resources

BOOKS

- American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders*. 4th ed., revised. Washington, D.C.: American Psychiatric Association, 2000.
- Beers, Mark H., MD, and Robert Berkow, MD, editors. “Psychiatry in Medicine.” Section 15, Chapter 1 85 *In The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

PERIODICALS

- Benton, Tami D., MD, and Jacqueline Lynch, MSW. “Adjustment Disorders.” *eMedicine* September 3, 2004. < <http://emedicine.com/med/topic3348.htm> > .
- Cosen-Binker, L. I., M. G. Binker, G. Negri, and O. Tiscornia. “Influence of Stress in Acute Pancreatitis and Correlation with Stress-Induced Gastric Ulcer.” *Pancreatology* 4 (July 2004): 470–484.
- Motzer, S. A., and V. Hertig. “Stress, Stress Response, and Health.” *Nursing Clinics of North America* 39 (March 2004): 1–17.
- Yates, William R., MD. “Anxiety Disorders.” *eMedicine* August 15, 2004. < <http://emedicine.com/med/topic152.htm> > .

ORGANIZATIONS

- American Institute of Stress. 124 Park Avenue, Yonkers, NY 10703 (914) 963-1200. Fax: (914) 965-6267. < <http://www.stress.org> > .
- American Psychiatric Association. 1400 K Street NW, Washington DC 20005. (888) 357-7924. < <http://www.psych.org> > .
- Canadian Institute of Stress/Hans Selye Foundation. Medcan Clinic Office, Suite 1500, 150 York Street, Toronto, Ontario, Canada M5H 3S5. (416) 236-4218. < <http://www.stresscanada.org> > .
- National Institute of Mental Health (NIMH). 6001 Executive Boulevard, Room 8184, MSC 9663, Bethesda, MD 20892-9663. (301) 443-4513. < <http://www.nimh.nih.gov> > .

OTHER

- “Stress management, General adaptation syndrome, GAS...” < http://www.holisticonline.com/stress/stress_GAS.htm > .

Michael V. Zuck, PhD
Rebecca J. Frey, PhD

General anesthetic see **Anesthesia, general**

General surgery

Definition

General surgery is the treatment of injury, deformity, and disease using operative procedures.

Purpose

General surgery is frequently performed to alleviate suffering when a cure is unlikely through medication alone. It can be used for routine procedures performed in a physician’s office, such as **vasectomy**, or for more complicated operations requiring a medical team in a

hospital setting, such as laparoscopic **cholecystectomy** (removal of the gallbladder). Areas of the body treated by general surgery include the stomach, liver, intestines, appendix, breasts, thyroid gland, salivary glands, some arteries and veins, and the skin. The brain, heart, eyes, and feet, to name only a few, are areas that require specialized surgical repair.

New methods and techniques are less invasive than previous practices, permitting procedures that were considered impossible in the past. For example, microsurgery has been used in reattaching severed body parts by successfully reconnecting small blood vessels and nerves.

Precautions

Patients who are obese, smoke, have bleeding tendencies, or are over 60, need to follow special precautions, as do patients who have recently experienced an illness such as **pneumonia** or a **heart attack**. Patients on medications such as heart and blood pressure medicine, blood thinners, **muscle relaxants**, tranquilizers, insulin, or sedatives, may require special lab tests prior to surgery and special monitoring during surgery. Special precautions may be necessary for patients using mind-altering drugs such as **narcotics**, psychedelics, hallucinogens, **marijuana**, sedatives, or **cocaine** since these drugs may interact with the anesthetic agents used during surgery.

Description

In earlier times, surgery was a dangerous and dirty practice. Until the middle of the 19th century, as many patients died of surgery as were cured. With the discovery and development of **general anesthesia** in the mid-1800s, surgery became more humane. And as knowledge about infections grew, surgery became more successful as sterile practices were introduced into the operating room. The last 50 years of the 20th century have seen continued advancements.

Types of General Surgery

General surgery experienced major advances with the introduction of the endoscope. This is an instrument for visualizing the interior of a body canal or a hollow organ. Endoscopic surgery relies on this pencil-thin instrument, capable of its own lighting system and small video camera. The endoscope is inserted through tiny incisions called portals. While viewing the procedure on a video screen, the surgeon then operates with various other small, precise instruments inserted through one or more of the portals. The specific area

of the body treated determines the type of endoscopic surgery performed. For example, **colonoscopy** uses an endoscope, which can be equipped with a device for obtaining tissue samples for visual examination of the colon. Gastroscopy uses an endoscope inserted through the mouth to examine the interior of the stomach. **Arthroscopy** refers to joint surgery, and abdominal procedures are called laparoscopies.

Endoscopy is used in both treatment and diagnosis especially involving the digestive and female reproductive systems. Endoscopy has advantages over many other surgical procedures, resulting in a quicker recovery and shorter hospital stay. This non-invasive technique is being used for appendectomies, gallbladder surgery, hysterectomies and the repair of shoulder and knee ligaments. However, endoscopy does not come without limitations such as complications and high operating expense. Also, endoscopy doesn't offer advantages over conventional surgery in all procedures. Some literature states that as general surgeons become more experienced in their prospective fields, additional non-invasive surgery will be a more common option to patients.

ONE-DAY SURGERY. One-day surgery is also termed same-day, or outpatient surgery. Surgical procedures usually take two hours or less and involve minimal blood loss and a short recovery time. In the majority of surgical cases, oral medications control postoperative **pain**. Cataract removal, **laparoscopy**, **tonsillectomy**, repair of broken bones, **hernia repair**, and a wide range of cosmetic procedures are common same-day surgical procedures. Many individuals prefer the convenience and atmosphere of one-day surgery centers, as there is less competition for attention with more serious surgical cases. These centers are accredited by the Joint Commission on Accreditation of Healthcare Organizations or the Accreditation Association for Ambulatory Health Care.

Preparation

The preparation of patients has advanced significantly with improved diagnostic techniques and procedures. Before surgery the patient may be asked to undergo a series of tests including blood and urine studies, x rays and specific heart studies if the patient's past medical history and/or physical exam warrants this testing. Before any general surgery the physician will explain the nature of the surgery needed, the reason for the procedure, and the anticipated outcome. The risks involved will be discussed along with the types of anesthesia utilized. The expected length of recovery and limitations imposed during the recovery period are also explained in detail before any general surgical procedure.

Surgical procedures most often require some type of anesthetic. Some procedures require only **local anesthesia**, produced by injecting the anesthetic agent into the skin near the site of the operation. The patient remains awake with this form of medication. Injecting anesthetic agents into a primary nerve located near the surgical site produces block anesthesia (also known as regional anesthesia), which is a more extensive local anesthesia. The patient remains conscious, but is usually sedated. General anesthesia involves injecting anesthetic agents into the blood stream and/or inhaling medicines through a mask placed over the patient's face. During general anesthesia, the patient is asleep and an airway tube is usually placed into the windpipe to help keep the airway open.

As part of the preoperative preparation, the patient will receive printed educational material and may be asked to review audio or videotapes. The patient will be instructed to shower or bathe the evening before or morning of surgery and may be asked to scrub the operative site with a special antibacterial soap. Instructions will also be given to the patient to ingest nothing by mouth for a determined period of time prior to the surgical procedure.

Aftercare

After surgery, blood studies and a laboratory examination of removed fluid or tissue are often performed especially in the case of **cancer** surgery. After the operation, the patient is brought to a recovery room and vital signs, fluid status, dressings and surgical drains are monitored. Pain medications are offered and used as necessary. Breathing exercises are encouraged to maximize respiratory function and leg exercises are encouraged to promote adequate circulation and prevent pooling of blood in the lower extremities. Patients must have a responsible adult accompany them home if leaving the same day as the surgery was performed.

Risks

One of the risks involved with general surgery is the potential for postoperative complications. These complications include—but are not limited to—pneumonia, internal bleeding, and wound infection as well as adverse reactions to anesthesia.

Normal results

Advances in diagnostic and surgical techniques have increased the success rate of general surgery by

KEY TERMS

Appendectomy—Removal of the appendix.

Endoscope—Instrument for examining visually the inside of a body canal or a hollow organ such as the stomach, colon, or bladder.

Hysterectomy—Surgical removal of part or all of the uterus.

Laparoscopic cholecystectomy—Removal of the gallbladder using a laparoscope, a fiberoptical instrument inserted through the abdomen.

Microsurgery—Surgery on small body structures or cells performed with the aid of a microscope and other specialized instruments.

Portal—An entrance or a means of entrance.

many times compared to the past. Today's less invasive surgical procedures have reduced the length of hospital stays, shortened recovery time, decreased postoperative pain and decreased the size of surgical incision. On the average, a conventional abdominal surgery requires a three to six-day hospital stay and three to six-week recovery time.

Abnormal results

Abnormal results from general surgery include persistent pain, swelling, redness, drainage or bleeding in the surgical area and surgical wound infection resulting in slow healing.

Resources

ORGANIZATIONS

American Medical Association. 515 N. State St., Chicago, IL 60612. (312) 464-5000. <<http://www.ama-assn.org>> .

Jeffrey P. Larson, RPT

Generalized anxiety disorder

Definition

Generalized **anxiety** disorder is a condition characterized by “free floating” anxiety or apprehension not linked to a specific cause or situation.

Description

Some degree of fear and anxiety is perfectly normal. In the face of real danger, fear makes people more alert and also prepares the body to fight or flee (the so-called “fight or flight” response). When people are afraid, their hearts beat faster and they breathe faster in anticipation of the physical activity that will be required of them. However, sometimes people can become anxious even when there is no identifiable cause, and this anxiety can become overwhelming and very unpleasant, interfering with their daily lives. People with debilitating anxiety are said to be suffering from **anxiety disorders**, such as **phobias**, panic disorders, and generalized anxiety disorder. The person with generalized anxiety disorder generally has chronic (officially, having more days with anxiety than not for at least six months), recurrent episodes of anxiety that can last days, weeks, or even months.

Causes and symptoms

Generalized anxiety disorder afflicts between 2–3% of the general population, and is slightly more common in women than in men. It accounts for almost one-third of cases referred to psychiatrists by general practitioners.

Generalized anxiety disorder may result from a combination of causes. Some people are genetically predisposed to developing it. Psychological traumas that occur during childhood, such as prolonged separation from parents, may make people more vulnerable as well. Stressful life events, such as a move, a major job change, the loss of a loved one, or a divorce, can trigger or contribute to the anxiety.

Psychologically, the person with generalized anxiety disorder may develop a sense of dread for no apparent reason—the irrational feeling that some nameless catastrophe is about to happen. Physical symptoms similar to those found with **panic disorder** may be present, although not as severe. They may include trembling, sweating, heart **palpitations** (the feeling of the heart pounding in the chest), **nausea**, and “butterflies in the stomach.”

According to the *Diagnostic and Statistical Manual of Mental Disorders*, 4th edition, a person must have at least three of the following symptoms, with some being present more days than not for at least six months, in order to be diagnosed with generalized anxiety disorder:

- restlessness or feeling on edge
- being easily fatigued

- difficulty concentrating
- irritability
- muscle tension
- sleep disturbance

While generalized anxiety disorder is not completely debilitating, it can compromise a person’s effectiveness and quality of life.

Diagnosis

Anyone with chronic anxiety for no apparent reason should see a physician. The physician may diagnose the condition based on the patient’s description of the physical and emotional symptoms. The doctor will also try to rule out other medical conditions that may be causing the symptoms, such as excessive **caffeine** use, thyroid disease, **hypoglycemia**, cardiac problems, or drug or alcohol withdrawal. Psychological conditions, such as depressive disorder with anxiety, will also need to be ruled out.

In June 2004, the Anxiety Disorders Association of America released follow-up guidelines to help primary care physicians better diagnose and manage patients with generalized anxiety disorder. They include considering the disorder when medical causes for general, vague physical complaints cannot be ruled out. Since generalized anxiety disorder often co-occurs with **mood disorders** and **substance abuse**, the clinician may have to treat these conditions as well, and therefore must consider them in making the diagnosis.

Treatment

Over the short term, a group of tranquilizers called **benzodiazepines**, such as clonazepam (Klonopin) may help ease the symptoms of generalized anxiety disorder. Sometimes **antidepressant drugs**, such as amitriptyline (Elavil), or **selective serotonin reuptake inhibitors** (SSRIs), such as paroxetine (Paxil), escitalopram (Lexapro), and venlafaxine (Effexor), which also has norepinephrine, may be preferred. Other SSRIs are fluoxetine (Prozac) and sertraline (Zoloft).

Psychotherapy can be effective in treating generalized anxiety disorder. The therapy may take many forms. In some cases, psychodynamically-oriented psychotherapy can help patients work through this anxiety and solve problems in their lives. Cognitive behavioral therapy aims to reshape the way people perceive and react to potential stressors in their lives. Relaxation techniques have also been used in treatment, as well as in prevention efforts.

KEY TERMS

Cognitive behavioral therapy—A psychotherapeutic approach that aims at altering cognitions—including thoughts, beliefs, and images—as a way of altering behavior.

Prognosis

When properly treated, most patients with generalized anxiety disorder experience improvement in their symptoms.

Prevention

While preventive measures have not been established, a number of techniques may help manage anxiety, such as relaxation techniques, breathing exercises, and distraction—putting the anxiety out of one's mind by focusing thoughts on something else.

Resources

PERIODICALS

“Guidelines to Assist Primary Care Physicians in Diagnosing GAD.” *Psychiatric Times* (July 1, 2004): 16.

Sherman, Carl. “GAD Patients Often Require Combined Therapy.” *Clinical Psychiatry News* (August 2004): 12–14.

ORGANIZATIONS

American Psychiatric Association. 1400 K Street NW, Washington DC 20005. (888) 357-7924. <<http://www.psych.org>> .

Anxiety Disorders Association of America. 11900 Park Lawn Drive, Ste. 100, Rockville, MD 20852. (800) 545-7367. <<http://www.adaa.org>> .

National Institute of Mental Health. Mental Health Public Inquiries, 5600 Fishers Lane, Room 15C-05, Rockville, MD 20857. (888) 826-9438. <<http://www.nimh.nih.gov>> .

Robert Scott Dinsmoor
Teresa G. Odle

Genetic counseling

Definition

Genetic counseling aims to facilitate the exchange of information regarding a person's genetic legacy. It attempts to:

- accurately diagnose a disorder
- assess the risk of recurrence in the concerned family members and their relatives
- provide alternatives for decision-making
- provide support groups that will help family members cope with the recurrence of a disorder.

Purpose

Genetic counselors work with people concerned about the risk of an inherited disease. The counselor does not prevent the incidence of a disease in a family, but can help family members assess the risk for certain hereditary diseases and offer guidance. Many couples seek genetic counseling because there is a family history of known genetic disorders, **infertility**, **miscarriage**, still births, or early infant mortality. Other reasons for participating in genetic counseling may be the influences of a job or lifestyle that exposes a potential parent to health risks such as radiation, chemicals, or drugs. Any family history of **mental retardation** can be of concern as is a strong family history of heart disease at an early age. Recent statistics show a 3% chance of delivering a baby with **birth defects**. An additional 2% chance of having a baby with **Down syndrome** is present for women in their late thirties and older.

Genetic counseling may take on new emphasis in the near future as genetic research continues to advance. In April 2003, the Human Genome Project announced completion of mapping the entire human genetic makeup. The project identified more than 1,400 disease genes and completed study of the ethical, legal, and social issues raised by this expanded knowledge of human genetics. As knowledge expands and scientists discover more methods to identify and treat various diseases, people will face more difficult decisions about their own genetic information.

Precautions

Amniocentesis, one of the specific tests used to gather information for genetic counseling, is best performed between weeks 15 and 17 of a **pregnancy** and an additional one to four weeks may be required to culture skin cells and analyze them. Thus, these test data are not available to assist prospective parents in decision-making until the second trimester of the pregnancy. Individuals who participate in genetic counseling and associated testing also must be aware that there are no cures or treatments for some of the disorders that may be identified.

Description

With approximately 2,000 genes identified and approximately 5,000 disorders caused by genetic defects, genetic counseling is important in the medical discipline of obstetrics. Genetic counselors, educated in the medical and the psychosocial aspects of genetic diseases, convey complex information to help people make life decisions. There are limitations to the power of genetic counseling, though, since many of the diseases that have been shown to have a genetic basis currently offer no cure (for example, Down syndrome or Huntington's disease). Although a genetic counselor cannot predict the future unequivocally, he or she can discuss the occurrence of a disease in terms of probability.

Genetic counseling also can help people with diseases they may face in their own lifetimes. A 2003 study in Great Britain found that women with a family history of **breast cancer** were less worried about getting the disease if they had genetic counseling.

A genetic counselor, with the aid of the patient or family, creates a detailed family pedigree that includes the incidence of disease in first-degree (parents, siblings, and children) and second-degree (aunts, uncles, and grandparents) relatives. Before or after this pedigree is completed, certain genetic tests are performed using DNA analysis, x ray, ultrasound, urine analysis, **skin biopsy**, and physical evaluation. For a pregnant woman, prenatal diagnosis can be made using amniocentesis or **chorionic villus sampling**.

Family pedigree

An important aspect of the genetic counseling session is the compilation of a family pedigree or medical history. To accurately assess the risk of inherited diseases, information on three generations, including health status and/or cause of **death**, usually is needed. If the family history is complicated, information from more distant relatives may be helpful, and medical records may be requested for any family members who have had a genetic disorder. Through an examination of the family history a counselor may be able to discuss the probability of future occurrence of genetic disorders. In all cases, the counselor provides information in a non-directive way that leaves the decision-making up to the client.

Family history questionnaire

As more detailed genetic information becomes available, physicians and genetic counselors may feel the need to dig more deeply than a family pedigree

allows. In 2004, physicians attending an American College of Medical Genetics meeting announced use of a structured questionnaire with 50 items to consistent, thorough gather family history data. Although the questionnaire's format and terminology were confusing to some patients, once a formula was applied to the answers, it still helped reviewers agree on a counseling plan 79% of the time.

Screening tests

Screening blood tests help identify individuals who carry genes for recessive genetic disorders. Screening tests usually are only done if:

- The disease is lethal or causes severe handicaps or disabilities.
- The person is likely to be a carrier due to family pedigree or membership in an at-risk ethnic, geographic or racial group.
- The disorder can be treated or reproductive options exist.
- A reliable test is available.

Genetic disorders such as **Tay-Sachs disease**, sickle-cell anemia, and **thalassemia** meet these criteria, and screening tests are commonly done to identify carriers of these diseases. In addition, screening tests may be done for individuals with family histories of Huntington's disease (a degenerative neurological disease) or **hemophilia** (a bleeding disorder). Such screening tests can eliminate the need for more invasive tests during a pregnancy.

Another screening test commonly used in the United States in the alpha-fetoprotein (AFP) test. This test is done on a sample of maternal blood around week 16 of a pregnancy. An elevation in the serum AFP level indicates that the fetus may have certain birth defects such as neural tube defects (including **spina bifida** and anencephaly). If the test yields an elevated result, it may be run again after seven days. If the level still is elevated after repeat testing, additional diagnostic tests (e.g. ultrasound and/or amniocentesis) are done in an attempt to identify the specific birth defect present.

Ultrasound

Ultrasound is a noninvasive procedure that uses sound waves to produce a reflected image of the fetus upon a screen. It is used to determine the age and position of the fetus, and the location of the placenta. Ultrasound also is useful in detecting visible birth defects such as spina bifida (a defect in the development of the vertebrae of the spinal column and/or the

spinal cord). It also is useful for detecting heart defects, and malformations of the head, face, body, and limbs. This procedure, however, cannot detect biochemical or chromosomal alterations in the fetus.

Amniocentesis

Amniocentesis is useful in determining genetic and developmental disorders not detectable by ultrasound. This procedure involves the insertion of a needle through the abdomen and into the uterus of a pregnant woman. A sample of amniotic fluid is withdrawn containing skin cells that have been shed by the fetus. The sample is sent to a laboratory where fetal cells contained in the fluid are isolated and grown in order to provide enough genetic material for testing. This takes about seven to 14 days. The material then is extracted and treated so that visual examination for defects can be made. For some disorders, like Tay-Sachs disease, the simple presence of a telltale chemical compound in the amniotic fluid is enough to confirm a diagnosis.

While it has been routine in recent years to suggest amniocentesis to every pregnant woman age 35 and older to screen for Down syndrome, evidence in 2003 began suggesting that it made more sense and was safer to offer blood test screening. The “triple screen” blood test can identify about three-fourths of Down syndrome cases by measuring certain chemicals in the mother’s blood.

Chorionic villus sampling

Chorionic villus sampling involves the removal of a small amount of tissue directly from the chorionic villi (minute vascular projections of the fetal chorion that combine with maternal uterine tissue to form the placenta). In the laboratory, the chromosomes of the fetal cells are analyzed for number and type. Extra chromosomes, such as are present in Down syndrome, can be identified. Additional laboratory tests can be performed to look for specific disorders and the results usually are available within a week after the sample is taken. The primary benefit of this procedure is that it is usually performed between weeks 10 and 12 of a pregnancy, allowing earlier detection of fetal disorders. A 2003 study reported that this test resulted in fewer cases of pregnancy loss, amniotic fluid leakage, or birth defects than early amniocentesis.

Preparation

Genetic diagnosis requires that a couple share information about inherited disorders in their background

with the genetic counselor, including details of any genetic diseases in either family. A couple undergoing genetic counseling also reports any past miscarriages and discusses the possibility of exposure to chemicals, radiation (including x rays), or other occupational environmental hazards. The couple also needs to disclose information about personal habits before or during pregnancy such as drug or alcohol **abuse** and the use of prescription or over-the-counter drugs taken by the mother since the beginning of pregnancy. The genetic counselor explains the procedures used in testing that will be done and describes what each test can and cannot reveal.

Aftercare

Genetic counseling provides couples with information that can help them make decisions about future pregnancies. It also gives couples additional time to emotionally prepare if a disorder is detected in the fetus. The counselor discusses the results of testing and informs the couple if a problem is apparent. The doctor or genetic counselor also discusses the treatment options available. Genetic counseling is done in a non-directive way, so that any treatment selected remains the personal choice of the individuals involved. Genetic counseling can provide information essential for family planning and pregnancy management, thus maximizing the chances of a positive outcome.

Risks

Because prenatal testing, such as amniocentesis and chorionic villus sampling, is invasive and carries a 1% risk of miscarriage it should never be considered routine.

Normal results

Screening tests and/or prenatal tests reveal no birth defects or genetic abnormalities.

Abnormal results

A birth defect or genetic disorder is detected. The early diagnosis of birth defects and genetic disorders allows a greater number of treatment options. Some disorders can be treated in utero (before birth while the fetus is still in the uterus), while others may require early delivery, immediate surgery, or **cesarean section** to minimize fetal trauma. Prior warning of fetal difficulties allows parents time to prepare emotionally for the birth of the child. In some instances, termination

KEY TERMS

Sickle-cell anemia—A chronic, inherited blood disorder characterized by crescent-shaped red blood cells. It occurs primarily in people of African descent, and produces symptoms including episodic pain in the joints, fever, leg ulcers, and jaundice.

Tay-Sachs disease—A hereditary disease affecting young children of eastern European Jewish descent. This disease is caused by an enzyme deficiency leading to the accumulation of gangliosides (galactose-containing cerebroside) found in the surface membranes of nerve cells in the brain and nerve tissue. This deficiency results in mental retardation, convulsions, blindness, and, finally, death.

Thalassemia—An inherited group of anemias occurring primarily among people of Mediterranean descent. It is caused by defective formation of part of the hemoglobin molecule.

of the pregnancy may be chosen. Whatever the test results, this information is essential for family planning and pregnancy management.

Resources

PERIODICALS

"Best Early Test." *Fit Pregnancy* (October-November, 2003): 37.

"Blood Test Screening Reduces Need for Amniocentesis." *Women's Health Weekly* (December 4, 2003): 51.

Bodenhorn, Nancy, and Gerald Lawson. "Genetic Counseling: Implications for Community Counselors." *Journal of Counseling and Development* (Fall 2003): 497-495.

"Genetic Counseling Questionnaire Helps Assess Family's Genetic History." *Internal Medicine News* (April 15, 2004): 45.

"Genetic Counseling Reduces Women's Fears." *Women's Health Weekly* (September 11, 2003): 23.

Wechsler, Jill. "From Genome Exploration to Drug Development." *Pharmaceutical Technology Europe* (June 2003): 18-23.

ORGANIZATIONS

American Medical Association. 515 N. State St., Chicago, IL 60612. (312) 464-5000. <<http://www.ama-assn.org>>.

American Society of Human Genetics. 9650 Rockville Pike, Bethesda, MD 20814-3998. (301) 571-1825. <<http://www.faseb.org/genetics/ashg/ashgmenu.htm>>.

March of Dimes Birth Defects Foundation. 1275 Mamaroneck Ave., White Plains, NY 10605. (914) 428-7100. resourcecenter@modimes.org. <<http://www.modimes.org>>.

Jeffrey P. Larson, RPT
Teresa G. Odle

Genetic studies see **Genetic testing**

Genetic testing

Definition

A genetic test examines the genetic information contained inside a person's cells, called DNA, to determine if that person has or will develop a certain disease or could pass a disease to his or her offspring. Genetic tests also determine whether or not couples are at a higher risk than the general population for having a child affected with a genetic disorder.

Purpose

Some families or ethnic groups have a higher incidence of a certain disease than the population as a whole. For example, individuals from Eastern European, Ashkenazi Jewish descent are at higher risk for carrying genes for rare conditions that occur much less frequently in populations from other parts of the world. Before having a child, a couple from such a family or ethnic group may want to know if their child would be at risk of having that disease. Genetic testing for this type of purpose is called genetic screening.

During **pregnancy**, a baby's cells can be studied for certain genetic disorders or chromosomal problems such as **Down syndrome**. Chromosome testing is most commonly offered when the mother is 35 years or older at the time of delivery. When there is a family medical history of a genetic disease or there are individuals in a family affected with developmental and physical delays, genetic testing also may be offered during pregnancy. Genetic testing during pregnancy is called prenatal diagnosis.

Prior to becoming pregnant, couples who are having difficulty conceiving a child or who have suffered multiple miscarriages may be tested to see if a genetic cause can be identified.

A genetic disease may be diagnosed at birth by doing a physical evaluation of the baby and observing

characteristics of the disorder. Genetic testing can help to confirm the diagnosis made by the physical evaluation. In addition, genetic testing is used routinely on all newborns to screen for certain genetic diseases that can affect a newborn baby's health shortly after birth.

There are several genetic diseases and conditions in which the symptoms do not occur until adulthood. One such example is Huntington's disease. This is a serious disorder affecting the way in which individuals walk, talk and function on a daily basis. Genetic testing may be able to determine if someone at risk will in fact develop the disease.

Genetic testing may take on new emphasis in the near future as genetic research continues to advance. In April 2003, the Human Genome Project announced completion of mapping the entire human genetic makeup. The project identified more than 1,400 disease genes and completed study of the ethical, legal, and social issues raised by this expanded knowledge of human genetics. As knowledge expands and scientists discover more methods to identify and treat various diseases, people will face more difficult decisions about their own genetic information. In fact, the amount of genetic testing was increasing internationally in 2003, especially for rare diseases.

Some genetic defects may make a person more susceptible to certain types of **cancer**. Testing for these defects can help predict a person's risk. Other types of genetic tests help diagnose and predict and monitor the course of certain kinds of cancer, particularly leukemia and lymphoma.

Precautions

Because genetic testing is not always accurate and because there are many concerns surrounding insurance and employment discrimination for the individual receiving a genetic test, **genetic counseling** should always be performed prior to genetic testing. A genetic counselor is an individual with a master's degree in genetic counseling. A medical geneticist is a physician specializing and board certified in genetics.

A genetic counselor reviews the person's family history and medical records and the reason for the test. The counselor explains the likelihood that the test will detect all possible causes of the disease in question (known as the sensitivity of the test), and the likelihood that the disease will develop if the test is positive (known as the positive predictive value of the test).

Learning about the disease in question, the benefits and risks of both a positive and a negative result, and what treatment choices are available if the result is

positive, will help prepare the person undergoing testing. During the genetic counseling session, the individual interested in genetic testing will be asked to consider how the test results will affect his or her life, family, and future decisions.

After this discussion, the person should have the opportunity to indicate in writing that he or she gave informed consent to have the test performed, verifying that the counselor provided complete and understandable information.

Description

Genes and chromosomes

Deoxyribonucleic acid (DNA) is a long molecule made up of two strands of genetic material coiled around each other in a unique double helix structure. This structure was discovered in 1953 by Francis Crick and James Watson.

DNA is found in the nucleus, or center, of most cells (Some cells, such as a red blood cell, don't have a nucleus). Each person's DNA is a unique blueprint, giving instructions for a person's physical traits, such as eye color, hair texture, height, and susceptibility to disease. DNA is organized into structures called chromosomes.

The instructions are contained in DNA's long strands as a code spelled out by pairs of bases, which are four chemicals that make up DNA. The bases occur as pairs because a base on one strand lines up with and is bound to a corresponding base on the other strand. The order of these bases form DNA's code. The order of the bases on a DNA strand is important to ensuring that we are not affected with any genetic diseases. When the bases are out of order, or missing, our cells often do not produce important proteins which can lead to a genetic disorder. While our genes are found in every cell of our body, not every gene is functioning all of the time. Some genes are turned on during critical points in development and then remain silent for the rest of our lives. Other genes remain active all of our lives so that our cells can produce important proteins that help us digest food properly or fight off the **common cold**.

The specific order of the base pairs on a strand of DNA is important in order for the correct protein to be produced. A grouping of three base pairs on the DNA strand is called a codon. Each codon, or three base pairs, comes together to spell a word. A string of many codons together can be thought of as a series of words all coming together to make a sentence. This sentence is what instructs our cells to make a protein that helps our bodies function properly.

Our DNA strands, containing a hundred to several thousand copies of genes, are found on structures called chromosomes. Each cell typically has 46 chromosomes arranged into 23 pairs. Each parent contributes one chromosome to each pair. The first 22 pairs are called autosomal chromosomes, or non-sex chromosomes, and are assigned a number from 1–22. The last pair are the sex chromosomes and include the X and Y chromosomes. If a child receives an X chromosome from each parent, the child is female. If a child receives an X from the mother, and a Y from the father, the child is male.

Just as each parent contributes one chromosome to each pair, so each parent contributes one gene from each chromosome. The pair of genes produces a specific trait in the child. In autosomal dominant conditions, it takes only one copy of a gene to influence a specific trait. The stronger gene is called dominant; the weaker gene, recessive. Two copies of a recessive gene are needed to control a trait while only one copy of a dominant gene is needed. Our sex chromosomes, the X and the Y, also contain important genes. Some genetic diseases are caused by missing or altered genes on one of the sex chromosomes. Males are most often affected by sex chromosome diseases when they inherit an X chromosome with missing or mutated genes from their mother.

TYPES OF GENETIC MUTATIONS. Genetic disease results from a change, or mutation, in a chromosome or in one or several base pairs on a gene. Some of us inherit these mutations from our parents, called hereditary or germline mutations, while other mutations can occur spontaneously, or for the first time in an affected child. For many of the adult on-set diseases, genetic mutations can occur over the lifetime of the individual. This is called acquired or somatic mutations and these occur while the cells are making copies of themselves or dividing in two. There may be some environmental effects, such as radiation or other chemicals, which can contribute to these types of mutations as well.

There are a variety of different types of mutations that can occur in our genetic code to cause a disease. And for each genetic disease, there may be more than one type of mutation to cause the disease. For some genetic diseases, the same mutation occurs in every individual affected with the disease. For example, the most common form of dwarfism, called **achondroplasia**, occurs because of a single base pair substitution. This same mutation occurs in all individuals affected with the disease. Other genetic diseases are caused by different types of genetic mutations that may occur anywhere along the length of a gene. For example, **cystic fibrosis**, the most common genetic disease in

the caucasian population is caused by over hundreds of different mutations along the gene. Individual families may carry the same mutation as each other, but not as the rest of the population affected with the same genetic disease.

Some genetic diseases occur as a result of a larger mutation which can occur when the chromosome itself is either rearranged or altered or when a baby is born with more than the expected number of chromosomes. There are only a few types of chromosome rearrangements which are possibly hereditary, or passed on from the mother or the father. The majority of chromosome alterations where the baby is born with too many chromosomes or missing a chromosome, occur sporadically or for the first time with a new baby.

The type of mutation that causes a genetic disease will determine the type of genetic test to be performed. In some situations, more than one type of genetic test will be performed to arrive at a diagnosis. The cost of genetic tests vary: chromosome studies can cost hundreds of dollars and certain gene studies, thousands. Insurance coverage also varies with the company and the policy. It may take several days or several weeks to complete a test. Research testing where the exact location of a gene has not yet been identified, can take several months to years for results.

Types of Genetic Testing

Direct DNA mutation analysis

Direct DNA sequencing examines the direct base pair sequence of a gene for specific gene mutations. Some genes contain more than 100,000 bases and a mutation of any one base can make the gene nonfunctional and cause disease. The more mutations possible, the less likely it is for a test to detect all of them. This test usually is done on white blood cells from a person's blood but also can be performed on other tissues. There are different ways in which to perform direct DNA mutation analysis. When the specific genetic mutation is known, it is possible to perform a complete analysis of the genetic code, also called direct sequencing. There are several different lab techniques used to test for a direct mutation. One common approach begins by using chemicals to separate DNA from the rest of the cell. Next, the two strands of DNA are separated by heating. Special enzymes (called restriction enzymes) are added to the single strands of DNA and then act like scissors, cutting the strands in specific places. The DNA fragments are then sorted by size through a process called electrophoresis. A special piece of DNA, called a probe, is added to the fragments. The probe is designed to bind to specific mutated portions of the gene. When bound to

the probe, the mutated portions appear on x-ray film with a distinct banding pattern.

Indirect DNA Testing

Family linkage studies are done to study a disease when the exact type and location of the genetic alteration is not known, but the general location on the chromosome has been identified. These studies are possible when a chromosome marker has been found associated with a disease. Chromosomes contain certain regions that vary in appearance between individuals. These regions are called polymorphisms and do not cause a genetic disease to occur. If a polymorphism is always present in family members with the same genetic disease, and absent in family members without the disease, it is likely that the gene responsible for the disease is near that polymorphism. The gene mutation can be indirectly detected in family members by looking for the polymorphism.

To look for the polymorphism, DNA is isolated from cells in the same way it is for direct DNA mutation analysis. A probe is added that will detect the large polymorphism on the chromosome. When bound to the probe, this region will appear on x-ray film with a distinct banding pattern. The pattern of banding of a person being tested for the disease is compared to the pattern from a family member affected by the disease.

Linkage studies have disadvantages not found in direct DNA mutation analysis. These studies require multiple family members to participate in the testing. If key family members choose not to participate, the incomplete family history may make testing other members useless. The indirect method of detecting a mutated gene also causes more opportunity for error.

Chromosome analysis

Various genetic syndromes are caused by structural chromosome abnormalities. To analyze a person's chromosomes, his or her cells are allowed to grow and multiply in the laboratory until they reach a certain stage of growth. The length of growing time varies with the type of cells. Cells from blood and bone marrow take one to two days; fetal cells from amniotic fluid take seven to 10 days.

When the cells are ready, they are placed on a microscope slide using a technique to make them burst open, spreading their chromosomes. The slides are stained: the stain creates a banding pattern unique to each chromosome. Under a microscope, the chromosomes are counted, identified, and analyzed based on their size, shape, and stained appearance.

A karyotype is the final step in the chromosome analysis. After the chromosomes are counted, a photograph is taken of the chromosomes from one or more cells as seen through the microscope. Then the chromosomes are cut out and arranged side-by-side with their partner in ascending numerical order, from largest to smallest. The karyotype is done either manually or using a computer attached to the microscope. Chromosome analysis also is called cytogenetics.

Applications for Genetic Testing

Newborn screening

Genetic testing is used most often for newborn screening. Every year, millions of newborn babies have their blood samples tested for potentially serious genetic diseases.

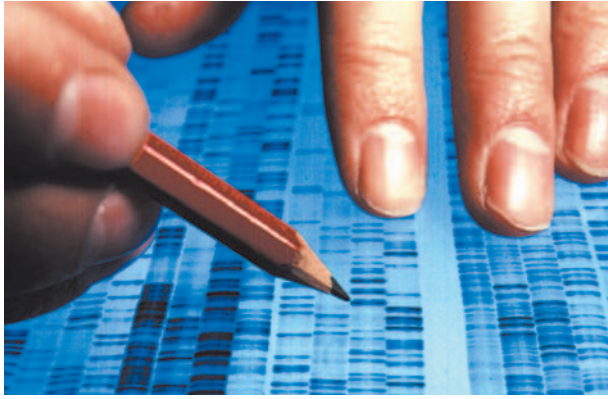
Carrier testing

An individual who has a gene associated with a disease but never exhibits any symptoms of the disease is called a carrier. A carrier is a person who is not affected by the mutated gene he or she possesses, but can pass the gene to an offspring. Genetic tests have been developed that tell prospective parents whether or not they are carriers of certain diseases. If one or both parents are a carrier, the risk of passing the disease to a child can be predicted.

To predict the risk, it is necessary to know if the gene in question is autosomal or sex-linked. If the gene is carried on any one of chromosomes 1–22, the resulting disease is called an autosomal disease. If the gene is carried on the X or Y chromosome, it is called a sex-linked disease.

Sex-linked diseases, such as the bleeding condition **hemophilia**, are usually carried on the X chromosome. A woman who carries a disease-associated mutated gene on one of her X chromosomes, has a 50% chance of passing the gene to her son. A son who inherits that gene will develop the disease because he does not have another normal copy of the gene on a second X chromosome to compensate for the mutated copy. A daughter who inherits the disease associated mutated gene from her mother on one of her X chromosomes will be at risk for having a son affected with the disease.

The risk of passing an autosomal disease to a child depends on whether the gene is dominant or recessive. A prospective parent carrying a dominant gene has a 50% chance of passing the gene to a child. A child needs to receive only one copy of the mutated gene to be affected by the disease.



A scientist examines a DNA sequencing autoradiogram on a light box. (Photo Researchers, Inc. Reproduced by permission.)

If the gene is recessive, a child needs to receive two copies of the mutated gene, one from each parent, to be affected by the disease. When both prospective parents are carriers, their child has a 25% chance of inheriting two copies of the mutated gene and being affected by the disease; a 50% chance of inheriting one copy of the mutated gene, and being a carrier of the disease but not affected; and a 25% chance of inheriting two normal genes. When only one prospective parent is a carrier, a child has a 50% chance of inheriting one mutated gene and being an unaffected carrier of the disease, and a 50% chance of inheriting two normal genes.

Cystic fibrosis is a disease that affects the lungs and pancreas and is discovered in early childhood. It is the most common autosomal recessive genetic disease found in the caucasian population: one in 25 people of Northern European ancestry are carriers of a mutated cystic fibrosis gene. The gene, located on chromosome 7, was identified in 1989.

The gene mutation for cystic fibrosis is detected by a direct DNA test. More than 600 mutations of the cystic fibrosis gene have been found; each of these mutations causes the same disease. Tests are available for the most common mutations. Tests that check for 86 of the most common mutations in the Caucasian population will detect 90% of carriers for cystic fibrosis. (The percentage of mutations detected varies according to the individual's ethnic background). If a person tests negative, it is likely, but not guaranteed that he or she does not have the gene. Both prospective parents must be carriers of the gene to have a child with cystic fibrosis.

Tay-Sachs disease, also autosomal recessive, affects children primarily of Ashkenazi Jewish descent. Children with this disease die between the ages

of two and five. This disease was previously detected by looking for a missing enzyme. The mutated gene has now been identified and can be detected using direct DNA mutation analysis.

Presymptomatic testing

Not all genetic diseases show their effect immediately at birth or early in childhood. Although the gene mutation is present at birth, some diseases do not appear until adulthood. If a specific mutated gene responsible for a late-onset disease has been identified, a person from an affected family can be tested before symptoms appear.

Huntington's disease is one example of a late-onset autosomal dominant disease. Its symptoms of mental confusion and abnormal body movements do not appear until middle to late adulthood. The chromosome location of the gene responsible for Huntington's chorea was located in 1983 after studying the DNA from a large Venezuelan family affected by the disease. Ten years later the gene was identified. A test now is available to detect the presence of the expanded base pair sequence responsible for causing the disease. The presence of this expanded sequence means the person will develop the disease.

The specific genetic cause of **Alzheimer's disease**, another late onset disease, is not as clear. Although many cases appear to be inherited in an autosomal dominant pattern, many other cases exist as single incidents in a family. Like Huntington's, symptoms of mental deterioration first appear in adulthood. Genetic research has found an association between this disease and genes on four different chromosomes. The validity of looking for these genes in a person without symptoms or without family history of the disease is still being studied.

CANCER SUSCEPTIBILITY TESTING. Cancer can result from an inherited (germline) mutated gene or a gene that mutated sometime during a person's lifetime (acquired mutation). Some genes, called tumor suppressor genes, produce proteins that protect the body from cancer. If one of these genes develops a mutation, it is unable to produce the protective protein. If the second copy of the gene is normal, its action may be sufficient to continue production, but if that gene later also develops a mutation, the person is vulnerable to cancer. Other genes, called oncogenes, are involved in the normal growth of cells. A mutation in an oncogene can cause too much growth, the beginning of cancer.

Direct DNA tests currently are available to look for gene mutations identified and linked to several kinds of cancer. People with a family history of these

cancers are those most likely to be tested. If one of these mutated genes is found, the person is more susceptible to developing the cancer. The likelihood that the person will develop the cancer, even with the mutated gene, is not always known because other genetic and environmental factors also are involved in the development of cancer.

Cancer susceptibility tests are most useful when a positive test result can be followed with clear treatment options. In families with **familial polyposis** of the colon, testing a child for a mutated APC gene can reveal whether or not the child needs frequent monitoring for the disease. In 2003, reports showed that genetic testing for high-risk **colon cancer** patients has improved risk assessment. In families with potentially fatal familial medullary **thyroid cancer** or multiple endocrine neoplasia type 2, finding a mutated RET gene in a child provides the opportunity for that child to have preventive removal of the thyroid gland. In the same way, MSH1 and MSH2 mutations can reveal which members in an affected family are vulnerable to familial colorectal cancer and would benefit from aggressive monitoring.

In 1994, a mutation linked to early-onset familial breast and **ovarian cancer** was identified. BRCA1 is located on chromosome 17. Women with a mutated form of this gene have an increased risk of developing breast and ovarian cancer. A second related gene, BRCA2, was later discovered. Located on chromosome 13, it also carries increased risk of breast and ovarian cancer. Although both genes are rare in the general population, they are slightly more common in women of Ashkenazi Jewish descent.

When a woman is found to have a mutation of one of these genes, the likelihood that she will get breast or ovarian cancer increases, but not to 100%. Other genetic and environmental factors influence the outcome.

Testing for these genes is most valuable in families where a mutation has already been found. BRCA1 and BRCA2 are large genes; BRCA1 includes 100,000 bases. More than 120 mutations to this gene have been discovered, but a mutation could occur in any one of the bases. Studies show tests for these genes may miss 30% of existing mutations. The rate of missed mutations, the unknown disease likelihood in spite of a positive result, and the lack of a clear preventive response to a positive result, make the value of this test for the general population uncertain.

Prenatal and postnatal chromosome analysis

Chromosome analysis can be done on fetal cells primarily when the mother is age 35 or older at the

time of delivery, experienced multiple miscarriages, or reports a family history of a genetic abnormality. Prenatal testing is done on the fetal cells from a **chorionic villus sampling** (from the baby's developing placenta) at 9–12 weeks or from the amniotic fluid (the fluid surrounding the baby) at 15–22 weeks of pregnancy. Cells from amniotic fluid grow for seven to 10 days before they are ready to be analyzed. Chorionic villi cells have the potential to grow faster and can be analyzed sooner.

Chromosome analysis using blood cells is done on a child who is born with or later develops signs of **mental retardation** or physical malformation. In the older child, chromosome analysis may be done to investigate developmental delays.

Extra or missing chromosomes cause mental and physical abnormalities. A child born with an extra chromosome 21 (trisomy 21) has Down syndrome. An extra chromosome 13 or 18 also produce well known syndromes. A missing X chromosome causes **Turner syndrome** and an extra X in a male causes **Klinefelter syndrome**. Other abnormalities are caused by extra or missing pieces of chromosomes. **Fragile X syndrome** is a sex-linked disease, causing mental retardation in males.

Chromosome material also may be rearranged, such as the end of chromosome 1 moved to the end of chromosome 3. This is called a chromosomal translocation. If no material is added or deleted in the exchange, the person may not be affected. Such an exchange, however, can cause **infertility** or abnormalities if passed to children.

Evaluation of a man and woman's infertility or repeated miscarriages will include blood studies of both to check for a chromosome translocation. Many chromosome abnormalities are incompatible with life; babies with these abnormalities often miscarry during the first trimester. Cells from a baby that died before birth can be studied to look for chromosome abnormalities that may have caused the **death**.

Cancer diagnosis and prognosis

Certain cancers, particularly leukemia and lymphoma, are associated with changes in chromosomes: extra or missing complete chromosomes, extra or missing portions of chromosomes, or exchanges of material (translocations) between chromosomes. Studies show that the locations of the chromosome breaks are at locations of tumor suppressor genes or oncogenes.

Chromosome analysis on cells from blood, bone marrow, or solid tumor helps diagnose certain kinds of

leukemia and lymphoma and often helps predict how well the person will respond to treatment. After treatment has begun, periodic monitoring of these chromosome changes in the blood and bone marrow gives the physician information as to the effectiveness of the treatment.

A well-known chromosome rearrangement is found in chronic myelogenous leukemia. This leukemia is associated with an exchange of material between chromosomes 9 and 22. The resulting smaller chromosome 22 is called the Philadelphia chromosome.

Preparation

Most tests for genetic diseases of children and adults are done on blood. To collect the 5–10 mL of blood needed, a healthcare worker draws blood from a vein in the inner elbow region. Collection of the sample takes only a few minutes.

Prenatal testing is done either on amniotic fluid or a chorionic villus sampling. To collect amniotic fluid, a physician performs a procedure called **amniocentesis**. An ultrasound is done to find the baby's position and an area filled with amniotic fluid. The physician inserts a needle through the woman's skin and the wall of her uterus and withdraws 5–10 mL of amniotic fluid. Placental tissue for a chorionic villus sampling is taken through the cervix. Each procedure takes approximately 30 minutes. A 2003 study comparing the two tests reported that chorionic villus sampling resulted in fewer cases of pregnancy loss, amniotic fluid leakage, and **birth defects**.

Bone marrow is used for chromosome analysis in a person with leukemia or lymphoma. The person is given **local anesthesia**. Then the physician inserts a needle through the skin and into the bone (usually the sternum or hip bone). One-half to 2 mL of bone marrow is withdrawn. This procedure takes approximately 30 minutes.

Aftercare

After blood collection the person can feel discomfort or bruising at the puncture site or may become dizzy or faint. Pressure to the puncture site until the bleeding stops reduces bruising. Warm packs to the puncture site relieve discomfort.

Chorionic villus sampling, amniocentesis and bone marrow procedures are done under a physician's supervision. The person is asked to rest after the procedure and is watched for weakness and signs of bleeding.

Risks

Collection of amniotic fluid and chorionic villus sampling, have the risk of **miscarriage**, infection, and bleeding; the risks are higher for the chorionic villus sampling. Because of the potential risks for miscarriage, 0.5% following the amniocentesis and 1% following the chorionic villus sampling procedure, both of these prenatal tests are offered to couples, but not required. A woman should tell her physician immediately if she has cramping, bleeding, fluid loss, an increased temperature, or a change in the baby's movement following either of these procedures.

After bone marrow collection, the puncture site may become tender and the person's temperature may rise. These are signs of a possible infection.

Genetic testing involves other nonphysical risks. Many people fear the possible loss of privacy about personal health information. Results of genetic tests may be reported to insurance companies and affect a person's insurability. Some people pay out-of-pocket for genetic tests to avoid this possibility. Laws have been proposed to deal with this problem. Other family members may be affected by the results of a person's genetic test. Privacy of the person tested and the family members affected is a consideration when deciding to have a test and to share the results.

A positive result carries a psychological burden, especially if the test indicates the person will develop a disease, such as Huntington's chorea. The news that a person may be susceptible to a specific kind of cancer, while it may encourage positive preventive measures, also may negatively shadow many decisions and activities.

A genetic test result may also be inconclusive, meaning no definitive result can be given to the individual or family. This may cause the individual to feel more anxious and frustrated and experience psychological difficulties.

Prior to undergoing genetic testing, individuals need to learn from the genetic counselor the likelihood that the test could miss a mutation or abnormality.

Normal results

A normal result for chromosome analysis is 46, XX or 46, XY. This means there are 46 chromosomes (including two X chromosomes for a female or one X and one Y for a male) with no structural abnormalities. A normal result for a direct DNA mutation analysis or linkage study is no gene mutation found.

KEY TERMS

Autosomal disease—A disease caused by a gene located on a chromosome other than a sex chromosome (autosomal chromosome).

Carrier—A person who possesses a gene for an abnormal trait without showing signs of the disorder. The person may pass the abnormal gene on to offspring.

Chromosome—A microscopic thread-like structure found within each cell of the body that consists of a complex of proteins and DNA. Humans have 46 chromosomes arranged into 23 pairs. Changes in either the total number of chromosomes or their shape and size (structure) may lead to physical or mental abnormalities.

Deoxyribonucleic acid (DNA)—The genetic material in cells that holds the inherited instructions for growth, development, and cellular functioning.

Dominant gene—A gene, whose presence as a single copy, controls the expression of a trait.

Enzyme—A protein that catalyzes a biochemical reaction or change without changing its own structure or function.

Gene—A building block of inheritance, which contains the instructions for the production of a particular protein, and is made up of a molecular

sequence found on a section of DNA. Each gene is found on a precise location on a chromosome.

Karyotype—A standard arrangement of photographic or computer-generated images of chromosome pairs from a cell in ascending numerical order, from largest to smallest.

Mutation—A permanent change in the genetic material that may alter a trait or characteristic of an individual, or manifest as disease, and can be transmitted to offspring.

Positive predictive value (PPV)—The probability that a person with a positive test result has, or will get, the disease.

Recessive gene—A type of gene that is not expressed as a trait unless inherited by both parents.

Sensitivity—The proportion of people with a disease who are correctly diagnosed (test positive based on diagnostic criteria). The higher the sensitivity of a test or diagnostic criteria, the lower the rate of ‘false negatives,’ people who have a disease but are not identified through the test.

Sex-linked disorder—A disorder caused by a gene located on a sex chromosome, usually the X chromosome.

There can be some benefits from genetic testing when the individual tested is not found to carry a genetic mutation. Those who learn with certainty they are no longer at risk for a genetic disease may choose not to undergo preventive therapies and may feel less anxious and relieved.

Abnormal results

An abnormal chromosome analysis report will include the total number of chromosomes and will identify the abnormality found. Tests for gene mutations will report the mutations found.

There are many ethical issues to consider with an abnormal prenatal test result. Many of the diseases tested for during a pregnancy cannot be treated or cured. In addition, some diseases tested for during pregnancy may have a late-onset of symptoms or have minimal effects on the affected individual.

Before making decisions based on an abnormal test result, the person should meet again with a genetic

counselor to fully understand the meaning of the results, learn what options are available based on the test result, and the risks and benefits of each of those options.

Resources

PERIODICALS

“Best Early Test.” *Fit Pregnancy* (October–November, 2003): 37.

Bodenhorn, Nancy, and Gerald Lawson. “Genetic Counseling: Implications for Community Counselors.” *Journal of Counseling and Development* (Fall 2003): 497–495.

“Genetic Testing for High-risk Colon Cancer Patients has Improved Risk Assessment.” *Genomics & Genetics Weekly* (August 1, 2003): 18.

“Genetic Testing Increasing Internationally.” *Health & Medicine Week* (September 29, 2003): 283.

Wechsler, Jill. “From Genome Exploration to Drug Development.” *Pharmaceutical Technology Europe* (June 2003): 18–23.

Yan, Hai. “Genetic Testing—Present and Future.” *Science* (September 15, 2000): 1890–1892.

ORGANIZATIONS

Alliance of Genetic Support Groups. 4301 Connecticut Ave. NW, Suite 404, Washington, DC 20008. (202) 966-5557. Fax: (202) 966-8553. <<http://www.geneticalliance.org>> .

American College of Medical Genetics. 9650 Rockville Pike, Bethesda, MD 20814-3998. (301) 571-1825. <<http://www.faseb.org/genetics/acmg/acmgmenu.htm>> .

American Society of Human Genetics. 9650 Rockville Pike, Bethesda, MD 20814-3998. (301) 571-1825. <<http://www.faseb.org/genetics/ashg/ashgmenu.htm>> .

Centers for Disease Control. GDP Office, 4770 Buford Highway NE, Atlanta, GA 30341-3724. (770) 488-3235. <<http://www.cdc.gov/genetics>> .

March of Dimes Birth Defects Foundation. 1275 Mamaroneck Ave., White Plains, NY 10605. (888) 663-4637. resourcecenter@modimes.org. <<http://www.modimes.org>> .

National Human Genome Research Institute. The National Institutes of Health, 9000 Rockville Pike, Bethesda, MD 20892. (301) 496-2433. <<http://www.nhgri.nih.gov>> .

National Society of Genetic Counselors. 233 Canterbury Dr., Wallingford, PA 19086-6617. (610) 872-1192. <<http://www.nsgc.org/GeneticCounselingYou.asp>> .

OTHER

Blazing a Genetic Trail. Online genetic tutorial. <<http://www.hhmi.org/GeneticTrail/>> .

The Gene Letter. Online newsletter. <<http://www.geneletter.org>> .

Online Mendelian Inheritance in Man. Online genetic testing information sponsored by National Center for Biotechnology Information. <<http://www.ncbi.nlm.nih.gov/Omim/>> .

Understanding Gene Testing. Online brochure produced by the U.S. Department of Health and Human Services. <<http://www.gene.com/ae/AE/AEPC/NIH/index.html>> .

Katherine S. Hunt, MS
Teresa G. Odle

Genital herpes

Definition

Genital herpes is a sexually transmitted disease caused by a herpes virus. The disease is characterized by the formation of fluid-filled, painful blisters in the genital area.

Description

Genital herpes (herpes genitalis, herpes progenitalis) is characterized by the formation of fluid-filled blisters on the genital organs of men and women. The word “herpes” comes from the Greek adjective *herpestes*, meaning *creeping*, which refers to the serpent-like pattern that the blisters may form. Genital herpes is a sexually transmitted disease which means that it is spread from person-to-person only by sexual contact. Herpes may be spread by vaginal, anal, and oral sexual activity. It is not spread by objects (such as a toilet seat or doorknob), swimming pools, hot tubs, or through the air.

Genital herpes is a disease resulting from an infection by a herpes simplex virus. There are eight different kinds of human herpes viruses. Only two of these, herpes simplex types 1 and 2, can cause genital herpes. It has been commonly believed that herpes simplex virus type 1 infects above the waist (causing cold sores) and herpes simplex virus type 2 infects below the waist (causing genital sores). This is not completely true. Both herpes virus type 1 and type 2 can cause herpes lesions on the lips or genitals, but recurrent cold sores are almost always type 1. The two viruses seem to have evolved to infect better at one site or the other, especially with regard to recurrent disease.

To determine the occurrence of herpes type 2 infection in the United States, the Centers for Disease Control and Prevention (CDC) used information from a survey called the National Health and **Nutrition** Examination Survey III (1988–1994). This survey of 40,000 noninstitutionalized people found that 21.9% of persons age 12 or older had antibodies to herpes type 2. This means that 45 million Americans have been exposed at some point in their lives to herpes simplex virus type 2. More women (25.6%) than men (17.8%) had antibodies. The racial differences for herpes type 2 antibodies were whites, 17.6%; blacks, 45.9%; and Mexican Americans, 22.3%. Interestingly, only 2.6% of adults reported that they have had genital herpes. Over half (50% to 60%) of the white adults in the United States have antibodies to herpes simplex virus type 1. The occurrence of antibodies to herpes type 1 is higher in blacks.

Viruses are different from bacteria. While bacteria are independent and can reproduce on their own, viruses cannot reproduce without the help of a cell. Viruses enter human cells and force them to make more virus. A human cell infected with herpes virus releases thousands of new viruses before it is killed. The cell death and resulting tissue damage causes the actual sores. The highest risk for spreading the virus is

the time period beginning with the appearance of blisters and ending with scab formation.

Herpes virus can also infect a cell and instead of making the cell produce new viruses, it hides inside the cell and waits. Herpes virus hides in cells of the nervous system called “neurons.” This is called “latency.” A latent virus can wait inside neurons for days, months, or even years. At some future time, the virus “awakens” and causes the cell to produce thousands of new viruses which causes an active infection. Sometimes an active infection occurs without visible sores. Therefore, an infected person can spread herpes virus to other people even in the absence of sores.

This process of latency and active infection is best understood by considering the genital sore cycle. An active infection is obvious because sores are present. The first infection is called the “primary” infection. This active infection is then controlled by the body’s immune system and the sores heal. In between active infections, the virus is latent. At some point in the future latent viruses become activated and once again cause sores. These are called “recurrent infections” or “outbreaks.” Genital sores caused by herpes type 1 recur much less frequently than sores caused by herpes type 2.

Although it is unknown what triggers latent viruses to activate, several conditions seem to bring on infections. These include illness, tiredness, exposure to sunlight, menstruation, skin damage, food allergy and hot or cold temperatures. Although many people believe that **stress** can bring on their genital herpes outbreaks, there is no scientific evidence that there is a link between stress and recurrences. However, at least one clinical study has shown a connection between how well people cope with stress and their belief that stress and recurrent infections are linked.

Newborn babies who are infected with herpes virus experience a very severe, and possibly fatal disease. This is called “neonatal herpes infection.” In the United States, one in 3,000–5,000 babies born will be infected with herpes virus. Babies can become infected during passage through the birth canal, but can become infected during the **pregnancy** if the membranes rupture early. Doctors will perform a **Cesarean section** on women who go into labor with active genital herpes.

Causes and symptoms

While anyone can be infected by herpes virus, not everyone will show symptoms. Risk factors for genital herpes include: early age at first sexual activity,

multiple sexual partners, and a medical history of other sexually-transmitted diseases.

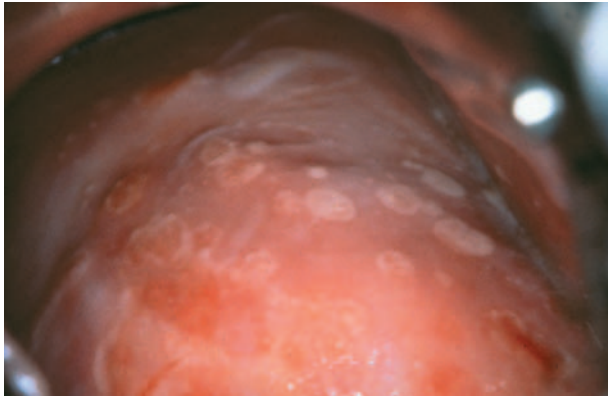
Most patients with genital herpes experience a prodrome (symptoms of oncoming disease) of **pain**, burning, **itching**, or **tingling** at the site where blisters will form. This prodrome stage may last anywhere from a few hours, to one to two days. The herpes infection prodrome can occur for both the primary infection and recurrent infections. The prodrome for recurrent infections may be severe and cause a severe burning or stabbing pain in the genital area, legs, or buttocks.

Primary genital herpes

The first symptoms of herpes usually occur within two to seven days after contact with an infected person but may take up to two weeks. Symptoms of the primary infection are usually more severe than those of recurrent infections. For up to 70% of the patients, the primary infection causes symptoms which affect the whole body (called “constitutional symptoms”) including tiredness, **headache**, **fever**, chills, muscle aches, loss of appetite, as well as painful, swollen lymph nodes in the groin. These symptoms are greatest during the first three to four days of the infection and disappear within one week. The primary infection is more severe in women than in men.

Following the prodrome come the herpes blisters, which are similar on men and women. First, small red bumps appear. These bumps quickly become fluid-filled blisters. In dry areas, the blisters become filled with pus and take on a white to gray appearance, become covered with a scab, and heal within two to three weeks. In moist areas, the fluid-filled blisters burst and form painful ulcers which drain before healing. New blisters may appear over a period of one week or longer and may join together to form very large ulcers. The pain is relieved within two weeks and the blisters and ulcers heal without scarring by three to four weeks.

Women can experience a very severe and painful primary infection. Herpes blisters first appear on the labia majora (outer lips), labia minora (inner lips), and entrance to the vagina. Blisters often appear on the clitoris, at the urinary opening, around the anal opening, and on the buttocks and thighs. In addition, women may get herpes blisters on the lips, breasts, fingers, and eyes. The vagina and cervix are almost always involved which causes a watery discharge. Other symptoms that occur in women are: painful or difficult urination (83%), swelling of the urinary tube (85%), **meningitis** (36%), and throat infection (13%). Most women develop painful, swollen lymph nodes (lymphadenopathy) in the groin and pelvis. About one



Female cervix covered with herpes lesions (Photo Researchers. Reproduced by permission.)

in ten women get a vaginal yeast infection as a complication of the primary herpes infection.

In men, the herpes blisters usually form on the penis but can also appear on the scrotum, thighs, and buttocks. Fewer than half of the men with primary herpes experience the constitutional symptoms. Thirty percent to 40% of men have a discharge from the urinary tube. Some men develop painful swollen lymph nodes (lymphadenopathy) in the groin and pelvis. Although less frequently than women, men too may experience painful or difficult urination (44%), swelling of the urinary tube (27%), meningitis (13%), and throat infection (7%).

Recurrent genital herpes

One or more outbreaks of genital herpes per year occur in 60–90% of those infected with herpes virus. About 40% of the persons infected with herpes simplex virus type 2 will experience six or more outbreaks each year. Genital herpes recurrences are less severe than the primary infection; however, women still experience more severe symptoms and pain than men. Constitutional symptoms are not usually present. Blisters will appear at the same sites during each outbreak. Usually there are fewer blisters, less pain, and the time period from the beginning of symptoms to healing is shorter than the primary infection. One out of every four women experience painful or difficult urination during recurrent infection. Both men and women may develop lymphadenopathy.

Diagnosis

Because genital herpes is so common, it is diagnosed primarily by symptoms. It can be diagnosed and treated by the family doctor, dermatologists (doctors

who specialize in skin diseases), urologists (doctors who specialize in the urinary tract diseases of men and women and the genital organs of men), gynecologists (doctors who specialize in the diseases of women's genital organs) and infectious disease specialists. The diagnosis and treatment of this infectious disease should be covered by most insurance providers.

Laboratory tests may be performed to look for the virus. Because healing sores do not shed much virus, a sample from an open sore would be taken for viral culture. A sterile cotton swab would be wiped over open sores and the sample used to infect human cells in culture. Cells which are killed by herpes virus have a certain appearance under microscopic examination. The results of this test are available within two to ten days. Other areas which may be sampled, depending upon the disease symptoms in a particular patient, include the urinary tract, vagina, cervix, throat, eye tissues, and cerebrospinal fluid.

Direct staining and microscopic examination of the lesion sample may also be used. A blood test may be performed to see if the patient has antibodies to herpes virus. The results of blood testing are available within one day. The disadvantage of this blood test is that it usually does not distinguish between herpes type 1 and 2, and only determines that the patient has had a herpes infection at some point in his or her life. Therefore, the viral culture test must be performed to be absolutely certain that the sores are caused by herpes virus.

Because genital sores can be symptoms of many other diseases, the doctor must determine the exact cause of the sores. The above mentioned tests are performed to determine that herpes virus is causing the genital sores. Other diseases which may cause genital sores are **syphilis**, **chancroid**, **lymphogranuloma venereum**, **granuloma inguinale**, herpes zoster, erythema multiform, Behçet's syndrome, inflammatory bowel disease, **contact dermatitis**, **candidiasis**, and **impetigo**.

Because most newborns who are infected with herpes virus were born to mothers who had no symptoms of infection it is important to check all newborn babies for symptoms. Any skin sore should be sampled to determine if it is caused by herpes simplex. Babies should be checked for sores in their mouth and for signs of herpes infection in their eyes.

Treatment

There is no cure for herpes virus infections. There are **antiviral drugs** available which have some effect in



A close-up view of a man's penis with a blister (center of image) caused by the herpes simplex virus. (Photograph by Dr. P. Marazzi, Custom Medical Stock Photo. Reproduced by permission.)

lessening the symptoms and decreasing the length of herpes outbreaks. There is evidence that some may also prevent future outbreaks. These antiviral drugs work by interfering with the replication of the viruses and are most effective when taken as early in the infection process as possible. For the best results, drug treatment should begin during the prodrome stage before blisters are visible. Depending on the length of the outbreak, drug treatment could continue for up to 10 days.

Acyclovir (Zovirax) is the drug of choice for herpes infection and can be given intravenously, taken by mouth (orally), or applied directly to sores as an ointment. Acyclovir has been in use for many years and only five out of 100 patients experience side effects. Side effects of acyclovir treatment include **nausea, vomiting**, itchy rash, and **hives**. Although acyclovir is the recommended drug for treating herpes infections, other drugs may be used including famciclovir (Famvir), valacyclovir (Valtrex), vidarabine (Vira-A), idoxuridine (Herplex Liquifilm, Stoxil), trifluorothymidine (Viroptic), and penciclovir (Denavir).

Acyclovir is effective in treating both the primary infection and recurrent outbreaks. When taken intravenously or orally, acyclovir reduces the healing time, virus shedding period, and duration of vesicles. The standard oral dose of acyclovir for primary herpes is 200 mg five times daily or 400 mg three times daily for

a period of 10 days. Recurrent herpes is treated with the same doses for a period of five days. Intravenous acyclovir is given to patients who require hospitalization because of severe primary infections or herpes complications such as aseptic meningitis or sacral ganglionitis (inflammation of nerve bundles).

Patients with frequent outbreaks (greater than six to eight per year) may benefit from long term use of acyclovir which is called "suppressive therapy." Patients on suppressive therapy have longer periods between herpes outbreaks. The specific dosage used for suppression needs to be determined for each patient and should be reevaluated every few years. Alternatively, patients may use short term suppressive therapy to lessen the chance of developing an active infection during special occasions such as weddings or holidays.

There are several things that a patient may do to lessen the pain of genital sores. Wearing loose fitting clothing and cotton underwear is helpful. Removing clothing or wearing loose pajamas while at home may reduce pain. Soaking in a tub of warm water and using a blow dryer on the "cool" setting to dry the infected area is helpful. Putting an ice pack on the affected area for 10 minutes, followed by five minutes off and then repeating this procedure may relieve pain. A zinc sulfate ointment may help to heal the sores. Application of a baking soda compress to sores may be soothing.

Neonatal herpes

Newborn babies with herpes virus infections are treated with intravenous acyclovir or vidarabine for 10 days. These drugs have greatly reduced deaths and increased the number of babies who appear normal at one year of age. However, because neonatal herpes infection is so serious, even with treatment babies may not survive, or may suffer nervous system damage. Infected babies may be treated with long term suppressive therapy.

Alternative treatment

An imbalance in the amino acids lysine and arginine is thought to be one contributing factor in herpes virus outbreaks. A ratio of lysine to arginine that is in balance (that is more lysine than arginine is present) seems to help the immune system work optimally. Thus, a diet that is rich in lysine may help prevent recurrences of genital herpes. Foods that contain high levels of lysine include most vegetables, legumes, fish, turkey, beef, lamb, cheese, and chicken. Patients may take 500 mg of lysine daily and increase to 1,000 mg three times a day during an outbreak. Intake of the

KEY TERMS

Groin—The region of the body that lies between the abdomen and the thighs.

Latent virus—A nonactive virus which is in a dormant state within a cell. Herpes virus is latent in cells of the nervous system.

Prodrome—Symptoms which warn of the beginning of disease. The herpes prodrome consists of pain, burning, tingling, or itching at a site before blisters are visible.

Recurrence—The return of an active herpes infection following a period of latency.

Ulcer—A painful, pus-draining, depression in the skin caused by an infection.

amino acid arginine should be reduced. Foods rich in arginine that should be avoided are chocolate, peanuts, almonds, and other nuts and seeds.

Clinical experience indicates a connection between high stress and herpes outbreaks. Some patients respond well to **stress reduction** and relaxation techniques. **Acupressure** and massage may relieve tiredness and stress. **Meditation, yoga, tai chi,** and **hypnotherapy** can also help relieve stress and promote relaxation.

Some herbs, including **echinacea** (*Echinacea* spp.) and garlic (*Allium sativum*), are believed to strengthen the body's defenses against viral infections. Red marine algae (family Dumontiaceae), both taken internally and applied topically, is thought to be effective in treating herpes type I and type II infections. Other topical treatments may be helpful in inhibiting the growth of the herpes virus, in minimizing the damage it causes, or in helping the sores heal. Zinc sulphate ointment seems to help sores heal and to fight recurrence. Lithium succinate ointment may interfere with viral replication. An ointment made with glycyrrhizic acid, a component of licorice (*Glycyrrhiza glabra*), seems to inactivate the virus. Topical applications of vitamin E or tea tree oil (*Melaleuca* spp.) help dry up herpes sores. Specific combinations of homeopathic remedies may also be helpful treatments for genital herpes.

Prognosis

Although physically and emotionally painful, genital herpes is usually not a serious disease. The

primary infection can be severe and may require hospitalization for treatment. Complications of the primary infection may involve the cervix, urinary system, anal opening, and the nervous system. Persons who have a decreased ability to produce an immune response to infection (called "immunocompromised") due to disease or medication are at risk for a very severe, and possibly fatal, herpes infection. Even with antiviral treatment, neonatal herpes infections can be fatal or cause permanent nervous system damage.

Prevention

The only way to prevent genital herpes is to avoid contact with infected persons. This is not an easy solution because many people aren't aware that they are infected and can easily spread the virus to others. Avoid all sexual contact with an infected person during a herpes outbreak. Because herpes virus can be spread at any time, **condom** use is recommended to prevent the spread of virus to uninfected partners. As of early 1998 there were no herpes vaccines available, although new herpes vaccines are being tested in humans.

Resources

BOOKS

Ebel, Charles. *Managing Herpes: How to Live and Love With a Chronic STD*. American Social Health Association, 1998.

Belinda Rowland, PhD

Genital warts

Definition

Genital **warts**, which are also called condylomata acuminata or venereal warts, are growths in the genital area caused by a sexually transmitted papillomavirus. A papillomavirus is a virus that produces papillomas, or benign growths on the skin and mucous membranes.

Description

Genital warts are the most common sexually transmitted disease (STD) in the general population. It is estimated that 1% of sexually active people between the ages of 18 and 45 have genital warts; however, polymerase chain reaction (PCR) testing indicates that as many as 40% of sexually active adults carry the human papillomavirus (HPV) that causes genital warts.

Genital warts vary somewhat in appearance. They may be either flat or resemble raspberries or cauliflower in appearance. The warts begin as small red or pink growths and grow as large as four inches across, interfering with intercourse and **childbirth**. The warts grow in the moist tissues of the genital areas. In women, they occur on the external genitals and on the walls of the vagina and cervix; in men, they develop in the urethra and on the shaft of the penis. The warts then spread to the area behind the genitals surrounding the anus.

Risk factors for genital warts include:

- multiple sexual partners
- infection with another STD
- pregnancy
- anal intercourse
- poor personal hygiene
- heavy perspiration

Causes and symptoms

There are about 80 types of human papilloma-virus. Genital warts are caused by HPV types 1, 2, 6, 11, 16, and 18. HPV is transmitted by sexual contact. The incubation period varies from one to six months.

The symptoms include bleeding, **pain**, and odor as well as the visible warts.

Diagnosis

The diagnosis is usually made by examining scrapings from the warts under a darkfield microscope. If the warts are caused by HPV, they will turn white when a 5% solution of white vinegar is added. If the warts reappear, the doctor may order a biopsy to rule out **cancer**.

Treatment

No treatment for genital warts is completely effective because therapy depends on destroying skin infected by the virus. There are no drugs that will kill the virus directly.

Medications

Genital warts were treated until recently with applications of podophyllum resin, a corrosive substance that cannot be given to pregnant patients. A milder form of podophyllum, podofilox (Condylox), has been introduced. Women are also treated with 5-fluorouracil cream, bichloroacetic



Man with genital warts. (Custom Medical Stock Photo. Reproduced by permission.)

KEY TERMS

Condylomata acuminata—Another name for genital warts.

Papilloma—A benign growth on the skin or mucous membrane. Viruses that cause these growths are called human papillomaviruses (HPVs).

Podophyllum resin—A medication derived from the May apple or mandrake and used to treat genital warts.

acid, or trichloroacetic acid. All of these substances irritate the skin and require weeks of treatment.

Genital warts can also be treated with injections of interferon. Interferon works best in combination with podofilox applications.

Surgery

Surgery may be necessary to remove warts blocking the patient's vagina, urethra, or anus. Surgical techniques include the use of liquid nitrogen, electro-surgery, and **laser surgery**.

Prognosis

Genital warts are benign growths and are not cancerous by themselves. Repeated HPV infection in women, however, appears to increase the risk of later **cervical cancer**. Women infected with HPV types 16 and 18 should have yearly cervical smears. Recurrence is common with all present methods of

treatment—including surgery—because HPV can remain latent in apparently normal surrounding skin.

Prevention

The only reliable method of prevention is sexual abstinence. The use of condoms minimizes but does not eliminate the risk of HPV transmission. The patient's sexual contacts should be notified and examined.

Resources

BOOKS

Foster, David C. "Vulvar and Vaginal Disease." In *Current Diagnosis*. Vol. 9. edited by Rex B. Conn, et al. Philadelphia: W. B. Saunders Co., 1997.

MacKay, H. Trent. "Gynecology." In *Current Medical Diagnosis and Treatment, 1998*, edited by Stephen McPhee, et al., 37th ed. Stamford: Appleton & Lange, 1997.

Rebecca J. Frey, PhD

Gentamicin see **Aminoglycosides**

German measles see **Rubella**

Gestalt therapy

Definition

Gestalt therapy is a humanistic therapy technique that focuses on gaining an awareness of emotions and behaviors in the present rather than in the past. The therapist does not interpret experiences for the patient. Instead, the therapist and patient work together to help the patient understand him/herself. This type of therapy focuses on experiencing the present situation rather than talking about what occurred in the past. Patients are encouraged to become aware of immediate needs, meet them, and let them recede into the background. The well-adjusted person is seen as someone who has a constant flow of needs and is able to satisfy those needs.

Purpose

In Gestalt therapy (from the German word meaning *form*), the major goal is self-awareness. Patients work on uncovering and resolving interpersonal issues during therapy. Unresolved issues are unable to fade into the background of consciousness because the needs they represent are never met. In

Gestalt therapy, the goal is to discover people connected with a patient's unresolved issues and try to engage those people (or images of those people) in interactions that can lead to a resolution. Gestalt therapy is most useful for patients open to working on self-awareness.

Precautions

The choice of a therapist is crucial. Some people who call themselves "therapists" have limited training in Gestalt therapy. It is important that the therapist be a licensed mental health professional. Additionally, some individuals may not be able to tolerate the intensity of this type of therapy.

Description

Gestalt therapy has developed into a form of therapy that emphasizes medium to large groups, although many Gestalt techniques can be used in one-on-one therapy. Gestalt therapy probably has a greater range of formats than any other therapy technique. It is practiced in individual, couples, and family therapies, as well as in therapy with children.

Ideally, the patient identifies current sensations and emotions, particularly ones that are painful or disruptive. Patients are confronted with their unconscious feelings and needs, and are assisted to accept and assert those repressed parts of themselves.

The most powerful techniques involve role-playing. For example, the patient talks to an empty chair as they imagine that a person associated with an unresolved issue is sitting in the chair. As the patient talks to the "person" in the chair, the patient imagines that the person responds to the expressed feelings. Although this technique may sound artificial and might make some people feel self-conscious, it can be a powerful way to approach buried feelings and gain new insight into them.

Sometimes patients use *battacca* bats, padded sticks that can be used to hit chairs or sofas. Using a *battacca* bat can help a patient safely express anger. A patient may also experience a Gestalt therapy marathon, where the participants and one or more facilitators have nonstop **group therapy** over a weekend. The effects of the intense emotion and the lack of sleep can eliminate many psychological defenses and allow significant progress to be made in a short time. This is true only if the patient has adequate psychological strength for a marathon and is carefully monitored by the therapist.

Preparation

Gestalt therapy begins with the first contact. There is no separate diagnostic or assessment period. Instead, assessment and screening are done as part of the ongoing relationship between patient and therapist. This assessment includes determining the patient's willingness and support for work using Gestalt methods, as well as determining the compatibility between the patient and the therapist. Unfortunately, some "encounter groups" led by poorly trained individuals do not provide adequate pre-therapy screening and assessment.

Aftercare

Sessions are usually held once a week. Frequency of sessions held is based on how long the patient can go between sessions without losing the momentum from the previous session. Patients and therapists discuss when to start sessions, when to stop sessions, and what kind of activities to use during a session. However, the patient is encouraged and required to make choices.

Risks

Disturbed people with severe mental illness may not be suitable candidates for Gestalt therapy. Facilities that provide Gestalt therapy and train Gestalt therapists vary. Since there are no national standards for these Gestalt facilities, there are no set national standards for Gestalt therapy or Gestalt therapists.

Normal results

Scientific documentation on the effectiveness of Gestalt therapy is limited. Evidence suggests that this type of therapy may not be reliably effective.

Abnormal results

This approach can be anti-intellectual and can discount thoughts, thought patterns, and beliefs. In the hands of an ineffective therapist, Gestalt procedures can become a series of mechanical exercises, allowing the therapist as a person to stay hidden. Moreover, there is a potential for the therapist to manipulate the patient with powerful techniques, especially in therapy marathons where **fatigue** may make a patient vulnerable.

Resources

ORGANIZATIONS

Association for the Advancement of Gestalt Therapy. 400 East 58th St., New York, NY 10022. (212) 486-1581. < <http://www.aagt.org> > .

David James Doermann

Gestational diabetes

Definition

Gestational diabetes is a condition that occurs during **pregnancy**. Like other forms of diabetes, gestational diabetes involves a defect in the way the body processes and uses sugars (glucose) in the diet. Gestational diabetes, however, has a number of characteristics that are different from other forms of diabetes.

Description

Glucose is a form of sugar that is present in many foods, including sweets, potatoes, pasta, and breads. The body uses glucose to provide energy. It is stored in the liver, muscles, and fatty tissue. The pancreas produces a hormone (a chemical produced in one part of the body, which travels to another part of the body in order to exert its effect) called insulin. Insulin is required to allow glucose to enter the liver, muscles, and fatty tissues, thus reducing the amount of glucose in the blood. In diabetes, blood levels of glucose remain abnormally high. In many forms of diabetes, this is because the pancreas does not produce enough insulin.

In gestational diabetes, the pancreas is not at fault. Instead, the problem is in the placenta. During pregnancy, the placenta provides the baby with nourishment. It also produces a number of hormones that interfere with the body's usual response to insulin. This condition is referred to as "insulin resistance." Most pregnant women do not suffer from gestational diabetes, because the pancreas works to produce extra quantities of insulin in order to compensate for **insulin resistance**. However, when a woman's pancreas cannot produce enough extra insulin, blood levels of glucose stay abnormally high, and the woman is considered to have gestational diabetes.

About 1–3% of all pregnant women develop gestational diabetes. Women at risk for gestational diabetes include those who:

- are overweight
- have a family history of diabetes
- have previously given birth to a very large, heavy baby
- have previously had a baby who was stillborn, or born with a birth defect
- have an excess amount of amniotic fluid (the cushioning fluid within the uterus that surrounds the developing fetus)
- are over 25 years of age
- belong to an ethnic group known to experience higher rates of gestational diabetes (in the United

States, these groups include Mexican-Americans, American Indians, African-Americans, as well as individuals from Asia, India, or the Pacific Islands)

- have a previous history of gestational diabetes during a pregnancy

Causes and symptoms

Most women with gestational diabetes have no recognizable symptoms. However, leaving gestational diabetes undiagnosed and untreated is risky to the developing fetus. Left untreated, a diabetic mother's blood sugar levels will be consistently high. This sugar will cross the placenta and pour into the baby's system through the umbilical cord. The unborn baby's pancreas will respond to this high level of sugar by constantly putting out large amounts of insulin. The insulin will allow the fetus's cells to take in glucose, where it will be converted to fat and stored. A baby who has been exposed to constantly high levels of sugar throughout pregnancy will be abnormally large. Such a baby will often grow so large that he or she cannot be born through the vagina, but will instead need to be born through a surgical procedure (**cesarean section**).

Furthermore, when the baby is born, the baby will still have an abnormally large amount of insulin circulating. After birth, when the mother and baby are no longer attached to each other via the placenta and umbilical cord, the baby will no longer be receiving the mother's high level of sugar. The baby's high level of insulin, however, will very quickly use up the glucose circulating in the baby's bloodstream. The baby is then at risk for having a dangerously low level of blood glucose (a condition called **hypoglycemia**).

Diagnosis

Since gestational diabetes most often exists with no symptoms detectable by the patient, and since its existence puts the developing baby at considerable risk, screening for the disorder is a routine part of pregnancy care. This screening is usually done between the 24th and 28th week of pregnancy. By this point in the pregnancy, the placental hormones have reached a sufficient level to cause insulin resistance. Screening for gestational diabetes involves the pregnant woman drinking a special solution that contains exactly 50 grams of glucose. An hour later, the woman's blood is drawn and tested for its glucose level. A level less than 140 mg/dl is considered normal.

When the screening glucose level is over 140 mg/dl, a special three-hour glucose tolerance test is performed. This involves following a special diet for three days

prior to the test. This diet is set up to contain at least 150 grams of carbohydrates each day. Just before the test, the patient is instructed to eat and drink nothing (except water) for 10–14 hours. A blood sample is then tested to determine the **fasting** glucose level. The patient then drinks a special solution containing exactly 100 grams of glucose, and her blood is tested every hour for the next three hours. If two or more of these levels are elevated over normal, then the patient is considered to have gestational diabetes.

Treatment

Treatment for gestational diabetes will depend on the severity of the diabetes. Mild forms can be treated with diet (decreasing the intake of sugars and fats, in particular). Many women are put on strict, detailed **diets**, and are asked to stay within a certain range of calorie intake. **Exercise** sometimes is used to keep blood sugar levels lower. Patients often are asked to regularly measure their blood sugar. This is done by poking a finger with a needle called a lancet, putting a drop of blood on a special type of paper, and feeding the paper into a meter that analyzes and reports the blood sugar level. In fact self-monitoring of blood glucose helps manage gestational diabetes and prevent complications. When diet and exercise do not keep blood glucose levels within an acceptable range, a patient may need to take regular shots of insulin.

Many babies born to women with gestational diabetes are large enough to cause more difficult deliveries, and they may require the use of forceps, suction, or cesarean section. Once the baby is born, it is important to carefully monitor its blood glucose levels. These levels may drop sharply and dangerously once the baby is no longer receiving large quantities of sugar from the mother. When this occurs, it is easily resolved by giving the baby glucose.

Prognosis

Prognosis for women with gestational diabetes, and their babies, is generally good. Almost all such women stop being diabetic after the birth of their baby. However, some research has shown that nearly 50% of these women will develop a permanent form of diabetes within 15 years. The child of a mother with gestational diabetes has a greater-than-normal chance of developing diabetes sometime in adulthood, also. A woman who has had gestational diabetes during one pregnancy has about a 66% chance of having it again during any subsequent pregnancies. Women who had gestational diabetes usually are tested for diabetes at the postpartum checkup or after stopping breastfeeding.

KEY TERMS

Glucose—A form of sugar. The final product of the breakdown of carbohydrates (starches).

Insulin—A hormone produced by the pancreas that is central to the processing of sugars and carbohydrates in the diet.

Placenta—An organ that is attached to the inside wall of the mother's uterus and to the fetus via the umbilical cord. The placenta allows oxygen and nutrients from the mother's bloodstream to pass into the unborn baby.

Prevention

There is no known way to actually prevent diabetes, particularly since gestational diabetes is due to the effects of normal hormones of pregnancy. However, the effects of insulin resistance can be best handled through careful attention to diet, avoiding becoming overweight throughout life, and participating in reasonable exercise. A 2003 report also linked **smoking** to increased risk of gestational diabetes.

Resources

PERIODICALS

"Self-monitoring of blood glucose Useful for Managing Gestational Diabetes." *Women's Health Weekly* (August 7, 2003): 42.

"Smoking Tied to Increased Risk of Gestational Diabetes; Dose Response Relationship." *OB GYN News* (September 1, 2003): 5.

ORGANIZATIONS

American Diabetes Association. 1701 North Beauregard Street, Alexandria, VA 22311. (800) 342-2383. <<http://www.diabetes.org>>.

Rosalyn Carson-DeWitt, MD
Teresa G. Odle

GI bleeding studies

Definition

GI bleeding studies uses radioactive materials in the investigation of bleeding from the gastrointestinal (GI) tract. These studies go under various names such as "GI bleeding scans" or "Tagged red blood cell scans." They

are performed and interpreted by radiologists (physicians who specialize in diagnosis and treatment of diseases by means of x rays or related substances).

Purpose

These studies are designed to find the source of blood loss from the GI tract; that is the stomach, small bowel, or colon. They work best when bleeding is either too slow, intermittent, or too rapid to be identified by other means, such as endoscopy, upper GI series, or **barium enema**.

They are particularly useful when other methods have not been able to determine the site or cause of bleeding.

Precautions

Because of the use of radioactive materials, these studies are best avoided in pregnant patients. Another important relates to the interpretation of these tests, whether normal or abnormal. Since these studies are far from perfect, they can only be used as "guides" as to the cause or site of bleeding. In most instances, further studies must be performed to confirm their findings.

Description

Bleeding scans are based on the accumulation of radioactive material as it exits from the vessels during a bleeding episode. Blood is first withdrawn from the patient. Then, the blood, along with a radioactive substance is injected into a vein and over several hours scans measuring radioactivity are performed. The studies were initially reported to be very sensitive and accurate; however, critical evaluation of these tests have shown them to be less accurate than originally believed.

Preparation

No preparation is needed for these tests. They are often done on an "emergency" basis.

Aftercare

No special care is needed after the exam.

Risks

Bleeding scans are free of any risks or side-effects, aside from the fact that they should best be avoided in **pregnancy**.



A clay model of the human digestive system. (Custom Medical Stock Photo. Reproduced by permission.)

KEY TERMS

Endoscope, Endoscopy—An endoscope as used in the field of gastroenterology is a thin flexible tube which uses a lens or miniature camera to view various areas of the gastrointestinal tract. The performance of an exam using an endoscope is referred to by the general term endoscopy. Diagnosis through biopsies or other means and therapeutic procedures can be done with these instruments.

Normal results

A normal exam would fail to show any evidence of accumulation of radioactive material on the scan. However, scans may be normal in as many as 70% of patients who later turn out to have significant causes

of bleeding. This is known as a false-negative result. A patient must be bleeding at the same time the scan is performed for it to be seen. Therefore, not finding evidence of a bleeding source during the study, can be misleading.

Abnormal results

The accumulation of radioactive material indicating a “leakage” of blood from the vessels is abnormal. The scan gives a rough, though not exact, guide as to the location of the bleeding. It can tell where the bleeding may be, but usually not the cause. Thus, extreme caution and skill is needed in interpreting these scans, and decisions involving surgery or other treatment should await more definitive tests.

Resources

BOOKS

Lane, Loren. “Radionuclide Scanning.” In *Sleisenger & Fordtran’s Gastrointestinal and Liver Disease*, edited by Mark Feldman, et al. Philadelphia: W. B. Saunders Co., 1997.

David Kaminstein, MD

Giant-cell arteritis see **Temporal arteritis**

Giardia lamblia infection see **Giardiasis**

Giardiasis

Definition

Giardiasis is a common intestinal infection spread by eating contaminated food, drinking contaminated water, or through direct contact with the organism that causes the disease, *Giardia lamblia*. Giardiasis is found throughout the world and is a common cause of traveler’s **diarrhea**. In the United States it is a growing problem, especially among children in childcare centers.

Description

Giardia is one of the most common intestinal parasites in the world, infecting as much as 20% of the entire population of the earth. It is common in overcrowded developing countries with poor sanitation and a lack of clean water. Recent tests have found *Giardia* in 7% of all stool samples tested nationwide, indicating that this disease is much more widespread

than was originally believed. It has been found not only in humans, but also in wild and domestic animals.

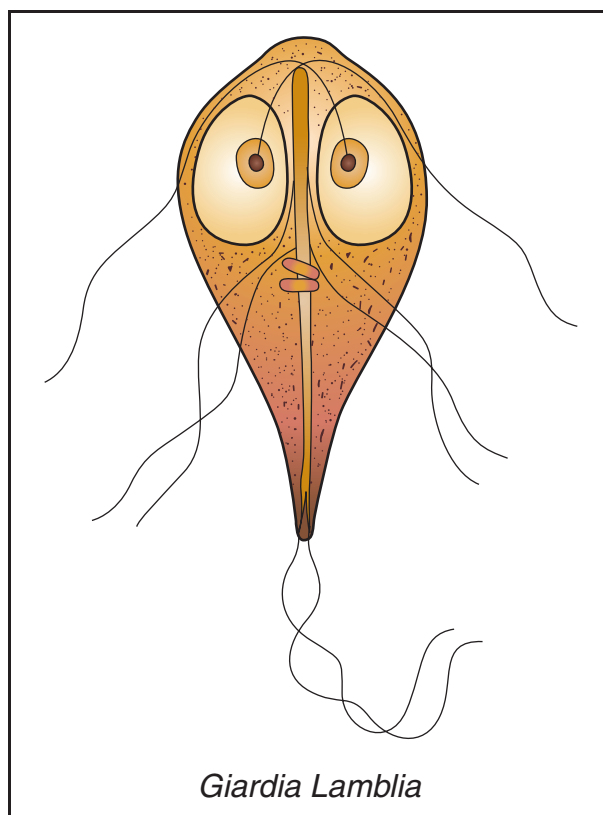
Giardiasis is becoming a growing problem in the United States, where it affects three times more children than adults. In recent years, giardiasis outbreaks have been common among people in schools or day-care centers and at catered affairs and large public picnic areas. Children can easily pass on the infection by touching contaminated toys, changing tables, utensils, or their own feces, and then touching other people. For this reason, infection spreads quickly through a daycare center or institution for the developmentally disabled.

Unfiltered streams or lakes that may be contaminated by human or animal wastes are a common source of infection. Outbreaks can occur among campers and hikers who drink untreated water from mountain streams. While 20 million Americans drink unfiltered city water from streams or rivers, giardiasis outbreaks from tainted city water have been rare. Most of these problems have occurred not due to the absence of filters, but because of malfunctions in city water treatment plants, such as a temporary drop in chlorine levels. It is possible to become infected in a public swimming pool, however, since *Giardia* can survive in chlorinated water for about 15 minutes. During that time, it is possible for an individual to swallow contaminated pool water and become infected.

Causes and symptoms

Giardiasis is spread by food or water contaminated by the *Giardia lamblia* protozoan organism found in the human intestinal tract and feces. When the cysts are ingested, the stomach acid degrades the cysts and releases the active parasite into the body. Once within the body, the parasites cling to the lining of the small intestine, reproduce, and are swept into the fecal stream. As the liquid content of the bowel dries up, the parasites form cysts, which are then passed in the feces. Once excreted, the cysts can survive in water for more than three months. The parasite is spread further by direct fecal-oral contamination, such as can occur if food is prepared without adequate hand-washing, or by ingesting the cysts in water or food.

Giardiasis is not fatal, and about two-thirds of infected people exhibit no symptoms. Symptoms will not occur until between one and two weeks after infection. When present, symptoms include explosive, watery diarrhea that can last for a week or more and, in chronic cases, may persist for months. Because the



Infection with the protozoan *Giardia lamblia*, shown above, causes diarrhea in humans. (Illustration by Electronic Illustrators Group).

infection interferes with the body's ability to absorb fats from the intestinal tract, the stool is filled with fat. Other symptoms include foul-smelling and greasy feces, stomach pains, gas and bloating, loss of appetite, **nausea and vomiting**. In cases in which the infection becomes chronic, lasting for months or years, symptoms might include poor digestion, problems digesting milk, intermittent diarrhea, **fatigue**, weakness, and significant weight loss.

Diagnosis

Diagnosis can be difficult because it can be easy to overlook the presence of the giardia cysts during a routine inspection of a stool specimen. In the past, the condition has been diagnosed by examining three stool samples for the presence of the parasites. However, because the organism is shed in some stool samples and not others, the infection may not be discovered using this method.

A newer, more accurate method of diagnosing the condition is the enzyme-linked immunosorbent assay

KEY TERMS

Antibody—A specific protein produced by the immune system in response to a specific foreign protein or particle called an antigen.

Antigen—A substance (usually a protein) identified as foreign by the body's immune system, triggering the release of antibodies as part of the body's defense mechanism.

Enzyme-linked immunosorbent assay (ELISA)—A laboratory technique used to detect specific antigens or antibodies. It can be used to diagnose giardiasis.

Giardia lamblia—A type of protozoa with a whip-like tail that infects the human intestinal tract, causing giardiasis. The protozoa will not spread to other parts of the body.

Immunocompromised—A state in which the immune system is suppressed or not functioning properly.

(ELISA) that detects cysts and antigen in stool, and is approximately 90% accurate. While slightly more expensive, it only needs to be done once and is therefore less expensive overall than the earlier test.

Treatment

Acute giardiasis can usually be allowed to run its natural course and tends to clear up on its own. **Antibiotics** are helpful, however, in easing symptoms and preventing the spread of infection. Medications include metronidazole, furazolidone and paromomycin. Healthy carriers with no symptoms do not need antibiotic treatment. If treatment should fail, the patient should wait two weeks and repeat the drug course. Anyone with an impaired immune system (immunocompromised), such as a person with **AIDS**, may need to be treated with a combination of medications.

Prognosis

Giardiasis is rarely fatal, and when treated promptly, antibiotics usually cure the infection. While most people respond quickly to treatment, some have lingering symptoms and suffer with diarrhea and cramps for long periods, losing weight and not growing well. Those most at-risk for a course like this are the elderly, people with a weakened immune

system, malnourished children, and anyone with low stomach acid.

Prevention

The best way to avoid giardiasis is to avoid drinking untreated surface water, especially from mountain streams. The condition also can be minimized by practicing the following preventive measures:

- thoroughly washing hands before handling food
- maintaining good personal cleanliness
- boiling any untreated water for at least three minutes
- properly disposing of fecal material

Children with severe diarrhea (and others who are unable to control their bowel habits) should be kept at home until the stool returns to normal. If an outbreak occurs in a daycare center, the director should notify the local health department. Some local health departments require a follow-up stool testing to confirm that the person is no longer contagious. People not in high-risk settings can return to their routine activities after recovery.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

OTHER

Centers for Disease Control. <<http://www.cdc.gov/ncidod/EID/eidtext.htm>>.

International Society of Travel Medicine.
<<http://www.istm.org>>.

Carol A. Turkington

Gigantism see **Acromegaly and gigantism**

Gilchrist's disease see **Blastomycosis**

Gilles de la Tourette's syndrome see **Tourette syndrome**

Gingivitis see **Periodontal disease**

Ginkgo biloba

Definition

Ginkgo biloba, known as the maidenhair tree, is one of the oldest trees on Earth, once part of the flora

of the Mesozoic period. The ginkgo tree is the only surviving species of the Ginkgoaceae family. This ancient deciduous tree may live for thousands of years. Ginkgo is indigenous to China, Japan, and Korea, but also thrived in North America and Europe prior to the Ice Age. This drastic climate change destroyed the wild ginkgo tree throughout much of the world. In China, ginkgo was cultivated in temple gardens as a sacred tree known as *bai gou*, thus assuring its survival there for over 200 million years. Ginkgo fossils found from the Permian period are identical to the living tree, which is sometimes called a living fossil.

Description

Ginkgo trees may grow to 122 ft (37.2 m) tall and measure 4 ft (1.2 m) in girth. The female trees have a somewhat pointed shape at the top, like a pyramid. The male trees are broader at the crown. The bark of the ornamental ginkgo tree is rough and fissured and may be an ash to dark-brown in color. Distinctive, fan-shaped leaves with long stalks emerge from a sheath on the stem. Leaves are bright green in spring and summer, and turn to golden yellow in the fall. Ginkgo trees may take as long as 30 years to flower. Ginkgo is dioecious, with male and female flowers blooming on separate trees. Blossoms grow singly from the axils of the leaf. The female flowers appear at the end of a leafless branch. The yellow, plum-shaped fruits develop an unpleasant scent as they ripen. They contain an edible inner seed that is available in Asian country marketplaces. Ginkgo's longevity may be due, in part, to its remarkable resistance to disease, pollution, and insect damage. Ginkgo trees are part of the landscape plan in many urban areas throughout the world. Millions of ginkgo trees, grown for harvest of the medicinal leaves, are raised on plantations in the United States, France, South Korea, and Japan, and are exported to Europe for pharmaceutical processing.

Purpose

Ginkgo leaves, fresh or dry, and seeds, separated from the outer layer of the fruit, are used medicinally. Ginkgo has remarkable healing virtues that have been recorded as far back as 2800 B.C. in the oldest Chinese materia medica. Ginkgo seeds were traditionally served to guests along with alcohol drinks in Japan. An enzyme present in the ginkgo seed has been shown in clinical research to speed up alcohol metabolism in the body, underscoring the wisdom of this folk custom. The leaf extract has been used in Asia for

thousands of years to treat **allergies**, **asthma**, and **bronchitis**. It is also valued in Chinese medicine as a heart tonic, helpful in the treatment of cardiac arrhythmia. Ginkgo was first introduced to Europe in 1730, and to North America in 1784 where it was planted as an exotic garden ornamental near Philadelphia. Ginkgo medicinal extracts are the primary prescription medicines used in France and Germany.

Ginkgo acts to increase blood flow throughout the body, particularly cerebral blood flow. It acts as a circulatory system tonic, stimulating greater tone in the venous system. The herb is a useful and proven remedy for numerous diseases caused by restricted blood flow. European physicians prescribe the extract for treatment of **Raynaud's disease**, a condition of impaired circulation to the fingers. It is also recommended to treat **intermittent claudication**, a circulatory condition that results in painful cramping of the calf muscles in the leg and impairs the ability to walk. German herbalists recommend ingesting the extract for treatment of leg ulcers, and large doses are used to treat **varicose veins**. Ginkgo is widely recommended in Europe for the treatment of **stroke**. The dried leaf extract may also act to prevent hemorrhagic stroke by strengthening the blood capillaries throughout the body. In studies of patients with atherosclerotic clogging of the penile artery, long-term therapy with ginkgo extract has provided significant improvement in erectile function. Ginkgo extract also acts to eliminate damaging free-radicals in the body, and has been shown to be effective in treatment of **premenstrual syndrome**, relieving tender or painful breasts.

Ginkgo extract has proven benefits to elderly persons. This ancient herb acts to enhance oxygen utilization and thus improves memory, concentration, and other mental faculties. The herbal extract is used to treat **Alzheimer's disease**. It has been shown to have beneficial effect on the hippocampus, an area of the brain affected by Alzheimer's disease. The herbal extract has also been shown to significantly improve long-distance vision and may reverse damage to the retina of the eye. Studies have also confirmed its value in the treatment of depression in elderly persons. The ginkgo extract may provide relief for persons with **headache**, **sinusitis**, and vertigo. It may also help relieve chronic ringing in the ears known as **tinnitus**.

The active constituents in the ginkgo tree, known as ginkgolides, interfere with a blood protein known as the platelet activating factor, or PAF. Other phytochemicals in ginkgo include flavonoids, biflavonoides, proanthocyanidins, trilactonic diterpenes (including the ginkgolides A, B, C, and M), and bilabolide, a trilactonic sesquiterpene. The therapeutic effects of



Ginkgo biloba leaves. (Photograph by Robert J. Huffman. Field Mark Publications. Reproduced by permission.)

this herb have not been attributed to a single chemical constituent; rather, the medicinal benefits are due to the synergy between the various chemical constituents. The standardized extract of ginkgo must be taken consistently to be effective. A period of at least 12 weeks of use may be required before the beneficial results are evident.

Preparations

Ginkgo's active principles are dilute in the leaves. The herb must be processed to extract the active phytochemicals before it is medicinally useful. It would take an estimated 50 fresh ginkgo leaves to yield one standard dose of the extract. Dry extracts of the leaf, standardized to a potency of 24% flavone glycosides and 6% terpenes, are commercially available. A standard dose is 40 mg, three times daily, though dosages as high as 240 mg daily are sometimes indicated.

Ginkgo extracts are widely used in Europe where they are sold in prescription form or over the counter as an approved drug. This is not the case in the United States, where ginkgo extract is sold as a food supplement in tablet and capsule form.

Precautions

Ginkgo is generally safe and non-toxic in therapeutic dosages. Exceeding a daily dose of 240 mg of the dried extract may result in restlessness, **diarrhea**, and mild gastrointestinal disorders. Those on anticoagulants should have their doctor adjust their dose or should avoid ginkgo in order to avoid over-thinning their blood and hemorrhaging. Ginkgo should be avoided two days before and one to two weeks after surgery to avoid bleeding complications.

Side effects

Severe allergic skin reactions, similar to those caused by poison ivy, have been reported after contact with the fruit pulp of ginkgo. Eating even a small amount of the fruit has caused severe gastrointestinal irritation in some persons. People with persistent headaches should stop taking ginkgo. Some patients on medications for nervous system disease should avoid ginkgo. It can interact with some other medicines, but clinical information is still emerging.

Interactions

The chemically active ginkgolides present in the extract, specifically the ginkgolide B component, act to reduce the clotting time of blood and may interact with antithrombotic medicines, including **aspirin**.

Resources

BOOKS

PDR for Herbal Medicines. Montvale, NJ: Medical Economics Company, 1998.

Clare Hanrahan

Ginseng, Korean

Definition

Korean ginseng is one of the most widely used and acclaimed herbs in the world. Its scientific name is *Panax ginseng*, which is the species from which Chinese, Korean, red, and white ginseng are produced. Chinese and Korean ginseng are the same plant cultivated in different regions, and have slightly different properties according to Chinese medicine. White ginseng is simply the dried or powdered root of Korean ginseng, while red ginseng is the same root that is steamed and dried in heat or sunlight. Red ginseng is said to be slightly stronger and more stimulating in the body than white, according to Chinese herbalism.

Description

Korean ginseng has had a long and illustrious history as an herb for health, and has been used for thousands of years throughout the Orient as a medicine and tonic. Early Chinese medicine texts written in the first century A.D. mention ginseng, and ginseng

has long been classified by Chinese medicine as a “superior” herb. This means it is said to promote longevity and vitality. Legends around the world have touted ginseng as an aphrodisiac and sexual tonic. Researchers have found a slight connection between sex drive and consuming ginseng, although a direct link and the mechanism of action are still researched and disputed.

Korean ginseng grows on moist, shaded mountainsides in China, Korea, and Russia. It is a perennial herb that reaches heights of two or more feet, and is distinguished by its dark green leaves and red clusters of berries. The root of the plant is the part valued for its medicinal properties. The root is long and slender and sometimes resembles the shape of the human body. Asian legends claim that this “man-root” has magical powers for those lucky enough to afford or find it, and the roots bearing the closest resemblance to the human body are still the most valuable ones. The word *ren shen* in Chinese means roughly “the essence of the earth in the shape of a man.”

Korean ginseng has historically been one of the most expensive of herbs, as it has been highly in demand in China and the Far East for centuries. Wars have been fought in Asia over lands where it grew wild. Wild Korean ginseng is now nearly extinct from many regions. Single roots of wild plants have recently been auctioned in China and New York City for sums approaching \$50,000. Most of the world's supply of Korean ginseng is cultivated by farmers in Korea and China.

Because of the number of herbs sold under the name of ginseng, there can be some confusion for the consumer. Korean ginseng is a member of the *Araliaceae* family of plants, which also includes closely related American ginseng (*Panax quinquefolius*) and Siberian ginseng (*Eleutherococcus senticosus*). Both American and Siberian ginseng are considered by Chinese herbalists to be different herbs than Korean ginseng, and are said to have different effects and healing properties in the body. To add more confusion, there are eight herbs in Chinese medicine which are sometimes called ginseng, including black ginseng, purple ginseng, and prince's ginseng, some of which are not at all botanically related to *Panax ginseng*, so consumers should choose ginseng products with awareness.

Purpose

The word *panax* is formed from Greek roots meaning “cure-all,” and *Panax ginseng* has long been considered to be one of the great healing and

strengthening herbs in natural medicine. Ginseng is classified as an *adaptogen*, which is a substance that helps the body adapt to **stress** and balance itself without causing major side effects. Korean ginseng is used as a tonic for improving overall health and stamina, and Chinese herbalists particularly recommend it for the ill, weak, or elderly. Korean ginseng has long been asserted to have longevity, anti-senility, and memory improvement effects in the aged population. As it helps the body to adapt to stress, athletes may use ginseng as herbal support during rigorous training. Korean ginseng generally increases physical and mental energy. It is a good tonic for the adrenal glands, and is used by those suffering from exhaustion, burn-out, or debilitation from chronic illness.

Traditional Chinese medicine also prescribes Korean ginseng to treat diabetes, and research has shown that it enhances the release of insulin from the pancreas and lowers blood sugar levels. Korean ginseng has been demonstrated to lower blood cholesterol levels. It has also been shown to have antioxidant effects and to increase immune system activity, which makes it a good herbal support for those suffering from **cancer** and **AIDS** and other chronic conditions that impair the immune system. Further uses of Korean ginseng in Chinese medicine include treatment of **impotence**, **asthma**, and digestive weakness.

Research

Scientists have isolated what they believe are the primary active ingredients in ginseng, chemicals termed *saponin triterpenoid glycosides*, or commonly called *ginsenosides*. There are nearly 30 ginsenosides in Korean ginseng. Much research on Korean ginseng has been conducted in China, but controlled human experiments with it have not been easily accessible to the English-speaking world. Recent research in China was summarized by Dr. C. Lui in the February 1992 issue of the *Journal of Ethnopharmacology*, where he wrote that *Panax ginseng* was found to contain 28 ginsenosides that “act on the central nervous system, cardiovascular system and endocrine secretion, promote immune function, and have effects on anti-aging and relieving stress.”

To summarize other research, Korean ginseng has been shown in studies to have significant effects for the following.

- Physical improvement and performance enhancement for athletes: A study performed over three years in Germany showed athletes given ginseng had favorable improvement in several categories over a control group who took a placebo. Another

1982 study showed that athletes given ginseng had improved oxygen intake and faster recovery time than those given placebos.

- **Mental performance improvement and mood enhancement:** In general, studies show that ginseng enhances mental performance, learning time, and memory. One study of sixteen volunteers showed improvement on a wide variety of mental tests, including mathematics. Another study showed that those performing intricate and mentally demanding tasks improved performance when given Korean ginseng. Finally, a study has shown improvement of mood in depression sufferers with the use of ginseng.
- **Antifatigue and antistress actions:** Patients with chronic **fatigue** who were given ginseng showed a statistically significant improvement in physical tests and in mental attention and concentration, when compared with those given placebos.
- **Lowering blood sugar:** Animal studies have shown that ginseng can facilitate the release of insulin from the pancreas and increase the number of insulin receptors in the body.
- **Antioxidant properties:** Scientific analysis of ginseng has shown that it has antioxidant effects, similar to the effects of **vitamins A, C, and E**. Thus, ginseng could be beneficial in combating the negative effects of pollution, radiation, and aging.
- **Cholesterol reduction:** Some studies have shown that Korean ginseng reduces total cholesterol and increases levels of good cholesterol in the body.
- **Anticancer effects and immune system stimulation:** Several tests have shown that Korean ginseng increases immune cell activity in the body, including the activity of T-cells and lymphocytes, which are instrumental in fighting cancer and other immune system disorders like AIDS. A Korean study indicates that taking ginseng may reduce the chances of getting cancer, as a survey of more than 1,800 patients in a hospital in Seoul showed that those who did not have cancer were more likely to have taken ginseng regularly than those patients who had contracted cancer.
- **Physical and mental improvement in the elderly:** One study showed significant improvement in an elderly test group in visual and auditory reaction time and cardiopulmonary function when given controlled amounts of Korean ginseng. Korean ginseng has also been shown to alleviate symptoms of menopause.
- **Impotence:** Studies of human sexual function and Korean ginseng have been generally inconclusive,



Dried Korean ginseng. (Custom Medical Stock Photo. Reproduced by permission.)

despite the wide acclaim of ginseng as a sexual tonic. Tests with lab animals and ginseng have shown some interesting results, indicating that Korean ginseng promotes the growth of male reproductive organs, increases sperm and testosterone levels, and increases sexual activity in laboratory animals. In general, scientists believe the link between ginseng and sex drive is due to ginseng's effect of strengthening overall health and balancing the hormonal system.

Preparations

Korean ginseng can be purchased as whole roots, powder, liquid extracts, and tea. Roots should be sliced and boiled in water for up to 45 minutes to extract all the beneficial nutrients. One to five grams of dry root is the recommended amount for one serving of tea. Herbalists recommend that ginseng not be boiled in metal pots, to protect its antioxidant properties. Ginseng should be taken between meals for best assimilation.

Some high quality Korean ginseng extracts and products are standardized to contain a specified amount of ginsenosides. The recommended dosage

KEY TERMS

Adaptogen—Substance that improves the body's ability to adapt to stress.

Ginsenoside—Active substances found in ginseng.

for extracts containing four to eight percent of ginsenosides is 100 mg once or twice daily. The recommended dosage for non-standardized root powder or extracts is 1–2 g daily, taken in capsules or as a tea. It is recommended that ginseng be taken in cycles and not continuously; after each week of taking ginseng, a few days without ingesting the herb should be observed. Likewise, Korean ginseng should not be taken longer than two months at a time, after which one month's rest period should be allowed before resuming the cycle again. Chinese herbalists recommend that ginseng be taken primarily in the autumn and winter months.

Precautions

Consumers should be aware of the different kinds of ginseng, and which type is best suited for them. Red Korean ginseng is considered stronger and more stimulating than white, wild ginseng is stronger than cultivated, and Korean ginseng is generally believed to be slightly stronger than Chinese. Furthermore, American and Siberian ginseng have slightly different properties than Korean ginseng, and consumers should make an informed choice as to which herb is best suited for them. Chinese herbalists do not recommend Korean ginseng for those people who have “heat” disorders in their bodies, such as ulcers, high blood pressure, tension headaches, and symptoms associated with high stress levels. Korean ginseng is generally not recommended for those with symptoms of nervousness, mental imbalance, inflammation, or **fever**. Korean ginseng is not recommended for pregnant or lactating women, and women of childbearing age should use ginseng sparingly, as some studies imply that it can influence estrogen levels. Also, Chinese herbalists typically only prescribe ginseng to older people or the weak, as they believe that younger and stronger people do not benefit as much from it and ginseng is “wasted on the young.”

Because of the number of and demand for ginseng products on the market, consumers should search for a reputable brand, preferably with a standardized percentage of active ingredients. To illustrate the mislabeling found with some ginseng products, *Consumer*

Reports magazine analyzed ten nationally-distributed ginseng products in 1995. They found that several of them lacked significant amounts of ginsenosides, despite claims on the packaging to the contrary. Ginseng fraud has led the American Botanical Council, publisher of *HerbalGram* magazine, to initiate the Ginseng Evaluation Program, a comprehensive study and standardization of ginseng products on the American market. This study and its labeling standards are still under development, and consumers should watch for it.

Side effects

Korean ginseng acts as a slight stimulant in the body, and in some cases can cause overstimulation, irritability, nervousness and **insomnia**, although strong side effects are generally rare. Taking too high a dosage of ginseng, or taking ginseng for too long without a break, can cause *ginseng intoxication*, for which symptoms might include headaches, insomnia, seeing spots, **dizziness**, shortage of breath and gastrointestinal discomfort. Long term use may cause menstrual abnormalities and breast tenderness in some women.

Interactions

Those taking hormonal drugs should use ginseng with care. Ginseng should not be taken with **caffeine** or other stimulants as these may increase its stimulatory effects and cause uncomfortable side effects.

Resources

BOOKS

Hobbs, Christopher. *Ginseng: The Energy Herb*. Loveland, CO: Botanica Press, 1996.

PERIODICALS

HerbalGram (a quarterly journal of the American Botanical Council and Herb Research Foundation). P.O. Box 144345, Austin, TX 78714-4345. (800) 373-7105.

Douglas Dupler, MA

Glaucoma

Definition

Glaucoma is a group of eye diseases characterized by damage to the optic nerve usually due to excessively high intraocular pressure (IOP). This increased pressure within the eye, if untreated can lead to optic nerve

damage resulting in progressive, permanent vision loss, starting with unnoticeable blind spots at the edges of the field of vision, progressing to tunnel vision, and then to blindness.

Description

Between two to three million people in the United States have glaucoma, and 120,000 of those are legally blind as a result. It is the leading cause of preventable blindness in the United States and the most frequent cause of blindness in African-Americans, who are at about a three-fold higher risk of glaucoma than the rest of the population. The risk of glaucoma increases dramatically with age, but it can strike any age group, even newborn infants and fetuses.

Glaucoma can be classified into two categories: open-angle glaucoma and narrow-angle glaucoma. To understand what glaucoma is and what these terms mean, it is useful to understand eye structure.

Eyes are sphere-shaped. A tough, non-leaky protective sheath (the sclera) covers the entire eye, except for the clear cornea at the front and the optic nerve at the back. Light comes into the eye through the cornea, then passes through the lens, which focuses it onto the retina (the innermost surface at the back of the eye). The rods and cones of the retina transform the light energy into electrical messages, which are transmitted to the brain by the bundle of nerves known as the optic nerve.

The iris, the colored part of the eye shaped like a round picture frame, is between the dome-shaped cornea and the lens. It controls the amount of light that enters the eye by opening and closing its central hole (pupil) like the diaphragm in a camera. The iris, cornea, and lens are bathed in a liquid called the aqueous humor, which is somewhat similar to plasma. This liquid is continually produced by nearby ciliary tissues and moved out of the eye into the bloodstream by a system of drainage canals (called the trabecular meshwork). The drainage area is located in front of the iris, in the angle formed between the iris and the point at which the iris appears to meet the inside of the cornea.

Glaucoma occurs if the aqueous humor is not removed rapidly enough or if it is made too rapidly, causing pressure to build-up. The high pressure distorts the shape of the optic nerve and destroys the nerve. Destroyed nerve cells result in blind spots in places where the image from the retina is not being transmitted to the brain.

Open-angle glaucoma accounts for over 90% of all cases. It is called “open-angle” because the angle

between the iris and the cornea is open, allowing drainage of the aqueous humor. It is usually chronic and progresses slowly. In narrow-angle glaucoma, the angle where aqueous fluid drainage occurs is narrow, and therefore may drain slowly or may be at risk of becoming closed. A closed-angle glaucoma attack is usually acute, occurring when the drainage area is blocked. This can occur, for example, if the iris and lens suddenly adhere to each other and the iris is pushed forward. In patients with very narrow angles, this can occur when the eyes dilate (e.g., when entering a dark room, or if taking certain medications).

Congenital glaucoma occurs in babies and is the result of incomplete development of the eye's drainage canals during embryonic development. Microsurgery can often correct the defects or they can be treated with a combination of medicine and surgery.

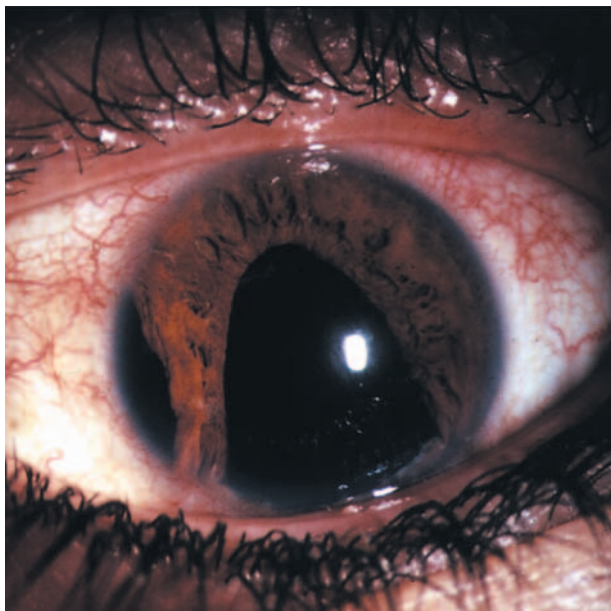
One rare form of open-angle glaucoma, normal tension glaucoma, is different. People with normal-tension glaucoma have optic nerve damage in the presence of normal IOP. As of 1998, the mechanism of this disease is a mystery but is generally detected after an examination of the optic nerve. Those at higher risk for this form of glaucoma are people with a familial history of normal tension glaucoma, people of Japanese ancestry, and people with a history of systemic heart disease such as irregular heart rhythm.

Glaucoma is also a secondary condition of over 60 widely diverse diseases and can also result from injury, inflammation, tumor, or in advanced cases of cataract or diabetes.

Causes and symptoms

Causes

The cause of vision loss in all forms of glaucoma is optic nerve damage. There are many underlying causes and forms of glaucoma. Most causes of glaucoma are not known, but it is clear that a number of different processes are involved, and a malfunction in any one of them could cause glaucoma. For example, trauma to the eye could result in the angle becoming blocked, or, as a person ages, the lens becomes larger and may push the iris forward. The cause of optic nerve damage in normal-tension glaucoma is also unknown, but there is speculation that the optic nerves of these patients are susceptible to damage at lower pressures than what is usually considered to be abnormally high.



A close-up view of an inflamed eye with acute glaucoma and an irregularly enlarged pupil. (Custom Medical Stock Photo. Reproduced by permission.)

It is probable that most glaucoma is inherited. At least ten defective genes that cause glaucoma have been identified.

Symptoms

At first, chronic open-angle glaucoma is without noticeable symptoms. The pressure build-up is gradual and there is no discomfort. Moreover, the vision loss is too gradual to be noticed and each eye fills-in the image where its partner has a blind spot. However, if it is not treated, vision loss becomes evident, and the condition can be very painful.

On the other hand, acute closed-angle glaucoma is obvious from the beginning of an attack. The symptoms are, blurred vision, severe **pain**, sensitivity to light, **nausea**, and halos around lights. The normally clear corneas may be hazy. This is an ocular emergency and needs to be treated immediately.

Similarly, congenital glaucoma is evident at birth. Symptoms are bulging eyes, cloudy corneas, excessive tearing, and sensitivity to light.

Diagnosis

Intraocular pressure, visual field defects, the angle in the eye where the iris meets the cornea, and the appearance of the optic nerve are all considered in the diagnosis of glaucoma. IOP is measured with an

instrument known as a tonometer. One type of tonometer involves numbing the eye with an eyedrop that has a yellow coloring in it and touching the cornea with a small probe. This quick test is a routine part of an **eye examination** and is usually included without extra charge in the cost of a visit to an ophthalmologist or optometrist.

Ophthalmoscopes, hand-held instruments with a light source, are used to detect optic nerve damage by looking through the pupil. The optic nerve is examined for changes; the remainder of the back of the eye can be examined as well. Other types of lenses that can be used to examine the back of the eye may also be used. A slit lamp will allow the doctor to examine the front of the eye (i.e., cornea, iris, and lens).

Visual field tests (perimetry) can detect blind spots in a patient's field of vision before the patient is aware of them. Certain defects may indicate glaucoma.

Another test, gonioscopy, can distinguish between narrow-angle and open-angle glaucoma. A gonioscope, which is a hand-held contact lens with a mirror, allows visualization of the angle between the iris and the cornea.

Intraocular pressure can vary throughout the day. For that reason, the doctor may have a patient return for several visits to measure the IOP at different times of the day.

Treatment

Medications

When glaucoma is diagnosed, drugs, typically given as eye drops, are usually tried before surgery. Several classes of medications are effective at lowering IOP and thus preventing optic nerve damage in chronic and neonatal glaucoma. **Beta blockers**, like Timoptic; carbonic anhydrase inhibitors, like acetazolamide; and alpha-2 agonists, such as Alphagan, inhibit the production of aqueous humor. Miotics, like pilocarpine, and prostaglandin analogues, like Xalatan, increase the outflow of aqueous humor. Cosopt is the first eyedrop that is a combined beta blocker (Timoptic) and carbonic anhydrase inhibitor and may be helpful for patients required to take more than one glaucoma medication each day. The Food and drug administration recently approved two new prostaglandin-related drugs, Travatan and Lumigan on March 16, 2001. These drugs work by decreasing intraocular pressure and may be considered for people with glaucoma that are unable to tolerate other IOP

lowering drugs. Additionally, Travatan may work best for African-Americans with glaucoma (a population at high risk for glaucoma).

It is important for patients to tell their doctors about any conditions they have or medications they are taking. Certain drugs used to treat glaucoma should not be prescribed for patients with pre-existing conditions. Some of these drugs mentioned have side effects, so patients taking them should be monitored closely, especially for cardiovascular, pulmonary, and behavioral symptoms. Different medications lower IOP by different amounts, and a combination of medications may be necessary. It is important that patients take their medications and that their regimens are monitored regularly, to be sure that the IOP is lowered sufficiently. IOP should be measured three to four times per year.

Normal-tension glaucoma is treated in the same way as chronic high-intraocular-pressure glaucoma. This reduces IOP to less-than-normal levels, on the theory that overly susceptible optic nerves are less likely to be damaged at lower pressures. Research underway may point to better treatments for this form of glaucoma.

Attacks of acute closed-angle glaucoma are medical emergencies. IOP is rapidly lowered by successive deployment of acetazolamide, hyperosmotic agents, a topical beta-blocker, and pilocarpine. Epinephrine should not be used because it exacerbates angle closure.

Surgery

There are several types of **laser surgery** used to treat glaucoma. Laser peripheral iridotomy makes an opening in the iris allowing the fluid to drain, argon laser trabeculoplasty is aimed at the fluid channel opening to help the drainage system function and laser cyclophotocoagulation is used to decrease the amount of fluid made. Microsurgery, also called “filtering surgery” has been used in many different types of glaucoma. A new opening is created in the sclera allowing the intraocular fluid to bypass the blocked drainage canals. The tissue over this opening forms a little blister or bleb on the clear conjunctiva that Doctors monitor ensuring that fluid is draining. These surgeries are usually successful, but the effects often last less than a year. Nevertheless, they are an effective treatment for patients whose IOP is not sufficiently lowered by drugs and for those who can’t tolerate the drugs. Because all surgeries have risks, patients should speak to their doctors about the procedure being performed.

Alternative treatment

Vitamin C, vitamin B₁ (thiamine), chromium, zinc, bilberry and rutin may reduce IOP.

There is evidence that medicinal **marijuana** lowers IOP, too. However, marijuana has serious side effects and contains carcinogens, and any IOP-lowering medication must be taken continually to avoid optic nerve damage. Although the Food and Drug Administration (FDA) and National Institutes of Health (NIH) currently recommend against treating glaucoma with marijuana, they are supporting research to learn more about it and to determine the feasibility of separating the components that lower IOP from components that produce side effects and carcinogens.

Any glaucoma patient using alternative methods to attempt to prevent optic nerve damage should also be under the care of a traditionally trained ophthalmologist or optometrist who is licensed to treat glaucoma, so that IOP and optic nerve damage can be monitored.

Prognosis

About half of the people stricken by glaucoma are not aware of it. For them, the prognosis is not good, and many of them will become blind. Sight lost due to glaucoma cannot be restored. On the other hand, the prognosis for treated glaucoma is excellent.

Prevention

Because glaucoma may not initially result in symptoms, the best form of prevention is to have regular eye exams.

Patients with narrow angles should avoid certain medications (even over-the-counter medications, such as some cold or allergy medications). Any person who is glaucoma-susceptible (i.e. narrow angles and borderline IOPs) should read the warning labels on over-the-counter medicines and inform their physicians of products they are considering taking. Steroids may also raise IOP, so patients may need to be monitored more frequently if it is necessary to use steroids for another medical condition.

Not enough is known about the underlying mechanisms of glaucoma to prevent the disease itself. However, prevention of optic nerve damage from glaucoma is essential and can be effectively accomplished when the condition is diagnosed and treated. As more is learned about the genes that cause glaucoma, it will become possible to test DNA and identify

KEY TERMS

Agonist—A drug that mimics one of the body's own molecules.

Alpha-2 agonist (alpha-2 adrenergic receptor agonist)—A class of drugs that bind to and stimulate alpha-2 adrenergic receptors, causing responses similar to those of adrenaline and noradrenaline. They inhibit aqueous humor production and have a wide variety of effects, including dry mouth, fatigue, and drowsiness.

Aqueous humor—A transparent liquid, contained within the eye, that is composed of water, sugars, vitamins, proteins, and other nutrients.

Betablocker (beta-adrenergic blocker)—A class of drugs that bind beta-adrenergic receptors and thereby decrease the ability of the body's own natural epinephrine to bind to those receptors, leading to inhibition of various processes in the body's sympathetic system. Betablockers can slow the heart rate, constrict airways in the lungs, lower blood pressure, and reduce aqueous secretion by ciliary tissues in the eye.

Carbonic anhydrase inhibitor—A class of diuretic drugs that inhibit the enzyme carbonic anhydrase, an enzyme involved in producing bicarbonate, which is required for aqueous humor production by the ciliary tissues in the eye. Thus, inhibitors of this enzyme inhibit aqueous humor production. Some side effects are urinary frequency, kidney stones, loss of the sense of taste, depression, and anemia.

Cornea—Clear, bowl-shaped structure at the front of the eye. It is located in front of the colored part of the eye (iris). The cornea lets light into the eye and partially focuses it.

Gonioscope—An instrument used to examine the trabecular meshwork; consists of a magnifier and a lens equipped with mirrors, which sits on the patient's cornea.

Hyperosmotic drugs—Refers to a class of drugs for glaucoma that increase the osmotic pressure in the

blood, which then pulls water from the eye into the blood.

Iris—The colored part of the eye just behind the cornea and in front of the lens that controls the amount of light sent to the retina.

Lens (the crystalline lens)—A transparent structure in the eye that focuses light onto the retina.

Laser cyclophotocoagulation—A procedure used for severe glaucoma in patients who have not responded well to previous treatments. The laser partially destroys the tissues that make the fluid of the eye.

Laser peripheral iridotomy—This procedure makes a drainage hole in the iris allowing the fluid to drain from the eye.

Laser Trabeculoplasty—In this procedure the laser attempts to open the normal drainage channels of the eye so fluid can drain more effectively.

Miotic—A drug that causes pupils to contract.

Ophthalmoscope—An instrument, with special lighting, designed to view structures in the back of the eye.

Optic nerve—The nerve that carries visual messages from the retina to the brain.

Prostaglandin—A group of molecules that exert local effects on a variety of processes including fluid balance, blood flow, and gastrointestinal function.

Prostaglandin analogue—A class of drugs that are similar in structure and function to prostaglandin.

Retina—The inner, light-sensitive layer of the eye containing rods and cones.

Sclera—The tough, fibrous, white outer protective covering that surrounds the eye.

Tonometry—The measurement of pressure.

Trabecular meshwork—A sponge-like tissue located near the cornea and iris that functions to drain the aqueous humor from the eye into the blood.

potential glaucoma victims, so they can be treated even before their IOP becomes elevated.

Bonny McClain

Glaucoma surgery see **Trabeculectomy**

Glioma see **Brain tumor**

Glipizide see **Antidiabetic drugs**

Glomerulonephritis

Definition

Acute glomerulonephritis is an inflammatory disease of both kidneys predominantly affecting children from ages two to 12. Chronic glomerulonephritis can develop over a period of 10–20 years and is most often associated with other systemic disease,

including diabetes, **malaria**, hepatitis, or **systemic lupus erythematosus**.

Description

Acute glomerulonephritis is an inflammation of the glomeruli, bundles of tiny vessels inside the kidneys. The damaged glomeruli cannot effectively filter waste products and excess water from the bloodstream to make urine. The kidneys appear enlarged, fatty, and congested.

Causes and symptoms

Acute glomerulonephritis most often follows a streptococcal infection of the throat or skin. In children, it is most often associated with an upper respiratory infection, **tonsillitis**, or **scarlet fever**. Kidney symptoms usually begin two to three weeks after the initial infection. Exposure to certain paints, glue or other organic solvents may also be the causative agent. It is thought that the kidney is damaged with exposure to the toxins that are excreted into the urine.

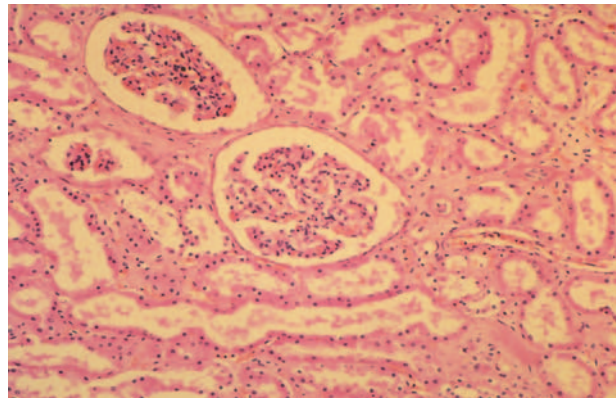
Mild glomerulonephritis may produce no symptoms, and diagnosis is made with laboratory studies of the urine and blood. Individuals with more severe cases of the disease may exhibit:

- fatigue
- nausea and **vomiting**
- shortness of breath
- disturbed vision
- high blood pressure
- swelling, especially noted in the face, hands, feet, and ankles
- blood and protein in the urine, resulting in a smoky or slightly red appearance

The individual with chronic glomerulonephritis may discover their condition with a routine physical exam revealing high blood pressure, or an eye exam showing vascular or hemorrhagic changes. The kidneys may be reduced to as little as one-fifth their normal size, consisting largely of fibrous tissues.

Diagnosis

Diagnosis of glomerulonephritis is established based on medical history, combined with laboratory studies. A “dipstick” test of urine will reveal increased protein levels. A 24 hour urine collection allows measurement of the excretion of proteins and creatinine. Creatinine clearance from the bloodstream by the



A close-up view of glomerulonephritis affecting the kidney.
(Custom Medical Stock Photo. Reproduced by permission.)

kidneys is considered an index of the glomerular filtration rate. Blood studies may reveal a low **blood count**, and may also be checked for the presence of a streptococcal antibody titer (a sophisticated blood test indicating presence of streptococcal infection). A **kidney biopsy** may also be performed, using ultrasound to guide the needle for obtaining the specimen.

Treatment

The main objectives in the treatment of acute glomerulonephritis are to:

- decrease the damage to the glomeruli
- decrease the metabolic demands on the kidneys
- improve kidney function

Bedrest helps in maintaining adequate blood flow to the kidney. If residual infection is suspected, antibiotic therapy may be needed. In the presence of fluid overload, **diuretics** may be used to increase output with urination. Iron and vitamin supplements may be ordered if anemia develops, and antihypertensives, if high blood pressure accompanies the illness. In order to rest the kidney during the acute phase, decreased sodium and protein intake may be recommended. The amount of protein allowed is dependent upon the amount lost in the urine, and the requirements of the individual patient. Sodium limitations depend on the amount of **edema** present. Fluid restrictions are adjusted according to the patient's urinary output and body weight.

An accurate daily record of the patient's weight, fluid intake and urinary output assist in estimating kidney function. The patient must be watched for signs of complications and recurrent infection. As edema is reduced and the urine becomes free of protein and red blood cells, the patient is allowed to increase

KEY TERMS

Dialysis—A process of filtering and removing waste products from the bloodstream. Two main types are hemodialysis and peritoneal dialysis. In hemodialysis, the blood flows out of the body into a machine that filters out the waste products and routes the cleansed blood back into the body. In peritoneal dialysis, the cleansing occurs inside the body. Dialysis fluid is injected into the peritoneal cavity and wastes are filtered through the peritoneum, the thin membrane that surrounds the abdominal organs.

Glomeruli—Groups of tiny blood vessels with very thin walls that function as filters in the kidney. Glomeruli become inflamed and are destroyed in the disease process of glomerulonephritis.

Renal—Relating to the kidneys, from the Latin word *renes*.

activity. A woman who has had glomerulonephritis requires special medical attention during **pregnancy**.

Prognosis

In acute glomerulonephritis, symptoms usually subside in two weeks to several months, with 90% of children recovering without complications and adults recovering more slowly. Chronic glomerulonephritis is a disease that tends to progress slowly, so that there are no symptoms until the kidneys can no longer function. The resultant renal failure may require dialysis or kidney transplant.

Prevention

Prevention of glomerulonephritis is best accomplished by avoiding upper respiratory infections, as well as other acute and chronic infections, especially those of a streptococcal origin. Cultures of the infection site, usually the throat, should be obtained and antibiotic sensibility of the offending organism determined. Prompt medical assessment for necessary antibiotic therapy should be sought when infection is suspected. The use of prophylactic immunizations is recommended as appropriate.

Resources

ORGANIZATIONS

American Association of Kidney Patients. 100 S. Ashley Dr., #280, Tampa, FL 33602. (800) 749-2257. <<http://www.aakp.org>>.

American Kidney Fund (AKF). Suite 1010, 6110 Executive Boulevard, Rockville, MD 20852. (800) 638-8299. <<http://216.248.130.102/Default.htm>>.

National Kidney Foundation. 30 East 33rd St., New York, NY 10016. (800) 622-9010. <<http://www.kidney.org>>.

National Kidney Foundation and Urologic Diseases Information Clearinghouse. 3 Information Way, Bethesda, MD 20892-3580. (800) 891-5390. <<http://www.niddk.nih.gov/health/kidney/nkudic.htm>>.

Kathleen D. Wright, RN

Glossopharyngeal neuralgia see **Neuralgia**

Glucose-6-phosphate dehydrogenase deficiency

Definition

Glucose-6-phosphate dehydrogenase deficiency is an inherited condition caused by a defect or defects in the gene that codes for the enzyme, glucose-6-phosphate dehydrogenase (G6PD). It can cause **hemolytic anemia**, varying in severity from life-long anemia, to rare bouts of anemia to total unawareness of the condition. The episodes of hemolytic anemia are usually triggered by oxidants, infection, or by eating fava beans.

Description

G6PD deficiency is the most common enzyme deficiency in the world, with about 400 million people living with it. It is most prevalent in people of African, Mediterranean, and Asian ancestry. The incidence in different populations varies from zero in South American Indians to less than 0.1% of Northern Europeans to about 50% of Kurdish males. In the United States, it is most common among African American males; about 11 to 14% are G6PD-deficient.

G6PD deficiency is a recessive sex-linked trait. Thus, males have only one copy of the G6PD gene, but females have two copies. Recessive genes are masked in the presence of a gene that encodes normal G6PD. Accordingly, females with one copy of the gene for G6PD deficiency are usually normal, while males with one copy have the trait.

G6PD is present in all human cells but is particularly important to red blood cells. It is required to make NADPH in red blood cells but not in other cells. It is also required to make glutathione.

Glutathione and NADPH both help protect red blood cells against oxidative damage. Thus, when G6PD is defective, oxidative damage to red blood cells readily occurs, and they break open as a result. This event is called hemolysis, and multiple hemolyses in a short time span constitute an episode of hemolytic anemia.

As of 1998, there are almost 100 different known forms of G6PD enzyme molecules encoded by defective G6PD genes, yet not one of them is completely inactive. This suggests that G6PD is indispensable. Many G6PD defective enzymes are deficient in their stability rather than their initial ability to function. Since red blood cells lack nuclei, they, unlike other cells, cannot synthesize new enzyme molecules to replace defective ones. Hence, we expect young red blood cells to have new, functional G6PD and older cells to have non-functioning G6PD. This explains why episodes of hemolytic anemia are frequently self-limiting; new red blood cells are generated with enzymes able to afford protection from oxidation.

The geographic distribution of G6PD deficiency, allowing for migration, coincides with the geographic distribution of **malaria**. This fact and survival statistics suggest that G6PD deficiency protects against malaria.

Glucose-6-phosphate dehydrogenase deficiency is also known as G6PD deficiency, favism, and primaquine sensitivity.

Causes and symptoms

Causes

G6PD deficiency is caused by one copy of a defective G6PD gene in males or two copies of a defective G6PD gene in females. Hemolytic anemic attacks can be caused by oxidants, infection, and or by eating fava beans.

Symptoms

The most significant consequence of this disorder is hemolytic anemia, which is usually episodic, but the vast majority of people with G6PD deficiency have no symptoms.

The many different forms of G6PD deficiency have been divided into five classes according to severity.

- Class 1—enzyme deficiency with chronic hemolytic anemia
- Class 2—severe enzyme deficiency with less than 10% of normal activity

- Class 3—moderate to mild enzyme deficiency with 10–60% of normal activity
- Class 4—very mild or no enzyme deficiency
- Class 5—increased enzyme activity Fortunately, only a small number of people fall into Class 1.

The major symptoms of hemolytic anemia are **jaundice**, dark urine, abdominal **pain**, back pain, lowered red blood cell count, and elevated bilirubin. People who suffer from severe and chronic forms of G6PD deficiency in addition may have **gallstones**, enlarged spleens, defective white blood cells, and **cataracts**.

Attacks of hemolytic anemia are serious for infants. Brain damage and **death** are possible but preventable outcomes. Newborns with G6PD deficiency are about 1.5 times as likely to get **neonatal jaundice** than newborns without G6PD deficiency.

Diagnosis

Blood tests can detect G6PD deficiency, either by measuring the G6PD enzyme activity between episodes or by measuring bilirubin during an episode. Such tests cost about \$50.00. Family histories are helpful, too.

Treatment

In a typical attack of hemolytic anemia, no treatment is needed; the patient will recover in about eight days. However, blood transfusions are necessary in severe cases. Recent success treating elevated bilirubin in newborns by exposing them to bright light has decreased the need for neonatal transfusions.

Alternative treatment

Vitamin E and **folic acid** (both anti-oxidants) may help decrease hemolysis in G6PD-deficient individuals.

Prognosis

The prognosis for almost everyone with G6PD deficiency is excellent. Large studies have shown that G6PD-deficient individuals do not acquire any illnesses more frequently than the rest of the population. In fact the opposite may be true for some diseases like ischemic heart disease and cerebrovascular disease.

KEY TERMS

Bilirubin—A breakdown product derived from hemoglobin; removed from the blood by the liver.

Enzyme—A protein catalyst; one of the two kinds of biological catalysts, which are exceedingly specific; each different enzyme only catalyzes one or two specific reactions.

Enzyme activity—A measure of the ability of an enzyme to catalyze a specific reaction.

Glutathione—A molecule that acts as a co-enzyme in cellular oxidation-reduction reactions.

Hemolysis—Lysis (opening) of red blood cells, with concomitant leakage of cell contents from the cells.

Hemolytic anemia—Anemia due to hemolysis.

Jaundice—Yellowish skin color due to liver disease.

Neonatal—Describes babies just after they are born.

Recessive trait—An inherited trait that is outwardly obvious only when two copies of the gene for that trait are present—as opposed to a dominant trait where one copy of the gene for the dominant trait is sufficient to display the trait. The recessive condition is said to be masked by the presence of the dominant gene when both are present; i.e., the recessive condition is seen only in the absence of the dominant gene.

Sex-linked—Refers to genes or traits carried on one of the sex chromosomes, usually the X.

X chromosome—One of the two types of sex chromosomes, present twice in female cells and once in male cells.

Prevention

Most episodes of hemolytic anemia can be prevented by avoiding fava beans, oxidant drugs, and oxidant chemicals. All of the following oxidants can trigger attacks: acetanilid, dapsone, doxorubicin, furazolidone, methylene blue, nalidixic acid, naphthalene, niridazole, nitrofurantoin, phenazopyridine, phenylhydrazine, primaquine, quinidine, quinine, sulfacetamide, sulfamethoxazole, sulfonamide, sulfapyridine, thiazolesulfone, toluidine blue, and trinitrotoluene. Since infections also trigger hemolytic attacks and have other dire consequences, sometimes it is advisable to use one of the listed drugs.

It is especially important to screen newborns who are likely to have G6PD deficiency to ensure that

G6PD-deficient babies won't be subjected to any of the triggers of hemolytic anemia. Pregnant women, especially in areas where G6PD deficiency is prevalent, should avoid eating fava beans.

Resources

ORGANIZATIONS

Alliance of Genetic Support Groups. 4301 Connecticut Ave. NW, Suite 404, Washington, D.C. 20008. (202) 966-5557. Fax: (202) 966-8553. <<http://www.geneticalliance.org>>.

OTHER

Favism Home Page. <<http://rialto.com/favism/index.htm>>.

Lorraine Lica, PhD

Glucosylcerebroside lipidoses see **Gaucher disease**

Gluten enteropathy see **Celiac disease**

Glyburide see **Antidiabetic drugs**

Glycogen storage diseases

Definition

Glycogen serves as the primary fuel reserve for the body's energy needs. Glycogen storage diseases, also known as glycogenoses, are genetically linked metabolic disorders that involve the enzymes regulating glycogen metabolism. Symptoms vary by the glycogen storage disease (GSD) type and can include **muscle cramps** and wasting, enlarged liver, and low blood sugar. Disruption of glycogen metabolism also affects other biochemical pathways as the body seeks alternative fuel sources. Accumulation of abnormal metabolic by-products can damage the kidneys and other organs. GSD can be fatal, but the risk hinges on the type of GSD.

Description

Most of the body's cells rely on glucose as an energy source. Glucose levels in the blood are very stringently controlled within a range of 70–100 mg/dL, primarily by hormones such as insulin and glucagon. Immediately after a meal, blood glucose levels rise and exceed the body's immediate energy requirements. In a process analogous to putting money in the bank, the body bundles up the extra glucose and stores

it as glycogen in the liver and muscles. Later, as the blood glucose levels begin to dip, the body makes a withdrawal from its glycogen savings.

The system for glycogen metabolism relies on a complex system of enzymes. These enzymes are responsible for creating glycogen from glucose, transporting the glycogen to and from storage areas within cells, and extracting glucose from the glycogen as needed. Both creating and tearing down the glycogen macromolecule are multistep processes requiring a different enzyme at each step. If one of these enzymes is defective and fails to complete its step, the process halts. Such enzyme defects are the underlying cause of GSDs.

The enzyme defect arises from an error in its gene. Since the error is in the genetic code, GSDs can be passed down from generation-to-generation. However, all but one GSD are linked to autosomal genes, which means a person inherits one copy of the gene from each parent. Following a Mendelian inheritance pattern, the normal gene is dominant and the defective gene is recessive. As long as a child receives at least one normal gene, there is no risk for a GSD. GSDs appear only if a person inherits a defective gene from both parents.

The most common forms of GSD are Types I, II, III, and IV, which may account for more than 90% of all cases. The most common form is Type I, or von Gierke's disease, which occurs in one out of every 100,000 births. Other forms, such as Types VI and IX, are so rare that reliable statistics are not available. The overall frequency of all forms of glycogen storage disease is approximately one in 20,000–25,000 live births.

Causes and symptoms

GSD symptoms depend on the enzyme affected. Since glycogen storage occurs mainly in muscles and the liver, those sites display the most prominent symptoms.

There are at least 10 different types of GSDs which are classified according to the enzyme affected:

- Type Ia, or von Gierke's disease, is caused by glucose-6-phosphatase deficiency in the liver, kidney, and small intestine. The last step in glycogenolysis, the breaking down of glycogen to glucose, is the transformation of glucose-6-phosphate to glucose. In GSD I, that step does not occur. As a result, the liver is clogged with excess glycogen and becomes enlarged and fatty. Other symptoms include low blood sugar and elevated levels of lactate, lipids, and uric acid in the blood. Growth is impaired, **puberty** is often delayed, and bones may be weakened by **osteoporosis**. Blood platelets are also affected and frequent nosebleeds and easy bruising are common. Primary symptoms improve with age, but after age 20–30, liver tumors, **liver cancer**, chronic renal disease, and **gout** may appear.
- Type Ib is caused by glucose-6-phosphatase translocase deficiency. In order to carry out the final step of glycogenolysis, glucose-6-phosphate has to be transported into a cell's endoplasmic reticulum. If translocase, the enzyme responsible for that movement, is missing or defective, the same symptoms occur as in Type Ia. Additionally, the immune system is weakened and victims are susceptible to bacterial infections, such as **pneumonia**, mouth and gum infections, and inflammatory bowel disease. Types Ic and Id are also caused by defects in the translocase system.
- Type II, or Pompe's disease or acid maltase deficiency, is caused by lysosomal alpha-D-glucosidase deficiency in skeletal and heart muscles. GSD II is subdivided according to the age of onset. In the infantile form, infants seem normal at birth, but within a few months they develop muscle weakness, trouble breathing, and an enlarged heart. Cardiac failure and **death** usually occur before age 2, despite medical treatment. The juvenile and adult forms of GSD II affect mainly the skeletal muscles in the body's limbs and torso. Unlike the infantile form, treatment can extend life, but there is no cure. **Respiratory failure** is the primary cause of death.
- Type III, or Cori's disease, is caused by glycogen debrancher enzyme deficiency in the liver, muscles, and some blood cells, such as leukocytes and erythrocytes. About 15% of GSD III cases only involve the liver. The glycogen molecule is not a simple straight chain of linked glucose molecules, but rather an intricate network of short chains that branch off from one another. In glycogenolysis, a particular enzyme is required to unlink the branch points. When that enzyme fails, symptoms similar to GSD I occur; in childhood, it may be difficult to distinguish the two GSDs by symptoms alone. In addition to the low blood sugar, retarded growth, and enlarged liver causing a swollen abdomen, GSD III also causes muscles prone to wasting, an enlarged heart, and heightened levels of lipids in the blood. The muscle wasting increases with age, but the other symptoms become less severe.
- Type IV, or Andersen's disease, is caused by glycogen brancher enzyme deficiency in the liver, brain, heart, skeletal muscles, and skin fibroblasts. The

glycogen constructed in GSD IV is abnormal and insoluble. As it accumulates in the cells, cell death leads to organ damage. Infants born with GSD IV appear normal at birth, but are diagnosed with enlarged livers and **failure to thrive** within their first year. Infants who survive beyond their first birthday develop **cirrhosis** of the liver by age 3–5 and die as a result of chronic liver failure.

- Type V, or McArdle's disease, is caused by glycogen phosphorylase deficiency in skeletal muscles. Under normal circumstances, muscles cells rely on oxidation of fatty acids during rest or light activity. More demanding activity requires that they draw on their glycogen stockpile. In GSD V, this form of glycogenolysis is disabled and glucose is not available. The main symptoms are muscle weakness and cramping brought on by **exercise**, as well as burgundy-colored urine after exercise due to myoglobin (a breakdown product of muscle) in the urine.
- Type VI, or Hers' disease, is caused by liver phosphorylase deficiency, which blocks the first step of glycogenolysis. In contrast to other GSDs, Type VI seems to be linked to the X chromosome. Low blood sugar is one of the key symptoms, but it is not as severe as in some other forms of GSD. An enlarged liver and mildly retarded growth also occur.
- Type VII, or Tarui's disease, is caused by muscle phosphofructokinase deficiency. Although glucose may be available as a fuel in muscles, the cells cannot metabolize it. Therefore, abnormally high levels of glycogen are stockpiled in the muscle cells. The symptoms are similar to GSD V, but also include anemia and increased levels of uric acid.
- Types VIII and XI are caused by defects of enzymes in the liver phosphorylase activating-deactivating cascade and have symptoms similar to GSD VI.
- Type IX is caused by liver glycogen phosphorylase kinase deficiency and, symptom-wise, is very similar to GSD VI. The main differences are that the symptoms may not be as severe and may also include exercise-related problems in the muscles, such as **pain** and cramps. The symptoms abate after puberty with proper treatment. Most cases of GSD IX are linked to the X chromosome and therefore affect males.
- Type X is caused by a defect in the cyclic adenosine monophosphate-dependent (AMP) kinase enzyme and presents symptoms similar to GSDs VI and IX.

Diagnosis

Diagnosis usually occurs in infancy or childhood, although some milder types of GSD go unnoticed well

into adulthood and old age. It is even conceivable that some of the milder GSDs are never diagnosed.

The four major symptoms that typically lead a doctor to suspect GSDs are low blood sugar, enlarged liver, retarded growth, and an abnormal blood biochemistry profile. A definitive diagnosis is obtained by biopsy of the affected organ or organs. The biopsy sample is tested for its glycogen content and assayed for enzyme activity. There are DNA-based techniques for diagnosing some GSDs from more easily available samples, such as blood or skin. These DNA techniques can also be used for prenatal testing.

Treatment

Some GSD types cannot be treated, while others are relatively easy to control through symptom management. In more severe cases, receiving an organ transplant is the only option. In the most severe cases, there are no available treatments and the victim dies within the first few years of life.

Of the treatable types of GSD, many are treated by manipulating the diet. The key to managing GSD I is to maintain consistent levels of blood glucose through a combination of nocturnal intragastric feeding (usually for infants and children), frequent high-carbohydrate meals during the day, and regular oral doses of cornstarch (people over age 2). Juvenile and adult forms of GSD II can be managed somewhat by a high protein diet, which also helps in cases of GSD III, GSD VI, and GSD IX. GSD V and GSD VII can also be managed with a high protein diet and by avoiding strenuous exercise.

For GSD cases in which dietary therapy is ineffective, organ transplantation may be the only viable alternative. Liver transplants have been effective in reversing the symptoms of GSD IV.

Advances in genetic therapy offer hope for effective treatment in the future. This therapy involves using viruses to deliver a correct form of the gene to affected cells. Another potential therapy utilizes transgenic animals to produce correct copies of the defective enzyme in their milk. In late 1997, a Dutch pharmaceutical company, Pharming Health Care Products, began clinical trials to treat GSD II with human alpha-glucosidase derived from the milk of transgenic rabbits. Researchers at Duke University in North Carolina are also focusing on a treatment for Pompe's disease and, aided by Synpac Pharmaceuticals Limited of the United Kingdom, plan to begin clinical trials of a recombinant form of the enzyme in 1998.

KEY TERMS

Amniocentesis—A medical test done during pregnancy in which a small sample of the amniotic fluid is taken from around the fetus. The fluid contains fetal cells that can be examined for genetic abnormalities.

Autosomal gene—A gene found on one of the 22 autosomal chromosome pairs; i.e., not on a sex (X or Y) chromosome.

Chorionic villus sampling—A medical test done during pregnancy in which a sample of the membrane surrounding the fetus is removed for examination. This examination can reveal genetic fetal abnormalities.

Glucose—A form of sugar that serves as the body's main energy source.

Glycogen—A macromolecule composed mainly of glucose that serves as the storage form of glucose that is not immediately needed by the body.

Glycogenolysis—The process of tearing-down a glycogen molecule to free up glucose.

Glycogenesis—An alternate term for glycogen storage disease. The plural form is glycogenoses.

Gout—A painful condition in which uric acid precipitates from the blood and accumulates in joints and connective tissues.

Mendelian inheritance—An inheritance pattern for autosomal gene pairs. The genetic trait displayed results from one parent's gene dominating over the gene inherited from the other parent.

Osteoporosis—A disease in which the bones become weak and brittle.

Renal disease—Kidney disease.

Transgenic animal—Animals that have had genes from other species inserted into their genetic code.

Prognosis

People with well-managed, treatable types of GSD can lead long, relatively normal lives. This goal is accomplished with the milder types of GSD, such as Types VI, IX, and X. As the GSD type becomes more severe, a greater level of vigilance against infections and other complications is required. Given current treatment options, complications such as **liver disease**, **heart failure**, and respiratory failure may not be warded-off indefinitely. Quality of life and life expectancy are substantially decreased.

Prevention

Because GSD is an inherited condition, it is not preventable. If both parents carry the defective gene, there is a one-in-four chance that their offspring will inherit the disorder. Other children may be carriers or they may miss inheriting the gene altogether.

Through chorionic villi sampling and **amniocentesis**, the disorder can be detected prior to birth. Some types of GSD can be detected even before conception occurs, if both parents are tested for the presence of the defective gene. Before undergoing such testing, the prospective parents should meet with a genetic counselor and other professionals in order to make an informed decision.

Resources

ORGANIZATIONS

Acid Maltase Deficiency Association. PO Box 700248, San Antonio, TX 78270-0248. (210) 494-6144. <<http://www.amda-pompe.org>> .

American Liver Foundation. 1425 Pompton Ave., Cedar Grove, NJ 07009. (800) 223-0179. <<http://www.liverfoundation.org>> .

Association for Glycogen Storage Disease. PO Box 896, Durant, Iowa 52747-9769. (319) 785-6038.

Julia Barrett

Glycosylated hemoglobin test

Definition

Glycosylated hemoglobin is a test that indicates how much sugar has been in a person's blood during the past two to four months. It is used to monitor the effectiveness of diabetes treatment.

Purpose

Diabetes is a disease in which a person cannot effectively use sugar in the blood. Left untreated, blood sugar levels can be very high. High sugar levels increase risk of complications, such as damage to eyes, kidneys, heart, nerves, blood vessels, and other organs.

A routine blood sugar test reveals how close to normal a sugar level is at the time of the test. The glycosylated **hemoglobin test** reveals how close to normal it has been during the past several months.

This information helps a physician evaluate how well a person is responding to diabetes treatment and to determine how long sugar levels have been high in a person newly diagnosed with diabetes.

Description

The Diabetes Control and Complications Trial (DCCT) demonstrated that people with diabetes who maintained blood glucose (sugar) and total **fasting** hemoglobin levels at or close to a normal range decreased their risk of complications by 50–75%. Based on results of this study, the American Diabetes Association (ADA) recommends routine glycosylated hemoglobin testing to measure long-term control of blood sugar.

Glycosylated hemoglobin measures the percentage of hemoglobin bound to glucose. Hemoglobin is a protein found in every red blood cell. As hemoglobin and glucose are together in the red blood cell, the glucose gradually binds to the A1c form of hemoglobin in a process called glycosylation. The amount bound reflects how much glucose has been in the blood during the past average 120-day lifespan of red cells.

Several methods are used to measure the amount of bound hemoglobin and glucose. They are electrophoresis, chromatography, and immunoassay. All are based on the separation of hemoglobin bound to glucose from that without glucose.

The ADA recommends glycosylated hemoglobin be done during a person's first diabetes evaluation, again after treatment is begun and sugar levels are stabilized, then repeated at least semiannually. If the person does not meet treatment goals or sugar levels have not stabilized, the test should be repeated quarterly.

Other names for the test include: Hemoglobin A1c, Diabetic control index, GHb, glycosylated hemoglobin, and glycated hemoglobin. The test is covered by insurance. Results usually are available the following day.

Preparation

A person does not need to fast before this test. A healthcare worker ties a tourniquet on the person's upper arm, locates a vein in the inner elbow region, and inserts a needle into the vein. Vacuum action draws the blood through the needle into an attached tube. Collection of the sample takes only a few minutes. This test requires 5 mL of blood.

KEY TERMS

Diabetes mellitus—A disease in which a person can't effectively use sugar in the blood to meet the needs of the body. It is caused by a lack of the hormone insulin.

Glucose—The main form of sugar used by the body for energy.

Glycosylated hemoglobin—A test that measures the amount of hemoglobin bound to glucose. It is a measure of how much glucose has been in the blood during the past two to four months.

In 2004, more convenient fingerstick methods of the test were being developed. An over-the-counter, at-home test kit was in the beginning stages of approval. People with diabetes could stick their own fingers, draw the blood and send the sample by mail for results within five to seven days.

Aftercare

Discomfort or bruising may occur at the puncture site, or the person may feel dizzy or faint. Pressure to the puncture site until bleeding stops reduces bruising. Warm packs relieve discomfort.

Normal results

Diabetes treatment should achieve glycosylated hemoglobin levels of less than 7.0%. Normal value for a non-diabetic person is 4.0–6.0%.

Because laboratories use different methods, results from different laboratories can not always be compared. The National Glycosylation Standardization Program gives a certification to laboratories using tests standardized to those used in the DCCT study.

Abnormal results

Results require interpretation by a physician with knowledge of the person's clinical condition, as well as the test method used. Some methods give false high or low results if the person has an abnormal hemoglobin, such as hemoglobin S or F.

Conditions that increase the lifespan of red cells, such as a **splenectomy** (removal of the spleen), falsely increase levels. Conditions that decrease the lifespan, such as hemolysis (disruption of the red blood cell membrane), falsely decrease levels.

Resources

PERIODICALS

"Simple Choice A1c." *Diabetes Forecast* January 2004: RG7.

ORGANIZATIONS

American Diabetes Association. 1701 North Beauregard Street, Alexandria, VA 22311. (800) 342-2383. <<http://www.diabetes.org>>.

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

National Diabetes Information Clearinghouse. 1 Information Way, Bethesda, MD 20892-3560. (800) 860-8747. <<http://www.niddk.nih.gov/health/diabetes/ndic.htm>>.

Nancy J. Nordenson
Teresa G. Odle



This woman's goiter may have been caused by an insufficient intake of iodine. (Custom Medical Stock Photo. Reproduced by permission.)

Goiter

Definition

Goiter refers to any visible enlargement of the thyroid gland.

Description

The thyroid gland sits astride the trachea (windpipe) and is shaped like a butterfly. It makes thyroxin, a hormone that regulates the metabolic activity of the body, rather like the gas pedal on a car. Too much thyroxin increases the metabolism, causing weight loss, temperature elevation, nervousness, and irritability. Too little thyroxin slows the metabolism down, deepens the voice, causes weight gain and water retention, and retards growth and mental development in children. Both conditions also alter hair and skin growth, bowel function, and menstrual flow.

Curiously, the thyroid gland is often enlarged whether it is making too much hormone, too little, or sometimes even when it is functioning normally. The thyroid is controlled by the pituitary gland, which secretes thyroid stimulating hormone (TSH) in response to the amount of thyroxin it finds in the blood. TSH increases the amount of thyroxin secreted by the thyroid and also causes the thyroid gland to grow.

- **Hyperthyroid goiter**—If the amount of stimulating hormone is excessive, the thyroid will both enlarge

and secrete too much thyroxin. The result—hyperthyroidism with a goiter. Graves' disease is the most common form of this disorder.

- **Euthyroid goiter**—The thyroid is the only organ in the body to use iodine. If dietary iodine is slightly inadequate, too little thyroxin will be secreted, and the pituitary will sense the deficiency and produce more TSH. The thyroid gland will enlarge enough to make sufficient thyroxin.
- **Hypothyroid goiter**—If dietary iodine is severely reduced, even an enlarged gland will not be able to make enough thyroxin. The gland will keep growing under the influence of TSH, but it may never be able to make enough thyroxin.

Causes and symptoms

Excess TSH (or similar hormones), cysts, and tumors will enlarge the thyroid gland. Of these, TSH enlarges the entire gland while cysts and tumors enlarge only a part of it.

The only symptom from a goiter is the large swelling just above the breast bone. Rarely, it may constrict the trachea (windpipe) or esophagus and cause difficulty breathing or swallowing. The rest of the symptoms come from thyroxin or the lack of it.

KEY TERMS

Cyst—A liquid-filled structure developing abnormally in the body.

Euthyroid—Having the right amount of thyroxin stimulation.

Hyperthyroid—Having too much thyroxin stimulation.

Hypothyroid—Having too little thyroxin stimulation.

Pituitary gland—The master gland, located in the middle of the head, that controls most of the other glands by secreting stimulating hormones.

Radiotherapy—The use of ionizing radiation, either as x rays or radioactive isotopes, to treat disease.

Thyroxin—The hormone secreted by the thyroid gland.

Diagnosis

The size, shape, and texture of the thyroid gland help the physician determine the cause. A battery of blood tests are required to verify the specific thyroid disease. Functional imaging studies using radioactive iodine determine how active the gland is and what it looks like.

Treatment

Goiters of all types will regress with treatment of the underlying condition. Dietary iodine may be all that is needed. However, if an iodine deficient thyroid that has grown in size to accommodate its deficiency is suddenly supplied an adequate amount of iodine, it could suddenly make large amounts of thyroxin and cause a thyroid storm, the equivalent of racing your car motor at top speed.

Hyperthyroidism can be treated with medications, therapeutic doses of radioactive iodine, or surgical reduction. Surgery is much less common now than it used to be because of progress in drugs and radiotherapy.

Prognosis

Although goiters diminish in size, the thyroid may not return to normal. Sometimes thyroid function does not return after treatment, but thyroxin is easy to take as a pill.

Prevention

Euthyroid goiter and hypothyroid goiter are common around the world because many regions have inadequate dietary iodine, including some places in the United States. International relief groups are providing iodized salt to many of these populations. Because **mental retardation** is a common result of **hypothyroidism** in children, this is an extremely important project.

Resources

ORGANIZATIONS

International Council for the Control of Iodine Deficiency Disorders. 43 Circuit Road, Chester Hill, MA, 02167. (207) 335-2221. <<http://www.tulane.edu/~icec/iccidhome.htm>> .

Micronutrient Initiative (c/o International Development Research Centre). 250 Albert St., Ottawa, Ontario, Canada K1G 3H9. (613) 236-6163, ext. 2050. <<http://www.idrc.ca/mi/index.htm>> .

J. Ricker Polsdorfer, MD

Gonadal dysgenesis see **Turner syndrome**

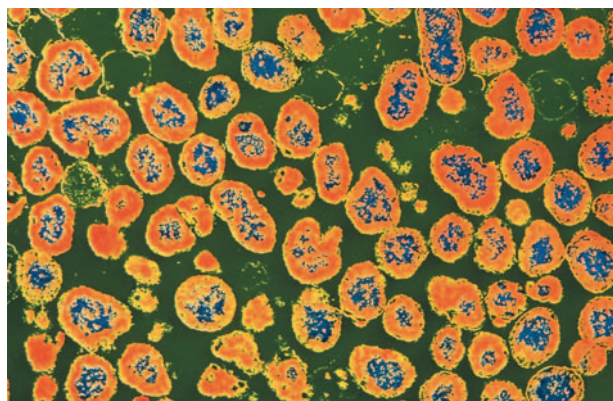
Gonorrhea

Definition

Gonorrhea is a highly contagious sexually transmitted disease that is caused by the bacterium *Neisseria gonorrhoeae*. The mucous membranes of the genital region may become inflamed without the development of any other symptoms. When symptoms occur, they are different in men and women. In men, gonorrhea usually begins as an infection of the vessel that carries urine and sperm (urethra). In women, it will most likely infect the narrow part of the uterus (cervix). If untreated, gonorrhea can result in serious medical complications.

Description

Gonorrhea is commonly referred to as “the clap.” The incidence of gonorrhea has steadily declined since the 1980s, largely due to increased public awareness campaigns and the risk of contracting other **sexually transmitted diseases**, such as **AIDS**. Still, current estimates range from 400,000 to as many as one million projected cases of gonorrhea in the United States each year. These estimates vary due to the private nature of the disease and the consequent underreporting that



A transmission electron microscopy (TEM) image of *Neisseria gonorrhoeae*. (Custom Medical Stock Photo. Reproduced by permission.)

occurs. The majority of reported cases of gonorrhea come from public health clinics.

The disease affects people of all ages, races, and socioeconomic levels, but some individuals are more at-risk than others. Adolescents and young adults are the highest risk group, with more than 80% of the reported cases each year occurring in the 15–29 age group. Those individuals with multiple sexual partners and who use no barrier **contraception**, such as condoms, are most at-risk. Reported rates vary among racial and ethnic groups.

The risk factors for gonorrhea are not unlike those for all sexually transmitted diseases. Both men and women can become infected through a variety of sexual contact behaviors, including oral, anal, or vaginal intercourse. The disease is transmitted very efficiently. In fact, women run a 60–90% chance of contracting the disease after just one sexual encounter with an infected male. The disease can also be transmitted from an infected mother to her infant during delivery.

Causes and symptoms

If treated early, gonorrhea can be cured. Unfortunately, many individuals with gonorrhea, particularly women, will experience no symptoms to alert them to the possibility that they have contracted gonorrhea, and therefore, many do not seek treatment. When present, the symptoms and complications of gonorrhea are primarily limited to the genital, urinary, and gastrointestinal systems and usually begin between one day and two weeks following infection. If left untreated, serious complications can result if the disease spreads to the bloodstream and infects the brain, heart valves, and joints. Untreated gonorrhea

can also result in severe damage to the reproductive system, making an individual unable to conceive a child (sterile).

Symptoms of gonorrhea in women

As many as 80% of women with gonorrhea show no symptoms. If present, symptoms may include the following:

- bleeding between menstrual periods
- chronic abdominal **pain**
- painful urination
- vaginal discharge, often cloudy and yellow
- in the case of oral infection, there may be no symptoms or only a sore throat.
- anal infection may cause rectal **itching** or discharge.

Because women often do not show any symptoms, complications are more likely to occur as the disease progresses. The most common complication is **pelvic inflammatory disease** (PID). PID can occur in up to 40% of women with gonorrhea and may result in damage to the fallopian tubes, a **pregnancy** developing outside the uterus (**ectopic pregnancy**), or sterility. If an infected woman is pregnant, gonorrhea can be passed on to her newborn through the birth canal during delivery. These infants may experience eye infections that could lead to blindness.

Symptoms of gonorrhea in men

Men are more likely to experience the following symptoms:

- thick and cloudy discharge from the penis
- burning or pain during urination
- more frequent urination
- in the case of oral infection, there may be no symptoms or only a sore throat
- anal infection may cause rectal itching or discharge

In men, complications can affect the prostate, testicles, and surrounding glands. Inflammation, tissue death and pus formation (abscesses), and scarring can occur and result in sterility.

Diagnosis

The diagnosis of gonorrhea can be made at a public health clinic or a family physician office. First, the doctor will discuss symptoms and the patient's known contact or at-risk behavior. There are three methods available to test for the presence of

Neisseria gonorrhoeae. These include a culture, a Gram stain, and an ELISA test. Culture of secretions from the infected area is the preferred method for gonorrhea screening in patients with or without symptoms. A cotton swab can be used to collect enough sample for a culture. The sample is incubated for up to two days, providing enough time for the bacteria to multiply and be accurately identified. This test is nearly 100% accurate.

Gram stains are more accurate in the diagnosis of gonorrhea in men than in women. To perform this test, a small amount of discharge from the infected area will be placed on a slide, stained with a special dye, and examined under a microscope for the presence of the gonococcus bacteria. The advantage to this test is that results can be obtained very quickly at the initial visit. Because it requires that the physician or technician be able to recognize and accurately identify the bacteria simply by looking at it under a microscope, however, this test is only about 70% accurate. As a result, one of the other methods may also be used to confirm the diagnosis.

ELISA, or enzyme-linked immunosorbent assay, has emerged as a rapid and sensitive test for gonorrhea. It is much more sensitive than the gram stain and is more convenient than the culture test, which involves the transport and storage of samples. As of late 1997, several other diagnostic tests were being researched with the goal of providing a cost-effective method of screening for a variety of sexually transmitted diseases. One of the most interesting of these is a home test that can be taken by the patient themselves, allowing for a degree of privacy and confidentiality.

When a patient suspects exposure to or experiences symptoms of gonorrhea, he or she may see a public health provider or family practice physician. Physicians trained in obstetrics or gynecology may also be involved, particularly if gynecological complications occur. Men who experience complications may be referred to a urologist. There are also infectious disease physicians who specialize in the treatment and research of all infectious diseases, including those transmitted sexually. All physicians must report this highly contagious disease to public health officials, and patients are asked to provide the names of sex partners during the suspected period of infection so that they can be notified of the risk.

Treatment

Gonorrhea has become more difficult and expensive to treat since the 1970s, due to the increased

resistance of gonorrhea to certain **antibiotics**. In fact, according to projections from the Centers for Disease Control and Prevention, 30% of the strains of gonorrhea were resistant to routine antibiotics in 1994, and resistance has been increasing steadily. Furthermore, many patients have both gonorrhea and chlamydial infections. Therefore, two drug treatment regimens are common. Medications used to treat gonorrhea include ceftriaxone, cefixime, spectinomycin, ciprofloxacin, and ofloxacin. Ceftriaxone and doxycycline or azithromycin are often given simultaneously to treat possible co-existing chlamydia (in pregnant women, erythromycin should be substituted for the aforementioned anti-chlamydial agents). In 2004, reports said that oral antibiotics were preferred over intramuscular forms of the drugs. Also, researchers reported that cefixime had not been available and that fluoroquinolone had been used by more physicians to treat gonorrhea. However, fluoroquinolone resistance was rising among patients with gonorrhea, and in June 2004 the Centers for Disease Control recommended that clinicians no longer prescribe the drug as first-line treatment for gonorrhea in men who have sex with men.

An extremely important consideration is to make sure that all of the prescribed medication is taken. If a course of antibiotics is not completed, the medication will only kill those organisms that are susceptible to the antibiotic, allowing those that are resistant to the effects of that particular antibiotic to multiply and possibly cause a new infection that will be more difficult to treat. Patients should refrain from sexual intercourse until treatment is complete and return for follow-up testing. Any sexual partners during the time of infection, even if those partners do not show symptoms, should be notified and treated when any sexually transmitted disease is involved.

Alternative treatment

Although there is no known alternative to antibiotics in the treatment of gonorrhea, there are herbs and **minerals** that may be used to supplement antibiotic treatment:

- *Lactobacillus acidophilus* or live-culture yogurts are helpful, while taking antibiotics, to replenish gastrointestinal flora.
- The following supplements may be used to improve the body's immune function: zinc, multivitamins and mineral complexes, vitamin C, and garlic (*Allium sativum*).

- Several herbs may reduce some symptoms or help speed healing: kelp has balanced **vitamins** and minerals. Calendula (*Calendula officinalis*), myrrh (*Commiphora molmol*), and thuja (*Thuja occidentalis*) may help reduce discharge and inflammation when used as a tea or douche.
- Hot baths may also help reduce pain and inflammation.
- A variety of herbs may help with symptoms of the reproductive and urinary systems.
- If a physician approves, **fasting**, combined with certain juices, may help cleanse the urinary and gastrointestinal systems.
- There may be **acupressure** and **acupuncture** points that will help with system cleansing. These exact pressure points can be provided and treated by an acupressurist or acupuncturist.

Prognosis

The prognosis for patients with gonorrhea varies based on how early the disease is detected and treated. If treated early and properly, patients can be entirely cured of the disease. Up to 40% of female patients who are not treated early may develop pelvic inflammatory disease (PID) and the possibility of resulting sterility. Although the risk of **infertility** is higher in women than in men, men may also become sterile if the urethra becomes inflamed (**urethritis**) as a result of an untreated gonorrhea infection. Following an episode of PID, a woman is six to 10 times more likely, should a pregnancy occur, to have a pregnancy develop outside the uterus (ectopic pregnancy), which can result in death. Liver infection may also occur in untreated women. In approximately 2% of patients with untreated gonorrhea, the gonococcal infection may spread throughout the body and can cause **fever**, arthritis-like joint pain, and **skin lesions**.

Prevention

Currently, there is no vaccine for gonorrhea, but several are under development. The best prevention is to abstain from having sex or to engage in sex only when in a mutually monogamous relationship in which both partners have been tested for gonorrhea, AIDS, and other sexually transmitted diseases. The next line of defense is the use of condoms, which have been shown to be highly effective in preventing disease (and unwanted pregnancies). To be 100% effective, condoms must be used properly. A female birth-control device that

KEY TERMS

Cervix—The narrow part or neck of the uterus.

Chlamydia—The most common bacterial sexually transmitted disease in the United States that often accompanies gonorrhea and is known for its lack of evident symptoms in the majority of women.

Ectopic pregnancy—A pregnancy that occurs outside the uterus, such as in the fallopian tubes. Although the fetus will not survive, in some cases, ectopic pregnancy can also result in the death of the mother.

ELISA—Enzyme-linked immunosorbent assay. This test has been used a screening test for AIDS for many years and has also been used to detect gonorrhea bacteria.

HIV—Human immunodeficiency virus, the virus that causes AIDS. The risk of acquiring AIDS is increased by the presence of gonorrhea or other sexually transmitted diseases.

Neisseria gonorrhoeae—The bacterium that causes gonorrhea. It cannot survive for any length of time outside the human body.

Pelvic inflammatory disease (PID)—An infection of the upper genital tract that is the most serious threat to a woman's ability to reproduce. At least 25% of women who contract the disease, which can be a complication of gonorrhea, will experience long-term consequences such as infertility or ectopic pregnancy.

Sexually transmitted diseases (STDs)—A group of diseases which are transmitted by sexual contact. In addition to gonorrhea, this groups generally includes chlamydia, HIV (AIDS), herpes, syphilis, and genital warts.

Sterile—Unable to conceive a child.

Urethra—The canal leading from the bladder, and in men, also a path for sperm fluid.

Urethritis—Inflammation of the urethra.

blocks the entry of sperm into the cervix (**diaphragm**) can also reduce the risk of infection. The risk of contracting gonorrhea increases with the number of sexual partners. Any man or woman who has sexual contact with more than one partner is advised to be tested regularly for gonorrhea and other sexually transmitted diseases.

Resources

BOOKS

Sparling, P. Fredrick. "Gonococcal Infections." In *Cecil Textbook of Medicine*, edited by Russel L. Cecil, et al. Philadelphia: W.B. Saunders Company, 2000.

PERIODICALS

"Fluoroquinolone-resistant Gonorrhea on the Rise: Exposure History is Critical." *Emergency Medicine Alert* July 2004: 11–12.

Georgia, Kristen. "Revised Approach to Gonorrhea Treatment." *Patient Care* July 2004: 11.

ORGANIZATIONS

American Foundation for the Prevention of Venereal Disease, Inc. 799 Broadway, Suite 638, New York, NY 10003. (212) 759-2069.

National Institute of Allergy and Infectious Diseases. National Institutes of Health, Bethesda, MD 20892.

Teresa G. Odle

Goodpasture's syndrome

Definition

An uncommon and life-threatening hypersensitivity disorder believed to be an autoimmune process related to antibody formation in the body. Goodpasture's syndrome is characterized by renal (kidney) disease and lung hemorrhage.

Description

The disorder is characterized by autoimmune reaction which deposits of antibodies in the membranes of both the lung and kidneys, causing both inflammation of kidney (**glomerulonephritis**) and lung bleeding. It is typically a disease of young males.

Causes and symptoms

The exact cause is unknown. It is an autoimmune disorder; that is, the immune system is fighting the body's own normal tissues through creating antibodies that attack the lungs and kidneys. Sometimes the disorder is triggered by a viral infection, or by the inhalation of gasoline or other hydrocarbon solvents. An association also exists between cigarette **smoking** and the syndrome. The target antigen of the Goodpasture's antibodies has been localized to a protein chain (type IV collagen).

Symptoms include foamy, bloody, or dark colored urine, decreased urine output, **cough** with

bloody sputum, difficulty breathing after exertion, weakness, **fatigue**, **nausea** or **vomiting**, weight loss, nonspecific chest **pain** and/or pale skin.

Diagnosis

The clinician will perform a battery of tests to confirm a diagnosis. These tests include a complete **blood count** (CBC) to confirm anemia, iron levels to check for blood loss and blood urea nitrogen (BUN) and creatinine levels to test the kidney function. A **urinalysis** will be done to check for damage to the kidneys. A sputum test will be done to look for specific antibodies. A **chest x ray** will be done to assess the amount of fluid in the lung tissues. A lung needle biopsy and a **kidney biopsy** will show immune system deposits. The kidney biopsy can also show the presence of the harmful antibodies that attack the lungs and kidneys.

Treatment

Treatment is focused on slowing the progression of the disease. Treatment is most effective when begun early, before kidney function has deteriorated to a point where the kidney is permanently damaged, and dialysis is necessary. **Corticosteroids**, such as prednisone, or other anti-inflammatory medications may be used to reduce the immune response. Immune suppressants such as cyclophosphamide or azathioprine are used aggressively to reduce immune system effects.

A procedure whereby blood plasma, which contains antibodies, is removed from the body and replaced with fluids or donated plasma (**plasmapheresis**) may be performed daily for two or more weeks to remove circulating antibodies. It is fairly effective in slowing or reversing the disorder. Dialysis to clean the blood of wastes may be required if kidney function is poor. A kidney transplant may be successful, especially if performed after circulating antibodies have been absent for several months.

Prognosis

The probable outcome is variable. Most cases progress to severe renal failure and end-stage renal disease within months. Early diagnosis and treatment makes the probable outcome more favorable.

Prevention

No known prevention of Goodpasture's syndrome exists. People should avoid glue sniffing and the siphoning gasoline. Stopping smoking, if a family history of renal failure exists, may prevent some cases. Early diagnosis and treatment may slow progression of the disorder.

KEY TERMS

Antibody—A protein molecule produced by the immune system in response to a protein that is not recognized as belonging in the body.

Antigen—Any substance that, as a result of coming in contact with appropriate cells, induces a state of sensitivity and/or immune responsiveness after a period of time and that reacts in a demonstrable way with antibodies.

Autoimmune disorder—An abnormality within the body whereby the immune system incorrectly attacks the body's normal tissues, thereby causing disease or organ dysfunction.

Blood urea nitrogen (BUN)—A test used to measure the blood level of urea nitrogen, a waste that is normally filtered from the kidneys.

Creatinine—A test used to measure the blood level of creatinine, a waste product filtered out of the blood by the kidneys. Higher than usual levels of this substance may indicate kidney disease.

Glomerulus (glomeruli)—A small tuft of blood capillaries in the kidney, responsible for filtering out waste products.

Resources

BOOKS

Tierney, Lawrence, et. al. *Current Medical Diagnosis and Treatment*. Los Altos: Lange Medical Publications, 2001.

ORGANIZATIONS

American Kidney Fund. 6110 Executive Boulevard, Suite 1010, Rockville, MD 20852. (800) 638-8299. <<http://www.akfinc.org>>.

National Kidney Foundation. 30 East 33rd Street, New York, NY 10016. (800) 622-9010. <<http://www.kidney.org>>.

Kim A. Sharp, M.Ln.

Gout

Definition

Gout is a form of acute arthritis that causes severe **pain** and swelling in the joints. It most commonly affects the big toe, but may also affect the heel,

ankle, hand, wrist, or elbow. It affects the spine often enough to be a factor in back pain. Gout usually comes on suddenly, goes away after 5–10 days, and can keep recurring. Gout is different from other forms of arthritis because it occurs when there are high levels of uric acid circulating in the blood, which can cause urate crystals to settle in the tissues of the joints.

Description

Uric acid, which is found naturally in the blood stream, is formed as the body breaks down waste products, mainly those containing purine, a substance that is produced by the body and is also found in high concentrations in some foods, including brains, liver, sardines, anchovies, and dried peas and beans. Normally, the kidneys filter uric acid out of the blood and excrete it in the urine. Sometimes, however, the body produces too much uric acid or the kidneys aren't efficient enough at filtering it from the blood, and it builds up in the blood stream, a condition known as hyperuricemia. A person's susceptibility to gout may increase because of the inheritance of certain genes or from being overweight and eating a rich diet. In some cases, another disease (such as lymphoma, leukemia, or **hemolytic anemia**) may be the underlying cause of the uric acid buildup that results in gout. An additional factor is occupational or environmental; it is now known that chronic exposure to high levels of lead decreases the body's excretion of urates, allowing uric acid to accumulate in the blood.

Hyperuricemia doesn't always cause gout. Over the course of years, however, sharp urate crystals build up in the synovial fluid of the joints. Often, some precipitating event, such as an infection, surgery, the **stress** of hospitalization, a stubbed toe, or even a heavy drinking binge can cause inflammation. White blood cells, mistaking the urate crystals for a foreign invader, flood into the joint and surround the crystals, causing inflammation—in other words, the redness, swelling, and pain that are the hallmarks of a gout attack.

Causes and symptoms

As a result of high levels of uric acid in the blood, needle-like urate crystals gradually accumulate in the joints. Urate crystals may be present in the joint for a long time without causing symptoms. Infection, injury to the joint, surgery, drinking too much, or eating the wrong kinds of foods may suddenly bring on the symptoms, which include pain, tenderness, redness, warmth, and swelling of the joint. In many cases, the gout attack begins in the middle of the night. The pain is often so excruciating that the sufferer cannot bear

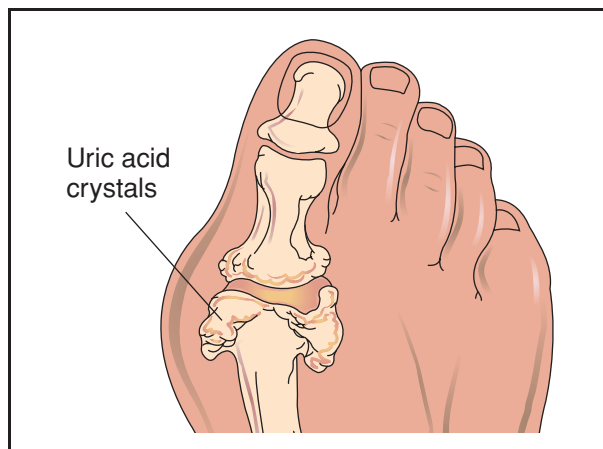
weight on the joint or tolerate the pressure of bed-covers. The inflamed skin over the joint may be red, shiny, and dry, and the inflammation may be accompanied by a mild **fever**. These symptoms may go away in about a week and disappear for months or years at a time. However, over the course of time, attacks of gout recur more and more frequently, last longer, and affect more joints. Eventually, stone-like deposits known as **tophi** may build up in the joints, ligaments, and tendons, leading to permanent joint deformity and decreased motion. (In addition to causing the tophi associated with gout, hyperuricemia can also cause **kidney stones**, also called renal calculi or uroliths.)

Gout affects an estimated one million Americans; according to the National Institutes of Health, it accounts for about 5% of all cases of arthritis. It occurs more often in men than in women; the sex ratio is about 4:1. Uric-acid levels tend to increase in men at **puberty**, and, because it takes 20 years of hyperuricemia to cause gout symptoms, men commonly develop gout in their late 30s or early 40s. Women more typically develop gout later in life, starting in their 60s. According to some medical experts, estrogen protects against hyperuricemia, and when estrogen levels fall during **menopause**, urate crystals can begin to build up in the joints. Excess body weight, regular excessive alcohol intake, the use of blood pressure medications called **diuretics**, and high levels of certain fatty substances in the blood (serum triglycerides) associated with an increased risk of heart disease can all increase a person's risk of developing gout.

Gout appears to be on the increase in the American population. According to a study published in November 2002, there was a twofold increase in the incidence of gout over the 20 years between 1977 and 1997. It is not yet known whether this increase is the result of improved diagnosis or whether it is associated with risk factors that have not yet been identified.

Diagnosis

Usually, physicians can diagnose gout based on the **physical examination** and medical history (the patient's description of symptoms and other information). Doctors can also administer a test that measures the level of uric acid in the blood. While normal uric acid levels don't necessarily rule out gout and high levels don't confirm it, the presence of hyperuricemia increases the likelihood of gout. The development of a tophus can confirm the diagnosis of gout. The most definitive way to diagnose gout is



Gout, a form of acute arthritis, most commonly occurs in the big toe. It is caused by high levels of uric acid in the blood, in which urate crystals settle in the tissues of the joints and produce severe pain and swelling. (Illustration by Electronic Illustrators Group.)

to take a sample of fluid from the joint and test it for urate crystals.

Treatment

The goals of treatment for gout consist of alleviating pain, avoiding severe attacks in the future, and preventing long-term joint damage. In addition to taking pain medications as prescribed by their doctors, people having gout attacks are encouraged to rest and to increase the amount of fluids that they drink.

Acute attacks of gout can be treated with nonaspirin, **nonsteroidal anti-inflammatory drugs** (NSAIDs) such as naproxen sodium (Aleve), ibuprofen (Advil), or indomethacin (Indocin). In some cases, these drugs can aggravate a peptic ulcer or existing **kidney disease** and cannot be used. Doctors sometimes also use colchicine (Colbenemid), especially in cases where nonsteroidal anti-inflammatory drugs cannot be used. Colchicine may cause **diarrhea**, which tends to go away once the patient stops taking it. **Corticosteroids** such as prednisone (Deltasone) and adrenocorticotrophic hormone (Acthar) may be given orally or may be injected directly into the joint for a more concentrated effect. While all of these drugs have the potential to cause side effects, they are used for only about 48 hours and are not likely to cause major problems. However, **aspirin** and closely related drugs (salicylates) should be avoided because they can ultimately worsen gout.

Once an acute attack has been successfully treated, doctors try to prevent future attacks of gout and

long-term joint damage by lowering uric acid levels in the blood. There are two types of drugs for correcting hyperuricemia. Such uricosuric drugs as probenecid (Benemid) and sulfinpyrazone (Anturane) lower the levels of urate in the blood by increasing its removal from the body (excretion) through the urine. These drugs may promote the formation of kidney stones, however, and they may not work for all patients, especially those with kidney disease. Allopurinol (Zyloprim), a type of drug called a xanthine-oxidase inhibitor, blocks the production of urate in the body, and can dissolve kidney stones as well as treating gout. The potential side effects of allopurinol include rash, a skin condition known as **dermatitis**, and liver dysfunction. In 2004, the FDA was seeking trial data on a new drug called oxypurinol (Oxyprim) for treating chronic gout. These medications may have to be taken for life to prevent further gout attacks.

New quality of care indicators were released in 2004 to improve care for patients with gout. The aim of the guidelines was to prevent repeat gout attacks and to reduce medication errors associated with intravenous colchicine in hospitals.

Alternative treatment

Alternative approaches to gout focus on correcting hyperuricemia by encouraging weight loss and limiting the intake of alcohol and purine-rich foods. In addition, consuming garlic (*Allium sativum*) has been recommended to help prevent gout. Increasing fluid intake, especially by drinking water, is also recommended. During an acute attack, contrast **hydrotherapy** (alternating three-minute hot compresses with 30-second cold compresses) can help dissolve the crystals and resolve the pain faster.

Prognosis

Gout cannot be cured but usually it can be managed successfully. As tophi dissolve, joint mobility generally improves. (In some cases, however, medicines alone do not dissolve the tophi and they must be removed surgically.) Lowering uric acid in the blood also helps to prevent or improve the kidney problems that may accompany gout.

Prevention

For centuries, gout has been known as a “rich man’s disease” or a disease caused by overindulgence in food and drink. While this view is perhaps a little overstated and oversimplified, lifestyle factors clearly influence a person’s risk of developing gout. Since

KEY TERMS

Allopurinol—A drug that corrects hyperuricemia by inhibiting urate production.

Colchicine—A drug used to treat painful flare-ups of gout.

Corticosteroids—Medications related to a natural body hormone called hydrocortisone, which are used to treat inflammation.

Hyperuricemia—High levels of a waste product called uric acid in the blood.

Probenecid—A drug that corrects hyperuricemia by increasing the urinary excretion of urate.

Purine—A substance found in foods that is broken down into urate and may contribute to hyperuricemia and gout.

Sulfinpyrazone—A drug that corrects hyperuricemia by increasing the urinary excretion of urate.

Synovial fluid—Fluid surrounding the joints which acts as a lubricant, reducing the friction between the joints.

Tophus (plural, tophi)—A chalky deposit of a uric acid compound found in gout. Tophi occur most frequently around joints and in the external ear.

Urate crystals—Crystals formed by high levels of uric acid in the blood.

obesity and excessive alcohol intake are associated with hyperuricemia and gout, losing weight and limiting alcohol intake can help ward off gout. **Dehydration** may also promote the formation of urate crystals, so people taking diuretics or “water pills” may be better off switching to another type of blood pressure medication. Everyone should be sure to drink at least six to eight glasses of water each day. Since purine is broken down in the body into urate, it may also be helpful to avoid foods high in purine, such as organ meats, sardines, anchovies, red meat, gravies, beans, beer, and wine. A 2004 study revealed that eating more low-fat dairy products could reduce risk of developing gout.

Resources

BOOKS

Parker, James N., M.D., and Philip M. Parker, Ph. D. *The 2002 Official Patient's Sourcebook on Gout*. San Diego, CA: ICON Health Publications, 2002.

PERIODICALS

Arromdee, E., C. J. Michet, C. S. Crowson, et al. “Epidemiology of Gout: Is the Incidence Rising?”

Journal of Rheumatology 29 (November 2002): 2403–2406.

Coghill, Kim. “FDA Panel Discusses Endpoints for Approval of Gout Products.” *Bioworld Today* (June 3, 2004).

“Dairy-rich Diet May Help Prevent Gout.” *Tufts University Health & Nutrition Letter* (June 2004): 2.

Hsu, C. Y., T. T. Shih, K. M. Huang, et al. “Tophaceous Gout of the Spine: MR Imaging Features.” *Clinical Radiology* 57 (October 2002): 919–925.

Lin, J. L., D. T. Tan, H. H. Ho, and C. C. Yu. “Environmental Lead Exposure and Urate Excretion in the General Population.” *American Journal of Medicine* 113 (November 2002): 563–568.

MacReady, Norma. “New Gout Quality-of-care Standards Take Aim at Medication-related Errors.” *Internal Medicine News* (June 1, 2004): 18.

Perez-Ruiz, F., M. Calabozo, G. G. Erauskin, et al. “Renal Underexcretion of Uric Acid is Present in Patients with Apparent High Urinary Uric Acid Output.” *Arthritis and Rheumatism* 47 (December 15, 2002): 610–613.

Raj, J. M., S. Sudhakar, K. Sems, and R. W. Carlson. “Arthritis in the Intensive Care Unit.” *Critical Care Clinics* 18 (October 2002): 767–780.

Shekarriz, B., and M. L. Stoller. “Uric Acid Nephrolithiasis: Current Concepts and Controversies.” *Journal of Urology* 168, no. 4, Part 1 (October 2002): 1307–1314.

ORGANIZATIONS

Arthritis Foundation. 1300 W. Peachtree St., Atlanta, GA 30309. (800) 283-7800. <<http://www.arthritis.org>>.

National Institute of Arthritis and Musculoskeletal and Skin Diseases (NIAMS). National Institutes of Health (NIH), 1 AMS Circle, Bethesda, MD 20892-3675. <www.niams.nih.gov>.

OTHER

National Institute of Arthritis and Musculoskeletal and Skin Diseases (NIAMS). *Questions and Answers About Gout*. Bethesda, MD: NIAMS, 2002. NIH Publication No. 02-5027. <www.niams.nih.gov/hi/topics/gout/gout/htm>.

Robert Scott Dinsmoor
Rebecca J. Frey, PhD
Teresa G. Odle

Gout drugs

Definition

Gout drugs are medicines that prevent or relieve the symptoms of gout, a disease that affects the joints and kidneys.

Purpose

Gout is a disease in which uric acid, a waste product that normally passes out of the body in urine, collects and forms crystals in the joints and the kidneys. When uric acid crystals build up in the joints, the tissue around the joint becomes inflamed, and nerve endings in the area become irritated, causing extreme **pain**. Uric acid crystals in the kidneys can lead to **kidney stones** and eventually to kidney failure.

The symptoms of gout—severe pain, usually in the hand or foot (often at the base of the big toe), but sometimes in the elbow or knee—should be reported to a health care professional. If not treated, gout can lead to high blood pressure, deformed joints, and even **death** from kidney failure. Fortunately, the condition is easily treated. For patients who have just had their first attack, physicians may prescribe only medicine to reduce the pain and inflammation, such as **nonsteroidal anti-inflammatory drugs**, **corticosteroids**, or colchicine. Patients may also be advised to change their eating and drinking habits, avoiding organ meats and other protein-rich foods, cutting out alcoholic beverages, and drinking more water. Some people never have another gout attack after the first. For those who do, physicians may prescribe additional drugs that either help the body get rid of uric acid or reduce the amount of uric acid the body produces. These drugs will not relieve gout attacks that already have started, but will help prevent attacks when taken regularly.

Description

Three main types of drugs are used in treating gout. Colchicine helps relieve the symptoms of gout by reducing inflammation. Allopurinol (Lopurin, Zyloprim) reduces the amount of uric acid produced in the body. Probenecid (Benemid, Probalan) and sulfinpyrazone (Anturane) help the body get rid of excess uric acid. Physicians may recommend that patients take more than one type of gout drug at the same time. Some of these medicines may also be prescribed for other medical conditions that are caused by too much uric acid in the body.

Recommended dosage

The recommended dosage depends on the type of gout drug. Check with the physician who prescribed the drug or the pharmacist who filled the prescription for the correct dosage.

Always take gout drugs exactly as directed. Never take larger or more frequent doses than recommended. Patients who are told to take more than one

gout drug should carefully follow the physician's directions for taking all medicines.

Gout drugs such as allopurinol, probenecid, and sulfinpyrazone must be taken regularly to prevent gout attacks. The medicine may take some time to begin working, so gout attacks may continue for awhile after starting to take the drug. Continuing to take the drug is important, even if it does not seem to be working at first.

Colchicine may be taken regularly in low doses to help prevent gout attacks or in high doses for only a few hours at a time to relieve an attack. The chance of serious side effects is greater when this medicine is taken in high doses for short periods.

Precautions

Seeing a physician regularly while taking gout drugs is important. The physician will check to make sure the medicine is working as it should and will watch for unwanted side effects. Blood tests may be ordered to help the physician monitor how well the drug is working.

Drinking alcohol, including beer and wine, may increase the amount of uric acid in the body and may interfere with the effects of gout medicine. People with gout (or other conditions that result from excess uric acid) may need to limit the amount of alcohol they drink or stop drinking alcohol altogether.

Some people feel drowsy or less alert when taking gout drugs. Anyone who takes this type of medicine should not drive, use machines or do anything else that might be dangerous until they have found out how the drugs affect them.

Some gout drugs may change the results of certain medical tests. Before having medical tests, anyone taking this medicine should alert the health care professional in charge.

Older people may be especially sensitive to the effects of colchicine. The drug may also stay in their bodies longer than it does in younger people. Both the increased sensitivity to the drug and the longer time for the drug to leave the body may increase the chance of side effects.

Special conditions

People who have certain medical conditions or who are taking certain other medicines can have problems if they take gout drugs. Before taking these drugs, be sure to let the physician know about any of these conditions:

ALLERGIES. Anyone who has ever had unusual reactions to gout drugs or to medicines used to relieve pain or inflammation should let his or her physician know before taking gout drugs. The physician should also be told about any **allergies** to foods, dyes, preservatives, or other substances.

DIABETES. Some gout drugs may cause false results on certain urine sugar tests, but not on others. Diabetic patients who take gout drugs should check with their physicians to find out if their medicine will affect the results of their urine sugar tests.

PREGNANCY. The effects of taking gout drugs during **pregnancy** are not fully understood. Women who are pregnant or who may become pregnant should check with their physicians before using gout drugs.

BREASTFEEDING. Gout drugs may pass into breast milk. Women who are taking this medicine and want to breastfeed their babies should check with their physicians.

OTHER MEDICAL CONDITIONS. Gout drugs may cause problems for people with certain medical conditions. For example, the risk of severe allergic reactions or other serious side effects is greater when people with these medical conditions take certain gout drugs:

- congestive heart disease
- high blood pressure
- blood disease
- diabetes
- kidney disease or kidney stones
- cancer being treated with drugs or radiation
- stomach or intestinal problems, including stomach ulcer (now or in the past)

Before using gout drugs, people with any of medical problems listed above should make sure their physicians are aware of their conditions.

USE OF CERTAIN MEDICINES. Taking gout drugs with certain other drugs may affect the way the drugs work or may increase the chance of side effects.

Side effects

A skin rash that develops during treatment with gout drugs may be a sign of a serious and possibly life-threatening reaction. If any of these symptoms occur, stop taking the medicine and check with a physician immediately:

- skin rash, **itching**, or **hives**
- scaly or peeling skin

- chills, **fever**, **sore throat**, **nausea and vomiting**, yellow skin or eyes, joint pain, muscle aches or pains—especially if these symptoms occur at the same time or shortly after a skin rash

Patients taking colchicine should stop taking it immediately if they have **diarrhea**, stomach pain, **nausea**, or **vomiting**. If these symptoms continue for 3 hours or more after the medicine is stopped, check with a physician.

Other side effects of may also need medical attention. If any of the following symptoms occur while taking gout drugs, check with the physician who prescribed the medicine as soon as possible:

- pain in the side or lower back
- painful urination
- blood in the urine

Less serious side effects, such as **headache**, loss of appetite, and joint pain and inflammation usually go away as the body adjusts to the drug and do not need medical treatment.

Other side effects may occur. Anyone who has unusual symptoms while taking gout difficult to interpret. Even so, each individual cell in the body carries thousands of genes coding for proteins, with some estimates as high as 150,000 genes. For **gene therapy** to advance to its full potential, scientists must discover the biological role of each of these individual genes and where the base pairs that make them up are located on DNA.

To address this issue, the National Institutes of Health initiated the Human Genome Project in 1990. Led by James D. Watson (one of the co-discoverers of the chemical makeup of DNA) the project's 15-year goal is to map the entire human genome (a combination of the words gene and chromosomes). A genome map would clearly identify the location of all genes as well as the more than three billion base pairs that make them up. With a precise knowledge of gene locations and functions, scientists may one day be able to conquer or control diseases that have plagued humanity for centuries.

Scientists participating in the Human Genome Project have identified an average of one new gene a day, but many expect this rate of discovery to increase. By the year 2005, their goal is to determine the exact location of all the genes on human DNA and the exact sequence of the base pairs that make them up. Some of the genes identified through this project include a gene that predisposes people to **obesity**, one associated with programmed cell death (apoptosis), a gene that guides HIV viral reproduction, and the genes of inherited

disorders like Huntington's disease, Lou Gehrig's disease, and some colon and breast cancers. In February of 2001, scientists published a rough draft of the complete human genome. With fewer than the anticipated number of genes found, between 30,000–40,000, the consequences of this announcement are enormous. Scientists caution however, that the initial publication is only a draft of the human genome and much more work is still ahead for the completion of the project. As the human genome is completed, there will be more information available for gene therapy research and implementation.

The future of gene therapy

Gene therapy seems elegantly simple in its concept: supply the human body with a gene that can correct a biological malfunction that causes a disease. However, there are many obstacles and some distinct questions concerning the viability of gene therapy. For example, viral vectors must be carefully controlled lest they infect the patient with a viral disease. Some vectors, like retroviruses, can also enter cells functioning properly and interfere with the natural biological processes, possibly leading to other diseases. Other viral vectors, like the adenoviruses, are often recognized and destroyed by the immune system so their therapeutic effects are short-lived. Maintaining gene expression so it performs its role properly after vector delivery is difficult. As a result, some therapies need to be repeated often to provide long-lasting benefits.

One of the most pressing issues, however, is gene regulation. Genes work in concert to regulate their functioning. In other words, several genes may play a part in turning other genes on and off. For example, certain genes work together to stimulate cell division and growth, but if these are not

Nancy Ross-Flanigan

Gouty arthritis see **Gout**

Graft-vs.-host disease

Definition

Graft-vs.-host disease is an immune attack on the recipient by cells from a donor.

Description

The main problem with transplanting organs and tissues is that the recipient host does not recognize the new tissue as its own. Instead, it attacks it as foreign in the same way it attacks germs, to destroy it.

If immunogenic cells from the donor are transplanted along with the organ or tissue, they will attack the host, causing graft vs. host disease.

The only transplanted tissues that house enough immune cells to cause graft vs. host disease are the blood and the bone marrow. Blood transfusions are used every day in hospitals for many reasons. Bone marrow transplants are used to replace blood forming cells and immune cells. This is necessary for patients whose **cancer** treatment has destroyed their own bone marrow. Because bone marrow cells are among the most sensitive to radiation and **chemotherapy**, it often must be destroyed along with the cancer. This is true primarily of leukemias, but some other cancers have also been treated this way.

Causes and symptoms

Even if the donor and recipient are well matched, graft-vs.-host disease can still occur. There are many different elements involved in generating immune reactions, and each person is different, unless they are identical twins. Testing can often find donors who match all the major elements, but there are many minor ones that will always be different. How good a match is found also depends upon the urgency of the need and some good luck.

Blood **transfusion** graft-vs.-host disease affects mostly the blood. Blood cells perform three functions: carrying oxygen, fighting infections, and clotting. All of these cell types are decreased in a transfusion graft-vs.-host reaction, leading to anemia (lack of red blood cells in the blood), a decrease in resistance to infections, and an increase in bleeding. The reaction occurs between four to 30 days after the transfusion.

The tissues most affected by bone marrow graft-vs.-host disease are the skin, the liver, and the intestines. One form or the other occurs in close to half of the patients who receive bone marrow transplants.

Bone marrow graft-vs.-host disease comes in an acute and a chronic form. The acute form appears within two months of the transplant; the chronic form usually appears within three months. The acute disease produces a skin rash, liver abnormalities, and **diarrhea** that can be bloody. The skin rash is primarily a patchy thickening of the skin. Chronic disease can produce a similar skin rash, a tightening or an

inflammation of the skin, lesions in the mouth, drying of the eyes and mouth, hair loss, liver damage, lung damage, and **indigestion**. The symptoms are similar to an autoimmune disease called **scleroderma**.

Both forms of graft-vs.-host disease bring with them an increased risk of infections, either because of the process itself or its treatment with cortisone-like drugs and immunosuppressives. Patients can die of liver failure, infection, or other severe disturbances of their system.

Treatment

Both the acute and the chronic disease are treated with cortisone-like drugs, immunosuppressive agents like cyclosporine, or with **antibiotics** and immune chemicals from donated blood (gamma globulin). Infection with one particular virus, called cytomegalovirus (CMV) is so likely a complication that some experts recommend treating it ahead of time.

Prognosis

Children with **acute leukemias** have greatly benefited from the treatment made possible by **bone marrow transplantation**. Survival rates have climbed by 15–50%. It is an interesting observation that patients who develop graft-vs.-host disease are less likely to have a recurrence of the leukemia that was being treated. This phenomenon is called graft-vs.-leukemia.

Bone marrow transplant patients who do not have a graft-vs.-host reaction gradually return to normal immune function in a year. A graft-vs.-host reaction may prolong the diminished immune capacity indefinitely, requiring supplemental treatment with immunoglobulins (gamma globulin).

Somehow the grafted cells develop a tolerance to their new home after six to 12 months, and the medications can be gradually withdrawn. Graft-vs.-host disease is not the only complication of blood transfusion or bone marrow transplantation. Host-vs.-graft or rejection is also common and may require a repeat transplant with another donor organ. Infections are a constant threat in bone marrow transplant because of the disease being treated, the prior radiation or chemotherapy and the medications used to treat the transplant.

Prevention

For recipients of blood transfusions who are especially likely to have graft-vs.-host reactions, the

KEY TERMS

Anemia—Too few red blood cells, or too little hemoglobin in them.

Immunoglobulin—Chemicals in the blood that defend against infections.

Immunosuppressive—A chemical which suppresses an immune response.

Inflammation—The body's immune reaction to presumed foreign substances like germs. Inflammation is characterized by increased blood supply and activation of defense mechanisms. It produces redness, swelling, heat, and pain.

Lesion—Localized disease or damage.

Scleroderma—Progressive disease of the connective tissue of the skin and internal organs.

red blood cells can safely be irradiated (using x rays) to kill all the immune cells. The red blood cells are less sensitive to radiation and are not harmed by this treatment.

Much current research is directed towards solving the problem of graft-vs.-host disease. There are efforts to remove the immunogenic cells from the donor tissue, and there are also attempts to extract and purify bone marrow cells from the patient before treating the cancer. These cells are then given back to the patient after treatment has destroyed all that were left behind.

Resources

BOOKS

Armitage, James O. "Bone Marrow Transplantation."
In *Harrison's Principles of Internal Medicine*, edited
by Anthony S. Fauci, et al. New York:
McGraw-Hill, 1997.

J. Ricker Polsdorfer, MD

Grafts and grafting see **Bone grafting;**
Coronary artery bypass graft surgery;
Graft-vs.-host disease; Skin grafting

Granular conjunctivitis see **Trachoma**

Granulocytic ehrlichiosis see **Ehrlichiosis**

Granulocytopenia see **Neutropenia**

Granuloma inguinale

Definition

Granuloma inguinale is a sexually transmitted infection that affects the skin and mucous membranes of the anal and genital areas. Its name is derived from granuloma, a medical term for a mass or growth of granulation tissue, and *inguinale*, a Latin word that means located in the groin. Granulation tissue is tissue formed during wound healing that is rich in blood capillaries and has a rough or lumpy surface.

Description

Granuloma inguinale is a chronic infection with frequent relapses caused by a rod-shaped bacterium. It occurs worldwide but is most common in tropical or subtropical countries, where it is associated with poverty and poor hygiene. As many as 20% of male patients with **sexually transmitted diseases** (STDs) in tropical countries have granuloma inguinale. The disease is less common in the United States, with fewer than 100 reported cases per year. Most patients are between the ages of 20 and 40 years, with a 2:1 male-to-female ratio.

Although granuloma inguinale is relatively uncommon in the United States in comparison with other STDs, it is still a significant public health problem. It can be acquired through casual sexual contacts when traveling abroad. Moreover, patients with granuloma inguinale are vulnerable to superinfection (infection by other disease agents) with other STDs, especially **syphilis**. Patients with granuloma inguinale are also a high-risk group for Acquired Immune Deficiency Syndrome (AIDS) transmission, because the disease causes open genital ulcers that can be easily invaded by the AIDS virus.

Granuloma inguinale is spread primarily through heterosexual and male homosexual contact; however, its occurrence in children and sexually inactive adults indicates that it may also be spread by contact with human feces. Granuloma inguinale is not highly contagious; however, persons with weakened immune systems are at greater risk of infection.

Causes and symptoms

Granuloma inguinale, which is sometimes called donovanosis, is caused by *Calymmatobacterium granulomatis*, a rod-shaped bacterium formerly called *Donovania granulomatis*. The bacterium has an incubation period ranging from eight days to 12 weeks,

with an average of two to four weeks. The disease has a slow and gradual onset, beginning with an inconspicuous pimple or lumpy eruption on the skin. In 90% of patients, the initial sign of infection is in the genital region, but a minority of patients will develop the sore in their mouth or anal area if their sexual contact involved those parts of the body. Many patients do not notice the sore because it is small and not usually painful. In some women, the first symptom of granuloma inguinale is bleeding from the genitals.

The initial pimple or sore is typically followed by three stages of disease. In the first stage, the patient develops a mass of pink or dull red granulation tissue in the area around the anus. In the second stage, the bacteria erode the skin to form shallow, foul-smelling ulcers which spread from the genital and anal areas to the thighs and lower abdomen. The edges of the ulcers are marked by granulation tissue. In the third stage, the ulcerated areas form deep masses of keloid or scar tissue that may spread slowly for many years.

Patients with long-term infections are at risk for serious complications. The ulcers in second-stage granuloma inguinale often become superinfected with syphilis or other STD organisms. Superinfected ulcers become painful to touch, filled with pus and dead tissue, and are much more difficult to treat. There may be sizable areas of tissue destruction in superinfected patients. In addition, the scar tissue produced by third-stage infection can grow until it closes off parts of the patient's urinary tract. It is also associated with a higher risk of genital **cancer**.

Diagnosis

The most important aspect of diagnosis is distinguishing between granuloma inguinale and other STDs, particularly since many patients will be infected with more than one STD. Public health officials recommend that patients tested for granuloma inguinale be given a blood test for syphilis as well. In addition, the doctor will need to distinguish between granuloma inguinale and certain types of skin cancer, **amebiasis**, fungal infections, and other bacterial ulcers. The most significant distinguishing characteristic of granuloma inguinale is the skin ulcer, which is larger than in most other diseases, painless, irregular in shape, and likely to bleed when touched.

The diagnosis of granuloma inguinale is made by finding Donovan bodies in samples of the patient's skin tissue. Donovan bodies are oval rod-shaped organisms that appear inside infected tissue cells under a microscope. The doctor obtains a tissue sample either by cutting a piece of tissue from the edge of

KEY TERMS

Donovan bodies—Rod-shaped oval organisms found in tissue samples from patients with granuloma inguinale. Donovan bodies appear deep purple when stained with Wright's stain.

Granulation tissue—A kind of tissue formed during wound healing, with a rough or irregular surface and a rich supply of blood capillaries.

Granuloma—An inflammatory swelling or growth composed of granulation tissue, as in granuloma inguinale.

Keloid—An unusual or abnormal growth of scar tissue, as in the third stage of granuloma inguinale.

Punch biopsy—A method of obtaining skin samples under local anesthesia using a surgical skin punch.

Superinfection—A condition in which a patient with a contagious disease acquires a second infection, as when a patient with granuloma inguinale is also infected with syphilis.

Wright's stain—A chemical used to stain tissue samples for laboratory analysis.

an skin ulcer with a scalpel or by taking a punch biopsy. To make a punch biopsy, the doctor will inject a local anesthetic into an ulcerated area and remove a piece of skin about 1/16 of an inch in size with a surgical skin punch. The tissue sample is then air-dried and stained with Wright's stain, a chemical that will cause the Donovan bodies to show up as dark purple safety pin-shaped objects inside lighter-staining capsules.

Treatment

Granuloma inguinale is treated with oral **antibiotics**. Three weeks of treatment with erythromycin, streptomycin, or tetracycline, or 12 weeks of treatment with ampicillin are standard forms of therapy. Although the skin ulcers will start to show signs of healing in about a week, the patient must take the full course of medication to minimize the possibility of relapse.

Prognosis

Most patients with granuloma inguinale recover completely, although superinfected ulcers may require lengthy courses of medication. Early treatment prevents the complications associated with second- and third-stage infection.

Prevention

Prevention of granuloma inguinale has three important aspects:

- Avoidance of casual sexual contacts, particularly among homosexual males, in countries with high rates of the disease
- Tracing and examination of an infected person's recent sexual contacts
- Monitoring the patient's ulcers or scar tissue for signs of reinfection for a period of six months after antibiotic treatment

Resources

BOOKS

Chambers, Henry F. "Infectious Diseases: Bacterial & Chlamydial." In *Current Medical Diagnosis and Treatment*, 1998. edited by Stephen McPhee, et al., 37th ed. Stamford: Appleton & Lange, 1997.

Rebecca J. Frey, PhD

Granulomatous ileitis see **Crohn's disease**

Graves' disease see **Hyperthyroidism**

Greenfield filter see **Vena cava**

Grippe see **Influenza**

Group A streptococcus infection see **Streptococcal infections**

Group B streptococcus infection see **Streptococcal infections**

Group therapy

Definition

Group therapy is a form of psychosocial treatment where a small group of patients meet regularly to talk, interact, and discuss problems with each other and the group leader (therapist).

Purpose

Group therapy attempts to give individuals a safe and comfortable place where they can work out problems and emotional issues. Patients gain insight into their own thoughts and behavior, and offer suggestions and support to others. In addition, patients who have a difficult time with interpersonal relationships

can benefit from the social interactions that are a basic part of the group therapy experience.

Precautions

Patients who are suicidal, homicidal, psychotic, or in the midst of a major acute crisis are typically not referred for group therapy until their behavior and emotional state have stabilized. Depending on their level of functioning, cognitively impaired patients (like patients with organic brain disease or a traumatic brain injury) may also be unsuitable for group therapy intervention. Some patients with sociopathic traits are not suitable for most groups.

Description

A psychologist, psychiatrist, social worker, or other healthcare professional typically arranges and conducts group therapy sessions. In some therapy groups, two co-therapists share the responsibility of group leadership. Patients are selected on the basis of what they might gain from group therapy interaction and what they can contribute to the group as a whole.

Therapy groups may be homogeneous or heterogeneous. Homogeneous groups have members with similar diagnostic backgrounds (for example, they may all suffer from depression). Heterogeneous groups have a mix of individuals with different emotional issues. The number of group members varies widely, but is typically no more than 12. Groups may be time limited (with a predetermined number of sessions) or indefinite (where the group determines when therapy ends). Membership may be closed or open to new members once sessions begin.

The number of sessions in group therapy depends on the makeup, goals, and setting of the group. For example, a therapy group that is part of a **substance abuse** program to rehabilitate inpatients would be called short-term group therapy. This term is used because, as patients, the group members will only be in the hospital for a relatively short period of time. Long-term therapy groups may meet for six months, a year, or longer. The therapeutic approach used in therapy depends on the focus of the group and the psychological training of the therapist. Some common techniques include psychodynamic, cognitive-behavioral, and **Gestalt therapy**.

In a group therapy session, group members are encouraged to openly and honestly discuss the issues that brought them to therapy. They try to help other group members by offering their own suggestions, insights, and empathy regarding their problems.



Group therapy is practiced in a variety of settings, including both inpatient and outpatient facilities, and is used to treat anxiety, mood, and personality disorders as well as psychoses. (Photo Researchers, Inc. Reproduced by permission.)

There are no definite rules for group therapy, only that members participate to the best of their ability. However, most therapy groups do have some basic ground rules that are usually discussed during the first session. Patients are asked not to share what goes on in therapy sessions with anyone outside of the group. This protects the confidentiality of the other members. They may also be asked not to see other group members socially outside of therapy because of the harmful effect it might have on the dynamics of the group.

The therapist's main task is to guide the group in self-discovery. Depending on the goals of the group and the training and style of the therapist, he or she may lead the group interaction or allow the group to take their own direction. Typically, the group leader does some of both, providing direction when the group gets off track while letting them set their own agenda. The therapist may guide the group by simply reinforcing the positive behaviors they engage in. For example, if a group member shows empathy to another member, or offers a constructive suggestion, the therapist will point this out and explain the value of these

actions to the group. In almost all group therapy situations, the therapist will attempt to emphasize the common traits among group members so that members can gain a sense of group identity. Group members realize that others share the same issues they do.

The main benefit group therapy may have over individual psychotherapy is that some patients behave and react more like themselves in a group setting than they would one-on-one with a therapist. The group therapy patient gains a certain sense of identity and social acceptance from their membership in the group. Suddenly, they are not alone. They are surrounded by others who have the same anxieties and emotional issues that they have. Seeing how others deal with these issues may give them new solutions to their problems. Feedback from group members also offers them a unique insight into their own behavior, and the group provides a safe forum in which to practice new behaviors. Lastly, by helping others in the group work through their problems, group therapy members can gain more self-esteem. Group therapy may also simulate family experiences of patients and will allow family dynamic issues to emerge.

KEY TERMS

Cognitive-behavioral—A therapy technique that focuses on changing beliefs, images, and thoughts in order to change maladjusted behaviors.

Gestalt—A humanistic therapy technique that focuses on gaining an awareness of emotions and behaviors in the present rather than in the past.

Psychodynamic—A therapy technique that assumes improper or unwanted behavior is caused by unconscious, internal conflicts and focuses on gaining insight into these motivations.

Self-help groups like Alcoholics Anonymous and Weight Watchers fall outside of the psychotherapy realm. These self-help groups do offer many of the same benefits of social support, identity, and belonging that make group therapy effective for many. Self-help group members meet to discuss a common area of concern (like **alcoholism**, eating disorders, **bereavement**, parenting). Group sessions are not run by a therapist, but by a nonprofessional leader, group member, or the group as a whole. Self-help groups are sometimes used in addition to psychotherapy or regular group therapy.

Preparation

Patients are typically referred for group therapy by a psychologist or psychiatrist. Some patients may need individual therapy first. Before group sessions begin, the therapist leading the session may conduct a short intake interview with the patient to determine if the group is right for the patient. This interview will also allow the therapist to determine if the addition of the patient will benefit the group. The patient may be given some preliminary information on the group before sessions begin. This may include guidelines for success (like being open, listening to others, taking risks), rules of the group (like maintaining confidentiality), and educational information on what group therapy is about.

Aftercare

The end of long-term group therapy may cause feelings of grief, loss, abandonment, anger, or rejection in some members. The group therapist will attempt to foster a sense of closure by encouraging members to explore their feelings and use newly acquired coping techniques to deal with them.

Working through this termination phase of group therapy is an important part of the treatment process.

Risks

Some very fragile patients may not be able to tolerate aggressive or hostile comments from group members. Patients who have trouble communicating in group situations may be at risk for dropping out of group therapy. If no one comments on their silence or makes an attempt to interact with them, they may begin to feel even more isolated and alone instead of identifying with the group. Therefore, the therapist usually attempts to encourage silent members to participate early on in treatment.

Normal results

Studies have shown that both group and individual psychotherapy benefit about 85% of the patients that participate in them. Optimally, patients gain a better understanding of themselves, and perhaps a stronger set of interpersonal and coping skills through the group therapy process. Some patients may continue therapy after group therapy ends, either individually or in another group setting.

Resources

ORGANIZATIONS

American Psychiatric Association. 1400 K Street NW, Washington DC 20005. (888) 357-7924. <<http://www.psych.org>>.

American Psychological Association (APA). 750 First St. NE, Washington, DC 20002-4242. (202) 336-5700. <<http://www.apa.org>>.

Paula Anne Ford-Martin

Growth hormone suppression test see

Growth hormone tests

Growth hormone tests

Definition

Growth hormone (hGH), or somatotropin, is a hormone responsible for normal body growth and development by stimulating protein production in muscle cells and energy release from the breakdown of fats. Tests for growth hormone include Somatotropin hormone test, Somatomedin C, Growth hormone suppression test (glucose loading

test), and Growth hormone stimulation test (Arginine test or Insulin tolerance test).

Purpose

Growth hormone tests are ordered for the following reasons:

- to identify growth deficiencies, including delayed **puberty** and small stature in adolescents that result from pituitary or thyroid malfunction
- to aid in the diagnosis of hyperpituitarism that is evident in gigantism or acromegaly
- to screen for inadequate or reduced pituitary gland function
- to assist in the diagnosis of **pituitary tumors** or tumors related to the hypothalamus, an area of the brain
- to evaluate hGH therapy

Precautions

Taking certain drugs such as amphetamines, dopamine, **corticosteroids**, and phenothiazines may increase and decrease growth hormone secretion, respectively. Other factors influencing hGH secretion include **stress**, **exercise**, diet, and abnormal glucose levels. These tests should not be done within a week of any radioactive scan.

Description

Several hormones play important roles in human growth. The major human growth hormone (hGH), or somatotropin, is a protein made up of 191 amino acids that is secreted by the anterior pituitary gland and coordinates normal growth and development. Human growth is characterized by two spurts, one at birth and the other at puberty. hGH plays an important role at both of these times. Normal individuals have measurable levels of hGH throughout life. Yet levels of hGH fluctuate during the day and are affected by eating and exercise. Receptors that respond to hGH exist on cells and tissues throughout the body. The most obvious effect of hGH is on linear skeletal development. But the metabolic effects of hGH on muscle, the liver, and fat cells are critical to its function. Surprisingly, a 2004 study reported that obese people have lower-than-normal levels of human growth hormone in their bodies. Humans have two forms of hGH, and the functional difference between the two is unclear. They are both formed from the same gene, but one lacks the amino acids in positions 32–46.

hGH is produced in the anterior portion of the pituitary gland by somatotrophs under the control of hormonal signals in the hypothalamus. Two hypothalamic hormones regulate hGH; they are growth hormone-releasing hormone (GHRH) and growth hormone—inhibiting hormone (GHIH). When blood glucose levels fall, GHRH triggers the secretion of stored hGH. As blood glucose levels rise, GHRH release is turned off. Increases in blood protein levels trigger a similar response. As a result of this hypothalamic feedback loop, hGH levels fluctuate throughout the day. Normal plasma hGH levels average 1–3 ng/mL with peaks as high as 60 ng/mL. In addition, plasma glucose and amino acid availability for growth is also regulated by the hormones adrenaline, glucagon, and insulin.

Most hGH is released at night. Peak spikes of hGH release occur around 10 P.M., midnight, and 2 A.M. The logic behind this night-time release is that most of hGH's effects are controlled by other hormones, including the somatomedins, IGH-I and IGH-II. As a result, the effects of hGH are spread out more evenly during the day.

A number of hormonal conditions can lead to excessive or diminished growth. Because of its critical role in producing hGH and other hormones, an abnormal pituitary gland will often yield altered growth. Dwarfism (very small stature) can be due to underproduction of hGH, lack of IGH-I, or a flaw in target tissue response to either of these growth hormones. Overproduction of hGH or IGH-I, or an exaggerated response to these hormones can lead to gigantism or acromegaly, both of which are characterized by a very large stature.

Gigantism is the result of hGH overproduction in early childhood leading to a skeletal height up to 8 feet (2.5m) or more. Acromegaly results when hGH is overproduced after the onset of puberty. In this condition, the epiphyseal plates of the long bone of the body do not close, and they remain responsive to additional stimulated growth by hGH. This disorder is characterized by an enlarged skull, hands and feet, nose, neck, and tongue.

Somatotropin

Somatotropin is used to identify hGH deficiency in adolescents with short stature, delayed sexual maturity, and other growth deficiencies. It also aids in documenting excess hGH production that is responsible for gigantism or acromegaly, and confirms underactivity or overproduction of the pituitary gland (**hypopituitarism** or hyperpituitarism). However, due

to the episodic secretion of hGH, as well as hGH production in response to stress, exercise, or other factors, random assays are not an adequate determination of hGH deficiency. To negate these variables and obtain more accurate readings, a blood sample can be drawn one to 1.5 hours after sleep (hGH levels increase during sleep), or strenuous exercise can be performed for 30 minutes before blood is drawn (hGH levels increase after exercise). The hGH levels at the end of an exercise period are expected to be maximal.

Somatomedin C

The somatomedin C test is usually ordered to detect pituitary abnormalities, hGH deficiency, and acromegaly. Also called insulin-like growth factor (IGF-1), somatomedin C is considered a more accurate reflection of the blood concentration of hGH because such variables as time of day, activity levels, or diet do not influence the results. Somatomedin C is part of a group of peptides, called somatomedins, through which hGH exerts its effects. Because it circulates in the bloodstream bound to long-lasting proteins, it is more stable than hGH. Levels of somatomedin C depend on hGH levels, however. As a result, somatomedin C levels are low when hGH levels are deficient. Abnormally low test results of somatomedin C require an abnormally reduced or absent hGH during an hGH stimulation test in order to diagnose hGH deficiency. Nonpituitary causes of reduced somatomedin C include **malnutrition**, severe chronic illness, severe **liver disease**, **hypothyroidism**, and Laron's dwarfism.

Growth hormone stimulation test

The hGH stimulation test, also called hGH Provocation test, Insulin Tolerance, or Arginine test, is performed to test the body's ability to produce human growth hormone, and to identify suspected hGH deficiency. A normal patient can have low hGH levels, but if hGH is still low after stimulation, a diagnosis can be more accurately made.

Insulin-induced **hypoglycemia** (via intravenous injection of insulin) stimulates hGH and corticotropin secretion as well. If such stimulation is unsuccessful, then there is a malfunction of the anterior pituitary gland. Blood samples may be obtained following an energetic exercise session lasting 20 minutes.

A substance called hGH-releasing factor has recently been used for hGH stimulation. This approach promises to be more accurate and specific for hGH deficiency caused by the pituitary. Growth

hormone deficiency is also suspected when x ray determination of bone age indicates retarded growth in comparison to chronologic age. At present, the best method to identify hGH-deficient patients is a positive stimulation test followed by a positive response to a therapeutic trial of hGH.

Growth hormone suppression test

Also called the glucose loading test, this procedure is used to evaluate excessive baseline levels of human growth hormone, and to confirm diagnosis of gigantism in children and acromegaly in adults. The procedure requires two different blood samples, one drawn before the administration of 100 g of glucose (by mouth), and a second sample two hours after glucose ingestion.

Normally, a glucose load suppresses hGH secretion. In a patient with excessive hGH levels, failure of suppression indicates anterior pituitary dysfunction and confirms a diagnosis of **acromegaly and gigantism**.

Preparation

Somatotropin: This test requires a blood sample. The patient should be **fasting** (nothing to eat or drink from midnight the night before the test). Stress and/or exercise increases hGH levels, so the patient should be at complete rest for 30 minutes before the blood sample is drawn. If the physician has requested two samples, they should be drawn on consecutive days at approximately the same time on both days, preferably between 6 A.M. and 8 A.M.

Somatomedin C: This test requires a blood sample. The patient should have nothing to eat or drink from midnight the night before the test.

Growth hormone stimulation: This test requires intravenous administration of medications and the withdrawal of frequent blood samples, which are obtained at 0, 60, and 90 minutes after injection of arginine and/or insulin. The patient should have nothing to eat or drink after midnight the night before the test.

Growth hormone suppression: This test requires two blood samples, one before the test and another two hours after administration of 100 g of glucose solution by mouth. The patient should have nothing to eat or drink after midnight, and physical activity should be limited for 10–12 hours before the test.

Risks

Growth hormone stimulation: Only minor discomfort is associated with this test, and results from the insertion of the IV line and the low blood sugar (hypoglycemia) induced by the insulin injection. Some patients may experience sleepiness, sweating and/or nervousness, all of which can be corrected after the test by ingestion of cookies, juice, or a glucose infusion. Severe cases of hypoglycemia may cause ketosis (excessive amounts of fatty acid byproducts in the body), acidosis (a disturbance of the body's acid-base balance), or **shock**. With the close observation required for the test, these are unlikely.

Growth hormone suppression: Some patients experience **nausea** after the administration of this amount of glucose. Ice chips can alleviate this symptom.

Normal results

Normal results may vary from laboratory to laboratory but are usually within the following ranges:

Somatotropin:

- men: 5 ng/ml
- women: less than 10 ng/ml
- children: 0–10 ng/ml
- newborn: 10–40 ng/ml.

Somatomedin C:

- adult: 42–110 ng/ml
- Child:
- 0–8 years: Girls 7–110 ng/ml; Boys 4–87 ng/ml
- 9–10 years: Girls 39–186 ng/ml; Boys 26–98 ng/ml
- 11–13 years: Girls 66–215 ng/ml; Boys 44–207 ng/ml
- 14–16 years: Girls 96–256 ng/ml; Boys 48–255 ng/ml.

Growth hormone stimulation: greater than 10 ng/ml.

Growth hormone suppression: Normally, glucose suppresses hGH to levels of undetectable to 3 ng/ml in 30 minutes to two hours. In children, rebound stimulation may occur after two to five hours.

Abnormal results

Somatotropin hormone: Excess hGH is responsible for the syndromes of gigantism and acromegaly. Excess secretion is stimulated by **anorexia nervosa**, stress, hypoglycemia, and exercise. Decreased levels

KEY TERMS

Acromegaly—A rare disease resulting from excessive growth hormone caused by a benign tumor. If such a tumor develops within the first 10 years of life, the result is gigantism (in which growth is accelerated) and not acromegaly. Symptoms include coarsening of the facial features, enlargement of the hands, feet, ears, and nose, jutting of the jaw, and a long face.

Dwarfism, pituitary—Short stature. When caused by inadequate amounts of growth hormone (as opposed to late growth spurt or genetics), hGH deficiency results in abnormally slow growth and short stature with normal proportions.

Gigantism—Excessive growth, especially in height, resulting from overproduction during childhood or adolescence of growth hormone by a pituitary tumor. Untreated, the tumor eventually destroys the pituitary gland, resulting in death during early adulthood. If the tumor develops after growth has stopped, the result is acromegaly, not gigantism.

Pituitary gland—The pituitary is the most important of the endocrine glands (glands that release hormones directly into the bloodstream). Sometimes referred to as the “master gland,” the pituitary regulates and controls the activities of other endocrine glands and many body processes.

are seen in hGH deficiency, dwarfism, hyperglycemia, **failure to thrive**, and delayed sexual maturity.

Somatomedin C: Increased levels contribute to the syndromes of gigantism and acromegaly. Stress, major surgery, hypoglycemia, **starvation**, and exercise stimulate hGH secretion, which in turn stimulates somatomedin C.

Growth hormone stimulation: Decreased levels are seen in pituitary deficiency and hGH deficiency. Diseases of the pituitary can result in failure of the pituitary to secrete hGH and/or all the pituitary hormones. As a result, the hGH stimulation test will fail to stimulate hGH secretion.

Growth hormone suppression: The acromegaly syndrome elevates base hGH levels to 75 ng/ml, which in turn are not suppressed to less than 5 ng/ml during the test. Excess hGH secretion may cause unchanged or rising hGH levels in response to glucose loading, confirming a diagnosis of acromegaly or

gigantism. In such cases, verification of results is required by repeating the test after a one-day rest.

Resources

PERIODICALS

“Weight-loss Hormone.” *Better Nutrition* (May 2004): 32.

Janis O. Flores
Teresa G. Odle

G6PD deficiency see **Glucose-6-phosphate dehydrogenase deficiency**

Guaifenesin see **Expectorants**

Guided imagery

Definition

Guided imagery is the use of relaxation and mental visualization to improve mood and/or physical well-being.

Purpose

The connection between the mind and physical health has been well documented and extensively studied. Positive mental imagery can promote relaxation and reduce **stress**, improve mood, control high blood pressure, alleviate **pain**, boost the immune system, and lower cholesterol and blood sugar levels. Through guided imagery techniques, patients can learn to control functions normally controlled by the autonomic nervous system, such as heart rate, blood pressure, respiratory rate, and body temperature.

One of the biggest benefits of using guided imagery as a therapeutic tool is its availability. Imagery can be used virtually anywhere, anytime. It is also an equal opportunity therapy. Although some initial training in the technique may be required, guided imagery is accessible to virtually everyone regardless of economic status, education, or geographical location.

Guided imagery also gives individuals a sense of empowerment, or control. The technique is induced by a therapist who guides the patient. The resulting mental imagery used is solely a product of the individual's imagination. Some individuals have difficulty imagining. They may not get actual clear images but perhaps vague feelings about the guided journey. However these individuals' brains and nervous systems

responses seem to be the same as those with more detailed imaginings.

Patients who feel uncomfortable “opening up” in a traditional therapist-patient session may feel more at ease with a self-directed therapy like guided imagery.

Description

Guided imagery is simply the use of one's imagination to promote mental and physical health. It can be self-directed, where the individual puts himself into a relaxed state and creates his own images, or directed by others. When directed by others, an individual listens to a therapist, video, or audiotaped **exercise** that leads him through a relaxation and imagery exercise. Some therapists also use guided imagery in group settings.

Guided imagery is a two-part process. The first component involves reaching a state of deep relaxation through breathing and muscle relaxation techniques. During the relaxation phase, the person closes her eyes and focuses on the slow, in and out sensation of breathing. Or, she might focus on releasing the feelings of tension from her muscles, starting with the toes and working up to the top of the head. Relaxation tapes often feature soft music or tranquil, natural sounds such as rolling waves and chirping birds in order to promote feelings of relaxation.

Once complete relaxation is achieved, the second component of the exercise is the imagery, or visualization, itself. There are a number of different types of guided imagery techniques, limited only by the imagination. Some commonly used types include relaxation imagery, healing imagery, pain control imagery, and mental rehearsal.

Relaxation imagery

Relaxation imagery involves conjuring up pleasant, relaxing images that rest the mind and body. These may be experiences that have already happened, or new situations.

Healing imagery

Patients coping with diseases and injuries can imagine **cancer** cells dying, **wounds** healing, and the body mending itself. Or, patients may picture themselves healthy, happy, and symptom-free. Another healing imagery technique is based on the idea of *qi*, or energy flow, an idea borrowed from **traditional Chinese medicine**. Chinese medicine practitioners believe that illness is the result of a blockage or slowing of energy flow in the body. Individuals may use guided

imagery to imagine energy moving freely throughout the body as a metaphor for good health.

Pain control imagery

Individuals can control pain through several imagery techniques. One method is to produce a mental image of the pain and then transform that image into something less frightening and more manageable. Another is to imagine the pain disappearing, and the patient as completely pain-free. Or, one may imagine the pain as something over which he has complete control. For example, patients with back problems may imagine their pain as a high voltage electric current surging through their spine. As they use guided imagery techniques, they can picture themselves reaching for an electrical switch and turning down the power on the current to alleviate the pain.

Mental rehearsal

Mental rehearsal involves imagining a situation or scenario and its ideal outcome. It can be used to reduce **anxiety** about an upcoming situation, such as labor and delivery, surgery, or even a critical life event such as an important competition or a job interview. Individuals picture themselves going through each step of the anxiety-producing event and then successfully completing it.

Preparations

For a successful guided imagery session, individuals should select a quiet, relaxing location where there is a comfortable place to sit or recline. If the guided imagery session is to be prompted with an audiotape or videotape, a stereo, VCR, or portable tape player should be available. Some people find that quiet background music improves their imagery sessions.

The session, which can last anywhere from a few minutes to an hour, should be uninterrupted. Taking the phone off the hook and asking family members for solitude can ensure a more successful and relaxing session.

Imagery combined with other relaxation techniques such as **yoga**, massage, or **aromatherapy** can greatly enhance the effects of these therapies. It can be done virtually anywhere.

Precautions

Because of the state of extreme relaxation involved in guided imagery, individuals should never

KEY TERMS

Aromatherapy—The therapeutic use of plant-derived, aromatic essential oils to promote physical and psychological well-being.

Autonomic nervous system—The part of the nervous system that controls so-called involuntary functions such as heart rate, salivary gland secretion, respiratory function, and pupil dilation.

attempt to use guided imagery while driving or operating heavy machinery.

Side effects

Guided imagery can induce sleepiness, and some individuals may fall asleep during a session. Other than this, there are no known adverse side effects to guided imagery.

Research and general acceptance

Use of guided imagery is a widely accepted practice among mental healthcare providers and is gaining acceptance as a powerful pain control tool across a number of medical disciplines. Results of a study conducted at The Cleveland Clinic Foundation and published in 1999 found that cardiac surgery patients who used a guided imagery tape prior to surgery experienced less pain and anxiety. These patients also left the hospital earlier following surgery than patients who used pain medication only.

Another study conducted by Harvard Medical School researchers found that for more than 200 patients undergoing invasive vascular or renal surgery, guided imagery controlled pain and anxiety more effectively than medication alone.

Resources

BOOKS

Battino, Rubin. *Guided Imagery and Other Approaches to Healing*. Carmarthen, United Kingdom: Crown House Publishing, 2000.

PERIODICALS

Lang, Elvira, et al. "Adjunctive non-pharmacological analgesia for invasive medical procedures: a randomized trial." *The Lancet*. 355, no. 9214, (April 2000): 1486-1490.

ORGANIZATIONS

The Academy for Guided Imagery. P.O. Box 2070, Mill Valley, CA 94942. (800) 726-2070.

OTHER

Brennan, Patricia. "Stress First Aid Kit." (Guided imagery audiotape set.) Available from Inside Out Publishing at (888) 727-3296 or <<http://www.facingthedawn.com>>.

Paula Anne Ford-Martin

Guillain-Barré syndrome

Definition

Guillain-Barré syndrome (GBS) causes progressive muscle weakness and **paralysis** (the complete inability to use a particular muscle or muscle group), which develops over days or up to four weeks, and lasts several weeks or even months.

Description

The classic scenario in GBS involves a patient who has just recovered from a typical, seemingly uncomplicated viral infection. Symptoms of muscle weakness appear one to four weeks later. The most common preceding infections are cytomegalovirus, herpes, Epstein-Barr virus, and viral hepatitis. A gastrointestinal infection with the bacteria *Campylobacter jejuni* is also common and may cause a severe type of GBS from which it is particularly difficult to recover. About 5% of GBS patients have a surgical procedure as a preceding event. Patients with lymphoma, **systemic lupus erythematosus**, or **AIDS** have a higher than normal risk of GBS. Other GBS patients have recently received an immunization, while still others have no known preceding event. In 1976–77, there was a vastly increased number of GBS cases among people who had been recently vaccinated against the Swine flu. The reason for this phenomenon has never been identified, and no other flu vaccine has caused such an increase in GBS cases.

Causes and symptoms

The cause of the weakness and paralysis of GBS is the loss of myelin, which is the material that coats nerve cells (the loss of myelin is called demyelination). Myelin is an insulating substance which is wrapped around nerves in the body, serving to speed conduction of nerve impulses. Without myelin, nerve

conduction slows or stops. GBS has a short, severe course. It causes inflammation and destruction of the myelin sheath, and it disturbs multiple nerves. Therefore, it is considered an acute inflammatory demyelinating polyneuropathy.

The reason for the destruction of myelin in GBS is unknown, although it is thought that the underlying problem is autoimmune in nature. An autoimmune disorder is one in which the body's immune system, trained to fight against such foreign invaders as viruses and bacteria, somehow becomes improperly programmed. The immune system becomes confused, and is not able to distinguish between foreign invaders and the body itself. Elements of the immune system are unleashed against areas of the body, resulting in damage and destruction. For some reason, in the case of GBS, the myelin sheath appears to become a target for the body's own immune system.

The first symptoms of GBS consist of muscle weakness (legs first, then arms, then face), accompanied by prickly, **tingling** sensations (paresthesias). Symptoms affect both sides of the body simultaneously, a characteristic that helps distinguish GBS from other causes of weakness and paresthesias. Normal reflexes are first diminished, then lost. The weakness eventually affects all the voluntary muscles, resulting in paralysis. When those muscles necessary for breathing become paralyzed, the patient must be placed on a mechanical ventilator which takes over the function of breathing. This occurs about 30% of the time. Very severely ill GBS patients may have complications stemming from other nervous system abnormalities which can result in problems with fluid balance in the body, severely fluctuating blood pressure, and heart rhythm irregularities.

Diagnosis

Diagnosis of GBS is made by looking for a particular cluster of symptoms (progressively worse muscle weakness and then paralysis), and by examining the fluid that bathes the brain and spinal canal through **cerebrospinal fluid (CSF) analysis**. This fluid is obtained by inserting a needle into the lower back (lumbar region). When examined in a laboratory, the CSF of a GBS patient will reveal a greater-than-normal quantity of protein, with normal numbers of white blood cells and a normal amount of sugar. Electrodiagnostic studies may show slowing or block of conduction in nerve endings in parts of the body other than the brain. Minor abnormalities will be present in 90% of patients.

KEY TERMS

Autoimmune—The body's immune system directed against the body itself.

Demyelination—Disruption or destruction of the myelin sheath, leaving a bare nerve. Results in a slowing or stopping of impulses traveling along that nerve.

Inflammatory—Having to do with inflammation, the body's response to either invading foreign substances (such as viruses or bacteria) or to direct injury of body tissue.

Myelin—The substance that is wrapped around nerves, and which is responsible for speed and efficiency of impulses traveling through those nerves.

Treatment

There is no direct treatment for GBS. Instead, treatments are used that support the patient with the disabilities caused by the disease. The progress of paralysis must be carefully monitored, in order to provide mechanical assistance for breathing if it becomes necessary. Careful attention must also be paid to the amount of fluid the patient is taking in by drinking and eliminating by urinating. Blood pressure, heart rate, and heart rhythm also must be monitored.

A procedure called **plasmapheresis**, performed early in the course of GBS, has been shown to shorten the course and severity of GBS. Plasmapheresis consists of withdrawing the patient's blood, passing it through an instrument that separates the different types of blood cells, and returning all the cellular components (red and white blood cells and platelets) along with either donor plasma or a manufactured replacement solution. This is thought to rid the blood of the substances that are attacking the patient's myelin.

It has also been shown that the use of high doses of immunoglobulin given intravenously (by drip through a needle in a vein) may be just as helpful as plasmapheresis. Immunoglobulin is a substance naturally manufactured by the body's immune system in response to various threats. It is interesting to note that corticosteroid medications (such as prednisone), often the mainstay of anti-autoimmune disease treatment, are not only unhelpful, but may in fact be harmful to patients with GBS.

Prognosis

About 85% of GBS patients make reasonably good recoveries. However, 30% of adult patients, and a greater percentage of children, never fully regain their previous level of muscle strength. Some of these patients suffer from residual weakness, others from permanent paralysis. About 10% of GBS patients begin to improve, then suffer a relapse. These patients suffer chronic GBS symptoms. About 5% of all GBS patients die, most from cardiac rhythm disturbances.

Patients with certain characteristics tend to have a worse outcome. These include people of older age, those who required breathing support with a mechanical ventilator, and those who had their worst symptoms within the first seven days.

Prevention

Because so little is known about what causes GBS to develop, there are no known methods of prevention.

Resources

ORGANIZATIONS

American Academy of Neurology. 1080 Montreal Ave., St. Paul, MN 55116. (612) 695-1940. <<http://www.aan.com>>.

Guillain-Barré Syndrome Foundation International. PO Box 262, Wynnewood, PA 19096. (610) 667-0131. (610) 667-0131. <<http://www.webmast.com/gbs>>.

Rosalyn Carson-DeWitt, MD

Guinea worm infection

Definition

Infection occurs when the parasitic guinea worm resides within the body. Infection is not apparent until a pregnant female worm prepares to expel embryos. The infection is rarely fatal, but the latter stage is painful. The infection is also referred to as dracunculiasis, and less commonly as dracontiasis.

Description

Before the early 1980s, guinea worms infected 10–15 million people annually in central Africa and parts of Asia. By 1996, worldwide incidence of infection fell to fewer than 153,000 cases per year.

Complete eradication of guinea worm infection is a goal of international water safety programs.

To survive, guinea worms require three things: water during the embryo stage, an intermediate host during early maturation, and a human host during adulthood. In bodies of water, such as ponds, guinea worm embryos are eaten by tiny, lobster-like water fleas. Once ingested, the embryos mature into larvae.

Humans become hosts by consuming water containing infected water fleas. Once in the human intestine, larvae burrow into surrounding tissue. After three to four, the worms mate. Males die soon after, but pregnant females continue to grow. As adults, each threadlike worm can be three feet long and harbor three million embryos. More than one guinea worm can infect a person at the same time.

About eight months later, the female prepares to expel mature embryos by migrating toward the skin surface. Until this point, most people are unaware that they are infected. Extreme **pain** occurs as the worm emerges from under the skin, often around the infected person's ankle. The pain is temporarily relieved by immersing the area in water, an act that contaminates the water and starts the cycle again.

Causes and symptoms

Dracunculus medinensis, or guinea worm, causes infection. Symptoms are commonly absent until a pregnant worm prepares to expel embryos. By secreting an irritating chemical, the worm causes a blister to form on the skin surface. This chemical also causes **nausea, vomiting, dizziness, and diarrhea**. The blister is accompanied by a burning, stabbing pain and can form anywhere on the body; but, the usual site is the lower leg or foot. Once the blister breaks, an open sore remains until the worm has expelled all the embryos.

Diagnosis

Guinea worm infection is identified by the symptoms.

Treatment

Most people infected with guinea worm rely on traditional medicine. The worm is extracted by gently and gradually pulling the worm out and winding it around a small strip of wood. Surgical removal is possible, but rarely done in rural areas. Extraction is complemented by herbs and oils to treat the wound site. Such treatment can ease extraction and may help prevent secondary infections.

KEY TERMS

Guinea worm embryo—The guinea worm at its earliest life stage prior to or shortly after being expelled from an adult female worm.

Guinea worm larvae—The guinea worm during its middle life stage as it matures within a water flea. The larvae can only grow to adulthood within a human host.

Host—With regard to guinea worm infection, either the water flea or human from which the worm gets nourishment and shelter as it matures.

Secondary infection—An illness—typically caused by bacteria—that follows from a guinea worm infection.

Modern medicine offers safe surgical removal of the guinea worm, and drug therapy can prevent infection and pain. Using drugs to combat the worms has had mixed results.

Prognosis

If the worm is completely removed, the wound heals in approximately two to four weeks. However, if a worm emerges from a sensitive area, such as the sole of a foot, or if several worms are involved, healing requires more time. Recovery is also complicated if the worm breaks during extraction. Serious secondary infections frequently occur in such situations. There is the risk of permanent disability in some cases, and having one guinea worm infection does not confer immunity against future infections.

Prevention

Guinea worm infection is prevented by disrupting transmission. Wells and other protected water sources are usually safe from being contaminated with worm embryos. In open water sources, poisons may be used to kill water fleas. Otherwise, water must be boiled or filtered.

Resources

OTHER

World Health Organization. June 7, 1998. <<http://www.who.int/home-page/index.en.shtml>>.

Julia Barrett

Gulf War syndrome

Definition

Gulf War syndrome describes a wide spectrum of illnesses and symptoms ranging from **asthma** to **sexual dysfunction** that have been reported by U.S. and U.S. allied soldiers who served in the Persian Gulf War in 1990–1991.

Description

Between 1994 and 1999, 145 federally funded research studies on Gulf War-related illnesses were undertaken at a cost of over \$133 million. Despite this investment and the data collected from over 100,000 veterans who have registered with the Department of Defense (DOD) and/or Veterans Administration (VA) as having Gulf War-related illnesses, there is still much debate over the origin and nature of Gulf War syndrome. As of early 2001, the DOD has failed to establish a definite cause for the disorder. Veterans who have the illness experience a wide range of debilitating symptoms that elude a single diagnosis. Common symptoms include **fatigue**, trouble breathing, headaches, disturbed sleep, memory loss, and lack of concentration. Similar experiences among Gulf War veterans have been reported in the United Kingdom and Canada.

Causes and symptoms

There is much current debate over a possible causative agent for Gulf War syndrome other than the **stress** of warfare. Intensive efforts by the Veterans Administration and other public and private institutions have investigated a wide range of potential factors. These include chemical and biological weapons, the immunizations and preventive treatments used to protect against them, smoke from oil well fires, exposure to depleted uranium, and diseases endemic to the Arabian peninsula. So far investigators have not approached a consensus. In its final report released in December 2000, the Presidential Special Oversight Board for Department of Defense Investigations of Gulf War Chemical and Biological Incidents cited combat stress as a possible causative factor, but called for further research. There is also a likelihood that U.S. and allied forces were exposed to low levels of sarin and/or cyclosarin (nerve gases) released during the destruction of Iraqi munitions at Kharnisiyah, Iraq, and that these chemicals might be linked to the syndrome. In July 1997, the VA informed approximately 100,000 U.S. servicemen of their possible exposure to the nerve agents.

In October 1999, the U.S. Pentagon released a report that hypothesized that an experimental drug known as pyriostigmine bromide (PB) might be linked to the physical symptoms manifested in Gulf War Syndrome. The experimental drug was given to U.S. and Canadian troops during the war to protect soldiers against the effects of the chemical nerve agent soman. It has also been suggested that botulinum toxoid and **anthrax** vaccinations administered to soldiers during the conflict may be responsible for some manifestations of the syndrome.

Some studies have shown that Gulf War veterans have a higher incidence of positive tests for *Mycoplasma fermentans*, a bacteria, in their bloodstream. However, other clinical studies have not found a link between the bacterial infection and Gulf War-related illnesses.

Statistical analysis tells us that the following symptoms are about twice as likely to appear in Gulf War veterans than in their non-combat peers: depression, posttraumatic stress disorder (PTSD), chronic fatigue, cognitive dysfunction (diminished ability to calculate, order thoughts, evaluate, learn, and remember), **bronchitis**, asthma, **fibromyalgia**, alcohol **abuse**, **anxiety**, and sexual discomfort. PTSD is the modern equivalent of shell shock (World War I) and battle fatigue (World War II). It encompasses most of the psychological symptoms of war veterans, including nightmares, panic at sudden loud noises, and inability to adjust to peacetime living. **Chronic fatigue syndrome** has a specific medical definition that attempts to separate common fatigue from a more disabling illness in hope of finding a specific cause. Fibromyalgia is another newly defined syndrome, and as such it has arbitrarily rigid defining characteristics. These include a certain duration of illness, a specified minimum number of joint and muscle **pain** located in designated areas of the body, sleep disturbances, and other associated symptoms and signs.

Researchers have identified three distinct syndromes and several variations in Gulf War veterans. Type one patients suffer primarily from impaired thinking. Type two patients have a greater degree of confusion and ataxia (loss of coordination). Type three patients were the most affected by joint pains, muscle pains, and extremity paresthesias (unnatural sensations like burning or tingling in the arms and legs). In each of the three types, researchers found different but measurable impairments on objective testing of neurological function. The business of the nervous system is much more complex and subtle than other body functions. Measuring it requires equally complex effort. The tests used in this study carefully

measured and compared localized nerve performance at several different tasks against the same values in normal subjects. Brain wave response to noise and touch, eye muscle response to spinning, and caloric testing (stimulation of the ear with warm and cold water, which causes vertigo) were clearly different between the normal and the test subjects. The researchers concluded that there was “a generalized injury to the nervous system.” Another research group concluded their study by stating that there was “a spectrum of neurologic injury involving the central, peripheral, and autonomic nervous systems.”

Diagnosis

Until there is a clear definition of the disease, diagnosis is primarily an **exercise** in identifying those Gulf War veterans who have undefined illness in an effort to learn more about them and their symptoms. Both the Department of Defense and the Veterans Administration currently have programs devoted to this problem. Both the DOD's Comprehensive Clinical Evaluation Program and the VA's Persian Gulf Registry provide free, in-depth medical evaluations to Gulf War veterans and their families. In addition to providing individual veterans with critical medical care, these organizations use the cumulative data from these programs to advance research on Gulf War Syndrome itself.

Treatment

Specific treatment awaits specific diagnosis and identification of a causative agent. Meanwhile, veterans can benefit from the wide variety of supportive and non-specific approaches to this and similar problems. There are many drugs available for symptomatic relief. Psychological counseling by those specializing in this area can be immensely beneficial, even life-saving for those contemplating **suicide**. Veterans' benefits are available for those who are impaired by their symptoms.

Alternative treatment

The symptoms can be worked with using many modalities of alternative health care. The key to working successfully with people living their lives with Gulf War syndrome is long-term, ongoing care, whether it be **hypnotherapy**, **acupuncture**, homeopathy, **nutrition**, vitamin/mineral therapy, or bodywork.

Experimental treatment with **antibiotics** is advocated by some healthcare professionals who believe that Gulf War illness is related to a *Mycoplasma*

KEY TERMS

Ataxia—Lack of coordination.

Caloric testing—Flushing warm and cold water into the ear stimulates the labyrinth and causes vertigo and nystagmus if all the nerve pathways are intact.

Endemic—Always there.

Paresthesia—An altered sensation often described as burning, tingling, or pin pricks.

Syndrome—Common features of a disease or features that appear together often enough to suggest they may represent a single, as yet unknown, disease entity. When a syndrome is first identified, an attempt is made to define it as strictly as possible, even to the exclusion of some cases, in order to separate out a pure enough sample to study. This process is most likely to identify a cause, a positive method of diagnosis, and a treatment. Later on, less typical cases can be considered.

fermentans bacterial infection. However, a conclusive link has not been clinically proven.

Prognosis

The outlook for Persian Gulf War veterans is unclear, but will hopefully improve as more information is gathered about the illness. Gradual return to a functioning life may take many years of work and much help. It is important to note that even in the absence of an identifiable and curable cause, recovery is possible.

Resources

BOOKS

Wheelwright, Jeff. *The Irritable Heart*. New York: W.W. Norton & Co., 2001.

ORGANIZATIONS

The American Legion. Gulf War Veteran Issues. <<http://www.legion.org/gulftoc.htm>>.

Office of the Special Assistant for Gulf War Illnesses. 5113 Leesburg Pike, Suite 901, Falls Church, VA, 22041. (703) 578-8518. <<http://www.gulflink.osd.mil>>.

U.S. Department of Defense. Comprehensive Clinical Evaluation Program (CCEP). (800) 796-9699.

Veterans Administration. Persian Gulf Medical Information Helpline. 400 South 18th Street, St. Louis, Missouri 63103-2271. (800)749-8387.

Veterans Administration. Persian Gulf Registry. (800)PGW-VETS (800) 749-8387. <<http://www.va.gov>>.

OTHER

Office of the Special Assistant for Gulf War Illnesses. *Fourth Annual Report: Office of the Special Assistant for Gulf War Illnesses*. Falls Church, VA: Office of the Special Assistant for Gulf War Illnesses, 2000. <http://www.gulflink.osd.mil/library/annual/4thannual_report_jan01.htm>.

Paula Anne Ford-Martin

Gum disease see **Periodontal disease**

Günther's disease see **Porphyrrias**

Gynecomastia

Definition

Gyne refers to female, and mastia refers to the breast. Gynecomastia is strictly a male disease and is any growth of the adipose (fatty) and glandular tissue in a male breast. Not all breast growth in men is considered abnormal, just excess growth.

Causes and symptoms

Breast growth is directed exclusively by female hormones—estrogens. Although men have some estrogen in their system, it is usually insufficient to cause much breast enlargement because it is counterbalanced by male hormones—androgens. Upsetting the balance, either by more of one or less of the other, results in the male developing female characteristics, breast growth being foremost.

At birth both male and female infants will have little breast buds from their mother's hormones. These recede until adolescence, when girls always, and boys sometimes, have breast growth. At this time, the boy's breast growth is minimal, often one-sided and temporary.

Extra or altered sex chromosomes can produce intersex problems of several kinds. Breast growth along with male genital development is seen in Klinefelter syndrome—the condition of having an extra X (female) chromosome—and a few other chromosomal anomalies. One of the several glands that produce hormones can malfunction for reasons other than chromosomes. Failure of androgen production is as likely to produce gynecomastia as overabundant

KEY TERMS

Androgen—Male sex hormone.

Cirrhosis—Diffuse scarring caused by alcohol or chronic hepatitis often leading to liver failure.

Estrogen—Sex hormone responsible for stimulating female sexual characteristics.

Klinefelter syndrome—A condition in a male characterized by having an extra X (female) chromosome and suffering from infertility and gynecomastia.

Thyroid—A gland in the neck that makes thyroxin. Thyroxin regulates the speed of metabolism.

estrogen production. Testicular failure and castration can also be a cause. Some cancers and some benign tumors can make estrogens. Lung **cancer** is known to increase estrogens.

If the hormone manufacturing organs are functioning properly, problems can still arise elsewhere. The liver is the principle chemical factory in the body. Other organs like the thyroid and kidneys also effect chemical processes. If any of these organs are diseased, a chemical imbalance can result that alters the manufacturing process. Men with **cirrhosis** of the liver will often develop gynecomastia from increased production of estrogens.

Finally, drugs can also cause breast enlargement. Estrogens are given to men to treat **prostate cancer** and a few other diseases. **Marijuana** and heroin, along with some prescription drugs, have estrogen effects in some men. On the list are methyldopa (for blood pressure), cimetidine (for peptic ulcers), diazepam (Valium), antidepressants, and spironolactone (a diuretic).

Diagnosis

Carefully feeling the area beneath the nipple of an adolescent boy with breast enlargement will reveal a discreet and sometimes tender lump the size of a fat nickel or quarter. For more serious gynecomastia, the underlying disease will require evaluation, if it is not already well understood.

Treatment

This condition is usually not treated. If it is the result of endocrine disease, hormone manipulations may reduce the effects of the imbalance. There are a

number of medical and surgical interventions possible. Radiation of misbehaving organs and cancers is considered an effective treatment.

Prognosis

The progress of gynecomastia is determined by its cause.

Resources

BOOKS

Wilson, Jean D. "Endocrine Disorders of the Breast." In *Harrison's Principles of Internal Medicine*, edited by Anthony S. Fauci, et al. New York: McGraw-Hill, 1997.

J. Ricker Polsdorfer, MD

H

H-2 blockers

Definition

Histamine H-2 receptor blockers act by stopping the pathway that leads to the secretion of stomach acid. There are two kinds of pathways that react to stimulation by histamine. Histamine is produced in the body and released by mast cells in response to some types of injury or to the presence of an antigen. When histamine reaches the H-1 receptors, the reaction results in dilation of capillaries, leading to redness and swelling, along with **itching**. These reactions can be controlled with traditional **antihistamines**.

Histamine that reaches the H-2 receptors causes increased secretion of stomach acid.

Purpose

H2 receptor blockers are used to treat conditions associated with excess amounts of stomach acid, although in some cases they have been replaced by the **proton pump inhibitors**, which have a greater effect on reducing acid secretions.

H2 receptor blockers are used to treat the following conditions:

- duodenal ulcer, as short term therapy and maintenance
- gastric ulcer, as short term therapy and maintenance
- gastroesophageal reflux disease (GERD), including endoscopically diagnosed erosive esophagitis
- pathological hypersecretory conditions such as Zollinger-Ellison syndrome, systemic **mastocytosis**, and multiple endocrine adenomas
- upper GI bleeding
- **heartburn**, acid **indigestion**, and sour stomach

None of the drugs in this class has been approved for use by children under the age of 12 years. However,

standard pediatric texts have reported on use by infants and children.

Description

There are four H2 receptor blockers on the market. Although they all work in the same manner and have similar effects, they are not all approved for the same uses.

Cimetidine (Tagamet) is available in both prescription and over-the-counter forms. The oldest of the group and the most studied, this drug is the least potent of the H2 receptor blockers, which means that higher dosages are required to provide comparable effects. There is no evidence that higher potency improves therapeutic results.

Cimetidine is the only drug in its class which is approved for prevention of upper gastro-intestinal bleeding. It has been reported on for a number of uses, with varying degrees of success. Cimetidine, like ranitidine, has shown some benefit in treatment of colorectal **cancer**. Although some claims have been made that cimetidine is useful in treatment of **acetaminophen** overdose, the evidence for this use is lacking, and cimetidine should not be used. Because cimetidine is a mild antiandrogen, it has been of some use in treatment of **hirsutism** (abnormal growth of hair on a woman's face and body).

The three other H2 receptor blockers, famotidine (Pepcid, Pepcid AC), nizatidine (Axid), and ranitidine (Zantac), are similar in their uses. All are approved for treatment of duodenal ulcer both acute treatment and maintenance therapy, gastro-esophageal reflux disease, including erosive esophagitis and gastric ulcer short term treatment, although in this group ranitidine alone is approved for maintenance treatment.

In their over-the-counter (non-prescription) forms, cimetidine and famotidine are approved for treatment of heartburn, acid indigestion, and sour stomach.

Drugs in this class are similar in other respects as well. Although study results vary, cimetidine will usually show its effects within one hour and last for about five hours after a single dose; famotidine and nizatidine also show effects within one hour but may act for up to 12 hours at maximum dosing. Ranitidine has a comparable onset of action and duration in adults but may be slower in the elderly. Onset and duration of action will vary with the individual, the dose of medication, and the presence or absence of food or **antacids** in the stomach.

When *Facts and Comparisons*, a widely used on-line drug information resource, compared the published reports on cure rates for duodenal ulcers, it found that after eight weeks of treatment, all drugs showed healing rates in the range of 82% to 95%. These results were based on comparing separate studies and did not represent comparative trials of the drugs against each other.

Recommended dosage

Cimetidine doses for patients over the age of 12 years, for oral administration.

- Short-term treatment of active duodenal ulcer: 800 mg at bedtime. Other dose regimens are sometimes used.
- Heartburn, acid indigestion, and sour stomach using the over-the-counter product: 100 to 200 mg with water when symptoms start. The dose may be repeated once in 24 hours.
- Prevention of heartburn, acid indigestion, and sour stomach using the over-the-counter product: 100 to 200 mg with water up to one hour before eating food or drinking beverages expected to cause symptoms. Dose should not exceed 400 mg in 24 hours.
- Treatment of hypersecretory conditions: 300 mg four times a day, with meals and at bedtime.
- Gastroesophageal reflux disease: 800 to 1600 mg a day, divided into smaller doses. Treatment usually lasts 12 weeks.

Famotidine doses for patients over the age of 12 years, for oral administration.

- Treatment of duodenal ulcers: 40 mg once a day at bedtime. If necessary, 20 milligrams two times a day may be used.
- Prevention of duodenal ulcers: 20 mg once a day at bedtime.
- Gastric ulcers: 40 mg once a day at bedtime.

- To treat heartburn, acid indigestion, and sour stomach using the over-the-counter product: 10 mg with water when symptoms start. The dose may be repeated once in 24 hours.
- Hypersecretory conditions: 20 mg every six hours.
- Gastroesophageal reflux disease: 20 mg two times a day, usually for up to six weeks.

Nizatidine doses for patients over the age of 12 years, for oral administration.

- Treatment of duodenal or gastric ulcers: 300 mg once a day at bedtime. Alternately, 150 mg two times a day.
- Prevention of duodenal ulcers: 150 mg once a day at bedtime.
- Prevention of heartburn, acid indigestion, and sour stomach: 75 mg taken thirty to sixty minutes before meals which may cause symptoms. The dose may be repeated once in 24 hours.
- Gastroesophageal reflux disease: 150 mg two times a day.

Ranitidine doses for patients over the age of 12 years, for oral administration.

- Duodenal ulcers, treatment: 150 mg two times a day. Alternately, 300 mg once a day at bedtime.
- Duodenal ulcers, prevention: 150 mg at bedtime.
- Gastric ulcers, treatment: 150 mg two times a day.
- Heartburn, acid indigestion, and sour stomach, treatment: 75 mg with water when symptoms start. The dose may be repeated once in 24 hours.
- Heartburn, acid indigestion, and sour stomach, prevention: 75 mg with water taken thirty to sixty minutes before meals or beverages which may cause symptoms. The dose may be repeated once in 24 hours.
- Hypersecretory conditions: 150 mg two times a day.
- Gastroesophageal reflux disease: 150 mg two times a day. The dose may be increased as needed.

Precautions

Overall, the histamine H2 receptor blockers are a safe class of drugs. However, some patients may be particularly susceptible to adverse effects of these drugs. H2 receptor blockers are metabolized in the liver and excreted through the kidneys. Therefore, patients with kidney or liver problems may require reduced doses in order to maintain safe blood levels of the drugs.

Although the safety and effectiveness of H₂ receptor blockers in patients over the age of 65 appears to be similar to that seen in younger patients, age-associated reductions in kidney function may lead to elevated blood levels.

Allergic reactions to these drugs are rare but have been reported.

The histamine H₂ receptor blockers are **Pregnancy** category B. There are no adequate and well-controlled studies with these agents in pregnant women. Women should use them only when clearly needed and when the potential benefits outweigh the potential hazards to the fetus. Cimetidine is known to cross the placenta.

All drugs in this class are excreted into breast milk and should not be taken by nursing women. Decide whether to discontinue nursing, or discontinue the drug, taking into account the importance of the drug to the mother.

These drugs may mask the symptoms of **stomach cancer**.

Side effects

Although side effects due to the H₂ receptor blockers are relatively rare and usually mild, a large number of adverse effects have been reported, in part because of the high use of these drugs. For example, the most common single adverse effect of cimetidine has been a 4% incidence of breast enlargement among males taking the drug in high doses for hypersecretory conditions. Similarly, the incidence of **headache** among high-dose cimetidine patients was 3.5%. Among patients taking lower doses, the frequency of headache was 2.1% compared to 2.3% in a placebo control group. Decreased white blood cell count was reported in 1 in 1,000,000 patients.

The reported side effects from the H₂ receptor blocks are:

- abdominal **pain**
- back, leg, or stomach pain
- bleeding or crusting sores on lips
- blistering, burning, redness, scaling, or tenderness of skin
- blisters on palms of hands and soles of feet
- changes in vision or blurred vision
- coughing or difficulty in swallowing
- dark-colored urine
- dizziness

- fainting
- fast, pounding, or irregular heartbeat
- fever and/or chills
- flu-like symptoms
- general feeling of discomfort or illness
- hives
- inflammation of blood vessels
- joint pain
- light-colored stools
- mood or mental changes, including **anxiety**, agitation, confusion, **hallucinations** (seeing, hearing, or feeling things that are not there), mental depression, nervousness, or severe mental illness
- muscle cramps or aches
- nausea, **vomiting**, or loss of appetite
- pain
- peeling or sloughing of skin
- red or irritated eyes
- shortness of breath
- skin rash or itching
- slow heartbeat
- sore throat
- sores, ulcers, or white spots on lips, in mouth, or on genitals
- sudden difficult breathing
- swelling of face, lips, mouth, tongue, or eyelids
- swelling of hands or feet
- swollen or painful glands
- tightness in chest
- troubled breathing, unusually slow or irregular breathing
- unusual bleeding or bruising
- unusual tiredness or weakness
- wheezing
- yellow eyes or skin

Less frequently reported are

- constipation
- decreased sexual ability (especially in patients with Zollinger-Ellison disease who have received high doses of cimetidine for at least 1 year)
- decrease in sexual desire

- diarrhea
- difficult urination
- dizziness
- drowsiness
- dryness of mouth or skin
- headache
- increased or decreased urination
- increased sweating
- loss of hair
- ringing or buzzing in ears
- runny nose
- swelling of breasts or breast soreness in females and males
- trouble in sleeping

Not all of these adverse effects have been reported with all of the H₂ receptor blockers, and some of the adverse effects may not have been drug related. However, because of the high similarity between drugs in this class, any of the reported adverse effects may be considered a possible result of therapy.

Interactions

Cimetidine and ranitidine are both metabolized in the liver using the cytochrome P450 oxidase enzyme system. Since the same enzymes metabolize many drugs, taking two or more drugs that affect the same group of enzymes may cause one of the drugs to be retained in the body longer than would have been expected. The following is a partial list of drugs which may interact with cimetidine, or to a lesser extent with ranitidine.

- benzodiazepines, including Valium, Librium and Xanax
- caffeine
- calcium channel blockers, including Adalat, Calan, Procardia, and others
- carbamazepine
- chloroquine
- labetolol
- lidocaine
- metoprolol
- metronidazole
- phenytoin
- propranolol

KEY TERMS

Duodenal—Pertaining to the first part of the small intestine.

Gastric—Pertaining to the stomach.

Hirsutism—Abnormal growth of hair on a woman's face and body.

Histamine—A physiologically active compound found in plant and animal tissue and released from mast cells as part of an allergic reaction in humans. It stimulates gastric secretion and causes dilation of capillaries, constriction of bronchial smooth muscle, and decreased blood pressure.

Hypersecretory—Excessive secretions, over production of stomach acid.

Mast cell—A cell found in connective tissue that releases substances such as heparin and histamine in response to injury or inflammation of bodily tissues.

Peptic—Induced by or associated with the action of digestive secretions.

Ulcer—A slow-healing sore on the surface of a mucous membrane, especially the membrane lining the stomach or other part of the digestive tract.

Zollinger-Ellison syndrome—Severe peptic ulceration from excessive stomach acid production stimulated by one or more tumors that produce a powerful acid secretion.

- quinidine
- quinine
- sulfonylureas (includes many of the drugs used to treat diabetes)
- theophyllines (used to treat **asthma**; Dyphylline, a member of this group, does not interact with cimetidine)
- triamterene (a diuretic drug rarely used alone but may be found in fixed combinations, including Dyazide and Maxzide)
- tricyclic antidepressants (a group that includes amitriptyline, imipramine, and others)
- valproic acid
- warfarin

Additional drugs may also interact with the H₂ receptor blockers, particularly those which might have a similar mechanism or action or adverse effects.

Resources

BOOKS

Beers, Mark H., ed. *Merck Manual of Medical Information: Home Edition*. Riverside, NJ: Simon & Schuster, 2004. *Physicians' Desk Reference 2005*. Montvale, NJ: Thomson Healthcare, 2004.

Robertson, Jason, et al. *The Harriet Lane Handbook: A Manual for Pediatric House Officers*. Orlando, FL: Mosby Inc., 2005.

PERIODICALS

Black, R. A., and D. A. Hill. "Over-the-counter medications in pregnancy." *American Family Physician* 67, no. 12 (June 15, 2003): 2517–24.

Chandramouli, J. "What is the most effective therapy for preventing NSAID-induced gastropathy?" *Journal of Pain and Palliative Care Pharmacotherapy* 16, no. 2 (2002): 23–36.

ORGANIZATION

American College of Gastroenterology. PO Box 342260, Bethesda, MD 20827–2260. (301)263-9000. <www.acg.gi.org>.

American Gastroenterological Association. 4930 Del Ray Avenue, Bethesda, MD 20814. (301)654-2055. <www.gastro.org>.

Samuel D. Uretsky, Pharm.D.

Habitual abortion see **Recurrent miscarriage**

Hair transplantation

Definition

Hair transplantation is a surgical procedure used to treat baldness or hair loss. Typically, tiny patches of scalp are removed from the back and sides of the head and implanted in the bald spots in the front and top of the head.

Purpose

Hair transplantation is a cosmetic procedure performed on men (and occasionally on women) who have significant hair loss, thinning hair, or bald spots where hair no longer grows. In men, hair loss and baldness are most commonly due to genetic factors (a tendency passed on in families) and age. Male pattern baldness, in which the hairline gradually recedes to expose more and more of the forehead, is the most common form. Men may also experience a gradual thinning of hair at the crown or very top of

the skull. For women, hair loss is more commonly due to hormonal changes and is more likely to be a thinning of hair from the entire head. An estimated 50,000 men get transplants each year. Transplants can also be done to replace hair lost due to **burns**, injury, or diseases of the scalp.

Precautions

Although hair transplantation is a fairly simple procedure, some risks are associated with any surgery. It is important to inform the physician about any medications currently being used and about previous allergic reactions to drugs or anesthetic agents. Patients with blood clotting disorders also need to inform their physician before the procedure is performed.

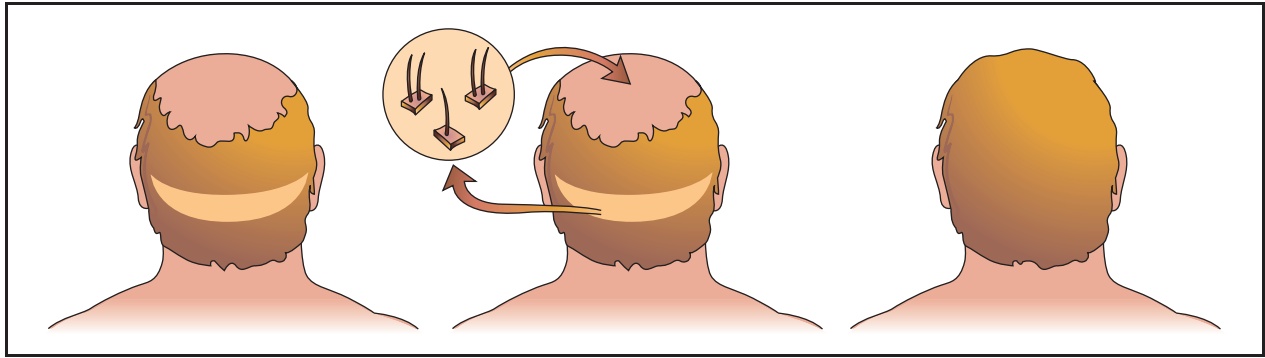
Description

Hair transplantation surgery is performed by a physician in an office, clinic, or hospital setting. Each surgery lasts two to three hours during which approximately 250 grafts will be transplanted. A moderately balding man may require up to 1,000 grafts to get good coverage of a bald area, so a series of surgeries scheduled three to four months apart is usually required. The patient may be completely awake during the procedure with just a local anesthetic drug applied to numb the areas of the scalp. Some patients may be given a drug to help them relax or may be given an anesthetic drug that puts them to sleep.

The most common transplant procedure uses a thin strip of hair and scalp from the back of the head. This strip is cut into smaller clumps of five or six hairs. Tiny cuts are made in the balding area of the scalp and a clump is implanted into each slit. The doctor performing the surgery will attempt to recreate a natural looking hairline along the forehead. Minigrafts, micrografts, or implants of single hair follicles can be used to fill in between larger implant sites and can provide a more natural-looking hairline. The implants will also be arranged so that thick and thin hairs are interspersed and the hair will grow in the same direction.

Another type of hair replacement surgery is called scalp reduction. This involves removing some of the skin from the hairless area and "stretching" some of the nearby hair-covered scalp over the cut-away area.

Health insurance will not pay for hair transplants that are done for cosmetic reasons. Insurance may pay for hair replacement surgery to correct hair loss due to accident, burn, or disease.



The most common hair transplant procedure involves taking small strips of scalp containing hair follicles from the donor area, usually at the sides or back of the head. These strips are then divided into several hundred smaller grafts. The surgeon relocates these grafts containing skin, follicle, and hair to tiny holes in the balding area by using microsurgical instruments or lasers. (Illustration by Electronic Illustrators Group.)

It is important to be realistic about what the final result of a hair transplant will look like. This procedure does not create new hair, it simply redistributes the hair that the patient still has. Some research has been conducted where chest hair has been transplanted to the balding scalp, but this procedure is not widely practiced.

Preparation

It is important to find a respected, well-established, experienced surgeon and discuss the expected results prior to the surgery. The patient may need blood tests to check for bleeding or clotting problems and may be asked not to take **aspirin** products before the surgery. The type of anesthesia used will depend on how extensive the surgery will be and where it will be performed. The patient may be awake during the procedure, but may be given medication to help them relax. A local anesthetic drug which numbs the area will be applied or injected into the skin at the surgery sites.

Aftercare

The area may need to be bandaged overnight. The patient can return to normal activities; however, strenuous activities should be avoided in the first few days after the surgery. On rare occasions, the implants can be “ejected” from the scalp during vigorous **exercise**. There may be some swelling, bruising, **headache**, and discomfort around the graft areas and around the eyes. These symptoms can usually be controlled with a mild **pain** reliever like aspirin. Scabs may form at the graft sites and should not be scraped off. There may be some numbness at the sites, but it will diminish within two to three months.

KEY TERMS

Anesthetic agents—Medication or drugs that can be injected with a needle or rubbed onto an area to make it numb before a surgical procedure. Anesthesia drugs may also be given by mouth, breathed in as a gas, or injected into a vein or muscle to make a patient relaxed or unconscious.

Hair follicle—A tube-like indentation in the skin from which a single hair grows.

Minigraft or micrograft—Transplantation of a small number of hair follicles, as few as one to three hairs, into a transplant site.

Transplantation—Surgically cutting out hair follicles and replanting them in a different spot on the head.

Risks

Although there are rare cases of infection or scarring, the major risk is probably that the grafted area does not look the way the patient expected it to look.

Normal results

The transplanted hair will fall out within a few weeks, however, new hair will start to grow in the graft sites within about three months. A normal rate of hair growth is about 0.25–0.5 in (6–13 mm) per month.

Abnormal results

Major complications as a result of hair transplantation are extremely rare. Occasionally, a patient may

have problems with delayed healing, infection, scarring, or rejection of the graft; but this is uncommon.

Resources

ORGANIZATIONS

American Academy of Cosmetic Surgery. 401 N. Michigan Ave., Chicago, IL 60611-4267. (313) 527-6713.

< <http://www.cosmeticsurgeryonline.com> > .

American Academy of Facial Plastic and Reconstructive Surgery. 1110 Vermont Avenue NW, Suite 220, Washington, DC 20005. (800) 332-3223.

OTHER

“Hair Transplant.” Ienhance. < <http://www.ienhance.com> > .

“Transplants; Flap Surgery; and The Perfect Candidate.” Transplant Network. < <http://www.hair-transplants.net> > .

Altha Roberts Edgren

Hairy cell leukemia

Definition

Hairy cell leukemia is a disease in which a type of white blood cell called the lymphocyte, present in the blood and bone marrow, becomes malignant and proliferates. It is called hairy cell leukemia because the cells have tiny hair-like projections when viewed under the microscope.

Description

Hairy cell leukemia (HCL) is a rare **cancer**. It was first described in 1958 as *leukemic reticuloendotheliosis*, erroneously referring to a red blood cell because researchers were unsure of the cell of origin. It became more easily identifiable in the 1970s. There are approximately 600 new cases diagnosed every year in the United States, making up about 2% of the adult cases of leukemia each year.

HCL is found in cells located in the blood. There are three types of cells found in the blood: the red blood cells that carry oxygen to all the parts of the body; the white blood cells that are responsible for fighting infection and protecting the body from diseases; and the platelets that help in the clotting of blood. Hairy cell leukemia affects a type of white blood cell called the lymphocyte. Lymphocytes are made in the bone marrow, spleen, lymph nodes, and other organs. It specifically affects B-lymphocytes,

which mature in the bone marrow. However, extremely rare variants of HCL have been discovered developing from T-lymphocytes, which mature in the thymus.

When hairy cell leukemia develops, the white blood cells become abnormal both in the way they appear (by acquiring hairy projections) and in the way they act (by proliferating without the normal control mechanisms). Further, the cells tend to accumulate in the spleen, causing it to become enlarged. The cells may also collect in the bone marrow and prevent it from producing normal blood cells. As a result, there may not be enough normal white blood cells in the blood to fight infection.

The median age at which people develop HCL is 52 years. Though it occurs in all ages, HCL more commonly develops in the older population. Men are four times more likely to develop HCL than women. There have been reports of familial aggregation of disease, with higher occurrences in Ashkenazi Jewish men. A potential genetic link is undergoing further investigation.

Causes and symptoms

The cause of hairy cell leukemia is not specifically known. However, exposure to radiation is a known cause of leukemia in general. Familial involvement is another theory, suggesting that there is a genetic component associated with this disease.

HCL is a chronic (slowly progressing) disease, and the patients may not show any symptoms for many years. As the disease advances, the patients may suffer from one or more of the following symptoms:

- weakness
- **fatigue**
- recurrent infections
- **fever**
- anemia
- bruising
- **pain** or discomfort in the abdominal area
- weight loss (uncommon)
- night sweats (uncommon)

Pain and discomfort are caused by an enlarged spleen, which results from the accumulation of the abnormal hairy cells in the spleen. Blood tests may show abnormal counts of all the different types of cells. This happens because the cancerous cells invade the bone marrow as well and prevent it from

producing normal blood cells. Because of the low white cell count in the blood, the patient may have frequent infections. Fever often accompanies the infections. The patient is most susceptible to bacterial infections, but infections of any kind are the major cause of **death**. The low red cell count may cause anemia, fatigue, and weakness, and the low **platelet count** may cause the person to bruise and bleed easily.

Diagnosis

When a patient suffers from the above symptoms, the doctor will palpate the abdomen and may order scans to see if the spleen is enlarged (splenomegaly). An enlarged spleen is present in 80% of patients. An enlarged liver is less common, but can occur.

If the spleen is enlarged, the doctor may order several blood tests. In these tests, the total numbers of each of the different types of blood cells (CBC) are reported. Sixty to eighty percent of patients suffer from pancytopenia, which is a dramatic reduction in the number of red blood cells, white blood cells, and platelets circulating in the blood.

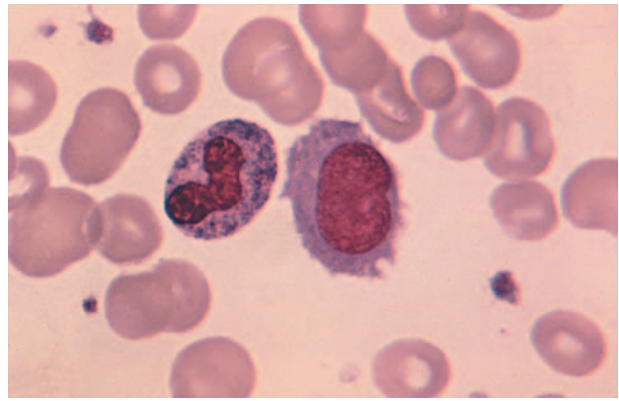
If the blood tests are abnormal, the doctor may order a bone marrow aspiration and biopsy. In order to establish a diagnosis, hairy cells must be present in the bone marrow.

Treatment

When physicians perform blood tests, they will determine the level of hemoglobin (the oxygen-transporting molecule of red blood cells). Serum hemoglobin levels and the size of the spleen, which can be measured on exam and by using an x ray, are proposed criteria for determining the stage of HCL. The following are the three proposed stages and their criteria:

- Stage I: Hemoglobin greater than 12 g/dL (1 g = approximately 0.02 pint and 1 dL = approximately 0.33 ounce) and spleen less than or equal to 10 cm (3.9 inches).
- Stage II: Hemoglobin between 8.5 and 12 g/dL and spleen greater than 10 cm (3.9 inches).
- Stage III: Hemoglobin less than 8.5 g/dL and spleen greater than 10 cm (3.9 inches).

Since there is generally no accepted staging system, another method for evaluating the progression of HCL is to group patients into two categories: untreated HCL and progressive HCL, in which hairy cells are present after therapy has been administered.



A magnified image of white blood cells with "hairy" projections. (Photograph by M. Abbey. Photo Researchers, Inc. Reproduced by permission.)

Some people with hairy cell leukemia have very few or no symptoms at all, and it is reasonable to expect that 10% of patients may not need any treatment. However, if the patient is symptomatic and needs intervention, HCL is especially responsive to treatment.

There are three main courses of treatment: chemotherapy, splenectomy (surgical removal of the spleen), and immunotherapy. Once a patient meets treatment criteria, purine analogues, particularly the drugs, pentostatin and cladribine, are the first-line therapy. Pentostatin is administered at 5mg/m² for two days every other week until total remission is achieved. Patients may experience side effects such as fever, **nausea**, **vomiting**, **photosensitivity**, and keratoconjunctivitis. However, follow-up studies estimate a relapse-free survival rate at 76%. Cladribine (2-CdA) taken at 0.1mg/kg/day for seven days also has an impressive response. Eighty-six percent of patients experience complete remission after treatment, while 16% experience partial remission. Fever is the principal side effect of 2-CdA.

Biological therapy or immunotherapy, where the body's own immune cells are used to fight cancer, is also being investigated in clinical trials for hairy cell leukemia. A substance called interferon that is produced by the white blood cells of the body was the first systemic treatment that showed consistent results in fighting HCL. The FDA approved interferon-alpha (INF-alpha) to fight HCL. The mechanism by which INF-alpha works is not clearly understood. However, it is known that interferon stimulates the body's natural killer cells that are suppressed during HCL. The standard dosage is 2 MU/m² three times a week for 12 months. Side

KEY TERMS

Anemia—A condition in which there is low iron in the blood due to a deficiency of red blood cells.

Bone marrow—The spongy tissue inside the large bones in the body that is responsible for making the red blood cells, white blood cells, and platelets.

Bone marrow aspiration and biopsy—A procedure in which a needle is inserted into the large bones of the hip or spine and a small piece of marrow is removed for microscopic examination.

Immunotherapy—A mode of cancer treatment in which the immune system is stimulated to fight the cancer.

Leukemia—A disease in which the cells that constitute the blood become cancerous or abnormal.

Lymph nodes—Oval-shaped organs that are the size of peas, located throughout the body, and contain clusters of cells called lymphocytes. They filter out and destroy the bacteria, foreign particles, and cancerous cells from the blood.

Malignant—Cells that have the ability to invade locally, cause destruction of surrounding tissue, and travel to other sites in the body.

Keratoconjunctivitis—Inflammation of the conjunctiva and cornea of the eye.

Spleen—An organ that lies next to the stomach. Its function is to remove the worn-out blood cells and foreign materials from the blood stream.

Splenectomy—A surgical procedure that involves the surgical removal of the spleen.

effects include fever, myalgia, malaise, **rashes**, and gastrointestinal complaints.

If the spleen is enlarged, it may be removed in a surgical procedure known as splenectomy. This usually causes a remission of the disease. However, 50% of patients that undergo splenectomy require some type of systemic treatment such as **chemotherapy** or immunotherapy. Splenectomy is not the most widely used course of treatment as it was many years ago. Although the spleen is not an indispensable organ, it is responsible for helping the body fight infection. Therefore, other therapies are preferred in order to salvage the spleen and its functions.

Most patients have excellent prognosis and can expect to live 10 years or longer. The disease may remain silent for years with treatment. Continual

follow-up is necessary to monitor the patient for relapse and determine true cure rates.

Alternative treatment

Many individuals choose to supplement traditional therapy with complementary methods. Often, these methods improve the tolerance of side effects and symptoms as well as enrich the quality of life. The American Cancer Society recommends that patients talk to their doctor to ensure that the methods they are using are safely supplementing traditional therapy. Some complementary treatments include the following:

- yoga
- meditation
- religious practices and prayer
- music therapy
- art therapy
- massage therapy
- aromatherapy

Prevention

Since the cause for the disease is unknown and there are no specific risk factors, there is no known prevention.

Resources

BOOKS

Bast, Robert C. *Cancer Medicine*. Lewiston, NY:B.C. Decker Inc., 2000.

Haskell, Charles M. *Cancer Treatment*. 5th ed. Philadelphia: W.B. Saunders Company, 2001.

ORGANIZATIONS

American Cancer Society. 1599 Clifton Road, N.E. Atlanta, Georgia 30329. (800) 227-2345. < <http://www.cancer.org> >.

Cancer Research Institute (National Headquarters). 681 Fifth Avenue, New York, N.Y. 10022. (800) 992-2623. < <http://www.cancerresearch.org> >.

Hairy Cell Leukemia Research Foundation. 2345 County Farm Lane, Schaumburg, IL 60194. (800) 693-6173.

Leukemia Society of America, Inc. 600 Third Ave, New York, NY 10016. (800) 955-4572.

National Cancer Institute. 9000 Rockville Pike, Building 31, Room 10A16, Bethesda, Maryland, 20892. (800) 422-6237. < <http://wwwicic.nci.nih.gov> >.

Oncolink. University of Pennsylvania Cancer Center. < <http://cancer.med.upenn.edu> >.

OTHER

“Coping With Side Effects.” National Cancer Institute. July 2, 2001. < <http://cancernet.nci.nih.gov/chemotherapy/chemoside.html> >.

NCI/PDQ Patient Statement, “Hairy cell leukemia.” National Cancer Institute, 2001.

Lata Cherath, PhD
Sally C. McFarlane-Parrott

Halitosis see **Bad breath**

Hallucinations

Definition

Hallucinations are false or distorted sensory experiences that appear to be real perceptions. These sensory impressions are generated by the mind rather than by any external stimuli, and may be seen, heard, felt, and even smelled or tasted.

Description

A hallucination occurs when environmental, emotional, or physical factors such as **stress**, medication, extreme **fatigue**, or mental illness cause the mechanism within the brain that helps to distinguish conscious perceptions from internal, memory-based perceptions to misfire. As a result, hallucinations occur during periods of consciousness. They can appear in the form of visions, voices or sounds, tactile feelings (known as haptic hallucinations), smells, or tastes.

Patients suffering from **dementia** and psychotic disorders such as **schizophrenia** frequently experience hallucinations. Hallucinations can also occur in patients who are not mentally ill as a result of stress overload or exhaustion, or may be intentionally induced through the use of drugs, **meditation**, or sensory deprivation. A 1996 report, published in the *British Journal of Psychiatry*, noted that 37% of 4,972 people surveyed experienced hypnagogic hallucinations (hallucinations that occur as a person is falling to sleep). Hypnopomic hallucinations (hallucinations that occur just upon waking) were reported by 12% of the sample.

Causes and symptoms

Common causes of hallucinations include:

- **Drugs.** Hallucinogenics such as ecstasy (3,4-methylenedioxymethamphetamine, or **MDMA**), **LSD** (lysergic acid diethylamide, or acid), mescaline



Ecstasy tablets. (Corbis. Reproduced by permission)

(3,4,5-trimethoxyphenethylamine, or peyote), and psilocybin (4-phosphoryloxy-N, N-dimethyltryptamine, or mushrooms) trigger hallucinations. Other drugs such as **marijuana** and PCP have hallucinatory effects. Certain prescription medications may also cause hallucinations. In addition, drug withdrawal may induce tactile and visual hallucinations; as in an alcoholic suffering from **delirium** tremens (DTs).

- **Stress.** Prolonged or extreme stress can impede thought processes and trigger hallucinations.
- **Sleep deprivation and/or exhaustion.** Physical and emotional exhaustion can induce hallucinations by blurring the line between sleep and wakefulness.
- **Meditation and/or sensory deprivation.** When the brain lacks external stimulation to form perceptions, it may compensate by referencing the memory and form hallucinatory perceptions. This condition is commonly found in blind and deaf individuals.
- **Electrical or neurochemical activity in the brain.** A hallucinatory sensation—usually involving touch—called an aura, often appears before, and gives warning of, a migraine. Also, auras involving smell and touch (tactile) are known to warn of the onset of an epileptic attack.
- **Mental illness.** Up to 75% of schizophrenic patients admitted for treatment report hallucinations.
- **Brain damage or disease.** Lesions or injuries to the brain may alter brain function and produce hallucinations.

Diagnosis

Aside from hypnagogic and hypnopompic hallucinations, more than one event suggests a person should seek evaluation. A general physician, psychologist, or psychiatrist will try to rule out possible organic, environmental, or psychological causes through a detailed medical examination and social history. If a psychological cause such as schizophrenia is suspected, a psychologist will typically conduct an interview with the patient and his family and administer one of several clinical inventories, or tests, to evaluate the mental status of the patient.

Occasionally, people who are in good mental health will experience a hallucination. If hallucinations are infrequent and transitory, and can be accounted for by short-term environmental factors such as sleep deprivation or meditation, no treatment may be necessary. However, if hallucinations are hampering an individual's ability to function, a general physician, psychologist, or psychiatrist should be consulted to pinpoint their source and recommend a treatment plan.

Treatment

Hallucinations that are symptomatic of a mental illness such as schizophrenia should be treated by a psychologist or psychiatrist. Antipsychotic medication such as thioridazine (Mellaril), haloperidol (Haldol), chlorpromazine (Thorazine), clozapine (Clozaril), or risperidone (Risperdal) may be prescribed.

Prognosis

In many cases, chronic hallucinations caused by schizophrenia or some other mental illness can be controlled by medication. If hallucinations persist, psychosocial therapy can be helpful in teaching the patient the coping skills to deal with them. Hallucinations due to sleep deprivation or extreme stress generally stop after the cause is removed.

Resources

ORGANIZATIONS

American Psychological Association (APA). 750 First St. NE, Washington, DC 20002-4242. (202) 336-5700. < <http://www.apa.org> >.

National Alliance for the Mentally Ill (NAMI). Colonial Place Three, 2107 Wilson Blvd., Ste. 300, Arlington, VA 22201-3042. (800) 950-6264. < <http://www.nami.org> >.

Paula Anne Ford-Martin

KEY TERMS

Aura—A subjective sensation or motor phenomenon that precedes and indicates the onset of a neurological episode, such as a migraine or an epileptic seizure.

Hypnagogic hallucination—A hallucination, such as the sensation of falling, that occurs at the onset of sleep.

Hypnopompic hallucination—A hallucination that occurs as a person is waking from sleep.

Sensory deprivation—A situation where an individual finds himself in an environment without sensory cues. Also, (used here) the act of shutting one's senses off to outside sensory stimuli to achieve hallucinatory experiences and/or to observe the psychological results.

Hallucinogen see **Lysergic acid diethylamide**

Hallux valgus see **Bunion**

Haloperidol see **Antipsychotic drugs**

Hammertoe

Definition

Hammertoe is a condition in which the toe is bent in a claw-like position. It can be present in more than one toe but is most common in the second toe.

Description

Hammertoe is described as a deformity in which the toes bend downward with the toe joint usually enlarged. Over time, the joint enlarges and stiffens as it rubs against shoes. Other foot structures involved include the overlying skin and blood vessels and nerves connected to the involved toes.

Causes and symptoms

The shortening of tendons responsible for the control and movement of the affected toe or toes cause hammertoe. Top portions of the toes become callused from the friction produced against the inside of shoes. This common foot problem often results

from improper fit of footwear. This is especially the case with high-heeled shoes placing pressure on the front part of the foot that compresses the smaller toes tightly together. The condition frequently stems from muscle imbalance, and usually leaves the affected individual with impaired balance.

Diagnosis

A thorough medical history and physical exam by a physician is always necessary for the proper diagnosis of hammertoe and other foot conditions. Because the condition involves bony deformity, x rays can help to confirm the diagnosis.

Treatment

Conservative

Wearing proper footwear and stockings with plenty of room in the toe region can provide treatment for hammertoe. Stretching exercises may be helpful in lengthening the excessively tight tendons.

Surgery

In advanced cases, where conservative treatment is unsuccessful, surgery may be recommended. The tendons that attach to the involved toes are located and an incision is made to free the connective tissue to the foot bones. Additional incisions are made so the toes no longer bend in a downward fashion. The middle joints of the affected toes are connected together permanently with surgical hardware such as pins and wire sutures. The incision is then closed with fine sutures. These sutures are removed approximately seven to ten days after surgery.

Alternative treatment

Various soft tissue and joint treatments offered by **chiropractic** and **massage therapy** may be useful to decrease the tightness of the affected structures.

Prognosis

If detected early, hammertoe can be treated non-surgically. If surgery becomes necessary, surgical risks are minimal with the overall outcome providing good results.

Prevention

Wearing comfortable shoes that fit well can prevent many foot ailments. Foot width may increase with



Hammertoe most commonly affects the second toe which, as shown, often develops a corn over the deformity. (Photograph by Dr. H.C. Robinson, Custom Medical Stock Photo. Reproduced by permission.)

age. Feet should always be measured before buying shoes. The upper part of the shoes should be made of a soft, flexible material to match the shape of the foot. Shoes made of leather can reduce the possibility of skin irritations. Soles should provide solid footing and not be slippery. Thick soles lessen pressure when walking on hard surfaces. Low-heeled shoes are more comfortable, safer, and less damaging than high-heeled shoes.

Resources

ORGANIZATIONS

American Orthopedic Foot and Ankle Society. 222 South Prospect, Park Ridge, IL 60068.

American Podiatry Medical Association. 9312 Old Georgetown Road, Bethesda, MD 20814.

Jeffrey P. Larson, RPT

Hand-foot-and-mouth disease

Definition

Hand-foot-and-mouth disease is an infection of young children in which characteristic fluid-filled blisters appear on the hands, feet, and inside the mouth.

Description

Coxsackie viruses belong to a family of viruses called enteroviruses. These viruses live in the gastrointestinal tract, and are therefore present in feces. They can be spread easily from one person to another when poor hygiene allows the virus within the feces to be passed from person to person. After exposure to the virus, development of symptoms takes only four to six days. Hand-foot-and-mouth disease can occur year-round, although the largest number of cases are in summer and fall months.

An outbreak of hand-foot-and-mouth disease occurred in Singapore in 2000, with more than 1,000 diagnosed cases, all in children, resulting in four deaths. A smaller outbreak occurred in Malaysia in 2000. In 1998, a serious outbreak of enterovirus 71 in Taiwan resulted in more than one million cases of hand-foot-and-mouth disease. Of these, there were 405 severe cases and 78 deaths, 71 of which were children younger than five years of age.

Hand-foot-and-mouth should not be confused with foot and mouth disease, which infects cattle but is extremely rare in humans. An outbreak of foot and mouth disease swept through Great Britain and into other parts of Europe and South America in 2001.

Causes and symptoms

Hand-foot-and-mouth disease is very common among young children, and often occurs in clusters of children who are in daycare together. It is spread when poor hand-washing after a diaper change or contact with saliva (drool) allows the virus to be passed from one child to another.

Within about four to six days of acquiring the virus, an infected child may develop a relatively low-grade **fever**, ranging from 99–102 °F (37.2–38.9 °C). Other symptoms include **fatigue**, loss of energy, decreased appetite, and a sore sensation in the mouth that may interfere with feeding. After one to two days, fluid-filled bumps (vesicles) appear on the inside of the mouth, along the surface of the tongue, on the roof of the mouth, and on the insides of the cheeks. These are tiny blisters, about 3–7 mm in diameter. Eventually, they may appear on the palms of the hands and on the soles of the feet. Occasionally, these vesicles may occur in the diaper region.

The vesicles in the mouth cause the majority of discomfort, and the child may refuse to eat or drink due to **pain**. This phase usually lasts for an average of a week. As long as the bumps have clear fluid within them, the disease is at its most contagious. The fluid



A child's foot with pustules on toes, indicating hand-foot-mouth disease. (Custom Medical Stock Photo. Reproduced by permission.)

within the vesicles contains large quantities of the causative viruses. Extra care should be taken to avoid contact with this fluid.

Diagnosis

Diagnosis is made by most practitioners solely on the basis of the unique appearance of blisters of the mouth, hands, and feet, in a child not appearing very ill.

Treatment

There are no treatments available to cure or decrease the duration of the disease. Medications like **acetaminophen** or ibuprofen may be helpful for decreasing pain, and allowing the child to eat and drink. It is important to try to encourage the child to take in adequate amounts of fluids, in the form of ice chips or popsicles if other foods or liquids are too uncomfortable.

Alternative treatment

There are no effective alternative treatments for hand-foot-and-mouth disease.

Prognosis

The prognosis for a child with hand-foot-and-mouth disease is excellent. The child is usually completely better within about a week of the start of the illness.

Prevention

Prevention involves careful attention to hygiene. Thorough, consistent hand-washing practices, and

KEY TERMS

Enteroviruses—Viruses which live in the gastrointestinal tract. Coxsackie viruses, viruses that cause hand-foot-mouth disease, are an enterovirus.

Vesicle—A bump on the skin filled with fluid.

discouraging the sharing of clothes, towels, and stuffed toys are all helpful. Virus continues to be passed in the feces for several weeks after infection, so good hygiene should be practiced long after all signs of infection have passed.

Resources

BOOKS

Morag, Abraham, and Pearay L. Ogra. "Viral Infections." In *Nelson Textbook of Pediatrics*. edited by Richard Behrman, 16th ed. Philadelphia: W.B. Saunders Co., 2000.

PERIODICALS

"New Tricks for an Old Enterovirus." *Infectious Disease Alert* (December 15, 1999): 45.

Rosalyn Carson-DeWitt, MD
Ken R. Wells

Hand-Schüller-Christian syndrome see

Histiocytosis X

Hansen's disease see **Leprosy**

Hantavirus infections

Definition

Hantavirus infection is caused by a group of viruses that can infect humans with two serious illnesses: hemorrhagic **fever** with renal syndrome (HFRS), and Hantavirus pulmonary syndrome (HPS).

Description

Hantaviruses are found without causing symptoms within various species of rodents and are passed to humans by exposure to the urine, feces, or saliva of those infected rodents. Ten different hantaviruses have been identified as important in humans. Each is found in specific geographic regions, and therefore is spread by different rodent carriers. Further, each type

of virus causes a slightly different form of illness in its human hosts:

- Hantaan virus is carried by the striped field mouse, and exists in Korea, China, Eastern Russia, and the Balkans. Hantaan virus causes a severe form of hemorrhagic fever with renal syndrome (HFRS).
- Puumala virus is carried by bank voles, and exists in Scandinavia, western Russia, and Europe. Puumala virus causes a milder form of HFRS, usually termed *nephropathia epidemica*.
- Seoul virus is carried by a type of rat called the Norway rat, and exists worldwide, but causes disease almost exclusively in Asia. Seoul virus causes a form of HFRS that is slightly milder than that caused by Hantaan virus, but results in liver complications.
- Prospect Hill virus is carried by meadow voles and exists in the United States, but has not been found to cause human disease.
- Sin Nombre virus, the most predominant strain in the United States, is carried by the deer mouse. This virus was responsible for severe cases of HPS that occurred in the Southwestern United States in 1993.
- Black Creek Canal virus has been found in Florida. It is predominantly carried by cotton rats.
- New York virus strain has been documented in New York State. The vectors for this virus seem to be deer mice and white-footed mice.
- Bayou virus has been reported in Louisiana and Texas and is carried by the marsh rice rat.
- Blue River virus has been found in Indiana and Oklahoma and seems to be associated with the white-footed mouse.
- Monongahela virus, discovered in 2000, has been found in Pennsylvania and is transmitted by the white-footed mouse.

Causes and symptoms

Hemorrhagic fever with renal syndrome (HFRS)

Hantaviruses that produce forms of hemorrhagic fever with renal syndrome (HFRS) cause a classic group of symptoms, including fever, malfunction of the kidneys, and low **platelet count**. Because platelets are blood cells important in proper clotting, low numbers of circulating platelets can result in spontaneous bleeding, or hemorrhage.

Patients with HFRS have **pain** in the head, abdomen, and lower back, and may report bloodshot eyes and blurry vision. Tiny pinpoint hemorrhages, called petechiae, may appear on the upper body and the soft

palate in the mouth. The patient's face, chest, abdomen, and back often appear flushed and red, as if sunburned.

After about five days, the patient may have a sudden drop in blood pressure; often it drops low enough to cause the clinical syndrome called **shock**. Shock is a state in which blood circulation throughout the body is insufficient to deliver proper quantities of oxygen. Lengthy shock can result in permanent damage to the body's organs, particularly the brain, which is very sensitive to oxygen deprivation.

Around day eight of HFRS, kidney involvement results in multiple derangements of the body chemistry. Simultaneously, the hemorrhagic features of the illness begin to cause spontaneous bleeding, as demonstrated by bloody urine, bloody vomit, and in very serious cases, brain hemorrhages with resulting changes in consciousness.

Day eleven often brings further chemical derangements, with associated confusion, **hallucinations**, seizures, and lung complications. Those who survive this final phase usually begin to turn the corner toward recovery at this time, although recovery takes approximately six weeks.

Hantavirus pulmonary syndrome (HPS)

Hantavirus pulmonary syndrome (HPS) develops in four stages. They are:

- The incubation period. This lasts from one to five weeks from exposure. Here, the patient may exhibit no symptoms.
- The prodrome, or warning signs, stage. Symptoms begin with a fever, muscle aches, **headache**, **dizziness**, and abdominal pain and upset. Sometimes there is **vomiting** and diarrhea.
- The cardiopulmonary stage. The patient slips into this stage rapidly, sometimes within a day or two of initial symptoms; sometimes as long as 10 days later. There is a drop in blood pressure, shock, and leaking of the blood vessels of the lungs, which results in fluid accumulation in the lungs, and subsequent **shortness of breath**. The fluid accumulation can be so rapid and so severe as to put the patient in **respiratory failure** within only a few hours. Some patients experience severe abdominal tenderness.
- The convalescent stage. If the patient survives the respiratory complications of the previous stage, there is a rapid recovery, usually within a day or two. However, abnormal liver and lung functioning may persist for six months.

Diagnosis

Serologic techniques help diagnose a hantavirus infection. The patient's blood is drawn, and the ELISA (enzyme-linked immunosorbent assay) is done in a laboratory to identify the presence of specific immune substances (antibodies)—substances which an individual's body would only produce in response to the hantavirus.

It is very difficult to demonstrate the actual virus in human tissue, or to grow cultures of the virus within the laboratory, so the majority of diagnostic tests use indirect means to demonstrate the presence of the virus.

Treatment

Treatment of hantavirus infections is primarily supportive, because there are no agents available to kill the viruses and interrupt the infection. Broad-spectrum **antibiotics** are given until the diagnosis is confirmed. Supportive care consists of providing treatment in response to the patient's symptoms. Because both HFRS and HPS progress so rapidly, patients must be closely monitored, so that treatment may be started at the first sign of a particular problem. Low blood pressure is treated with medications. Blood transfusions are given for both hemorrhage and shock states. Hemodialysis is used in kidney failure. (Hemodialysis involves mechanically cleansing the blood outside of the body, to replace the kidney's normal function of removing various toxins from the blood.) Rapid respiratory assistance is critical, often requiring intubation.

The anti-viral agent ribavirin has been approved for use in early treatment of hantavirus infections.

Prognosis

The diseases caused by hantaviruses are extraordinarily lethal. About 6–15% of people who contract HFRS have died. Almost half of all people who contract HPS will die. This gives HPS one of the highest fatality rates of any acute viral disease. It is essential that people living in areas where the hantaviruses exist seek quick medical treatment should they begin to develop an illness that might be due to a hantavirus.

Prevention

There are no immunizations currently available against any of the hantaviruses. In 2003, developments in genetic science were helping researchers work on a

KEY TERMS

Hemodialysis—A method of mechanically cleansing the blood outside of the body, in order to remove various substances that would normally be cleared by the kidneys. Hemodialysis is used when an individual is in relative, or complete, kidney failure.

Hemorrhagic—A condition resulting in massive, difficult-to-control bleeding.

Petechiae—Pinpoint size red spots caused by hemorrhaging under the skin.

Platelets—Circulating blood cells that are crucial to the mechanism of clotting.

Prodrome—Early symptoms or warning signs

Pulmonary—Referring to the lungs.

Renal—Referring to the kidneys.

Shock—Shock is a state in which blood circulation is insufficient to deliver adequate oxygen to vital organs.

possible **vaccination** and therapy for several versions of hantavirus, including the Sin Nombre virus that causes HPS. With further work, a gene-based vaccine could become available in the future. However, the only known forms of hantavirus prevention involve rodent control within the community and within individual households. The following is a list of preventive measures:

- Avoiding areas known to be infested by rodents is essential.
- Keeping a clean home and keeping food in rodent-proof containers.
- Disposing of garbage and emptying pet food dishes at night.
- Setting rodent traps around baseboards and in tight places. Disposing of dead animals with gloves and disinfecting the area with bleach.
- Using rodenticide as necessary.
- Sealing any entry holes 0.25 inch wide or wider around foundations with screen, cement, or metal flashing.
- Clearing brush and junk from house foundations.
- Putting metal flashing around house foundations.
- Elevating hay, woodpiles, and refuse containers.

- Airing out all sealed outbuildings or cabins 30 minutes before cleaning for the season.
- When camping, avoiding sleeping on the bare ground. It is advised to sleep on a cot or in a tent with a floor.

Resources

BOOKS

Harper, David R., and Andrea S. Meyer. *Of Mice, Men, and Microbes: Hantavirus*. San Diego: Academic Press, 1999.

PERIODICALS

“DNA Vaccine Protects Against Hantavirus Pulmonary Syndrome.” *Heart Disease Weekly* November 2, 2003: 31.

Jones, Amy. “Setting a Trap for Hantavirus.” *Nursing* September 2000: 20.

Monroe, Martha C., Sergey P. Morzunov, Angela M. Johnson, Michael De. Bowen, et al. “Genetic Diversity and Distribution of Peromyscus-Borne Hantaviruses in North America.” *Nursing* January-February 1999: 75–86.

Naughton, Laurie. “Hantavirus Infection in the United States: Are We Prepared?” *Physician Assistant* May 2000: 33.

Rhodes III, Luther V., Cinnia Huang, Angela J. Sanchez, Stuart T. Nichol, et al. “Hantavirus Pulmonary Syndrome Associated with Monongahela Virus, Pennsylvania.” *Emerging Infectious Diseases* November 2000: 616.

Van Bevern, Pamela A. “Hantavirus Pulmonary Syndrome.” *Clinician Reviews* July 2000: 108.

Janie F. Franz
Teresa G. Odle

Haptoglobin test

Definition

This test is done to help evaluate a person for **hemolytic anemia**.

Purpose

Haptoglobin is a blood protein made by the liver. The haptoglobin levels decrease in hemolytic anemia. Hemolytic **anemias** include a variety of conditions that result in hemolyzed, or burst, red blood cells.

Decreased values can also indicate a slower type of red cell destruction unrelated to anemia. For example, destruction can be caused by mechanical heart

valves or abnormal hemoglobin, such as **sickle cell disease** or **thalassemia**.

Haptoglobin is known as an acute phase reactant. Its level increases during acute conditions such as infection, injury, tissue destruction, some cancers, **burns**, surgery, or trauma. Its purpose is to remove damaged cells and debris and rescue important material such as iron. Haptoglobin levels can be used to monitor the course of these conditions.

Description

Hemoglobin is the protein in the red blood cell that carries oxygen throughout the body. Iron is an essential part of hemoglobin; without iron, hemoglobin can not function. Haptoglobin's main role is to save iron by attaching itself to any hemoglobin released from a red cell.

When red blood cells are destroyed, the hemoglobin is released. Haptoglobin is always present in the blood waiting to bind to released hemoglobin. White blood cells (called macrophages) bring the haptoglobin-hemoglobin complex to the liver, where the haptoglobin and hemoglobin are separated and the iron is recycled.

In hemolytic anemia, so many red cells are destroyed that most of the available haptoglobin is needed to bind the released hemoglobin. The more severe the hemolysis, the less haptoglobin remains in the blood.

Haptoglobin is measured in several different ways. One way is called rate nephelometry. A person's serum is mixed with a substance that will bind to haptoglobin. The amount of bound haptoglobin is measured using a rate nephelometer, which measures the amount of light scattered by the bound haptoglobin. Another way of measuring haptoglobin is to measure it according to how much hemoglobin it can bind.

Preparation

This test requires 5 mL of blood. The person being tested should avoid taking **oral contraceptives** or androgens before this test. A healthcare worker ties a tourniquet on the person's upper arm, locates a vein in the inner elbow region, and inserts a needle into that vein. Vacuum action draws the blood through the needle into an attached tube. Collection of the sample takes only a few minutes.

Aftercare

Discomfort or bruising may occur at the puncture site or the person may feel dizzy or faint. Pressure to the

KEY TERMS

Acute phase reactant—A substance in the blood that increases as a response to an acute condition such as infection, injury, tissue destruction, some cancers, burns, surgery, or trauma.

Haptoglobin—A blood protein made by the liver. Its main role is to save iron by attaching itself to any hemoglobin released from a red cell.

Hemoglobin—The protein in the red blood cell that carries oxygen.

Hemolytic anemia—A variety of conditions that result in hemolyzed, or burst, red blood cells.

puncture site until the bleeding stops reduces bruising. Warm packs to the puncture site relieve discomfort.

Normal results

Normal results vary based on the laboratory and test method used. Haptoglobin is not present in newborns at birth, but develop adult levels by 6 months.

Abnormal results

Decreased haptoglobin levels usually indicates hemolytic anemia. Other causes of red cell destruction also decrease haptoglobin: a blood **transfusion** reaction; mechanical heart valve; abnormally shaped red cells; or abnormal hemoglobin, such as thalassemia or sickle cell anemia.

Haptoglobin levels are low in **liver disease**, because the liver can not manufacture normal amounts of haptoglobin. Low levels may also indicate an inherited lack of haptoglobin, a condition found particularly in African Americans.

Haptoglobin increases as a reaction to illness, trauma, or rheumatoid disease. High haptoglobin values should be followed-up with additional tests. Drugs can also effect haptoglobin levels.

Normal results vary widely from person to person. Unless the level is very high or very low, haptoglobin levels are most valuable when the results of several tests done on different days are compared.

Nancy J. Nordenson

Hardening of the arteries see **Atherosclerosis**

Harelip see **Cleft lip and palate**

Hartnup disease

Definition

Hartnup disease is an inherited nutritional disorder with primary symptoms including a red, scaly rash and sensitivity to sunlight.

Description

Hartnup disease was first identified in the 1950s in the Hartnup family in London. A defect in intestines and kidneys makes it difficult to break down and absorb protein in the diet. This causes a condition very similar to pellagra (niacin deficiency). The condition occurs in about one of every 26,000 live births.

Causes and symptoms

Hartnup disease is an in-born error of metabolism, that is, a condition where certain nutrients cannot be digested and absorbed properly. The condition is passed on genetically in families. It occurs when a person inherits two recessive genes for the disease, one from each parent. People with Hartnup disease are not able to absorb some of the amino acids (the smaller building blocks that make up proteins) in their intestines. One of the amino acids that is not well absorbed is tryptophan, which the body uses to make its own form of niacin.

The majority of people with this disorder do not show any symptoms. About 10–20% of people with Hartnup disease do have symptoms. The most prominent symptom is a red, scaly rash that gets worse when the patient is exposed to sunlight. **Headache, fainting, and diarrhea** may also occur. Mental retardation, cerebral ataxia (muscle weakness), and **delirium** (a confused, agitated, delusional state) are some of the more serious complications that can occur. Short stature has also been noted in some patients. Although this is an inherited disease, the development of symptoms depends on a variety of factors including diet, environment, and other genetic traits controlling amino acid levels in the body. Symptoms can be brought on by exposure to sunlight, fever, drugs, or other stresses. Poor **nutrition** frequently precedes an attack of symptoms. The frequency of attacks usually decreases as the patient gets older.

Diagnosis

The symptoms of this disease suggest a deficiency of a B vitamin called niacin. A detailed diet history can

KEY TERMS

Amino acids—Proteins are made up of organic compounds called amino acids. The human body uses amino acids to build and repair body tissue. The body can make some of its own amino acids from other nutrients in the diet; these are called non-essential amino acids. Essential amino acids are those that cannot be made by the body but must be consumed in the diet. Animal proteins (like meat, eggs, fish, and milk) provide all of the amino acids.

Aminoaciduria—A condition confirmed by laboratory tests where high levels of amino acids are found in the urine.

Pellagra—A condition caused by a dietary deficiency of one of the B vitamins, called niacin.

Tryptophan—An essential amino acid that has to be consumed in the diet because it cannot be manufactured by the body. Tryptophan is converted by the body to niacin, one of the B vitamins.

be used to assess if there is adequate protein and **vitamins** in the diet. The diagnosis of Hartnup disease is confirmed by a laboratory test of the urine which will contain an abnormally high amount of amino acids (aminoaciduria).

Treatment

The vitamin niacin is given as a treatment for Hartnup disease. The typical dosage ranges from 40–200 mg of nicotinamide (a form of niacin) per day to prevent pellagra-like symptoms. Some patients may require dietary supplements of tryptophan.

Eating a healthy, high protein diet can relieve the symptoms and prevent them from recurring.

Prognosis

The prognosis for a healthy life is good once the condition has been identified and treated.

Prevention

Hartnup disease is an inherited condition. Parents may not have the disease themselves, but may pass the genes responsible for it on to their children. Genetic testing can be used to identify carriers of the genes. Symptoms can usually be controlled with a high

protein diet, vitamin supplements of niacin, and by avoiding the stresses that contribute to attacks of symptoms.

Resources

ORGANIZATIONS

National Organization for Rare Disorders. PO Box 8923, New Fairfield, CT 06812-8923. (800) 999-6673. <<http://www.rarediseases.org>>.

NIH/National Institute of Diabetes, Digestive and Kidney Diseases. Building 31, Room 9A04, 31 Center Drive, Bethesda MD 20892-2560. (301) 496-3583.

OTHER

“Hartnup disorder.” OMIM Homepage, Online Mendelian Inheritance in Man. <<http://www.ncbi.nlm.nih.gov/Omim>>.

“Nephrology: Hartnup disease.” Medstudents.com. <<http://www.medstudents.com>>.

Altha Roberts Edgren

Hashimoto’s disease see **Thyroiditis**

Hatha yoga

Definition

Hatha **yoga** is the most widely practiced form of yoga in America. It is the branch of yoga which concentrates on physical health and mental well-being. Hatha yoga uses bodily postures (*asanas*), breathing techniques (*pranayama*), and **meditation** (*dyana*) with the goal of bringing about a sound, healthy body and a clear, peaceful mind. There are nearly 200 hatha yoga postures, with hundreds of variations, which work to make the spine supple and to promote circulation in all the organs, glands, and tissues. Hatha yoga postures also stretch and align the body, promoting balance and flexibility.

Purpose

In a celebrated 1990 study, *Dr. Dean Ornish’s Program for Reversing Heart Disease* (Random House), a cardiologist showed that yoga and meditation combined with a low-fat diet and group support could significantly reduce the blockage of coronary arteries. Other studies have shown yoga’s benefit in reducing stress-related problems such as high blood pressure and cholesterol. Meditation has been adopted by medical schools and clinics as an effective **stress** management technique. Hatha yoga is also used

by physical therapists to improve many injuries and disabilities, as the gentleness and adaptability of yoga make it an excellent **rehabilitation** program.

Yoga has been touted for its ability to reduce problems with such varying conditions as **asthma**, backaches, diabetes, **constipation**, **menopause**, **multiple sclerosis**, **varicose veins**, and **carpal tunnel syndrome**. A vegetarian diet is the dietary goal of yoga, and this change of lifestyle has been shown to significantly increase longevity and reduce heart disease.

Yoga as a daily **exercise** program can improve fitness, strength, and flexibility. People who practice yoga correctly every day report that it can promote high levels of overall health and energy. The mental component of yoga can clarify and discipline the mind, and yoga practitioners say its benefits can permeate all facets of a person’s life and attitude, raising self-esteem and self-understanding.

Description

Origins

Yoga was developed in ancient India as far back as 5,000 years ago; sculptures detailing yoga positions have been found in India which date back to 3000 B.C. Yoga is derived from a Sanskrit word which means “union.” The goal of classical yoga is to bring self-transcendence, or enlightenment, through physical, mental and spiritual health. Many people in the West mistakenly believe yoga to be a religion, but its teachers point out that it is a system of living designed to promote health, peace of mind, and deeper awareness of ourselves. There are several branches of yoga, each of which is a different path and philosophy toward self-improvement. Some of these paths include service to others, pursuit of wisdom, non-violence, devotion to God, and observance of spiritual rituals. Hatha yoga is the path which has physical health and balance as a primary goal, for its practitioners believe that greater mental and spiritual awareness can be brought about with a healthy and pure body.

The origins of hatha yoga have been traced back to the eleventh century A.D. The Sanskrit word *ha* means “sun” and *tha* means “moon,” and thus hatha, or literally sun-moon yoga, strives to balance opposing parts of the physical body, the front and back, left and right, top and bottom. Some yoga masters (*yogis*) claim that hatha yoga was originally developed by enlightened teachers to help people survive during the Age of Kali, or the spiritual dark ages, in which Hindus believe we are now living.

The original philosophers of yoga developed it as an eight-fold path to complete health. These eight steps include moral and ethical considerations (such as honesty, non-aggression, peacefulness, non-stealing, generosity, and sexual propriety), self-discipline (including purity, simplicity, devotion to God, and self-knowledge), posture, breath control, control of desires, concentration, meditation, and happiness. According to yogis, if these steps are followed diligently, a person can reach high levels of health and mental awareness.

As it has subsequently developed, hatha yoga has concentrated mainly on two of the eight paths, breathing and posture. Yogis believe breathing to be the most important metabolic function; we breathe roughly 23,000 times per day and use about 4,500 gallons of air, which increases during exercise. Thus, breathing is extremely important to health, and *prana*, or life-force, is found most abundantly in the air and in the breath. If we are breathing incorrectly, we are hampering our potential for optimal health. *Pranayama*, literally the “science of breathing” or “control of life force,” is the yogic practice of breathing correctly and deeply.

In addition to breathing, hatha yoga utilizes asanas, or physical postures, to bring about flexibility, balance and strength in the body. Each of these postures has a definite form and precise steps for achieving the desired position and for exiting it. These postures, yogis maintain, have been scientifically developed to increase circulation and health in all parts of the body, from the muscular tissues to the glands and internal organs. Yogis claim that although hatha yoga can make the body as strong and fit as any exercise program, its real benefits come about because it is a system of maintenance and balance for the whole body.

Yoga was brought to America in the late 1800s, when Swami Vivekananda, an Indian yogi, presented a lecture on yoga in Chicago. Hatha yoga captured the imagination of the Western mind, because accomplished yogis could demonstrate incredible levels of fitness, flexibility, and control over their bodies and metabolism. Yoga has flourished in the West. Americans have brought to yoga their energy and zest for innovation, which troubles some Indian yogis and encourages others, as new variations and schools of yoga have developed. For instance, power yoga is a recent Americanized version of yoga which takes hatha yoga principles and speeds them up into an extremely rigorous aerobic workout, and many strict hatha yoga teachers oppose this sort of change to their philosophy. Other variations of hatha yoga

in America now include Iyengar, Ashtanga, Kripalu, Integral, Viniyoga, Hidden Language, and Bikram yoga, to name a few. Sivananda yoga was practiced by Lilius Folen, who was responsible for introducing many Americans to yoga through public television.

Iyengar yoga was developed by B.K.S. Iyengar, who is widely accepted as one of the great living yogis. Iyengar uses classical hatha yoga asanas and breathing techniques, but emphasizes great precision and strict form in the poses, and uses many variations on a few postures. Iyengar allows the use of props such as belts, ropes, chairs, and blocks to enable students to get into postures they otherwise couldn't. In this respect, Iyengar yoga is good for physical therapy because it assists in the manipulation of inflexible or injured areas.

Ashtanga yoga, made popular by yogi K. Patabhi Jois, also uses hatha yoga asanas, but places an emphasis on the sequences in which these postures are performed. Ashtanga routines often unfold like long dances with many positions done quickly one after the other. Ashtanga is thus a rigorous form of hatha yoga, and sometimes can resemble a difficult aerobic workout. Ashtanga teachers claim that this form of yoga uses body heat, sweating, and deep breathing to purify the body.

Kripalu yoga uses hatha yoga positions but emphasizes the mental and emotional components of each asana. Its teachers believe that tension and long-held emotional problems can be released from the body by a deep and meditative approach to the yoga positions. Integral yoga seeks to combine all the paths of yoga, and is generally more meditative than physical, emphasizing spirituality and awareness in everyday life. Viniyoga tries to adapt hatha yoga techniques to each individual body and medical problem. Hidden Language yoga was developed by Swami Sivananda Radha, a Western man influenced by Jungian psychology. It emphasizes the symbolic and psychological parts of yoga postures and techniques. Its students are encouraged to write journals and participate in group discussions as part of their practice. Bikram yoga has become very popular in the late 1990s, as its popular teacher, Bikram Choudhury, began teaching in Beverly Hills and has been endorsed by many famous celebrities. Bikram yoga uses the repetition of 26 specific poses and two breathing techniques to stretch and tone the whole body.

A hatha yoga routine consists of a series of physical postures and breathing techniques. Routines can take anywhere from 20 minutes to two hours, depending on the needs and ability of the practitioner. Yoga should

always be adapted to one's state of health; that is, a shorter and easier routine should be used when a person is fatigued. Yoga is ideally practiced at the same time every day, to encourage the discipline of the practice. It can be done at any time of day; some prefer it in the morning as a wake-up routine, while others like to wind down and de-stress with yoga at the end of the day.

Yoga asanas consist of three basic movements: backward bends, forward bends, and twisting movements. These postures are always balanced; a back bend should be followed with a forward bend, and a leftward movement should be followed by one to the right. Diaphragm breathing is important during the poses, where the breath begins at the bottom of the lungs. The stomach should move outward with the inhalation and relax inward during exhalation. The breath should be through the nose at all times during hatha asanas. Typically, one inhales during backward bends and exhales during forward bending movements.

The mental component in yoga is as important as the physical movements. Yoga is not a competitive sport, but a means to self-awareness and self-improvement. An attitude of attention, care, and non-criticism is important; limitations should be acknowledged and calmly improved. Patience is important, and yoga stretches should be slow and worked up to gradually. The body should be worked with, and never against, and a person should never overexert. A yoga stretch should be done only so far as proper form and alignment of the whole body can be maintained. Some yoga stretches can be uncomfortable for beginners, and part of yoga is learning to distinguish between sensations that are beneficial and those that can signal potential injury. A good rule is that positions should be stopped when there is sharp **pain** in the joints, muscles, or tendons.

Preparations

All that is needed to perform hatha yoga is a flat floor and adequate space for stretching out. A well-ventilated space is preferable, for facilitating proper breathing technique. Yoga mats are available which provide non-slip surfaces for standing poses. Loose, comfortable clothing should be worn. Yoga should be done on an empty stomach; a general rule is to wait three hours after a meal.

Yoga is an exercise that can be done anywhere and requires no special equipment. Yoga uses only gravity and the body itself as resistance, so it is a low-impact activity excellent for those who don't do well with other types of exercise. The mental component of yoga can appeal to those who get bored easily with

KEY TERMS

Asana—Yoga posture or stance.

Diaphragm breathing—Method of deep breathing using the entire lungs.

Dyana—Yoga meditation.

Meditation—Technique of mental relaxation.

Prana—Yoga term for life-enhancing nutrient found in air, food and water.

Pranayama—Yoga method of breathing.

exercise. By the same token, yoga can be a good stress management tool for those who prefer movement to sitting meditation.

Precautions

As with any exercise program, people should check with their doctors before starting yoga practice for the first time. Those with medical conditions, injuries or spinal problems should find a yoga teacher familiar with their conditions before beginning yoga. Pregnant women, particularly after the third month of **pregnancy**, should only perform a few yoga positions with the supervision of an experienced teacher. Some yoga asanas can be very difficult, and potentially injurious, for beginners, so teachers should always be consulted as preparation for advanced yoga positions. Certain yoga positions should not be performed by those with fevers, or during menstruation.

Side effects

Those just beginning hatha yoga programs often report **fatigue** and soreness throughout the body, as yoga stretches and exercises muscles and tendons which are often long-neglected. Some yogic breathing and meditation techniques can be difficult for beginners and can cause **dizziness** or disorientation; these are best performed under the guidance of a teacher.

Resources

BOOKS

Feuerstein, Georg. *Yoga for Dummies*. New York: IDG Books, 1999.

PERIODICALS

Yoga International Magazine. R.R. 1 Box 407, Honesdale, PA 18431. <<http://www.yimag.com>>.

Yoga Journal. P.O. Box 469088, Escondido, CA 92046. <<http://www.yogajournal.com>>.

ORGANIZATIONS

International Association of Yoga Therapists (IAYT). 4150 Tivoli Ave., Los Angeles, CA 90066.

Douglas Dupler, MA

Haverhill fever see **Rat-bite fever**

Hay fever see **Allergic rhinitis**

HBF test see **Fetal hemoglobin test**

HCG see **Infertility drugs**

Head and neck cancer

Definition

The term head and neck cancers refers to a group of cancers found in the head and neck region. This includes tumors found in:

- The oral cavity (mouth). The lips, the tongue, the teeth, the gums, the lining inside the lips and cheeks, the floor of the mouth (under the tongue), the roof of the mouth and the small area behind the wisdom teeth are all included in the oral cavity.
- The oropharynx (which includes the back one-third of the tongue, the back of the throat and the tonsils).
- Nasopharynx (which includes the area behind the nose).
- Hypopharynx (lower part of the throat).
- The larynx (voice box, located in front of the neck, in the region of the Adam's apple). In the larynx, the **cancer** can occur in any of the three regions: the glottis (where the vocal cords are); the supraglottis (the area above the glottis); and the subglottis (the area that connects the glottis to the windpipe).

The most frequently occurring cancers of the head and neck area are oral cancers and laryngeal cancers. Almost half of all the head and neck cancers occur in the oral cavity, and a third of the cancers are found in the larynx. By definition, the term "head and neck cancers" usually excludes tumors that occur in the brain.

Description

Head and neck cancers involve the respiratory tract and the digestive tract; and they interfere with the functions of eating and breathing. Laryngeal cancers affect speech. Loss of any of these functions is

significant. Hence, early detection and appropriate treatment of head and neck cancers is of utmost importance.

Roughly 10% of all cancers are related to the head and the neck. It is estimated that more than 55,000 Americans will develop cancer of the head and neck in 1998, and nearly 13,000 will die from the disease. The American Cancer Society estimates that in 1998, approximately, 11,100 new cases of **laryngeal cancer** alone will be diagnosed and 4,300 people will die of this disease. Oral cancer is the sixth most common cancer in the United States. Approximately 40,000 new cases are diagnosed each year and it causes at least 8,000 deaths. Among the major cancers, the survival rate for head and neck cancers is one of the poorest. Less than 50% of the patients survive five years or more after initial diagnosis. This is because the early signs of head and neck cancers are frequently ignored. Hence, when it is first diagnosed, it is often in an advanced stage and not very amenable to treatment.

The risk for both oral cancer and laryngeal cancer seems to increase with age. Most of the cases occur in individuals over 40 years of age, the average age at diagnosis being 60. While oral cancer strikes men twice as often as it does women, laryngeal cancer is four times more common in men than in women. Both diseases are more common in black Americans than among whites.

Causes and symptoms

Although the exact cause for these cancers is unknown, tobacco is regarded as the single greatest risk factor: 75–80% of the oral and laryngeal cancer cases occur among smokers. Heavy alcohol use has also been included as a risk factor. A combination of tobacco and alcohol use increases the risk for oral cancer by 6–15 times more than for users of either substance alone. In rare cases, irritation to the lining of the mouth, due to jagged teeth or ill-fitting dentures, has been known to cause oral cancer. Exposure to asbestos appears to increase the risk of developing laryngeal cancer.

In the case of lip cancer, just like skin cancer, exposure to sun over a prolonged period has been shown to increase the risk. In the Southeast Asian countries (India and Sri Lanka), chewing of betel nut has been associated with cancer of the lining of the cheek. An increased incidence of nasal cavity cancer has been observed among furniture workers, probably due to the inhalation of wood dust. A virus (Epstein-Barr) has been shown to cause nasopharyngeal cancer.

Head and neck cancers are one of the easiest to detect. The early signs can be both seen and felt. The signs and symptoms depend on the location of the cancer:

- Mouth and oral cavity: a sore that does not heal within two weeks, unusual bleeding from the teeth or gums, a white or red patch in the mouth, a lump or thickening in the mouth, throat, or tongue.
- Larynx: persistent hoarseness or **sore throat**, difficulty breathing, or **pain**.
- Hypopharynx and oropharynx: difficulty in swallowing or chewing food, ear pain.
- Nose, sinuses, and nasopharyngeal cavity: pain, bloody discharges from the nose, blocked nose, and frequent sinus infections that do not respond to standard **antibiotics**.

When detected early and treated appropriately, head and neck cancers have an excellent chance of being cured completely.

Diagnosis

Specific diagnostic tests used depend on the location of the cancer. The standard tests are:

Physical examination

The first step in diagnosis is a complete and thorough examination of the oral and nasal cavity, using mirrors and other visual aids. The tongue and the back of the throat are examined as well. Any suspicious looking lumps or lesions are examined with fingers (palpation). In order to look inside the larynx, the doctor may sometimes perform a procedure known as **laryngoscopy**. In indirect laryngoscopy, the doctor looks down the throat with a small, long handled mirror. Sometimes the doctor inserts a lighted tube (laryngoscope or a fiberoptic scope) through the patient's nose or mouth. As the tube goes down the throat, the doctor can observe areas that cannot be seen by a simple mirror. This procedure is called a direct laryngoscopy. Sometimes patients may be given a mild sedative to help them relax, and a local anesthetic to ease any discomfort.

Blood tests

The doctor may order blood or other immunological tests. These tests are aimed at detecting antibodies to the Epstein-Barr virus, which has been known to cause cancer of the nasopharynx.

Imaging tests

X rays of the mouth, the sinuses, the skull, and the chest region may be required. A computed tomography scan (CT scan), a procedure in which a computer takes a series of x-ray pictures of areas inside the body, may be done. Ultrasonograms (images generated using sound waves) or an MRI (**magnetic resonance imaging**, a procedure in which a picture is created using magnets linked to a computer), are alternate procedures which a doctor may have done to get detailed pictures of the areas inside the body.

Biopsy

When a sore does not heal or a suspicious patch or lump is seen in the mouth, larynx, nasopharynx, or throat, a biopsy may be performed to rule out the possibility of cancer. The biopsy is the most definitive diagnostic tool for detecting the cancer. If cancerous cells are detected in the biopsied sample, the doctor may perform more extensive tests in order to find whether, and to where, the cancer may have spread.

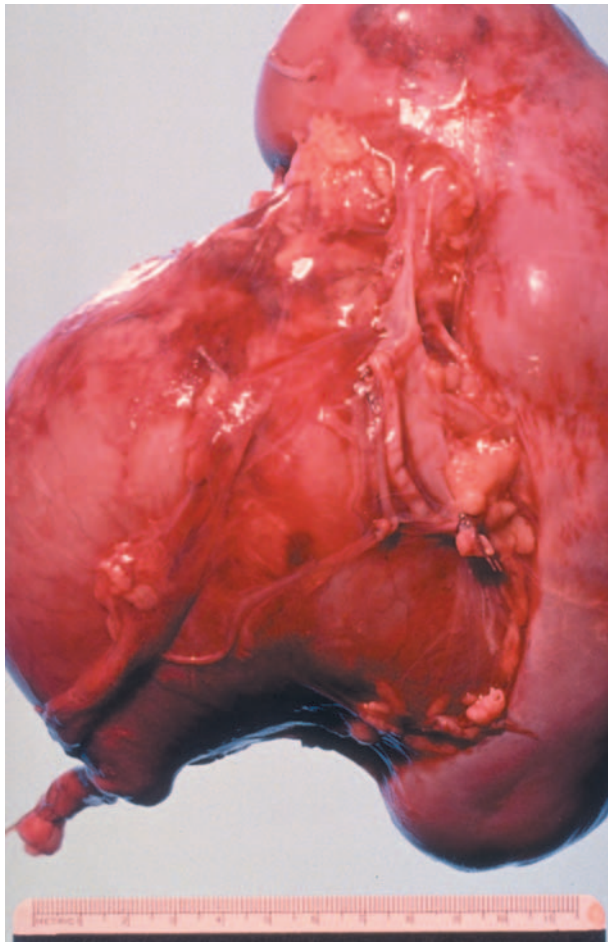
Treatment

The cancers can be treated successfully if diagnosed early. The choice of treatment depends on the size of the tumor, its location, and whether it has spread to other parts of the body.

In the case of lip and mouth cancers, sometimes surgery is performed to remove the cancer. Radiation therapy, which destroys the cancerous cells, is also one of the primary modes of treatment, and may be used alone or in combination with surgery. If lip surgery is drastic, **rehabilitation** cosmetic or reconstructive surgery may have to be considered.

Cancers of the nasal cavity are often diagnosed late because they have no specific symptoms in their early stages, or the symptoms may just resemble chronic **sinusitis**. Hence, treatment is often complex, involving a combination of radiotherapy and surgery. Surgery is generally recommended for small tumors. If the cancer cannot be removed by surgery, radiotherapy is used alone.

Treatment of oropharynx cancers (cancers that are either in the back of the tongue, the throat, or the tonsils) generally involves **radiation therapy** and/or surgery. After aggressive surgery and radiation, rehabilitation is often necessary and is an essential part of the treatment. The patient may experience difficulties with swallowing, chewing, and speech and may require a team of health care workers, including speech therapists, prosthodontists, occupational therapists, etc.



A specimen of a squamous cell carcinoma of the tongue and jaw. (Custom Medical Stock Photo. Reproduced by permission.)

Cancers of the nasopharynx are different from the other head and neck cancers in that there does not appear to be any association between alcohol and tobacco use and the development of the cancer. In addition, the incidence is seen primarily in two age groups: young adults and 50–70 year-olds. The Epstein-Barr virus has been implicated as the causative agent in most patients. While 80–90% of small tumors are curable by radiation therapy, advanced tumors that have spread to the bone and cranial nerves are difficult to control. Surgery is not very helpful and, hence, is rarely attempted. Radiation remains the only treatment of choice to treat the cancer that has metastasized (traveled) to the lymph nodes in the neck.

In the case of cancer of the larynx, radiotherapy is the first choice to treat small lesions. This is done in an attempt to preserve the voice. If the cancer recurs later, surgery may be attempted. If the cancer is limited to one of the two vocal cords, laser excision surgery is

used. In order to treat advanced cancers, a combination of surgery and radiation therapy is often used. Because the chances of a cure in the case of advanced laryngeal cancers are rather low with current therapies, the patient may be advised to participate in clinical trials so they may get access to new experimental drugs and procedures, such as chemotherapy, that are being evaluated.

When only part of the larynx is removed, a relatively slight change in the voice may occur—the patient may sound slightly hoarse. However, in a total **laryngectomy**, the entire voice box is removed. The patients then have to re-learn to speak using different approaches, such as esophageal speech, tracheo-esophageal (TE) speech, or by means of an artificial larynx.

In esophageal speech, the patients are taught how to create a new type of voice by forcing air through the esophagus (food pipe) into the mouth. This method has a high success rate of approximately 65% and patients are even able to go back to jobs that require a high level of verbal communication, such as telephone operators and salespersons.

In the second approach, TE speech, a small opening, called a **fistula**, is created surgically between the trachea (breathing tube to the lungs) and the esophagus (tube into the stomach) to carry air into the throat. A small tube, known as the “voice prosthesis,” is placed in the opening of the fistula to keep it open and to prevent food and liquid from going down into the trachea. In order to talk, the stoma (or the opening made at the base of the neck) must be covered with one’s thumb during exhalation. As the air is forced out from the trachea into the esophagus, it vibrates the walls of the esophagus. This produces a sound that is then modified by the lips and tongue to produce normal sounding speech.

In the third approach, an artificial larynx, a battery driven vibrator, is placed on the outside of the throat. Sound is created as air passes through the stoma (opening made at the base of the neck) and the mouth forms words.

Prognosis

Oral Cavity

With early detection and immediate treatment, survival rates can be dramatically improved. For lip and oral cancer, if detected at its early stages, almost 80% of the patients survive five years or more. However, when diagnosed at the advanced stages, the five year survival rate drops to a mere 18%.

Nose and sinuses

Cancers of the nasal cavity often go undetected until they reach an advanced stage. If diagnosed at the early stages, the five-year survival rates are 60–70%. However, if cancers are more advanced, only 10–30% of the patients survive five years or more.

Oropharynx

In cancer of the oropharynx, 60–80% of the patients survive five years or more if the cancer is detected in the early stages. As the cancer advances, the survival rate drops to 15–30%.

Nasopharynx

Patients who are diagnosed with early stage cancers that have originated in the nasopharynx have an excellent chance of a complete cure (almost 95%). Unfortunately, most of the time, the patients are in an advanced stage at the time of initial diagnosis. With the new **chemotherapy** drugs, the five year survival rate has improved and 5–40% of the patients survive five years or longer.

Larynx

Small cancers of the larynx have an excellent five-year survival rate of 75–95%. However, as with most of the head and neck cancers, the survival rates drop dramatically as the cancer advances. Only 15–25% of the patients survive five years or more after being initially diagnosed with advanced laryngeal cancer.

Prevention

Refraining from the use of all tobacco products (cigarettes, cigars, pipe tobacco, chewing tobacco), consuming alcohol in moderation, and practicing good oral hygiene are some of the measures that one can take to prevent head and neck cancers. Since there is an association between excessive exposure to the sun and lip cancer, people who spend a lot of time outdoors in the sun should protect themselves from the sun's harmful rays. Regular physical examinations, or mouth examination by the patient himself, or by the patient's doctor or dentist, can help detect oral cancer in its very early stages.

Since working with asbestos has been shown to increase one's risk of getting cancer of the larynx, asbestos workers should follow safety rules to avoid inhaling asbestos fibers. Also, malnutrition and vitamin deficiencies have been shown to have some association with an increased incidence of head and neck

KEY TERMS

Biopsy—The surgical removal and microscopic examination of living tissue for diagnostic purposes.

Chemotherapy—Treatment of cancer with synthetic drugs that destroy the tumor either by inhibiting the growth of the cancerous cells or by killing the cancer cells.

Clinical trials—Highly regulated and carefully controlled patient studies, where either new drugs to treat cancer or novel methods of treatment are investigated.

Computerized tomography scan (CT scan)—A medical procedure where a series of X-rays are taken and put together by a computer in order to form detailed pictures of areas inside the body.

Laryngoscopy—A medical procedure that uses flexible, lighted, narrow tubes inserted through the mouth or nose to examine the larynx and other areas deep inside the neck.

Magnetic resonance imaging (MRI)—A medical procedure used for diagnostic purposes where pictures of areas inside the body can be created using a magnet linked to a computer.

Radiation therapy—Treatment using high energy radiation from x-ray machines, cobalt, radium, or other sources.

Stoma—When the entire larynx must be surgically removed, an opening is surgically created in the neck so that the windpipe can be brought out to the neck. This opening is called the stoma.

Ultrasonogram—A procedure where high-frequency sound waves that cannot be heard by human ears are bounced off internal organs and tissues. These sound waves produce a pattern of echoes which are then used by the computer to create sonograms, or pictures of areas inside the body.

X rays—High energy radiation used in high doses, either to diagnose or treat disease.

cancers. The American Cancer Society, therefore, recommends eating a healthy diet, consisting of at least five servings of fruits and vegetables every day, and six servings of food from other plant sources such as cereals, breads, grain products, rice, pasta and beans. Reducing one's intake of high-fat food from animal sources is advised.

Resources

ORGANIZATIONS

American Association of Oral and Maxillofacial Surgeons.
9700 West Bryn Mawr Ave., Rosemont, IL 60018-5701.
(847) 678-6200. <<http://www.aaoms.org>>.

International Association of Laryngectomees (IAL). 7440
North Shadeland Ave., Suite 100, Indianapolis, IN
46250. <<http://www.larynxlink.com/>>.

National Cancer Institute. Building 31, Room 10A31, 31
Center Drive, MSC 2580, Bethesda, MD 20892-2580.
(800) 422-6237. <<http://www.nci.nih.gov>>.

National Oral Health Information ClearingHouse; 1 NOHIC
Way, Bethesda, MD 20892-3500. (301)-402-7364.

Oral Health Education Foundation, Inc. 5865 Colonist Drive,
P.O. Box 396, Fairburn, GA 30213. (770) 969-7400.

Lata Cherath, PhD

Head injury

Definition

Injury to the head may damage the scalp, skull or brain. The most important consequence of head trauma is traumatic brain injury. Head injury may occur either as a closed head injury, such as the head hitting a car's windshield, or as a penetrating head injury, as when a bullet pierces the skull. Both may cause damage that ranges from mild to profound. Very severe injury can be fatal because of profound brain damage.

Description

External trauma to the head is capable of damaging the brain, even if there is no external evidence of damage. More serious injuries can cause skull fracture, **blood clots** between the skull and the brain, or bruising and tearing of the brain tissue itself.

Injuries to the head can be caused by traffic accidents, sports injuries, falls, workplace accidents, assaults, or bullets. Most people have had some type of head injury at least once in their lives, but rarely do they require a hospital visit.

However, each year about two million people suffer from a more serious head injury, and up to 750,000 of them are severe enough to require hospitalization. Brain injury is most likely to occur in males between ages 15 and 24, usually as a result of car and motorcycle accidents. About 70% of all accidental deaths

are due to head injuries, as are most of the disabilities that occur after trauma.

A person who has had a head injury and who is experiencing the following symptoms should seek medical care immediately:

- serious bleeding from the head or face
- loss of consciousness, however brief
- confusion and lethargy
- lack of pulse or breathing
- clear fluid drainage from the nose or ear

Causes and symptoms

A head injury may cause damage both from the direct physical injury to the brain and from secondary factors, such as lack of oxygen, brain swelling, and disturbance of blood flow. Both closed and penetrating head injuries can cause swirling movements throughout the brain, tearing nerve fibers and causing widespread bleeding or a blood clot in or around the brain. Swelling may raise pressure within the skull (intracranial pressure) and may block the flow of oxygen to the brain.

Head trauma may cause a **concussion**, in which there is a brief loss of consciousness without visible structural damage to the brain. In addition to loss of consciousness, initial symptoms of brain injury may include:

- memory loss and confusion
- **vomiting**
- **dizziness**
- partial **paralysis** or **numbness**
- shock
- anxiety

After a head injury, there may be a period of impaired consciousness followed by a period of confusion and impaired memory with disorientation and a breakdown in the ability to store and retrieve new information. Others experience temporary amnesia following head injury that begins with memory loss over a period of weeks, months, or years before the injury (retrograde **amnesia**). As the patient recovers, memory slowly returns. Post-traumatic amnesia refers to loss of memory for events during and after the accident.

Epilepsy occurs in 2–5% of those who have had a head injury; it is much more common in people who have had severe or penetrating injuries. Most cases of epilepsy appear right after the accident or within the

first year, and become less likely with increased time following the accident.

Closed head injury

Closed head injury refers to brain injury without any penetrating injury to the brain. It may be the result of a direct blow to the head; of the moving head being rapidly stopped, such as when a person's head hits a windshield in a car accident; or by the sudden deceleration of the head without its striking another object. The kind of injury the brain receives in a closed head injury is determined by whether or not the head was unrestrained upon impact and the direction, force, and velocity of the blow. If the head is resting on impact, the maximum damage will be found at the impact site. A moving head will cause a "contrecoup injury" where the brain damage occurs on the side opposite the point of impact, as a result of the brain slamming into that side of the skull. A closed head injury also may occur without the head being struck, such as when a person experiences **whiplash**. This type of injury occurs because the brain is of a different density than the skull, and can be injured when delicate brain tissues hit against the rough, jagged inner surface of the skull.

Penetrating head injury

If the skull is fractured, bone fragments may be driven into the brain. Any object that penetrates the skull may implant foreign material and dirt into the brain, leading to an infection.

Skull fracture

A skull fracture is a medical emergency that must be treated promptly to prevent possible brain damage. Such an injury may be obvious if blood or bone fragments are visible, but it's possible for a fracture to have occurred without any apparent damage. A skull fracture should be suspected if there is:

- blood or clear fluid leaking from the nose or ears
- unequal pupil size
- bruises or discoloration around the eyes or behind the ears
- swelling or depression of part of the head

Intracranial hemorrhage

Bleeding (hemorrhage) inside the skull may accompany a head injury and cause additional damage to the brain. A blood clot (hematoma) may occur if a blood vessel between the skull and the brain ruptures; when the blood leaks out and forms a clot, it

can press against brain tissue, causing symptoms from a few hours to a few weeks after the injury. If the clot is located between the bones of the skull and the covering of the brain (dura), it is called an **epidural hematoma**. If the clot is between the dura and the brain tissue itself, the condition is called a **subdural hematoma**. In other cases, bleeding may occur deeper inside the brain. This condition is called **intracerebral hemorrhage** or **intracerebral contusion** (from the word for bruising).

In any case, if the blood flow is not stopped, it can lead to unconsciousness and **death**. The symptoms of bleeding within the skull include:

- nausea and vomiting
- **headache**
- loss of consciousness
- unequal pupil size
- lethargy

Postconcussion syndrome

If the head injury is mild, there may be no symptoms other than a slight headache. There also may be confusion, dizziness, and blurred vision. While the head injury may seem to have been quite mild, in many cases symptoms persist for days or weeks. Up to 60% of patients who sustain a mild brain injury continue to experience a range of symptoms called "postconcussion syndrome," as long as six months or a year after the injury.

The symptoms of postconcussion syndrome can result in a puzzling interplay of behavioral, cognitive, and emotional complaints that can be difficult to diagnose, including:

- headache
- dizziness
- mental confusion
- behavior changes
- memory loss
- cognitive deficits
- depression
- emotional outbursts

Diagnosis

The extent of damage in a severe head injury can be assessed with computed tomography (CT) scan, **magnetic resonance imaging (MRI)**, **positron emission tomography (PET)** scans, electroencephalograms



A three-dimensional computed tomography (CT) scan of a human skull showing a depressed skull fracture above the right eye. (Custom Medical Stock Photo. Reproduced by permission.)

(EEG), and routine neurological and neuropsychological evaluations.

Doctors use the Glasgow **Coma** Scale to evaluate the extent of brain damage based on observing a patient's ability to open his or her eyes, respond verbally, and respond to stimulation by moving (motor response). Patients can score from three to 15 points on this scale. People who score below eight when they are admitted usually have suffered a severe brain injury and will need rehabilitative therapy as they recover. In general, higher scores on the Glasgow Coma Scale indicate less severe brain injury and a better prognosis for recovery.

Patients with a mild head injury who experience symptoms are advised to seek out the care of a specialist; unless a family physician is thoroughly familiar with medical literature in this newly emerging area,

experts warn that there is a good chance that patient complaints after a mild head injury will be downplayed or dismissed. In the case of mild head injury or postconcussion syndrome, CT and MRI scans, electroencephalograms (EEG), and routine neurological evaluations all may be normal because the damage is so subtle. In many cases, these tests can't detect the microscopic damage that occurs when fibers are stretched in a mild, diffuse injury. In this type of injury, the axons lose some of their covering and become less efficient. This mild injury to the white matter reduces the quality of communication between different parts of the brain. A PET scan, which evaluates cerebral blood flow and brain metabolism, may be of help in diagnosing mild head injury.

Patients with continuing symptoms after a mild head injury should call a local chapter of a head-injury foundation that can refer patients to the best nearby expert.

Treatment

If a concussion, bleeding inside the skull, or skull fracture is suspected, the patient should be kept quiet in a darkened room, with head and shoulders raised slightly on pillow or blanket.

After initial emergency treatment, a team of specialists may be needed to evaluate and treat the problems that result. A penetrating wound may require surgery. Those with severe injuries or with a deteriorating level of consciousness may be kept hospitalized for observation. If there is bleeding inside the skull, the blood may need to be surgically drained; if a clot has formed, it may need to be removed. Severe skull **fractures** also require surgery. Supportive care and specific treatments may be required if the patient experiences further complications. People who experience seizures, for example, may be given **anticonvulsant drugs**, and people who develop fluid on the brain (**hydrocephalus**) may have a shunt inserted to drain the fluid.

In the event of long-term disability as a result of head injury, there are a variety of treatment programs available, including long-term **rehabilitation**, coma treatment centers, transitional living programs, behavior management programs, life-long residential or day treatment programs and independent living programs.

Prognosis

Prompt, proper diagnosis and treatment can help alleviate some of the problems after a head injury.

KEY TERMS

Computed tomography scan (CT)—A diagnostic technique in which the combined use of a computer and x rays produce clear cross-sectional images of tissue. It provides clearer, more detailed information than x rays alone.

Electroencephalogram (EEG)—A record of the tiny electrical impulses produced by the brain's activity. By measuring characteristic wave patterns, the EEG can help diagnose certain conditions of the brain.

Magnetic resonance imaging (MRI)—A diagnostic technique that provides high quality cross-sectional images of organs within the body without x rays or other radiation.

Positron emission tomography (PET) scan—A computerized diagnostic technique that uses radioactive substances to examine structures of the body. When used to assess the brain, it produces a three-dimensional image that reflects the metabolic and chemical activity of the brain.

However, it usually is difficult to predict the outcome of a brain injury in the first few hours or days; a patient's prognosis may not be known for many months or even years.

The outlook for someone with a minor head injury generally is good, although recovery may be delayed and symptoms such as headache, dizziness, and cognitive problems can persist for up to a year or longer after an accident. This can limit a person's ability to work and cause strain in personal relationships.

Serious head injuries can be devastating, producing permanent mental and physical disability. Epileptic seizures may occur after a severe head injury, especially a penetrating brain injury, a severe skull fracture, or a serious brain hemorrhage. Recovery from a severe head injury can be very slow, and it may take five years or longer to heal completely. Risk factors associated with an increased likelihood of memory problems or seizures after head injury include age, length and depth of coma, duration of post-traumatic and retrograde amnesia, presence of focal brain injuries, and initial Glasgow Coma Scale score.

As researchers learn more about the long-term effects of head injuries, they have begun to uncover links to later conditions. A 2003 report found that

mild brain injury during childhood could speed up expression of **schizophrenia** in those who were already likely to get the disorder because of genetics. Those with a history of a childhood brain injury, even a minor one, were more likely to get familial schizophrenia than a sibling and to have earlier onset. Another study in 2003 found that people who had a history of a severe head injury were four times more likely to develop Parkinson's disease than the average population. Those requiring hospitalization for their head injuries were 11 times as likely. The risk did not increase for people receiving mild head injuries.

Prevention

Many severe head injuries could be prevented by wearing protective helmets during certain sports, or when riding a bike or motorcycle. Seat belts and airbags can prevent many head injuries that result from car accidents. Appropriate protective headgear always should be worn on the job where head injuries are a possibility.

Resources

PERIODICALS

"Childhood Head Injury Tied to Later Schizophrenia." *The Brown University Child and Adolescent Behavior Letter* June 2003: 5.

"Link to Head Injury Found." *Pain & Central Nervous System Week* June 9, 2003: 3.

ORGANIZATIONS

American Epilepsy Society. 342 North Main Street, West Hartford, CT 06117-2507. (860) 586-7505. <<http://www.aesnet.org>>.

Brain Injury Association. 1776 Massachusetts Ave. NW, Ste. 100, Washington, DC 20036. (800) 444-6443.

Family Caregiver Alliance. 425 Bush St., Ste. 500, San Francisco, CA 94108. (800) 445-8106. <<http://www.caregiver.org>>.

Head Injury Hotline. PO Box 84151, Seattle WA 98124. (206) 621- 8558. <<http://www.headinjury.com>>.

Head Trauma Support Project, Inc. 2500 Marconi Ave., Ste. 203, Sacramento, CA 95821. (916) 482-5770.

National Head Injury Foundation. 333 Turnpike Rd., Southboro, MA 01722. (617) 485-9950.

Carol A. Turkington
Teresa G. Odle

Head lice see **Lice infestation**

Head trauma see **Head injury**

Headache

Definition

A headache involves **pain** in the head which can arise from many disorders or may be a disorder in and of itself.

Description

There are three types of primary headaches: tension-type (muscular contraction headache), migraine (vascular headaches), and cluster. Virtually everyone experiences a tension-type headache at some point. An estimated 18% of American women suffer migraines, compared to 6% of men. Cluster headaches affect fewer than 0.5% of the population, and men account for approximately 80% of all cases. Headaches caused by illness are secondary headaches and are not included in these numbers.

Approximately 40–45 million people in the United States suffer chronic headaches. Headaches have an enormous impact on society due to missed workdays and productivity losses.

Causes and symptoms

Traditional theories about headaches link tension-type headaches to muscle contraction, and migraine and cluster headaches to blood vessel dilation (swelling). Pain-sensitive structures in the head include blood vessel walls, membranous coverings of the brain, and scalp and neck muscles. Brain tissue itself has no sensitivity to pain. Therefore, headaches may result from contraction of the muscles of the scalp, face or neck; dilation of the blood vessels in the head; or brain swelling that stretches the brain's coverings. Involvement of specific nerves of the face and head may also cause characteristic headaches. Sinus inflammation is a common cause of headache. Keeping a headache diary may help link headaches to stressful occurrences, menstrual phases, food triggers, or medication.

Tension-type headaches are often brought on by stress, overexertion, loud noise, and other external factors. The typical tension-type headache is described as a tightening around the head and neck, and an accompanying dull ache.

Migraines are intense throbbing headaches occurring on one or both sides of the head, usually on one side. The pain is accompanied by other symptoms such as **nausea**, **vomiting**, blurred vision, and aversion to light, sound, and movement. Migraines often are

triggered by food items, such as red wine, chocolate, and aged cheeses. For women, a hormonal connection is likely, since headaches occur at specific points in the menstrual cycle, with use of oral contraceptives, or the use of hormone replacement therapy after **menopause**. Research shows that a complex interaction of nerves and neurotransmitters in the brain act to cause migraine headaches.

Cluster headaches cause excruciating pain. The severe, stabbing pain centers around one eye, and eye tearing and nasal congestion occur on the same side. The headache lasts from 15 minutes to four hours and may recur several times in a day. Heavy smokers are more likely to suffer cluster headaches, which also are associated with alcohol consumption.

Diagnosis

Since headaches arise from many causes, a physical exam assesses general health and a neurologic exam evaluates the possibility of neurologic disease as a cause for the headache. If the headache is the primary illness, the doctor asks for a thorough history of the headache. Questions revolve around its frequency and duration, when it occurs, pain intensity and location, possible triggers, and any prior symptoms. This information aids in classifying the headache.

Warning signs that should point out the need for prompt medical intervention include:

- “Worst headache of my life.” This may indicate subarachnoid hemorrhage from a ruptured aneurysm (swollen blood vessel) in the head or other neurological emergency.
- Headache accompanied by one-sided weakness, **numbness**, visual loss, speech difficulty, or other signs. This may indicate a **stroke**. Migraines may include neurological symptoms.
- Headache that becomes worse over a period of 6 months, especially if most prominent in the morning or if accompanied by neurological symptoms. This may indicate a **brain tumor**.
- Sudden onset of headache. If accompanied by **fever** and stiff neck, this can indicate **meningitis**.

Headache diagnosis may include neurological imaging tests such as computed tomography scan (CT scan) or **magnetic resonance imaging** (MRI).

Treatment

Headache treatment is divided into two forms: abortive and prophylactic. Abortive treatment

addresses a headache in progress, and prophylactic treatment prevents headache occurrence.

Tension-type headaches can be treated with **aspirin**, **acetaminophen**, ibuprofen, or naproxen. In early 1998, the FDA approved extra-strength Excedrin, which includes **caffeine**, for mild migraines. Physicians continue to investigate and monitor the best treatment for migraines and generally prefer a stepped approach, depending on headache severity, frequency and impact on the patient's quality of life. A group of drugs called triptans are usually preferred for abortive treatment. About seven triptans are available in the United States and the pill forms are considered most effective. They should be taken as early as possible during the typical migraine attack. The most common prophylactic therapies include antidepressants, **beta blockers**, **calcium channel blockers** and antiseizure medications. Antiseizure medications have proven particularly effective at blocking the actions of neurotransmitters that start migraine attacks. Topiramate (Topamax) was shown effective in several combined clinical trials in 2004 at 50 to 200 mg per day.

In 2004, a new, large study added evidence to show the effectiveness of botulinum toxin type A (Botox) treatment to prevent headache pain for those with frequent, untreatable tension and migraine headaches. Patients were treated every three months, with two to five injections each time. They typically received relief within two to three weeks.

Cluster headaches may also be treated with ergotamine and sumatriptan, as well as by inhaling pure oxygen. Prophylactic treatments include prednisone, calcium channel blockers, and methysergide.

Alternative treatment

Alternative headache treatments include:

- acupuncture or **acupressure**
- biofeedback
- chiropractic
- herbal remedies using feverfew (*Chrysanthemum parthenium*), valerian (*Valeriana officinalis*), white willow (*Salix alba*), or skullcap (*Scutellaria lateriflora*), among others
- homeopathic remedies chosen specifically for the individual and his/her type of headache
- hydrotherapy
- massage
- magnesium supplements

KEY TERMS

Abortive—Referring to treatment that relieves symptoms of a disorder.

Analgesics—A class of pain-relieving medicines, including aspirin and Tylenol.

Biofeedback—A technique in which a person is taught to consciously control the body's response to a stimulus.

Chronic—Referring to a condition that occurs frequently or continuously or on a regular basis.

Prophylactic—Referring to treatment that prevents symptoms of a disorder from appearing.

Transcutaneous electrical nerve stimulation—A method that electrically stimulates nerve and blocks the transmission of pain signals, called TENS.

- regular physical **exercise**
- relaxation techniques, such as **meditation** and **yoga**
- transcutaneous **electrical nerve stimulation** (TENS) (A procedure that electrically stimulates nerves and blocks the signals of pain transmission.)

Prognosis

Headaches are typically resolved through the use of **analgesics** and other treatments. Research in 2004 showed that people who have migraine headaches more often than once a month may be at increased risk for stroke.

Prevention

Some headaches may be prevented by avoiding triggering substances and situations, or by employing alternative therapies, such as yoga and regular exercise. Since **food allergies** often are linked with headaches, especially cluster headaches, identification and elimination of the allergy-causing food(s) from the diet can be an important preventive measure.

Resources

PERIODICALS

Kruit, Mark C., et al. "Migraine as a Risk Factor for Subclinical Brain Lesions." *JAMA, Journal of the American Medical Association* January 28, 2004: 427–435.

Norton, Patrice G. W. "Botox Stops Headache Pain in Recalcitrant Cases." *Clinical Psychiatry News* March 2004: 72.

Taylor, Frederick, et al. "Diagnosis and Management of Migraine in Family Practice." *Journal of Family Practice* January 2004: S3–S25.

ORGANIZATIONS

American Council for Headache Education (ACHE). 19 Mantua Road, Mt. Royal, NJ 08061. (800) 255-2243. <<http://www.achenet.org>>.

National Headache Foundation. 428 W. St. James Place, Chicago, IL 60614. (800) 843-2256. <<http://www.headaches.org>>.

Julia Barrett
Teresa G. Odle

Hearing aids

Definition

A hearing aid is a device that can amplify sound waves in order to help a deaf or hard-of-hearing person hear sounds more clearly.

Purpose

Recent technology can help most people with hearing loss understand speech better and achieve better communication.

Precautions

It's important that a person being fitted for a hearing aid understand what an aid can and can't do. An aid can help a person hear better, but it won't return hearing to normal levels. Hearing aids boost all sounds, not just those the person wishes to hear. Especially when the source of sound is far away (such as up on a stage), environmental noise can interfere with good speech perception. And while the aid amplifies sound, it doesn't necessarily improve the clarity of the sound. A hearing aid is a machine, and can never duplicate the true sound that people with normal hearing experience, but it will help the person take advantage of the hearing that remains.

Description

More than 1,000 different models are available in the United States. All of them include a microphone (to pick up sound), amplifier (to boost sound strength), a receiver or speaker (to deliver sound to the ear), and are powered by a battery. Depending

on the style, it's possible to add features to filter or block out background noise, minimize feedback, lower sound in noisy settings, or boost power when needed.

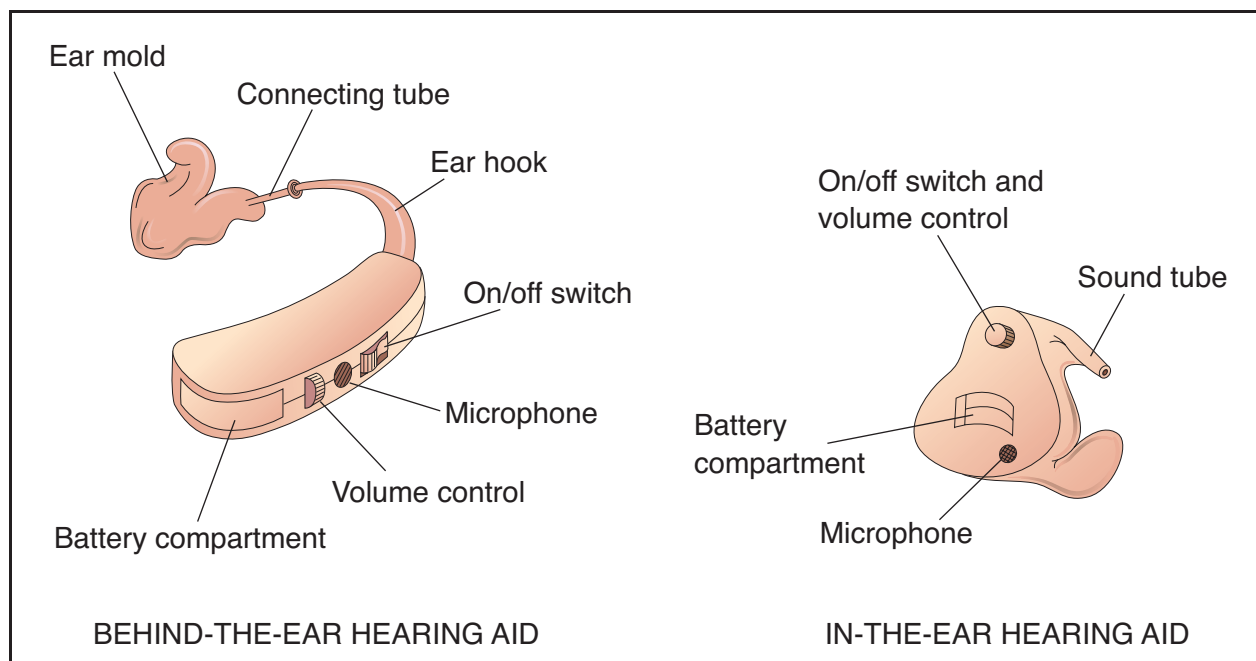
Hearing aids are either "monaural" (a hearing aid for one ear), or "binaural" (for two ears); more than 65% of all users have binaural aids. Hearing aids are divided into several different types:

- digital
- in-the-ear
- in-the-canal
- behind-the-ear
- on-the-body

Digital aids are sophisticated, very expensive aids that borrow computer technology to allow a person to tailor an aid to a specific **hearing loss** pattern. Using miniature computer chips, the aids can selectively boost certain frequencies while leaving others alone. This means a person could wear such an aid to a loud party, and screen out unwanted background noise, while tuning in on one-on-one conversations. The aid is programmed by the dealer to conform to the patient's specific hearing loss. Some models can be programmed to allow the wearer to choose different settings depending on the noise of the environment.

In-the-ear aids are lightweight devices whose custom-made housings contain all the components; this device fits into the ear canal with no visible wires or tubes. It's possible to control tone but not volume with these aids, so they are helpful only for people with mild hearing loss. Some people find these aids are easier to put on and take off than behind-the-ear aids. However, because they are custom-fit to a person's ear, it is not possible to try on before ordering. Some people find them uncomfortable in hot weather.

In-the-canal aids fit far into the ear canal, with only a small bit extending into the external ear. The smallest is the MicroCanal, which fits out of sight down next to the eardrum and is removed with a small transparent wire. These are extremely expensive, but they are not visible, offer better acoustics, and are easier to maintain. They can more closely mimic natural sound because of the position of the microphone; this position also cuts down on wind noise. But their small size makes them harder to handle, and their battery is especially small and difficult to insert. Adjusting the volume may be hard, since a person must stick a finger down into the ear to adjust volume, and this very tiny aid doesn't have the power of other, larger, aids.



Hearing aids are devices that can amplify sound waves to help a deaf or hard-of-hearing person hear sounds more clearly.
(Illustration by Electronic Illustrators Group.)

Behind-the-ear aids include a microphone, amplifier and receiver inside a small curved case worn behind the ear; the case is connected to the earmold by a short plastic tube. The earmold extends into the ear canal. Some models have both tone and volume control, plus a telephone pickup device. However, many users think them unattractive and out of date; and people who wear glasses find that the glasses interfere with the aid's fit. Others don't have space behind the ear for the mold to fit comfortably. However, they do offer a few advantages.

Behind-the-ear aids:

- don't require as much maintenance
- are easily interchangeable if they need to be serviced
- are more powerful
- are easier to handle than smaller aids
- can provide better sound quality
- tend to be more reliable

Eyeglass models are the same as behind-the-ear devices, except that the case fits into an eyeglass frame instead of resting behind the ears. Not many people buy this type of aid, but those who do believe it's less obvious, although there is a tube that travels from the temple of the glasses to the earmold. But it can be hard to fit this type of aid, and repairs can be problematic.

Also, if the aid breaks, the person also loses the benefit of the glasses.

CROS or the crossover system type of hearing aid is often used in conjunction with the eyeglass model. The CROS (contralateral routing of signal) system features a microphone behind the ear that feeds the amplified signal to the better ear, eliminating "head shadow," which occurs when the head blocks sound from the better ear. This type may help make speech easier to understand for people with a high-frequency loss in both ears.

A BI-CROS system uses two microphones (one above each ear) that send signals to a single amplifier. Sound then travels to a single receiver, which transfers it to the better ear via a conventional earmold.

On-the-body aids feature a larger microphone, amplifier, and power supply inside a case carried inside the pocket, or attached to clothing. The receiver attaches directly to the earmold; its power comes through a flexible wire from the amplifier. Although larger than other aids, the on-the-body aids are more powerful and easier to adjust than other devices. While not popular for everyone, they are often used by those with a profound hearing loss, or by very young children. Some people who are almost totally deaf find they need the extra power boost available only from a body aid.

The latest aids on the market may eliminate the amplifier and speaker in favor of a tiny magnet mounted on a silicone disk, similar to a contact lens, which rests right on the eardrum. Called the Earlens, it is designed to be held in place by a thin film of oil. Users wear a wireless microphone, either in the ear or on a necklace, that picks up sounds and converts them into magnetic signals, making the magnet vibrate. As the Earlens vibrates, so does the eardrum, transmitting normal-sounding tones to the middle and inner ears.

Other researchers are bypassing the middle ear completely; they surgically implant a tiny magnet in the inner ear. By attaching a magnet to the round window, they open a second pathway to the inner ear. An electromagnetic coil implanted in bone behind the ear vibrates the implanted magnet. Unlike the Earlens, this magnetic implant would not block the normal hearing pathway.

Preparation

The first step in getting a hearing aid is to have a medical exam and a hearing evaluation. (Most states prohibit anyone selling a hearing aid until the patient has been examined by a physician to rule out medical problems.) After performing a hearing evaluation, an audiologist should be able to determine whether a hearing aid will help, and which one will do the most good. This is especially important because aids can be very expensive (between \$500 and \$4,000), and are often not covered by health insurance. Hearing aids come in a wide range of styles and types, requiring careful testing to make sure the aid is the best choice for a particular hearing loss.

Some audiologists sell aids; others can make a recommendation, or give one a list of competent dealers in one's area. Patients should shop around and compare prices. In all but three states, hearing aids must be fitted and sold only by licensed specialists called dealers, specialists, dispensers, or dispensing audiologists.

The hearing aid dealer will make an impression of the consumer's ears using a putty-like material, from which a personalized earmold will be created. It's the dealer's job to make sure the aid fits properly. The person may need several visits to find the right hearing aid and learn how to use it. The dealer will help the consumer learn how to put the aid on, adjust the controls, and maintain the device. The dealer should be willing to service the aid and provide information about what to do if sensitivity to the earmold develops. (Some people are allergic to the materials in the mold.)

KEY TERMS

Audiologist—A person with a degree and/or certification in the areas of identification and measurement of hearing impairments and rehabilitation of those with hearing problems.

Eardrum—A paper-thin covering stretching across the ear canal that separates the middle and outer ears.

Middle ear—The small cavity between the eardrum and the oval window that houses the three tiny bones of hearing.

Oval window—A tiny opening at the entrance to the inner ear.

Aftercare

Within several weeks, the wearer should return to the dealer to have the aid checked, and to discuss the progress in wearing the aid. About 40% of all aids need some modification or adjustment in the beginning.

Within the first month of getting an aid, the patient should make an appointment for a full hearing examination to determine if the aid is functioning properly.

Risks

While there are no medical risks to hearing aids, there is a risk associated with hearing aids: many people end up not wearing their aids because they say everything seems loud when wearing them. This is because they have lived for so long with a hearing problem that they have forgotten how loud "normal" sound can be. Other potential problems with hearing aids include earmold discomfort, and a build up of excess ear wax after getting a hearing aid.

Normal results

A hearing aid will boost the loudness of sound, which can improve a person's ability to understand speech.

Resources

ORGANIZATIONS

American Academy of Otolaryngology-Head and Neck Surgery, Inc. One Prince St., Alexandria VA 22314-3357. (703) 836-4444. <<http://www.entnet.org>>.

Better Hearing Institute. 515 King Street, Suite 420, Alexandria, VA 22314. (703) 684-3391.

Hearing Industries Association. 1800 M St. NW, Washington, DC 20036. (202) 651-5258.

Hear Now. 9745 E. Hampden Ave., Ste. 300, Denver, CO 80231. (800) 648-HEAR. (202) 651-5258.

National Hearing Aid Society. 20361 Middlebelt, Livonia, MI 48152. (800) 521-5247 or (313) 478-2610.

National Information Center on Deafness. Gallaudet College, 800 Florida Ave. NE, Washington, DC 20002. (202) 651-5051; (202) 651-5052 (TDD).

Carol A. Turkington

Hearing loss

Definition

Hearing loss is any degree of impairment of the ability to apprehend sound.

Description

Sound can be measured accurately. The term decibel (dB) refers to an amount of energy moving sound from its source to our ears or to a microphone. A drop of more than 10 dB in the level of sound a person can hear is significant.

Sound travels through a medium like air or water as waves of compression and rarefaction. These waves are collected by the external ear and cause the tympanic membrane (ear drum) to vibrate. The chain of ossicles connected to the ear drum—the incus, malleus, and stapes—carries the vibration to the oval window, increasing its amplitude 20 times on the way. There the energy causes a standing wave in the watery liquid (endolymph) inside the Organ of Corti. (A standing wave is one that does not move. A vibrating cup of coffee will demonstrate standing waves.) The configuration of the standing wave is determined by the frequency of the sound. Many thousands of tiny nerve fibers detect the highs and lows of the standing wave and transmit their findings to the brain, which interprets the signals as sound.

To summarize, sound energy passes through the air of the external ear, the bones of the middle ear and the liquid of the inner ear. It is then translated into nerve impulses, sent to the brain through nerves and understood there as sound. It follows that there are five steps in the hearing process:

- air conduction through the external ear to the ear drum

Decibel Ratings And Hazardous Levels Of Noise

Decibel Level	Example Of Sounds
30	Soft whisper
35	Noise may prevent the listener from falling asleep
40	Quiet office noise level
50	Quiet conversation
60	Average television volume, sewing machine, lively conversation
70	Busy traffic, noisy restaurant
80	Heavy city traffic, factory noise, alarm clock
90	Cocktail party, lawn mower
100	Pneumatic drill
120	Sandblasting, thunder
140	Jet airplane
180	Rocket launching pad

Above 110 decibels, hearing may become painful

Above 120 decibels is considered deafening

Above 135 decibels, hearing will become extremely painful and hearing loss may result if exposure is prolonged Above 180 decibels, hearing loss is almost certain with any exposure

- bone conduction through the middle ear to the inner ear
- water conduction to the Organ of Corti
- nerve conduction into the brain
- interpretation by the brain.

Hearing can be interrupted in several ways at each of the five steps.

The external ear canal can be blocked with ear wax, foreign objects, infection, and tumors. Overgrowth of the bone, a condition that occurs when the ear canal has been flushed with cold water repeatedly for years, can also narrow the passageway, making blockage and infection more likely. This condition occurs often in Northern Californian surfers and is therefore called “surfer’s ear.”

The ear drum is so thin a physician can see through it into the middle ear. Sharp objects, pressure from an infection in the middle ear, even a firm cuffing or slapping of the ear, can rupture it. It is also susceptible to pressure changes during scuba diving.

Several conditions can diminish the mobility of the ossicles (small bones) in the middle ear. **Otitis media** (an infection in the middle ear) occurs when fluid cannot escape into the throat because of blockage of the eustachian tube. The fluid that accumulates, whether it be pus or just mucus and dampens the motion of the ossicles. A disease called **otosclerosis** can bind the stapes in the oval window and thereby cause deafness.



An Oto-Acoustic Emission (OAE) hearing test being performed on a newborn baby. The probe emits harmless sound into the baby's ear, and the response of the inner ear is detected and registered on a computer. Early diagnosis of a hearing disorder is important in young children, who may experience difficulties in speech and language development. (Photograph by James King-Holmes, Photo Researchers, Inc. Reproduced by permission.)

All the conditions mentioned so far, those that occur in the external and middle ear, are causes of conductive hearing loss. The second category, sensory hearing loss, refers to damage to the Organ of Corti and the acoustic nerve. Prolonged exposure to loud noise is the leading cause of sensory hearing loss. A million people have this condition, many identified during the military draft and rejected as being unfit for duty. The cause is often believed to be prolonged exposure to rock music. Occupational noise exposure is the other leading cause of noise induced hearing loss (NIHL) and is ample reason for wearing ear protection on the job. A third of people over 65 have presbycusis—sensory hearing loss due to **aging**. Both NIHL and presbycusis are primarily high frequency losses. In most languages, it is the high frequency sounds that define speech, so these people hear plenty of noise, they just cannot easily make out what it means. They have particular trouble selecting out speech from background noise. Brain infections like **meningitis**, drugs such as the aminoglycoside **antibiotics** (streptomycin, gentamycin, kanamycin, tobramycin),

and Meniere's disease also cause permanent sensory hearing loss. Meniere's disease combines attacks of hearing loss with attacks of vertigo. The symptoms may occur together or separately. High doses of salicylates like **aspirin** and quinine can cause a temporary high-frequency loss. Prolonged high doses can lead to permanent deafness. There is an hereditary form of sensory deafness and a congenital form most often caused by rubella (German **measles**).

Sudden hearing loss—at least 30dB in less than three days—is most commonly caused by cochleitis, a mysterious viral infection.

The final category of hearing loss is neural. Damage to the acoustic nerve and the parts of the brain that perform hearing are the most likely to produce permanent hearing loss. Strokes, multiple sclerosis, and acoustic neuromas are all possible causes of neural hearing loss.

Hearing can also be diminished by extra sounds generated by the ear, most of them from the same kinds of disorders that cause diminished hearing.

These sounds are referred to as **tinnitus** and can be ringing, blowing, clicking, or anything else that no one but the patient hears.

Diagnosis

An examination of the ears and nose combined with simple hearing tests done in the physician's office can detect many common causes of hearing loss. An audiogram often concludes the evaluation, since these simple means often produce a diagnosis. If the defect is in the brain or the acoustic nerve, further neurological testing and imaging will be required.

The audiogram has many uses in diagnosing hearing deficits. The pattern of hearing loss across the audible frequencies gives clues to the cause. Several alterations in the testing procedure can give additional information. For example, speech is perceived differently than pure tones. Adequate perception of sound combined with inability to recognize words points to a brain problem rather than a sensory or conductive deficit. Loudness perception is distorted by disease in certain areas but not in others. Acoustic neuromas often distort the perception of loudness.

Treatment

Conductive hearing loss can almost always be restored to some degree, if not completely.

- matter in the ear canal can be easily removed with a dramatic improvement in hearing.
- surfer's ear gradually regresses if cold water is avoided or a special ear plug is used. In advanced cases, surgeons can grind away the excess bone.
- middle ear infection with fluid is also simple to treat. If medications do not work, surgical drainage of the ear is accomplished through the ear drum, which heals completely after treatment.
- traumatically damaged ear drums can be repaired with a tiny skin graft.
- surgical repair of otosclerosis through an operating microscope is one of the most intricate of procedures, substituting tiny artificial parts for the original ossicles.

Sensory and neural hearing loss, on the other hand, cannot readily be cured. Fortunately it is not often complete, so that **hearing aids** can fill the deficit.

In-the-ear hearing aids can boost the volume of sound by up to 70 dB. (Normal speech is about 60 dB.) Federal law now requires that they be dispensed only

KEY TERMS

Decibel—A unit of the intensity of sound, a measure of loudness.

Meniere's disease—The combination of vertigo and decreased hearing caused by abnormalities in the inner ear.

Multiple sclerosis—A progressive disease of brain and nerve tissue.

Otosclerosis—A disease that scars and limits the motion of the small conducting bones in the middle ear.

Stroke—Sudden loss of blood supply to part of the brain.

upon a physician's prescription. For complete conduction hearing loss there are now available bone conduction hearing aids and even devices that can be surgically implanted in the cochlea.

Tinnitus can sometimes be relieved by adding white noise (like the sound of wind or waves crashing on the shore) to the environment.

Decreased hearing is such a common problem that there are legions of organizations to provide assistance. Special language training, both in lip reading and signing, special schools and special camps for children are all available in most regions of the United States.

Alternative treatment

Conductive hearing loss can be treated with alternative therapies that are specific to the particular condition. Sensory hearing loss may be helped by homeopathic therapies. Oral supplementation with essential fatty acids such as flax oil and omega 3 oil can help alleviate the accumulation of wax in the ear.

Prevention

Prompt treatment and attentive follow-up of middle ear infections in children will prevent this cause of conductive hearing loss. Control of infectious childhood diseases such as measles has greatly reduced sensory hearing loss as a complication of epidemic diseases. Laws that require protection from loud noise in the workplace have achieved substantial reduction in noise induced hearing loss. Surfers should use the right kind of ear plugs.

Resources

ORGANIZATIONS

Alexander Graham Bell Association for the Deaf. 3417 Volta Place NW, Washington, DC 20007. (202) 337-5220. <<http://www.agbell.org>>.

Auditory-Verbal International. 2121 Eisenhower Ave., Suite 402, Alexandria, VA 22314. (703) 739-1049. <avi@auditory-verbal.org> <<http://www.auditory-verbal.org/contact.htm>>.

Better Hearing Institute. 515 King Street, Suite 420, Alexandria, VA 22314. (703) 684-3391.

Central Institute for the Deaf. Washington University. St. Louis, Missouri. <<http://cidmac.wustl.edu>>.

League for the Hard of Hearing. 71 West 23rd St., New York, New York 10010-4162. (212) 741-7650. <<http://www.lhh.org>>.

National Association of the Deaf. 814 Thayer Ave., Silver Spring, MD, 20910. (301) 587-1788. <<http://nad.policy.net>>.

Self Help for Hard of Hearing People, Inc. 7800 Wisconsin Ave., Bethesda, MD 20814. (301) 657-2248. <<http://www.shhh.org>>.

The Sight & Hearing Association (SHA). <<http://www.sightandhearing.org>>.

World Recreation Association of the Deaf (WRAD). <<http://www.wrad.org>>.

J. Ricker Polsdorfer, MD

Hearing test with an audiometer see

Audiometry

Hearing tests with a tuning fork

Definition

A tuning fork is a metal instrument with a handle and two prongs or tines. Tuning forks, made of steel, aluminum, or magnesium-alloy will vibrate at a set frequency to produce a musical tone when struck. The vibrations produced can be used to assess a person's ability to hear various sound frequencies.

Purpose

A vibrating tuning fork held next to the ear or placed against the skull will stimulate the inner ear to vibrate, and can help determine if there is **hearing loss**.

Precautions

No special precautions are necessary when tuning forks are used to conduct a hearing test.

Description

Two types of hearing tests with tuning forks are typically conducted. In the Rinne test, the vibrating tuning fork is held against the skull, usually on the bone behind the ear (mastoid process) to cause vibrations through the bones of the skull and inner ear. It is also held next to, but not touching, the ear, to cause vibrations in the air next to the ear. The patient is asked to determine which sound is louder, the sound heard through the bone or through the air. A second hearing test using a tuning fork is the Weber test. For this test, the stem or handle of the vibrating tuning fork is placed at various points along the midline of the skull and face. The patient is then asked to identify which ear hears the sound created by the vibrations. Tuning forks of different sizes produce different frequencies of vibrations and can be used to establish the range of hearing for an individual patient.

Preparation

No special preparation is required for a hearing test with tuning forks.

Aftercare

No special aftercare is required. If hearing loss is revealed during testing with tuning forks, the patient may require further testing to determine the extent of the hearing loss.

Risks

There are no risks associated with the use of tuning forks to screen for hearing loss.

Normal results

With the Rinne test, a person will hear the tone of the vibration longer and louder when the tuning fork is held next to the ear, rather than when it is held against the mastoid bone. For the Weber test, the tone produced when the tuning fork is placed along the center of the skull, or face, sounds about the same volume in each ear.

Abnormal results

The Rinne test detects a hearing loss when a patient hears a louder and longer tone when the

KEY TERMS

Mastoid process—The protrusions of bone behind the ears at the base of the skull.

Rinne test—A hearing test using a vibrating tuning fork which is held near the ear and held at the back of the skull.

Weber test—A hearing test using a vibrating tuning fork which is held at various points along the midline of the skull and face.

vibrating tuning fork is held against the mastoid bone than when it is held next to the ear. The volume of sound vibrations conducted through parts of the skull and face in the Weber test can indicate which ear may have a hearing loss.

Resources

ORGANIZATIONS

American Academy of Otolaryngology-Head and Neck Surgery, Inc. One Prince St., Alexandria VA 22314-3357. (703) 836-4444. <<http://www.entnet.org>>.

Ear Foundation. 1817 Patterson St., Nashville, TN 37203. (800) 545-4327. <<http://www.earfoundation.org>>.

Altha Roberts Edgren

Heart arrest see **Sudden cardiac death**

Heart arrhythmias see **Arrhythmias**

Heart attack

Definition

A heart attack is the death of, or damage to, part of the heart muscle because the supply of blood to the heart muscle is severely reduced or stopped.

Description

Heart attack is the leading cause of death in the United States. More than 1.5 million Americans suffer a heart attack every year, and almost half a million die, according to the American Heart Association. Most heart attacks are the end result of years of silent but progressive **coronary artery disease**, which can be prevented in many people. A heart attack

often is the first symptom of coronary artery disease. According to the American Heart Association, 63% of women and 48% of men who died suddenly of coronary artery disease had no previous symptoms. Heart attacks also are called myocardial infarctions (MIs).

A heart attack occurs when one or more of the coronary arteries that supply blood to the heart are completely blocked and blood to the heart muscle is cut off. The blockage usually is caused by **atherosclerosis**, the build-up of plaque in the artery walls, and/or by a blood clot in a coronary artery. Sometimes, a healthy or atherosclerotic coronary artery has a spasm and the blood flow to part of the heart decreases or stops. Why this happens is unclear, but it can result in a heart attack.

About half of all heart attack victims wait at least two hours before seeking help. This increases their chance of sudden death or being disabled. The longer the artery remains blocked during a heart attack, the more damage will be done to the heart. If the blood supply is cut off severely or for a long time, muscle cells suffer irreversible injury and die. This can cause the patient to die. That is why it is important to recognize the signs of a heart attack and seek prompt medical attention at the nearest hospital with 24-hour emergency cardiac care.

About one-fifth of all heart attacks are silent, that is, the victim does not know one has occurred. Although the victim feels no **pain**, silent heart attacks still can damage the heart.

The outcome of a heart attack also depends on where the blockage is, whether the heart rhythm is disturbed, and whether another coronary artery supplies blood to that part of the heart. Blockages in the left coronary artery usually are more serious than in the right coronary artery. Blockages that cause an arrhythmia, an irregular heartbeat, can cause sudden death.

Causes and symptoms

Heart attacks generally are caused by severe coronary artery disease. Most heart attacks are caused by **blood clots** that form on atherosclerotic plaque. This blocks a coronary artery from supplying oxygen-rich blood to part of the heart. A number of major and contributing risk factors increase the risk of developing coronary artery disease. Some of these can be changed and some cannot. People with more risk factors are more likely to develop coronary artery disease.

Major risk factors

Major risk factors significantly increase the risk of coronary artery disease. Those which cannot be changed are:

- **Heredity.** People whose parents have coronary artery disease are more likely to develop it. African Americans also are at increased risk, due to their higher rate of severe **hypertension** than whites.
- **Sex.** Men under the age of 60 years of age are more likely to have heart attacks than women of the same age.
- **Age.** Men over the age of 45 and women over the age of 55 are considered at risk. Older people (those over 65) are more likely to die of a heart attack. Older women are twice as likely to die within a few weeks of a heart attack as a man. This may be because of other co-existing medical problems.

Major risk factors that can be changed are:

- **Smoking.** Smoking greatly increases both the chance of developing coronary artery disease and the chance of dying from it. Smokers have two to four times the risk of non-smokers of **sudden cardiac death** and are more than twice as likely to have a heart attack. They also are more likely to die within an hour of a heart attack. Second-hand smoke also may increase risk.
- **High cholesterol.** Cholesterol is a soft, waxy substance that is produced by the body, as well as obtained from eating foods such as meat, eggs, and other animal products. Cholesterol level is affected by age, sex, heredity, and diet. Risk of developing coronary artery disease increases as blood cholesterol levels increase. When combined with other factors, the risk is even greater. Total cholesterol of 240 mg/dL and over poses a high risk, and 200–239 mg/dL a borderline high risk. In LDL cholesterol, high risk starts at 130–159 mg/dL, depending on other risk factors. HDL (healthy cholesterol) can lower or raise the coronary risk also.
- **High blood pressure.** High blood pressure makes the heart work harder, and over time, weakens it. It increases the risk of heart attack, **stroke**, kidney failure, and congestive **heart failure**. A blood pressure of 140 over 90 or above is considered high. As the numbers increase, high blood pressure goes from Stage 1 (mild) to Stage 4 (very severe). When combined with obesity, smoking, high cholesterol, or diabetes, the risk of heart attack or stroke increases several times.
- **Lack of physical activity.** This increases the risk of coronary artery disease. Even modest physical activity is beneficial if done regularly.
- **Use of certain drugs or supplements.** Extreme caution is advised in the use of the herbal supplement ephedra. The supplement, which was marketed for weight loss and to improve athletic performance, was found to contribute to heart attack, seizure, stroke and death. In April 2003, the U.S. Food and Drug Administration (FDA) investigating controlling or banning the substance. While it was once believed that **hormone replacement therapy** (HRT) helped prevent heart disease in women, a large clinical trial called the Women's Health Initiative found the opposite to be true. In 2003, the FDA began requiring manufacturers of HRT to place warnings on the box listing adverse effects of estrogen, including increased risk of heart attack, stroke and blood clots. The labels also must mention that HRT should not be used as a preventive medicine for heart disease.

Contributing risk factors

Contributing risk factors have been linked to coronary artery disease, but their significance or prevalence cannot always be demonstrated. Contributing risk factors are:

- **Diabetes mellitus.** The risk of developing coronary artery disease is seriously increased for diabetics. More than 80% of diabetics die of some type of heart or blood vessel disease.
- **Obesity.** Excess weight increases the strain on the heart and increases the risk of developing coronary artery disease, even if no other risk factors are present. Obesity increases both blood pressure and blood cholesterol, and can lead to diabetes.
- **Stress and anger.** Some scientists believe that stress and anger can contribute to the development of coronary artery disease. Stress, the mental and physical reaction to life's irritations and challenges, increases the heart rate and blood pressure, and can injure the lining of the arteries. Evidence shows that anger increases the risk of dying from heart disease and more than doubles the risk of having a heart attack right after an episode of anger.
- **Rheumatoid arthritis** in women. A report released in 2003 noted that women with rheumatoid arthritis has a higher risk of heart attack than those without the condition. The reason is most likely the inflammation arthritis causes.

More than 60% of heart attack victims experience symptoms before the heart attack occurs. These sometimes occur days or weeks before the heart attack. Sometimes, people do not recognize the symptoms of a heart attack or are in denial that they are having one. Symptoms are:

- Uncomfortable pressure, fullness, squeezing, or pain in the center of the chest. This lasts more than a few minutes, or may go away and return.
- Pain that spreads to the shoulders, neck, or arms.
- Chest discomfort accompanied by lightheadedness, fainting, sweating, **nausea**, or **shortness of breath**.

All of these symptoms do not occur with every heart attack. Sometimes, symptoms disappear and then reappear. A person with any of these symptoms should immediately call an emergency rescue service or be driven to the nearest hospital with a 24-hour cardiac care unit, whichever is quicker.

Diagnosis

Experienced emergency care personnel usually can diagnose a heart attack simply by looking at the patient. To confirm this diagnosis, they talk with the patient, check heart rate and blood pressure, perform an electrocardiogram, and take a blood sample. The electrocardiogram shows which coronary artery is blocked. Electrodes covered with conducting jelly are placed on the patient's chest, arms, and legs. They send impulses of the heart's activity through an oscilloscope (a monitor) to a recorder, which traces them on paper. The blood test shows the leak of enzymes or other biochemical markers from damaged cells in the heart muscle. In 2003, the FDA cleared a new test for ruling out heart attacks in people who come to emergency rooms with severe chest pains. It is the first new blood test for evaluation of heart attacks since 1994 and is used along with an electrocardiogram.

Treatment

Heart attacks are treated with cardiopulmonary resuscitation (CPR) when necessary to start and keep the patient breathing and his heart beating. Additional treatment can include close monitoring, electric shock, drug therapy, re-vascularization procedures, percutaneous transluminal coronary **angioplasty** and coronary artery bypass surgery. Upon arrival at the hospital, the patient is closely monitored. An electrical-shock device, a defibrillator, may be used to restore a normal rhythm if the heartbeat is fluttering uncontrollably. Oxygen often is used to ease the heart's workload or to help victims of severe heart attack breathe easier. If oxygen is used within hours of the heart attack, it may help limit damage to the heart.

Drugs to stabilize the patient and limit damage to the heart include thrombolytics, aspirin, anticoagulants, painkillers and tranquilizers, beta-blockers, ace-inhibitors, nitrates, rhythm-stabilizing drugs, and

diuretics. Drugs that limit damage to the heart work only if given within a few hours of the heart attack. Thrombolytic drugs that break up blood clots and enable oxygen-rich blood to flow through the blocked artery increase the patient's chance of survival if given as soon as possible after the heart attack. Thrombolytics given within a few hours after a heart attack are the most effective. Injected intravenously, these include anisoylated plasminogen streptokinase activator complex (APSAC) or anistreplase (Eminase), recombinant tissue-type plasminogen activator (r-tPA, Retevase, or Activase), and streptokinase (Streptase, Kabikinase).

To prevent additional heart attacks, **aspirin** and an anticoagulant drug often follow the thrombolytic drug. These prevent new blood clots from forming and existing blood clots from growing. **Anticoagulant drugs** help prevent the blood from clotting. The most common anticoagulants are heparin and warfarin. Heparin is given intravenously while the patient is in the hospital. Warfarin, taken orally, often is given later. Aspirin helps to prevent the dissolved blood clots from reforming.

To relieve pain, a nitroglycerine tablet taken under the tongue may be given. If the pain continues, morphine sulfate may be prescribed. Tranquilizers such as diazepam (Valium) and alprazolam (Ativan) may be prescribed to lessen the trauma of a heart attack.

To slow down the heart rate and give the heart a chance to heal, beta-blockers often are given intravenously right after the heart attack. These can also help prevent sometimes fatal ventricular fibrillation. Beta-blockers include atenolol (Tenormin), metoprolol (Lopressor), nadolol, pindolol (Visken), propranolol (Inderal), and timolol (Blocadren).

Nitrates, a type of vasodilator, also are given right after a heart attack to help improve the delivery of blood to the heart and ease heart failure symptoms. Nitrates include isosorbide mononitrate (Imdur), isosorbide dinitrate (Isordil, Sorbitrate), and nitroglycerin (Nitrostat).

When a heart attack causes an abnormal heart-beat, arrhythmia drugs may be given to restore the heart's normal rhythm. These include: amiodarone (Cordarone), atropine, bretylium, disopyramide (Norpace), lidocaine (Xylocaine), procainamide (Procan), propafenone (Rythmol), propranolol (Inderal), quinidine, and sotalol (Betapace). Angiotensin-converting enzyme (ACE) inhibitors reduce the resistance against which the heart beats and are used to manage and prevent heart failure.

They are used to treat heart attack patients whose hearts do not pump well or who have symptoms of heart failure. Taken orally, they include Altace, Capoten, Lotensin, Monopril, Prinivil, Vasotec, and Zestril. Angiotensin receptor blockers, such as losartan (Cozaar) may substitute. **Diuretics** can help get rid of excess fluids that sometimes accumulate when the heart is not pumping effectively. Usually taken orally, they cause the body to dispose of fluids through urination. Common diuretics include: bumetanide (Bumex), chlorthalidone (Hygroton), chlorothiazide (Diuril), furosemide (Lasix), hydrochlorothiazide (HydroDIRUIL, Esidrix), spironolactone (Aldactone), and triamterene (Dyrenium).

Percutaneous transluminal coronary angioplasty and coronary artery bypass surgery are invasive revascularization procedures that open blocked coronary arteries and improve blood flow. They usually are performed only on patients for whom clot-dissolving drugs do not work, or who have poor **exercise** stress tests, poor left ventricular function, or **ischemia**. Generally, angioplasty is performed before coronary artery bypass surgery.

Percutaneous transluminal coronary angioplasty, usually called coronary angioplasty, is a non-surgical procedure in which a catheter (a tiny plastic tube) tipped with a balloon is threaded from a blood vessel in the thigh or arm into the blocked artery. The balloon is inflated and compresses the plaque to enlarge the blood vessel and open the blocked artery. The balloon is then deflated and the catheter is removed. Coronary angioplasty is performed in a hospital and generally requires a two-day stay. It is successful about 90% of the time. For one third of patients, the artery narrows again within six months after the procedure. The procedure can be repeated. It is less invasive and less expensive than coronary artery bypass surgery.

In coronary artery bypass surgery, called bypass surgery, a detour is built around the coronary artery blockage with a healthy leg or chest wall artery or vein. The healthy vein then supplies oxygen-rich blood to the heart. Bypass surgery is major surgery appropriate for patients with blockages in two or three major coronary arteries or severely narrowed left main coronary arteries, as well as those who have not responded to other treatments. It is performed in a hospital under **general anesthesia** using a heart-lung machine to support the patient while the healthy vein is attached to the coronary artery. About 70% of patients who have bypass surgery experience full relief from **angina**; about 20% experience partial relief. Long term, symptoms recur in only about three or

four percent of patients per year. Five years after bypass surgery, survival expectancy is 90%, at 10 years it is about 80%, at 15 years it is about 55%, and at 20 years it is about 40%.

There are several experimental surgical procedures for unblocking coronary arteries under study including: **atherectomy**, where the surgeon shaves off and removes strips of plaque from the blocked artery; laser angioplasty, where a catheter with a laser tip is inserted to burn or break down the plaque; and insertion of a metal coil called a stent that can be implanted permanently to keep a blocked artery open.

Prognosis

The aftermath of a heart attack is often severe. Two-thirds of heart attack patients never recover fully. Within one year, 27% of men and 44% of women die. Within six years, 23% of men and 31% of women have another heart attack, 13% of men and 6% of women experience sudden death, and about 20% have heart failure. People who survive a heart attack have a chance of sudden death that is four to six times greater than others and a chance of illness and death that is two to nine times greater. Older women are more likely than men to die within a few weeks of a heart attack. In 2003, a new drug showed some promise in helping patients who have had a heart attack and developed heart failure. Called eplerenone, it lowered the death rate and risk of sudden death among patients tested.

Prevention

Many heart attacks can be prevented through a healthy lifestyle, which can reduce the risk of developing coronary artery disease. For patients who have already had a heart attack, a healthy lifestyle and carefully following doctor's orders can prevent another heart attack. A heart healthy lifestyle includes eating right, regular exercise, maintaining a healthy weight, no smoking, moderate drinking, no illegal drugs, controlling hypertension, and managing stress.

A healthy diet includes a variety of foods that are low in fat (especially saturated fat), low in cholesterol, and high in fiber; plenty of fruits and vegetables; and limited sodium. Some foods are low in fat but high in cholesterol, and some are low in cholesterol but high in fat. Saturated fat raises cholesterol, and, in excessive amounts, it increases the amount of the proteins in blood that form blood clots. Polyunsaturated and monounsaturated fats are relatively good for the

heart. Fat should comprise no more than 30 percent of total daily calories.

Cholesterol, a waxy, lipid-like substance, comes from eating foods such as meat, eggs, and other animal products. It also is produced in the liver. Soluble fiber can help lower cholesterol. Cholesterol should be limited to about 300 mg per day. Many popular lipid-lowering drugs can reduce LDL-cholesterol by an average of 25–30% when combined with a low-fat, low-cholesterol diet. Fruits and vegetables are rich in fiber, **vitamins**, and **minerals**. They are also low calorie and nearly fat free. Vitamin C and beta-carotene, found in many fruits and vegetables, keep LDL-cholesterol from turning into a form that damages coronary arteries. Excess sodium can increase the risk of high blood pressure. Many processed foods contain large amounts of sodium, which should be limited to a daily intake of 2,400 mg—about the amount in a teaspoon of salt.

The “Food Guide” Pyramid developed by the U.S. Departments of Agriculture and Health and Human Services provides easy to follow guidelines for daily heart-healthy eating: six to 11 servings of bread, cereal, rice, and pasta; three to five servings of vegetables; two to four servings of fruit; two to three servings of milk, yogurt, and cheese; and two to three servings of meat, poultry, fish, dry beans, eggs, and nuts. Fats, oils, and sweets should be used sparingly.

Regular aerobic exercise can lower blood pressure, help control weight, and increase HDL (“good”) cholesterol. It may keep the blood vessels more flexible. Moderate intensity aerobic exercise lasting about 30 minutes four or more times per week is recommended for maximum heart health, according to the Centers for Disease Control and Prevention and the American College of Sports Medicine. Three 10-minute exercise periods also are beneficial. Aerobic exercise—activities such as walking, jogging, and cycling—uses the large muscle groups and forces the body to use oxygen more efficiently. It also can include everyday activities such as active gardening, climbing stairs, or brisk housework.

Maintaining a desirable body weight also is important. About one-fourth of all Americans are overweight, and nearly one-tenth are obese, according to the Surgeon General’s Report on Nutrition and Health. People who are 20% or more over their ideal body weight have an increased risk of developing coronary artery disease. Losing weight can help reduce total and LDL cholesterol, reduce triglycerides, and boost relative levels of HDL cholesterol. It also may reduce blood pressure.

Smoking has many adverse effects on the heart. It increases the heart rate, constricts major arteries, and can create irregular heartbeats. It also raises blood pressure, contributes to the development of plaque, increases the formation of blood clots, and causes blood platelets to cluster and impede blood flow. Heart damage caused by smoking can be repaired by quitting—even heavy smokers can return to heart health. Several studies have shown that ex-smokers face the same risk of heart disease as non-smokers within five to 10 years of quitting.

Drinking should be done in moderation. Modest consumption of alcohol can actually protect against coronary artery disease. This is believed to be because alcohol raises HDL cholesterol levels. The American Heart Association defines moderate consumption as one ounce of alcohol per day—roughly one cocktail, one 8-ounce glass of wine, or two 12-ounce glasses of beer. A study released in 2003 reported that risk of heart attack in men was reduced 30% to 35 % if they drank moderate amounts of alcoholic beverages three or four times a week. In some people, however, moderate drinking can increase risk factors for heart disease, such as raising blood pressure. Excessive drinking is always bad for the heart. It usually raises blood pressure, and can poison the heart and cause abnormal heart rhythms or even heart failure. Illegal drugs, like **cocaine**, can seriously harm the heart and should never be used.

High blood pressure, one of the most common and serious risk factors for coronary artery disease, can be completely controlled through lifestyle changes and medication. People with moderate hypertension may be able to control it through lifestyle changes such as reducing sodium and fat, exercising regularly, managing stress, quitting smoking, and drinking alcohol in moderation. If these changes do not work, and for people with severe hypertension, there are eight types of drugs that provide effective treatment.

Stress management means controlling mental and physical reactions to life’s irritations and challenges. Techniques for controlling stress include: taking life more slowly, spending time with family and friends, thinking positively, getting enough sleep, exercising, and practicing relaxation techniques.

Daily aspirin therapy has been proven to help prevent blood clots associated with atherosclerosis. It also can prevent heart attacks from recurring, prevent heart attacks from being fatal, and lower the risk of strokes.

KEY TERMS

Angina—Chest pain that happens when diseased blood vessels restrict the flow of blood to the heart. Angina often is the first symptom of coronary artery disease.

Atherosclerosis—A process in which the walls of the coronary arteries thicken due to the accumulation of plaque in the blood vessels. Atherosclerosis is the cause of coronary artery disease.

Coronary arteries—The two arteries that provide blood to the heart. The coronary arteries surround the heart like a crown, coming out of the aorta, arching down over the top of the heart, and dividing into two branches. These are the arteries where coronary artery disease occurs.

Myocardial infarction—The technical term for heart attack. Myocardial means heart muscle and infarction means death of tissue from lack of oxygen.

Plaque—A deposit of fatty and other substances that accumulate in the lining of the artery wall.

Resources

PERIODICALS

“First New Blood Test to Evaluate Heart Attacks.” *Biomedical Market Newsletter* January-February 2003: 42.

“Heart Attacks Reduced by Alcohol.” *The Lancet* January 11, 2003: 149.

Kirn, Timothy F. “FDA Probes Ephedra, Proposes Warning Label (Risk of Heart Attack, Seizure, Stroke).” *Clinical Psychiatry News* April 2003: 49.

Pitt, Bertram, et al. “Eplerenone, a Selective Aldosterone Blocker, in Patients with Left Ventricular Dysfunction after Myocardial Infarction.” *The New England Journal of Medicine* April 3, 2003: 1309-1313.

Stephenson, Joan. “FDA Orders Estrogen Safety Warnings: Agency Offers Guidance for HRT Use.” *JAMA* February 5, 2003: 537.

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

National Heart, Lung and Blood Institute. PO Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

Texas Heart Institute. Heart Information Service. PO Box 20345, Houston, TX 77225-0345. <<http://www.tmc.edu/thi>>.

Lori De Milto
Teresa G. Odle

Heart block

Definition

Heart block refers to a delay in the normal flow of electrical impulses that cause the heart to beat. They are further classified as first-, second-, or third-degree block.

Description

The muscles of the heart contract in a rhythmic order for each heart beat, because electrical impulses travel along a specific route called the conduction system. The main junction of this system is called the atrioventricular node (AV node). Just as on a highway, there are occasionally some delays getting the impulse from one point to another. These delays are classified according to their severity.

In first-degree heart block, the signal is just slowed down a little as it travels along the defective part of the conduction system so that it arrives late traveling from the atrium to the ventricle.

In second-degree heart block, not every impulse reaches its destination. The block may affect every other beat, every second or third beat, or be very rare. If the blockage is frequent, it results in an overall slowing of the heart called bradycardia.

Third-degree block, also called complete heart block, is the most serious. When no signals can travel through the AV node, the heart uses its backup impulse generator in the lower portion of the heart. Though this impulse usually keeps the heart from stopping entirely, it is too slow to be an effective pump.

Causes and symptoms

First-degree heart block is fairly common. It is seen in teenagers, young adults and in well-trained athletes. The condition may be caused by rheumatic fever, some types of heart disease and by some drugs. First-degree heart block produces no symptoms.

Some cases of second-degree heart block may benefit from an artificial pace-maker. Second-degree block can occasionally progress to third-degree.

Third-degree heart block is a serious condition that affects the heart's ability to pump blood effectively. Symptoms include **fainting**, **dizziness** and sudden **heart failure**. If the ventricles beat more than 40 times per minute, symptoms are not as severe, but include tiredness, low blood pressure on standing, and **shortness of breath**.

KEY TERMS

Atrioventricular node (AV node)—Highly specialized area of the heart muscle which transmits electrical impulses.

Bradycardia—A slow heart rate, usually under 60 beats per minute.

Young children who have received a forceful blunt chest injury, can experience first-, or second-degree heart block.

Diagnosis

Diagnosis of first-, and second-degree heart block is made by observing it on an electrocardiograph (ECG).

Third-degree heart block usually results in symptoms such as fainting, dizziness and sudden heart failure, which require immediate medical care. A physical exam and ECG confirm the presence of heart block.

Treatment

Some second- and almost all third-degree heart blocks require an artificial pacemaker. In an emergency, a temporary pacemaker can be used until an implanted device is advisable. Most people need the pacemaker for the rest of their lives.

Prognosis

Most people with first- and second-degree heart block don't even know they have it. For people with third-degree block, once the heart has been restored to its normal, dependable rhythm, most people live full and comfortable lives.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

Dorothy Elinor Stonely

Heart catheterization see **Cardiac catheterization**

Heart failure

Definition

Heart failure is a condition in which the heart has lost the ability to pump enough blood to the body's tissues. With too little blood being delivered, the organs and other tissues do not receive enough oxygen and nutrients to function properly.

Description

According to the American Heart Association, about 4.9 million Americans are living with congestive heart failure. Of these, 2.5 million are males and 2.4 million are females. Ten of every 1,000 people over age 65 have this condition. There are about 400,000 new cases each year.

Heart failure happens when a disease affects the heart's ability to deliver enough blood to the body's tissues. Often, a person with heart failure may have a buildup of fluid in the tissues, called **edema**. Heart failure with this kind of fluid buildup is called congestive heart failure. Where edema occurs in the body depends on the part of the heart that is affected by heart failure. Heart failure caused by abnormality of the lower left chamber of the heart (left ventricle) means that the left ventricle cannot pump blood out to the body as fast as it returns from the lungs. Because blood cannot get back to the heart, it begins to back up in the blood vessels of the lungs. Some of the fluid in the blood is forced into the breathing space of the lungs, causing **pulmonary edema**. A person with pulmonary edema has shortness of breath, which may be acute, severe and life threatening. A person with congestive heart failure feels tired because not enough blood circulates to supply the body's tissues with the oxygen and nutrients they need. Abnormalities of the heart structure and rhythm also can be responsible for left ventricular congestive heart failure.

In right-sided heart failure, the lower right chamber of the heart (right ventricle) cannot pump blood to the lungs as fast as it returns from the body through the veins. Blood then engorges the right side of the heart and the veins. Fluid backed up in the veins is forced out into the tissues, causing swelling (edema), usually in the feet and legs. Congestive heart failure of the right ventricle often is caused by abnormalities of the heart valves and lung disorders.

When the heart cannot pump enough blood, it tries to make up for this by becoming larger. By becoming enlarged (hypertrophic) the ventricle can

contract more strongly and pump more blood. When this happens, the heart chamber becomes larger and the muscle in the heart wall becomes thicker. The heart also compensates by pumping more often to improve blood output and circulation. The kidneys try to compensate for a failing heart by retaining more salt and water to increase the volume of blood. This extra fluid also can cause edema. Eventually, as the condition worsens over time these measures are not enough to keep the heart pumping enough blood needed by the body. Kidneys often weaken under these circumstances, further aggravating the situation and making therapy more difficult.

For most people, heart failure is a chronic disease with no cure. However, it can be managed and treated with medicines and changes in diet, exercise, and lifestyle habits. **Heart transplantation** is considered in some cases.

Causes and symptoms

The most common causes of heart failure are:

- **coronary artery disease** and heart attack (which may be “silent”)
- **cardiomyopathy**
- high blood pressure (**hypertension**)
- heart valve disease
- congenital heart disease
- **alcoholism** and drug **abuse**

The most common cause of heart failure is coronary artery disease. In coronary artery disease, the arteries supplying blood to the heart become narrowed or blocked. When blood flow to an area of the heart is completely blocked, the person has a **heart attack**. Some heart attacks go unrecognized. The heart muscle suffers damage when its blood supply is reduced or blocked. If the damage affects the heart’s ability to pump blood, heart failure develops.

Cardiomyopathy is a general term for disease of the heart muscle. Cardiomyopathy may be caused by coronary artery disease and various other heart problems. Sometimes the cause of cardiomyopathy cannot be found. In these cases the heart muscle disease is called **idiopathic cardiomyopathy**. Whatever the cause, cardiomyopathy can weaken the heart, leading to heart failure.

High blood pressure is another common cause of heart failure. High blood pressure makes the heart work harder to pump blood. After a while, the heart

cannot keep up and the symptoms of heart failure develop.

Defects of the heart valves, congenital heart diseases, alcoholism, and drug abuse cause damage to the heart that can all lead to heart failure.

A person with heart failure may experience the following:

- **shortness of breath**
- frequent coughing, especially when lying down
- swollen feet, ankles, and legs
- abdominal swelling and **pain**
- fatigue
- **dizziness** or **fainting**
- sudden **death**

A person with left-sided heart failure may have shortness of breath and coughing caused by the fluid buildup in the lungs. Pulmonary edema may cause the person to **cough** up bubbly phlegm that contains blood. With right-sided heart failure, fluid build-up in the veins and body tissues causes swelling in the feet, legs, and abdomen. When body tissues, such as organs and muscles, do not receive enough oxygen and nutrients they cannot function as well, leading to tiredness and dizziness.

Diagnosis

Diagnosis of heart failure is based on:

- symptoms
- medical history
- **physical examination**
- **chest x ray**
- electrocardiogram (ECG; also called EKG)
- other imaging tests
- **cardiac catheterization**

A person’s symptoms can provide important clues to the presence of heart failure. Shortness of breath while engaging in activities and episodes of shortness of breath that wake a person from sleep are classic symptoms of heart failure. During the physical examination, the physician listens to the heart and lungs with a stethoscope for telltale signs of heart failure. Irregular heart sounds, “gallops,” a rapid heart rate, and murmurs of the heart valves may be heard. If there is fluid in the lungs a crackling sound may be heard. Rapid breathing or other changes in breathing may also be present. Patients with heart failure also may have a rapid pulse.

By pressing on the abdomen, the physician can feel if the liver is enlarged. The skin of the fingers and toes may have a bluish tint and feel cool if not enough oxygen is reaching them.

A chest x ray can show if there is fluid in the lungs and if the heart is enlarged. Abnormalities of heart valves and other structures also may be seen on chest x ray.

An electrocardiogram gives information on the heart rhythm and the size of the heart. It can show if the heart chamber is enlarged and if there is damage to the heart muscle from blocked arteries.

Besides chest x ray, other imaging tests may help make a diagnosis. **Echocardiography** uses sound waves to make images of the heart. These images can show if the heart wall or chambers are enlarged and if there are any abnormalities of the heart valves. An echocardiogram also can be used to find out how much blood the heart is pumping. It determines the amount of blood in the ventricle (ventricular volume) and the amount of blood the ventricle pumps each time it beats (called the ejection fraction). A healthy heart pumps at least one-half the amount of blood in the left ventricle with each heartbeat. Radionuclide ventriculography also measures the ejection fraction by imaging with very low doses of an injected radioactive substance as it travels through the heart.

A new test that measures the level of a particular hormone in the blood was introduced in 2003 and researchers said the test may be useful for testing for heart failure in physicians' offices because it could provide results in 15 minutes.

Cardiac catheterization involves using a small tube (catheter) that is inserted through a blood vessel into the heart. It is used to measure pressure in the heart and the amount of blood pumped by the heart. This test can help find abnormalities of the coronary arteries, heart valves, and heart muscle, and other blood vessels. Combined with echocardiography and other tests, cardiac catheterization can help find the cause of heart failure. It is not always necessary, however.

Treatment

Heart failure usually is treated with lifestyle changes and medicines. Sometimes surgery is needed to correct abnormalities of the heart or heart valves. Heart transplantation is a last resort to be considered in certain cases.

Dietary changes to maintain proper weight and reduce salt intake may be needed. Reducing salt intake

helps to lessen swelling in the legs, feet, and abdomen. Appropriate **exercise** also may be recommended, but it is important that heart failure patients only begin an exercise program with the advice of their doctors. Walking, bicycling, swimming, or low-impact aerobic exercises may be recommended. There are good heart **rehabilitation** programs at most larger hospitals.

Other lifestyle changes that may reduce the symptoms of heart failure include stopping smoking or other tobacco use, eliminating or reducing alcohol consumption, and not using harmful drugs.

One or more of the following types of medicines may be prescribed for heart failure:

- **diuretics**
- digitalis
- **vasodilators**
- beta blockers
- angiotensin converting enzyme inhibitors (ACE inhibitors)
- angiotensin receptor blockers (ARBs)
- calcium channel blockers

Diuretics help eliminate excess salt and water from the kidneys by making patients urinate more often. This helps reduce the swelling caused by fluid buildup in the tissues. Digitalis helps the heart muscle to have stronger pumping action. Vasodilators, ACE inhibitors, ARBs, and calcium channel blockers lower blood pressure and expand the blood vessels so blood can move more easily through them. This action makes it easier for the heart to pump blood through the vessels. Cholesterol-lowering drugs called statins can help prevent death from heart failure. A 2003 study showed a 62% drop in mortality rate among patients with severe heart failure who took statin therapy.

In 2003, a new noninvasive procedure was being tested for patients with congestive heart failure. Called enhanced external counterpulsation (EECP), it consisted of inflating three sets of pneumatic cuffs attached to the patient's legs. The therapy had positive effects on the blood pressure and reduced frequency of episodes of **angina** (pain) in a clinical trial by as much as 70%.

Surgery is used to correct certain heart conditions that cause heart failure. Congenital heart defects and abnormal heart valves can be repaired with surgery. Blocked coronary arteries usually can be treated with **angioplasty** or coronary artery bypass surgery.

KEY TERMS

Angioplasty—A technique for treating blocked coronary arteries by inserting a catheter with a tiny balloon at the tip into the artery and inflating it.

Angiotensin-converting enzyme (ACE) inhibitor—A drug that relaxes blood vessel walls and lowers blood pressure.

Arrhythmias—Abnormal heartbeat.

Atherosclerosis—Buildup of a fatty substance called a plaque inside blood vessels.

Calcium channel blocker—A drug that relaxes blood vessels and lowers blood pressure.

Cardiac catheterization—A diagnostic test for evaluating heart disease; a catheter is inserted into an artery and passed into the heart.

Cardiomyopathy—Disease of the heart muscle.

Catheter—A thin, hollow tube.

Congenital heart defects—Abnormal formation of structures of the heart or of its major blood vessels present at birth.

Congestive heart failure—A condition in which the heart cannot pump enough blood to supply the body's tissues with sufficient oxygen and nutrients; back up of blood in vessels and the lungs causes buildup of fluid (congestion) in the tissues.

Coronary arteries—Arteries that supply blood to the heart muscle.

Coronary artery bypass—Surgical procedure to reroute blood around a blocked coronary artery.

Coronary artery disease—Narrowing or blockage of coronary arteries by atherosclerosis.

Digitalis—A drug that helps the heart muscle to have stronger pumping action.

Diuretic—A type of drug that helps the kidneys eliminate excess salt and water.

Edema—Swelling caused by fluid buildup in tissues.

Ejection fraction—A measure of the portion of blood that is pumped out of a filled ventricle.

Heart valves—Valves that regulate blood flow into and out of the heart chambers.

Hypertension—High blood pressure.

Hypertrophic—Enlarged.

Idiopathic cardiomyopathy—Cardiomyopathy without a known cause.

Pulmonary edema—Buildup of fluid in the tissue of the lungs.

Vasodilator—Any drug that relaxes blood vessel walls.

Ventricles—The two lower chambers of the heart.

With severe heart failure, the heart muscle may become so damaged that available treatments do not help. Patients with this stage of heart failure are said to have end-stage heart failure. Heart transplant usually is considered for patients with end-stage heart failure when all other treatments have stopped working.

Prognosis

Most patients with mild or moderate heart failure can be successfully treated with dietary and exercise programs and the right medications. In fact, in 2003, the American Heart Association said that even those awaiting heart transplants could benefit from exercise. Many people are able to participate in normal daily activities and lead relatively active lives.

Patients with severe heart failure may eventually have to consider heart transplantation. Approximately 50% of patients diagnosed with congestive heart

failure live for five years with the condition. Women with heart failure usually live longer than men with heart failure.

Prevention

Heart failure usually is caused by the effects of some type of heart disease. The best way to try to prevent heart failure is to eat a healthy diet and get regular exercise, but many causes of heart failure cannot be prevented. People with risk factors for coronary disease (such as high blood pressure and **high cholesterol** levels) should work closely with their physician to reduce likelihood of heart attack and heart failure.

Heart failure sometimes can be avoided by identifying and treating any conditions that might lead to heart disease. These include high blood pressure, alcoholism, and coronary artery disease. Regular blood pressure checks and obtaining immediate medical care for symptoms of coronary artery disease, such

as chest pain, will help to get these conditions found and treated early, before they can damage the heart muscle.

A 2003 initiative called OPTIMIZE H-F was aimed at preventing severe heart failure and deaths among patients discharge from hospitals. The project created a registry or database of patients with heart failure that could be shared among hospitals. Finally, diagnosing and treating heart failure before the heart becomes severely damaged can improve the prognosis. With proper treatment, many patients may continue to lead active lives for a number of years.

Resources

PERIODICALS

"Even Heart Failure Patients Should Exercise." *Clinician Reviews* April 2003: 50-52.

Jancin, Bruce. "Noninvasive Procedure Eyed for Heart Failure: Enhanced External Counterpulsation." *Family Practice News* June 1, 2003: 12.

"New Care Initiative to Improve Outcomes for Heart Failure Patients." *Heart Disease Weekly* April 20, 2003: 45.

"Rapid Congestive Heart Failure Test a Useful Tool in Physician Offices." *Heart Disease Weekly* June 15, 2003: 19.

Zoler, Michael N. "Heart Failure Deaths Plunge with Statins." *Internal Medicine News* April 15, 2003: 35-41.

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

National Heart, Lung and Blood Institute. PO Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

Texas Heart Institute. Heart Information Service. PO Box 20345, Houston, TX 77225-0345. <<http://www.tmc.edu/thi>>.

Toni Rizzo
Teresa G. Odle

Heart murmurs

Definition

A heart murmur is an abnormal, extra sound during the heartbeat cycle made by blood moving through the heart and its valves. It is detected by the physician's examination using a stethoscope.

Description

A heart which is beating normal makes two sounds, "lubb" when the valves between the atria and ventricles close, and "dupp" when the valves between the ventricles and the major arteries close. A heart murmur is a series of vibratory sounds made by turbulent blood flow. The sounds are longer than normal heart sounds and can be heard between the normal sounds of the heart.

Heart murmurs are common in children and can also result from heart or valve defects. Nearly two thirds of heart murmurs in children are produced by a normal heart and are harmless. This type of heart murmur is usually called an "innocent" heart murmur. It can also be called "functional" or "physiologic." Innocent heart murmurs are usually very faint, intermittent, and occur in a small area of the chest. Pathologic heart murmurs may indicate the presence of a serious heart defect. They are louder, continual, and may be accompanied by a click or gallop.

Some heart murmurs are continually present; others happen only when the heart is working harder than usual, including during **exercise** or certain types of illness. Heart murmurs can be diastolic or systolic. Those which occur during relaxation of the heart between beats are called diastolic murmurs. Those which occur during contraction of the heart muscle are called systolic murmurs. The characteristics of the murmur may suggest specific alterations in the heart or its valves.

Causes and symptoms

Innocent heart murmurs are caused by blood flowing through the chambers and valves of the heart or the blood vessels near the heart. Sometimes anxiety, stress, **fever**, anemia, overactive thyroid, and **pregnancy** will cause innocent murmurs that can be heard by a physician using a stethoscope. Pathologic heart murmurs, however, are caused by structural abnormalities of the heart. These include defective heart valves or holes in the walls of the heart. Valve problems are more common. Valves that do not open completely cause blood to flow through a smaller opening than normal, while those that do not close properly may cause blood to go back through the valve. A hole in the wall between the left and right sides of the heart, called a septal defect, can cause heart murmurs. Some septal defects close on their own; others require surgery to prevent progressive damage to the heart.

The symptoms of heart murmurs differ depending on the cause of the heart murmur. Innocent heart

murmurs and those which do not impair the function of the heart have no symptoms. Murmurs that are due to severe abnormalities of a heart valve may cause shortness of breath, **dizziness**, chest pains, **palpitations**, and lung congestion.

Diagnosis

Heart murmurs can be heard when a physician listens to the heart through a stethoscope during a regular check-up. Very loud heart murmurs and those with clicks or extra heart sounds should be evaluated further. Infants with heart murmurs who do not thrive, eat, or breath properly and older children who lose consciousness suddenly or are intolerant to exercise should also be evaluated. If the murmur sounds suspicious, the physician may order a chest x ray, an electrocardiogram, and an echocardiogram.

An electrocardiogram (ECG) shows the heart's activity and may reveal muscle thickening, damage, or a lack of oxygen. Electrodes covered with conducting jelly are placed on the patient's chest, arms, and legs. They send impulses of the heart's activity through a monitor (oscilloscope) to a recorder which traces them on paper. The test takes about 10 minutes and is commonly performed in a physician's office. An exercise ECG can reveal additional information.

An echocardiogram (cardiac ultrasound), may be ordered to identify a structural problem that is causing the heart murmur. An echocardiogram uses sound waves to create an image of the heart's chambers and valves. The technician applies gel to a hand-held transducer then presses it against the patient's chest. The sound waves are converted into an image that can be displayed on a monitor. Performed in a cardiology outpatient diagnostic laboratory, the test takes 30 minutes to an hour.

Treatment

Innocent heart murmurs do not affect the patient's health and require no treatment. Heart murmurs due to septal defects may require surgery. Those due to valvular defects may require antibiotics to prevent infection during certain surgical or dental procedures. Severely damaged or diseased valves can be repaired or replaced through surgery.

Alternative treatment

If a heart murmur requires surgical treatment, there are no alternative treatments, although there are alternative therapies that are helpful for pre- and

KEY TERMS

Atria—The upper two chambers of the heart.

Echocardiogram—A non-invasive ultrasound test that shows an image of the inside of the heart. An echocardiogram can be performed to identify any structural problems which cause a heart murmur.

Electrocardiogram—A test that shows the electrical activity of the heart by placing electronic sensors on the patient. This test can be used to confirm the presence of a heart murmur.

Pathologic—Characterized by disease or the structural and functional changes due to disease. Pathologic heart murmurs may indicate a heart defect.

Ventricles—The lower two chambers of the heart.

post-surgical support of the patient. If the heart murmur is innocent, heart activity can be supported using the herb hawthorn (*Crataegus laevigata* or *C. oxyacantha*) or coenzyme Q10. These remedies improve heart contractility and the heart's ability to use oxygen. If the murmur is valvular in origin, herbs that act like **antibiotics** as well as options that build resistance to infection in the valve areas may be considered.

Prognosis

Most children with innocent heart murmurs grow out of them by the time they reach adulthood. Severe causes of heart murmurs may progress to severe symptoms and **death**.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

Texas Heart Institute. Heart Information Service. PO Box 20345, Houston, TX 77225-0345. <<http://www.tmc.edu/thi>>.

Lori De Milto

Heart muscle infection see **Myocarditis**

Heart scan see **Echocardiography**

Heart septal defect see **Atrial septal defect**

Heart sonogram see **Echocardiography**

Heart surgery for congenital defects

Definition

A variety of surgical procedures that are performed to repair the many types of heart defects that may be present at birth.

Purpose

Heart surgery for congenital defects is performed to repair a defect as much as possible and improve the flow of blood and oxygen to the body. While congenital heart defects vary in their severity, most require surgery. Surgery is recommended for congenital heart defects that result in a lack of oxygen, a poor quality of life, or a patient who does not thrive. Some types of congenital heart defects that don't cause symptoms are treated surgically because they can lead to serious complications.

Precautions

There are many types of surgery for congenital heart defects and many considerations in the decision to operate. The patient's cardiologist or surgeon will discuss these issues on an individual basis.

Description

There are many types of congenital heart defects. Most obstruct the flow of blood in the heart, or the vessels near it, or cause an abnormal flow of blood through the heart. Rarer types include newborns born with one ventricle, one side of the heart that is not completely formed, or the pulmonary artery and the aorta coming out of the same ventricle. Most congenital heart defects require surgery during infancy or childhood. Recommended ages for surgery for the most common congenital heart defects are:

- **atrial septal defects:** during the preschool years
- **patent ductus arteriosus:** between ages one and two
- **coarctation of the aorta:** in infancy, if it's symptomatic, at age four otherwise
- **tetralogy of Fallot:** age varies, depending on the patient's signs and symptoms
- **transposition of the great arteries:** often in the first weeks after birth, but before the patient is 12 months old

Surgical procedures seek to repair the defect as much as possible and restore circulation to as close to normal as possible. Sometimes, multiple, serial, surgical procedures are necessary. Smaller congenital heart defects can now be repaired in a cardiac catheterization lab instead of an operating room. Catheterization procedures include balloon atrial septostomy and **balloon valvuloplasty**. Surgical procedures include arterial switch, Damus-Kaye-Stansel procedure, Fontan procedure, Ross procedure, shunt procedure, and venous switch or intra-atrial baffle.

Catheterization procedures

Balloon atrial septostomy and balloon valvuloplasty are cardiac catheterization procedures. **Cardiac catheterization** procedures can save the lives of critically ill neonates and in some cases eliminate or delay more invasive surgical procedures. It is expected that catheterization procedures will continue to replace more types of surgery for congenital heart defects in the future. A thin tube called a catheter is inserted into an artery or vein in the leg, groin, or arm and threaded into the area of the heart which needs repair. The patient receives a local anesthetic at the insertion site and is awake but sedated during the procedure.

BALLOON ATRIAL SEPTOSTOMY. Balloon atrial septostomy is the standard procedure for correcting transposition of the great arteries; it is sometimes used in patients with mitral, pulmonary, or tricuspid atresia (atresia is a defect that causes the blood to carry too little oxygen to the body). Balloon atrial septostomy enlarges the atrial opening. A special balloon-tipped catheter is inserted into the right atrium and inflated to create a large opening in the atrial septum.

BALLOON VALVULOPLASTY. Balloon valvuloplasty uses a balloon-tipped catheter to open a narrowed heart valve, improving the flow of blood. It is the procedure of choice in pulmonary stenosis and is sometimes used in aortic stenosis. Balloons made of plastic polymers are placed at the end of the catheter and inflated to relieve the obstruction in the heart valve. Long-term results are excellent in most cases. The operative **death** rate is 2–4%.

Surgical procedures

These procedures are performed under **general anesthesia**. Some require the use of a heart-lung machine, which cools the body to reduce the need for oxygen and takes over for the heart and lungs during the procedure.

ARTERIAL SWITCH. Arterial switch is performed to correct transposition of the great arteries, where the position of the pulmonary artery and the aorta are reversed. The procedure involves connecting the aorta to the left ventricle and the pulmonary artery to the right ventricle.

DAMUS-KAYE-STANSEL PROCEDURE. Transposition of the great arteries can also be corrected by the Damus-Kaye-Stansel procedure, in which the pulmonary artery is cut in two and connected to the ascending aorta and right ventricle.

FONTAN PROCEDURE. For tricuspid atresia and pulmonary atresia, the Fontan procedure connects the right atrium to the pulmonary artery directly or with a conduit, and the atrial defect is closed. Survival is over 90%.

PULMONARY ARTERY BANDING. Pulmonary artery banding is narrowing the pulmonary artery with a band to reduce blood flow and pressure in the lungs. It is used for ventricular septal defect, atrioventricular canal defect, and tricuspid atresia. Later, the band can be removed and the defect corrected with open heart surgery.

ROSS PROCEDURE. To correct aortic stenosis, the Ross procedure grafts the pulmonary artery to the aorta.

SHUNT PROCEDURE. For Tetralogy of Fallot, tricuspid atresia, or pulmonary atresia, the shunt procedure creates a passage between blood vessels, sending blood into parts of the body that need it.

VENOUS SWITCH. For transposition of the great arteries, venous switch creates a tunnel inside the atria to re-direct oxygen-rich blood to the right ventricle and aorta and venous blood to the left ventricle and pulmonary artery.

OTHER TYPES OF SURGERY. These surgical procedures are also used to treat common congenital heart defects. A medium to large ventricular or **atrial septal defect** can be closed by suturing it or covering it with a Dacron patch. For patent ductus arteriosus, surgery consists of dividing the ductus into two and tying off the ends. If performed within the patient's first few years, there is practically no risk associated with this operation. Surgery for coarctation of the aorta involves opening the chest wall, removing the defect, and reconnecting the ends of the aorta. If the defect is too long to be reconnected, a Dacron graft is used to replace the missing piece. In uncomplicated cases, the risk of the operation is 1–2%.

Preparation

Before surgery for congenital heart defects, the patient will receive a complete evaluation, which includes a physical exam, a detailed family history, a chest x ray, an electrocardiogram, an echocardiogram, and usually cardiac catheterization. For six to eight hours before the surgery, the patient cannot eat or drink anything. An electrocardiogram shows the heart's activity and may reveal a lack of oxygen. Electrodes covered with conducting jelly are placed on the patient's chest, arms, and legs and the heart's impulses are traced on paper. An echocardiogram uses sound waves to create an image of the heart's chambers and valves. Gel is applied to a hand-held transducer and then pressed against the patient's chest. Cardiac catheterization is an invasive diagnostic technique used to evaluate the heart in which a long tube is inserted into a blood vessel and guided into the heart. A contrast solution is injected to make the heart visible on x rays.

Aftercare

After heart surgery for congenital defects, the patient goes to an intensive care ward where he or she is connected to a variety of tubes and monitors, including a ventilator. Patients are monitored every 15 minutes until vital signs are stable. Heart sounds, oxygenation, and the electrocardiogram are monitored. Chest tubes will be checked to ensure that they're draining properly and there is no hemorrhage. **Pain** medications will be administered. Complications such as **stroke**, lung **blood clots**, and reduced blood flow to the kidneys will be watched for. After the ventilator and breathing tube are removed, **chest physical therapy** and exercises to improve circulation will be started.

Risks

Complications from heart surgery for congenital defects can be severe. They include shock, congestive **heart failure**, lack of oxygen or too much carbon dioxide in the blood, irregular heartbeat, stroke, infection, kidney damage, lung blood clot, low blood pressure, hemorrhage, cardiac arrest, and death.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

KEY TERMS

Atresia—A congenital defect in which the blood pumped through the body has too little oxygen. In tricuspid atresia, the baby lacks a tricuspid valve. In pulmonary atresia, a pulmonary valve is missing.

Coarctation of the aorta—A congenital defect in which severe narrowing or constriction of the aorta obstructs the flow of blood.

Congenital heart defects—Congenital means conditions which are present at birth. Congenital heart disease includes a variety of defects that babies are born with.

Patent ductus arteriosus—A congenital defect in which the temporary blood vessel connecting the left pulmonary artery to the aorta in the fetus doesn't close in the newborn.

Septal defects—These are holes in the septum, the muscle wall separating the right and left sides of the heart. Atrial septal defects are openings between the two upper heart chambers and ventricular septal defects are openings between the two lower heart chambers.

Stenosis—A narrowing of the heart's valves. This congenital defect can occur in the pulmonary (lung) or aortic (the main heart artery) valve.

Tetralogy of Fallot—A cyanotic defect in which the blood pumped through the body has too little oxygen. Tetralogy of Fallot includes four defects: a large hole between the ventricles, narrowing at or beneath the pulmonary valve, an overly muscular right ventricle, and an aorta over the large hole.

Transposition of the great arteries—A cyanotic defect in which the blood pumped through the body has too little oxygen. The pulmonary artery and the aorta are reversed.

Children's Health Information Network. 1561 Clark Drive, Yardley, PA 19067. (215) 493-3068. <<http://www.tchin.org>>.

Congenital Heart Anomalies Support, Education & Resources, Inc. 2112 North Wilkins Road, Swanton, OH 43558. (419) 825-5575. <<http://www.csun.edu/~hfmth006/chaser>>.

Texas Heart Institute. Heart Information Service. PO Box 20345, Houston, TX 77225-0345. <<http://www.tmc.edu/thi>>.

Lori De Milto

Heart transplantation

Definition

Heart transplantation, also called cardiac transplantation, is the replacement of a patient's diseased or injured heart with a healthy donor heart.

Purpose

Heart transplantation is performed on patients with end-stage heart failure or some other life-threatening heart disease. Before a doctor recommends heart transplantation for a patient, all other possible treatments for his or her disease must have been tried. The purpose of heart transplantation is to extend and improve the life of a person who would otherwise die from **heart failure**. Most patients who receive a new heart were so sick before transplantation that they could not live a normal life. Replacing a patient's diseased heart with a healthy, functioning donor heart often allows the recipient to return to normal daily activities.

Precautions

Because healthy donor hearts are in short supply, strict rules dictate who should or should not get a heart transplant. Patients who have conditions that might cause the new heart to fail should not have a heart transplant. Similarly, patients who may be too sick to survive the surgery or the side effects of the drugs they must take to keep their new heart working would not be good transplant candidates.

Patients who have any of the following conditions may not be eligible for heart transplantation:

- active infection
- pulmonary **hypertension**
- chronic lung disease with loss of more than 40% of lung function
- untreatable liver or **kidney disease**
- diabetes that has caused serious damage to vital organs
- disease of the blood vessels in the brain, such as a stroke
- serious disease of the arteries
- mental illness or any condition that would make a patient unable to take the necessary medicines on schedule
- continuing alcohol or drug abuse

Description

Patients with end-stage heart disease that threatens their life even after medical treatment may be considered for heart transplantation. Potential candidates must have a complete medical examination before they can be put on the transplant waiting list. Many types of tests are done, including blood tests, x rays, and tests of heart, lung, and other organ function. The results of these tests indicate to doctors how serious the heart disease is and whether or not a patient is healthy enough to survive the transplant surgery.

Organ waiting list

A person approved for heart transplantation is placed on the heart transplant waiting list of a heart transplant center. All patients on a waiting list are registered with the United Network for Organ Sharing (UNOS). UNOS has organ transplant specialists who run a national computer network that connects all the transplant centers and organ-donation organizations.

When a donor heart becomes available, information about it is entered into the UNOS computer and compared to information from patients on the waiting list. The computer program produces a list of patients ranked according to blood type, size of the heart, and how urgently they need a heart. Because the heart must be transplanted as quickly as possible, the list of local patients is checked first for a good match. After that, a regional list, and then a national list, are checked. The patient's transplant team of heart and transplant specialists makes the final decision as to whether a donor heart is suitable for the patient.

The transplant procedure

When a heart becomes available and is approved for a patient, it is packed in a sterile cold solution and rushed to the hospital where the recipient is waiting.

Heart transplant surgery involves the following basic steps:

- A specialist in cardiovascular anesthesia gives the patient **general anesthesia**.
- Intravenous **antibiotics** are usually given to prevent bacterial wound infections.
- The patient is put on a heart/lung machine, which performs the functions of the heart and lungs and pumps the blood to the rest of the body during surgery. This procedure is called cardiopulmonary bypass.

National Transplant Waiting List By Organ Type (June 2000)

Organ Needed	Number Waiting
Kidney	48,349
Liver	15,987
Heart	4,139
Lung	3,695
Kidney-Pancreas	2,437
Pancreas	942
Heart-Lung	212
Intestine	137

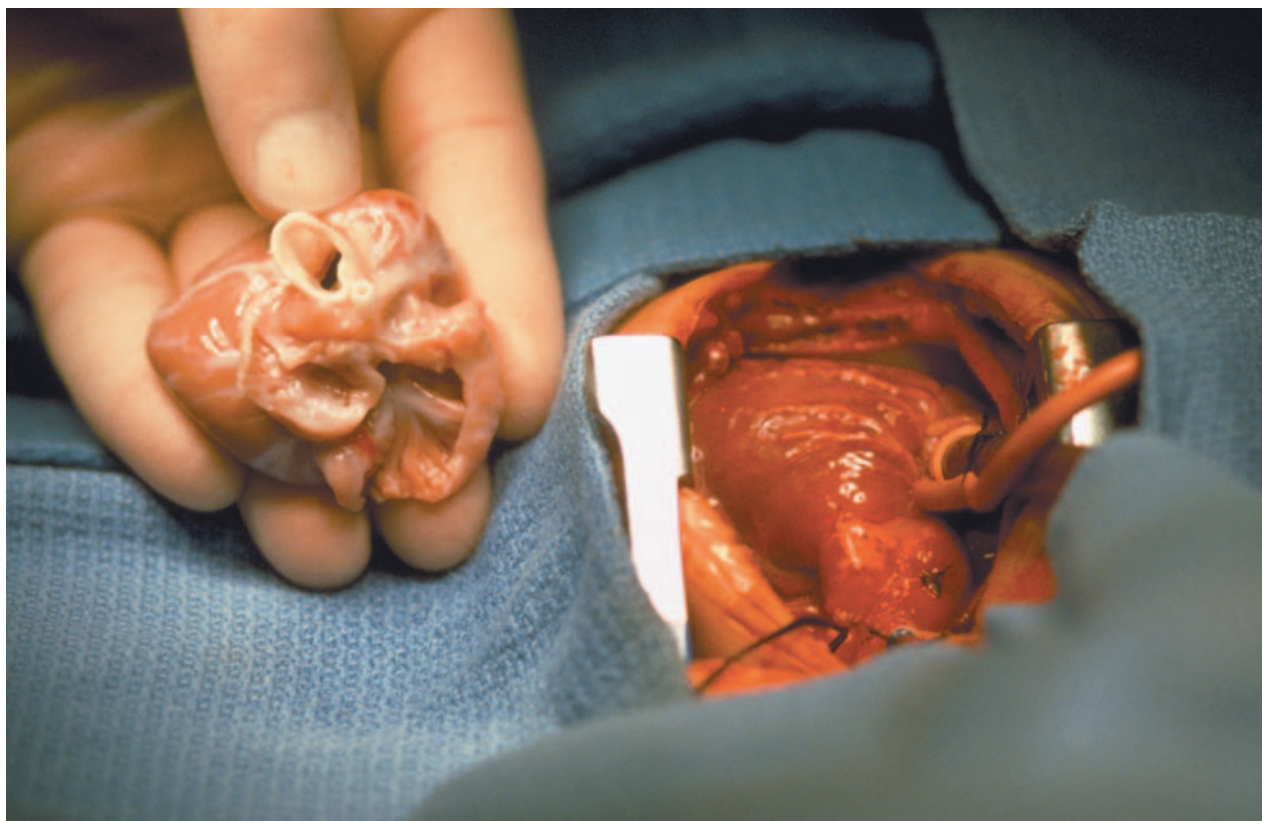
- After adequate blood circulation is established, the patient's diseased heart is removed.
- The donor heart is attached to the patient's blood vessels.
- After the blood vessels are connected, the new heart is warmed up and begins beating. If the heart does not begin to beat immediately, the surgeon may start it with an electrical shock.
- The patient is taken off the heart/lung machine.
- The new heart is stimulated to maintain a regular beat with medications for two to five days after surgery, until the new heart functions normally on its own.

Heart transplant recipients are given immunosuppressive drugs to prevent the body from rejecting the new heart. These drugs are usually started before or during the heart transplant surgery. Immunosuppressive drugs keep the body's immune system from recognizing and attacking the new heart as foreign tissue. Normally, immune system cells recognize and attack foreign or abnormal cells, such as bacteria, **cancer** cells, and cells from a transplanted organ. The drugs suppress the immune cells and allow the new heart to function properly. However, they can also allow infections and other adverse effects to occur to the patient.

Because the chance of rejection is highest during the first few months after the transplantation, recipients are usually given a combination of three or four immunosuppressive drugs in high doses during this time. Afterwards, they must take maintenance doses of immunosuppressive drugs for the rest of their lives.

Cost and insurance coverage

The total cost for heart transplantation varies, depending on where it is performed, whether transportation and lodging are needed, and on whether there are any complications. The costs for the surgery and



A comparison of the old and new hearts of Dylan Stork, the smallest heart transplant recipient in the world. Dylan was seven weeks old and weighed 5.5 pounds (2.5 kg) at the time of the operation. (Photograph by Alexander Tsiaras, Photo Researchers, Inc. Reproduced by permission.)

first year of care are estimated to be about \$250,000. The medical tests and medications after the first year cost about \$21,000 per year.

Insurance coverage for heart transplantation varies depending on the policy. Most commercial insurance companies pay a certain percentage of heart transplant costs. Medicare pays for heart transplants if the surgery is performed at Medicare-approved centers. Medicaid pays for heart transplants in 33 states and in the District of Colombia.

Preparation

Before patients are put on the transplant waiting list, their blood type is determined so a compatible donor heart can be found. The heart must come from a person with the same blood type as the patient, unless it is blood type O. A blood type O heart can be transplanted into a person with any type of blood.

A panel reactive antibodies (PRA) test is also done before heart transplantation. This test tells doctors whether or not the patient is at high risk for having a

hyperacute reaction against a donor heart. A hyperacute reaction is a strong immune response against the new heart that happens within minutes to hours after the new heart is transplanted. If the PRA shows that a patient has a high risk for this kind of reaction, then a crossmatch is done between a patient and a donor heart before transplant surgery. A crossmatch checks how close the match is between the patient's tissue type and the tissue type of the donor heart.

Most people are not high risk and a crossmatch usually is not done before the transplant because the surgery must be done as quickly as possible after a donor heart is found.

While waiting for heart transplantation, patients are given treatment to keep the heart as healthy as possible. They are regularly checked to make sure the heart is pumping enough blood. Intravenous medications may be used to improve cardiac output. If these drugs are not effective, a mechanical pump can help keep the heart functioning until a donor heart becomes available. Inserted through an artery into the aorta, the pump assists the heart in pumping blood.

Aftercare

Immediately following surgery, patients are monitored closely in the intensive care unit (ICU) of the hospital for 24–72 hours. Most patients need to receive oxygen for four to 24 hours following surgery. Blood pressure, heart function, and other organ functions are carefully monitored during this time.

Heart transplant patients start taking immunosuppressive drugs before or during surgery to prevent immune rejection of the heart. High doses of immunosuppressive drugs are given at this time, because rejection is most likely to happen within the first few months after the surgery. A few months after surgery, lower doses of immunosuppressive drugs usually are given and must be taken for the rest of the patient's life.

For six to eight weeks after the transplant surgery, patients usually come back to the transplant center twice a week for physical examinations and medical tests. These tests check for any signs of infection, rejection of the new heart, or other complications.

In addition to **physical examination**, the following tests may be done during these visits:

- laboratory tests to check for infection
- chest x ray to check for early signs of lung infection
- electrocardiogram (ECG) to check heart function
- echocardiogram to check the function of the ventricles in the heart
- blood tests to check liver and kidney function
- complete blood counts (CBC) to check the numbers of blood cells
- taking of a small tissue sample from the donor heart (endomyocardial biopsy) to check for signs of rejection.

During the physical examination, the blood pressure is checked and the heart sounds are listened to with a stethoscope to determine if the heart is beating properly and pumping enough blood. Kidney and liver function are checked because these organs may lose function if the heart is being rejected.

An endomyocardial biopsy is the removal of a small sample of the heart muscle. This is done with a very small instrument that is inserted through an artery or vein and into the heart. The heart muscle tissue is examined under a microscope for signs that the heart is being rejected. Endomyocardial biopsy is usually done weekly for the first four to eight weeks after transplant surgery and then at longer intervals after that.

Risks

The most common and dangerous complications of heart transplant surgery are organ rejection and infection. Immunosuppressive drugs are given to prevent rejection of the heart. Most heart transplant patients have a rejection episode soon after transplantation, but doctors usually diagnose it immediately when it will respond readily to treatment. Rejection is treated with combinations of immunosuppressive drugs given in higher doses than maintenance immunosuppression. Most of these rejection situations are successfully treated.

Infection can result from the surgery, but most infections are a side effect of the immunosuppressive drugs. Immunosuppressive drugs keep the immune system from attacking the foreign cells of the donor heart. However, the suppressed immune cells are also unable to adequately fight bacteria, viruses, and other microorganisms. Microorganisms that normally do not affect persons with healthy immune systems can cause dangerous infections in transplant patients taking immunosuppressive drugs.

Patients are given antibiotics during surgery to prevent bacterial infection. Patients may also be given an antiviral drug to prevent virus infections. Patients who develop infections may need to have their immunosuppressive drugs changed or the dose adjusted. Infections are treated with antibiotics or other drugs, depending on the type of infection.

Other complications that can happen immediately after surgery are:

- bleeding
- pressure on the heart caused by fluid in the space surrounding the heart (pericardial tamponade)
- irregular heart beats
- reduced cardiac output
- increased amount of blood in the circulatory system
- decreased amount of blood in the circulatory system

About half of all heart transplant patients develop coronary artery disease 1–5 years after the transplant. The coronary arteries supply blood to the heart. Patients with this problem develop chest pains called **angina**. Other names for this complication are coronary allograft vascular disease and chronic rejection.

Outcomes

Heart transplantation is an appropriate treatment for many patients with end-stage heart failure.

KEY TERMS

Anesthesia—Loss of the ability to feel pain, caused by administration of an anesthetic drug.

Angina—Characteristic chest pain which occurs during exercise or stress in certain kinds of heart disease.

Cardiopulmonary bypass—Mechanically circulating the blood with a heart/lung machine that bypasses the heart and lungs.

Cardiovascular—Having to do with the heart and blood vessels.

Complete blood count (CBC)—A blood test to check the numbers of red blood cells, white blood cells, and platelets in the blood.

Coronary artery disease—Blockage of the arteries leading to the heart.

Crossmatch—A test to determine if patient and donor tissues are compatible.

Donor—A person who donates an organ for transplantation.

Echocardiogram—A test that visualizes and records the position and motion of the walls of the heart using ultrasound waves.

Electrocardiogram (ECG)—A test that measures electrical conduction of the heart.

End-stage heart failure—Severe heart disease that does not respond adequately to medical or surgical treatment.

Endomyocardial biopsy—Removal of a small sample of heart tissue to check it for signs of damage caused by organ rejection.

Fatigue—Loss of energy; tiredness.

Graft—A transplanted organ or other tissue.

Immunosuppressive drug—Medication used to suppress the immune system.

Inotropic drugs—Medications used to stimulate the heart beat.

Pulmonary hypertension—An increase in the pressure in the blood vessels of the lungs.

Recipient—A person who receives an organ transplant.

The outcomes of heart transplantation depend on the patient's age, health, and other factors. About 73% of heart transplant patients are alive four years after surgery.

After transplant, most patients regain normal heart function, meaning the heart pumps a normal amount of blood. A transplanted heart usually beats slightly faster than normal because the heart nerves are cut during surgery. The new heart also does not increase its rate as quickly during exercise. Even so, most patients feel much better and their capacity for **exercise** is dramatically improved from before they received the new heart. About 85% of patients return to work and other daily activities. Many are able to participate in sports.

Resources

ORGANIZATIONS

American Council on Transplantation. P.O. Box 1709, Alexandria, VA 22313. 1-800-ACT-GIVE.

Health Services and Resources Administration, Division of Organ Transplantation. Room 11A-22, 5600 Fishers Lane, Rockville, MD 20857.

United Network for Organ Sharing (UNOS). 1100 Boulders Parkway, Suite 500, P.O. Box 13770, Richmond, VA 23225-8770. (804) 330-8500. <<http://www.unos.org>>.

OTHER

“What Every Patient Needs to Know.” *United Network for Organ Sharing (UNOS)*. <http://www.unos.org/frame_Default.asp?Category=Patients>.

Toni Rizzo

Heart tumors see **Myxoma**

Heart valve repair

Definition

Heart valve repair is a surgical procedure used to correct a malfunctioning heart valve. Repair usually involves separating the valve leaflets (the one-way “doors” of the heart valve which open and close to pump blood through the heart) or forcing them open with a balloon catheter, a technique known as *balloon valvuloplasty*.

Purpose

To correct damage to the mitral, aortic, pulmonary, or tricuspid heart valves caused by a systemic infection, **endocarditis**, rheumatic heart disease, a congenital heart defect, or mitral and/or aortic valve disease. Damaged valves may not open properly (stenosis) or they may not close adequately (valve regurgitation, insufficiency, or incompetence).

Precautions

Patients who have a diseased heart valve that is badly scarred or calcified may be better candidates for valve replacement surgery.

Description

Heart valve repair is performed in a hospital setting by a cardiac surgeon. During valve repair surgery, the patient's heart is stopped, and his/her blood is circulated outside of the body through an *extracorporeal bypass circuit*, also called heart-lung machine or just "the pump." The extracorporeal circuit consists of tubing and medical devices that take over the function of the patient's heart and lungs during the procedure. As blood passes through the circuit, carbon dioxide is removed from the bloodstream and replaced with oxygen. The oxygenated blood is then returned to the body. Other components may also be added to the circuit to filter fluids from the blood or concentrate red blood cells.

In cases of valve disease where the leaflets have become fused together, a procedure known as a valvulotomy is performed. In valvulotomy, the leaflets of the valves are surgically separated, or partially resected, with an incision to increase the size of the valve opening. The surgeon may also make adjustments to the chordae, the cord-like tissue that connects the valve leaflets to the ventricle muscles, to improve valve function.

Another valve repair technique, **balloon valvuloplasty**, is used in patients with pulmonary, aortic, and **mitral valve stenosis** to force open the valve. Valvuloplasty is similar to a cardiac **angioplasty** procedure in that it involves the placement of a balloon-tipped catheter into the heart. Once inserted into the valve, the balloon is inflated and the valve dilates, or opens. Valvuloplasty does not require a bypass circuit.

Preparation

A number of diagnostic tests may be administered prior to valve repair surgery. **Magnetic resonance imaging** (MRI), echocardiogram, angiogram, and/or scintigram are used to help the surgeon get an accurate picture of the extent of damage to the heart valve and the status of the coronary arteries.

Aftercare

The patient's blood pressure and vital signs will be carefully monitored following a valve repair procedure, and he or she watched closely for signs of edema or congestive **heart failure**.

KEY TERMS

Angiogram—An angiogram uses a radiopaque substance, or dye, to make the blood vessels or arteries visible under x ray.

Calcified—Hardened by calcium deposits.

Catheter—A long, thin, flexible tube used in valvuloplasty to widen the valve opening.

Echocardiogram—Ultrasound of the heart; generates a picture of the heart through the use of soundwaves.

Edema—Fluid accumulation in the body.

Scintigram—A nuclear angiogram; a scintigram involves injection of a radioactive substance into the patient's circulatory system. As the substance travels through the body, a special scanning camera takes pictures.

Stenosis—Narrowing of the heart valve opening.

Echocardiography or other diagnostic tests are ordered for the patient at some point during or after surgery to evaluate valvular function. A **cardiac rehabilitation** program may also be recommended to assist the patient in improving **exercise** tolerance after the procedure.

Risks

As with any invasive surgical procedure, hemorrhage, infarction, **stroke**, **heart attack**, and infection are all possible complications of heart valve repair. The overall risks involved with the surgery depend largely on the complexity of the procedure and physical condition of the patient.

Normal results

Ideally, a successful heart valve repair procedure will return heart function to age-appropriate levels. If valvuloplasty is performed, a follow-up valve repair or replacement surgery may be necessary at a later date.

Resources

BOOKS

DeBakey, Michael E., and Antonio Gotto, Jr. *The New Living Heart*. Holbrook, MA: Adams Media Corporation, 1997.

Paula Anne Ford-Martin

Heart valve replacement

Definition

Heart valve replacement is a surgical procedure during which surgeons remove a damaged valve from the heart and substitute a healthy one.

Purpose

Four valves direct blood to and from the body through the heart: the aortic valve, the pulmonic valve, the tricuspid valve, and the mitral valve. Any of these valves may malfunction because of a birth defect, infection, disease, or trauma. When the malfunction is so severe that it interferes with blood flow, an individual will have heart **palpitations**, **fainting** spells, and/or difficulty breathing. These symptoms will progressively worsen and cause **death** unless the damaged valve is replaced surgically.

Precautions

Abnormal tricuspid valves usually are not replaced because they do not cause serious symptoms. Mildly or even moderately diseased mitral valves may not need to be replaced because their symptoms are tolerable or they can be treated with such drugs as beta blockers or calcium antagonists, which slow the heart rate. However, a severely diseased mitral valve should be repaired or replaced unless the person is too ill to tolerate the operation because of another condition or illness.

Description

After cutting through and separating the breastbone and ribs, surgeons place the patient on a cardiopulmonary bypass machine, which will perform the functions of the heart and lungs during the operation. They then open the heart and locate the faulty valve. Slicing around the edges of the valve, they loosen it from the tendons that connect it to the rest of the heart and withdraw it. The new valve is inserted and sutured into place. The patient is then taken off the bypass machine and the chest is closed. The surgery takes three to five hours and is covered by most insurance plans.

There are three types of replacement valves. One class is made from animal tissue, usually a pig's aortic valve. Another is mechanical and is made of metal and plastic. The third, includes human valves that have been removed from an organ donor or that, rarely, are the patient's own pulmonic valve.

There is no single ideal replacement valve. The choice between an animal valve or a mechanical valve depends largely on the age of the patient. Because valves obtained from animals have a life expectancy of 7–15 years, they usually are given to older patients. Mechanical valves are used in younger patients because they are more durable. Because mechanical valves are made of foreign material, however, **blood clots** can form on their surface. Therefore, patients who receive these valves must take anticoagulants the rest of their lives.

Donor or pulmonic valves are given only to those patients who will deteriorate rapidly because of a narrowing of the passageway between the aorta and the left ventricle (aortic stenosis). These valves are limited in their use because of the small supply available from donors and the strain that could be caused by removing and transferring a patient's own pulmonic valve.

Preparation

Before patients undergo heart valve replacement, they must be evaluated carefully for any signs that they may not tolerate the surgery.

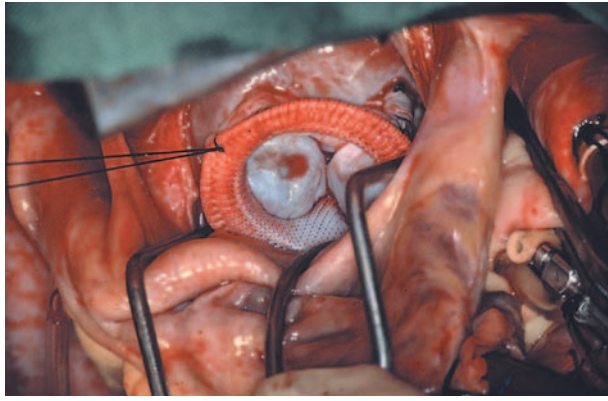
Preoperative tests include:

- electrocardiography, which assesses the electrical activity of the heart
- echocardiography, which uses sound waves to show the extent of the obstruction of blood flow through the heart and determine the degree of loss of heart function due to the malfunctioning valve
- chest x ray, which provides an overall view of the anatomy of the heart and the lungs

Cardiac catheterization may also be performed to further assess the valve and to determine if coronary bypass surgery should also be done.

Aftercare

A patient usually spends one to three days in the hospital intensive care unit (ICU) after heart valve replacement so that the working of his or her heart and circulation can be monitored closely. When first brought to the ICU after surgery, the patient undergoes a neurological examination to be sure he or she has not suffered a **stroke**. The patient continues to breathe by means of a tube inserted in the trachea at the time of surgery. This mechanical ventilation is not withdrawn until the patient is fully awake from anesthesia, shows signs that he or she can breathe



Open heart surgery showing replacement of a valve.
(Photograph by David Leah. Photo Researchers. Reproduced by permission.)

satisfactorily without mechanical support, and has steadfast circulation.

Once stabilized, the patient is transferred to a standard medical/surgical unit where he or she receives drugs that will prevent excess fluid from building up around the heart. As soon as possible, the patient begins walking and exercising to regain strength. He or she is also placed on a diet that is low in salt and cholesterol.

After being released from the hospital, the patient continues a daily **exercise** program that includes vigorous walking, and he or she may also join a recommended **cardiac rehabilitation** program. He or she usually can return to work or other normal activities within two months of the surgery.

Risks

Complications following heart valve replacement are not common, but can be serious. All valves made from animal tissue will develop calcium deposits over time. If these deposits hamper the function of the valve, it must be replaced. Valves may become dislodged. Blood clots may form on the surface of the substitute valve, break off into the general circulation, and become wedged in an artery supplying blood to the brain, kidneys, or legs. These blood clots may cause fainting spells, stroke, kidney failure, or loss of circulation to the legs. These blood clots can be treated with drugs or surgery.

Infection of heart muscle affects up to 2% of patients who have heart valve replacement. Such an infection is treated with intravenous **antibiotics**. If the infection persists, the new valve may have to be replaced.

KEY TERMS

Anticoagulants—Drugs that prevent blood clots from forming.

Aortic valve—A fold in the channel leading from the aorta to the left ventricle of the heart. The aortic valve directs blood flow that has received oxygen from the lungs to the aorta which transmits blood to the rest of the body.

Cardiac catheterization—A thin tube called a catheter is inserted into an artery or vein in the leg, groin or arm. The catheter tube is carefully threaded into the area of the heart needing surgical repair. A local anaesthesia is used at the insertion sites.

Cardiopulmonary bypass machine—A mechanical instrument that takes over the circulation of the body while heart surgery is taking place.

Echocardiography—A diagnostic instrument that assesses the structure of the heart using sound waves.

Electrocardiography—A diagnostic instrument that evaluates the function of the heart by measuring the electrical activity generated by the beating of the heart.

Mitral valve—A fold in between the left atrium and the left ventricle of the heart that directs blood that has received oxygen from the lungs to the aortic valve and the aorta.

Pulmonic valve—A fold in the pulmonary artery that directs blood to the lungs. It may be transferred to replace a severely diseased aortic valve during heart valve replacement surgery for aortic stenosis.

Tricuspid valve—A fold in between the right atrium and the right ventricle of the heart that directs blood that needs oxygen to the lungs.

Normal results

Few patients die as a result of the surgery. Approximately 3% of all patients die during or immediately after heart valve replacement, and less than 1% of patients below the age of 65 die because of the operation. The vast majority of patients who have heart valve replacement return to normal activity after the surgery. Depending on the type of valve they receive, these patients will have no symptoms of valve abnormality for at least seven years. Also, their quality of life will improve because they may no longer have difficulty breathing, fainting spells, or palpitations.

Resources

BOOKS

American Heart Association. *American Heart Association's Your Heart: An Owner's Manual*. Englewood Cliffs, NJ: Prentice Hall, 1991.

ORGANIZATIONS

The American College of Cardiology. 9111 Old Georgetown Road, Bethesda, MD 20814. (800) 253-4636. <<http://www.acc.org>>.

American College of Surgeons. 55 E. Erie St., Chicago, IL 60611. (312) 202-5000. <<http://www.facs.org>>.

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

Karen Marie Sandrick

Heartburn

Definition

Heartburn is a burning sensation in the chest that can extend to the neck, throat, and face; it is worsened by bending or lying down. It is the primary symptom of gastroesophageal reflux, which is the movement of stomach acid into the esophagus. On rare occasions, it is due to gastritis (stomach lining inflammation).

Description

More than one-third of the population is afflicted by heartburn, with about one-tenth afflicted daily. Infrequent heartburn is usually without serious consequences, but chronic or frequent heartburn (recurring more than twice per week) can have severe consequences. Accordingly, early management is important.

Understanding heartburn depends on understanding the structure and action of the esophagus. The esophagus is a tube connecting the throat to the stomach. It is about 10 in (25 cm) long in adults, lined with squamous (plate-like) epithelial cells, coated with mucus, and surrounded by muscles that push food to the stomach by sequential waves of contraction (peristalsis). The lower esophageal sphincter (LES) is a thick band of muscles that encircles the esophagus just above the uppermost part of the stomach. This sphincter is usually tightly closed and normally opens only when food passes from the esophagus into the stomach. Thus, the contents of the stomach are normally kept from moving back into the esophagus.

The stomach has a thick mucous coating that protects it from the strong acid it secretes into its interior when food is present, but the much thinner esophageal coating doesn't provide protection against acid. Thus, if the LES opens inappropriately or fails to close completely, and stomach contents leak into the esophagus, the esophagus can be burned by acid. The resulting burning sensation is called heartburn.

Occasional heartburn has no serious long-lasting effects, but repeated episodes of gastroesophageal reflux can ultimately lead to esophageal inflammation (esophagitis) and other damage. If episodes occur more frequently than twice a week, and the esophagus is repeatedly subjected to acid and digestive enzymes from the stomach, ulcerations, scarring, and thickening of the esophagus walls can result. This thickening of the esophagus wall causes a narrowing of the interior of the esophagus. Such narrowing affects swallowing and peristaltic movements. Repeated irritation can also result in changes in the types of cells that line the esophagus. The condition associated with these changes is termed Barrett's syndrome and can lead to **esophageal cancer**.

Causes and symptoms

Causes

A number of different factors may contribute to LES malfunction with its consequent gastroesophageal acid reflux:

- The eating of large meals that distend the stomach can cause the LES to open inappropriately.
- Lying down within two to three hours of eating can cause the LES to open.
- Obesity, **pregnancy**, and tight clothing can impair the ability of the LES to stay closed by putting pressure on the abdomen.
- Certain drugs, notably nicotine, alcohol, diazepam (Valium), meperidine (Demerol), theophylline, morphine, prostaglandins, calcium channel blockers, nitrate heart medications, anticholinergic and adrenergic drugs (drugs that limit nerve reactions), including dopamine, can relax the LES.
- Progesterone is thought to relax the LES.
- Greasy foods and some other foods such as chocolate, coffee, and peppermint can relax the LES.
- Paralysis and **scleroderma** can cause the LES to malfunction.
- Hiatus **hernia** may also cause heartburn according to some gastroenterologists. (Hiatus hernia is a

protrusion of part of the stomach through the diaphragm to a position next to the esophagus.)

Symptoms

Heartburn itself is a symptom. Other symptoms also caused by gastroesophageal reflux can be associated with heartburn. Often heartburn sufferers salivate excessively or regurgitate stomach contents into their mouths, leaving a sour or bitter taste. Frequent gastroesophageal reflux leads to additional complications including difficult or painful swallowing, sore throat, hoarseness, coughing, **laryngitis**, **wheezing**, **asthma**, **pneumonia**, gingivitis, **bad breath**, and earache.

Diagnosis

Gastroenterologists and internists are best equipped to diagnose and treat gastroesophageal reflux. Diagnosis is usually based solely on patient histories that report heartburn and other related symptoms. Additional diagnostic procedures can confirm the diagnosis and assess damage to the esophagus, as well as monitor healing progress. The following diagnostic procedures are appropriate for anyone who has frequent, chronic, or difficult-to-treat heartburn or any of the complicating symptoms noted in the previous paragraph.

X rays taken after a patient swallows a barium suspension can reveal esophageal narrowing, ulcerations or a reflux episode as it occurs. However, this procedure cannot detect the structural changes associated with different degrees of esophagitis. This diagnostic procedure has traditionally been called the “upper GI series” or “barium swallow” and costs about \$250.00.

Esophagoscopy is a newer procedure that uses a thin flexible tube to view the inside of the esophagus directly. It should be done by a gastroenterologist or gastrointestinal endoscopist and costs about \$700. It gives an accurate picture of any damage present and gives the physician the ability to distinguish between different degrees of esophagitis.

Other tests may also be used. They include pressure measurements of the LES; measurements of esophageal acidity (pH), usually throughout a 24-hour period; and microscopic examination of biopsied tissue from the esophageal wall (to inspect esophageal cell structure for Barrett’s syndrome and malignancies).

New technology introduced by 2003 allows for continuous monitoring of pH levels to help determine the cause. A tiny wireless capsule can be delivered to the lining of the esophagus through a catheter and

data recorder on a device the size of a pager that is clipped to the patient’s belt or purse for 48 hours. The capsule eventually sloughs off and passes harmlessly through the gastrointestinal tract in seven to 10 days.

Note: A burning sensation in the chest is usually heartburn and is not associated with the heart. However, chest **pain** that radiates into the arms and is not accompanied by regurgitation is a warning of a possible serious heart problem. Anyone with these symptoms should contact a doctor immediately.

Treatment

Drugs

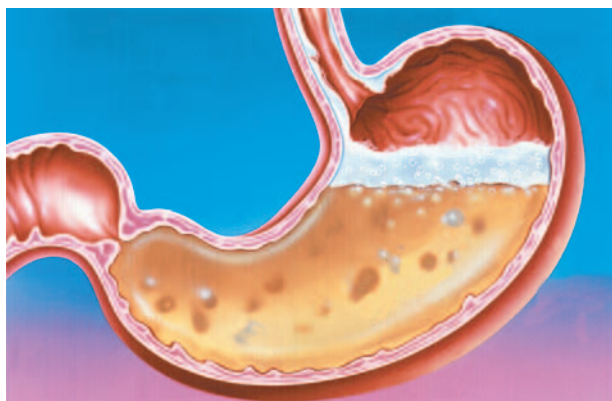
Occasional heartburn is probably best treated with over-the-counter **antacids**. These products go straight to the esophagus and immediately begin to decrease acidity. However, they should not be used as the sole treatment for heartburn sufferers who either have two or more episodes per week or who suffer for periods of more than three weeks. There is a risk of kidney damage and other metabolic changes.

H2 blockers (histamine receptor blockers, such as Pepcid AC, Zantac, Tagamet) decrease stomach acid production and are effective against heartburn. H2 blocker treatment also allows healing of esophageal damage but is not very effective when there is a high degree of damage. It takes 30–45 minutes for these drugs to take effect, so they must be taken prior to an episode. Thus, they should be taken daily, usually two to four times per day for several weeks. Six to 12 weeks of standard-dose treatment relieves symptoms in about one-half the patients. Higher doses relieve symptoms in a greater fraction of the population, but at least 25% of heartburn sufferers are not helped by H2 blockers.

Proton-pump inhibitors also inhibit acid production by the stomach, but are much more effective than H2 blockers for some people. They are also more effective in aiding the healing process. Esophagitis is healed in about 90% of the patients undergoing proton-pump inhibitor treatment.

The long-term effects of inhibiting stomach acid production are unknown. Without the antiseptic effects of a consistently very acidic stomach environment, users of H2 blockers or proton-pump inhibitors may become more susceptible to bacterial and viral infection. Absorption of some drugs is also lowered by this less-acidic environment.

Prokinetic agents (also known as motility drugs) act on the LES, stimulating it to close more tightly, thereby keeping stomach contents out of the



An illustration of foaming antacid on top of the contents of a human stomach. Heartburn is caused by a backflow of the stomach's acidic contents into the esophagus, causing inflammation and a sense of pain that can rise to the throat. (Illustration by John Bavosi, Custom Medical Stock Photo. Reproduced by permission.)

esophagus. It is not known how effectively these drugs promote healing. Some of the early motility drugs had serious neurological side effects, but a newer drug, cisapride, seems to act only on digestive system nerve connections.

Surgery

Fundoplication, a surgical procedure to increase pressure on the LES by stretching and wrapping the upper part of the stomach around the sphincter, is a treatment of last resort. About 10% of heartburn sufferers undergo this procedure. It is not always effective and its effectiveness may decrease over time, especially several years after surgery. Dr. Robert Marks and his colleagues at the University of Alabama reported in 1997 on the long-term outcome of this procedure. They found that 64% of the patients in their study who had fundoplication between 1992 and 1995 still suffered from heartburn and reported an impaired quality of life after the surgery.

However, **laparoscopy** (an examination of the interior of the abdomen by means of the laparoscope) now provides hope for better outcomes. Fundoplication performed with a laparoscope is less invasive. Five small incisions are required instead of one large incision. Patients recover faster, and it is likely that studies will show they suffer from fewer surgical complications.

Alternative treatment

Prevention, as outlined below, is a primary feature for heartburn management in alternative medicine

and traditional medicine. Dietary adjustments can eliminate many causes of heartburn.

Herbal remedies include bananas, aloe vera gel, chamomile (*Matricaria recutita*), ginger (*Zingiber officinale*), and citrus juices, but there is little agreement here. For example, ginger, which seems to help some people, is claimed by other practitioners to *cause* heartburn and is thought to relax the LES. There are also many recommendations to *avoid* citrus juices, which are themselves acidic. Licorice (*Glycyrrhiza uralensis*) can help relieve the symptoms of heartburn by reestablishing balance in the acid output of the stomach.

Several homeopathic remedies are useful in treating heartburn symptoms. Among those most often recommended are *Nux vomica*, *Carbo vegetabilis*, and *Arsenicum album*. **Acupressure** and acupuncture may also be helpful in treating heartburn.

Sodium bicarbonate (baking soda) is an inexpensive alternative to use as an antacid. It reduces esophageal acidity immediately, but its effect is not long-lasting and should not be used by people on sodium-restricted **diets**.

Prognosis

The prognosis for people who get heartburn only occasionally or people without esophageal damage is excellent. The prognosis for people with esophageal damage who become involved in a treatment program that promotes healing is also excellent. The prognosis for anyone with esophageal **cancer** is very poor. There is a strong likelihood of a painful illness and a less than 5% chance of surviving more than five years.

Prevention

Given the lack of completely satisfactory treatments for heartburn or its consequences and the lack of a cure for esophageal cancer, prevention is of the utmost importance. Proponents of traditional *and* alternative medicine agree that people disposed to heartburn should:

- avoid eating large meals
- avoid alcohol, **caffeine**, fatty foods, fried foods, hot or spicy foods, chocolate, peppermint, and nicotine
- avoid drugs known to contribute to heartburn, such as nitrates (heart medications such as Isonate and Nitrocap), **calcium channel blockers** (e.g., Cardizem and Procardia), and anticholinergic drugs (e.g., Probanthine and Benty), and check with their doctors about any drugs they are taking

KEY TERMS

Barrett's syndrome—Also called Barrett's esophagus or Barrett's epithelia, this is a condition where the squamous epithelial cells that normally line the esophagus are replaced by thicker columnar epithelial cells.

Digestive enzymes—Molecules that catalyze the breakdown of large molecules (usually food) into smaller molecules.

Esophagitis—Inflammation of the esophagus.

Fundoplication—A surgical procedure that increases pressure on the LES by stretching and wrapping the upper part of the stomach around the sphincter.

Gastroesophageal reflux—The flow of stomach contents into the esophagus.

Hiatus hernia—A protrusion of part of the stomach through the diaphragm to a position next to the esophagus.

Metabolic—Refers to the chemical reactions in living things.

Mucus—Thick, viscous, gel-like material that functions to moisten and protect inner body surfaces.

Peristalsis—A sequence of muscle contractions that progressively squeeze one small section of the digestive tract and then the next to push food along the tract, something like pushing toothpaste out of its tube.

Scleroderma—An autoimmune disease with many consequences, including esophageal wall thickening.

Squamous epithelial cells—Thin, flat cells found in layers or sheets covering surfaces such as skin and the linings of blood vessels and esophagus.

Ulceration—An open break in surface tissue.

- avoid clothing that fits tightly around the abdomen
- control body weight
- wait about three hours after eating before going to bed or lying down
- elevate the head of the bed 6–9 inches to alleviate heartburn at night. This can be done with bricks under the bed or with a wedge designed for this purpose.

Preventing heartburn's switch to cancer begins with preventing heartburn in the first place. A study

in Great Britain in 2004 also looked at using a combination of **aspirin** and an anti-ulcer drug to try to prevent Barrett's esophagus from forming in patients with long-term heartburn. Aspirin has been found in previous studies to reduce cases of esophageal cancer. However, since one of its side effects is an increased risk of stomach ulcers, the researchers were including an effective anti-ulcer drug for participants.

Resources

PERIODICALS

"Aspirin Trial Launched to Block Heartburn's Switch to Cancer." *Drug Week* January 23, 2004:188.

Bealfsky, Peter C., and William Halsey. "An Endoscopic View of a Wireless pH-Monitoring Capsule." *Ear, Nose and Throat Journal* April 2003: 254.

ORGANIZATIONS

The American College of Gastroenterology (ACG). PO Box 3099, Alexandria, VA 22302. (800) HRT-BURN. < <http://www.healthtouch.com> >.

The American Gastroenterological Association (AGA). 7910 Woodmont Ave., 7th Floor, Bethesda, MD 20814. (310) 654-2055. < <http://www.gastro.org/index.html> >.

American Society for Gastrointestinal Endoscopy. 13 Elm St., Manchester, MA 01944. (508) 526-8330. < <http://www.asge.org/doc/201> >.

National Digestive Diseases Information Clearinghouse. 2 Information Way, Bethesda, MD 20892-3570. (800) 891-5389. < <http://www.niddk.nih.gov/health/digest/nddic.htm> >.

Lorraine Lica, PhD
Teresa G. Odle

Heat cramps see **Heat disorders**

Heat disorders

Definition

Heat disorders are a group of physically related illnesses caused by prolonged exposure to hot temperatures, restricted fluid intake, or failure of temperature regulation mechanisms of the body. Disorders of heat exposure include heat cramps, heat exhaustion, and heat **stroke** (also called sunstroke). Hyperthermia is the general name given to heat-related illnesses. The two most common forms of hyperthermia are heat exhaustion and heat stroke, which is especially dangerous and requires immediate medical attention.

Description

Heat disorders are harmful to people of all ages, but their severity is likely to increase as people age. Heat cramps in a 16-year-old may be heat exhaustion in a 45-year-old and heat stroke in a 65-year-old. The body's temperature regulating mechanisms rely on the thermal regulating centers in the brain. Through these complex centers, the body tries to adapt to high temperatures by adjusting the amount of salt in the perspiration. Salt helps the cells in body tissues retain water. In hot weather, a healthy body will lose enough water to cool the body while creating the lowest level of chemical imbalance. Regardless of extreme weather conditions, the healthy human body keeps a steady temperature of approximately 98.6°F (37°C). In hot weather, or during vigorous activity, the body perspires. As perspiration evaporates from the skin, the body is cooled. If the body loses too much salt and fluids, the symptoms of **dehydration** can occur.

Heat cramps

Heat cramps are the least severe of the heat-related illnesses. This heat disorder is often the first signal that the body is having difficulty with increased temperature. Individuals exposed to excessive heat should think of heat cramps as a warning sign to a potential heat-related emergency.

Heat exhaustion

Heat exhaustion is a more serious and complex condition than heat cramps. Heat exhaustion can result from prolonged exposure to hot temperatures, restricted fluid intake, or failure of temperature regulation mechanisms of the body. It often affects athletes, firefighters, construction workers, factory workers, and anyone who wears heavy clothing in hot humid weather.

Heat stroke

Heat exhaustion can develop rapidly into heat stroke. Heat stroke can be life threatening and because the percentage of victims dying from heat stroke is very high, immediate medical attention is critical when problems first begin. Heat stroke, like heat exhaustion, is also a result of prolonged exposure to hot temperatures, restricted fluid intake, or failure of temperature regulation mechanisms of the body. However, the severity of impact on the body is much greater with heat stroke.

Causes and symptoms

Heat cramps

Heat cramps are painful **muscle spasms** caused by the excessive loss of salts (electrolytes), due to heavy perspiration. The muscle tissue becomes less flexible, causing **pain**, difficult movement, and involuntary tightness. Heavy exertion in extreme heat, restricted fluid intake, or failure of temperature regulation mechanisms of the body may lead to heat cramps. This disorder occurs more often in the legs and abdomen than in other areas of the body. Individuals at higher risk are those working in extreme heat, elderly people, young children, people with health problems, and those who are unable to naturally and properly cool their bodies. Individuals with poor circulation and who take medications to reduce excess body fluids can be at risk when conditions are hot and humid.

Heat exhaustion

Heat exhaustion is caused by exposure to high heat and humidity for many hours, resulting in excessive loss of fluids and salts through heavy perspiration. The skin may appear cool, moist, and pale. The individual may complain of **headache** and nausea with a feeling of overall weakness and exhaustion. Dizziness, faintness, and mental confusion are often present, as is rapid and weak pulse. Breathing becomes fast and shallow. Fluid loss reduces blood volume and lowers blood pressure. Yellow or orange urine often is a result of inadequate fluid intake, along with associated intense thirst. Insufficient water and salt intake or a deficiency in the production of sweat place an individual at high risk for heat exhaustion.

Heat stroke

Heat stroke is caused by overexposure to extreme heat, resulting in a breakdown in the body's heat regulating mechanisms. The body's temperature reaches a dangerous level, as high as 106°F (41.1°C). An individual with heat stroke has a body temperature higher than 104°F (40°C). Other symptoms include mental confusion with possible combativeness and bizarre behavior, staggering, and faintness.

The pulse becomes strong and rapid (160–180 beats per minute) with the skin taking on a dry and flushed appearance. There is often very little perspiration. The individual can quickly lose consciousness or have convulsions. Before heat stroke, an individual suffers from heat exhaustion and the associated symptoms. When the body can no longer maintain a normal temperature, heat exhaustion becomes heat stroke.

Heat stroke is a life-threatening medical emergency that requires immediate initiation of life-saving measures.

Diagnosis

The diagnosis of heat cramps usually involves the observation of individual symptoms such as muscle cramping and thirst. Diagnosis of heat exhaustion or heat stroke, however, may require a physician to review the medical history, document symptoms, and obtain a blood pressure and temperature reading. The physician may also take blood and urine samples for further laboratory testing. A test to measure the body's electrolytes can also give valuable information about chemical imbalances caused by the heat-related illness.

Treatment

Heat cramps

The care of heat cramps includes placing the individual at rest in a cool environment, while giving cool water with a teaspoon of salt per quart, or a commercial sports drink. Usually rest and liquids are all that is needed for the patient to recover. Mild stretching and massaging of the muscle area follows once the condition improves. The individual should not take salt tablets, since this may actually worsen the condition. When the cramps stop, the person can usually start activity again if there are no other signs of illness. The individual needs to continue drinking fluids and should be watched carefully for further signs of heat-related illnesses.

Heat exhaustion

The individual suffering from heat exhaustion should stop all physical activity and move immediately to a cool place out of the sun, preferably a cool, air-conditioned location. She or he should then lay down with feet slightly elevated, remove or loosen clothing, and drink cold (but not iced), slightly salty water or commercial sports drink. Rest and replacement of fluids and salt is usually all the treatment that is needed, and hospitalization is rarely required. Following rehydration, the person usually recovers rapidly.

Heat stroke

Simply moving the individual afflicted with heat stroke to a cooler place is not enough to reverse the internal overheating. Emergency medical assistance

KEY TERMS

Convulsions—Also termed seizures; a sudden violent contraction of a group of muscles.

Electrolytes—An element or compound that when melted or dissolved in water dissociates into ions and is able to conduct an electrical current. Careful and regular monitoring of electrolytes and intravenous replacement of fluid and electrolytes are part of the acute care in many illnesses.

Rehydration—The restoration of water or fluid to a body that has become dehydrated.

should be called immediately. While waiting for help to arrive, quick action to lower body temperature must take place. Treatment involves getting the victim to a cool place, loosening clothes or undressing the heat stroke victim, and allowing air to circulate around the body. The next important step is wrapping the individual in wet towels or clothing, and placing ice packs in areas with the greatest blood supply. These areas include the neck, under the arm and knees, and in the groin. Once the patient is under medical care, **cooling treatments** may continue as appropriate. The victim's body temperature will be monitored constantly to guard against overcooling. Breathing and heart rate will be monitored closely, and fluids and electrolytes will be replaced intravenously. Anti-convulsant drugs may be given. After severe heat stroke, bed rest may be recommended for several days.

Prognosis

Prompt treatment for heat cramps is usually very effective with the individual returning to activity thereafter. Treatment of heat exhaustion usually brings full recovery in one to two days. Heatstroke is a very serious condition and its outcome depends upon general health and age. Due to the high internal temperature of heat stroke, permanent damage to internal organs is possible.

Prevention

Because heat cramps, heat exhaustion, and heat stroke have a cascade effect on each other, the prevention of the onset of all heat disorders is similar. Avoid strenuous **exercise** when it is very hot. Individuals exposed to extreme heat conditions should drink plenty of fluids. Wearing light and loose-fitting

clothing in hot weather is important, regardless of the activity. It is important to consume water often and not to wait until thirst develops. If perspiration is excessive, fluid intake should be increased. When urine output decreases, fluid intake should also increase. Eating lightly salted foods can help replace salts lost through perspiration. Ventilation in any working areas in warm weather must be adequate. This can be achieved as simply as opening a window or using an electric fan. Proper ventilation will promote adequate sweat evaporation to cool the skin. Sunblocks and **sunscreens** with a protection factor of 15 (SPF 15) can be very helpful when one is exposed to extreme direct sunlight.

Resources

OTHER

Griffith, H. Winter. "Complete Guide to Symptoms, Illness & Surgery." ThriveOnline. <<http://thriveonline.oxygen.com>>.

Jeffrey P. Larson, RPT

Heat exhaustion see **Heat disorders**

Heat treatments

Definition

Heat treatments are applications of therapeutic thermal agents to specific body areas experiencing injury or dysfunction.

Purpose

The general purpose of a heat treatment is to increase the extensibility of soft tissues, remove toxins from cells, enhance blood flow, increase function of the tissue cells, encourage muscle relaxation, and help relieve **pain**. There are two types of heat treatments: superficial and deep. Superficial heat treatments apply heat to the outside of the body. Deep heat treatments direct heat toward specific inner tissues through ultrasound or by electric current. Heat treatments are beneficial prior to exercise, providing a warm-up effect to the soft tissues involved.

Precautions

Heat treatments should not be used on individuals with circulation problems, heat intolerance, or lack of

sensation in the affected area. Low blood circulation may contribute to heat-related injuries. Heat treatments also should not be used on individuals afflicted with heart, lung, or kidney diseases. Deep heat treatments should not be used on areas above the eye, heart, or on a pregnant patient. Deep heat treatments over areas with metal surgical implants should be avoided in case of rapid temperature increase and subsequent injury.

Description

There are four different ways to convey heat:

- Conduction is the transfer of heat between two objects in direct contact with each other.
- Conversion is the transition of one form of energy to heat.
- Radiation involves the transmission and absorption of electromagnetic waves to produce a heating effect.
- Convection occurs when a liquid or gas moves past a body part creating heat.

Hot packs, water bottles, and heating pads

Hot packs are a very common form of heat treatment utilizing conduction as a form of heat transfer. Moist heat packs are readily available in most hospitals, physical therapy centers, and athletic training rooms. Treatment temperature should not exceed 131 °F (55 °C). The pack is used over multiple layers of toweling to achieve a comfortable warming effect for approximately 30 minutes. More recently, several manufacturers have developed packs that may be warmed in a microwave over a specified amount of time prior to use.

Hot-water bottles are another form of superficial heat treatment. The bottles are filled half way with hot water between 115–125 °F (46.1–52 °C). Covered by a protective toweling, the hot-water bottle is placed on the treatment area and left until the water has cooled off.

Electrical heating pads continue to be used, however because of the need for an electrical outlet, safety and convenience become an issue.

Paraffin

Paraffin, a conductive form of superficial heat, is often used for heating uneven surfaces of the body such as the hands. It consists of melted paraffin wax and mineral oil. Paraffin placed in a small bath unit becomes solid at room temperature and is used as a liquid heat treatment when heated at 126–127.4 °F (52–53 °C). The most common form of paraffin

application is called the dip and wax method. In this technique, the patient will dip eight to 12 times and then the extremity will be covered with a plastic bag and a towel for insulation. Most treatment sessions are about 20 minutes.

Hydrotherapy

Hydrotherapy is used in a form of heat treatment for many musculoskeletal disorders. The hydrotherapy tanks and pools are all generally set at warm temperatures, never exceeding 150 °F (65.6 °C). Because the patient often performs resistance exercises while in the water, higher water temperatures become a concern as the treatment becomes more physically draining. Because of this, many hydrotherapy baths are now being set at 95–110 °F (35–43.3 °C). There are also units available with moveable turbine jets, which provide a light massage effect. Hydrotherapy is helpful as a warm-up prior to **exercise**.

Fluidotherapy

Fluidotherapy is a form of heat treatment developed in the 1970s. It is a dry heat modality consisting of cellulose particles suspended in air. Units come in different sizes and some are restricted to only treating a hand or foot. The turbulence of the gas-solid mixture provides thermal contact with objects that are immersed in the medium. Temperatures of this treatment range from 110–123 °F (43.3–50.5 °C). Fluidotherapy allows the patient to exercise the limb during the treatment, and also massages the limb, increasing blood flow.

Ultrasound

Ultrasound heat treatments penetrate the body to provide relief to inner tissue. Ultrasound energy comes from the acoustic or sound spectrum and is undetectable to the human ear. By using conducting agents such as gel or mineral oil, the ultrasound transducer warms areas of the musculoskeletal system. Some areas of the musculoskeletal system absorb ultrasound better than others. Muscle tissue and other connective tissue such as ligaments and tendons absorb this form of energy very well, however fat absorbs to a much lesser degree. Ultrasound has a relatively longlasting effect, continuing up to one hour.

Diathermy

Diathermy is another deep heat treatment. An electrode drum is used to apply heat to an affected area. It consists of a wire coil surrounded by dead space and other insulators such as a plastic housing. Plenty of toweling must be layered between the unit

and the patient. This device is unique in that it utilizes the basis of a magnetic field on connective tissues. One advantage of diathermy over various other heat treatments is that fat does resist an electrical field, which is not the case with a magnetic field. It is found to be helpful with those experiencing chronic **low back pain** and **muscle spasms**. Prior to ultrasound technology, diathermy was a popular heat therapy of the 1940s–1960s.

Preparation

Before administering any form of heat treatment, heat sensitivity is accessed and the skin over the affected area is cleansed. When a patient is undergoing any form of heat treatment, supervision should always be present especially in the treatment of hydrotherapy.

Aftercare

Once the heat treatment has been completed, any symptoms of **dizziness** and **nausea** should be noted and documented along with any skin irritations or discoloring not present prior to the heat treatment. A one hour interval between treatments should be adhered to in order to avoid restriction of blood flow.

Risks

All heat treatments have the potential of tissue damage resulting from excessive temperatures. Proper insulation and treatment duration should be carefully administered for each method. Overexposure during a superficial heat treatment may result in redness, blisters, burns, or reduced blood circulation. During ultrasound therapy, excessive treatment over bony areas with little soft tissue (such as hand, feet, and elbow) can cause excessive heat resulting in pain and possible tissue damage. Exposure to the electrode drum during diathermy may produce hot spots.

Resources

ORGANIZATIONS

American Physical Therapy Association. 1111 North Fairfax St., Alexandria, Virginia 22314. (800) 999-2782. <<https://www.apta.org>>.

Jeffrey P. Larson, RPT

Heatstroke see **Heat disorders**

Heavy menstruation see **Dysfunctional uterine bleeding**

Heavy metal poisoning

Definition

Heavy metal **poisoning** is the toxic accumulation of heavy metals in the soft tissues of the body.

Description

Heavy metals are chemical elements that have a specific gravity (a measure of density) at least five times that of water. The heavy metals most often implicated in accidental human poisoning are lead, mercury, arsenic, and cadmium. More recently, thallium has gained some attention in the media as the poison used in several murder cases in the 1990s. Some heavy metals, such as zinc, copper, chromium, iron, and manganese, are required by the body in small amounts, but these same elements can be toxic in larger quantities.

Heavy metals may enter the body in food, water, or air, or by absorption through the skin. Once in the body, they compete with and displace essential **minerals** such as zinc, copper, magnesium, and calcium, and interfere with organ system function. People may come in contact with heavy metals in industrial work, pharmaceutical manufacturing, and agriculture. Children may be poisoned as a result of playing in contaminated soil. **Lead poisoning** in adults has been traced to the use of lead-based glazes on pottery vessels intended for use with food, and contamination of Ayurvedic and other imported herbal remedies. Arsenic and thallium have been mixed with food or beverages to attempt **suicide** or poison others.

Another form of mercury poisoning that is seen more and more frequently in the United States is self-injected mercury under the skin. Some boxers inject themselves with mercury in the belief that it adds muscle bulk. Metallic mercury is also used in folk medicine or religious rituals in various cultures. These practices increase the risk of mercury poisoning of children in these ethnic groups or subcultures.

Causes and symptoms

Symptoms will vary, depending on the nature and the quantity of the heavy metal ingested. Patients may complain of **nausea**, **vomiting**, **diarrhea**, stomach pain, **headache**, sweating, and a metallic taste in the mouth. Depending on the metal, there may be blue-black lines in the gum tissues. In severe cases, patients exhibit obvious impairment of cognitive, motor, and language skills. The expression “mad as a hatter” comes from the mercury poisoning prevalent in 17th-century

France among hatmakers who soaked animal hides in a solution of mercuric nitrate to soften the hair.

Diagnosis

Heavy metal poisoning may be detected using blood and urine tests, hair and tissue analysis, or x ray. The diagnosis is often overlooked, however, because many of the early symptoms of heavy metal poisoning are nonspecific. The doctor should take a thorough patient history with particular emphasis on the patient's occupation.

In childhood, blood lead levels above 80 ug/dL generally indicate lead poisoning, however, significantly lower levels (>30 ug/dL) can cause **mental retardation** and other cognitive and behavioral problems in affected children. The Centers for Disease Control and Prevention considers a blood lead level of 10 ug/dL or higher in children a cause for concern. In adults, symptoms of lead poisoning are usually seen when blood lead levels exceed 80 ug/dL for a number of weeks.

Blood levels of mercury should not exceed 3.6 ug/dL, while urine levels should not exceed 15 ug/dL. Symptoms of mercury poisoning may be seen when mercury levels exceed 20 ug/dL in blood and 60 ug/dL in urine. Mercury levels in hair may be used to gauge the severity of chronic mercury exposure.

Since arsenic is rapidly cleared from the blood, blood arsenic levels may not be very useful in diagnosis. Arsenic in the urine (measured in a 24-hour collection following 48 hours without eating seafood) may exceed 50 ug/dL in people with arsenic poisoning. If acute arsenic or thallium poisoning is suspected, an x ray may reveal these substances in the abdomen (since both metals are opaque to x rays). Arsenic may also be detected in the hair and nails for months following exposure.

Cadmium toxicity is generally indicated when urine levels exceed 10 ug/dL of creatinine and blood levels exceed 5 ug/dL.

Thallium poisoning often causes hair loss (**alopecia**), **numbness**, and a burning sensation in the skin as well as nausea, vomiting, and **dizziness**. As little as 15–20 mg of thallium per kilogram of body weight is fatal in humans; however, smaller amounts can cause severe damage to the nervous system.

Treatment

When heavy metal poisoning is suspected, it is important to begin treatment as soon as possible

to minimize long-term damage to the patient's nervous system and digestive tract. Heavy metal poisoning is considered a medical emergency, and the patient should be taken to a hospital emergency room.

The treatment for most heavy metal poisoning is chelation therapy. A chelating agent specific to the metal involved is given either orally, intramuscularly, or intravenously. The three most common chelating agents are calcium disodium edetate, dimercaprol (BAL), and penicillamine. The chelating agent encircles and binds to the metal in the body's tissues, forming a complex; that complex is then released from the tissue to travel in the bloodstream. The complex is filtered out of the blood by the kidneys and excreted in the urine. This process may be lengthy and painful, and typically requires hospitalization. Chelation therapy is effective in treating lead, mercury, and arsenic poisoning, but is not useful in treating cadmium poisoning. To date, no treatment has been proven effective for cadmium poisoning. Thallium poisoning is treated with a combination of Prussian blue (potassium ferric hexacyanoferrate) and a diuretic, because about 35% of it is excreted in the urine; however, if treatment is not started within 72 hours of ingesting the poisoning, damage to the patient's nervous system may be permanent.

In cases of acute mercury, arsenic, or thallium ingestion, vomiting may be induced. **Activated charcoal** may be given in cases of thallium poisoning. Washing out the stomach (gastric lavage) may also be useful. The patient may also require treatment such as intravenous fluids for such complications of poisoning as shock, anemia, and kidney failure.

Patients who have taken arsenic, thallium, or mercury in a suicide attempt will be seen by a psychiatrist as part of emergency treatment.

Prognosis

The chelation process can only halt further effects of the poisoning; it cannot reverse neurological damage already sustained.

Prevention

Because arsenic and thallium were commonly used in rat and insect poisons at one time, many countries have tried to lower the rate of accidental poisonings by banning the use of heavy metals in pest control products. Thallium was banned in the United States as a rodent poison in 1984. As a result, almost all recent cases of arsenic and thallium

KEY TERMS

Alopecia—Loss of hair.

Chelation—The process by which a molecule encircles and binds to a metal and removes it from tissue.

Heavy metal—One of 23 chemical elements that has a specific gravity (a measure of density) at least five times that of water.

Prussian blue—The common name of potassium ferric hexacyanoferrate, a compound approved in the United States for treatment of thallium poisoning. Prussian blue gets its name from the fact that it was first used by artists in 1704 as a dark blue pigment for oil paints. It has also been used in laundry bluing and fabric printing.

poisoning in the United States were deliberate rather than accidental.

Because exposure to heavy metals is often an occupational hazard, protective clothing and respirators should be provided and worn on the job. Protective clothing should then be left at the work site and not worn home, where it could carry toxic dust to family members. Industries are urged to reduce or replace the heavy metals in their processes wherever possible. Exposure to environmental sources of lead, including lead-based paints, plumbing fixtures, vehicle exhaust, and contaminated soil, should be reduced or eliminated.

People who use Ayurvedic or traditional Chinese herbal preparations as alternative treatments for various illnesses should purchase them only from reliable manufacturers.

Resources

BOOKS

- Beers, Mark H., MD, and Robert Berkow, MD., editors. "Poisoning." Section 23, Chapter 307 In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.
- Beers, Mark H., MD, and Robert Berkow, MD., editors. "Psychiatric Emergencies." Section 15, Chapter 194 In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.
- Wilson, Billie A., Margaret T. Shannon, and Carolyn L. Stang. *Nurses Drug Guide 2000*. Stamford, CT: Appleton & Lange, 2000.

PERIODICALS

- Boyarsky, Igor, DO, and Adrain D. Crisan, MD. "Toxicity, Thallium." *eMedicine* August 3, 2004. <<http://www.emedicine.com/emerg/topic926.htm>> .
- Centers for Disease Control and Prevention (CDC). "Adult Blood Lead Epidemiology and Surveillance—United States, 2002." *Morbidity and Mortality Weekly Report* 53 (July 9, 2004): 578–582.
- Counter, S. A., and L. H. Buchanan. "Mercury Exposure in Children: A Review." *Toxicology and Applied Pharmacology* 198 (July 15, 2004): 209–230.
- Ferner, David J., MD. "Toxicity, Heavy Metals." *eMedicine* May 25, 2001. <<http://www.emedicine.com/EMERG/topic237.htm>> .
- Prasad, V. L. "Subcutaneous Injection of Mercury: 'Warding Off Evil'." *Environmental Health Perspectives* 111 (September 2004): 1326–1328.
- Schilling, U., R. Muck, and E. Heidemann. "Lead Poisoning after Ingestion of Ayurvedic Drugs." [in German] *Medizinische Klinik* 99 (August 15, 2004): 476–480.
- Thompson, D. F., and E. D. Callen. "Soluble or Insoluble Prussian Blue for Radiocesium and Thallium Poisoning?" *Annals of Pharmacotherapy* 38 (September 2004): 1509–1514.

ORGANIZATIONS

- American Society of Health-System Pharmacists (ASHP). 7272 Wisconsin Avenue, Bethesda, MD 20814. (301) 657-3000. <<http://www.ashp.org>> .
- Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>> .
- Food and Drug Administration. Office of Inquiry and Consumer Information. 5600 Fisher Lane, Room 12-A-40, Rockville, MD 20857. (301) 827-4420. <<http://www.fda.gov/fdahomepage.html>> .
- National Institutes of Health. National Institute of Environmental Health Sciences Clearinghouse. EnviroHealth, 2605 Meridian Parkway, Suite 115, Durham, NC 27713. (919) 361-9408.

Bethany Thivierge
Rebecca J. Frey, PhD

Heel spurs

Definition

A heel spur is a bony projection on the sole (plantar) region of the heel bone (also known as the calcaneus). This condition may accompany or result from severe cases of inflammation to the structure called plantar fascia. This associated plantar fascia is a

fibrous band of connective tissue on the sole of the foot, extending from the heel to the toes.

Description

Heel spurs are a common foot problem resulting from excess bone growth on the heel bone. The bone growth is usually located on the underside of the heel bone, extending forward to the toes. One explanation for this excess production of bone is a painful tearing of the plantar fascia connected between the toes and heel. This can result in either a heel spur or an inflammation of the plantar fascia, medically termed plantar fasciitis. Because this condition is often correlated to a decrease in the arch of the foot, it is more prevalent after the age of six to eight years, when the arch is fully developed.

Causes and symptoms

One frequent cause of heel spurs is an abnormal motion and mal-alignment of the foot called pronation. For the foot to function properly, a certain degree of pronation is required. This motion is defined as an inward action of the foot, with dropping of the inside arch as one plants the heel and advances the weight distribution to the toes during walking. When foot pronation becomes extreme from the foot turning in and dropping beyond the normal limit, a condition known as excessive pronation creates a mechanical problem in the foot. In some cases the sole or bottom of the foot flattens and becomes unstable because of this excess pronation, especially during critical times of walking and athletic activities. The portion of the plantar fascia attached into the heel bone or calcaneus begins to stretch and pull away from the heel bone.

At the onset of this condition, **pain** and swelling become present, with discomfort particularly noted as pushing off with the toes occurs during walking. This movement of the foot stretches the fascia that is already irritated and inflamed. If this condition is allowed to continue, pain is noticed around the heel region because of the newly formed bone, in response to the **stress**. This results in the development of the heel spur. It is common among athletes and others who run and jump a significant amount.

An individual with the lower legs angulating inward, a condition called genu valgum or "knock knees," can have a tendency toward excessive pronation. As a result, this too can lead to a fallen arch resulting in plantar fasciitis and heel spurs.



Illustration of bony projection, a spur, which developed from chronic irritation. (Photo Researchers. Reproduced by permission.)

Women tend to have more genu valgum than men do. Heel spurs can also result from an abnormally high arch.

Other factors leading to heel spurs include a sudden increase in daily activities, an increase in weight, or a change of shoes. Dramatic increase in training intensity or duration may cause plantar fasciitis. Shoes that are too flexible in the middle of the arch or shoes that bend before the toe joints will cause an increase in tension in the plantar fascia and possibly lead to heel spurs.

The pain this condition causes forces an individual to attempt walking on his or her toes or ball of the foot to avoid pressure on the heel spur. This can lead to other compensations during walking or running that in turn cause additional problems to the ankle, knee, hip, or back.

Diagnosis

A thorough medical history and physical exam by a physician is always necessary for the proper diagnosis of heel spurs and other foot conditions. X rays of the heel area are helpful, as excess bone production will be visible.

Treatment

Conservative

Heel spurs and plantar fasciitis are usually controlled with conservative treatment. Early intervention includes stretching the calf muscles while avoiding re-injuring the plantar fascia. Decreasing or changing activities, losing excess weight, and improving the proper fitting of shoes are all important measures to decrease this common source of foot pain. Modification of footwear includes shoes with a raised heel and better arch support. Shoe orthotics recommended by a healthcare professional are often very helpful in conjunction with exercises to increase strength of the foot muscles and arch. The orthotic prevents excess pronation and lengthening of the plantar fascia and continued tearing of this structure. To aid in this reduction of inflammation, applying ice for 10–15 minutes after activities and use of anti-inflammatory medication can be helpful. Physical therapy can be beneficial with the use of heat modalities, such as ultrasound that creates a deep heat and reduces inflammation. If the pain caused by inflammation is constant, keeping the foot raised above the heart and/or compressed by wrapping with an ace bandage will help.

Corticosteroid injections are also frequently used to reduce pain and inflammation. Taping can help speed the healing process by protecting the fascia from reinjury, especially during stretching and walking.

Heel surgery

When chronic heel pain fails to respond to conservative treatment, surgical treatment may be necessary. Heel surgery can provide relief of pain and restore mobility. The type of procedure used is based on examination and usually consists of releasing the excessive tightness of the plantar fascia, called a plantar fascia release. Depending on the presence of excess bony build up, the procedure may or may not include removal of heel spurs. Similar to other surgical interventions, there are various modifications and surgical enhancements regarding surgery of the heel.

KEY TERMS

Calcaneous—The heel bone.

Genu valgum—Deformity in which the legs are curved inward so that the knees are close together, nearly or actually knocking as a person walks with ankles widely apart of each other.

Plantar fascia—A tough fibrous band of tissue surrounding the muscles of the sole of the foot. Also called plantar aponeurosis.

Pronation—The lowering or descending of the inner edge of the foot by turning the entire foot outwards.

Alternative treatment

Acupuncture and accupressure have been used to address the pain of heel spurs, in addition to using friction massage to help break up scar tissue and delay onset of bony formations.

Prognosis

Usually, heel spurs are curable with conservative treatment. If not, heel spurs are curable with surgery. About 10% of those that continue to see a physician for plantar fascitis have it for more than a year. If there is limited success after approximately one year of conservative treatment, patients are often advised to have surgery.

Prevention

To prevent this condition, wearing shoes with proper arches and support is very important. Proper stretching is always a necessity, especially when there is an increase in activities or a change in running technique. It is not recommended to attempt working through the pain, as this can change a mild case of heel spurs and plantar fascitis into a long lasting and painful episode of this condition.

Resources

ORGANIZATIONS

American Orthopedic Foot and Ankle Society. 222 South Prospect, Park Ridge, IL 60068.

American Podiatry Medical Association. 9312 Old Georgetown Road, Bethesda, MD 20814.

Jeffrey P. Larson, RPT

Heimlich maneuver

Definition

The Heimlich maneuver is an emergency procedure for removing a foreign object lodged in the airway that is preventing a person from breathing.

Purpose

Every year about 3,000 adults die because they accidentally inhale rather than swallow food. The food gets stuck and blocks their trachea, making breathing impossible. **Death** follows rapidly unless the food or other foreign material can be displaced from the airway. This condition is so common it has been nicknamed the “cafe coronary.”

In 1974 Dr. Henry Heimlich first described an emergency technique for expelling foreign material blocking the trachea. This technique, now called the Heimlich maneuver or abdominal thrusts, is simple enough that it can be performed immediately by anyone trained in the maneuver. The Heimlich maneuver is a standard part of all first aid courses.

The theory behind the Heimlich maneuver is that by compressing the abdomen below the level of the diaphragm, air is forced under pressure out of the lungs dislodging the obstruction in the trachea and bringing the foreign material back up into the mouth.

The Heimlich maneuver is used mainly when solid material like food, coins, vomit, or small toys are blocking the airway. There has been some controversy about whether the Heimlich maneuver is appropriate to use routinely on **near-drowning** victims. After several studies of the effectiveness of the Heimlich maneuver on reestablishing breathing in near-drowning victims, the American Red Cross and the American Heart Association both recommend that the Heimlich maneuver be used only as a last resort after traditional airway clearance techniques and **cardiopulmonary resuscitation (CPR)** have been tried repeatedly and failed or if it is clear that a solid foreign object is blocking the airway.

Precautions

Incorrect application of the Heimlich maneuver can damage the chest, ribs, and internal organs of the person on whom it is performed. People may also vomit after being treated with the Heimlich maneuver.

Description

The Heimlich maneuver can be performed on all people. Modifications are necessary if the **choking** victim is very obese, pregnant, a child, or an infant.

Indications that a person's airway is blocked include:

- The person can not speak or cry out.
- The person's face turns blue from lack of oxygen.
- The person desperately grabs at his or her throat.
- The person has a weak **cough**, and labored breathing produces a high-pitched noise.
- The person does all of the above, then becomes unconscious.

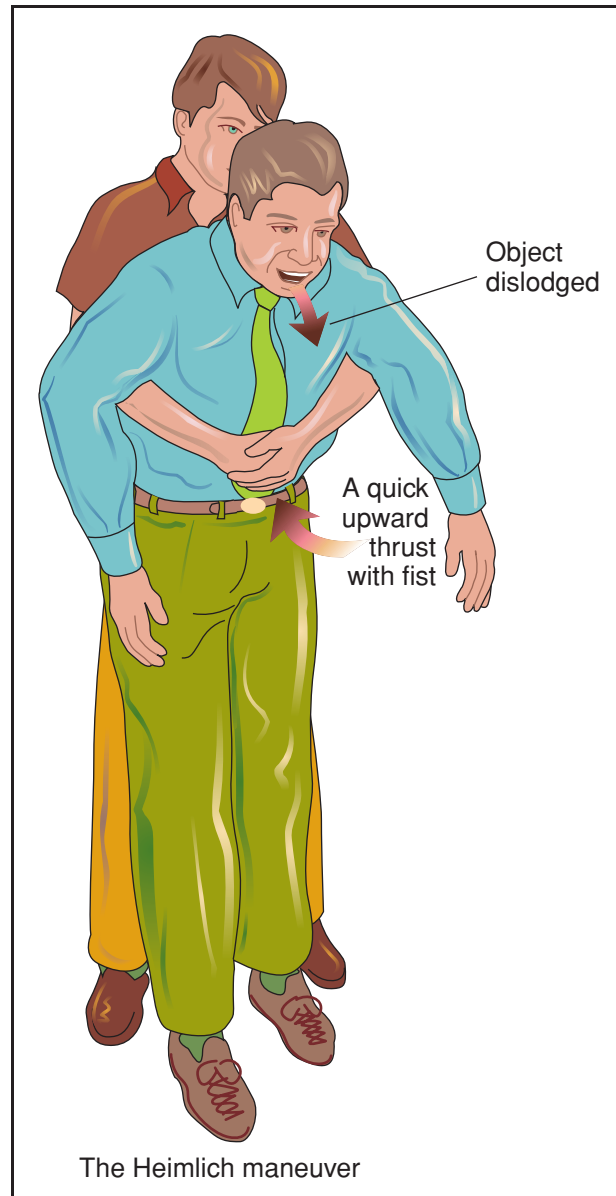
Performing the Heimlich maneuver on adults

To perform the Heimlich maneuver on a conscious adult, the rescuer stands behind the victim. The victim may either be sitting or standing. The rescuer makes a fist with one hand, and places it, thumb toward the victim, below the rib cage and above the waist. The rescuer encircles the victim's waist, placing his other hand on top of the fist.

In a series of 6–10 sharp and distinct thrusts upward and inward, the rescuer attempts to develop enough pressure to force the foreign object back up the trachea. If the maneuver fails, it is repeated. It is important not to give up if the first attempt fails. As the victim is deprived of oxygen, the muscles of the trachea relax slightly. Because of this loosening, it is possible that the foreign object may be expelled on a second or third attempt.

If the victim is unconscious, the rescuer should lay him or her on the floor, bend the chin forward, make sure the tongue is not blocking the airway, and feel in the mouth for **foreign objects**, being careful not to push any farther into the airway. The rescuer kneels astride the victim's thighs and places his fists between the bottom of the victim's breastbone and the navel. The rescuer then executes a series of 6–10 sharp compressions by pushing inward and upward.

After the abdominal thrusts, the rescuer repeats the process of lifting the chin, moving the tongue, feeling for and possibly removing the foreign material. If the airway is not clear, the rescuer repeats the abdominal thrusts as often as necessary. If the foreign object has been removed, but the victim is not breathing, the rescuer starts **CPR**.



To perform the Heimlich maneuver on a conscious adult (as illustrated above), the rescuer stands behind the victim and encircles his waist. The rescuer makes a fist with one hand and places the other hand on top, positioned below the rib cage and above the waist. The rescuer then applies pressure by a series of upward and inward thrusts to force the foreign object back up the victim's trachea. (Illustration by Electronic Illustrators Group.)

Performing the Heimlich maneuver under special circumstances

OBVIOUSLY PREGNANT AND VERY OBESE PEOPLE. The main difference in performing the Heimlich maneuver on this group of people is in the placement of the fists. Instead of using abdominal

thrusts, chest thrusts are used. The fists are placed against the middle of the breastbone, and the motion of the chest thrust is in and downward, rather than upward. If the victim is unconscious, the chest thrusts are similar to those used in CPR.

CHILDREN. The technique in children over one year of age is the same as in adults, except that the amount of force used is less than that used with adults in order to avoid damaging the child's ribs, breastbone, and internal organs.

INFANTS UNDER ONE YEAR OLD. The rescuer sits down and lays the infant along his or her forearm with the infant's face pointed toward the floor. The rescuer's hand supports the infant's head, and his or her forearm rests on his or her own thigh for additional support. Using the heel of the other hand, the rescuer administers four or five rapid blows to the infant's back between the shoulder blades.

After administering the back blows, the rescuer sandwiches the infant between his or her arms, and turns the infant over so that the infant is lying face up supported by the opposite arm. Using the free hand, the rescuer places the index and middle finger on the center of the breastbone and makes four sharp chest thrusts. This series of back blows and chest thrusts is alternated until the foreign object is expelled.

SELF-ADMINISTRATION OF THE HEIMLICH MANEUVER. To apply the Heimlich maneuver to oneself, one should make a fist with one hand and place it in the middle of the body at a spot above the navel and below the breastbone, then grasp the fist with the other hand and push sharply inward and upward. If this fails, the victim should press the upper abdomen over the back of a chair, edge of a table, porch railing or something similar, and thrust up and inward until the object is dislodged.

Preparation

Any lay person can be trained to perform the Heimlich maneuver. Knowing how may save someone's life. Before doing the maneuver, it is important to determine if the airway is completely blocked. If the person choking can talk or cry, Heimlich maneuver is not appropriate. If the airway is not completely blocked, the choking victim should be allowed to try to cough up the foreign object on his or her own.

Aftercare

Many people vomit after being treated with the Heimlich maneuver. Depending on the length and

KEY TERMS

Diaphragm—The thin layer of muscle that separates the chest cavity containing the lungs and heart from the abdominal cavity containing the intestines and digestive organs.

Trachea—The windpipe. A tube extending from below the voice box into the chest where it splits into two branches, the bronchi, that lead to each lung.

severity of the choking episode, the choking victim may need to be taken to a hospital emergency room.

Risks

Incorrectly applied, the Heimlich maneuver can break bones or damage internal organs. In infants, the rescuer should never attempt to sweep the baby's mouth without looking to remove foreign material. This is likely to push the material farther down the trachea.

Normal results

In many cases the foreign material is dislodged from the throat, and the choking victim suffers no permanent effects of the episode. If the foreign material is not removed, the person dies from lack of oxygen.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

Tish Davidson, A.M.

Helicobacter pylori infection see
Helicobacteriosis

Helicobacteriosis

Definition

Helicobacteriosis refers to infection of the gastrointestinal tract with the bacteria, *Helicobacter pylori* (*H. pylori*). While there are other rarer strains of *Helicobacter* species that can infect humans, only

H. pylori has been convincingly shown to be a cause of disease in humans. The organism was first documented to cause injury to the stomach in 1983, by two researchers in Australia, who ingested the organism to prove their theory. Since then, *H. pylori* has been shown to be the main cause of ulcer disease, and has revolutionized the treatment of peptic ulcer disease. It also is believed to be linked to various cancers of the stomach.

Description

H. pylori is a gram-negative, spiral-shaped organism, that contains flagella (tail-like structures) and other properties. In addition to flagella, which help the organism to move around in the liquid mucous layer of the stomach, *H. pylori* also produces an enzyme called urease, that protects it from gastric acid present in the stomach. As the production of this enzyme is relatively unusual, new diagnostic tests have enabled rapid identification of the bacteria.

H. pylori also produces two other chemicals: a cytotoxin called vacA, and a protein known as cagA. Patients with ulcer disease are more likely to produce the cytotoxin (vacA). The cagA protein not only occurs frequently in ulcer disease but also in cancer. It is still not known how these substances enable *H. pylori* to cause disease.

Causes and symptoms

Infection with *H. pylori* is largely dependent on two factors; age and income status. The bacteria is acquired mainly in childhood, especially in areas of poor hygiene or overcrowding. *H. pylori* is two to three times more prevalent in developing, non-industrialized countries. In the United States for example, the organism is believed to be present in about one third of the population.

The exact way in which *H. pylori* gets passed from one individual to another is uncertain, but person to person transmission is most likely. In most cases, children are felt to be the source of spread. Reinfection of those who have been cured has been documented, especially in areas of overcrowding.

The bacteria is well adapted to survival within the stomach. Not only does it survive there for years, but once infection begins, a form of chronic inflammation (chronic **gastritis**) always develops. In most individuals, initial infection causes little or no symptoms; however, some individuals such as the original researchers who ingested the bacteria, wind up with abdominal **pain** and **nausea**.

In about 15% of infected persons, ulcer disease develops either in the stomach or duodenum. Why some develop ulcer disease and others do not remains unclear. Ulcer symptoms are characterized by upper abdominal pain that is typically of a burning or “gnawing” type, and usually is rapidly relieved by **antacids** or food.

Acid secretion increases in most patients with duodenal ulcers. This increase returns to normal once *H. pylori* is eliminated. It is now known that elimination of the bacteria will substantially decrease the risk of recurrent bouts of ulcer disease in the vast majority of patients. In fact, a 2003 report showed that by eradicating *H. pylori*, ulcer bleeding rarely recurs.

In the last decade it has been shown that *H. pylori* is not only the prime cause of ulcer disease of the stomach and duodenum, but is also strongly associated with various tumors of the stomach. Bacterial infection is nine times more common in patients with cancer of the stomach, and seven times more common in those with lymphoma of the stomach (tumor of the lymphatic tissue), called a MALT tumor. It is believed that the prolonged inflammation leads to changes in cell growth and tumors. Eliminating *H. pylori* can lead to regression of some tumors.

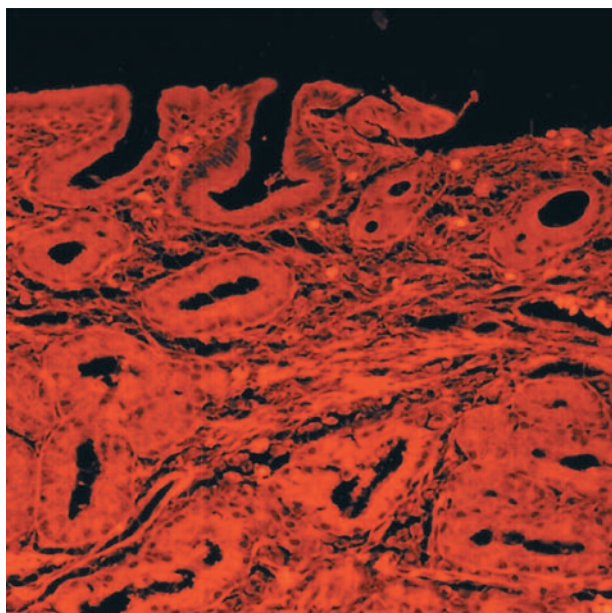
In addition to the above damage caused by *H. pylori*, some individuals lose normal gastric function, such as the ability to absorb vitamin B₁₂.

Diagnosis

There are basically two types of tests to identify infection: one group is “invasive” in that it involves the use of an endoscopy to obtain biopsy specimens for evaluation, while the other “noninvasive” methods depend on blood or breath samples. Invasive tests can be less accurate because of technical limitations: the biopsy may miss the area where the bacteria hides.

Invasive studies make use of tissue obtained by endoscopic biopsy to identify the organism. The bacteria can be searched for in pieces of biopsy tissue or grown (cultured) from the specimen. However, *H. pylori* is not easy to culture. Another method uses the bacteria’s production of the enzyme urease. Biopsy specimens are placed on a card that changes color if urease is present. Results often are available within a few minutes, but can take up to 24 hours.

Noninvasive tests are of two types: blood tests and breath test. Blood tests measure antibodies to make a diagnosis accurately within minutes. This can be done immediately in the doctor’s office. In addition,



A light microscopic image of a stomach ulcer. Gastric and duodenal ulcers are usually caused by infection with the bacteria *Helicobacter pylori*. This bacterium is also believed to be a cause of various cancers of the stomach. (Photograph by J.L. Carson, Custom Medical Stock Photo. Reproduced by permission.)

antibody levels can be measured several months after treatment, to see if *H. pylori* has been eradicated.

The breath test uses radioactive or non-radioactive forms of a compound called urea, which the patient drinks. The method that uses a radioactive form urea is easier to perform, as the equipment is commonly available in x-ray departments. Radiation exposure is less than that of a chest x ray. The test that uses non-radioactive urea is safer for children. A 2003 study in Brazil showed that the urea breath test and *H. pylori* stool antigen test also worked well to detect the bacteria in children. The breath test is the best way to be sure of elimination of *H. pylori*. The test can be used within 30 days after treatment. This is an advantage over following antibody levels that take six months or longer to diminish.

Treatment

H. pylori peptic ulcers are treated with drugs to kill the bacteria, drugs to reduce stomach acid, and drugs to protect the lining of the stomach. The antibiotics most commonly used to kill the bacteria are: amoxicillin, clarithromycin, metronidazole, and tetracycline. Drugs used to reduce stomach acid may be histamine blockers or **proton pump inhibitors**. The most commonly used histamine blockers are:

KEY TERMS

Antibiotic—A medication that is designed to kill or weaken bacteria.

Endoscope, Endoscopy—An Endoscope as used in the field of Gastroenterology is a thin flexible tube that uses a lens or miniature camera to view various areas of the gastrointestinal tract. When the procedure is performed to examine certain organs such as the bile ducts or pancreas, the organs are not viewed directly, but rather indirectly through the injection of x ray. The performance of an exam using an endoscope is referred by the general term endoscopy. Diagnosis through biopsies or other means and therapeutic procedures can be done with these instruments.

Gram-negative—Refers to the property of many bacteria in which they do not take or color with Gram's stain, a method which is used to identify bacteria. Gram-positive bacteria that take up the stain turn purple, while Gram-negative bacteria which do not take up the stain turn red.

cimetidine, famotidine, nizatidine, and ranitidine. The most commonly used proton pump inhibitors are: lansoprazole and omeprazole. The drug bismuth subsalicylate (a component of Pepto-Bismol) is used to protect the stomach lining.

The most common drug treatment is a two-week course of treatment called triple therapy. This treatment regimen involves taking two **antibiotics** to kill the bacteria and either an acid reducer or a stomach-lining shield. This therapy has been shown to kill the bacteria, reduce ulcer symptoms, and prevent ulcer recurrence in over 90% of patients.

The main drawback of triple therapy is that some patients find it difficult to follow because it often requires taking as many as 20 pills a day. The antibiotics also may cause unpleasant side effects that may make certain patients less likely to follow the treatment protocol. These side effects include: dark stools, **diarrhea**, **dizziness**, **headache**, a metallic taste in the mouth, nausea, **vomiting**, and yeast infections in women.

Prognosis

The elimination of *H. pylori* and cure of ulcer disease is now possible in more than 90% of those infected. The finding that most ulcers are due to an

infectious agent has brought a dramatic change in treatment and outlook for those suffering from the disease. Some patients will wind up with repeated infection, but this is most common in overcrowded areas.

Prevention

Attempts to develop a vaccine to protect against infection may be worthwhile in areas where the *H. pylori* infection rate and occurrence of **cancer** of the stomach is high. Research has shown such a vaccine would likely be safe in humans, but a vaccine has yet to be fully identified and developed as of mid-2003.

Resources

PERIODICALS

"Urea Breath, Stool Antigen Tests Work Well to Detect *H. Pylori* in Children." *Health & Medicine Week* September 22, 2003: 315.

"Vaccination Against *H. Pylori* Is an Achievable Goal." *Drug Week* July 18, 2003: 153.

Worcester, Sharon. "Eradicating *H. Pylori* May Prevent Bleeding Ulcers: No [Histamine. Sub2] Blockers Needed." *Internal Medicine News* September 15, 2003: 33.

OTHER

"*H. Pylori* and Peptic Ulcer." National Institutes of Health. <<http://www.niddk.nih.gov/health/digest/pubs/hpylori/hpylori.htm>>.

"Management Strategies for *Helicobacter pylori* Seropositive Patients with Dyspepsia." <<http://www.acponline.org/journals/annals/15feb97/treatcounsel.htm>>.

"Moving closer to an ulcer vaccine." <<http://www.msnbc.com/news/161712.asp>>.

"Treating Stomach Ulcers and *H. pylori* Infection." <<http://www.aafp.org/patientinfo/ulcers.html>>.

"What Is *Helicobacter pylori* Infection?" Centers for Disease Control. <http://www.cdc.gov/ncidod/aip/aip_a2b.htm>.

Paul A. Johnson, Ed.M.
Teresa G. Odle

Hellerwork

Definition

Hellerwork is a system of bodywork that combines deep tissue massage, body movement education, and verbal dialogue. It is designed to realign the

body's structure for overall health, improvement of posture, and reduction of physical and mental **stress**.

Purpose

Hellerwork improves posture and brings the body's natural structure into proper balance and alignment. This realignment can bring relief from general aches and pains; improve breathing; and relieve physical and mental stress. Hellerwork has also been used to treat such specific physical problems as chronic back, neck, shoulder, and joint **pain** as well as repetitive stress injuries, including **carpal tunnel syndrome**. Hellerwork is also used to treat and prevent athletic injuries.

Description

Origins

Joseph Heller (1940–) developed Hellerwork, a system of structural integration patterned after **Rolfing**. Although Heller received a degree in engineering and worked for NASA's Jet Propulsion Laboratory in Pasadena, CA, he became interested in humanistic psychology in the 1970s. He spent two years studying bioenergetics and **Gestalt therapy** as well as studying under architect and futurist Buckminster Fuller (1895–1983), flotation tank therapy developer John Lilly, family therapist Virginia Satir, and body movement pioneer **Judith Aston**.

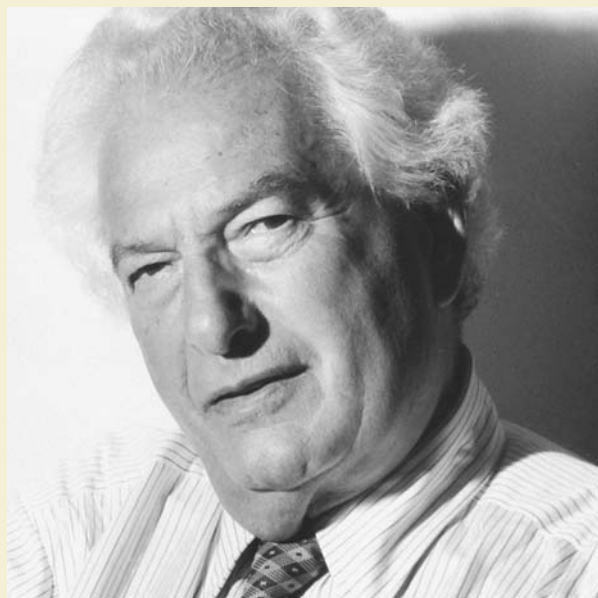
During this period, he trained for six years with Dr. Ida P. Rolf (1896–1979), the founder of Rolfing, and became a certified Rolfer in 1972. After Heller developed his own system of bodywork, he founded Hellerwork in 1979 and established a training facility in Mt. Shasta, California, where he continues his work.

Hellerwork is based largely on the principles of Rolfing, in which the body's connective tissue is manipulated or massaged to realign and balance the body's structure. Because Heller believes that physical realignment is insufficient, however, he expanded his system to include movement education and verbal dialogue as well as deep tissue massage.

Connective tissue massage

The **massage therapy** aspect of Hellerwork is designed to release the tension that exists in the deep connective tissue, called fascia, and return it to a normal alignment. The fascia is plastic and highly adaptable; it can tighten and harden in response to the general effects of gravity on the body, other ongoing

JOSEPH HELLER(1940–)



(AP/Wide World Photos. Reproduced by permission.)

Born in Poland, Joseph Heller attended school in Europe until age 16, when he immigrated to the United States. Living in Los Angeles, he attended the California Institute of Technology in Pasadena and graduated in 1962 with a degree in engineering. He worked for 10 years at the National Aeronautics and Space Administration's Jet Propulsion Laboratory (JPL) in Pasadena as an aerospace engineer. During his service at JPL, Heller became interested in humanistic psychology. After leaving JPL in 1972, he became director of Kairos, a center for human development in Los Angeles. He spent two years studying bioenergetics and gestalt. He also trained under Buckminster Fuller, flotation tank therapy developer John Lilly, self-esteem trainer Virginia Satir, and body movement pioneer Judith Aston.

He became a certified Rolfer in 1972 and spent the next six years studying structural integration under Rolfing founder Ida P. Rolf. He became the first president of the Rolf Institute in 1975. During his training with Rolf, Heller began developing his own system of bodywork. He left the institute in 1978 and moved to Northern California where he founded Hellerwork. He conducts classes and continues his work today at his headquarters, 406 Berry St., Mt. Shasta, CA 96067.

physical stresses, negative attitudes and emotions, and periodic physical traumas. One example of ongoing physical stress is carrying a briefcase, which pulls down the shoulder on one side of the body. Over time, the connective tissue becomes hard and stiff; the body becomes adapted to that position even when the person is not carrying a briefcase. In trying to adjust to the uneven weight distribution, the rest of the body becomes unbalanced and out of proper alignment.

Heller believes that as people age, more of these stress and trauma patterns become ingrained in the connective tissue, further throwing the body out of alignment. As stress accumulates, the body shortens and stiffens, a process commonly attributed to **aging**. Hellerwork seeks to recondition the body and make the connective tissue less rigid.

Movement education

The second component of Hellerwork, movement education, trains patients in the proper physical movements needed to keep the body balanced and correctly aligned. Movement education focuses on common actions, such as sitting, standing, and walking.

Hellerwork practitioners also teach better patterns of movement for activities that are specific to each individual, such as their job and favorite sports or social activities.

Verbal dialogue

Verbal dialogue is the third aspect of Hellerwork. It is designed to teach awareness of the relationships among emotions, life attitudes, and the body. Hellerwork practitioners believe that as patients become responsible for their attitudes, their body movements and patterns of self-expression improve. Dialogue focuses on the theme of each session and the area of the body that is worked on during that session.

Hellerwork consists of eleven 90-minute sessions costing about \$90–100 each. The first three sessions focus on the surface layers of the fascia and on developmental issues of infancy and childhood. The next four sessions are the core sessions and work on the deep layers and on adolescent developmental issues. The final four treatments are the integrative sessions, and build upon all the previous ones, while also looking at questions of maturity.

KEY TERMS

Bioenergetics—A system of therapy that combines breathing and body exercises, psychological therapy, and the free expression of emotions to release blocked physical and psychic energy.

Bodywork—A term that covers a variety of therapies that include massage, realignment of the body, and similar techniques to treat deeply ingrained stresses and traumas carried in the tissues of the body.

Chronic—A disease or condition that progresses slowly but persists or reoccurs over time.

Fascia—The sheet of connective tissue that covers the body under the skin and envelops the muscles and various organs.

Gestalt therapy—A form of therapy that focuses on helping patients reconnect with their bodies and their feelings directly, as contrasted with verbal intellectual analysis.

Kinesiology—The study of the anatomy and physiology of body movement, particularly in relation to therapy.

Rolfing—A deep-tissue therapy that involves manipulating the body's fascia to realign and balance the body's structure.

Preparations

No advance preparations are required to begin Hellerwork treatment. The treatment is usually done on a massage table with the patient wearing only undergarments.

Precautions

Since Hellerwork involves vigorous deep tissue massage, it is often described as uncomfortable and sometimes painful, especially during the first several sessions. As it requires the use of hands, it may be a problem for people who do not like or are afraid of being touched. It is not recommended as a treatment for any disease or a chronic inflammatory condition such as arthritis, and can worsen such a condition. Anyone with a serious medical condition, including heart disease, diabetes, or respiratory problems, should consult a medical practitioner before undergoing Hellerwork.

Side effects

There are no reported serious side effects associated with Hellerwork when delivered by a certified practitioner to adults and juveniles.

Research and general acceptance

As most alternative or holistic treatments, there is little mainstream scientific research documenting the effectiveness of Hellerwork therapy. Since the deep tissue massage aspect of Hellerwork is similar to Rolfing, however, several scientific studies of Rolfing may be useful in evaluating Hellerwork. A 1988 study published in the *Journal of the American Physical Therapy Association* indicated that Rolfing stimulates the parasympathetic nervous system, which can help speed the recovery of damaged tissue. A 1997 article in *The Journal of Orthopaedic and Sports Physical Therapy* reported that Rolfing can provide effective and sustained pain relief from lower back problems.

Resources

BOOKS

Goldten, Roger. *The Owner's Guide to the Body*. London: Thorsons, 1999.

Levine, Andrew S., and Valerie J. Levine. *The Bodywork and Massage Sourcebook*. Lincolnwood, IL: Lowell House, 1999.

ORGANIZATIONS

Hellerwork. 406 Berry St. Mt. Shasta, CA 96067.
(530) 926-2500. <<http://www.hellerwork.com>>.

Ken R. Wells

HELLP syndrome see **Preeclampsia and eclampsia**

Hemangiomas see **Birthmarks**

Hematocrit

Definition

The hematocrit measures how much space in the blood is occupied by red blood cells. It is useful when evaluating a person for anemia.

Purpose

Blood is made up of red and white blood cells, and plasma. A decrease in the number or size of red cells

also decreases the amount of space they occupy, resulting in a lower hematocrit. An increase in the number or size of red cells increases the amount of space they occupy, resulting in a higher hematocrit. **Thalassemia** is a condition which can cause an increased number of red blood cells but a decreased size and hematocrit.

The hematocrit is usually done on a person with symptoms of anemia. An anemic person has fewer or smaller than normal red cells. A low hematocrit, combined with other abnormal blood tests, confirms the diagnosis.

Some conditions, such as polycythemia, cause an overproduction of red blood cells, resulting in an increased hematocrit.

Transfusion decisions are based on the results of laboratory tests, including hematocrit. Transfusion is not considered if the hematocrit level is reasonable. The level differs for each person, depending on his or her clinical condition.

Description

Blood drawn from a fingerstick is often used for hematocrit testing. The blood fills a small tube, which is then spun in a small centrifuge. As the tube spins, the red blood cells go to the bottom of the tube, the white blood cells cover the red in a thin layer called the buffy coat, and the liquid plasma rises to the top. The spun tube is examined for the line that divides the red cells from the buffy coat and plasma. The height of the red cell column is measured as a percent of the total blood column. The higher the column of red cells, the higher the hematocrit.

The hematocrit test can also be done on an automated instrument as part of a complete blood count. It is also called Packed Red Cell Volume or Packed Cell Volume, or abbreviated as Hct or Crit. The test is covered by insurance when medically necessary. Results are usually available the same or following day.

Preparation

To collect the blood by fingerstick, a healthcare worker punctures a finger with a lancet and allows the blood to fill a small tube held to the puncture site.

Tests done on an automated instrument require 5–7 mL of blood. A healthcare worker ties a tourniquet on the person's upper arm, locates a vein in the inner elbow region, and inserts a needle into that vein. Vacuum action draws the blood through the needle into an attached tube. Collection of the sample takes only a few minutes.

KEY TERMS

Anemia—A condition where a person has fewer or smaller than normal red blood cells.

Hemoglobin—The percentage of space in blood occupied by red blood cells.

Aftercare

Discomfort or bruising may occur at the puncture site or the person may feel dizzy or faint. Pressure to the puncture site until the bleeding stops reduces bruising. Warm packs to the puncture site relieve discomfort.

Normal results

Normal values vary with age and sex. Adult male range is 42–52%, adult female 36–48%.

Abnormal results

Hematocrit values decrease when the size or number of red cells decrease. This is most common in anemia, but other conditions have similar effects: excessive bleeding, damaged cells due to a mechanical heart valve, **liver disease**, and cancers affecting the bone marrow. Additional tests, and the person's symptoms and medical history help distinguish these conditions or diagnose a specific type of anemia. Hematocrit values increase when the size or number of red cells increase, such as in polycythemia.

Fluid volume in the blood affects the hematocrit. Pregnant women have extra fluid, which dilutes the blood, decreasing the hematocrit. Dehydration concentrates the blood, increasing the hematocrit.

Nancy J. Nordenson

Hemiplegia see **Paralysis**

Hemochromatosis

Definition

Hemochromatosis is an inherited blood disorder that causes the body to retain excessive amounts of iron. This iron overload can lead to serious health consequences, most notably **cirrhosis** of the liver.

Description

Hemochromatosis is also known as iron overload, bronze diabetes, hereditary hemochromatosis and familial hemochromatosis. The inherited disorder causes increased absorption of intestinal iron, well beyond that needed to replace the body's loss of iron. Iron overload diseases afflict as many as 1.5 million persons in the United States. The most common of these, as well as one of the most common genetic disorders in the United States, is hereditary hemochromatosis. Men and women are equally affected by hemochromatosis, but women are diagnosed later in life because of blood loss from menstruation and **childbirth**. It most commonly appears in patients between the ages of 40–60 years, since it takes many years for the body to accumulate excessive iron. Symptoms appear later in females than in males—usually after **menopause**.

Hemochromatosis causes excess iron storage in several organs of the body including the liver, pancreas, endocrine glands, heart, skin, joints, and intestinal lining. The buildup of iron in these organs can lead to serious complications, including **heart failure**, **liver cancer**, and cirrhosis of the liver. It is estimated that about 5% of cirrhosis cases are caused by hereditary hemochromatosis.

Idiopathic pulmonary hemosiderosis, a disorder afflicting children and young adults, is a similar overload disorder characterized by abnormal accumulation of hemosiderin. Hemosiderin is a protein found in most tissues, especially the liver. It is produced by digestion of hematin, an iron-related substance.

Hemochromatosis is one of the most common genetic disorders in the United States. Approximately one in nine individuals have one abnormal hemochromatosis gene (11% of the population). Since everyone has two copies of each gene, these individuals have an abnormal *HFE* gene and a normal gene. They are called carriers. Between 1/200–1/400 individuals have two abnormal genes for hemochromatosis and no normal gene.

With most autosomal recessive conditions, an affected person's parents are carriers. If more than one family member has the condition, they are siblings. Hemochromatosis is so common, however, that families are seen in which both parents are affected, or one parent is affected and the other parent is a carrier. More than one generation may be affected, which is not usually seen in rare autosomal recessive conditions.

Causes and symptoms

Hereditary hemochromatosis is an autosomal recessive condition. This means that individuals with hemochromatosis have inherited an altered (mutated) gene from both of their parents. Affected individuals have two abnormal hemochromatosis genes and no normal hemochromatosis gene.

The gene that causes hemochromatosis has been identified, and the most common abnormalities of the gene have been described. The gene is on chromosome 6; it is called *HFE*. Scientists have not confirmed the function of the normal gene product; they do know that it interacts with the cell receptor for transferrin. Transferrin binds and transports iron in the blood.

Because it is an autosomal recessive condition, siblings of individuals who have hemochromatosis are at a 25% risk to also be affected. However, the likelihood that an individual will develop symptoms depends on which gene mutation he or she has as well as environmental factors. The two most common changes in the *HFE* gene are *C282Y* and *H63D*. The age at which symptoms begin is variable, even within the same family.

The symptoms of hemochromatosis include **fatigue**, weight loss, weakness, **shortness of breath**, heart **palpitations**, chronic abdominal **pain**, and impaired sexual performance. The patient may also show symptoms commonly connected with heart failure, diabetes or cirrhosis of the liver. Changes in the pigment of the skin may appear, such as grayness in certain areas, or a tanned or yellow (**jaundice**) appearance. The age of onset and initial symptoms vary.

Idiopathic pulmonary hemosiderosis may first, and only, appear as paleness of the skin. Sometimes, the patient will experience spitting of blood from the lungs or bronchial tubes.

Diagnosis

The most common diagnostic methods for hemochromatosis are blood studies of iron, genetic blood studies, **magnetic resonance imaging** (MRI), and **liver biopsy**. Blood studies of transferrin-iron saturation and ferritin concentration are often used to screen for iron overload. Ferritin is a protein that transports iron and liver enzymes. Additional studies are performed to confirm the diagnosis.

Blood studies used to confirm the diagnosis include additional iron studies and/or genetic blood studies. Genetic blood studies became available in the late 1990s. **Genetic testing** is a reliable method of diagnosis. However, in the year 2001 scientists and

physicians are studying how accurately having a hemochromatosis mutation predicts whether a person will develop symptoms. Most individuals affected with hemochromatosis (87%) have two identifiable gene mutations; that is, genetic testing will confirm the diagnosis in most individuals. Genetic studies are also be used to determine whether the affected person's family members are at risk for hemochromatosis. The results of genetic testing are the same whether or not a person has developed symptoms.

MRI scans and/or liver biopsy may be necessary to confirm the diagnosis. MRI studies of the liver (or other iron absorbing organs), with quantitative assessment of iron concentration, may reveal abnormal iron deposits. For the liver biopsy, a thin needle is inserted into the liver while the patient is under **local anesthesia**. The needle will extract a small amount of liver tissue, which can be analyzed microscopically to measure its iron content and other signs of hemochromatosis. Diagnosis of idiopathic pulmonary hemosiderosis begins with blood tests and x-ray studies of the chest.

Treatment

Patients who show signs of iron overload will often be treated with **phlebotomy**. Phlebotomy is a procedure that involves drawing blood from the patient, just like **blood donation**. Its purpose as a treatment is to rid the body of excess iron storage. Patients may need these procedures one or two times a week for a year or more. Less frequent phlebotomy may be continued in subsequent years to keep excess iron from accumulating. Patients who cannot tolerate phlebotomy due to other medical problems can be treated with Desferal (desferrioxamine). Diet restrictions may also be prescribed to limit the amount of iron ingested. Complications from hemochromatosis, such as cirrhosis or diabetes, may also require treatment. Treatment for idiopathic pulmonary hemosiderosis is based on symptoms.

Diet restrictions may help lower the amount of iron in the body, but do not prevent or treat hemochromatosis. Individuals who are affected or who know they have two *C282Y* and/or *H63D* genes may reduce iron intake by avoiding iron and mineral supplements, excess vitamin C, and uncooked seafood. If a patient is symptomatic, he/she may be advised to abstain from drinking alcohol.

Prognosis

With early detection and treatment, the prognosis is usually good. All potential symptoms are prevented

KEY TERMS

Autosomal—Relating to any chromosome besides the X and Y sex chromosomes. Human cells contain 22 pairs of autosomes and one pair of sex chromosomes.

Cirrhosis—A chronic degenerative disease of the liver, in which normal cells are replaced by fibrous tissue. Cirrhosis is a major risk factor for the later development of liver cancer.

Diabetes mellitus—The clinical name for common diabetes. It is a chronic disease characterized by inadequate production or use of insulin.

Phlebotomy—The taking of blood from the body through an incision in the vein, usually in the treatment of disease.

if iron levels are kept within the normal range, which is possible if the diagnosis is made before an individual is symptomatic. If a patient is symptomatic but treated successfully before he/she develops liver cirrhosis, the patient's life expectancy is near normal. However, if left untreated, complications may arise which can be fatal. These include liver **cancer**, liver cirrhosis, **diabetes mellitus**, congestive heart failure, and difficulty depleting iron overload through phlebotomy. Liver biopsy can be helpful in determining prognosis of more severely affected individuals. Genetic testing may also be helpful, as variable severity has been noted in patients who have two *C282Y* genes compared to patients with two *H63D* genes or one of each. Men are two times more likely than women to develop severe complications. The prognosis for patients with idiopathic pulmonary hemosiderosis is fair, depending on detection and complications.

Prevention

Screening for hemochromatosis is cost effective, particularly for certain groups of people. Relatives of patients with hemochromatosis—including children, siblings, and parents—should be tested by the most appropriate method. The best screening method may be iron and ferritin studies or genetic testing. If the affected person's diagnosis has been confirmed by genetic testing, relatives may have genetic testing to determine whether or not they have the genetic changes present in the affected individual. Many medical groups oppose genetic testing of children. Relatives who are affected but do not have symptoms can reduce iron intake and/or begin phlebotomy prior

to the onset of symptoms, possibly preventing ever becoming symptomatic.

In the winter of 2000, population screening for hereditary hemochromatosis is being widely debated. Many doctors and scientists want population screening because hemochromatosis is easily and cheaply treated, and quite common. Arguments against treatment include the range of symptoms seen (and not seen) with certain gene mutations, and the risk of discrimination in health and life insurance. Whether or not population screening becomes favored by a majority, the publicity is beneficial. Hemochromatosis is a common, easily and effectively treated condition. However, diagnosis may be difficult because the presenting symptoms are the same as those seen with many other medical problems. The screening debate has the positive effect of increasing awareness and suspicion of hemochromatosis. Increased knowledge leads to earlier diagnosis and treatment of symptomatic individuals, and increased testing of their asymptomatic at-risk relatives.

Resources

BOOKS

Barton, James C., and Corwin Q. Edwards, editors.
Hemochromatosis: Genetics, Pathophysiology, Diagnosis and Treatment. Cambridge: Cambridge University Press, 2000.

PERIODICALS

Motulsky, A.G., and E. Beutler. "Population Screening for Hemochromatosis." *Annual Review of Public Health* 21 (2000): 65-79.

ORGANIZATIONS

American Hemochromatosis Society, Inc. 777 E. Atlantic Ave., PMB Z-363, Delray Beach, FL 33483-5352. (561) 266-9037 or (888) 655-IRON. ahs@emi.net. <<http://www.americanhs.org>>.

American Liver Foundation. 1425 Pompton Ave., Cedar Grove, NJ 07009. (800) 223-0179. <<http://www.liverfoundation.org>>.

Hemochromatosis Foundation, Inc. PO Box 8569, Albany, NY 12208-0569. (518) 489-0972. s.kleiner@shiva.hunter.cuny.edu. <<http://www.hemochromatosis.org>>.

Iron Disorders Institute, Inc. PO Box 3021, Greenville, SC 29602. (864) 241-0111. irondis@aol.com. <<http://www.irondisorders.org>>.

Iron Overload Diseases Association, Inc. 433 Westwind Dr., North Palm Beach, FL 33408. (561) 840-8512. iod@ironoverload.org.

OTHER

"Hemochromatosis." *GeneClinics*. <<http://www.geneclinics.org/profiles/hemochromatosis/>>.

Hemochromatosis Information Sheet. National Institute of Diabetes & Digestive & Kidney Diseases (NIDDK). <<http://www.niddk.nih.gov/health/digest/pubs/hemochrom/hemochromatosis.htm>>.

Hereditary Hemochromatosis. Lecture by Richard Fass, MD, hematologist, Advanced Oncology Associates, given April 25, 1999. <<http://www.advancedoncology.org/listen.htm>>.

Michelle Q. Bosworth, MS, CGC

Hemodialysis see **Dialysis, kidney**

Hemoglobin electrophoresis

Definition

Hemoglobin electrophoresis (also called Hgb electrophoresis), is a test that measures the different types of hemoglobin in the blood. The method used is called electrophoresis, a process that causes movement of particles in an electric field, resulting in formation of "bands" that separate toward one end or the other in the field.

Purpose

Hgb electrophoresis is performed when a disorder associated with abnormal hemoglobin (hemoglobinopathy) is suspected. The test is used primarily to diagnose diseases involving these abnormal forms of hemoglobin, such as sickle cell anemia and thalassemia.

Precautions

Blood transfusions within the previous 12 weeks may alter test results.

Description

Hemoglobin (Hgb) is comprised of many different types, the most common being A₁, A₂, F, S, and C.

Hgb A₁ is the major component of hemoglobin in the normal red blood cell. Hgb A₂ is a minor component of normal hemoglobin, comprising approximately 2–3% of the total.

Hgb F is the major hemoglobin component in the fetus, but usually exists only in minimal quantities in the normal adult. Levels of Hgb F greater than 2% in patients over three years of age are considered abnormal.

Hgb S is an abnormal form of hemoglobin associated with the disease of sickle cell anemia, which occurs predominantly in African-Americans. A distinguishing characteristic of sickle cell disease is the crescent-shaped red blood cell. Because the survival rate of this type of cell is limited, patients with **sickle cell disease** also have anemia.

Hgb C is another hemoglobin variant found in African Americans. Red blood cells containing Hgb C have a decreased life span and are more readily destroyed than normal red blood cells, resulting in mild to severe **hemolytic anemia**.

Each of the major hemoglobin types has an electrical charge of a different degree, so the most useful method for separating and measuring normal and abnormal hemoglobins is electrophoresis. This process involves subjecting hemoglobin components from dissolved red blood cells to an electric field. The components then move away from each other at different rates, and when separated form a series of distinctly pigmented bands. The bands are then compared with those of a normal sample. Each band can be further assessed as a percentage of the total hemoglobin, thus indicating the severity of any abnormality.

Preparation

This test requires a blood sample. No special preparation is needed before the test.

Risks

Risks for this test are minimal, but may include slight bleeding from the blood-drawing site, fainting or feeling lightheaded after venipuncture, or hematoma (blood accumulating under the puncture site).

Normal results

Normal reference values can vary by laboratory, but are generally within the following ranges.

Adults:

- Hgb A₁: 95–98%
- Hgb A₂: 2–3%
- Hgb F: 0.8–2.0%
- Hgb S: 0%
- Hgb C: 0%.

Child (Hgb F):

- 6 months: 8%
- greater than 6 months: 1–2%
- newborn (Hgb F): 50–80%

KEY TERMS

Hemoglobin C disease—A disease of abnormal hemoglobin, occurring in 2–3% of African-Americans. Only those who have two genes for the disease develop anemia, which varies in severity. Symptoms include episodes of abdominal and joint pain, an enlarged spleen and mild jaundice.

Hemoglobin H disease—A thalassemia-like syndrome causing moderate anemia and red blood cell abnormalities.

Heterozygous—Two different genes controlling a specified inherited trait.

Homozygous—Identical genes controlling a specified inherited trait.

Thalassemias—The name for a group of inherited disorders resulting from an imbalance in the production of one of the four chains of amino acids that make up hemoglobin. Thalassemias are categorized according to the amino acid chain affected. The two main types are alpha-thalassemia and beta-thalassemia. The disorders are further characterized by the presence of one defective gene (thalassemia minor) or two defective genes (thalassemia major). Symptoms vary, but include anemia, jaundice, skin ulcers, gallstones, and an enlarged spleen.

Abnormal results

Abnormal reference values can vary by laboratory, but when they appear within these ranges, results are usually associated with the conditions that follow in parentheses.

Hgb A₂:

- 4–5.8% (β -thalassemia minor)
- under 2% (Hgb H disease)

Hgb F:

- 2–5% (β -thalassemia minor)
- 10–90% (β -thalassemia major)
- 5–35% (Heterozygous hereditary persistence of fetal hemoglobin, or HPFH)
- 100% (Homozygous HPFH)
- 15% (Homozygous Hgb S)

Homozygous Hgb S:

- 70–98% (Sickle cell disease).

Homozygous Hgb C:

- 90–98% (Hgb C disease)

Resources

BOOKS

Pagana, Kathleen Deska. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis: Mosby, Inc., 1998.

Janis O. Flores

Hemoglobin F test see **Fetal hemoglobin test**

Hemoglobin test

Definition

Hemoglobin is a protein inside red blood cells that carries oxygen throughout the body. A hemoglobin test reveals how much hemoglobin is in a person's blood, helping to diagnose and monitor anemia and **polycythemia vera**.

Purpose

A hemoglobin test is done when a person is ill or during a general **physical examination**. Good health requires an adequate amount of hemoglobin. The amount of oxygen in the body tissues depends on how much hemoglobin is in the red cells. Without enough hemoglobin, the tissues lack oxygen and the heart and lungs must work harder to try to compensate.

If the test indicates a “less than” or “greater than” normal amount of hemoglobin, the cause of the decrease or increase must be discovered. A low hemoglobin usually means the person has anemia. Anemia results from conditions that decrease the number or size of red cells, such as excessive bleeding, a dietary deficiency, destruction of cells because of a **transfusion** reaction or mechanical heart valve, or an abnormally formed hemoglobin.

A high hemoglobin may be caused by polycythemia vera, a disease in which too many red blood cells are made.

Hemoglobin levels also help determine if a person needs a blood transfusion. Usually a person's hemoglobin must be below 8 gm/dl before a transfusion is considered.

Description

Hemoglobin is made of heme, an iron compound, and globin, a protein. The iron gives blood its red color. Hemoglobin tests make use of this red color. A chemical is added to a sample of blood to make the red blood cells burst. When they burst, the red cells release hemoglobin into the surrounding fluid, coloring it clear red. By measuring the color using an instrument called a spectrophotometer, the amount of hemoglobin is determined.

Hemoglobin is often ordered as part of a complete blood count (CBC), a test that includes other blood cell measurements.

Some people inherit hemoglobin with an abnormal structure. These abnormal hemoglobins cause diseases, such as sickle cell or Hemoglobin C disease. Special tests, using a process called hemoglobin electrophoresis, identify abnormal hemoglobins.

Preparation

This test requires 5 mL of blood. A healthcare worker ties a tourniquet on the person's upper arm, locates a vein in the inner elbow region, and inserts a needle into that vein. Vacuum action draws the blood through the needle into an attached tube. Collection of the sample takes only a few minutes.

The person should avoid **smoking** before this test as smoking can increase hemoglobin levels.

Aftercare

Discomfort or bruising may occur at the puncture site or the person may feel dizzy or faint. Pressure to the puncture site until the bleeding stops reduces bruising. Warm packs to the puncture site relieve discomfort.

Normal results

Normal values vary with age and sex. Women generally have lower hemoglobin values than men. Men have 14.0–18.0 g/dL, while women have levels of 12.0–16.0 g/dL.

Abnormal results

A low hemoglobin usually indicates the person has anemia. Further tests are done to discover the cause and type of anemia. Dangerously low hemoglobin levels put a person at risk of a **heart attack**, congestive heart failure, or **stroke**.

KEY TERMS

Anemia—A condition characterized by a decrease in the size or number of red blood cells.

Hemoglobin—A protein inside red blood cells that carries oxygen to body tissues.

Polycythemia vera—A disease in which the bone marrow makes too many red blood cells.

A high hemoglobin indicates the body is making too many red cells. Further tests are done to see if this is caused by polycythemia vera, or as a reaction to illness, high altitudes, heart failure, or lung disease.

Fluid volume in the blood affects hemoglobin values. Pregnant women and people with cirrhosis have extra fluid, which dilutes the blood, decreasing the hemoglobin. **Dehydration** concentrates the blood, increasing the hemoglobin.

Resources

PERIODICALS

Hsia, Connie C. W. "Respiratory Function of Hemoglobin." *New England Journal of Medicine* 338 (January 1998): 239-247.

Nancy J. Nordenson

Hemoglobinopathies

Definition

Hemoglobinopathies are genetic (inherited) disorders of hemoglobin, the oxygen-carrying protein of the red blood cells.

Description

The hemoglobin molecule is composed of four separate polypeptide chains of amino acids, two alpha chains and two beta chains, as well as four iron-bearing heme groups that bind oxygen. The alpha chains are coded for by two similar genes on chromosome 16; the beta chains by a single gene on chromosome 11. Mutations and deletions in these genes cause one of the many hemoglobinopathies.

In general, hemoglobinopathies are divided into those in which the gene abnormality results in a

qualitative change in the hemoglobin molecule and those in which the change is quantitative. Sickle cell anemia (**sickle cell disease**) is the prime example of the former, and the group of disorders known as the thalassemias constitute the latter. It has been estimated that one third of a million people worldwide are seriously affected by one of these genetic disorders.

Causes and symptoms

Sickle cell anemia (SSA), an autosomal recessive disorder more common in the Black population, is caused by a single mutation in the gene that codes for the beta polypeptide. Approximately 1/400 to 1/600 African-Americans are born with the disorder, and, one in ten is a carrier of one copy of the mutation. In certain parts of the African continent, the prevalence of the disease reaches one in fifty individuals.

The sickle cell mutation results in the substitution of the amino acid valine for glutamic acid in the sixth position of the beta polypeptide. In turn, this alters the conformation of the hemoglobin molecule and causes the red blood cells to assume a characteristic sickle shape under certain conditions. These sickle-shaped cells, no longer able to pass smoothly through small capillaries, can block the flow of blood. This obstruction results in symptoms including growth retardation, severe **pain** crises, tissue and organ damage, splenomegaly, and strokes. Individuals with SSA are anemic and prone to infections, particularly **pneumonia**, a significant cause of **death** in this group. Some or all of these symptoms are found in individuals who have the sickle mutation in both copies of their beta-globin gene. Persons with one abnormal gene and one normal gene are said to be carriers of the sickle cell trait. Carriers are unaffected because of the remaining normal copy of the gene.

The thalassemias are a diverse group of disorders characterized by the fact that the causative mutations result in a decrease in the amount of normal hemoglobin. Thalassemias are common in Mediterranean populations as well as in Africa, India, the Mideast, and Southeast Asia. The two main types of thalassemias are alpha-thalassemia due to mutations in the alpha polypeptide and beta-thalassemia resulting from beta chain mutations.

Since individuals possess a total of four genes for the alpha polypeptide (two genes on each of their two chromosomes 16), disease severity depends on how many of the four genes are abnormal. A defect in one or two of the genes has no clinical effect. Abnormalities of three results in a mild to moderately severe anemia (hemoglobin H disease) and splenomegaly. Loss of

function of all four genes usually causes such severe oxygen deprivation that the affected fetus does not survive. A massive accumulation of fluid in the fetus (hydrops fetalis) results in **stillbirth** or neonatal death.

Beta thalassemias can range from mild and clinically insignificant (beta **thalassemia** minor) to severe and life-threatening (beta thalassemia major, also known as Cooley's anemia), depending on the exact nature of the gene mutation and whether one or both copies of the beta gene are affected. While the milder forms may only cause slight anemia, the more severe types result in growth retardation, skeletal changes, splenomegaly, vulnerability to infections, and death as early as the first decade of life.

Diagnosis

Many countries, including the United States, have made concerted efforts to screen for sickle cell anemia at birth because of the potential for beginning early treatment and counseling parents about their carrier status. Diagnosis is traditionally made by blood tests including **hemoglobin electrophoresis**. Similar tests are used to determine whether an individual is a sickle cell or thalassemia carrier. In certain populations with a high prevalence of one of the mutations, carrier testing is common. If both members of a couple are carriers of one of these conditions, it is possible through prenatal **genetic testing** to determine if the fetus will be affected, although the severity of the disease cannot always be predicted.

Treatment

Treatment of SSA has improved greatly in recent years with a resulting increase in life expectancy. The use of prophylactic (preventative) antibiotic therapy has been particularly successful. Other treatments include fluid therapy to prevent **dehydration**, oxygen supplementation, pain relievers, blood transfusions, and several different types of medications. Recent interest has focused on **bone marrow transplantation**, which has been successful in selected patients.

Since the clinically important thalassemias are characterized by severe anemia, the traditional treatment has been blood **transfusion**, but the multiple transfusions needed to sustain life lead to an iron overload throughout the tissues of the body and eventual destruction of the heart and other organs. For this reason, transfusion therapy must also include infusions of medications such as deferoxamine (desferoxamine) to rid the body of excess iron. **Phlebotomy** is another technique that has been used with some

success to lower the concentration of iron in the patient's blood. As with sickle cell anemia, bone marrow therapy has been successful in some cases.

Until very recently, patients being treated with bone marrow transplants had to find a sibling or other closely related donor in order to avoid rejection of the transplant. Advances in the preparation of the transplanted cells, however, have made the use of bone marrow from unrelated donors (URD) an option for patients with hemoglobinopathies. As of 2003, the National Marrow Donor Program reports that about 40% of bone marrow transplants involve a patient in the United States receiving marrow from an international donor or an international patient receiving marrow from a donor in the United States.

Emphasis is also being placed on developing drugs that treat sickle cell anemia directly. The most promising of these drugs in the late 1990s is hydroxyurea, a drug that was originally designed for anticancer treatment. Hydroxyurea has been shown to reduce the frequency of painful crises and acute chest syndrome in adults, and to lessen the need for blood transfusions. Hydroxyurea seems to work by inducing a higher production of fetal hemoglobin. The major side effects of the drug include decreased production of platelets, red blood cells, and certain white blood cells. The effects of long-term hydroxyurea treatment are unknown; however, a nine-year follow-up study of 299 adults with frequent painful crises reported in 2003 that taking hydroxyurea was associated with a 40% reduction in mortality.

Another promising development for the treatment of hemoglobinopathies is **gene therapy**, which has interested researchers since the early 1990s. In late 2001, genetic scientists reported that they had designed a gene that might lead to a future treatment of sickle cell anemia. Although the gene had not been tested in humans, early results showed that the injected gene protected cells from sickling. As of 2003, experiments in gene therapy for sickle cell disease have been carried out in mice, using lentiviral vectors to transfer the corrective gene into the mouse's stem cells. This technique, however, has not yet been attempted in human subjects as of late 2003.

Prognosis

Hemoglobinopathies are life-long disorders. The prognosis depends upon the exact nature of the mutation, the availability of effective treatment, as well as the individual's compliance with therapies. Hemoglobinopathies significantly complicate **pregnancy**, and increase the risk of infant mortality.

KEY TERMS

Amino acids—Organic compounds that form the building blocks of protein. There are 20 different amino acids.

Autosomal recessive—A pattern of inheritance in which both copies of an autosomal gene must be abnormal for a genetic condition or disease to occur. An autosomal gene is a gene that is located on one of the autosomes or non-sex chromosomes. When both parents have one abnormal copy of the same gene, they have a 25% chance with each pregnancy that their offspring will have the disorder.

Hemoglobin—Protein-iron compound in the blood that carries oxygen to the cells and carries carbon dioxide away from the cells.

Hydroxyurea—A drug that has been shown to induce production of fetal hemoglobin. Fetal hemoglobin has a pair of gamma-globin molecules in place of the typical beta-globins of adult hemoglobin. Higher-than-normal levels of fetal hemoglobin can prevent sickling from occurring.

Phlebotomy—Drawing blood from a vein for diagnosis or treatment. Phlebotomy is sometimes used in the treatment of hemoglobinopathies to lower the iron concentration of the blood.

Sickle cell—A red blood cell that has assumed an elongated shape due to the presence of hemoglobin S.

Splenomegaly—Enlargement of the spleen.

Prevention

Because the hemoglobinopathies are inherited diseases, primary prevention involves carriers making reproductive decisions to prevent passage of the abnormal gene to their offspring. At present, most prevention is targeted toward the symptoms using treatments such as those described above.

Resources

BOOKS

- Beers, Mark H., MD, and Robert Berkow, MD., editors. "Anemias Caused by Excessive Hemolysis: Sickle Cell Diseases." Section 11, Chapter 127 In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

- Beers, Mark H., MD, and Robert Berkow, MD., editors. "Pregnancy Complicated by Disease: Hemoglobinopathies." Section 18, Chapter 251 In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.
- Behrman, Richard E., et al. *Nelson Textbook of Pediatrics*. 16th ed. Philadelphia, W. B. Saunders, 2000.
- Jorde, Lynn B., et al. *Medical Genetics*. 2nd ed. New York: Mosby, 1999.

PERIODICALS

- Davies, S. C., and A. Gilmore. "The Role of Hydroxyurea in the Management of Sickle Cell Disease." *Blood Reviews* 17 (June 2003): 99–109.
- Koduri, P. R. "Iron in Sickle Cell Disease: A Review Why Less Is Better." *American Journal of Hematology* 73 (May 2003): 59–63.
- Krishnamurti, L., S. Abel, M. Maiers, and S. Flesch. "Availability of Unrelated Donors for Hematopoietic Stem Cell Transplantation for Hemoglobinopathies." *Bone Marrow Transplantation* 31 (April 2003): 547–550.
- Markham, M. J., R. Lottenberg, and M. Zumberg. "Role of Phlebotomy in the Management of Hemoglobin SC Disease: Case Report and Review of the Literature." *American Journal of Hematology* 73 (June 2003): 121–125.
- Nienhuis, A. W., H. Hanawa, N. Sawai, et al. "Development of Gene Therapy for Hemoglobin Disorders." *Annals of the New York Academy of Science* 996 (May 2003): 101–111.
- Olivieri, Nancy F. "The Beta-Thalassemias." *The New England Journal of Medicine* 341, no. 2 (July 1999): 99–109.
- Steinberg, M. H., F. Barton, O. Castro, et al. "Effect of Hydroxyurea on Mortality and Morbidity in Adult Sickle Cell Anemia: Risks and Benefits up to 9 Years of Treatment." *Journal of the American Medical Association* 289 (April 2, 2003): 1645–1651.

ORGANIZATIONS

- American Sickle Cell Anemia Association. <<http://www.ascaa.org>>.
- National Marrow Donor Program (NMDP). Suite 500, 3001 Broadway Street Northeast, Minneapolis, MN 55413. (800) 627-7692. <<http://www.marrow-donor.org>>.
- Sickle Cell Disease Association of America, Inc. 200 Corporate Point Suite 495, Culver City, CA 90230-8727. (800) 421-8453. Scdaa@sicklecelldisease.org. <<http://sicklecelldisease.org/>>.
- Sickle Cell Information Center. PO Box 109, Grady Memorial Hospital, 80 Bulter Street, SE, Atlanta, GA 30303. (404) 616- 3572. <<http://www.emory.edu>>.

Sallie Boineau Freeman, PhD
Rebecca J. Frey, PhD

Hemolytic anemia

Definition

Red blood cells have a normal life span of approximately 90–120 days, at which time the old cells are destroyed and replaced by the body's natural processes. Hemolytic anemia is a disorder in which the red blood cells are destroyed prematurely. The cells are broken down at a faster rate than the bone marrow can produce new cells. Hemoglobin, the component of red blood cells that carries oxygen, is released when these cells are destroyed.

Description

As a group, **anemias** (conditions in which the number of red blood cells or the amount of hemoglobin in them is below normal) are the most common blood disorders. Hemolytic anemias, which result from the increased destruction of red blood cells, are less common than anemias caused by excessive blood loss or by decreased hemoglobin or red cell production.

Since a number of factors can increase red blood cell destruction, hemolytic anemias are generally identified by the disorder that brings about the premature destruction. Those disorders are classified as either inherited or acquired. Inherited hemolytic anemias are caused by inborn defects in components of the red blood cells—the cell membrane, the enzymes, or the hemoglobin. Acquired hemolytic anemias are those that result from various other causes. With this type, red cells are produced normally, but are prematurely destroyed because of damage that occurs to them in the circulation.

Causes and symptoms

Inherited hemolytic anemias involve conditions that interfere with normal red blood cell production. Disorders that affect the red blood cell membrane include hereditary spherocytosis, in which the normally disk-shaped red cells become spherical, and hereditary elliptocytosis, in which the cells are oval, rather than disk-shaped. Other hereditary conditions that cause hemolytic anemia include disorders of the hemoglobin, such as sickle cell anemia and **thalassemia**, and red blood cell enzyme deficiencies, such as G6PD deficiency.

The causes of acquired hemolytic anemias vary, but the most common are responses to certain medications and infections. Medications may cause the body to develop antibodies that bind to the red blood cells

and cause their destruction in the spleen. Immune hemolytic anemia most commonly involves antibodies that react against the red blood cells at body temperature (warm-antibody hemolytic anemia), which can cause premature destruction of the cells. About 20% of hemolytic anemias caused by warm antibodies come from diseases such as lymphocytic leukemia, 10% from an autoimmune disease, and others are drug-induced. Cold-antibody hemolytic anemia is a condition in which the antibodies react with the red blood cells at a temperature below that of normal body temperature. Red blood cells can also receive mechanical damage as they circulate through the blood vessels. Aneurysms, artificial heart valves, or very high blood pressure can cause the red cells to break up and release their contents. In addition, hemolytic anemia may be caused by a condition called **hypersplenism**, in which a large, overactive spleen rapidly destroys red blood cells.

Major symptoms of hemolytic anemias are similar to those for all anemias, including shortness of breath; noticeable increase in heart rate, especially with exertion; **fatigue**; pale appearance; and dark urine. A yellow tint, or **jaundice**, may be seen in the skin or eyes of hemolytic anemia patients. Examination may also show an enlarged spleen. A more emergent symptom of hemolytic anemia is **pain** in the upper abdomen. Severe anemia is indicated if there are signs of **heart failure** or an enlarged liver.

Diagnosis

In order to differentiate hemolytic anemia from others, physicians will examine the blood for the number of young red blood cells, since the number of young cells is increased in hemolytic anemia. The physician will also examine the abdominal area to check for spleen or liver enlargement. If the physician knows the duration of hemolysis, it may also help differentiate between types of anemia. There are a number of other indications that can be obtained from blood samples that will help a physician screen for hemolytic anemia. An antiglobulin (Coomb's) test may be performed as the initial screening exam after determining hemolysis. In the case of immune hemolytic anemia, a direct Coomb's test is almost always positive.

Treatment

Treatment will depend on the cause of the anemia, and may involve treatment of the underlying cause. If the hemolytic anemia was brought on by hereditary spherocytosis, the spleen may be removed. Corticosteroid medications, or adrenal steroids, may

KEY TERMS

Antibody—Antibodies are parts of the immune system which counteract or eliminate foreign substances or antigens.

Erythrocyte—The name for red blood cells or red blood corpuscles. These components of the blood are responsible for carrying oxygen to tissues and removing carbon dioxide from tissues.

Hemolysis—The process of breaking down of red blood cells. As the cells are destroyed, hemoglobin, the component of red blood cells which carries the oxygen, is liberated.

Thalassemia—One of a group of inherited blood disorders characterized by a defect in the metabolism of hemoglobin, or the portion of the red blood cells that transports oxygen throughout the blood stream.

be effective, especially in hemolytic anemia due to antibodies. If the cause of the disorder is a medication, the medication should be stopped. When anemia is severe in conditions such as sickle cell anemia and thalassemia, blood transfusions may be indicated.

Prognosis

Hemolytic anemias are seldom fatal. However, if left untreated, hemolytic anemia can lead to heart failure or liver complications.

Prevention

Hemolytic anemia due to inherited disorders can not be prevented. Acquired hemolytic anemia may be prevented if the underlying disorder is managed properly.

Resources

ORGANIZATIONS

American Autoimmune Related Diseases Association, Inc.

Focus: A quarterly newsletter of the AARDA. Detroit, MI. (313) 371-8600. <<http://www.aarda.org>>.

The American Society of Hematology. 1200 19th Street NW, Suite 300, Washington, DC 20036-2422. (202) 857-1118. <<http://www.hematology.org>>.

National Heart, Lung and Blood Institute. PO Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

Teresa Odle

Hemolytic-uremic syndrome

Definition

Hemolytic-uremic syndrome (HUS) is a rare condition that affects mostly children under the age of 10, but also may affect the elderly as well as persons with other illnesses. HUS, which most commonly develops after a severe bowel infection with certain toxic strains of a bacteria, is characterized by destruction of red blood cells, damage to the lining of blood vessel walls, and in severe cases, kidney failure.

Description

Most cases of HUS occur after an infection in the digestive system that has been caused by toxin-producing strains of the bacterium *Escherichia coli*. About 75% of HUS cases in the United States are caused by the strain referred to as *E. coli* O157:H7, which is found in the intestinal tract of cattle, while the remaining cases are caused by non-O157 strains. Some children infected with *E. coli* O157:H7 will develop HUS. HUS also can follow respiratory infection episodes in young children. In the United States, there are about 20,000 infections and 250 deaths annually that are caused by *E. coli* O157:H7. HUS has also been known to occur in persons using drugs such as **oral contraceptives**, immunosuppressors, and antineoplastics, and in women during the postpartum period.

E. coli O157:H7, first identified in 1982, and isolated with increasing frequency since then, is found in contaminated foods such as meat, dairy products, and juices. Infection with *E. coli* O157:H7 causes severe **gastroenteritis**, which can include abdominal **pain**, **vomiting**, and bloody **diarrhea**. For most children, the vomiting and diarrhea stop within two to three days. However, about 5 to 10% of the children will develop HUS and will become pale, tired, and irritable. Toxins produced by the bacteria enter the blood stream, where they destroy red blood cells and platelets, which contribute to the clotting of blood. The damaged red blood cells and platelets clog tiny blood vessels in the kidneys, or form lesions to occur in the kidneys, making it difficult for the kidneys to remove wastes and extra fluid from the body, resulting in **hypertension**, fluid accumulation, and reduced production of urine.

Causes and symptoms

The most common way an *E. coli* O157:H7 infection is contracted is through the consumption of undercooked ground beef (e.g., eating hamburgers

that are still pink inside). Healthy cattle carry *E. coli* within their intestines. During the slaughtering process, the meat can become contaminated with the *E. coli* from the intestines. When contaminated beef is ground up, the *E. coli* are spread throughout the meat. Additional ways to contract an *E. coli* infection include drinking contaminated water and unpasteurized milk and juices, eating contaminated fruits and vegetables, and working with cattle. The infection is also easily transmitted from an infected person to others in settings such as day care centers and nursing homes when improper sanitary practices are used.

Symptoms of an *E. coli* O157:H7 infection start about seven days after infection with the bacteria. The first symptom is sudden onset of severe abdominal cramps. After a few hours, watery diarrhea starts, causing loss of fluids and electrolytes (**dehydration**), which causes the person to feel tired and ill. The watery diarrhea lasts for about a day, and then changes to bright red bloody stools, as the infection causes sores to form in the intestines. The bloody diarrhea lasts for two to five days, with as many as ten bowel movements a day. Additional symptoms may include **nausea and vomiting**, without a **fever**, or with only a mild fever. After about five to ten days, HUS can develop, which is characterized by paleness, irritability, and **fatigue**, as well as reduced urine production.

Diagnosis

The diagnosis of an *E. coli* infection is made through a **stool culture**. The culture must be taken within the first 48 hours after the start of the bloody diarrhea. If a positive culture is obtained, the patient should be monitored for the development of HUS, with treatment initiated as required.

Children should not go to day care until they have had two negative stool cultures. Older people in nursing homes should stay in bed until two stool cultures are negative.

Treatment

Treatment of HUS is supportive, with particular attention to management of fluids and electrolytes. Treatment generally is provided in a hospital setting. Blood transfusions may be required. In about 50% of the cases, short term replacement of kidney function is required in the form of dialysis. Most patients will recover kidney function and be able to discontinue dialysis.

Some studies have shown that the use of **antibiotics** and antimotility agents during an *E. coli* infection

KEY TERMS

Antineoplastics—Agents that inhibit or prevent the development, maturation, and proliferation of malignant cells.

Gastroenteritis—An acute inflammation of the lining of the stomach and intestines, characterized by nausea, diarrhea, abdominal pain and weakness, which has various causes, including food poisoning due to infection with such organisms as *Escherichia coli*, *Staphylococcus aureus* and *Salmonella* species, consumption of irritating food or drink, or psychological factors such as anger, stress and fear.

may worsen the course of the infection and should be avoided. However, other studies have been less definitive. Physicians should stay informed so that clinical practices matches medical advances on this aspect of treatment.

Alternative treatment

Persons with HUS must be under the care of health care professionals skilled in the treatment of HUS.

Prognosis

Ninety percent of children with HUS who receive careful supportive care survive the initial acute stages of the condition, with most having no long-term effects. However, between 10 and 30 percent of the survivors will have kidney damage that will lead to kidney failure immediately or within several years. These children with kidney failure require on-going dialysis to remove wastes and extra fluids from their bodies, or may require a kidney transplant.

Prevention

Prevention of HUS caused by ingestion of foods contaminated with *E. coli* O157:H7 and other toxin-producing bacteria is accomplished through practicing hygienic food preparation techniques, including adequate handwashing, cooking of meat thoroughly, defrosting meats safely, vigorous washing of fruits and vegetables, and handling leftovers properly. Irradiation of meat has been approved by the United States Food and Drug Administration and the United States Department of Agriculture in order to decrease bacterial contamination of consumer meat supplies.

Resources

OTHER

National Kidney and Urologic Diseases Information Clearinghouse. Fact Sheet: Hemolytic Uremic Syndrome. NIH Publication No. 99-4570. March 2000. <<http://www.niddk.nih.gov/health/kidney/summary/hus/>>.

Judith Sims

Hemophilia

Definition

Hemophilia is a genetic disorder—usually inherited—of the mechanism of blood clotting. Depending on the degree of the disorder present in an individual, excess bleeding may occur only after specific, predictable events (such as surgery, dental procedures, or injury), or occur spontaneously, with no known initiating event.

Description

The normal mechanism for blood clotting is a complex series of events involving the interaction of the injured blood vessel, blood cells (called platelets), and over 20 different proteins which also circulate in the blood.

When a blood vessel is injured in a way that causes bleeding, platelets collect over the injured area, and form a temporary plug to prevent further bleeding. This temporary plug, however, is too disorganized to serve as a long-term solution, so a series of chemical events occur, resulting in the formation of a more reliable plug. The final plug involves tightly woven fibers of a material called fibrin. The production of fibrin requires the interaction of several chemicals, in particular a series of proteins called clotting factors. At least thirteen different clotting factors have been identified.

The clotting cascade, as it is usually called, is the series of events required to form the final fibrin clot. The cascade uses a technique called amplification to rapidly produce the proper sized fibrin clot from the small number of molecules initially activated by the injury.

In hemophilia, certain clotting factors are either decreased in quantity, absent, or improperly formed. Because the clotting cascade uses amplification to rapidly plug up a bleeding area, absence or inactivity of just one clotting factor can greatly increase **bleeding time**.

Hemophilia A is the most common type of bleeding disorder and involves decreased activity of factor VIII. There are three levels of factor VIII deficiency: severe, moderate, and mild. This classification is based on the percentage of normal factor VIII activity present:

- Individuals with less than 1% of normal factor VIII activity level have severe hemophilia. Half of all people with hemophilia A fall into this category. Such individuals frequently experience spontaneous bleeding, most frequently into their joints, skin, and muscles. Surgery or trauma can result in life-threatening hemorrhage, and must be carefully managed.
- Individuals with 1–5% of normal factor VIII activity level have moderate hemophilia, and are at risk for heavy bleeding after seemingly minor traumatic injury.
- Individuals with 5–40% of normal factor VIII activity level have mild hemophilia, and must prepare carefully for any surgery or dental procedures.

Individuals with hemophilia B have symptoms very similar to those of hemophilia A, but the deficient factor is factor IX. This type of hemophilia is also known as Christmas disease.

Hemophilia C is very rare, and much more mild than hemophilia A or B; it involves factor XI.

Hemophilia A affects between one in 5,000 to one in 10,000 males in most populations.

One recent study estimated the prevalence of hemophilia was 13.4 cases per 100,000 U.S. males (10.5 hemophilia A and 2.9 hemophilia B). By race/ethnicity, the prevalence was 13.2 cases/100,000 among white, 11.0 among African-American, and 11.5 among Hispanic males.

Causes and symptoms

Hemophilia A and B are both caused by a genetic defect present on the X chromosome. (Hemophilia C is inherited in a different fashion.) About 70% of all people with hemophilia A or B inherited the disease. The other 30% develop from a spontaneous genetic mutation.

The following concepts are important to understanding the inheritance of these diseases. All humans have two chromosomes determining their gender: females have XX, males have XY. Because the trait is carried only on the X chromosome, it is called “sex-linked.” The chromosome’s flawed unit is referred to as the gene.

Both factors VIII and IX are produced by a genetic defect of the X chromosome, so hemophilia

A and B are both sex-linked diseases. Because a female child always receives two X chromosomes, she nearly always will receive at least one normal X chromosome. Therefore, even if she receives one flawed X chromosome, she will still be capable of producing a sufficient quantity of factors VIII and IX to avoid the symptoms of hemophilia. Such a person who has one flawed chromosome, but does not actually suffer from the disease, is called a carrier. She carries the flaw that causes hemophilia and can pass it on to her offspring. If, however, she has a son who receives her flawed X chromosome, he will be unable to produce the right quantity of factors VIII or IX, and he will suffer some degree of hemophilia. (Males inherit one X and one Y chromosome, and therefore have only one X chromosome.)

In rare cases, a hemophiliac father and a carrier mother can pass on the right combination of parental chromosomes to result in a hemophiliac female child. This situation, however, is rare. The vast majority of people with either hemophilia A or B are male.

About 30% of all people with hemophilia A or B are the first member of their family to ever have the disease. These individuals have had the unfortunate occurrence of a spontaneous mutation; meaning that in their early development, some random genetic accident befell their X chromosome, resulting in the defect causing hemophilia A or B. Once such a spontaneous genetic mutation takes place, offspring of the affected person can inherit the newly created, flawed chromosome.

In the case of severe hemophilia, the first bleeding event usually occurs prior to eighteen months of age. In some babies, hemophilia is suspected immediately, when a routine **circumcision** (removal of the foreskin of the penis) results in unusually heavy bleeding. Toddlers are at particular risk, because they fall frequently, and may bleed into the soft tissue of their arms and legs. These small bleeds result in bruising and noticeable lumps, but don't usually need treatment. As a child becomes more active, bleeding may occur into the muscles; a much more painful and debilitating problem. These muscle bleeds result in **pain** and pressure on the nerves in the area of the bleed. Damage to nerves can cause **numbness** and decreased ability to use the injured limb.

Some of the most problematic and frequent bleeds occur into the joints, particularly into the knees and elbows. Repeated bleeding into joints can result in scarring within the joints and permanent deformities. Individuals may develop arthritis in joints that have suffered continued irritation from the presence of

blood. Mouth injuries can result in compression of the airway, and, therefore, can be life-threatening. A blow to the head, which might be totally insignificant in a normal individual, can result in bleeding into the skull and brain. Because the skull has no room for expansion, the hemophiliac individual is at risk for brain damage due to blood taking up space and exerting pressure on the delicate brain tissue.

People with hemophilia are at very high risk of hemorrhage (severe, heavy, uncontrollable bleeding) from injuries such as motor vehicle accidents and also from surgery.

Some other rare clotting disorders such as Von Willebrand disease present similar symptoms but are not usually called hemophilia.

Diagnosis

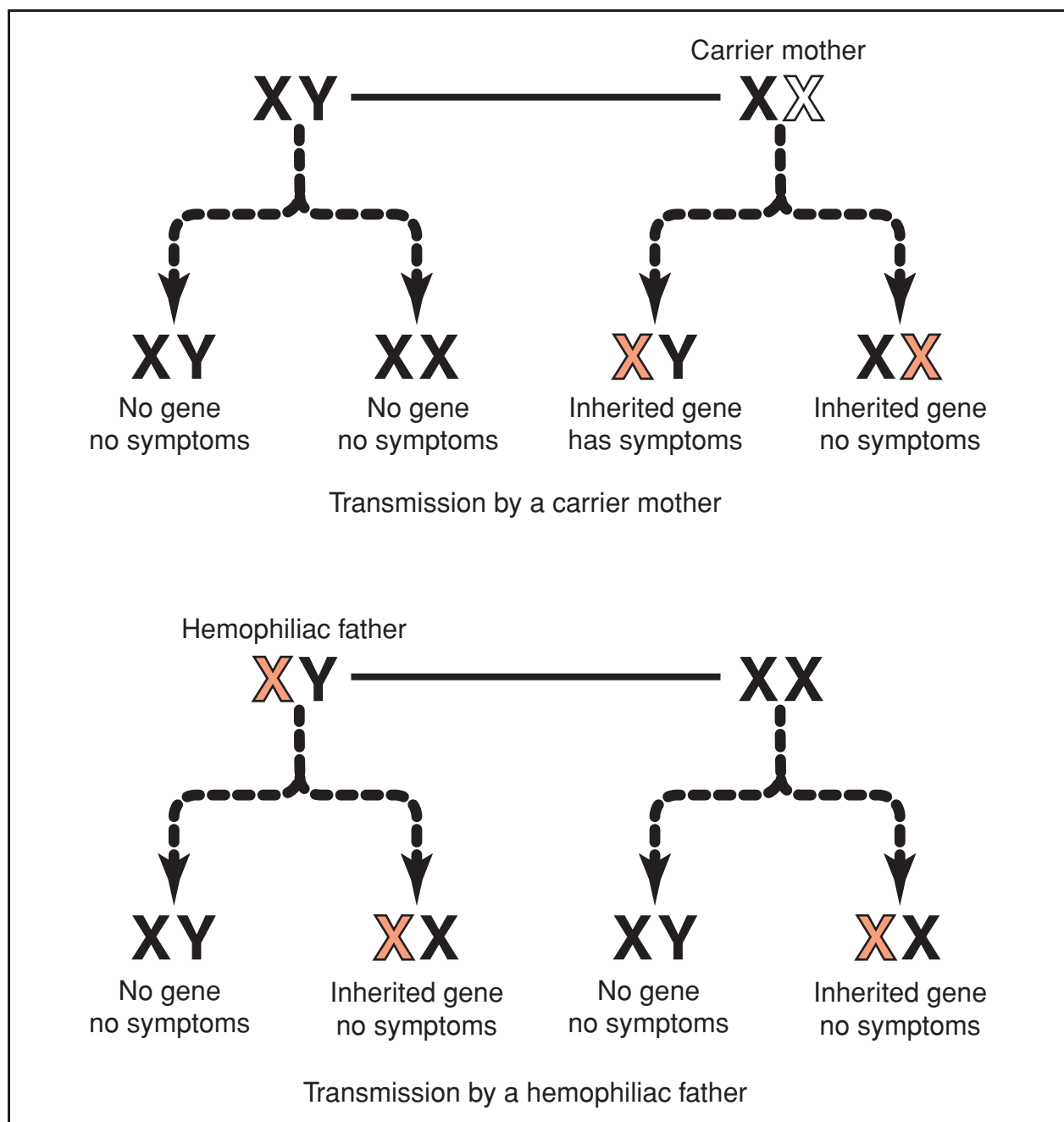
Various tests are available to measure, under very carefully controlled conditions, the length of time it takes to produce certain components of the final fibrin clot. Tests called assays can also determine the percentage of factors VIII and IX present compared to normal percentages. This information can help in demonstrating the type of hemophilia present, as well as the severity.

Individuals with a family history of hemophilia may benefit from genetic counseling before deciding to have a baby. Families with a positive history of hemophilia can also have tests done during a **pregnancy** to determine whether the fetus is a hemophiliac. The test called chorionic villous sampling examines proteins for the defects that lead to hemophilia. This test, which is associated with a 1% risk of miscarriage, can be performed at 10–14 weeks. The test called **amniocentesis** examines the DNA of fetal cells shed into the amniotic fluid for genetic mutations. Amniocentesis, which is associated with a one in 200 risk of miscarriage, is performed at 15–18 weeks gestation.

Treatment

The most important thing that individuals with hemophilia can do to prevent complications of this disease is to avoid injury. Those individuals who require dental work or any surgery may need to be pre-treated with an infusion of factor VIII to avoid hemorrhage. Also, hemophiliacs should be vaccinated against hepatitis. Medications or drugs that promote bleeding, such as aspirin, should be avoided.

Various types of factors VIII and IX are available to replace a patient's missing factors. These are administered intravenously (directly into the patient's veins



Hemophilia A and B are both caused by a genetic defect present on the X chromosome. Approximately 70% of people with hemophilia A or B inherited the disease, while the remaining 30% have hemophilia due to a spontaneous genetic mutation.
(Illustration by Electronic Illustrators Group.)

by needle). These factor preparations may be obtained from a single donor, by pooling the donations of as many as thousands of donors, or by laboratory creation through highly advanced genetic techniques.

The frequency of treatment with factors depends on the severity of the individual patient's disease. Patients with relatively mild disease will only require

treatment in the event of injury, or to prepare for scheduled surgical or dental procedures. Patients with more severe disease will require regular treatment to avoid spontaneous bleeding.

While appropriate treatment of hemophilia can both decrease suffering and be life-saving, complications associated with treatment can also be quite

KEY TERMS

Amplification—A process by which something is made larger. In clotting, only a very few chemicals are released by the initial injury; they result in a cascade of chemical reactions which produces increasingly larger quantities of different chemicals, resulting in an appropriately-sized, strong fibrin clot.

Factors—Coagulation factors are substances in the blood, such as proteins and minerals, that are necessary for clotting. Each clotting substance is designated with roman numerals I through XIII.

Fibrin—The final substance created through the clotting cascade, which provides a strong, reliable plug to prevent further bleeding from the initial injury.

Hemorrhage—Very severe, massive bleeding that is difficult to control. Hemorrhage can occur in hemophiliacs after what would be a relatively minor injury to a person with normal clotting factors.

Mutation—A permanent change in the genetic material that may alter a trait or characteristic of an individual, or manifest as disease, and can be transmitted to offspring.

Platelets—Small disc-shaped structures that circulate in the blood stream and participate in blood clotting.

Trauma—Injury.

serious. About 20% of all patients with hemophilia A begin to produce chemicals in their bodies which rapidly destroy infused factor VIII. The presence of such a chemical may greatly hamper efforts to prevent or stop a major hemorrhage.

Individuals who receive factor prepared from pooled donor blood are at risk for serious infections that may be passed through blood. Hepatitis, a severe and potentially fatal viral liver infection, may be contracted from pooled factor preparations. Recently, a good deal of concern has been raised about the possibility of hemophiliacs contracting a fatal slow virus infection of the brain (**Creutzfeldt-Jakob disease**) from blood products. Unfortunately, pooled factor preparations in the early 1980s were contaminated with human immunodeficiency virus (HIV), the virus which causes **AIDS**. A large number of hemophiliacs were infected with HIV and some

statistics show that HIV is still the leading cause of **death** among hemophiliacs. Currently, careful methods of donor testing, as well as methods of inactivating viruses present in donated blood, have greatly lowered this risk.

The most exciting new treatments currently being researched involve efforts to transfer new genes to hemophiliacs. These new genes would have the ability to produce the missing factors. As yet, these techniques are not being performed on humans, but there is great hope that eventually this type of **gene therapy** will be available.

Prognosis

Prognosis is very difficult to generalize. Because there are so many variations in the severity of hemophilia, and because much of what befalls a hemophiliac patient will depend on issues such as physical activity level and accidental injuries, statistics on prognosis are not generally available.

Resources

BOOKS

Khoury, Muin J., Wylie Burke, and Elizabeth J. Thomson, editors. *Genetics and Public Health in the 21st Century: Using Genetic Information to Improve Health and Prevent Disease*. New York: Oxford University Press, 2000.

Resnick, Susan. *Blood Saga: Hemophilia, AIDS, and the Survival of a Community*. Berkeley: University of California Press, 1999.

PERIODICALS

Stephenson, J. "New Therapies Show Promise for Patients with Leukemia, Hemophilia, and Heart Disease." *JAMA* 285 (January 1, 2001): 153+.

ORGANIZATIONS

National Hemophilia Foundation. 116 West 32nd St., 11th Floor, New York, NY 10001. (800) 42-HANDI. <<http://www.info@hemophilia.org>>.

National Organization for Rare Disorders (NORD). PO Box 8923, New Fairfield, CT 06812-8923. (203) 746-6518 or (800) 999-6673. Fax: (203) 746-6481. <<http://www.rarediseases.org>>.

OTHER

March of Dimes. <www.modimes.org>.

National Organization for Rare Disorders. <www.rarediseases.org>.

Jennifer F. Wilson, MS

Hemophilus ducreyi infection see **Chancroid**

Hemophilus infections

Definition

Hemophilus infections, most of which are due to *Haemophilus influenzae* infections, are a group of contagious diseases that are caused by a gram-negative bacterium, and affect only humans. Some hemophilus infections are potentially fatal.

Description

H. influenzae is a common organism worldwide; it has been found in the nasal secretions of as many as 90% of healthy individuals in the general population. Hemophilus infections are characterized by acute inflammation with a discharge (exudate). They may affect almost any organ system, but are most common in the respiratory tract. The organism can be transmitted by person-to-person contact, or by contact with nasal discharges and other body fluids. Hemophilus infections in the United States are most likely to spread in the late winter or early spring.

The primary factor influencing the rate of infection is age; children between the ages of six months and four years are most vulnerable to *H. influenzae*. In previous years, about 50% of children would acquire a hemophilus infection before reaching one year of age; almost all children would develop one before age three. These figures are declining, however, as a result of the increasing use of hemophilus vaccines for children.

Adults are also susceptible to hemophilus diseases. *H. influenzae* **pneumonia** is a common nosocomial infection (illnesses contracted in hospitals). The rate of hemophilus infections in the adult population has increased over the past 40 years. The reasons for this change are unclear, but some researchers speculate that the overuse of **antibiotics** has led to the development of drug-resistant strains of *H. influenzae*. The risk factors for hemophilus infections among adults include:

- smoking
- alcoholism
- chronic lung disease
- old age
- living in a city or institutional housing with a large group of people
- poor **nutrition** and hygiene
- HIV infection, or other immune system disorder

Causes and symptoms

Hemophilus infections are primarily caused by *Haemophilus influenzae*, a gram-negative bacterium that is capable of spreading from the nasal tissues and upper airway, where it is usually found, to the chest, throat, or middle ear. The organism sometimes invades localized areas of tissue, producing **meningitis**, **infectious arthritis**, **conjunctivitis**, **cellulitis**, **epiglottitis**, or inflammation of the membrane surrounding the heart. The most serious infections are caused by a strain called *H. influenzae* b (Hib). Before routine **vaccination**, Hib was the most common cause of bacterial meningitis, and responsible for most of the cases of acquired mental retardation in the United States.

Hemophilus infections in children

BACTERIAL SEPSIS IN THE NEWBORN. Bacterial **sepsis** (sepsis is the presence of illness-causing microorganisms, or their poisons, in the blood) is a potentially fatal illness in newborn infants. The child may acquire the disease organism as it passes through the mother's birth canal, or from the hospital environment. *H. influenzae* can also produce inflammations of the eye (conjunctivitis) in newborn children. The signs of sepsis may include fever, crankiness, feeding problems, breathing difficulties, pale or mottled skin, or drowsiness. Premature birth is the most significant risk factor for hemophilus infections in newborns.

EPIGLOTTITIS. Epiglottitis is a potentially fatal hemophilus infection. Although children are more likely to develop epiglottitis, it can occur in adults as well. When the epiglottis (a piece of cartilage behind the tongue which protects the opening to the windpipe by opening and closing) is infected, it can swell to the point where it blocks the windpipe. The symptoms of epiglottitis include a sudden high **fever**, drooling, the feeling of an object stuck in the throat, and **stridor**. The epiglottis will look swollen and bright red if the doctor examines the patient's throat with a laryngoscope (a viewing device).

MENINGITIS. Meningitis caused by Hib is most common in children between nine months and four years of age. The child usually develops upper respiratory symptoms followed by fever, loss of appetite, vomiting, **headache**, and a stiff or sore neck or back. In severe cases, the child may have convulsions or go into **shock** or coma.

OTHER INFECTIONS. Hib is the second most common cause of middle ear infection and **sinusitis** in

children. The symptoms of sinusitis include fever, **pain, bad breath**, and coughing. Children may also develop infectious arthritis from Hib. The joints most frequently affected are the large weight-bearing joints.

Hemophilus infections in adults

PNEUMONIA. Hib pneumonia is the most common hemophilus infection in adults. The symptoms include empyema (sputum containing pus), and fever. The hemophilus organism can usually be identified from sputum samples. Hib pneumonia is increasingly common in the elderly.

MENINGITIS. Meningitis caused by Hib can develop in adults as a complication of an ear infection or sinusitis. The symptoms are similar to those in children but are usually less severe in adults.

Diagnosis

The diagnosis is usually based on a combination of the patient's symptoms and the results of blood counts, cultures, or antigen detection tests.

Laboratory tests

Laboratory tests can be used to confirm the diagnosis of hemophilus infections. The bacterium can be grown on chocolate agar, or identified by blood cultures or Gram stain of body fluids. Antigen detection tests can be used to identify hemophilus infections in children. These tests include latex agglutination and electrophoresis.

Other laboratory findings that are associated with hemophilus infections include anemia (low red blood cell count), and a drop in the number of white blood cells in children with severe infections. Adults often show an abnormally high level of white blood cells; cell counts of 15,000–30,000/mm³ are not unusual.

Treatment

Because some hemophilus infections are potentially fatal, treatment is started without waiting for the results of laboratory tests.

Medications

Hemophilus infections are treated with antibiotics. Patients who are severely ill are given ampicillin or a third-generation cephalosporin, such as cefotaxime or ceftriaxone, intravenously. Patients with milder infections are given oral antibiotics, including

amoxicillin, cefaclor, erythromycin, or trimethoprim-sulfamethoxazole. Patients who are allergic to penicillin are usually given cefaclor or trimethoprim-sulfamethoxazole.

Patients with Hib strains that are resistant to ampicillin may be given chloramphenicol. Chloramphenicol is not a first-choice drug because of its side effects, including interference with bone marrow production of blood cells.

The duration of antibiotic treatment depends on the location and severity of the hemophilus infection. Adults with respiratory tract infections, or Hib pneumonia, are usually given a 10–14 day course of antibiotics. Meningitis is usually treated for 10–14 days, but a seven-day course of treatment with ceftriaxone appears to be sufficient for infants and children. Ear infections are treated for seven to 10 days.

Supportive care

Patients with serious hemophilus infections require bed rest and a humidified environment (such as a **croup** tent) if the respiratory tract is affected. Patients with epiglottitis frequently require intubation (insertion of a breathing tube) or a **tracheotomy** to keep the airway open. Patients with inflammation of the heart membrane, pneumonia, or arthritis may need surgical treatment to drain infected fluid from the chest cavity or inflamed joints.

Supportive care also includes monitoring of blood cell counts for patients using chloramphenicol, ampicillin, or other drugs that may affect production of blood cells by the bone marrow.

Prognosis

The most important factors in the prognosis are the severity of the infection and promptness of treatment. Untreated hemophilus infections—particularly meningitis, sepsis, and epiglottitis—have a high mortality rate. Bacterial sepsis of the newborn has a mortality rate between 13–50%. The prognosis is usually good for patients with mild infections who are treated without delay. Children who develop Hib arthritis sometimes have lasting problems with joint function.

Prevention

Hemophilus vaccines

There are three different vaccines for hemophilus infections used to immunize children in the United States: PRP-D, HBOC, and PRP-OMP. PRP-D is used only in children older than 15 months. HBOC is

KEY TERMS

Bacterium—A microscopic one-celled organism. *Haemophilus influenzae* is a specific bacterium.

Epiglottitis—Inflammation of the epiglottis. The epiglottis is a piece of cartilage behind the tongue that closes the opening to the windpipe when a person swallows. An inflamed epiglottis can swell and close off the windpipe, thus causing the patient to suffocate.

Exudate—A discharge produced by the body. Some exudates are caused by infections.

Gram-negative—A term that means that a bacterium will not retain the violet color when stained with Gram's dye. *Haemophilus influenzae* is a gram-negative bacterium.

Intubation—The insertion of a tube into the patient's airway to protect the airway from collapsing. Intubation is sometimes done as an emergency procedure for patients with epiglottitis.

Nosocomial—Contracted in a hospital. Pneumonia caused by *H. influenzae* is an example of a nosocomial infection.

Sepsis—Invasion of body tissues by disease organisms or their toxins. Sepsis may be either localized or generalized. *Haemophilus influenzae* can cause bacterial sepsis in newborns.

Stridor—A harsh or crowing breath sound caused by partial blockage of the patient's upper airway.

Tracheotomy—An emergency procedure in which the surgeon cuts directly through the patient's neck into the windpipe in order to keep the airway open.

administered to infants at two, four, and six months after birth, with a booster dose at 15–18 months. PRP-OMP is administered to infants at two and four months, with the third dose at the child's first birthday. All three vaccines are given by intramuscular injection. About 5% of children may develop fever or soreness in the area of the injection.

Other measures

Other preventive measures include isolating patients with respiratory hemophilus infections; treating appropriate contacts of infected patients with rifampin; maintaining careful standards of cleanliness in hospitals, including proper disposal of soiled tissues; and washing hands properly.

Resources

BOOKS

Chambers, Henry F. "Infectious Diseases: Bacterial & Chlamydial." In *Current Medical Diagnosis and Treatment*, 1998, edited by Stephen McPhee, et al., 37th ed. Stamford: Appleton & Lange, 1997.

Rebecca J. Frey, PhD

Haemophilus influenzae infections see
Hemophilus infections

Hemoptysis

Definition

Hemoptysis is the coughing up of blood or bloody sputum from the lungs or airway. It may be either self-limiting or recurrent. Massive hemoptysis is defined as 200–600 mL of blood coughed up within a period of 24 hours or less.

Description

Hemoptysis can range from small quantities of bloody sputum to life-threatening amounts of blood. The patient may or may not have chest **pain**.

Causes and symptoms

Hemoptysis can be caused by a range of disorders:

- Infections. These include **pneumonia**; **tuberculosis**; **aspergillosis**; and parasitic diseases, including ascariasis, **amebiasis**, and paragonimiasis.
- Tumors that erode blood vessel walls.
- Drug **abuse**. **Cocaine** can cause massive hemoptysis.
- Trauma. Chest injuries can cause bleeding into the lungs.
- Vascular disorders, including aneurysms, **pulmonary embolism**, and malformations of the blood vessels.
- Bronchitis. Its most common cause is long-term **smoking**.
- Foreign object(s) in the airway.
- Blood clotting disorders.
- Bleeding following such surgical procedures as bronchial biopsies and heart catheterization.

Diagnosis

The diagnosis of hemoptysis is complicated by the number of possible causes.

Patient history

It is important for the doctor to distinguish between blood from the lungs and blood coming from the nose, mouth, or digestive tract. Patients may aspirate, or breathe, blood from the nose or stomach into their lungs and **cough** it up. They may also swallow blood from the chest area and then vomit. The doctor will ask about stomach ulcers, repeated **vomiting**, **liver disease**, **alcoholism**, smoking, tuberculosis, mitral valve disease, or treatment with anticoagulant medications.

Physical examination

The doctor will examine the patient's nose, throat, mouth, and chest for bleeding from these areas and for signs of chest trauma. The doctor also listens to the patient's breathing and heartbeat for indications of heart abnormalities or lung disease.

Laboratory tests

Laboratory tests include blood tests to rule out clotting disorders, and to look for food particles or other evidence of blood from the stomach. Sputum can be tested for fungi, bacteria, or parasites.

X ray and bronchoscopy

Chest x rays and **bronchoscopy** are the most important studies for evaluating hemoptysis. They are used to evaluate the cause, location, and extent of the bleeding. The bronchoscope is a long, flexible tube used to identify tumors or remove **foreign objects**.

Imaging and other tests

Computed tomography scans (CT scans) are used to detect aneurysms and to confirm x-ray results. Ventilation-perfusion scanning is used to rule out pulmonary **embolism**. The doctor may also order an angiogram to rule out pulmonary embolism, or to locate a source of bleeding that could not be seen with the bronchoscope.

In spite of the number of diagnostic tests, the cause of hemoptysis cannot be determined in 20–30% of cases.

Treatment

Massive hemoptysis is a life-threatening emergency that requires treatment in an intensive care

KEY TERMS

Aneurysm—A sac formed by the dilation of the wall of an artery, vein, or heart; it is filled with clotted blood or fluid.

Angiography—A technique for imaging the blood vessels by injecting a substance that is opaque to x rays.

Aspergillosis—A lung infection caused by the mold *Aspergillus fumigatus*.

Intubation—The insertion of a tube into a body canal or hollow organ, as into the trachea or stomach.

Pulmonary embolism—The blocking of an artery in the lung by a blood clot.

unit. The patient will be intubated (the insertion of a tube to help breathing) to protect the airway, and to allow evaluation of the source of the bleeding. Patients with lung **cancer**, bleeding from an aneurysm (blood clot), or persistent traumatic bleeding require chest surgery.

Patients with tuberculosis, aspergillosis, or bacterial pneumonia are given **antibiotics**.

Foreign objects are removed with a bronchoscope.

If the cause cannot be determined, the patient is monitored for further developments.

Prognosis

The prognosis depends on the underlying cause. In cases of massive hemoptysis, the mortality rate is about 15%. The rate of bleeding, however, is not a useful predictor of the patient's chances for recovery.

Resources

BOOKS

Stauffer, John L. "Lung." In *Current Medical Diagnosis and Treatment*, 1998, edited by Stephen McPhee, et al., 37th ed. Stamford: Appleton & Lange, 1997.

Rebecca J. Frey, PhD

Hemorrhagic colitis see ***Escherichia coli***

Hemorrhagic fever with renal syndrome see **Hantavirus infections**

Hemorrhagic fevers

Definition

Hemorrhagic fevers are caused by viruses that exist throughout the world. However, they are most common in tropical areas. Early symptoms, such as muscle aches and **fever**, can progress to a mild illness or to a more debilitating, potentially fatal disease. In severe cases, a prominent symptom is bleeding, or hemorrhaging, from orifices and internal organs.

Description

Although hemorrhagic fevers are regarded as emerging diseases, they probably have existed for many years. This designation isn't meant to imply that they are newly developing, but rather that human exposure to the causative viruses is increasing to the point of concern.

These viruses are maintained in nature in arthropod (insects, spiders and other invertebrates with external hard skeletons) or animal populations—so-called disease reservoirs. Individuals within these populations become infected with a virus but do not die from it. In many cases, they don't even develop symptoms. Then the viruses are transmitted from a reservoir population to humans by vectors—either members of the reservoir population or an intervening species, such as mosquitoes.

Hemorrhagic fevers are generally either endemic or linked to specific locations. If many people reside in an endemic area, the number of cases may soar. For example, dengue fever, a type of hemorrhagic fever, affects approximately 100 million people annually. A large percentage of those infected live in densely populated southeast Asia; an area in which the disease vector, a mosquito, thrives. Some hemorrhagic fevers are exceedingly rare, because people very infrequently encounter the virus. Marburg hemorrhagic fever, which has affected fewer than 40 people since its discovery in 1967, provides one such example. Fatality rates are also variable. In cases of dengue hemorrhagic fever–dengue **shock** syndrome, 1–5% of the victims perish. On the other end of the spectrum is Ebola, an African hemorrhagic fever, that kills 30–90% of those infected.

The onset of hemorrhagic fevers may be sudden or gradual, but all of them are linked by the potential for hemorrhaging. However, not all cases progress to this very serious symptom. Hemorrhaging may be attributable to the destruction of blood coagulating factors

or to increased permeability of body tissues. The severity of bleeding ranges from petechiae, which are pinpoint hemorrhages under the skin surface, to distinct bleeding from such body orifices as the nose or vagina.

Causes and symptoms

The viruses that cause hemorrhagic fevers are found most commonly in tropical locations; however, some are found in cooler climates. Typical disease vectors include rodents, ticks, or mosquitoes, but person-to-person transmission in health care settings or through sexual contact can also occur.

Filoviruses

Ebola is the most famous of the Filoviridae, a virus family that also includes the Marburg virus. Ebola is endemic to Africa, particularly the Republic of the Congo and Sudan; the Marburg virus is found in sub-Saharan Africa. The natural reservoir of filoviruses is unknown. The incubation period, or time between infection and appearance of symptoms, is thought to last three to eight days, possibly longer.

Symptoms appear suddenly, and include severe headache, fever, chills, muscle aches, malaise, and appetite loss. These symptoms may be accompanied by nausea, **vomiting**, **diarrhea**, and abdominal **pain**. Victims become apathetic and disoriented. Severe bleeding commonly occurs from the gastrointestinal tract, nose and throat, and vagina. Other bleeding symptoms include petechiae and oozing from injection sites. Ebola is fatal in 30–90% of cases.

Arenaviruses

Viruses of the Arenaviridae family cause the Argentinian, Brazilian, Bolivian, and Venezuelan hemorrhagic fevers. Lassa fever, which occurs in west Africa, also arises from an arenavirus. Infected rodents, the natural reservoir, shed virus particles in their urine and saliva, which humans may inhale or otherwise come in contact with.

Fever, muscle aches, malaise, and appetite loss gradually appear one to two weeks after infection with the South American viruses. Initial symptoms are followed by **headache**, back pain, dizziness, and gastrointestinal upset. The face and chest appear flushed and the gums begin to bleed. In about 30% of cases, the disease progresses to bleeding under the skin and from the mucous membranes, and/or to effects on the nervous system, such as delirium, **coma**, and convulsions. Untreated, South American hemorrhagic fevers have a 10–30% fatality rate.

Lassa fever also begins gradually, following an 8–14 day incubation. Initial symptoms resemble those of the South American hemorrhagic fevers, followed by a **sore throat**, muscle and joint pain, severe headache, pain above the stomach, and a dry **cough**. The face and neck become swollen, and fluid may accumulate in the lungs. Bleeding occurs in 15–20% of infected individuals, mostly from the gums and nose. Overall, the fatality rate is lower than 2%, but hospitals may encounter 20% fatality rates, treating typically the most serious of cases.

Flaviviruses

The Flaviviridae family includes the viruses that cause yellow and dengue fevers.

Yellow fever occurs in tropical areas of the Americas and Africa and is transmitted from monkeys to humans by mosquitoes. The virus may produce a mild, possibly unnoticed illness, but some individuals are suddenly stricken with a fever, weakness, low back pain, muscle pain, **nausea**, and vomiting. This phase lasts one to seven days, after which the symptoms recede for one to two days. Symptoms then return with greater intensity, along with **jaundice**, **delirium**, seizures, stupor, and coma. Bleeding occurs from the mucous membranes and under the skin surface, and dark blood appears in stools and vomit.

Mosquitoes also transmit the dengue virus. **Dengue fever** is endemic in southeast Asia and areas of the Americas. Cases have also been reported in the Caribbean, Saudi Arabia, and northern Australia. In 2004 several cases were reported along the border between Texas and Mexico in the southwestern United States. This virus causes either the mild dengue fever or the more serious dengue hemorrhagic fever–dengue shock syndrome (DHF-DSS).

In children, dengue fever is characterized by a sore throat, runny nose, slight cough, and a fever lasting for a week or less. Older children and adults experience more severe symptoms: fever, headache, muscle and joint pain, loss of appetite, and a rash. The skin appears flushed, and intense pain occurs in the bones and limbs. After nearly a week, the fever subsides for one to two days before returning. Minor hemorrhaging, such as from the gums, or more serious gastrointestinal bleeding may occur.

DHF-DSS primarily affects children younger than 15 years. The symptoms initially resemble those of dengue fever in adults, without the bone and limb pain. As the fever begins to abate, the individual's condition worsens and hemorrhaging occurs from the nose, gums, and injection sites. Bleeding is also

seen from the gastrointestinal, genitourinary, and respiratory tracts.

Bunyaviruses

The Bunyaviridae family includes several hundred viruses but only a few are responsible for hemorrhagic fevers in humans.

Rift Valley fever is caused by the phlebovirus, found in sub-Saharan Africa and the Nile delta. Natural reservoirs are wild and domestic animals, and transmission occurs through contact with infected animals or through mosquito bites. The incubation period lasts 3–12 days. Most cases of Rift Valley fever are mild and may be symptomless. If symptoms develop, they include fever, backache, muscle and joint pain, and headache. Hemorrhagic symptoms occur rarely; while **death**, which occurs in fewer than 3% of cases, is attributable to massive liver damage.

Crimean-Congo hemorrhagic fever is caused by hantavirus and occurs in central and southern Africa, Asia, Eurasia, and the Middle East. The virus is found in hares, birds, ticks, and domestic animals and may be transmitted by ticks or by contact with infected animals. The hantavirus incubation period is three to 12 days; after which an individual experiences fever, chills, headache, severe muscle pain, pain above the stomach, nausea, vomiting, and appetite loss. Bleeding under the skin and gastrointestinal and vaginal bleeding may develop in the most severe cases. Death rates range from 10% in southern Russia to 50% in parts of Asia.

Hemorrhagic fever with renal (kidney) syndrome is caused by the hantaviruses: Hantaan, Seoul, Puumala, and Dobrava. Hantaan virus occurs in northern Asia, the Far East, and the Balkans; Seoul virus is found worldwide; Puumala virus is found in Scandinavia and northern Europe; while Dobrava virus occurs in the Balkans. Wild rodents are the natural reservoirs and transmit the virus via their excrement or body fluids or through direct contact. Initial symptoms develop within 10–40 days and include fever, headache, muscle pain, and **dizziness**. Other symptoms are blurry vision, abdominal and back pain, nausea, and vomiting. High levels of protein in the urine signal kidney damage; hemorrhaging may also occur. Death rates range from 0–10%.

Diagnosis

Since the hemorrhagic fevers share symptoms with many other diseases, positive identification of

the disease relies on evidence of the viruses in the bloodstream—such as detection of antigens and antibodies—or isolation of the virus from the body. Disruptions in the normal levels of bloodstream components may be helpful in determining some, but not all, hemorrhagic fevers.

Treatment

Lassa fever, and possibly other hemorrhagic fevers, respond to ribavirin, an antiviral medication. However, most of the hemorrhagic fever viruses can only be treated with supportive care. Interferon is not useful and may in fact complicate management. Such care centers around maintaining correct fluid and electrolyte balances in the body and protecting the patient against secondary infections. Heparin and vitamin K administration, coagulation factor replacement, and blood transfusions may be effective in lessening or stopping hemorrhage in some cases.

Some researchers are investigating the possibility of targeting tissue factor (TF) as a way of treating viral hemorrhagic fevers. TF is a protein that activates the coagulation process in these illnesses, and experimental models suggest that a blockade of tissue factor assists the body's immune response to hemorrhagic fever viruses.

Prognosis

Recovery from some hemorrhagic fevers is more certain than from others. The filoviruses are among the most lethal; fatality rates for Ebola range from 30–90%, while DHF-DSS cases result in a 1–5% fatality rate. Whether a case occurs during an epidemic or as an isolated case also has a bearing on the outcome. For example, isolated cases of yellow fever have a 5% mortality rate, but 20–50% of epidemic cases may be fatal.

Permanent disability can occur with some types of hemorrhagic fever. About 10% of severely ill Rift Valley fever victims suffer retina damage and may be permanently blind, and 25% of South American hemorrhagic fever victims suffer potentially permanent deafness.

Proper treatment is vital. In cases of DHF-DSS, fatality can be reduced from 40–50% to less than 2% with adequate medical care. For individuals who survive hemorrhagic fevers, prolonged convalescence is usually inevitable. However, survivors seem to gain lifelong immunity against the virus that made them ill.

KEY TERMS

Antibody—A molecule created by the body's immune system to combat a specific infectious agent, such as a virus or bacteria.

Antigen—A specific feature, such as a protein, on an infectious agent. Antibodies use this feature as a means of identifying infectious intruders.

Coagulating factors—Components within the blood that help form clots.

Endemic—Referring to a specific geographic area in which a disease may occur.

Hemorrhage—As a noun, this refers to the point at which blood is released. As a verb, this refers to bleeding.

Incubation—The time period between exposure to an infectious agent, such as a virus or bacteria, and the appearance of symptoms of illness.

Petechiae—Pinpoint hemorrhages that appear as reddish dots beneath the surface of the skin.

Reservoir—A population in which a virus is maintained without causing serious illness to the infected individuals.

Ribavirin—A drug that is used to combat viral infections.

Tissue factor—A glycoprotein involved in blood coagulation.

Vector—A member of the reservoir population or an intervening species that can transmit a virus to a susceptible victim. Mosquitoes are common vectors, as are ticks and rodents.

Prevention

Hemorrhagic fevers can be prevented through vector control and personal protection measures. Attempts have been made in urban and settled areas to destroy mosquito and rodent populations. In areas where such measures are impossible, individuals can use insect repellents, mosquito netting, and other methods to minimize exposure.

Vaccines have been developed against yellow fever, Argentinian hemorrhagic fever, and Crimean-Congo hemorrhagic fever. Vaccines against other hemorrhagic fevers are being researched. Another possible preventive measure is increasing the number of natural killer (NK) cells in the body. These cells appear to be an important innate source of protection against Ebola and other filoviruses.

Prevention of epidemics of hemorrhagic fevers has acquired a new importance in the early 2000s from concern that the causative viruses might be used as weapons of bioterrorism. These viruses can be transmitted in aerosol form as well as having a high mortality rate.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD., editors. "Viral Diseases." Section 13, Chapter 162 In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

PERIODICALS

- Izadi, S., K. H. Naieni, S. R. Madjdzadeh, and A. Nadim. "Crimean-Congo Hemorrhagic Fever in Sistan and Baluchestan Province of Iran, A Case-Control Study on Epidemiological Characteristics." *International Journal of Infectious Diseases* 8 (September 2004): 299–306.
- Mahanty, S., and M. Bray. "Pathogenesis of Filoviral Haemorrhagic Fevers." *Lancet Infectious Diseases* 4 (August 2004): 487–498.
- Ruf, W. "Emerging Roles of Tissue Factor in Viral Hemorrhagic Fever." *Trends in Immunology* 25 (September 2004): 461–464.
- Salvaggio, M. R., and J. W. Baddley. "Other Viral Bioweapons: Ebola and Marburg Hemorrhagic Fever." *Dermatologic Clinics* 22 (July 2004): 291–302.
- Setlick, R. F., D. Ouellette, J. Morgan, et al. "Pulmonary Hemorrhage Syndrome Associated with an Autochthonous Case of Dengue Hemorrhagic Fever." *Southern Medical Journal* 97 (July 2004): 688–691.
- Warfield, K. L., J. G. Perkins, D. L. Swenson, et al. "Role of Natural Killer Cells in Innate Protection against Lethal Ebola Virus Infection." *Journal of Experimental Medicine* 200 (July 19, 2004): 169–179.

ORGANIZATIONS

- Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.
- Infectious Diseases Society of America (IDSA). 66 Canal Center Plaza, Suite 600, Alexandria, VA 22314. (703) 299-0200. Fax: (703) 299-0204. <<http://www.idsociety.org>>.
- World Health Organization (WHO). <<http://www.who.int/en/>>.

OTHER

- Centers for Disease Control and Prevention, Special Pathogens Branch. "Ebola Hemorrhagic Fever," August 23, 2004. <<http://www.cdc.gov/ncidod/dvrd/spb/mnpages/dispages/ebola.htm>>.

Centers for Disease Control and Prevention, Special Pathogens Branch. "Marburg Hemorrhagic Fever," August 23, 2004. <<http://www.cdc.gov/ncidod/dvrd/spb/mnpages/dispages/marburg.htm>>.

Julia Barrett
Rebecca J. Frey, PhD

Hemorrhoids

Definition

Hemorrhoids are enlarged veins in the anus or lower rectum. They often go unnoticed and usually clear up after a few days, but can cause long-lasting discomfort, bleeding and be excruciatingly painful. Effective medical treatments are available, however.

Description

Hemorrhoids (also called piles) can be divided into two kinds, internal and external. Internal hemorrhoids lie inside the anus or lower rectum, beneath the anal or rectal lining. External hemorrhoids lie outside the anal opening. Both kinds can be present at the same time.

Hemorrhoids are a very common medical complaint. More than 75% of Americans have hemorrhoids at some point in their lives, typically after age 30. Pregnant women often develop hemorrhoids, but the condition usually clears up after childbirth. Men are more likely than women to suffer from hemorrhoids that require professional medical treatment.

Causes and symptoms

Precisely why hemorrhoids develop is unknown. Researchers have identified a number of reasons to explain hemorrhoidal swelling, including the simple fact that people's upright posture places a lot of pressure on the anal and rectal veins. Aging, obesity, **pregnancy**, chronic constipation or **diarrhea**, excessive use of **enemas** or **laxatives**, straining during bowel movements, and spending too much time on the toilet are considered contributing factors. Heredity may also play a part in some cases. There is no reason to believe that hemorrhoids are caused by jobs requiring, for instance, heavy lifting or long hours of sitting,



Clinical photo of a thrombosed external hemorrhoid. (Custom Medical Stock Photo. Reproduced by permission.)

although activities of that kind may make existing hemorrhoids worse.

The commonest symptom of internal hemorrhoids is bright red blood in the toilet bowl or on one's feces or toilet paper. When hemorrhoids remain inside the anus they are almost never painful, but they can prolapse (protrude outside the anus) and become irritated and sore. Sometimes, prolapsed hemorrhoids move back into the anal canal on their own or can be pushed back in, but at other times they remain permanently outside the anus until treated by a doctor.

Small external hemorrhoids usually do not produce symptoms. Larger ones, however, can be painful and interfere with cleaning the anal area after a bowel movement. When, as sometimes happens, a blood clot forms in an external hemorrhoid (creating what is called a thrombosed hemorrhoid), the skin around the anus becomes inflamed and a very painful lump develops. On rare occasions the clot will begin to bleed after a few days and leave blood on the underwear. A thrombosed hemorrhoid will not cause an **embolism**.

Diagnosis

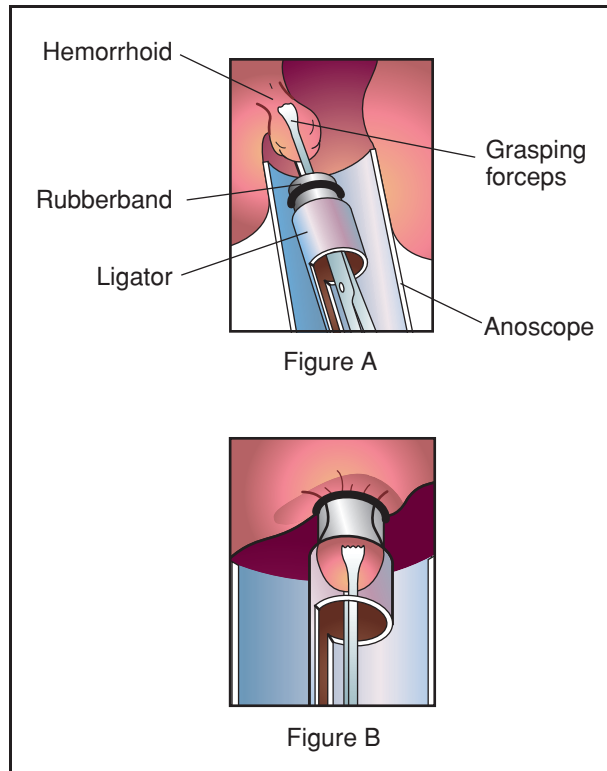
Diagnosis begins with a visual examination of the anus, followed by an internal examination during which the doctor carefully inserts a gloved and lubricated finger into the anus. The doctor may also use an anoscope, a small tube that allows him or her to see into the anal canal. Under some circumstances the doctor may wish to check for other problems by using a sigmoidoscope or colonoscope, a flexible instrument that allows inspection of the lower colon (in the case of the sigmoidoscope) or the entire colon (in the case of the colonoscope).

Treatment

Hemorrhoids can often be effectively dealt with by dietary and lifestyle changes. Softening the feces and avoiding **constipation** by adding fiber to one's diet is important, because hard feces lead to straining during defecation. Fruit, leafy vegetables, and whole-grain breads and cereals are good sources of fiber, as are bulk laxatives and fiber supplements such as Metamucil or Citrucel. Exercising, losing excess weight, and drinking six to eight glasses a day of water or another liquid (not alcohol) also helps. Soap or toilet paper that is perfumed may irritate the anal area and should be avoided, as should excessive cleaning, rubbing, or wiping of that area. Reading in the bathroom is also considered a bad idea, because it adds to the time one spends on the toilet and may increase the strain placed on the anal and rectal veins. After each bowel movement, wiping with a moistened tissue or pad sold for that purpose helps lessen irritation. Hemorrhoid **pain** is often eased by sitting in a tub of warm water for about 10 or 15 minutes two to four times a day (**sitz bath**). A cool compress or ice pack to reduce swelling is also recommended (the ice pack should be wrapped in a cloth or towel to prevent direct contact with the skin). Many people find that over-the-counter hemorrhoid creams and foams bring relief, but these medications do not make hemorrhoids disappear.

When painful hemorrhoids do not respond to home-based remedies, professional medical treatment is necessary. The choice of treatment depends on the type of hemorrhoid, what medical equipment is available, and other considerations.

Rubber band ligation is probably the most widely used of the many treatments for internal hemorrhoids (and the least costly for the patient). This procedure is performed in the office of a family doctor or specialist, or in a hospital on an outpatient basis. An applicator is used to place one or two small rubber bands around the base of the hemorrhoid, cutting off its blood supply. After three to 10 days in the bands, the hemorrhoid falls off, leaving a sore that heals in a week or two. Because internal hemorrhoids are located in a part of the anus that does not sense pain, anesthetic is unnecessary and the procedure is painless in most cases. Although there can be minor discomfort and bleeding for a few days after the bands are applied, complications are rare and most people are soon able to return to work and other activities. If more than one hemorrhoid exists or if banding is not entirely effective the first time (as occasionally happens), the procedure may need to be repeated a few weeks later. After five years, 15–20% of patients experience a recurrence of



Rubber band ligation is probably the most widely used treatment for internal hemorrhoids. An applicator is used to place one or two small rubber bands around the base of the hemorrhoid, cutting off its blood supply (figures A and B). After 3-10 days, the rubber bands and the hemorrhoid fall off, leaving a scab which disappears within a week or two. (Illustration by Electronic Illustrators Group.)

internal hemorrhoids, but in most cases all that is needed is another banding.

External hemorrhoids, and some prolapsed internal hemorrhoids, are removed by conventional surgery in a hospital. Depending on the circumstances, this requires a local, regional, or general anesthetic. Surgery does cause a fair amount of discomfort, but an overnight hospital stay is usually not necessary. Full healing takes two to four weeks, but most people are able to resume normal activities at the end of a week. Hemorrhoids rarely return after surgery.

Alternative treatment

Like mainstream practitioners, alternative practitioners **stress** the importance of a high-fiber diet. To prevent hemorrhoids by strengthening the veins of the anus, rectum, and colon, they recommend blackberries, blueberries, cherries, vitamin C, butcher's broom (*Ruscus aculeatus*), and flavonoids (plant pigments

KEY TERMS

Anus—The opening at the lower end of the rectum. The anus and rectum are both part of the large intestine, a digestive system organ.

Colon—The major part of the large intestine, a digestive system organ.

Defecation—Passage of feces through the anus.

Embolism—Obstruction of blood flow in an artery by a blood clot or other substance arising from another site. An untreated embolism can endanger health and even cause death.

Enema—The introduction of water or another liquid into the bowels through a tube inserted into the anus. Enemas are used to treat constipation and for other purposes.

Feces—Undigested food and other waste that is eliminated through the anus. Also called stools.

Rectum—The lower section of the large intestine, a digestive system organ. After food has passed through the stomach and intestines and been digested, the leftover material, in the form of feces, enters the rectum, where it stays until defecation.

Suppository—A medicinal substance that slowly dissolves after being inserted into the rectum (or other body cavity).

found in fruit and fruit products, tea, and soy). Herbal teas, ointments, and suppositories, and other kinds of herbal preparations, are suggested for reducing discomfort and eliminating hemorrhoids. In particular, pilewort (*Ranunculus ficaria*), applied in an ointment or taken as a tea, can reduce the pain of external hemorrhoids. **Acupuncture**, **acupressure**, **aromatherapy**, and homeopathy are also used to treat hemorrhoids.

Prognosis

Hemorrhoids do not cause **cancer** and are rarely dangerous or life threatening. Most clear up after a few days without professional medical treatment. However, because colorectal cancer and other digestive system diseases can cause anal bleeding and other hemorrhoid-like symptoms, people should always consult a doctor when those symptoms occur.

Prevention

A high-fiber diet and the other lifestyle changes recommended for coping with existing hemorrhoids

also help to prevent hemorrhoids. Not straining during bowel movements is essential.

Resources

ORGANIZATIONS

National Digestive Diseases Information Clearinghouse.
2 Information Way, Bethesda, MD 20892-3570. (800)
891-5389. <<http://www.niddk.nih.gov/health/digest/nddic.htm>>.

Howard Baker

Henoch-Schönlein purpura see **Allergic purpura**

Hepatic carcinoma see **Liver cancer, primary**

Hepatic encephalopathy see **Liver encephalopathy**

Hepatitis-associated antigen (HAA) test see **Hepatitis virus tests**

Hepatitis A

Definition

Hepatitis A is an inflammation of the liver caused by a virus, the hepatitis A virus (HAV). It varies in severity, running an acute course, generally starting within two to six weeks after contact with the virus, and lasting no longer than two or three months. HAV may occur in single cases after contact with an infected relative or sex partner. Alternately, epidemics may develop when food or drinking water is contaminated by the feces of an infected person.

Description

Hepatitis A was previously known as infectious hepatitis because it spread relatively easily from those infected to close household contacts. Once the infection ends, there is no lasting, chronic phase of illness. However it is not uncommon to have a second episode of symptoms about a month after the first; this is called a relapse, but it is not clear that the virus persists when symptoms recur. Both children and adults may be infected by HAV. Children are the chief victims, but very often have no more than a flu-like illness or no symptoms at all (so-called “subclinical” infection),

whereas adults are far likelier to have more severe symptoms.

Epidemics of HAV infection can infect dozens and even hundreds (or, on rare occasions, thousands) of persons. In the public’s mind, outbreaks of hepatitis A usually are linked with the eating of contaminated food at a restaurant. It is true that food-handlers, who may themselves have no symptoms, can start an alarming, widespread epidemic. Many types of food can be infected by sewage containing HAV, but shellfish, such as clams and oysters, are common culprits.

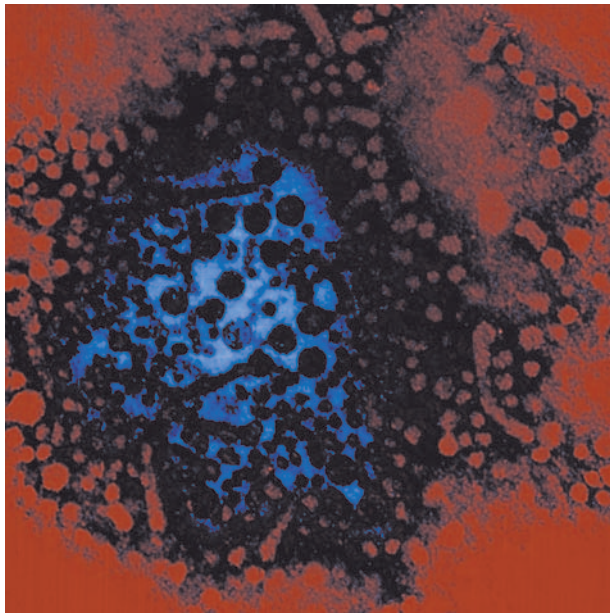
Apart from contaminated food and water, certain groups are at increased risk of getting infectious hepatitis:

- Children at day care centers make up an estimated 14–40% of all cases of HAV infection in the United States. Changing diapers transmits infection through fecal-oral contact. Toys and other objects may remain contaminated for some time. Often a child without symptoms brings the infection home to siblings and parents.
- Troops living under crowded conditions at military camps or in the field. During World War II there were an estimated five million cases in German soldiers and civilians.
- Anyone living in heavily populated and squalid conditions, such as the very poor and those placed in refugee or prisoner-of-war camps.
- Homosexual men are increasingly at risk of HAV infection from oral-anal sexual contact.
- Travelers visiting an area where hepatitis A is common are at risk of becoming ill.

Causes and symptoms

The time from exposure to HAV and the onset of symptoms ranges from two to seven weeks and averages about a month. The virus is passed in the feces, especially late during this incubation period, before symptoms first appear. Infected persons are most contagious starting a week or so before symptoms develop, and remain so up until the time **jaundice** (yellowing of the skin) is noted.

Often the first symptoms to appear are **fatigue**, aching all over, **nausea**, and a loss of appetite. Those who like drinking coffee and **smoking** cigarettes may lose their taste for them. Mild **fever** is common; it seldom is higher than 101 °F (38.3 °C). The liver often enlarges, causing **pain** or tenderness in the right upper part of the abdomen. Jaundice then develops, typically lasting seven to ten days. Many patients do



Hepatitis A virus magnified 225,000 times. (*Custom Medical Stock Photo. Reproduced by permission.*)

not visit the doctor until their skin turns yellow. As many as three out of four children have no symptoms of HAV infection, but about 85% of adults will have symptoms. Besides jaundice, the commonest are abdominal pain, loss of appetite, and feeling generally poorly.

Special situations

An occasional patient with hepatitis A will remain jaundiced for a month, two months or even longer, but eventually the jaundice will pass. Very rarely, a patient will develop such severe hepatitis that the liver fails. HAV infection causes about 100 deaths each year in the United States. In developed countries, a pregnant woman who contracts hepatitis A can be expected to do well although a different form of viral hepatitis (**hepatitis E**) can cause severe infection in pregnant women. In developing countries, however, the infection may prove fatal, probably because **nutrition** is not adequate.

Diagnosis

The early, flu-like symptoms and jaundice, as well as rapid recovery, suggest infectious hepatitis without special tests being done. If there is any question, a specialist in gastrointestinal disorders or infectious diseases can confirm the diagnosis—the detection of a specific antibody, called hepatitis A IgM antibody, that develops when HAV is present in the body. This

KEY TERMS

Antibody—A substance made by the body in response to a foreign body, such as a virus, which is able to attack and destroy the invading virus.

Contamination—The process by which an object or body part becomes exposed to an infectious agent such as a virus.

Epidemic—A situation where a large number of infections by a particular agent, such as a virus, develops in a short time. The agent is rapidly transmitted to many individuals.

Incubation period—The interval from initial exposure to an infectious agent, such as a virus, and the first symptoms of illness.

Jaundice—Yellowing of the skin (and whites of the eyes) when pigments normally eliminated by the liver collect in high amounts in the blood.

Vaccine—A substance prepared from a weakened or killed virus which, when injected, helps the body to form antibodies that will prevent infection by the natural virus.

test always registers positive when a patient has symptoms, and should continue to register positive for four to six months. However, hepatitis A IgM antibody will persist lifelong in the blood and is protective against reinfection.

Treatment

Once symptoms appear, no **antibiotics** or other medicines will shorten the course of infectious hepatitis. Patients should rest in bed as needed, take a healthy diet, and avoid drinking alcohol and/or any medications that could further damage the liver. If a patient feels well it is all right to return to school or work even if some jaundice remains.

Prognosis

Most patients with acute hepatitis, even when severe, begin feeling better in two to three weeks, and recover completely in four to eight weeks. After recovering from hepatitis A, a person no longer carries the virus and remains immune for life. In the United States, serious complications are infrequent and deaths are very rare. In the United States, as many as 75% of adults over 50 years of age will have blood test evidence of previous hepatitis A.

Prevention

The single best way to keep from spreading hepatitis A infection is to wash the hands carefully after using the toilet. Those who are infected should not share items that might carry infection. Special care should be taken to avoid transmitting infection to a sex partner. Travelers should avoid water and ice if unsure of their purity, or they can boil water for one minute before drinking it. All foods eaten should be packaged, well cooked or, in the case of fresh fruit, peeled.

If exposure is a possibility, infection may be prevented by an injection of a serum fraction containing antibody against HAV. This material, called immune serum globulin (ISG), is 90% protective even when injected after exposure—providing it is given within two weeks. Anyone living with an infected patient should receive ISG. For long-term protection, a killed virus hepatitis A vaccine became available in 1995. More than 95% of those vaccinated will develop an adequate amount of anti-HAV antibody. Those who should consider being vaccinated include healthcare professionals, those working at day care and similar facilities, frequent travelers to areas with poor sanitation, those with any form of chronic **liver disease**, and those who are very sexually active. Starting in 2000, routine immunization with the hepatitis A vaccine was recommended for children born in states where the rate of hepatitis A was two or more times the national average (Alaska, Arizona, California, Idaho, Nevada, New Mexico, Oklahoma, Oregon, South Dakota, Utah, and Washington) and suggested in states where the rate was 1.5 times the national average (Arkansas, Colorado, Missouri, Montana, Texas and Wyoming).

Larry I. Lutwick, MD

Hepatitis, alcoholic

Definition

Alcoholic hepatitis is an inflammation of the liver caused by alcohol.

Description

Irritation, be it from toxins or infections, causes a similar response in body organs. The response is known as inflammation and consists of:

- an increase in the blood to the affected organ
- redness and swelling of the organ
- influx of immune agents like white blood cells and their arsenal of chemical weapons
- **pain**

As the acute process subsides, there is either healing or lingering activity. Lingering activity—chronic disease—has a milder presentation with similar ingredients. Healing often takes the form of scarring, wherein normal functioning tissue is replaced by tough, fibrous, and non-productive scar. Both chronic disease and healing can happen simultaneously, so that scar tissue progressively replaces normal tissue. This leads to cirrhosis, a liver so scarred it is unable to do its job adequately.

Alcohol can cause either an acute or a chronic disease in the liver. The acute disease can be severe, even fatal, and can bring with it hemolysis—blood cell destruction. Alcohol can also cause a third type of liver disease, fatty liver, in which the continuous action of alcohol turns the liver to useless fat. This condition eventually progresses to **cirrhosis** if the **poisoning** continues.

Causes and symptoms

Inflammation of the liver can be caused by a great variety of agents—poisons, drugs, viruses, bacteria, protozoa, and even larger organisms like worms. Alcohol is a poison if taken in more than modest amounts. It favors destroying stomach lining, liver, heart muscle, and brain tissue. The liver is a primary target because alcohol travels to the liver after leaving the intestines. Those who drink enough to get alcohol poisoning have a tendency to be undernourished, since alcohol provides ample calories but little **nutrition**. It is suspected that both the alcohol and the poor nutrition produce alcoholic hepatitis.

Diagnosis

Hepatitis of all kinds causes notable discomfort, loss of appetite, **nausea**, pain in the liver, and usually **jaundice** (turning yellow). Blood test abnormalities are unmistakably those of hepatitis, but selecting from so many the precise cause may take additional diagnostic work.

Treatment

As with all poisonings, removal of the offending agent is primary. There is no specific treatment for

KEY TERMS

Cirrhosis—Disruption of normal liver structure and function caused by any type of chronic disease such as hepatitis and alcohol abuse.

Fatty liver—An abnormal amount of fat tissue in the liver caused by alcohol abuse.

Hemolysis—Disintegration of red blood cells.

Protozoa—One celled microscopic organisms like amoeba.

alcohol poisoning. General supportive measures must see the patient through until the liver has healed by itself. In the case of fulminant (sudden and severe) disease, the liver may be completely destroyed and have to be replaced by a transplant.

Prognosis

The liver is robust. It can heal without scarring after one or a few episodes of hepatitis that resolve without lingering. It can, moreover, regrow from a fragment of its former self, provided there is not disease or poison still inhibiting it.

Prevention

Alcohol is lethal in many ways when ingested in excess. Research suggests that the maximum healthy dose of alcohol per day is roughly one pure ounce—the amount in two cocktails, two glasses of wine, or two beers.

Resources

ORGANIZATIONS

American Liver Foundation. 1425 Pompton Ave., Cedar Grove, NJ 07009. (800) 223-0179. <<http://www.liverfoundation.org>>.

Local chapters of Alcoholics Anonymous.

J. Ricker Polsdorfer, MD

Hepatitis, autoimmune

Definition

A form of liver inflammation in which the body's immune system attacks liver cells.

Description

Autoimmunity causes the body's defense mechanisms to turn against itself. Many of the tissues in the body can be the target of such an attack. While one tissue type predominates, others may be involved in a general misdirection of immune activity, perhaps because the specific target antigen is present in differing quantities in each of the affected tissues. There seem to be hereditary causes for autoimmunity, since these diseases tend to run in families and have genetic markers. Among the more common diseases believed to fall within this category are **rheumatoid arthritis**, **systemic lupus erythematosus**, **multiple sclerosis**, and **psoriasis**.

The process of autoimmune disease is very similar to infectious disease and allergy, so that great caution is observed in placing a disorder in this class. Germs were found to cause several diseases originally thought to be autoimmune. Allergens cause others. Many more may be uncovered. Autoimmunity is often believed to originate with a virus infection. A chemical in the virus resembles a body chemical so closely that the immune system attacks both.

Autoimmune hepatitis is similar to viral hepatitis, a disease of the liver. It can be an acute disease that kills over a third of its victims within six months, it can persist for years, or it can return periodically. Some patients develop **cirrhosis** of the liver which, over time, causes the liver to cease functioning.

Causes and symptoms

Symptoms of autoimmune hepatitis resemble those of other types of hepatitis. Patients who develop autoimmune hepatitis experience **pain** under the right ribs, **fatigue** and general discomfort, loss of appetite, **nausea**, sometimes **vomiting** and **jaundice**. In addition, other parts of the body may be involved and contribute their own symptoms.

Diagnosis

Extensive laboratory testing may be required to differentiate this disease from viral hepatitis. The distinction may not even be made during the initial episode. There are certain markers of autoimmune disease in the blood that can lead to the correct diagnosis if they are sought. In advanced or chronic cases a **liver biopsy** may be necessary.

Treatment

Autoimmune hepatitis is among the few types of hepatitis that can be treated effectively. Since treatment itself introduces problems in at least 20%

KEY TERMS

Allergen—Any chemical that causes an immune reaction only in people sensitive to it.

Antigen—Any chemical that can be the target of an immune response.

Biopsy—Surgical removal of a piece of tissue for examination.

Jaundice—A yellow color to the skin from bile that backs up into the circulation.

of patients, it is reserved for the more severe cases. Up to 80% of patients improve with cortisone treatment, although a cure is unlikely. Another drug—azathioprine—is sometimes used concurrently. Treatment continues for over a year and may be restarted during a relapse. At least half the patients relapse at some point, and most will still continue to have progressive liver scarring.

If the liver fails, transplant is the only recourse.

Prognosis

In spite of treatment autoimmune hepatitis can re-erupt at any time, and may continue to damage and scar the liver. The rate of progression varies considerably from patient to patient.

Resources

ORGANIZATIONS

American Liver Foundation. 1425 Pompton Ave., Cedar Grove, NJ 07009. (800) 223-0179. <<http://www.liverfoundation.org>>.

J. Ricker Polsdorfer, MD

Hepatitis B

Definition

Hepatitis B is a potentially serious form of liver inflammation due to infection by the hepatitis B virus (HBV). It occurs in both rapidly developing (acute) and long-lasting (chronic) forms, and is one of the most common chronic infectious diseases worldwide. An effective vaccine is available that will prevent the disease in those who are later exposed.

Description

Commonly called “serum hepatitis,” hepatitis B ranges from mild to severe. Some people who are infected by HBV develop no symptoms and are totally unaware of the fact, but they may carry HBV in their blood and pass the infection on to others. In its chronic form, HBV infection may destroy the liver through a scarring process, called **cirrhosis**, or it may lead to **cancer** of the liver.

When a person is infected by HBV, the virus enters the bloodstream and body fluids, and is able to pass through tiny breaks in the skin, mouth, or the male or female genital area. There are several ways of getting the infection:

- During birth, a mother with hepatitis B may pass HBV on to her infant.
- Contact with infected blood is a common means of transmitting hepatitis B. One way this may happen is by being stuck with a needle. Both health care workers and those who inject drugs into their veins are at risk in this way.
- Having sex with a person infected by HBV is an important risk factor (especially anal sex).

Although there are many ways of passing on HBV, the virus actually is not very easily transmitted. There is no need to worry that casual contact, such as shaking hands, will expose one to hepatitis B. There is no reason not to share a workplace or even a restroom with an infected person.

More than 300 million persons throughout the world are infected by HBV. While most who become chronic carriers of the virus live in Asia and Africa, there are no fewer than 1.5 million carriers in the United States. Because carriers represent a constant threat of transmitting the infection, the risk of hepatitis B is always highest where there are many carriers. Such areas are said to be endemic for hepatitis B. When infants or young children living in an endemic area are infected, their chance of becoming a chronic hepatitis B carrier is at least 90%. This probably is because their bodies are not able to make the substances (antibodies) that destroy the virus. In contrast, no more than 5% of infected teenagers and adults develop chronic infection.

Causes and symptoms

With the exception of HBV, all the common viruses that cause hepatitis are known as RNA viruses because they contain ribonucleic acid or RNA as their genetic material. HBV is the only deoxyribonucleic

acid or DNA virus that is a major cause of hepatitis. HBV is made up of several fragments, called antigens, that stimulate the body's immune system to produce the antibodies that can neutralize or even destroy the infecting virus. It is, in fact, the immune reaction, not the virus, that seems to cause the liver inflammation.

Acute hepatitis B

In the United States, a majority of acute HBV infections occur in teenagers and young adults. Half of these youth never develop symptoms, and only about 20%—or one in five infected patients—develop severe symptoms and yellowing of the skin (**jaundice**). Jaundice occurs when the infected liver is unable to get rid of certain colored substances, or pigments, as it normally does. The remaining 30% of patients have only “flu-like” symptoms and will probably not even be diagnosed as having hepatitis unless certain tests are done.

The most common symptoms of acute hepatitis B are loss of appetite, **nausea**, generally feeling poorly, and **pain** or tenderness in the right upper part of the abdomen (where the liver is located). Compared to patients with **hepatitis A** or C, those with HBV infection are less able to continue their usual activities and require more time resting in bed.

Occasionally patients with HBV infection will develop joint swelling and pain (arthritis) as well as **hives** or a skin rash before jaundice appears. The joint symptoms usually last no longer than three to seven days.

Typically the symptoms of acute hepatitis B do not persist longer than two or three months. If they continue for four months, the patient has an abnormally long-lasting acute infection. In a small number of patients—probably fewer than 3%—the infection keeps getting worse as the liver cells die off. Jaundice deepens, and patients may bleed easily when the levels of coagulation factors (normally made by the liver) decrease. Large amounts of fluid collect in the abdomen and beneath the skin (**edema**). The least common outcome of acute HBV infection, seen in fewer than 1% of patients, is fulminant hepatitis, when the liver fails entirely. Only about half of these patients can be expected to live.

Chronic hepatitis B

HBV infection lasting longer than six months is said to be chronic. After this time it is much less likely for the infection to disappear. Not all carriers of the virus develop chronic **liver disease**; in fact, a majority have no symptoms. But, about one in every four HBV carriers develop liver disease that gets worse over time,

as the liver becomes more and more scarred and less able to carry out its normal functions. A badly scarred liver is called cirrhosis. Patients are likely to have an enlarged liver and spleen, as well as tiny clusters of abnormal blood vessels in the skin that resemble spiders.

The most serious complication of chronic HBV infection is **liver cancer**. Worldwide this is the most common cancer to occur in men. Nevertheless, the overall chance that liver cancer will develop at any time in a patient's life is probably much lower than 10%. Patients with chronic hepatitis B who drink or smoke are more likely to develop liver cancer. It is not unusual for a person to simultaneously have both HBV infection and infection by HIV (human **immuno-deficiency** virus, the cause of **AIDS**). A study released in 2003 reported that men infected with both HIV and HBV were more likely to die from liver disease than people infected with just one of the diseases.

Diagnosis

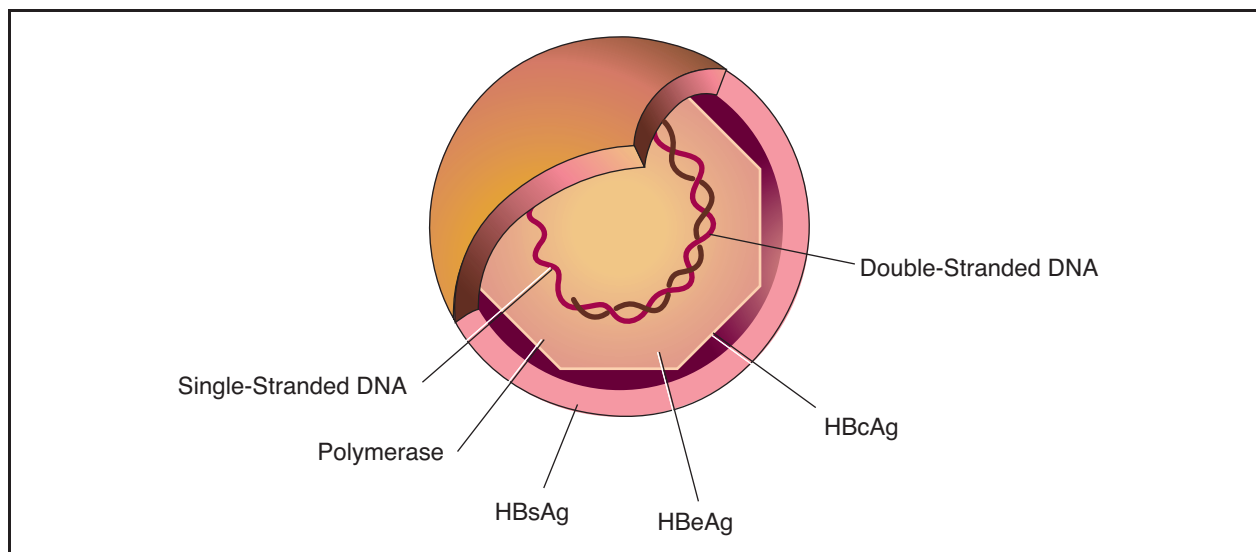
Hepatitis B is diagnosed by detecting one of the viral antigens—called hepatitis B surface antigen (HBsAg)—in the blood. Later in the acute disease, HBsAg may no longer be present, in which case a test for antibodies to a different antigen—hepatitis B core antigen—is used. If HBsAg can be detected in the blood for longer than six months, chronic hepatitis B is diagnosed. A number of tests can be done to learn how well, or poorly, the liver is working. They include blood clotting tests and tests for enzymes that are found in abnormally high amounts when any form of hepatitis is present.

Treatment

In the past, there was no treatment available for hepatitis B. But developments have been made in recent years on drugs that suppress the virus and its symptoms. In early 2003, a drug called adefovir was reported as an effective treatment. Another drug called tenofovir was demonstrated as effective in patients infected with both hepatitis B and HIV. Two studies also reported on the effectiveness of a drug called Preveon, which was more expensive than others. Patients also should rest in bed as needed, continue to eat a healthy diet, and avoid alcohol. Any non-critical surgery should be postponed.

Prognosis

Each year an estimated 150,000 persons in the United States get hepatitis B. More than 10,000 will



Hepatitis B virus (HBV) is composed of an inner protein core and an outer protein capsule. The outer capsule contains the hepatitis B surface antigen (HBsAg). The inner core contains HBV core antigen (HBcAg) and hepatitis B e-antigen (HBeAg). This cell also contains polymerase, which catalyzes the formation of the cell's DNA. HBV is the only hepatitis-causing virus that has DNA, instead of RNA. (Illustration by Electronic Illustrators Group.)

require hospital care, and as many as 5,000 will die from complications of the infection. About 90% of all those infected will have acute disease only. A large majority of these patients will recover within three months. It is the remaining 10%, with chronic infection, who account for most serious complications and deaths from HBV infection. In the United States, perhaps only 2% of all those who are infected will become chronically ill. The course of chronic HBV infection in any particular patient is unpredictable. Some patients who do well at first may later develop serious complications. Even when no symptoms of liver disease develop, chronic carriers remain a threat to others by serving as a source of infection.

Prevention

The best way to prevent any form of viral hepatitis is to avoid contact with blood and other body fluids of infected individuals. The use of condoms during sex also is advisable.

If a person is exposed to hepatitis B, a serum preparation containing a high level of antibody against HBV may prevent infection if given within three to seven days of exposure. Babies born of a mother with HBV should receive the vaccine within 24 hours. An effective and safe vaccine is available that reliably prevents hepatitis B. **Vaccination** is suggested for most infants and for children aged 10 and younger whose parents are from a place where hepatitis B is

KEY TERMS

Antibody—A substance formed in the body in response to a foreign body, such as a virus, which can then attack and destroy the invading virus.

Antigen—Part of an invading microorganism, such as a virus, that causes tissue damage (in hepatitis, to the liver), and that also stimulates the body's immune system to produce antibodies.

Cirrhosis—The end result of many forms of liver disease, the condition of the liver when its cells have been damaged or destroyed and are replaced by scar tissue.

Vaccine—A substance prepared from a weakened or killed virus which, when injected, helps the body to form antibodies that will attack an invading virus and may prevent infection altogether.

common. Teenagers not vaccinated as children and all adults at risk of exposure also should be vaccinated against hepatitis B. Three doses are recommended.

Those at increased risk of getting hepatitis B, and who therefore should be vaccinated, include:

- household contacts of a person carrying HBV
- healthcare workers who often come in contact with patients' blood or other body fluids

- patients with **kidney disease** who periodically undergo hemodialysis
- homosexual men who are sexually active, and heterosexuals who have multiple sex partners
- persons coming from areas where HBV infection is a major problem
- prisoners and others living in crowded institutions
- drug abusers who use needles to inject drugs into their veins

Studies released in 2003 showed increased risk of nonresponse to hepatitis B vaccines among adults over age 30. This may be related to age-associated changes in the immune system.

Resources

PERIODICALS

“Antiviral Effective Against Hepatitis B Virus in HIV-coinfected.” *Virus Weekly* January 28, 2003: 16.

Bauer, Jeff. “Co-infection with Hepatitis B and HIV Increases Men’s Risks of Death from Liver Disease.” *RN* March 2003: 97.

Elliott, William T. “Warfarin Effectively Prevents Venous Thromboembolism (Pharmacology Watch).” *Critical Care Alert* April 2003.

“Hepatitis B Vaccine Loses Effectiveness in Older Adults.” *Vaccine Weekly* January 29, 2003: 23.

ORGANIZATIONS

Hepatitis B Foundation. 101 Greenwood Ave., Suite 570, Jenkintown, PA 19046. (215) 884-8786.
<info@hepb.org>.

David A. Cramer, MD
Teresa G. Odle

Hepatitis C

Definition

Hepatitis C is a form of liver inflammation that causes primarily a long-lasting (chronic) disease. Acute (newly developed) hepatitis C is rarely observed as the early disease is generally quite mild. Spread mainly by contact with infected blood, the hepatitis C virus (HCV) causes most cases of viral liver infection not due to the A and B hepatitis viruses. In fact, before other viral types were found, hepatitis C was referred to as “non-A, non-B hepatitis.” It is not a new infection, just newly diagnosable and has been widely present in the U.S. population for decades.

Description

HCV is a blood-borne virus that is and always was the major cause of “transfusion hepatitis,” which can develop in patients who are given blood or most blood products except for gamma-globulin. The existence of a third hepatitis virus (besides the A and B viruses) became clear in 1974, but HCV was first identified in 1989. Thereafter, tests were devised to detect the virus in blood units before transfusing them. As a result, since the early 1990s transfused blood is less commonly the cause of hepatitis C.

The hepatitis C form of hepatitis is generally mild in its early, acute stage, but it is much likelier than **hepatitis B** (85% as compared to 10%) to produce chronic **liver disease**. Therefore, more than two of every three persons who are infected by HCV may continue to have the virus in their blood and so become carriers, who can transmit the infection to others.

The most common way of transmitting hepatitis C is when blood containing the virus enters another person’s circulation through a break in the skin or the mucosa (inner lining) of the mouth or genitals. HCV also can be passed (although uncommonly) from an infected mother to the infant she is carrying. (The risk of infection from breast milk is very low.) Also, HCV can be rarely spread through sexual intercourse. Usually, however, the sexual contacts of chronic carriers of hepatitis C are not infected.

Those at increased risk of developing hepatitis C include:

- healthcare workers who come in contact with infected blood from a cut or bruise, or from a device or instrument that has been infected (“contaminated”)
- persons who inject illicit drugs into their veins and skin, especially if they share needles and syringes with other users
- anyone who gets a tattoo or has his or her skin pierced with an infected needle
- persons with **hemophilia** (who because they bleed very easily may require large amounts of blood and blood products over time)
- patients with **kidney disease** who have periodic dialysis—a treatment that rids their blood of toxic substances—and often requires the patient to have blood transfusions

About one-fourth of patients with hepatitis C do not belong to any of these high-risk groups. Although blood **transfusion** is a much less common cause of HCV infection than in earlier years, cases still occur. Also, sexual transmission is possible, and may take place with either heterosexual or homosexual behavior.

Causes and symptoms

More than half of all patients who develop hepatitis C have no symptoms or signs of liver disease. Some, however, may have a minor illness with flu-like symptoms. Any form of hepatitis may keep the liver from eliminating certain colored (pigmented) substances as it normally does. These pigments collect in the skin, turning it yellow, and also may cause yellowing of the whites of the eyes. About one in four patients with hepatitis C will develop this yellowing of the skin called jaundice (or yellow **jaundice**). Some patients lose their appetite and frequently feel tired. Patients may also feel nauseous or even vomit.

In most patients, HCV can still be found in the blood six months after the start of acute infection, and these patients are considered to be carriers. If the virus persists for one year, it is very unlikely to disappear. About 20% of chronic carriers develop **cirrhosis** (scarring) of the liver when the virus damages or destroys large numbers of liver cells, which are then replaced by scar tissue. Cirrhosis may develop only after a long period of time (as long as 20 years) and often even more has passed. Most (four in five) patients will not develop cirrhosis and instead have a mild, chronic form of infection called chronic persistent hepatitis and when they die, will die with, not of, the infection.

Patients with chronic HCV infection are at risk of developing certain very serious complications:

- Patients with hepatitis C who develop cirrhosis may go on to have liver cancer—called hepatocellular carcinoma. Patients with liver cancer have an average life expectancy measured in months unless the tumor is totally removed.
- Patients also are at risk of developing a combination of joint **pain**, weakness, and areas of bleeding into the skin. The kidneys and brain also may be affected. Perhaps 5% of patients with chronic HCV infection develop this condition, called cryoglobulinemia.
- Patients with porphyria (metabolic disturbances characterized by extreme sensitivity to light) develop blisters in areas of their skin that are exposed to sunlight. The skin also may be easily bruised, and, in time, can become discolored.

Diagnosis

Hepatitis C should be suspected if a patient develops jaundice and reports recent contact with the blood of a person who may have been infected. There is a blood test to detect HCV IgG antibody, a substance that the body makes to combat HCV. Care is required,

as the test often does not show positive for up to two to three months after infection. Also, the test only shows whether a person has ever been infected by HCV, not whether the virus is still present. A less available and more expensive test measuring HCV RNA (the viral gene) can be found in early infection before the antibody is measurable. Simpler blood tests can be done to show how much jaundice-causing pigment is in a patient's blood, or to measure the levels of certain proteins made by the liver. High levels of these "liver enzymes" (called ALT and AST) indicate that the liver is inflamed. Rising levels could suggest that the infection is getting worse.

Treatment

Patients who fail to recover promptly may be advised to see a specialist in gastrointestinal disorders (which include liver disease) or infectious diseases. A balanced diet with little fat is best, and patients should limit their alcohol intake, or, better, avoid alcohol altogether. Any medication that can cause liver damage should be avoided. The amount of time in bed depends on how poorly a particular patient feels.

A natural body protein, interferon alpha, now can be made in large amounts by genetic engineering, and improves the outlook for many patients who have chronic hepatitis C. The protein can lessen the symptoms of infection and improve liver function. Not all patients respond, however, and others get less benefit the longer they take interferon. **Fever** and flu-like symptoms are frequent side effects of this treatment. Using a high dose for six months, nearly half of patients have responded positively. Half the patients who do respond well will relapse after the drug is stopped. A newer medication called ribavirin is now commonly used with interferon and, if tolerated, does increase response rates. A newer form of interferon, called pegylated interferon, is also being used for treatment. Because of the problems with treatment, many people have sought alternative medications such as milk thistle or certain Asian herbs.

When hepatitis destroys most or all of the liver, the only hope may be a liver transplant. Unfortunately the new liver usually becomes infected by HCV. On the other hand, total liver failure is less frequent than in patients with hepatitis B.

Prognosis

In roughly one-fifth of patients who develop hepatitis C, the acute infection will subside, and they will recover completely within four to eight weeks and

KEY TERMS

Antibody—A substance formed in the body in response to a foreign body, such as a virus, which can attack and destroy the invading foreign body or virus.

Carrier—A person who, after recovering from a viral infection, continues to “carry” the virus in the blood and can pass it on to others who then may develop infection.

Contamination—Passage of an infectious organism, such as a virus, from an infected person to an object such as a needle, which then, when used, may pass infection to another person.

Hepatocellular carcinoma—A dangerous cancer of the liver that may develop in patients who have had hepatitis, sometimes as long as 20 or 30 years earlier.

Porphyrria—Any of a group of disturbances of porphyrin metabolism characterized by excess porphyrins (various biologically active compounds with a distinct structure) in the urine and by extreme sensitivity to light.

have no later problems. Other patients face two risks: they themselves may develop chronic liver infection and possibly serious complications such as **liver cancer**, and, also, they will continue carrying the virus and may pass it on to others. The overall risk of developing cirrhosis, or liver scarring, is about 15% of all patients infected by HCV. Acute liver failure is less frequent in patients with chronic hepatitis C than in those with other forms of hepatitis.

Prevention

No vaccine has yet been developed to prevent hepatitis C in persons exposed to the virus. In addition, there is no role of gamma-globulin in the prevention of the infection. There are, however, many ways in which infection may be avoided:

- Those who inject drugs should never share needles, syringes, swabs, spoons, or anything else that comes in contact with bodily fluids. They should always use clean equipment.
- Hands should be washed before and after contact with another person’s blood or if the skin is penetrated.

- The sharing of personal items should be avoided, particularly those that can puncture the skin or inside of the mouth, such as razors, nail files and scissors, and even toothbrushes.
- Condoms should be used for either vaginal or oral sex.

If a person does develop hepatitis C, its spread may be prevented by:

- not donating blood
- not sharing personal items with others
- wiping up any spilled blood while using gloves, household bleach, and disposable paper towels
- carefully covering any cut or wound with a bandaid or dressing
- practicing safe sex, especially during the acute phase of the infection

Larry I. Lutwick, MD

Hepatitis D

Definition

Hepatitis D (or delta, the Greek letter “D”), is a form of liver inflammation that occurs only in patients who also are infected by the **hepatitis B** virus. Infection by the hepatitis delta virus (HDV) either occurs at the same time as hepatitis B develops, or develops later when infection by hepatitis B virus (HBV) has entered the chronic (long-lasting) stage.

Description

Delta hepatitis can be quite severe, but it is seen only in patients already infected by HBV. In the late 1970s, Italian physicians discovered that some patients with hepatitis B had another type of infectious agent in their liver cells. Later the new virus—HDV—was confirmed by experimentally infecting chimpanzees. When both viruses are present, acute infection tends to be more severe. Furthermore, patients with both infections are likelier than those with HBV alone to develop chronic **liver disease**, and, when it occurs, it is more severe.

About 300 million persons worldwide carry HBV. Of them, at least 5% probably also have delta hepatitis. In North America HDV infection appears to be less frequent: 4% of all patients with acute hepatitis B

have HDV infection. The delta virus causes an estimated 2% of all cases of acute viral hepatitis in the United States. The rate of HDV infection varies widely in different parts of the world; it is a very serious infection in some countries and quite mild in others. Chronic delta hepatitis is a more serious disease than either chronic hepatitis B alone or **hepatitis C**.

Certain individuals—the same ones who are at increased risk of developing hepatitis B—are the prime candidates to be infected by HDV. For example:

- Not infrequently, HDV infection occurs in patients with chronic HBV infection who also have **hemophilia**, a bleeding disease. These patients are at risk because they require large amounts of transfused blood and blood products that may contain HDV.
- In some areas, one-fourth to one-half of patients with chronic HBV infection who inject themselves with illicit drugs become infected by HDV as well. Drug abusers who share contaminated needles are likely to infect one another.
- Patients who get HBV infection by sexual contact may also be infected by HDV, although the delta virus is less often spread in this way than is HBV itself. Between 10–25% of homosexual men with chronic HBV infection harbor the delta virus.
- Like hepatitis B, HDV infection may develop in health-care workers who are victims of a needle stick, and it also can be spread within households when personal items such as a razor or toothbrush are shared.

Causes and symptoms

The delta virus is a small and incomplete viral particle. Perhaps this is why it cannot cause infection on its own. Its companion virus, HBV, actually forms a covering over the HDV particle. In chronically ill patients (those whose virus persists longer than six months), the combined viruses cause inflammation throughout the liver and eventually destroy the liver cells, which are then replaced by scar tissue. This scarring is called **cirrhosis**.

When HBV and HDV infections develop at the same time, a condition called coinfection, recovery is the rule. Only 2–5% of patients become chronic carriers (have the virus remain in their blood more than six months after infection). It may be that HDV actually keeps HBV from reproducing as rapidly as it would if it were alone, so chronic infection is less likely.

When HBV infection occurs first and is followed by HDV infection, the condition is called superinfection. This is a more serious situation. Between half and

two-thirds of patients with superinfection develop severe acute hepatitis. Once the liver cells contain large numbers of HBV viruses, HDV tends to reproduce more actively. Massive infection and liver failure are more common in superinfection. The risk of **liver cancer**, however, is no greater than from hepatitis B alone.

As with other forms of hepatitis, the earliest symptoms are **nausea**, loss of appetite, joint pains, and tiredness. There may be **fever** (not marked) and an enlarged liver may cause discomfort or actual **pain** in the right upper part of the abdomen. Later, **jaundice** (a yellowing of the skin and whites of the eyes that occurs when the liver is no longer able to eliminate certain pigmented substances) may develop.

Diagnosis

HDV infection may be diagnosed by detecting the antibody against the virus. Unfortunately this test cannot detect acute coinfection or superinfection as early as when symptoms first develop. Antibody against HDV usually is found no sooner than 30 days after symptoms appear. Until recently, the virus itself could only be identified by testing a small sample of liver tissue. Scientists now are developing a blood test for HDV that should make diagnosis faster and easier. When HDV is present, liver enzymes (proteins made by the liver) are present in abnormally high amounts. In some patients with coinfection, the enzyme levels peak twice, once when HBV infection starts and again at the time of HDV infection.

Treatment

As in any form of hepatitis, patients in the acute stage should rest in bed as needed, eat a balanced diet, and avoid alcohol. Alpha-interferon, the natural body substance which helps control hepatitis C, has generally not been found helpful in treating hepatitis D. If the liver is largely destroyed and has stopped functioning, **liver transplantation** is an option. Even when the procedure is successful, disease often recurs and cirrhosis may actually develop more rapidly than before.

Prognosis

A large majority of patients with coinfection of HBV and HDV recover from an episode of acute hepatitis. However, about two-thirds of patients chronically infected by HDV go on to develop cirrhosis of the liver. In one long-term study, just over half of patients who became carriers of HDV had moderate or severe liver disease, and one-fourth of them died. If very severe liver failure develops, the chance of

KEY TERMS

Alpha-interferon—A natural body substance that now can be made in large quantities and is an effective treatment for some types of viral inflammatory disease, including hepatitis C.

Antibody—A substance formed in the body in response to an invading microorganism, such as a virus, which can attack and destroy the invading virus.

Coinfection—Invasion of the body by two viruses at about the same time.

Hemophilia—A bleeding disease that may call for the transfusion of large amounts of blood and blood products.

Superinfection—Infection by a second virus after a previous infection by a different virus has become well established.

a patient surviving is no better than 50%. A liver transplant may improve this figure to 70%. When transplantation is done for cirrhosis, rather than for liver failure, nearly 90% of patients live five years or longer. The major concern with transplantation is infection of the transplanted liver; this may occur in as many as 40% of transplant patients.

When a child with viral hepatitis develops cirrhosis, HDV infection is commonly responsible. A woman who develops delta hepatitis while pregnant will do as well as if she were not pregnant; and there is no increased risk that the newborn will be malformed in any way.

Prevention

The vaccine against hepatitis B also prevents delta hepatitis, since it cannot occur unless HBV infection is present. Hopefully, a vaccine can be developed that will keep delta infection from developing in chronic HBV carriers. However, if a person already has HBV infection, any exposure to blood should be strictly avoided. A high level of sexual activity with multiple partners is also a risk factor for delta hepatitis.

Resources

ORGANIZATIONS

American Liver Foundation. 1425 Pompton Ave., Cedar Grove, NJ 07009. (800) 223-0179. <<http://www.liverfoundation.org>>.

David A. Cramer, MD

Hepatitis, drug-induced

Definition

Inflammation of the liver due to an adverse reaction with a drug.

Description

The liver is a very important organ to the body. It is a large internal organ weighing more than three pounds in the average adult. It performs over 100 functions including formation of bile, detoxification of harmful substances, vitamin storage and metabolism of carbohydrates, fats and proteins. Serious complications could arise when the liver becomes inflamed due to hepatitis when it is not able to perform these tasks. A virus most often causes hepatitis but certain drugs can also induce it.

Drug-induced hepatitis (also called toxic hepatitis) occurs in eight in every 10,000 people because the liver reacts abnormally during drug exposure, leading to liver damage. This pathology causes the liver not to function properly and the symptoms can begin to be seen. Women tend to be affected almost twice as often as men. Older people are more prone to this type of hepatitis because their bodies aren't able to repair themselves as fast as younger people. Drugs that can be associated with drug-induced hepatitis include acetaminophen, vitamin A, and PTU (a drug treatment for **tuberculosis**).

Causes and symptoms

There are three general types of drug-induced hepatitis: toxic, metabolic idiosyncrasy and immunologic idiosyncrasy. With toxic hepatitis liver damage as the result of a drug complication with hepatotoxins happens to everyone who takes that particular drug. On the other hand, hepatitis resulting from a metabolic or immunologic idiosyncrasy only happens to certain people, those predisposed to particular idiosyncrasy.

In patients with a metabolic idiosyncrasy the person metabolizes the drug differently than most people causing a harmful by-product that damages the liver. A metabolic idiosyncrasy is seen in 0.1-2% of people and it is complicated by use of alcohol.

With an immunologic idiosyncrasy the patient's body recognizes the metabolized drug by-products as foreign. This leads to the destruction of liver cells containing the by-product via the immune system

KEY TERMS

Hepatitis—General inflammation of the liver.

Hepatomegaly—General swelling of the liver.

Hepatotoxin—A substance that is toxic to the liver.

Idiosyncrasy—A defect in that particular pathway resulting in an abnormality.

resulting in hepatitis. An immunologic idiosyncrasy is seen in less than one person per 10,000 (0.01%) people and is more than twice as common in women.

The symptoms of drug-induced hepatitis are similar to viral hepatitis. Drug induced hepatitis tends to be acute. If it is not caught soon enough the damage could be permanent resulting in chronic hepatitis. Some of the common symptoms are:

- nausea
- vomiting
- headache
- anorexia
- jaundice
- clay color stools
- dark urine
- hepatomegaly

Diagnosis

Diagnosis is typically made through a physical exam along with a patient history to identify any possible hepatotoxins. Blood tests are usually done as well. An increased white blood cell count is typical.

Treatment

There isn't any specific treatment other than immediate discontinuance of the causative agent. Rest during the acute phase of the disease is vital along with the intake of fluids to maintain hydration.

Prognosis

Usually the symptoms will go away after the drug has been eliminated due to the liver repairing itself. A full recovery is typically expected unless it wasn't treated quickly resulting in more liver damage being done than normal.

Prevention

If there is a history of liver damage certain medications should not be taken. Doctors will be familiar with these.

Resources

BOOKS

Feldman, Sleisenger, and Scharschmidt. "Liver Diseases Caused by Drugs, Anesthetics, and Toxins." In *Gastrointestinal and Liver Disease*. Philadelphia: W.B. Saunders Company, 2001, pp.1232-1237.

Holmes, Nancy H. "Hepatobiliary Disorders." In *Diseases*. Pennsylvania: Springhouse Corporation, 2001, pp.744-53.

PERIODICALS

Hautekeete, Horsmans, Van Waeyenberge, Demanet, Henrion, Verbist, Brenard, Sempoux, Michielsen, Yap, Rahier, and Geubel. "HLA Association of Amoxicillin-Clavulanate—Induced Hepatitis." *Gastroenterology* November 1999: 1181-86.

Thomas Scott Eagan
Ronald Watson, PhD

Hepatitis E

Definition

The hepatitis E virus (HEV) is a common cause of hepatitis that is transmitted via the intestinal tract, and is not caused by the **hepatitis A** virus. Spread most often by contaminated drinking water, HEV infection occurs mainly in developing countries.

Description

Hepatitis E is also known as epidemic non-A, non-B hepatitis. Like hepatitis A, it is an acute and short-lived illness that can sometimes cause liver failure. HEV, discovered in 1987, is spread by the fecal-oral route. It is constantly present (endemic) in countries where human waste is allowed to get into drinking water without first being purified. Large outbreaks (epidemics) have occurred in Asian and South American countries where there is poor sanitation. In the United States and Canada no outbreaks have been reported, but persons traveling to an endemic region may return with HEV.

Causes and symptoms

There are at least two strains of HEV, one found in Asia and another in Mexico. The virus may start dividing in the gastrointestinal tract, but it grows mostly in the liver. After an incubation period (the time from when a person is first infected by a virus until the appearance of the earliest symptoms) of two to eight weeks, infected persons develop **fever**, may feel nauseous, lose their appetite, and often have discomfort or actual **pain** in the right upper part of the abdomen where the liver is located. Some develop yellowing of the skin and the whites of the eyes (**jaundice**). Most often the illness is mild and disappears within a few weeks with no lasting effects. Children younger than 14 years and persons over age 50 seldom have jaundice or show other clinical signs of hepatitis.

Hepatitis E never becomes a chronic (long-lasting) illness, but on rare occasions the acute illness damages and destroys so many liver cells that the liver can no longer function. This is called fulminant liver failure, and may cause **death**. Pregnant women are at much higher risk of dying from fulminant liver failure; this increased risk is not true of any other type of viral hepatitis. The great majority of patients who recover from acute infection do not continue to carry HEV and cannot pass on the infection to others.

Diagnosis

HEV can be found by microscopically examining a stool sample, but this is not a reliable test, as the virus often dies when stored for a short time. Like other hepatitis viruses, HEV stimulates the body's immune system to produce a substance called an antibody, which can swallow up and destroy the virus. Blood tests can determine elevated antibody levels, which indicate the presence of HEV virus in the body. Unfortunately, such antibody blood tests are not widely available.

Treatment

There is no way of effectively treating the symptoms of any acute hepatitis, including hepatitis E. During acute infection, a patient should take a balanced diet and rest in bed as needed.

Prognosis

In the United States hepatitis E is not a fatal illness, but elsewhere about 1–2% of those infected

KEY TERMS

Antibody—A substance made by the body's immune system in response to an invading virus, the antibodies then attack and destroy the virus.

Incubation period—The time from when a person is first infected by a virus until the appearance of the earliest symptoms.

Jaundice—Yellowing of the skin that occurs when pigments normally eliminated by the liver collect in high amounts in the blood.

Sanitation—The process of keeping drinking water, foods, or any anything else with which people come into contact free of microorganisms such as viruses.

Vaccine—A substance prepared from a weakened or killed virus which, when injected, stimulates the immune system to produce antibodies that can prevent infection by the natural virus.

die of advanced liver failure. In pregnant women the death rate is as high as 20%. It is not clear whether having hepatitis E once guarantees against future HEV infection.

Prevention

Most attempts to use blood serum containing HEV antibody to prevent hepatitis in those exposed to HEV have failed. Hopefully, this approach can be made to work so that pregnant women living in endemic areas can be protected. No vaccine is available, though several are being tested. It also is possible that effective anti-viral drugs will be found. The best ways to prevent hepatitis E are to provide safe drinking water and take precautions to use sterilized water and beverages when traveling.

Resources

ORGANIZATIONS

American Liver Foundation. 1425 Pompton Ave., Cedar Grove, NJ 07009. (800) 223-0179. <<http://www.liverfoundation.org>>.

OTHER

King, J. W. *Bug Bytes*. Louisiana State University Medical Center. <<http://www.ccm.lsumc.edu/bugbytes>>.

David A. Cramer, MD

Hepatitis G

Definition

Hepatitis G is a newly discovered form of liver inflammation caused by hepatitis G virus (HGV), a distant relative of the **hepatitis C** virus.

Description

HGV, also called hepatitis GB virus, was first described early in 1996. Little is known about the frequency of HGV infection, the nature of the illness, or how to prevent it. What is known is that transfused blood containing HGV has caused some cases of hepatitis. For this reason, patients with **hemophilia** and other bleeding conditions who require large amounts of blood or blood products are at risk of hepatitis G. HGV has been identified in between 1–2% of blood donors in the United States. Also at risk are patients with **kidney disease** who have blood exchange by hemodialysis, and those who inject drugs into their veins. It is possible that an infected mother can pass on the virus to her newborn infant. Sexual transmission also is a possibility.

Often patients with hepatitis G are infected at the same time by the **hepatitis B** or C virus, or both. In about three of every thousand patients with acute viral hepatitis, HGV is the only virus present. There is some indication that patients with hepatitis G may continue to carry the virus in their blood for many years, and so might be a source of infection in others.

Causes and symptoms

Some researchers believe that there may be a group of GB viruses, rather than just one. Others remain doubtful that HGV actually causes illness. If it does, the type of acute or chronic (long-lasting) illness that results is not clear. When diagnosed, acute HGV infection has usually been mild and brief. There is no evidence of serious complications, but it is possible that, like other hepatitis viruses, HGV can cause severe liver damage resulting in liver failure. The virus has been identified in as many as 20% of patients with long-lasting viral hepatitis, some of whom also have hepatitis C.

Diagnosis

The only method of detecting HGV is a complex and costly DNA test that is not widely available.

KEY TERMS

Antibody—A substance made by the body's immune system in response to an invading virus; antibodies then attack and destroy the virus.

Hemophilia—A bleeding disorder that often makes it necessary to give patients dozens or even hundreds of units of blood and blood products over time.

Efforts are under way, however, to develop a test for the HGV antibody, which is formed in response to invasion by the virus. Once antibody is present, however, the virus itself generally has disappeared, making the test too late to be of use.

Treatment

There is no specific treatment for any form of acute hepatitis. Patients should rest in bed as needed, avoid alcohol, and be sure to eat a balanced diet.

Prognosis

What little is known about the course of hepatitis G suggests that illness is mild and does not last long. When more patients have been followed up after the acute phase, it will become clear whether HGV can cause severe liver damage.

Prevention

Since hepatitis G is a blood-borne infection, prevention relies on avoiding any possible contact with contaminated blood. Drug users should not share needles, syringes, or other equipment.

Resources

ORGANIZATIONS

American Liver Foundation. 1425 Pompton Ave.,
Cedar Grove, NJ 07009. (800) 223-0179.
<<http://www.liverfoundation.org>>.

David A. Cramer, MD

Hepatitis virus studies see **Hepatitis virus tests**

Hepatitis virus tests

Definition

Viral hepatitis is any type of liver inflammation caused by a viral infection. The three most common viruses now recognized to cause **liver disease** are hepatitis A, **hepatitis B**, and hepatitis non-A, non-B (also called hepatitis C). Several other types have been recognized: **hepatitis D**, **hepatitis E**, and the recently identified **hepatitis G**. A seventh type (hepatitis F) is suspected but not yet confirmed.

Purpose

The different types of viral hepatitis produce similar symptoms, but they differ in terms of transmission, course of treatment, prognosis, and carrier status. When the clinical history of a patient is insufficient for differentiation, hepatitis virus tests are used as an aid in diagnosis and in monitoring the course of the disease. These tests are based primarily on antigen-antibody reactions—an antigen being a protein foreign to the body, and an antibody another type of protein manufactured by lymphocytes (a type of white blood cell) to neutralize the antigen.

Description

There are five major types of viral hepatitis. The diseases, along with the antigen-antibody tests available to aid in diagnosis, are described below.

Hepatitis A

Commonly called infectious hepatitis, this is caused by the **hepatitis A virus (HAV)**. It is usually a mild disease, most often spread by food and water contamination, but sometimes through sexual contact. Immunologic tests are not commercially available for the HAV antigen, but two types of antibodies to HAV can be detected. IgM antibody (anti-HAV/IgM), appears approximately three to four weeks after exposure and returns to normal within several months. IgG (anti-HAV/IgG) appears approximately two weeks after the IgM begins to increase and remains positive. Acute hepatitis is suspected if IgM is elevated; conversely, if IgG is elevated without IgM, a convalescent stage of HAV is presumed. IgG antibody can remain detectable for decades after infection.

Hepatitis B

Commonly known as serum hepatitis, this is caused by the hepatitis B virus (HBV). The disease

can be mild or severe, and it can be acute (of limited duration) or chronic (ongoing). It is usually spread by sexual contact with another infected person, through contact with infected blood, by intravenous drug use, or from mother to child at birth.

HBV, also called the Dane particle, is composed of an inner protein core surrounded by an outer protein capsule. The outer capsule contains the hepatitis B surface antigen (HBsAg), formerly called the Australia antigen. The inner core contains HBV core antigen (HBcAg), and the hepatitis B e-antigen (HBeAg). Antibodies to these antigens are called anti-HBs, anti-HBc, and anti-HBe. Testing for these antigens and antibodies is as follows:

- **Hepatitis B surface antigen (HBsAg).** This is the first test for hepatitis B to become abnormal. HBsAg begins to elevate before the onset of clinical symptoms, peaks during the first week of symptoms, and usually disappears by the time the accompanying **jaundice** (yellowing of the skin and other tissues) begins to subside. HBsAg indicates an active HBV infection. A person is considered to be a carrier if this antigen persists in the blood for six or more months.
- **Hepatitis B surface antibody (anti-HBs).** This appears approximately one month after the disappearance of the HBsAg, signaling the end of the acute infection period. Anti-HBs is the antibody that demonstrates immunity after administration of the hepatitis B vaccine. Its presence also indicates immunity to subsequent infection.
- **Hepatitis B core antigen (HBcAg).** No tests are commercially available to detect this antigen.
- **Hepatitis B core antibody (anti-HBc).** This appears just before acute hepatitis develops and remains elevated (although it slowly declines) for years. It is also present in chronic hepatitis. The hepatitis B core antibody is elevated during the time lag between the disappearance of the hepatitis B surface antigen and the appearance of the hepatitis B surface antibody in an interval called the “window.” During this time, the hepatitis B core antibody is the only detectable marker of a recent hepatitis B infection.
- **Hepatitis B e-antigen (HBeAg).** This is more useful as an index of infection than for diagnostic purposes. The presence of this antigen correlates with early and active disease, as well as with high infectivity in patients with acute HBV infection. When HBeAg levels persist in the blood, the development of chronic HBV infection is suspected.
- **Hepatitis B e-antibody (anti-HBe).** In the bloodstream, this indicates a reduced risk of infectivity in

patients who have previously been HBeAg positive. Chronic hepatitis B surface antigen carriers can be positive for either HBeAg or anti-HBe, but are less infectious when anti-HBe is present. Antibody to e antigen can persist for years, but usually disappears earlier than anti-HBs or anti-HBc.

Hepatitis C

Previously known as non-A, non-B hepatitis, this disease is primarily caused by the hepatitis C virus (HCV). It is generally mild, but more likely than hepatitis B to lead to chronic liver disease, possible liver failure, and the eventual need for transplant. Chronic carrier states develop in more than 80% of patients, and chronic liver disease is a major problem. As many as 20% of patients with chronic **hepatitis C** will develop liver failure or liver cancer. HCV is spread through sexual contact, as well as through sharing drug needles, although nearly half of infections can't be traced as to origin.

Hepatitis C is detected by HCV serology (tests on blood sera). A specific type of assay called enzyme-linked immunosorbent assay (ELISA) was developed to detect antibody to hepatitis C for diagnostic purposes, as well as for screening blood donors. Most cases of post-transfusion non-A, non-B hepatitis are caused by HCV, but application of this test has virtually eliminated post-transfusion hepatitis. An HCV viral titer to detect HCV RNA in the blood is now available, and recently, IgM anti-HCV core is proving to be a useful acute marker for HCV infection.

Hepatitis D

Also called delta hepatitis, this is caused by the hepatitis D virus (HDV). The disease occurs only in those who have HBV in the blood from a past or simultaneously occurring infection. Experts believe transmission may occur through sexual contact, but further research is needed to confirm that. Most cases occur among those who are frequently exposed to blood and blood products. Many cases also occur among drug users who share contaminated needles. Hepatitis D virus (HDV) antigen can be detected by radioimmunoassay within a few days after infection, together with IgM and total antibodies to HDV.

Hepatitis E

Caused by the hepatitis E virus (HEV), this is actually another type of non-A, non-B hepatitis. The virus is most often spread through fecally contaminated water, but the role of person-to-person transmission is unclear. This form of hepatitis is quite

rare in the United States. There are currently no antigen or antibody tests widely available to accurately detect HEV.

Preparation

Hepatitis virus tests require a blood sample. It is not necessary for the patient to withhold food or fluids before any of these tests, unless requested to do so by the physician.

Risks

Risks for these tests are minimal for the patient, but may include slight bleeding from the blood-drawing site, **fainting** or feeling lightheaded after venipuncture, or hematoma (blood accumulating under the puncture site).

Normal results

Reference ranges for the antigen/antibody tests are as follows:

- hepatitis A antibody, IgM: Negative
- hepatitis B core antibody: Negative
- hepatitis B e antibody: Negative
- hepatitis B e-antigen: Negative
- hepatitis B surface antibody: Varies with clinical circumstance (Note: As the presence of anti-HBs indicates past infection with resolution of previous hepatitis B infection, or **vaccination** against hepatitis B, additional patient history may be necessary for diagnosis.)
- hepatitis B surface antigen: Negative
- hepatitis C serology: Negative
- hepatitis D serology: Negative.

Abnormal results

Hepatitis A: A single positive anti-HAV test may indicate previous exposure to the virus, but due to the antibody persisting so long in the bloodstream, only evidence of a rising anti-HAV titer confirms hepatitis A. Determining recent infection rests on identifying the antibody as IgM (associated with recent infection). A negative anti-HAV test rules out hepatitis A.

Hepatitis B: High levels of HBsAg that continue for three or more months after onset of acute infection suggest development of chronic hepatitis or carrier status. Detection of anti-HBs signals late convalescence

or recovery from infection. This antibody remains in the blood to provide immunity to reinfection.

Hepatitis C (non-A, non-B hepatitis): Anti-HBc develops after exposure to hepatitis B. As an early indicator of acute infection, antibody (IgM) to core antigen (anti-HBc IgM) is rarely detected in chronic infection, so it is useful in distinguishing acute from chronic infection, and hepatitis B from non-A, non-B.

Resources

BOOKS

Pagana, Kathleen Deska. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis: Mosby, Inc., 1998.

Janis O. Flores

Hepatobiliary scan see **Gallbladder nuclear medicine scan**

Hepatocellular carcinoma see **Liver cancer, primary**

Hepatolenticular degeneration see **Wilson's disease**

Hepatoma see **Liver cancer, primary**

Herbal medicine see **Herbalism, western**

Herbalism, traditional Chinese

Definition

Chinese herbalism is one of the major components of traditional Chinese medicine (TCM), or Oriental medicine (OM). In TCM, herbs are often used in conjunction with other techniques, such as **acupuncture** or massage. Chinese herbalism is a holistic medical system, meaning that it looks at treating a patient as a whole person, looking at the mental and spiritual health, as well as the physical health, of the individual. Illness is seen as a disharmony or imbalance among these aspects of the individual. Chinese herbalism has been practiced for over 4,000 years.

One of the earliest and certainly the most important Chinese herbal text is the *Huang Ti Nei Ching*, or *Yellow Emperor's Classic of Internal Medicine*. It is believed to be authored by Huang Ti during his reign over China, which started about 2697 B.C. Since that time, herbal practices have been more extensively documented and refined. In modern China, traditional Chinese herbalism is taught alongside conventional

Western pharmacology. Chinese herbal remedies have been used in the West only relatively recently, over the past two decades. These remedies are more gentle and natural than conventional medicines. In addition, they have fewer unpleasant side effects. Individuals with chronic disorders in particular are increasingly drawn to the holistic aspect of Chinese herbalism and TCM in general.

Purpose

Because it is a safe, inexpensive solution to health problems of all kinds, Chinese herbalism is very popular in China. In recent years, herbalism has been modernized with the introduction of quality control. For example, herbs are subjected to absorption spectrometry to determine levels of heavy metals found in some. Because they are standardized, Chinese herbs are safer for self-treatment. This puts the individual, not the physician, in charge of the individual's health; that is a basic goal of Chinese herbalism.

Chinese herbalism offers unique advice regarding what foods can help and what can hinder, and an herbalist can help an individual discover what he is allergic to. In addition, Chinese herbs stimulate the immune system and provide beneficial nutrients, aside from their role in curing illness.

At M.D. Anderson Hospital in Texas, medical research has confirmed that patients undergoing **chemotherapy** were shown to have an improved degree of immune function when they took the tonic herb astragalus (*huang qi*). (It is well known that chemotherapy suppresses the immune system.) Research also showed that T-cell and macrophage activity and interferon production was increased in patients using the Chinese herbs ganoderma, lentinus, and polyporous, helping the body fight cancer cells. Agents also found in ganoderma were found to inhibit platelet aggregation and thrombocyte formation, which would be helpful to counter circulation and heart problems.

An ingredient of ginseng was found to promote adrenal function, which would give the herb properties of enhancing many hormone functions in the body.

Description

Origins

HISTORICAL BACKGROUND. **Traditional Chinese medicine** originated in the region of eastern Asia that today includes China, Tibet, Vietnam, Korea, and Japan. Tribal shamans and holy men who lived as

Five Popular Chinese Herbs Used In The U.S.

Herb	Purpose
Astragalus (huang chi)	Builds immune system; offsets side effects of chemotherapy and radiation treatments
Don Quai (dang qui)	Stimulates the production of red blood cells and bone marrow; increases cardiovascular endurance; regulates menstrual disorders
Ginseng (ren shen)	Increases physical stamina; general tonic
Reishi mushroom (ling zhi)	Eliminates toxins; increases physical stamina
Schisandra (wu wei zu)	Prevents fluid loss, e.g., excessive sweating, runny nose, incontinence

hermits in the mountains of China as early as 3500 B.C. practiced what was called the “Way of Long Life.” This regimen included a diet based on herbs and other plants; kung-fu exercises; and special breathing techniques that were thought to improve vitality and life expectancy.

After the Han dynasty, the next great age of Chinese medicine was under the Tang emperors, who ruled from A.D. 608 to 906. The first Tang emperor established China’s first medical school in A.D. 629. Under the Song (A.D. 960–1279) and Ming (A.D. 1368–1644) dynasties, new medical schools were established, their curricula and qualifying examinations were standardized, and the traditional herbal prescriptions were written down and collected into encyclopedias. One important difference between the development of medicine in China and in the West is the greater interest in the West in surgical procedures and techniques.

PHILOSOPHICAL BACKGROUND: THE COSMIC AND NATURAL ORDER. In Taoist thought, the Tao, or universal first principle, generated a duality of opposing principles that underlie all the patterns of nature. These principles, yin and yang, are mutually dependent as well as polar opposites. They are basic concepts in traditional Chinese medicine. Yin represents everything that is cold, moist, dim, passive, slow, heavy, and moving downward or inward; while yang represents heat, dryness, brightness, activity, rapidity, lightness, and upward or outward motion. Both forces are equally necessary in nature and in human well-being, and neither force can exist without the other. The dynamic interaction of these two principles is reflected in the cycles of the seasons, the human life cycle, and other natural phenomena. One objective of traditional Chinese medicine is to keep yin and yang in harmonious balance within a person.

In addition to yin and yang, Taoist teachers also believed that the Tao produced a third force, primordial energy or qi (also spelled chi or ki). The interplay between yin, yang, and qi gave rise to the Five Elements of water, metal, earth, wood, and fire. These entities are all reflected in the structure and functioning of the human body.

THE HUMAN BEING. Traditional Chinese physicians did not learn about the structures of the human body from dissection because they thought that cutting open a body insulted the person’s ancestors. Instead they built up an understanding of the location and functions of the major organs over centuries of observation, and then correlated them with the principles of yin, yang, qi, and the Five Elements. Thus wood is related to the liver (yin) and the gall bladder (yang); fire to the heart (yin) and the small intestine (yang); earth to the spleen (yin) and the stomach (yang); metal to the lungs (yin) and the large intestine (yang); and water to the kidneys (yin) and the bladder (yang). The Chinese also believed that the body contains Five Essential Substances, which include blood, spirit, vital essence (a principle of growth and development produced by the body from qi and blood), Fluids (all body fluids other than blood, such as saliva, spinal fluid, sweat, etc.), and qi.

Chinese herbal treatment differs from **Western herbalism** in several respects. In Chinese practice, several different herbs may be used, according to each plant’s effect on the individual’s Qi and the Five Elements. There are many formulas used within traditional Chinese medicine to treat certain common imbalance patterns. These formulas can be modified to fit specific individuals more closely.

A traditional Chinese herbal formula typically contains four classes of ingredients, arranged in a hierarchical order: a chief (the principal ingredient, chosen for the patient’s specific illness); a deputy (to reinforce the chief’s action or treat a coexisting condition); an assistant (to counteract side effects of the first two ingredients); and an envoy (to harmonize all the other ingredients and convey them to the parts of the body that they are to treat).

Methods of diagnosis

A Chinese herbalist will not prescribe a particular herb on the strength of symptoms only, but will take into consideration the physical condition, emotional health, and mental state of the patient. He or she may look at the condition of the patient’s hair, skin, and tongue, as well as the appearance of the eyes, lips, and general complexion. The practitioner then listens to

the sounds the body makes when breathing. He or she may smell the breath, body odor, or sputum in diagnosis.

TCM practitioners take an extensive medical history of a patient. He or she may ask about dietary habits, lifestyle, and sleep patterns. The patient will be questioned about chief medical complaints, as well as on his or her particular emotional state and sexual practices.

Chinese herbalists employ touch as a diagnostic tool. They may palpate the body or use light massage to assess the patient's physical health. Another chief component of Chinese medical diagnosis is pulse diagnosis, or sphygmology. This is a very refined art that takes practitioners years to master. Some practitioners can detect 12 different pulse points that correspond to the 12 major organs in Chinese medicine. There are over 30 pulse qualities that practitioners are able to detect on each point. The strength, speed, quality, and rhythm of the pulse, to name a few, will be determined before a diagnosis is given.

Herbs

Chinese herbs may be used alone or in combination. Relatively few are used alone for medicinal purposes. Practitioners believe that illness can be effectively treated by combining herbs based on their various characteristics and the patient's overall health. Every herb has four basic healing properties: nature, taste, affinity, and effect.

An herb's nature is described according to its yin or yang characteristics. Yang, or warming, herbs treat cold deficiencies. They are frequently used in the treatment of the upper respiratory tract, skin, or extremities. Yin, or cooling, herbs, treat hot excess conditions. They are most often used to treat internal conditions and problems with organs. Herbs can also be neutral in nature.

An herb's taste does not refer to its flavor, but to its effect on qi, blood, fluids, and phlegm. Sour herbs have a concentrating action. They are prescribed to treat bodily excess conditions, such as **diarrhea**, and concentrate qi. Bitter herbs have an eliminating or moving downward action. They are used to treat coughs, **constipation**, and heart problems. Sweet or bland herbs have a harmonizing action. They are used as restorative herbs and to treat **pain**. Spicy herbs have a stimulating action. They are prescribed to improve blood and qi circulation. Salty herbs have a softening action. They are used to treat constipation and other digestion problems.

An herb's affinity describes its action on a specific bodily Organ. (Note that Chinese medicine does not have the anatomical correlation for organ names. They correspond more closely to the organ's function.) Sour herbs have an affinity for the Liver and Gallbladder. Bitter herbs act on the Heart and Small Intestine. Sweet and bland herbs affect the Stomach and Spleen. Spicy herbs have an affinity for Lungs and Large Intestine, whereas salty herbs act on the Kidneys and Bladder.

Chinese herbs are lastly classified according to their specific actions, which are divided into four effects. Herbs that dispel are used to treat an accumulation, sluggishness, or spasm by relaxing or redistributing. Herbs with an astringent action are used to consolidate or restrain a condition characterized by discharge or excessive elimination. Herbs that purge treat an obstruction or "poison" by encouraging elimination and **detoxification**. Tonifying herbs nourish, support, and calm where there is a deficiency.

Treatment of diabetes

The incidence of diabetes has increased quite dramatically in recent years, especially in the United States, where in general people take less **exercise**, and food is taken in greater quantity with a general reduction in quality. This has led to a scramble to find new solutions to the problem, and many researchers have focused their interest on Chinese herbal remedies. In the search for more effective and more convenient treatments, the alkaloid berberine has come under close scrutiny for its many uses, among them the treatment of diabetes. In trials, rats given a mixture of berberine and alloxan showed less likelihood of incurring a rise in blood sugar. Patients suffering from type II diabetes who were given between 300 and 600 mg of berberine daily for between one and three months, showed a reduction in blood sugar levels, when taken in conjunction with a controlled diet.

Treatment of AIDS and cancer

Independent researchers are investigating indications that Chinese herbalism can reduce the toxicity of chemotherapy and other medications, in addition to stimulating immune responses.

Preparations

Those who are unfamiliar with Chinese herbs and their uses should consult a practitioner before starting any treatment. Once a remedy is prescribed, it may be found at Oriental markets or health food stores. The

KEY TERMS

Absorption spectrometry—A scientific procedure to determine chemical makeup of samples.

Interferon—A substance proved to be necessary in the body to help fight cancer cells.

Immune function—The body's defense system against bacteria, viruses and fungi, and any malfunction of the organism.

Pharmaco-dynamics—The study of the relationships and interactions of herbs.

Platelet aggregation—The clumping together of blood cells, possibly forming a clot.

Thrombocyte—Another name for platelet.

remedies used in Chinese herbalism are standardized and sold prepared for use, with instructions for dosage. A Chinese herbalist may prescribe herbs to be made into tea, or taken as capsules.

Precautions

When treating a patient, the herbalist will aim to gently “nudge” the system into shape, rather than producing any immediate reaction. A return to health, therefore, may take time, and it is important that the patient realizes the principle of the treatment. Some practitioners estimate that treatment will take a month for every year that a chronic condition has existed. The advantage of the slow pace is that if there is a bad reaction to any herb, which is rare, it will be mild because the treatment itself is gentle.

As with most naturopathic therapies, Chinese herbal remedies work best when taken in conjunction with a healthy lifestyle and program of exercise.

Side effects

Some Chinese herbs are incompatible with certain prescription drugs, certain foods, or should not be taken during **pregnancy**. To be certain, a Chinese herbalist should be consulted.

Research and general acceptance

At present, there is renewed interest in the West in traditional Chinese medicine and Chinese herbalism. Of the 700 herbal remedies used by traditional Chinese practitioners, over 100 have been tested and found effective by the standards of Western science. Several

United States agencies, including the National Institutes of Health, the Office of Alternative Medicine, and the Food and Drug Administration are currently investigating Chinese herbal medicine as well as acupuncture and *Tui na* massage. In general, however, Western studies of Chinese medicine focus on the effects of traditional treatments and the reasons for those effects, thus attempting to fit traditional Chinese medicine within the Western framework of precise physical measurements and scientific hypotheses.

Resources

ORGANIZATIONS

The California Association of Acupuncture and Oriental Medicine. <<http://www.CAAOM.ORG/medicine/overview.htm>>.

For help with herbs and a list of practitioners. <<http://www.craneherb.com>>.

Institute of Chinese Materia Medica. China Academy of Traditional Chinese Medicine. *Beijing, 100700*.

National Center for Complementary and Alternative Medicine. <<http://nccam.nih.gov/nccam>>.

Patricia Skinner

Herbalism, Western

Definition

Western herbalism is a form of the healing arts that draws from herbal traditions of Europe and the Americas, and that emphasizes the study and use of European and Native American herbs in the treatment and prevention of illness. Western herbalism is based on physicians' and herbalists' clinical experience and traditional knowledge of medicinal plant remedies preserved by oral tradition and in written records over thousands of years. Western herbalism, like the much older system of **traditional Chinese medicine**, relies on the synergistic and curative properties of the plant to treat symptoms and disease and maintain health.

Western herbalism is based upon pharmacognosy, the study of natural products. Pharmacognosy includes the identification, extraction methods, and applications of specific plant constituents responsible for specific therapeutic actions, such as the use of digoxin from *Digitalis* leaf for **heart failure**. These constituents are extracted, purified and studied in vitro, in vivo, and in clinical research. They may be

concentrated to deliver standardized, set doses. Sometimes, the natural constituent can be synthesized in the lab, or changed and patented. Practitioners may choose to use fresh medicinal plants, simple extracts, or standardized extracts.

In standardized extracts, a specific quantity of a constituent is called a marker compound, and it may or may not be the active constituent(s) in the plant medicine. There are preparations with standardized active constituent quantities, and preparations with greater emphasis on quality of crude plant material and traditional preparation methodology than on finalized total quantity of marker compounds. The preference between the two for precision dosing is philosophical, practical and variable. When using plant extracts in which the active constituents and their cofactors are well established, or the therapeutic and lethal dose are close, standardized products are often preferred. When using plant extracts whose active constituents remain obscure, or the active constituents when purified produce weaker therapeutic results or more undesirable side effects, the products produced under good manufacturing processes and according to the traditional *National Formulary U. S. Dispensatory* or *U. S. Pharmacopeia* are preferred.

Purpose

The benefits of botanical medicine may be subtle or dramatic, depending on the remedy used and the symptom or problem being addressed. Herbal remedies usually have a much slower effect than pharmaceutical drugs. Some herbal remedies have a cumulative effect and work slowly over time to restore balance, and others are indicated for short-term treatment of acute symptoms. When compared to the pharmaceutical drugs, herbal remedies prepared from the whole plant have relatively few side effects. This is due to the complex chemistry and synergistic action of the full range of phytochemicals present in the whole plant, and the relatively lower concentrations. They are generally safe when used in properly designated therapeutic dosages, and less costly than the isolated chemicals or synthetic prescription drugs available from western pharmaceutical corporations.

Description

Origins

Over 2,500 years ago Hippocrates wrote, "In medicine one must pay attention not to plausible theorizing but to experience and reason together." This Greek physician and herbalist from the fourth

century B.C. is considered the father of western medicine. He stressed the importance of diet, water quality, climate, and social environment in the development of disease. Hippocrates believed in treating the whole person, rather than merely isolating and treating symptoms. He recognized the innate capacity of the body to heal itself, and emphasized the importance of keen observation in the medical practice. He recommended simple herbal remedies to assist the body in restoring health.

Ancient Greek medicine around the fifth century B.C. was a fertile ground for contrasting philosophies and religions. Greek physicians were influenced by the accumulated medical knowledge from Egypt, Persia, and Babylon. Medical advances flourished and practitioners and scholars were free to study and practice without religious and secular constraints. In the fourth century B.C., Theophrastus wrote the *Historia Plantarum*, considered to be the founding text in the science of botany.

During the first century A.D. Dioscorides, a Greek physician who traveled with the Roman legions, produced five medical texts. His herbal text, known as the *De Materia Medica* is considered to be among the most influential of all western herbal texts. It became a standard reference for practitioners for the next 1,500 years. This influential book also included information on medicinal herbs and treatments that had been used for centuries in Indian **Ayurvedic medicine**. Galen of Pergamon, who also lived in the first century A.D., was a Roman physician and student of anatomy and physiology. He authored a recipe book containing 130 antidotes and medicinal preparations. These elaborate mixtures, known as galenicals, sometimes included up to one hundred herbs and other substances. This complex approach to herbal medicine was a dramatic change from the simple remedies recommended by Hippocrates and employed by traditional folk healers. Galen developed a rigid system of medicine in which the physician, with his specialized knowledge of complex medical formulas, was considered the ultimate authority in matters of health care. The Galenic system, relying on theory and scholarship rather than observation, persisted throughout the Middle Ages. The galenical compounds, along with bloodletting, and purging, were among the drastic techniques practiced by the medical professionals during those times; however, traditional herbal healers persisted outside the mainstream medical system.

During the eighth century a medical school was established in Salerno, Italy, where the herbal knowledge accumulated by Arab physicians was preserved. The Arabian Muslims conducted extensive

research on medicinal herbs found in Europe, Persia, India, and the Far East. Arab businessmen opened the first herbal pharmacies early in the ninth century. The *Leech Book of Bald*, the work of a Christian monk, was compiled in the tenth century. It preserved important medical writings that had survived from the work of physicians in ancient Greece and Rome.

The Middle Ages in Europe were a time of widespread **death** by plagues and pestilence. The **Black Plague** of 1348, particularly, and other health catastrophes in later years, claimed so many lives that survivors began to lose faith in the dominant Galenic medical system. Fortunately, the knowledge of traditional herbal medicine had not been lost. Medieval monks who cultivated extensive medicinal gardens on the monastery grounds, also patiently copied the ancient herbal and medical texts. Folk medicine as practiced in Europe by traditional healers persisted, even though many women herbalists were persecuted as witches and enemies of the Catholic church and their herbal arts were suppressed.

The growing spice trade and explorations to the New World introduced exotic plants, and a whole new realm of botanical medicines became available to Europeans. Following the invention of the printing press in the fifteenth century, a large number of herbal texts, also simply called herbals, became available for popular use. Among them were the beautifully illustrated works of the German botanists Otto Brunfels and Leonhard Fuchs published in 1530, and the Dutch herbal of Belgian physician Rembert Dodoens, a popular work that was later reproduced in English. In 1597, the physician and gardener John Gerard published one of the most famous of the English herbals, still in print today. Gerard's herbal, known as *The Herball or General Historie of Plantes* was not an original work. Much of the content was taken from the translated text of his Belgian predecessor Dodoens. Gerard did, however, include descriptions of some of the more than one thousand species of rare and exotic plants and English flora from his own garden.

The correspondence of astrology with herbs was taught by Arab physicians who regarded astrology as a science helpful in the selection of medicines and in the treatment of diseases. This approach to western herbalism was particularly evident in the herbal texts published in the sixteenth and seventeenth centuries. One of the most popular and controversial English herbals is *The English Physician Enlarged* published in 1653. The author, Nicholas Culpeper, was an apothecary by trade. He also published a translation of the Latin language *London Pharmacopoeia* into

English. Culpeper was a nonconformist in loyalist England, and was determined to make medical knowledge more accessible to the apothecaries, the tradesmen who prescribed most of the herbal remedies. Culpeper's herbal was criticized by the medical establishment for its mix of magic and astrology with botanical medicine, but it became one of the most popular compendiums of botanical medicine of its day. Culpeper also accepted the so-called "Doctrine of Signatures," practiced by medieval monks in their medicinal gardens. This theory teaches that the appearance of plants is the clue to their curative powers. Plants were chosen for treatment of particular medical conditions based on their associations with the four natural elements and with a planet or sign. The place where the plant grows, its dominant physical feature, and the smell and taste of an herb determined the plant's signature. Culpeper's herbal is still in print in facsimile copies, and some pharmacognosists and herbalists in the twenty-first century voice the same criticisms that Culpeper's early critics did.

European colonists brought their herbal knowledge and plant specimens to settlements in North America where they learned from the indigenous Americans how to make use of numerous nutritive and medicinal plants, native to the New World. Many European medicinal plants escaped cultivation from the early settlements and have become naturalized throughout North America. The first record of Native American herbalism is found in the manuscript of the native Mexican Indian physician, Juan Badianus published in 1552. The American Folk tradition of herbalism developed as a blend of traditional European medicine and Native American herbalism. The pioneer necessity for self-reliance contributed to the perseverance of folk medicine well into the twentieth century.

In Europe in the seventeenth century, the alchemist Paracelsus changed the direction of western medicine with the introduction of chemical and mineral medicines. He was the son of a Swiss chemist and physician. Paracelsus began to apply chemicals, such as arsenic, mercury, sulfur, iron, and copper sulfate to treat disease. His chemical approach to the treatment of disease was a forerunner to the reliance in the twentieth century on chemical medicine as the orthodox treatment prescribed in mainstream medical practice.

The nineteenth and twentieth centuries brought a renewed interest in the practice of western herbalism and the development of natural therapies and health care systems that ran counter to the mainstream methods of combating disease symptoms with synthetic pharmaceuticals.



A selection of Western herbal medical equipment and traditional herbs, including foxglove (upper right), ginger (center right), and periwinkle (lower left). (Photo Researchers, Inc. Reproduced by permission.)

In the late eighteenth century, the German physician **Samuel Hahnemann** developed a system of medicine known as homeopathy. This approach to healing embraces the philosophy of “like cures like.” Homeopathy uses extremely diluted solutions of herbs, animal products, and chemicals that are believed to hold a “trace memory” or energetic imprint of the substance used. Homeopathic remedies are used to amplify the patient’s symptoms with remedies that would act to produce the same symptom in a healthy person. Homeopathy holds that the symptoms of illness are evidence of the body’s natural process of healing and eliminating the cause of the disease.

In 1895, the European medical system known as Naturopathy was introduced to the North America. Like homeopathy, this medical approach is based on the Hippocratic idea of eliminating disease by assisting the body’s natural healing abilities. The naturopath uses nontoxic methods to assist the body’s natural healing processes, including **nutritional supplements**, herbal remedies, proper diet, and **exercise** to restore health.

Western herbalism is regaining popularity at a time when the world is assaulted by the **stress** of overpopulation and development that threatens the natural biodiversity necessary for these valuable medicinal plants to survive. The American herb market is growing rapidly and increasing numbers of individuals are choosing alternative therapies over the mainstream allopathic western medicine. It is projected that by the year 2002 consumers will spend more than seven billion dollars a year on herbal products.

An estimated 2,400 acres of native plant habitat are lost to development every day. As much as 29% of all plant life in North America is in danger of extinction, including some of the most important native medicinal plants, according to the 1997 World Conservation Union Red List of Threatened Plants.

Though research into the efficacy and safety of traditional herbal remedies is increasing, it has been limited by the high costs of clinical studies and laboratory research, and by the fact that whole plants and their constituents are not generally patentable (therefore, there is no drug profit after market introduction). Outside the United States, herbalism has successfully combined with conventional medicine, and in some countries is fully integrated into the nations’ health care systems. At the beginning of the twenty-first century, 80% of the world’s population continues to rely on herbal treatments. The World Health Organization, an agency of the United Nations, promotes traditional herbal medicine for treatment of many local health problems, particularly in the third world where it is affordable and already well-integrated into the cultural fabric.

In the United States, the re-emergence in interest in holistic approaches to health care is evident. Citizens are demanding access to effective, safe, low-cost, natural medicine. Legislative and societal change is needed, however, before natural therapies can be fully integrated into the orthodox allopathic health care system and provide citizens with a wide range of choices for treatment. If the current trend continues, U. S. citizens will benefit from a choice among a variety of safe and effective medical treatments.

Herbs are generally defined as any plant or plant part that may be used for medicinal, nutritional, culinary, or other beneficial purposes. The active constituents of plants (if known) may be found in varying amounts in the root, stem, leaf, flower, and fruit, etc. of the plant. Herbs may be classified into many different categories. Some western herbalists categorize herbal remedies according to their strength, action, and characteristics. Categories may include sedatives, stimulants, **laxatives**, febrifuges (to reduce **fever**), and many others. One system of classification is based on a principle in traditional Chinese medicine that categorizes herbs into four classes: tonics, specifics, heroics, or cleansers and protectors. Within these broad classifications are the numerous medicinal actions of the whole herb which may be due to a specific chemical or combination of chemicals in the plant.

- **Tonics.** Herbs in this classification are also known as alteratives in western herbalism. They are

generally mild in their action and act slowly in the body, providing gentle stimulation and **nutrition** to specific organs and systems. Tonic herbs act over time to strengthen and nourish the whole body. These herbs are generally safe and may be used regularly, even in large quantities. These tonic herbs are known as “superior” remedies in traditional Chinese medicine. The therapeutic dose of tonic remedies is far removed from the possible toxic dose. American ginseng is an example of a tonic herb.

- **Specifics.** Herbs in this classification are strong and specific in their therapeutic action. They are generally used for short periods of time in smaller dosages to treat acute conditions. Herbs classified as specifics are not used beyond the therapeutic treatment period. **Echinacea** is a specific herb.
- **Heroic.** These herbs offer high potency but are potentially toxic, and should not be used in self-treatment. Because the therapeutic dosage may be close to the lethal dosage, these herbs are presented cautiously and closely monitored or avoided by trained clinicians. They should not be used continuously or without expert supervision. Poke (*Phytolacca americana*) is an example of a heroic remedy.
- **Cleansers and protectors.** These herbs, plants, and plant tissues remove wastes and pollutants, while minimally affecting regular body processes. An example of a cleanser is pectin. Pectins are the water soluble substances that bind cell walls in plant tissues, and some believe that they help remove heavy metals and environmental toxins from the body.

Preparations

Herbal preparations are commercially available in a variety of forms including tablets or capsules, tinctures, teas, fluid extracts, douches, washes, suppositories, dried herbs, and many other forms. The medicinal properties of herbs are extracted from the fresh or dried plant parts by the use of solvents appropriate to the particular herb. Alcohol, oil, water, vinegar, glycerin, and propylene glycol are some of the solvents used to extract and concentrate the medicinal properties. Steam distillation and cold-pressing techniques are used to extract the essential oils. The quality of any herbal remedy and the potency of the phytochemicals found in the herb depends greatly on the conditions of weather and soil where the herb was grown, the timing and care in harvesting, and the manner of preparation and storage.

KEY TERMS

In vitro—A biological reaction occurring in a laboratory apparatus.

In vivo—Occurring in a living organism.

Phyto-, as in phytochemical, phytomedicinal, and phytotherapy—Meaning, or pertaining to, a plant or plants.

Wildcrafting—Gathering of herbs or other natural materials.

Precautions

Herbal remedies prepared by infusion, decoction, or alcohol tincture from the appropriate plant part, such as the leaf, root, or flower are generally safe when ingested in properly designated therapeutic dosages. However, many herbs have specific contraindications for use when certain medical conditions are present. Not all herbal remedies may be safely administered to infants or small children. Many herbs are not safe for use by pregnant or lactating women. Some herbs are toxic, even deadly, in large amounts, and there is little research on the chronic toxicity that may result from prolonged use. Herbal remedies are sold in the United States as dietary supplements and are not regulated for content or efficacy. Self-diagnosis and treatment with botanical medicinals may be risky. A consultation with a clinical herbalist, Naturopathic physician, or certified clinical herbalist is prudent before undertaking a course of treatment.

Essential oils are highly concentrated and should not be ingested as a general rule. They should also be diluted in water or in a non-toxic carrier oil before application to the skin to prevent **contact dermatitis** or photo-sensitization. The toxicity of the concentrated essential oil varies depending on the chemical constituents of the herb.

The American Professor of Pharmacognosy, Varro E. Tyler, believes that “herbal chaos” prevails in the United States with regard to herbs and phytomedicinals. In part he blames the herb producers and marketers of crude herbs and remedies for what he terms unproven hyperbolic, poor quality control, deceptive labeling, resistance to standardization of dosage forms, and continued sale of herbs determined to be harmful.

Side effects

Herbs have a variety of complex phytochemicals that act on the body as a whole or on specific organs

and systems. Some of these chemical constituents are mild and safe, even in large doses. Other herbs contain chemicals that act more strongly and may be toxic in large doses or when taken continuously. **Drug interactions** are possible with certain herbs when combined with certain pharmaceutical drugs. Some herbs are tonic in a small amount and toxic in larger dosages.

Research and general acceptance

Western herbalism is experiencing a revival of popular and professional interest. The number of training schools and qualified herbal practitioners is growing to meet the demand. Western herbalism is incorporated into the medical practice of licensed Naturopathic doctors, who receive special training in clinical herbalism. Folk herbalists, heir to the continuing oral traditions passed from generation to generation in many rural areas, as well as amateur, self-taught herbalists, keep the practice of botanical medicine alive at the grassroots level. Traditional western herbalism relies on traditional use and materia medica, folk wisdom, and recent clinical research and advances in the extraction processes. These advances provide increased quality control on the concentration and potency of the active ingredients. Western physicians, educated in allopathic medicine, typically receive no training in the use of herbs. These doctors rely on pharmaceutical drugs for their patients, and some cite the following reasons for continuing to do so: lack of standardized dosages, lack of quality control in the preparation of herbal medicinals, and the dearth of clinical research verifying the safety and effectiveness of many traditional herbal remedies.

Herbalism is widely practiced throughout Europe, particularly in England, France, Italy, and Germany where phytomedicinals are available in prescription form and as over-the-counter remedies. In Germany, plant medicines are regulated by a special government body known as the Commission E. In the United States, however, despite increasing popularity, traditional herbalism is not integrated into the allopathic medical system. Phytomedicinals are sold as dietary supplements rather than being adequately researched and recognized as safe and effective drugs. The Dietary Supplement Health and Education Act of 1994 circumvented a U. S. Food and Drug Administration (FDA) effort to effectively remove botanicals from the marketplace and implement regulations restricting sale. Massive popular outcry against the proposed regulations on the sale of herbs and phytomedicinals resulted in this Congressional action. In 2000, U.S. President Bill Clinton, by executive order, created the White House Commission on Alternative Medicine in an effort to hold alternative

medicine therapies “to the same standard of scientific rigor as more traditional health care interventions.” That Commission is charged with recommending federal guidelines and legislation regarding the use of alternative medical therapies in the twenty-first century.

Resources

PERIODICALS

Deneen, Sally, with Rembert, Tracey. “Uprooted, The Worldwide Plant Crisis Is Accelerating.” *E Magazine* July-August 1999: 36-40.

Liebmann, Richard, N.D. “United Plant Savers—Planting the Future.” *PanGaia* 22 (Winter 1999-2000): 23- 26.

ORGANIZATIONS

National Center for the Preservation of Medicinal Herbs.
3350 Beech Grove Road, Rutland, Ohio 45775.
(740)742- 4401.

United Plant Savers. P.O. Box 98, East Barre, Vermont 05649. (802) 479-9825. < <http://www.plantsavers.org> >.

OTHER

Hobbs, Christopher. “Specific and Tonic Immune Herbs: Exploring a Practical System of Western Herbalism.” *Health World*. < <http://www.healthy.net> >.

Oracle Tree New Age Mall. “Western Medical Astrology: A Brief History.” < <http://www.oracletree.com/avalonphysics/wesmedas.html> >.

Wicke, Roger, Ph.D. “A World History of Herbology and Herbalism: Oppressed Arts.” Rocky Mountain Herbal Institute. < <http://www.rmhiherbalorg/a/f.ahrl.hist.html> >.

Clare Hanrahan

Herbs see **Echinacea; Ginkgo biloba; Ginseng; Saw palmetto; St. John's wort**

Hereditary cerebral hemorrhage with amyloidosis see **Cerebral amyloid angiopathy**

Hereditary chorea see **Huntington's disease**

Hereditary fructose intolerance

Definition

Hereditary fructose intolerance is an inherited condition where the body does not produce the chemical needed to break down fructose (fruit sugar).

Description

Fructose is a sugar found naturally in fruits, vegetables, honey, and table sugar. Fructose intolerance is a disorder caused by the body's inability to produce an enzyme called aldolase B (also called fructose 1-phosphate aldolase) that is necessary for absorption of fructose. The undigested fructose collects in the liver and kidneys, eventually causing liver and kidney failure. One person in about 20,000 is born with this disorder. It is reported more frequently in the United States and Northern European countries than in other parts of the world. It occurs with equal frequency in males and females.

Causes and symptoms

Fructose intolerance is an inherited disorder passed on to children through their parents' genes. Both the mother and father have the gene that causes the condition, but may not have symptoms of fructose intolerance themselves. (This is called an autosomal recessive pattern of inheritance.) The disorder will not be apparent until the infant is fed formula, juice, fruits, or baby foods that contain fructose. Initial symptoms include **vomiting**, **dehydration**, and unexplained **fever**. Other symptoms include extreme thirst and excessive urination and sweating. There will also be a loss of appetite and a failure to grow. **Tremors** and seizures caused by low blood sugar can occur. The liver becomes swollen and the patient becomes jaundiced with yellowing of the eyes and skin. Left untreated, this condition can lead to **coma** and **death**.

Diagnosis

Urine tests can be used to detect fructose sugar in the urine. Blood tests can also be used to detect *hyperbilirubinemia* and high levels of liver enzymes in the blood. A **liver biopsy** may be performed to test for levels of enzymes present and to evaluate the extent of damage to the liver. A fructose-loading test where a dose of fructose is given to the patient in a well-controlled hospital or clinical setting may also be used to confirm fructose intolerance. Both the biopsy and the loading test can be very risky, particularly in infants that are already sick.

Treatment

Once diagnosed, fructose intolerance can be successfully treated by eliminating fructose from the diet. Patients usually respond within three to four weeks and can make a complete recovery if fructose-containing foods are avoided. Early recognition and

KEY TERMS

Aldolase B—Also called fructose 1-phosphate aldolase, this chemical is produced in the liver, kidneys, and brain. It is needed for the breakdown of fructose, a sugar found in fruits, vegetables, honey, and other sweeteners.

Hyperbilirubinemia—A condition where there is a high level of bilirubin in the blood. Bilirubin is a natural by-product of the breakdown of red blood cells, however, a high level of bilirubin may indicate a problem with the liver.

Liver biopsy—A surgical procedure where a small piece of the liver is cut out for examination. A needle or narrow tube may be inserted either directly through the skin and muscle or through a small incision and passed into the liver for collection of a sample of liver tissue.

treatment of the disease is important to avoid damage to the liver, kidneys, and small intestine.

Prognosis

If the condition is not recognized and the diet is not well controlled, death can occur in infants or young children. With a well-controlled diet, the child can develop normally.

Prevention

Carriers of the gene for hereditary fructose intolerance can be identified through DNA analysis. Anyone who is known to carry the disease or who has the disease in his or her family can benefit from **genetic counseling**. Since this is a hereditary disorder, there is currently no known way to prevent it other than assisting at-risk individuals with family planning and reproductive decisions.

Resources

ORGANIZATIONS

National Institutes of Health. National Institute of Diabetes, Digestive and Kidney Diseases. Building 31, Room 9A04, 31 Center Drive, Bethesda, MD 20892-2560. (301) 496-3583.

OTHER

"What Is Hereditary Fructose Intolerance?" Hereditary Fructose Intolerance & Aldolase Homepage.
< <http://www.bu.edu/aldolase> >.

Altha Roberts Edgren

Hereditary hemorrhagic telangiectasia

Definition

Hereditary hemorrhagic telangiectasia is an inherited condition characterized by abnormal blood vessels which are delicate and prone to bleeding. Hereditary hemorrhagic telangiectasia is also known as Rendu-Osler-Weber disease.

Description

The term telangiectasia refers to a spot formed, usually on the skin, by a dilated capillary or terminal artery. In hereditary hemorrhagic telangiectasia these spots occur because the blood vessel is fragile and bleeds easily. The bleeding may appear as small, red or red-dish-violet spots on the face, lips, inside the mouth and nose or the tips of the fingers and toes. Other small telangiectasias may occur in the digestive tract.

Unlike **hemophilia**, where bleeding is caused by an ineffective clotting mechanism in the blood, bleeding in hereditary hemorrhagic telangiectasia is caused by fragile blood vessels. However, like hemophilia, bleeding may be extensive and can occur without warning.

Causes and symptoms

Hereditary hemorrhagic telangiectasia, an autosomal dominant inherited disorder, occurs in one in 50,000 people.

Recurrent nosebleeds are a nearly universal symptom in this condition. Usually the nosebleeds begin in childhood and become worse with age. The skin changes begin at **puberty**, and the condition becomes progressively worse until about 40 years of age, when it stabilizes.

Diagnosis

The physician will look for red spots on all areas of the skin, but especially on the upper half of the body, and in the mouth and nose and under the tongue.

Treatment

There is no specific treatment for hereditary hemorrhagic telangiectasia. The bleeding resulting from the condition can be stopped by applying compresses or direct pressure to the area. If necessary, a laser can be used to destroy the vessel. In severe cases, the leaking artery can be plugged or covered with a graft from normal tissue.

KEY TERMS

Autosomal dominant—A pattern of inheritance in which the dominant gene on any non-sex chromosome carries the defect.

Chromosome—A threadlike structure in the cell which transmits genetic information.

Prognosis

In most people, recurrent bleeding results in an iron deficiency. It is usually necessary to take iron supplements.

Prevention

Hereditary hemorrhagic telangiectasia is an inherited disorder and cannot be prevented.

Resources

ORGANIZATIONS

American Medical Association. 515 N. State St., Chicago, IL 60612. (312) 464-5000. <<http://www.ama-assn.org>>. Association of Birth Defect Children. 3526 Emerywood Lane, Orlando, FL, 32806. (305) 859-2821.

Dorothy Elinor Stonely

Hereditary hyperuricemia see **Lesch-Nyhan syndrome**

Hereditary spinocerebellar ataxia see **Friedreich's ataxia**

Hermaphroditism see **Intersex states**

Hernia

Definition

Hernia is a general term used to describe a bulge or protrusion of an organ through the structure or muscle that usually contains it.

Description

There are many different types of hernias. The most familiar type are those that occur in the abdomen, in which part of the intestines protrude through the abdominal wall. This may occur in different areas

and, depending on the location, the hernia is given a different name.

An inguinal hernia appears as a bulge in the groin and may come and go depending on the position of the person or their level of physical activity. It can occur with or without **pain**. In men, the protrusion may descend into the scrotum. Inguinal hernias account for 80% of all hernias and are more common in men.

Femoral hernias are similar to inguinal hernias but appear as a bulge slightly lower. They are more common in women due to the strain of **pregnancy**.

A ventral hernia is also called an incisional hernia because it generally occurs as a bulge in the abdomen at the site of an old surgical scar. It is caused by thinning or stretching of the scar tissue, and occurs more frequently in people who are obese or pregnant.

An umbilical hernia appears as a soft bulge at the navel (umbilicus). It is caused by a weakening of the area or an imperfect closure of the area in infants. This type of hernia is more common in women due to pregnancy, and in Chinese and black infants. Some umbilical hernias in infants disappear without treatment within the first year.

A hiatal or diaphragmatic hernia is different from abdominal hernias in that it is not visible on the outside of the body. With a hiatal hernia, the stomach bulges upward through the muscle that separates the chest from the abdomen (the diaphragm). This type of hernia occurs more often in women than in men, and it is treated differently from other types of hernias.

Causes and symptoms

Most hernias result from a weakness in the abdominal wall that either develops or that an infant is born with (congenital). Any increase in pressure in the abdomen, such as coughing, straining, heavy lifting, or pregnancy, can be a considered causative factor in developing an abdominal hernia. **Obesity** or recent excessive weight loss, as well as **aging** and previous surgery, are also risk factors.

Most abdominal hernias appear suddenly when the abdominal muscles are strained. The person may feel tenderness, a slight burning sensation, or a feeling of heaviness in the bulge. It may be possible for the person to push the hernia back into place with gentle pressure, or the hernia may disappear by itself when the person reclines. Being able to push the hernia back is called reducing it. On the other hand, some hernias cannot be pushed back into place, and are termed incarcerated or irreducible.

A hiatal hernia may also be caused by obesity, pregnancy, aging, or previous surgery. About 50% of all people with hiatal hernias do not have any symptoms. If symptoms exist they will include **heartburn**, usually 30–60 minutes following a meal. There may be some mid chest pain due to gastric acid from the stomach being pushed up into the esophagus. The pain and heartburn are usually worse when lying down. Frequent belching and feelings of abdominal fullness may also be present.

Diagnosis

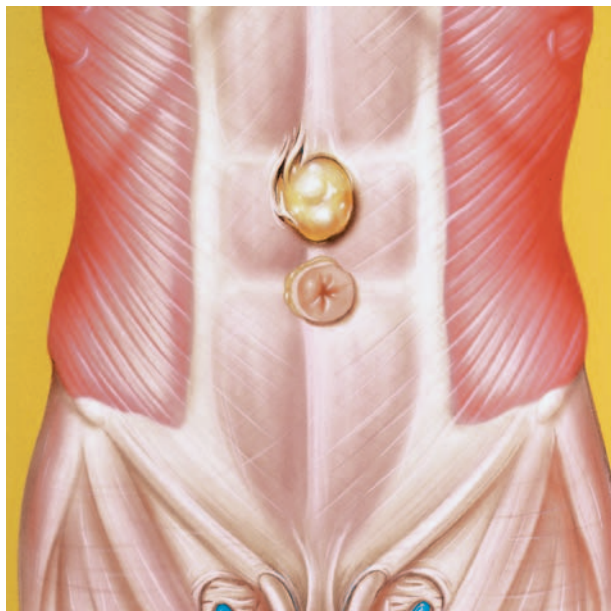
Generally, abdominal hernias need to be seen and felt to be diagnosed. Usually the hernia will increase in size with an increase in abdominal pressure, so the doctor may ask the person to cough while he or she feels the area. Once a diagnosis of an abdominal hernia is made, the doctor will usually send the person to a surgeon for a consultation. Surgery provides the only cure for a hernia through the abdominal wall.

With a hiatal hernia, the diagnosis is based on the symptoms reported by the person. The doctor may then order tests to confirm the diagnosis. If a barium swallow is ordered, the person drinks a chalky white barium solution, which will help any protrusion through the diaphragm show up on the x ray that follows. Currently, a diagnosis of hiatal hernia is more frequently made by endoscopy. This procedure is done by a gastroenterologist (a specialist in digestive diseases). During an endoscopy the person is given an intravenous sedative and a small tube is inserted through the mouth, then into the esophagus and stomach where the doctor can visualize the hernia. The procedure takes about 30 minutes and usually causes no discomfort. It is done on an outpatient basis.

Treatment

Once an abdominal hernia occurs it tends to increase in size. Some patients with abdominal hernias wait and watch for a while prior to choosing surgery. In these cases, they must avoid strenuous physical activity such as heavy lifting or straining with **constipation**. They may also wear a truss, which is a support worn like a belt to keep a small hernia from protruding. People can tell if their hernia is getting worse if they develop severe constant pain, **nausea and vomiting**, or if the bulge does not return to normal when lying down or when they try to gently push it back in place. In these cases they should consult with their doctor immediately. But, ultimately, surgery is the treatment in almost all cases.

There are risks to not repairing a hernia surgically. Left untreated, a hernia may become incarcerated,



An illustration of an epigastric (abdominal) hernia in an adult male. The torso is shown with its skin removed. Epigastric hernia is caused commonly by a congenital weakness in muscles of the central upper abdomen; the intestine bulges out through the muscle at a point between the navel and breastbone. (Photograph by John Bavosi, Photo Researchers, Inc. Reproduced by permission.)

which means it can no longer be reduced or pushed back into place. With an incarcerated hernia the intestines become trapped outside the abdomen. This could lead to a blockage in the intestine. If it is severe enough it may cut off the blood supply to the intestine and part of the intestine might actually die.

When the blood supply is cut off, the hernia is termed “strangulated.” Because of the risk of tissue death (necrosis) and **gangrene**, and because the hernia can block food from moving through the bowel, a strangulated hernia is a medical emergency requiring immediate surgery. Repairing a hernia before it becomes incarcerated or strangulated is much safer than waiting until complications develop.

Surgical repair of a hernia is called a **herniorrhaphy**. The surgeon will push the bulging part of the intestine back into place and sew the overlying muscle back together. When the muscle is not strong enough, the surgeon may reinforce it with a synthetic mesh.

Surgery can be done on an outpatient basis. It usually takes 30 minutes in children and 60 minutes in adults. It can be done under either local or **general anesthesia** and is frequently done with a laparoscope. In this type of surgery, a tube that allows visualization of the abdominal cavity is inserted through a small

puncture wound. Several small punctures are made to allow surgical instruments to be inserted. This type of surgery avoids a larger incision.

A hiatal hernia is treated differently. Medical treatment is preferred. Treatments include:

- avoiding reclining after meals
- avoiding spicy foods, acidic foods, alcohol, and tobacco
- eating small, frequent, bland meals
- eating a high-fiber diet.

There are also several types of medications that help to manage the symptoms of a hiatal hernia. **Antacids** are used to neutralize gastric acid and decrease heartburn. Drugs that reduce the amount of acid produced in the stomach (H2 blockers) are also used. This class of drugs includes famotidine (sold under the name Pepcid), cimetidine (Tagamet), and ranitidine (Zantac). Omeprazole (Prilosec) is not an H2 blocker, but is another drug that suppresses gastric acid secretion and is used for hiatal hernias. Another option may be metoclopramide (Reglan), a drug that increases the tone of the muscle around the esophagus and causes the stomach to empty more quickly.

Alternative treatment

There are alternative therapies for hiatal hernia. Visceral manipulation, done by a trained therapist, can help replace the stomach to its proper positioning. Other options in addition to H2 blockers are available to help regulate stomach acid production and balance. One of them, deglycyrrhizinated licorice (DGL), helps balance stomach acid by improving the protective substances that line the stomach and intestines and by improving blood supply to these tissues. DGL does not interrupt the normal function of stomach acid.

As with traditional therapy, dietary modifications are important. Small, frequent meals will keep pressure down on the esophageal sphincter. Also, raising the head of the bed several inches with blocks or books can help with both the quality and quantity of sleep.

Prognosis

Abdominal hernias generally do not recur in children but can recur in up to 10% of adult patients. Surgery is considered the only cure, and the prognosis is excellent if the hernia is corrected before it becomes strangulated.

Hiatal hernias are treated successfully with medication and diet modifications 85% of the time.

KEY TERMS

Endoscopy—A diagnostic procedure in which a tube is inserted through the mouth, into the esophagus and stomach. It is used to visualize various digestive disorders, including hiatal hernias.

Herniorrhaphy—Surgical repair of a hernia.

Incarcerated hernia—A hernia that can not be reduced, or pushed back into place inside the intestinal wall.

Reducible hernia—A hernia that can be gently pushed back into place or that disappears when the person lies down.

Strangulated hernia—A hernia that is so tightly incarcerated outside the abdominal wall that the intestine is blocked and the blood supply to that part of the intestine is cut off.

The prognosis remains excellent even if surgery is required in adults who are in otherwise good health.

Prevention

Some hernias can be prevented by maintaining a reasonable weight, avoiding heavy lifting and constipation, and following a moderate **exercise** program to maintain good abdominal muscle tone.

Resources

BOOKS

Bare, Brenda G., and Suzanne C. Smeltzer. *Brunner and Suddarth's Textbook of Medical-Surgical Nursing*. 8th ed. Philadelphia: Lippincott-Raven Publishers, 1996.

Joyce S. Siok, RN

Hernia repair

Definition

Hernia repair is a surgical procedure to return an organ that protrudes through a weak area of muscle to its original position.

Purpose

Hernias occur when a weakness in the wall of the abdomen allows an organ, usually the intestines, to

bulge out of place. Hernias may result from a genetic predisposition toward this weakness. They can also be the result of weakening the muscle through improper **exercise** or poor lifting techniques. Both children and adults get hernias. Some are painful, while others are not.

There are three levels of hernias. An uncomplicated hernia is one where the intestines bulge into the peritoneum (the membrane lining the abdomen), but they can still be manipulated back into the body (although they don't stay in place without corrective surgery). This is termed a reducible hernia.

If the intestines bulge through the hernia defect and become trapped, this is called an incarcerated hernia. If the blood supply to an incarcerated hernia is shut off, the hernia is called a strangulated hernia. Strangulated hernias can result in **gangrene**.

Both incarcerated and strangulated hernias are medical emergencies and require emergency surgery to correct. For this reason, doctors generally recommend the repair of an uncomplicated hernia, even if it causes no discomfort to the patient.

Precautions

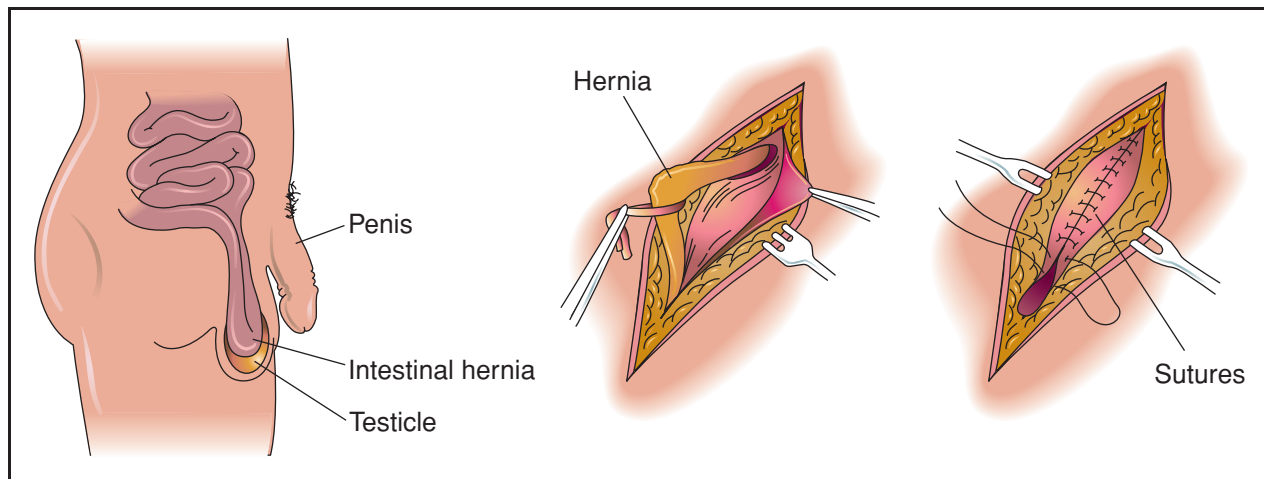
Hernia repair can be performed under local, regional, or **general anesthesia**. The choice depends on the age and health of the patient and the type of hernia. Generally hernia repair is very safe surgery, but—as with any surgery—the risk of complications increases if the patient smokes, is obese, is very young or very old, uses alcohol heavily, or uses illicit drugs.

Description

Hernia repairs are performed in a hospital or outpatient surgical facility by a general surgeon. Depending on the patient's age, health, and the type of hernia, patients may be able to go home the same day or may remain hospitalized for up to three to five days.

There are two types of hernia repair. A herniorrhaphy is used for simpler hernias. The intestines are returned to their proper place and the defect in the abdominal wall is mended. A hernioplasty is used for larger hernias. In this procedure, plastic or steel mesh is added to the abdominal wall to repair and reinforce the weak spot.

There are five kinds of common hernia repairs. They are named for the part of the body closest to the hernia, or bulge.



In this inguinal hernia repair, an incision is made in the abdomen. The hernia is located, and the intestines are returned to the abdomen. The abdominal wall is then sutured together to close any space and reinforce the weak area. (Illustration by Electronic Illustrators Group.)

Femoral hernia repair

This procedure repairs a hernia that occurs in the groin where the thigh meets the abdomen. It is called a femoral hernia repair because it is near the spot where the femoral artery and vein pass from the leg into the trunk of the body. Sometimes this type of hernia creates a noticeable bulge.

An incision is made in the groin area. The tissues are separated from the hernia sac, and the intestines are returned to the abdomen. The area is often reinforced with webbing before it is sewn shut. The skin is closed with sutures or metal clips that can be removed in about one week.

Inguinal hernia repair

Inguinal hernia repair closes a weakness in the abdominal wall that is near the inguinal canal, the spot where the testes descend from the body into the scrotum. This type of hernia occurs in about two percent of adult males.

An incision is made in the abdomen, then the hernia is located and repaired. The surgeon must be alert not to injure the spermatic cord, the testes, or the blood supply to the testes. If the hernia is small, it is simply repaired. If it is large, the area is reinforced with mesh to prevent a recurrence. External skin sutures can be removed in about a week. Patients should not resume sexual activity until being cleared by their doctor.

Umbilical hernia repair

This procedure repairs a hernia that occurs when the intestines bulge through the abdomen wall near

the navel. Umbilical hernias are most common in infants.

An incision is made near the navel. The hernia is located and the intestines are returned to the abdomen. The peritoneum is closed, then the large abdominal muscle is pulled over the weak spot in such a way as to reinforce the area. External sutures or skin clips can be removed in about 10 days.

Incisional hernia repair

Incisional hernias occur most frequently at the site of a scar from earlier abdominal surgery. Once again, the abdomen is opened and the intestines returned to their proper place. The area is reinforced with mesh, and the abdominal wall is reconstructed to prevent another hernia from developing. External sutures can be removed in about a week.

Hiatal hernia

A hiatal hernia repair is slightly different from the other hernias described here, because it corrects a weakness or opening in the diaphragm, the muscle that separates the chest cavity from the abdominal cavity. This surgery is done to prevent the stomach from shifting up into the chest cavity and to prevent the stomach from spilling gastric juices into the esophagus, causing **pain** and scarring.

An incision is made in the abdomen or chest, and the hole or weakness in the diaphragm is located and repaired. The top of the stomach is wrapped around the bottom of the esophagus, and they are sutured together to hold the stomach in place. Sometimes the

KEY TERMS

Endoscopy—A procedure in which an instrument containing a camera is inserted into the gastrointestinal tract so that the doctor can visually inspect the gastrointestinal system.

Gangrene—Death and decay of body tissue because the blood supply is cut off. Tissues that have died in this way must be surgically removed.

Peritoneum—The transparent membrane lining the abdominal cavity that holds organs such as the intestines in place.

vagus nerve is cut in order to decrease the amount of acid the stomach produces. External sutures can be removed in about one week. This type of hernia repair often requires a longer hospital stay than the other types, although techniques are being improved that reduce invasiveness of the surgery and the length of the hospital stay.

Preparation

Before the operation, the patient will have blood and urine collected for testing. X rays are taken of the affected area. In a hiatal hernia, an endoscopy (a visual inspection of the organs) is done.

Patients should meet with the anesthesiologist before the operation to discuss any medications or conditions that might affect the administration of anesthesia. Patients may be asked to temporarily discontinue certain medications. The day of the operation, patients should not eat or drink anything. They may be given an enema to clear the bowels.

Aftercare

Patients should eat a clear liquid diet until the gastrointestinal tract begins functioning again. Normally this is a short period of time. After that, they are free to eat a healthy, well-balanced diet of their choice. They may bathe normally, using a gentle, unscented soap. An antibiotic ointment may be prescribed for the incision. After the operation, a hard ridge will form along the incision line. With time, this ridge softens and becomes less noticeable. Patients who remain in the hospital will have blood drawn for follow-up studies.

Patients should begin easy activities, such as walking, as soon as they are comfortable, but should avoid

strenuous exercise for four to six weeks, and especially avoid heavy lifting. Learning and practicing proper lifting techniques is an important part of patient education after the operation. Patients may be given a laxative or stool softener so that they will not strain to have bowel movements. They should discuss with their doctor when to resume driving and sexual activity.

Risks

As with any surgery, there exists the possibility of excessive bleeding and infection after the surgery. In inguinal and femoral hernia repair, a slight risk of damage to the testicles or their blood supply exists for male patients. Accidental damage may be caused to the intestinal tract, but generally complications are few.

Normal results

The outcome of surgery depends on the age and health of the patient and on the type of hernia. Although most hernias can be repaired without complications, hernias recur in 10–20% of people who have had hernia surgery.

Resources

OTHER

“Hernia Repair.” *ThriveOnline*.
< <http://thriveonline.oxygen.com> >.

Tish Davidson, A.M.

Herniated disk

Definition

Disk herniation is a rupture of fibrocartilagenous material (annulus fibrosis) that surrounds the intervertebral disk. This rupture involves the release of the disk's center portion containing a gelatinous substance called the nucleus pulposus. Pressure from the vertebrae above and below may cause the nucleus pulposus to be forced outward, placing pressure on a spinal nerve and causing considerable **pain** and damage to the nerve. This condition most frequently occurs in the lumbar region and is also commonly called herniated nucleus pulposus, prolapsed disk, ruptured intervertebral disk, or slipped disk.

Description

The spinal column is made up of 26 vertebrae that are joined together and permit forward and backward bending, side bending, and rotation of the spine. Five distinct regions comprise the spinal column, including the cervical (neck) region, thoracic (chest) region, lumbar (low back) region, sacral and coccygeal (tailbone) region. The cervical region consists of seven vertebrae, the thoracic region includes 12 vertebrae, and the lumbar region contains five vertebrae. The sacrum is composed of five fused vertebrae, which are connected to four fused vertebrae forming the coccyx. Intervertebral disks lie between each adjacent vertebra.

Each disk is composed of a gelatinous material in the center, called the nucleus pulposus, surrounded by rings of a fibrous tissue (annulus fibrosus). In disk herniation, an intervertebral disk's central portion herniates or slips through the surrounding annulus fibrosus into the spinal canal, putting pressure on a nerve root. Disk herniation most commonly affects the lumbar region between the fifth lumbar vertebra and the first sacral vertebra. However, disk herniation can also occur in the cervical spine. The incidence of cervical disk herniation is most common between the fifth and sixth cervical vertebrae. The second most common area for cervical disk herniation occurs between the sixth and seventh cervical vertebrae. Disk herniation is less common in the thoracic region.

Predisposing factors associated with disk herniation include age, gender, and work environment. The peak age for occurrence of disk herniation is between 20–45 years of age. Studies have shown that males are more commonly affected than females in lumbar disk herniation by a 3:2 ratio. Prolonged exposure to a bent-forward work posture is correlated with an increased incidence of disk herniation.

There are four classifications of disk pathology:

- A protrusion may occur where a disk bulges without rupturing the annulus fibrosis.
- The disk may prolapse where the nucleus pulposus migrates to the outermost fibers of the annulus fibrosis.
- There may be a disk extrusion, which is the case if the annulus fibrosis perforates and material of the nucleus moves into the epidural space.
- The sequestered disk may occur as fragments from the annulus fibrosis and nucleus pulposus are outside the disk proper.

Causes and symptoms

Any direct, forceful, and vertical pressure on the lumbar disks can cause the disk to push its fluid contents into the vertebral body. Herniated nucleus pulposus may occur suddenly from lifting, twisting, or direct injury, or it can occur gradually from degenerative changes with episodes of intensifying symptoms. The annulus may also become weakened over time, allowing stretching or tearing and leading to a disk herniation. Depending on the location of the herniation, the herniated material can also press directly on nerve roots or on the spinal cord, causing a shock-like pain (sciatica) down the legs, weakness, **numbness**, or problems with bowels, bladder, or sexual function.

Diagnosis

Several radiographic tests are useful for confirming a diagnosis of disk herniation and locating the source of pain. These tests also help the surgeon indicate the extent of the surgery needed to fully decompress the nerve. X rays show structural changes of the lumbar spine. Myelography is a special x ray of the spine in which a dye or air is injected into the patient's spinal canal. The patient lies strapped to a table as the table tilts in various directions and spot x rays are taken. X rays showing a narrowed dye column in the intervertebral disk area indicate possible disk herniation.

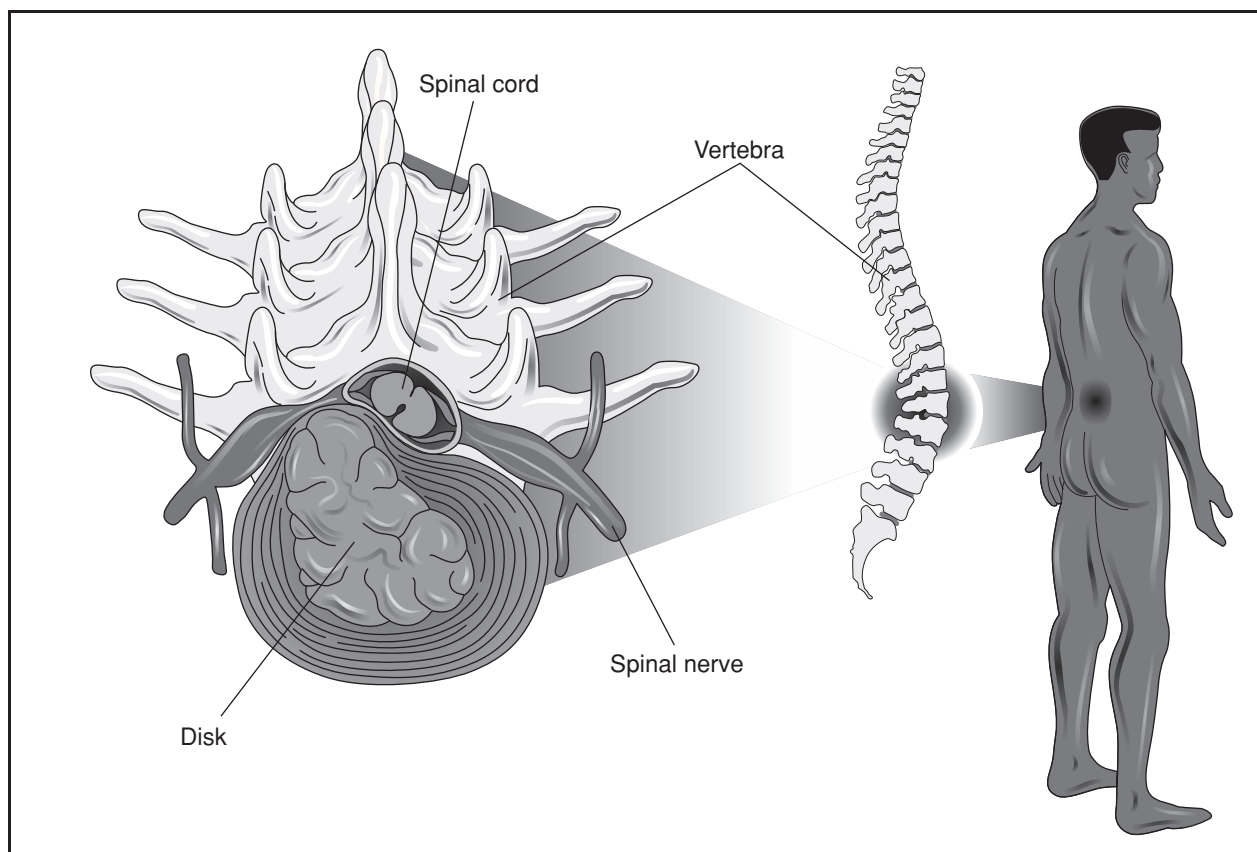
Computed tomography scan (CT or CAT scans) exhibit the details of pathology necessary to obtain consistently good surgical results. Magnetic resonance imaging (MRI) analysis of the disks can accurately detect the early stages of disk aging and degeneration. Electromyograms (EMGs) measure the electrical activity of the muscle contractions and possibly show evidence of nerve damage. An EMG is a powerful tool for assessing muscle **fatigue** associated with muscle impairment with **low back pain**.

Treatment

Drugs

Unless serious neurologic symptoms occur, herniated disks can initially be treated with pain medication and up to 48 hours of bed rest. There is no proven benefit from resting more than 48 hours. Patients are then encouraged to gradually increase their activity. Pain medications, including antiinflammatories, muscle relaxers, or in severe cases, **narcotics**, may be continued if needed.

Epidural steroid injections have been used to decrease pain by injecting an antiinflammatory drug,



A herniated disk refers to the rupture of fibrocartilagenous material, called the annulus fibrosis, that surrounds the intervertebral disk. When this occurs, pressure from the vertebrae above and below may force the disk's center portion, a gel-like substance, outward, placing additional pressure on the spinal nerve and causing pain and damage to the nerve. (Illustration by Electronic Illustrators Group.)

usually a corticosteroid, around the nerve root to reduce inflammation and edema (swelling). This partly relieves the pressure on the nerve root as well as resolves the inflammation.

Physical therapy

Physical therapists are skilled in treating acute back pain caused by the disk herniation. The physical therapist can provide noninvasive therapies, such as ultrasound or diathermy to project heat deep into the tissues of the back or administer manual therapy, if mobility of the spine is impaired. They may help improve posture and develop an exercise program for recovery and long-term protection. Appropriate **exercise** can help take pressure off inflamed nerve structures, while improving overall posture and flexibility. **Traction** can be used to try to decrease pressure on the disk. A lumbar support can be helpful for a herniated disk at this level as a temporary measure to reduce pain and improve posture.

Surgery

Surgery is often appropriate for conditions that do not improve with the usual treatment. In this event, a strong, flexible spine is important for a quick recovery after surgery. There are several surgical approaches to treating a herniated disk, including the classic discectomy, microdiscectomy, or percutaneous discectomy. The basic differences among these procedures are the size of the incision, how the disk is reached surgically, and how much of the disk is removed.

Discectomy is the surgical removal of the portion of the disk that is putting pressure on a nerve causing the back pain. In the classic discectomy, the surgeon first enters through the skin and then removes a bony portion of the vertebra called the lamina, hence the term laminectomy. The surgeon removes the disk material that is pressing on a nerve. Rarely is the entire lamina or disk entirely removed. Often, only one side is removed and the surgical procedure is termed hemi-laminectomy.

In microdiscectomy, through the use of an operating microscope, the surgeon removes the offending bone or disk tissue until the nerve is free from compression or stretch. This procedure is possible using **local anesthesia**. Microsurgery techniques vary and have several advantages over the standard discectomy, such as a smaller incision, less trauma to the musculature and nerves, and easier identification of structures by viewing into the disk space through microscope magnification.

Percutaneous disk excision is performed on an outpatient basis, is less expensive than other surgical procedures, and does not require a **general anesthesia**. The purpose of percutaneous disk excision is to reduce the volume of the affected disk indirectly by partial removal of the nucleus pulposus, leaving all the structures important to stability practically unaffected. In this procedure, large incisions are avoided by inserting devices that have cutting and suction capability. Suction is applied and the disk is sliced and aspirated.

Arthroscopic microdiscectomy is similar to percutaneous discectomy, however it incorporates modified arthroscopic instruments, including scopes and suction devices. A suction irrigation of saline solution is established through two entry sites. A video discoscope is introduced from one site and the deflecting instruments from the opposite side. In this way, the surgeon is able to search and extract the nuclear fragments under direct visualization.

Laser disk decompression is performed using similar means as percutaneous excision and arthroscopic microdiscectomy, however laser energy is used to remove the disk tissue. Here, laser energy is percutaneously introduced through a needle to vaporize a small volume of nucleus pulposus, thereby dropping the pressure of the disk and decompressing the involved neural tissues. One disadvantage of this procedure is the high initial cost of the laser equipment. It is important to realize that only a very small percentage of people with herniated lumbar disks go on to require surgery. Further, surgery should be followed by appropriate **rehabilitation** to decrease the chance of reinjury.

Chemoneucleolysis

Chemoneucleolysis is an alternative to surgical excision. Chymopapain, a purified enzyme derived from the papaya plant, is injected percutaneously into the disk space to reduce the size of the herniated disks. It hydrolyses proteins, thereby decreasing water-binding capacity, when injected into the nucleus pulposus inner disk material. The reduction in size of the disk relieves pressure on the nerve root.

KEY TERMS

Annulus fibrosis—The outer portion of the intervertebral disk made primarily of fibrocartilage rings.

Epidural space—The space immediately surrounding the outermost membrane of the spinal cord.

Excision—The process of excising, removing, or amputating.

Fibrocartilage—Cartilage that consists of dense fibers.

Nucleus pulposus—The center portion of the intervertebral disk that is made up of a gelatinous substance.

Percutaneous—Performed through the skin.

Spinal fusion

Spinal fusion is the process by which bone grafts harvested from the iliac crest (thick border of the ilium located on the pelvis) are placed between the intervertebral bodies after the disk material is removed. This approach is used when there is a need to reestablish the normal bony relationship between the vertebrae. A total discectomy may be needed in some cases because lumbar spinal fusion can help prevent recurrent lumbar disk herniation at a particular level.

Alternative treatment

Acupuncture involves the use of fine needles inserted along the pathway of the pain to move energy locally and relieve the pain. An acupuncturist determines the location of the nerves affected by the herniated disk and positions the needles appropriately. Massage therapists may also provide short-term relief from a herniated disk. Following manual examination and x-ray diagnosis, **chiropractic** treatment usually includes manipulation to correct muscle and joint malfunctions, while care is taken not to place an additional strain on the injured disk. If a full trial of conservative therapy fails, or if neurologic problems (weakness, bowel or bladder problems, and sensory loss) develop, the next step is usually evaluation by an orthopedic surgeon.

Prognosis

Only 5–10% of patients with unrelenting **sciatica** and neurological involvement, leading to chronic pain of the lumbar spine, need to have a surgical procedure performed. This strongly suggests that many patients

with herniated disks at the lumbar level respond well to conservative treatment. For those patients who do require surgery for lumbar disk herniation, the reviewed procedures of nerve root decompression caused by disk herniation is favorable. Results of studies varied from 60–90% success rates. Disk surgery has progressively evolved in the direction of decreasing invasiveness. Each surgical procedure is not without possible complications, which can lead to chronic low back pain and restricted lifestyle.

Prevention

Proper exercises to strengthen the lower back and abdominal muscles are key in preventing excess **stress** and compressive forces on lumbar disks. Good posture will help prevent problems on cervical, thoracic, and lumbar disks. A good flexibility program is critical for prevention of muscle and spasm that can cause an increase in compressive forces on disks at any level. Proper lifting of heavy objects is important for all muscles and levels of the individual disks. Good posture in sitting, standing, and lying down is helpful for the spine. Losing weight, if needed, can prevent weakness and unnecessary stress on the disks caused by obesity. Choosing proper footwear may also be helpful to reduce the impact forces to the lumbar disks while walking on hard surfaces. Wearing special back support devices may be helpful if heavy lifting is required with combinations of twisting.

Resources

OTHER

“Back Pain.” Healthtouch Online Page.
< <http://www.healthtouch.com> >.

Jeffrey P. Larson, RPT

Hernioplasty see **Hernia repair**

Herniorrhaphy see **Hernia repair**

Herpes see **Cold sore**

Herpes encephalitis see **Encephalitis**

Herpes genitalis see **Genital herpes**

Herpes simplex see **Cold sore**

Herpes simplex type 2 see **Genital herpes**

Herpes type 2 see **Genital herpes**

Herpes zoster infection see **Shingles**

Heterotopic transplant see **Liver transplantation**

Heterotropia see **Strabismus**

HFRS see **Hantavirus infections**

Hiccups

Definition

Hiccups are the result of an involuntary, spasmodic contraction of the diaphragm followed by the closing of the throat.

Description

Hiccups are one of the most common, but thankfully mildest, disorders to which humans are prey. Virtually everyone experiences them at some point, but they rarely last long or require a doctor's care. Occasionally, a bout of hiccups will last longer than two days, earning it the name “persistent hiccups.” Very few people will experience intractable hiccups, in which hiccups last longer than one month.

A hiccup involves the coordinated action of the diaphragm and the muscles that close off the windpipe (trachea). The diaphragm is a dome-shaped muscle separating the chest and abdomen, normally responsible for expanding the chest cavity for inhalation. Sensation from the diaphragm travels to the spinal cord through the phrenic nerve and the vagus nerve, which pass through the chest cavity and the neck. Within the spinal cord, nerve fibers from the brain monitor sensory information and adjust the outgoing messages that control contraction. These messages travel along the phrenic nerve.

Irritation of any of the nerves involved in this loop can cause the diaphragm to undergo involuntary contraction, or spasm, pulling air into the lungs. When this occurs, it triggers a reflex in the throat muscles. Less than a tenth of a second afterward, the trachea is closed off, making the characteristic “hic” sound.

Causes and symptoms

Hiccups can be caused by central nervous system disorders, injury or irritation to the phrenic and vagus nerves, and toxic or metabolic disorders affecting the central or peripheral nervous systems. They may be of unknown cause or may be a symptom of psychological **stress**. Hiccups often occur after drinking carbonated beverages or alcohol. They may also follow overeating or rapid temperature changes.

Persistent or intractable hiccups may be caused by any condition which irritates or damages the relevant nerves, including:

- overstretching of the neck
- laryngitis
- heartburn (gastroesophageal reflux)
- irritation of the eardrum (which is innervated by the vagus nerve)
- general anesthesia
- surgery
- bloating
- tumor
- infection
- diabetes

Diagnosis

Hiccups are diagnosed by observation, and by hearing the characteristic sound. Diagnosing the cause of intractable hiccups may require imaging studies, blood tests, pH monitoring in the esophagus, and other tests.

Treatment

Most cases of hiccups will disappear on their own. Home remedies which interrupt or override the spasmodic nerve circuitry are often effective. Such remedies include:

- holding one's breath for as long as possible
- breathing into a paper bag
- swallowing a spoonful of sugar
- bending forward from the waist and drinking water from the wrong side of a glass

Treating any underlying disorder will usually cure the associated hiccups. Chlorpromazine (Thorazine) relieves intractable hiccups in 80% of cases. Metoclopramide (Reglan), carbamazepam, valproic acid (Depakene), and phenobarbital are also used. As a last resort, surgery to block the phrenic nerve may be performed, although it may lead to significant impairment of respiration.

Prognosis

Most cases of hiccups last no longer than several hours, with or without treatment.

KEY TERMS

Nerve—Fibers that carry sensory information, movement stimuli, or both from the brain and spinal cord to other parts of the body and back again. Some nerves, including the vagus nerve, innervate distantly separated parts of the body.

Prevention

Some cases of hiccups can be avoided by drinking in moderation, avoiding very hot or very cold food, and avoiding cold showers. Carbonated beverages when drunk through a straw deliver more gas to the stomach than when sipped from a container; therefore, avoid using straws.

Resources

BOOKS

Hurst, J. Willis. *Medicine for the Practicing Physician*. Stamford: Appleton & Lange, 1988.

Richard Robinson

High-altitude sickness see **Altitude sickness**

High blood phosphate level see **Phosphorus imbalance**

High blood pressure see **Pulmonary hypertension**

High calcium blood level see **Hypercalcemia**

High cholesterol see **Cholesterol, high**

High potassium blood level see **Hyperkalemia**

High-risk pregnancy

Definition

A high risk **pregnancy** is one in which some condition puts the mother, the developing fetus, or both at higher-than-normal risk for complications during or after the pregnancy and birth.

Description

A pregnancy can be considered a high-risk pregnancy for a variety of reasons. Factors can be divided into maternal and fetal. Maternal factors include age (younger than age 15, older than age 35); weight (pre-pregnancy weight under 100 lb or obesity); height (under five feet); history of complications during previous pregnancies (including stillbirth, fetal loss, preterm labor and/or delivery, small-for-gestational age baby, large baby, pre-eclampsia or **eclampsia**); more than five previous pregnancies; bleeding during the third trimester; abnormalities of the reproductive tract; **uterine fibroids**; **hypertension**; Rh incompatibility; **gestational diabetes**; infections of the vagina and/or cervix; kidney infection; **fever**; acute surgical emergency (appendicitis, gallbladder disease, bowel obstruction); post-term pregnancy; pre-existing chronic illness (such as asthma, autoimmune disease, **cancer**, sickle cell anemia, **tuberculosis**, herpes, **AIDS**, heart disease, kidney disease, **Crohn's disease**, ulcerative colitis, diabetes). Fetal factors include exposure to infection (especially herpes simplex, viral hepatitis, **mumps**, **rubella**, varicella, **syphilis**, **toxoplasmosis**, and infections caused by coxsackievirus); exposure to damaging medications (especially phenytoin, **folic acid** antagonists, lithium, streptomycin, tetracycline, thalidomide, and warfarin); exposure to addictive substances (cigarette **smoking**, alcohol intake, and illicit or abused drugs). A pregnancy is also considered high-risk when prenatal tests indicate that the baby has a serious health problem (for example, a heart defect). In such cases, the mother will need special tests, and possibly medication, to carry the baby safely through to delivery. Furthermore, certain maternal or fetal problems may prompt a physician to deliver a baby early, or to choose a surgical delivery (cesarean section) rather than a vaginal delivery.

Most women will see one healthcare provider during pregnancy, either an obstetrician, a midwife, or a nurse practitioner. Women who have a medical problem may need to see a medical specialist as well. Women diagnosed with a high-risk pregnancy may also need the expert advice and care of a perinatologist. A perinatologist is a medical doctor (obstetrician) who specializes in the care of women who are at high risk for having problems during pregnancy. Perinatologists care for women who have pre-existing medical problems as well as women who develop complications during pregnancy.

Diagnosis

A woman with a high-risk pregnancy will need closer monitoring than the average pregnant woman.

Such monitoring may include more frequent visits with the primary caregiver, tests to monitor the medical problem, blood tests to check the levels of medication, **amniocentesis**, serial ultrasound examination, and fetal monitoring. These tests are designed to track the original condition, survey for complications, verify that the fetus is growing adequately, and make decisions regarding whether labor may need to be induced to allow for early delivery of the fetus.

Treatment

Treatment varies widely with the type of disease, the effect that pregnancy has on the disease, and the effect that the disease has on pregnancy. Additional tests may help determine the need for changes in medication or additional treatment.

Prognosis

The prognosis depends in large part on the specific medical condition. Some medical conditions make it difficult to get pregnant and lead to a higher risk of problems in the baby. An example of this type of condition is thyroid disease. In thyroid disease, the thyroid gland (located in the neck) may produce too much or too little thyroid hormone. Abnormal levels of thyroid hormone can cause problems in pregnancy and affect the health of the baby. Fortunately, thyroid disease can be treated with medication. As long as the level of thyroid hormone is controlled throughout pregnancy, there should be no problems for mother or baby.

There are many medical conditions that usually do not interfere with pregnancy, but are themselves affected by pregnancy. This group includes **asthma**, **epilepsy**, and **ulcerative colitis**. For example, some women with **ulcerative colitis** experience a worsening of their symptoms during pregnancy, while others will have no change or may get better during pregnancy. The same is true of asthma; some women notice that their asthma symptoms are better during pregnancy, some find their asthma worse, and some women notice no change in symptoms during pregnancy. No one understands why this is so, but due to this unpredictability, all women with chronic illnesses should be monitored carefully throughout pregnancy.

There is also a group of medical conditions that can have a major impact on pregnancy. Women with lupus (disease caused by alterations in the immune system that result in inflammation of connective tissue and organs) or **kidney disease** face real risks during pregnancy. Pregnancy can cause their symptoms to

worsen significantly and can lead to serious illness. Because these diseases can affect the mother's ability to supply oxygen and nutrients to the baby through the placenta, they can cause problems for the baby as well. These babies may not be able to grow and gain weight properly (**intrauterine growth retardation**). There is also an increased risk of **stillbirth**.

Diabetes is a medical condition that is both affected by pregnancy and affects pregnancy. Diabetes can lead to miscarriages, **birth defects**, and stillbirths. When a woman monitors her blood sugar carefully and treats high levels with insulin, the risk of these negative outcomes drops a great deal. Unfortunately, pregnancy makes diabetes much harder to control. In general, blood sugar and the need for insulin to control it rise throughout pregnancy.

Most medical conditions do not lead to complications in pregnancy. With frequent visits to healthcare providers, and careful attention to medication, women with medical problems usually enjoy healthy, successful pregnancies. There are a few medical conditions that can cause health risks to both mother and baby during pregnancy. Women with these medical problems should consider these risks before deciding to become pregnant. Many of these women will benefit from the care of a perinatologist during pregnancy. Only rarely (in the case of severe heart disease, for example) are the risks to the mother so high that she should not consider pregnancy at all.

Prevention

A pre-pregnancy visit with a healthcare provider is especially important for a woman who has a medical problem. The doctor will discuss how women with this condition usually fare during pregnancy. For some diseases (such as lupus), pregnancy can mean increased risk of health problems for mother and baby.

Sometimes, the medication a woman needs to control a medical condition can cause problems for the baby. There may be another medication available that is safer for use in pregnancy. In some cases there is no other medication, and a woman must weigh the risks to the baby when deciding whether or not to become pregnant.

A woman who has not had a pre-pregnancy visit should contact a healthcare provider as soon as she learns she is pregnant. Often, the provider will schedule the first prenatal visit within a day or two, instead of waiting until eight to 10 weeks of pregnancy. This is because certain medical conditions can increase the risk of **miscarriage**. The provider will want to be sure

KEY TERMS

Gestational diabetes—Diabetes of pregnancy leading to increased levels of blood sugar. Unlike diabetes mellitus, gestational diabetes is caused by pregnancy and goes away when pregnancy ends. Like diabetes mellitus, gestational diabetes is treated with a special diet and insulin, if necessary.

Preeclampsia—A disease that only affects pregnant women. The most common signs and symptoms are increased blood pressure, swelling in the hands and feet, and abnormal results on special blood and urine tests.

Premature labor—Labor beginning before 36 weeks of pregnancy.

that any medication is adjusted properly to increase the chance of having a successful pregnancy.

Resources

BOOKS

Beers, Mark H., et al., editors. "High-Risk Pregnancies." In *The Merck Manual of Diagnosis and Therapy*. Rahway, NJ: Merck Research Laboratories, 2004.

Rosalyn Carson-DeWitt, MD

High sodium blood level see **Hypernatremia**

Hindu medicine see **Ayurvedic medicine**

Hip bath see **Sitz bath**

Hip replacement see **Joint replacement**

Hirschsprung's disease

Definition

Hirschsprung's disease, also known as congenital megacolon or aganglionic megacolon, is an abnormality in which certain nerve fibers are absent in segments of the bowel, resulting in severe bowel obstruction. It was first identified in 1886 by a physician named Harold Hirschsprung.

Description

Hirschsprung's disease is caused when certain nerve cells (called parasympathetic ganglion cells) in

the wall of the large intestine (colon) do not develop before birth. Without these nerves, the affected segment of the colon lacks the ability to relax and move bowel contents along. This causes a constriction and as a result, the bowel above the constricted area dilates due to stool becoming trapped, producing megacolon (dilation of the colon). The disease can affect varying lengths of bowel segment, most often involving the region around the rectum. In up to 10% of children, however, the entire colon and part of the small intestine are involved. This condition is known as total colonic aganglionosis, or TCA.

Hirschsprung's disease occurs once in every 5,000 live births, and it is about four times more common in males than females. Between 4% and 50% of siblings are also afflicted. The wide range for recurrence is due to the fact that the recurrence risk depends on the gender of the affected individual in the family (i.e., if a female is affected, the recurrence risk is higher) and the length of the aganglionic segment of the colon (i.e., the longer the segment that is affected, the higher the recurrence risk).

Causes and symptoms

Hirschsprung's disease occurs early in fetal development when, for unknown reasons, there is either failure of nerve cell development, failure of nerve cell migration, or arrest in nerve cell development in a segment of the bowel. The absence of these nerve fibers, which help control the movement of bowel contents, is what results in intestinal obstruction accompanied by other symptoms.

There is a genetic basis to Hirschsprung's disease, and it is believed that it may be caused by different genetic factors in different subsets of families. Proof that genetic factors contribute to Hirschsprung's disease is that it is known to run in families, and it has been seen in association with some chromosome abnormalities. For example, about 10% of children with the disease have Down syndrome (the most common chromosomal abnormality). Molecular diagnostic techniques have identified many genes that cause susceptibility to Hirschsprung's disease. As of the early 2000s, a total of six genes have been identified: the RET gene, the glial cell line-derived neurotrophic factor gene, the endothelin-B receptor gene, endothelin converting enzyme, the endothelin-3 gene, and the Sry-related transcription factor SOX10. Mutations that inactivate the RET gene are the most frequent, occurring in 50% of familial cases (cases which run in families) and 15–20% of sporadic (non-familial) cases. Mutations in these genes do not

cause the disease, but they make the chance of developing it more likely. Mutations in other genes or environmental factors are required to develop the disease, and these other factors are not understood. As of 2004, at least three chromosomes are known to be involved: 13q22, 21q22, and 10q. Hirschsprung's disease has also been reported in association with abnormal forms of chromosome 18.

For persons with a **ganglion** growth beyond the sigmoid segment of the colon, the inheritance pattern is autosomal dominant with reduced penetrance (risk closer to 50%). For persons with smaller segments involved, the inheritance pattern is multifactorial (caused by an interaction of more than one gene and environmental factors, risk lower than 50%) or autosomal recessive (one disease gene inherited from each parent, risk closer to 25%) with low penetrance.

The initial symptom is usually severe, continuous **constipation**. A newborn may fail to pass meconium (the first stool) within 24 hours of birth, may repeatedly vomit yellow- or green-colored bile and may have a distended (swollen, uncomfortable) abdomen. Occasionally, infants may have only mild or intermittent constipation, often with **diarrhea**.

While two-thirds of cases are diagnosed in the first three months of life, Hirschsprung's disease may also be diagnosed later in infancy or childhood. Occasionally, even adults are diagnosed with a variation of the disease. In older infants, symptoms and signs may include anorexia (lack of appetite or inability to eat), lack of the urge to move the bowels or empty the rectum on **physical examination**, distended abdomen, and a mass in the colon that can be felt by the physician during examination. It should be suspected in older children with abnormal bowel habits, especially a history of constipation dating back to infancy and ribbon-like stools.

Occasionally, the presenting symptom may be a severe intestinal infection called enterocolitis, which is life-threatening. The symptoms are usually explosive, watery stools and fever in a very ill-appearing infant. It is important to diagnose the condition before the intestinal obstruction causes an overgrowth of bacteria that evolves into a medical emergency. Enterocolitis can lead to severe diarrhea and massive fluid loss, which can cause **death** from **dehydration** unless surgery is done immediately to relieve the obstruction.

Hirschsprung's disease sometimes occurs in children with other disorders of the autonomic nervous system, such as congenital central hypoventilation syndrome, a breathing disorder. Other syndromes associated with Hirschsprung disease include congenital

deafness and Waardenburg syndrome, a genetic disorder characterized by facial abnormalities and the loss of normal pigmentation in the hair, skin, and the iris of the eye.

Diagnosis

Hirschsprung's disease in the newborn must be distinguished from other causes of intestinal obstruction. The diagnosis is suspected by the child's medical history and physical examination, especially the rectal exam. The diagnosis is confirmed by a barium enema x ray, which shows a picture of the bowel. The x ray will indicate if a segment of bowel is constricted, causing dilation and obstruction. A biopsy of rectal tissue will reveal the absence of the nerve fibers. Adults may also undergo manometry, a balloon study (device used to enlarge the anus for the procedure) of internal anal sphincter pressure and relaxation.

Treatment

Hirschsprung's disease is treated surgically. The goal is to remove the diseased, nonfunctioning segment of the bowel and restore bowel function. This is often done in two stages. The first stage relieves the intestinal obstruction by performing a **colostomy**. This is the creation of an opening in the abdomen (stoma) through which bowel contents can be discharged into a waste bag. When the child's weight, age, or condition is deemed appropriate, surgeons close the stoma, remove the diseased portion of bowel, and perform a "pull-through" procedure, which repairs the colon by connecting functional bowel to the anus. The pull-through operation usually establishes fairly normal bowel function.

Children with total colonic aganglionosis occasionally fail to benefit from a pull-through procedure. One option in treating these patients is the construction of an ileoanal S-pouch.

The surgeon may recommend a permanent **ostomy** if the child has **Down syndrome** in addition to Hirschsprung disease, as these children usually have more difficulty with bowel control.

Prognosis

Overall, prognosis is very good. Most infants with Hirschsprung's disease achieve good bowel control after surgery, but a small percentage of children may have lingering problems with soilage or constipation. These infants are also at higher risk for an overgrowth of bacteria in the intestines, including subsequent

KEY TERMS

Anus—The opening at the end of the intestine that carries waste out of the body

Barium enema x ray—A procedure that involves the administration of barium into the intestines by a tube inserted into the rectum. Barium is a chalky substance that enhances the visualization of the gastrointestinal tract on x ray.

Colostomy—The creation of an artificial opening into the colon through the skin for the purpose of removing bodily waste. Colostomies are usually required because key portions of the intestine have been removed.

Enterocolitis—Severe inflammation of the intestines that affects the intestinal lining, muscle, nerves and blood vessels.

Manometry—A balloon study of internal anal sphincter pressure and relaxation.

Meconium—The first waste products to be discharged from the body in a newborn infant, usually greenish in color and consisting of mucus, bile and so forth.

Megacolon—Dilation of the colon.

Parasympathetic ganglion cell—Type of nerve cell normally found in the wall of the colon.

episodes of enterocolitis, and should be closely followed by a physician. Mortality from enterocolitis or surgical complications in infancy is 25–30%.

Prevention

Hirschsprung's disease is a congenital abnormality that has no known means of prevention. It is important to diagnose the condition early in order to prevent the development of enterocolitis. Genetic counseling can be offered to a couple with a previous child with the disease or to an affected individual considering pregnancy to discuss recurrence risks and treatment options. Prenatal diagnosis is not available as of the early 2000s.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD., editors. "Congenital Anomalies." Section 19, Chapter 261 In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

PERIODICALS

- Chen, M. L., and T. G. Keens. "Congenital Central Hypoventilation Syndrome: Not Just Another Rare Disorder." *Paediatric Respiratory Reviews* 5 (September 2004): 182–189.
- Lal, D. R., P. F. Nichol, B. A. Harms, et al. "Ileo-Anal S-Pouch Reconstruction in Patients with Total Colonic Aganglionosis after Failed Pull-Through Procedure." *Journal of Pediatric Surgery* 39 (July 2004): e7–e9.
- Martucciello, G., et al. "Pathogenesis of Hirschsprung's Disease." *Journal of Pediatric Surgery* 35 (2000): 1017–1025.
- Munnes, M., et al. "Familial Form of Hirschsprung Disease: Nucleotide Sequence Studies Reveal Point Mutations in the RET Proto-oncogene in Two of Six Families But Not in Other Candidate Genes." *American Journal of Medical Genetics* 94 (2000): 19–27.
- Neville, Holly, MD, and Charles S. Cox, Jr., MD. "Hirschsprung Disease." *eMedicine* July 15, 2003. < <http://www.emedicine.com/ped/topic1010.htm> > .
- Prabhakara, K., H. E. Wyandt, X. L. Huang, et al. "Recurrent Proximal 18p Monosomy and 18q Trisomy in a Family with a Maternal Pericentric Inversion of Chromosome 18." *Annales de génétique* 47 (July–September 2004): 297–303.

ORGANIZATIONS

- American Pseudo-Obstruction & Hirschsprung's Society.
158 Pleasant St., North Andover, MA 01845.
(978)685-4477.
- National Organization for Rare Disorders (NORD).
55 Kenosia Avenue, P. O. Box 1968, Danbury, CT 06813-1968. (203) 744-0100. Fax: (203) 798-2291.
< <http://www.rarediseases.org> > .
- Pull-thru Network. 316 Thomas St., Bessemer, AL 35020.
(205) 428-5953.

Amy Vance, MS, CGC
Rebecca J. Frey, PhD

Hirsutism

Definition

Excessive growth of facial or body hair in women is called hirsutism.

Description

Hirsutism is not a disease. The condition usually develops during **puberty** and becomes more pronounced as the years go by. However, an inherited tendency,

over-production of male hormones (androgens), medication, or disease, can cause it to appear at any age.

Women who have hirsutism usually have irregular menstrual cycles. They sometimes have small breasts and deep voices, and their muscles and genitals may become larger than women without the condition.

Types of hirsutism

Idiopathic hirsutism is probably hereditary, because there is usually a family history of the disorder. Women with idiopathic hirsutism have normal menstrual cycles and no evidence of any of the conditions associated with secondary hirsutism.

Secondary hirsutism is most often associated with polycystic ovary syndrome (an inherited hormonal disorder characterized by menstrual irregularities, biochemical abnormalities, and **obesity**). This type of hirsutism may also be caused by:

- malfunctions of the pituitary or adrenal glands
- use of male hormones or **minoxidil** (Loniten), a drug used to widen blood vessels
- adrenal or ovarian tumors.

Causes and symptoms

Hirsutism is rarely caused by a serious underlying disorder. **Pregnancy** occasionally stimulates its development. Hirsutism triggered by tumors is very unusual.

Hair follicles usually become enlarged, and the hairs themselves become larger and darker. A woman whose hirsutism is caused by an increase in male hormones has a pattern of hair growth similar to that of a man. A woman whose hirsutism is not hormone-related has long, fine hairs on her face, arms, chest, and back.

Diagnosis

Diagnosis is based on a family history of hirsutism, a personal history of menstrual irregularities, and masculine traits. Laboratory tests are not needed to assess the status of patients whose menstrual cycles are normal and who have mild, gradually progressing hirsutism.

A family physician or endocrinologist may order blood tests to measure hormone levels in women with long-standing menstrual problems or more severe hirsutism. **Computed tomography scans** (CT scans) are sometimes performed to evaluate diseases of the adrenal glands. Additional diagnostic procedures may be used to confirm or rule out underlying diseases or disorders.

KEY TERMS

Idiopathic—A term for a disease with no known cause, from the Greek stems *idio* (peculiar or separate) and *pathy* (disease).

Treatment

Primary hirsutism can be treated mechanically. Mechanical treatment involves bleaching or physically removing unwanted hair by:

- cutting
- electrolysis
- shaving
- tweezing
- waxing
- using hair-removing creams (depilatories)

Low-dose dexamethasone (a synthetic adrenocortical steroid), birth-control pills, or medications that suppress male hormones (for example, spironolactone) may be prescribed for patients whose condition stems from high androgen levels.

Treatment of secondary hirsutism is determined by the underlying cause of the condition.

Prognosis

Birth-control pills alone cause this condition to stabilize in one of every two patients and to improve in one of every 10.

When spironolactone (Aldactone) is prescribed to suppress hair growth, 70% of patients experience improvement within six months. When women also take birth-control pills, menstrual cycles become regular and hair growth is suppressed even more.

Resources

ORGANIZATIONS

American Society for Reproductive Medicine. 1209 Montgomery Highway, Birmingham, AL 35216-2809. (205) 978-5000. <<http://www.asrm.com>>.

Maureen Haggerty

Hispanic American health see **Minority health**

Histamine see **Antilucer drugs**

Histamine headache see **Cluster headache**

Histiocytosis X

Definition

Histiocytosis X is a generic term that refers to an increase in the number of histiocytes, a type of white blood cell, that act as scavengers to remove foreign material from the blood and tissues. Since recent research demonstrated Langerhan cell involvement as well as histiocytes, this led to a proposal that the term Langerhans Cell Histiocytosis (LCH) be used in place of histiocytosis X. Either term refers to three separate illnesses (listed in order of increasing severity): eosinophilic granuloma, Hand-Schuller-Christian disease and Letterer-Siwe disease.

Description

Epidermal (skin) Langerhans cells (a form of dendritic cell) accumulate with other immune cells in various parts of the body and cause damage by the release of chemicals. Normally, Langerhans cells recognize foreign material, including bacteria, and stimulate the immune system to react to them. Langerhans cells are usually found in skin, lymph nodes, lungs, and the gastrointestinal tract. Under abnormal conditions these cells affect skin, bone, and the pituitary gland as well as the lungs, intestines, liver, spleen, bone marrow, and brain. Therefore, the disease is not confined to areas where Langerhans cells are normally found. The disease is more common in children than adults and tends to be most severe in very young children.

Histiocytosis X or LCH is a family of related conditions characterized by a distinct inflammatory and proliferative process but differs from each other in which parts of the body are involved. The least severe of the histiocytosis X/LCH family is eosinophilic granuloma. Approximately 60–80% of all diagnosed cases are in this classification, which usually occurs in children aged 5–10 years. The bones are involved 50–75% of the time, which includes the skull or mandible, and the long bones. If the bone marrow is involved, anemia can result. With skull involvement, growths can occur behind the eyes, bulging them forward. One recent case study involved swelling of the eyes caused by histiocytosis in a three-year-old girl. The lungs are involved less than 10% of the time, and this involvement signals the worst prognosis.

Next in severity is Hand-Schuller-Christian disease, a chronic, scattered form of histiocytosis. It occurs most commonly from the age of one to three years and is a slowly progressive disease that affects the softened areas of the skull, other flat bones, the eyes, and skin.

Letterer-Siwe disease is the acute form of this series of diseases. It is generally found from the time of birth to one year of age. It causes an enlarged liver, bruising and skin lesions, anemia, enlarged lymph glands, other organ involvement, and extensive skull lesions.

Causes and symptoms

This is a rare disorder affecting approximately 1 in 200,000 children or adults each year. The International Histiocyte Society formed a registry in 2000 that has registered a total of 274 adults from 13 countries as of 2003. Because histiocytic disorders are so rare, little research has been done to determine their cause. Over time, histiocytosis may lessen in its assault on the body but there are still problems from damage to the tissues. There are no apparent inheritance patterns in these diseases with the exception of a form involving the lymphatic system; of the 274 adults in the international registry, only one came from a family with a history of the disease.

The symptoms of histiocytosis are caused by substances called cytokines and prostaglandins, which are normally produced by histiocytes and act as messengers between cells. When these chemicals are produced in excess amounts and in the wrong places, they cause tissue swelling and abnormal growth. Thus, symptoms may include painful lumps in the skull and limbs as well as **rashes** on the skin. General symptoms may include: poor appetite, failure to gain weight, recurrent **fever**, and irritability. Symptoms from other possible sites of involvement include:

- gums: swelling, usually without significant discomfort
- ear: chronic discharge
- liver or spleen: abdominal discomfort or swelling
- pituitary: This gland at the base of the brain is affected at some stage in approximately 20%–30% of children causing a disturbance in water balance to produce thirst and frequent urination.
- eyes: Due to the bony disease, behind-the-eye bulging may occur (exophthalmos)
- lungs: breathing problems

Diagnosis

The diagnosis can be made only by performing a biopsy, that is, taking a tissue sample under anesthesia from a site in the patient thought to be involved. Blood and urine tests, chest and other x rays, **magnetic resonance imaging** (MRI) and **computed tomography scans** (CAT scans) (to check the extent of involvement), and

KEY TERMS

Anemia—Abnormally low level of red blood cells in the blood.

Biopsy—Surgical removal of tissue for examination.

CT or CAT—Computed tomography, a radiologic imaging that uses computer processing to generate an image of tissue density in slices through the patient's body.

Cytokines—The term used to include all protein messengers that regulate immune responses.

Dendritic—Branched like a tree.

Eosinophils—A leukocyte with coarse, round granules present.

Epidermal—The outermost layer of the skin.

Inflammatory—A localized protective response of the body caused by injury or destruction of tissues.

MRI—Magnetic resonance imaging, a noninvasive nuclear procedure that uses electromagnetic energy to create images of structures inside the body.

Pituitary gland—The master gland located in the middle of the head that controls the endocrine glands and affects most bodily functions.

Prostaglandins—A group of nine naturally occurring chemicals in the body that affect smooth muscles.

Serous—Thin and watery, like serum.

possibly bone marrow or breathing tests may be required to confirm the diagnosis.

Treatment

Although this disease is not **cancer**, most patients diagnosed with it are treated in cancer clinics. There are two reasons for this:

- Historically, cancer specialists treated it before the cause was known.
- The treatment requires the use of drugs typically required to treat cancer.

Any cancer drugs utilized are usually given in smaller doses, which diminishes the severity of their side effects. **Radiation therapy** is rarely used, and special drugs may be prescribed for skin symptoms. If there is only one organ affected, steroids may be injected locally, or a drug called indomethacin may be used. Indomethacin is an anti-inflammatory medication

that may achieve a similar response with less severe side effects.

Prognosis

The disease fluctuates markedly. If only one system is involved, the disease often resolves by itself. Multisystem disease usually needs treatment although it may disappear spontaneously. The disease is not normally fatal unless organs vital to life are damaged. In general, the younger the child at diagnosis and the more organs involved, the poorer the outlook. If the condition resolves, there could still be long-term complications because of the damage done while the disease was active.

Resources

BOOKS

- Beers, Mark H., MD, and Robert Berkow, MD., editors. "Histiocytic Syndromes." Section 11, Chapter 137 In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.
- Behrman, Richard E., Robert Kliegman, and Hal B. Jenson, editors. *Nelson Textbook of Pediatrics*. Philadelphia: W. B. Saunders, 2000.

PERIODICALS

- Arico, M., M. Girschikofsky, T. Genereau, et al. "Langerhans Cell Histiocytosis in Adults. Report from the International Registry of the Histiocyte Society." *European Journal of Cancer* 39 (November 2003): 2341–2348.
- Eckhardt, A., and A. Schulze. "Maxillofacial Manifestations of Langerhans Cell Histiocytosis: A Clinical and Therapeutic Analysis of 10 Patients." *Oral Oncology* 39 (October 2003): 687–694.
- Kobyahsi, M., O. Yamamoto, Y. Suenaga, and M. Asahi. "Electron Microscopic Study of Langerhans Cell Histiocytosis." *Journal of Dermatology* July 27, 2000: 453–7.
- Levy, J., T. Monos, J. Kapelushnik, et al. "Langerhans Cell Histiocytosis with Periorbital Cellulitis." *American Journal of Ophthalmology* 136 (November 2003): 939–942.

ORGANIZATIONS

- Histiocytosis Association of America. 302 North Broadway, Pitman, NJ 08071. (800) 548–2758 (USA and Canada). <<http://www.histio.org>>.

OTHER

- "Immunity Disorders." *NurseMinerva*. June 26, 2001. <<http://nurseminerva.co.uk/immunity.htm>>.

Linda K. Bennington, CNS
Rebecca J. Frey, PhD

Histoplasmosis

Definition

Histoplasmosis is an infectious disease caused by inhaling the microscopic spores of the fungus *Histoplasma capsulatum*. The disease exists in three forms. Acute or primary histoplasmosis causes flu-like symptoms. Most people who are infected recover without medical intervention. Chronic histoplasmosis affects the lungs and can be fatal. Disseminated histoplasmosis affects many organ systems in the body and is often fatal, especially to people with acquired **immunodeficiency syndrome (AIDS)**.

Description

Histoplasmosis is an airborne infection. The spores that cause this disease are found in soil that has been contaminated with bird or bat droppings. In the United States, the disease is most common in eastern and midwestern states and is widespread in the upper Mississippi, Ohio, Missouri, and St. Lawrence river valleys. Sometimes histoplasmosis is called Ohio Valley disease, Central Mississippi River Valley disease, Appalachian Mountain disease, Darling's disease, or *Histoplasma capsulatum* infection.

Anyone can get histoplasmosis, but people who come in contact with bird and bat excrement are more likely to be infected. This includes farmers, gardeners, bridge inspectors and painters, roofers, chimney cleaners, demolition and construction workers, people installing or servicing heating and air conditioning units, people restoring old or abandoned buildings, and people who explore caves.

The very young and the elderly, especially if they have a pre-existing lung disease or are heavy smokers, are more likely to develop symptoms that are more severe. People who have a weakened immune system, either from diseases such as AIDS or leukemia, or as the result of medications they take (**corticosteroids**, **chemotherapy** drugs), are more likely to develop chronic or disseminated histoplasmosis.

Causes and symptoms

When the spores of *H. capsulatum* are inhaled, they lodge in the lungs where they divide and cause lesions. This is known as acute or primary histoplasmosis. It is not contagious.

Many otherwise healthy people show no symptoms of infection at all. When symptoms do occur, they appear 3–17 days after exposure (average time is 10 days).

The symptoms are usually mild and resemble those of a cold or flu; **fever**, dry cough, enlarged lymph glands, tiredness, and a general feeling of ill health. A small number of people develop bronchopneumonia. About 95% of people who are infected either experience no symptoms or have symptoms that clear up spontaneously. These people then have partial immunity to re-infection.

In some people, the spores that cause the disease continue to live in the lungs. In about 5% of people who are infected, usually those with chronic lung disease, diabetes mellitus, or weakened immune systems, the disease progresses to chronic histoplasmosis. This can take months or years. Symptoms of chronic histoplasmosis resemble those of **tuberculosis**. Cavities form in the lung tissue, parts of the lung may collapse, and the lungs fill with fluid. Chronic histoplasmosis is a serious disease that can result in death.

The rarest form of histoplasmosis is disseminated histoplasmosis. Disseminated histoplasmosis is seen almost exclusively in patients with AIDS or other immune defects. In disseminated histoplasmosis the infection may move to the spleen, liver, bone marrow, or adrenal glands. Symptoms include a worsening of those found in chronic histoplasmosis, as well as weight loss, **diarrhea**, the development of open sores in the mouth and nose, and enlargement of the spleen, liver, and adrenal gland.

Diagnosis

A simple skin test similar to that given for tuberculosis will tell if a person has previously been infected by the fungus *H. capsulatum*. Chest x rays often show lung damage caused by the fungus, but do not lead to a definitive diagnosis because the damage caused by other diseases has a similar appearance on the x ray. Diagnosis of chronic or disseminated histoplasmosis can be made by culturing a sample of sputum or other body fluids in the laboratory to isolate the fungus. The urine, blood serum, washings from the lungs, or cerebrospinal fluid can all be tested for the presence of an antigen produced in response to the infection. Most cases of primary histoplasmosis go undiagnosed.

Treatment

Acute primary histoplasmosis generally requires no treatment other than rest. Non-prescription drugs such as **acetaminophen** (Tylenol) may be used to treat **pain** and relieve fever. Avoiding smoke and using a cool air humidifier may ease chest pain.

Patients with an intact immune system who develop chronic histoplasmosis are treated with the

drug ketoconazole (Nizoral) or amphotericin B (Fungizone). Patients with suppressed immune systems are treated with amphotericin B, which is given intravenously. Because of its potentially toxic side effects, hospitalization is often required. The patient may also receive other drugs to minimize the side effects of the amphotericin B.

Patients with AIDS must continue to take the drug itraconazole (Sporonox) orally for the rest of their lives in order to prevent a relapse. If the patient can not tolerate itraconazole, the drug fluconazole (Diflucan) can be substituted.

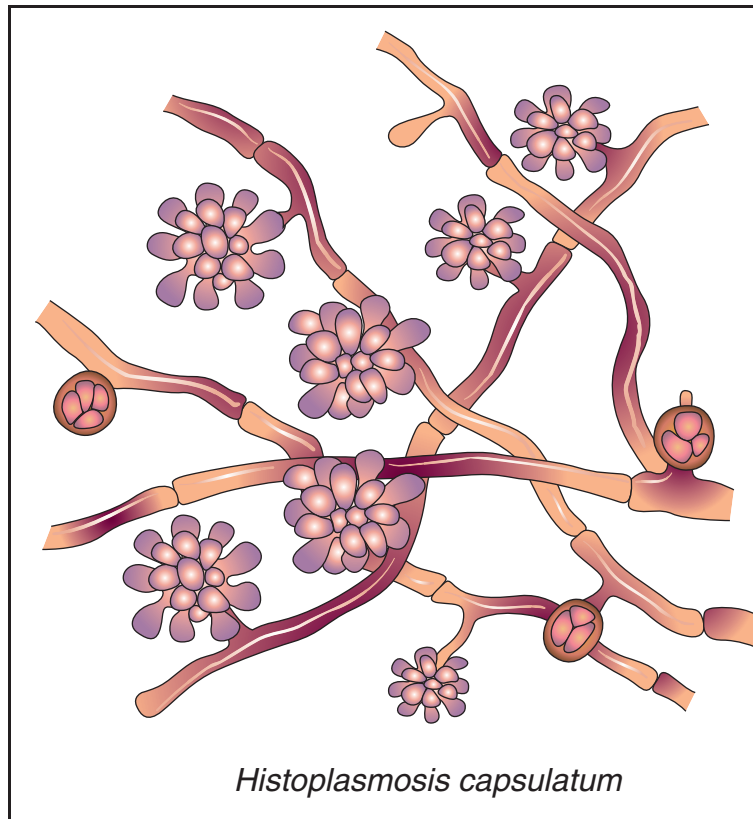
Alternative treatment

In non-immunocompromised patients, alternative therapies can be very successful. Alternative treatment for fungal infections focuses on creating an environment where the fungus cannot survive. This is accomplished by maintaining good health and eating a diet low in dairy products, sugars, including honey and fruit juice, and foods like beer that contain yeast. This is complemented by a diet high in raw food. Supplements of antioxidant **vitamins** C, E, and A, along with B complex, may also be added to the diet. *Lactobacillus acidophilus* and *Bifidobacteria* will replenish the good bacteria in the intestines. Antifungal herbs, like garlic, can be consumed in relatively large doses and for an extended period of time in order to be most effective.

Prognosis

Most people recover from primary histoplasmosis in a few weeks without medical intervention. Patients with chronic histoplasmosis who are treated with antifungal drugs generally recover rapidly if they do not have an underlying serious disease. When left untreated, or if serious disease is present, histoplasmosis can be fatal.

AIDS patients with disseminated histoplasmosis vary in their response to amphotericin B, depending on their general health and how well they tolerate the side effects of the drug. Treatment often suppresses the infection temporarily, but patients with AIDS are always in danger of a relapse and must continue to take medication for the rest of their lives to keep the infection at bay. New combinations of therapies and new drugs are constantly being evaluated, making hard statistics on prognosis difficult to come by. AIDS patients have problems with multiple opportunistic infections, making it difficult to isolate **death** rates due to any one particular fungal infection.



***Histoplasma capsulatum*.** If a person inhales the spores of this fungus, they may contract histoplasmosis, an infectious disease which can exist in three forms: acute or primary histoplasmosis, which causes flu-like symptoms; chronic histoplasmosis, which affects the lungs and can be fatal; and disseminated histoplasmosis, which can affect multiple body systems and is often fatal. (Illustration by Electronic Illustrators Group.)

KEY TERMS

Acidophilus—The bacteria *Lactobacillus acidophilus*, usually found in yogurt.

Adrenal gland—A pair of organs located above the kidneys. The outer tissue of the gland produces the hormones epinephrine (adrenaline) and norepinephrine, while the inner tissue produces several steroid hormones.

Antigen—A foreign protein to which the body reacts by making antibodies.

Bifidobacteria—A group of bacteria normally present in the intestine. Commercial supplements are available.

Corticosteroids—A group of hormones produced naturally by the adrenal gland or manufactured synthetically. They are often used to treat inflammation. Examples include cortisone and prednisone.

Prevention

Since the spores of *H. capsulatum* are so widespread, it is almost impossible to prevent exposure in endemic areas. Dust suppression measures when working with contaminated soil may help limit exposure. Individuals who are at risk of developing the more severe forms of the disease should avoid situations where they will be exposed to bat and bird droppings.

Resources

ORGANIZATIONS

American Lung Association. 1740 Broadway, New York, NY 10019. (800) 586-4872. <<http://www.lungusa.org>>.

Histoplasmosis: Protecting Workers at Risk. Centers for Disease Control and Prevention. <<http://www.cdc.gov/niosh/97146eng.html>>.

National Center for Infectious Diseases. Atlanta, Georgia. (404) 639-3158. <<http://www.cdc.gov/ncidod/ncid/ncid.htm>>.

National Institute for Occupational Safety and Health.
Cincinnati, Ohio. (800) 356-4674.

OTHER

Histoplasmosis: Protecting Workers at Risk. Centers for Disease Control and Prevention.
< <http://www.cdc.gov/niosh/97146eng.html> >.

Tish Davidson, A.M.

HIV infection see **AIDS**

Hives

Definition

Hives is an allergic skin reaction causing localized redness, swelling, and **itching**.

Description

Hives is a reaction of the body's immune system that causes areas of the skin to swell, itch, and become reddened (wheals). When the reaction is limited to small areas of the skin, it is called "urticaria." Involvement of larger areas, such as whole sections of a limb, is called "angioedema."

Causes and symptoms

Causes

Hives is an allergic reaction. The body's immune system is normally responsible for protection from foreign invaders. When it becomes sensitized to normally harmless substances, the resulting reaction is called an allergy. An attack of hives is set off when such a substance, called an allergen, is ingested, inhaled, or otherwise contacted. It interacts with immune cells called mast cells, which reside in the skin, airways, and digestive system. When mast cells encounter an allergen, they release histamine and other chemicals, both locally and into the bloodstream. These chemicals cause blood vessels to become more porous, allowing fluid to accumulate in tissue and leading to the swollen and reddish appearance of hives. Some of the chemicals released sensitize pain nerve endings, causing the affected area to become itchy and sensitive.

A wide variety of substances may cause hives in sensitive people, including foods, drugs, and insect bites or stings. Common culprits include:



Hives on the back of a young woman's legs. The accompanying inflammation develops as an allergic reaction which ranges in size from small spots to patches measuring several inches across. (Photograph by John Radcliffe, Custom Medical Stock Photo. Reproduced by permission.)

- nuts, especially peanuts, walnuts, and Brazil nuts
- fish, mollusks, and shellfish
- eggs
- wheat
- milk
- strawberries
- food additives and preservatives
- penicillin or other **antibiotics**
- flu vaccines
- tetanus toxoid vaccine
- gamma globulin
- bee, wasp, and hornet stings
- bites of mosquitoes, fleas, and scabies

Symptoms

Urticaria is characterized by redness, swelling, and itching of small areas of the skin. These patches usually grow and recede in less than a day, but may be replaced by hives in other locations. Angioedema is characterized by more diffuse swelling. Swelling of the airways may cause **wheezing** and respiratory distress. In severe cases, airway obstruction may occur.

KEY TERMS

Allergen—A substance capable of producing an immediate type of hypersensitivity, or allergy.

Wheal—A smooth, slightly elevated area on the body surface, which is redder or paler than the surrounding skin.

Diagnosis

Hives are easily diagnosed by visual inspection. The cause of hives is usually apparent, but may require a careful medical history in some cases.

Treatment

Mild cases of hives are treated with **antihistamines**, such as diphenhydramine (Benadryl) or desloratadine (Clarinet). Clarinet is non-sedating, meaning it will not make patients drowsy. More severe cases may require oral **corticosteroids**, such as prednisone. Topical corticosteroids are not effective. Airway swelling may require emergency injection of epinephrine (adrenaline).

Alternative treatment

An alternative practitioner will try to determine what allergic substance is causing the reaction and help the patient eliminate or minimize its effects. To deal with the symptoms of hives, an oatmeal bath may help to relieve itching. Chickweed (*Stellaria media*), applied as a poultice (crushed or chopped herbs applied directly to the skin) or added to bath water, may also help relieve itching. Several homeopathic remedies, including *Urtica urens* and *Apis* (*Apis mellifica*), may help relieve the itch, redness, or swelling associated with hives.

Prognosis

Most cases of hives clear up within one to seven days without treatment, providing the cause (allergen) is found and avoided.

Prevention

Preventing hives depends on avoiding the allergen causing them. Analysis of new items in the diet or new drugs taken may reveal the likely source of the reaction. Chronic hives may be aggravated by **stress**, caffeine, alcohol, or tobacco; avoiding these may reduce the frequency of reactions.

Resources

PERIODICALS

Kirn, F. Timothy. "Desloratadine Improves Urticaria in Clinical Setting." *Skin & Allergy News* September 2004: 41.

Richard Robinson
Teresa G. Odle

HLA-B27 antigen test see **Tissue typing**

HLA test see **Human leukocyte antigen test**

HMG-CoA reductase inhibitors see
Cholesterol-reducing drugs

Hodgkin's disease

Hodgkin's disease is a rare lymphoma, a **cancer** of the lymphatic system.

Hodgkin's disease, or Hodgkin's lymphoma, was first described in 1832 by Thomas Hodgkin, a British physician. Hodgkin clearly differentiated between this disease and the much more common non-Hodgkin's lymphomas. Prior to 1970, few individuals survived Hodgkin's disease. Now, however, the majority of individuals with this cancer can be cured.

The lymphatic system is part of the body's immune system, for fighting disease, and a part of the blood-producing system. It includes the lymph vessels and nodes, and the spleen, bone marrow, and thymus. The narrow lymphatic vessels carry lymphatic fluid from throughout the body. The lymph nodes are small organs that filter the lymphatic fluid and trap foreign substances, including viruses, bacteria, and cancer cells. The spleen, in the upper left abdomen, removes old cells and debris from the blood. The bone marrow, the tissue inside the bones, produces new red and white blood cells.

Lymphocytes are white blood cells that recognize and destroy disease-causing organisms. Lymphocytes are produced in the lymph nodes, spleen, and bone marrow. They circulate throughout the body in the blood and lymphatic fluid. Clusters of immune cells also exist in major organs.

Hodgkin's disease is a type of lymphoma in which antibody-producing cells of the lymphatic system begin to grow abnormally. It usually begins in a lymph node and progresses slowly, in a fairly

DOROTHY MENDENHALL (1874–1964)

Dorothy Reed Mendenhall, the last of three children, was born September 22, 1874, in Columbus, Ohio, to William Pratt Reed, a shoe manufacturer, and Grace Kimball Reed, both of whom had descended from English settlers who came to America in the seventeenth century. Mendenhall attended Smith College and obtained a baccalaureate degree. Although she initially contemplated a career in journalism, Mendenhall's interest in medicine was inspired by a biology course she attended.

Dorothy Reed Mendenhall was a well-respected researcher, obstetrician, and pioneer in methods of child-birth. She was the first to discover that Hodgkin's disease was not a form of tuberculosis, as had been thought. This finding received international acclaim. As a result of her work, the cell type characteristic of Hodgkin's disease bears her name. The loss of her first child due to poor obstetrics changed her research career to a lifelong effort to reduce infant mortality rates. Mendenhall's efforts paid off with standards being set for weight and height for children ages birth to six and also in programs that stressed the health of both the mother and child in the birthing process.

predictable way, spreading via the lymphatic vessels from one group of lymph nodes to the next. Sometimes it invades organs that are adjacent to the lymph nodes. If the cancer cells spread to the blood, the disease can reach almost any site in the body. Advanced cases of Hodgkin's disease may involve the spleen, liver, bone marrow, and lungs.

There are different subtypes of Hodgkin's disease:

- nodular sclerosis (30–60% of cases)
- mixed cellularity (20–40% of cases)
- lymphocyte predominant (5–10% of cases)
- lymphocyte depleted (less than 5% of cases)
- unclassified

The American Cancer Society estimates that there will be 7,400 new cases of Hodgkin's disease in the United States in 2001—3,500 in females and 3,900 in males. It is estimated that 700 men and 600 women in the United States will die of the disease in 2001.

Hodgkin's disease can occur at any age. However, the majority of cases develop in early adulthood (ages 15–40) and late adulthood (after age 55). Approximately 10–15% of cases are in children under age 17. It is more common in boys than in girls

under the age of 10. The disease is very rare in children under five.

The cause of Hodgkin's disease is not known. It is suspected that some interaction between an individual's genetic makeup, environmental exposures, and infectious agents may be responsible. Immune system deficiencies also may be involved.

Early symptoms of Hodgkin's disease may be similar to those of the flu:

- fevers, night sweats, chills
- **fatigue**
- loss of appetite
- weight loss
- itching
- pain after drinking alcoholic beverages
- swelling of one or more lymph nodes

Sudden or emergency symptoms of Hodgkin's disease include:

- sudden high fever
- loss of bladder and/or bowel control
- numbness in the arms and legs and a loss of strength

As lymph nodes swell, they may push on other structures, causing a variety of symptoms:

- pain due to pressure on nerve roots
- loss of function in muscle groups served by compressed nerves
- coughing or **shortness of breath** due to compression of the windpipe and/or airways, by swollen lymph nodes in the chest
- kidney failure from compression of the ureters, the tubes that carry urine from the kidneys to the bladder
- swelling in the face, neck, or legs, due to pressure on veins
- paralysis in the legs due to pressure on the spinal cord

As Hodgkin's disease progresses, the immune system becomes less effective at fighting infection. Thus, patients with Hodgkin's lymphoma become more susceptible to both common infections caused by bacteria and unusual (opportunistic) infections. Later symptoms of Hodgkin's disease include the formation of tumors.

Significantly, as many as 75% of individuals with Hodgkin's disease do not have any typical symptoms.

As with many forms of cancer, diagnosis of Hodgkin's disease has two major components.

- identification of Hodgkin's lymphoma as the cause of the patient's disease
- staging of the disease to determine how far the cancer has spread

The initial diagnosis of Hodgkin's disease often results from abnormalities in a **chest x ray** that was performed because of nonspecific symptoms. The physician then takes a medical history to check for the presence of symptoms and conducts a complete **physical examination**.

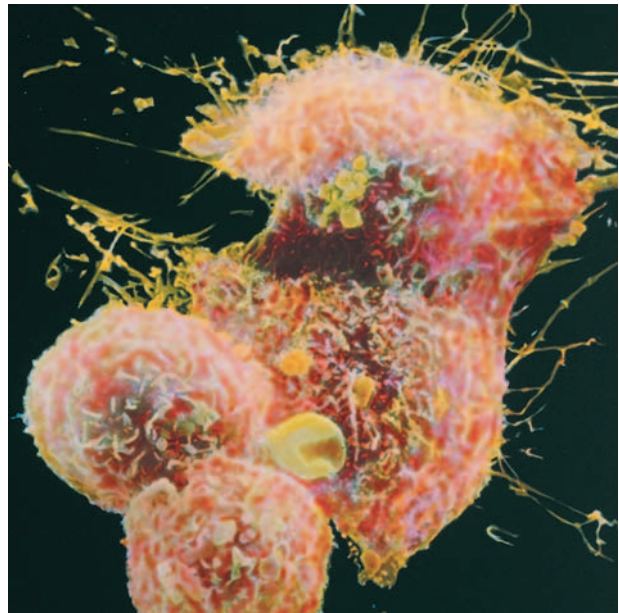
The size, tenderness, firmness, and location of swollen lymph nodes are determined and correlated with any signs of infection. In particular, lymph nodes that do not shrink after treatment with **antibiotics** may be a cause for concern. The lymph nodes that are most often affected by Hodgkin's disease include those of the neck, above the collarbone, under the arms, and in the chest above the diaphragm.

Diagnosis of Hodgkin's disease requires either the removal of an entire enlarged lymph node (an excisional biopsy) or an incisional biopsy, in which only a small part of a large tumor is removed. If the node is near the skin, the biopsy is performed with a local anesthetic. However, if it is inside the chest or abdomen, **general anesthesia** is required.

The sample of biopsied tissue is examined under a microscope. Giant cells called Reed-Sternberg cells must be present to confirm a diagnosis of Hodgkin's disease. These cells, which usually contain two or more nuclei, are named for the two pathologists who discovered them. Normal cells have only one nucleus (the organelle within the cell that contains the genetic material). Affected lymph nodes may contain only a few Reed-Sternberg cells and they may be difficult to recognize. Characteristics of other types of cells in the biopsied tissue help to diagnose the subtype of Hodgkin's disease.

A fine needle aspiration (FNA) biopsy, in which a thin needle and syringe are used to remove a small amount of fluid and bits of tissue from a tumor, has the advantage of not requiring surgery. An FNA may be performed prior to an excisional or incisional biopsy, to check for infection or for the spread of cancer from another organ. However an FNA biopsy does not provide enough tissue to diagnose Hodgkin's disease.

Occasionally, additional biopsies are required to diagnose Hodgkin's disease. In rare instances, other tests, that detect certain substances on the surfaces of cancer cells or changes in the DNA of cells, are used to distinguish Hodgkin's disease from non-Hodgkin's lymphoma.



A scanning electron micrograph (SEM) image of dividing Hodgkin's cells from the pleural effusions (abnormal accumulations of fluid in the lungs) of a 55-year-old male patient. (Photograph by Dr. Andrejs Liepins, Photo Researchers, Inc. Reproduced by permission.)

Staging is very important in Hodgkin's disease. This is because the cancer usually spreads in a predictable pattern, without skipping sets of lymph nodes until late in the progression of the disease.

Imaging of the abdomen, chest, and pelvis is used to identify areas of enlarged lymph nodes and abnormalities in the spleen or other organs. Computerized axial tomography (CT or CAT) scans use a rotating x-ray beam to obtain pictures. Magnetic resonance imaging (MRI) uses magnetic fields and radio waves to produce images of the body. Chest x rays also may be taken. These images will reveal rounded lumps called nodules in the affected lymph nodes and other organs.

Another imaging technique for Hodgkin's disease is a gallium scan, in which the radioactive element gallium is injected into a vein. The cancer cells take up the gallium and a special camera that detects the gallium is used to determine the location and size of tumors. Gallium scans are used when Hodgkin's disease is in the chest and may be hard to detect by other methods. Gallium scans also are used to monitor progress during treatment.

A lymphangiogram, a radiograph of the lymphatic vessels, involves injecting a dye into a lymphatic vessel in the foot. Tracking of the dye locates the

disease in the abdomen and pelvis. This method is used less frequently and is usually not used with children.

Positron emission tomography (PET) scans are an extremely accurate method for staging Hodgkin's disease. A very low dose of radioactive glucose, a sugar, is injected into the body. The glucose travels to metabolically active sites, including cancerous regions that require large amounts of glucose. The PET scan detects the radioactivity and produces images of the entire body that distinguish between cancerous and non-cancerous tissues.

Anemia (a low red-blood-cell count), fevers, or night sweats are indications that Hodgkin's disease may be in the bone marrow. In these cases, a bone-marrow biopsy, in which a large needle is used to remove a narrow, cylindrical piece of bone, may be necessary to determine the spread of the cancer. Alternatively, an aspiration, in which a needle is used to remove small bits of bone marrow, may be used. The marrow usually is removed from the back of the hip or other large bone.

Sometimes further staging, called pathological staging or a staging laparotomy, is used for Hodgkin's disease. In this operation, a surgeon checks the abdominal lymph nodes and other organs for cancer and removes small pieces of tissue. A pathologist examines the tissue samples for Hodgkin's disease cells. Usually the spleen is removed (a **splenectomy**) during the laparotomy. The splenectomy helps with staging Hodgkin's disease, as well as removing a disease site.

All of the available treatments for Hodgkin's disease have serious side effects, both short and long-term. However, with accurate staging, physicians and patients often can choose the minimum treatment that will cure the disease. The staging system for Hodgkin's disease is the Ann Arbor Staging Classification, also called the Cotswold System or the Revised Ann Arbor System.

Hodgkin's disease is divided into four stages, with additional substages:

- Stage I: The disease is confined to one lymph node area
- Stage IE: The disease extends from the one lymph node area to adjacent regions
- Stage II: The disease is in two or more lymph node areas on one side of the diaphragm (the muscle below the lungs)
- Stage IIE: The disease extends to adjacent regions of at least one of these nodes

- Stage III: The disease is in lymph node areas on both sides of the diaphragm
- Stage IIIE/IIISE: The disease extends into adjacent areas or organs (IIIE) and/or the spleen (IIISE)
- Stage IV: The disease has spread from the lymphatic system to one or more other organs, such as the bone marrow or liver

Treatment for Hodgkin's disease depends both on the stage of the disease and whether or not symptoms are present. Stages are labeled with an A if no symptoms are present. If symptoms are present, the stage is labeled with a B. These symptoms include:

- loss of more than 10% of body weight over the previous six months
- fevers above 100 (37.70 C) degrees F
- drenching night sweats

Radiation therapy and/or **chemotherapy** (drug therapy) are the standard treatments for Hodgkin's disease. If the disease is confined to one area of the body, radiotherapy is usually used. This treatment, with x rays or other high-energy rays, also is used when the disease is in bulky areas such as the chest, where chemotherapeutic drugs cannot reach all of the cancer. External-beam radiation, a focused beam from an external machine, is used to irradiate only the affected lymph nodes. This procedure is called involved field radiation.

More advanced stages of Hodgkin's disease may be treated with mantle field radiation, in which the lymph nodes of the neck, chest, and underarms are irradiated. Inverted Y field radiation is used to irradiate the spleen and the lymph nodes in the upper abdomen and pelvis. Total nodal irradiation includes both mantle field and inverted Y field radiation.

Since external-beam radiation damages healthy tissue near the cancer cells, the temporary side effects of radiotherapy can include sunburn-like skin damage, fatigue, **nausea**, and **diarrhea**. Other temporary side effects may include a sore throat and difficulty swallowing. Long-term side effects depend on the dose and the location of the radiation and the age of the patient. Since radiation of the ovaries causes permanent sterility (the inability to have offspring), the ovaries of girls and young women are protected during radiotherapy. Sometimes the ovaries are surgically moved from the region to be irradiated.

If the Hodgkin's disease has progressed to additional lymph nodes or other organs, or if there is a recurrence of the disease within two years of radiation treatment, chemotherapy is used.

Chemotherapy utilizes a combination of drugs, each of which kills cancer cells in a different way. The most common chemotherapy regimens for Hodgkin's disease are MOPP (either mechlorethamine or methotrexate with Oncovin, procarbazine, prednisone) and ABVD (Adriamycin or doxorubicin, bleomycin, vincristine, dacarbazine). Each of these consists of four different drugs. ABVD is used more frequently than MOPP because it has fewer severe side effects. However MOPP is used for individuals who are at risk for heart failure. The chemotherapeutic drugs may be injected into a vein or muscle, or taken orally, as a pill or liquid.

Children who are sexually mature when they develop Hodgkin's disease, and whose muscle and bone mass are almost completely developed, usually receive the same treatment as adults. Younger children usually are treated with chemotherapy, since radiation will adversely affect bone and muscle growth. However, radiation may be used in low dosages, in combination with chemotherapy. The chemotherapy for children with Hodgkin's disease usually includes more drugs than ABVD and MOPP.

The side effects of chemotherapy for Hodgkin's disease depend on the dose of drugs and the length of time they are taken. Since these drugs target rapidly dividing cancer cells, they also affect normal cells that grow rapidly. These include the cells of the bone marrow, the linings of the mouth and intestines, and hair follicles. Damage to bone marrow leads to lower white blood cell counts and lower resistance to infection. It also leads to lower red blood cell counts, which can result in fatigue and easy bleeding and bruising. Damage to intestinal cells leads to a loss of appetite, nausea, and **vomiting**. Mouth sores and hair loss also are common side effects of chemotherapy. These side effects disappear when the chemotherapy is discontinued. Some drugs can reduce or prevent the **nausea and vomiting**.

Chemotherapy for Hodgkin's disease may lead to long-term complications. The drugs may damage the heart, lungs, kidneys, and liver. In children, growth may be impeded. Some chemotherapy can cause sterility, so men may choose to have their sperm frozen prior to treatment. Women may stop ovulating and menstruating during chemotherapy. This may or may not be permanent.

Treatment for higher-stage Hodgkin's disease often involves a combination of radiotherapy and chemotherapy. Following three or four chemotherapy regimens, involved field radiation may be directed at the most affected areas of the body. The long-term side

effects often are more severe when radiation and chemotherapy are used in combination.

The development of a second type of cancer is the most serious risk from radiation and chemotherapy treatment for Hodgkin's disease. In particular, there is a risk of developing leukemia, **breast cancer**, bone cancer, or thyroid cancer. Chemotherapy, particularly MOPP, or chemotherapy in conjunction with radiotherapy, significantly increases the risk for leukemia.

Following treatment, the original diagnostic tests for Hodgkin's disease are repeated, to determine whether all traces of the cancer have been eliminated and to check for long-term side effects of treatment. In resistant Hodgkin's disease, some cancer cells remain following treatment. If the cancer continues to spread during treatment, it is called progressive Hodgkin's disease. If the disease returns after treatment, it is known as recurrent Hodgkin's disease. It may recur in the area where it first started or elsewhere in the body. It may recur immediately after treatment or many years later.

Additional treatment is necessary with these types of Hodgkin's disease. If the initial treatment was radiation therapy alone, chemotherapy may be used, or vice versa. Chemotherapy with different drugs, or higher doses, may be used to treat recurrent Hodgkin's. However, radiation to the same area is never repeated.

An autologous bone marrow and/or a peripheral blood **stem cell transplantation** (PBSCT) often is recommended for treating resistant or recurrent Hodgkin's disease, particularly if the disease recurs within a few months of a chemotherapy-induced remission. These transplants are autologous because they utilize the individual's own cells. The patient's bone marrow cells or peripheral blood stem cells (immature bone marrow cells found in the blood) are collected and frozen prior to high-dosage chemotherapy, which destroys bone marrow cells. A procedure called leukapheresis is used to collect the stem cells. Following the high-dosage chemotherapy, and possibly radiation, the bone marrow cells or stem cells are reinjected into the individual.

Most complementary therapies for Hodgkin's disease are designed to stimulate the immune system to destroy cancer cells and repair normal cells that have been damaged by treatment. These therapies are used in conjunction with standard treatment.

Immunologic therapies, also known as immunotherapies, biological therapies, or biological response modifier therapies, utilize substances that are produced by the immune system. These include interferon (an immune system protein), monoclonal antibodies

KEY TERMS

Antibody—An immune system protein that recognizes a specific foreign molecule.

Biopsy—The removal of a small sample of tissue for examination under a microscope; used for the diagnosis of cancer and to check for infection.

Bone marrow—Tissue inside the bones that produce red and white blood cells.

Chemotherapy—Treatment with various combinations of chemicals or drugs, particularly for the treatment of cancer.

Epstein-Barr virus (EBV)—Very common virus that infects immune cells and can cause mononucleosis.

Interferon—A potent immune-defense protein produced by viral-infected cells; used as an anti-cancer and anti-viral drug.

Interleukins—A family of potent immune-defense molecules; used in various medical therapies.

Laparotomy—A surgical incision of the abdomen.

Leukapheresis—A technique that uses a machine to remove stem cells from the blood; the cells are frozen and then returned to the patient following treatment that has destroyed the bone marrow.

Lymph nodes—Small round glands, located throughout the body and containing lymphocytes that remove foreign organisms and debris from the lymphatic fluid.

Lymphatic system—The vessels, lymph nodes, and organs, including the bone marrow, spleen, and

thymus, that produce and carry white blood cells to fight disease.

Lymphocyte—White blood cells that produce antibodies and other agents for fighting disease.

PBSCT—Peripheral blood stem cell transplant; a method for replacing blood-forming cells that are destroyed by cancer treatment.

Radiotherapy—Disease treatment involving exposure to x rays or other types of radiation.

Reed-Sternberg cells—An abnormal lymphocyte that is characteristic of Hodgkin's disease.

Spleen—An organ of the lymphatic system, on the left side of the abdomen near the stomach; it produces and stores lymphocytes, filters the blood, and destroys old blood cells.

Splenectomy—Surgical removal of the spleen.

Staging—The use of various diagnostic methods to accurately determine the extent of disease; used to select the appropriate type and amount of treatment and to predict the outcome of treatment.

Stem cells—The cells from which all blood cells are derived.

Thymus—An organ of the lymphatic system, located behind the breast bone, that produces the T lymphocytes of the immune system.

Thyroid—A gland in the throat that produces hormones that regulate growth and metabolism.

(specially engineered antibodies), colony-stimulating (growth) factors (such as filgrastim), and vaccines. Many immunotherapies for Hodgkin's disease are experimental and available only through clinical trials. These biological agents may have side effects.

Coenzyme Q10 (CoQ10) and polysaccharide K (PSK) are being evaluated for their ability to stimulate the immune system and protect healthy tissue, as well as possible anti-cancer activities. Camphor, also known as 714-X, green tea, and hoxsey (which is a mixture of a number of substances), have been promoted as immune system enhancers. However there is no evidence that they are effective against Hodgkin's disease. Hoxsey, in particular, can produce serious side effects.

Hodgkin's disease, particularly in children, is one of the most curable forms of cancer. Approximately 90% of individuals are cured of the disease with chemotherapy and/or radiation.

The one-year relative survival rate following treatment for Hodgkin's disease is 93%. Relative survival rates do not include individuals who die of causes other than Hodgkin's disease. The percentage of individuals who have not died of Hodgkin's disease within five years of diagnosis is 90–95% for those with stage I or stage II disease. The figure is 85–90% for those diagnosed with stage III Hodgkin's and approximately 80% for those diagnosed with stage IV disease. The 15-year relative survival rate is 63%. Approximately 75% of children are alive and cancer free 20 years after the original diagnosis of Hodgkin's.

Acute myelocytic leukemia, a very serious cancer, may develop in as many as 2–6% of individuals receiving certain types of treatment for Hodgkin's disease. Women under the age of 30 who are treated with radiation to the chest have a much higher risk for developing breast cancer. Both men and women are

at higher risk for developing lung or thyroid cancers as a result of chest irradiation.

Individuals with the type of Hodgkin's disease known as nodular lymphocytic predominance have a 2% chance of developing non-Hodgkin's lymphoma. Apparently, this is a result of the Hodgkin's disease itself and not the treatment.

Rosalyn Carson-DeWitt, MD
Margaret Alic, Ph.D.

Holistic medicine

Definition

Holistic medicine is a term used to describe therapies that attempt to treat the patient as a whole person. That is, instead of treating an illness, as in orthodox allopathy, holistic medicine looks at an individual's overall physical, mental, spiritual, and emotional well-being before recommending treatment. A practitioner with a holistic approach treats the symptoms of illness as well as looking for the underlying cause of the illness. Holistic medicine also attempts to prevent illness by placing a greater emphasis on optimizing health. The body's systems are seen as interdependent parts of the person's whole being. Its natural state is one of health, and an illness or disease is an imbalance in the body's systems. Holistic therapies tend to emphasize proper nutrition and avoidance of substances—such as chemicals—that pollute the body. Their techniques are non-invasive.

Some of the world's health systems that are holistic in nature include **naturopathic medicine**, homeopathy, and traditional Chinese medicine. Many alternative or natural therapies have a holistic approach, although that is not always the case. The term complementary medicine is used to refer to the use of both allopathic and holistic treatments. It is more often used in Great Britain, but is gaining acceptance in the United States.

There are no limits to the range of diseases and disorders that can be treated in a holistic way, as the principle of holistic healing is to balance the body, mind, spirit, and emotions so that the person's whole being functions smoothly. When an individual seeks holistic treatment for a particular illness or condition, other health problems improve without direct treatment, due to improvement in the performance of the immune system, which is one of the goals of holistic medicine.

Origins

The concept of holistic medicine is not new. In the 4th century B.C., Socrates warned that treating one part of the body only would not have good results. Hippocrates considered that many factors contribute to the health or otherwise of a human being, weather, **nutrition**, emotional factors, and in our time, a host of different sources of pollution can interfere with health. And of course, holistic medicine existed even before ancient Greece in some ancient healing traditions, such as those from India and China, which date back over 5,000 years. However, the term "holistic" only became part of everyday language in the 1970s, when Westerners began seeking an alternative to allopathic medicine.

Interestingly, it was only at the beginning of the twentieth century that the principles of holistic medicine fell out of favor in Western societies, with the advent of major advances in what we now call allopathic medicine. Paradoxically, many discoveries of the twentieth century have only served to confirm many natural medicine theories. In many cases, researchers have set out to debunk holistic medicine, only to find that their research confirms it, as has been the case, for example, with many herbal remedies.

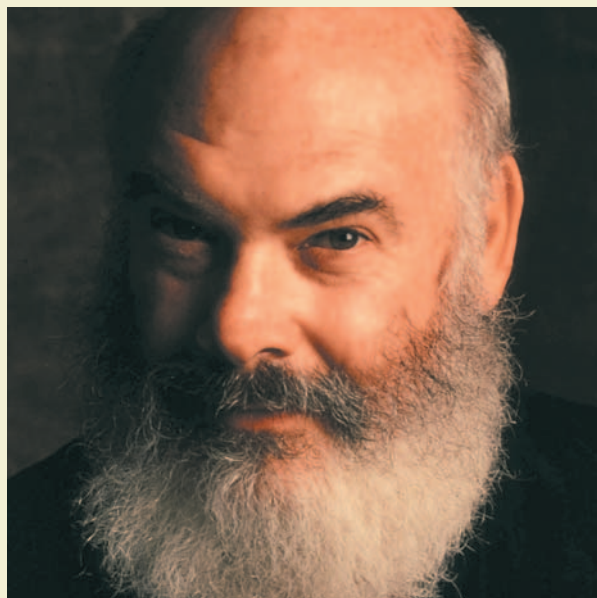
Purpose

Many people are now turning to holistic medicine, often when suffering from chronic ailments that have not been successfully treated by allopathic means. Although many wonderful advances and discoveries have been made in modern medicine, surgery and drugs alone have a very poor record for producing optimal health because they are designed to attack illness. Holistic medicine is particularly helpful in treating chronic illnesses and maintaining health through proper nutrition and **stress** management.

Description

There are a number of therapies that come under the umbrella of "holistic medicine." They all use basically the same principles, promoting not only physical health, but also mental, emotional, and spiritual health. Most emphasize quality nutrition. Refined foods typically eaten in modern America contain chemical additives and preservatives, are high in fat, cholesterol, and sugars, and promote disease. Alternative nutritionists counter that by recommending whole foods whenever possible and minimizing the amount of meat—especially red meat—that is consumed. Many alternative therapies promote vegetarianism as a method of **detoxification**.

ANDREW WEIL (1942–)



(Photograph by M. Greenberg. Gamma Liaison. Reproduced by permission.)

Dr. Andrew Weil, a Harvard-educated physician, adds credibility and expertise to the natural healing methods he espouses in his best-selling books, on his Internet Web site, in his talk show appearances, and in his popular audio CD of music and **meditation**. Weil's *Spontaneous Healing* spent more than a year on the best-seller list, and his 1997 book, *Eight Weeks to Optimum Health*, also was

a runaway best-seller. Perhaps the best-known proponent of naturalistic healing methods, Weil has been trying to establish a field he calls integrative medicine. He is director of Tucson's Center for Integrative Medicine, which he founded in 1993. In 1997, he began training doctors in the discipline at the University of Arizona, where he teaches.

After getting his bachelor's degree in botany from Harvard University, Weil applied for admission to Harvard Medical School in 1964. During his second year, he led a group of students who argued they could succeed better studying on their own than going to classes; in fact, the group got higher scores on their final exams than their classmates. After graduating from Harvard Medical School, he volunteered at the notorious counter-cultural Haight-Asbury Free Clinic in San Francisco, CA. Later in 1969, Weil got a job in Washington, DC, with the National Institute of Mental Health's Drug Studies Division. From 1971 to 1975, he traveled extensively in South America and Africa, soaking up information about medicinal plants, **shamanism**, and natural healing techniques. He never returned to the practice of conventional medicine.

His approach to alternative medicine is eclectic, mingling traditional medicine with herbal therapy, **acupuncture**, **homeopathy**, **chiropractic**, hypnotism, cranial manipulation, and other alternative healing methods. Though his books discuss the benefits of everything from healing touch to herbal cures, Weil doesn't dismiss the benefits of standard Western medicine when appropriate.

The aim of holistic medicine is to bring all areas of an individual's life, and most particularly the energy flowing through the body, back into harmony. Ultimately, of course, only the patient can be responsible for this, for no practitioner can make the necessary adjustments to diet and lifestyle to achieve health. The practice of holistic medicine does not rule out the practice of allopathic medicine; the two can complement each other.

A properly balanced holistic health regimen, which takes into consideration all aspects of human health and includes noninvasive and nonpharmaceutical healing methods, can often completely eradicate even acute health conditions safely. If a patient is being treated with allopathic medicine, holistic therapies may at least support the body during treatment, and alleviate the symptoms that often come with drug treatments and surgery. In addition, holistic therapies aim at the underlying source of the illness, to prevent recurrence.

Here are some of the major holistic therapies:

- herbal medicine
- homeopathy
- naturopathic medicine
- **traditional Chinese medicine**
- Ayurvedic medicine
- nutritional therapies
- **chiropractic**
- stress reduction
- psychotherapy
- massage

Because holistic medicine aims to treat the whole person, holistic practitioners sometimes may advise treatment from more than one type of practitioner. This is to ensure that all aspects of health are

addressed. Some practitioners also specialize in more than one therapy, and so may be able to offer more comprehensive assistance.

Preparations

How to choose a holistic practitioner

- How did you hear of this therapist? A personal referral can sometimes be more reliable than a professional one. What do other professionals say about this therapist? What qualifications, board certification, or affiliations does this practitioner have?
- How do you feel personally about this practitioner? Do you feel comfortable in his/her office and with his/her staff? Is your sense of well being increased? Are you kept waiting for appointments?
- Do you have confidence in this practitioner, does he/she respect you as a person? Does he/she show an interest in your family, lifestyle, and diet? Are various treatment options explained to you?
- Is your personal dignity respected?
- Do you feel that this practitioner is sensitive to your feelings and fears regarding treatment?
- Is this practitioner a good advertisement for his/her profession? Signs of stress or ill health may mean that you would be better off choosing another practitioner.
- Do you feel that you are rushed into decisions, or do you feel that you are allowed time to make an informed choice regarding treatment?
- Are future health goals outlined for you? And do you feel that the practitioner is taking your progress seriously?
- Do you feel unconditionally accepted by this practitioner?
- Would you send your loved ones to this practitioner?

If you answered yes to all the above, then you have found a suitable practitioner. The cost of treatment by a holistic therapist varies widely, depending on the level of qualification and the discipline, so it is best to discuss how much treatment can be expected to cost with a practitioner before beginning a course. Some forms of holistic treatment may be covered by health insurance.

Precautions

Many people who try holistic therapies focus on one area of their health only, often detoxification and nutrition. However, practitioners stress that it is only when all areas of a person's potential well being are

tackled that total health and happiness can be achieved. They stress that the spiritual and emotional health contribute just as much as physical and mental health to a person's overall state of well-being.

When seeking treatment from a holistic practitioner, it is important to ensure that they are properly qualified. Credentials and reputation should always be checked. In addition, it is important that allopathic physicians and alternative physicians communicate about a patient's care.

Side effects

One of the main advantages of holistic therapies is that they have few side effects when used correctly. If a reputable practitioner is chosen, and guidelines are adhered to, the worst that typically happens is that when lifestyle is changed, and fresh nutrients are provided, the body begins to eliminate toxins that may have accumulated in the cells over a lifetime.

Often this results in what is known in alternative medicine circles as a "healing crisis." This comes about when the cells eliminate poisons into the blood stream all at the same time, throwing the system into a state of toxic overload until it can clear the "backlog." Symptoms such as **nausea**, headaches, or sensitivities to noise and other stimulations may be experienced.

The answer to most otherwise healthy patients is often just to lie quietly in a darkened room and take herbal teas. However, in the case of someone who has a serious illness, such as arthritis, colitis, diabetes, or **cancer**, (the list is much longer than this), it is strongly advised that they seek the help of a qualified practitioner. Therapists can help patients achieve detoxification in a way that causes the least stress to their bodies.

Research and general acceptance

Traditionally, holistic medicine, in all its different forms, has been regarded with mistrust and skepticism on the part of the allopathic medical profession. This situation is gradually changing. As of the year 2000, many insurance companies will provide for some form of alternative, or complementary treatment.

In addition, many allopathic physicians, recognizing the role alternative medicine can play in overall health and well being, are actually referring patients to reputable practitioners, particularly chiropractic and relaxation therapists, for help with a varied range of complaints.

Training and certification

Holistic or alternative medicine practitioners are usually affiliated with an organization in their field. Training varies tremendously with the category, and ranges from no qualifications at all—experience only—to holding a Ph.D. from an accredited university. Again, credentials and memberships should be checked by prospective patients.

An excellent source for qualified practitioners is the American Board of Holistic Medicine, (AHBM), which was incorporated in 1996. Also, the American Holistic Medicine Association has a comprehensive list of practitioners in all types of therapies across the United States, which they call “the holistic doctor finder.” However, they stress that it is the responsibility of the patient to check each practitioner’s credentials prior to treatment.

The ABHM has established the core curriculum upon which board certification for holistic medicine will be based. It includes the following twelve categories:

Body

Physical and environmental health

- nutritional medicine
- exercise medicine
- environmental medicine

Mind

Mental and emotional health

- behavioral medicine

Spirit

Spiritual health

- spiritual attunement
- social health

The six specialized areas:

- biomolecular diagnosis and therapy
- botanical medicine
- energy medicine
- ethno-medicine—including traditional Chinese medicine, Ayurveda, and Native American medicine
- homeopathy
- manual medicine

Founded in 1978 for the purpose of uniting practitioners of holistic medicine, membership of the AHMA is open to licensed medical doctors (MDs)

KEY TERMS

Detoxification—Treating the body in such a way that it eliminates poisons accumulated in the cells.

Healing crisis—When the body begins to eliminate toxins at an accelerated rate, unpleasant sensations may be experienced.

and doctors of osteopathic medicine (DOs) from every specialty, and to medical students studying for those degrees. Associate membership is open to health care practitioners who are certified, registered or licensed in the state in which they practice. The mission of the AHMA is to support practitioners in their personal and professional development as healers, and to educate physicians about holistic medicine.

Resources

ORGANIZATIONS

American Holistic Medicine Association, 4101 Lake Boone Trail, Suite 201, Raleigh, NC 27607.

< <http://www.holisticmedicine.org/index.html> >.

Holistic medicine Website. < <http://www.holisticmed.com/whatis.html> >.

Patricia Skinner

Holter monitoring

Definition

Holter monitoring is continuous monitoring of the electrical activity of a patient’s heart muscle (**electrocardiography**) for 24 hours, using a special portable device called a Holter monitor. Patients wear the Holter monitor while carrying out their usual daily activities.

Purpose

Holter monitoring is used to help determine whether someone has an otherwise undetected heart disease, such as abnormal heart rhythm (cardiac arrhythmia), or inadequate blood flow through the heart. Specifically, it can detect abnormal electrical activity in the heart that may occur randomly or only under certain circumstances, such as during sleep or periods of physical activity or **stress**, which may or may not be picked up by standard, short-term electrocardiography performed in a doctor’s office.

Traditionally, an **exercise** stress test has been used to screen people for “silent” heart disease (heart disease with none of the usual symptoms). However, an exercise **stress test** is not completely foolproof, often producing false negative results (indicating no heart disease when heart disease is actually present) and false positives (indicating heart disease when there is none). Furthermore, some people cannot undergo exercise stress testing because of other medical conditions, such as arthritis.

Holter monitoring, also known as ambulatory or 24-hour electrocardiography, offers an alternate means of testing people for heart disease. By monitoring electrocardiographic activity throughout the day, Holter monitoring can uncover heart problems that occur during the patient’s everyday activities. It can also help to recognize any activities that may be causing the heart problems. And it can define and correlate symptoms that may be caused by irregularities of the heart.

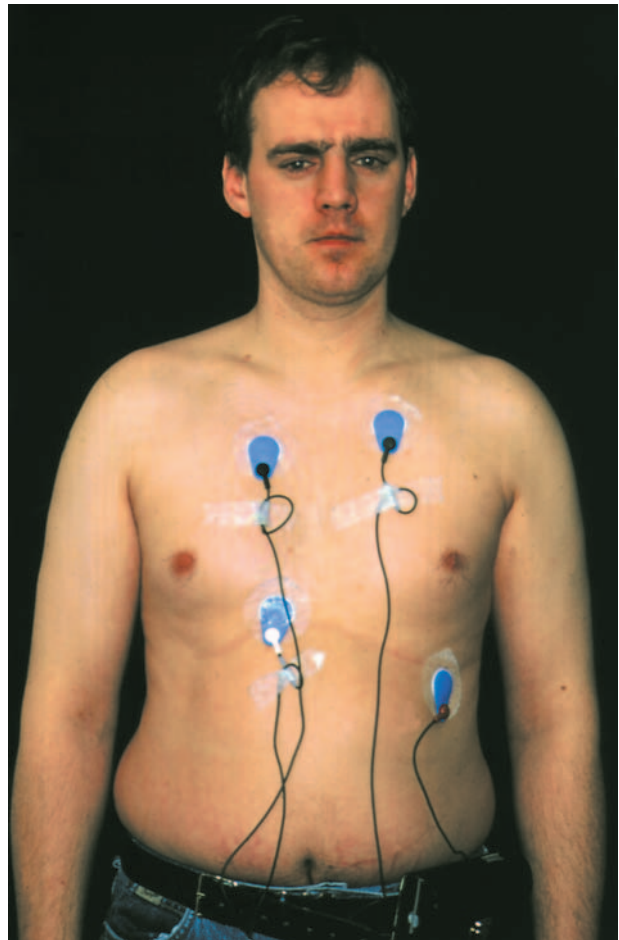
Precautions

Holter monitoring is an extremely safe procedure and no special precautions are required.

Description

The technician affixes electrodes on the surface of the skin at specific areas of the patient’s chest, using adhesive patches with special gel that conducts electrical impulses. Typically, electrodes are placed under each collarbone and each bottom rib, and several electrodes are placed across the chest in a rough outline of the heart. The electrodes are attached to a portable electrocardiographic device called a Holter monitor, which records the electrical activity of the heart over 24–48 hours. The device is worn over the patient’s shoulder or attached to a belt around the waist.

The Holter monitor records the continuous electrical activity throughout the course of the day, while the patient carries out his or her daily activities. During this time, the patient also keeps a detailed log or diary, recording his or her various activities, such as exercise, eating, sleeping, straining, breathing too hard (hyperventilating), and any stressful situations. The patient also notes the time and circumstances of any symptoms—especially chest pain, dizziness, **shortness of breath**, heart palpitations, and any other signs of heart trouble. Some Holter monitors allow patients to record their symptoms electronically, highlighting the portion of the electrocardiogram recorded while the symptoms are occurring.



A male patient wears electrodes attached to his chest, which is connected to a Holter monitor at his waist. (Photograph by Dr. P. Marazzi, Photo Researchers, Inc. Reproduced by permission.)

After 24–48 hours, the Holter monitor is removed. A computer-assisted analysis is performed on the electrocardiographic recording, and the doctor compares the recording against the patient’s log to see if there is any correlation between electrocardiographic abnormalities and any of the patient’s activities or symptoms. The physician makes a final interpretation.

Preparation

In the doctor’s office, electrodes are attached to the patient’s chest. In some cases, the patient’s chest hair may have to be shaved to facilitate attaching the electrodes. The patient then begins carrying the monitor on a shoulder harness, in a pocket, or on the belt while carrying out his or her usual daily routine. The patient should inform the doctor of any drugs he or she may be taking, because certain drugs can alter heart rhythms and may affect the results of the test.

KEY TERMS

Cardiac arrhythmia—An abnormal heart rhythm.

Electrocardiography—Recording of the electrical activity of the heart muscle, used to diagnose certain heart abnormalities.

Aftercare

The patient returns to the doctor's office to have the monitor and electrodes removed. No special measures need to be taken following Holter monitoring. The test results are usually available within a few days after the monitor is removed.

Risks

There are no known risks associated with Holter monitoring. The main complaint that people have with Holter monitoring is that the monitor may be cumbersome and interfere with certain activities, especially sleeping. Bathing and showering are not allowed during the study.

Normal results

A normal Holter monitoring test shows relatively normal electrical activity in the heart around the clock and no evidence of silent **ischemia** (deprivation of oxygen-rich blood).

Abnormal results

An abnormal result on Holter monitoring may indicate ischemia to the heart muscle or heart rhythm disturbances. Abnormalities are especially likely to show up during periods of stress or heavy activity, but sometimes serious abnormalities are recorded while the patient is sleeping.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

National Heart, Lung and Blood Institute. PO Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

Robert Scott Dinsmoor

Holtzman ink blot test

Definition

The Holtzman Inkblot Technique (HIT) is a projective personality assessment test for persons ages five and up.

Purpose

The HIT is used to assess the personality structure of a test subject. It is sometimes used as a diagnostic tool in assessing **schizophrenia**, depression, **addiction**, and character disorders.

Precautions

Psychometric testing requires a clinically trained examiner. The HIT should be administered and interpreted by a trained psychologist, psychiatrist, or appropriately trained mental health professional.

Some consider projective tests to be less reliable than objective personality tests. If the examiner is not well-trained in psychometric evaluation, subjective interpretations may affect the outcome of the test.

Description

The HIT, developed by psychologist Wayne Holtzman and colleagues, was introduced in 1961. The test was designed to overcome some of the deficiencies of its famous predecessor, the Rorschach Inkblot Test.

Unlike the Rorschach, the Holtzman is a standardized measurement with clearly defined objective scoring criteria. The HIT consists of 45 inkblots. The test administrator, or examiner, has a stack of 47 cards with inkblots (45 test cards and 2 practice cards) face down in front of him or her. The examiner hands each card to the subject and asks the test subject what he or she sees in the inkblot. Only one response per inkblot is requested. Occasionally, the examiner may ask the test subject to clarify or elaborate on a response. The Administration of the HIT typically takes 50–80 minutes. The HIT is then scored against 22 personality-related characteristics.

The HIT can also be administered in a group setting. In group testing, 30–45 inkblots are projected onto a screen and test subjects provide written responses to each inkblot.

The 1997 Medicare reimbursement rate for psychological and neuropsychological testing is \$58.35 an

KEY TERMS

Projective personality assessment—A test in which the subject is asked to interpret ambiguous stimuli, such as an inkblot. The subject's responses provide insight into his or her thought processes.

Standardization—The process of determining established norms and procedures for a test to act as a standard reference point for future test results.

hour. Billing time typically includes test administration, scoring and interpretation, and reporting. Many insurance plans cover all or a portion of diagnostic psychological testing.

Normal results

Because of the complexity of the scoring process and the projective nature of the test, results for the HIT should only be interpreted by a clinically trained psychologist, psychiatrist, or appropriately trained mental health professional.

Resources

ORGANIZATIONS

American Psychological Association (APA). 750 First St. NE, Washington, DC 20002-4242. (202) 336-5700.
< <http://www.apa.org> >.

Paula Anne Ford-Martin

Homeopathic medicine

Definition

Homeopathy, or homeopathic medicine, is a holistic system of treatment that originated in the late eighteenth century. The name homeopathy is derived from two Greek words that mean “like disease.” The system is based on the idea that substances that produce symptoms of sickness in healthy people will have a curative effect when given in very dilute quantities to sick people who exhibit those same symptoms. Homeopathic remedies are believed to stimulate the body's own healing processes. Homeopaths use the term “allopathy,” or “different than disease,” to describe the use of drugs used in conventional medicine to oppose or counteract the symptom being treated.

Purpose

Homeopathic physicians seek to cure their patients on the physical, mental and emotional levels, and each treatment is tailored to a patient's individual needs. Homeopathy is generally a safe treatment, as it uses medicines in extremely diluted quantities, and there are usually minimal side effects. Its non-toxicity makes it a good choice for the treatment of children. Another benefit of homeopathy is the cost of treatments; homeopathic remedies are inexpensive, often a fraction of the cost of conventional drugs.

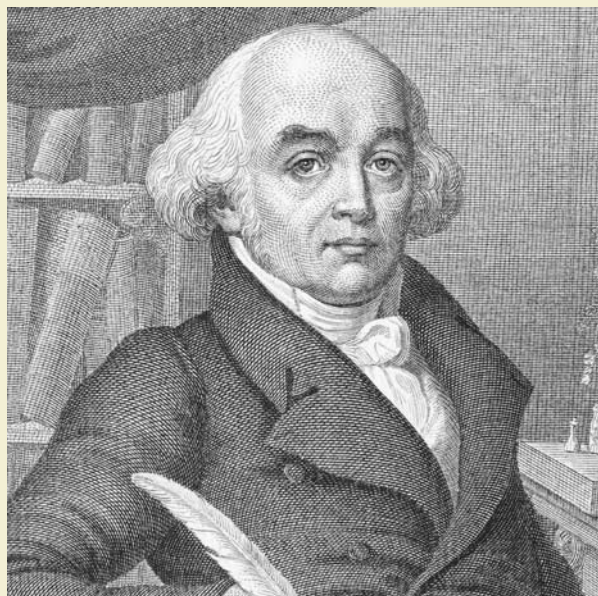
Homeopathic treatment has been shown effective in treating many conditions. Colds and flu may be effectively treated with aconite and bryonia. Influenza suffers in a double-blind study found that they were twice as likely to recover in 48 hours when they took homeopathic remedies. Studies have been published in British medical journals confirming the efficacy of homeopathic treatment for rheumatoid arthritis. Homeopathic remedies are effective in treating infections, circulatory problems, respiratory problems, heart disease, depression and nervous disorders, migraine headaches, **allergies**, arthritis, and diabetes. Homeopathy is a good treatment to explore for acute and chronic illnesses, particularly if these are found in the early stages and where there is not severe damage. Homeopathy can be used to assist the healing process after surgery or **chemotherapy**.

Description

Origins

Homeopathy was founded by German physician **Samuel Hahnemann** (1755–1843), who was much disturbed by the medical system of his time, believing that its cures were crude and some of its strong drugs and treatments did more harm than good to patients. Hahnemann performed experiments on himself using Peruvian bark, which contains quinine, a **malaria** remedy. He concluded that in a healthy person, quinine creates the same symptoms as malaria, including fevers and chills, which is the reason why it is effective as a remedy. He then began to analyze the remedies available in nature by what he called provings. Provings of homeopathic remedies are still compiled by dosing healthy adults with various substances and documenting the results, in terms of the dose needed to produce the symptoms and the length of the dose's effectiveness. The provings are collected in large homeopathic references called *materia medica* or materials of medicine.

SAMUEL HAHNEMANN (1755–1843)



(Corbis Corporation. Reproduced by permission.)

Samuel Christian Hahnemann created and developed the system called homeopathy. It is also known as *similia similibus curentur* or like cures like. Although his new methods initially met with ridicule and criticism, by the time of his death they were accepted the world over as a result of the great success he had with his new cure.

Hahnemann was born in Meissen, Saxony (now part of Germany) into a financially challenged middle class family. His parents initially educated him at home, where his father taught him never to accept anything he learned without first questioning it. He graduated as a physician at Erlangen in 1779 after studying at Leipzig and Vienna. He was also fluent in English, German, Italian, French, Greek, Arabic, Latin and Hebrew.

At age 27 he married his first wife, Johanna Henriette Kuchler, the daughter of an apothecary, with whom he had 11 children.

Living in poverty, Hahnemann began practicing medicine in 1781 and translating scientific texts to supplement his income. However, disillusioned with medicine, he eventually gave it up entirely.

He discovered the concept of homeopathy when considering the effect of quinine on malaria, and went on to cure soldiers and then sufferers of a typhus epidemic with astounding success. He documented his discoveries in the *Organon*, a treatise on his work. Homeopathy also proved its worth in 1831 when there was an outbreak of cholera. Hahnemann used homeopathic treatment with a 96% success rate, compared to the 41% of allopathic medicine. He also wrote his *Materia Medica Pura*.

In 1834, Hahnemann met his second wife, Marie Melanie d'Hervilly. Despite a great difference in age, they were happily married until his death in Paris on July 2, 1843, at the age of 88.

Hahnemann formulated these principles of homeopathy:

- Law of Similars (like cures like)
- Law of the Infinitesimal Dose (The more diluted a remedy is, the more potent it is.)
- illness is specific to the individual

Hahnemann's Law of Similars was based on thinking that dated back to Hippocrates in the fourth century B.C. It is the same thinking that provided the basis for vaccinations created by Edward Jenner and Louis Pasteur. These vaccines provoke a reaction in the individual that protects against the actual disease. Allergy treatments work the same way. By exposing a person to minute quantities of the allergen, the person's tolerance levels are elevated.

The Law of the Infinitesimal Dose has always caused controversy among those outside the field of homeopathy. Hahnemann contended that as he

diluted his remedies with water and alcohol and succussed, or shook, them, the remedies actually worked more effectively. In fact, diluted homeopathic remedies may have no chemical trace of the original substance. Practitioners believe that the electromagnetic energy of the original substance is retained in the dilution, but toxic side effects of the remedy are not. It is this electrochemical "message" that stimulates the body to heal itself.

Homeopathic practitioners believe that illness is specific to an individual. In other words, two people with severe headaches may not receive the same remedies. The practitioner will ask the patient questions about lifestyle, dietary habits, and personality traits, as well as specific questions about the nature of the **headache** and when it occurs. This information gathering is called profiling or case-taking.

In the early 1900s, homeopathy was popular in America, with over 15 percent of all doctors being homeopathic. There were 22 major homeopathic

Homeopathic Remedies That Work

Name	Description
Aconite	Commonly known as monkshood, aconite is highly toxic. A nontoxic, diluted extract of aconite is used in homeopathy to treat symptoms similar to that of poison.
Allium cepa	Commonly known as red onion, homeopathic physicians use a dilute extract of red onion to treat symptoms similar to that of red onion—watery eyes, burning, etc.
Apis	Commonly known as the honeybee, apis as a homeopathic remedy is made from the body of the bee. It is used to treat symptoms similar to that of a bee sting—redness, swelling, etc.
Arnica	Commonly known as the mountain daisy, arnica is used by homeopathic physicians to treat bruises, sprains, and strains.
Arsenicum album	Also known as ars alb, arsenicum album is a diluted form of arsenic, a metallic poison. It is used by homeopathic physicians to treat symptoms similar to the effects of arsenic poisoning—dehydration, burning pain, etc.
Belladonna	Commonly known as deadly knightshade, belladonna is used in homeopathy to treat symptoms of dry mouth, nausea, delirium, etc.
Bryonia	Commonly known as wild hops, bryonia is used in homeopathy to treat vomiting, diarrhea, inflammation, etc.
Calcarea carbonica	Also known as calcium carbonate or calc carb, it is used in homeopathy to treat symptoms of exhaustion, depression, and anxiety.
Cantharis	Commonly known as Spanish fly, cantharis is used in homeopathy to treat conditions with symptoms of abdominal cramps, vomiting, diarrhea, convulsions, etc.
Chamomilla	Derived from German chamomile, it is used in homeopathy to treat irritability, impatience, etc. It is most often prescribed to children.
Ferrum phosphoricum	Also known as ferrum phos or iron phosphate, it is used to treat symptoms of low energy and anemia.
Gelsemium	Also known as yellow jasmine, it is used to treat conditions that effect vision, balance, thought, and locomotion.
Hepar sulphuris	Derived from the inner layer of oyster shells, hepar sulphuris is used to treat infection.
Hypericum	Commonly known as St. John's wort, hypericum is used to treated nerve damage.
Ignatia	Derived from seeds of a plant, this homeopathic remedy is prescribed to treat conditions with symptoms such as headache, cramping, and tremors.
Ipecac	Ipecac induces vomiting and causes gastrointestinal distress. Homeopaths prescribe it to treat similar symptoms.
Kali bichromicum	Commonly known as potassium bichromate, kali bichromicum is a poison used also in textile dyes, wood stain, etc. Homeopaths use it to treat localized pain.
Lachesis	Derived from the venom of the bushmaster snake, this homeopathic remedy is used to treat conditions that cause the same symptoms as the venom itself.
Ledum	Also known as marsh tea, ledum is used to treat infections, most often from animal bites, stings, cuts, etc.
Lycopodium	Commonly known as club moss, lycopodium is used to treat diarrhea, digestive upset, etc.
Mercurius vivus	Also known as quicksilver, it is used to treat symptoms of sweats, shaking, nausea, etc.
Natrum muriaticum	Commonly known as salt, it is used to treat conditions that cause excessive thirst and salt cravings.
Nux vomica	It is used to treat symptoms caused by overeating and too much caffeine or alcohol.
Phosphorus	It is used to treat symptoms of excessive thirst, fatigue, and nervousness.

[Continued]

Homeopathic Remedies That Work [Continued]

Name	Description
Pulsatilla	It is used to treat conditions that are accompanied by discharge, such as bedwetting, sinusitis, etc.
Rhus toxicodendron	Commonly known as poison ivy, homeopaths use it to treat conditions with symptoms of fever, swollen glands, and restlessness.
Ruta	It is used to treat conditions with bruising, such as tennis elbow, sciatica, etc.
Sepia	Sepia is the discharge used by the cuttlefish to disappear from a predator. Homeopaths use sepia to treat symptoms of apathy and weakness.
Silica	Also called flint, silica is used by homeopaths to treat conditions that cause weakness, sweating, and sensitivity to cold.
Sulphur	It is used to treat conditions with symptoms of itching, burning pains, and odor.

medical schools, including Boston University and the University of Michigan. However, with the formation of the American Medical Association, which restricted and closed down alternative practices, homeopathy declined for half a century. When the 1960s invigorated back-to-nature trends and distrust of artificial drugs and treatments, homeopathy began to grow again dramatically through the next decades. In 1993, *The New England Journal of Medicine* reported that 2.5 million Americans used homeopathic remedies and 800,000 patients visited homeopaths in 1990, and it has continued to grow. Homeopathy is much more popular in Europe than in the United States. French pharmacies are required to make homeopathic remedies available along with conventional medications. Homeopathic hospitals and clinics are part of the national health system in Britain. It is also practiced in India and Israel, among other countries.

A visit to a homeopath can be a different experience than a visit to a regular physician. Surveys have shown that homeopathic doctors spend much more time during initial consultations than conventional doctors spend. This is because a homeopath does a complete case-taking to get a complete picture of a person's general health and lifestyle, as well as particular symptoms, on the physical, mental and emotional levels. Some symptoms can be so subtle that the patient is not always completely aware of them, and the doctor must spend time getting to know the patient.

The initial visit often includes a long questionnaire about a patient's medical and family history, and then a long interview with the doctor, who prompts the patient with many questions. Sometimes

a homeopathic doctor will use lab tests to establish a patient's general level of health. The initial interview usually lasts between one and two hours.

The purpose of homeopathy is the restoration of the body to homeostasis, or healthy balance, which is its natural state. The symptoms of a disease are regarded as the body's own defensive attempt to correct its imbalance, rather than as enemies to be defeated. Because a homeopath regards symptoms as positive evidence of the body's inner intelligence, he or she will prescribe a remedy designed to stimulate this internal curative process, rather than suppress the symptoms.

In homeopathy, the curative process extends beyond the relief of immediate symptoms of illness. Healing may come in many stages, as the practitioner treats layers of symptoms that are remnants of traumas or chronic disease in the patient's past. This is part of Hering's Laws of Cure, named for Constantine Hering, the father of homeopathy in America. Hering believed that healing starts from the deepest parts of the body to the extremities, and from the upper parts of the body to the lower parts. Hering's Laws also state that homeopaths should treat disease symptoms in reverse chronological order, from the most recent to the oldest, restoring health in stages. Sometimes, the patient may feel worse before feeling better. This is called a healing crisis.

When prescribing a remedy, homeopaths will match a patient's symptoms with the proper remedy in a repertory or *materia medica* that has been compiled throughout the history of homeopathy. Classical homeopaths prescribe only one remedy at a time. However, it is becoming more common, especially in Europe, to use combination formulas of several remedies for the treatment of some combinations of symptoms.

The cost of homeopathic care can vary. The cost of visits will be comparable to conventional medicine, with initial visits ranging from \$50 to \$300. Non-M.D. homeopaths can charge from \$50 to \$250. Follow-up visits are less, at about \$35 to \$100. Homeopathic medicine is significantly cheaper than pharmaceuticals, and most remedies cost between \$2 and \$10. Some doctors provide remedies without charge. Homeopaths rarely use lab tests, which reduces the cost of treatment further. In general, homeopathy is much more economical than conventional medicine. In 1991, the French government did a study on the cost of homeopathic medicine, and found that it costs half as much to treat patients, considering all costs involved.

KEY TERMS

Aggravation—Temporary increase in symptoms due to homeopathic remedy.

Antidote—Substance which cancels the effect of homeopathic remedies

Homeopath—A homeopathic physician.

Proving—Case study of the effect of a homeopathic medicine.

Repertory—Reference manual of homeopathic remedies.

Vital force—Innate wisdom and energy of the body.

When homeopaths are licensed professionals, most insurance companies will pay for their fees. Consumers should consult their insurance policies to determine individual regulations. Insurance usually will not cover homeopathic medicine, because it is sold over-the-counter.

Precautions

Although homeopathic remedies sometimes use substances that are toxic, they are diluted and prescribed in non-toxic doses. Remedies should be prescribed by a homeopathic practitioner. Those preparing to take homeopathic remedies should also avoid taking *antidotes*, substances which homeopathic doctors believe cancel the effects of their remedies. These substances include alcohol, coffee, prescription drugs, peppermint (in toothpaste and mouthwash), camphor (in salves and lotions), and very spicy foods. Homeopathic medicine should also be handled with care, and should not be touched with the hands or fingers, which can contaminate it.

Side effects

A homeopathic *aggravation* sometimes occurs during initial treatment with homeopathic remedies. This means that symptoms can temporarily worsen during the process of healing. Although this is usually mild, the aggravation can sometimes be severe. Homeopaths see aggravation as a positive sign that the remedy is a good match for the patient's symptoms. The healing crisis, which happens when the patient is undergoing treatment for layers of symptoms, may also cause the patient to feel worse before feeling better. Some patients can experience emotional

disturbances like weeping or depression, if suppressed emotional problems led to the illness in the first place.

Research and general acceptance

Since the early 1900s, when the American Medical Association and pharmacists waged a battle against it, homeopathy has been neglected and sometimes ridiculed by mainstream medicine. Aside from politics, part of the reason for this is that there are some aspects of homeopathy which have not been completely explained scientifically. For instance, homeopaths have found that the more they dilute and succuss a remedy, the greater effect it seems to have on the body. Some homeopathic remedies are so diluted that not even a single molecule of the active agent remains in a solution, yet it still works; studies have demonstrated this paradox, yet can't explain it. Also, homeopathy puts an emphasis on analyzing symptoms and then applying remedies to these symptoms, rather than working by classifying diseases. Thus, some people with the same disease may require different homeopathic medicines and treatments. Furthermore, conventional medicine strives to find out how medicines work in the body before they use them; homeopathy is less concerned with the intricate biochemistry involved than with whether a remedy ultimately works and heals holistically. For all these reasons, conventional medicine claims that homeopathy is not scientific, but homeopaths are quick to reply that homeopathy has been scientifically developed and studied for centuries, with much documentation and success.

There continue to be many studies that affirm the effectiveness of homeopathic treatments. Among the most celebrated, the *British Medical Journal* in 1991 published a large analysis of homeopathic treatments that were given over the course of 25 years. This project involved over 100 studies of patients with problems ranging from vascular diseases, respiratory problems, infections, stomach problems, allergies, recovery from surgeries, arthritis, trauma, psychological problems, diabetes, and others. The study found improvement with homeopathic treatment in most categories of problems, and concluded that the evidence was "sufficient for establishing homeopathy as a regular treatment for certain indications."

Resources

PERIODICALS

Homeopathy Today. 801 N. Fairfax St. #306, Alexandria, VA 22314. (703) 548-7790.

Similimum. P.O. Box 69565, Portland, OR 97201. (503) 795-0579.

OTHER

Ayurvedic Institute. < <http://www.ayurveda.com> >.

National Center for Homeopathy. < <http://www.healthy.net/nch> >.

North American Society of Homeopaths.

< <http://www.homeopathy.org> >.

Homeopathic medicine, acute prescribing

Definition

Acute homeopathic prescribing is that part of homeopathy that treats illness which has an abrupt onset and needs immediate attention. In homeopathic medicine, acute refers primarily to the speed of onset and self-limiting character of the disorder rather than its seriousness. Colds, **influenza**, sore throats, insect stings, cuts, **bruises**, **vomiting**, **diarrhea**, **fever**, muscle aches, and short-term insomnia are all examples of conditions that are treated by acute prescribing. The remedies given in acute homeopathic prescribing are intended to stimulate the body's internal ability to heal itself; they do not kill germs or suppress symptoms. Acute prescribing can be done—within limits—by patients at home, as well as by homeopathic practitioners. Study courses, self-treatment guides, and homeopathic home medicine kits are now available by mail order from homeopathic pharmacies and educational services.

Purpose

Homeopathic physicians seek to cure their patients on physical, mental, and emotional levels, and each treatment is tailored to a patient's individual needs. Homeopathy is generally a safe treatment, as it uses medicines in extremely diluted quantities, and there are usually minimal side effects. Its non-toxicity makes it a good choice for the treatment of children. Another benefit of homeopathy is the cost of treatments; homeopathic remedies are inexpensive, often a fraction of the cost of conventional drugs.

Acute homeopathic prescribing is thought to benefit a wide range of ailments. These include **altitude sickness**, Bell's palsy, the **common cold**, **allergies**, coughing, **dengue fever**, dysentery, earaches, migraine headaches, fever, **food poisoning**, grief, influenza, **motion sickness**, shock, sore throat, surgical complications, and reactions to vaccinations and drug therapy. Acute remedies may also be prescribed to treat insect

stings, animal bites, and problems related to poison oak and poison ivy. It may be further employed in treating injuries including black eyes, **burns**, bruises, concussions, cuts, damaged tendons and ligaments, **dislocations**, **fractures**, herniated discs, nosebleeds, puncture **wounds**, **sprains**, and **strains**.

Description

Origins

Homeopathy is a gentle, painless, holistic system of healing developed during the 1790s by Samuel Hahnemann, a German physician. Experimenting on himself with the anti-malarial drug quinine, Hahnemann noticed that large doses of the medicine actually caused malaria-like symptoms, while smaller doses cured the symptoms. From this, he advanced his concept of *Similia similibus curentur*, or “let like be cured with like.” Hahnemann then developed an extensive system of medicine based on this concept. He named it homeopathy, from the Greek words *homoios* (the same) and *pathos* (suffering).

Homeopathic remedies are almost always made from natural materials—plant, animal, or mineral substances—that have been treated to form mother tinctures or nonsoluble powders. Liquid extracts are then potentized, or increased in power, by a series of dilutions and succussions, or shakings. It is thought that succussion is necessary to transfer the energy of the natural substance to the solution. In addition, the potency of the remedy is regarded as increasing with each dilution. After the tincture has been diluted to the prescribed potency, the resulting solution is added to a bottle of sucrose/lactose tablets, which are stored in a cool, dark place. If the remedy is not soluble in water, it is ground to a fine powder and triturated with powdered lactose to achieve the desired potency.

Proponents of homeopathy over the years have included Louisa May Alcott, Charles Dickens, Benjamin Disraeli, Johann Wolfgang Goethe, Nathaniel Hawthorne, William James, Henry Wadsworth Longfellow, Pope Pius X, John D. Rockefeller, Harriet Beecher Stowe, William Thackeray, Daniel Webster, and W. B. Yeats. England's Royal Family has employed homeopathic practitioners since the 1830s.

Homeopathic prescribing differs in general from allopathic medicine in its tailoring of remedies to the patient's overall personality type and totality of symptoms, rather than to the disease. Whereas a conventional physician would prescribe the same medication

or treatment regimen to all patients with the common cold, for example, a homeopathic practitioner would ask detailed questions about each patient's symptoms and the modalities, or factors, that make them better or worse. As a result, the homeopath might prescribe six different remedies for six different patients with the same illness. In acute prescribing homeopathy, consultations are more brief compared to constitutional homeopathic prescribing. A typical patient might spend just 10–15 minutes with the practitioner, compared to more than an hour for constitutional prescribing.

Homeopathic classification of symptoms

Homeopathic practitioners use the word symptom in a more inclusive fashion than traditional medicine. In homeopathy, symptoms include any change that the patient experiences during the illness, including changes in emotional or mental patterns.

Homeopaths classify symptoms according to a hierarchy of four categories for purposes of acute prescribing:

- Peculiar symptoms. These are symptoms unique to the individual that do not occur in most persons with the acute disease. Homeopaths make note of peculiar symptoms because they often help to determine the remedy.
- Mental and emotional symptoms. These are important general symptoms that inform the homeopath about the patient's total experience of the disorder.
- Other general symptoms. These are physical symptoms felt throughout the patient's body, such as tiredness, changes in appetite, or restlessness.
- Particular symptoms. Particular symptoms are localized in the body; they include such symptoms as **nausea**, skin **rashes**, **headache**, etc.

During homeopathic case-taking, the practitioner will evaluate the intensity of the patient's symptoms, assess their depth within the patient's body, note any peculiar symptoms, evaluate the modalities of each symptom, and make a list of key symptoms to guide the selection of the proper medicine.

Homeopathic remedies

There are several hundred homeopathic remedies. Homeopathic medicines are usually formulated from diluted or triturated natural substances, including plants, **minerals**, or even venom from snakes or stinging insects. Some remedies may be given in a spray, ointment, or cream, but the most common

forms of administration are liquid dilutions and two sizes of pellets, or cylindrical tablets (for triturated remedies). A dose consists of one drop of liquid; 10–20 small pellets; or 1–3 large pellets. Since the remedies are so dilute, the exact size of the dose is not of primary importance. The frequency of dosing is considered critical, however; patients are advised not to take further doses until the first has completed its effect.

Homeopathic remedies can be kept indefinitely with proper handling. Proper handling includes storing the remedies in the original bottles and discarding them if they become contaminated by sunlight or other intense light; temperatures over 100°F (37.8°C); vapors from camphor, mothballs, or perfume; or from other homeopathic remedies being opened in the same room at the same time.

Preparations

Case-taking

The first step in acute prescribing is a lengthy interview with the patient, known as case-taking. In addition to noting the character, location, and severity of the patient's symptoms, the homeopath will ask about their modalities. The modalities are the circumstances or factors (e.g., weather, time of day, body position, behavior or activity, etc.) that make the symptoms either better or worse. Case-taking can be done by the patient or a family member as well as by a homeopath.

Selection and administration of a remedy

The choice of a specific remedy is guided by the patient's total symptom profile rather than by the illness. Homeopathic remedies are prescribed according to the law of similars, which holds that a substance that produces specific symptoms in healthy people cures those symptoms in sick people when given in highly diluted forms. For example, a patient with influenza who is irritable, headachy, and suffering from joint or muscle pains is likely to be given *bryonia* (wild hops), because this plant extract would cause this symptom cluster in a healthy individual.

Patients are instructed to avoid touching homeopathic medicines with their fingers. The dose can be poured onto a piece of white paper or the bottle's cap and tipped directly into the mouth. Homeopathic remedies are not taken with water; patients should not eat or drink anything for 15–20 minutes before or after taking the dose.

Precautions

Homeopathic acute prescribing is not recommended for the treatment of chronic conditions requiring constitutional prescribing, for severe infections requiring antibiotic treatment, or for conditions requiring major surgery. It is also not recommended for the treatment of mental health problems.

Persons who are treating themselves with homeopathic remedies should follow professional guidelines regarding the limitations of home treatment. Most homeopathic home treatment guides include necessary information regarding symptoms and disorders that require professional attention.

Homeopathic remedies may lose their potency if used at the same time as other products. Some homeopathic practitioners recommend the avoidance of mint and mentholated products (toothpastes, candies, chewing gum, mouth rinses), as well as camphor and camphorated products (including eucalyptus and Tiger Balm), patchouli and other essential oils, moth balls, strong perfumes, aftershaves, scented soaps, **stress**, x rays, coffee, nicotine, recreational drugs (**marijuana**) and certain therapeutic drugs (most notably cortisone and prednisone) during treatment. Patients are also advised to avoid electric blankets and dental work, as these are thought to adversely affect homeopathic therapy. Homeopathic remedies should never be placed near magnets.

Practitioners caution that high-potency preparations should be used only under the supervision of a homeopathic practitioner.

Side effects

Homeopathic medicines are so diluted that sometimes no trace of the original substance can be detected. These medicines are therefore considered non-toxic and generally free of harmful side effects. There may, however, be individual reactions to **homeopathic medicine**.

An intensified healing response may occur as treatment begins, which causes symptoms to worsen, but the phenomenon is temporary. In some patients, old symptoms may re-appear from past conditions from which recovery was not complete. Such phenomena are taken as positive indications that the healing process has commenced.

Research and general acceptance

As Samuel Hahnemann's healing system grew in popularity during the 1800s, it quickly attracted

vehement opposition from the medical and apothecary professions. Since the early 1900s, when the American Medical Association and pharmacists waged a battle against it, homeopathy has been neglected and sometimes ridiculed by mainstream medicine. Aside from politics, part of the reason for this is that there are some aspects of homeopathy which have not been completely explained scientifically. For instance, homeopaths have found that the more they dilute and succuss a remedy, the greater effect it seems to have on the body. Some homeopathic remedies are so diluted that not even a single molecule of the active agent remains in a solution, yet homeopaths maintain it still works; some studies have demonstrated this paradox, yet cannot explain it. Also, homeopathy puts an emphasis on analyzing symptoms and then applying remedies to these symptoms, rather than working by classifying diseases. Thus, some people with the same disease may require different homeopathic medicines and treatments. Furthermore, conventional medicine strives to find out how medicines work in the body before they use them; homeopathy is less concerned with the intricate biochemistry involved than with whether a remedy ultimately works and heals holistically. For all these reasons, conventional medicine claims that homeopathy is not scientific, but homeopaths are quick to reply that homeopathy has been scientifically developed and studied for centuries, with much documentation and success.

There continue to be many studies that affirm the effectiveness of homeopathic treatments. Among the most celebrated, the *British Medical Journal* in 1991 published a large analysis of homeopathic treatments that were given over the course of 25 years. This project involved over 100 studies of patients with problems ranging from vascular diseases, respiratory problems, infections, stomach problems, allergies, recovery from surgeries, arthritis, trauma, psychological problems, diabetes, and others. The study found improvement with homeopathic treatment in most categories of problems, and concluded that the evidence was “sufficient for establishing homeopathy as a regular treatment for certain indications.”

In the United Kingdom and other countries where homeopathy is especially popular, some medical doctors incorporate aspects of acute prescribing homeopathy into their practices. Countries in which homeopathy is popular include France, India, Pakistan, Sri Lanka, Brazil, and Argentina. Large homeopathic hospitals exist in London and

KEY TERMS

Acute prescribing—Homeopathic treatment for self-limiting illnesses with abrupt onset.

Allopathy—Conventional medical treatment of disease symptoms that uses substances or techniques to oppose or suppress the symptoms.

Law of similars—The basic principle of homeopathic medicine that governs the selection of a specific remedy. It holds that a substance of natural origin that produces certain symptoms in a healthy person will cure those same symptoms in a sick person.

Modalities—The factors and circumstances that cause a patient’s symptoms to improve or worsen.

Mother tincture—The first stage in the preparation of a homeopathic remedy, made by soaking a plant, animal, or mineral product in a solution of alcohol.

Potentization—The process of increasing the power of homeopathic preparations by successive dilutions and succussions of a mother tincture.

Succussion—The act of shaking diluted homeopathic remedies as part of the process of potentization.

Trituration—The process of diluting a nonsoluble substance for homeopathic use by grinding it to a fine powder and mixing it with lactose powder.

Glasgow, and homeopathic medical centers can be found in India and South America.

Resources

BOOKS

Strohecker, James. *Alternative Medicine: The Definitive Guide*. Tiburon, Calif.: Future Medicine Publishing, Inc., 1999.

ORGANIZATIONS

- The American Institute of Homeopathy. 1585 Glencoe, Denver, CO 80220. (303) 898-5477.
- Council for Homeopathic Certification. P.O. Box 157, Corte Madera, CA 94976.
- International Foundation for Homeopathy. 2366 Eastlake Avenue East, #301, Seattle, WA 98102. (425) 776-4147.
- National Center for Homeopathy. 801 North Fairfax Street, Suite 306, Alexandria, VA 22134. (703) 548-7790.
- North American Society of Homeopaths. 10700 Old County Rd. 15, #350, Minneapolis, MN 55441. (612) 593-9458.

Homeopathic medicine, constitutional prescribing

Definition

Constitutional homeopathic prescribing, also called classical prescribing, is a holistic system of medicine that has been practiced for more than 200 years. Unlike acute homeopathic prescribing, constitutional prescribing refers to the selection and administration of homeopathic preparations over a period of time for treatment related to what practitioners call miasmatic disorders, those caused by an inherited predisposition to a disease. The term *miasm* comes from a Greek word meaning stain or pollution. As in acute prescribing, constitutional prescribing is holistic in that it is intended to treat the patient on the emotional and spiritual levels of his or her being as well as the physical. Constitutional prescribing is also aimed at eventual cure of the patient, not just suppression or relief of immediate symptoms.

Purpose

Homeopathic physicians seek to cure their patients on physical, mental, and emotional levels, and each treatment is tailored to a patient's individual needs. Homeopathy is generally a safe treatment, as it uses medicines in extremely diluted quantities, and there are usually minimal side effects. Its non-toxicity makes it a good choice for the treatment of children. Another benefit of homeopathy is the cost of treatments; homeopathic remedies are inexpensive, often a fraction of the cost of conventional drugs.

Classical homeopathy has been used to treat a wide range of diseases and conditions, most of which tend to be long-term. These include: alcoholism, allergies, **anxiety**, arthritis, asthma, bladder conditions, chronic fatigue syndrome, depression, drug dependencies, gastrointestinal problems, Gulf War sickness, **headache**, hearing problems, herpes, hypersensitivity, immune disorders, **insomnia**, joint problems, kidney conditions, liver problems, **Lyme disease**, lower back problems, **malaria**, **menopause**, menstrual problems, migraine, **multiple sclerosis**, **paralysis**, **phobias**, **shingles**, sinus problems, skin disorders, repetitive **stress** injury, rheumatism, vertigo, vision problems, and yeast infections.

Description

Origins

Homeopathy was developed during the 1790s by **Samuel Hahnemann**, a German physician.

Experimenting on himself with the anti-malarial drug quinine, Hahnemann noticed that large doses of the medicine actually caused malaria-like symptoms, while smaller doses cured the symptoms. From this, he advanced his concept of *Similia similibus curentur*, or "let like be cured with like." Hahnemann then developed an extensive system of medicine based on this concept. He named it homeopathy, from the Greek words *homoios* (the same) and *pathos* (suffering).

There are several hundred homeopathic remedies. They are almost always made from natural materials—plant, animal, or mineral substances—that have been treated to form mother tinctures or nonsoluble powders. Liquid extracts are then potentized, or increased in power, by a series of dilutions and succussions, or shakings. It is thought that succussion is necessary to transfer the energy of the natural substance to the solution. In addition, the potency of the remedy is regarded as increasing with each dilution. After the tincture has been diluted to the prescribed potency, the resulting solution is added to a bottle of sucrose/lactose tablets, which are stored in a cool, dark place. If the remedy is not soluble in water, it is ground to a fine powder and triturated with powdered lactose to achieve the desired potency.

Proponents of homeopathy over the years have included Louisa May Alcott, Charles Dickens, Benjamin Disraeli, Johann Wolfgang Goethe, Nathaniel Hawthorne, William James, Henry Wadsworth Longfellow, Pope Pius X, John D. Rockefeller, Harriet Beecher Stowe, William Thackeray, Daniel Webster, and W. B. Yeats. England's Royal Family has employed homeopathic practitioners since the 1830s.

Constitutional prescribing is based on the patient's symptom profile and specific aspects of homeopathic theory.

Homeopathic classification of symptoms

Homeopathic practitioners use the word symptom in a more inclusive fashion than traditional medicine. In homeopathy, symptoms include any change that the patient experiences during the illness, including changes in emotional or mental patterns.

Homeopaths classify symptoms according to a hierarchy of four categories:

- Peculiar symptoms. These are symptoms unique to the individual that do not occur in most persons. Homeopaths make note of peculiar symptoms because they often help to determine the remedy.

- Mental and emotional symptoms. These are important general symptoms that inform the homeopath about the patient's total experience of the disorder.
- Other general symptoms. These are physical symptoms felt throughout the patient's body, such as tiredness, changes in appetite, or restlessness.
- Particular symptoms. Particular symptoms are localized in the body; they include such symptoms as **nausea**, skin **rashes**, or headaches.

Miasms

Homeopaths regard the patient's symptom profile as a systemic manifestation of an underlying chronic disorder called a **miasm**. Miasms are serious disturbances of what homeopaths call the patient's vital force that are inherited from parents at the time of conception. Hahnemann believed that the parents' basic lifestyle, their emotional condition and habitual diet, and even the atmospheric conditions at the time of conception would affect the number and severity of miasms passed on to the child. Hahnemann himself distinguished three miasms: the psoric, which he considered the most universal source of chronic disease in humans; the syphilitic; and the sycotic, which he attributed to **gonorrhea**. Later homeopaths identified two additional miasms, the canceric and the tuberculinic. The remaining major source of miasms is allopathic medicine. It is thought that specific allopathic treatments—particularly **smallpox** vaccinations, cortisone preparations, major tranquilizers, and antibiotics—can produce additional layers of miasms in the patient's constitution. Constitutional prescribing evaluates the person's current state or miasmatic picture, and selects a remedy intended to correct or balance that state. The homeopath may prescribe a different remedy for each miasmatic layer over time, but gives only one remedy at a time directed at the person's current state. The basic principle governing the prescription of each successive remedy is the law of similars, or "like cures like."

Hering's laws of cure

The homeopathic laws of cure were outlined by Constantine Hering, a student of Hahnemann who came to the United States in the 1830s. Hering enunciated three laws or principles of the patterns of healing that are used by homeopaths to evaluate the effectiveness of specific remedies and the overall progress of constitutional prescribing:

- Healing progresses from the deepest parts of the organism to the external parts. Homeopaths consider the person's mental and emotional dimensions,

together with the brain, heart, and other vital organs, as a person's deepest parts. The skin, hands, and feet are considered the external parts.

- Symptoms appear or disappear in the reverse of their chronological order of appearance. In terms of constitutional treatment, this law means that miasms acquired later in life will resolve before earlier ones.
- Healing proceeds from the upper to the lower parts of the body.

Healing crises

Homeopaths use Hering's laws to explain the appearance of so-called healing crises, or aggravations, in the course of homeopathic treatment. It is not unusual for patients to experience temporary worsening of certain symptoms after taking their first doses of homeopathic treatment. For example, a person might notice that arthritic pains in the shoulders are better but that the hands feel worse. Hering's third law would indicate that the remedy is working because the symptoms are moving downward in the body. In constitutional prescribing, a remedy that removes one of the patient's miasmatic layers will then allow the symptoms of an older miasm to emerge. Thus the patient may find that a physical disease is followed by a different set of physical problems or by emotional symptoms.

Preparations

The most important aspects of preparation for constitutional prescribing are the taking of a complete patient history and careful patient education.

Case-taking

Homeopathic case-taking for constitutional prescribing is similar to that for acute prescribing, but more in-depth. The initial interview generally takes one to two hours. The practitioner is concerned with recording the totality of the patient's symptoms and the modalities that influence their severity. Also included are general characteristics about the patient and his or her lifestyle choices. For example, a practitioner might ask the patient if he or she likes being outside or is generally hot or cold. There is also an emphasis on the patient's lifetime medical history, particularly records of allopathic treatments.

Patient education

Homeopaths regard patients as equal partners in the process of recovery. They will take the time to explain the theories underlying constitutional

prescribing to the patient as well as taking the history. Patient education is especially important in constitutional prescribing in order to emphasize the need for patience with the slowness of results and length of treatment, and to minimize the possibility of self-treatment with allopathic drugs if the patient has a healing crisis.

Homeopathic remedies

In constitutional prescribing, one dose of the selected remedy is given. Patients then wait two to six weeks before following up with the homeopath, while the body begins the healing process. At the follow-up visit, the remedy may be repeated, or a different remedy prescribed. The preparation, selection, administration, and storage of remedies for constitutional prescribing are the same as for acute prescribing. These procedures are described more fully in the article on acute prescribing.

Precautions

Constitutional homeopathic prescribing is not appropriate for diseases or health crises requiring emergency treatment, whether medical, surgical, or psychiatric. In addition, constitutional prescribing should not be self-administered. Although home treatment kits of homeopathic remedies are available for acute self-limited disorders, the knowledge of homeopathic theory and practice required for constitutional evaluation is beyond the scope of most patients.

Patients are instructed to avoid touching homeopathic medicines with their fingers. The dose can be poured onto a piece of white paper or the bottle's cap and tipped directly into the mouth. Homeopathic remedies are not taken with water; patients should not eat or drink anything for 15–20 minutes before or after taking the dose.

Homeopathic remedies may lose their potency if used at the same time as other products. Some homeopathic practitioners recommend the avoidance of mint and mentholated products (toothpastes, candies, chewing gum, mouth rinses), as well as camphor and camphorated products (including eucalyptus and Tiger Balm), patchouli and other essential oils, moth balls, strong perfumes, aftershaves, scented soaps, stress, x rays, coffee, nicotine, recreational drugs (marijuana) and certain therapeutic drugs (most notably cortisone and prednisone) during treatment. Patients are also advised to avoid electric blankets and dental work, as these are thought to adversely affect homeopathic therapy. Homeopathic remedies should never be placed near magnets.

Side effects

Homeopathic medicines are so diluted that sometimes no trace of the original substance can be detected. These medicines are therefore considered non-toxic and generally free of harmful side effects. The primary risks to the patient from constitutional homeopathic treatment are the symptoms of the healing crisis and individual reactions to **homeopathic medicine**. The complexity of constitutional prescribing requires homeopaths to have detailed knowledge of the *materia medica* and the repertories, and to take careful and extensive case notes.

An intensified healing response may occur as treatment begins, which causes symptoms to worsen, but the phenomenon is temporary. In some patients, old symptoms may re-appear from past conditions from which recovery was not complete. Such phenomena are taken as positive indications that the healing process has commenced.

Research and general acceptance

As Samuel Hahnemann's healing system grew in popularity during the 1800s, it quickly attracted vehement opposition from the medical and apothecary professions. Since the early 1900s, when the American Medical Association and pharmacists waged a battle against it, homeopathy has been neglected and sometimes ridiculed by mainstream medicine. Aside from politics, part of the reason for this is that there are some aspects of homeopathy which have not been completely explained scientifically. For instance, homeopaths have found that the more they dilute and succuss a remedy, the greater effect it seems to have on the body. Some homeopathic remedies are so diluted that not even a single molecule of the active agent remains in a solution, yet homeopaths maintain it still works; some studies have demonstrated this paradox, yet cannot explain it. Also, homeopathy puts an emphasis on analyzing symptoms and then applying remedies to these symptoms, rather than working by classifying diseases. Thus, some people with the same disease may require different homeopathic medicines and treatments. Furthermore, conventional medicine strives to find out how medicines work in the body before they use them; homeopathy is less concerned with the intricate biochemistry involved than with whether a remedy ultimately works and heals holistically. For all these reasons, conventional medicine claims that homeopathy is not scientific, but homeopaths are quick to reply that homeopathy has been scientifically developed and studied for centuries, with much documentation and success.

KEY TERMS

Aggravation—Another term used by homeopaths for the healing crisis.

Allopathy—Conventional medical treatment of disease symptoms that uses substances or techniques to oppose or suppress the symptoms.

Constitutional prescribing—Homeopathic treatment for long-term or chronic disorders related to inherited predispositions to certain types of illnesses.

Healing crisis—A temporary worsening of the patient's symptoms during successive stages of homeopathic treatment.

Law of similars—The basic principle of homeopathic medicine that governs the selection of a specific remedy. It holds that a substance of natural origin that produces certain symptoms in a healthy person will cure those same symptoms in a sick person.

Laws of cure—A set of three rules used by homeopaths to assess the progress of a patient's recovery.

Materia medica—In homeopathy, reference books compiled from provings of the various natural remedies.

Miasm—In homeopathic theory, a general weakness or predisposition to chronic disease that is transmitted down the generational chain.

Modalities—The factors and circumstances that cause a patient's symptoms to improve or worsen, including weather, time of day, effects of food, and similar factors.

Repertories—Homeopathic reference books consisting of descriptions of symptoms. The process of selecting a homeopathic remedy from the patient's symptom profile is called repertorizing.

There continue to be many studies that affirm the effectiveness of homeopathic treatments. Among the most celebrated, the *British Medical Journal* in 1991 published a large analysis of homeopathic treatments that were given over the course of 25 years. This project involved over 100 studies of patients with problems ranging from vascular diseases, respiratory problems, infections, stomach problems, **allergies**, recovery from surgeries, arthritis, trauma, psychological problems, diabetes, and others. The study found improvement with homeopathic treatment in most

categories of problems, and concluded that the evidence was “sufficient for establishing homeopathy as a regular treatment for certain indications.”

In the United Kingdom and other countries where homeopathy is especially popular, some medical doctors incorporate aspects of acute prescribing homeopathy into their practices. Countries in which homeopathy is popular include France, India, Pakistan, Sri Lanka, Brazil, and Argentina. Large homeopathic hospitals exist in London and Glasgow, and homeopathic medical centers can be found in India and South America.

Resources

BOOKS

Strohecker, James. *Alternative Medicine: The Definitive Guide*. Tiburon, Calif.: Future Medicine Publishing, Inc., 1999.

ORGANIZATIONS

American Institute of Homeopathy. 1585 Glencoe, Denver, CO 80220. (303) 898-5477.

Council for Homeopathic Certification. P.O. Box 157, Corte Madera, CA 94976.

International Foundation for Homeopathy. 2366 Eastlake Avenue East, #301, Seattle, WA 98102. (425) 776-4147.

National Center for Homeopathy. 801 North Fairfax Street, Suite 306, Alexandria, VA 22134. (703) 548-7790.

North American Society of Homeopaths. 10700 Old County Rd. 15, #350, Minneapolis, MN 55441. (612) 593-9458.

Homocysteine

Definition

Homocysteine is a naturally occurring amino acid found in blood plasma. High levels of homocysteine in the blood are believed to increase the chance of heart disease, **stroke**, **Alzheimer's disease**, and **osteoporosis**.

Description

Homocysteine is a sulfur-containing amino acid that occurs naturally in all humans. It is broken down in the body through two metabolic pathways. The chemical changes that must occur to break down homocysteine require the presence of **folic acid** (also called folate) and **vitamins B⁶** and **B¹²**. The level of homocysteine in the blood is influenced by the presence of these substances.

Homocystinuria is a rare genetic disorder that occurs in about one in every 200,000 individuals. This congenital metabolic disorder causes large amounts of homocysteine to be excreted in the urine. Homocystinuria is associated **mental retardation** and the development of heart disease before age 30.

In the late 1960s, doctors documented that individuals with homocystinuria developed narrowing of the arteries at a very early age, sometimes even in childhood. Although homocystinuria is rare, this finding stimulated research on whether people who did not have homocystinuria but who did have unusually high levels of homocysteine in their blood were at greater risk of developing heart disease or stroke.

Many risk factors, including family history of heart disease, **smoking**, **obesity**, lack of **exercise**, diabetes, high levels of low-density lipoprotein cholesterol (LDL or “bad” cholesterol), low levels of high-density lipoprotein cholesterol (HDL or “good” cholesterol), and high blood pressure have been documented to increase the risk of stroke and heart disease. With so many other risk factors, it has been difficult to determine whether high levels of homocysteine are an independent risk factor for the development these diseases. However, a substantial number of controlled, well-designed, and well-documented studies have shown that individuals who have high levels of homocysteine in the blood are at increased risk of developing blocked blood vessels, a condition known as occlusive arterial disease or at risk to worsen **atherosclerosis** (“hardening of the arteries”).

In the 2000s, studies also suggested that high levels of homocysteine were associated with poorer mental functioning, leading to ongoing investigations into the role of homocysteine in Alzheimer’s disease. Additional studies have also suggested that high levels of homocysteine can lead to osteoporosis and an increased risk of broken bones in the elderly. As of 2005, homocysteine was being tested in half a dozen clinical trials to determine its role in these and several other conditions. Information on clinical trials that are enrolling patients can be found on-line at <<http://www.clinicaltrials.gov>>.

Causes and symptoms

Homocysteine is thought to irritate the lining of the blood vessels causing them to become scarred, hardened, and narrowed. This increases the work the heart must do, leading to heart disease. High levels of homocysteine also cause increased blood clotting. **Blood clots** can decrease or block the flow of blood through blood vessels, resulting in strokes and heart

attacks. If and how homocysteine directly plays a role in osteoporosis and Alzheimer’s disease is not clear as of 2005.

The level of homocysteine in the blood naturally varies with age, gender, diet, hereditary factors, and general health, but it is estimated that 5–10% of the population has homocysteine levels that are considered high. With the exception of rare individuals who have congenital homocystinuria, people with high blood levels of homocysteine do not have any obvious signs or symptoms.

Diagnosis

The American Heart Association and the American College of Cardiology do not recommend routine screening of homocysteine levels, but they do recommend screening as part of a cardiac risk assessment for individuals who have a family history of **coronary artery disease** but no obvious symptoms of heart disease. The level of homocysteine in the blood can be measured with a simple blood test that is often, but not always, done after **fasting**. Homocysteine levels of 12 mmol/L are considered normal and levels below 10 mmol/L are considered desirable.

Treatment

Lowering homocysteine blood levels is linked to increasing the intake of folic acid and vitamins B₆ and B₁₂. The healthiest way to increase intake is by eating more foods that are high in these substances. Good sources of folic acid, vitamin B₆ and B₁₂ include green leafy vegetables, fortified breakfast cereals, lentils, chickpeas, asparagus, spinach, and most beans. Taking a daily multivitamin is also a way to increase the levels of these substances. However, megadoses of folic acid, vitamin B₆ and B₁₂ are not recommended. Individuals should discuss dosage with their doctor before beginning any supplements. It is important to note that a direct link between increased intake of folic acid, vitamin B₆, and vitamin B₁₂ and decreased incidence stroke and **heart attack** has not been proved. However, one study published in the *Journal of the American Medical Association* found that women whose folic acid levels were in the lowest 25% were 69% more likely to die of coronary problems than women whose folic acid levels were in the top 25%.

Individuals with homocystinuria are treated with the drug betaine (Cystadane). This is a powder dissolved in water, juice, or milk and drunk usually twice a day with meals. This drug is not normally used

KEY TERMS

Alzheimer's disease—A degenerative brain disease of unknown origin that is a common cause of dementia in older individuals.

Amino acid—A nitrogen-containing building block of protein molecules.

Androgens—Male sex hormones.

Congenital—Present at birth.

Osteoporosis—Loss of minerals from the bones, causing bones to break more easily.

Plasma—The liquid part of the blood in which cells are suspended.

simply to lower high levels of homocysteine in the absence of congenital disease.

Prognosis

Individuals who increase the folic acid, vitamin B₆ and B₁₂ in their diet are expected to see a decrease in blood levels of homocysteine and as a result decrease their risk of heart disease and stroke.

Prevention

Certain drugs are suspected of increasing the level of homocysteine in the blood. People using these drugs should discuss with their doctor the advisability of increasing their intake of folic acid, vitamin B₆, and vitamin B₁₂. These drugs include:

- lipid-lowering drugs such as fenofibrate (Tricor) and bezafibrate (Bezalip)
- metformin (Glucophage), a drug to modify insulin resistance
- anti-epileptic drugs such as phenobarbital, phenytoin (Dilantin), primidone (Mysoline) and carbamazepine (Tegretol)
- levodopa (Sinemet) for treatment of Parkinson's disease
- methotrexate (Rheumatrex, Trexall) for treatment of **cancer, psoriasis, rheumatoid arthritis**, and systemic lupus erythematosus
- androgen treatment
- nitrous oxide ("laughing gas"), a mild anesthetic

Resources

BOOKS

McCully, Kilmer S. *The Homocysteine Revolution*. 2nd ed. New York: McGraw Hill 1999.

PERIODICALS

McLean, Robert R., et al. "Homocysteine as a Predictive Factor for Hip Fracture in Older Persons." *New England Journal of Medicine*, 350 (May 13, 2004): 2042–49 [cited 23 March 2005]. < <http://content.nejm.org/cgi/content/abstract/350/20/2042> >.

Morey, Sharon S. "Practice Guidelines: AHA and ACC Outline Approaches to Coronary Disease Risk Assessment." *American Family Physician*, 61, no. 8 (April 15, 2000): 2534–44 [cited March 23, 2005]. < <http://www.aafp.org/afp/2000415/practice.html> >.

ORGANIZATIONS

American Heart Association. 7272 Greenville Ave., Dallas, TX 75231. (800)242-8721.
< <http://www.americanheart.org> >.

OTHER

Homocysteine.net, May 10, 2004. [cited March 23, 2005]. < <http://www.homocysteine.net> >

Tish Davidson, A.M.

Hong Kong flu see **Influenza**

Hookworm disease

Definition

Hookworm disease is an illness caused by one of two types of S-shaped worms that infect the intestine of humans (the worm's host).

Description

Two types of hookworm are responsible for hookworm disease in humans. *Necator americanus* and *Ancylostoma duodenale* have similar life cycles and similar methods of causing illness. The adult worm of both *Necator americanus* and *Ancylostoma duodenale* is about 10 mm long, pinkish-white in color, and curved into an S-shape or double hook.

Both types of hookworm have similar life cycles. The females produce about 10,000–20,000 eggs per day. These eggs are passed out of the host's body in feces. The eggs enter the soil, where they incubate. After about 48 hours, the immature larval form hatches out of the eggs. These larvae take about six weeks to develop into the mature larval form that is capable of causing human infection. If exposed to human skin at this point (usually bare feet walking in the dirt or bare hands digging in the dirt), the larvae will bore through the skin and ride through the lymph circulation to the right side of the heart. The larvae are then pumped into



A micrograph image of the head of the hookworm *Ancylostoma* spp. (Photo Researchers, Inc. Reproduced by permission.)

the lungs. There they bore into the tiny air sacs (alveoli) of the lungs. Their presence within the lungs usually causes enough irritation to produce coughing. The larvae are coughed up into the throat and mouth, and are then swallowed and passed into the small intestine. It is within the intestine that they develop into the adult worm, producing illness in their human host.

Ancylostoma duodenale is found primarily in the Mediterranean, the Middle East, and throughout Asia. *Necator americanus* is common in tropical areas including Asia, parts of the Americas, and throughout Africa. Research suggests that at least 25% of all people in the world have hookworm disease. In the United States, 700,000 people are believed to be infected with hookworms at any given time.

Causes and symptoms

Hookworms cause trouble for their human host when the worms attach their mouths to the lining of the small intestine and suck the person's blood.

An itchy, slightly raised rash called "ground itch" may appear around the area where the larvae first bored through the skin. The skin in this area may become red and swollen. This lasts for several days and commonly occurs between the toes.

When the larvae are in the lungs, the patient may have a **fever**, **cough**, and some **wheezing**. Some people, however, have none of these symptoms.

KEY TERMS

Alveoli—The small air sacs clustered at the ends of the bronchioles, in the lungs in which oxygen-carbon dioxide exchange takes place.

Anemia—Any condition where the oxygen-carrying capacity of the red blood cells is reduced; symptoms often include fatigue.

Host—The organism (like a human) in which another infecting organism (like a worm) is living.

Larva—An immature form of an organism, occurring early in that organism's development.

Once established within the intestine, the adult worms can cause abdominal **pain**, decreased appetite, **diarrhea**, and weight loss. Most importantly, the worms suck between 0.03–0.2 ml of blood per day. When a worm moves from one area of the intestine to another, it detaches its mouth from the intestinal lining, leaving an irritated area that may continue to bleed for some time. This results in even further blood loss. A single adult worm can live for up to 14 years in a patient's intestine. Over time, the patient's blood loss may be very significant. Anemia is the most serious complication of hookworm disease, progressing over months or years. Children are particularly harmed by such anemia, and can suffer from heart problems, mental retardation, slowed growth, and delayed sexual development. In infants, hookworm disease can be deadly.

Diagnosis

Diagnosis of hookworm disease involves collecting a stool sample for examination under a microscope. Hookworm eggs have a characteristic appearance. Counting the eggs in a specific amount of feces allows the healthcare provider to estimate the severity of the infection.

Treatment

Minor infections are often left untreated, especially in areas where hookworm is very common. If treatment is required, the doctor will prescribe a three-day dose of medication. One to two weeks later, another stool sample will be taken to see if the infection is still present.

Anemia is treated with iron supplements. In severe cases, blood **transfusion** may be necessary. Two

medications, pyrantel pamoate and mebendazole, are frequently used with good results.

Prognosis

The prognosis for patients with hookworm disease is generally good. However, reinfection rates are extremely high in countries with poor sanitation.

Prevention

Prevention of hookworm disease involves improving sanitation and avoiding contact with soil in areas with high rates of hookworm infection. Children should be required to wear shoes when playing outside in such areas, and people who are gardening should wear gloves.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Rosalyn Carson-DeWitt, MD

Hormone replacement therapy

Definition

Hormone replacement therapy (HRT) is the use of synthetic or natural female hormones to make up for the decline or lack of natural hormones produced in a woman's body. HRT is sometimes referred to as estrogen replacement therapy (ERT), because the first medications that were used in the 1960s for female hormone replacement were estrogen compounds.

Estrogens

In order to understand how HRT works and the controversies surrounding it, women should know that there are different types of estrogen medications commonly prescribed in the United States and Europe. These drugs are given in a variety of prescription strengths and methods of administration. There are at present three estrogen compounds used in Western countries. Only the first two are readily available in the United States.

- **Estrone.** Estrone is the form of estrogen present in women after **menopause**. It is available as tablets under the brand name Ogen. The most commonly prescribed estrogen in the United States, Premarin, is a so-called conjugated estrogen that is a mixture of estrone and other estrogens.
- **Estradiol.** This is the form of estrogen naturally present in perimenopausal women. It is available as tablets (Estrace), skin patches (Estraderm), or vaginal creams (Estrace).
- **Estriol.** Estriol is a weaker form of estrogen produced by the breakdown of other forms of estrogen in the body. This is the form of estrogen most commonly given in Europe, under the brand name Estriol. It is the only form that is thought not to cause **cancer**.

In addition to pills taken by mouth, skin patches, and vaginal creams, estrogen preparations can be given by injection or by pellets implanted under the skin. Estrogen implants, however, are used less and less frequently.

Progestins

Most HRT programs include progestin treatment with estrogen compounds. Progestins—sometimes called progestogens—are synthetic forms of progesterone that are given to reduce the possibility that estrogen by itself will cause cancer of the uterus. Progestins are commonly prescribed under the brand names Provera and **Depo-Provera**. Other common brand names are Norlutate, Norlutin, and Aygestin.

Estrogen/testosterone combinations

Women's ovaries secrete small amounts of a male sex hormone (testosterone) throughout their lives. Women who have had both ovaries removed by surgery are sometimes given testosterone along with estrogen as part of HRT. Combinations of these hormones are available as tablets under the brand name Estratest or as vaginal creams. Women who cannot take estrogens can use 1% testosterone cream alone for problems with vaginal soreness.

Estrogen/tranquilizer combinations

There are several medications that combine estrogen with a tranquilizer like chlordiazepoxide (sold under the trade name Menrium) or meprobamate (sold under the trade name PMB). Many doctors warn against these combination drugs because the tranquilizers can be habit-forming.

Purpose

Hormone replacement therapy has been prescribed for two primary purposes; preventive treatment against **osteoporosis** and heart disease; and relief of physical symptoms associated with menopause.

Menopausal symptoms

Women in midlife enter a stage of development called menopause, when their menstrual periods become irregular and finally stop. The early phase of this transition is called the perimenopause. In the United States, the average age at menopause is presently 50 or 51, but some women begin menopause as early as 40 and others as late as 55. It can take as long as 10 years for a woman to complete the process. Women who have had their ovaries removed surgically are said to have undergone surgical menopause.

Doctors have not always agreed on definitions of menopause. Some use age as the baseline. Others define menopause as the point when a woman has had no menstrual periods for a full calendar year. Still others define menopause as the end of ovulation. It is not always clear, however, when a woman has had her last period or when she has stopped ovulating. In addition, women who take **oral contraceptives** can have breakthrough bleeding long after they have stopped ovulating. As a result, some doctors now measure the level of follicle-stimulating hormone (FSH) in a woman's blood to estimate whether the woman has entered menopause. During perimenopause, the FSH levels in a woman's blood rise as her body attempts to stimulate the release of ripe ova. An FSH level over 40 is considered an indicator of menopause.

During the menopausal transition, the levels of estrogen in the woman's body drop. The lowered estrogen level is responsible for a group of symptoms that include hot flashes (or flushes), weight gain, changes in skin texture, mood swings, heart **palpitations**, sleep disturbances, a need to urinate more frequently, and loss of sexual desire. The estrogen that is given in HRT has been shown to eliminate hot flashes, night sweats, lack of vaginal lubrication, and urinary tract problems. HRT will not prevent weight gain or wrinkles. It also does not cure depression in most women.

Preventive care

HRT has been recommended by some doctors to protect women against two serious midlife health problems, including osteoporosis and heart disease.

While clinical trials have continued to demonstrate HRT's effectiveness in preventing osteoporosis, women must weigh the risk of the therapy with the benefits. The trials also showed that HRT actually increased rather than decreased risk of heart disease.

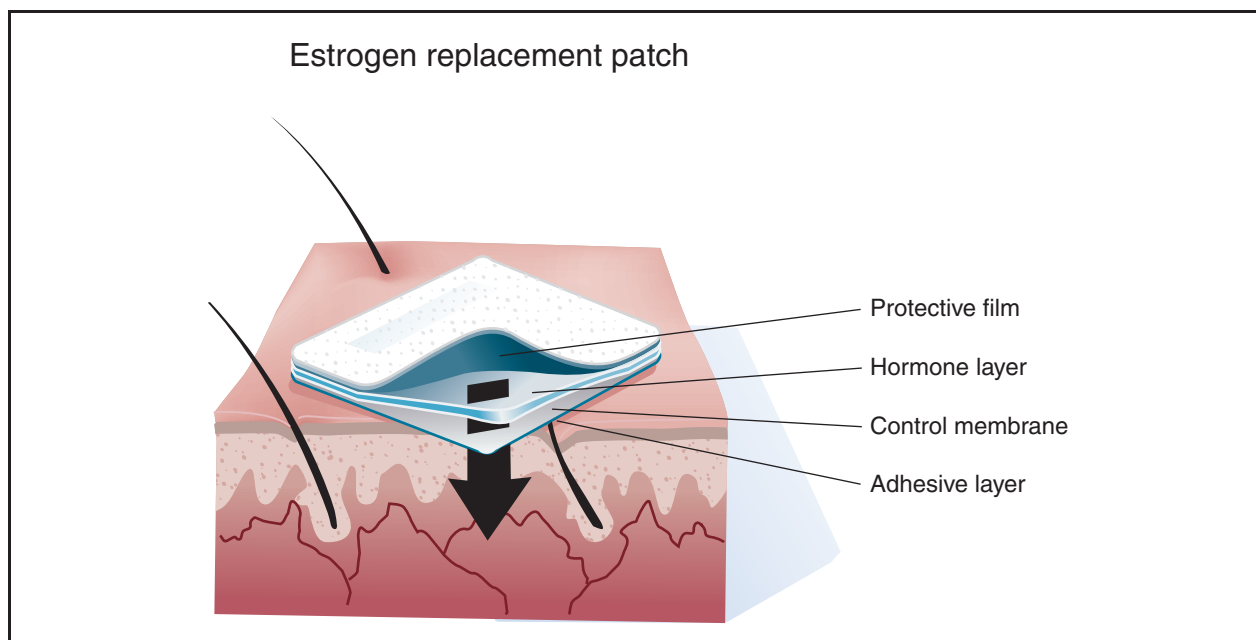
OSTEOPOROSIS. Osteoporosis is a disorder in which the bones become more brittle and more easily fractured. It is a particular problem for postmenopausal women because the lower levels of estrogen in the blood lead to weakening of the bone. About 25% of Caucasian women will develop severe osteoporosis; Asian women have a slightly lower risk level; Latino and African American women are least at risk.

In addition to race, there are other factors that put some women at higher risk of developing osteoporosis. Women in any of the following groups should take bone loss into account when considering HRT:

- family history of osteoporosis
- menopause before age 40
- kidney disease and dialysis
- thin body build or being underweight
- history of colitis, **Crohn's disease**, or chronic **diarrhea**
- thyroid medications
- childlessness
- chronic use of **antacids**
- lack of **exercise**
- poor food choices, including high salt intake, lack of vitamin D, high **caffeine** consumption, and low calcium intake
- **smoking** and alcohol **abuse**
- cortisone therapy

HEART DISEASE. Heart disease is a major health concern of women in midlife. It is the leading cause of **death** in women over 60. The primary disorders of the circulatory system in postmenopausal women are **stroke**, **hypertension**, and **coronary artery disease**. While doctors once believed that HRT helped decrease heart disease and stroke among postmenopausal women, a major clinical trial discovered the opposite to be true. In 2002, the Women's Health Initiative (WHI) stopped giving HRT to the women enrolled in the study because of adverse effects. Among these effects was a 29% increase in coronary heart disease and 41% increase in stroke in postmenopausal women taking HRT.

Other major factors that are known to increase the risk of heart disease include:



Estrogen replacement patches adhere to a patient's skin and slowly administer estrogen to the body. (Illustration by Argosy Inc.)

- history of smoking
- being overweight
- high-fat **diets**
- alcohol abuse
- family history of heart disease
- high blood pressure
- high blood cholesterol levels
- diabetes.

Less important risk factors include being African American, having a sedentary lifestyle, undergoing menopause before age 45, and having high levels of family- or job-related **stress**.

Precautions

Medical conditions

The findings of the WHI presented new problems for the women relying on hormones to ease their transition to menopause and postmenopausal years and for doctors prescribing HRT. The combination of estrogen and progesterone also was found to increase risk of invasive **breast cancer** by 26%, which was the reason researchers halted the study. In addition, while some clinicians have thought that HRT helped delay **dementia** (a disorder of the mind that affects memory and perception), the

WHI also found that combined estrogen/progesterone increased the risk of probable dementia in women age 65 and older. Physicians and women were advised not to panic about HRT, however. Short-term use of the therapy may not produce these risks. Women have been advised to meet with their physicians and weigh the benefits against the risks on an individual basis.

Certain groups of women should not use HRT. They include women with:

- breast cancer
- cancer of the uterus
- heart disease
- abnormal vaginal bleeding that has not been diagnosed
- high blood pressure that rises when HRT is used
- liver disease
- **gallstones** or diseases of the gallbladder

Drug interactions

HRT can interact with other prescription medications that a woman may be taking. Women who are taking **corticosteroids**, drugs to slow the clotting of blood (anticoagulants), and rifampin should ask their doctor about possible interactions.

Combining estrogens with certain other medicines can cause liver damage. Among the drugs that may cause liver damage when taken with estrogens are:

- acetaminophen (Tylenol), when used in high doses over long periods
- anabolic steroids such as nandrolone (Anabolin) or oxymetholone (Anadrol)
- medicine for infections
- antiseizure medicines such as divalproex (Depakote), valproic acid (Depakene), or phenytoin (Dilantin)
- antianxiety drugs, including chlorpromazine (Thorazine), prochlorperazine (Compazine), and thioridazine (Mellaril).

In addition, estrogens may interfere with the effects of bromocriptine (Parlodel), used to treat Parkinson's disease and other conditions; they may also increase the chance of toxic side effects when taken with cyclosporine (Sandimmune), a drug that helps prevent organ transplant rejection.

Description

HRT medications come in several different forms, including tablets, stick-on patches, injections, and creams that are worn inside the vagina. The form prescribed depends on the purpose of the hormone replacement therapy. Women who want relief from vaginal dryness, for example, would be given a cream or vaginal ring. Women using HRT to relieve hot flashes or to prevent osteoporosis and heart disease often prefer oral medications or patches. All HRT medications used in the United States are available only with a doctor's prescription.

HRT treatment regimens

One of the complications of HRT is the number of treatment options, including combinations of types of estrogen; dosage levels; forms of administration; and whether or not progestins are used with the estrogen to offset the risk of uterine cancer. This variety, however, means that a woman who wants to use HRT while minimizing side effects can try different forms of medication or dosage schedules when she consults her doctor. It is vital, however, for women to follow their doctor's directions exactly and not change dosages themselves.

At present, women who are taking a combination of estrogens and progestins are placed on one of three dosage schedules:

- Estrogen pills taken daily from the first through the 25th day of each month, with a progestin pill taken daily during the last 10–14 days of the cycle. Both

drugs are then stopped for the next five to six days to allow the uterus to shed its lining.

- Estrogen pills taken on a daily basis with low-dose progestin pills, also on a daily basis. Both medications are taken continuously with no days off.
- Estrogen pills and low-dose progestins taken on a daily basis for five days each week, with both medications stopped on the last two days of each week.

TIMING AND LENGTH OF TREATMENT. One of the disagreements about HRT concerns the best time to begin using it. Some doctors think that women should begin using HRT while they are still in perimenopause. Others think that there is no harm in a woman's waiting to decide. Either way, the question of timing means that a woman should keep track of changes in her periods and other signs of perimenopause so that her doctor can evaluate her readiness for HRT.

The other question of timing concerns length of treatment. Some women use HRT only as long as they need it to relieve the symptoms of menopause. Others regard it as a lifetime commitment because of concerns about osteoporosis. One study found that the average length of time that women stay on HRT is 23 months. Information from the WHI released in 2002 and 2003 would indicate that long-term HRT use produced too many risks for the expected benefits.

UNWANTED SIDE EFFECTS. In addition to the identified health risks mentioned above, much of the disagreement about unwanted side effects from HRT concerns the role of progestins in the estrogen/progestin combinations that are commonly prescribed. Many women who find that estrogen relieves hot flashes and other symptoms of menopause have the opposite experience with progestin. Progestin frequently causes moodiness, depression, sore breasts, weight gain, and severe headaches.

Other treatment approaches

Women who are uncertain about HRT, or who should not take estrogens, should know about other treatment options, such as natural progesterone. Progestins, which are synthetic hormones, were developed because natural progesterone cannot be absorbed in the body when taken in pill form. A new technique called micronization has made it possible for women to take natural progesterone by mouth. Many women prefer this form of hormone because it lacks the side effects of the synthetic progestins even though it is somewhat more expensive. The most common form of natural progesterone is called Prometrium, which is available by prescription only. Another form of natural progesterone consists of the hormone suspended in

vitamin E oil. It is absorbed through the skin and is available without a prescription.

Alternative therapies also are available. Many mainstream as well as alternative practitioners recommend changes in diet and **nutrition** as helpful during menopause. Women who limit their intake of fats and salts, increase their use of fresh fruits and vegetables, quit smoking, and drink only in moderation often find that these dietary changes help them feel better. Naturopaths typically recommend vitamin and mineral supplements for general well-being as well as for relief from hot flashes and leg cramps. In addition, herbal teas and tonics are helpful to some women in treating water retention, **insomnia**, **constipation**, or moodiness.

Women who find menopause emotionally stressful because of negative social attitudes toward older women are often helped by **meditation**, **biofeedback**, therapeutic massage, and other relaxation techniques. **Yoga** and **tai chi** provide physical exercise as well as **stress reduction**. Exercise is an important safeguard against osteoporosis.

Preparation

Women who are considering HRT should visit their doctor for a series of tests to make sure that they do not have any serious health disorders. They should have a Pap smear and breast examination to rule out cancer. They also should have a **urinalysis**, a **bone density test**, and blood tests to measure their red blood cell level, blood sugar level, cholesterol level, and liver and thyroid function.

In addition to these tests, most doctors will also give a progesterone challenge test. It consists of doses of progesterone given over a 10-day period to see if the woman is still producing her own estrogen. If she bleeds at the end of the test, she is still producing estrogen.

Aftercare

Aftercare is a very important part of HRT. Women who are taking HRT will need to see their doctor more frequently. At a minimum, they should be checked twice a year with a blood pressure test and breast examination. They should have a complete physical on a yearly basis. Any abnormal bleeding must be reported to the doctor as soon as it occurs. The doctor will need to order a tissue biopsy or dilation and curettage (D & C) in order to rule out cancer of the uterus.

Women who are taking HRT and decide to stop should taper their dosage over a period of several months rather than discontinuing abruptly. A gradual reduction minimizes the possibility of hot flashes and other side effects.

KEY TERMS

Dilation and curettage (D & C)—A surgical procedure in which the patient's cervix is widened (dilated) and the endometrium is scraped with a scoop-shaped instrument (curette).

Estrogen—The primary sex hormone that controls normal sexual development in females. During the menstrual cycle, estrogen helps prepare the body for possible pregnancy.

Follicle-stimulating hormone (FSH)—A hormone produced by the pituitary gland that stimulates the follicles in the ovaries to swell and release ripe ova. Doctors sometimes use its levels in a woman's blood to evaluate whether she is in menopause.

Hormone—A substance secreted by an endocrine gland that is carried by blood or other body fluids to its target tissues or organs.

Hot flash—A warm or hot sensation on the face, neck and upper body, sometimes accompanied by flushing and sweating. Some women refer to hot flashes as hot flushes.

Osteoporosis—A bone disorder in which the bones become brittle, porous, and easily broken. It is a major health concern for postmenopausal women.

Ovary—The female sex gland that produces eggs and female reproductive hormones.

Ovulation—The cyclical process of egg maturation and release from the ovary.

Progesterone—A female hormone produced by the ovary. It functions to prepare the lining of the uterus to receive a fertilized ovum.

Progesterone challenge test—A test that is given to see if a woman is still secreting estrogen. It consists of doses of progesterone given over a 10-day period.

Progestin—Synthetic progesterone available as an oral medication.

Testosterone—A male sex hormone that is sometimes given as part of HRT to women whose ovaries have been removed. Testosterone helps with problems of sexual desire.

Uterus—The hollow organ in women in which fertilized eggs develop during pregnancy. The uterus is sometimes called the womb.

Risks

The short-term risks associated with HRT include a range of physical side effects. Common side effects include fluid retention, bloating, weight gain, sore

breasts, leg cramps, vaginal discharges, migraine headaches, hair loss, **nausea and vomiting**, **acne**, depression, **shortness of breath**, and **dizziness**. Potentially serious side effects include tissue growths in the uterus (fibroids), gallstones, **thrombophlebitis**, **hypoglycemia**, abnormal growth (hyperplasia) of uterine tissue, thyroid disorders, high blood pressure, and cancer.

Long-term risks should be discussed with a woman's physicians before considering hormone replacement therapy. Identified risks for combined (estrogen plus progestin) HRT use include increased incidence of invasive breast cancer, stroke, heart disease, and **pulmonary embolism**.

Normal results

Normal results of HRT include relief of hot flashes, night sweats, vaginal dryness, and urinary symptoms associated with menopause.

Resources

BOOKS

Goldman, Lee, et al, editors. *Cecil Textbook of Medicine*. 21st ed. W. B. Saunders, 2000.

Goroll, Alan H. *Primary Care Medicine*. 4th ed. Lippincott Williams & Wilkins, 2000.

PERIODICALS

Doering, Paul L. "Treatment of Menopause Post-WHI: What Now?" *Drug Topics* April 21, 2003: 85.

Elliott, William T. "HRT, Estrogen, and Postmenopausal Women: Year-old WHI Study Continues to Raise Questions." *Critical Care Alert* July 2003: 1.

OTHER

Menopausal Hormone Replacement Therapy. Fact sheet. National Cancer Institute. <<http://rex.nci.nih.gov>>.

Laith Farid Gulli, M.D.
Teresa G. Odle

Hospital-acquired infections

Definition

A hospital-acquired infection is usually one that first appears three days after a patient is admitted to a hospital or other health care facility. Infections acquired in a hospital are also called nosocomial infections.

Description

About 5–10% of patients admitted to hospitals in the United States develop a nosocomial infection. The Centers for Disease Control and Prevention (CDC) estimate that more than two million patients develop hospital-acquired infections in the United States each year. About 90,000 of these patients die as a result of their infections. Hospital-acquired infections usually are related to a procedure or treatment used to diagnose or treat the patient's illness or injury. About 25% of these infections can be prevented by healthcare workers taking proper precautions when caring for patients.

Hospital-acquired infections can be caused by bacteria, viruses, fungi, or parasites. These microorganisms may already be present in the patient's body or may come from the environment, contaminated hospital equipment, health care workers, or other patients. Depending on the causal agents involved, an infection may start in any part of the body. A localized infection is limited to a specific part of the body and has local symptoms. For example, if a surgical wound in the abdomen becomes infected, the area of the wound becomes red, hot, and painful. A generalized infection is one that enters the bloodstream and causes general systemic symptoms such as **fever**, chills, low blood pressure, or mental confusion.

Hospital-acquired infections may develop from surgical procedures, catheters placed in the urinary tract or blood vessels, or from material from the nose or mouth that is inhaled into the lungs. The most common types of hospital-acquired infections are urinary tract infections (UTIs), pneumonia, and surgical wound infections.

Causes and symptoms

All hospitalized patients are susceptible to contracting a nosocomial infection. Some patients are at greater risk than others—young children, the elderly, and persons with compromised immune systems are more likely to get an infection. Other risk factors for getting a hospital-acquired infection are a long hospital stay, the use of indwelling catheters, failure of healthcare workers to wash their hands, and overuse of **antibiotics**.

Any type of invasive procedure can expose a patient to the possibility of infection. Common causes of hospital-acquired infections include:

- urinary bladder catheterization
- respiratory procedures
- surgery and **wounds**
- intravenous (IV) procedures

Urinary tract infection (UTI) is the most common type of hospital-acquired infection. Most hospital-acquired UTIs happen after urinary catheterization. Catheterization is the placement of a catheter through the urethra into the urinary bladder. This procedure is done to empty urine from the bladder, relieve pressure in the bladder, measure urine in the bladder, put medicine into the bladder, or for other medical reasons.

The healthy urinary bladder is sterile, which means it doesn't have any bacteria or other microorganisms in it. There may be bacteria in or around the urethra but they normally cannot enter the bladder. A catheter can pick up bacteria from the urethra and allow them into the bladder, causing an infection to start.

Bacteria from the intestinal tract are the most common type to cause UTIs. Patients with poorly functioning immune systems or who are taking antibiotics are also at risk for infection by a fungus called *Candida*.

Pneumonia is the second most common type of hospital-acquired infection. Bacteria and other microorganisms are easily brought into the throat by respiratory procedures commonly done in the hospital. The microorganisms come from contaminated equipment or the hands of health care workers. Some of these procedures are respiratory intubation, suctioning of material from the throat and mouth, and mechanical ventilation. The introduced microorganisms quickly colonize the throat area. This means that they grow and form a colony, but do not yet cause an infection. Once the throat is colonized, it is easy for a patient to inhale the microorganisms into the lungs.

Patients who cannot **cough** or gag very well are most likely to inhale colonized microorganisms into their lungs. Some respiratory procedures can keep patients from gagging or coughing. Patients who are sedated or who lose consciousness may also be unable to cough or gag. The inhaled microorganisms grow in the lungs and cause an infection that can lead to pneumonia.

Surgical procedures increase a patient's risk of getting an infection in the hospital. Surgery directly invades the patient's body, giving bacteria a way into normally sterile parts of the body. An infection can be acquired from contaminated surgical equipment or from healthcare workers. Following surgery, the surgical wound can become infected. Other wounds from trauma, burns, and ulcers may also become infected.

Many hospitalized patients need a steady supply of medications or nutrients delivered to their

bloodstream. An intravenous (IV) catheter is placed in a vein and the medication or other substance is infused into the vein. Bacteria transmitted from the surroundings, contaminated equipment, or healthcare workers' hands can invade the site where the catheter is inserted. A local infection may develop in the skin around the catheter. The bacteria also can enter the blood through the vein and cause a generalized infection. The longer a catheter is in place, the greater the risk of infection.

Other hospital procedures that put patients at risk for nosocomial infection are gastrointestinal procedures, obstetric procedures, and kidney dialysis.

Fever is often the first sign of infection. Other symptoms and signs of infection are rapid breathing, mental confusion, low blood pressure, reduced urine output, and a high white blood cell count.

Patients with a UTI may have **pain** when urinating and blood in the urine. Symptoms of pneumonia may include difficulty breathing and coughing. A localized infection causes swelling, redness, and tenderness at the site of infection.

Diagnosis

An infection is suspected any time a hospitalized patient develops a fever that cannot be explained by a known illness. Some patients, especially the elderly, may not develop a fever. In these patients, the first signs of infection may be rapid breathing or mental confusion.

Diagnosis of a hospital-acquired infection is based on:

- symptoms and signs of infection
- examination of wounds and catheter entry sites
- review of procedures that might have led to infection
- laboratory test results

A complete **physical examination** is conducted in order to locate symptoms and signs of infection. Wounds and the skin where catheters have been placed are examined for redness, swelling, or the presence of pus or an **abscess**. The physician reviews the patient's record of procedures performed in the hospital to determine if any posed a risk for infection.

Laboratory tests are done to look for signs of infection. A complete **blood count** can reveal if the white blood cell count is high. White blood cells are immune system cells that increase in numbers in response to an infection. White blood cells or blood may be present in the urine when there is a UTI.

Cultures of blood, urine, sputum, other body fluids, or tissue are done to look for infectious microorganisms. If an infection is present, it is necessary to identify the microorganism so the patient can be treated with the correct medication. A sample of the fluid or tissue is placed in a special medium that bacteria will grow in. Other tests can also be done on blood and body fluids to look for and identify bacteria, fungi, viruses, or other microorganisms responsible for an infection.

If a patient has symptoms suggestive of pneumonia, a chest x ray is done to look for infiltrates of white blood cells and other inflammatory substances in the lung tissue. Samples of sputum can be studied with a microscope or cultured to look for bacteria or fungi.

Treatment

Once the source of the infection is identified, the patient is treated with antibiotics or other medication that kills the responsible microorganism. Many different antibiotics are available that are effective against different bacteria. Some common antibiotics are penicillin, **cephalosporins**, **tetracyclines**, and erythromycin. More and more commonly, some types of bacteria are becoming resistant to the standard antibiotic treatments. When this happens, a different, more powerful antibiotic must be used. Two strong antibiotics that have been effective against resistant bacteria are vancomycin and imipenem, although some bacteria are developing resistance to these antibiotics as well.

Fungal infections are treated with antifungal medications. Examples of these are amphotericin B, nystatin, ketoconazole, itraconazole, and fluconazole.

A number of **antiviral drugs** have been developed that slow the growth or reproduction of viruses. Acyclovir, ganciclovir, foscarnet, and amantadine are examples of antiviral medications.

Prognosis

Hospital-acquired infections are serious illnesses that cause **death** in about 1% of cases. Rapid diagnosis and identification of the responsible microorganism is necessary, so treatment can be started as soon as possible.

Prevention

Hospitals and other healthcare facilities have developed extensive **infection control** programs to

KEY TERMS

Abscess—A localized pocket of pus at a site of infection.

Candida—A yeast-like fungal organism.

Catheter—A thin, hollow tube inserted into the body at specific points in order to inject or withdraw fluids from the body.

Generalized infection—An infection that has entered the bloodstream and has general systemic symptoms such as fever, chills, and low blood pressure.

Localized infection—An infection that is limited to a specific part of the body and has local symptoms.

Nosocomial infection—An infection acquired in the hospital.

prevent nosocomial infections. These programs focus on identifying high risk procedures and other possible sources of infection. High risk procedures such as **urinary catheterization** should be performed only when necessary and catheters should be left in for as little time as possible. Medical instruments and equipment must be properly sterilized to ensure they are not contaminated. Frequent handwashing by healthcare workers and visitors is necessary to avoid passing infectious microorganisms to hospitalized patients. In 2003, the Joint Commission on Accreditation of Healthcare Organizations (JCAHO) announced it would make prevention of nosocomial infections a major goal in 2004 and the coming years. JCAHO, the body that inspects hospitals for quality and accredits them accordingly, issued an alert stating that hospital-acquired infections are seriously underreported. The problem has become more serious for hospitals to address as many bacteria are becoming resistant to antibiotics.

Antibiotics should be used only when necessary. Use of antibiotics creates favorable conditions for infection with the fungal organism *Candida*. Overuse of antibiotics is also responsible for the development of bacteria that are resistant to antibiotics.

Resources

PERIODICALS

Burke, John P. "Infection Control—A Problem for Patient Safety." *The New England Journal of Medicine* February 13, 2003: 651–656.

“Hospital-Acquired Infections are Being Underreported.”
RN March 2003: 16.
 “Nosocomial Infection (From the Editor).” *Health Care Food & Nutrition Focus* June 2003: 2.

Toni Rizzo
 Teresa G. Odle

Hot-spot imaging see **Technetium heart scan**

HRT see **Hormone replacement therapy**

HTLV-1 associated myelopathy see **Tropical spastic paraparesis**

HTLV-1 infection see **Tropical spastic paraparesis**

Human-potential movement

Definition

The human-potential movement is a term used for humanistic psychotherapies that first became popular in the 1960s and early 1970s. The movement emphasized the development of individuals through such techniques as encounter groups, sensitivity training, and primal therapy. Although the human-potential movement and humanistic therapy are sometimes used as synonyms, in reality, humanistic therapy preceded the human-potential movement and provided the movement's theoretical base. Humanistic therapy flourished in the 1940s and 1950s. Its theorists were mostly psychologists rather than medical doctors. They included Gordon Allport, Abraham Maslow, Everett Shostrom, Carl Rogers, and Fritz Perls.

The human-potential movement and humanistic therapy is distinguished by the following emphases:

- A concern for what is uniquely human rather than what humans share with other animals.
- A focus on each person's open-ended growth rather than reshaping individuals to fit society's demands.
- An interest in the here-and-now rather than in a person's childhood history or supposed unconscious conflicts.
- A holistic approach concerned with all levels of human being and functioning—not just the intellectual—including creative and spiritual functioning.
- A focus on psychological health rather than disturbance.

Purpose

The purpose of humanistic therapy is to allow a person to make full use of his or her personal capacities leading to self-actualization. Self-actualization requires the integration of all the components of one's unique personality. These elements or components of personality include the physical, emotional, intellectual, behavioral, and spiritual. The marks of a self-actualized person are maturity, self-awareness, and authenticity. Humanistic therapists think that most people—not only those with obvious problems—can benefit from opportunities for self-development. Humanistic therapy uses both individual and group approaches.

Precautions

Psychotic patients, substance abusers, and persons with severe **personality disorders** or disorders of impulse control may not be appropriate for treatment with humanistic methods.

Description

Humanistic approaches to individual treatment usually follow the same format as other forms of outpatient counseling. Therapists may be medical doctors, nurses, psychologists, social workers, or clergy. Humanistic group treatment formats are flexible, and a wide range of treatment methods are used, ranging from encounter groups and therapy groups to assertiveness training and consciousness-raising groups. In addition, the humanistic tradition has fostered the publication of self-help books for people interested in psychological self-improvement.

Risks

The chief risks include the reinforcement of self-centered tendencies in some patients and the dangers resulting from encounter groups led by persons without adequate training. Poorly led encounter groups can be traumatic to persons with low tolerance for confrontation or “uncovering” of private issues.

Normal results

The anticipated outcome of humanistic therapy is a greater degree of personal wholeness, self-acceptance, and exploration of one's potential. In group treatment, participants are expected to grow in interpersonal empathy and relationship skills. However, there have been few controlled studies to determine the reasonableness of these expectations.

KEY TERMS

Encounter group—A form of humanistic therapy in which participants meet with a trained leader to increase self-awareness and social skills through emotional sharing and confrontation.

Humanistic therapy—An approach to psychotherapy that emphasizes human uniqueness, positive qualities, and individual potential. It is sometimes used as a synonym for the human potential movement.

Primal therapy—A form of humanistic therapy that originated in the 1970s. Participants were encouraged to relive painful events and release feelings through screaming or crying rather than analysis.

Sensitivity training—A form of humanistic group therapy that began in the 1950s. Members participated in unstructured discussions in order to improve understanding of themselves and others.

Resources

BOOKS

Severin, Frank T. "Humanistic Psychology." In *The Encyclopedia of Psychiatry, Psychology, and Psychoanalysis*, edited by Benjamin B. Wolman. New York: Henry Holt and Co., 1996.

Rebecca J. Frey, PhD

Human bite infections

Definition

Human bite infections are potentially serious infections caused by rapid growth of bacteria in broken skin.

Description

Bites—animal and human—are responsible for about 1% of visits to emergency rooms. Bite injuries are more common during the summer months.

Closed-fist injury

In adults, the most common form of human bite is the closed-fist injury, sometimes called the "fight bite." These injuries result from the breaking of the skin over the knuckle joint when a person's fist strikes someone's teeth during a fight.

Causes and symptoms

In children, bite infections result either from accidents during play or from fighting. Most infected bites in adults result from fighting.

The infection itself can be caused by a number of bacteria that live in the human mouth. These include streptococci, staphylococci, anaerobic organisms, and *Eikenella corrodens*. Infections that begin less than 24 hours after the injury are usually produced by a mixture of organisms and can cause a necrotizing infection (causing the death of a specific area of tissue), in which tissue is rapidly destroyed. If a bite is infected, the skin will be sore, red, swollen, and warm to the touch.

Diagnosis

In most cases the diagnosis is made by an emergency room physician on the basis of the patient's history.

Because the human mouth contains a variety of bacteria, the physician will order a laboratory culture to choose the most effective antibiotic.

Treatment

Treatment involves surgical attention as well as medications. Because bites cause puncturing and tearing of skin rather than clean-edged cuts, they must be carefully cleansed. The doctor will wash the wound with water under high pressure and debride it. **Debridement** is the removal of dead tissue and **foreign objects** from a wound to prevent infection. If the bite is a closed-fist injury, the doctor will look for torn tendons or damage to the spaces between the joints. Examination includes x rays to check for bone **fractures** or foreign objects in the wound.

Doctors do not usually suture a bite wound because the connective tissues and other structures in the hand form many small closed spaces that make it easy for infection to spread. Emergency room doctors often consult surgical specialists if a patient has a deep closed-fist injury or one that appears already infected.

The doctor will make sure that the patient is immunized against **tetanus**, which is routine procedure for any open wound. A study released in June 2004 showed that routine use of **antibiotics** for human bites may not be necessary, as physicians try to minimize overuse of antibiotics. Superficial **wounds** in low-risk areas may no longer need antibiotic treatment, but more serious human bites to high-risk areas such as the hands should be treated with antibiotics to prevent serious infection. Patients with closed-fist injuries

KEY TERMS

Closed-fist injury—A hand wound caused when the skin of the fist is torn open by contact with teeth.

Debridement—The surgical removal of dead tissue and/or foreign bodies from a wound or cut.

“Fight bite”—Another name for closed-fist injury.

Necrotizing—Causing the death of a specific area of tissue. Human bites frequently cause necrotizing infections.

may need inpatient treatment in addition to an intravenous antibiotic.

Prognosis

The prognosis depends on the location of the bite and whether it was caused by a child or an adult. Bites caused by children rarely become infected because they are usually shallow. Between 15–30% of bites caused by adults become infected, with a higher rate for closed-fist injuries.

Prevention

Prevention of human bite infections depends upon prompt treatment of any bite caused by a human being, particularly a closed-fist injury.

Resources

PERIODICALS

“Do All Human Bite Wounds Need Antibiotics?”
Emergency Medicine Alert June 2004: 3.

Rebecca J. Frey, PhD
Teresa G. Odle

Human chorionic gonadotropin see
Infertility drugs

Human chorionic gonadotropin pregnancy test

Definition

The most common test of **pregnancy** involves the detection of a hormone known as human chorionic gonadotropin (hCG) in a sample of blood or urine.

Purpose

To determine whether or not a woman is pregnant.

Description

Shortly after a woman's egg is fertilized by her male partner's sperm and is implanted in the lining of the womb (uterus), a placenta begins to form. This organ will help nourish the developing new life. The placenta produces hCG, whose presence, along with other hormones, helps maintain the early stages of pregnancy. Because hCG is produced only by placental tissue and the hormone can be found in the blood or urine of a pregnant woman, it has become a convenient chemical test of pregnancy.

After implantation, the level of detectable hCG rises very rapidly, approximately doubling in quantity every two days until a peak is reached between the sixth and eighth week. Over the next ten or more weeks, the quantity of hCG slowly decreases. After this point, a much lower level is sustained for the duration of the pregnancy. Detectable levels of this hormone may even persist for a month or two after delivery.

Blood tests for hCG are the most sensitive and can detect a pregnancy earlier than urine tests. Blood tests for hCG can also distinguish normal pregnancies from impending miscarriages or pregnancies that occur outside of the uterus (ectopic pregnancies).

If a woman misses her menstrual period and wants to know if she may be pregnant, she can purchase one of many home pregnancy test kits that are currently available. Although each of these products may look slightly different and provide a different set of directions for use, each one detects the presence of hCG. This indicator contains chemical components called antibodies that are sensitive to a certain quantity of this hormone.

Precautions

Although home pregnancy tests may be advertised as having an accuracy of 97% or better, studies indicate that, in practice, pregnancy tests performed in the home may incorrectly indicate that a woman is not pregnant (a false positive result) between 25–50% of the time. Studies also indicate that the false negative results usually result from failing to follow the package directions or testing too soon after a missed menstrual period. Waiting a few days

KEY TERMS

Ectopic pregnancy—A pregnancy that develops outside of the mother's uterus. Ectopic pregnancies often cause severe pain in the lower abdomen and are potentially life-threatening because of the massive blood loss that may occur as the developing embryo/fetus ruptures and damages the tissues in which it has implanted.

Embryo—In humans, the developing individual from the time of implantation to about the end of the second month after conception. From the third month to the point of delivery, the individual is called a fetus.

Hormone—A chemical produced by a specific organ or tissue of the body that is released into the bloodstream in order to exert an effect in another part of the body.

Human chorionic gonadotropin (hCG)—A hormone produced by the placenta of a developing pregnancy.

Hydatidiform mole—A rare, generally benign grape-like mass that grows in the uterus from the remains of an abnormally developed embryo and surrounding tissue. In extremely rare cases, the mole develops into a choriocarcinoma, a malignant tumor whose cells can invade the wall of the uterus.

Implantation—The attachment of the fertilized egg or embryo to the wall of the uterus.

Menstrual cycle—A hormonally regulated series of monthly events that occur during the reproductive years of the human female to ensure that the proper internal environment exists for fertilization, implantation, and development of a baby. Each month, a mature egg is released from the follicle of an ovary. If an egg is released, fertilized, and implanted, the lining of the uterus continues to build. If fertilization and/or implantation does not occur, the egg and all of the excess uterine lining are shed from the body during menstruation.

Miscarriage—Loss of the embryo or fetus and other products of pregnancy before the middle of the second trimester. Often, early in a pregnancy, if the condition of the baby and/or the mother's uterus is not compatible with sustaining life, the pregnancy stops, and the contents of the uterus are expelled. For this reason, miscarriage is also referred to as spontaneous abortion.

Placenta—The organ that unites the developing new life (first called an embryo and later a fetus) to the mother's uterus. The placenta produces hCG, among other hormones, to help maintain the pregnancy. After delivery, the placenta, known at this point as afterbirth, is expelled.

after the missed period was expected can increase the accuracy of the test. Blood and urine tests performed by a laboratory are from 97–100% accurate in detecting pregnancy.

Preparation

Generally, no preparation is required for a pregnancy test given in a doctor's office.

Home pregnancy test kits can be divided into two basic types. One type involves the use of a wand-like device that a woman must place into her urine stream for a brief period of time. The other type of kit involves the use of a cup, a dropper, and a wand or stick with a small well. The cup is used to collect the urine, and the dropper is used to transfer a specific number of drops into the well. Results are displayed by a color change. It's important to follow the package directions very carefully (the techniques vary from brand to brand) and to read the results in the time specified.

Aftercare

No special care is required after a urine test for hCG. Women who feel faint or who continue to bleed after a blood test should be observed until the condition goes away.

Risks

Tests for hCG levels pose no direct risk to a woman's health. The main risk with a home pregnancy test is a false negative result, which may be lessened by following the manufacturer's instructions carefully and waiting at least several days after the expected menstrual period to test. A false negative result can cause a delay in seeking prenatal care, which can pose a risk to both the woman and the baby.

Abnormal results

In most cases, a positive result is an indication of pregnancy. However, false positive results may

also occur. If a pregnancy test is performed within a month or two of a recent birth or **miscarriage**, it is possible to test positive for pregnancy since hCG may still be detected in a woman's urine. Sometimes positive pregnancy tests provide clues of an early miscarriage that might have otherwise gone unrecognized because it occurred before or just after a missed period. An **ectopic pregnancy** (one in which an embryo implants outside the uterus), certain types of masses (such as an ovarian tumor or a **hydatidiform mole**), and the use of some fertility drugs that contain hCG are among other possibilities behind false positive results.

Normal results

A woman should notify her physician immediately if her home pregnancy test is positive. Pregnancy can then be confirmed with hCG urine or blood tests taken in the doctor's office and evaluated by laboratory personnel. If performed accurately, home pregnancy tests have been found to be highly reliable. However, the versions of these tests performed by qualified laboratory technologists are considered to be definitive. Often, such a test will produce positive results before a woman experiences symptoms or before a doctor's exam reveals signs of pregnancy.

Resources

PERIODICALS

Bastian, L. A., et al. "Is This Patient Pregnant?" *The Journal of the American Medical Association* 278, no. 7: 586-591.

Perey, T. R., and R. D. Powers. "Bedside Diagnostic Testing of Body Fluids." *American Journal of Emergency Medicine* 15, no. 4: 404-405.

Betty Mishkin

Human herpes see **Roseola**

Human leukocyte antigen test

Definition

The human leukocyte antigen test, also known as HLA, is a test that detects antigens (genetic markers) on white blood cells. There are four types of human leukocyte antigens: HLA-A, HLA-B, HLA-C, and HLA-D.

Purpose

The HLA test is used to provide evidence of tissue compatibility typing of tissue recipients and donors. It is also an aid in **genetic counseling** and in paternity testing.

Precautions

This test may have to be postponed if the patient has recently undergone a **transfusion**.

Description

Human leukocyte antigen (leukocyte is the name for white blood cell, while antigen refers to a genetic marker) is a substance that is located on the surface of white blood cells. This substance plays an important role in the body's immune response.

Because the HLA antigens are essential to immunity, identification aids in determination of the degree of tissue compatibility between transplant recipients and donors. Testing is done to diminish the likelihood of rejection after transplant, and to avoid graft-versus-host disease (GVHD) following major organ or **bone marrow transplantation**. It should be noted that risk of GVHD exists even when the donor and recipient share major antigens. As an example, it was recently discovered that a mismatch of HA-1 (a minor antigen) was a cause of GVHD in bone marrow grafts from otherwise HLA-identical donors.

HLA can aid in paternity exclusion testing, a highly specialized area of forensic medicine. To resolve cases of disputed paternity, a man who demonstrates a phenotype (two haplotypes: one from the father and one from the mother) with no haplotype or antigen pair identical to one of the child's is excluded as the father. Conversely, a man who has one haplotype identical to one of the child's may be the father (the probability varies with the appearance of that particular haplotype in the population). Because of the issues involved, this type of testing is referred to experts.

Certain HLA types have been linked to diseases, such as **rheumatoid arthritis**, **multiple sclerosis**, serum lupus erythematosus, and other **autoimmune disorders**. By themselves, however, none of the HLA types are considered definitive. Because the clinical significance of many of the marker antigens has not yet been well defined, definitive diagnosis of disease is obtained by the use of more specific tests.

KEY TERMS

Autoimmune disorders—A disorder caused by a reaction of an individual's immune system against the organs or tissues of the body. Autoimmune processes can have different results: slow destruction of a particular type of cell or tissue, stimulation of an organ into excessive growth, or interference in function.

Haplotype—A set of alleles (an alternative form of a gene that can occupy a particular place on a chromosome) of a group of closely linked genes which are usually inherited as a unit.

Phenotype—1) The entire physical, biochemical, and physiologic makeup of an individual, as opposed to genotype. 2) The expression of a single gene or gene pair.

Preparation

The HLA test requires a blood sample. There is no need for the patient to be **fasting** (having nothing to eat or drink) before the test.

Risks

Risks for this test are minimal, but may include slight bleeding from the blood-drawing site, **fainting** or feeling lightheaded after venipuncture, or hematoma (blood accumulating under the puncture site).

Normal results

Identification of specific leukocyte antigens, HLA-A, HLA-B, HLA-C and HLA-D.

Abnormal results

Incompatible groups between organ donors and recipients may cause unsuccessful tissue transplantation.

Certain diseases have a strong association with certain types of HLAs, which may aid in genetic counseling. For example, Hashimoto's **thyroiditis** (an autoimmune disorder involving underproduction by the thyroid gland) is associated with HLA-DR5, while B8 and Dw3 are allied with Graves' disease (another autoimmune disorder, but with overproduction by the thyroid gland). Hereditary **hemochromatosis** (too much iron in the blood) is associated with HLA-A3,

B7, and B14. HLA-A3 is found in approximately 70% of patients with hemochromatosis, but as is the case with other HLA-associated disorders, the expense of HLA typing favors use of other tests. In cases of suspected hemochromatosis, for example, diagnosis is better aided by two tests called transferrin saturation and serum ferritin.

Resources

BOOKS

Pagana, Kathleen Deska. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis: Mosby, Inc., 1998.

Janis O. Flores

Humanistic therapy see **Gestalt therapy; Human-potential movement**

Humpback see **Kyphosis**

Hunchback see **Kyphosis**

Hunter's syndrome see **Mucopolysaccharidoses**

Huntington disease

Definition

Huntington disease (HD) is a progressive neurodegenerative disease causing uncontrolled physical movements and mental deterioration. The disease was discovered by George Sumner Huntington (1850–1916), an Ohio doctor who first described the hereditary movement disorder in 1872.

Description

Huntington disease is also called Huntington chorea or hereditary chorea. The word chorea comes from the Greek word for “dance” and refers to the involuntary movements of the patient's feet, lower arms, and face that develop as the disease progresses. It is occasionally referred to as “Woody Guthrie's disease” for the American folk singer who died from it. Huntington disease (HD) causes progressive loss of cells in areas of the brain responsible for certain aspects of movement control and mental abilities. A person with HD gradually develops abnormal movements and changes in cognition (thinking), behavior and personality.

The onset of symptoms of HD usually occurs between the ages of 30 and 50; although in 10% of cases, onset is in late childhood or early adolescence. Approximately 30,000 people in the United States are affected by HD, with another 150,000 at risk for developing this disorder. The frequency of HD is 4–7 cases per 100,000 persons.

Causes and symptoms

Huntington disease is caused by a defect in the HD gene (an inherited unit which contains a code for a protein), which is located on the short arm of chromosome 4. The gene codes for a protein called huntingtin, whose function is not known as of early 2005. The nucleotide codes (building blocks of genes arranged in a specific code which chemically forms into proteins), contain CAG repeats (40 or more of these repeat sequences). The extra building blocks in the huntingtin gene cause the protein that is made from it to contain an extra section as well. It is currently thought that this extra protein section, or portion, interacts with other proteins in brain cells where it occurs, and that this interaction ultimately leads to cell death.

The HD gene is a dominant gene, meaning that only one copy of it is needed to develop the disease. HD affects both males and females. The gene may be inherited from either parent, who will also be affected by the disease. A parent with the HD gene has a 50% chance of passing it on to each offspring. The chances of passing on the HD gene are not affected by the results of previous pregnancies.

The symptoms of HD fall into three categories: motor or movement symptoms; personality and behavioral changes; and cognitive decline. The severity and rate of progression of each type of symptom can vary from person to person.

Early motor symptoms include restlessness, twitching and a desire to move about. Handwriting may become less controlled, and coordination may decline. Later symptoms include:

- dystonia, or sustained abnormal postures, including facial grimaces, a twisted neck, or an arched back
- chorea, in which involuntary jerking, twisting or writhing motions become pronounced
- slowness of voluntary movements, inability to regulate the speed or force of movements, inability to initiate movement and slowed reactions
- difficulty speaking and swallowing due to involvement of the throat muscles

- localized or generalized weakness and impaired balance ability
- rigidity, especially in late-stage disease

Personality and behavioral changes include depression, irritability, **anxiety** and apathy. The person with HD may become impulsive, aggressive or socially withdrawn.

Cognitive changes include loss of ability to plan and execute routine tasks, slowed thought, and impaired or inappropriate judgment. Short-term memory loss usually occurs, although long-term memory is usually not affected. The person with late-stage HD usually retains knowledge of his environment and recognizes family members or other loved ones, despite severe cognitive decline.

Diagnosis

Diagnosis of HD begins with a detailed medical history, and a thorough physical and neurological examination. The family's medical history is very important. **Magnetic resonance imaging** (MRI) or computed tomography scan (CT scan) imaging may be performed to look for degeneration in the basal ganglia and cortex, the brain regions most affected in HD.

Physicians have recently developed a Uniform Huntington's Disease Rating Scale, or UHDRS, to assess a patient's symptoms and the speed of progression of the disease.

A genetic test is available for confirmation of the clinical diagnosis. In this test, a small blood sample is taken, and DNA from it is analyzed to determine the CAG repeat number. A person with a repeat number of 30 or below will not develop HD. A person with a repeat number between 35 and 40 may not develop the disease within their normal lifespan. A person with a very high number of repeats (70 or above) is likely to develop the juvenile-onset form. An important part of **genetic testing** is extensive **genetic counseling**.

Prenatal testing is available. A person at risk for HD (a child of an affected person) may obtain fetal testing without determining whether she herself carries the gene. This test, also called a linkage test, examines the pattern of DNA near the gene in both parent and fetus, but does not analyze for the triple nucleotide repeat (CAG). If the DNA patterns do not match, the fetus can be assumed not to have inherited the HD gene, even if present in the parent. A pattern match indicates the fetus probably has the same genetic makeup of the at-risk parent.

Treatment

There is no cure for HD, nor any treatment that can slow the rate of progression. Treatment is aimed at reducing the disability caused by the motor impairments, and treating behavioral and emotional symptoms.

Physical therapy is used to maintain strength and compensate for lost strength and balance. Stretching and range of motion exercises help minimize contraction, or muscle shortening, a result of weakness and disuse. The physical therapist also advises on the use of mobility aids such as walkers or wheelchairs.

Motor symptoms may be treated with drugs, although some studies suggest that anti-chorea treatment rarely improves function. Chorea (movements caused by abnormal muscle contractions) can be suppressed with drugs that deplete dopamine, an important brain chemical regulating movement. As HD progresses, natural dopamine levels fall, leading to loss of chorea and an increase in rigidity and movement slowness. Treatment with L-dopa (which resupplies dopamine) may be of some value. Frequent reassessment of the effectiveness and appropriateness of any drug therapy is necessary.

Occupational therapy is used to design compensatory strategies for lost abilities in the activities of daily living, such as eating, dressing, and grooming. The occupational therapist advises on modifications to the home that improve safety, accessibility, and comfort.

Difficulty swallowing may be lessened by preparation of softer foods, blending food in an electric blender, and taking care to eat slowly and carefully. Use of a straw for all liquids can help. The potential for **choking** on food is a concern, especially late in the disease progression. Caregivers should learn the use of the **Heimlich maneuver**. In addition, passage of food into the airways increases the risk for **pneumonia**. A gastric feeding tube may be needed, if swallowing becomes too difficult or dangerous.

Speech difficulties may be partially compensated by using picture boards or other augmentative communication devices. Loss of cognitive ability affects both speech production and understanding. A speech-language pathologist can work with the family to develop simplified and more directed communication strategies, including speaking slowly, using simple words, and repeating sentences exactly.

Early behavioral changes, including depression and anxiety, may respond to drug therapy. Maintaining a calm, familiar, and secure environment

is useful as the disease progresses. Support groups for both patients and caregivers form an important part of treatment.

Experimental transplant of fetal brain tissue has been attempted in a few HD patients. Early results show some promise, but further trials are needed to establish the effectiveness of this treatment.

Tetrabenazine (Nitoman), a drug that has been considered investigational in the United States, appears to benefit some patients with HD by controlling the involuntary movements of chorea. It works by lowering the levels of dopamine and other neurotransmitters in the brain. The Food and Drug Administration (FDA) granted tetrabenazine fast-track and orphan drug status in the United States as of 2004. It is not yet manufactured in the United States but can be obtained from the United Kingdom.

In 2004 the Food and Drug Administration (FDA) also approved deep brain stimulation (DBS) as an acceptable treatment for HD and other **movement disorders**. In DBS, the surgeon implants a battery-operated medical device called a neurostimulator, which delivers electrical impulses to the areas of the brain that govern movement. In the case of Huntington's patients, the part of the brain that is targeted is a structure called the globus pallidus. It is thought that DBS works by increasing the flow of blood to this area. A group of Canadian researchers reported in 2004 that some patients with HD are better able to control their movements after DBS of the globus pallidus.

Psychotherapy is often recommended for individuals who know themselves to be at risk for the disease. Some persons want to know their risk status while others prefer not to be tested. Psychotherapy may be useful in helping at-risk persons decide about testing as well as coping with the results of the test.

Prognosis

The person with Huntington disease may be able to maintain a job for several years after diagnosis, despite the increase in disability. Loss of cognitive functions and increase in motor and behavioral symptoms eventually prevent the person with HD from continuing employment. Ultimately, severe motor symptoms prevent mobility. Death usually occurs between 10 and 30 years after disease onset, typically as the result of pneumonia or a fall. Progressive weakness of respiratory and swallowing muscles leads to increased risk of respiratory infection and choking, the most common causes of death. Future research in this area is currently focusing on nerve cell transplantation.

KEY TERMS

Basal ganglia—A structure at the base of the brain composed of four groups of nerve cells, responsible for body movements and coordination.

Chorea—Brief and purposeless involuntary movements of the lower arms, feet, and face.

Chromosome—The structures that carry genetic information. Chromosomes are located within every cell, and are responsible for directing the development and functioning of all the cells in the body. The normal number is 46 (23 pairs).

Cortex—The part of the brain responsible for thought, memory, and sensory perception.

Dystonia—A movement disorder characterized by sustained muscle contractions that result in writhing or twisting movements and unusual body postures.

Globus pallidus—A pale-colored spherical structure within the basal ganglia. Deep brain stimulation of this area is helpful in controlling the chorea of some patients with HD.

Huntingtin—A protein of unknown function encoded by the HD gene. The repeated CAG sequence in a defective HD gene causes the body to produce an abnormal form of huntingtin. It is not yet known why the abnormal form of huntingtin affects only certain regions of the brain.

Orphan drug—A term for a drug that treats a rare disease, defined by the Food and Drug Administration (FDA) as one that affects fewer than 200,000 Americans. The FDA has an Office of Orphan Products Development (OOPD), which offers grants to researchers to develop these products.

Tetrabenazine—A drug given to control chorea that appears to benefit HD patients.

Revilla, Fredy J., MD, and Jaime Grutzendler, MD. "Huntington Disease." *eMedicine* November 3, 2004. < <http://www.emedicine.com/NEURO/topic81.htm> > .

Richartz, E. R., and C. Frank. "A Psychodynamic Approach in Counselling Vulnerable Persons for Chorea Huntington—A Case Report." [in German] *Psychiatrische Praxis* 31 (July 2004): 255–258.

Seneca, S., D. Fagnart, K. Keymolen, et al. "Early-Onset Huntington Disease: A Neuronal Degeneration Syndrome." *European Journal of Pediatrics* 26 (August 2004): e-pub.

ORGANIZATIONS

Huntington Disease Society of America. 140 W. 22nd St. New York, NY 10011. (800) 345-HDSA.

National Institute of Neurological Disorders and Stroke (NINDS). NIH Neurological Institute, P. O. Box 5801, Bethesda, MD 20824. (800) 352-9424 or (301) 496-5751. < <http://www.ninds.nih.gov> > .

National Organization for Rare Disorders (NORD). 55 Kenosia Avenue, P. O. Box 1968, Danbury, CT 06813-1968. (203) 744-0100. Fax: (203) 798-2291. < <http://www.rarediseases.org> > .

United States Food and Drug Administration (FDA). 5600 Fishers Lane, Rockville, MD 20857-0001. (888) INFO-FDA. < <http://www.fda.gov> > .

OTHER

Food and Drug Administration (FDA). "Grants Awarded by the OOPD Program." < <http://www.fda.gov/orphan/grants/previous.htm> > .

National Institute of Neurological Disorders and Stroke (NINDS). "Huntington's Disease: Hope Through Research." NIH Publication No. 98-49. Bethesda, MD: NINDS, 2005. < http://www.ninds.nih.gov/disorders/huntington/detail_huntington.htm > .

National Institute of Neurological Disorders and Stroke (NINDS). "NINDS Deep Brain Stimulation for Parkinson's Disease Information Page." Bethesda, MD: NINDS, 2004. < http://www.ninds.nih.gov/disorders/deep_brain_stimulation/deep_brain_stimulation.htm > .

Laith Farid Gulli, M.D.
Rebecca J. Frey, PhD

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD., editors. "Disorders of Movement." Section 14, Chapter 179 In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

PERIODICALS

Montgomery, E. B., Jr. "Deep Brain Stimulation for Hyperkinetic Disorders." *Neurosurgical Focus* 17 (July 15, 2004): E1.

Moro, E., A. E. Lang, A. P. Strafella, et al. "Bilateral Globus Pallidus Stimulation for Huntington's Disease." *Annals of Neurology* 56 (August 2004): 290–294.

Huntington's chorea see **Huntington's disease**

Hurler's syndrome see **Mucopolysaccharidoses**

HUS see **Hemolytic-uremic syndrome**

Hyaline see **Respiratory distress syndrome**

Hydatid see **Echinococcosis**

Hydatidiform mole

Definition

A hydatidiform mole is a relatively rare condition in which tissue around a fertilized egg that normally would have developed into the placenta instead develops as an abnormal cluster of cells. (This is also called a molar pregnancy.) This grape-like mass forms inside of the uterus after fertilization instead of a normal embryo. A hydatidiform mole triggers a positive **pregnancy** test and in some cases can become cancerous.

Description

A hydatidiform mole (“hydatid” means “drop of water” and “mole” means “spot”) occurs in about 1 out of every 1,500 (1/1,500) pregnancies in the United States. In some parts of Asia, however, the incidence may be as high as 1 in 200 (1/200). Molar pregnancies are most likely to occur in younger and older women (especially over age 45) than in those between ages 20–40. About 1–2% of the time a woman who has had a molar pregnancy will have a second one.

A molar pregnancy occurs when cells of the chorionic villi (tiny projections that attach the placenta to the lining of the uterus) don’t develop correctly. Instead, they turn into watery clusters that can’t support a growing baby. A partial molar pregnancy includes an abnormal embryo (a fertilized egg that has begun to grow) that does not survive. In a complete molar pregnancy there is a small cluster of clear blisters or pouches that don’t contain an embryo.

If not removed, about 15% of **moles** can become cancerous. They burrow into the wall of the uterus and cause serious bleeding. Another 5% will develop into fast-growing cancers called choriocarcinomas. Some of these tumors spread very quickly outside the uterus in other parts of the body. Fortunately, **cancer** developing from these moles is rare and highly curable.

Causes and symptoms

The cause of hydatidiform mole is unclear; some experts believe it is caused by problems with the chromosomes (the structures inside cells that contain genetic information) in either the egg or sperm, or both. It may be associated with poor **nutrition**, or a problem with the ovaries or the uterus. A mole sometimes can develop from placental tissue that is left behind in the uterus after a **miscarriage** or **childbirth**.

Women with a hydatidiform mole will have a positive pregnancy test and often believe they have a normal pregnancy for the first three or four months. However, in these cases the uterus will grow abnormally fast. By the end of the third month, if not earlier, the woman will experience vaginal bleeding ranging from scant spotting to excessive bleeding. She may have **hyperthyroidism** (overproduction of **thyroid hormones** causing symptoms such as weight loss, increased appetite, and intolerance to heat). Sometimes, the grapelike cluster of cells itself will be shed with the blood during this time. Other symptoms may include severe **nausea and vomiting** and high blood pressure. As the pregnancy progresses, the fetus will not move and there will be no fetal heartbeat.

Diagnosis

The physician may not suspect a molar pregnancy until after the third month or later, when the absence of a fetal heartbeat together with bleeding and severe nausea and vomiting indicates something is amiss.

First, the physician will examine the woman’s abdomen, feeling for any strange lumps or abnormalities in the uterus. A tubal pregnancy, which can be life threatening if not treated, will be ruled out. Then the physician will check the levels of human chorionic gonadotropin (hCG), a hormone that is normally produced by a placenta or a mole. Abnormally high levels of hCG together with the symptoms of vaginal bleeding, lack of fetal heartbeat, and an unusually large uterus all indicate a molar pregnancy. An ultrasound of the uterus to make sure there is no living fetus will confirm the diagnosis.

Treatment

It is extremely important to make sure that all of the mole is removed from the uterus, since it is possible that the tissue is potentially cancerous. Often, the tissue is naturally expelled by the fourth month of pregnancy. In some instances, the physician will give the woman a drug called oxytocin to trigger the release of the mole that is not spontaneously aborted.

If this does not happen, however, a vacuum aspiration can be performed to remove the mole. In a procedure similar to a **dilatation and curettage** (D & C), a woman is given an anesthetic (to deaden feeling during the procedure), her cervix (the structure at the bottom of the uterus) is dilated and the contents of the uterus is gently suctioned out. After the mole has been mostly removed, gentle scraping of the uterus lining is usually performed.

KEY TERMS

Dilatation and curettage (D & C)—Dilating the cervix and scraping the lining of the uterus with an instrument called a curette.

Placenta—The circular, flat organ that connects the fetus via the umbilical cord to the uterus for oxygen, food, and elimination of wastes.

If the woman is older and does not want any more children, the uterus can be surgically removed (**hysterectomy**) instead of a vacuum aspiration because of the higher risk of cancerous moles in this age group.

Because of the cancer risk, the physician will continue to monitor the patient for at least two months after the end of a molar pregnancy. Since invasive disease is usually signaled by high levels of hCG that don't go down after the pregnancy has ended, the woman's hCG levels will be checked every two weeks. If the levels don't return to normal by that time, the mole may have become cancerous.

If the hCG level is normal, the woman's hCG will be tested each month for six months, and then every two months for a year.

If the mole has become cancerous, treatment includes removal of the cancerous issue and **chemotherapy**. If the cancer has spread to other parts of the body, radiation will be added. Specific treatment depends on how advanced the cancer is.

Women should make sure not to become pregnant within a year after hCG levels have returned to normal. If a woman were to become pregnant sooner than that, it would be difficult to tell whether the resulting high levels of hCG were caused by the pregnancy or a cancer from the mole.

Prognosis

A woman with a molar pregnancy often goes through the same emotions and sense of loss as does a woman who has a miscarriage. Most of the time, she truly believed she was pregnant and now has suffered a loss of the baby she thought she was carrying. In addition, there is the added worry that the tissue left behind could become cancerous.

In the unlikely case that the mole is cancerous the cure rate is almost 100%. As long as the uterus was not removed, it would still be possible to have a child at a later time.

Resources

BOOKS

Carlson, Karen J., Stephanie A. Eisenstat, and Terra Ziporyn. *The Harvard Guide to Women's Health*. Cambridge, MA: Harvard University Press, 1996.

Carol A. Turkington

Hydrocelectomy

Definition

Hydrocelectomy is a surgical procedure to remove a hydrocele. A hydrocele is collected fluid in the membrane surrounding the testes.

Purpose

Hydrocelectomy is performed to relieve the **pain** or reoccurrence of a hydrocele. Normally, hydroceles are not very painful. They tend to be a soft swelling in the membrane surrounding the testes. As the hydrocele grows, the scrotum gets larger. Hydroceles do not damage the testes. The main symptom is scrotal swelling. There are two types of hydroceles depending on how they form. One type is seen in children, generally shortly after birth. It is caused by a failure of the processus vaginalis to close. Usually, surgery isn't used to treat hydrocele until after two years of age because the processus vaginalis frequently closes by itself if given extra time. In adults, hydroceles develop slowly. Most hydroceles develop because of blocked lymphatic flow. Hydroceles also develop after infection, injury, or local **cancer** tumors. Generally, hydroceles are treated by aspiration of the collected fluid. To do this, a needle is inserted into the scrotum and directed toward the hydrocele. Once there, as much fluid as possible is removed. Hydroceles can reoccur. Rarely, hydroceles grow larger and cause pain. Surgery is used to remove large or painful hydroceles. It is also the recommended procedure to remove hydroceles that reoccur after aspiration. Hydroceles are distinguished from other testicular problems by transillumination and **scrotal ultrasound** examinations.

Precautions

No special precautions are required for hydrocelectomy. It is typically performed on an outpatient basis.

KEY TERMS

Aspiration—The process of removing fluids or gases from the body by suction.

Hernia—The protrusion of an organ or tissue through a wall that normally contains it.

Hydrocele—An accumulation of fluid in the membrane surrounding the testes (tunica vaginalis testis).

Description

Aspiration of the fluid in a hydrocele is usually successful. However, aspiration may be only a temporary solution because of the potential that the hydrocele will reoccur. Generally, surgical repair of a hydrocele will eliminate the hydrocele. The extent of the surgery depends on whether other factors are present. If the hydrocele is uncomplicated, an incision is made in the scrotum. The hydrocele is cut out, removing the tissues involved in the hydrocele. If there are complications present, such as a **hernia**, an incision is made in the inguinal (groin) area. This approach allows repair of hernias and other complicating factors at the same time. Patients are placed under **general anesthesia** for these operations.

Preparation

A physician or nurse will explain the procedure and, in some cases, the need for a temporary drain to be inserted. The drain lessens the chance of infection and prevents fluid build-up.

Aftercare

Following surgery, the patient usually only needs a follow-up examination several weeks after the surgery to examine the incision and to check for signs of infection.

Risks

There is a slight risk of infection and internal hemorrhage as well as a chance of excessive bleeding from the surgical incision.

Normal results

There may be swelling of the scrotum for up to a month. The patient is able to resume most activities within 7–10 days, although heavy lifting and sexual

activities may be delayed for up to six weeks. The hydrocele does not grow back.

Abnormal results

Swelling that lasts for several months is sometimes a complication of hydrocelectomy. Infection can also occur.

Resources

BOOKS

Way, Lawrence W., editor. *Current Surgical Diagnosis and Treatment*. 10th ed. Stamford: Appleton & Lange, 1994.

Hydrocephalus

Definition

Hydrocephalus is an abnormal expansion of cavities (ventricles) within the brain that is caused by the accumulation of cerebrospinal fluid. Hydrocephalus comes from two Greek words: *hydros* means water and *cephalus* means head.

There are two main varieties of hydrocephalus: congenital and acquired. An obstruction of the cerebral aqueduct (aqueductal stenosis) is the most frequent cause of congenital hydrocephalus. Acquired hydrocephalus may result from spina bifida, intraventricular hemorrhage, **meningitis**, head trauma, tumors, and cysts.

Description

Hydrocephalus is the result of an imbalance between the formation and drainage of cerebrospinal fluid (CSF). Approximately 500 milliliters (about a pint) of CSF is formed within the brain each day, by epidermal cells in structures collectively called the choroid plexus. These cells line chambers called ventricles that are located within the brain. There are four ventricles in a human brain. Once formed, CSF usually circulates among all the ventricles before it is absorbed and returned to the circulatory system. The normal adult volume of circulating CSF is 150 ml. The CSF turn-over rate is more than three times per day. Because production is independent of absorption, reduced absorption causes CSF to accumulate within the ventricles.

There are three different types of hydrocephalus. In the most common variety, reduced absorption occurs when one or more passages connecting the ventricles become blocked. This prevents the movement of CSF to its drainage sites in the subarachnoid space just inside the skull. This type of hydrocephalus is called “noncommunicating.” In a second type, a reduction in the absorption rate is caused by damage to the absorptive tissue. This variety is called “communicating hydrocephalus.”

Both of these types lead to an elevation of the CSF pressure within the brain. This increased pressure pushes aside the soft tissues of the brain. This squeezes and distorts them. This process also results in damage to these tissues. In infants whose skull bones have not yet fused, the intracranial pressure is partly relieved by expansion of the skull, so that symptoms may not be as dramatic. Both types of elevated-pressure hydrocephalus may occur from infancy to adulthood.

A third type of hydrocephalus, called “normal pressure hydrocephalus,” is marked by ventricle enlargement without an apparent increase in CSF pressure. This type affects mainly the elderly.

Hydrocephalus has a variety of causes including:

- congenital brain defects
- hemorrhage, either into the ventricles or the subarachnoid space
- infection of the central nervous system (**syphilis**, herpes, meningitis, **encephalitis**, or mumps)
- tumor

Hydrocephalus is believed to occur in approximately one to two of every 1,000 live births. The incidence of adult onset hydrocephalus is not known. There is no known way to prevent hydrocephalus.

Causes and symptoms

Hydrocephalus that is congenital (present at birth) is thought to be caused by a complex interaction of genetic and environmental factors. Aqueductal stenosis, an obstruction of the cerebral aqueduct, is the most frequent cause of congenital hydrocephalus. As of 2001, the genetic factors are not well understood. According to the British Association for **Spina Bifida** and Hydrocephalus, in very rare circumstances, hydrocephalus is due to hereditary factors, which might affect future generations.

Signs and symptoms of elevated-pressure hydrocephalus include:

- headache
- nausea and **vomiting**, especially in the morning

- lethargy
- disturbances in walking (gait)
- double vision
- subtle difficulties in learning and memory
- delay in children achieving developmental milestones

Irritability is the most common sign of hydrocephalus in infants. If this is not treated, it may lead to lethargy. Bulging of the fontanelles, or the soft spots between the skull bones, may also be an early sign. When hydrocephalus occurs in infants, fusion of the skull bones is prevented. This leads to abnormal expansion of the skull.

Symptoms of normal pressure hydrocephalus include **dementia**, gait abnormalities, and incontinence (involuntary urination or bowel movements).

Diagnosis

Imaging studies—x ray, computed tomography scan (CT scan), ultrasound, and especially **magnetic resonance imaging** (MRI)—are used to assess the presence and location of obstructions, as well as changes in brain tissue that have occurred as a result of the hydrocephalus. Lumbar puncture (spinal tap) may be performed to aid in determining the cause when infection is suspected.

Treatment

The primary method of treatment for both elevated and normal pressure hydrocephalus is surgical installation of a shunt. A shunt is a tube connecting the ventricles of the brain to an alternative drainage site, usually the abdominal cavity. A shunt contains a one-way valve to prevent reverse flow of fluid. In some cases of non-communicating hydrocephalus, a direct connection can be made between one of the ventricles and the subarachnoid space, allowing drainage without a shunt.

Installation of a shunt requires lifelong monitoring by the recipient or family members for signs of recurring hydrocephalus due to obstruction or failure of the shunt. Other than monitoring, no other management activity is usually required.

Some drugs may postpone the need for surgery by inhibiting the production of CSF. These include acetazolamide and furosemide. Other drugs that are used to delay surgery include glycerol, digoxin, and isosorbide.

Some cases of elevated pressure hydrocephalus may be avoided by preventing or treating the

KEY TERMS

Cerebral ventricles—Spaces in the brain that are located between portions of the brain and filled with cerebrospinal fluid.

Cerebrospinal fluid—Fluid that circulates throughout the cerebral ventricles and around the spinal cord within the spinal canal.

Choroid plexus—Specialized cells located in the ventricles of the brain that produce cerebrospinal fluid.

Fontanelle—One of several “soft spots” on the skull where the developing bones of the skull have yet to fuse.

Shunt—A small tube placed in a ventricle of the brain to direct cerebrospinal fluid away from the blockage into another part of the body.

Stenosis—The constricting or narrowing of an opening or passageway.

Subarachnoid space—The space between two membranes surrounding the brain, the arachnoid and pia mater.

infectious diseases which precede them. Prenatal diagnosis of congenital brain malformation is often possible, offering the option of family planning.

Prognosis

The prognosis for elevated-pressure hydrocephalus depends on a wide variety of factors, including the cause, age of onset, and the timing of surgery. Studies indicate that about half of all children who receive appropriate treatment and follow-up will develop IQs greater than 85. Those with hydrocephalus at birth do better than those with later onset due to meningitis. For individuals with normal pressure hydrocephalus, approximately half will benefit by the installation of a shunt.

Resources

BOOKS

Toporek, Chuck, and Kellie Robinson. *Hydrocephalus: A Guide for Patients, Families & Friends*. Cambridge, Mass.: O'Reilly & Associates, 1999.

PERIODICALS

“Hydrocephalus.” *Review of Optometry* 137, no. 8 (August 15, 2000): 56A.

ORGANIZATIONS

Association for Spina Bifida and Hydrocephalus. 42 Park Rd., Peterborough, PE1 2UQ. UK 0173 355 5988. Fax: 017 3355 5985. postmaster@asbah.org. <<http://www.asbah.demon.co.uk>>.

Hydrocephalus Foundation, Inc., (HyFI). 910 Rear Broadway, Saugus, MA 01906. (781) 942-1161. HyFI1@netscape.net. <<http://www.hydrocephalus.org>>.

OTHER

“Hydrocephalus.” *American Association of Neurological Surgeons/Congress of Neurological Surgeons*. <<http://www.neurosurgery.org/pubpages/patres/hydrobroch.html>>.

“Hydrocephalus.” *Institute for Neurology and Neurosurgery*. Beth Israel Medical Center, New York, NY. <<http://nyneurosurgery.org/child/hydrocephalus/hydrocephalus.htm>>.

“Hydrocephalus.” National Library of Medicine. *MEDLINEplus*. <<http://www.nlm.nih.gov/medlineplus/hydrocephalus.html>>.

L. Fleming Fallon, MD, PhD, DrPH

Hydrochlorothiazide see **Diuretics**

Hydrocodone see **Analgesics, opioid**

Hydrogen peroxide see **Antiseptics**

Hydronephrosis

Definition

Hydronephrosis is the swelling of the kidneys when urine flow is obstructed in any of part of the urinary tract. Swelling of the ureter, which always accompanies hydronephrosis, is called hydroureter. Hydronephrosis implies that a ureter and the renal pelvis (the connection of the ureter to the kidney) are overfilled with urine.

Description

The kidneys filter urine out of the blood as a waste product. It collects in the renal pelvis and flows down the ureters into the bladder. The ureters are not simple tubes, but muscular passages that actively propel urine into the bladder. At their lower end is a valve (the ureterovesical junction) that prevents urine from flowing backward into the ureter. The bladder stores urine. The prostate gland surrounds the bladder outlet in males. Urine then flows through the urethra and out of the body as a waste product.

Because the urinary tract is closed save for the one opening at the bottom, urine cannot escape. Instead, the parts distend. Rupture is rare unless there is violent trauma like an automobile accident.

Obstructed flow anywhere along the drainage route can cause swelling of the upper urinary tract, but if the obstruction is below the bladder, the ureterovesical valve will protect the upper tract to a certain extent. Even then, with no place to go, the urine will back up all the way to its source. Eventually, the back pressure causes kidney function to deteriorate.

Obstruction need not be complete for problems to arise. Intermittent or partial obstruction is far more common than complete blockage, allowing time for the parts to enlarge gradually. Furthermore, if a ureterovesical valve is absent or incompetent, the pressure generated by bladder emptying will force urine backward into the ureter and kidney, causing dilation even without mechanical obstruction.

Causes and symptoms

Causes are numerous. Various congenital deformities of the ureter may sooner or later produce back pressure. **Kidney stones** are a common cause. They form in the renal pelvis and become lodged in the kidney, usually at the ureterovesical junction. In older men, the continued growth of the prostate gland leads commonly to restricted urine flow out of the bladder. **Prostate cancer**, and **cancer** anywhere else along the urine pathways, can obstruct flow. **Pregnancy** normally causes ureteral obstruction from the pressure of the enlarged uterus (womb) on the ureters.

Symptoms relate to the passage of urine. Sometimes, urine may be difficult to pass, irregular, or uncontrolled. **Pain** from distension of the structures is present. Blood in the urine may be visible, but it is usually microscopic.

In all cases where bodily fluids cannot flow freely, infection is inevitable. Symptoms of urinary infection may include:

- painful, burning urine
- cloudy urine
- pain in the back, flank, or groin
- fever, sweats, chills, and generalized discomfort

Patients often mistake a serious urinary infection for the flu.

Diagnosis

If the bladder is significantly distended, it can be felt through the abdomen. An analysis of the urine

may reveal blood (if there is a stone), infection, or chemical changes suggesting kidney damage. Blood tests may also detect a decrease in kidney function.

All urinary obstructions will undergo imaging of some sort. Beginning with standard x rays to look for stones, radiologists, physicians specializing in the use of radiant energy for diagnostic purposes, will select from a wide array of tests. Ultrasound is simple, inexpensive, and very useful for these conditions. Standard x rays can be enhanced with contrast agents in several ways. If the kidneys are functioning, they will filter an x ray dye out of the blood and concentrate it in the urine, giving excellent pictures and also an assessment of kidney function. For better images of the lower urinary tract, contrast agents can be instilled from below. This is usually done with a cystoscope placed in the bladder. Through the cystoscope, a small tube can be threaded into the ureter through the ureterovesical valve, allowing dye to be injected all the way up to the kidney. CT and MRI scanning provide miraculous detail, more than is often needed for this condition.

Treatment

The obstruction must be relieved, even if it is partial or functional, as in the case of reflux from the bladder. If not, the kidney will ultimately be damaged, infection will appear, or both. The task may be as simple as placing a catheter through a restricting prostate or as complicated as removing a cancerous bladder and rebuilding a new one with a piece of bowel. In some cases, a badly damaged kidney may have to be removed.

Alternative treatment

Catheters or other urinary diversions may be better for weak or ill patients who cannot tolerate more extensive procedures. There is support using botanical medicine that can help the patient using a catheter avoid infections. Consultation with a trained health care practitioner is necessary.

Prognosis

After relief of the obstruction, a kidney may react with a brief flood of urine, but if the obstruction has been of short duration, normal kidney function will return. If one kidney is destroyed, the other will compensate for the lost organ.

Prevention

Kidney stones can be prevented by dietary changes and medication. Prompt evaluation of infections and

KEY TERMS

Catheter—A tube placed into the body that allows fluids to pass through it.

Contrast agent—Substances that cast shadows on x rays or other imaging methods.

CT and MRI—Two high technology methods of creating images of internal organs. Computerized axial tomography (CT or CAT) uses x rays, while magnetic resonance imaging (MRI) uses magnet fields and radio-frequency signals. Both construct images using a computer.

Cystoscope—A pencil-thin instrument that allows viewing and operating inside the urinary system.

Renal pelvis—The middle section of the kidney where urine first collects after filtration from the blood.

urinary complaints will usually detect problems early enough to prevent long-term complications.

Resources

ORGANIZATIONS

American Association of Kidney Patients. 100 S. Ashley Dr., #280, Tampa, FL 33602. (800) 749-2257. <<http://www.aakp.org>>.

American Kidney Foundation. 6110 Executive Boulevard, #1010, Rockville, MD 20852. (800) 638-8299.

National Kidney Foundation. 30 East 33rd St., New York, NY 10016. (800) 622-9010. <<http://www.kidney.org>>.

J. Ricker Polsdorfer, MD

Hydrotherapy

Definition

Hydrotherapy, or water therapy, is the use of water (hot, cold, steam, or ice) to relieve discomfort and promote physical well-being.

Purpose

Hydrotherapy can soothe sore or inflamed muscles and joints, rehabilitate injured limbs, lower fevers, soothe headaches, promote relaxation, treat **burns** and frostbite, ease labor pains, and clear up skin problems. The temperature of water used affects the therapeutic

VINZENZ PRIESSNITZ (1799–1851)

Hydrotherapy inventor Vinzenz Priessnitz was the son of a Silesian farmer from a remote Austrian territory in the Jeseniky Mountains. From the age of 12, Priessnitz dutifully provided for his blind father, his elderly mother, and his sister. His formal education was sporadic at best. However, Priessnitz possessed a level head and a high degree of intelligence along with a keen and active mind. As he matured he became extremely aware of his surroundings in nature.

At age 16, Priessnitz fell from a horse and was seriously hooved by the animal. He received the morbid prognosis that he might be crippled at best, or might die at worst. He set to treating his own chest wound with cold packs, in emulation of a doe that he had once observed bathing a wound in a cool mountain stream. The hydrotherapy regimen proved highly effective and drew considerable attention to his small hometown of Gräfenberg. In 1822 he rebuilt the family home, renovating its wooden frame into a solid brick spa structure. The spa, known as the castle, housed as many as 1,500 guests each year by 1939. Among the guests were medical professionals who were intent upon exposing the therapy as a sham.

Detractors notwithstanding, word of the simple and effective treatment spread to Vienna, where Priessnitz traveled on occasion to provide counsel at the emperor's court. Priessnitz, for his remarkable discovery, received the Austrian Gold Civil Merit Medal First Class, the highest civilian honor of the Austrian government.

Priessnitz died on November 28, 1851. He was survived by a wife, Zofie Priessnitz, and a young son, Vinzenz Pavel. Joseph Schindler took over the operation of the spa at Gräfenberg following the death of its founder.

properties of the treatment. Hot water is chosen for its relaxing properties. It is also thought to stimulate the immune system. Tepid water can also be used for stress reduction, and may be particularly relaxing in hot weather. Cold water is selected to reduce inflammation. Alternating hot and cold water can stimulate the circulatory system and improve the immune system. Adding herbs and essential oils to water can enhance its therapeutic value. Steam is frequently used as a carrier for essential oils that are inhaled to treat respiratory problems.

Since the late 1990s, hydrotherapy has been used in critical care units to treat a variety of serious conditions, including such disorders of the nervous system as **Guillain-Barré syndrome**.

Description

Origins

The therapeutic use of water has a long history. Ruins of an ancient bath were unearthed in Pakistan and date as far back as 4500 B.C. Bathhouses were an essential part of ancient Roman culture. The use of steam, baths, and aromatic massage to promote well being is documented since the first century. Roman physicians Galen and Celsus wrote of treating patients with warm and cold baths in order to prevent disease.

By the seventeenth and eighteenth centuries, bathhouses were extremely popular with the public throughout Europe. Public bathhouses made their first American appearance in the mid 1700s.

In the early nineteenth century, Sebastien Kneipp, a Bavarian priest and proponent of water healing, began treating his parishioners with cold water applications after he himself was cured of tuberculosis through the same methods. Kneipp wrote extensively on the subject, and opened a series of hydrotherapy clinics known as the Kneipp clinics, which are still in operation today. Around the same time in Austria, Vincenz Priessnitz was treating patients with baths, packs, and showers of cold spring water. Priessnitz also opened a spa that treated over 1,500 patients in its first year of operation, and became a model for physicians and other specialists to learn the techniques of hydrotherapy.

Water can be used therapeutically in a number of ways. Common forms of hydrotherapy include:

- Whirlpools, jacuzzis, and hot tubs. These soaking tubs use jet streams to massage the body. They are frequently used by physical therapists to help injured patients regain muscle strength and to soothe joint and muscle **pain**. Some midwives and obstetricians also approve of the use of hot tubs to soothe the pain of labor.
- Pools and Hubbard tanks. Physical therapists and **rehabilitation** specialists may prescribe underwater pool exercises as a low-impact method of rebuilding muscle strength in injured patients. The buoyancy experienced during pool immersion also helps ease pain in conditions such as arthritis.
- Baths. Tepid baths are prescribed to reduce a **fever**. Baths are also one of the oldest forms of relaxation therapy. Aromatherapists often recommend adding essential oils of lavender (*Lavandula angustifolia*) to a warm to hot bath to promote relaxation and **stress reduction**. Adding Epsom salts (magnesium sulfate) or Dead Sea salts to a bath can also promote relaxation and soothe rheumatism and arthritis.
- Showers. Showers are often prescribed to stimulate the circulation. Water jets from a shower head are also used to massage sore muscles. In addition, showering hydrotherapy has been shown to be preferable to immersion hydrotherapy for treating burn patients.
- Moist compresses. Cold, moist compresses can reduce swelling and inflammation of an injury. They can also be used to cool a fever and treat a headache. Hot or warm compresses are useful for soothing muscle aches and treating abscesses.
- Steam treatments and saunas. Steam rooms and saunas are recommended to open the skin pores and cleanse the body of toxins. Steam inhalation is prescribed to treat respiratory infections. Adding botanicals to the steam bath can increase its therapeutic value.
- Internal hydrotherapy. **Colonic irrigation** is an enema that is designed to cleanse the entire bowel. Proponents of the therapy say it can cure a number of digestive problems. Douching, another form of internal hydrotherapy, directs a stream of water into the vagina for cleansing purposes. The water may or may not contain medications or other substances. Douches can be self-administered with kits available at most drug stores.

Preparations

Because of the expense of the equipment and the expertise required to administer effective treatment, hydrotherapy with pools, whirlpools, Hubbard tanks, and saunas is best taken in a professional healthcare facility, and/or under the supervision of a healthcare professional. However, baths, steam inhalation treatments, and compresses can be easily administered at home.

Bath preparations

Warm to hot bath water should be used for relaxation purposes, and a tepid bath is recommended for reducing fevers. Herbs can greatly enhance the therapeutic value of the bath for a variety of illnesses and minor discomforts.

Herbs for the bath can be added to the bath in two ways—as essential oils or whole herbs and flowers. Whole herbs and flowers can be placed in a muslin or cheesecloth bag that is tied at the top to make an herbal bath bag. The herbal bath bag is then soaked in the warm tub, and can remain there throughout the bath. When using essential oils, add five to 10 drops of oil to a full tub. Oils can be combined to enhance their

therapeutic value. Marjoram (*Origanum marjorana*) is good for relieving sore muscles; juniper (*Juniperus communis*) is recommended as a detoxifying agent for the treatment of arthritis; lavender, ylang ylang (*Conanga odorata*), and chamomile (*Chamaemelum nobilis*) are recommended for stress relief; cypress (*Cupressus sempervirens*), yarrow (*Achillea millefolium*), geranium (*Pelargonium graveolens*), clary sage (*Savlia sclaria*), and myrtle (*Myrtus communis*) can promote healing of **hemorrhoids**; and spike lavender and juniper (*Juniperus communis*) are recommended for rheumatism.

To prepare salts for the bath, add one or two handfuls of epsom salts or Dead Sea salts to boiling water until they are dissolved, and then add them to the tub.

A **sitz bath**, or hip bath, can also be taken at home to treat hemorrhoids and promote healing of an **epi-siotomy**. There is special apparatus available for taking a seated sitz bath, but it can also be taken in a regular tub partially filled with warm water.

Steam inhalation

Steam inhalation treatments can be easily administered with a bowl of steaming water and a large towel. For colds and other conditions with nasal congestion, aromatherapists recommend adding five drops of an essential oil that has decongestant properties, such as peppermint (*Mentha piperita*) and eucalyptus blue gum (*Eucalyptus globulus*). Oils that act as **expectorants**, such as myrtle (*Myrtus communis*) or rosemary (*Rosmarinus officinalis*), can also be used. After the oil is added, the individual should lean over the bowl of water and place the towel over head to trap the steam. After approximately three minutes of inhaling the steam, with eyes closed, the towel can be removed.

Other herbs and essential oils that can be beneficial in steam inhalation include:

- tea tree oil (*Melaleuca alternifolia*) for **bronchitis** and sinus infections
- sandalwood (*Santalum album*), virginian cedarwood (*Juniperus virginiana*), and frankincense (*Boswellia carteri*) for sore throat
- lavender (*Lavandula angustifolia*) and thyme (*Thymus vulgaris*) for cough

Compresses

A cold compress is prepared by soaking a cloth or cotton pad in cold water and then applying it to the area of injury or distress. When the cloth reaches room



This patient is treating his injured left leg with a whirlpool bath. (Custom Medical Stock Photo. Reproduced by permission.)

temperature, it should be resoaked and reapplied. Applying gentle pressure to the compress with the hand may be useful. Cold compresses are generally used to reduce swelling, minimize bruising, and to treat headaches and **sprains**.

Warm or hot compresses are used to treat abscesses and muscle aches. A warm compress is prepared in the same manner as a cold compress, except steaming water is used to wet the cloth instead of cold water. Warm compresses should be refreshed and reapplied after they cool to room temperature.

Essential oils may be added to moist compresses to increase the therapeutic value of the treatment. Peppermint, a cooling oil, is especially effective when added to cold compresses. To add oils to compresses, place five drops of the oil into the bowl of water the compress is to be soaked in. Never apply essential oils directly to a cloth, as they may irritate the skin in undiluted form.

Precautions

Individuals with **paralysis**, **frostbite**, or other conditions that impair the nerve endings and cause reduced sensation should only take hydrotherapy treatments under the guidance of a trained hydrotherapist, physical therapist, or other appropriate healthcare professional. Because these individuals cannot accurately sense temperature changes in the water, they run the risk of being seriously burned without proper supervision. Diabetics and people with hypertension should also consult their healthcare professional before using hot tubs or other heat hydrotherapies.

Hot tubs, jacuzzis, and pools can become breeding grounds for bacteria and other infectious organisms if they are not cleaned regularly, maintained properly, kept at the appropriate temperatures, and treated with the proper chemicals. Individuals should check with their healthcare provider to ensure that the hydrotherapy equipment they are using is sanitary. Those who are using hot tubs and other hydrotherapy equipment in their homes should follow the directions for use and maintenance provided by the original equipment manufacturer.

Certain essential oils should not be used by pregnant or nursing women or by people with specific illnesses or physical conditions. Individuals suffering from any chronic or acute health condition should inform their healthcare provider before starting treatment with any essential oil.

Essential oils such as cinnamon leaf, juniper, lemon, eucalyptus blue gum, peppermint, and thyme can be extremely irritating to the skin if applied in full concentration. Oils used in hydrotherapy should always be diluted in water before they are applied to the skin. Individuals should never apply essential oils directly to the skin unless directed to do so by a trained healthcare professional and/or aromatherapist.

Colonic irrigation should only be performed by a healthcare professional. Pregnant women should never douche, as the practice can introduce bacteria into the vagina and uterus. They should also avoid using hot tubs without the consent of their healthcare provider.

The vagina is self-cleansing, and douches have been known to upset the balance of vaginal pH and flora, promoting vaginitis and other infections. Some studies have linked excessive vaginal douching to increased incidence of pelvic inflammatory disease (PID).

Side effects

Most forms of hydrotherapy are well tolerated. There is a risk of allergic reaction (also known as

KEY TERMS

Contact dermatitis—Skin irritation as a result of contact with a foreign substance.

Episiotomy—An incision made in the perineum during labor to assist in delivery and to avoid abnormal tearing of the perineum.

Essential oil—A volatile oil extracted from the leaves, fruit, flowers, roots, or other components of a plant and used in aromatherapy, perfumes, and foods and beverages.

Hubbard tank—A large water tank or tub used for underwater exercises.

contact dermatitis) for some patients using essential oils and herbs in their bath water. These individuals may want to test for allergic sensitization to herbs by performing a skin patch test (i.e., rubbing a small amount of diluted herb on the inside of their elbow and observing the spot for redness and irritation). People who experience an allergic reaction to an essential oil should discontinue its use and contact their healthcare professional for further guidance.

The most serious possible side effect of hydrotherapy is overheating, which may occur when an individual spends too much time in a hot tub or jacuzzi. However, when properly supervised, this is a minimal risk.

Research and general acceptance

Hydrotherapy treatments are used by both allopathic and complementary medicine to treat a wide variety of discomforts and disorders. Not as well accepted are invasive hydrotherapy techniques, such as colonic irrigation, **enemas**, and douching. These internal cleansing techniques can actually harm an individual by upsetting the natural balance of the digestive tract and the vagina. Most conventional medical professionals agree that vaginal douches are not necessary to promote hygiene in most women, and can actually do more harm than good.

Resources

BOOKS

Chaitow, Leon. *Hydrotherapy: Water Therapy for Health and Beauty*. Boston, MA: Element Books, 1999.

Pelletier, Dr. Kenneth R. *The Best Alternative Medicine, Part I: Naturopathic Medicine*. New York: Simon and Schuster, 2002.

PERIODICALS

- Baird, Carol L. "First-Line Treatment for Osteoarthritis: Part 2: Nonpharmacologic Interventions and Evaluation." *Orthopaedic Nursing* 20 (November-December 2001): 13–20.
- Barker, K. L., H. Dawes, P. Hansford, and D. Shamley. "Perceived and Measured Levels of Exertion of Patients with Chronic Back Pain Exercising in a Hydrotherapy Pool." *Archives of Physical Medicine and Rehabilitation* 84 (September 2003): 1319–1323.
- Cider, A., M. Schauffelberger, K. S. Sunnerhagen, and B. Andersson. "Hydrotherapy—A New Approach to Improve Function in the Older Patient with Chronic Heart Failure." *European Journal of Heart Failure* 5 (August 2003): 527–535.
- Johnson, Kate. "Hydrotherapy Greatly Eases Delivery Stress, Pain." *OB GYN News* 34 (November 1999): 27.
- Keegan, L. "Therapies to Reduce Stress and Anxiety." *Critical Care Nursing Clinics of North America* 15 (September 2003): 321–327.
- Mayhall, C. G. "The Epidemiology of Burn Wound Infections: Then and Now." *Clinical Infectious Diseases* 37 (August 15, 2003): 543–550.
- Molter, N. C. "Creating a Healing Environment for Critical Care." *Critical Care Nursing Clinics of North America* 15 (September 2003): 295–304.
- Taylor, S. "The Ventilated Patient Undergoing Hydrotherapy: A Case Study." *Australian Critical Care* 16 (August 2003): 111–115.

ORGANIZATIONS

- The American Association of Naturopathic Physicians. 8201 Greensboro Drive, Suite 300, McLean, Virginia 22102. (206) 298-0126. <<http://naturopathic.org>>.
- Canadian Naturopathic Association/Association canadienne de naturopathie. 1255 Sheppard Avenue East at Leslie, North York, ON M2K 1E2. (800) 551-4381 or (416) 496-8633. <<http://www.naturopathicassoc.ca>>.

Paula Anne Ford-Martin
Rebecca J. Frey, PhD

Hydroxyzine see **Anti-itch drugs**

Hyperactivity see **Attention-deficit/
Hyperactivity disorder (ADHD)**

Hyperaldosteronism

Definition

Hyperaldosteronism is a disorder which is defined by the body's overproduction of aldosterone, a hormone that controls sodium and potassium levels in the blood. Its overproduction leads to retention of salt

and loss of potassium, which leads to **hypertension** (high blood pressure).

Description

Also known as Conn's syndrome, primary aldosteronism, and secondary aldosteronism, this disorder takes several forms. It often begins with a tumor that produces aldosterone. In fact, approximately 60–70% of the cases of primary aldosteronism result from tumors in the adrenal gland area. Aldosterone is normally produced by the adrenal cortex, or the outer portion of the gland that rests on top of each kidney. Primary aldosteronism is due to adenoma, a typically benign tumor in which the cells form to act as glands or cause the glands on which they rest to overproduce. It can cause a number of problems, most notably hypertension. In secondary aldosteronism, factors outside the adrenal gland may cause overproduction of aldosterone, or overproduction of renin, an enzyme stored in the kidney area that stimulates aldosterone and raises blood pressure. Obstructive renal artery disease may also cause hypertension from elevated renin stimulating aldosterone. **Oral contraceptives** have been known to increase the secretion of aldosterone in some patients. This disorder is more common in women.

Causes and symptoms

Hyperaldosteronism is most often caused by the invasion of adenoma. Other adrenal cancers and hyperplasia, or the increase in the bulk of an organ due to increased cell production, may also cause hyperaldosteronism. Those diseases and factors influencing the adrenal and kidney functions may lead to secondary aldosteronism. The primary symptom of hyperaldosteronism is moderate hypertension, or high blood pressure. In addition, a patient may experience **orthostatic hypotension**, or reduced blood pressure when a person stands after lying down. **Constipation**, muscle weakness (sometimes to the point of **periodic paralysis**), excessive urination, excessive thirst, headache, and personality changes are also possible symptoms. Some patients will show no obvious symptoms.

Diagnosis

Screening tests can be conducted to pinpoint a diagnosis of hyperaldosteronism. If a patient is taking drugs to reduce high blood pressure, the physician may order these drugs stopped for a time period before conducting tests, since these drugs will affect results.

Blood and urine tests may be conducted to check for levels of aldosterone, potassium levels, or renin activity. A computed tomography scan (CT scan) may be ordered to detect tumors as small as five to seven mm. These combined tests approach 95% accuracy for detecting aldosterone-producing adenoma. Laboratory findings recording blood pressure, **edema**, and aldosterone and plasma renin activity can help the physician differentiate between primary aldosteronism and secondary aldosteronism.

Treatment

Once the physician has made a diagnosis of hyperaldosteronism, the adrenal glands should be checked for possible adenomas. This can be done through imaging or with a surgical dissection of the gland. Surgical or ablative treatment will vary depending on the number of tumors found. Since more than 60% of hyperaldosteronism cases are caused by these tumors, treatment of the tumors will help eliminate the resulting high blood pressure in many patients. Some patients will receive **antihypertensive drugs**, like calcium channel blockers, to control high blood pressure. The use of **diuretics** can help control hypertension by reducing volume. Potassium levels should be considered in the type of diuretic ordered and the levels should be checked throughout treatment. The most widely used drug for treatment of hyperaldosteronism is spironolactone. This drug helps control aldosterone, but should not be prescribed for some patients, especially those with certain kidney diseases. Spironolactone has several possible adverse effects, depending on the dosage. In all cases of hyperaldosteronism, the treatment should be carefully based on the specific type or underlying cause of the disorder.

Alternative treatment

Patients may choose to work with their physician or alternative provider to control hypertension with diet, **stress reduction** (including massage, meditation, **biofeedback**, and yoga), and other remedies. Blood pressure elevation needs to be controlled and monitored by frequent blood pressure measurements. There is no alternative treatment known for the underlying adenoma.

Prognosis

Hyperaldosteronism carries with it all the possible complications of high blood pressure, including thickening of arterial walls and a higher risk of angina,

KEY TERMS

Ablative—Used to describe a procedure involving removal of a tissue or body part, or destruction of its function.

Adenoma—A growth of cells, usually a benign tumor, that forms a gland or gland-like substance. These tumors can secrete hormones or cause changes in hormone production in nearby glands.

Adrenal—Refers to the glands which sit on top of each kidney and that secrete various hormones.

Antihypertensive—Used to describe drugs or treatments designed to control hypertension, or high blood pressure.

Diuretic—A substance or drug that is taken to promote the formation and release of urine. In the treatment of high blood pressure, diuretics can help reduce the overall fluid volume in the body.

Renal—Relating to the kidney. The renal artery is one of two branches of the large blood vessel in the stomach area that serves the kidneys, ureters (tubes that carry urine from the kidney to the bladder) and adrenal glands.

kidney failure, **stroke**, or heart attack. Another possible, and less reversible complication than hypertension, is kidney damage. When primary aldosteronism is caused by a solitary adenoma, the prognosis is good. Once this tumor is removed, blood pressure will drop, and 70% of these patients have full remission. Patients whose hyperaldosteronism results from adrenal hyperplasia will remain hypertensive. However, in up to 70% of patients, blood pressure can be reduced somewhat with drug therapy. Many patients will be faced with the prospect of controlling their hypertension for the remainder of their lives.

Prevention

There is no known prevention for most causes of hyperaldosteronism.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

American Society of Hypertension. 515 Madison Ave., Suite 1212, New York, NY 10022. (212) 644-0650. <<http://www.ash-us.org>>.

National Heart, Lung and Blood Institute. PO Box 30105,
Bethesda, MD 20824-0105. (301) 251-1222.
< <http://www.nhlbi.nih.gov> >.

OTHER

Hypertension Network. < <http://www.bloodpressure.com> >.

Teresa Odle

Hyperbaric Chamber

Definition

A hyperbaric chamber is a room that allows an individual to breathe 100% pure oxygen at greater than 1 standard atmosphere of pressure.

Purpose

Hyperbaric chambers are used to deliver hyperbaric **oxygen therapy** (HBOT). HBOT was developed to treat underwater divers suffering from **decompression sickness** (the bends). It has since been approved by the Undersea and Hyperbaric Medical Society for 13 conditions including:

- air or gas embolism
- carbon monoxide (CO) **poisoning**
- smoke inhalation
- gas **gangrene** caused by certain bacteria
- decompression sickness
- radiation tissue damage
- thermal burns
- non-healing skin grafts
- crush injuries
- wounds that fail to heal through conventional treatment
- serious blood loss
- intracranial abscess.

Although hyperbaric therapy has become increasingly popular for other uses, especially in sports medicine, its use is controversial. Terrell Owens of the Philadelphia Eagles used HBOT for an ankle injury prior to playing in the Super Bowl in 2005, but medical professionals questioned the appropriateness of this treatment.

Precautions

Individuals who have lung disease including **asthma**, **emphysema**, obstructive lung disease, or any condition in which air is trapped in the lungs, are poor candidates for this therapy and should discuss the relative benefits and drawbacks of HBOT with their doctor. Individuals who have had chest surgery or who have had a central venous catheter implanted are also at higher risk for complications. People with seizure disorders should be carefully monitored, as this treatment may increase the risk and severity of seizures. People with colds or clogged ears may want to wait to undergo HBOT, as they may experience difficulties with pressure equalization that can cause damage to the middle or inner ear. HBOT in **pregnancy** is controversial. Individuals with diabetes may need to adjust their glucose and insulin balance, since HBOT slows the absorption of insulin.

Description

At normal atmospheric pressure, oxygen binds with a molecule in red blood cells called hemoglobin. The oxygen is carried through the body to tissues where it is needed as the blood circulates. Under normal conditions, almost all (about 97%) of the available hemoglobin carries oxygen. Increasing the atmospheric pressure does little to increase the oxygen-carrying capacity of the blood. However, under normal conditions, only a small amount of oxygen is dissolved in the fluid that carries the red blood cells (blood plasma). Increasing the atmospheric pressure to two to three times normal and breathing 100% oxygen forces more oxygen to dissolve into the blood plasma. In this way, hyperbaric chambers increase the amount of oxygen circulating in the body. This can promote healing in areas that are not receiving adequate oxygen. The extra oxygen can also help to cure certain infections caused by anerobic bacteria that can live only in the absence of oxygen.

There are two types of hyperbaric chambers—monoplace and multiplace. Monoplace chambers accommodate a single person. The patient enters the chamber, then it is closed and the pressure is increased. The advantages of a monoplace chamber are that the patient does not have to wear a mask or a hood to receive the oxygen and the treatment regimen is designed specifically for each individual. The major disadvantages are that the patient is inaccessible to the staff during treatment should an emergency arise, and the pure oxygen atmosphere creates an increased fire hazard. In multiplace chambers, several patients use the same chamber simultaneously. Each person is

given oxygen through a face mask or hood, but all patients receive the same treatment. A staff member remains in the chamber throughout the procedure.

Hyperbaric chambers can be associated with hospitals, but are increasingly part of free-standing clinics. Insurance may cover the cost of treatment for approved indications such as **carbon monoxide poisoning**, but may reject payment for uses that are considered experimental or controversial. The American Board of Medical Specialists certifies physician competency in the undersea medicine, including the use of hyperbaric chambers. The Baromedical Nurses Association offers three levels of certification for hyperbaric nurses, and the National Board of Hyperbaric Medicine Technology certifies hyperbaric technicians. Individuals considering hyperbaric therapy should seek facilities run by health care providers credentialed by these organizations.

Preparation

No special preparation is needed to use a hyperbaric chamber other than educating patients about what to expect during treatment.

Aftercare

After HBOT is complete, a period of decompression in the chamber is required until the pressure in the chamber is equal to the pressure outside. Serious complications can occur if decompression occurs suddenly.

Risks

Hyperbaric chambers, because of their use of 100% oxygen, present a potential fire risk. In addition, although hyperbaric oxygen therapy is very safe when used correctly, complications can occur. Oxygen poisoning, also called oxygen toxicity, can occur when an individual is exposed to high doses of oxygen for a prolonged period. Excess oxygen causes chemical changes in the body that negatively affect cells and metabolic processes. Symptoms of oxygen poisoning include **nausea**, **vomiting**, dry **cough**, seizures, chest **pain**, sweating, muscle twitching, ringing of the ears, **hallucinations**, **dizziness**, **shortness of breath** and a decreased level of consciousness.

Other complications can occur as the result of increased pressure within the chamber. These include pain and bloody discharge from congested sinuses, ear pain, rupture of the eardrum, and bleeding from the ear if the Eustachian tube that connects the ear to the back of the throat is clogged and pressure on either

KEY TERMS

Embolism—An obstruction in a blood vessel, often caused by gas or a blood clot.

Eustachian tube—A tube of cartilage that connects the middle ear to the back of the throat. Its purpose is to equalize the pressure on either side of the eardrum.

side of the eardrum is not equalized. Teeth that are infected or have been repaired may become painful or explode if gas is trapped within them. A few individuals develop **pneumothorax**. This is a serious condition where air is trapped between the lungs and the chest cavity.

Normal results

HBOT is expected to promote healing and improve the health of individuals with conditions for which it is approved.

Abnormal results

Under some conditions HBOT fails to cause improvement or complications occur.

Resources

ORGANIZATION

Undersea and Hyperbaric Medicine Society. 10531 Metropolitan Avenue, Kensington, MD 20895. 310-942-7804. < www.uhms.org > .

OTHER

Moder, Cheryl. *Hyperbaric Oxygen Therapy: Where Medicine Meets the Deep Blue Sea*, 20 February 2005 [cited 20 February 2005]. < <http://www.emedicine.com/plastic/topic526.htm> > .

Neumeister, Michael. *Hyperbaric Oxygen Therapy*, 11 November 2004 [cited 16 February 2005]. < <http://www.emedicine.com/plastic/topic526.htm> > .

Prince, Mark. *Hyperbaric Oxygen*, 27 October 2004 [cited 16 February 2005]. < <http://www.emedicine.com/ent/topic733.htm> > .

Tish Davidson, A. M.

Hyperbaric oxygenation see **Oxygen/ozone therapy**

Hyperbilirubinemia see **Neonatal jaundice**

Hypercalcemia

Definition

Hypercalcemia is an abnormally high level of calcium in the blood, usually more than 10.5 milligrams per deciliter of blood.

Description

Calcium plays an important role in the development and maintenance of bones in the body. It is also needed in tooth formation and is important in other body functions. Normally, the body maintains a balance between the amount of calcium in food sources and the calcium already available in the body's tissues. The balance can be upset if excess amounts of calcium are eaten or if the body is unable to process the mineral because of disease.

Calcium is one of the most important and most abundant **minerals** in the human body. Dairy products are the major source of calcium. Eggs, green leafy vegetables, broccoli, legumes, nuts, and whole grains provide smaller amounts. Only about 10–30% of the calcium in food is absorbed into the body. Most calcium is found in combination with other dietary components and must be broken down by the digestive system before it can be used. Calcium is absorbed into the body in the small intestine. Its absorption is influenced by such factors as the amount of vitamin D hormone available to aid the process and the levels of calcium already present in the body. As much as 99% of the body's calcium is stored in bone tissue. A healthy person experiences a constant turnover of calcium as bone tissue is built and reshaped. The remaining 1% of the body's calcium circulates in the blood and other body fluids. Circulating calcium plays an important role in the control of many body functions, such as blood clotting, transmission of nerve impulses, muscle contraction, and other metabolic activities. In the bloodstream, calcium maintains a constant balance with another mineral, phosphate.

Two main control agents are vital in maintaining calcium levels, vitamin D hormone and parathyroid hormone. A hormone is a chemical substance that is formed in one organ or part of the body and carried in the blood to another organ. It can alter the function, and sometimes the structure, of one or more organs.

- **Parathyroid hormone (PTH).** The four parathyroid glands are endocrine glands located next to the thyroid gland in the neck. A gland is a cell or group

of cells that produces a material substance (secretion). When the level of calcium circulating in the blood drops, the parathyroid gland releases its hormone. PTH then acts in three ways to restore the normal blood calcium level. It stimulates the absorption of more calcium in the intestine; it takes more calcium from the bone tissue, and it causes the kidneys to excrete more phosphate.

- **Vitamin D hormone.** This hormone works with parathyroid hormone to control calcium absorption and affects the deposit of calcium and phosphate in the bone tissue.

The kidneys also help to control calcium levels. Healthy kidneys can increase calcium excretion almost fivefold to maintain normal concentrations in the body. Hypercalcemia can occur when the concentration of calcium overwhelms the ability of the kidneys to maintain balance.

Causes and symptoms

Causes of hypercalcemia

Many different conditions can cause hypercalcemia; the most common are **hyperparathyroidism** and **cancer**.

PRIMARY HYPERPARATHYROIDISM. Primary hyperparathyroidism is the excessive secretion of parathyroid hormone by one or more of the parathyroid glands. It is the most common cause of hypercalcemia in the general population. Women have this condition more frequently than men do, and it is more common in older people. It can appear thirty or more years after radiation treatments to the neck. Ninety percent of the cases of primary hyperparathyroidism are caused by a non-malignant growth on the gland.

Hyperparathyroidism can also occur as part of a rare hereditary disease called multiple endocrine neoplasia. In this disease, tumors develop on the parathyroid gland.

CANCER. People with cancer often have hypercalcemia. In fact, it is the most common life-threatening metabolic disorder associated with cancer. Ten to twenty percent of all persons with cancer have hypercalcemia. Cancers of the breast, lung, head and neck, and kidney are frequently associated with hypercalcemia. It also occurs frequently in association with certain cancers of the blood, particularly malignant myeloma. It is seen most often in patients with tumors of the lung (25–35%) and breast (20–40%), according to the National Cancer Institute. Cancer causes hypercalcemia in two ways. When a tumor grows into the bone, it destroys bony tissue (osteolysis). When the

bone is not involved, factors secreted by cancer cells can increase calcium levels (humoral hypercalcemia of malignancy). The two mechanisms may operate at the same time.

Because immobility causes an increase in the loss of calcium from bone, cancer patients who are weak and spend most of their time in bed are more prone to hypercalcemia. Cancer patients are often dehydrated because they take in inadequate amounts of food and fluids and often suffer from **nausea and vomiting**. **Dehydration** reduces the ability of the kidneys to remove excess calcium from the body. Hormones and **diuretics** that increase the amount of fluid released by the body can also trigger hypercalcemia.

OTHER CAUSES. Other conditions can cause hypercalcemia. Excessive intake of vitamin D increases intestinal absorption of calcium. During therapy for peptic ulcers, abnormally high amounts of calcium **antacids** are sometimes taken. Over use of antacids can cause milk-alkali syndrome and hypercalcemia. Diseases such as Paget's, in which bone is destroyed or reabsorbed, can also cause hypercalcemia. As in cancer or **paralysis** of the arms and legs, any condition in which the patient is immobilized for long periods of time can lead to hypercalcemia due to bone loss.

Common symptoms

Many patients with mild hypercalcemia have no symptoms and the condition is discovered during routine laboratory screening. Gastrointestinal symptoms include loss of appetite, nausea, vomiting, **constipation**, and abdominal **pain**. There may be a blockage in the bowel. If the kidneys are involved, the individual will have to urinate frequently during both the day and night and will be very thirsty. As the calcium levels rise, the symptoms become more serious. Stones may form in the kidneys and waste products can build up. Blood pressure rises. The heart rhythm may change. Muscles become increasingly weak. The individual may experience mood swings, confusion, **psychosis**, and eventually, **coma** and **death**.

Diagnosis

High levels of calcium in the blood are a good indication of hypercalcemia, but these levels may fluctuate. Calcium levels are influenced by other compounds in the blood that may combine with calcium. Higher calcium and lower phosphate levels may suggest primary hyperparathyroidism. The blood levels of protein (serum albumin) and parathyroid hormone (PTH) are also measured in the diagnosis

of hypercalcemia. Too much PTH in the blood may indicate primary hyperparathyroidism. Levels of calcium and phosphate in the urine should also be measured. The medical history and physical condition of the individual must be taken into consideration, especially in the early stages of hypercalcemia when symptoms are mild.

Treatment

The treatment of hypercalcemia depends on how high the calcium level is and what is causing the elevation. Hypercalcemia can be life-threatening and rapid reduction may be necessary. If the patient has normal kidney function, fluids can be given by vein (intravenously) to clear the excess calcium. The amount of fluid taken in and eliminated must be carefully monitored. If the patient's kidneys are not working well, acute hemodialysis is probably the safest and most effective method to reduce dangerous calcium levels. In this procedure, blood is circulated through tubes made of semi-permeable membranes against a special solution that filters out unwanted substances before returning the blood to the body.

Drugs such as furosemide, called loop diuretics, can be given after adequate fluid intake is established. These drugs inhibit calcium reabsorption in the kidneys and promote urine production. Drugs that inhibit bone loss, such as calcitonin, biphosphates, and plicamycin, are helpful in achieving long-term control. Phosphate pills help lower high calcium levels caused by a deficiency in phosphate. Anti-inflammatory agents such as steroids are helpful with some cancers and toxic levels of vitamin D.

Treatment of the underlying cause of the hypercalcemia will also correct the imbalance. Hyperparathyroidism is usually treated by surgical removal of one or more of the parathyroid glands and any tissue, other than the glands themselves, that is producing excessive amounts of the hormone.

The hypercalcemia caused by cancer is difficult to treat without controlling the cancer. Symptoms can be alleviated with fluids and drug therapy as outlined above.

Prognosis

Surgery to remove the parathyroid glands and any misplaced tissue that is producing excessive amounts of hormone succeeds in about 90% of all cases. Outcome is also influenced by whether any damage to the kidneys can be reversed.

KEY TERMS

Calcium—A silvery-yellow metal that is the basic element of lime and makes up about 3% of the earth's crust. It is the most abundant mineral in the human body. Calcium and phosphorous combine as calcium phosphate, the hard material of bones and teeth.

Hormone—A chemical substance that is carried through the blood to another part of the body, stimulating it to change its function or structure. Many hormones are produced by glands.

Metabolism—All the physical and chemical changes that take place within an organism.

Milk-alkali syndrome—A chronic disorder of the kidneys caused by the ingestion of large amounts of calcium and alkali in the treatment of peptic ulcer. The disorder is reversible in its early stages but can progress to kidney failure.

Mineral—A substance that does not contain carbon (inorganic) and is widely distributed in nature. Minerals play an important role in human metabolism.

Parathyroid hormone (PTH)—A chemical substance produced by the parathyroid glands. This hormone is a major element in regulating calcium in the body.

Vitamin D hormone—Vitamin D is a vitamin that also acts as a hormone. Vitamin D hormone acts with parathyroid hormone to regulate calcium levels in the blood and to supply appropriate amounts of calcium to all cells.

Mild hypercalcemia can be controlled through good fluid intake and the use of effective drugs.

Hypercalcemia generally develops as a late complication of cancer and the expected outlook is grim without effective anticancer therapy.

Prevention

People with cancer who are at risk of developing hypercalcemia should be familiar with early symptoms and know when to see a doctor. Good fluid intake (up to four quarts of liquid a day if possible), controlling nausea and vomiting, paying attention to fevers, and keeping physically active as much as possible can help prevent problems. Dietary calcium restriction is not necessary because hypercalcemia reduces absorption of calcium in the intestine.

Resources

OTHER

"Hypercalcemia." *National Cancer Institute Page*. <<http://www.nci.nih.gov>>.

Karen Ericson, RN

Hypercholesterolemia

Definition

Hypercholesterolemia refers to levels of cholesterol in the blood that are higher than normal.

Description

Cholesterol circulates in the blood stream. It is an essential molecule for the human body. Cholesterol is a molecule from which hormones and steroids are made. It is also used to maintain nerve cells. Between 75 and 80% of the cholesterol that circulates in a person's bloodstream is made in that person's liver. The remainder is acquired from outside sources. Cholesterol is found in animal sources of food. It is not found in plants.

Normal blood cholesterol level is a number derived by laboratory analysis. A normal or desirable cholesterol level is defined as less than 200 mg of cholesterol per deciliter of blood (mg/dL). Blood cholesterol is considered to be borderline when it is in the range of 200 to 239 mg/dL. Elevated cholesterol level is 240 mg/dL or above. Elevated blood cholesterol is considered to be hypercholesterolemia.

Cholesterol has been divided into two major categories: low-density lipoprotein (LDL), the so-called "bad" cholesterol, and high-density lipoprotein (HDL), the so-called "good" cholesterol. Diet, **exercise**, **smoking**, alcohol, and certain illnesses can affect the levels of both types of cholesterol. Eating a high fat diet will increase one's level of LDL cholesterol. Exercising and reducing one's weight will both increase HDL cholesterol and lower LDL cholesterol.

The most common cause of elevated serum cholesterol is eating foods that are rich in saturated fats or contain high levels of cholesterol. Elevated cholesterol also can be caused by an underlying disease that raises blood cholesterol levels such as **diabetes mellitus**, **kidney disease**, **liver disease**, or **hypothyroidism**. It also can be caused by an inherited disorder in which cholesterol is not metabolized properly by the body.

Obesity, which generally results from eating a diet high in fat, also can lead to elevated cholesterol levels in the blood. This is because obesity itself leads the body to produce excessive amounts of cholesterol.

Hypercholesterolemia increases the risk of heart disease. Elevated levels of circulating cholesterol cause deposits to form inside blood vessels. These deposits, called plaque, are composed of fats deposited from the bloodstream. When the deposits become sufficiently large, they block blood vessels and decrease the flow of blood. These deposits result in a disease process called **atherosclerosis**, which can cause **blood clots** to form that will ultimately stop blood flow. If this happens in the arteries supplying the heart, a **heart attack** will occur. If it happens in the brain, the result is a **stroke** where a portion of brain tissue dies. Atherosclerosis causes more deaths from heart disease than any other single condition. Heart disease has been the leading cause of **death** in the United States for the past half century.

There is a syndrome called familial hypercholesterolemia. Affected persons have consistently high levels of LDL. This leads to early clogging of the coronary arteries. In turn this leads to a heart attack. Among affected males, a first heart attack typically occurs in their 40s to 50s. Approximately 85% of men with this disorder have experienced a heart attack by the time they reach 60 years of age. The incidence of heart attacks among women with this disorder also is increased. However, it is delayed 10 years compared to men. The incidence of familial hypercholesterolemia is seven out of 1,000 people.

Causes and symptoms

Hypercholesterolemia is silent. There are no symptoms that are obvious to the naked eye. It is diagnosed by a blood test or after a heart attack or stroke occurs.

Diagnosis

Hypercholesterolemia is diagnosed by using a blood test. A blood specimen is obtained after the patient does not eat or drink anything (except water) for 12 hours. The **fasting** is done to measure the LDL and HDL cholesterol, which can only be determined accurately in a fasting state. Some experts agree that an acceptable limit for LDL cholesterol as 130 mg/dL, though the National Cholesterol Education Program Adult Treatment Panel III recommended a goal of less than 100mg/dL. Total cholesterol of under 200 mg/dL is thought to be an acceptable range.

Treatment

If an individual's cholesterol is elevated, discussions with a physician should be scheduled to determine what course of treatment may be needed. Initial treatment for hypercholesterolemia usually requires dietary changes to reduce the intake of total fat, saturated fat, and cholesterol. Most health care professionals will recommend that a person's weight and height be proportionate. In addition to diet, guidelines recommend exercise to help bring weight and cholesterol to acceptable levels. Further, experts counsel persons with elevated blood cholesterol levels to increase their intake of soluble fiber. Sources of soluble fiber include bran, foods containing whole grains and other sources of indigestible fiber such as lignin. Physicians also recommend that patients with **high cholesterol** stop smoking as part of first-line therapy for hypercholesterolemia.

The reason for treating elevated cholesterol is to reduce an individual's risk of complications. If a diet low in cholesterol and saturated fats doesn't significantly reduce a person's cholesterol level, medication may be required. For every 1 percent reduction in cholesterol level, the risk of heart disease is reduced by 2 percent. It also is possible to partially reverse atherosclerosis that has already occurred by aggressively lowering cholesterol levels with diet and medications.

Prescription drugs are available to help lower cholesterol levels in the blood. These may be used as first-line therapy in high-risk patients or after about three months of dietary and lifestyle therapy. Cholestyramine, cholestipol, lovastatin, simvastatin, pravastatin, fluvastatin, rosuvastatin, and gemfibrozil are some of the drugs approved for use in the United States. The most often prescribed group of drugs are the statins, which also have been shown in some studies to reduce risk of depression and **dementia**.

Alternative treatment

There are advocates of treatment using **vitamins**, **minerals** and antioxidant substances in relatively high amounts. These amounts generally exceed those provided by the Food and Drug Administration in its Minimum Daily Requirements (MDR). Advocates of such therapies also emphasize increased levels of exercise, attaining an ideal body weight and increasing levels of fiber in one's diet.

Some people have advocated the use of garlic, soy and isoflavones to lower serum cholesterol levels. In 2003, enriched green tea was found to be an effective addition to a low-fat diet for lowering LDL cholesterol in adults.

Prognosis

The prognosis is in direct proportion to serum cholesterol levels. People with hypercholesterolemia are at high risk of dying from heart disease.

Many studies have looked at the relationship between elevated cholesterol levels, increased risk for heart attack and death. In one investigation of relatively young males who had no known heart disease, cholesterol levels were measured and participants were followed for six years. During this time, all heart attacks and deaths that occurred among participants were recorded. As serum cholesterol levels increased, so did the risk of experiencing a fatal heart attack. The risk of a fatal heart attack was approximately five times higher among persons having cholesterol levels of 300 mg/dL or more compared to those with cholesterol levels below 200 mg/dL.

The Framingham Heart Study is an ongoing research effort. Cholesterol levels, smoking habits, heart attack rates, and deaths in the population of an entire town have been recorded for over 40 years. After 30 years, more than 85% of persons with cholesterol levels of 180 mg/dL or less were still alive; almost a third of those with cholesterol levels greater than 260 mg/dL had died.

Prevention

Experts suggest the following steps to maintain serum cholesterol within normal limits: an important component is to maintain a normal weight for height and to reduce one's weight if it is inappropriate for height. Changing dietary habits by reducing the amount of fat and cholesterol consumed is advised. Doctors recommend avoiding smoking by not starting or quitting if currently a smoker. Increasing levels of fiber in the diet by including foods such as beans, raw fruits, whole grains and vegetables is recommended. It is important to exercise on a regular basis. Aerobic exercise is especially helpful in reducing serum cholesterol levels.

People from families with a strong history of early heart attacks should be evaluated with a lipid screen. Proper diet, exercise and the use of effective drugs can reduce serum lipid levels.

Nutrition and cardiac experts offer the following suggestions:

- purchasing low-fat or fat-free dairy products such as milk, cheese, sour cream, and yogurt
- eating lean red meats, chicken without skin, and fish
- reducing consumption of foods high in saturated fat such as french fries

KEY TERMS

Atherosclerosis—A disease process whereby plaques of fatty substances are deposited inside arteries, reducing the inside diameter of the vessels and eventually causing damage to the tissues located beyond the site of the blockage.

Coronary artery—One of five vessels that supply blood to the heart.

Deciliter (dL)—100 cubic centimeters (cc).

High density lipoprotein (HDL)—A fraction of total serum lipids, the so called “good” cholesterol.

Low density lipoprotein (LDL)—A fraction of total serum lipids, the so called “bad” cholesterol.

- avoid foods that are rich sources of cholesterol such as eggs, liver, cheese, and bacon
- eating smaller servings
- keeping a food journal and writing down everything eaten each day
- prepare food by microwaving, boiling, broiling, or baking food instead of frying
- trimming the fat from meat before cooking it.

Resources

BOOKS

- Braunwald, Eugene, Douglas Zipes, and Perter Libby. *Heart Disease: A Textbook of Cardiovascular Medicine*. 6th ed. Philadelphia: Saunders, 2001.
- Foody, J.M., and Eugene Braunwald. *Preventive Cardiology: Strategies for the Prevention and Treatment of Coronary Artery Disease*. Totowa, NJ: Humana Press, 2001.
- Hiatt, William R. “Atherosclerotic peripheral arterial disease.” In *Cecil Textbook of Medicine*, edited by Lee Goldman and J. Claude Bennett, 21st ed. Philadelphia: W.B. Saunders, 2000, pp. 357-362.
- Hiatt, William R., Judith Regensteiner, and Alan T. Hirsch. *Peripheral Arterial Disease Handbook*. Boca Raton, FL: CRC Press, 2001.
- Silver, Malcom, and Schoen Gottlieb. *Cardiovascular Pathology*. 3rd ed. Boston, Churchill Livingstone, 2001.

PERIODICALS

- Aronow, Wilbert S. “Hypercholesterolemia: The Evidence Supports Use of Statins.” *Geriatrics* August 2003: 18.
- “Cholesterol-lowering Effect of Green Tea.” *Nutraceuticals International* September 2003.
- Jackson, P.R. “Cholesterol-lowering Therapy for Smokers.” *The Lancet* 357, no. 9260 (2001): 960-961.
- “Link to Cholesterol Drugs Disputed.” *Cardiovascular Week* September 29, 2003: 73.

Mechcatie, Elizabeth. "FDA Okays Rosuvastatin for Hypercholesterolemia: Most Potent Statin to Date." *Internal Medicine News* September 1, 2003: 30–31.

Shamir R., A. Lerner, and E. A. Fisher.

"Hypercholesterolemia in Children." *Israel Medical Association Journal* 2, no. 10 (2000): 767–771.

ORGANIZATIONS

American Heart Association, National Center. 7272 Greenville Avenue, Dallas, Texas 75231. (877) 242-4277. <<http://www.americanheart.org>>.

American Medical Association. 515 N. State Street, Chicago, IL 60610. (312) 464-5000. <<http://www.ama-assn.org/>>.

American Society of Nuclear Cardiology. 9111 Old Georgetown Road, Bethesda, MD 20814-1699. (301) 493-2360. Fax: (301) 493-2376, <<http://www.asnc.org>>.

OTHER

American Academy of Family Practice: <<http://www.aafp.org/afp/20000201/675.html>>.

American Heart Association. <<http://www.americanheart.org/Scientific/pubs/hyperchol/>>.

Merck Manual. <<http://www.merck.com/pubs/mmanual/section2/chapter15/15c.htm>>.

National Library of Medicine: <<http://www.nlm.nih.gov/medlineplus/ency/article/000403.htm>>.

L. Fleming Fallon, Jr., MD, DrPH
Teresa G. Odle

Hypercoagulation disorders

Definition

Hypercoagulation disorders (or hypercoagulable states or disorders) have the opposite effect of the more common **coagulation disorders**. In hypercoagulation, there is an increased tendency for clotting of the blood, which may put a patient at risk for obstruction of veins and arteries (phlebitis or pulmonary embolism).

Description

In normal hemostasis, or the stoppage of bleeding, clots form at the site of the blood vessel's injury. The difference between that sort of clotting and the clotting present in hypercoagulation is that these clots develop in circulating blood.

This disorder can cause clots throughout the body's blood vessels, sometimes creating a condition known as thrombosis. Thrombosis can lead to infarction, or death of tissue, as a result of blocked blood

supply to the tissue. However, hypercoagulability does not always lead to thrombosis. In **pregnancy**, and other hypercoagulable states, the incidence of thrombosis is higher than that of the general population, but is still under 10%. However, in association with certain genetic disorders, hypercoagulation disorders may be more likely to lead to thrombosis. Hypercoagulation disorders may also be known as hyperhomocystinemia, antithrombin III deficiency, factor V leiden, and protein C or protein S deficiency.

Causes and symptoms

Hypercoagulation disorders may be acquired or hereditary. Some of the genetic disorders that lead to hypercoagulation are abnormal clotting factor V, variations in fibrinogen, and deficiencies in proteins C and S. Other body system diseases may also lead to these disorders, including diabetes, sickle cell anemia, congenital heart disease, lupus, **thalassemia**, polycythemia rubra vera, and others. Antithrombin III deficiency is a hereditary hypercoagulation disorder that affects both sexes. Symptoms include obstruction of a blood vessel by a clot (thromboembolic disease), vein inflammation (phlebitis), and ulcers of the lower parts of the legs. The role of proteins C and S is a complex one. In order for coagulation to occur, platelets (small, round fragments in the blood) help contract blood vessels to lessen blood loss and also to help plug damaged blood vessels. However, the conversion of platelets into actual clots is a complicated web involving proteins that are identified clotting factors. The factors are carried in the plasma, or liquid portion of the blood. Proteins C and S are two of the clotting factors that are present in the plasma to help regulate or activate parts of the clotting process. Protein C is considered an anticoagulant. Mutation defects in the proteins may decrease their concentrations in the blood, and may or may not affect their resulting anticoagulant activity. Factor V is an unstable clotting factor also present in plasma. Abnormal factor V resists the changes that normally occur through the influence of protein C, which can also lead to hypercoagulability. Prothrombin, a glycoprotein that converts to thrombin in the early stage of the clotting process, is affected by the presence of these proteins, as well as other clotting factors.

Diagnosis

The diagnosis of hypercoagulation disorders is completed with a combination of **physical examination**, medical history, and blood tests. An accurate medical history is important to determine possible symptoms

KEY TERMS

Antithrombin—Any substance that counters the effect of thrombin, an enzyme that converts fibrinogen into fibrin, leading to blood coagulation.

Congenital—Refers to a condition or disorder present at birth.

Hemostasis—The arrest of bleeding.

Heparin—An anticoagulant, or blood clot “dissolver.”

Polycythemia—A condition characterized by an overabundance of red blood cells.

Thalassemia—One of a group of inherited blood disorders characterized by a defect in the metabolism of hemoglobin, or the portion of the red blood cells that transports oxygen throughout the blood stream.

Thrombosis—Formation of a clot in the blood that either blocks, or partially blocks, a blood vessel. The thrombus may lead to infarction, or death of tissue due to a blocked blood supply.

and causes of hypercoagulation disorders. There are a number of blood tests that can determine the presence or absence of proteins, clotting factors, and platelet counts in the blood. Among the tests used to detect hypercoagulation is the Antithrombin III assay. Protein C and Protein S concentrations can be diagnosed with immunoassay or plasma antigen level tests.

Treatment

Coumadin and heparin anticoagulants may be administered to reduce the clotting effects and maintain fluidity in the blood. Heparin is an anticoagulant that prevents thrombus formation and is used primarily for liver and lung clots.

Prognosis

The prognosis for patients with hypercoagulation disorders varies depending on the severity of the clotting and thrombosis. If undetected and untreated, thrombosis could lead to recurrent thrombosis and **pulmonary embolism**, a potentially fatal problem.

Prevention

Hereditary hypercoagulation disorders may not be prevented. Genetic and blood testing may help

determine a person's tendency to develop these disorders.

Resources

ORGANIZATIONS

National Heart, Lung and Blood Institute. PO Box 30105, Bethesda, MD 20824-0105. (301) 251-1222.

< <http://www.nhlbi.nih.gov> >.

National Hemophilia Foundation. 116 West 32nd St., 11th Floor, New York, NY 10001. 800-424-2634.

< <http://www.hemophilia.org/home.htm> >.

Teresa Odle

Hyperemesis gravidarum

Definition

Hyperemesis gravidarum means excessive **vomiting** during **pregnancy**.

Description

In pregnant women, **nausea and vomiting** (morning sickness) are common, affecting up to 80% of pregnancies. Hyperemesis, or extreme nausea and excessive vomiting, occur in about 1% of pregnancies. This condition causes uncontrollable vomiting, severe dehydration, and weight loss for the mother. However, hyperemesis gravidarum rarely causes problems for the unborn baby.

Causes and symptoms

The cause of nausea and vomiting during pregnancy is unknown but may be related to the level of certain hormones produced during pregnancy. Hyperemesis is seen more often in first pregnancies and multiple pregnancies (twins, triplets, etc.). The main symptom of hyperemesis is severe vomiting, which causes **dehydration** and weight loss.

Diagnosis

Although many women with morning sickness feel like they are vomiting everything they eat, they continue to gain weight and are not dehydrated; they do not have hyperemesis gravidarum. Women with this condition will start to show signs of starvation, including weight loss. **Physical examination** and laboratory tests of blood and urine samples will be

used to help diagnose the condition. One of the most common tests used to help diagnosis and monitor hyperemesis gravidarum is a test for ketones in the urine. Excessive ketones in the urine (ketonuria) indicate that the body is not using carbohydrates from food as fuel and is inadequately trying to break down fat as fuel. Ketonuria is a sign that the body is beginning to operate in **starvation** mode.

Treatment

Hospitalization is often required. Intravenous fluids with substances that help the body conduct nerve signals (electrolytes) may be given to correct the dehydration and excessive acid in the blood (acidosis). Anti-nausea or sedative medications may be given by injection to stop the vomiting. In some cases, oral medication may be prescribed to control the nausea and vomiting while food is reintroduced. If food cannot be tolerated at all, intravenous nutritional supplements may be necessary. Injections of vitamin B₆, in particular, may help overcome nutritional deficiencies that often occur.

Alternative treatment

The severe vomiting associated with hyperemesis gravidarum requires medical attention. Milder episodes of nausea or vomiting may be reduced with deep breathing and relaxation exercises. The use of herbal remedies should be done with extreme caution during pregnancy, especially in the first trimester. Natural remedies to reduce nausea include a teaspoon of cider vinegar in a cup of warm water, or tea made from anise (*Pinpinella anisum*), fennel seed (*Foeniculum vulgare*), red raspberry (*Rubus idaeus*), or ginger (*Zingiber officinale*). Wristbands can be positioned over **acupressure** points on both wrists. **Aromatherapy** with lavender, rose, or chamomile can be soothing, as can smelling ground ginger. Homeopathic remedies—which use extremely diluted solutions as treatments—can be safe and effective for controlling symptoms in some women.

Prognosis

In virtually all cases, the pregnancy can continue to the successful delivery of a healthy baby.

Prevention

Although there is no evidence that hyperemesis gravidarum can be prevented, vomiting during pregnancy sometimes may be lessened. Maintaining a

KEY TERMS

Ketonuria—The presence of large amount of ketones in the urine. These byproducts of inadequate breakdown of nutrients indicate that the body is in starvation.

healthy diet, getting adequate sleep, and controlling **stress** may contribute to prevention or improvement of symptoms. Several strategies may help lessen the nausea and vomiting. Eating dry foods and limiting fluid intake may also be helpful. Small meals should be eaten frequently throughout the day, with a protein snack at night. Eating soda crackers before rising from bed in the morning may help prevent early morning nausea. Iron supplements may cause nausea and can be eliminated until the nausea is controlled. Sitting upright for 45 minutes after meals may also help.

Resources

OTHER

Levy, B. T., and P. L. Brown. “Nausea and Vomiting in Pregnancy.” *The Virtual Hospital Page*. University of Iowa. <<http://www.vh.org>>.

“Natural Remedies During Pregnancy: Frequently Asked Questions.” *Childbirth.Org*. <<http://www.childbirth.org/articles/remedy.html>>.

Altha Roberts Edgren

Hyperhidrosis

Definition

Hyperhidrosis is a disorder marked by excessive sweating. It usually begins at **puberty** and affects the palms, soles, and armpits.

Description

Sweating is the body’s way of cooling itself and is a normal response to a hot environment or intense **exercise**. However, excessive sweating unrelated to these conditions can be a problem for some people. Those with constantly moist hands may feel uncomfortable shaking hands or touching, while others with sweaty armpits and feet may have to contend with the unpleasant odor that results from the bacterial

breakdown of sweat and cellular debris (bromhidrosis). People with hyperhidrosis often must change their clothes at least once a day, and their shoes can be ruined by the excess moisture. Hyperhidrosis may also contribute to such skin diseases as **athlete's foot** (tinea pedis) and **contact dermatitis**.

In addition to excessive sweat production, the texture and color of the skin itself may be affected by hyperhidrosis. The skin may turn pink or bluish white. Severe hyperhidrosis of the soles of the feet may produce cracks, fissures, and scaling of the skin.

Hyperhidrosis in general and axillary hyperhidrosis (excessive sweating in the armpits) in particular are more common in the general population than was previously thought. A group of dermatologists in Virginia reported in 2004 that 2.8% of the United States population, or about 7.8 million persons, have hyperhidrosis. Of this group, slightly more than half (4 million persons) have axillary hyperhidrosis. One-third of the latter group, or about 1.3 million persons, find that the condition significantly interferes with daily activities and is barely tolerable. Only 38%, however, had ever discussed their excessive sweating with their doctor.

Causes and symptoms

There are three basic forms of hyperhidrosis: emotionally induced; localized; and generalized. Emotionally induced hyperhidrosis typically affects the palms of the hands, soles of the feet, and the armpits. Localized hyperhidrosis typically affects the palms, armpits, groin, face, and the area below the breasts in women, while generalized hyperhidrosis may affect the entire body.

Hyperhidrosis may be either idiopathic (of unknown cause) or secondary to **fever**, metabolic disorders, **alcoholism**, **menopause**, **Hodgkin's disease**, **tuberculosis**, various types of **cancer**, or the use of certain medications. The medications most commonly associated with hyperhidrosis are propranolol, venlafaxine, **tricyclic antidepressants**, pilocarpine, and physostigmine.

Most cases of hyperhidrosis begin during childhood or adolescence. Hyperhidrosis that begins in adult life should prompt the doctor to look for a systemic illness, medication side effect, or metabolic disorder.

Hyperhidrosis affects both sexes equally and may occur in any age group. People of any race may be affected; however, for some reason unknown as of the early 2000s, Japanese are affected 20 times more frequently than members of other ethnic groups.

Diagnosis

Hyperhidrosis is diagnosed by patient report and a **physical examination**. In many cases the physician can directly observe the excessive sweating.

The doctor may also perform an iodine starch test, which involves spraying the affected areas of the patient's body with a mixture of 500 g of water-soluble starch and 1 g iodine crystals. Areas of the skin producing sweat will turn black.

The doctor will order other laboratory or imaging tests if he or she suspects that the sweating is associated with another disease or disorder.

Treatment

Most over-the-counter antiperspirants are not strong enough to effectively prevent hyperhidrosis. To treat the disorder, doctors usually prescribe 20% aluminum chloride hexahydrate solution (Drysol), which the patient applies at night to the affected areas that are then wrapped in a plastic film until morning. Drysol works by blocking the sweat pores. Formaldehyde- and glutaraldehyde-based solutions can also be prescribed; however, formaldehyde may trigger an allergic reaction and glutaraldehyde can stain the skin (for this reason it is primarily applied to the soles). Anticholinergic drugs may also be given. These drugs include such medications as propantheline, oxybutynin, and benztropine.

Injections of botulinum toxin (Botox) given under the skin work well for some patients. Botox works to stop the excessive sweating by preventing the transmission of nerve impulses to the sweat glands. These injections must be repeated every 4–12 months, however.

In addition, an electrical device that emits low-voltage current can be held against the skin to reduce sweating. These treatments are usually conducted in a doctor's office on a daily basis for several weeks, followed by weekly visits. Dermatologists also recommend that patients wear clothing made of natural or absorbent fabrics also may help, avoid high-buttoned collars, use talc or cornstarch, and keep underarms shaved.

The only permanent cure for hyperhidrosis of the palms is a surgical procedure known as a **sympathectomy**. To treat severe excessive sweating, a surgeon can remove a portion of the nerve near the top of the spine that controls palm sweat. However, not very many neurosurgeons in the United States will perform the procedure, because it often results in compensatory sweating in other regions of the body. Alternatively,

KEY TERMS

Anticholinergic drugs—Drugs that block the action of the neurotransmitter acetylcholine.

Axilla (plural, axillae)—The medical term for the armpit.

Bromhidrosis—Bacterial breakdown of sweat and cellular debris resulting in a foul odor.

Contact dermatitis—Skin inflammation that occurs when the skin is exposed to a substance originating outside of the body.

Idiopathic—Of spontaneous origin or unknown cause. Many cases of hyperhidrosis are idiopathic.

Sympathectomy—Surgical cutting or interruption of any of the pathways in the sympathetic nervous system. It may be performed to control hyperhidrosis that does not respond to medications.

Sympathetic nervous system—The part of the nervous system that originates in the lumbar and thoracic portions of the spinal cord. It regulates involuntary reactions to stress, including sweating as well as heart rate, breathing rate, and digestive secretions.

Tinea pedis—Fungal infection of the feet of the skin characterized by dry, scaly lesions.

it is possible to surgically remove the sweat gland-bearing skin of the armpits, but this is a major procedure that may require skin grafts.

More recently, **liposuction** under the armpits has been successfully used to treat hyperhidrosis in this region of the body. The liposuction removes some of the excess sweat glands responsible for axillary hyperhidrosis. The procedure also has the advantage of leaving smaller **scars** and being less disruptive of the overlying skin.

Prognosis

Hyperhidrosis is not associated with increased mortality; it primarily affects the patient's quality of life rather than longevity. While the condition cannot be cured without radical surgery, it can usually be controlled effectively.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD., editors. "Disorders of Sweating." Section 10, Chapter 124 In

The Merck Manual of Diagnosis and Therapy. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

PERIODICALS

Altman, Rachel, MD, George Kihiczak, MD, and Robert Schwartz, MD. "Hyperhidrosis." *eMedicine* August 18, 2004. <<http://www.emedicine.com/derm/topic893.htm>>.

Licht, P. B., and H. K. Pilegaard. "Severity of Compensatory Sweating after Thoracoscopic Sympathectomy." *Annals of Thoracic Surgery* 78 (August 2004): 427–431.

Strutton, D. R., J. W. Kowalski, D. A. Glaser, and P. E. Stang. "US Prevalence of Hyperhidrosis and Impact on Individuals with Axillary Hyperhidrosis: Results from a National Survey." *Journal of the American Academy of Dermatology* 51 (August 2004): 241–248.

ORGANIZATIONS

American Academy of Dermatology (AAD). P. O. Box 4014, Schaumburg, IL 60168-4014. (847) 330-0230. <<http://www.aad.org>>.

Carol A. Turkington
Rebecca J. Frey, PhD

Hyperhomocystinemia see

Hypercoagulation disorders

Hypericum perforatum see **St. Johns wort**

Hyperkalemia

Definition

The normal concentration of potassium in the serum is in the range of 3.5 to 5.0 mM. Hyperkalemia refers to serum or plasma levels of potassium ions above 5.0 mM. The concentration of potassium is often expressed in units of milliequivalents per liter (mEq/L), rather than in units of millimolarity (mM). Both units mean the same thing when applied to concentrations of potassium ions.

Description

A normal adult who weighs about 70 kg contains a total of about 3.6 moles of potassium ions in the body. Most of this potassium (about 98%) occurs inside various cells and organs, where its concentration is about 150 mM. This level is in contrast to the much lower concentration found in the blood serum, where only about 0.4% of the body's potassium resides. Hyperkalemia can be caused by an overall

excess of body potassium, or by a shift from inside to outside cells. For example, hyperkalemia can be caused by the sudden release of potassium ions from muscle into the surrounding fluids.

In a normal person, hyperkalemia from too much potassium in the diet is prevented by at least three types of regulatory processes. First, various cells and organs act to prevent hyperkalemia by taking up potassium from the blood. It is also prevented by the action of the kidneys, which excrete potassium into the urine. A third protective mechanism is **vomiting**. Consumption of a large dose of potassium ions, such as potassium chloride, induces a vomiting reflex to expel most of the potassium before it can be absorbed.

Causes and symptoms

Hyperkalemia can occur from a variety of causes, including the consumption of too much of a potassium salt; the failure of the kidneys to normally excrete potassium ions into the urine; the leakage of potassium from cells and tissues into the bloodstream; and from acidosis. The most common cause of hyperkalemia is kidney (or renal) disease, which accounts for about three quarters of all cases. Kidney function is measured by the glomerular filtration rate, the rate at which each kidney performs its continual processing and cleansing of blood. The normal glomerular filtration rate is about 100 ml/min. If the kidney is damaged so that the glomerular filtration rate is only 5 ml/min or less, hyperkalemia may result, especially if high-potassium foods are consumed. The elderly are at particular risk, since many regulatory functions of the body do not work well in this population. Elderly patients who are being treated with certain drugs for high blood pressure, such as spironolactone (Aldactone) and triamterene (Dyazide), must especially be monitored for possible hyperkalemia, as these medications promote the retention of potassium by the kidneys.

Hyperkalemia can also be caused by a disease of the adrenal gland called **Addison's disease**. The adrenal gland produces the hormone aldosterone that promotes the excretion of potassium into the urine by the kidney.

Hyperkalemia can also result from injury to muscle or other tissues. Since most of the potassium in the body is contained in muscle, a severe trauma that crushes muscle cells results in an immediate increase in the concentration of potassium in the blood. Hyperkalemia may also result from severe **burns** or infections.

Acidic blood plasma, or acidosis, is an occasional cause of hyperkalemia. Acidosis, which occurs in a

number of diseases, is defined as an increase in the concentration of hydrogen ions in the bloodstream. In the body's attempt to correct the situation, hydrogen is taken up by muscle cells out of the blood in an exchange mechanism involving the transfer of potassium ions into the bloodstream. This can abnormally elevate the plasma's concentration of potassium ions. When acidosis is the cause of hyperkalemia, treating the patient for acidosis has two benefits: a reversal of both the acidosis and the hyperkalemia.

Symptoms of hyperkalemia include abnormalities in the behavior of the heart. Heart abnormalities of mild hyperkalemia (5.0 to 6.5 mM potassium) can be detected by an electrocardiogram (ECG or EKG). With severe hyperkalemia (over 8.0 mM potassium), the heart may beat at a dangerously rapid rate (fibrillation) or stop beating entirely (cardiac arrest). Patients with moderate or severe hyperkalemia may also develop nervous symptoms such as **tingling** of the skin, **numbness** of the hands or feet, weakness, or a flaccid **paralysis**, which is characteristic of both hyperkalemia and **hypokalemia** (low plasma potassium).

Diagnosis

Hyperkalemia can be measured by acquiring a sample of blood, preparing blood serum, and using a potassium sensitive electrode for measuring the concentration of potassium ions. Alternatively, atomic absorption spectroscopy can be used for measuring potassium. Since high or low potassium levels result in abnormalities in heart function, the electrocardiogram is usually the method of choice for the diagnosis of both hyperkalemia and hypokalemia.

Treatment

Insulin injections are used to treat hyperkalemia in emergency situations. Insulin is a hormone well known for its ability to stimulate the entry of sugar (glucose) into cells. It also provokes the uptake of potassium ions by cells, decreasing potassium ion concentration in the blood. When insulin is used to treat hyperkalemia, glucose is also injected. Serum potassium levels begin to decline within 30 to 60 minutes and remain low for several hours. In non-emergency situations, hyperkalemia can be treated with a low potassium diet. If this does not succeed, the patient can be given a special resin to bind potassium ions. One such resin, sodium polystyrene sulfonate (Kayexalate), remains in the intestines, where it absorbs potassium and forms a complex of resin and potassium. Eventually this complex is excreted in the feces. A typical dose of resin is 15 grams, taken one to

KEY TERMS

Acidosis—An abnormally high acid (hydrogen ion) concentration in blood plasma. The unit of acid content is pH, with a lower value indicating more acidic conditions. Blood plasma normally has a pH of 7.35–7.45. Alkaline blood has a pH value greater than pH 7.45. When the blood pH value is less than 7.35, the patient is in acidosis.

four times per day. The correction of hyperkalemia with resin treatment takes at least 24 hours.

Prognosis

The prognosis for specifically correcting hyperkalemia is excellent. However, hyperkalemia is usually caused by kidney failure, an often irreversible and eventually fatal condition.

Prevention

Healthy people are not at risk for hyperkalemia. Patients with renal disease and those on certain diuretic medications must be monitored to prevent its occurrence.

Resources

PERIODICALS

Greenberg, A. "Hyperkalemia: treatment options." *Seminars in Nephrology* 18 (1998): 46-57.

Tom Brody, PhD

Hyperkinetic disorder see **Attention-deficit/Hyperactivity disorder (ADHD)**

Hyperlipemia see **Hyperlipoproteinemia**

Hyperlipidemia see **Hyperlipoproteinemia**

Hyperlipoproteinemia

Definition

Hyperlipoproteinemia occurs when there is too much lipid (fat) in the blood. Shorter terms that mean the same thing are hyperlipidemia and hyperlipemia. Dyslipidemia refers to a redistribution of cholesterol from one place to another that increases

the risk of vascular disease without increasing the total amount of cholesterol. When more precise terms are needed, hypercholesterolemia and hypertriglyceridemia are used.

Description

It is commonly known that oil and water do not mix unless another substance like a detergent is added. Yet the body needs to transport both lipids (fats) and water-based blood within a single circulatory system. There must be a way to mix the two, so that essential fatty nutrients can be transported in the blood and so that fatty waste products can be carried away from tissues. The solution is to combine the lipids with protein to form water-soluble packages that can be transported in the blood.

These packages of fats are called lipoproteins. They are a complex mixture of triglycerides, cholesterol, phospholipids and special proteins. Some of these chemicals are fatty nutrients absorbed from the intestines on their way to being made part of the body. Cholesterol is a waste product on its way out of the body through the liver, the bile, and ultimately the bowel for excretion. The proteins and phospholipids make the packages water-soluble.

There are five different sizes of these chemical packages. Each package needs all four chemicals in it to hold everything in solution. They differ in how much of each they contain. If blood serum is spun very rapidly in an ultracentrifuge, these five packages will layer out according to their density. They have, therefore, been named according to their densities—high-density lipoproteins (HDL), low-density lipoproteins (LDL), intermediate-density lipoproteins (IDL), very low density lipoproteins (VLDL), and chylomicrons. Only the HDLs and the LDLs will be discussed in the rest of this article.

If there is not enough detergent in the laundry, the oily stains will remain in the clothes. In the same way, if the balance of chemicals in these packages is not right, cholesterol will stay in tissues rather than being excreted from the body. What is even worse, if the chemical composition of these packages changes, the cholesterol can fall out of the blood and stay where it lands. On the other hand, a different change in the balance can remove cholesterol from tissues where there is too much. This appears to be exactly what is going on in **atherosclerosis**. The lesions contain lots of cholesterol.

The LDLs are overloaded with cholesterol. A minor change in the other chemicals in this package

will leave cholesterol behind. The HDLs have a third to a half as much cholesterol. They seem to be able to pick up cholesterol left behind by the LDLs. It seems that atherosclerosis begins with tiny tears at stressed places in the walls of the arteries. Low density lipoproteins from the blood enter these tears, where their chemistry changes enough to leave cholesterol behind. The cholesterol causes irritation; the body responds with inflammation; damage and scarring follow. Eventually the artery gets so diseased blood cannot flow through it. Strokes and heart attacks are the result.

But if there are lots of HDLs in the blood, the cholesterol is rapidly picked up and not allowed to cause problems. Women before **menopause** have estrogen (the female hormone), which encourages the formation of HDLs. This is the reason they have so little vascular disease, and why they rapidly catch up to men after menopause, when estrogen levels fall. Replacement of estrogen after menopause has been prescribed to for protection through the later years. However, in 2003, the Women's Health Initiative, a large clinical trial involving postmenopausal women, was halted in July 2002 because of the many detrimental effects of combined estrogen and progesterone therapy (called **hormone replacement therapy**). Among the effects was increased risk of heart disease, sometimes within the first year of use.

Cholesterol is the root of the problem, but like any other root it cannot just be eliminated. Ninety percent of the cholesterol in the body is created there as a waste product of necessary processes. The solution lies in getting it out to the body without clogging the arteries.

Of course the story is much more complex. The body has dozens of chemical processes that make up, break down, and reconfigure all these chemicals. It is these processes that are the targets of intervention in the effort to cure vascular disease.

Diseases

Near the dawn of concern over cholesterol and vascular disease a family of hereditary diseases was identified, all of which produced abnormal quantities of blood fats. These diseases were called dyslipoproteinemias and came in both too many and too little varieties. The hyperlipoproteinemias found their way into five categories, depending on which chemical was in excess.

- Type 1 has a pure elevation of triglycerides in the chylomicron fraction. These people sometimes get **pancreatitis** and abdominal pains, but they do not seem to have an increase in vascular disease.
- Type 2 appears in two distinct genetic patterns and a third category, which is by far the most important

kind, because everyone is at risk for it. All Type 2s have elevated cholesterol. Some have elevated triglycerides also. The familial (genetic) versions of Type 2 often develop xanthomas, which are yellow fatty deposits under the skin of the knuckles, elbows, buttocks or heels. They also may have xanthelasmas, smaller yellow patches on the eyelids.

- Type 3 appears in one in 10,000 people and elevates both triglycerides and cholesterol with consequent vascular disease. In 2003, researchers discovered the molecular mechanism that contributes to high triglycerides in those with this type of hyperlipoproteinemia.
- Type 4 elevates only triglycerides and does not increase the risk of vascular disease.
- Type 5 is similar to Type 1.
- Dyslipidemia refers to a normal amount of cholesterol that is mostly in LDLs, where it causes problems.

All but Type 2 are rare and of interest primarily because they give insight into the chemistry of blood fats.

In addition to the above genetic causes of blood fat disorders, a number of acquired conditions can raise lipoprotein levels.

- Diabetes mellitus, because it alters the way the body handles its energy needs, and also affects the way it handles fats. The result is elevated triglycerides and reduced HDL cholesterol. This effect is amplified by **obesity**.
- Hypothyroidism is a common cause of lipid abnormalities. The thyroid hormone affects the rate of many chemical processes in the body, including the clearing of fats from the blood. The consequence usually is an elevation of cholesterol.
- Kidney disease affects the blood's proteins and consequently the composition of the fat packages. It usually raises the LDLs.
- Liver disease, depending on its stage and severity, can raise or lower any of the blood fats.
- Alcohol raises triglycerides. In moderate amounts (if they are very moderate) it raises HDLs and can be beneficial.
- Cigarette **smoking** lowers HDL cholesterol, as does malnutrition and obesity.

Certain medications elevate blood fat levels. Because some of these medications are used to treat heart disease, it has been necessary to reevaluate their usefulness:

- Thiazides, water pills used to treat high blood pressure, can raise both cholesterol and triglycerides.
- Beta-blockers, another class of medication used to treat high blood pressure, cortisone-like drugs, and estrogen can raise triglycerides.
- Progesterone, the **pregnancy** hormone, raises cholesterol.

Not all of these effects are necessarily bad, nor are they necessarily even significant. For instance, estrogen is clearly beneficial. Each effect must be considered in the overall goal of treatment.

Causes and symptoms

A combination of heredity and diet is responsible for the majority of fat disorders. It is not so much the cholesterol in the diet that is the problem, because that accounts for only 10% of the body's store. It is the other fats in the diet that alter the way the body handles its cholesterol. There is a convincing relation between fats in the diet and the incidence of atherosclerosis. The guilty fats are mostly the animal fats, but palm and coconut oil also are harmful. These fats are called saturated fats for the chemical reason that most of their carbon atoms have as many hydrogen atoms attached as they can accommodate. More important than the kind of fat is the amount of fat. For many people, fat is half of their diet. One-fifth to one-fourth is a much healthier fraction, the rest of the diet being made up of complex carbohydrates and protein.

This disease is silent for decades, until the first episode of heart disease or **stroke**.

Diagnosis

It would be easier if simple cholesterol and triglyceride tests were all it took to assess the risk of atherosclerosis. But the important information is which package the cholesterol is in—the LDLs or the HDLs. That takes a more elaborate testing process. To complicate matters further, the amount of fats in the blood varies greatly in relation to the last meal—how long ago it was and what kind of food was eaten. A true estimate of the risk comes from several tests several weeks apart, each done after at least twelve hours of **fasting**.

Treatment

Diet and lifestyle change are the primary focus for most cholesterol problems. It is a mistake to think that a pill will reverse the effects of a bad diet, obesity, smoking, excess alcohol, stress, and inactivity. Reducing the amount of fat in the diet by at least

half is the most important move to make. Much of the food eaten to satisfy a “sweet tooth” is higher in fat than in sugar. A switch away from saturated fats is the next step, but the rush to polyunsaturated fats was ill-conceived. These, particularly the hydrogenated fats in margarine, have problems of their own. They raise the risk of **cancer** and are considered more dangerous than animal fat by many experts. Theory supports population studies that suggest monounsaturated olive oil may be the healthiest of all.

There was a tremendous push at the end of the 20th century to use lipid-lowering medications. The most popular and most expensive agents, the “statins,” hinder the body's production of cholesterol and sometimes damage the liver as a side effect. Their full name is 3-hydroxy-3-methylglutaryl-coenzyme A (*HMG-CoA*) reductase inhibitors. Their generic names are cerivastatin, fluvastatin, lovastatin, pravastatin, simvastatin, and the newest and most powerful to date, rosuvastatin. Studies show that these drugs lower cholesterol. Only recently, though, has any evidence appeared that this affects health and longevity. Earlier studies showed, in fact, an increased **death** rate among users of the first class of lipid-altering agents—the fibric acid derivatives. The chain of events connecting raised HDL and lowered LDL cholesterol to longer, healthier lives is still to be forged.

High-tech methods of rapidly reducing very high blood fat levels are performed for those rare disorders that require it. There are resins that bind cholesterol in the intestines. They taste awful, feel like glue and routinely cause gas, bloating, and **constipation**. For acute cases, there is a filtering system that takes fats directly out of the blood.

Niacin (nicotinic acid) lowers cholesterol effectively and was the first medication proven to improve overall life expectancy. It also can be liver toxic, and the usual formulation causes a hot flash in many people. This can be overcome by taking a couple of aspirins 30 minutes before the niacin, or by taking a special preparation called “flush free,” “inositol-bound” or inositol hexanicotinate.

Alternative treatment

Omega-3 oil is a special kind found mostly in certain kinds of fish. It is beneficial in lowering cholesterol. An herbal alternative called guggulipid, *Commiphora mukul*, an extract of an Indian plant, has been touted as working the same way as the expensive and liver toxic cholesterol-lowering medications. However, a 2003 clinical trial found that the

KEY TERMS

Atherosclerosis—Hardening of the arteries due to fat (cholesterol) deposits in their walls. Also known as *arteriosclerosis*.

Genetic—Refers to the genes and characteristics inherited from parents.

Inflammation—The body's response to irritation, by releasing chemicals that attack germs and tissues and also repair the damage done.

Lesion—Localized disease or damage.

Pancreatitis—Inflammation of the pancreas.

Serum—The liquid part of blood, from which all the cells have been removed.

supplement did not meet these claims. In fact, guggul did not lower total cholesterol, LDL cholesterol, or triglycerides. Most patients tolerated the supplement, but some developed a hypersensitivity rash.

To lower cholesterol, **naturopathic medicine**, **traditional Chinese medicine**, and ayurvedic medicine may be considered. Some herbal therapies include alfalfa (*Medicago sativa*), Asian ginseng (*Panax ginseng*), and fenugreek (*Trigonella foenum-graecum*). Garlic (*Allium sativum*) and onions are also reported to have cholesterol-lowering effects. In naturopathic medicine, the liver is considered to be an organ that needs cleansing and rebalancing. The liver often is treated with a botanical formula that will act as a bitter to stimulate bile flow in the liver. Before initiating alternative therapies, medical consultation is strongly advised.

Prognosis

The prognosis is good for Type 1 hyperlipoproteinemia with treatment; without treatment, death may result. For Type 2 the prognosis is poor even with treatment. The prognosis for type 3 is good when the prescribed diet is strictly followed. For types 4 and 5 the prognosis is uncertain, due to the risk of developing premature **coronary artery disease** and pancreatitis, respectively.

Prevention

Genetic inheritance cannot be changed, but its effects may be modified with proper treatment. Family members of an individual with hyperlipoproteinemia should consider having their blood lipids

assessed. The sooner any problems are identified, the better the chances of limiting or preventing the associated health risks. Anyone with a family history of disorders leading to hyperlipoproteinemia also may benefit from genetic testing and counseling to assist them in making reproductive decisions.

Resources

PERIODICALS

Brunk, Doug. "Three Studies Further Confirm Ill Effects of HRT: Heart Disease Risk Rises First Year of Use: Continuing Analysis of WHI Data." *Family Practice News* 33, no. 17 (September 1, 2003): 1–2.

Dowhower Karp, Kelly. "New Statin Said to be More Powerful than Others." *Drug Topics* 147, no. 17 (September 1, 2003): 27.

"Herbal Extract Not Effective in Treating High Cholesterol." *Drug Week* August 29, 2003: 197.

Kyperos, Kyriakos E., et al. "Molecular Mechanisms of Type III Hyperlipoproteinemia: the Contribution of the Carboxy-terminal Domain of ApoE Can Account for the Dyslipidemia that is Associated With the E2/E2 Phenotype." *Biochemistry* 42, no. 33 (August 26, 2003): 9841–9853.

ORGANIZATIONS

Inherited High Cholesterol Foundation. 410 Chipeta Way, Room 167, Salt Lake City, UT 84104. (888) 244-2465.

J. Ricker Polsdorfer, MD
Teresa G. Odle

Hypermagnesemia see **Magnesium imbalance**

Hypermenorrhea see **Dysfunctional uterine bleeding**

Hypermetropia see **Hyperopia**

Hypernatremia

Definition

The normal concentration of sodium in the blood plasma is 136–145 mM. Hypernatremia is defined as a serum sodium level over 145 mM. Severe hypernatremia, with serum sodium above 152 mM, can result in seizures and **death**.

Description

Sodium is an atom, or ion, that carries a single positive charge. The sodium ion may be abbreviated

as Na^+ or as simply Na. Sodium can occur as a salt in a crystalline solid. Sodium chloride (NaCl), sodium phosphate (Na_2HPO_4) and sodium bicarbonate (NaHCO_3) are commonly occurring salts. These salts can be dissolved in water or in juices of various foods. Dissolving involves the complete separation of ions, such as sodium and chloride in common table salt (NaCl).

About 40% of the body's sodium is contained in bone. Approximately 2–5% occurs within organs and cells and the remaining 55% is in blood plasma and other extracellular fluids. The amount of sodium in blood plasma is typically 140 mM, a much higher amount than is found in intracellular sodium (about 5 mM). This asymmetric distribution of sodium ions is essential for human life. It makes possible proper nerve conduction, the passage of various nutrients into cells, and the maintenance of blood pressure.

The body continually regulates its handling of sodium. When dietary sodium is too high or low, the intestines and kidneys respond to adjust concentrations to normal. During the course of a day, the intestines absorb dietary sodium while the kidneys excrete a nearly equal amount of sodium into the urine. If a low sodium diet is consumed, the intestines increase their efficiency of sodium absorption, and the kidneys reduce its release into urine.

The concentration of sodium in the blood plasma depends on two things: the total amount of sodium and water in arteries, veins, and capillaries (the circulatory system). The body uses separate mechanisms to regulate sodium and water, but they work together to correct blood pressure when it is too high or too low. Too high a concentration of sodium, or hypernatremia, can be corrected either by decreasing sodium or by increasing body water. The existence of separate mechanisms that regulate sodium concentration account for the fact that there are numerous diseases that can cause hypernatremia, including diseases of the kidney, pituitary gland, and hypothalamus.

Causes and symptoms

Vasopressin, also called anti-diuretic hormone, is made by the hypothalamus and released by the pituitary gland into the bloodstream. There it travels to the kidney where it reduces the release of water into the urine. With less vasopressin production, the body fails to conserve water, and the result is a trend toward higher plasma sodium concentrations. Hypernatremia may occur in diabetes insipidus, a disease that causes excessive urine production. (It is not the same disease as **diabetes mellitus**, a disease resulting from impaired insulin production.) The defect involves either the

failure of the hypothalamus to make vasopressin or the failure of the kidney to respond to vasopressin. In either case, the kidney is able to conserve and regulate the body's sodium levels, but is unable to conserve and retain the body's water. Hypernatremia does not occur in **diabetes insipidus** if the patient is able to drink enough water to keep up with urinary loss, which may be as high as 10 liters per day.

Hypernatremia may occur in unconscious (or comatose) patients due to the inability to drink water. Water is continually lost by evaporation from the lungs and in the urine. If the patient is not given water via infusion, the sodium concentration in the blood may increase and hypernatremia could develop. Hypernatremia can also occur in rare diseases in which the thirst impulse is impaired.

Hypernatremia can also occur accidentally in the hospital when patients are infused with solutions containing sodium, such as sodium bicarbonate for the treatment of acidosis (acidic blood). It can also be accidentally induced with sodium chloride infusions, especially in elderly patients with impaired kidney function.

Hypernatremia can cause neurological damage due to shrinkage of brain cells. Neurological symptoms include confusion, **coma**, **paralysis** of the lung muscles, and death. The severity of the symptoms is related to how rapidly the hypernatremia developed. Hypernatremia that comes on rapidly does not allow the cells of the brain time to adapt to their new high-sodium environment. Hypernatremia is especially dangerous for children and the elderly.

Diagnosis

Hypernatremia is diagnosed by acquiring a blood sample, preparing plasma, and using a sodium-sensitive electrode for measuring the concentration of sodium ions.

Treatment

Hypernatremia is treated with infusions of a solution of water containing 0.9% sodium chloride (0.9 grams NaCl /100 ml water), which is the normal concentration of sodium chloride in the blood plasma. The infusion is performed over many hours or days to prevent abrupt and dangerous changes in brain cell volume. In emergencies, such as when hypernatremia is causing neurological symptoms, infusions may be conducted with salt solutions containing 0.45% sodium chloride, which is half the normal physiologic level.

KEY TERMS

Blood plasma and serum—Blood plasma, or plasma, is prepared by obtaining a sample of blood and removing the blood cells. The red blood cells and white blood cells are removed by spinning with a centrifuge. Chemicals are added to prevent the blood's natural tendency to clot. If these chemicals include sodium, then a false measurement of plasma sodium content will result. Serum is prepared by obtaining a blood sample, allowing formation of the blood clot, and removing the clot using a centrifuge. Both plasma and serum are light yellow in color.

Prognosis

The prognosis for treating hypernatremia is excellent, except if neurological symptoms are severe or if overly rapid attempts are made to treat and reverse the condition.

Prevention

Hypernatremia occurs only in unusual circumstances that are not normally under a person's control.

Resources

PERIODICALS

Fried, L. F., and P. M. Palevsky. "Hyponatremia and hypernatremia." *Medical Clinics of North America* 81 (1997): 585-609.

Tom Brody, PhD

Hypernephroma see **Kidney cancer**

Hyperopia

Definition

Hyperopia (farsightedness) is the condition of the eye where incoming rays of light reach the retina before they converge into a focused image.

Description

When light goes through transparent but dense material like the materials of the eye's lens system (the lens and cornea), its velocity decreases. If the surface of

the dense material is not perpendicular to the incoming light, as is the case with the curved surfaces on lenses and corneas, the direction of the light changes. The greater the curvature of the lens system, the greater the change in the direction of the light.

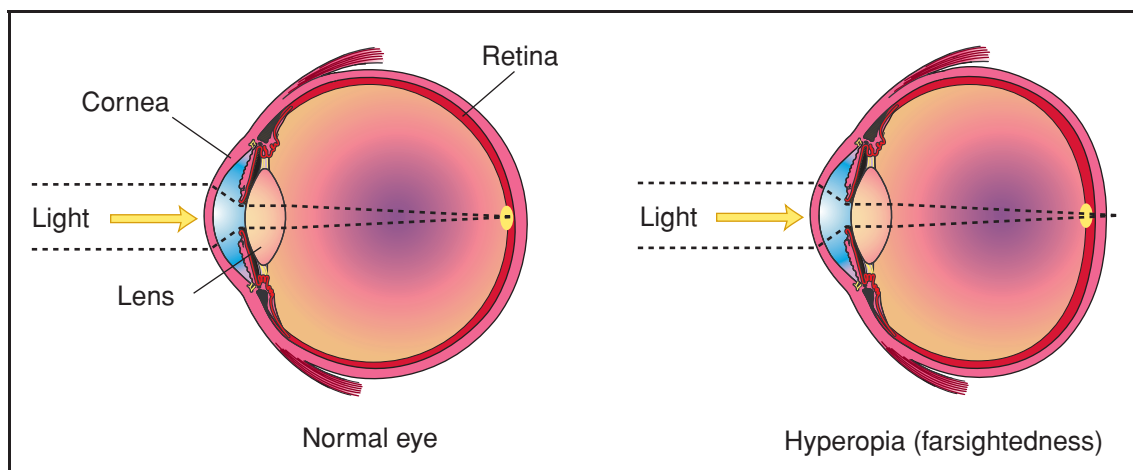
When parallel light rays from an object go through the lens system of the eye, they are bent so they converge at a point some distance behind the lens. With perfect vision this point of convergence, where the light rays are focused, is on the retina. This happens when the cumulative curvature of the lens plus cornea and the distance from the lens to the retina are just right for each other. The condition where the point of focus of parallel light rays from an object is behind the retina is called hyperopia. This condition exists when the combined curvature of the lens and cornea is insufficient (e.g., flatter than needed for the length of the eyeball). This condition can be equivalently described by saying hyperopia exists when the eyeball is too short for the curvature of its lens system.

There is a connection between the focusing of the lens of the eye (accommodation) and convergence of the eyes (the two eyes turning in to point at a close object). The best example is during reading. The lens accommodates to make the close-up material clear and the eyes turn in to look at the print and keep it single. Because of this connection between accommodation and convergence, if the lens needs to accommodate to focus for distance (to bring the image back onto the retina) the eyes may appear to turn in even when looking at the distance. This can cause a condition known as accommodative esotropia in children. The eyes turn in and the cause is accommodation because of hyperopia.

Causes and symptoms

Babies are generally born slightly hyperopic. This tends to decrease with age. There is normal variation in eyeball length and curvature of the lens and cornea. Some combinations of these variables give rise to eyes where the cornea is too flat for the distance between the cornea and the retina. If the hyperopia is not too severe the lens may be able to accommodate and bring the image back onto the retina. This would result in clear distance vision, but the constant focusing might result in headaches or eyestrain. If the lens cannot accommodate for the full amount of the hyperopia the distance image would be blurry.

If the eyes are focusing for distance and now the person is looking at a near object, the eyes need to accommodate further. This may result in blurry near objects or headaches during near work.



Hyperopia, or farsightedness, is a condition of the eye where incoming rays of light impinge on the retina before converging into a focused image, resulting in difficulty seeing nearby objects clearly. (Illustration by Electronic Illustrators Group.)

Depending upon the amount of hyperopia, symptoms can range from none to clear distance vision but blurry near vision, to blurry distance and near vision. Headaches and eyestrain may also occur, particularly when doing near tasks. An eye turned in (esotropia) may be a result of hyperopia, particularly in children. However, because a turned eye may be a result of more serious causes it is very important to have it checked out.

Diagnosis

Because it is possible to have good visual acuity with some degree of hyperopia it is important to relax accommodation before the eye exam. This is done with the use of eyedrops and is called a cycloplegic exam or cycloplegic refraction. The drops relax the accommodation (thus making reading blurry until the drops wear off). Patients will usually be asked to have someone drive them home because of the blurriness. The doctor can then determine the patient's visual status with a hand-held instrument called a retinoscope and/or have the patient read from an eye chart while placing different lenses in front of the patient's eyes. Refractive error is measured in units called diopters (D).

Treatment

The usual treatment for hyperopia is corrective lenses (spectacles or contact lenses).

Different surgical methods to correct hyperopia are under investigation. One approach is to implant

KEY TERMS

Cornea—The clear, dome-shaped outer covering of the front of the eye. It lies in front of the iris and pupil.

Iris—The colored ring just behind the cornea and in front of the lens that controls the amount of light sent to the retina.

Pupil—The black hole in the center of the iris. Light enters here on the way to the lens and retina.

Refraction—Method of determining the optical status of the eyes. Lenses are placed before the patient's eyes while reading from an eye chart. The result is the eyeglass or contact lens prescription.

Retina—The inner, light-sensitive layer of the eye containing rods and cones; transforms the image it receives into electrical messages sent to the brain via the optic nerve.

corrective contact lenses behind the patient's iris. The first experimental implantable contact lenses were implanted in 1997. Another approach is to surgically increase the curvature of the eye's existing cornea or lens. Although there have been many reports of success using different kinds of lasers to increase corneal curvature, as of 1998 there are still problems with stability and predictability. The introduction of light-activated biologic tissue glue in 1997 holds promise for improvements in those areas.

Prognosis

The prognosis for fully corrected vision is excellent for patients with low to moderate amounts of hyperopia. Patients with very high hyperopia (+10.00D or more) may not achieve full correction. Moreover, surgery to correct hyperopia will probably be perfected and approved in the near future.

Hyperopia increases the chances of chronic **glaucoma**, but vision loss from glaucoma is preventable.

Prevention

Hyperopia is usually present at birth, and there is no known way to prevent it.

Resources

ORGANIZATIONS

American Academy of Ophthalmology. 655 Beach Street, PO Box 7424, San Francisco, CA 94120-7424. <<http://www.eyenet.org>>.

American Optometric Association. 243 North Lindbergh Blvd., St. Louis, MO 63141. (314) 991-4100. <<http://www.aoanet.org>>.

OTHER

Edmiston, Dave. "Hyperopia." Lasikpatient.org. <<http://www.lasikpatient.org>>.

Lorraine Lica, PhD

Hyperparathyroidism

Definition

Parathyroid glands are four pea-sized glands located just behind the thyroid gland in the front of the neck. The function of parathyroid glands is to produce a hormone called parathyroid hormone (parathormone), which helps regulate calcium and phosphate in the body. Hyperparathyroidism is the overproduction of this hormone.

Description

Thyroid glands and parathyroid glands, despite their similar name and proximity, are entirely separate, and each produces hormones with different functions. Hyperparathyroidism may be primary or secondary. It most often occurs in those over age 30, and most commonly in patients 50 to 60 years old. It rarely occurs in children or the elderly. Women are

affected by the disease up to three times more often than men. It is estimated that 28 of every 100,000 people in the United States will develop hyperparathyroidism each year.

Normally, parathyroid glands produce the parathormone as calcium levels drop and lower to meet the demands of a growing skeleton, **pregnancy**, or **lactation**. However, when one or more parathyroid glands malfunction, it can lead to overproduction of the hormone and elevated calcium level in the blood. Therefore, a common result of hyperparathyroidism is **hypercalcemia**, or an abnormally high level of calcium in the blood. Primary hyperparathyroidism occurs as a malfunction of one of the glands, usually as a result of a benign tumor, called adenoma. Secondary hyperparathyroidism occurs as the result of a metabolic abnormality outside the parathyroid glands, which causes a resistance to the function of the parathyroid hormones. Primary hyperparathyroidism is one of the most common endocrine disorders, led only by diabetes and **hyperthyroidism**.

Causes and symptoms

Often, there are no obvious symptoms or suspicion of hyperparathyroidism, and it is first diagnosed when a patient is discovered to be hypercalcemic during a routine blood chemistry profile. Patients may believe they have felt fine, but realize improvements in sleep, irritability, and memory following treatment. When symptoms are present, they may include development of gastric ulcers or **pancreatitis** because high calcium levels can cause inflammation and **pain** in the linings of the stomach and pancreas.

Most of the symptoms of hyperparathyroidism are those present as a result of hypercalcemia, such as **kidney stones**, **osteoporosis**, or bone degradation resulting from the bones giving up calcium. Muscle weakness, central nervous system disturbances such as depression, psychomotor and personality disturbances, and rarely, even **coma** can occur. Patients may also experience **heartburn**, **nausea**, **constipation**, or abdominal pain. In secondary hyperparathyroidism, patients may show signs of calcium imbalance such as deformities of the long bones. Symptoms of the underlying disease may also be present.

Most commonly, hyperparathyroidism occurs as the result of a single adenoma, or benign tumor, in one of the parathyroid glands. About 90% of all cases of hyperparathyroidism are caused by an adenoma. The tumors are seldom cancerous. They will grow to a much larger size than the parathyroid glands, often to the size of a walnut. Genetic disorders or multiple

endocrine tumors can also cause a parathyroid gland to enlarge and oversecrete hormone. In 10% or fewer of patients with primary hyperparathyroidism, there is enlargement of all four parathyroid glands. This condition is called parathyroid hyperplasia.

Diagnosis

Diagnosis of hyperparathyroidism is most often made when a blood test (radioimmunoassay) reveals high levels of parathyroid hormone and calcium. A blood test that specifically measures the amount of parathyroid hormone has made diagnosis simpler. X-ray examinations may be performed to look for areas of diffuse bone demineralization, bone cysts, outer bone absorption and erosion of the long bones of the fingers and toes. Hypercalcemia is mild or intermittent in some patients, but is an excellent indicator of primary hyperparathyroidism. Dual energy x-ray absorptiometry (DEXA or DXA), a tool used to diagnose and measure osteoporosis, is used to show reduction in bone mass for primary hyperparathyroidism patients. Once a diagnosis of hyperparathyroidism is reached, the physician will probably order further tests to evaluate complications. For example, abdominal radiographs might reveal kidney stones.

For secondary hyperparathyroidism, normal or slightly decreased calcium levels in the blood and variable phosphorous levels may be visible. Patient history of familial **kidney disease** or convulsive disorders may suggest a diagnosis of secondary hyperparathyroidism. Other tests may reveal a disease or disorder, which is causing the secondary hyperparathyroidism.

Treatment

Hyperparathyroidism cases will usually be referred to an endocrinologist, a physician specializing in hormonal problems, or a nephrologist, who specializes in kidney and mineral disorders.

Patients with mild cases of hyperparathyroidism may not need immediate treatment if they have only slight elevations in blood calcium level and normal kidneys and bones. These patients should be regularly checked, probably as often as every six months, by **physical examination** and measurement of kidney function and calcium levels. A bone densitometry measurement should be performed every one or two years. After several years with no worsened symptoms, the length of time between exams may be increased.

KEY TERMS

Demineralization—A loss or decrease of minerals in the bones.

Endocrine—Glands and hormone secretions in the body circulation.

Phosphate—An organic compound necessary for mineralization of bone and other key cellular processes.

Patients with more advanced hyperparathyroidism will usually have all or half of the affected parathyroid gland or glands surgically removed. This surgery is relatively safe and effective. The primary risks are those associated with **general anesthesia**. There are some instances when the surgery can be performed with the patient under regional, or cervical block, anesthesia. Often studies such as ultrasonography prior to surgery help pinpoint the affected areas.

Alternative treatment

Forcing fluids and reducing intake of calcium-rich foods can help decrease calcium levels prior to surgery or if surgery is not necessary.

Prognosis

Removal of the enlarged parathyroid gland or glands cures the disease 95% of the time and relief of bone pain may occur in as few as three days. In up to 5% of patients undergoing surgery, chronically low calcium levels may result, and these patients will require calcium supplement or vitamin D treatment. Damage to the kidneys as a result of hyperparathyroidism is often irreversible. Prognosis is generally good, however complications of hyperparathyroidism such as osteoporosis, bone **fractures**, kidney stones, peptic ulcers, pancreatitis, and nervous system difficulties may worsen prognosis.

Prevention

Secondary hyperparathyroidism may be prevented by early treatment of the disease causing it. Early recognition and treatment of hyperparathyroidism may prevent hypercalcemia. Since the cause of primary hyperparathyroidism, or the adenoma which causes parathyroid enlargement, is largely unknown, there are not prescribed prevention methods.

Resources

ORGANIZATIONS

Osteoporosis and Related Bone Diseases—National Resource Center. 1150 17th S. NW, Ste. 500, Washington, DC 20036. (800) 624-2663.

Paget Foundation. 200 Varick St., Ste. 1004. New York, NY 10014-4810. (800) 23-PAGET.

OTHER

“Endocrine Disorder and Endocrine Surgery.” Endocrine Web Page. <<http://www.endocrineweb.com>>.

Teresa Odle

Hyperpigmentation

Definition

Hyperpigmentation is the increase in the natural color of the skin.

Description

Melanin, a brown pigment manufactured by certain cells in the skin called melanocytes, is responsible for skin color. Melanin production is stimulated by a pituitary hormone called melanocyte stimulating hormone (MSH). Other pigments appear in the skin much less often.

Causes and symptoms

Darkened spots on the skin come in several varieties. The most ominous is **malignant melanoma**, a very aggressive cancer that begins as an innocent mole. The majority of **moles** (nevus), however, are and remain benign (harmless). The average person has several dozen, and certain people with a hereditary excess may have hundreds. Freckles, age spots, and cafe au lait spots, known as ephelides, are always flat and not as dark. Cafe au lait spots are seen mostly in people with another hereditary disorder called neurofibromatosis. “Port wine stains” are congenital dark red blotches on the skin. Other common dark colorations on the skin are called keratosis and consist of locally overgrown layers of skin that are dark primarily because there is more tissue than normal. A few of these turn into skin cancers of a much less dangerous kind than melanoma.

Darkened regions of the skin occur as a result of abnormal tanning when the skin is sensitive to

sunlight. Several diseases and many drugs can cause **photosensitivity**. Among the common drugs responsible for this uncommon reaction are birth control pills, **antibiotics** (**sulfonamides** and **tetracyclines**), **diuretics**, nonsteroidal anti-inflammatory drugs (NSAID), **pain** relievers, and some psychoactive medications. Some of the same drugs may also cause patches of discolored skin known as localized drug reactions and representing an allergy to that drug. Sunlight darkens an abnormal chemical in the skin of patients with porphyria cutanea tarda. Several endocrine diseases, some cancers, and several drugs abnormally stimulate melanocytes, usually through an overproduction of MSH. Arsenic **poisoning** and Addison’s disease are among these causes. A condition known as acanthosis nigricans is a velvety darkening of skin in folded areas (arm pits, groin, and neck) that can signal a **cancer** or hormone imbalance.

Of particular note is a condition called melasma (dark pigmentation of the skin), caused by the female hormone estrogen. Normal in **pregnancy**, this brownish discoloration of the face can also happen with birth control pills that contain estrogen.

Overall darkening of the skin may be due to pigmented chemicals in the skin. Silver, gold, and iron each have a characteristic color when visible in the skin. Several drugs and body chemicals, like bilirubin, can end up as deposits in the skin and discolor it.

There are a number of other rare entities that color the skin, each in its own peculiar way. Among these are strange syndromes that seem to be birth defects and vitamin and nutritional deficiencies.

Diagnosis

The pattern of discoloration is immediately visible to the trained dermatologist, a physician specializing in skin diseases, and may be all that is required to name and characterize the discoloration. Many of these pigment changes are signs of internal disease that must be identified. Pigmentation changes may also be caused by medication, and the drug responsible for the reaction must be identified and removed.

Treatment

Skin sensitive to sunlight must be protected by shade or **sunscreens** with an SPF of 15 or greater. Skin cancers must be, and unsightly benign lesions may be, surgically removed. **Laser surgery** is an effective removal technique for many localized lesions.

KEY TERMS

Addison's disease—A degenerative disease that is characterized by weight loss, low blood pressure, extreme weakness, and dark brown pigmentation of the skin.

Dermatologist—A physician specializing in the study of skin conditions and diseases

Diuretic—A cause of increased urine flow.

Keratosis—A skin disease characterized by an overgrowth of skin, which usually appears discolored.

Lesion—Any localized abnormality.

Melasma—Dark pigmentation of the skin.

Neurofibromatosis—Otherwise known as von Recklinghausen's disease, consists of pigmented skin spots and numerous soft tumors all over the body.

Nevus—Birthmark or mole.

NSAID—Nonsteroidal anti-inflammatory drugs— aspirin, ibuprofen, naproxen, and many others.

Porphyria cutanea tarda—An inherited disease that results in the overproduction of porphyrins.

Syndrome—Common features of a disease or features that appear together often enough to suggest they may represent a single, as yet unknown, disease entity.

Because it spreads so rapidly, melanoma should be immediately removed, as well as some of the surrounding tissue to prevent regrowth.

Prevention

Sunlight is the leading cause of dark spots on the skin, so shade and sunscreens are necessary preventive strategies, especially in people who burn easily.

Resources

PERIODICALS

Bernstein L. J., et al. "The Short- and Long-term Side Effects of Carbon Dioxide Laser Resurfacing." *Dermatologic Surgery* 23 (July 1997): 519-525.

J. Ricker Polsdorfer, MD

Hyperprolactation see **Galactorrhea**

Hypersensitivity pneumonitis

Definition

Hypersensitivity pneumonitis refers to an inflammation of the lungs caused by repeated breathing in of a foreign substance, such as an organic dust, a fungus, or a mold. The body's immune system reacts to these substances, called antigens, by forming antibodies, molecules that attack the invading antigen and try to destroy it. The combination of antigen and antibody produces acute inflammation, or pneumonitis (a hypersensitivity reaction), which later can develop into chronic lung disease that impairs the lungs' ability to take oxygen from the air and eliminate carbon dioxide.

Description

Hypersensitivity pneumonitis (HP) is sometimes called "allergic alveolitis." "Allergic" refers to the antigen-antibody reaction, and "alveolitis" means an inflammation of the tiny air sacs in the lungs where oxygen and CO₂ are exchanged, the alveoli. It also is known as "extrinsic" allergic alveolitis, meaning that the antigen that sets up the allergic reaction (also called an allergen) comes from the outside. Most of the antigens that cause this disease come from plant or animal proteins or microorganisms, and many of those affected are exposed either at work or in the course of some hobby or other activity. The first known type of HP, farmer's lung, is caused by antigens from tiny microorganisms living on moldy hay. An example of disease connected with a hobby is pigeon breeder's lung, caused by inhaling protein material from bird droppings or feathers. After a time, very little of the allergenic material is needed to set off a reaction in the lungs.

Roughly one in every 10,000 persons develops some form of HP. A mysterious aspect of this condition is that, even though many persons may be exposed to a particular antigen, only a small number of them will develop the disease. Genetic differences may determine who becomes ill; this remains unclear. Probably between 5% and 15% of all persons who are regularly exposed to organic materials develop HP. Most of those who do get it do *not* smoke (**smoking** may create the type of cells that take up antigens and neutralize them). The amount of antigen is an important factor in whether HP will develop and what form it will take. Sudden heavy exposure can produce symptoms in a matter of hours, whereas mild but frequent exposures tend to produce a long-lasting, "smoldering" illness. HP may be more likely

to develop in persons exposed to polluted air or industrial fumes.

Typical changes occur in the lungs of persons with HP. In the acute stage, large numbers of inflammatory cells are found throughout the lungs and the air sacs may be filled by a thick fluid mixed with these cells. In the subacute stage, disease extends into the small breathing tubes, or bronchioles, and the inflammatory cells collect into tiny granules called granulomas. Finally, in the chronic stage of HP, the previously inflamed parts of the lungs become scarred and unable to function, as in **pulmonary fibrosis**.

Causes and symptoms

A number of different types of HP are known, since a wide range of allergens may produce an allergic reaction in the lungs. Many of them produce similar symptoms and abnormal physical findings, but some have their own typical features. Some of the more common forms are:

- **Farmer's lung.** Can affect any farmer who works with wet hay or other moldy dust. Small farmers who have to directly thresh and handle their hay are most at risk, as are those living in cold and humid areas where damp weather is common.
- **Pigeon breeder's lung.** Also called "bird fancier's lung," it is second to farmer's lung as the best known type of HP. A substance has been found in pigeon droppings that may cause the allergic reaction, but there may be more than one such substance. Besides pigeons, the disorder may follow exposure to ducks, geese, pheasants, and even canaries. Parakeets produce an especially severe form of disease. Most patients are middle-aged women, who usually care for birds either at home or on bird breeding farms.
- **Bagassosis.** Caused by bagasse, a substance produced when juice is extracted from sugar cane and is used in making paper and explosives. A fungus is probably responsible. Young and middle-aged men who work in the sugar industry are at risk.
- **Byssinosis.** A similar condition affecting workers who inhale dust from cotton, flax, or hemp.
- **Humidifier lung.** An acute form of HP caused by inhaling actinomycetes, the same organisms that cause farmer's lung, which grow in contaminated humidifier vents, air conditioners, heating systems, and even saunas.
- **Other antigens.** HP has been seen in persons working with detergents, silicone, mushrooms, cheese, wood dust, maple bark, coffee, and furs.

In the acute stage, patients with HP begin coughing, develop **fever**, and note tightness in the chest as well as extreme tiredness and aching, four to eight hours after the most recent exposure. Most patients are well aware of the connection between their work (or an activity) and their symptoms. After a time, patients may have trouble breathing. They also may lose their appetite, lose weight, and generally feel ill. Finally, in the chronic stage, the patient will have increasing trouble breathing and may sometimes wheeze. With advanced disease, the skin may appear bluish (because too little oxygen is getting into the blood). When the physician listens to the patient's chest with a stethoscope, there may be crackling sounds or loud **wheezing**. In the late stages, club-shaped fingertips are a sign that the patient has not been getting enough oxygen for an extended period of time.

Diagnosis

No single test can make a definite diagnosis of HP. The key is to relate some specific exposure or activity to episodes of symptoms. The **chest x ray** may be normal in the acute stage, but later may show a hazy appearance that looks like "ground glass." There may be linear or rounded shadows in the central parts of the lungs. Studies of lung function in the acute stage typically show abnormally small lung volume. The ability to breathe at a fast rate is impaired. Blood from an artery typically has a low level of oxygen. Later, when the lungs have begun to scar, the airways (breathing tubes) are obstructed and the rate of air flow is reduced.

Some experts believe that skin testing can help diagnose HP and show which particular antigen is causing the symptoms. Small amounts of several suspect antigens are injected just beneath the surface of the skin, usually on the arm or back, and the reactions compared to that caused by injecting a harmless salt solution. Another diagnostic test is to place a thin tube into the airways, inject a small amount of fluid, and draw it back up (bronchoalveolar lavage). A very large number of cells called lymphocytes is typical of HP, and mast cells, which are part of the immune system, may also be seen. Rarely, a tissue sample (biopsy) of lung tissue may be taken through a tube placed in the airways and examined under a microscope. Finally, a patient may be "challenged" by actually inhaling a particular antigen in the form of an aerosol and noting whether lung function suddenly becomes worse. This test is usually not necessary.

KEY TERMS

Allergen—An outside substance, such as dust or a mold, that, when inhaled, sets off an allergic (hypersensitivity) reaction in the lungs.

Fibrosis—A result of long-standing inflammatory disease in which normal tissue is replaced by scar tissue that is functionally useless.

Granuloma—A collection of inflammatory cells forming a microscopic lesion, many of which are scattered throughout the lung tissue in patients who have had numerous acute episodes of HP.

Hypersensitivity—After the body's immune system attacks an outside invader (such as organic dust or a fungus) many times, exposure to even a tiny amount of this allergen can provoke a strong inflammatory response.

Pneumonitis—Inflammation of the lung tissues.

Steroid—A natural body substance may be given orally or by injection, and serves to dampen or even halt inflammation anywhere in the body, including the lungs.

Treatment

Treatment of HP requires identifying the offending antigen and avoiding further exposure. Although it may sometimes be necessary for a patient to find a totally different type of work, often it is possible to simply perform different duties or switch to a work site where exposure is minimal. In some cases, (like pigeon breeder's lung), wearing a mask can prevent exposure. If acute symptoms are severe, the patient may be treated with a steroid hormone for two to six weeks. This often suppresses the inflammatory response and allows the lungs a chance to recover. In the chronic stage, steroid treatment can delay further damage to the lungs and help preserve their function.

Prognosis

In general, most of the symptoms of HP disappear when the patient is no longer exposed to the causative allergen. The actual chances of complete recovery depend in part on what form of HP is present. Older patients and those exposed repeatedly for long periods after initially developing symptoms tend to have a poorer long-term outlook. The worst outcome is that long repeated episodes of exposure will cause chronic lung inflammation, scar the lungs, and permanently make

then unable to properly provide oxygen to the blood. Rarely, a patient will become permanently disabled.

Prevention

It is often not possible to prevent initial episodes of HP, because there is no way of predicting which individuals (such as farmers) will have an allergic reaction to a particular allergen. Once the connection is made between a type of exposure and definite hypersensitivity symptoms, prevention of further episodes is simple as long as further exposure can be avoided.

Exactly how to avoid exposure depends on a person's work or activities and what he or she is reacting to. People with farmer's lung can dry hay thoroughly before storing it. For pigeon breeder's lung (and many other types of HP), a mask can be worn. In many industrial settings, it is possible to take precautions that will limit the amount of allergen that workers will inhale. If it is not possible to avoid exposure altogether, exposure can be timed and strictly minimized.

Resources

ORGANIZATIONS

American Lung Association. 1740 Broadway, New York, NY 10019. (800) 586-4872. <<http://www.lungusa.org>>.

Asthma and Allergy Foundation of America. 1233 20th Street, NW, Suite 402, Washington, DC 20036. (800) 727-8462. <<http://www.aafa.org>>.

David A. Cramer, MD

Hypersomnia see **Sleep disorders**

Hypersplenism

Definition

Hypersplenism is a type of disorder which causes the spleen to rapidly and prematurely destroy blood cells.

Description

The spleen is located in the upper left area of the abdomen. One of this organ's major functions is to remove blood cells from the body's bloodstream. In hypersplenism, its normal function accelerates, and it begins to automatically remove cells that may still be normal in function. Sometimes, the spleen will temporarily hold onto up to 90% of the body's platelets and 45% of the red blood cells. Hypersplenism may

occur as a primary disease, leading to other complications, or as a secondary disease, resulting from an underlying disease or disorder. Hypersplenism is sometimes referred to as enlarged spleen (splenomegaly). An enlarged spleen is one of the symptoms of hypersplenism. What differentiates hypersplenism is its premature destruction of blood cells.

Causes and symptoms

Hypersplenism may be caused by a variety of disorders. Sometimes, it is brought on by a problem within the spleen itself and is referred to as primary hypersplenism. Secondary hypersplenism results from another disease such as chronic **malaria**, **rheumatoid arthritis**, **tuberculosis**, or **polycythemia vera**, a blood disorder. Spleen disorders in general are almost always secondary in nature. Hypersplenism may also be caused by tumors.

Symptoms of hypersplenism include easy bruising, easy contracting of bacterial diseases, fever, weakness, heart palpitations, and ulcerations of the mouth, legs and feet. Individuals may also bleed unexpectedly and heavily from the nose or other mucous membranes, and from the gastrointestinal or urinary tracts. Most patients will develop an enlarged spleen, anemia, leukopenia, or abnormally low white blood cell counts, or **thrombocytopenia**, a deficiency of circulating platelets in the blood. Other symptoms may be present that reflect the underlying disease that has caused hypersplenism.

An enlarged spleen can be caused by a variety of diseases, including **hemolytic anemia**, liver **cirrhosis**, leukemia, malignant lymphoma and other infections and inflammatory diseases. Splenomegaly occurs in about 10% of **systemic lupus erythematosus** patients. Sometimes, it is caused by recent viral infection, such as mononucleosis. An enlarged spleen may cause pain in the upper left side of the abdomen and a premature feeling of fullness at meals.

Diagnosis

Diagnosis of hypersplenism begins with review of symptoms and patient history, and careful feeling (palpation) of the spleen. Sometimes, a physician can feel an enlarged spleen. X-ray studies, such as ultrasound and computed tomography scan (CT scan), may help diagnose an enlarged spleen and possible underlying causes, such as tumors. Blood tests indicate decreases in white blood cells, red blood cells, or platelets. Another test measures red blood cells in the liver and spleen after injection of a radioactive substance,

KEY TERMS

Cirrhosis—Hardening of an organ, usually the liver. Cirrhosis of the liver is a progressive disease which leads to destruction of liver cells, interference with blood flow in the liver, and interference with the function of the liver.

Palpitations—Throbbing or pulsation. Heart palpitations usually infer an irregular or rapid rhythm.

Polycythemia vera—A chronic disorder characterized by increased red blood cell mass and other malfunctions of the blood system. It most commonly occurs in males of Jewish ancestry between the ages of 40 and 60.

Systemic—Relating to a system, or especially the entire system.

Systemic lupus erythematosus—A connective tissue disease that results in fever, weakness, fatigue, joint pain and arthritis.

Ulcerations—Breaks in skin or mucous membranes that are often accompanied by loss of tissue on the surface.

and indicates areas where the spleen is holding on to large numbers of red cells or is destroying them.

Enlarged spleens are diagnosed using a combination of patient history, **physical examination**, including palpation of the spleen, if possible, and diagnostic tests. A history of **fever** and systemic symptoms may be present because of infection, malaria, or an inflammatory disorder. A complete **blood count** is taken to check counts of young red blood cells. Liver function tests, CT scans, and ultrasound exams can also help to detect an enlarged spleen.

Treatment

In secondary hypersplenism, the underlying disease must be treated to prevent further sequestration or destruction of blood cells, and possible spleen enlargement. Those therapies will be tried prior to removal of the spleen (**splenectomy**), which is avoided if possible. In severe cases, the spleen must be removed. Splenectomy will correct the effects of low blood cell concentrations in the blood.

Prognosis

Prognosis depends on the underlying cause and progression of the disease. Left untreated, spleen

enlargement can lead to serious complications. Hypersplenism can also lead to complications due to decreased blood cell counts.

Prevention

Some of the underlying causes of hypersplenism or enlarged spleen can be prevented, such as certain forms of anemia and cirrhosis of the liver due to alcohol. In other cases, the hypersplenism may not be preventable, as it is a complication to an underlying disorder.

Resources

ORGANIZATIONS

American Liver Foundation. 1425 Pompton Ave., Cedar Grove, NJ 07009. (800) 223-0179. <<http://www.liverfoundation.org>>.

American Society of Hematology. 1200 19th Street NW, Suite 300, Washington, DC 20036-2422. (202) 857-1118. <<http://www.hematology.org>>.

National Heart, Lung and Blood Institute. PO Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

Teresa Norris, RN

Hypertension

Definition

Hypertension is high blood pressure. Blood pressure is the force of blood pushing against the walls of arteries as it flows through them. Arteries are the blood vessels that carry oxygenated blood from the heart to the body's tissues.

Description

As blood flows through arteries it pushes against the inside of the artery walls. The more pressure the blood exerts on the artery walls, the higher the blood pressure will be. The size of small arteries also affects the blood pressure. When the muscular walls of arteries are relaxed, or dilated, the pressure of the blood flowing through them is lower than when the artery walls narrow, or constrict.

Blood pressure is highest when the heart beats to push blood out into the arteries. When the heart relaxes to fill with blood again, the pressure is at its lowest point. Blood pressure when the heart beats is

called systolic pressure. Blood pressure when the heart is at rest is called diastolic pressure. When blood pressure is measured, the systolic pressure is stated first and the diastolic pressure second. Blood pressure is measured in millimeters of mercury (mm Hg). For example, if a person's systolic pressure is 120 and diastolic pressure is 80, it is written as 120/80 mm Hg. The American Heart Association has long considered blood pressure less than 140 over 90 normal for adults. However, the National Heart, Lung, and Blood Institute in Bethesda, Maryland released new clinical guidelines for blood pressure in 2003, lowering the standard normal readings. A normal reading was lowered to less than 120 over less than 80.

Hypertension is a major health problem, especially because it has no symptoms. Many people have hypertension without knowing it. In the United States, about 50 million people age six and older have high blood pressure. Hypertension is more common in men than women and in people over the age of 65 than in younger persons. More than half of all Americans over the age of 65 have hypertension. It also is more common in African-Americans than in white Americans.

Hypertension is serious because people with the condition have a higher risk for heart disease and other medical problems than people with normal blood pressure. Serious complications can be avoided by getting regular blood pressure checks and treating hypertension as soon as it is diagnosed.

If left untreated, hypertension can lead to the following medical conditions:

- arteriosclerosis, also called **atherosclerosis**
- **heart attack**
- **stroke**
- enlarged heart
- kidney damage.

Arteriosclerosis is hardening of the arteries. The walls of arteries have a layer of muscle and elastic tissue that makes them flexible and able to dilate and constrict as blood flows through them. High blood pressure can make the artery walls thicken and harden. When artery walls thicken, the inside of the blood vessel narrows. Cholesterol and fats are more likely to build up on the walls of damaged arteries, making them even narrower. **Blood clots** also can get trapped in narrowed arteries, blocking the flow of blood.

Arteries narrowed by arteriosclerosis may not deliver enough blood to organs and other tissues.

Reduced or blocked blood flow to the heart can cause a heart attack. If an artery to the brain is blocked, a stroke can result.

Hypertension makes the heart work harder to pump blood through the body. The extra workload can make the heart muscle thicken and stretch. When the heart becomes too enlarged it cannot pump enough blood. If the hypertension is not treated, the heart may fail.

The kidneys remove the body's wastes from the blood. If hypertension thickens the arteries to the kidneys, less waste can be filtered from the blood. As the condition worsens, the kidneys fail and wastes build up in the blood. Dialysis or a kidney transplant are needed when the kidneys fail. About 25% of people who receive **kidney dialysis** have kidney failure caused by hypertension.

Causes and symptoms

Many different actions or situations can normally raise blood pressure. Physical activity can temporarily raise blood pressure. Stressful situations can make blood pressure go up. When the **stress** goes away, blood pressure usually returns to normal. These temporary increases in blood pressure are not considered hypertension. A diagnosis of hypertension is made only when a person has multiple high blood pressure readings over a period of time.

The cause of hypertension is not known in 90 to 95 percent of the people who have it. Hypertension without a known cause is called primary or essential hypertension.

When a person has hypertension caused by another medical condition, it is called secondary hypertension. Secondary hypertension can be caused by a number of different illnesses. Many people with kidney disorders have secondary hypertension. The kidneys regulate the balance of salt and water in the body. If the kidneys cannot rid the body of excess salt and water, blood pressure goes up. Kidney infections, a narrowing of the arteries that carry blood to the kidneys, called **renal artery stenosis**, and other kidney disorders can disturb the salt and water balance.

Cushing's syndrome and tumors of the pituitary and adrenal glands often increase levels of the adrenal gland hormones cortisol, adrenalin, and aldosterone, which can cause hypertension. Other conditions that can cause hypertension are blood vessel diseases, thyroid gland disorders, some prescribed drugs, **alcoholism**, and **pregnancy**.

Even though the cause of most hypertension is not known, some people have risk factors that give them a greater chance of getting hypertension. Many of these risk factors can be changed to lower the chance of developing hypertension or as part of a treatment program to lower blood pressure.

Risk factors for hypertension include:

- age over 60
- male sex
- race
- heredity
- salt sensitivity
- obesity
- inactive lifestyle
- heavy alcohol consumption
- use of **oral contraceptives**

Some risk factors for getting hypertension can be changed, while others cannot. Age, male sex, and race are risk factors that a person can't do anything about. Some people inherit a tendency to get hypertension. People with family members who have hypertension are more likely to develop it than those whose relatives are not hypertensive. People with these risk factors can avoid or eliminate the other risk factors to lower their chance of developing hypertension. A 2003 report found that the rise in incidence of high blood pressure among children is most likely due to an increase in the number of overweight and obese children and adolescents.

Diagnosis

Because hypertension doesn't cause symptoms, it is important to have blood pressure checked regularly. Blood pressure is measured with an instrument called a sphygmomanometer. A cloth-covered rubber cuff is wrapped around the upper arm and inflated. When the cuff is inflated, an artery in the arm is squeezed to momentarily stop the flow of blood. Then, the air is let out of the cuff while a stethoscope placed over the artery is used to detect the sound of the blood spurting back through the artery. This first sound is the systolic pressure, the pressure when the heart beats. The last sound heard as the rest of the air is released is the diastolic pressure, the pressure between heart beats. Both sounds are recorded on the mercury gauge on the sphygmomanometer.

Normal blood pressure is defined by a range of values. Blood pressure lower than 120/80 mm Hg is considered normal. A number of factors such as **pain**,

stress or anxiety can cause a temporary increase in blood pressure. For this reason, hypertension is not diagnosed on one high blood pressure reading. If a blood pressure reading is 120/80 or higher for the first time, the physician will have the person return for another blood pressure check. Diagnosis of hypertension usually is made based on two or more readings after the first visit.

Systolic hypertension of the elderly is common and is diagnosed when the diastolic pressure is normal or low, but the systolic is elevated, e.g. 170/70 mm Hg. This condition usually co-exists with hardening of the arteries (atherosclerosis).

Blood pressure measurements are classified in stages, according to severity:

- normal blood pressure: less than less than 120/80 mm Hg
- pre-hypertension: 120–129/80–89 mm Hg
- Stage 1 hypertension: 140–159/90–99 mm Hg
- Stage 2 hypertension: at or greater than 160–179/100–109 mm Hg

A typical **physical examination** to evaluate hypertension includes:

- medical and family history
- physical examination
- ophthalmoscopy: Examination of the blood vessels in the eye
- chest x ray
- electrocardiograph (ECG)
- blood and urine tests.

The medical and family history help the physician determine if the patient has any conditions or disorders that might contribute to or cause the hypertension. A family history of hypertension might suggest a genetic predisposition for hypertension.

The physical exam may include several blood pressure readings at different times and in different positions. The physician uses a stethoscope to listen to sounds made by the heart and blood flowing through the arteries. The pulse, reflexes, and height and weight are checked and recorded. Internal organs are palpated, or felt, to determine if they are enlarged.

Because hypertension can cause damage to the blood vessels in the eyes, the eyes may be checked with a instrument called an ophthalmoscope. The physician will look for thickening, narrowing, or hemorrhages in the blood vessels.



The effects of hypertension on the heart and kidney. Hypertension has caused renal atrophy and scarring, and left ventricular hypertrophy in the sectioned heart (at right). (Photograph by Dr. E. Walker, Photo Researchers, Inc. Reproduced by permission.)

A chest x ray can detect an enlarged heart, other vascular (heart) abnormalities, or lung disease.

An electrocardiogram (ECG) measures the electrical activity of the heart. It can detect if the heart muscle is enlarged and if there is damage to the heart muscle from blocked arteries.

Urine and blood tests may be done to evaluate health and to detect the presence of disorders that might cause hypertension.

Treatment

There is no cure for primary hypertension, but blood pressure can almost always be lowered with the correct treatment. The goal of treatment is to lower blood pressure to levels that will prevent heart disease and other complications of hypertension. In secondary hypertension, the disease that is responsible for the hypertension is treated in addition to the hypertension itself. Successful treatment of the underlying disorder may cure the secondary hypertension.

Guidelines advise that clinicians work with patients to agree on blood pressure goals and develop a treatment plan for the individual patient. Actual combinations of medications and lifestyle changes will vary from one person to the next. Treatment to lower blood pressure may include changes in diet, getting regular exercise, and taking antihypertensive medications. Patients falling into the pre-hypertension range who don't have damage to the heart or kidneys often are advised to make needed lifestyle changes only. A 2003 report of a clinical trial showed that adults with elevated blood pressures lowered them

as much as 38% by making lifestyle changes and participating in the DASH diet, which encourages eating more fruit and vegetables.

Lifestyle changes that may reduce blood pressure by about 5 to 10 mm Hg include:

- reducing salt intake
- reducing fat intake
- losing weight
- getting regular **exercise**
- quitting **smoking**
- reducing alcohol consumption
- managing stress

Patients whose blood pressure falls into the Stage 1 hypertension range may be advised to take antihypertensive medication. Numerous drugs have been developed to treat hypertension. The choice of medication will depend on the stage of hypertension, side effects, other medical conditions the patient may have, and other medicines the patient is taking.

If treatment with a single medicine fails to lower blood pressure enough, a different medicine may be tried or another medicine may be added to the first. Patients with more severe hypertension may initially be given a combination of medicines to control their hypertension. Combining antihypertensive medicines with different types of action often controls blood pressure with smaller doses of each drug than would be needed for just one.

Antihypertensive medicines fall into several classes of drugs:

- **diuretics**
- beta-blockers
- **calcium channel blockers**
- angiotensin converting enzyme inhibitors (ACE inhibitors)
- alpha-blockers
- alpha-beta blockers
- **vasodilators**
- peripheral acting adrenergic antagonists
- centrally acting agonists

Diuretics help the kidneys eliminate excess salt and water from the body's tissues and the blood. This helps reduce the swelling caused by fluid buildup in the tissues. The reduction of fluid dilates the walls of arteries and lowers blood pressure. New guidelines released in 2003 suggest diuretics as the first drug of

choice for most patients with high blood pressure and as part of any multi-drug combination.

Beta-blockers lower blood pressure by acting on the nervous system to slow the heart rate and reduce the force of the heart's contraction. They are used with caution in patients with heart failure, **asthma**, diabetes, or circulation problems in the hands and feet.

Calcium channel blockers block the entry of calcium into muscle cells in artery walls. Muscle cells need calcium to constrict, so reducing their calcium keeps them more relaxed and lowers blood pressure.

ACE inhibitors block the production of substances that constrict blood vessels. They also help reduce the build-up of water and salt in the tissues. They often are given to patients with heart failure, **kidney disease**, or diabetes. ACE inhibitors may be used together with diuretics.

Alpha-blockers act on the nervous system to dilate arteries and reduce the force of the heart's contractions.

Alpha-beta blockers combine the actions of alpha and beta blockers.

Vasodilators act directly on arteries to relax their walls so blood can move more easily through them. They lower blood pressure rapidly and are injected in hypertensive emergencies when patients have dangerously high blood pressure.

Peripheral acting adrenergic antagonists act on the nervous system to relax arteries and reduce the force of the heart's contractions. They usually are prescribed together with a diuretic. Peripheral acting adrenergic antagonists can cause slowed mental function and lethargy.

Centrally acting agonists also act on the nervous system to relax arteries and slow the heart rate. They are usually used with other antihypertensive medicines.

Prognosis

There is no cure for hypertension. However, it can be well controlled with the proper treatment. Therapy with a combination of lifestyle changes and antihypertensive medicines usually can keep blood pressure at levels that will not cause damage to the heart or other organs. The key to avoiding serious complications of hypertension is to detect and treat it before damage occurs. Because antihypertensive medicines control blood pressure, but do not cure it, patients must continue taking the medications to maintain reduced blood pressure levels and avoid complications.

KEY TERMS

Arteries—Blood vessels that carry blood to organs and other tissues of the body.

Arteriosclerosis—Hardening and thickening of artery walls.

Cushing's syndrome—A disorder in which too much of the adrenal hormone, cortisol, is produced; it may be caused by a pituitary or adrenal gland tumor.

Diastolic blood pressure—Blood pressure when the heart is resting between beats.

Hypertension—High blood pressure.

Renal artery stenosis—Disorder in which the arteries that supply blood to the kidneys constrict.

Sphygmomanometer—An instrument used to measure blood pressure.

Systolic blood pressure—Blood pressure when the heart contracts (beats).

Vasodilator—Any drug that relaxes blood vessel walls.

Ventricle—One of the two lower chambers of the heart.

Prevention

Prevention of hypertension centers on avoiding or eliminating known risk factors. Even persons at risk because of age, race, or sex or those who have an inherited risk can lower their chance of developing hypertension.

The risk of developing hypertension can be reduced by making the same changes recommended for treating hypertension:

- reducing salt intake
- reducing fat intake
- losing weight
- getting regular exercise
- quitting smoking
- reducing alcohol consumption
- managing stress

Resources

PERIODICALS

McNamara, Damian. "Obesity Behind Rise in Incidence of Primary Hypertension." *Family Practice News* April 1, 2003: 45-51.

McNamara, Damian. "Trial Shows Efficacy of Lifestyle Changes for BP: More Intensive Than Typical Office Visit." *Family Practice News* July 1, 2003: 1-2.

"New BP Guidelines Establish Diagnosis of Pre-hypertension: Level Seeks to Identify At-risk Individuals Early." *Case Management Advisor* July 2003: S1.

"New Hypertension Guidelines: JNC-7." *Clinical Cardiology Alert* July 2003: 54-63.

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

National Heart, Lung and Blood Institute. PO Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

Texas Heart Institute. Heart Information Service. PO Box 20345, Houston, TX 77225-0345. <<http://www.tmc.edu/thi>>.

Toni Rizzo
Teresa G. Odle

Hyperthermia see **Fever**

Hyperthyroidism

Definition

Hyperthyroidism is the overproduction of thyroid hormones by an overactive thyroid.

Description

Located in the front of the neck, the thyroid gland produces the hormones thyroxine (T₄) and triiodothyronine (T₃) that regulate the body's metabolic rate by helping to form protein ribonucleic acid (RNA) and increasing oxygen absorption in every cell. In turn, the production of these hormones are controlled by thyroid-stimulating hormone (TSH) that is produced by the pituitary gland. When production of the thyroid hormones increases despite the level of TSH being produced, hyperthyroidism occurs. The excessive amount of **thyroid hormones** in the blood increases the body's metabolism, creating both mental and physical symptoms.

The term hyperthyroidism covers any disease which results in overabundance of thyroid hormone. Other names for hyperthyroidism, or specific diseases within the category, include Graves' disease, diffuse toxic **goiter**, Basedow's disease, Parry's disease, and thyrotoxicosis. The disease is 10 times more common

in women than in men, and the annual incidence of hyperthyroidism in the United States is about one per 1,000 women. Although it occurs at all ages, hyperthyroidism is most likely to occur after the age of 15. There is a form of hyperthyroidism called Neonatal Grave's disease, which occurs in infants born of mothers with Graves' disease. Occult hyperthyroidism may occur in patients over 65 and is characterized by a distinct lack of typical symptoms. Diffuse toxic goiter occurs in as many as 80% of patients with hyperthyroidism.

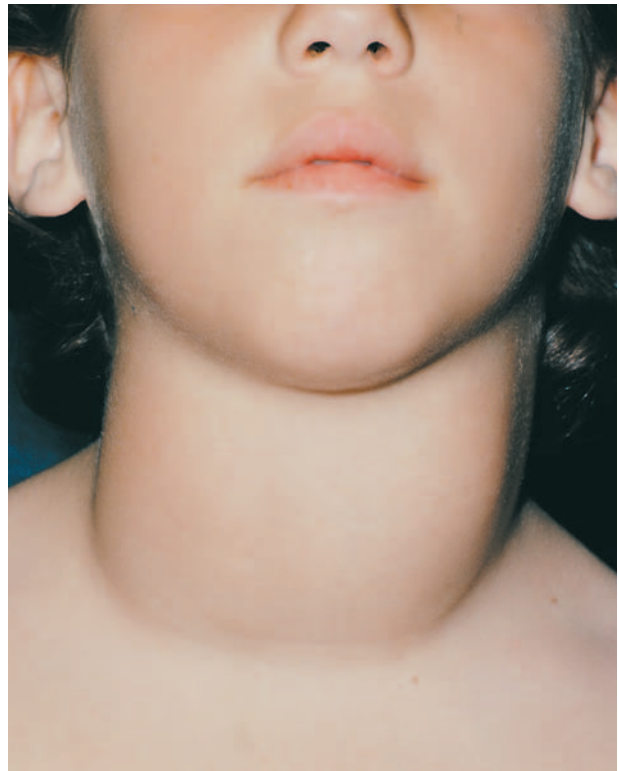
Causes and symptoms

Hyperthyroidism is often associated with the body's production of autoantibodies in the blood which cause the thyroid to grow and secrete excess thyroid hormone. This condition, as well as other forms of hyperthyroidism, may be inherited. Regardless of the cause, hyperthyroidism produces the same symptoms, including weight loss with increased appetite, **shortness of breath** and **fatigue**, intolerance to heat, heart **palpitations**, increased frequency of bowel movements, weak muscles, **tremors**, **anxiety**, and difficulty sleeping. Women may also notice decreased menstrual flow and irregular menstrual cycles.

Patients with Graves' disease often have a goiter (visible enlargement of the thyroid gland), although as many as 10% do not. These patients may also have bulging eyes. Thyroid storm, a serious form of hyperthyroidism, may show up as sudden and acute symptoms, some of which mimic typical hyperthyroidism, as well as the addition of fever, substantial weakness, extreme restlessness, confusion, emotional swings or **psychosis**, and perhaps even **coma**.

Diagnosis

Physicians will look for physical signs and symptoms indicated by patient history. On inspection, the physician may note symptoms such as a goiter or eye bulging. Other symptoms or family history may be clues to a diagnosis of hyperthyroidism. An elevated body temperature (basal body temperature) above 98.6°F (37°C) may be an indication of a heightened metabolic rate (basal metabolic rate) and hyperthyroidism. A simple blood test can be performed to determine the amount of thyroid hormone in the patient's blood. The diagnosis is usually straightforward with this combination of clinical history, physical examination, and routine blood hormone tests. Radioimmunoassay, or a test to show concentrations of thyroid hormones with the use of a radioisotope



A symptom of hyperthyroidism is the enlargement of the thyroid gland. (Photograph by Lester V. Bergman, Corbis Images. Reproduced by permission.)

mixed with fluid samples, helps confirm the diagnosis. A thyroid scan is a nuclear medicine procedure involving injection of a radioisotope dye which will tag the thyroid and help produce a clear image of inflammation or involvement of the entire thyroid. Other tests can determine thyroid function and thyroid-stimulating hormone levels. Ultrasonography, computed tomography scans (CT scan), and magnetic resonance imaging (MRI) may provide visual confirmation of a diagnosis or help to determine the extent of involvement.

Treatment

Treatment will depend on the specific disease and individual circumstances such as age, severity of disease, and other conditions affecting a patient's health.

Antithyroid drugs

Antithyroid drugs are often administered to help the patient's body cease overproduction of thyroid hormones. This medication may work for young

adults, pregnant women, and others. Women who are pregnant should be treated with the lowest dose required to maintain thyroid function in order to minimize the risk of **hypothyroidism** in the infant.

Radioactive iodine

Radioactive iodine is often prescribed to damage cells that make thyroid hormone. The cells need iodine to make the hormone, so they will absorb any iodine found in the body. The patient may take an iodine capsule daily for several weeks, resulting in the eventual shrinkage of the thyroid in size, reduced hormone production and a return to normal blood levels. Some patients may receive a single larger oral dose of radioactive iodine to treat the disease more quickly. This should only be done for patients who are not of reproductive age or are not planning to have children, since a large amount can concentrate in the reproductive organs (gonads).

Surgery

Some patients may undergo surgery to treat hyperthyroidism. Most commonly, patients treated with **thyroidectomy**, in the form of partial or total removal of the thyroid, suffer from large goiter and have suffered relapses, even after repeated attempts to address the disease through drug therapy. Some patients may be candidates for surgery because they were not good candidates for iodine therapy, or refused iodine administration. Patients receiving thyroidectomy or iodine therapy must be carefully monitored for years to watch for signs of hypothyroidism, or insufficient production of thyroid hormones, which can occur as a complication of thyroid production suppression.

Alternative treatment

Consumption of foods such as broccoli, brussel sprouts, cabbage, cauliflower, kale, rutabagas, spinach, turnips, peaches, and pears can help naturally suppress thyroid hormone production. Caffeinated drinks and dairy products should be avoided. Under the supervision of a trained physician, high dosages of certain vitamin/mineral combinations can help alleviate hyperthyroidism.

Prognosis

Hyperthyroidism is generally treatable and carries a good prognosis. Most patients lead normal lives with proper treatment. Thyroid storm, however, can be life-

KEY TERMS

Goiter—Chronic enlargement of the thyroid gland.

Gonads—Organs that produce sex cells—the ovaries and testes.

Palpitations—Rapid and forceful heartbeat.

Radioisotope—A chemical tagged with radioactive compounds that is injected during a nuclear medicine procedure to highlight organ or tissue.

Thyroidectomy—Removal of the thyroid gland.

threatening and can lead to heart, liver, or kidney failure.

Prevention

There are no known prevention methods for hyperthyroidism, since its causes are either inherited or not completely understood. The best prevention tactic is knowledge of family history and close attention to symptoms and signs of the disease. Careful attention to prescribed therapy can prevent complications of the disease.

Resources

ORGANIZATIONS

Thyroid Foundation of America. 350 Ruth Sleeper Hall - RSL 350, Parkman St., Boston, MA. 02114. (800) 832-8321. <<http://www.clark.net/pub/tfa>>.

OTHER

"Endocrine Disorder and Endocrine Surgery." *Endocrine Web Page*. <<http://www.endocrineweb.com>>.

Teresa Odle

Hypertrophic cardiomyopathy

Definition

Cardiomyopathy is an ongoing disease process that damages the muscle wall of the lower chambers of the heart. Hypertrophic cardiomyopathy is a form of cardiomyopathy in which the walls of the heart's chambers thicken abnormally. Other names for hypertrophic cardiomyopathy are idiopathic

hypertrophic subaortic stenosis and asymmetrical septal hypertrophy.

Description

Hypertrophic cardiomyopathy usually appears in young people, often in athletes. For this reason it is sometimes called athletic heart muscle disease. However, people of any age can develop hypertrophic cardiomyopathy. Often there are no symptoms of hypertrophic cardiomyopathy. Sudden **death** can occur, caused by a heart arrhythmia. The American Heart Association reports that 36% of young athletes who die suddenly have probable or definite hypertrophic cardiomyopathy.

Hypertrophic cardiomyopathy is the result of abnormal growth of the heart muscle cells. The wall between the heart's chambers (the septum) may become so thickened that it blocks the flow of blood through the lower left chamber (left ventricle). The thickened wall may push on the heart valve between the two left heart chambers (mitral valve), making it leaky. The thickened muscle walls also prevent the heart from stretching as much as it should to fill with blood.

Causes and symptoms

The cause of hypertrophic cardiomyopathy is not known. In about one-half of cases, the disease is inherited. An abnormal gene has been identified in these patients. In cases that are not hereditary, a gene that was normal at birth may later become abnormal.

Often people with hypertrophic cardiomyopathy have no symptoms. Unfortunately, the first sign of the condition may be sudden death caused by an abnormal heart rhythm. When symptoms do appear, they include **shortness of breath** on exertion, **dizziness**, **fainting**, **fatigue**, and chest **pain**.

Diagnosis

The diagnosis is based on the patient's symptoms (if any), a complete **physical examination**, and tests that detect abnormalities of the heart chambers. Usually, there is an abnormal heart murmur that worsens with the Valsalva maneuver. The electrocardiogram (ECG), which provides a record of electrical changes in the heart muscle during the heartbeat, also is typically abnormal.

Sometimes, a routine **chest x ray** may show that the heart is enlarged. **Echocardiography**, a procedure that produces images of the heart's structure, is



This illustration shows hypertrophic muscle in the heart. The abnormally thick wall of muscle prevents the chambers from stretching to fill up with blood, making the heart less efficient. The extra tissue may also push on the heart valve (center), causing it to leak. (Illustration by Bryson Biomedical Illustrations, Custom Medical Stock Photo. Reproduced by permission.)

usually done. These images can show if the heart wall is thickened and if there are any abnormalities of the heart valves.

Treatment

Treatment of hypertrophic cardiomyopathy usually consists of taking medicines and restricting strenuous **exercise**. Drugs called beta blockers and **calcium channel blockers** are usually prescribed. **Beta blockers** reduce the force of the heart's contractions. Calcium channel blockers can help improve the flexibility of the heart muscle walls, allowing them to stretch more. Antiarrhythmic drugs may also be given to prevent abnormal heart rhythms.

Patients with hypertrophic cardiomyopathy are also told to avoid strenuous exercise to reduce the risk of passing out or sudden death.

In some cases, if the medications do not help relieve symptoms, surgery may help. In an operation called myotomy-myectomy a piece of the septum is removed to improve blood flow through the heart chamber.

KEY TERMS

Arrhythmias—Abnormal heartbeat.

Calcium channel blocker—A drug that relaxes blood vessels and lowers blood pressure.

Mitral valve—The heart valve that controls blood flow between the heart's left upper chamber (atrium) and left lower chamber (ventricle).

Septum—The muscular wall dividing the left and right heart chambers.

Ventricles—The two lower chambers of the heart.

Some patients have **pacemakers** and/or defibrillators implanted to help control the heart rate and rhythm. Pacemakers and defibrillators provide electrical impulses to the heart, which can return the heart beat to a normal rhythm.

If these treatment methods fail and a patient develops heart failure, a heart transplant may be necessary.

Prognosis

Some people with hypertrophic cardiomyopathy may not have obstructed blood flow and may never experience symptoms. Others may only experience mild symptoms. With treatment, symptoms may improve. In some patients, the disease may progress to **heart failure**.

Prevention

While hypertrophic cardiomyopathy cannot be prevented, precautionary measures may prevent sudden deaths. Anyone planning to take part in a program of strenuous competitive exercise should have a checkup by a physician first. A physical examination before athletic participation can usually, but not always, detect conditions like hypertrophic cardiomyopathy. Anyone who experiences symptoms of shortness of breath, tiredness, or fainting with exercise should see a physician.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

National Heart, Lung and Blood Institute. PO Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

Texas Heart Institute. Heart Information Service. PO Box 20345, Houston, TX 77225-0345. <<http://www.tmc.edu/thi>>.

Toni Rizzo

Hypervitaminosis see **Vitamin toxicity**

Hyphema

Definition

A hyphema is an accumulation of blood in the front (anterior) chamber of the eye. It is usually caused by blunt eye trauma.

Description

The anterior chamber (AC) is located behind the front of the eye. The AC is filled with a fluid called aqueous humor. This fluid helps form a cushion for the eye and provides an important route for nutrient and waste transport. Contusive forces from high velocity projectiles (approximately 34% of emergency room cases) such as a rock, crab apples, ice balls, badminton birds, and bungee cords can tear local blood vessels in the eye. Blunt impact from a basketball or racketball accounts for about 62% of cases. Tearing a small blood vessel can cause seepage of blood into a visible layer portion of the AC, causing the affected person to have red eye.

Causes and symptoms

Hyphema is caused by blunt, projectile, or explosion (about 4% of cases) injuries. These injuries cause a local blood vessel in the eye to tear, filling the front portion of the AC with blood. The initial complaint is a dramatic decrease in vision that eventually gets better as blood seeps towards the back of the eye. Patients will have extreme pain, an increase in intraocular pressure (the pressure inside the eye), and **nausea**. Patients usually will show a red eye and a recent history of trauma. Patients are vulnerable to more bleeding three to five days post injury.

Diagnosis

All persons with hyphema must be examined by an ophthalmologist (a physician who specializes in the medical and surgical care of the eye). Usually the clinician will use an ophthalmoscope to visualize

KEY TERMS

Microhyphema—Small bleed in the anterior chamber of the eye.

Ophthalmologist—A physician with specialized training in the medical and surgical treatment of eye diseases.

Optic Nerve—A cranial nerve that carries visual impulses to the brain for processing

the internal structures and damage. In some cases there may be small microscopic bleeds that may form clots (microhyphema) and require specialized instrumentation (a slit lamp) for visualization.

Treatment

Bloodthinners, such as **aspirin** and nonsteroidal anti-inflammatory drugs, should be avoided. In most cases the affected person can be medically managed on an outpatient basis. The eye should be shielded, but not patched. The patient should be placed at bed rest with the head elevated 45°. This position allows blood to leave the AC allowing for better vision. Several studies suggest administering medications (aminocaproic acid) that stabilize clot formation, reducing the possibility of increased bleeding.

Prognosis

The outcome depends on the severity of the trauma. Most cases progress well with conservative treatment. Some cases may develop an increase in the pressure within the eye (glaucoma). If this develops the hyphema must be surgically removed by an ophthalmologist. In patients who have a preexisting blood disorder, surgical evacuation should be considered to prevent damage to the optic nerve (the nerve that transmits impulses for processing in the brain).

Prevention

The American Academy of Ophthalmology recommends special eyewear made of polycarbonate lenses when at risk of eye injury. This type of lens has sufficient impact resistance.

Resources

BOOKS

Behrman, Richard E., et al, editors. *Nelson Textbook of Pediatrics*. 16th ed. W. B. Saunders Company, 2000.

Yanoff, Myron, et al, editors. *Ophthalmology*. Mosby International Ltd., 1999.

Laith Farid Gulli, M.D.

Hypnosis see **Hypnotherapy**

Hypnotherapy

Definition

Hypnotherapy is the treatment of a variety of health conditions by hypnotism or by inducing prolonged sleep.

Pioneers in this field, such as James Braid and James Esdaile discovered that hypnosis could be used to successfully anesthetize patients for surgeries. James Braid accidentally discovered that one of his patients began to enter a hypnotic state while staring at a fixed light as he waited for his **eye examination** to begin. Since mesmerism had fallen out of favor, Braid coined the term hypnotism, which is derived from the Greek word for sleep. Braid also used the techniques of monotony, rhythm, and imitation to assist in inducing a hypnotic state. As of 2000, these techniques are still in use.

Around 1900, there were very few preoperative anesthetic drugs available. Patients were naturally apprehensive when facing surgery. One out of four hundred patients would die, not from the surgical procedure, but from the anesthesia. Dr. Henry Munro was one of the first physicians to use hypnotherapy to alleviate patient fears about having surgery. He would get his patients into a hypnotic state and discuss their fears with them, telling them they would feel a lot better following surgery. Ether was the most common anesthetic at that time, and Dr. Munro found that he was able to perform surgery using only about 10% of the usual amount of ether.

Purpose

Hypnotherapy is used in a number of fields including psychotherapy, surgery, dentistry, research, and medicine. Hypnotherapy is commonly used as an alternative treatment for a wide range of health conditions, including weight control, pain management, and **smoking** cessation. It is also used to control **pain** in a variety of conditions such as **headache**, facial **neuralgia**, arthritis, **burns**, musculoskeletal disorders, childbirth, and many more. Hypnotherapy is being

used in place of anesthesia, particularly in patients who prove to be allergic to anesthetic drugs, for surgeries such as hysterectomies, cesarean sections, certain cardiovascular procedures, **thyroidectomy**, and others. Dentistry is using hypnotherapy with success on patients who are allergic to all types of novocaine drugs. Hypnotherapy is also useful in helping patients overcome **phobias**.

Hypnotherapy is used for nonmedical patients as well as those who wish to overcome bad habits. Hypnotherapy has been shown to help those who suffer from performance **anxiety**, such as in sports, and speaking in public. In academic applications, it has also been shown to help with learning, participating in the classroom, concentrating, studying, focusing attention span, improving memory, and helping remove mental blocks about particular subjects.

In more general areas, hypnotherapy has been found to be beneficial for problems such as motivation, procrastination, decision making, personal achievement and development, job performance, buried or repressed memories, relaxation, and stress management.

Description

Origins

Hypnotherapy is thought to date back to the healing practices of ancient Greece and Egypt. Many religions such as Judaism, Christianity, Islam, and others have attributed trance-like behavior to spiritual or divine possession.

Austrian physician, Franz Mesmer (1734–1815), is credited with being the first person to scientifically investigate the idea of hypnotherapy, in 1779, to treat a variety of health conditions. Mesmer studied medicine at the University of Vienna and received his medical degree in 1766. Mesmer is believed to have been the first doctor to understand the relationship of psychological trauma to illness. He induced a trance-like state, which became known as mesmerism, in his patients to successfully treat nervous disorders. These techniques became the foundation for modern-day hypnotherapy.

Mesmer's original interest was in the effect of celestial bodies on human lives. He later became interested in the effects of magnetism, and found that magnets could have tremendous healing effects on the human body. Mesmer believed that the human body contained a magnetic fluid that promoted health and well being. It was thought that any blockage to the normal flow of this magnetic fluid would result in

illness, and that the use of the mesmerism technique could restore the normal flow.

Mesmer performed his technique by passing his hands up and down the patient's body. The technique was supposed to transmit magnetic fluid from his hands to the bodies of his patients. During this time period, there was no clear delineation between health conditions that were physical or psychological in nature. Although Mesmer did not realize it at that time, his treatments were most effective for those conditions that were primarily psychosomatic.

Mesmer's technique appeared to be quite successful in the treatment of his patients, but he was the subject of scorn and ridicule from the medical profession. Because of all the controversy surrounding mesmerism, and because Mesmer's personality was quite eccentric, a commission was convened to investigate his techniques and procedures. A very distinguished panel of investigators included Benjamin Franklin, the French chemist Antoine-Laurent Lavoisier, and physician Jacques Guillotin. The commission acknowledged that patients did seem to obtain noticeable relief from their conditions, but the whole idea was dismissed as being medical quackery.

It took more than two hundred years for hypnotherapy to become incorporated into medical treatment. In 1955, the British Medical Association approved the use of hypnotherapy as a valid medical treatment, with the American Medical Association (AMA) giving its approval in 1958.

Hypnotherapy involves achieving a psychological state of awareness that is different from the ordinary state of consciousness. While in a hypnotic state, a variety of phenomena can occur. These phenomena include alterations in memory, heightened susceptibility to suggestion, paralysis, sweating, and blushing. All of these changes can be produced or removed in the hypnotic state. Many studies have shown that roughly 90% of the population is capable of being hypnotized.

This state of awareness can be achieved by relaxing the body, focusing on breathing, and shifting attention away from the external environment. In this state, the patient has a heightened receptivity to suggestion. The usual procedure for inducing a hypnotic trance in another person is by direct command repeated in a soothing, monotonous tone of voice.

Preparations

Ideally, the following conditions should be present to successfully achieve a state of hypnosis:

- willingness to be hypnotized
- rapport between the patient or client and the hypnotherapist
- a comfortable environment that is conducive to relaxation

Precautions

Hypnotherapy can have negative outcomes. When used as entertainment, people have been hypnotized to say or do things that would normally embarrass them. There have been instances where people already dangerously close to psychological breakdown have been pushed into an emotional crisis during what was supposed to be a harmless demonstration of hypnosis. A statement from the World Hypnosis Organization (WHO) warns against performing hypnosis on patients suffering from **psychosis**, organic psychiatric conditions, or antisocial **personality disorders**. Because there are no standard licensing requirements, in the wrong hands, there is a risk that the hypnotist will have difficulty in controlling or ending a hypnotic state that has been induced in the patient.

There is a commonly held belief that a person cannot be coerced into doing things that they would not normally do while under hypnosis. The hypnotherapist should take care however, not to give suggestions during hypnosis that are contrary to the patient's moral code.

Many religions do not condone the practice of hypnotherapy. Leaders of the Jehovah's Witnesses and Christian Science religions oppose the use of hypnotherapy and advise their members to avoid it completely, whether for entertainment or therapy. The Church of Jesus Christ of Latter-Day Saints approves it for medical purposes, but cautions members against allowing themselves to be hypnotized for entertainment or demonstration purposes.

In 1985, The AMA convened a commission that warned against using hypnotherapy to aid in recollection of events. The commission cited studies that showed the possibility of hypnotic recall resulting on confabulation or an artificial sense of certainty about the course of events. As a result, many states limit or prohibit testimony of hypnotized witnesses or victims.

Side effects

Experiments have been conducted to determine any side effects of hypnotherapy. Some subjects have reported side effects such as headache, stiff neck, drowsiness, cognitive distortion or confusion, **dizziness**, and

anxiety. However, most of these effects cleared up within several hours of the hypnotherapy session.

Research and general acceptance

Research on the effectiveness of hypnotherapy on a variety of medical conditions is extensive. In one study, the use of hypnotherapy did not seem to alter the core symptoms in the treatment of attention-deficit hyperactivity disorder (ADHD); however, it did seem to be useful in managing the associated symptoms including sleep disturbances and tics.

Hypnotherapy is being studied in children who have common, chronic problems and to aid in relieving pain. Children are particularly good candidates for hypnotherapy because their lack of worldly experience enables them to move easily between the rational world and their imagination. Studies with children have shown responses to hypnotherapy ranging from diminished pain and anxiety during a number of medical procedures, a 50% range in reduction of symptoms or a complete resolution of a medical condition, and a reduction in use of anti-nausea medication and vomiting during **chemotherapy** for childhood cancers.

The use of hypnotherapy with **cancer** patients is another area being investigated. A meta-analysis of 116 studies showed very positive results of using hypnotherapy with cancer patients. Ninety-two percent showed a positive effect on depression; 93% showed a positive effect on physical well-being; 81% showed a positive effect on vomiting; and 92% showed a positive effect on pain.

Resources

PERIODICALS

- Baumgaertel, Anna. "Attention-Deficit/Hyperactivity Disorder: Alternative and Controversial Treatments for Attention-Deficit/Hyperactivity Disorder." *Pediatric Clinics of North America* October 1999.
- Margolis, Clorinda G. "Hypnotic Trance: The Old and the New." *Primary Care Clinics in Office Practice*.
- Newell, Sallie, and Rob W. Sanson-Fisher. "Australian on Bologists' self-reported knowledge and attitudes about non-traditional therapies used by cancer patients." *Medical Journal of Australia* February 7, 2000.

ORGANIZATIONS

- American Board of Hypnotherapy. 16842 Von Karman Avenue, Suite 476, Irvine, CA 92714. <<http://www.hypnosis.com>>.
- American Psychotherapy & Medical Hypnosis Association. 210 S. Sierra, Reno, NV 89501. <<http://members.xoom.com/Hypnosis>>.
- American Society of Clinical Hypnosis. 200 E. Devon Avenue, Des Plaines, IL 60018.

International Council for Medical and Clinical Therapists.
7361 McWhorter Place, Suite 300, Annandale, VA
22003-5469. <<http://www.ultradept.com/ICMCT.htm>>.

International Medical and Dental Hypnotherapy
Association. 4110 Edgeland, Suite 800, Royal Oak, MI
48073-2285. <<http://www.infinityinst.com>>.

National Board for Hypnotherapy and Hypnotic
Anaesthesiology. 7841 West Ludlow Drive, Suite A,
Peoria, AZ 85381. <<http://www.nbha-medicine.com/index.html>>.

National Guild of Hypnotists. PO Box 308, Merrimack,
NH. <<http://www.ngh.net>>.

Society for Clinical and Experimental Hypnosis. 6728 Old
McLean Village Drive, McLean, VA 22101.

World Hypnosis Organization, Inc. 2521 W. Montrose
Avenue, Chicago, IL 60618. <<http://www.worldhypnosis.org/about.html>>.

Kim A. Sharp, M.Ln.

Hypocalcemia

Definition

Hypocalcemia, a low blood calcium level, occurs when the concentration of free calcium ions in the blood falls below 4.0 mg/dL (dL=one tenth of a liter). The normal concentration of free calcium ions in the blood serum is 4.0–6.0 mg/dL.

Description

Calcium is an important mineral for maintaining human health. It is not only a component of bones and teeth, but is also essential for normal blood clotting and necessary for normal muscle and nerve functions. The calcium ion (Ca^{2+}) has two positive charges. In bone, calcium ions occur as a complex with phosphate to form crystals of calcium phosphate. In the bloodstream, calcium ions also occur in complexes, and here calcium is found combined with proteins and various nutrients. However, in the bloodstream, calcium also occurs in a free form. Normally, about 47% of the calcium in the blood plasma is free, while 53% occurs in a complexed form. Although all of the calcium in the bloodstream serves a useful purpose, it is only the concentration of free calcium ions which has a direct influence on the functioning of our nerves and muscles. For this reason, the measurement of the concentration of free calcium is more important, in the diagnosis of disease, than measuring the level of total

calcium or of complexed calcium. The level of total calcium in the blood serum is normally 8.5–10.5 mg/dL, while the level of free calcium is normally 4–5 mg/dL.

Causes and symptoms

Hypocalcemia can be caused by **hypoparathyroidism**, by failure to produce 1,25-dihydroxyvitamin D, by low levels of plasma magnesium, or by failure to get adequate amounts of calcium or vitamin D in the diet. Hypoparathyroidism involves the failure of the parathyroid gland to make parathyroid hormone. Parathyroid hormone controls and maintains plasma calcium levels. The hormone exerts its effect on the kidneys, where it triggers the synthesis of 1,25-dihydroxyvitamin D. Thus, hypocalcemia can be independently caused by damage to the parathyroid gland or to the kidneys. 1,25-Dihydroxyvitamin D stimulates the uptake of calcium from the diet and the mobilization of calcium from the bone. Bone mobilization means the natural process by which the body dissolves part of the bone in the skeleton in order to maintain or raise the levels of plasma calcium ions.

Low plasma magnesium levels (hypomagnesia) can result in hypocalcemia. Hypomagnesemia can occur with **alcoholism** or with diseases characterized by an inability to properly absorb fat. Magnesium is required for parathyroid hormone to play its part in maintaining plasma calcium levels. For this reason, any disease that results in lowered plasma magnesium levels may also cause hypocalcemia.

Hypocalcemia may also result from the consumption of toxic levels of phosphate. Phosphate is a constituent of certain enema formulas. An enema is a solution that is used to cleanse the intestines via a hose inserted into the rectum. Cases of hypocalcemia have been documented where people swallowed enema formulas, or where an enema has been administered to an infant.

Symptoms of severe hypocalcemia include **numbness** or **tingling** around the mouth or in the feet and hands, as well as in **muscle spasms** in the face, feet, and hands. Hypocalcemia can also result in depression, memory loss, or **hallucinations**. Severe hypocalcemia occurs when serum free calcium is under 3 mg/dL. Chronic and moderate hypocalcemia can result in **cataracts** (damage to the eyes). In this case, the term “chronic” means lasting one year or longer.

Diagnosis

Hypocalcemia is diagnosed by acquiring a sample of blood serum and measuring the concentration of

KEY TERMS

Plasma—Plasma is blood with the cells removed.

Serum—Serum is blood plasma with the blood clotting proteins removed.

free calcium using a calcium-sensitive electrode. Hypocalcemia has several causes, and hence a full diagnosis requires assessment of health of the parathyroid gland, kidneys, and of plasma magnesium concentration.

Treatment

The method chosen for treatment depends on the exact cause and on the severity of the hypocalcemia. Severe hypocalcemia requires injection of calcium ions, usually in the form of calcium gluconate. Oral calcium supplements are prescribed for long term treatment (non-emergency) of hypocalcemia. The oral supplements may take the form of calcium carbonate, calcium chloride, calcium lactate, or calcium gluconate. Where hypocalcemia results from kidney failure, treatment includes injections of 1,25-dihydroxyvitamin D. Oral vitamin D supplements can increase gastrointestinal absorption of calcium. Where hypocalcemia results from hypoparathyroidism, treatment may include oral calcium, 1,25-dihydroxyvitamin D, or other drugs. Where low serum magnesium levels occur, concurrently with hypocalcemia, the magnesium deficiency must be corrected to effectively treat the hypocalcemia.

Prognosis

The prognosis for correcting hypocalcemia is excellent. However, the eye damage that may result from chronic hypocalcemia cannot be reversed.

Prevention

The first, and most obvious, way to help prevent hypocalcemia is to ensure that adequate amounts of calcium and vitamin D are consumed each day, either in the diet or as supplements. The hypocalcemia that may occur with damage to the parathyroid gland or to the kidneys cannot be prevented. Hypocalcemia resulting from overuse of **enemas** can be prevented by reducing enema usage. Hypocalcemia resulting from magnesium deficiency tends to occur in chronic alcoholics, and this type of hypocalcemia can be prevented

by reducing alcohol consumption and increasing the intake of healthful food.

Resources

BOOKS

Brody, Tom. *Nutritional Biochemistry*. San Diego: Academic Press, 1998.

Tom Brody, PhD

Hypochondriac see **Hypochondriasis**

Hypochondriasis

Definition

Hypochondriasis is a mental disorder characterized by excessive fear of or preoccupation with a serious illness, despite medical testing and reassurance to the contrary. It was formerly called hypochondriacal neurosis.

Description

Although hypochondriasis is often considered a disorder that primarily affects adults, it is now increasingly recognized in children and adolescents. In addition, hypochondriasis may develop in elderly people without previous histories of health-related fears. The disorder accounts for about 5% of psychiatric patients and is equally common in men and women.

Causes and symptoms

The causes of hypochondriasis are not precisely known. Children may have physical symptoms that resemble or mimic those of other family members. In adults, hypochondriasis may sometimes reflect a self-centered character structure or a wish to be taken care of by others; it may also have been copied from a parent's behavior. In elderly people, hypochondriasis may be associated with depression or grief. It may also involve biologically based hypersensitivity to internal stimuli.

Most hypochondriacs are worried about being physically sick, although some express fear of insanity. The symptoms reported can range from general descriptions of a specific illness to unusual complaints. In many instances the symptoms reflect intensified awareness of ordinary body functions, such as heart-beat, breathing, or stomach noises. It is important to

understand that a hypochondriac's symptoms are not "in the head" in the sense of being delusional. The symptoms are real, but the patient misinterprets bodily functions and attributes them to a serious or even lethal cause.

Diagnosis

The diagnosis is often complicated by the patient's detailed understanding of symptoms and medical terminology from previous contacts with doctors. If a new doctor suspects hypochondriasis, he or she will usually order a complete medical workup in order to rule out physical disease.

Psychological evaluation is also necessary to rule out other disorders that involve feelings of **anxiety** or complaints of physical illness. These disorders include depression, **panic disorder**, and **schizophrenia** with somatic (physical) **delusions**. The following features are characteristic of hypochondriasis:

- The patient is not psychotic (out of touch with reality or hallucinating).
- The patient gets upset or blames the doctor when told there is "nothing wrong," or that there is a psychological basis for the problem.
- There is a correlation between episodes of hypochondriacal behavior and stressful periods in the patient's life.
- The behavior has lasted at least six months.

Evaluation of children and adolescents with hypochondriasis should include the possibility of **abuse** by family members.

Treatment

The goal of therapy is to help the patient (and family) live with the symptoms and to modify thinking and behavior that reinforces hypochondriacal symptoms. This treatment orientation is called supportive, as distinct from insight-oriented, because hypochondriacs usually resist psychological interpretations of their symptoms. Supportive treatment may include medications to relieve anxiety. Some clinicians look carefully for "masked" depression and treat with antidepressants.

Follow-up care includes regular physical check-ups, because about 30% of patients with hypochondriasis will eventually develop a serious physical illness. The physician also tries to prevent unnecessary medical testing and "doctor shopping" on the patient's part.

KEY TERMS

Somatoform disorder—A category of psychiatric disorder characterized by conversion of emotional distress into physical symptoms or by symptoms of physical illness that have no discernible organic cause. Hypochondriasis is classified as a somatoform disorder.

Supportive therapy—Any form of treatment intended to relieve symptoms or help the patient live with them rather than attempt changes in character structure.

Prognosis

From 33–50% of patients with hypochondriasis can expect significant improvement from the current methods of treatment.

Resources

BOOKS

Eisendrath, Stuart J. "Psychiatric Disorders." In *Current Medical Diagnosis and Treatment*, 1998, edited by Stephen McPhee, et al., 37th ed. Stamford: Appleton & Lange, 1997.

Rebecca J. Frey, PhD

Hypoesthesias see **Numbness and tingling**

Hypoglycemia

Definition

The condition called hypoglycemia is literally translated as low blood sugar. Hypoglycemia occurs when blood sugar (or blood glucose) concentrations fall below a level necessary to properly support the body's need for energy and stability throughout its cells.

Description

Carbohydrates are the main dietary source of the glucose that is manufactured in the liver and absorbed into the bloodstream to fuel the body's cells and organs. Glucose concentration is controlled by hormones, primarily insulin and glucagon. Glucose concentration also is controlled by epinephrine

(adrenalin) and norepinephrine, as well as growth hormone. If these regulators are not working properly, levels of blood sugar can become either excessive (as in hyperglycemia) or inadequate (as in hypoglycemia). If a person has a blood sugar level of 50 mg/dl or less, he or she is considered hypoglycemic, although glucose levels vary widely from one person to another.

Hypoglycemia can occur in several ways.

Drug-induced hypoglycemia

Drug-induced hypoglycemia, a complication of diabetes, is the most commonly seen and most dangerous form of hypoglycemia.

Hypoglycemia occurs most often in diabetics who must inject insulin periodically to lower their blood sugar. While other diabetics also are vulnerable to low blood sugar episodes, they have a lower risk of a serious outcome than insulin-dependant diabetics. Unless recognized and treated immediately, severe hypoglycemia in the insulin-dependent diabetic can lead to generalized convulsions followed by **amnesia** and unconsciousness. Death, though rare, is a possible outcome.

In insulin-dependent diabetics, hypoglycemia known as an insulin reaction or insulin shock can be caused by several factors. These include overmedicating with manufactured insulin, missing or delaying a meal, eating too little food for the amount of insulin taken, exercising too strenuously, drinking too much alcohol, or any combination of these factors.

Ideopathic or reactive hypoglycemia

Ideopathic or reactive hypoglycemia (also called postprandial hypoglycemia) occurs when some people eat. A number of reasons for this reaction have been proposed, but no single cause has been identified.

In some cases, this form of hypoglycemia appears to be associated with malfunctions or diseases of the liver, pituitary, adrenals, liver, or pancreas. These conditions are unrelated to diabetes. Children intolerant of a natural sugar (fructose) or who have inherited defects that affect digestion also may experience hypoglycemic attacks. Some children with a negative reaction to aspirin also experience reactive hypoglycemia. It sometimes occurs among people with an intolerance to the sugar found in milk (galactose), and it also often begins before diabetes strikes later on.

Fasting hypoglycemia

Fasting hypoglycemia sometimes occurs after long periods without food, but it also happens occasionally

following strenuous **exercise**, such as running in a marathon.

Other factors sometimes associated with hypoglycemia include:

- pregnancy
- a weakened immune system
- a poor diet high in simple carbohydrates
- prolonged use of drugs, including antibiotics
- chronic physical or mental **stress**
- heartbeat irregularities (arrhythmias)
- **allergies**
- breast **cancer**
- high blood pressure treated with beta-blocker medications (after strenuous exercise)
- upper gastrointestinal tract surgery.

Causes and symptoms

When carbohydrates are eaten, they are converted to glucose that goes into the bloodstream and is distributed throughout the body. Simultaneously, a combination of chemicals that regulate how our body's cells absorb that sugar is released from the liver, pancreas, and adrenal glands. These chemical regulators include insulin, glucagon, epinephrine (adrenalin), and norepinephrine. The mixture of these regulators released following digestion of carbohydrates is never the same, since the amount of carbohydrates that are eaten is never the same.

Interactions among the regulators are complicated. Any abnormalities in the effectiveness of any one of the regulators can reduce or increase the body's absorption of glucose. Gastrointestinal enzymes such as amylase and lactase that break down carbohydrates may not be functioning properly. These abnormalities may produce hyperglycemia or hypoglycemia, and can be detected when the level of glucose in the blood is measured.

Cell sensitivity to these regulators can be changed in many ways. Over time, a person's stress level, exercise patterns, advancing age, and dietary habits influence cellular sensitivity. For example, a diet consistently overly rich in carbohydrates increases insulin requirements over time. Eventually, cells can become less receptive to the effects of the regulating chemicals, which can lead to glucose intolerance.

Diet is both a major factor in producing hypoglycemia as well as the primary method for controlling it. **Diets** typical of western cultures contain excess

carbohydrates, especially in the form of simple carbohydrates such as sweeteners, which are more easily converted to sugar. In poorer parts of the world, the typical diet contains even higher levels of carbohydrates. Fewer dairy products and meats are eaten, and grains, vegetables, and fruits are consumed. This dietary trend is balanced, however, since people in these cultures eat smaller meals and usually use carbohydrates more efficiently through physical labor.

Early symptoms of severe hypoglycemia, particularly in the drug-induced type of hypoglycemia, resemble an extreme **shock** reaction. Symptoms include:

- cold and pale skin
- numbness around the mouth
- apprehension
- heart **palpitations**
- emotional outbursts
- hand tremors
- mental cloudiness
- dilated pupils
- sweating
- fainting

Mild attacks, however, are more common in reactive hypoglycemia and are characterized by extreme tiredness. Patients first lose their alertness, then their muscle strength and coordination. Thinking grows fuzzy, and finally the patient becomes so tired that he or she becomes “zombie-like,” awake but not functioning. Sometimes the patient will actually fall asleep. Unplanned naps are typical of the chronic hypoglycemic patient, particularly following meals.

Additional symptoms of reactive hypoglycemia include headaches, double vision, staggering or inability to walk, a craving for salt and/or sweets, abdominal distress, premenstrual tension, chronic colitis, allergies, ringing in the ears, unusual patterns in the frequency of urination, skin eruptions and inflammations, **pain** in the neck and shoulder muscles, memory problems, and sudden and excessive sweating.

Unfortunately, a number of these symptoms mimic those of other conditions. For example, the depression, **insomnia**, irritability, lack of concentration, crying spells, **phobias**, forgetfulness, confusion, unsocial behavior, and suicidal tendencies commonly seen in nervous system and psychiatric disorders also may be hypoglycemic symptoms. It is very important that anyone with symptoms that may suggest reactive hypoglycemia see a doctor.

Because all of its possible symptoms are not likely to be seen in any one person at a specific time, diagnosing hypoglycemia can be difficult. One or more of its many symptoms may be due to another illness. Symptoms may persist in a variety of forms for long periods of time. Symptoms also can change over time within the same person. Some of the factors that can influence symptoms include physical or mental activities, physical or mental state, the amount of time passed since the last meal, the amount and quality of sleep, and exercise patterns.

Diagnosis

Drug-induced hypoglycemia

Once diabetes is diagnosed, the patient then monitors his or her blood sugar level with a portable machine called a glucometer. The diabetic places a small blood sample on a test strip that the machine can read. If the test reveals that the blood sugar level is too low, the diabetic can make a correction by eating or drinking an additional carbohydrate.

Reactive hypoglycemia

Reactive hypoglycemia only can be diagnosed by a doctor. Symptoms usually improve after the patient has gone on an appropriate diet. Reactive hypoglycemia was diagnosed more frequently 10–20 years ago than today. Studies have shown that most people suffering from its symptoms test normal for blood sugar, leading many doctors to suggest that actual cases of reactive hypoglycemia are quite rare. Some doctors think that people with hypoglycemic symptoms may be particularly sensitive to the body's normal postmeal release of the hormone epinephrine, or are actually suffering from some other physical or mental problem. Other doctors believe reactive hypoglycemia actually is the early onset of diabetes that occurs after a number of years. There continues to be disagreement about the cause of reactive hypoglycemia.

A common test to diagnose hypoglycemia is the extended oral glucose tolerance test. Following an overnight fast, a concentrated solution of glucose is drunk and blood samples are taken hourly for five to six hours. Though this test remains helpful in early identification of diabetes, its use in diagnosing chronic reactive hypoglycemia has lost favor because it can trigger hypoglycemic symptoms in people with otherwise normal glucose readings. Some doctors now recommend that blood sugar be tested at the actual time a person experiences hypoglycemic symptoms.

Treatment

Treatment of the immediate symptoms of hypoglycemia can include eating sugar. For example, a patient can eat a piece of candy, drink milk, or drink fruit juice. Glucose tablets can be used by patients, especially those who are diabetic. Effective treatment of hypoglycemia over time requires the patient to follow a modified diet. Patients usually are encouraged to eat small, but frequent, meals throughout the day, avoiding excess simple sugars (including alcohol), fats, and fruit drinks. Those patients with severe hypoglycemia may require fast-acting glucagon injections that can stabilize their blood sugar within approximately 15 minutes.

Alternative treatment

A holistic approach to reactive hypoglycemia is based on the belief that a number of factors may create the condition. Among them are heredity, the effects of other illnesses, emotional stress, too much or too little exercise, bad lighting, poor diet, and environmental pollution. Therefore, a number of alternative methods have been proposed as useful in treating the condition. Homeopathy, **acupuncture**, and **applied kinesiology**, for example, have been used, as have herbal remedies. One of the herbal remedies commonly suggested for hypoglycemia is a decoction (an extract made by boiling) of gentian (*Gentiana lutea*). It should be drunk warm 15–30 minutes before a meal. Gentian is believed to help stimulate the endocrine (hormone-producing) glands.

In addition to the dietary modifications recommended above, people with hypoglycemia may benefit from supplementing their diet with chromium, which is believed to help improve blood sugar levels. Chromium is found in whole grain breads and cereals, cheese, molasses, lean meats, and brewer's yeast. Hypoglycemics should avoid alcohol, **caffeine**, and cigarette smoke, since these substances can cause significant swings in blood sugar levels.

Prevention

Drug-induced hypoglycemia

Preventing hypoglycemic insulin reactions in diabetics requires taking glucose readings through frequent blood sampling. Insulin then can be regulated based on those readings. Continuous glucose monitoring sensors have been developed to help diabetics remain more aware of possible hypoglycemic episodes. These monitors even can check for episodes while the patient sleeps, when many will experience

severe hypoglycemia but not know it. Those who don't pay attention to severe hypoglycemia events or who have had previous severe hypoglycemia are the most likely to have future severe hypoglycemia. An audible alert can let the patient know immediately that he or she needs to take care of his or her blood sugar level. Continuous monitoring has proved particularly helpful in pediatric patients with Type 1 diabetes.

Maintaining proper diet also is a factor. Programmable insulin pumps implanted under the skin have proven useful in reducing the incidence of hypoglycemic episodes for insulin-dependent diabetics. As of late 1997, clinical studies continue to seek additional ways to control diabetes and drug-induced hypoglycemia. Tests of a substance called pramlintide indicate that it may help improve glycemic control in diabetics.

Reactive hypoglycemia

The onset of reactive hypoglycemia can be avoided or at least delayed by following the same kind of diet used to control it. While not as restrictive as the diet diabetics must follow to keep tight control over their disease, it is quite similar.

There are a variety of diet recommendations for the reactive hypoglycemic. Patients should:

- avoiding overeating
- never skipping breakfast
- including protein in all meals and snacks, preferably from sources low in fat, such as the white meat of chicken or turkey, most fish, soy products, or skim milk
- restricting intake of fats (particularly saturated fats, such as animal fats), and avoiding refined sugars and processed foods
- being aware of the differences between some vegetables, such as potatoes and carrots. These vegetables have a higher sugar content than others (like squash and broccoli). Patients should be aware of these differences and note any reactions they have to them.
- being aware of differences found in grain products. White flour is a carbohydrate that is rapidly absorbed into the bloodstream, while oats take much longer to break down in the body.
- keeping a "food diary." Until the diet is stabilized, a patient should note what and how much he/she eats and drinks at every meal. If symptoms appear following a meal or snack, patients should note them and look for patterns.

KEY TERMS

Adrenal glands—Two organs that sit atop the kidneys; these glands make and release hormones such as epinephrine.

Epinephrine—Also called adrenalin, a secretion of the adrenal glands (along with norepinephrine) that helps the liver release glucose and limits the release of insulin. Norepinephrine is both a hormone and a neurotransmitter, a substance that transmits nerve signals.

Fructose—A type of natural sugar found in many fruits, vegetables, and in honey.

Glucagon—A hormone produced in the pancreas that raises the level of glucose in the blood. An injectable form of glucagon, which can be bought in a drug store, is sometimes used to treat insulin shock.

Postprandial—After eating or after a meal.

- eat fresh fruits, but restrict the amount they eat at one time. Patients should remember to eat a source of protein whenever they eat high sources of carbohydrate like fruit. Apples make particularly good snacks because, of all fruits, the carbohydrate in apples is digested most slowly.
- following a diet that is high in fiber. Fruit is a good source of fiber, as are oatmeal and oat bran. Fiber slows the buildup of sugar in the blood during digestion.

A doctor can recommend a proper diet, and there are many cookbooks available for diabetics. Recipes found in such books are equally effective in helping to control hypoglycemia.

Prognosis

Like diabetes, there is no cure for reactive hypoglycemia, only ways to control it. While some chronic cases will continue through life (rarely is there complete remission of the condition), others will develop into type II (age onset) diabetes. Hypoglycemia appears to have a higher-than-average incidence in families where there has been a history of hypoglycemia or diabetes among their members, but whether hypoglycemia is a controllable warning of oncoming diabetes has not yet been determined by clinical research.

A condition known as hypoglycemia unawareness can develop in those who do not control their blood glucose, particularly in people with Type 1 diabetes.

These people may lose notice of the automatic warning symptoms of hypoglycemia that normally occur as their bodies become so used to frequent periods of hypoglycemia. It is not a permanent event, but can be treated by careful avoidance of hypoglycemia for about two weeks.

Resources

BOOKS

Ruggiero, Roberta. *The Do's and Don'ts of Low Blood Sugar*. Hollywood, FL: Frederick Fell Publishers.

PERIODICALS

Brauker, James, et al. "Use of Continuous Glucose Monitoring Alerts to Better Predict, Prevent and Treat Postprandial Hyperglycemia." *Diabetes* June 2003: 90-91.

Gertzman, Jerilyn, et al. "Severity of Hypoglycemia and Hypoglycemia Unawareness Are Associated with the Extent of Unsuspected Nocturnal Hypoglycemia." *Diabetes* June 2003:146-151.

Kumar, Rajeev, and Miles Fisher. "Impaired Hypoglycemia Awareness: Are we Aware?" *Diabetes and Primary Care* Summer 2004: 33-38.

Ludvigsson, Johnny, and Ragnar Hanas. "Continuous Subcutaneous Glucose Monitoring Improved Metabolic Control in Pediatric Patients With Type 1 Diabetes: A Controlled Crossover Study." *Pediatrics* May 2003: 933-936.

ORGANIZATIONS

Hypoglycemia Association, Inc. 18008 New Hampshire Ave., PO Box 165, Ashton, MD 20861-0165.

National Hypoglycemia Association, Inc. PO Box 120, Ridgewood, NJ 07451. (201) 670-1189.

Martin W. Dodge, PhD
Teresa G. Odle

Hypogonadism

Definition

Hypogonadism is the condition more prevalent in males in which the production of sex hormones and germ cells are inadequate.

Description

Gonads are the organs of sexual differentiation—in the female, they are ovaries; in the male, the testes. Along with producing eggs and sperm, they produce sex hormones that generate all the differences between men and women. If they produce too little sex

hormone, then either the growth of the sexual organs or their function is impaired.

The gonads are not independent in their function, however. They are closely controlled by the pituitary gland. The pituitary hormones are the same for males and females, but the gonadal hormones are different. Men produce mostly androgens, and women produce mostly estrogens. These two hormones regulate the development of the embryo, determining whether it is a male or a female. They also direct the adolescent maturation of sex organs into their adult form. Further, they sustain those organs and their function throughout the reproductive years. The effects of estrogen reach beyond that to sustain bone strength and protect the cardiovascular system from degenerative disease.

Hormones can be inadequate during or after each stage of development—embryonic and adolescent. During each stage, inadequate hormone stimulation will prevent normal development. After each stage, a decrease in hormone stimulation will result in failed function and perhaps some shrinkage. The organs affected principally by sex hormones are the male and female genitals, both internal and external, and the female breasts. Body hair, fat deposition, bone and muscle growth, and some brain functions are also influenced.

Causes and symptoms

Sex is determined at the moment of conception by sex chromosomes. Females have two X chromosomes, while males have one X and one Y chromosome. If the male sperm with the Y chromosome fertilizes an egg, the baby will be male. This is true throughout the animal kingdom. Genetic defects sometimes result in changes in the chromosomes. If sex chromosomes are involved, there is a change in the development of sexual characteristics.

Female is the default sex of the embryo, so most of the sex organ deficits at birth occur in boys. Some, but not all, are due to inadequate androgen stimulation. The penis may be small, the testicles undescended (cryptorchidism) or various degrees of “feminization” of the genitals may be present.

After birth, sexual development does not occur until **puberty**. Hypogonadism most often shows up as an abnormality in boys during puberty. Again, not every defect is due to inadequate hormones. Some are due to too much of the wrong ones. Kallmann’s syndrome is a birth defect in the brain that prevents release of hormones and appears as failure of male puberty. Some boys have adequate amounts of androgen in their system but fail to respond to them, a condition known as androgen resistance.

KEY TERMS

Biopsy—Surgical removal of pieces of tissue for examination.

Embryo—Refers to life before birth, specifically the first two months after conception.

Fetus—The unborn person or animal, still in the womb.

Hypothalamus—Part of the brain just above the pituitary that stimulates pituitary gland function.

Ionizing radiation—X rays. Diagnostic x rays are too weak to do damage under normal circumstances, but x rays used to treat cancer must be used with great care.

Undescended testicle—A testicle that is still in the groin and has not made its way into the scrotum.

Female problems in puberty are not caused by too little estrogen. Even female reproductive problems are rarely related to a simple lack of hormones, but rather to complex cycling rhythms gone wrong. All the problems with too little hormone happen during **menopause**, which is a normal hypogonadism.

A number of adverse events can damage the gonads and result in decreased hormone levels. The childhood disease **mumps**, if acquired after puberty, can infect and destroy the testicles—a disease called viral **orchitis**. Ionizing radiation and **chemotherapy**, trauma, several drugs (spironolactone, a diuretic and ketoconazole, an antifungal agent), alcohol, **marijuana**, heroin, **methadone**, and environmental toxins can all damage testicles and decrease their hormone production. Severe diseases in the liver or kidneys, certain infections, sickle cell anemia, and some cancers also affect gonads. To treat some male cancers, it is necessary to remove the testicles, thereby preventing the androgens from stimulating **cancer** growth. This procedure, still called castration or **orchiectomy**, removes androgen stimulation from the whole body.

For several reasons the pituitary can fail. It happens rarely after **pregnancy**. It used to be removed to treat advanced breast or **prostate cancer**. Sometimes the pituitary develops a tumor that destroys it. Failure of the pituitary is called **hypopituitarism** and, of course, leaves the gonads with no stimulation to produce hormones.

Besides the tissue changes generated by hormone stimulation, the only other symptoms relate to sexual desire and function. Libido is enhanced by testosterone,

and male sexual performance requires androgens. The role of female hormones in female sexual activity is less clear, although hormones strengthen tissues and promote healthy secretions, facilitating sexual activity.

Diagnosis

Presently, there are accurate blood tests for most of the hormones in the body, including those from the pituitary and even some from the hypothalamus. Chromosomes can be analyzed, and gonads can, but rarely are, biopsied.

Treatment

Replacement of missing body chemicals is much easier than suppressing excesses. Estrogen replacement is recommended for nearly all women after menopause for its many beneficial effects. Estrogen can be taken by mouth, injection, or skin patch. It is strongly recommended that the other female hormone, progesterone, be taken as well, because it prevents overgrowth of uterine lining and uterine cancer. Testosterone replacement is available for males who are deficient.

Resources

BOOKS

Carr, Bruce R., and Karen D. Bradshaw. "Disorders of the Ovary and Female Reproductive Tract." In *Harrison's Principles of Internal Medicine*, edited by Anthony S. Fauci, et al. New York: McGraw-Hill, 1998.

J. Ricker Polsdorfer, MD

Hypokalemia

Definition

Hypokalemia is a condition of below normal levels of potassium in the blood serum. Potassium, a necessary electrolyte, facilitates nerve impulse conduction and the contraction of skeletal and smooth muscles, including the heart. It also facilitates cell membrane function and proper enzyme activity. Levels must be kept in a proper (homeostatic) balance for the maintenance of health. The normal concentration of potassium in the serum is in the range of 3.5–5.0 mM. Hypokalemia means serum or plasma levels of potassium ions that fall below 3.5 mM. (Potassium concentrations are often expressed in units of milliequivalents per liter [mEq/L], rather than in units of

millimolarity [mM], however, both units are identical and mean the same thing when applied to concentrations of potassium ions.)

Hypokalemia can result from two general causes: either from an overall depletion in the body's potassium or from excessive uptake of potassium by muscle from surrounding fluids.

Description

A normal adult weighing about 154 lbs (70 kg) has about 3.6 moles of potassium ions in his body. Most of this potassium (about 98%) occurs inside various cells and organs, where normal concentration are about 150 mM. Blood serum concentrations are much lower—only about 0.4% of the body's potassium is found in blood serum. As noted above, hypokalemia can be caused by the sudden uptake of potassium ions from the bloodstream by muscle or other organs or by an overall depletion of the body's potassium. Hypokalemia due to overall depletion tends to be a chronic phenomenon, while hypokalemia due to a shift in location tends to be a temporary disorder.

Causes and symptoms

Hypokalemia is most commonly caused by the use of **diuretics**. Diuretics are drugs that increase the excretion of water and salts in the urine. Diuretics are used to treat a number of medical conditions, including **hypertension** (high blood pressure), congestive heart failure, **liver disease**, and **kidney disease**. However, diuretic treatment can have the side effect of producing hypokalemia. In fact, the most common cause of hypokalemia in the elderly is the use of diuretics. The use of furosemide and thiazide, two commonly used diuretic drugs, can lead to hypokalemia. In contrast, spironolactone and triamterene are diuretics that do not provoke hypokalemia.

Other common causes of hypokalemia are excessive **diarrhea** or **vomiting**. Diarrhea and vomiting can be produced by infections of the gastrointestinal tract. Due to a variety of organisms, including bacteria, protozoa, and viruses, diarrhea is a major world health problem. It is responsible for about a quarter of the 10 million infant deaths that occur each year. Although nearly all of these deaths occur in the poorer parts of Asia and Africa, diarrheal diseases are a leading cause of infant **death** in the United States. Diarrhea results in various abnormalities, such as dehydration (loss in body water), **hyponatremia** (low sodium level in the blood), and hypokalemia.

Because of the need for potassium to control muscle action, hypokalemia can cause the heart to stop

beating. Young infants are especially at risk for death from this cause, especially where severe diarrhea continues for two weeks or longer. Diarrhea due to laxative **abuse** is an occasional cause of hypokalemia in the adolescent or adult. Enema abuse is a related cause of hypokalemia. Laxative abuse is especially difficult to diagnose and treat, because patients usually deny the practice. Up to 20% of persons complaining of chronic diarrhea practice laxative abuse. Laxative abuse is often part of eating disorders, such as anorexia nervosa or **bulimia nervosa**. Hypokalemia that occurs with these eating disorders may be life-threatening.

Surprisingly, the potassium loss that accompanies vomiting is only partly due to loss of potassium from the vomit. Vomiting also has the effect of provoking an increase in potassium loss in the urine. Vomiting expels acid from the mouth, and this loss of acid results in alkalization of the blood. (Alkalization of the blood means that the pH of the blood increases slightly.) An increased blood pH has a direct effect on the kidneys. Alkaline blood provokes the kidneys to release excessive amounts of potassium in the urine. So, severe and continual vomiting can cause excessive losses of potassium from the body and hypokalemia.

A third general cause of hypokalemia is prolonged fasting and **starvation**. In most people, after three weeks of **fasting**, blood serum potassium levels will decline to below 3.0 mM and result in severe hypokalemia. However, in some persons, serum potassium may be naturally maintained at about 3.0 mM, even after 100 days of fasting. During fasting, muscle is naturally broken down, and the muscle protein is converted to sugar (glucose) to supply to the brain the glucose which is essential for its functioning. Other organs are able to survive with a mixed supply of fat and glucose. The potassium within the muscle cell is released during the gradual process of muscle breakdown that occurs with starvation, and this can help counteract the trend to hypokalemia during starvation. Eating an unbalanced diet does not cause hypokalemia because most foods, such as fruits (especially bananas, oranges, and melons), vegetables, meat, milk, and cheese, are good sources of potassium. Only foods such as butter, margarine, vegetable oil, soda water, jelly beans, and hard candies are extremely poor in potassium.

Alcoholism occasionally results in hypokalemia. About one half of alcoholics hospitalized for withdrawal symptoms experience hypokalemia. The hypokalemia of alcoholics occurs for a variety of reasons, usually poor **nutrition**, vomiting, and diarrhea. Hypokalemia can also be caused by hyperaldosteronism; Cushing's syndrome; hereditary kidney defects

such as Liddle's syndrome, Bartter's syndrome, and Franconi's syndrome; and eating too much licorice.

Symptoms

Mild hypokalemia usually results in no symptoms, while moderate hypokalemia results in confusion, disorientation, weakness, and discomfort of muscles. On occasion, moderate hypokalemia causes cramps during **exercise**. Another symptom of moderate hypokalemia is a discomfort in the legs that is experienced while sitting still. The patient may experience an annoying feeling that can be relieved by shifting the positions of the legs or by stomping the feet on the floor. Severe hypokalemia results in extreme weakness of the body and, on occasion, in **paralysis**. The paralysis that occurs is "flaccid paralysis," or limpness. Paralysis of the muscles of the lungs results in death. Another dangerous result of severe hypokalemia is abnormal heart beat (arrhythmia) that can lead to death from cardiac arrest (cessation of heart beat). Moderate hypokalemia may be defined as serum potassium between 2.5 and 3.0 mM, while severe hypokalemia is defined as serum potassium under 2.5 mM.

Diagnosis

Hypokalemia can be measured by acquiring a sample of blood, preparing blood serum, and using a potassium sensitive electrode for measuring the concentration of potassium ions. Atomic absorption spectroscopy can also be used to measure the potassium ions. Since hypokalemia results in abnormalities in heart behavior, the electrocardiogram is usually used in the diagnosis of hypokalemia. The diagnosis of the cause of hypokalemia can be helped by measuring the potassium content of the urine. Where urinary potassium is under 25 mmoles per day, it means that the patient has experienced excessive losses of potassium due to diarrhea. The urinary potassium test is useful in cases where the patient is denying the practice of laxative or enema abuse. In contrast, where hypokalemia is due to the use of diuretic drugs, the content of potassium in the urine will be high—over 40 mmoles per day.

Treatment

In emergency situations, when severe hypokalemia is suspected, the patient should be put on a cardiac monitor, and respiratory status should be assessed. If laboratory test results show potassium levels below 2.5 mM, intravenous potassium should be given. In less urgent cases, potassium can be given orally in the pill form. Potassium supplements take the form of pills

KEY TERMS

Diuretics—A class of drugs that cause the kidneys to excrete excess sodium, water, and potassium.

pH—The unit of acid content is pH. The blood plasma normally has a pH of 7.35–7.45. Acidic blood has a pH value slightly less than pH 7.35. Alkaline blood has a pH value slightly greater than pH 7.45.

Potassium—An electrolyte necessary to proper functioning of the body.

containing potassium chloride (KCl), potassium bicarbonate (KHCO₃), and potassium acetate. Oral potassium chloride is the safest and most effective treatment for hypokalemia. Generally, the consumption of 40–80 mmoles of KCl per day is sufficient to correct the hypokalemia that results from diuretic therapy. For many people taking diuretics, potassium supplements are not necessary as long as they eat a balanced diet containing foods rich in potassium.

Prognosis

The prognosis for correcting hypokalemia is excellent. However, in emergency situations, where potassium is administered intravenously, the physician must be careful not to give too much potassium. The administration of potassium at high levels, or at a high rate, can lead to abnormally high levels of serum potassium.

Prevention

Hypokalemia is not a concern for healthy persons, since potassium is present in a great variety of foods. For patients taking diuretics, however, the American Dietetic Association recommends use of a high potassium diet. The American Dietetic Association states that if hypokalemia has already occurred, use of the high potassium diet alone may not reverse hypokalemia. Useful components of a high potassium diet include bananas, tomatoes, cantaloupes, figs, raisins, kidney beans, potatoes, and milk.

Resources

BOOKS

Brody, Tom. *Nutritional Biochemistry*. San Diego:Academic Press, 1998.

Tom Brody, PhD

Hypolipoproteinemia

Definition

Hypolipoproteinemia (or hypolipidemia) is the lack of fat in the blood.

Description

Although quite rare, hypolipoproteinemia is a serious condition. Blood absorbs fat from food in the intestine and transports it as a combined package with proteins and other chemicals like cholesterol. Much of the fat goes straight into the liver for processing. The cholesterol, a waste product, ends up in the bile. The proteins act as vessels, carrying the other chemicals around. These packages of fat, cholesterol, and proteins are called lipoproteins.

Causes and symptoms

Low blood fats can be the result of several diseases, or they can be a primary genetic disease with other associated abnormalities.

- **Malnutrition** is a lack of food, including fats, in the diet.
- Malabsorption is the inability of the bowel to absorb food, causing malnutrition.
- Anemia (too few red blood cells) and **hyperthyroidism** (too much thyroid hormone) also reduce blood fats.
- Rare genetic conditions called hypobetalipoproteinemia and abetalipoproteinemia cause malabsorption plus nerve, eye, and skin problems in early childhood.
- Tangier disease, causes only the cholesterol to be low. It also produces nerve and eye problems in children.

Symptoms are associated more closely with the cause rather than the actual low blood fats.

Diagnosis

Blood studies of the various fat particles help identify both the low and high fat diseases. These tests are often done after an overnight fast to prevent interference from fat just being absorbed from food. Fats and proteins are grouped together and described by density—high-density lipoproteins (HDL), low-density lipoproteins (LDL), and very low-density lipoproteins (VLDL). There are also much bigger particles

KEY TERMS

Cholesterol—A steroid alcohol found in animal cells and fluids.

Lipoprotein—Class of proteins that contain protein and lipid. The fundamental component of living cells.

called chylomicrons. Each contain different proportions of cholesterol, fats, and protein.

Treatment

Supplemental vitamin E helps children with the betalipoprotein deficiencies. There is no known treatment for Tangier disease. Treatment of the causes of the other forms of low blood fats reverses the condition.

Resources

BOOKS

Ginsberg, Henry N., and Ira J. Goldberg. "Disorders Of Lipoprotein Metabolism." In *Harrison's Principles of Internal Medicine*, edited by Anthony S. Fauci, et al. New York: McGraw-Hill, 1997.

J. Ricker Polsdorfer, MD

Hypomagnesemia see **Magnesium imbalance**

Hyponatremia

Definition

The normal concentration of sodium in the blood plasma is 136–145 mM. Hyponatremia occurs when sodium falls below 130 mM. Plasma sodium levels of 125 mM or less are dangerous and can result in seizures and **coma**.

Description

Sodium is an atom, or ion, that carries a single positive charge. The sodium ion may be abbreviated as Na^+ or as simply Na. Sodium can occur as a salt in a crystalline solid. Sodium chloride (NaCl), sodium phosphate (Na_2HPO_4) and sodium bicarbonate

(NaHCO_3) are commonly occurring salts. These salts can be dissolved in water or in juices of various foods. Dissolving involves the complete separation of ions, such as sodium and chloride in common table salt (NaCl).

About 40% of the body's sodium is contained in bone. Approximately 2–5% occurs within organs and cells and the remaining 55% is in blood plasma and other extracellular fluids. The amount of sodium in blood plasma is typically 140 mM, a much higher amount than is found in intracellular sodium (about 5 mM). This asymmetric distribution of sodium ions is essential for human life. It makes possible proper nerve conduction, the passage of various nutrients into cells, and the maintenance of blood pressure.

The body continually regulates its handling of sodium. When dietary sodium is too high or low, the intestines and kidneys respond to adjust concentrations to normal. During the course of a day, the intestines absorb dietary sodium while the kidneys excrete a nearly equal amount of sodium into the urine. If a low sodium diet is consumed, the intestines increase their efficiency of sodium absorption, and the kidneys reduce its release into urine.

The concentration of sodium in the blood plasma depends on two things: the total amount of sodium and water in arteries, veins, and capillaries (the circulatory system). The body uses separate mechanisms to regulate sodium and water, but they work together to correct blood pressure when it is too high or too low. Too low a concentration of sodium, or hyponatremia, can be corrected either by increasing sodium or by decreasing body water. The existence of separate mechanisms that regulate sodium concentration account for the fact that there are numerous diseases that can cause hyponatremia, including diseases of the kidney, pituitary gland, and hypothalamus.

Causes and symptoms

Hyponatremia can be caused by abnormal consumption or excretion of dietary sodium or water and by diseases that impair the body's ability to regulate them. Maintenance of a low salt diet for many months or excessive sweat loss during a race on a hot day can present a challenge to the body to conserve adequate sodium levels. While these conditions alone are not likely to cause hyponatremia, it can occur under special circumstances. For example, hyponatremia often occurs in patients taking diuretic drugs who maintain a low sodium diet. This is especially of concern in elderly patients, who have a reduced ability to regulate the concentrations of various nutrients in the

bloodstream. Diuretic drugs that frequently cause hyponatremia include furosemide (Lasix), bumetanide (Bumex), and most commonly, the thiazides. Diuretics enhance the excretion of sodium into the urine, with the goal of correcting high blood pressure. However, too much sodium excretion can result in hyponatremia. Usually only mild hyponatremia occurs in patients taking **diuretics**, but when combined with a low sodium diet or with the excessive drinking of water, severe hyponatremia can develop.

Severe and prolonged **diarrhea** can also cause hyponatremia. Severe diarrhea, causing the daily output of 8–10 liters of fluid from the large intestines, results in the loss of large amounts of water, sodium, and various nutrients. Some diarrheal diseases release particularly large quantities of sodium and are therefore most likely to cause hyponatremia.

Drinking excess water sometimes causes hyponatremia, because the absorption of water into the bloodstream can dilute the sodium in the blood. This cause of hyponatremia is rare, but has been found in psychotic patients who compulsively drink more than 20 liters of water per day. Excessive drinking of beer, which is mainly water and low in sodium, can also produce hyponatremia when combined with a poor diet.

Marathon running, under certain conditions, leads to hyponatremia. Races of 25–50 miles can result in the loss of great quantities (8 to 10 liters) of sweat, which contains both sodium and water. Studies show that about 30% of marathon runners experience mild hyponatremia during a race. But runners who consume only pure water during a race can develop severe hyponatremia because the drinking water dilutes the sodium in the bloodstream. Such runners may experience neurological disorders as a result of the severe hyponatremia and require emergency treatment.

Hyponatremia also develops from disorders in organs that control the body's regulation of sodium or water. The adrenal gland secretes a hormone called aldosterone that travels to the kidney, where it causes the kidney to retain sodium by not excreting it into the urine. **Addison's disease** causes hyponatremia as a result of low levels of aldosterone due to damage to the adrenal gland. The hypothalamus and pituitary gland are also involved in sodium regulation by making and releasing vasopressin, known as anti-diuretic hormone, into the bloodstream. Like aldosterone, vasopressin acts in the kidney, but it causes it to reduce the amount of water released into urine. With more vasopressin production, the body conserves water, resulting in a lower concentration of plasma sodium. Certain types of **cancer** cells produce vasopressin, leading to hyponatremia.

KEY TERMS

Blood plasma and serum—Blood plasma, or plasma, is prepared by obtaining a sample of blood and removing the blood cells. The red blood cells and white blood cells are removed by spinning with a centrifuge. Chemicals are added to prevent the blood's natural tendency to clot. If these chemicals include sodium, then a false measurement of plasma sodium content will result. Serum is prepared by obtaining a blood sample, allowing formation of the blood clot, and removing the clot using a centrifuge. Both plasma and serum are light yellow in color.

Symptoms of moderate hyponatremia include tiredness, disorientation, **headache**, **muscle cramps**, and **nausea**. Severe hyponatremia can lead to seizures and coma. These neurological symptoms are thought to result from the movement of water into brain cells, causing them to swell and disrupt their functioning.

In most cases of hyponatremia, doctors are primarily concerned with discovering the underlying disease causing the decline in plasma sodium levels. **Death** that occurs during hyponatremia is usually due to other features of the disease rather than to the hyponatremia itself.

Diagnosis

Hyponatremia is diagnosed by acquiring a blood sample, preparing plasma, and using a sodium-sensitive electrode for measuring the concentration of sodium ions. Unless the cause is obvious, a variety of tests are subsequently run to determine if sodium was lost from the urine, diarrhea, or from **vomiting**. Tests are also used to determine abnormalities in aldosterone or vasopressin levels. The patient's diet and use of diuretics must also be considered.

Treatment

Severe hyponatremia can be treated by infusing a solution of 5% sodium chloride in water into the bloodstream. Moderate hyponatremia due to use of diuretics or an abnormal increase in vasopressin is often treated by instructions to drink less water each day. Hyponatremia due to adrenal gland insufficiency is treated with hormone injections.

Prognosis

Hyponatremia is just one manifestation of a variety of disorders. While hyponatremia can easily be corrected, the prognosis for the underlying condition that causes it varies.

Prevention

Patients who take diuretic medications must be checked regularly for the development of hyponatremia.

Resources

PERIODICALS

Fried, L. F., and P. M. Palevsky. "Hyponatremia and Hyponatremia." *Medical Clinics of North America* 81 (1997): 585-609.

Tom Brody, PhD

Hypoparathyroidism

Definition

Hypoparathyroidism is the result of a decrease in production of parathyroid hormones by the parathyroid glands located behind the thyroid glands in the neck. The result is a low level of calcium in the blood.

Description

Parathyroid glands consist of four pea-shaped glands located on the back and side of the thyroid gland. The gland produces parathyroid hormone which, along with vitamin D and calcitonin, are important for the regulation of the calcium level in the body. Hypoparathyroidism affects both males and females of all ages.

Causes and symptoms

The accidental removal of the parathyroid glands during neck surgery is the most frequent cause of hypoparathyroidism. Complications of surgery on the parathyroid glands is another common cause of this disorder. There is the possibility of autoimmune genetic disorders causing hypoparathyroidism such as Hashimoto's **thyroiditis**, **pernicious anemia**, and **Addison's disease**. The destruction of the gland by radiation is a rare cause of hypoparathyroidism. Occasionally, the parathyroids are absent at birth

causing low calcium levels and possible convulsions in the newborn. Symptoms in the advanced and continuous stages of hypoparathyroidism include splitting of the nails, inadequate tooth development and **mental retardation** in children, and seizures.

Abnormal low levels of calcium result in irritability of nerves, causing **numbness and tingling** of the hands and feet, with painful-cramp like muscle spasms known as tetany. Laryngeal spasms may also occur causing respiratory obstruction.

Diagnosis

Diagnostic measures begin with the individual's own observation of symptoms. A thorough medical history and **physical examination** by a physician is always required for an accurate diagnosis. The general practitioner may refer the individual to an endocrinologist, a medical specialist who studies the function of the parathyroid glands as well as other hormone producing glands. Laboratory studies include blood and urine tests to help determine phosphate and calcium levels. X rays are useful to determine any abnormalities in bone density associated with abnormal calcium levels. These autoimmune disorders may accompany hypoparathyroidism, but are not an actual cause of it.

Treatment

In the event of severe **muscle spasms**, hospitalization may be warranted for calcium injections. Raising carbon-dioxide levels in the blood, which can decrease muscle spasms, may be achieved in immediate situations by placing a paper bag over the mouth and blowing into it to "reuse" each breath. It is critical to obtain timely periodic laboratory tests to check calcium levels. A high calcium, low-phosphorous diet may be of significance and is directed by the physician or dietitian.

Prognosis

Presently hypoparathyroidism is considered incurable. The disorder requires lifelong replacement therapy to control symptoms. Medical research however, continues to search for a cure.

Prevention

There are no specific preventive measures for hypoparathyroidism. However, careful surgical techniques are critical to reduce the risk of damage to the gland during surgery.

KEY TERMS

Addison's disease—A disease caused by partial or total failure of adrenocortical (relating to, or derived from the adrenal gland) function, which is characterized by a bronze-like pigmentation of the skin and mucous membranes, anemia, weakness, and low blood pressure.

Autoimmunity—A condition by which the body's defense mechanism attacks itself.

Calcitonin—A hormone produced by the thyroid gland in human beings that lowers plasma calcium and phosphate levels without increasing calcium accumulation.

Hashimoto's thyroiditis—The self destruction of the thyroid cells from an autoimmune disorder.

Hormones—A substance produced by one tissue and conveyed by the bloodstream to another to affect physiological activity, such as growth or metabolism.

Pernicious anemia—A severe anemia most often affecting older adults, caused by failure of the stomach to absorb vitamin B12 and characterized by abnormally large red blood cells, gastrointestinal disturbances, and lesions of the spinal cord.

Resources

ORGANIZATIONS

American Medical Association. 515 N. State St., Chicago, IL 60612. (312) 464-5000. <<http://www.ama-assn.org>>.

Jeffrey P. Larson, RPT

Hypophysectomy

Definition

Hypophysectomy or hypophysis is the removal of the pituitary gland.

Purpose

The pituitary gland is in the middle of the head. Removing this master gland is a drastic step that was taken in the extreme circumstance of two cancers that had escaped all other forms of treatment. Cancers of the female breast and male prostate grow faster in the presence of sex hormones. It used to be that sex hormones

could be suppressed only by removing their source, the glands that made them. After the gonads were removed, some cancers continued to grow, so other stimulants to their growth had to be removed. At this point, some cancer specialists turned to the pituitary.

With the development of new therapeutic agents and methods, especially new ways to manipulate hormones without removing their source, this type of endocrine surgery has been largely relegated to history. However, tumors develop in the pituitary gland that require removal. Here, the idea is to remove the tumor but partially preserve the gland.

Description

There are several surgical approaches to the pituitary. The surgeon will choose the best one for the specific procedure. The pituitary lies directly behind the nose, and access through the nose or the sinuses is often the best approach. Opening the skull and lifting the frontal lobe of the brain will expose the delicate neck of the pituitary gland. This approach works best if tumors have extended above the pituitary fossa (the cavity in which the gland lies).

Newer surgical methods using technology have made other approaches possible. Stereotaxis is a three-dimensional aiming technique using x rays or scans for guidance. Instruments can be placed in the brain with pinpoint accuracy through tiny holes in the skull. These instruments can then manipulate brain tissue, either to destroy it or remove it. Stereotaxis is also used to direct radiation with similar precision using a gamma knife. Access to some brain lesions can be gained through the blood vessels using tiny tubes and wires guided by x rays.

Preparation

Pituitary surgery is performed by neurosurgeons deep inside the skull. All the patient can do to prepare is keep as healthy as possible and trust that the surgeon will do his usual excellent job. Informed surgical consent is important so that the patient is fully confident of the need for surgery and the expected outcome.

Aftercare

Routine post-operative care is required. In addition, pituitary function will be assessed.

Risks

The risks of surgery are multiple. Procedures are painstakingly selected to minimize risk and maximize benefit. Unique to surgery on the pituitary is the risk of

KEY TERMS

Endocrine system—Group of glands and parts of glands that control metabolic activity. Pituitary, thyroid, adrenals, ovaries, and testes are all part of the endocrine system.

Hormone—A chemical made in one place that has effects in distant places in the body. Hormone production is usually triggered by the pituitary gland.

destroying the entire gland and leaving the entire endocrine system without guidance. This used to be the whole purpose of hypophysectomy. After the procedure, the endocrinologist, a physician specializing in the study and care of the endocrine system, would provide the patient with all the hormones needed. Patients with no pituitary function did and still do quite well because of the available hormone replacements.

Normal results

Complete removal of the pituitary was the goal for cancer treatment. Today, removal of tumors with preservation of the gland is the goal.

Abnormal results

Tumors may not be completely removed, due to their attachment to vital structures.

Resources

BOOKS

Biller, Beverly M. K., and Gilbert H. Daniels. "Neuroendocrine Regulation and Diseases of the Anterior Pituitary and Hypothalamus." In *Harrison's Principles of Internal Medicine*, edited by Anthony S. Fauci, et al. New York: McGraw-Hill, 1997.

J. Ricker Polsdorfer, MD

Hypopigmentation see **Albinism; Vitiligo**

Hypopituitarism

Definition

Hypopituitarism is loss of function in an endocrine gland due to failure of the pituitary gland to secrete hormones which stimulate that gland's

function. The pituitary gland is located at the base of the brain. Patients diagnosed with hypopituitarism may be deficient in one single hormone, several hormones, or have complete pituitary failure.

Description

The pituitary is a pea-sized gland located at the base of the brain, and surrounded by bone. The hypothalamus, another endocrine organ in the brain, controls the function of the pituitary gland by providing "hormonal orders." In turn, the pituitary gland regulates the many hormones that control various functions and organs within the body. The posterior pituitary acts as a sort of storage area for the hypothalamus and passes on hormones that control function of the muscles and kidneys. The anterior pituitary produces its own hormones which help to regulate several endocrine functions.

In hypopituitarism, something interferes with the production and release of these hormones, thus affecting the function of the target gland. Commonly affected hormones may include:

Gonadotropin deficiency

Gonadotropin deficiency involves two distinct hormones affecting the reproductive system. Luteinizing hormone (LH) stimulates the testes in men and the ovaries in women. This deficiency can affect fertility in men and women and menstruation in women. Follicle-stimulating hormone (FSH) has similar effects to LH.

Thyroid stimulating hormone deficiency

Thyroid stimulating hormone (TSH) is involved in stimulation of the thyroid gland. A lack of stimulation in the gland leads to **hypothyroidism**.

Adrenocorticotrophic hormone deficiency

Also known as corticotropin, adrenocorticotrophic hormone (ACTH) stimulates the adrenal gland to produce a hormone similar to cortisone, called cortisol. The loss of this hormone can lead to serious problems.

Growth hormone deficiency

Growth hormone (GH) regulates the body's growth. Patients who lose supply of this hormone before physical maturity will suffer impaired growth. Loss of the hormone can also affect adults.

Other hormone deficiencies

Prolactin stimulates the female breast to produce milk. A hormone produced by the posterior pituitary, antidiuretic hormone (ADH), controls the function of the kidneys. When this hormone is deficient, **diabetes insipidus** can result. However, patients with hypopituitarism rarely suffer ADH deficiency, unless the hypopituitarism is the result of hypothalamus disease.

Multiple hormone deficiencies

Deficiency of a single pituitary hormone occurs less commonly than deficiency of more than one hormone. Sometimes referred to as progressive pituitary hormone deficiency or partial hypopituitarism, there is usually a predictable order of hormone loss. Generally, growth hormone is lost first, then luteinizing hormone deficiency follows. The loss of follicle-stimulating hormone, thyroid stimulating hormone and adrenocorticotrophic hormones follow much later. The progressive loss of pituitary hormone secretion is usually a slow process, which can occur over a period of months or years. Hypopituitarism does occasionally start suddenly with rapid onset of symptoms.

Panhypopituitarism

This condition represents the loss of all hormones released by the anterior pituitary gland. Panhypopituitarism is also known as complete pituitary failure.

Causes and symptoms

There are three major mechanisms which lead to the development of hypopituitarism. The first involves decreased release of hypothalamic hormones that stimulate pituitary function. The cause of decreased hypothalamic function may be congenital or acquired through interference such as tumors, inflammation, infection, mass lesions or interruption of blood supply. A second category of causes is any event or mass which interrupts the delivery of hormones from the hypothalamus. These may include particular tumors and aneurysms. Damage to the pituitary stalk from injury or surgery can also lead to hypopituitarism.

The third cause of hypopituitarism is damage to the pituitary gland cells. Destroyed cells can not produce the pituitary hormones that would normally be secreted by the gland. Cells may be destroyed by a number of tumors and diseases. Hypopituitarism is often caused by tumors, the most common of which is pituitary adenoma.

Symptoms of hypopituitarism vary with the affected hormones and severity of deficiency.

Frequently, patients have had years of symptoms that were nonspecific until a major illness or stress occurred. Overall symptoms may include **fatigue**, sensitivity to cold, weakness, decreased appetite, weight loss and abdominal **pain**. Low blood pressure, **headache** and visual disturbances are other associated symptoms.

Gonadotropin deficiency

Symptoms specific to this hormone deficiency include decreased interest in sex for women and **infertility** in women and men. Women may also have premature cessation of menstruation, hot flashes, vaginal dryness and pain during intercourse. Women who are postmenopausal will not have obvious symptoms such as these and may first present with headache or loss of vision. Men may also suffer sexual dysfunction as a result of gonadotropin deficiency. In acquired gonadotropin deficiency, both men and women may notice loss of body hair.

Thyroid stimulating hormone deficiency

Intolerance to cold, fatigue, weight gain, **constipation** and pale, waxy and dry skin indicate thyroid hormone deficiency.

Adrenocorticotrophic hormone deficiency

Symptoms of ACTH deficiency include fatigue, weakness, weight loss and low blood pressure. **Nausea**, pale skin and loss of pubic and armpit hair in women may also indicate deficiency of ACTH.

Growth hormone deficiency

In children, growth hormone deficiency will result in short stature and growth retardation. Symptoms such as **obesity** and skin wrinkling may or may not show in adults and normal release of growth hormone normally declines with age.

Other hormone deficiencies

Prolactin deficiency is rare and is the result of partial or generalized anterior pituitary failure. When present, the symptom is absence of milk production in women. There are no known symptoms for men. ADH deficiency may produce symptoms of diabetes insipidus, such as excessive thirst and frequent urination.

Multiple hormone deficiencies

Patients with multiple hormone deficiencies will show symptoms of one or more specific hormone

deficiencies or some of the generalized symptoms listed above.

Panhypopituitarism

The absence of any pituitary function should show symptoms of one or all of the specific hormone deficiencies. In addition to those symptoms, patients may have dry, pale skin that is finely textured. The face may appear finely wrinkled and contain a disinterested expression.

Diagnosis

Once the diagnosis of a single hormone deficiency is made, it is strongly recommended that tests for other hormone deficiencies be conducted.

Gonadotropin deficiency

The detection of low levels of gonadotropin can be accomplished through simple blood tests which measure luteinizing hormone and follicle-stimulating hormone, simultaneously with gonadal steroid levels. The combination of results can indicate to a physician if the cause of decreased hormone levels or function belongs to hypopituitarism or some sort of primary gonadal failure. Diagnosis will vary among men and women.

Thyroid stimulating hormone deficiency

Laboratory tests measuring thyroid function can help determine a diagnosis of TSH deficiency. The commonly used tests are T4 and TSH measurement done simultaneously to determine the reserve, or pool, of thyroid-stimulating hormone.

Adrenocorticotopic hormone deficiency

An insulin tolerance test may be given to determine if cortisol levels rise when **hypoglycemia** is induced. If they do not rise, there is insufficient reserve of cortisol, indicating an ACTH deficiency. If the insulin tolerance test is not safe for a particular patient, a glucagon test offers similar results. A CRH (corticotropin-releasing hormone) test may also be given. It involves injection of CRH to measure, through regularly drawn blood samples, a resulting rise in ACTH and cortisol. Other tests which stimulate ACTH may be ordered.

Growth hormone deficiency

Growth hormone deficiency is measured through the use of insulin-like growth factor I tests, which

measure growth factors that are dependent on growth hormones. Sleep and **exercise** studies may also be used to test for growth hormone deficiency, since these activities are known to stimulate growth hormone secretion. Several drugs also induce secretion of growth hormone and may be given to measure hormone response. The standard test for growth hormone deficiency is the insulin-induced hypoglycemia test. This test does carry some risk from the induced hypoglycemia. Other tests include an arginine infusion test, clonidine test and growth-hormone releasing hormone test.

Other hormone deficiencies

If a test calculates normal levels of prolactin, deficiency of the hormone is eliminated as a diagnosis. A TRH (thyrotropin-releasing hormone) simulation test can determine prolactin levels. A number of tests are available to detect ADH levels and to determine diagnosis of diabetes insipidus.

Multiple and general hypopituitarism tests

Physicians should be aware that nonspecific symptoms can indicate deficiency of one or more hormones and should conduct a thorough clinical history. In general, diagnosis of hypopituitarism can be accomplished with a combination of dynamic tests and simple blood tests, as well as imaging exams. Most of these tests can be conducted in an outpatient lab or radiology facility. **Magnetic resonance imaging (MRI)** exams with gadolinium contrast enhancement are preferred imaging exams to study the region of the hypothalamus and pituitary gland. When MRI is not available, a properly conducted computed tomography scan (CT scan) exam can take its place. These exams can demonstrate a tumor or other mass, which may be interfering with pituitary function.

Panhypopituitarism

The insulin-induced hypoglycemia, or insulin tolerance test, which is used to determine specific hormone deficiencies, is an excellent test to diagnose panhypopituitarism. This test can reveal levels of growth hormone, ACTH (cortisol) and prolactin deficiency. The presence of insufficient levels of all of these hormones is a good indication of complete pituitary failure. Imaging studies and clinical history are also important.

Treatment

Treatment differs widely, depending on the age and sex of the patient, severity of the deficiency, the number of hormones involved, and even the

underlying cause of the hypopituitarism. Immediate hormone replacement is generally administered to replace the specific deficient hormone. Patient education is encouraged to help patients manage the impact of their hormone deficiency on daily life. For instance, certain illnesses, accidents or surgical procedures may have adverse complications due to hypopituitarism.

Gonadotropin deficiency

Replacement of gonadal steroids is common treatment for LH and FSH deficiency. Estrogen for women and testosterone for men will be prescribed in the lowest effective dosage possible, since there can be complications to this therapy. To correct women's loss of libido, small doses of androgens may be prescribed. To restore fertility in men, regular hormone injections may be required. Male and female patients whose hypopituitarism results from hypothalamic disease may be successfully treated with a hypothalamic releasing hormone (GnRH), which can restore gonadal function and fertility.

Thyroid stimulating hormone deficiency

In patients who have hypothyroidism, the function of the adrenal glands will be tested and treated with steroids before administering thyroid hormone replacement.

Adrenocorticotopic hormone deficiency

Hydrocortisone or cortisone in divided doses may be given to replace this hormone deficiency. Most patients require 20 mg or less of hydrocortisone per day.

Growth hormone deficiency

It is essential to treat children suffering from growth hormone deficiency. The effectiveness of growth hormone therapy in adults, particularly elderly adults, is not as well documented. It is thought to help restore normal muscle to fat ratios. Growth hormone is an expensive and cautiously prescribed treatment.

Treatment of multiple deficiencies and panhypopituitarism

The treatment of hypopituitarism is usually very straightforward, but must normally continue for the remainder of the patient's life. Some patients may receive treatment with GnRH, the hypothalamic hormone. In most cases, treatment will be based on the specific deficiency demonstrated. Patients with hypopituitarism should be followed regularly to measure treatment effectiveness and to avoid overtreatment with

KEY TERMS

Adenoma—A benign (not threatening or cancerous) tumor that originates in a gland.

Androgen—A hormone that usually stimulates the sex hormones of the male.

Congenital—Present at birth.

Diabetes insipidus—A disorder originating in the pituitary gland which is characterized by excessive thirst and urination.

Endocrine—Refers to the system of internal secretion of substances into the body system from glands.

Hypoglycemia—Abnormal decrease of sugar in the blood.

Hypothyroidism—Deficient activity of the thyroid gland and resulting loss of energy.

hormone therapy. If the cause of the disorder is a tumor or lesion, radiation or surgical removal are treatment options. Successful removal may reverse the hypopituitarism. However, even after removal of the mass, hormone replacement therapy may still be necessary.

Prognosis

The prognosis for most patients with hypopituitarism is excellent. As long as therapy is continued, many experience normal life spans. However, hypopituitarism is usually a permanent condition and prognosis depends on the primary cause of the disorder. It can be potentially life threatening, particularly when acute hypopituitarism occurs as a result of a large pituitary tumor. Morbidity from the disease has increased, although the cause is not known. It is possible that increased morbidity and **death** are due to overtreatment with hormones. Any time that recovery of pituitary function can occur is preferred to lifelong hormone therapy.

Prevention

There is no known prevention of hypopituitarism, except for prevention of damage to the pituitary/hypothalamic area from injury.

Resources

ORGANIZATIONS

Alliance for Genetic Support Groups. 35 Wisconsin Circle, Suite 440. Chevy Chase, MD 20815-7015. <<http://www.medhelp.org/geneticalliance>>.

Human Growth Foundation. 997 Glen Cove Ave., Glen Head, NY 11545. (800) 451-6434. <http://www.hgfound.org>.

OTHER

HealthAnswers.com <http://www.healthanswers.com>.

Teresa Norris, RN

Hypoplastic anemia see **Aplastic anemia**

Hypospadias and epispadias

Definition

Hypospadias is a congenital defect, primarily of males, in which the urethra opens on the underside (ventrum) of the penis. It is one of the most common congenital abnormalities in the United States, occurring in about one of every 125 live male births. The corresponding defect in females is an opening of the urethra into the vagina and is rare.

Epispadias (also called bladder exstrophy) is a congenital defect of males in which the urethra opens on the upper surface (dorsum) of the penis. The corresponding defect in females is a fissure in the upper wall of the urethra and is quite rare.

Description

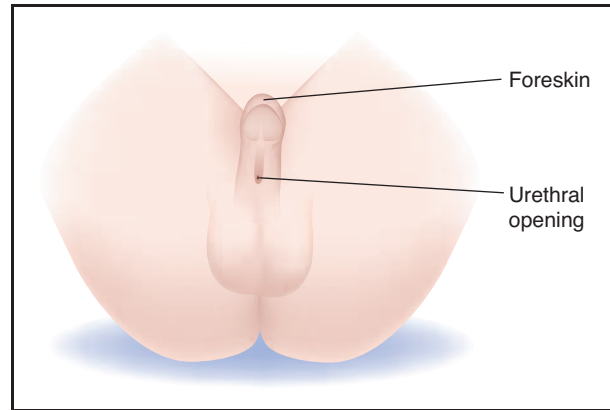
In a male, the external opening of the urinary tract (external meatus) is normally located at the tip of the penis. In a female, it is normally located between the clitoris and the vagina.

In males with hypospadias, the urethra opens on the inferior surface or underside of the penis. In females with hypospadias, the urethra opens into the cavity of the vagina.

In males with epispadias, the urethra opens on the superior surface or upper side of the penis. In females with epispadias, there is a crack or fissure in the wall of the urethra and out of the body through an opening in the skin above the clitoris.

During the embryological development of males, a groove of tissue folds inward and then fuses to form a tube that becomes the urethra. Hypospadias occurs when the tube does not form or does not fuse completely. Epispadias is due to a defect in the tissue that folds inward to form the urethra.

During the development of a female, similar processes occur to form the urethra. The problem usually



In hypospadias, the urethra opens along the penile shaft rather than at the penile tip. (Illustration by Argosy Inc.)

is insufficient length of the tube that becomes the urethra. As a result, the urethra opens in an abnormal location, resulting in a hypospadias. Occasionally, fissures form in the bladder. These may extend to the surface of the abdomen and fuse with the adjacent skin. This most often is identified as a defect in the bladder although it is technically an epispadias.

Hypospadias in males generally occur alone. Female hypospadias may be associated with abnormalities of the genital tract, since the urinary and genital tracts are formed in the same embryonic process.

Because it represents incomplete development of the penis, some experts think that insufficient male hormone may be responsible for hypospadias.

In males, the incidence of hypospadias is approximately one per 250 to 300 live births. Epispadias is much less common, having an incidence of about one per 100,000 live male births.

In females, hypospadias is much less common than in males. It appears about once in every 500,000 live female births. Epispadias is even rarer. Reliable estimates of the prevalence of epispadias in females are not available. Epispadias in females is often diagnosed and recorded as a bladder anomaly.

Causes and symptoms

Hypospadias and epispadias are congenital defects of the urinary tract. This means they occur during intrauterine development. There is no genetic basis for the defects. Specific causes for hypospadias are not known. This means that blood relatives do not have increased chances of developing them. Reports have shown some rise in prevalence of hypospadias among offspring of mothers who work in certain

occupations where they may be exposed to chemicals that disrupt the endocrine system. However, a large trial ending in 2003 showed that aside from a slight increased risk among women who were hairdressers from 1992-1996, there is no evidence that maternal occupation or certain chemical exposure increases risk of hypospadias. The role of chemicals in the development of the defect remains uncertain.

Concern was once raised that use of the antihistamine loratadine (Claritin) early in **pregnancy** might cause hypospadias. However, a national clinical trial revealed in 2004 that there was no link between the drug and risk of second- or third-degree hypospadias.

Hypospadias usually is not associated with other defects of the penis or urethra. In males, it can occur at any site along the underside of the penis. In females, the urethra exits the body in an abnormal location. This usually is due to inadequate length of the urethra.

Epispadias is associated with bladder abnormalities. In females, the front wall of the bladder does not fuse or close. The bladder fissure may extend to the external abdominal wall. In such a rare case, the front of the pelvis also is widely separated. In males, the bladder fissure extends into the urethra and simply becomes an opening somewhere along the upper surface of the penis.

Hypospadias is associated with difficulty in assigning gender to babies. This occurs when gender is not obvious at birth because of deformities in the sex organs.

Diagnosis

Male external urinary tract defects are discovered at birth during the first detailed examination of the newborn. Female urethral defects may not be discovered for some time due to the difficulty in viewing the infant vagina.

Treatment

Surgery is the treatment of choice for both hypospadias and epispadias. All surgical repairs should be undertaken early and completed without delay. This minimizes psychological trauma.

In males with hypospadias, one surgery usually is sufficient to repair the defect. With more complicated hypospadias (more than one abnormally situated urethral opening), multiple surgeries may be required. In females with hypospadias, surgical repair technically is more complicated but can usually be completed in a brief interval of time.

Repairing an epispadias is more difficult. In males, this may involve other structures in the penis. Males should not be circumcised since the foreskin often is needed for the repair. Unfortunately, choices may be required that affect the ability to inseminate a female partner. Reproduction requires that the urethral meatus be close to the tip of the penis. Cosmetic appearance and ability to urinate (urinary continence) usually are the primary goals. Surgery for these defects is successful 70 to 80% of the time. Modern treatment of complete male epispadias allows for an excellent genital appearance and achievement of urinary continence.

In females, repair of epispadias may require multiple surgical procedures. Urinary continence and cosmetic appearance are the usual primary considerations. Urinary continence usually is achieved although cosmetic appearance may be somewhat compromised. Fertility is not usually affected. Repair rates that are similar or better than those for males usually can be achieved for females.

Hypospadias in both males and females is more of a nuisance and hindrance to reproduction than a threat to health. If surgery is not an option, the condition may be allowed to persist. This usually leads to an increased risk of infections in the lower urinary tract.

Prognosis

With adequate surgical repair, most males with simple hypospadias can lead normal lives with a penis that appears and functions in a normal manner. This includes fathering children. Females with simple hypospadias also have normal lives, including conceiving and bearing children.

The prognosis for epispadias depends on the extent of the defect. Most males with relatively minor epispadias lead normal lives, including fathering children. As the extent of the defect increases, surgical reconstruction generally is acceptable. However, many of these men are unable to conceive children. Most epispadias in females can be surgically repaired. The chances of residual disfigurement increase as the extent of the epispadias increases. Fertility in females is not generally affected by epispadias.

In hypospadias, the urethra opens along the penile shaft rather than at the penile tip.

Resources

BOOKS

Nelson, Waldo E., et al., editors. "Anomalies of the Bladder" In *Nelson Textbook of Pediatrics*. Philadelphia: W. B. Saunders, 2000, pp. 1639-1642.

KEY TERMS

Bladder—This is the organ that stores urine after it flows out of the kidneys and through the ureters.

Circumcision—The surgical removal of the foreskin of the penis.

Continence—Normal function of the urinary bladder and urethra, allowing fluid flow during urination and completely stopping flow at other times.

External meatus—The external opening through which urine and seminal fluid (in males only) leave the body.

Genital tract—The organs involved in reproduction. In a male, they include the penis, testicles, prostate and various tubular structures to transport seminal fluid and sperm. In a female, they include the clitoris, vagina, cervix, uterus, fallopian tubes and ovaries.

Urethra—The tubular portion of the urinary tract connecting the bladder and external meatus through which urine passes. In males, seminal fluid and sperm also pass through the urethra.

Nelson, Waldo E., et al., editors. "Anomalies of the penis and urethra" In *Nelson Textbook of Pediatrics*. Philadelphia: W. B. Saunders, 2000. pp. 1645-1650.

PERIODICALS

Kubetin, Sally Koch. "Hypospadias, Loratidine Use in Pregnancy: No Link." *Pediatric News* July 2004.

"Molecular Epidemiology of Hypospadias: Genetic and Environmental Risk Factors." *Health & Medicine Week* December 15, 2003: 424.

Vrijheid, M., et al. "Risk of Hypospadias in Relation to Maternal Occupational Exposure to Potential Endocrine Disrupting Chemicals." *Occupational and Environmental Medicine* August 2003: 543-548.

ORGANIZATIONS

Association for the Bladder Exstrophy Community. PO Box 1472, Wake Forest, NC 27588-1472. (919) 624-9447. <<http://www.bladderexstrophy.com/support.htm>>.

Hypospadias Association of America. 4950 S. Yosemite Street, Box F2-156, Greenwood Village, CO 80111. hypospadiasasn@yahoo.com. <<http://www.hypospadias.net>>.

Support for Parents with Hypospadias Boys. <<http://clubs.yahoo.com/clubs/mumswithhypospadiaskids>>.

University of California - San Francisco. <<http://itsa.ucsf.edu/~uroweb/Uro/hypospadias/index.html>>.

OTHER

Hatch, David A., MD. "Abnormal Development of the Penis and Male Urethra." *Genitourinary Development*.

<<http://www.meddean.luc.edu/lumen/MedEd/urol-ogy/abnpendv.htm>>.

"Hypospadias." *Atlas of Congenital Deformities of the External Genitalia*. <<http://www.atlasperovic.com/contents/9.htm>>.

"Hypospadias." Columbia Presbyterian Hospital. <<http://cpmcnet.cpmc.columbia.edu/dept/urology/pediatric/hypospadias.html>>.

"Hypospadias." University of Michigan. <<http://www.urology.med.umich.edu/clinic/pediatric/hypospadias.html>>.

Johns Hopkins University Pediatric Urology Center.

"Epispadias." Johns Hopkins Exstrophy Database. <<http://www.med.jhu.edu/pediurology/pediatric/exstrophy/database/web4d.html>>.

The Penis.com. <<http://www.the-penis.com/hypospadias.html>>.

Society for Pediatric Urology. <<http://www.spu.org/>>.

L. Fleming Fallon, Jr., MD, PhD, DrPH
Teresa G. Odle

Hypotension

Definition

Hypotension is the medical term for low blood pressure.

Description

The pressure of the blood in the arteries rises and falls as the heart and muscles handle demands of daily living, such as **exercise**, sleep and **stress**. Some healthy people have blood pressure well below the average for their age, even though they have a completely normal heart and blood vessels. This is often true of athletes who are in superior shape. The term "hypotension" is usually used only when blood pressure has fallen so far that enough blood can no longer reach the brain, causing **dizziness** and **fainting**.

Causes and symptoms

Postural hypotension is the most common type of low blood pressure. In this condition, symptoms appear after a person sits up or stands quickly. In normal people, the cardiovascular system must make a quick adjustment to raise blood pressure slightly to account for the change in position. For those with postural hypotension, the blood pressure adjustment is not adequate or it doesn't happen. Postural hypotension may occur if someone is taking certain drugs

or medicine for high blood pressure. It also happens to diabetics when nerve damage has disrupted the reflexes that control blood pressure.

Many people have a chronic problem with low blood pressure that is not particularly serious. This may include people who require certain medications, who are pregnant, have bad veins, or have arteriosclerosis (hardening of the arteries).

The most serious problem with low blood pressure occurs when there is a sudden drop, which can be life-threatening due to widespread ischemia (insufficient supply of blood to an organ due to blockage in an artery). This type of low blood pressure may be due to a wide variety of causes, including:

- trauma with extensive blood loss
- serious burns
- shock from various causes (e.g. anaphylaxis)
- heart attack
- adrenal failure (Addisonian crisis)
- cancer
- severe fever
- serious infection (septicemia)

Diagnosis

Blood pressure is a measure of the pressure in the arteries created by the heart contracting. During the day, a normal person's blood pressure changes constantly, depending on activity. Low blood pressure can be diagnosed by taking the blood pressure with a sphygmomanometer. This is a device with a soft rubber cuff that is inflated around the upper arm until it's tight enough to stop blood flow. The cuff is then slowly deflated until the health care worker, listening to the artery in the arm with a stethoscope, can hear the blood first as a beat forcing its way along the artery. This is the systolic pressure. The cuff is then deflated more until the beat disappears and the blood flows steadily through the open artery; this gives the diastolic pressure.

Blood pressure is recorded as systolic (higher) and diastolic (lower) pressures. A healthy young adult has a blood pressure of about 110/75, which typically rises with age to about 140/90 by age 60 (a reading now considered mildly elevated).

Treatment

Treatment of low blood pressure depends on the underlying cause, which can usually be resolved. For

KEY TERMS

Arteriosclerosis—A group of disorders that causes thickening and loss of elasticity in artery walls.

those people with postural hypotension, a medication adjustment may help prevent the problem. These individuals may find that rising more slowly, or getting out of bed in slow stages, helps the problem. Low blood pressure with no other symptoms does not need to be treated.

Prognosis

Low blood pressure as a result of injury or other underlying condition can usually be successfully treated if the trauma is not too extensive or is treated in time. Less serious forms of chronic low blood pressure have a good prognosis and do not require treatment.

Resources

BOOKS

Smeltzer, Suzanne C., and Brenda G. Bare. *Brunner and Suddarth's Textbook of Medical and Surgical Nursing*. 8th ed. Philadelphia:Lippincott-Raven Publishers, 1996.

Carol A. Turkington

Hypothermia

Definition

Hypothermia, a potentially fatal condition, occurs when body temperature falls below 95°F (35°C).

Description

Although hypothermia is an obvious danger for people living in cold climates, many cases have occurred when the air temperature is well above the freezing mark. Elderly people, for instance, have succumbed to hypothermia after prolonged exposure to indoor air temperatures of 50–65°F (10–18.3°C). In the United States, hypothermia is primarily an urban phenomenon associated with **alcoholism**, drug **addiction**, mental illness, and cold—water immersion

accidents. The victims are often homeless male alcoholics. Officially, 11,817 deaths were attributed to hypothermia in the United States from 1979 to 1994, but experts suspect that many fatal cases go unrecognized. Nearly half the victims were 65 or older, with males dominating every age group. Nonwhites were also overrepresented in the statistics. Among males 65 and older, nonwhites outnumbered whites by more than four to one.

Causes and symptoms

Measured orally, a healthy person's body temperature can fluctuate between 97°F (36.1°C) and 100°F (37.8°C). Survival depends on maintaining temperature stability within this range by balancing the heat produced by metabolism with the heat lost to the environment through (for the most part) the skin and lungs. When environmental or other changes cause heat loss to outpace heat production, the brain triggers physiological and behavioral responses to restore the balance. The involuntary muscular activity of shivering, for example, aids heat production by accelerating metabolism. But if the cold **stress** is too great and the body's defenses are overwhelmed, body temperature begins to fall. Hypothermia is considered to begin once body temperature reaches 95°F (35°C), though even smaller drops in temperature can have an adverse effect.

Hypothermia is divided into two types: primary and secondary. Primary hypothermia occurs when the body's heat-balancing mechanisms are working properly but are subjected to extreme cold, whereas secondary hypothermia affects people whose heat-balancing mechanisms are impaired in some way and cannot respond adequately to moderate or perhaps even mild cold. Primary hypothermia typically involves exposure to cold air or immersion in cold water. The cold air variety usually takes at least several hours to develop, but immersion hypothermia will occur within about an hour of entering the water, since water draws heat away from the body much faster than air does. In secondary hypothermia, the body's heat-balancing mechanisms can fail for any number of reasons, including strokes, diabetes, **malnutrition**, bacterial infection, thyroid disease, spinal cord injuries (which prevent the brain from receiving crucial temperature-related information from other parts of the body), and the use of medications and other substances that affect the brain or spinal cord. Alcohol is one such substance. In smaller amounts it can put people at risk by interfering with their ability to recognize and avoid cold-weather dangers. In larger amounts it shuts down the body's heat-balancing mechanisms.

Secondary hypothermia is often a threat to the elderly, who may be on medications or suffering from illnesses that affect their ability to conserve heat. Malnutrition and immobility can also put the elderly at risk. Some medical research suggests as well that shivering and blood vessel narrowing—two of the body's defenses against cold—may not be triggered as quickly in older people. For these and other reasons, the elderly can, over a period of days or even weeks, fall victim to hypothermia in poorly insulated homes or other surroundings that family, friends, and caregivers may not recognize as life threatening. Another risk for the elderly is the fact that hypothermia can easily be misdiagnosed as a **stroke** or some other common illness of old age.

The signs and symptoms of hypothermia follow a typical course, though the body temperatures at which they occur vary from person to person depending on age, health, and other factors. The impact of hypothermia on the nervous system often becomes apparent quite early. Coordination, for instance, may begin to suffer as soon as body temperature reaches 95°F (35°C). The early signs of hypothermia also include cold and pale skin and intense shivering; the latter stops between 90°F (32.2°C) and 86°F (30°C). As body temperature continues to fall, speech becomes slurred, the muscles go rigid, and the victim becomes disoriented and experiences eyesight problems. Other harmful consequences include **dehydration** as well as liver and kidney failure. Heart rate, respiratory rate, and blood pressure rise during the first stages of hypothermia, but fall once the 90°F (32.2°C) mark is passed. Below 86°F (30°C) most victims are comatose, and below 82°F (27.8°C) the heart's rhythm becomes dangerously disordered. Yet even at very low body temperatures, people can survive for several hours and be successfully revived, though they may appear to be dead.

Diagnosis

Information on the patient's prior health and activities often helps doctors establish a correct diagnosis and treatment plan. Pulse, blood pressure, temperature, and respiration require immediate monitoring. Because the temperature of the mouth is not an accurate guide to the body's core temperature, readings are taken at one or two other sites, usually the ear, rectum, or esophagus. Other diagnostic tools include **electrocardiography**, which is used to evaluate heart rhythm, and blood and urine tests, which provide several kinds of key information; a **chest x ray** is also required. A computed tomography scan (CT scan) or **magnetic resonance imaging** (MRI) may be needed to check for head and other injuries.

Treatment

Emergency medical help should be summoned whenever a person appears hypothermic. The danger signs include intense shivering; stiffness and **numbness** in the arms and legs; stumbling and clumsiness; sleepiness, confusion, disorientation, **amnesia**, and irrational behavior; and difficulty speaking. Until emergency help arrives, a victim of outdoor hypothermia should be brought to shelter and warmed by removing wet clothing and footwear, drying the skin, and wrapping him or her in warm blankets or a sleeping bag. Gentle handling is necessary when moving the victim to avoid disturbing the heart. Rubbing the skin or giving the victim alcohol can be harmful, though warm drinks such as clear soup and tea are recommended for those who can swallow. Anyone who aids a victim of hypothermia should also look for signs of **frostbite** and be aware that attempting to rewarm a frostbitten area of the body before emergency help arrives can be extremely dangerous. For this reason, frostbitten areas must be kept away from heat sources such as campfires and car heaters.

Rewarming is the essence of hospital treatment for hypothermia. How rewarming proceeds depends on the body temperature. Different approaches are used for patients who are mildly hypothermic (the patient's body temperature is 90–95 °F [32.2–35 °C]), moderately hypothermic (86–90 °F [30–32.2 °C]), or severely hypothermic (less than 86 °F [30 °C]). Other considerations, such as the patient's age or the condition of the heart, can also influence treatment choices.

Mild hypothermia is reversed with passive rewarming. This technique relies on the patient's own metabolism to rewarm the body. Once wet clothing is removed and the skin is dried, the patient is covered with blankets and placed in a warm room. The goal is to raise the patient's temperature by 0.5–2 °C an hour.

Moderate hypothermia is often treated first with active external rewarming and then with passive rewarming. Active external rewarming involves applying heat to the skin, for instance by placing the patient in a warm bath or wrapping the patient in electric heating blankets.

Severe hypothermia requires active internal rewarming, which is recommended for some cases of moderate hypothermia as well. There are several types of active internal rewarming. Cardiopulmonary bypass, in which the patient's blood is circulated through a rewarming device and then returned to the body, is considered the best, and can raise body temperature by 1–2 °C every 3–5 minutes. However, many hospitals are not equipped to offer this treatment. The alternative is to introduce warm oxygen or fluids into the body.

KEY TERMS

Antibiotics—Substances used against microorganisms that cause infection.

Computed tomography—A process that uses x rays to create three-dimensional images of structures inside the body.

Esophagus—A muscular tube through which food and liquids pass on their way to the stomach.

Insulin—A substance that regulates blood glucose levels. Glucose is a sugar.

Magnetic resonance imaging—The use of electromagnetic energy to create images of structures inside the body.

Metabolism—The chemical changes by which the body breaks down food and other substances and builds new substances necessary for life.

Nervous system—The system that transmits information, in the form of electrochemical impulses, throughout the body. It comprises the brain, spinal cord, and nerves.

Rectum—The lower section of the large intestine. The intestines are part of the digestive system.

Stroke—A condition involving loss of blood flow to the brain.

Thyroid—A gland (fluid-secreting structure) in the neck. It plays an important role in metabolism.

Hypothermia treatment can also include, among other things, insulin, **antibiotics**, and fluid replacement therapy. When the heart has stopped, both **cardiopulmonary resuscitation (CPR)** and rewarming are necessary. Once a patient's condition has stabilized, he or she may need treatment for an underlying problem such as alcoholism or thyroid disease.

Prognosis

Victims of mild or moderate hypothermia usually enjoy a complete recovery. In regard to severely hypothermic patients, the prognosis for survival varies due to differences in people's physiological responses to cold.

Prevention

People who spend time outdoors in cold weather can reduce heat loss by wearing their clothing

loosely and in layers and by keeping their hands, feet, and head well covered (30–50% of body heat is lost through the head). Because water draws heat away from the body so easily, staying dry is important, and wet clothing and footwear should be replaced as quickly as possible. Wind- and water-resistant outer garments are also crucial. Alcohol should be avoided because it promotes heat loss by expanding the blood vessels that carry body heat to the skin.

Preventing hypothermia among the elderly requires vigilance on the part of family, friends, and caregivers. An elderly person's home should be properly insulated and heated, with living areas kept at a temperature of 70°F (21.1°C). Warm clothing and bedding are essential, as are adequate food, rest, and **exercise**; warming the bed and bedroom before going to sleep is also recommended. Older people who live alone should be visited regularly—at least once a day during very cold weather—to ensure that their health remains sound and that they are taking good care of themselves. For help and advice, family members and others can turn to government and social service agencies. Meals on wheels and visiting nurse programs, for instance, may be available, and it may be possible to obtain financial aid for winterizing and heating homes.

Resources

BOOKS

Petty, Kevin J. "Hypothermia." In *Harrison's Principles of Internal Medicine*, edited by Anthony S. Fauci, et al. New York: McGraw-Hill, 1997.

Howard Baker

Hypothyroidism

Definition

Hypothyroidism, or underactive thyroid, develops when the thyroid gland fails to produce or secrete as much thyroxine (T₄) as the body needs. Because T₄ regulates such essential functions as heart rate, digestion, physical growth, and mental development, an insufficient supply of this hormone can slow life-sustaining processes, damage organs and tissues in every part of the body, and lead to life-threatening complications.

Description

Hypothyroidism is one of the most common chronic diseases in the United States. Symptoms may not appear until years after the thyroid has stopped functioning and they are often mistaken for signs of other illnesses, **menopause**, or **aging**. Although this condition is believed to affect as many as 11 million adults and children, as many as two of every three people with hypothyroidism may not know they have the disease.

Nicknamed "Gland Central" because it influences almost every organ, tissue, and cell in the body, the thyroid is shaped like a butterfly and located just below the Adam's apple. The thyroid stores iodine the body gets from food and uses this mineral to create T₄. Low T₄ levels can alter weight, appetite, sleep patterns, body temperature, sex drive, and a variety of other physical, mental, and emotional characteristics.

There are three types of hypothyroidism. The most common is primary hypothyroidism, in which the thyroid doesn't produce an adequate amount of T₄. Secondary hypothyroidism develops when the pituitary gland does not release enough of the thyroid-stimulating hormone (TSH) that prompts the thyroid to manufacture T₄. Tertiary hypothyroidism results from a malfunction of the hypothalamus, the part of the brain that controls the endocrine system. Drug-induced hypothyroidism, an adverse reaction to medication, occurs in two of every 10,000 people, but rarely causes severe hypothyroidism.

Hypothyroidism is at least twice as common in women as it is in men. Although hypothyroidism is most common in women who are middle-aged or older, the disease can occur at any age. Newborn infants are tested for congenital thyroid deficiency (cretinism) using a test that measures the levels of thyroxine in the infant's blood. Treatment within the first few months of life can prevent **mental retardation** and physical abnormalities. Older children who develop hypothyroidism suddenly stop growing.

Factors that increase a person's risk of developing hypothyroidism include age, weight, and medical history. Women are more likely to develop the disease after age 50; men, after age 60. Obesity also increases risk. A family history of thyroid problems or a personal history of **high cholesterol** levels or such autoimmune diseases as lupus, **rheumatoid arthritis**, or diabetes can make an individual more susceptible to hypothyroidism.

Causes and symptoms

Hypothyroidism is most often the result of Hashimoto's disease, also known as chronic thyroiditis (inflammation of the thyroid gland). In this disease, the immune system fails to recognize that the thyroid gland is part of the body's own tissue and attacks it as if it were a foreign body. The attack by the immune system impairs thyroid function and sometimes destroys the gland. Other causes of hypothyroidism include:

- **Radiation.** Radioactive iodine used to treat hyperthyroidism (overactive thyroid) or radiation treatments for head or neck cancers can destroy the thyroid gland.
- **Surgery.** Removal of the thyroid gland because of **cancer** or other thyroid disorders can result in hypothyroidism.
- **Viruses and bacteria.** Infections that depress thyroid hormone production usually cause permanent hypothyroidism.
- **Medication.** Nitroprusside, lithium, or iodides can induce hypothyroidism. Because patients who use these medications are closely monitored by their doctors, this side effect is very rare.
- **Pituitary gland malfunction.** This is a rare condition in which the pituitary gland fails to produce enough TSH to activate the thyroid's production of T₄.
- **Congenital defect.** One of every 4,000 babies is born without a properly functioning thyroid gland.
- **Diet.** Because the thyroid makes T₄ from iodine drawn from food, an iodine-deficient diet can cause hypothyroidism. Adding iodine to table salt and other common foods has eliminated iodine deficiency in the United States. Certain foods (cabbage, rutabagas, peanuts, peaches, soybeans, spinach) can interfere with thyroid hormone production.
- **Environmental contaminants.** Certain man-made chemicals—such as PCBs—found in the local environment at high levels may also cause hypothyroidism.

Hypothyroidism is sometimes referred to as a “silent” disease because early symptoms may be so mild that no one realizes anything is wrong. Untreated symptoms become more noticeable and severe, and can lead to confusion and mental disorders, breathing difficulties, heart problems, fluctuations in body temperature, and **death**.

Someone who has hypothyroidism will probably have more than one of the following symptoms:

- progressive **hearing loss**
- weight gain
- problems with memory and concentration
- depression
- goiter (enlarged thyroid gland)
- muscle **pain** or weakness
- loss of interest in sex
- numb, **tingling** hands
- dry skin
- swollen eyelids
- dryness, loss, or premature graying of hair
- extreme sensitivity to cold
- **constipation**
- irregular menstrual periods
- hoarse voice

Hypothyroidism usually develops gradually. When the disease results from surgery or other treatment for **hyperthyroidism**, symptoms may appear suddenly and include severe muscle cramps in the arms, legs, neck, shoulders, and back.

It's important to see a doctor if any of these symptoms appear unexpectedly. People whose hypothyroidism remains undiagnosed and untreated may eventually develop myxedema. Symptoms of this rare but potentially deadly complication include enlarged tongue, swollen facial features, hoarseness, and physical and mental sluggishness.

Myxedema **coma** can cause unresponsiveness; irregular, shallow breathing; and a drop in blood pressure and body temperature. The onset of this medical emergency can be sudden in people who are elderly or have been ill, injured, or exposed to very cold temperatures; who have recently had surgery; or who use sedatives or anti-depressants. Without immediate medical attention, myxedema coma can be fatal.

Diagnosis

Diagnosis of hypothyroidism is based on the patient's observations, medical history, physical examination, and thyroid function tests. Doctors who specialize in treating thyroid disorders (endocrinologists) are most apt to recognize subtle symptoms and physical indications of hypothyroidism. A blood test known as a thyroid-stimulating hormone (TSH) assay, thyroid nuclear medicine scan, and **thyroid ultrasound** are used to confirm the diagnosis. A woman being tested for hypothyroidism should let

her doctor know if she is pregnant or breastfeeding and all patients should be sure their doctors are aware of any recent procedures involving radioactive materials or contrast media.

The TSH assay is extremely accurate, but some doctors doubt the test's ability to detect mild hypothyroidism. They advise patients to monitor their basal (resting) body temperature for below-normal readings that could indicate the presence of hypothyroidism.

Treatment

Natural or synthetic **thyroid hormones** are used to restore normal (euthyroid) thyroid hormone levels. Synthetic hormones are more effective than natural substances, but it may take several months to determine the correct dosage. Patients start to feel better within 48 hours, but symptoms will return if they stop taking the medication.

Most doctors prescribe levothyroxine sodium tablets, and most people with hypothyroidism will take the medication for the rest of their lives. Aging, other medications, and changes in weight and general health can affect how much replacement hormone a patient needs, and regular TSH tests are used to monitor hormone levels. Patients should not switch from one brand of thyroid hormone to another without a doctor's permission.

Regular **exercise** and a high-fiber diet can help maintain thyroid function and prevent constipation.

Alternative treatment

Alternative treatments are primarily aimed at strengthening the thyroid and will not eliminate the need for thyroid hormone medications. Herbal remedies to improve thyroid function and relieve symptoms of hypothyroidism include bladder wrack (*Fucus vesiculosus*), which can be taken in capsule form or as a tea. Some foods, including cabbage, peaches, radishes, soybeans, peanuts, and spinach, can interfere with the production of thyroid hormones. Anyone with hypothyroidism may want to avoid these foods. The Shoulder Stand **yoga** position (at least once daily for 20 minutes) is believed to improve thyroid function.

Prognosis

Thyroid **hormone replacement therapy** generally maintains normal thyroid hormone levels unless treatment is interrupted or discontinued.

KEY TERMS

Cretinism—Severe hypothyroidism that is present at birth.

Endocrine system—The network of glands that produce hormones and release them into the bloodstream. The thyroid gland is part of the endocrine system.

Hypothalamus—The part of the brain that controls the endocrine system.

Myxedema—A condition that can result from a thyroid gland that produces too little of its hormone. In addition to a decreased metabolic rate, symptoms may include anemia, slow speech, an enlarged tongue, puffiness of the face and hands, loss of hair, coarse and thickened skin, and sensitivity to cold.

Pituitary gland—Small, oval endocrine gland attached to the hypothalamus. The pituitary gland releases TSH, the hormone that activates the thyroid gland.

Thyroid-stimulating hormone (TSH)—A hormone secreted by the pituitary gland that controls the release of T_4 by the thyroid gland.

Thyroxine (T_4)—Thyroid hormone that regulates many essential body processes.

Prevention

Primary hypothyroidism can't be prevented, but routine screening of adults could detect the disease in its early stages and prevent complications.

Resources

ORGANIZATIONS

American Thyroid Association. Montefiore Medical Center, 111 E. 210th St., Bronx, NY 10467.

Endocrine Society. 4350 East West Highway, Suite 500, Bethesda, MD 20814-4410. (301) 941-0200.

Thyroid Foundation of America, Inc. Ruth Sleeper Hall, RSL 350, Boston, MA 02114-2968. (800) 832-8321 or (617) 726-8500.

Thyroid Society for Education and Research. 7515 S. Main St., Suite 545, Houston, TX 77030. (800) THYROID or (713) 799-9909.

Maureen Haggerty

Hypotonic duodenography

Definition

Hypotonic duodenography is an x-ray procedure that produces images of the duodenum. The duodenum is the first part of the small intestine.

Purpose

Hypotonic duodenography may be ordered to detect tumors of the head of the pancreas or the area where the pancreatic and bile ducts meet the small intestine. Lesions causing upper abdominal **pain** may be demonstrated by duodenography, and the procedure can aid in the diagnosis of chronic **pancreatitis**.

Precautions

Some patients with narrowing of the tubes in the upper gastrointestinal tract should not receive duodenography. Patients with certain heart disorders and **glaucoma** are cautioned against receiving an agent called anticholinergic, which is administered during the procedure to lessen intestinal **muscle spasms**. A hormone called glucagon may also be used to relax the intestines, but its use is not recommended in patients with most forms of diabetes.

Description

Hypotonic duodenography is also referred to as x ray of the duodenum or simply as duodenography. The patient is seated while the radiologist places a catheter in the nose and down into the stomach. Then the patient lies down and the tube is continued to the duodenum. The radiologist is guided in this placement by a fluoroscopic image. (Fluoroscopic equipment shows an immediate x ray. In this case, the x ray shows the location of the catheter as it is moved into the stomach and duodenum.) Next, either the glucagon is administered intravenously or anticholinergic is injected into the patient to relax the muscles of the intestine.

After several minutes, the physician will administer barium through the catheter. Barium is a contrast agent that will help highlight the area on the fluoroscopy screen and x rays. After a few films are taken, some of the barium is withdrawn and air is sent in through the catheter. Additional images are acquired and the catheter is then removed. The procedure takes from 30–60 minutes.

Preparation

Patients are required to fast from midnight before the test until after the test, or about 6–12 hours. Just prior to the exam, patients should remove dentures, glasses, and other objects that may interfere with the procedure. The patient may be instructed to empty his or her bladder just prior to duodenography.

Aftercare

The barium should be expelled within two to three days. Extra fluids and/or an agent given by the physician to help encourage bowel movement may aid in barium elimination. Physicians and patients should watch for possible reactions to the anticholinergic or glucagon. If an anticholinergic is used, patients are advised to empty their bladder within a few hours after the exam and to wait two hours for clearing of vision or have someone drive them home. Patients will notice that their stools are chalky white from the barium for one to three days following the procedure.

Risks

Abdominal cramping may occur when the physician instills air into the duodenum, but aside from the discomfort, there are few risks associated with this procedure. Side effects from the contrast, hormones or agents may occur. Those patients with diabetes, heart disease, or glaucoma run the highest risk of reaction and should not receive anticholinergic or glucagon, depending on their specific conditions. Elderly patients or those who are extremely ill, must be closely monitored during the procedure for possible return of fluid, or gastric reflux.

Normal results

The linings of the duodenum and surrounding tissues will look smooth and even. The shape of the head of the pancreas will appear normal and near the duodenal wall.

Abnormal results

Any masses or irregular nodules on the wall of the duodenum may indicate tumors or abnormality of tissue. Tumors of the head of the pancreas or of the opening into the intestine from the pancreatic and bile ducts may be seen. Chronic pancreatitis may be indicated on the x rays. In many instances, follow-up laboratory or imaging studies may be ordered to further study the abnormal findings and confirm a diagnosis.

KEY TERMS

Anticholinergic—A drug that lessens muscle spasms in the intestines, lungs, bladder, and eye muscles.

Fluoroscopic (fluoroscopy)—An x-ray procedure that produces immediate images and motion on a screen. The images look like those seen at airport baggage security stations.

Glucagon—A hormone that changes glycogen, a carbohydrate stored in muscles and the liver, into glucose. It can be used to relax muscles for a procedure such as duodenography.

Pancreas—A five-inch-long gland that lies behind the stomach and next to the duodenum. The pancreas releases glucagon, insulin and some of the enzymes which aid digestion. Pancreatitis is the swelling of the pancreas which can cause nausea, jaundice, and severe pain and may be fatal.

Resources

ORGANIZATIONS

American College of Radiology. 1891 Preston White Drive, Reston, VA 22091. (800) 227-5463. <<http://www.acr.org>>.

National Cancer Institute. Building 31, Room 10A31, 31 Center Drive, MSC 2580, Bethesda, MD 20892-2580. (800) 422-6237. <<http://www.nci.nih.gov>>.

Teresa Odle

Hypovolemic shock see **Shock**

Hysterectomy

Definition

Hysterectomy is the surgical removal of the uterus. In a total hysterectomy, the uterus and cervix are removed. In some cases, the fallopian tubes and ovaries are removed along with the uterus (called hysterectomy with bilateral salpingo-oophorectomy). In a subtotal hysterectomy, only the uterus is removed. In a radical hysterectomy, the uterus, cervix, ovaries, oviducts, lymph nodes, and lymph channels are removed. The type of hysterectomy performed depends on the reason for the procedure. In all cases,

menstruation stops and a woman loses the ability to bear children.

Purpose

Hysterectomy is the second most common operation performed in the United States. About 556,000 of these surgeries are done annually. By age 60, approximately one out of every three American women will have had a hysterectomy. Yet it's estimated that 30 percent of hysterectomies are unnecessary.

About 10% of hysterectomies are performed to treat cancer of the cervix, ovaries, or uterus. Women with **cancer** in one or more of these organs almost always have the organ(s) removed as one part of their cancer treatment.

The most frequent reason for hysterectomy in the United States is to remove fibroid tumors, accounting for 30% of these surgeries. Fibroid tumors are non-cancerous (benign) growths in the uterus, which can cause pelvic and **low back pain** and heavy or lengthy menstrual periods. They occur in 30–40% of women over age 40, and are three times more likely to be present in African-American women than in Caucasian women. Fibroids do not need to be removed unless they are causing symptoms that interfere with a woman's normal activities.

Treatment of **endometriosis** is the reason for 20% of hysterectomies. The endometrium is the lining of the uterus. Endometriosis is a condition that occurs when the cells from the endometrium begin growing outside the uterus. The outlying endometrial cells respond to the hormones that control the menstrual cycle, bleeding each month the way the lining of the uterus does. This causes irritation of the surrounding tissue, leading to pain and scarring.

Another 20% percent of hysterectomies are done because of heavy or abnormal vaginal bleeding that can not be linked to any specific cause and cannot be controlled by other means. The remaining 20% of hysterectomies are performed to treat prolapsed uterus, **pelvic inflammatory disease**, and endometrial hyperplasia, a potentially precancerous condition.

Alternatives

There are several alternatives to hysterectomy today. They include:

Embolization

Uterine artery embolization is not a surgical procedure. Instead, interventional radiologists put a

catherter into the artery that leads to the uterus and inject polyvinyl alcohol particles right where the artery leads to the blood vessels that nourish the fibroids. By killing off those blood vessels, the fibroids have no more blood supply, and they die off. Severe cramping and **pain** after the procedure is common, but serious complications are less than 5 percent and it may protect fertility.

Myomectomy

A **myomectomy** is a surgery used to remove fibroids, thus avoiding a hysterectomy. Hysteroscopic myomectomy, in which a surgical “telescope,” or laparoscope, is inserted into the uterus through the vagina can be done on an outpatient basis. If there are large fibroids, however, an abdominal incision is required. Then women typically are hospitalized for two to three days, and require up to six weeks recovery. However, laparoscopic myomectomies are also being done more often. They only require three small incisions in the abdomen, and have a much shorter hospitalization and recovery time.

Once the fibroids have been removed, the surgeon must repair the wall of the uterus to eliminate future bleeding or infection.

Endometrial ablation

In this surgical procedure, recommended for women with small fibroids, the entire lining of the uterus is removed. Women are no longer fertile, however. The uterine cavity is filled with fluid and a **hysteroscopy**, or telescope, inserted to provide a clear view of the uterus. Then the uterus is destroyed using a laser beam or electric voltage. The procedure is typically done under anesthesia, although women can go home the same day as the surgery. Another, newer procedure involves using a balloon, which is filled with superheated liquid and inflated until it fills the uterus. The liquid kills the lining, and after 8 minutes the balloon is removed.

Endometrial resection

Like endometrial ablation, the uterine lining is also destroyed during this procedure, only instead of a laser, an electrosurgical wire loop is used.

Total hysterectomy

A total hysterectomy, sometimes called a simple hysterectomy, removes the entire uterus and the cervix. The ovaries are not removed and continue to secrete hormones. Total hysterectomies are always

performed in the case of uterine and cervical cancer. This is the most common kind of hysterectomy.

Sometimes, in addition to a total hysterectomy a procedure called a bilateral salpingo-oophorectomy is performed. This surgery removes the ovaries and the fallopian tubes. Removal of the ovaries eliminates the main source of the hormone estrogen, so **menopause** occurs immediately. Removal of the ovaries and fallopian tubes is performed in about one-third of hysterectomy operations, often to reduce the risk of **ovarian cancer**.

Subtotal hysterectomy

If the reason for the hysterectomy is to remove uterine fibroids, treat abnormal bleeding, or relieve pelvic pain, it may be possible to remove only the uterus and leave the cervix. This procedure, called a subtotal hysterectomy (or partial hysterectomy), removes the least amount of tissue. The opening to the cervix is left in place. Some women feel that leaving the cervix intact aids in their achieving sexual satisfaction. This procedure, which used to be rare, is now performed more frequently when requested.

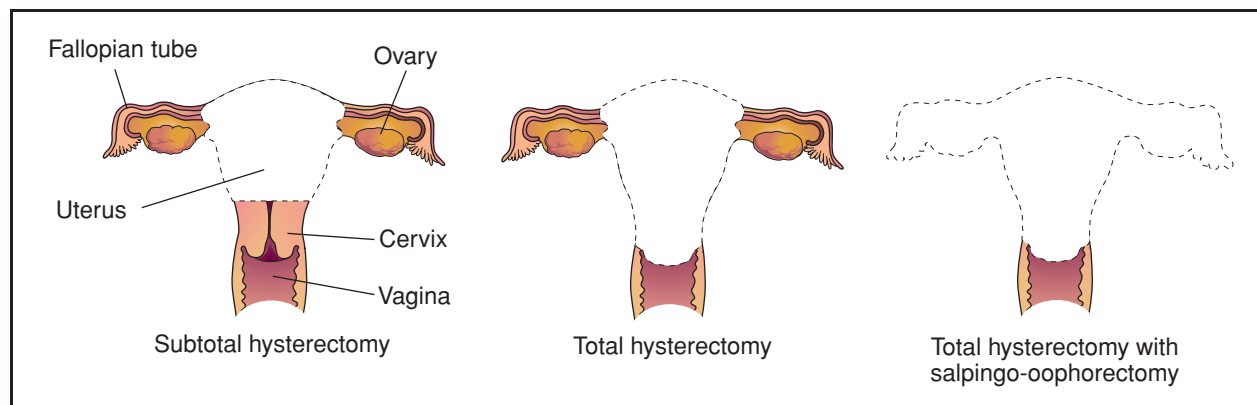
Subtotal hysterectomy is easier to perform than a total hysterectomy, but leaves a woman at risk for **cervical cancer**. She will still need to get yearly pap smears.

Radical hysterectomy

Radical hysterectomies are performed on women with cervical cancer or **endometrial cancer** that has spread to the cervix. A radical hysterectomy removes the uterus, cervix, top part of the vagina, ovaries, fallopian tubes, lymph nodes, lymph channels, and tissue in the pelvic cavity that surrounds the cervix. This type of hysterectomy removes the most tissue and requires the longest hospital stay and longer recovery period.

Precautions

The frequency with which hysterectomies are performed in the United States has been questioned in recent years. It has been suggested that a large number of hysterectomies are performed unnecessarily. The United States has the highest rate of hysterectomies (number of hysterectomies per thousand women) of any country in the world. Also, the frequency of this surgery varies across different regions of the United States. Rates are highest in the South and Midwest, and are higher for African American women. In recent years, although the number of hysterectomies



Three types of hysterectomies: subtotal, total, and total with salpingo-oophorectomy. (Illustration by Electronic Illustrators Group.)

performed has declined, the number of hysterectomies performed on younger women in their 30s and 40s is increasing, and 55 percent of all hysterectomies are performed on women 35 to 49.

Women for whom a hysterectomy is recommended should discuss possible alternatives with their doctor and consider getting a second opinion, since this is major surgery with life-changing implications. Alternative treatments exist for many conditions. Whether these alternatives are appropriate for any individual woman is a decision she and her doctor should make together.

As in all major surgery, the health of the patient affects the risk of the operation. Women who have chronic heart or lung diseases, diabetes, or iron-deficiency anemia may not be good candidates for this operation. Heavy **smoking**, **obesity**, use of steroid drugs, and use of illicit drugs add to the surgical risk.

Description

There are two ways that hysterectomies can be performed. The choice of method depends on the type of hysterectomy, the doctor's experience, and the reason for the hysterectomy.

Abdominal hysterectomy

About 75% of hysterectomies performed in the United States are abdominal hysterectomies. The surgeon makes a four to six inch incision either horizontally across the pubic hair line from hip bone to hip bone or vertically from navel to pubic bone. Horizontal incisions leave a less noticeable scar, but vertical incisions give the surgeon a better view of the abdominal cavity. The blood vessels, fallopian tubes, and ligaments are cut away from the uterus, which is lifted out.

Abdominal hysterectomies take from one to three hours. The hospital stay is three to five days, and it takes four to eight weeks to return to normal activities.

The advantages of an abdominal hysterectomy are that the uterus can be removed even if a woman has internal scarring (**adhesions**) from previous surgery or her fibroids are large. The surgeon has a good view of the abdominal cavity and more room to work. Also, surgeons have the most experience with this type of hysterectomy. The abdominal incision is more painful than with vaginal hysterectomy and the recovery period is longer.

Vaginal hysterectomy

With a vaginal hysterectomy, the surgeon makes an incision near the top of the vagina. The surgeon then reaches through this incision to cut and tie off the ligaments, blood vessels, and fallopian tubes. Once the uterus is cut free, it is removed through the vagina. The operation takes one to two hours. The hospital stay is usually one to three days, and return to normal activities takes about four weeks.

The advantages of this procedure are that it leaves no visible scar and is less painful. The disadvantage is that it is more difficult for the surgeon to see the uterus and surrounding tissue. This makes complications more common. Large fibroids cannot be removed using this technique. It is very difficult to remove the ovaries during a vaginal hysterectomy, so this approach may not be possible if the ovaries are involved.

Vaginal hysterectomy can also be performed using a laparoscopic technique. With this surgery, a tube containing a tiny camera is inserted through an incision in the navel. This allows the surgeon to see the uterus on a video monitor. The surgeon then inserts

two slender instruments through small incisions in the abdomen and uses them to cut and tie off the blood vessels, fallopian tubes, and ligaments. When the uterus is detached, it is removed through a small incision at the top of the vagina.

This technique, called laparoscopic-assisted vaginal hysterectomy, allows surgeons to perform a vaginal hysterectomy that might be too difficult otherwise. The hospital stay is usually only one day. Recovery time is about two weeks. The disadvantage is that this operation is relatively new and requires great skill by the surgeon.

Any vaginal hysterectomy may have to be converted to an abdominal hysterectomy during surgery if complications develop.

Preparation

Before surgery the doctor will order blood and urine tests. The woman may also meet with the anesthesiologist to evaluate any special conditions that might affect the administration of anesthesia. On the evening before the operation, the woman should eat a light dinner and then avoid eating or drinking anything.

Aftercare

After surgery a woman will feel pain. The degree of discomfort varies, and is generally greatest in abdominal hysterectomies because of the incision. Hospital stays vary from about two days (laparoscopic-assisted vaginal hysterectomy) to five or six days (abdominal hysterectomy with bilateral **salpingo-oophorectomy**). During the hospital stay, the doctor will probably order more blood tests.

Return to normal activities such as driving and working takes anywhere from two to eight weeks, again depending on the type of surgery. Some women have emotional changes following a hysterectomy. Women who have had their ovaries removed will probably start taking **hormone replacement therapy**.

Risks

Hysterectomy is a relatively safe operation, although like all major surgery it carries risks. These include unanticipated reaction to anesthesia, internal bleeding, **blood clots**, damage to other organs such as the bladder, and post-surgery infection. The risk of **death** is about one in every 1,000 (1/1,000) women having the operation.

KEY TERMS

Cervix—The lower part of the uterus extending into the vagina.

Fallopian tubes—Slender tubes that carry eggs (ova) from the ovaries to the uterus.

Lymph nodes—Small, compact structures lying along the channels that carry lymph, a yellowish fluid. Lymph nodes produce white blood cells (lymphocytes), which are important in forming antibodies that fight disease.

Prolapsed uterus—A uterus that has slipped out of place, sometimes protruding down through the vagina.

Other complications sometimes reported after a hysterectomy include changes in sex drive, weight gain, **constipation**, and pelvic pain. Hot flashes and other symptoms of menopause can occur if the ovaries are removed. Women who have both ovaries removed and who do not take estrogen replacement therapy run an increased risk for heart disease and **osteoporosis** (a condition that causes bones to be brittle). Women with a history of psychological and emotional problems before the hysterectomy are more likely to experience psychological difficulties after the operation.

Normal results

Although there is some concern that hysterectomies may be performed unnecessarily, there are many conditions for which the operation improves a woman's quality of life. In the Maine Woman's Health Study, 71% of women who had hysterectomies to correct moderate or severe painful symptoms reported feeling better mentally, physically, and sexually after the operation.

Resources

ORGANIZATIONS

American Cancer Society. (800) 227-2345. <<http://www.cancer.org>>.

National Cancer Institute. (800) 4-CANCER. <<http://www.nci.nih.gov>>.

OTHER

Parker, William H. "A Gynecologist's Second Opinion." <<http://www.gynsecondopinion.com>>.

Debra Gordon

Hysteria

Definition

The term “hysteria” has been in use for over 2,000 years and its definition has become broader and more diffuse over time. In modern psychology and psychiatry, hysteria is a feature of hysterical disorders in which a patient experiences physical symptoms that have a psychological, rather than an organic, cause; and histrionic personality disorder characterized by excessive emotions, dramatics, and attention-seeking behavior.

Description

Hysterical disorders

Patients with hysterical disorders, such as conversion and somatization disorder experience physical symptoms that have no organic cause. Conversion disorder affects motor and sensory functions, while somatization affects the gastrointestinal, nervous, cardiopulmonary, or reproductive systems. These patients are not “faking” their ailments, as the symptoms are very real to them. Disorders with hysteric features typically begin in adolescence or early adulthood.

Histrionic personality disorder

Histrionic personality disorder has a prevalence of approximately 2–3% of the general population. It begins in early adulthood and has been diagnosed more frequently in women than in men. Histrionic personalities are typically self-centered and attention seeking. They operate on emotion, rather than fact or logic, and their conversation is full of generalizations and dramatic appeals. While the patient’s enthusiasm, flirtatious behavior, and trusting nature may make them appear charming, their need for immediate gratification, mercurial displays of emotion, and constant demand for attention often alienates them from others.

Causes and symptoms

Hysterical disorders

Hysteria may be a defense mechanism to avoid painful emotions by unconsciously transferring this distress to the body. There may be a symbolic function for this, for example a **rape** victim may develop paralyzed legs. Symptoms may mimic a number of physical and neurological disorders which must be ruled out before a diagnosis of hysteria is made.

Histrionic personality disorder

According to the *Diagnostic and Statistical Manual of Mental Disorders*, Fourth Edition (DSM-IV), individuals with histrionic personality possess at least five of the following symptoms or personality features:

- a need to be the center of attention
- inappropriate, sexually seductive, or provocative behavior while interacting with others
- rapidly changing emotions and superficial expression of emotions
- vague and impressionistic speech (gives opinions without any supporting details)
- easily influenced by others
- believes relationships are more intimate than they are.

Diagnosis

Hysterical disorders frequently prove to be actual medical or neurological disorders, which makes it important to rule these disorders out before diagnosing a patient with hysterical disorders. In addition to a patient interview, several clinical inventories may be used to assess the patient for hysterical tendencies, such as the Minnesota Multiphasic Personality Inventory-2 (MMPI-2) or the Millon Clinical Multiaxial Inventory-III (MCMI-III). These tests may be administered in an outpatient or hospital setting by a psychiatrist or psychologist.

Treatment

Hysterical disorders

For people with hysterical disorders, a supportive healthcare environment is critical. Regular appointments with a physician who acknowledges the patient’s physical discomfort are important. Psychotherapy may be attempted to help the patient gain insight into the cause of their distress. Use of behavioral therapy can help to avoid reinforcing symptoms.

Histrionic personality disorder

Psychotherapy is generally the treatment of choice for histrionic personality disorder. It focuses on supporting the patient and on helping develop the skills needed to create meaningful relationships with others.

JEAN MARTIN CHARCOT (1825–1893)



(Library of Congress.)

Jean Martin Charcot was born to a carriage maker on November 29, 1825, in Paris, France. Charcot attended the University of Paris, earning his medical degree in 1853. In 1860, he accepted a position at the university as

a professor of pathological anatomy until 1862, when he was named senior physician at the Salpêtrière, a hospital for the treatment of mental illness.

Charcot's research and work on psychoneuroses and hysterical disorders ultimately helped to dispell the belief that hysteria was a disorder found only in women. Charcot also explored the possibility that physiological abnormalities of the nervous system played a part when behavioral problems were exhibited. He became known for his ability to diagnose and locate these abnormalities of the central nervous system. Finally, Charcot's most notable contribution to the field of psychiatry was his successful use of hypnotism in the diagnosis and treatment of hysteria. He found that, while hypnotized, the patient recalled details, which were not readily available to the individual in a conscious state. In addition, Charcot found that the therapist could more easily influence the hypnotized patient during therapy. In 1882, Charcot presented his research findings to the French Academy of Sciences with favorable results.

Charcot was a prolific writer and a talented artist. Between 1888 and 1894, his complete works were compiled into nine volumes. His most noted work *Lectures on the Diseases of the Nervous System* was published in 1877. Charcot died on August 16, 1893.

KEY TERMS

Conversion disorder—A psychological disorder that alters motor or sensory functions. Paralysis, blindness, anesthesia (lack of feeling), coordination or balance problems, and seizures are all common symptoms of the disorder.

Somatization disorder—The appearance of physical symptoms in the gastrointestinal system, the nervous system, the cardiopulmonary system, or the reproductive system that have no organic cause.

Prognosis

Hysterical disorders

The outcome for hysterical disorders varies by type. Somatization is typically a lifelong disorder, while conversion disorder may last for months or years. Symptoms of hysterical disorders may suddenly disappear, only to reappear in another form later.

Histrionic personality disorder

Individuals with histrionic personality disorder may be at a higher risk for suicidal gestures, attempts, or threats in an effort to gain attention. Providing a supportive environment for patients with both hysterical disorders and histrionic personality disorder is key to helping these patients.

Resources

ORGANIZATIONS

American Psychiatric Association. 1400 K Street NW, Washington DC 20005. (888) 357-7924. < <http://www.psych.org> >.

American Psychological Association (APA). 750 First St. NE, Washington, DC 20002-4242. (202) 336-5700. < <http://www.apa.org> >.

National Alliance for the Mentally Ill (NAMI). Colonial Place Three, 2107 Wilson Blvd., Ste. 300, Arlington, VA 22201-3042. (800) 950-6264. < <http://www.nami.org> >.

Paula Anne Ford-Martin

Hysterosalpingography

Definition

Hysterosalpingography is a procedure where x rays are taken of a woman's reproductive tract after a dye is injected. *Hystero* means uterus and *salpingo* means tubes, so hysterosalpingography literally means to take pictures of the uterus and fallopian tubes. This procedure may also be called hystero-graphy (or HSG).

Purpose

Hysterosalpingography is used to determine if the fallopian tubes are open, or if there are any apparent abnormalities or defects in the uterus. It can be used to detect tumors, scar tissue, or tears in the lining of the uterus. This procedure is often used to help diagnose **infertility** in women. The fallopian tubes are the location where an egg from the ovary joins with sperm to produce a fertilized ovum. If the fallopian tubes are blocked or deformed, the egg may not be able to descend or the sperm may be blocked from moving up to meet the egg. Up to 30% of all cases of infertility are due to damaged or blocked fallopian tubes.

Precautions

This procedure should not be done on women who suspect they might be pregnant or who may have a pelvic infection. Women who have had an allergic reaction to dye used in previous x-ray procedures should inform their doctor.

Description

As with other types of pelvic examinations, the woman will lie on her back on an examination table with her legs sometimes raised in stirrups. The x-ray equipment is placed above the abdomen.

A speculum is inserted into the vagina and a catheter (a thin tube) is inserted into the uterus through the cervix (the opening to the uterus). A small balloon in the catheter is inflated to hold it in place. A liquid water-based or oil-based dye is then injected through the catheter into the uterus. This process can cause cramping, **pain**, and uterine spasms.

As the dye spreads through the reproductive tract, the doctor may watch for blockages or abnormalities on an x-ray monitor. Several x rays will also be taken.



A hysterosalpingogram of the abdomen of a woman whose fallopian tubes are blocked. The fallopian tube (right on image) is blocked near the uterus, the triangular shape at center. The other fallopian tube is obstructed at a point further from the uterus where dilatation has occurred.
(Photo Researchers, Inc. Reproduced by permission.)

The procedure takes approximately 15–30 minutes. The x rays will be developed while the patient waits, but the final reading and interpretation of the x rays by a radiologist (a doctor who specializes in x rays) may not be available for a few days.

Interestingly, sometimes the hysterosalpingography procedure itself can be considered a treatment. The dye used can sometimes open up small blockages in the fallopian tubes. The need for additional test procedures or surgical treatments to deal with infertility should be discussed with the doctor.

Preparation

This procedure is generally done in the x-ray department of a hospital or large clinic. **General anesthesia** is not needed. A pain reliever may be taken prior to the procedure to lessen the severity of cramping.

Aftercare

While no special aftercare is required after a hysterosalpingography, the woman may be observed for some period after the procedure to ensure that she does not have any allergic reactions to the dye. A sanitary napkin may be worn after

KEY TERMS

Catheter—A thin tube, usually made of plastic, that is inserted into the body to allow the passage of fluid into or out of a site.

Fallopian tubes—The narrow ducts leading from a woman's ovaries to the uterus. After an egg is released from the ovary during ovulation, fertilization (the union of sperm and egg) normally occurs in the fallopian tubes.

HysteroGRAPHY—Another term for the x-ray procedure of the uterus and fallopian tubes.

Hysterosalpingogram—The term for the x ray taken during a hysterosalpingography procedure.

Speculum—A plastic or stainless steel instrument that is inserted into the opening of the vagina so the cervix (the opening of the uterus) and interior of the vagina can be examined.

the procedure to absorb dye that will flow out through the vaginal opening. If a blockage is seen in a tube, the patient may be given an antibiotic. A woman should notify her doctor if she experiences excessive bleeding, extensive pelvic pain, **fever**, or an unpleasant vaginal odor after the procedure. These symptoms may indicate a pelvic infection. Counseling may be necessary to interpret the results of the x rays, and to discuss any additional procedures to treat tubal blockages or uterine abnormalities found.

Risks

Cramps during the procedure are common. Complications associated with hysterosalpingography include abdominal pain, pelvic infection, and allergic reactions. It is also possible that abnormalities of the fallopian tubes and uterus will not be detected by this procedure.

Normal results

A normal hysterosalpingography will show a healthy, normally shaped uterus and unblocked fallopian tubes.

Abnormal results

Blockage of one or both of the fallopian tubes or abnormalities of the uterus may be detected.

Resources

ORGANIZATIONS

American Society for Reproductive Medicine. 1209 Montgomery Highway, Birmingham, AL 35216-2809. (205) 978-5000. <<http://www.asrm.com>>.

Altha Roberts Edgren

Hysteroscopy

Definition

Hysteroscopy is a procedure that allows a physician to look through the vagina and neck of the uterus (cervix) to inspect the cavity of the uterus. A telescope-like instrument called a hysteroscope is used. Hysteroscopy is used as both a diagnostic and a treatment tool.

Purpose

Diagnostic hysteroscopy may be used to evaluate the cause of **infertility**, to determine the cause of repeated miscarriages, or to help locate polyps and fibroids.

The procedure is also used to treat gynecological conditions, often instead of or in addition to **dilatation and curettage (D&C)**. A D&C is a procedure for scraping the lining of the uterus. A D&C can be used to take a sample of the lining of the uterus for analysis. Hysteroscopy is an advance over D&C because the doctor can take tissue samples of specific areas or actually see fibroids, polyps, or structural abnormalities.

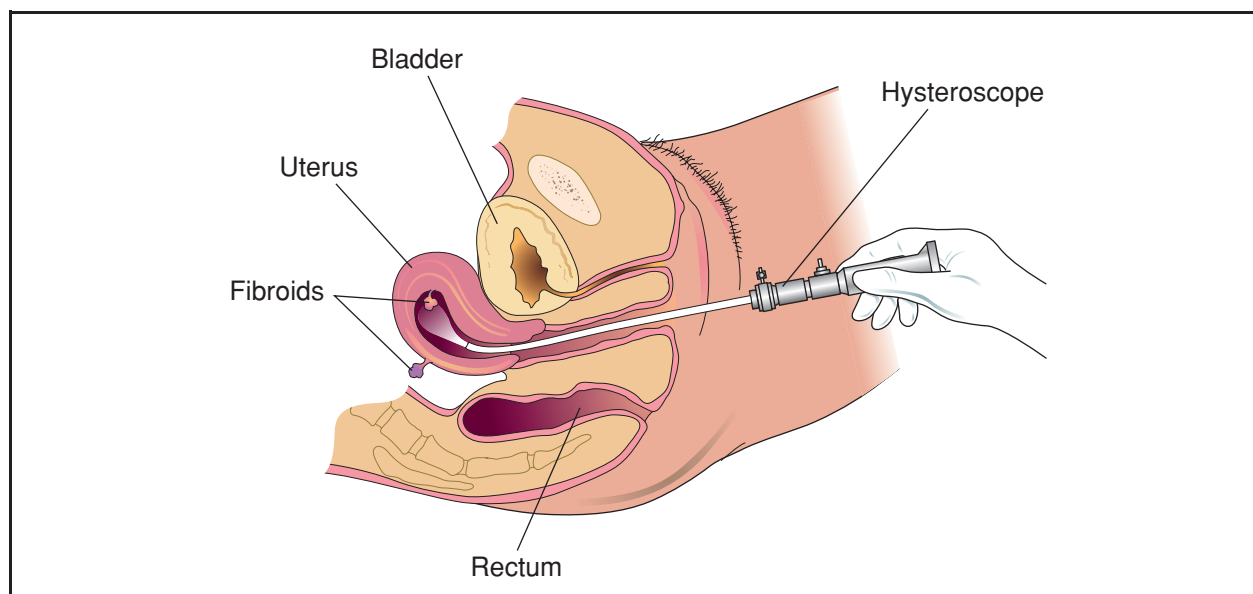
When used for treatment, the hysteroscope is used with other devices to remove polyps, fibroids, or IUDs that have become embedded in the wall of the uterus.

Precautions

The procedure is not performed on women with **cervical cancer**, **endometrial cancer**, or acute pelvic inflammation.

Description

Diagnostic hysteroscopy is performed in either a doctor's office or hospital. Before inserting the hysteroscope, the doctor injects a local anesthetic around the cervix. Once it has taken effect, the doctor dilates the cervix and then inserts a narrow lighted tube (the



Hysteroscopy is a procedure that allows inspection of the uterus by using a telescope-like instrument called a hysteroscope.
(Illustration by Electronic Illustrators Group.)

hysteroscope) through the cervix to reveal the inside of the uterus. Ordinarily, the walls of the uterus are touching each other. In order to get a better view, the uterus is inflated with carbon dioxide gas or fluid. Hysteroscopy takes about 30 minutes, and can cost anywhere from \$750 to \$4,000 depending on the extent of the procedure.

Treatment involving the use of hysteroscopy is usually performed as a day surgical procedure with regional or **general anesthesia**. Tiny surgical instruments are inserted through the hysteroscope, and are used to remove polyps or fibroids. A small sample of tissue lining the uterus is often removed for examination, especially if there is any abnormal bleeding.

Preparation

If the procedure is done in the doctor's office, the patient will be given a mild **pain** reliever before the procedure to ease cramping. The doctor will wash the vagina and cervix with an antiseptic solution.

If the procedure is done in the hospital under general anesthesia, the patient should not eat or drink anything (not even water) after midnight the night before the procedure.

Aftercare

Many women experience light bleeding for several days after surgical hysteroscopy. Mild cramping or

pain is common after operative hysteroscopy, but usually fades away within eight hours. If carbon dioxide gas was used, there may also be some shoulder pain. Nonprescription pain relievers may help ease discomfort. Women may want to take the day off and relax after having hysteroscopy.

Risks

Diagnostic hysteroscopy is a fairly safe procedure that only rarely causes complications. The primary risk is prolonged bleeding or infection, usually following surgical hysteroscopy to remove a growth.

Very rare complications include perforation of the uterus, bowel, or bladder. Surgery under general anesthesia causes the additional risks typically associated with anesthesia.

Patients should alert their health care provider if they develop any of these symptoms:

- abnormal discharge
- heavy bleeding
- fever over 101 °F (38.3 °C)
- severe lower abdominal pain

Normal results

A normal, healthy uterus with no fibroids or other growths.

KEY TERMS

Fibroid—A benign tumor of the uterus

Polyp—A growth that projects from the lining of the cervix, the nose, or any other mucus membrane.

Septum—A condition present at birth in which there is an extra fold of tissue down the center of the uterus that can cause infertility. This tissue can be removed with a wire electrode and a hysteroscope.

Abnormal results

Using hysteroscopy, the doctor may find **uterine fibroids** or polyps (often the cause of abnormal bleeding) or a septum (extra fold of tissue down the center of the uterus) that can cause infertility. Sometimes, precancerous or malignant growths are discovered.

Resources

PERIODICALS

Anon. "Looking Inside the Uterus." *Harvard Women's Health Watch* 4, no. 5 (January 1997): 4-5.

Carol A. Turkington

Hysterosonography

Definition

Hysterosonography, which is also called sonohysterography, is a new noninvasive technique that involves the slow infusion of sterile saline solution into a woman's uterus during ultrasound imaging. Hysterosonography allows the doctor to evaluate abnormal growths inside the uterus; abnormalities of the tissue lining the uterus (the endometrium); or disorders affecting deeper tissue layers. Hysterosonography does not require either radiation or contrast media, or invasive surgical procedures

Purpose

Hysterosonography is used to evaluate patients in the following groups:

- peri- or postmenopausal women with unexplained vaginal bleeding

- women whose endometrium appears abnormal during baseline ultrasound imaging
- women with fertility problems. **Infertility** is sometimes related to polyps, leiomyomas (fibroids), or **adhesions** inside the uterus. Adhesions are areas of tissue that have grown together to form bands or membranes across the inside of the uterus.
- women receiving tamoxifen therapy for **breast cancer**

Hysterosonography is useful as a screening test to minimize the use of more invasive diagnostic procedures, such as tissue biopsies and dilation and curettage (D&C). Hysterosonography can also be used as a follow-up after uterine surgery to evaluate its success.

Precautions

Hysterosonography is difficult to perform in patients with certain abnormalities:

- Cervical stenosis. Cervical stenosis means that the lower end of the uterus is narrowed or tightened. It complicates the insertion of a tube (catheter).
- Adhesions or large fibroids. These growths sometimes block the flow of saline fluid into the uterus.

Patients with active **pelvic inflammatory disease** (PID) should not be tested with hysterosonography until the disease is brought under control. Women with chronic PID or heart problems are given **antibiotics** before the procedure.

Description

A hysterosonography is preceded by a baseline ultrasound examination performed through the vagina. This allows the doctor to detect an unsuspected **pregnancy** and to assess the thickness and possible abnormalities of the patient's endometrium. The doctor then inserts a catheter into the uterus and injects sterile saline fluid while ultrasound imaging is recorded on film or videotape. The procedure takes about 10 to 15 minutes.

Preparation

Patients do not require special preparation apart from the timing of the procedure. Patients with fertility problems are examined during the first 10 days of the menstrual cycle. Patients who may have polyps are usually examined at a later phase in the cycle. The best time for examining women with fibroids is still under discussion.

KEY TERMS

Adhesion—An abnormal union or attachment of two areas of tissue.

Contrast medium—A chemical substance used to make an organ or body part opaque on x ray.

Dilation and curettage (D&C)—A surgical procedure in which the patient's cervix is widened (dilated) and the endometrium is scraped with a scoop-shaped knife (curette).

Endometrium—The tissue that lines the uterus.

Fibroid—Another word for leiomyoma.

Leiomyoma—A benign tumor composed of muscle tissue. Leiomyomas in the uterus are sometimes called fibroids.

Pelvic inflammatory disease (PID)—An inflammation of the fallopian tubes, usually caused by bacterial infection.

Polyp—A growth projecting from the lining of the uterus. Polyps can cause fertility problems or abnormal vaginal bleeding.

Saline solution—A solution of sterile water and salt used in a variety of medical procedures. In hysterosonography, saline solution is used to fill the uterus for diagnostic imaging.

Transvaginal ultrasound (US)—The diagnostic imaging procedure that serves as the baseline for a hysterosonographic examination.

Aftercare

Aftercare consists of advising the patient to contact her doctor in case of abnormal bleeding, **fever**, or abdominal **pain**. Some spotting or cramping is common, however, and can usually be treated with **non-steroidal anti-inflammatory drugs**, such as ibuprofen.

Risks

The chief risks are mild spotting and cramping after the procedure.

Normal results

Normal findings include a symmetrical uterus with a normal endometrium and no visible masses or tumors.

Abnormal results

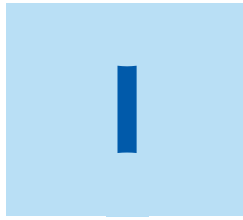
Abnormal findings include adhesions; polyps; leiomyomas; abnormal thickening of the endometrium; or tissue changes related to tamoxifen (Nolvadex), which is a drug given for breast **cancer**.

Resources

PERIODICALS

Cullinan, Joanne, et al. "Sonohysterography: A Technique for Endometrial Evaluation." *RadioGraphics* 15 (May 1995): 501-514.

Rebecca J. Frey, PhD



IBS see **Irritable bowel syndrome**

Ibuprophen see **Nonsteroidal anti-inflammatory drugs**

Ichthyosis

Definition

Derived from two Greek words meaning “fish” and “disease,” ichthyosis is a congenital (meaning present at birth) dermatological (skin) disease that is represented by thick, scaly skin.

Description

The ichthyoses are a group of skin diseases caused by an abnormality in skin growth that results in drying and scaling. There are at least 20 types of ichthyosis. Ichthyosis can be more or less severe, sometimes accumulating thick scales and cracks that are painful and bleed. Ichthyosis is not contagious. Some forms of ichthyosis are inherited while others are acquired in later life as a symptom of systemic disorders.

The most common form of ichthyosis, accounting for 95% of all cases of ichthyosis, is called ichthyosis vulgaris (*vulgaris* is the Latin word for “common”), and occurs in approximately one person in every 250. It is inherited in an autosomal dominant manner. The most rare types of ichthyosis occur in fewer than one person in one million and are inherited in an autosomal recessive manner. Ichthyosis occurs regardless of the part of the world the child is from, or the ethnic background of the parents.

Acquired ichthyosis is extremely rare; neither its incidence in the United States nor its incidence in the rest of the world are known as of 2003.

Both inherited and acquired ichthyoses affect males and females equally.

Causes and symptoms

Inherited ichthyoses

Depending on the specific type of ichthyosis, the inheritance can be autosomal recessive, autosomal dominant, X-linked recessive, X-linked dominant, or sporadic. Autosomal recessive means that the altered gene for the disease or trait is located on one of the first 22 pairs of chromosomes, which are also called “autosomes.” Males and females are equally likely to have an autosomal recessive disease or trait. Recessive means that two copies of the altered gene are necessary to express the condition. Therefore, a child inherits one copy of the altered gene from each parent, who are called carriers (because they have only one copy of the altered gene). Since carriers do not express the altered gene, parents usually do not know they carry the altered gene that causes ichthyosis until they have an affected child. Carrier parents have a 1-in-4 chance (or 25%) with each **pregnancy**, to have a child with ichthyosis.

Autosomal dominant inheritance also means that both males and females are equally likely to have the disease but only one copy of the altered gene is necessary to have the condition. An individual with ichthyosis has a 50/50 chance to pass the condition to his or her child.

The skin is made up of several layers, supported underneath by a layer of fat that is thicker or thinner depending on location. The lower layers contain blood vessels, the middle layers contain actively growing cells, and the upper layer consists of dead cells that serve as a barrier to the outside world. This barrier is nearly waterproof and highly resistant to infection. Scattered throughout the middle layers are hair

follicles, oil and sweat glands, and nerve endings. The upper layer is constantly flaking off and being replaced from beneath by new tissue. In ichthyosis, the skin's natural shedding process is slowed or inhibited; and in some types, skin cells are produced too rapidly.

The abnormality in skin growth and hydration called ichthyosis may present with symptoms at birth or in early childhood. Ichthyosis can itch relentlessly, leading to such complications of scratching as lichen simplex (**dermatitis** characterized by raw patches of skin). Either the cracking or the scratching can introduce infection, bringing with it discomfort and complications.

Acquired ichthyoses

The mildest form of acquired ichthyosis is called xeroderma, or dry flaky skin. It is not associated with any systemic diseases. Xeroderma occurs most often on the lower legs of middle-aged and elderly adults during cold weather, or on the lower legs of people who bathe too often. It is characterized chiefly by mild or moderate **itching**.

Ichthyosis may also be an early symptom of such disorders as **AIDS**, lymphoma, **hypothyroidism**, or **leprosy**. In these cases the ichthyosis is most noticeable on the patient's trunk and legs.

A few rare cases of acquired ichthyosis have been attributed to the use of certain drugs, specifically cimetidine (Tagamet), triparanol (Metasqualene), dixyrazine (a phenothiazine derivative used as an antipsychotic), nicotinic acid (vitamin B₃, butyrophe-none antipsychotics (Haldol, Inapsine, Orap), and clofazimine (Lamprene).

Diagnosis

A dermatologist will often make the diagnosis of ichthyosis based on findings from a clinical examination. However, a **skin biopsy**, or DNA study (from a small blood sample) is necessary to confirm the diagnosis. Evaluation for associated problems is done by a complete physical medical examination.

For some types of ichthyosis, the abnormal gene has been identified and prenatal testing is available. At present this is true for the autosomal recessive congenital ichthyoses, which include: lamellar ichthyosis (LI), autosomal recessive lamellar ichthyosis (ARLI), congenital ichthyosiform erythroderma (CIE), and non-bullous congenital ichthyosiform erythroderma (NBCIE).

There are four different genes that have been located for the autosomal recessive congenital ichthyoses. Testing, however, is available for only one gene, known as transglutaminase-1 (TGM1). This gene is located on chromosome 14. Once a couple has had a child with ichthyosis, and they have had the genetic cause identified by DNA studies (performed from a small blood sample), prenatal testing for future pregnancies may be considered. (Note that prenatal testing may not be possible if both mutations cannot be identified.) Prenatal diagnosis is available via either **chorionic villus sampling** (CVS) or **amniocentesis**. CVS is a biopsy of the placenta performed in the first trimester of pregnancy under ultrasound guidance. Ultrasound is the use of sound waves to visualize the developing fetus. The genetic makeup of the placenta is identical to the fetus and therefore the TGM1 gene can be studied from this tissue. There is approximately a one in 100 chance for **miscarriage** with CVS. Amniocentesis is a procedure done under ultrasound guidance in which a long thin needle is inserted through the mother's abdomen into the uterus, to withdraw a couple of tablespoons of amniotic fluid (fluid surrounding the developing baby) to study. The TGM1 gene can be studied using cells from the amniotic fluid. Other genetic tests, such as a chromosome analysis, may also be performed through either CVS or amniocentesis.

Acquired ichthyosis is usually diagnosed in the course of identifying the underlying disorder. With the exception of acquired ichthyosis related to lymphoma, a doctor cannot tell the difference between inherited and acquired ichthyosis by examining skin samples through a microscope.

Treatment

Most treatments for ichthyosis are topical, which means that they are applied directly to the skin, not taken internally. Xeroderma is easily treated by minimizing bathing and applying an emollient cream or mineral oil after bathing while the skin is still moist. Some forms of ichthyosis require two forms of treatment—a reduction in the amount of scale buildup and moisturizing of the underlying skin. Several agents are available for each purpose. Reduction in the amount of scale is achieved by keratolytics. Among this class of drugs are urea, lactic acid, and salicylic acid. Petrolatum, 60% propylene glycol, and glycerin are successful moisturizing agents, as are many commercially available products. Increased humidity of the ambient air is also helpful in preventing skin dryness.

KEY TERMS

Amniocentesis—A procedure performed at 16–18 weeks of pregnancy in which a needle is inserted through a woman's abdomen into her uterus to draw out a small sample of the amniotic fluid from around the baby. Either the fluid itself or cells from the fluid can be used for a variety of tests to obtain information about genetic disorders and other medical conditions in the fetus.

Amniotic fluid—The fluid that surrounds a developing baby during pregnancy.

Autosomal dominant—A pattern of genetic inheritance where only one abnormal gene is needed to display the trait or disease.

Autosomal recessive inheritance—A pattern of genetic inheritance where two abnormal genes are needed to display the trait or disease.

Dermatologist—A physician that specializes in diagnosing and treating disorders of the skin.

Emollients—Petroleum or lanolin-based skin lubricants.

Keratin—A tough, nonwater-soluble protein found in the nails, hair, and the outermost layer of skin. Human hair is made up largely of keratin.

Keratinocytes—Skin cells.

Keratolytic—An agent that dissolves or breaks down the outer layer of skin (keratins).

Retinoids—A derivative of synthetic Vitamin A.

Sporadic—Isolated or appearing occasionally with no apparent pattern.

X-linked dominant inheritance—The inheritance of a trait by the presence of a single gene on the X chromosome in a male or female, passed from an affected female who has the gene on one of her X chromosomes.

X-linked recessive inheritance—The inheritance of a trait by the presence of a single gene on the X chromosome in a male, passed from a female who has the gene on one of her X chromosomes, and who is referred to as an unaffected carrier.

Because the skin acts as a barrier to the outside environment, medicines have a hard time penetrating, especially through the thick skin of the palms of the hands and the soles of the feet. This resistance is diminished greatly by maceration (softening the skin). Soaking hands in water macerates skin so that it looks like prune skin. Occlusion (covering) with

rubber gloves or plastic wrap will also macerate skin. Applying medicines and then covering the skin with an occlusive dressing will facilitate entrance of the medicine and greatly magnify its effect.

Secondary treatments are necessary to control pruritus (itching) and infection. Commercial products containing camphor, menthol, eucalyptus oil, aloe, and similar substances are very effective as antipruritics. If the skin cracks deeply enough, a pathway for infection is created. **Topical antibiotics** like bacitracin are effective in prevention and in the early stages of these skin infections. Cleansing with hydrogen peroxide inhibits infection as well.

Finally, there are topical and internal derivatives of vitamin A called retinoids that improve skin growth and are used for severe cases of **acne**, ichthyosis, and other skin conditions. Tazarotene (Tazorac), a retinoid that was originally developed to treat **psoriasis** and acne, appears to give good results in treating ichthyosis with fewer side effects than other retinoids.

Prognosis

This condition requires continuous care throughout a lifetime. Properly treated, in most cases it is a cosmetic problem. There are a small number of lethal forms, such as harlequin fetus.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD, editors. "Ichthyosis." Section 10, Chapter 121. In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

PERIODICALS

Fleckman, P. "Management of the Ichthyoses." *Skin Therapy Letter* 8 (September 2003): 3–7.

Hatsell, S. J., H. Stevens, A. P. Jackson, et al. "An Autosomal Recessive Exfoliative Ichthyosis with Linkage to Chromosome 12q13." *British Journal of Dermatology* 149 (July 2003): 174–180.

Lefevre, C., S. Audebert, F. Jobard, et al. "Mutations in the Transporter ABCA12 Are Associated with Lamellar Ichthyosis Type 2." *Human Molecular Genetics* 12 (September 15, 2003): 2369–2378.

Marulli, G. C., E. Campione, M. S. Chimenti, et al. "Type I Lamellar Ichthyosis Improved by Tazarotene 0.1% Gel." *Clinical and Experimental Dermatology* 28 (July 2003): 391–393.

Okulicz, Jason F., MD, and Robert A. Schwartz, MD, MPH. "Ichthyosis Vulgaris, Hereditary and Acquired." *eMedicine* November 1, 2001. <<http://www.emedicine.com/derm/topic678.htm>>.

ORGANIZATIONS

Alliance of Genetic Support Groups. 4301 Connecticut Ave. NW, Suite 404, Washington, DC 20008. (202) 966-5557. Fax: (202) 966-8553. <<http://www.geneticalliance.org>>.
 Foundation for Ichthyosis and Related Skin Types (FIRST). 650 N. Cannon Ave., Suite 17, Landsdale, PA 19446. (215) 631-1411 or (800) 545-3286. Fax: (215) 631-1413. <<http://www.scalyskin.org>>.

National Organization for Rare Disorders, Inc. (NORD). 55 Kenosia Avenue, P. O. Box 1968, Danbury, CT 06813. (800) 999-6673 or (203) 744-0100. <<http://www.rarediseases.org>>.

OTHER

Immune Deficiency Foundation Website.
 <www.primaryimmune.org>.

Catherine L. Tesla, MS, CGC
 Rebecca J. Frey, PhD

Icterus *see* **jaundice**

Idiopathic hypertrophic subaortic stenosis
see **Hypertrophic cardiomyopathy**

Idiopathic infiltrative lung diseases

Definition

The term *idiopathic* means “cause unknown.” The idiopathic infiltrative lung diseases, also known as interstitial lung diseases, are a group of more than a hundred disorders seen in both adults and (less often) in children, whose cause is unknown but which tend to spread, or “infiltrate” through much or all of the lung tissue. They range from mild conditions that respond well to treatment, to progressive, nonresponsive disease states that severely limit lung function and may cause **death**.

Description

The body produces inflammatory cells in response to a variety of conditions, including a number of different diseases, pollutants, certain infections, exposure to organic dust or toxic fumes and vapors, and various drugs and poisons. When white blood cells and tissue fluid rich in protein collect in the small air sacs of the lungs, or alveoli, the sacs become inflamed (alveolitis). In time, the fluid may solidify and cause scar formation that replaces the normal lung tissue. This process is known as **pulmonary fibrosis**. In about half of all

patients, no specific cause is ever found; they are said to have idiopathic pulmonary fibrosis.

Some patients have special types of interstitial lung disease that may occur in certain types of patients, or feature typical pathological changes when a sample of lung tissue is examined under a microscope. They include:

- Usual interstitial pneumonitis. Disease occurs in a patchy form throughout the lungs. Parts of the lungs can appear normal while others have dense scar tissue and lung cysts, often the end result of pulmonary fibrosis. This disease progresses quite slowly. Both children and adults may be affected.
- Desquamative interstitial pneumonitis. Similar-appearing lesions are present throughout the lungs. Both inflammatory cells and cells that have separated from the air sac linings (“desquamated”) are present. Some researchers believe this is an early form of usual interstitial pneumonitis.
- Lymphocytic interstitial pneumonitis. Most of the cells infiltrating the lungs are the type of white blood cells called lymphocytes. Both the breathing tubes (bronchi) and blood vessels of the lungs become thickened. In children, this condition tends to occur when the immune system is not operating properly as occurs with Acquired Immune Deficiency Syndrome (AIDS).

Causes and symptoms

By definition, the causes of *idiopathic* infiltrative lung diseases are not known. Some forms of pulmonary fibrosis, however, do have specific causes and these may provide a clue as to what may cause idiopathic diseases. Known causes of pulmonary fibrosis include diseases that impair the body’s immune function; infection by viruses and the bacterium causing **tuberculosis**; and exposure to such mineral dusts as silica or asbestos, or such organic materials as bird droppings. Other cases of pulmonary fibrosis result from exposure to fumes and vapors, radiation (in industry or medically), and certain drugs used to treat disease.

Patients with interstitial lung disease usually have labored breathing when exerting themselves. Often they **cough** and feel overly tired (“no stamina”). **Wheezing** is uncommon. When the physician listens to the patient’s chest with a stethoscope, dry, crackling sounds may be heard. Some patients have vague chest **pain**. When disease progresses, the patient may breathe very rapidly, have mottled blue skin (because of getting too little oxygen), and lose weight. The fingertips may appear thick or club-shaped.

Diagnosis

Both **scars** in the lung and cysts (air-filled spaces) can be seen on a **chest x ray**. Up to 10% of patients, however, may have normal x rays even if their symptoms are severe. A special type of x ray, high-resolution computed tomography scan (CT scan), often is helpful in adult patients. Tests of lung function will show that the lungs cannot hold enough air with each breath, and there is too little oxygen in the blood, especially after exercising. In a procedure called bronchoalveolar lavage, a tube is placed through the nose and windpipe into the bronchi and a small amount of saline is released and then withdrawn. This fluid can then be analyzed for cells. A tiny piece of lung tissue can be sampled using the same instrument. If necessary, a larger sample (a biopsy) is taken through an incision in the chest wall and examined under a microscope.

Treatment

The first medication given, providing scarring is not too extensive, is usually a steroid drug such as prednisone. An occasional patient will improve dramatically if steroid therapy stops the inflammation. Most patients, however, improve to a limited extent. It may take 6–12 weeks for a patient to begin to respond. Patients must be watched closely for a gain in body weight, high blood pressure, and depression. Steroids can also result in diabetes, ulcer disease, and cataract. Patients treated with steroids are at risk of contracting serious infection. If steroids have not proved effective or have caused serious side effects, other anti-inflammatory drugs, such as cyclophosphamide (Cytoxan) or azathioprine (Imuran), can be tried. Cytoxan sometimes is combined with a steroid, but it carries its own risks, which include bladder inflammation and suppression of the bone marrow. Some patients will benefit from a bronchodilator drug that relaxes the airway and makes breathing easier.

Some patients with interstitial lung disease, especially children, will need **oxygen therapy**. Usually oxygen is given during sleep or **exercise**, but if the blood oxygen level is very low it may be given constantly. A program of conditioning, training in how to breathe efficiently, energy-saving tips, and a proper diet will help patients achieve the highest possible level of function given the state of their illness. All patients should be vaccinated each year against **influenza**. A last resort for those with very advanced disease who do not respond to medication is **lung transplantation**. This operation is being done more widely, and it is even possible to replace both lungs.

KEY TERMS

Bronchoalveolar lavage—A way of obtaining a sample of fluid from the airways by inserting a flexible tube through the windpipe. Used to diagnose the type of lung disease.

Desquamation—Shedding of the cells lining the insides of the air sacs. A feature of desquamative interstitial pneumonitis.

Idiopathic—A disease whose cause is unknown.

Immune system—A set of body chemicals and specialized cells that attack an invading agent (such as a virus) by forming antibodies that can engulf and destroy it.

Infiltrative—A process whereby inflammatory or other types of disease spread throughout an organ such as the lungs.

Interstitial—Refers to the connective tissue that supports the “working parts” of an organ, in the case of the lungs the air sacs.

Pulmonary fibrosis—A scarring process that is the end result of many forms of long-lasting lung disease.

Prognosis

A scoring system based on lung function and x ray appearances has been designed to help monitor a patient's course. In general, idiopathic forms of interstitial lung disease cause a good deal of illness, and a significant number of deaths. A majority of patients get worse over time, although survival for many years is certainly possible. An estimated one in five affected children fail to survive. In different series, survival times average between four and ten years. Early diagnosis gives the best chance of a patient recovering or at least stabilizing. Once the lungs are badly scarred, nothing short of lung transplantation offers hope of restoring lung function. Patients with desquamative interstitial pneumonitis tend to respond well to steroid treatment, and live longer than those with other types of infiltrative lung disease.

Prevention

Since we do not understand what causes idiopathic interstitial lung diseases, there is no way to prevent them. What can be done is to prevent extensive scarring of the lungs by making the diagnosis shortly after the first symptoms develop, and trying steroids or other drugs in hope of

suppressing lung inflammation. Every effort should be made to avoid exposure to dusts, gases, chemicals, and even pets. Keeping fit and learning how to breathe efficiently will help maintain lung function as long as possible.

Resources

ORGANIZATIONS

American Lung Association. 1740 Broadway, New York, NY 10019. (800) 586-4872. <<http://www.lungusa.org>>.

David A. Cramer, MD

Idiopathic primary renal hematuric/proteinuric syndrome

Definition

This syndrome includes a group of disorders characterized by blood and protein in the urine and by damage to the kidney glomeruli (filtering structures) that may lead to kidney failure.

Description

This syndrome, also known as Berger's disease or IgA nephropathy, arises when internal kidney structures called glomeruli become inflamed and injured. It can occur at any age, but the great majority of patients are 16–35 when diagnosed. Males seem to be affected more often than females, and whites are more often affected than blacks. Blood in the urine (hematuria), either indicated by a visible change in the color of the urine or detected by laboratory testing, is a hallmark of this syndrome, and it may occur continuously or sporadically. The pattern of occurrence is not indicative of the severity of kidney damage.

Causes and symptoms

The glomeruli are the kidney structures that filter the blood and extract waste, which is then excreted as urine. The barrier between the blood and the urine side of the filter mechanism is a membrane only one cell layer thick. Anything that damages the membrane will result in hematuria. Symptoms of idiopathic primary renal hematuric/proteinuric syndrome are caused by inflammation of the glomeruli and deposit of IgA antibodies in kidney tissue. Although a genetic basis

for this syndrome is suspected, this has not been proven. Symptoms often appear 24–48 hours after an upper respiratory or gastrointestinal infection. Symptoms of the syndrome include:

- blood in the urine (hematuria)
- protein in the urine (proteinuria)
- pain in the lower back or kidney area
- elevated blood pressure (20–30% of cases)
- nephrotic syndrome (less than 10% of cases)
- swelling (occasionally)

This condition usually does not get worse with time, although renal failure occasionally results. In patients with large amounts of IgA deposits in their glomeruli, the long-term prognosis may not be favorable. The syndrome can go into remission spontaneously, although this is more common in children than in adults.

Diagnosis

One of the objectives of diagnosis is to distinguish glomerular from non-glomerular kidney diseases. Idiopathic primary hematuric/proteinuric syndrome involves the glomeruli. The presence of fragmented or distorted red blood cells in the urine is evidence of glomerular disease. A high concentration of protein in the urine is also evidence for glomerular disease. The hematuria associated with this syndrome must be distinguished from that caused by urinary tract diseases, which can also cause a loss of blood into the urine. Biopsy of the patient's kidney shows deposits of IgA antibodies. Detecting IgA-antibody deposits rules out thin membrane disease as the cause of the hematuria and proteinuria. Test values are normal for ASO, complement, rheumatoid factor, antinuclear antibodies, anti-DNase, and cryoglobulins, all of which are associated with different types of **kidney disease**. A diagnosis of idiopathic primary renal hematuric/proteinuric syndrome is largely made by ruling out other diseases and their causes, leaving this syndrome as the remaining possible diagnosis.

Treatment

Many patients do not need specific treatment, except for those who have symptoms indicating a poor prognosis. Oral doses of **corticosteroids** are effective in patients with mild proteinuria and good kidney function. Other treatments, such as medications to lower blood pressure, are aimed at slowing or

KEY TERMS

Glomeruli (singular, glomerulus)—Filtering structures in the kidneys.

Hematuria—The presence of hemoglobin or red blood cells in the urine.

Idiopathic—Refers to a disease that arises from an obscure or unknown cause.

Nephrotic syndrome—A kidney disorder characterized by fluid retention (edema) and proteinuria. It is caused by damage to the kidney glomeruli.

Proteinuria—The presence of protein in the urine exceeding normal levels.

preventing kidney damage. If kidney failure develops, dialysis or **kidney transplantation** is necessary.

Prognosis

Idiopathic primary renal hematuric/proteinuric syndrome progresses slowly and in many cases does not progress at all. Risk for progression of the disorder is considered higher if there is:

- high blood pressure
- large amounts of protein in the urine
- increased levels of urea and creatinine in the blood (indications of kidney function)

About 25–35% of patients may develop kidney failure within about 25 years.

Prevention

Since the underlying causes of this syndrome are so poorly understood, there is no known prevention.

Resources

BOOKS

Greenberg, A. *Primer on Kidney Diseases*. San Diego: Academic Press, 1998.

ORGANIZATIONS

IgA Nephropathy Support Network. 964 Brown Ave., Huntingdon Valley, PA 19006. (215) 663–0536.

National Kidney Foundation. 30 East 33rd St., New York, NY 10016. (800) 622-9010. <<http://www.kidney.org>>.

John T. Lohr, PhD

Idiopathic thrombocytopenic purpura

Definition

Idiopathic thrombocytopenic purpura, or ITP, is a bleeding disorder caused by an abnormally low level of platelets in the patient's blood. Platelets are small plate-shaped bodies in the blood that combine to form a plug when a blood vessel is injured. The platelet plug then binds certain proteins in the blood to form a clot that stops bleeding. ITP's name describes its cause and two symptoms. Idiopathic means that the disorder has no apparent cause. ITP is now often called immune thrombocytopenic purpura rather than idiopathic because of recent findings that ITP patients have autoimmune antibodies in their blood. **Thrombocytopenia** is another word for a decreased number of blood platelets. Purpura refers to a purplish or reddish-brown skin rash caused by the leakage of blood from broken capillaries into the skin. Other names for ITP include purpura hemorrhagica and essential thrombocytopenia.

Description

ITP may be either acute or chronic. The acute form is most common in children between the ages of two and six years; the chronic form is most common in adult females between 20 and 40. Between 10% and 20% of children with ITP have the chronic form. ITP does not appear to be related to race, lifestyle, climate, or environmental factors.

ITP is a disorder that affects the overall *number* of blood platelets rather than their function. The normal platelet level in adults is between 150,000 and 450,000/mm³. Platelet counts below 50,000 mm³ increase the risk of dangerous bleeding from trauma; counts below 20,000/mm³ increase the risk of spontaneous bleeding.

Causes and Symptoms

In adults, ITP is considered an autoimmune disorder, which means that the body produces antibodies that damage some of its own products—in this case, blood platelets. Some adults with chronic ITP also have other immune system disorders, such as **systemic lupus erythematosus (SLE)**. In children, ITP is usually triggered by a virus infection, most often **rubella**, **chickenpox**, **measles**, cytomegalovirus, or Epstein-Barr virus. It usually begins about two or three weeks after the infection.

Acute ITP

Acute ITP is characterized by bleeding into the skin or from the nose, mouth, digestive tract, or urinary tract. The onset is usually sudden. Bleeding into the skin takes the form of purpura or petechiae. Purpura is a purplish or reddish-brown rash or discoloration of the skin; petechiae are small round pinpoint hemorrhages. Both are caused by the leakage of blood from tiny capillaries under the skin surface. In addition to purpura and petechiae, the patient may notice that he or she **bruises** more easily than usual. In extreme cases, patients with ITP may bleed into the lungs, brain, or other vital organs.

Chronic ITP

Chronic ITP has a gradual onset and may have minimal or no external symptoms. The low **platelet count** may be discovered in the course of a routine blood test. Most patients with chronic ITP, however, will consult their primary care doctor because of the purpuric skin rash, nosebleeds, or bleeding from the digestive or urinary tract. Women sometimes go to their gynecologist for unusually heavy or lengthy menstrual periods.

The risk factors for the development of chronic ITP include:

- female sex
- age over 10 years at onset of symptoms
- slow onset of bruising
- presence of other autoantibodies in the blood

Diagnosis

ITP is usually considered a diagnosis of exclusion, which means that the doctor arrives at the diagnosis by a process of ruling out other possible causes. If the patient belongs to one or more of the risk groups for chronic ITP, the doctor may order a blood test for autoantibodies in the blood early in the diagnostic process.

Physical examination

If the doctor suspects ITP, he or she will examine the patient's skin for bruises, purpuric areas, or petechiae. If the patient has had nosebleeds or bleeding from the mouth or other parts of the body, the doctor will examine these areas for other possible causes of bleeding. Patients with ITP usually look and feel healthy except for the bleeding.

The most important features that the doctor will be looking for during the **physical examination** are the

condition of the patient's spleen and the presence of **fever**. Patients with ITP do not have fever, whereas patients with lupus and some other types of thrombocytopenia are usually feverish. The doctor will have the patient lie flat on the examining table in order to feel the size of the spleen. If the spleen is noticeably enlarged, ITP is not absolutely ruled out but is a less likely diagnosis.

Laboratory testing

The doctor will order a complete **blood count** (CBC), a test of clotting time, a bone marrow test, and a test for antiplatelet antibodies if it is available in the hospital laboratory. Patients with ITP usually have platelet counts below 20,000/mm³ and prolonged **bleeding time**. The size and appearance of the platelets may be abnormal. The red blood cell count (RBC) and white blood cell count (WBC) are usually normal, although about 10% of patients with ITP are also anemic. The blood marrow test yields normal results. Detection of antiplatelet antibodies in the blood is considered to confirm the diagnosis of ITP.

Treatment

General care and monitoring

There is no specific treatment for ITP. In most cases, the disorder will resolve without medications or surgery within two to six weeks. Nosebleeds can be treated with ice packs when necessary.

General care includes explaining ITP to the patient and advising him or her to watch for bruising, petechiae, or other signs of recurrence. Children should be discouraged from rough contact sports or other activities that increase the risk of trauma. Patients are also advised to avoid using **aspirin** or ibuprofen (Advil, Motrin) as **pain** relievers because these drugs lengthen the clotting time of blood.

Emergency treatment

Patients with acute ITP who are losing large amounts of blood or bleeding into their central nervous system require emergency treatment. This includes transfusions of platelets, intravenous immunoglobulins, or prednisone. Prednisone is a steroid medication that decreases the effects of antibody on platelets and eventually lowers antibody production. If the patient has a history of ITP that has not responded to prednisone or immunoglobulins, the surgeon may remove the patient's spleen. This operation is called a **splenectomy**. The reason for removing the spleen when ITP does not respond to other forms of

KEY TERMS

Autoimmune disorder—A disorder in which the patient's immune system produces antibodies that destroy some of the body's own products. ITP in adults is thought to be an autoimmune disorder.

Idiopathic—Of unknown cause. Idiopathic refers to a disease that is not preceded or caused by any known dysfunction or disorder in the body.

Petechiae—Small pinpoint hemorrhages in skin or mucous membranes caused by the rupture of capillaries.

Platelet—A blood component that helps to prevent blood from leaking from broken blood vessels. ITP is a bleeding disorder caused by an abnormally low level of platelets in the blood.

Prednisone—A corticosteroid medication that is used to treat ITP. Prednisone works by decreasing the effects of antibody on blood platelets. Long-term treatment with prednisone is thought to decrease antibody production.

Purpura—A skin discoloration of purplish or brownish red spots caused by bleeding from broken capillaries.

Splenectomy—Surgical removal of the spleen.

Thrombocytopenia—An abnormal decline in the number of platelets in the blood.

treatment is that the spleen sometimes keeps platelets out of the general blood circulation.

Medications and transfusions

Patients with chronic ITP can be treated with prednisone, immune globulin, or large doses of intravenous gamma globulin. Although 90% of patients respond to immunoglobulin treatment, it is very expensive. About 80% of patients respond to prednisone therapy. Platelet transfusions are not recommended for routine treatment of ITP. If the patient's platelet level does not improve within one to four months, or requires high doses of prednisone, the doctor may recommend splenectomy. All medications for ITP are given either orally or intravenously; intramuscular injection is avoided because of the possibility of causing bleeding into the skin.

Surgery

Between 80% and 85% of adults with ITP have a remission of the disorder after the spleen is removed.

Splenectomy is usually avoided in children younger than five years because of the increased risk of a severe infection after the operation. In older children, however, splenectomy is recommended if the child has been treated for 12 months without improvement; if the ITP is very severe or the patient is getting worse; if the patient begins to bleed into the head or brain; and if the patient is an adolescent female with extremely heavy periods.

Prognosis

The prognosis for recovery from acute ITP is good; 80% of patients recover without special treatment. The prognosis for chronic ITP is also good; most patients experience long-term remissions. In rare instances, however, ITP can cause life-threatening hemorrhage or bleeding into the central nervous system.

Resources

BOOKS

Linker, Charles A. "Blood." In *Current Medical Diagnosis and Treatment*, 1998, edited by Stephen McPhee, et al., 37th ed. Stamford: Appleton & Lange, 1997.

Rebecca J. Frey, PhD

IHSS see **Hypertrophic cardiomyopathy**

Ileal conduit see **Urinary diversion surgery**

Ileocol see **Crohn's disease**

Ileostomy see **Enterostomy**

Ileus

Definition

Ileus is a partial or complete non-mechanical blockage of the small and/or large intestine. The term "ileus" comes from the Latin word for **colic**.

Description

There are two types of **intestinal obstructions**, mechanical and non-mechanical. Mechanical obstructions occur because the bowel is physically blocked and its contents can not pass the point of the obstruction. This happens when the bowel twists on itself (volvulus) or as the result of hernias, impacted feces,

abnormal tissue growth, or the presence of foreign bodies in the intestines.

Unlike mechanical obstruction, non-mechanical obstruction, called ileus or paralytic ileus, occurs because peristalsis stops. Peristalsis is the rhythmic contraction that moves material through the bowel. Ileus is most often associated with an infection of the peritoneum (the membrane lining the abdomen). It is one of the major causes of bowel obstruction in infants and children.

Another common cause of ileus is a disruption or reduction of the blood supply to the abdomen. Handling the bowel during abdominal surgery can also cause peristalsis to stop, so people who have had abdominal surgery are more likely to experience ileus. When ileus results from abdominal surgery the condition is often temporary and usually lasts only 48–72 hours.

Ileus sometimes occurs as a complication of surgery on other parts of the body, including **joint replacement** or chest surgery.

Ileus can also be caused by kidney diseases, especially when potassium levels are decreased. Heart disease and certain **chemotherapy** drugs, such as vinblastine (Velban, Velsar) and vincristine (Oncovin, Vincasar PES, Vincrex), also can cause ileus. Infants with **cystic fibrosis** are more likely to experience meconium ileus (a dark green material in the intestine). Over all, the total rate of bowel obstruction due both to mechanical and non-mechanical causes is one in one thousand people (1/1,000).

Causes and symptoms

When the bowel stops functioning, the following symptoms occur:

- abdominal cramping
- abdominal distention
- nausea and vomiting
- failure to pass gas or stool

Diagnosis

When a doctor listens with a stethoscope to the abdomen there will be few or no bowel sounds, indicating that the intestine has stopped functioning. Ileus can be confirmed by x rays of the abdomen, **computed tomography scans** (CT scans), or ultrasound. It may be necessary to do more invasive tests, such as a **barium enema** or upper GI series, if the obstruction is mechanical. Blood tests also are useful in diagnosing paralytic ileus.

Barium studies are used in cases of mechanical obstruction, but may cause problems by increasing pressure or intestinal contents if used in ileus. Also, in cases of suspected mechanical obstruction involving the gastrointestinal tract (from the small intestine downward) use of barium x rays are contraindicated, since they may contribute to the obstruction. In such cases a barium enema should always be done first.

Treatment

Patients may be treated with supervised bed rest in a hospital and bowel rest. Bowel rest means that nothing is taken by mouth and patients are fed intravenously or through the use of a nasogastric tube. A nasogastric tube is a tube inserted through the nose, down the throat, and into the stomach. A similar tube can be inserted in the intestine. The contents are then suctioned out. In some cases, especially where there is a mechanical obstruction, surgery may be necessary.

Drug therapies that promote intestinal motility (ability of the intestine to move spontaneously), such as cisapride and vasopressin (Pitressin), are sometimes prescribed.

Alternative treatment

Alternative practitioners offer few treatment suggestions, but focus on prevention by keeping the bowels healthy through eating a good diet, high in fiber and low in fat. If the case is not a medical emergency, homeopathic treatment and **traditional Chinese medicine** can recommend therapies that may help to reinstate peristalsis.

Prognosis

The outcome of ileus varies depending on its cause.

Prevention

Most cases of ileus are not preventable. Surgery to remove a tumor or other mechanical obstruction will help prevent a recurrence.

Some measures that have been recommended to minimize the severity of postoperative ileus or shorten its duration include making sure that any electrolyte imbalances are corrected, and using nonopioid medications to relieve **pain**, as opioid drugs (including morphine, oxycodone, and codeine) tend to cause **constipation**. One group of drugs that shows promise for treating abdominal pain is a class of medications known as kappa-opioid agonists. As of 2004, however,

KEY TERMS

Computed tomography scan (or CT scan)—A computer enhanced x-ray study performed to detect abnormalities that do not show up on normal x rays.

Meconium—A greenish fecal material that forms the first bowel movement of an infant.

Peritoneum—The transparent membrane lining the abdominal cavity that holds internal organs in place.

these drugs are still under investigation for controlling visceral pain in humans.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD, editors. "Ileus." Section 3, Chapter 25. In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

PERIODICALS

- Baig, M. K., and S. D. Wexner. "Postoperative Ileus: A Review." *Diseases of the Colon and Rectum* 47 (April 2004): 516–526.
- Lassandro, F., N. Gagliardi, M. Scuderi, et al. "Gallstone Ileus Analysis of Radiological Findings in 27 Patients." *European Journal of Radiology* 50 (April 2004): 23–29.
- Pavone, P., T. Johnson, P. S. Saulog, et al. "Perioperative Morbidity in Bilateral One-Stage Total Knee Replacements." *Clinical Orthopaedics and Related Research* 421 (April 2004): 155–161.
- Riviere, P. J. "Peripheral Kappa-Opioid Agonists for Visceral Pain." *British Journal of Pharmacology* 141 (April 2004): 1331–1334.

OTHER

- "Bowel Paralysis." *Trigan Oncology Associates Page*. <<http://www.trigan.com/ileus.htm>>.
- "Intestinal Obstruction." *HealthAnswers.com*. <<http://www.healthanswers.com/database/ami/converted/000260.html>>.

Tish Davidson, A. M.
Rebecca J. Frey, PhD

Immobilization

Definition

Immobilization refers to the process of holding a joint or bone in place with a splint, cast, or brace.

This is done to prevent an injured area from moving while it heals.

Purpose

Splints, casts, and braces support and protect broken bones, dislocated joints, and such injured soft tissue as tendons and ligaments. Immobilization restricts motion to allow the injured area to heal. It can help reduce **pain**, swelling, and muscle spasm. In some cases, splints and casts are applied after surgical procedures that repair bones, tendons, or ligaments. This allows for protection and proper alignment early in the healing phase.

Precautions

There are no special precautions for immobilization.

Description

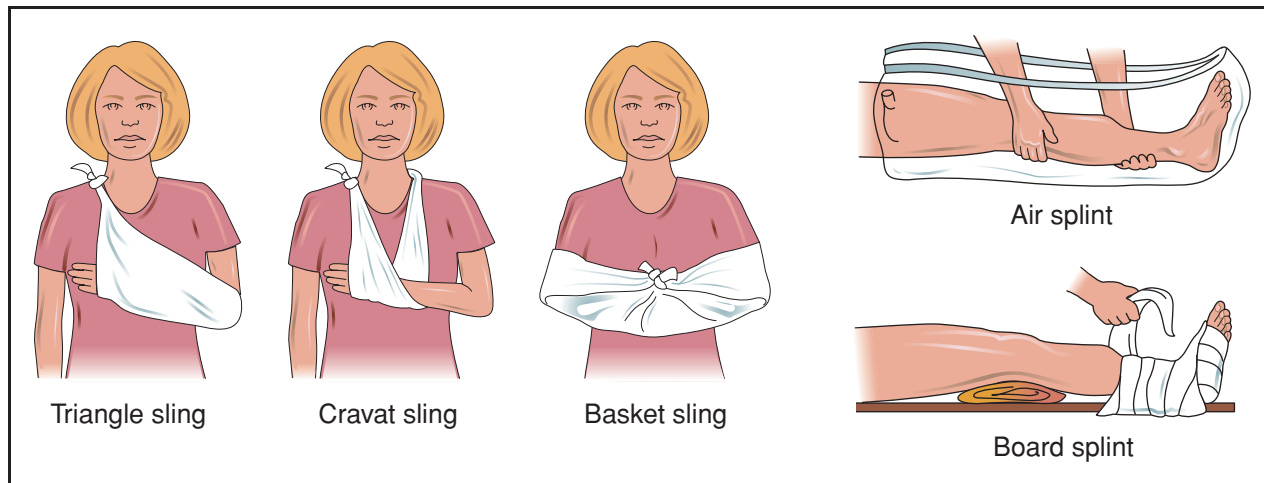
When an arm, hand, leg, or foot requires immobilization, the cast, splint, or brace will generally extend from the joint above the injury to the joint below the injury. For example, an injury to the mid-calf requires immobilization from the knee to the ankle and foot. Injuries of the hip and upper thigh or shoulder and upper arm require a cast that encircles the body and extends down the injured leg or arm.

Casts and splints

Casts are generally used for immobilization of a broken bone. Once the doctor makes sure the two broken ends of the bone are aligned, a cast is put on to keep them in place until they are rejoined through natural healing. Casts are applied by a physician, a nurse, or an assistant. They are custom-made to fit each person, and are usually made of plaster or fiberglass. Fiberglass weighs less than plaster, is more durable, and allows the skin more adequate airflow than plaster. A layer of cotton or synthetic padding is first wrapped around the skin to cover the injured area and protect the skin. The plaster or fiberglass is then applied over this.

Most casts should not be gotten wet. However, some types of fiberglass casts use Gore-tex padding that is waterproof and allows the person to completely immerse the cast in water when taking a shower or bath. There are some circumstances when this type of cast material can not be used.

A splint is often used to immobilize a dislocated joint while it heals. Splints are also often used for finger injuries, such as **fractures** or baseball finger. Baseball finger is an injury in which the tendon at the



Immobilization refers to the process of immobilizing or fixating the position of a joint, bone, extremity, or torso with a splint, cast, or brace. Immobilization can help reduce pain, swelling, and muscle spasms. The illustrations above feature several types of immobilization techniques. (Illustration by Electronic Illustrators Group.)

end of the finger is separated from the bone as a result of trauma. Splinting also is used to immobilize an injured arm or leg immediately after an injury. Before moving a person who has injured an arm or leg some type of temporary splint should be applied to prevent further injury to the area. Splints may be made of acrylic, polyethylene foam, plaster of paris, or aluminum. In an emergency, a splint can be made from a piece of wood or rolled magazine.

Slings

Slings are often used to support the arm after a fracture or other injury. They are generally used along with a cast or splint, but sometimes are used alone as a means of immobilization. They can be used in an emergency to immobilize the arm until the person can be seen by a doctor. A triangular bandage is placed under the injured arm and then tied around the neck.

Braces

Braces are used to support, align, or hold a body part in the correct position. Braces are sometimes used after a surgical procedure is performed on an arm or leg. They can also be used for an injury. Since some braces can be easily taken off and put back on, they are often used when the person must have physical therapy or **exercise** the limb during the healing process. Many braces can also be adjusted to allow for a certain amount of movement.

Braces can be custom-made, or a ready-made brace can be used. The off-the-shelf braces are made in a variety of shapes and sizes. They generally have

Velcro straps that make the brace easy to adjust, and to put on and take off. Both braces and splints offer less support and protection than a cast and may not be a treatment option in all circumstances.

Collars

A collar is generally used for neck injuries. A soft collar can relieve pain by restricting movement of the head and neck. They also transfer some of the weight of the head from the neck to the chest. Stiff collars are generally used to support the neck when there has been a fracture in one of the bones of the neck. Cervical collars are widely used by emergency personnel at the scene of injuries when there is a potential neck or **head injury**.

Traction

Immobilization may also be secured by **traction**. Traction involves using a method for applying tension to correct the alignment of two structures (such as two bones) and hold them in the correct position. For example, if the bone in the thigh breaks, the broken ends may have a tendency to overlap. Use of traction will hold them in the correct position for healing to occur. The strongest form of traction involves inserting a stainless steel pin through a bony prominence attached by a horseshoe-shaped bow and rope to a pulley and weights suspended over the end of the patient's bed.

Traction must be balanced by countertraction. This is obtained by tilting the bed and allowing the patient's body to act as a counterweight. Another technique involves applying weights pulling in the opposite direction.

Traction for neck injuries may be in the form of a leather or cotton cloth halter placed around the chin and lower back of the head. For very severe neck injuries that require maximum traction, tongs that resemble ice tongs are inserted into small holes drilled in the outer skull.

All traction requires careful observation and adjustment by doctors and nurses to maintain proper balance and alignment of the traction with free suspension of the weights.

Immobilization can also be secured by a form of traction called skin traction. This is a combination of a splint and traction that is applied to the arms or legs by strips of adhesive tape placed over the skin of the arm or leg. Adhesive strips, moleskin, or foam rubber traction strips are applied on the skin. This method is effective only if a moderate amount of traction is required.

Preparation

There are many reasons for immobilization using splints, casts, and braces. Each person should understand his or her diagnosis clearly.

Aftercare

After a cast or splint has been put on, the injured arm or leg should be elevated for 24 to 72 hours. It is recommended that the person lie or sit with the injured arm or leg raised above the level of the heart. Rest combined with elevation will reduce pain and speed the healing process by minimizing swelling.

Fingers or toes can be exercised as much as can be tolerated after casting. This has been found to decrease swelling and prevent stiffness. If excessive swelling is noted, the application of ice to the splint or cast may be helpful.

After the cast, splint, or brace is removed, gradual exercise is usually performed to regain muscle strength and motion. The doctor may also recommend **hydrotherapy**, **heat treatments**, and other forms of physical therapy.

Risks

For some people, such as those in traction, immobilization will require long periods of bedrest. Lying in one position in bed for an extended period of time can result in sores on the skin (decubitus ulcers) and skin infection. Long periods of bedrest can also cause a buildup of fluid in the lungs or an infection in the lungs (**pneumonia**). Urinary infection can also be a result of extended bedrest.

KEY TERMS

Decubitus ulcers — A pressure sore resulting from ulceration of the skin occurring in persons confined to bed for long periods of time

Ligament—Ligaments are structures that hold bones together and prevent excessive movement of the joint. They are tough, fibrous bands of tissue.

Pneumonia — An acute or chronic disease characterized by inflammation of the lungs and caused by viruses, bacteria, or other microorganisms.

Tendon—Tendons are structures that attach bones to muscles and muscles to other muscles.

People who have casts, splints, or braces on their arms or legs will generally spend several weeks not using the injured arm or leg. This lack of use can result in decreased muscle tone and shrinkage of the muscle (atrophy). Much of this loss can usually be regained, however, through **rehabilitation** after the injury has healed.

Immobility can also cause psychological **stress**. An individual restricted to a bed with a traction device may become frustrated and bored, and perhaps even depressed, irritable, and withdrawn.

There is the possibility of decreased circulation if the cast, splint, or brace fits too tightly. Excessive pressure over a nerve can cause irritation or possible damage if not corrected. If the cast, splint, or brace breaks or malfunctions, the healing process of the bone or soft tissue can be disrupted and lead to deformity.

Normal results

Normally, the surgical or injured area heals appropriately with the help of immobilization. The form of immobilization can be discontinued, which is followed by an appropriate rehabilitation program under the supervision of a physical therapist to regain range of motion and strength.

Resources

OTHER

“Casts & Splints.” *The Center for Orthopaedics and Sports Medicine*. <<http://www.arthroscopy.com>>.

Jeffrey P. Larson, RPT

Immune complex detection see **Immune complex test**

Immune complex test

Definition

These tests evaluate the immune system, whose function is to defend the body against such invaders as bacteria and viruses. The immune system also plays a role in the control of **cancer**, and is responsible for the phenomena of allergy, hypersensitivity, and rejection problems when organs or tissue are transplanted.

One of the ways the immune system protects the body is by producing proteins called antibodies. Antibodies are formed in response to another type of protein called an antigen (anything foreign or different from a natural body protein). Immune complex reactions occur when large numbers of antigen-antibody complexes accumulate in the body.

Purpose

The purpose of the immune complex test is to demonstrate circulating immune complexes in the blood, to estimate the severity of immune complex disease, and to monitor response to therapy.

Precautions

Because this test is requested when the physician suspects that a patient's immune system is not functioning properly, special care should be taken during and after blood is drawn. For example, the venipuncture site should be kept clean and dry to avoid any chance of infection.

Description

Immune complexes are normally not detected in the blood. However, when immune complexes are produced faster than they can be cleared by the system, immune complex disease may occur. Examples of such disorders are drug sensitivity, **rheumatoid arthritis**, and a disease called **systemic lupus erythematosus**, or SLE.

The method generally used for detecting immune complexes is examination of a tissue obtained by biopsy (removal and examination of tissue sample) and the subsequent use of different staining techniques with specific antibodies. However, since tissue biopsies do not provide information about the level of complexes still in the circulatory system, serum assays obtained from blood samples which indirectly detect circulating immune complexes are useful. However, due to the variability of these complexes, several test

KEY TERMS

Antibody—A (immunoglobulin) molecule that interacts with a specific antigen. Antibodies provide protection from microscopic invaders like bacteria.

Antigen—Any substance that is capable under certain circumstances of producing an immune response either from antibodies or T-cells; bacteria are often antigens.

Autoimmune disorder—A disorder caused by a reaction of an individual's immune system against the organs or tissues of the body. Autoimmune processes can have different results: slow destruction of a particular type of cell or tissue, stimulation of an organ into excessive growth, or interference in function.

Biopsy—The removal and examination, usually under a microscope, of tissue from the living body. Used for diagnosis.

Systemic lupus erythematosus—A chronic disease of the connective tissues in the body; characterized by involvement of the skin, joints, kidneys, and serosal membranes (membranes that form the outer covering of organs in the abdomen or chest).

methods may be used. Also, as most immune complex assays have not been standardized, more than one test may be required to achieve accurate results.

Preparation

This test requires a blood sample. It is not necessary for the patient to be in a **fasting** (nothing to eat or drink) state before the test.

Risks

Risks for this test are minimal, but may include slight bleeding from the blood-drawing site, **fainting** or feeling lightheaded after venipuncture, or hematoma (blood accumulating under the puncture site).

Normal results

Normally, immune complexes are not detected in the blood.

Abnormal results

The presence of detectable immune complexes in the blood is important in the diagnosis of autoimmune

diseases, such as SLE and rheumatoid arthritis. However, for definitive diagnosis, the results of other studies must be considered with the presence of any immune complex. For example, immune complexes are associated with high numbers of a component called antinuclear antibodies in the diagnosis of systemic lupus erythematosus. A different example are the kidneys. Because of their filtering functions, elements in the kidneys called renal glomeruli can be affected by immune complexes. In such cases, renal biopsy is used to provide conclusive evidence for immune complex.

Resources

BOOKS

Jacobs, David S., et al. *Laboratory Test Handbook*. 4th ed. New York: Lexi-Comp Inc., 1996.

Janis O. Flores

Immunodeficiency

Definition

Immunodeficiency disorders are a group of disorders in which part of the immune system is missing or defective. Therefore, the body's ability to fight infections is impaired. As a result, the person with an immunodeficiency disorder will have frequent infections that are generally more severe and last longer than usual.

Description

The immune system is the body's main method for fighting infections. Any defect in the immune system decreases a person's ability to fight infections. A person with an immunodeficiency disorder may get more frequent infections, heal more slowly, and have a higher incidence of some cancers.

The normal immune system involves a complex interaction of certain types of cells that can recognize and attack "foreign" invaders, such as bacteria, viruses, and fungi. It also plays a role in fighting **cancer**. The immune system has both innate and adaptive components. Innate immunity is made up of immune protections people are born with. Adaptive immunity develops throughout life. It adapts to fight off specific invading organisms. Adaptive immunity is

divided into two components: humoral immunity and cellular immunity.

The innate immune system is made up of the skin (which acts as a barrier to prevent organisms from entering the body), white blood cells called phagocytes, a system of proteins called the complement system, and chemicals called interferons. When phagocytes encounter an invading organism, they surround and engulf it to destroy it. The complement system also attacks bacteria. The elements in the complement system create a hole in the outer layer of the target cell, which leads to the death of the cell.

The adaptive component of the immune system is extremely complex, and is still not entirely understood. Basically, it has the ability to recognize an organism or tumor cell as not being a normal part of the body, and to develop a response to attempt to eliminate it.

The humoral response of adaptive immunity involves a type of cell called B lymphocytes. B lymphocytes manufacture proteins called antibodies (which are also called immunoglobulins). Antibodies attach themselves to the invading foreign substance. This allows the phagocytes to begin engulfing and destroying the organism. The action of antibodies also activates the complement system. The humoral response is particularly useful for attacking bacteria.

The cellular response of adaptive immunity is useful for attacking viruses, some parasites, and possibly cancer cells. The main type of cell in the cellular response is T lymphocytes. There are helper T lymphocytes and killer T lymphocytes. The helper T lymphocytes play a role in recognizing invading organisms, and they also help killer T lymphocytes to multiply. As the name suggests, killer T lymphocytes act to destroy the target organism.

Defects can occur in any component of the immune system or in more than one component (combined immunodeficiency). Different immunodeficiency diseases involve different components of the immune system. The defects can be inherited and/or present at birth (congenital) or acquired.

Congenital immunodeficiency disorders

Congenital immunodeficiency is present at the time of birth, and is the result of genetic defects. These immunodeficiency disorders are also called primary immunodeficiencies. Even though more than 70 different types of congenital immunodeficiency disorders have been identified, they rarely occur. About 50,000 new cases are diagnosed in the United States each year. Congenital immunodeficiencies may occur as a result of

defects in B lymphocytes, T lymphocytes, or both. They also can occur in the innate immune system.

HUMORAL IMMUNITY DISORDERS. Bruton's agammaglobulinemia, also known as **X-linked agammaglobulinemia**, a congenital immunodeficiency disorder. The defect results in a decrease or absence of B lymphocytes, and therefore a decreased ability to make antibodies. People with this disorder are particularly susceptible to infections of the throat, skin, middle ear, and lungs. It is seen only in males because it is caused by a genetic defect on the X chromosome. Since males have only one X chromosome, they always have the defect if the gene is present. Females can have the defective gene, but since they have two X chromosomes, there will be a normal gene on the other X chromosome to counter it. Women may pass the defective gene on to their male children.

B LYMPHOCYTE DEFICIENCIES. If there is an abnormality in either the development or function of B lymphocytes, the ability to make antibodies will be impaired. This allows the body to be susceptible to recurrent infections.

A type of B lymphocyte deficiency involves a group of disorders called selective immunoglobulin deficiency syndromes. Immunoglobulin is another name for antibody, and there are five different types of immunoglobulins (called IgA, IgG, IgM, IgD, and IgE). The most common type of immunoglobulin deficiency is selective IgA deficiency, occurring in about one in every 500 white persons. The amounts of the other antibody types are normal. Some patients with selective IgA deficiency experience no symptoms, while others have occasional lung infections and **diarrhea**. In another immunoglobulin disorder, IgG and IgA antibodies are deficient and there is increased IgM. People with this disorder tend to get severe bacterial infections.

Common variable immunodeficiency is another type of B lymphocyte deficiency. In this disorder, the production of one or more of the immunoglobulin types is decreased and the antibody response to infections is impaired. It generally develops around the age of 10-20. The symptoms vary among affected people. Most people with this disorder have frequent infections, and some also will experience anemia and **rheumatoid arthritis**. Many people with common variable immunodeficiency develop cancer.

T LYMPHOCYTE DEFICIENCIES. Severe defects in the ability of T lymphocytes to mature results in impaired immune responses to infections with viruses, fungi, and certain types of bacteria. These infections are usually severe and can be fatal.

DiGeorge syndrome is a T lymphocyte deficiency that starts during fetal development and is the result of a deletion in a particular chromosome. Children with DiGeorge syndrome either do not have a thymus or have an underdeveloped thymus. Since the thymus is a major organ that directs the production of T-lymphocytes, these patients have very low numbers of T-lymphocytes. They are susceptible to recurrent infections, and usually have physical abnormalities as well. For example, they may have low-set ears, a small receding jawbone, and wide-spaced eyes. People with DiGeorge syndrome are particularly susceptible to viral and fungal infections.

In some cases, no treatment is required for DiGeorge syndrome because T lymphocyte production improves. Either an underdeveloped thymus begins to produce more T lymphocytes or organ sites other than the thymus compensate by producing more T lymphocytes.

COMBINED IMMUNODEFICIENCIES. Some types of immunodeficiency disorders affect both B lymphocytes and T lymphocytes. For example, **severe combined immunodeficiency** disease (SCID) is caused by the defective development or function of these two types of lymphocytes. It results in impaired humoral and cellular immune responses. SCID usually is recognized during the first year of life. It tends to cause a fungal infection of the mouth (thrush), diarrhea, **failure to thrive**, and serious infections. If not treated with a bone marrow transplant, a person with SCID will generally die from infections before age two. In 2003, a report showed a new form of severe SCID with severe mutation of T receptor cells.

DISORDERS OF INNATE IMMUNITY. Disorders of innate immunity affect phagocytes or the complement system. These disorders also result in recurrent infections.

Acquired immunodeficiency disorders

Acquired immunodeficiency is more common than congenital immunodeficiency. It is the result of an infectious process or other disease. For example, the human immunodeficiency virus (HIV) is the virus that causes acquired immunodeficiency syndrome (AIDS). However, this is not the most common cause of acquired immunodeficiency.

Acquired immunodeficiency often occurs as a complication of other conditions and diseases. For example, the most common causes of acquired immunodeficiency are **malnutrition**, some types of cancer, and infections. People who weigh less than 70% of the average weight of persons of the same age and gender

are considered to be malnourished. Examples of types of infections that can lead to immunodeficiency are **chickenpox**, cytomegalovirus, German **measles**, measles, **tuberculosis**, **infectious mononucleosis** (Epstein-Barr virus), chronic hepatitis, lupus, and bacterial and fungal infections.

In 2003, a new infection emerged that produces immunodeficiency. **Severe acute respiratory syndrome (SARS)** mysteriously appeared in a hospital in China. It eventually affected 8,000 people in Asia and Canada, killing 800 altogether. The virus is characterized by **fever**, lower respiratory tract symptoms, and abnormal chest x rays. However, it also produces immunodeficiency. No cases of the disease were reported from July 2003 through December 2003, but scientists feared it would reappear.

Sometimes, acquired immunodeficiency is brought on by drugs used to treat another condition. For example, patients who have an organ transplant are given drugs to suppress the immune system so the body will not reject the organ. Also, some **chemotherapy** drugs, which are given to treat cancer, have the side effect of killing cells of the immune system. During the period of time that these drugs are being taken, the risk of infection increases. It usually returns to normal after the person stops taking the drugs.

Causes and symptoms

Congenital immunodeficiency is caused by genetic defects, which generally occur while the fetus is developing in the womb. These defects affect the development and/or function of one or more of the components of the immune system. Acquired immunodeficiency is the result of a disease process, and it occurs later in life. The causes, as described above, can be diseases, infections, or the side effects of drugs given to treat other conditions.

People with an immunodeficiency disorder tend to become infected by organisms that do not usually cause disease in healthy persons. The major symptoms of most immunodeficiency disorders are repeated infections that heal slowly. These chronic infections cause symptoms that persist for long periods of time. People with chronic infection tend to be pale and thin. They may have skin **rashes**. Their lymph nodes tend to be larger than normal and their liver and spleen also may be enlarged. The lymph nodes are small organs that house antibodies and lymphocytes. Broken blood vessels, especially near the surface of the skin, may be seen. This can result in black-and-blue marks in the skin. The person may lose hair from their head.

Sometimes, a red inflammation of the lining of the eye (**conjunctivitis**) is present. They may have a crusty appearance in and on the nose from chronic nasal dripping.

Diagnosis

Usually, the first sign that a person might have an immunodeficiency disorder is that they do not improve rapidly when given **antibiotics** to treat an infection. Strong indicators that an immunodeficiency disorder may be present is when rare diseases occur or the patient gets ill from organisms that do not normally cause diseases, especially if the patient gets repeatedly infected. If this happens in very young children it is an indication that a genetic defect may be causing an immunodeficiency disorder. When this situation occurs in older children or young adults, their medical history will be reviewed to determine if childhood diseases may have caused an immunodeficiency disorder. Other possibilities will then be considered, such as recently acquired infections—for example, HIV, hepatitis, tuberculosis, etc.

Laboratory tests are used to determine the exact nature of the immunodeficiency. Most tests are performed on blood samples. Blood contains antibodies, lymphocytes, phagocytes, and complement components—all of the major immune components that might cause immunodeficiency. A blood cell count will determine if the number of phagocytic cells or lymphocytes is below normal. Lower than normal counts of either of these two cell types correlates with immunodeficiencies. The blood cells also are checked for their appearance. Sometimes a person may have normal cell counts, but the cells are structurally defective. If the lymphocyte cell count is low, further testing is usually done to determine whether any particular type of lymphocyte is lower than normal. A lymphocyte proliferation test is done to determine if the lymphocytes can respond to stimuli. The failure to respond to stimulants correlates with immunodeficiency. Antibody levels can be measured by a process called electrophoresis. Complement levels can be determined by immunodiagnostic tests.

Treatment

There is no cure for immunodeficiency disorders. Therapy is aimed at controlling infections and, for some disorders, replacing defective or absent components.

Patients with Bruton's agammaglobulinemia must be given periodic injections of a substance called

gamma globulin throughout their lives to make up for their decreased ability to make antibodies. The gamma globulin preparation contains antibodies against common invading bacteria. If left untreated, the disease usually is fatal.

Common variable immunodeficiency also is treated with periodic injections of gamma globulin throughout life. Additionally, antibiotics are given when necessary to treat infections.

Patients with selective IgA deficiency usually do not require any treatment. Antibiotics can be given for frequent infections.

In some cases, no treatment is required for DiGeorge syndrome because T lymphocyte production improves on its own. Either an underdeveloped thymus begins to produce more T lymphocytes or organ sites other than the thymus compensate by producing more T lymphocytes. In some severe cases, a bone marrow transplant or thymus transplant can be done to correct the problem.

For patients with SCID, **bone marrow transplantation** is necessary. In this procedure, healthy bone marrow from a donor who has a similar type of tissue (usually a relative, like a brother or sister) is removed. The bone marrow is a substance that resides in the cavity of bones. It is the factory that produces blood, including some of the white blood cells that make up the immune system. The bone marrow of the person receiving the transplant is destroyed, and is then replaced with marrow from the donor.

Treatment of the HIV infection that causes AIDS consists of drugs called antiretrovirals. These drugs attempt to inhibit the process that the virus goes through to kill T lymphocytes. Several of these drugs used in various combinations with one another can prolong the period of time before the disease becomes apparent. However, this is not a cure. Other treatments for people with AIDS are aimed at the particular infections and conditions that arise as a result of the impaired immune system. SARS is a relatively new acquired disease. Treatment to date involves combination therapy with steroids and interferon and supplemental oxygen for breathing difficulties. In 2004, reports in the United States said that a drug called octagam 5%, an intravenous immunoglobulin, was used to treat primary immunodeficiency diseases. The drug has been used in Europe for the same purpose.

In most cases, immunodeficiency caused by malnutrition is reversible. The health of the immune system is directly linked to the nutritional status of the

patient. Among the essential nutrients required by the immune system are proteins, **vitamins**, iron, and zinc.

For people being treated for cancer, periodic relief from chemotherapy drugs can restore the function of the immune system.

In general, people with immunodeficiency disorders should maintain a healthy diet. This is because malnutrition can aggravate immunodeficiencies. They also should avoid being near people who have colds or are sick because they can easily acquire new infections. For the same reason, they should practice good personal hygiene, especially dental care. People with immunodeficiency disorders also should avoid eating undercooked food because it might contain bacteria that could cause infection. This food would not cause infection in normal persons, but in someone with an immunodeficiency, food is a potential source of infectious organisms. People with immunodeficiency should be given antibiotics at the first indication of an infection.

Prognosis

The prognosis depends on the type of immunodeficiency disorder. People with Bruton's agammaglobulinemia who are given injections of gamma globulin generally live into their 30s or 40s. They often die from chronic infections, usually of the lung. People with selective IgA deficiency generally live normal lives. They may experience problems if given a blood **transfusion**, and therefore they should wear a Medic Alert bracelet or have some other way of alerting any physician who treats them that they have this disorder.

SCID is the most serious of the immunodeficiency disorders. If a bone marrow transplant is not successfully performed, the child usually will not live beyond two years old.

People with HIV/AIDS are living longer than in the past because of the **antiretroviral drugs** that became available in the mid 1990s. However, AIDS still is a fatal disease. People with AIDS usually die of opportunistic infections, which are infections that occur because the impaired immune system is unable to fight them.

Prevention

There is no way to prevent a congenital immunodeficiency disorder. However, someone with a congenital immunodeficiency disorder might want to consider getting **genetic counseling** before having children to find out if there is a chance they will pass the defect on to their children.

KEY TERMS

Agammaglobulinemia—The lack of gamma globulins in the blood. Antibodies are the main gamma globulins of interest, so this term means a lack of antibodies.

Some of the infections associated with acquired immunodeficiency can be prevented or treated before they cause problems. For example, there are effective treatments for tuberculosis and most bacterial and fungal infections. HIV infection can be prevented by practicing “safe sex” and not using illegal intravenous drugs. These are the primary routes of transmitting the virus. For people who do not know the HIV status of the person with whom they are having sex, safe sex involves using a **condom**.

Malnutrition can be prevented by getting adequate **nutrition**. Malnutrition tends to be more of a problem in developing countries.

Resources

PERIODICALS

“2003 Begins With SARS, Ends With Flu.” *Medical Letter on the CDC & FDA* January 11, 2004: 24.

Cooper, Megan A., Thomas L. Pommering, and Katalin Koranyi. “Primary Immunodeficiencies.” *American Family Physician* November 15, 2003: 2001.

Fischer, Alain. “Have We Seen the Last Variant of Severe Combined Immunodeficiency?” *The New England Journal of Medicine* November 6, 2003: 1789.

Low, Donald E., and Allison McGreer. “SARS—One Year Later.” *The New England Journal of Medicine* December 18, 2003: 2381.

“Octagam is Efficacious for Treating Primary Immunodeficiency Diseases.” *Medical Letter on the CDC & FDA* July 11, 2004: 52.

“Preliminary Report Suggests Combination Therapy May Help Treat SARS.” *Drug Week* January 9, 2004: 557.

John T. Lohr, PhD
Teresa G. Odle

Immunoelectrophoresis

Definition

Immunoelectrophoresis, also called gamma globulin electrophoresis, or immunoglobulin electrophoresis,

is a method of determining the blood levels of three major immunoglobulins: immunoglobulin M (IgM), immunoglobulin G (IgG), and immunoglobulin A (IgA).

Purpose

Immunoelectrophoresis is a powerful analytical technique with high resolving power as it combines separation of antigens by electrophoresis with immunodiffusion against an antiserum. The increased resolution is of benefit in the immunological examination of serum proteins. Immunoelectrophoresis aids in the diagnosis and evaluation of the therapeutic response in many disease states affecting the immune system. It is usually requested when a different type of electrophoresis, called a serum **protein electrophoresis**, has indicated a rise at the immunoglobulin level. Immunoelectrophoresis is also used frequently to diagnose **multiple myeloma**, a disease affecting the bone marrow.

Precautions

Drugs that may cause increased immunoglobulin levels include therapeutic gamma globulin, hydralazine, isoniazid, phenytoin (Dilantin), procainamide, **oral contraceptives**, **methadone**, steroids, and **tetanus** toxoid and antitoxin. The laboratory should be notified if the patient has received any vaccinations or immunizations in the six months before the test. This is mainly because prior immunizations lead to the increased immunoglobulin levels resulting in false positive results.

It should be noted that, because immunoelectrophoresis is not quantitative, it is being replaced by a procedure called immunofixation, which is more sensitive and easier to interpret.

Description

Serum proteins separate in agar gels under the influence of an electric field into albumin, alpha 1, alpha 2, and beta and gamma globulins. Immunoelectrophoresis is performed by placing serum on a slide containing a gel designed specifically for the test. An electric current is then passed through the gel, and immunoglobulins, which contain an electric charge, migrate through the gel according to the difference in their individual electric charges. Antiserum is placed alongside the slide to identify the specific type of immunoglobulin present. The results are used to identify different disease entities, and to aid in monitoring the course of the disease and the therapeutic response of the patient to such conditions as immune deficiencies, autoimmune disease, chronic

infections, chronic viral infections, and intrauterine fetal infections.

There are five classes of antibodies: IgM, IgG, IgA, IgE, and IgD.

IgM is produced upon initial exposure to an antigen. For example, when a person receives the first tetanus **vaccination**, antitetanus antibodies of the IgM class are produced 10 to 14 days later. IgM is abundant in the blood but is not normally present in organs or tissues. IgM is primarily responsible for ABO blood grouping and rheumatoid factor, yet is involved in the immunologic reaction to other infections, such as hepatitis. Since IgM does not cross the placenta, an elevation of this immunoglobulin in the newborn indicates intrauterine infection such as **rubella**, cytomegalovirus (CMV) or a sexually transmitted disease (STD).

IgG is the most prevalent type of antibody, comprising approximately 75% of the serum immunoglobulins. IgG is produced upon subsequent exposure to an antigen. As an example, after receiving a second tetanus shot, or booster, a person produces IgG antibodies in five to seven days. IgG is present in both the blood and tissues, and is the only antibody to cross the placenta from the mother to the fetus. Maternal IgG protects the newborn for the first months of life, until the infant's immune system produces its own antibodies.

IgA constitutes approximately 15% of the immunoglobulins within the body. Although it is found to some degree in the blood, it is present primarily in the secretions of the respiratory and gastrointestinal tract, in saliva, colostrum (the yellowish fluid produced by the breasts during late **pregnancy** and the first few days after **childbirth**), and in tears. IgA plays an important role in defending the body against invasion of germs through the mucous membrane-lined organs.

IgE is the antibody that causes acute allergic reactions; it is measured to detect allergic conditions. IgD, which constitutes the smallest portion of the immunoglobulins, is rarely evaluated or detected, and its function is not well understood.

Preparation

This test requires a blood sample.

Aftercare

Because this test is ordered when either very low or very high levels of immunoglobulins are suspected, the patient should be alert for any signs of infection

after the test, including **fever**, chills, rash, or skin ulcers. Any bone **pain** or tenderness should also be immediately reported to the physician.

Risks

Risks for this test are minimal, but may include slight bleeding from the blood-drawing site, **fainting** or feeling lightheaded after venipuncture, or bruising.

Normal results

Reference ranges vary from laboratory to laboratory and depend upon the method used. For adults, normal values are usually found within the following ranges (1 mg = approximately 0.000035 oz. and 1 dL = approximately 0.33 oz.):

- IgM: 60–290 mg/dL
- IgG: 700–1,800 mg/dL
- IgA: 70–440 mg/dL

Abnormal results

Increased IgM levels can indicate Waldenström's macroglobulinemia, a malignancy caused by secretion of IgM at high levels by malignant lymphoplasma cells. Increased IgM levels can also indicate chronic infections, such as hepatitis or mononucleosis and autoimmune diseases, like **rheumatoid arthritis**.

Decreased IgM levels can be indicative of **AIDS**, immunosuppression caused by certain drugs like steroids or dextran, or leukemia.

Increased levels of IgG can indicate chronic **liver disease**, autoimmune diseases, hyperimmunization reactions, or certain chronic infections, such as **tuberculosis** or **sarcoidosis**.

Decreased levels of IgG can indicate **Wiskott-Aldrich syndrome**, a genetic deficiency caused by inadequate synthesis of IgG and other immunoglobulins. Decreased IgG can also be seen with **AIDS** and leukemia.

Increased levels of IgA can indicate chronic liver disease, chronic infections, or inflammatory bowel disease.

Decreased levels of IgA can be found in ataxia, a condition affecting balance and gait, limb or eye movements, speech, and telangiectasia, an increase in the size and number of the small blood vessels in an area of skin, causing redness. Decreased

KEY TERMS

Antibody—A protein manufactured by the white blood cells to neutralize an antigen in the body. In some cases, excessive formation of antibodies leads to illness, allergy, or autoimmune disorders.

Antigen—A substance that can cause an immune response, resulting in production of an antibody, as part of the body's defense against infection and disease. Many antigens are foreign proteins not found naturally in the body, and include germs, toxins, and tissues from another person used in organ transplantation.

Autoimmune disorder—A condition in which antibodies are formed against the body's own tissues; for example, in some forms of arthritis.

IgA levels are also seen in conditions of low blood protein (hypoproteinemia), and drug immunosuppression.

Resources

BOOKS

Fischbach, Frances T. *A Manual of Laboratory Diagnostic Tests*. Philadelphia: Lippincott Williams & Wilkins, 1999.

Pagana, Kathleen D., and Timothy J. Pagana. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis, MO: Mosby, Inc., 1999.

Janis O. Flores

Immunoglobulin see **Gammaglobulin**

Immunoglobulin deficiency syndromes

Definition

Immunoglobulin deficiency syndromes are a group of **immunodeficiency** disorders in which the patient has a reduced number of or lack of antibodies.

Description

Immunoglobulins (Ig) are antibodies. There are five major classes of antibodies: IgG, IgM, IgA, IgD, and IgE.

- IgG is the most abundant of the classes of immunoglobulins. It is the antibody for viruses, bacteria, and antitoxins. It is found in most tissues and plasma.
- IgM is the first antibody present in an immune response.
- IgA is an early antibody for bacteria and viruses. It is found in saliva, tears, and all other mucous secretions.
- IgD activity is not well understood.
- IgE is present in the respiratory secretions. It is an antibody for parasitic diseases, **Hodgkin's disease**, hay fever, **atopic dermatitis**, and allergic **asthma**.

All antibodies are made by B-lymphocytes (B-cells). Any disease that harms the development or function of B-cells will cause a decrease in the amount of antibodies produced. Since antibodies are essential in fighting infectious diseases, people with immunoglobulin deficiency syndromes become ill more often. However, the cellular immune system is still functional, so these patients are more prone to infection caused by organisms usually controlled by antibodies. Most of these invading germs (microbes) make capsules, a mechanism used to confuse the immune system. In a healthy body, antibodies can bind to the capsule and overcome the bacteria's defenses. The bacteria that make capsules include the streptococci, meningococci, and *Haemophilus influenzae*. These organisms cause such diseases as otitis, **sinusitis**, **pneumonia**, **meningitis**, **osteomyelitis**, septic arthritis, and **sepsis**. Patients with immunoglobulin deficiencies are also prone to some viral infections, including echovirus, enterovirus, and **hepatitis B**. They may also have a bad reaction to the attenuated version of the **polio** virus vaccine.

There are two types of immunodeficiency diseases: primary and secondary. Secondary disorders occur in normally healthy bodies that are suffering from an underlying disease. Once the disease is treated, the immunodeficiency is reversed. Immunoglobulin deficiency syndromes are primary immunodeficiency diseases, occurring because of defective B-cells or antibodies. They account for 50% of all primary immunodeficiencies, and they are, therefore, the most prevalent type of immunodeficiency disorders.

- **X-linked agammaglobulinemia** is an inherited disease. The defect is on the X chromosome and, consequently, this disease is seen more frequently in males than females. The defect results in a failure of B-cells to mature. Mature B-cells are capable of making antibodies and developing "memory," a feature in which the B-cell will rapidly recognize and

respond to an infectious agent the next time it is encountered. All classes of antibodies are decreased in agammaglobulinemia.

- Selective IgA deficiency is an inherited disease, resulting from a failure of B-cells to switch from making IgM, the early antibody, to IgA. Although the B-cell numbers are normal, and the B-cells are otherwise normal (they can still make all other classes of antibodies), the amount of IgA produced is limited. This results in more infections of mucosal surfaces, such as the nose, throat, lungs, and intestines.
- Transient hypogammaglobulinemia of infancy is a temporary disease of unknown cause. It is believed to be caused by a defect in the development of T-helper cells (cells that recognize foreign antigens and activate T- and B-cells in an immune response). As the child ages, the number and condition of T-helper cells improves and this situation corrects itself. Hypogammaglobulinemia is characterized by low levels of **gammaglobulin** (antibodies) in the blood. During the disease period, patients have decreased levels of IgG and IgA antibodies. In lab tests, the antibodies that are present do not react well with infectious bacteria.
- Common variable immunodeficiency is a defect in both B cells and T-lymphocytes. It results in a near complete lack of antibodies in the blood.
- Ig heavy chain deletions is a genetic disease in which part of the antibody molecule is not produced. It results in the loss of several antibody classes and subclasses, including most IgG antibodies and all IgA and IgE antibodies. The disease occurs because part of the gene for the heavy chain has been lost.
- Selective IgG subclass deficiencies is a group of genetic diseases in which some of the subclasses of IgG are not made. There are four subclasses in the IgG class of antibodies. As the B-cell matures, it can switch from one subclass to another. In these diseases there is a defect in the maturation of the B-cells that results in a lack of switching.
- IgG deficiency with hyper-IgM is a disease that results when the B-cell fails to switch from making IgM to IgG. This produces an increase in the amount of IgM antibodies present and a decrease in the amount of IgG antibodies. This disease is the result of a genetic mutation.

Causes and symptoms

Immunoglobulin deficiencies are the result of congenital defects affecting the development and function

KEY TERMS

Antibody—Another term for immunoglobulin. A protein molecule that specifically recognizes and attaches to infectious agents.

T-helper cell—A type of cell that recognizes foreign antigens and activates T- and B-cells in an immune response.

of B lymphocytes (B-cells). There are two main points in the development of B-cells when defects can occur. First, B-cells can fail to develop into antibody-producing cells. X-linked agammaglobulinemia is an example of this disease. Secondly, B-cells can fail to make a particular type of antibody or fail to switch classes during maturation. Initially, when B-cells start making antibodies for the first time, they make IgM. As they mature and develop memory, they switch to one of the other four classes of antibodies. Failures in switching or failure to make a subclass of antibody leads to immunoglobulin deficiency diseases. Another mechanism that results in decreased antibody production is a defect in T-helper cells. Generally, defects in T-helper cells are listed as severe combined immunodeficiencies.

Symptoms are persistent and frequent infections, **diarrhea, failure to thrive**, and malabsorption (of nutrients).

Diagnosis

An immunodeficiency disease is suspected when children become ill frequently, especially from the same organisms. The profile of organisms that cause infection in patients with immunoglobulin deficiency syndrome is unique and is preliminary evidence for this disease. Laboratory tests are performed to verify the diagnosis. Antibodies can be found in the blood. Blood is collected and analyzed for the content and types of antibodies present. Depending on the type of immunoglobulin deficiency the laboratory tests will show a decrease or absence of antibodies or specific antibody subclasses.

Treatment

Immunodeficiency diseases cannot be cured. Patients are treated with **antibiotics** and immune serum. Immune serum is a source of antibodies. Antibiotics are useful for fighting bacteria infections. There are some

drugs that are effective against fungi, but very few drugs that are effective against viral diseases.

Bone marrow transplantation can, in most cases, completely correct the immunodeficiency.

Prognosis

Patients with immunoglobulin deficiency syndromes must practice impeccable health maintenance and care, paying particular attention to optimal dental care, in order to stay in good health.

Resources

BOOKS

Berkow, Robert, editor. *Merck Manual of Medical Information*. Whitehouse Station, NJ: Merck Research Laboratories, 1997.

Jacqueline L. Longe

Immunoglobulin electrophoresis see

Immuno-electrophoresis

Immunoglobulins G, A, and M test see

Immuno-electrophoresis

Immunologic therapies

Definition

Immunologic therapy is the treatment of disease using medicines that boost the body's natural immune response.

Purpose

Immunologic therapy is used to improve the immune system's natural ability to fight diseases such as **cancer**, hepatitis and **AIDS**. These drugs may also be used to help the body recover from immunosuppression resulting from treatments such as **chemotherapy** or **radiation therapy**.

Description

Most drugs in this category are synthetic versions of substances produced naturally in the body. In their natural forms, these substances help defend the body against disease. For example, aldesleukin (Proleukin) is an artificially made form of interleukin-2, which helps white blood cells work. Aldesleukin is administered to patients with kidney cancers and skin cancers that have

spread to other parts of the body. Filgrastim (Neupogen) and sargramostim (Leukine) are versions of natural substances called colony stimulating factors, which drive the bone marrow to make new white blood cells. Another type of drug, epoetin (Epogen, Procrit), is a synthetic version of human erythropoietin that stimulates the bone marrow to make new red blood cells. Thrombopoietin stimulates the production of platelets, disk-shaped bodies in the blood that are important in clotting. Interferons are substances the body produces naturally using immune cells to fight infections and tumors. The synthetic interferons carry brand names such as Alferon, Roferon or Intron A. Some of the interferons that are currently in use as drugs are Recombinant Interferon Alfa-2a, Recombinant Interferon Alfa-2b, interferon alfa-n1 and Interferon Alfa-n3. Alfa interferons are used to treat **hairy cell leukemia**, **malignant melanoma** and **AIDs-related Kaposi's sarcoma**. In addition interferons are also used for other conditions such as laryngeal papillomatosis, **genital warts** and certain types of hepatitis.

Recommended dosage

The recommended dosage depends on the type of immunologic therapy. For some medicines, the physician will decide the dosage for each patient, taking into account a patient's weight and whether he/she is taking other medicines. Some drugs used in immunologic therapy are given only in a hospital, under a physician's supervision. For those that patients may give themselves, check with the physician who prescribed the medicine or the pharmacist who filled the prescription for the correct dosage.

Most of these drugs come in injectable form. These drugs are generally administered by the cancer care provider.

Precautions

Aldesleukin

This medicine may temporarily increase the chance of getting infections. It may also lower the number of platelets in the blood, and thus possibly interfering with the blood's ability to clot. Taking these precautions may reduce the chance of such problems:

- Avoid people with infections, if possible.
- Be alert to signs of infection, such as **fever**, chills, **sore throat**, **pain** in the lower back or side, **cough**, hoarseness, or painful or difficulty with urination. If any of these symptoms occur, get in touch with a physician immediately.

- Be alert to signs of bleeding problems, such as black, tarry stools, tiny red spots on the skin, blood in the urine or stools, or any other unusual bleeding or bruising.
- Take care to avoid cuts or other injuries. Be especially careful when using knives, razors, nail clippers and other sharp objects. Check with a dentist for the best ways to clean the teeth and mouth without injuring the gums. Do not have dental work done without checking with a physician.
- Wash hands frequently, and avoid touching the eyes or inside of the nose unless the hands have just been washed.

Aldesleukin may make some medical conditions worse, such as **chickenpox**, **shingles** (herpes zoster), **liver disease**, lung disease, heart disease, underactive thyroid, **psoriasis**, immune system problems and mental problems. The medicine may increase the chance of seizures (convulsions) in people who are prone to having them. Also, the drug's effects may be greater in people with **kidney disease**, because their kidneys are slow to clear the medicine from their bodies.

Colony stimulating factors

Certain drugs used in treating cancer reduce the body's ability to fight infections. Although colony stimulating factors help restore the body's natural defenses, the process takes time. Getting prompt treatment for infections is important, even while taking this medicine. Call the physician at the first sign of illness or infection, such as a sore throat, fever or chills.

People with certain medical conditions could have problems if they take colony stimulating factors. People who have kidney disease, liver disease or conditions caused by inflammation or immune system problems can worsen these problems with colony stimulating factors. Those who have heart disease may be more likely to experience side effects such as water retention and heart rhythm problems while taking these drugs. Finally, patients who have lung disease might increase their chances of suffering from **shortness of breath**. Those who have any of these medical conditions should check with their personal physicians before using colony stimulating factors.

Epoetin

Epoetin is a medicine that may cause seizures (convulsions), especially in people who are prone to having them. No one who takes these drugs should drive, use machines or do anything considered dangerous in case of a seizure.

Epoetin helps the body make new red blood cells, but it is not effective unless there is adequate iron in the body. The physician may recommend taking iron supplements or certain **vitamins** that help supply the body with iron. It is necessary to follow the physician's advice in this instance—recommendations for iron in this case, as with any supplements should only come from a physician.

In studies of laboratory animals, epoetin taken during **pregnancy** caused **birth defects**, including damage to the bones and spine. However, the drug has not been reported to cause problems in human babies whose mothers take it. Women who are pregnant or who may become pregnant should check with their physicians for the most up-to-date information on the safety of taking this medicine during pregnancy.

People with certain medical conditions may have problems if they take this medicine. For example, the chance of side effects may be greater in people with high blood pressure, heart or blood vessel disease or a history of **blood clots**. Epoetin may not work properly in people who have bone problems or sickle cell anemia.

Interferons

Interferons can add to the effects of alcohol and other drugs that slow down the central nervous system, such as **antihistamines**, cold medicine, allergy medicine, sleep aids, medicine for seizures, tranquilizers, some pain relievers, and **muscle relaxants**. They may also add to the effects of anesthetics, including those used for dental procedures. Those taking interferons should check with their physicians before taking any of the above.

Some people experience **dizziness**, unusual **fatigue**, or become less alert than usual while being treated with these drugs. Because of these possible problems, anyone who takes these drugs should not drive, use machines or do anything else considered dangerous until they have determined how the drugs affect them.

Interferons often cause flu-like symptoms, including fever and chills. The physician who prescribes this medicine may recommend taking **acetaminophen** (Tylenol) before—and sometimes after—each dose to keep the fever from getting too high. If the physician recommends this, follow instructions carefully.

Like aldesleukin, interferons may temporarily increase the chance of getting infections and lower the number of platelets in the blood, leading to

clotting problems. To help prevent these problems, follow the precautions for reducing the risk of infection and bleeding listed for aldesleukin.

People who have certain medical conditions may have problems if they take interferons. For example, the drugs may worsen some medical conditions, including heart disease, kidney disease, liver disease, lung disease, diabetes, bleeding problems and mental problems. In people who have overactive immune systems, these drugs can even increase the activity of the immune system. People who have shingles or chickenpox, or who have recently been exposed to chickenpox may increase their risk of developing severe problems in other parts of the body if they take interferons. People with a history of seizures or mental problems could at risk if taking interferon.

In teenage women, interferons may cause changes in the menstrual cycle. Young women should discuss this possibility with their physicians. Older people may be more sensitive to the effects of interferons. This may increase the chance of side effects.

These drugs are not known to cause fetal **death**, birth defects or other problems in humans when taken during pregnancy. Women who are pregnant or who may become pregnant should ask their physicians for the latest information on the safety of taking these drugs during pregnancy.

Women who are breastfeeding their babies may need to stop while taking this medicine. Whether interferons pass into breast milk is not known. Because of the chance of serious side effects to the baby, breastfeeding while taking interferon is discouraged. Check with a physician for advice.

General precautions for all types of immunologic therapy

Regular physician visits are necessary during immunologic therapy treatment. This gives the physician a chance to make sure the medicine is working and to check for unwanted side effects.

Anyone who has had unusual reactions to drugs used in immunologic therapy should let the physician know before resuming the drugs. Any **allergies** to foods, dyes, preservatives, or other substances should also be reported.

Side effects

Aldesleukin

In addition to its helpful effects, this medicine may cause serious side effects. Generally, it is given only in

a hospital, where medical professionals can watch for early signs of problems. Medical tests might be performed to check for unwanted effects.

Anyone who has breathing problems, fever or chills while being given aldesleukin should check with a physician immediately.

Other side effects should be brought to a physician's attention as soon as possible:

- dizziness
- drowsiness
- confusion
- agitation
- depression
- **nausea and vomiting**
- **diarrhea**
- sores in the mouth and on the lips
- **tingling** of hands or feet
- decrease in urination
- unexplained weight gain of five or more pounds

Some side effects are usually temporary and do not need medical attention unless they are bothersome. These include dry skin; itchy or burning skin rash or redness followed by peeling; loss of appetite; and a general feeling of illness or discomfort.

Colony stimulating factors

As this medicine starts to work, the patient might experience mild pain in the lower back or hips. This is nothing to cause undue concern, and will usually go away within a few days. If the pain is intense or causes discomfort, the physician may prescribe a painkiller.

Other possible side effects include **headache**, joint or muscle pain and skin rash or **itching**. These side effects tend to disappear as the body adjusts to the medicine, and do not need medical treatment. If they continue, or they interfere with normal activities, check with a physician.

Epoetin

This medicine may cause flu-like symptoms, such as muscle aches, bone pain, fever, chills, shivering, and sweating, within a few hours after it is taken. These symptoms usually go away within 12 hours. If they do not, or if they are troubling, check with a physician. Other possible side effects that do not need medical attention are diarrhea, **nausea** or **vomiting** and fatigue or weakness.

Certain side effects should be brought to a physician's attention as soon as possible. These include headache, vision problems, increased blood pressure, fast heartbeat, weight gain and swelling of the face, fingers, lower legs, ankles or feet.

Anyone who has chest pain or seizures after taking epoetin should seek professional emergency medical attention immediately.

Interferons

This medicine may cause temporary hair loss (**alopecia**). While upsetting, it is not a sign that something is seriously wrong. The hair should grow back normally after treatment ends.

As the body adjusts to the medicine many other side effects usually go away during treatment. These include flu-like symptoms, taste alteration, loss of appetite (anorexia), nausea and vomiting, skin rash, and unusual fatigue. If these problems persist, or if they interfere with normal life, check with a physician.

A few more serious side effects should be brought to a physician's attention as soon as possible:

- confusion
- difficulty thinking or concentrating
- nervousness
- depression
- sleep problems
- numbness or tingling in the fingers, toes and face

General caution regarding side effects for all types of immunologic therapy

Other side effects are possible with any type of immunologic therapy. Anyone who has unusual symptoms during or after treatment with these drugs should contact the physician immediately.

Interactions

Anyone who has immunologic therapy should let the physician know all other medicines being taken. Some combinations of drugs may interact, that can increase or decrease the effects of one or both drugs or can increase the likelihood of side effects. Consultation with a physician is highly recommended to get the insight on whether the possible interactions can interfere with drug therapy or cause harmful effects.

Immunoprevention

Considering that most of the biological modifiers such as cytokines elicit immune response that inhibit incipient tumors before they are clinically evident, immunoprevention has been proposed as a recent strategy for combating cancer. Treatment involving immune molecules (such as cytokines) prepared synthetically or that are not produced by the patients themselves is called as passive immunotherapy. Conversely, a vaccine is a form of active immune therapy because it elicits an immune response in patients. A cancer vaccine may be made of whole tumor cell or of substances or fragments contained in the tumor called antigens.

Newer types of immunologic therapy that are still considered investigational as of 2003 include cell-based therapies. Instead of using synthetic chemicals that resemble substances produced by the body, cell-based therapies use modified stem cells or dendritic cells as vaccines against cancer. Stem cells are undifferentiated cells whose daughter cells can develop into various types of specialized cells, while dendritic cells are cells that are able to initiate and modify the immune system's responses to cancer by activating B cells and T cells. Dendritic cells appear to offer a promising new form of immunotherapy for cancer.

Another investigational form of treatment is the development of cell-free tumor-specific peptide vaccines. Peptides are subunits of protein molecules that contain two or more amino acids. Peptide vaccines are intended to induce responses in the patient's T cells that inhibit tumor growth. As of late 2003, however, peptide-based tumor vaccines have been shown to shrink cancerous tumors only in patients with limited disease.

Adoptive immunotherapy

Adoptive immunotherapy involves stimulating T lymphocytes by exposing them to tumor antigens. These modified cells are grown in the laboratory and then injected into patients. Since the cells taken from a different individual for this purpose often results in rejection, patients serve both as donor and recipient of their own T cells. Adoptive immunotherapy is particularly effective in patients who have received massive doses of radiation and chemotherapy. In such patients, therapy results in immunosuppression (weakened immune systems), making them vulnerable to viral infections. For example, CMV-specific T cells can reduce the risk of cytomegalovirus (CMV) infection in transplant patients.

KEY TERMS

AIDS—Acquired immune deficiency syndrome. A disease caused by infection with the human immunodeficiency virus (HIV). In people with this disease, the immune system breaks down, increasing vulnerability to other infections and some types of cancer.

Bone marrow—Soft tissue that fills the hollow centers of bones. Blood cells and platelets (disk-shaped bodies in the blood that are important in clotting) are produced in the bone marrow.

Chemotherapy—Treatment of an illness with chemical agents. The term usually is used to describe the treatment of cancer with drugs.

Clot—A hard mass that forms when blood coagulates.

Fetus—A developing baby inside the womb.

Hepatitis—Inflammation of the liver caused by a virus, chemical, or drug.

Immune response—The body's natural protective reaction to disease and infection.

Immune system—The system that protects the body against disease and infection through immune responses.

Inflammation—Pain, redness, swelling, and heat that usually develop in response to injury or illness.

Psoriasis—A skin disease that manifests itself with itchy, scaly, red patches on the skin.

Seizure—A sudden attack, spasm, or convulsion.

Shingles—A disease caused by an infection with the Herpes zoster virus—the same virus that causes chickenpox. Symptoms of shingles include pain and blisters along one nerve, usually on the face, chest, stomach, or back.

Sickle cell anemia—An inherited disorder in which red blood cells contain an abnormal form of hemoglobin, a protein that carries oxygen. The abnormal form of hemoglobin causes the red cells to become sickle-shaped. The misshapen cells may clog blood vessels, preventing oxygen from reaching tissues and leading to pain, blood clots and other problems. Sickle cell anemia is most common in people of African descent and in people from Italy, Greece, India, and the Middle East.

Resources

PERIODICALS

Fishman, M. N., and S. J. Antonia. "Cell-Based Immune Therapy for Metastatic Renal Cancer." *Expert Review of Anticancer Therapy* 3 (December 2003): 837–849.

"Immunoprevention of Cancer: Is the time Ripe?" *Cancer Research* 60 (May 15, 2000): 2571–2575.

National Cancer Institute. *Treating Cancer with Vaccine Therapy*. June 29, 2001. <<http://cancertrials.nci.nih.gov/news/features/vaccine/html/page05.htm> 2000 >.

Nieda, M., M. Tomiyama, and K. Egawa. "Ex vivo Enhancement of Antigen-Presenting Function of Dendritic Cells and Its Application for DC-Based Immunotherapy." *Human Cell* 16 (December 2003): 199–204.

Paczesny, S., H. Ueno, J. Fay, et al. "Dendritic Cells as Vectors for Immunotherapy of Cancer." *Seminars in Cancer Biology* 13 (December 2003): 439–447.

Rosenberg, S. A. "Progress in Human Tumor Immunology and Immunotherapy." *Nature* 411, no. 6835 (2001): 380–385.

Scheibenbogen, C., A. Letsch, A. Schmitt, et al. "Rational Peptide-Based Tumour Vaccine Development and T Cell Monitoring." *Seminars in Cancer Biology* 13 (December 2003): 423–429.

Nancy Ross-Flanigan
Kausalya Santhanam, PhD
Teresa G. Odle
Rebecca J. Frey, PhD

Immunosuppressant drugs

Definition

Immunosuppressant drugs, also called **anti-rejection drugs**, are used to prevent the body from rejecting a transplanted organ.

Purpose

When an organ, such as a liver, a heart or a kidney, is transplanted from one person (the donor) into another (the recipient), the immune system of the recipient triggers the same response against the new organ it would have to any foreign material, setting off a chain of events that can damage the transplanted organ. This process is called rejection and it can occur rapidly (acute rejection), or over a long period of time (chronic rejection). Rejection can occur despite close matching of the donated organ and the transplant patient. Immunosuppressant drugs greatly decrease

the risks of rejection, protecting the new organ and preserving its function. These drugs act by blocking the immune system so that it is less likely to react against the transplanted organ. A wide variety of drugs are available to achieve this aim but work in different ways to reduce the risk of rejection.

In addition to being used to prevent organ rejection, immunosuppressant drugs are also used to treat such severe skin disorders as **psoriasis** and such other diseases as **rheumatoid arthritis**, **Crohn's disease** (chronic inflammation of the digestive tract) and patchy hair loss (**alopecia areata**). Some of these conditions are termed "autoimmune" diseases, indicating that the immune system is acting against the body itself.

Description

Immunosuppressant drugs can be classified according to their specific molecular mode of action. The three main immunosuppressant drugs currently used in organ transplantations are the following:

- Cyclosporins (Neoral, Sandimmune, SangCya). These drugs act by inhibiting T-cell activation, thus preventing T-cells from attacking the transplanted organ.
- Azathioprine (Imuran). These drugs disrupt the synthesis of DNA and RNA and cell division.
- **Corticosteroids** such as prednisolone (Deltasone, Orasone). These drugs suppress the inflammation associated with transplant rejection.

Most patients are prescribed a combination of drugs after their transplant, one from each of the above main groups; for example cyclosporin, azathioprine and prednisolone. Over a period of time, the doses of each drug and the number of drugs taken may be reduced as the risks of rejection decrease. However, most patients need to take at least one immunosuppressive for the rest of their lives.

Immunosuppressants can also be classified depending on the specific transplant:

- basiliximab (Simulect) is also used in combination with such other drugs as cyclosporin and corticosteroids, in kidney transplants
- daclizumab (Zenapax) is also used in combination with such other drugs as cyclosporin and corticosteroids, in kidney transplants
- muromonab CD3 (Orthoclone OKT3) is used, along with cyclosporin, in kidney, liver and heart transplants

- tacrolimus (Prograf) is used in liver transplants and is under study for kidney, bone marrow, heart, pancreas, pancreatic island cell, and small bowel transplantation

Some immunosuppressants are also used to treat a variety of autoimmune diseases:

- Azathioprine (Imuran) is used not only to prevent organ rejection in kidney transplants, but also in treatment of rheumatoid arthritis. It has been used to treat chronic **ulcerative colitis**, but it has been of limited value for this use.
- Cyclosporin (Sandimmune, Neoral) is used in heart, liver, kidney, pancreas, bone marrow and heart/lung transplantation. The Neoral form has been used to treat psoriasis and rheumatoid arthritis. The drug has also been used for many other conditions including **multiple sclerosis**, diabetes and myasthenia gravis.
- Glatiramer acetate (Copaxone) is used in treatment of relapsing-remitting multiple sclerosis. In one study, glatiramer reduced the frequency of multiple sclerosis attacks by 75% over a two-year period.
- Mycophenolate (CellCept) is used along with cyclosporin in kidney, liver and heart transplants. It has also been used to prevent the kidney problems associated with lupus erythematosus.
- Sirolimus (Rapamune) is used in combination with other drugs including cyclosporin and corticosteroids, in kidney transplants. The drug is also used for the treatment of psoriasis.

Recommended dosage

Immunosuppressant drugs are available only with a physician's prescription. They come in tablet, capsule, liquid and injectable forms.

The recommended dosage depends on the type and form of immunosuppressant drug and the purpose for which it is being used. Doses may be different for different patients. The prescribing physician or the pharmacist who filled the prescription will advise on correct dosage.

Taking immunosuppressant drugs exactly as directed is very important. Smaller, larger or more frequent doses should never be taken, and the drugs should never be taken for longer than directed. The physician will decide exactly how much of the medicine each patient needs. Blood tests often are necessary to monitor the action of the drug.

The prescribing physician should be consulted before stopping an immunosuppressant drug.

Precautions

Seeing a physician regularly while taking immunosuppressant drugs is important. These regular check-ups will allow the physician to make sure the drug is working as it should and to watch for unwanted side effects. These drugs are very powerful and can cause serious side effects, such as high blood pressure, kidney problems and liver problems. Some side effects may not show up until years after the medicine is used. Anyone who has been advised to take immunosuppressant drugs should thoroughly discuss the risks and benefits with the prescribing physician.

Immunosuppressant drugs lower a person's resistance to infection and can make infections harder to treat. The drugs can also increase the chance of uncontrolled bleeding. Anyone who has a serious infection or injury while taking immunosuppressant drugs should get prompt medical attention and should make sure that the treating physician knows about the immunosuppressant prescription. The prescribing physician should be immediately informed if signs of infection, such as **fever** or chills, **cough** or hoarseness, **pain** in the lower back or side, or painful or difficult urination, bruising or bleeding, blood in the urine, bloody or black, tarry stools occur. Other ways of preventing infection and injury include washing the hands frequently, avoiding sports in which injuries may occur, and being careful when using knives, razors, fingernail clippers or other sharp objects. Avoiding contact with people who have infections is also important. In addition, people who are taking or have been taking immunosuppressant drugs should not have immunizations, such as **smallpox** vaccinations, without checking with their physicians. Because of their low resistance to infection, people taking these drugs might get the disease that the vaccine is designed to prevent. People taking immunosuppressant drugs also should avoid contact with anyone who has taken the oral **polio** vaccine, as there is a chance the virus could be passed on to them. Other people living in their home should not take the oral polio vaccine.

Immunosuppressant drugs may cause the gums to become tender and swollen or to bleed. If this happens, a physician or dentist should be notified. Regular brushing, flossing, cleaning and gum massage may help prevent this problem. A dentist can provide advice on how to clean the teeth and mouth without causing injury.

Special conditions

People who have certain medical conditions or who are taking certain other medicines may have problems if they take immunosuppressant drugs. Before

taking these drugs, the prescribing physician should be informed about any of these conditions:

ALLERGIES. Anyone who has had unusual reactions to immunosuppressant drugs in the past should let his or her physician know before taking the drugs again. The physician should also be told about any **allergies** to foods, dyes, preservatives, or other substances.

PREGNANCY. Azathioprine may cause **birth defects** if used during **pregnancy**, or if either the male or female is using it at time of conception. Anyone taking this medicine should use a barrier method of birth control, such as a **diaphragm** or condoms. Birth control pills should not be used without a physician's approval. Women who become pregnant while taking this medicine should check with their physicians immediately.

The medicine's effects have not been studied in humans during pregnancy. Women who are pregnant or who may become pregnant and who need to take this medicine should check with their physicians.

BREASTFEEDING. Immunosuppressant drugs pass into breast milk and may cause problems in nursing babies whose mothers take it. Breastfeeding is not recommended for women taking this medicine.

OTHER MEDICAL CONDITIONS. People who have certain medical conditions may have problems if they take immunosuppressant drugs. For example:

- People who have **shingles** (herpes zoster) or **chickenpox**, or who have recently been exposed to chickenpox, may develop severe disease in other parts of their bodies when they take these medicines.
- The medicine's effects may be greater in people with **kidney disease** or **liver disease**, because their bodies are slow to get rid of the medicine.
- The effects of oral forms of this medicine may be weakened in people with intestinal problems, because the medicine cannot be absorbed into the body.

Before using immunosuppressant drugs, people with these or other medical problems should make sure their physicians are aware of their conditions.

USE OF CERTAIN MEDICINES. Taking immunosuppressant drugs with certain other drugs may affect the way the drugs work or may increase the chance of side effects.

Side effects

Increased risk of infection is a common side effect of all the immunosuppressant drugs. The immune system protects the body from infections and when the immune system is suppressed, infections are more likely. Taking

KEY TERMS

Antibody—Protein produced by the immune system in response to the presence in the body of an antigen.

Antigen—Any substance or organism that is foreign to the body. Examples of antigens are: bacteria, bacterial toxins, viruses, or other cells or proteins.

Autoimmune disease—A disease in which the immune system is overactive and has lost the ability to distinguish between self and non-self.

Chronic—A word used to describe a long-lasting condition. Chronic conditions often develop gradually and involve slow changes.

Corticosteroids—A class of drugs that are synthetic versions of the cortisone produced by the body. They rank among the most powerful anti-inflammatory agents.

Cortisone—Glucocorticoid produced by the adrenal cortex in response to stress. Cortisone is a steroid with anti-inflammatory and immunosuppressive properties.

Inflammation—A process occurring in body tissues, characterized by increased circulation and the accumulation of white blood cells. Inflammation also occurs in such disorders as arthritis and causes harmful effects.

Inflammatory—Pertaining to inflammation.

Immune response—Physiological response of the body controlled by the immune system that involves the production of antibodies to fight off specific foreign substances or agents (antigens).

Immune system—The network of organs, cells, and molecules that work together to defend the body

from foreign substances and organisms causing infection and disease such as: bacteria, viruses, fungi and parasites.

Immunosuppressant—Any chemical substance that suppresses the immune response.

Immunosuppressive—Any agent that suppresses the immune response of an individual.

Immunosuppressive cytotoxic drugs—A class of drugs that function by destroying cells and suppressing the immune response.

Lymphocyte—Lymphocytes are white blood cells that participate in the immune response. The two main groups are the B cells that have antibody molecules on their surface and T cells that destroy antigens.

Psoriasis—A skin disease characterized by itchy, scaly, red patches on the skin.

Rejection—Rejection occurs when the body recognizes a new transplanted organ as “foreign” and turns on the immune system of the body.

T cells—Any of several lymphocytes that have specific antigen receptors, and that are involved in cell-mediated immunity and destruction of antigen-bearing cells.

Transplantation—The removal of tissue from one part of the body for implantation to another part of the body; or the removal of tissue or an organ from one individual and its implantation in another individual by surgery.

such **antibiotics** as co-trimoxazole prevents some of these infections. Immunosuppressant drugs are also associated with a slightly increased risk of **cancer** because the immune system also plays a role in protecting the body against some forms of cancer. For example, long-term use of immunosuppressant drugs carries an increased risk of developing skin cancer as a result of the combination of the drugs and exposure to sunlight.

Other side effects of immunosuppressant drugs are minor and usually go away as the body adjusts to the medicine. These include loss of appetite, **nausea** or **vomiting**, increased hair growth, and trembling or shaking of the hands. Medical attention is not necessary unless these side effects continue or cause problems.

The treating physician should be notified immediately if any of the following side effects occur:

- unusual tiredness or weakness
- fever or chills
- frequent need to urinate

Interactions

Immunosuppressant drugs may interact with other medicines. When this happens, the effects of one or both drugs may change or the risk of side effects may be greater. Other drugs may also have an adverse effect on immunosuppressant therapy. This is particularly important for patients taking cyclosporin or tacrolimus. For example, some drugs can cause the blood levels to rise, while others can cause the blood levels to fall and it is important to avoid such contraindicated combinations. Other examples are:

- The effects of azathioprine may be greater in people who take allopurinol, a medicine used to treat gout.
- A number of drugs, including female hormones (estrogens), male hormones (androgens), the antifungal drug ketoconazole (Nizoral), the ulcer drug cimetidine (Tagamet) and the **erythromycins** (used to treat infections), may increase the effects of cyclosporine.
- When sirolimus is taken at the same time as cyclosporin, the blood levels of sirolimus may be increased to a level where there are severe side effects. Although these two drugs are usually used together, the sirolimus should be taken four hours after the dose of cyclosporin.
- Tacrolimus is eliminated through the kidneys. When the drug is used with other drugs that may harm the kidneys, such as cyclosporin, the antibiotics gentamicin and amikacin, or the antifungal drug amphotericin B, blood levels of tacrolimus may be increased. Careful kidney monitoring is essential when tacrolimus is given with any drug that might cause kidney damage.
- The risk of cancer or infection may be greater when immunosuppressant drugs are combined with certain other drugs which also lower the body's ability to fight disease and infection. These drugs include corticosteroids such as prednisone; the **anticancer drugs** chlorambucil (Leukeran), cyclophosphamide (Cytoxan) and mercaptopurine (Purinethol); and the monoclonal antibody muromonab-CD3 (Orthoclone), which also is used to prevent transplanted organ rejection.

Not every drug that may interact with immunosuppressant drugs is listed here. Anyone who takes immunosuppressant drugs should let the physician know all other medicines he or she is taking and should ask whether the possible interactions can interfere with treatment.

Resources

BOOKS

- Abbas, A. K., and A. H. Lichtman. *Basic Immunology: Functions and Disorders of the Immune System*. Philadelphia: W. B. Saunders Co., 2001.
- Sompayrac, L. M. *How the Immune System Works*. Boston: Blackwell Science, 1999.
- Travers, P. *Immunobiology: The Immune System in Health and Disease*. 5th edition. New York: Garland Publishers, 2001.

Nancy Ross-Flanigan

Immunotherapy see **Immunologic therapies**

Impacted tooth

Definition

An impacted tooth is any tooth that is prevented from reaching its normal position in the mouth by tissue, bone, or another tooth.

Description

The teeth that most commonly become impacted are the third molars, also called wisdom teeth. These large teeth are the last to develop, beginning to form when a person is about nine years old, but not breaking through the gum tissue until the late teens or early twenties. By this time, the jaws have stopped growing and may be too small to accommodate these four additional teeth. As the wisdom teeth continue to move, one or more may become impacted, either by running into the teeth next to them or becoming blocked within the jawbone or gum tissue. An impacted tooth can cause further dental problems, including infection of the gums, displacement of other teeth, or decay. At least one wisdom tooth becomes impacted in nine of every ten people.

Causes and Symptoms

The movement of an erupting wisdom tooth and any subsequent impaction may produce **pain** at the back of the jaw. Pain may also be the result of infection, either from decay in any exposed portion of the tooth or from trapped food and plaque in the surrounding gum tissue. Infection typically produces an unpleasant taste when biting down and **bad breath**. Another source of pain may be pericoronitis, a gum condition in which the crown of the incompletely erupted tooth produces inflammation, redness, and tenderness of the gums. Less common symptoms of an impacted tooth are swollen lymph nodes in the neck, difficulty opening the mouth, and prolonged **headache**.

Diagnosis

Upon visual examination, the dentist may find signs of infection or swelling in the area where the tooth is present or only partially erupted. Dental x rays are necessary to confirm tooth impaction.

Treatment

Because impacted teeth may cause dental problems with few if any symptoms to indicate damage,

KEY TERMS

Dry socket—A painful condition following tooth extraction in which a blood clot does not properly fill the empty socket, leaving the bone underneath exposed to air and food.

Eruption—The process of a tooth breaking through the gum tissue to grow into place in the mouth.

Extraction—The removal of a tooth from its socket in the bone.

Pericoronitis—A gum condition in which irritation and inflammation are produced by the crown of an incompletely erupted tooth.

Wisdom tooth—One of the four last teeth on the top and bottom rows of teeth. Also called a third molar.

dentists commonly recommend the removal of all wisdom teeth, preferably while the patient is still a young adult. A dentist may perform an extraction with forceps and local anesthetic if the tooth is exposed and appears to be easily removable in one piece. However, he or she may refer a difficult extraction to an oral surgeon, a specialist who administers either nitrous oxide-oxygen (commonly called “laughing gas”), an intravenous sedative, or a general anesthetic to alleviate any pain or discomfort during the surgical procedure. Extracting an impacted tooth typically requires cutting through gum tissue to expose the tooth, and may require removing portions of bone to free the tooth. The tooth may have to be removed in pieces to minimize destruction to the surrounding structures. The extraction site may or may not require one or more stitches to help the incision heal.

Prognosis

The prognosis is very good when impacted teeth are removed from young healthy adults without complications. Potential complications include postoperative infection, temporary **numbness** from nerve irritation, jaw fracture, and jaw joint pain. An additional condition which may develop is called dry socket: when a blood clot does not properly form in the empty tooth socket, or is disturbed by an oral vacuum (such as from drinking through a straw or **smoking**), the bone beneath the socket is painfully exposed to air and food, and the extraction site heals more slowly.

Resources

ORGANIZATIONS

American Association of Oral and Maxillofacial Surgeons.
9700 West Bryn Mawr Ave., Rosemont, IL 60018-5701.
(847) 678-6200. <<http://www.aaoms.org>>.

Bethany Thivierge

Impedance phlebography

Definition

Impedance phlebography is a noninvasive test that uses electrical monitoring to measure blood flow in veins of the leg. Information from this test helps a doctor to detect **deep vein thrombosis** (**blood clots** or **thrombophlebitis**).

Purpose

Impedance phlebography may be done in order to:

- detect blood clots lodged in the deep veins of the leg
- screen patients who are likely to have blood clots in the leg
- detect the source of blood clots in the lungs (pulmonary emboli)

Blood clots in the legs can lead to more serious problems. If a clot breaks loose from a leg vein, it may travel to the lungs and lodge in a blood vessel in the lungs. Blood clots are more likely to occur in people who have recently had leg injuries, surgery, **cancer**, or a long period of bed rest.

Precautions

Because this test is not invasive, it can be done on all patients. However, the accuracy of the results will be affected if the patient does not breathe normally or keep the leg muscles relaxed. Compression of the veins because of pelvic tumors or decreased blood flow, due to **shock** or any condition that reduces the amount of blood the heart pumps, may also change the test results.

Description

Impedance phlebography works by measuring the resistance to the transmission of electrical energy (impedance). This resistance changes depending on

the volume of blood flowing through the veins. By graphing the impedance, a doctor or technician can tell whether a clot is obstructing blood flow.

Using conductive jelly, the examiner puts electrodes on the patient's calf. These electrodes are connected to an instrument called a plethysmograph, which records the changes in electrical resistance that occur during the test.

The patient lies down and raises one leg at a 30° angle, so that the calf is above the level of the heart. The examiner wraps a pressure cuff around the patient's thigh and inflates it to a pressure of 45–60 cm of water for 45 seconds. The plethysmograph records the electrical changes that correspond to changes in the volume of blood in the vein at the time the pressure is exerted and again three seconds after the cuff is deflated. This procedure is repeated several times in both legs.

This test takes 30 to 45 minutes. Impedance phlebography is also called an impedance test of blood flow or impedance plethysmography.

Preparation

Patients undergoing this test do not need to alter their diet, change their normal activities, or stop taking any medications. They will wear a surgical gown during the test, and be asked to urinate before the test starts. If keeping the legs elevated causes discomfort, mild **pain** medication will be given.

Aftercare

The patient may resume normal or postoperative activities after the test.

Risks

Impedance phlebography is painless and safe. It presents no risk to the patient.

Normal results

Normally, inflating the pressure cuff will cause a sharp rise in the pressure in the veins of the calf because blood flow is blocked. When the cuff is released, the pressure decreases rapidly as the blood flows away.

Abnormal results

If a clot is present, the pressure in the calf veins will already be high. It does not become sharply higher

KEY TERMS

Thrombophlebitis—Inflammation of a vein, associated with the formation of a blood clot.

when the pressure cuff is tightened. When the pressure cuff is deflated, the clot blocks the flow of blood out of the calf vein. The decrease in pressure is not as rapid as when no clot is present, and the shape of the resulting graph is different.

Resources

OTHER

Griffith, H. Winter. "Complete Guide to Medical Tests." *ThriveOnline*. <<http://thriveonline.oxygen.com>>.

Tish Davidson, A.M.

Impedance plethysmography see **Impedance phlebography**

Impedance test of blood flow see **Impedance phlebography**

Impetigo

Definition

Impetigo refers to a very localized bacterial infection of the skin. There are two types, bullous and epidemic.

Description

Impetigo is a skin infection that tends primarily to afflict children. Impetigo caused by the bacterium *Staphylococcus aureus* (also known as staph) affects children of all ages. Impetigo caused by the bacteria called group A streptococci (also known as strep) are most common in children ages two to five.

The bacteria that cause impetigo are very contagious. They can be spread by a child from one part of his or her body to another by scratching, or contact with a towel, clothing, or stuffed animal. These same methods can pass the bacteria on from one person to another.

Impetigo tends to develop in areas of the skin that have already been damaged through some other

mechanism (a cut or scrape, burn, insect bite, or vesicle from **chickenpox**).

Causes and symptoms

The first sign of bullous impetigo is a large bump on the skin with a clear, fluid-filled top (called a vesicle). The bump develops a scab-like, honey-colored crust. There is usually no redness or **pain**, although the area may be quite itchy. Ultimately, the skin in this area will become dry and flake away. Bullous impetigo is usually caused by staph bacteria.

Epidemic impetigo can be caused by staph or strep bacteria, and (as the name implies) is very easily passed among children. Certain factors, such as heat and humidity, crowded conditions, and poor hygiene increase the chance that this type of impetigo will spread rapidly among large groups of children. This type of impetigo involves the formation of a small vesicle surrounded by a circle of reddened skin. The vesicles appear first on the face and legs. When a child has several of these vesicles close together, they may spread to one another. The skin surface may become eaten away (ulcerated), leaving irritated pits. When there are many of these deep, pitting ulcers, with pus in the center and brownish-black scabs, the condition is called **ecthyma**. If left untreated, the type of bacteria causing this type of impetigo has the potential to cause a serious **kidney disease** called **glomerulonephritis**. Even when impetigo is initially caused by strep bacteria, the vesicles are frequently secondarily infected with staph bacteria.

Impetigo is usually an uncomplicated skin condition. Left untreated, however, it may develop into a serious disease, including **osteomyelitis** (bone infection), septic arthritis (joint infection), or **pneumonia**. If large quantities of bacteria are present and begin circulating in the bloodstream, the child is in danger of developing an overwhelming systemic infection known as **sepsis**.

Diagnosis

Characteristic appearance of the skin is the usual method of diagnosis, although fluid from the vesicles can be cultured and then examined in an attempt to identify the causative bacteria.

Treatment

Uncomplicated impetigo is usually treated with a topical antibiotic cream called mupirocin. In more serious, widespread cases of impetigo, or when the child has a **fever** or swollen glands, **antibiotics** may be



Impetigo is a contagious bacterial skin infection that mostly affects the area around the nose and mouth. Usually caused by staphylococci, this person's impetigo was triggered by herpes simplex. (Photo Researchers, Inc. Reproduced by permission.)

KEY TERMS

Systemic—Involving the whole body; the opposite of localized.

Ulcer—An irritated pit in the surface of a tissue.

Vesicle—A bump on the skin filled with fluid.

given by mouth or even through a needle placed in a vein (intravenously).

Prognosis

Prognosis for a child with impetigo is excellent. The vast majority of children recover quickly, completely, and uneventfully.

Prevention

Prevention involves good hygiene. Handwashing; never sharing towels, clothing, or stuffed animals; and keeping fingernails well-trimmed are easy precautions to take to avoid spreading the infection from one person to another.

Resources

PERIODICALS

"Bullous Impetigo." *Archives of Pediatrics and Adolescent Medicine* 151, no. 11 (November 1997): 1168+.

Rosalyn Carson-DeWitt, MD

Implant therapy see **Radioactive implants**

Implantable cardioverter-defibrillator

Definition

The implantable cardioverter-defibrillator is an electronic device to treat life-threatening heartbeat irregularities. It is surgically implanted.

Purpose

The implantable cardioverter-defibrillator is used to detect and stop serious ventricular **arrhythmias** and restore a normal heartbeat in people who are at high risk of sudden **death**. The American Heart Association recommends that implantable cardioverter-defibrillators only be considered for patients who have a life-threatening arrhythmia. A recent study by the National Heart, Lung, and Blood Institute demonstrated that implantable cardioverter-defibrillators are the treatment of choice instead of drug therapy for patients who have had a cardiac arrest or **heart attack** and are at risk for developing **ventricular tachycardia**, which is a very rapid heartbeat, or **ventricular fibrillation**, which is an ineffective, irregular heart activity. Other studies suggest that 20% of these high risk patients would die within two years without an implantable cardioverter-defibrillator. With the device, the five-year risk of sudden death drops to five percent.

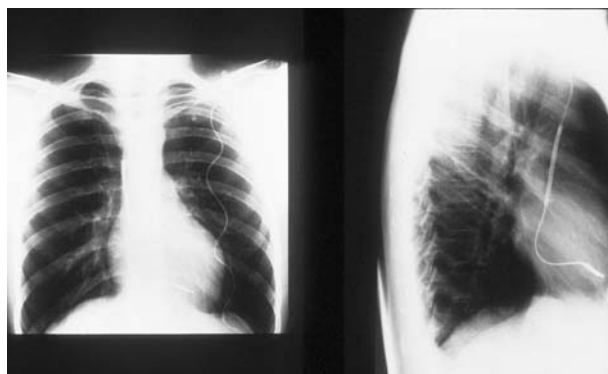
Precautions

The implantable cardioverter-defibrillator should not be used on patients who faint from causes other than a known life-threatening ventricular arrhythmia, to treat slow heart rates, or during an emergency.

Description

According to the American College of Cardiology, more than 80,000 Americans currently have an implantable cardioverter-defibrillator; 17,000 of these were implanted in 1995 alone. The battery-powered device rescues the patient from a life-threatening arrhythmia by rapid pacing and/or delivering electrical shock(s) to suspend heart activity and then allow it to initiate a normal rhythm. Before the development of the implantable cardioverter-defibrillator, most people who experienced ventricular fibrillation and were not near a hospital with a well equipped emergency team died within minutes.

The implantable cardioverter-defibrillator is like a mini computer connected to the patient's heart. Newer



X ray of implanted cardioverter-defibrillator (Custom Medical Stock Photo. Reproduced by permission.)

models weigh less than 10 ounces and can be implanted beneath the skin of the chest in the pectoral region, without major surgery. A lead from the device is then inserted into the heart through a vein. The procedure is performed in an operating room under **general anesthesia**. Earlier versions of implantable cardioverter-defibrillators were implanted in the abdomen and required open-chest surgery to connect the electrodes to the left and right ventricles.

The implantable cardioverter-defibrillator is set above the patient's **exercise** heart rate. Once the device is in place, many tests will be conducted to ensure that the device is sensing and defibrillating properly. The newer implantable cardioverter-defibrillators last seven or eight years. Technology and procedures continue to evolve.

Preparation

Before the procedure, a complete medical history and physical exam will be done. **Electrocardiography**, special electrophysiologic testing, **chest x ray**, **urinalysis**, and a blood test are usually also required.

Aftercare

The patient is monitored for arrhythmias and to ensure that the implantable cardioverter-defibrillator is working properly. The physician also watches for signs of infection. Before the patient leaves the hospital, the device is tested again. Anti-arrhythmia drug therapy is necessary in more than half of all patients with implantable cardioverter-defibrillators, but the number of drugs and the dosages are usually reduced. Any time a significant change in anti-arrhythmia medication is made, the device will be tested again.

KEY TERMS

Arrhythmia—A variation of the normal rhythm of the heartbeat.

Cardioverter—A device to apply electric shock to the chest to convert an abnormal heartbeat into a normal heartbeat.

Defibrillation—An electronic process which helps re-establish a normal heart rhythm.

Ventricles—The two large lower chambers of the heart which pump blood to the lungs and the rest of the human body.

Ventricular fibrillation—An arrhythmia in which the heart beats very fast but blood is not pumped out to the body. Ventricular fibrillation can quickly become fatal if not corrected.

Ventricular tachycardia—An arrhythmia in which the heart rate is more than 100 beats per minute.

The patient is taught how the device works, and that the shock it delivers will feel like a punch or kick in the chest. The patient is told to notify his/her physician when the implantable cardioverter-defibrillator delivers a shock, and to go to the emergency room if multiple shocks are sent within a short period of time.

Although most patients with implantable cardioverter-defibrillators are glad that they have the device and feel that it has extended their lives, they do experience fear and **anxiety**. This stems from the sensation of the shock(s), the unpredictable circumstances under which shock(s) occurs, and unknown outcomes.

Risks

There can be serious complications to the implantation of a cardioverter-defibrillator. These include inflammation of the pericardium, the sac that surrounds the heart; heart attack; congestive **heart failure**; and post-operative **stroke**. Serious infections can develop in the area around the device while the patient is initially hospitalized or up to several months later. Death due to the device's failure while being tested during surgery is an uncommon risk. The risk of death from the implantation procedure is about the same as that for a pacemaker, less than one percent. There are also potentially serious risks associated with the device's improper functioning once it is in place.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

Texas Heart Institute. Heart Information Service. PO Box 20345, Houston, TX 77225-0345. <<http://www.tmc.edu/thi>>.

Lori De Milto

Impotence

Definition

Impotence, often called **erectile dysfunction**, is the inability to achieve or maintain an erection long enough to engage in sexual intercourse.

Description

Under normal circumstances, when a man is sexually stimulated, his brain sends a message down the spinal cord and into the nerves of the penis. The nerve endings in the penis release chemical messengers, called neurotransmitters, that signal the corpora cavernosa (the two spongy rods of tissue that span the length of the penis) to relax and fill with blood. As they expand, the corpora cavernosa close off other veins that would normally drain blood from the penis. As the penis becomes engorged with blood, it enlarges and stiffens, causing an erection. Problems with blood vessels, nerves, or tissues of the penis can interfere with an erection.

Causes and symptoms

It is estimated that up to 20 million American men frequently suffer from impotence and that it strikes up to half of all men between the ages of 40 and 70. Doctors used to think that most cases of impotence were psychological in origin, but they now recognize that, at least in older men, physical causes may play a primary role in 60% or more of all cases. In men over the age of 60, the leading cause is **atherosclerosis**, or narrowing of the arteries, which can restrict the flow of blood to the penis. Injury or disease of the connective tissue, such as **Peyronie's disease**, may prevent the corpora cavernosa from completely expanding. Damage to the nerves of the penis, from certain types of surgery or neurological conditions, such as

Parkinson's disease or **multiple sclerosis**, may also cause impotence. Men with diabetes are especially at risk for impotence because of their high risk of both atherosclerosis and a nerve disease called **diabetic neuropathy**.

Certain types of blood pressure medications, anti-ulcer drugs, **antihistamines**, tranquilizers (especially before intercourse), antifungals (hetoconazole), antipsychotics, **antianxiety drugs**, and antidepressants, known as **selective serotonin reuptake inhibitors** (SSRIs, including Prozac and Paxil), can interfere with erectile function. **Smoking**, excessive alcohol consumption, and illicit drug use may also contribute. In rare cases, low levels of the male hormone testosterone may contribute to erectile failure. Finally, psychological factors, such as **stress**, guilt, or **anxiety**, may also play a role, even when the impotence is primarily due to organic causes.

Diagnosis

The doctor also obtains a thorough medical history to find out about past pelvic surgery, diabetes, cardiovascular disease, **kidney disease**, and any medications the man may be taking. The **physical examination** should include a genital examination, a measurement of blood flow through the penis, hormone tests, and a glucose test for diabetes.

In some cases, nocturnal penile tumescence testing is performed to find out whether the man has erections while asleep. Healthy men usually have about four or five erections throughout the night. The man applies a device to the penis called a Rigiscan before going to bed at night, and the device can determine whether he has had erections. (If a man is able to have normal erections at night, this suggests a psychological cause for his impotence.)

Treatment

Years ago, the standard treatment for impotence was an implantable penile prosthesis or long-term psychotherapy. Although physical causes are now more readily diagnosed and treated, individual or marital counseling is still an effective treatment for impotence when emotional factors play a role. Fortunately, other approaches are now available to treat the physical causes of impotence.

Medications

The first line and by far the most common treatment today is with the prescription drug **sildenafil citrate**, sold under the brand name Viagra. An

estimated 20 million prescriptions for the pill have been filled since it was approved by the FDA in March 1998. It is also the most effective treatment with a success rate of more than 60%. The drug boosts levels of a substance called cyclic GMP, which is responsible for widening the blood vessels of the penis. In clinical studies, Viagra produced headaches in 16% of men who took it, and other side effects included flushing, **indigestion**, and stuffy nose.

The primary drawback to Viagra, which works about an hour after it is taken, is that the FDA cautions men with heart disease or low blood pressure to be thoroughly examined by a physician before obtaining a prescription.

In the summer of 2002, two investigational drugs were announced to become available in the near future to also treat erectile dysfunction. Vardenafil and tadalafil both helped men who also had such conditions as diabetes, high blood pressure and benign prostatic hypertrophy. The drugs are awaiting final FDA approval.

Vardenafil and tadalafil belong to the same group of chemical compounds as sildenafil, namely phosphodiesterase type 5 (PDE-5) inhibitors. Some men cannot benefit from sildenafil or the two newer PDE-5 inhibitors because they have low levels of nitric oxide. British investigators reported in late 2002 that three different types of compounds are being studied as possible medications for men with low levels of nitric oxide. They are Rho-kinase inhibitors, soluble guanylate cyclase activators, and nitric oxide-releasing PDE-5 inhibitors.

Other medications under investigation as treatments for impotence are topical agents. Topical means that they are applied externally to the skin rather than being injected or taken by mouth. If approved, these drugs would provide a noninvasive alternative for men who cannot take sildenafil or other oral medications for impotence.

Injection therapy involves injecting a substance into the penis to enhance blood flow and cause an erection. The Food and Drug Administration (FDA) approved a drug called alprostadil (Caverject) for this purpose in July of 1995. Alprostadil relaxes smooth muscle tissue to enhance blood flow into the penis. It must be injected shortly before intercourse. Another, similar drug that is sometimes used is papaverine—not yet been approved by the FDA for this use. Either drug may sometimes cause painful erections or **priapism** (uncomfortable, prolonged erections) that must be treated with a shot of epinephrine.

Alprostadil may also be administered into the urethral opening of the penis. In MUSE (medical urethral system for erection), the man inserts a thin tube the width of a vermicelli noodle into his urethral opening and presses down on a plunger to deliver a tiny pellet containing alprostadil into his penis. The drug takes about 10 minutes to work and the erection lasts about an hour. The main side effect is a sensation of **pain** and burning in the urethra, which can last about five to 15 minutes.

Mechanical and surgical treatments

Another approach is vacuum therapy. The man inserts his penis into a clear plastic cylinder and uses a pump to force air out of the cylinder. This forms a partial vacuum around the penis, which helps to draw blood into the corpora cavernosa. The man then places a special ring over the base of the penis to trap the blood inside it. The only side effect with this type of treatment is occasional bruising if the vacuum is left on too long.

Implantable **penile prostheses** are usually considered a last resort for treating impotence. They are implanted in the corpora cavernosa to make the penis rigid without the need for blood flow. The semi-rigid type of prosthesis consists of a pair of flexible silicone rods that can be bent up or down. This type of device has a low failure rate but, unfortunately, it causes the penis to always be erect, which can be difficult to conceal under clothing.

The inflatable type of device consists of cylinders that are implanted in the corpora cavernosa, a fluid reservoir implanted in the abdomen, and a pump placed in the scrotum. The man squeezes the pump to move fluid into the cylinders and cause them to become rigid. (He reverses the process by squeezing the pump again.) While these devices allow for intermittent erections, they have a slightly higher malfunction rate than the silicon rods.

Men can return to sexual activity six to eight weeks after implantation surgery. Since implants affect the corpora cavernosa, they permanently take away a man's ability to have a natural erection.

In rare cases, if narrowed or diseased veins are responsible for impotence, surgeons may reroute the blood flow into the corpus cavernosa or remove leaking vessels. However, the success rate with these procedures has been very low, and they are still considered experimental.

Gene therapy

A newer investigational approach to the treatment of erectile dysfunction is **gene therapy**. As of

late 2002, several preclinical studies have shown promise, but none of the gene-based strategies so far have yet been tested for safety.

Alternative treatment

A number of herbs have been promoted for treating impotence. The most widely touted herbs for this purpose are *Coryanthe yohimbe* (available by prescription as yohimbine, with the trade name Yocon) and ginkgo (*Ginkgo biloba*), although neither has been conclusively shown to help the condition in controlled studies. In addition, ginkgo carries some risk of abnormal blood clotting and should be avoided by men taking blood thinners such as coumadin. Other herbs promoted for treating impotence include true unicorn root (*Aletriurus farinosa*), **saw palmetto** (*Serenoa repens*), ginseng (*Panax ginseng*), and Siberian ginseng (*Eleuthrococcus senticosus*). *Strychnos Nuxvomica* has been recommended, especially when impotence is caused by excessive alcohol, cigarettes, or dietary indiscretions, but it can be very toxic if taken improperly, so it should be used only under the strict supervision of a physician trained in its use.

Prognosis

With proper diagnosis, impotence can nearly always be treated or managed successfully. Unfortunately, fewer than 10% of impotent men seek treatment.

Prevention

There is no specific treatment to prevent impotence. Perhaps the most important measure is to maintain general good health and avoid atherosclerosis by exercising regularly, controlling weight, controlling **hypertension** and **high cholesterol** levels, and avoiding smoking. Avoiding excessive alcohol intake may also help.

Resources

BOOKS

- Beers, Mark H., MD, and Robert Berkow, MD, editors. "Erectile Dysfunction." Section 17, Chapter 220. In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.
- Miller, Lucinda G., and Wallace J. Murray, editors. *Herbal Medicinals: A Clinician's Guide*. Binghamton, N.Y.: Haworth Press, 1999.
- Pelletier, Kenneth R., MD. *The Best Alternative Medicine*, Part II. "CAM Therapies for Specific Conditions: Impotence." New York: Simon & Schuster, 2002.

KEY TERMS

Alprostadil—A smooth muscle relaxant sometimes injected into the penis or applied to the urethral opening to treat impotence.

Atherosclerosis—A disorder in which plaques of cholesterol, lipids, and other debris build up on the inner walls of arteries, narrowing them.

Corpus cavernosum (plural, corpora cavernosa)—One of two rods of spongy tissue in the penis that become engorged with blood in order to produce an erection.

Gene therapy—A method of treating a disorder by replacing damaged or abnormal genes with normal ones. Some researchers think that gene therapy may offer a new way to treat impotence.

Neurotransmitters—Chemicals that modify or help transmit impulses between nerve synapses.

Papaverine—A smooth muscle relaxant sometimes injected into the penis as a treatment for impotence.

Peyronie's disease—A disease resulting from scarring of the corpus cavernosa, causing painful erections.

Topical—A type of medication that is applied to a specific and limited area of skin, and affects only the area to which it is applied.

Urethra—The small tube that drains urine from the bladder, as well as serving as a conduit for semen during ejaculation in men.

Viagra—Trade name of an orally administered drug for erectile failure first cleared for marketing in the United States in March 1998. Its generic name is sildenafil citrate.

PERIODICALS

Campbell, Adam. "Soft Science: The Exclusive World on Which Sex Supplements may Help and Which Won't." *Men's Health* May 2002: 100.

Cellek, S., R. W. Rees, and J. Kalsi. "A Rho-Kinase Inhibitor, Soluble Guanylate Cyclase Activator and Nitric Oxide-Releasing PDE5 Inhibitor: Novel Approaches to Erectile Dysfunction." *Expert Opinion on Investigational Drugs* 11 (November 2002): 1563–1573.

Christ, G. J. "Gene Therapy for Erectile Dysfunction: Where Is It Going?" *Current Opinion in Urology* 12 (November 2002): 497–501.

Cowley, Geoffrey. "Looking Beyond Viagra." *Newsweek* April 24, 2000: 77.

Gresser, U., and C. H. Gleiter. "Erectile Dysfunction: Comparison of Efficacy and Side Effects of the PDE-5 Inhibitors Sildenafil, Vardenafil and Tadalafil—Review of the Literature." *European Journal of Medical Research* 7 (October 29, 2002): 435–446.

"Is Viagra Safe?" *Internal Medicine Alert* June 29, 2002: 90.

Norton, Patrice G.W. "Investigational Drugs in Erectile Dysfunction. (Vardenafil, Tadalafil)." *Internal Medicine News* June 1, 2002: 50.

Yap, R. L., and K. T. McVary. "Topical Agents and Erectile Dysfunction: Is There a Place?" *Current Urology Reports* 3 (December 2002): 471–476.

"Yohimbe Tree Bark: Herbal Viagra Better Gotten by Rx." *Environmental Nutrition* February 1999: 8.

ORGANIZATIONS

American Foundation for Urologic Disease. 1128 North Charles Street, Baltimore, MD 21201. (410) 468-1800.

American Urological Association (AUA). 1120 North Charles Street, Baltimore, MD 21201. (410) 727-1100. <www.auanet.org>.

Center for Biologics Evaluation and Research (CBER), U. S. Food and Drug Administration (FDA). 1401 Rockville Pike, Rockville, MD 20852-1448. (800) 835-4709 or (301) 827-1800. <www.fda.gov/cber>.

Impotence Institute of America, Impotents Anonymous. 10400 Little Patuxent Parkway, Suite 485, Columbia, MD 21044-3502. (800) 669-1603.

National Kidney and Urologic Diseases Information Clearinghouse. 3 Information Way, Bethesda, MD 20892-3580. (800) 891-5390.

Ken R. Wells
Rebecca J. Frey, PhD

Impulse control disorders

Definition

Impulse control disorders are characterized by an inability to resist the impulse to perform an action that is harmful to one's self or others. This is a relatively new class of **personality disorders**, and the most common of these are **intermittent explosive disorder**, kleptomania, pyromania, compulsive gambling disorder, and trichotillomania.

Description

All of these impulse control disorders involve the loss or lack of control in certain specific situations. The hallmark of these disorders is the individual's inability to stop impulses that may cause harm to themselves or others. Affected individuals often feel

anxiety or tension in considering these behaviors. This anxiety or tension is relieved or diminished once the action is performed.

Intermittent explosive disorder is more common among men, and involves aggressive outbursts that lead to assaults on others or destruction of property. These outbursts are unprovoked or seem to be out of proportion to the event that precedes them.

Kleptomania is more common among women, and involves the theft of objects that are seemingly worthless. The act of stealing relieves tension and is seen by the individual to be rewarding. The actual stealing is not preplanned, and the concept of punishment for the crime does not occur to these individuals, although they are aware that what they are doing is wrong.

Pyromania is more common among men, and involves setting fires in order to feel pleasure and relieve tension.

Pathological gambling occurs in roughly 1-3% of the population, and involves excessive gambling despite heavy monetary losses. These losses actually act as a motivating factor in continuing gambling in order to recoup some of what was lost.

Trichotillomania involves pulling hair from one's own scalp, face, or body, and is more common in women. It often begins in childhood, and is often associated with major depression or **attention-deficit/hyperactivity disorder**.

Causes and symptoms

The exact causes of impulse control disorders are not fully understood as of 2004. Individuals who have had serious head injuries, however, can be at a higher risk for developing impulse control disorders, as are those with epilepsy.

Some cases of impulse control disorders appear to be side effects of general medical conditions. As of 2004, several groups of researchers have noted that some older adults with Parkinson's disease become compulsive gamblers as the disease progresses. It is thought that this gambling behavior is a side effect of dopaminergic drugs, as it does not respond to standard treatments for compulsive gambling but only to changes in the patient's medication.

Another medical condition that is associated with impulse control disorders is carcinoid syndrome. In one group of 20 consecutive patients with the syndrome, 75% met DSM-IV diagnostic criteria for one or another impulse control disorder. The researchers

attribute the connection to the high levels of serotonin (a neurotransmitter) produced by carcinoid tumors.

Diagnosis

A diagnosis of any of these impulse control disorders can be made only after other medical and psychiatric disorders that may cause the same symptoms have been ruled out.

Some doctors may administer questionnaires or similar psychiatric screeners as part of the differential diagnosis. Two instruments that have been devised in the early 2000s to specifically target impulsive behavior are the Gambling Urge Scale (GUS) and the Lifetime History of Impulsive Behaviors (LHIB) Interview.

Intermittent explosive disorder involves severe acts of assault or destruction of property. The aggression seen during these acts is vastly out of proportion to events that may seem to have precipitated the acts.

Kleptomania involves stealing objects that are unnecessary and of little monetary value. The act of stealing is not an expression of anger or vengeance. Again, there is an increased tension before the act is committed, and this is resolved or relieved once the object is stolen.

Pyromania is classified by the deliberate setting of fires more than once. The individual will exhibit a fascination and attraction to fire and any objects associated with it. Before the fire is set, there is tension, with a resolving relief once the fire is set. Acts of true pyromania are not done for monetary gain, to express anger, to conceal criminal behavior, or in response to hallucination.

Pathological gambling is a disorder to gamble despite continuing losses and monetary insufficiency. This disorder typically begins in youth, and affected individuals are often competitive, easily bored, restless, and generous.

For a diagnosis of pathological gambling, five or more of the following symptoms must be present:

- a preoccupation with gambling
- a need to gamble with more money to achieve the thrill of winning
- repeated attempts to control or stop gambling
- irritability or restlessness due to repeated attempts of control
- gambling as an escape from stress
- lying to cover up gambling

KEY TERMS

Carcinoid syndrome—The pattern of symptoms (often including asthma and diarrhea) associated with carcinoid tumors of the digestive tract or lungs.

Compulsive gambling disorder—An impulse control disorder in which an individual cannot resist gambling despite repeated losses.

Intermittent explosive disorder—A personality disorder in which an individual is prone to intermittent explosive episodes of aggression during which he or she causes bodily harm or destroys property.

Kleptomania—An impulse control disorder in which one steals objects that are of little or no value.

Pyromania—An impulse control disorder in which one sets fires.

Trichotillomania—An impulse or compulsion to pull out one's own hair.

- conducting illegal activities, such as embezzling or fraud, to finance gambling
- losing a job or personal relationship due to gambling
- borrowing money to fund gambling

Trichotillomania is the continuous pulling out of one's own hair. Again, there is an increased sense of tension before pulling the hair, which is relieved once it is pulled out. Recurrent pulling out of one's hair resulting in noticeable hair loss. Affected individuals can undergo significant distress and impaired social, occupational, and functional behavior.

Treatment

A combination of psychological counseling and medication are the preferred treatments for the impulse control disorders. For kleptomania, pyromania, and trichotillomania, behavior modification is usually the treatment of choice. Children with trichotillomania are often helped by antidepressant medication. For pathological gambling, treatment usually involves an adaptation of the model set forth by Alcoholics Anonymous. Individuals are counseled with the goal of eventual response to appropriate social limits. In the case of intermittent explosive disorder, anger management and medication may be used in extreme cases of aggression.

Prognosis

These disorders can usually be controlled with medication, although it may need to be continued long-term to help prevent further aggressive outbursts. Long-term counseling is usually necessary as well. Support groups and meetings may also help these individuals.

The prognosis for intermittent explosive disorder, kleptomania, and pyromania is fair. Little is known about the prognosis for trichotillomania, and studies have shown that the condition can disappear for long periods (months to years) without any psychological counseling. For pathological gambling, the prognosis varies greatly from person to person. While total cure for this condition is unlikely, much like **alcoholism**, long periods of abstinence or continuous abstinence are possible.

Prevention

There are no known preventive treatments or measures for impulse control disorders.

Resources

BOOKS

American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders*. 4th ed., revised. Washington, D.C.: American Psychiatric Association, 2000.

PERIODICALS

Avanzi, M., E. Uber, and F. Bonfa. "Pathological Gambling in Two Patients on Dopamine Replacement Therapy for Parkinson's Disease." *Neurological Sciences* 25 (June 2004): 98–101.

Kurlan, R. "Disabling Repetitive Behaviors in Parkinson's Disease." *Movement Disorders* 19 (April 2004): 433–437.

Raylu, N., and T. P. Oei. "The Gambling Urge Scale: Development, Confirmatory Factor Validation, and Psychometric Properties." *Psychology of Addictive Behaviors* 18 (June 2004): 100–105.

Russo, S., J. C. Boon, I. P. Kema, et al. "Patients with Carcinoid Syndrome Exhibit Symptoms of Aggressive Impulse Dysregulation." *Psychosomatic Medicine* 66 (May-June 2004): 422–425.

Schmidt, C. A., A. E. Fallon, and E. F. Coccaro. "Assessment of Behavioral and Cognitive Impulsivity: Development and Validation of the Lifetime History of Impulsive Behaviors Interview." *Psychiatry Research* 126 (April 30, 2004): 107–121.

Tay, Y. K., M. L. Levy, and D. W. Metry. "Trichotillomania in Childhood: Case Series and Review." *Pediatrics* 113 (May 2004): 494–498.

ORGANIZATIONS

American Psychiatric Association. 1400 K Street, NW, Washington, DC 20005. <<http://www.psych.org>>.

Gamblers Anonymous International Service Office. PO Box 17173, Los Angeles, CA 90017. (213) 386-8789, Fax: (213) 386-0030. <<http://www.gamblersanonymous.org/>>.

National Institute of Mental Health. 6001 Executive Boulevard, Room 8184, MSC 9663, Bethesda, MD 20892-9663. (301) 443-4513. <<http://www.nimh.nih.gov>>.

Trichotillomania Learning Center, Inc. 1215 Mission Street, Suite 2, Santa Cruz, CA 95060. (831) 457-1004, Fax: (831) 426-4383. <<http://www.trich.org>>.

Liz Meszaros
Rebecca Frey, PhD

In vitro fertilization

Definition

In vitro fertilization (IVF) is a procedure in which eggs (ova) from a woman's ovary are removed. They are fertilized with sperm in a laboratory procedure, and then the fertilized egg (embryo) is returned to the woman's uterus.

Purpose

IVF is one of several assisted reproductive techniques (ART) used to help infertile couples to conceive a child. If after one year of having sexual intercourse without the use of birth control a woman is unable to get pregnant, **infertility** is suspected. Some of the reasons for infertility are damaged or blocked fallopian tubes, hormonal imbalance, or **endometriosis** in the woman. In the man, low sperm count or poor quality sperm can cause infertility.

IVF is one of several possible methods to increase the chance for an infertile couple to become pregnant. Its use depends on the reason for infertility. IVF may be an option if there is a blockage in the fallopian tube or endometriosis in the woman or low sperm count or poor quality sperm in the man. There are other possible treatments for these conditions, such as surgery for blocked tubes or endometriosis, which may be tried before IVF.

IVF will not work for a woman who is not capable of ovulating or a man who is not able to produce at least a few healthy sperm.

Precautions

The screening procedures and treatments for infertility can become a long, expensive, and sometimes, disappointing process. Each IVF attempt takes at least an entire menstrual cycle and can cost \$5,000-\$10,000, which may or may not be covered by health insurance. The **anxiety** of dealing with infertility can challenge both individuals and their relationship. The added **stress** and expense of multiple clinic visits, testing, treatments, and surgical procedures can become overwhelming. Couples may want to receive counseling and support throughout the process.

Description

In vitro fertilization is a procedure where the joining of egg and sperm takes place outside of the woman's body. A woman may be given fertility drugs before this procedure so that several eggs mature in the ovaries at the same time. Eggs (ova) are removed from a woman's ovaries using a long, thin needle. The physician gains access to the ovaries using one of two possible procedures. One procedure involves inserting the needle through the vagina (transvaginally). The physician guides the needle to the location of the ovaries with the help of an ultrasound machine. In the other procedure, called **laparoscopy**, a small thin tube with a viewing lens is inserted through an incision in the navel. This allows the physician to see inside the patient, and locate the ovaries, on a video monitor.

Once the eggs are removed, they are mixed with sperm in a laboratory dish or test tube. (This is where the term *test tube baby* comes from.) The eggs are monitored for several days. Once there is evidence that fertilization has occurred and the cells begin to divide, they are then returned to the woman's uterus.

In the procedure to remove eggs, enough may be gathered to be frozen and saved (either fertilized or unfertilized) for additional IVF attempts. A 2004 study from the Mayo Clinic found that frozen sperm was as effective as fresh sperm for IVF.

IVF has been used successfully since 1978, when the first child to be conceived by this method was born in England. Over the past 20 years, thousands of couples have used this method of ART or similar procedures to conceive.

Other types of assisted reproductive technologies might be used to achieve **pregnancy**. A procedure called intracytoplasmic sperm injection (ICSI) uses a manipulation technique that must be performed using a microscope to inject a single sperm into each egg. The

PATRICK CHRISTOPHER STEPTOE (1913–1988)



(Archive. Reproduced by permission.)

Patrick Christopher Steptoe was born in Oxfordshire, England, on June 9, 1913. His mother was a social worker and his father was a church organist. Steptoe entered the

University of London's St. George Hospital Medical School, earning his physician's license in 1939 and becoming a member of the Royal College of Surgeons. When Steptoe volunteered as a naval surgeon during World War II, he was captured and held as a prisoner until his release in 1943. Following his release, Steptoe studied obstetrics and gynecology and moved to Manchester to start a private practice in 1948. In 1951, Steptoe accepted a position at Oldham General and District Hospital in England.

During his time at Oldham, Steptoe continued his study of fertility problems. Using a laparoscope, he developed a method to remove eggs from a woman's ovaries. In 1966, Steptoe teamed with physiologist Robert G. Edwards who had successfully fertilized eggs outside of the body. In 1968, the pair had a breakthrough when Edwards successfully fertilized an egg that Steptoe had removed, but their attempts to implant the embryo failed repeatedly. However, Steptoe and Edwards experienced success when a fertilized egg was implanted into the uterus of Leslie Brown. Brown gave birth to a healthy baby girl on July 25, 1978.

Steptoe retired and built a clinic in Cambridge. He and Edwards were named Commanders of the British Empire, and Steptoe was honored with fellowship in the Royal Society. He and his wife had two children. Steptoe died on March 21, 1988.

fertilized eggs can then be returned to the uterus, as in IVF. In gamete intrafallopian tube transfer (GIFT) the eggs and sperm are mixed in a narrow tube and then deposited in the fallopian tube, where fertilization normally takes place. Another variation on IVF is zygote intrafallopian tube transfer (ZIFT). As in IVF, the fertilization of the eggs occurs in a laboratory dish. And, similar to GIFT, the embryos are placed in the fallopian tube (rather than the uterus as with IVF).

Preparation

Once a woman is determined to be a good candidate for in vitro fertilization, she will generally be given "fertility drugs" to stimulate ovulation and the development of multiple eggs. These drugs may include gonadotropin releasing hormone agonists (GnRHa), Pergonal, Clomid, or human chorionic gonadotropin (hcg). The maturation of the eggs is then monitored with ultrasound tests and frequent blood tests. If enough eggs mature, the physician will perform the procedure to remove them. The woman

may be given a sedative prior to the procedure. A local anesthetic agent may also be used to reduce discomfort during the procedure.

Aftercare

After the IVF procedure is performed the woman can resume normal activities. A pregnancy test can be done approximately 12–14 days later to determine if the procedure was successful.

Risks

The risks associated with in vitro fertilization include the possibility of **multiple pregnancy** (since several embryos may be implanted) and **ectopic pregnancy** (an embryo that implants in the fallopian tube or in the abdominal cavity outside the uterus). There is a slight risk of ovarian rupture, bleeding, infections, and complications of anesthesia. If the procedure is successful and pregnancy is achieved, the pregnancy would carry the same risks as any pregnancy achieved without assisted technology.

KEY TERMS

Fallopian tubes—In a woman's reproductive system, a pair of narrow tubes that carry the egg from the ovary to the uterus.

GIFT—Stands for gamete intrafallopian tube transfer. This is a process where eggs are taken from a woman's ovaries, mixed with sperm, and then deposited into the woman's fallopian tube.

ICSI—Stands for intracytoplasmic sperm injection. This process is used to inject a single sperm into each egg before the fertilized eggs are put back into the woman's body. The procedure may be used if the male has a low sperm count.

ZIFT—Stands for zygote intrafallopian tube transfer. In this process of in vitro fertilization, the eggs are fertilized in a laboratory dish and then placed in the woman's fallopian tube.

Normal results

Success rates vary widely between clinics and between physicians performing the procedure and implantation does not guarantee pregnancy. Therefore, the procedure may have to be repeated more than once to achieve pregnancy. However, success rates have improved in recent years, up from 20% in 1995 to 27% in 2001.

Abnormal results

An ectopic or multiple pregnancy may abort spontaneously or may require termination if the health of the mother is at risk. The number of multiple pregnancies has decreased in recent years as technical advances and professional guidelines have led to implanting of fewer embryos per attempt.

Resources

PERIODICALS

"Frozen, Fresh Sperm Both Effective for In Vitro Fertilization." *Obesity, Fitness & Wellness Week* June 5, 2004: 1059.

"Multiple Births Via In Vitro Fertilization Are Declining." *Women's Health Weekly* May 6, 2004: 16.

ORGANIZATIONS

American Society for Reproductive Medicine. 1209 Montgomery Highway, Birmingham, AL 35216-2809. (205) 978-5000. <asrm@asrm.com> <<http://www.asrm.com>>.

Center for Fertility and In Vitro Fertilization Loma Linda University. 11370 Anderson St., Loma Linda, CA

92354. (909) 796-4851. <<http://www.llu.edu/llumc/fertility>>.

Resolve. 1310 Broadway, Somerville, MA 02144-1731. (617) 623-0744. <<http://www.resolve.org>>.

OTHER

"Infertility." *HealthWorld Online Page*. <<http://www.healthy.net>>.

"In vitro Fertilization: A Teacher's Guide from Newton's Apple." *PBS Page*. <<http://www.pbs.org/ktca/newtons/11/invitro.html>>.

Altha Roberts Edgren
Teresa G. Odle

Inclusion blennorrhea see **Inclusion conjunctivitis**

Inclusion conjunctivitis

Definition

Inclusion **conjunctivitis** is an inflammation of the conjunctiva (the membrane that lines the eyelids and covers the white part, or sclera, of the eyeball) by *Chlamydia trachomatis*. Chlamydia is a sexually transmitted organism.

Description

Inclusion conjunctivitis, known as neonatal inclusion conjunctivitis in the newborn and adult inclusion conjunctivitis in the adult, is also called inclusion blennorrhea, chlamydial conjunctivitis, or swimming pool conjunctivitis. This disease affects four of 1,000 (0.4%) live births. Approximately half of the infants born to untreated infected mothers will develop the disease.

Causes and symptoms

Inclusion conjunctivitis in the newborn results from passage through an infected birth canal and develops 5–14 days after birth. Both eyelids and conjunctivae are swollen. There may be a discharge of pus from the eyes.

Most instances of adult inclusion conjunctivitis result from exposure to infected genital secretions. It is transmitted to the eye by fingers and occasionally by the water in swimming pools, poorly chlorinated hot tubs, or by sharing makeup. In adult inclusion conjunctivitis, one eye is usually involved, with a stringy discharge of mucus and pus. There may be little bumps

called follicles inside the lower eyelid and the eye is red. Occasionally, the condition damages the cornea, causing cloudy areas and a growth of new blood vessels (neovascularization).

Diagnosis

Inclusion conjunctivitis is usually considered when the patient has a follicular conjunctivitis that will not go away, even after using **topical antibiotics**. Diagnosis depends upon tests performed on the discharge from the eye. Gram stains determine the type of microorganism, while culture and sensitivity tests determine which antibiotic will kill the harmful microorganism. Conjunctival scraping determines whether chlamydia is present in cells taken from the conjunctiva.

Treatment

Treatment in the newborn consists of administration of tetracycline ointment to the conjunctiva and erythromycin orally or through intravenous therapy for fourteen days. The mother should be treated for **cervicitis** and the father for **urethritis**, even if they do not have symptoms of these diseases.

In adults, tetracycline ointment or drops should be applied to the conjunctiva and oral tetracycline, amoxicillin, or erythromycin should be taken for three weeks, or doxycycline for one week.

Patients should have weekly checkups so the doctor can monitor the healing.

Oral tetracycline should not be administered to children whose permanent teeth have not erupted. It should also not be given to nursing or pregnant women.

Prognosis

Untreated inclusion conjunctivitis in the newborn persists for 3–12 months and usually heals; however, there may be scarring or neovascularization. In the adult, if left untreated, the disease may continue for months and cause corneal neovascularization. Even if treated, **antibiotics** usually do not reverse damage that may have occurred, but they may help prevent it if given early enough.

Prevention

The neonatal infection may be prevented by instilling erythromycin ointment in the conjunctival cul-de-sac at birth. It is not prevented by silver nitrate.

Chlamydia is a contagious, sexually transmitted disease. Some systemic symptoms include a history of

KEY TERMS

Cervicitis—Cervicitis is an inflammation of the cervix or neck of the uterus.

Conjunctiva—The conjunctiva is the membrane that lines the eyelids and covers the white part of the eyeball (sclera).

Cornea—The clear dome-shaped structure that covers the colored part of the eye (iris).

Neovascularization—Neovascularization is the growth of new blood vessels.

Urethritis—Urethritis is an inflammation of the urethra, the canal for the discharge of urine that extends from the bladder to the outside of the body.

vaginitis, **pelvic inflammatory disease**, or urethritis. Patients with symptoms of these diseases should be treated by a physician.

Resources

BOOKS

Newell, Frank W. *Ophthalmology: Principles and Concepts*. 8th ed. St. Louis: Mosby, 1996.

ORGANIZATIONS

American Academy of Ophthalmology. 655 Beach Street, PO Box 7424, San Francisco, CA 94120-7424. <<http://www.eyenet.org>>.

American Optometric Association. 243 North Lindbergh Blvd., St. Louis, MO 63141. (314) 991-4100. <<http://www.aoanet.org>>.

Lorraine Steefel, RN

Incompetent cervix

Definition

A cervix (the structure at the bottom of the uterus) that is incompetent is abnormally weak, and therefore it can gradually widen during **pregnancy**. Left untreated, this can result in repeated pregnancy losses or premature delivery.

Description

Incompetent cervix is the result of an anatomical abnormality. Normally, the cervix remains closed

throughout pregnancy until labor begins. An incompetent cervix gradually opens due to the pressure from the developing fetus after about the 13th week of pregnancy. The cervix begins to thin out and widen without any contractions or labor. The membranes surrounding the fetus bulge down into the opening of the cervix until they break, resulting in the loss of the baby or a very premature delivery.

Causes and symptoms

Some factors that can contribute to the chance of a woman having an incompetent cervix include trauma to the cervix, physical abnormality of the cervix, or having been exposed to the drug diethylstilbestrol (DES) in the mother's womb. Some women have cervical incompetence for no obvious reason.

Diagnosis

Incompetent cervix is suspected when a woman has three consecutive spontaneous pregnancy losses during the second trimester (the fourth, fifth and sixth months of the pregnancy). The likelihood of this happening by random chance is less than 1%. Spontaneous losses due to incompetent cervix account for 20–25% of all second trimester losses. A spontaneous second trimester pregnancy loss is different from a **miscarriage**, which usually happens during the first three months of pregnancy.

The physician can check for abnormalities in the cervix by performing a manual examination or by an ultrasound test. The physician can also check to see if the cervix is prematurely widened (dilated). Because incompetent cervix is only one of several potential causes for this, the patient's past history of pregnancy losses must also be considered when making the diagnosis.

Treatment

Treatment for incompetent cervix is a surgical procedure called cervical cerclage. A stitch (suture) is used to tie the cervix shut to give it more support. It is most effective if it is performed somewhere between 14–16 weeks into the pregnancy. The stitch is removed near the end of pregnancy to allow for a normal birth.

Cervical cerclage can be performed under spinal, epidural, or **general anesthesia**. The patient will need to stay in the hospital for one or more days. The procedure to remove the suture is done without the need for anesthesia. The vagina is held open with an instrument called a speculum and the stitch is cut and removed.

KEY TERMS

Diethylstilbestrol (DES)—DES is a drug given to women a generation ago to prevent miscarriage. At that time it was not known that female children born of women who had been given DES would show a higher rate for cervical and other reproductive abnormalities, as well as a rare form of vaginal cancer, when they reached reproductive age.

Effacement—The thinning out of the cervix that normally occurs along with dilation shortly before delivery.

Preterm labor—Labor before the thirty-seventh week of pregnancy.

This may be slightly uncomfortable, but should not be painful.

Some possible risks of cerclage are premature rupture of the amniotic membranes, infection of the amniotic sac, and preterm labor. The risk of infection of the amniotic sac increases as the pregnancy progresses. For a cervix that is dilated 3 centimeters (cm), the risk is 30%.

After cerclage, a woman will be monitored for any preterm labor. The woman needs to consult her obstetrician immediately if there are any signs of contractions.

Cervical cerclage can not be performed if a woman is more than 4 cm dilated, if the fetus has already died in her uterus, or if her amniotic membranes are torn and her water has broken.

Prognosis

The success rate for cerclage correction of incompetent cervix is good. About 80-90% of the time women deliver healthy infants. The success rate is higher for cerclage done early in pregnancy.

Resources

OTHER

Weiss, Robin. "Incompetent Cervix." *The Mining Co. Guide to Pregnancy/Childbirth Page*. March 1998. <<http://www.pregnancy.miningco.com/library/weekly/aa011298.htm>>.

Tish Davidson, A.M.

Incontinence see **Urinary incontinence**

Indigestion

Definition

Indigestion, which is sometimes called **dyspepsia**, is a general term covering a group of nonspecific symptoms in the digestive tract. It is often described as a feeling of fullness, bloating, **nausea**, **heartburn**, or gassy discomfort in the chest or abdomen. The symptoms develop during meals or shortly afterward. In most cases, indigestion is a minor problem that often clears up without professional treatment.

Description

Indigestion or dyspepsia is a widespread condition, estimated to occur in 25% of the adult population of the United States. Most people with indigestion do not feel sick enough to see a doctor; nonetheless, it is a common reason for office visits. About 3% of visits to primary care doctors are for indigestion.

Causes and symptoms

Physical causes

The symptoms associated with indigestion have a variety of possible physical causes, ranging from commonplace food items to serious systemic disorders:

- Diet. Milk, milk products, alcoholic beverages, tea, and coffee cause indigestion in some people because they stimulate the stomach's production of acid.
- Medications. Certain prescription drugs as well as over-the-counter medications can irritate the stomach lining. These medications include **aspirin**, NSAIDs, some **antibiotics**, digoxin, theophylline, **corticosteroids**, iron (ferrous sulfate), **oral contraceptives**, and tricyclic antidepressants.
- Disorders of the pancreas and gallbladder. These include inflammation of the gallbladder or pancreas, **cancer** of the pancreas, and **gallstones**.
- Intestinal parasites. Parasitic infections that cause indigestion include **amebiasis**, fluke and tapeworm infections, **giardiasis**, and strongyloidiasis.
- Systemic disorders, including diabetes, thyroid disease, collagen vascular disease.
- Cancers of the digestive tract.
- Conditions associated with women's reproductive organs. These conditions include menstrual cramps, **pregnancy**, and pelvic inflammatory disease.

Psychologic and emotional causes

Indigestion often accompanies an emotional upset, because the part of the nervous system involved in the so-called "fight-or-flight" response also affects the digestive tract. People diagnosed with **anxiety** or **somatoform disorders** frequently have problems with indigestion. Many people in the general population, however, will also experience heartburn, "butterflies in the stomach," or stomach cramps when they are in upsetting situations—such as school examinations, arguments with family members, crises in their workplace, and so on. Some people's digestive systems appear to react more intensely to emotional **stress** due to hypersensitive nerve endings in their intestinal tract.

Specific gastrointestinal disorders

In some cases, the patient's description of the symptoms suggests a specific digestive disorder as the cause of the indigestion. Some doctors classify these cases into three groups:

ESOPHAGITIS TYPE. Esophagitis is an inflammation of the tube that carries food from the throat to the stomach (the esophagus). The tissues of the esophagus can become irritated by the flow (reflux) of stomach acid backward into the lower part of the esophagus. If the patient describes the indigestion in terms of frequent or intense heartburn, the doctor will consider gastroesophageal reflux disease (GERD) as a possible cause. GERD is a common disorder in the general population, affecting about 30% of adults.

PEPTIC ULCER TYPE. Patients who smoke and are over 45 are more likely to have indigestion of the peptic ulcer type. This group also includes people who find that their indigestion is relieved by taking **antacids** or eating a small amount of food. Patients in this category are often found to have *Helicobacter pylori* infections. *H. pylori* is a rod-shaped bacterium that lives in the tissues of the stomach and causes irritation of the mucous lining of the stomach walls. Most people with *H. pylori* infections do not develop chronic indigestion, but the organism appears to cause peptic ulcer disease (PUD) in a vulnerable segment of the population.

NONULCER TYPE. Most cases of chronic indigestion—as many as 65%—fall into this third category. Nonulcer dyspepsia is sometimes called functional dyspepsia because it appears to be related to abnormalities in the way that the stomach empties its contents into the intestine. In some people, the stomach empties either too slowly or too rapidly. In others, the stomach's muscular contractions are irregular and

uncoordinated. These disorders of stomach movement (motility) may be caused by hypersensitive nerve endings in the stomach tissues. Patients in this group are likely to be younger than 45 and have a history of taking medications for anxiety or depression.

Diagnosis

Patient history

Because indigestion is a nonspecific set of symptoms, patients who feel sick enough to seek medical attention are likely to go to their primary care doctor. The history does not always point to an obvious diagnosis. The doctor can, however, use the process of history-taking to evaluate the patient's mood or emotional state in order to assess the possibility of a psychiatric disturbance. In addition, asking about the location, intensity, timing, and recurrence of the indigestion can help the doctor weigh the different diagnostic possibilities.

An important part of the history-taking is asking about symptoms that may indicate a serious illness. These warning symptoms include:

- Weight loss
- Persistent vomiting
- Difficulty or **pain** in swallowing
- Vomiting blood or passing blood in the stools
- Anemia.

Imaging studies

If the doctor thinks that the indigestion should be investigated further, he or she will order an endoscopic examination of the stomach. An endoscope is a slender tube-shaped instrument that allows the doctor to look at the lining of the patient's stomach. If the patient has indigestion of the esophagitis type or nonulcer type, the stomach lining will appear normal. If the patient has PUD, the doctor will be able to see breaks or ulcerated areas in the tissue. He or she may also order ultrasound imaging of the abdomen, or a radio-nuclide scan to evaluate the motility of the stomach.

Laboratory tests

BLOOD TESTS. If the patient is over 45, the doctor will have the patient's blood analyzed for a complete blood cell count, measurements of liver enzyme levels, electrolyte and serum calcium levels, and thyroid function.

TESTS FOR *HELICOBACTER PYLORI*. Doctors can now test patients for the presence of *H. pylori* without

having to take a tissue sample from the stomach. One of these noninvasive tests is a blood test and the other is a breath test.

Treatment

Since most cases of indigestion are not caused by serious disorders, many doctors prefer to try medications and other treatment measures before ordering an endoscopy.

Diet and stress management

Many patients benefit from the doctor's reassurance that they do not have a serious or fatal disorder. Cutting out alcoholic beverages and drinks containing **caffeine** often helps. The patient may also be asked to keep a record of food intake, daily schedule, and symptom severity. Food diaries sometimes reveal psychologic or dietary factors that influence indigestion.

Medications

Patients with the esophagitis type of indigestion are often treated with H₂ antagonists. H₂ antagonists are drugs that block the secretion of stomach acid. They include ranitidine (Zantac) and famotidine (Pepcid).

Patients with motility disorders may be given prokinetic drugs. Prokinetic medications speed up the emptying of the stomach and increase intestinal motility. They include metoclopramide (Reglan) and cisapride (Propulsid). These drugs relieve symptoms in 60–80% of patients.

*Removal of *H. pylori**

It is not clear that patients with *H. pylori* infections who have *not* developed gastric ulcers need to have the bacterium removed. Some studies indicate, however, that these patients may benefit from antibiotic therapy.

Alternative treatment

Herbal medicine

Practitioners of Chinese traditional herbal medicine might recommend medicines derived from peony (*Paeonia lactiflora*), hibiscus (*Hibiscus sabdariffa*), or hare's ear (*Bupleurum chinense*) to treat indigestion. Western herbalists are likely to prescribe fennel (*Foeniculum vulgare*), lemon balm (*Melissa officinalis*), or peppermint (*Mentha piperita*) to relieve stomach cramps and heartburn.

KEY TERMS

Dyspepsia—Another name for indigestion.

Endoscope—A slender tubular instrument used to examine the inside of the stomach.

Gastroesophageal reflux disease (GERD)—A disorder of the lower end of the esophagus, caused by stomach acid flowing backward into the esophagus and irritating the tissues.

H₂ antagonist—A type of drug that relieves indigestion by reducing the production of stomach acid.

Heartburn—A popular term for an uncomfortable burning sensation in the stomach and lower esophagus, sometimes caused by the reflux of small amounts of stomach acid.

Helicobacter pylori—A gram-negative rod-shaped bacterium that lives in the tissues of the stomach and causes inflammation of the stomach lining.

Motility—The movement or capacity for movement of an organism or body organ. Indigestion is sometimes caused by abnormal patterns in the motility of the stomach.

Peptic ulcer disease (PUD)—A stomach disorder marked by corrosion of the stomach lining due to the acid in the digestive juices.

Prokinetic—A drug that works to speed up the emptying of the stomach and the motility of the intestines.

Reflux—The backward flow of a body fluid or secretion. Indigestion is sometimes caused by the reflux of stomach acid into the esophagus.

Homeopathy

Homeopaths tailor their remedies to the patient's overall personality profile as well as the specific symptoms. Depending on the patient's reaction to the indigestion and some of its likely causes, the homeopath might choose *Gelsemium* (*Gelsemium sempervirens*), *Carbo vegetalis*, *Nux vomica*, or *Pulsatilla* (*Pulsatilla nigricans*).

Other treatments

Some alternative treatments are aimed at lowering the patient's stress level or changing attitudes and beliefs that contribute to indigestion. These therapies and practices include **Reiki**, **reflexology**, **hydrotherapy**, therapeutic massage, **yoga**, and **meditation**.

Prognosis

Most cases of mild indigestion do not need medical treatment. For patients who consult a doctor and are given an endoscopic examination, 5-15% are diagnosed with GERD and 15-25% with PUD. About 1% of patients who are endoscoped have **stomach cancer**. Most patients with functional dyspepsia do well on either H₂ antagonists or prokinetic drugs, depending on the cause of their indigestion.

Prevention

Indigestion can often be prevented by attention to one's diet, general stress level, and ways of managing stress. Specific preventive measures include:

- Stop **smoking**.
- Cutting down on or eliminating alcohol, tea, or coffee.
- Avoiding foods that are highly spiced or loaded with fat.
- Eating slowly and keeping mealtimes relaxed.
- Practicing yoga or meditation.
- Not taking aspirin or other medications on an empty stomach.
- Keeping one's weight within normal limits.

Resources

OTHER

"Indigestion." *ThriveOnline*. April 6, 1998. <<http://thriveonline.oxygen.com>>.

Rebecca J. Frey, PhD

Indinavir see **Protease inhibitors**

Indirect Coombs' test see **Coombs' tests**

Indium scan of the body

Definition

A scanning procedure in which a patient's white blood cells are first labeled with the radioactive substance indium, and then the patient's body is scanned as a way of tracking the white blood cells at the site of possible infection.

KEY TERMS

Indium—A silvery metallic element with some nonmetallic chemical properties used to label white blood cells prior to scanning.

Leukocyte—A white blood cell protects the body against infection and fight infection when it occurs. They are bigger than red blood cells.

Purpose

The procedure is used to detect inflammatory processes in the body such as infections. By labelling the leukocytes (white blood cells), radiologists or nuclear medicine specialists can then watch their migration toward an **abscess** or other infection.

Description

A nuclear medicine technologist withdraws about 50 ml. of blood. White blood cells are collected, exposed to indium, and reinjected by IV back into the patient.

The scan is scheduled for between 18 and 24 hours after the white blood cells have been labelled with indium. (In some cases, more scanning may be scheduled 48 hours after labelling).

For the scan, the patient lies on a special scanning table, as either a single camera passing underneath the table or two cameras (one above the table and one underneath) are placed as close as possible to the body, slowly scanning the person's body.

The radiologist may need extra pictures, but these take only a few minutes each.

While the patient must remain perfectly still during the scan, there should be no discomfort.

Aftercare

After the scan, the patient should be able to continue with normal daily activities with no problems.

Risks

The only risk during this scanning procedure could be to a patient who is pregnant, as with any type of injectable radioactive substance. If the woman is pregnant, the radiologist must be notified; if the scan is cleared, the radiologist may use a lower dosage of indium.

Normal results

The scan should reveal no infection or pathology.

Abnormal results

The scan will reveal details, such as location, about an infection in the patient's body.

Carol A. Turkington

Induction of labor

Definition

Induction of labor involves using artificial means to assist the mother in delivering her baby.

Purpose

Labor is brought on, or induced, when the **pregnancy** has extended significantly beyond the expected delivery date and the mother shows no signs of going into labor. Generally, if the unborn baby is more than two weeks past due, labor will be induced. In most cases, a mother delivers her baby between 38–42 weeks of pregnancy. This usually means that labor is induced if the pregnancy has lasted more than 42 weeks. Labor is also induced if the mother is suffering from diseases (**preeclampsia**, chronic **hypertension**), if there is an Rh blood incompatibility between the baby and the mother, or if the mother or baby has a medical problem that requires delivery of the baby (like a premature rupture of the membranes).

Description

The uterus is the hollow female organ that supports the development and nourishment of the unborn baby during pregnancy. Sometimes labor is induced by the rupturing the amniotic membrane to release amniotic fluid. This is an attempt to mimic the normal process of “breaking water” that occurs early in the normal birth process. This method is sometimes enough stimulation to induce contractions in the mother's uterus. If labor fails to start, drugs are used.

Most labor is induced by using the drug Pitocin, a synthetic form of oxytocin. Oxytocin is a natural hormone produced in the body by the pituitary gland. During normal labor, oxytocin causes contractions. When labor does not occur naturally, the doctor may

KEY TERMS

Cesarean section—Delivery of a baby through an incision in the mother's abdomen instead of through the vagina; also called a C-section.

Preeclampsia—Hypertension (high blood pressure) experienced during pregnancy.

Rh blood incompatibility—A blood type problem between mother (who is Rh negative) and baby (who is Rh positive), making the immune system of the mother attack her unborn baby. During delivery of the first pregnancy, the mother's immune system becomes sensitive to the Rh positive blood of the baby. The mother's system may then attack later pregnancies and cause severe illness or death to those babies.

Vasoconstriction—Constriction of a blood vessel.

give the mother Pitocin to start the contractions. Pitocin makes the uterus contract with strength and force almost immediately. This drug is given through a vein in a steady flow that allows the doctor to control the amount the mother is given.

Sometimes vaginal gels are used to induce labor. Normally, the baby will pass through the opening of the uterus (the cervix) into the birth canal during delivery. Because of this, the cervix softens and begins to enlarge (dilate) during the early part of labor to make room for the baby to pass through. The cervix will continue to dilate, and the contractions will eventually push the baby out of the mother's body. When labor needs to be induced, the cervix is often small, hard, and not ready for the process. The doctor may need to prepare or "ripen" the cervix to induce labor. The hormone prostaglandin in a gel form may be applied high in the vagina to soften and dilate the cervix, making the area ready for labor. This may be enough to stimulate contractions on its own. More often, prostaglandin gel is used in conjunction with Pitocin.

If all attempts to induce labor fail, a **cesarean section** is performed.

Risks

Once labor has been induced, the unborn baby is monitored to guard against a reduction in its oxygen supply, or hypoxia. The drugs used to induce labor cause vasoconstriction, which can decrease blood supply to the unborn baby. Throughout the process, the

baby's heart rate is monitored by an electronic device placed on top of the mother's abdomen. The heart rate is one sign that the unborn baby is getting enough oxygen and remains healthy. Once the membranes are broken, prolonged labor may result in infection to either the newborn or the mother.

Normal results

Once labor is induced and the cervix has dilated, labor usually proceeds normally. When performed properly, induced labor is a safe procedure for both mother and baby.

Resources

BOOKS

Berkow, Robert, et al., editors. *The Merck Manual of Medical Information*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

John T. Lohr, PhD

Infant massage

Definition

Infant massage refers to **massage therapy** as specifically applied to infants. In most cases, oil or lotion is used as it would be on an adult subject by a trained and licensed massage therapist. Medical professionals caring for infants might also use massage techniques on infants born prematurely, on those with motor or gastrointestinal problems, or on those who have been exposed to **cocaine** in utero.

Purpose

Research from experiments conducted at the Touch Research Institutes at the University of Miami School of Medicine and Nova Southeastern University has been cited for the clinical benefits massage has on infants and children. Tiffany Field, Ph. D., director, noted that the research "... suggests that touch is as important to infants and children as eating and sleeping. Touch therapy triggers many physiological changes that help infants and children grow and develop. For example, massage can stimulate nerves in the brain which facilitate food absorption, resulting in faster weight gain. It also lowers level of **stress** hormones, resulting in improved immune function."

The benefits of infant massage include:

- Relaxation
- relief from stress
- interaction with adults
- stimulation of the nervous system

The results of several studies showed that infant massage alleviates the stress that newborns experience as a result of the enormous change that birth brings about in their lives after the 6–9 months they have spent in the womb. Both premature infants and full-term babies need the relaxation that comes from massaging and moving their limbs and muscles. In infants with **colic**, massage provides the relief necessary to disperse gas, ease muscle spasm, tone the digestive system and help it work efficiently. Some techniques even help bring relief from teething and emotional stress. The stimulation an infant receives from massage can aid circulation, strengthen muscles, help digestion, and relieve **constipation**. The bonding that occurs with massage between a parent and child enhances the entire process of bonding that comes with contact through all of the senses, including touch, voice, and sight. It affords a physical experience of quality time between the parents and the child as well as with any significant others in a baby's life.

Description

Origins

The practice of massaging infants dates back to ancient times, particularly in Asian and Pacific Island cultures; that is, massage was a component of the baby's regular bath routine among the Maoris and Hawaiians. Touch in these cultures is considered healthful both physically and spiritually. In the West, however, infant massage has received more attention in recent years in conjunction with the popularity of natural **childbirth** and midwife-assisted births. Dr. Frédéric Leboyer, a French physician who was one of the leaders of the natural childbirth movement, helped to popularize infant massage through his photojournalistic book on the Indian art of baby massage.

Infant massage was introduced formally into the United States in 1978 when Vimala Schneider McClure, a **yoga** practitioner who served in an orphanage in Northern India, developed a training program for instructors at the request of childbirth educators. An early research study by R. Rice in 1976 had showed that premature babies who were massaged surged

ahead in weight gain and neurological development over those who were not massaged. From McClure's training in India, her knowledge of Swedish massage and **reflexology**, along with her knowledge of yoga postures that she had already adapted for babies, she became the foremost authority on infant massage. In 1986 she founded the International Association of Infant Massage (IAIM), which has 27 chapters worldwide as of 2000.

Various techniques are used in infant massage, with the different strokes specific to a particular therapy. Special handling is used for treating a baby with gas and colic. Some of the strokes are known as "Indian milking," which is a gentle stroking of the child's legs; and the "twist and squeeze" stroke, a gentle squeeze of the muscles in the thigh and calf. The light "feather" strokes often employed in regular Swedish massage are applied at the end of a massage. The procedure is not unlike certain forms of adult massage, but with extra care taken for the fragility of the infant.

There are also specific Chinese techniques of pediatric massage, including massage of children with special needs. In China, these forms of massage can be given by medical professionals, but parents are often taught how to do the simpler forms for home treatment of their children.

Preparations

If lotions or oils are used, care is taken to ensure their safety on a baby's delicate skin. The most important consideration is to use vegetable oils rather than mineral oils, which can clog the pores in the skin. The oil that is used should be warmed in the caregiver's hands before applying it to the baby's skin. The environment in which the massage is given to an infant should be comfortably warm, and as calm and non-threatening as possible.

Precautions

Extreme caution is necessary when performing infant massage. Strokes are made with the greatest delicacy in order not to harm the infant in any way. Proper techniques are taught by licensed massage therapists ensuring that the infant is treated with appropriate physical touch. Anyone who is unfamiliar with handling a baby should receive appropriate instruction before beginning infant massage.

Side effects

No adverse side effects have been reported when infant massage is done properly after careful

instruction, or by a licensed massage therapist who specializes in infant care.

Research and general acceptance

In addition to the study already noted regarding touch therapy, a website devoted to infant massage lists research published as early as 1969, and cites hundreds of individual projects that have been conducted throughout the world focusing on infant massage. Many of the studies are related to the benefits of massage and touch for premature infants and others born with such risk factors as drug dependence. Conclusions regarding the benefits are overwhelmingly positive. The proliferation of therapists licensed in infant massage across the United States and worldwide indicates that infant massage is increasingly recognized as a legitimate health care treatment.

Resources

BOOKS

Cline, Kyle. *Chinese Massage for Infants and Children: Traditional Techniques for Alleviating Colic, Teething Pain, Earache, and Other Common Childhood Conditions*. Inner Traditions International, Limited, 1999.

Fan, Ya-Li. *Chinese Pediatric Massage Therapy: Traditional Techniques for Alleviating Colic, Colds, Earaches, and Other Common Childhood Conditions*, edited by Bob Flaws. Blue Poppy Enterprises, 1999.

Gordon, Jay, and Brenda Adderly. *Brighter Baby: Boosting Your Child's Intelligence, Health and Happiness through Infant Therapeutic Massage*. New York: Regnery Publishing, Inc. 1999.

ORGANIZATIONS

International Association of Infant Massage. P.O. Box 1045. Oak View, CA 93022.

International Institute of Infant Massage. 605 Bledsoe Rd. NW. Albuquerque, NM 87107. (505) 341-9381. Fax: (505) 341-9386. <<http://www.infantmassage.com>>.

Jane Spehar

Infant respiratory distress syndrome see

Respiratory distress syndrome

Infantile paralysis see **Polio**

Infarct avid imaging see **Technetium heart scan**

Infarction see **Stroke**

Infection control

Definition

Infection control refers to policies and procedures used to minimize the risk of spreading infections, especially in hospitals and human or animal health care facilities.

Purpose

The purpose of infection control is to reduce the occurrence of infectious diseases. These diseases are usually caused by bacteria or viruses and can be spread by human to human contact, animal to human contact, human contact with an infected surface, airborne transmission through tiny droplets of infectious agents suspended in the air, and, finally, by such common vehicles as food or water. Diseases that are spread from animals to humans are known as zoonoses; animals that carry disease agents from one host to another are known as vectors.

Infection control in hospitals and other health care settings

Infections contracted in hospitals are also called nosocomial infections. They occur in approximately 5% of all hospital patients. These infections result in increased time spent in the hospital and, in some cases, **death**. There are many reasons nosocomial infections are common, one of which is that many hospital patients have a weakened immune system which makes them more susceptible to infections. This weakened immune system can be caused either by the patient's diseases or by treatments given to the patient. Second, many medical procedures can increase the risk of infection by introducing infectious agents into the patient. Thirdly, many patients are admitted to hospitals because of infectious disease. These infectious agents can then be transferred from patient to patient by hospital workers or visitors.

Infection control has become a formal discipline in the United States since the 1950s, due to the spread of **staphylococcal infections** in hospitals. Because there is both the risk of health care providers acquiring infections themselves, and of them passing infections on to patients, the Centers for Disease Control and Prevention (CDC) established guidelines for infection control procedures. In addition to hospitals, infection control is important in nursing homes, clinics, child care centers, and restaurants, as well as in the home.

ELIZABETH LEE HAZEN (1885–1975)

Elizabeth Lee Hazen was born on August 24, 1885, in Rich, Mississippi. Hazen, born the middle of three children to Maggie (Harper) and William Edgar Hazen, was orphaned before she turned four. She and her sister went to live with their aunt and uncle shortly after her younger brother died. Hazen attended the Mississippi Industrial Institute and College at Columbus, receiving her B.S. degree in 1910. During college, Hazen became interested in science and she studied biology at Columbia University, earning her M.S. in 1917. After working in the U.S. Army laboratories during World War I, she returned to Columbia where she received her Ph.D. in microbiology in 1927. Following her work as an instructor at Columbia, Hazen accepted a position with the New York Department of Health where she researched bacterial diseases.

In 1948, Hazen and Rachel Brown began researching fungal infections found in humans due to antibiotic treatments and diseases. Some of the antibiotics they discovered did indeed kill the fungus; however, they also killed the test mice. Finally, Hazen located a microorganism on a farm in Virginia, and Brown's tests indicated that the microorganism produced two antibiotics, one of which proved effective for treating fungus and candidiasis in humans. Brown purified the antibiotic which was patented under the name *nystatin*. In 1954, the antibiotic became available in pill form. Hazen and Brown continued their research and discovered two other antibiotics. Hazen received numerous awards individually and with her research partner, Rachel Brown. Elizabeth Hazen passed away on June 24, 1975.

To lower the risk of nosocomial infections, the CDC began a national program of hospital inspection in 1970 known as the National Nosocomial Infections Surveillance system, or NNIS. The CDC reported that over 300 hospitals participate in the NNIS system as of the early 2000s. Data collected from the participating hospitals show that infection control programs can significantly improve patient safety, lower infection rates, and lower patient mortality.

Dental health care settings are similar to hospitals in that both personnel and equipment can transmit infection if proper safeguards are not observed. The CDC issued new guidelines in 2003 for the proper maintenance and sterilization of dental equipment, hand hygiene for dentists and dental hygienists, dental radiology, medications, and oral surgery, environmental infection control, and standards for dental laboratories.

Selected Infectious Diseases And Corresponding Treatment

Disease	Symptoms	Transmittal	Treatment
Chicken pox	Rash, low-grade fever	Person to person	None
Common cold/ Influenza	Runny nose, sore throat , cough , fever, headache , muscle aches	Person to person	None
Hepatitis	Jaundice, flu-like symptoms	Sexual contact with an infected person, contaminated blood, food, or water	None
Legionnaire's Disease	Flu symptoms, pneumonia, diarrhea , vomiting , kidney failure, respiratory failure	Air conditioning or water systems	Antibiotics
Measles	Skin rash, runny nose and eyes, fever, cough	Person to person	None
Meningitis	Neck pain , headache, pain caused by exposure to light, fever, nausea , drowsiness	Person to person	Antibiotics for bacterial meningitis, hospital care for viral meningitis
Mumps	Swelling of salivary glands	Person to person	Anti-inflammatory drugs
Ringworm	Skin rash	Contact with infected animal or person	Antifungal drugs applied topically
Tetanus	Lockjaw, other spasms	Soil infection of wounds	Antibiotics, antitoxins, muscle relaxers

The newest addition to the infection control specialist's resources is molecular typing, which speeds up the identification of a disease agent. Rapid identification in turn allows for timely containment of a disease outbreak.

Threat of emerging infectious diseases

Due to constant changes in our lifestyles and environments, new diseases are constantly appearing that people are susceptible to, making protection from the threat of infectious disease urgent. Many new contagious diseases have been identified in the past 30 years, such as **AIDS**, Ebola, and hantavirus. Increased travel between continents makes the worldwide spread of disease a bigger concern than it once

was. Additionally, many common infectious diseases have become resistant to known treatments.

The emergence of the **severe acute respiratory syndrome (SARS)** epidemic in Asia in February 2003 was a classic instance of an emerging disease that spread rapidly because of the increased frequency of international and intercontinental travel. In addition, the SARS outbreak demonstrated the vulnerability of hospitals and health care workers to emerging diseases. Clusters of cases within hospitals occurred in the early weeks of the epidemic when the disease had not yet been recognized and the first SARS patients were admitted without **isolation** precautions.

The SARS epidemic also raised a number of ethical and legal questions regarding current attitudes toward infection control.

Problems of antibiotic resistance

Because of the overuse of **antibiotics**, many bacteria have developed a resistance to common antibiotics. This means that newer antibiotics must continually be developed in order to treat an infection. However, further resistance seems to come about almost simultaneously. This indicates to many scientists that it might become more and more difficult to treat infectious diseases. The use of antibiotics outside of medicine also contributes to increased antibiotic resistance. One example of this is the use of antibiotics in animal husbandry. These negative trends can only be reversed by establishing a more rational use of antibiotics through treatment guidelines.

Bioterrorism

The events of September 11, 2001, and the **anthrax** scare that followed in October 2001 alerted public health officials as well as the general public to the possible use of infectious disease agents as weapons of terrorism. The Centers for Disease Control and Prevention (CDC) now has a list of topics and resources related to bioterrorism on its web site.

Description

The goals of infection control programs are: immunizing against preventable diseases, defining precautions that can prevent exposure to infectious agents, and restricting the exposure of health care workers to an infectious agent. An infection control practitioner is a specially trained professional, oftentimes a nurse, who oversees infection control programs.

Commonly recommended precautions to avoid and control the spread of infections include:

- Vaccinate people and pets against diseases for which a vaccine is available. As of 2003, the vaccines used against infectious diseases are very safe compared to most drugs.
- Wash hands often.
- Cook food thoroughly.
- Use antibiotics only as directed.
- See a doctor for infections that do not heal.
- Avoid areas with a lot of insects.
- Be cautious around wild or unfamiliar animals, or any animals that are unusually aggressive. Do not purchase exotic animals as pets.
- Do not engage in unprotected sex or in intravenous drug use.
- Find out about infectious diseases when you make travel plans. Travelers' advisories and adult **vaccination** recommendations are available on the CDC web site or by calling the CDC's telephone service at 404-332-4559.

Because of the higher risk of spreading infectious disease in a hospital setting, higher levels of precautions are taken there. Typically, health care workers wear gloves with all patients, since it is difficult to know whether a transmittable disease is present or not. Patients who have a known infectious disease are isolated to decrease the risk of transmitting the infectious agent to another person. Hospital workers who come in contact with infected patients must wear gloves and gowns to decrease the risk of carrying the infectious agent to other patients. All articles of equipment that are used in an isolation room are decontaminated before reuse. Patients who are immunocompromised may be put in protective isolation to decrease the risk of infectious agents being brought into their room. Any hospital worker with infections, including colds, are restricted from that room.

Hospital infections can also be transmitted through the air. Thus care must be taken when handling infected materials so as to decrease the numbers of infectious agents that become airborne. Special care should also be taken with hospital ventilation systems to prevent recirculation of contaminated air.

Resources

BOOKS

- Beers, Mark H., MD, and Robert Berkow, MD, editors.
 "Immunizations for Adults." Section 13, Chapter 152. In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

KEY TERMS

Acquired immune deficiency syndrome (AIDS)—A disease that weakens the body's immune system. It is also known as HIV infection.

Antibiotic—A substance, such as a drug, that can stop a bacteria from growing or destroy the bacteria.

Antibiotic resistance—The ability of infectious agents to change their biochemistry in such a way as to make an antibiotic no longer effective.

Bioterrorism—The intentional use of disease-causing microbes or other biologic agents to intimidate or terrorize a civilian population for political or military reasons.

Ebola—The disease caused by the newly described and very deadly Ebola virus found in Africa.

Epidemiology—The branch of medicine that deals with the transmission of infectious diseases in large populations and with detection of the sources and causes of epidemics.

Hantavirus—A group of arboviruses that cause hemorrhagic fever (characterized by sudden onset, fever, aching and bleeding in the internal organs).

Immunization—Immunity refers to the body's ability to protect itself from a certain disease after it has been exposed to that disease. Through immunization, also known as vaccination, a small amount of an infectious agent is injected into the body to stimulate the body to develop immunity.

Immunocompromized—Refers to the condition of having a weakened immune system. This can happen due to genetic factors, drugs, or disease.

Nosocomial infection—An infection acquired in a hospital setting.

Staphylococcal infection—An infection caused by the organism *Staphylococcus*. Infection by this agent is common and is often resistant to antibiotics.

Vector—An animal carrier that transfers an infectious organism from one host to another.

Zoonosis (plural, zoonoses)—Any disease of animals that can be transmitted to humans under natural conditions. Lyme disease, rabies, psittacosis (parrot fever), cat-scratch fever, and monkeypox are examples of zoonoses.

PERIODICALS

Ashford, D. A., R. M. Kaiser, M. E. Bales, et al. "Planning Against Biological Terrorism: Lessons from Outbreak

Investigations." *Emerging Infectious Diseases* 9 (May 2003): 515–519.

Gostin, L. O., R. Bayer, and A. L. Fairchild. "Ethical and Legal Challenges Posed by Severe Acute Respiratory Syndrome: Implications for the Control of Severe Infectious Disease Threats." *Journal of the American Medical Association* 290 (December 24, 2003): 3229–3237.

Ho, P. L., X. P. Tang, and W. H. Seto. "SARS: Hospital Infection Control and Admission Strategies." *Respirology* 8, Supplement (November 2003): S41–S45.

Jacobson, R. M., K. S. Zabel, and G. A. Poland. "The Overall Safety Profile of Currently Available Vaccines Directed Against Infectious Diseases." *Expert Opinion on Drug Safety* 2 (May 2003): 215–223.

Jarvis, W. R. "Benchmarking for Prevention: the Centers for Disease Control and Prevention's National Nosocomial Infections Surveillance (NNIS) System Experience." *Infection* 31, Supplement 2 (December 2003): 44–48.

Kohn, W. G., A. S. Collins, J. L. Cleveland, et al. "Guidelines for Infection Control in Dental Health-Care Settings—2003." *Morbidity and Mortality Weekly Reports: Reports and Recommendations* 52, RR-17 (December 19, 2003): 1–61.

Peng, P. W., D. T. Wong, D. Bevan, and M. Gardam. "Infection Control and Anesthesia: Lessons Learned from the Toronto SARS Outbreak." *Canadian Journal of Anaesthesiology* 50 (December 2003): 989–997.

Petrak, R. M., D. J. Sexton, M. L. Butera, et al. "The Value of an Infectious Diseases Specialist." *Clinical Infectious Diseases* 36 (April 15, 2003): 1013–1017.

Schulster, L., and R. Y. Chinn. "Guidelines for Environmental Infection Control in Health-Care Facilities. Recommendations of CDC and the Healthcare Infection Control Practices Advisory Committee (HICPAC)." *Morbidity and Mortality Recommendations and Reports* 52, RR-10 (June 6, 2003): 1–42.

Subramanian, D., J. A. Sandoe, V. Keer, and M. H. Wilcox. "Rapid Spread of Penicillin-Resistant *Streptococcus pneumoniae* Among High-Risk Hospital Inpatients and the Role of Molecular Typing in Outbreak Confirmation." *Journal of Hospital Infection* 54 (June 2003): 99–103.

ORGANIZATIONS

American College of Epidemiology. 1500 Sunday Drive, Suite 102, Raleigh, NC 27607. (919) 861-5573. <<http://www.acepidemiology.org>>.

American Public Health Association (APHA). 800 I Street NW, Washington, DC 20001-3710. (202) 777-APHA. <<http://www.apha.org>>.

American Veterinary Medical Association (AVMA). 1931 North Meacham Road, Suite 100, Schaumburg, IL 60173-4360. <<http://www.avma.org>>.

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

National Institute of Allergy and Infectious Diseases
(NIAID). 31 Center Drive, Room 7A50 MSC 2520,
Bethesda, MD, 20892. (301) 496-5717. <<http://www.niaid.nih.gov>>.

Cindy L. A. Jones, PhD
Rebecca J. Frey, PhD

Infectious arthritis

Definition

Infectious arthritis, which is sometimes called septic arthritis or pyogenic arthritis, is a serious infection of the joints characterized by **pain**, **fever**, occasional chills, inflammation and swelling in one or more joints, and loss of function in the affected joints. It is considered a medical emergency.

Description

Infectious arthritis can occur in any age group, including newborns and children. In adults, it usually affects the wrists or one of the patient's weight-bearing joints—most often the knee—although about 20% of adult patients have symptoms in more than one joint. Multiple joint infection is common in children and typically involves the shoulders, knees, and hips.

Some groups of patients are at greater risk for developing infectious arthritis. These high-risk groups include:

- Patients with chronic **rheumatoid arthritis**.
- Patients with certain systemic infections, including **gonorrhea** and HIV infection. Women and male homosexuals are at greater risk for gonorrheal arthritis than are male heterosexuals.
- Patients with certain types of **cancer**.
- IV drug abusers and alcoholics.
- Patients with artificial (prosthetic) joints.
- Patients with diabetes, sickle cell anemia, or **systemic lupus erythematosus** (SLE).
- Patients with recent joint injuries or surgery, or patients receiving medications injected directly into a joint.

Causes and symptoms

In general, infectious arthritis is caused by the spread of a bacterial, viral, or fungal infection through

the bloodstream to the joint. The disease agents may enter the joint directly from the outside as a result of an injury or a surgical procedure, or they may be carried to the joint by the blood from infections elsewhere in the body. The specific organisms vary somewhat according to age group. Newborns are most likely to acquire gonococcal infections of the joints from a mother with gonorrhea. Children may also acquire infectious arthritis from a hospital environment, often as a result of catheter placement. The organisms involved are usually either *Haemophilus influenzae* (in children under two years of age) or *Staphylococcus aureus*. In older children or adults, the infectious organisms include *Streptococcus pyogenes* and *Streptococcus viridans* as well as *Staphylococcus aureus*. *Staphylococcus epidermidis* is usually involved in joint infections related to surgery. Sexually active teenagers and adults frequently develop infectious arthritis from *Neisseria gonorrhoeae* infections. Older adults are often vulnerable to joint infections caused by gram-negative bacilli, including *Salmonella* and *Pseudomonas*.

Infectious arthritis often has a sudden onset, but symptoms sometimes develop over a period of three to 14 days. The symptoms include swelling in the infected joint and pain when the joint is moved. Infectious arthritis in the hip may be experienced as pain in the groin area that becomes much worse if the patient tries to walk. In 90% of cases, there is some leakage of tissue fluid into the affected joint. The joint is sore to the touch; it may or may not be warm to the touch, depending on how deep the infection lies within the joint. In most cases the patient will have fever and chills, although the fever may be only low-grade. Children sometimes develop **nausea and vomiting**.

Septic arthritis is considered a medical emergency because of the damage it causes to bone as well as cartilage, and its potential for creating **septic shock**, which is a potentially fatal condition. *Staphylococcus aureus* is capable of destroying cartilage in one or two days. Destruction of cartilage and bone in turn leads to **dislocations** of the joints and bones. If the infection is caused by bacteria, it can spread to the blood and surrounding tissues, causing abscesses or even blood **poisoning**. The most common complication of infectious arthritis is **osteoarthritis**.

Diagnosis

The diagnosis of infectious arthritis depends on a combination of laboratory testing with careful history-taking and **physical examination** of the affected

joint. It is important to keep in mind that infectious arthritis can coexist with other forms of arthritis, **gout**, **rheumatic fever**, **Lyme disease**, or other disorders that can cause a combination of joint pain and fever. In some cases, the doctor may consult a specialist in orthopedics or rheumatology to avoid misdiagnosis.

Patient history

The patient's history will tell the doctor whether he or she belongs to a high-risk group for infectious arthritis. Sudden onset of joint pain is also important information.

Physical examination

The doctor will examine the affected joint for swelling, soreness, warmth, and other signs of infection. Location is sometimes a clue to diagnosis; infection of an unusual joint, such as the joints between the breastbone and collarbone, or the pelvic joints, often occurs in drug abusers.

Laboratory tests

Laboratory testing is necessary to confirm the diagnosis of infectious arthritis. The doctor will perform an arthrocentesis, which is a procedure that involves withdrawing a sample of synovial fluid (SF) from the joint with a needle and syringe. SF is a lubricating fluid secreted by tissues surrounding the joints. Patients should be warned that arthrocentesis is a painful procedure. The fluid sample is sent for culture in the sealed syringe. SF from infected joints is usually streaked with pus or looks cloudy and watery. Cell counts usually indicate a high level of white cells; a level higher than 100,000 cells/mm³ or a neutrophil proportion greater than 90% suggests septic arthritis. A Gram's stain of the culture obtained from the SF is usually positive for the specific disease organism.

Doctors sometimes order a biopsy of the synovial tissue near the joint if the fluid sample is negative. Cultures of other body fluids, such as urine, blood, or cervical mucus, may be taken in addition to the SF culture.

Diagnostic imaging

Diagnostic imaging is not helpful in the early stages of infectious arthritis. Destruction of bone or cartilage does not appear on x rays until 10–14 days after the onset of symptoms. Imaging studies are sometimes useful if the infection is in a deep-seated joint.

Treatment

Infectious arthritis usually requires several days of treatment in a hospital, with follow-up medication and physical therapy lasting several weeks or months.

Medications

Because of the possibility of serious damage to the joint or other complications if treatment is delayed, the patient will be started on intravenous **antibiotics** before the specific organism is identified. After the disease organism has been identified, the doctor may give the patient a drug that targets the specific bacterium or virus. **Nonsteroidal anti-inflammatory drugs** are usually given for viral infections.

Intravenous antibiotics are given for about two weeks, or until the inflammation has disappeared. The patient may then be given a two- to four-week course of oral antibiotics.

Surgery

In some cases, surgery is necessary to drain fluid from the infected joint. Patients who need surgical drainage include those who have not responded to antibiotic treatment, those with infections of the hip or other joints that are difficult to reach with arthrocentesis, and those with joint infections related to gunshot or other penetrating **wounds**.

Patients with severe damage to bone or cartilage may need **reconstructive surgery**, but it cannot be performed until the infection is completely gone.

Monitoring and supportive treatment

Infectious arthritis requires careful monitoring while the patient is in the hospital. The doctor will drain the joint on a daily basis and remove a small sample of fluid for culture to check the patient's response to the antibiotic.

Infectious arthritis often causes intense pain. Patients are given medications to relieve pain, together with hot compresses or ice packs on the affected joint. In some cases the patient's arm or leg is put in a splint to protect the sore joint from accidental movement. Recovery can be speeded up, however, if the patient practices range-of-motion exercises to the extent that the pain allows.

Prognosis

The prognosis depends on prompt treatment with antibiotics and drainage of the infected

KEY TERMS

Arthrocentesis—A procedure in which the doctor inserts a needle into the patient's joint to withdraw fluid for diagnostic testing or to drain infected fluid from the joint.

Pyogenic arthritis—Another name for infectious arthritis. Pyogenic means that pus is formed during the disease process.

Sepsis—Invasion of the body by disease organisms or their toxins. Generalized sepsis can lead to shock and eventual death.

Septic arthritis—Another name for infectious arthritis.

Synovial fluid (SF)—A fluid secreted by tissues surrounding the joints that lubricates the joints.

joint. About 70% of patients will recover without permanent joint damage. However, many patients will develop osteoarthritis or deformed joints. Children with infected hip joints sometimes suffer damage to the growth plate. If treatment is delayed, infectious arthritis has a mortality rate between 5% and 30% due to septic **shock** and **respiratory failure**.

Prevention

Some cases of infectious arthritis are preventable by lifestyle choices. These include avoidance of self-injected drugs; sexual abstinence or monogamous relationships; and prompt testing and treatment for suspected cases of gonorrhea. Patients receiving corticosteroid injections into the joints for osteoarthritis may want to weigh this treatment method against the increased risk of infectious arthritis.

Resources

BOOKS

Hellman, David B., "Arthritis & Musculoskeletal Disorders." In *Current Medical Diagnosis and Treatment*, 1998, edited by Stephen McPhee, et al., 37th ed. Stamford: Appleton & Lange, 1997.

Rebecca J. Frey, PhD

Infectious hepatitis see **Hepatitis A**

Infectious mononucleosis

Definition

Infectious mononucleosis is a contagious illness caused by the Epstein-Barr virus that can affect the liver, lymph nodes, and oral cavity. While mononucleosis is not usually a serious disease, its primary symptoms of **fatigue** and lack of energy can linger for several months.

Description

Infectious mononucleosis, frequently called "mono" or the "kissing disease," is caused by the Epstein-Barr virus (EBV) found in saliva and mucus. The virus affects a type of white blood cell called the B lymphocyte producing characteristic atypical lymphocytes that may be useful in the diagnosis of the disease.

While anyone, even young children, can develop mononucleosis, it occurs most often in young adults between the ages of 15 and 35 and is especially common in teenagers. The mononucleosis infection rate among college students who have not previously been exposed to EBV has been estimated to be about 15%. In younger children, the illness may not be recognized.

The disease typically runs its course in four to six weeks in people with normally functioning immune systems. People with weakened or suppressed immune systems, such as **AIDS** patients or those who have had organ transplants, are particularly vulnerable to the potentially serious complications of infectious mononucleosis.

Causes and symptoms

The EBV that causes mononucleosis is related to a group of herpes viruses, including those that cause cold sores, chicken pox, and **shingles**. Most people are exposed to EBV at some point during their lives. Mononucleosis is most commonly spread by contact with virus-infected saliva through coughing, sneezing, kissing, or sharing drinking glasses or eating utensils.

In addition to general weakness and fatigue, symptoms of mononucleosis may include any or all of the following:

- **Sore throat** and/or swollen tonsils
- Fever and chills
- Nausea and **vomiting**, or decreased appetite
- Swollen lymph nodes in the neck and armpits
- Headaches or joint **pain**

- Enlarged spleen
- Jaundice
- Skin rash.

Complications that can occur with mononucleosis include a temporarily enlarged spleen or inflamed liver. In rare instances, the spleen may rupture, producing sharp pain on the left side of the abdomen, a symptom that warrants immediate medical attention. Additional symptoms of a ruptured spleen include light headedness, rapidly beating heart, and difficulty breathing. Other rare, but potentially life-threatening, complications may involve the heart or brain. The infection may also cause significant destruction of the body's red blood cells or platelets.

Symptoms do not usually appear until four to seven weeks after exposure to EBV. An infected person can be contagious during this incubation time period and for as many as five months after the disappearance of symptoms. Also, the virus will be excreted in the saliva intermittently for the rest of their lives, although the individual will experience no symptoms. Contrary to popular belief, the EBV is not highly contagious. As a result, individuals living in a household or college dormitory with someone who has mononucleosis have a very small risk of being infected unless they have direct contact with the person's saliva.

Diagnosis

If symptoms associated with a cold persist longer than two weeks, mononucleosis is a possibility; however, a variety of other conditions can produce similar symptoms. If mononucleosis is suspected, a physician will typically conduct a **physical examination**, including a "Monospot" antibody blood test that can indicate the presence of proteins or antibodies produced in response to infection with the EBV. These antibodies may not be detectable, however, until the second or third weeks of the illness. Occasionally, when this test is inconclusive, other blood tests may be conducted.

Treatment

The most effective treatment for infectious mononucleosis is rest and a gradual return to regular activities. Individuals with mild cases may not require bed rest but should limit their activities. Any strenuous activity, athletic endeavors, or heavy lifting should be avoided until the symptoms completely subside, since excessive activity may cause the spleen to rupture.

KEY TERMS

Antibody—A specific protein produced by the immune system in response to a specific foreign protein or particle called an antigen.

Herpes viruses—A group of viruses that can cause cold sores, shingles, chicken pox, and congenital abnormalities. The Epstein-Barr virus which causes mononucleosis belongs to this group of viruses.

Reye's syndrome—A very serious, rare disease, most common in children, which involves an upper respiratory tract infection followed by brain and liver damage.

The sore throat and **dehydration** that usually accompany mononucleosis may be relieved by drinking water and fruit juices. Gargling salt water or taking throat lozenges may also relieve discomfort. In addition, taking over-the-counter medications, such as **acetaminophen** or ibuprofen, may relieve symptoms, but **aspirin** should be avoided because mononucleosis has been associated with **Reye's syndrome**, a serious illness aggravated by aspirin.

While **antibiotics** do not affect EBV, the sore throat accompanying mononucleosis can be complicated by a streptococcal infection, which can be treated with antibiotics. Cortisone anti-inflammatory medications are also occasionally prescribed for the treatment of severely swollen tonsils or throat tissues.

Prognosis

While the severity and length of illness varies, most people diagnosed with mononucleosis will be able to return to their normal daily routines within two to three weeks, particularly if they rest during this time period. It may take two to three months before a person's usual energy levels return. One of the most common problems in treating mononucleosis, particularly in teenagers, is that people return to their usual activities too quickly and then experience a relapse of symptoms. Once the disease has completely run its course, the person cannot be re-infected.

Prevention

Although there is no way to avoid becoming infected with EBV, paying general attention to good hygiene and avoiding sharing beverage glasses or

having close contact with people who have mononucleosis or cold symptoms can help prevent infection.

Resources

ORGANIZATIONS

National Institute of Allergy and Infectious Disease.
Building 31, Room 7A-50, 31 Center Drive MSC 2520,
Bethesda, MD 20892-2520. (301) 496-5717.
< <http://www.niaid.nih.gov/default.htm> >.

OTHER

“Communicable Disease Fact Sheet.” New York State
Department of Health.
“Mononucleosis: A Tiresome Disease.” *Mayo Clinic Online*.
< <http://www.mayo.ivl.com/mayo/9701/htm/mono.htm> >.

Susan J. Montgomery

Infertility

Definition

Infertility is the failure of a couple to conceive a **pregnancy** after trying to do so for at least one full year. In primary infertility, pregnancy has never occurred. In secondary infertility, one or both members of the couple have previously conceived, but are unable to conceive again after a full year of trying.

Description

Currently, in the United States, about 20% of couples struggle with infertility at any given time. Infertility has increased as a problem over the last 30 years. Some studies pin the blame for this increase on social phenomena, including the tendency for marriage to occur at a later age, which means that couples are trying to start families at a later age. It is well known that fertility in women decreases with increasing age, as illustrated by the following statistics:

- Infertility in married women ages 16–20 = 4.5%
- Infertility in married women ages 35–40 = 31.8%
- Infertility in married women over the age of 40 = 70%.

Today, individuals often have multiple sexual partners before they marry and try to have children. This increase in numbers of sexual partners has led to an increase in **sexually transmitted diseases**. Scarring from these infections, especially from **pelvic inflammatory disease** (a serious infection of the female

reproductive organs, most commonly caused by **gonorrhea**) seems to be in part responsible for the increase in infertility noted. Furthermore, use of some forms of a contraceptive called the intrauterine device (**IUD**) contributed to an increased rate of pelvic inflammatory disease, with subsequent scarring. However, newer IUDs do not lead to this increased rate of infection.

To understand issues of infertility, it is first necessary to understand the basics of human reproduction. Fertilization occurs when a sperm from the male merges with an egg (ovum) from the female, creating a zygote that contains genetic material (DNA) from both the father and the mother. If pregnancy is then established, the zygote will develop into an embryo, then a fetus, and ultimately a baby will be born.

The male contribution to fertilization and the establishment of pregnancy is the sperm. Sperm are small cells that carry the father's genetic material. This genetic material is contained within the oval head of the sperm. The sperm are mixed into a fluid called semen, which is discharged from the penis during sexual intercourse. The whip-like tail of the sperm allows the sperm to swim up the female reproductive tract, in search of the egg it will try to fertilize.

The female makes many contributions to fertilization and the establishment of pregnancy. The ovum is the cell that carries the mother's genetic material. These ova develop within the ovaries. Once a month, a single mature ovum is produced, and leaves the ovary in a process called ovulation. This ovum enters a tube leading to the uterus (the fallopian tube). The ovum needs to meet up with the sperm in the fallopian tube if fertilization is to occur.

When fertilization occurs, the resulting cell (which now contains genetic material from both the mother and the father) is called the zygote. This single cell will divide into multiple other cells within the fallopian tube, and the resulting cluster of cells (called a blastocyst) will then move into the womb (uterus). The uterine lining (endometrium) has been preparing itself to receive a pregnancy by growing thicker. If the blastocyst successfully reaches the inside of the uterus and attaches itself to the wall of the uterus, then implantation and pregnancy have been achieved.

Causes and symptoms

Unlike most medical problems, infertility is an issue requiring the careful evaluation of two separate individuals, as well as an evaluation of their

interactions with each other. In about 3–4% of couples, no cause for their infertility will be discovered. About 40% of the time, the root of the couple's infertility is due to a problem with the male partner; about 40% of the time, the root of the infertility is due to the female partner; and about 20% of the time, there are fertility problems with both the man and the woman. Recently, a study in Great Britain reported that **smoking** adds to infertility problems for both men and women. In addition, men and women who smoke are less likely to respond to infertility treatment.

The main factors involved in causing infertility, listing from the most to the least common, include:

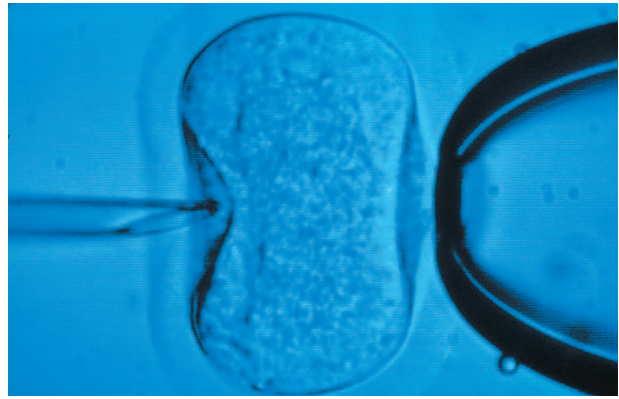
- Male problems: 35%
- Ovulation problems: 20%
- Tubal problems: 20%
- **Endometriosis**: 10%
- Cervical factors: 5%.

Male factors

Male infertility can be caused by a number of different characteristics of the sperm. To check for these characteristics, a sample of semen is obtained and examined under the microscope (**semen analysis**). Four basic characteristics are usually evaluated:

- Sperm count refers to the number of sperm present in a semen sample. The normal number of sperm present in just one milliliter (ml) of semen is more than 20 million. An individual with only 5–20 million sperm per ml of semen is considered subfertile, an individual with less than 5 million sperm per ml of semen is considered infertile.
- Sperm are also examined to see how well they swim (sperm motility) and to be sure that most have normal structure.
- Not all sperm within a specimen of semen will be perfectly normal. Some may be immature, and some may have abnormalities of the head or tail. A normal semen sample will contain no more than 25% abnormal forms of sperm.
- Volume of the semen sample is important. An abnormal amount of semen could affect the ability of the sperm to successfully fertilize an ovum.

Another test can be performed to evaluate the ability of the sperm to penetrate the outer coat of the ovum. This is done by observing whether sperm in a semen sample can penetrate the outer coat of a guinea pig ovum; fertilization cannot occur, of course, but this test is useful in predicting the ability of the individual's sperm to penetrate a human ovum.



A microscopic image of a needle (left) injecting sperm cells directly into a human egg (center). The broad object at right is a pipette used to hold the ovum steady. (Phototake NYC. Reproduced by permission.)

Any number of conditions result in abnormal findings in the semen analysis. Men can be born with testicles that have not descended properly from the abdominal cavity (where testicles develop originally) into the scrotal sac, or may be born with only one instead of the normal two testicles. Testicle size can be smaller than normal. Past infection (including **mumps**) can affect testicular function, as can a past injury. The presence of abnormally large veins (varicocele) in the testicles can increase testicular temperature, which decreases sperm count. History of having been exposed to various toxins, drug use, excess alcohol use, use of anabolic steroids, certain medications, diabetes, thyroid problems, or other endocrine disturbances can have direct effects on the formation of sperm (spermatogenesis). Problems with the male anatomy can cause sperm to be ejaculated not out of the penis, but into the bladder, and scarring from past infections can interfere with ejaculation.

Treatment of male infertility includes addressing known reversible factors first; for example, discontinuing any medication known to have an effect on spermatogenesis or ejaculation, as well as decreasing alcohol intake, and treating thyroid or other endocrine disease. Varicoceles can be treated surgically. Testosterone in low doses can improve sperm motility.

Other treatments of male infertility include collecting semen samples from multiple ejaculations, after which the semen is put through a process that allows the most motile sperm to be sorted out. These motile sperm are pooled together to create a concentrate that can be deposited into the female partner's uterus at a time that coincides with ovulation. In cases

where the male partner's sperm is proven to be absolutely unable to cause pregnancy in the female partner, and with the consent of both partners, donor sperm may be used for this process. Depositing the male partner's sperm or donor sperm by mechanical means into the female partner are both forms of artificial insemination.

Ovulatory problems

The first step in diagnosing ovulatory problems is to make sure that an ovum is being produced each month. A woman's morning body temperature is slightly higher around the time of ovulation. A woman can measure and record her temperatures daily and a chart can be drawn to show whether or not ovulation has occurred. Luteinizing hormone (LH) is released just before ovulation. A simple urine test can be done to check if LH has been released around the time that ovulation is expected.

Treatment of ovulatory problems depends on the cause. If a thyroid or pituitary problem is responsible, simply treating that problem can restore fertility. (The thyroid and pituitary glands release hormones that also are involved in regulating a woman's menstrual cycle.) Medication can also be used to stimulate fertility. The most commonly used of these are called Clomid and Pergonal. These drugs increase the risk of multiple births (twins, triplets, etc.). Other possible medications include gonadotropin medications, which are injected medications made up of hormones produced in the pituitary glands. They may directly stimulate the ovaries to produce eggs. Follicle stimulating hormone (FSH) has a 95% chance of simulating ovulation in women with an ovulatory problem. However, its use does not guarantee a successful pregnancy and may lead to multiple pregnancies.

Pelvic adhesions and endometriosis

Pelvic **adhesions** and endometriosis can cause infertility by preventing the sperm from reaching the egg or interfering with fertilization.

Pelvic adhesions are fibrous **scars**. These scars can be the result of past infections, such as pelvic inflammatory disease, or infections following abortions or prior births. Previous surgeries can also leave behind scarring.

Endometriosis may lead to pelvic adhesions. Endometriosis is the abnormal location of uterine tissue outside of the uterus. When uterine tissue is planted elsewhere in the pelvis, it still bleeds on a monthly basis with the start of the normal menstrual period. This leads

to irritation within the pelvis around the site of this abnormal tissue and bleeding, and may cause scarring.

Pelvic adhesions cause infertility by blocking the fallopian tubes. The ovum may be prevented from traveling down the fallopian tube from the ovary or the sperm may be prevented from traveling up the fallopian tube from the uterus.

A hysterosalpingogram (HSG) can show if the fallopian tubes are blocked. This is an x-ray exam that tests whether dye material can travel through the patient's fallopian tubes. A few women become pregnant following this x-ray exam. It is thought that the dye material in some way helps flush out the tubes, decreasing any existing obstruction. Scarring also can be diagnosed by examining the pelvic area through the use of a scope that can be inserted into the abdomen through a tiny incision made near the naval. This scoping technique is called **laparoscopy**.

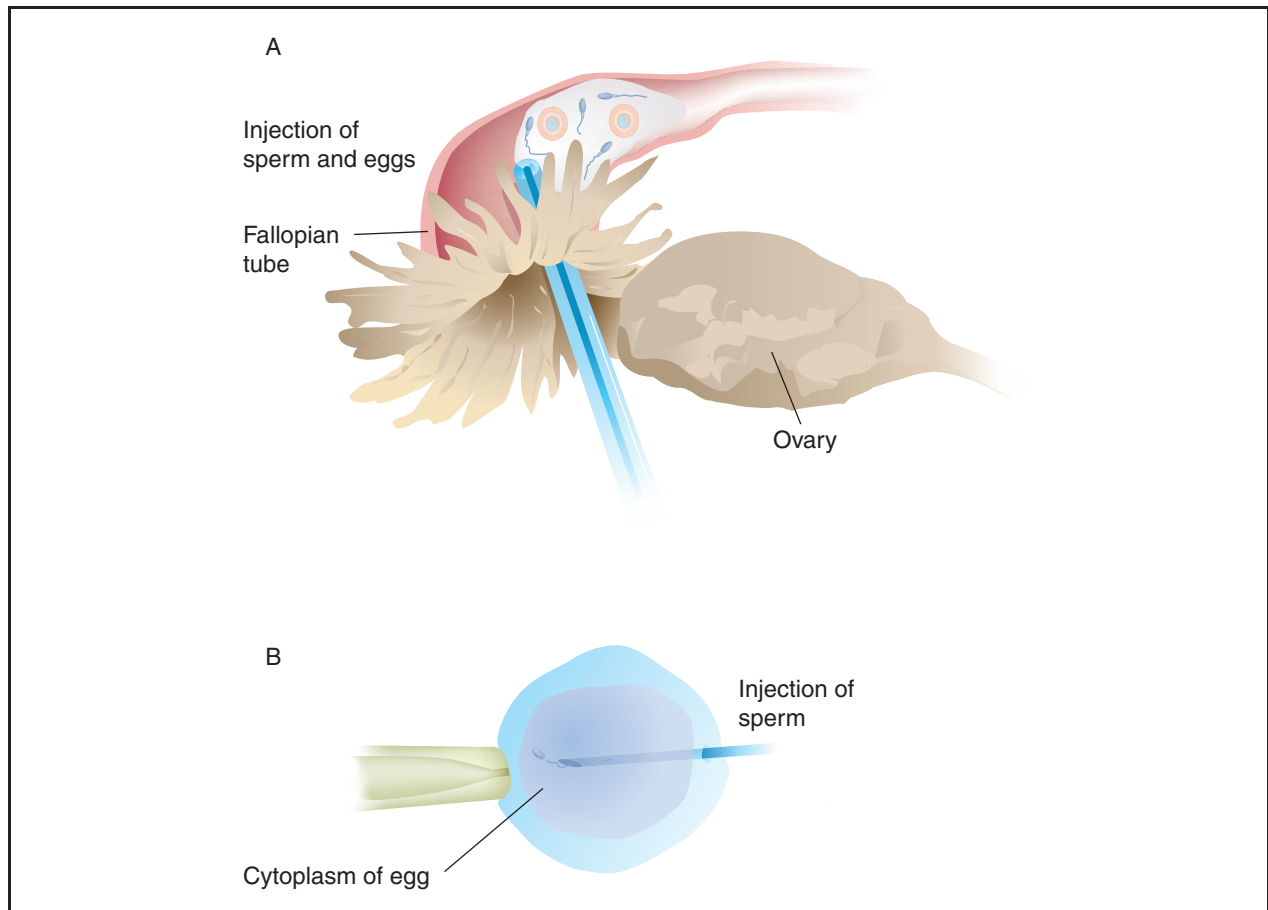
Pelvic adhesions can be treated during laparoscopy. The adhesions are cut using special instruments. Endometriosis can be treated with certain medications, but may also require surgery to repair any obstruction caused by adhesions.

Cervical factors

The cervix is the opening from the vagina into the uterus through which the sperm must pass. Mucus produced by the cervix helps to transport the sperm into the uterus. Injury to the cervix or scarring of the cervix after surgery or infection can result in a smaller than normal cervical opening, making it difficult for the sperm to enter. Injury or infection can also decrease the number of glands in the cervix, leading to a smaller amount of cervical mucus. In other situations, the mucus produced is the wrong consistency (perhaps too thick) to allow sperm to travel through. In addition, some women produce antibodies (immune cells) that are specifically directed to identify sperm as foreign invaders and to kill them.

Cervical mucus can be examined under a microscope to diagnose whether cervical factors are contributing to infertility. The interaction of a live sperm sample from the male partner and a sample of cervical mucus from the female partner can also be examined. This procedure is called a post-coital test.

Treatment of cervical factors includes **antibiotics** in the case of an infection, steroids to decrease production of anti-sperm antibodies, and artificial insemination techniques to completely bypass the cervical mucus.



A. An egg and sperm are injected into the fallopian tube to encourage natural fertilization in a procedure called gamete intrafallopian transfer (GIFT). **B.** An alternative to GIFT is the injection of sperm directly into an egg using microscopic needles. (Illustration by Argosy Inc.)

Treatment

Assisted reproductive techniques include **in vitro fertilization** (IVF), gamete intrafallopian transfer (GIFT), and zygote intrafallopian tube transfer (ZIFT). These are usually used after other techniques to treat infertility have failed.

In vitro fertilization involves the use of a drug to induce the simultaneous release of many eggs from the female's ovaries, which are retrieved surgically. Meanwhile, several semen samples are obtained from the male partner, and a sperm concentrate is prepared. The ova and sperm are then combined in a laboratory, where several of the ova may be fertilized. Cell division is allowed to take place up to the embryo stage. While this takes place, the female may be given drugs to ensure that her uterus is ready to receive an embryo. Three or four of the embryos are transferred to the female's uterus, and the wait begins to see if any or all of them implant and result in an actual pregnancy.

Success rates of IVF are still rather low. Most centers report pregnancy rates between 10–20%. Since most IVF procedures put more than one embryo into the uterus, the chance for a multiple birth (twins or more) is greatly increased in couples undergoing IVF.

GIFT involves retrieval of both multiple ova and semen, and the mechanical placement of both within the female partner's fallopian tubes, where one hopes that fertilization will occur. ZIFT involves the same retrieval of ova and semen, and fertilization and growth in the laboratory up to the zygote stage, at which point the zygotes are placed in the fallopian tubes. Both GIFT and ZIFT seem to have higher success rates than IVF.

Prognosis

It is very hard to obtain statistics regarding the prognosis of infertility because many different problems may exist within an individual or couple trying to conceive. In general, it is believed that of all couples who

KEY TERMS

Blastocyst—A cluster of cells representing multiple cell divisions that have occurred in the fallopian tube after successful fertilization of an ovum by a sperm. This is the developmental form which must leave the fallopian tube, enter the uterus, and implant itself in the uterus to achieve actual pregnancy.

Cervix—The opening from the vagina, which leads into the uterus.

Embryo—The stage of development of a baby between the second and eighth weeks after conception.

Endometrium—The lining of the uterus.

Fallopian tube—The tube leading from the ovary into the uterus. Just as there are two ovaries, there are two Fallopian tubes.

Fetus—A baby developing in the uterus from the third month to birth.

Ovary—The female organ in which eggs (ova) are stored and mature.

Ovum (plural: ova)—The reproductive cell of the female, which contains genetic information and participates in the act of fertilization. Also popularly called the egg.

Semen—The fluid that contains sperm, which is ejaculated by the male.

Sperm—The reproductive cell of the male, which contains genetic information and participates in the act of fertilization of an ovum.

Spermatogenesis—The process by which sperm develop to become mature sperm, capable of fertilizing an ovum.

Zygote—The result of the sperm successfully fertilizing the ovum. The zygote is a single cell that contains the genetic material of both the mother and the father.

undergo a complete evaluation of infertility followed by treatment, about half will ultimately have a successful pregnancy. Of those couples who do not choose to undergo evaluation or treatment, about 5% will go on to conceive after a year or more of infertility.

Resources

BOOKS

Martin, Mary C. "Infertility." In *Current Obstetric and Gynecologic Diagnosis and Treatment*, edited by Alan H. Cecherney and Martin L. Pernoll. Norwalk, CT: 1994.

PERIODICALS

"Infertility; Treatment." *NWHRC Health Center* March 10, 2004.

Kmietowicz, Zosia. "Smoking is Causing Impotence, Miscarriages, and Infertility." *British Medical Journal* February 14, 2004: 364.

ORGANIZATIONS

American Society for Reproductive Medicine. 1209 Montgomery Highway, Birmingham, AL 35216-2809. (205) 978-5000. <<http://www.asrm.com>>.

International Center for Infertility Information Dissemination. <<http://www.inciid.org>>.

Rosalyn Carson-DeWitt, MD
Teresa G. Odle

Infertility drugs

Definition

Infertility drugs are medicines that help bring about **pregnancy**.

Purpose

Infertility is the inability of a man and woman to achieve pregnancy after at least a year of having regular sexual intercourse without any type of birth control. There are many possible reasons for infertility, and finding the most effective treatment for a couple may involve many tests to find the problem. For pregnancy to occur, the woman's reproductive system must release eggs regularly—a process called ovulation. The man must produce healthy sperm that are able to reach and unite with an egg. And once an egg is fertilized, it must travel to the woman's uterus (womb), become implanted and remain there to be nourished.

If a couple is infertile because the woman is not ovulating, infertility drugs may be prescribed to stimulate ovulation. The first step usually is to try a drug such as clomiphene. If that does not work, human chorionic gonadotropin (HCG) may be tried, usually in combination with other infertility drugs.

Clomiphene and HCG may also be used to treat other conditions in both males and females.

Description

Clomiphene (Clomid, Serophene) comes in tablet form and is available only with a physician's

prescription. Human chorionic gonadotropin is given as an injection, only under a physician's supervision.

Clomiphene citrate is used to increase the natural production of the hormones that stimulate ovulation in otherwise healthy women. When clomiphene is administered, the body produces higher levels of luteinizing hormone (LH), follicle stimulating hormone (FSH), and gonadotropins. These hormones induce ovulation.

Human chorionic gonadotropin (hCG) is sold under many brand names including Gonic, Pregnyl and Profasi. This hormone stimulates the gonads in both men and women. In men, hCG increases androgen production. In women, it increases the levels of progesterone. Human chorionic gonadotropin can help stimulate ovulation in women.

Although some people believe that hCG can help lose weight, there is no evidence that this hormone offers any benefit in weight loss programs. It should not be used for this purpose.

A number of other natural and synthetic hormones are used to induce ovulation. Urofollitropins (Fertinex) is a concentrated preparation of human hormones, while follitropin alfa (Gonal-F) and follitropin beta (Follistim) are human FSH preparations of recombinant DNA origin. Developments in this field are continuous. For example, in June 2004, the U.S. Food and Drug Administration (FDA) approved a follitropin beta injection in individualized doses for women to self-inject.

Menotropins (Pergonal, Humegon, Repronex) are given with human chorionic gonadotropin to stimulate ovulation in women and sperm production in men.

Recommended dosage

The dosage may be different for different patients. The physician who prescribed the drug or the pharmacist who filled the prescription will recommend the correct dosage.

Clomiphene must be taken at certain times during the menstrual cycle and patients should follow directions exactly.

Precautions

Seeing a physician regularly while taking infertility drugs is important.

Treatment with infertility drugs increases the chance of multiple births. Although this may seem like a good thing to couples who want children very badly,

multiple fetuses can cause problems during pregnancy and delivery and can even threaten the babies' survival.

Having intercourse at the proper time in the woman's menstrual cycle helps increase the chance of pregnancy. The physician may recommend using an ovulation prediction test kit to help determine the best times for intercourse.

Some people feel dizzy or lightheaded, or less alert when using clomiphene. The medicine may also cause blurred vision and other vision changes. Anyone who takes clomiphene should not drive, use machines or do anything else that might be dangerous until they have found out how the drugs affect them.

Questions remain about the safety of long-term treatment with clomiphene. Women should not have more than six courses of treatment with this drug and should ask their physicians for the most up-to-date information about its use.

Special conditions

People who have certain medical conditions or who are taking certain other medicines may have problems if they take infertility drugs. Before taking these drugs, patients should tell the physician about any of these conditions:

ALLERGIES. Anyone who has had unusual reactions to infertility drugs in the past should let his or her physician know before taking the drugs again. The physician should also be told about any **allergies** to foods, dyes, preservatives, or other substances.

PREGNANCY. Clomiphene may cause **birth defects** if taken during pregnancy. Women who think they have become pregnant while taking clomiphene should stop taking the medicine immediately and check with their physicians.

OTHER MEDICAL CONDITIONS. Infertility drugs may make some medical conditions worse. Before using infertility drugs, people with any of these medical problems should make sure their physicians are aware of their conditions:

- Endometriosis
- Fibroid tumors of the uterus
- Unusual vaginal bleeding
- Ovarian cyst
- Enlarged ovaries
- Inflamed veins caused by **blood clots**
- Liver disease, now or in the past
- Depression.

KEY TERMS

Endometriosis—A condition in which tissue like that normally found in the lining of the uterus is present outside the uterus. The condition often causes pain and bleeding.

Fetus—A developing baby inside the womb.

Fibroid tumor—A noncancerous tumor formed of fibrous tissue.

Ovary—A reproductive organ in females that produces eggs and hormones.

USE OF CERTAIN MEDICINES. Taking infertility drugs with certain other medicines may affect the way the drugs work or may increase the chance of side effects.

Side effects

When used in low doses for a short time, clomiphene and HCG rarely cause side effects. However, anyone who has stomach or pelvic **pain** or bloating while taking either medicine should check with a physician immediately. Infertility drugs may also cause less serious symptoms such as hot flashes, breast tenderness or swelling, heavy menstrual periods, bleeding between menstrual periods, **nausea** or **vomiting**, **dizziness**, lightheadedness, irritability, nervousness, restlessness, **headache**, tiredness, sleep problems, or depression. These problems usually go away as the body adjusts to the drug and do not require medical treatment unless they continue or they interfere with normal activities.

Other side effects are possible. Anyone who has unusual symptoms after taking infertility drugs should get in touch with a physician.

Interactions

Infertility drugs may interact with other medicines. When this happens, the effects of one or both of the drugs may change or the risk of side effects may be greater. Anyone who takes infertility drugs should let the physician know all other medicines she is taking.

Resources

PERIODICALS

Georgi, Kristen. "News from the FDA." *Patient Care* June 2004: 10.

Nancy Ross-Flanigan
Teresa G. Odle

Infertility therapies

Definition

Infertility is the inability of a man and a woman to conceive a child through sexual intercourse. There are many possible reasons for the problem, which can involve the man, the woman, or both partners. Various treatments are available that enable a woman to become pregnant; the correct one will depend on the specific cause of the infertility.

Purpose

Infertility treatment is aimed at enabling a woman to have a baby by treating the man, the woman, or both partners. During normal conception of a child, the man's sperm will travel to the woman's fallopian tubes, where, if conditions are right, it will encounter an egg that has been released from the ovary. The sperm will fertilize the egg, which will enter the uterus where it implants and begins to divide, forming an embryo. The embryo will develop during **pregnancy** into a baby.

Infertility treatment attempts to correct or compensate for any abnormalities in this process that prevent the fertilization of an egg or development of an embryo.

Precautions

It is important for a couple contemplating infertility treatment to examine their own ideas and feelings about the process and consider ethical objections before the woman becomes pregnant from such treatment.

Description

About 90% of women who are trying to get pregnant and use no birth control will do so within one year. If, after one year of having frequent sexual intercourse with no **contraception** a couple has not conceived, they should seek the advice of a physician. Tests can be performed to look for possible infertility problems.

Treating an underlying infection or illness is the first step in infertility treatment. The physician may also suggest improving general health, dietary changes, reducing **stress**, and counseling.

Treatment

Low sperm count treatments

The most common cause of male infertility is failure to produce enough healthy sperm. For

fertilization to happen, the number of sperm cells in the man's semen (the fluid ejected during sexual intercourse) must be sufficient, and the sperm cells must have the right shape, appearance, and activity (motility).

Defects in the sperm can be caused by an infection resulting from a sexually transmitted disease, a blockage caused by a varicose vein in the scrotum (varicocele), an endocrine imbalance, or problems with other male reproductive organs (such as the testicles, prostate gland, or seminal vesicles).

If a low sperm count is the problem, it is possible to restore fertility by:

- Treating underlying infections.
- Timing sex to coincide with the time the woman is ovulating, which means that the egg is released from the ovary and is beginning to travel down the fallopian tube (the site of fertilization).
- Having sex less often to build up the number of sperm in the semen.
- Treating any endocrine imbalance with drugs.
- Having a surgical procedure to remove a varicocele (varicocelectomy).

Fertility drugs

If infertility is due to a woman's failure to release eggs from the ovary (ovulate), fertility drugs can help bring hormone levels into balance, stimulating the ovaries and triggering egg production.

Surgical repair

In some women, infertility is due to blocked fallopian tubes. The egg is released from the ovary, but the sperm is prevented from reaching it because of a physical obstruction in the fallopian tube. If this is the case, surgery may help repair the damage. Microsurgery can sometimes repair the damage to scarred fallopian tubes if it is not too severe. Not all tube damage can be repaired, however, and most tubal problems are more successfully treated with **in vitro fertilization**.

Fibroid tumors in the uterus also may cause infertility, and they can be surgically treated. **Endometriosis**, a condition in which parts of the lining of the uterus become imbedded on other internal organs (such as the ovaries or fallopian tubes) may contribute to infertility. It may be necessary to surgically remove the endometrial tissue to improve fertility.

Artificial insemination

Artificial insemination may be tried if sperm count is low, the man is impotent, or the woman's vagina creates a hostile environment for the sperm. The procedure is not always successful. In this procedure, the semen is collected and placed into the woman's cervix with a small syringe at the time of ovulation. From the cervix, it can travel to the fallopian tube where fertilization takes place. If the partner's sperm count is low, it can be mixed with donor sperm before being transferred into the uterus.

If there is no sperm in the male partner's semen, then artificial insemination can be performed using a donor's sperm obtained from a sperm bank.

Assisted reproductive technologies

Some fertility treatments require removal of the eggs and/or sperm and manipulation of them in certain ways in a laboratory to assist fertilization. These techniques are called assisted reproductive technologies.

IN VITRO FERTILIZATION (IVF). When infertility cannot be treated by other means or when the cause is not known, it is still possible to become pregnant through in vitro fertilization (IVF), a costly, complex procedure that achieves pregnancy 20% of the time.

In this procedure, a woman's eggs are removed by withdrawing them with a special needle. Attempts are then made to fertilize the eggs with sperm from her partner or a donor. This fertilization takes place in a petri dish in a laboratory. The fertilized egg (embryo) is then returned to the woman's uterus.

Often, three to six fertilized eggs are returned at the same time into the uterus. Usually one or two of the embryos survive and grow into fetuses, but sometimes three or more fetuses result.

A child born in this method is popularly known as a "test tube baby," but in fact the child actually develops inside the mother. Only the fertilization of the egg takes place in the laboratory.

INTRACYTOPLASMIC SPERM INJECTION (ICSI). In a variation of IVF called intracytoplasmic sperm injection (ICSI), single sperm cells are injected directly into each egg. This may be helpful for men with severe infertility.

GAMETE INTERFALLOPIAN TRANSFER (GIFT). In this technique, sperm and eggs are placed directly into the woman's fallopian tubes to encourage fertilization to occur naturally. This procedure is done with the help of **laparoscopy**. In laparoscopy, a small tube with a viewing lens at one end is inserted into the

KEY TERMS

Gamete—An egg (ovum) from the female or a mature sperm from the male.

Laparoscopy—A procedure in which a viewing tube is inserted through the abdominal wall to examine a woman's reproductive organs.

Ovulation—The release of an egg from the ovary. Fertilization can occur within a day or two of ovulation.

Zygote—A fertilized egg.

abdomen through a small incision. The lens allows the physician to see inside the patient on a video monitor.

ZYGOTE INTRAFALLOPIAN TRANSFER (ZIFT). If infertility is caused by a low sperm count, zygote intrafallopian transfer (ZIFT) can be tried. This technique combines GIFT and IVF. This procedure is also called a “tubal embryo transfer.”

In this technique, in-vitro fertilization is first performed, so that the actual fertilization takes place and is confirmed in the laboratory. Two days later, instead of placing the embryo in the uterus, the physician performs laparoscopy to place the embryos in the fallopian tube, much like the GIFT procedure.

A woman must have at least one functioning fallopian tube in order to participate in ZIFT.

Preparation

Couples who are having fertility problems may want to limit or avoid:

- Tobacco
- Alcohol
- Caffeine
- Stress
- Tight-fitting undershorts (men)
- Hot tubs, saunas and steam rooms (high temperatures can kill sperm).

Risks

Women who take fertility drugs have a higher likelihood of getting pregnant with more than one child at once. There are also rare but serious side effects to fertility drugs.

Normal results

Typically, at least half of all couples who are infertile will respond to treatment with a successful pregnancy. For those who cannot become pregnant with treatment or insemination, surrogate parenting or adopting may be a workable option.

Resources

ORGANIZATIONS

American Society for Reproductive Medicine. 1209 Montgomery Highway, Birmingham, AL 35216. (205) 978-5000.
Resolve. 1310 Broadway, Somerville, MA 02144-1731. (617) 623-0744. <<http://www.resolve.org>>.

Carol A. Turkington

Influenza

Definition

Usually referred to as the flu or gripe, influenza is a highly infectious respiratory disease. The disease is caused by certain strains of the influenza virus. When the virus is inhaled, it attacks cells in the upper respiratory tract, causing typical flu symptoms such as **fatigue**, **fever** and chills, a hacking **cough**, and body aches. Influenza victims are also susceptible to potentially life-threatening secondary infections. Although the stomach or intestinal “flu” is commonly blamed for stomach upsets and **diarrhea**, the influenza virus rarely causes gastrointestinal symptoms. Such symptoms are most likely due to other organisms such as rotavirus, *Salmonella*, *Shigella*, or *Escherichia coli*.

Description

The flu is considerably more debilitating than the **common cold**. Influenza outbreaks occur suddenly, and infection spreads rapidly. The annual **death** toll attributable to influenza and its complications averages 20,000 in the United States alone. In the 1918-1919 Spanish flu pandemic, the death toll reached a staggering 20–40 million worldwide. Approximately 500,000 of these fatalities occurred in America.

Influenza outbreaks occur on a regular basis. The most serious outbreaks are pandemics, which affect millions of people worldwide and last for several months. The 1918–1919 influenza outbreak serves as

the primary example of an influenza pandemic. Pandemics also occurred in 1957 and 1968 with the Asian flu and Hong Kong flu, respectively. The Asian flu was responsible for 70,000 deaths in the United States, while the Hong Kong flu killed 34,000.

Epidemics are widespread regional outbreaks that occur every two to three years and affect 5–10% of the population. The Russian flu in the winter of 1977 is an example of an epidemic. A regional epidemic is shorter lived than a pandemic, lasting only several weeks. Finally, there are smaller outbreaks each winter that are confined to specific locales.

The earliest existing descriptions of influenza were written nearly 2500 years ago by the ancient Greek physician Hippocrates. Historically, influenza was ascribed to a number of different agents, including “bad air” and several different bacteria. In fact, its name comes from the Italian word for “influence,” because people in eighteenth-century Europe thought that the disease was caused by the influence of bad weather. It was not until 1933 that the causative agent was identified as a virus.

There are three types of influenza viruses, identified as A, B, and C. Influenza A can infect a range of animal species, including humans, pigs, horses, and birds, but only humans are infected by types B and C. Influenza A is responsible for most flu cases, while infection with types B and C virus are less common and cause a milder illness.

In the United States, 90% of all deaths from influenza occur among persons older than 65. Flu-related deaths have increased substantially in the United States since the 1970s, largely because of the **aging** of the American population. In addition, elderly persons are vulnerable because they are often reluctant to be vaccinated against flu.

A new concern regarding influenza is the possibility that hostile groups or governments could use the virus as an agent of bioterrorism. A report published in early 2003 noted that Type A influenza virus has a high potential for use as such an agent because of the virulence of the Type A strain that broke out in Hong Kong in 1997 and the development of laboratory methods for generating large quantities of the virus. The report recommended the stockpiling of present **antiviral drugs** and speeding up the development of new ones.

Causes and symptoms

Approximately one to four days after infection with the influenza virus, the victim is hit with an array of symptoms. “Hit” is an appropriate term,

because symptoms are sudden, harsh, and unmistakable. Typical influenza symptoms include the abrupt onset of a **headache**, dry cough, and chills, rapidly followed by overall achiness and a fever that may run as high as 104°F (40°C). As the fever subsides, nasal congestion and a **sore throat** become noticeable. Flu victims feel extremely tired and weak and may not return to their normal energy levels for several days or even a couple of weeks.

Influenza complications usually arise from bacterial infections of the lower respiratory tract. Signs of a secondary respiratory infection often appear just as the victim seems to be recovering. These signs include high fever, intense chills, chest pains associated with breathing, and a productive cough with thick yellowish green sputum. If these symptoms appear, medical treatment is necessary. Other secondary infections, such as sinus or ear infections, may also require medical intervention. Heart and lung problems, and other chronic diseases, can be aggravated by influenza, which is a particular concern with elderly patients.

With children and teenagers, it is advisable to be alert for symptoms of **Reye’s syndrome**, a rare, but serious complication. Symptoms of Reye’s syndrome are **nausea and vomiting**, and more seriously, neurological problems such as confusion or **delirium**. The syndrome has been associated with the use of **aspirin** to relieve flu symptoms.

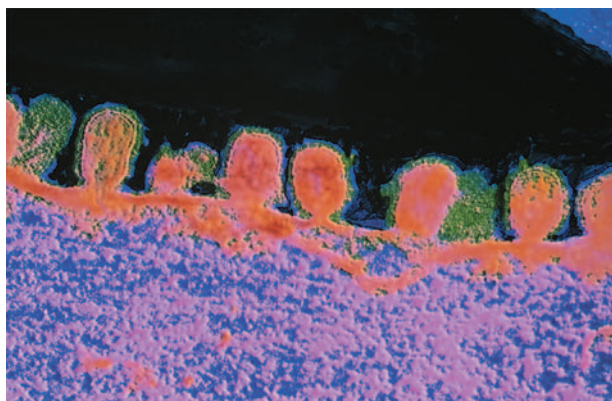
Diagnosis

Although there are specific tests to identify the flu virus strain from respiratory samples, doctors typically rely on a set of symptoms and the presence of influenza in the community for diagnosis. Specific tests are useful to determine the type of flu in the community, but they do little for individual treatment. Doctors may administer tests, such as throat cultures, to identify secondary infections.

Since 1999, however, seven rapid diagnostic tests for flu have become commercially available. These tests appear to be especially useful in diagnosing flu in children, allowing doctors to make more accurate treatment decisions in less time.

Treatment

Essentially, a bout of influenza must be allowed to run its course. Symptoms can be relieved with bed rest and by keeping well hydrated. A steam vaporizer may make breathing easier, and **pain** relievers will take care of the aches and pain. Food may not seem very appetizing, but an effort should be made to consume



A transmission electron microscopy (TEM) image of influenza viruses budding from the surface of an infected cell. (Photo Researchers, Inc. Reproduced by permission.)

nourishing food. Recovery should not be pushed too rapidly. Returning to normal activities too quickly invites a possible relapse or complications.

Drugs

Since influenza is a viral infection, **antibiotics** are useless in treating it. However, antibiotics are frequently used to treat secondary infections.

Over-the-counter medications are used to treat flu symptoms, but it is not necessary to purchase a medication marketed specifically for flu symptoms. Any medication that is designed to relieve symptoms, such as pain and coughing, will provide some relief. Medications containing alcohol, however, should be avoided because of the dehydrating effects of alcohol. The best medicine for symptoms is simply an analgesic, such as aspirin, **acetaminophen**, or naproxen. Without a doctor's approval, aspirin is generally not recommended for people under 18 owing to its association with Reye's syndrome, a rare aspirin-associated complication seen in children recovering from the flu. To be on the safe side, children should receive acetaminophen or ibuprofen to treat their symptoms.

There are four antiviral drugs marketed for treating influenza as of 2003. To be effective, treatment should begin no later than two days after symptoms appear. Antivirals may be useful in treating patients who have weakened immune systems or who are at risk for developing serious complications. They include amantadine (Symmetrel, Symadine) and rimantadine (Flumandine), which work against Type A influenza, and zanamavir (Relenza) and oseltamavir phosphate (Tamiflu), which work against both Types A and B influenza. Amantadine and rimantadine can cause side effects

such as nervousness, **anxiety**, lightheadedness, and **nausea**. Severe side effects include seizures, delirium, and hallucination, but are rare and are nearly always limited to people who have kidney problems, seizure disorders, or psychiatric disorders. The new drugs zanamavir and oseltamavir phosphate have few side effects but can cause **dizziness**, jitters, and **insomnia**.

Alternative treatments

There are several alternative treatments that may help in fighting off the virus and recovering from the flu, in addition to easing flu symptoms.

- **Acupuncture** and **acupressure**. Both are said to stimulate natural resistance, relieve nasal congestion and headaches, fight fever, and calm coughs, depending on the acupuncture and acupressure points used.
- **Aromatherapy**. Aromatherapists recommend gargling daily with one drop each of the essential oils of tea tree (*Melaleuca* spp.) and lemon mixed in a glass of warm water. If already suffering from the flu, two drops of tea tree oil in a hot bath may help ease the symptoms. Essential oils of eucalyptus (*Eucalyptus globulus*) or peppermint (*Mentha piperita*) added to a steam vaporizer may help clear chest and nasal congestion.
- **Herbal remedies**. Herbal remedies can be used stimulate the immune system (**echinacea**), as antivirals (*Hydrastis canadensis*) goldenseal and garlic (*Allium sativum*), or directed at whatever symptoms arise as a result of the flu. For example, an infusion of boneset (*Eupatatorium perfoliatum*) may counteract aches and fever, and yarrow (*Achillea millefolium*) or elderflower tinctures may combat chills.
- **Homeopathy**. To prevent flu, a homeopathic remedy called *Oscillococcinum* may be taken at the first sign of flu symptoms and repeated for a day or two. Although *oscillococcinum* is a popular flu remedy in Europe, however, a research study published in 2003 found it to be ineffective. Other homeopathic remedies recommended vary according to the specific flu symptoms present. *Gelsemium* (*Gelsemium sempervirens*) is recommended to combat weakness accompanied by chills, a headache, and nasal congestion. *Bryonia* (*Bryonia alba*) may be used to treat muscle aches, headaches, and a dry cough. For restlessness, chills, hoarseness, and achy joints, poison ivy (*Rhus toxicodendron*) is recommended. Finally, for achiness and a dry cough or chills, *Eupatorium perfoliatum* is suggested.
- **Hydrotherapy**. A bath to induce a fever will speed recovery from the flu by creating an environment in

the body where the flu virus cannot survive. The patient should take a bath as hot as he/she can tolerate and remain in the bath for 20–30 minutes. While in the bath, the patient drinks a cup of yarrow or elderflower tea to induce sweating. During the bath, a cold cloth is held on the forehead or at the nape of the neck to keep the temperature down in the brain. The patient is assisted when getting out of the bath (he/she may feel weak or dizzy) and then gets into bed and covers up with layers of blankets to induce more sweating.

- **Traditional Chinese medicine (TCM).** Practitioners of TCM recommend mixtures of herbs to prevent flu as well as to relieve symptoms once a person has fallen ill. There are several different recipes for these remedies, but most contain ginger and Japanese honeysuckle in addition to other ingredients.
- **Vitamins.** For adults, 2–3 grams of vitamin C daily may help prevent the flu. Increasing the dose to 5–7 grams per day during the flu can help fight the infection. (The dose should be reduced if diarrhea develops.)

Prognosis

Following proper treatment guidelines, healthy people under the age of 65 usually suffer no long-term consequences associated with flu infection. The elderly and the chronically ill are at greater risk for secondary infection and other complications, but they can also enjoy a complete recovery.

Most people recover fully from an influenza infection, but it should not be viewed complacently. Influenza is a serious disease, and approximately 1 in 1,000 cases proves fatal.

Prevention

The Centers for Disease Control and Prevention recommend that people get an influenza vaccine injection each year before flu season starts. In the United States, flu season typically runs from late December to early March. Vaccines should be received two to six weeks prior to the onset of flu season to allow the body enough time to establish immunity. Adults only need one dose of the yearly vaccine, but children under nine years of age who have not previously been immunized should receive two doses with a month between each dose.

Each season's flu vaccine contains three virus strains that are the most likely to be encountered in the coming flu season. When there is a good match between the anticipated flu strains and the strains used in the vaccine, the vaccine is 70–90% effective in

people under 65. Because immune response diminishes somewhat with age, people over 65 may not receive the same level of protection from the vaccine, but even if they do contract the flu, the vaccine diminishes the severity and helps prevent complications.

The virus strains used to make the vaccine are inactivated and will not cause the flu. In the past, flu symptoms were associated with vaccine preparations that were not as highly purified as modern vaccines, not to the virus itself. In 1976, there was a slightly increased risk of developing **Guillain-Barré syndrome**, a very rare disorder, associated with the swine flu vaccine. This association occurred only with the 1976 swine flu vaccine preparation and has never recurred.

Serious side effects with modern vaccines are extremely unusual. Some people experience a slight soreness at the point of injection, which resolves within a day or two. People who have never been exposed to influenza, particularly children, may experience one to two days of a slight fever, tiredness, and muscle aches. These symptoms start within 6–12 hours after the **vaccination**.

It should be noted that certain people should not receive an influenza vaccine. Infants six months and younger have immature immune systems and will not benefit from the vaccine. Since the vaccines are prepared using hen eggs, people who have severe **allergies** to eggs or other vaccine components should not receive the influenza vaccine. As an alternative, they may receive a course of amantadine or rimantadine, which are also used as a protective measure against influenza. Other people who might receive these drugs are those that have been immunized after the flu season has started or who are immunocompromised, such as people with advanced HIV disease. Amantadine and rimantadine are 70–90% effective in preventing influenza.

Certain groups are strongly advised to be vaccinated because they are at increased risk for influenza-related complications:

- All people 65 years and older
- Residents of nursing homes and chronic-care facilities, regardless of age
- Adults and children who have chronic heart or lung problems, such as **asthma**
- Adults and children who have chronic metabolic diseases, such as diabetes and renal dysfunction, as well as severe anemia or inherited hemoglobin disorders
- Children and teenagers who are on long-term aspirin therapy

KEY TERMS

Bioterrorism—The intentional use of disease-causing microbes or other biologic agents to intimidate or terrorize a civilian population for political or military reasons. Type A influenza virus could be used as an agent of bioterrorism.

Common cold—A mild illness caused by a upper respiratory viruses. Usual symptoms include nasal congestion, coughing, sneezing, throat irritation, and a low-grade fever.

Epidemic—A widespread regional disease outbreak.

Guillain-Barré syndrome—Also called acute idiopathic polyneuritis, this condition is a neurologic syndrome that can cause numbness in the limbs and muscle weakness following certain viral infections.

Pandemic—Worldwide outbreak of an infection, afflicting millions of victims.

- Women who will be in their second or third trimester during flu season or women who are nursing
- Anyone who is immunocompromised, including HIV-infected persons, **cancer** patients, organ transplant recipients, and patients receiving steroids, **chemotherapy**, or **radiation therapy**
- Anyone in contact with the above groups, such as teachers, care givers, health-care personnel, and family members
- Travelers to foreign countries.

A person need not be in one of the at-risk categories listed above, however, to receive a flu vaccination. Anyone who wants to forego the discomfort and inconvenience of an influenza attack may receive the vaccine.

As of early 2003, researchers are working on developing an intranasal flu vaccine in aerosol form. An aerosol vaccine using a weakened form of Type A influenza virus has been tested in pilot studies and awaits further clinical trials.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD, editors. "Respiratory Viral Diseases: Influenza." Section 13, Chapter 162. In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

Pelletier, Kenneth R., MD. *The Best Alternative Medicine*, Part II. "CAM Therapies for Specific Conditions: Colds/Flu." New York: Simon & Schuster, 2002.

PERIODICALS

- Elkins, Rita. "Combat Colds and Flu." *Let's Live*. 68 (January 2000): 81 + .
- Jonas, W. B., T. J. Kaptchuk, and K. Linde. "A Critical Overview of Homeopathy." *Annals of Internal Medicine* 138 (March 4, 2003): 393–399.
- Krug, R. M. "The Potential Use of Influenza Virus as an Agent for Bioterrorism." *Antiviral Research* 57 (January 2003): 147–150.
- Oxford, J. S., S. Bossuyt, S. Balasingam, et al. "Treatment of Epidemic and Pandemic Influenza with Neuraminidase and M2 Proton Channel Inhibitors." *Clinical Microbiology and Infection* 9 (January 2003): 1–14.
- Roth, Y., J. S. Chapnik, and P. Cole. "Feasibility of Aerosol Vaccination in Humans." *Annals of Otolaryngology, Rhinology, and Laryngology* 112 (March 2003): 264–270.
- Shortridge, K. F., J. S. Peiris, and Y. Guan. "The Next Influenza Pandemic: Lessons from Hong Kong." *Journal of Applied Microbiology* 94, Supplement (2003): 70S–79S.
- Storch, G. A. "Rapid Diagnostic Tests for Influenza." *Current Opinion in Pediatrics* 15 (February 2003): 77–84.
- Thompson, W. W., D. K. Shay, E. Weintraub, et al. "Mortality Associated with Influenza and Respiratory Syncytial Virus in the United States." *Journal of the American Medical Association* 289 (January 8, 2003): 179–186.

ORGANIZATIONS

- Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.
- National Institute of Allergy and Infectious Diseases (NIAID). 31 Center Drive, MSC 2520, Bethesda, MD 20892-2520. <<http://www.niaid.nih.gov>>.

OTHER

- NIAID Fact Sheet: Flu. Bethesda, MD: NIAID, January 2003. <<http://www.niaid.nih.gov/factsheets/flu.htm>>.

Julia Barrett
Rebecca J. Frey, PhD

Infrequent menstruation see **Oligomenorrhea**

Inhalation therapies

Definition

Inhalation therapies are a group of respiratory, or breathing, treatments designed to help restore or

improve breathing function in patients with a variety of diseases, conditions, or injuries. The treatments range from at-home **oxygen therapy** for patients with chronic obstructive pulmonary disease to mechanical ventilation for patients with acute **respiratory failure**. Inhalation therapies usually include the following categories:

- Oxygen therapy
- Incentive spirometry
- Continuous positive airway pressure (CPAP)
- Oxygen chamber therapy
- Mechanical ventilation
- Newborn **life support**.

Purpose

Inhalation therapies are ordered for various stages of diseases which are causing progressive or sudden respiratory failure. Although physicians generally follow guidelines to assign specific therapy according to the type and stage of a disease, the ultimate decision is based on a number of tests indicating pulmonary function and the presence or absence of oxygen in body organs and tissues.

Oxygen therapy

Oxygen therapy is most commonly ordered to support patients with **emphysema** and other chronic obstructive pulmonary disease (COPD). The oxygen therapy is usually ordered once decreased oxygen saturation in the blood or tissues is demonstrated. Oxygen therapy may also be used in the hospital setting to help return a patient's breathing and oxygen levels to normal.

Incentive spirometry

Spirometry is a diagnostic method for measuring gases and respiratory function. Incentive spirometry may be ordered to help patients practice and improve controlled breathing. It may be ordered after surgery to the abdomen, lungs, neck, or head.

Continuous positive airway pressure (CPAP)

Common uses of continuous positive airway pressure include **sleep apnea**, **respiratory distress syndrome** in infants, and **adult respiratory distress syndrome**. Signs of **atelectasis** (absence of gas from the lungs) or abnormalities of the lower airways may also indicate CPAP.

Oxygen chamber therapy

Oxygen chamber therapy is ordered for various causes that indicate immediate need for oxygen saturation in the blood. Divers with decompression illness, climbers in high altitude, patients suffering from severe carbon dioxide **poisoning**, and children or adults in acute respiratory distress may require oxygen chamber therapy. In recent years, physicians have also used the forced pressure of oxygen chambers to help heal **burns** and other **wounds**, since the pressure under which the oxygen is delivered can reach areas that are blocked off or suffering from poor circulation.

Mechanical ventilation

Mechanical ventilation is ordered for patients in acute respiratory distress, and is often used in an intensive care situation. In some cases, mechanical ventilation is a final attempt to continue the breathing function in a patient and may be considered "life-sustaining."

Newborn life support

Newborn babies, particularly those who were premature, may require inhalation therapies immediately upon birth, since the lungs are among the last organs to fully develop. Some newborns suffer from serious respiratory problems or birth complications, such as respiratory distress syndrome, neonatal wet lung syndrome, apnea of **prematurity** or persistent fetal circulation, which may require inhalation therapies.

Precautions

There are numerous indications for not prescribing various inhalation therapies.

Oxygen therapy

Patients and family members who smoke should not have oxygen prescribed or should avoid **smoking** in the area to prevent combustion. Sedatives should be avoided for patients on oxygen therapy.

Incentive spirometry

Patients who are unable or unwilling to properly and consistently practice incentive spirometry as prescribed should not receive this form of treatment.

Continuous positive airway pressure (CPAP)

Patients unable or unwilling to comply with the physician's instructions for use of CPAP are not likely to have it prescribed. Extremely obese patients may

have less success with this form of therapy for the treatment of sleep apnea.

Oxygen chamber therapy

Complications may arise from this form of treatment and during transport to or from the oxygen chamber. Therefore, some patients may not receive enough benefit to outweigh possible complications. All patients, particularly children, must be carefully monitored.

Mechanical ventilation

Use of mechanical ventilation will be carefully weighed against benefit and possible risks. Some patients will require **sedation** to prevent fighting of the ventilator, which can increase the risk of complications.

Newborn life support

Not all infants with breathing problems will require measures as severe as mechanical ventilation. The physician will make the determination based on weight and condition of the infant. Newborns with patent ductus arteriosus, a handicap affecting the pulmonary artery, are more likely to suffer pulmonary hemorrhage from mechanical ventilation.

Description

Oxygen therapy

Once a patient shows hypoxemia, or decreased oxygen in arterial blood, supplemental oxygen may be ordered. The main purpose of the oxygen is to prevent damage to vital organs resulting from inadequate oxygen supply. The lowest possible saturation will be given to keep the patient's measurements at a minimum acceptable level. The oxygen is administered through a mask or nasal tube, or sometimes directly into the trachea. The amount of oxygen prescribed is measured in liters of flow per minute. Patients with chronic hypoxemia, most likely in late stages of COPD, will often receive long-term oxygen therapy.

Most patients will receive their long-term oxygen therapy through home oxygen use. A physician must prescribe home oxygen and levels will be monitored to ensure that the correct amount of oxygen is administered. Some patients will receive oxygen therapy only at night or when exercising.

The choice of type of home oxygen systems will vary depending on availability, cost considerations, and the mobility of the patient. Those patients who

are ambulatory, especially those who work, will need a system with a small portable tank. Depending on the system chosen, frequent deliveries of oxygen and filling of portable tanks will be necessary.

In the case of respiratory distress in newborns or adults, oxygen therapy may be attempted before mechanical ventilation since it is a noninvasive and less expensive choice. Oxygen has been found effective in treating patients with other diseases such as **cystic fibrosis**, chronic congestive **heart failure**, or other lung diseases.

Incentive spirometry

Incentive spirometry is also referred to as sustained maximal inspiration. It is designed to mimic natural sighs and yawns. A device provides positive feedback when a patient inhales at a predetermined rate and sustains the breath for a specific period of time. This helps teach the patient to take long, slow, and deep breaths. A spirometer, or equipment that measures pulmonary function, is provided to the patient and a respiratory therapist will work with the patient to demonstrate and explain the technique. Once patients show mastery of the technique, they are instructed to practice the exercises frequently on their own.

Continuous positive airway pressure (CPAP)

Patients with sleep apnea will receive continuous positive airway pressure to prevent upper airway collapse. It is usually administered through a tight-fitting mask as humidified oxygen. The pressure of flow is constant during both exhaling and inhaling and the level of pressure is determined based on each individual. Most patients undergoing CPAP in a hospital setting will receive continuous monitoring of some vital signs and periodic sampling of blood gas values.

Oxygen chamber therapy

Also known as hyperbaric oxygen chamber or hyperbaric oxygen therapy (HBO), this treatment delivers pure oxygen under pressure equal to that of 2–3 times normal atmospheric pressure. For years, this treatment has been especially effective on scuba divers who suffer from the “bends,” or decompression illness. The patient enters the chamber, a plastic cylinder-shaped structure that is normally transparent. In most cases, just one patient will enter by being rolled into the chamber on a type of stretcher. Once inside, the oxygen will be delivered under forced pressure and the patient is free to read, nap, or listen to the radio. The therapy usually lasts one hour, although it can take up to five

hours in serious decompression cases. Before exiting the chamber, the pressure will eventually be lowered to normal atmospheric level.

Mechanical ventilation

In general, mechanical ventilation replaces or supports the normal ventilatory lung function of a patient. Although normally delivered in a hospital, often to treat serious illness, mechanical ventilation may be performed at home under the order and supervision of a physician and home health agency. The patient will usually be intubated and the ventilator machine “takes over” the breathing function.

There are several modes and methods of mechanical ventilation, each offering different advantages and disadvantages. In assist/control ventilation, the oldest mode of ventilation, the physician predetermines settings and the ventilator delivers a breath each time the patient makes an effort to inhale. In synchronized intermittent mandatory ventilation, the machine senses a patient’s effort to inhale and delivers the preset amount. The amount cannot be increased by the patient’s effort. Pressure-control ventilation involves the physician’s selection of a peak pressure and this method is most useful for patients suffering from obstructive airways disease. In cases of severe hypoventilation, an endotracheal tube must be inserted. If a patient will be on mechanical ventilation for more than two weeks, a tracheostomy, or surgical incision, will be performed for placement of the breathing tubes.

There are other modes of ventilation that may be used, including high-frequency ventilation, a newer technique that delivers 100 to 200 breaths per minute to the patient. The breaths are delivered through a humidified, high-pressure gas jet. High-frequency ventilation may be ordered when a patient does not respond to conventional mechanical ventilation or for certain conditions and circumstances.

Newborn life support

Premature infants, especially those born before the 28th week of gestation, have underdeveloped breathing muscles and immature structures within the lungs. These infants will require breathing support, often in the form of mechanical ventilation. The support delivers warm, humidified, oxygen-enriched gases either by oxygen hood or through mechanical ventilation. In serious cases, the infant may require mechanical ventilation with CPAP or positive-end expiratory pressure (PEEP) through a tightly fitting face mask or even by endotracheal intubation.

Need for continued resuscitation for newborns depends not only on gestational age, but on signs indicating ineffective breathing, including color, heart rate, and respiratory effort. CPAP will be delivered through nasal or endotracheal tubes with a continuous-flow ventilator specifically designed for infants. An alarm system alerts the neonatal staff to problems and monitoring of breathing and other vital functions will accompany the therapy. As respiratory distress syndrome begins to resolve, usually in four or five days, the type of support will be reduced accordingly and the infant may be weaned from the ventilator and moved to only CPAP or an oxygen hood.

Preparation

Preparation for any of these treatments is normally not necessary, and in fact, these therapies may be administered as a result of an emergency situation. Some of the methods, particularly incentive spirometry, or at-home oxygen or ventilation, will require education and cooperation with a home health agency or respiratory therapist. Pretreatment testing of various indicators of respiratory function and oxygen saturation will be performed to determine exact needs of individual patients.

Aftercare

Pulmonary function tests and other tests will be performed to verify that treatments have been successful or to monitor and adjust treatments. Mechanical ventilation will require weaning from the equipment and may also require care for the area surrounding the intubation.

Risks

Inhalation therapies may carry risks, complications or side effects including:

Oxygen therapy

At-home oxygen therapy carries risk if improper care is taken to follow instructions when handling the oxygen. Patients are cautioned not to smoke near the oxygen supply and to keep the supply away from other sources that may cause electrical spark, flames, or intense heat. Patients on home oxygen therapy should avoid use of sedatives.

Incentive spirometry

The major risk associated with incentive spirometry relates to improper use. Patients must be

carefully instructed in the technique and monitored periodically for compliance and improvement. Barotrauma, injury to the middle ear or sinuses caused by imbalance between the affected cavity and the outside, or ambient pressure, can result from incentive spirometry. A patient may also suffer discomfort or **fatigue**.

Continuous positive airway pressure (CPAP)

The effectiveness of CPAP may be limited if patients do not cooperate. Possible side effects of CPAP include skin abrasions from the mask, leakage from the tube or mask, nasal congestion, nasal or oral dryness, or discomfort from the pressure of delivery.

Oxygen chamber therapy

Hyperbaric oxygen therapy is painless. The only risk would be associated with improper administration of the pressure levels, which should not occur, since respiratory staff and the supervising physician should be thoroughly trained in performance of this therapy. The drawback to hyperbaric oxygen treatment is the limited availability of chambers. Many cities do not have readily available chambers.

Mechanical ventilation

The biggest risk of mechanical ventilation is sometimes considered to be a patient's dependence on the machine and the difficulty of weaning the patient. The physician will carefully select and monitor the mode of ventilation, the machine's settings, and the patient's progress to prevent this complication. A patient may therefore be left on a ventilator after sufficient progress is made to gradually wean breathing dependence.

Intubation and mechanical ventilation are frightening and uncomfortable for many patients and they may fight the ventilator. If this occurs, the physician may order a sedative to ensure cooperation and effectiveness of the therapy. Intubation often results in irritation to the trachea and larynx. Tracheostomy is associated with risk of bleeding, **pneumothorax**, local infection, and increased incidence of aspiration.

Newborn life support

Infants are continuously monitored to determine even small changes in breathing function. Mechanical ventilation can result in increases in respiratory distress or other complications. It is possible for the ventilator to be accidentally disconnected and staff is trained to watch for signs or alarms indicating

disconnection. Mechanical ventilation increases risk of infection in premature babies. Complications of PEEP or CPAP may include pneumothorax or decreased cardiac output.

Normal results

Oxygen therapy

In the case of COPD, oxygen therapy does not treat the disease but can prolong life, quality of life, and onset of more serious symptoms. Effective oxygen therapy for any patient should lead to improved or sustained levels of oxygen in arterial blood.

Incentive spirometry

With proper use of incentive spirometry, the physician should observe improved pulse rate, decreased respiratory rate, improved respiratory muscle performance, and other indicators of improved function. Lung function following lung resection should show marked improvement following incentive spirometry.

Continuous positive airway pressure

Successful CPAP will result in reduction in apnea for those suffering from sleep apnea. A study reported on in 1998 demonstrated that CPAP was effective in the majority of patients with sleep apnea, with the exception of significantly obese patients with blood gas values that were worse during waking hours at rest and at **exercise**. Hospitalized patients on CPAP therapy should show improvement in blood gas and other pulmonary measurements as expected by the treating physician.

Oxygen chamber therapy

Divers undergoing emergency treatment in a **hyperbaric chamber** should show immediate improvement in oxygen levels throughout the body, regardless of blood flow restrictions, after one or two treatments. Those patients receiving oxygen chamber therapy for difficult wounds may continue to receive treatments daily for several weeks before satisfactory results are reached. Patients with carbon dioxide poisoning should show improvement in or recovery of neurologic function. Results of hyperbaric chamber therapy depend largely on how quickly the patient was brought to the chamber, as well as the severity of the initial condition.

Mechanical ventilation

Successful mechanical ventilation will result in gradual decrease in dependence on the ventilator

KEY TERMS

Aspiration—Accidental suction of fluids or vomit into the respiratory system.

Cannula—A tube inserted into a cavity to serve as a channel for the transport of fluid.

Endotracheal—Placed within the trachea.

Hypoventilation—Reduced ventilation in the lungs' air sacs resulting in above normal carbon dioxide pressure.

Hypoxemia—A condition in which there is deficient oxygen supply in the blood.

Hypoxia—Low levels of oxygen in blood, tissue, or air.

Intubation—Placement of a tube into a hollow organ (such as the trachea).

Pneumothorax—Presence of gas or air in the hollow space around the lungs.

Trachea—The windpipe, or main by which air passes to and from the lungs.

and weaning from the machine. Reduction of therapy to another form, such as CPAP or oxygen therapy, indicates that ventilation has worked as expected. In the case of COPD, exacerbation may be successfully treated with mechanical ventilation and the patient may return to home oxygen therapy. Pediatric patients will demonstrate normal growth and development as a normal result of long-term mechanical ventilation at home. Some patients, particularly those in a hospital intensive care unit, will not be able to breathe again without the ventilator and families and physicians will face tough choices about continued life support.

Newborn life support

Neonates will be constantly monitored to measure lung function. Those measurements will help caregivers determine if and when mechanical ventilation can be reduced and CPAP or oxygen mask begun. CPAP is considered successful when the infant's respiratory rate is reduced by 30–40%, a chest radiograph shows improved lung volume and appearance, stabilization of oxygen levels is documented and caregivers observe improvement in the infant's comfort. Evidence that there is no infection from ventilation is also considered normal. In some cases, inhalation therapy, including mechanical

ventilation, will not work and the infant's parents and physicians will face tough decisions about invasive procedures with associated high risks or cessation of life support.

Resources

ORGANIZATIONS

American Association for Respiratory Care. 11030 Ables Lane, Dallas, TX 75229. (972) 243-2272, Fax (972) 484-2720.

American Lung Association. 1740 Broadway, New York, NY 10019. (800) 586-4872. <<http://www.lungusa.org>>.

National Heart, Lung and Blood Institute. P.O. Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

OTHER

Hyperbaric Research and Treatment Center Page. <<http://www.hyperbaricrx.com>>.

Teresa Odle

Inner ear infection see **Labyrinthitis**

Insecticide poisoning

Definition

Insecticide **poisoning** is exposure to a group of chemicals designed to eradicate insects that cause affected persons to develop clinical signs that can progress to **death**.

Description

Insecticides belong to a group of chemicals called organophosphates used to protect against insects. Their use is popular since they are effective and do not remain in the environment, disintegrating within a few days. Organophosphates act to inhibit an enzyme in humans called acetyl cholinesterase. This enzyme functions to degrade a chemical called acetylcholine, which excites nerve cells. The resultant effect of organophosphates would be an increase in acetylcholine, thus causing initial excitation of nerve cells.

Poisoning can occur with a broad range of symptoms affecting the functioning of nerves and initial symptoms similar to the flu such as **vomiting**, abdominal **pain**, **dizziness**, and **headache**. Common names for insecticides include dichlorvos, chlorpyrifos, diazinon, fenthion, malathion, parathion, and carbamate. A special type of insecticide called paraquat is

very lethal and responsible for approximately 1,000 deaths per year just in Japan. Paraquat poisoning releases oxygen free radicals that destroy lung and kidney tissues. When poisoning is suspected, a comprehensive management and assessment plan should be performed. This initial assessment should include:

- Description of toxins: names of chemical(s).
- Magnitude of exposure: determination of amount of exposure.
- Progression of symptoms: determining the progression of symptoms can provide information concerning **life support** and overall outcome.
- Time of exposure: knowing the time of exposure is vital since symptoms may be delayed, and it may assist to develop a management plan.
- Medical history: underlying diseases and therapeutic mediations may worsen toxic manifestations.

Causes and symptoms

Exposure to insecticides can occur by ingestion, inhalation, or exposure to skin or eyes. The chemicals are absorbed through the skin, lungs, and gastrointestinal tract and then widely distributed in tissues. Symptoms cover a broad spectrum and affect several organ systems:

- Gastrointestinal: **nausea**, vomiting, cramps, excess salivation, and loss of bowel movement control
- Lungs: increases in bronchial mucous secretions, coughing, **wheezing**, difficulty breathing, and water collection in the lungs (this can progress to breathing cessation)
- Skin: sweating
- Eyes: blurred vision, smaller sized pupil, and increased tearing
- Heart: slowed heart rate, block of the electrical conduction responsible of heartbeat, and lowered blood pressure
- Urinary system: urinary frequency and lack of control
- Central nervous system: convulsions, confusion, **paralysis**, and **coma**

Diagnosis

The confirmatory diagnosis for insecticide poisoning is the measurement of blood acetyl cholinesterase less than 50% of normal. The chemicals can also be detected by specific urine testing. Signs and

KEY TERMS

Acetylcholine—A chemical called a neurotransmitter that functions to excite nerve cells.

Acetylcholinesterase—An enzyme that breaks down acetylcholine.

Central nervous system—Consists of the brain and spinal cord and integrates and processes information.

Enzyme—A protein that speeds up a chemical reaction, but is not consumed during the process.

Oxygen free radicals—Reactive molecules containing oxygen and can cause cell damage.

symptoms in addition to a comprehensive poisoning assessment are essential for diagnosis. Carbamate insecticide poisoning exhibits symptoms similar to organophosphate poisoning but without central nervous system signs.

Treatment

Decontaminate exposed clothing and wash with soap and water immediately. Emergency measures may focus on ventilator support and heart monitoring. If inhalation is suspected, the patient should be removed from the site of exposure. If the eyes were the entry site, they should be flushed with large amounts of water. If the chemicals were ingested, the stomach may be washed out and **activated charcoal** may be administered. Atropine or glycopyrrolate (Robinul) is the drug of choice for carbamate insecticide poisoning. It reverses many symptoms, but is only partially effective for central nervous symptom effects such as coma and convulsions. A medication called Pralidoxime is also commonly indicated to reactivate acetylcholinesterase and to reverse typical symptoms due to organophosphate poisoning. Additionally, the patient is monitored for heart, lung, liver functioning, specific blood tests, and oxygen levels in blood.

Prognosis

Prognosis depends on the specific chemical of exposure, magnitude and time of exposure, progression of symptoms (severity), and onset for medical attention.

Prevention

Adherence to accepted guidelines for handling and management is the key to preventing insecticide

poisoning. These may include masks, gowns, gloves, goggles, respiratory breathing machines, or hazardous material suits.

Resources

BOOKS

- Behrman, Richard E., et al, editors. *Nelson Textbook*. 16th ed. W. B. Saunders Company, 2000.
- Goldman, Lee, et al. *Cecil's Textbook of Medicine*. 21st ed. W. B. Saunders Company, 2000.
- Rakel, Robert E., et al. *Conn's Current Therapy*. 53rd ed. W. B. Saunders Company, 2001.

PERIODICALS

- Blain, P. G. "Effects of Insecticides." *Lancet* 357 (5 May 2001).

OTHER

- Material Safety Data Sheets. <<http://oshweb.me.tut.fi>>.
- National Toxicology Program. <http://ntp-server.niehs.nih.gov/main_Pages/NTP_ALL_STDY_PG.html>.

Laith Farid Gulli, M.D.

Insomnia

Definition

Insomnia is the inability to obtain an adequate amount or quality of sleep. The difficulty can be in falling asleep, remaining asleep, or both. People with insomnia do not feel refreshed when they wake up. Insomnia is a common symptom affecting millions of people that may be caused by many conditions, diseases, or circumstances.

Description

Sleep is essential for mental and physical restoration. It is a cycle with two separate states: rapid eye movement (REM), the stage in which most dreaming occurs; and non-REM (NREM). Four stages of sleep take place during NREM: stage I, when the person passes from relaxed wakefulness; stage II, an early stage of light sleep; stages III and IV, which are increasing degrees of deep sleep. Most stage IV sleep (also called delta sleep), occurs in the first several hours of sleep. A period of REM sleep normally follows a period of NREM sleep.

Insomnia is more common in women and older adults. People who are divorced, widowed, or separated are more likely to have the problem than those

who are married, and it is more frequently reported by those with lower socioeconomic status. Short-term, or transient, insomnia is a common occurrence and usually lasts only a few days. Long-term, or chronic insomnia lasts more than three weeks and increases the risk for injuries in the home, at the workplace, and while driving because of daytime sleepiness and decreased concentration. Chronic insomnia can also lead to **mood disorders** like depression.

Causes and symptoms

Transient insomnia is often caused by a temporary situation in a person's life, such as an argument with a loved one, a brief medical illness, or **jet lag**. When the situation is resolved or the precipitating factor disappears, the condition goes away, usually without medical treatment.

Chronic insomnia usually has different causes, and there may be more than one. These include:

- A medical condition or its treatment, including **sleep apnea**
- Use of substances such as **caffeine**, alcohol, and nicotine
- Psychiatric conditions such as mood or **anxiety disorders**
- **Stress**, such as sadness caused by the loss of a loved one or a job
- Disturbed sleep cycles caused by a change in work shift
- Sleep-disordered breathing, such as **snoring**
- Periodic jerky leg movements (*nocturnal myoclonus*), which happen just as the individual is falling asleep
- Repeated nightmares or panic attacks during sleep.

Another cause is excessive worrying about whether or not a person will be able to go to sleep, which creates so much **anxiety** that the individual's bedtime rituals and behavior actually trigger insomnia. The more one worries about falling asleep, the harder it becomes. This is called psychophysiological insomnia.

Symptoms of insomnia

People who have insomnia do not start the day refreshed from a good night's sleep. They are tired. They may have difficulty falling asleep, and commonly lie in bed tossing and turning for hours. Or the individual may go to sleep without a problem but wakes in the early hours of the morning and is either unable to go back to sleep, or drifts into a restless unsatisfying

sleep. This is a common symptom in the elderly and in those suffering from depression. Sometimes sleep patterns are reversed and the individual has difficulty staying awake during the day and takes frequent naps. The sleep at night is fitful and frequently interrupted.

Diagnosis

The diagnosis of insomnia is made by a physician based on the patient's reported signs and symptoms. It can be useful for the patient to keep a daily record for two weeks of sleep patterns, food intake, use of alcohol, medications, **exercise**, and any other information recommended by the physician. If the patient has a bed partner, information can be obtained about whether the patient snores or is restless during sleep. This, together with a medical history and **physical examination**, can help confirm the doctor's assessment.

A wide variety of healthcare professionals can recognize and treat insomnia, but when a patient with chronic insomnia does not respond to treatment, or the condition is not adequately explained by the patient's physical, emotional, or mental circumstances, then more extensive testing by a specialist in **sleep disorders** may be warranted.

Treatment

Treatment of insomnia includes alleviating any physical and emotional problems that are contributing to the condition and exploring changes in lifestyle that will improve the situation.

Changes in behavior

Patients can make changes in their daily routine that are simple and effective in treating their insomnia. They should go to bed only when sleepy and use the bedroom only for sleep. Other activities like reading, watching television, or snacking should take place somewhere else. If they are unable to go to sleep, they should go into another room and do something that is relaxing, like reading. Watching television should be avoided because it has an arousing effect. The person should return to bed only when they feel sleepy. Patients should set the alarm and get up every morning at the same time, no matter how much they have slept, to establish a regular sleep-wake pattern. Naps during the day should be avoided, but if absolutely necessary, than a 30 minute nap early in the afternoon may not interfere with sleep at night.

Another successful technique is called sleep-restriction therapy, which restricts the amount of

time spent in bed to the actual time spent sleeping. This approach allows a slight sleep debt to build up, which increases the individual's ability to fall asleep and stay asleep. If a patient is sleeping five hours a night, the time in bed is limited to 5–5 1/2 hours. The time in bed is gradually increased in small segments, with the individual rising at the same time each morning; at least 85% of the time in bed must be spent sleeping.

Drug therapy

Medications given for insomnia include sedatives, tranquilizers, and **antianxiety drugs**. All require a doctor's prescription and may become habit-forming. They can lose effectiveness over time and can reduce alertness during the day. The medications should be taken two to four times daily for approximately three to four weeks, though this will vary with the physician and patient. If the insomnia is related to depression, then an antidepressant medication may be helpful. Over-the-counter drugs such as **antihistamines** are not very effective in bringing about sleep and can affect the quality of sleep.

Other measures

Relaxing before going to bed will help a person fall asleep faster. Learning to substitute pleasant thoughts for unpleasant ones (imagery training) is a technique that can be very helpful in reducing worry. Another effective measure is the use of audiotapes which combine the sounds of nature with soft relaxing music. These, alone or in combination with other relaxation techniques, can safely promote sleepiness.

Changes in diet and exercise routines can also have a beneficial effect. Dietary items to be avoided include drinks that contain caffeine such as coffee, tea and colas, chocolate (which contains a stimulant), and alcohol, which initially makes a person sleepy but a few hours later can have the opposite effect. Maintaining a comfortable bedroom temperature, reducing noise and eliminating light are also helpful. Regularly scheduled morning or afternoon exercise can relax the body. This should be done 3–4 times a week and be sufficient to produce a light sweat.

Alternative treatments

Many alternative treatments are effective in treating both the symptom of insomnia and its underlying causes. Incorporating relaxation techniques into

KEY TERMS

Biofeedback—A training technique that enables an individual to gain some element of control over involuntary body functions.

Mood disorder—A group of mental disorders involving a disturbance of mood, along with either a full or partial excessively happy (manic) or extremely sad (depressive) syndrome not caused by any other physical or mental disorder. Mood refers to a prolonged emotion.

Sleep apnea—A condition in which a person stops breathing while asleep. These periods can last up to a minute or more, and can occur many times each hour. In order to start breathing again, the person must become semi-awake. The episodes are not remembered, but the following day the client feels tired and sleepy. If severe, sleep apnea can cause other medical problems.

Sleep disorder—Any condition that interferes with sleep. At least 84 have been identified, according to the American Sleep Disorders Association.

bedtime rituals will help a person go to sleep faster, as well as improve the quality of sleep. These methods include **meditation**, massage, breathing exercises, and a warm bath, scented with rose, lavender (*Lavendula officinalis*), marjoram, or chamomile (*Matricaria recutita*). Eating a healthy diet rich in calcium, magnesium, and the **B vitamins** is also beneficial. A high protein snack like yogurt before going to bed is recommended, or a cup of herb tea made with chamomile, hops (*Humulus lupulus*), passionflower (*Passiflora incarnata*), or St John's Wort (*Hypericum perforatum*) to encourage relaxation. **Acupuncture** and **biofeedback** have also proven useful.

Prevention

Prevention of insomnia centers around promotion of a healthy lifestyle. A balance of rest, recreation and exercise in combination with stress management, regular physical examinations, and a healthy diet can do much to reduce the risk.

Resources

ORGANIZATIONS

American Sleep Disorders Association. 1610 14th St. NW, Ste. 300, Rochester, MN 55901. (507) 287-6006.
< <http://www.asda.org> >.

OTHER

"What to Do When You Can't Sleep." *The Virtual Hospital* Page. University of Iowa. < <http://www.vh.org> >.

Donald G. Barstow, RN

Insulin see **Antidiabetic drugs**

Insulin resistance

Definition

Insulin resistance is not a disease as such but rather a state or condition in which a person's body tissues have a lowered level of response to insulin, a hormone secreted by the pancreas that helps to regulate the level of glucose (sugar) in the body. As a result, the person's body produces larger quantities of insulin to maintain normal levels of glucose in the blood. There is considerable individual variation in sensitivity to insulin within the general population, with the most insulin-sensitive persons being as much as six times as sensitive to the hormone as those identified as most resistant. Some doctors use an arbitrary number, defining insulin resistance as a need for 200 or more units of insulin per day to control blood sugar levels. Various researchers have estimated that 3–16 percent of the general population in the United States and Canada is insulin-resistant; another figure that is sometimes given is 70–80 million Americans.

Insulin resistance can be thought of as a set of metabolic dysfunctions associated with or contributing to a range of serious health problems. These disorders include type 2 diabetes (formerly called adult-onset or non-insulin-dependent diabetes), the metabolic syndrome (formerly known as syndrome X), **obesity**, and **polycystic ovary syndrome**. Some doctors prefer the term "insulin resistance syndrome" to "metabolic syndrome."

Description

To understand insulin resistance, it may be helpful for the reader to have a brief account of the way insulin works in the body. After a person eats a meal, digestive juices in the small intestine break down starch or complex sugars in the food into glucose, a simple sugar. The glucose then passes into the bloodstream. When the concentration of glucose in the blood reaches a certain point, the pancreas is stimulated to release insulin into the blood. As the insulin reaches cells in muscle and fatty (adipose) tissues, it

attaches itself to molecules called insulin receptors on the surface of the cells. The activation of the insulin receptors sets in motion a series of complex biochemical signals within the cells that allow the cells to take in the glucose and convert it to energy. If the pancreas fails to produce enough insulin or the insulin receptors do not function properly, the cells cannot take in the glucose and the level of glucose in the blood remains high.

The insulin may fail to bind to the insulin receptors for any of several reasons. Some persons inherit a gene mutation that leads to the production of a defective form of insulin that cannot bind normally to the insulin receptor. Others may have one of two types of abnormalities in the insulin receptors themselves. In type A, the insulin receptor is missing from the cell surface or does not function properly. In type B, the person's immune system produces autoantibodies to the insulin receptor.

In the early stages of insulin resistance, the pancreas steps up its production of insulin in order to control the increased levels of glucose in the blood. As a result, it is not unusual for patients to have high blood sugar levels and high blood insulin levels (a condition known as hyperinsulinemia) at the same time. If insulin resistance is not detected and treated, however, the islets of Langerhans (the insulin-secreting groups of cells) in the pancreas may eventually shut down and decrease in number.

Causes & symptoms

Causes

The reasons for the development of insulin resistance are not completely understood as of the early 2000s, but several factors that contribute to it have been identified:

- **Genetic factors.** Insulin resistance is known to run in families. Genetic mutations may affect the insulin receptor, the signaling proteins within cells, or the mechanisms of glucose transport.
- **Obesity.** Being overweight keeps the muscles from using insulin properly, as it decreases the number of insulin receptors on cell surfaces.
- **Low level of physical activity.** Because muscle tissue takes up 95 percent of the glucose that insulin helps the body utilize (brain cells and blood cells do not depend on insulin to help them use glucose), inactivity further reduces the muscles ability to use insulin effectively.
- **Aging.** The aging process affects the efficiency of glucose transport.

- **Other diseases and disorders.** Some disorders—most notably Cushing syndrome and cirrhosis—and such stresses on the body as trauma, surgery, **malnutrition**, or severe infections speed up the breakdown of insulin or interfere with its effects.
- **Certain medications.** Some drugs, including cyclosporine, niacin, and the **protease inhibitors** used to treat HIV infection, may contribute to insulin resistance.

Symptoms

The symptoms of insulin resistance vary considerably from person to person. Some people may have no noticeable symptoms until they develop signs of heart disease or are diagnosed with high blood pressure during a routine checkup. Other patients may come to the doctor with extremely high levels of blood sugar (hyperglycemia) and such classical symptoms of diabetes as thirst, frequent urination, and weight loss. A small percentage of patients—most commonly women with polycystic ovary syndrome—develop a velvet-textured blackish or dark brown discoloration of the skin known as acanthosis nigricans. This symptom, which is most commonly found on the neck, groin, elbows, knees, knuckles, or armpits, is thought to appear when high levels of insulin in the blood spill over into the skin. This spillover activates insulin receptors in the skin and causes it to develop an abnormal texture and color. Acanthosis nigricans occurs more frequently in Hispanic and African American patients than in Caucasians.

Disorders associated with insulin resistance

Insulin resistance became an important field of research in the late 1980s, when doctors first began to understand it as a precondition of several common but serious threats to health. As of the early 2000s, insulin resistance is associated with the following disorders:

- **Obesity.** Obesity is not only the most common cause of insulin resistance but is a growing health concern in its own right. According to the National Institutes of Health (NIH), the percentage of American adults who meet the criteria for obesity rose from 25 percent to 33 percent between 1990 and 2000—an increase of a third within the space of a decade. Obesity is a risk factor for the development of type 2 diabetes, high blood pressure, and coronary artery disease.
- **Pre-diabetes and type 2 diabetes.** The NIH estimates that about 6.3 percent of the American population has diabetes. Of these 18.3 million people, 5.2 million are undiagnosed. Type 2 diabetes is much more

common than type 1, accounting for 90–95 percent of patients with diabetes. Diabetes increases a person's risk of blindness, **kidney disease**, heart disease and **stroke**, disorders of the nervous system, complications during **pregnancy**, and dental problems; it also worsens the prognosis for such infectious diseases as **influenza** or **pneumonia**. About 41 million Americans are thought to have pre-diabetes, which is a condition marked by elevated levels of blood glucose after **fasting** or after a 2-hour test for glucose tolerance. According to the NIH, a majority of pre-diabetic people will develop type 2 diabetes within 10 years unless they lose between 5 and 7 percent of their body weight.

- Heart disease. Insulin resistance has been linked to a group of risk factors for heart disease and stroke known as the metabolic syndrome (formerly called syndrome X). The metabolic syndrome, like obesity, has become increasingly prevalent in the United States since the 1990s; as of the early 2000s, about a quarter of the general adult population is thought to have it, with the rate rising to 40 percent for adults over the age of 60. To be diagnosed with the metabolic syndrome, a person must have three or more of the following risk factors: a waist circumference greater than 40 in (102 cm) in men or 35 in (88 cm) in women; a level of blood triglycerides of 150 milligrams per deciliter (mg/dL) or higher; blood pressure of 130/85 Hg or higher; fasting blood sugar level of 110 mg/dL or higher; and a blood level of high-density lipoprotein (HDL) cholesterol (the so-called “good” cholesterol) lower than 50 mg/dL for men or 40 mg/dL for women.
- Polycystic ovary syndrome (PCOS). PCOS is an endocrine disorder that develops in 3–10 percent of premenopausal women as a result of the formation of cysts (small fluid-filled sacs) in the ovaries. Women with PCOS do not have normal menstrual periods; they are often infertile and may develop **hirsutism** (excess body hair) or other indications of high levels of androgens (male sex hormones) in the blood. This condition is called hyperandrogenism, and has been linked to insulin resistance in women with PCOS. Weight loss in these patients usually corrects hyperandrogenism and often restores normal ovulation patterns and fertility.

Diagnosis

Patient history and physical examination

Because insulin resistance is a silent condition in many people, the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK)

recommends that all adults over the age of 45 be tested for type 2 diabetes. People younger than 45 who are overweight and have one or more of the following risk factors should also visit their doctor to be tested:

- One or more family members with diabetes.
- High levels of triglycerides and low levels of HDL cholesterol as defined by the criteria for metabolic syndrome.
- Hypertension (high blood pressure).
- A history of **smoking**.
- A history of diabetes during pregnancy (**gestational diabetes**).
- Giving birth to a baby weighing more than 9 pounds. In addition to increasing the mother's risk of developing type 2 diabetes, children who are large for their gestational age (LGA) at birth have an increased risk of developing insulin resistance and metabolic syndrome in later life.
- Having African American, Hispanic, Native American, or Asian American/Pacific Islander heritage.

Some signs and symptoms associated with insulin resistance can be detected by a primary care physician during a routine office visit. Blood pressure, weight, body shape, and the condition of the skin can be checked, as well as determining whether the patient meets the criteria for obesity or is less severely overweight. Obesity is determined by the patient's body mass index, or BMI. The BMI, which is an indirect measurement of the amount of body fat, is calculated in English units by multiplying a person's weight in pounds by 703.1, and dividing that number by the person's height in inches squared. A BMI between 19 and 24 is considered normal; 25–29 is overweight; 30–34 is moderately obese; 35–39 is severely obese; and 40 or higher is defined as morbidly obese. The doctor may also evaluate the patient for obesity in the office by measuring the thickness of the skinfold at the back of the upper arm.

The distribution of the patient's weight is also significant, as insulin resistance is associated with a so-called “apple-shaped” figure, in which much of the excess weight is carried around the abdomen. People whose excess weight is carried on the hips (the “pear-shaped” figure) or distributed more evenly on the body are less likely to develop insulin resistance. One way of measuring weight distribution is the patient's waist-to-hip ratio; a ratio greater than 1.0 in men or 0.8 in women is strongly correlated with insulin resistance.

Laboratory tests

There is no single laboratory test that can be used to diagnose insulin resistance by itself as of 2005. Doctors usually evaluate individual patients on the basis of specific symptoms or risk factors. The tests most commonly used include the following:

- **Blood glucose tests.** A high level of blood glucose may indicate either that the body is not producing enough insulin or is not using it effectively. Two common tests used to screen for insulin resistance are the fasting glucose test and the glucose tolerance test. In the fasting glucose test, the person takes no food after midnight and has their blood glucose level measured early in the morning. Normal blood glucose levels after several hours without food should be below 100 milligrams per deciliter (mg/dL). If the level is between 100 and 125 mg/dL, the person has impaired fasting glucose (IFG) or pre-diabetes. If the level is over 126 and is confirmed by a second test, the person has diabetes. In the glucose tolerance test, the person is given a sugar solution to drink and their blood glucose level is measured 2 hours later. A normal level is 140 mg/dL; 140–199 mg/dL indicates impaired glucose tolerance (IGT) or pre-diabetes, while a level of 200 mg/dL or higher indicates diabetes.
- **Tests of blood insulin levels.** These help to determine whether high blood glucose levels are the result of insufficient production of insulin or inefficient use of insulin.
- **Lipid profile test.** This test measures the amount of total cholesterol, high-density lipoprotein (HDL) cholesterol, low-density lipoprotein (LDL) cholesterol, and triglycerides. Patients with insulin resistance will have high levels of LDL cholesterol and triglycerides with low levels of HDL cholesterol.
- **Measurement of blood electrolytes and uric acid.** Many patients with the metabolic syndrome have high blood levels of uric acid.

A highly accurate technique for measuring insulin resistance is called the euglycemic clamp technique. The patient's blood insulin level is kept ("clamped") at a high but steady level by continual insulin infusion while the blood glucose level is monitored at frequent intervals. Glucose concentrations in the blood are maintained at a normal level by an adjustable-rate glucose drip. The amount of glucose needed to maintain a normal blood glucose level over a given unit of time indicates the degree of insulin resistance. This test, however, requires complex equipment and careful monitoring; it is considered too cumbersome to use in routine screening and is used mostly by researchers.

Treatment

Lifestyle modifications

Lifestyle modifications are the first line of treatment in dealing with insulin resistance:

- **Weight reduction.** Losing weight increases the body's sensitivity to insulin. It is not necessary, however, for patients to reduce their weight to the ideal levels listed on life insurance charts. In recent years, researchers have found that even a modest weight loss—usually defined as 10 percent of the patient's pretreatment weight—is enough to control or at least improve insulin resistance and other health complications of obesity. Weight reduction is usually accomplished by a combination of reduced calorie intake and increased physical activity. Insulin sensitivity is reported to improve within a few days of lowered calorie intake, even before the patient loses a measurable amount of weight.
- **Exercise.** Regular exercise improves the body's sensitivity to insulin by increasing the muscles' uptake of glucose from the bloodstream, by increasing the efficiency of the circulatory system and glucose transport, and by reducing the amount of fat around the patient's abdomen. The American Academy of Family Practice (AAFP) recommends 30 minutes of moderately intense physical activity on most or all days of the week for people diagnosed with insulin resistance. Walking is a very good form of exercise because it does not require any special equipment other than comfortable walking shoes, can be combined with doing errands, and can be done either alone or with a group of friends. Riding a bicycle is another form of exercise recommended for weight control.
- **Adding foods high in fiber to the diet.** A diet high in natural fiber, found in whole grains and vegetables, lowers the levels of blood insulin as well as lowering the patient's risk of developing high blood pressure.
- **Quitting smoking.** Giving up smoking lowers the risk of heart disease, stroke, or lung **cancer** as well as increasing the body's sensitivity to insulin.
- **Limiting alcohol consumption.** Alcohol is a source of "empty" calories with little nutritional value of its own.

Medications

There are several different types of medications that can be used to treat patients with abnormal blood sugar or insulin levels:

- **Biguanides.** Biguanides are drugs that improve the body's sensitivity to insulin by lowering the absorption of glucose in the small intestine, decreasing the liver's production of glucose, and increasing the uptake of glucose in muscle and fatty tissues. Metformin (Glucophage), a drug used in the treatment of type 2 diabetes, is the most commonly used biguanide in treating insulin resistance. It has also been studied as a possible treatment in preventing or delaying the onset of type 2 diabetes.
- **Thiazolidinediones.** These drugs stimulate glucose uptake in the muscles and fatty tissues by activating specific receptors in the cell nucleus. They also lower blood insulin levels in patients with hyperinsulinemia. The thiazolidinediones include pioglitazone (Actos) and rosiglitazone (Avandia).
- **Glucocorticoids.** These drugs may be given to patients with insulin resistance caused by anti-insulin antibodies produced by their immune system. Prednisone (Deltasone) is the most commonly used glucocorticoid.
- **Insulin itself.** Some patients with insulin resistance benefit from injectable insulin to reduce their blood sugar levels.

As of early 2005, however, the Food and Drug Administration (FDA) has not approved any drugs for the treatment of insulin resistance by itself. For this reason, the American Diabetes Association does not recommend treating insulin resistance with medications unless the patient has already been diagnosed with diabetes.

The patient's doctor may also prescribe medications to treat specific health problems associated with insulin resistance. These drugs may include **diuretics** and other medications to lower blood pressure; **aspirin** to reduce the risk of **heart attack**; medications to lower the levels of triglycerides and LDL cholesterol in the blood; and weight-control drugs. The drugs most frequently prescribed in the early 2000s to help patients lose weight are orlistat (Xenical) and sibutramine (Meridia).

Acanthosis nigricans may be treated with topical preparations containing Retin-A, 20% urea, or salicylic acid; however, many patients find that the skin disorder improves by itself following weight loss.

Surgery

Insulin resistance by itself does not require surgical treatment; however, patients who have already developed heart disease may require coronary artery

bypass surgery. In addition, very obese patients—those with a BMI of 40 or higher—may benefit from **bariatric surgery**. Bariatric surgery includes such procedures as vertical banded gastroplasty and gastric bypass, which limit the amount of food that the stomach can contain.

Alternative treatment

Some alternative treatments for insulin resistance and type 2 diabetes have been studied by the Agency for Healthcare Research and Quality (AHRQ). One study reported in 2004 that **omega-3 fatty acids**, a dietary supplement commonly derived from fish, canola, or soybean oil, did not appear to have any significant effect on blood sugar levels or blood insulin levels in patients diagnosed with type 2 diabetes or the metabolic syndrome. An earlier study of **Ayurvedic medicine**, the traditional medical system of India, reported in 2001 that certain herbs used to make Ayurvedic medicines, such as fenugreek, holy basil, *Coccinia indica*, and *Gymnema sylvestre* appear to be effective in lowering blood sugar levels and merit further study. The AHRQ report also noted that the Ayurvedic practice of combining herbal medicines with **yoga** and other forms of physical activity should be investigated further.

Other alternative treatments for insulin resistance and type 2 diabetes include chromium supplements, ginseng, **biofeedback**, and **acupuncture**. The connection between chromium supplementation and insulin resistance is that the body needs chromium to produce a substance called glucose tolerance factor, which increases the effectiveness of insulin. Further studies need to be done, however, before recommendations about dietary chromium as a treatment for insulin resistance can be made.

Prognosis

Since insulin resistance is a condition that precedes the appearance of symptoms of a number of different disorders, its prognosis depends in part on the patient's age, ethnicity, family history, and severity of any current health problems. Some patients diagnosed with insulin resistance eventually develop type 2 diabetes, but it is not yet known why the others do not; for example, some patients do not develop diabetes in spite of a high degree of insulin resistance. What is known at present is that weight reduction and exercise can control or even reverse insulin resistance in many people.

KEY TERMS

Acanthosis nigricans—A dark brownish or blackish discoloration of the skin related to overweight and high levels of insulin in the blood. Acanthosis nigricans is most likely to develop in the groin or armpits, or around the back of the neck.

Bariatrics—The branch of medicine that deals with the prevention and treatment of obesity and related disorders.

Body mass index (BMI)—A measurement that has replaced weight as the preferred determinant of obesity. The BMI can be calculated (in English units) as 703.1 times a person's weight in pounds divided by the square of the person's height in inches.

Glucose—A simple sugar produced when carbohydrates are broken down in the small intestine. It is the primary source of energy for the body. Various tests that measure blood glucose levels are used in diagnosing insulin resistance.

Hyperandrogenism—Excessive secretion of androgens (male sex hormones).

Hyperinsulinemia—The medical term for high levels of insulin in the blood.

Insulin—A protein hormone secreted by the islets of Langerhans in the pancreas in response to eating. Insulin carries glucose and amino acids to muscle and adipose cells and promotes their efficient use and storage.

Islets of Langerhans—Special structures in the pancreas responsible for insulin secretion among other functions. They are named for Paul Langerhans, the German researcher who first identified them in 1869.

Lipids—A group of fats and fat-like substances that are not soluble in water, are stored in the body, and serve as a source of fuel for the body.

Metabolic syndrome—A group of risk factors for heart disease, diabetes, and stroke. It includes abdominal obesity, high blood pressure, high blood glucose levels, and low levels of high-density lipoprotein (HDL) cholesterol. The metabolic syndrome is sometimes called the insulin resistance syndrome.

Metabolism—The sum of an organism's physical and chemical processes that produce and maintain living tissue, and make energy available to the organism. Insulin resistance is a disorder of metabolism.

Obesity—Excessive weight gain due to accumulation of fat in the body, sometimes defined as a BMI of 30 or higher, or body weight greater than 30 percent above one's desirable weight on standard height-weight tables.

Pancreas—A large gland located behind the stomach near the spleen that secretes digestive enzymes into the small intestine and insulin into the bloodstream.

Syndrome—In general, a set of symptoms that occur together as signs of a disease or disorder.

Syndrome X—A term that was sometimes used for metabolic syndrome when the syndrome was first identified in the 1960s.

Triglycerides—Fatty compounds synthesized from carbohydrates during the process of digestion and stored in the body's adipose (fat) tissues. High levels of triglycerides in the blood are associated with insulin resistance.

Type 2 diabetes mellitus—One of the two major types of diabetes mellitus, characterized by late age of onset (30 years or older), insulin resistance, high levels of blood sugar, and little or no need for supplemental insulin. It was formerly known as adult-onset or non-insulin-dependent diabetes.

Prevention

Genetic factors contributing to insulin resistance cannot be changed as of the early 2000s.

With regard to lifestyle factors, the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) reported the findings of a study of the effects of lifestyle changes or metformin on the incidence of diabetes in a group of over 3200 overweight people with impaired glucose tolerance,

which is a risk factor for developing type 2 diabetes. The researchers found that the subjects in the lifestyle modification group, who lowered their food intake and took 30-minute walks five days a week, had a 58-percent lower incidence of diabetes. The subjects who received metformin had a 31-percent lower incidence of diabetes. Lifestyle changes were most effective in volunteers over the age of 60, while metformin was most effective in younger subjects. In short, the 2002 study confirmed the

beneficial effects of lowered food intake and increased activity as preventive measures against type 2 diabetes.

Another important part of preventing insulin resistance is patient education. A number of resources on weight control and exercise written for the general public are available from the Weight-Control Information Network (WIN) on the NIDDK website at <http://win.niddk.nih.gov/publications/physical.htm>. Some pamphlets are available in Spanish as well as English. Patient education materials on insulin resistance in relation to heart disease and diabetes can be downloaded free of charge from the American Heart Association and American Diabetes Association websites.

Resources

BOOKS

- “Diabetes Mellitus.” Section 2, Chapter 13 in *The Merck Manual of Diagnosis and Therapy*, edited by Mark H. Beers, MD, and Robert Berkow, MD. Whitehouse Station, NJ: Merck Research Laboratories, 2004.
- Flancbaum, Louis, MD, with Erica Manfred and Deborah Biskin. *The Doctor's Guide to Weight Loss Surgery*. West Hurley, NY: Fredonia Communications, 2001.
- “Nutritional Disorders: Obesity.” Section 1, Chapter 5 in *The Merck Manual of Diagnosis and Therapy*, edited by Mark H. Beers, MD, and Robert Berkow, MD. Whitehouse Station, NJ: Merck Research Laboratories, 2004.
- Pelletier, Kenneth R., MD. *The Best Alternative Medicine*. New York: Simon & Schuster, 2002.

PERIODICALS

- Boney, C. M., A. Verma, R. Tucker, and B. R. Vohr. “Metabolic Syndrome in Childhood: Association with Birth Weight, Maternal Obesity, and Gestational Diabetes Mellitus.” *Pediatrics* 115 (March 2005): 290–296.
- Diabetes Prevention Program Research Group. “Reduction in the Incidence of Type 2 Diabetes with Lifestyle Intervention or Metformin.” *New England Journal of Medicine* 346 (February 7, 2002): 393–403.
- Ford, Earl S., MD, MPH, Wayne H. Giles, MD, MSc, and William H. Dietz, MD, PhD. “Prevalence of the Metabolic Syndrome Among US Adults.” *Journal of the American Medical Association* 287 (January 16, 2002): 356–359.
- Litonjua, P., A. Pinero-Pilona, L. Aviles-Santa, and P. Raskin. “Prevalence of Acanthosis Nigricans in Newly-Diagnosed Type 2 Diabetes.” *Endocrine Practice* 10 (March–April 2004): 101–106.
- Olatunbosun, Samuel, MD, and Samuel Dagogo-Jack, MD. “Insulin Resistance.” *eMedicine*, 3 June 2004. <<http://www.emedicine.com/med/topic1173.htm>>.
- Rao, Goutham, MD. “Insulin Resistance Syndrome.” *American Family Physician* 63 (March 15, 2001): 1159–1166.

Scheinfeld, N. S. “Obesity and Dermatology.” *Clinical Dermatology* 22 (July–August 2004): 303–309.

Sivitz, William I., MD. “Understanding Insulin Resistance: What Are the Clinical Implications?” *Postgraduate Medicine* 116 (July 2004): 41–48.

ORGANIZATIONS

- American Academy of Dermatology (AAD). P. O. Box 4014, Schaumburg, IL 60168-4014. (847) 330-0230. Fax: (847) 330-0050. <<http://www.aad.org>>.
- American Diabetes Association. 1701 North Beauregard Street, Alexandria, VA 22311. (800) 342-2383. <<http://www.diabetes.org>>.
- American Heart Association National Center. 7272 Greenville Avenue, Dallas, TX 75231. (800) 242-8721. <<http://www.americanheart.org>>.
- American Obesity Association (AOA). 1250 24th Street NW, Suite 300, Washington, DC 20037. (202) 776-7711 or (800) 98-OBESE. <www.obesity.org>.
- National Diabetes Information Clearinghouse (NDIC). 1 Information Way, Bethesda, MD 20892-3560. (800) 860-8747. Fax: (703) 738-4929.

OTHER

- Agency for Healthcare Research and Quality (AHRQ). Evidence Report/Technology Assessment: Number 41. *Ayurvedic Interventions for Diabetes Mellitus*. Rockville, MD: AHRQ, 2001. <<http://www.ahrq.gov/clinic/epcsums/ayurvedsum.htm>>.
- Agency for Healthcare Research and Quality (AHRQ). Evidence Report/Technology Assessment: Number 89. *Effects of Omega-3 Fatty Acids and Glycemic Control in Type II Diabetes and the Metabolic Syndrome and on Inflammatory Bowel Disease, Rheumatoid Arthritis, Renal Disease, Systemic Lupus Erythematosus, and Osteoporosis*. Rockville, MD: AHRQ, 2004. <<http://www.ahrq.gov/clinic/epcsums/o3lipidsum.htm>>.
- Mayo Clinic Staff. *Metabolic Syndrome*. <<http://www.mayoclinic.com/invoke.cfm?id=DS00522>>.
- National Diabetes Information Clearinghouse (NDIC). *Insulin Resistance and Pre-Diabetes*. NIH Publication No. 04-4893. Bethesda, MD: National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK), 2004. <<http://diabetes.niddk.nih.gov/dm/pubs/insulinresistance>>.
- National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) News Brief, 6 February 2002. “Diet and Exercise Delay Diabetes and Normalize Blood Glucose.” <<http://www.niddk.nih.gov/welcome/releases/02-06-02.htm>>.

Rebecca Frey, PhD

Intelligence tests see **Stanford-Binet intelligence scales**; **Wechsler intelligence test**

Intention tremor see **Tremors**

Interferon see **Antiviral drugs; Immunologic therapies**

Interleukin-2 see **Immunologic therapies**

Intermittent claudication

Definition

Intermittent claudication is a **pain** in the leg that a person experiences when walking or exercising. The pain is intermittent and goes away when the person rests.

Description

Claudication comes from the Latin word that means “to limp,” and the condition is characterized by intermittent pain in the leg muscles. Poor circulation produces the pain. The legs do not receive the oxygen-rich blood supply needed for activities like walking and exercising. The decreased blood flow is caused by the narrowing of arteries that bring blood to the leg and foot.

The leg pain produced by claudication is usually experienced as cramping in the thighs, calves, hips, and feet. The pain stops several minutes after the person rests, but returns when the person performs activities that use the leg muscles. If untreated, claudication is no longer intermittent and occurs when a person is resting.

People age 50 or older are at risk of intermittent claudication. The risk increases with age. Statistically, 5% of men and 2.5% of women experience symptoms of this condition, according to the Vascular Disease Foundation (VDF). Many people with this condition have cholesterol plaque build-up in the leg’s arteries. **Smoking** raises the risk, as do high blood pressure, **obesity**, lack of **exercise**, and diabetes.

Intermittent claudication is the main symptom of peripheral arterial disease (PAD), which is also known as **peripheral vascular disease**, and or occlusive arterial disease. Intermittent claudication is an early symptom of the condition that affects peripheral arteries, those blood vessels located outside the heart. PAD is caused by arteriosclerosis, the narrowing and hardening of the arteries.

Women with diabetes are 7.6 times as likely to experience PAD as those without diabetes, according to the American Diabetes Association (ADA). Furthermore, intermittent claudication is related to a

two to three times increased risk of coronary heart disease, **stroke**, or cardiac failure in men with diabetes, according to the association.

Causes & symptoms

Intermittent claudication is caused by poor circulation and is experienced in a person’s muscle groups. It affects the peripheral arteries that convey oxygen-rich blood from the heart to the legs. A person with this condition feels pain shortly after beginning to exercise. Walking may trigger the pain in an inactive person. Activities such as climbing stairs, walking rapidly or dancing can cause the pain to increase.

The person feels the pain as tightness in the calf, thigh, or buttocks. The pain stops after the person rests for several minutes. However, it returns when the person reaches the exertion level that previously produced the pain.

Intermittent claudication is the primary symptom of PAD, the condition causing reduced flow of blood and oxygen to tissues. If the intermittent condition is not treated, the person will find that resting does not relieve pain. As arteries become more clogged, the person could feel pain even when not exercising. Symptoms include cold or numb feet and toes, poor balance when standing, a drop in leg strength, sores that heal slowly, and **erectile dysfunction (impotence)**.

In the advanced stages of PAD, the person experiences pain when resting. This condition, ischemic rest pain, is characterized by symptoms visible on the feet and toes. These include ulcers, loss of hair, and the change to red color when feet are suspended. Other symptoms include blue or purple markings on the legs, feet and toes. The coloring is a sign that less oxygen is reaching these areas. Furthermore, black skin on the legs and feet is a sign of **gangrene** infection.

Diagnosis

People experiencing symptoms of intermittent claudication should contact their doctor immediately. The doctor will review the patient’s medical history and examine the person for signs of the condition. This examination includes checking for a lower pulse or the absence of a pulse behind the knee, on the ankle, foot, and groin.

The doctor may order an ankle-brachial index (ABI) test to determine whether arteries are blocked. This procedure will verify if the person has PAD and provides information on the severity of the condition. The ABI measures blood pressure in the arms and

ankle. Readings are taken when the person is at rest and after exercising lightly by walking on a treadmill. The ABI index is found by dividing the ankle blood pressure by the pressure for the arm. An ABI below 0.90 in a person at rest is a sign of PAD.

The physician may also order a Doppler ultrasound exam to measure the flow of blood through the arteries. Cuffs are placed on four places on each leg. The doctor then moves an ultrasound probe over arteries in the foot. The probe detects signals from the artery.

Testing can last from 20 minutes to an hour. Costs vary by location and facility. In March of 2005, the ABI test was priced from \$45 to \$150. Some facilities offered a package of tests for around \$200. The procedures are usually covered by medical insurance.

Treatment

Lifestyle changes are the primary form of treating intermittent claudication. Physicians advise people to quit smoking, exercise, and to follow the American Heart Association's healthy diet guidelines. Diabetics need to control blood sugar levels. The patient may need to lower blood pressure and cholesterol levels. Those measurements should drop when a person makes lifestyle changes.

The goal of treatment is stop development of PAD. By exercising, eating a diet that includes fiber and low-fat foods, and not smoking, a person could also reverse the build-up that clogs arteries. After several months of this regimen, many people experience a lessening of leg pain. If pain continues, the doctor may prescribe medication. Furthermore, surgery may be needed in some cases.

Walking

Walking is frequently an important treatment for intermittent claudication. A person experiencing the pain of intermittent claudication may not feel like walking. However, walking can increase the capacity to exercise. Before starting an exercise program, the person should consult with a doctor.

The physician reviews the patient's medical history, does a physical and may order an exercise **stress test** on a treadmill. The test shows how long a person walks before claudication starts. Information such as blood pressure is used to evaluate the person's ability to walk. The findings are also used to develop a medically supervised exercise program.

At the beginning, the treadmill is set to cause claudication symptoms in three to five minutes,

according to the Vascular Disease Foundation. The person walks until pain is moderately severe. A rest period is scheduled after the person walks eight to 10 minutes.

The person walks and rests, with the goal of walking for a total of 35 minutes. The person walks at least three times weekly. If a treadmill is not available, VDF recommends that people walk on a track. Generally, a person walking three times weekly will be able to walk longer after three to 12 months.

Medications

People diagnosed with PAD are at a high risk for a stroke and **heart attack**. They should take **aspirin** to reduce this risk. Clopidogrel, a drug marketed as Plavix, worked even better than aspirin in a study. However, it was more costly, with 30 tablets selling for about \$119 in 2005. The dosage is one tablet daily.

A doctor may also prescribe cilostazol—sold under the name Pletal—which extends the distance people can walk without pain. One tablet is taken twice daily. The medication cost around \$112 for 60 tablets.

Surgical procedures

Surgical procedures may be necessary in cases where intermittent claudication is disabling. The person experiences pain when resting, has open sores that do not heal, or symptoms of gangrene like dying skin in the leg or foot.

Bypass surgery directs blood through a grafted blood vessel, bypassing the damaged artery. The grafted vessel is either a healthy artery or vein or an artificial vessel.

Angioplasty is a procedure to open blocked blood vessels. A catheter (tube) is inserted in the groin and moved to the artery. Then a tiny balloon is inflated to open the artery. Another angioplasty procedure involves the insertion of a stent, a metal device that keeps the vessel open.

Angioplasty is a minimally invasive procedure. **Local anesthesia** is used, and a person is able to resume normal activities within one to two days. The cost averaged \$6,502, according to a 2001 Dartmouth Medical School study. Bypass surgery cost \$12,422 on average, according to the study. A non-invasive procedure is not as risky as surgery. However, a bypass may be needed when multiple sections of blood vessels are blocked.

Once a person is diagnosed with intermittent claudication, health plans usually cover part of

KEY TERMS

Arthrosclerosis—A condition where fatty deposits cause the arteries to narrow.

Gangrene—The decay of tissue in the body; it is caused by an obstruction to the blood supply.

treatment costs. A study on the overall costs of this condition from diagnosis through treatment should be completed by the end of 2005, according to a Sheryl Benjamin, VDF executive director.

Alternative treatment

Ginkgo biloba extract, an herbal remedy, has been used by people with intermittent claudication. The extract made from the dried leaves of the Ginkgo tree is thought to improve blood flow, allowing people to walk longer without pain.

However, herbal remedies are not regulated the U.S. Food and Drug Administration, and people should consult with their doctors before taking Ginkgo. Furthermore, use of this remedy could interact adversely when taken with Vitamin E and some medications.

Prognosis

If untreated, intermittent claudication will advance and eventually restrict a person's mobility. In later stages, people feel pain when resting. The leg or foot may feel cold. In the extreme stage, the person might need a cane, walker, or wheelchair. There is more risk of gangrene developing, and **amputation** might be necessary. Diabetics face an increased risk of amputation.

PAD also increases the risk of heart attacks and strokes.

Prevention

A healthy lifestyle is the best method for preventing intermittent claudication. Cigarette smokers should quit smoking. Regular exercise and a healthy diet help reduce the risk of this condition. If necessary, people should work to lower cholesterol and blood pressure. Diabetics should strive to manage that condition, obese people should lose weight.

The methods of preventing intermittent claudication are also the means for managing the risks associated with a diagnosis of PAD.

People can learn more about peripheral vascular disease through public education programs like the free Legs for Life screenings held at sites across the nation. The program started the Society of Interventional Radiology features a free ABI testing.

Resources

PERIODICALS

- "New Treatment for Leg Pain—Conference News Update." *Clinician Reviews*, April 2002 [cited March 25, 2005]. <http://www.findarticles.com/p/articles/mi_m0BUY/is_4_12/ai_86050499>.
- Warkentin, Donald L. "Intermittent Claudication." *Clinical Reference Systems*. January 1, 2004, 1813.

ORGANIZATIONS

- American Diabetes Association. 1701 North Beauregard Street, Alexandria, VA 22311. 800-342-2383. <<http://www.diabetes.org>>.
- Society of Interventional Radiology. 10201 Lee Highway, Suite 500, Fairfax, VA 22030. 703-691-1805. <<http://www.sirweb.org/index.shtml>>.
- Vascular Disease Foundation. 1075 S. Yukon Street, Suite 320, Lakewood, CO 80226. 866-723-4636. <http://www.vdf.org/Contact_Frame.htm>.

OTHER

- "Intermittent Claudication: 8 Ways to Ease the Pain." The Doctors Book of Home Remedies, Rodale. [cited March 25, 2005]. <<http://www.mothernature.com/Library/Bookshelf/Books/47/85.cfm>>.
- "Peripheral Arterial Disease of the Legs." Kaiser Permanente. Updated November 3, 2003 [cited March 25, 2005]. <<http://prospectivemembers.kaiserpermanente.org>>.

Liz Swain

Intermittent explosive disorder

Definition

Intermittent explosive disorder (IED) is a mental disturbance that is characterized by specific episodes of violent and aggressive behavior that may involve harm to others or destruction of property. IED is discussed in the *Diagnostic and Statistical Manual of Mental Disorders* fourth edition (DSM-IV) under the heading of "Impulse-Control Disorders Not Elsewhere Classified." As such, it is grouped together with kleptomania, pyromania, and pathological gambling.

A person must meet certain specific criteria to be diagnosed with IED:

- There must be several separate episodes of failure to restrain aggressive impulses that result in serious assaults against others or property destruction.
- The degree of aggression expressed must be out of proportion to any provocation or other stressor prior to the incidents.
- The behavior cannot be accounted for by another mental disorder, **substance abuse**, medication side effects, or such general medical conditions as epilepsy or head injuries.

The reader should note that DSM-IV's classification of IED is not universally accepted. Many psychiatrists do not place intermittent explosive disorder into a separate clinical category but consider it a symptom of other psychiatric and mental disorders. In many cases individuals diagnosed with IED do in fact have a dual psychiatric diagnosis. IED is frequently associated with mood and **anxiety disorders**, **substance abuse** and eating disorders, and narcissistic, paranoid, and antisocial **personality disorders**.

Description

People diagnosed with IED sometimes describe strong impulses to act aggressively prior to the specific incidents reported to the doctor and/or the police. They may experience racing thoughts or a heightened energy level during the aggressive episode, with **fatigue** and depression developing shortly afterward. Some report various physical sensations, including tightness in the chest, **tingling** sensations, tremor, hearing echoes, or a feeling of pressure inside the head.

Many people diagnosed with IED appear to have general problems with anger or other impulsive behaviors between explosive episodes. Some are able to control aggressive impulses without acting on them while others act out in less destructive ways, such as screaming at someone rather than attacking them physically.

Although the editors of DSM-IV stated in 2000 that IED "is apparently rare," a group of researchers in Chicago reported in 2004 that it is more common than previously thought. They estimate that 1.4 million persons in the United States currently meet the criteria for IED, with a total of 10 million meeting the lifetime criteria for the disorder.

With regard to sex and age group, 80% of individuals diagnosed with IED in the United States are adolescent and adult males. Women do experience

IED, however, and have reported it as part of **premenstrual syndrome (PMS)**.

Causes and symptoms

Causes

As with other impulse-control disorders, the cause of IED has not been determined. As of 2004, researchers disagree as to whether it is learned behavior, the result of biochemical or neurological abnormalities, or a combination of factors. Some scientists have reported abnormally low levels of serotonin, a neurotransmitter that affects mood, in the cerebrospinal fluid of some anger-prone persons, but the relationship of this finding to IED is not clear. Similarly, some individuals diagnosed with IED have a medical history that includes migraine headaches, seizures, attention-deficit hyperactivity disorder, or developmental problems of various types, but it is not clear that these cause IED, as most persons with migraines, learning problems, or other neurological disorders do not develop IED.

Some psychiatrists who take a cognitive approach to mental disorders believe that IED results from rigid beliefs and a tendency to misinterpret other people's behavior in accordance with these beliefs. According to Dr. Aaron Beck, a pioneer in the application of cognitive therapy to violence-prone individuals, most people diagnosed with IED believe that other people are basically hostile and untrustworthy, that physical force is the only way to obtain respect from others, and that life in general is a battlefield. Beck also identifies certain characteristic errors in thinking that go along with these beliefs:

- **Personalizing.** The person interprets others' behavior as directed specifically against him.
- **Selective perception.** The person notices only those features of situations or interactions that fit his negative view of the world rather than taking in all available information.
- **Misinterpreting the motives of others.** The person tends to see neutral or even friendly behavior as either malicious or manipulative.
- **Denial.** The person blames others for provoking his violence while denying or minimizing his own role in the fight or other outburst.

Symptoms

The symptoms of IED are described by the DSM-IV criteria for diagnosing the disorder.

Diagnosis

The diagnosis of IED is basically a diagnosis of exclusion, which means that the doctor will eliminate such other possibilities as neurological disorders, mood or substance abuse disorders, **anxiety** syndromes, and personality disorders before deciding that the patient meets the DSM-IV criteria for IED. In addition to taking a history and performing a **physical examination** to rule out general medical conditions, the doctor may administer one or more psychiatric inventories or screeners to determine whether the person meets the criteria for other mental disorders.

In some cases the doctor may order imaging studies or refer the person to a neurologist to rule out brain tumors, traumatic injuries of the nervous system, epilepsy, or similar physical conditions.

Treatment

Emergency room treatment

A person brought to a hospital emergency room by family members, police, or other emergency personnel after an explosive episode will be evaluated by a psychiatrist to see whether he can safely be released after any necessary medical treatment. If the patient appears to be a danger to himself or others, he may be committed against his will for further treatment. In terms of legal issues, a doctor is required by law to notify the specific individuals as well as the police if the patient threatens to harm particular persons. In most states, the doctor is also required by law to report suspected abuse of children, the elderly, or other vulnerable family members.

The doctor will perform a thorough medical examination to determine whether the explosive outburst was related to substance abuse, withdrawal from drugs, head trauma, **delirium**, or other physical conditions. If the patient becomes assaultive inside the hospital, he may be placed in restraints or given a tranquilizer (usually either lorazepam [Ativan] or diazepam [Valium]), most often by injection. In addition to the physical examination, the doctor will obtain as detailed a history as possible from the family members or others who accompanied the patient.

Medications

Medications that have been shown to be beneficial in treating IED in nonemergency situations include lithium, carbamazepine (Tegretol), propranolol (Inderal), and such **selective serotonin reuptake inhibitors** as fluoxetine (Prozac) and sertraline (Zoloft).

Adolescents diagnosed with IED have been reported to respond well to clozapine (Clozaril), a drug normally used to treat **schizophrenia** and other psychotic disorders.

Psychotherapy

Some persons with IED benefit from cognitive therapy in addition to medications, particularly if they are concerned about the impact of their disorder on their education, employment, or interpersonal relationships. Psychoanalytic approaches are not useful in treating IED.

Prognosis

The prognosis of IED depends on several factors that include the individual's socioeconomic status, the stability of his or her family, the values of the surrounding neighborhood, and his or her motivation to change. One reason why the Chicago researchers think that IED is more common than previously thought is that most people who meet the criteria for the disorder do not seek help for the problems in their lives that result from it. The researchers found that although 88% of the 253 individuals with IED that they studied were upset by the results of their explosive outbursts, only 13% had ever asked for treatment in dealing with it.

Prevention

Since the cause(s) of IED are not fully understood as of the early 2000s, preventive strategies should focus on treatment of young children (particularly boys) who may be at risk for IED before they enter adolescence.

Resources

BOOKS

- American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders*, 4th ed., revised. Washington, D.C.: American Psychiatric Association, 2000.
- Beck, Aaron T., MD. *Prisoners of Hate: The Cognitive Basis of Anger, Hostility, and Violence*. New York: HarperCollins Publishers, 1999.
- Beers, Mark H., MD., and Robert Berkow, MD, editors. "Psychiatric Emergencies." Section 15, Chapter 194. In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

PERIODICALS

- Citrome, Leslie L., MD, MPH, and Jan Volavka, MD. "Aggression." *eMedicine*. February 8, 2002. <<http://www.emedicine.com/Med/topic3005.htm>>.

KEY TERMS

Cognitive therapy—A form of short-term psychotherapy that focuses on changing people's patterns of emotional reaction by correcting distorted patterns of thinking and perception.

Delirium—An acute but temporary disturbance of consciousness marked by confusion, difficulty paying attention, delusions, hallucinations, or restlessness. Delirium may be caused by drug intoxication, high fever related to infection, head trauma, brain tumors, kidney or liver failure, or various metabolic disturbances.

Kleptomania—A mental disorder characterized by impulsive stealing.

Neurotransmitter—Any of a group of chemicals that transmit nerve impulses across the gap (synapse) between two nerve cells.

Pyromania—A mental disorder characterized by setting fires.

Serotonin—A neurotransmitter or brain chemical that is responsible for transporting nerve impulses.

Coccaro, E. F., C. A. Schmidt, J. F. Samuels, and G. Nestadt. "Lifetime and 1-Month Prevalence Rates of Intermittent Explosive Disorder in a Community Sample." *Journal of Clinical Psychiatry* 65 (June 2004): 820–824.

Grant, J. E., and M. N. Potenza. "Impulse Control Disorders: Clinical Characteristics and Pharmacological Management." *Annals of Clinical Psychiatry* 16 (January–March 2004): 27–34.

Kant, R., R. Chalansani, K. N. Chengappa, and M. F. Dieringer. "The Off-Label Use of Clozapine in Adolescents with Bipolar Disorder, Intermittent Explosive Disorder, or Posttraumatic Stress Disorder." *Journal of Child and Adolescent Psychopharmacology* 14 (Spring 2004): 57–63.

McElroy, Susan L. "Recognition and Treatment of DSM-IV Intermittent Explosive Disorder." *Journal of Clinical Psychiatry* (1999): 12–16.

ORGANIZATIONS

American Academy of Child and Adolescent Psychiatry. 3615 Wisconsin Avenue, NW, Washington, DC 20016-3007. (202) 966-7300. Fax: (202) 966-2891. <<http://www.aacap.org>>.

American Psychiatric Association. 1400 K Street, NW, Washington, DC 20005. <<http://www.psych.org>>.

National Institute of Mental Health. 6001 Executive Boulevard, Room 8184, MSC 9663, Bethesda, MD 20892-9663. (301) 443-4513. <<http://www.nimh.nih.gov>>.

OTHER

Padgitt, Steven T. "Treating Intermittent Explosive Disorder with Neurofeedback" Behavenet.com. May 7, 2001. <<http://www.behavenet.com/capsules/disorders/explosivedis.htm>>.

Janie F. Franz
Rebecca Frey, PhD

Internal fetal monitoring see **Electronic fetal monitoring**

Internuclear ophthalmoplegia see **Ophthalmoplegia**

Interpositional reconstruction see **Arthroplasty**

Intersex states

Definition

Intersex states are conditions where a newborn's sex organs (genitals) look unusual, making it impossible to identify the sex of the baby from its outward appearance.

Description

All developing babies start out with external sex organs that look female. If the baby is male, the internal sex organs mature and begin to produce the male hormone testosterone. If the hormones reach the tissues correctly, the external genitals that looked female change into the scrotum and penis. Sometimes, the genetic sex (as indicated by chromosomes) may not match the appearance of the external sex organs. About 1 in every 2,000 births results in a baby whose sex organs look unusual.

Patients with intersex states can be classified as a true hermaphrodite, a female pseudohermaphrodite, or a male pseudohermaphrodite. This is determined by examining the internal and external structures of the child.

A true hermaphrodite is born with both ovaries and testicles. They also have mixed male and female external genitals. This condition is extremely rare.

A female pseudohermaphrodite is a genetic female. However, the external sex organs have been masculinized and look like a penis. This may occur if the mother takes the hormone progesterone to prevent



This infant was born with female and male genitalia.
(Photography by Mike Peres, Custom Medical Stock Photo.
Reproduced by permission.)

a **miscarriage**, but more often it is caused by an overproduction of certain hormones.

A male pseudohermaphrodite is a genetic male. However, the external sex organs fail to develop normally. Intersex males may have testes and a female-like vulva, or a very small penis.

Causes and symptoms

Any abnormality in chromosomes or sex hormones, or in the unborn baby's response to the hormones, can lead to an intersex state in a newborn.

Intersex states may also be caused by a condition called **congenital adrenal hyperplasia**, which occurs in about 1 out of every 5,000 newborns. This disease blocks the baby's metabolism and can cause a range of symptoms, including abnormal genitals.

Diagnosis

When doctors are uncertain about a newborn's sex, a specialist in infant hormonal problems is consulted as soon as possible. Ultrasound can locate a uterus behind the bladder and can determine if there is a cervix or uterine canal. Blood tests can check the levels of sex hormones in the baby's blood, and chromosome analysis (called **karyotyping**) can determine sex. Explorative surgery or a biopsy of reproductive tissue may be necessary. Only after thorough testing can a correct diagnosis and determination of sex be made.

Treatment

Treatment of intersex states is controversial. Traditional treatment assigns sex according to test

KEY TERMS

Chromosomes—Spaghetti-like structures located within the nucleus (or central portion) of each cell. Chromosomes contain the genetic information necessary to direct the development and functioning of all cells and systems in the body. They pass on hereditary traits from parents to child (like eye color) and determine whether the child will be male or female.

results, the potential for the child to identify with a sex, and the ease of genital surgery to make the organs look more normal. Treatment may then include **reconstructive surgery** followed by hormone therapy. Babies born with congenital adrenal hyperplasia can be treated with cortisone-type drugs and sometimes surgery.

Counseling should be given to the entire family of an intersex newborn. Families should explore all available medical and surgical options. Counseling should also be provided to the child when he or she is old enough.

Prognosis

Since the mid-1950s, doctors have typically assigned a sex to an intersex infant based on how easy reconstructive surgery would be. The American Academy of Pediatrics states that children with these types of genitals can be raised successfully as members of either sex, and recommends surgery within the first 15 months of life.

Some people are critical of this approach, including intersex adults who were operated on as children. The remolded genitals do not function sexually and can be the source of lifelong **pain**. They suggest that surgery be delayed until the patient can make informed choices about surgery and intervention.

Resources

ORGANIZATIONS

Ambiguous Genitalia Support Network. P.O. Box 313,
Clements, CA 95227. (209) 727-0313.

Intersex Society. P.O. Box 31791, San Francisco, CA 94131.

Carol A. Turkington

Interstitial microwave thermal therapy

Definition

Interstitial microwave thermal therapy is a type of hyperthermia treatment for **cancer**, in which heat produced by microwaves (which are a non-ionizing form of radiation) is used in conjunction with other cancer treatments, such as radiation or **chemotherapy**, to kill cancer cells associated with tumors located deep within the body.

Purpose

The purpose of interstitial microwave thermal therapy is to damage and kill cancer cells associated with tumors that are deep within the body. Interstitial microwave therapy is a type of hyperthermia cancer treatment procedure (also called thermal therapy or thermotherapy) in which body tissue and the cancerous tumor are exposed to high temperatures (up to 113°F). Hyperthermia kills cancer cells with usually only minimal injury to normal tissues by damaging proteins and structures within the cells. Thermal therapy is usually used with other forms of cancer therapy, such as radiation and chemotherapy. The increased temperatures may make some cancer cells more sensitive to radiation or may harm some cancer cells that radiation cannot damage.

Precautions

If a patient has become insensitive to **pain** due to disease, radiation, surgery, anesthetics, or other conditions, hyperthermia treatment cannot be used to treat tumors. Also the excessive heating of normal surrounding tissue is prevented by normal blood perfusion, so hyperthermia should not be used in patients with known circulatory problems in the heated areas or in patients who are taking vasoconstrictive drugs.

Description

Interstitial microwave thermal therapy is used to treat tumors that are deep within the body, such as brain, cervical, breast, prostate, and neck tumors. This technique allows the tumor to be heated to higher temperatures than external thermal therapy techniques allow. Probes or needles are inserted into the tumor, guided by the use of imaging techniques, such as ultrasound, to make sure that the probe is properly located within the tumor. A new type of microwave generator includes electronic phase control that allows the

operator to electronically direct and shape the pattern of hyperthermia treatments based on the positioning of the microwave antennae array that is used in treating the tumor. The treatment pattern can be electronically targeted on the tumor position, shape, and size.

Tissues are heated as the electromagnetic energy produced by the microwave treatment results in heating through molecular excitation. This energy is dissipated in normal living tissue by the blood that perfuses through the tissue. However, since large solid malignant tumors have less blood perfusion than the surrounding normal tissue, for a given absorbed thermal dose, the tumor reaches a higher temperature than the normal tissue. Tumors present within normal tissue will therefore be preferentially heated and will reach higher temperatures than the surrounding tissue.

As cancerous tumors grow rapidly, their need for blood quickly begins to exceed the available blood supply, and major portions of the tumors become blood-starved. These blood-starved tumors are resistant to both radiation and chemotherapy. Chemotherapy drugs carried through the blood cannot effectively penetrate tumors that have poor blood flow. Poor blood flow also means that the tumors are oxygen-starved (hypoxic), making it difficult for **radiation therapy** to make the oxygen radicals that are needed to destroy the DNA of cancer cells. Hypoxic cancer cells, which are an especially dangerous type of cancer, for they have a tendency to metastasize and spread the cancer to other parts of the body, are three times more resistant to ionizing radiation than are normal cells.

When the tumor is heated to **fever** levels through the use of microwave thermal therapy, its blood vessels expand so that more blood can flow into the tumor in order to carry away the excess heat. With the increased blood flow, more blood-borne chemotherapy drugs can be carried into the tumor. Blood is also the source of oxygen for tumors, so with increased blood flow due to the thermal therapy, radiation therapy can form the necessary oxygen radicals to kill the cancer cells. The increased temperature also acts as a drug activator, accelerating chemical reactions and pulling increased oxygen molecules into the tumor tissue for chemical reactions with the chemotherapy drug. **Hypothermia** also enhances the effectiveness of chemotherapy drugs encapsulated in liposomes, increasing the penetration of the drug into the tumor.

When tumors are heated to higher temperatures for at least an hour, the tumors in some cases have been shown to decrease in size and to exhibit necrosis

(death of the tumor cells). Therefore, hyperthermia by itself also tends to shrink tumors, often dramatically, due to the collapse of dead cancer cells, making it easier to remove the tumor by surgical techniques. For tumors of the head and neck, a smaller tumor due to hyperthermia treatment may reduce the disfigurement associated with surgical removal of the tumor.

Hyperthermia is being studied as a means of enhancing **gene therapy** by acting as an activator to turn on new biological therapies, by speeding up gene production by thousands of times. In addition, hyperthermia is used as an essential tool to turn on anti-tumor vaccines that are based on heat shock proteins. Hyperthermia has been shown to prevent a cancerous tumor from growing new blood vessels to expand its blood supply. With regards to quality of life, hyperthermia lessens pain and stimulates the immune system, thus helping patients recover from toxic cancer therapies such as ionizing radiation and chemotherapy. Even in patients with terminal cancer, hyperthermia can provide benefits through the alleviation of bleeding, pain, and infection.

As of 2005, the use of hyperthermia alone and in conjunction with radiation therapy has been approved by the United States Food and Drug Administration for the treatment of advanced, recurrent, and persistent tumors, upon authorization of a licensed practitioner. When used with radiation, the treatment regimen usually consists of 10 hyperthermia treatments delivered twice a week, at 72 hour intervals. The prescribed radiation is administered within 30 minutes of the hyperthermia treatment. During each heat treatment, the temperature within the tumor is usually maintained at 42.5 degrees centigrade for 60 minutes. The use of hyperthermia in conjunction with chemotherapy is presently investigational in the United States as of 2005.

The effectiveness of treatment is related to the temperature achieved during the treatment process, the length of the treatment, and cell and tissue characteristics. The temperature within the tumor must be monitored to ensure that the appropriate temperature is achieved, but not exceeded, in the treatment area. Monitoring is accomplished by inserting small needles or tubes with small thermometers into the treatment area. Imaging techniques such as computed tomography (CT) are used to make sure that the temperature probes are positioned appropriately.

Preparation

The safety and effectiveness of hyperthermia treatment is dependent on careful placement of the

temperature probes and careful monitoring of tissue temperatures during treatment.

Aftercare

During the treatment period, which may last for weeks, the patient must be instructed in proper care of implanted catheters and temperature probe sites to avoid the risk of infection.

Risks

Excessive heating of normal tissues may result in areas of thermal aseptic necrosis that will require medical care. The electromagnetic radiation from the microwave equipment can interfere with electronic devices such as cardiac **pacemakers** or other implanted electronic devices. Thermal treatment of tumors in the neck or head may result in inadvertent heating of thermoregulatory centers in the brain stem, thus resulting in overheating of the body, beyond levels that the patient can tolerate. Metallic implants, such as joint prostheses or dental braces may become excessively and preferentially overheated and adversely affect the patient.

Normal results

The effectiveness of interstitial microwave thermal therapy varies among cancer patients. For example, studies have shown in Phase III clinical trials, when hyperthermia was used with ionizing radiation treatment, that the following treatment improvements were seen, as compared to the use of ionizing radiation therapy alone:

- complete response for recurrent **breast cancer** increased from 38% to 60%
- 2-year survival for glioblastoma (brain cancer) increased from 15% to 31%
- complete response for advanced **cervical cancer** increased from 57% to 83%

Known side effects of hypothermia are associated with direct effects of heat on tissues and indirect effects of tumor necrosis. These side effects, as determined in various medical studies, include:

- surface **burns** and blistering in the area of application of heat by the microwave applicators; experienced in about 10% of the tumor sites studied.
- localized and temporary pain in the area of and during the delivery of the heat by the microwave applicators; experienced in about 8% of the tumor sites studied.

KEY TERMS

Blood perfusion—A physiological term that refers to the process of nutritive delivery of arterial blood to a capillary bed in the biological tissue.

Hyperthermia therapy—A type of treatment in which body tissue is exposed to high temperatures to damage and kill cancer cells or to make cancer cells more sensitive to the effects of radiation and certain anticancer drugs.

Non-ionizing radiation—Rays of energy that move in long, slow wave patterns and do not penetrate cells.

Ionizing radiation—Any electromagnetic or particulate radiation capable of producing ions, directly or indirectly, in its passage through matter.

- ulceration from rapid tumor necrosis following successful hyperthermia treatment, resulting in fever from toxemia and patient discomfort through drainage and bleeding; experienced in about 4% of the tumor sites studied.
- local and systematic (rarely) infections from placement of the temperature probes and from ulceration related to tumor necrosis; experienced in about 2% of the tumor sites studied.

Abnormal results

Hyperthermia has the potential for producing the following adverse reactions as a result of exposure to electromagnetic radiation:

- cataracts
- permanent or temporary male sterility
- exacerbation of existing diseases due to additional systemic stress
- enhanced drug activity
- thermal stress

Resources

BOOKS

Rosenbaum, Ernest, Dollinger, Malin, Tempero, Margaret, and Mulvihill, Sean. *Everyone's Guide to Cancer Therapy*. 4th Edition. Kansas City, KS: Andrews McMeel Publishing, 2002

PERIODICALS

Falk, M.H., and Issels, R.D. "Hyperthermia in Oncology." *International Journal of Hyperthermia*. 2001, (17) 1–18.

van der Zee, J. "Heating the Patient: A Promising Approach?" *Annals of Oncology* 2002, (13) 1173–1184.

Wust, P., Hildebrandt, B., Sreenivasa, G. et al. "Hyperthermia in Combined Treatment of Cancer." *The Lancet Oncology* 2002, (3) 487–497.

ORGANIZATION

Hyperthermia in Cancer Treatment. National Cancer Institute. <www.cis.nci.gov/fact/7_3.htm>

Society for Thermal Medicine. <www.thermaltherapy.org>.

Judith L. Sims

Intestinal culture see **Stool culture**

Intestinal lymphangiectasia see **Malabsorption syndrome**

Intestinal obstructions

Definition

Intestinal obstruction refers to the partial or complete mechanical or nonmechanical blockage of the small or large intestine.

Description

There are two types of intestinal obstructions, mechanical and nonmechanical. Mechanical obstructions occur because the bowel is physically blocked and its contents cannot get past the obstruction. Mechanical obstructions can occur for several reasons. Sometimes the bowel twists on itself (volvulus) or telescopes into itself (**intussusception**). Mechanical obstruction can also result from hernias, impacted feces, abnormal tissue growth, the presence of foreign bodies in the intestines (including **gallstones**), or inflammatory bowel disease (**Crohn's disease**). Nonmechanical obstruction, called **ileus**, occurs because the wavelike muscular contractions of the intestine (peristalsis) that ordinarily move food through the digestive tract stop.

Mechanical obstruction in infants

Infants under one year of age are most likely to have intestinal obstruction caused by meconium ileus, volvulus, and intussusception. Meconium ileus, which is the inability to pass the first fecal excretion after birth (meconium), is a disorder of newborns. It is an early clue that the infant has **cystic fibrosis**, but may also occur in very low birth weight (VLBW) infants. In meconium ileus, the material that is blocking the

intestine is thick and stringy, rather than the collection of mucus and bile that is passed by normal infants. The abnormal meconium must be removed with an enema or through surgery.

Volvulus is the medical term for twisting of either the small or large bowel. The twisting may cut off the blood supply to the bowel, leading to tissue **death (gangrene)**. This development is called a strangulating obstruction.

In intussusception, the bowel telescopes into itself like a radio antenna folding up. Intussusception is most common in children between the ages of three and nine months, although it also occurs in older children. Almost twice as many boys suffer intussusception as girls. It is, however, difficult for doctors to predict which infants will suffer from intestinal obstruction.

Mechanical obstruction in adults

Obstructions in adults are usually caused by tumors, trauma, volvulus, the presence of foreign bodies such as gallstones, or hernias, although they have also been reported in adults with cystic fibrosis. Volvulus occurs most often in elderly adults and psychiatrically disturbed patients. Intussusception in adults is usually associated with tumors in the bowel, whether benign or malignant.

More recently, gastroenterologists have described a postsurgical complication known as early postoperative small bowel obstruction, or EPSBO. Although this condition was at one time confused with postoperative ileus, it is now known to be caused by mechanical obstructions resulting from **radiation therapy for cancer** or laparoscopic surgery. Most cases can be successfully treated within 10–14 days of surgery.

Causes and symptoms

One of the earliest signs of mechanical intestinal obstruction is abdominal **pain** or cramps that come and go in waves. Infants typically pull up their legs and cry in pain, then stop crying suddenly. They will then behave normally for as long as 15–30 minutes, only to start crying again when the next cramp begins. The cramping results from the inability of the muscular contractions of the bowel to push the digested food past the obstruction.

Vomiting is another symptom of intestinal obstruction. The speed of its onset is a clue to the location of the obstruction. Vomiting follows shortly after the pain if the obstruction is in the small intestine but is delayed if it is in the large intestine. The vomited

material may be fecal in character. When the patient has a mechanical obstruction, the doctor will first hear active, high-pitched gurgling and splashing bowel sounds while listening with a stethoscope. Later these sounds decrease, then stop. If the blockage is complete, the patient will not pass any gas or feces. If the blockage is only partial, however, the patient may have **diarrhea**. Initially there is little or no **fever**.

When the material in the bowel cannot move past the obstruction, the body reabsorbs large amounts of fluid and the abdomen becomes sore to the touch and swollen. The balance of certain important chemicals (electrolytes) in the blood is upset. Persistent vomiting can cause the patient to become dehydrated. Without treatment, the patient can suffer **shock** and kidney failure.

Strangulation occurs when a loop of the intestine is cut off from its blood supply. Strangulation occurs in about 25% of cases of small bowel obstruction. It is a serious condition that can progress to gangrene within six hours.

Diagnosis

Imaging studies

If the doctor suspects intestinal obstruction based on the **physical examination** and patient history, he or she will order x rays, a computed tomography scan (CT scan), or an ultrasound evaluation of the abdomen. In many cases the patient is given a **barium enema**. Barium sulfate, which is a white powder, is inserted through the rectum and the intestinal area is photographed. Barium acts as a contrast material and allows the location of the obstruction to be pinpointed on film.

Laboratory tests

The first blood test of a patient with an intestinal obstruction usually gives normal results, but later tests indicate electrolyte imbalances. There is no way to determine if an obstruction is simple or strangulated except surgery.

Treatment

Initial assessment

All patients with suspected intestinal obstruction are hospitalized. Treatment must be rapid, because strangulating obstructions can be fatal. The first step in treatment is inserting a nasogastric tube to suction out the contents of the stomach and intestines. The patient is then given intravenous fluids to prevent **dehydration** and correct electrolyte imbalances.

KEY TERMS

Electrolytes—Salts and minerals that ionize in body fluids. Electrolytes control the body's fluid balance as well as performing other important functions.

Gangrene—The death of soft tissue in any part of the body when the blood supply is obstructed.

Ileus—Obstruction of the intestines caused by the absence of peristalsis.

Intussusception—The slipping or telescoping of one part of the intestine into the section next to it.

Meconium—A greenish fecal material that forms the first bowel movement of an infant.

Peristalsis—The waves of muscular contraction in the intestines that push the food along during the process of digestion.

Strangulated obstruction—An obstruction in which a loop of the intestine has its blood supply cut off.

Volvulus—A twisting of the intestine that causes an obstruction.

Nonsurgical approaches

Surgery can be avoided for some patients. In some cases of volvulus, guiding a rectal tube into the intestines will straighten the twisted bowels. In infants, a barium enema may reverse intussusception in 50–90%. An air enema is sometimes used instead of a barium enema. This treatment successfully relieves the obstruction in many infants. The children are usually hospitalized for observation for two to three days after these procedures. In patients with only partial obstruction, a barium enema may dissolve the blockage.

Surgical treatment

If these efforts fail, surgery is necessary. Strangulated obstructions require emergency surgery. The obstructed area is removed and part of the bowel is cut away. If the obstruction is caused by tumors, polyps, or scar tissue, they are removed. Hernias, if present, are repaired. **Antibiotics** are given to reduce the possibility of infection.

Alternative treatment

Alternative practitioners offer few suggestions for treatment. They focus on preventive strategies,

particularly the use of high-fiber **diets** to keep the bowels healthy through regular elimination.

Prognosis

Mortality

Untreated intestinal obstructions can be fatal. Delayed diagnosis of volvulus in infants has a mortality rate of 23–33% with prompt diagnosis and treatment the mortality rate is 3–9%. The bowel either strangulates or perforates, causing massive infection. With prompt treatment, however, most patients recover without complications.

Recurrence

As many as 80% of patients whose volvulus is treated without surgery have recurrences. Recurrences in infants with intussusception are most likely to happen during the first 36 hours after the blockage has been cleared. The mortality rate for unsuccessfully treated infants is 1–2%.

Prevention

Most cases of intestinal obstruction are not preventable. Surgery to remove tumors, polyps, or gallstones helps prevent recurrences.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD, editors. "Mechanical Intestinal Obstruction." Section 3, Chapter 25. In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

PERIODICALS

Chahine, A. Alfred, MD. "Intussusception." *eMedicine* June 10, 2004. <<http://www.emedicine.com/ped/topic1208.htm>>.

Dray, X., T. Bienvenu, N. Desmazes-Dufeu, et al. "Distal Intestinal Obstruction Syndrome in Adults with Cystic Fibrosis." *Clinical Gastroenterology and Hepatology* 2 (June 2004): 498–503.

Emil, S., T. Nguyen, J. Sills, and G. Padilla. "Meconium Obstruction in Extremely Low-Birth-Weight Neonates: Guidelines for Diagnosis and Management." *Journal of Pediatric Surgery* 39 (May 2004): 731–737.

Hebra, Andre, MD, and Melissa Miller, MD. "Intestinal Volvulus." *eMedicine* February 25, 2004. <<http://www.emedicine.com/ped/topic1205.htm>>.

Sajja, S. B., and M. Schein. "Early Postoperative Small Bowel Obstruction." *British Journal of Surgery* 91 (June 2004): 683–691.

OTHER

HealthAnswers.com. “Intestinal Obstruction.” <<http://www.healthanswers.com/database/ami/converted/000260.html>>.

“Intussusception: A Case Study for Nurses.” *University of New Brunswick Nursing Faculty Page*. <<http://www.unb.ca/courses/nur4284/intu.htm>>.

Tish Davidson, A.M.
Rebecca Frey, PhD

Intestinal polyps

Definition

The word polyp refers to any overgrowth of tissue from the surface of mucous membranes. Intestinal polyps grow out of the lining of the small and large bowels. Polyps come in a variety of shapes—round, droplet, and irregular being the most common.

Description

Polyps are one of many forms of tissue overproduction that can occur in the body. Cells in many body tissues sometimes keep growing beyond their usual limits. Medical scientists call this process *neoplasia*, which means simply “new growth.” An individual overgrowth is called a neoplasm. In most cases these growths are limited, and the result is a benign swelling or mass of cells called a tumor. If the new growth occurs on the surface of the tissue instead of inside an organ it is often called a polyp. **Cancer** is another type of neoplasm marked by unlimited tissue growth. The essential feature that distinguishes cancer from nonmalignant neoplasms is that it does not stop growing.

Intestinal polyps are a common form of neoplasm. All intestinal polyps arise from the inner lining of the intestinal wall. This layer of mucosal tissue does the work of digestion. About 30% of the general population will develop intestinal polyps at some point in life, with the likelihood increasing with age. Most of these polyps are never noticed during a person’s lifetime because they cause no problems. They are often discovered accidentally at **autopsy**. The primary importance of intestinal polyps is that 1% of them become cancerous. Because the polyps that eventually turn malignant cannot be identified in advance, they are all suspect.

Location of intestinal polyps

The chances of a polyp’s becoming cancerous depend to some extent on its location within the digestive tract.

COLON. Ninety-five percent of all intestinal polyps develop inside the large bowel. There are several hereditary diseases that produce large numbers of intestinal polyps. These disorders include:

- Familial polyposis of the colon.
- Gardner’s syndrome.
- Lynch’s syndrome.
- Turcot’s syndrome.
- Peutz-Jeghers syndrome.
- Juvenile polyposis.

All of these disorders are inherited in what is called an autosomal dominant pattern. This pattern means that the disorders are not sex-linked and that a child can inherit the disorder from either parent. In all of these hereditary disorders, the intestinal polyps appear during or after **puberty**. The first four diseases on the list have such a high rate of cancer of the large bowel (colon)—virtually 100% by the age of 40—that persons diagnosed with any of them should have the colon removed surgically in early adulthood.

STOMACH. The stomach’s lining is host to polyps of a similar appearance, but there is no agreement as to their potential for becoming **stomach cancer**.

SMALL INTESTINE. Polyps in the small bowel do not seem to have malignant potential. Instead they can produce obstruction in either of two ways. A large polyp can obstruct the bowel by its sheer size. Smaller polyps can be picked up by the rhythmic contractions (peristalsis) of the intestines and pull the part of the bowel to which they are attached into the adjoining section. The result is a telescoping of one section of bowel into another, called **intussusception**.

Causes and symptoms

Population studies of **colon cancer** suggest that diet plays an important role in the disease, and by implication in the formation of colon polyps. The most consistent interpretation of these data is that animal fats—though not vegetable fats—are the single most important dietary factor. Lack of fiber in the diet may also contribute to polyp formation. Other types of polyps are too rare to produce enough data for evaluation.

Most polyps cause no symptoms. Large ones eventually cause intestinal obstruction, which produces

KEY TERMS

Autosomal dominance—A pattern of heredity in which a trait is inherited without respect to sex and from either parent. The hereditary diseases associated with intestinal polyps are all autosomal dominant.

Colectomy—Surgical removal of the large bowel.

Intussusception—The slipping of one section of the intestine inside an adjoining section. Intussusception can be caused by small intestinal polyps.

Mucosal—Refers to tissues that produce mucus, such as the digestive, genital and urinary tracts.

Neoplasm—A new growth of abnormal tissue.

Peristalsis—The rhythmic contractions of muscular tubes like the intestines that carry the contents along the tube.

Sigmoid—The S-shaped curve of the large intestine where the colon joins the rectum.

cramping abdominal **pain** with **nausea and vomiting**. As colon polyps evolve into cancers, they begin to produce symptoms that include bleeding and altered bowel habits.

Diagnosis

Routine screening for bowel cancer is recommended for everyone over the age of 40. Screening may be as simple as testing the stool for blood or as elaborate as **colonoscopy**. Colonoscopy is a procedure in which the doctor threads an instrument called a colonoscope up through the entire large bowel. Most polyps are in the lower segment of the colon, called the sigmoid colon. These polyps can be seen with a shorter scope called a sigmoidoscope. X ray imaging can also be used to look for polyps. For x rays, the colon is first filled with barium, which is a white substance that shows up as a shadowed area on the film. The colon can also be filled with barium and air, which is called a double contrast study.

Because polyps take about five years to turn into cancers, routine examinations are recommended every three years.

Treatment

All polyps should be removed as preventive care. Most of them can be taken out through a colonoscope. Complications like obstruction and intussusception are surgical emergencies.

Prevention

Patients with hereditary disorders associated with polyps must undergo total colectomy early in adult life. All children of parents with these disorders should be screened early in adulthood, because half of them will have the same disease. For the bulk of the population, increased dietary fiber and decreased animal fat are the best preventives known at present. For the occasional intestinal polyp that arises in spite of good dietary habits, routine screening should prevent it from becoming cancerous.

Resources

BOOKS

Silverstein, Fred E. "Gastrointestinal Endoscopy." In *Harrison's Principles of Internal Medicine*, edited Anthony S. Fauci, et al. New York: McGraw-Hill, 1997.

J. Ricker Polsdorfer, MD

Intestinal strangulation see **Intestinal obstructions**

Intoxication confusional state see **Delirium**

Intracavity therapy see **Radioactive implants**

Intracranial abscess see **Brain abscess**

Intrapartum monitoring see **Electronic fetal monitoring**

Intrauterine device see **IUD**

Intrauterine growth retardation

Definition

Intrauterine growth retardation (IUGR) occurs when the unborn baby is at or below the 10th weight percentile for his or her age (in weeks).

Description

There are standards or averages in weight for unborn babies according their age in weeks. When the baby's weight is at or below the 10th percentile for his or her age, it is called intrauterine growth retardation or fetal growth restriction. These babies are smaller than they should be for their age. How much a baby weighs at birth depends not only on how many weeks old it is, but the rate at which it has

grown. This growth process is complex and delicate. There are three phases associated with the development of the baby. During the first phase, cells multiply in the baby's organs. This occurs from the beginning of development through the early part of the fourth month. During the second phase, cells continue to multiply and the organs grow. In the third phase (after 32 weeks of development), growth occurs quickly and the baby may gain as much as 7 ounces per week. If the delicate process of development and weight gain is disturbed or interrupted, the baby can suffer from restricted growth.

IUGR is usually classified as symmetrical or asymmetrical. In symmetrical IUGR, the baby's head and body are proportionately small. In asymmetrical IUGR, the baby's brain is abnormally large when compared to the liver. In a normal infant, the brain weighs about three times more than the liver. In asymmetrical IUGR, the brain can weigh five or six times more than the liver.

Causes and symptoms

Doctors think that the two types of IUGR may be linked to the time during development that the problem occurs. Symmetrical IUGR may occur when the unborn baby experiences a problem during early development. Asymmetrical IUGR may occur when the unborn baby experiences a problem during later development. While not true for all asymmetrical cases, doctors think that sometimes the placenta may allow the brain to get more oxygen and **nutrition** while the liver gets less.

There are many IUGR risk factors involving the mother and the baby. A mother is at risk for having a growth restricted infant if she:

- Has had a previous baby who suffered from IUGR
- Is small in size
- Has poor weight gain and nutrition during **pregnancy**
- Is socially deprived
- Uses substances (like tobacco, **narcotics**, alcohol) that can cause abnormal development or **birth defects**
- Has a vascular disease (like **preeclampsia**)
- Has chronic **kidney disease**
- Has a low total blood volume during early pregnancy
- Is pregnant with more than one baby
- Has an antibody problem that can make successful pregnancy difficult (antiphospholipid antibody syndrome).

KEY TERMS

Preeclampsia—Hypertension (high blood pressure) during pregnancy.

Additionally, an unborn baby may suffer from IUGR if it has:

- Exposure to an infection, including German **measles (rubella)**, cytomegalovirus, **tuberculosis**, **syphilis**, or **toxoplasmosis**
- A birth defect (like a severe cardiovascular defect)
- A chromosome defect, especially trisomy 18 (**Edwards' syndrome**)
- A primary disorder of bone or cartilage
- A chronic lack of oxygen during development (hypoxia)
- Placenta or umbilical cord defects
- Developed outside of the uterus.

Diagnosis

IUGR can be difficult to diagnose and in many cases doctors are not able to make an exact diagnosis until the baby is born. A mother who has had a growth restricted baby is at risk of having another during a later pregnancy. Such mothers are closely monitored during pregnancy. The length in weeks of the pregnancy must be carefully determined so that the doctor will know if development and weight gain are appropriate. Checking the mother's weight and abdomen measurements can help diagnose cases when there are no other risk factors present. Measuring the girth of the abdomen is often used as a tool for diagnosing IUGR. During pregnancy, the healthcare provider will use a tape measure to record the height of the upper portion of the uterus (the uterine fundal height). As the pregnancy continues and the baby grows, the uterus stretches upward in the direction of the mother's head. Between 18 and 30 weeks of gestation, the uterine fundal height (in cm.) equals the weeks of gestation. If the uterine fundal height is more than 2–3 cm below normal, then IUGR is suspected. Ultrasound is used to evaluate the growth of the baby. Usually, IUGR is diagnosed after week 32 of pregnancy. This is during the phase of rapid growth when the baby should be gaining more weight. IUGR caused by genetic factors or infection may sometimes be detected earlier.

Treatment

There is no treatment that improves fetal growth, but IUGR babies who are at or near term have the best outcome if delivered promptly. If IUGR is caused by a problem with the placenta and the baby is otherwise healthy, early diagnosis and treatment of the problem may reduce the chance of a serious outcome.

Prognosis

Babies who suffer from IUGR are at an increased risk for **death**, low blood sugar (**hypoglycemia**), low body temperature (**hypothermia**), and abnormal development of the nervous system. These risks increase with the severity of the growth restriction. The growth that occurs after birth cannot be predicted with certainty based on the size of the baby when it is born. Infants with asymmetrical IUGR are more likely to catch up in growth after birth than are infants who suffer from prolonged symmetrical IUGR. However, as of 1998, doctors cannot reliably predict an infant's future progress. Each case is unique. Some infants who have IUGR will develop normally, while others will have complications of the nervous system or intellectual problems like **learning disorders**. If IUGR is related to a disease or a genetic defect, the future of the infant is related to the severity and the nature of that disorder.

Resources

BOOKS

Cunningham, F. Gary, et al. *Williams Obstetrics*. 20th ed. Stamford: Appleton & Lange, 1997.

Linda Jones

Intravenous nutrition see **Nutrition through an intravenous line**

Intravenous pyelography see **Intravenous urography**

Intravenous rehydration

Definition

Sterile water solutions containing small amounts of salt or sugar, are injected into the body through a tube attached to a needle which is inserted into a vein.

KEY TERMS

Intravenous—Into a vein; a needle is inserted into a vein in the back of the hand, inside the elbow, or some other location on the body. Fluids, nutrients, and drugs can be injected.

Purpose

Fever, **vomiting**, and **diarrhea** can cause a person to become dehydrated fairly quickly. Infants and children are especially vulnerable to **dehydration**. Patients can become dehydrated due to an illness, surgery, or accident. Athletes who have overexerted themselves may also require rehydration with IV fluids. An IV for rehydration can be used for several hours to several days, and is generally used if a patient cannot drink fluids.

Precautions

Patients receiving IV therapy need to be monitored to ensure that the IV solutions are providing the correct amounts of fluids and **minerals** needed. People with kidney and heart disease are at increased risk for **overhydration**, so they must be carefully monitored when receiving IV therapy.

Description

Basic IV solutions are sterile water with small amounts of sodium (salt) or dextrose (sugar) supplied in bottles or thick plastic bags that can hang on a stand mounted next to the patient's bed. Additional minerals like potassium and calcium, **vitamins**, or drugs can be added to the IV solution by injecting them into the bottle or bag with a needle.

Preparation

A doctor orders the IV solution and any additional nutrients or drugs to be added to it. The doctor also specifies the rate at which the IV will be infused. The IV solutions are prepared under the supervision of a doctor, pharmacist, or nurse, using sanitary techniques that prevent bacterial contamination. Just like a prescription, the IV is clearly labeled to show its contents and the amounts of any additives. The skin around the area where the needle is inserted is cleaned and disinfected. Once the needle is in place, it will be taped to the skin to prevent it from dislodging.

Aftercare

Patients need to take fluids by mouth before an IV solution is discontinued. After the IV needle is removed, the site should be inspected for any signs of bleeding or infection.

Risks

There is a small risk of infection at the injection site. It is possible that the IV solution may not provide all of the nutrients needed, leading to a deficiency or an imbalance. If the needle becomes dislodged, it is possible that the solution may flow into tissues around the injection site rather than into the vein.

Resources

OTHER

Martinez-Bianchi, Viviana, Michelle Rejman-Peterson, and Mark A. Graber. "Pediatrics: Vomiting, Diarrhea, and Dehydration." *Family Practice Handbook*. University of Iowa. <<http://www.vh.org/Providers/ClinRef/FPHandbook/Chapter10/17-10.html>>.

Toth, Peter P. "Gastroenterology: Acute Diarrhea." *Family Practice Handbook*. University of Iowa. <<http://www.vh.org/Providers/ClinRef/FPHandbook/Chapter04/01-4.html>>.

Altha Roberts Edgren

Intravenous urography

Definition

Intravenous urography is a test which x rays the urinary system using intravenous dye for diagnostic purposes.

Of the many ways to obtain images of the urinary system, the intravenous injection of a contrast agent has been traditionally considered the best. The kidneys excrete the dye into the urine. X rays can then create pictures of every structure through which the urine passes.

The procedure has several variations and many names.

- Intravenous pyelography (IVP).
- Urography.
- Pyelography.
- Antegrade pyelography differentiates this procedure from "retrograde pyelography," which injects dye

into the lower end of the system, therefore flowing backward or "retrograde." Retrograde pyelography is better able to define problems in the lower parts of the system and is the only way to get x rays if the kidneys are not working well.

- Nephrotomography is somewhat different in that the x rays are taken by a moving x ray source onto a film moving in the opposite direction. By accurately coordinating the movement, all but a single plane of tissue is blurred, and that plane is seen without overlying shadows.

Every method available gives good pictures of this system, and the question becomes one of choosing among many excellent alternatives. Each condition has special requirements, while each technique has distinctive benefits and drawbacks.

- Nuclear scans rely on the radiation given off by certain atoms. Chemicals containing such atoms are injected into the bloodstream. They reach the kidneys, where images are constructed by measuring the radiation emitted. The radiation is no more dangerous than standard x rays. The images require considerable training to interpret, but unique information is often available using this technology. Different chemicals can concentrate the radiation in different types of tissue. This technique may require several days for the chemical to concentrate at its destination. It also requires a special detector to create the image.
- Ultrasound is a quick, safe, simple, and inexpensive way to obtain views of internal organs. Although less detailed than other methods, it may be sufficient.
- Retrograde pyelography is better able to define problems in the lower parts of the system and is the only way to get x rays if the kidneys are not working well. Dye is usually injected through an instrument (cystoscope) passed into the bladder through the urethra.
- Computed tomography scans (CT or CAT scanning) uses the same kind of radiation used in x rays, but it collects information by computer in such a way that three dimensional images can be constructed, eliminating interference from nearby structures. CT scanning requires a special apparatus.
- Magnetic resonance imaging (MRI) uses magnetic fields and radio frequency signals, instead of ionizing radiation, to create computerized images. This form of energy is entirely safe as long as the patient has no metal on board. The technique is far more versatile than CT scanning. MRI requires special apparatus and, because of the powerful magnets needed, even a special building all by itself. It is quite expensive.

Purpose

Most diseases of the kidneys, ureters, and bladder will yield information to this procedure, which actually has two phases. First, it requires a functioning kidney to filter the dye out of the blood into the urine. The time required for the dye to appear on x rays correlates accurately with kidney function. The second phase gives detailed anatomical images of the urinary tract. Within the first few minutes the dye “lights up” the kidneys, a phase called the nephrogram. Subsequent pictures follow the dye down the ureters and into the bladder. A final film taken after urinating reveals how well the bladder empties.

IVPs are most often done to assess structural abnormalities or obstruction to urine flow. If kidney function is at issue, more films are taken sooner to catch the earliest phase of the process.

- Stones, tumors and congenital malformations account for many of the findings.
- Kidney cysts and cancers can be seen.
- Displacement of a kidney or ureter suggests a space-occupying lesion like a **cancer** pushing it out of the way.
- Bad valves where the ureters enter the bladder will often show up.
- Bladder cancers and other abnormalities are often outlined by the dye in the bladder.
- An **enlarged prostate** gland will show up as incomplete bladder emptying and a bump at the bottom of the bladder.

Precautions

The only serious complication of an IVP is allergy to the iodine-containing dye that is used. Such an allergy is rare, but it can be dramatic and even lethal. Emergency measures taken immediately are usually effective.

Description

IVPs are usually done in the morning. In the x ray suite, the patient will undress and lie down. There are two methods of injecting the dye. An intravenous line can be established, through which the dye will be consistently fed through the body during the procedure. The other method is to give the dye all at once through a needle that is immediately withdrawn. X rays are taken until the dye has reached the bladder, an interval of half an hour or less. The patient will be asked to empty the bladder before one last x ray.

KEY TERMS

Contrast agent—Any substance that causes shadows on x rays, also known as contrast dye or medium.

Intravenous—Into a vein.

Preparation

Emptying the bowel with **laxatives** or **enemas** prevents bowel shadows from obscuring the details of the urinary system. An empty stomach prevents the complications of **vomiting**, a rare effect of the contrast agent. Therefore, the night before the IVP the patient will be asked to evacuate the bowels and to drink sparingly.

Risks

Allergy to the contrast agent is the only risk. Anyone with a possible iodine allergy or a previous reaction to x ray dye must be particularly careful to inform the x ray personnel.

Resources

BOOKS

Merrill, Vinta. *Atlas of Roentgenographic Positions and Standard Radiologic Procedures*. Saint Louis: The C.V. Mosby Co., 1975.

J. Ricker Polsdorfer, MD

Intussusception

Definition

Intussusception is the enfolding of one segment of the intestine within another. It is characterized and initially presents with recurring attacks of cramping abdominal **pain** that gradually become more painful.

Description

Intussusception occurs when part of the bowel or intestine is wrapped around itself producing a mass-like object on the right side of the abdomen during palpation (a procedure used during a **physical examination**, when the examiner touches the abdomen with

his/her hand, usually feeling for mass, pain, or discomfort). The number of new cases of intussusception is approximately 1.5 to four cases per 1,000 live births. The onset of abdominal pain is usually abrupt and severe. Just as fast as the onset of pain appears, it disappears and the child resumes activity normally. This process of sudden severe abdominal pain appearing out of the blue then disappearing is repeated with duration of painful attacks. The pain usually increases after approximately five hours of recurrent cycles of severe abdominal pain followed by relaxation. **Vomiting** and **diarrhea** occur in about 90% of cases with six to 12 hours after initial onset of symptoms.

Physical examination and palpation usually reveal a sausage shaped mass of enfolded bowel in the right upper mid portion of the abdomen. Within a few hours approximately 50% of cases have bloody, mucus filled bowel movements. At about this time the child is visibly very ill with **fever**, tenderness, and distended abdomen. Intussusception is the most frequent cause of intestinal obstruction during the first two years of life and commonly affects children between three to 12 months of age. The disease is three times more common in males than in females. In about 85% of cases the cause is idiopathic (meaning unknown). The remaining 15% of cases can be caused by a variety of other diseases such as tumors of the lymph nodes (lymphoma), fat tumors (lipomas), foreign bodies/objects, or from infections that mobilize immune cells to the area causing an inflammatory reaction and intestinal blockage. Most cases of intussusception do not strangulate the affected bowel within the first 24 hours. If the disease is not treated after this time, the possibility of intestinal **gangrene**, **shock**, and **death** increases.

Causes and symptoms

The major symptom of intussusception is when a healthy child suddenly and without warning experiences severe abdominal pain that subsides and usually results in continuation of normal activities such as playing. The duration of the painful attacks increases as the hours go by. Usually, the child develops **nausea**, vomiting, and diarrhea soon afterwards in about 90% of all cases. The child becomes weak, exhausted, and develops a fever. The affected child may also expel bloody, mucus-like bowel movements. These blood filled bowel movements are usually due to impaired blood flow to the obstructed area. During palpation there may be a sausage-shaped mass located on the upper right mid portion of the abdomen. If the disease progresses and

KEY TERMS

Barium—A chemical used in certain radiological studies to enhance visualization of anatomical structures.

Obstruction—A blockage that prevents movement.

is undetected, the child may develop necrosis death of cells within the affected area. Additionally, there may be perforation or hole in the intussusception bowel that can cause a life threatening infection in the peritoneum (a layer of tissue that protects the organs and intestines within the abdominal cavity). This infection of the peritoneum is called **peritonitis**. Some patients may exhibit altered states of consciousness or seizures.

Diagnosis

A presumed diagnosis can be made by history alone. If the clinician suspects intussusception x-ray films should be performed, which may reveal a mass in the right upper mid abdominal region. Two classical clinical signs are mucus-blood filled stools and a “coiled string” appearance in the affected bowel as visualized during an x ray with a **barium enema**. Blood chemistry analysis is not specific for intussusception. Depending on vomiting and blood loss through the stools, blood chemistry may reflect signs of **dehydration** and anemia.

Treatment

Treating intussusception by reduction (alleviating the source of blockage) is an emergency procedure. The barium examination is not only the diagnostic tool of choice, but also frequently curative. Infusion by gravity from a catheter placed in the rectum will tend to relieve pressure buildup. If this does not relieve the area, then air can be pumped into the colon to clear blockage. If these procedures are unsuccessful then surgery is required. Approximately 25% of affected children require surgical intervention. Surgery in the affected bowel is advantageous since the actual cause can be removed, and the procedure decreases the possibility of recurrences. In general without surgical correction of the affected bowel, there is a 5–10% chance of recurrence. Recurrence usually appears within the first 24 to 48 hours after barium procedure.

Prognosis

The outcome of intussusception depends on the duration of symptoms before treatment initiation. Most infants will recover if treatment is initiated within the first 24 hours. Untreated intussusception is almost always fatal. Overall even with treatment, approximately 1–2% of affected children will die.

Prevention

Prevention of death can be accomplished with immediate medical care, within the first 24 hours. Once intussusception is suspected, emergency measures should be initiated. Untreated intussusception is almost always fatal. There is an increased chance for death if the disorder is not treated within 48 hours.

Resources

BOOKS

Behrman, Richard E., et al, editors. *Nelson Textbook of Pediatrics*. 16th ed. W. B. Saunders Company, 2000.
Townsend, Courtney M., et al. *Sabiston Textbook of Surgery*. 16th ed. W. B. Saunders Company, 2001.

OTHER

University of Maryland. 2001. <<http://www.umm.edu>>.

Laith Farid Gulli, M.D.

Intussusception see **Intestinal obstructions**

Iodine see **Antiseptics**

Iodine uptake test see **Thyroid nuclear medicine scan**

Ipecac

Definition

Ipecac is a medicine commonly used to induce **vomiting** in cases of accidental **poisoning**. It is also a homeopathic remedy.

Purpose

Treatment of poisoning

At one time, standard medical practice recommended syrup of ipecac to cause vomiting in cases of poisoning in order to remove the toxic substance from the stomach before absorption occurs. More recently,

however, doctors are discouraging the use of ipecac in emergency treatment of poisoning. The reason for this change in practice is that ipecac has not been shown to be more effective than **activated charcoal** or gastric lavage; in addition, its use may delay the administration of these other methods of treatment. In 2004 the American College of Toxicology updated a 1997 position paper on the use of syrup of ipecac to treat poisoning with the following statement: “There is no evidence from clinical studies that ipecac improves the outcome of poisoned patients and its routine administration in the emergency department should be abandoned.”

Ipecac should *never* be used to induce vomiting if the poison is one of the following:

- strychnine
- alkalis (lye)
- strong acids
- kerosene
- fuel oil
- gasoline
- coal oil
- paint thinner
- cleaning fluid

In cases of poisoning, it is always best to contact a local poison control center, local hospital emergency room, or the family doctor for instructions before using syrup of ipecac.

Ipecac’s reputation for inducing vomiting has encouraged some bulimics to take it on a regular basis in order to purge the contents of the stomach after an eating binge. This misuse of ipecac is extremely dangerous; it can cause heart problems, tears in the esophagus or stomach lining, vomiting blood, seizures, or even **death**.

Homeopathy

The homeopathic remedy made from ipecac is called *Ipecacuanha*. Homeopathic preparations are given for a reason completely opposite from that of standard allopathic treatment. In homeopathy, ipecac is given to stop vomiting rather than to induce it. According to Hahnemann’s law of similars, a substance that would cause vomiting in large doses when given to a healthy person will stimulate a sick person’s natural defenses when given in extremely dilute and carefully prepared doses. *Ipecacuanha* is a favorite homeopathic remedy for morning sickness associated with **pregnancy**. It is also given to stop **nausea** that is not relieved by vomiting; when the vomitus is



Ipecac plant (*Cephaelis ipecacuanha*). (PlantaPhile Germany. Reproduced by permission.)

slimy and white; when there is gagging and heavy salivation; when the tongue is clean despite the patient's feelings of nausea; and when the patient is not thirsty. The nausea may be accompanied by a **headache, cough,** or heavy menstrual bleeding. The modalities (circumstances) that suggest *Ipecacuanha* as the appropriate homeopathic remedy is that the patient feels worse lying down; in dry weather; in winter; and when exercising or moving about.

A homeopathic practitioner would not necessarily prescribe ipecac for all cases of nausea. *Arsenicum* would be given when the nausea is caused by **food poisoning** and accompanied by strong thirst, *Nux vomica* when the nausea is the result of overindulgence in food or alcohol and accompanied by gas or **heartburn**. A sick child might be given *Pulsatilla*, particularly if rich foods have been eaten.

On the other hand, a homeopathic practitioner may prescribe ipecac for any of the following conditions that are not related to **nausea and vomiting**.

- Nosebleeds producing bright red blood.
- Dental bleeding.
- Diarrhea with cramping abdominal **pain**. The stools are green with froth or foam.
- Asthma of sudden onset. The patient has to sit up in order to breathe, but cannot bring up any mucus in spite of violent coughing.
- Hoarseness or loss of voice following a cold.
- Physical or mental exhaustion.

Description

The medicinal effects of ipecac were recognized centuries ago by the Portuguese who settled in South

America. They found a plant that can make people vomit and appropriately named it *Caephalis ipecacuanha*, meaning sick-making plant. Syrup of ipecac is now considered the safest drug to treat poisoning and is often the most effective. There are different types of ipecac preparations that vary greatly in strength. Syrup of ipecac is best for use at home to treat accidental poisoning. Ipecac fluid extract and ipecac tincture should be avoided as they are much stronger compounds and can be toxic.

Ipecacuanha is a homeopathic remedy made from ipecac by a process of dilution and succussion (shaking). In contrast to syrup of ipecac, it is given to relieve vomiting.

Recommended dosage

Syrup of ipecac

Syrup of ipecac is made from the dried roots and rhizomes (underground stems) of *Cephaelis ipecacuanha*. It is available over the counter in 0.5–1 oz bottles. Larger bottles require a doctor's prescription. The dosage for infants under 6 months old should be prescribed by the family doctor or poison control center. For children six months to one year, the usual dose is 5–10 ml or 1–2 tsp. One-half or one full glass (4–8 oz) of water should be taken immediately before or after the dose. The dose may be repeated once after 20–30 minutes if vomiting does not occur. For children one to 12 years of age, the usual dose is 15 ml (1 tbsp) to be taken with one full glass (8 oz) of water. Adults and teenagers should take 15–30 ml of ipecac with at least 1 full glass of water. Syrup of ipecac should not be taken with milk or soda drinks as these foods may prevent it from working properly. If vomiting does not occur within 20–30 minutes after the first dose, a second dose may be needed. If the second dose fails to induce vomiting, the patient should be taken to a hospital emergency room.

If both activated charcoal and syrup of ipecac are recommended to treat poison, ipecac must be used first. Activated charcoal should not be taken until 30 minutes after taking syrup of ipecac, or until the vomiting caused by ipecac stops.

Homeopathic preparations

Ipecacuanha is available as an over-the-counter remedy in 30x potency. This is a decimal potency, which means that one part of ipecac has been mixed with nine parts of alcohol or water; 30x means that this decimal dilution has been repeated 30 times. The dilute

solution of ipecac is then added to sugar tablets so that the remedy can be taken in tablet form.

Precautions

Syrup of ipecac

For inducing vomiting in cases of accidental poisoning, only the syrup form of ipecac should be used. Syrup of ipecac should not be mixed with milk or carbonated drinks as they may prevent vomiting.

Syrup of ipecac should not be used in the following situations (contact poison control center or family doctor for alternative treatments).

- Poisoning caused by strychnine; sustained-release theophylline; such corrosive substances as strong alkalis (lye); strong acids (such as toilet bowl cleaner); and such petroleum products as kerosene, gasoline, coal oil, fuel oil, paint thinner, or cleaning fluids.
- Overdoses of medications given for depression.
- Excessive vomiting.
- A serious heart condition.
- Timing. Do not give ipecac more than 4–6 hours after the poison was ingested.
- Pregnancy.
- Very young children (less than six months old). Infants and very young children may choke on their own vomit or get vomit into their lungs.
- Drowsy or unconscious patients.
- Seizures.

Homeopathic preparations

Ipecacuanha should not be given after *Arsenicum* or *Tabac* because these remedies will counteract it.

Side effects

The following side effects have been associated with the use of syrup of ipecac:

- Loose bowel movements.
- Diarrhea.
- Fast irregular heartbeat.
- Inhaling or **choking** on vomit.
- Stomach cramps or pains.
- Coughing.
- Weakness.
- Aching.

KEY TERMS

Bulimia nervosa—An eating disorder characterized by episodic binge eating followed by self-induced vomiting or laxative abuse.

Cephaeline—A chemical compound found in ipecac that irritates the stomach lining and triggers the vomiting reflex.

Fluid extract—A concentrated preparation of a drug.

Law of similars—A principle of homeopathic treatment according to which substances that cause specific symptoms in healthy people are given to sick people with similar symptoms.

Modality—A factor or circumstance that makes a patient's symptoms better or worse. Modalities include such factors as time of day, room temperature, the patient's level of activity, sleep patterns, etc.

Tincture—An alcoholic solution of a chemical or drug.

- Muscle stiffness.
- Severe heart problems often occur in cases of ipecac **abuse**. Because ipecac stays in the body for a long time, damage to the heart frequently occurs in persons who repeatedly take ipecac to induce vomiting.
- Seizures. These are most likely to occur in patients who accidentally swallow ipecac or in ipecac abusers.
- Death. Deaths have been reported due to ipecac abuse in bulimic persons.

Homeopathic *Ipecacuanha* has been highly diluted and is relatively nontoxic.

Interactions

If used to induce vomiting, ipecac should not be given together with other drugs because it can decrease their effectiveness and increase their toxicity. If both syrup of ipecac and activated charcoal are needed to treat suspected poisons, ipecac should be given first. Activated charcoal should not be given until vomiting induced by ipecac has stopped. Carbonated beverages should also be avoided because they can cause the stomach to swell. The person should lie on the stomach or side in case vomiting occurs.

Homeopathic *Ipecacuanha* is considered complementary to *Arnica* and *Cuprum*. It is counteracted by *Arsenicum* and *Tabac*.

Resources

BOOKS

- Beers, Mark H., MD, and Robert Berkow, MD, editors. "Bulimia Nervosa." Section 15, Chapter 196. In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.
- Beers, Mark H., MD, and Robert Berkow, MD, editors. "Poisoning." Section 23, Chapter 307. In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.
- PDR Nurse's Drug Handbook*. Montvale, NJ: Delmar Publishers, 2000.

PERIODICALS

- "Position Paper: Ipecac Syrup." *Journal of Toxicology: Clinical Toxicology* 42 (February 2004): 133–143.

ORGANIZATIONS

- American College of Toxicology (ACT). 9650 Rockville Pike, Bethesda, MD 20814. (301) 634-7840. Fax: (301) 634-7852. <<http://www.actox.org>>.
- American Foundation for Homeopathy. 1508 S. Garfield, Alhambra, CA 91801.
- Homeopathic Educational Services. 2124B Kittredge St. Berkeley, CA 94704. (510) 649-0294. Fax: (510) 649-1955.

Rebecca J. Frey, PhD

Ipratropium see **Bronchodilators**

I.Q. tests see **Stanford-Binet intelligence scales; Wechsler intelligence test**

Iridocyclitis see **Uveitis**

Iritis see **Uveitis**

Iron-binding capacity test see **Iron tests**

Iron-utilization anemias see **Sideroblastic anemia**

Iron deficiency anemia

Definition

Anemia can be caused by iron deficiency, folate deficiency, vitamin B₁₂ deficiency, and other causes. The term iron deficiency anemia means anemia that is due to iron deficiency. Iron deficiency anemia is

characterized by the production of small red blood cells. When examined under a microscope, the red blood cells also appear pale or light colored. For this reason, the anemia that occurs with iron deficiency is also called hypochronic microcytic anemia.

Description

Iron deficiency anemia is the most common type of anemia throughout the world. In the United States, iron deficiency anemia occurs to a lesser extent than in developing countries because of the higher consumption of red meat and the practice of food fortification (addition of iron to foods by the manufacturer). Anemia in the United States is caused by a variety of sources, including excessive losses of iron in menstrual fluids and excessive bleeding in the gastrointestinal tract. In developing countries located in tropical climates, the most common cause of iron deficiency anemia is infestation with hookworm.

Causes and symptoms

Infancy is a period of increased risk for iron deficiency. The human infant is born with a built-in supply of iron, which can be tapped during periods of drinking low-iron milk or formula. Both human milk and cow milk contain rather low levels of iron (0.5–1.0 mg iron/liter). However, the iron in human milk is about 50% absorbed by the infant, while the iron of cow milk is only 10% absorbed. During the first six months of life, growth of the infant is made possible by the milk in the diet and by the infant's built-in supply. However, premature infants have a lower supply of iron and, for this reason, it is recommended that preterm infants (beginning at 2 months of age) be given oral supplements of 7 mg iron/day, as ferrous sulfate. Iron deficiency can be provoked where infants are fed formulas that are based on unfortified cow milk. For example, unfortified cow milk is given free of charge to mothers in Chile. This practice has the fortunate result of preventing general **malnutrition**, but the unfortunate result of allowing the development of mild iron deficiency.

The normal rate of blood loss in the feces is 0.5–1.0 ml per day. These losses can increase with colorectal **cancer**. About 60% of colorectal cancers result in further blood losses, where the extent of blood loss is 2–10 ml/day. Cancer of the colon and rectum can provoke losses of blood, resulting in iron deficiency anemia. The fecal blood test is widely used to screen for the presence of cancer of the colon or rectum. In the absence of testing, colorectal cancer may be first

detected because of the resulting iron deficiency anemia.

Infection with hookworm can provoke iron deficiency and iron deficiency anemia. The hookworm is a parasitic worm. It thrives in warm climates, including in the southern United States. The hookworm enters the body through the skin, as through bare feet. The hookworm then migrates to the small intestines where it attaches itself to the villi (small sausage-shaped structures in the intestines that are used for the absorption of all nutrients). The hookworm provokes damage to the villi, resulting in blood loss, and they produce anti-coagulants which promote continued bleeding. Each worm can provoke the loss of up to 0.25 ml of blood per day.

Bleeding and blood losses through gastrointestinal tract can be provoked by colorectal cancer and hookworms, as mentioned above, but also by **hemorrhoids**, anal fissures, **irritable bowel syndrome**, aspirin-induced bleeding, blood clotting disorders, and **diverticulosis** (a condition caused by an abnormal opening from the intestine or bladder). Several genetic diseases exist which lead to bleeding disorders, and these include **hemophilia A**, hemophilia B, and von Willebrand's disease. Of these, only von Willebrand's disease leads to gastrointestinal bleeding.

The symptoms of iron deficiency anemia include weakness and **fatigue**. These symptoms result because of the lack of function of the red blood cells, and the reduced ability of the red blood cells to carry iron to exercising muscles. Iron deficiency can also affect other tissues, including the tongue and fingernails. Prolonged iron deficiency can result in changes of the tongue, and it may become smooth, shiny, and reddened. This condition is called glossitis. The fingernails may grow abnormally, and acquire a spoon-shaped appearance.

Decreased iron intake is a contributing factor in iron deficiency and iron deficiency anemia. The iron content of cabbage, for example, is about 1.6 mg/kg food, while that of spinach (33 mg/kg), lima beans (15 mg/kg), potato (14 mg/kg), tomato (3 mg/kg), apples (1.5 mg/kg), raisins (20 mg/kg), whole wheat bread (43 mg/kg), eggs (20 mg/kg), canned tuna (13 mg/kg), chicken (11 mg/kg), beef (28 mg/kg), corn oil (0.6 mg/kg), and peanut butter (6.0 mg/kg), are indicated. One can see that apples, tomatoes, and vegetable oil are relatively low in iron, while whole wheat bread and beef are relatively high in iron. The assessment of whether a food is low or high in iron can also be made by comparing the amount of that food eaten per day

with the recommended dietary allowance (RDA) for iron. The RDA for iron for the adult male is 10 mg/day, while that for the adult woman is 15 mg/day. The RDA during **pregnancy** is 30 mg/day. The RDA for infants of 0–0.5 years of age is 6 mg/day, while that for infants of 0.5–1.0 years of age is 10 mg/day. The RDA values are based on the assumption that the consumer eats a mixture of plant and animal foods.

The above list of iron values alone may be deceptive, since the availability of iron in fruits, vegetables, and grains is very low, while that the availability from meat is much higher. The availability of iron in plants ranges from only 1–10%, while that in meat, fish, chicken, and liver is 20–30%. The term availability means the percent of dietary iron that is absorbed via the gastrointestinal tract to the bloodstream. Non-absorbed iron is lost in the feces.

Interactions between various foods can influence the absorption of dietary iron. Vitamin C can increase the absorption of dietary iron. Orange juice is a rich source of vitamin C. Thus, if a plant food, such as rice, is consumed with orange juice, then the orange juice can enhance the absorption of the iron of the rice. Vitamin C is also added to infant formulas, and the increased use of formulas fortified with both iron and vitamin C have led to a marked decline in anemia in infants and young children in the United States (Dallman, 1989). In contrast, if rice is consumed with tea, certain chemicals in the tea (tannins) can reduce the absorption of the iron. Phytic acid is a chemical that naturally occurs in legumes, cereals, and nuts. Phytic acid, which can account for 1–5% of the weight of these foods, is a potent inhibitor of iron absorption. The increased availability of the iron in meat products is partly due to the fact that heme-iron is absorbed to a greater extent than free iron salts, and to a greater extent than iron in the phytic acid/iron complex. Nearly all of the iron in plants is nonheme-iron. Much of the iron in meat is nonheme-iron as well. The nonheme-iron in meat, fish, chicken and liver may be about 20% available. The heme-iron of meat may be close to 30% available. The most available source of iron is human milk (50% availability).

Diagnosis

Iron deficiency anemia in infants is defined as a hemoglobin level below 109 mg/ml of whole blood, and a **hematocrit** of under 33%. Anemia in adult males is defined as a hemoglobin under 130 mg/ml and a hematocrit of under 38%. Anemia in adult females is defined as hemoglobin under 120 mg/ml

and a hematocrit of under 32%. Anemia in pregnant women is defined as hemoglobin of under 110 mg/ml and hematocrit of under 31%.

When an abnormally high presence of blood is found in the feces during a **fecal occult blood test**, the physician needs to examine the gastrointestinal tract to determine the cause of bleeding. Here, the diagnosis for iron deficiency anemia includes the examination using a sigmoidoscope. The sigmoidoscope is an instrument that consists of a flexible tube that permits examination of the colon to a distance of 60 cm. A **barium enema**, with an x ray, may also be used to detect abnormalities that can cause bleeding.

The diagnosis of iron deficiency anemia should include a test for oral iron absorption, where evidence suggests that oral iron supplements fail in treating anemia. The oral iron absorption test is conducted by eating 64 mg iron (325 mg ferrous sulfate) in a single dose. Blood samples are then taken after 2 hours and 4 hours. The iron content of the blood serum is then measured. The concentration of iron should rise by an increment of about 22 micromolar, where iron absorption is normal. Lesser increases in concentration mean that iron absorption is abnormal, and that therapy should involve injections or infusions of iron.

Treatment

Oral iron supplements (pills) may contain various iron salts. These iron salts include ferrous sulfate, ferrous gluconate, or ferrous fumarate. Injections and infusions of iron can be carried out with a preparation called iron dextran. In patients with poor iron absorption (by the gut), therapy with injection or infusion is preferable over oral supplements. Treatment of iron deficiency anemia sometimes requires more than therapy with iron. Where iron deficiency was provoked by hemorrhoids, surgery may prove essential to prevent recurrent iron deficiency anemia. Where iron deficiency is provoked by bleeding due to **aspirin** treatment, aspirin should be discontinued. Where iron deficiency is provoked by hookworm infections, therapy for this parasite should be used, along with protection of the feet by wearing shoes whenever walking in hookworm-infested soil.

Prognosis

The prognosis for treating and curing iron deficiency anemia is excellent. Perhaps the main problem is failure to take iron supplements. In cases of pregnant women, the health care worker may recommend taking 100–200 mg iron/day. This dose is rather

KEY TERMS

Hematocrit—The proportion of whole blood in the body, by volume, that is composed of red blood cells.

Hemoglobin—Hemoglobin is an iron-containing protein that resides within red blood cells. Hemoglobin accounts for about 95% of the protein in the red blood cell.

Protoporphyrin IX—Protoporphyrin IX is a protein. The measurement of this protein is useful for the assessment of iron status. Hemoglobin consists of a complex of a protein plus heme. Heme consists of iron plus protoporphyrin IX. Normally, during the course of red blood cell formation, protoporphyrin IX acquires iron, to generate heme, and the heme becomes incorporated into hemoglobin. However, in iron deficiency, protoporphyrin IX builds up.

Recommended Dietary Allowance (RDA)—The Recommended Dietary Allowances (RDAs) are quantities of nutrients of the diet that are required to maintain human health. RDAs are established by the Food and Nutrition Board of the National Academy of Sciences and may be revised every few years.

high, and can lead to **nausea**, **diarrhea**, or abdominal **pain** in 10–20% of women taking this dose. The reason for using this high dose is to effect a rapid cure for anemia, where the anemia is detected at a mid-point during the pregnancy. The above problems of side-effects and noncompliance can be avoided by taking iron doses (100–200 mg) only once a week, where supplements are initiated some time prior to conception, or continuously throughout the fertile period of life. The problem of compliance is not an issue where infusions are used, however a fraction of patients treated with iron infusions experience side effects, such as flushing, **headache**, nausea, **anaphylaxis**, or seizures. A number of studies have shown that iron deficiency anemia in infancy can result in reduced intelligence, where intelligence was measured in early childhood. It is not certain if iron supplementation of children with reduced intelligence, due to iron-deficiency anemia in infancy, has any influence in allowing a “catch-up” in intellectual development.

Prevention

In the healthy population, all of the mineral deficiencies can be prevented by the consumption of

inorganic nutrients at levels defined by the RDA. Iron deficiency anemia in infants and young children can be prevented by the use of fortified foods. Liquid cow milk-based infant formulas are generally supplemented with iron (12 mg/L). The iron in liquid formulas is added as ferrous sulfate or ferrous gluconate. Commercial infant cereals are also fortified with iron, and here small particles of elemental iron are added. The levels used are about 0.5 gram iron/kg dry cereal. This amount of iron is about 10-fold greater than that of the iron naturally present in the cereal.

Resources

PERIODICALS

Walter, T., P. Pino, F. Pizarro, and B. Lozoff. "Prevention of Iron-deficiency Anemia: Comparison of High- and Low-iron Formulas in Term Healthy Infants after Six Months of Life." *Journal of Pediatrics* 132 (1998): 635-640.

Tom Brody, PhD

Iron overload see **Hemochromatosis**

Iron tests

Definition

Iron tests are a group of blood tests that are done to evaluate the iron level in blood serum, the body's capacity to absorb iron, and the amount of iron actually stored in the body. Iron is an essential trace element; it is necessary for the formation of red blood cells and certain enzymes. At the other extreme, high levels of iron can be poisonous.

Purpose

There are four different types of tests that measure the body's iron levels and storage. They are called iron level tests, total iron-binding capacity (TIBC) tests, ferritin tests, and transferrin tests. These tests are given for several reasons:

- To help in the differential diagnosis of different types of anemia.
- To assess the severity of anemia and monitor the treatment of patients with chronic anemia.
- To evaluate protein depletion and other forms of **malnutrition**.
- To check for certain liver disorders.

- To evaluate the possibility of chronic gastrointestinal bleeding. Blood loss from the digestive tract is a common cause of **iron deficiency anemia**.
- To help diagnose certain unusual disorders, including iron **poisoning**, **thalassemia**, hemosiderosis, and **hemochromatosis**.

A serum iron test can be used without the others to evaluate cases of iron poisoning.

Precautions

Patients should not have their blood tested for iron within four days of a blood **transfusion** or tests and treatments that use radioactive materials. Recent high **stress** levels or sleep deprivation are additional reasons for postponing iron tests.

Blood samples for iron tests should be taken early in the morning because serum iron levels vary during the day. This precaution is especially important in evaluating the results of iron replacement therapy.

Description

Iron tests are performed on samples of the patient's blood, withdrawn from a vein into a vacuum tube. The amount of blood taken is between 6 mL and 10 mL (1/3 of a fluid ounce). The procedure, which is called a venipuncture, takes about five minutes.

Iron level test

The iron level test measures the amount of iron in the blood serum that is being carried by a protein (transferrin) in the blood plasma.

Medications and substances that can cause *increased* iron levels include chloramphenicol, estrogen preparations, dietary iron supplements, alcoholic beverages, methyl dopa, and birth control pills.

Medications that can cause *decreased* iron levels include ACTH, colchicine, deferoxamine, methicillin, and testosterone.

Total iron-binding capacity (TIBC) test

The TIBC test measures the amount of iron that the blood would carry if the transferrin were fully saturated. Since transferrin is produced by the liver, the TIBC can be used to monitor liver function and **nutrition**.

Medications that can cause *increased* TIBC levels include fluorides and birth control pills.

Medications that can cause *decreased* TIBC levels include chloramphenicol and ACTH.

Transferrin test

The transferrin test is a direct measurement of transferrin—which is also called siderophilin—levels in the blood. Some laboratories prefer this measurement to the TIBC. The saturation level of the transferrin can be calculated by dividing the serum iron level by the TIBC.

Ferritin test

The ferritin test measures the level of a protein in the blood that stores iron for later use by the body.

Medications that can cause *increased* ferritin levels include dietary iron supplements. In addition, some diseases that do not directly affect the body's iron storage can cause artificially high ferritin levels. These disorders include infections, late-stage cancers, lymphomas, and severe inflammations. Alcoholics often have high ferritin levels.

Preparation

Patient history

Before patients are tested for iron, they should be checked for any of the following factors:

- Prescription medications that affect iron levels, absorption, or storage
- Blood transfusion or radioactive medications within the last four days
- Recent extreme stress or sleep deprivation
- Recent eating habits. Test results can be affected by eating large amounts of iron-rich foods shortly before the blood test.

Fasting

Patients scheduled for an iron level, TIBC, or transferrin test should fast for 12 hours before the blood is drawn. They are allowed to drink water. Patients scheduled for a ferritin test do not need to fast but they should not have any alcoholic beverages before the test.

Aftercare

Aftercare consists of routine care of the area around the venipuncture.

Risks

The primary risk is the possibility of a bruise or swelling in the area of the venipuncture. The patient

can apply moist warm compresses if there is any discomfort.

Normal results

Iron level test

Normal serum iron values are as follows:

- Adult males: 75–175 micrograms/dL
- Adult females: 65–165 micrograms/dL
- Children: 50–120 micrograms/dL
- Newborns: 100–250 micrograms/dL.

TIBC test

Normal TIBC values are as follows:

- Adult males: 300–400 micrograms/dL
- Adult females: 300–450 micrograms/dL.

Transferrin test

Normal transferrin values are as follows:

- Adult males: 200–400 mg/dL
- Adult females: 200–400 mg/dL
- Children: 203–360 mg/dL
- Newborns: 130–275 mg/dL.

Normal transferrin saturation values are between 30–40%.

Ferritin test

Normal ferritin values are as follows:

- Adult males: 20–300 ng/mL
- Adult females: 20–120 ng/mL
- Children (one month): 200–600 ng/mL
- Children (two to five months): 50–200 ng/mL
- Children (six months to 15 years): 7–140 ng/mL
- Newborns: 25–200 ng/mL.

Abnormal results

Iron level test

Serum iron level is *increased* in thalassemia, hemochromatosis, severe hepatitis, **liver disease**, **lead poisoning**, acute leukemia, and **kidney disease**. It is also increased by multiple blood transfusions and intramuscular iron injections.

KEY TERMS

Anemia—A disorder marked by low hemoglobin levels in red blood cells, which leads to a deficiency of oxygen in the blood.

Ferritin—A protein found in the liver, spleen, and bone marrow that stores iron.

Hemochromatosis—A disorder of iron absorption characterized by bronze-colored skin. It can cause painful joints, diabetes, and liver damage if the iron concentration is not lowered.

Hemosiderosis—An overload of iron in the body resulting from repeated blood transfusions. Hemosiderosis occurs most often in patients with thalassemia.

Iron poisoning—A potentially fatal condition caused by swallowing large amounts of iron dietary supplements. Most cases occur in children who have taken adult-strength iron formulas. The symptoms of iron poisoning include vomiting, bloody diarrhea, convulsions, low blood pressure, and turning blue.

Plasma—The liquid part of blood.

Siderophilin—Another name for transferrin.

Thalassemia—A hereditary form of anemia that occurs most frequently in people of Mediterranean origin.

Transferrin—A protein in blood plasma that carries iron derived from food intake to the liver, spleen, and bone marrow.

Iron levels above 350–500 micrograms/dL are considered toxic; levels over 1000 micrograms/dL indicate severe iron poisoning.

Serum iron level is *decreased* in iron deficiency anemia, chronic blood loss, chronic diseases (lupus, **rheumatoid arthritis**), late **pregnancy**, chronically heavy menstrual periods, and thyroid deficiency.

TIBC test

The TIBC is *increased* in iron deficiency anemia, **polycythemia vera**, pregnancy, blood loss, severe hepatitis, and the use of birth control pills.

The TIBC is *decreased* in malnutrition, severe **burns**, hemochromatosis, anemia caused by infections and chronic diseases, **cirrhosis** of the liver, and kidney disease.

Transferrin test

Transferrin is *increased* in iron deficiency anemia, pregnancy, **hormone replacement therapy** (HRT), and the use of birth control pills.

Transferrin is *decreased* in protein deficiency, liver damage, malnutrition, severe burns, kidney disease, chronic infections, and certain genetic disorders.

Ferritin test

Ferritin is *increased* in liver disease, iron overload from hemochromatosis, certain types of anemia, acute leukemia, **Hodgkin's disease**, **breast cancer**, thalassemia, infections, inflammatory diseases, and hemosiderosis. Ferritin levels may be normal or slightly above normal in patients with kidney disease.

Ferritin is *decreased* in chronic iron deficiency and severe protein depletion.

Resources

BOOKS

Pagana, Kathleen Deska. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis: Mosby, Inc., 1998.

Rebecca J. Frey, PhD

Irregular bite see **Malocclusion**

Irritable bowel syndrome

Definition

Irritable bowel syndrome (IBS) is a common intestinal condition characterized by abdominal **pain** and cramps; changes in bowel movements (**diarrhea**, **constipation**, or both); gassiness; bloating; **nausea**; and other symptoms. There is no cure for IBS. Much about the condition remains unknown or poorly understood; however, dietary changes, drugs, and psychological treatment are often able to eliminate or substantially reduce its symptoms.

Description

IBS is the name people use today for a condition that was once called—among other things—colitis, mucous colitis, spastic colon, nervous colon, spastic bowel, and functional bowel disorder. Some of these names reflected the now outdated belief that IBS is a purely psychological disorder, a product of the patient's imagination. Although modern medicine

recognizes that **stress** can trigger IBS attacks, medical specialists agree that IBS is a genuine physical disorder—or group of disorders—with specific identifiable characteristics.

No one knows for sure how many Americans suffer from IBS. Surveys indicate a range of 10–20%, with perhaps as many as 30% of Americans experiencing IBS at some point in their lives. IBS normally makes its first appearance during young adulthood, and in half of all cases symptoms begin before age 35. Women with IBS outnumber men by two to one, for reasons that are not yet understood. IBS is responsible for more time lost from work and school than any medical problem other than the **common cold**. It accounts for a substantial proportion of the patients seen by specialists in diseases of the digestive system (gastroenterologists). Yet only half—possibly as few as 15%—of IBS sufferers ever consult a doctor.

Causes and symptoms

Symptoms

The symptoms of IBS tend to rise and fall in intensity rather than growing steadily worse over time. They always include abdominal pain, which may be relieved by defecation; diarrhea or constipation; or diarrhea alternating with constipation. Other symptoms—which vary from person to person—include cramps; gassiness; bloating; nausea; a powerful and uncontrollable urge to defecate (urgency); passage of a sticky fluid (mucus) during bowel movements; or the feeling after finishing a bowel movement that the bowels are still not completely empty. The accepted diagnostic criteria—known as the Rome criteria—require at least three months of continuous or recurrent symptoms before IBS can be confirmed. According to Christine B. Dalton and Douglas A. Drossman in the *American Family Physician*, an estimated 70% of IBS cases can be described as “mild;” 25% as “moderate;” and 5% as “severe.” In mild cases the symptoms are slight. As a general rule, they are not present all the time and do not interfere with work and other normal activities. Moderate IBS occasionally disrupts normal activities and may cause some psychological problems. People with severe IBS often find living a normal life impossible and experience crippling psychological problems as a result. For some the physical pain is constant and intense.

Causes

Researchers remain unsure about the cause or causes of IBS. It is called a functional disorder because it is thought to result from changes in the

activity of the major part of the large intestine (the colon). After food is digested by the stomach and small intestine, the undigested material passes in liquid form into the colon, which absorbs water and salts. This process may take several days. In a healthy person the colon is quiet during most of that period except after meals, when its muscles contract in a series of wavelike movements called peristalsis. Peristalsis helps absorption by bringing the undigested material into contact with the colon wall. It also pushes undigested material that has been converted into solid or semisolid feces toward the rectum, where it remains until defecation. In IBS, however, the normal rhythm and intensity of peristalsis is disrupted. Sometimes there is too little peristalsis, which can slow the passage of undigested material through the colon and cause constipation. Sometimes there is too much, which has the opposite effect and causes diarrhea. A Johns Hopkins University study found that healthy volunteers experienced 6–8 contractions of the colon each day, compared with up to 25 contractions a day for volunteers suffering from IBS with diarrhea, and an almost complete absence of contractions among constipated IBS volunteers. In addition to differences in the number of contractions, many of the IBS volunteers experienced powerful spasmodic contractions affecting a larger-than-normal area of the colon—“like having a Charlie horse in the gut,” according to one of the investigators.

DIET. Some kinds of food and drink appear to play a key role in triggering IBS attacks. Food and drink that healthy people can ingest without any trouble may disrupt peristalsis in IBS patients, which probably explains why IBS attacks often occur shortly after meals. Chocolate, milk products, **caffeine** (in coffee, tea, colas, and other drinks), and large quantities of alcohol are some of the chief culprits. Other kinds of food have also been identified as problems, however, and the pattern of what can and cannot be tolerated is different for each person. Characteristically, IBS symptoms rarely occur at night and disrupt the patient’s sleep.

STRESS. Stress is an important factor in IBS because of the close nervous system connections between the brain and the intestines. Although researchers do not yet understand all of the links between changes in the nervous system and IBS, they point out the similarities between mild digestive upsets and IBS. Just as healthy people can feel nauseated or have an upset stomach when under stress, people with IBS react the same way, but to a greater degree. Finally, IBS symptoms sometimes intensify during

menstruation, which suggests that female reproductive hormones are another trigger.

Diagnosis

Diagnosing IBS is a fairly complex task because the disorder does not produce changes that can be identified during a **physical examination** or by laboratory tests. When IBS is suspected, the doctor (who can be either a family doctor or a specialist) needs to determine whether the patient's symptoms satisfy the Rome criteria. The doctor must rule out other conditions that resemble IBS, such as **Crohn's disease** and **ulcerative colitis**. These disorders are ruled out by questioning the patient about his or her physical and mental health (the medical history), performing a physical examination, and ordering laboratory tests. Normally the patient is asked to provide a stool sample that can be tested for blood and intestinal parasites. In some cases x rays or an internal examination of the colon using a flexible instrument inserted through the anus (a sigmoidoscope or colonoscope) is necessary. The doctor also may ask the patient to try a lactose-free diet for two or three weeks to see whether **lactose intolerance** is causing the symptoms.

Treatment

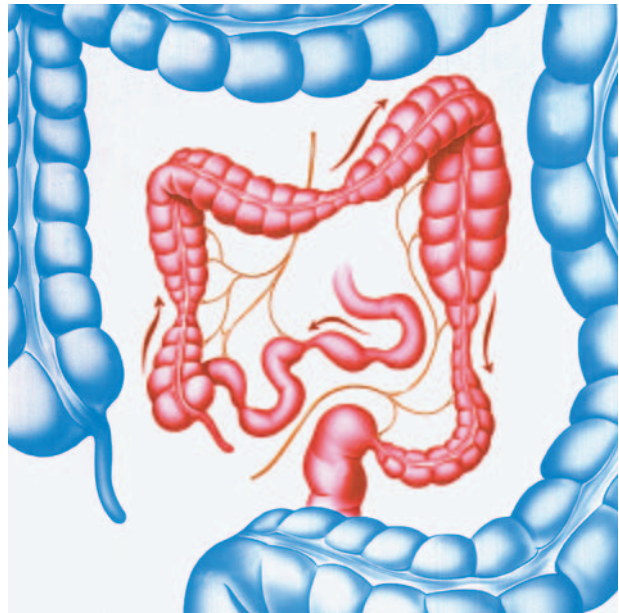
Dietary changes, sometimes supplemented by drugs or psychotherapy, are considered the key to successful treatment. The following approach, offered by Dalton and Drossman, is typical of the advice found in the medical literature on IBS. The authors tie their approach to the severity of the patient's symptoms:

Mild symptoms

Dalton and Drossman recommend a low-fat, high-fiber diet. Problem-causing substances such as lactose, caffeine, beans, cabbage, cucumbers, broccoli, fatty foods, alcohol, and medications should be identified and avoided. Bran or 15-25 grams a day of an over-the-counter psyllium laxative (Metamucil or Fiberall) may also help both constipation and diarrhea. The patient can still have milk or milk products if lactose intolerance is not a problem. People with irregular bowel habits—particularly constipated patients—may be helped by establishing set times for meals and bathroom visits.

Moderate symptoms

The advice given by Dalton and Drossman in mild cases applies here as well. They also suggest that



Normal and diseased (center) colons. Areas of constriction in the colon cause constipation, while areas of distention cause diarrhea. (John Bavosi/Science Photo Library. Custom Medical Stock Photo. Reproduced by permission.)

patients keep a diary of symptoms for two or three weeks, covering daily activities including meals, and emotional responses to events. The doctor can then review the diary with the patient to identify possible problem areas.

Although a high-fiber diet remains the standard treatment for constipated patients, such **laxatives** as lactulose (Chronulac) or sorbitol may be prescribed. Loperamide (Imodium) and cholestyramine (Questran) are suggested for diarrhea. Abdominal pain after meals can be reduced by taking **antispasmodic drugs** such as hyoscyamine (Anaspaz, Cystospaz, or Levsin) or dicyclomine (Bemote, Bentyl, or Di-Spaz) before eating.

Dalton and Drossman also suggest psychological counseling or behavioral therapy for some patients to reduce **anxiety** and to learn to cope with the pain and other symptoms of IBS. Relaxation therapy, hypnosis, **biofeedback**, and **cognitive-behavioral therapy** are examples of behavioral therapy.

Severe symptoms

When IBS produces constant pain that interferes with everyday life, **antidepressant drugs** can help by blocking pain transmission from the nervous system. Dalton and Drossman also underscore the importance

KEY TERMS

Anus—The opening at the lower end of the rectum.

Crohn's disease—A disease characterized by inflammation of the intestines. Its early symptoms may resemble those of IBS.

Defecation—Passage of feces through the anus.

Feces—Undigested food and other waste that is eliminated through the anus. Feces are also called fecal matter or stools.

Lactose—A sugar found in milk and milk products. Some people are lactose intolerant, meaning they have trouble digesting lactose. Lactose intolerance can produce symptoms resembling those of IBS.

Peristalsis—The periodic waves of muscular contractions that move food through the intestines during the process of digestion.

Ulcerative colitis—A disease that inflames and causes breaks (ulcers) in the colon and rectum, which are parts of the large intestine.

of an ongoing and supportive doctor-patient relationship.

Alternative treatment

Alternative and mainstream approaches to IBS treatment overlap to a certain extent. Like mainstream doctors, alternative practitioners advise a high-fiber diet to reduce digestive system irritation. They also suggest avoiding alcohol, caffeine, and fatty, gassy, or spicy foods. Recommended stress management techniques include **yoga**, **meditation**, hypnosis, bio-feedback, and **reflexology**. Reflexology is a technique of foot massage that is thought to relieve diarrhea, constipation, and other IBS symptoms.

Alternative medicine also emphasizes such herbal remedies as ginger (*Zingiber officinale*), buckthorn (*Rhamnus purshiana*), and enteric-coated peppermint oil. Enteric coating prevents digestion until the peppermint oil reaches the small intestine, thus avoiding irritation of the upper part of the digestive tract. Chamomile (*Matricaria recutita*), valerian (*Valeriana officinalis*), rosemary (*Rosemarinus officinalis*), lemon balm (*Melissa officinalis*), and other herbs are recommended for their antispasmodic properties. The list of alternative treatments for IBS is in fact quite long. It includes **aromatherapy**, homeopathy, **hydrotherapy**,

juice therapy, **acupuncture**, **chiropractic**, **osteopathy**, **naturopathic medicine**, and Chinese traditional herbal medicine.

Prognosis

IBS is not a life-threatening condition. It does not cause intestinal bleeding or inflammation, nor does it cause other bowel diseases or **cancer**. Although IBS can last a lifetime, in up to 30% of cases the symptoms eventually disappear. Even if the symptoms cannot be eliminated, with appropriate treatment they can usually be brought under control to the point where IBS becomes merely an occasional inconvenience. Treatment requires a long-term commitment, however; six months or more may be needed before the patient notices substantial improvement.

Resources

ORGANIZATIONS

International Foundation for Functional Gastrointestinal Disorders. P.O. Box 17864, Milwaukee, WI 53217. (888) 964-2001. <<http://www.iffgd.org>>.

National Digestive Diseases Information Clearinghouse. 2 Information Way, Bethesda, MD 20892-3570. (800) 891-5389. <<http://www.niddk.nih.gov/health/digest/nddic.htm>>.

Howard Baker

Ischemia

Definition

Ischemia is an insufficient supply of blood to an organ, usually due to a blocked artery.

Description

Myocardial ischemia is an intermediate condition in **coronary artery disease** during which the heart tissue is slowly or suddenly starved of oxygen and other nutrients. Eventually, the affected heart tissue will die. When blood flow is completely blocked to the heart, ischemia can lead to a **heart attack**. Ischemia can be silent or symptomatic. According to the American Heart Association, up to four million Americans may have silent ischemia and be at high risk of having a heart attack with no warning.

Symptomatic ischemia is characterized by chest **pain** called **angina** pectoris. The American Heart Association estimates that nearly seven million Americans have angina pectoris, usually called angina. Angina occurs more frequently in women than in men, and in blacks and Hispanics more than in whites. It also occurs more frequently as people age—25% of women over the age of 85 and 27% of men who are 80–84 years old have angina.

People with angina are at risk of having a heart attack. Stable angina occurs during exertion, can be quickly relieved by resting or taking nitroglycerine, and lasts from three to twenty minutes. Unstable angina, which increases the risk of a heart attack, occurs more frequently, lasts longer, is more severe, and may cause discomfort during rest or light exertion.

Ischemia can also occur in the arteries of the brain, where blockages can lead to a **stroke**. About 80–85% of all strokes are ischemic. Most blockages in the cerebral arteries are due to a blood clot, often in an artery narrowed by plaque. Sometimes, a blood clot in the heart or aorta travels to a cerebral artery. A **transient ischemic attack** (TIA) is a “mini-stroke” caused by a temporary deficiency of blood supply to the brain. It occurs suddenly, lasts a few minutes to a few hours, and is a strong warning sign of an impending stroke. Ischemia can also effect intestines, legs, feet and kidneys. Pain, malfunctions, and damage in those areas may result.

Causes and symptoms

Ischemia is almost always caused by blockage of an artery, usually due to atherosclerotic plaque. Myocardial ischemia is also caused by **blood clots** (which tend to form on plaque), artery spasms or contractions, or any of these factors combined. Silent ischemia is usually caused by emotional or mental **stress** or by exertion, but there are no symptoms. Angina is usually caused by increased oxygen demand when the heart is working harder than usual, for example, during **exercise**, or during mental or physical stress. According to researchers at Harvard University, physical stress is harder on the heart than mental stress. A TIA is caused by a blood clot briefly blocking a cerebral artery.

Risk factors

The risk factors for myocardial ischemia are the same as those for coronary artery disease. For TIA, coronary artery disease is also a risk factor.

- **Heredity.** People whose parents have coronary artery disease are more likely to develop it. African Americans are also at higher risk.
- **Sex.** Men are more likely to have heart attacks than women, and to have them at a younger age.
- **Age.** Men who are 45 years of age and older and women who are 55 years of age and older are considered to be at risk.
- **Smoking.** Smoking increases both the chance of developing coronary artery disease and the chance of dying from it. Second hand smoke may also increase risk.
- **High cholesterol.** Risk of developing coronary artery disease increases as blood cholesterol levels increase. When combined with other factors, the risk is even greater.
- **High blood pressure.** High blood pressure makes the heart work harder, and with time, weakens it. When combined with **obesity**, smoking, high cholesterol, or diabetes, the risk of heart attack or stroke increases several times.
- **Lack of physical activity.** Lack of exercise increases the risk of coronary artery disease.
- **Diabetes mellitus.** The risk of developing coronary artery disease is seriously increased for diabetics.
- **Obesity.** Excess weight increases the strain on the heart and increases the risk of developing coronary artery disease, even if no other risk factors are present. Obesity increases blood pressure and blood cholesterol, and can lead to diabetes.
- **Stress and anger.** Some scientists believe that stress and anger can contribute to the development of coronary artery disease. Stress increases the heart rate and blood pressure and can injure the lining of the arteries. Angina attacks often occur after anger, as do many heart attacks and strokes.

Angina symptoms include:

- A tight, squeezing, heavy, burning, or **choking** pain that is usually beneath the breastbone—the pain may spread to the throat, jaw, or one arm
- A feeling of heaviness or tightness that is not painful
- A feeling similar to gas or indigestion
- Attacks brought on by exertion and relieved by rest.

If the pain or discomfort continues or intensifies, immediate medical help should be sought, ideally within 30 minutes.

TIA symptoms include:

- Sudden weakness, **tingling**, or **numbness**, usually in one arm or leg or both the arm and leg on the same side of the body, as well as sometimes in the face
- Sudden loss of coordination
- Loss of vision or double vision
- Difficulty speaking
- Vertigo and loss of balance.

Diagnosis

Diagnostic tests for myocardial ischemia include: resting, exercise, or ambulatory electrocardiograms; scintigraphic studies (radioactive heart scans); **echocardiography**; coronary **angiography**; and, rarely, **positron emission tomography**. Diagnostic tests for TIA include physician review of symptoms, **computed tomography scans** (CT scans), carotid artery ultrasound (**Doppler ultrasonography**), and **magnetic resonance imaging**. Angiography is the best test for ischemia of any organ.

An electrocardiogram (ECG) shows the heart's activity and may reveal a lack of oxygen. Electrodes covered with conducting jelly are placed on the patient's chest, arms, and legs. Impulses of the heart's activity are recorded on paper. The test takes about 10 minutes and is performed in a physician's office. About 25% of patients with angina have normal electrocardiograms. Another type of electrocardiogram, the exercise **stress test**, measures response to exertion when the patient is exercising on a treadmill or a stationary bike. It is performed in a physician's office or an exercise laboratory and takes 15 to 30 minutes. This test is more accurate than a resting ECG in diagnosing ischemia. Sometimes an ambulatory ECG is ordered. For this test, the patient wears a portable ECG machine called a Holter monitor for 12, 24, or 48 hours.

Myocardial perfusion scintigraphy and radionuclide angiography are nuclear studies involving the injection of a radioactive material (e.g., thallium) which is absorbed by healthy tissue. A gamma scintillation camera displays and records a series of images of the radioactive material's movement through the heart. Both tests are usually performed in a hospital's nuclear medicine department and take about 30 minutes to an hour. A perfusion scan is sometimes performed at the end of a stress test.

An echocardiogram uses sound waves to create an image of the heart's chambers and valves. The technician applies gel to a handheld transducer then presses

it against the patient's chest. The heart's sound waves are converted into an image on a monitor. Performed in a cardiology outpatient diagnostic laboratory, the test takes 30 minutes to an hour. It can reveal abnormalities in the heart wall that indicate ischemia, but it doesn't evaluate the coronary arteries directly.

Coronary angiography is the most accurate diagnostic technique, but it is also the most invasive. It shows the heart's chambers, great vessels, and coronary arteries by using a contrast solution and x-ray technology. A moving picture is recorded of the blood flow through the coronary arteries. The patient is awake, but sedated, and connected to ECG electrodes and an intravenous line. A local anesthetic is injected. The cardiologist then inserts a catheter into a blood vessel and guides it into the heart. Coronary angiography is performed in a **cardiac catheterization** laboratory and takes from half an hour to two hours.

Positron emission tomography (PET) is a non-invasive nuclear test used to evaluate the heart tissue. A PET scanner traces high-energy gamma rays released from radioactive particles to provide three-dimensional images of the heart tissue. Performed at a hospital, it usually takes from one hour to one hour and 45 minutes. PET is very expensive and not widely available.

Computed tomography scans (CT scans) and magnetic resonance imaging (MRI) are computerized scanning methods. CT scanning uses a thin x-ray beam to show three-dimensional views of soft tissues. It is performed at a hospital or clinic and takes less than a minute. MRI uses a magnetic field to produce clear, cross-sectional images of soft tissues. The patient lies on a table which slides into a tunnel-like scanner. It is usually performed at a hospital and takes about 30 minutes.

Treatment

Angina is treated with drug therapy and surgery. Drugs such as nitrates, beta-blockers, and **calcium channel blockers** relieve chest pain, but they cannot clear blocked arteries. **Aspirin** helps prevent blood clots. Surgical procedures include percutaneous transluminal coronary **angioplasty** and **coronary artery bypass graft surgery**.

Nitroglycerin is the classic treatment for angina. It quickly relieves pain and discomfort by opening the coronary arteries and allowing more blood to flow to the heart. **Beta blockers** reduce the amount of oxygen required by the heart during stress. Calcium channel blockers help keep the arteries open and reduce blood



This patient's foot is affected with ischemia. Ischemia occurs when there is an insufficient supply of blood to a specific organ or tissue. (Photograph by Dr. P. Marazzi, Photo Researchers, Inc. Reproduced by permission.)

pressure. Aspirin helps prevent blood clots from forming on plaques.

Percutaneous transluminal coronary angioplasty and coronary artery bypass graft surgery are invasive procedures which improve blood flow in the coronary arteries. Percutaneous transluminal coronary angioplasty is a non-surgical procedure in which a catheter tipped with a balloon is threaded from a blood vessel in the thigh into the blocked artery. The balloon is inflated, compressing the plaque to enlarge the blood vessel and open the blocked artery. The balloon is deflated and the catheter is removed. The procedure is performed by a cardiologist in a hospital and generally requires a two-day stay. Sometimes a metal stent is placed in the artery to prevent closing of the artery.

In coronary artery bypass graft, called bypass surgery, a detour is built around the coronary artery blockage with a healthy leg vein or chest wall artery. The healthy vein or artery then supplies oxygen-rich blood to the heart. Bypass surgery is major surgery appropriate for patients with blockages in two or three major coronary arteries or severely narrowed left main coronary arteries, as well as those who have not responded to other treatments. It is performed in a hospital under **general anesthesia** using a heart-lung machine to support the patient while the healthy vein or artery is attached to the coronary artery.

There are several experimental surgical procedures: **atherectomy**, where the surgeon shaves off and removes strips of plaque from the blocked artery; laser angioplasty, where a catheter with a laser tip is inserted to burn or break down the plaque; and insertion of a metal coil, called a stent, that can be implanted permanently to keep a blocked artery

open. This stenting procedure is becoming more common. Another experimental procedure uses a laser to drill channels in the heart muscle to increase blood supply.

TIAs are treated by drugs that control high blood pressure and reduce the likelihood of blood clots and surgery. Aspirin is commonly used and anticoagulants are sometimes used to prevent blood clots. In some cases, carotid **endarterectomy** surgery is performed to help prevent further TIAs. The procedure involves removing arterial plaque from inside blood vessels.

The use of **chelation therapy**, a long-term injection by a physician of a cocktail of synthetic amino acid, ethylenediaminetetracetic acid, and **anticoagulant drugs** and nutrients, is controversial.

Alternative treatment

Ischemia can be life-threatening. Although there are alternative treatments for angina, traditional medical care may be necessary. Prevention of the cause of ischemia, primarily **atherosclerosis**, is primary. This becomes even more important for people with a family history of heart disease. Dietary modifications, especially the reduction or elimination of saturated fats (primarily found in meat), are essential. Increased fiber (found in fresh fruits and vegetables, grains, and beans) can help the body eliminate excessive cholesterol through the stools. Exercise, particularly aerobic exercise, is essential for circulation health. Not smoking will prevent damage from smoke and the harmful substances it contains.

Abana, a mixture of herbs and **minerals** used in **Ayurvedic medicine**, can reduce the frequency and severity of angina attacks. Western herbal medicine recommends hawthorn (*Crataegus laevigata* or *C. oxyacantha*) to relieve long-term angina, since it strengthens the contractility of the heart muscles. **Nutritional supplements** and botanical medicines that act as antioxidants, for example, **vitamins C and E**, selenium, ginkgo (*Ginkgo biloba*), bilberry (*Vaccinium myrtillus*), and hawthorn, can help prevent initial arterial injury that can lead to the formation of plaque deposits. Cactus (*Cactus grandiflorus*) is a homeopathic remedy used for pain relief during an attack. Mind/body relaxation techniques such as **yoga** and **biofeedback** can help control strong emotions and stress.

Prognosis

In many cases, ischemia can be successfully treated, but the underlying disease process of

atherosclerosis is usually not “cured.” New diagnostic techniques enable doctors to identify ischemia earlier. New technologies and surgical procedures can prevent angina from leading to a heart attack or TIA from resulting in a stroke. The outcome for patients with silent ischemia has not been well established.

Prevention

A healthy lifestyle, including eating right, getting regular exercise, maintaining a healthy weight, not smoking, drinking in moderation, not using illegal drugs, controlling **hypertension**, and managing stress are practices that can reduce the risk of ischemia progressing to a heart attack or stroke.

A healthy diet includes a variety of foods that are low in fat, especially saturated fat; low in cholesterol; and high in fiber. Plenty of fruits and vegetables should be eaten and sodium should be limited. Fat should comprise no more than 30% of total daily calories. Cholesterol should be limited to about 300 mg and sodium to about 2,400 mg per day.

Moderate aerobic exercise lasting about 30 minutes four or more times per week is recommended for maximum heart health, according to the Centers for Disease Control and Prevention and the American College of Sports Medicine. Three 10-minute exercise periods are also beneficial. If any risk factors are present, a physician’s clearance should be obtained before starting exercise.

Maintaining a desirable body weight is also important. People who are 20% or more over their ideal body weight have an increased risk of developing coronary artery disease or stroke.

Smoking has many adverse effects on the heart and arteries, so should be avoided. Heart damage caused by smoking can be improved by quitting. Several studies have shown that ex-smokers face the same risk of heart disease as non-smokers within five to ten years of quitting.

Excessive drinking can increase risk factors for heart disease. Modest consumption of alcohol, however, can actually protect against coronary artery disease. The American Heart Association defines moderate consumption as one ounce of alcohol per day—roughly one cocktail, one 8-ounce glass of wine, or two 12-ounce glasses of beer.

Commonly used illegal drugs can seriously harm the heart and should never be used. Even stimulants like ephedra and **decongestants** like pseudoephedrine can be harmful to patients with hypertension or heart disease.

KEY TERMS

Atherosclerosis—A process in which the walls of the arteries thicken due to the accumulation of plaque in the blood vessels. Atherosclerosis is the cause of most coronary artery disease.

Coronary artery disease—A narrowing or blockage, due to atherosclerosis, of the arteries that provide oxygen and nutrients to the heart. When blood flow is cut-off, the result is a heart attack.

Plaque—A deposit of fatty and other substances that accumulate in the lining of the artery wall.

Stroke—A sudden decrease or loss of consciousness caused by rupture or blockage of a blood vessel by a blood clot or hemorrhage in the brain. Ischemic strokes are caused by blood clots in a cerebral artery.

Treatment should be sought for hypertension. High blood pressure can be completely controlled through lifestyle changes and medication. Stress, which can increase the risk of a heart attack or stroke, should also be managed. While it cannot always be avoided, it can be controlled.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

National Heart, Lung and Blood Institute. P.O. Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

Texas Heart Institute. Heart Information Service. P.O. Box 20345, Houston, TX 77225-0345. <<http://www.tmc.edu/thi>>.

Lori De Milto

Isocarboxazid see **Monoamine oxidase inhibitors**

Isolation

Definition

Isolation refers to the precautions that are taken in the hospital to prevent the spread of an infectious

agent from an infected or colonized patient to susceptible persons.

Purpose

Isolation practices are designed to minimize the transmission of infection in the hospital, using current understanding of the way infections can transmit. Isolation should be done in a user friendly, well-accepted, inexpensive way that interferes as little as possible with patient care, minimizes patient discomfort, and avoids unnecessary use.

Precautions

The type of precautions used should be viewed as a flexible scale that may range from the least to the most demanding methods of prevention. These methods should always take into account that differences exist in the way that diseases are spread. Recognition and understanding of these differences will avoid use of insufficient or unnecessary interventions.

Description

Isolation practices can include placement in a private room or with a select roommate, the use of protective barriers such as masks, gowns and gloves, a special emphasis on handwashing (which is always very important), and special handling of contaminated articles. Because of the differences among infectious diseases, more than one of these precautions may be necessary to prevent spread of some diseases but may not be necessary for others.

The Centers for Disease Control and Prevention (CDC) and the Hospital Infection Control Practice Advisory Committee (HICPAC) have led the way in defining the guidelines for hospital-based infection precautions. The most current system recommended for use in hospitals consists of two levels of precautions. The first level is Standard Precautions which apply to all patients at all times because signs and symptoms of infection are not always obvious and therefore may unknowingly pose a risk for a susceptible person. The second level is known as Transmission-Based Precautions which are intended for individuals who have a known or suspected infection with certain organisms.

Frequently, patients are admitted to the hospital without a definite diagnosis, but with clues to suggest an infection. These patients should be isolated with the appropriate precautions until a definite diagnosis is made.

Standard Precautions

Standard Precautions define all the steps that should be taken to prevent spread of infection from person to person when there is an anticipated contact with:

- Blood
- Body fluids
- Secretions, such as phlegm
- Excretions, such as urine and feces (not including sweat) whether or not they contain visible blood
- Nonintact skin, such as an open wound
- Mucous membranes, such as the mouth cavity.

Standard Precautions includes the use of one or combinations of the following practices. The level of use will always depend on the anticipated contact with the patient:

- Handwashing, the most important **infection control** method
- Use of latex or other protective gloves
- Masks, eye protection and/or face shield
- Gowns
- Proper handling of soiled patient care equipment
- Proper environmental cleaning
- Minimal handling of soiled linen
- Proper disposal of needles and other sharp equipment such as scalpels
- Placement in a private room for patients who cannot maintain appropriate cleanliness or contain body fluids.

Transmission Based Precautions

Transmission Based Precautions may be needed in addition to Standard Precautions for selected patients who are known or suspected to harbor certain infections. These precautions are divided into three categories that reflect the differences in the way infections are transmitted. Some diseases may require more than one isolation category.

AIRBORNE PRECAUTIONS. Airborne Precautions prevent diseases that are transmitted by minute particles called droplet nuclei or contaminated dust particles. These particles, because of their size, can remain suspended in the air for long periods of time; even after the infected person has left the room. Some examples of diseases requiring these precautions are **tuberculosis**, **measles**, and **chickenpox**.

KEY TERMS

Colonized—This occurs when a microorganism is found on or in a person without causing a disease.

Disinfected—Decreased the number of microorganisms on or in an object.

Latex—A rubber material which gloves and condoms are made from.

Phlegm—Another word for sputum; material coughed up from a person's airways.

Stethoscope—A medical instrument for listening to a patient's heart and lungs.

A patient needing Airborne Precautions should be assigned to a private room with special ventilation requirements. The door to this room must be closed at all possible times. If a patient must move from the isolation room to another area of the hospital, the patient should be wearing a mask during the transport. Anyone entering the isolation room to provide care to the patient must wear a special mask called a respirator.

DROPLET PRECAUTIONS. Droplet Precautions prevent the spread of organisms that travel on particles much larger than the droplet nuclei. These particles do not spend much time suspended in the air, and usually do not travel beyond a several foot range from the patient. These particles are produced when a patient coughs, talks, or sneezes. Examples of disease requiring droplet precautions are meningococcal **meningitis** (a serious bacterial infection of the lining of the brain), **influenza**, **mumps**, and German measles (**rubella**).

Patients who require Droplet Precautions should be placed in a private room or with a roommate who is infected with the same organism. The door to the room may remain open. Health care workers will need to wear masks within 3 ft of the patient. Patients moving about the hospital away from the isolation room should wear a mask.

CONTACT PRECAUTIONS. Contact Precautions prevent spread of organisms from an infected patient through direct (touching the patient) or indirect (touching surfaces or objects that that been in contact with the patient) contact. Examples of patients who might be placed in Contact Precautions are those infected with:

- Antibiotic-resistant bacteria
- Hepatitis A

- Scabies
- Impetigo
- Lice.

This type of precaution requires the patient to be placed in a private room or with a roommate who has the same infection. Health care workers should wear gloves when entering the room. They should change their gloves if they touch material that contains large volumes of organisms such as soiled dressings. Prior to leaving the room, health care workers should remove the gloves and wash their hands with medicated soap. In addition, they may need to wear protective gowns if there is a chance of contact with potentially infective materials such as **diarrhea** or wound drainage that cannot be contained or if there is likely to be extensive contact with the patient or environment.

Patient care items, such as a stethoscope, that are used for a patient in Contact Precautions should not be shared with other patients unless they are properly cleaned and disinfected before reuse. Patients should leave the isolation room infrequently.

Resources

BOOKS

Edmond, M. "Isolation." In *A Practical Handbook for Hospital Epidemiologists*, edited by L. A. Herwaldt and M. D. Decker. Thorofare, NJ: Slack Inc., 1998.

Suzanne M. Lutwick, MPH

Isoniazid see **Antituberculosis drugs**

Isosorbide dinitrate see **Antiangina drugs**

Isotretinoin see **Antiacne drugs**

Isradipine see **Calcium channel blockers**

Itching

Definition

Itching is an intense, distracting irritation or tickling sensation that may be felt all over the skin's surface, or confined to just one area. The medical term for itching is pruritus.

Description

Itching instinctively leads most people to scratch the affected area. Different people can tolerate different amounts of itching, and anyone's threshold of

tolerance can be changed due to **stress**, emotions, and other factors. In general, itching is more severe if the skin is warm, and if there are few distractions. This is why people tend to notice itching more at night.

Causes and symptoms

The biology underlying itching is not fully understood. It is believed that itching results from the interactions of several different chemical messengers. Although itching and **pain** sensations were at one time thought to be sent along the same nerve pathways, researchers reported the discovery in 2003 of itch-specific nerve pathways. Nerve endings that are specifically sensitive to itching have been named pruriceptors.

Research into itching has been helped by the recent invention of a mechanical device called the Matcher, which electrically stimulates the patient's left hand. When the intensity of the stimulation equals the intensity of itching that the patient is experiencing elsewhere in the body, the patient stops the stimulation and the device automatically records the measurement. The Matcher was found to be sensitive to immediate changes in the patient's perception of itching as well as reliable in its measurements.

Stress and emotional upset can make itching worse, no matter what the underlying cause. If emotional problems are the primary reason for the itch, the condition is known as psychogenic itching. Some people become convinced that their itch is caused by a parasite; this conviction is often linked to burning sensations in the tongue, and may be caused by a major psychiatric disorder.

Generalized itching

Itching that occurs all over the body may indicate a medical condition such as **diabetes mellitus**, **liver disease**, kidney failure, **jaundice**, thyroid disorders (and rarely, **cancer**). Blood disorders such as leukemia, and lymphatic conditions such as **Hodgkin's disease** may sometimes cause itching as well.

Some people may develop an itch without a rash when they take certain drugs (such as **aspirin**, codeine, **cocaine**); others may develop an itchy red "drug rash" or **hives** because of an allergy to a specific drug. Some medications given to cancer patients may also cause itching.

Itching also may be caused when any of the family of hookworm larvae penetrate the skin. This includes swimmer's itch and creeping eruption caused by cat or dog hookworm, and ground itch caused by the "true" hookworm.

Many skin conditions cause an itchy rash. These include:

- **Atopic dermatitis**
- Chickenpox
- **Contact dermatitis**
- **Dermatitis** herpetiformis (occasionally)
- Eczema
- Fungus infections (such as **athlete's foot**)
- Hives (urticaria)
- Insect bites
- Lice
- Lichen planus
- Neurodermatitis (**lichen simplex chronicus**)
- **Psoriasis** (occasionally)
- Scabies.

On the other hand, itching all over the body can be caused by something as simple as bathing too often, which removes the skin's natural oils and may make the skin too dry and scaly.

Localized itching

Specific itchy areas may occur if a person comes in contact with soap, detergents, and wool or other rough-textured, scratchy material. Adults who have **hemorrhoids**, anal fissure, or persistent **diarrhea** may notice itching around the anus (called "pruritus ani"). In children, itching in this area is most likely due to worms.

Intense itching in the external genitalia in women ("pruritus vulvae") may be due to **candidiasis**, hormonal changes, or the use of certain spermicides or vaginal suppositories, ointments, or deodorants.

It is also common for older people to suffer from dry, itchy skin (especially on the back) for no obvious reason. Younger people also may notice dry, itchy skin in cold weather. Itching is also a common complaint during **pregnancy**.

Diagnosis

Itching is a symptom that is quite obvious to its victim. Someone who itches all over should seek medical care. Because itching can be caused by such a wide variety of triggers, a complete physical exam and medical history will help diagnose the underlying problem. A variety of blood and stool tests may help determine the underlying cause.

Treatment

Antihistamines such as diphenhydramine (Benadryl) can help relieve itching caused by hives, but will not affect itching from other causes. Most antihistamines also make people sleepy, which can help patients sleep who would otherwise be awake from the itch.

Specific treatment of itching depends on the underlying condition that causes it. In general, itchy skin should be treated very gently. While scratching may temporarily ease the itch, in the long run scratching just makes it worse. In addition, scratching can lead to an endless cycle of itch–scratch–more itching.

To avoid the urge to scratch, a person can apply a cooling or soothing lotion or cold compress when the urge to scratch occurs. Soaps are often irritating to the skin, and can make an itch worse; they should be avoided, or used only when necessary.

Creams or ointments containing cortisone may help control the itch from insect bites, contact dermatitis or eczema. Cortisone cream should not be applied to the face unless a doctor prescribes it.

Probably the most common cause of itching is dry skin. There are a number of simple things a person can do to ease the annoying itch:

- Do not wear tight clothes
- Avoid synthetic fabrics
- Do not take long baths
- Wash the area in lukewarm water with a little baking soda
- For generalized itching, take a lukewarm shower
- Try a lukewarm oatmeal (or Aveeno) bath for generalized itching
- Apply bath oil or lotion (without added colors or scents) right after bathing.

Itching may also be treated with whole-body medications. In addition to antihistamines, some of these systemic treatments include:

- tricyclic antidepressants
- sedatives or tranquilizers
- such selective serotonin reuptake inhibitors as paroxetine (Paxil) and sertraline (Zoloft)
- binding agents (such as cholestyramine which relieves itching associated with kidney or liver disease).
- aspirin
- cimetidine

People who itch as a result of mental problems or stress should seek help from a mental health expert.

Alternative and complementary therapies

A well-balanced diet that includes carbohydrates, fats, **minerals**, proteins, **vitamins**, and liquids will help to maintain skin health. Capsules that contain eicosapentaenoic acid, which is obtained from herring, mackerel, or salmon, may help to reduce itching. Vitamin A plays an important role in skin health. Vitamin E (capsules or ointment) may reduce itching. Patients should check with their treating physician before using supplements.

Homeopathy has been reported to be effective in treating systemic itching associated with hemodialysis.

Baths containing oil with milk or oatmeal are effective at relieving localized itching. Evening primrose oil may soothe itching and may be as effective as **corticosteroids**. Calendula cream may relieve short-term itching. Other herbal treatments that have been recently reported to relieve itching include sangre de drago, a preparation made with sap from a South American tree; and a mixture of honey, olive oil, and beeswax.

Distraction, **music therapy**, relaxation techniques, and visualization may be useful in relieving itching. Ultraviolet light therapy may relieve itching associated with conditions of the skin, kidneys, blood, and gallbladder. There are some reports of the use of **acupuncture** and transcutaneous electrical nerve stimulators (TENS) to relieve itching.

Prognosis

Most cases of itching go away when the underlying cause is treated successfully.

Prevention

There are certain things people can do to avoid itchy skin. Patients who tend toward itchy skin should:

- Avoid a daily bath
- Use only lukewarm water when bathing
- Use only gentle soap
- Pat dry, not rub dry, after bathing, leaving a bit of water on the skin
- Apply a moisture-holding ointment or cream after the bath
- Use a humidifier in the home.

Patients who are allergic to certain substances, medications, and so on can avoid the resulting itch if they

KEY TERMS

Atopic dermatitis—An intensely itchy inflammation often found on the face of people prone to allergies. In infants and early childhood, it is called infantile eczema.

Creeping eruption—Itchy irregular, wandering red lines on the foot made by burrowing larvae of the hookworm family and some roundworms.

Dermatitis herpetiformis—A chronic very itchy skin disease with groups of red lesions that leave spots behind when they heal. It is sometimes associated with cancer of an internal organ.

Eczema—A superficial type of inflammation of the skin that may be very itchy and weeping in the early stages; later, the affected skin becomes crusted, scaly, and thick. There is no known cause.

Hodgkin's disease—A type of cancer characterized by a slowly-enlarging lymph tissue; symptoms include generalized itching.

Lichen planus—A noncancerous, chronic itchy skin disease that causes small, flat purple plaques on wrists, forearm, ankles.

Neurodermatitis—An itchy skin disease (also called lichen simplex chronicus) found in nervous, anxious people.

Pruriceptors—Nerve endings specialized to perceive itching sensations.

Pruritus—The medical term for itching.

Psoriasis—A common, chronic skin disorder that causes red patches anywhere on the body. Occasionally, the lesions may itch.

Scabies—A contagious parasitic skin disease characterized by intense itching.

Swimmer's itch—An allergic skin inflammation caused by a sensitivity to flatworms that die under the skin, causing an itchy rash.

Manual of Diagnosis and Therapy. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

PERIODICALS

- Al-Waili, N. S. "Topical Application of Natural Honey, Beeswax and Olive Oil Mixture for Atopic Dermatitis or Psoriasis: Partially Controlled, Single-Blinded Study." *Complementary Therapies in Medicine* 11 (December 2003): 226–234.
- Browning, J., B. Combes, and M. J. Mayo. "Long-Term Efficacy of Sertraline as a Treatment for Cholestatic Pruritus in Patients with Primary Biliary Cirrhosis." *American Journal of Gastroenterology* 98 (December 2003): 2736–2741.
- Cavalcanti, A. M., L. M. Rocha, R. Carillo Jr., et al. "Effects of Homeopathic Treatment on Pruritus of Haemodialysis Patients: A Randomised Placebo-Controlled Double-Blind Trial." *Homeopathy* 92 (October 2003): 177–181.
- Ikoma, A., R. Rukwied, S. Stander, et al. "Neurophysiology of Pruritus: Interaction of Itch and Pain." *Archives of Dermatology* 139 (November 2003): 1475–1478.
- Jones, K. "Review of Sangre de Drago (*Croton lechleri*)—A South American Tree Sap in the Treatment of Diarrhea, Inflammation, Insect Bites, Viral Infections, and Wounds: Traditional Uses to Clinical Research." *Journal of Alternative and Complementary Medicine* 9 (December 2003): 877–896.
- Ochoa, J. G. "Pruritus, a Rare but Troublesome Adverse Reaction of Topiramate." *Seizure* 12 (October 2003): 516–518.
- Stener-Victorin, E., T. Lundeborg, J. Kowalski, et al. "Perceptual Matching for Assessment of Itch; Reliability and Responsiveness Analyzed by a Rank-Invariant Statistical Method." *Journal of Investigative Dermatology* 121 (December 2003): 1301–1305.
- Zylicz, Z., M. Krajnik, A. A. Sorge, and M. Costantini. "Paroxetine in the Treatment of Severe Non-Dermatological Pruritus: A Randomized, Controlled Trial." *Journal of Pain and Symptom Management* 26 (December 2003): 1105–1112.

Carol A. Turkington
Rebecca J. Frey, PhD

avoid contact with the allergen. Avoiding insect bites, bee stings, poison ivy and so on can prevent the resulting itch. Treating sensitive skin carefully, avoiding overdrying of the skin, and protecting against diseases that cause itchy rashes are all good ways to avoid itching.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD, editors. "Pruritus." Section 10, Chapter 109. In *The Merck*

IUD

Definition

An IUD is an intrauterine device made of plastic and/or copper that is inserted into the womb (uterus) by way of the vaginal canal. One type releases a hormone (progesterone), and is replaced each year. The second type is made of copper and can be left in place

for five years. The most common shape in current use is a plastic “T” which is wrapped with copper wire.

Purpose

IUDs are used to prevent **pregnancy** and are considered to be 95-98% effective. It should be noted that IUDs offer no protection against the acquired immune deficiency syndrome (**AIDS**) virus or other **sexually transmitted diseases** (STDs).

Precautions

IUDs are placed in the uterus by physicians. Prior to placement the doctor will take a medical history, do a **physical examination**, and take a **Pap test**. Women who have had tubal pregnancies, an abnormal Pap smear, or abnormal vaginal bleeding are generally disqualified from using this form of **contraception**. Also, women who have STDs, an allergy to copper, severe **pain** with periods (menstruation), sex with multiple partners, or who are currently pregnant are not eligible for an IUD. There are no age restrictions.

Description

There is continuing controversy over exactly how IUDs prevent pregnancy. Some researchers think pregnancy is controlled by preventing conception (fertilization), while others believe that the devices prevent embryo attachment to the uterine wall (implantation).

IUDs which release a hormone may prevent pregnancy in several ways. Since one hormonal response is a thickening of the mucous at the entrance to the uterus, it is more difficult for the sperm to gain entry. This prevents the sperm from reaching an ovum. At the same time, the lining of the uterus becomes thinner, making it more difficult for a fertilized egg to implant itself in the uterus. The copper device slowly releases copper which is believed to weaken and perhaps kill sperm. An alternate explanation is that these objects “sweep” the uterus, dislodging any fertilized egg that attempts to implant itself. In addition, both devices tend to cause a mild inflammatory reaction in the lining of the uterus which also has an adverse impact on implantation.

Preparation

After the physician approves the use of an IUD, the woman’s genital area is washed thoroughly with soap and water in preparation of IUD insertion. The opening into the uterus (cervix) will also be cleaned with an antiseptic such as an iodine solution. Actual IUD insertion takes about five minutes, during which

KEY TERMS

Antiseptic—An antiseptic is a chemical that prevents the growth of germs.

Hormone—Hormones are chemicals that are produced in an organ or gland and then are carried by the blood to another part of the body where they produce a special effect for which they were designed.

Pap test—This is a procedure by which cells are collected from the cervix and vagina by inserting a swab into the vaginal canal. These cells are then examined under a microscope in order to detect signs of early cancer.

a **local anesthesia** is used to reduce any discomfort associated with the procedure. A plastic string connected to the IUD will hang out of the uterus into the vagina. The string is used to periodically check the position of the IUD.

Aftercare

The woman will be taught to watch for the signs and symptoms of potential complications and how to check the string, which should be done at least once a week. To check the string, the woman should first wash her hands with soap and water. From a squatting position, or with one foot elevated (such as on a chair), she should gently insert her finger into the vagina until she nears the cervix. If she cannot feel the string, if the string feels longer than it should, or if she can feel part of the IUD, she should notify her physician immediately. Additional information that needs to be reported includes painful intercourse and unusual discharge from the vagina.

Risks

Serious risks are rare, but include heavy bleeding, pain, infection, cramps, **pelvic inflammatory disease**, perforation of the uterus, and **ectopic pregnancy**.

Resources

ORGANIZATIONS

Planned Parenthood Federation of America, Inc. 810 Seventh Ave., New York, NY, 10019. (800) 669-0156. < <http://www.plannedparenthood.org> >.

Donald G. Barstow, RN

Ivory bones see **Osteopetroses**

Ivy method see **Bleeding time**



Japanese encephalitis

Definition

Japanese **encephalitis** is an infection of the brain caused by a virus. The virus is transmitted to humans by mosquitoes.

Description

The virus that causes Japanese encephalitis is called an arbovirus, which is an arthropod-borne virus. Mosquitoes are a type of arthropod. Mosquitoes in a number of regions carry this virus and are responsible for passing it along to humans. Many of these areas are in Asia, including Japan, Korea, China, India, Thailand, Indonesia, Malaysia, Vietnam, Taiwan, and the Philippines. Areas where the disease-causing arbovirus is always present are referred to as being endemic for the disease. In such areas, blood tests will reveal that more than 70% of all adults have been infected at some point with the arbovirus.

Because the virus that causes Japanese encephalitis is carried by mosquitoes, the number of people infected increases during those seasons when mosquitoes are abundant. This tends to be in the warmest, rainiest months. In addition to humans, other animals like wild birds, pigs, and horses are susceptible to infection with this arbovirus. Because the specific type of mosquito carrying the Japanese encephalitis arbovirus frequently breeds in rice paddies, the disease is considered to be primarily a rural problem.

Causes and symptoms

The virus is transferred to a human when an infected mosquito sucks that person's blood. Once in the body, the virus travels to various glands where it multiplies. The virus can then enter the bloodstream.

Ultimately, the virus settles in the brain, where it causes serious problems.

Japanese encephalitis begins with **fever**, severe **headache**, **nausea**, and **vomiting**. As the tissue covering the brain and spinal cord (the meninges) becomes infected and swollen, the patient will develop a stiff and painful neck. By day two or three, the patient begins to suffer the effects of swelling in the brain. These effects include:

- Problems with balance and coordination
- Paralysis of some muscle groups
- Tremors
- Seizures
- Lapses in consciousness
- A stiff, mask-like appearance of the face.

The patient becomes dehydrated and loses weight. If the patient survives the illness, the fever will decrease by about day seven and the symptoms will begin to improve by about day 14. Other patients will continue to have extremely high fevers and their symptoms will get worse. In these cases, **coma** and then **death** occur in 7-14 days. Many patients who recover have permanent disabilities due to brain damage.

Diagnosis

Most diagnostic techniques for Japanese encephalitis do not yield results very quickly. The diagnosis is made primarily on the basis of the patient's symptoms and the knowledge of the kinds of illnesses endemic to a particular geographic region.

Immunofluorescence tests, where special viral markers react with human antibodies that have been tagged with a fluorescent chemical, are used to verify the disease. However, these results tend to be unavailable until week two of the infection. Other tests involve comparing the presence and quantity of particular

KEY TERMS

Antibody—A type of cell made by the immune system that has the ability to recognize markers (antigens) on the surface of invading organisms, like bacteria and viruses.

Encephalitis—A swelling of the brain, potentially causing serious brain damage.

Endemic—Naturally and consistently present in a certain geographical region.

antibodies in the blood or spinal fluid during week one with those present during week two of the illness.

Treatment

There are no treatments available to stop or slow the progression of Japanese encephalitis. Only the symptoms of each patient can be treated. Fluids are given to decrease **dehydration** and medications are given to decrease fever and **pain**. Medications are available to attempt to decrease brain swelling. Patients in a coma may require mechanical assistance with breathing.

Prognosis

While the majority of people infected with arbo-virus never become sick, those who develop Japanese encephalitis become very ill. Some outbreaks have a 50% death rate. A variety of long-term problems may haunt those who recover from the illness. These problems include:

- Movement difficulties where the arms, legs, or body jerks or writhes involuntarily
- Shaking
- Paralysis
- Inability to control emotions
- Loss of mental abilities
- Mental disturbances, including **schizophrenia** (which may affect as many as 75% of Japanese encephalitis survivors).

Young children are most likely to have serious, long-term problems after an infection.

Prevention

A three-dose vaccine is available for Japanese encephalitis and is commonly given to young children

in areas where the disease is endemic. Travelers to these regions can also receive the vaccine.

Controlling the mosquito population with insecticides is another preventive measure. Visitors to regions with high rates of Japanese encephalitis should take precautions (like using mosquito repellents and sleeping under a bed net) to avoid contact with mosquitoes.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Rosalyn Carson-DeWitt, MD

Jaundice

Definition

Jaundice is a condition in which a person's skin and the whites of the eyes are discolored yellow due to an increased level of bile pigments in the blood resulting from **liver disease**. Jaundice is sometimes called icterus, from a Greek word for the condition.

Description

In order to understand jaundice, it is useful to know about the role of the liver in producing bile. The most important function of the liver is the processing of chemical waste products like cholesterol and excreting them into the intestines as bile. The liver is the premier chemical factory in the body—most incoming and outgoing chemicals pass through it. It is the first stop for all nutrients, toxins, and drugs absorbed by the digestive tract. The liver also collects chemicals from the blood for processing. Many of these outward-bound chemicals are excreted into the bile. One particular substance, bilirubin, is yellow. Bilirubin is a product of the breakdown of hemoglobin, which is the protein inside red blood cells. If bilirubin cannot leave the body, it accumulates and discolors other tissues. The normal total level of bilirubin in blood serum is between 0.2mg/dL and 1.2 mg/dL. When it rises to 3 mg/dL or higher, the person's skin and the whites of the eyes become noticeably yellow.

Bile is formed in the liver. It then passes into the network of hepatic bile ducts, which join to form a single tube. A branch of this tube carries bile to the gallbladder, where it is stored, concentrated, and released on a signal from the stomach. Food entering the stomach is the signal that stimulates the gallbladder to release the bile. The tube, which is called the common bile duct, continues to the intestines. Before the common bile duct reaches the intestines, it is joined by another duct from the pancreas. The bile and the pancreatic juice enter the intestine through a valve called the ampulla of Vater. After entering the intestine, the bile and pancreatic secretions together help in the process of digestion.

Causes and symptoms

There are many different causes for jaundice, but they can be divided into three categories based on where they start—before, in, or after the liver (pre-hepatic, hepatic and post-hepatic). When bilirubin begins its life cycle, it cannot be dissolved in water. The liver changes it so that it is soluble in water. These two types of bilirubin are called unconjugated (insoluble) and conjugated (soluble). Blood tests can easily distinguish between these two types of bilirubin.

Hemoglobin and bilirubin formation

Bilirubin begins as hemoglobin in the blood-forming organs, primarily the bone marrow. If the production of red blood cells (RBCs) falls below normal, the extra hemoglobin finds its way into the bilirubin cycle and adds to the pool.

Once hemoglobin is in the red cells of the blood, it circulates for the life span of those cells. The hemoglobin that is released when the cells die is turned into bilirubin. If for any reason the RBCs die at a faster rate than usual, bilirubin can accumulate in the blood and cause jaundice.

Hemolytic disorders

Many disorders speed up the death of red blood cells. The process of red blood cell destruction is called hemolysis, and the diseases that cause it are called hemolytic disorders. If red blood cells are destroyed faster than they can be produced, the patient develops anemia. Hemolysis can occur in a number of diseases, disorders, conditions, and medical procedures:

- **Malaria.** The malaria parasite develops inside red blood cells. When it is mature it breaks the cell apart and swims off in the blood. This process happens to most of the parasites simultaneously, causing

the intermittent symptoms of the disease. When enough cells burst at once, jaundice may result from the large amount of bilirubin formed from the hemoglobin in the dead cells. The pigment may reach the urine in sufficient quantities to cause “blackwater fever,” an often lethal form of malaria.

- Side effects of certain drugs. Some common drugs can cause hemolysis as a rare but sudden side effect. These medications include some antibiotic and anti-tuberculosis medicines; drugs that regulate the heart-beat; and levodopa, a drug used to treat Parkinson’s disease.
- Certain drugs in combination with a hereditary enzyme deficiency known as glucose-6-phosphate dehydrogenase (G6PD). G6PD is a deficiency that affects more than 200 million people in the world. Some of the drugs listed above are more likely to cause hemolysis in people with G6PD. Other drugs cause hemolysis only in people with this disorder. Most important among these drugs are anti-malarial medications such as quinine, and **vitamins C and K**.
- Poisons. Snake and spider venom, certain bacterial toxins, copper, and some organic industrial chemicals directly attack the membranes of red blood cells.
- Artificial heart valves. The inflexible moving parts of heart valves damage RBCs as they flutter back and forth. This damage is one reason to recommend pig valves and valves made of other organic materials.
- Hereditary RBC disorders. There are a number of hereditary defects that affect the blood cells. There are many genetic mutations that affect the hemoglobin itself, the best-known of which is **sickle cell disease**. Such hereditary disorders as spherocytosis weaken the outer membrane of the red cell. There are also inherited defects that involve the internal chemistry of RBCs.
- Enlargement of the spleen. The spleen is an organ that is located near the upper end of the stomach and filters the blood. It is supposed to filter out and destroy only worn-out RBCs. If it has become enlarged, it filters out normal cells as well. Malaria, other infections, cancers and leukemias, some of the hereditary **anemias** mentioned above, obstruction of blood flow from the spleen—all these and many more diseases can enlarge the spleen to the point where it removes too many red blood cells.
- Diseases of the small blood vessels. Hemolysis that occurs in diseased small blood vessels is called microangiopathic hemolysis. It results from damage caused by rough surfaces on the inside of the capillaries. The RBCs squeeze through capillaries one at a

time and can easily be damaged by scraping against the vessel walls.

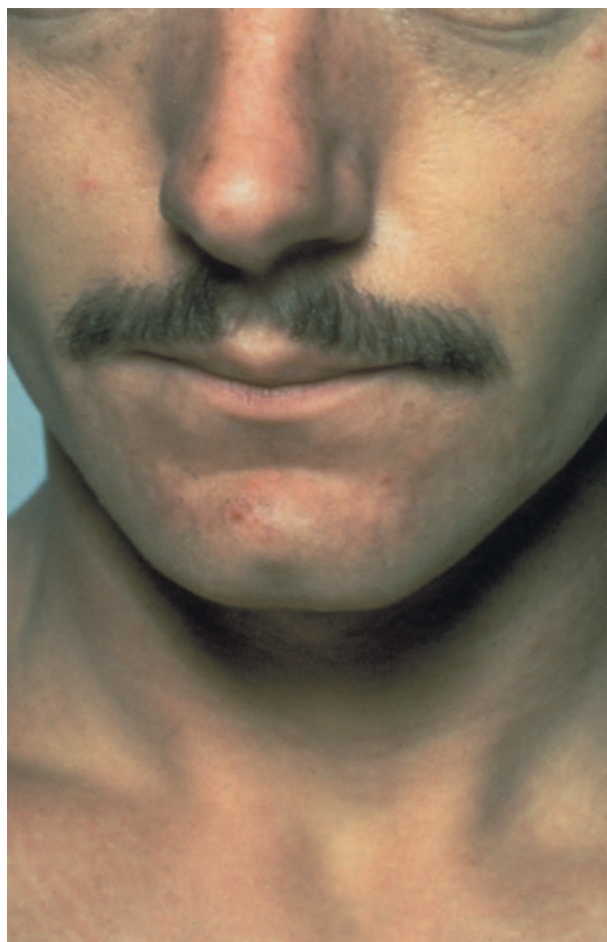
- Immune reactions to RBCs. Several types of **cancer** and immune system diseases produce antibodies that react with RBCs and destroy them. In 75% of cases, this reaction occurs all by itself, with no underlying disease to account for it.
- Transfusions. If a patient is given an incompatible blood type, hemolysis results.
- Kidney failure and other serious diseases. Several diseases are characterized by defective blood coagulation that can destroy red blood cells.
- **Erythroblastosis fetalis**. Erythroblastosis fetalis is a disease of newborns marked by the presence of too many immature red blood cells (erythroblasts) in the baby's blood. When a baby's mother has a different blood type, antibodies from the mother may leak into the baby's circulation and destroy blood cells. This reaction can produce severe hemolysis and jaundice in the newborn. Rh factor incompatibility is the most common cause.
- High bilirubin levels in newborns. Even in the absence of blood type incompatibility, the newborn's bilirubin level may reach threatening levels.

Normal jaundice in newborns

Normal newborn jaundice is the result of two conditions occurring at the same time—a pre-hepatic and a hepatic source of excess bilirubin. First of all, the baby at birth immediately begins converting hemoglobin from a fetal type to an adult type. The fetal type of hemoglobin was able to extract oxygen from the lower levels of oxygen in the mother's blood. At birth the infant can extract oxygen directly from his or her own lungs and does not need the fetal hemoglobin any more. So fetal hemoglobin is removed from the system and replaced with adult hemoglobin. The resulting bilirubin loads the system and places demands on the liver to clear it. But the liver is not quite ready for the task, so there is a period of a week or so when the liver has to catch up. During that time the baby is jaundiced.

Hepatic jaundice

Liver diseases of all kinds threaten the organ's ability to keep up with bilirubin processing. **Starvation**, circulating infections, certain medications, hepatitis, and **cirrhosis** can all cause hepatic jaundice, as can certain hereditary defects of liver chemistry, including Gilbert's syndrome and Crigler-Najjar syndrome.



This patient suffers from obstructive jaundice, which is often caused by gallstones. (Custom Medical Stock Photo. Reproduced by permission.)

Post-hepatic jaundice

Post-hepatic forms of jaundice include the jaundices caused by failure of soluble bilirubin to reach the intestines after it has left the liver. These disorders are called obstructive jaundices. The most common cause of obstructive jaundice is the presence of **gallstones** in the ducts of the biliary system. Other causes have to do with **birth defects** and infections that damage the bile ducts; drugs; infections; cancers; and physical injury. Some drugs—and **pregnancy** on rare occasions—simply cause the bile in the ducts to stop flowing.

Symptoms and complications associated with jaundice

Certain chemicals in bile may cause **itching** when too much of them end up in the skin. In newborns, insoluble bilirubin may get into the brain and do permanent damage. Long-standing jaundice may upset

the balance of chemicals in the bile and cause stones to form. Apart from these potential complications and the discoloration of skin and eyes, jaundice by itself is inoffensive. Other symptoms are determined by the disease producing the jaundice.

Diagnosis

Physical examination

In many cases the diagnosis of jaundice is suggested by the appearance of the patient's eyes and complexion. The doctor will ask the patient to lie flat on the examining table in order to feel (palpate) the liver and spleen for enlargement and to evaluate any abdominal **pain**. The location and severity of abdominal pain and the presence or absence of **fever** help the doctor to distinguish between hepatic and obstructive jaundice.

Laboratory tests

Disorders of blood formation can be diagnosed by more thorough examination of the blood or the bone marrow, where blood is made. Occasionally a **bone marrow biopsy** is required, but usually the blood itself will reveal the diagnosis. The spleen can be evaluated by an ultrasound examination or a nuclear scan if the **physical examination** has not yielded enough information.

Liver disease is usually assessed from blood studies alone, but again a biopsy may be necessary to clarify less obvious conditions. A **liver biopsy** is performed at the bedside. The doctor uses a thin needle to take a tiny core of tissue from the liver. The tissue sample is sent to the laboratory for examination under a microscope.

Assessment of jaundice in newborns

Newborns are more likely to have problems with jaundice if:

- They are premature.
- They are Asian or Native Americans.
- They have been bruised during the birth process.
- They have lost too much weight during the first few days.
- They are born at high altitude.
- The mother has diabetes.
- Labor had to be induced.

In 2003, research was continuing to find noninvasive methods to determine bilirubin levels in newborns



A newborn baby undergoes phototherapy with visible blue light to treat his jaundice. (Photograph by Ron Sutherland, Photo Researchers, Inc. Reproduced by permission.)

so that physicians did not have to rely on visual examination alone to determine which infants should receive blood tests. Once these measurements of skin pigment can be shown effective and cost-effective in clinical practice, they may become more available. Another study used this measurement method incorporated into home health visits to monitor babies within 24 hours of discharge from the hospital following birth.

Imaging studies

Disease in the biliary system can be identified by imaging techniques, of which there are many. X rays are taken a day after swallowing a contrast agent that is secreted into the bile. This study gives functional as well as anatomical information. There are several ways of injecting contrast dye directly into the bile ducts. It can be done through a thin needle pushed

straight into the liver or through a scope passed through the stomach that can inject dye into the Ampulla of Vater. CT and MRI scans are very useful for imaging certain conditions like cancers in and around the liver or gallstones in the common bile duct.

Treatment

Jaundice in newborns

Newborns are the only major category of patients in whom the jaundice itself requires attention. Because the insoluble bilirubin can get into the brain, the amount in the blood must not go over certain levels. If there is reason to suspect increased hemolysis in the newborn, the bilirubin level must be measured repeatedly during the first few days of life. If the level of bilirubin shortly after birth threatens to go too high, treatment must begin immediately. Exchanging most of the baby's blood was the only way to reduce the amount of bilirubin until a few decades ago. Then it was discovered that bright blue light rendered the bilirubin harmless. Now jaundiced babies are fitted with eye protection and placed under bright fluorescent lights. The light chemically alters the bilirubin in the blood as it passes through the baby's skin. In 2003, researchers were testing a new drug called Stanate that showed promise in blocking bilirubin production. However, debate concerning the use of the drug for treatment of only those infants with jaundice or as a preventive measure was delaying its FDA approval and widespread use.

Hemolytic disorders

Hemolytic diseases are treated, if at all, with medications and blood transfusions, except in the case of a large spleen. Surgical removal of the spleen (**splenectomy**) can sometimes cure **hemolytic anemia**. Drugs that cause hemolysis or arrest the flow of bile must be stopped immediately.

Hepatic jaundice

Most liver diseases have no specific cure, but the liver is so robust that it can heal from severe damage and regenerate itself from a small remnant of its original tissue.

Post-hepatic jaundice

Obstructive jaundice frequently requires a surgical cure. If the original passageways cannot be restored, surgeons have several ways to create alternate routes. A popular technique is to sew an open piece of intestine over a bare patch of liver. Tiny bile ducts in that part of the liver will begin to discharge

KEY TERMS

Ampulla of Vater—The widened portion of the duct through which the bile and pancreatic juices enter the intestine. Ampulla is a Latin word for a bottle with a narrow neck that opens into a wide body.

Anemia—A condition in which the blood does not contain enough hemoglobin.

Biliary system/Bile ducts—The gall bladder and the system of tubes that carries bile from the liver into the intestines.

Bilirubin—A reddish pigment excreted by the liver into the bile as a breakdown product of hemoglobin.

Crigler-Najjar syndrome—A moderate to severe form of hereditary jaundice.

Erythroblastosis fetalis—A disorder of newborn infants marked by a high level of immature red blood cells (erythroblasts) in the infant's blood.

Gilbert's syndrome—A mild hereditary form of jaundice.

Glucose-6-phosphate dehydrogenase (G6PD) deficiency—A hereditary disorder that can lead to episodes of hemolytic anemia in combination with certain medications.

Hemoglobin—The red chemical in blood cells that carries oxygen.

Hemolysis—The destruction or breakdown of red blood cells.

Hepatic—Refers to the liver.

Icterus—Another name for jaundice.

Microangiopathic—Pertaining to disorders of the small blood vessels.

Pancreas—The organ beneath the stomach that produces digestive juices, insulin, and other hormones.

Sickle cell disease—A hereditary defect in hemoglobin synthesis that changes the shape of red cells and makes them more fragile.

Splenectomy—Surgical removal of the spleen.

their bile into the intestine, and pressure from the obstructed ducts elsewhere will find release in that direction. As the flow increases, the ducts grow to accommodate it. Soon all the bile is redirected through the open pathways.

Prevention

Erythroblastosis fetalis can be prevented by giving an Rh negative mother a gamma globulin solution called RhoGAM whenever there is a possibility that she is developing antibodies to her baby's blood. G6PD hemolysis can be prevented by testing patients before giving them drugs that can cause it. Medication side effects can be minimized by early detection and immediate cessation of the drug. Malaria can often be prevented by certain precautions when traveling in tropical or subtropical countries. These precautions include staying in after dark; using prophylactic drugs such as mefloquine; and protecting sleeping quarters with mosquito nets treated with insecticides and mosquito repellents. In 2003, new studies showed promise for a possible vaccine against malaria. Early trials showed that **vaccination** combination might stimulate T-cell activity against malaria, the best type of protection that researchers can hope to find. However, further studies will have to be done.

Resources

PERIODICALS

- Grimm, David. "Baby Pigment Peril." *U.S. News & World Report* July 28, 2003: 39.
- Lawrence, David. "Combination Malaria Vaccine Shows Early Promise in Human Trials." *The Lancet* May 31, 2003: 1875.
- Morantz, Carrie, and Brian Torrey. "AHRQ Report on Neonatal Jaundice." *American Family Physician* June 1, 2003: 2417.
- Richmond, Glenn, Melissa Brown, and Patricia Wagstaff. "Using a Home Care Model to Monitor Bilirubin Levels in Early Discharged Infants." *Topics in Health Information Management* January–March 2003: 39–43.

ORGANIZATIONS

- American Liver Foundation. 1425 Pompton Ave., Cedar Grove, NJ 07009. (800) 223-0179. <<http://www.liverfoundation.org>>.

J. Ricker Polsdorfer, MD
Teresa G. Odle

Jaw wiring

Definition

Jaw wiring, also known as maxillomandibular fixation, is a surgical procedure where metal pins and wires are anchored into the jaw bones and surrounding tissues to keep the jaw from moving.

Purpose

Sports injuries, automobile accidents, falls, or fist-fights are a few of the situations where the jaw might be fractured or broken. In these cases, jaw wiring may be necessary to keep the bones aligned and stable while the jaw heals. The presence of **cancer** or other diseased tissues may make removal and reconstruction of the jaw necessary. Wiring the jaws shut has been used in the past as a weight loss aid in cases of extreme **obesity** where other treatments had failed, although this procedure is rarely used for that purpose today.

Precautions

Traumatic injuries to the face can cause damage to facial nerves and salivary glands and ducts. These injuries can also leave **scars** that may require additional surgery to correct.

Description

Jaw wiring surgery can be performed by an oral or maxillofacial surgeon (a specially trained dentist), or by an otolaryngologist (a doctor specializing in surgeries of the head and neck). The procedure may be done in a medical or dental office if the office is staffed and equipped to handle this type of surgery. More often, this surgery is performed in a hospital or medical center surgical area. If jaw wiring is required due to an injury, the surgeon may set the fracture immediately before swelling sets in. It is also possible to wait (up to several weeks) until the swelling goes down and some of the soft tissue injuries have healed, prior to wiring the jaw fracture.

The surgeon realigns the fractured bones. Every effort is made to restore the shape and appearance of the original jaw line. If any teeth were damaged, repair or replacement may be done at the same time. Small incisions may be made through the skin and surrounding tissue so the pins and wires can be set into the jawbone to hold the fracture together. To prevent the lower jaw from moving during healing, pins and wires may be inserted into the top jaw, as well. The upper and lower jaws are then wired together in order to stabilize the fracture.

As with other types of bone **fractures**, the jaw may take several weeks to heal. Another type of jaw **immobilization** that has been developed more recently, rigid fixation uses small metal plates and screws rather than pins and wires to secure the jaw bones. The main benefit of this technique is that the jaws do not have to be wired shut, allowing the patient to return to a more normal lifestyle sooner.

KEY TERMS

Oral and maxillofacial surgeon—A dentist who is trained to perform surgery to correct injuries, defects, or conditions of the mouth, teeth, jaws, and face.

Otolaryngologist—A doctor who is trained to treat injuries, defects, or conditions of the head and neck.

Preparation

X rays of the fractured area may be taken prior to surgery. Depending on the extent of the facial injury or condition to be corrected, the patient may receive a sedative for relaxation, a local anesthetic drug to numb the area, and/or an anesthetic agent to induce unconsciousness prior to the surgery.

Aftercare

A patient whose jaw has been wired will not be able to eat solid foods for several weeks. In order for the bone and surrounding tissues to heal, it is important to maintain adequate **nutrition**. A liquid diet that can be consumed through a straw, will be required. Soft, precooked foods can be liquefied in a blender, however, it may be difficult for the patient to consume adequate calories, protein, **vitamins**, and **minerals** with this type of diet. Liquid diet formulas may be a good alternative. The patient will also have to be taught how to care for the mouth, teeth, and injured area while the wires are in place.

Risks

It is possible that scarring may occur due to the need to make small incisions in the skin in order to insert the wires. With any surgical procedure, there are risks associated with the anesthetic drugs used and the possibility of infection. If there is a risk that the patient may vomit, the jaw wiring may pose a **choking** hazard. It may be recommended that wire cutters be kept available in case the wires need to be cut in an emergency situation.

Resources

ORGANIZATIONS

American Association of Oral & Maxillofacial Surgeons.
9700 West Bryn Mawr Avenue; Rosemont, IL
60018-5701; (847) 678-6200.

American Dental Association. 211 E. Chicago Ave., Chicago, IL 60611. (312) 440-2500. <<http://www.ada.org>>.

OTHER

“Know the Score on Facial Sports Injuries” *The Virtual Hospital Page*. University of Iowa. <<http://www.vh.org>>.

“Topic: Maxillofacial Trauma.” *Connecticut Maxillofacial Surgeons Page*. <<http://www.cmsllc.com/toptm.html>>.

Altha Roberts Edgren

JC virus infection see **Progressive multifocal leukoencephalopathy**

Jejunostomy see **Enterostomy**

Jet lag

Definition

Jet lag is a condition marked by **fatigue**, **insomnia**, and irritability that is caused by air travel through changing time zones. It is commonplace: a 2002 study of international business travelers (IBTs) found that jet lag was one of the most common health problems reported, affecting as many as 74% of IBTs.

Description

Living organisms are accustomed to periods of night and day alternating at set intervals. Most of the human body's regulating hormones follow this cycle, known as circadian rhythm. The word circadian comes from the Latin, *circa*, meaning about, and *dies*, meaning day. These cycles are not exactly 24 hours long, hence the “circa.” Each chemical has its own cycle of highs and lows, interacting with and influencing the other cycles. Body temperature, sleepiness, thyroid function, growth hormone, metabolic processes, adrenal hormones, and the sleep hormone melatonin all cycle with daylight. There is a direct connection between the retina (where light hits the back of the eye) and the part of the brain that controls all these hormones. Artificial light has some effect, but sunlight has much more.

When people are without clocks in a compartment that is completely closed to sunlight, most of them fall into a circadian cycle of about 25 hours. Normally, all the regulating chemicals follow one another in order like threads in a weaving pattern. Every morning the sunlight resets the cycle, stimulating the leading

chemicals and thus compensating for the difference between the 24-hour day and the 25-hour innate rhythm.

When traveling through a number of time zones, most people reset their rhythms within a few days, demonstrating the adaptability of the human species. Some people, however, have upset rhythms that last indefinitely.

Causes and symptoms

Traveling through a few time zones at a time is not as disruptive to circadian rhythms as traveling around the world can be. The foremost symptom of jet lag is altered sleep pattern—sleepiness during the day, and insomnia during the night. Jet lag may also include **indigestion** and trouble concentrating. Individuals afflicted by jet lag will alternate in and out of a normal day-night cycle.

Treatment

Current treatments

In cases of short-term insomnia triggered by jet lag, a physician may recommend sleeping pills or prescription medication. Such medication should only be taken under the guidance of a health care professional.

Investigational treatments

In 2002, a team from Flanders University invented new jet lag sunglasses equipped with a vision device that used light to stimulate travelers' brains. They believed that wearing the glasses before and during flights could help the human clock adjust more easily to changing time zones. The researchers were looking for a commercial partner to help them further study the glasses and make them widely available. The effectiveness of glasses or other head-mounted light devices is still uncertain, however. A team of researchers at Columbia University reported in the fall of 2002 that the use of a head-mounted light visor yielded only modest improvement in the test subjects' symptoms of jet lag.

A newer medication that is considered investigational is a melatonin agonist presently known as LY 156735. An agonist is a drug that stimulates activity at cell receptors that are normally stimulated by such naturally occurring substances as melatonin. LY 156735 was found to speed up the readaptation time of volunteer subjects following a simulated 9-hour time shift.

Another new area of research involves the genes that encode the proteins governing circadian rhythms. It is known as of late 2002 that differences among individuals in adaptability to time zone changes are to some extent genetically determined. Targeting the genes that affect this adaptability may yield new treatments for jet lag and other disorders of circadian rhythm.

Alternative treatment

Exposure to bright morning sunlight cures jet lag after a few days in most people. A few will have prolonged sleep phase difficulties. For these, there is a curious treatment that has achieved success. By forcing one's self into a 27 hour day, complete with the appropriate stimulation from bright light, all the errant chemical cycles will be able to catch up during one week.

When selecting an international flight, individuals should try to arrange an early evening arrival in their destination city. When an individual is traveling to a destination in the east, he or she can try going to bed and waking up a few hours earlier several days before their flight. If travel is to the west, going to bed and waking up later than usual can help the body start to adjust to the upcoming time change. More specific recommendations are available as of 2002, tailored to whether the person is traveling through six time zones, 7–9 zones, or 10 or more.

The following precautions taken during an international flight can help to limit or prevent jet lag:

- Stay hydrated. Drink plenty of water and juices to prevent **dehydration**. Beverages and foods with **caffeine** should be avoided because of their stimulant properties. Alcohol should also be avoided.
- Stretch and walk. As much movement as possible during a flight helps circulation, which moves nutrients and waste through the body and aids in elimination.
- Stay on time. Set watches and clocks ahead to the time in the destination city to start adjusting to the change.
- Sleep smart. Draw the shade and sleep during the evening hours in the destination city, even if it is still daylight outside of the airplane. Earplugs and sleep masks may be helpful in blocking noise and light. Many airlines provide these items on international flights.
- Dress comfortably. Wear or bring comfortable clothes and slippers that will make sleeping during the flight easier.

KEY TERMS

Agonist—A medication that has an affinity for and stimulates the activity of cell receptors that are normally stimulated by naturally occurring substances, including melatonin.

Circadian—Pertaining to biological rhythms occurring at approximately 24-hour intervals. Jet lag is caused by a disruption of the human body's circadian clock.

Hormone—A chemical made in one part of the body that has an effect on another part.

Melatonin—A hormone that helps to regulate circadian rhythms.

Once arriving in their destination city, individuals should spend as much time outdoors in the sunlight as possible during the day to reset their internal clock and lessen the symptoms of jet lag. Bedtime should be postponed until at least 10 P.M., with no daytime naps. If a daytime nap is absolutely necessary, it should be limited to no more than two hours.

To promote a restful sleeping environment in a hotel setting, individuals should request that the hotel desk hold all phone calls. Because sleeping in too late can also prolong jet lag, an early wake up call should be requested if an alarm clock is not available. If the hotel room is noisy, a portable white noise machine can help to block outside traffic and hallway noises. A room air conditioner or fan can serve the same purpose. The temperature in the room should also be adjusted for sleeping comfort.

All antioxidants help to decrease the effects of jet lag. Extra doses of **vitamins A, C, and E**, as well as zinc and selenium, two days before and two days after a flight help to alleviate jet lag. Melatonin, a hormone which helps to regulate circadian rhythms, can also help to combat jet lag. Melatonin is available as an over-the-counter supplement in most health food stores and pharmacies, but no more than 3 mg should be used in a 24-hour period.

If weather prevents an individual from spending time in the sunlight, light therapy may be beneficial in decreasing jet lag symptoms. Light therapy, or **phototherapy**, uses a device called a light box, which contains a set of fluorescent or incandescent lights in front of a reflector. Typically, the patient sits for 30 minutes next to a 10,000-lux box (which is about 50 times as bright as an ordinary indoor light). Light therapy is

safe for most people, but those with eye diseases should consult a healthcare professional before undergoing the treatment.

Prognosis

Jet lag usually lasts 24–48 hours after travel has taken place. In that short time period, the body adjusts to the time change, and with enough rest and daytime exposure to sunlight, it returns to normal circadian rhythm.

Prevention

Eating a high protein diet that is low in calories before intended travel may help reduce the effects of jet lag.

Resources

PERIODICALS

- Boulos, Z., M. M. Macchi, M. P. Sturchler, et al. "Light Visor Treatment for Jet Lag After Westward Travel Across Six Time Zones." *Aviation, Space, and Environmental Medicine* 73 (October 2002): 953–963.
- "Jet Lag Sunglasses Help Body Clock Tick." *Optician* August 2, 2002: 1.
- Monson, Nancy. "What Really Works for Jet Lag." *Shape* August 2002: 78.
- Nickelsen, T., A. Samel, M. Vejvoda, et al. "Chronobiotic Effects of the Melatonin Agonist LY 156735 Following a Simulated 9h Time Shift: Results of a Placebo-Controlled Trial." *Chronobiology International* 19 (September 2002): 915–936.
- Parry, B. L. "Jet Lag: Minimizing Its Effects with Critically Timed Bright Light and Melatonin Administration." *Journal of Molecular Microbiology and Biotechnology* 4 (September 2002): 463–466.
- Rogers, H. L., and S. M. Reilly. "A Survey of the Health Experiences of International Business Travelers. Part One—Physiological Aspects." *Journal of the American Association of Occupational Health Nurses* 50 (October 2002): 449–459.
- Wisor, J. P. "Disorders of the Circadian Clock: Etiology and Possible Therapeutic Targets." *Current Drug Targets: Cns and Neurological Disorders* 1 (December 2002): 555–566.

ORGANIZATIONS

- American Sleep Disorders Association. 1610 14th Street NW, Suite 300. Rochester, MN 55901. (507) 287–6006.
- National Sleep Foundation. 1367 Connecticut Avenue NW, Suite 200. Washington, DC 20036. (202) 785–2300.

Paula Anne Ford-Martin
Rebecca J. Frey, PhD

Jock itch

Definition

Also known as *Tinea cruris*, jock itch is a growth of fungus in the warm, moist area of the groin.

Description

Although there are many causes of jock itch, this term has become synonymous with *tinea cruris*, a common fungal infection that affects the groin and inner thighs of men and woman. *Tinea* is the name of the fungus; *cruris* comes from the Latin word for leg. Jock itch can develop when tight garments trap moisture and heat. This creates an environment in which fungi multiply and flourish. Athletes often get jock itch but non-athletic men who sweat a lot can also get it. Jock itch occurs more commonly in men, but can affect women as well. The jock itch fungus may cause a rash on the upper and inner thighs, the armpits, and the area just underneath the breasts. Many people with *tinea cruris* also have **athlete's foot**. Athlete's foot is called *tinea pedis*.

Causes & symptoms

The rash of jock itch starts in the groin fold usually on both sides. If the rash advances, it usually advances down the inner thigh. The advancing edge is redder and more raised than areas that have been infected longer. The advancing edge is usually scaly and very easily distinguished or well demarcated. The skin within the border turns a reddish-brown and loses much of its scale. Jock itch can spread to the pubic and genital regions and sometimes to the buttocks.

Jock itch caused by *T. rubrum* does not involve the scrotum or penis. If those areas are involved, the most likely agent is *Candida albicans*, the same type of yeast that causes vaginal yeast infections.

Diagnosis

Often a case of jock itch can be identified based on the characteristic description previously described. If assessed by a conventional doctor, the area of affected skin may be scraped onto a glass slide for definitive diagnosis under the microscope. In order to determine the exact type of fungus present, a small piece of affected skin maybe sent off to the lab for further study.

KEY TERMS

Fungus—A single-celled or multi-celled organism without chlorophyll that reproduces by spores and lives by absorbing nutrients from organic matter.

Scrotum—The external pouch of skin and muscle containing the testes (testicles).

Vaginal yeast infection—An overgrowth of fungus in the vaginal area.

Treatment

Typical conventional treatment for jock itch involves the use of an anti-fungal cream, spray, or powder twice a day for about two weeks. Three commonly used, over-the-counter anti-fungals are *miconazole* (Micatin), *clotrimazole* (Lotrimin) and *tolnaftate* (Tinactin). While the tendency to discontinue treatment once **itching** disappears is common, it is important to use the anti-fungal for a full two-week course in order to prevent recurrence of the infection.

Alternative treatment

Topical treatments include poultices of peppermint, oregano, or lavender. Tea tree oil, diluted with a carrier oil of almond oil, can be applied to the rash several times per day. Cedarwood and jasmine oils can relieve itching when applied in the same manner. Grapefruit seed extract can be taken as a strong solution of 15 drops in 1 oz of water.

Another alternative remedy for jock itch is to wash the groin area with the diluted juice of a freshly squeezed lemon, which can help dry up the rash. A hair dryer on the cool setting can also be used on the area after showering to dry it thoroughly. A warm bath relieves itching in many patients. The affected area should kept clean and dry, and loose-fitting, cotton underwear is recommended.

Prognosis

Treatment for jock itch is quick and usually effective, but the condition often comes back. With treatment, jock itch improves in two or three days and is completely gone in three or four weeks. The following people should be especially vigilant to prevent the problem from returning:

- Athletes
- People with fungal infections that affect other parts of the body (such as athlete's foot)

- People who wear tight clothing
- People with damaged or altered immune systems, including people with HIV or AIDS

Prevention

The best prevention of jock itch is cleanliness and sanitation. This includes keeping the groin area dry, wearing loose-fitting rather than tight clothing, wear boxer shorts rather than briefs, change sweat-covered clothes as soon as possible, showering immediately after working out or playing a sport and then applying talc, and washing workout clothes or sports uniforms after each use.

Resources

BOOKS

Icon Health Publications. *Jock Itch—A Medical Dictionary, Bibliography, and Annotated Research Guide to Internet References* San Diego: Icon Health Publications, 2003.

PERIODICALS

Grin, Caron. "Tinea: Diagnostic Clues, Treatment Keys." *Consultant* (February 2004): 214-216.

"Jock Itch." *Clinical Reference Systems* (January 1, 2004): 1859.

Schmitt, B. D. "Jock Itch for Teenagers." *Clinical Reference Systems* (January 1, 2004): 1858.

ORGANIZATIONS

American Academy of Dermatology. 1350 I St. NW, Suite 870, Washington, DC 20005. (202) 842-3555. <<http://www.aad.org>>.

OTHER

August 5, 2003. National Institutes of Health Medical Encyclopedia *Jock Itch* <<http://www.nlm.nih.gov/medlineplus/print/ency/article/000876.htm>>. (Accessed March 30, 2005).

Ken R. Wells,

Jock itch see **Ringworm**

Joint aspiration see **Joint fluid analysis**

Joint biopsy

Definition

A joint or synovial membrane biopsy refers to a procedure where a sample of the joint lining or synovial membrane is taken.

KEY TERMS

Joint—The point where two bones meet.

Pathology—The branch of medicine that looks at abnormal changes in cells and tissues which signal disease.

Synovial membrane—Membrane lining a joint.

Trocar—A sharp pointed tube through which a needle can be inserted.

Purpose

A joint biopsy is performed to determine why a joint is painful or swollen. It is usually reserved for more difficult cases where the diagnosis is not clear. The test can be used to diagnose bacterial or fungal infections, an abnormal buildup of iron, **cancer**, or other diseases.

Precautions

The procedure must be done under very sterile conditions to reduce the risk of infection.

Description

The test is performed either in the doctor's office, clinic, or hospital by a surgeon. There are many different ways to perform this biopsy: through an incision in the joint; with a scope inserted in the joint; or, more typically, by the insertion of a sharp instrument through the skin. The procedure can be taken from any joint, but the most common joint requiring biopsy is the knee. A sharp instrument (trocar) is pushed into the joint space. A needle with an attached syringe is inserted into the joint to withdraw fluid for laboratory analysis. The surgeon may instill numbing medicine into the joint and along the needle track before the needle is withdrawn. The trocar and then the biopsy needle is inserted and specimens taken. After the specimen is taken, both the trocar and the biopsy needle are removed, a bandage is placed over the joint, and the samples are sent to pathology for analysis.

Preparation

Blood tests will be done to check that blood clots properly. A mild sedative may be given before the procedure. With the patient lying down, the skin over the joint is disinfected and a local anesthetic is injected into the skin and tissue just below the skin.

Aftercare

The joint will need rest for at least one day. Normal activity can resume if there is no increased **pain** or swelling.

Risks

There is a chance of joint swelling or tenderness. Rarely, bleeding and infection can occur in the joint, or the biopsy needle could break off or strike a nerve or blood vessel. The risk of infection is higher if the patient has an immune deficiency.

Resources

BOOKS

Schumacher, H. Ralph, Jr. "Synovial Fluid Analysis and Synovial Biopsy." In *Textbook of Rheumatology*, edited by William N. Kelley, et al. Philadelphia: W. B. Saunders Co., 1997.

Jeanine Barone, Physiologist

Joint endoscopy see **Arthroscopy**

Joint fluid analysis

Definition

Joint fluid analysis, also called synovial fluid analysis, or arthrocentesis, is a procedure used to assess joint-related abnormalities, such as in the knee or elbow.

Purpose

The purpose of a joint fluid analysis is to identify the cause of swelling in the joints, to relieve **pain** and distention from fluid accumulation in the joint, and to diagnose certain types of arthritis and inflammatory joint diseases. The test is also a method to determine whether an infection, either bacterial or fungal, exists within the joint.

Precautions

Joint fluid analysis should not be performed on any patient who is uncooperative, especially if the patient cannot or will not keep the joint immobile throughout the procedure. Patients with certain infections should be excluded from the procedure, particularly those who have a local infection along the proposed needle track. The joint space should be

accessible. Therefore, a poorly accessible joint space, such as in hip aspiration in an obese patient, should not be subject to this procedure.

Description

The test is also called arthrocentesis, joint tap, and closed joint aspiration. Normal synovial fluid is a clear or pale-yellow fluid found in small amounts in joints, bursae (fluid-filled sac found on points of friction, like joints), and tendon sheaths. The procedure is done by passing a needle into a joint space and sucking out (aspirating) synovial fluid for diagnostic analysis. When the sample is sent to the laboratory, the fluid is analyzed for color, clarity, quantity, and chemical composition. It is also examined microscopically to check for the presence of bacteria and other cells.

The procedure takes about 10 minutes. Prior to the procedure, any risks that are involved should be explained to the patient. No intravenous pain medications or sedatives are required, although the patient will be given a local anesthetic.

The patient is asked to lie on their back and remain relaxed. The local anesthetic, typically an injection of lidocaine, is then administered. The clinician is usually seated next to the patient. Then the clinician marks exactly where the needle is to enter. As the needle enters the joint, a "pop" may be felt or heard. This is normal. Correct placement of the needle in the joint space is normally painless. At this point, the clinician slowly drains some of the fluid into the syringe. The needle is then withdrawn and adhesive tape is placed over the needle site.

Preparation

Glucose, or sugar, in the joint can be a signal of arthritis. If the clinician will be doing a glucose test, the patient will be asked to fast for 6-12 hours preceding the procedure. If not, there is no special preparation required for a joint fluid analysis.

Aftercare

Some post-procedural pain may be experienced. For this reason, the patient should arrange to be driven home by someone else. Aftercare of the joints will depend on the results of the analysis.

Risks

While joint fluid analysis is generally a safe procedure, especially when performed on a large, easily accessible joint, such as the knee, some risks are

KEY TERMS

Aspirate—The removal by suction of a fluid from a body cavity using a needle.

Bursae—A closed sac lined with a synovial membrane and filled with fluid, usually found in areas subject to friction, such as where a tendon passes over a bone.

Hematoma—A localized mass of blood that is confined within an organ or tissue.

Synovial fluid—A transparent lubricating fluid secreted in a sac to protect an area where a tendon passes over a bone.

possible. Some of the complications to the procedure, although rare, include infection at the site of the needle stick, an accumulation of blood (hematoma) formation, local pain, injury to cartilage, tendon rupture, and nerve damage.

Normal results

The results of a normal joint fluid analysis include fluid of a clear or pale-yellow color and the absence of bacteria, fungus, and other cells, such as white blood cells.

Abnormal results

The results of an abnormal joint fluid analysis include fluid that is turbid, or cloudy. Also, white blood cells and other blood cells may be found, from which the clinician can make a diagnosis and arrive at a treatment for the joint problem. An abnormal result can indicate an infection caused by a bacteria, or **tuberculosis**. Or, there might be inflammation that is caused by **gout**, **rheumatoid arthritis**, or **osteoarthritis**.

Resources

BOOKS

Arnold, William J., and Robert W. Ike. "Specialized Procedures in the Management of Patients with Rheumatic Diseases." In *Cecil Textbook of Medicine*, edited by James B. Wyngaarden, et al. Philadelphia: W. B. Saunders Co., 1996.

Ron Gasbarro, PharmD

Joint infection see **Infectious arthritis**

Joint radiography see **Arthrography**

Joint replacement

Definition

Joint replacement is the surgical replacement of a joint with an artificial prosthesis.

Purpose

Great advances have been made in joint replacement since the first hip replacement was performed in the United States in 1969. Improvements have been made in the endurance and compatibility of materials used and the surgical techniques to install artificial joints. Custom joints can be made using a mold of the original joint that duplicate the original with a very high degree of accuracy.

The most common joints to be replaced are hips and knees. There is ongoing work on elbow and shoulder replacement, but some joint problems are still treated with joint resection (the surgical removal of the joint in question) or interpositional reconstruction (the reassembly of the joint from constituent parts).

Seventy percent of joint replacements are performed because arthritis has caused the joint to stiffen and become painful to the point where normal daily activities are no longer possible. If the joint does not respond to conservative treatment such medication, weight loss, activity restriction, and use of walking aids such as a cane, joint replacement is considered appropriate.

Patients with **rheumatoid arthritis** or other connective tissue diseases may also be candidates for joint replacement, but the results are usually less satisfactory in those patients. Elderly people who fall and break their hip often undergo hip replacement when the probability of successful bone healing is low.

More than 170,000 hip replacements are performed in the United States each year. Since the lifetime of the artificial joint is limited, the best candidates for joint replacement are over age 60.

Precautions

Joint replacements are performed successfully on an older-than-average group of patients. People with diseases that interfere with blood clotting are not good candidates for joint replacement. Joint replacement surgery should not be done on patients with infection, or any heart, kidney or lung problems that would make it risky to undergo **general anesthesia**.

Description

Joint replacements are performed under general or regional anesthesia in a hospital by an orthopedic surgeon. Some medical centers specialize in joint replacement, and these centers generally have a higher success rate than less specialized facilities. The specific techniques of joint replacement vary depending on the joint involved.

Hip Replacement

The surgeon makes an incision along the top of the thigh bone (femur) and pulls the thigh bone away from the socket of the hip bone (the acetabulum). An artificial socket made of metal coated with polyethylene (plastic) to reduce friction is inserted in the hip. The top of the thigh bone is cut, and a piece of artificial thigh made of metal is fitted into the lower thigh bone on one end and the new socket on the other.

The artificial hip can either be held in place by a synthetic cement or by natural bone in-growth. The cement is an acrylic polymer. It assures good locking of the prosthesis to the remaining bone. However, bubbles left in the cement after it cures may act as weak spots, causing the development of cracks. This promotes loosening of the prosthesis later in life. If additional surgery is needed, all the cement must be removed before surgery can be performed.

An artificial hip fixed by natural bone in-growth requires more precise surgical techniques to assure maximum contact between the remaining natural bone and the prosthesis. The prosthesis is made so that it contains small pores that encourage the natural bone to grow into it. Growth begins 6 to 12 weeks after surgery. The short term outcome with non-cemented hips is less satisfactory, with patients reporting more thigh **pain**, but the long term outlook is better, with fewer cases of hip loosening in non-cemented hips. The trend is to use the non-cemented technique. Hospital stays last from four to eight days.

Knee Replacement

The doctor puts a tourniquet above the knee, then makes a cut to expose the knee joint. The ligaments surrounding the knee are loosened, then the shin bone and thigh bone are cut and the knee removed. The artificial knee is then cemented into place on the remaining stubs of those bones. The excess cement is removed, and the knee is closed. Hospital stays range from three to six days.

In both types of surgery, preventing infection is very important. **Antibiotics** are given intravenously



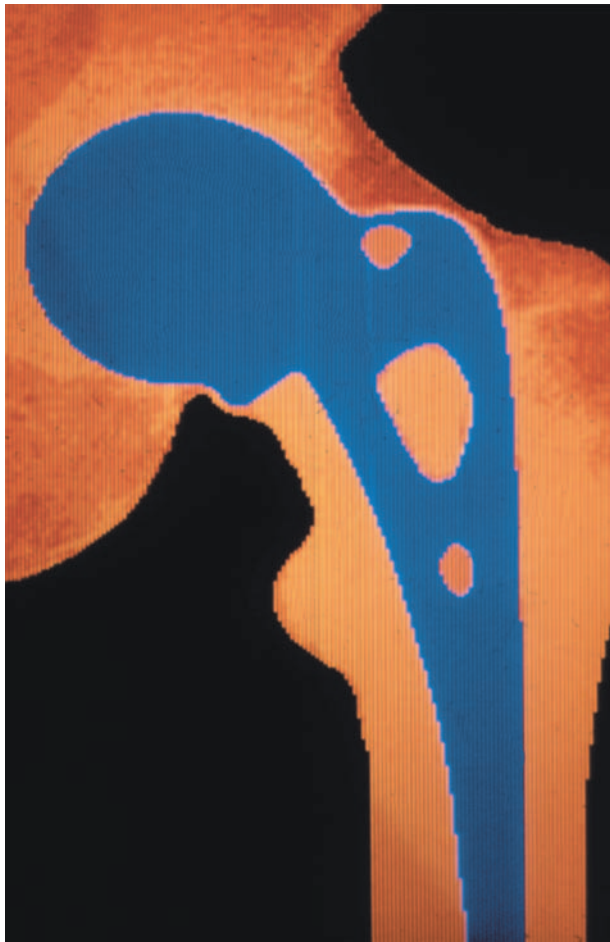
The components of a prosthetic hip joint, removed due to loosening. On the right is the metal shaft encased in the cement which fixed it to the inside of the femur. On the left is the plastic socket. (Custom Medical Stock Photo. Reproduced by permission.)

and continued in pill form after the surgery. Fluid and blood loss can be great, and sometimes blood transfusions are needed.

Preparation

Many patients choose to donate their own blood for **transfusion** during the surgery. This prevents any blood incompatibility problems or the transmission of bloodborne diseases.

Prior to surgery, all the standard preoperative blood and urine tests are performed, and the patient meets with the anesthesiologist to discuss any special conditions that affect the administration of anesthesia. Patients receiving general anesthesia should not eat or drink for ten hours prior to the operation.



A false color x-ray image of the human pelvis showing a prosthetic hip joint. (Custom Medical Stock Photo. Reproduced by permission.)

KEY TERMS

Catheterization—Inserting a tube into the bladder so that a patient can urinate without leaving the bed.

Prosthesis—A synthetic replacement for a missing part of the body, such as a knee or a hip.

Rheumatoid arthritis—A joint disease of unknown origins that may begin at an early age causing deformity and loss of function in the joints.

Aftercare

Immediately after the operation the patient will be catheterized so that he or she will not have to get

out of bed to urinate. The patient will be monitored for infection. Antibiotics are continued and pain medication is prescribed. Physical therapy begins (first passive exercises, then active ones) as soon as possible using a walker, cane, or crutches for additional support. Long term care of the artificial joint involves refraining from heavy activity and heavy lifting, and learning how to sit, walk, how to get out of beds, chairs, and cars so as not to dislocate the joint.

Risks

The immediate risks during and after surgery include the development of **blood clots** that may come loose and block the arteries, excessive loss of blood, and infection. Blood thinning medication is usually given to reduce the risk of clots forming. Some elderly people experience short term confusion and disorientation from the anesthesia.

Although joint replacement surgery is highly successful, there is an increased risk of nerve injury. Dislocation or fracture of the hip joint is also a possibility. Infection caused by the operation can occur as long as a year later and can be difficult to treat. Some doctors add antibiotics directly to the cement used to fix the replacement joint in place. Loosening of the joint is the most common cause of failure in hip joints that are not infected. This may require another joint replacement surgery in about 12% of patients within a 15-year period following the first procedure.

Normal results

More than 90% of patients receiving hip replacements achieve complete relief from pain and significant improvement in joint function. The success rate is slightly lower in knee replacements, and drops still more for other joint replacement operations.

Resources

PERIODICALS

Siopack, Jorge, and Harry Jergensen. "Total Hip Arthroplasty." In *Western Journal of Medicine* March 1995: 43–50.

Tish Davidson, A.M.

Joint resection see **Arthroplasty**

Joint x rays see **Arthrography**

Juvenile arthritis

Definition

Juvenile arthritis (JA), also called juvenile **rheumatoid arthritis** (JRA), refers to a number of different conditions, all of which strike children, and all of which have immune-mediated joint inflammation as their major manifestation. JRA is also known as juvenile idiopathic arthritis or JIA. The European League Against Rheumatism, or EULAR, refers to the disorder as juvenile chronic arthritis, or JCA.

Description

The skeletal system of the body is made up of different types of the strong, fibrous tissue known as connective tissue. Bone, cartilage, ligaments, and tendons are all forms of connective tissue which have different compositions, and thus different characteristics.

The joints are structures which hold two or more bones together. Some joints (synovial joints) allow for movement between the bones being joined (called articulating bones). The simplest model of a synovial joint involves two bones, separated by a slight gap called the joint cavity. The ends of each articular bone are covered by a layer of cartilage. Both articular bones and the joint cavity are surrounded by a tough tissue called the articular capsule. The articular capsule has two components: the fibrous membrane on the outside, and the synovial membrane (or synovium) on the inside. The fibrous membrane may include tough bands of fibrous tissue called ligaments, which are responsible for providing support to the joints. The synovial membrane has special cells and many capillaries (tiny blood vessels). This membrane produces a supply of synovial fluid which fills the joint cavity, lubricates it, and helps the articular bones move smoothly about the joint.

In JA, the synovial membrane becomes intensely inflamed. Usually thin and delicate, the synovium becomes thick and stiff, with numerous infoldings on its surface. The membrane becomes invaded by white blood cells, which produce a variety of destructive chemicals. The cartilage along the articular surfaces of the bones may be attacked and destroyed, and the bone, articular capsule, and ligaments may begin to be worn away (eroded). These processes severely interfere with movement in the joint.

JA specifically refers to chronic arthritic conditions which affect a child under the age of 16 years,

and which last for a minimum of three to six months. JA is often characterized by a waxing and waning course, with flares separated by periods of time during which no symptoms are noted (remission). Some literature refers to JA as juvenile rheumatoid arthritis, although most types of JA differ significantly from the adult disease called rheumatoid arthritis, in terms of symptoms, progression, and prognosis.

Causes and symptoms

A number of different causes have been sought to explain the onset of JA. There seems to be some genetic link, based on the fact that the tendency to develop JA sometimes runs in a particular family, and based on the fact that certain genetic markers are more frequently found in patients with JA and other related diseases. Many researchers have looked for some infectious cause for JA, but no clear connection to a particular organism has ever been made. JA is considered by some to be an autoimmune disorder. **Autoimmune disorders** occur when the body's immune system mistakenly identifies the body's own tissue as foreign, and goes about attacking those tissues, as if trying to rid the body of an invader (such as a bacteria, virus, or fungi). While an autoimmune mechanism is strongly suspected, certain markers of such a mechanism (such as rheumatoid factor, often present in adults with such disorders) are rarely present in children with JA.

Joint symptoms of arthritis may include stiffness, **pain**, redness and warmth of the joint, and swelling. Bone in the area of an affected joint may grow too quickly, or too slowly, resulting in limbs which are of different lengths. When the child tries to avoid moving a painful joint, the muscle may begin to shorten from disuse. This is called a contracture.

Symptoms of JA depend on the particular subtype. According to criteria published by the American College of Rheumatology (ACR) in 1973 and modified in 1977, JRA is classified by the symptoms that appear within the first six months of the disorder:

- **Pauciarticular JA:** This is the most common and the least severe type of JA, affecting about 40–60% of all JA patients. This type of JA affects fewer than four joints, usually the knee, ankle, wrist, and/or elbow. Other more general (systemic) symptoms are usually absent, and the child's growth usually remains normal. Very few children (less than 15%) with pauciarticular JA end up with deformed joints. Some children with this form of JA experience painless swelling of the joint. Some children with JA have a serious inflammation of structures within the eye,

which if left undiagnosed and untreated could even lead to blindness. This condition is known as **uveitis**, and affects about 20% of children diagnosed with JRA. While many children have cycles of flares and remissions, in some children the disease completely and permanently resolves within a few years of diagnosis.

- **Polyarticular JA:** About 40% of all cases of JA are of this type. More girls than boys are diagnosed with this form of JA. This type of JA is most common in children up to age three, or after the age of 10. Polyarticular JA affects five or more joints simultaneously. This type of JA usually affects the small joints of both hands and both feet, although other large joints may be affected as well. Some patients with arthritis in their knees will experience a different rate of growth in each leg. Ultimately, one leg will grow longer than the other. About half of all patients with polyarticular JA have arthritis of the spine and/or hip. Some patients with polyarticular JA will have other symptoms of a systemic illness, including anemia (low red blood cell count), decreased growth rate, low appetite, low-grade **fever**, and a slight rash. The disease is most severe in those children who are diagnosed in early adolescence. Some of these children will test positive for a marker present in other autoimmune disorders, called rheumatoid factor (RF). RF is found in adults who have rheumatoid arthritis. Children who are positive for RF tend to have a more severe course, with a disabling form of arthritis which destroys and deforms the joints. This type of arthritis is thought to be the adult form of rheumatoid arthritis occurring at a very early age.
- **Systemic onset JA:** Sometimes called Still disease (after a physician who originally described it), this type of JA occurs in about 10–20% of all patients with JA. Boys and girls are equally affected, and diagnosis is usually made between the ages of 5–10 years. The initial symptoms are not usually related to the joints. Instead, these children have high fevers; a rash; decreased appetite and weight loss; severe joint and muscle pain; swollen lymph nodes, spleen, and liver; and serious anemia. Some children experience other complications, including inflammation of the sac containing the heart (**pericarditis**); inflammation of the tissue lining the chest cavity and lungs (**pleuritis**); and inflammation of the heart muscle (**myocarditis**). The eye inflammation often seen in pauciarticular JA is uncommon in systemic onset JA. Symptoms of actual arthritis begin later in the course of systemic onset JA, and they often involve the wrists and ankles. Many of these children

continue to have periodic flares of fever and systemic symptoms throughout childhood. Some children will go on to develop a polyarticular type of JA.

- **Spondyloarthropathy:** This type of JA most commonly affects boys older than eight years of age. The arthritis occurs in the knees and ankles, moving over time to include the hips and lower spine. Inflammation of the eye may occur occasionally, but usually resolves without permanent damage.
- **Psoriatic JA:** This type of arthritis usually shows up in fewer than four joints, but goes on to include multiple joints (appearing similar to polyarticular JA). Hips, back, fingers, and toes are frequently affected. A skin condition called **psoriasis** accompanies this type of arthritis. Children with this type of JA often have pits or ridges in their fingernails. The arthritis usually progresses to become a serious, disabling problem.

As of 2003, there is some disagreement among specialists about the classification of JRA. Some prefer the EULAR classification, also introduced in 1977, to the ACR system. In 1997, the World Health Organization (WHO) met in Durban and issued a new classification system for JRA known as the Durban criteria, in an attempt to standardize definitions of the various subtypes of JRA. None of the various classification systems, however, are considered fully satisfactory as of early 2004.

Diagnosis

Diagnosis of JA is often made on the basis of the child's collection of symptoms. Laboratory tests often show normal results. Some nonspecific indicators of inflammation may be elevated, including white blood cell count, **erythrocyte sedimentation rate**, and a marker called **C-reactive protein**. As with any chronic disease, anemia may be noted. Children with an extraordinarily early onset of the adult type of rheumatoid arthritis will have a positive test for rheumatoid factor.

Treatment

Treating JA involves efforts to decrease the amount of inflammation, in order to preserve movement. Medications which can be used for this include nonsteroidal anti-inflammatory agents (such as ibuprofen and naproxen). Oral (by mouth) steroid medications are effective, but have many serious side effects with long-term use. Injections of steroids into an affected joint can be helpful. Steroid eye drops are used to treat eye inflammation. Other drugs which

have been used to treat JA include methotrexate, sulfasalazine, penicillamine, and hydroxychloroquine. Physical therapy and exercises are often recommended in order to improve joint mobility and to strengthen supporting muscles. Occasionally, splints are used to rest painful joints and to try to prevent or improve deformities.

The Food and Drug Administration (FDA) approved a new drug, etanercept, marketed under the brand name Enbrel, in 1999. It is the most dramatic advancement in treating JRA in recent years. A study by Children's Hospital Medical Center in Cincinnati, Ohio, released in 1999, showed the drug was effective in 75% of children with severe JRA. The drug eases joint pain, reduces swelling, and improves mobility.

In 2003, a group of Japanese researchers noted that the blood serum of patients with JRA contains elevated levels of interleukin-6, a cytokine (nonantibody protein) that is critical to regulation of the immune system and blood cell formation. Because interleukin-6 is also associated with inflammation, the researchers think that compounds inhibiting the formation of interleukin-6 might provide new treatment options for JRA.

Alternative treatment

Alternative treatments that have been suggested for arthritis include juice therapy, which can work to detoxify the body, helping to reduce JA symptoms. Some recommended fruits and vegetables to include in the juice are carrots, celery, cabbage, potatoes, cherries, lemons, beets, cucumbers, radishes, and garlic. Tomatoes and other vegetables in the nightshade (potatoes, eggplant, red and green peppers) are discouraged. As an adjunct therapy, **aromatherapy** preparations utilize cypress, fennel, and lemon. Massage oils include rosemary, benzoin, chamomile, camphor, juniper, and lavender. Other types of therapy which have been used include **acupuncture**, **acupressure**, and body work. **Nutritional supplements** that may be beneficial include large amounts of antioxidants (**vitamins** C, A, E, zinc, selenium, and flavonoids), as well as B vitamins and a full complement of **minerals** (including boron, copper, manganese). Other nutrients that assist in detoxifying the body, including methionine, cysteine, and other amino acids, may also be helpful. A number of autoimmune disorders, including JA, seem to have a relationship to **food allergies**. Identification and elimination of reactive foods may result in a decrease in JA symptoms. Constitutional homeopathy can also work to

KEY TERMS

Articular bones—Two or more bones which are connected with each other via a joint.

Cytokine—A general term for nonantibody proteins released by a specific type of cell as part of the body's immune response.

Idiopathic—Of unknown cause or spontaneous origin. JRA is sometimes called juvenile idiopathic arthritis or JIA because its causes are still not fully known.

Joint—A structure that holds two or more bones together.

Rheumatology—The branch of medicine that specializes in the diagnosis and treatment of disorders affecting the muscles and joints.

Synovial joint—A particular type of joint that allows for movement in the articular bones.

Synovial membrane—The membrane that lines the inside of the articular capsule of a joint and produces a lubricating fluid called synovial fluid.

Uveitis—Inflammation of the pigmented vascular covering of the eye, which includes the choroid, iris, and ciliary body. Uveitis is a common complication of JRA.

quiet the symptoms of JA and bring about balance to the whole person.

Prognosis

The prognosis for pauciarticular JA is quite good, as is the prognosis for spondyloarthropathy. Polyarticular JA carries a slightly worse prognosis. RF-positive polyarticular JA carries a difficult prognosis, often with progressive, destructive arthritis and joint deformities. Systemic onset JA has a variable prognosis, depending on the organ systems affected, and the progression to polyarticular JA. About 1-5% of all JA patients die of such complications as infection, inflammation of the heart, or **kidney disease**.

Prevention

Because so little is known about what causes JA, there are no recommendations available for how to avoid developing it.

Resources

BOOKS

- Beers, Mark H., MD, and Robert Berkow, MD., editors. "Juvenile Rheumatoid Arthritis." Section 19, Chapter 270 In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2002.
- Behrman, Richard, et al., editors. *Nelson Textbook of Pediatrics*. 16th ed. Philadelphia: W. B. Saunders Co., 2000.
- Kredich, Deborah Welt. "Juvenile Rheumatoid Arthritis." In *Rudolph's Pediatrics*, edited by Abraham M. Rudolph. Stamford: McGraw-Hill, 2002.
- Peacock, Judith. *Juvenile Arthritis*. Mankato, MN: LifeMatters Books, 2000.

PERIODICALS

- de Boer, J., N. Wulffraat, and A. Rothova. "Visual Loss in Uveitis of Childhood." *British Journal of Ophthalmology* 87 (July 2003): 879–884.
- Henderson, Charles W. "Etanercept a Dramatic Advancement in Treatment, Say Researchers." *Immunotherapy Weekly* April 2, 2000.
- Kotaniemi, K., A. Savolainen, A. Karma, and K. Aho. "Recent Advances in Uveitis of Juvenile Idiopathic Arthritis." *Survey of Ophthalmology* 48 (September–October 2003): 489–502.
- Larkin, Marilynn. "Juvenile Arthritis Helped by Resistance Exercise." *Lancet* November 20, 1999: 1797.
- Manners, P., J. Lesslie, D. Speldewinde, and D. Tunbridge. "Classification of Juvenile Idiopathic Arthritis: Should

Family History Be Included in the Criteria?" *Journal of Rheumatology* 30 (August 2003): 1857–1863.

- Moran, M. "Autoimmune Diseases Could Share Common Genetic Etiology." *American Medical News* 44, no. 38 (October 8, 2001): 38.
- Yokota, S. "Interleukin 6 as a Therapeutic Target in Systemic-Onset Juvenile Idiopathic Arthritis." *Current Opinion in Rheumatology* 15 (September 2003): 581–586.

ORGANIZATIONS

- American College of Rheumatology. 1800 Century Place, Suite 250, Atlanta, GA 30345. (404) 633-3777. <<http://www.rheumatology.org>>.
- Arthritis Foundation. 1330 West Peachtree St., Atlanta, GA 30309. (404) 872-7100. <<http://www.arthritis.org>>.
- National Arthritis and Musculoskeletal and Skin Diseases Information Clearinghouse. National Institutes of Health, 1 AMS Circle, Bethesda, MD 20892. (301) 495-4484. <<http://www.nih.gov/niams>>.

OTHER

- National Institute of Arthritis and Musculoskeletal and Skin Diseases (NIAMS). *Questions and Answers About Juvenile Rheumatoid Arthritis*. NIH Publication No. 01-4942. Bethesda, MD: NIAMS, 2001. <http://www.niams.nih.gov/hi/topics/juvenile_arthritis/juvarthr.htm>.

Rosalyn Carson-DeWitt, MD
Rebecca J. Frey, PhD

K

Kala-azar see **Leishmaniasis**

Kaposi's sarcoma

Definition

Kaposi's sarcoma is a form of skin **cancer** that can involve internal organs. It most often is found in patients with acquired **immunodeficiency** syndrome (**AIDS**), and can be fatal.

Description

Kaposi's sarcoma (KS) was once a very rare form of cancer, primarily affecting elderly men of Mediterranean and eastern European background, until the 1980s, when it began to appear among AIDS patients. It manifests in four distinct forms. The first form, called classic KS, was described by the Austrian dermatologist Moricz Kaposi more than a century ago. Classic KS usually affects older men of Mediterranean or eastern European backgrounds by producing tumors on the lower legs. Though at times painful and disfiguring, they generally are not life-threatening. The second form of the disease, African endemic KS, primarily affects boys and men. It can appear as classic KS, or in a more deadly form that quickly spreads to tissues below the skin, the bones and lymph system, leading to **death** within a few years of diagnosis. Another form of KS, iatrogenic KS, is observed in kidney and liver transplant patients who take immunosuppressive drugs to prevent rejection of their organ transplant. Iatrogenic KS usually reverses after the immunosuppressive drug is stopped. The fourth form of KS, AIDS-related KS, emerged as one of the first illnesses observed among those with AIDS. Unlike classic KS, AIDS-related KS

tumors generally appear on the upper body, including the head, neck, and back. The tumors also can appear on the soft palate and gum areas of the mouth. In more advanced cases, they can be found in the stomach and intestines, the lymph nodes, and the lungs.

Incidence of Kaposi's sarcoma has been reported as high as 20% in homosexual men who have HIV, 3% in heterosexual intravenous drug users, 3% in women and children, 3% in **transfusion** recipients, and 1% in hemophiliacs. Once regarded as only a defining illness for AIDS, KS has proven to be a progressive, fatal disease on its own, especially when the disease becomes systemic. Yet involvement throughout the body is not the only factor in patient mortality. Research in 2000 found that patients with KS in oral mucosa had a higher risk of death than those with KS appearing only on the skin.

Causes and symptoms

A variety of factors appear to contribute to the development of KS. One of the first avenues offered as causal agents was genetic predisposition. People with classic KS, and those who develop the tumors after transplantation, are more likely than others to possess a genetically determined immune factor called HLA-DR. Cases of KS that run in families, however, are rare.

The fact that the disease is more likely to afflict men than women suggests sex hormones, such as testosterone in men, may stimulate the growth of KS tumors, and that estrogen in women may retard their growth.

Immune suppression was the next likely cause since liver, kidney, and bone marrow patients who take immunosuppressive drugs to prevent transplant rejection frequently develop KS lesions. Similarly,

MORIZ KAPOSI, (1837–1902)

Moriz Kohn Kaposi was born in 1837 to very poor Hungarian parents. He studied dermatology under Ferdinand von Hebra at the University of Vienna, earning his medical degree in 1861. Kaposi took a position in Hebra's clinic and ultimately became a lecturer where he was responsible for educating numerous dermatologists. Kaposi and Hebra coauthored *The Handbook of Diseases of the Skin* which had great success. Kaposi married Hebra's daughter and the couple had one son, Hermann (1872). When Hebra died, Kaposi filled the vacant spot as director of the skin clinic and as Vienna's most renowned dermatologist.

Between 1872 and 1887, Kaposi discovered nine skin diseases that had not been previously documented. His discovery of a malignant disease that strikes the lymph nodes and skin (Kaposi's sarcoma 1872) has been documented as the most noteworthy. This disease was seen relatively rarely in the United States until the 1980s when it was tied to AIDS. This sarcoma has been the most common tumor found in AIDS patients. In 1872, Kaposi also studied cases of lupus erythematosus, which had no previous documentation. Kaposi was a prolific writer who published *Pathology and Treatment of Diseases of the Skin* in addition to numerous other publications, which he completed individually and with other authors. Kaposi died in 1902.

KS has been observed in patients receiving systemic treatment with high-dose **corticosteroids**, which also suppresses the immune system. Immune suppression is the hallmark of AIDS.

The current theory is the discovery of an infectious agent. A number of viruses have been proposed as possible causes, including cytomegalovirus and human papilloma virus, fragments of which have been found in KS tumor specimens. A more likely candidate, however, is a new herpes virus that has been called human herpes virus 8 (HHV-8) or KS-associated herpes virus (KSHV). Since fragments of the virus were first disclosed in KS samples in 1994, they have since been found in KS samples taken from patients with classic KS, African endemic KS, and KS in transplant patients. Fragments of HHV-8, however, also have been found in patients who have other skin diseases but who do not have KS.

Studies in 2000 showed that HHV-8 was indeed the culprit behind KS. Nevertheless, it does not work alone. In combination with a patient's altered response to cytokines (regulatory proteins produced



This HIV-positive patient is afflicted with Kaposi's sarcoma inside the mouth. (Custom Medical Stock Photo. Reproduced by permission.)

by the immune system) and the HIV-1 transactivating protein Tat which promotes the growth of endothelial cells, HHV-8 can then encode interleukin 6 viral proteins. These specific cytokines stimulate cell growth in the skin, becoming KS.

HHV-8 destroys the immune system further by directing a cell to remove the major histocompatibility complex (MHC-1) proteins that protect it from invasion. These proteins then are transferred to the interior of the cell and are destroyed. This leaves the cell unguarded and vulnerable to invaders which normally would be targeted for attack by the immune system.

Research in early 2001 showed that transmission of HHV-8 virus can be more casual than once was thought, giving rise to incidence among women and children. Women who are intravenous drug users and who also have had a sexually transmitted disease have been found to harbor HHV-8. This evidence shows that women can contract HHV-8 through blood. In addition, researchers in 2000 found that HHV-8 could be transmitted orally through kissing. This study found more HHV-8 virus in oral samples than in genital secretions. In fact, HHV-8 was difficult to find in genital samples. This may indicate why children and women who were not intravenous drug users have had KS.

Kaposi's sarcoma produces pink, purple, or brown tumors on the skin, mucous membranes, or internal organs.

Diagnosis

Many physicians will diagnose KS based on the appearance of the skin tumors and the patient's medical history. Unexplained **cough** or chest **pain**, as well as unexplained stomach or intestinal pain or bleeding, could suggest that the disease has moved beyond the skin. The most certain diagnosis can be achieved by taking a biopsy sample of a suspected KS lesion and examining it under high-power magnification. For suspected involvement of internal organs, physicians will use a bronchoscope to examine the lungs or an endoscope to view the stomach and intestinal tract.

Treatment

Treatment goals for KS are simple: to reduce the severity of symptoms, shrink tumors, and prevent disease progression. Unfortunately, there is no single best treatment plan that can achieve all of these goals. Treatments range from topical agents for mild disease with few tumors to more aggressive systemic **chemotherapy** for more serious KS that has spread to large areas of skin or the internal organs. Physicians will frequently combine topical, radiation, and various systemic chemotherapy drugs, depending on the sites of the body affected, the speed at which it is progressing, and the patient's overall health, among other considerations.

Local therapy

When the number of KS tumors is small and the disease appears to be progressing slowly, physicians have had great success with the application, by the patient, of a topical gel containing alitretinoin. This product is a naturally occurring retinoid (a derivative of vitamin A) that can inhibit cell growth and activate apoptosis (cell death). Patients tolerate the product well with only mild to moderate skin irritation at the site of application in some individuals. Duration of treatment is long term, with the patient seeing results after four to eight weeks of therapy. Treatment slows the progress of the disease and reduces the size of the lesions.

Other local treatments include **cryotherapy** (using a liquid nitrogen spray or probe to freeze the tumor), injections of vinblastine (a drug also used for systemic chemotherapy) directly into the tumor, laser therapy, or **radiation therapy** targeted at the tumor sites. These methods have some success, but they also have unpleasant side effects. Vinblastine injections are about 70% effective, but they do not resolve the lesions completely.



Kaposi's sarcoma usually appears on the lower extremities, as evidenced on this patient's hip. (Custom Medical Stock Photo. Reproduced by permission.)

Systemic chemotherapy

With widespread KS lesions over the body surface, or evidence of spread to other parts of the body, physicians will consider systemic chemotherapy drugs. A new class of chemotherapy drugs, called liposomal anthracyclines, appears to produce good results with fewer toxic side effects than do more conventional chemotherapy drugs. Two of these drugs, liposomal doxorubicin (Doxil) and liposomal daunorubicin (DaunoXone) have become the treatment of choice. These drugs last longer in the human body, demonstrate higher concentrations of the drug in tumors, and have fewer toxic side effects.

Paclitaxel (Taxol) is the newest drug in the KS arsenal. It has a 75% effective rate and is very effective in patients who are resistant to anthracycline drugs. The 3-hour infusion time and the incidence of bone marrow suppression, hair loss, and joint and muscle pain make it less attractive to patients.

Antiretroviral therapy

Evidence suggests that for some individuals, the class of AIDS drugs called **protease inhibitors**, in combination with other anti-HIV drugs, can reduce the levels of detectable HIV in the blood to nearly zero, and in some patients stabilize or reverse KS tumors. A study late in 2003 showed that highly active antiretroviral therapy (HAART) containing a protease inhibitor helped block KS tumor growth, invasion and distant spread. HAART is a treatment used to treat HIV patients. Since the discovery of HHV-8, interest in an antiviral approach to KS has increased. There is no evidence, however, that two **antiviral drugs** commonly prescribed for herpes, acyclovir and ganciclovir, have any effect on the disease. One study of 20,000 patients with HIV and AIDS found that those who took foscarnet, another antiretroviral medication that works in a different way than acyclovir and ganciclovir, were less likely to develop KS tumors.

Another treatment source is interferon-alpha, which is made by the body and has powerful effects on the immune system. Investigators have tried injecting it directly into lesions, and also in combination with other anti-HIV drugs such as zidovudine, with some success. It has been used with patients who have KS limited only to the skin and who have little immunosuppression. Interferon-alpha has had poor tumor response and significant toxic effects in patients, especially those with seriously-depressed immune systems.

Still other avenues of therapy being researched are sex hormones, thalidomide, SU5516 (an endothelial growth factor inhibitor), and angiogenesis inhibitors, which prevent the growth of blood vessels within a cell that supplies oxygen and nutrients. There also is some research involving the oral administration of alitretinoin.

Alternative treatment

The Bastyr University AIDS Research Study has been investigating and collecting data on treatment for KS and other opportunistic conditions that are AIDS-related. Among the treatments under investigation are nutritional and herbal therapies (both internal and external). Bastyr University is located in Seattle, Washington.

Prognosis

The prognosis for patients with classic KS is good. Tumors can frequently be controlled and patients frequently die of other causes before any serious spread. African endemic KS can progress rapidly and lead to

KEY TERMS

African endemic Kaposi's sarcoma—Affects men and boys; can appear like classic KS or in a more lethal form.

AIDS-related Kaposi's sarcoma—Emerged as one of the first illnesses associated with AIDS patients. These tumors usually appear on the upper body, the soft palate and gum areas, and, as the disease advances, in the lymph nodes, stomach, intestines, and lungs.

Apoptosis—Cell death.

Classic Kaposi's sarcoma—Usually affects older men of Mediterranean or eastern European backgrounds, and produces tumors on the lower legs.

Cytokines—Regulatory proteins that are produced by the immune system.

Human herpesvirus 8—Also called Kaposi's sarcoma-associated herpesvirus (KSHV). Thought to be a viral cause for KS.

Iatrogenic Kaposi's sarcoma—Develops in transplant patients who take immunosuppressive drugs to prevent rejection of their organ transplant.

MCH-1—Major histocompatibility complex proteins that protect cells from invasion.

premature death, despite treatment. In AIDS-related KS, milder cases can frequently be controlled; the prognosis for more advanced and rapidly progressing cases is less certain and dependent on the patient's overall medical condition. There are indications that KS can be stabilized or reversed in patients whose level of HIV in the blood is reduced to undetectable levels via combined antiretroviral therapy.

Prevention

Safer sex practices may help to prevent AIDS-related KS by decreasing the risk of transmission of HHV-8 through sexual means. However, the addition of avoidance of deep kissing to those precautions may be necessary. Intravenous drug users should still be urged not to share needles. Treatment with antiretrovirals may help to preserve the function of the immune system in HIV patients and delay the appearance and progression of KS lesions. In fact, since the introduction of HAART in those infected with HIV, KS has decreased substantially. However, it still remains the most common cancer among those infected with HIV.

Large clinical trials underway in 2003 were showing some promise for preventing infection with HHV-8 through **prophylaxis** (preventive medication) with antiherpes drugs.

Resources

PERIODICALS

- “Alitretinoin Gel Effective for Treating Lesions.” *AIDS Weekly* February 12, 2001.
- Cannon, Michael J., A. Scott Laney, and Philip E. Pellett. “Human Herpesvirus 8: Current Issues.” *Clinical Infectious Diseases* July 1, 2003: 82–86.
- Dezube, Bruce J. “AIDS-Related Kaposi Sarcoma.” *Archives of Dermatology* 136, no. 2 (December 2000): 1554.
- Henderson, Charles W. “Kissing May Spread Cause of Kaposi’s Sarcoma.” *Cancer Weekly* November 21, 2000: NA.
- Mann, Arnold. “Kaposi’s Lesions Benefit from Long-Term Alitretinoin Gel Use.” *Family Practice News* 30, no. 19 (October 1, 2000): 12.
- “Protease Inhibitors Used for Treating HIV also Block Kaposi Sarcoma.” *Cancer Weekly* 30, no. 19 (November 4, 2003): 125.
- Rohrmus, Bettina, Eva M. Thoma-Greber, Hohannes R. Bogner, and Martin Rocken. “Outlook in Oral and Cutaneous Kaposi’s Sarcoma.” *The Lancet* 356, no. 9248 (December 23, 2000): 2160.

Janie F. Franz
Teresa G. Odle

Kawasaki syndrome

Definition

Kawasaki syndrome is a potentially fatal inflammatory disease that affects several organ systems in the body, including the heart, circulatory system, mucous membranes, skin, and immune system. It occurs primarily in infants and children but has also been identified in adults as old as 34 years. Its cause is unknown.

Description

Kawasaki syndrome, also called mucocutaneous lymph node syndrome (MLNS), is an inflammatory disorder with potentially fatal complications affecting the heart and its larger arteries. Nearly twice as many males are affected as females. Although persons of Asian descent are affected more frequently than either

black or white individuals, there does not appear to be a distinctive geographic pattern of occurrence. Eighty percent of cases involve children under the age of four. Although the disease usually appears in individuals, it sometimes affects several members of the same family and occasionally occurs in small epidemics.

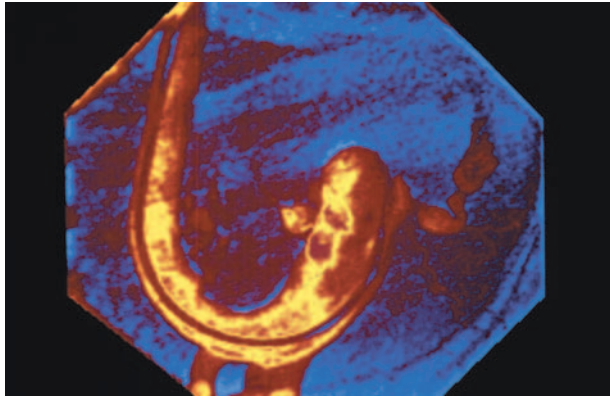
Causes and symptoms

The specific cause of Kawasaki syndrome is unknown, although the disease resembles infectious illnesses in many ways. It has been suggested that Kawasaki syndrome represents an allergic reaction or other unusual response to certain types of infections. Some researchers think that the syndrome may be caused by the interaction of an immune cell, called the T cell, with certain poisons (toxins) secreted by bacteria.

Kawasaki syndrome has an abrupt onset, with **fever** as high as 104°F (40°C) and a rash that spreads over the patient’s chest and genital area. The fever is followed by a characteristic peeling of the skin beginning at the fingertips and toenails. In addition to the body rash, the patient’s lips become very red, with the tongue developing a “strawberry” appearance. The palms, soles, and mucous membranes that line the eyelids and cover the exposed portion of the eyeball (conjunctivae) become purplish-red and swollen. The lymph nodes in the patient’s neck may also become swollen. These symptoms may last from two weeks to three months, with relapses in some patients.

In addition to the major symptoms, about 30% of patients develop joint pains or arthritis, usually in the large joints of the body. Others develop **pneumonia**, **diarrhea**, dry or cracked lips, **jaundice**, or an inflammation of the membranes covering the brain and spinal cord (**meningitis**). A few patients develop symptoms of inflammation in the liver (hepatitis), gallbladder, lungs, or tonsils.

About 20% of patients with Kawasaki syndrome develop complications of the cardiovascular system. These complications include inflammation of the heart tissue (**myocarditis**), disturbances in heartbeat rhythm (**arrhythmias**), and areas of blood vessel dilation (aneurysms) in the coronary arteries. Other patients may develop inflammation of an artery (arteritis) in their arms or legs. Complications of the heart or arteries begin to develop around the tenth day after the illness begins, when the fever and rash begin to subside. A few patients may develop **gangrene**, or the death of soft tissue, in their hands and feet. The specific causes of these complications are not yet known.



An angiogram showing abnormal coronary arteries in a child suffering from Kawasaki's disease. The coronary arteries are abnormal and weakened in that they bulge into balloon shapes, or aneurysms, along their lengths. This illness afflicts children between the ages of 1–2 years. (Photograph by Mehau Kulyk, Photo Researchers, Inc. Reproduced by permission.)

Diagnosis

Because Kawasaki syndrome is primarily a disease of infants and young children, the disease is most likely to be diagnosed by a pediatrician. The physician will first consider the possible involvement of other diseases that cause fever and skin **rashes**, including **scarlet fever**, **measles**, **Rocky Mountain spotted fever**, **toxoplasmosis** (a disease carried by cats), juvenile **rheumatoid arthritis**, and a blistering and inflammation of the skin caused by reactions to certain medications (Stevens-Johnson syndrome).

Once other diseases have been ruled out, the patient's symptoms will be compared with a set of diagnostic criteria. The patient must have a fever lasting five days or longer that does not respond to **antibiotics**, together with four of the following five symptoms:

- Inflammation of the conjunctivae of both eyes with no discharge
- At least one of the following changes in the mucous membranes of the mouth and throat: "strawberry" tongue; cracked lips; or swollen throat tissues
- At least one of the following changes in the hands or feet: swelling caused by excess fluid in the tissues; peeling of the skin; or abnormal redness of the skin
- A skin eruption or rash associated with fever (exanthem) on the patient's trunk
- Swelling of the lymph nodes in the neck to a size greater than 1.5 cm.

Since the cause of Kawasaki syndrome is unknown, there are no laboratory tests that can confirm the diagnosis. The following test results, however, are associated with the disease:

- Blood tests show a high white blood cell count, high **platelet count**, a high level of protein in the blood serum, and mild anemia
- Chest x ray may show enlargement of the heart (cardiomegaly)
- Urine may show the presence of pus or an abnormally high level of protein
- An electrocardiogram may show changes in the heartbeat rhythm

In addition to these tests, it is important to take a series of echocardiograms during the course of the illness because 20% of Kawasaki patients will develop coronary aneurysms or arteritis that will not appear during the first examination.

Treatment

Kawasaki syndrome is usually treated with a combination of **aspirin**, to control the patient's fever and skin inflammation, and high doses of intravenous immune globulin to reduce the possibility of coronary artery complications. Some patients with heart complications may be treated with drugs that reduce blood clotting or may receive corrective surgery.

Follow-up care includes two to three months of monitoring with chest x rays, **electrocardiography**, and **echocardiography**. Treatment with aspirin is often continued for several months.

Prognosis

Most patients with Kawasaki syndrome will recover completely, but about 1–2% will die as a result of **blood clots** forming in the coronary arteries or as a result of a **heart attack**. Deaths are sudden and unpredictable. Almost 95% of fatalities occur within six months of infection, but some have been reported as long as 10 years afterward. Long-term follow-up of patients with aneurysms indicates that about half show some healing of the aneurysm. The remaining half has a high risk of heart complications in later life.

Resources

BOOKS

Shandera, Wayne X., and Maria E. Carlini. "Infectious Diseases: Viral & Rickettsial." In *Current Medical*

KEY TERMS

Aneurysm—Dilation of an artery caused by thinning and weakening of the vessel wall.

Arrhythmia—Abnormal heart rhythm.

Arteritis—Inflammation of an artery.

Cardiomegaly—An enlarged heart.

Conjunctivae—The mucous membranes that cover the exposed area of the eyeball and line the inner surface of the eyelids.

Exanthem—A skin eruption associated with a disease, usually one accompanied by fever as in Kawasaki syndrome.

Gangrene—The death of soft tissue in a part of the body, usually caused by obstructed circulation.

Hepatitis—Inflammation of the liver.

Meningitis—Inflammation of the membranes, called the meninges, covering the brain and spinal cord.

Mucocutaneous lymph node syndrome (MLNS)—Mucocutaneous lymph node syndrome, another name for Kawasaki syndrome. The name comes from the key symptoms of the disease, which involve the mucous membranes of the mouth and throat, the skin, and the lymph nodes.

Myocarditis—Inflammation of the heart muscle.

Stevens-Johnson syndrome—A severe inflammatory skin eruption that occurs as a result of an allergic reaction or respiratory infection.

T cell—A type of white blood cell that develops in the thymus gland and helps to regulate the immune system's response to infections or malignancy.

Diagnosis and Treatment, 1998, edited by Stephen McPhee, et al., 37th ed. Stamford: Appleton & Lange, 1997.

Rebecca J. Frey, PhD

Keloids

Definition

Keloids are overgrowths of fibrous tissue or **scars** that can occur after an injury to the skin. These heavy

scars are also called cheloid or hypertrophic scars. In individuals prone to keloids, even minor traumas to the skin, such as ear **piercing**, can cause keloids. The word “keloid” itself comes from the Greek word for a crab’s claw; it was first used by a French physician to describe the way that keloids grow sideways into normal skin.

Description

Keloids can occur anywhere on the body, but they are most common on the earlobes, upper back, shoulders, and chest. The pattern of distribution of keloids differs according to race, with facial keloids more common in Caucasians and relatively uncommon in Asians. African Americans are more likely to develop keloids on the legs or feet than either Asians or Caucasians. In general, keloids consist of hard, raised scars that may be slightly pink or whitish. These may itch and be painful, and some keloids can grow to be quite large.

Causes and symptoms

Although the cause of keloids is unknown, it is thought that they are due to the body’s failure to turn off the healing process needed to repair skin. When this occurs, extra collagen forms at the site of the scar, and keeps forming because it is not shut off. This results in keloid formation.

Keloids occur most frequently in individuals of African-American descent and in those with darker skin. They are more common in Polynesians and Chinese than in people from India or Malaysia. Caucasians are the least frequently affected by keloids. Other risk factors include a family history of keloids, surgery, **acne**, **burns**, ear piercing, vaccinations, or even insect bites. Spontaneous keloids have been reported occasionally in siblings. In addition, women and young people under the age of 30 are more prone to develop them. Keloids are infrequent among the elderly.

Although the association of keloids with darker skin pigmentation suggests a genetic linkage of some sort, no specific genes have been identified in connection with keloids as of the early 2000s.

Initially, keloids will begin as a small lump where the skin has been injured. This lump grows and can eventually become very large and cosmetically unacceptable.

Diagnosis

A dermatologist can usually make the diagnosis of a keloid based on looking at the scar. In some cases,

however, a biopsy may be necessary to rule out other types of **skin lesions**, such as tumors.

Treatment

The treatment of choice for keloids is usually an injection of corticosteroid drugs such as cortisone directly into the lesion. These injections cause the keloid to become atrophic, or thinner, and are repeated every three to four weeks until the keloid has been resolved to the individual's satisfaction. Other therapies include laser treatment or **radiation therapy**, and topical treatments are undergoing study.

Surgery is often used in combination with corticosteroid injections. The injections are given for several weeks, and then the keloid is surgically removed. The injections are then continued for several weeks. Surgical removal of the keloid may also be used in conjunction with radiation therapy, which delivers small amounts of radiation to the affected area.

Another surgical option is cryosurgery, in which liquid nitrogen is used to freeze the tissues in the keloid. The treatment may need to be repeated to remove as much of the keloid as possible; however, cryosurgery prevents keloids from recurring in about 70% of patients.

Newer approaches include silastic gel sheeting, which makes use of pressure to flatten the keloid. The gel is applied and kept securely in place with tape, cloth, or an Ace bandage. The dressing is to be changed every seven to 10 days for as long as 12 months.

Finally, researchers are now studying a type of tape that has been soaked with steroids, which are released slowly into the keloid, causing it to thin over time.

Newer treatments include injections of interferon directly into the keloids, and local application of 5% imiquimod cream, which induces the skin where it is applied to produce interferon. The imiquimod cream is reported to significantly lower the risk of keloid recurrence.

Prognosis

Although keloids are unsightly, they are not life-threatening. Keloids do not have a tendency to develop into malignancies, but they can become cosmetically unacceptable. Keloids can gradually lessen after treatment, but many recur. And just as they can occur spontaneously, they can also resolve spontaneously.

KEY TERMS

Atrophy—A wasting away of, becoming thinner, less strong.

Corticosteroids—Any of several steroid medications used to suppress inflammation, allergic, or immune responses of the body.

Cryosurgery—The use of extreme cold to kill or remove tissue.

Prevention

Preventive measures include avoiding any trauma to the skin, and compression pressure dressing for high-risk patients who have suffered burns to their skin. Patients with a tendency to form keloids should avoid any sort of elective surgery. Individuals who are prone to develop keloids or who have a history of keloids should immediately care for any cuts or abrasions they may sustain.

To lower the risk of keloids, surgeons are advised to close incisions with as little tension on the sutures as possible, and to use buried sutures whenever possible.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD., editors. "Benign Tumors." Section 10, Chapter 125 In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

PERIODICALS

- Berman, Brian, MD, PhD, and Sonia Kapoor, MBBS. "Keloid and Hypertrophic Scar." *eMedicine* November 30, 2001. <<http://www.emedicine.com/derm/topic205.htm>>.
- Dinh, Q., M. Veness, and S. Richards. "Role of Adjuvant Radiotherapy in Recurrent Earlobe Keloids." *Australasian Journal of Dermatology* 45 (August 2004): 162–166.
- Food and Drug Administration (FDA). "General and Plastic Surgery Devices; Classification of Silicone Sheeting. Final Rule." *Federal Register* 69 (August 9, 2004): 48146–48148.
- Mandal, A., D. Imran, and G. S. Rao. "Spontaneous Keloids in Siblings." *Irish Medical Journal* 97 (September 2004): 250–251.

ORGANIZATIONS

American Academy of Dermatology. 930 N. Meacham Road, PO Box 4014, Schaumburg, IL 60168-4014. (847) 330-0230. Fax: (847) 330-0050. <<http://www.aad.org>>.

United States Food and Drug Administration (FDA). 5600 Fishers Lane, Rockville, MD 20857-0001. (888) INFO-FDA. <<http://www.fda.gov>>.

OTHER

<<http://www.skincare.com>>.

“Keloids.” Black Women’s Health. <<http://www.black-womenshealth.com/Keloids.htm>>.

Liz Meszaros
Rebecca J. Frey, PhD

Keratitis

Definition

Keratitis is an inflammation of the cornea, the transparent membrane that covers the colored part of the eye (iris) and pupil of the eye.

Description

There are many types and causes of keratitis. Keratitis occurs in both children and adults. Organisms cannot generally invade an intact, healthy cornea. However, certain conditions can allow an infection to occur. For example, a scratch can leave the cornea open to infection. A very dry eye can also decrease the cornea’s protective mechanisms.

Risk factors that increase the likelihood of developing this condition include:

- poor contact lens care; overuse of contact lenses
- illnesses or other factors that reduce the body’s ability to overcome infection
- cold sores, **genital herpes**, and other viral infections
- crowded, dirty living conditions; poor hygiene
- poor **nutrition** (especially a deficiency of Vitamin A, which is essential for normal vision)

Some common types of keratitis are listed below, however there are many other forms

Herpes simplex keratitis

A major cause of adult eye disease, herpes simplex keratitis may lead to:

- chronic inflammation of the cornea
- development of tiny blood vessels in the eye
- scarring

- loss of vision
- glaucoma

This infection generally begins with inflammation of the membrane lining the eyelid (conjunctiva) and the portion of the eyeball that comes into contact with it. It usually occurs in one eye. Subsequent infections are characterized by a pattern of lesions that resemble the veins of a leaf. These infections are called dendritic keratitis and aid in the diagnosis.

Recurrences may be brought on by **stress**, **fatigue**, or ultraviolet light (UV) exposure (e.g., skiing or boating increase the exposure of the eye to sunlight; the sunlight reflects off of the surfaces). Repeated episodes of dendritic keratitis can cause sores, permanent scarring, and **numbness** of the cornea.

Recurrent dendritic keratitis is often followed by disciform keratitis. This condition is characterized by clouding and deep, disc-shaped swelling of the cornea and by inflammation of the iris.

It is very important not to use topical **corticosteroids** with herpes simplex keratitis as it can make it much worse, possibly leading to blindness.

Bacterial keratitis

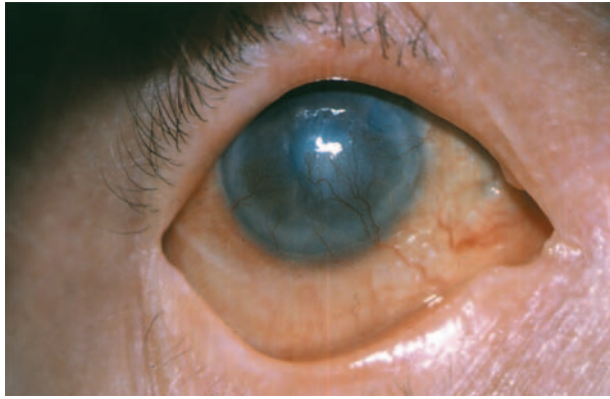
People who have bacterial keratitis wake up with their eyelids stuck together. There can be **pain**, sensitivity to light, redness, tearing, and a decrease in vision. This condition, which is usually aggressive, can be caused by wearing soft contact lenses overnight. One study found that overnight wear can increase risk by 10-15 times more than if wearing daily wear contact lenses. Improper lens care is also a factor. Contaminated makeup can also contain bacteria.

Bacterial keratitis makes the cornea cloudy. It may also cause abscesses to develop in the stroma, which is located beneath the outer layer of the cornea.

Fungal keratitis

Usually a consequence of injuring the cornea in a farm-like setting or in a place where plant material is present, fungal keratitis often develops slowly. This condition:

- usually affects people with weakened immune systems
- often results in infection within the eyeball
- may cause stromal abscesses



Close-up of a damaged cornea due to complications following cataract surgery. (Custom Medical Stock Photo. Reproduced by permission.)

Peripheral ulcerative keratitis

Peripheral ulcerative keratitis is also called marginal keratolysis or peripheral rheumatoid ulceration. This condition is often associated with active or chronic:

- rheumatoid arthritis
- relapsing polychondritis (connective-tissue inflammation)
- Wegener's granulomatosis, a rare condition characterized by **kidney disease** and development of nodules in the respiratory tract

Superficial punctate keratitis

Often associated with the type of viruses that cause upper respiratory infection (adenoviruses), superficial punctate keratitis is characterized by destruction of pinpoint areas in the outer layer of the cornea (epithelium). One or both eyes may be affected.

Acanthamoeba keratitis

This pus-producing condition is very painful. It is a common source of infection in people who wear soft or rigid contact lenses. It can be found in tap water, soil, and swimming pools.

Photokeratitis

Photokeratitis or snowblindness is caused by excess exposure to UV light. This can occur with sunlight, tanning lamps, or a welding arc. It is called snowblindness because the sunlight is reflected off of the snow. It therefore can occur in water sports as well, because of the reflection of light off of the water. It is

very painful and may occur several hours after exposure. It may last one to two days.

Interstitial keratitis

Also called parenchymatous keratitis, interstitial keratitis is a chronic inflammation of tissue deep within the cornea. Interstitial keratitis is rare in the United States. Interstitial keratitis affects both eyes and usually occurs as a complication of congenital or acquired **syphilis**. In congenital syphilis it can occur between age two and **puberty**. It may also occur in people with **tuberculosis**, **leprosy**, or other diseases.

Causes and symptoms

In summary, keratitis can be caused by:

- bacterial, viral, or fungal infections
- dry eyes resulting from disorders of the eyelid or diminished ability to form tears
- exposure to very bright light
- **foreign objects** that injure or become lodged in the eye
- sensitivity or allergic reactions to eye makeup, dust, pollen, pollution, or other irritants
- vitamin A deficiency, which people with normal **diets** rarely develop

Symptoms of keratitis include, but are not limited to:

- tearing
- pain
- sensitivity to light
- inflammation of the eyelid
- decrease in vision
- redness

Diagnosis

A case history will be taken and the vision will be tested. Examination with a slit lamp, an instrument that's a microscope and focuses a beam of light on the eye is important for diagnosis. The cornea can be examined with fluorescein, a yellow dye which will highlight defects in the cornea. Deeper layers of the cornea can also be examined with the slit lamp. Infiltrates, hazy looking areas in the cornea, can be seen by the doctor and will aid in the diagnosis. Samples of infectious matter removed from the eye will be sent for laboratory analysis.

KEY TERMS

Abscess—A collection of pus.

Glaucoma—An eye disease characterized by an increase of pressure in the eye. Left untreated, blindness may result.

Infiltrate—A collection of cells not usually present in that area. In the cornea, infiltrates may be a collection of white blood cells.

Inflammation—A localized response to an injury. May include swelling, redness, and pain.

Treatment

Antibiotics, antifungals, and antiviral medication will be used to treat the appropriate organism. Broad spectrum antibiotics will be used immediately, but once the lab analysis determines the offending organism, the medication may be changed. Sometimes more than one medication is necessary. It depends upon the infection, but the patient should be clear on how often and how to use the medications.

A sterile, cotton-tipped applicator may be used to gently remove infected tissue and allow the eye to heal more rapidly. **Laser surgery** is sometimes performed to destroy unhealthy cells, and some severe infections require corneal transplants.

Antifungal, antibiotic, or antiviral eyedrops or ointments are usually prescribed to cure keratitis, but they should be used only by patients under a doctor's care. Inappropriate prescriptions or over-the-counter preparations can make symptoms more severe and cause tissue deterioration. Topical corticosteroids can cause great harm to the cornea in patient's with herpes simplex keratitis.

A patient with keratitis may wear a patch to protect the healing eye from bright light, foreign objects, the lid rubbing against the cornea, and other irritants. Sometimes a patch can make it worse, so again, the patient must discuss with the doctor whether or not a patch is necessary. The patient will probably return every day to the eye doctor to check on the progress.

Although early detection and treatment can cure most forms of keratitis, the infection can cause:

- glaucoma
- permanent scarring
- ulceration of the cornea
- blindness

Prevention

Children and adults who wear contact lenses should always use sterile lens-cleaning and disinfecting solutions. Tap water is not sterile and should not be used to clean contact lenses. It is important to go for follow-up checkups because small defects in the cornea can occur without the patient being aware of it. Do not overwear contact lenses. Remove them if the eyes become red or irritated. Replace contact lenses when scheduled to do so. Proteins and other things can deposit on the contacts, leading to an increased risk of infection. Rinse contact lens cases in hot water every night, if possible, and let them air dry. Replace contact lens cases every three months. Organisms have been cultured from contact lens cases.

Eating a well-balanced diet and wearing protective glasses when working or playing in potentially dangerous situations can reduce anyone's risk of developing keratitis. Protective goggles can even be worn mowing the lawn so that if twigs are tossed up they can't hurt the eye. Goggles or sunglasses with UV coatings can help protect against damage from UV light.

Resources

ORGANIZATIONS

American Academy of Ophthalmology. 655 Beach Street, P.O. Box 7424, San Francisco, CA 94120-7424. <<http://www.eyenet.org>>.

American Optometric Association. 243 North Lindbergh Blvd., St. Louis, MO 63141. (314) 991-4100. <<http://www.aoanet.org>>.

National Eye Institute. 2020 Vision Place, Bethesda, MD 20892-3655. (301) 496-5248. <<http://www.nei.nih.gov>>.

Prevent Blindness America. 500 East Remington Road, Schaumburg, IL 60173. (800) 331-2020. <<http://www.preventblindness.org>>.

Maureen Haggerty

Keratitis pilaris

Definition

Keratitis pilaris is a common skin condition that looks like small goose bumps, which are actually dead skin cells that build up around the hair follicle.

Description

Keratosis pilaris is a disorder that occurs around the hair follicles of the upper arms, thighs, and sometimes the buttocks. It presents as small, benign bumps or papules that are actually waxy build-ups of keratin. Normally skin sloughs off. However, around the hair follicle where the papules form, the keratinized skin cells slough off at a slower rate, clogging the follicles.

This is generally thought to be genetic disorder, although the symptoms of keratosis pilaris are often seen with **ichthyosis** and allergic **dermatitis**. It can also be observed in people of all ages who have either inherited it or have a **vitamin A deficiency** or have dry skin. Keratosis pilaris is a self-limiting disorder that disappears as the person ages. It can become more severe when conditions are dry such as during the winter months or in dry climates.

Causes and symptoms

The specific causes of this disorder are unknown. Since this disorder runs in families, it is thought to be hereditary. Keratosis pilaris is not a serious disorder and is not contagious.

The symptoms of keratosis pilaris are based on the development of small white papules the size of a grain of sand on the upper arms, thighs, and occasionally the buttocks and face. The papules occur around a hair follicle and are firm and white. They feel a little like coarse sandpaper, but they are not painful and there usually is no **itching** associated with them. They are easily removed and the material inside the papule usually contains a small, coiled hair.

Diagnosis

A dermatologist or a general practitioner can easily diagnose this disorder. A **physical examination** is all that is necessary to diagnose keratosis pilaris. Special tests are not needed.

Treatment

To treat keratosis pilaris patients can try several strategies to lessen the bumps. First, the patient can supplement the natural removal of dry skin and papules by using a loofah or another type of scrub showering or bathing. A variety of different over-the-counter (OTC) lotions, ointments, and creams can also be applied after showering while the skin is still moist and then several times a day to keep the area moist. Medicated lotions with urea, 15% alpha-hydroxy acids, or Retin A can also be prescribed by

KEY TERMS

Benign—Not cancerous.

Dermatologist—A physician that specializes in diseases and disorders of the skin.

Ichthyosis—A group of congenital disorders of keratinization characterized by dryness and scaling of the skin.

Keratin—The hard, waxy material that is made by the outer layer of skin cells.

the dermatologist and applied one to two times daily. Systemic (oral) medications are not prescribed for keratosis pilaris. However if papules are opened and become infected, **antibiotics** may be necessary to treat the infection.

Prognosis

Unfortunately, the treatment for keratosis pilaris is often disappointing. Although extreme cases of keratosis pilaris can occasionally be unsightly, the disorder is not life threatening and usually begins to disappear as the patient ages.

Prevention

Since keratosis pilaris is thought to be a genetic disorder and is observed in several members of the same family, there is nothing that can be done to prevent this disorder. Following the treatment advice above can alleviate the outward characteristics of keratosis pilaris.

Sally C. McFarlane-Parrott

Kidney biopsy

Definition

Kidney biopsy is a medical procedure in which a small piece of tissue is removed from the kidney for microscopic examination.

Purpose

The test is usually done to diagnose **kidney disease** and to evaluate the extent of damage to the kidney.

A biopsy is also frequently ordered to detect the reason for acute renal failure when normal office procedures and tests fail to establish the cause. In addition, information regarding the progression of the disease and how it is responding to medical treatment can be obtained from a biopsy. Occasionally a biopsy may be done to confirm a diagnosis of **kidney cancer**, to determine its aggressiveness, and decide on the mode of treatment.

Precautions

The biopsy is not recommended for patients who have any uncontrollable bleeding disorders. Platelets are blood cells that play an important role in the blood clotting process. If the bleeding disorder is caused by a low **platelet count** (less than 50,000 per cubic millimeter of blood), then a platelet **transfusion** can be done just before performing the biopsy.

Description

The kidneys, a pair of organs that are shaped like beans, lie on either side of the backbone, just above the waist. The periphery (parenchyma) of the kidney is made up of tiny tubes. These tubes filter and clean the blood by taking out the waste products and making urine. The urine is collected in the central portion of the kidney. Tubes called ureters drain the urine from the kidney into the bladder, where it is held until it is voided from the body.

A kidney specialist (nephrologist) performs the biopsy. It can be done either in the doctor's office or in a local hospital. The patient may be given a calming drug before the procedure to help him relax. The skin and muscles on the back overlying the site that is to be biopsied may be numbed with **local anesthesia**.

The patient will be asked to lie face down and a pad or a rolled towel may be placed under the stomach. Either the left or the right kidney may be biopsied depending on the results of the imaging tests: x rays, **computed tomography scans** (CT scans), **magnetic resonance imaging** (MRI), and ultrasound. The area that will be biopsied is cleaned with an antiseptic solution and sterile drapes are placed on it. The skin is numbed with local anesthesia. A small incision is made on the skin with a scalpel blade. Using a long needle, the physician injects local anesthesia into the incision so that it infiltrates down to the kidney. The biopsy needle is then advanced slowly through the incision. The patient is asked to hold his or her breath each time the needle is pushed forward. Once the wall (capsule) of the kidney has been penetrated, the patient can

breathe normally. The tissue is collected for examination and the needle is withdrawn. The needle may be re-inserted into another part of the kidney so that tissue is collected from at least three different areas. The tissue samples are sent to the laboratory for examination. The entire procedure may last about an hour.

Preparation

Before performing the biopsy, the doctor should be made aware of all the medications that the patient is taking. The doctor should also be told whether the patient is allergic to any medications. The procedure and the risks of the procedure are explained to the patient and the necessary consent forms are obtained. The patient should be told that a kidney biopsy requires a 24-hour stay in the hospital after the biopsy.

Some doctors order blood tests to check for clotting problems before performing the biopsy. The patient's blood type may also be determined in case a transfusion becomes necessary.

Aftercare

Immediately after the biopsy, pulse, respiration, and temperature (vital signs) are measured. If they are stable, the patient is instructed to lie flat in bed for at least 12 hours. The pulse and blood pressure are checked at regular intervals by the nursing staff. All urine voided by the patient in the first 12–24 hours is examined in the laboratory for blood cells.

If bleeding is severe, iron levels in the blood drop significantly, or the patient complains of severe **pain** at the biopsy site, the physician should be contacted immediately. After the patient goes home, he should avoid heavy lifting, vigorous **exercise**, and contact sports for at least one or two weeks.

Risks

The risks of a kidney biopsy are very small. Severe bleeding may occur after the procedure. There is also a slight chance that an infection or a lump of blood under the skin that looks black and blue (hematoma) may develop. In most cases, the hematoma disappears by itself and does not cause any pain. However, severe pain or a drop in blood pressure and iron levels in the blood indicates that the hematoma is expanding. This condition could lead to complications and should be reported immediately to the doctor.

Very rarely, the patient may develop high blood pressure (**hypertension**), and the bleeding may be severe enough to require a transfusion. In extremely

KEY TERMS

Biopsy—The surgical removal and microscopic examination of living tissue for diagnostic purposes.

Computed tomography (CT) scan—A medical procedure in which a series of x rays are taken and put together by a computer in order to form detailed pictures of areas inside the body.

Magnetic resonance imaging (MRI)—A medical procedure used for diagnostic purposes in which pictures of areas inside the body can be created using a magnet linked to a computer.

Nephrologist—A doctor who specializes in the diseases and disorders of the kidneys.

Renal ultrasound—A painless and non-invasive procedure in which sound waves are bounced off the kidneys. These sound waves produce a pattern of echoes that are then used by the computer to create pictures of areas inside the kidney (sonograms).

rare circumstances, the kidney may rupture, or the surrounding organs (pancreas, bowel, spleen, and liver) may be punctured. **Death** occurs in about one in 3000 cases.

Normal results

The results are normal if no abnormalities can be seen in the tissue samples with the naked eye, with an electron microscope or through staining with a fluorescent dye (immunofluorescence).

Abnormal results

Any abnormalities in the size, color, and consistency of the sample will be reported as an abnormal result. In addition, any change in the structure of the renal tubules, the presence of red blood cells, or abnormalities in the cells are considered an abnormal result. If cancerous changes are detected in the kidney cells, they are further characterized in order to determine the stage of the tumor and decide on the appropriate mode of treatment.

Resources

ORGANIZATIONS

National Kidney Cancer Association. 1234 Sherman Ave., Suite 203, Evanston, IL 60202-1375. (800) 850-9132.

National Kidney Foundation. 30 East 33rd St., New York, NY 10016. (800) 622-9010. <<http://www.kidney.org>>.

Lata Cherath, PhD

Kidney cancer

Definition

Kidney **cancer** is a disease in which the cells in certain tissues of the kidney start to grow uncontrollably and form tumors. Renal cell carcinoma, which occurs in the cells lining the kidneys (epithelial cells), is the most common type of kidney cancer. Eighty-five percent of all kidney tumors are renal cell carcinomas. **Wilms' tumor** is a rapidly developing cancer of the kidney most often found in children under four years of age.

Description

The kidneys are a pair of organs shaped like kidney beans that lie on either side of the spine just above the waist. Inside each kidney are tiny tubes (tubules) that filter and clean the blood, taking out the waste products and making urine. The urine that is made by the kidney passes through a tube called the ureter into the bladder. Urine is held in the bladder until it is discharged from the body. Renal cell carcinoma generally develops in the lining of the tubules that filter and clean the blood. Cancer that develops in the central portion of the kidney (where the urine is collected and drained into the ureters) is known as transitional cell cancer of the renal pelvis. Transitional cell cancer is similar to **bladder cancer**.

Kidney cancer accounts for approximately 2–3% of all cancers. In the United States, kidney cancer is the tenth most common cancer and the incidence has increased by 43% since 1973; the **death** rate has increased by 16%. According to the American Cancer Society, 35,710 Americans were diagnosed with kidney cancer in 2004, and 12,480 died from the disease. RCC accounts for 90–95% of malignant neoplasms that originate from the kidney.

Causes and symptoms

The causes of kidney cancer are unknown, but men seem to have a greater risk than women of contracting the disease; the male:female ratio in the United States and Canada is 3:2 as of the early

2000s. There is a strong association between cigarette **smoking** and kidney cancer. Cigarette smokers are twice as likely as non-smokers are to develop kidney cancer. Working around coke ovens has been shown to increase people's risk of developing this cancer. Certain types of painkillers that contain the chemical phenacetin are associated with kidney cancer. The United States government discontinued use of **analgesics** containing phenacetin about 20 years ago. **Obesity** may be yet another risk factor for kidney cancer. Some studies show a loose association between kidney cancer and occupational exposure to cadmium, petroleum products, lead, dry-cleaning solvents, trichloroethylene (TCE), and asbestos. Other risk factors for the development of kidney cancer include Hispanic heritage and preexisting von Hippel-Lindau disease.

The most common symptom of kidney cancer is blood in the urine (hematuria). Other symptoms include painful urination, **pain** in the lower back or on the sides, abdominal pain, a lump or hard mass that can be felt in the kidney area, unexplained weight loss, **fever**, weakness, **fatigue**, and high blood pressure.

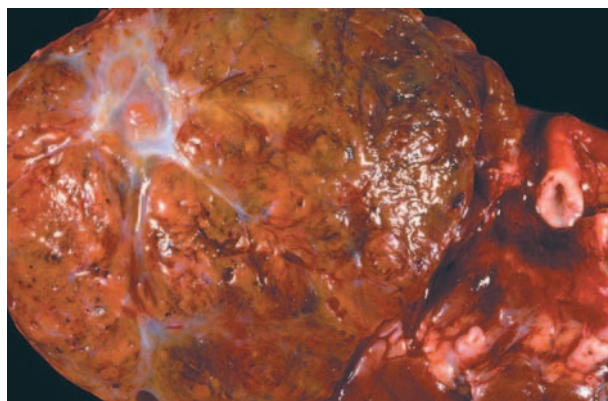
Other symptoms may occur if the cancer has spread beyond its original location. Spread of kidney cancer most commonly occurs to the lung (55%), liver (33%), bone (33%), adrenal (20%), and opposite kidney (10%). Lymph node spread is also common, occurring in about 25% of patients).

Diagnosis

A diagnostic examination for kidney cancer includes taking a thorough medical history and making a complete **physical examination** in which the doctor will probe (palpate) the abdomen for lumps. Blood tests will be ordered to check for changes in blood chemistry caused by substances released by the tumor. Laboratory tests may show abnormal levels of iron in the blood. Either a low red blood cell count (anemia) or a high red blood cell count (erythrocytosis) may accompany kidney cancer. Occasionally, patients will have high calcium levels.

If the doctor suspects kidney cancer, an intravenous pyelogram (IVP) may be ordered. An IVP is an x-ray test in which a dye is injected into a vein in the arm. The dye travels through the body, and when it is concentrated in the urine to be discharged, it outlines the kidneys, ureters, and the urinary bladder. On an x-ray image, the dye will reveal any abnormalities of the urinary tract. The IVP may miss small kidney cancers.

Renal ultrasound is a diagnostic test in which sound waves are used to form an image of the kidneys.



An extracted cancerous kidney. (Custom Medical Stock Photo. Reproduced by permission.)

Ultrasound is a painless and non-invasive procedure that can be used to detect even very small kidney tumors. Imaging tests such as **computed tomography scans** (CT scans) and **magnetic resonance imaging** (MRI) can be used to evaluate the kidneys and the surrounding organs. These tests are used to check whether the tumor has spread outside the kidney to other organs in the abdomen. If the patient complains of bone pain, a special x ray called a bone scan may be ordered to rule out spread to the bones. A **chest x ray** may be taken to rule out spread to the lungs.

A **kidney biopsy** is used to positively confirm the diagnosis of kidney cancer. During this procedure, a small piece of tissue is removed from the tumor and examined under a microscope. The biopsy will give information about the type of tumor, the cells that are involved, and the aggressiveness of the tumor (tumor stage).

Treatment

Each person's treatment is different and depends on several factors. The location, size, and extent of the tumor have to be considered in addition to the patient's age, general health, and medical history. In addition, much has changed in the treatment and management of kidney cancer since the 1980s, including new surgical techniques, new **anticancer drugs**, and the development of effective treatments for advanced disease.

The primary treatment for kidney cancer that has not spread to other parts of the body, which is a Stage I, II, or III tumor, is surgical removal of the diseased kidney (**nephrectomy**). Because most cancers affect only one kidney, the patient can function well with the remaining one. Two types of surgical procedure

are used. Radical nephrectomy removes the entire kidney and the surrounding tissue. Sometimes, the lymph nodes surrounding the kidney are also removed. Partial nephrectomy removes only part of the kidney along with the tumor. This procedure is used either when the tumor is very small or when it is not practical to remove the entire kidney. It is not practical to remove a kidney when the patient has only one kidney or when both kidneys have tumors. There is a small (5%) chance of missing some of the cancer. Nephrectomy can also be useful for Stage IV cancers, but alternative surgical procedures such as transarterial angioinfarction may be used.

The rapid development and widespread use of laparoscopic techniques has made it possible for surgeons to remove small tumors while sparing the rest of the kidney. Most tumors removed by **laparoscopy** are 4 cm (1.6 in) in size or smaller. Laparoscopy also allows the surgeon to remove small tumors with cryoablation (destroying the tumor by freezing it) rather than cutting.

Radiation therapy, which consists of exposing the cancer cells to high-energy gamma rays from an external source, generally destroys cancer cells with minimal damage to the normal tissue. Side effects are **nausea**, fatigue, and stomach upsets. These symptoms disappear when the treatment is over. In kidney cancer, radiation therapy has been shown to alleviate pain and bleeding, especially when the cancer is inoperable. However, it has not proven to be of much use in destroying the kidney cancer cells. Therefore radiation therapy is not used very often as a treatment for cancer or as a routine adjuvant to nephrectomy. Radiotherapy, however, is used to manage metastatic kidney cancer.

Treatment of kidney cancer with anticancer drugs (**chemotherapy**) has not produced good results. However, new drugs and new combinations of drugs continue to be tested in clinical trials. One new drug, semaxanib (SU5416), is reported to have good results in treating patients with kidney cancer. As of 2004, however, semaxanib is still undergoing clinical trials in the United States.

Immunologic therapy (or immunotherapy), a form of treatment in which the body's immune system is harnessed to help fight the cancer, is a new mode of therapy that is being tested for kidney cancer. Clinical trials with substances produced by the immune cells (aldesleukin and interferon) have shown some promise in destroying kidney cancer cells. These substances have been approved for use but they can be very toxic and produce severe side effects. The benefits

KEY TERMS

Biopsy—The surgical removal and microscopic examination of living tissue for diagnostic purposes.

Bone scan—An x-ray study in which patients are given an intravenous injection of a small amount of a radioactive material that travels in the blood. When it reaches the bones, it can be detected by x ray to make a picture of their internal structure.

Chemotherapy—Treatment with anticancer drugs.

Computed tomography (CT) scan—A medical procedure in which a series of x-ray images are made and put together by a computer to form detailed pictures of areas inside the body.

Cryoablation—A technique for removing tissue by destroying it with extreme cold.

Hematuria—Blood in the urine.

Immunotherapy—Treatment of cancer by stimulating the body's immune defense system.

Intravenous pyelogram (IVP)—A procedure in which a dye is injected into a vein in the arm. The dye travels through the body and concentrates in the urine to be discharged. It outlines the kidneys, ureters, and the urinary bladder. An x-ray image is then made and any abnormalities of the urinary tract are revealed.

Magnetic resonance imaging (MRI)—A medical procedure used for diagnostic purposes in which pictures of areas inside the body can be created using a magnet linked to a computer.

Nephrectomy—A medical procedure in which the kidney is surgically removed.

Radiation therapy—Treatment with high-energy radiation from x-ray machines, cobalt, radium, or other sources.

Renal ultrasound—A painless and non-invasive procedure in which sound waves are bounced off the kidneys. These sound waves produce a pattern of echoes that are then used by the computer to create pictures of areas inside the kidney (sonograms).

derived from the treatment have to be weighed very carefully against the side effects in each case. Immunotherapy is the most promising systemic therapy for metastatic kidney cancer.

Prognosis

Because kidney cancer is often caught early and sometimes progresses slowly, the chances of a surgical cure are good. Length of survival depends on the size of the original tumor, the aggressiveness of the specific cells making up the tumor, and whether the cancer cells spread from the kidney to surrounding or distant tissues.

Kidney cancer is also one of the few cancers for which there are well-documented cases of spontaneous remission without therapy. Unfortunately, recurrences can occur even as long as ten years after the original diagnosis and treatment, and cancer can also crop up in the other, previously unaffected kidney.

Prevention

The exact cause of kidney cancer is not known, so it is not possible to prevent all cases. However, because a strong association between kidney cancer and tobacco has been shown, avoiding tobacco is the best way to lower one's risk of developing this cancer. Using care when working with cancer-causing agents such as asbestos and cadmium and eating a well-balanced diet may also help prevent kidney cancer.

Resources

BOOKS

- Beers, Mark H., MD, and Robert Berkow, MD., editors. "Renal Cell Carcinoma (Hypernephroma; Adenocarcinoma of the Kidney)." Section 17, Chapter 233 In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.
- Quek, Marcus L., and John P. Stein. "Malignant Tumors of the Urogenital Tract." In *Conn's Current Therapy 2001*. 53th ed. Philadelphia: W.B. Saunders Company, 2001.

PERIODICALS

- Brauch, H., G. Weirich, B. Klein, et al. "VHL Mutations in Renal Cell Cancer: Does Occupational Exposure to Trichloroethylene Make a Difference?" *Toxicology Letters* 151 (June 15, 2004): 301–310.
- Dutcher, J.P. "Immunotherapy: Are We Making a Difference?" *Current Opinion in Urology* September 2000: 435–9.
- Godley, P.A., and K.I. Ataga. "Renal Cell Carcinoma." *Current Opinion in Oncology* May 2000: 260–4.
- Griffiths, T. R., and J. K. Mellon. "Evolving Immunotherapeutic Strategies in Bladder and Renal Cancer." *Postgraduate Medical Journal* 80 (June 2004): 320–327.
- Jennens, R. R., M. A. Rosenthal, G. J. Lindeman, and M. Michael. "Complete Radiological and Metabolic Response of Metastatic Renal Cell Carcinoma to

SU5416 (Semaxanib) in a Patient with Probable von Hippel-Lindau Syndrome." *Urologic Oncology* 22 (May–June 2004): 193–196.

- Lam, J. S., O. Svarts, and A. J. Pantuck. "Changing Concepts in the Surgical Management of Renal Cell Carcinoma." *European Urology* 45 (June 2004): 692–705.
- Lotan, Y., D. A. Duchene, J. A. Cadeddu, et al. "Changing Management of Organ-Confined Renal Masses." *Journal of Endourology* 18 (April 2004): 263–268.
- Moon, T. D., F. T. Lee, Jr., S. P. Hedican, et al. "Laparoscopic Cryoablation under Sonographic Guidance for the Treatment of Small Renal Tumors." *Journal of Endourology* 18 (June 2004): 436–440.

ORGANIZATIONS

- American Cancer Society. 1599 Clifton Road, N.E., Atlanta, GA 30329. (800) 227-2345. <<http://www.cancer.org>>.
- Cancer Research Institute (National Headquarters). 681 Fifth Avenue, New York, NY 10022. (800) 992-2623. <<http://www.cancerresearch.org>>.
- National Cancer Institute (NCI). 9000 Rockville Pike, Building 31, Room 10A16, Bethesda, MD 20892. (800) 422-6237. <<http://www.nci.nih.gov>>.
- National Kidney Cancer Association. 1234 Sherman Avenue, Suite 203, Evanston, IL 60202-1375. (800) 850-9132.
- National Kidney Foundation. 30 East 33rd Street, New York, NY 10016. (800) 622-9010. <<http://www.kidney.org>>.

Rosalyn Carson-DeWitt, MD
Rebecca Frey, PhD

Kidney dialysis see **Dialysis, kidney**

Kidney disease

Definition

Kidney disease is a general term for any damage that reduces the functioning of the kidney. Kidney disease is also called renal disease.

Description

The kidneys are a pair bean-shaped, fist-sized organs that are located below the rib cage near the middle of the back. In adults they filter about 200 quarts (190 L) of blood every day to remove waste products that result from the normal activities of tissues in the body. These wastes circulate in the blood, and if not removed they would damage the body. The kidneys also play a crucial role in regulating the amount of water and chemicals (electrolytes) in

the body such as sodium, potassium and phosphorous.

Inside the kidneys are about one million tiny units called nephrons. Inside each nephron is a very thin blood vessel called a capillary that twists around a very thin tube called a tubule. This combination of capillary and tubule inside the nephron is called a glomerulus and it is here that the blood is filtered. Water, electrolytes, and waste products (but not red blood cells) can pass across the capillary wall and into the tubule. The kidney then regulates how much water and which other substances can pass back into the blood in the capillary to keep the body in balance. Waste products, excess water, and excess electrolytes remain in the tubule and eventually leave the body as urine.

The kidneys also release three regulatory chemicals—erythropoietin, renin, and calcitriol—that affect other functions in the body. Erythropoietin stimulates the bone marrow to produce new red blood cells. Renin helps regulate blood pressure, and calcitriol is a form of vitamin D and is important in maintaining bones and the level of calcium in the body.

Because the kidney has many functions, there are many types of kidney disease. Congenital kidney diseases are disorders that are present at birth. **Polycystic kidney disease (PKD)** is a rare disorder in which children inherit defective genes from both parents that cause cysts full of fluid to develop in the kidneys and replace the blood filtering units. As a result, the kidneys cannot adequately remove wastes from the body. There are two other types of PKD. One is inherited, but does not appear until adulthood, and the other develops as a result of long-term kidney damage. In total, about half a million people in the United States have some form of PKD. Hereditary disease and **birth defects** are the most common causes of kidney disease in children up to age 14.

Acute kidney diseases are problems that develop suddenly. Many acute kidney diseases can be cured, but some may cause permanent damage. Common acute kidney diseases include kidney infection, hemolytic uremic syndrome, **nephrotic syndrome** in children, and damage caused by injury to the kidney or **poisoning**. Hemolytic uremic syndrome is a rare disease that usually affects children under age ten and is caused by eating food contaminated with bacteria. The bacteria release a poison that damages the kidney and causes **acute kidney failure**. Most children who develop this disease recover and their kidney function returns to normal.

Chronic kidney disease is disease that is slow to develop and usually does not show any symptoms

until kidney damage is permanent. The National Kidney and Urologic Disease Information Clearinghouse, a federal agency, estimates that about 4.5% of people over age 20 have chronic kidney disease as indicated by tests that measure kidney function. The most common cause of chronic kidney disease in the United States is diabetes. It accounts for between 33% and 40% of all new cases of chronic kidney disease in the United States. In diabetes, the body cannot break down glucose (sugar). This extra glucose in the blood damages the nephrons so that they no longer filter blood effectively.

High or uncontrolled blood pressure (**hypertension**) is the second leading cause of chronic kidney disease. It accounts for between 27% and 30% of all new cases of chronic kidney disease. High blood pressure damages the capillaries in the nephron, so that they can no longer work with the tubules to filter the blood. Glomerulonephritis is a term for several different chronic kidney diseases where damage to the nephrons causes protein or red blood cells pass into the urine. **Kidney cancer** is uncommon, accounting for only 2% of **cancer** cases.

Over-the-counter **analgesics** (pain medications) such as **aspirin**, **acetaminophen** (Tylenol), ibuprofen (Advil), naxopren sodium (Aleve), and similar medications that can be bought without a prescription may make kidney disease worse in individuals who already have kidney damage or cause kidney damage in healthy individuals who take these medications daily for several years. The chance of damage is increased when these pain medications are taken in combination with each other or with **caffeine** or codeine (Some painkilling tablets are a combination of pain medications and caffeine or codeine). Individuals who take these painkillers regularly or who have been told they have kidney damage should discuss the risk of these medications with their physician.

Chronic kidney disease can lead to end-stage renal disease (ESRD), in which there is almost total failure of the kidneys. If renal function is reduced to only 25% of normal, serious illness results. When this drops to 10–15% of normal, **death** occurs unless the individual receives dialysis or a kidney transplant. In 2002, there were over 100,300 new cases of ESRD, 44% of which were caused by diabetes. Treatment of ESRD in the United States cost about \$25.2 billion in 2002.

Causes & symptoms

Causes of kidney disease are many and varied. Leading causes are diabetes, high blood pressure, inherited disease, and infection. Acute kidney disease

is often marked by a lack of urination and increased fluid build up in the body. Chronic kidney disease is often called a “silent” killer, because no obvious symptoms develop until the kidneys are permanently damaged. The National Kidney Foundation estimated in 2005 that 20 million Americans had undetected moderate chronic kidney disease. Chronic kidney disease most often results from other diseases such as diabetes or hypertension.

Diagnosis

Simple blood and urine tests can indicate kidney disease, but more extensive testing may be needed to determine the exact nature of the disease. A blood test that measures serum creatinine, a waste product, can indicate how well the kidneys are working. Although normal levels of creatinine vary (an average range is 0.6–1.2 mg/dL), a higher than expected level in the blood may indicate kidney damage. A blood urea nitrogen (BUN) blood test measures waste products circulating in the blood. Normal levels range from 7%–20 mg/dL. The less well the kidney is working, the higher the BUN.

A 24-hour urine collection test will accurately measure how much urine the kidneys are producing in a day. A **urinalysis** can determine if protein or red blood cells are leaking into the urine indicating abnormal kidney function. A creatinine clearance test compares the amount of creatinine in a 24-hour urine sample with the amount of creatinine in the blood to determine how much blood the kidneys are filtering each minute.

Based on the results of blood and urine tests, other tests such as a CT scan, MRI, or **kidney biopsy** may be ordered.

Treatment

Most treatment for kidney disease involves treating the underlying cause of the disease, such as controlling high blood pressure or diabetes. Diuretic medication (“water pills”) may be given to help relieve fluid accumulation. **Antibiotics** are used to treat kidney infections. Other drugs may be given to treat specific kidney diseases.

Diet and lifestyle changes are an important part of controlling kidney disease. **Obesity** increases blood pressure, so losing weight can help limit kidney damage, as can stopping **smoking**. Reducing sodium (salt) in the diet also helps control blood pressure. In certain kinds of kidney disease, potassium is removed in abnormally large quantities by the kidneys and

excreted in urine. Eating more foods such as bananas, dried beans and peas, nuts, and potatoes that are high in potassium or taking a potassium supplement pill help reverse this effect. When protein is found in the urine, some physicians recommend reducing the amount of protein (mainly found in meat) in the diet.

When kidneys fail completely in ESRD, there are only two alternatives: dialysis or kidney transplant. There are two types of dialysis. Peritoneal dialysis uses a membrane in the individual’s abdomen to filter waste products. The most common kind of peritoneal dialysis is continuous ambulatory peritoneal dialysis (CAPD), in which the individual is hooked up to a bag of dialysis fluid that he carries with him, allowing continuous dialysis. The fluid is changed four times a day. In another form of peritoneal dialysis, the abdomen is filled with dialysis fluid. Wastes filter into the fluid for several hours often while the individual is asleep, then the fluid is drained from the body. Peritoneal dialysis can be done at home without the need for a health care professional.

In hemodialysis, the individual must go to a dialysis center about three times a week. His blood is sent through a machine that filters out the waste and then returns the cleansed blood to his body. The process takes three to four hours and is done by a health care professional.

Kidney transplants can come from either a living donor or a deceased donor. Donors are matched with recipients based on blood type and must take drugs to prevent their immune system from rejecting the kidney after transplantation. The United Network for Organ Sharing (UNOS) coordinates matching donor kidneys with appropriate recipients. As of 2005, over 100 clinical trials were enrolling patients with various types of kidney disease. Information on current clinical trials can be found at <<http://www.clinicaltrials.gov>>.

Alternative treatment

Alternative treatments tend to focus on removing excess water from the body, but have limited effect on serious disease. Asparagus (*Asparagus officinalis*) birch tea (*Betula* species), goldenrod infusion (*Solidago* species), horsetail (*Equisetum arvense*), and stinging nettle (*Urtica dioica*) all are used to stimulate urine production.

Prognosis

Many individuals recover normal kidney function after developing acute kidney disease, although in some cases, such as poisoning and injury, kidney

KEY TERMS

Congenital—Present at birth.

Diuretic—A substance that stimulates the kidney to excrete water.

Glomerulosis—A twisted mass of blood capillaries and urine tubules in the kidney where filtering of waste products occurs.

Hormone—A chemical produced by living cells that travels through the circulatory system and affects tissue at some distance from where it was released.

Hypertension—High blood pressure.

Nephron—The smallest functional unit of the kidney involved in the removal of waste products and excess water from the blood.

damage may be permanent. Chronic kidney disease tends to get progressively worse as the individual ages. More than 15,000 kidney transplants are done each year, and there is a often long waiting list for donated kidneys. As of 2001 (last year for which statistics were available), 80.6% of individuals receiving a transplant from a deceased donor survived for at least 5 years, and 90.4% of individuals receiving a kidney donated from a living donor survived for at least 5 years.

Prevention

Maintaining a healthy body weight, getting regular **exercise**, and not smoking all promote kidney health. Controlling underlying diseases such as diabetes and high blood pressure are important in preventing chronic kidney diseases.

Resources

ORGANIZATIONS

American Association of Kidney Patients. 3505 East Frontage Road, Suite 315, Tampa, FL 33607. Telephone: 800-749-2257. < <http://www.aakp.org> >.

American Foundation for Urologic Disease. 1128 North Charles Street, Baltimore, MD 21201 Telephone: 800-242-2383. < <http://www.afud.org> >.

National Kidney Foundation. 30 East 33rd Street, New York, NY 10016 Telephone: 800-622-9010. < <http://www.kidney.org> >.

National Kidney and Urologic Disease Clearinghouse. 3 Information Way, Bethesda, MD 20892-3580. Telephone: 800-891-5390. < <http://www.niddk.nih.gov> >.

United Network for Organ Sharing (UNOS). P. O. Box 13770, Richmond, VA 23225. Telephone: 804-330-8500. < <http://www.unos.org> >.

OTHER

“Your Kidneys and How They Work.” National Kidney and Urologic Disease Clearinghouse. July 2003 [cited 23 March 2005]. < <http://www.kidney.niddke.nih.gov/kudiseases/pubs/yourkidneys/index.htm#rate> >.

“Kidney Disease Overview.” National Kidney Disease Education Program. 9 March 2005 [cited 23 March 2005]. < <http://www.nkdep.nih.gov/patientspublic/kindneydiseaseoverview.htm#4> >.

Tish Davidson, A.M.

Kidney failure see **Acute kidney failure**;
Chronic kidney failure

Kidney function tests

Definition

Kidney function tests is a collective term for a variety of individual tests and procedures that can be done to evaluate how well the kidneys are functioning.

Purpose

The kidneys, the body's natural filtration system, perform many vital functions, including removing metabolic waste products from the bloodstream, regulating the body's water balance, and maintaining the pH (acidity/alkalinity) of the body's fluids. Approximately one and a half quarts of blood per minute are circulated through the kidneys, where waste chemicals are filtered out and eliminated from the body (along with excess water) in the form of urine. Kidney function tests help to determine if the kidneys are performing their tasks adequately.

Precautions

A complete history should be taken prior to kidney function tests to assess the patient's food and drug intake. A wide variety of prescription and over-the-counter medications can affect blood and urine kidney function test results, as can some food and beverages.

Description

Many conditions can affect the ability of the kidneys to carry-out their vital functions. Some lead to a

rapid (acute) decline in kidney function; others lead to a gradual (chronic) decline in function. Both result in a build-up of toxic waste substances in the blood. A number of clinical laboratory tests that measure the levels of substances normally regulated by the kidneys can help determine the cause and extent of kidney dysfunction. These tests are done on urine samples, as well as on blood samples.

Urine tests

There are a variety of urine tests that assess kidney function. A simple, inexpensive screening test, called a routine **urinalysis**, is often the first test administered if kidney problems are suspected. A small, randomly collected urine sample is examined physically for things like color, odor, appearance, and concentration (specific gravity); chemically for substances such as protein, glucose, and pH (acidity/ alkalinity); and microscopically for the presence of cellular elements (red blood cells, white blood cells, and epithelial cells), bacteria, crystals, and casts (structures formed by the deposit of protein, cells, and other substances in the kidneys' tubules). If results indicate a possibility of disease or impaired kidney function, one or more of the following additional tests is usually performed to more specifically diagnose the cause and the level of decline in kidney function.

- **Creatinine clearance test.** This test evaluates how efficiently the kidneys clear a substance called creatinine from the blood. Creatinine, a waste product of muscle energy metabolism, is produced at a constant rate that is proportional to the muscle mass of the individual. Because the body does not recycle it, all of the creatinine filtered by the kidneys in a given amount of time is excreted in the urine, making creatinine clearance a very specific measurement of kidney function. The test is performed on a timed urine specimen—a cumulative sample collected over a two to twenty-four hour period. Determination of the blood creatinine level is also required to calculate the urine clearance.
- **Urea clearance test.** Urea is a waste product that is created by protein metabolism and excreted in the urine. The urea clearance test requires a blood sample to measure the amount of urea in the bloodstream and two urine specimens, collected one hour apart, to determine the amount of urea that is filtered, or cleared, by the kidneys into the urine.
- **Urine osmolality test.** Urine osmolality is a measurement of the number of dissolved particles in urine. It is a more precise measurement than specific gravity for evaluating the ability of the kidneys to

concentrate or dilute the urine. Kidneys that are functioning normally will excrete more water into the urine as fluid intake is increased, diluting the urine. If fluid intake is decreased, the kidneys excrete less water and the urine becomes more concentrated. The test may be done on a urine sample collected first thing in the morning, on multiple timed samples, or on a cumulative sample collected over a twenty-four hour period. The patient will typically be prescribed a high-protein diet for several days before the test and asked to drink no fluids the night before the test.

- **Urine protein test.** Healthy kidneys filter all proteins from the bloodstream and then reabsorb them, allowing no protein, or only slight amounts of protein, into the urine. The persistent presence of significant amounts of protein in the urine, then, is an important indicator of **kidney disease**. A positive screening test for protein (included in a routine urinalysis) on a random urine sample is usually followed-up with a test on a 24-hour urine sample that more precisely measures the quantity of protein.

Blood tests

There are also several blood tests that can aid in evaluating kidney function. These include:

- **Blood urea nitrogen test (BUN).** Urea is a by-product of protein metabolism. This waste product is formed in the liver, then filtered from the blood and excreted in the urine by the kidneys. The BUN test measures the amount of nitrogen contained in the urea. High BUN levels can indicate kidney dysfunction, but because blood urea nitrogen is also affected by protein intake and liver function, the test is usually done in conjunction with a blood creatinine, a more specific indicator of kidney function.
- **Creatinine test.** This test measures blood levels of creatinine, a by-product of muscle energy metabolism that, like urea, is filtered from the blood by the kidneys and excreted into the urine. Production of creatinine depends on an individual's muscle mass, which usually fluctuates very little. With normal kidney function, then, the amount of creatinine in the blood remains relatively constant and normal. For this reason, and because creatinine is affected very little by liver function, an elevated blood creatinine is a more sensitive indication of impaired kidney function than the BUN.
- **Other blood tests.** Measurement of the blood levels of other elements regulated in part by the kidneys can also be useful in evaluating kidney function. These include sodium, potassium, chloride, bicarbonate,

KEY TERMS

Blood urea nitrogen (BUN)—The nitrogen portion of urea in the bloodstream. Urea is a waste product of protein metabolism in the body.

Creatinine—The metabolized by-product of creatine, an organic acid that assists the body in producing muscle contractions. Creatinine is found in the bloodstream and in muscle tissue. It is removed from the blood by the kidneys and excreted in the urine.

Osmolality—A measurement of urine concentration that depends on the number of particles dissolved in it. Values are expressed as milliosmols per kilogram (mOsm/kg) of water.

Urea—A by-product of protein metabolism that is formed in the liver. Because urea contains ammonia, which is toxic to the body, it must be quickly filtered from the blood by the kidneys and excreted in the urine.

calcium, magnesium, phosphorus, protein, uric acid, and glucose.

Preparation

Patients will be given specific instructions for collection of urine samples, depending on the test to be performed. Some timed urine tests require an extended collection period of up to 24 hours, during which time the patient collects all urine voided and transfers it to a specimen container. Refrigeration and/or preservatives are typically required to maintain the integrity of such urine specimens. Certain dietary and/or medication restrictions may be imposed for some of the blood and urine tests. The patient may also be instructed to avoid **exercise** for a period of time before a test.

Aftercare

If medication was discontinued prior to a urine kidney function test, it may be resumed once the test is completed.

Risks

Risks for these tests are minimal, but may include slight bleeding from a blood-drawing site, hematoma (accumulation of blood under a puncture site), or **fainting** or feeling light-headed after venipuncture. In addition, suspension of medication or dietary changes

imposed in preparation for some blood or urine tests may trigger side-effects in some individuals.

Normal results

Normal values for many tests are determined by the patient's age and sex. Reference values can also vary by laboratory, but are generally within the ranges that follow.

Urine tests

- **Creatinine clearance.** For a 24-hour urine collection, normal results are 90-139 ml/min for adult males less than 40 years old, and 80-125 ml/min for adult females less than 40 years old. For people over 40, values decrease by 6.5 ml/min for each decade of life.
- **Urea clearance.** With maximum clearance, normal is 64–99 ml/min.
- **Urine osmolality.** With restricted fluid intake (concentration testing), osmolality should be greater than 800 mOsm/kg of water. With increased fluid intake (dilution testing), osmolality should be less than 100 mOsm/kg in at least one of the specimens collected.
- **Urine protein.** A 24-hour urine collection should contain no more than 150 mg of protein.

Blood tests

- **Blood urea nitrogen (BUN).** 8–20 mg/dl.
- **Creatinine.** 0.8–1.2 mg/dl for males, and 0.6–0.9 mg/dl for females.

Abnormal results

Low clearance values for creatinine and urea indicate diminished ability of the kidneys to filter these waste products from the blood and excrete them in the urine. As clearance levels decrease, blood levels of creatinine and urea nitrogen increase. Since it can be affected by other factors, an elevated BUN, by itself, is suggestive, but not diagnostic, for kidney dysfunction. An abnormally elevated blood creatinine, a more specific and sensitive indicator of kidney disease than the BUN, is diagnostic of impaired kidney function.

Inability of the kidneys to concentrate the urine in response to restricted fluid intake, or to dilute the urine in response to increased fluid intake during osmolality testing may indicate decreased kidney function. Because the kidneys normally excrete almost no protein in the urine, its persistent presence, in amounts that exceed the normal 24-hour urine value, usually indicates some type of kidney disease as well.

Resources

ORGANIZATIONS

National Kidney Foundation. 30 East 33rd St., New York, NY 10016. (800) 622-9010. <<http://www.kidney.org>>.

Paula Anne Ford-Martin

Kidney nuclear medicine scan

Definition

A kidney nuclear medicine scan, or study, is a simple outpatient test that involves administering small amounts of radioactive substances, called tracers, into the body and then imaging the kidneys and bladder with a special camera. The images obtained can help in the diagnosis and treatment of certain kidney diseases.

Purpose

While many tests, such as x rays, ultrasound exams, or **computed tomography scans** (CT scans), can reveal the structure of the kidneys (its anatomy), the kidney nuclear medicine scan is unique in that it reveals how the kidneys are functioning. This is valuable information in helping a doctor make a diagnosis. Therefore, the kidney nuclear medicine scan is performed primarily to see how well the kidneys are working and, at the same time, they can identify some of the various structures that make up the kidney.

Precautions

If a patient is pregnant, it is generally recommended that she not have a kidney nuclear medicine scan. The unborn baby is more sensitive to radiation than an adult. If a woman thinks she might be pregnant, she should inform her doctor of this too.

Women who are breastfeeding should also inform their doctor. The doctor may recommend the woman stop breastfeeding for a day or two after a kidney nuclear medicine scan, depending on the particular tracer that was used since the tracer can accumulate in breast milk.

Description

Nuclear medicine is a branch of radiology that uses radioactive materials to diagnose or treat various

diseases. These radioactive materials (tracers) may also be called radiopharmaceuticals, and they accumulate (collect) in specific organs in the body. Radiopharmaceuticals are able to yield valuable information about the particular organ being studied.

Whether outside the body or inside the body, tracers emit radioactive signals, called gamma rays, which can be collected and counted by a special device, called a gamma camera. The images of the kidney that the camera produces are called renal scans.

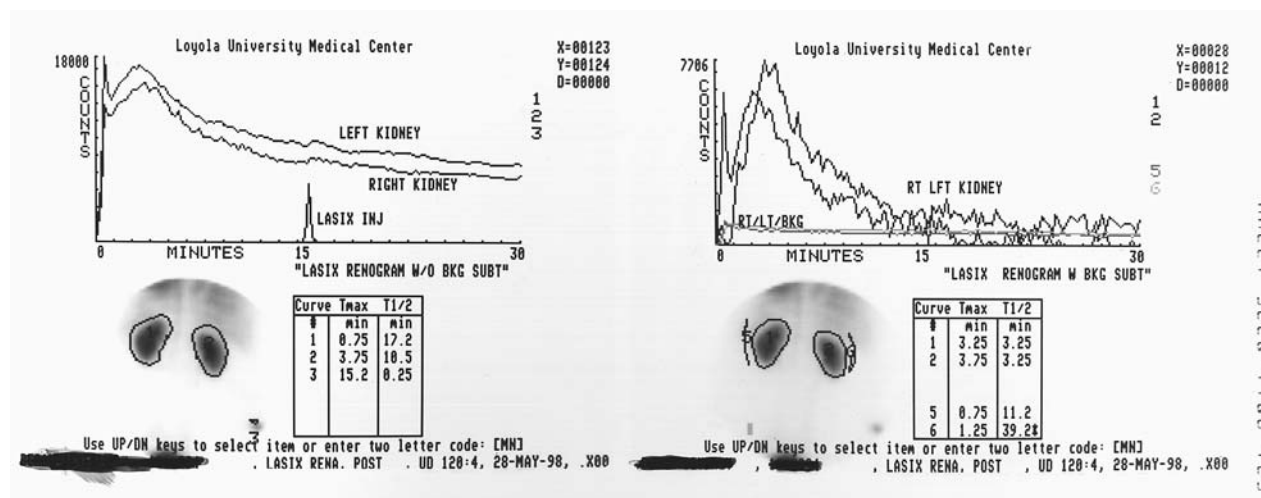
The kidney nuclear medicine scan can be performed on an outpatient basis, usually by a nuclear medicine technologist. The technologist helps prepare the patient for the exam by positioning him or her on an exam table or cart in the imaging area. The patient's position is usually flat on the back. The patient must lie still during imaging to prevent blurring of the images that will be taken. The technologist positions the camera as close to the kidney (or kidneys) as possible to obtain the best images.

In the next step of the procedure, the technologist injects the radiopharmaceutical into the patient. This may be done with one single injection or through an intravenous (IV) line. Immediately after the tracer is injected, imaging begins. It is important to obtain images right away because the tracer's radioactivity begins to diminish (decay). The time required for one-half of the tracer's activity to decay is called the tracer's half-life ($T_{1/2}$). The half-life is unique to each radiopharmaceutical. Also, it is important to see the kidney in its immediate state.

Serial pictures are taken with the gamma camera and may be seen on a computer or TV-like screen. The camera doesn't emit radiation, it only records it. The images then are stored on film.

A kidney nuclear medicine scan ranges from 45 minutes to three hours in length, depending on the goals of the test. But the test typically takes about an hour to an hour-and-a-half.

Once the images and curves are obtained, the nuclear medicine physician or radiologist analyzes, or reads, them. Various information can be provided to the doctor through these, depending on the test that was performed. A variety of kidney nuclear medicine studies are available for a doctor to help in making diagnoses. It is important to understand that kidney nuclear medicine scans are good at identifying when there is an abnormality, but they do not always identify the specific problem. They are very useful in providing information about how the various parts of the kidneys function, which, in turn, can assist in making a diagnosis.



A computer-generated time activity curve generated from a renal scan. This time activity curve looks at the radiation count over a period of time. (Photograph by Collette Placek. Reproduced by permission.)

Studies may be performed to determine the rate at which the kidney's are filtering a patient's blood. These studies use a radiopharmaceutical, called Technetium DTPA (Tc 99m DTPA). This radiopharmaceutical also can identify obstruction (blockage) in the collecting system. To study how well the tubules and ducts of the kidney are functioning, the radiopharmaceutical Technetium MAG3 is used. Studying tubular function is a good indicator of overall renal function. In many renal diseases, one of the first things that disappears or diminishes is the tubular function.

Candidates for a kidney nuclear medicine scan are patients who have:

- Renal failure or chronic renal failure
- Obstruction in their urine collection systems
- **Renal artery stenosis**
- A kidney transplant.

Preparation

No preparation is necessary for a kidney nuclear medicine scan. The doctor may ask the patient to refrain from certain medications, however, before the scan if the medications might interfere with the test. For example, if a scan is being performed to study renal artery stenosis, the patient may have to refrain from taking medications for **hypertension**.

Aftercare

Patients can resume their normal daily activities immediately after the test. Most tracers are passed

KEY TERMS

Intravenous pyelogram (IVP)—X ray technique using dye to image the kidneys, ureters, and bladder.

Renal—Having to do with the kidneys.

Renal artery stenosis—Narrowing or constriction of the artery that supplies the kidney with blood.

naturally from the body, though drinking fluids after a kidney nuclear medicine scan can help flush the tracer into the urine and out of the body more quickly.

Risks

Nuclear medicine procedures are very safe. Unlike some of the dyes that may be used in x-ray studies, radioactive tracers rarely cause side effects. There are no long-lasting effects of the tracers themselves, because they have no functional effects on the body's tissues.

Normal results

The test reveals normal kidney function for age and medical situation.

Abnormal results

The test reveals a change in function that may be attributable to a disease process, such as obstruction

or a malfunctioning kidney. If the test is abnormal, the patient may be recalled another day for a repeat study, performed differently, to narrow the list of causes.

Resources

ORGANIZATIONS

Society of Nuclear Medicine. 1850 Samuel Morse Dr.,
Reston, VA 10016. (703) 708-9000. <<http://www.snm.org>>.

Collette L. Placek

Kidney removal see **Nephrectomy**

Kidney stones

Definition

Kidney stones are solid accumulations of material that form in the tubal system of the kidney. Kidney stones cause problems when they block the flow of urine through or out of the kidney. When the stones move along the ureter, they cause severe **pain**.

Description

Urine is formed by the kidneys. Blood flows into the kidneys, and specialized tubes (nephrons) within the kidneys allow a certain amount of fluid from the blood, and certain substances dissolved in that fluid, to flow out of the body as urine. Sometimes, a problem causes the dissolved substances to become solid again. Tiny crystals may form in the urine, meet, and cling together to create a larger solid mass called a kidney stone.

Many people do not ever find out that they have stones in their kidneys. These stones are small enough to allow the kidney to continue functioning normally, never causing any pain. These are called “silent stones.” Kidney stones cause problems when they interfere with the normal flow of urine. They can block (obstruct) the flow down the tube (the ureter) that carries urine from the kidney to the bladder. The kidney is not accustomed to experiencing any pressure. When pressure builds from backed-up urine, the kidney may swell (**hydronephrosis**). If the kidney is subjected to this pressure for some time, it may cause damage to the delicate kidney structures. When the kidney stone is lodged further down the ureter, the backed-up urine may also cause the ureter to swell (hydroureter). Because the ureters are muscular

tubes, the presence of a stone will make these muscular tubes spasm, causing severe pain.

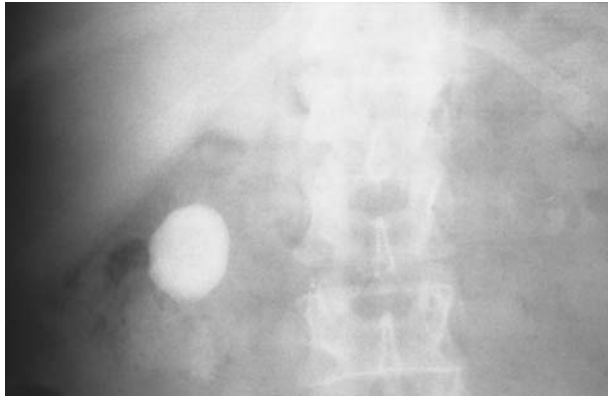
About 10% of all people will have a kidney stone in his or her lifetime. Kidney stones are most common among:

- Caucasians
- Males
- People over the age of 30
- People who have had kidney stones previously
- Relatives of kidney stone patients

Causes and symptoms

Kidney stones can be composed of a variety of substances. The most common types of kidney stones include:

- Calcium stones. About 80% of all kidney stones fall into this category. These stones are composed of either calcium and phosphate, or calcium and oxalate. People with calcium stones may have other diseases that cause them to have increased blood levels of calcium. These diseases include primary parathyroidism, **sarcoidosis**, **hyperthyroidism**, **renal tubular acidosis**, **multiple myeloma**, hyperoxaluria, and some types of **cancer**. A diet heavy in meat, fish, and poultry can cause calcium oxalate stones.
- Struvite stones. About 10% of all kidney stones fall into this category. This type of stone is composed of magnesium ammonium phosphate. These stones occur most often when patients have had repeated urinary tract infections with certain types of bacteria. These bacteria produce a substance called urease, which increases the urine pH and makes the urine more alkaline and less acidic. This chemical environment allows struvite to settle out of the urine, forming stones.
- Uric acid stones. About 5% of all kidney stones fall into this category. Uric acid stones occur when increased amounts of uric acid circulate in the bloodstream. When the uric acid content becomes very high, it can no longer remain dissolved and solid bits of uric acid settle out of the urine. A kidney stone is formed when these bits of uric acid begin to cling to each other within the kidney, slowly growing into a solid mass. About half of all patients with this type of stone also have deposits of uric acid elsewhere in their body, commonly in the joint of the big toe. This painful disorder is called **gout**. Other causes of uric acid stones include **chemotherapy** for cancer, certain bone marrow disorders where blood cells



X ray of kidney stone. (Custom Medical Stock Photo. Reproduced by permission.)

are over-produced, and an inherited disorder called **Lesch-Nyhan syndrome**.

- **Cystine stones.** About 2% of all kidney stones fall into this category. Cystine is a type of amino acid, and people with this type of kidney stone have an abnormality in the way their bodies process amino acids in the diet.

Patients who have kidney stones usually do not have symptoms until the stones pass into the ureter. Prior to this, some people may notice blood in their urine. Once the stone is in the ureter, however, most people will experience bouts of very severe pain. The pain is crampy and spasmodic, and is referred to as “colic.” The pain usually begins in the flank region, the area between the lower ribs and the hip bone. As the stone moves closer to the bladder, a patient will often feel the pain radiating along the inner thigh. In women, the pain may be felt in the vulva. In men, the pain may be felt in the testicles. **Nausea, vomiting,** extremely frequent and painful urination, and obvious blood in the urine are common. **Fever** and chills usually means that the ureter has become obstructed, allowing bacteria to become trapped in the kidney causing a kidney infection (**pyelonephritis**).

Diagnosis

Diagnosing kidney stones is based on the patient’s history of the very severe, distinctive pain associated with the stones. Diagnosis includes laboratory examination of a urine sample and an x-ray examination. During the passage of a stone, examination of the urine almost always reveals blood. A number of x-ray tests are used to diagnose kidney stones. A plain x ray of the kidneys, ureters, and bladder may or may not reveal the stone. A series of x rays taken

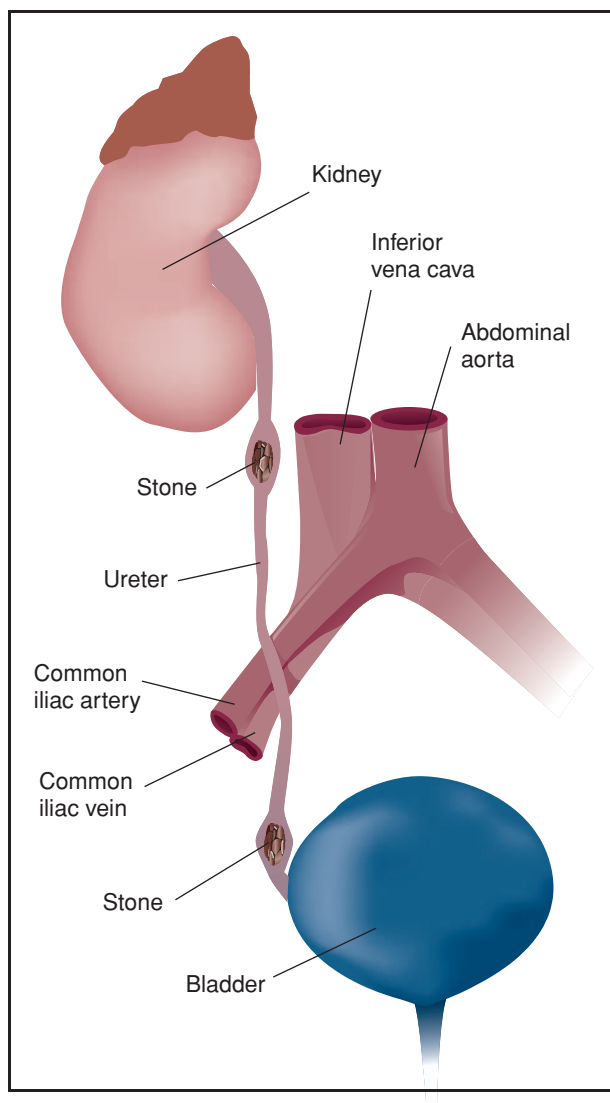
after injecting iodine dye into a vein is usually a more reliable way of seeing a stone. This procedure is called an intravenous pyelogram (IVP). The dye “lights up” the urinary system as it travels. In the case of an obstruction, the dye will be stopped by the stone or will only be able to get past the stone at a slow trickle.

When a patient is passing a kidney stone, it is important that all of his or her urine is strained through a special sieve. This is to make sure that the stone is caught. The stone can then be sent to a special laboratory for analysis so that the chemical composition of the stone can be determined. After the kidney stone has been passed, other tests will be required in order to understand the underlying condition that may have caused the stone to form. Collecting urine for 24 hours, followed by careful analysis of its chemical makeup, can often determine a number of reasons for stone formation.

Treatment

A patient with a kidney stone will say that the most important aspect of treatment is adequate pain relief. Because the pain of passing a kidney stone is so severe, narcotic pain medications (like morphine) are usually required. It is believed that stones may pass more quickly if the patient is encouraged to drink large amounts of water (2–3 quarts per day). If the patient is vomiting or unable to drink because of the pain, it may be necessary to provide fluids through a vein. If symptoms and urine tests indicate the presence of infection, **antibiotics** will be required.

Although most kidney stones will pass on their own, some will not. Surgical removal of a stone may become necessary when a stone appears too large to pass. Surgery may also be required if the stone is causing serious obstructions, pain that cannot be treated, heavy bleeding, or infection. Several alternatives exist for removing stones. One method involves inserting a tube into the bladder and up into the ureter. A tiny basket is then passed through the tube, and an attempt is made to snare the stone and pull it out. Open surgery to remove an obstructing kidney stone was relatively common in the past, but current methods allow the stone to be crushed with shock waves (called **lithotripsy**). These shock waves may be aimed at the stone from outside of the body by passing the necessary equipment through the bladder and into the ureter. The shock waves may be aimed at the stone from inside the body by placing the instrument through a tiny incision located near the stone. The stone fragments may then pass on their own or may be removed through the incision. All of these methods



Kidney stones can occur in the ureter near the bladder or kidney. (Illustration by Argosy Inc.)

reduce the patient's recovery time considerably when compared to the traditional open operation.

Alternative treatment

Alternative treatments for kidney stones include the use of herbal medicine, homeopathy, **acupuncture**, **acupressure**, hypnosis, or **guided imagery** to relieve pain. Starfruit (*Averrhoa carambola*) is recommended to increase the amount of urine a patient passes and to relieve pain. Dietary changes can be made to reduce the risk of future stone formation and to facilitate the resorption of existing stones. Supplementation with magnesium, a smooth muscle relaxant, can help reduce pain and facilitate stone passing. Homeopathy and herbal

medicine, both western and Chinese, recommend a number of remedies that may help prevent kidney stones.

Prognosis

A patient's prognosis depends on the underlying disorder causing the development of kidney stones. In most cases, patients with uncomplicated calcium stones will recover very well. About 60% of these patients, however, will have other kidney stones. Struvite stones are particularly dangerous because they may grow extremely large, filling the tubes within the kidney. These are called staghorn stones and will not pass out in the urine. They will require surgical removal. Uric acid stones may also become staghorn stones.

Prevention

Prevention of kidney stones depends on the type of stone and the presence of an underlying disease. In almost all cases, increasing fluid intake so that a person consistently drinks several quarts of water a day is an important preventative measure. Patients with calcium stones may benefit from taking a medication called a diuretic, which has the effect of decreasing the amount of calcium passed in the urine. Eating less meat, fish, and chicken may be helpful for patients with calcium oxalate stones. Other items in the diet that may encourage calcium oxalate stone formation include beer, black pepper, berries, broccoli, chocolate, spinach, and tea. Uric acid stones may require treatment with a medication called allopurinol. Struvite stones will require removal and the patient should receive an antibiotic. When a disease is identified as the cause of stone formation, treatment specific to that disease may lessen the likelihood of repeated stones.

Resources

ORGANIZATIONS

American Foundation for Urologic Disease. 300 West Pratt St., Baltimore, MD 21201-2463. (800) 242-2383.

National Kidney Foundation. 30 East 33rd St., New York, NY 10016. (800) 622-9010. <<http://www.kidney.org>>.

Rosalyn Carson-DeWitt, MD

Kidney transplantation

Definition

Kidney transplantation is a surgical procedure to remove a healthy, functioning kidney from a living or

National Transplant Waiting List By Organ Type (June 2000)

Organ Needed	Number Waiting
Kidney	48,349
Liver	15,987
Heart	4,139
Lung	3,695
Kidney-Pancreas	2,437
Pancreas	942
Heart-Lung	212
Intestine	137

brain-dead donor and implant it into a patient with non-functioning kidneys.

Purpose

Kidney transplantation is performed on patients with **chronic kidney failure**, or end-stage renal disease (ESRD). ESRD occurs when a disease or disorder damages the kidneys so that they are no longer capable of adequately removing fluids and wastes from the body or of maintaining the proper level of certain kidney-regulated chemicals in the bloodstream. Without long-term dialysis or a kidney transplant, ESRD is fatal.

Precautions

Patients with a history of heart disease, lung disease, **cancer**, or hepatitis may not be suitable candidates for receiving a kidney transplant.

Description

Kidney transplantation involves surgically attaching a functioning kidney, or graft, from a brain-dead organ donor (a cadaver transplant) or from a living donor, to a patient with ESRD. Living donors may be related or unrelated to the patient, but a related donor has a better chance of having a kidney that is a stronger biological “match” for the patient.

The surgical procedure to remove a kidney from a living donor is called a *nephrectomy*. The kidney donor is administered **general anesthesia** and an incision is made on the side or front of the abdomen. The blood vessels connecting the kidney to the donor are cut and clamped, and the ureter is also cut between the bladder and kidney and clamped. The kidney and an attached section of ureter is removed from the donor. The vessels and ureter in the donor are then tied off and

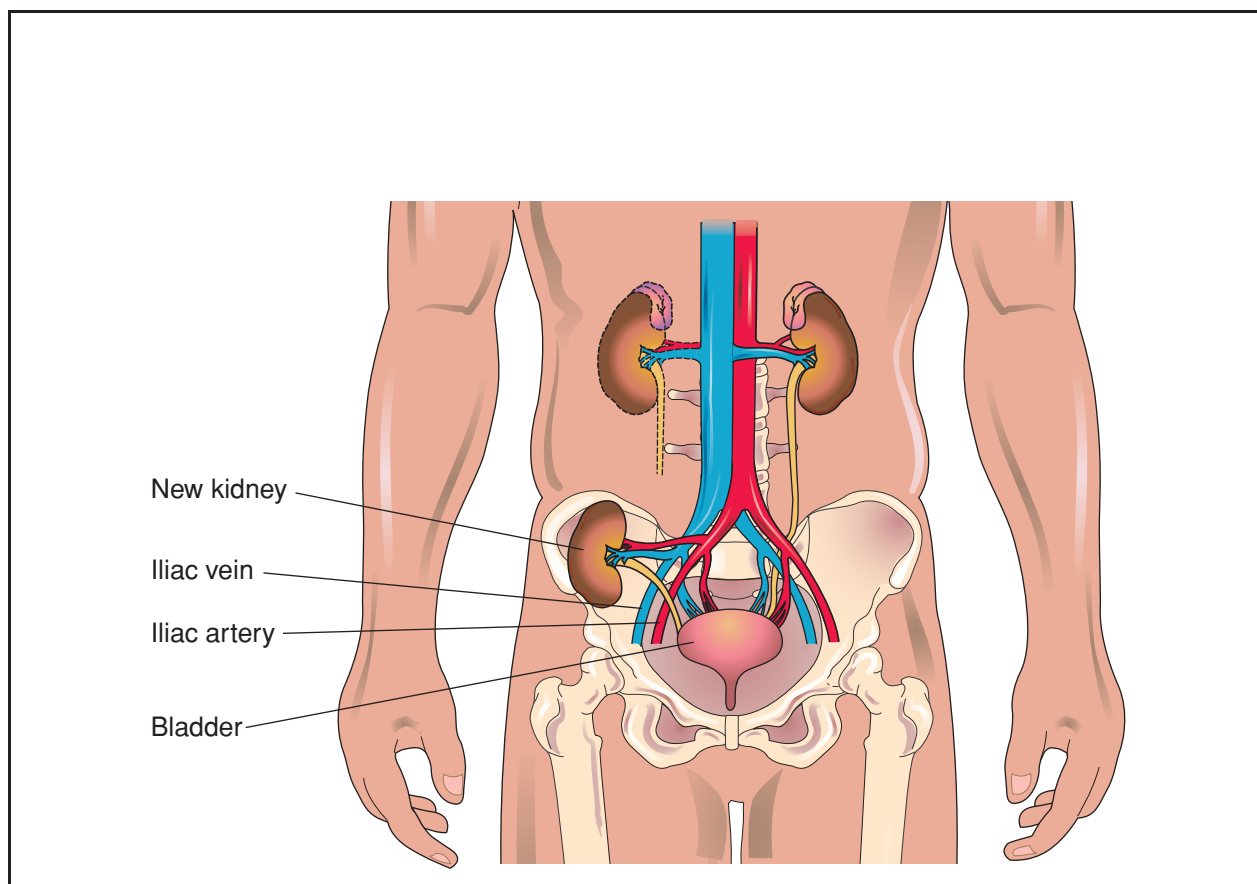
the incision is sutured together again. A similar procedure is used to harvest cadaver kidneys, although both kidneys are typically removed at once, and blood and cell samples for **tissue typing** are also taken.

Laparoscopic **nephrectomy** is a form of minimally-invasive surgery using instruments on long, narrow rods to view, cut, and remove the donor kidney. The surgeon views the kidney and surrounding tissue with a flexible videoscope. The videoscope and surgical instruments are maneuvered through four small incisions in the abdomen. Once the kidney is freed, it is secured in a bag and pulled through a fifth incision, approximately 3 in (7.6 cm) wide, in the front of the abdominal wall below the navel. Although this surgical technique takes slightly longer than a traditional nephrectomy, preliminary studies have shown that it promotes a faster recovery time, shorter hospital stays, and less post-operative **pain** for kidney donors.

Once removed, kidneys from live donors and cadavers are placed on ice and flushed with a cold preservative solution. The kidney can be preserved in this solution for 24–48 hours until the transplant takes place. The sooner the transplant takes place after harvesting the kidney, the better the chances are for proper functioning.

During the transplant operation, the kidney recipient patient is typically under general anesthesia and administered **antibiotics** to prevent possible infection. A catheter is placed in the bladder before surgery begins. An incision is made in the flank of the patient and the surgeon implants the kidney above the pelvic bone and below the existing, non-functioning kidney by suturing the kidney artery and vein to the patient’s iliac artery and vein. The ureter of the new kidney is attached directly to the bladder of the kidney recipient. Once the new kidney is attached, the patient’s existing, diseased kidneys may or may not be removed, depending on the circumstances surrounding the kidney failure.

Since 1973, Medicare has picked up 80% of ESRD treatment costs, including the costs of transplantation for both the kidney donor and recipient. Medicare also covers 80% of immunosuppressive medication costs for up to three years, although federal legislation was under consideration in early 1998 that may remove the time limit on these benefits. To qualify for Medicare ESRD benefits, a patient must be insured or eligible for benefits under Social Security, or be a spouse or child of an eligible American. Private insurance and state Medicaid programs often cover the remaining 20% of treatment costs.



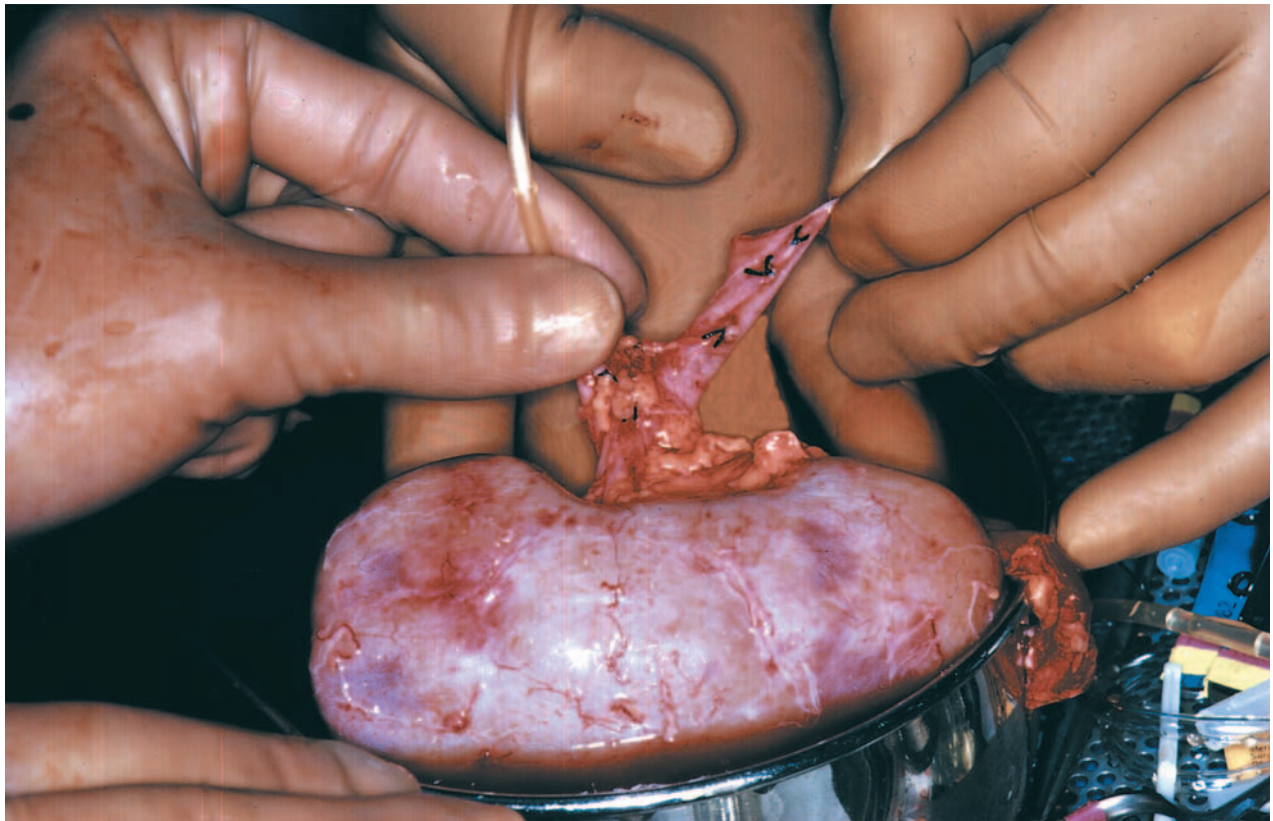
Kidney transplantation involves the surgical attachment of a functioning kidney, or graft, from a donor to a patient with end-stage renal disease (ESRD). During the procedure, the surgeon makes an incision in the patient's flank and implants the new kidney above the pelvic bone and below the non-functioning kidney by suturing the kidney artery and vein to the patient's iliac artery and vein. The ureter of the new kidney is then attached directly to the bladder of the patient. (Illustration by Electronic Illustrators Group.)

Preparation

Patients with chronic renal disease who need a transplant and do not have a living donor register with United Network for Organ Sharing (UNOS) will be placed on a waiting list for a cadaver kidney transplant. UNOS is a non-profit organization that is under contract with the federal government to administer the Organ Procurement and Transplant Network (OPTN) and the national Scientific Registry of Transplant Recipients (SR). Kidney availability is based on the patient's health status. The most important factor is that the kidney be compatible to the patient's body. A human kidney has a set of six antigens, substances that stimulate the production of antibodies. (Antibodies then attach to cells they recognize as foreign and attack them.) Donors are tissue-matched for 0 to 6 of the antigens, and compatibility is determined by the number and

strength of those matched pairs. Patients with a living donor who is a close relative have the best chance of a close match.

Potential kidney donors undergo a complete medical history and **physical examination** to evaluate their suitability for donation. Extensive blood tests are performed on both donor and recipient. The blood samples are used to tissue type for antigen matches, and confirm that blood types are compatible. A panel of reactive antibody (PRA) is performed by mixing white blood cells from the donor and serum from the recipient to ensure that the recipient antibodies will not have a negative reaction to the donor antigens. A urine test is performed on the donor to evaluate his kidney function. In some cases, a special dye that shows up on x rays is injected into an artery, and x rays are taken to show the blood supply of the donor kidney (a procedure called an arteriogram).



A human kidney is being prepped by medical personnel prior to transplantation. (Photograph by Brad Nelson, Custom Medical Stock Photo. Reproduced by permission.)

Once compatibility is confirmed and the physical preparations for kidney transplantation are complete, both donor and recipient may undergo a psychological or psychiatric evaluation to ensure that they are emotionally prepared for the transplant procedure and aftercare regimen.

Aftercare

Kidney donors and recipients will experience some discomfort in the area of the incision. Pain relievers are administered following the transplant operation. Patients may also experience **numbness**, caused by severed nerves, near or on the incision.

A regimen of immunosuppressive, or anti-rejection, medication is prescribed to prevent the body's immune system from rejecting the new kidney. Common immunosuppressants include cyclosporine, prednisone, and azathioprine. The kidney recipient will be required to take immunosuppressants for the life span of the new kidney. Intravenous antibodies may also be administered after transplant surgery. Daclizumab, a monoclonal antibody, is a promising new therapy that

can be used in conjunction with standard immunosuppressive medications to reduce the incidence of organ rejection.

Transplant recipients may need to adjust their dietary habits. Certain immunosuppressive medications cause increased appetite or sodium and protein retention, and the patient may have to adjust his or her intake of calories, salt, and protein to compensate.

Risks

As with any surgical procedure, the kidney transplantation procedure carries some risk for both a living donor and a graft recipient. Possible complications include infection and bleeding (hemorrhage). The most common complication for kidney recipients is a urine leak. In approximately 5% of kidney transplants, the ureter suffers some damage, which results in the leak. This problem is usually correctable with follow-up surgery.

The biggest risk to the recovering transplant recipient is not from the operation or the kidney itself, but from the immunosuppressive medication he or she

KEY TERMS

Arteriogram—A diagnostic test that involves viewing the arteries and/or attached organs by injecting a contrast medium, or dye, into the artery and taking an x ray.

Dialysis—A blood filtration therapy that replaces the function of the kidneys, filtering fluids and waste products out of the bloodstream. There are two types of dialysis treatment—hemodialysis, which uses an artificial kidney, or dialyzer, as a blood filter; and peritoneal dialysis, which uses the patient's abdominal cavity (peritoneum) as a blood filter.

Iliac artery—Large blood vessel in the pelvis that leads into the leg.

Immunosuppressive medication—Drugs given to a transplant recipient to prevent his or her immune system from attacking the transplanted organ.

Rejection—The process in which the immune system attacks tissue it sees as foreign to the body.

Videoscope—A surgical camera.

must take. Because these drugs suppress the immune system, the patient is susceptible to infections such as cytomegalovirus (CMV) and varicella (**chickenpox**). The immunosuppressants can also cause a host of possible side effects, from high blood pressure to **osteoporosis**. Prescription and dosage adjustments can lessen side effects for some patients.

Normal results

The new kidney may start functioning immediately, or may take several weeks to begin producing urine. Living donor kidneys are more likely to begin functioning earlier than cadaver kidneys, which frequently suffer some reversible damage during the kidney transplant and storage procedure. Patients may have to undergo dialysis for several weeks while their new kidney establishes an acceptable level of functioning.

The success of a kidney transplant graft depends on the strength of the match between donor and recipient and the source of the kidney. Cadaver kidneys have a four-year survival rate of 66%, compared to an 80.9% survival rate for living donor kidneys. However, there have been cases of cadaver and living, related donor kidneys functioning well for over 25 years.

Studies have shown that after they recover from surgery, kidney donors typically have no long-term

complications from the loss of one kidney, and their remaining kidney will increase its functioning to compensate for the loss of the other.

Abnormal results

A transplanted kidney may be rejected by the patient. Rejection occurs when the patient's immune system recognizes the new kidney as a foreign body and attacks the kidney. It may occur soon after transplantation, or several months or years after the procedure has taken place. Rejection episodes are not uncommon in the first weeks after transplantation surgery, and are treated with high-dose injections of **immunosuppressant drugs**. If a rejection episode cannot be reversed and kidney failure continues, the patient will typically go back on dialysis. Another transplant procedure can be attempted at a later date if another kidney becomes available.

Resources

ORGANIZATIONS

American Association of Kidney Patients. 100 S. Ashley Drive, #280, Tampa, FL 33602. (800) 749-2257. <<http://www.aakp.org>>.

American Kidney Fund (AKF). Suite 1010, 6110 Executive Boulevard, Rockville, MD 20852. (800) 638-8299. <<http://www.arbon.com/kidney>>.

National Kidney Foundation. 30 East 33rd St., New York, NY 10016. (800) 622-9010. <<http://www.kidney.org>>.

United Network for Organ Sharing (UNOS). 1100 Boulders Parkway, Suite 500, P.O. Box 13770, Richmond, VA 23225-8770. (804) 330-8500. <<http://www.unos.org>>.

United States Renal Data System (USRDS). The University of Michigan, 315 W. Huron, Suite 240, Ann Arbor, MI 48103. (734) 998-6611. <<http://www.med.umich.edu/usrds>>.

OTHER

Transweb. <<http://www.transweb.org>>.

Paula Anne Ford-Martin

Kidney ultrasound see **Abdominal ultrasound**

Kidney, ureter, and bladder x-ray study

Definition

A kidney, ureter, and bladder (KUB) x-ray study is an abdominal x ray. Despite its name, KUB does not show the ureters and only sometimes shows the kidneys and bladder and, even then, with uncertainty.



An x-ray image of a human torso and abdomen showing a blocked ureter. (Custom Medical Stock Photo. Reproduced by permission.)

Purpose

The KUB study is a diagnostic test used to detect **kidney stones** and to diagnose some gastrointestinal disorders. The KUB is also used as a follow-up procedure after the placement of devices such as ureteral stents and nasogastric or nasointestinal tubes (feeding tubes) to verify proper positioning.

Precautions

Because of the risks of radiation exposure to the fetus, pregnant women are advised to avoid this x-ray procedure.

A KUB study is a preliminary screening test for kidney stones, and should be followed by a more sophisticated series of diagnostic tests [such as an **abdominal ultrasound**, **intravenous urography**, or computed tomography scan (CT scan)] if kidney stones are suspected.

KEY TERMS

Ureteral stent—A surgical device implanted in patients with damaged ureters that holds the ureter open so that urine can flow freely from the kidneys to the bladder.

Description

A KUB is typically a single x-ray procedure. The patient lies flat on his back on an x-ray table. An x-ray plate is placed underneath him near the small of the back, and the x-ray camera is aimed at his abdomen. The patient is asked to hold his breath and lie still while the x ray is taken. Sometimes a second KUB will be ordered, with the patient standing, or if unable to do so, lying on his side.

Preparation

A KUB study requires no special diet, fluid restrictions, medications, or other preparation. The patient is typically required to wear a hospital gown or similar attire and to remove all jewelry so the x-ray camera has an unobstructed view of the abdomen. A lead apron may be placed over the abdominal areas of the body not being x rayed to shield the patient from unnecessary radiation.

Aftercare

No special aftercare treatment or regimen is required for a KUB study.

Risks

Because the KUB study is an x-ray procedure, it does involve minor exposure to radiation.

Normal results

Normal KUB x-ray films show two kidneys of a similar size and shape. A normal amount of intestinal gas is seen.

Abnormal results

Abnormal KUB films may show calculi (kidney stones). If both kidneys are visible, it may be possible to diagnose renal size discrepancies. The films may also show too much bowel gas indicating possible obstruction or soft tissue masses.

Resources

BOOKS

Pagana, Kathleen Deska. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis: Mosby, Inc., 1998.

Paula Anne Ford-Martin

Kinesiology, applied

Definition

Kinesiology is a series of tests that locate weaknesses in specific muscles reflecting imbalances throughout the body. Then specific massages or **acupuncture** techniques are used in an attempt to rebalance what has been revealed by the kinesiology tests. Thus, kinesiology is used as both an assessment tool and as a limited therapeutic modality.

Purpose

Kinesiology claims to be a healing system that detects and corrects imbalances in the body before they develop into a disease, and which restores overall system balance and harmony. It is used to alleviate muscle, bone, and joint problems, treat all manner of aches and pains, and correct many areas of imbalance and discomfort.

Precautions

Since interpretation of the muscle tests is both complex and subjective, it should only be performed by a licensed health professional trained to look for “subclinical” symptoms (those which have not yet become a major problem). Kinesiology, itself, is more of a diagnostic technique and should not be thought of as a cure for any particular problem.

Description

Traditionally, the word “kinesiology” refers simply to the study of muscles and body movement. In 1964, however, American chiropractor George J. Goodheart founded what has become known as applied kinesiology when he linked oriental ideas about energy flow in the body with western techniques of muscle testing. First, Goodheart noted that all muscles are related to other muscles. He observed that for each movement a muscle makes, there is another muscle or group of muscles involved with

that movement; one muscle contracts while another one relaxes. So when he was presented with a painful, overly-tight muscle, he would observe and treat the opposite, and necessarily weak, muscle to restore balance. This was then a very new technique.

Further, Goodheart argued that there is a definite and real connection between muscles, glands, and organs, and that by testing the strength of certain muscles he could learn about the health or condition of the gland or organ to which it was related.

Applied kinesiology is based on the idea that the body is an interacting unit made of different parts that interconnect and affect each other. Everything we do affects the body as a whole; therefore, a problem in one area can cause trouble in another area. According to kinesiology, the muscles eventually register and reflect anything that is wrong with any part of the body, whether physical or mental. Thus, a particular digestive problem might show up in the related and corresponding muscles of the legs. By testing the strength of certain muscles, the kinesiologist claims to be able to gain access to the body's communication system, and, thus, to read the health status of each of the body's major components.

The manual testing of muscles or muscle strength is not new, and was used in the late 1940s to evaluate muscle function and strength and to assess the extent of an injury. Applied kinesiology measures whether a muscle is stuck in the “on” position, acting like a tense muscle spasm, or is stuck “off,” appearing weak or flaccid. It is called manual testing because it is done without instruments, using only the kinesiologist's fingertip pressure. During the first and longest appointment, which lasts about an hour, the kinesiologist conducts a complete consultation, asking about the patient's history and background. During the **physical examination**, patients sit or lie down, then the kinesiologist holds the patient's leg or arm to isolate a particular muscle. The practitioner then touches a point on the body which he believes is related to that muscle, and, with quick, gentle, and painless pressure, pushes down on the limb. Patients are asked to resist this pressure, and, if they cannot, an imbalance is suspected in the related organ, gland, or body part. This diagnostic technique uses muscles to find the cause of a problem, and is based on **traditional Chinese medicine** and its idea that the body has common energy meridians, or channels, for both organs and muscles. Kinesiologists also claim that they are able to locate muscle weaknesses that stem from a variety of causes such as **allergies**, mineral and vitamin deficiencies, as well as from problems with the lymph system. Once the exact cause is determined, the

KEY TERMS

Acupressure—A form of acupuncture in which certain points of the body are pressed with the fingers and hands to release energy blocks.

Alleviate—To make something easier to be endured.

Complementary—Something that serves to fill out or complete something else.

Deficiency—A shortage of something necessary for health.

Diagnostic—The art or act of identifying a disease from its signs and symptoms.

Flaccid—Flabby, limp, weak.

Meridian—In traditional Chinese medicine, the channels which run beneath the skin through which the body's energy flows.

Spasm—An involuntary, sudden, violent contraction of a muscle or a group of muscles.

kinesiologist uses his fingertips to work the appropriate corresponding acupressure points in order to rebalance the flow of energy and restore health. Often he will recommend a complementary program of **nutrition** therapy.

Risks

There are no major risks associated with this gentle, noninvasive therapy. It is generally safe for people of all ages and has no side effects.

Normal results

If applied kinesiology does what it claims, patients should expect muscle testing to discover the cause of their physical complaint and to be told how to correct it.

Resources

ORGANIZATIONS

International College of Applied Kinesiology. 6405 Metcalf Ave., Suite 503, Shawnee Mission, KS 66202. (913) 384-5336. <<http://www.icakusa.com>> and <<http://www.icak.com>>.

Leonard C. Bruno, PhD

Kleine-Levin syndrome see **Sleep disorders**

Klinefelter syndrome

Definition

Klinefelter syndrome is a chromosomal disorder that affects only males. People with this condition are born with at least one extra X chromosome. The syndrome was first identified and described in 1942 by Harry Fitch Klinefelter, Jr., an American physician.

Description

Klinefelter syndrome is a condition in which one or more extra X chromosomes are present in a male. Boys with this condition appear normal at birth. They enter **puberty** normally, but by mid puberty have low levels of testosterone causing small testicles and the inability to make sperm. Affected males may also have learning disabilities and behavior problems such as **shyness** and immaturity, and an increased risk for certain other health problems.

Klinefelter syndrome is one of the most common chromosomal abnormalities. About 1 in every 500 to 800 males is born with this disorder; approximately 3000 affected boys are born each year in the United States. About 3% of the infertile male population have Klinefelter syndrome. The condition appears to affect all racial and ethnic groups equally.

Causes and symptoms

Chromosomes are found in the cells in the body. Chromosomes contain genes, structures that tell the body how to grow and develop. Chromosomes are responsible for passing on hereditary traits from parents to child. Chromosomes also determine whether the child will be male or female. Normally, a person has a total of 46 chromosomes in each cell, two of which are responsible for determining that individual's sex. These two sex chromosomes are called X and Y. The combination of these two types of chromosomes determines the sex of a child. Females have two X chromosomes (the XX combination); males have one X and one Y chromosome (the XY combination).

In Klinefelter syndrome, a problem very early in development results in an abnormal number of chromosomes. About 60% of embryos with Klinefelter syndrome do not survive the fetal period. Most commonly, a male with Klinefelter syndrome will be born with 47 chromosomes in each cell, rather than the normal number of 46. The extra chromosome is an X chromosome. This means that rather than having

the normal XY combination, the male has an XXY combination. Because people with Klinefelter syndrome have a Y chromosome, they are all male.

Approximately 1/3 of all males with Klinefelter syndrome have other chromosomal abnormalities involving an extra X chromosome. Mosaic Klinefelter syndrome occurs when some of the cells in the body have an extra X chromosome and the others have normal male chromosomes. These males can have the same or milder symptoms than non-mosaic Klinefelter syndrome. Males with more than one additional extra X chromosome, such as 48,XXXY, are usually more severely affected than males with 47,XXY.

Klinefelter syndrome is not considered an inherited condition. The risk of Klinefelter syndrome reoccurring in another **pregnancy** is not increased above the general population risk.

The symptoms of Klinefelter syndrome are variable and not every affected person will have all of the features of the condition. Males with Klinefelter syndrome appear normal at birth and have normal male genitalia. From childhood, males with Klinefelter syndrome are taller than average with long limbs. Approximately 20–50% have a mild intention tremor, an uncontrolled shaking. Many males with Klinefelter syndrome have poor upper body strength and can be clumsy. Klinefelter syndrome does not cause homosexuality. Approximately 1/3 of males with Klinefelter syndrome have **gynecomastia** or breast growth, some requiring **breast reduction** surgery.

Most boys enter puberty normally, though some can be delayed. The Leydig cells in the testicles usually produce testosterone. With Klinefelter syndrome, the Leydig cells fail to work properly causing the testosterone production to slow. By mid-puberty, testosterone production is decreased to approximately half of normal. This can lead to decreased facial and pubic hair growth. The decreased testosterone also causes an increase in two other hormones, follicle stimulating hormone (FSH) and luteinizing hormone (LH). Normally, FSH and LH help the immature sperm cells grow and develop. In Klinefelter syndrome, there are few or no sperm cells. The increased amount of FSH and LH causes hyalinization and fibrosis, the growth of excess fibrous tissue, in the seminiferous tubules, where the sperm are normally located. As a result, the testicles appear smaller and firmer than normal. With rare exception, men with Klinefelter syndrome are infertile because they can not make sperm.

While it was once believed that all boys with Klinefelter syndrome are mentally retarded, doctors

now know that the disorder can exist without retardation. However, children with Klinefelter syndrome frequently have difficulty with language, including learning to speak, read, and write. Approximately 50% of males with Klinefelter syndrome are dyslexic.

Some people with Klinefelter syndrome have difficulty with social skills and tend to be more shy, anxious, or immature than their peers. They can also have poor judgment and do not handle stressful situations well. As a result, they often do not feel comfortable in large social gatherings. Some people with Klinefelter syndrome can also have **anxiety**, nervousness and/or depression.

The greater the number of X chromosomes present, the greater the disability; each extra X chromosome lowers the child's IQ by about 15 points. Boys with several extra X-chromosomes have distinctive facial features, more severe retardation, deformities of bony structures, and even more disordered development of male features.

Diagnosis

Diagnosis of Klinefelter syndrome is made by examining chromosomes for evidence of more than one X chromosome present in a male. This can be done in pregnancy with prenatal testing such as a **chorionic villus sampling** or **amniocentesis**. Chorionic villus sampling is a procedure done early in pregnancy (approximately 10–12 weeks) to obtain a small sample of the placenta for testing. An amniocentesis is done further along in pregnancy (from approximately 16–18 weeks) to obtain a sample of fluid surrounding the baby for testing. Both procedures have a risk of **miscarriage**. Usually these procedures are done for a reason other than diagnosing Klinefelter syndrome. For example, a prenatal diagnostic procedure may be done on an older woman to determine if her baby has **Down syndrome**. If the diagnosis of Klinefelter syndrome is suspected in a young boy or adult male, chromosome testing can also be on a small blood or skin sample after birth.

Many men with Klinefelter syndrome go through life without being diagnosed. The two most common complaints leading to diagnosis of the condition are gynecomastia and **infertility**.

Treatment

There is no treatment available as of the early 2000s to change a person's chromosomal makeup. Children with Klinefelter syndrome may benefit from speech therapy for speech problems or other

KEY TERMS

Chromosome—A microscopic thread-like structure found within each cell of the body and consists of a complex of proteins and DNA. Humans have 46 chromosomes arranged into 23 pairs. Changes in either the total number of chromosomes or their shape and size (structure) may lead to physical or mental abnormalities.

Gonadotrophin—Hormones that stimulate the ovary and testicles.

Gynecomastia—Excessive growth of breast tissue in males.

Leydig cells—Cells that make up the endocrine tissue of the testis and produce testosterone. They are named for Franz von Leydig (1821–1908), the German professor of anatomy who first identified them.

Testosterone—Hormone produced in the testicles that is involved in male secondary sex characteristics.

educational interventions for learning disabilities. Testosterone injections started around the time of puberty may help to produce more normal development including more muscle mass, hair growth and increased sex drive. Testosterone supplementation will not increase testicular size, decrease breast growth or correct infertility. Psychiatric consultation may be helpful when the boy reaches adolescence.

Some doctors recommend **mastectomy** as a surgical treatment for gynecomastia, on the grounds that the enlarged breasts are often socially stressful for affected males and significantly increase their risk of **breast cancer**.

Prognosis

While many men with Klinefelter syndrome go on to live normal lives, nearly 100% of these men will be sterile (unable to produce a child). However, a few men with Klinefelter syndrome have been reported who have fathered a child through the use of assisted fertility services.

Males with Klinefelter syndrome have an increased risk of several systemic conditions, including epilepsy, **osteoporosis**, such **autoimmune disorders** as

lupus and arthritis, diabetes, and breast and germ cell tumors. One Danish study reported in 2004 that men with Klinefelter's syndrome have a slightly shortened life span, dying about 2.1 years earlier than men without the syndrome.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD., editors. "Chromosomal Abnormalities." Section 19, Chapter 261 In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

Beers, Mark H., MD, and Robert Berkow, MD., editors. "Infertility." Section 18, Chapter 245 In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

Probasco, Teri, and Gretchen A. Gibbs. *Klinefelter Syndrome*. Richmond, IN: Print Press, 1999.

PERIODICALS

Bojesen, A., S. Juul, N. Birkebaek, and C. H. Gravholt. "Increased Mortality in Klinefelter Syndrome." *Journal of Clinical Endocrinology and Metabolism* 89 (August 2004): 3830–3834.

Chen, Harold, MD. "Klinefelter Syndrome." *eMedicine* December 17, 2004. <<http://emedicine.com/ped/topic1252.htm>>.

Diamond, M., and L. A. Watson. "Androgen Insensitivity Syndrome and Klinefelter's Syndrome: Sex and Gender Considerations." *Child and Adolescent Psychiatric Clinics of North America* 13 (July 2004): 623–640.

Grosso, S., M. A. Farnetani, R. M. Di Bartolo, et al. "Electroencephalographic and Epileptic Patterns in X Chromosome Anomalies." *Journal of Clinical Neurophysiology* 21 (July-August 2004): 249–253.

Lanfranco, F., A. Kamischke, M. Zitzmann, and E. Nieschlag. "Klinefelter's Syndrome." *Lancet* 364 (July 17, 2004): 273–283.

Tyler, C., and J. C. Edman. "Down Syndrome, Turner Syndrome, and Klinefelter Syndrome: Primary Care throughout the Life Span." *Primary Care* 31 (September 2004): 627–648.

ORGANIZATIONS

Klinefelter's Organization. PO Box 60, Orpington, BR68ZQ. UK. <<http://hometown.aol.com/KSCUK/index.htm>>.

Klinefelter Syndrome and Associates, Inc. PO Box 119, Roseville, CA 95678-0119. (916) 773-2999 or (888) 999-9428. Fax: (916) 773-1449. ksinfo@genetic.org. <<http://www.genetic.org/ks>>.

National Organization for Rare Disorders (NORD). 55 Kenosia Avenue, P. O. Box 1968, Danbury, CT 06813-1968. (203) 744-0100. Fax: (203) 798-2291. <<http://www.rarediseases.org>>.

OTHER

Klinefelter Syndrome Support Group Home Page. <<http://klinefeltersyndrome.org/index.html>>.

Carin Lea Beltz, MS
Rebecca J. Frey, PhD

Knee injuries

Definition

The five most common knee problems are arthritis, tendonitis, **bruises**, cartilage tears, and damaged ligaments. Knee injuries can be caused by accidents, impact, sudden or awkward movements, and gradual wear and tear of the knee joint.

Description

Because the knee joint is both vulnerable and used extensively in many activities, it is prone to injuries. The American Academy of Orthopaedic Surgeons estimates that 19.5 million visits to doctors' offices in 2002 were for knee problems. In some sports including football, skiing, and gymnastics, and racket sports, injury rates to avid practitioners can near 50 percent, and knee injuries are the most common reason patients visit orthopedic doctors. An estimated one in five runners gets a knee injury. The majority of knee injuries, however, are minor and do not require intensive treatment.

The knee, the largest joint in the body, connects the thighbone (femur) to the lower leg (tibia). It is a complex and efficient joint consisting of ligaments, cartilage, and the bone of the kneecap (patella). All of these parts can be injured. Inside the knee joint is synovial fluid that protects and lubricates the parts, which during injuries may increase and cause swelling. The bursa are sacs in the knee that contain synovial fluid and provide cushioning and lubrication.

Four ligaments comprise the knee joint. The medial collateral ligament (MCL) runs along the inside of the knee, while the lateral collateral ligament (LCL) is on the outside of the knee. The cruciate ligaments cross inside the knee. The anterior cruciate ligament (ACL) is deep inside the knee and limits rotation of the joint. The posterior cruciate ligament (PCL) is also inside the knee and limits the backward movement of the joint. Ligaments in the knee can be partially or completely torn, depending on the extent of the injury.

The minisci cartilage are two thin, oval-shaped tissues that act as cushions between the ends of the leg bones. The medial miniscus is the cartilage closest to the other leg while the lateral miniscus is nearer the outside of the knee. Injuries to the minisci include tears from injuries and impact and degenerative wearing away of the structure. The minisci can be partially or completely torn during injury.

The bones around the knee, including the kneecap, can be broken, fractured, or chipped. The patellar tendon connects the kneecap to the shinbone, while the quadriceps tendon connects the quadriceps muscle to the patella. The patellar tendon can be torn or can develop injury and **pain** from degeneration. It can also be fully dislocated or partially dislocated (called subluxation). The tendons in the knee may develop pain and inflammation known as tendonitis.

The bones of the knee joint are covered with tissue known as articular cartilage. This cartilage can be injured or fractured, and can also develop a degenerative condition called chondromalacia. **Osteoarthritis** is the pain associated with the wearing down of this cartilage.

Causes & symptoms

Arthritis may develop from an auto-immune disorder, known as **rheumatoid arthritis**, or may be caused by the gradual wear and tear of the joint, known as osteoarthritis. Symptoms of arthritis in the knee include pain ranging from dull aches to severe pain, and may be accompanied by swelling and range of movement loss. Arthritic symptoms may tend to be worst in the morning and decrease throughout the day as the knee is used. Arthritis can be caused by lupus, **Lyme disease**, and other infections. Knee swelling also may be caused by **bursitis**, the inflammation of the bursa.

Cartilage injuries may include chondromalacia, with symptoms including dull pain and pain while climbing stairs. Damage to the minisci cartilage often occurs from sudden twists, forceful plants, and awkward movements. A torn cartilage may make a popping sound, and may be accompanied by mild to severe pain, particularly while straightening the leg. Swelling, stiffening, and loss of movement are also symptoms of cartilage tears, as are clicking sounds and friction in the knee during movement.

Ligament injuries may cause dull or severe pain, swelling, loss of the range of movement of the joint, and loss of the stability and strength of the knee. Ligament injuries typically occur from strong blows

and forces applied to the knee. Injuries to the MCL are the most common, often caused by impact to the side of the knee joint. Of the cruciate ligaments, the PCL is less commonly injured than the ACL. Typically, the PCL is injured by forceful blows to the knee, such as during car accidents, while the ACL can be injured by impacts and by sudden twists. Torn ligaments may be accompanied by a popping sound indicating the rupture, and may not always cause pain, so that some of them go unnoticed. Torn ligaments may weaken the knee and cause buckling or folding under weight.

Tendon injuries range from tendonitis to torn tendons. Symptoms of tendonitis include pain and swelling, while ruptured tendons can cause more intense pain, swelling, and loss of movement.

Osgood-Shlatter disease is a condition common in young boys who play running and jumping sports. Symptoms include inflammation of the patellar tendon and pain in the front of the knee during and after strenuous activity.

Iliotibial band syndrome is common in running and other repetitive sports, characterized by pain at the side of the knee caused by **stress** on the band of tendons there. Sometimes this condition causes a snapping sensation when the knee is straightened.

Diagnosis

Depending on the severity of the condition, family physicians or orthopedic physicians who specialize in the knee joint may be consulted. If arthritis is suspected, a rheumatologist may be consulted. The diagnosis process includes taking a complete patient history with details of the pain and the circumstances of the injury. The physician will give the patient a thorough **physical examination**, including utilizing several manual techniques of moving the knee joint and legs in various positions to help determine the type of injury. An experienced practitioner can often make an accurate diagnosis of injuries by performing a sequence of manual diagnostic tests.

Laboratory tests may be ordered to further or clarify the diagnosis. X rays can show damage to the bones as well as the narrowing of the knee space that may imply cartilage problems. For more in-depth diagnosis, a computerized axial tomographic (CAT) scan is an x-ray technique that can provide three-dimensional views of the bones in the knee. A **magnetic resonance imaging (MRI)** scan gives a computerized portrait of the interior of the knee, and may show damage to the ligaments and cartilage. **Arthroscopy** is a form of minor surgery that inserts a

tiny camera into the knee and gives a very accurate view of the joint. Radionuclide scanning (bone scans) use radioactive material injected into the bloodstream to monitor the blood flow in particular areas. If infection or rheumatoid arthritis is suspected, a physician may order blood tests for diagnosis. Biopsies, in which pieces of tissue are laboratory tested, may also be used for diagnosis.

Treatment

When a person suspects a knee injury, the first treatment recommended is R.I.C.E., which stands for rest, ice, compression, and elevation. First, the person should cease the activity which caused the injury and immediately rest and immobilize the joint. Ice may be applied to reduce pain and swelling, and compression, such as wraps and braces, may be used to immobilize the knee. Elevating the leg is also helpful in reducing swelling and aiding circulation. Immediate care will prevent the worsening of the injury.

Treatment options for knee injuries can range from rest and light activity, to physical therapy, to surgery. Most knee injuries are treated with proper rest, **exercise**, and strengthening programs recommended by physicians. For injuries that require surgery or deeper diagnosis, arthroscopy is the least invasive technique and has a quick recovery time associated with the procedure. Arthroscopy is commonly used to repair cartilage and partially torn ligaments. For severe knee injuries, **reconstructive surgery** or open knee surgery may be required. Full knee replacements may also be performed for severely damaged knees. After surgery, physical therapy programs for **rehabilitation** are recommended. Treatment for osteoarthritis includes over-the-counter painkillers, exercise, and weight reduction. For rheumatoid arthritis, more powerful prescription medications, such as steroids and stronger painkillers, and intensive physical therapy may be ordered. Knee injuries associated with infection may require **antibiotics**.

Alternative treatment

Alternative therapies for knee injuries focus on supporting the body's ability to heal itself. Various therapies may include bodywork and postural adjustments such as **chiropractic** and **Rolfing** work, in addition to physical therapy and gentle exercise routines. Herbal remedies and **nutritional supplements** may be used to aid the healing process and reduce symptoms. **Acupuncture** may be used for pain relief, and **yoga** is a low-impact exercise routine that increases flexibility, good alignment, and strength.

KEY TERMS

Lupus—Chronic inflammatory disease caused by immune system disorder.

Lyme disease—Bacteria infection spread by ticks.

Orthopedist—Physician specializing in the diagnosis and treatment of problems involving the skeletal system.

Rheumatologist—Physician specializing in the diagnosis and treatment of arthritis and related conditions.

Prevention

The best prevention for knee injuries is being aware of activities that carry high risks for knee injuries and acting carefully. The knees can be strengthened by evenly building the muscles in the quadriceps and hamstrings. Increasing flexibility in the body through stretching can also help reduce injuries. Properly fitting shoes and other sports equipment are essential for preventing injury as well. Finally, before engaging in activities that stress the knee, a thorough and gradual warm-up routine, including aerobic activity and stretching, will lessen the chances of knee injury.

Resources

BOOKS

Grelsamer, Ronald, M.D. *What Your Doctor May Not Tell You about Knee Pain and Surgery*. Warner Books, 2002.

Halpern, Brian. *The Knee Crisis Handbook*. Rodale Books, 2003.

Scott, W. Norman. *Dr. Scott's Knee Book*. Fireside Books, 1996.

ORGANIZATIONS

American Academy of Orthopaedic Surgeons. P.O. Box 2058, Des Plaines, IL 60017. 800-824-BONE. < <http://www.aaos.org> >.

National Athletic Trainers' Association. 2952 Stemmons Freeway, Dallas, TX 75247-6196. (214) 637-6282. < <http://www.nata.org> >.

American Physical Therapy Association. 1111 North Fairfax St., Alexandria, VA 22314-1488. (703) 684-2782. < <http://www.apta.org> >.

Douglas Dupler

Knee replacement see **Joint replacement**

Kneecap removal

Definition

Kneecap removal, or patellectomy, is the surgical removal of the patella, commonly called the kneecap.

Purpose

Kneecap removal is done under three circumstances:

- When the kneecap is fractured or shattered
- When the kneecap dislocates easily and repeatedly
- When degenerative arthritis of the kneecap causes extreme **pain**

A person of any age can break a kneecap in an accident. When the bone is shattered beyond repair, the kneecap is removed. No prosthesis or artificial replacement part is put in its place.

Dislocation of the kneecap is most common in young girls between the ages of 10-14. Initially, the kneecap will pop back into place of its own accord, but pain may continue. If dislocation occurs too often, or the kneecap doesn't go back into place correctly, the patella may rub the other bones in the knee, causing an arthritis-like condition. Some people are born with **birth defects** that cause the kneecap to dislocate frequently.

Degenerative arthritis of the kneecap, also called patellar arthritis or *chondromalacia patellae*, can cause enough pain that it is necessary to remove the kneecap. As techniques of **joint replacement** have improved, arthritis in the knee is more frequently treated with total knee replacement.

Precautions

People who have had their kneecap removed for degenerative arthritis and then later have to have a total knee replacement are more likely to have problems with the stability of their artificial knee than those who only have total knee replacement. This is because the realigned muscles and tendons provide less support once the kneecap is removed.

Description

Kneecap removal is performed under either general or **local anesthesia** at a hospital or freestanding surgical center, by an orthopedic surgeon. The surgeon makes an incision around the kneecap. Then, the muscles and tendons attached to the kneecap are

KEY TERMS

Degenerative arthritis, or osteoarthritis—A noninflammatory type of arthritis, usually occurring in older people, characterized by degeneration of cartilage, enlargement of the margins of the bones, and changes in the membranes in the joints.

cut and the kneecap is removed. Next, the muscles are sewed back together, and the skin is closed with sutures or clips that stay in place about one week. Any hospital stay is generally brief.

Preparation

Prior to surgery, x rays and other diagnostic tests are done on the knee to determine if removing the kneecap is the appropriate treatment. Pre-operative blood and urine tests are also done.

Aftercare

Pain relievers may be prescribed for a few days. The patient will initially need to use a cane, or crutches, to walk. Physical therapy exercises to strengthen the knee should be begun immediately. Driving should be avoided for several weeks. Full recovery can take months.

Risks

Risks involved with kneecap removal are similar to those that occur in any surgical procedure, mainly allergic reaction to anesthesia, excessive bleeding, and infection.

Normal results

People who have kneecap removal because of a broken bone or repeated **dislocations** have the best chance for complete recovery. Those who have this operation because of arthritis may have less successful results, and later need a total knee replacement.

Resources

BOOKS

“Kneecap Removal.” In *The Complete Guide to Symptoms, Illness and Surgery*, edited by H. Winter Griffith, et al, 3rd ed. New York: Berkeley Publishing, 1995.

Tish Davidson, A.M.

KOH test

Definition

The KOH test takes its name from the chemical formula for potassium hydroxide (KOH), which is the substance used in the test. The test, which is also called a potassium hydroxide preparation, is done to rapidly diagnose fungal infections of the hair, skin, or nails. A sample of the infected area is analyzed under a microscope following the addition of a few drops of potassium hydroxide.

Purpose

The primary purpose of the KOH test is the differential diagnosis of infections produced by dermatophytes and *Candida albicans* from other skin disorders. Dermatophytes are a type of fungus that invade the top layer of the skin, hair, or nails, and produce an infection commonly known as **ringworm**, technically known as *tinea*. It can appear as “jock itch” in the groin or inner thighs (*tinea cruris*); on the feet (*tinea pedis*); on the scalp and hair (*tinea capitis*); and on the nails (*tinea unguium*). *Tinea versicolor* appears anywhere on the skin and produces characteristic unpigmented patches. *Tinea unguium* affects the nails.

Similar symptoms of redness, scaling, and **itching** can be caused by other conditions, such as eczema and **psoriasis**. The KOH test is a quick, inexpensive test—often done in a physician’s office—to see if these symptoms are caused by a dermatophyte. If a dermatophyte is found, treatment is started immediately; further tests are seldom necessary.

A yeast (candidal) infection of the skin or a mucous membrane, such as the mouth, often produces a white cheesy material at the infection site. This type of infection, known as thrush, is also identified with the KOH test.

Description

The KOH test involves the preparation of a slide for viewing under the laboratory microscope. KOH mixed with a blue-black dye is added to a sample from the infected tissues. This mixture makes it easier to see the dermatophytes or yeast under the microscope. The KOH dissolves skin cells, hair, and debris; the dye adds color. The slide is gently heated to speed up the action of the KOH. Finally the slide is examined under a microscope.

Dermatophytes are easily recognized under the microscope by their long branch-like structures.

KEY TERMS

Dermatophyte—A type of fungus that causes diseases of the skin, including tinea or ringworm.

KOH—The chemical formula for potassium hydroxide, which is used to perform the KOH test. The tests is also called a potassium hydroxide preparation.

Thrush—A disease of the mouth, caused by *Candida albicans* and characterized by a whitish growth and ulcers. It can be diagnosed with the KOH test.

Tinea—A superficial infection of the skin, hair, or nails, caused by a fungus and commonly known as ringworm.

Yeast cells look round or oval. The dermatophyte that causes tinea versicolor has a characteristic spaghetti-and-meatballs appearance.

If the KOH test is done in the doctor's office, the results are usually available while the person waits. If the test is sent to a laboratory, the results will be ready the same or following day. The KOH test is covered by insurance when medically necessary.

Preparation

The physician selects an infected area from which to collect the sample. Scales and cells from the area are scraped using a scalpel. If the test is to be analyzed immediately, the scrapings are placed directly onto a microscope slide. If the test will be sent to a laboratory, the scrapings are placed in a sterile covered container.

Normal results

A normal, or negative, KOH test shows no fungi (no dermatophytes or yeast).

Abnormal results

Dermatophytes or yeast seen on a KOH test indicate the person has a fungal infection. Follow-up tests are usually unnecessary.

Resources

PERIODICALS

Crissey, John Thorne. "Common Dermatophyte Infections. A Simple Diagnostic Test and Current Management."

Postgraduate Medicine February 1998: 191–192, 197–198, 200, 205.

Nancy J. Nordenson

Korsakoff's psychosis see **Korsakoff's syndrome**

Korsakoff's syndrome

Definition

Korsakoff's syndrome is a memory disorder which is caused by a deficiency of vitamin B₁, also called thiamine.

Description

In the United States, the most common cause of thiamine deficiency is **alcoholism**. Other conditions which cause thiamine deficiency occur quite rarely, but can be seen in patients undergoing dialysis (a procedure used primarily for patients suffering from kidney failure, during which the patient's blood circulates outside of the body, is mechanically cleansed, and then is circulated back into the body), pregnant women with a condition called **hyperemesis gravidarum** (a condition of extreme morning sickness, during which the woman vomits up nearly all fluid and food intake), and patients after surgery who are given vitamin-free fluids for a prolonged period of time. Thiamine deficiency is an important cause of disability in developing countries where the main source of food is polished rice (rice with the more nutritious outer husk removed).

An associated disorder, Wernicke's syndrome, often precedes Korsakoff's syndrome. In fact, they so often occur together that the spectrum of symptoms produced during the course of the two diseases is frequently referred to as Wernicke-Korsakoff syndrome. The main symptoms of Wernicke's syndrome include ataxia (difficulty in walking and maintaining balance), **paralysis** of some of the muscles responsible for movement of the eyes, and confusion. Untreated Wernicke's will lead to **coma** and then **death**.

Causes and symptoms

One of the main reasons that alcoholism leads to thiamine deficiency has to do with the high-calorie nature of alcohol. A person with a large alcohol intake often, in essence, substitutes alcohol for other, more

nutritive calorie sources. Food intake drops off considerably, and multiple vitamin deficiencies develop. Furthermore, it is believed that alcohol increases the body's requirements for **B vitamins**, at the same time interfering with the absorption of thiamine from the intestine and impairing the body's ability to store and use thiamine. Direct neurotoxic (poisonous damage to the nerves) effects of alcohol may also play some role.

Thiamine is involved in a variety of reactions which provide energy to the neurons (nerve cells) of the brain. When thiamine is unavailable, these reactions cannot be carried out, and the important end-products of the reactions are not produced. Furthermore, certain other substances begin to accumulate, and are thought to cause damage to the vulnerable neurons. The area of the brain believed to be responsible for the symptoms of Korsakoff's syndrome is called the diencephalon, specifically the structures called the mamillary bodies and the thalamus.

An individual with Korsakoff's syndrome displays much difficulty with memory. The main area of memory affected is the ability to learn new information. Usually, intelligence and memory for past events is relatively unaffected, so that an individual may remember what occurred 20 years previously, but is unable to remember what occurred 20 minutes ago. This memory defect is referred to as anterograde **amnesia**, and leads to a peculiar symptom called "confabulation," in which a person suffering from Korsakoff's fills in the gaps in his or her memory with fabricated or imagined information. For instance, a person may insist that a doctor to whom he or she has just been introduced is actually an old high school classmate, and may have a lengthy story to back this up. When asked, as part of a memory test, to remember the name of three objects which the examiner listed ten minutes earlier, a person with Korsakoff's may list three entirely different objects and be completely convincing in his or her certainty. In fact, one of the hallmarks of Korsakoff's is the person's complete unawareness of the memory defect, and complete lack of worry or concern when it is pointed out.

Diagnosis

Whenever someone has a possible diagnosis of alcoholism, and then has the sudden onset of memory difficulties, it is important to seriously consider the diagnosis of Korsakoff's syndrome. While there is no specific laboratory test to diagnose Korsakoff's syndrome in a patient, a careful exam of the individual's mental state should be rather revealing. Although the patient's ability to confabulate answers may be convincing, checking the patient's retention of factual

KEY TERMS

Amnesia—Inability to remember events or experiences. Memory loss. Includes: 1) Anterograde amnesia: inability to retain the memory of events occurring after the time of the injury or disease which brought about the amnesic state. 2) Retrograde amnesia: inability to recall the memory of events which occurred prior to the time of the injury or disease which brought about the amnesic state.

Confabulation—An attempt to fill in memory gaps by fabricating information or details.

Diencephalon—A part of the brain that binds the mesencephalon to the cerebral hemispheres. Considered by some as part of the brain stem.

information (asking, for example, for the name of the current president of the United States), along with the patient's ability to learn new information (repeating a series of numbers, or recalling the names of three objects ten minutes after having been asked to memorize them) should point to the diagnosis. Certainly a patient known to have just begun recovery from Wernicke's syndrome, who then begins displaying memory difficulties, would be very likely to have developed Korsakoff's syndrome. A **physical examination** may also show signs of Wernicke's syndrome, such as **peripheral neuropathy**.

Treatment

Treatment of both Korsakoff's and Wernicke's syndromes involves the immediate administration of thiamine. In fact, any individual who is hospitalized for any reason and who is suspected of being an alcoholic, should receive thiamine. The combined Wernicke-Korsakoff syndrome has actually been precipitated in alcoholic patients hospitalized for other medical illnesses, due to the administration of thiamine-free intravenous fluids (intravenous fluids are those fluids containing vital sugars and salts which are given to the patient through a needle inserted in a vein). Also, the vitamin therapy may be impaired by the feeding of carbohydrates prior to the giving of thiamine; since carbohydrates cannot be metabolized with thiamine.

Prognosis

Fifteen to twenty percent of all patients hospitalized for Wernicke's syndrome will die of the disorder.

Although the degree of ataxia nearly always improves with treatment, half of those who survive will continue to have some permanent difficulty walking. The paralysis of the eye muscles almost always resolves completely with thiamine treatment. Recovery from Wernicke's begins to occur rapidly after thiamine is given. Improvement in the symptoms of Korsakoff's syndrome, however, can take months and months of thiamine replacement. Furthermore, patients who develop Korsakoff's syndrome are almost universally memory-impaired for the rest of their lives. Even with thiamine treatment, the memory deficits tend to be irreversible, with less than 20% of patients even approaching recovery. The development of Korsakoff's syndrome often results in an individual requiring a supervised living situation.

Prevention

Prevention depends on either maintaining a diet with a sufficient intake of thiamine, or supplementing an inadequate diet with vitamin preparations. Certainly, one of the most important forms of prevention involves treating the underlying alcohol **addiction**.

Resources

ORGANIZATIONS

National Institute on Alcoholism Abuse and Alcoholism.
6000 Executive Boulevard, Willco Building, Bethesda,
Maryland 20892-7003. < <http://www.niaaa.nih.gov> >.

Rosalyn Carson-DeWitt, MD

KUB see **Kidney, ureter, and bladder x-ray study**

Kuru see **Creutzfeldt-Jakob disease**

Kwashiorkor see **Protein-energy malnutrition**

Kyphosis

Definition

Kyphosis is the extreme curvature of the upper back also known as a hunchback.

Description

The upper back bone (thoracic region), is normally curved forward. If the curve exceeds 50° it is considered abnormal (kyphotic).

Causes and symptoms

Kyphosis can be divided into three ages of acquisition—birth, old age, and the time in between.

- Spinal **birth defects** can result in a fixed, exaggerated curve. Vertebrae can be fused together, shaped wrong, extraneous, or partially missing. Congenital and hereditary defects in bone growth weaken bone and result in exaggerated curves wherever gravity or muscles pull on them. Dwarfism is such a defect.
- During life, several events can distort the spine. Because the natural tendency of the thoracic spine is to curve forward, any weakness of the supporting structures will tend in that direction. A diseased thoracic vertebra (a spine bone) will ordinarily crumble its forward edge first, increasing the kyphotic curve. Conditions that can do this include **cancer**, **tuberculosis**, Scheuermann's disease, and certain kinds of arthritis. Healthy vertebra will fracture forward with rapid deceleration injuries, such as in car crashes when the victim is not wearing a seat belt.
- Later in life, kyphosis is caused from **osteoporosis**, bone weakness, and crumbling forward.

The **stress** caused by kyphosis produces such symptoms as an increase in musculoskeletal pains, tension headaches, back aches, and joint pains.

Diagnosis

A quick look at the back will usually identify kyphosis. X rays of the spine will confirm the diagnosis and identify its cause.

Treatment

Congenital defects have to be repaired surgically. The procedures are delicate, complicated, and lengthy. Often orthopedic hardware must be placed to stabilize the back bone. At other times, a device called a Milwaukee brace can hold the back in place from the outside. Fitting Milwaukee braces comfortably is difficult because they tend to rub and cause sores.

Kyphosis acquired during the younger years requires treatment directed at the cause, such as medications for tuberculosis. Surgical reconstruction or bracing may also be necessary.

Kyphosis induced by osteoporosis is generally not treated except to prevent further bone softening.



This patient's spine shows excessive backward curvature at the level of the upper chest. (Custom Medical Stock Photo. Reproduced by permission.)

Prognosis

Congenital kyphosis may be alleviated to some extent by surgery and bracing. Kyphosis occurring later in life may worsen over time.

KEY TERMS

Congenital—Present at birth.

Dwarfism—A congenital disease of bone growth that results in short stature and weak bones.

Orthopedic—Refers to surgery on the supporting structures of the body—bones, joints, ligaments, muscles.

Osteoporosis—A weakening of bones due to calcium loss that affects post-menopausal women.

Scheuermann's disease—Juvenile kyphosis due to damaged bone in the spinal vertebrae.

Prevention

Preventing osteoporosis is within the grasp of modern medicine. Menopausal women must start early with estrogen replacement, calcium supplementation, and appropriate **exercise**. The treatment must continue through the remainder of life. Evidence suggests that a high calcium intake even during younger years delays the onset of symptomatic osteoporosis. Dairy products are the major dietary sources of calcium.

Resources

ORGANIZATIONS

Arthritis Foundation. 1300 W. Peachtree St., Atlanta, GA 30309. (800) 283-7800. <<http://www.arthritis.org>>.

National Osteoporosis Foundation. 1150 17th St., Suite 500 NW, Washington, DC 20036-4603. (800) 223-9994. <<http://www.nof.org>>.

Osteoporosis and Related Bone Diseases-National Resource Center. 1150 17th S. NW, Ste. 500, Washington, DC 20036. (800) 624-2663.

J. Ricker Polsdorfer, MD

L

Labor and delivery see **Childbirth**

Labor induction see **Induction of labor**

Labyrinthitis

Definition

Labyrinthitis is an inflammation of the inner ear that is often a complication of **otitis media**. It is caused by the spread of bacterial or viral infections from the head or respiratory tract into the inner ear.

Description

Labyrinthitis is characterized by **dizziness** or feelings of **motion sickness** caused by disturbance of the sense of balance.

Causes and symptoms

Causes

The disease agents that cause labyrinthitis may reach the inner ear by one of three routes:

- Bacteria may be carried from the middle ear or the membranes that cover the brain.
- The viruses that cause **mumps**, **measles**, **influenza**, and colds may reach the inner ear following an upper respiratory infection.
- The **rubella** virus can cause labyrinthitis in infants prior to birth.

Labyrinthitis can also be caused by toxic drugs.

Symptoms

The primary symptoms of labyrinthitis are vertigo (dizziness), accompanied by **hearing loss** and a

sensation of ringing in the ears called **tinnitus**. Vertigo occurs because the inner ear controls the sense of balance as well as hearing. Some patients also experience **nausea and vomiting** and spontaneous eye movements in the direction of the unaffected ear. Bacterial labyrinthitis may produce a discharge from the infected ear.

Diagnosis

The diagnosis of labyrinthitis is based on a combination of the patient's symptoms and history—especially a history of a recent upper respiratory infection. The doctor will test the patient's hearing, and order a laboratory culture to identify the organism if the patient has a discharge.

If there is no history of a recent infection, the doctor will order extra tests in order to exclude injuries to the brain or Meniere's disease.

Treatment

Medication

Patients with labyrinthitis are given **antibiotics**, either by mouth or intravenously to clear up the infection. They may also be given meclizine (Antivert, Bonine) for vertigo and **nausea**.

Surgery

Some patients require surgery to drain the inner and middle ear.

Supportive care

Patients with labyrinthitis should rest in bed for three to five days until the acute dizziness subsides. Patients who are dehydrated by repeated **vomiting** may need intravenous fluid replacement. In addition, patients are advised to avoid driving or similar

KEY TERMS

Labyrinth—The bony cavity of the inner ear.

Meniere's syndrome—A disease of the inner ear marked by recurrent episodes of vertigo and roaring in the ears lasting several hours. Its cause is unknown.

Otitis media—Inflammation of the middle ear. It can lead to labyrinthitis.

Vertigo—A sensation of dizziness marked by the feeling that one's self or surroundings are spinning or whirling.

activities for four to six weeks after the acute symptoms subside, because they may have occasional dizzy spells during that period.

Prognosis

Most patients with labyrinthitis recover completely, although it often takes five to six weeks for the vertigo to disappear completely and the patient's hearing to return to normal. In a few cases the hearing loss is permanent.

Prevention

The most effective preventive strategy includes prompt treatment of middle ear infections, as well as monitoring of patients with mumps, measles, influenza, or colds for signs of dizziness or hearing problems.

Resources

BOOKS

Jackler, Robert K., and Michael J. Kaplan. "Ear, Nose, & Throat." In *Current Medical Diagnosis and Treatment*, 1998 edited by Stephen McPhee, et al., 37th ed. Stamford: Appleton & Lange, 1997.

Rebecca J. Frey, PhD

Laceration repair

Definition

A laceration is a wound caused by a sharp object producing edges that may be jagged, dirty, or bleeding. Lacerations most often affect the skin, but any



Eleven sutures are necessary to close up the laceration on this person's forehead. (Custom Medical Stock Photo. Reproduced by permission.)

tissue may be lacerated, including subcutaneous fat, tendon, muscle, or bone.

Purpose

A laceration should be repaired if it:

- Continues to bleed after application of pressure for ten to fifteen minutes
- Is more than one-eighth to one-fourth inch deep
- Exposes fat, muscle, tendon, or bone
- Causes a change in function surrounding the area of the laceration
- Is dirty or has visible debris in it
- Is located in an area where an unsightly scar is undesirable.

KEY TERMS

Debridement—The act of removing any foreign material and damaged or contaminated tissue from a wound to expose surrounding healthy tissue.

Precautions

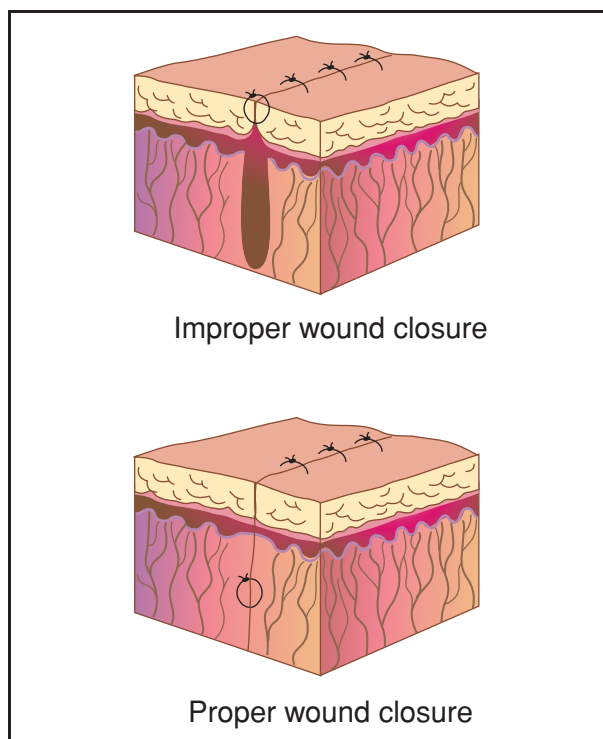
Lacerations are less likely to become infected if they are repaired soon after they occur. Many physicians will not repair a laceration that is more than eight hours old because the risk of infection is too great.

Description

Laceration repair mends a tear in the skin or other tissue. The procedure is similar to repairing a tear in clothing. Primary care physicians, emergency room physicians, and surgeons usually repair lacerations. The four goals of laceration repair are to stop bleeding, prevent infection, preserve function, and restore appearance. Insurance companies do pay for the procedure. Cost depends upon the severity and size of the laceration.

Before repairing the laceration, the physician thoroughly examines the wound and the underlying tendons or nerves. If nerves or tendons have been injured, a surgeon may be needed to complete the repair. The laceration is cleaned by removing any foreign material or debris. Removing **foreign objects** from penetrating **wounds** can sometimes cause bleeding, so this type of wound must be cleaned very carefully. The wound is then irrigated with saline solution and a disinfectant. The disinfecting agent may be mild soap or a commercial preparation. An antibacterial agent may be applied.

Once the wound has been cleansed, the physician anesthetizes the area of the repair by injecting a local anesthetic. The physician may trim edges that are jagged or extremely uneven. Tissue that is too damaged to heal must be removed (**debridement**) to prevent infection. If the laceration is deep, several absorbable stitches (sutures) are placed in the tissue under the skin to help bring the tissue layers together. Suturing also helps eliminate any pockets where tissue fluid or blood can accumulate. The skin wound is closed with sutures. Suture material used on the surface of a wound is usually non-absorbable and will have to be removed later. A light dressing or an adhesive bandage is applied for 24-48 hours. In areas where a dressing is not feasible, an antibiotic ointment can be applied. If the laceration is the result of a human or



A laceration is a traumatic break in the skin caused by a sharp object producing edges that may be jagged, dirty, or bleeding. The underlying tissue may also be severed. In such instances, the physician may place absorbable sutures in the tissue to help bring the edges together before the skin is sutured close. (Illustration by Electronic Illustrators Group.)

animal bite, if it is very dirty, or if the patient has a medical condition that alters wound healing, oral **anti-biotics** may be prescribed.

Aftercare

The laceration is kept clean and dry for at least 24 hours after the repair. Light bathing is generally permitted after 24 hours if the wound is not soaked. The physician will provide directions for any special wound care. Sutures are removed 3-14 days after the repair is completed. Timing of suture removal depends on the location of the laceration and physician preference.

The repair should be observed frequently for signs of infection, which include redness, swelling, tenderness, drainage from the wound, red streaks in the skin surrounding the repair, chills, or **fever**. If any of these occur, the physician should be contacted immediately.

Risks

The most common complication of any laceration repair is infection. Risk of infection can be minimized

by cleansing the wound thoroughly. Wounds from bites or dirty objects or wounds that have a large amount of dirt in them are most likely to become infected.

All lacerations will heal with a scar. Wounds that are repaired with sutures are less likely to develop **scars** that are unsightly, but no one can predict how wounds will heal and who will develop unsightly scars. **Plastic surgery** can improve the appearance of many scars.

Resources

OTHER

"Caring for Cuts and Scrapes at Home." *Mayo Clinic Online*. <http://www.mayohealth.org/mayo/9611/htm/cuts_sb.htm>.

"Laceration Repair." *ThriveOnline*. <<http://thriveonline.oxygen.com>>.

Mary Jeanne Krob, MD, FACS

Lacerations *see* **Wounds**

Lacrimal duct obstruction

Definition

A lacrimal duct obstruction is blockage of the tear duct, the thin channel that normally drains tears from the surface of the eye.

Description

The lacrimal glands, located above each eyeball, produce tears. The tears flow over the eye, then drain through the nasolacrimal ducts. A tiny hole at the inner edge of each eyelid marks the opening of the ducts, which lead to the lacrimal sacs located on the side of the nose. The tears pass from the sacs into the nasolacrimal ducts and then into the nose.

When a tear duct becomes obstructed, tears may spill over the eyelids and run down the face. Stagnant tears within the system can become infected, leading to recurrent red eyes and infections. Excessive tearing can also produce secondary skin changes on the lower eyelids.

Causes and symptoms

An obstructed lacrimal tear duct can result in inflammation and infection of the lacrimal sac. The area beneath the eyes next to the nose can become red, inflamed, and sensitive to the touch. The area usually

is swollen, and there may be a mucous discharge from the opening of the nasal corner of the eye. Common complaints include **itching**, irritation, burning, redness, foreign body sensation, and tearing.

Children frequently have a congenital lacrimal duct obstruction. Six to ten percent of all children are born before their tear ducts are open.

In adults, a common cause of lacrimal duct obstruction is involution, which is progressive degeneration occurring naturally with advancing age, resulting in shrivelling of organs or tissues. Other causes include **eyelid disorders**, infections by bacteria, viruses, fungi, and parasites, inflammations, the use of eye drops or excessive nasal spray, systemic **chemotherapy**, trauma from previous surgeries, injury to the bone at the side of the nose, foreign bodies, sinus disease, **nasal polyps**, and malignant or benign tumors.

Diagnosis

If the primary symptom is excessive tearing, the first step is for the health care professional to determine if the overflow of tears is due to an increase in tear production or a decrease in tear drainage. Causes of increased tear production may include trichiasis, a disease in which the eyelashes produce constant irritation, and eyelid malpositions and diseases. If abnormal tear production is ruled out, then obstructions in tear drainage is the most likely cause of the excessive tearing. Additional observations of swollen lacrimal sac area and purulent eye discharge indicate that there may be a lacrimal duct infection present. To further define the diagnosis, the lacrimal discharge may be cultured to determine possible infective agents, while various imaging techniques may be used to detect the type of obstruction. Dye tracer tests are also used to test for blockages.

Treatment

Lacrimal duct obstructions in children often resolve spontaneously, with 95% showing resolution before the child is one year old. Daily massaging of the lacrimal sac may help open the blockage. A topical antibiotic ointment may be applied if infection is present. If the blockage is not resolved after several weeks to months of this therapy, a physician may attempt forceful irrigation. Surgical probing to open up the duct under **general anesthesia** is a last resort, after a year or so of less invasive treatments.

In adults, conservative treatments are usually recommended. The infected or inflamed area may be massaged, with warm compresses applied to provide relief and speed the healing process. The health care

KEY TERMS

Lacrimal duct—A short canal leading from a small orifice at the medial angle of each eyelid to the lacrimal sac.

Lacrimal gland—An almond-shaped gland that secretes tears.

Lacrimal sac—The dilated upper end of the nasolacrimal duct in which the lacrimal ducts empty.

Nasolacrimal duct—A channel that transmits tears from the lacrimal sac to the nose.

Purulent—Consisting of or containing pus

Tear—A drop of the clear, salty fluid secreted by the lacrimal gland.

Trichiasis—A disease of the eye, in which the eyelashes, being turned in upon the eyeball, produce constant irritation by the motion of the lids.

provider may also massage or irrigate the infected area. Topical antibiotic ointments and oral **antibiotics** may also be used reduce infection. The use of **analgesics** such as **aspirin** may be recommended to control discomfort and reduce swelling. Severe cases may require surgical intervention to prevent future recurrences. Surgical approaches include insertion of a probe or catheter to remove an obstruction or creation of an artificial duct to bypass the obstruction.

Prognosis

If more conservative approaches fail to clear the obstruction, surgical procedures are available, with success rates greater than 90%.

Prevention

In many cases, the cause of a lacrimal duct obstruction is not known. However, in some cases, lacrimal duct obstruction may be caused by **smoking** and **abuse** of nasal sprays.

Resources

PERIODICALS

Camara, Jorge G., and Alfonso U. Bengzon. "Nasolacrimal Duct Obstruction." *eMedicine Journal* August 24, 2000. < <http://www.emedicine.com/OPH/topic465.htm> >.

Judith Sims

Lacrimal sac infection see **Dacryocystitis**

Lactate dehydrogenase isoenzymes test

Definition

The enzyme lactate dehydrogenase (also known as lactic dehydrogenase, or LDH) is found in the cells of almost all body tissues. The enzyme is especially concentrated in the heart, liver, red blood cells, kidneys, muscles, brain, and lungs. The total LDH can be further separated into five components or fractions labeled by number: LDH-1, LDH-2, LDH-3, LDH-4, and LDH-5. Each of these fractions, called isoenzymes, is used mainly by a different set of cells or tissues in the body. For this reason, the relative amounts of a particular isoenzyme of LDH in the blood can provide valuable diagnostic information.

Purpose

The LDH isoenzymes test assists in differentiating **heart attack**, anemia, lung injury, or **liver disease** from other conditions that may cause the same symptoms (differential diagnosis).

Precautions

Strenuous **exercise** may raise levels of total LDH, specifically the isoenzymes LDH-1, LDH-2, and LDH-5. Alcohol, anesthetics, **aspirin**, **narcotics**, procainamide, fluorides, and mithramycin may also raise levels of LDH. Ascorbic acid (vitamin C) can lower levels of LDH.

Description

LDH is found in the cells of almost all body tissues. When certain conditions injure cells in tissues containing LDH, it is released into the bloodstream. Because LDH is so widely distributed throughout the body, analysis of total LDH will not help make a diagnosis of a particular disease. Because this enzyme is actually composed of five different isoenzymes, however, analysis of the different LDH isoenzyme levels in the blood can help in the diagnosis of some diseases.

The five LDH isoenzymes are: LDH-1, LDH-2, LDH-3, LDH-4, and LDH-5. In general, each isoenzyme is used mostly by the cells in a specific tissue. LDH-1 is found mainly in the heart. LDH-2 is primarily associated with the system in the body that defends against infection (reticuloendothelial system). LDH-3 is found in the lungs and other

tissues, LDH-4 in the kidney, placenta, and pancreas, and LDH-5 in liver and striated (skeletal) muscle. Normally, levels of LDH-2 are higher than those of the other isoenzymes.

Certain diseases have classic patterns of elevated LDH isoenzyme levels. For example, an LDH-1 level higher than that of LDH-2 is indicative of a heart attack or injury; elevations of LDH-2 and LDH-3 indicate lung injury or disease; elevations of LDH-4 and LDH-5 indicate liver or muscle disease or both. A rise of all LDH isoenzymes at the same time is diagnostic of injury to multiple organs. For example, a heart attack with congestive **heart failure** may cause symptoms of lung and liver congestion. Advanced **cancer** and autoimmune diseases such as lupus can also cause this pattern.

One of the most important diagnostic uses for the LDH isoenzymes test is in the differential diagnosis of myocardial infarction or heart attack. The total LDH level rises within 24–48 hours after a heart attack, peaks in two to three days, and returns to normal in approximately five to ten days. This pattern is a useful tool for a delayed diagnosis of heart attack. The LDH-1 isoenzyme level, however, is more sensitive and specific than the total LDH. Normally, the level of LDH-2 is higher than the level of LDH-1. An LDH-1 level higher than that of LDH-2, a phenomenon known as “flipped LDH,” is strongly indicative of a heart attack. The flipped LDH usually appears within 12–24 hours after a heart attack. In about 80% of cases, flipped LDH is present within 48 hours of the incident. A normal LDH-1/LDH-2 ratio is considered reliable evidence that a heart attack has not occurred.

It should be noted that two conditions might cause elevated LDH isoenzymes at the same time and that one may confuse the other. For example, a patient with **pneumonia** may also be having an acute heart attack. In this instance, the LDH-1 level would rise with the LDH-2 and LDH-3. Because of this complication, some laboratories measure only the LDH-1 and consider an elevated LDH level with LDH-1 higher than 40% to be diagnostic of heart damage. LDH isoenzymes test is not used much anymore for diagnosis of heart attack. Tests for the protein troponin, which is found in myocardial cells, have been found to be more accurate.

Preparation

This test requires a blood sample. The patient need not fast (nothing to eat or drink) before the test unless requested to do so by the physician.

KEY TERMS

Differential diagnosis—Comparing and contrasting the signs, symptoms, and laboratory findings of two or more diseases to determine which is causing the patient’s condition.

Enzyme—A protein that regulates the rate of a chemical reaction in the body, increasing the speed at which the change occurs.

Isoenzyme—One of a group of enzymes that bring about the same reaction but are vary in their physical properties.

Risks

Risks for this test are minimal. The patient may experience slight bleeding from the blood-drawing site, **fainting** or feeling lightheaded after the vein is punctured (venipuncture), or an accumulation of blood under the puncture site (hematoma).

Normal results

Reference values for normal levels of LDH isoenzymes vary from laboratory to laboratory but can generally be found within the following ranges:

- LDH-1: 17-27%
- LDH-2: 27-37%
- LDH-3: 18-25%
- LDH-4: 8-16%
- LDH-5: 6-16%.

Abnormal results

Increased levels of LDH-1 are seen in myocardial infarction, red blood cell diseases like **hemolytic anemia**, **kidney disease** including **kidney transplantation** rejection, and testicular tumors. Increased levels of LDH-2 are found in lung diseases such as pneumonia and congestive heart failure, as well as in lymphomas and other tumors. Elevations of LDH-3 are significant in lung disease and certain tumors. Elevations of LDH-4 are greatly increased in **pancreatitis**. High levels of LDH-5 are found in liver disease, intestinal problems, and skeletal muscle disease and injury, such as **muscular dystrophy** and recent muscular trauma.

Diffuse disease or injury (for example, collagen disease, **shock**, low blood pressure) and advanced

solid-tumor cancers cause significant elevations of all LDH isoenzymes at the same time.

Resources

BOOKS

Pagana, Kathleen Deska. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis: Mosby, Inc., 1998.

Janis O. Flores

Lactate dehydrogenase test

Definition

Lactate dehydrogenase, also called lactic dehydrogenase, or LDH, is an enzyme found in the cells of many body tissues, including the heart, liver, kidneys, skeletal muscle, brain, red blood cells, and lungs. It is responsible for converting muscle lactic acid into pyruvic acid, an essential step in producing cellular energy.

Purpose

Lactic dehydrogenase is present in almost all body tissues, so the LDH test is used to detect tissue alterations and as an aid in the diagnosis of **heart attack**, anemia, and **liver disease**. Newer injury markers are becoming more useful than LDH for heart attack diagnosis.

Precautions

Because the LDH enzyme is so widely distributed throughout the body, cellular damage causes an elevation of the total serum LDH. As a result, the diagnostic usefulness of this enzyme by itself is not as valuable as determination of the five fractions that comprise the LDH. These fractions are called isoenzymes and are better indicators of disease than is the total LDH. The fractions are LDH-1, LDH-2, LDH-3, LDH-4, and LDH-5. A normal total LDH level does not mean that individual isoenzyme levels should not be measured. Individual isoenzyme ranges can help differentiate a diagnosis.

Description

When disease or injury affects tissues containing LDH, the cells release LDH into the bloodstream, where it is identified in higher than normal levels.

KEY TERMS

Enzyme—A protein that regulates the rate of a chemical reaction in the body, increasing the speed at which the change occurs.

Isoenzyme—One of a group of enzymes that catalyze the same reaction but are differentiated by variations in physical properties.

For example, when a person has a heart attack, the LDH level begins to rise about 12 hours after the attack and usually returns to normal within 5-10 days. The LDH is also elevated in diseases of the liver, in certain types of anemia, and in cases of excessive destruction of cells, as in **fractures**, trauma, muscle damage, and **shock**.

Cancers can also elevate LDH level. Additionally, some patients have chronically elevated LDH with no identifiable cause and no apparent consequence.

Preparation

This test requires a blood sample. It is not necessary for the patient to fast (nothing to eat or drink) before the test unless the physician requests it.

Risks

Risks for this test are minimal, but may include slight bleeding from the blood-drawing site, **fainting** or feeling lightheaded after venipuncture, or hematoma (blood accumulating under the puncture site).

Normal results

Reference ranges for total LDH vary from laboratory to laboratory. Normal values are also higher in childhood. For adults, in most laboratories, the range can be up to approximately 200 units/L, but is usually found within 45-90 U/L.

Abnormal results

Due to the fact that many common disease processes cause elevations in the total LDH level, a breakdown of the five different isoenzymes that make up the total LDH is often helpful for diagnosis. In certain disorders, the total LDH may be within normal limits, but individual isoenzyme elevations can indicate specific organ or tissue damage. For example, the LDH-2 fraction is normally greater than LDH-1 in the blood.

After an acute heart attack, however, the LDH-1 rises over the LDH-2 in what is known as a “flipped LDH.”

Certain diagnoses can be assisted by determination of the total LDH. One example is **infectious mononucleosis**, in which the LDH is usually more elevated than a liver enzyme called AST. Conversely, in cases of viral hepatitis, the liver enzymes AST and ALT are greatly increased over the LDH.

Resources

BOOKS

Pagana, Kathleen Deska. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis: Mosby, Inc., 1998.

Janis O. Flores

Lactation

Definition

Lactation is the medical term for yielding of milk by the mammary glands which leads to breastfeeding. Human milk contains the ideal amount of nutrients for the infant, and provides important protection from diseases through the mother's natural defenses.

Description

Early in a woman's **pregnancy** her milk-producing glands begin to prepare for her baby's arrival, and by the sixth month of pregnancy the breasts are ready to produce milk. Immediately after the baby is born, the placenta is delivered. This causes a hormone in the woman's body (prolactin) to activate the milk-producing glands. By the third to fifth day, the woman's breasts fill with milk.

Then, as the baby continues to suck each day, nursing triggers the continuing production of milk. The baby's sucking stimulates nerve endings in the nipple, which signal the mother's pituitary gland to release oxytocin, a hormone that causes the mammary glands to release milk to the nursing baby. This is called the “let-down reflex.” While the baby's sucking is the primary stimulus for this reflex, a baby's cry, thoughts of the baby, or the sound of running water also may trigger the response. Frequent nursing will lead to increased milk production.

Breast milk cannot be duplicated by commercial baby food formulas, although both contain protein, fat, and carbohydrates. In particular, breast milk

changes to meet the specific needs of a baby. The composition of breast milk changes as the baby grows to meet the baby's changing needs. Most important, breast milk contains substances called antibodies from the mother that can protect the child against illness and **allergies**. Antibodies are part of the body's natural defense system against infections and other agents that can cause disease. Breast milk also helps a baby's own immune system mature faster. As a result, breast-fed babies have less **diarrhea** and fewer ear infections, **rashes**, allergies, and other medical problems than bottle-fed babies.

There are many other benefits to breast milk. Because it is easily digested, babies do not get constipated. Breast-fed babies may have fewer speech impediments, and breastfeeding can improve cheekbone development and jaw alignment.

Breastfeeding is also good for the mother. The act of breastfeeding releases hormones that stimulate the uterus to contract, helping it to return to normal size after delivery and reducing the risk of bleeding. The act of producing milk is thought to burn more calories, helping the mother to lose excess weight gained during pregnancy. However, research in 2004 disputed this belief when body composition changes of lactation and non-lactating women were compared at intervals for six months postpartum. Breastfeeding may be related to a lower risk of **breast cancer**, **ovarian cancer**, or **cervical cancer**. This benefit is stronger the younger a woman is when she breastfeeds; women who breastfeed before age 20 and nurse for at least six months have a 50% drop in the risk for breast **cancer**.

In addition, breastfeeding does not involve any formulas, bottles and nipples, or sterilizing equipment. Breast milk is free, and saves money by eliminating the need to buy formula, bottles, and nipples. Because breast-fed babies are healthier, health care costs for breast-fed infants are lower.

Procedure

Breastfeeding should begin as soon as possible after birth, and should continue every two to three hours. However, all babies are different; some need to nurse almost constantly at first, while others can go much longer between feedings. A baby should be fed at least 8-12 times in 24 hours. Because breast milk is easily digested, a baby may be hungry again as soon as one and one-half hours after the last meal.

Mothers should wear comfortable, loose, front-opening clothes and a good nursing bra. Mothers should find a comfortable chair with lots of pillows, supporting the arm and back. Feet should rest on a

low footstool, with knees raised slightly. The baby should be level with the breast. The new mother may have to experiment with different ways of holding the baby before finding one that is comfortable for both the mother and baby.

Some babies have no trouble breastfeeding, while others may need some assistance. Once the baby begins to suck, the mother should make sure that the entire dark area around the nipple is in the baby's mouth. This will help stimulate milk flow, allowing the baby to get enough milk. It will also prevent nipple soreness.

Breastfeeding mothers will usually offer the baby both breasts at each feeding. Breastfeeding takes about 15-20 minutes on each side. After stopping the feeding on one side, the mother should burp the baby before beginning the feeding on the other breast. If the baby falls asleep at the breast, the next feeding should begin with the breast that was not nursed.

Mothers can tell if the baby is getting enough milk by checking diapers; a baby who is wetting between four to six disposable diapers (six to eight cloth) and who has three or four bowel movements in 24 hours is getting enough milk.

Nursing problems

New mothers may experience nursing problems, including:

- Engorged breasts. Breasts that are too full can prevent the baby from sucking. Expressing milk manually or with a breast pump can help.
- Sore nipples. In the early weeks nipples may become sore; a nipple shield can ease discomfort.
- Infection. Soreness and inflammation on the breast surface or a **fever** in the mother, may be an indication of breast infection. **Antibiotics** and continued nursing on the affected side may solve the problem.

Prognosis

There are no rules about when to stop breastfeeding. A baby needs breast milk for at least the first year of life; as long as a baby eats age-appropriate solid food, the mother may nurse for several years.

Prevention

Most common illnesses can not be transmitted via breast milk. However, some viruses, including HIV (the virus that causes **AIDS**) can be passed in breast milk; for this reason, women who are HIV-positive should not breastfeed.

KEY TERMS

Bromocriptine—A drug used to treat Parkinson's disease that can decrease a woman's milk supply.

Ergotamine—A drug used to prevent or treat migraine headaches. This can cause vomiting, diarrhea, and convulsions in infants.

Lithium—A drug used to treat manic depression (bipolar disorder) that can be transmitted in breast milk.

Methotrexate—An anticancer drug also used to treat arthritis that can suppress an infant's immune system when taken by a nursing mother.

Many medications have not been tested in nursing women, so it is not known if these drugs can affect a breast-fed child. A nursing woman should always check with her doctor before taking any medications, including over-the-counter drugs.

These drugs are not safe to take while nursing:

- Radioactive drugs for some diagnostic tests
- Chemotherapy drugs for cancer
- Bromocriptine
- Ergotamine
- Lithium
- Methotrexate
- Street drugs (including **marijuana**, heroin, amphetamines)
- Tobacco.

Resources

PERIODICALS

Wosje, Karen S., and Heidi J. Kalkwarf. "Lactation, Weaning, and Calcium Supplementation: Effects on Body Composition in Postpartum Women." *American Journal of Clinical Nutrition* August 2004: 676.

ORGANIZATIONS

International Lactation Consultants Assoc. 201 Brown Ave., Evanston, IL 60202. (708) 260-8874.

La Leche League International. 1400 North Meacham Rd., Schaumburg, IL 60173. (800) LA-LECHE.

National Alliance for Breastfeeding Advocacy. 254 Conant Rd., Weston, MA 02193. (617) 893-3553.

Carol A. Turkington
Teresa G. Odle

Lactic acid test

Definition

Lactic acid is an acid produced by cells during chemical processes in the body that do not require oxygen (anaerobic metabolism). Anaerobic metabolism occurs only when too little oxygen is present for the more usual aerobic metabolism (oxygen requiring). Lactic acid is a contributing factor in **muscle cramps**. It is also produced in tissues when conditions such as **heart attack** or **shock** reduce the blood supply responsible for carrying oxygen. Normally, lactic acid is removed from the blood by the liver. When an excess of lactic acid accumulates for any reason, the result is a condition called lactic acidosis.

Purpose

The lactic acid test is used as an indirect assessment of the oxygen level in tissues and to determine the cause and course of lactic acidosis.

Precautions

During blood collection, the patient should be instructed to relax the hand. Clenching and unclenching the fist will cause a build-up of potassium and lactic acid from the hand muscles that will falsely elevate the levels.

Description

The degree of acidity is an important chemical property of blood and other body fluids. Acidity is expressed on a pH scale where 7.0 is neutral, above 7.0 is basic (alkaline), and below 7.0 is acidic. A strong acid has a very low pH (near 1.0). A strong base has a very high pH (near 14.0). Blood is normally slightly alkaline or basic. It has a pH range of 7.35-7.45. The balance of acid to base in blood is precisely controlled. Even a minor deviation from the normal range can severely affect many organs.

Lactic acid (present in the blood as lactate ion) is a product of the breakdown of glucose to generate energy. It is found primarily in muscle cells and red blood cells. The lactate ion concentration in the blood depends on the rates of energy production and metabolism. Levels may increase significantly during **exercise**.

Together, lactic acid and another chemical (pyruvate) form a reversible reaction regulated by the oxygen supply to the blood and tissues. When oxygen

KEY TERMS

Acidosis—A disturbance of the balance of acid to base in the body causing an accumulation of acid or loss of alkali (base). There are two types of acidosis: metabolic and respiratory. One of the most common causes of metabolic acidosis is an overdose of aspirin. Respiratory acidosis is caused by impaired breathing caused by conditions such as severe chronic bronchitis, bronchial asthma, or airway obstruction.

levels are low, pyruvate converts to lactic acid; when oxygen levels are adequate, lactic acid converts to pyruvate. When the liver fails to metabolize lactose sufficiently or when too much pyruvate converts to lactate, lactic acidosis occurs. Measurement of blood lactate levels is recommended for all patients with symptoms of lactic acidosis. Testing is generally indicated if the blood pH level falls below 7.25-7.35.

Because of the close relationship between pyruvate and lactic acid, comparison of blood levels of the two substances can provide reliable information about tissue oxidation. However, pyruvate measurement is technically difficult and seldom performed. Lactic acid is measured more often, in either venous or arterial blood samples.

Preparation

This test requires a blood sample. The patient should have nothing to eat or drink (**fasting**) from midnight the night before the test. Because lactic acid is produced by exertion, the patient should rest for at least one hour before the test.

Risks

Risks for this test are minimal. The patient may experience slight bleeding from the blood-drawing site, **fainting** or feeling lightheaded after puncture of the vein (venipuncture), or an accumulation of blood under the puncture site (hematoma).

Normal results

Reference values vary from laboratory to laboratory but can be found within the following ranges:

- Venous blood: 4.5-19.8 mg/dL
- Arterial blood: 4.5-14.4 mg/dL.

Abnormal results

High blood lactate levels, together with decreased oxygen in tissues, may be caused by strenuous muscle exercise, shock, hemorrhage, severe infection in the blood stream, heart attack, or cardiac arrest. When tissue oxygenation is low for no apparent reason, increased lactate levels may be caused by systemic disorders like diabetes, leukemia, **liver disease**, or kidney failure. Defects in enzymes may also be responsible, as in glycogen storage disease (von Gierke's disease). Lactate is also increased in certain instances of intestinal obstruction.

Lactic acidosis can be caused by taking large doses of **acetaminophen** and alcohol and by intravenous infusion of epinephrine, glucagon, fructose, or sorbitol. Antifreeze **poisoning** can also cause lactic acidosis. In rare instances, a diabetic medication, metformin (Glucophage), causes lactic acidosis. People with weak kidneys should not take metformin.

Resources

BOOKS

Pagana, Kathleen Deska, and Timothy James Pagana.
Mosby's Manual of Diagnostic and Laboratory Tests.
St. Louis: Mosby, Inc., 1998.

Paul A. Johnson, Ed.M.

Lactic acidosis see **Metabolic acidosis**

Lactogen test see **Prolactin test**

Lactogenic hormone test see **Prolactin test**

Lactose intolerance

Definition

Lactose intolerance refers to the inability of the body to digest lactose.

Description

Lactose is the form of sugar present in milk. The enzyme lactase, which is normally produced by cells lining the small intestine, breaks down lactose into substances that can be absorbed into the bloodstream. When dairy products are ingested, the lactose reaches the digestive system and is broken down by lactase into the simpler sugars glucose and galactose. The liver changes the galactose into glucose, which then

enters the bloodstream and raises the blood glucose level. Lactose intolerance occurs when, due to a deficiency of lactase, lactose is not completely broken down and the glucose level does not rise. While not usually dangerous, lactose intolerance can cause severe discomfort.

From 30 to 50 million Americans suffer from the symptoms of lactose intolerance, but not everyone who is deficient in lactase experiences symptoms. Experts believe that 75% of the adult population worldwide does not produce enough lactase and is at risk for some or all of the symptoms of lactose intolerance.

Causes and symptoms

Lactose intolerance can be caused by some diseases of the digestive system and by injuries to the small intestine that result in a decreased production of lactase. While rare, some children are also born unable to produce the enzyme. For many, however, lactase deficiency develops naturally because, after about two years of age, the body produces less lactase.

Symptoms include **nausea**, cramps, **diarrhea**, bloating and gas. The symptoms usually occur between 30 minutes to two hours after eating or drinking lactose-containing foods.

Diagnosis

Usually, health care professionals measure the absorption of lactose in the digestive system by using the lactose tolerance test, hydrogen breath test or stool acidity test. Each of these can be performed outpatient, through a hospital, clinic or doctor's office.

People taking the lactose tolerance test must fast before being tested. They then drink a lactose-containing liquid for the test and medical personnel take blood samples during the next two hours to measure the patient's blood glucose level. The blood glucose level, or blood sugar level, indicates how well the body is digesting the lactose. A diagnosis of lactose intolerance is confirmed when blood glucose level does not rise. This test is not administered to infants and very young children because they are more prone to **dehydration**, which can result from diarrhea from the liquid.

Health care professionals measure the amount of hydrogen in the breath using the hydrogen breath test. Hydrogen is usually detected only in small amounts in the breath. However when undigested lactose found in the colon is fermented by bacteria, hydrogen in the breath is produced in greater quantities. The hydrogen is exhaled after being absorbed from the intestines and

carried through the bloodstream to the lungs. The hydrogen breath test involves having the patient drink a lactose-containing beverage. Health care professionals monitor the breath at regular intervals to see if the hydrogen levels rise, which indicates improper lactose digestion. People taking the test who have had certain foods, medications or cigarettes before the test may get inaccurate results. While the test is available to children and adults, newborns and young children should not have it because of the risk of dehydration from drinking the beverage that can cause diarrhea in those who are lactose intolerant.

A stool acidity test measures the amount of acid in the stool. This is a safe test for newborns and young children. The test detects lactic acid and other short-chain fatty acids from undigested lactose fermented by bacteria in the colon. Glucose might also be in the stool sample, resulting from unabsorbed lactose in the colon.

Treatment

Pediatricians might recommend that parents of newborns and very young children who are suspected of having lactose intolerance simply change from cow's milk to a soya formula. Since there is no treatment that can improve the body's ability to produce lactase, lactose deficiency treatments instead, are focused on controlling the diet.

Most people affected by lactose intolerance do well if they limit their intake of lactose foods and drinks. People differ in the amounts they can handle before experiencing symptoms. Some have to stop lactose completely. People who are sensitive after ingesting small amounts of lactose can take lactase enzymes, which are available without a prescription. Using the liquid form, people can add a few drops in their milk, put the milk in the refrigerator and drink it after 24 hours, when the lactase enzymes have worked to reduce the lactose content by 70%. If the milk is heated first and double the amount of lactase liquid is added, the milk will be 90 percent lactose free. Recently, researchers have developed a chewable lactase enzyme tablet. By taking three to six tablets just before eating, the tablets help people digest lactose-containing solid foods. Supermarkets also carry lactose-reduced milk and other products, which contain the needed nutrients found in the regular products but without the lactose.

Foods that contain lactose are milk, low-fat milk, skim milk, chocolate milk, buttermilk, sweetened condensed milk, dried whole milk, instant nonfat dry milk, low-fat yogurts, frozen yogurts ice cream, ice

KEY TERMS

Galactose—Simple sugar derived from milk sugar.

Glucose—A simple sugar and the chief energy source in the body.

Lactase enzyme—The enzyme produced by cells that line the small intestine which allows the body to break down lactose.

Lactose—The primary sugar in milk.

milk, sherbet, cheese, cottage cheese, low-fat cottage cheese, cream and butter. Other foods that may contain hidden lactose are: nondairy creamers, powdered artificial sweeteners, foods containing milk powder or nonfat milk solids, bread, cake, margarine, creamed soups, pancakes, waffles, processed breakfast cereals, salad dressings, lunch meats, puddings, custards, confections and some meat products.

Prognosis

Lactose intolerance is easy to manage. People of all ages however, especially children, have to replace the calcium lost by cutting back on milk products by taking supplements and eating calcium-rich foods, such as broccoli, kale, canned salmon with bones, calcium-fortified foods and tofu. Many people who suffer with lactose intolerance will be able to continue eating some milk products. The condition is not considered dangerous.

Prevention

Often, lactose intolerance is a natural occurrence that cannot be avoided. However, people can prevent symptoms by managing the condition with diet and lactase supplements.

Resources

ORGANIZATIONS

American Dietetic Association. (800) 366-1655. <<http://www.eatright.org/nfs/nfs43.html>>.

OTHER

"Lactose Intolerance." Onebody.com. <<http://www.onebody.com>>.

Lisette Hilton

Lamblasis see **Giardiasis**

Laminectomy see **Disk removal**

Language disturbance see **Aphasia**

Laparoscopic cholecystectomy see **Cholecystectomy**

Laparoscopy

Definition

Laparoscopy is a type of surgical procedure in which a small incision is made, usually in the navel, through which a viewing tube (laparoscope) is inserted. The viewing tube has a small camera on the eyepiece. This allows the doctor to examine the abdominal and pelvic organs on a video monitor connected to the tube. Other small incisions can be made to insert instruments to perform procedures. Laparoscopy can be done to diagnose conditions or to perform certain types of operations. It is less invasive than regular open abdominal surgery (laparotomy).

Purpose

Since the late 1980s, laparoscopy has been a popular diagnostic and treatment tool. The technique dates back to 1901, when it was reportedly first used in a gynecologic procedure performed in Russia. In fact, gynecologists were the first to use laparoscopy to diagnose and treat conditions relating to the female reproductive organs: uterus, fallopian tubes, and ovaries.

Laparoscopy was first used with **cancer** patients in 1973. In these first cases, the procedure was used to observe and biopsy the liver. Laparoscopy plays a role in the diagnosis, staging, and treatment for a variety of cancers.

As of 2001, the use of laparoscopy to completely remove cancerous growths and surrounding tissues (in place of open surgery) is controversial. The procedure is being studied to determine if it is as effective as open surgery in complex operations. Laparoscopy is also being investigated as a screening tool for **ovarian cancer**.

Laparoscopy is widely used in procedures for noncancerous conditions that in the past required open surgery, such as removal of the appendix (**appendectomy**) and gallbladder removal (**cholecystectomy**).

Diagnostic procedure

As a diagnostic procedure, laparoscopy is useful in taking biopsies of abdominal or pelvic growths, as well as lymph nodes. It allows the doctor to examine the abdominal area, including the female organs, appendix, gallbladder, stomach, and the liver.

Laparoscopy is used to determine the cause of pelvic **pain** or gynecological symptoms that cannot be confirmed by a physical exam or ultrasound. For example, **ovarian cysts**, **endometriosis**, **ectopic pregnancy**, or blocked fallopian tubes can be diagnosed using this procedure. It is an important tool when trying to determine the cause of **infertility**.

Operative procedure

While laparoscopic surgery to completely remove cancerous tumors, surrounding tissues, and lymph nodes is used on a limited basis, this type of operation is widely used in noncancerous conditions that once required open surgery. These conditions include:

- **Tubal ligation.** In this procedure, the fallopian tubes are sealed or cut to prevent subsequent pregnancies.
- **Ectopic pregnancy.** If a fertilized egg becomes embedded outside the uterus, usually in the fallopian tube, an operation must be performed to remove the developing embryo. This often can be done with laparoscopy.
- **Endometriosis.** This is a condition in which tissue from inside the uterus is found outside the uterus in other parts of (or on organs within) the pelvic cavity. This can cause cysts to form. Endometriosis is diagnosed with laparoscopy, and in some cases the cysts and other tissue can be removed during laparoscopy.
- **Hysterectomy.** This procedure to remove the uterus can, in some cases, be performed using laparoscopy. The uterus is cut away with the aid of the laparoscopic instruments and then the uterus is removed through the vagina.
- **Ovarian masses.** Tumors or cysts in the ovaries can be removed using laparoscopy.
- **Appendectomy.** This surgery to remove an inflamed appendix required open surgery in the past. It is now routinely performed with laparoscopy.
- **Cholecystectomy.** Like appendectomy, this procedure to remove the gall bladder used to require open surgery. Now it can be performed with laparoscopy, in some cases.

In contrast to open abdominal surgery, laparoscopy usually involves less pain, less risk, less scarring, and faster recovery. Because laparoscopy is so much

less invasive than traditional abdominal surgery, patients can leave the hospital sooner.

Cancer staging

Laparoscopy can be used in determining the spread of certain cancers. Sometimes it is combined with ultrasound. Although laparoscopy is a useful staging tool, its use depends on a variety of factors, which are considered for each patient. Types of cancers where laparoscopy may be used to determine the spread of the disease include:

- **Liver cancer.** Laparoscopy is an important tool for determining if cancer is present in the liver. When a patient has non-liver cancer, the liver is often checked to see if the cancer has spread there. Laparoscopy can identify up to 90% of malignant lesions that have spread to that organ from a cancer located elsewhere in the body. While computerized tomography (CT) can find cancerous lesions that are 0.4 in (10 mil) in size, laparoscopy is capable of locating lesions that are as small as 0.04 in (1 millimeter).
- **Pancreatic cancer.** Laparoscopy has been used to evaluate pancreatic cancer for years. In fact, the first reported use of laparoscopy in the United States was in a case involving pancreatic cancer.
- **Esophageal and stomach cancers.** Laparoscopy has been found to be more effective than **magnetic resonance imaging** (MRI) or computerized tomography (CT) in diagnosing the spread of cancer from these organs.
- **Hodgkin's disease.** Some patients with Hodgkin's disease have surgical procedures to evaluate lymph nodes for cancer. Laparoscopy is sometimes selected over laparotomy for this procedure. In addition, the spleen may be removed in patients with Hodgkin's disease. Laparoscopy is the standard surgical technique for this procedure, which is called a splenectomy.
- **Prostate cancer.** Patients with prostate cancer may have the nearby lymph nodes examined. Laparoscopy is an important tool in this procedure.

Cancer treatment

Laparoscopy is sometimes used as part of a palliative cancer treatment. This type of treatment is not a cure, but can often lessen the symptoms. An example is the feeding tube, which cancer patients may have if they are unable to take in food by mouth. The feeding tube provides **nutrition** directly into the stomach. Inserting the tube with a laparoscopy saves the patient the ordeal of open surgery.



This surgeon is performing a laparoscopic procedure on a patient. (Photo Researchers, Inc. Reproduced by permission.)

Precautions

As with any surgery, patients should notify their physician of any medications they are taking (prescription, over-the-counter, or herbal) and of any **allergies**. Precautions vary due to the several different purposes for laparoscopy. Patients should expect to rest for several days after the procedure, and should set up a comfortable environment in their home (with items such as pain medication, heating pads, feminine products, comfortable clothing, and food readily accessible) prior to surgery.

Description

Laparoscopy is a surgical procedure that is done in the hospital under anesthesia. For diagnosis and biopsy, **local anesthesia** is sometimes used. In operative procedures, such as abdominal surgery, **general anesthesia** is required. Before starting the procedure, a catheter is inserted through the urethra to empty the bladder, and the skin of the abdomen is cleaned.

After the patient is anesthetized, a hollow needle is inserted into the abdomen in or near the navel, and carbon dioxide gas is pumped through the needle to expand the abdomen. This allows the surgeon a better view of the internal organs. The laparoscope is then inserted through this incision to look at the internal organs. The image from the camera attached to the end of the laparoscope is seen on a video monitor.

Sometimes, additional small incisions are made to insert other instruments that are used to lift the tubes and ovaries for examination or to perform surgical procedures.

Preparation

Patients should not eat or drink after midnight on the night before the procedure.

Aftercare

After the operation, nurses will check the vital signs of patients who had general anesthesia. If there are no complications, the patient may leave the hospital within four to eight hours. (Traditional abdominal surgery requires a hospital stay of several days).

There may be some slight pain or throbbing at the incision sites in the first day or so after the procedure. The gas that is used to expand the abdomen may cause discomfort under the ribs or in the shoulder for a few days. Depending on the reason for the laparoscopy in gynecological procedures, some women may experience some vaginal bleeding. Many patients can return to work within a week of surgery and most are back to work within two weeks.

Risks

Laparoscopy is a relatively safe procedure, especially if the physician is experienced in the technique. The risk of complication is approximately 1%.

The procedure carries a slight risk of puncturing a blood vessel or organ, which could cause blood to seep into the abdominal cavity. Puncturing the intestines could allow intestinal contents to seep into the cavity. These are serious complications and major surgery may be required to correct the problem. For operative procedures, there is the possibility that it may become apparent that open surgery is required. Serious complications occur at a rate of only 0.2%.

Rare complications include:

- hemorrhage
- inflammation of the abdominal cavity lining
- abscess
- problems related to general anesthesia

Laparoscopy is generally not used in patients with certain heart or lung conditions, or in those who have some intestinal disorders, such as bowel obstruction.

KEY TERMS

Biopsy—Microscopic evaluation of a tissue sample. The tissue is closely examined for the presence of abnormal cells.

Cancer staging—Determining the course and spread of cancer.

Cyst—An abnormal lump or swelling that is filled with fluid or other material.

Palliative treatment—A type treatment that does not provide a cure, but eases the symptoms.

Tumor—A growth of tissue, benign or malignant, often referred to as a mass.

Normal results

In diagnostic procedures, normal results would indicate no abnormalities or disease of the organs or lymph nodes that were examined.

Abnormal results

A diagnostic laparoscopy may reveal cancerous or benign masses or lesions. Abnormal findings include tumors or cysts, infections (such as **pelvic inflammatory disease**), **cirrhosis**, endometriosis, fibroid tumors, or an accumulation of fluid in the cavity. If a doctor is checking for the spread of cancer, the presence of malignant lesions in areas other than the original site of malignancy is an abnormal finding.

Resources

BOOKS

Kurtz, Robert C., and Robert J. Ginsberg. "Cancer Diagnosis: Endoscopy." In *Cancer: Principles & Practice of Oncology*, edited by Vincent T. DeVita, Jr. Philadelphia: Lippincott, Williams & Wilkins, 2001, pp. 725-27.

Lefor, Alan T. "Specialized Techniques in Cancer Management." In *Cancer: Principles & Practice of Oncology*, edited by Vincent T. DeVita Jr., et al., 6th ed. Philadelphia: Lippincott, Williams & Wilkins, 2001, pp. 739-57.

OTHER

Iannitti, David A. "The Role of Laparoscopy in the Management of Pancreatic Cancer." *Home Journal Library Index*. March 23, 2001. [cited June 27, 2001]. <<http://bioscience.org/1998/v3/e/iannitti/e181-185.htm>>.

Carol A. Turkington
Rhonda Cloos, R.N.

Laryngeal cancer

Definition

Laryngeal **cancer** is cancer of the larynx or voice box.

Description

The larynx is located where the throat divides into the esophagus and the trachea. The esophagus is the tube that takes food to the stomach. The trachea, or windpipe, takes air to the lungs. The area where the larynx is located is sometimes called the Adam's apple.

The larynx has two main functions. It contains the vocal cords, cartilage, and small muscles that make up the voice box. When a person speaks, small muscles tighten the vocal cords, narrowing the distance between them. As air is exhaled past the tightened vocal cords, it creates sounds that are formed into speech by the mouth, lips, and tongue.

The second function of the larynx is to allow air to enter the trachea and to keep food, saliva, and foreign material from entering the lungs. A flap of tissue called the epiglottis covers the trachea each time a person swallows. This blocks foreign material from entering the lungs. When not swallowing, the epiglottis retracts, and air flows into the trachea. During treatment for cancer of the larynx, both of these functions may be lost.

Cancers of the larynx develop slowly. About 95% of these cancers develop from thin, flat cells similar to skin cells called squamous epithelial cells. These cells line the larynx. Gradually, the squamous epithelial cells begin to change and are replaced with abnormal cells. These abnormal cells are not cancerous but are pre-malignant cells that have the potential to develop into cancer. This condition is called dysplasia. Most people with dysplasia never develop cancer. The condition simply goes away without any treatment, especially if the person with dysplasia stops **smoking** or drinking alcohol.

The larynx is made up of three parts, the glottis, the supraglottis, and the subglottis. Cancer can start in any of these regions. Treatment and survival rates depend on which parts of the larynx are affected and whether the cancer has spread to neighboring areas of the neck or distant parts of the body.

The glottis is the middle part of the larynx. It contains the vocal cords. Cancers that develop on the vocal cords are often diagnosed very early because even small vocal cord tumors cause hoarseness. In

addition, the vocal cords have no connection to the lymphatic system. This means that cancers on the vocal cord do not spread easily. When confined to the vocal cords without any involvement of other parts of the larynx, the cure rate for this cancer is 75% to 95%.

The supraglottis is the area above the vocal cords. It contains the epiglottis, which protects the trachea from foreign materials. Cancers that develop in this region are usually not found as early as cancers of the glottis because the symptoms are less distinct. The supraglottis region has many connections to the lymphatic system, so cancers in this region tend to spread easily to the lymph nodes and may spread to other parts of the body (lymph nodes are small bean-shaped structures that are found throughout the body; they produce and store infection-fighting cells). In 25% to 50% of people with cancer in the supraglottal region, the cancer has already spread to the lymph nodes by the time they are diagnosed. Because of this, survival rates are lower than for cancers that involve only the glottis.

The subglottis is the region below the vocal cords. Cancer starting in the subglottis region is rare. When it does, it is usually detected only after it has spread to the vocal cords, where it causes obvious symptoms such as hoarseness. Because the cancer has already begun to spread by the time it is detected, survival rates are generally lower than for cancers in other parts of the larynx.

About 12,000 new cases of cancer of the larynx develop in the United States each year. Each year, about 3,900 die of the disease. Laryngeal cancer is between four and five times more common in men than in women. Almost all men who develop laryngeal cancer are over age 55. Laryngeal cancer is about 50% more common among African-American men than among other Americans.

It is thought that older men are more likely to develop laryngeal cancer than women because the two main risk factors for acquiring the disease are lifetime habits of smoking and alcohol **abuse**. More men are heavy smokers and drinkers than women, and more African-American men are heavy smokers than other men in the United States. However, as smoking becomes more prevalent among women, it seems likely that more cases of laryngeal cancer in females will be seen.

Causes and symptoms

Laryngeal cancer develops when the normal cells lining the larynx are replaced with abnormal cells

(dysplasia) that become malignant and reproduce to form tumors. The development of dysplasia is strongly linked to life-long habits of smoking and heavy use of alcohol. The more a person smokes, the greater the risk of developing laryngeal cancer. It is unusual for someone who does not smoke or drink to develop cancer of the larynx. Occasionally, however, people who inhale asbestos particles, wood dust, paint or industrial chemical fumes over a long period of time develop the disease.

The symptoms of laryngeal cancer depend on the location of the tumor. Tumors on the vocal cords are rarely painful, but cause hoarseness. Anyone who is continually hoarse for more than two weeks or who has a **cough** that does not go away should be checked by a doctor.

Tumors in the supraglottal region above the vocal cords often cause more, but less distinct symptoms. These include:

- persistent **sore throat**
- **pain** when swallowing
- difficulty swallowing or frequent **choking** on food
- bad breath
- lumps in the neck
- persistent ear pain (called referred pain; the source of the pain is not the ear)
- change in voice quality

Tumors that begin below the vocal cords are rare, but may cause noisy or difficult breathing. All the symptoms above can also be caused other cancers as well as by less seriousness illnesses. However, if these symptoms persist, it is important to see a doctor and find their cause, because the earlier cancer treatment begins, the more successful it is.

Diagnosis

On the first visit to a doctor for symptoms that suggest laryngeal cancer, the doctor first takes a complete medical history, including family history of cancer and lifestyle information about smoking and alcohol use. The doctor also does a **physical examination**, paying special attention to the neck region for lumps, tenderness, or swelling.

The next step is examination by an otolaryngologist, or ear, nose, and throat (ENT) specialist. This doctor also performs a physical examination, but in addition will also want to look inside the throat at the larynx. Initially, the doctor may spray a local anesthetic on the back of the throat to prevent gagging,

then use a long-handled mirror to look at the larynx and vocal cords. This examination is done in the doctor's office. It may cause gagging but is usually painless.

A more extensive examination involves a **laryngoscopy**. In a laryngoscopy, a lighted fiberoptic tube called a laryngoscope that contains a tiny camera is inserted through the patient's nose and mouth and snaked down the throat so that the doctor can see the larynx and surrounding area. This procedure can be done with a sedative and local anesthetic in a doctor's office. More often, the procedure is done in an outpatient surgery clinic or hospital under **general anesthesia**. This allows the doctor to use tiny clips on the end of the laryngoscope to take biopsies (tissue samples) of any abnormal-looking areas.

Laryngoscopies are normally painless and take about one hour. Some people find their throat feels scratchy after the procedure. Since laryngoscopies are done under **sedation**, patients should not drive immediately after the procedure, and should have someone available to take them home. Laryngoscopy is a standard procedure that is covered by insurance.

The locations of the samples taken during the laryngoscopy are recorded, and the samples are then sent to the laboratory where they are examined under the microscope by a pathologist who specializes in diagnosing diseases through cell samples and laboratory tests. It may take several days to get the results. Based on the findings of the pathologist, cancer can be diagnosed and staged.

Once cancer is diagnosed, other tests will probably be done to help determine the exact size and location of the tumors. This information is helpful in determining which treatments are most appropriate. These tests may include:

- Endoscopy. Similar to a laryngoscopy, this test is done when it appears that cancer may have spread to other areas, such as the esophagus or trachea.
- Computed tomography (CT or CAT) scan. Using x-ray images taken from several angles and computer modeling, CT scans allow parts of the body to be seen as a cross section. This helps locate and size the tumors, and provides information on whether they can be surgically removed.
- Magnetic resonance imaging (MRI). MRI uses magnets and radio waves to create more detailed cross-sectional scans than computed tomography. This detailed information is needed if surgery on the larynx area is planned.

- Barium swallow. Barium is a substance that, unlike soft tissue, shows up on x rays. Swallowed barium coats the throat and allows x-ray pictures to be made of the tissues lining the throat.
- Chest x ray. Done to determine if cancer has spread to the lungs. Since most people with laryngeal cancer are smokers, the risk of also having lung cancer or **emphysema** is high.
- Fine needle aspiration (FNA) biopsy. If any lumps on the neck are found, a thin needle is inserted into the lump, and some cells are removed for analysis by the pathologist.
- Additional blood and urine tests. These tests do not diagnose cancer, but help to determine the patient's general health and provide information to determine which cancer treatments are most appropriate.

Treatment

Staging

Once cancer of the larynx is found, more tests will be done to find out if cancer cells have spread to other parts of the body. This is called staging. A doctor needs to know the stage of the disease to plan treatment. In cancer of the larynx, the definitions of the early stages depend on where the cancer started.

STAGE I. The cancer is only in the area where it started and has not spread to lymph nodes in the area or to other parts of the body. The exact definition of stage I depends on where the cancer started, as follows:

- Supraglottis: The cancer is only in one area of the supraglottis and the vocal cords can move normally.
- Glottis: The cancer is only in the vocal cords and the vocal cords can move normally.
- Subglottis: The cancer has not spread outside of the subglottis.

STAGE II. The cancer is only in the larynx and has not spread to lymph nodes in the area or to other parts of the body. The exact definition of stage II depends on where the cancer started, as follows:

- Supraglottis: The cancer is in more than one area of the supraglottis, but the vocal cords can move normally.
- Glottis: The cancer has spread to the supraglottis or the subglottis or both. The vocal cords may or may not be able to move normally.
- Subglottis: The cancer has spread to the vocal cords, which may or may not be able to move normally.

STAGE III. Either of the following may be true:

- The cancer has not spread outside of the larynx, but the vocal cords cannot move normally, or the cancer has spread to tissues next to the larynx.
- The cancer has spread to one lymph node on the same side of the neck as the cancer, and the lymph node measures no more than 3 centimeters (just over 1 inch).

STAGE IV. Any of the following may be true:

- The cancer has spread to tissues around the larynx, such as the pharynx or the tissues in the neck. The lymph nodes in the area may or may not contain cancer.
- The cancer has spread to more than one lymph node on the same side of the neck as the cancer, to lymph nodes on one or both sides of the neck, or to any lymph node that measures more than 6 centimeters (over 2 inches).
- The cancer has spread to other parts of the body.

RECURRENT. Recurrent disease means that the cancer has come back (recurred) after it has been treated. It may come back in the larynx or in another part of the body.

Treatment

Treatment is based on the stage of the cancer as well as its location and the health of the individual. Generally, there are three types of treatments for cancer of the larynx. These are surgery, radiation, and **chemotherapy**. They can be used alone or in combination based in the stage of the cancer. Getting a second opinion after the cancer has been staged can be very helpful in sorting out treatment options and should always be considered.

SURGERY. The goal of surgery is to cut out the tissue that contains malignant cells. There are several common surgeries to treat laryngeal cancer.

Stage III and stage IV cancers are usually treated with total **laryngectomy**. This is an operation to remove the entire larynx. Sometimes other tissues around the larynx are also removed. Total laryngectomy removes the vocal cords. Alternate methods of voice communication must be learned with the help of a speech pathologist. Laryngectomy is treated in depth as a separate entry in this volume.

Smaller tumors are sometimes treated by partial laryngectomy. The goal is to remove the cancer but save as much of the larynx (and corresponding speech capability) as possible. Very small tumors or cancer in situ are sometimes successfully treated with laser

excision surgery. In this type of surgery, a narrowly-targeted beam of light from a laser is used to remove the cancer.

Advanced cancer (Stages III and IV) that has spread to the lymph nodes often requires an operation called a neck dissection. The goal of a neck dissection is to remove the lymph nodes and prevent the cancer from spreading. There are several forms of neck dissection. A **radical neck dissection** is the operation that removes the most tissue.

Several other operations are sometimes performed because of laryngeal cancer. A **tracheotomy** is a surgical procedure in which an artificial opening is made in the trachea (windpipe) to allow air into the lungs. This operation is necessary if the larynx is totally removed. A **gastrostomy** tube is a feeding tube placed through skin and directly into the stomach. It is used to give **nutrition** to people who cannot swallow or whose esophagus is blocked by a tumor. People who have a total laryngectomy usually do not need a gastrostomy tube if their esophagus remains intact.

RADIATION. Radiation therapy uses high-energy rays, such as x rays or gamma rays, to kill cancer cells. The advantage of radiation therapy is that it preserves the larynx and the ability to speak. The disadvantage is that it may not kill all the cancer cells. Radiation therapy can be used alone in early stage cancers or in combination with surgery. Sometimes it is tried first with the plan that if it fails to cure the cancer, surgery still remains an option. Often, radiation therapy is used after surgery for advanced cancers to kill any cells the surgeon might not have removed.

There are two types of radiation therapy. External beam radiation therapy focuses rays from outside the body on the cancerous tissue. This is the most common type of radiation therapy used to treat laryngeal cancer. With internal radiation therapy, also called brachytherapy, radioactive materials are placed directly on the cancerous tissue. This type of radiation therapy is a much less common treatment for laryngeal cancer.

External radiation therapy is given in doses called fractions. A common treatment involves giving fractions five days a week for seven weeks. Clinical trials are underway to determine the benefits of accelerating the delivery of fractions (accelerated fractionation) or dividing fractions into smaller doses given more than once a day (hyperfractionation). Side effects of radiation therapy include **dry mouth**, sore throat, hoarseness, skin problems, trouble swallowing, and diminished ability to taste.

CHEMOTHERAPY. Chemotherapy is the use of drugs to kill cancer cells. Unlike radiation therapy, which is targeted to a specific tissue, chemotherapy drugs are either taken by mouth or intravenously (through a vein) and circulate throughout the whole body. They are used mainly to treat advanced laryngeal cancer that is inoperable or that has metastasized to a distant site. Chemotherapy is often used after surgery or in combination with radiation therapy. Clinical trials are underway to determine the best combination of treatments for advanced cancer.

The two most common chemotherapy drugs used to treat laryngeal cancer are cisplatin and 5-fluorouracil (5-FU). There are many side effects associated with chemotherapy drugs, including **nausea and vomiting**, loss of appetite, hair loss, **diarrhea**, and mouth sores. Chemotherapy can also damage the blood-producing cells of the bone marrow, which can result in low blood cell counts, increased chance of infection, and abnormal bleeding or bruising.

Alternative treatment

Alternative and complementary therapies range from herbal remedies, vitamin supplements, and special **diets** to spiritual practices, **acupuncture**, massage, and similar treatments. When these therapies are used in addition to conventional medicine, they are called complementary therapies. When they are used instead of conventional medicine, they are called alternative therapies.

Complementary or alternative therapies are widely used by people with cancer. One large study published in the *Journal of Clinical Oncology* in July, 2000 found that 83% of all cancer patients studied used some form of complementary or alternative medicine as part of their cancer treatment. No specific alternative therapies have been directed toward laryngeal cancer. However, good nutrition and activities that reduce **stress** and promote a positive view of life have no unwanted side-effects and appear to be beneficial in boosting the immune system in fighting cancer.

Unlike traditional pharmaceuticals, complementary and alternative therapies are not evaluated by the United States Food and Drug Administration (FDA) for either safety or effectiveness. These therapies may have interactions with traditional pharmaceuticals. Patients should be wary of “miracle cures” and notify their doctors if they are using herbal remedies, vitamin supplements or other unprescribed treatments. Alternative and experimental treatments normally are not covered by insurance.

KEY TERMS

Dysplasia—The abnormal change in size, shape or organization of adult cells.

Lymph—Clear, slightly yellow fluid carried by a network of thin tubes to every part of the body. Cells that fight infection are carried in the lymph.

Lymphatic system—Primary defense against infection in the body. The lymphatic system consists of tissues, organs, and channels (similar to veins) that produce, store, and transport lymph and white blood cells to fight infection.

Lymph nodes—Small, bean-shaped collections of tissue found in a lymph vessel. They produce cells and proteins that fight infection, and also filter lymph. Nodes are sometimes called lymph glands.

Metastasis—Spread of cells from the original site of the cancer to other parts of the body where secondary tumors are formed.

Malignant—Cancerous. Cells tend to reproduce without normal controls on growth and form tumors or invade other tissues.

Prognosis

Cure rates and survival rates can predict group outcomes, but can never precisely predict the outcome for a single individual. However, the earlier laryngeal cancer is discovered and treated, the more likely it will be cured.

Cancers found in stage 0 and stage 1 have a 75% to 95% cure rate depending on the site. Late stage cancers that have metastasized have a very poor survival rate, with intermediate stages falling somewhere in between. People who have had laryngeal cancer are at greatest risk for recurrence (having cancer come back), especially in the head and neck, during the first two to three years after treatment. Check-ups during the first year are needed every other month, and four times a year during the second year. It is rare for laryngeal cancer to recur after five years of being cancer-free.

Prevention

By far, the most effective way to prevent laryngeal cancer is not to smoke. Smokers who quit smoking also significantly decrease their risk of developing the disease. Other ways to prevent laryngeal cancer include limiting the use of alcohol, eating a well-

balanced diet, seeking treatment for prolonged **heartburn**, and avoiding inhaling asbestos and chemical fumes.

Resources

PERIODICALS

Ahmad, I., B. N. Kumar, K. Radford, J. O'Connell, and A. J. Batch. "Surgical Voice Restoration Following Ablative Surgery for Laryngeal and Hypopharyngeal Carcinoma." *Journal of Laryngology and Otolaryngology* 114 (July 2000): 522–5.

ORGANIZATIONS

American Cancer Society. 1599 Clifton Rd. NE, Atlanta, GA 30329. 800 (ACS)-2345. <<http://www.cancer.org>>.

National Cancer Institute. Cancer Information Service. Bldg. 31, Room 10A19, 9000 Rockville Pike, Bethesda, MD 20892. (800) 4-CANCER. <<http://www.nci.nih.gov/cancerinfo/index.html>>.

National Cancer Institute Office of Cancer Complementary and Alternative Medicine. <<http://occam.nci.nih.gov>>.

National Center for Complementary and Alternative Medicine. P. O. Box 8218, Silver Spring, MD 20907-8218. (888) 644-6226. <<http://nccam.nih.gov>>.

OTHER

"Laryngeal Cancer." *CancerNet*. July 19, 2001. <http://www.graylab.ac.uk/cancernet/201519.html#3_STAGEEXPLANATION>.

"What you Need to Know About Cancer of the Larynx." *CancerNet* November 2000. [cited July 19, 2001]. <<http://www.cancernet.nci.nih.gov>>.

Tish Davidson, A.M.

Laryngeal cancer see **Head and neck cancer**

Laryngectomy

Definition

Laryngectomy is the partial or complete surgical removal of the larynx, usually as a treatment for **cancer** of the larynx.

Purpose

Normally a laryngectomy is performed to remove tumors or cancerous tissue. In rare cases, it may be done when the larynx is badly damaged by gunshot, automobile injuries, or similar violent accidents. Laryngectomies can be total or partial. Total laryngectomies are done when cancer is advanced. The

entire larynx is removed. Often if the cancer has spread, other surrounding structures in the neck, such as lymph nodes, are removed at the same time. Partial laryngectomies are done when cancer is limited to one spot. Only the area with the tumor is removed. Laryngectomies may also be performed when other cancer treatment options, such as radiation or **chemotherapy**, fail.

Precautions

Laryngectomy is done only after cancer of the larynx has been diagnosed by a series of tests that allow the otolaryngologist (a specialist often called an ear, nose, and throat doctor) to look into the throat and take tissue samples (biopsies) to confirm and stage the cancer. People need to be in good general health to undergo a laryngectomy, and will have standard pre-operative blood work and tests to make sure they are able to safely withstand the operation.

Description

The larynx is located slightly below the point where the throat divides into the esophagus, which takes food to the stomach, and the trachea (windpipe), which takes air to the lungs. Because of its location, the larynx plays a critical role in normal breathing, swallowing, and speaking. Within the larynx, vocal folds (often called vocal cords) vibrate as air is exhaled past, thus creating speech. The epiglottis protects the trachea, making sure that only air gets into the lungs. When the larynx is removed, these functions are lost.

Once the larynx is removed, air can no longer flow into the lungs. During this operation, the surgeon removes the larynx through an incision in the neck. The surgeon also performs a **tracheotomy**. He makes an artificial opening called a stoma in the front of the neck. The upper portion of the trachea is brought to the stoma and secured, making a permanent alternate way for air to get to the lungs. The connection between the throat and the esophagus is not normally affected, so after healing, the person whose larynx has been removed (called a laryngectomee) can eat normally. However, normal speech is no longer possible. Several alternate means of vocal communication can be learned with the help of a speech pathologist.

Preparation

As with any surgical procedure, the patient will be required to sign a consent form after the procedure is thoroughly explained. Many patients prefer a second opinion, and some insurers require it.

Blood and urine studies, along with **chest x ray** and EKG may be ordered as the doctor deems necessary. The patient also has a pre-operative meeting with an anesthesiologist. If a complete laryngectomy is planned, it may be helpful to meet with a speech pathologist and/or an established laryngectomee for discussion of post-operative expectations and support.

Aftercare

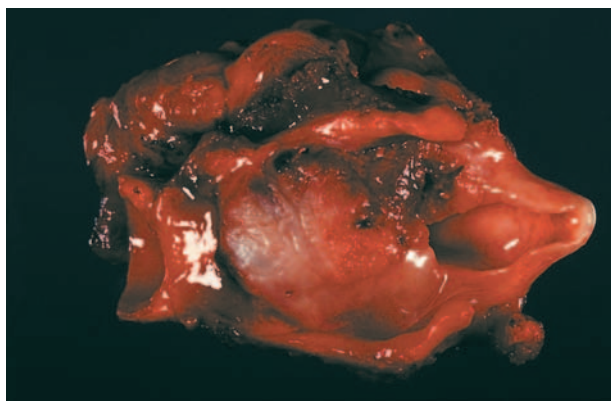
A person undergoing a laryngectomy spends several days in intensive care (ICU) and receives intravenous (IV) fluids and medication. As with any major surgery, the blood pressure, pulse, and respirations are monitored regularly. The patient is encouraged to turn, **cough**, and deep breathe to help mobilize secretions in the lungs. One or more drains are usually inserted in the neck to remove any fluids that collect. These drains are removed after several days.

It takes two to three weeks for the tissues of the throat to heal. During this time, the laryngectomee cannot swallow food and must receive **nutrition** through a tube inserted through the nose and down the throat into the stomach. During this time, even people with partial laryngectomies are unable to speak.

When air is drawn in normally through the nose, it is warmed and moistened before it reaches the lungs. When air is drawn in through the stoma, it does not have the opportunity to be warmed and humidified. In order to keep the stoma from drying out and becoming crusty, laryngectomees are encouraged to breathe artificially humidified air. The stoma is usually covered with a light cloth to keep it clean and to keep unwanted particles from accidentally entering the lungs. Care of the stoma is extremely important, since it is the person's only way to get air to the lungs. After a laryngectomy, a healthcare professional will teach the laryngectomee and his or her caregivers how to care for the stoma.

Immediately after a laryngectomy, an alternate method of communication such as writing notes, gesturing, or pointing must be used. A partial laryngectomy patient will gradually regain some speech several weeks after the operation, but the voice may be hoarse, weak, and strained. A speech pathologist will work with a complete laryngectomee to establish new ways of communicating.

There are three main methods of vocalizing after a total laryngectomy. In esophageal speech the



A pathology photograph of an extracted tumor found on the larynx. (Custom Medical Stock Photo. Reproduced by permission.)

laryngectomy learns how to “swallow” air down into the esophagus and creates sounds by releasing the air. This method requires quite a bit of coordination and learning, and produces short bursts (7 or 8 syllables) of low-volume sound.

Tracheoesophageal speech diverts air through a hole in the trachea made by the surgeon. The air then passes through an implanted artificial voice prosthesis (a small tube that makes a sound when air goes through it). Recent advances have been made in implanting voice prostheses that produce good voice quality.

The third method of artificial sound communication involves using a hand-held electronic device that translates vibrations into sounds. There are several different styles of these devices, but all require the use of at least one hand to hold the device to the throat. The choice of which method to use depends on many things including the age and health of the laryngectomy, and whether other parts of the mouth, such as the tongue, have also been removed.

Many patients resume daily activities after surgery. Special precautions must be taken during showering or shaving. Special instruction and equipment is also required for those who wish to swim or water ski, as it is dangerous for water to enter the windpipe and lungs through the stoma.

Regular follow-up visits are important following treatment for cancer of the larynx because there is a higher-than-average risk of developing a new cancer in the mouth, throat, or other regions of the head or neck. Many self-help and support groups are available to help patients meet others who face similar problems.

KEY TERMS

Larynx—Also known as the voice box, the larynx is composed of cartilage that contains the apparatus for voice production. This includes the vocal cords and the muscles and ligaments that move the cords.

Lymph nodes—Accumulations of tissue along a lymph channel, which produce cells called lymphocytes that fight infection.

Tracheostomy—A surgical procedure in which an artificial opening is made in the trachea (windpipe) to allow air into the lungs.

Risks

Laryngectomy is often successful in curing early stage cancers. However it does cause lifestyle changes. Laryngectomees must learn new ways of speaking. They must be continually concerned about the care of their stoma. Serious infections can occur if water or other foreign material enters the lungs through an unprotected stoma. Also, women who undergo partial laryngectomy or who learn some types of artificial speech will have a deep voice similar to that of a man. For some women this presents psychological challenges.

Normal results

Ideally, removal of the larynx will remove all cancerous material. The person will recover from the operation, make lifestyle adjustments, and return to an active life.

Abnormal results

Sometimes cancer has spread to surrounding tissues and it is necessary to remove lymph nodes, parts of the tongue, or other cancerous tissues. As with any major operation, post-surgical infection is possible. Infection is of particular concern to laryngectomees who have chosen to have a voice prosthesis implanted, and is one of the major reasons for having to remove the device.

Resources

ORGANIZATIONS

American Cancer Society. 1599 Clifton Road NE, Atlanta, GA 30329. (800) ACS -2345. <<http://www.cancer.org>>.

Cancer Information Service. National Cancer Institute, Building 31, Room 10A19, 9000 Rockville Pike, Bethesda, MD 20892. (800) 4-CANCER. <<http://www.nci.nih.gov/cancerinfo/index.html>>.

International Association of Laryngectomees (IAL). 7440 North Shadeland Ave., Suite 100, Indianapolis, IN 46250. <<http://www.larynxlink.com/>>.

National Institute on Deafness and Other Communication Disorders. National Institutes of Health, 31 Center Drive, MSC 2320, Bethesda, MD 20892-2320. <<http://www.nidcd.nih.gov>>.

The Voice Center at Eastern Virginia Medical School. Norfolk, VA 23507. <<http://www.voice-center.com>>.

Kathleen D. Wright, RN
Tish Davidson, AM

Laryngitis

Definition

Laryngitis is caused by inflammation of the larynx, resulting in hoarseness of the voice.

Description

When air is breathed in (inspired), it passes through the nose and the nasopharynx or through the mouth and the oropharynx. These are both connected to the larynx, a tube made of cartilage. The vocal cords, responsible for setting up the vibrations necessary for speech, are located within the larynx. The air continues down the larynx to the trachea. The trachea then splits into two branches, the left and right bronchi (bronchial tubes). These bronchi branch into smaller air tubes which run within the lungs, leading to the small air sacs of the lungs (alveoli).

Either food, liquid, or air may be taken in through the mouth. While air goes into the larynx and the respiratory system, food and liquid are directed into the tube leading to the stomach, the esophagus. Because food or liquid in the bronchial tubes or lungs could cause a blockage or lead to an infection, the airway must be protected. The epiglottis is a leaf-like piece of cartilage extending upwards from the larynx. The epiglottis can close down over the larynx when someone is eating or drinking, preventing these substances from entering the airway.

In laryngitis, the tissues below the level of the epiglottis are swollen and inflamed. This causes swelling around the area of the vocal cords, so that they cannot vibrate normally. A hoarse sound to the voice is very characteristic of laryngitis. Laryngitis is a very

common problem, and often occurs during the course of an upper respiratory tract infection (cold).

Causes and symptoms

Laryngitis is caused almost 100% of the time by a virus. The same viruses which cause the majority of simple upper respiratory infections (colds, etc.) are responsible for laryngitis. These include parainfluenzae virus, **influenza** virus, respiratory syncytial virus, rhinovirus, coronavirus, and echovirus. Extremely rarely, bacteria such as Group A streptococcus, *M. catarrhalis*, or that which causes **tuberculosis** may cause laryngitis. In people with faulty immune systems (particular due to acquired **immunodeficiency** syndrome, or **AIDS**), infections with fungi may be responsible for laryngitis.

Symptoms usually begin along with, or following, symptoms of a cold. A sore, scratchy throat, **fever**, runny nose, achiness, and **fatigue** may all occur. Difficulty swallowing sometimes occurs with **streptococcal infections**. The patient may **cough** and wheeze. Most characteristically, the patient's voice will sound strained, hoarse, and raspy.

In extremely rare cases, the swelling of the larynx may cause symptoms of airway obstruction. This is more common in infants, because the diameter of their airways is so small. In that case, the baby may have a greatly increased respiratory rate, and exhibit loud high-pitched sounds with breathing (called **stridor**).

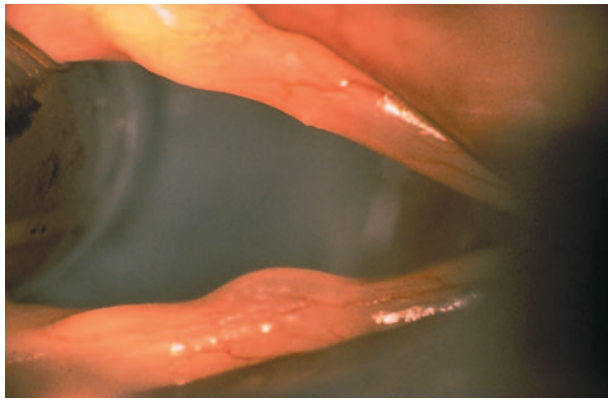
Diagnosis

Diagnosis is usually made by learning the history of a cold followed by hoarseness. The throat usually appears red and somewhat swollen. Listening to the chest and back with a stethoscope may reveal some harsh **wheezing** sounds with inspiration (breathing in).

In long-standing (chronic laryngitis), tuberculosis may be suspected. Using a scope called a laryngoscope, examination of the airway will show redness, swelling, small bumps of tissue called nodules, and irritated pits in the tissue called ulcerations. Special skin testing (TB testing) will reveal that the individual has been exposed to the bacteria causing TB.

Treatment

Treatment of a simple, viral laryngitis simply addresses the symptoms. Gargling with warm salt water, **pain** relievers such as **acetaminophen**, the use of vaporizers to create moist air, and rest will help the illness resolve within a week.



An endoscopic view of a patient's vocal cords with laryngitis.
(Custom Medical Stock Photo. Reproduced by permission.)

KEY TERMS

Epiglottis—A leaf-like piece of cartilage extending upwards from the larynx, which can close like a lid over the trachea to prevent the airway from receiving any food or liquid being swallowed.

Larynx—The part of the airway lying between the pharynx and the trachea.

Nasopharynx—The part of the airway into which the nose leads.

Oropharynx—The part of the airway into which the mouth leads.

Trachea—The part of the airway which leads into the bronchial tubes.

In an infant who is clearly struggling for air, it may be necessary to put in an artificial airway for a short period of time. This is very rarely needed.

An individual with tubercular laryngitis is treated with a combination of medications used to treat classic TB. In people with fungal laryngitis, a variety of anti-fungal medications are available.

Alternative treatment

Alternative treatments include **aromatherapy** inhalations made with benzoin, lavender, frankincense, thyme, and sandalwood. Decoctions (extracts made by boiling an herb in water) or infusions (extracts made by steeping an herb in boiling water) can be made with red sage (*Salvia officinalis* var. *rubra*) and yarrow (*Achillea millefolium*) or with licorice

(*Glycyrrhiza glabra*). These are used for gargling, and are said to reduce pain. **Echinacea** (*Echinacea* spp.) tincture taken in water every hour for 48 hours is recommended to boost the immune system. Antiviral herbs, including usnea (*Usnea* spp.), lomatium (*Lomatium dissectum*), and ligusticum (*Ligusticum porteri*), may help hasten recovery from laryngitis. Homeopathic remedies are recommended based on the patient's symptoms. Some people may get relief from placing cold compresses on the throat.

Prognosis

Prognosis for laryngitis is excellent. Recovery is complete, and usually occurs within a week's time.

Prevention

Prevention of laryngitis is the same as for any upper respiratory infections. The only way to even attempt to prevent such illnesses is by good handwashing, and by avoiding situations where one might come in contact with people who might be sick. However, even with relatively good hygiene practices, most people will get about five to six colds per year. It is unpredictable which of these may lead to laryngitis.

Resources

ORGANIZATIONS

American Academy of Otolaryngology-Head and Neck Surgery, Inc. One Prince St., Alexandria VA 22314-3357. (703) 836-4444. <<http://www.entnet.org>>.

Rosalyn Carson-DeWitt, MD

Laryngoscopy

Definition

Laryngoscopy refers to a procedure used to view the inside of the larynx (the voice box).

Description

The purpose and advantage of seeing inside the larynx is to detect tumors, foreign bodies, nerve or structural injury, or other abnormalities. Two methods allow the larynx to be seen directly during the examination. In one, a flexible tube with a fiber-optic device is threaded through the nasal passage and down into the throat. The other method uses a rigid viewing tube

KEY TERMS

Endoscopic tube—a tube that is inserted into a hollow organ permitting a physician to see the inside it.

passed directly from the mouth, through the throat, into the larynx. A light and lens affixed to the endoscope are used in both methods. The endoscopic tube may also be equipped to suction debris or remove material for biopsy. **Bronchoscopy** is a similar, but more extensive procedure in which the tube is continued through the larynx, down into the trachea and bronchi.

Preparation

Laryngoscopy is done in the hospital with a local anesthetic spray to minimize discomfort and suppress the gag reflex. Patients are requested not to eat for several hours before the examination.

Aftercare

If the throat is sore, soothing liquids or lozenges will probably relieve any temporary discomfort.

Risks

This procedure carries no serious risks, although the patient may experience soreness of the throat or **cough** up small amounts of blood until the irritation subsides.

Normal results

A normal result would be the absence of signs of disease or damage.

Abnormal results

An abnormal finding, such as a tumor or an object lodged in the tissue, would either be removed or described for further medical attention.

Jill S. Lasker

Larynx removal see **Laryngectomy**

Laser-assisted in-situ keratomileusis see **Photorefractive keratectomy and laser-assisted in-**

Laser surgery

Definition

Laser (light amplification by stimulated emission of radiation) surgery uses an intensely hot, precisely focused beam of light to remove or vaporize tissue and control bleeding in a wide variety of non-invasive and minimally invasive procedures.

Purpose

Laser surgery is used to:

- Cut or destroy tissue that is abnormal or diseased without harming healthy, normal tissue
- Shrink or destroy tumors and lesions
- Caulterize (seal) blood vessels to prevent excessive bleeding.

Precautions

Anyone who is thinking about having laser surgery should ask his doctor to:

- Explain why laser surgery is likely to be more beneficial than traditional surgery
- Describe his experience in performing the laser procedure the patient is considering.

Because some lasers can temporarily or permanently discolor the skin of Blacks, Asians, and Hispanics, a dark-skinned patient should make sure that his surgeon has successfully performed laser procedures on people of color.

Some types of laser surgery should not be performed on pregnant women or on patients with severe cardiopulmonary disease or other serious health problems.

Description

The first working laser was introduced in 1960. The device was initially used to treat diseases and disorders of the eye, whose transparent tissues gave ophthalmic surgeons a clear view of how the narrow, concentrated beam was being directed. Dermatologic surgeons also helped pioneer laser surgery, and developed and improved upon many early techniques and more refined surgical procedures.

Types of lasers

The three types of lasers most often used in medical treatment are the:

- Carbon dioxide (CO₂) laser. Primarily a surgical tool, this device converts light energy to heat strong enough to minimize bleeding while it cuts through or vaporizes tissue.
- Neodymium:yttrium-aluminum-garnet (Nd:YAG) laser. Capable of penetrating tissue more deeply than other lasers, the Nd:YAG makes blood clot quickly and can enable surgeons to see and work on parts of the body that could otherwise be reached only through open (invasive) surgery.
- Argon laser. This laser provides the limited penetration needed for eye surgery and superficial skin disorders. In a special procedure known as photodynamic therapy (PDT), this laser uses light-sensitive dyes to shrink or dissolve tumors.

Laser applications

Sometimes described as “scalpels of light,” lasers are used alone or with conventional surgical instruments in a diverse array of procedures that:

- improve appearance
- relieve **pain**
- restore function
- save lives

Laser surgery is often standard operating procedure for specialists in:

- cardiology
- dentistry
- dermatology
- gastroenterology (treatment of disorders of the stomach and intestines)
- gynecology
- neurosurgery
- oncology (**cancer** treatment)
- ophthalmology (treatment of disorders of the eye)
- orthopedics (treatment of disorders of bones, joints, muscles, ligaments, and tendons)
- otolaryngology (treatment of disorders of the ears, nose, and throat)
- pulmonary care (treatment of disorders of the respiratory system)
- urology (treatment of disorders of the urinary tract and of the male reproductive system)

Routine uses of lasers include erasing **birthmarks**, skin discoloration, and skin changes due to **aging**, and removing benign, precancerous, or cancerous tissues or tumors. Lasers are used to stop **snoring**, remove

tonsils, remove or transplant hair, and relieve pain and restore function in patients who are too weak to undergo major surgery. Lasers are also used to treat:

- angina (chest pain)
- cancerous or non-cancerous tumors that cannot be removed or destroyed
- cold and **canker sores**, gum disease, and tooth sensitivity or decay
- ectopic **pregnancy** (development of a fertilized egg outside the uterus)
- endometriosis
- fibroid tumors
- gallstones
- glaucoma, mild-to-moderate nearsightedness and **astigmatism**, and other conditions that impair sight
- migraine headaches
- non-cancerous enlargement of the prostate gland
- nosebleeds
- ovarian cysts
- ulcers
- varicose veins
- warts
- numerous other conditions, diseases, and disorders

Advantages of laser surgery

Often referred to as “bloodless surgery,” laser procedures usually involve less bleeding than conventional surgery. The heat generated by the laser keeps the surgical site free of germs and reduces the risk of infection. Because a smaller incision is required, laser procedures often take less time (and cost less money) than traditional surgery. Sealing off blood vessels and nerves reduces bleeding, swelling, scarring, pain, and the length of the recovery period.

Disadvantages of laser surgery

Although many laser surgeries can be performed in a doctor’s office rather than in a hospital, the person guiding the laser must be at least as thoroughly trained and highly skilled as someone performing the same procedure in a hospital setting. The American Society for Laser Medicine and Surgery, Inc. urges that:

- All operative areas be equipped with oxygen and other drugs and equipment required for **cardiopulmonary resuscitation (CPR)**



Cosmetic laser surgery in progress. The wavelengths of the laser's light can be matched to a specific target, enabling the physician to destroy the capillaries near the skin's surface without damaging the surrounding tissue.
(Photograph by Will & Deni McIntyre, Photo Researchers, Inc. Reproduced by permission.)

- Non-physicians performing laser procedures be properly trained, licensed, and insured
- A qualified and experienced supervising physician be able to respond to and manage unanticipated events or other emergencies within five minutes of the time they occur
- Emergency transportation to a hospital or other acute-care facility be available whenever laser surgery is performed in a non-hospital setting.
- Determines the patient's general health and current medical status
- Describes how the patient has responded to other illnesses, hospital stays, and diagnostic or therapeutic procedures
- Clarifies what the patient expects the outcome of the procedure to be.

Imprecisely aimed lasers can burn or destroy healthy tissue.

Preparation

Because laser surgery is used to treat so many dissimilar conditions, the patient should ask his physician for detailed instructions about how to prepare for a specific procedure. Diet, activities, and medications may not have to be limited prior to surgery, but some procedures require a **physical examination** and a medical history that:

Aftercare

Most laser surgeries can be performed on an out-patient basis, and patients are usually permitted to leave the hospital or medical office when their vital signs have stabilized. A patient who has been sedated should not be discharged:

- Until he has recovered from the anesthesia and knows who and where he is
- Unless he is accompanied by a responsible adult.

The doctor may prescribe analgesic (pain-relieving) medication, and should provide easy-to-understand written instructions that describe how the

patient's recovery should progress and what to do in case complications or emergency arise.

Risks

Like traditional surgery, laser surgery can be complicated by:

- hemorrhage
- infection
- perforation (piercing) of an organ or tissue

Laser surgery can also involve risks that are not associated with traditional surgical procedures. Being careless or not practicing safe surgical techniques can severely burn the patient's lungs or even cause them to explode. Patients must wear protective eye shields while undergoing laser surgery on any part of the face near the eyes or eyelids, and the United States Food and Drug Administration (FDA) has said that both doctors and patients must use special protective eyewear whenever a CO₂ laser is used.

Laser beams can burn or destroy healthy tissue, cause injuries that are painful and sometimes permanent, and actually compound problems they are supposed to solve. Errors or inaccuracies in laser surgery can worsen a patient's vision, for example, and lasers can scar and even change the skin color of some patients.

Normal results

The nature and severity of the problem, the skill of the surgeon performing the procedure, and the patient's general health and realistic expectations are among the factors that influence the outcome of laser surgery. Successful procedures can enable patients to:

- feel better
- look younger
- enjoy longer, fuller, more active lives

A patient who is considering any kind of laser surgery should ask his doctor to provide detailed information about what the outcome of the surgery is expected to be, what the recovery process will involve, and how long it will probably be before he regains a normal appearance and can resume his normal activities.

Abnormal results

A person who is considering any type of laser surgery should ask his doctor to provide specific and detailed information about what could go wrong during the procedure and what the negative impact on the patient's health or appearance might be.

KEY TERMS

Argon—A colorless, odorless gas.

Astigmatism—A condition in which one or both eyes cannot filter light properly and images appear blurred and indistinct.

Canker sore—A blister-like sore on the inside of the mouth that can be painful but is not serious.

Carbon dioxide—A heavy, colorless gas that dissolves in water.

Cardiopulmonary resuscitation—An emergency procedure used to restore circulation and prevent brain death to a person who has collapsed, is unconscious, is not breathing, and has no pulse.

Cauterize—To use heat or chemicals to stop bleeding, prevent the spread of infection, or destroy tissue.

Cornea—The outer, transparent lens that covers the pupil of the eye and admits light.

Endometriosis—An often painful gynecologic condition in which endometrial tissue migrates from the inside of the uterus to other organs inside and beyond the abdominal cavity.

Glaucoma—A disease of the eye in which increased pressure within the eyeball can cause gradual loss of vision.

Invasive surgery—A form of surgery that involves making an incision in the patient's body and inserting instruments or other medical devices into it.

Nearsightedness—A condition in which one or both eyes cannot focus normally, causing objects at a distance to appear blurred and indistinct. Also called myopia.

Ovarian cyst—A benign or malignant growth on an ovary. An ovarian cyst can disappear without treatment or become extremely painful and have to be surgically removed.

Vaporize—To dissolve solid material or convert it into smoke or gas.

Varicose veins—Swollen, twisted veins, usually occurring in the legs, that occur more often in women than in men.

Lighter or darker skin may appear, for example, when a laser is used to remove sun damage or age spots from an olive-skinned or dark-skinned individual. This abnormal pigmentation may or may not disappear in time.

Scarring or rupturing of the cornea is uncommon, but laser surgery on one or both eyes can:

- increase sensitivity to light or glare
- reduce night vision
- permanently cloud vision, or cause sharpness of vision to decline throughout the day

Signs of infection following laser surgery include:

- burning
- crusting of the skin
- itching
- pain
- scarring
- severe redness
- swelling

Resources

ORGANIZATIONS

American Society for Dermatologic Surgery. 930 N. Meacham Road, P.O. Box 4014, Schaumburg, IL 60168-4014. (847) 330-9830. <<http://www.asds-net.org>>.

American Society for Laser Medicine and Surgery. 2404 Stewart Square, Wausau, WI 54401. (715) 845-9283. <<http://www.aslms.org>>.

Cancer Information Service. 9000 Rockville Pike, Building 31, Suite 10A18, Bethesda, MD 20892. 1-800-4-CANCER. <<http://www.wicic.nci.nih.gov>>. 7.

National Cancer Institute. Building 31, Room 10A31, 31 Center Drive, MSC 2580, Bethesda, MD 20892-2580. (800) 422-6237. <<http://www.nci.nih.gov>>.

Maureen Haggerty

LASIK see **Photorefractive keratectomy and laser-assisted in-**

Lassa fever see **Hemorrhagic fevers**

Laxatives

Definition

Laxatives are products that promote bowel movements.

Purpose

Laxatives are used to treat constipation—the passage of small amounts of hard, dry stools, usually

fewer than three times a week. Before recommending use of laxatives, differential diagnosis should be performed. Prolonged **constipation** may be evidence of a significant problem, such as localized **peritonitis** or **diverticulitis**. Complaints of constipation may be associated with **obsessive-compulsive disorder**. Use of laxatives should be avoided in these cases. Patients should be aware that patterns of defecation are highly variable, and may vary from two to three times daily to two to three times weekly.

Laxatives may also be used prophylactically for patients, such as those recovering from a myocardial infarction or those who have had recent surgery, who should not strain during defecation.

Laxatives are also used to cleanse the lower bowel before a **colonoscopy** or similar diagnostic procedure.

Description

Laxatives may be grouped by mechanism of action.

Saline cathartics include dibasic sodium phosphate (Phospo-Soda), magnesium citrate, magnesium hydroxide (milk of magnesia), magnesium sulfate (Epsom salts), sodium biphosphate, and others. They act by attracting and holding water in the intestinal lumen, and may produce a watery stool. Magnesium sulfate is the most potent of the laxatives in this group.

Stimulant and irritant laxatives increase the peristaltic movement of the intestine. Examples include cascara and bisacodyl (Dulcolax.) Castor oil works in a similar fashion.

Bulk-producing laxatives increase the volume of the stool, and will both soften the stool and stimulate intestinal motility. Psyllium (Metamucil, Konsil) and methylcellulose (Citrucel) are examples of this type. The overall effect is similar to that of eating high-fiber foods, and this class of laxative is most suitable for regular use. Many primary care physicians suggest that patients try laxatives in this category before using saline or stimulant laxatives.

Docusate (Colace) is the only representative example of the stool softener class. It holds water within the fecal mass, providing a larger, softer stool. Docusate has no effect on acute constipation, since it must be present before the fecal mass forms to have any effect, but may be useful for prevention of constipation in patients with recurrent problems, or those who are about to take a constipating drug, such as narcotic **analgesics**.

Mineral oil is an emollient laxative. It acts by retarding intestinal absorption of fecal water, thereby softening the stool.

The hyperosmotic laxatives are glycerin and lactulose (Chronulac, Duphalac), both of which act by holding water within the intestine. Lactulose may also increase peristaltic action of the intestine.

Some newer options for the treatment of chronic constipation are being developed by various groups of researchers. These include such alternative therapies as **biofeedback**; newer drugs like tegaserod (Zelnorm) and prucalopride, which stimulate peristalsis; a nerve growth factor known as neurotrophin-3; and electrical stimulation of the colon.

Recommended dosage

See specific products.

Precautions

Short-term use of laxatives is generally safe except in **appendicitis**, fecal impaction, or intestinal obstruction. Lactulose is composed of two sugar molecules; galactose and fructose, and should not be administered to patients who require a low galactose diet.

Chronic use of laxatives may result in fluid and electrolyte imbalances, steatorrhea, osteomalacia, **diarrhea**, cathartic colon, and **liver disease**. Excessive intake of mineral oil may cause impaired absorption of oil soluble **vitamins**, particularly A and D. Excessive use of magnesium salts may cause hypermagnesemia.

Lactulose and magnesium sulfate are **pregnancy category B**. Casanthranol, cascara sagrada, danthron, docusate sodium, docusate calcium, docusate potassium, mineral oil and senna are category C. Casanthranol, cascara sagrada and danthron are excreted in breast milk, resulting in a potential increased incidence of diarrhea in the nursing infant.

The American College of Toxicology states that cathartics should *not* be used as a means of clearing poisons from the digestive tract of a **poisoning** victim. Although some physicians have administered these laxatives along with **activated charcoal** in order to reduce the body's absorption of the poison, this treatment is no longer recommended.

Interactions

Mineral oil and docusate should not be used in combination. Docusate is an emulsifying agent which will increase the absorption of mineral oil.

KEY TERMS

Carbohydrates—Compounds, such as cellulose, sugar, and starch, that contain only carbon, hydrogen, and oxygen, and are a major part of the diets of people and other animals.

Cathartic colon—A poorly functioning colon, resulting from the chronic abuse of stimulant cathartics.

Colon—The large intestine.

Diverticulitis—Inflammation of the part of the intestine known as the diverticulum.

Fiber—Carbohydrate material in food that cannot be digested.

Hyperosmotic—Hypertonic, containing a higher concentration of salts or other dissolved materials than normal tissues.

Osteomalacia—A disease of adults, characterized by softening of the bone. Similar to rickets which is seen in children.

Pregnancy category—A system of classifying drugs according to their established risks for use during pregnancy. Category A: Controlled human studies have demonstrated no fetal risk. Category B: Animal studies indicate no fetal risk, but no human studies, or adverse effects in animals, but not in well-controlled human studies. Category C: No adequate human or animal studies, or adverse fetal effects in animal studies, but no available human data. Category D: Evidence of fetal risk, but benefits outweigh risks. Category X: Evidence of fetal risk. Risks outweigh any benefits.

Steatorrhea—An excess of fat in the stool.

Stool—The solid waste that is left after food is digested. Stool forms in the intestines and passes out of the body through the anus.

Bisacodyl tablets are enteric coated, and so should not be used in combination with **antacids**. The antacids will cause premature rupture of the enteric coating.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD., editors. "Diarrhea and Constipation." In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

Karch, A. M. *Lippincott's Nursing Drug Guide*. Springhouse, PA: Lippincott Williams & Wilkins, 2003.

ORGANIZATIONS

American Society of Health-System Pharmacists (ASHP). 7272 Wisconsin Avenue, Bethesda, MD 20814. (301) 657-3000. <<http://www.ashp.org>>.

National Digestive Diseases Information Clearinghouse. 2 Information Way, Bethesda, MD 20892-3570. nddic@erie.com. <<http://www.niddk.nih.gov/Brochures/NDDIC.htm>>.

United States Food and Drug Administration (FDA). 5600 Fishers Lane, Rockville, MD 20857-0001. (888) INFO-FDA. <<http://www.fda.gov>>.

Samuel D. Uretsky, PharmD
Rebecca J. Frey, PhD

Lazy eye see **Amblyopia**

LCM see **Lymphocytic choriomeningitis**

LDH isoenzymes test see **Lactate dehydrogenase isoenzymes test**

LDH test see **Lactate dehydrogenase test**

Lead poisoning

Definition

Lead **poisoning** occurs when a person swallows, absorbs, or inhales lead in any form. The result can be damaging to the brain, nerves, and many other parts of the body. Acute lead poisoning, which is somewhat rare, occurs when a relatively large amount of lead is taken into the body over a short period of time. Chronic lead poisoning—a common problem in children—occurs when small amounts of lead are taken in over a longer period. The Centers for Disease Control and Prevention (CDC) defines childhood lead poisoning as a whole-blood lead concentration equal to or greater than 10 micrograms/dL.

Description

Lead can damage almost every system in the human body, and it can also cause high blood pressure (**hypertension**). It is particularly harmful to the developing brain of fetuses and young children. The higher the level of lead in a child's blood, and the longer this elevated level lasts, the greater the chance of ill effects. Over the long term, lead poisoning in a child can lead to learning disabilities, behavioral problems, and even

mental retardation. At very high levels, lead poisoning can cause seizures, **coma**, and even **death**. According to the National Center for Environmental Health, there were about 200 deaths from lead poisoning in the United States between 1979 and 1998. Most of the deaths were among males (74%), African Americans (67%), adults over the age of 45 (76%), and Southerners (70%).

About one out of every six children in the United States has a high level of lead in the blood, according to the Agency for Toxic Substances and Disease Registry. Many of these children are exposed to lead through peeling paint in older homes. Others are exposed through dust or soil that has been contaminated by old paint or past emissions of leaded gasoline. Since children between the ages of 12–36 months are apt to put things in their mouths, they are more likely than older children to take in lead. Pregnant women who come into contact with lead can pass it along to the fetus.

Over 80% of American homes built before 1978 have lead-based paint in them, according to the Centers for Disease Control and Prevention (CDC). The older the home, the more likely it is to contain lead paint, and the higher the concentration of lead in the paint is apt to be. Some homes also have lead in the water pipes or plumbing. People may have lead in the paint, dust, or soil around their homes or in their drinking water without knowing it, since lead can't be seen, smelled, or tasted. Because lead doesn't break down naturally, it can continue to cause problems until it is removed.

Causes and symptoms

Before scientists knew how harmful it could be, lead was widely used in paint, gasoline, water pipes, and many other products. Today house paint is almost lead-free, gasoline is unleaded, and household plumbing is no longer made with lead materials. Still, remnants of the old hazards remain. Following are some sources of lead exposure:

- **Lead-based paint.** This is the most common source of exposure to large amounts of lead among preschoolers. Children may eat paint chips from older homes that have fallen into disrepair. They may also chew on painted surfaces such as windowsills. In addition, paint may be disturbed during remodeling.
- **Dust and soil.** These can be contaminated with lead from old paint or past emissions of leaded gasoline. In addition, pollution from operating or abandoned industrial sites and smelters can find its way into the soil, resulting in soil contamination.

- **Drinking water.** Exposure may come from lead water pipes, found in many homes built before 1930. Even newer copper pipes may have lead solder. Also, some new homes have brass faucets and fittings that can leach lead.
- **Jobs and hobbies.** A number of activities can expose participants to lead. These include making pottery or stained glass, refinishing furniture, doing home repairs, and using indoor firing ranges. When adults take part in such activities, they may inadvertently expose children to lead residue that is on their clothing or on scrap materials.
- **Food.** Imported food cans often have lead solder. Lead may also be found in leaded crystal glassware and some imported ceramic or old ceramic dishes (e.g., ceramic dishes from Mexico). A 2003 study of cases of lead poisoning in pregnant women found that 70% of the patients were Hispanics, most of whom had absorbed the lead from their pottery. In addition, food may be contaminated by lead in the water or soil.
- **Folk medicines.** Certain folk medicines (for example, alarcon, alkohl, azarcon, bali goli, coral, ghasard, greta, liga, pay-loo-ah, and rueda) and traditional cosmetics (kohl, for example) contain large amounts of lead.
- **Moonshine whiskey.** Lead poisoning from drinking illegally distilled liquor is still a cause of death among adults in the southern United States.
- **Gunshot wounds.** Toxic amounts of lead can be absorbed from bullets or bullet fragments that remain in the body after emergency surgery.

Chronic lead poisoning

New evidence suggests that lead may be harmful to children even at low levels that were once thought to be safe, and the risk of damage rises as blood levels of lead increase. The symptoms of chronic lead poisoning take time to develop, however. Children can appear healthy despite having high levels of lead in their blood. Over time, though, problems such as the following may arise:

- learning disabilities
- hyperactivity
- mental retardation
- slowed growth
- hearing loss
- headaches

It is also known that certain genetic factors increase the harmful effects of lead poisoning in susceptible children; however, these factors are not completely understood as of 2003.

Lead poisoning is also harmful to adults, in whom it can cause high blood pressure, digestive problems, nerve disorders, memory loss, and muscle and joint **pain**. In addition, it can lead to difficulties during **pregnancy**, as well as cause reproductive problems in both men and women.

More recently, chronic exposure to lead in the environment has been found to speed up the progression of kidney disorders in patients without diabetes.

Acute lead poisoning

Acute lead poisoning, while less common, shows up more quickly and can be fatal. Symptoms such as the following may occur:

- severe abdominal pain
- diarrhea
- nausea and **vomiting**
- weakness of the limbs
- seizures
- coma

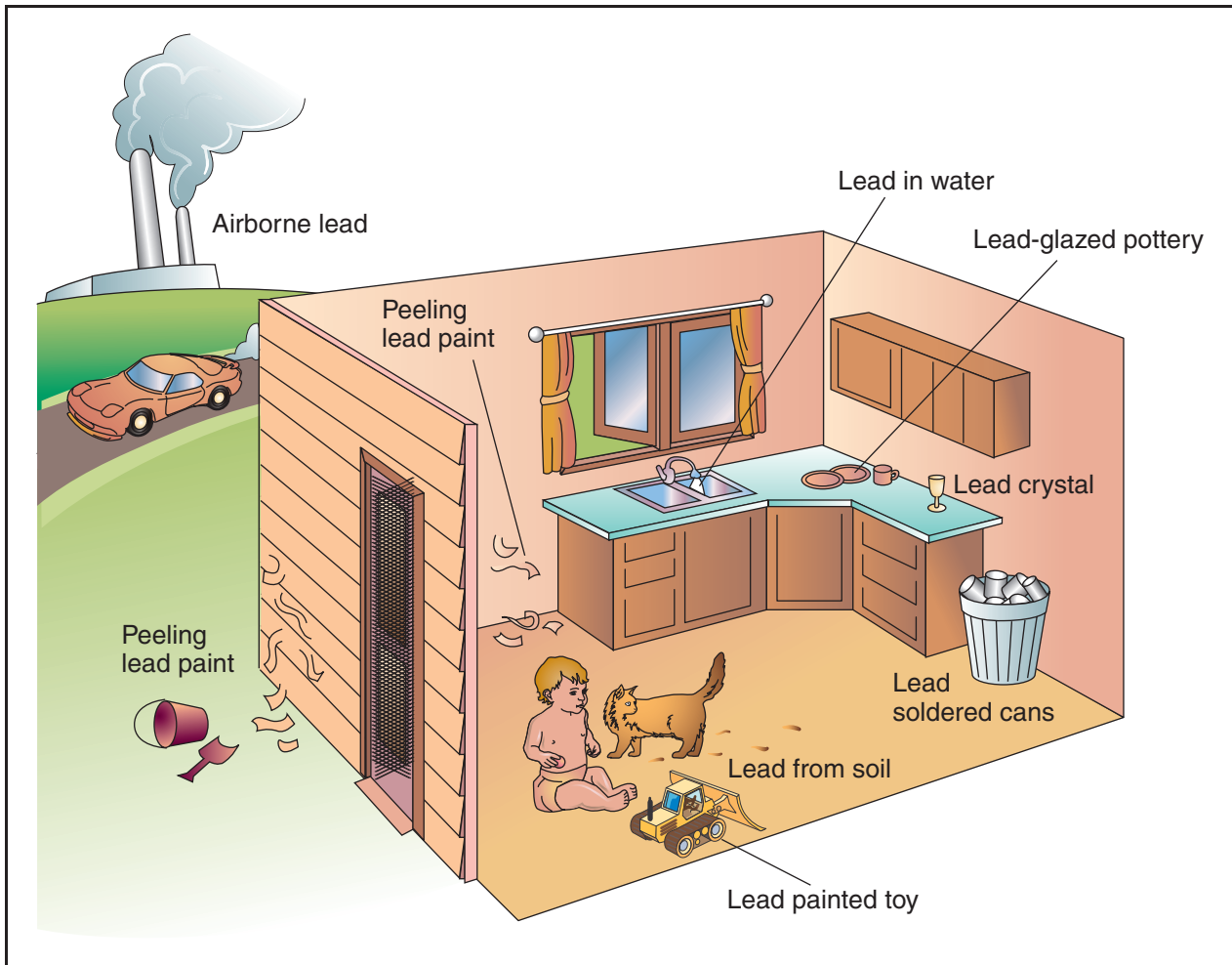
Diagnosis

A high level of lead in the blood can be detected with a simple blood test. In fact, testing is the only way to know for sure if children without symptoms have been exposed to lead, since they can appear healthy even as long-term damage occurs. The CDC recommends testing all children at 12 months of age and, if possible, again at 24 months. Testing should start at six months for children at risk for lead poisoning. Based on these test results and a child's risk factors, the doctor will then decide whether further testing is needed and how often. In some states, more frequent testing is required by law.

Children at risk

Children with an increased risk of lead poisoning include those who:

- Live in or regularly visit a house built before 1978 in which chipped or peeling paint is present.
- Live in or regularly visit a house that was built before 1978 where remodeling is planned or underway.



Continuous exposure to lead can damage nearly every system in the human body and is particularly harmful to the developing brain of fetuses and young children. Common sources of lead exposure include lead-based paint, dust and soil, drinking water, food from cans, and eating utensils, such as plates and drinking glasses, that are lead-based. (Illustration by Electronic Illustrators Group.)

- Have a brother or sister, housemate, or playmate who has been diagnosed with lead poisoning.
- Have the habit of eating dirt, or have been diagnosed with pica.
- Live with an adult whose job or hobby involves exposure to lead.
- Live near an active lead smelter, battery-recycling plant, or other industry that can create lead pollution.
- glazed pottery or stained glass making
- furniture refinishing
- home renovation
- target shooting at indoor firing ranges
- battery reclamation
- precious metal refining
- radiator repair
- art restoration

Adults at risk

Testing is also important for adults whose job or hobby puts them at risk for lead poisoning. This includes people who take part in the following activities:

Treatment

The first step in treating lead poisoning is to avoid further contact with lead. For adults, this usually

means making changes at work or in hobbies. For children, it means finding and removing sources of lead in the home. In most states, the public health department can help assess the home and identify lead sources.

If the problem is lead paint, a professional with special training should remove it. Removal of lead paint is not a do-it-yourself project. Scraping or sanding lead paint creates large amounts of dust that can poison people in the home. This dust can stay around long after the work is completed. In addition, heating lead paint can release lead into the air. For these reasons, lead paint should only be removed by someone who knows how to do the job safely and has the equipment to clean up thoroughly. Occupants, especially children and pregnant women, should leave the home until the cleanup is finished.

Medical professionals should take all necessary steps to remove bullets or bullet fragments from patients with gunshot injuries.

Chelation therapy

If blood levels of lead are high enough, the doctor may also prescribe **chelation therapy**. This refers to treatment with chemicals that bind to the lead and help the body pass it in urine at a faster rate. There are four chemical agents that may be used for this purpose, either alone or in combination. Edetate calcium disodium (EDTA calcium) and dimercaprol (BAL) are given through an intravenous line or in shots, while succimer (Chemet) and penicillamine (Cuprimine, Depen) are taken by mouth. (Although many doctors prescribe penicillamine for lead poisoning, this use of the drug has not been approved by the Food and Drug Administration.)

Alternative treatment

Changes in diet are no substitute for medical treatment. However, getting enough calcium, zinc, and protein may help reduce the amount of lead the body absorbs. Iron is also important, since people who are deficient in this nutrient absorb more lead. Garlic and thiamine, a B-complex vitamin, have been used to treat lead poisoning in animals. However, their usefulness in humans for this purpose has not been proved. Nutritional, botanical, and homeopathic medicines can be administered once the source is removed, to help correct any imbalances brought on by lead toxicity.

KEY TERMS

Chelation therapy—Treatment with chemicals that bind to a poisonous metal and help the body pass it in urine at a faster rate.

Dimercaprol (BAL)—A chemical agent used to remove excess lead from the body.

Edetate calcium disodium (EDTA calcium)—A chemical agent used to remove excess lead from the body.

Penicillamine (Cuprimine, Depen)—A drug used to treat medical problems (such as excess copper in the body and rheumatoid arthritis) and to prevent kidney stones. It is also sometimes prescribed to remove excess lead from the body.

Pica—An abnormal appetite or craving for non-food items, often such substances as chalk, clay, dirt, laundry starch, or charcoal.

Succimer (Chemet)—A drug used to remove excess lead from the body.

Prognosis

If acute lead poisoning reaches the stage of seizures and coma, there is a high risk of death. Even if the person survives, there is a good chance of permanent brain damage. The long-term effects of lower levels of lead can also be permanent and severe. However, if chronic lead poisoning is caught early, these negative effects can be limited by reducing future exposure to lead and getting proper medical treatment.

Prevention

Many cases of lead poisoning can be prevented. These steps can help:

- Keep the areas where children play as clean and dust-free as possible.
- Wash pacifiers and bottles when they fall to the floor, and wash stuffed animals and toys often.
- Make sure children wash their hands before meals and at bedtime.
- Mop floors and wipe windowsills and other chewable surfaces, such as cribs, twice a week with a solution of powdered dishwasher detergent in warm water.
- Plant bushes next to an older home with painted exterior walls to keep children at a distance.

- Plant grass or another ground cover in soil that is likely to be contaminated, such as soil around a home built before 1960 or located near a major highway.
- Have household tap water tested to find out if it contains lead.
- Use only water from the cold-water tap for drinking, cooking, and making baby formula, since hot water is likely to contain higher levels of lead.
- If the cold water hasn't been used for six hours or more, run it for several seconds, until it becomes as cold as it will get, before using it for drinking or cooking. The more time water has been sitting in the pipes, the more lead it may contain.
- If you work with lead in your job or hobby, change your clothes before you go home.
- Do not store food in open cans, especially imported cans.
- Do not store or serve food in pottery meant for decorative use.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD., editors. "Poisoning: Lead Poisoning." In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

PERIODICALS

Gavaghan, Helen. "Lead, Unsafe at Any Level." *Bulletin of the World Health Organization* January 2002: 82.

Kaufmann, R. B., C. J. Staes, and T. D. Matte. "Deaths Related to Lead Poisoning in the United States, 1979-1998." *Environmental Research* 91 (February 2003): 78-84.

Lanphear, B. P., K. N. Dietrich, and O. Berger. "Prevention of Lead Toxicity in US Children." *Ambulatory Pediatrics* 3 (January-February 2003): 27-36.

Lidsky, T. I., and J. S. Schneider. "Lead Neurotoxicity in Children: Basic Mechanisms and Clinical Correlates." *Brain* 126, Part 1 (January 2003): 5-19.

Lin, J. L., D. T. Lin-Tan, K. H. Hsu, and C. C. Yu. "Environmental Lead Exposure and Progression of Chronic Renal Diseases in Patients Without Diabetes." *New England Journal of Medicine* 348 (January 23, 2003): 277-286.

"National Campaign to Promote New 24/7 Poison Hotline." *Medical Letter on the CDC & FDA* March 10, 2002: 12.

Shannon, M. "Severe Lead Poisoning in Pregnancy." *Ambulatory Pediatrics* 3 (January-February 2003): 37-39.

Tarkin, I. S., A. Hatzidakis, S. C. Hoxie, et al. "Arthroscopic Treatment of Gunshot Wounds to the Shoulder." *Arthroscopy* 19 (January 2003): 85-89.

"Tofu May Lower Lead Levels in Blood." *Townsend Letter for Doctors and Patients* February-March 2002: 23.

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

National Lead Information Center, National Safety Council. 1025 Connecticut Ave. N.W., Suite 1200, Washington, DC 20036. (800) 532-3394. <<http://www.nsc.org/ehc/lead.htm>>.

Office of Water Resources Center, Environmental Protection Agency. Mail Code (4100), Room 2615 East Tower Basement, 401 M St. S.W., Washington, DC 20460. (800) 426-4791. <<http://www.epa.gov/ow/>>.

Linda Wasmer Smith
Rebecca J. Frey, PhD

Learning disorders

Definition

Learning disorders are academic difficulties experienced by children and adults of average to above-average intelligence. People with learning disorders have difficulty with reading, writing, mathematics, or a combination of the three. These difficulties significantly interfere with academic achievement or daily living.

Description

Learning disorders, or disabilities, affect approximately 2 million children between the ages of six and 17 (5% of public school children), although some experts think the figure may be as high as 15%. These children have specific impairments in acquiring, retaining, and processing information. Standardized tests place them well below their IQ range in their area of difficulty. The three main types of learning disorders are reading disorders, mathematics disorders, and disorders of written expression. The male: female ratio for learning disorders is about 5:1.

Reading disorders

Reading disorders are the most common type of learning disorder. Children with reading disorders have difficulty recognizing and interpreting letters and words (**dyslexia**). They aren't able to recognize

and decode the sounds and syllables (phonetic structure) behind written words and language in general. This condition lowers accuracy and comprehension in reading.

Mathematics disorders

Children with mathematics disorders (dyscalculia) have problems recognizing and counting numbers correctly. They have difficulty using numbers in everyday settings. Mathematics disorders are typically diagnosed in the first few years of elementary school when formal teaching of numbers and basic math concepts begins. Children with mathematics disorders usually have a co-existing reading disorder, a disorder of written expression, or both.

Disorders of written expression

Disorders of written expression typically occur in combination with reading disorders or mathematics disorders or both. The condition is characterized by difficulty with written compositions (dysgraphia). Children with this type of learning disorder have problems with spelling, punctuation, grammar, and organizing their thoughts in writing.

Causes and symptoms

Learning disorders are thought to be caused by neurological abnormalities that trigger impairments in the regions of the brain that control visual and language processing and attention and planning. These traits may be genetically linked. Children from families with a history of learning disorders are more likely to develop disorders themselves. In 2003 a team of Finnish researchers reported finding a candidate gene for developmental dyslexia on human chromosome 15q21.

Learning difficulties may also be caused by such medical conditions as a traumatic brain injury or brain infections such as **encephalitis** or **meningitis**.

The defining symptom of a learning disorder is academic performance that is markedly below a child's age and grade capabilities and measured IQ. Children with a reading disorder may confuse or transpose words or letters and omit or add syllables to words. The written homework of children with disorders of written expression is filled with grammatical, spelling, punctuation, and organizational errors. The child's handwriting is often extremely poor. Children with mathematical disorders are often unable to count in the correct sequence, to name numbers, and to understand numerical concepts.

KEY TERMS

Dyslexia—An inability to read, write, or spell words in spite of the ability to see and recognize letters. Dyslexia is an autosomal dominant disorder that occurs more frequently in males.

IQ—Intelligence quotient; a measure of intellectual functioning determined by performance on standardized intelligence tests.

Phonics—A system to teach reading by teaching the speech sounds associated with single letters, letter combinations, and syllables.

Diagnosis

Problems with vision or hearing, mental disorders (depression, **attention-deficit/hyperactivity disorder**), **mental retardation**, cultural and language differences, and inadequate teaching may be mistaken for learning disorders or complicate a diagnosis. A comprehensive medical, psychological, and educational assessment is critical to making a clear and correct diagnosis.

A child thought to have a learning disorder should undergo a complete medical examination to rule out an organic cause. If none is found, a psychoeducational assessment should be performed by a psychologist, psychiatrist, neurologist, neuropsychologist, or learning specialist. A complete medical, family, social, and educational history is compiled from existing medical and school records and from interviews with the child and the child's parents and teachers. A series of written and verbal tests are then given to the child to evaluate his or her cognitive and intellectual functioning. Commonly used tests include the Wechsler Intelligence Scale for Children (WISC-III), the Woodcock-Johnson Psychoeducational Battery, the Peabody Individual Achievement Test-Revised (PIAT-R) and the California Verbal Learning Test (CVLT). Federal legislation mandates that this testing is free of charge within the public school system.

Treatment

Once a learning disorder has been diagnosed, an individual education plan (IEP) is developed for the child in question. IEPs are based on psychoeducational test findings. They provide for annual retesting to measure a child's progress. Learning-disordered students may receive special instruction within a regular general education class or they may be taught in a special education or learning center for a portion of the day.

Common strategies for the treatment of reading disorders focus first on improving a child's recognition of the sounds of letters and language through phonics training. Later strategies focus on comprehension, retention, and study skills. Students with disorders of written expression are often encouraged to keep journals and to write with a computer keyboard instead of a pencil. Instruction for students with mathematical disorders emphasizes real-world uses of arithmetic, such as balancing a checkbook or comparing prices.

Prognosis

Resources

BOOKS

- American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders*. 4th ed., revised. Washington, DC: American Psychiatric Association, 2000.
- Beers, Mark H., MD, and Robert Berkow, MD., editors. "Learning Disorders." In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

PERIODICALS

- Galaburda, D. M., and B. C. Duchaine. "Developmental Disorders of Vision." *Neurologic Clinics* 21 (August 2003): 687–707.
- Gillberg, C., and H. Soderstrom. "Learning Disability." *Lancet* 362 (September 6, 2003): 811–821.
- Taipale, M., N. Kaminen, J. Nopola-Hemmi, et al. "A Candidate Gene for Developmental Dyslexia Encodes a Nuclear Tetratricopeptide Repeat Domain Protein Dynamically Regulated in Brain." *Proceedings of the National Academy of Sciences in the USA* 100 (September 30, 2003): 11553–11558.
- Taylor, K. E., and J. Walter. "Occupation Choices of Adults With and Without Symptoms of Dyslexia." *Dyslexia* 9 (August 2003): 177–185.
- Witt, W. P., A. W. Riley, and M. J. Coiro. "Childhood Functional Status, Family Stressors, and Psychosocial Adjustment Among School-Aged Children with Disabilities in the United States." *Archives of Pediatric and Adolescent Medicine* 157 (July 2003): 687–695.

ORGANIZATIONS

- The Interactive Guide to Learning Disabilities for Parents, Teachers, and Children. <<http://www.ldonline.org>>.
- Learning Disabilities Association of America. 4156 Library Road, Pittsburgh, PA 15234. (412) 341-1515. <<http://www.ldanatl.org>>.
- National Center for Learning Disabilities (NCLD). 381 Park Avenue South, Suite 1401, New York, NY 10016. (410) 296-0232. <<http://www.nclld.org>>.

OTHER

LD Online Page. <<http://www.ldonline.org>>.

Paula Anne Ford-Martin
Rebecca J. Frey, PhD

Leeches

Definition

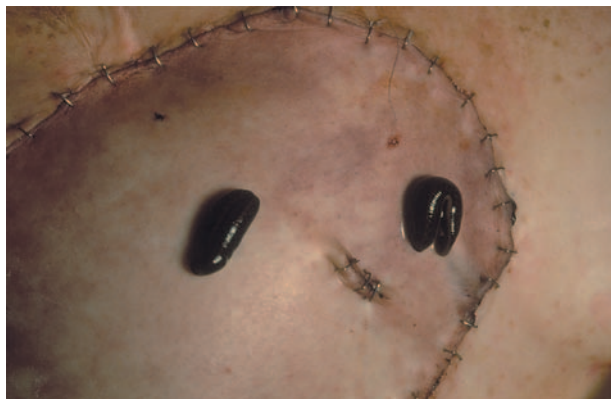
Leeches are bloodsucking worms with segmented bodies. They belong to the same large classification of worms as earthworms and certain oceanic worms.

Leeches can primarily be found in freshwater lakes, ponds, or rivers. They range in size from 0.2 in (5 mm) to nearly 18 in (45 cm) and have two characteristic suckers located at either end of their bodies. Leeches consume the blood of a wide variety of animal hosts, ranging from fish to humans. To feed, a leech first attaches itself to the host using the suckers. One of these suckers surrounds the leech's mouth, which contains three sets of jaws that bite into the host's flesh, making a Y-shaped incision. As the leech begins to feed, its saliva releases chemicals that dilate blood vessels, thin the blood, and deaden the **pain** of the bite. Because of the saliva's effects, a person bitten by a leech may not even be aware of it until afterwards, when he or she sees the incision and the trickle of blood that is difficult to stop.

For centuries, leeches were a common tool of doctors, who believed that many diseases were the result of "imbalances" in the body that could be stabilized by releasing blood. For example, leeches were sometimes attached to veins in the temples to treat headaches. Advances in medical knowledge led doctors to abandon bloodletting and the use of leeches in the mid-nineteenth century. In recent years, however, doctors have found a new purpose for leeches—helping to restore blood circulation to grafted or severely injured tissue.

Purpose

There are many occasions in medicine, mostly in surgery and trauma care, when blood accumulates and causes trouble. Leeches can be used to reduce the swelling of any tissue that is holding too much blood. This problem is most likely to occur in two situations:



These leeches are being used to reduce venous congestion, or excessive amounts of blood in the blood vessels.
(Photograph by Michael English, M.D., Custom Medical Stock Photo. Reproduced by permission.)

- Trauma. Large **blood clots** resulting from trauma can threaten tissue survival by their size and pressure. Blood clots can also obstruct the patient's airway.
- Surgical procedures involving reattachment of severed body parts or tissue reconstruction following **burns**. In these situations it is difficult for the surgeon to make a route for blood to leave the affected part and return to the circulation. The hardest part of reattaching severed extremities like fingers, toes and ears is to reconnect the tiny veins. If the veins are not reconnected, blood will accumulate in the injured area. A similar situation occurs when plastic surgeons move large flaps of skin to replace skin lost to burns, trauma or radical surgery. The skin flaps often drain blood poorly, get congested, and begin to die. Leeches have come to the rescue in both situations.

Precautions

It is important to use only leeches that have been raised in the laboratory under sterile conditions in order to protect patients from infection. Therapeutic leeches belong to one of two species—*Hirudo michaelsoni* or *Hirudo medicinalis*.

Description

One or more leeches are applied to the swollen area, depending on the size of the graft or injury, and left on for several hours. The benefits of the treatment lie not in the amount of blood that the leeches ingest, but in the anti-bloodclotting (anticoagulant) enzymes in the saliva that allow blood to flow from the bite for up to six hours after the animal is detached, effectively draining away blood that could otherwise accumulate

KEY TERMS

Anemia—A blood disorder marked by low hemoglobin levels in red blood cells, which leads to a deficiency of oxygen in the blood.

Anticoagulant—A chemical or medication that prevents blood from clotting.

and cause tissue death. Leech saliva has been described as a better anticoagulant than many currently available to treat strokes and heart attacks. Active investigation of the chemicals in leech saliva is currently under way, and one anticoagulant drug, hirudin, is derived from the tissues of *Hirudo medicinalis*.

Aftercare

The leeches are removed by pulling them off or by loosening their grip with **cocaine**, heat, or acid. The used leeches are then killed by placing them in an alcohol solution and disposed of as a biohazard. Proper care of the patient's sore is important, as is monitoring the rate at which it bleeds after the leech is removed. Any clots that form at the wound site during treatment should be removed to ensure effective blood flow.

Risks

Infection is a constant possibility until the sore heals. It is also necessary to monitor the amount of blood that the leeches have removed from the patient, since a drop in red blood cell counts could occur in rare cases of prolonged bleeding.

Resources

PERIODICALS

Daane, S., et al. "Clinical Use of Leeches in Reconstructive Surgery." *American Journal of Orthopedics* 26, no. 8 (August 1997): 528-532.

J. Ricker Polsdorfer, MD

Left ventricular failure see **Heart failure**

Leg veins x ray see **Venography**

Legg-Calvé see **Osteochondroses**

Legionella pneumophila infection see **Legionnaires' disease**

Legionellosis see **Legionnaires' disease**

Legionnaires' disease

Definition

Legionnaires' disease is a type of **pneumonia** caused by *Legionella* bacteria. The bacterial species responsible for Legionnaires' disease is *L. pneumophila*. Major symptoms include **fever**, chills, muscle aches, and a **cough** that is initially nonproductive. Definitive diagnosis relies on specific laboratory tests for the bacteria, bacterial antigens, or antibodies produced by the body's immune system. As with other types of pneumonia, Legionnaires' disease poses the greatest threat to people who are elderly, ill, or immunocompromised.

Description

Legionella bacteria were first identified as a cause of pneumonia in 1976, following an outbreak of pneumonia among people who had attended an American Legion convention in Philadelphia, Pennsylvania. This eponymous outbreak prompted further investigation into *Legionella* and it was discovered that earlier unexplained pneumonia outbreaks were linked to the bacteria. The earliest cases of Legionnaires' disease were shown to have occurred in 1965, but samples of the bacteria exist from 1947.

Exposure to the *Legionella* bacteria doesn't necessarily lead to infection. According to some studies, an estimated 5-10% of the American population show serologic evidence of exposure, the majority of whom do not develop symptoms of an infection. *Legionella* bacteria account for 2-15% of the total number of pneumonia cases requiring hospitalization in the United States.

There are at least 40 types of *Legionella* bacteria, half of which are capable of producing disease in humans. A disease that arises from infection by *Legionella* bacteria is referred to as legionellosis. The *L. pneumophila* bacterium, the root cause of Legionnaires' disease, causes 90% of legionellosis cases. The second most common cause of legionellosis is the *L. micdadei* bacterium, which produces the Philadelphia pneumonia-causing agent.

Approximately 10,000-40,000 people in the United States develop Legionnaires' disease annually. The people who are the most likely to become ill are over age 50. The risk is greater for people who suffer from health conditions such as malignancy, diabetes, lung disease, or **kidney disease**. Other risk factors include immunosuppressive therapy and cigarette **smoking**. Legionnaires' disease has occurred in children, but typically it has been confined to newborns

receiving respiratory therapy, children who have had recent operations, and children who are immunosuppressed. People with HIV infection and **AIDS** do not seem to contract Legionnaires' disease with any greater frequency than the rest of the population, however, if contracted, the disease is likely to be more severe compared to other cases.

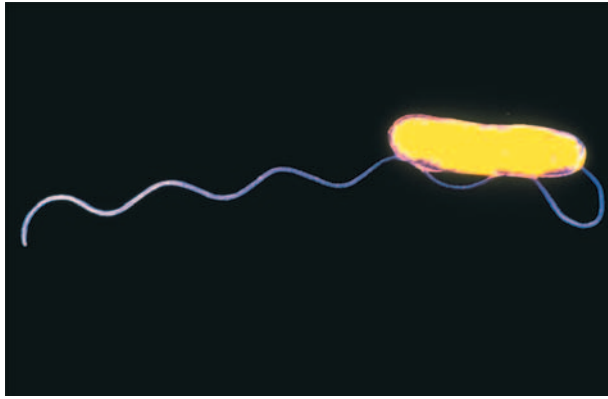
Cases of Legionnaires' disease that occur in conjunction with an outbreak, or epidemic, are more likely to be diagnosed quickly. Early diagnosis aids effective and successful treatment. During epidemic outbreaks, fatalities have ranged from 5% for previously healthy individuals to 24% for individuals with underlying illnesses. Sporadic cases (that is, cases unrelated to a wider outbreak) are harder to detect and treatment may be delayed pending an accurate diagnosis. The overall fatality rate for sporadic cases ranges from 10-19%. The outlook is bleaker in severe cases that require respiratory support or dialysis. In such cases, fatality may reach 67%.

Causes and symptoms

Legionnaires' disease is caused by inhaling *Legionella* bacteria from the environment. Typically, the bacteria are dispersed in aerosols of contaminated water. These aerosols are produced by devices in which warm water can stagnate, such as air-conditioning cooling towers, humidifiers, shower heads, and faucets. There have also been cases linked to whirlpool spa baths and water misters in grocery store produce departments. Aspiration of contaminated water is also a potential source of infection, particularly in hospital-acquired cases of Legionnaires' disease. There is no evidence of person-to-person transmission of Legionnaires' disease.

Once the bacteria are in the lungs, cellular representatives of the body's immune system (alveolar macrophages) congregate to destroy the invaders. The typical macrophage defense is to phagocytose the invader and demolish it in a process analogous to swallowing and digesting it. However, the *Legionella* bacteria survive being phagocytosed. Instead of being destroyed within the macrophage, they grow and replicate, eventually killing the macrophage. When the macrophage dies, many new *Legionella* bacteria are released into the lungs and worsen the infection.

Legionnaires' disease develops 2-10 days after exposure to the bacteria. Early symptoms include lethargy, headaches, fever, chills, muscle aches, and a lack of appetite. Respiratory symptoms such as coughing or congestion are usually absent. As the disease progresses, a dry, hacking cough develops



A transmission electron microscopy (TEM) image of *Legionella pneumophila*, the bacteria which causes Legionnaires' disease. (Custom Medical Stock Photo. Reproduced by permission.)

and may become productive after a few days. In about a third of Legionnaires' disease cases, blood is present in the sputum. Half of the people who develop Legionnaires' disease suffer **shortness of breath** and a third complain of breathing-related chest **pain**. The fever can become quite high, reaching 104 °F (40 °C) in many cases, and may be accompanied by a decreased heart rate.

Although the pneumonia affects the lungs, Legionnaires' disease is accompanied by symptoms that affect other areas of the body. About half the victims experience **diarrhea** and a quarter have **nausea and vomiting** and abdominal pain. In about 10% of cases, acute renal failure and scanty urine production accompany the disease. Changes in mental status, such as disorientation, confusion, and **hallucinations**, also occur in about a quarter of cases.

In addition to Legionnaires' disease, *L. pneumophila* legionellosis also includes a milder disease, Pontiac fever. Unlike Legionnaires' disease, Pontiac fever does not involve the lower respiratory tract. The symptoms usually appear within 36 hours of exposure and include fever, **headache**, muscle aches, and lethargy. Symptoms last only a few days and medical intervention is not necessary.

Diagnosis

The symptoms of Legionnaires' disease are common to many types of pneumonia and diagnosis of sporadic cases can be difficult. The symptoms and chest x rays that confirm a case of pneumonia are not useful in differentiating between Legionnaires' disease and other pneumonias. If a pneumonia case involves multisystem symptoms, such as diarrhea and

vomiting, and an initially dry cough, laboratory tests are done to definitively identify *L. pneumophila* as the cause of the infection.

If Legionnaires' disease is suspected, several tests are available to reveal or indicate the presence of *L. pneumophila* bacteria in the body. Since the immune system creates antibodies against infectious agents, examining the blood for these indicators is a key test. The level of immunoglobulins, or antibody molecules, in the blood reveals the presence of infection. In microscopic examination of the patient's sputum, a fluorescent stain linked to antibodies against *L. pneumophila* can uncover the presence of the bacteria. Other means of revealing the bacteria's presence from patient sputum samples include isolation of the organism on culture media or detection of the bacteria by DNA probe. Another test detects *L. pneumophila* antigens in the urine.

Treatment

Most cases of *Legionella* pneumonia show improvement within 12-48 hours of starting antibiotic therapy. The antibiotic of choice has been erythromycin, sometimes paired with a second antibiotic, rifampin. Tetracycline, alone or with rifampin, is also used to treat Legionnaires' disease, but has had more mixed success in comparison to erythromycin. Other **antibiotics** that have been used successfully to combat *Legionella* include doxycycline, clarithromycin, fluorinated quinolones, and trimethoprim/sulfamethoxazole.

The type of antibiotic prescribed by the doctor depends on several factors including the severity of infection, potential **allergies**, and interaction with previously prescribed drugs. For example, erythromycin interacts with warfarin, a blood thinner. Several drugs, such as **penicillins** and **cephalosporins**, are ineffective against the infection. Although they may be deadly to the bacteria in laboratory tests, their chemical structure prevents them from being absorbed into the areas of the lung where the bacteria are present.

In severe cases with complications, antibiotic therapy may be joined by respiratory support. If renal failure occurs, dialysis is required until renal function is recovered.

Prognosis

Appropriate medical treatment has a major impact on recovery from Legionnaires' disease. Outcome is also linked to the victim's general health and absence of complications. If the patient survives

KEY TERMS

Antibody—A molecule created by the immune system in response to the presence of an antigen. It serves to recognize the invader and help defend the body from infection.

Antigen—A molecule, such as a protein, which is associated with a particular infectious agent. The immune system uses this molecule as the identifying characteristic of the infectious invader.

Culture—A laboratory system for growing bacteria for further study.

DNA probe—An agent that binds directly to a pre-defined sequence of nucleic acids.

Immunocompromised—Refers to conditions in which the immune system is not functioning properly and cannot adequately protect the body from infection.

Immunoglobulin—The protein molecule that serves as the primary building block of antibodies.

Immunosuppressive therapy—Medical treatment in which the immune system is purposefully thwarted. Such treatment is necessary, for example, to prevent organ rejection in transplant cases.

Legionellosis—A disease caused by infection with a *Legionella* bacterium.

Media—Substance which contains all the nutrients necessary for bacteria to grow in a culture.

Phagocytosis—The “ingestion” of a piece of matter by a cell.

the infection, recovery from Legionnaires’ disease is complete. Similar to other types of pneumonia, severe cases of Legionnaires’ disease may cause scarring in the lung tissue as a result of the infection. Renal failure, if it occurs, is reversible and renal function returns as the patient’s health improves. Occasionally, **fatigue** and weakness may linger for several months after the infection has been successfully treated.

Prevention

Since the bacteria thrive in warm stagnant water, regularly disinfecting ductwork, pipes, and other areas that may serve as breeding areas is the best method for preventing outbreaks of Legionnaires’ disease. Most outbreaks of Legionnaires’ disease can be traced to specific points of exposure, such as hospitals, hotels, and other places where people gather. Sporadic cases

are harder to determine and there is insufficient evidence to point to exposure in individual homes.

Resources

PERIODICALS

Shuman, H. A., et al. “Intracellular Multiplication of *Legionella pneumophila*: Human Pathogen of Accidental Tourist?” *Current Topics in Microbiology and Immunology* 225 (1998): 99.

Julia Barrett

Leiomyomas see **Uterine fibroids**

Leishmaniasis

Definition

Leishmaniasis refers to several different illnesses caused by infection with an organism called a protozoan.

Description

Protozoa are considered to be the most simple organisms in the animal kingdom. They are all single-celled. The types of protozoa that cause leishmaniasis are carried by the blood-sucking sandfly. The sandfly is referred to as the disease vector, simply meaning that the infectious agent (the protozoan) is carried by the sandfly and passed on to other animals or humans in whom the protozoan will set up residence and cause disease. The animal or human in which the protozoan then resides is referred to as the host.

Once the protozoan is within the human host, the human’s immune system is activated to try to combat the invader. Specialized immune cells called macrophages work to swallow up the protozoa. Usually, this technique kills a foreign invader, but these protozoa can survive and flourish within macrophages. The protozoa multiply within the macrophages, ultimately causing the macrophage to burst open. The protozoa are released, and take up residence within other neighboring cells.

At this point, the course of the disease caused by the protozoa is dependent on the specific type of protozoa, and on the type of reaction the protozoa elicits from the immune system. There are several types of protozoa that cause leishmaniasis, and they cause different patterns of disease progression.

At any one time, about 20 million people throughout the world are infected with leishmaniasis. Between one million and one and one-half million cases of cutaneous leishmaniasis are reported yearly worldwide. While leishmaniasis exists as a disease in 88 countries on five continents, some countries are hit harder than others. These include Bangladesh, India, Nepal, Sudan, Afghanistan, Brazil, Iran, Peru, Saudi Arabia, and Syria. Other areas that harbor the causative protozoa include China, many countries throughout Africa, Mexico, Central and South America, Turkey, and Greece. Although less frequent, cases have occurred in the United States, in Texas.

As Americans travel to these countries, they will come in contact the protozoa that cause forms of leishmaniasis. Also, physicians were advised in 2004 to suspect cutaneous leishmaniasis in military personnel who were deployed to areas where the infection is present. From August 2002 to February 2004, staff from the U.S. Department of Defense identified 522 confirmed cases of the disease in American military personnel.

In some areas of southern Europe, leishmaniasis is becoming an important disease that infects people with weakened immune systems. In particular, individuals with acquired **immunodeficiency** syndrome (**AIDS**) are at great risk of this infection.

Causes and symptoms

There are a number of types of protozoa that can cause leishmaniasis. Each type exists in specific locations, and there are different patterns to the kind of disease each causes. The overall species name is *Leishmania* (commonly abbreviated *L.*). The specific types include: *L. Donovan*i, *L. Infantum*, *L. Chagasi*, *L. Mexicana*, *L. Amazonensis*, *L. Tropica*, *L. Major*, *L. Aethiopica*, *L. Brasiliensis*, *L. Guyaensis*, *L. Panamensis*, *L. Peruviana*. Some of the names are reflective of the locale in which the specific protozoa is most commonly found, or in which it was first discovered.

Localized cutaneous leishmaniasis

This type of disease occurs most commonly in China, India, Asia Minor, Africa, the Mediterranean Basin, and Central America. It has occurred in an area ranging from northern Argentina all the way up to southern Texas. It is called different names in different locations, including chiclero ulcer, bush **yaws**, uta, oriental sore, Aleppo boil, and Baghdad sore.

This is perhaps the least drastic type of disease caused by any of the *Leishmania*. Several weeks or

months after being bitten by an infected sandfly, the host may notice an itchy bump (lesion) on an arm, leg, or face. Lymph nodes in the area of this bump may be swollen. Within several months, the bump develops a crater (ulceration) in the center, with a raised, reddened ridge around it. There may be several of these lesions near each other, and they may spread into each other to form one large lesion. Although localized cutaneous leishmaniasis usually heals on its own, it may take as long as one year. A depressed, light-colored scar usually remains behind. Some lesions never heal, and may invade and destroy the tissue below. For example, lesions on the ears may slowly, but surely, invade and destroy the cartilage that supports the outer ear.

Diffuse cutaneous leishmaniasis

This type of disease occurs most often in Ethiopia, Brazil, Dominican Republic, and Venezuela.

The lesions of diffuse cutaneous leishmaniasis are very similar to those of localized cutaneous leishmaniasis, except they are spread all over the body. The body's immune system apparently fails to battle the protozoa, which are free to spread throughout. The characteristic lesions resemble those of the dread biblical disease, **leprosy**.

Mucocutaneous leishmaniasis

This form of leishmaniasis occurs primarily in the tropics of South America. The disease begins with the same sores noted in localized cutaneous leishmaniasis. Sometimes these primary lesions heal, other times they spread and become larger. Some years after the first lesion is noted (and sometimes several years after that lesion has totally healed), new lesions appear in the mouth and nose, and occasionally in the area between the genitalia and the anus (the perineum). These new lesions are particularly destructive and painful. They erode underlying tissue and cartilage, frequently eating through the septum (the cartilage that separates the two nostrils). If the lesions spread to the roof of the mouth and the larynx (the part of the wind pipe which contains the vocal cords), they may prevent speech. Other symptoms include **fever**, weight loss, and anemia (low red blood cell count). There is always a large danger of bacteria infecting the already open sores.

Visceral leishmaniasis

This type of leishmaniasis occurs in India, China, the southern region of Russia, and throughout Africa,



This condition, also called an oriental sore, is caused by the bacterium *L. tropica*. (Photograph by Lester V. Bergman, Corbis Images. Reproduced by permission.)

the Mediterranean, and South and Central America. It /is frequently called Kala-Azar or Dumdum fever.

In this disease, the protozoa use the bloodstream to travel to the liver, spleen, lymph nodes, and bone marrow. Fever may last for as long as eight weeks, disappear, and then reappear again. The lymph nodes, spleen, and liver are often quite enlarged. Weakness, **fatigue**, loss of appetite, **diarrhea**, and weight loss are common. Kala-azar translates to mean “black fever.” The name kala-azar comes from a characteristic of this form of leishmaniasis. Individuals with light-colored skin take on a darker, grayish skin tone, particularly of their face and hands. A variety of lesions appear on the skin.

Diagnosis

Diagnosis for each of these types of leishmaniasis involves taking a scraping from a lesion, preparing it in a laboratory, and examining it under a microscope to demonstrate the causative protozoan. Other methods that have been used include culturing a sample piece of tissue in a laboratory to allow the protozoa to multiply for easier microscopic identification; injecting a mouse or hamster with a solution made of scrapings from a patient’s lesion to see if the animal develops a leishmaniasis-like disease; and demonstrating the presence in macrophages of the characteristic-appearing protozoan, called Leishman-Donovan bodies.

In some forms of leishmaniasis, a skin test (similar to that given for TB) may be used. In this test, a solution containing a small bit of the protozoan

antigen (cell marker that causes the human immune system to react) is injected or scratched into a patient’s skin. In a positive reaction, cells from the immune system will race to this spot, causing a characteristic skin lesion. Not all forms of leishmaniasis cause a positive skin test, however.

Treatment

The treatment of choice for all forms of leishmaniasis is a type of drug containing the element antimony. These include sodium stibogluconate, and meglumine antimonate. When these types of drugs do not work, other medications with anti-protozoal activity are utilized, including amphotericin B, pentamidine, flagyl, and allopurinol. In 2004, it was reported that the world’s first non-profit drug company was seeking approval in India for a drug to cure visceral leishmaniasis. An estimated 200,000 people die annually from the disease in that country. The company, called OneWorld Health, hoped to offer the drug called paromomycin for a day for a three-week treatment course.

Prognosis

The prognosis for leishmaniasis is quite variable, and depends on the specific strain of infecting protozoan, as well as the individual patient’s immune system response to infection. Localized cutaneous leishmaniasis may require no treatment. Although it may take many months, these lesions usually heal themselves completely. Only rarely do these lesions fail to heal and become more destructive.

Disseminated cutaneous leishmaniasis may smolder on for years without treatment, ultimately causing **death** when the large, open lesions become infected with bacteria.

Mucocutaneous leishmaniasis is often relatively resistant to treatment. Untreated visceral leishmaniasis has a 90% death rate, but only a 10% death rate with treatment.

Prevention

Prevention involves protecting against sandfly bites. Insect repellents used around homes, on clothing, on skin, and on bednets (to protect people while sleeping) are effective measures.

Reducing the population of sandflies is also an important preventive measure. In areas where leishmaniasis is very common, recommendations include clearing the land of trees and brush for at least 984 ft

KEY TERMS

Host—The organism (such as a monkey or human) in which another organism (such as a virus or bacteria) is living.

Larynx—The part of the airway lying between the pharynx and the trachea.

Leishman-Donovan body—A body of a (trypanosomatid) protozoa at a particular and characteristic stage in its life cycle; the infectious (trypanosomatid) protozoa can cause leishmaniasis, and is relatively easy to identify at that stage.

Lesion—A disruption of the normal structure and function of a tissue by some disease process.

Macrophage—A cell of the immune system that engulfs and digests foreign invaders such as bacteria and viruses in an attempt to stop them from causing disease within the body.

Protozoa—A group of organisms which are the smallest members of the animal kingdom, consisting of a single cell.

Ulceration—An area of pitting and irritation.

Vector—A carrier organism (such as a fly or mosquito) that delivers a virus (or other agent of infection) to a host.

(300 m) around all villages, and regularly spraying the area with insecticides. Because rodents often carry the protozoan that causes leishmaniasis, careful rodent control should be practiced. Dogs, which also carry the protozoan, can be given a simple blood test.

Resources

PERIODICALS

MacReady, Norma. "Leishmaniasis Hits Military Hot Spots." *Internal Medicine News* June 15, 2004: 58.

"Seeking First-time Approval." *Chemist & Druggist* May 8, 2004: 12.

"Treatment of Cutaneous Leishmaniasis Among Travelers Reviewed." *Vaccine Weekly* April 28, 2004: 58.

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Rosalyn Carson-DeWitt, MD
Teresa G. Odle

Leprosy

Definition

Leprosy is a slowly progressing bacterial infection that affects the skin, peripheral nerves in the hands and feet, and mucous membranes of the nose, throat, and eyes. Destruction of the nerve endings causes the the affected areas to lose sensation. Occasionally, because of the loss of feeling, the fingers and toes become mutilated and fall off, causing the deformities that are typically associated with the disease.

Description

Leprosy is also known as Hansen's disease after G. A. Hansen, who in 1878 identified the bacillus *Mycobacterium leprae* that causes the disease.

The infection is characterized by abnormal changes of the skin. These changes, called lesions, are at first flat and red. Upon enlarging, they have irregular shapes and a characteristic appearance. The lesions are typically darker in color around the edges with discolored pale centers. Because the organism grows best at lower temperatures the leprosy bacillus has a preference for the skin, the mucous membranes and the nerves. Infection in and destruction of the nerves leads to sensory loss. The loss of sensation in the fingers and toes increases the risk of injury. Inadequate care causes infection of open **wounds**. **Gangrene** may also follow, causing body tissue to die and become deformed.

Because of the disabling deformities associated with it, leprosy has been considered one of the most dreaded diseases since biblical times, though much of what was called leprosy in the Old Testament most likely was not the same disease. Its victims were often shunned by the community, kept at arm's length, or sent to a leper colony. Many people still have misconceptions about the disease. Contrary to popular belief, it is not highly communicable and is extremely slow to develop. Household contacts of most cases and the medical personnel caring for Hansen's disease patients are not at particular risk. It is very curable, although the treatment is long-term, requiring multiple medications.

The World Health Organization (WHO) puts the number of identified leprosy cases in the world at about 600,000 as of the early 2000s. Seventy percent of all cases are found in just three countries: India, Indonesia, and Myanmar (Burma). The infection can be acquired, however, in the Western Hemisphere as well. There are about 5000 reported cases in the United States as of 2004, almost all of which involve

immigrants from developing countries. Cases also occur in some areas of the Caribbean. Although it was thought for many years that only humans are affected by the disease, 15% of wild armadillos in southern Texas and Louisiana have been found to be infected with *M. leprae*.

Causes and symptoms

The organism that causes leprosy is a rod-shaped bacterium called *Mycobacterium leprae*. This bacterium is related to *Mycobacterium tuberculosis*, the causative agent of **tuberculosis**. Because special staining techniques involving acids are required to view these bacteria under the microscope, they are referred to as acid-fast bacilli (AFB).

When *Mycobacterium leprae* invades the body, one of two reactions can take place. In tuberculoid leprosy (TT), the milder form of the disease, the body's immune cells attempt to seal off the infection from the rest of the body by surrounding the offending pathogen. Because this response by the immune system occurs in the deeper layers of the skin, the hair follicles, sweat glands, and nerves can be destroyed. As a result, the skin becomes dry and discolored and loses its sensitivity. Involvement of nerves on the face, arms, or legs can cause them to enlarge and become easily felt by the doctor. This finding is highly suggestive of TT. The scarcity of bacteria in this type of leprosy leads to it being referred to as paucibacillary (PB) leprosy. Seventy to eighty percent of all leprosy cases are of the tuberculoid type.

In lepromatous (LL) leprosy, which is the second and more contagious form of the disease, the body's immune system is unable to mount a strong response to the invading organism. Hence, the organism multiplies freely in the skin. This type of leprosy is also called the multibacillary (MB) leprosy, because of the presence of large numbers of bacteria. The characteristic feature of this disease is the appearance of large nodules or lesions all over the body and face. Occasionally, the mucous membranes of the eyes, nose, and throat may be involved. Facial involvement can produce a lion-like appearance (leonine facies). This type of leprosy can lead to blindness, drastic change in voice, or mutilation of the nose. Leprosy can strike anyone; however, children seem to be more susceptible than adults.

Well-defined **skin lesions** that are numb are the first symptoms of tuberculoid leprosy. Lepromatous leprosy is characterized by a chronic stuffy nose due to invasion of the mucous membranes, and the presence of nodules and lesions all over the body and face.

Although patients with leprosy are commonly thought not to suffer **pain**, neuroapthic pain caused by inflammation of peripheral nerve endings is increasingly recognized as a major complication of the disease in many patients. **Corticosteroids** may be given to reduce the inflammation.

The incubation period of the leprosy bacillus varies anywhere from six months to ten years. On an average, it takes four years for the symptoms of tuberculoid leprosy to develop. Probably because of the slow growth of the bacillus, lepromatous leprosy develops even more slowly, taking an average of eight years for the initial lesions to appear.

It is still not very clear how the leprosy bacillus is transmitted from person to person; about 50% of patients diagnosed with the disease have a history of close contact with an infected family member. Since untreated patients have a large number of *M. leprae* bacilli in their nasal secretions, it is thought that transmission may take place via nasal droplets. The milder tubercular form of leprosy may be transmitted by insect carriers or by contact with infected soil.

The incidence of leprosy is highest in the poverty belt of the globe. Therefore, environmental factors such as unhygienic living conditions, overpopulation, and **malnutrition** may also be contributing factors favoring the infection.

It is also possible that genetic factors are involved in susceptibility to leprosy. In 2003, scientists conducting a genome scan of a large Vietnamese family with many cases of leprosy found that susceptibility to the disease was linked to region q25 on the long arm of chromosome 6. Further study indicated that the leprosy susceptibility gene lies within a region shared by two genes for Parkinson's disease. Further research may confirm that the emergence of leprosy in certain individuals is related to inheritance of genes for Parkinson's disease.

Diagnosis

One of the hallmarks of leprosy is the presence of AFB in smears taken from the skin lesions, nasal scrapings, or tissue secretions. In patients with LL leprosy, the bacilli are easily detected; however, in TT leprosy the bacteria are very few and almost impossible to find. In such cases, a diagnosis is made based on the clinical signs and symptoms, the type and distribution of skin lesions, and history of having lived in an endemic area.

The signs and symptoms characteristic of leprosy can be easily identified by a health worker after a short



Lesions such as these are characteristic of leprosy. (Phototake NYC. Reproduced by permission.)

training period. There is no need for a laboratory investigation to confirm a leprosy diagnosis, except in very rare circumstances.

In an endemic area, if smears from an individual show the presence of AFB, or if he has typical skin lesions, he should definitely be regarded as having leprosy. Usually, there is slight discoloration of the skin and loss of skin sensitivity. Thickened nerves accompanied by weakness of muscles supplied by the affected nerve are very typical of the disease. One characteristic occurrence is a foot drop where the foot cannot be flexed upwards, affecting the ability to walk.

Treatment

The most widely used drug for leprosy is dapsone (DDS). However, the emergence of dapsone-resistant strains prompted the introduction of multidrug therapy, or MDT. MDT combines dapsone, rifampin (Rifadin; also known as rifampicin), and clofazimine (Lamprene), all of which are powerful antibacterial drugs. Patients with MB leprosy are usually treated with all three drugs, while patients with PB leprosy are only given rifampin and dapsone. Usually three months after starting treatment, a patient ceases being infectious, though not everyone with this disease is necessarily infectious before treatment. Depending on the type of leprosy, the time required for treatment may vary from six months to two years or more.

Each of the drugs has minor side effects. Dapsone can cause **nausea, dizziness, palpitations, jaundice** and rash. A doctor should be contacted immediately if a rash develops. Dapsone also interacts with the second drug, rifampin. Rifampin increases the metabolizing of dapsone in the body, requiring an adjustment of the dapsone dosage. rifampin may also cause **muscle cramps**, or nausea. If jaundice, flu-like symptoms or a rash appear, a doctor should be contacted immediately. The third drug, clofazimine may cause severe abdominal pain and **diarrhea**, as well as discoloration of the skin. Red to brownish black discoloration of the skin and bodily fluids, including sweat, may persist for months to years after use.

Thalidomide, the most famous agent of **birth defects** in the twentieth century, is now being used to treat complications of leprosy and similar diseases. Thalidomide regulates the immune response by suppressing a protein, tumor necrosis factor alpha.

Leprosy patients should be aware that treatment itself can cause a potentially serious immune system response called a lepra reaction. When **antibiotics** kill *M. leprae*, antigens (the proteins on the surface of the organism that initiate the body's immune system response) are released from the dying bacteria. In some people, when the antigens combine with the antibodies to *M. Leprae* in the bloodstream, a reaction called **erythema nodosum leprosum** may occur, resulting in new lesions and peripheral nerve damage. Cortisone-type medications and, increasingly, thalidomide are used to minimize the effects of lepra reactions.

In some cases, severe ulcers caused by leprosy may be treated surgically with small skin grafts.

Prognosis

Leprosy is curable; however, the deformities and nerve damage associated with leprosy are often irreversible. Preventions or rehabilitation of these defects is an integral part of management of the disease. **Reconstructive surgery**, aimed at preventing and correcting deformities, offers the greatest hope for disabled patients. Sometimes, the deformities are such that the patients will not benefit from this type of surgery.

Comprehensive care involves teaching patients to care for themselves. If the patients have significant nerve damage or are at high risk of developing deformities, they must be taught to take care of their insensitive limbs, similar to diabetics with lower leg nerve damage. Lacking the sensation of pain in many cases, the patients should constantly check themselves

KEY TERMS

Endemic area—A geographical area where a particular disease is prevalent.

Gangrene—Death of tissue due to loss of blood supply followed by bacterial invasion and putrefaction.

Incubation period—The time it takes for symptoms to develop after initial exposure to a disease-causing organism.

Lesion—Any visible, local abnormality of the tissues of the skin, such as a wound, sore, rash, or boil.

Mucous membranes—The inner tissue that covers or lines body cavities or canals open to the outside, such as nose and mouth. These membranes secrete mucus and absorb water and salts.

Nasal scraping—Pathological material obtained for clinical study by scratching the inner surface of the nose with a clinical instrument.

Nodules—A small mass of tissue in the form of a protuberance or a knot that is solid and can be detected by touch.

Pathogen—Any disease-producing agent or microorganism.

Smear—A specimen prepared for microscopic study by spreading the material across a slide and treating it with a specific stain.

to identify cuts and **bruises**. If adequate care is not taken, these wounds become festering sores and a source of dangerous infection. Physiotherapy exercises are taught to the patients to maintain a range of movement in finger joints and prevent the deformities from worsening. Prefabricated standardized splints are available and are extremely effective in correcting and preventing certain common deformities in leprosy. Special kinds of footwear have been designed for patients with insensitive feet in order to prevent or minimize the progression of foot ulcers.

Prevention

By early diagnosis and appropriate treatment of infected individuals, even a disease as ancient as leprosy can be controlled. People who are in immediate contact with the leprosy patient should be tested for leprosy. Annual examinations should also be conducted on these people for a period of five years following their last contact with an infectious patient.

Some physicians have advocated dapsone treatment for people in close household contact with leprosy patients.

The WHO Action Program for the Elimination of Leprosy adopted a resolution calling for the elimination of leprosy around the world by the year 2005. This goal is not likely to be reached, however; a computer simulation performed for WHO by a team of Dutch researchers in 2004 indicates that leprosy is likely to persist in some parts of the world until 2020, although its incidence will continue to decline.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD., editors. "Infectious Diseases Caused by Mycobacteria." In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

PERIODICALS

- Buschman, E., and E. Skamene. "Linkage of Leprosy Susceptibility to Parkinson's Disease Genes." *International Journal of Leprosy and Other Mycobacterial Diseases* 72 (June 2004): 169–170.
- Jayaseelan, A., and W. Aithal. "Pinch Skin Grafting in Non-Healing Leprous Ulcers." *International Journal of Leprosy and Other Mycobacterial Diseases* 72 (June 2004): 139–142.
- Meima, A., W. C. Smith, G. J. van Oortmarssen, et al. "The Future Incidence of Leprosy: A Scenario Analysis." *Bulletin of the World Health Organization* 82 (May 2004): 373–380.
- Stump, P. R., R. Baccarelli, L. H. Marciano, et al. "Neuropathic Pain in Leprosy Patients." *International Journal of Leprosy and Other Mycobacterial Diseases* 72 (June 2004): 134–148.

ORGANIZATIONS

- American Leprosy Missions. 1 ALM Way, Greenville, SC 29601. (1-800-LEPROSY).
- British Leprosy Relief Association, LEPROA. Fairfax House, Causton Road, Colchester, Essex CO1 1PU, UK.
- INFOLEP, Leprosy Information Services. Postbus 95005, 1090 HA, Amsterdam, Netherlands. Infolep@antenna.nl.
- WHO/LEP, Action Programme for the Elimination of Leprosy. 20 Avenue Appia CH-1211, Geneva 27, Suisse. <http://www.who.ch/programmes/lep/lep_home.htm>.

OTHER

- Cherath, Lata. "Leprosy." *A Healthy Me Page*. <<http://www.ahealthyme.com/topic/topic100587076>>.

Lata Cherath, PhD
Rebecca J. Frey, PhD

Leptospirosis

Definition

Leptospirosis is a febrile disease (**fever**) caused by infection with the bacterium *Leptospira interrogans*. *L. interrogans* is sometimes classified as a spirochete because it has a spiral shape. The disease can range from very mild and symptomless to a more serious, even life threatening form, that may be associated with kidney (renal) failure.

Description

An infection by the bacterium *Leptospira interrogans* goes by different names in different regions. Alternate names for leptospirosis include mud fever, swamp fever, cane cutter's fever, rice field fever, Stuttgart disease, swineherd's disease, and Fort Bragg fever. More severe cases of leptospirosis are called Weil's syndrome or icterohemorrhagic fever. This disease is commonly found in tropical and subtropical climates but occurs worldwide.

According to the Centers for Disease Control and Prevention (CDC), between 100 and 200 cases of leptospirosis are reported in the United States each year as of the early 2000s. Almost 75% of cases of leptospirosis in North America occur in males. About 50% of these cases occur in Hawaii, followed by the southern Atlantic, Gulf, and Pacific coastal states. However, because of the nonspecific symptoms of leptospirosis, it is believed that the occurrence in the United States is actually much higher. Leptospirosis occurs year-round in North America, but about half of the cases occur between July and October.

Leptospirosis is called a **zoonosis** because it is a disease of animals that can be transmitted to humans. It can be a very serious problem in the livestock industry. *Leptospira* bacteria have been found in dogs, rats, livestock, mice, voles, rabbits, hedgehogs, skunks, possums, frogs, fish, snakes, and certain birds and insects. Infected animals will pass the bacteria in their urine for months, or even years. In the United States, rats and dogs are more commonly linked with human leptospirosis than other animals.

Humans are considered accidental hosts and become infected with *Leptospira interrogans* by coming into contact with urine from infected animals. Transmission of the organism occurs either through direct contact with urine, or through contact with soil, water, or plants that have been contaminated by animal urine. *Leptospira interrogans* can survive for

as long as six months outdoors under favorable conditions. *Leptospira* bacteria can enter the body through cuts or other skin damage or through mucous membranes (such as the inside of the mouth and nose). It is believed that the bacteria may be able to pass through intact skin, but this is not known.

Once past the skin barrier, the bacteria enter the blood stream and rapidly spread throughout the body. The infection causes damage to the inner lining of blood vessels. The liver, kidneys, heart, lungs, central nervous system, and eyes may be affected.

There are two stages in the disease process. The first stage is during the active *Leptospira* infection and is called the bacteremic or septicemic phase. The bacteremic phase lasts from three to seven days and presents as typical flu-like symptoms. During this phase, bacteria can be found in the patient's blood and cerebrospinal fluid. The second stage, or immune phase, occurs either immediately after the bacteremic stage or after a 1–3 day symptom-free period. The immune phase can last up to one month. During the immune phase, symptoms are milder but **meningitis** (inflammation of spinal cord and brain tissues) is common. Bacteria can be isolated only from the urine during this second phase.

Causes and symptoms

Leptospirosis is caused by an infection with the bacterium *Leptospira interrogans*. The bacteria are spread through contact with urine from infected animals. Persons at an increased risk for leptospirosis include farmers, miners, animal health care workers, fish farmers and processors, sewage and canal workers, cane harvesters, and soldiers. High-risk activities include care of pets, hunting, trail biking, freshwater swimming, rafting, canoeing, kayaking, and participating in sports in muddy fields. One recent outbreak occurred in Ireland following a canoe competition on a local river.

Symptoms of *Leptospira* infection occur within 7–12 days following exposure to the bacteria. Because the symptoms can be nonspecific, most people who have antibodies to *Leptospira* do not remember having had an illness. Eighty-five to 90% of the cases are not serious and clear up on their own. Symptoms of the first stage of leptospirosis last three to seven days and are: fever (100–105 °F [37.8–40.6 °C]), severe **headache**, muscle **pain**, stomach pain, chills, **nausea**, **vomiting**, back pain, joint pain, neck stiffness, and extreme exhaustion. **Cough** and body rash sometimes occur.

Following the first stage of disease, a brief symptom-free period occurs for most patients. The symptoms

of the second stage vary in each patient. Most patients have a low-grade fever, headache, vomiting, and rash. Aseptic meningitis is common in the second stage, symptoms of which include headache and **photosensitivity** (sensitivity of the eye to light). *Leptospira* can affect the eyes and make them cloudy and yellow to orange colored. Vision may be blurred.

Ten percent of the persons infected with *Leptospira* develop a serious disease called Weil's syndrome. The symptoms of Weil's syndrome are more severe than those described above and there is no distinction between the first and second stages of disease. The hallmark of Weil's syndrome is liver, kidney, and blood vessel disease. The signs of severe disease are apparent after 3–7 days of illness. In addition to those listed above, symptoms of Weil's syndrome include **jaundice** (yellow skin and eyes), decreased or no urine output, **hypotension** (low blood pressure), rash, anemia (decreased number of red blood cells), **shock**, and severe mental status changes. Red spots on the skin, "blood shot" eyes, and bloody sputum signal that blood vessel damage and hemorrhage have occurred.

Diagnosis

Leptospirosis can be diagnosed and treated by doctors who specialize in infectious diseases. During the bacteremic phase of the disease, the symptoms are relatively nonspecific. This often causes an initial misdiagnosis because many diseases have similar symptoms to leptospirosis. The later symptoms of jaundice and kidney failure together with the bacteremic phase symptoms suggest leptospirosis. Blood samples will be tested to look for antibodies to *Leptospira interrogans*. Blood samples taken over a period of a few days would show an increase in the number of antibodies. Isolating *Leptospira* bacteria from blood, cerebrospinal fluid (performed by spinal tap), and urine samples is diagnostic of leptospirosis. It may take six weeks for *Leptospira* to grow in laboratory media. Most insurance companies would cover the diagnosis and treatment of this infection.

Several diagnostic tests for leptospirosis have been devised in the early 2000s that are more accurate as well as faster than standard culture. One test uses flow cytometry light scatter analysis; this method can evaluate a sample of infected serum in as little as 90 minutes. A second technique is an IgM-enzyme-linked immunosorbent assay (ELISA), which detects the presence of IgM antibodies to *L. interrogans* in blood serum samples.

Treatment

Leptospirosis is treated with **antibiotics**, penicillin (Bicillin, Wycillin), doxycycline (Monodox), ibramycin, or erythromycin (E-mycin, Ery-Tab). As of the early 2000s, however, many doctors prefer to treat patients with ceftriaxone, which is easier to use than intravenous penicillin. Ciprofloxacin may be combined with other drugs in treating patients who develop **uveitis**. It is generally agreed that antibiotic treatment during the first few days of illness is helpful. However, leptospirosis is often not diagnosed until the later stages of illness. The benefit of antibiotic treatment in the later stages of disease, however, is controversial. A rare complication of antibiotic therapy for leptospirosis is the occurrence of the Jarisch-Herxheimer reaction, which is characterized by fever, chills, headache, and muscle pain.

Patients with severe illness will require hospitalization for treatment and monitoring. Medication or other treatment for pain, fever, vomiting, fluid loss, bleeding, mental changes, and low blood pressure may be provided. Patients with kidney failure will require hemodialysis to remove waste products from the blood.

Prognosis

The majority of patients infected with *Leptospira interrogans* experience a complete recovery. Ten percent of the patients will develop eye inflammation (uveitis) up to one year after the illness. In the United States, about one out of every 100 patients will die from leptospirosis. **Death** is usually caused by kidney failure, but has also been caused by **myocarditis** (inflammation of heart tissue), **septic shock** (reduced blood flow to the organs because of the bacterial infection), organ failure, and/or poorly functioning lungs. Mortality is highest in patients over 60 years of age.

Prevention

Persons who are at an extremely high risk (such as soldiers who are training in wetlands) can be pretreated with 200 mg of doxycycline once a week. As of the early 2000s, there are no vaccines available to prevent leptospirosis in humans, although such vaccines have been formulated by veterinarians for dogs, swine, and cattle.

There are many ways to decrease the chances of being infected by *Leptospira*. These include:

- Avoid swimming or wading in freshwater ponds and slowly moving streams, especially those located near farms.

KEY TERMS

Hemodialysis—The removal of waste products from the blood stream in patients with kidney failure. Blood is removed from a vein, passed through a dialysis machine, and then put back into a vein.

Jarisch-Herxheimer reaction—A rare reaction to the dead bacteria in the blood stream following antibiotic treatment.

Meningitis—Inflammation of tissues in the brain and spinal cord. Aseptic meningitis refers to meningitis with no bacteria present in the cerebral spinal fluid.

Spirochete—Any of a family of spiral- or coil-shaped bacteria known as Spirochetes. *L. interrogans* is a spirochete, as well as the organisms that cause syphilis and relapsing fever.

Zoonosis (plural, zoonoses)—Any disease of animals that can be transmitted to humans. Leptospirosis is an example of a zoonosis.

- Do not conduct canoe or kayak capsizing drills in freshwater ponds. Use a swimming pool instead.
- Boil or chemically treat pond or stream water before drinking it or cooking with it.
- Control rats and mice around the home.
- Have pets and farm animals vaccinated against *Leptospira*.
- Wear protective clothing (gloves, boots, long pants, and long-sleeved shirts) when working with wet soil or plants.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD., editors. "Infectious Diseases Caused by Spirochetes." In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

PERIODICALS

Boland, M., G. Sayers, T. Coleman, et al. "A Cluster of Leptospirosis Cases in Canoeists following a Competition on the River Liffey." *Epidemiology and Infection* 132 (April 2004): 195–200.

Faucher, J. F., B. Hoen, and J. M. Estavoyer. "The Management of Leptospirosis." *Expert Opinion in Pharmacotherapy* 5 (April 2004): 819–827.

Vitale, G., C. La Russa, A. Galioto, et al. "Evaluation of an IgM-ELISA Test for the Diagnosis of Human

Leptospirosis." *New Microbiologica* 27 (April 2004): 149–154.

Yitzhaki, S., A. Barnea, A. Keysary, and E. Zahavy. "New Approach for Serological Testing for Leptospirosis by Using Detection of *Leptospira* Agglutination by Flow Cytometry Light Scatter Analysis." *Journal of Clinical Microbiology* 42 (April 2004): 1680–1685.

ORGANIZATIONS

American Veterinary Medical Association (AVMA).
1931 North Meacham Road, Suite 100, Schaumburg,
IL 60173-4360. <<http://www.avma.org>>.

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Belinda Rowland, PhD
Rebecca J. Frey, PhD

Lesch-Nyhan syndrome

Definition

Lesch-Nyhan syndrome, which is also known as HPRT deficiency or Kelley-Seegmiller syndrome, is a rare genetic disorder that affects males. Males with this syndrome develop physical handicaps, **mental retardation**, and kidney problems. It is caused by a total absence of a key enzyme that affects the level of uric acid in the body. Self-injury or **self-mutilation** is a distinctive feature of this genetic disease.

Description

Lesch-Nyhan syndrome was first described in 1964 by Drs. Michael Lesch and William Nyhan. The enzyme deficiency that causes the disorder was discovered in 1967 by a researcher named Seegmiller. The syndrome is caused by a severe change (mutation) in a gene that encodes an enzyme known as hypoxanthine-guanine phosphoribosyl transferase, or HPRT. This gene was identified and sequenced by Friedmann and colleagues in 1985. HPRT catalyzes a reaction that is necessary to prevent the buildup of uric acid, a nitrogenous waste product that is ordinarily excreted from the body through the kidneys. A severe mutation in the HPRT gene leads to an absence of HPRT enzyme activity which, in turn, leads to markedly elevated uric acid levels in the blood (hyperuricemia). This buildup of uric acid is toxic to the body and is related to the symptoms associated with the disease. Absence of the HPRT enzyme activity is also thought to alter

the chemistry of certain parts of the brain, such as the basal ganglia, affecting neurotransmitters (chemicals used for communication between nerve cells), acids, and other chemicals. This change in the nervous system is also related to the symptoms associated with Lesch-Nyhan syndrome.

Males with Lesch-Nyhan syndrome develop neurologic problems during infancy. Infants with Lesch-Nyhan syndrome have weak muscle tone (hypotonia) and are unable to develop normally. Affected males develop uncontrollable writhing movements (athetosis) and muscle stiffness (spasticity) over time. Lack of speech is also a common feature of Lesch-Nyhan syndrome. The most dramatic symptom of Lesch-Nyhan syndrome, however, is the compulsive self-injury seen in 85% of affected males. This self-injury involves the biting of their own lips, tongue, and finger tips, as well as head banging. This behavior leads to serious injury and scarring.

Lesch-Nyhan syndrome affects approximately one in 380,000 live births. It occurs evenly among races. Almost always, only male children are affected, although a few cases of the disorder in girls have been reported. Women carriers usually do not have any symptoms. Women carriers can occasionally develop inflammation of the joints (**gout**) as they get older.

Causes and symptoms

Severe changes (mutations) in the HPRT gene completely halt the activity of the enzyme HPRT. There have been many different severe mutations identified in the HPRT gene. These mutations may be different between families. The HPRT gene is located on the X chromosome. Since the HPRT gene is located on the X chromosome, Lesch-Nyhan syndrome is considered X-linked. This means that it only affects males.

A person's sex is determined by their chromosomes. Males have one X chromosome and one Y chromosome. Females, on the other hand, have two X chromosomes. Males who possess a severe mutation in their HPRT gene will develop Lesch-Nyhan syndrome. Females who possess a severe mutation in their HPRT gene will not. They are considered to be carriers. This is because females have another X chromosome without the mutation that prevents them from getting this disease. If a woman is a carrier, she has a 50% risk with any **pregnancy** to pass on her X chromosome with the mutation. Therefore, with every male pregnancy she has a 50% risk to have an affected son, and with every female pregnancy she has a 50% risk to have a daughter who is a carrier.

At birth, males with Lesch-Nyhan syndrome appear completely normal. Development is usually normal for the first few months. Symptoms develop between three to six months of age. Sand-like crystals of uric acid in the diapers may be one of the first symptoms of the disease. The baby may be unusually irritable. Typically, the first sign of nervous system impairment is the inability to lift their head or sit up at an appropriate age. Many patients with Lesch-Nyhan will never learn to walk. By the end of the first year, writhing motions (athetosis), and spasmodic movements of the limbs and facial muscles (chorea) are clear evidence of defective motor development.

The compulsive self-injury associated with Lesch-Nyhan syndrome begins, on average, at three years. The self-injury begins with biting of the lips and tongue. As the disease progresses, affected individuals frequently develop finger biting and head banging. The self-injury can increase during times of **stress**.

Males with Lesch-Nyhan disease may also develop kidney damage due to **kidney stones**. Swollen and tender joints (gout) is another common problem.

Diagnosis

The diagnosis of Lesch-Nyhan syndrome is based initially on the distinctive pattern of the child's symptoms, most commonly involuntary muscle movements or failure to crawl and walk at the usual ages. In some cases the first symptom is related to overproduction of uric acid; the parents notice "orange sand" in the child's diapers. The "sand" is actually crystals of uric acid tinged with blood.

Measuring the amount of uric acid in a person's blood or urine can not definitively diagnose Lesch-Nyhan syndrome. It is diagnosed by measuring the activity of the HPRT enzyme through a blood test. When the activity of the enzyme is very low it is diagnostic of Lesch-Nyhan syndrome. It can also be diagnosed by DNA testing. This is also a blood test. DNA testing checks for changes (mutations) in the HPRT gene. Results from DNA testing are helpful in confirming the diagnosis and also when the child's family is interested in prenatal testing for future pregnancies.

Prenatal diagnosis is possible by DNA testing of fetal tissue drawn by **amniocentesis** or **chorionic villus sampling (CVS)**. Fetuses should be tested if the mother is a carrier of a change (mutation) in her HPRT gene. A woman is at risk of being a carrier if she has a son with Lesch-Nyhan syndrome or someone in her family has Lesch-Nyhan syndrome. Any woman at risk of being a carrier should have DNA testing through a blood test.

KEY TERMS

Amniocentesis—A procedure performed at 16-18 weeks of pregnancy in which a needle is inserted through a woman's abdomen into her uterus to draw out a small sample of the amniotic fluid from around the baby. Either the fluid itself or cells from the fluid can be used for a variety of tests to obtain information about genetic disorders and other medical conditions in the fetus.

Athetosis—A condition marked by slow, writhing, involuntary muscle movements.

Basal ganglia—A section of the brain responsible for smooth muscular movement.

Chorea—Involuntary, rapid, jerky movements.

Chorionic villus sampling (CVS)—A procedure used for prenatal diagnosis at 10-12 weeks gestation. Under ultrasound guidance a needle is inserted either through the mother's vagina or abdominal wall and a sample of cells is collected from around the early embryo. These cells are then tested for chromosome abnormalities or other genetic diseases.

Enzyme—A protein that catalyzes a biochemical reaction or change without changing its own structure or function.

Mutation—A permanent change in the genetic material that may alter a trait or characteristic of an individual, or manifest as disease, and can be transmitted to offspring.

Neurotransmitter—Chemical in the brain that transmits information from one nerve cell to another.

Palsy—Uncontrollable tremors.

Spasticity—Increased muscle tone, or stiffness, which leads to uncontrolled, awkward movements.

Treatment

There are no known treatments for the neurological defects of Lesch-Nyhan. Allopurinol (Aloprim, Zyloprim), a drug usually prescribed to lower the risk of gout attacks, can lower blood uric acid levels. This medication is a preventive; it does not correct many of the symptoms of Lesch-Nyhan. Other drugs that are given to manage spasticity include baclofen (Lioresal), which is a muscle relaxant, and benzodiazepine tranquilizers.

Some patients with Lesch-Nyhan syndrome have their teeth removed to prevent self-injury. Restraints may be recommended to reduce self-destructive behaviors, although some patients can be managed with a

combination of behavioral modification therapy and medications.

Prognosis

With strong supportive care, infants born with Lesch-Nyhan can live into adulthood with symptoms continuing throughout life. Few live beyond 40, however, with **death** usually resulting either from kidney failure or from aspiration **pneumonia**. Sudden unexpected death from **respiratory failure** is common in these patients.

At present, there are no preventive measures for Lesch-Nyhan syndrome. However, recent studies have indicated that this genetic disorder may be a good candidate for treatment with gene replacement therapy. Unfortunately, the technology necessary to implement this therapy has not yet been perfected.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD., editors. "Cerebral Palsy Syndromes." In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

Jinnah, H. A., and Theodore Friedmann. "Lesch-Nyhan Disease and Its Variants." *The Metabolic and Molecular Bases of Inherited Disease*. New York: McGraw-Hill, 2001.

PERIODICALS

Jinnah, Hyder, MD, PhD. "Lesch-Nyhan Syndrome." *eMedicine* January 25, 2002. <<http://www.emedicine.com/neuro/topic630.htm>>.

Mak, B. S., et al. "New Mutations of the HPRT Gene in Lesch-Nyhan Syndrome." *Pediatric Neurology* October 2000: 332-335.

Visser, J.E., et al. "Lesch-Nyhan Disease and the Basal Ganglia." *Brain Research Reviews* November 1999: 450-469.

Willers, I. "Germline Mosaicism Complicates Molecular Diagnosis of Lesch-Nyhan Syndrome." *Prenatal Diagnosis* 24 (September 2004): 737-740.

ORGANIZATIONS

Alliance of Genetic Support Groups. 4301 Connecticut Ave. NW, Suite 404, Washington, DC 20008. (202) 966-5557. Fax: (202) 966-8553. <<http://www.geneticalliance.org>>.

International Lesch-Nyhan Disease Association. 114 Winchester Way, Shamong, NJ 08088-9398. (215) 677-4206.

Lesch-Nyhan Syndrome Registry. New York University School of Medicine, Department of Psychiatry, 550 First Ave., New York, NY 10012. (212) 263-6458.

National Organization for Rare Disorders (NORD). 55 Kenosia Avenue, P. O. Box 1968, Danbury,

CT 06813-1968. (203) 744-0100. Fax: (203) 798-2291.
< <http://www.rarediseases.org> >.

OTHER

GeneClinics. < <http://www.geneclinics.org/profiles/lms/details.html> >.

Pediatric Database(PEDBASE). < <http://www.icodata.com/health/pedbase/files/LESCH-NY.HTM> >.

Holly Ann Ishmael, M.S.
Rebecca J. Frey, PhD

Leukemia stains

Definition

Leukemia stains are laboratory tests done on bone marrow or blood samples to help diagnose specific types of leukemia.

Purpose

Leukemia stains are done to diagnose and classify leukemia. Blood contains red cells, several varieties of white cells, and platelets. Cancerous overproduction of any one type of cell produces one of many types of leukemia. A patient's specific type of leukemia must be classified in order to provide the best treatment and most accurate prognosis.

The type and maturity of the cells involved are identified by analyzing blood and bone marrow under a microscope. Often, however, the abnormality or immaturity of the cells make it difficult to identify the cell types with certainty. Special leukemia stains help to distinguish one cell type from another.

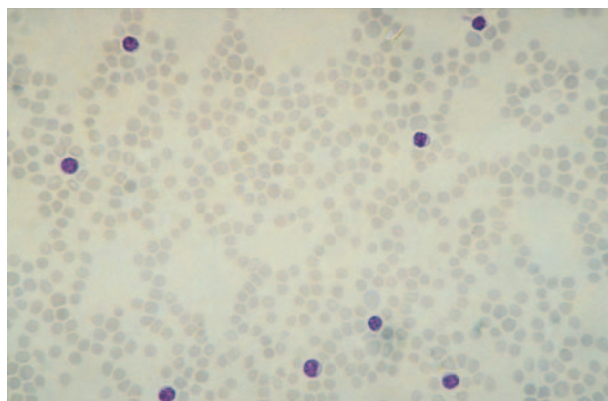
Description

Special stains are added to bone marrow or blood that has been smeared on a microscope slide. Cell types react differently to the chemicals in the stains.

If the patient has few white cells, a buffy coat smear is made. A tube of blood is spun in a centrifuge. Red cells fall, plasma rises, and white cells settle in a thin middle layer called the buffy coat. The smear is made from this layer.

Sudan black B stain

This stain distinguishes between acute lymphoblastic leukemia (cells stain positive) and acute myeloblastic leukemia (cells stain negative).



A magnified stain of chronic lymphocytic leukemia cells.
(Custom Medical Stock Photo. Reproduced by permission.)

Periodic acid-Schiff stain (PAS)

The PAS stain is primarily used to identify erythroleukemia, a leukemia of immature red blood cells. These cells stain a bright fuchsia.

Terminal deoxynucleotidyl transferase stain (TdT)

The TdT stain differentiates between acute lymphoblastic leukemia (cells stain positive) and acute myelogenous leukemia (cells stain negative).

Leukocyte alkaline phosphatase (LAP)

The LAP stain is used to determine if an increase of cells is due to chronic myelogenous leukemia or a noncancerous reaction to an infection or similar conditions. Cells from a noncancerous reaction stain positive with many intense blue granules; cells from chronic myelogenous leukemia have few blue granules.

Tartrate-resistant acid phosphatase stain (TRAP)

The TRAP stain is primarily used to identify **hairy cell leukemia** cells. These cells stain with purple to dark red granules.

Myeloperoxidase stain

The myeloperoxidase stain distinguishes between the immature cells in acute myeloblastic leukemia (cells stain positive) and those in acute lymphoblastic leukemia (cells stain negative).

Leukocyte specific esterase

This stain identifies granulocytes, which show red granules.

KEY TERMS

Bone marrow—The spongy tissue inside large bones where blood cells are formed.

Buffy coat—The thin layer of concentrated white blood cells that forms when a tube of blood is spun in a centrifuge.

Leukemia—Any of several cancers of the bone marrow characterized by the abnormal increase of a type of blood cell.

Leukemia stains—Special stains added to smears of blood or bone marrow, performed to diagnose and classify leukemia.

Leukocyte nonspecific esterase

Nonspecific esterase stain identifies monocytes and immature platelets (megakaryocytes), which show positive black granules.

Preparation

Leukemia stains are done on smears of blood or bone marrow. To collect blood, a healthcare worker draws blood from a vein in the inner elbow region. Collection of the sample takes only a few minutes.

When bone marrow is needed, the person is given **local anesthesia**. Then the physician inserts a needle through the skin and into the bone—usually the breast bone or hip bone—and 0.5-2 mL of bone marrow is withdrawn. This procedure takes approximately 30 minutes.

Aftercare

Patients sometimes feel discomfort or bruising at the puncture site after blood collection. They may also become dizzy or faint. Pressure to the puncture site until the bleeding stops reduces bruising. Warm packs to the puncture site relieve discomfort.

Collection of bone marrow is done under a physician's supervision. The patient is asked to rest after the procedure and is watched for weakness and signs of bleeding.

Normal results

A normal blood or bone marrow smear shows no evidence of leukemic cells. The expected reaction of cells varies with the type of stain.

Abnormal results

Leukemia stain results that help diagnosis and classify leukemia are supported by the results of other laboratory tests and the person's clinical condition.

Resources

BOOKS

Fischbach, Francis. *Manual of Laboratory and Diagnostic Tests*. Philadelphia: Lippincott, 1996.

Nancy J. Nordenson

Leukemias, acute

Definition

Leukemia is a **cancer** that starts in the organs that make blood, namely the bone marrow and the lymph system. Depending on their characteristics, leukemias can be divided into two broad types. Acute leukemias are the rapidly progressing leukemias, while the **chronic leukemias** progress more slowly. The vast majority of the childhood leukemias are of the acute form.

Description

The cells that make up blood are produced in the bone marrow and the lymph system. The bone marrow is the spongy tissue found in the large bones of the body. The lymph system includes the spleen (an organ in the upper abdomen), the thymus (a small organ beneath the breastbone), and the tonsils (an organ in the throat). In addition, the lymph vessels (tiny tubes that branch like blood vessels into all parts of the body) and lymph nodes (pea-shaped organs that are found along the network of lymph vessels) are also part of the lymph system. The lymph is a milky fluid that contains cells. Clusters of lymph nodes are found in the neck, underarm, pelvis, abdomen, and chest.

The cells found in the blood are the red blood cells (RBCs), which carry oxygen and other materials to all tissues of the body; white blood cells (WBCs) that fight infection; and the platelets, which play a part in the clotting of the blood. The white blood cells can be further subdivided into three main types: granulocytes, monocytes, and lymphocytes.

The granulocytes, as their name suggests, have particles (granules) inside them. These granules

CHARLOTTE FRIEND (1921–1987)



(Library of Congress.)

Charlotte Friend was born to Russian immigrants, Morris Friend and Cecilia (Wolpin), on March 11, 1921, in New York City. At three years of age, her father died of a

heart condition. Friend's decision to pursue a career in medicine may well have been influenced by her father's death and by her mother's occupation as a pharmacist. As a child, Friend read books about bacteriologists and, by age ten, she knew that she wanted to study bacteriology. She attended Hunter College, enlisting in the U.S. Navy after her graduation in 1944.

Friend attended Yale University, earning her Ph.D. in bacteriology in 1950. After working for the Memorial Sloan-Kettering Institute for Cancer Research, she became an associate professor at Cornell University in 1952. Friend began researching cancer and became particularly interested in leukemia and its cause. She believed that a virus caused the disease and confirmed this theory by using an electron microscope to photograph the virus in mice. Her findings were initially met with much skepticism but she was able to develop a vaccine that was used successfully with mice, which added credibility to her theory. Her breakthroughs have led medical researchers to new methods of treating cancer and to a greater understanding of the disease.

Friend was a prolific writer who published 113 original papers, 49 abstracts, book chapters, and reviews, many of which she completed individually. She was diagnosed with lymphoma and died on January 13, 1987.

contain special proteins (enzymes) and several other substances that can break down chemicals and destroy microorganisms, such as bacteria. Monocytes are the second type of white blood cell. They are also important in defending the body against pathogens.

The lymphocytes form the third type of white blood cell. There are two main types of lymphocytes: T lymphocytes and B lymphocytes. They have different functions within the immune system. The B cells protect the body by making "antibodies." Antibodies are proteins that can attach to the surfaces of bacteria and viruses. This "attachment" sends signals to many other cell types to come and destroy the antibody-coated organism. The T cells protect the body against viruses. When a virus enters a cell, it produces certain proteins that are projected onto the surface of the infected cell. The T cells recognize these proteins and make certain chemicals that are capable of destroying the virus-infected cells. In addition, the T cells can destroy some types of cancer cells.

The bone marrow makes stem cells, which are the precursors of the different blood cells. These stem cells mature through stages into either RBCs, WBCs, or

platelets. In acute leukemias, the maturation process of the white blood cells is interrupted. The immature cells (or "blasts") proliferate rapidly and begin to accumulate in various organs and tissues, thereby affecting their normal function. This uncontrolled proliferation of the immature cells in the bone marrow affects the production of the normal red blood cells and platelets as well.

Acute leukemias are of two types: acute lymphocytic leukemia and acute myelogenous leukemia. Different types of white blood cells are involved in the two leukemias. In acute lymphocytic leukemia (ALL), it is the T or the B lymphocytes that become cancerous. The B cell leukemias are more common than T cell leukemias. Acute myelogenous leukemia, also known as acute nonlymphocytic leukemia (ANLL), is a cancer of the monocytes and/or granulocytes.

Leukemias account for 2% of all cancers. Because leukemia is the most common form of childhood cancer, it is often regarded as a disease of childhood. However, leukemias affect nine times as many adults as children. Half of the cases occur in people who are 60 years of age or older. The incidence of acute and chronic leukemias is

about the same. According to the estimates of the American Cancer Society (ACS), approximately 29,000 new cases of leukemia were diagnosed in 1998.

Causes and symptoms

Leukemia strikes both sexes and all ages. The human T-cell leukemia virus (HTLV-I) is believed to be the causative agent for some kinds of leukemias. However, the cause of most leukemias is not known. Acute lymphoid leukemia (ALL) is more common among Caucasians than among African-Americans, while acute myeloid leukemia (AML) affects both races equally. The incidence of acute leukemia is slightly higher among men than women. People with Jewish ancestry have a higher likelihood of getting leukemia. A higher incidence of leukemia has also been observed among persons with **Down syndrome** and some other genetic abnormalities.

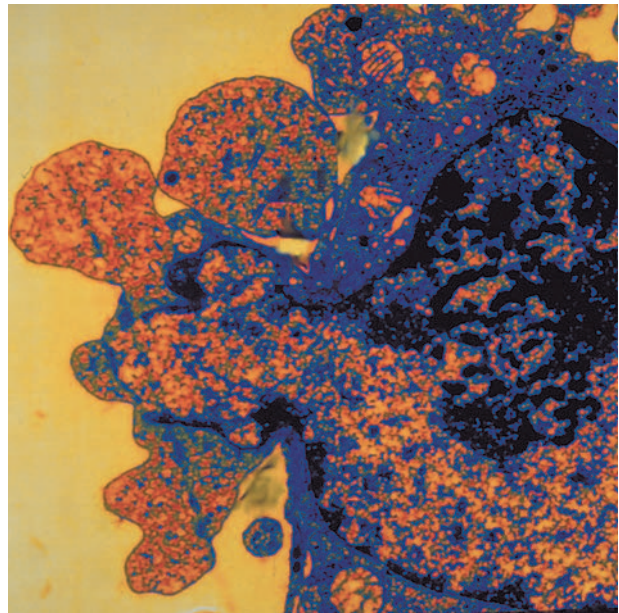
Exposure to ionizing radiation and to certain organic chemicals, such as benzene, is believed to increase the risk of getting leukemia. Having a history of diseases that damage the bone marrow, such as **aplastic anemia**, or a history of cancers of the lymphatic system puts people at a high risk for developing acute leukemias. Similarly, the use of anticancer medications, immunosuppressants, and the antibiotic chloramphenicol are also considered risk factors for developing acute leukemias.

The symptoms of leukemia are generally vague and non-specific. A patient may experience all or some of the following symptoms:

- weakness or chronic **fatigue**
- fever of unknown origin
- weight loss that is not due to dieting or **exercise**
- frequent bacterial or viral infections
- headaches
- skin rash
- non-specific bone **pain**
- easy bruising
- bleeding from gums or nose
- blood in urine or stools
- enlarged lymph nodes and/or spleen
- abdominal fullness

Diagnosis

Like all cancers, acute leukemias are best treated when found early. There are no screening tests available.

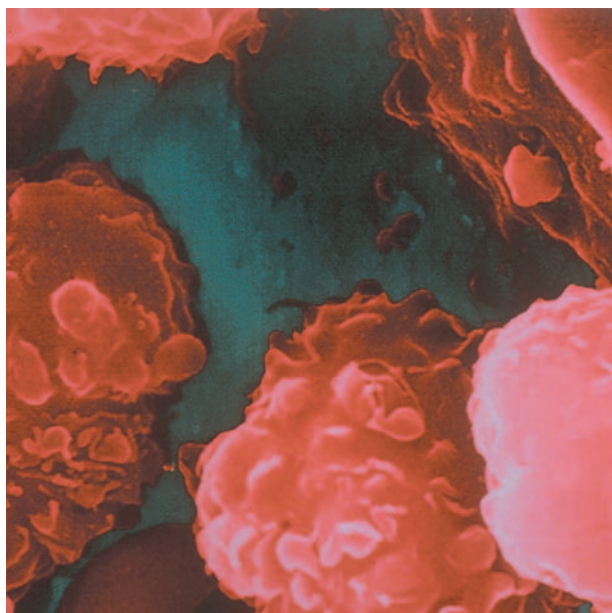


An enhanced transmission electron microscopy (TEM) image of acute myelogenous leukemia cells. (Photograph by Robert Becker, Ph.D., Custom Medical Stock Photo. Reproduced by permission.)

If the doctor has reason to suspect leukemia, he or she will conduct a very thorough **physical examination** to look for enlarged lymph nodes in the neck, underarm, and pelvic region. Swollen gums, enlarged liver or spleen, **bruises**, or pinpoint red **rashes** all over the body are some of the signs of leukemia. Urine and blood tests may be ordered to check for microscopic amounts of blood in the urine and to obtain a complete differential **blood count**. This count will give the numbers and percentages of the different cells found in the blood. An abnormal blood test might suggest leukemia; however, the diagnosis has to be confirmed by more specific tests.

The doctor may perform a **bone marrow biopsy** to confirm the diagnosis of leukemia. During the biopsy, a cylindrical piece of bone and marrow is removed. The tissue is generally taken out of the hipbone. These samples are sent to the laboratory for examination. In addition to diagnosis, the biopsy is also repeated during the treatment phase of the disease to see if the leukemia is responding to therapy.

A spinal tap (lumbar puncture) is another procedure that the doctor may order to diagnose leukemia. In this procedure, a small needle is inserted into the spinal cavity in the lower back to withdraw some cerebrospinal fluid and to look for leukemic cells.



An enhanced scanning electron microscopy (SEM) image of acute myelogenous leukemia cells. (Photograph by Robert Becker, Ph.D., Custom Medical Stock Photo. Reproduced by permission.)

Standard imaging tests, such as x rays, **computed tomography scans** (CT scans), and **magnetic resonance imaging** (MRI) may be used to check whether the leukemic cells have invaded other areas of the body, such as the bones, chest, kidneys, abdomen, or brain. A gallium scan or bone scan is a test in which a radioactive chemical is injected into the body. This chemical accumulates in the areas of cancer or infection, allowing them to be viewed with a special camera.

Treatment

There are two phases of treatment for leukemia. The first phase is called “induction therapy.” As the name suggests, during this phase, the main aim of the treatment is to reduce the number of leukemic cells as far as possible and induce a remission in the patient. Once the patient shows no obvious signs of leukemia (no leukemic cells are detected in blood tests and bone marrow biopsies), the patient is said to be in remission. The second phase of treatment is then initiated. This is called continuation or maintenance therapy, and the aim in this case is to kill any remaining cells and to maintain the remission for as long as possible.

Chemotherapy is the use of drugs to kill cancer cells. It is usually the treatment of choice and is used to relieve symptoms and achieve long-term remission

of the disease. Generally, combination chemotherapy, in which multiple drugs are used, is more efficient than using a single drug for the treatment. Some drugs may be administered intravenously through a vein in the arm; others may be given by mouth in the form of pills. If the cancer cells have invaded the brain, then chemotherapeutic drugs may be put into the fluid that surrounds the brain through a needle in the brain or back. This is known as intrathecal chemotherapy.

Because leukemia cells can spread to all the organs via the blood stream and the lymph vessels, surgery is not considered an option for treating leukemias.

Radiation therapy, which involves the use of x rays or other high-energy rays to kill cancer cells and shrink tumors, may be used in some cases. For acute leukemias, the source of radiation is usually outside the body (external radiation therapy). If the leukemic cells have spread to the brain, radiation therapy can be given to the brain.

Bone marrow transplantation is a process in which the patient’s diseased bone marrow is replaced with healthy marrow. There are two ways of doing a bone marrow transplant. In an allogeneic bone marrow transplant, healthy marrow is taken from a donor whose tissue is either the same as or very closely resembles the patient’s tissues. The donor may be a twin, a brother or sister (sibling), or a person who is not related at all. First, the patient’s bone marrow is destroyed with very high doses of chemotherapy and radiation therapy. Healthy marrow from the donor is then given to the patient through a needle in a vein to replace the destroyed marrow.

In the second type of bone marrow transplant, called an autologous bone marrow transplant, some of the patient’s own marrow is taken out and treated with a combination of **anticancer drugs** to kill all the abnormal cells. This marrow is then frozen to save it. The marrow remaining in the patient’s body is destroyed with high-dose chemotherapy and radiation therapy. The marrow that was frozen is then thawed and given back to the patient through a needle in a vein. This mode of bone marrow transplant is currently being investigated in clinical trials.

Biological therapy or immunotherapy is a mode of treatment in which the body’s own immune system is harnessed to fight the cancer. Substances that are routinely made by the immune system (such as growth factors, hormones, and disease-fighting proteins) are either synthetically made in a laboratory or their effectiveness is boosted and they are then put back into the patient’s body. This treatment mode is also being

KEY TERMS

Antibodies—Proteins made by the B lymphocytes in response to the presence of infectious agents, such as bacteria or viruses, in the body.

Biopsy—The surgical removal and microscopic examination of living tissue for diagnostic purposes.

Chemotherapy—Treatment with drugs that act against cancer.

Computerized tomography (CT) scan—A series of x rays put together by a computer in order to form detailed pictures of areas inside the body.

Cytokines—Chemicals made by the cells that act on other cells to stimulate or inhibit their function. Cytokines that stimulate growth are called “growth factors.”

Immunotherapy—Treatment of cancer by stimulating the body’s immune defense system.

Lumbar puncture—A procedure in which the doctor inserts a small needle into the spinal cavity in the lower back to withdraw some spinal fluid for testing. Also known as a “spinal tap.”

Magnetic resonance imaging (MRI)—A medical procedure using a magnet linked to a computer to picture areas inside the body.

Maturation—The process by which stem cells transform from immature cells without a specific function into a particular type of blood cell with defined functions.

Radiation therapy—Treatment using high-energy radiation from x-ray machines, cobalt, radium, or other sources.

Remission—A disappearance of a disease as a result of treatment. Complete remission means that all disease is gone. Partial remission means that the disease is significantly improved by treatment, but residual traces of the disease are still present.

investigated in clinical trials all over the country at major cancer centers.

Prognosis

Like all cancers, the prognosis for leukemia depends on the patient’s age and general health. According to statistics, more than 60% of the patients with leukemia survive for at least a year after diagnosis. Acute myelocytic leukemia (AML) has a poorer prognosis rate than acute lymphocytic leukemias

(ALL) and the chronic leukemias. In the last 15 to 20 years, the five-year survival rate for patients with ALL has increased from 38% to 57%.

Interestingly enough, since most childhood leukemias are of the ALL type, chemotherapy has been highly successful in their treatment. This is because chemotherapeutic drugs are most effective against actively growing cells. Due to the new combinations of anticancer drugs being used, the survival rates among children with ALL have improved dramatically. Eighty percent of the children diagnosed with ALL now survive for five years or more, as compared to 50% in the late 1970s.

Prevention

Most cancers can be prevented by changes in lifestyle or diet, which will reduce the risk factors. However, in leukemias, there are no such known risk factors. Therefore, at the present time, no way is known to prevent leukemias from developing. People who are at an increased risk for developing leukemia because of proven exposure to ionizing radiation or exposure to the toxic liquid benzene, and people with Down syndrome, should undergo periodic medical checkups.

Resources

ORGANIZATIONS

American Cancer Society. 1599 Clifton Rd., NE, Atlanta, GA 30329-4251. (800) 227-2345. <<http://www.cancer.org>>.

Cancer Research Institute. 681 Fifth Ave., New York, N.Y. 10022. (800) 992-2623. <<http://www.cancerresearch.org>>.

Leukemia Society of America, Inc. 600 Third Ave., New York, NY 10016. (800) 955-4572. <<http://www.leukemia.org>>.

National Cancer Institute. Building 31, Room 10A31, 31 Center Drive, MSC 2580, Bethesda, MD 20892-2580. (800) 422-6237. <<http://www.nci.nih.gov>>.

Oncolink. University of Pennsylvania Cancer Center. <<http://cancer.med.upenn.edu>>.

Lata Cherath, PhD

Leukemias, chronic

Definition

Chronic leukemia is a disease in which too many white blood cells are made in the bone marrow.

Depending on the type of white blood cell that is involved, chronic leukemia can be classified as chronic lymphocytic leukemia or chronic myeloid leukemia.

Description

Chronic leukemia is a **cancer** that starts in the blood cells made in the bone marrow. The bone marrow is the spongy tissue found in the large bones of the body. The bone marrow makes precursor cells called “blasts” or “stem cells” that mature into different types of blood cells. Unlike **acute leukemias**, in which the process of maturation of the blast cells is interrupted, in chronic leukemias, the cells do mature and only a few remain as immature cells. However, even though the cells appear normal, they do not function as normal cells.

The different types of cells that are produced in the bone marrow are red blood cells (RBCs), which carry oxygen and other materials to all tissues of the body; white blood cells (WBCs), which fight infection; and platelets, which play a part in the clotting of the blood. The white blood cells can be further subdivided into three main types: the granulocytes, monocytes, and the lymphocytes.

The granulocytes, as their name suggests, have granules (particles) inside them. These granules contain special proteins (enzymes) and several other substances that can break down chemicals and destroy microorganisms such as bacteria.

Monocytes are the second type of white blood cell. They are also important in defending the body against pathogens.

The lymphocytes form the third type of white blood cell. There are two main types of lymphocytes: T lymphocytes and B lymphocytes. They have different functions within the immune system. The B cells protect the body by making “antibodies.” Antibodies are proteins that can attach to the surfaces of bacteria and viruses. This attachment sends signals to many other cell types to come and destroy the antibody-coated organism. The T cell protects the body against viruses. When a virus enters a cell, it produces certain proteins that are projected onto the surface of the infected cell. The T cells can recognize these proteins and produce certain chemicals (cytokines) that are capable of destroying the virus-infected cells. In addition, the T cells can destroy some types of cancer cells.

Chronic leukemias develop very gradually. The abnormal lymphocytes multiply slowly, but in a poorly regulated manner. They live much longer and thus their numbers build up in the body. The two types

of chronic leukemias can be easily distinguished under the microscope. Chronic lymphocytic leukemia (CLL) involves the T or B lymphocytes. B cell abnormalities are more common than T cell abnormalities. T cells are affected in only 5% of the patients. The T and B lymphocytes can be differentiated from the other types of white blood cells based on their size and by the absence of granules inside them. In chronic myelogenous leukemia (CML), the cells that are affected are the granulocytes.

Chronic lymphocytic leukemia (CLL) often has no symptoms at first and may remain undetected for a long time. Chronic myelogenous leukemia (CML), on the other hand, may progress to a more acute form.

Chronic leukemias account for 1.2% of all cancers. Because leukemia is the most common form of childhood cancer, it is often regarded as a disease of childhood. However, leukemias affect nine times as many adults as children. In chronic lymphoid leukemia, 90% of the cases are seen in people who are 50 years or older, with the average age at diagnosis being 65. The incidence of the disease increases with age. It is almost never seen in children. Chronic myeloid leukemias are generally seen in people in their mid-40s. It accounts for about 4% of childhood leukemia cases. According to the estimates of the American Cancer Society (ACS), approximately 29,000 new cases of leukemia will be diagnosed in 1998.

Causes and symptoms

Leukemia strikes both sexes and all ages. Although the cause is unknown, chronic leukemia is linked to genetic abnormalities and environmental factors. For example, exposure to ionizing radiation and to certain organic chemicals, such as benzene, is believed to increase the risks for getting leukemia. Chronic leukemia occurs in some people who are infected with two human retroviruses (HTLV-I and HTLV-II). An abnormal chromosome known as the Philadelphia chromosome is seen in 90% of those with CML. The incidence of chronic leukemia is slightly higher among men than women.

The symptoms of chronic leukemia are generally vague and non-specific. In chronic lymphoid leukemia (CLL), a patient may experience all or some of the following symptoms:

- swollen lymph nodes
- an enlarged spleen, which could make the patient complain of abdominal fullness
- chronic **fatigue**

- a general feeling of ill-health
- fever of unknown origin
- night sweats
- weight loss that is not due to dieting or **exercise**
- frequent bacterial or viral infections

In the early stages of chronic myeloid leukemia (CML), the symptoms are more or less similar to CLL. In the later stages of the disease, the patient may experience these symptoms:

- non-specific bone **pain**
- bleeding problems
- mucus membrane irritation
- frequent infections
- a pale color due to a low red blood cell count (anemia)
- swollen lymph glands
- fever
- night sweats

Diagnosis

There are no screening tests available for chronic leukemias. The detection of these diseases may occur by chance during a routine **physical examination**.

If the doctor has reason to suspect leukemia, he or she will conduct a very thorough physical examination to look for enlarged lymph nodes in the neck, underarm, and pelvic region. Swollen gums, an enlarged liver or spleen, **bruises**, or pinpoint red **rashes** all over the body are some of the signs of leukemia. Urine and blood tests may be ordered to check for microscopic amounts of blood in the urine and to obtain a complete differential **blood count**. This count will give the numbers and percentages of the different cells found in the blood. An abnormal blood test might suggest leukemia; however, the diagnosis has to be confirmed by more specific tests.

The doctor may perform a **bone marrow biopsy** to confirm the diagnosis of leukemia. During the bone marrow biopsy, a cylindrical piece of bone and marrow is removed. The tissue is generally taken out of the hipbone. These samples are sent to the laboratory for examination. In addition to diagnosis, bone marrow biopsy is also done during the treatment phase of the disease to see if the leukemia is responding to therapy.

Standard imaging tests such as x rays, **computed tomography scans** (CT scans), and **magnetic resonance imaging** (MRI) may be used to check whether the

leukemic cells have invaded other organs of the body, such as the bones, chest, kidneys, abdomen, or brain.

Treatment

The treatment depends on the specific type of chronic leukemia and its stage. In general, **chemotherapy** is the standard approach to both CLL and CML. **Radiation therapy** is occasionally used. Because leukemia cells can spread to all the organs via the blood stream and the lymph vessels, surgery is not considered an option for treating leukemias.

Bone marrow transplantation (BMT) is becoming the treatment of choice for CML because it has the possibility of curing the illness. BMT is generally not considered an option in treating CLL because CLL primarily affects older people, who are not considered to be good candidates for the procedure.

In BMT, the patient's diseased bone marrow is replaced with healthy marrow. There are two ways of doing a bone marrow transplant. In an allogeneic bone marrow transplant, healthy marrow is taken from another person (donor) whose tissue is either the same or very closely resembles the patient's tissues. The donor may be a twin, a sibling, or a person who is not related at all. First, the patient's bone marrow is destroyed with very high doses of chemotherapy and radiation therapy. To replace the destroyed marrow, healthy marrow from the donor is given to the patient through a needle in the vein.

In the second type of bone marrow transplant, called an autologous bone marrow transplant, some of the patient's own marrow is taken out and treated with a combination of **anticancer drugs** to kill all the abnormal cells. This marrow is then frozen to save it. The marrow remaining in the patient's body is then destroyed with high dose chemotherapy and radiation therapy. Following that, the patient's own marrow that was frozen is thawed and given back to the patient through a needle in the vein. This mode of bone marrow transplant is currently being investigated in clinical trials.

In chronic lymphoid leukemia (CLL), chemotherapy is generally the treatment of choice. Depending on the stage of the disease, single or multiple drugs may be given. Drugs commonly prescribed include steroids, chlorambucil, fludarabine, and cladribine. Low dose radiation therapy may be given to the whole body, or it may be used to alleviate the symptoms and discomfort due to an enlarged spleen and lymph nodes. The spleen may be removed in a procedure called a **splenectomy**.

KEY TERMS

Antibodies—Proteins made by the B lymphocytes in response to the presence of infectious agents, such as bacteria or viruses, in the body.

Biopsy—The surgical removal and microscopic examination of living tissue for diagnostic purposes.

Chemotherapy—Treatment with drugs that act against cancer.

Computerized tomography (CT) scan—A series of x rays put together by a computer in order to form detailed pictures of areas inside the body.

Cytokines—Chemicals made by the cells that act on other cells to stimulate or inhibit their function. Cytokines that stimulate growth are called “growth factors.”

Immunotherapy—Treatment of cancer by stimulating the body’s immune defense system.

Lumbar puncture—A procedure in which the doctor inserts a small needle into the spinal cavity in the lower back to withdraw some spinal fluid for testing. Also known as a “spinal tap.”

Magnetic resonance imaging (MRI)—A medical procedure using a magnet linked to a computer to picture areas inside the body.

Maturation—The process by which stem cells transform from immature cells without a specific function into a particular type of blood cell with defined functions.

Radiation therapy—Treatment using high-energy radiation from x-ray machines, cobalt, radium, or other sources.

Remission—A disappearance of a disease as a result of treatment. Complete remission means that all disease is gone. Partial remission means that the disease is significantly improved by treatment, but residual traces of the disease are still present.

In chronic myeloid leukemia (CML), the treatment of choice is bone marrow transplantation. During the slow progress (chronic phase) of the disease, chemotherapy may be given to try to improve the cell counts. Radiation therapy, which involves the use of x rays or other high-energy rays to kill cancer cells and shrink tumors, may be used in some cases to reduce the discomfort and pain due to an enlarged

spleen. For chronic leukemias, the source of radiation is usually outside the body (external radiation therapy). If the leukemic cells have spread to the brain, radiation therapy can be directed at the brain. As the disease progresses, the spleen may be removed in an attempt to try to control the pain and to improve the blood counts.

In the acute phase of CML, aggressive chemotherapy is given. Combination chemotherapy, in which multiple drugs are used, is more efficient than using a single drug for the treatment. The drugs may either be administered intravenously through a vein in the arm or by mouth in the form of pills. If the cancer cells have invaded the central nervous system (CNS), chemotherapeutic drugs may be put into the fluid that surrounds the brain through a needle in the brain or back. This is known as intrathecal chemotherapy.

Biological therapy or immunotherapy is a mode of treatment in which the body’s own immune system is harnessed to fight the cancer. Substances that are routinely made by the immune system (such as growth factors, hormones, and disease-fighting proteins) are either synthetically made in a laboratory, or their effectiveness is boosted and they are then put back into the patient’s body. This treatment mode is also being investigated in clinical trials all over the country at major cancer centers.

Prognosis

The prognosis for leukemia depends on the patient’s age and general health. According to statistics, in chronic lymphoid leukemia, the overall survival for all stages of the disease is nine years. Most of the deaths in people with CLL are due to infections or other illnesses that occur as a result of the leukemia.

In CML, if bone marrow transplantation is performed within one to three years of diagnosis, 50-60% of the patients survive three years or more. If the disease progresses to the acute phase, the prognosis is poor. Less than 20% of these patients go into remission.

Prevention

Most cancers can be prevented by changes in lifestyle or diet, which will reduce the risk factors. However, in leukemias, there are no known risk factors. Therefore, at the present time, there is no way known to prevent the leukemias from developing. People who are at an increased risk for developing leukemia because of proven exposure to ionizing radiation, the organic liquid benzene, or people who

have a history of other cancers of the lymphoid system (Hodgkin's lymphoma) should undergo periodic medical checkups.

Resources

ORGANIZATIONS

American Cancer Society. 1599 Clifton Rd., NE, Atlanta, GA 30329-4251. (800) 227-2345. <<http://www.cancer.org>>.

Cancer Research Institute. 681 Fifth Ave., New York, N.Y. 10022. (800) 992-2623. <<http://www.cancerresearch.org>>.

Leukemia Society of America, Inc. 600 Third Ave., New York, NY 10016. (800) 955 4572. <<http://www.leukemia.org>>.

National Cancer Institute. Building 31, Room 10A31, 31 Center Drive, MSC 2580, Bethesda, MD 20892-2580. (800) 422-6237. <<http://www.nci.nih.gov>>.

Oncolink. University of Pennsylvania Cancer Center. <<http://cancer.med.upenn.edu>>.

Lata Cherath, PhD

Leukocytosis

Definition

Leukocytosis is a condition characterized by an elevated number of white cells in the blood.

Description

Leukocytosis is a condition that affects all types of white blood cells. Other illnesses, such as neutrophilia, lymphocytosis, and granulocytosis, target specific types of white blood cells. Normal white blood cell counts are 4,300-10,800 white blood cells per microliter. Leukocyte or white blood cell levels are considered elevated when they are between 15,000-20,000 per microliter. The increased number of leukocytes can occur abnormally as a result of an infection, **cancer**, or drug intake; however, leukocytosis can occur normally after eating a large meal or experiencing **stress**.

Causes and symptoms

Leukemias can cause white blood cell counts to increase to as much as 100,000. Each kind of white cell can produce a leukemia. Apart from leukemias, nearly all leukocytosis is due to one type of white blood cell, the polymorphonuclear leukocyte (PMN). These conditions are more accurately referred to as neutrophilia.

KEY TERMS

Biopsy—Surgical removal of tissue for examination.

Inflammation—Heat, swelling, redness, and pain caused by tissue injury.

Ketoacidosis—A severe stage of diabetes where acids and ketones accumulate in the body.

NSAID—Non-steroidal anti-inflammatory drug such as ibuprofen.

The most common and important cause of neutrophilia is infection, and most infections cause neutrophilia. The degree of elevation often indicates the severity of the infection. Tissue damage from other causes raises the white count for similar reasons. **Burns**, infarction (cutting off the blood supply to a region of the body so that it dies), crush injuries, inflammatory diseases, poisonings, and severe diseases, like kidney failure and **diabetic ketoacidosis**, all cause neutrophilia.

Counts almost as high occur in leukemoid (leukemia-like) reactions caused by infection and non-infectious inflammation.

Drugs can also cause leukocytosis. Cortisone-like drugs (prednisone), lithium, and NSAIDs are the most common offenders.

Non-specific stresses also cause white blood cells to increase in the blood. Extensive testing of medical students reveals that neutrophilia accompanies every examination. Vigorous **exercise** and intense excitement also cause elevated white blood cell counts.

Diagnosis

A complete **blood count** (CBC) is one of the first tests obtained in any medical setting. More than 11,000 white cells in a cubic millimeter of blood is considered high. **Bone marrow biopsy** may help clarify the cause.

Treatment

Relieving the underlying cause returns the count to normal.

Prognosis

By treating the underlying condition, white blood cell counts usually return to normal.

Resources

BOOKS

Holland, Steven M., and John I. Gallin. "Disorders of Granulocytes and Monocytes." In *Harrison's Principles of Internal Medicine*, edited by Anthony S. Fauci, et al. New York: McGraw-Hill, 1997.

J. Ricker Polsdorfer, MD

Leukotriene inhibitors

Definition

Leukotriene inhibitors are prescription medications that treat **asthma** and some **allergies** by blocking the formation or activity of leukotrienes—small mediator chemicals produced by cells in the body.

Purpose

More than 50 million Americans suffer from asthma and allergies. Asthma is one of the most prevalent chronic diseases in the United States, affecting 9 million (12.7%) of children. Seasonal allergies affect 20–40 million (20%) of Americans, about 40% of them children. It is estimated that 60–70% of those with asthma also suffer from **allergic rhinitis**, allergies affecting the mucous membranes of the nose.

Asthma, an inflammation of the bronchial airways, and seasonal allergies and allergic **rhinitis** involve several chemical mediators including histamine and leukotrienes. Leukotrienes are a class of unsaturated fatty-acid chains containing 20 carbon atoms.

During an asthma attack or within minutes of exposure to an allergen such as dust or pollen, leukotrienes are released by a type of blood cell in the lungs, causing the following responses:

- contraction of the bronchial airway muscles
- inflammation of the airway linings
- swelling and narrowing of the airways
- production of mucus and fluid
- **wheezing** and shortness of breath
- nasal congestion

Leukotriene inhibitors may decrease the symptoms of mild to moderate allergen-induced asthma, improve nighttime symptoms, and reduce the number of acute asthma attacks. Taken daily on a long-term

basis they may help to prevent or control the symptoms of persistent asthma—asthma with symptoms that last at least two days per week or two nights per month. They also are prescribed for children with frequent or more severe asthma attacks and for those who dislike or have difficulty using asthma inhalers. Although leukotriene inhibitors may decrease the need for inhaled beta-agonists or **corticosteroids**, they are not used to treat asthma attacks. Leukotriene inhibitors also may be used to treat symptoms of allergic rhinitis or short-term seasonal allergies, including sneezing, runny nose, **itching**, and wheezing.

Description

Leukotriene inhibitors are often called leukotriene:

- blockers
- modifiers
- antagonists
- pathway modifiers

When they were first introduced in 1996, leukotriene inhibitors represented the first new class of asthma medication in two decades. Classified as anti-inflammatories, they were originally developed to improve lung function in asthmatics by relaxing the smooth muscles around the bronchial airways and by reducing lung inflammation.

Types of leukotriene inhibitors

The available leukotriene inhibitors are: montelukast (Singulair), zafirlukast (Accolate), and zileuton (Zyflo).

Montelukast and zafirlukast are leukotriene-receptor antagonists that prevent leukotriene from binding to cell receptors and initiating the chain of events leading to symptoms of allergy and asthma. Montelukast works rapidly. It is the only leukotriene inhibitor that has been approved by the U.S. Food and Drug Administration (FDA) for use in children as young as two, as well as for the treatment of seasonal allergies.

Zafirlukast is a synthetic peptide that inhibits the receptor binding of three leukotrienes (LTC₄, LTD₄, and LTE₄) that cause smooth muscle constriction. It is used for mild to moderate persistent asthma, exercise-induced asthma, and the management of allergic rhinitis in those aged seven and older.

Zileuton is a 5-lipoxygenase pathway inhibitor that interferes with the synthesis of LTA₄, LTC₄,

LTD4, and LTE4. It is used to treat chronic asthma in adolescents and adults.

Effectiveness

Leukotriene inhibitors may be prescribed along with **inhaled corticosteroids** for control of mild to moderate, persistent asthma. Used alone they are less effective than low-dose inhaled corticosteroids. However, they enable some people to reduce their doses of inhaled corticosteroids. Leukotriene inhibitors may be an option for people with mild asthma who want to avoid corticosteroids, which can cause serious side effects with long-term use. When used in conjunction with beta-agonists, leukotriene inhibitors reduce symptoms and may lower the beta-agonist usage.

Leukotriene inhibitors appear to decrease the symptoms of seasonal allergic rhinitis. Although they may relieve nasal congestion better than **antihistamines**, they are less effective than corticosteroid nasal sprays. A leukotriene inhibitor combined with an antihistamine may be more effective than either drug alone.

Leukotriene inhibitors have helped some children who suffer from nocturnal asthma, exercise- and aspirin-induced asthma, allergic rhinitis, and seasonal allergies.

Clinical studies

Montelukast appears to be an effective asthma controller in about one-third of patients. Another one-third receives no benefit. However, most long-term studies have found that standard inhaled corticosteroids are more effective for controlling asthma than either beta-agonists or leukotriene inhibitors.

A 2003 analysis of 13 clinical studies found that Singulair and Accolate resulted in 60% more asthma flare-ups and other symptoms as compared with traditional asthma treatments. Patients using inhaled corticosteroids had fewer daytime symptoms and night awakenings than those taking Singulair or Accolate. The researchers advised against switching to a leukotriene inhibitor unless the dosage of inhaled medication is less than 400 micrograms per day.

A 2005 study sponsored by Merck, the maker of Singulair, found that a one-year course of Singulair was useful for treating two- to five-year-olds with occasional asthma attacks that were triggered by respiratory infections. Singulair reduced this type of

asthma flare-up by 32% as compared with a control group receiving a placebo. Singulair also delayed the onset of the first asthma flare-up and reduced the need for inhaled medication. However, it did not reduce the length or severity of the flare-up once it had begun. The researchers suggested that children with infection-triggered asthma should begin taking a leukotriene inhibitor before the start of the flu season or at the onset of an upper-respiratory-tract infection.

Another 2005 study found that children whose asthma improved with montelukast alone were younger and had had asthma for a shorter period of time as compared with children whose asthma improved only with inhaled corticosteroids. Among the children whose lung function improved by at least 7.5%, 5% took montelukast alone, 23% were on inhaled corticosteroids only, and 17% were on both medications.

Other uses

Leukotriene inhibitors have been used successfully to treat inflammations of the esophagus (esophagitis) or stomach and intestines (**gastroenteritis**) that are caused by white blood cells called eosinophils that are involved in allergic reactions. Montelukast has been used to successfully treat symptoms of interstitial **cystitis**, a chronic inflammation of the bladder.

Recommended dosage

Montelukast is taken once per day in the evening so as to relieve morning allergy symptoms. Although dosing may vary, average daily doses of montelukast for asthma and seasonal allergies are: children aged 1–5: one 4-mg chewable tablet or 4-mg oral granules (one packet), swallowed whole or mixed in a spoonful of soft food; children aged 6–14: one 5-mg chewable tablet; children over 14 and adults: one 10-mg tablet.

The average doses of zafirlukast for children aged 7–11 are 10-mg tablets twice a day. Children aged 12 and older and adults usually take 20-mg tablets twice a day. Zafirlukast is taken one hour before or two hours after a meal, since food reduces its bioavailability by about 49%.

The average dose of zileuton is a 600-mg tablet four times per day for children aged 12 and older and adults.

Leukotriene inhibitors are expensive. Missed doses should be taken as soon as possible unless it is almost time for the next dose, in which case the dose should be skipped.

Precautions

Although leukotriene inhibitors are considered safe, they can raise the levels of liver enzymes. The FDA recommends **liver function tests** monthly for the first three months on medication, followed by quarterly monitoring for the next year, and continued interim testing. Zileuton is contraindicated for those with elevated liver enzymes, active **alcoholism**, or **liver disease**. Increased levels of liver enzymes may be detectable in the blood within two months of starting zileuton. Zileuton can affect liver function and, on rare occasions, can damage the liver.

It is unclear whether leukotriene inhibitors should be taken during **pregnancy**. Zafirlukast and zileuton should not be used by a woman who is breastfeeding. Both medications have been found to increase the risk of mild to moderate respiratory tract infections in patients aged 55 and older.

Medical conditions that may interfere with the use of montelukast include: allergies to **aspirin** or non-steroidal anti-inflammatories (NSAIDs); liver disease, which can increase the blood levels of the drug; and **phenylketonuria** because chewable tablets may contain aspartame.

A healthcare provider should be contacted if an increased number of short-acting bronchodilator inhalations are needed to relieve an acute asthma attack or if more than the maximum number of daily inhalations are required while using zileuton.

To be effective montelukast and zafirlukast must be taken at the same time every day. Zileuton must be taken at regularly spaced intervals every day, even if asthma symptoms appear to improve. Montelukast should be continued through an acute asthma attack in addition to rescue medication.

Side effects

Although leukotriene inhibitors generally have few side effects and those may subside as the body adjusts to the drug, headaches are common with these medications. Headaches occur in 18–19% of those taking montelukast and in 25% of those taking zileuton. Among 7 to 11 year olds on zafirlukast, 4.5% suffer from headaches, as do 12.9% of those aged 12 and over.

Other less common side effects of leukotriene inhibitors include:

- rash
- fatigue
- dizziness

- abdominal pain
- nausea and vomiting
- diarrhea

Montelukast appears to cause fewer side effects than other leukotriene inhibitors and is less likely to affect the liver. Side effects occurring in less than 4.2% of patients include:

- heartburn
- weakness
- fever
- nasal congestion
- cough
- dental pain
- rarely, pus in the urine

Rare side effects of zileuton include:

- itching
- flu-like symptoms
- upper right abdominal pain
- yellow eyes or skin (jaundice)

Interactions

Drugs that may interact with montelukast include:

- aspirin
- NSAIDs
- phenobarbital
- rifampin

Zafirlukast and zileuton can raise the blood levels of the asthma medication theophylline (Theo Dur and others) and the blood thinner warfarin (Coumarin). Theophylline levels and blood-clotting times should be monitored frequently.

Medications that may interact with zafirlukast include:

- aspirin
- blood pressure medications
- some seizure medications

Medications that may interact with zileuton include:

- the beta-blocker propranolol
- beta-agonists
- terfenadine (Seldane and others)

KEY TERMS

Allergic rhinitis—Nasal symptoms caused by an allergic reaction.

Asthma—A disease that causes the bronchial airways to narrow, swell, and produce mucus, making breathing difficult.

Beta-agonist—Beta2-agonist; beta-adrenergic agonist; a bronchodilator medication—inhaled or oral—that relaxes the muscles surrounding the airways to relieve asthma symptoms.

Corticosteroids—Inhaled medications for long-term control of asthma.

Leukotrienes—A class of small molecules produced by cells in response to allergen exposure; they contribute to allergy and asthma symptoms.

Montelukast (Singulair)—An inhibitor that prevents leukotrienes from binding to cell receptors; taken over time, montelukast can reduce or prevent symptoms of asthma and allergies.

Zafirlukast (Accolate)—An inhibitor that prevents leukotrienes from binding to cell receptors; taken over time, zafirlukast can help reduce or prevent asthma symptoms.

Zileuton (Zyflo)—A medication that interferes with the biosynthetic pathway that produces leukotrienes; used to help prevent asthma attacks.

Evidence.” *British Medical Journal* 236, no. 7390 (March 22, 2003): 621–6.

ORGANIZATIONS

American Academy of Allergy, Asthma and Immunology.
555 East Wells Street, Suite 1100, Milwaukee, WI
53202-3823. (800)822-2762. <<http://www.aaaai.org>>.
Childhood Asthma Research and Education (CARE)
Network. National Heart, Lung, and Blood Institute.
6701 Rockledge Drive, MSC 7952, Bethesda, MD
20892-7952. (301) 435-0202. <<http://www.asthma-care.net>>.

OTHER

Lehnert, Paul. “Leukotriene Modifiers for Allergic Rhinitis.” *Health Guide A-Z*. WebMD. October 8, 2003 [cited March 14, 2005]. <<http://my.webmed.com/hw/allergies/ug2390.asp>>.

———. “Leukotriene Pathway Modifiers for Asthma.” *Health Guide A-Z*. WebMD. March 5, 2004 [cited March 14, 2005]. <<http://my.webmed.com/hw/asthma/hw163553.asp>>.

Mayo Clinic Staff. “Medications and Immunotherapy for Asthma.” *Asthma Center*. MayoClinic.com. August 13, 2004 [cited March 14, 2005]. <<http://www.mayoclinic.com/invoke.cfm?objectid=58BCFCCC-BB00-4E67-A29876A9D30B65ED>>.

Margaret Alic, Ph.D.

Levodopa see **Antiparkinson drugs**

Levothyroxine see **Thyroid hormones**

LGV see **Lymphogranuloma venereum**

Resources

BOOKS

Haughney, John. *Leukotriene Receptor Antagonists: Evidence and Experience Examined*. London: Royal Society of Medicine Press, 2001.

Sampson, A. P., and S. T. Holgate. *Leukotriene Modifiers in Asthma Treatment*. London: Martin Dunitz, 1999.

PERIODICALS

“Asthma: Corticosteroids Are More Effective than Beta-Agonists or Leukotriene Antagonists.” *Health & Medicine Week* (December 20, 2004): 77.

“Asthma: Journal Publishes Research on Tailoring Asthma Treatment in Children.” *Health & Medicine Week* (February 28, 2005): 75.

Banasiak, Nancy Cantey, and Mikki Meadows-Oliver. “Leukotrienes: Their Role in the Treatment of Asthma and Seasonal Allergic Rhinitis.” *Pediatric Nursing* 31, no. 1 (January/February 2005): 35–8.

Ducharme, F. M. “Inhaled Glucocorticoids Versus Leukotriene Receptor Antagonists as Single Agent Asthma Treatment: Systematic Review of Current

Lice infestation

Definition

Lice infestations (pediculoses) are infections of the skin, hair, or genital region caused by lice living directly on the body or in hats or other garments. Lice are small wingless insect-like parasites with sucking mouthparts that feed on human blood and lay their eggs on body hairs or in clothing. The name pediculosis comes from the Latin word for louse (singular) or lice (plural). Some anthropologists believe that the different species of head and body lice developed in response to humans' invention and use of clothing about 50,000 years ago.

Description

Lice infestations are not dangerous infections by themselves. It is, however a serious public health

problem because some lice can carry organisms that cause other diseases, including **relapsing fever**, **trench fever**, and epidemic **typhus**. Although trench fever is self-limiting, the other two diseases have mortality rates of 5%–10%. Pubic lice are often associated with other **sexually transmitted diseases** (STDs) but do not spread them.

Lice infestations are frequent occurrences in areas of overcrowding or inadequate facilities for bathing and laundry. They are often associated with homelessness in the general population or with military, refugee, or prisoner camps in war-torn areas. All humans are equally susceptible to louse infestation; the elderly, however, are more vulnerable to typhus and other diseases carried by lice.

Causes and symptoms

The symptoms of lice infestations vary somewhat according to body location, although all are characterized by intense **itching**, usually with injury to the skin caused by scratching or scraping. The itching is an allergic reaction to a toxin in the saliva of the lice. Repeated bites can lead to a generalized skin eruption or inflammation.

Head lice

This type of infestation is caused by *Pediculus humanus capitis*, the head louse. Head lice can be transmitted from one person to another by the sharing of hats, combs, or hair brushes. Epidemics of head lice are common among school-age children from all class backgrounds in all parts of the United States. The head louse is about 1/16 of an inch in length. The adult form may be visible on the patient's scalp, especially around the ears; or its grayish-white nits (eggs) may be visible at the base of the hairs close to the scalp. It takes between three and 14 days for the nits to hatch. After the nits hatch, the louse must feed on blood within a day or die.

Head lice can spread from the scalp to the eyebrows, eyelashes, and beard in adults, although they are more often limited to the scalp in children. The itching may be intense, and may be followed by bacterial infection of skin that has been scratched open. Another common complication is swelling or inflammation of the neck glands. Head lice do not spread typhus or other systemic diseases.

Body lice

Infestations of body lice are caused by *Pediculus humanus corporis*, an organism that is similar in size to



This woman's eyelashes are infested with nits, or eggs, of a body louse. (Custom Medical Stock Photo. Reproduced by permission.)

head lice. Body lice, however, are rarely seen on the skin itself because they come to the skin only to feed. They should be looked for in the seams of the patient's clothing. This type of infestation is associated by wearing the same clothing for long periods of time without laundering, as may happen in wartime or in cold climates; or with poor personal hygiene. It can be spread by close personal contact or shared bedding.

Patients with body lice often have intense itching with deep scratches around the upper shoulders, flanks, or neck. The bites first appear as small red pimples but may cause a generalized skin rash. If the infestation is not treated, the patient may develop complications that include **headache**, fever, and bacterial infection with scarring. Body lice can spread systemic typhus or other infections.

Pubic lice

Pubic lice are sometimes called "crabs." This type of infestation is caused by *Phthirus pubis* and is commonly spread by intimate contact. People can also get pubic lice from using the bedding, towels, or clothes of an infected person.

Pubic lice usually appear first on pubic hair, but may spread to other parts of the body, particularly if the patient is very hairy. Pubic lice are also sometimes seen on the eyelashes of children born to infected mothers. It is usually easier for the doctor to see marks from the patient's scratching than the bites from the lice, but pubic lice sometimes produce small bluish spots called maculae ceruleae on the patient's trunk or thighs. Pubic lice also sometimes leave small dark brown specks from their own excreted matter on the parts of the patient's underwear that cover the anal or genital areas.

Diagnosis

Doctors can diagnose lice infestations from looking closely at the parts of the body where the patient has been scratching. Lice are large enough to be easily seen with the naked eye or a magnifying glass. The eggs of pubic lice as well as head lice can often be found by looking at the base of the patient's hairs. Pediatricians are most likely to diagnose lice in school-age children.

It is important for doctors to rule out other diseases that can cause scratching and skin inflammation because the medications used to kill lice are very strong and can have bothersome side effects. The doctor will need to distinguish between head lice and dandruff; between body lice and **scabies** (a disease caused by skin mites); and between pubic lice and eczema. Blood tests or other laboratory tests are not useful in diagnosing lice infestations.

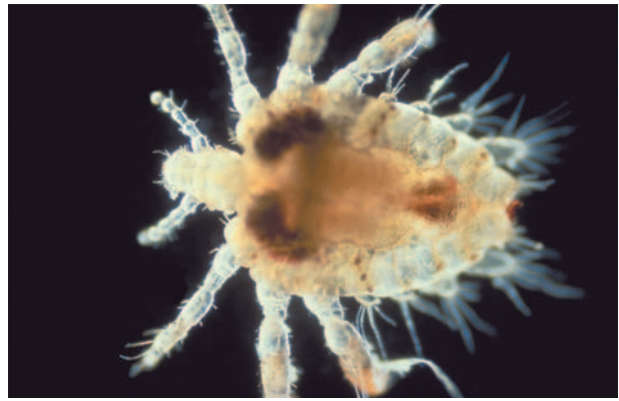
Treatment

Cases of head lice are usually treated with shampoos or rinses containing either lindane (Kwell) or permethrin (Nix). Permethrin is considered preferable, however, because lindane is absorbed through the skin and may produce symptoms of neurotoxicity. The person applying the treatment should wear rubber gloves and rinse the patient's hair or body completely after use. Following the treatment, nits should be removed from the hair with a fine-toothed comb or tweezers. Lindane is also effective for treating infestations of body or pubic lice, but it should not be used by pregnant women. In most cases one treatment is sufficient, but the medication can be reapplied a week later if living lice have reappeared.

Another drug that appears to be effective in treating lice is ivermectin (Stromectol), a strong antiparasite drug that is usually given to treat intestinal worms. Two doses of the drug, however, cured 99% of cases of head lice as well as intestinal worms in a poor population with high rates of infestation with both types of parasites.

Infestations of body lice can also be treated by washing the patient's clothes or bedding in boiling water, ironing seams with an iron on a high setting, or treating the clothes with 1% malathion powder or 10% DDT powder.

If the patient's eyelashes have been infested, the only safe treatments are either a thick coating of petroleum jelly (Vaseline) applied twice daily for eight days, or 1% yellow oxide of mercury applied four times a day for two weeks. Any remaining nits should be removed with tweezers.



A close-up view of a body louse. (Custom Medical Stock Photo. Reproduced by permission.)

Patients with pubic lice should be examined and tested for other STDs.

Alternative treatment

For pubic lice, some practitioners of **holistic medicine** recommend a mixture of 25% oil of pennyroyal (*Mentha pulegium*), 25% garlic (*Allium sativum*) oil, and 50% distilled water applied three times in a three-day period, followed by removal of dormant eggs to prevent reinfestation.

Another alternative treatment for head lice is tea tree oil, sometimes called melaleuca oil. Tea tree oil appears to work well in treating head lice that have developed resistance to other preparations.

Prognosis

Lice can be successfully eradicated in almost all cases, although a growing number of cases of drug-resistant lice have been reported. As of 2004, some researchers are concerned about the rapid but unpredictable spread of these resistant lice. Ovide, a newer medication containing malathion, appears to be effective in treating patients with permethrin-resistant lice.

In general, patients are more at risk from typhus, trench fever, rickettsial infections, and other diseases spread by lice than from the lice themselves.

Prevention

There are no vaccines or skin treatments that will protect a person against lice prior to contact. In addition, lice infestation does not provide immunity against reinfection; recurrences are in fact quite common.

KEY TERMS

Crabs—An informal or slang term for pubic lice.

Lindane—A benzene compound that is used to kill body and pubic lice. Lindane works by being absorbed into the louse's central nervous system, causing seizures and death.

Maculae ceruleae—Bluish or blue-grey skin eruptions often seen on the trunk or thighs of patients with pubic lice. The Latin words mean blue spots.

Malathion—An insecticide that can be used in 1% powdered form to disinfect the clothes of patients with body lice.

Nits—The eggs produced by head or pubic lice, usually grayish-white in color and visible at the base of hair shafts.

Pediculosis (plural, pediculoses)—The medical term for infestation with lice.

Permethrin—A medication used to rid the scalp of head lice. Permethrin works by paralyzing the lice, so that they cannot feed after hatching within the 24 hours required for survival.

Prevention depends on adequate personal hygiene at the individual level and the following public health measures:

- Teaching school-age children the basics of good personal hygiene, including the importance of not lending or borrowing combs, brushes, or hats.
- Notifying and treating an adult patient's close personal and sexual contacts.
- Examining homeless people, elderly patients incapable of self-care, and other high-risk individuals prior to hospital admission for signs of louse infestation. This measure is necessary to protect other hospitalized people from the spread of lice.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD., editors. "Pediculosis." In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

PERIODICALS

Downs, A. M. "Managing Head Lice in an Era of Increasing Resistance to Insecticides." *American Journal of Clinical Dermatology* 5 (March 2004): 169–177.

Foucault, C., D. Raoult, and P. Brouqui. "Randomized Open Trial of Gentamicin and Doxycycline for

Eradication of *Bartonella quintana* from Blood in Patients with Chronic Bacteremia." *Antimicrobial Agents and Chemotherapy* 47 (July 2003): 2204–2207.

Heukelbach, J., and H. Feldmeier. "Ectoparasites—The Underestimated Realm." *Lancet* 363 (March 13, 2004): 889–891.

Heukelbach, J., T. Wilcke, B. Winter, et al. "Efficacy of Ivermectin in a Patient Population Concomitantly Infected with Intestinal Helminths and Ectoparasites." *Arzneimittelforschung* 54 (2004): 416–421.

Hunter, J. A., and S. C. Barker. "Susceptibility of Head Lice (*Pediculus humanus capitis*) to Pediculicides in Australia." *Parasitology Research* 90 (August 2003): 476–478.

Kittler, R., M. Kayser, and M. Stoneking. "Molecular Evolution of *Pediculus humanus* and the Origin of Clothing." *Current Biology* 13 (August 19, 2003): 1414–1417.

Mills, C., B. J. Cleary, J. F. Gilmer, and J. J. Walsh. "Inhibition of Acetylcholinesterase by Tea Tree Oil." *Journal of Pharmacy and Pharmacology* 56 (March 2004): 375–379.

Yoon, K. S., J. R. Gao, S. H. Lee, et al. "Permethrin-Resistant Human Head Lice, *Pediculus capitis*, and Their Treatment." *Archives of Dermatology* 139 (August 2003): 1061–1064.

ORGANIZATIONS

American Academy of Dermatology (AAD). 930 East Woodfield Road, Schaumburg, IL 60173. (847) 330-0230. <<http://www.aad.org>>.

Rebecca J. Frey, PhD

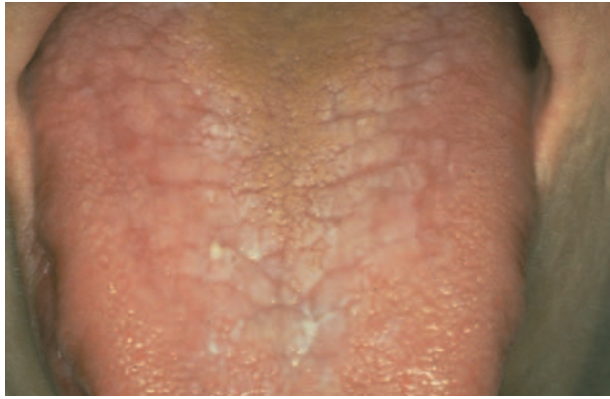
Lichen planus

Definition

Lichen planus is a skin condition of unknown origin that produces small, shiny, flat-topped, itchy pink or purple raised spots on the wrists, forearms or lower legs, especially in middle-aged patients.

Description

Lichen planus affects between 1-2% of the population, most of whom are middle-aged women. The condition is less common in the very young and the very old. The lesions are found on the skin, genitals, and in the mouth. Most cases resolve spontaneously within two years. Lichen planus is found throughout the world and is equally distributed among races.



One example of lichen planus on the tongue. (Custom Medical Stock Photo. Reproduced by permission.)

Causes and symptoms

No one knows what causes lichen planus, although some experts suspect that it is an abnormal immune reaction following a viral infection, probably aggravated by **stress**. The condition is similar to symptoms caused by exposure to arsenic, bismuth, gold, or developers used in color photography. Occasionally, lichen planus in the mouth appears to be an allergic reaction to medications, filling material, dental hygiene products, chewing gum or candy.

Symptoms can appear suddenly, or they may gradually develop, usually on the arms or legs. The lesions on the skin may be preceded by a dryness and metallic taste or burning in the mouth.

Once the lesions appear, they change over time into flat, glistening, purple lesions marked with white lines or spots. Mild to severe **itching** is common. White, lacy lesions are usually painless, but eroded lesions often burn and can be painful. As the lesions clear up, they usually leave a brown discoloration behind, especially in dark skinned people.

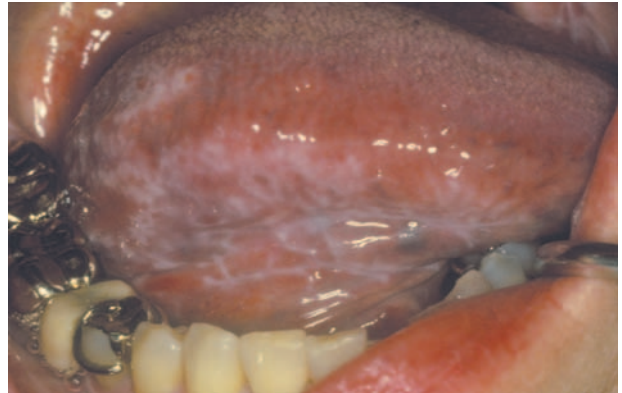
Lichen planus in the mouth occurs in six different forms with a variety of symptoms, appearing as lacy-white streaks, white plaques, or eroded ulcers. Often the gums are affected, so that the surface of the gum peels off, leaving the gums red and raw.

Diagnosis

A doctor can probably diagnose the condition simply from looking at the characteristic lesions, but a **skin biopsy** may be needed to confirm the diagnosis.

KEY TERMS

PUVA—A type of phototherapy that combines the oral or topical photosensitizing chemical psoralen, plus long-wave ultraviolet light-A (UVA).



Lichen planus appearing under the tongue. (Custom Medical Stock Photo. Reproduced by permission.)

Treatment

Treatment is aimed at easing symptoms. Itching can be treated with steroid creams and oral **antihistamines**. Severe lesions can be treated with **corticosteroids** by mouth, or combinations of photochemotherapy (PUVA) and griseofulvin.

Patients with lesions in the mouth may find that regular professional cleaning of the teeth and conscientious dental care improve the condition. Using milder toothpastes instead of tartar control products also seems to lessen the number of ulcers and makes them less sensitive.

Prognosis

While lichen planus can be annoying, it is usually fairly benign and clears up on its own. It may take months to reach its peak, but it usually clears up within 18 months.

Resources

OTHER

Lichen Planus Self-Help. Baylor College of Dentistry.
< <http://www.tamcd.edu/lichen> >.

Carol A. Turkington

Lichen simplex chronicus

Definition

Lichen simplex chronicus is a chronic inflammation of the skin (**dermatitis**) characterized by small, round itchy spots that thicken and become leathery as a result of scratching.

Description

Also termed neurodermatitis, lichen simplex chronicus is the result of chronic skin irritation. It occurs in 4-5 out of every thousand people. Initial irritation causes **itching**, and in turn, itching causes scratching. Scratching leads to further irritation, which damages the skin. The possibility of infection is greatly increased when the outer layer of protective skin is broken. Skin usually repairs itself quickly; however, in the case of lichen simplex chronicus, healing skin causes more itching and more scratching causes a thickening of the skin (lichen). The small skin patches are usually 1–10 in (2.54–25.4 cm) in diameter.

Causes and symptoms

Lichen simplex chronicus is often caused by constant rubbing of the skin. The rubbing begins the chain of events that leads from itching to scratching and then to the presence of leather-like skin patches.

Symptoms are chronic itching which is often accompanied by nervous tension. The appearance of scratch marks and the leathery skin patches can be found anywhere on the body. A prolonged lichen simplex chronicus can result in brown-colored pigmentation at the site of irritation.

Diagnosis

A dermatologist, a physician specializing in the study and treatment of skin disorders, can make a diagnosis after a visual exam.

Treatment

Treatment of the itching is necessary to stop the scratching and resulting skin damage. There are a number of ways to stop itching. Perhaps the most important is to cut fingernails very short. Ice can substitute for the relief of scratching. Heat and fuzzy clothing worsen itching; cold and smooth clothing pacify it. If the itching is persistent, dressings may be applied to the affected areas.

KEY TERMS

Antihistamine—A chemical that interferes with the action of histamine. Histamine is part of an inflammatory response and helps to cause itching.

Callus—Thickened skin due to chronic rubbing or irritation.

Lesion—Abnormal change in tissue caused by localized disease.

Among the topical medications that relieve itching are a number of commercial preparations containing menthol, camphor, eucalyptus oil, and aloe. Topical cortisone is also available without a prescription. Some preparations also contain **antihistamines**, which penetrate intact skin poorly. All these medicines work better under occlusion, which means putting a waterproof barrier like a rubber glove or plastic wrap over them. For broken skin, **topical antibiotics** like bacitracin help prevent infection. These should be used early to forestall further damage to the skin.

Reducing the buildup of thick skin may require medicines that dissolve or melt keratin, the major chemical in skin's outer layer. These keratolytics include urea, lactic acid, and salicylic acid.

Resistant cases of lichen simplex chronicus will often respond to cortisone-like drugs injected directly into the lesions.

Sedatives or tranquilizers may be prescribed to combat the nervous tension and **anxiety** that often accompanies the condition.

Prognosis

Diligent adherence to treatment is usually rewarded with a resolution of the condition. The original cause of itching may be gone, or it may reappear. Preventive treatment in its early stages will arrest the process.

Prevention

Early, gentler substitutes for scratching can entirely prevent lichen simplex chronicus.

Resources

BOOKS

Swerlick, Robert A., and Thomas J. Lawley. "Eczema, Psoriasis, Cutaneous Infections, Acne, and Other

Common Skin Disorders.” In *Harrison’s Principles of Internal Medicine*, edited by Anthony S. Fauci, et al. New York: McGraw-Hill, 1997.

J. Ricker Polsdorfer, MD

Life support

Definition

Life support refers to a spectrum of techniques used to maintain life after the failure of one or more vital organs.

Purpose

A patient requires life support when one or more vital organs fail, due to causes such as trauma, infection, **cancer**, **heart attack**, or chronic disease. Among the purposes of life support are to:

- Establish and maintain the ABC’s of resuscitation—airway, breathing, and circulation.
- Restore the patient’s homeostasis—the internal chemical and physical balance of the body.
- Protect the patient from complications of the underlying disease and its treatment.

Precautions

Patients and families need to recognize that life support is an extremely painful, expensive, and emotionally wrenching experience. Life support exposes a patient to vast risks of further medical complications, and offers no guarantee of a positive outcome. Even in successful cases, recovery may be slow and frustrating.

Description

Successful life support begins with establishing the ABC’s of resuscitation—airway, breathing, and circulation.

The airway refers to a clear passageway for air to enter the lungs from outside the body. The patient’s airway may become blocked by:

- Foreign body obstruction, as by food or dentures
- Injury-related damage and swelling, as from a wound or surgery
- Loss of protective reflexes due to **coma** of any origin

Life support may begin with basic **cardiopulmonary resuscitation (CPR)**, as in cases of cardiac arrest. Thereafter, the most common technique used to create a secure airway is insertion of an endotracheal (ET) tube through the mouth or nose into the windpipe (trachea). An alternative method of securing an airway is by **tracheotomy**, a surgical procedure in which a tube is inserted into the trachea through an incision made in the base of the throat. Of the two options, placement of an ET tube is usually quicker and more convenient, and thus occurs much more commonly. Doctors perform a tracheotomy when they cannot establish an ET airway, or when the patient will require an artificial airway for more than a week or two.

Breathing refers to the movement of air in and out of the lungs. Inadequate breathing may result from:

- Heart disease, as in congestive heart failure
- Primary disease of the lungs, such as **pneumonia**, **asthma**, or emphysema
- Coma of any cause, such as narcotic overdose or stroke
- Muscle **fatigue** or neuromuscular disease (**spinal cord injury** or polio)
- Pain, from rib **fractures** or surgery on the chest

When the patient cannot breathe sufficiently, the physician will use a ventilator, a machine that pumps air in and out of the patient’s lungs. For many doctors and members of the public, the term “life support” calls up the image of an ET tube and ventilator.

Circulation refers to the flow of blood around the body from the heart to vital organs. Circulation can fail due to:

- Primary disease of the heart (heart attack)
- Blood loss (trauma or internal bleeding of any cause)
- Severe infection (sepsis)
- Drug reactions or overdoses
- Extreme allergic reaction
- Severe **dehydration** (**gastroenteritis** or heat-related illness)

In order to ensure adequate circulation, the patient will require one or more intravenous (IV) tubes (catheters). The IVs may include both the short needle and tube commonly used in the hand or forearm, and longer catheters inserted into the larger and more central veins of the body. Catheters inserted into these larger veins are known as central lines. Through the IVs the patient receives fluids, drugs, and blood transfusions as needed to support the circulation.

Once the ABC's are secure, life support is directed at maintaining homeostasis, the body's delicate chemical and physical balance. In a healthy person, the body keeps precise control over many components of its makeup, such as its fluids, nutrients, and pressures. When vital organs fail, the body can no longer regulate these components, and the doctor must take steps to restore the normal state.

Preserving the body's internal equilibrium requires careful monitoring of innumerable indicators of the patient's well-being. These indicators include:

- Vital signs (heartbeats per minute, breaths per minute, blood pressure, body temperature, and weight)
- Fluids (input and output of the body)
- Blood cell counts
- Chemical substances of the body (sodium, potassium, sugar, and many others)
- Pressures in the circulation, lungs, and perhaps even the brain
- Presence of germs (bacteria, fungi) causing infection in body systems (lungs, blood, urine)

This intensive monitoring usually takes place in an intensive care unit (ICU) or critical care unit (CCU) and requires:

- Specialized physicians, such as cardiologists, intensivists, and surgeons
- Highly-skilled nursing care, often one nurse per patient around-the-clock
- Extensive support staff, such as respiratory therapists, laboratory technicians, radiology technicians, dieticians, and pharmacists
- Constant measurement of basics such as pulse, heart rhythm, and oxygen level in the blood
- Frequent inspection of the patient's alertness, color, and level of pain
- Use of catheters in the veins and arteries to withdraw blood samples and measure pressures in the circulation
- Use of tubes in the bladder (Foley catheter), stomach (nasogastric tube), and other body cavities
- Frequent laboratory tests on blood, urine, drainage from **wounds**, and other body specimens
- X-ray, ultrasound, computerized tomography (CT), and other imaging procedures
- Electrocardiograms

The treatments of life support include:

- Oxygen
- Intravenous fluids with sugar and basic salts

- Drugs to improve circulation and other body functions
- Antibiotics
- Transfusions
- Surgery
- Nutritional supplements by vein or stomach tube
- Tubes in body cavities (chest or abdomen) to relieve fluid buildup
- Dialysis
- Pacemaker
- Electrical defibrillation
- Various machines to assist heart or lung function
- Transplantation of organs or mechanical substitutes (artificial heart)
- Sedation or even temporary **paralysis** to enable the patient to tolerate these procedures

Preparation

The need for life support may arise suddenly and with little warning. All people should discuss in advance with family and doctor their wishes for the use of life support should a medical crisis develop. The doctor will note the preferences in the patient's record. Patients should sign documents such as an Advance Directive and Durable Power of Attorney for Health Care to express their wishes and designate a surrogate decision-maker in case of incapacitation.

Physicians and medical care providers must anticipate the possibility that a patient will require life support, perhaps suddenly. In preparation, doctors and medical staff must:

- Receive training in resuscitation skills
- Monitor patients carefully
- Maintain proper supplies and equipment
- Discuss in advance with patients and patients' families whether or not to begin life support

Aftercare

If a patient survives life support treatments, doctors will cautiously try to wean the patient from the support systems. Being able to breathe adequately without the ventilator is one major hurdle. Patients commonly fail in their first attempts to breathe on their own, often tiring out after a few hours. Thus, the doctor will reconnect the ventilator, give the patient a rest, and try again in a day or two.

As the patient regains organ function, there is less need for monitors, tests, and treatments that require an intensive care setting. The doctor may transfer the patient to a lower level of hospital care, a skilled nursing facility (SNF), or perhaps directly to home. Physical and occupational therapists may help the patient improve strength and endurance. The patient will receive continuing care from the primary doctor and specialists as needed. The patient may require prescription drugs, assist devices, and psychological therapists.

Risks

The risks and consequences of life support are enormous. These risks include:

- Physical dangers
- Emotional suffering
- Financial costs
- Societal discord

The physical dangers of life support encompass all the hazards of the patient's underlying disease and treatments. Among these risks are:

- Permanent damage to the brain, kidneys, and other vital organs caused by poor circulation or low oxygen content of the blood
- Direct damage to organs from use of medical instruments and procedures
- Infections, often with organisms that are highly resistant to antibiotics
- Abnormal blood clots
- Skin ulcers from lying immobilized for long periods
- Extreme pain
- Exposure of medical personnel to communicable diseases

The emotional consequences of life support touch patients, families, and medical caregivers. These repercussions arise from:

- The frightening environment of an ICU
- The need to make life-and-death decisions
- The anger, guilt, and grief that relate to life-threatening illness
- The fact that many lengthy and difficult treatments will end in failure

The financial costs of life support are huge. A single day of life support costs many thousands of dollars. These expenses fall on individual payers, insurance companies, health plans, and governments.

KEY TERMS

Cardiopulmonary—Relating to the heart and lungs.

Central line—A tube placed by needle into a large, central vein of the body.

Coma—Unconsciousness.

Defibrillation—Use of an electric shock to restore a normal heartbeat.

Endotracheal tube—A tube placed into the windpipe through the nose or mouth.

Foley catheter—A tube that drains urine from the bladder.

Homeostasis—The internal chemical and physical balance of the body.

Nasogastric tube—A tube placed through the nose into the stomach.

Neuromuscular—Relating to nerves and muscles.

Resuscitation—Treatments to restore an adequate airway, breathing, and circulation.

Sepsis—An overwhelming infection with effects throughout the body.

Tracheotomy—A surgical procedure in which a tube is inserted into the trachea through an incision made in the base of the throat.

Trauma—Serious physical injury.

Ventilator—A machine that pumps air in and out of the lungs.

Vital signs—Basic indicators of body function, usually meaning heartbeats per minute, breaths per minute, blood pressure, body temperature, and weight.

All such payers face difficult decisions regarding the allotment of money for such treatment, especially in cases that are likely to be futile.

Society as a whole faces difficult decisions surrounding life support. Some governments have enacted regulations that establish priorities for the spending of health care resources. Patients who do not receive treatment under such rules may feel victimized by society's choices.

Resources

BOOKS

Irwin, Richard S., Frank B. Cerra, and James M. Rippe, editors. *Irwin and Rippe's Intensive Care Medicine*. Philadelphia: Lippincott-Raven, 1999.

Luce, John M., "Approach to the Patient In a Critical Care Setting." In *Textbook of Medicine*, edited by Lee Goldman and J. Claude Bennett, 21st ed. Vol. 1. Philadelphia: W.B. Saunders Company, 2000, pp. 483-4.

Tintinalli, Judith E., et al, editors. *Emergency Medicine: a comprehensive study guide*. New York: McGraw-Hill, 2000.

Isaac R. Berniker

Light sensitivity see **Photosensitivity**

Light therapy see **Phototherapy**

Light treatment see **Ultraviolet light treatment**

Lipase test

Definition

The lipase test is a blood test performed to determine the serum level of a specific protein (enzyme) involved in digestion. Lipase is an enzyme produced by the pancreas, which is a large gland situated near the stomach. Lipase works to break down a certain type of blood lipid (triglycerides) into fatty acids.

Lipase appears in the blood together with another enzyme called amylase following damage to or diseases affecting the pancreas. It was once thought that abnormally high lipase levels were associated only with diseases of the pancreas. Other conditions are now known to be associated with high lipase levels, especially kidney failure and intestinal obstruction. Diseases involving the pancreas, however, produce much higher lipase levels than diseases of other organs. Lipase levels in pancreatic disorders are often 5-10 times higher than normal.

Purpose

The lipase test is most often used in evaluating inflammation of the pancreas (**pancreatitis**), but it is also useful in diagnosing kidney failure, intestinal obstruction, **mumps**, and peptic ulcers. Doctors often order amylase and lipase tests at the same time to help distinguish pancreatitis from ulcers and other disorders in the abdomen. If the patient has acute (sudden onset) pancreatitis, the lipase level usually rises somewhat later than the amylase level—about 24-48 hours after onset of symptoms—and remains abnormally high for 5-7 days. Because the lipase level

KEY TERMS

Amylase—A digestive enzyme that breaks down starch.

Lipid—A greasy organic compound that cannot be dissolved in water. Triglycerides, which are broken down by lipase, are one type of blood lipid.

Pancreas—An elongated gland situated across the back of the abdomen behind the stomach. It secretes both digestive enzymes and hormones. Pancreatic hormones regulate the level of sugar in the blood.

Pancreatitis—Inflammation of the pancreas, frequently caused by gallstones, alcohol abuse, viral infection, or injury.

Turbidimetry—A technique of measurement that analyzes the amount of sediment in a liquid.

peaks later and remains elevated longer, its determination is more useful in late diagnosis of acute pancreatitis. Conversely, however, lipase levels are not as useful in diagnosing chronic pancreatic disease.

Precautions

Patients should be asked whether they are taking certain prescription drugs that can affect the accuracy of the lipase test. Drugs that can cause elevated lipase levels include bethanechol, cholinergics, codeine, indomethacin, meperidine, methacholine, and morphine. Drugs that may decrease levels include calcium ions.

Description

A lipase test is performed on a sample of the patient's blood, withdrawn from a vein into a vacuum tube. The procedure, which is called a venipuncture, takes about five minutes.

Preparation

The patient should have nothing to eat or drink for 12 hours before the lipase test.

Risks

Risks for this test are minimal, but may include slight bleeding from the puncture site, a small bruise or swelling in the area, **fainting**, or feeling lightheaded.

Normal results

Reference values for lipase determination are laboratory- and method-specific. In general, normal results are usually less than 200 units/L (triolein methods by titration or turbidimetry).

Abnormal results

Increased lipase levels are found in acute pancreatitis, chronic relapsing pancreatitis, and pancreatic **cancer**. High lipase levels also occur in certain liver diseases, kidney failure, bowel obstruction, peptic ulcer disease, and tumors or inflammation of the salivary glands.

Resources

BOOKS

Pagana, Kathleen Deska. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis: Mosby, Inc., 1998.

Janis O. Flores

Lipidoses

Definition

Lipidoses are heredity disorders, passed from parents to their children, characterized by defects of the digestive system that impair the way the body uses fat from the diet. When the body is unable to properly digest fats, lipids accumulate in body tissues in abnormal amounts.

Description

The digestion, storage, and use of fats from foods is a complex process that involves hundreds of chemical reactions in the body. In most people, the body is already programmed by its genetic code to produce all of the enzymes and chemicals necessary to carry out these functions. These genetic instructions are passed from parents to their offspring during reproduction.

People with lipidoses are born without the genetic codes needed to tell their bodies how to complete a particular part of the fat digestion process. In most of these disorders, the body does not produce a certain enzyme or chemical. Over 30 different disorders of fat metabolism are related to genetic defects. Although the defects are passed from parents

to children, the parents often do not have the disorders themselves.

The symptoms, available treatments, and long-term consequences of these conditions vary greatly. Some of the conditions become apparent shortly after the infant is born; in others, symptoms may not develop until adulthood. For most of the lipidoses, diagnosis is suspected based on the symptoms and family history. Blood tests, urine tests, and tissue tests can be used to confirm the diagnosis. **Genetic testing** can be used, in some cases, to identify the defective gene. Some of these disorders can be controlled with changes in the diet, medications, or enzyme supplements. For many, no treatment is available. Some may cause **death** in childhood or contribute to a shortened life expectancy. Some of the most common or most serious lipidoses are discussed below.

Fabry's disease

Causes and symptoms

Approximately 1 in every 40,000 males is born with Fabry's disease. This condition has an X-linked, recessive pattern of inheritance, meaning that the defective gene is carried on the X chromosome. A female who carries a defective recessive gene on one of her two X chromosomes has a 50% chance of passing the defective gene to her sons who will develop the disorder associated with the defective gene (a male receives one X chromosome from his mother and one Y chromosome from his father). She also has a 50% chance of passing the defective recessive gene to her daughters who will be carriers of the disorder (like their mother). Some female carriers of Fabry's disease show mild signs of the disorder, especially cloudiness of the cornea.

The gene that is defective in Fabry's disease causes a deficiency of the enzyme alpha-galactosidase A. Without this enzyme, fatty compounds start to line the blood vessels. The collection of fatty deposits eventually affects blood vessels in the skin, heart, kidneys, and nervous system. The first symptoms in childhood are **pain** and discomfort in the hands and feet brought on by **exercise**, **fever**, **stress**, or changes in the weather. A raised rash of dark red-purple spots is common, especially on skin between the waistline and the knees. Other symptoms include a decreased ability to sweat and changes in the cornea or outer layer of the eye. Although the disease begins in childhood, it progresses very slowly. Kidney and heart problems develop in adulthood.

Diagnosis

The diagnosis can be confirmed by a blood test to measure for alpha-galactosidase A. Women who are carriers of the defective gene can also be identified by a blood test.

Treatment

Treatment focuses on prevention of symptoms and long-term complications. Daily doses of diphenylhydantoin (Dilantin) or carbamazepine (Tegretol) can prevent or reduce the severity of pain in the hands and feet associated with the condition. A low sodium, low protein diet may be beneficial to those patients who have some kidney complications. If kidney problems progress, **kidney dialysis** or **kidney transplantation** may be required. Enzyme replacement therapy is currently being explored.

Prognosis

Although patients with Fabry's disease usually survive to adulthood, they are at increased risk for **stroke**, heart attacks, and kidney damage.

Gaucher disease

Causes and symptoms

Gaucher (pronounced go-shay) disease is the most common of the lipid storage disorders. It is found in populations all over the world (20,000 to 40,000 people have a type of the disease), and it occurs with equal frequency in males and females. **Gaucher disease** has a recessive pattern of inheritance, meaning that a person must inherit a copy of the defective gene from both parents in order to have the disease. The genetic defect causes a deficiency of the enzyme glucocerebrosidase that is responsible for breaking down a certain type of fat and releasing it from fat cells. These fat cells begin to crowd out healthy cells in the liver, spleen, bones, and nervous system. Symptoms of Gaucher disease can start in infancy, childhood, or adulthood.

Three types of Gaucher disease have been identified, but there are many variations in how symptoms develop. Type 1 is the most common and affects both children and adults. It occurs much more often in people of Eastern European and Russian Jewish (Ashkenazi) ancestry, affecting 1 out of every 450 live births. The first signs of the disease include an enlarged liver and spleen, causing the abdomen to swell. Children with this condition may be shorter than normal. Other symptoms include tiredness, pain, bone deterioration, broken bones, anemia, and

increased bruising. Type 2 Gaucher disease is more serious, beginning within the first few months after birth. Symptoms, which are similar to those in Type 1, progress rapidly, but also include nervous system damage. Symptoms of Type 3 Gaucher disease begin during early childhood with symptoms like Type 1. Unlike Type 2, the progress of the disease is slower, although it also includes nervous system damage.

Diagnosis

Gaucher disease may be suspected based on symptoms and is confirmed with a blood test for levels of the enzyme. Samples of tissue from an affected area may also be used to confirm a diagnosis of the disease.

Treatment

The symptoms of Gaucher disease can be stopped and even reversed by treatment with injections of enzyme replacements. Two enzyme drugs currently available are alglucerase (Ceredase) and imiglucerase (Cerezyme). Other treatments address specific symptoms such as anemia, broken bones, or pain.

Prognosis

The pain and deformities associated with symptoms can make coping with this illness very challenging for individuals and families. With treatment and control of symptoms, people with Type 1 Gaucher disease may lead fairly long and normal lives. Most infants with Type 2 die before the age of 2. Children with Type 3 Gaucher disease may survive to adolescence and early adulthood.

Krabbe's disease

Causes and symptoms

Krabbe's disease is caused by a deficiency of the enzyme galactoside beta-galactosidase. It has a recessive pattern of inheritance and is believed to occur in 1 of 40,000 births in the United States. This condition, which is also called globoid cell leukodystrophy or Krabbe leukodystrophy, is characterized by acute nervous system degeneration. It develops in early infancy with initial symptoms of irritability, **vomiting** and episodes of partial unconsciousness. Symptoms progress rapidly to seizures, difficulty swallowing, blindness, deafness, **mental retardation**, and **paralysis**.

Treatment

No treatment is available.

Prognosis

Children born with Krabbe's disease die in infancy.

Niemann-pick disease

Causes and symptoms

At least five different forms of Niemann-Pick disease (NPD) have been identified. The different types seem to be related to the activity level of the enzyme sphingomyelinase. In patients with Types A and B NPD, there is a build up of sphingomyelin in cells of the brain, liver, spleen, kidney and lung. Type A is the most common form of NPD and the most serious, with death usually occurring by the age of 18 months. Symptoms develop within the first few months of life and include poor appetite, failure to grow, enlarged liver and spleen, and the appearance of cherry red spots in the retina of the eye. Type B develops in infancy or childhood with symptoms of mild liver or spleen enlargement and lung problems. Some adults with this form (Type E) may also show a loss of muscle coordination. Types C or D NPD are related to cholesterol transfer out of cells. Children with Types C or D grow normally in early childhood, but eventually develop difficulty in walking and loss of muscle coordination. Ultimately, the nervous system becomes severely damaged and these patients die. Type C occurs in any population, while Type D has been identified only in patients from Nova Scotia, Canada.

Diagnosis

Diagnosis is confirmed by analyzing a sample of tissue. Prenatal diagnosis of Types A and B of NPD can be done with **amniocentesis** or **chorionic villus sampling**.

Treatment

Treatment consists of supportive care to deal with symptoms and the development of complications. **Bone marrow transplantation** is being investigated as a possible treatment. Low-cholesterol **diets** may be helpful for patients with Types C and D.

Prognosis

Patients with Type A NPD usually die within the first year and a half of life. Type B patients generally live to adulthood but suffer from significant liver and lung problems. With Types C and D NPD, there is significant nervous system damage leading to severe **muscle spasms**, seizures, and eventually, to **coma** and death. Some patients with Types C and D die in childhood, while less severely affected patients may survive to adulthood.

Refsum's disease

Causes and symptoms

Refsum's disease has a recessive pattern of inheritance and affects populations from Northern Europe, particularly Scandinavians most frequently. It is due to a deficiency of phytanic acid hydroxylase, an enzyme that breaks down a fatty acid called phytanic acid. This condition affects the nervous system, eyes, bones, and skin. Symptoms, which usually appear by age 20, include vision problems [retinitis pigmentosa and rhythmic eye movements (nystagmus)], loss of muscle coordination, loss of sense of smell (**anosmia**), pain, **numbness**, and elevated protein in the cerebrospinal fluid.

Treatment

A diet free of phytanic acid (found in dairy products, tuna, cod, haddock, lamb, stewed beef, white bread, white rice, boiled potatoes, and egg yolk) can reduce some of the symptoms. **Plasmapheresis**, a process where whole blood is removed from the body, processed through a filtering system, and then return to the body, may be used to filter phytanic acid from the blood.

Tay-sachs disease

Causes and symptoms

Tay-Sachs disease (TSD) is a fatal condition caused by a deficiency of the enzyme hexosaminidase A (Hex-A). The defective gene that causes this disorder is found in roughly 1 in 250 people in the general population. However, certain populations have significantly higher rates of TSD. French-Canadians living near the St. Lawrence River and in the Cajun regions of Louisiana are at higher risk of having a child with TSD. The highest risk seems to be in people of Eastern European and Russian Jewish (Ashkenazi) descent. Tay-Sachs disease has a recessive pattern of inheritance, and approximately 1 in every 27 people of Jewish ancestry in the United States carries the TSD gene. Symptoms develop in infancy and are due to the accumulation of a fatty acid compound in the nervous system. Early symptoms include loss of vision and physical coordination, seizures, and mental retardation. Eventually, the child develops problems with breathing and swallowing. Blindness, paralysis, and death follow.

Diagnosis

Carriers of the Tay-Sachs related gene can be identified with a blood test. Amniocentesis or chorionic villi sampling can be used to determine if the fetus has Tay-Sachs disease.